

Shu Hotta

Mathematical Physical Chemistry

Practical and Intuitive Methodology

Third Edition



Springer

Mathematical Physical Chemistry

Shu Hotta

Mathematical Physical Chemistry

Practical and Intuitive Methodology

Third Edition

 Springer

Shu Hotta
Takatsuki, Osaka, Japan

ISBN 978-981-99-2511-7 ISBN 978-981-99-2512-4 (eBook)
<https://doi.org/10.1007/978-981-99-2512-4>

© Springer Nature Singapore Pte Ltd. 2018, 2020, 2023

This work is subject to copyright. All rights are reserved by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.

The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.

The publisher, the authors, and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, expressed or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publisher remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

This Springer imprint is published by the registered company Springer Nature Singapore Pte Ltd.
The registered company address is: 152 Beach Road, #21-01/04 Gateway East, Singapore 189721, Singapore

*To the memory of
my brother Kei Hotta
and
Emeritus Professor Yûsaku Hamada*

Preface to the Third Edition

This book is the third edition of *Mathematical Physical Chemistry*. Although the main concept and main theme remain unchanged, this book's third edition contains the introductory quantum theory of fields. The major motivation for including the relevant topics in the present edition is as follows: The Schrödinger equation has prevailed as a quantum-mechanical fundamental equation up to the present date in many fields of natural science. The Dirac equation was formulated as a relativistically covariant expression soon after the Schrödinger equation was discovered. In contrast to the Schrödinger equation, however, the Dirac equation is less prevailing even nowadays.

One of the major reasons for this is that the Dirac equation has been utilized almost exclusively by elementary particle physicists and related researchers. And what is more, a textbook of the elementary particle physics or quantum theory of fields cannot help but be thick because it has to deal with so many topics that require a detailed explanation, starting with, e.g., renormalization and regularization as well as the underlying gauge theory. Such a situation inevitably leads to significant reduction of descriptions about fundamental properties of the Dirac equation. Under these circumstances, the author has aimed to explain the fundamental properties of the Dirac equation in a plain style so that scientific audience of various fields of specialization can readily understand the essence. In particular, the author described in detail the transformation properties of the Dirac equation (or Dirac operator) and its plane wave solutions in terms of standard matrix algebra.

Another motivation for writing the present edition lies in extending the concepts of vector spaces. The theory of linear vector spaces has been developed in previous editions, in which the inner product space is intended as a vector space. The Dirac equation was constructed so as to be consistent with the requirement of the special theory of relativity that is closely connected to the Minkowski space, a type of vector space. In relation to the Minkowski space, moreover, various kinds of vector spaces play a role in the quantum theory of fields. Examples include tensor space and spinor space. In particular, the tensor spaces are dealt with widely in different fields of

natural science. In this book, therefore, the associated theory of tensor spaces has been treated rather systematically along with related concepts.

As is the case with the previous editions of this book, the author has disposed the mathematical topics at the last chapter of individual parts (Parts I–V). In the present edition, we added a chapter that dealt with advanced topics of Lie algebra at the end of the book. The description has been improved in several parts so that readers can gain clear understanding. Errors found in the previous editions have been corrected. As in the case of the previous editions, readers benefit from going freely back and forth across the whole topics of this book.

The author owes sincere gratitude to the late Emeritus Professor Yūsaku Hamada in Kyoto Institute of Technology for a course of excellent lectures delivered by him on applied mathematics in the undergraduate days of the author. The author is deeply grateful as well to Emeritus Professor Chiaki Tsukamoto in Kyoto Institute of Technology for his helpful discussions and suggestions on exponential functions of matrices (Chap. 15). Once again, the author wishes to thank many students for valuable discussions and Dr. Shin'ichi Koizumi, Springer, for giving him an opportunity to write this book. Finally, the author is most grateful to his wife, Kazue Hotta, for continually encouraging him to write the book.

Takatsuki, Osaka, Japan
March 2023

Shu Hotta

Preface to the Second Edition

This book is the second edition of *Mathematical Physical Chemistry*. Mathematics is a common language of natural science including physics, chemistry, and biology. Although the words mathematical physics and physical chemistry (or chemical physics) are commonly used, mathematical physical chemistry sounds rather uncommon. Therefore, it might well be reworded as the mathematical physics for chemists. The book title could have been, for instance, “*The Mathematics Of Physics And Chemistry*” accordingly, in tribute to the famous book that was written three-quarters of a century ago by H. Margenau and G. M. Murphy. Yet, the word mathematical physical chemistry is expected to be granted citizenship, considering that chemistry and related interdisciplinary fields such as materials science and molecular science are becoming increasingly mathematical.

The main concept and main theme remain unchanged, but this book’s second edition contains the theory of analytic functions and the theory of continuous groups. Both the theories are counted as one of the most elegant theories of mathematics. The mathematics of these topics is of a somewhat advanced level and something like a “sufficient condition” for chemists, whereas that of the first edition may be a prerequisite (or a necessary condition) for them. Therefore, chemists (or may be physicists as well) can creatively use the two editions. In association with these major additions to the second edition, the author has disposed the mathematical topics (the theory of analytic functions, Green’s functions, exponential functions of matrices, and the theory of continuous groups) at the last chapter of individual parts (Parts I–IV).

At the same time, the author has also made several specific revisions including the introductory discussion on the perturbation method and variational method, both of which can be effectively used for gaining approximate solutions of various quantum-mechanical problems. As another topic, the author has presented the recent progress on organic lasers. This topic is expected to help develop high-performance light-emitting devices, one of important fields of materials science. As in the case of the first edition, readers benefit from going freely back and forth across the whole topics of this book.

Once again, the author wishes to thank many students for valuable discussions and Dr. Shin'ichi Koizumi, Springer, for giving him an opportunity to write this book.

Takatsuki, Osaka, Japan
October 2019

Shu Hotta

Preface to the First Edition

The contents of this book are based upon manuscripts prepared for both undergraduate courses of Kyoto Institute of Technology by the author entitled “Polymer Nanomaterials Engineering” and “Photonics Physical Chemistry” and a master’s course lecture of Kyoto Institute of Technology by the author entitled “Solid-State Polymers Engineering.”

This book is intended for graduate and undergraduate students, especially those who major in chemistry and, at the same time, wish to study mathematical physics. Readers are supposed to have basic knowledge of analysis and linear algebra. However, they are not supposed to be familiar with the theory of analytic functions (i.e., complex analysis), even though it is desirable to have relevant knowledge about it.

At the beginning, mathematical physics looks daunting to chemists, as used to be the case with myself as a chemist. The book introduces the basic concepts of mathematical physics to chemists. Unlike other books related to mathematical physics, this book makes a reasonable selection of material so that students majoring in chemistry can readily understand the contents in spontaneity. In particular, we stress the importance of practical and intuitive methodology. We also expect engineers and physicists to benefit from reading this book.

In Parts I and II, the book describes quantum mechanics and electromagnetism. Relevance between the two is well considered. Although quantum mechanics covers the broad field of modern physics, in Part I we focus on a harmonic oscillator and a hydrogen (like) atom. This is because we can study and deal with many of fundamental concepts of quantum mechanics within these restricted topics. Moreover, knowledge acquired from the study of the topics can readily be extended to practical investigation of, e.g., electronic states and vibration (or vibronic) states of molecular systems. We describe these topics by both analytic method (that uses differential equations) and operator approach (using matrix calculations). We believe that the basic concepts of quantum mechanics can be best understood by contrasting the analytical and algebraic approaches. For this reason, we give matrix representations

of physical quantities whenever possible. Examples include energy eigenvalues of a quantum-mechanical harmonic oscillator and angular momenta of a hydrogen-like atom. At the same time, these two physical systems supply us with a good opportunity to study classical polynomials, e.g., Hermite polynomials, (associated) Legendre polynomials, Laguerre polynomials, and Gegenbauer polynomials and special functions, more generally. These topics constitute one of important branches of mathematical physics. One of the basic concepts of quantum mechanics is that a physical quantity is represented by a Hermitian operator or matrix. In this respect, the algebraic approach gives a good opportunity to get familiar with this concept. We present tangible examples for this. We also emphasize the importance of the notion of Hermiticity of a differential operator. We often encounter unitary operator or unitary transformation alongside of the notion of Hermitian operator. We show several examples of the unitary operators in connection with transformation of vectors and coordinates.

Part II describes Maxwell equations and their applications to various phenomena of electromagnetic waves. These include their propagation, reflection, and transmission in dielectric media. We restrict ourselves to treating those phenomena in dielectrics without charge. Yet, we cover a wide range of important topics. In particular, when two (or more) dielectrics are in contact with each other at a plane interface, reflection and transmission of light are characterized by various important parameters such as reflection and transmission coefficients, Brewster angles, and critical angles. We should have their proper understanding not only from a point of view of basic study, but also to make use of relevant knowledge in optical device applications such as a waveguide. In contrast to the concept of electromagnetic waves, light possesses a characteristic of light quanta. We present semiclassical and statistical approach to blackbody radiation occurring in a simplified system in relation to Part I. The physical processes are well characterized by a notion of two-level atoms. In this context, we outline the dipole radiation within the framework of the classical theory. We briefly describe how the optical processes occurring in a confined dielectric medium are related to a laser that is of great importance in fundamental science and its applications. Many of basic equations of physics are described as second-order linear differential equations (SOLDEs). Different methods were developed and proposed to seek their solutions. One of the most important methods is that of Green's functions. We present the introductory theory of Green's functions accordingly. In this connection, we rethink the Hermiticity of a differential operator.

In Parts III and IV, we describe algebraic structures of mathematical physics. Their understanding is useful to studies of quantum mechanics and electromagnetism whose topics are presented in Parts I and II. Part III deals with theories of linear vector spaces. We focus on the discussion on vectors and their transformations in finite-dimensional vector spaces. Generally, we consider the vector transformations among the vector spaces of different dimensions. In this book, however, we restrict ourselves to the case of the transformation between the vector spaces of same dimension, i.e., endomorphism of the space ($V^n \rightarrow V^n$). This is not only because this is most often the case with many of physical applications, but because the

relevant operator is represented by a square matrix. Canonical forms of square matrices hold an important position in algebra. These include a triangle matrix, diagonalizable matrix as well as a nilpotent matrix and idempotent matrix. The most general form will be Jordan canonical form. We present its essential parts in detail taking a tangible example. Next to the general discussion, we deal with an inner product space. Once an inner product is defined between any couple of vectors, the vector space is given a fruitful structure. An example is a norm (i.e., “length”) of a vector. Also, we gain a clear relationship between Parts III and I. We define various operators or matrices that are important in physical applications. Examples include normal operators (or matrices) such as Hermitian operators, projection operators, and unitary operators. Once again, we emphasize the importance of Hermitian operators. In particular, two commutable Hermitian matrices share simultaneous eigenvectors (or eigenstates) and, in this respect, such two matrices occupy a special position in quantum mechanics.

Finally, Part IV describes the essence of group theory and its chemical applications. Group theory has a broad range of applications in solid-state physics, solid-state chemistry, molecular science, etc. Nonetheless, the knowledge of group theory does not seem to have fully prevailed among chemists. We can discover an adequate reason for this in a preface to the first edition of “*Chemical Applications of Group Theory*” written by F. A. Cotton. It might well be natural that definition and statement of abstract algebra, especially group theory, sound somewhat pretentious for chemists, even though the definition of group is quite simple. Therefore, we present various examples for readers to get used to notions of group theory. Notion of mapping is important as in the case of linear vector spaces. Aside from being additive with calculation for a vector space and multiplicative for a group, the fundamentals of calculation regulations are pretty much the same regarding the vector space and group. We describe the characteristics of symmetry groups in detail partly because related knowledge is useful for molecular orbital (MO) calculations that are presented in the last section of the book. Representation theory is probably one of the most daunting notions for chemists. Practically, however, the representation is just homomorphism that corresponds to a linear transformation in a vector space. In this context, the representation is merely denoted by a number or a matrix. Basis functions of representation correspond to basis vectors in a vector space. Grand orthogonality theorem (GOT) is a “nursery bed” of the representation theory. Therefore, readers are encouraged to understand its essence apart from the rigorous proof of the theorem. In conjunction with Part III, we present a variety of projection operators. These are very useful to practical applications in, e.g., quantum mechanics and molecular science. The final parts of the book are devoted to applications of group theory to problems of physical chemistry, especially those of quantum chemistry, more specifically molecular orbital calculations. We see how symmetry consideration, particularly use of projection operators, saves us a lot of labor. Examples include aromatic hydrocarbons and methane.

The previous sections sum up the contents of this book. Readers may start with any part and go freely back and forth. This is because contents of many parts are interrelated. For example, we stress the importance of Hermiticity of differential

operators and matrices. Also, projection operators and nilpotent matrices appear in many parts along with their tangible applications to individual topics. Hence, readers are recommended to carefully examine and compare the related contents throughout the book. We believe that readers, especially chemists, benefit from the writing style of this book, since it is suited to chemists who are good at intuitive understanding.

The author would like to thank many students for their valuable suggestions and discussions at the lectures. The author also wishes to thank many students for valuable discussions and Dr. Shin'ichi Koizumi, Springer, for giving him an opportunity to write this book.

Kyoto, Japan
October 2017

Shu Hotta

Contents

Part I Quantum Mechanics

1	Schrödinger Equation and Its Application	3
1.1	Early-Stage Quantum Theory	3
1.2	Schrödinger Equation	8
1.3	Simple Applications of Schrödinger Equation	14
1.4	Quantum-Mechanical Operators and Matrices	21
1.5	Commutator and Canonical Commutation Relation	28
	Reference	30
2	Quantum-Mechanical Harmonic Oscillator	31
2.1	Classical Harmonic Oscillator	31
2.2	Formulation Based on an Operator Method	33
2.3	Matrix Representation of Physical Quantities	42
2.4	Coordinate Representation of Schrödinger Equation	45
2.5	Variance and Uncertainty Principle	53
	References	58
3	Hydrogen-Like Atoms	59
3.1	Introductory Remarks	59
3.2	Constitution of Hamiltonian	60
3.3	Separation of Variables	69
3.4	Generalized Angular Momentum	75
3.5	Orbital Angular Momentum: Operator Approach	79
3.6	Orbital Angular Momentum: Analytic Approach	94
	3.6.1 Spherical Surface Harmonics and Associated Legendre Differential Equation	94
	3.6.2 Orthogonality of Associated Legendre Functions	105
3.7	Radial Wave Functions of Hydrogen-Like Atoms	109
	3.7.1 Operator Approach to Radial Wave Functions	110
	3.7.2 Normalization of Radial Wave Functions	114

3.7.3	Associated Laguerre Polynomials	118
3.8	Total Wave Functions	124
	References	125
4	Optical Transition and Selection Rules	127
4.1	Electric Dipole Transition	127
4.2	One-Dimensional System	130
4.3	Three-Dimensional System	134
4.4	Selection Rules	144
4.5	Angular Momentum of Radiation	150
	References	152
5	Approximation Methods of Quantum Mechanics	155
5.1	Perturbation Method	155
5.1.1	Quantum State and Energy Level Shift Caused by Perturbation	157
5.1.2	Several Examples	160
5.2	Variational Method	176
	References	183
6	Theory of Analytic Functions	185
6.1	Set and Topology	185
6.1.1	Basic Notions and Notations	186
6.1.2	Topological Spaces and Their Building Blocks	189
6.1.3	T_1 -Space	199
6.1.4	Complex Numbers and Complex Plane	202
6.2	Analytic Functions of a Complex Variable	205
6.3	Integration of Analytic Functions: Cauchy's Integral Formula	213
6.4	Taylor's Series and Laurent's Series	223
6.5	Zeros and Singular Points	229
6.6	Analytic Continuation	232
6.7	Calculus of Residues	234
6.8	Examples of Real Definite Integrals	236
6.9	Multivalued Functions and Riemann Surfaces	256
6.9.1	Brief Outline	256
6.9.2	Examples of Multivalued Functions	264
	References	274

Part II Electromagnetism

7	Maxwell's Equations	277
7.1	Maxwell's Equations and Their Characteristics	277
7.2	Equation of Wave Motion	284
7.3	Polarized Characteristics of Electromagnetic Waves	289
7.4	Superposition of Two Electromagnetic Waves	294
	References	303

8	Reflection and Transmission of Electromagnetic Waves in Dielectric Media	305
8.1	Electromagnetic Fields at an Interface	305
8.2	Basic Concepts Underlying Phenomena	307
8.3	Transverse Electric (TE) Waves and Transverse Magnetic (TM) Waves	313
8.4	Energy Transport by Electromagnetic Waves	319
8.5	Brewster Angles and Critical Angles	322
8.6	Total Reflection	326
8.7	Waveguide Applications	331
8.7.1	TE and TM Waves in a Waveguide	331
8.7.2	Total Internal Reflection and Evanescent Waves	338
8.8	Stationary Waves	343
	References	350
9	Light Quanta: Radiation and Absorption	351
9.1	Blackbody Radiation	351
9.2	Planck's Law of Radiation and Mode Density of Electromagnetic Waves	353
9.3	Two-Level Atoms	358
9.4	Dipole Radiation	361
9.5	Lasers	367
9.5.1	Brief Outlook	367
9.5.2	Organic Lasers	372
9.6	Mechanical System	396
	References	399
10	Introductory Green's Functions	401
10.1	Second-Order Linear Differential Equations (SOLDEs)	401
10.2	First-Order Linear Differential Equations (FOLDEs)	406
10.3	Second-Order Differential Operators	411
10.4	Green's Functions	417
10.5	Construction of Green's Functions	424
10.6	Initial-Value Problems (IVPs)	431
10.6.1	General Remarks	431
10.6.2	Green's Functions for IVPs	433
10.6.3	Estimation of Surface Terms	436
10.6.4	Examples	440
10.7	Eigenvalue Problems	448
	References	453
 Part III Linear Vector Spaces		
11	Vectors and Their Transformation	457
11.1	Vectors	457
11.2	Linear Transformations of Vectors	463

11.3	Inverse Matrices and Determinants	473
11.4	Basis Vectors and Their Transformations	478
	References	484
12	Canonical Forms of Matrices	485
12.1	Eigenvalues and Eigenvectors	485
12.2	Eigenspaces and Invariant Subspaces	495
12.3	Generalized Eigenvectors and Nilpotent Matrices	500
12.4	Idempotent Matrices and Generalized Eigenspaces	505
12.5	Decomposition of Matrix	512
12.6	Jordan Canonical Form	515
	12.6.1 Canonical Form of Nilpotent Matrix	515
	12.6.2 Jordan Blocks	521
	12.6.3 Example of Jordan Canonical Form	528
12.7	Diagonalizable Matrices	539
	References	548
13	Inner Product Space	549
13.1	Inner Product and Metric	549
13.2	Gram Matrices	552
13.3	Adjoint Operators	561
13.4	Orthonormal Basis	566
	References	570
14	Hermitian Operators and Unitary Operators	571
14.1	Projection Operators	571
14.2	Normal Operators	578
14.3	Unitary Diagonalization of Matrices	580
14.4	Hermitian Matrices and Unitary Matrices	588
14.5	Hermitian Quadratic Forms	592
14.6	Simultaneous Eigenstates and Diagonalization	599
	References	605
15	Exponential Functions of Matrices	607
15.1	Functions of Matrices	607
15.2	Exponential Functions of Matrices and Their Manipulations	611
15.3	System of Differential Equations	618
	15.3.1 Introduction	618
	15.3.2 System of Differential Equations in a Matrix Form: Resolvent Matrix	621
	15.3.3 Several Examples	626
15.4	Motion of a Charged Particle in Polarized Electromagnetic Wave	637
	References	644

Part IV Group Theory and Its Chemical Applications

16	Introductory Group Theory	647
16.1	Definition of Groups	647
16.2	Subgroups	649
16.3	Classes	651
16.4	Isomorphism and Homomorphism	653
16.5	Direct-Product Groups	657
	Reference	659
17	Symmetry Groups	661
17.1	A Variety of Symmetry Operations	661
17.2	Successive Symmetry Operations	669
17.3	O and T_d Groups	680
17.4	Special Orthogonal Group $SO(3)$	689
	17.4.1 Rotation Axis and Rotation Matrix	690
	17.4.2 Euler Angles and Related Topics	695
	References	704
18	Representation Theory of Groups	705
18.1	Definition of Representation	705
18.2	Basis Functions of Representation	709
18.3	Schur's Lemmas and Grand Orthogonality Theorem (GOT)	717
18.4	Characters	723
18.5	Regular Representation and Group Algebra	726
18.6	Classes and Irreducible Representations	734
18.7	Projection Operators: Revisited	737
18.8	Direct-Product Representation	746
18.9	Symmetric Representation and Antisymmetric Representation	750
	References	753
19	Applications of Group Theory to Physical Chemistry	755
19.1	Transformation of Functions	755
19.2	Method of Molecular Orbitals (MOs)	761
19.3	Calculation Procedures of Molecular Orbitals (MOs)	768
19.4	MO Calculations Based on π -Electron Approximation	773
	19.4.1 Ethylene	773
	19.4.2 Cyclopropenyl Radical	783
	19.4.3 Benzene	791
	19.4.4 Allyl Radical	797
19.5	MO Calculations of Methane	805
	References	826

20	Theory of Continuous Groups	827
20.1	Introduction: Operators of Rotation and Infinitesimal Rotation	827
20.2	Rotation Groups: $SU(2)$ and $SO(3)$	834
20.2.1	Construction of $SU(2)$ Matrices	837
20.2.2	$SU(2)$ Representation Matrices: Wigner Formula	840
20.2.3	$SO(3)$ Representation Matrices and Spherical Surface Harmonics	843
20.2.4	Irreducible Representations of $SU(2)$ and $SO(3)$	853
20.2.5	Parameter Space of $SO(3)$	861
20.2.6	Irreducible Characters of $SO(3)$ and Their Orthogonality	868
20.3	Clebsch–Gordan Coefficients of Rotation Groups	873
20.3.1	Direct-Product of $SU(2)$ and Clebsch–Gordan Coefficients	874
20.3.2	Calculation Procedures of Clebsch–Gordan Coefficients	880
20.3.3	Examples of Calculation of Clebsch–Gordan Coefficients	890
20.4	Lie Groups and Lie Algebras	896
20.4.1	Definition of Lie Groups and Lie Algebras: One-Parameter Groups	896
20.4.2	Properties of Lie Algebras	899
20.4.3	Adjoint Representation of Lie Groups	905
20.5	Connectedness of Lie Groups	915
20.5.1	Several Definitions and Examples	915
20.5.2	$O(3)$ and $SO(3)$	918
20.5.3	Simply Connected Lie Groups: Local Properties and Global Properties	920
	References	927

Part V Introduction to the Quantum Theory of Fields

21	The Dirac Equation	931
21.1	Historical Background	931
21.2	Several Remarks on the Special Theory of Relativity	934
21.2.1	Minkowski Space and Lorentz Transformation	934
21.2.2	Event and World Interval	939
21.3	Constitution and Solutions of the Dirac Equation	944
21.3.1	General Form of the Dirac Equation	944
21.3.2	Plane Wave Solutions of the Dirac Equation	949
21.3.3	Negative-Energy Solution of the Dirac Equation	955
21.3.4	One-Particle Hamiltonian of the Dirac Equation	960

21.4	Normalization of the Solutions of the Dirac Equation	964
21.5	Charge Conjugation	967
21.6	Characteristics of the Gamma Matrices	971
	References	976
22	Quantization of Fields	977
22.1	Lagrangian Formalism of the Fields	977
22.2	Introductory Fourier Analysis	985
22.2.1	Fourier Series Expansion	986
22.2.2	Fourier Integral Transforms: Fourier Transform and Inverse Fourier Transform	987
22.3	Quantization of the Scalar Field	991
22.3.1	Lagrangian Density and Action Integral	991
22.3.2	Equal-Time Commutation Relation and Field Quantization	993
22.3.3	Hamiltonian and Fock Space	999
22.3.4	Invariant Delta Functions of the Scalar Field	1005
22.3.5	Feynman Propagator of the Scalar Field	1015
22.3.6	General Consideration on the Field Quantization	1023
22.4	Quantization of the Dirac Field	1028
22.4.1	Lagrangian Density and Hamiltonian Density of the Dirac Field	1028
22.4.2	Quantization Procedure of the Dirac Field	1035
22.4.3	Antiparticle: Positron	1039
22.4.4	Invariant Delta Functions of the Dirac Field	1040
22.4.5	Feynman Propagator of the Dirac Field	1043
22.5	Quantization of the Electromagnetic Field	1049
22.5.1	Relativistic Formulation of the Electromagnetic Field	1049
22.5.2	Lagrangian Density and Hamiltonian Density of the Electromagnetic Field	1055
22.5.3	Polarization Vectors of the Electromagnetic Field	1059
22.5.4	Canonical Quantization of the Electromagnetic Field	1061
22.5.5	Hamiltonian and Indefinite Metric	1065
22.5.6	Feynman Propagator of Photon Field	1071
	References	1074
23	Interaction Between Fields	1075
23.1	Lorentz Force and Minimal Coupling	1075
23.2	Lagrangian and Field Equation of the Interacting Fields	1079
23.3	Local Phase Transformation and $U(1)$ Gauge Field	1082
23.4	Interaction Picture	1085
23.5	S -Matrix and S -Matrix Expansion	1091

23.6	N-Product and T-Product	1095
23.6.1	Example of Two Field Operators	1095
23.6.2	Calculations Including Both the N-Products and T-Products	1099
23.7	Feynman Rules and Feynman Diagrams in QED	1103
23.7.1	Zeroth- and First-Order S -Matrix Elements	1104
23.7.2	Second-Order S -Matrix Elements and Feynman Diagrams	1108
23.8	Example: Compton Scattering	1113
23.8.1	Introductory Remarks	1113
23.8.2	Feynman Amplitude and Feynman Diagrams of Compton Scattering	1115
23.8.3	Scattering Cross-Section	1120
23.8.4	Spin and Photon Polarization Sums	1124
23.8.5	Detailed Calculation Procedures of Feynman Amplitude	1126
23.8.6	Experimental Observations	1136
23.9	Summary	1139
	References	1140
24	Basic Formalism	1141
24.1	Extended Concepts of Vector Spaces	1141
24.1.1	Bilinear Mapping	1142
24.1.2	Tensor Product	1149
24.1.3	Bilinear Form	1155
24.1.4	Dual Vector Space	1155
24.1.5	Invariants	1165
24.1.6	Tensor Space and Tensors	1171
24.1.7	Euclidean Space and Minkowski Space	1177
24.2	Lorentz Group and Lorentz Transformations	1187
24.2.1	Lie Algebra of the Lorentz Group	1189
24.2.2	Successive Lorentz Transformations	1193
24.3	Covariant Properties of the Dirac Equation	1197
24.3.1	General Consideration on the Physical Equation	1197
24.3.2	The Klein—Gordon Equation and the Dirac Equation	1200
24.4	Determination of General Form of Matrix $S(\Lambda)$	1204
24.5	Transformation Properties of the Dirac Spinors	1207
24.6	Transformation Properties of the Dirac Operators	1212
24.6.1	Case I: Single Lorentz Boost	1212
24.6.2	Case II: Non-Coaxial Lorentz Boosts	1221
24.7	Projection Operators and Related Operators for the Dirac Equation	1229

24.8	Spectral Decomposition of the Dirac Operators	1235
24.9	Connectedness of the Lorentz Group	1241
24.9.1	Polar Decomposition of a Non-Singular Matrix	1242
24.9.2	Special Linear Group: $SL(2, \mathbb{C})$	1250
24.10	Representation of the Proper Orthochronous Lorentz Group $SO_0(3, 1)$	1258
	References	1269
25	Advanced Topics of Lie Algebra	1271
25.1	Differential Representation of Lie Algebra	1271
25.1.1	Overview	1271
25.1.2	Adjoint Representation of Lie Algebra	1273
25.1.3	Differential Representation of Lorentz Algebra	1278
25.2	Cartan–Weyl Basis of Lie Algebra	1287
25.2.1	Complexification	1288
25.2.2	Coupling of Angular Momenta: Revisited	1294
25.3	Decomposition of Lie Algebra	1298
25.4	Further Topics of Lie Algebra	1301
25.5	Closing Remarks	1304
	References	1304
	Index	1305

Part I

Quantum Mechanics

Quantum mechanics is clearly distinguished from classical physics whose major pillars are Newtonian mechanics and electromagnetism established by Maxwell. Quantum mechanics was first established as a theory of atomic physics that handled the microscopic world. Later on, quantum mechanics was applied to the macroscopic world, i.e., cosmos. The question on how exactly quantum mechanics describes the natural world and on how far the theory can go remains yet problematic and is in dispute to this day.

Such an ultimate question is irrelevant to this monograph. Our major aim is to study a standard approach to applying Schrödinger equation to selected topics. The topics include a particle confined within a potential well, a harmonic oscillator, and hydrogen-like atoms. Our major task rests on solving eigenvalue problems of these topics. To this end, we describe both an analytical method and algebraic (or operator) method. Focusing on these topics, we will be able to acquire various methods to tackle a wide range of quantum-mechanical problems. These problems are usually posed as an analytical equation (i.e., differential equation) or an algebraic equation. A Hamiltonian is constructed analytically or algebraically accordingly. Besides Hamiltonian, physical quantities are expressed as a differential operator or a matrix operator. In both analytical and algebraic approaches, Hermitian property (or Hermiticity) of an operator and matrix is of crucial importance. This feature will, therefore, be highlighted not only in this part but also throughout this book along with a unitary operator and matrix.

Optical transition and associated selection rules are dealt with in relation to the above topics. Those subjects are closely related to electromagnetic phenomena that are considered in Part II.

Unlike the eigenvalue problems of the above-mentioned topics, it is difficult to get exact analytical solutions in most cases of quantum-mechanical problems. For this reason, we need appropriate methods to obtain approximate solutions with respect to various problems including the eigenvalue problems. In this context, we deal with approximation techniques of a perturbation method and variational method.

In the last part, we study the theory of analytic functions, one of the most elegant theories of mathematics. This approach not only helps cultivate a broad view of pure mathematics, but also leads to the acquisition of practical methodology. The last part deals with the introductory set theory and topology as well.

Chapter 1

Schrödinger Equation and Its Application



Quantum mechanics is an indispensable research tool of modern natural science that covers cosmology, atomic physics, molecular science, materials science, and so forth. The basic concept underlying quantum mechanics rests upon Schrödinger equation. The Schrödinger equation is described as a second-order linear differential equation (SOLDE). The equation is analytically solved accordingly. Alternatively, equations of the quantum mechanics are often described in terms of operators and matrices and physical quantities are represented by those operators and matrices. Normally, they are non-commutative. In particular, the quantum-mechanical formalism requires the canonical commutation relation between position and momentum operators. One of the great characteristics of the quantum mechanics is that physical quantities must be Hermitian. This aspect is deeply related to the requirement that these quantities should be described by real numbers. We deal with the Hermiticity from both an analytical point of view (or coordinate representation) relevant to the differential equations and an algebraic viewpoint (or matrix representation) associated with the operators and matrices. Including these topics, we briefly survey the origin of Schrödinger equation and consider its implications. To get acquainted with the quantum-mechanical formalism, we deal with simple examples of the Schrödinger equation.

1.1 Early-Stage Quantum Theory

The Schrödinger equation is a direct consequence of discovery of quanta. It stemmed from the hypothesis of energy quanta propounded by Max Planck (1900). This hypothesis was further followed by photon (light quantum) hypothesis propounded by Albert Einstein (1905). He claimed that light is an aggregation of light quanta and that individual quanta carry an energy E expressed as Planck constant h multiplied by frequency of light ν , i.e.,

$$E = h\nu = \hbar\omega, \quad (1.1)$$

where $\hbar \equiv h/2\pi$ and $\omega = 2\pi\nu$. The quantity ω is called angular frequency with ν being frequency. The quantity $\hbar$ is said to be a reduced Planck constant.

Also Einstein (1917) concluded that momentum of light quantum p is identical to the energy of light quantum divided by light velocity in vacuum c . That is, we have

$$p = E/c = \hbar\omega/c = \hbar k, \quad (1.2)$$

where $k \equiv 2\pi/\lambda$ (λ is wavelength of light in vacuum) and k is called wavenumber. Using vector notation, we have

$$\mathbf{p} = \hbar \mathbf{k}, \quad (1.3)$$

where $\mathbf{k} \equiv \frac{2\pi}{\lambda} \mathbf{n}$ ($\mathbf{n}$: a unit vector in the direction of propagation of light) is said to be a wavenumber vector.

Meanwhile, Arthur Compton (1923) conducted various experiments where he investigated how an incident X-ray beam was scattered by matter (e.g., graphite, copper, etc.). As a result, Compton found out a systematical redshift in X-ray wavelengths as a function of scattering angles of the X-ray beam (Compton effect). Moreover he found that the shift in wavelengths depended only on the scattering angle regardless of quality of material of a scatterer. The results can be summarized in a simple equation described as

$$\Delta\lambda = \frac{h}{m_e c} (1 - \cos \theta), \quad (1.4)$$

where $\Delta\lambda$ denotes a shift in wavelength of the scattered beam; m_e is a rest mass of an electron; θ is a scattering angle of the X-ray beam (see Fig. 1.1). A quantity $\frac{h}{m_e c}$ has a dimension of length and denoted by λ_e . That is,

$$\lambda_e \equiv h/m_e c. \quad (1.5)$$

In other words, λ_e is equal to the maximum shift in the wavelength of the scattered beam; this shift is obtained when $\theta = \pi/2$. The quantity λ_e is called an electron Compton wavelength and has an approximate value of 2.426×10^{-12} (m).

Let us derive (1.4) on the basis of conservation of energy and momentum. To this end, in Fig. 1.1 we assume that an electron is originally at rest. An X-ray beam is incident to the electron. Then the X-ray is scattered and the electron recoils as shown. The energy conservation reads as

$$\hbar\omega + m_e c^2 = \hbar\omega' + \sqrt{p^2 c^2 + m_e^2 c^4}, \quad (1.6)$$

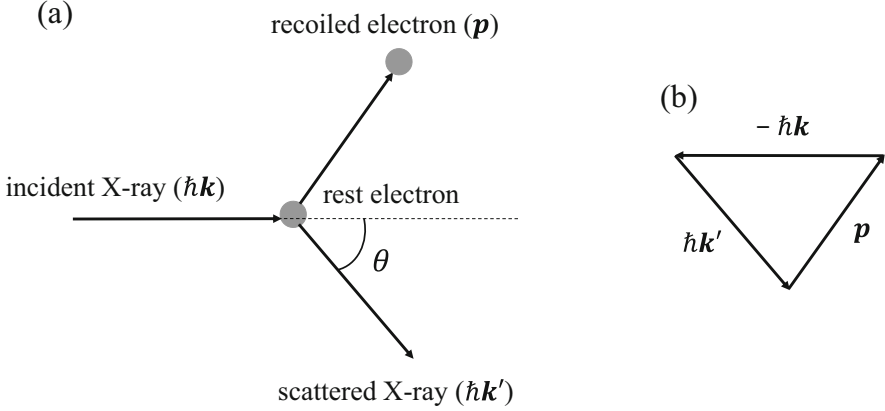


Fig. 1.1 Scattering of an X-ray beam by an electron. (a) θ denotes a scattering angle of the X-ray beam. (b) Conservation of momentum

where ω and ω' are initial and final angular frequencies of the X-ray; the second term of RHS is an energy of the electron in which p is a magnitude of momentum after recoil. Meanwhile, conservation of the momentum as a vector quantity reads as

$$\hbar\mathbf{k} = \hbar\mathbf{k}' + \mathbf{p}, \quad (1.7)$$

where $\mathbf{k}$ and $\mathbf{k}'$ are wavenumber vectors of the X-ray before and after being scattered; $\mathbf{p}$ is a momentum of the electron after recoil. Note that an initial momentum of the electron is zero since the electron is originally at rest. Here $\mathbf{p}$ is defined as

$$\mathbf{p} \equiv m\mathbf{u}, \quad (1.8)$$

where $\mathbf{u}$ is a velocity of an electron and m is given by [1]

$$m = m_e / \sqrt{1 - |\mathbf{u}|^2/c^2}. \quad (1.9)$$

Figure 1.1 shows that $-\hbar\mathbf{k}$, $\hbar\mathbf{k}'$, and $\mathbf{p}$ form a closed triangle. From (1.6), we have

$$[m_e c^2 + \hbar(\omega - \omega')]^2 = p^2 c^2 + m_e^2 c^4. \quad (1.10)$$

Hence, we get

$$2m_e c^2 \hbar(\omega - \omega') + \hbar^2(\omega - \omega')^2 = p^2 c^2. \quad (1.11)$$

From (1.7), we have

$$\begin{aligned}
p^2 &= \hbar^2 (\mathbf{k} - \mathbf{k}')^2 = \hbar^2 (k^2 + k'^2 - 2kk' \cos \theta) \\
&= \frac{\hbar^2}{c^2} (\omega^2 + \omega'^2 - 2\omega\omega' \cos \theta),
\end{aligned} \tag{1.12}$$

where we used the relations $\omega = ck$ and $\omega' = ck'$ with the third equality. Therefore, we get

$$p^2 c^2 = \hbar^2 (\omega^2 + \omega'^2 - 2\omega\omega' \cos \theta). \tag{1.13}$$

From (1.11) and (1.13), we have

$$2m_e c^2 \hbar (\omega - \omega') + \hbar^2 (\omega - \omega')^2 = \hbar^2 (\omega^2 + \omega'^2 - 2\omega\omega' \cos \theta). \tag{1.14}$$

Equation (1.14) is simplified to the following:

$$2m_e c^2 \hbar (\omega - \omega') - 2\hbar^2 \omega\omega' = -2\hbar^2 \omega\omega' \cos \theta.$$

That is,

$$m_e c^2 (\omega - \omega') = \hbar \omega\omega' (1 - \cos \theta). \tag{1.15}$$

Thus, we get

$$\frac{\omega - \omega'}{\omega\omega'} = \frac{1}{\omega'} - \frac{1}{\omega} = \frac{1}{2\pi c} (\lambda' - \lambda) = \frac{\hbar}{m_e c^2} (1 - \cos \theta), \tag{1.16}$$

where λ and λ' are wavelengths of the initial and final X-ray beams, respectively. Since $\lambda' - \lambda = \Delta\lambda$, we have (1.4) from (1.16) accordingly.

We have to mention another important person, Louis-Victor de Broglie (1924) in the development of quantum mechanics. Encouraged by the success of Einstein and Compton, he propounded the concept of matter wave, which was referred to as the de Broglie wave afterward. Namely, de Broglie reversed the relationship of (1.1) and (1.2) such that

$$\omega = E/\hbar, \tag{1.17}$$

and

$$k = p/\hbar \quad \text{or} \quad \lambda = h/p, \tag{1.18}$$

where p equals $|\mathbf{p}|$ and λ is a wavelength of a corpuscular beam. This is said to be the de Broglie wavelength. In (1.18), de Broglie thought that a particle carrying an

energy E and momentum p is accompanied by a wave that is characterized by an angular frequency ω and wavenumber k (or a wavelength $\lambda = 2\pi/k$). Equation (1.18) implies that if we are able to determine the wavelength of the corpuscular beam experimentally, we can decide a magnitude of momentum accordingly.

In turn, from squares of both sides of (1.8) and (1.9) we get

$$u = \frac{p}{m_e \sqrt{1 + (p/m_e c)^2}}. \quad (1.19)$$

This relation represents a velocity of particles of the corpuscular beam. If we are dealing with an electron beam, (1.19) gives the velocity of the electron beam. As a non-relativistic approximation (i.e., $p/m_e c \ll 1$), we have

$$p \approx m_e u.$$

We used a relativistic relation in the second term of RHS of (1.6), where an energy of an electron E_e is expressed by

$$E_e = \sqrt{p^2 c^2 + m_e^2 c^4}. \quad (1.20)$$

In the meantime, deleting u^2 from (1.8) and (1.9) we have

$$m c^2 = \sqrt{p^2 c^2 + m_e^2 c^4}.$$

Namely, we get [1]

$$E_e = m c^2. \quad (1.21)$$

The relation (1.21) is due to Einstein (1905, 1907) and is said to be the equivalence theorem of mass and energy.

If an electron is accompanied by a matter wave, that wave should be propagated with a certain phase velocity v_p and a group velocity v_g . Thus, using (1.17) and (1.18) we have

$$\begin{aligned} v_p &= \omega/k = E_e/p = \sqrt{p^2 c^2 + m_e^2 c^4}/p > c, \\ v_g &= \partial \omega / \partial k = \partial E_e / \partial p = c^2 p / \sqrt{p^2 c^2 + m_e^2 c^4} < c, \\ v_p v_g &= c^2. \end{aligned} \quad (1.22)$$

Notice that in the above expressions, we replaced E of (1.17) with E_e of (1.20). The group velocity is thought to be a velocity of a wave packet and, hence, a propagation velocity of a matter wave should be identical to v_g . Thus, v_g is considered as a particle velocity as well. In fact, v_g given by (1.22) is identical to u expressed in (1.19). Therefore, a particle velocity must not exceed c . As for

photons (or light quanta), $v_p = v_g = c$ and, hence, once again we get $v_p v_g = c^2$. We will encounter the last relation of (1.22) in Part II as well.

The above discussion is a brief historical outlook of early-stage quantum theory before Erwin Schrödinger (1926) propounded his equation.

1.2 Schrödinger Equation

First we introduce a wave equation expressed by

$$\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}, \quad (1.23)$$

where ψ is an arbitrary function of a physical quantity relevant to propagation of a wave; v is a phase velocity of wave; ∇^2 called Laplacian (or Laplace operator) is defined below

$$\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}. \quad (1.24)$$

One of the special solutions for (1.24) called a plane wave is well-studied and expressed as

$$\psi = \psi_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}. \quad (1.25)$$

In (1.25), $\mathbf{x}$ denotes a position vector of a three-dimensional Cartesian coordinate and is described as

$$\mathbf{x} = (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad (1.26)$$

where $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ denote basis vectors of an orthonormal base pointing to positive directions of x -, y -, and z -axes, respectively. Here we make it a rule to represent basis vectors by a *row vector* and represent a coordinate or a component of a vector by a *column vector*; see Sect. 11.1.

The other way around, now we wish to seek a basic equation whose solution is described as (1.25). Taking account of (1.1)–(1.3) as well as (1.17) and (1.18), we rewrite (1.25) as

$$\psi = \psi_0 e^{i(\frac{p}{\hbar}x - \frac{E}{\hbar}t)}, \quad (1.27)$$

where we redefine $\mathbf{p} = (e_1 \ e_2 \ e_3) \begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix}$ and E as quantities associated with those of matter (electron) wave. Taking partial differentiation of (1.27) with respect to x , we obtain

$$\frac{\partial \psi}{\partial x} = \frac{i}{\hbar} p_x \psi_0 e^{i(\frac{p}{\hbar}x - \frac{E}{\hbar}t)} = \frac{i}{\hbar} p_x \psi. \quad (1.28)$$

Rewriting (1.28), we have

$$\frac{\hbar}{i} \frac{\partial \psi}{\partial x} = p_x \psi. \quad (1.29)$$

Similarly we have

$$\frac{\hbar}{i} \frac{\partial \psi}{\partial y} = p_y \psi \quad \text{and} \quad \frac{\hbar}{i} \frac{\partial \psi}{\partial z} = p_z \psi. \quad (1.30)$$

Comparing both sides of (1.29), we notice that we may relate a differential operator $\frac{\hbar}{i} \frac{\partial}{\partial x}$ to p_x . From (1.30), similar relationship holds with the y and z components. That is, we have the following relations:

$$\frac{\hbar}{i} \frac{\partial}{\partial x} \leftrightarrow p_x, \quad \frac{\hbar}{i} \frac{\partial}{\partial y} \leftrightarrow p_y, \quad \frac{\hbar}{i} \frac{\partial}{\partial z} \leftrightarrow p_z. \quad (1.31)$$

Taking partial differentiation of (1.28) once more,

$$\frac{\partial^2 \psi}{\partial x^2} = \left(\frac{i}{\hbar} p_x \right)^2 \psi_0 e^{i(\frac{p}{\hbar}x - \frac{E}{\hbar}t)} = -\frac{1}{\hbar^2} p_x^2 \psi. \quad (1.32)$$

Hence,

$$-\hbar^2 \frac{\partial^2 \psi}{\partial x^2} = p_x^2 \psi. \quad (1.33)$$

Similarly we have

$$-\hbar^2 \frac{\partial^2 \psi}{\partial y^2} = p_y^2 \psi \quad \text{and} \quad -\hbar^2 \frac{\partial^2 \psi}{\partial z^2} = p_z^2 \psi. \quad (1.34)$$

As in the above cases, we have

$$-\hbar^2 \frac{\partial^2}{\partial x^2} \leftrightarrow p_x^2, \quad -\hbar^2 \frac{\partial^2}{\partial y^2} \leftrightarrow p_y^2, \quad -\hbar^2 \frac{\partial^2}{\partial z^2} \leftrightarrow p_z^2. \quad (1.35)$$

Summing both sides of (1.33) and (1.34) and then dividing by $2m$, we have

$$-\frac{\hbar^2}{2m} \nabla^2 \psi = \frac{\mathbf{p}^2}{2m} \psi \quad (1.36)$$

and the following correspondence

$$-\frac{\hbar^2}{2m} \nabla^2 \leftrightarrow \frac{\mathbf{p}^2}{2m}, \quad (1.37)$$

where m is the mass of a particle.

Meanwhile, taking partial differentiation of (1.27) with respect to t , we obtain

$$\frac{\partial \psi}{\partial t} = -\frac{i}{\hbar} E \psi_0 e^{i(\frac{p}{\hbar}x - \frac{E}{\hbar}t)} = -\frac{i}{\hbar} E \psi. \quad (1.38)$$

That is,

$$i\hbar \frac{\partial \psi}{\partial t} = E \psi. \quad (1.39)$$

As the above, we get the following relationship:

$$i\hbar \frac{\partial}{\partial t} \leftrightarrow E. \quad (1.40)$$

Thus, we have relationships between c -numbers (classical numbers) and q -numbers (quantum numbers, namely, operators) in (1.35) and (1.40). Subtracting (1.36) from (1.39), we get

$$i\hbar \frac{\partial \psi}{\partial t} + \frac{\hbar^2}{2m} \nabla^2 \psi = \left(E - \frac{\mathbf{p}^2}{2m} \right) \psi. \quad (1.41)$$

Invoking the relationship on energy

$$(\text{Total energy}) = (\text{Kinetic energy}) + (\text{Potential energy}), \quad (1.42)$$

we have

$$E = \frac{\mathbf{p}^2}{2m} + V, \quad (1.43)$$

where V is a potential energy. Thus, (1.41) reads as

$$i\hbar \frac{\partial \psi}{\partial t} + \frac{\hbar^2}{2m} \nabla^2 \psi = V\psi. \quad (1.44)$$

Rearranging (1.44), we finally get

$$\left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi = i\hbar \frac{\partial \psi}{\partial t}. \quad (1.45)$$

This is the Schrödinger equation, a fundamental equation of quantum mechanics. In (1.45), we define a following Hamiltonian operator H as

$$H \equiv -\frac{\hbar^2}{2m} \nabla^2 + V. \quad (1.46)$$

Then we have a shorthand representation such that

$$H\psi = i\hbar \frac{\partial \psi}{\partial t}. \quad (1.47)$$

On going from (1.25) to (1.27), we realize that quantities $\mathbf{k}$ and ω pertinent to a field have been converted to quantities $\mathbf{p}$ and E related to a particle. At the same time, whereas $\mathbf{x}$ and t represent a whole space-time in (1.25), those in (1.27) are characterized as localized quantities.

From a historical point of view, we have to mention a great achievement accomplished by Werner Heisenberg (1925) who propounded matrix mechanics. The matrix mechanics is often contrasted with the wave mechanics Schrödinger initiated. Schrödinger and Paul Dirac (1926) demonstrated that wave mechanics and matrix mechanics are mathematically equivalent. Note that the Schrödinger equation is described as a non-relativistic expression based on (1.43). In fact, kinetic energy K of a particle is given by [1]

$$K = \frac{m_e c^2}{\sqrt{1 - (u/c)^2}} - m_e c^2.$$

As a non-relativistic approximation, we get

$$K \approx m_e c^2 \left[1 + \frac{1}{2} \left(\frac{u}{c} \right)^2 \right] - m_e c^2 = \frac{1}{2} m_e u^2 \approx \frac{p^2}{2m_e},$$

where we used $p \approx m_e u$ again as a non-relativistic approximation; also, we used

$$\frac{1}{\sqrt{1-x}} \approx 1 + \frac{1}{2}x$$

when $x (>0)$ corresponding to $\left(\frac{u}{c}\right)^2$ is enough small than 1. This implies that in the above case the group velocity u of a particle is supposed to be well below light velocity c . Dirac (1928), however, formulated an equation that describes relativistic quantum mechanics of electron (the Dirac equation). The detailed description with respect to the Dirac equation will be found in Part V.

In (1.45) ψ varies as a function of $\mathbf{x}$ and t . Suppose, however, that a potential V depends only upon $\mathbf{x}$. Then we have

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right] \psi(\mathbf{x}, t) = i\hbar \frac{\partial \psi(\mathbf{x}, t)}{\partial t}. \quad (1.48)$$

Now, let us assume that separation of variables can be done with (1.48) such that

$$\psi(\mathbf{x}, t) = \phi(\mathbf{x}) \xi(t). \quad (1.49)$$

Then, we have

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right] \phi(\mathbf{x}) \xi(t) = i\hbar \frac{\partial \phi(\mathbf{x}) \xi(t)}{\partial t}. \quad (1.50)$$

Accordingly, (1.50) can be recast as

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right] \phi(\mathbf{x}) / \phi(\mathbf{x}) = i\hbar \frac{\partial \xi(t)}{\partial t} / \xi(t). \quad (1.51)$$

For (1.51) to hold, we must equate both sides to a constant E . That is, for a certain fixed point $\mathbf{x}_0$ we have

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}_0) \right] \phi(\mathbf{x}_0) / \phi(\mathbf{x}_0) = i\hbar \frac{\partial \xi(t)}{\partial t} / \xi(t), \quad (1.52)$$

where $\phi(\mathbf{x}_0)$ of a numerator should be evaluated after operating ∇^2 , while with $\phi(\mathbf{x}_0)$ in a denominator, $\phi(\mathbf{x}_0)$ is evaluated simply replacing $\mathbf{x}$ in $\phi(\mathbf{x})$ with $\mathbf{x}_0$. Now, let us define a function $\Phi(\mathbf{x})$ such that

$$\Phi(\mathbf{x}) \equiv \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right] \phi(\mathbf{x}) / \phi(\mathbf{x}). \quad (1.53)$$

Then, we have

$$\Phi(\mathbf{x}_0) = i\hbar \frac{\partial \xi(t)}{\partial t} / \xi(t). \quad (1.54)$$

If RHS of (1.54) varied depending on t , $\Phi(\mathbf{x}_0)$ would be allowed to have various values, but this must not be the case with our present investigation. Thus, RHS of (1.54) should take a constant value E . For the same reason, LHS of (1.51) should take a constant.

Thus, (1.48) or (1.51) should be separated into the following equations:

$$H\phi(\mathbf{x}) = E\phi(\mathbf{x}), \quad (1.55)$$

$$i\hbar \frac{\partial \xi(t)}{\partial t} = E\xi(t). \quad (1.56)$$

Equation (1.56) can readily be solved. Since (1.56) depends on a sole variable t , we have

$$\frac{d\xi(t)}{\xi(t)} = \frac{E}{i\hbar} dt \quad \text{or} \quad d \ln \xi(t) = \frac{E}{i\hbar} dt. \quad (1.57)$$

Integrating (1.57) from zero to t , we get

$$\ln \frac{\xi(t)}{\xi(0)} = \frac{Et}{i\hbar}. \quad (1.58)$$

That is,

$$\xi(t) = \xi(0) \exp(-iEt/\hbar). \quad (1.59)$$

Comparing (1.59) with (1.38), we find that the constant E in (1.55) and (1.56) represents an energy of a particle (electron).

Thus, the next task we want to do is to solve an eigenvalue equation of (1.55). After solving the problem, we get a solution

$$\psi(\mathbf{x}, t) = \phi(\mathbf{x}) \exp(-iEt/\hbar), \quad (1.60)$$

where the constant $\xi(0)$ has been absorbed in $\phi(\mathbf{x})$. Normally, $\phi(\mathbf{x})$ is to be normalized after determining the functional form (vide infra).

1.3 Simple Applications of Schrödinger Equation

The Schrödinger equation has been expressed as (1.48). The equation is a second-order linear differential equation (SOLDE). In particular, our major interest lies in solving an eigenvalue problem of (1.55). Eigenvalues consist of points in a complex plane. Those points sometimes form a continuous domain, but we focus on the eigenvalues that comprise discrete points in the complex plane. Therefore, in our studies the eigenvalues are countable and numbered as, e.g., λ_n ($n = 1, 2, 3, \dots$). An example is depicted in Fig. 1.2. Having this common belief as a background, let us first think of a simple form of SOLDE.

Example 1.1 Let us think of a following differential equation:

$$\frac{d^2 y(x)}{dx^2} + \lambda y(x) = 0, \quad (1.61)$$

where x is a real variable; y may be a complex function of x with λ possibly being a complex constant as well. Suppose that $y(x)$ is defined within a domain $[-L, L]$ ($L > 0$). We set boundary conditions (BCs) for (1.61) such that

$$y(L) = 0 \quad \text{and} \quad y(-L) = 0 \quad (L > 0). \quad (1.62)$$

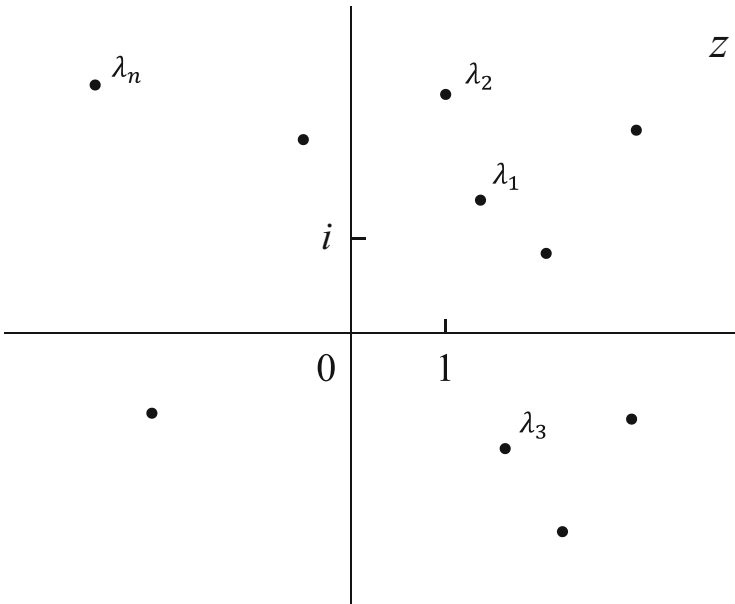


Fig. 1.2 Eigenvalues λ_n ($n = 1, 2, 3, \dots$) on a complex plane

The BCs of (1.62) are called Dirichlet conditions. We define the following differential operator D described as

$$D \equiv -\frac{d^2}{dx^2}. \quad (1.63)$$

Then rewriting (1.61), we have

$$Dy(x) = \lambda y(x). \quad (1.64)$$

According to a general principle of SOLDE, it has two linearly independent solutions. In the case of (1.61), we choose exponential functions for those solutions described by

$$e^{ikx} \quad \text{and} \quad e^{-ikx} \quad (k \neq 0). \quad (1.65)$$

This is because the above functions do not change a functional form with respect to the differentiation and we ascribe solving a differential equation to solving an algebraic equation among constants (or parameters). In the present case, λ and k are such constants.

The parameter k could be a complex variable, because λ is allowed to take a complex value as well. Linear independence of these functions is ensured from a non-vanishing Wronskian, W . That is,

$$W = \begin{vmatrix} e^{ikx} & e^{-ikx} \\ (e^{ikx})' & (e^{-ikx})' \end{vmatrix} = \begin{vmatrix} e^{ikx} & e^{-ikx} \\ ike^{ikx} & -ike^{-ikx} \end{vmatrix} = -ik - ik = -2ik.$$

If $k \neq 0$, $W \neq 0$. Therefore, as a general solution, we get

$$y(x) = ae^{ikx} + be^{-ikx} \quad (k \neq 0), \quad (1.66)$$

where a and b are (complex) constant. We call two linearly independent solutions e^{ikx} and e^{-ikx} ($k \neq 0$) a fundamental set of solutions of a SOLDE. Inserting (1.66) into (1.61), we have

$$(\lambda - k^2)(ae^{ikx} + be^{-ikx}) = 0. \quad (1.67)$$

For (1.67) to hold with any x , we must have

$$\lambda - k^2 = 0, \quad \text{i.e.,} \quad \lambda = k^2. \quad (1.68)$$

Using BCs (1.62), we have

$$ae^{ikL} + be^{-ikL} = 0 \quad \text{and} \quad ae^{-ikL} + be^{ikL} = 0. \quad (1.69)$$

Rewriting (1.69) in a matrix form, we have

$$\begin{pmatrix} e^{ikL} & e^{-ikL} \\ e^{-ikL} & e^{ikL} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (1.70)$$

For a and b in (1.70) to have non-vanishing solutions, we must have

$$\begin{vmatrix} e^{ikL} & e^{-ikL} \\ e^{-ikL} & e^{ikL} \end{vmatrix} = 0, \quad \text{i.e.,} \quad e^{2ikL} - e^{-2ikL} = 0. \quad (1.71)$$

It is because if (1.71) were not zero, we would have $a = b = 0$ and $y(x) \equiv 0$. Note that with an eigenvalue problem we must avoid having a solution that is identically zero. Rewriting (1.71), we get

$$(e^{ikL} + e^{-ikL})(e^{ikL} - e^{-ikL}) = 0. \quad (1.72)$$

That is, we have either

$$e^{ikL} + e^{-ikL} = 0 \quad (1.73)$$

or

$$e^{ikL} - e^{-ikL} = 0. \quad (1.74)$$

In the case of (1.73), inserting this into (1.69) we have

$$e^{ikL}(a - b) = 0. \quad (1.75)$$

Therefore,

$$a = b, \quad (1.76)$$

where we used the fact that e^{ikL} is a non-vanishing function for any ikL (either real or complex). Similarly, in the case of (1.74), we have

$$a = -b. \quad (1.77)$$

For (1.76), from (1.66) we have

$$y(x) = a(e^{ikx} + e^{-ikx}) = 2a \cos kx. \quad (1.78)$$

With (1.77), in turn, we get

$$y(x) = a(e^{ikx} - e^{-ikx}) = 2ia \sin kx. \quad (1.79)$$

Thus, we get two linearly independent solutions (1.78) and (1.79).

Inserting BCs (1.62) into (1.78), we have

$$\cos kL = 0. \quad (1.80)$$

Hence,

$$kL = \frac{\pi}{2} + m\pi \quad (m = 0, \pm 1, \pm 2, \dots). \quad (1.81)$$

In (1.81), for instance, we have $k = \frac{\pi}{2L}$ for $m = 0$ and $k = -\frac{\pi}{2L}$ for $m = -1$. Also, we have $k = \frac{3\pi}{2L}$ for $m = 1$ and $k = -\frac{3\pi}{2L}$ for $m = -2$. These cases, however, individually give linearly dependent solutions for (1.78). Therefore, to get a set of linearly independent eigenfunctions we may define k as positive. Correspondingly, from (1.68) we get eigenvalues of

$$\lambda = (2m + 1)^2 \pi^2 / 4L^2 \quad (m = 0, 1, 2, \dots). \quad (1.82)$$

Also, inserting BCs (1.62) into (1.79), we have

$$\sin kL = 0. \quad (1.83)$$

Hence,

$$kL = n\pi \quad (n = 1, 2, 3, \dots). \quad (1.84)$$

From (1.68) we get

$$\lambda = n^2 \pi^2 / L^2 = (2n)^2 \pi^2 / 4L^2 \quad (n = 1, 2, 3, \dots), \quad (1.85)$$

where we chose positive numbers n for the same reason as the above. With the second equality of (1.85), we made eigenvalues easily comparable to those of (1.82). Figure 1.3 shows the eigenvalues given in both (1.82) and (1.85) in a unit of $\pi^2/4L^2$.

From (1.82) and (1.85), we find that λ is *positive definite* (or strictly positive), and so from (1.68) we have

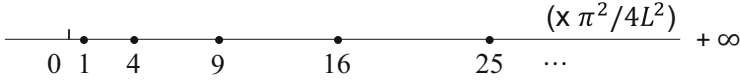


Fig. 1.3 Eigenvalues of a differential equation (1.61) under boundary conditions given by (1.62). The eigenvalues are given in a unit of $\pi^2/4L^2$ on a real axis

$$k = \sqrt{\lambda}. \quad (1.86)$$

The next step is to normalize eigenfunctions. This step corresponds to appropriate choice of a constant a in (1.78) and (1.79) so that we can have

$$I = \int_{-L}^L y(x)^* y(x) dx = \int_{-L}^L |y(x)|^2 dx = 1. \quad (1.87)$$

That is,

$$\begin{aligned} I &= 4|a|^2 \int_{-L}^L \cos^2 kx dx = 4|a|^2 \int_{-L}^L \frac{1}{2} (1 + \cos 2kx) dx \\ &= 2|a|^2 \left[x + \frac{1}{2k} \sin 2kx \right]_{-L}^L = 4L|a|^2. \end{aligned} \quad (1.88)$$

Combining (1.87) and (1.88), we get

$$|a| = \frac{1}{2} \sqrt{\frac{1}{L}}. \quad (1.89)$$

Thus, we have

$$a = \frac{1}{2} \sqrt{\frac{1}{L}} e^{i\theta}, \quad (1.90)$$

where θ is any real number and $e^{i\theta}$ is said to be a phase factor. We usually set $e^{i\theta} \equiv 1$. Then, we have $a = \frac{1}{2} \sqrt{\frac{1}{L}}$. Thus for a normalized cosine eigenfunctions, we get

$$y(x) = \sqrt{\frac{1}{L}} \cos kx \quad \left[kL = \frac{\pi}{2} + m\pi \quad (m = 0, 1, 2, \dots) \right] \quad (1.91)$$

that corresponds to an eigenvalue $\lambda = (2m+1)^2 \pi^2/4L^2$ ($m = 0, 1, 2, \dots$). For another series of normalized sine functions, similarly we get

$$y(x) = \sqrt{\frac{1}{L}} \sin kx \quad [kL = n\pi \quad (n = 1, 2, 3, \dots)] \quad (1.92)$$

that corresponds to an eigenvalue $\lambda = (2n)^2 \pi^2 / 4L^2$ ($n = 1, 2, 3, \dots$).

Notice that arranging λ in ascending order, we have even functions and odd functions alternately as eigenfunctions corresponding to λ . Such a property is said to be *parity*. We often encounter it in quantum mechanics and related fields. From (1.61) we find that if $y(x)$ is an eigenfunction, so is $cy(x)$. That is, we should bear in mind that the eigenvalue problem is always accompanied by an indeterminate constant and that normalization of an eigenfunction does not mean the uniqueness of the solution (see Chap. 10).

Strictly speaking, we should be careful to assure that (1.81) holds on the basis of (1.80). It is because we have yet the possibility that k is a complex number. To see it, we examine zeros of a cosine function that is defined in a complex domain. Here the zeros are (complex) numbers to which the function takes zero. That is, if $f(z_0) = 0$, z_0 is called a zero (i.e., one of zeros) of $f(z)$. Now we have

$$\cos z \equiv \frac{1}{2} (e^{iz} + e^{-iz}); \quad z = x + iy \quad (x, y : \text{real}). \quad (1.93)$$

Inserting $z = x + iy$ in $\cos z$ and rearranging terms, we get

$$\cos z = \frac{1}{2} [\cos x(e^y + e^{-y}) + i \sin x(e^{-y} - e^y)]. \quad (1.94)$$

For $\cos z$ to vanish, both its real and imaginary parts must be zero. Since $e^y + e^{-y} > 0$ for all real numbers y , we must have $\cos x = 0$ for the real part to vanish; i.e.,

$$x = \frac{\pi}{2} + m\pi \quad (m = 0, \pm 1, \pm 2, \dots). \quad (1.95)$$

Note in this case that $\sin x = \pm 1 (\neq 0)$. Therefore, for the imaginary part to vanish, $e^{-y} - e^y = 0$. That is, we must have $y = 0$. Consequently, the zeros of $\cos z$ are real numbers. In other words, with respect to z_0 that satisfies $\cos z_0 = 0$ we have

$$z_0 = \frac{\pi}{2} + m\pi \quad (m = 0, \pm 1, \pm 2, \dots). \quad (1.96)$$

The above discussion equally applies to a sine function as well.

Thus, we ensure that k is a non-zero real number. Eigenvalues λ are positive definite from (1.68) accordingly. This conclusion is not fortuitous but a direct consequence of the form of a differential equation we have dealt with in combination with the BCs we imposed, i.e., the Dirichlet conditions. Detailed discussion will follow in Sects. 1.4, 10.3, and 10.4 in relation to Hermiticity of a differential operator.

Example 1.2 A particle confined within a potential well The results obtained in Example 1.1 can immediately be applied to dealing with a particle (electron) in a one-dimensional infinite potential well. In this case, (1.55) reads as

$$\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + E\psi(x) = 0, \quad (1.97)$$

where m is a mass of a particle and E is an energy of the particle. A potential V is expressed as

$$V(x) = \begin{cases} 0 & (-L \leq x \leq L), \\ \infty & (-L > x; \quad x > L). \end{cases}$$

Rewriting (1.97), we have

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0 \quad (1.98)$$

with BCs

$$\psi(L) = \psi(-L) = 0. \quad (1.99)$$

If we replace λ of (1.61) with $\frac{2mE}{\hbar^2}$, we can follow the procedures of Example 1.1. That is, we put

$$E = \frac{\hbar^2\lambda}{2m} \quad (1.100)$$

with $\lambda = k^2$ in (1.68). For k we use the values of (1.81) and (1.84). Therefore, with energy eigenvalues we get either

$$E = \frac{\hbar^2}{2m} \cdot \frac{(2l+1)^2\pi^2}{4L^2} \quad (l=0, 1, 2, \dots),$$

to which

$$\psi(x) = \sqrt{\frac{1}{L}} \cos kx \left[kL = \frac{\pi}{2} + l\pi \quad (l=0, 1, 2, \dots) \right] \quad (1.101)$$

corresponds or

$$E = \frac{\hbar^2}{2m} \cdot \frac{(2n)^2 \pi^2}{4L^2} \quad (n = 1, 2, 3, \dots),$$

to which

$$\psi(x) = \sqrt{\frac{1}{L}} \sin kx \quad [kL = n\pi \quad (n = 1, 2, 3, \dots)] \quad (1.102)$$

corresponds.

Since the particle behaves as a free particle within the potential well ($-L \leq x \leq L$) and $p = \hbar k$, we obtain

$$E = \frac{p^2}{2m} = \frac{\hbar^2}{2m} k^2,$$

where

$$k = \begin{cases} (2l+1)\pi/2L & (l = 0, 1, 2, \dots), \\ 2n\pi/2L & (n = 1, 2, 3, \dots). \end{cases}$$

The energy E is a kinetic energy of the particle.

Although in (1.97), $\psi(x) \equiv 0$ trivially holds, such a function may not be regarded as a solution of the eigenvalue problem. In fact, considering that $|\psi(x)|^2$ represents existence probability of a particle, $\psi(x) \equiv 0$ corresponds to a situation where a particle in question does not exist. Consequently, such a trivial case has physically no meaning.

1.4 Quantum-Mechanical Operators and Matrices

As represented by (1.55), a quantum-mechanical operator corresponds to a physical quantity. In (1.55), we connect a Hamiltonian operator to an energy (eigenvalue). Let us rephrase the situation as follows:

$$P\Psi = p\Psi. \quad (1.103)$$

In (1.103), we are viewing P as an operation or measurement on a physical system that is characterized by the quantum state Ψ . Operating P on the physical system (or state), we obtain a physical quantity p relevant to P as a result of the operation (or measurement).

A way to effectively achieve the above is to use a matrix and vector to represent the operation and physical state, respectively. Let us glance a little bit of matrix calculation to get used to the quantum-mechanical concept and, hence, to obtain

clear understanding about it. In Part III, we will deal with matrix calculation in detail from a point of view of a general principle. At present, a (2, 2) matrix suffices. Let A be a (2, 2) matrix expressed as

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \quad (1.104)$$

Let $|\psi\rangle$ be a (2, 1) matrix, i.e., a column vector such that

$$|\psi\rangle = \begin{pmatrix} e \\ f \end{pmatrix}. \quad (1.105)$$

Note that operating (2, 2) matrix on a (2, 1) matrix produces another (2, 1) matrix. Furthermore, we define an adjoint matrix $A^\dagger$ such that

$$A^\dagger = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}, \quad (1.106)$$

where a^* is a complex conjugate of a . That is, $A^\dagger$ is a complex conjugate transposed matrix of A . Also, we define an adjoint vector $\langle\psi|$ or $|\psi\rangle^\dagger$ such that

$$\langle\psi| \equiv |\psi\rangle^\dagger = (e^* \ f^*). \quad (1.107)$$

In this case, $|\psi\rangle^\dagger$ also denotes a complex conjugate transpose of $|\psi\rangle$. The notation $|\psi\rangle$ and $\langle\psi|$ are due to Dirac. He named $\langle\psi|$ and $|\varphi\rangle$ a *bra* vector and *ket* vector, respectively. This naming or equivoque comes from that $\langle\psi| \cdot |\varphi\rangle = \langle\psi|\varphi\rangle$ forms a bracket. This is a $(1, 2) \times (2, 1) = (1, 1)$ matrix, i.e., a *c-number* (including a complex number) and $\langle\psi|\varphi\rangle$ represents an inner product. These notations are widely used nowadays in the field of mathematics and physics.

Taking another vector $|\xi\rangle = \begin{pmatrix} g \\ h \end{pmatrix}$ and using a matrix calculation rule, we have

$$A^\dagger |\psi\rangle = |A^\dagger \psi\rangle = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix} \begin{pmatrix} e \\ f \end{pmatrix} = \begin{pmatrix} a^*e + c^*f \\ b^*e + d^*f \end{pmatrix}. \quad (1.108)$$

According to the definition (1.107), we have

$$|A^\dagger \psi\rangle^\dagger = \langle A^\dagger \psi| = (ae^* + cf^* \quad be^* + df^*). \quad (1.109)$$

Thus, we get

$$\langle A^\dagger \psi | \xi \rangle = (ae^* + cf^* \quad be^* + df^*) \begin{pmatrix} g \\ h \end{pmatrix} = (ag + bh)e^* + (cg + dh)f^*. \quad (1.110)$$

Similarly, we have

$$\langle \psi | A\xi \rangle = (e^* \quad f^*) \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} g \\ h \end{pmatrix} = (ag + bh)e^* + (cg + dh)f^*. \quad (1.111)$$

Comparing (1.110) and (1.111), we get

$$\langle A^\dagger \psi | \xi \rangle = \langle \psi | A\xi \rangle. \quad (1.112)$$

Also, we have

$$\langle \psi | A\xi \rangle^* = \langle A\xi | \psi \rangle. \quad (1.113)$$

Replacing A with $A^\dagger$ in (1.112), we get

$$\langle (A^\dagger)^\dagger \psi | \xi \rangle = \langle \psi | A^\dagger \xi \rangle. \quad (1.114)$$

From (1.104) and (1.106), obviously we have

$$(A^\dagger)^\dagger = A. \quad (1.115)$$

Then, from (1.114) and (1.115) we have

$$\langle A\psi | \xi \rangle = \langle \psi | A^\dagger \xi \rangle = \langle \xi | A\psi \rangle^*, \quad (1.116)$$

where the second equality comes from (1.113) obtained by exchanging ψ and ξ there. Moreover, we have a following relation:

$$(AB)^\dagger = B^\dagger A^\dagger. \quad (1.117)$$

The proof is left for readers. Using this relation, we have

$$\langle A\psi | = |A\psi\rangle^\dagger = [A|\psi\rangle]^\dagger = |\psi\rangle^\dagger A^\dagger = \langle \psi | A^\dagger = \langle \psi A^\dagger |. \quad (1.118)$$

Making an inner product by multiplying $|\xi\rangle$ from the right of the leftmost and rightmost sides of (1.118) and using (1.116), we get

$$\langle A\psi|\xi\rangle = \langle \psi A^\dagger|\xi\rangle = \langle \psi|A^\dagger\xi\rangle.$$

This relation may be regarded as the associative law with regard to the symbol “|” of the inner product. This is equivalent to the associative law with regard to the matrix multiplication.

The results obtained above can readily be extended to a general case where (n, n) matrices are dealt with.

Now, let us introduce an Hermitian operator (or matrix) H . When we have

$$H^\dagger = H, \quad (1.119)$$

H is called an Hermitian matrix. Then, applying (1.112) to the Hermitian matrix H , we have

$$\langle H^\dagger\psi|\xi\rangle = \langle \psi|H\xi\rangle = \langle \psi|H^\dagger\xi\rangle \quad \text{or} \quad \langle H\psi|\xi\rangle = \langle \psi|H^\dagger\xi\rangle = \langle \psi|H\xi\rangle. \quad (1.120)$$

Also let us introduce a norm of a vector $|\psi\rangle$ such that

$$\|\psi\| = \sqrt{\langle \psi|\psi\rangle}. \quad (1.121)$$

A norm is a natural extension for a notion of a “length” of a vector. The norm $\|\psi\|$ is zero, if and only if $|\psi\rangle = 0$ (zero vector). For, from (1.105) and (1.107), we have

$$\langle \psi|\psi\rangle = |e|^2 + |f|^2.$$

Therefore, $\langle \psi|\psi\rangle = 0 \Leftrightarrow e = f = 0$ in (1.105), i.e., $|\psi\rangle = 0$.

Let us further consider an eigenvalue problem represented by our newly introduced notation. The eigenvalue equation is symbolically written as

$$H|\psi\rangle = \lambda|\psi\rangle, \quad (1.122)$$

where H represents an Hermitian operator and $|\psi\rangle$ is an eigenfunction that belongs to an eigenvalue λ . Operating $\langle\psi|$ on (1.122) from the left, we have

$$\langle \psi|H|\psi\rangle = \langle \psi|\lambda|\psi\rangle = \lambda\langle \psi|\psi\rangle = \lambda, \quad (1.123)$$

where we assume that $|\psi\rangle$ is normalized; namely $\langle \psi|\psi\rangle = 1$ or $\|\psi\| = 1$. Notice that the symbol “|” in an inner product is of secondary importance. We may disregard this notation as in the case where a product notation “ $\times$ ” is omitted by denoting ab instead of $a \times b$.

Taking a complex conjugate of (1.123), we have

$$\langle \psi | H \psi \rangle^* = \lambda^*. \quad (1.124)$$

Using (1.116) and (1.124), we have

$$\lambda^* = \langle \psi | H \psi \rangle^* = \langle \psi | H^\dagger \psi \rangle = \langle \psi | H \psi \rangle = \lambda, \quad (1.125)$$

where with the third equality we used the definition (1.119) for the Hermitian operator. Hence, the relation $\lambda^* = \lambda$ obviously shows that any eigenvalue λ is real, if H is Hermitian. The relation (1.125) immediately tells us that even though $|\psi\rangle$ is not an eigenfunction, $\langle \psi | H \psi \rangle$ is real as well, if H is Hermitian. The quantity $\langle \psi | H \psi \rangle$ is said to be an expectation value. This value is interpreted as the most probable or averaged value of H obtained as a result of operation of H on a physical state $|\psi\rangle$. We sometimes denote the expectation value as

$$\langle H \rangle \equiv \langle \psi | H \psi \rangle, \quad (1.126)$$

where $|\psi\rangle$ is normalized. Unless $|\psi\rangle$ is normalized, it can be normalized on the basis of (1.121) by choosing $|\Phi\rangle$ such that

$$|\Phi\rangle = |\psi\rangle / \|\psi\|. \quad (1.127)$$

Thus, we have an important consequence; if an Hermitian operator has an eigenvalue, it must be real. An expectation value of an Hermitian operator is real as well. The real eigenvalue and expectation value are a prerequisite for a physical quantity.

As discussed above, the Hermitian matrices play a central role in quantum physics. Taking a further step, let us extend the notion of Hermiticity to a function space.

In Example 1.1, we have remarked that we have finally reached a solution where λ is a real (and positive) number, even though at the beginning we set no restriction on λ . This is because the SOLDE form (1.61) accompanied by BCs (1.62) is Hermitian, and so eigenvalues λ are real.

In this context, we give a little bit of further consideration. We define an inner product between two functions as follows:

$$\langle g | f \rangle \equiv \int_a^b g(x)^* f(x) dx, \quad (1.128)$$

where $g(x)^*$ is a complex conjugate of $g(x)$; x is a real variable and an integration range can be either bounded or unbounded. If a and b are real definite numbers, $[a, b]$ is the bounded case. With the unbounded case, we have, e.g., $(-\infty, \infty)$, $(-\infty, c)$, and (c, ∞) , etc. where c is a definite number. This notation will appear again in Chap. 10. In (1.128) we view functions f and g as vectors in a function space, often

referred to as a Hilbert space. We assume that any function f is square-integrable; i.e., $\int_a^b |f|^2$ is finite. That is,

$$\int_a^b |f(x)|^2 dx < \infty. \quad (1.129)$$

Using the above definition, let us calculate $\langle g | Df \rangle$, where D was defined in (1.63). Then, using the integration by parts, we have

$$\begin{aligned} \langle g | Df \rangle &= \int_a^b g(x)^* \left[-\frac{d^2 f(x)}{dx^2} \right] dx = -[g^* f']_a^b + \int_a^b g^* f' dx \\ &= -[g^* f']_a^b + [g'^* f]_a^b - \int_a^b g^{*''} f dx = [g'^* f - g^* f']_a^b + \int_a^b (-g^{*''} f) dx \\ &= [g'^* f - g^* f']_a^b + \langle Dg | f \rangle. \end{aligned} \quad (1.130)$$

If we have BCs such that

$$\begin{aligned} f(b) = f(a) = 0 \quad \text{and} \\ g(b)^* = g(a)^* = 0 \quad \text{i.e.,} \quad g(b) = g(a) = 0, \end{aligned} \quad (1.131)$$

we get

$$\langle g | Df \rangle = \langle Dg | f \rangle. \quad (1.132)$$

In light of (1.120), (1.132) implies that D is Hermitian. In (1.131), notice that the functions f and g satisfy the same BCs. Normally, for an operator to be Hermitian has this property. Thus, the Hermiticity of a differential operator is closely related to BCs of the differential equation.

Next, we consider a following inner product:

$$\langle f | Df \rangle = - \int_a^b f^* f'' dx = -[f^* f']_a^b + \int_a^b f^{*'} f' dx = -[f^{*'} f]_a^b + \int_a^b |f'|^2 dx. \quad (1.133)$$

Note that the definite integral of (1.133) cannot be negative. There are two possibilities for D to be Hermitian according to different BCs.

1. Dirichlet conditions: $f(b) = f(a) = 0$. If we could have $f' = 0$, $\langle f | Df \rangle$ would be zero. But, in that case f should be constant. If so, $f(x) \equiv 0$ according to BCs. We must exclude this trivial case. Consequently, to avoid this situation we must have

$$\int_a^b |f'|^2 dx > 0 \quad \text{or} \quad \langle f | Df \rangle > 0. \quad (1.134)$$

In this case, the operator D is said to be positive definite. Suppose that such a positive-definite operator has an eigenvalue λ . Then, for a corresponding eigenfunction $y(x)$ we have

$$Dy(x) = \lambda y(x). \quad (1.135)$$

In this case, we state that $y(x)$ is an eigenfunction or eigenvector that corresponds (or belongs) to an eigenvalue λ . Taking an inner product of both sides, we have

$$\langle y | Dy \rangle = \langle y | \lambda y \rangle = \lambda \langle y | y \rangle = \lambda \|y\|^2 \quad \text{or} \quad \lambda = \langle y | Dy \rangle / \|y\|^2. \quad (1.136)$$

Both $\langle y | Dy \rangle$ and $\|y\|^2$ are positive and, hence, we have $\lambda > 0$. Thus, if D has an eigenvalue, it must be positive. In this case, λ is said to be positive definite as well; see Example 1.1.

2. Neumann conditions: $f'(b) = f'(a) = 0$. From (1.130), D is Hermitian as well. Unlike the condition (1), however, f may be a non-zero constant in this case. Therefore, we are allowed to have

$$\int_a^b |f'|^2 dx = 0 \quad \text{or} \quad \langle f | Df \rangle = 0. \quad (1.137)$$

For any function, we have

$$\langle f | Df \rangle \geq 0. \quad (1.138)$$

In this case, the operator D is said to be *non-negative* (or *positive semi-definite*). The eigenvalue may be zero from (1.136) and, hence, is called non-negative accordingly.

3. Periodic conditions: $f(b) = f(a)$ and $f'(b) = f'(a)$. We are allowed to have $\langle f | Df \rangle \geq 0$ as in the case of the condition (2). Then, the operator and eigenvalues are non-negative.

Thus, in spite of being formally the same operator, that operator behaves differently according to the different BCs. In particular, for a differential operator to be associated with an eigenvalue of zero produces a special interest. We will encounter another illustration in Chap. 3.

1.5 Commutator and Canonical Commutation Relation

In quantum mechanics it is important whether two operators A and B are commutable. In this context, a commutator between A and B is defined such that

$$[A, B] \equiv AB - BA. \quad (1.139)$$

If $[A, B] = 0$ (zero matrix), A and B are said to be commutable (or commutative). If $[A, B] \neq 0$, A and B are non-commutative. Such relationships between two operators are called commutation relation.

We have canonical commutation relation as an underlying concept of quantum mechanics. This is defined between a (canonical) coordinate q and a (canonical) momentum p such that

$$[q, p] = i\hbar, \quad (1.140)$$

where the presence of a unit matrix E is implied. Explicitly writing it, we have

$$[q, p] = i\hbar E. \quad (1.141)$$

The relations (1.140) and (1.141) are called the canonical commutation relation. On the basis of a relation $p = \frac{\hbar}{i} \frac{\partial}{\partial q}$, a brief proof for this is as follows:

$$\begin{aligned} [qp]|\psi\rangle &= (qp - pq)|\psi\rangle \\ &= \left(q \frac{\hbar}{i} \frac{\partial}{\partial q} - \frac{\hbar}{i} \frac{\partial}{\partial q} q \right) |\psi\rangle = q \frac{\hbar}{i} \frac{\partial}{\partial q} |\psi\rangle - \frac{\hbar}{i} \frac{\partial}{\partial q} (q|\psi\rangle) \\ &= q \frac{\hbar}{i} \frac{\partial}{\partial q} |\psi\rangle - \frac{\hbar}{i} \frac{\partial q}{\partial q} |\psi\rangle - \frac{\hbar}{i} q \frac{\partial}{\partial q} |\psi\rangle = -\frac{\hbar}{i} |\psi\rangle = i\hbar |\psi\rangle. \end{aligned} \quad (1.142)$$

Since $|\psi\rangle$ is an arbitrarily chosen vector, we have (1.140).

Using (1.117), we have

$$[A, B]^\dagger = (AB - BA)^\dagger = B^\dagger A^\dagger - A^\dagger B^\dagger. \quad (1.143)$$

If in (1.143) A and B are both Hermitian, we have

$$[A, B]^\dagger = BA - AB = -[A, B]. \quad (1.144)$$

If we have an operator G such that

$$G^\dagger = -G, \quad (1.145)$$

G is said to be anti-Hermitian. Therefore, $[A, B]$ is anti-Hermitian, if both A and B are Hermitian. If an anti-Hermitian operator has an eigenvalue, that eigenvalue is zero or pure imaginary. To show this, suppose that

$$G | \psi \rangle = \lambda | \psi \rangle, \quad (1.146)$$

where G is an anti-Hermitian operator and $|\psi\rangle$ has been normalized. As in the case of (1.123) we have

$$\langle \psi | G | \psi \rangle = \lambda \langle \psi | \psi \rangle = \lambda. \quad (1.147)$$

Taking a complex conjugate of (1.147), we have

$$\langle \psi | G | \psi \rangle^* = \lambda^*. \quad (1.148)$$

Using (1.116) and (1.145) again, we have

$$\lambda^* = \langle \psi | G | \psi \rangle^* = \langle \psi | G^\dagger | \psi \rangle = - \langle \psi | G | \psi \rangle = -\lambda. \quad (1.149)$$

This shows that λ is zero or pure imaginary.

Therefore, (1.142) can be viewed as an eigenvalue equation to which any physical state $|\psi\rangle$ has a pure imaginary eigenvalue $i\hbar$ with respect to $[q, p]$. Note that both q and p are Hermitian (see Sect. 10.2, Example 10.3), and so $[q, p]$ is anti-Hermitian as mentioned above. The canonical commutation relation given by (1.140) is believed to underpin the uncertainty principle.

In quantum mechanics, it is of great importance whether a quantum operator is Hermitian or not. A position operator and momentum operator along with an angular momentum operator are particularly important when we constitute Hamiltonian. Let f and g be arbitrary functions. Let us consider, e.g., a following inner product with the momentum operator.

$$\langle g | pf \rangle = \int_a^b g(x)^* \frac{\hbar}{i} \frac{\partial}{\partial x} [f(x)] dx, \quad (1.150)$$

where the domain $[a, b]$ depends on a physical system; this can be either bounded or unbounded. Performing integration by parts, we have

$$\begin{aligned} \langle g | pf \rangle &= \frac{\hbar}{i} [g(x)^* f(x)]_a^b - \int_a^b \frac{\partial}{\partial x} [g(x)^*] \frac{\hbar}{i} f(x) dx \\ &= \frac{\hbar}{i} [g(b)^* f(b) - g(a)^* f(a)] + \int_a^b \left[\frac{\hbar}{i} \frac{\partial}{\partial x} g(x) \right]^* f(x) dx. \end{aligned} \quad (1.151)$$

If we require $f(b) = f(a)$ and $g(b) = g(a)$, the first term vanishes and we get

$$\langle g|pf\rangle = \int_a^b \left[\frac{\hbar}{i} \frac{\partial}{\partial x} g(x) \right]^* f(x) dx = \langle pg|f\rangle. \quad (1.152)$$

Thus, as in the case of (1.120), the momentum operator p is Hermitian. Note that a position operator q of (1.142) is Hermitian as a priori assumption.

Meanwhile, the z -component of angular momentum operator L_z is described in a polar coordinate as follows:

$$L_z = \frac{\hbar}{i} \frac{\partial}{\partial \phi}, \quad (1.153)$$

where ϕ is an azimuthal angle varying from 0 to 2π . The notation and implication of L_z will be mentioned in Chap. 3. Similarly as the above, we have

$$\langle g|L_z f\rangle = \frac{\hbar}{i} [g(2\pi)^* f(2\pi) - g(0)^* f(0)] + \int_0^{2\pi} \left[\frac{\hbar}{i} \frac{\partial}{\partial \phi} g(x) \right]^* f(x) d\phi. \quad (1.154)$$

Requiring an arbitrary function f to satisfy a BC $f(2\pi) = f(0)$, we reach

$$\langle g|L_z f\rangle = \langle L_z g|f\rangle. \quad (1.155)$$

Note that we must have the above BC, because $\phi = 0$ and $\phi = 2\pi$ are spatially the same point. Thus, we find that L_z is Hermitian as well on this condition.

On the basis of aforementioned argument, let us proceed to quantum-mechanical studies of a harmonic oscillator. Regarding the angular momentum, we will study their basic properties in Chap. 3.

Reference

1. Møller C (1952) The theory of relativity. Oxford University Press, London

Chapter 2

Quantum-Mechanical Harmonic Oscillator



Quantum-mechanical treatment of a harmonic oscillator has been a well-studied topic from the beginning of the history of quantum mechanics. This topic is a standard subject in classical mechanics as well. In this chapter, first we briefly survey characteristics of a classical harmonic oscillator. From a quantum-mechanical point of view, we deal with features of a harmonic oscillator through matrix representation. We define creation and annihilation operators using position and momentum operators. A Hamiltonian of the oscillator is described in terms of the creation and annihilation operators. This enables us to easily determine energy eigenvalues of the oscillator. As a result, energy eigenvalues are found to be positive definite. Meanwhile, we express the Schrödinger equation by the coordinate representation. We compare the results with those of the matrix representation and show that the two representations are mathematically equivalent. Thus, the treatment of the quantum-mechanical harmonic oscillator supplies us with a firm ground for studying basic concepts of the quantum mechanics.

2.1 Classical Harmonic Oscillator

Classical Newtonian equation of a one-dimensional harmonic oscillator is expressed as

$$m \frac{d^2 x(t)}{dt^2} = -sx(t), \quad (2.1)$$

where m is a mass of an oscillator and s is a spring constant. Putting $s/m = \omega^2$, we have

$$\frac{d^2x(t)}{dt^2} + \omega^2 x(t) = 0. \quad (2.2)$$

In (2.2) we set ω positive, namely,

$$\omega = \sqrt{s/m}, \quad (2.3)$$

where ω is called an angular frequency of the oscillator.

If we replace ω^2 with λ , we have formally the same equation as (1.61). Two linearly independent solutions of (2.2) are the same as before (see Example 1.1); we have $e^{i\omega t}$ and $e^{-i\omega t}$ ($\omega \neq 0$) as such. Note, however, that in Example 1.2 we were dealing with a quantum state related to existence probability of a particle in a potential well. In (2.2), however, we are examining a position of harmonic oscillator undergoing a force of a spring. We are thus considering a different situation.

As a general solution we have

$$x(t) = ae^{i\omega t} + be^{-i\omega t}, \quad (2.4)$$

where a and b are suitable constants. Let us consider BCs different from those of Example 1.1 or 1.2 this time. That is, we set BCs such that

$$x(0) = 0 \quad \text{and} \quad x'(0) = v_0 \quad (v_0 > 0). \quad (2.5)$$

Notice that (2.5) gives initial conditions (ICs). Mathematically, ICs are included in BCs (see Chap. 10). From (2.4) we have

$$x(t) = a + b = 0 \quad \text{and} \quad x'(0) = i\omega(a - b) = v_0. \quad (2.6)$$

Then, we get $a = -b = v_0/2i\omega$. Thus, we get a simple harmonic motion as a solution expressed as

$$x(t) = \frac{v_0}{2i\omega} (e^{i\omega t} - e^{-i\omega t}) = \frac{v_0}{\omega} \sin \omega t. \quad (2.7)$$

From this, we have

$$E = K + V = \frac{1}{2}mv_0^2. \quad (2.8)$$

In particular, if $v_0 = 0$, $x(t) \equiv 0$. This is a solution of (2.1) that has the meaning that the particle is eternally at rest. It is physically acceptable as well. Notice also that unlike Examples 1.1 and 1.2, the solution has been determined uniquely. This is due to the different BCs.

From a point of view of a mechanical system, mathematical formulation of the classical harmonic oscillator resembles that of electromagnetic fields confined within a cavity. We return this point later in Sect. 9.6.

2.2 Formulation Based on an Operator Method

Now let us return to our task to find quantum-mechanical solutions of a harmonic oscillator. Potential V is given by

$$V(q) = \frac{1}{2}sq^2 = \frac{1}{2}m\omega^2q^2, \quad (2.9)$$

where q is used for a one-dimensional position coordinate. Then, we have a classical Hamiltonian H expressed as

$$H = \frac{p^2}{2m} + V(q) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2q^2. \quad (2.10)$$

Following the formulation of Sect. 1.2, the Schrödinger equation as an eigenvalue equation related to energy E is described as

$$H\psi(q) = E\psi(q)$$

or

$$-\frac{\hbar^2}{2m}\nabla^2\psi(q) + \frac{1}{2}m\omega^2q^2\psi(q) = E\psi(q). \quad (2.11)$$

This is a SOLDE and it is well known that the SOLDE can be solved by a power series expansion method.

In the present studies, however, let us first use an operator method to solve the eigenvalue equation (2.11) of a one-dimensional oscillator. To this end, we use a quantum-mechanical Hamiltonian where a momentum operator p is explicitly represented. Thus, the Hamiltonian reads as

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2q^2. \quad (2.12)$$

Equation (2.12) is formally the same as (2.10). Note, however, that in (2.12) p and q are expressed as quantum-mechanical operators.

As in (1.126), we first examine an expectation value $\langle H \rangle$ of H . It is given by

$$\begin{aligned}
\langle H \rangle &= \langle \psi | H \psi \rangle = \left\langle \psi \left| \frac{p^2}{2m} \right| \psi \right\rangle + \left\langle \psi \left| \frac{1}{2} m \omega^2 q^2 \right| \psi \right\rangle \\
&= \frac{1}{2m} \langle p^\dagger \psi | p \psi \rangle + \frac{1}{2} m \omega^2 \langle q^\dagger \psi | q \psi \rangle = \frac{1}{2m} \langle p \psi | p \psi \rangle + \frac{1}{2} m \omega^2 \langle q \psi | q \psi \rangle \\
&= \frac{1}{2m} \|p \psi\|^2 + \frac{1}{2} m \omega^2 \|q \psi\|^2 \geq 0,
\end{aligned} \tag{2.13}$$

where again we assumed that $|\psi\rangle$ has been normalized. In (2.13) we used the notation (1.126) and the fact that both q and p are Hermitian. In this situation, $\langle H \rangle$ takes a non-negative value.

In (2.13), the equality holds if and only if $|p \psi\rangle = 0$ and $|q \psi\rangle = 0$. Let us specify a vector $|\psi_0\rangle$ that satisfies these conditions such that

$$|p \psi_0\rangle = 0 \quad \text{and} \quad |q \psi_0\rangle = 0. \tag{2.14}$$

Multiplying q from the left on the first equation of (2.14) and multiplying p from the left on the second equation, we have

$$qp |\psi_0\rangle = 0 \quad \text{and} \quad pq |\psi_0\rangle = 0. \tag{2.15}$$

Subtracting the second equation of (2.15) from the first equation, we get

$$(qp - pq) |\psi_0\rangle = i\hbar |\psi_0\rangle = 0, \tag{2.16}$$

where with the first equality we used (1.140). Therefore, we would have $|\psi_0(q)\rangle \equiv 0$. This leads to the relations (2.14). That is, if and only if $|\psi_0(q)\rangle \equiv 0$, $\langle H \rangle = 0$. But, since it has no physical meaning, $|\psi_0(q)\rangle \equiv 0$ must be rejected as unsuitable for the solution of (2.11). Regarding a physically acceptable solution of (2.13), $\langle H \rangle$ must take a *positive definite* value accordingly. Thus, on the basis of the canonical commutation relation, we restrict the range of the expectation values.

Instead of directly dealing with (2.12), it is well known to introduce the following operators [1]:

$$a \equiv \sqrt{\frac{m\omega}{2\hbar}} q + \frac{i}{\sqrt{2m\hbar\omega}} p \tag{2.17}$$

and its adjoint (complex conjugate) operator

$$a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} q - \frac{i}{\sqrt{2m\hbar\omega}} p. \tag{2.18}$$

Notice here again that both q and p are Hermitian. Using a matrix representation for (2.17) and (2.18), we have

$$\begin{pmatrix} a \\ a^\dagger \end{pmatrix} = \begin{pmatrix} \sqrt{\frac{m\omega}{2\hbar}} & \frac{i}{\sqrt{2m\hbar\omega}} \\ \sqrt{\frac{m\omega}{2\hbar}} & -\frac{i}{\sqrt{2m\hbar\omega}} \end{pmatrix} \begin{pmatrix} q \\ p \end{pmatrix}. \quad (2.19)$$

Then we have

$$\begin{aligned} a^\dagger a &= (2m\hbar\omega)^{-1} (m\omega q - ip)(m\omega q + ip) \\ &= (2m\hbar\omega)^{-1} [m^2\omega^2 q^2 + p^2 + im\omega(qp - pq)] \\ &= (\hbar\omega)^{-1} \left(\frac{1}{2} m\omega^2 q^2 + \frac{1}{2} p^2 + \frac{1}{2} i\omega i\hbar \right) = (\hbar\omega)^{-1} \left(H - \frac{1}{2} \hbar\omega \right), \end{aligned} \quad (2.20)$$

where the second last equality comes from (1.140). Rewriting (2.20), we get

$$H = \hbar\omega a^\dagger a + \frac{1}{2} \hbar\omega. \quad (2.21)$$

Similarly we get

$$H = \hbar\omega a a^\dagger - \frac{1}{2} \hbar\omega. \quad (2.22)$$

Subtracting (2.22) from (2.21), we have

$$0 = \hbar\omega a^\dagger a - \hbar\omega a a^\dagger + \hbar\omega. \quad (2.23)$$

That is,

$$[a, a^\dagger] = 1 \text{ or } [a, a^\dagger] = E, \quad (2.24)$$

where E represents an identity operator.

Furthermore, using (2.21) we have

$$[H, a^\dagger] = \hbar\omega \left[a^\dagger a + \frac{1}{2}, a^\dagger \right] = \hbar\omega (a^\dagger a a^\dagger - a^\dagger a^\dagger a) = \hbar\omega a^\dagger [a, a^\dagger] = \hbar\omega a^\dagger. \quad (2.25)$$

Similarly, we get

$$[H, a] = -\hbar\omega a. \quad (2.26)$$

Next, let us calculate an expectation value of H . Using a normalized function $|\psi\rangle$, from (2.21) we have

$$\begin{aligned}
\langle \psi | H | \psi \rangle &= \left\langle \psi \left| \hbar \omega a^\dagger a + \frac{1}{2} \hbar \omega \right| \psi \right\rangle = \hbar \omega \langle \psi | a^\dagger a | \psi \rangle + \frac{1}{2} \hbar \omega \langle \psi | \psi \rangle \\
&= \hbar \omega \langle a \psi | a \psi \rangle + \frac{1}{2} \hbar \omega = \hbar \omega \|a \psi\|^2 + \frac{1}{2} \hbar \omega \geq \frac{1}{2} \hbar \omega.
\end{aligned} \tag{2.27}$$

Thus, the expectation value is equal to or larger than $\frac{1}{2} \hbar \omega$. This is consistent with that an energy eigenvalue is positive definite as mentioned above. Equation (2.27) also tells us that if we have

$$|a \psi_0\rangle = 0, \tag{2.28}$$

we get

$$\langle \psi_0 | H | \psi_0 \rangle = \frac{1}{2} \hbar \omega. \tag{2.29}$$

Equation (2.29) means that the smallest expectation value is $\frac{1}{2} \hbar \omega$ on the condition of (2.28). On the same condition, using (2.21) we have

$$H | \psi_0 \rangle = \hbar \omega a^\dagger a | \psi_0 \rangle + \frac{1}{2} \hbar \omega | \psi_0 \rangle = \frac{1}{2} \hbar \omega | \psi_0 \rangle. \tag{2.30}$$

Thus, $|\psi_0\rangle$ is an eigenfunction corresponding to an eigenvalue $\frac{1}{2} \hbar \omega \equiv E_0$, which is identical with the smallest expectation value of (2.29). Since this is the lowest eigenvalue, $|\psi_0\rangle$ is said to be a ground state. We ensure later that $|\psi_0\rangle$ is certainly an eligible function for a ground state.

The above method is consistent with the variational principle [2] which stipulates that under appropriate BCs an expectation value of Hamiltonian estimated with any arbitrary function is always larger than or equal to the smallest eigenvalue corresponding to the ground state.

Next, let us evaluate energy eigenvalues of the oscillator. First we have

$$H | \psi_0 \rangle = \frac{1}{2} \hbar \omega | \psi_0 \rangle = E_0 | \psi_0 \rangle. \tag{2.31}$$

Operating $a^\dagger$ on both sides of (2.31) from the left, we have

$$a^\dagger H | \psi_0 \rangle = a^\dagger E_0 | \psi_0 \rangle. \tag{2.32}$$

Meanwhile, using (2.25), we have

$$a^\dagger H | \psi_0 \rangle = (H a^\dagger - \hbar \omega a^\dagger) | \psi_0 \rangle. \tag{2.33}$$

Equating RHSs of (2.32) and (2.33), we get

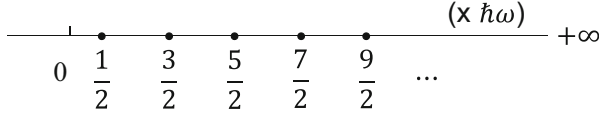


Fig. 2.1 Energy eigenvalues of a quantum-mechanical harmonic oscillator on a real axis

$$Ha^\dagger |\psi_0\rangle = (E_0 + \hbar\omega)a^\dagger |\psi_0\rangle. \quad (2.34)$$

This implies that $a^\dagger |\psi_0\rangle$ belongs to an eigenvalue $(E_0 + \hbar\omega)$, which is larger than E_0 as expected. Again multiplying $a^\dagger$ on both sides of (2.34) from the left and using (2.25), we get

$$H(a^\dagger)^2 |\psi_0\rangle = (E_0 + 2\hbar\omega)(a^\dagger)^2 |\psi_0\rangle. \quad (2.35)$$

This implies that $(a^\dagger)^2 |\psi_0\rangle$ belongs to an eigenvalue $(E_0 + 2\hbar\omega)$. Thus, repeatedly taking the above procedures, we get

$$H(a^\dagger)^n |\psi_0\rangle = (E_0 + n\hbar\omega)(a^\dagger)^n |\psi_0\rangle. \quad (2.36)$$

Thus, $(a^\dagger)^n |\psi_0\rangle$ belongs to an eigenvalue

$$E_n \equiv E_0 + n\hbar\omega = \left(n + \frac{1}{2}\right)\hbar\omega, \quad (2.37)$$

where E_n denotes an energy eigenvalue of the n -th excited state. The energy eigenvalues are plotted in Fig. 2.1.

Our next task is to seek normalized eigenvectors of the n -th excited state. Let c_n be a normalization constant of that state. That is, we have

$$|\psi_n\rangle = c_n (a^\dagger)^n |\psi_0\rangle, \quad (2.38)$$

where $|\psi_n\rangle$ is a normalized eigenfunction of the n -th excited state. To determine c_n , let us calculate $a |\psi_n\rangle$. This includes a factor $a(a^\dagger)^n$. We have

$$\begin{aligned}
a(a^\dagger)^n &= (aa^\dagger - a^\dagger a)(a^\dagger)^{n-1} + a^\dagger a(a^\dagger)^{n-1} \\
&= [a, a^\dagger](a^\dagger)^{n-1} + a^\dagger a(a^\dagger)^{n-1} = (a^\dagger)^{n-1} + a^\dagger a(a^\dagger)^{n-1} \\
&= (a^\dagger)^{n-1} + a^\dagger [a, a^\dagger](a^\dagger)^{n-2} + (a^\dagger)^2 a(a^\dagger)^{n-2} \\
&= 2(a^\dagger)^{n-1} + (a^\dagger)^2 a(a^\dagger)^{n-2} \\
&= 2(a^\dagger)^{n-1} + (a^\dagger)^2 [a, a^\dagger](a^\dagger)^{n-3} + (a^\dagger)^3 a(a^\dagger)^{n-3} \\
&= 3(a^\dagger)^{n-1} + (a^\dagger)^3 a(a^\dagger)^{n-3} \\
&= \cdots.
\end{aligned} \tag{2.39}$$

In the above procedures, we used $[a, a^\dagger] = 1$. What is implied in (2.39) is that a coefficient of $(a^\dagger)^{n-1}$ increased one by one with a transferred toward the right one by one in the second term of RHS. Notice that in the second term a is sandwiched such that $(a^\dagger)^m a (a^\dagger)^{n-m}$ ($m = 1, 2, \cdots$). Finally, we have

$$a(a^\dagger)^n = n(a^\dagger)^{n-1} + (a^\dagger)^n a. \tag{2.40}$$

Meanwhile, operating a on both sides of (2.38) from the left and using (2.40), we get

$$\begin{aligned}
a|\psi_n\rangle &= c_n a(a^\dagger)^n |\psi_0\rangle = c_n [n(a^\dagger)^{n-1} + (a^\dagger)^n a] |\psi_0\rangle = c_n n(a^\dagger)^{n-1} |\psi_0\rangle \\
&= n \frac{c_n}{c_{n-1}} c_{n-1} (a^\dagger)^{n-1} |\psi_0\rangle = n \frac{c_n}{c_{n-1}} |\psi_{n-1}\rangle,
\end{aligned} \tag{2.41}$$

where the third equality comes from (2.28).

Next, operating a on (2.40) we obtain

$$\begin{aligned}
a^2(a^\dagger)^n &= na(a^\dagger)^{n-1} + a(a^\dagger)^n a \\
&= n[(n-1)(a^\dagger)^{n-2} + (a^\dagger)^{n-1} a] + a(a^\dagger)^n a \\
&= n(n-1)(a^\dagger)^{n-2} + n(a^\dagger)^{n-1} a + a(a^\dagger)^n a.
\end{aligned} \tag{2.42}$$

Operating another a on (2.42), we get

$$\begin{aligned}
a^3(a^\dagger)^n &= n(n-1)a(a^\dagger)^{n-2} + na(a^\dagger)^{n-1}a + a^2(a^\dagger)^na \\
&= n(n-1)\left[(n-2)(a^\dagger)^{n-3} + (a^\dagger)^{n-2}a\right] + na(a^\dagger)^{n-1}a + a^2(a^\dagger)^na \\
&= n(n-1)(n-2)(a^\dagger)^{n-3} + n(n-1)(a^\dagger)^{n-2}a + na(a^\dagger)^{n-1}a + a^2(a^\dagger)^na \\
&= \dots
\end{aligned} \tag{2.43}$$

To generalize the above procedures, operating a on (2.40) m ($< n$) times we get

$$a^m(a^\dagger)^n = n(n-1)(n-2)\cdots(n-m+1)(a^\dagger)^{n-m} + f(a, a^\dagger)a, \tag{2.44}$$

where $m < n$ and $f(a, a^\dagger)$ is a polynomial of $a^\dagger$ that has a power of a as a coefficient. Further operating $\langle\psi_0|$ and $|\psi_0\rangle$ from the left and right on both sides of (2.44), respectively, we have

$$\begin{aligned}
\langle\psi_0|a^m(a^\dagger)^n|\psi_0\rangle &= n(n-1)(n-2)\cdots(n-m+1)\langle\psi_0|(a^\dagger)^{n-m}|\psi_0\rangle \\
&\quad + \langle\psi_0|f(a, a^\dagger)a|\psi_0\rangle \\
&= n(n-1)(n-2)\cdots(n-m+1)\langle\psi_0|(a^\dagger)^{n-m}|\psi_0\rangle.
\end{aligned} \tag{2.45}$$

Note that in (2.45) $\langle\psi_0|f(a, a^\dagger)a|\psi_0\rangle$ vanishes because of (2.28). Meanwhile, taking adjoint of (2.28), we have

$$\langle\psi_0|a^\dagger = 0. \tag{2.46}$$

Operating $a^\dagger(n-m-1)$ times from the right of LHS of (2.46), we have

$$\langle\psi_0|(a^\dagger)^{n-m} = 0.$$

Further operating $|\psi_0\rangle$ from the right of the above equation, we obtain

$$\langle\psi_0|(a^\dagger)^{n-m}|\psi_0\rangle = 0.$$

Therefore, from (2.45) we get

$$\langle\psi_0|a^m(a^\dagger)^n|\psi_0\rangle = 0. \tag{2.47}$$

Taking adjoint of (2.47) once again, we have

$$\langle \psi_0 | a^n (a^\dagger)^m | \psi_0 \rangle = 0. \quad (2.48)$$

Equation (2.48) can be obtained by repeatedly using (1.117). From (2.47) and (2.48), we get

$$\langle \psi_0 | a^m (a^\dagger)^n | \psi_0 \rangle = 0, \quad (2.49)$$

when $m \neq n$. If $m = n$, from (2.45) we obtain

$$\langle \psi_0 | a^n (a^\dagger)^n | \psi_0 \rangle = n! \langle \psi_0 | \psi_0 \rangle. \quad (2.50)$$

If we assume that $|\psi_0\rangle$ is normalized; i.e., $\langle \psi_0 | \psi_0 \rangle = 1$, (2.50) is expressed as

$$\langle \psi_0 | a^n (a^\dagger)^n | \psi_0 \rangle = n!. \quad (2.51)$$

From (2.51), if we put

$$| \psi_n \rangle = \frac{1}{\sqrt{n!}} (a^\dagger)^n | \psi_0 \rangle, \quad (2.52)$$

then we have

$$\langle \psi_n | = \frac{1}{\sqrt{n!}} \langle \psi_0 | a^n.$$

Thus, from (2.49) and (2.52) we get

$$\langle \psi_m | \psi_n \rangle = \delta_{mn}. \quad (2.53)$$

At the same time, for c_n of (2.38) we get

$$c_n = \frac{1}{\sqrt{n!}}. \quad (2.54)$$

Notice here that an undetermined phase factor $e^{i\theta}$ (θ : real) is intended such that

$$c_n = \frac{1}{\sqrt{n!}} e^{i\theta}.$$

But $e^{i\theta}$ is usually omitted for the sake of simplicity. Thus, we have constructed a series of orthonormal eigenfunctions $|\psi_n\rangle$.

Furthermore, using (2.41) and (2.54) we obtain

$$a | \psi_n \rangle = \sqrt{n} | \psi_{n-1} \rangle. \quad (2.55)$$

From (2.36), we get

$$H | \psi_n \rangle = (E_0 + n\hbar\omega) | \psi_n \rangle. \quad (2.56)$$

Meanwhile, from (2.21) we have

$$H | \psi_n \rangle = (\hbar\omega a^\dagger a + E_0) | \psi_n \rangle. \quad (2.57)$$

Equating RHSs of (2.56) and (2.57), we get

$$a^\dagger a | \psi_n \rangle = n | \psi_n \rangle. \quad (2.58)$$

Thus, we find that an integer n is an eigenvalue of $a^\dagger a$ when it is evaluated with respect to $|\psi_n\rangle$. For this reason $a^\dagger a$ is called a number operator. Notice that $a^\dagger a$ is Hermitian because we have

$$(a^\dagger a)^\dagger = a^\dagger (a^\dagger)^\dagger = a^\dagger a, \quad (2.59)$$

where we used (1.115) and (1.117).

Moreover, from (2.55) and (2.58),

$$a^\dagger a | \psi_n \rangle = \sqrt{n} a^\dagger | \psi_{n-1} \rangle = n | \psi_n \rangle. \quad (2.60)$$

Thus,

$$a^\dagger | \psi_{n-1} \rangle = \sqrt{n} | \psi_n \rangle. \quad (2.61)$$

Or replacing n with $n + 1$, we get

$$a^\dagger | \psi_n \rangle = \sqrt{n+1} | \psi_{n+1} \rangle. \quad (2.62)$$

As implied in (2.55) and (2.62), we find that operating a on $|\psi_n\rangle$ lowers an energy level by one and that operating $a^\dagger$ on $|\psi_n\rangle$ raises an energy level by one. For this reason, a and $a^\dagger$ are said to be an annihilation operator and creation operator, respectively.

2.3 Matrix Representation of Physical Quantities

Equations (2.55) and (2.62) clearly represent the relationship between an operator and eigenfunction (or eigenvector). The relationship is characterized by

$$(\text{Matrix}) \times (\text{Vector}) = (\text{Vector}). \quad (2.63)$$

Thus, we are now in a position to construct this relation using matrices. From (2.53), we should be able to construct basis vectors using a *column vector* such that

$$|\psi_0\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ \vdots \end{pmatrix}, \quad |\psi_1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ \vdots \end{pmatrix}, \quad |\psi_2\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ \vdots \end{pmatrix}, \dots \quad (2.64)$$

Notice that these vectors form a vector space of an infinite dimension. The orthonormal relation (2.53) can easily be checked using (2.64). We represent a and $a^\dagger$ so that (2.55) and (2.62) can be satisfied. We obtain

$$a = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & \cdots \\ 0 & 0 & \sqrt{2} & 0 & 0 & \cdots \\ 0 & 0 & 0 & \sqrt{3} & 0 & \cdots \\ 0 & 0 & 0 & 0 & 2 & \cdots \\ 0 & 0 & 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \quad (2.65)$$

Similarly,

$$a^\dagger = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \cdots \\ 1 & 0 & 0 & 0 & 0 & \cdots \\ 0 & \sqrt{2} & 0 & 0 & 0 & \cdots \\ 0 & 0 & \sqrt{3} & 0 & 0 & \cdots \\ 0 & 0 & 0 & 2 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \quad (2.66)$$

Note that neither a nor $a^\dagger$ is Hermitian.

Since a determinant of the matrix of (2.19) is $-i/\hbar \neq 0$, using its inverse matrix we get

$$\begin{pmatrix} q \\ p \end{pmatrix} = \begin{pmatrix} \sqrt{\frac{\hbar}{2m\omega}} & \sqrt{\frac{\hbar}{2m\omega}} \\ \frac{1}{i}\sqrt{\frac{m\hbar\omega}{2}} & -\frac{1}{i}\sqrt{\frac{m\hbar\omega}{2}} \end{pmatrix} \begin{pmatrix} a \\ a^\dagger \end{pmatrix}. \quad (2.67)$$

That is, we have

$$q = \sqrt{\frac{\hbar}{2m\omega}}(a + a^\dagger) \quad \text{and} \quad p = \frac{1}{i}\sqrt{\frac{m\hbar\omega}{2}}(a - a^\dagger). \quad (2.68)$$

With the inverse matrix, we will deal with it in Part III. Note that q and p are both Hermitian. Inserting (2.65) and (2.66) into (2.68), we get

$$q = \sqrt{\frac{\hbar}{2m\omega}} \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & \cdots \\ 1 & 0 & \sqrt{2} & 0 & 0 & \cdots \\ 0 & \sqrt{2} & 0 & \sqrt{3} & 0 & \cdots \\ 0 & 0 & \sqrt{3} & 0 & 2 & \cdots \\ 0 & 0 & 0 & 2 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \quad (2.69)$$

$$p = \sqrt{\frac{m\hbar\omega}{2}}i \begin{pmatrix} 0 & -1 & 0 & 0 & 0 & \cdots \\ 1 & 0 & -\sqrt{2} & 0 & 0 & \cdots \\ 0 & \sqrt{2} & 0 & -\sqrt{3} & 0 & \cdots \\ 0 & 0 & \sqrt{3} & 0 & -2 & \cdots \\ 0 & 0 & 0 & 2 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \quad (2.70)$$

Equations (2.69) and (2.70) obviously show that q and p are Hermitian. We can derive various physical quantities from these equations. For instance, following matrix algebra Hamiltonian H can readily be calculated. The result is expressed as

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2 = \frac{\hbar\omega}{2} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 3 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 5 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 7 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 9 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \quad (2.71)$$

Looking at (2.69)–(2.71), we immediately realize that although neither q or p is diagonalized, H is diagonalized. The matrix representation of (2.71) is said to be a representation that diagonalizes H . This representation is of great practical use. In fact, using (2.64) we get, e.g.,

$$\begin{aligned} H|\psi_2\rangle &= \frac{\hbar\omega}{2} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 3 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 5 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 7 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 9 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ \vdots \end{pmatrix} = \frac{\hbar\omega}{2} \begin{pmatrix} 0 \\ 0 \\ 5 \\ 0 \\ 0 \\ \vdots \end{pmatrix} = \frac{5\hbar\omega}{2} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ \vdots \end{pmatrix} \\ &= \frac{5\hbar\omega}{2} |\psi_2\rangle = \left(\frac{1}{2} + 2\right)\hbar\omega |\psi_2\rangle. \end{aligned} \quad (2.72)$$

This clearly means that the second-excited state $|\psi_2\rangle$ has an eigenenergy $(\frac{1}{2} + 2)\hbar\omega$. More generally, we find that $|\psi_n\rangle$ has an eigenenergy $(\frac{1}{2} + n)\hbar\omega$ as already shown in (2.36) and (2.56).

Furthermore, let us confirm the canonical commutation relation of (1.140). Using (2.68), we have

$$\begin{aligned} qp - pq &= \frac{1}{i} \sqrt{\frac{\hbar}{2m\omega}} \sqrt{\frac{m\hbar\omega}{2}} [(a + a^\dagger)(a - a^\dagger) - (a - a^\dagger)(a + a^\dagger)] \\ &= \frac{\hbar}{2i} \cdot (-2) \cdot (aa^\dagger - a^\dagger a) = i\hbar[a, a^\dagger] = i\hbar = i\hbar E, \end{aligned} \quad (2.73)$$

where with the second last equality we used (2.24) and the identity matrix E of an infinite dimension is given by

$$E = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \quad (2.74)$$

Thus, the canonical commutation relation holds with the quantum-mechanical harmonic oscillator. This can be confirmed directly from (2.69) and (2.70). The proof is left for readers as an exercise.

2.4 Coordinate Representation of Schrödinger Equation

The Schrödinger equation has been given in (1.55) or (2.11) as a SOLDE form. In contrast to the matrix representation, (1.55) and (2.11) are said to be coordinate representation of Schrödinger equation. Now, let us derive a coordinate representation of (2.28) to obtain an analytical solution.

On the basis of (1.31) and (2.17), a is expressed as

$$a = \sqrt{\frac{m\omega}{2\hbar}}q + \frac{i}{\sqrt{2m\hbar\omega}}p = \sqrt{\frac{m\omega}{2\hbar}}q + \frac{i}{\sqrt{2m\hbar\omega}} \frac{\hbar}{i} \frac{\partial}{\partial q} = \sqrt{\frac{m\omega}{2\hbar}}q + \sqrt{\frac{\hbar}{2m\omega}} \times \frac{\partial}{\partial q}. \quad (2.75)$$

Thus, (2.28) reads as a following first-order linear differential equation (FOLDE):

$$\left(\sqrt{\frac{m\omega}{2\hbar}}q + \sqrt{\frac{\hbar}{2m\omega}} \frac{\partial}{\partial q} \right) \psi_0(q) = 0 \quad (2.76)$$

or

$$\sqrt{\frac{m\omega}{2\hbar}}q\psi_0(q) + \sqrt{\frac{\hbar}{2m\omega}} \frac{\partial \psi_0(q)}{\partial q} = 0. \quad (2.77)$$

This is further reduced to

$$\frac{\partial \psi_0(q)}{\partial q} + \frac{m\omega}{\hbar} q \psi_0(q) = 0. \quad (2.78)$$

From this FOLDE form, we anticipate the following solution:

$$\psi_0(q) = N_0 e^{-\alpha q^2}, \quad (2.79)$$

where N_0 is a normalization constant and α is a constant coefficient. Putting (2.79) into (2.78), we have

$$\left(-2\alpha q + \frac{m\omega}{\hbar} q\right) N_0 e^{-\alpha q^2} = 0. \quad (2.80)$$

Hence, we get

$$\alpha = \frac{m\omega}{2\hbar}. \quad (2.81)$$

Thus,

$$\psi_0(q) = N_0 e^{-\frac{m\omega}{2\hbar} q^2}. \quad (2.82)$$

The constant N_0 can be determined by the normalization condition given by

$$\int_{-\infty}^{\infty} |\psi_0(q)|^2 dq = 1 \text{ or } N_0^2 \int_{-\infty}^{\infty} e^{-\frac{m\omega}{\hbar} q^2} dq = 1. \quad (2.83)$$

Recalling the following formula:

$$\int_{-\infty}^{\infty} e^{-cq^2} dq = \sqrt{\frac{\pi}{c}} \quad (c > 0), \quad (2.84)$$

we have

$$\int_{-\infty}^{\infty} e^{-\frac{m\omega}{\hbar} q^2} dq = \sqrt{\frac{\pi\hbar}{m\omega}}. \quad (2.85)$$

To get (2.84), putting $I \equiv \int_{-\infty}^{\infty} e^{-cq^2} dq$, we have

$$\begin{aligned}
 I^2 &= \left(\int_{-\infty}^{\infty} e^{-cq^2} dq \right) \left(\int_{-\infty}^{\infty} e^{-cs^2} ds \right) = \int_{-\infty}^{\infty} e^{-c(q^2+s^2)} dq ds \\
 &= \int_0^{\infty} e^{-cr^2} r dr \int_0^{2\pi} d\theta = \frac{1}{2} \int_0^{\infty} e^{-cR} dR \int_0^{2\pi} d\theta = \frac{\pi}{c},
 \end{aligned} \tag{2.86}$$

where with the third equality we converted two-dimensional Cartesian coordinate to polar coordinate; take $q = r \cos \theta$, $s = r \sin \theta$ and convert an infinitesimal area element $dq ds$ to $dr \cdot r d\theta$. With the second last equality of (2.86), we used the variable transformation of $r^2 \rightarrow R$. Hence, we get $I = \sqrt{\pi/c}$.

Thus, we get

$$N_0 = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} \quad \text{and} \quad \psi_0(q) = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-\frac{m\omega}{2\hbar} q^2}. \tag{2.87}$$

Also, we have

$$\begin{aligned}
 a^\dagger &= \sqrt{\frac{m\omega}{2\hbar}} q - \frac{i}{\sqrt{2m\hbar\omega}} p = \sqrt{\frac{m\omega}{2\hbar}} q - \frac{i}{\sqrt{2m\hbar\omega}} \frac{\hbar}{i} \frac{\partial}{\partial q} = \sqrt{\frac{m\omega}{2\hbar}} q - \sqrt{\frac{\hbar}{2m\omega}} \\
 &\quad \times \frac{\partial}{\partial q}.
 \end{aligned} \tag{2.88}$$

From (2.52), we get

$$\begin{aligned}
 \psi_n(q) &= \frac{1}{\sqrt{n!}} (a^\dagger)^n | \psi_0 \rangle = \frac{1}{\sqrt{n!}} \left(\sqrt{\frac{m\omega}{2\hbar}} q - \sqrt{\frac{\hbar}{2m\omega}} \frac{\partial}{\partial q} \right)^n \psi_0(q) \\
 &= \frac{1}{\sqrt{n!}} \left(\frac{m\omega}{2\hbar} \right)^{n/2} \left(q - \frac{\hbar}{m\omega} \frac{\partial}{\partial q} \right)^n \psi_0(q).
 \end{aligned} \tag{2.89}$$

Putting

$$\beta \equiv \sqrt{m\omega/\hbar} \quad \text{and} \quad \xi = \beta q, \tag{2.90}$$

we rewrite (2.89) as

$$\begin{aligned}
\psi_n(q) &= \psi_n\left(\frac{\xi}{\beta}\right) = \frac{1}{\sqrt{n!}} \left(\frac{m\omega}{2\hbar\beta^2}\right)^{\frac{n}{2}} \left(\xi - \frac{\partial}{\partial \xi}\right)^n \psi_0(q) \\
&= \frac{1}{\sqrt{n!}} \left(\frac{1}{2}\right)^{n/2} \left(\xi - \frac{\partial}{\partial \xi}\right)^n \psi_0(q) \\
&= \sqrt{\frac{1}{2^n n!}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \left(\xi - \frac{\partial}{\partial \xi}\right)^n e^{-\frac{1}{2}\xi^2}.
\end{aligned} \tag{2.91}$$

Comparing (2.81) and (2.90), we have

$$\alpha = \frac{\beta^2}{2}. \tag{2.92}$$

Moreover, putting

$$N_n \equiv \sqrt{\frac{1}{2^n n!}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} = \sqrt{\frac{\beta}{\pi^{1/2} 2^n n!}}, \tag{2.93}$$

we get

$$\psi_n(\xi/\beta) = N_n \left(\xi - \frac{\partial}{\partial \xi}\right)^n e^{-\frac{1}{2}\xi^2}. \tag{2.94}$$

We have to normalize (2.94) with respect to a variable ξ . Since $\psi_n(q)$ has already been normalized as in (2.53), we have

$$\int_{-\infty}^{\infty} |\psi_n(q)|^2 dq = 1. \tag{2.95}$$

Changing a variable q to ξ , we have

$$\frac{1}{\beta} \int_{-\infty}^{\infty} |\psi_n(\xi/\beta)|^2 d\xi = 1. \tag{2.96}$$

Let us define $\widetilde{\psi}_n(\xi)$ as being normalized with ξ . In other words, $\psi_n(q)$ is converted to $\widetilde{\psi}_n(\xi)$ by means of variable transformation and concomitant change in normalization condition. Then, we have

$$\int_{-\infty}^{\infty} |\widetilde{\psi}_n(\xi)|^2 d\xi = 1. \tag{2.97}$$

Comparing (2.96) and (2.97), if we define $\widetilde{\psi}_n(\xi)$ as

$$\widetilde{\psi}_n(\xi) \equiv \sqrt{\frac{1}{\beta}} \psi_n(\xi/\beta), \quad (2.98)$$

$\widetilde{\psi}_n(\xi)$ should be a proper normalized function. Thus, we get

$$\widetilde{\psi}_n(\xi) = \widetilde{N}_n \left(\xi - \frac{\partial}{\partial \xi} \right)^n e^{-\frac{1}{2}\xi^2} \quad \text{with} \quad \widetilde{N}_n \equiv \sqrt{\frac{1}{\pi^{1/2} 2^n n!}}. \quad (2.99)$$

Meanwhile, according to a theory of classical orthogonal polynomial, the Hermite polynomials $H_n(x)$ are defined as [3]

$$H_n(x) \equiv (-1)^n e^{x^2} \frac{d^n}{dx^n} (e^{-x^2}) \quad (n \geq 0), \quad (2.100)$$

where $H_n(x)$ is a n -th order polynomial. We wish to show the following relation on the basis of mathematical induction:

$$\widetilde{\psi}_n(\xi) = \widetilde{N}_n H_n(\xi) e^{-\frac{1}{2}\xi^2}. \quad (2.101)$$

Comparing (2.87), (2.98), and (2.99), we make sure that (2.101) holds with $n = 0$. When $n = 1$, from (2.99) we have

$$\begin{aligned} \widetilde{\psi}_1(\xi) &= \widetilde{N}_1 \left(\xi - \frac{\partial}{\partial \xi} \right) e^{-\frac{1}{2}\xi^2} = \widetilde{N}_1 \left[\xi e^{-\frac{1}{2}\xi^2} - (-\xi) e^{-\frac{1}{2}\xi^2} \right] = \widetilde{N}_1 \cdot 2\xi e^{-\frac{1}{2}\xi^2} \\ &= \widetilde{N}_1 \left[(-1)^1 e^{\xi^2} \frac{d}{d\xi} (e^{-\xi^2}) \right] e^{-\frac{1}{2}\xi^2} = \widetilde{N}_1 H_1(\xi) e^{-\frac{1}{2}\xi^2}. \end{aligned} \quad (2.102)$$

Then, (2.101) holds with $n = 1$ as well.

Next, from supposition of mathematical induction we assume that (2.101) holds with n . Then, we have

$$\begin{aligned}
\widetilde{\psi}_{n+1}(\xi) &= \widetilde{N}_{n+1} \left(\xi - \frac{\partial}{\partial \xi} \right)^{n+1} e^{-\frac{1}{2}\xi^2} = \frac{1}{\sqrt{2(n+1)}} \left(\xi - \frac{\partial}{\partial \xi} \right) \widetilde{N}_n \left(\xi - \frac{\partial}{\partial \xi} \right)^n e^{-\frac{1}{2}\xi^2} \\
&= \frac{1}{\sqrt{2(n+1)}} \left(\xi - \frac{\partial}{\partial \xi} \right) \left[\widetilde{N}_n H_n(x) e^{-\frac{1}{2}\xi^2} \right] \\
&= \frac{1}{\sqrt{2(n+1)}} \widetilde{N}_n \left(\xi - \frac{\partial}{\partial \xi} \right) \left\{ \left[(-1)^n e^{\xi^2} \frac{d^n}{d\xi^n} \left(e^{-\xi^2} \right) \right] e^{-\frac{1}{2}\xi^2} \right\} \\
&= \frac{1}{\sqrt{2(n+1)}} \widetilde{N}_n (-1)^n \left(\xi - \frac{\partial}{\partial \xi} \right) \left[e^{\frac{1}{2}\xi^2} \frac{d^n}{d\xi^n} \left(e^{-\xi^2} \right) \right] \\
&= \frac{1}{\sqrt{2(n+1)}} \widetilde{N}_n (-1)^n \left\{ \xi e^{\frac{1}{2}\xi^2} \frac{d^n}{d\xi^n} \left(e^{-\xi^2} \right) - \frac{\partial}{\partial \xi} \left[e^{\frac{1}{2}\xi^2} \frac{d^n}{d\xi^n} \left(e^{-\xi^2} \right) \right] \right\} \\
&= \frac{1}{\sqrt{2(n+1)}} \widetilde{N}_n (-1)^n \left\{ \xi e^{\frac{1}{2}\xi^2} \frac{d^n}{d\xi^n} \left(e^{-\xi^2} \right) - \xi e^{\frac{1}{2}\xi^2} \frac{d^n}{d\xi^n} \left(e^{-\xi^2} \right) \right. \\
&\quad \left. - e^{\frac{1}{2}\xi^2} \frac{d^{n+1}}{d\xi^{n+1}} \left(e^{-\xi^2} \right) \right\} \\
&= \frac{1}{\sqrt{2(n+1)}} \widetilde{N}_n (-1)^{n+1} e^{\frac{1}{2}\xi^2} \frac{d^{n+1}}{d\xi^{n+1}} \left(e^{-\xi^2} \right) \\
&= \widetilde{N}_{n+1} \left[(-1)^{n+1} e^{\xi^2} \frac{d^{n+1}}{d\xi^{n+1}} \left(e^{-\xi^2} \right) \right] e^{-\frac{1}{2}\xi^2} = \widetilde{N}_{n+1} H_{n+1}(x) e^{-\frac{1}{2}\xi^2}.
\end{aligned}$$

This means that (2.101) holds with $n+1$ as well. Thus, it follows that (2.101) is true of n that is zero or any positive integer.

Orthogonal relation reads as

$$\int_{-\infty}^{\infty} \widetilde{\psi}_m(\xi)^* \widetilde{\psi}_n(\xi) d\xi = \delta_{mn}, \quad (2.103)$$

where δ_{mn} is Kronecker delta to which

$$\delta_{mn} = \begin{cases} 1 & (m=n), \\ 0 & (m \neq n). \end{cases} \quad (2.104)$$

Placing (2.98) back into the function form $\psi_n(q)$, we have

$$\psi_n(q) = \sqrt{\beta} \widetilde{\psi}_n(\beta q). \quad (2.105)$$

Using (2.101) and explicitly rewriting (2.105), we get

Table 2.1 First six Hermite polynomials

$H_0(x) = 1$
$H_1(x) = 2x$
$H_2(x) = 4x^2 - 2$
$H_3(x) = 8x^3 - 12x$
$H_4(x) = 16x^4 - 48x^2 + 12$
$H_5(x) = 32x^5 - 160x^3 + 120x$

$$\psi_n(q) = \left(\frac{m\omega}{\hbar}\right)^{1/4} \sqrt{\frac{1}{\pi^{1/2} 2^n n!}} H_n\left(\sqrt{\frac{m\omega}{\hbar}} q\right) e^{-\frac{m\omega}{2\hbar} q^2} \quad (n = 0, 1, 2, \dots). \quad (2.106)$$

We tabulate first several Hermite polynomials $H_n(x)$ in Table 2.1, where the index n represents the highest order of the polynomials. In Table 2.1 we see that even functions and odd functions appear alternately (i.e., parity). This is the case with $\psi_n(q)$ as well, because $\psi_n(q)$ is a product of $H_n(x)$ and an even function $e^{-\frac{m\omega}{2\hbar} q^2}$.

Combining (2.101) and (2.103) as well as (2.99), the orthogonal relation between $\widetilde{\psi}_n(\xi)$ ($n = 0, 1, 2, \dots$) can be described alternatively as [3]

$$\int_{-\infty}^{\infty} e^{-\xi^2} H_m(\xi) H_n(\xi) d\xi = \sqrt{\pi} 2^n n! \delta_{mn}. \quad (2.107)$$

Note that $H_m(\xi)$ is a real function, and so $H_m(\xi)^* = H_m(\xi)$. The relation (2.107) is well known as the orthogonality of Hermite polynomials with $e^{-\xi^2}$ taken as a weight function [3]. Here the weight function is a real and non-negative function within the domain considered [e.g., $(-\infty, +\infty)$ in the present case] and independent of indices m and n . We will deal with it again in Sect. 10.4.

The relation (2.101) and the orthogonality relationship described as (2.107) can more explicitly be understood as follows: From (2.11) we have the Schrödinger equation of a one-dimensional quantum-mechanical harmonic oscillator such that

$$-\frac{\hbar^2}{2m} \frac{d^2 u(q)}{dq^2} + \frac{1}{2} m\omega^2 q^2 u(q) = E u(q). \quad (2.108)$$

Changing a variable as in (2.90), we have

$$-\frac{d^2 u(\xi)}{d\xi^2} + \xi^2 u(\xi) = \frac{2E}{\hbar\omega} u(\xi). \quad (2.109)$$

Defining a dimensionless parameter

$$\lambda \equiv \frac{2E}{\hbar\omega} \quad (2.110)$$

and also defining a differential operator D such that

$$D \equiv -\frac{d^2}{d\xi^2} + \xi^2, \quad (2.111)$$

we have a following eigenvalue equation:

$$Du(\xi) = \lambda u(\xi). \quad (2.112)$$

We further consider a following function $v(\xi)$ such that

$$u(\xi) = v(\xi)e^{-\xi^2/2}. \quad (2.113)$$

Then, (2.109) is converted as follows:

$$\left[-\frac{d^2 v(\xi)}{d\xi^2} + 2\xi \frac{dv(\xi)}{d\xi} \right] e^{-\xi^2/2} = (\lambda - 1) v(\xi) e^{-\xi^2/2}. \quad (2.114)$$

Since $e^{-\xi^2/2}$ does not vanish with any ξ , we have

$$-\frac{d^2 v(\xi)}{d\xi^2} + 2\xi \frac{dv(\xi)}{d\xi} = (\lambda - 1) v(\xi). \quad (2.115)$$

If we define another differential operator $\tilde{D}$ such that

$$\tilde{D} \equiv -\frac{d^2}{d\xi^2} + 2\xi \frac{d}{d\xi}, \quad (2.116)$$

we have another eigenvalue equation

$$\tilde{D}v(\xi) = (\lambda - 1) v(\xi). \quad (2.117)$$

Meanwhile, we have a following well-known differential equation:

$$\frac{d^2 H_n(\xi)}{d\xi^2} - 2\xi \frac{dH_n(\xi)}{d\xi} + 2nH_n(\xi) = 0. \quad (2.118)$$

This equation is said to be Hermite differential equation. Using (2.116), (2.118) can be recast as an eigenvalue equation such that

$$\tilde{D}H_n(\xi) = 2nH_n(\xi). \quad (2.119)$$

Therefore, comparing (2.115) and (2.118) and putting

$$\lambda = 2n + 1, \quad (2.120)$$

we get

$$v(\xi) = cH_n(\xi), \quad (2.121)$$

where c is an arbitrary constant. Thus, using (2.113), for a solution of (2.109) we get

$$u_n(\xi) = cH_n(\xi)e^{-\xi^2/2}, \quad (2.122)$$

where the solution $u(\xi)$ is indexed with n . From (2.110), as an energy eigenvalue we have

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega.$$

Thus, (2.37) is recovered. A normalization constant c of (2.122) can be decided as in (2.106).

As discussed above, the operator representation and coordinate representation are fully consistent.

2.5 Variance and Uncertainty Principle

Uncertainty principle is one of the most fundamental concepts of quantum mechanics. To think of this conception on the basis of a quantum harmonic oscillator, let us introduce a variance operator [4]. Let A be a physical quantity and let $\langle A \rangle$ be an expectation value as defined in (1.126). We define a variance operator as

$$\langle (\Delta A)^2 \rangle,$$

where we have

$$\Delta A \equiv A - \langle A \rangle. \quad (2.123)$$

In (2.123), we assume that $\langle A \rangle$ is obtained by operating A on a certain physical state $|\psi\rangle$. Then, we have

$$\langle (\Delta A)^2 \rangle = \langle (A - \langle A \rangle)^2 \rangle = \langle A^2 - 2\langle A \rangle A + \langle A \rangle^2 \rangle = \langle A^2 \rangle - \langle A \rangle^2. \quad (2.124)$$

If A is Hermitian, ΔA is Hermitian as well. This is because

$$(\Delta A)^\dagger = A^\dagger - \langle A \rangle^* = A - \langle A \rangle = \Delta A, \quad (2.125)$$

where we used the fact that an expectation value of an Hermitian operator is real. Then, $\langle (\Delta A)^2 \rangle$ is non-negative as in the case of (2.13). Moreover, if $|\psi\rangle$ is an eigenstate of A , $\langle (\Delta A)^2 \rangle = 0$. Therefore, $\langle (\Delta A)^2 \rangle$ represents a measure of how large measured values are dispersed when A is measured in reference to a quantum state $|\psi\rangle$. Also, we define a standard deviation δA as

$$\delta A \equiv \sqrt{\langle (\Delta A)^2 \rangle}. \quad (2.126)$$

We have a following important theorem on a standard deviation δA [4].

Theorem 2.1 Let A and B be Hermitian operators. If A and B satisfy

$$[A, B] = ik \quad (k : \text{non-zero real number}), \quad (2.127)$$

then we have

$$\delta A \cdot \delta B \geq |k|/2 \quad (2.128)$$

in reference to any quantum state $|\psi\rangle$.

Proof We have

$$[\Delta A, \Delta B] = [A - \langle \psi|A|\psi\rangle, B - \langle \psi|B|\psi\rangle] = [A, B] = ik. \quad (2.129)$$

In (2.129), we used the fact that $\langle \psi|A|\psi\rangle$ and $\langle \psi|B|\psi\rangle$ are just real numbers and those commute with any operator. Next, we calculate a following quantity in relation to a real number λ :

$$\begin{aligned} \|(\Delta A + i\lambda\Delta B)|\psi\rangle\|^2 &= \langle \psi|(\Delta A - i\lambda\Delta B)(\Delta A + i\lambda\Delta B)|\psi\rangle \\ &= \langle \psi|(\Delta A)^2|\psi\rangle - k\lambda + \langle \psi|(\Delta B)^2|\psi\rangle\lambda^2, \end{aligned} \quad (2.130)$$

where we used the fact that ΔA and ΔB are Hermitian. For the above quadratic form to hold with any real number λ , we have

$$(-k)^2 - 4\langle \psi|(\Delta A)^2|\psi\rangle\langle \psi|(\Delta B)^2|\psi\rangle \leq 0. \quad (2.131)$$

Thus, (2.128) will follow. This completes the proof.

On the basis of Theorem 2.1, we find that both δA and δB are positive on condition that (2.127) holds. We have another important theorem.

Theorem 2.2 Let A be an Hermitian operator. The necessary and sufficient condition for a physical state $|\psi_0\rangle$ to be an eigenstate of A is $\delta A = 0$.

Proof Suppose that $|\psi_0\rangle$ is a normalized eigenstate of A that belongs to an eigenvalue a . Then, we have

$$\begin{aligned}\langle\psi_0|A^2\psi_0\rangle &= a\langle\psi_0|A\psi_0\rangle = a^2\langle\psi_0|\psi_0\rangle = a^2, \\ \langle\psi_0|A\psi_0\rangle^2 &= [a\langle\psi_0|\psi_0\rangle]^2 = a^2.\end{aligned}\tag{2.132}$$

From (2.124) and (2.126), we have

$$\langle\psi_0|(\Delta A)^2\psi_0\rangle = 0 \quad \text{i.e.,} \quad \delta A = 0.\tag{2.133}$$

Note that δA is measured in reference to $|\psi_0\rangle$. Conversely, suppose that $\delta A = 0$. Then,

$$\delta A = \sqrt{\langle\psi_0|(\Delta A)^2\psi_0\rangle} = \sqrt{\langle\Delta A\psi_0|\Delta A\psi_0\rangle} = \|\Delta A\psi_0\|,\tag{2.134}$$

where we used the fact that ΔA is Hermitian. From the definition of norm of (1.121), for $\delta A = 0$ to hold, we have

$$\Delta A\psi_0 = (A - \langle A \rangle)\psi_0 = 0 \quad \text{i.e.,} \quad A\psi_0 = \langle A \rangle\psi_0.\tag{2.135}$$

This indicates that ψ_0 is an eigenstate of A that belongs to an eigenvalue $\langle A \rangle$. This completes the proof.

Theorem 2.1 implies that (2.127) holds with *any* physical state $|\psi\rangle$. That is, we must have $\delta A > 0$ and $\delta B > 0$, if δA and δB are evaluated in reference to any $|\psi\rangle$ on condition that (2.127) holds. From Theorem 2.2, in turn, it follows that eigenstates cannot exist with A or B under the condition of (2.127).

To explicitly show this, we take an inner product using (2.127). That is, with Hermitian operators A and B , consider the following inner product:

$$\langle\psi|[A, B]|\psi\rangle = \langle\psi|ik|\psi\rangle \quad \text{i.e.,} \quad \langle\psi|AB - BA|\psi\rangle = ik,\tag{2.136}$$

where we assumed that $|\psi\rangle$ is arbitrarily chosen normalized vector. Suppose now that $|\psi_0\rangle$ is an eigenstate of A that belongs to an eigenvalue a . Then, we have

$$A|\psi_0\rangle = a|\psi_0\rangle.\tag{2.137}$$

Taking an adjoint of (2.137), we get

$$\langle \psi_0 | A^\dagger = \langle \psi_0 | A = \langle \psi_0 | a^* = a \langle \psi_0 | , \quad (2.138)$$

where the last equality comes from the fact that A is Hermitian. From (2.138), we would have

$$\begin{aligned} \langle \psi_0 | AB - BA | \psi_0 \rangle &= \langle \psi_0 | AB | \psi_0 \rangle - \langle \psi_0 | BA | \psi_0 \rangle \\ &= \langle \psi_0 | aB | \psi_0 \rangle - \langle \psi_0 | Ba | \psi_0 \rangle = a \langle \psi_0 | B | \psi_0 \rangle - a \langle \psi_0 | B | \psi_0 \rangle = 0. \end{aligned}$$

This would imply that (2.136) does not hold with $|\psi_0\rangle$, in contradiction to (2.127), where $ik \neq 0$. Namely, we conclude that any physical state cannot be an eigenstate of A on condition that (2.127) holds. Equation (2.127) is rewritten as

$$\langle \psi | BA - AB | \psi \rangle = -ik. \quad (2.139)$$

Suppose now that $|\varphi_0\rangle$ is an eigenstate of B that belongs to an eigenvalue b . Then, we can similarly show that any physical state cannot be an eigenstate of B .

Summarizing the above, we restate that once we have a relation $[A, B] = ik$ ($k \neq 0$), their representation matrix does not diagonalize A or B . Or, once we postulate $[A, B] = ik$ ($k \neq 0$), we must abandon an effort to have a representation matrix that diagonalizes A and B . In the quantum-mechanical formulation of a harmonic oscillator, we have introduced the canonical commutation relation (see Sect. 2.3) described by $[q, p] = i\hbar$; see (1.140) and (2.73). Indeed, neither q nor p is diagonalized as shown in (2.69) or (2.70).

Example 2.1 Taking a quantum harmonic oscillator as an example, we consider variance of q and p in reference to $|\psi_n\rangle$ ($n = 0, 1, \dots$). We have

$$\langle (\Delta q)^2 \rangle = \langle \psi_n | q^2 | \psi_n \rangle - \langle \psi_n | q | \psi_n \rangle^2. \quad (2.140)$$

Using (2.55) and (2.62) as well as (2.68), we get

$$\begin{aligned} \langle \psi_n | q | \psi_n \rangle &= \sqrt{\frac{\hbar}{2m\omega}} \langle \psi_n | a + a^\dagger | \psi_n \rangle \\ &= \sqrt{\frac{\hbar n}{2m\omega}} \langle \psi_n | \psi_{n-1} \rangle + \sqrt{\frac{\hbar(n+1)}{2m\omega}} \langle \psi_n | \psi_{n+1} \rangle = 0, \end{aligned} \quad (2.141)$$

where the last equality comes from (2.53). We have

$$q^2 = \frac{\hbar}{2m\omega} (a + a^\dagger)^2 = \frac{\hbar}{2m\omega} [a^2 + E + 2a^\dagger a + (a^\dagger)^2], \quad (2.142)$$

where E denotes an identity operator and we used (2.24) along with the following relation:

$$aa^\dagger = aa^\dagger - a^\dagger a + a^\dagger a = [a, a^\dagger] + a^\dagger a = E + a^\dagger a. \quad (2.143)$$

Using (2.55) and (2.62), we have

$$\langle \psi_n | q^2 | \psi_n \rangle = \frac{\hbar}{2m\omega} [\langle \psi_n | \psi_n \rangle + 2\langle \psi_n | a^\dagger a \psi_n \rangle] = \frac{\hbar}{2m\omega} (2n + 1), \quad (2.144)$$

where we used (2.60) with the last equality. Thus, we get

$$\langle (\Delta q)^2 \rangle = \langle \psi_n | q^2 | \psi_n \rangle - \langle \psi_n | q | \psi_n \rangle^2 = \frac{\hbar}{2m\omega} (2n + 1).$$

Following similar procedures to those mentioned above, we get

$$\langle \psi_n | p | \psi_n \rangle = 0 \quad \text{and} \quad \langle \psi_n | p^2 | \psi_n \rangle = \frac{m\hbar\omega}{2} (2n + 1). \quad (2.145)$$

Thus, we get

$$\langle (\Delta p)^2 \rangle = \langle \psi_n | p^2 | \psi_n \rangle = \frac{m\hbar\omega}{2} (2n + 1).$$

Accordingly, we have

$$\delta q \cdot \delta p = \sqrt{\langle (\Delta q)^2 \rangle} \cdot \sqrt{\langle (\Delta p)^2 \rangle} = \frac{\hbar}{2} (2n + 1) \geq \frac{\hbar}{2}. \quad (2.146)$$

The quantity $\delta q \cdot \delta p$ is equal to $\frac{\hbar}{2}$ for $n = 0$ and becomes larger with increasing n .

The above example gives a good illustration for Theorem 2.1. Note that putting $A = q$ and $B = p$ along with $k = \hbar$ in Theorem 2.1, from (1.140) we should have

$$\delta q \cdot \delta p \geq \frac{\hbar}{2}. \quad (2.147)$$

Thus, the quantum-mechanical harmonic oscillator gives us an interesting example with respect to the standard deviation of the position operator and momentum operator. That is, (2.146) is a special case of (2.147). The general relation represented by (2.147) is well-known as uncertainty principle.

In relation to the aforementioned argument, we might well wonder if Examples 1.1 and 1.2 have an eigenstate of a fixed momentum. Suppose that we chose for an eigenstate $y(x) = ce^{ikx}$, where c is a constant. Then, we would have $\frac{\hbar}{i} \frac{\partial y(x)}{\partial x} = \hbar k y(x)$ and get an eigenvalue $\hbar k$ for a momentum. Nonetheless, such $y(x)$ does not satisfy the proper BCs; i.e., $y(L) = y(-L) = 0$. This is because e^{ikx} never vanishes with any real numbers of k or x (any complex numbers of k or x , more generally). Thus, we

cannot obtain a proper solution that has an eigenstate with a fixed momentum in a confined physical system.

References

1. Messiah A (1999) Quantum mechanics. Dover, New York, NY
2. Stakgold I (1998) Green's functions and boundary value problems, 2nd edn. Wiley, New York, NY
3. Lebedev NN (1972) Special functions and their applications. Dover, New York, NY
4. Shimizu A (2003) Upgrade foundation of quantum theory. Saiensu-sha, Tokyo. (in Japanese)

Chapter 3

Hydrogen-Like Atoms



In a history of quantum mechanics, it was first successfully applied to the motion of an electron in a hydrogen atom along with a harmonic oscillator. Unlike the case of a one-dimensional harmonic oscillator we dealt with in Chap. 2, however, with a hydrogen atom we have to consider three-dimensional motion of an electron. Accordingly, it takes somewhat elaborate calculations to constitute the Hamiltonian. The calculation procedures themselves, however, are worth following to understand underlying basic concepts of the quantum mechanics. At the same time, this chapter is a treasure of special functions. In Chap. 2, we have already encountered one of them, i.e., Hermite polynomials. Here we will deal with Legendre polynomials, associated Legendre polynomials, etc. These special functions arise when we deal with a physical system having, e.g., the spherical symmetry. In a hydrogen atom, an electron is moving in a spherically-symmetric Coulomb potential field produced by a proton. This topic provides us with a good opportunity to study various special functions. The related Schrödinger equation can be separated into an angular part and a radial part. The solutions of angular parts are characterized by spherical (surface) harmonics. The (associated) Legendre functions are correlated to them. The solutions of the radial part are connected to the (associated) Laguerre polynomials. The exact solutions are obtained by the product of the (associated) Legendre functions and (associated) Laguerre polynomials accordingly. Thus, to study the characteristics of hydrogen-like atoms from the quantum-mechanical perspective is of fundamental importance.

3.1 Introductory Remarks

The motion of the electron in hydrogen is well-known as a two-particle problem (or two-body problem) in a central force field. In that case, the coordinate system of the physical system is separated into the relative coordinates and center-of-mass coordinates. To be more specific, the coordinate separation is true of the case where

two particles are moving under control only by a force field between the two particles without other external force fields [1].

In the classical mechanics, equation of motion is separated into two equations related to the relative coordinates and center-of-mass coordinates accordingly. Of these, a term of the potential field is only included in the equation of motion with respect to the relative coordinates.

The situation is the same with the quantum mechanics. Namely, the Schrödinger equation of motion with the relative coordinates is expressed as an eigenvalue equation that reads as

$$\left[-\frac{\hbar^2}{2\mu} \nabla^2 + V(r) \right] \psi = E\psi, \quad (3.1)$$

where μ is a reduced mass of two particles [1], i.e., an electron and a proton; $V(r)$ is a potential with r being a distance between the electron and proton. In (3.1), we assume the spherically-symmetric potential; i.e., the potential is expressed only as a function of the distance r . Moreover, if the potential is coulombic,

$$\left(-\frac{\hbar^2}{2\mu} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r} \right) \psi = E\psi, \quad (3.2)$$

where ϵ_0 is permittivity of vacuum and e is an elementary charge.

If we think of hydrogen-like atoms such as He^+ , Li^{2+} , Be^{3+} , etc., we have an equation described as

$$\left(-\frac{\hbar^2}{2\mu} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} \right) \psi = E\psi, \quad (3.3)$$

where Z is an atomic number and μ is a reduced mass of an electron and a nucleus pertinent to the atomic (or ionic) species. We start with (3.3) in this chapter.

3.2 Constitution of Hamiltonian

As explicitly described in (3.3), the coulombic potential has a spherical symmetry. In such a case, it will be convenient to recast (3.3) in a spherical coordinate (or polar coordinate). As the physical system is of three-dimensional, we have to consider orbital angular momentum L in Hamiltonian.

We have

$$\mathbf{L} = (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} L_x \\ L_y \\ L_z \end{pmatrix}, \quad (3.4)$$

where $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ denote orthonormal basis vectors in a three-dimensional Cartesian space ($\mathbb{R}^3$); L_x , L_y , and L_z represent each component of $\mathbf{L}$. The angular momentum $\mathbf{L}$ is expressed in a form of determinant as

$$\mathbf{L} = \mathbf{x} \times \mathbf{p} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ x & y & z \\ p_x & p_y & p_z \end{vmatrix},$$

where $\mathbf{x}$ denotes a position vector with respect to the relative coordinates x , y , and z . That is,

$$\mathbf{x} = (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} x \\ y \\ z \end{pmatrix}. \quad (3.5)$$

The quantity $\mathbf{p}$ denotes a momentum of an electron (as a particle carrying a reduced mass μ) with p_x , p_y , and p_z being their components; $\mathbf{p}$ is denoted similarly to the above.

As for each component of $\mathbf{L}$, we have, e.g.,

$$L_x = yp_z - zp_y. \quad (3.6)$$

To calculate $\mathbf{L}^2$, we estimate L_x^2 , L_y^2 , and L_z^2 separately. We have

$$\begin{aligned} L_x^2 &= (yp_z - zp_y) \cdot (yp_z - zp_y) \\ &= yp_z yp_z - yp_z zp_y - zp_y yp_z - zp_y zp_y \\ &= y^2 p_z^2 - yp_z zp_y - zp_y yp_z + z^2 p_y^2 \\ &= y^2 p_z^2 - y(zp_z - i\hbar)p_y - z(yp_y - i\hbar)p_z + z^2 p_y^2 \\ &= y^2 p_z^2 + z^2 p_y^2 - yzp_z p_y - zyp_y p_z + i\hbar(yp_y + zp_z), \end{aligned}$$

where we have used canonical commutation relation (1.140) in the second to the last equality. In the above calculations, we used commutability of, e.g., y and p_z ; z and p_y . For example, we have

$$[p_z, y] | \psi \rangle = \frac{\hbar}{i} \left(\frac{\partial}{\partial z} y - y \frac{\partial}{\partial z} \right) | \psi \rangle = \frac{\hbar}{i} \left(y \frac{\partial}{\partial z} | \psi \rangle - y \frac{\partial}{\partial z} | \psi \rangle \right) = 0.$$

Since $|\psi\rangle$ is arbitrarily chosen, this relation implies that p_z and y commute. We obtain similar relations regarding L_y^2 and L_z^2 as well. Thus, we have

$$\begin{aligned} \mathbf{L}^2 &= L_x^2 + L_y^2 + L_z^2 \\ &= (y^2 p_z^2 + z^2 p_y^2 + z^2 p_x^2 + x^2 p_z^2 + x^2 p_y^2 + y^2 p_x^2) \\ &\quad + (x^2 p_x^2 - x^2 p_x^2 + y^2 p_y^2 - y^2 p_y^2 + z^2 p_z^2 - z^2 p_z^2) - (yz p_z p_y \\ &\quad + zy p_y p_z + zx p_x p_z + xz p_z p_x + xy p_y p_x + yx p_x p_y) \\ &\quad + i\hbar (yp_y + zp_z + zp_z + xp_x + xp_x + yp_y) \\ &= \left(y^2 p_z^2 + z^2 p_y^2 + z^2 p_x^2 + x^2 p_z^2 + x^2 p_y^2 + y^2 p_x^2 + x^2 p_x^2 + y^2 p_y^2 \right. \\ &\quad \left. + z^2 p_z^2 \right) - \left(x^2 p_x^2 + y^2 p_y^2 + z^2 p_z^2 + yz p_z p_y + zy p_y p_z + zx p_x p_z \right. \\ &\quad \left. + xz p_z p_x + xy p_y p_x + yx p_x p_y \right) + i\hbar (yp_y + zp_z + zp_z + xp_x + xp_x + yp_y) \\ &= \mathbf{r}^2 \cdot \mathbf{p}^2 - \mathbf{r}(\mathbf{r} \cdot \mathbf{p}) \cdot \mathbf{p} + 2i\hbar(\mathbf{r} \cdot \mathbf{p}). \end{aligned} \tag{3.7}$$

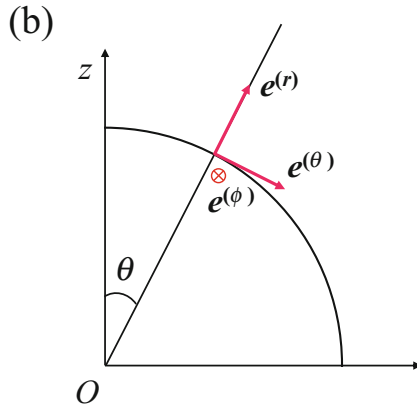
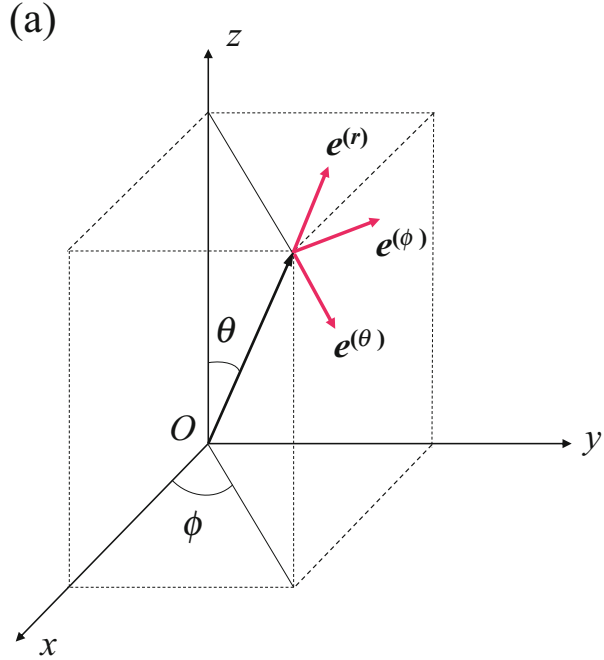
In (3.7), we are able to ease the calculations by virtue of putting a term $(x^2 p_x^2 - x^2 p_x^2 + y^2 p_y^2 - y^2 p_y^2 + z^2 p_z^2 - z^2 p_z^2)$. As a result, for the second term after the second to the last equality we have

$$\begin{aligned} &- \left(x^2 p_x^2 + y^2 p_y^2 + z^2 p_z^2 + yz p_z p_y + zy p_y p_z + zx p_x p_z + xz p_z p_x \right. \\ &\quad \left. + xy p_y p_x + yx p_x p_y \right) \\ &= - \left[x(xp_x + yp_y + zp_z)p_x + y(xp_x + yp_y + zp_z)p_y \right. \\ &\quad \left. + z(xp_x + yp_y + zp_z)p_z \right] \\ &= - \mathbf{r}(\mathbf{r} \cdot \mathbf{p}) \cdot \mathbf{p}. \end{aligned}$$

The calculations of $\mathbf{r}^2 \cdot \mathbf{p}^2$ [the first term of (3.7)] and $\mathbf{r} \cdot \mathbf{p}$ (in the third term) are straightforward.

In a spherical coordinate, momentum $\mathbf{p}$ is expressed as

Fig. 3.1 Spherical coordinate system and orthonormal basis set. (a) Orthonormal basis vectors $\mathbf{e}^{(r)}$, $\mathbf{e}^{(\theta)}$, and $\mathbf{e}^{(\phi)}$ in $\mathbb{R}^3$. (b) The basis vector $\mathbf{e}^{(\phi)}$ is perpendicular to the plane shaped by the z -axis and a straight line of $y = x \tan \phi$



$$\mathbf{p} = p_r \mathbf{e}^{(r)} + p_\theta \mathbf{e}^{(\theta)} + p_\phi \mathbf{e}^{(\phi)}, \quad (3.8)$$

where p_r , p_θ , and p_ϕ are components of $\mathbf{p}$; $\mathbf{e}^{(r)}$, $\mathbf{e}^{(\theta)}$, and $\mathbf{e}^{(\phi)}$ are orthonormal basis vectors of $\mathbb{R}^3$ in the direction of increasing r , θ , and ϕ , respectively (see Fig. 3.1). In Fig. 3.1b, $\mathbf{e}^{(\phi)}$ is perpendicular to the plane shaped by the z -axis and a straight line of $y = x \tan \phi$. Notice that the said plane is spanned by $\mathbf{e}^{(r)}$ and $\mathbf{e}^{(\theta)}$. Meanwhile, the momentum operator is expressed as [2]

$$\begin{aligned}\mathbf{p} &= \frac{\hbar}{i} \nabla \\ &= \frac{\hbar}{i} \left[\mathbf{e}^{(r)} \frac{\partial}{\partial r} + \mathbf{e}^{(\theta)} \frac{1}{r} \frac{\partial}{\partial \theta} + \mathbf{e}^{(\phi)} \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \right].\end{aligned}\quad (3.9)$$

The vector notation of (3.9) corresponds to (1.31). That is, in the Cartesian coordinate, we have

$$\mathbf{p} = \frac{\hbar}{i} \nabla = \frac{\hbar}{i} \left(\mathbf{e}_1 \frac{\partial}{\partial x} + \mathbf{e}_2 \frac{\partial}{\partial y} + \mathbf{e}_3 \frac{\partial}{\partial z} \right), \quad (3.10)$$

where ∇ is said to be nabla (or del), a kind of differential vector operator.

Noting that

$$\mathbf{r} = r \mathbf{e}^{(r)},$$

and using (3.9), we have

$$\mathbf{r} \cdot \mathbf{p} = \mathbf{r} \cdot \frac{\hbar}{i} \nabla = r \left(\frac{\hbar}{i} \frac{\partial}{\partial r} \right). \quad (3.11)$$

Hence,

$$\mathbf{r}(\mathbf{r} \cdot \mathbf{p}) \cdot \mathbf{p} = r \left[r \left(\frac{\hbar}{i} \frac{\partial}{\partial r} \right) \right] \left(\frac{\hbar}{i} \frac{\partial}{\partial r} \right) = -\hbar^2 r^2 \frac{\partial^2}{\partial r^2}. \quad (3.12)$$

Thus, we have

$$\mathbf{L}^2 = r^2 \mathbf{p}^2 + \hbar^2 r^2 \frac{\partial^2}{\partial r^2} + 2\hbar^2 r \frac{\partial}{\partial r} = r^2 \mathbf{p}^2 + \hbar^2 \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right). \quad (3.13)$$

Therefore,

$$\mathbf{p}^2 = -\frac{\hbar^2}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{\mathbf{L}^2}{r^2}. \quad (3.14)$$

Notice here that $\mathbf{L}^2$ does not contain r (vide infra); i.e., $\mathbf{L}^2$ commutes with r^2 , and so it can freely be divided by r^2 . Thus, the Hamiltonian H is represented by

$$\begin{aligned}
H &= \frac{\mathbf{p}^2}{2\mu} + V(r) \\
&= \frac{1}{2\mu} \left[-\frac{\hbar^2}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{\mathbf{L}^2}{r^2} \right] - \frac{Ze^2}{4\pi\epsilon_0 r}.
\end{aligned} \tag{3.15}$$

Thus, the Schrödinger equation can be expressed as

$$\left\{ \frac{1}{2\mu} \left[-\frac{\hbar^2}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{\mathbf{L}^2}{r^2} \right] - \frac{Ze^2}{4\pi\epsilon_0 r} \right\} \psi = E\psi. \tag{3.16}$$

Now, let us describe $\mathbf{L}^2$ in a polar coordinate. The calculation procedures are somewhat lengthy, but straightforward. First we have

$$\left. \begin{aligned} x &= r \sin \theta \cos \phi, \\ y &= r \sin \theta \sin \phi, \\ z &= r \cos \theta, \end{aligned} \right\} \tag{3.17}$$

where we have $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq 2\pi$. Rewriting (3.17) with respect to r , θ , and ϕ , we get

$$\left. \begin{aligned} r &= (x^2 + y^2 + z^2)^{1/2}, \\ \theta &= \tan^{-1} \frac{(x^2 + y^2)^{1/2}}{z}, \\ \phi &= \tan^{-1} \frac{y}{x}. \end{aligned} \right\} \tag{3.18}$$

Thus, we have

$$L_z = xp_y - yp_x = -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right), \tag{3.19}$$

$$\frac{\partial}{\partial x} = \frac{\partial r}{\partial x} \frac{\partial}{\partial r} + \frac{\partial \theta}{\partial x} \frac{\partial}{\partial \theta} + \frac{\partial \phi}{\partial x} \frac{\partial}{\partial \phi}. \tag{3.20}$$

In turn, we have

$$\begin{aligned}
\frac{\partial r}{\partial x} &= \frac{x}{r} = \sin \theta \cos \phi, \\
\frac{\partial \theta}{\partial x} &= \frac{1}{1 + (x^2 + y^2)/z^2} \cdot \frac{(x^2 + y^2)^{-1/2} \cdot 2x}{2z} = \frac{z}{x^2 + y^2 + z^2} \cdot \frac{x}{(x^2 + y^2)^{1/2}} \\
&= \frac{\cos \theta \cos \phi}{r}, \\
\frac{\partial \phi}{\partial x} &= \frac{1}{1 + (y^2/x^2)} \cdot y \left(-\frac{1}{x^2} \right) = -\frac{\sin \phi}{r \sin \theta}.
\end{aligned} \tag{3.21}$$

In calculating the last two equations of (3.21), we used the differentiation of an arc tangent function along with a composite function. Namely,

$$(\tan^{-1}x)' = \frac{1}{1+x^2}.$$

Inserting (3.21) into (3.20), we get

$$\frac{\partial}{\partial x} = \sin \theta \cos \phi \frac{\partial}{\partial r} + \frac{\cos \theta \cos \phi}{r} \frac{\partial}{\partial \theta} - \frac{\sin \phi}{r \sin \theta} \frac{\partial}{\partial \phi}. \tag{3.22}$$

Similarly, we have

$$\frac{\partial}{\partial y} = \sin \theta \sin \phi \frac{\partial}{\partial r} + \frac{\cos \theta \sin \phi}{r} \frac{\partial}{\partial \theta} + \frac{\cos \phi}{r \sin \theta} \frac{\partial}{\partial \phi}. \tag{3.23}$$

Inserting (3.22) and (3.23) together with (3.17) into (3.19), we get

$$L_z = -i\hbar \frac{\partial}{\partial \phi}. \tag{3.24}$$

In a similar manner, we have

$$\frac{\partial}{\partial z} = \frac{\partial r}{\partial z} \frac{\partial}{\partial r} + \frac{\partial \theta}{\partial z} \frac{\partial}{\partial \theta} = \cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta}.$$

Combining this relation with either (3.23) or (3.22), we get

$$L_x = i\hbar \left(\sin \phi \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right), \tag{3.25}$$

$$L_y = -i\hbar \left(\cos \phi \frac{\partial}{\partial \theta} - \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right). \tag{3.26}$$

Now, we introduce the following operators:

$$L^{(+)} \equiv L_x + iL_y \quad \text{and} \quad L^{(-)} \equiv L_x - iL_y. \quad (3.27)$$

Then, we have

$$L^{(+)} = \hbar e^{i\phi} \left(\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right) \quad \text{and} \quad L^{(-)} = \hbar e^{-i\phi} \left(-\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right). \quad (3.28)$$

Thus, we get

$$\begin{aligned} L^{(+)}L^{(-)} &= \hbar^2 e^{i\phi} \left(\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right) e^{-i\phi} \left(-\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right) \\ &= \hbar^2 e^{i\phi} \left\{ e^{-i\phi} \left[-\frac{\partial^2}{\partial \theta^2} + i \left(-\frac{1}{\sin^2 \theta} \right) \frac{\partial}{\partial \phi} + i \cot \theta \frac{\partial^2}{\partial \theta \partial \phi} \right] \right. \\ &\quad \left. + e^{-i\phi} \cot \theta \left(-\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right) + i e^{-i\phi} \cot \theta \left(-\frac{\partial^2}{\partial \phi \partial \theta} + i \cot \theta \frac{\partial^2}{\partial \phi^2} \right) \right\} \\ &= -\hbar^2 \left(\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + i \frac{\partial}{\partial \phi} + \cot^2 \theta \frac{\partial^2}{\partial \phi^2} \right). \end{aligned} \quad (3.29)$$

In the above calculation procedure, we used differentiation of a product function. For instance, we have

$$\begin{aligned} \frac{\partial}{\partial \theta} \left(i \cot \theta \frac{\partial}{\partial \phi} \right) &= i \left(\frac{\partial \cot \theta}{\partial \theta} \frac{\partial}{\partial \phi} + \cot \theta \frac{\partial^2}{\partial \theta \partial \phi} \right) \\ &= i \left[\left(-\frac{1}{\sin^2 \theta} \right) \frac{\partial}{\partial \phi} + \cot \theta \frac{\partial^2}{\partial \theta \partial \phi} \right]. \end{aligned}$$

Note also that $\frac{\partial^2}{\partial \theta \partial \phi} = \frac{\partial^2}{\partial \phi \partial \theta}$. This is because we are dealing with continuous and differentiable functions.

Meanwhile, we have the following commutation relations:

$$[L_x, L_y] = i\hbar L_z, \quad [L_y, L_z] = i\hbar L_x, \quad \text{and} \quad [L_z, L_x] = i\hbar L_y. \quad (3.30)$$

This can easily be confirmed by requiring canonical commutation relations. The derivation can routinely be performed, but we show it because the procedures include several important points. For instance, we have

$$\begin{aligned}
[L_x, L_y] &= L_x L_y - L_y L_x \\
&= (y p_z - z p_y)(z p_x - x p_z) - (z p_x - x p_z)(y p_z - z p_y) \\
&= y p_z z p_x - y p_z x p_z - z p_y z p_x + z p_y x p_z \\
&\quad - z p_x y p_z + z p_x z p_y + x p_z y p_z - x p_z z p_y \\
&= (y p_x z p_z - z p_x y p_z) + (z p_y x p_z - x p_z z p_y) \\
&\quad + (x p_z y p_z - y p_z x p_z) + (z p_x z p_y - z p_y z p_x) \\
&= -y p_x (z p_z - p_z z) + x p_y (z p_z - p_z z) = i\hbar (x p_y - y p_x) = i\hbar L_z.
\end{aligned}$$

In the above calculations, we used the canonical commutation relation as well as commutability of, e.g., y and p_x ; y and z ; p_x and p_y . For example, we get

$$[p_x, p_y] |\psi\rangle = -\hbar^2 \left(\frac{\partial}{\partial x} \frac{\partial}{\partial y} - \frac{\partial}{\partial y} \frac{\partial}{\partial x} \right) |\psi\rangle = -\hbar^2 \left(\frac{\partial^2}{\partial x \partial y} |\psi\rangle - \frac{\partial^2}{\partial y \partial x} |\psi\rangle \right) = 0.$$

In the above equation, we assumed that the order of differentiation with respect to x and y can be switched. It is because we are dealing with continuous and differentiable normal functions. Thus, p_x and p_y commute.

For other important commutation relations, we have

$$[L_x, L^2] = 0, \quad [L_y, L^2] = 0, \quad \text{and} \quad [L_z, L^2] = 0. \quad (3.31)$$

With the derivation, use

$$[A, B + C] = [A, B] + [A, C].$$

The derivation is straightforward and it is left for readers. The relations (3.30) and (3.31) imply that a *simultaneous eigenstate* exists for L^2 and one of L_x , L_y , and L_z . This is because L^2 commute with them from (3.31), whereas L_z does not commute with L_x or L_y . The detailed argument about the simultaneous eigenstate can be seen in Part III.

Thus, we have

$$\begin{aligned}
L^{(+)} L^{(-)} &= L_x^2 + L_y^2 + i(L_y L_x - L_x L_y) = L_x^2 + L_y^2 + i[L_y, L_x] \\
&= L_x^2 + L_y^2 + \hbar L_z.
\end{aligned}$$

Notice here that $[L_y, L_x] = -[L_x, L_y] = -i\hbar L_z$. Hence,

$$L^2 = L^{(+)} L^{(-)} + L_z^2 - \hbar L_z. \quad (3.32)$$

From (3.24), we have

$$L_z^2 = -\hbar^2 \frac{\partial^2}{\partial \phi^2}. \quad (3.33)$$

Finally, we get

$$L^2 = -\hbar^2 \left(\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right)$$

or

$$L^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]. \quad (3.34)$$

Replacing L^2 in (3.15) with that of (3.34), we have

$$\begin{aligned} H = & -\frac{\hbar^2}{2\mu r^2} \left[\frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \\ & - \frac{Ze^2}{4\pi\epsilon_0 r}. \end{aligned} \quad (3.35)$$

Thus, the Schrödinger equation of (3.3) takes a following form:

$$\begin{aligned} & \left\{ -\frac{\hbar^2}{2\mu r^2} \left[\frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] - \frac{Ze^2}{4\pi\epsilon_0 r} \right\} \psi \\ & = E\psi. \end{aligned} \quad (3.36)$$

3.3 Separation of Variables

If the potential is spherically symmetric (e.g., a Coulomb potential), it is well-known that the Schrödinger equations of (3.1)–(3.3) can be solved by a method of separation of variables. More specifically, (3.36) can be separated into two differential equations one of which only depends on a radial component r and the other of which depends only upon angular components θ and ϕ .

To apply the method of separation of variables to (3.36), let us first return to (3.15). Considering that L^2 is expressed as (3.34), we assume that L^2 has eigenvalues γ (at any rate if any) and takes eigenfunctions $Y(\theta, \phi)$ (again, if any as well) corresponding to γ . That is,

$$\mathbf{L}^2 Y(\theta, \phi) = \gamma Y(\theta, \phi), \quad (3.37)$$

where $Y(\theta, \phi)$ is assumed to be normalized. Meanwhile,

$$\mathbf{L}^2 = L_x^2 + L_y^2 + L_z^2. \quad (3.38)$$

From (3.6), we have

$$L_x^\dagger = (yp_z - zp_y)^\dagger = p_z^\dagger y^\dagger - p_y^\dagger z^\dagger = p_z y - p_y z = yp_z - zp_y = L_x. \quad (3.39)$$

Note that p_z and y commute, so do p_y and z . Therefore, L_x is Hermitian, so is L_x^2 . More generally if an operator A is Hermitian, so is A^n (n : a positive integer); readers, please show it. Likewise, L_y and L_z are Hermitian as well. Thus, $\mathbf{L}^2$ is Hermitian, too.

Next, we consider an expectation value of $\mathbf{L}^2$, i.e., $\langle \mathbf{L}^2 \rangle$. Let $|\psi\rangle$ be an arbitrary normalized non-zero vector (or function). Then,

$$\begin{aligned} \langle \mathbf{L}^2 \rangle &\equiv \langle \psi | \mathbf{L}^2 \psi \rangle \\ &= \langle \psi | L_x^2 \psi \rangle + \langle \psi | L_y^2 \psi \rangle + \langle \psi | L_z^2 \psi \rangle \\ &= \langle L_x^\dagger \psi | L_x \psi \rangle + \langle L_y^\dagger \psi | L_y \psi \rangle + \langle L_z^\dagger \psi | L_z \psi \rangle \\ &= \langle L_x \psi | L_x \psi \rangle + \langle L_y \psi | L_y \psi \rangle + \langle L_z \psi | L_z \psi \rangle \\ &= ||L_x \psi||^2 + ||L_y \psi||^2 + ||L_z \psi||^2 \geq 0. \end{aligned} \quad (3.40)$$

Notice that the second last equality comes from that L_x , L_y , and L_z are Hermitian. An operator that satisfies (3.40) is said to be non-negative (see Sects. 1.4, 2.2, etc. where we saw the calculation routines). Note also that in (3.40) the equality holds only when the following relations hold:

$$|L_x \psi\rangle = |L_y \psi\rangle = |L_z \psi\rangle = 0. \quad (3.41)$$

On this condition, we have

$$\begin{aligned} |\mathbf{L}^2 \psi\rangle &= |(L_x^2 + L_y^2 + L_z^2) \psi\rangle = |L_x^2 \psi\rangle + |L_y^2 \psi\rangle + |L_z^2 \psi\rangle \\ &= L_x |L_x \psi\rangle + L_y |L_y \psi\rangle + L_z |L_z \psi\rangle = 0. \end{aligned} \quad (3.42)$$

The eigenfunction that satisfies (3.42) and the next relation (3.43) is a simultaneous eigenstate of L_x , L_y , L_z , and $\mathbf{L}^2$. This could seem to be in contradiction to the fact that L_z does not commute with L_x or L_y . However, this is an exceptional case. Let $|\psi_0\rangle$ be the eigenfunction that satisfies both (3.41) and (3.42). Then, we have

$$|L_x \psi_0\rangle = |L_y \psi_0\rangle = |L_z \psi_0\rangle = |\mathbf{L}^2 \psi_0\rangle = 0. \quad (3.43)$$

As can be seen from (3.24) to (3.26) along with (3.34), the operators L_x , L_y , L_z , and L^2 are differential operators. Therefore, (3.43) implies that $|\psi_0\rangle$ is a constant. We will come back this point later. In spite of this exceptional situation, it is impossible that all L_x , L_y , and L_z as well as L^2 take a whole set of eigenfunctions as simultaneous eigenstates. We briefly show this as follows.

In Chap. 2, we mention that if $[A, B] = ik$, any physical state cannot be an eigenstate of A or B . The situation is different, on the other hand, if we have a following case

$$[A, B] = iC, \quad (3.44)$$

where A , B , and C are Hermitian operators. The relation (3.30) is a typical example for this. If $C|\psi\rangle = 0$ in (3.44), $|\psi\rangle$ might well be an eigenstate of A and/or B . However, if $C|\psi\rangle = c|\psi\rangle$ ($c \neq 0$), $|\psi\rangle$ cannot be an eigenstate of A or B . This can readily be shown in a fashion similar to that described in Sect. 2.5. Let us think of, e.g., $[L_x, L_y] = i\hbar L_z$. Suppose that for $|\psi_0\rangle$ we have $L_z|\psi_0\rangle = 0$. Taking an inner product using $|\psi_0\rangle$, from (3.30) we have

$$\langle\psi_0|(L_x L_y - L_y L_x)\psi_0\rangle = 0.$$

In this case, moreover, even if we have $L_x|\psi_0\rangle = 0$ and $L_y|\psi_0\rangle = 0$, we have no inconsistency. If, on the other hand, $L_z|\psi\rangle = m|\psi\rangle$ ($m \neq 0$), $|\psi\rangle$ cannot be an eigenstate of L_x or L_y as mentioned above. Thus, we should be careful to deal with a general situation where we have $[A, B] = iC$.

In the case where $[A, B] = 0$; $AB = BA$, namely A and B commute, we have a different situation. This relation is equivalent to that an operator $AB - BA$ has an eigenvalue zero for any physical state $|\psi\rangle$. Yet, this statement is of less practical use. Again, regarding details we wish to make a discussion in Sect. 14.6 of Part III.

Returning to (3.40), let us replace ψ with a particular eigenfunction $Y(\theta, \phi)$. Then, we have

$$\langle Y|L^2 Y\rangle = \langle Y|\gamma Y\rangle = \gamma\langle Y|Y\rangle = \gamma \geq 0. \quad (3.45)$$

Again, if L^2 has an eigenvalue, the eigenvalue should be non-negative. Taking account of the coefficient $\hbar^2$ in (3.34), it is convenient to put

$$\gamma = \hbar^2 \lambda \quad (\lambda \geq 0). \quad (3.46)$$

On ground that the solution of (3.36) can be described as

$$\psi(r, \theta, \phi) = R(r)Y(\theta, \phi), \quad (3.47)$$

the Schrödinger equation (3.16) can be rewritten as

$$\left\{ \frac{1}{2\mu} \left[-\frac{\hbar^2}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{L^2}{r^2} \right] - \frac{Ze^2}{4\pi\epsilon_0 r} \right\} R(r) Y(\theta, \phi) = ER(r) Y(\theta, \phi). \quad (3.48)$$

That is,

$$\frac{1}{2\mu} \left[-\frac{\hbar^2}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R(r)}{\partial r} \right) Y(\theta, \phi) + \frac{L^2 Y(\theta, \phi)}{r^2} R(r) \right] - \frac{Ze^2}{4\pi\epsilon_0 r} R(r) Y(\theta, \phi) = ER(r) Y(\theta, \phi). \quad (3.49)$$

Recalling (3.37) and (3.46), we have

$$\frac{1}{2\mu} \left[-\frac{\hbar^2}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R(r)}{\partial r} \right) Y(\theta, \phi) + \frac{\hbar^2 \lambda Y(\theta, \phi)}{r^2} R(r) \right] - \frac{Ze^2}{4\pi\epsilon_0 r} R(r) Y(\theta, \phi) = ER(r) Y(\theta, \phi). \quad (3.50)$$

Dividing both sides by $Y(\theta, \phi)$, we get a SOLDE of a radial component as

$$\frac{1}{2\mu} \left[-\frac{\hbar^2}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R(r)}{\partial r} \right) + \frac{\hbar^2 \lambda}{r^2} \right] R(r) - \frac{Ze^2}{4\pi\epsilon_0 r} R(r) = ER(r). \quad (3.51)$$

Regarding angular components θ and ϕ , using (3.34), (3.37), and (3.46), we have

$$\begin{aligned} L^2 Y(\theta, \phi) &= -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] Y(\theta, \phi) \\ &= \hbar^2 \lambda Y(\theta, \phi). \end{aligned} \quad (3.52)$$

Dividing both sides by $\hbar^2$, we get

$$-\left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] Y(\theta, \phi) = \lambda Y(\theta, \phi). \quad (3.53)$$

Notice in (3.53) that the angular part of SOLDE does not depend on a specific form of the potential.

Now, we further assume that (3.53) can be separated into a zenithal angle part θ and azimuthal angle part ϕ such that

$$Y(\theta, \phi) = \Theta(\theta) \Phi(\phi). \quad (3.54)$$

Then we have

$$- \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \Theta(\theta)}{\partial \theta} \right) \Phi(\phi) + \frac{1}{\sin^2 \theta} \frac{\partial^2 \Phi(\phi)}{\partial \phi^2} \right] \Theta(\theta) = \lambda \Theta(\theta) \Phi(\phi). \quad (3.55)$$

Multiplying both sides by $\sin^2 \theta / \Theta(\theta) \Phi(\phi)$ and arranging both the sides, we get

$$- \frac{1}{\Phi(\phi)} \frac{\partial^2 \Phi(\phi)}{\partial \phi^2} = \frac{\sin^2 \theta}{\Theta(\theta)} \left\{ \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left[\sin \theta \frac{\partial \Theta(\theta)}{\partial \theta} \right] + \lambda \Theta(\theta) \right\}. \quad (3.56)$$

Since LHS of (3.56) depends only upon ϕ and RHS depends only on θ , we must have

$$\text{LHS of (3.56)} = \text{RHS of (3.56)} = \eta(\text{constant}). \quad (3.57)$$

Thus, we have a following relation of LHS of (3.56):

$$- \frac{1}{\Phi(\phi)} \frac{d^2 \Phi(\phi)}{d\phi^2} = \eta. \quad (3.58)$$

Putting $D \equiv - \frac{d^2}{d\phi^2}$, we get

$$D\Phi(\phi) = \eta\Phi(\phi). \quad (3.59)$$

The SOLDEs of (3.58) and (3.59) are formally the same as (1.61) of Sect. 1.3, where boundary conditions (BCs) are Dirichlet conditions. Unlike (1.61), however, we have to consider different BCs, i.e., the periodic BCs.

As in Example 1.1, we adopt two linearly independent solutions. That is, we have

$$e^{im\phi} \quad \text{and} \quad e^{-im\phi} \quad (m \neq 0).$$

As their linear combination, we have

$$\Phi(\phi) = ae^{im\phi} + be^{-im\phi}. \quad (3.60)$$

As BCs, we consider $\Phi(0) = \Phi(2\pi)$ and $\Phi'(0) = \Phi'(2\pi)$; i.e., we have

$$a + b = ae^{i2\pi m} + be^{-i2\pi m}. \quad (3.61)$$

Meanwhile, we have

$$\Phi'(\phi) = aime^{im\phi} - bime^{-im\phi}. \quad (3.62)$$

Therefore, from BCs we have

$$aim - bim = aime^{i2\pi m} - bime^{-i2\pi m}.$$

Then,

$$a - b = ae^{i2\pi m} - be^{-i2\pi m}. \quad (3.63)$$

From (3.61) and (3.63), we have

$$2a(1 - e^{i2\pi m}) = 0 \quad \text{and} \quad 2b(1 - e^{-i2\pi m}) = 0.$$

If $a \neq 0$, we must have $m = 0, \pm 1, \pm 2, \dots$. If $a = 0$, we must have $b \neq 0$ to avoid having $\Phi(\phi) \equiv 0$ as a solution. In that case, we have $m = 0, \pm 1, \pm 2, \dots$ as well. Thus, it suffices to put

$$\Phi(\phi) = ce^{im\phi} \quad (m = 0, \pm 1, \pm 2, \dots).$$

Therefore, as a normalized function $\tilde{\Phi}(\phi)$, we get

$$\tilde{\Phi}(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi} \quad (m = 0, \pm 1, \pm 2, \dots). \quad (3.64)$$

Inserting it into (3.58), we have

$$m^2 e^{im\phi} = \eta e^{im\phi}.$$

Therefore, we get

$$\eta = m^2 \quad (m = 0, \pm 1, \pm 2, \dots). \quad (3.65)$$

From (3.56) and (3.65), we have

$$-\frac{1}{\sin \theta} \frac{d}{d\theta} \left[\sin \theta \frac{d\Theta(\theta)}{d\theta} \right] + \frac{m^2 \Theta(\theta)}{\sin^2 \theta} = \lambda \Theta(\theta) \quad (m = 0, \pm 1, \pm 2, \dots). \quad (3.66)$$

In (3.64) putting $m = 0$ as an eigenvalue, we have $\Phi(\phi) = 1/\sqrt{2\pi}$ as a corresponding eigenfunction. Unlike Examples 1.1 and 1.2, this reflects that the differential operator $-d^2/d\phi^2$ accompanied by the periodic BCs is a non-negative operator that allows an eigenvalue of zero. Yet, we are uncertain of a range of m . To clarify this point, we consider generalized angular momentum in the next section.

3.4 Generalized Angular Momentum

We obtained commutation relations of (3.30) among individual angular momentum components L_x , L_y , and L_z . In an opposite way, we may start with (3.30) to define angular momentum. Such a quantity is called generalized angular momentum.

Let $\tilde{\mathbf{J}}$ be a generalized angular momentum as in the case of (3.4) such that

$$\tilde{\mathbf{J}} = (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} \tilde{J}_x \\ \tilde{J}_y \\ \tilde{J}_z \end{pmatrix}. \quad (3.67)$$

For the sake of simple notation, let us define $\mathbf{J}$ as follows so that we can eliminate $\hbar$ and deal with dimensionless quantities in the present discussion:

$$\begin{aligned} \mathbf{J} &\equiv \tilde{\mathbf{J}}/\hbar = (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} \tilde{J}_x/\hbar \\ \tilde{J}_y/\hbar \\ \tilde{J}_z/\hbar \end{pmatrix} = (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} J_x \\ J_y \\ J_z \end{pmatrix}, \\ \mathbf{J}^2 &= J_x^2 + J_y^2 + J_z^2. \end{aligned} \quad (3.68)$$

Then, we require the following commutation relations:

$$[J_x, J_y] = iJ_z, \quad [J_y, J_z] = iJ_x, \quad \text{and} \quad [J_z, J_x] = iJ_y. \quad (3.69)$$

Also, we require J_x , J_y , and J_z to be Hermitian. The operator $\mathbf{J}^2$ is Hermitian accordingly. The relations (3.69) lead to

$$[J_x, \mathbf{J}^2] = 0, \quad [J_y, \mathbf{J}^2] = 0, \quad \text{and} \quad [J_z, \mathbf{J}^2] = 0. \quad (3.70)$$

This can be confirmed as in the case of (3.30).

As noted above, again a simultaneous eigenstate exists for $\mathbf{J}^2$ and one of J_x , J_y , and J_z . According to the convention, we choose $\mathbf{J}^2$ and J_z for the simultaneous eigenstate. Then, designating the eigenstate by $|\zeta, \mu\rangle$, we have

$$\mathbf{J}^2 |\zeta, \mu\rangle = \zeta |\zeta, \mu\rangle \quad \text{and} \quad J_z |\zeta, \mu\rangle = \mu |\zeta, \mu\rangle. \quad (3.71)$$

The implication of (3.71) is that $|\zeta, \mu\rangle$ is the simultaneous eigenstate and that μ is an eigenvalue of J_z which $|\zeta, \mu\rangle$ belongs to with ζ being an eigenvalue of $\mathbf{J}^2$ which $|\zeta, \mu\rangle$ belongs to as well.

Since J_z and $\mathbf{J}^2$ are Hermitian, both μ and ζ are real (see Sect. 1.4). Of these, $\zeta \geq 0$ as in the case of (3.45). We define the following operators $\mathbf{J}^{(+)}$ and $\mathbf{J}^{(-)}$ as in the case of (3.27):

$$J^{(+)} \equiv J_x + iJ_y \quad \text{and} \quad J^{(-)} \equiv J_x - iJ_y. \quad (3.72)$$

Then, from (3.69) and (3.70), we get

$$[J^{(+)}, J^2] = [J^{(-)}, J^2] = 0. \quad (3.73)$$

Also, we obtain the following commutation relations:

$$[J_z, J^{(+)}] = J^{(+)}; \quad [J_z, J^{(-)}] = -J^{(-)}; \quad [J^{(+)}, J^{(-)}] = 2J_z. \quad (3.74)$$

From (3.70) to (3.72), we get

$$\begin{aligned} J^2 J^{(+)} | \zeta, \mu \rangle &= J^{(+)} J^2 | \mu \rangle = \zeta J^{(+)} | \zeta, \mu \rangle, \\ J^2 J^{(-)} | \zeta, \mu \rangle &= J^{(-)} J^2 | \zeta, \mu \rangle = \zeta J^{(-)} | \zeta, \mu \rangle. \end{aligned} \quad (3.75)$$

Equation (3.75) indicates that both $J^{(+)} | \zeta, \mu \rangle$ and $J^{(-)} | \zeta, \mu \rangle$ are eigenvectors of J^2 that correspond to an eigenvalue ζ .

Meanwhile, from (3.74) we get

$$\begin{aligned} J_z J^{(+)} | \zeta, \mu \rangle &= J^{(+)} (J_z + 1) | \zeta, \mu \rangle = (\mu + 1) J^{(+)} | \zeta, \mu \rangle, \\ J_z J^{(-)} | \zeta, \mu \rangle &= J^{(-)} (J_z - 1) | \zeta, \mu \rangle = (\mu - 1) J^{(-)} | \zeta, \mu \rangle. \end{aligned} \quad (3.76)$$

The relation (3.76) means that $J^{(+)} | \zeta, \mu \rangle$ is an eigenvector of J_z corresponding to an eigenvalue $(\mu + 1)$, while $J^{(-)} | \zeta, \mu \rangle$ is an eigenvector of J_z corresponding to an eigenvalue $(\mu - 1)$. This implies that $J^{(+)}$ and $J^{(-)}$ function as raising and lowering operators (or ladder operators) that have been introduced in this chapter. Thus, using undetermined constants (or phase factors) $a_\mu^{(+)}$ and $a_\mu^{(-)}$, we describe

$$J^{(+)} | \zeta, \mu \rangle = a_\mu^{(+)} | \zeta, \mu + 1 \rangle \quad \text{and} \quad J^{(-)} | \zeta, \mu \rangle = a_\mu^{(-)} | \zeta, \mu - 1 \rangle. \quad (3.77)$$

Next, let us characterize eigenvalues μ . We have

$$J_x^2 + J_y^2 = J^2 - J_z^2. \quad (3.78)$$

Therefore,

$$(J_x^2 + J_y^2) | \zeta, \mu \rangle = (J^2 - J_z^2) | \zeta, \mu \rangle = (\zeta - \mu^2) | \zeta, \mu \rangle. \quad (3.79)$$

Since $(J_x^2 + J_y^2)$ is a non-negative operator, its eigenvalues are non-negative as well, as can be seen from (3.40) and (3.45). Then, we have

$$\zeta - \mu^2 \geq 0. \quad (3.80)$$

Thus, for a fixed value of non-negative ζ , μ is bounded both upwards and downwards. We define then a maximum of μ as j and a minimum of μ as j' . Consequently, on the basis of (3.77), we have

$$J^{(+)} | \zeta, j \rangle = 0 \quad \text{and} \quad J^{(-)} | \zeta, j' \rangle = 0. \quad (3.81)$$

This is because we have no quantum state corresponding to $|\zeta, j+1\rangle$ or $|\zeta, j'-1\rangle$. From (3.75) and (3.81), possible numbers of μ are

$$j, j-1, j-2, \dots, j'. \quad (3.82)$$

From (3.69) and (3.72), we get

$$J^{(-)}J^{(+)} = \mathbf{J}^2 - J_z^2 - J_z, \quad J^{(+)}J^{(-)} = \mathbf{J}^2 - J_z^2 + J_z. \quad (3.83)$$

Operating these operators on $|\zeta, j\rangle$ or $|\zeta, j'\rangle$ and using (3.81) we get

$$\begin{aligned} J^{(-)}J^{(+)} | \zeta, j \rangle &= (\mathbf{J}^2 - J_z^2 - J_z) | \zeta, j \rangle = (\zeta - j^2 - j) | \zeta, j \rangle = 0, \\ J^{(+)}J^{(-)} | \zeta, j' \rangle &= (\mathbf{J}^2 - J_z^2 + J_z) | \zeta, j' \rangle = (\zeta - j'^2 + j') | \zeta, j' \rangle = 0. \end{aligned} \quad (3.84)$$

Since $|\zeta, j\rangle \neq 0$ and $|\zeta, j'\rangle \neq 0$, we have

$$\zeta - j^2 - j = \zeta - j'^2 + j' = 0. \quad (3.85)$$

This means that

$$\zeta = j(j+1) = j'(j'-1). \quad (3.86)$$

Moreover, from (3.86) we get

$$j(j+1) - j'(j'-1) = (j+j')(j-j'+1) = 0. \quad (3.87)$$

As $j \geq j'$, $j - j' + 1 > 0$. From (3.87), therefore, we get

$$j + j' = 0 \quad \text{or} \quad j = -j'. \quad (3.88)$$

Then, we conclude that the minimum of μ is $-j$. Accordingly, possible values of μ are

$$\mu = j, j-1, j-2, \dots, -j-1, -j. \quad (3.89)$$

That is, the number μ can take is $(2j+1)$. The relation (3.89) implies that taking a positive integer k ,

$$j-k = -j \quad \text{or} \quad j = k/2. \quad (3.90)$$

In other words, j is permitted to take a number zero, a positive integer, or a positive half-integer (or more precisely, half-odd-integer). For instance, if $j = 1/2$, μ can be $1/2$ or $-1/2$. When $j = 1$, μ can be $1, 0$, or -1 .

Finally, we have to decide undetermined constants $a_\mu^{(+)}$ and $a_\mu^{(-)}$. To this end, multiplying $\langle \zeta, \mu - 1 |$ on both sides of the second equation of (3.77) from the left, we have

$$\langle \zeta, \mu - 1 | J^{(-)} | \zeta, \mu \rangle = a_\mu^{(-)} \langle \zeta, \mu - 1 | \zeta, \mu - 1 \rangle = a_\mu^{(-)}, \quad (3.91)$$

where the second equality comes from that $|\zeta, \mu - 1\rangle$ has been normalized; i.e., $\| |\zeta, \mu - 1\rangle \| = 1$. Meanwhile, taking adjoint of both sides of the first equation of (3.77), we have

$$\langle \zeta, \mu | [J^{(+)}]^\dagger = [a_\mu^{(+)}]^* \langle \zeta, \mu + 1 |. \quad (3.92)$$

However, from (3.72) and the fact that J_x and J_y are Hermitian,

$$[J^{(+)}]^\dagger = J^{(-)}. \quad (3.93)$$

Using (3.93) and replacing μ in (3.92) with $\mu - 1$, we get

$$\langle \zeta, \mu - 1 | J^{(-)} = [a_{\mu-1}^{(+)}]^* \langle \zeta, \mu |. \quad (3.94)$$

Furthermore, multiplying $|\zeta, \mu\rangle$ on (3.94) from the right, we have

$$\langle \zeta, \mu - 1 | J^{(-)} | \zeta, \mu \rangle = [a_{\mu-1}^{(+)}]^* \langle \zeta, \mu | \zeta, \mu \rangle = [a_{\mu-1}^{(+)}]^*, \quad (3.95)$$

where again $|\zeta, \mu\rangle$ is assumed to be normalized. Comparing (3.91) and (3.95), we get

$$a_\mu^{(-)} = [a_{\mu-1}^{(+)}]^*. \quad (3.96)$$

Taking an inner product regarding the first equation of (3.77) and its adjoint,

$$\langle \zeta, \mu | J^{(-)} J^{(+)} | \zeta, \mu \rangle = \left[a_{\mu}^{(+)} \right]^* a_{\mu}^{(+)} \langle \zeta, \mu + 1 | \zeta, \mu + 1 \rangle = |a_{\mu}^{(+)}|^2. \quad (3.97)$$

Once again, the second equality of (3.97) results from the normalization of the vector.

Using (3.83) as well as (3.71) and (3.86), (3.97) can be rewritten as

$$\begin{aligned} \langle \zeta, \mu | J^2 - J_z^2 - J_z | \zeta, \mu \rangle \\ = \langle \zeta, \mu | j(j+1) - \mu^2 - \mu | \zeta, \mu \rangle = \langle \zeta, \mu | \zeta, \mu \rangle (j - \mu)(j + \mu + 1) = |a_{\mu}^{(+)}|^2. \end{aligned} \quad (3.98)$$

Thus, we get

$$a_{\mu}^{(+)} = e^{i\delta} \sqrt{(j - \mu)(j + \mu + 1)} \quad (\delta : \text{an arbitrary real number}), \quad (3.99)$$

where $e^{i\delta}$ is a phase factor. From (3.96) we also get

$$a_{\mu}^{(-)} = e^{-i\delta} \sqrt{(j - \mu + 1)(j + \mu)}. \quad (3.100)$$

In (3.99) and (3.100), we routinely put $\delta = 0$ so that $a_{\mu}^{(+)}$ and $a_{\mu}^{(-)}$ can be positive numbers. Explicitly rewriting (3.77), we get

$$\begin{aligned} J^{(+)} | \zeta, \mu \rangle &= \sqrt{(j - \mu)(j + \mu + 1)} | \zeta, \mu + 1 \rangle, \\ J^{(-)} | \zeta, \mu \rangle &= \sqrt{(j - \mu + 1)(j + \mu)} | \zeta, \mu - 1 \rangle, \end{aligned} \quad (3.101)$$

where j is a fixed given number chosen from among zero, positive integers, and positive half-integers (or half-odd-integers).

As discussed above, we have derived various properties and relations with respect to the generalized angular momentum on the basis of (1) the relation (3.30) or (3.69) and (2) the fact that J_x , J_y , and J_z are Hermitian operators. This notion is very useful in dealing with various angular momenta of different origins (e.g., orbital angular momentum, spin angular momentum, etc.) from a unified point of view. In Chap. 20, we will revisit this issue in more detail.

3.5 Orbital Angular Momentum: Operator Approach

In Sect. 3.4 we have derived various important results on angular momenta on the basis of the commutation relations (3.69) and the assumption that J_x , J_y , and J_z are Hermitian. Now, let us return to the discussion on orbital angular momenta we dealt with in Sects. 3.2 and 3.3. First, we treat the orbital angular momenta via operator approach. This approach enables us to understand why a quantity j introduced in

Sect. 3.4 takes a value zero or positive integers with the orbital angular momenta. In the next section (Sect. 3.6) we will deal with the related issues by an analytical method.

In (3.28) we introduced differential operators $L^{(+)}$ and $L^{(-)}$. According to Sect. 3.4, we define the following operators to eliminate $\hbar$ so that we can deal with dimensionless quantities:

$$\begin{aligned} \mathbf{M} &\equiv \mathbf{L}/\hbar = (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} M_x \\ M_y \\ M_z \end{pmatrix}, \\ \mathbf{M}^2 &= \mathbf{L}^2/\hbar^2 = M_x^2 + M_y^2 + M_z^2. \end{aligned} \quad (3.102)$$

Hence, we have

$$M_x = L_x/\hbar, \quad M_y = L_y/\hbar, \quad \text{and} \quad M_z = L_z/\hbar. \quad (3.103)$$

Moreover, we define the following operators [3]:

$$M^{(+)} \equiv M_x + iM_y = L^{(+)}/\hbar = e^{i\phi} \left(\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right), \quad (3.104)$$

$$M^{(-)} \equiv M_x - iM_y = L^{(-)}/\hbar = e^{-i\phi} \left(-\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right). \quad (3.105)$$

Then we have

$$\mathbf{M}^2 = - \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]. \quad (3.106)$$

Here we execute variable transformation such that

$$\xi = \cos \theta \quad (0 \leq \theta \leq \pi) \quad \text{or} \quad \sin \theta = \sqrt{1 - \xi^2}. \quad (3.107)$$

Noting, e.g., that

$$\begin{aligned} \frac{\partial}{\partial \theta} &= \frac{\partial \xi}{\partial \theta} \frac{\partial}{\partial \xi} = -\sin \theta \frac{\partial}{\partial \xi} = -\sqrt{1 - \xi^2} \frac{\partial}{\partial \xi}, \quad \sin \theta \frac{\partial}{\partial \theta} = \\ &= -\sin^2 \theta \frac{\partial}{\partial \xi} = -(1 - \xi^2) \frac{\partial}{\partial \xi}, \end{aligned} \quad (3.108)$$

we get

$$\begin{aligned}
M^{(+)} &= e^{i\phi} \left(-\sqrt{1-\xi^2} \frac{\partial}{\partial \xi} + i \frac{\xi}{\sqrt{1-\xi^2}} \frac{\partial}{\partial \phi} \right), \\
M^{(-)} &= e^{-i\phi} \left(\sqrt{1-\xi^2} \frac{\partial}{\partial \xi} + i \frac{\xi}{\sqrt{1-\xi^2}} \frac{\partial}{\partial \phi} \right), \\
M^2 &= -\frac{\partial}{\partial \xi} \left[(1-\xi^2) \frac{\partial}{\partial \xi} \right] - \frac{1}{1-\xi^2} \frac{\partial^2}{\partial \phi^2}.
\end{aligned} \tag{3.109}$$

Although we showed in Sect. 3.3 that $m = 0, \pm 1, \pm 2, \dots$, the range of m was unclear. The relationship between m and λ in (3.66) remains unclear so far as well. On the basis of a general approach developed in Sect. 3.4, however, we have known that the eigenvalue μ of the dimensionless z -component angular momentum J_z is bounded with its maximum and minimum being j and $-j$, respectively [see (3.89)], where j can be zero, a positive integer, or a positive half-odd-integer. Concomitantly, the eigenvalue ζ of $\mathbf{J}^2$ equals $j(j+1)$.

In the present section, let us reconsider the relationship between m and λ in (3.66) in light of the knowledge obtained in Sect. 3.4. According to the custom, we replace μ in (3.89) with m to have

$$m = j, j-1, j-2, \dots, j-1, -j. \tag{3.110}$$

At the moment, we assume that m can be a half-odd-integer besides zero or an integer [3].

Now, let us define notation of $Y(\theta, \phi)$ that appeared in (3.37). This function is eligible for a simultaneous eigenstate of M^2 and M_z and can be indexed with j and m as in (3.110). Then, let $Y(\theta, \phi)$ be described accordingly as

$$Y_j^m(\theta, \phi) \equiv Y(\theta, \phi). \tag{3.111}$$

From (3.54) and (3.64), we have

$$Y_j^m(\theta, \phi) \propto e^{im\phi}.$$

Therefore, we get

$$\begin{aligned}
M^{(+)} Y_j^m(\theta, \phi) &= e^{i\phi} \left(-\sqrt{1-\xi^2} \frac{\partial}{\partial \xi} - \frac{m\xi}{\sqrt{1-\xi^2}} \right) Y_j^m(\theta, \phi) \\
&= -e^{i\phi} \left(\sqrt{1-\xi^2} \right)^{m+1} \frac{\partial}{\partial \xi} \left[\left(\sqrt{1-\xi^2} \right)^{-m} Y_j^m(\theta, \phi) \right],
\end{aligned} \tag{3.112}$$

where we used the following equations:

$$\begin{aligned}
\frac{\partial}{\partial \xi} \left[\left(\sqrt{1 - \xi^2} \right)^{-m} \right] &= (-m) \left(\sqrt{1 - \xi^2} \right)^{-m-1} \cdot \frac{1}{2} \left(\sqrt{1 - \xi^2} \right)^{-1} (-2\xi) \\
&= m\xi \left(\sqrt{1 - \xi^2} \right)^{-m-2}, \\
\frac{\partial}{\partial \xi} \left[\left(\sqrt{1 - \xi^2} \right)^{-m} Y_j^m(\theta, \phi) \right] &= m\xi \left(\sqrt{1 - \xi^2} \right)^{-m-2} Y_j^m(\theta, \phi) \\
&\quad + \left(\sqrt{1 - \xi^2} \right)^{-m} \frac{\partial Y_j^m(\theta, \phi)}{\partial \xi}.
\end{aligned} \tag{3.113}$$

Similarly, we get

$$\begin{aligned}
M^{(-)} Y_j^m(\theta, \phi) &= e^{-i\phi} \left(\sqrt{1 - \xi^2} \frac{\partial}{\partial \xi} - \frac{m\xi}{\sqrt{1 - \xi^2}} \right) Y_j^m(\theta, \phi) \\
&= e^{-i\phi} \left(\sqrt{1 - \xi^2} \right)^{-m+1} \frac{\partial}{\partial \xi} \left[\left(\sqrt{1 - \xi^2} \right)^m Y_j^m(\theta, \phi) \right].
\end{aligned} \tag{3.114}$$

Let us derive the relations where $M^{(+)}$ or $M^{(-)}$ is successively operated on $Y_j^m(\theta, \phi)$. In the case of $M^{(+)}$, using (3.109) we have

$$\begin{aligned}
\left[M^{(+)} \right]^n Y_j^m(\theta, \phi) &= (-1)^n e^{in\phi} \left(\sqrt{1 - \xi^2} \right)^{m+n} \frac{\partial^n}{\partial \xi^n} \\
&\quad \times \left[\left(\sqrt{1 - \xi^2} \right)^{-m} Y_j^m(\theta, \phi) \right].
\end{aligned} \tag{3.115}$$

We confirm this relation by mathematical induction. We have (3.112) by replacing n with 1 in (3.115). Namely, (3.115) holds when $n = 1$. Next, suppose that (3.115) holds with n . Then, using the first equation of (3.109) and noting (3.64), we have

$$\begin{aligned}
[M^{(+)}]^{n+1} Y_j^m(\theta, \phi) &= M^{(+)} \left\{ [M^{(+)}]^n Y_j^m(\theta, \phi) \right\} \\
&= e^{i\phi} \left(-\sqrt{1-\xi^2} \frac{\partial}{\partial \xi} + i \frac{\xi}{\sqrt{1-\xi^2}} \frac{\partial}{\partial \phi} \right) \\
&\quad \times (-1)^n e^{in\phi} \left(\sqrt{1-\xi^2} \right)^{m+n} \frac{\partial^n}{\partial \xi^n} \left[\left(\sqrt{1-\xi^2} \right)^{-m} Y_j^m(\theta, \phi) \right] \\
&= (-1)^n e^{i(n+1)\phi} \left[-\sqrt{1-\xi^2} \frac{\partial}{\partial \xi} - \frac{\xi(n+m)}{\sqrt{1-\xi^2}} \right] \\
&\quad \times \left(\sqrt{1-\xi^2} \right)^{m+n} \frac{\partial^n}{\partial \xi^n} \left[\left(\sqrt{1-\xi^2} \right)^{-m} Y_j^m(\theta, \phi) \right] \\
&= (-1)^n e^{i(n+1)\phi} \left\{ -\sqrt{1-\xi^2} \left[(m+n) \left(\sqrt{1-\xi^2} \right)^{m+n-1} \right. \right. \\
&\quad \times \frac{1}{2} \left(\sqrt{1-\xi^2} \right)^{-1} (-2\xi) \left. \frac{\partial^n}{\partial \xi^n} \left[\left(\sqrt{1-\xi^2} \right)^{-m} Y_j^m(\theta, \phi) \right] \right. \\
&\quad - \sqrt{1-\xi^2} \left(\sqrt{1-\xi^2} \right)^{m+n} \frac{\partial^{n+1}}{\partial \xi^{n+1}} \left[\left(\sqrt{1-\xi^2} \right)^{-m} Y_j^m(\theta, \phi) \right] \\
&\quad \left. \left. - \frac{\xi(n+m)}{\sqrt{1-\xi^2}} \left(\sqrt{1-\xi^2} \right)^{m+n} \frac{\partial^n}{\partial \xi^n} \left[\left(\sqrt{1-\xi^2} \right)^{-m} Y_j^m(\theta, \phi) \right] \right] \right\} \\
&= (-1)^{n+1} e^{i(n+1)\phi} \left(\sqrt{1-\xi^2} \right)^{m+(n+1)} \frac{\partial^{n+1}}{\partial \xi^{n+1}} \left[\left(\sqrt{1-\xi^2} \right)^{-m} Y_j^m(\theta, \phi) \right].
\end{aligned} \tag{3.116}$$

Notice that the first and third terms in the second last equality canceled each other. Thus, (3.115) certainly holds with $(n+1)$.

Similarly, we have [3]

$$\begin{aligned}
[M^{(-)}]^n Y_j^m(\theta, \phi) &= e^{-in\phi} \left(\sqrt{1-\xi^2} \right)^{-m+n} \frac{\partial^n}{\partial \xi^n} \\
&\quad \times \left[\left(\sqrt{1-\xi^2} \right)^m Y_j^m(\theta, \phi) \right].
\end{aligned} \tag{3.117}$$

Proof of (3.117) is left for readers.

From the second equation of (3.81) and (3.114) where m is replaced with $-j$, we have

$$\begin{aligned}
M^{(-)} Y_j^{-j}(\theta, \phi) &= e^{-i\phi} \left(\sqrt{1-\xi^2} \right)^{j+1} \frac{\partial}{\partial \xi} \left[\left(\sqrt{1-\xi^2} \right)^{-j} Y_j^{-j}(\theta, \phi) \right] \\
&= 0.
\end{aligned} \tag{3.118}$$

This implies that $\left(\sqrt{1-\xi^2} \right)^{-j} Y_j^{-j}(\theta, \phi)$ is a constant with respect to ξ . We describe this as

$$\left(\sqrt{1-\xi^2} \right)^{-j} Y_j^{-j}(\theta, \phi) = c \quad (c : \text{constant with respect to } \xi). \tag{3.119}$$

Meanwhile, putting $m = -j$ and $n = 2j + 1$ in (3.115) and taking account of the first equation of (3.77) and the first equation of (3.81), we get

$$\begin{aligned}
\left[M^{(+)} \right]^{2j+1} Y_j^{-j}(\theta, \phi) &= (-1)^{2j+1} e^{i(2j+1)\phi} \left(\sqrt{1-\xi^2} \right)^{j+1} \frac{\partial^{2j+1}}{\partial \xi^{2j+1}} \\
&\times \left[\left(\sqrt{1-\xi^2} \right)^j Y_j^{-j}(\theta, \phi) \right] = 0.
\end{aligned} \tag{3.120}$$

This means that

$$\left(\sqrt{1-\xi^2} \right)^j Y_j^{-j}(\theta, \phi) = (\text{at most a } 2j\text{-degree polynomial with } \xi). \tag{3.121}$$

Replacing $Y_j^{-j}(\theta, \phi)$ in (3.121) with that of (3.119), we get

$$\begin{aligned}
c \left(\sqrt{1-\xi^2} \right)^j \left(\sqrt{1-\xi^2} \right)^j &= c(1-\xi^2)^j \\
&= (\text{at most a } 2j\text{-degree polynomial with } \xi).
\end{aligned} \tag{3.122}$$

Here if j is a half-odd-integer, $c(1-\xi^2)^j$ of (3.122) cannot be a polynomial. If on the other hand j is zero or a positive integer, $c(1-\xi^2)^j$ is certainly a polynomial and, to top it all, a $2j$ -degree polynomial with respect to ξ ; so is $\left(\sqrt{1-\xi^2} \right)^j Y_j^{-j}(\theta, \phi)$.

According to the custom, henceforth we use l as zero or a positive integer instead of j . That is,

$$Y(\theta, \phi) \equiv Y_l^m(\theta, \phi) \quad (l : \text{zero or a positive integer}). \tag{3.123}$$

At the same time, so far as the orbital angular momentum is concerned, from (3.71) and (3.86) we can identify ζ in (3.71) with $l(l+1)$. Namely, we have

$$\zeta = l(l+1).$$

Concomitantly, m in (3.110) is determined as

$$m = l, l-1, l-2, \dots, 1, 0, -1, \dots, -l+1, -l. \quad (3.124)$$

Thus, as expected m is zero or a positive or negative integer. Considering (3.37) and (3.46), ζ is identical with λ in (3.46). Finally, we rewrite (3.66) such that

$$-\frac{1}{\sin\theta} \frac{d}{d\theta} \left[\sin\theta \frac{d\Theta(\theta)}{d\theta} \right] + \frac{m^2 \Theta(\theta)}{\sin^2\theta} = l(l+1) \Theta(\theta), \quad (3.125)$$

where l is equal to zero or positive integers and m is given by (3.124).

On condition of $\xi = \cos\theta$ (3.107), defining the following function:

$$P_l^m(\xi) \equiv \Theta(\theta), \quad (3.126)$$

and considering (3.109) along with (3.54), we arrive at the next SOLDE described as

$$\frac{d}{d\xi} \left[(1-\xi^2) \frac{dP_l^m(\xi)}{d\xi} \right] + \left[l(l+1) - \frac{m^2}{1-\xi^2} \right] P_l^m(\xi) = 0. \quad (3.127)$$

The SOLDE of (3.127) is well-known as the associated Legendre differential equation. The solutions $P_l^m(\xi)$ are called associated Legendre functions.

In the next section, we characterize the said equation and functions by an analytical method. Before going into details, however, we further seek characteristics of $P_l^m(\xi)$ by the operator approach.

Adopting the notation of (3.123) and putting $m = l$ in (3.112), we have

$$M^{(+)} Y_l^l(\theta, \phi) = -e^{i\phi} \left(\sqrt{1-\xi^2} \right)^{l+1} \frac{\partial}{\partial \xi} \left[\left(\sqrt{1-\xi^2} \right)^{-l} Y_l^l(\theta, \phi) \right]. \quad (3.128)$$

Corresponding to (3.81), we have $M^{(+)} Y_l^l(\theta, \phi) = 0$. This implies that

$$\left(\sqrt{1-\xi^2} \right)^{-l} Y_l^l(\theta, \phi) = c \quad (c : \text{constant with respect to } \xi). \quad (3.129)$$

From (3.64) and (3.107), we get

$$Y_l^l(\theta, \phi) = \kappa_l \sin^l \theta e^{il\phi}, \quad (3.130)$$

where κ_l is another constant that depends on l , but is independent of θ and ϕ . Let us seek κ_l by normalization condition. That is,

$$\int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta |Y_l^l(\theta, \phi)|^2 = 2\pi \cdot |\kappa_l|^2 \int_0^\pi \sin^{2l+1} \theta d\theta = 1, \quad (3.131)$$

where the integration is performed on a unit sphere. Note that an infinitesimal area element on the unit sphere is represented by $\sin \theta d\theta d\phi$.

We evaluate the above integral denoted as

$$I \equiv \int_0^\pi \sin^{2l+1} \theta d\theta. \quad (3.132)$$

Using integration by parts,

$$\begin{aligned} I &= \int_0^\pi (-\cos \theta)' \sin^{2l} \theta d\theta \\ &= [(-\cos \theta) \sin^{2l} \theta]_0^\pi + \int_0^\pi (\cos \theta) \cdot 2l \cdot \sin^{2l-1} \theta \cos \theta d\theta \\ &= 2l \int_0^\pi \sin^{2l-1} \theta d\theta - 2l \int_0^\pi \sin^{2l+1} \theta d\theta. \end{aligned} \quad (3.133)$$

Thus, we get a recurrence relation with respect to I (3.132) such that

$$I = \frac{2l}{2l+1} \int_0^\pi \sin^{2l-1} \theta d\theta. \quad (3.134)$$

Repeating the above process, we get

$$I = \frac{2l}{2l+1} \cdot \frac{2l-2}{2l-1} \cdots \frac{2}{3} \int_0^\pi \sin \theta d\theta = \frac{2^{2l+1} (l!)^2}{(2l+1)!}. \quad (3.135)$$

Then,

$$|\kappa_l| = \frac{1}{2^l l!} \sqrt{\frac{(2l+1)!}{4\pi}} \quad \text{or} \quad \kappa_l = \frac{e^{i\chi}}{2^l l!} \sqrt{\frac{(2l+1)!}{4\pi}} \quad (\chi : \text{real}), \quad (3.136)$$

where $e^{i\chi}$ is an undetermined constant (phase factor) that is to be determined below. Thus we get

$$Y_l^l(\theta, \phi) = \frac{e^{i\chi}}{2^l l!} \sqrt{\frac{(2l+1)!}{4\pi}} \sin^l \theta e^{il\phi}. \quad (3.137)$$

Meanwhile, in the second equation of (3.101) replacing $J^{(-)}$, j , and μ in (3.101) with $M^{(-)}$, l , and m , respectively, and using $Y_l^m(\theta, \phi)$ instead of $|\zeta, \mu\rangle$, we get

$$Y_l^{m-1}(\theta, \phi) = \frac{1}{\sqrt{(l-m+1)(l+m)}} M^{(-)} Y_l^m(\theta, \phi). \quad (3.138)$$

Replacing m with l in (3.138), we have

$$Y_l^{l-1}(\theta, \phi) = \frac{1}{\sqrt{2l}} M^{(-)} Y_l^l(\theta, \phi). \quad (3.139)$$

Operating $M^{(-)}(l-m)$ times in total on $Y_l^l(\theta, \phi)$ of (3.139), we have

$$\begin{aligned} Y_l^m(\theta, \phi) &= \frac{1}{\sqrt{2l(2l-1)\cdots(l+m+1)}\sqrt{1\cdot 2\cdots(l-m)}} [M^{(-)}]^{l-m} Y_l^l(\theta, \phi) \\ &= \sqrt{\frac{(l+m)!}{(2l)!(l-m)!}} [M^{(-)}]^{l-m} Y_l^l(\theta, \phi). \end{aligned} \quad (3.140)$$

Putting $m = l$, $n = l - m$, and $j = l$ in (3.117), we have

$$\begin{aligned} [M^{(-)}]^{l-m} Y_l^l(\theta, \phi) &= e^{-i(l-m)\phi} \left(\sqrt{1-\xi^2} \right)^{-m} \frac{\partial^{l-m}}{\partial \xi^{l-m}} \\ &\times \left[\left(\sqrt{1-\xi^2} \right)^l Y_l^l(\theta, \phi) \right]. \end{aligned} \quad (3.141)$$

Further replacing $[M^{(-)}]^{l-m} Y_l^l(\theta, \phi)$ in (3.140) with that of (3.141), we get

$$\begin{aligned} Y_l^m(\theta, \phi) &= \sqrt{\frac{(l+m)!}{(2l)!(l-m)!}} e^{-i(l-m)\phi} \left(\sqrt{1-\xi^2} \right)^{-m} \frac{\partial^{l-m}}{\partial \xi^{l-m}} \\ &\times \left[\left(\sqrt{1-\xi^2} \right)^l Y_l^l(\theta, \phi) \right]. \end{aligned} \quad (3.142)$$

Finally, replacing $Y_l^l(\theta, \phi)$ in (3.142) with that of (3.137) and converting θ to ξ , we arrive at the following equation:

$$Y_l^m(\theta, \phi) = \frac{e^{i\chi}}{2^l l!} \sqrt{\frac{(2l+1)(l+m)!}{4\pi(l-m)!}} e^{im\phi} (1-\xi^2)^{-m/2} \frac{\partial^{l-m}}{\partial \xi^{l-m}} [(1-\xi^2)^l]. \quad (3.143)$$

Now, let us decide $e^{i\chi}$. Putting $m = 0$ in (3.143), we have

$$Y_l^0(\theta, \phi) = \frac{e^{i\chi}(-1)^l}{2^l l! (-1)^l} \sqrt{\frac{2l+1}{4\pi}} \frac{\partial^l}{\partial \xi^l} \left[(1 - \xi^2)^l \right], \quad (3.144)$$

where we put $(-1)^l$ on both the numerator and denominator. In RHS of (3.144),

$$\frac{(-1)^l}{2^l l!} \frac{\partial^l}{\partial \xi^l} \left[(1 - \xi^2)^l \right] \equiv P_l(\xi). \quad (3.145)$$

Equation (3.145) is well-known as Rodrigues formula of Legendre polynomials. We mention characteristics of Legendre polynomials in the next section. Thus,

$$Y_l^0(\theta, \phi) = \frac{e^{i\chi}}{(-1)^l} \sqrt{\frac{2l+1}{4\pi}} P_l(\xi). \quad (3.146)$$

According to the custom [2], we require $Y_l^0(0, \phi)$ to be positive. Noting that $\theta = 0$ corresponds to $\xi = 1$, we have

$$Y_l^0(0, \phi) = \frac{e^{i\chi}}{(-1)^l} \sqrt{\frac{2l+1}{4\pi}} P_l(1) = \frac{e^{i\chi}}{(-1)^l} \sqrt{\frac{2l+1}{4\pi}}, \quad (3.147)$$

where we used $P_l(1) = 1$. For this important relation, see Sect. 3.6.1. Also noting that $\left| \frac{e^{i\chi}}{(-1)^l} \right| = 1$, we must have

$$\frac{e^{i\chi}}{(-1)^l} = 1 \quad \text{or} \quad e^{i\chi} = (-1)^l \quad (3.148)$$

so that $Y_l^0(0, \phi)$ can be positive. Thus, (3.143) is rewritten as

$$\begin{aligned} Y_l^m(\theta, \phi) &= \frac{(-1)^l}{2^l l!} \sqrt{\frac{(2l+1)(l+m)!}{4\pi(l-m)!}} e^{im\phi} (1 - \xi^2)^{-m/2} \frac{\partial^{l-m}}{\partial \xi^{l-m}} \\ &\quad \times \left[(1 - \xi^2)^l \right]. \end{aligned} \quad (3.149)$$

In Sect. 3.3, we mentioned that $|\psi_0\rangle$ in (3.43) is a constant. In fact, putting $l = m = 0$ in (3.149), we have

$$Y_0^0(\theta, \phi) = \sqrt{1/4\pi}. \quad (3.150)$$

Thus, as a simultaneous eigenstate of all L_x, L_y, L_z , and L^2 corresponding to $l = 0$ and $m = 0$, we have

$$|\psi_0\rangle \equiv Y_0^0(\theta, \phi).$$

The normalized functions $Y_l^m(\theta, \phi)$ described as (3.149) define simultaneous eigenfunctions of L^2 (or M^2) and L_z (or M_z). Those functions are called spherical surface harmonics and frequently appear in various fields of mathematical physics.

As in the case of Sect. 2.3, matrix representation enables us to intuitively grasp the relationship between angular momentum operators and their eigenfunctions (or eigenvectors). Rewriting the relations of (3.101) so that they can meet the present purpose, we have

$$\begin{aligned} M^{(-)} |l, m\rangle &= \sqrt{(l-m+1)(l+m)} |l, m-1\rangle, \\ M^{(+)} |l, m\rangle &= \sqrt{(l-m)(l+m+1)} |l, m+1\rangle, \end{aligned} \quad (3.151)$$

where we used l instead of ζ [or λ in (3.46)] to designate the eigenstate.

Now, we know that m takes $(2l+1)$ different values that correspond to each l . This implies that the operators can be expressed with $(2l+1, 2l+1)$ matrices. As implied in (3.151), $M^{(-)}$ takes the following form:

$$M^{(-)} = \begin{pmatrix} 0 & \sqrt{2l \cdot 1} & & & & \\ & 0 & \sqrt{(2l-1) \cdot 2} & & & \\ & & 0 & \ddots & & \\ & & & 0 & \ddots & \\ & & & & 0 & \sqrt{(2l-k+1) \cdot k} \\ & & & & & 0 \\ & & & & & & \ddots \\ & & & & & & & 0 & \sqrt{1 \cdot 2l} \\ & & & & & & & & 0 \end{pmatrix} \quad (3.152)$$

where diagonal elements are zero and a $(k, k+1)$ element is $\sqrt{(2l-k+1) \cdot k}$. That is, non-zero elements are positioned just *above* the zero diagonal elements.

Correspondingly, we have

$$M^{(+)} = \begin{pmatrix} 0 & & & & & \\ \sqrt{2l \cdot 1} & 0 & & & & \\ & \sqrt{(2l-1) \cdot 2} & 0 & & & \\ & & \ddots & & & \\ & & & 0 & & \\ & & & \sqrt{(2l-k+1) \cdot k} & 0 & \\ & & & & \ddots & \\ & & & & & 0 \\ & & & & & \ddots & 0 \\ & & & & & & \sqrt{1 \cdot 2l} & 0 \end{pmatrix}, \quad (3.153)$$

where again diagonal elements are zero and a $(k+1, k)$ element is $\sqrt{(2l-k+1) \cdot k}$. In this case, non-zero elements are positioned just *below* the zero diagonal elements. Notice also that $M^{(-)}$ and $M^{(+)}$ are adjoint to each other and that these notations correspond to (2.65) and (2.66).

Basis functions $Y_l^m(\theta, \phi)$ can be represented by a column vector, as in the case of Sect. 2.3. These are denoted as follows:

$$\begin{aligned} |l, -l\rangle &= \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}, & |l, -l+1\rangle &= \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}, & \dots, \\ & & |l, l-1\rangle &= \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ 0 \end{pmatrix}, & |l, l\rangle &= \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ 1 \end{pmatrix}, \end{aligned} \quad (3.154)$$

where the first number l in $|l, -l\rangle$, $|l, -l+1\rangle$, etc. denotes the quantum number associated with $\lambda = l(l+1)$ given in (3.46) and is kept constant; the latter number denotes m . Note from (3.154) that the column vector whose k -th row is 1 corresponds to m such that

$$m = -l + k - 1. \quad (3.155)$$

For instance, if $k = 1$, $m = -l$; if $k = 2l + 1$, $m = l$, etc.

The operator $M^{(-)}$ converts the column vector whose $(k + 1)$ -th row is 1 to that whose k -th row is 1. The former column vector corresponds to $|l, m + 1\rangle$ and the latter corresponding to $|l, m\rangle$. Therefore, using (3.152), we get the following representation:

$$M^{(-)} |l, m + 1\rangle = \sqrt{(2l - k + 1) \cdot k} |l, m\rangle = \sqrt{(l - m)(l + m + 1)} \times |l, m\rangle, \quad (3.156)$$

where the second equality is obtained by replacing k with that of (3.155), i.e., $k = l + m + 1$. Changing m to $(m - 1)$, we get the first equation of (3.151). Similarly, we obtain the second equation of (3.151) as well. That is, we have

$$M^{(+)} |l, m\rangle = \sqrt{(2l - k + 1) \cdot k} |l, m + 1\rangle = \sqrt{(l - m)(l + m + 1)} \times |l, m + 1\rangle. \quad (3.157)$$

From (3.32) we have

$$\mathbf{M}^2 = M^{(+)}M^{(-)} + M_z^2 - M_z.$$

In the above, $M^{(+)}M^{(-)}$ and M_z are diagonal matrices and, hence, M_z^2 and $\mathbf{M}^2$ are diagonal matrices as well such that

$$M_z = \begin{pmatrix} -l & & & & \\ & -l+1 & & & \\ & & -l+2 & & \\ & & & \ddots & \\ & & & & k-l \\ & & & & & \ddots \\ & & & & & & l \end{pmatrix},$$

$$M^{(+)}M^{(-)} = \begin{pmatrix} 0 & & & & \\ & 2l \cdot 1 & & & \\ & & (2l-1) \cdot 2 & & \\ & & & \ddots & \\ & & & & (2l-k+1) \cdot k \\ & & & & & \ddots \\ & & & & & & 1 \cdot 2l \end{pmatrix}, \quad (3.158)$$

where $k-l$ and $(2l-k+1) \cdot k$ represent $(k+1, k+1)$ elements of M_z and $M^{(+)}M^{(-)}$, respectively. Therefore, $(k+1, k+1)$ element of M^2 is calculated as

$$(2l-k+1) \cdot k + (k-l)^2 - (k-l) = l(l+1).$$

As expected, M^2 takes a constant value $l(l+1)$. A matrix representation is shown in the following equation such that

$$M^2 = \begin{pmatrix} l(l+1) & & & & \\ & l(l+1) & & & \\ & & \ddots & & \\ & & & l(l+1) & \\ & & & & \ddots \\ & & & & & l(l+1) \\ & & & & & & l(l+1) \end{pmatrix}. \quad (3.159)$$

These expressions are useful to understand how the vectors of (3.154) constitute simultaneous eigenstates of M^2 and M_z . In this situation, the matrix representation is

said to diagonalize both $\mathbf{M}^2$ and M_z . In other words, the quantum states represented by (3.154) are simultaneous eigenstates of $\mathbf{M}^2$ and M_z .

The matrices (3.152) and (3.153) that represent $M^{(-)}$ and $M^{(+)}$, respectively, are said to be ladder operators or raising and lowering operators, because operating column vectors those operators convert $|m\rangle$ to $|m \mp 1\rangle$ as mentioned above. The operators $M^{(-)}$ and $M^{(+)}$ correspond to a and $a^\dagger$ given in (2.65) and (2.66), respectively. All these operators are characterized by that the corresponding matrices have diagonal elements of zero and that non-vanishing elements are only positioned on “right above” or “right below” relative to the diagonal elements.

These matrices are a kind of triangle matrices and all their diagonal elements are zero. The matrices are characteristic of *nilpotent* matrices. That is, if a suitable power of a matrix is zero as a matrix, such a matrix is said to be a nilpotent matrix (see Part III). In the present case, $(2l+1)$ -th power of $M^{(-)}$ and $M^{(+)}$ becomes zero as a matrix.

The operator $M^{(-)}$ and $M^{(+)}$ can be described by the following shorthand representations:

$$\left[M^{(-)} \right]_{kj} = \sqrt{(2l-k+1) \cdot k} \delta_{k+1,j} \quad (1 \leq k \leq 2l). \quad (3.160)$$

If $l = 0$, $M_z = M^{(+)}M^{(-)} = \mathbf{M}^2 = 0$. This case corresponds to $Y_0^0(\theta, \phi) = \sqrt{1/4\pi}$ and we do not need the matrix representation. Defining

$$a_k \equiv \sqrt{(2l-k+1) \cdot k},$$

we have, for instance,

$$\begin{aligned} \left\{ \left[M^{(-)} \right]^2 \right\}_{kj} &= \sum_p a_k \delta_{k+1,p} a_p \delta_{p+1,j} = a_k a_{k+1} \delta_{k+2,j} \\ &= \sqrt{(2l-k+1) \cdot k} \sqrt{[2l-(k+1)+1] \cdot (k+1)} \delta_{k+2,j}, \end{aligned} \quad (3.161)$$

where the summation is non-vanishing only if $p = k+1$. The factor $\delta_{k+2,j}$ implies that the elements are shifted by one toward upper right by being squared. Similarly we have

$$\left[M^{(+)} \right]_{kj} = a_{k-1} \delta_{k,j+1} \quad (1 \leq k \leq 2l+1). \quad (3.162)$$

In (3.158), $M^{(+)}M^{(-)}$ is represented as follows:

$$\begin{aligned}
\left[M^{(+)} M^{(-)} \right]_{kj} &= \sum_p a_{k-1} \delta_{k,p+1} a_p \delta_{p+1,j} = a_{k-1} a_{j-1} \delta_{k,j} = (a_{k-1})^2 \delta_{k,j} \\
&= [2l - (k-1) + 1] \cdot (k-1) \delta_{k,j} = (2l - k + 2)(k-1) \delta_{k,j}.
\end{aligned} \tag{3.163}$$

Notice that although a_0 is not defined, $\delta_{1,j+1} = 0$ for any j , and so this causes no inconvenience. Hence, $[M^{(+)} M^{(-)}]_{kj}$ of (3.163) is well-defined with $1 \leq k \leq 2l+1$.

Important properties of angular momentum operators examined above are based upon the fact that those operators are ladder operators and represented by nilpotent matrices. These characteristics will further be studied in Parts III–V.

3.6 Orbital Angular Momentum: Analytic Approach

In this section, our central task is to solve the associated Legendre differential equation expressed by (3.127) by an analytical method. Putting $m = 0$ in (3.127), we have

$$\frac{d}{dx} \left[(1-x^2) \frac{dP_l^0(x)}{dx} \right] + l(l+1)P_l^0(x) = 0, \tag{3.164}$$

where we use a variable x instead of ξ . Equation (3.164) is called Legendre differential equation and its characteristics and solutions have been widely investigated. Hence, we put

$$P_l^0(x) \equiv P_l(x), \tag{3.165}$$

where $P_l(x)$ is said to be Legendre polynomials. We first start with Legendre differential equation and Legendre polynomials.

3.6.1 Spherical Surface Harmonics and Associated Legendre Differential Equation

Let us think of a following identity according to Byron and Fuller [4]:

$$(1-x^2) \frac{d}{dx} (1-x^2)^l = -2lx(1-x^2)^l, \tag{3.166}$$

where l is a positive integer. We differentiate both sides of (3.166) $(l+1)$ times. Here we use the Leibniz rule about differentiation of a product function that is described by

$$d^n(uv) = \sum_{m=0}^n \frac{n!}{m!(n-m)!} d^m u d^{n-m} v, \quad (3.167)$$

where

$$d^m u / dx^m \equiv d^m u.$$

The above shorthand notation is due to Byron and Fuller [4]. We use this notation for simplicity from place to place.

Noting that the third order and higher order differentiations of $(1 - x^2)$ vanish in LHS of (3.166), we have

$$\begin{aligned} \text{LHS} &= d^{l+1} \left[(1 - x^2) d(1 - x^2)^l \right] \\ &= (1 - x^2) d^{l+2} (1 - x^2)^l - 2(l+1)x d^{l+1} (1 - x^2)^l \\ &\quad - l(l+1) d^l (1 - x^2)^l. \end{aligned}$$

Also noting that the second order and higher order differentiations of $2lx$ vanish in LHS of (3.166), we have

$$\begin{aligned} \text{RHS} &= -d^{l+1} \left[2lx(1 - x^2)^l \right] \\ &= -2lx d^{l+1} (1 - x^2)^l - 2l(l+1) d^l (1 - x^2)^l. \end{aligned}$$

Therefore,

$$\text{LHS} - \text{RHS} = (1 - x^2) d^{l+2} (1 - x^2)^l - 2x d^{l+1} (1 - x^2)^l + l(l+1) d^l (1 - x^2)^l = 0.$$

We define $P_l(x)$ as

$$P_l(x) \equiv \frac{(-1)^l}{2^l l!} \frac{d^l}{dx^l} \left[(1 - x^2)^l \right], \quad (3.168)$$

where a constant $\frac{(-1)^l}{2^l l!}$ is multiplied according to the custom so that we can explicitly represent Rodrigues formula of Legendre polynomials. Thus, from (3.164) $P_l(x)$ defined above satisfies Legendre differential equation. Rewriting it, we get

$$(1 - x^2) \frac{d^2 P_l(x)}{dx^2} - 2x \frac{dP_l(x)}{dx} + l(l+1) P_l(x) = 0. \quad (3.169)$$

Or equivalently, we have

$$\frac{d}{dx} \left[(1-x^2) \frac{dP_l(x)}{dx} \right] + l(l+1)P_l(x) = 0. \quad (3.170)$$

Returning to (3.127) and using x as a variable, we rewrite (3.127) as

$$\frac{d}{dx} \left[(1-x^2) \frac{dP_l^m(x)}{dx} \right] + \left[l(l+1) - \frac{m^2}{1-x^2} \right] P_l^m(x) = 0, \quad (3.171)$$

where l is a non-negative integer and m is an integer that takes following values:

$$m = l, \quad l-1, \quad l-2, \quad \dots, \quad 0, \quad -1, \quad \dots, \quad -l+1, \quad -l.$$

Differential equations expressed as

$$\frac{d}{dx} \left[p(x) \frac{dy(x)}{dx} \right] + c(x)y(x) = 0$$

are of particular importance. We will come back to this point in Sect. 10.3.

Since m can be either positive or negative, from (3.171) we notice that $P_l^m(x)$ and $P_l^{-m}(x)$ must satisfy the same differential equation (3.171). This implies that $P_l^m(x)$ and $P_l^{-m}(x)$ are connected, i.e., linearly dependent. First, let us assume that $m \geq 0$. In the case of $m < 0$, we will examine it later soon.

According to Dennery and Krzywicki [5], we assume

$$P_l^m(x) = \kappa (1-x^2)^{m/2} C(x), \quad (3.172)$$

where κ is a constant. Inserting (3.172) into (3.171) and rearranging the terms, we obtain

$$(1-x^2) \frac{d^2 C}{dx^2} - 2(m+1)x \frac{dC}{dx} + (l-m)(l+m+1)C = 0 \quad (0 \leq m \leq l). \quad (3.173)$$

Recall once again that if $m = 0$, the associated Legendre differential equation given by (3.127) and (3.171) is exactly identical to Legendre differential equation of (3.170). Differentiating (3.170) m times, we get

$$\begin{aligned} (1-x^2) \frac{d^2}{dx^2} \left(\frac{d^m P_l}{dx^m} \right) - 2(m+1)x \frac{d}{dx} \left(\frac{d^m P_l}{dx^m} \right) + (l-m)(l+m+1) \\ \times \frac{d^m P_l}{dx^m} = 0, \end{aligned} \quad (3.174)$$

where we used the Leibniz rule about differentiation of (3.167). Comparing (3.173) and (3.174), we find that

$$C(x) = \kappa' \frac{d^m P_l}{dx^m},$$

where κ' is a constant. Inserting this relation into (3.172) and setting $\kappa\kappa' = 1$, we get

$$P_l^m(x) = (1-x^2)^{m/2} \frac{d^m P_l(x)}{dx^m} \quad (0 \leq m \leq l). \quad (3.175)$$

Using Rodrigues formula of (3.168), we have

$$P_l^m(x) \equiv \frac{(-1)^l}{2^l l!} (1-x^2)^{m/2} \frac{d^{l+m}}{dx^{l+m}} \left[(1-x^2)^l \right]. \quad (3.176)$$

Equations (3.175) and (3.176) define the associated Legendre functions. Note, however, that the function form differs from literature to literature [2, 5, 6].

Among classical orthogonal polynomials, Gegenbauer polynomials $C_n^\lambda(x)$ often appear in the literature. The relevant differential equation is defined by

$$\begin{aligned} (1-x^2) \frac{d^2}{dx^2} C_n^\lambda(x) - (2\lambda+1)x \frac{d}{dx} C_n^\lambda(x) + n(n+2\lambda) C_n^\lambda(x) \\ = 0 \quad \left(\lambda > -\frac{1}{2} \right). \end{aligned} \quad (3.177)$$

Setting $n = l - m$ and $\lambda = m + \frac{1}{2}$ in (3.177) [5], we have

$$\begin{aligned} (1-x^2) \frac{d^2}{dx^2} C_{l-m}^{m+\frac{1}{2}}(x) - 2(m+1)x \frac{d}{dx} C_{l-m}^{m+\frac{1}{2}}(x) + (l-m) \\ \times (l+m+1) C_{l-m}^{m+\frac{1}{2}}(x) = 0. \end{aligned} \quad (3.178)$$

Once again comparing (3.174) and (3.178), we obtain

$$\frac{d^m P_l(x)}{dx^m} = \text{constant} \cdot C_{l-m}^{m+\frac{1}{2}}(x) \quad (0 \leq m \leq l). \quad (3.179)$$

Next, let us determine the constant appearing in (3.179). To this end, we consider a following generating function of the polynomials $C_n^\lambda(x)$ defined by [7, 8]

$$(1-2tx+t^2)^{-\lambda} \equiv \sum_{n=0}^{\infty} C_n^\lambda(x) t^n \quad \left(\lambda > -\frac{1}{2} \right). \quad (3.180)$$

To calculate (3.180), let us think of a following expression for x and λ :

$$(1+x)^{-\lambda} = \sum_{m=0}^{\infty} \binom{-\lambda}{m} x^m, \quad (3.181)$$

where λ is an arbitrary real number and we define $\binom{-\lambda}{m}$ as

$$\binom{-\lambda}{m} \equiv -\lambda(-\lambda-1)(-\lambda-2)\cdots(-\lambda-m+1)/m! \quad \text{and} \quad \binom{-\lambda}{0} \equiv 1. \quad (3.182)$$

Notice that (3.181) with (3.182) is an extension of binomial theorem (generalized binomial theorem). Putting $-\lambda = n$, we have

$$\binom{n}{m} = \frac{n!}{(n-m)!m!} \quad \text{and} \quad \binom{n}{0} = 1.$$

We rewrite (3.181) using gamma functions $\Gamma(z)$ such that

$$(1+x)^{-\lambda} = \sum_{m=0}^{\infty} \frac{\Gamma(-\lambda+1)}{m!\Gamma(-\lambda-m+1)} x^m, \quad (3.183)$$

where $\Gamma(z)$ is defined by integral representation as

$$\Gamma(z) = \int_0^{\infty} e^{-t} t^{z-1} dt \quad (\Re z > 0). \quad (3.184)$$

In (3.184), $\Re$ denotes the real part of a number z . Changing variables such that $t = u^2$, we have

$$\Gamma(z) = 2 \int_0^{\infty} e^{-u^2} u^{2z-1} du \quad (\Re z > 0). \quad (3.185)$$

Note that the above expression is associated with the following fundamental feature of the gamma functions:

$$\Gamma(z+1) = z\Gamma(z), \quad (3.186)$$

where z is any complex number.

Replacing x with $-t(2x-t)$ and rewriting (3.183), we have

$$(1 - 2tx + t^2)^{-\lambda} = \sum_{m=0}^{\infty} \frac{\Gamma(-\lambda + 1)}{m! \Gamma(-\lambda - m + 1)} (-t)^m (2x - t)^m. \quad (3.187)$$

Assuming that x is a real number belonging to an interval $[-1, 1]$, (3.187) holds with t satisfying $|t| < 1$ [8]. The discussion is as follows: When x satisfies the above condition, solving $1 - 2tx + t^2 = 0$ we get solution $t_{\pm}$ such that

$$t_{\pm} = x \pm i\sqrt{1 - x^2}.$$

Defining r as

$$r \equiv \min\{|t_+|, |t_-|\},$$

$(1 - 2tx + t^2)^{-\lambda}$, which is regarded as a function of t , is analytic in the disk $|t| < r$. However, we have

$$|t_{\pm}| = 1.$$

Thus, $(1 - 2tx + t^2)^{-\lambda}$ is analytic within the disk $|t| < 1$ and, hence, it can be expanded in a Taylor's series (see Chap. 6).

Continuing the calculation of (3.187), we have

$$\begin{aligned} (1 - 2tx + t^2)^{-\lambda} &= \sum_{m=0}^{\infty} \frac{\Gamma(-\lambda + 1)}{m! \Gamma(-\lambda - m + 1)} (-1)^m t^m \left[\sum_{k=0}^m \frac{m!}{k! (m-k)!} 2^{m-k} x^{m-k} (-1)^k t^k \right] \\ &= \sum_{m=0}^{\infty} \sum_{k=0}^m \frac{(-1)^{m+k}}{k! (m-k)!} \frac{\Gamma(-\lambda + 1)}{\Gamma(-\lambda - m + 1)} 2^{m-k} x^{m-k} t^{m+k} \\ &= \sum_{m=0}^{\infty} \sum_{k=0}^m \frac{(-1)^{m+k}}{k! (m-k)!} \frac{(-1)^m \Gamma(\lambda + m)}{\Gamma(\lambda)} 2^{m-k} x^{m-k} t^{m+k}, \end{aligned} \quad (3.188)$$

where the last equality results from that we rewrote gamma functions using (3.186). Replacing $(m + k)$ with n , we get

$$(1 - 2tx + t^2)^{-\lambda} = \sum_{n=0}^{\infty} \sum_{k=0}^{[n/2]} \frac{(-1)^k 2^{n-2k}}{k! (n-2k)!} \frac{\Gamma(\lambda + n - k)}{\Gamma(\lambda)} x^{n-2k} t^n, \quad (3.189)$$

where $[n/2]$ represents an integer that does not exceed $n/2$. This expression comes from a requirement that an order of x must satisfy the following condition:

$$n - 2k \geq 0 \quad \text{or} \quad k \leq n/2. \quad (3.190)$$

That is, if n is even, the maximum of $k = n/2$. If n is odd, the maximum of $k = (n - 1)/2$. Comparing (3.180) and (3.189), we get [8]

$$C_n^\lambda(x) = \sum_{k=0}^{[n/2]} \frac{(-1)^k 2^{n-2k}}{k!(n-2k)!} \frac{\Gamma(\lambda + n - k)}{\Gamma(\lambda)} x^{n-2k}. \quad (3.191)$$

Comparing (3.164) and (3.177) and putting $\lambda = 1/2$, we immediately find that the two differential equations are identical [7]. That is,

$$C_n^{1/2}(x) = P_n(x). \quad (3.192)$$

Hence, we further have

$$P_n(x) = \sum_{k=0}^{[n/2]} \frac{(-1)^k 2^{n-2k}}{k!(n-2k)!} \frac{\Gamma(\frac{1}{2} + n - k)}{\Gamma(\frac{1}{2})} x^{n-2k}. \quad (3.193)$$

Using (3.186) once again, we get [8]

$$P_n(x) = \sum_{k=0}^{[n/2]} \frac{(-1)^k (2n-2k)!}{2^n k! (n-k)! (n-2k)!} x^{n-2k}. \quad (3.194)$$

It is convenient to make a formula about a gamma function. In (3.193), $n - k > 0$, and so let us think of $\Gamma(\frac{1}{2} + m)$ (m : positive integer). Using (3.186), we have

$$\begin{aligned} \Gamma\left(\frac{1}{2} + m\right) &= \left(m - \frac{1}{2}\right) \Gamma\left(m - \frac{1}{2}\right) = \left(m - \frac{1}{2}\right) \left(m - \frac{3}{2}\right) \Gamma\left(m - \frac{3}{2}\right) = \cdots \\ &= \left(m - \frac{1}{2}\right) \left(m - \frac{3}{2}\right) \cdots \left(\frac{1}{2}\right) \Gamma\left(\frac{1}{2}\right) \\ &= 2^{-m} (2m-1)(2m-3) \cdots 3 \cdot 1 \cdot \Gamma\left(\frac{1}{2}\right) \\ &= 2^{-m} \frac{(2m-1)!}{2^{m-1} (m-1)!} \Gamma\left(\frac{1}{2}\right) = 2^{-2m} \frac{(2m)!}{m!} \Gamma\left(\frac{1}{2}\right). \end{aligned} \quad (3.195)$$

Notice that (3.195) still holds even if $m = 0$. Inserting $n - k$ into m of (3.195), we get

$$\Gamma\left(\frac{1}{2} + n - k\right) = 2^{-2(n-k)} \frac{(2n-2k)!}{(n-k)!} \Gamma\left(\frac{1}{2}\right). \quad (3.196)$$

Replacing $\Gamma(\frac{1}{2} + n - k)$ of (3.193) with RHS of the above equation, (3.194) will follow. A gamma function $\Gamma(\frac{1}{2})$ often appears in mathematical physics. According to (3.185), we have

$$\Gamma\left(\frac{1}{2}\right) = 2 \int_0^\infty e^{-u^2} du = \sqrt{\pi}.$$

For the derivation of the above definite integral, see (2.86) of Sect. 2.4. From (3.184), we also have

$$\Gamma(1) = 1.$$

In relation of the discussion of Sect. 3.5, let us derive an important formula about Legendre polynomials. From (3.180) and (3.192), we get

$$(1 - 2tx + t^2)^{-1/2} \equiv \sum_{n=0}^{\infty} P_n(x) t^n. \quad (3.197)$$

Assuming $|t| < 1$, when we put $x = 1$ in (3.197), we have

$$(1 - 2t + t^2)^{-1/2} = \frac{1}{1-t} = \sum_{n=0}^{\infty} t^n = \sum_{n=0}^{\infty} P_n(1) t^n. \quad (3.198)$$

Comparing individual coefficients of t^n in (3.198), we get

$$P_n(1) = 1.$$

See the related parts of Sect. 3.5.

Now, we are in the position to determine the constant in (3.179). Differentiating (3.194) m times, we have

$$\begin{aligned} d^m P_l(x) / dx^m &= \sum_{k=0}^{[(l-m)/2]} \frac{(-1)^k (2l-2k)! (l-2k)(l-2k-1) \cdots (l-2k-m+1)}{2^l k! (l-k)! (l-2k)!} x^{l-2k-m} \\ &= \sum_{k=0}^{[(l-m)/2]} \frac{(-1)^k (2l-2k)!}{2^l k! (l-k)! (l-2k-m)!} x^{l-2k-m}. \end{aligned} \quad (3.199)$$

Meanwhile, we have

$$C_{l-m}^{m+\frac{1}{2}}(x) = \sum_{k=0}^{[(l-m)/2]} \frac{(-1)^k 2^{l-2k-m}}{k! (l-2k-m)!} \frac{\Gamma(l+\frac{1}{2}-k)}{\Gamma(m+\frac{1}{2})} x^{l-2k-m}. \quad (3.200)$$

Using (3.195) and (3.196), we have

$$\frac{\Gamma(l+\frac{1}{2}-k)}{\Gamma(m+\frac{1}{2})} = 2^{-2(l-k-m)} \frac{(2l-2k)!}{(l-k)!} \frac{m!}{(2m)!}.$$

Therefore, we get

$$C_{l-m}^{m+\frac{1}{2}}(x) = \sum_{k=0}^{[(l-m)/2]} \frac{(-1)^k (2l-2k)!}{2^l k! (l-k)! (l-2k-m)!} \frac{2^m \Gamma(m+1)}{\Gamma(2m+1)} x^{l-2k-m}, \quad (3.201)$$

where we used $m! = \Gamma(m+1)$ and $(2m)! = \Gamma(2m+1)$. Comparing (3.199) and (3.201), we get

$$\frac{d^m P_l(x)}{dx^m} = \frac{\Gamma(2m+1)}{2^m \Gamma(m+1)} C_{l-m}^{m+\frac{1}{2}}(x). \quad (3.202)$$

Thus, we find that the constant appearing in (3.179) is $\frac{\Gamma(2m+1)}{2^m \Gamma(m+1)}$. Putting $m=0$ in (3.202), we have $P_l(x) = C_l^{1/2}(x)$. Therefore, (3.192) is certainly recovered. This gives an easy checkup to (3.202).

Meanwhile, Rodrigues formula of Gegenbauer polynomials [5] is given by

$$C_n^\lambda(x) = \frac{(-1)^n \Gamma(n+2\lambda) \Gamma(\lambda + \frac{1}{2})}{2^n n! \Gamma(n+\lambda+\frac{1}{2}) \Gamma(2\lambda)} (1-x^2)^{-\lambda+\frac{1}{2}} \frac{d^n}{dx^n} \left[(1-x^2)^{n+\lambda-\frac{1}{2}} \right]. \quad (3.203)$$

Hence, we have

$$\begin{aligned} C_{l-m}^{m+\frac{1}{2}}(x) &= \frac{(-1)^{l-m} \Gamma(l+m+1) \Gamma(m+1)}{2^{l-m} (l-m)! \Gamma(l+1) \Gamma(2m+1)} (1-x^2)^{-m} \frac{d^{l-m}}{dx^{l-m}} \\ &\quad \times \left[(1-x^2)^l \right]. \end{aligned} \quad (3.204)$$

Inserting (3.204) into (3.202), we have

$$\begin{aligned} \frac{d^m P_l(x)}{dx^m} &= \frac{(-1)^{l-m} \Gamma(l+m+1)}{2^l (l-m)! \Gamma(l+1)} (1-x^2)^{-m} \frac{d^{l-m}}{dx^{l-m}} \left[(1-x^2)^l \right] \\ &= \frac{(-1)^{l-m} (l+m)!}{2^l l! (l-m)!} (1-x^2)^{-m} \frac{d^{l-m}}{dx^{l-m}} \left[(1-x^2)^l \right]. \end{aligned} \quad (3.205)$$

Further inserting this into (3.175), we finally get

$$P_l^m(x) = \frac{(-1)^{l-m} (l+m)!}{2^l l! (l-m)!} (1-x^2)^{-m/2} \frac{d^{l-m}}{dx^{l-m}} \left[(1-x^2)^l \right]. \quad (3.206)$$

When $m=0$, we have

$$P_l^0(x) = \frac{(-1)^l}{2^l l!} \frac{d^l}{dx^l} \left[(1-x^2)^l \right] = P_l(x). \quad (3.207)$$

Thus, we recover the functional form of Legendre polynomials. The expression (3.206) is also meaningful for negative m , provided $|m| \leq l$, and permits an extension of the definition of $P_l^m(x)$ given by (3.175) to negative numbers of m [5].

Changing m to $-m$ in (3.206), we have

$$P_l^{-m}(x) = \frac{(-1)^{l+m}(l-m)!}{2^l l! (l+m)!} (1-x^2)^{m/2} \frac{d^{l+m}}{dx^{l+m}} \left[(1-x^2)^l \right]. \quad (3.208)$$

Meanwhile, from (3.168) and (3.175),

$$P_l^m(x) = \frac{(-1)^l}{2^l l!} (1-x^2)^{m/2} \frac{d^{l+m}}{dx^{l+m}} \left[(1-x^2)^l \right] \quad (0 \leq m \leq l). \quad (3.209)$$

Comparing (3.208) and (3.209), we get

$$P_l^{-m}(x) = \frac{(-1)^m (l-m)!}{(l+m)!} P_l^m(x) \quad (-l \leq m \leq l). \quad (3.210)$$

Thus, as expected earlier, $P_l^m(x)$ and $P_l^{-m}(x)$ are linearly dependent.

Now, we return back to (3.149). From (3.206) we have

$$(1-x^2)^{-m/2} \frac{d^{l-m}}{dx^{l-m}} \left[(1-x^2)^l \right] = \frac{(-1)^{l-m} 2^l l! (l-m)!}{(l+m)!} P_l^m(x). \quad (3.211)$$

Inserting (3.211) into (3.149) and changing the variable x to ξ , we have

$$Y_l^m(\theta, \phi) = (-1)^m \sqrt{\frac{(2l+1)(l-m)!}{4\pi(l+m)!}} P_l^m(\xi) e^{im\phi} \quad (\xi = \cos \theta; 0 \leq \theta \leq \pi). \quad (3.212)$$

The coefficient $(-1)^m$ appearing (3.212) is well-known as Condon–Shortley phase [7]. Another important expression obtained from (3.210) and (3.212) is

$$Y_l^{-m}(\theta, \phi) = (-1)^m [Y_l^m(\theta, \phi)]^*. \quad (3.213)$$

Since (3.208) or (3.209) involves higher order differentiations, it would somewhat be inconvenient to find their functional forms. Here we try to seek the convenient representation of spherical harmonics using familiar cosine and sine functions. Starting with (3.206) and applying Leibniz rule there, we have

$$\begin{aligned}
P_l^m(x) &= \frac{(-1)^{l-m}(l+m)!}{2^l l!(l-m)!} (1+x)^{-m/2} (1-x)^{-m/2} \\
&\quad \times \sum_{r=0}^{l-m} \frac{(l-m)!}{r!(l-m-r)!} \frac{d^r}{dx^r} [(1+x)^l] \frac{d^{l-m-r}}{dx^{l-m-r}} [(1-x)^l] \\
&= \frac{(-1)^{l-m}(l+m)!}{2^l l!(l-m)!} \sum_{r=0}^{l-m} \frac{(l-m)!}{r!(l-m-r)!} \frac{l!(1+x)^{l-r-\frac{m}{2}} (-1)^{l-m-r} l!(1-x)^{m+r-\frac{m}{2}}}{(l-r)! (m+r)!} \\
&= \frac{l!(l+m)!}{2^l} \sum_{r=0}^{l-m} \frac{(-1)^r}{r!(l-m-r)!} \frac{(1+x)^{l-r-\frac{m}{2}} (1-x)^{r+\frac{m}{2}}}{(l-r)! (m+r)!}.
\end{aligned} \tag{3.214}$$

Putting $x = \cos \theta$ in (3.214) and using a trigonometric formula, we have

$$P_l^m(x) = l!(l+m)! \sum_{r=0}^{l-m} \frac{(-1)^r}{r!(l-m-r)!} \frac{\cos^{2l-2r-m}(\frac{\theta}{2})}{(l-r)!} \frac{\sin^{2r+m}(\frac{\theta}{2})}{(m+r)!}. \tag{3.215}$$

Inserting this into (3.212), we get

$$\begin{aligned}
Y_l^m(\theta, \phi) &= l! \sqrt{\frac{(2l+1)(l+m)!(l-m)!}{4\pi}} e^{im\phi} \sum_{r=0}^{l-m} \frac{(-1)^{r+m} \cos^{2l-2r-m}(\frac{\theta}{2}) \sin^{2r+m}(\frac{\theta}{2})}{r!(l-m-r)!(l-r)!(m+r)!}.
\end{aligned} \tag{3.216}$$

Summation domain of r must be determined so that factorials of negative integers can be avoided [6]. That is,

1. If $m \geq 0$, $0 \leq r \leq l-m$; $(l-m+1)$ terms,
2. If $m < 0$, $|m| \leq r \leq l$; $(l-|m|+1)$ terms.

For example, if we choose l for m , putting $r = 0$ in (3.216) we have

$$\begin{aligned}
Y_l^l(\theta, \phi) &= \sqrt{\frac{(2l+1)!}{4\pi}} e^{il\phi} \frac{(-1)^l \cos^l(\frac{\theta}{2}) \sin^l(\frac{\theta}{2})}{l!} \\
&= \sqrt{\frac{(2l+1)!}{4\pi}} e^{il\phi} \frac{(-1)^l \sin^l \theta}{2^l l!}.
\end{aligned} \tag{3.217}$$

In particular, we have $Y_0^0(\theta, \phi) = \sqrt{\frac{1}{4\pi}}$ to recover (3.150). When $m = -l$, putting $r = l$ in (3.216) we get

$$Y_l^{-l}(\theta, \phi) = \sqrt{\frac{(2l+1)!}{4\pi}} e^{-il\phi} \frac{\cos^l\left(\frac{\theta}{2}\right) \sin^l\left(\frac{\theta}{2}\right)}{l!} = \sqrt{\frac{(2l+1)!}{4\pi}} e^{-il\phi} \frac{\sin^l\theta}{2^l l!}. \quad (3.218)$$

For instance, choosing $l = 3$ and $m = \pm 3$ and using (3.217) or (3.218), we have

$$Y_3^3(\theta, \phi) = -\sqrt{\frac{35}{64\pi}} e^{i3\phi} \sin^3\theta, \quad Y_3^{-3}(\theta, \phi) = \sqrt{\frac{35}{64\pi}} e^{-i3\phi} \sin^3\theta.$$

For the minus sign appearing in $Y_3^3(\theta, \phi)$ is due to the Condon–Shortley phase. For $l = 3$ and $m = 0$, moreover, we have

$$\begin{aligned} Y_3^0(\theta, \phi) &= 3! \sqrt{\frac{7 \cdot 3! \cdot 3!}{4\pi}} \sum_{r=0}^3 \frac{(-1)^r \cos^{6-2r}\left(\frac{\theta}{2}\right) \sin^{2r}\left(\frac{\theta}{2}\right)}{r!(3-r)!(3-r)!r!} \\ &= 18\sqrt{\frac{7}{\pi}} \left[\frac{\cos^6\left(\frac{\theta}{2}\right)}{0!3!3!0!} - \frac{\cos^4\left(\frac{\theta}{2}\right) \sin^2\left(\frac{\theta}{2}\right)}{1!2!2!1!} + \frac{\cos^2\left(\frac{\theta}{2}\right) \sin^4\left(\frac{\theta}{2}\right)}{2!1!1!2!} \right. \\ &\quad \left. - \frac{\sin^6\left(\frac{\theta}{2}\right)}{3!0!0!3!} \right] \\ &= 18\sqrt{\frac{7}{\pi}} \left\{ \frac{\cos^6\left(\frac{\theta}{2}\right) - \sin^6\left(\frac{\theta}{2}\right)}{36} + \frac{\cos^2\left(\frac{\theta}{2}\right) \sin^2\left(\frac{\theta}{2}\right) \left[\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \right]}{4} \right\} \\ &= \sqrt{\frac{7}{\pi}} \left(\frac{5}{4} \cos^3\theta - \frac{3}{4} \cos\theta \right), \end{aligned}$$

where in the last equality we used formulae of elementary algebra and trigonometric functions. At the same time, we get

$$Y_3^0(0, \phi) = \sqrt{\frac{7}{4\pi}}.$$

This is consistent with (3.147) in that $Y_3^0(0, \phi)$ is positive.

3.6.2 Orthogonality of Associated Legendre Functions

Orthogonality relation of functions is important. Here we deal with it, regarding the associated Legendre functions.

Replacing m with $(m - 1)$ in (3.174) and using the notation introduced before, we have

$$(1 - x^2)d^{m+1}P_l - 2mx d^m P_l + (l + m)(l - m + 1)d^{m-1}P_l = 0. \quad (3.219)$$

Multiplying both sides by $(1 - x^2)^{m-1}$, we have

$$(1 - x^2)^m d^{m+1}P_l - 2mx(1 - x^2)^{m-1} d^m P_l + (l + m)(l - m + 1)(1 - x^2)^{m-1} d^{m-1}P_l = 0.$$

Rewriting the above equation, we get

$$d[(1 - x^2)^m d^m P_l] = -(l + m)(l - m + 1)(1 - x^2)^{m-1} d^{m-1}P_l. \quad (3.220)$$

Now, let us define $f(m)$ as follows:

$$f(m) \equiv \int_{-1}^1 (1 - x^2)^m (d^m P_l) d^m P_{l'} dx \quad (0 \leq m \leq l, l'). \quad (3.221)$$

Rewriting (3.221) as follows and integrating it by parts, we have

$$\begin{aligned} f(m) &= \int_{-1}^1 (1 - x^2)^m d(d^{m-1}P_l) d^m P_{l'} dx \\ &= [(d^{m-1}P_l)(1 - x^2)^m d^m P_{l'}]_{-1}^1 - \int_{-1}^1 (d^{m-1}P_l) d[(1 - x^2)^m d^m P_{l'}] dx \\ &= - \int_{-1}^1 (d^{m-1}P_l) d[(1 - x^2)^m d^m P_{l'}] dx \\ &= \int_{-1}^1 (d^{m-1}P_l) (l' + m)(l' - m + 1)(1 - x^2)^{m-1} d^{m-1}P_{l'} dx \\ &= (l' + m)(l' - m + 1)f(m - 1), \end{aligned} \quad (3.222)$$

where with the second equality the first term vanished and with the second last equality we used (3.220). Equation (3.222) gives a recurrence formula regarding $f(m)$. Further performing the calculation, we get

$$\begin{aligned}
f(m) &= (l' + m)(l' + m - 1) \cdot (l' - m + 2)(l' - m + 1)f(m - 2) \\
&= \cdots \\
&= (l' + m)(l' + m - 1) \cdots (l' + 1) \cdot l' \cdots (l' - m + 2)(l' - m + 1)f(0) \\
&= \frac{(l' + m)!}{(l' - m)!} f(0),
\end{aligned} \tag{3.223}$$

where we have

$$f(0) = \int_{-1}^1 P_l(x) P_{l'}(x) dx. \tag{3.224}$$

Note that in (3.223) a coefficient of $f(0)$ comprises $2m$ factors.

In (3.224), $P_l(x)$ and $P_{l'}(x)$ are Legendre polynomials defined in (3.165). Then, using (3.168) we have

$$f(0) = \frac{(-1)^l}{2^l l!} \frac{(-1)^{l'}}{2^{l'} l'!} \int_{-1}^1 \left[d^l (1 - x^2)^l \right] \left[d^{l'} (1 - x^2)^{l'} \right] dx. \tag{3.225}$$

To evaluate (3.224), we have two cases; i.e., (1) $l \neq l'$ and (2) $l = l'$. With the first case, assuming that $l > l'$ and taking partial integration, we have

$$\begin{aligned}
I &= \int_{-1}^1 \left[d^l (1 - x^2)^l \right] \left[d^{l'} (1 - x^2)^{l'} \right] dx \\
&= \left[\left\{ d^{l-1} (1 - x^2)^l \right\} \left\{ d^{l'} (1 - x^2)^{l'} \right\} \right]_{-1}^1 \\
&\quad - \int_{-1}^1 \left[d^{l-1} (1 - x^2)^l \right] \left[d^{l'+1} (1 - x^2)^{l'} \right] dx.
\end{aligned} \tag{3.226}$$

In the above, we find that the first term vanishes because it contains $(1 - x^2)$ as a factor. Integrating (3.226) another l' times as before, we get

$$I = (-1)^{l'+1} \int_{-1}^1 \left[d^{l-l'-1} (1 - x^2)^l \right] \left[d^{2l'+1} (1 - x^2)^{l'} \right] dx. \tag{3.227}$$

In (3.227) we have $0 \leq l - l' - 1 \leq 2l$, and so $d^{l-l'-1} (1 - x^2)^l$ does not vanish, but $(1 - x^2)^{l'}$ is an at most $2l'$ -degree polynomial. Hence, $d^{2l'+1} (1 - x^2)^{l'}$ vanishes. Therefore,

$$f(0) = 0. \tag{3.228}$$

If $l < l'$, changing $P_l(x)$ and $P_{l'}(x)$ in (3.224), we get $f(0) = 0$ as well.

In the second case of $l = l'$, we evaluate the following integral:

$$I = \int_{-1}^1 \left[d^l (1-x^2)^l \right]^2 dx. \quad (3.229)$$

Similarly integrating (3.229) by parts l times, we have

$$I = (-1)^l \int_{-1}^1 (1-x^2)^l \left[d^{2l} (1-x^2)^l \right] dx = (-1)^{2l} (2l)! \int_{-1}^1 (1-x^2)^l dx. \quad (3.230)$$

In (3.230), changing x to $\cos\theta$, we have

$$\int_{-1}^1 (1-x^2)^l dx = \int_0^\pi \sin^{2l+1} \theta d\theta. \quad (3.231)$$

We have already estimate this integral of (3.132) to have $\frac{2^{2l+1}(l!)^2}{(2l+1)!}$ in (3.135). Therefore,

$$f(0) = \frac{(-1)^{2l} (2l)!}{2^{2l} (l!)^2} \frac{2^{2l+1} (l!)^2}{(2l+1)!} = \frac{2}{2l+1}. \quad (3.232)$$

Thus, we get

$$f(m) = \frac{(l+m)!}{(l-m)!} f(0) = \frac{(l+m)!}{(l-m)!} \frac{2}{2l+1}. \quad (3.233)$$

From (3.228) and (3.233), we have

$$\int_{-1}^1 P_l^m(x) P_l^m(x) dx = \frac{(l+m)!}{(l-m)!} \frac{2}{2l+1} \delta_{ll'}. \quad (3.234)$$

Accordingly, putting

$$\widetilde{P}_l^m(x) \equiv \sqrt{\frac{(2l+1)(l-m)!}{2(l+m)!}} P_l^m(x), \quad (3.235)$$

we get

$$\int_{-1}^1 \widetilde{P}_l^m(x) \widetilde{P}_l^m(x) dx = \delta_{ll'}. \quad (3.236)$$

Normalized Legendre polynomials immediately follow. This is given by

$$\tilde{P}_l(x) \equiv \sqrt{\frac{2l+1}{2}} P_l(x) = \frac{(-1)^l}{2^l l!} \sqrt{\frac{2l+1}{2}} \frac{d^l}{dx^l} [(1-x^2)^l]. \quad (3.237)$$

Combining a normalized function (3.235) with $\frac{1}{\sqrt{2\pi}} e^{im\phi}$, we recover

$$Y_l^m(\theta, \phi) = \sqrt{\frac{(2l+1)(l-m)!}{4\pi(l+m)!}} P_l^m(x) e^{im\phi} \quad (x = \cos \theta; 0 \leq \theta \leq \pi). \quad (3.238)$$

Notice in (3.238), however, we could not determine Condon–Shortley phase $(-1)^m$; see (3.212).

Since $P_l^m(x)$ and $P_l^{-m}(x)$ are linearly dependent as noted in (3.210), the set of the associated Legendre functions cannot define a complete set of orthonormal system. In fact, we have

$$\int_{-1}^1 P_l^m(x) P_l^{-m}(x) dx = \frac{(-1)^m (l-m)!}{(l+m)!} \frac{(l+m)!}{(l-m)!} \frac{2}{2l+1} = \frac{2(-1)^m}{2l+1}. \quad (3.239)$$

This means that $P_l^m(x)$ and $P_l^{-m}(x)$ are *not* orthogonal. Thus, we need $e^{im\phi}$ to constitute the complete set of orthonormal system. In other words,

$$\int_0^{2\pi} d\phi \int_{-1}^1 d(\cos \theta) [Y_l^{m'}(\theta, \phi)]^* Y_l^m(\theta, \phi) = \delta_{ll'} \delta_{mm'}. \quad (3.240)$$

3.7 Radial Wave Functions of Hydrogen-Like Atoms

In Sect. 3.1 we have constructed Hamiltonian of hydrogen-like atoms. If the physical system is characterized by the central force field, the method of separation of variables into the angular part (θ, ϕ) and radial (r) part is successfully applied to the problem and that method allows us to deal with the Schrödinger equation separately. The spherical surface harmonics play a central role in dealing with the differential equations related to the angular part. We studied important properties of the special functions such as Legendre polynomials and associated Legendre functions, independent of the nature of the specific central force fields such as Coulomb potential and Yukawa potential. With the Schrödinger equation pertinent to the radial part, on the other hand, its characteristics differ depending on the nature of individual force fields. Of these, the differential equation associated with the Coulomb potential gives exact (or analytical) solutions. It is well-known that the second-order differential equations are often solved by an operator representation method. Examples include its application to a quantum-mechanical harmonic oscillator and angular momenta of a particle placed in a central force field. Nonetheless, the

corresponding approach to the radial equation for the electron has been less popular to date. The initial approach, however, was made by Sunakawa [3]. The purpose of this chapter rests upon further improvement of that approach.

3.7.1 Operator Approach to Radial Wave Functions [3]

In Sect. 3.2, the separation of variables led to the radial part of the Schrödinger equation described as

$$\frac{1}{2\mu} \left[-\frac{\hbar^2}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R(r)}{\partial r} \right) + \frac{\hbar^2 \lambda}{r^2} \right] R(r) - \frac{Ze^2}{4\pi\epsilon_0 r} R(r) = ER(r). \quad (3.51)$$

We identified λ with $l(l+1)$; see Sect. 3.5. Thus, rewriting (3.51) and indexing $R(r)$ with l , we have

$$-\frac{\hbar^2}{2\mu r^2} \frac{d}{dr} \left[r^2 \frac{dR_l(r)}{dr} \right] + \left[\frac{\hbar^2 l(l+1)}{2\mu r^2} - \frac{Ze^2}{4\pi\epsilon_0 r} \right] R_l(r) = ER_l(r), \quad (3.241)$$

where $R_l(r)$ is a radial wave function parametrized with l ; μ , Z , ϵ_0 , and E denote a reduced mass of hydrogen-like atom, atomic number, permittivity of vacuum, and eigenvalue of energy, respectively. Otherwise we follow conventions.

Now, we are in position to solve (3.241). As in the cases of Chap. 2 of a quantum-mechanical harmonic oscillator and the previous section of the angular momentum operator, we present the operator formalism in dealing with radial wave functions of hydrogen-like atoms. The essential point rests upon that the radial wave functions can be derived by successively operating lowering operators on a radial wave function having a maximum allowed orbital angular momentum quantum number. The results agree with the conventional coordinate representation method based upon power series expansion that is related to associated Laguerre polynomials.

Sunakawa [3] introduced the following differential equation by suitable transformations of a variable, parameter, and function:

$$-\frac{d^2\psi_l(\rho)}{d\rho^2} + \left[\frac{l(l+1)}{\rho^2} - \frac{2}{\rho} \right] \psi_l(\rho) = \epsilon\psi_l(\rho), \quad (3.242)$$

where $\rho = \frac{Zr}{a}$, $\epsilon = \frac{2\mu}{\hbar^2} \left(\frac{a}{Z} \right)^2 E$, and $\psi_l(\rho) = \rho R_l(r)$ with $a (\equiv 4\pi\epsilon_0 \hbar^2 / \mu e^2)$ being Bohr radius of a hydrogen-like atom. Note that ρ and ϵ are dimensionless quantities. The related calculations are as follows: We have

$$\frac{dR_l}{dr} = \frac{d(\psi_l/\rho)}{d\rho} \frac{d\rho}{dr} = \left(\frac{d\psi_l}{d\rho} \frac{1}{\rho} - \frac{\psi_l}{\rho^2} \right) \frac{Z}{a}.$$

Thus, we get

$$r^2 \frac{dR_l}{dr} = \frac{d\psi_l}{d\rho} r - \frac{a}{Z} \psi_l, \quad \frac{d}{dr} \left(r^2 \frac{dR_l}{dr} \right) = \frac{d^2 \psi_l(\rho)}{d\rho^2} \rho.$$

Using the above relations we arrive at (3.242).

Here we define the following operators:

$$b_l \equiv \frac{d}{d\rho} + \left(\frac{l}{\rho} - \frac{1}{l} \right). \quad (3.243)$$

Hence,

$$b_l^\dagger = -\frac{d}{d\rho} + \left(\frac{l}{\rho} - \frac{1}{l} \right), \quad (3.244)$$

where the operator $b_l^\dagger$ is an adjoint operator of b_l . Notice that these definitions are different from those of Sunakawa [3]. The operator $\frac{d}{d\rho}$ ($\equiv A$) is *formally* an anti-Hermitian operator. We have mentioned such an operator in Sect. 1.5. The second terms of (3.243) and (3.244) are Hermitian operators, which we define as H . Thus, we foresee that b_l and $b_l^\dagger$ can be denoted as follows:

$$b_l = A + H \quad \text{and} \quad b_l^\dagger = -A + H.$$

These representations are analogous to those appearing in the operator formalism of a quantum-mechanical harmonic oscillator. Special care, however, should be taken in dealing with the operators b_l and $b_l^\dagger$. First, we should carefully examine whether $\frac{d}{d\rho}$ is in fact an anti-Hermitian operator. This is because for $\frac{d}{d\rho}$ to be anti-Hermitian, the solution $\psi_l(\rho)$ must satisfy boundary conditions in such a way that $\psi_l(\rho)$ vanishes or takes the same value at the endpoints $\rho \rightarrow 0$ and ∞ . Second, the coordinate system we have chosen is not Cartesian coordinate but the polar (spherical) coordinate, and so ρ is defined only on a domain $\rho > 0$. We will come back to this point later.

Let us proceed on calculations. We have

$$\begin{aligned}
b_l b_l^\dagger &= \left[\frac{d}{d\rho} + \left(\frac{l}{\rho} - \frac{1}{l} \right) \right] \cdot \left[-\frac{d}{d\rho} + \left(\frac{l}{\rho} - \frac{1}{l} \right) \right] \\
&= -\frac{d^2}{d\rho^2} + \frac{d}{d\rho} \left(\frac{l}{\rho} - \frac{1}{l} \right) - \left(\frac{l}{\rho} - \frac{1}{l} \right) \frac{d}{d\rho} + \frac{l^2}{\rho^2} - \frac{2}{\rho} + \frac{1}{l^2} \\
&= -\frac{d^2}{d\rho^2} - \frac{l}{\rho^2} + \frac{l^2}{\rho^2} - \frac{2}{\rho} + \frac{1}{l^2} = -\frac{d^2}{d\rho^2} + \frac{l(l-1)}{\rho^2} - \frac{2}{\rho} + \frac{1}{l^2}.
\end{aligned} \tag{3.245}$$

Also, we have

$$b_l^\dagger b_l = -\frac{d^2}{d\rho^2} + \frac{l(l+1)}{\rho^2} - \frac{2}{\rho} + \frac{1}{l^2}. \tag{3.246}$$

We further define an operator $H^{(l)}$ as follows:

$$H^{(l)} \equiv -\frac{d^2}{d\rho^2} + \left[\frac{l(l+1)}{\rho^2} - \frac{2}{\rho} \right].$$

Then, from (3.243) and (3.244) as well as (3.245) and (3.246) we have

$$H^{(l)} = b_{l+1} b_{l+1}^\dagger + \varepsilon^{(l)} \quad (l \geq 0), \tag{3.247}$$

where $\varepsilon^{(l)} \equiv -\frac{1}{(l+1)^2}$. Alternatively,

$$H^{(l)} = b_l^\dagger b_l + \varepsilon^{(l-1)} \quad (l \geq 1). \tag{3.248}$$

If we put $l = n - 1$ in (3.247) with n being a fixed given integer larger than l , we obtain

$$H^{(n-1)} = b_n b_n^\dagger + \varepsilon^{(n-1)}. \tag{3.249}$$

We evaluate the following inner product of both sides of (3.249):

$$\begin{aligned}
\langle \chi | H^{(n-1)} | \chi \rangle &= \langle \chi | b_n b_n^\dagger | \chi \rangle + \varepsilon^{(n-1)} \langle \chi | \chi \rangle = \langle b_n^\dagger \chi | b_n^\dagger \chi \rangle + \varepsilon^{(n-1)} \langle \chi | \chi \rangle \\
&= \|b_n^\dagger \chi\|^2 + \varepsilon^{(n-1)} \langle \chi | \chi \rangle \geq \varepsilon^{(n-1)}.
\end{aligned} \tag{3.250}$$

Here we assume that χ is normalized (i.e., $\langle \chi | \chi \rangle = 1$). On the basis of the variational principle [9], the above expected value must take a minimum $\varepsilon^{(n-1)}$ so that χ can be an eigenfunction. To satisfy this condition, we have

$$|b_n^\dagger \chi\rangle = 0. \quad (3.251)$$

In fact, if (3.251) holds, from (3.249) we have

$$H^{(n-1)}\chi = \epsilon^{(n-1)}\chi. \quad (3.252)$$

We define such a function as follows:

$$\psi_{n-1}^{(n)} \equiv \chi. \quad (3.253)$$

From (3.247) and (3.248), we have the following relationship:

$$H^{(l)}b_{l+1} = b_{l+1}H^{(l+1)} \quad (l \geq 0). \quad (3.254)$$

Meanwhile we define the functions as shown in the following:

$$\psi_{n-s}^{(n)} \equiv b_{n-s+1}b_{n-s+2}\cdots b_{n-1}\psi_{n-1}^{(n)} \quad (2 \leq s \leq n). \quad (3.255)$$

With these functions $(s-1)$ operators have been operated on $\psi_{n-1}^{(n)}$. Note that if s took 1, no operation of b_l would take place. Thus, we find that b_l functions upon the l -state to produce the $(l-1)$ -state. That is, b_l acts as an annihilation operator. For the sake of convenience we express

$$H^{(n,s)} \equiv H^{(n-s)}. \quad (3.256)$$

Using this notation and (3.254), we have

$$\begin{aligned} H^{(n,s)}\psi_{n-s}^{(n)} &= H^{(n,s)}b_{n-s+1}b_{n-s+2}\cdots b_{n-1}\psi_{n-1}^{(n)} \\ &= b_{n-s+1}H^{(n,s-1)}b_{n-s+2}\cdots b_{n-1}\psi_{n-1}^{(n)} \\ &= b_{n-s+1}b_{n-s+2}H^{(n,s-2)}\cdots b_{n-1}\psi_{n-1}^{(n)} \\ &\quad \dots\dots\dots \\ &= b_{n-s+1}b_{n-s+2}\cdots H^{(n,2)}b_{n-1}\psi_{n-1}^{(n)} \\ &= b_{n-s+1}b_{n-s+2}\cdots b_{n-1}H^{(n,1)}\psi_{n-1}^{(n)} \\ &= b_{n-s+1}b_{n-s+2}\cdots b_{n-1}\epsilon^{(n-1)}\psi_{n-1}^{(n)} \\ &= \epsilon^{(n-1)}b_{n-s+1}b_{n-s+2}\cdots b_{n-1}\psi_{n-1}^{(n)} \\ &= \epsilon^{(n-1)}\psi_{n-s}^{(n)}. \end{aligned} \quad (3.257)$$

Thus, total n functions $\psi_{n-s}^{(n)}$ ($1 \leq s \leq n$) belong to the same eigenvalue $\epsilon^{(n-1)}$. Notice that the eigenenergy E_n corresponding to $\epsilon^{(n-1)}$ is given by

$$E_n = -\frac{\hbar^2}{2\mu} \left(\frac{Z}{a}\right)^2 \frac{1}{n^2}. \quad (3.258)$$

If we define $l \equiv n - s$ and take account of (3.252), total n functions $\psi_l^{(n)}$ ($l = 0, 1, 2, \dots, n - 1$) belong to the same eigenvalue $\epsilon^{(n-1)}$.

The quantum state $\psi_l^{(n)}$ is associated with the operators $H^{(l)}$. Thus, the solution of (3.242) has been given by functions $\psi_l^{(n)}$ parametrized with n and l on condition that (3.251) holds. As explicitly indicated in (3.255) and (3.257), b_l lowers the parameter l by one from l to $l - 1$, when it operates on $\psi_l^{(n)}$. The operator b_0 cannot be defined as indicated in (3.243), and so the lowest number of l should be zero. Operators such as b_l are known as a ladder operator (lowering operator or annihilation operator in the present case). The implication is that the successive operations of b_l on $\psi_{n-1}^{(n)}$ produce various parameters l as a subscript down to zero, while retaining the same integer parameter n as a superscript.

3.7.2 Normalization of Radial Wave Functions [10]

Next we seek normalized eigenfunctions. Coordinate representation of (3.251) takes

$$-\frac{d\psi_{n-1}^{(n)}}{d\rho} + \left(\frac{n}{\rho} - \frac{1}{n}\right)\psi_{n-1}^{(n)} = 0. \quad (3.259)$$

The solution can be obtained as

$$\psi_{n-1}^{(n)} = c_n \rho^n e^{-\rho/n}, \quad (3.260)$$

where c_n is a normalization constant. This can be determined as follows:

$$\int_0^\infty \left| \psi_{n-1}^{(n)} \right|^2 d\rho = 1. \quad (3.261)$$

Namely,

$$|c_n|^2 \int_0^\infty \rho^{2n} e^{-2\rho/n} d\rho = 1. \quad (3.262)$$

Consider the following definite integral:

$$\int_0^\infty e^{-2\rho\xi} d\rho = \frac{1}{2\xi}.$$

Differentiating the above integral $2n$ times with respect to ξ gives

$$\int_0^\infty \rho^{2n} e^{-2\rho\xi} d\rho = \left(\frac{1}{2}\right)^{2n+1} (2n)! \xi^{-(2n+1)}. \quad (3.263)$$

Substituting $1/n$ into ξ , we obtain

$$\int_0^\infty \rho^{2n} e^{-2\rho/n} d\rho = \left(\frac{1}{2}\right)^{2n+1} (2n)! n^{(2n+1)}. \quad (3.264)$$

Hence,

$$c_n = \left(\frac{2}{n}\right)^{n+\frac{1}{2}} / \sqrt{(2n)!}. \quad (3.265)$$

To further normalize the other wave functions, we calculate the following inner product:

$$\langle \psi_l^{(n)} | \psi_l^{(n)} \rangle = \langle \psi_{n-1}^{(n)} b_{n-1}^\dagger \cdots b_{l+2}^\dagger b_{l+1}^\dagger | b_{l+1} b_{l+2} \cdots b_{n-1} \psi_{n-1}^{(n)} \rangle. \quad (3.266)$$

From (3.247) and (3.248), we have

$$b_l^\dagger b_l + \varepsilon^{(l-1)} = b_{l+1} b_{l+1}^\dagger + \varepsilon^{(l)} \quad (l \geq 1). \quad (3.267)$$

Applying (3.267)–(3.266) repeatedly and considering (3.251), we reach the following relationship:

$$\begin{aligned} \langle \psi_l^{(n)} | \psi_l^{(n)} \rangle &= [\varepsilon^{(n-1)} - \varepsilon^{(n-2)}] [\varepsilon^{(n-1)} - \varepsilon^{(n-3)}] \cdots [\varepsilon^{(n-1)} - \varepsilon^{(l)}] \\ &\quad \times \langle \psi_{n-1}^{(n)} | \psi_{n-1}^{(n)} \rangle. \end{aligned} \quad (3.268)$$

To show this, we use mathematical induction. We have already normalized $\psi_{n-1}^{(n)}$ in (3.261). Next, we calculate $\langle \psi_{n-2}^{(n)} | \psi_{n-2}^{(n)} \rangle$ such that

$$\begin{aligned}
\langle \psi_{n-2}^{(n)} | \psi_{n-2}^{(n)} \rangle &= \langle \psi_{n-1}^{(n)} b_{n-1}^\dagger | b_{n-1} \psi_{n-1}^{(n)} \rangle = \langle \psi_{n-1}^{(n)} b_{n-1}^\dagger b_{n-1} \psi_{n-1}^{(n)} \rangle \\
&= \langle \psi_{n-1}^{(n)} [b_n b_n^\dagger + \varepsilon^{(n-1)} - \varepsilon^{(n-2)}] \psi_{n-1}^{(n)} \rangle \\
&= \langle \psi_{n-1}^{(n)} b_n b_n^\dagger \psi_{n-1}^{(n)} \rangle + [\varepsilon^{(n-1)} - \varepsilon^{(n-2)}] \langle \psi_{n-1}^{(n)} | \psi_{n-1}^{(n)} \rangle \\
&= [\varepsilon^{(n-1)} - \varepsilon^{(n-2)}] \langle \psi_{n-1}^{(n)} | \psi_{n-1}^{(n)} \rangle,
\end{aligned} \tag{3.269}$$

where with the third equality we used (3.267) with $l = n - 1$; with the last equality we used (3.251). Therefore, (3.268) holds with $l = n - 2$. Then, it suffices to show that assuming that (3.268) holds with $\langle \psi_{l+1}^{(n)} | \psi_{l+1}^{(n)} \rangle$, it holds with $\langle \psi_l^{(n)} | \psi_l^{(n)} \rangle$ as well.

Let us calculate $\langle \psi_l^{(n)} | \psi_l^{(n)} \rangle$, starting with (3.266) as follows:

$$\begin{aligned}
\langle \psi_l^{(n)} | \psi_l^{(n)} \rangle &= \langle \psi_{n-1}^{(n)} b_{n-1}^\dagger \cdots b_{l+2}^\dagger b_{l+1}^\dagger | b_{l+1} b_{l+2} \cdots b_{n-1} \psi_{n-1}^{(n)} \rangle \\
&= \langle \psi_{n-1}^{(n)} b_{n-1}^\dagger \cdots b_{l+2}^\dagger (b_{l+1}^\dagger b_{l+1}) b_{l+2} \cdots b_{n-1} \psi_{n-1}^{(n)} \rangle \\
&= \langle \psi_{n-1}^{(n)} b_{n-1}^\dagger \cdots b_{l+2}^\dagger [b_{l+2} b_{l+2}^\dagger + \varepsilon^{(l+1)} - \varepsilon^{(l)}] b_{l+2} \cdots b_{n-1} \psi_{n-1}^{(n)} \rangle \\
&= \langle \psi_{n-1}^{(n)} b_{n-1}^\dagger \cdots b_{l+2}^\dagger b_{l+2} (b_{l+2}^\dagger b_{l+2}) \cdots b_{n-1} \psi_{n-1}^{(n)} \rangle \\
&\quad + [\varepsilon^{(l+1)} - \varepsilon^{(l)}] \langle \psi_{n-1}^{(n)} b_{n-1}^\dagger \cdots b_{l+2}^\dagger b_{l+2} \cdots b_{n-1} \psi_{n-1}^{(n)} \rangle \\
&= \langle \psi_{n-1}^{(n)} b_{n-1}^\dagger \cdots b_{l+2}^\dagger b_{l+2} (b_{l+2}^\dagger b_{l+2}) \cdots b_{n-1} \psi_{n-1}^{(n)} \rangle \\
&\quad + [\varepsilon^{(l+1)} - \varepsilon^{(l)}] \langle \psi_{l+1}^{(n)} | \psi_{l+1}^{(n)} \rangle.
\end{aligned} \tag{3.270}$$

In the next step, using $b_{l+2}^\dagger b_{l+2} = b_{l+3} b_{l+3}^\dagger + \varepsilon^{(l+2)} - \varepsilon^{(l+1)}$, we have

$$\begin{aligned}
\langle \psi_l^{(n)} | \psi_l^{(n)} \rangle &= \langle \psi_{n-1}^{(n)} b_{n-1}^\dagger \cdots b_{l+2}^\dagger b_{l+2} b_{l+3} b_{l+3}^\dagger b_{l+3} \cdots b_{n-1} \psi_{n-1}^{(n)} \rangle \\
&\quad + [\varepsilon^{(l+1)} - \varepsilon^{(l)}] \langle \psi_{l+1}^{(n)} | \psi_{l+1}^{(n)} \rangle + [\varepsilon^{(l+2)} - \varepsilon^{(l+1)}] \langle \psi_{l+1}^{(n)} | \psi_{l+1}^{(n)} \rangle.
\end{aligned} \tag{3.271}$$

Thus, we find that in the first term the index number of $b_{l+3}^\dagger$ has been increased by one with itself transferred toward the right side. On the other hand, we notice that with the second and third terms, $\varepsilon^{(l+1)} \langle \psi_{l+1}^{(n)} | \psi_{l+1}^{(n)} \rangle$ cancels out. Repeating the above processes, we reach a following expression:

$$\begin{aligned}
\langle \psi_l^{(n)} | \psi_l^{(n)} \rangle &= \langle \psi_{n-1}^{(n)} b_{n-1}^\dagger \cdots b_{l+2}^\dagger b_{l+2} \cdots b_{n-1} (b_n b_n^\dagger) \psi_{n-1}^{(n)} \rangle \\
&+ [\varepsilon^{(n-1)} - \varepsilon^{(n-2)}] \langle \psi_{l+1}^{(n)} | \psi_{l+1}^{(n)} \rangle + [\varepsilon^{(n-2)} - \varepsilon^{(n-3)}] \langle \psi_{l+1}^{(n)} | \psi_{l+1}^{(n)} \rangle + \cdots \\
&+ [\varepsilon^{(l+2)} - \varepsilon^{(l+1)}] \langle \psi_{l+1}^{(n)} | \psi_{l+1}^{(n)} \rangle + [\varepsilon^{(l+1)} - \varepsilon^{(l)}] \langle \psi_{l+1}^{(n)} | \psi_{l+1}^{(n)} \rangle \\
&= [\varepsilon^{(n-1)} - \varepsilon^{(l)}] \langle \psi_{l+1}^{(n)} | \psi_{l+1}^{(n)} \rangle.
\end{aligned} \tag{3.272}$$

In (3.272), the first term of RHS vanishes because of (3.251); the subsequent terms produce $\langle \psi_{l+1}^{(n)} | \psi_{l+1}^{(n)} \rangle$ whose coefficients have canceled out one another except for $[\varepsilon^{(n-1)} - \varepsilon^{(l)}]$.

Meanwhile, from assumption of the mathematical induction we have

$$\begin{aligned}
\langle \psi_{l+1}^{(n)} | \psi_{l+1}^{(n)} \rangle &= [\varepsilon^{(n-1)} - \varepsilon^{(n-2)}] [\varepsilon^{(n-1)} - \varepsilon^{(n-3)}] \cdots [\varepsilon^{(n-1)} - \varepsilon^{(l+1)}] \\
&\times \langle \psi_{n-1}^{(n)} | \psi_{n-1}^{(n)} \rangle.
\end{aligned}$$

Inserting this equation into (3.272), we arrive at (3.268). In other words, we have shown that if (3.268) holds with $l = l + 1$, (3.268) holds with $l = l$ as well. This completes the proof to show that (3.268) is true of $\langle \psi_l^{(n)} | \psi_l^{(n)} \rangle$ with l down to 0.

The normalized wave functions $\tilde{\psi}_l^{(n)}$ are expressed from (3.255) as

$$\tilde{\psi}_l^{(n)} = \kappa(n, l)^{-\frac{1}{2}} b_{l+1} b_{l+2} \cdots b_{n-1} \tilde{\psi}_{n-1}^{(n)}, \tag{3.273}$$

where $\kappa(n, l)$ is defined such that

$$\kappa(n, l) \equiv [\varepsilon^{(n-1)} - \varepsilon^{(n-2)}] \cdot [\varepsilon^{(n-1)} - \varepsilon^{(n-3)}] \cdots [\varepsilon^{(n-1)} - \varepsilon^{(l)}], \tag{3.274}$$

with $l \leq n - 2$. More explicitly, we get

$$\kappa(n, l) = \frac{(2n-1)!(n-l-1)!(l!)^2}{(n+l)!(n!)^2(n^{n-l-2})^2}. \tag{3.275}$$

In particular, from (3.265) we have

$$\tilde{\psi}_{n-1}^{(n)} = \left(\frac{2}{n}\right)^{n+\frac{1}{2}} \frac{1}{\sqrt{(2n)!}} \rho^n e^{-\rho/n}. \tag{3.276}$$

From (3.272), we define the following operator:

$$\tilde{b}_l \equiv \left[\varepsilon^{(n-1)} - \varepsilon^{(l-1)} \right]^{-\frac{1}{2}} b_l. \quad (3.277)$$

Then (3.273) becomes

$$\tilde{\psi}_l^{(n)} = \tilde{b}_{l+1} \tilde{b}_{l+2} \cdots \tilde{b}_{n-1} \tilde{\psi}_{n-1}^{(n)}. \quad (3.278)$$

3.7.3 Associated Laguerre Polynomials

It will be of great importance to compare the functions $\tilde{\psi}_l^{(n)}$ with conventional wave functions that are expressed using associated Laguerre polynomials. For this purpose we define the following functions $\Phi_l^{(n)}(\rho)$ such that

$$\Phi_l^{(n)}(\rho) \equiv \left(\frac{2}{n}\right)^{l+\frac{3}{2}} \sqrt{\frac{(n-l-1)!}{2n(n+l)!}} e^{-\frac{\rho}{n}} \rho^{l+1} L_{n-l-1}^{2l+1}\left(\frac{2\rho}{n}\right). \quad (3.279)$$

The associated Laguerre polynomials are described as

$$L_n^\nu(x) = \frac{1}{n!} x^{-\nu} e^x \frac{d^n}{dx^n} (x^{n+\nu} e^{-x}), (\nu > -1). \quad (3.280)$$

In a form of power series expansion, the polynomials are expressed for integer $k \geq 0$ as

$$L_n^k(x) = \sum_{m=0}^n \frac{(-1)^m (n+k)!}{(n-m)!(k+m)!m!} x^m. \quad (3.281)$$

Notice that “Laguerre polynomials” $L_n(x)$ are defined as

$$L_n(x) \equiv L_n^0(x).$$

Hence, instead of (3.280) and (3.281), the Rodrigues formula and power series expansion of $L_n(x)$ are given by [2, 5]

$$\begin{aligned} L_n(x) &= \frac{1}{n!} e^x \frac{d^n}{dx^n} (x^n e^{-x}), \\ L_n(x) &= \sum_{m=0}^n \frac{(-1)^m n!}{(n-m)!(m!)^2} x^m. \end{aligned}$$

The function $\Phi_l^{(n)}(\rho)$ contains multiplication factors $e^{-\frac{\rho}{n}}$ and ρ^{l+1} . The function $L_{n-l-1}^{2l+1}(\frac{2\rho}{n})$ is a polynomial of ρ with the highest order of ρ^{n-l-1} . Therefore, $\Phi_l^{(n)}(\rho)$ consists of summation of terms containing $e^{-\frac{\rho}{n}}\rho^t$, where t is an integer equal to 1 or larger. Consequently, $\Phi_l^{(n)}(\rho) \rightarrow 0$ when $\rho \rightarrow 0$ and $\rho \rightarrow \infty$ (vide supra). Thus, we have confirmed that $\Phi_l^{(n)}(\rho)$ certainly satisfies proper BCs mentioned earlier and, hence, the operator $\frac{d}{d\rho}$ is indeed an anti-Hermitian. To show this more explicitly, we define $D \equiv \frac{d}{d\rho}$. An inner product between arbitrarily chosen functions f and g is

$$\begin{aligned}\langle f|Dg\rangle &\equiv \int_0^\infty f^* Dg d\rho = [f^* g]_0^\infty - \int_0^\infty (Df^*) g d\rho \\ &= [f^* g]_0^\infty + \langle -D^* f|g\rangle,\end{aligned}\quad (3.282)$$

where f^* is a complex conjugate of f . Meanwhile, from (1.112) we have

$$\langle f|Dg\rangle = \langle D^\dagger f|g\rangle. \quad (3.283)$$

Therefore if the functions f and g vanish at $\rho \rightarrow 0$ and $\rho \rightarrow \infty$, we obtain $D^\dagger = -D^*$ by equating (3.282) and (3.283). However, since D is a real operator, $D^* = D$. Thus we get

$$D^\dagger = -D.$$

This means that D is anti-Hermitian. The functions $\Phi_l^{(n)}(\rho)$ we are dealing with certainly satisfy the required boundary conditions. The operator $H^{(l)}$ appearing in (3.247) and (3.248) is Hermitian accordingly. This is because

$$\begin{aligned}b_l^\dagger b_l &= (-A + H)(A + H) = H^2 - A^2 - AH + HA, \\ (b_l^\dagger b_l)^\dagger &= (H^2)^\dagger - (A^2)^\dagger - H^\dagger A^\dagger + A^\dagger H^\dagger = (H^\dagger)^2 - (A^\dagger)^2 - H^\dagger A^\dagger + A^\dagger H^\dagger \\ &= H^2 - (-A)^2 - H(-A) + (-A)H = H^2 - A^2 + HA - AH = b_l^\dagger b_l.\end{aligned}\quad (3.284)$$

$$(3.285)$$

The Hermiticity is true of $b_l b_l^\dagger$ as well. Thus, the eigenvalue and eigenstate (or wave function) which belongs to that eigenvalue are physically meaningful.

Next, consider the following operation:

$$\begin{aligned} \tilde{b}_l \Phi_l^{(n)}(\rho) &= \left[\epsilon^{(n-1)} - \epsilon^{(l-1)} \right]^{-\frac{1}{2}} \left(\frac{2}{n} \right)^{l+\frac{3}{2}} \sqrt{\frac{(n-l-1)!}{2n(n+l)!}} \left[\frac{d}{d\rho} + \left(\frac{l}{\rho} - \frac{1}{l} \right) \right] \\ &\times \left\{ e^{-\frac{\rho}{n}} \rho^{l+1} L_{n-l-1}^{2l+1} \left(\frac{2\rho}{n} \right) \right\}, \end{aligned} \quad (3.286)$$

where

$$\left[\epsilon^{(n-1)} - \epsilon^{(l-1)} \right]^{-\frac{1}{2}} = \frac{nl}{\sqrt{(n+l)(n-l)}}. \quad (3.287)$$

Rewriting $L_{n-l-1}^{2l+1} \left(\frac{2\rho}{n} \right)$ in a power series expansion form using (3.280) and rearranging the result, we obtain

$$\begin{aligned} \tilde{b}_l \Phi_l^{(n)}(\rho) &= \left(\frac{2}{n} \right)^{l+\frac{3}{2}} \frac{nl}{\sqrt{(n+l)(n-l)}} \sqrt{\frac{1}{2n(n+l)!(n-l-1)!}} \left[\frac{d}{d\rho} + \left(\frac{l}{\rho} - \frac{1}{l} \right) \right] \left[e^{\frac{\rho}{n}} \rho^{-l} \frac{d^{n-l-1}}{d\rho^{n-l-1}} \left(\rho^{n+l} e^{-\frac{2\rho}{n}} \right) \right] \\ &= \left(\frac{2}{n} \right)^{l+\frac{3}{2}} \frac{nl}{\sqrt{(n+l)(n-l)}} \sqrt{\frac{1}{2n(n+l)!(n-l-1)!}} \left[\frac{d}{d\rho} + \left(\frac{l}{\rho} - \frac{1}{l} \right) \right] \\ &\times e^{-\rho/n} \sum_{m=0}^{n-l-1} \frac{(-1)^m (n+l)!}{(2l+m+1)!} \rho^{l+m+1} \left(\frac{2}{n} \right)^m \frac{(n-l-1)!}{m!(n-l-m-1)!}, \end{aligned} \quad (3.288)$$

where we used well-known Leibniz rule of higher order differentiation of a product function; i.e., $\frac{d^{n-l-1}}{d\rho^{n-l-1}} \left(\rho^{n+l} e^{-\frac{2\rho}{n}} \right)$. To perform further calculation, notice that $\frac{d}{d\rho}$ does not change a functional form of $e^{-\rho/n}$, whereas it lowers the order of ρ^{l+m+1} by one. Meanwhile, operation of $\frac{1}{\rho}$ lowers the order of ρ^{l+m+1} by one as well. The factor $\frac{n+l}{2l}$ in the following equation (3.289) results from these calculation processes. Considering these characteristics of the operator, we get

$$\begin{aligned} \tilde{b}_l \Phi_l^{(n)}(\rho) &= \left(\frac{2}{n} \right)^{l+\frac{3}{2}} \frac{n+l}{2l} \frac{nl}{\sqrt{(n+l)(n-l)}} \sqrt{\frac{1}{2n(n+l)!(n-l-1)!}} e^{-\rho/n} \rho^l \\ &\times \left\{ \sum_{m=0}^{n-l-1} \frac{(-1)^{m+1} (n+l)!}{(2l+m+1)!} \frac{(n-l-1)!}{m!(n-l-m-1)!} \rho^{m+1} \left(\frac{2}{n} \right)^{m+1} \right. \\ &\left. + 2l \sum_{m=0}^{n-l-1} \frac{(-1)^m (n+l-1)!}{(2l+m)!} \frac{(n-l-1)!}{m!(n-l-m-1)!} \rho^m \left(\frac{2}{n} \right)^m \right\}. \end{aligned} \quad (3.289)$$

In (3.289), calculation of the part $\{\cdots\}$ next to the multiplication sign for RHS is somewhat complicated, and so we describe the outline of the calculation procedure below. We have

$\{\cdots\}$ of RHS of (3.289)

$$\begin{aligned}
&= \sum_{m=1}^{n-l} \frac{(-1)^m (n+l)!}{(2l+m)!} \frac{(n-l-1)!}{(m-1)!(n-l-m)!} \rho^m \left(\frac{2}{n}\right)^m \\
&\quad + 2l \sum_{m=0}^{n-l-1} \frac{(-1)^m (n+l-1)!}{(2l+m)!} \frac{(n-l-1)!}{m!(n-l-m-1)!} \rho^m \left(\frac{2}{n}\right)^m \\
&= \sum_{m=1}^{n-l-1} \frac{(-1)^m \left(\frac{2\rho}{n}\right)^m (n-l-1)!(n+l-1)!}{(2l+m)!} \left[\frac{n+l}{(m-1)!(n-l-m)!} + \frac{2l}{m!(n-l-m-1)!} \right] \\
&\quad + (-1)^{n-l} \left(\frac{2\rho}{n}\right)^{n-l} + \frac{(n+l-1)!}{(2l-1)!} \\
&= \sum_{m=1}^{n-l-1} \frac{(-1)^m \left(\frac{2\rho}{n}\right)^m (n-l-1)!(n+l-1)!}{(2l+m)!} \cdot \frac{(m-1)!(n-l-m-1)!(2l+m)(n-l)}{(m-1)!(n-l-m)!m!(n-l-m-1)!} \\
&\quad + (-1)^{n-l} \left(\frac{2\rho}{n}\right)^{n-l} + \frac{(n+l-1)!}{(2l-1)!} \\
&= \sum_{m=1}^{n-l-1} \frac{(-1)^m \left(\frac{2\rho}{n}\right)^m (n-l-1)!(n+l-1)!}{(2l+m-1)!(n-l-m)!m!} + (-1)^{n-l} \left(\frac{2\rho}{n}\right)^{n-l} + \frac{(n+l-1)!}{(2l-1)!} \\
&= (n-l)! \left[\sum_{m=1}^{n-l-1} \frac{(-1)^m \left(\frac{2\rho}{n}\right)^m (n+l-1)!}{(2l+m-1)!(n-l-m)!m!} + (-1)^{n-l} \frac{1}{(n-l)!} \left(\frac{2\rho}{n}\right)^{n-l} + \frac{(n+l-1)!}{(n-l)!(2l-1)!} \right] \\
&= (n-l)! \sum_{m=0}^{n-l} \frac{(-1)^m \left(\frac{2\rho}{n}\right)^m (n+l-1)!}{(2l+m-1)!(n-l-m)!m!} = (n-l)! L_{n-l}^{2l-1} \left(\frac{2\rho}{n}\right).
\end{aligned} \tag{3.290}$$

Notice that with the second equality of (3.290), the summation is divided into three terms; i.e., $1 \leq m \leq n-l-1$, $m = n-l$ (the highest-order term), and $m = 0$ (the lowest-order term). Note that with the second last equality of (3.290), the highest-order $(n-l)$ term and the lowest-order term (i.e., a constant) have been absorbed in a single equation, namely, an associated Laguerre polynomial. Correspondingly, with the second last equality of (3.290) the summation range is extended to $0 \leq m \leq n-l$.

Summarizing the above results, we get

$$\begin{aligned}
\tilde{b}_l \Phi_l^{(n)}(\rho) &= \left(\frac{2}{n}\right)^{l+\frac{3}{2}} \frac{n+l}{2l} \frac{nl(n-l)!}{\sqrt{(n+l)(n-l)}} \sqrt{\frac{1}{2n(n+l)!(n-l-1)!}} e^{-\rho/n} \rho^l L_{n-l}^{2l-1} \left(\frac{2\rho}{n}\right) \\
&= \left(\frac{2}{n}\right)^{l+\frac{1}{2}} \sqrt{\frac{(n-l)!}{2n(n+l-1)!}} e^{-\rho/n} \rho^l L_{n-l}^{2l-1} \left(\frac{2\rho}{n}\right) \\
&= \left(\frac{2}{n}\right)^{(l-1)+\frac{3}{2}} \sqrt{\frac{[n-(l-1)-1]!}{2n[n+(l-1)]!}} e^{-\frac{\rho}{n}} \rho^{(l-1)+1} L_{n-(l-1)-1}^{2(l-1)+1} \left(\frac{2\rho}{n}\right) \\
&\equiv \Phi_{l-1}^{(n)}(\rho).
\end{aligned} \tag{3.291}$$

Thus, we find out that $\Phi_l^{(n)}(\rho)$ behaves exactly like $\tilde{\psi}_l^{(n)}$. Moreover, if we replace l in (3.279) with $n-1$, we find

$$\Phi_{n-1}^{(n)}(\rho) = \tilde{\psi}_{n-1}^{(n)}. \tag{3.292}$$

Operating $\tilde{b}_{n-1}$ on both sides of (3.292),

$$\Phi_{n-2}^{(n)}(\rho) = \tilde{\psi}_{n-2}^{(n)}. \tag{3.293}$$

Likewise successively operating $\tilde{b}_l$ ($1 \leq l \leq n-1$),

$$\Phi_l^{(n)}(\rho) = \tilde{\psi}_l^{(n)}(\rho), \tag{3.294}$$

with all allowed numbers of l (i.e., $0 \leq l \leq n-1$). This permits us to identify

$$\Phi_l^{(n)}(\rho) \equiv \tilde{\psi}_l^{(n)}(\rho). \tag{3.295}$$

Consequently, it is clear that the parameter n introduced in (3.249) is identical to a principal quantum number and that the parameter l ($0 \leq l \leq n-1$) is an orbital angular momentum quantum number (or azimuthal quantum number). The functions $\Phi_l^{(n)}(\rho)$ and $\tilde{\psi}_l^{(n)}(\rho)$ are identical up to the constant c_n expressed in (3.265). Note, however, that a complex constant with an absolute value of 1 (phase factor) remains undetermined, as is always the case with the eigenvalue problem.

The radial wave functions are derived from the following relationship as described earlier:

$$R_l^{(n)}(r) = \tilde{\psi}_l^{(n)}/\rho. \tag{3.296}$$

To normalize $R_l^{(n)}(r)$, we have to calculate the following integral:

$$\begin{aligned} \int_0^\infty |R_l^{(n)}(r)|^2 r^2 dr &= \int_0^\infty \frac{1}{\rho^2} |\tilde{\psi}_l^{(n)}|^2 \left(\frac{a}{Z}\rho\right)^2 \frac{a}{Z} d\rho = \left(\frac{a}{Z}\right)^3 \int_0^\infty |\tilde{\psi}_l^{(n)}|^2 d\rho \\ &= \left(\frac{a}{Z}\right)^3. \end{aligned} \quad (3.297)$$

Accordingly, we choose the following functions $\tilde{R}_l^{(n)}(r)$ for the normalized radial wave functions:

$$\tilde{R}_l^{(n)}(r) = \sqrt{(Z/a)^3} R_l^{(n)}(r). \quad (3.298)$$

Substituting (3.296) into (3.298) and taking account of (3.279) and (3.280), we obtain

$$\tilde{R}_l^{(n)}(r) = \sqrt{\left(\frac{2Z}{an}\right)^3 \cdot \frac{(n-l-1)!}{2n(n+l)!}} \left(\frac{2Zr}{an}\right)^l \exp\left[-\left(\frac{Zr}{an}\right)\right] L_{n-l-1}^{2l+1}\left(\frac{2Zr}{an}\right). \quad (3.299)$$

Equation (3.298) is exactly the same as the normalized radial wave functions that can be obtained as the solution of (3.241) through the power series expansion. All these functions belong to the same eigenenergy $E_n = -\frac{\hbar^2}{2\mu} \left(\frac{Z}{a}\right)^2 \frac{1}{n^2}$; see (3.258).

Returning back to the quantum states sharing the same eigenenergy, we have such n states that belong to the principal quantum number n with varying azimuthal quantum number l ($0 \leq l \leq n-1$). Meanwhile, from (3.158), (3.159), and (3.171), $2l+1$ quantum states possess an eigenvalue of $l(l+1)$ with the quantity M^2 . As already mentioned in Sects. 3.5 and 3.6, the integer m takes the $2l+1$ different values of

$$m = l, l-1, l-2, \dots, 1, 0, -1, \dots, -l+1, -l.$$

The quantum states of a hydrogen-like atoms are characterized by a set of integers (m, l, n) . On the basis of the above discussion, the number of those sharing the same energy (i.e., the principal quantum number n) is

$$\sum_{l=0}^{n-1} (2l+1) = n^2.$$

Regarding this situation, we say that the quantum states belonging to the principal quantum number n , or the energy eigenvalue $E_n = -\frac{\hbar^2}{2\mu} \left(\frac{Z}{a}\right)^2 \frac{1}{n^2}$, are *degenerate* n^2 -fold. Note that we ignored the freedom of spin state.

In summary of this section, we have developed the operator formalism in dealing with radial wave functions of hydrogen-like atoms and seen how the operator formalism features the radial wave functions. The essential point rests upon that the radial wave functions can be derived by successively operating the lowering

operators b_l on $\tilde{\psi}_{n-1}^{(n)}$ that is parametrized with a principal quantum number n and an orbital angular momentum quantum number $l = n - 1$. This is clearly represented by (3.278). The results agree with the conventional coordinate representation method based upon the power series expansion that leads to associated Laguerre polynomials. Thus, the operator formalism is again found to be powerful in explicitly representing the mathematical constitution of quantum-mechanical systems.

3.8 Total Wave Functions

Since we have obtained angular wave functions and radial wave functions, we describe normalized total wave functions $\tilde{\Lambda}_{l,m}^{(n)}$ of hydrogen-like atoms as a product of the angular part and radial part such that

$$\tilde{\Lambda}_{l,m}^{(n)} = Y_l^m(\theta, \phi) \tilde{R}_l^{(n)}(r). \quad (3.300)$$

Let us seek several tangible functional forms of hydrogen ($Z = 1$) including angular and radial parts. For example, we have

$$\phi(1s) \equiv Y_0^0(\theta, \phi) \tilde{R}_0^{(1)}(r) = \sqrt{\frac{1}{4\pi}} a^{-3/2} \left(\frac{\tilde{\psi}_{n-1}^{(n)}}{\rho} \right) = \sqrt{\frac{1}{\pi}} a^{-3/2} e^{-r/a}, \quad (3.301)$$

where we used (3.276) and (3.295).

For $\phi(2s)$, using (3.277) and (3.278) we have

$$\phi(2s) \equiv Y_0^0(\theta, \phi) \tilde{R}_0^{(2)}(r) = \frac{1}{4\sqrt{2\pi}} a^{-\frac{3}{2}} e^{-\frac{r}{2a}} \left(2 - \frac{r}{a} \right). \quad (3.302)$$

For $\phi(2p_z)$, in turn, we express it as

$$\begin{aligned} \phi(2p_z) &\equiv Y_1^0(\theta, \phi) \tilde{R}_1^{(2)}(r) = \sqrt{\frac{3}{4\pi}} (\cos \theta) \frac{1}{2\sqrt{6}} a^{-\frac{3}{2}} \frac{r}{a} e^{-\frac{r}{2a}} \\ &= \frac{1}{4\sqrt{2\pi}} a^{-\frac{3}{2}} \frac{r}{a} e^{-\frac{r}{2a}} \cos \theta = \frac{1}{4\sqrt{2\pi}} a^{-\frac{3}{2}} \frac{r}{a} e^{-\frac{r}{2a}} \frac{z}{r} = \frac{1}{4\sqrt{2\pi}} a^{-\frac{5}{2}} e^{-\frac{r}{2a}} z. \end{aligned} \quad (3.303)$$

For $\phi(2p_x + iy)$, using (3.217) we get

$$\begin{aligned}
\phi(2p_{x+iy}) &\equiv Y_1^1(\theta, \phi) \tilde{R}_1^{(2)}(r) = -\frac{1}{8\sqrt{\pi}} a^{-\frac{3}{2}} \frac{r}{a} e^{-\frac{r}{2a}} \sin \theta e^{i\phi} \\
&= -\frac{1}{8\sqrt{\pi}} a^{-\frac{3}{2}} \frac{r}{a} e^{-\frac{r}{2a}} \frac{x+iy}{r} = -\frac{1}{8\sqrt{\pi}} a^{-\frac{5}{2}} e^{-\frac{r}{2a}} (x+iy).
\end{aligned} \tag{3.304}$$

In (3.304), the minus sign comes from the Condon–Shortley phase. Furthermore, we have

$$\begin{aligned}
\phi(2p_{x-iy}) &\equiv Y_1^{-1}(\theta, \phi) \tilde{R}_1^{(2)}(r) = \frac{1}{8\sqrt{\pi}} a^{-\frac{3}{2}} \frac{r}{a} e^{-\frac{r}{2a}} \sin \theta e^{-i\phi} \\
&= \frac{1}{8\sqrt{\pi}} a^{-\frac{3}{2}} \frac{r}{a} e^{-\frac{r}{2a}} \frac{x-iy}{r} = \frac{1}{8\sqrt{\pi}} a^{-\frac{5}{2}} e^{-\frac{r}{2a}} (x-iy).
\end{aligned} \tag{3.305}$$

Notice that the above notations $\phi(2p_{x+iy})$ and $\phi(2p_{x-iy})$ differ from the custom that uses, e.g., $\phi(2p_x)$ and $\phi(2p_y)$. We will come back to this point in Sect. 4.3.

References

1. Schiff LI (1955) Quantum mechanics, 2nd edn. McGraw-Hill, New York, NY
2. Arfken GB (1970) Mathematical methods for physicists, 2nd edn. Academic Press, Waltham, MA
3. Sunakawa S (1991) Quantum mechanics. Iwanami, Tokyo. (in Japanese)
4. Byron FW Jr, Fuller RW (1992) Mathematics of classical and quantum physics. Dover, New York, NY
5. Dennery P, Krzywicki A (1996) Mathematics for physicists. Dover, New York, NY
6. Riley KF, Hobson MP, Bence SJ (2006) Mathematical methods for physics and engineering, 3rd edn. Cambridge University Press, Cambridge
7. Arfken GB, Weber HJ, Harris FE (2013) Mathematical methods for physicists, 7th edn. Academic Press, Waltham, MA
8. Lebedev NN (1972) Special functions and their applications. Dover, New York, NY
9. Stakgold I (1998) Green's functions and boundary value problems, 2nd edn. Wiley, New York, NY
10. Hotta S (2017) Operator representations for radial wave functions of hydrogen-like atoms. Bull Kyoto Inst Technol 9:1–12

Chapter 4

Optical Transition and Selection Rules



In Sect. 1.2 we showed the Schrödinger equation as a function of space coordinates and time. In subsequent sections, we dealt with the time-independent eigenvalue problems of a harmonic oscillator and hydrogen-like atoms. This implies that the physical system is isolated from the outside world and that there is no interaction between the outside world and physical system we are considering. However, by virtue of the interaction, the system may acquire or lose energy, momentum, angular momentum, etc. As a consequence of the interaction, the system changes its quantum state as well. Such a change is said to be a transition. If the interaction takes place as an optical process, we are to deal with an optical transition. Of various optical transitions, the electric dipole transition is common and the most important. In this chapter, we study the optical transition of a particle confined in a potential well, a harmonic oscillator, and a hydrogen using a semiclassical approach. A question of whether the transition is allowed or forbidden is of great importance. We have a selection rule to judge it.

4.1 Electric Dipole Transition

We have a time-dependent Schrödinger equation described as

$$H\psi = i\hbar \frac{\partial \psi}{\partial t}. \quad (1.47)$$

Using the method of separation of variables, we obtained two equations expressed below:

$$H\phi(x) = E\phi(x), \quad (1.55)$$

$$i\hbar \frac{\partial \xi(t)}{\partial t} = E\xi(t). \quad (1.56)$$

Equation (1.55) is an eigenvalue equation of energy and (1.56) is an equation with time. So far we have focused our attention upon (1.55) taking a particle confined within a potential well, a one-dimensional harmonic oscillator, and hydrogen-like atoms as an example. In this chapter we deal with a time-evolved Schrödinger equation and its relevance to an optical transition. The optical transition takes place according to selection rules. We mention their significance as well.

We showed that after solving the eigenvalue equation, the Schrödinger equation is expressed as

$$\psi(\mathbf{x}, t) = \phi(\mathbf{x}) \exp(-iEt/\hbar). \quad (1.60)$$

The probability density of the system (i.e., normally a particle such as an electron, a harmonic oscillator, etc.) residing at a certain place $\mathbf{x}$ at a certain time t is expressed as

$$\psi^*(\mathbf{x}, t)\psi(\mathbf{x}, t).$$

If the Schrödinger equation is described as a form of separated variables as in the case of (1.60), the exponential factors including t cancel out and we have

$$\psi^*(\mathbf{x}, t)\psi(\mathbf{x}, t) = \phi^*(\mathbf{x})\phi(\mathbf{x}). \quad (4.1)$$

This means that the probability density of the system depends only on spatial coordinate and is constant in time. Such a state is said to be a stationary state. That is, the system continues residing in a quantum state described by $\phi(\mathbf{x})$ and remains unchanged independent of time.

Next, we consider a linear combination of functions described by (1.60). That is, we have

$$\psi(\mathbf{x}, t) = c_1\phi_1(\mathbf{x}) \exp(-iE_1t/\hbar) + c_2\phi_2(\mathbf{x}) \exp(-iE_2t/\hbar), \quad (4.2)$$

where the first term is pertinent to the state 1 and second term to the state 2 and c_1 and c_2 are complex constants with respect to the spatial coordinates but may be weakly time-dependent. The state described by (4.2) is called a *coherent state*. The probability distribution of that state is described as

$$\psi^*(\mathbf{x}, t)\psi(\mathbf{x}, t) = |c_1|^2|\phi_1|^2 + |c_2|^2|\phi_2|^2 + c_1^*c_2\phi_1^*\phi_2e^{-i\omega t} + c_2^*c_1\phi_2^*\phi_1e^{i\omega t}, \quad (4.3)$$

where ω is expressed as

$$\omega = (E_2 - E_1)/\hbar. \quad (4.4)$$

This equation shows that the probability density of the system undergoes a sinusoidal oscillation with time. The angular frequency equals the energy difference between the two states divided by the reduced Planck constant. If the system is a charged particle such as an electron and proton, the sinusoidal oscillation is accompanied by an oscillating electromagnetic field. Thus, the coherent state is associated with the optical transition from one state to another, when the transition is related to the charged particle.

The optical transitions result from various causes. Of these, the electric dipole transition yields the largest transition probability and the dipole approximation is often chosen to represent the transition probability. From the point of view of optical measurements, the electric dipole transition gives the strongest absorption or emission spectral lines. The matrix element of the electric dipole, more specifically a square of an absolute value of the matrix element, is a measure of the optical transition probability. Labeling the quantum states as a , b , etc. and describing the corresponding state vector as $|a\rangle$, $|b\rangle$, etc., the matrix element P_{ba} is given by

$$P_{ba} \equiv \langle b | \boldsymbol{\epsilon}_e \cdot \mathbf{P} | a \rangle, \quad (4.5)$$

where $\boldsymbol{\epsilon}_e$ is a unit polarization vector of the electric field of an electromagnetic wave (i.e., light). Equation (4.5) describes the optical transition that takes place as a result of the interaction between electrons and radiation field in such a way that the interaction causes electrons in the system to change the state from $|a\rangle$ to $|b\rangle$. That interaction is represented by $\boldsymbol{\epsilon}_e \cdot \mathbf{P}$. The quantum states $|a\rangle$ and $|b\rangle$ are referred to as an initial state and final state, respectively.

The quantity $\mathbf{P}$ is the electric dipole moment of the system, which is defined as

$$\mathbf{P} \equiv e \sum_j \mathbf{x}_j, \quad (4.6)$$

where e is an elementary charge ($e < 0$) and $\mathbf{x}_j$ is a position vector of the j -th electron. Detailed description of $\boldsymbol{\epsilon}_e$ and $\mathbf{P}$ can be seen in Chap. 7. The quantity P_{ba} is said to be transition dipole moment, or more precisely, transition electric dipole moment with respect to the states $|a\rangle$ and $|b\rangle$. We assume that the optical transition occurs from a quantum state $|a\rangle$ to another state $|b\rangle$. Since P_{ba} is generally a complex number, $|P_{ba}|^2$ represents the transition probability.

If we adopt the coordinate representation, (4.5) is expressed by

$$P_{ba} = \int \phi_b^* \boldsymbol{\epsilon}_e \cdot \mathbf{P} \phi_a d\tau, \quad (4.7)$$

where τ denotes an integral range of a space.

4.2 One-Dimensional System

Let us apply the aforementioned general description to individual cases of Chaps. 1–3.

Example 4.1 A particle confined in a square-well potential This example was treated in Chap. 1. As before, we assume that a particle (i.e., electron) is confined in a one-dimensional system $[-L \leq x \leq L]$ ($L > 0$).

We consider the optical transition from the ground state $\phi_1(x)$ to the first excited state $\phi_2(x)$. Here, we put $L = \pi/2$ for convenience. Then, the normalized coherent state $\psi(x)$ is described as

$$\psi(x, t) = \frac{1}{\sqrt{2}} [\phi_1(x) \exp(-iE_1 t/\hbar) + \phi_2(x) \exp(-iE_2 t/\hbar)], \quad (4.8)$$

where we put $c_1 = c_2 = \frac{1}{\sqrt{2}}$ in (4.2). In (4.8), we have

$$\phi_1(x) = \sqrt{\frac{2}{\pi}} \cos x \quad \text{and} \quad \phi_2(x) = \sqrt{\frac{2}{\pi}} \sin 2x. \quad (4.9)$$

Following (4.3), we have a following real function called a probability distribution density:

$$\psi^*(x, t)\psi(x, t) = \frac{1}{\pi} [\cos^2 x + \sin^2 2x + (\sin 3x + \sin x) \cos \omega t], \quad (4.10)$$

where ω is given by (4.4) as

$$\omega = 3\hbar/2m, \quad (4.11)$$

where m is a mass of an electron. Rewriting (4.10), we have

$$\psi^*(x, t)\psi(x, t) = \frac{1}{\pi} \left[1 + \frac{1}{2} (\cos 2x - \cos 4x) + (\sin 3x + \sin x) \cos \omega t \right]. \quad (4.12)$$

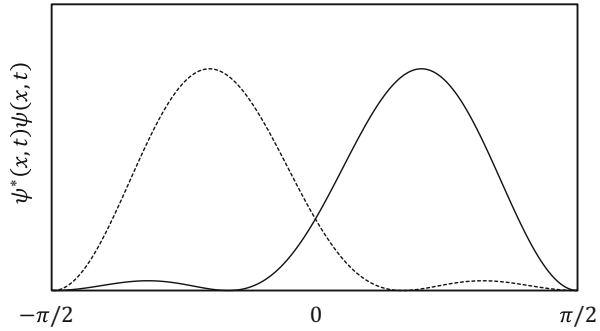
Integrating (4.12) over $[-\frac{\pi}{2}, \frac{\pi}{2}]$, a contribution from only the first term is non-vanishing to give 1, as anticipated (because of the normalization).

Putting $t = 0$ and integrating (4.12) over a positive domain $[0, \frac{\pi}{2}]$, we have

$$\int_0^{\pi/2} \psi^*(x, 0)\psi(x, 0)dx = \frac{1}{2} + \frac{4}{3\pi} \approx 0.924. \quad (4.13)$$

Similarly, integrating (4.12) over a negative domain $[-\frac{\pi}{2}, 0]$, we have

Fig. 4.1 Probability distribution density $\psi^*(x, t)\psi(x, t)$ of a particle confined in a square-well potential. The solid curve and broken curve represent the density of $t = 0$ and $t = \pi/\omega$ (i.e., half period), respectively



$$\int_{-\pi/2}^0 \psi^*(x, 0)\psi(x, 0)dx = \frac{1}{2} - \frac{4}{3\pi} \approx 0.076. \quad (4.14)$$

Thus, 92% of a total charge (as a probability density) is concentrated in the positive domain. Differentiation of $\psi^*(x, 0)\psi(x, 0)$ gives five extremals including both edges. Of these, a major maximum is located at 0.635 radian that corresponds to about 40% of $\pi/2$. This can be a measure of the transition moment. Figure 4.1 demonstrates these results (see a solid curve). Meanwhile, putting $t = \pi/\omega$ (i.e., half period), we plot $\psi^*(x, \pi/\omega)\psi(x, \pi/\omega)$. The result shows that the graph is obtained by folding back the solid curve of Fig. 4.1 with respect to the ordinate axis. Thus, we find that the charge (or the probability density) exerts a sinusoidal oscillation with an angular frequency $3\hbar/2m$ along the x -axis around the origin.

Let e_1 be a unit vector in the positive direction of the x -axis. Then, the electric dipole $\mathbf{P}$ of the system is

$$\mathbf{P} = e\mathbf{x} = exe_1, \quad (4.15)$$

where $\mathbf{x}$ is a position vector of the electron. Let us define the matrix element of the electric dipole transition as

$$P_{21} \equiv \langle \phi_2(x) | e_1 \cdot \mathbf{P} | \phi_1(x) \rangle = \langle \phi_2(x) | ex | \phi_1(x) \rangle. \quad (4.16)$$

Notice that we only have to consider that the polarization of light is parallel to the x -axis. With the coordinate representation, we have

$$\begin{aligned}
P_{21} &= \int_{-\pi/2}^{\pi/2} \phi_2^*(x) \exp_1(x) dx = \int_{-\pi/2}^{\pi/2} \sqrt{\frac{2}{\pi}} (\cos x) \exp \sqrt{\frac{2}{\pi}} \sin 2x dx \\
&= e \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} x \cos x \sin 2x dx = \frac{e}{\pi} \int_{-\pi/2}^{\pi/2} x (\sin x + \sin 3x) dx \\
&= \frac{e}{\pi} \int_{-\pi/2}^{\pi/2} \left[x(-\cos x)' + x \left(-\frac{1}{3} \cos 3x \right)' \right] dx = \frac{16e}{9\pi},
\end{aligned} \tag{4.17}$$

where we used a trigonometric formula and integration by parts. The factor $16/9\pi$ in (4.17) is about 36% of $\pi/2$. This number is pretty good agreement with 40% that is estimated above from the major maximum of $\psi^*(x, 0)\psi(x, 0)$.

Note that the transition moment vanishes if the two states associated with the transition have the same parity. In other words, if these are both described by sine functions or cosine functions, the integral vanishes.

Example 4.2 One-dimensional harmonic oscillator Second, let us think of an optical transition regarding a harmonic oscillator that we dealt with in Chap. 2. We denote the state of the oscillator as $|n\rangle$ in place of $|\psi_n\rangle$ ($n = 0, 1, 2, \dots$) of Chap. 2. Then, a general expression (4.5) can be written as

$$P_{kl} = \langle k | \boldsymbol{\epsilon}_e \cdot \mathbf{P} | l \rangle. \tag{4.18}$$

Since we are considering the sole one-dimensional oscillator, we have

$$\boldsymbol{\epsilon}_e = \tilde{\mathbf{q}} \quad \text{and} \quad \mathbf{P} = e\mathbf{q}, \tag{4.19}$$

where $\tilde{\mathbf{q}}$ is a unit vector in the positive direction of the coordinate q . Therefore, similar to the above, we have

$$\boldsymbol{\epsilon}_e \cdot \mathbf{P} = eq. \tag{4.20}$$

That is,

$$P_{kl} = e \langle k | q | l \rangle. \tag{4.21}$$

Since q is an Hermitian operator, we have

$$P_{kl}^* = e \langle l | q^\dagger | k \rangle = e \langle l | q | k \rangle = P_{lk}, \tag{4.22}$$

where we used (1.116). Using (2.68), we have

$$P_{kl} = e\sqrt{\frac{\hbar}{2m\omega}} \langle k|a + a^\dagger|l \rangle = e\sqrt{\frac{\hbar}{2m\omega}} [\langle k|a|l \rangle + \langle k|a^\dagger|l \rangle]. \quad (4.23)$$

Taking the adjoint of (2.62) and modifying the notation, we have

$$\langle k | a = \sqrt{k+1} \langle k+1 |. \quad (4.24)$$

Using (2.62) once again, we get

$$P_{kl} = e\sqrt{\frac{\hbar}{2m\omega}} [\sqrt{k+1} \langle k+1|l \rangle + \sqrt{l+1} \langle k|l+1 \rangle]. \quad (4.25)$$

Using orthonormal conditions between the state vectors, we have

$$P_{kl} = e\sqrt{\frac{\hbar}{2m\omega}} [\sqrt{k+1} \delta_{k+1, l} + \sqrt{l+1} \delta_{k, l+1}]. \quad (4.26)$$

Exchanging k and l in the above, we get

$$P_{kl} = P_{lk}.$$

The matrix element P_{kl} is symmetric with respect to indices k and l . Notice that the first term does not vanish only when $k+1 = l$. The second term does not vanish only when $k = l+1$. Therefore we get

$$P_{k,k+1} = e\sqrt{\frac{\hbar(k+1)}{2m\omega}} \quad \text{and} \quad P_{l+1,l} = e\sqrt{\frac{\hbar(l+1)}{2m\omega}}. \quad (4.27)$$

Meanwhile, we find that the transition matrix P is expressed as

$$P = eq = e\sqrt{\frac{\hbar}{2m\omega}} (a + a^\dagger) = e\sqrt{\frac{\hbar}{2m\omega}} \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & \cdots \\ 1 & 0 & \sqrt{2} & 0 & 0 & \cdots \\ 0 & \sqrt{2} & 0 & \sqrt{3} & 0 & \cdots \\ 0 & 0 & \sqrt{3} & 0 & 2 & \cdots \\ 0 & 0 & 0 & 2 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \quad (4.28)$$

where we used (2.68). Note that a real Hermitian matrix is a symmetric matrix.

Practically, it is a fast way to construct a transition matrix (4.28) using (2.65) and (2.66). It is an intuitively obvious and straightforward task. Having a glance at the matrix form immediately tells us that the transition matrix elements are

non-vanishing with only $(k, k + 1)$ and $(k + 1, k)$ positions. Whereas the $(k, k + 1)$ -element represents transition from the k -th excited state to $(k - 1)$ -th excited state accompanied by photoemission, the $(k + 1, k)$ -element implies the transition from $(k - 1)$ -th excited state to k -th excited state accompanied by photoabsorption. The two transitions give the same transition moment. Note that zeroth excited state means the ground state; see (2.64) for basis vector representations.

We should be careful about “addresses” of the matrix accordingly. For example, $P_{0, 1}$ in (4.27) represents a $(1, 2)$ element of the matrix (4.28); $P_{2, 1}$ stands for a $(3, 2)$ element.

Suppose that we seek the transition dipole moments using coordinate representation. Then, we need to use (2.106) and perform definite integration. For instance, we have

$$e \int_{-\infty}^{\infty} \psi_0(q) q \psi_1(q) dq$$

that corresponds to $(1, 2)$ element of (4.28). Indeed, the above integral gives $e \sqrt{\frac{\hbar}{2m\omega}}$. The confirmation is left for the readers. Nonetheless, to seek a definite integral of product of higher excited-state wave functions becomes increasingly troublesome. In this respect, the operator method described above provides us with a much better insight into complicated calculations.

Equations (4.26)–(4.28) imply that the electric dipole transition is allowed to occur only when the quantum number changes by one. Notice also that the transition takes place between the even function and odd function; see Table 2.1 and (2.101). Such a condition or restriction on the optical transition is called a selection rule. The former equation of (4.27) shows that the transition takes place from the upper state to the lower state accompanied by the photon emission. The latter equation, on the other hand, shows that the transition takes place from the lower state to the upper accompanied by the photon absorption.

4.3 Three-Dimensional System

The hydrogen-like atoms give us a typical example. Since we have fully investigated the quantum states of those atoms, we make the most of the related results.

Example 4.3 An electron in a hydrogen atom Unlike the one-dimensional system, we have to take account of an angular momentum in the three-dimensional system. We have already obtained explicit wave functions. Here we focus on $1s$ and $2p$ states of a hydrogen. For their normalized states, we have

$$\phi(1s) = \sqrt{\frac{1}{\pi a^3}} e^{-r/a}, \quad (4.29)$$

$$\phi(2p_z) = \frac{1}{4} \sqrt{\frac{1}{2\pi a^3}} \frac{r}{a} e^{-r/2a} \cos \theta, \quad (4.30)$$

$$\phi(2p_{x+iy}) = -\frac{1}{8} \sqrt{\frac{1}{\pi a^3}} \frac{r}{a} e^{-r/2a} \sin \theta e^{i\phi}, \quad (4.31)$$

$$\phi(2p_{x-iy}) = \frac{1}{8} \sqrt{\frac{1}{\pi a^3}} \frac{r}{a} e^{-r/2a} \sin \theta e^{-i\phi}, \quad (4.32)$$

where a denotes Bohr radius of a hydrogen. Note that a minus sign of $\phi(2p_{x+iy})$ is due to the Condon–Shortley phase. Even though the transition probability is proportional to a square of the matrix element and so the phase factor cancels out, we describe the state vector faithfully. The energy eigenvalues are

$$E(1s) = -\frac{\hbar^2}{2\mu a^2}, \quad E(2p_z) = E(2p_{x+iy}) = E(2p_{x-iy}) = -\frac{\hbar^2}{8\mu a^2}, \quad (4.33)$$

where μ is a reduced mass of a hydrogen. Note that the latter three states are degenerate.

First, we consider a transition between $\phi(1s)$ and $\phi(2p_z)$ states. Suppose that the normalized coherent state is described as

$$\psi(\mathbf{x}, t) = \frac{1}{\sqrt{2}} \left\{ \phi(1s) \exp \left[-\frac{iE(1s)t}{\hbar} \right] + \phi(2p_z) \exp \left[-\frac{iE(2p_z)t}{\hbar} \right] \right\}. \quad (4.34)$$

As before, we have

$$\begin{aligned} \psi^*(\mathbf{x}, t) \psi(\mathbf{x}, t) &= |\psi(\mathbf{x}, t)|^2 \\ &= \frac{1}{2} \left\{ [\phi(1s)]^2 + [\phi(2p_z)]^2 + 2\phi(1s)\phi(2p_z) \cos \omega t \right\}, \end{aligned} \quad (4.35)$$

where ω is given by

$$\omega = [E(2p_z) - E(1s)]/\hbar = 3\hbar/8\mu a^2. \quad (4.36)$$

In virtue of the third term of (4.35) that contains a $\cos \omega t$ factor, the charge distribution undergoes a sinusoidal oscillation along the z -axis with an angular frequency described by (4.36). For instance, $\omega t = 0$ gives +1 factor to (4.35) when $t = 0$, whereas it gives -1 factor when $\omega t = \pi$, i.e., $t = 8\pi\mu a^2/3\hbar$.

Integrating (4.35), we have

$$\begin{aligned}
\int \psi^*(\mathbf{x}, t) \psi(\mathbf{x}, t) d\tau &= \int |\psi(\mathbf{x}, t)|^2 d\tau \\
&= \frac{1}{2} \int \left\{ [\phi(1s)]^2 + [\phi(2p_z)]^2 \right\} d\tau \\
&\quad + \cos \omega t \int \phi(1s) \phi(2p_z) d\tau \\
&= \frac{1}{2} + \frac{1}{2} = 1,
\end{aligned}$$

where we used normalized functional forms of $\phi(1s)$ and $\phi(2p_z)$ together with orthogonality of them. Note that both of the functions are real.

Next, we calculate the matrix element. For simplicity, we denote the matrix element simply as $P^{(\mathbf{e}_e)}$ only by designating the unit polarization vector $\mathbf{e}_e$. Then, we have

$$P^{(\mathbf{e}_e)} = \langle \phi(1s) | \mathbf{e}_e \cdot \mathbf{P} | \phi(2p_z) \rangle, \quad (4.37)$$

where

$$\mathbf{P} = e\mathbf{x} = e(\mathbf{e}_1 \mathbf{e}_2 \mathbf{e}_3) \begin{pmatrix} x \\ y \\ z \end{pmatrix}. \quad (4.38)$$

We have three possibilities of choosing $\mathbf{e}_e$ out of $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$. Choosing $\mathbf{e}_3$, we have

$$\begin{aligned}
P_{z,|p_z\rangle}^{(\mathbf{e}_3)} &= e \langle \phi(1s) | z | \phi(2p_z) \rangle \\
&= \frac{e}{4\sqrt{2}\pi a^4} \int_0^\infty r^4 e^{-3r/2a} dr \int_0^\pi \cos^2 \theta \sin \theta d\theta \int_0^{2\pi} d\phi = \frac{2^7 \sqrt{2}}{3^5} ea \approx 0.745ea.
\end{aligned} \quad (4.39)$$

In (4.39), we express the matrix element as $P_{z,|p_z\rangle}^{(\mathbf{e}_3)}$ to indicate the z -component of position vector and to explicitly show that $\phi(2p_z)$ state is responsible for the transition. In (4.39), we used $z = r \cos \theta$. We also used a radial part integration such that

$$\int_0^\infty r^4 e^{-3r/2a} dr = 24 \left(\frac{2a}{3} \right)^5.$$

Also, we changed a variable $\cos \theta \rightarrow t$ to perform the integration with respect to θ . We see that a “leverage” length of the transition moment is comparable to Bohr radius a .

With the notation $P_{z,|p_z\rangle}^{(e_3)}$, we need some explanation for consistency with the latter description. Equation (4.39) represents the transition from $|\phi(2p_z)\rangle$ to $|\phi(1s)\rangle$ that is accompanied by the photon emission. Thus, $|p_z\rangle$ in the notation means that $|\phi(2p_z)\rangle$ is the initial state. In the notation, in turn, (e_3) denotes the polarization vector and z represents the electric dipole.

In the case of photon absorption where the transition occurs from $|\phi(1s)\rangle$ to $|\phi(2p_z)\rangle$, we use the following notation:

$$P_{z,|p_z\rangle}^{(e_3)} = e \langle \phi(2p_z) | z | \phi(1s) \rangle. \quad (4.40)$$

Since all the functions related to the integration are real, we have

$$P_{z,|p_z\rangle}^{(e_3)} = P_{z,|p_z\rangle}^{(e_3)},$$

where $\langle p_z |$ means that $\phi(2p_z)$ is designated as the final state. Meanwhile, if we choose e_1 for e_e to evaluate the matrix element P_x , we have

$$\begin{aligned} P_{x,|p_z\rangle}^{(e_1)} &= e \langle \phi(1s) | x | \phi(2p_z) \rangle \\ &= \frac{e}{4\sqrt{2}\pi a^4} \int_0^\infty r^4 e^{-3r/2a} dr \int_0^\pi \sin^2 \theta \cos \theta d\theta \int_0^{2\pi} \cos \phi d\phi = 0, \end{aligned} \quad (4.41)$$

where $\cos \phi$ comes from $x = r \sin \theta \cos \phi$ and an integration of $\cos \phi$ gives zero. In a similar manner, we have

$$P_{y,|p_z\rangle}^{(e_2)} = e \langle \phi(1s) | y | \phi(2p_z) \rangle = 0. \quad (4.42)$$

Next, we estimate the matrix elements associated with $2p_x$ and $2p_y$. For this purpose, it is convenient to introduce the following complex coordinates by a unitary transformation:

$$\begin{aligned}
(\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{i}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \\
&= \begin{pmatrix} \frac{1}{\sqrt{2}}(\mathbf{e}_1 - i\mathbf{e}_2) & \frac{1}{\sqrt{2}}(\mathbf{e}_1 + i\mathbf{e}_2) & \mathbf{e}_3 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}(x + iy) \\ \frac{1}{\sqrt{2}}(x - iy) \\ z \end{pmatrix}, \tag{4.43}
\end{aligned}$$

where a unitary transformation is represented by a unitary matrix defined as

$$U^\dagger U = U U^\dagger = E. \tag{4.44}$$

We will investigate details of the unitary transformation and matrix in Parts III and IV.

We define $\mathbf{e}_+$ and $\mathbf{e}_-$ as follows [1]:

$$\mathbf{e}_+ \equiv \frac{1}{\sqrt{2}}(\mathbf{e}_1 + i\mathbf{e}_2) \quad \text{and} \quad \mathbf{e}_- \equiv \frac{1}{\sqrt{2}}(\mathbf{e}_1 - i\mathbf{e}_2), \tag{4.45}$$

where complex vectors $\mathbf{e}_+$ and $\mathbf{e}_-$ represent the left-circularly polarized light and right-circularly polarized light that carry an angular momentum $\hbar$ and $-\hbar$, respectively. We will revisit the characteristics and implication of these complex vectors in Sect. 7.4.

We have

$$(\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = (\mathbf{e}_- \ \mathbf{e}_+ \ \mathbf{e}_3) \begin{pmatrix} \frac{1}{\sqrt{2}}(x + iy) \\ \frac{1}{\sqrt{2}}(x - iy) \\ z \end{pmatrix}. \tag{4.46}$$

Note that $\mathbf{e}_+$, $\mathbf{e}_-$, and $\mathbf{e}_3$ are orthonormal. That is,

$$\langle \mathbf{e}_+ | \mathbf{e}_+ \rangle = 1, \quad \langle \mathbf{e}_+ | \mathbf{e}_- \rangle = 0, \quad \text{etc.} \tag{4.47}$$

In this situation, $\mathbf{e}_+$, $\mathbf{e}_-$, and $\mathbf{e}_3$ are said to form an orthonormal basis in a three-dimensional complex vector space (see Sect. 11.4).

Now, choosing $\mathbf{e}_+$ for $\mathbf{e}_e$, we have [2]

$$P_{x-iy,|p_+}^{(e_+)} \equiv e \left\langle \phi(1s) \left| \frac{1}{\sqrt{2}}(x-iy) \right| \phi(2p_{x+iy}) \right\rangle, \quad (4.48)$$

where $|p_+\rangle$ is a shorthand notation of $\phi(2p_{x+iy})$ that is designated as the initial state; $x-iy$ represents a complex electric dipole. Equation (4.48) represents an optical process in which an electron causes transition from $\phi(2p_{x+iy})$ to $\phi(1s)$ to lose an angular momentum $\hbar$, whereas the radiation field gains that angular momentum to conserve a total angular momentum $\hbar$. The notation $P_{x-iy,|p_+}^{(e_+)}$ reflects this situation. Using the coordinate representation, we rewrite (4.48) as

$$P_{x-iy,|p_+}^{(e_+)} = -\frac{e}{8\sqrt{2}\pi a^4} \int_0^\infty r^4 e^{-3r/2a} dr \int_0^\pi \sin^3 \theta d\theta \int_0^{2\pi} e^{-i\phi} e^{i\phi} d\phi = -\frac{2^7 \sqrt{2}}{3^5} ea, \quad (4.49)$$

where we used

$$x-iy = r \sin \theta e^{-i\phi}. \quad (4.50)$$

In the definite integral of (4.49), $e^{-i\phi}$ comes from $x-iy$, while $e^{i\phi}$ comes from $\phi(2p_{x+iy})$. Note that from (3.24) $e^{i\phi}$ is an eigenfunction corresponding to an angular momentum eigenvalue $\hbar$. Notice that in (4.49) exponents $e^{-i\phi}$ and $e^{i\phi}$ cancel out and that an azimuthal integral is non-vanishing.

If we choose $\mathbf{e}_-$ for $\mathbf{e}_e$, we have

$$P_{x+iy,|p_+}^{(e_-)} = e \left\langle \phi(1s) \left| \frac{1}{\sqrt{2}}(x+iy) \right| \phi(2p_{x+iy}) \right\rangle \quad (4.51)$$

$$= \frac{e}{8\sqrt{2}\pi a^4} \int_0^\infty r^4 e^{-3r/2a} dr \int_0^\pi \sin^3 \theta d\theta \int_0^{2\pi} e^{2i\phi} d\phi = 0, \quad (4.52)$$

where we used

$$x+iy = r \sin \theta e^{i\phi}.$$

With (4.52), a factor $e^{2i\phi}$ results from the product $\phi(2p_{x+iy})(x+iy)$, which renders the integral (4.51) vanishing. Note that the only difference between (4.49) and (4.52) is about the integration of ϕ factor. For the same reason, if we choose $\mathbf{e}_3$ for $\mathbf{e}_e$, the matrix element vanishes. Thus, with the $\phi(2p_{x+iy})$ -related matrix element, only $P_{x-iy,|p_+}^{(e_+)}$ survives. Similarly, with the $\phi(2p_{x-iy})$ -related matrix element, only $P_{x+iy,|p_-}^{(e_-)}$ survives. Notice that $|p_-\rangle$ is a shorthand notation of $\phi(2p_{x-iy})$ that is designated as the initial state. That is, we have, e.g.,

$$\begin{aligned}
P_{x+iy,|p_{-}\rangle}^{(e_{-})} &= e \left\langle \phi(1s) \left| \frac{1}{\sqrt{2}}(x+iy) \right| \phi(2p_{x-iy}) \right\rangle = \frac{2^7\sqrt{2}}{3^5} ea, \\
P_{x-iy,|p_{-}\rangle}^{(e_{+})} &= e \left\langle \phi(1s) \left| \frac{1}{\sqrt{2}}(x-iy) \right| \phi(2p_{x-iy}) \right\rangle = 0.
\end{aligned} \tag{4.53}$$

Taking complex conjugate of (4.48), we have

$$\left[P_{x-iy,|p_{+}\rangle}^{(e_{+})} \right]^* = e \left\langle \phi(2p_{x+iy}) \left| \frac{1}{\sqrt{2}}(x+iy) \right| \phi(1s) \right\rangle = -\frac{2^7\sqrt{2}}{3^5} ea. \tag{4.54}$$

Here recall (1.116) and $(x-iy)^\dagger = x+iy$. Also note that since $P_{x-iy,|p_{+}\rangle}^{(e_{+})}$ is real, $\left[P_{x-iy,|p_{+}\rangle}^{(e_{+})} \right]^*$ is real as well so that we have

$$\left[P_{x-iy,|p_{+}\rangle}^{(e_{+})} \right]^* = P_{x-iy,|p_{+}\rangle}^{(e_{+})} = P_{x+iy,|p_{+}\rangle}^{(e_{-})}, \tag{4.55}$$

where $|p_{+}\rangle$ means that $\phi(2p_{x+iy})$ is designated as the final state.

Comparing (4.48) and (4.55), we notice that the polarization vector has been switched from e_{+} to e_{-} with the allowed transition, even though the matrix element remains the same. This can be explained as follows: In (4.48) the photon emission is occurring, while the electron is causing a transition from $\phi(2p_{x+iy})$ to $\phi(1s)$. As a result, the radiation field has gained an angular momentum by $\hbar$ during the process in which the electron has lost an angular momentum $\hbar$. In other words, $\hbar$ is transferred from the electron to the radiation field and this process results in the generation of left-circularly polarized light in the radiation field.

In (4.54), on the other hand, the reversed process takes place. That is, the photon absorption is occurring in such a way that the electron is excited from $\phi(1s)$ to $\phi(2p_{x+iy})$. After this process has been completed, the electron has gained an angular momentum by $\hbar$, whereas the radiation field has lost an angular momentum by $\hbar$. As a result, the positive angular momentum $\hbar$ is transferred to the electron from the radiation field that involves left-circularly polarized light. This can be translated into the statement that the radiation field has gained an angular momentum by $-\hbar$. This is equivalent to the generation of right-circularly polarized light (characterized by e_{-}) in the radiation field. In other words, the electron gains the angular momentum by $\hbar$ to compensate the change in the radiation field.

The implication of the first equation of (4.53) can be interpreted in a similar manner. Also we have

$$\begin{aligned} \left[P_{x+iy,|p_-}^{(e_-)} \right]^* &= P_{x+iy,|p_-}^{(e_-)} = P_{x-iy,|p_-}^{(e_+)} = e \left\langle \phi(2p_{x-iy}) \left| \frac{1}{\sqrt{2}}(x-iy) \right| \phi(1s) \right\rangle \\ &= \frac{2^7 \sqrt{2}}{3^5} ea. \end{aligned}$$

In the above relation, $\langle p_- |$ implies that $\phi(2p_{x-iy})$ is present as the final state. Notice that the inner products of (4.49) and (4.53) are real, even though neither $x+iy$ nor $x-iy$ is Hermitian. Also note that $P_{x-iy,|p_+}^{(e_+)}$ of (4.49) and $P_{x+iy,|p_-}^{(e_-)}$ of (4.53) have the same absolute value with minus and plus signs, respectively. The minus sign of (4.49) comes from the Condon–Shortley phase. The difference, however, is not essential, because the transition probability is proportional to $\left| P_{x+iy,|p_-}^{(e_-)} \right|^2$ or $\left| P_{x-iy,|p_+}^{(e_+)} \right|^2$. Some literature [3, 4] uses $-(x+iy)$ instead of $x+iy$. This is simply because of the inclusion of the Condon–Shortley phase; see (3.304).

Let us think of the coherent state that is composed of $\phi(1s)$ and $\phi(2p_{x+iy})$ or $\phi(2p_{x-iy})$. Choosing $\phi(2p_{x+iy})$, the state $\psi(\mathbf{x}, t)$ can be given by

$$\begin{aligned} \psi(\mathbf{x}, t) &= \frac{1}{\sqrt{2}} \\ &\times \left\{ \phi(1s) \exp[-iE(1s)t/\hbar] + \phi(2p_{x+iy}) \exp[-iE(2p_{x+iy})t/\hbar] \right\}, \end{aligned} \quad (4.56)$$

where $\phi(1s)$ is described by (3.301) and $\phi(2p_{x+iy})$ is expressed as (3.304). Then we have

$$\begin{aligned} \psi^*(\mathbf{x}, t) \psi(\mathbf{x}, t) &= |\psi(\mathbf{x}, t)|^2 \\ &= \frac{1}{2} \left\{ |\phi(1s)|^2 + |\phi(2p_{x+iy})|^2 + \phi(1s) \Re \epsilon(2p_{x+iy}) \left[e^{i(\phi-\omega t)} + e^{-i(\phi-\omega t)} \right] \right\} \\ &= \frac{1}{2} \left\{ |\phi(1s)|^2 + |\phi(2p_{x+iy})|^2 + 2\phi(1s) \Re \epsilon(2p_{x+iy}) \cos(\phi - \omega t) \right\}, \end{aligned} \quad (4.57)$$

where using $\Re \epsilon(2p_{x+iy})$, we denote $\phi(2p_{x+iy})$ as follows:

$$\phi(2p_{x+iy}) \equiv \Re \epsilon(2p_{x+iy}) e^{i\phi}. \quad (4.58)$$

That is, $\Re \epsilon(2p_{x+iy})$ represents a real part of $\phi(2p_{x+iy})$ that depends only on r and θ . For example, from (3.304) we have

$$\Re \epsilon(2p_{x+iy}) = -\frac{1}{8\sqrt{\pi}} a^{-\frac{3}{2}} \frac{r}{a} e^{-\frac{r}{2a}} \sin \theta.$$

The third term of (4.57) implies that the existence probability density of an electron represented by $|\psi(\mathbf{x}, t)|^2$ is rotating counterclockwise around the z -axis with an angular frequency of ω . Similarly, in the case of $\phi(2p_x - iy)$, the existence probability density of an electron is rotating clockwise around the z -axis with an angular frequency of ω .

Integrating (4.57), we have

$$\begin{aligned} \int \psi^*(\mathbf{x}, t) \psi(\mathbf{x}, t) d\tau &= \int |\psi(\mathbf{x}, t)|^2 d\tau \\ &= \frac{1}{2} \int \left\{ |\phi(1s)|^2 + |\phi(2p_{x+iy})|^2 \right\} d\tau \\ &\quad + \int_0^\infty r^2 dr \int_0^\pi \sin \theta d\theta \phi(1s) \Re e(2p_{x+iy}) \int_0^{2\pi} d\phi \cos(\phi - \omega t) \\ &= \frac{1}{2} + \frac{1}{2} = 1, \end{aligned}$$

where we used normalized functional forms of $\phi(1s)$ and $\phi(2p_x + iy)$; the last term vanishes because

$$\int_0^{2\pi} d\phi \cos(\phi - \omega t) = 0.$$

This is easily shown by suitable variable transformation.

In relation to the above discussion, we often use real numbers to describe wave functions. For this purpose, we use the following unitary transformation to transform the orthonormal basis of $e^{\pm im\phi}$ to $\cos m\phi$ and $\sin m\phi$. That is, we have

$$\begin{aligned} \begin{pmatrix} \frac{1}{\sqrt{\pi}} \cos m\phi & \frac{1}{\sqrt{\pi}} \sin m\phi \end{pmatrix} &= \begin{pmatrix} \frac{(-1)^m}{\sqrt{2\pi}} e^{im\phi} & \frac{1}{\sqrt{2\pi}} e^{-im\phi} \end{pmatrix} \\ &\times \begin{pmatrix} \frac{(-1)^m}{\sqrt{2}} & -\frac{(-1)^m i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{pmatrix}, \end{aligned} \quad (4.59)$$

where we assume that m is positive so that we can appropriately take into account the Condon–Shortley phase. Alternatively, we describe it via unitary transformation as follows:

$$\begin{aligned} \left(\frac{(-1)^m}{\sqrt{2\pi}} e^{im\phi} \quad \frac{1}{\sqrt{2\pi}} e^{-im\phi} \right) &= \left(\frac{1}{\sqrt{\pi}} \cos m\phi \quad \frac{1}{\sqrt{\pi}} \sin m\phi \right) \\ &\times \begin{pmatrix} \frac{(-1)^m}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{(-1)^m i}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \end{pmatrix}. \end{aligned} \quad (4.60)$$

In this regard, we have to be careful about normalization constants; for trigonometric functions the constant should be $\frac{1}{\sqrt{\pi}}$, whereas for the exponential representation, the constant is $\frac{1}{\sqrt{2\pi}}$. At the same time, trigonometric functions are expressed as a linear combination of $e^{im\phi}$ and $e^{-im\phi}$, and so if we use the trigonometric functions, information of a magnetic quantum number is lost.

In Sect. 3.7, we showed normalized functions $\tilde{\Lambda}_{l,m}^{(n)} = Y_l^m(\theta, \phi) \tilde{R}_l^{(n)}(r)$ of the hydrogen-like atom. Noting that $Y_l^m(\theta, \phi)$ is proportional to $e^{\pm im\phi}$, $\tilde{\Lambda}_{l,m}^{(n)}$ can be described using $\cos m\phi$ and $\sin m\phi$ for the basis vectors. We denote two linearly independent vectors by $\tilde{\Lambda}_{l, \cos m\phi}^{(n)}$ and $\tilde{\Lambda}_{l, \sin m\phi}^{(n)}$. Then, these vectors are expressed as

$$\begin{pmatrix} \tilde{\Lambda}_{l, \cos m\phi}^{(n)} & \tilde{\Lambda}_{l, \sin m\phi}^{(n)} \end{pmatrix} = \begin{pmatrix} \tilde{\Lambda}_{l,m}^{(n)} & \tilde{\Lambda}_{l,-m}^{(n)} \end{pmatrix} \begin{pmatrix} \frac{(-1)^m}{\sqrt{2}} & -\frac{(-1)^m i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{pmatrix}, \quad (4.61)$$

where we again assume that m is positive. In chemistry and materials science, we normally use real functions of $\tilde{\Lambda}_{l, \cos m\phi}^{(n)}$ and $\tilde{\Lambda}_{l, \sin m\phi}^{(n)}$. In particular, we use the notations of, e.g., $\phi(2p_x)$ and $\phi(2p_y)$ instead of $\tilde{\Lambda}_{1, \cos \phi}^{(2)}$ and $\tilde{\Lambda}_{1, \sin \phi}^{(2)}$, respectively. In that case, we explicitly have a following form:

$$\begin{aligned} \begin{pmatrix} \phi(2p_x) & \phi(2p_y) \end{pmatrix} &= \begin{pmatrix} \tilde{\Lambda}_{1,1}^{(2)} & \tilde{\Lambda}_{1,-1}^{(2)} \end{pmatrix} \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{pmatrix} \\ &= \begin{pmatrix} \phi(2p_{x+iy}) & \phi(2p_{x-iy}) \end{pmatrix} \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{4\sqrt{2\pi}} a^{-\frac{3}{2}} \frac{r}{a} e^{-\frac{r}{2a}} \sin \theta \cos \phi & \frac{1}{4\sqrt{2\pi}} a^{-\frac{3}{2}} \frac{r}{a} e^{-\frac{r}{2a}} \sin \theta \sin \phi \end{pmatrix}. \end{aligned} \quad (4.62)$$

Thus, the Condon–Shortley phase factor has been removed.

Using this expression, we calculate matrix elements of the electric dipole transition. We have

$$\begin{aligned} P_{x,|p_x\rangle}^{(e_1)} &= e\langle\phi(1s)|x|\phi(2p_x)\rangle \\ &= \frac{e}{4\sqrt{2}\pi a^4} \int_0^\infty r^4 e^{-3r/2a} dr \int_0^\pi \sin^3\theta d\theta \int_0^{2\pi} \cos^2\phi d\phi = \frac{2^7\sqrt{2}}{3^5} ea. \end{aligned} \quad (4.63)$$

Thus, we obtained the same result as (4.49) apart from the minus sign. Since a square of an absolute value of the transition moment plays a role, the minus sign is again of secondary importance. With $P_y^{(e_2)}$, similarly we have

$$\begin{aligned} P_{y,|p_y\rangle}^{(e_2)} &= e\langle\phi(1s)|y|\phi(2p_y)\rangle \\ &= \frac{e}{4\sqrt{2}\pi a^4} \int_0^\infty r^4 e^{-3r/2a} dr \int_0^\pi \sin^3\theta d\theta \int_0^{2\pi} \sin^2\phi d\phi = \frac{2^7\sqrt{2}}{3^5} ea. \end{aligned} \quad (4.64)$$

Comparing (4.39), (4.63), and (4.64), we have

$$P_{z,|p_z\rangle}^{(e_3)} = P_{x,|p_x\rangle}^{(e_1)} = P_{y,|p_y\rangle}^{(e_2)} = \frac{2^7\sqrt{2}}{3^5} ea.$$

In the case of $P_{z,|p_z\rangle}^{(e_3)}$, $P_{x,|p_x\rangle}^{(e_1)}$, and $P_{y,|p_y\rangle}^{(e_2)}$, the optical transition is said to be polarized along the z -, x -, and y -axis, respectively, and so linearly polarized lights are relevant. Note moreover that operators z , x , and y in (4.39), (4.63), and (4.64) are Hermitian and that $\phi(2p_z)$, $\phi(2p_x)$, and $\phi(2p_y)$ are real functions.

4.4 Selection Rules

In a three-dimensional system such as hydrogen-like atoms, quantum states of particles (i.e., electrons) are characterized by three quantum numbers: principal quantum numbers, orbital angular momentum quantum numbers (or azimuthal quantum numbers), and magnetic quantum numbers. In this section, we examine the *selection rules* for the electric dipole approximation.

Of the three quantum numbers mentioned above, angular momentum quantum numbers are denoted by l and magnetic quantum numbers by m . First, we examine the conditions on m . With the angular momentum operator $\mathbf{L}$ and its corresponding operator $\mathbf{M}$, we get the following commutation relations:

$$\begin{aligned}
[M_z, x] &= iy, & [M_y, z] &= ix, & [M_x, y] &= iz; \\
[M_z, iy] &= x, & [M_y, ix] &= z, & [M_x, iz] &= y; \quad \text{etc.}
\end{aligned}
\tag{4.65}$$

Notice that in the upper line the indices change cyclic like (z, x, y) , whereas in the lower line they change anti-cyclic such as (z, y, x) . The proof of (4.65) is left for the reader. Thus, we have, e.g.,

$$[M_z, x + iy] = x + iy, \quad [M_z, x - iy] = -(x - iy), \text{ etc.} \tag{4.66}$$

Putting

$$Q^+ \equiv x + iy \quad \text{and} \quad Q^- \equiv x - iy,$$

we have

$$[M_z, Q^+] = Q^+, \quad [M_z, Q^-] = -Q^-. \tag{4.67}$$

Taking an inner product of both sides of (4.67), we have

$$\begin{aligned}
\langle m' | [M_z, Q^+] | m \rangle &= \langle m' | M_z Q^+ - Q^+ M_z | m \rangle = m' \langle m' | Q^+ | m \rangle - m \langle m' | Q^+ | m \rangle \\
&= \langle m' | Q^+ | m \rangle,
\end{aligned}
\tag{4.68}$$

where the quantum state $|m\rangle$ is identical to $|l, m\rangle$ in (3.151). Here we need no information about l , and so it is omitted. Thus, we have, e.g., $M_z |m\rangle = m |m\rangle$. Taking its adjoint, we have $\langle m | M_z^\dagger = \langle m | M_z = m \langle m |$, where M_z is Hermitian. These results lead to (4.68). From (4.68), we get

$$(m' - m - 1) \langle m' | Q^+ | m \rangle = 0. \tag{4.69}$$

Therefore, for the matrix element $\langle m' | Q^+ | m \rangle$ not to vanish, we must have

$$m' - m - 1 = 0 \quad \text{or} \quad \Delta m = 1 \quad (\Delta m \equiv m' - m).$$

This represents the selection rule with respect to the coordinate Q^+ . Similarly, we get

$$(m' - m + 1) \langle m' | Q^- | m \rangle = 0. \tag{4.70}$$

In this case, for the matrix element $\langle m' | Q^- | m \rangle$ not to vanish, we have

$$m' - m + 1 = 0 \quad \text{or} \quad \Delta m = -1.$$

To derive (4.70), we can alternatively use the following: Taking the adjoint of (4.69), we have

$$(m' - m - 1) \langle m | Q^- | m' \rangle = 0.$$

Exchanging m' and m , we have

$$(m - m' - 1) \langle m' | Q^- | m \rangle = 0 \quad \text{or} \quad (m' - m + 1) \langle m' | Q^- | m \rangle = 0.$$

Thus, (4.70) is recovered.

Meanwhile, we have a commutation relation

$$[M_z, z] = 0. \quad (4.71)$$

Similarly, taking an inner product of both sides of (4.71), we have

$$(m' - m) \langle m' | z | m \rangle = 0.$$

Therefore, for the matrix element $\langle m' | z | m \rangle$ not to vanish, we must have

$$m' - m = 0 \quad \text{or} \quad \Delta m = 0. \quad (4.72)$$

These results are fully consistent with Example 4.3 of Sect. 4.3. That is, if circularly polarized light takes part in the optical transition, $\Delta m = \pm 1$. For instance, using the present notation, we rewrite (4.48) as

$$\left\langle \phi(1s) \left| \frac{1}{\sqrt{2}} (x - iy) \right| \phi(2p_{x+iy}) \right\rangle = \frac{1}{\sqrt{2}} \langle 0 | Q^- | 1 \rangle = -\frac{2^7 \sqrt{2}}{3^5} a.$$

If linearly polarized light is related to the optical transition, we have $\Delta m = 0$.

Next, we examine the conditions on l . To this end, we calculate a following commutator [5]:

$$\begin{aligned}
[\mathbf{M}^2, z] &= [M_x^2 + M_y^2 + M_z^2, z] = [M_x^2, z] + [M_y^2, z] \\
&= M_x(M_x z - z M_x) + M_x z M_x - z M_x^2 \\
&\quad + M_y(M_y z - z M_y) + M_y z M_y - z M_y^2 \\
&= M_x[M_x, z] + [M_x, z]M_x + M_y[M_y, z] + [M_y, z]M_y \\
&= i(M_y x + x M_y - M_x y - y M_x) \\
&= i(M_x y - y M_x - M_y x + x M_y + 2M_y x - 2M_x y) \\
&= i(2iz + 2M_y x - 2M_x y) = 2i(M_y x - M_x y + iz).
\end{aligned} \tag{4.73}$$

In the above calculations, (1) we used $[M_z, z] = 0$ (with the second equality); (2) RHS was modified so that the commutation relations can be used (the third equality); (3) we used $-M_{xy} = M_{yx} - 2M_{xy}$ and $M_{yx} = -M_{yx} + 2M_{yx}$ so that we can use (4.65) (the second to the last equality). Moreover, using (4.65), (4.73) can be written as

$$[\mathbf{M}^2, z] = 2i(xM_y - M_x y) = 2i(M_y x - yM_x).$$

Similar results on the commutator can be obtained with $[\mathbf{M}^2, x]$ and $[\mathbf{M}^2, y]$. For further use, we give alternative relations such that

$$\begin{aligned}
[\mathbf{M}^2, x] &= 2i(yM_z - M_y z) = 2i(M_z y - zM_y), \\
[\mathbf{M}^2, y] &= 2i(zM_x - M_z x) = 2i(M_x z - xM_z).
\end{aligned} \tag{4.74}$$

Using (4.73), we calculate another commutator such that

$$\begin{aligned}
[\mathbf{M}^2, [\mathbf{M}^2, z]] &= 2i\{[\mathbf{M}^2, M_y x] - [\mathbf{M}^2, M_x y] + i[\mathbf{M}^2, z]\} \\
&= 2i\{M_y[\mathbf{M}^2, x] - M_x[\mathbf{M}^2, y] + i[\mathbf{M}^2, z]\} \\
&= 2i\{2iM_y(yM_z - M_y z) - 2iM_x(M_x z - xM_z) + i[\mathbf{M}^2, z]\} \\
&= -2\left\{2(M_{xx} + M_{yy} + M_{zz})M_z - 2(M_x^2 + M_y^2 + M_z^2)z + \mathbf{M}^2 z - z\mathbf{M}^2\right\} \\
&= 2(\mathbf{M}^2 z + z\mathbf{M}^2).
\end{aligned} \tag{4.75}$$

In the above calculations, (1) we used $[\mathbf{M}^2, M_y] = [\mathbf{M}^2, M_x] = 0$ (with the second equality); (2) we used (4.74) (the third equality); (3) RHS was modified so that we can use the relation $\mathbf{M} \perp \mathbf{x}$ from the definition of the angular momentum operator, i.e., $M_{xx} + M_{yy} + M_{zz} = 0$ (the second to the last equality). We used $[M_z, z] = 0$ as well. Similar results are obtained with x and y . That is, we have

$$[\mathbf{M}^2, [\mathbf{M}^2, x]] = 2(\mathbf{M}^2 x + x \mathbf{M}^2), \quad (4.76)$$

$$[\mathbf{M}^2, [\mathbf{M}^2, y]] = 2(\mathbf{M}^2 y + y \mathbf{M}^2). \quad (4.77)$$

Rewriting, e.g., (4.75), we have

$$\mathbf{M}^4 z - 2\mathbf{M}^2 z \mathbf{M}^2 + z \mathbf{M}^4 = 2(\mathbf{M}^2 z + z \mathbf{M}^2). \quad (4.78)$$

Using the relation (4.78) and taking inner products of both sides, we get, e.g.,

$$\langle l' | \mathbf{M}^4 z - 2\mathbf{M}^2 z \mathbf{M}^2 + z \mathbf{M}^4 | l \rangle = \langle l' | 2(\mathbf{M}^2 z + z \mathbf{M}^2) | l \rangle. \quad (4.79)$$

That is,

$$\langle l' | \mathbf{M}^4 z - 2\mathbf{M}^2 z \mathbf{M}^2 + z \mathbf{M}^4 | l \rangle - \langle l' | 2(\mathbf{M}^2 z + z \mathbf{M}^2) | l \rangle = 0.$$

Considering that both terms of LHS contain a factor $\langle l' | z | l \rangle$ in common, we have

$$\begin{aligned} & \left[l'^2(l' + 1)^2 - 2l'l(l' + 1)(l + 1) + l^2(l + 1)^2 - 2l'(l' + 1) - 2l(l + 1) \right] \\ & \times \langle l' | z | l \rangle = 0, \end{aligned} \quad (4.80)$$

where the quantum state $|l\rangle$ is identical to $|l, m\rangle$ in (3.151) with m omitted.

To factorize the first factor of LHS of (4.80), we view it as a quartic equation with respect to l' . Replacing l' with $-l$, we find that the first factor vanishes, and so the first factor should have a factor $(l' + l)$. Then, we factorize the first factor of LHS of (4.80) such that

[the first factor of LHS of (4.80)]

$$\begin{aligned} & = (l' + l)^2(l' - l)^2 + 2(l' + l)(l'^2 - l'l + l^2) - 2l'l(l' + l) - 2(l' + l) - (l' + l)^2 \\ & = (l' + l) \left[(l' + l)(l' - l)^2 + 2(l'^2 - l'l + l^2) - 2l'l - 2 - (l' + l) \right] \\ & = (l' + l) \left[(l' + l)(l' - l)^2 + 2(l'^2 - 2l'l + l^2) - (l' + l + 2) \right] \\ & = (l' + l) \left[(l' + l)(l' - l)^2 + 2(l' - l)^2 - (l' + l + 2) \right] \\ & = (l' + l) \left\{ (l' - l)^2[(l' + l) + 2] - (l' + l + 2) \right\} \\ & = (l' + l)(l' + l + 2)(l' - l + 1)(l' - l - 1). \end{aligned} \quad (4.81)$$

Thus rewriting (4.80), we get

$$(l' + l)(l' + l + 2)(l' - l + 1)(l' - l - 1)\langle l' | z | l \rangle = 0. \quad (4.82)$$

We have similar relations with respect to $\langle l' | x | l \rangle$ and $\langle l' | y | l \rangle$ because of (4.76) and (4.77). For the electric dipole transition to be allowed, among $\langle l' | x | l \rangle$, $\langle l' | y | l \rangle$, and $\langle l' | z | l \rangle$, at least one term must be non-vanishing. For this, at least one of the four factors of (4.81) should be zero. Since $l' + l + 2 > 0$, this factor is excluded.

For $l' + l$ to vanish, we should have $l' = l = 0$; notice that both l' and l are nonnegative integers. We must then examine this condition. This condition is equivalent to that the spherical harmonics related to the angular variables θ and ϕ take the form of $Y_0^0(\theta, \phi) = 1/\sqrt{4\pi}$, i.e., a constant. Therefore, the θ -related integral for the matrix element $\langle l' | z | l \rangle$ only consists of a following factor:

$$\int_0^\pi \cos \theta \sin \theta d\theta = \frac{1}{2} \int_0^\pi \sin 2\theta d\theta = -\frac{1}{4} [\cos 2\theta]_0^\pi = 0,$$

where $\cos \theta$ comes from a polar coordinate $z = r \cos \theta$ and $\sin \theta$ is due to an infinitesimal volume of space, i.e., $r^2 \sin \theta dr d\theta d\phi$. Thus, we find that $\langle l' | z | l \rangle$ vanishes on condition that $l' = l = 0$. As a polar coordinate representation, $x = r \sin \theta \cos \phi$ and $y = r \sin \theta \sin \phi$, and so the ϕ -related integrals $\langle l' | x | l \rangle$ and $\langle l' | y | l \rangle$ vanish as well. That is,

$$\int_0^{2\pi} \cos \phi d\phi = \int_0^{2\pi} \sin \phi d\phi = 0.$$

Therefore, the matrix elements relevant to $l' = l = 0$ vanish with all the coordinates, i.e., we have

$$\langle l' | x | l \rangle = \langle l' | y | l \rangle = \langle l' | z | l \rangle = 0. \quad (4.83)$$

Consequently, we exclude $(l' + l)$ -factor as well, when we consider a condition of the allowed transition. Thus, regarding the condition that should be satisfied with the allowed transition, from (4.82) we get

$$l' - l + 1 = 0 \quad \text{or} \quad l' - l - 1 = 0. \quad (4.84)$$

Or defining $\Delta l \equiv l' - l$, we get

$$\Delta l = \pm 1. \quad (4.85)$$

Thus, for the transition to be allowed, the azimuthal quantum number must change by one.

4.5 Angular Momentum of Radiation [6]

In Sect. 4.3 we mentioned circularly polarized light. If the circularly polarized light acts on an electron, what can we anticipate? Here we deal with this problem within a framework of a semiclassical theory.

Let $\mathbf{E}$ and $\mathbf{H}$ be an electric and magnetic field of a left-circularly polarized light, respectively. They are expressed as

$$\mathbf{E} = \frac{1}{\sqrt{2}} E_0 (\mathbf{e}_1 + i\mathbf{e}_2) \exp i(kz - \omega t), \quad (4.86)$$

$$\mathbf{H} = \frac{1}{\sqrt{2}} H_0 (\mathbf{e}_2 - i\mathbf{e}_1) \exp i(kz - \omega t) = \frac{1}{\sqrt{2}} \frac{E_0}{\mu\nu} (\mathbf{e}_2 - i\mathbf{e}_1) \exp i(kz - \omega t). \quad (4.87)$$

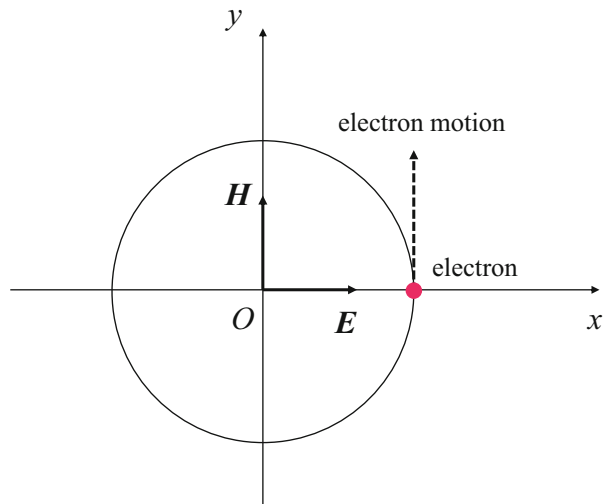
Here we assume that the light is propagating in the direction of the positive z -axis. The electric and magnetic fields described by (4.86) and (4.87) represent the left-circularly polarized light. A synchronized motion of an electron is expected, if the electron exerts a circular motion in such a way that the motion direction of the electron is always perpendicular to the electric field and parallel to the magnetic field (see Fig. 4.2). In this situation, magnetic Lorentz force does not affect the electron motion.

Here, the Lorentz force $\mathbf{F}(t)$ is described by

$$\mathbf{F}(t) = e\mathbf{E}(\mathbf{x}(t)) + e\dot{\mathbf{x}}(t) \times \mathbf{B}(\mathbf{x}(t)), \quad (4.88)$$

where the first term is electric Lorentz force and the second term represents the magnetic Lorentz force.

Fig. 4.2 Synchronized motion of an electron under a left-circularly polarized light



The quantity $\mathbf{B}$ called magnetic flux density is related to $\mathbf{H}$ as $\mathbf{B} = \mu_0 \mathbf{H}$, where μ_0 is permeability of vacuum; see (7.10) and (7.11) of Sect. 7.1. In (4.88) $\mathbf{E}$ and $\mathbf{B}$ are measured at a position where the electron is situated at a certain time t . Equation (4.88) universally describes the motion of a charged particle in the presence of electromagnetic fields. We consider another related example in Chap. 15.

Equation (4.86) can be rewritten as

$$\begin{aligned} \mathbf{E} &= \frac{1}{\sqrt{2}} E_0 [\mathbf{e}_1 \cos(kz - \omega t) - \mathbf{e}_2 \sin(kz - \omega t)] \\ &+ i \frac{1}{\sqrt{2}} E_0 [\mathbf{e}_2 \cos(kz - \omega t) + \mathbf{e}_1 \sin(kz - \omega t)]. \end{aligned} \quad (4.89)$$

Suppose that the electron exerts the circular motion in a region narrow enough around the origin and that the said electron motion is confined within the xy -plane that is perpendicular to the light propagation direction. Then, we can assume that $z \approx 0$ in (4.89). Ignoring kz in (4.89) accordingly and taking a real part, we have

$$\mathbf{E} = \frac{1}{\sqrt{2}} E_0 (\mathbf{e}_1 \cos \omega t + \mathbf{e}_2 \sin \omega t).$$

Thus, a force $\mathbf{F}$ exerting the electron is described by

$$\mathbf{F} = e\mathbf{E}, \quad (4.90)$$

where e is an elementary charge ($e < 0$). Accordingly, an equation of motion of the electron is approximated such that

$$m\ddot{\mathbf{x}} = e\mathbf{E}, \quad (4.91)$$

where m is a mass of an electron and $\mathbf{x}$ is a position vector of the electron. With individual components of the coordinate, we have

$$m\ddot{x} = \frac{1}{\sqrt{2}} eE_0 \cos \omega t \quad \text{and} \quad m\ddot{y} = \frac{1}{\sqrt{2}} eE_0 \sin \omega t. \quad (4.92)$$

Integrating (4.92) two times, we get

$$mx = -\frac{eE_0}{\sqrt{2}\omega^2} \cos \omega t + Ct + D, \quad (4.93)$$

where C and D are integration constants. Setting $x(0) = -\frac{eE_0}{\sqrt{2}m\omega^2}$ and $x'(0) = 0$, we have $C = D = 0$. Similarly, we have

$$my = -\frac{eE_0}{\sqrt{2}\omega^2} \sin \omega t + C't + D', \quad (4.94)$$

where C' and D' are integration constants. Setting $y(0) = 0$ and $y'(0) = -\frac{eE_0}{\sqrt{2}m\omega}$, we have $C' = D' = 0$. Thus, making t a parameter, we get

$$x^2 + y^2 = \left(\frac{eE_0}{\sqrt{2}m\omega^2} \right)^2. \quad (4.95)$$

This implies that the electron is exerting a counterclockwise circular motion with a radius $-\frac{eE_0}{\sqrt{2}m\omega^2}$ under the influence of the electric field. This is consistent with a motion of an electron in the coherent state of $\phi(1s)$ and $\phi(2p_x + iy)$ as expressed in (4.57).

An angular momentum the electron has acquired is

$$\mathbf{L} = \mathbf{x} \times \mathbf{p} = xp_y - yp_x = \left(-\frac{eE_0}{\sqrt{2}m\omega^2} \right) \left(-\frac{meE_0}{\sqrt{2}m\omega} \right) = \frac{e^2 E_0^2}{2m\omega^3}. \quad (4.96)$$

Identifying this with $\hbar$, we have

$$\frac{e^2 E_0^2}{2m\omega^3} = \hbar. \quad (4.97)$$

In terms of energy, we have

$$\frac{e^2 E_0^2}{2m\omega^2} = \hbar\omega. \quad (4.98)$$

Assuming a wavelength of the light is 600 nm, we need a left-circularly polarized light whose electric field is about 1.5×10^{10} [V/m].

A radius α of a circular motion of the electron is given by

$$\alpha = \frac{eE_0}{\sqrt{2}m\omega^2}. \quad (4.99)$$

Under the same condition as the above, α is estimated to be $\sim 2 \text{ \AA}$.

References

1. Jackson JD (1999) Classical electrodynamics, 3rd edn. Wiley, New York, NY
2. Fowles GR (1989) Introduction to modern optics, 2nd edn. Dover, New York, NY

3. Arfken GB, Weber HJ, Harris FE (2013) Mathematical methods for physicists, 7th edn. Academic Press, Waltham, MA
4. Chen JQ, Ping J, Wang F (2002) Group representation theory for physicists, 2nd edn. World Scientific, Singapore
5. Sunakawa S (1991) Quantum mechanics. Iwanami, Tokyo. (in Japanese)
6. Rossi B (1957) Optics. Addison-Wesley, Reading, MA

Chapter 5

Approximation Methods of Quantum Mechanics



In Chaps. 1–3 we have focused on solving eigenvalue equations with respect to a particle confined within a one-dimensional potential well or a harmonic oscillator along with an electron of a hydrogen-like atom. In each example we obtained exact analytical solutions with the quantum-mechanical states and corresponding eigenvalues (energy, angular momentum, etc.). In most cases of quantum-mechanical problems, however, we are not able to get such analytical solutions or accurately determine the corresponding eigenvalues. Under these circumstances, we need appropriate approximation methods of those problems. Among those methods, the perturbation method and variational method are widely used.

In terms of usefulness, we provide several examples concerning physical systems that have already appeared in Chaps. 1–3. In this chapter, we examine how these physical systems change their quantum states and corresponding energy eigenvalues as a result of undergoing influence from the external field. We assume that the change results from the application of external electric field. For simplicity, we focus on the change in eigenenergy and corresponding eigenstate with respect to the non-degenerate quantum state. As specific cases in these examples, we happen to be able to get perturbed physical quantities accurately. Including such cases, for later purposes we take in advance important concepts of a complete orthonormal system (CONS) and projection operator.

5.1 Perturbation Method

In Chaps. 1–3, we considered a situation where no external field is exerted on a physical system. In Chap. 4, however, we studied the optical transition that takes place as a consequence of the interaction between the physical system and electromagnetic wave. In this chapter, we wish to examine how the physical system changes its quantum state, when an external electric field is applied to the system. We usually assume that the external field is weak and the corresponding change in

the quantum state or energy is small. The applied external field causes the change in Hamiltonian of a quantum system and results in the change in that quantum state usually accompanied by the energy change of the system. We describe this process as a following eigenvalue equation:

$$H | \Psi_i \rangle = E_i | \Psi_i \rangle, \quad (5.1)$$

where H represents the total Hamiltonian, E_i is an energy eigenvalue, and Ψ_i is an eigenstate corresponding to E_i . Equation (5.1) implies that there should be a series of eigenvalues and corresponding eigenvectors belonging to those eigenvalues. Strictly speaking, however, we can get such combinations of the eigenstate and eigenvalue only by solving (5.1) analytically. This is some kind of circular argument. Yet, at present we do not have to go into detail about that issue. Instead we advance to discussion of the approximation methods from a practical point of view.

We assume that H is divided into two terms such that

$$H = H_0 + \lambda V, \quad (5.2)$$

where H_0 is the Hamiltonian of the unperturbed system without the external field; V is an additional Hamiltonian due to the applied external field, or interaction between the physical system and the external field. The second term includes a parameter λ that can be modulated according to the change in the strength of the applied field. The parameter λ should be real so that H of (5.2) can be Hermitian. The parameter λ can be the applied field itself or can merely be a dimensionless parameter that can be set at $\lambda = 1$ after finishing the calculation. We will come back to this point later in examples.

In (5.2) we assume that the following equation holds:

$$H_0 | i \rangle = E_i^{(0)} | i \rangle, \quad (5.3)$$

where $|i\rangle$ is the i -th quantum state. We designate $|0\rangle$ as the ground state. We assume the excited states $|i\rangle$ ($i = 1, 2, \dots$) to be numbered in order of increasing energies. These functions (or vectors) $|i\rangle$ (including $|0\rangle$) can be a quantum state that appeared as various eigenfunctions in the previous chapters. Of these, two or more functions may have the same eigenenergy (i.e., degenerate). If we are to deal with such degenerate states, those states $|i\rangle$ have to be distinguished by, e.g., $|i_d\rangle$ ($d = 1, 2, \dots, s$), where s denotes the degeneracy (or degree of degeneracy). For simplicity, however, in this chapter we are going to examine only the change in energy and physical states with respect to non-degenerate states. We disregard the issue on notation of the degenerate states accordingly. We pay attention to each individual case later, when necessary.

Regarding a one-dimensional physical system, e.g., a particle confined within potential well and quantum-mechanical harmonic oscillator, for example, all the quantum states are non-degenerate as we have already seen in Chaps. 1 and 2. As a

special case we also consider the ground state of a hydrogen-like atom. This is because that state is non-degenerate, and so the problem can be dealt with in parallel to the cases of the one-dimensional physical systems. We assume that the quantum states $|i\rangle$ ($i = 0, 1, 2, \dots$) constitute the orthonormal eigenvectors such that

$$\langle j|i\rangle = \delta_{ji}. \quad (5.4)$$

Remember that the notation has already appeared in (2.53); see Chap. 13 as well.

5.1.1 Quantum State and Energy Level Shift Caused by Perturbation

One of the most important applications of the perturbation method is to evaluate the shift in quantum state and energy level caused by the applied external field. To this end, we expand both the quantum state and energy as a power series of λ . That is, in (5.1) we expand $|\Psi_i\rangle$ and E_i such that

$$|\Psi_i\rangle = |i\rangle + \lambda |\phi_i^{(1)}\rangle + \lambda^2 |\phi_i^{(2)}\rangle + \lambda^3 |\phi_i^{(3)}\rangle + \dots, \quad (5.5)$$

$$E_i = E_i^{(0)} + \lambda E_i^{(1)} + \lambda^2 E_i^{(2)} + \lambda^3 E_i^{(3)} + \dots, \quad (5.6)$$

where $|\phi_i^{(1)}\rangle$, $|\phi_i^{(2)}\rangle$, $|\phi_i^{(3)}\rangle$, etc. are chosen as correction terms for the state $|i\rangle$ that is associated with the unperturbed system. Once again, we assume that the state $|i\rangle$ ($i = 0, 1, 2, \dots$) represents the non-degenerate normalized eigenvector that belongs to the eigenenergy $E_i^{(0)}$ of the unperturbed state. Unknown state vectors $|\phi_i^{(1)}\rangle$, $|\phi_i^{(2)}\rangle$, $|\phi_i^{(3)}\rangle$, etc. as well as $E_i^{(1)}$, $E_i^{(2)}$, $E_i^{(3)}$, etc. are to be determined after the calculation procedures carried out from now. These states $|\Psi_i\rangle$ and energies E_i result from the perturbation term λV and represent the deviation from $|i\rangle$ and $E_i^{(0)}$ of the unperturbed system.

With the normalization condition we impose the following condition upon $|\Psi_i\rangle$ such that

$$\langle i|\Psi_i\rangle = 1. \quad (5.7)$$

This condition, however, does not necessarily mean that $|\Psi_i\rangle$ has been normalized (vide infra). From (5.4) and (5.7) we have

$$\langle i|\phi_i^{(1)}\rangle = \langle i|\phi_i^{(2)}\rangle = \dots = 0. \quad (5.8)$$

Let us calculate $\langle \Psi_i|\Psi_i\rangle$ on condition of (5.7) and (5.8). From (5.5) we have

$$\begin{aligned}
\langle \Psi_i | \Psi_i \rangle &= \langle i | i \rangle + \lambda \left[\langle i | \phi_i^{(1)} \rangle + \langle \phi_i^{(1)} | i \rangle \right] \\
&+ \lambda^2 \left[\langle i | \phi_i^{(2)} \rangle + \langle \phi_i^{(2)} | i \rangle + \langle \phi_i^{(1)} | \phi_i^{(1)} \rangle \right] + \cdots \\
&= \langle i | i \rangle + \lambda^2 \langle \phi_i^{(1)} | \phi_i^{(1)} \rangle + \cdots \approx 1 + \lambda^2 \langle \phi_i^{(1)} | \phi_i^{(1)} \rangle,
\end{aligned}$$

where we used (5.4) and (5.8). Thus, $\langle \Psi_i | \Psi_i \rangle$ is normalized if we ignore the factor of λ^2 .

Now, inserting (5.5) and (5.6) into (5.1) and comparing the same power factors with respect to λ , we obtain

$$\left[E_i^{(0)} - H_0 \right] | i \rangle = 0, \quad (5.9)$$

$$\left[E_i^{(0)} - H_0 \right] | \phi_i^{(1)} \rangle + E_i^{(1)} | i \rangle = V | i \rangle, \quad (5.10)$$

$$\left[E_i^{(0)} - H_0 \right] | \phi_i^{(2)} \rangle + E_i^{(1)} | \phi_i^{(1)} \rangle + E_i^{(2)} | i \rangle = V | \phi_i^{(1)} \rangle. \quad (5.11)$$

Equation (5.9) is identical with (5.3). Operating $\langle j |$ on both sides of (5.10) from the left, we get

$$\begin{aligned}
\langle j | \left[E_i^{(0)} - H_0 \right] | \phi_i^{(1)} \rangle + \langle j | E_i^{(1)} | i \rangle &= \langle j | \left[E_i^{(0)} - E_j^{(0)} \right] | \phi_i^{(1)} \rangle + E_i^{(1)} \langle j | i \rangle \\
&= \left[E_i^{(0)} - E_j^{(0)} \right] \langle j | \phi_i^{(1)} \rangle + E_i^{(1)} \delta_{ji} = \langle j | V | i \rangle.
\end{aligned} \quad (5.12)$$

Putting $j = i$ on (5.12), we have

$$E_i^{(1)} = \langle i | V | i \rangle. \quad (5.13)$$

This represents the first-order correction term with the energy eigenvalue. Meanwhile, assuming $j \neq i$ on (5.12), we have

$$\langle j | \phi_i^{(1)} \rangle = \frac{\langle j | V | i \rangle}{\left[E_i^{(0)} - E_j^{(0)} \right]} \quad (j \neq i), \quad (5.14)$$

where we used $E_i^{(0)} \neq E_j^{(0)}$ on the assumption that the quantum state $|i\rangle$ that belongs to eigenenergy $E_i^{(0)}$ is non-degenerate. Here, we emphasize that the eigenstates that correspond to $E_j^{(0)}$ may or may not be degenerate. In other words, the quantum state that we are making an issue of with respect to the perturbation is non-degenerate. In this context, we do not question whether or not other quantum states [represented by $|j\rangle$ in (5.14)] are degenerate.

Here we postulate that the eigenvectors, i.e., $|i\rangle$ ($i = 0, 1, 2, \dots$) of H_0 form a complete orthonormal system (CONS) such that

$$P \equiv \sum_k |k\rangle\langle k| = E, \quad (5.15)$$

where $|k\rangle\langle k|$ is said to be a projection operator and E is an identity operator. As remarked just above, (5.15) holds regardless of whether the eigenstates are degenerate. The word of complete system implies that any vector (or function) can be expanded by a linear combination of the basis vectors of the said system. The formal definition of the projection operator can be seen in Chaps. 14 and 18.

Thus, we have

$$\begin{aligned} |\phi_i^{(1)}\rangle &= E |\phi_i^{(1)}\rangle = \sum_k |k\rangle\langle k|\phi_i^{(1)}\rangle = \sum_{k \neq i} |k\rangle\langle k|\phi_i^{(1)}\rangle \\ &= \sum_{k \neq i} |k\rangle \frac{\langle k|V|i\rangle}{[E_i^{(0)} - E_k^{(0)}]}, \end{aligned} \quad (5.16)$$

where with the third equality we used (5.8) and with the last equality we used (5.14). Operating $\langle i|$ from the left on (5.16) and using (5.4), we recover (5.8). Hence, using (5.5) the approximated quantum state to the first order of λ is described by

$$|\Psi_i\rangle \approx |i\rangle + \lambda |\phi_i^{(1)}\rangle = |i\rangle + \lambda \sum_{k \neq i} |k\rangle \frac{\langle k|V|i\rangle}{[E_i^{(0)} - E_k^{(0)}]}. \quad (5.17)$$

For a while, let us think of a case where we deal with the change in the eigenenergy and corresponding eigenstate with respect to the degenerate quantum state. In that case, regarding the state $|i\rangle$ in question on (5.12) we have ${}^3 |j\rangle$ ($j \neq i$) that satisfies $E_i^{(0)} = E_j^{(0)}$. Therefore, we would have $\langle j|V|i\rangle = 0$ from (5.12). Generally, it is not the case, however, and so we need a relation different from (5.12) to deal with the degenerate case. However, once again we do not get into details about this issue in this book.

Next let us seek the second-order correction term $E_i^{(2)}$ of the eigenenergy. Operating $\langle j|$ on both sides of (5.11) from the left, we get

$$[E_i^{(0)} - E_j^{(0)}] \langle j|\phi_i^{(2)}\rangle + E_i^{(1)} \langle j|\phi_i^{(1)}\rangle + E_i^{(2)} \delta_{ji} = \langle j|V|\phi_i^{(1)}\rangle.$$

Putting $j = i$ in the above relation as before, we have

$$E_i^{(2)} = \langle i|V|\phi_i^{(1)}\rangle. \quad (5.18)$$

Notice that we used (5.8) to derive (5.18). Using (5.16) furthermore, we get

$$\begin{aligned}
E_i^{(2)} &= \sum_{k \neq i} \langle i | V | k \rangle \frac{\langle k | V | i \rangle}{[E_i^{(0)} - E_k^{(0)}]} = \sum_{k \neq i} \langle k | V^\dagger | i \rangle^* \frac{\langle k | V | i \rangle}{[E_i^{(0)} - E_k^{(0)}]} \\
&= \sum_{k \neq i} \langle k | V | i \rangle^* \frac{\langle k | V | i \rangle}{[E_i^{(0)} - E_k^{(0)}]} = \sum_{k \neq i} \frac{1}{[E_i^{(0)} - E_k^{(0)}]} |\langle k | V | i \rangle|^2,
\end{aligned} \tag{5.19}$$

where with the second equality we used (1.116) in combination with (1.118) and with the third equality we used the fact that V is an Hermitian operator; i.e., $V^\dagger = V$. The state $|k\rangle$ is sometimes called an intermediate state.

Considering the expression of (5.16), we find that the approximated state of $|i\rangle + \lambda |\phi_i^{(1)}\rangle$ in (5.5) is not an eigenstate of energy, because the approximated state contains a linear combination of different eigenstates $|k\rangle$ that have eigenenergies different from $E_i^{(0)}$ of $|i\rangle$. Substituting (5.13) and (5.19) for (5.6), with the energy correction terms up to the second order we have

$$E_i \approx E_i^{(0)} + \lambda E_i^{(1)} + \lambda^2 E_i^{(2)} = E_i^{(0)} + \lambda \langle i | V | i \rangle + \lambda^2 \sum_{k \neq i} \frac{1}{[E_i^{(0)} - E_k^{(0)}]} |\langle k | V | i \rangle|^2. \tag{5.20}$$

If we think of the ground state $|0\rangle$ for $|i\rangle$, we get

$$\begin{aligned}
E_0 &\approx E_0^{(0)} + \lambda E_0^{(1)} + \lambda^2 E_0^{(2)} \\
&= E_0^{(0)} + \lambda \langle 0 | V | 0 \rangle + \lambda^2 \sum_{k \neq 0} \frac{1}{[E_0^{(0)} - E_k^{(0)}]} |\langle k | V | 0 \rangle|^2.
\end{aligned} \tag{5.21}$$

Since $E_0^{(0)} < E_k^{(0)}$ ($k = 1, 2, \dots$) for any $|k\rangle$, the second-order correction term is always negative. This contributes to the energy stabilization.

5.1.2 Several Examples

To deepen the understanding of the perturbation method, we show several examples. We focus on the change in the eigenenergy and corresponding eigenstate with respect to specific quantum states. We assume that the change is caused by the applied electric field as a perturbation.

Example 5.1 Suppose that the charged particle is confined within a one-dimensional potential well. This problem has already appeared in Example 1.2. Now let us consider how an energy of a particle carrying a charge e is shifted by the applied electric field. This situation could experimentally be achieved using a

“nano capacitor” where, e.g., an electron is confined within the capacitor while applying a voltage between the two electrodes.

We adopt the coordinate system the same as that of Example 1.2. That is, suppose that we apply an electric field F between the electrodes that are positioned at $x = \pm L$ and that the electron is confined between the two electrodes. Then, the Hamiltonian of the system is described as

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} - eFx, \quad (5.22)$$

where m is a mass of the particle. Note that we may choose $-eF$ for the parameter λ of Sect. 5.1.1. Then, the coordinate representation of the Schrödinger equation is given by

$$-\frac{\hbar^2}{2m} \frac{d^2\psi_i(x)}{dx^2} - eFx\psi_i(x) = E_i\psi_i(x), \quad (5.23)$$

where E_i is the energy of the i -th state from the bottom (i.e., the ground state); ψ_i is its corresponding eigenstate. We wish to seek the perturbation energy up to the second order and the quantum state up to the first order. We are particularly interested in the change in the ground state $|0\rangle$. Notice that the ground state is obtained in (1.101) by putting $l = 0$.

According to (5.21) we have

$$E_0 \approx E_0^{(0)} + \lambda\langle 0|V|0\rangle + \lambda^2 \sum_{k \neq 0} \frac{1}{[E_0^{(0)} - E_k^{(0)}]} |\langle k|V|0\rangle|^2, \quad (5.24)$$

where $V = -eFx$. Since the field F is thought to be an adjustable parameter, in (5.24) we may replace λ with $-eF$ (vide supra). Then, in (5.24) V is replaced by x in turn. In this way, (5.24) can be rewritten as

$$E_0 \approx E_0^{(0)} - eF\langle 0|x|0\rangle + (-eF)^2 \sum_{k \neq 0} \frac{1}{[E_0^{(0)} - E_k^{(0)}]} |\langle k|x|0\rangle|^2. \quad (5.25)$$

Considering that $|0\rangle$ is an even function [i.e., a cosine function in (1.101)] with respect to x and that x is an odd function, we find that the second term vanishes. Notice that the explicit coordinate representation of $|0\rangle$ is

$$|0\rangle = \sqrt{\frac{1}{L}} \cos \frac{\pi}{2L} x, \quad (5.26)$$

Fig. 5.1 Notation of the quantum states that distinguish cosine and sine functions.

$E_k^{(0)}$ ($k = 0, 1, 2, \dots$) represents energy eigenvalues of the unperturbed states

$$\begin{array}{c}
 \vdots \\
 E_4^{(0)} = \frac{25\pi^2\hbar^2}{8mL^2} \text{ ————— } |2_{\cos}\rangle \equiv |4\rangle \\
 \\
 E_3^{(0)} = \frac{16\pi^2\hbar^2}{8mL^2} \text{ ————— } |2_{\sin}\rangle \equiv |3\rangle \\
 \\
 E_2^{(0)} = \frac{9\pi^2\hbar^2}{8mL^2} \text{ ————— } |1_{\cos}\rangle \equiv |2\rangle \\
 \\
 E_1^{(0)} = \frac{4\pi^2\hbar^2}{8mL^2} \text{ ————— } |1_{\sin}\rangle \equiv |1\rangle \\
 E_0^{(0)} = \frac{\pi^2\hbar^2}{8mL^2} \text{ ————— } |0_{\cos}\rangle \equiv |0\rangle
 \end{array}$$

which is obtained by putting $l = 0$ in $|l_{\cos}\rangle \equiv \sqrt{\frac{1}{L}} \cos\left(\frac{\pi}{2L} + \frac{l\pi}{L}\right)x$ ($l = 0, 1, 2, \dots$) in (1.101).

By the same token, $\langle kl|x|0\rangle$ in the third term of RHS of (5.25) vanishes if $|k\rangle$ denotes a cosine function in (1.101). If, however, $|k\rangle$ denotes a sine function, $\langle kl|x|0\rangle$ does not vanish. To distinguish the sine functions from cosine functions, we denote

$$|n_{\sin}\rangle \equiv \sqrt{\frac{1}{L}} \sin \frac{n\pi}{L} x \quad (n = 1, 2, 3, \dots) \quad (5.27)$$

that is identical with (1.102); see Fig. 5.1 with the notation of the quantum states.

Now, putting $E_0^{(0)} = \frac{\hbar^2}{2m} \cdot \frac{\pi^2}{4L^2}$ and $E_{2n-1}^{(0)} = \frac{\hbar^2}{2m} \cdot \frac{(2n)^2\pi^2}{4L^2}$ ($n = 1, 2, 3, \dots$) in (5.25), we rewrite it as

$$\begin{aligned}
E_0 &\approx E_0^{(0)} - eF \langle 0|x|0 \rangle + (-eF)^2 \sum_{k=1}^{\infty} \frac{1}{[E_0^{(0)} - E_k^{(0)}]} |\langle k|x|0 \rangle|^2 \\
&= E_0^{(0)} - eF \langle 0|x|0 \rangle + (-eF)^2 \sum_{n=1}^{\infty} \frac{1}{[E_0^{(0)} - E_{2n-1}^{(0)}]} |\langle n_{\sin}|x|0 \rangle|^2,
\end{aligned}$$

where with the second equality n denotes the number appearing in (5.27) and $|n_{\sin}\rangle$ belongs to $E_{2n-1}^{(0)}$. Referring to, e.g., (4.17) with regard to the calculation of $\langle k|x|0 \rangle$, we have arrived at the following equation described by

$$E_0 \approx \frac{\hbar^2}{2m} \cdot \frac{\pi^2}{4L^2} - (-eF)^2 \frac{32^2 \cdot 8 \cdot mL^4}{\pi^6 \hbar^2} \sum_{n=1}^{\infty} \frac{n^2}{(4n^2 - 1)^5}. \quad (5.28)$$

The series of the second term in (5.28) rapidly converges. Defining the first N partial sum of the series as

$$S(N) \equiv \sum_{n=1}^N \frac{n^2}{(4n^2 - 1)^5},$$

we obtain

$$S(1) = 1/243 \approx 0.004115226 \quad \text{and} \quad S(6) \approx S(1000) \approx 0.004120685$$

as well as

$$\frac{[S(1000) - S(1)]}{S(1000)} = 0.001324653.$$

That is, only the first term of $S(N)$ occupies $\sim 99.9\%$ of the infinite series. Thus, the stabilization energy $\lambda^2 E_0^{(2)}$ due to the perturbation is satisfactorily given by

$$\lambda^2 E_0^{(2)} \approx (eF)^2 \frac{32^2 \cdot 8 \cdot mL^4}{243 \pi^6 \hbar^2}, \quad (5.29)$$

where the notation $\lambda^2 E_0^{(2)}$ is due to (5.6). Meanwhile, the corrected term of the quantum state is given by

$$\begin{aligned}
|\Psi_0\rangle &\approx |0\rangle + \lambda |\phi_0^{(1)}\rangle \approx |0\rangle - eF \sum_{k \neq 0} |k\rangle \frac{\langle k|x|0\rangle}{[E_0^{(0)} - E_k^{(0)}]} \\
&= |0\rangle + eF \frac{32 \cdot 8 \cdot mL^3}{\pi^4 \hbar^2} \sum_{n=1}^{\infty} |n_{\sin}\rangle \frac{(-1)^{n-1} n}{(4n^2 - 1)^3}. \quad (5.30)
\end{aligned}$$

For a reason similar to the above, the approximation is satisfactory, if we adopt only $n = 1$ in the second term of RHS of (5.30). That is, we obtain

$$|\Psi_0\rangle \approx |0\rangle + eF \frac{32 \cdot 8 \cdot mL^3}{27\pi^4 \hbar^2} |1_{\sin}\rangle. \quad (5.31)$$

Thus, we have roughly estimated the stabilization energy and the corresponding quantum state as in (5.29) and (5.31), respectively. It may well be worth estimating rough numbers of L and F in the actual situation (or perhaps in an actual “nano device”). The estimation is left for readers.

Example 5.2 [1] In Chap. 2 we dealt with a quantum-mechanical harmonic oscillator. Here let us suppose that a charged particle (with its charge e) is performing harmonic oscillation under an applied electric field. First we consider the coordinate representation of Schrödinger equation. Without the external field, the Schrödinger equation as an eigenvalue equation reads as

$$-\frac{\hbar^2}{2m} \frac{d^2 u(q)}{dq^2} + \frac{1}{2} m \omega^2 q^2 u(q) = E u(q). \quad (2.108)$$

Under the applied electric field, the perturbation is expressed as $-eFq$ and, hence, the equation can be written as

$$-\frac{\hbar^2}{2m} \frac{d^2 u(q)}{dq^2} + \frac{1}{2} m \omega^2 q^2 u(q) - eF(q)qu(q) = E u(q). \quad (5.32)$$

For simplicity we assume that the electric field is uniform independent of q so that we can deal with it as a constant. Then, (5.32) can be rewritten as

$$-\frac{\hbar^2}{2m} \frac{d^2 u(q)}{dq^2} + \frac{1}{2} m \omega^2 \left(q - \frac{eF}{m\omega^2} \right)^2 u(q) = \left[E + \frac{1}{2} m \omega^2 \left(\frac{eF}{m\omega^2} \right)^2 \right] u(q). \quad (5.33)$$

Changing the variable such that

$$Q \equiv q - \frac{eF}{m\omega^2}, \quad (5.34)$$

we have

$$-\frac{\hbar^2}{2m} \frac{d^2 u(q)}{dQ^2} + \frac{1}{2} m \omega^2 Q^2 u(q) = \left[E + \frac{1}{2} m \omega^2 \left(\frac{eF}{m\omega^2} \right)^2 \right] u(q). \quad (5.35)$$

Taking account of the change in functional form that results from the variable transformation, we rewrite (5.35) as

$$-\frac{\hbar^2}{2m} \frac{d^2 \tilde{u}(Q)}{dQ^2} + \frac{1}{2} m \omega^2 Q^2 \tilde{u}(Q) = \left[E + \frac{1}{2} m \omega^2 \left(\frac{eF}{m\omega^2} \right)^2 \right] \tilde{u}(Q), \quad (5.36)$$

where

$$u(q) = u\left(Q + \frac{eF}{m\omega^2}\right) \equiv \tilde{u}(Q). \quad (5.37)$$

Defining $\tilde{E}$ as

$$\tilde{E} \equiv E + \frac{1}{2} m \omega^2 \left(\frac{eF}{m\omega^2} \right)^2 = E + \frac{e^2 F^2}{2m\omega^2}, \quad (5.38)$$

we get

$$-\frac{\hbar^2}{2m} \frac{d^2 \tilde{u}(Q)}{dQ^2} + \frac{1}{2} m \omega^2 Q^2 \tilde{u}(Q) = \tilde{E} \tilde{u}(Q). \quad (5.39)$$

Thus, we recover exactly the same form as (2.108). Equation (5.39) can be solved analytically as already shown in Sect. 2.4. From (2.110) and (2.120), we have

$$\lambda \equiv \frac{2\tilde{E}}{\hbar\omega}, \quad \lambda = 2n + 1.$$

Then, we have

$$\tilde{E} = \frac{\hbar\omega}{2} \lambda = \frac{\hbar\omega(2n+1)}{2}. \quad (5.40)$$

From (5.38), we get

$$E = \tilde{E} - \frac{e^2 F^2}{2m\omega^2} = \hbar\omega \left(n + \frac{1}{2} \right) - \frac{e^2 F^2}{2m\omega^2}. \quad (5.41)$$

This implies that the energy stabilization due to the applied electric field is $-\frac{e^2 F^2}{2m\omega^2}$. Note that the system is always stabilized regardless of the sign of e .

Next, we wish to consider the problem on the basis of the matrix (or operator) representation. Let us evaluate the perturbation energy using (5.20). Conforming the notation of (5.20) to the present case, we have

$$E_n \approx E_n^{(0)} - eF \langle n|q|n \rangle + (-eF)^2 \sum_{k \neq n} \frac{1}{[E_n^{(0)} - E_k^{(0)}]} |\langle k|q|n \rangle|^2, \quad (5.42)$$

where $E_n^{(0)}$ is given by

$$E_n^{(0)} = \hbar\omega \left(n + \frac{1}{2} \right). \quad (5.43)$$

Taking account of (2.55), (2.68), and (2.62), the second term of (5.42) vanishes. With the third term, using (2.68) as well as (2.55) and (2.61) we have

$$\begin{aligned} |\langle k|q|n \rangle|^2 &= \sqrt{\frac{\hbar}{2m\omega}} \langle k|a + a^\dagger|n \rangle \sqrt{\frac{\hbar}{2m\omega}} \langle k|a + a^\dagger|n \rangle^* \\ &= \frac{\hbar}{2m\omega} [\sqrt{n} \langle k|n-1 \rangle + \sqrt{n+1} \langle k|n+1 \rangle]^2 \\ &= \frac{\hbar}{2m\omega} [\sqrt{n} \delta_{k,n-1} + \sqrt{n+1} \delta_{k,n+1}]^2 \\ &= \frac{\hbar}{2m\omega} [n \delta_{k,n-1} + (n+1) \delta_{k,n+1} + 2 \delta_{k,n-1} \delta_{k,n+1} \sqrt{n(n+1)}] \\ &= \frac{\hbar}{2m\omega} [n \delta_{k,n-1} + (n+1) \delta_{k,n+1}]. \end{aligned} \quad (5.44)$$

Notice that with the last equality of (5.44) there is no k that satisfies $k = n - 1 = n + 1$ at once. Also note that $\langle k|q|n \rangle$ is a real number. Hence, as the third term of (5.42) we get

$$\begin{aligned} \sum_{k \neq n} \frac{1}{[E_n^{(0)} - E_k^{(0)}]} |\langle k|q|n \rangle|^2 &= \sum_{k \neq n} \frac{1}{\hbar\omega(n-k)} \cdot \frac{\hbar}{2m\omega} [n \delta_{k,n-1} + (n+1) \delta_{k,n+1}] \\ &= \frac{1}{2m\omega^2} \left[\frac{n}{n-(n-1)} + \frac{n+1}{n-(n+1)} \right] = -\frac{1}{2m\omega^2}. \end{aligned} \quad (5.45)$$

Thus, from (5.42) we obtain

$$E_n \approx \hbar\omega \left(n + \frac{1}{2} \right) - \frac{e^2 F^2}{2m\omega^2}. \quad (5.46)$$

Consequently, we find that (5.46) obtained by the perturbation method is consistent with (5.41) that was obtained as the exact solution. As already pointed out in

Sect. 5.1.1, however, (5.46) does not represent an eigenenergy. In fact, using (5.17) together with (4.26) and (4.28), we have

$$\begin{aligned}
 |\Psi_0\rangle &\approx |0\rangle + \lambda |\phi_0^{(1)}\rangle = |0\rangle + \lambda \sum_{k \neq 0} |k\rangle \frac{\langle k|V|0\rangle}{[E_0^{(0)} - E_k^{(0)}]} \\
 &= |0\rangle + \lambda |1\rangle \frac{\langle 1|V|0\rangle}{[E_0^{(0)} - E_1^{(0)}]} \\
 &= |0\rangle - eF |1\rangle \frac{\langle 1|x|0\rangle}{[E_0^{(0)} - E_1^{(0)}]} = |0\rangle + \sqrt{\frac{e^2 F^2}{2m\omega^3 \hbar}} |1\rangle.
 \end{aligned} \tag{5.47}$$

Thus, $|\Psi_0\rangle$ is not an eigenstate because $|\Psi_0\rangle$ contains both $|0\rangle$ and $|1\rangle$ that have different eigenenergies. Hence, $|\Psi_0\rangle$ does not possess a fixed eigenenergy. Notice also that in (5.47) the factors $\langle k|x|0\rangle$ vanish except for $\langle 1|x|0\rangle$; see Chaps. 2 and 4.

To think of this point further, let us come back to the coordinate representation of Schrödinger equation. From (5.39) and (2.106), we have

$$\tilde{u}_n(Q) = \left(\frac{m\omega}{\hbar}\right)^{\frac{1}{4}} \sqrt{\frac{1}{\pi^{\frac{1}{2}} 2^n n!}} H_n\left(\sqrt{\frac{m\omega}{\hbar}} Q\right) e^{-\frac{m\omega}{2\hbar} Q^2} \quad (n=0, 1, 2, \dots),$$

where $H_n\left(\sqrt{\frac{m\omega}{\hbar}} Q\right)$ is the Hermite polynomial of the n -th order. Using (5.34) and (5.37), we replace Q with $q - \frac{eF}{m\omega^2}$ to obtain the following form described by

$$\begin{aligned}
 u_n(q) &= \tilde{u}_n\left(q - \frac{eF}{m\omega^2}\right) \\
 &= \left(\frac{m\omega}{\hbar}\right)^{\frac{1}{4}} \sqrt{\frac{1}{\pi^{\frac{1}{2}} 2^n n!}} H_n\left[\sqrt{\frac{m\omega}{\hbar}} \left(q - \frac{eF}{m\omega^2}\right)\right] e^{-\frac{m\omega}{2\hbar} \left(q - \frac{eF}{m\omega^2}\right)^2} \quad (n=0, 1, 2, \dots).
 \end{aligned} \tag{5.48}$$

Since $\psi_n(q)$ of (2.106) forms the (CONS) [2], $u_n(q)$ should be expanded using $\psi_n(q)$. That is, as a solution of (5.32) we get

$$u_n(q) = \sum_k c_{nk} \psi_k(q), \tag{5.49}$$

where a set of c_{nk} are appropriate coefficients. Since the functions $\psi_k(q)$ are non-degenerate, $u_n(q)$ expressed by (5.49) lacks a definite eigenenergy.

Example 5.3 [1] In the previous examples we studied the perturbation in one-dimensional systems, where all the energy eigenstates are non-degenerate. Here we deal with the change in the energy and quantum states of a hydrogen atom (i.e., a three-dimensional system). Since the energy eigenstates are generally degenerate (see Chap. 3), for simplicity we consider only the ground state that is

non-degenerate. As in the previous two cases, we assume that the perturbation is caused by the applied electric field. As the simplest example, we deal with the properties including the polarizability of the hydrogen atom in its ground state. The polarizability of atom, molecule, etc. is one of the most fundamental properties in materials science.

We assume that the electric field is applied along the direction of the z -axis; in this case the perturbation term V is expressed as

$$V = -eFz, \quad (5.50)$$

where e is a charge of an electron ($e < 0$). Then, the total Hamiltonian H is described by

$$\tilde{H} = H_0 + V, \quad (5.51)$$

$$H_0 = -\frac{\hbar^2}{2\mu r^2} \left[\frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] - \frac{e^2}{4\pi\epsilon_0 r}, \quad (5.52)$$

where H_0 is identical with the Hamiltonian H of (3.35), in which $Z = 1$.

In this example, we discuss topics on the polarization and energy shift (Stark effect) caused by the applied electric field [1, 3].

1. Polarizability of a hydrogen atom in the ground state

As in (5.5) and (5.16), the first-order perturbed state is described by

$$|\Psi_0\rangle \approx |0\rangle + \sum_{k \neq 0} |k\rangle \frac{\langle k|V|0\rangle}{[E_0^{(0)} - E_k^{(0)}]} = |0\rangle - eF \sum_{k \neq 0} |k\rangle \frac{\langle k|z|0\rangle}{[E_0^{(0)} - E_k^{(0)}]}, \quad (5.53)$$

where $|0\rangle$ denotes the ground state expressed as (3.301). Since the quantum states $|k\rangle$ represent all the quantum states of the hydrogen atom as implied in (5.3), in the second term of RHS in (5.53) we are considering all those states except for $|0\rangle$. The eigenenergy $E_0^{(0)}$ is identical with $E_1 = -\frac{\hbar^2}{2\mu a^2}$ obtained from (3.258), where $n = 1$. As already noted, we simply numbered the states $|k\rangle$ in order of increasing energies. In the case of $k \neq j$ ($k, j \neq 0$), we may have either $E_k^{(0)} = E_j^{(0)}$ or $E_k^{(0)} \neq E_j^{(0)}$ accordingly.

Using (5.51), let us calculate the polarizability of a hydrogen atom in a ground state. In Sect. 4.1 we gave a definition of the electric dipole moment such that

$$\mathbf{P} \equiv e \sum_j \mathbf{x}_j, \quad (4.6)$$

where $\mathbf{x}_j$ is a position vector of the j -th charged particle. Placing the proton of hydrogen at the origin, by use of the notation (3.5) $\mathbf{P}$ is expressed as

$$\mathbf{P} = (e_1 \ e_2 \ e_3) \begin{pmatrix} P_x \\ P_y \\ P_z \end{pmatrix} = e\mathbf{x} = (e_1 \ e_2 \ e_3) \begin{pmatrix} ex \\ ey \\ ez \end{pmatrix}.$$

Hence, the z -component of the electric dipole moment P_z of electron is given by

$$P_z = ez. \quad (5.54)$$

The quantum-mechanical analogue of (5.54) is given by

$$\langle P_z \rangle = e \langle \Psi_0 | z | \Psi_0 \rangle, \quad (5.55)$$

where $|\Psi_0\rangle$ is taken from (5.53). Substituting (5.53) for (5.55), we get

$$\begin{aligned} \langle \Psi_0 | z | \Psi_0 \rangle &= \left\langle \langle 0 | -eF \sum_{k \neq 0} \langle k | \frac{\langle k | z | 0 \rangle}{[E_0^{(0)} - E_k^{(0)}]} \right| z \left| \langle 0 | -eF \sum_{k \neq 0} | k \rangle \frac{\langle k | z | 0 \rangle}{[E_0^{(0)} - E_k^{(0)}]} \right\rangle \\ &= \langle 0 | z | 0 \rangle - eF \sum_{k \neq 0} \langle 0 | z | k \rangle \frac{\langle k | z | 0 \rangle}{[E_0^{(0)} - E_k^{(0)}]} \\ &\quad - eF \sum_{k \neq 0} \langle 0 | z^\dagger | k \rangle \frac{\langle k | z | 0 \rangle}{[E_0^{(0)} - E_k^{(0)}]} \\ &\quad + (eF)^2 \sum_{k \neq 0} \langle k | z | k \rangle \frac{|\langle k | z | 0 \rangle|^2}{[E_0^{(0)} - E_k^{(0)}]^2}. \end{aligned} \quad (5.56)$$

In (5.56) readers are referred to the computation rule of inner product described in Chap. 1 or in Chap. 13 for further details. Since z is Hermitian (i.e., $z^\dagger = z$), the second and third terms of RHS of (5.56) equal. Neglecting the second-order perturbation factor on the fourth term, we have

$$\langle \Psi_0 | z | \Psi_0 \rangle \approx \langle 0 | z | 0 \rangle - 2eF \sum_{k \neq 0} \frac{|\langle k | z | 0 \rangle|^2}{E_0^{(0)} - E_k^{(0)}}.$$

Thus, from (5.55) we obtain

$$\langle P_z \rangle \approx e \langle 0 | z | 0 \rangle - 2e^2 F \sum_{k \neq 0} \frac{|\langle k | z | 0 \rangle|^2}{E_0^{(0)} - E_k^{(0)}}. \quad (5.57)$$

Using (3.301), the coordinate representation of $\langle 0 | z | 0 \rangle$ is described by

$$\begin{aligned} \langle 0 | z | 0 \rangle &= \frac{a^{-3}}{\pi} \int_{-\infty}^{\infty} z e^{-2r/a} dx dy dz \\ &= \frac{a^{-3}}{\pi} \int_0^{\infty} r^3 e^{-2r/a} dr \int_0^{2\pi} d\phi \int_0^{\pi} \cos \theta \sin \theta d\theta \\ &= \frac{a^{-3}}{2\pi} \int_0^{\infty} r^3 e^{-2r/a} dr \int_0^{2\pi} d\phi \int_0^{\pi} \sin 2\theta d\theta = 0, \end{aligned} \quad (5.58)$$

where we used $\int_0^{\pi} \sin 2\theta d\theta = 0$. Then, we have

$$\langle P_z \rangle \approx -2e^2 F \sum_{k \neq 0} \frac{|\langle k | z | 0 \rangle|^2}{E_0^{(0)} - E_k^{(0)}}. \quad (5.59)$$

On the basis of classical electromagnetism, we define the polarizability α as [4]

$$\alpha \equiv \langle P_z \rangle / F. \quad (5.60)$$

Here we have an important relationship between the polarizability and electric dipole moment. That is,

$$(\text{polarizability}) \times (\text{electric field}) = (\text{electric dipole moment}).$$

From (5.59) and (5.60) we have

$$\alpha = -2e^2 \sum_{k \neq 0} \frac{|\langle k | z | 0 \rangle|^2}{E_0^{(0)} - E_k^{(0)}}. \quad (5.61)$$

At the first glance, to evaluate (5.61) appears to be formidable, but making the most of the fact that the total wave functions of a hydrogen atom form the CONS, the evaluation is straightforward. The discussion is as follows:

First, let us seek an operator G that satisfies [1]

$$z | 0 \rangle = (GH_0 - H_0G) | 0 \rangle. \quad (5.62)$$

To this end, we take account of all the quantum states $|k\rangle$ of a hydrogen atom (including $|0\rangle$) that are solutions (or eigenstates of energy) of (3.36). In Sect. 3.8 we have obtained the total wave functions described by

$$\tilde{\Lambda}_{l,m}^{(n)} = Y_l^m(\theta, \phi) \tilde{R}_l^{(n)}(r), \quad (3.300)$$

where $\tilde{\Lambda}_{l,m}^{(n)}$ is expressed as a product of spherical surface harmonics and radial wave functions. The latter functions are described by associated Laguerre functions. It is well-known [2] that the spherical surface harmonics constitute the CONS on a unit sphere and that associated Laguerre functions form the CONS in the real one-dimensional space. Thus, their product expressed as (3.300), i.e., aforementioned collection of $|k\rangle$ constitutes the CONS in the real three-dimensional space. This is equivalent to

$$\sum_j |j\rangle \langle j| = E, \quad (5.63)$$

where E is an identity operator.

The implications of (5.63) are as follows: Take any function $|f(\mathbf{r})\rangle$ and operate (5.63) from the left. Then, we get

$$Ef(\mathbf{r}) = f(\mathbf{r}) = \sum_k |k\rangle \langle k|f(\mathbf{r})\rangle = \sum_k f_k |k\rangle. \quad (5.64)$$

In other words, (5.64) implies that any function $f(\mathbf{r})$ can be expanded into a series of $|k\rangle$. The coefficients f_k are defined as

$$f_k \equiv \langle k|f(\mathbf{r})\rangle. \quad (5.65)$$

Those are the so-called Fourier coefficients. Related discussion is given in Sect. 10.4 as well as Chaps. 14, 18, 20, and 22.

We further impose the conditions on the operator G defined in (5.62). (1) G commutes with r ($=|\mathbf{r}|$). (2) G has a functional form described by $G = G(r, \theta)$. (3) G does not contain a differential operator. On those conditions, we have $\partial G / \partial \phi = 0$. From (3.301), we have $\partial |0\rangle / \partial \phi = 0$. Thus, we get

$$\begin{aligned} (GH_0 - H_0G) | 0 \rangle \\ = -\frac{\hbar^2}{2\mu} \left[G \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) G - \frac{1}{r^2} \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) G \right] | 0 \rangle. \end{aligned} \quad (5.66)$$

This calculation is somewhat complicated, and so we calculate (5.66) termwise. With the second term of (5.66), we have

$$\begin{aligned}
 -\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) G |0\rangle &= -\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial G}{\partial r} \right) |0\rangle - \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 G \frac{\partial |0\rangle}{\partial r} \right) \\
 &= -\frac{1}{r^2} \cdot 2r \frac{\partial G}{\partial r} |0\rangle - \frac{\partial^2 G}{\partial r^2} |0\rangle - \frac{\partial G}{\partial r} \frac{\partial |0\rangle}{\partial r} - \frac{G}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial |0\rangle}{\partial r} \right) - \frac{\partial G}{\partial r} \frac{\partial |0\rangle}{\partial r} \\
 &= -\frac{2}{r} \frac{\partial G}{\partial r} |0\rangle - \frac{\partial^2 G}{\partial r^2} |0\rangle - 2 \frac{\partial G}{\partial r} \frac{\partial |0\rangle}{\partial r} - \frac{G}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial |0\rangle}{\partial r} \right) \\
 &= -\frac{2}{r} \frac{\partial G}{\partial r} |0\rangle - \frac{\partial^2 G}{\partial r^2} |0\rangle + \frac{2}{a} \frac{\partial G}{\partial r} |0\rangle - \frac{G}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial |0\rangle}{\partial r} \right),
 \end{aligned} \tag{5.67}$$

where with the last equality we used (3.301), namely $|0\rangle \propto e^{-r/a}$, where a is Bohr radius (Sect. 3.7.1). The last term of (5.67) cancels the first term of (5.66). Then, we get

$$\begin{aligned}
 (GH_0 - H_0G)|0\rangle &= \frac{\hbar^2}{2\mu} \left[2 \left(\frac{1}{r} - \frac{1}{a} \right) \frac{\partial G}{\partial r} |0\rangle \right. \\
 &\quad \left. + \frac{\partial^2 G}{\partial r^2} |0\rangle + \frac{1}{r^2} \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial G}{\partial \theta} \right) |0\rangle \right] \\
 &= \frac{\hbar^2}{2\mu} \left[2 \left(\frac{1}{r} - \frac{1}{a} \right) \frac{\partial G}{\partial r} + \frac{\partial^2 G}{\partial r^2} + \frac{1}{r^2} \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial G}{\partial \theta} \right) \right] |0\rangle.
 \end{aligned} \tag{5.68}$$

From (5.62) we obtain

$$\langle \mathbf{r} | z | 0 \rangle = \langle \mathbf{r} | (GH_0 - H_0G) | 0 \rangle \tag{5.69}$$

so that we can have the coordinate representation. As a result, we get

$$\begin{aligned}
 r \cos \theta \phi(1s) &= \frac{\hbar^2}{2\mu} \\
 &\times \left[2 \left(\frac{1}{r} - \frac{1}{a} \right) \frac{\partial G}{\partial r} + \frac{\partial^2 G}{\partial r^2} + \frac{1}{r^2} \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial G}{\partial \theta} \right) \right] \phi(1s),
 \end{aligned} \tag{5.70}$$

where $\phi(1s)$ is given by (3.301). Dividing (5.70) by $\phi(1s)$ and rearranging it, we have

$$\frac{\partial^2 G}{\partial r^2} + 2\left(\frac{1}{r} - \frac{1}{a}\right) \frac{\partial G}{\partial r} + \frac{1}{r^2} \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial G}{\partial \theta} \right) = \frac{2\mu}{\hbar^2} r \cos \theta. \quad (5.71)$$

Now, we assume that G has a functional form of

$$G(r, \theta) = g(r) \cos \theta. \quad (5.72)$$

Inserting (5.72) into (5.71), we have

$$\frac{d^2 g}{dr^2} + 2\left(\frac{1}{r} - \frac{1}{a}\right) \frac{dg}{dr} - \frac{2g}{r^2} = \frac{2\mu}{\hbar^2} r. \quad (5.73)$$

Equation (5.73) has a regular singular point at the origin and resembles an Euler equation whose general form of a homogeneous equation is described by [5]

$$\frac{d^2 g}{dr^2} + \frac{a}{r} \frac{dg}{dr} + \frac{b}{r^2} g = 0, \quad (5.74)$$

where a and b are arbitrary constants. In particular, if we have a following differential equation described by

$$\frac{d^2 g}{dr^2} + \frac{a}{r} \frac{dg}{dr} - \frac{a}{r^2} g = 0, \quad (5.75)$$

we immediately see that one of the particular solutions is $g = r$. However, (5.73) differs from the general second-order Euler equation by the presence of $-\frac{2}{a} \frac{dg}{dr}$. Then, let us assume that a particular solution g has a form of

$$g = pr^2 + qr. \quad (5.76)$$

Inserting (5.76) into (5.73), we get

$$4p - \frac{4pr}{a} - \frac{2q}{a} = \frac{2\mu}{\hbar^2} r. \quad (5.77)$$

Comparing coefficient of r and constant term, we obtain

$$p = -\frac{a\mu}{2\hbar^2} \quad \text{and} \quad q = -\frac{a^2\mu}{\hbar^2}. \quad (5.78)$$

Hence, we get

$$g(r) = -\frac{a\mu}{\hbar^2} \left(\frac{r}{2} + a \right) r. \quad (5.79)$$

Accordingly, we have

$$G(r, \theta) = g(r) \cos \theta = -\frac{a\mu}{\hbar^2} \left(\frac{r}{2} + a \right) r \cos \theta = -\frac{a\mu}{\hbar^2} \left(\frac{r}{2} + a \right) z. \quad (5.80)$$

This is a coordinate representation of $G(r, \theta)$.

Returning back to (5.62) and operating $\langle j|$ from the left on its both sides, we have

$$\begin{aligned} \langle j|z|0\rangle &= \langle j|(GH_0 - H_0G)|0\rangle = \langle j|GH_0|0\rangle - \langle j|H_0G|0\rangle \\ &= [E_0^{(0)} - E_j^{(0)}] \langle j|G|0\rangle. \end{aligned} \quad (5.81)$$

Notice here that

$$H_0 |0\rangle = E_0^{(0)} |0\rangle \quad (5.82)$$

and that

$$\langle j|H_0 = [H_0^\dagger|j]\dagger = [H_0|j]\dagger = [E_j^{(0)}|j]\dagger = E_j^{(0)}\langle j|, \quad (5.83)$$

where we used a computation rule of (1.118) and with the second equality we used the fact that H_0 is Hermitian, i.e., $H_0^\dagger = H_0$. Rewriting (5.81), we have

$$\langle j|G|0\rangle = \frac{\langle j|z|0\rangle}{E_0^{(0)} - E_j^{(0)}}. \quad (5.84)$$

Multiplying $\langle 0|z|j\rangle$ on both sides of (5.84) and summing over j ($\neq 0$), we obtain

$$\sum_{j \neq 0} \langle 0|z|j\rangle \langle j|G|0\rangle = \sum_{j \neq 0} \frac{\langle 0|z|j\rangle \langle j|z|0\rangle}{[E_0^{(0)} - E_j^{(0)}]}. \quad (5.85)$$

Adding $\langle 0|z|0\rangle \langle 0|G|0\rangle$ on both sides of (5.85), we have

$$\sum_j \langle 0|z|j\rangle \langle j|G|0\rangle = \sum_{j \neq 0} \frac{\langle 0|z|j\rangle \langle j|z|0\rangle}{[E_0^{(0)} - E_j^{(0)}]} + \langle 0|z|0\rangle \langle 0|G|0\rangle. \quad (5.86)$$

But, from (5.58) $\langle 0|z|0\rangle = 0$. Moreover, using the completeness of $|j\rangle$ described by (5.63) for LHS of (5.86), we get

$$\begin{aligned}
\langle 0|zG|0\rangle &= \sum_j \langle 0|z|j\rangle \langle j|G|0\rangle = \sum_{j \neq 0} \frac{\langle 0|z|j\rangle \langle j|z|0\rangle}{[E_0^{(0)} - E_j^{(0)}]} \\
&= \sum_{j \neq 0} \frac{|\langle j|z|0\rangle|^2}{[E_0^{(0)} - E_j^{(0)}]} = -\frac{\alpha}{2e^2},
\end{aligned} \tag{5.87}$$

where with the second equality we used (5.84) and with the last equality we used (5.61). Also, we used the fact that z is Hermitian.

Meanwhile, using (5.80), LHS of (5.87) is expressed as

$$\langle 0|zG|0\rangle = -\frac{a\mu}{\hbar^2} \left\langle 0|z\left(\frac{r}{2} + a\right)z|0\right\rangle = -\frac{a\mu}{2\hbar^2} \langle 0|zrz|0\rangle - \frac{a^2\mu}{\hbar^2} \langle 0|z^2|0\rangle. \tag{5.88}$$

Noting that $|0\rangle$ is spherically symmetric and that z and r are commutative, (5.88) can be rewritten as

$$\langle 0|zG|0\rangle = -\frac{a\mu}{6\hbar^2} \langle 0|r^3|0\rangle - \frac{a^2\mu}{3\hbar^2} \langle 0|r^2|0\rangle, \tag{5.89}$$

where we used $\langle 0|x^2|0\rangle = \langle 0|y^2|0\rangle = \langle 0|z^2|0\rangle = \langle 0|r^2|0\rangle/3$.

Returning back to the coordinate representation and taking account of variables change from the Cartesian coordinate to the polar coordinate as in (5.58), we get

$$\begin{aligned}
\langle 0|zG|0\rangle &= -\frac{a\mu}{6\hbar^2} \frac{4\pi}{\pi a^3} \frac{120a^6}{64} - \frac{a^2\mu}{3\hbar^2} \frac{4\pi}{\pi a^3} \frac{3a^5}{4} = -\frac{a\mu}{\hbar^2} \frac{5a^3}{4} - \frac{a^2\mu}{\hbar^2} \frac{4a^3}{4} \\
&= -\frac{a\mu}{\hbar^2} \frac{9a^3}{4}.
\end{aligned} \tag{5.90}$$

To perform the definite integral calculations of (5.89) and obtain the result of (5.90), modify (3.263) and use the formula described below. That is, differentiate

$$\int_0^\infty e^{-r\xi} dr = \frac{1}{\xi}$$

four (or five) times with respect to ξ and replace ξ with $2/a$ to get the result of (5.90) appropriately.

Using (5.87) and (5.90), as the polarizability α we finally get

$$\begin{aligned}\alpha &= -2e^2 \langle 0 | z G | 0 \rangle = -2e^2 \cdot \left(-\frac{a\mu}{\hbar^2} \frac{9a^3}{4} \right) = \frac{e^2 a \mu}{\hbar^2} \frac{9a^3}{2} = 4\pi\epsilon_0 \frac{9a^3}{2} \\ &= 18\pi\epsilon_0 a^3,\end{aligned}\tag{5.91}$$

where we used

$$a = 4\pi\epsilon_0 \hbar^2 / \mu e^2.\tag{5.92}$$

See Sect. 3.7.1 for this relation.

2. Stark effect of a hydrogen atom in a ground state

The energy shift with the ground state of a hydrogen atom up to the second order is given by

$$E_0 \approx E_0^{(0)} - eF \langle 0 | z | 0 \rangle + (-eF)^2 \sum_{j \neq 0} \frac{1}{[E_0^{(0)} - E_j^{(0)}]} |\langle j | z | 0 \rangle|^2.\tag{5.93}$$

Using (5.58) and (5.87) obtained above, we readily get

$$E_0 \approx E_0^{(0)} - \frac{\alpha e^2 F^2}{2e^2} = E_0^{(0)} - \frac{\alpha F^2}{2} = E_0^{(0)} - 9\pi\epsilon_0 a^3 F^2.\tag{5.94}$$

Experimentally, the energy shift by $-9\pi\epsilon_0 a^3 F^2$ is well-known as the Stark shift.

In the above examples, we have seen how energy levels and related properties are changed by the applied electric field. The energies of the system that result from the applied external field should not be considered as an energy eigenvalue but should be thought to be an expectation value.

The perturbation theory has a wide application in physical problems. Examples include the evaluation of transition probability between quantum states. The theory also deals with scattering of particles. Including the treatment of the degenerate case, interested readers are referred to appropriate literature [1, 3].

5.2 Variational Method

Another approximation method is a variational method. This method also has a wide range of applications in mathematical physics.

A major application of the variational method lies in seeking an eigenvalue that is appropriately approximated. Suppose that we have an Hermitian differential operator D that satisfies an eigenvalue equation

$$Dy_n = \lambda_n y_n \quad (n = 1, 2, 3, \dots), \quad (5.95)$$

where λ_n is a real eigenvalue and y_n is a corresponding eigenfunction. We have already encountered one of the typical differential operators and eigenvalue equations in (1.63) and (1.64). In (5.95) we assume that

$$\lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \dots, \quad (5.96)$$

where corresponding eigenstates may be degenerate.

We also assume that a collection $\{y_n; n = 1, 2, 3, \dots\}$ constitutes the CONS. Now suppose that we have a function u that satisfies the boundary conditions (BCs) the same as those for y_n that is a solution of (5.95). Then, we have

$$\langle y_n | Du \rangle = \langle Dy_n | u \rangle = \langle \lambda_n y_n | u \rangle = \lambda_n^* \langle y_n | u \rangle = \lambda_n \langle y_n | u \rangle. \quad (5.97)$$

In (5.97) we used (1.132) and the computation rule of the inner product (see Sect. 13.1). Also, we used the fact that any eigenvalue of an Hermitian operator is real (Sect. 1.4). From the assumption that the functions $\{y_n\}$ constitute the CONS, u can be expanded in a series described by

$$u = \sum_j c_j | y_j \rangle \quad (5.98)$$

with

$$c_j \equiv \langle y_j | u \rangle. \quad (5.99)$$

This relation is in parallel with (5.64). Similarly, Du can be expanded such that

$$Du = \sum_j d_j | y_j \rangle = \sum_j \langle y_j | Du \rangle | y_j \rangle = \sum_j \lambda_j \langle y_j | u \rangle | y_j \rangle = \sum_j \lambda_j c_j | y_j \rangle, \quad (5.100)$$

where d_j is an expansion coefficient and with the third equality we used (5.95) and with the last equality we used (5.99). Comparing the individual coefficient of $|y_j\rangle$ of (5.100), we get

$$d_j = \lambda_j c_j. \quad (5.101)$$

Thus, we obtain

$$Du = \sum_j \lambda_j c_j | y_j \rangle. \quad (5.102)$$

Comparing (5.98) and (5.102), we find that we may termwise operate D on (5.98). Meanwhile, taking an inner product $\langle u | Du \rangle$ again termwise with (5.100), we get

$$\begin{aligned}
\langle u|Du\rangle &= \left\langle u \left| \sum_j \lambda_j c_j |y_j\rangle \right. \right\rangle = \sum_j \lambda_j c_j \langle u|y_j\rangle = \sum_j \lambda_j c_j c_j^* = \sum_j \lambda_j |c_j|^2 \\
&\geq \sum_j \lambda_1 |c_j|^2 = \lambda_1 \sum_j |c_j|^2,
\end{aligned} \tag{5.103}$$

where with the third equality we used (5.99) in combination with $\langle u|y_j\rangle = \langle y_j|u\rangle^* = c_j^*$. With $\langle u|u\rangle$, we have

$$\langle u|u\rangle = \left\langle \sum_j c_j \langle y_j| \right| \sum_j c_j |y_j\rangle \right\rangle = \sum_j c_j^* c_j \langle y_j|y_j\rangle = \sum_j |c_j|^2, \tag{5.104}$$

where with the last equality we used the fact that the functions $\{y_n\}$ are normalized. Comparing once again (5.103) and (5.104), we finally get

$$\langle u|Du\rangle \geq \lambda_1 \langle u|u\rangle \quad \text{or} \quad \lambda_1 \leq \frac{\langle u|Du\rangle}{\langle u|u\rangle}. \tag{5.105}$$

If we choose y_1 for u , we have an equality in (5.105). Thus, we reach the following important theorem.

Theorem 5.1 Suppose that we have a linear differential equation described by

$$Dy = \lambda y, \tag{5.106}$$

where D is a suitable Hermitian differential operator. Suppose also that under appropriate boundary conditions (BCs), (5.106) has a series of (real) eigenvalues. Then, the smallest eigenvalue is equal to the minimum of $\langle u|Du\rangle/\langle u|u\rangle$.

Using this powerful theorem, we are able to find a suitably approximated smallest eigenvalue. We have simple examples next. As before, let us focus on the change in the energies and quantum states caused by the applied electric field for a particle in a one-dimensional potential well and a harmonic oscillator. The latter problem will be left for readers as an exercise. As a specific case, for simplicity, we consider only the ground state and first excited state to choose a trial function. First, we deal with the problem in common with both the cases. Then, we discuss the feature individually.

As a general case, suppose that H is the Hamiltonian of the system (either a particle in a one-dimensional potential well or a harmonic oscillator). We calculate an expectation value of energy $\langle H \rangle$ when the system undergoes an influence of the electric field. The Hamiltonian is described by

$$H = H_0 - eFx, \tag{5.107}$$

where H_0 is the Hamiltonian without the external field F . We suppose that the quantum state $|u\rangle$ is expressed as

$$|u\rangle \approx |0\rangle + a|1\rangle, \quad (5.108)$$

where $|0\rangle$ and $|1\rangle$ denote the ground state and first excited state, respectively; a is an unknown parameter to be decided after the analysis based on the variational method. In the present case, $|u\rangle$ is a trial function. Then, we have

$$\langle H \rangle \equiv \frac{\langle u|Hu\rangle}{\langle u|u\rangle}. \quad (5.109)$$

With the numerator, we have

$$\begin{aligned} \langle u|Hu\rangle &= \langle \langle 0| + a\langle 1| | H_0 - eFx | | 0\rangle + a|1\rangle \rangle \\ &= \langle 0|H_0|0\rangle + a\langle 0|H_0|1\rangle - eF\langle 0|x|0\rangle - eFa\langle 0|x|1\rangle \\ &\quad + a^*\langle 1|H_0|0\rangle + a^*a\langle 1|H_0|1\rangle - eFa^*\langle 1|x|0\rangle - eFa^*a\langle 1|x|1\rangle \\ &= \langle 0|H_0|0\rangle - eFa\langle 0|x|1\rangle + a^*a\langle 1|H_0|1\rangle - eFa^*\langle 1|x|0\rangle. \end{aligned} \quad (5.110)$$

Note that in (5.110) four terms vanish because of the symmetry requirement of the symmetric potential well and harmonic oscillator with respect to the origin as well as because of the orthogonality between the states $|0\rangle$ and $|1\rangle$. Here x is Hermitian and we assume that the coordinate representation of $|0\rangle$ and $|1\rangle$ is real as in (1.101) and (1.102) or in (2.106). Then, we have

$$\langle 1|x|0\rangle^* = \langle 1|x|0\rangle = \langle 0|x^\dagger|1\rangle = \langle 0|x|1\rangle.$$

Also, assuming that a is a real number, we rewrite (5.110) as

$$\langle u|Hu\rangle = \langle 0|H_0|0\rangle - 2eFa\langle 0|x|1\rangle + a^2\langle 1|H_0|1\rangle. \quad (5.111)$$

As for the denominator of (5.109), we have

$$\langle u|u\rangle = \langle \langle 0| + a\langle 1| | | 0\rangle + a|1\rangle \rangle = 1 + a^2, \quad (5.112)$$

where we used the normalized conditions $\langle 0|0\rangle = \langle 1|1\rangle = 1$ and the orthogonal conditions $\langle 0|1\rangle = \langle 1|0\rangle = 0$. Thus, we get

$$\langle H \rangle = \frac{\langle 0|H_0|0\rangle - 2eFa\langle 0|x|1\rangle + a^2\langle 1|H_0|1\rangle}{1 + a^2}. \quad (5.113)$$

Here, defining the ground state energy and first excited energy as E_0 and E_1 , respectively, we put

$$H_0 | 0 \rangle = E_0 | 0 \rangle \quad \text{and} \quad H_0 | 1 \rangle = E_1 | 1 \rangle, \quad (5.114)$$

where $E_0 \neq E_1$. As already shown in Chaps. 1 and 2, both the systems have non-degenerate energy eigenstates. Then, we have

$$\langle H \rangle = \frac{E_0 - 2eFa\langle 0|x|1 \rangle + a^2 E_1}{1 + a^2}. \quad (5.115)$$

Now we wish to determine the condition where $\langle H \rangle$ has an extremum. That is, we are seeking a that satisfies

$$\frac{\partial \langle H \rangle}{\partial a} = 0. \quad (5.116)$$

Calculating LHS of (5.116) by use of (5.115), we have

$$\frac{\partial \langle H \rangle}{\partial a} = \frac{2eF\langle 0|x|1 \rangle a^2 + 2(E_1 - E_0)a - 2eF\langle 0|x|1 \rangle}{(1 + a^2)^2}. \quad (5.117)$$

For $\frac{\partial \langle H \rangle}{\partial a}$ to be zero, we must have

$$eF\langle 0|x|1 \rangle a^2 + (E_1 - E_0)a - eF\langle 0|x|1 \rangle = 0. \quad (5.118)$$

Then, solving the quadratic equation (5.118), we obtain

$$a = \frac{E_0 - E_1 \pm (E_1 - E_0) \sqrt{1 + \frac{[2eF\langle 0|x|1 \rangle]^2}{(E_1 - E_0)^2}}}{2eF\langle 0|x|1 \rangle}. \quad (5.119)$$

Taking the plus sign in the numerator of (5.119) and using

$$\sqrt{1 + \Delta} \approx 1 + \frac{\Delta}{2} \quad (5.120)$$

for small Δ , we obtain

$$a \approx \frac{eF\langle 0|x|1 \rangle}{E_1 - E_0}. \quad (5.121)$$

Thus, with the optimized state of (5.108) we get

$$|u\rangle \approx |0\rangle + \frac{eF\langle 0|x|1\rangle}{E_1 - E_0} |1\rangle. \quad (5.122)$$

If one compares (5.122) with (5.17), one should recognize that these two equations are the same within the first-order approximation. Notice that putting $i = 0$ and $k = 1$ in (5.17) and replacing V with $-eFx$, we have the expression similar to (5.122). Notice once again that $\langle 0|x|1\rangle = \langle 1|x|0\rangle$ as x is an Hermitian operator and that we are considering real inner products.

Inserting (5.121) into (5.115) and approximating

$$\frac{1}{1+a^2} \approx 1 - a^2, \quad (5.123)$$

we can estimate $\langle H \rangle$. The estimation depends upon the nature of the system we choose. The next examples deal with these specific characteristics.

Example 5.4 We adopt the same problem as Example 5.1. That is, we consider how an energy of a particle carrying a charge e confined within a one-dimensional potential well is changed by the applied electric field. Here we deal with it using the variational method.

We deal with the same Hamiltonian as in the case of Example 5.1. It is described as

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} - eFx. \quad (5.22)$$

By putting

$$H_0 \equiv -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}, \quad (5.124)$$

we rewrite the Hamiltonian as

$$H = H_0 - eFx.$$

Expressing the ground state as $|0\rangle$ and the first excited state as $|1\rangle$, we adopt a trial function $|u\rangle$ as

$$|u\rangle = |0\rangle + a |1\rangle. \quad (5.125)$$

In the coordinate representation, we have

$$|0\rangle = \sqrt{\frac{1}{L}} \cos \frac{\pi}{2L} x \quad (5.26)$$

and

$$|1\rangle = \sqrt{\frac{1}{L}} \sin \frac{\pi}{L} x. \quad (5.126)$$

From Example 1.2 of Sect. 1.3, we have

$$\begin{aligned} \langle 0|H_0|0\rangle &= E_0 = \frac{\hbar^2}{2m} \cdot \frac{\pi^2}{4L^2}, \\ \langle 1|H_0|1\rangle &= E_1 = \frac{\hbar^2}{2m} \cdot \frac{4\pi^2}{4L^2} = 4E_0. \end{aligned} \quad (5.127)$$

Inserting these results into (5.121), we immediately get

$$a \approx eF \frac{\langle 0|x|1\rangle}{3E_0}. \quad (5.128)$$

As in Example 5.1 of Sect. 5.1, we have

$$\langle 0|x|1\rangle = \frac{32L}{9\pi^2}. \quad (5.129)$$

For this calculation, use the coordinate representation of (5.26) and (5.126). Thus, we get

$$a \approx eF \frac{32 \cdot 8 \cdot mL^3}{27\pi^4 \hbar^2}. \quad (5.130)$$

Meanwhile, from (5.125) we obtain

$$|u\rangle \approx |0\rangle + eF \frac{\langle 0|x|1\rangle}{3E_0} |1\rangle. \quad (5.131)$$

The resulting $|u\rangle$ in (5.131) is the same as (5.31), which was obtained by the first-order approximation of (5.30). Inserting (5.130) into (5.113) and approximating the denominator as before such that

$$\frac{1}{1+a^2} \approx 1-a^2, \quad (5.123)$$

and further using (5.130) for a , we get

$$\langle H \rangle \approx E_0 - (eF)^2 \frac{32^2 \cdot 8 \cdot mL^4}{243\pi^6 \hbar^2}. \quad (5.132)$$

Once again, this is the same result as that of (5.28) and (5.29) where only $n = 1$ is taken into account.

Example 5.5 We adopt the same problem as Example 5.2. The calculation procedures are almost the same as those of Example 5.2. We use

$$\langle 0|q|1 \rangle = \sqrt{\frac{\hbar}{2m\omega}} \quad \text{and} \quad E_1 - E_0 = \hbar\omega. \quad (5.133)$$

The detailed procedures are left for readers as an exercise. We should get the same results as those obtained in Example 5.2.

As discussed in the above five simple examples, we showed the calculation procedures of the perturbation method and variational method. These methods not only supply us with suitable approximation techniques, but also provide physical and mathematical insight in many fields of natural science.

References

1. Sunakawa S (1991) Quantum mechanics. Iwanami, Tokyo. (in Japanese)
2. Byron FW Jr, Fuller RW (1992) Mathematics of classical and quantum physics. Dover, New York, NY
3. Schiff LI (1955) Quantum mechanics, 2nd edn. McGraw-Hill, New York, NY
4. Jackson JD (1999) Classical electrodynamics, 3rd edn. Wiley, New York, NY
5. Coddington EA (1989) An introduction to ordinary differential equations. Dover, New York, NY

Chapter 6

Theory of Analytic Functions



Theory of analytic functions is one of the major fields of modern mathematics. Its application covers a broad range of topics of natural science. A complex function $f(z)$, or a function that takes a complex number z as a variable, has various properties that often differ from those of functions that take a real number x as a variable. In particular, the analytic functions hold a paramount position in the complex analysis. In this chapter we explore various features of the analytic functions accordingly. From a practical point of view, the theory of analytic functions is very frequently utilized for the calculation of real definite integrals. For this reason, we describe the related topics together with tangible examples.

The complex plane (or Gaussian plane) can be dealt with as a topological space where the metric (or distance function) is defined. Since the complex plane has a two-dimensional extension, we can readily imagine and investigate its topological feature. Set theory allows us to make an axiomatic approach along with the topology. Therefore, we introduce basic notions and building blocks of the set theory and topology.

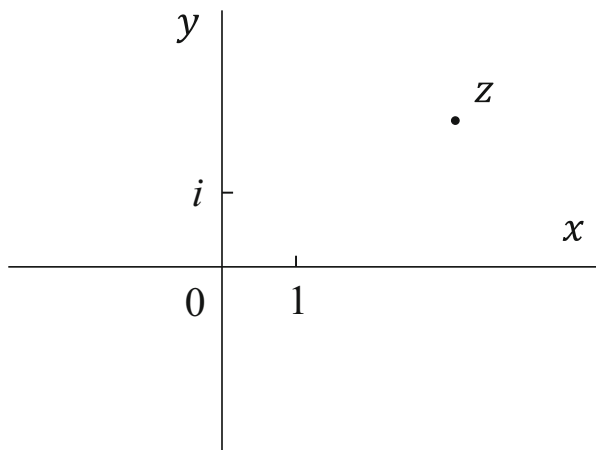
6.1 Set and Topology

A complex number z is usually expressed as

$$z = x + iy, \quad (6.1)$$

where x and y are the real numbers and i is an imaginary unit. Graphically, the number z is indicated as a point in a complex plane where the real axis is drawn as abscissa and the imaginary axis is depicted as ordinate (see Fig. 6.1). Since the complex plane has a two-dimensional extension, we can readily imagine the domain of variability of z and make a graphical object for it on the complex plane. A disk-like diagram that is enclosed with a closed curve C is frequently dealt with in the

Fig. 6.1 Complex plane where a complex number z is depicted



theory of analytic functions. Such a diagram provides a good subject for study of the set theory and topology. Before developing the theory of complex numbers and analytic functions, we briefly mention the basic notions as well as notations and meanings of sets and topology.

6.1.1 Basic Notions and Notations

A set comprises numbers (either real or complex) or mathematical objects, more generally. The latter include, e.g., vectors, their transformation, etc., as we shall see various illustrations in Parts III and IV. The set A is described such that

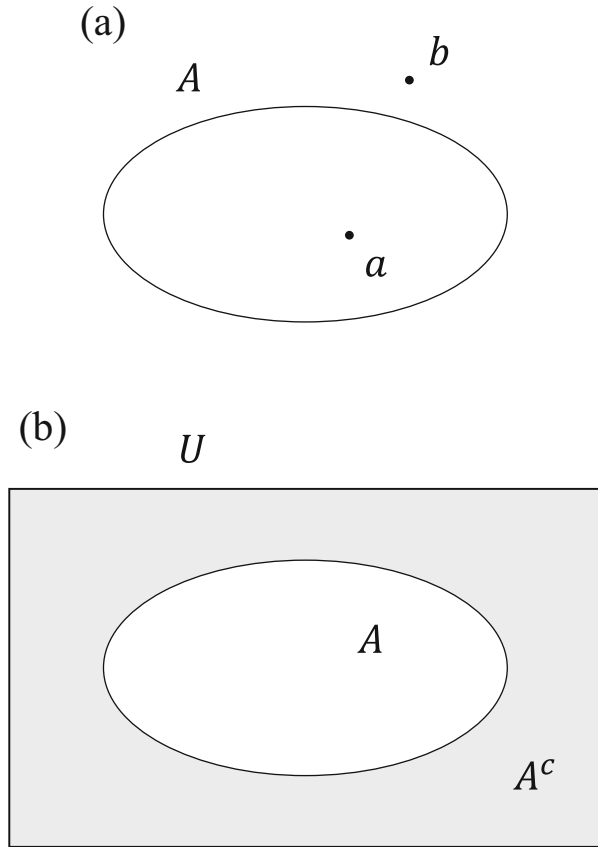
$$A = \{1, 2, 3, \dots, 10\} \text{ or } B = \{x; a < x < b, \ a, b \in \mathbb{R}\}, \quad (6.2)$$

where A contains ten elements in the former case and uncountably infinite elements are contained in the latter set B . In the latter case (sometimes in the former case as well), the elements are usually called points.

When we speak of the sets, a *universal set* [1] is implied in the background. This set contains all the sets in consideration of the study and is usually clearly defined in the context of the discussion. For example, with the real analysis the universal set is $\mathbb{R}$ and with the complex analysis it is $\mathbb{C}$ (an entire complex plane). In the latter case a two-dimensional complex plane (see Fig. 6.1) is frequently used as the universal set. We study various characteristics of sets (or subsets) that are contained in the universal set and use a Venn diagram to represent them. Figure 6.2a illustrates an example of Venn diagram that shows a subset A . Naturally, every set contained in the universal set is its subset. As is often the case, the universal set U is omitted in Fig. 6.2a accordingly.

Fig. 6.2 Venn diagrams.

(a) An example of the Venn diagram. (b) A subset A and its complement A^c . U shows the universal set



To show a (sub)set A we usually depict it with a closed curve. If an element a is contained in A , we write

$$a \in A$$

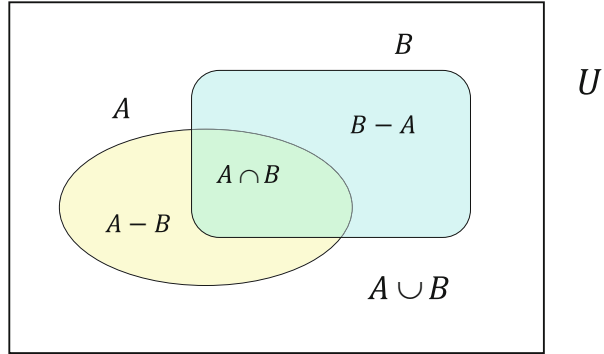
and depict it as a point inside the closed curve. To show that b is not contained in A , we write

$$b \notin A.$$

In that case, we depict b outside the closed curve of Fig. 6.2a. To show that a set A is contained in another set B , we write

$$A \subset B. \tag{6.3}$$

Fig. 6.3 Two different subsets A and B in a universal set U . The diagram shows the sum ($A \cup B$), intersection ($A \cap B$), and differences ($A - B$ and $B - A$)



If (6.3) holds, A is called a subset of B . If, however, the set B is contained in A , B is a subset of A and we write it as $B \subset A$. When both the relations $A \subset B$ and $B \subset A$ simultaneously hold, we describe it as

$$A = B. \quad (6.4)$$

Equation (6.4) indicates the equality of A and B as (sub)sets. The simultaneous establishment of $A \subset B$ and $B \subset A$ is often used to prove (6.4).

We need to explicitly show a set that has no element. The relevant set is called an empty set and denoted by $\emptyset$. Examples are given as follows:

$$\{x; x^2 < 0, x \in \mathbb{R}\} = \emptyset, \{x; x \neq x\} = \emptyset, x \notin \emptyset, \text{ etc.}$$

The subset A may have different cardinal numbers (or cardinalities) depending on the nature of A . To explicitly show this, elements are sometimes indexed as, e.g., a_i ($i = 1, 2, \dots$) or a_λ ($\lambda \in \Lambda$), where Λ can be a finite set or infinite set with different cardinalities; (6.2) is an example. A complement (or complementary set) of A is defined as a difference $U - A$ and denoted by $A^c (\equiv U - A)$. That is, the set A^c is said to be a complement of A with respect to U and indicated with a shaded area in Fig. 6.2b.

Let A and B be two different subsets in U (Fig. 6.3). Figure 6.3 represents a sum (or union, or union of sets) $A \cup B$, an intersection $A \cap B$, and differences $A - B$ and $B - A$. More specifically, $A - B$ is defined as a subset of A to which B is subtracted from A and referred to as a set difference of A relative to B . We have

$$A \cup B = (A - B) \cup (B - A) \cup (A \cap B). \quad (6.5)$$

If A has no element in common with B , $A - B = A$ and $B - A = B$ with $A \cap B = \emptyset$. Then, (6.5) trivially holds. Notice also that $A \cup B$ is the smallest set that contains both A and B . In (6.5) the elements must not be doubly counted. Thus, we have

$$A \cup A = A.$$

The well-known de Morgan's law can be understood graphically using Fig. 6.3.

$$(A \cup B)^c = A^c \cap B^c \text{ and } (A \cap B)^c = A^c \cup B^c. \quad (6.6)$$

This can be shown as follows: With $\forall u \in U$ we have

$$u \in (A \cup B)^c \Leftrightarrow u \notin A \cup B \Leftrightarrow u \notin A \text{ and } u \notin B$$

$$\Leftrightarrow u \in A^c \text{ and } u \in B^c \Leftrightarrow u \in A^c \cap B^c.$$

Furthermore, we have

$$u \in (A \cap B)^c \Leftrightarrow u \notin A \cap B \Leftrightarrow u \notin A \text{ or } u \notin B$$

$$\Leftrightarrow u \in A^c \text{ or } u \in B^c \Leftrightarrow u \in A^c \cup B^c.$$

We have other important relations such as the distributive law described by

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C), \quad (6.7)$$

$$(A \cap B) \cup C = (A \cup C) \cap (B \cup C). \quad (6.8)$$

The confirmation of (6.7) and (6.8) is left for readers. The Venn diagrams are intuitively acceptable, but we need more rigorous discussion to characterize and analyze various aspects of sets (vide infra).

6.1.2 Topological Spaces and Their Building Blocks

On the basis of the aforementioned discussion, we define a topology and topological space next. Let T be a universal set. Let τ be a collection of subsets of T . If the collection τ satisfies the following axioms, τ is called a topology on T . The relevant axioms are as follows:

$$(O1) \quad T \in \tau \text{ and } \emptyset \in \tau, \text{ where } \emptyset \text{ is an empty set.}$$

(O2) If $O_1, O_2, \dots, O_n \in \tau, O_1 \cap O_2 \dots \cap O_n \in \tau$.

(O3) If $O_\lambda (\lambda \in \Lambda) \in \tau, \bigcup_{\lambda \in \Lambda} O_\lambda \in \tau$.

In the above axioms, T is called an underlying set. The coupled object of T and τ is said to be a topological space and denoted by (T, τ) . The members (i.e., subsets) of τ are defined as *open sets*. (If we do not assume the topological space, the underlying set is equivalent to the universal set. Henceforth, however, we do not have to be too strict with the difference in this kind of terminology.)

Let (T, τ) be a topological space with a subset $O' \subset T$. Let τ' be defined such that

$$\tau' \equiv \{O' \cap O; O \in \tau\}.$$

Then, (O', τ') can be regarded as a topological space. In this case τ' is called a relative topology for O' [2, 3] and O' is referred to as a subspace of T . The topological space (O', τ') shares the same topological feature as that of (T, τ) . Notice that O' does not necessarily need to be an open set of T . However, if O' is an open set of T , then any open set belonging to O' is an open set of T as well. This is evident from the definition of the relative topology and Axiom (O2).

The above-mentioned Axioms (O1) to (O3) sound somewhat pretentious and the definition of the open sets seems to be “descent from heaven.” Nonetheless, these axioms and terminology turn out useful soon. Once a topological space (T, τ) is given, subsets of various types arise and a variety of relationships among them ensue as well. Note that the above axioms mention nothing about a set that is different from an open set. Let S be a subset (maybe an open set or maybe not) of T and let us think of the properties of S .

Definition 6.1 Let (T, τ) be a topological space and S be an open set of τ ; i.e., $S \in \tau$. Then, a subset defined as a complement of S in (T, τ) ; namely, $S^c = T - S$ is called a closed set in (T, τ) .

Replacing S with S^c in Definition 6.1, we have $(S^c)^c = S = T - S^c$. The above definition may be rephrased as follows: Let (T, τ) be a topological space and let A be a subset of T . Then, if $A^c = T - A \in \tau$, then A is a closed set in (T, τ) .

In parallel with the axioms (O1) to (O3), we have the following axioms related to the closed sets of (T, τ) : Let $\tilde{\tau}$ be a collection of all the closed sets of (T, τ) .

(C1) $T \in \tilde{\tau}$ and $\emptyset \in \tilde{\tau}$, where $\emptyset$ is an empty set.

(C2) If $C_1, C_2, \dots, C_n \in \tilde{\tau}, C_1 \cup C_2 \dots \cup C_n \in \tilde{\tau}$.

(C3) If $C_\lambda (\lambda \in \Lambda) \in \tilde{\tau}, \bigcap_{\lambda \in \Lambda} C_\lambda \in \tilde{\tau}$.

These axioms are obvious from those of (O1) to (O3) along with de Morgan's law (6.6).

Next, we classify and characterize a variety of elements (or points) and subsets as well as relationships among them in terms of the open sets and closed sets. We make the most of these notions and properties to study various aspects of topological spaces and their structures. In what follows, we assume the presence of a topological space (T, τ) .

(a) Neighborhoods [4]

Definition 6.2 Let (T, τ) be a topological space and S be a subset of T . If S contains an open set $\exists O$ that contains an element (or point) $a \in T$, i.e.,

$$a \in O \subset S,$$

then S is called a neighborhood of a .

Figure 6.4 shows a Venn diagram that represents a neighborhood of a . This simple definition occupies an important position in the study of the topological spaces and the theory of analytic functions.

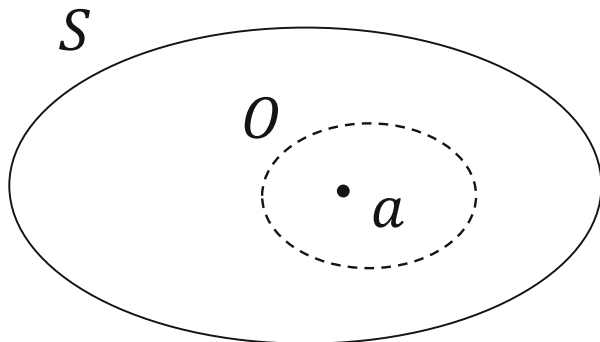
(b) Interior and Closure [4]

With the notion of neighborhoods at the core, we shall see further characterization of sets and elements of the topological spaces.

Definition 6.3 Let x be a point of T and S be a subset of T . If S is a neighborhood of x , x is said to be in the interior of S . In this case, x is called an interior point of S . The interior of S is denoted by S° .

The interior as a technical term can be understood as follows: Suppose that $x \notin S$. Then, by Definition 6.2 S is not a neighborhood of x . By Definition 6.3, this leads to $x \notin S^\circ$. This statement can be translated into $x \in S^c \Rightarrow x \in (S^\circ)^c$. That is, $S^c \subset (S^\circ)^c$. This is equivalent to

Fig. 6.4 Venn diagram that represents a neighborhood S of a . S contains an open set O that contains a



$$S^\circ \subset S. \quad (6.9)$$

Definition 6.4 Let x be a point of T and S be a subset of T . If for any neighborhood ${}^\forall N$ of x we have $N \cap S \neq \emptyset$, x is said to be in the closure of S . In this case, x is called an adherent point of S . The closure of S is denoted by $\bar{S}$.

According to the above definition, with the adherent point x of S we explicitly write $x \in \bar{S}$. Think of negation of the statement of Definition 6.4. That is, with some neighborhood ${}^\exists N$ of x we have (i) $N \cap S = \emptyset \Leftrightarrow x \notin \bar{S}$. Meanwhile, (ii) $N \cap S = \emptyset \Rightarrow x \notin S$. Combining (i) and (ii), we have $x \notin \bar{S} \Rightarrow x \notin S$. Similarly to the above argument about the interior, we get

$$S \subset \bar{S}. \quad (6.10)$$

Combining (6.9) and (6.10), we get

$$S^\circ \subset S \subset \bar{S}. \quad (6.11)$$

In relation to the interior and closure, we have the following important lemmas.

Lemma 6.1 Let S be a subset of T and O be any open set contained in S . Then, we have $O \subset S^\circ$.

Proof Suppose that with an arbitrary point $x, x \in O$. Since we have $x \in O \subset S$, S is a neighborhood of x (due to Definition 6.2). Then, from Definition 6.3 we have $x \in S^\circ$. Then, we have $x \in O \Rightarrow x \in S^\circ$. This implies $O \subset S^\circ$. This completes the proof.

Lemma 6.2 Let S be a subset of T . If $x \in S^\circ$, then there is some open set ${}^\exists O$ that satisfies $x \in O \subset S$.

Proof Suppose that with a point $x, x \in S^\circ$. Then by Definition 6.3, S is a neighborhood of x . Meanwhile, by Definition 6.2 there is some open set ${}^\exists O$ that satisfies $x \in O \subset S$. This completes the proof.

Lemmas 6.1 and 6.2 teach us further implications. From Lemma 6.1 and (6.11) we have

$$O \subset S^\circ \subset S \quad (6.12)$$

with an arbitrary open set ${}^\forall O$ contained in S° . From Lemma 6.2 we have $x \in S^\circ \Rightarrow x \in O$; i.e., $S^\circ \subset O$ with some open set ${}^\exists O$ containing S° . Expressing this specific O as $\tilde{O}$, we have $S^\circ \subset \tilde{O}$. Meanwhile, using Lemma 6.1 once again we must get

$$\tilde{O} \subset S^\circ \subset S. \quad (6.13)$$

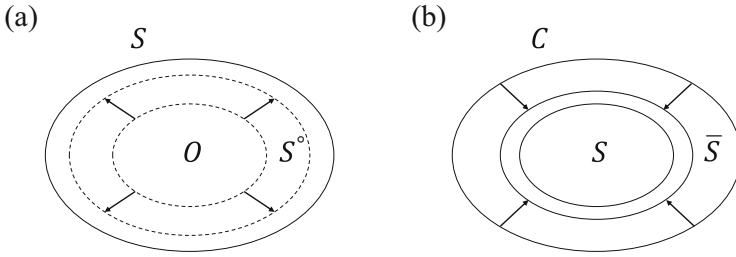


Fig. 6.5 (a) Largest open set S° contained in S . O denotes an open set. (b) Smallest closed set $\bar{S}$ containing S . C denotes a closed set

That is, we have $S^\circ \subset \tilde{O}$ and $\tilde{O} \subset S^\circ$ at once. This implies that with this specific open set $\tilde{O}$, we get $\tilde{O} = S^\circ$. That is,

$$\tilde{O} = S^\circ \subset S. \quad (6.14)$$

Relation (6.14) obviously shows that S° is an open set and moreover that S° is the largest open set contained in S . Figure 6.5a depicts this relation. In this context we have a following important theorem.

Theorem 6.1 Let S be a subset of T . The subset S is open if and only if $S = S^\circ$.

Proof From (6.14) S° is open. Therefore, if $S = S^\circ$, then S is open. Conversely, suppose that S is open. In that event S itself is an open set contained in S . Meanwhile, S° has been characterized as the largest open set contained in S . Consequently, we have $S \subset S^\circ$. At the same time, from (6.11) we have $S^\circ \subset S$. Thus, we get $S = S^\circ$. This completes the proof.

In contrast to Lemmas 6.1 and 6.2, we have the following lemmas with respect to closures.

Lemma 6.3 Let S be a subset of T and C be any closed set containing S . Then, we have $\bar{S} \subset C$.

Proof Since C is a closed set, C^c is an open set. Suppose that with a point x , $x \notin C$. Then, $x \in C^c$. Since $C \supset S$, $C^c \subset S^c$. This implies $C^c \cap S = \emptyset$. As $x \in C^c \subset C^c$, by Definition 6.2 C^c is a neighborhood of x . Then, by Definition 6.4 x is not in $\bar{S}$; i.e., $x \notin \bar{S}$. This statement can be translated into $x \in C^c \Rightarrow x \in (\bar{S})^c$; i.e., $C^c \subset (\bar{S})^c$. That is, we get $\bar{S} \subset C$.

Lemma 6.4 Let S be a subset of T and suppose that a point x does not belong to $\bar{S}$; i.e., $x \notin \bar{S}$. Then, $x \notin C$ for some closed set C containing S .

Proof If $x \notin \bar{S}$, then by Definition 6.4 we have some neighborhood 3N and some open set 3O contained in that N such that $x \in O \subset N$ and $N \cap S = \emptyset$. Therefore, we

must have $O \cap S = \emptyset$. Let $C = O^c$. Then C is a closed set with $C \supset S$. Since $x \in O$, $x \notin C$. This completes the proof.

From Lemmas 6.3 and 6.4 we obtain further implications. From Lemma 6.3 and (6.11) we have

$$S \subset \bar{S} \subset C \quad (6.15)$$

with an arbitrary closed set $\forall C$ containing S . From Lemma 6.4 we have $x \in (\bar{S})^c \Rightarrow x \in C^c$; i.e., $C \subset \bar{S}$ with $\exists C$ containing S . Expressing this specific C as $\tilde{C}$, we have $\tilde{C} \subset \bar{S}$. Meanwhile, using Lemma 6.3 once again we must get

$$S \subset \bar{S} \subset \tilde{C}. \quad (6.16)$$

That is, we have $\tilde{C} \subset \bar{S}$ and $\bar{S} \subset \tilde{C}$ at once. This implies that with this specific closed set $\tilde{C}$, we get $\tilde{C} = \bar{S}$. Hence, we get

$$S \subset \bar{S} = \tilde{C}. \quad (6.17)$$

Relation (6.17) obviously shows that $\bar{S}$ is a closed set and moreover that $\bar{S}$ is the smallest closed set containing S . Figure 6.5b depicts this relation. In this context we have a following important theorem.

Theorem 6.2 Let S be a subset of T . The subset S is closed if and only if $S = \bar{S}$.

Proof The proof can be done analogously to that of Theorem 6.1. From (6.17) $\bar{S}$ is closed. Therefore, if $S = \bar{S}$, then S is closed. Conversely, suppose that S is closed. In that event S itself is a closed set containing S . Meanwhile, $\bar{S}$ has been characterized as the smallest closed set containing S . Then, we have $S \supset \bar{S}$. At the same time, from (6.11) we have $\bar{S} \supset S$. Thus, we get $S = \bar{S}$. This completes the proof.

(c) Boundary [4]

Definition 6.5 Let (T, τ) be a topological space and S be a subset of T . If a point $x \in T$ is in both the closure of S and the closure of the complement of S , i.e., $\bar{S}^c$, x is said to be in the boundary of S . In this case, x is called a boundary point of S . The boundary of S is denoted by S^b .

By this definition S^b can be expressed as

$$S^b = \bar{S} \cap \bar{S}^c. \quad (6.18)$$

Replacing S with S^c , we get

$$(S^c)^b = \overline{S^c} \cap \overline{(S^c)^c} = \overline{S^c} \cap \overline{S}. \quad (6.19)$$

Thus, we have

$$S^b = (S^c)^b. \quad (6.20)$$

This means that S and S^c have the same boundary. Since both $\overline{S}$ and $\overline{S^c}$ are closed sets and the boundary is defined as their intersection, from Axiom (C3) the boundary is a closed set. From Definition 6.1 a closed set is defined as a complement of an open set, and vice versa. Thus, the open set and closed set do not make a confrontation concept but a complementation concept. In this respect we have the following lemma and theorem.

Lemma 6.5 Let S be an arbitrary subset of T . Then, we have

$$\overline{S^c} = (S^\circ)^c.$$

Proof Since $S^\circ \subset S$, $(S^\circ)^c \supset S^c$. As S° is open, $(S^\circ)^c$ is closed. Hence, $(S^\circ)^c \supset \overline{S^c}$, because $\overline{S^c}$ is the smallest closed set containing S^c . Next let C be an arbitrary closed set that contains S^c . Then C^c is an open set that is contained in S , and so we have $C^c \subset S^\circ$, because S° is the largest open set contained in S . Therefore, $C \supset (S^\circ)^c$. Then, if we choose $\overline{S^c}$ for C , we must have $\overline{S^c} \supset (S^\circ)^c$. Combining this relation with $(S^\circ)^c \supset \overline{S^c}$ obtained above, we get $\overline{S^c} = (S^\circ)^c$.

Theorem 6.3[4] A necessary and sufficient condition for S to be both open and closed at once is $S^b = \emptyset$.

Proof

1. Necessary condition: Let S be open and closed. Then, $S = \overline{S}$ and $S = S^\circ$. Suppose that $S^b \neq \emptyset$. Then, from Definition 6.5 we would have $\exists a \in \overline{S} \cap \overline{S^c}$ for some a . Meanwhile, from Lemma 6.5 we have $\overline{S^c} = (S^\circ)^c$. This leads to $a \in \overline{S} \cap (S^\circ)^c$. However, from the assumption we would have $a \in S \cap S^c$. We have no such a , however. Thus, using the proof by contradiction we must have $S^b = \emptyset$.
2. Sufficient condition: Suppose that $S^b = \emptyset$. Starting with $S \supset S^\circ$ and $S \neq S^\circ$, let us assume that there is a such that $a \in S$ and $a \notin S^\circ$. That is, we have $a \in (S^\circ)^c \Rightarrow a \in S \cap (S^\circ)^c \subset \overline{S} \cap (S^\circ)^c = \overline{S} \cap \overline{S^c} = S^b$, in contradiction to the supposition. Thus, we must not have such a , implying that $S = S^\circ$. Next, starting with $\overline{S} \supset S$ and $S \neq \overline{S}$, we assume that there is a such that $a \in \overline{S}$ and $a \notin S$. That is, we have $a \in S^c \Rightarrow a \in \overline{S} \cap S^c \subset \overline{S} \cap \overline{S^c} = S^b$, in contradiction to the supposition. Thus, we must not have such a , implying that $S = \overline{S}$. Combining this relation with $S = S^\circ$ obtained above, we get $\overline{S} = S = S^\circ$. In other words, S is both open and closed at once. These complete the proof.

The above-mentioned set is sometimes referred to as a closed–open set or a *clopen* set as a portmanteau word. An interesting example can be seen in Chap. 20 in relation to topological groups (or continuous groups).

(d) Accumulation Points and Isolated Points

We have another important class of elements and sets that include accumulation points and isolated points.

Definition 6.6 Let (T, τ) be a topological space and S be a subset of T . Suppose that we have a point $p \in T$. If for any neighborhood N of p we have $N \cap (S - \{p\}) \neq \emptyset$, then p is called an accumulation point of S . A set comprising all the accumulation points of S is said to be a derived set of S and denoted by S^d .

Note that from Definition 6.4 we have

$$N \cap (S - \{p\}) \neq \emptyset \Leftrightarrow p \in \overline{S - \{p\}}, \quad (6.21)$$

where N is an arbitrary neighborhood of p .

Definition 6.7 Let S be a subset of T . Suppose that we have a point $p \in S$. If for some neighborhood N of p we have $N \cap (S - \{p\}) = \emptyset$, then p is called an isolated point of S . A set comprising only isolated points of S is said to be a discrete set (or subset) of S and we denote it by S^{dsc} .

Note that if p is an isolated point, $N \cap (S - \{p\}) = N \cap S - N \cap \{p\} = N \cap S - \{p\} = \emptyset$. That is, $\{p\} = N \cap S$. Therefore, any isolated points of S are contained in S . Contrary to this, the accumulation points are not necessarily contained in S .

Comparing Definitions 6.6 and 6.7, we notice that the latter definition is obtained by negation of the statement of the former definition. This implies that any points of a set are classified into mutually exclusive two alternatives, i.e., the accumulation points or isolated points.

We divide S^d into a direct sum of two sets such that

$$S^d = (S \cup S^c) \cap S^d = (S^d \cap S) \cup (S^d \cap S^c), \quad (6.22)$$

where with the second equality we used (6.7).

Now, we define

$$(S^d)^+ \equiv S^d \cap S \text{ and } (S^d)^- \equiv S^d \cap S^c.$$

Then, (6.22) means that S^d is divided into two sets consisting of the accumulation points, i.e., $(S^d)^+$ that belongs to S and $(S^d)^-$ that does not belong to S . Thus, as another direct sum we have

$$S = (S^d \cap S) \cup S^{dsc} = (S^d)^+ \cup S^{dsc}. \quad (6.23)$$

Equation (6.23) represents the direct sum consisting of a part of the derived set and the whole discrete set. Here we have the following theorem.

Theorem 6.4 Let S be a subset of T . Then we have a following equation:

$$\bar{S} = S^d \cup S^{dsc} \quad (S^d \cap S^{dsc} = \emptyset). \quad (6.24)$$

Proof Let us assume that p is an accumulation point of S . Then, we have $p \in \overline{S - \{p\}} \subset \bar{S}$. Meanwhile, $S^{dsc} \subset S \subset \bar{S}$. This implies that $\bar{S}$ contains both S^d and S^{dsc} . That is, we have

$$\bar{S} \supset S^d \cup S^{dsc}.$$

It is natural to ask how to deal with other points that do not belong to $S^d \cup S^{dsc}$. Taking account of the above discussion on the accumulation points and isolated points, however, such remainder points should again be classified into accumulation and isolated points. Since all the isolated points are contained in S , they can be taken into $\bar{S}$.

Suppose that among the accumulation points, some points p are not contained in S ; i.e., $p \notin S$. In that case, we have $S - \{p\} = S$, namely $\overline{S - \{p\}} = \bar{S}$. Thus, $p \in \overline{S - \{p\}} \Leftrightarrow p \in \bar{S}$. From (6.21), in turn, this implies that in case p is not contained in S , for p to be an accumulation point of S is equivalent to that p is an adherent point of S . Then, those points p can be taken into $\bar{S}$ as well. Thus, finally, we get (6.24). From Definitions 6.6 and 6.7, obviously we have $S^d \cap S^{dsc} = \emptyset$. This completes the proof.

Rewriting (6.24), we get

$$\bar{S} = (S^d)^- \cup (S^d)^+ \cup S^{dsc} = (S^d)^- \cup S. \quad (6.25)$$

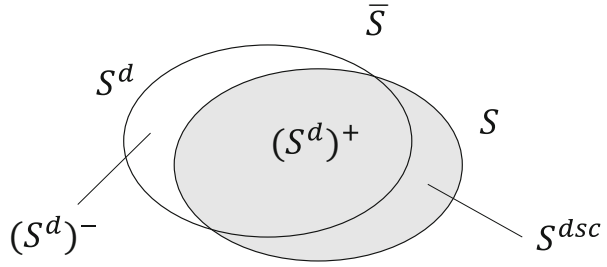
We have other important theorems.

Theorem 6.5 Let S be a subset of T . Then we have a following relation:

$$\bar{S} = S^d \cup S. \quad (6.26)$$

Proof Let us assume that $p \in \bar{S}$. Then, we have the following two cases: (i) if $p \in S$, we have trivially $p \in S \subset S^d \cup S$. (ii) Suppose $p \notin S$. Since $p \in \bar{S}$, from Definition 6.4 any neighborhood of p contains a point of S (other than p). Then, from Definition 6.6, p is an accumulation point of S . Thus, we have $p \in S^d \subset S^d \cup S$ and, hence, get $\bar{S} \subset S^d \cup S$. Conversely, suppose that $p \in S^d \cup S$. Then, (i) if $p \in S$, obviously we have $p \in \bar{S}$. (ii) Suppose $p \in S^d$. Then, from Definition 6.6 any neighborhood of

Fig. 6.6 Relationship among S , $\bar{S}$, S^d , and S^{dsc} . Regarding the symbols and notations, see text



$$\begin{aligned}\bar{S} &= S^d \cup S = S^d \cup S^{dsc}, \\ S^d &= (S^d)^+ \cup (S^d)^-\end{aligned}$$

p contains a point of S (other than p). Thus, from Definition 6.4, $p \in \bar{S}$. Taking into account the above two cases, we get $S^d \cup S \subset \bar{S}$. Combining this with $\bar{S} \subset S^d \cup S$ obtained above, we get (6.26). This completes the proof.

Theorem 6.6 Let S be a subset of T . A necessary and sufficient condition for S be a closed set is that S contains all the accumulation points of S .

Proof From Theorem 6.2, that S is a closed set is equivalent to $S = \bar{S}$. From Theorem 6.5 this is equivalent to $S = S^d \cup S$. Obviously, this is equivalent to $S^d \subset S$. In other words, S contains all the accumulation points of S . This completes the proof.

As another proof, taking account of (6.25), we have

$$(S^d)^- = \emptyset \Leftrightarrow \bar{S} = S \Leftrightarrow S \text{ is a closed set (from Theorem 6.2)}.$$

This relation is equivalent to the statement that S contains all the accumulation points of S .

In Fig. 6.6 we show the relationship among S , $\bar{S}$, S^d , and S^{dsc} . From Fig. 6.6, we can readily show that

$$S^{dsc} = \bar{S} - S^d = S - S^d.$$

We have a further example of direct sum decomposition. Using Lemma 6.5, we have

$$S^b = \bar{S} \cap \bar{S}^c = \bar{S} \cap (S^\circ)^c = \bar{S} - S^\circ. \quad (6.27)$$

Since $\bar{S} \supset S^\circ$, we get

$$\bar{S} = S^\circ \cup S^b; S^\circ \cap S^b = \emptyset.$$

Notice that (6.27) is a succinct expression of Theorem 6.3. That is,

$$S^b = \emptyset \Leftrightarrow \bar{S} = S^\circ \Leftrightarrow S \text{ is a clopen set.}$$

(e) Connectedness

In the precedent several paragraphs we studied various building blocks and their properties of the topological space and sets contained in the space. In this paragraph, we briefly mention the relationship between the sets. The connectedness is an important concept in the topology. The concept of the connectedness is associated with the subspaces defined earlier in this section. In terms of the connectedness, a topological space can be categorized into two main types; a subspace of the one type is *connected* and that of the other is *disconnected*.

Let A be a subspace of T . Intuitively, for A to be connected is defined as follows: Let z_1 and z_2 be any two points of A . If z_1 and z_2 can be joined by a continuous line that is contained entirely within A (see Fig. 6.7a), then A is said to be connected. Of the connected subspaces, suppose that the subspace A has the property that all the points inside any closed path (i.e., continuous closed line) C within A contain only the points that belong to A (Fig. 6.7b). Then, that subspace A is said to be *simply connected*. A subspace that is not simply connected but connected is said to be *multiply connected*. A typical example for the latter is a torus. If a subspace is given as a union of two (or more) disjoint non-empty (open) sets, that subspace is said to be disconnected. Notice that if two (or more) sets have no element in common, those sets are said to be disjoint. Fig. 6.7c shows an example for which a disconnected subspace A is a union of two disjoint sets B and C .

Interesting examples can be seen in Chap. 20 in relation to the connectedness.

6.1.3 T_I -Space

We can “enter” a variety of topologies τ into a set T to get a topological space (T, τ) . We have two extremes of them. One is an indiscrete topology (or trivial topology) and the other is a discrete topology [2]. For the former the topology is characterized as

$$\tau_* = \{\emptyset, T\} \tag{6.28}$$

and the latter case is

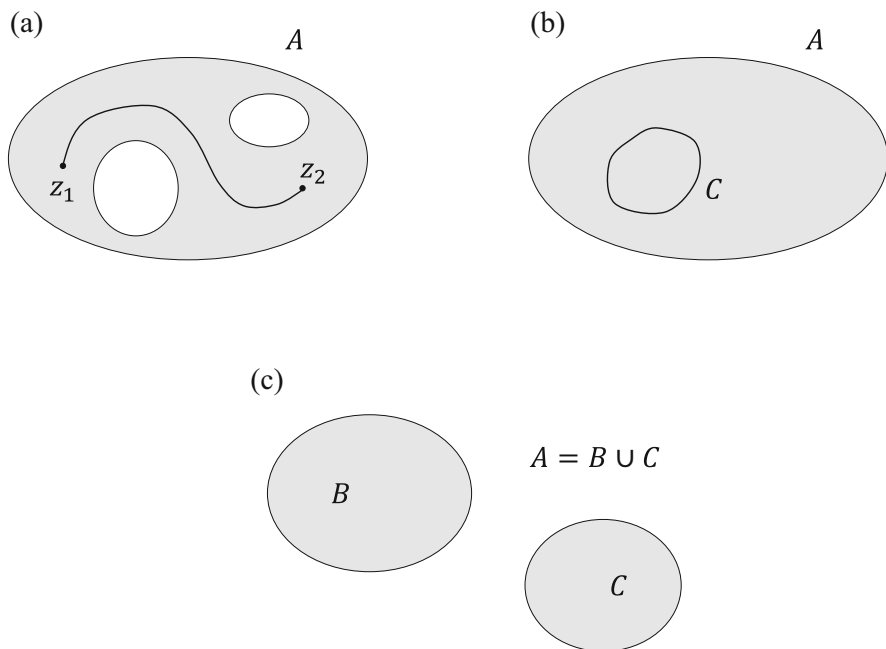


Fig. 6.7 Connectedness of subspaces. (a) Connected subspace A . Any two points z_1 and z_2 of A can be connected by a continuous line that is contained entirely within A . (b) Simply connected subspace A . All the points inside any closed path C within A contain only the points that belong to A . (c) Disconnected subspace A that comprises two disjoint sets B and C

$$\tau^* = \{\emptyset, \text{all the subsets of } T, T\}. \quad (6.29)$$

With τ^* all the subsets are open and complements to the individual subsets are closed by Definition 6.1. However, such closed subsets are contained in τ^* , and so they are again open. Thus, all the subsets are clopen sets. Practically speaking, the aforementioned two extremes are of less interest and value. For this reason, we need some moderate separation conditions with topological spaces. These conditions are well-established as the separation axioms. We will not get into details about the discussion [2], but we mention the first separation axiom (Fréchet axiom) that produces the T_1 -space.

Definition 6.8 Let (T, τ) be a topological space. Suppose that with respect to $\forall x, y (x \neq y) \in T$ there is a neighborhood N in such a way that

$$x \in N \text{ and } y \notin N.$$

The topological space that satisfies the above separation condition is called a T_1 -space.

We mention two important theorems of the T_1 -space.

Theorem 6.7 A necessary and sufficient condition for (T, τ) to be a T_1 -space is that for each point $x \in T$, $\{x\}$ is a closed set of T .

Proof

1. Necessary condition: Let (T, τ) be a T_1 -space. Choose any pair of elements x, y ($x \neq y$) $\in T$. Then, from Definition 6.8 there is a neighborhood $\exists N$ of y ($\neq x$) that does not contain x . Then, we have $y \in N$ and $N \cap \{x\} = \emptyset$. From Definition 6.4 this implies that $y \notin \overline{\{x\}}$. Thus, we have $y \neq x \Rightarrow y \notin \overline{\{x\}}$. This means that $y \in \overline{\{x\}} \Rightarrow y = x$. Namely, $\overline{\{x\}} \subset \{x\}$, but from (6.11) we have $\{x\} \subset \overline{\{x\}}$. Thus, $\overline{\{x\}} = \{x\}$. From Theorem 6.2 this shows that $\{x\}$ is a closed set of T .
2. Sufficient condition: Suppose that $\{x\}$ is a closed set of T . Choose any x and y such that $y \neq x$. Since $\{x\}$ is a closed set, $N = T - \{x\}$ is an open set that contains y ; i.e., $y \in T - \{x\}$ and $x \notin T - \{x\}$. Since $y \in T - \{x\} \subset T - \{x\}$ where $T - \{x\}$ is an open set, from Definition 6.2 $N = T - \{x\}$ is a neighborhood of y , and moreover N does not contain x . Then, from Definition 6.8 (T, τ) is a T_1 -space. These complete the proof.

A set comprising a unit element is called a singleton or a unit set.

Theorem 6.8 Let S be a subset of a T_1 -space. Then S^d is a closed set in the T_1 -space.

Proof Let p be a point of S . Suppose $p \in \overline{S^d}$. Suppose, furthermore, $p \notin S^d$. Then, from Definition 6.6 there is some neighborhood $\exists N$ of p such that $N \cap (S - \{p\}) = \emptyset$. Since $p \in \overline{S^d}$, from Definition 6.4 we must have $N \cap S^d \neq \emptyset$ at once for this special neighborhood N of p . Then, as we have $p \notin S^d$, we can take a point q such that $q \in N \cap S^d$ and $q \neq p$. Meanwhile, we may choose an open set N for the above neighborhood of p (i.e., an open neighborhood), because $p \in N (=N^\circ) \subset N$. Since $N \cap (T - \{p\}) = N - \{p\}$ is an open set from Theorem 6.7 as well as Axiom (O2) and Definition 6.1, $N - \{p\}$ is an open neighborhood of q which does not contain p . Namely, we have $q \in N - \{p\} [= (N - \{p\})^\circ] \subset N - \{p\}$. Note that $q \in N - \{p\}$ and $p \notin N - \{p\}$, in agreement with Definition 6.8. As $q \in S^d$, again from Definition 6.6 we have

$$(N - \{p\}) \cap (S - \{q\}) \neq \emptyset. \quad (6.30)$$

This implies that $N \cap (S - \{q\}) \neq \emptyset$ as well. However, it is in contradiction to the original relation of $N \cap (S - \{p\}) = \emptyset$, which is obtained from $p \notin S^d$. Then, we must have $p \in S^d$. Thus, from the original supposition we get $\overline{S^d} \subset S^d$. Meanwhile, we have $\overline{S^d} \supset S^d$ from (6.11). We have $\overline{S^d} = S^d$ accordingly. From Theorem 6.2, this means that S^d is a closed set. Hence, we have proven the theorem.

The above two theorems are intuitively acceptable. For, in a one-dimensional Euclidean space $\mathbb{R}$ we express a singleton $\{x\}$ as $[x, x]$ and a closed interval as $[x, y]$ ($x \neq y$). With the latter, $[x, y]$ might well be expressed as $(x, y)^d$. Such subsets are well-known as closed sets. According to the separation axioms, various topological

spaces can be obtained by imposing stronger constraints upon the T_1 -space [2]. A typical example for this is a metric space. In this context, the T_1 -space has acquired primitive notions of metric that can be shared with the metric space. Definitions of the metric (or distance function) and metric space will briefly be summarized in Chap. 13 in reference to an inner product space.

In the above discussion, we have described a brief outline of the set theory and topology. We use the results in this chapter and in Part IV as well.

6.1.4 Complex Numbers and Complex Plane

Returning to (6.1), we further investigate how to represent a complex number. In Fig. 6.8 we redraw a complex plane and a complex number on it. The complex plane is a natural extension of a real two-dimensional orthogonal coordinate plane (Cartesian coordinate plane) that graphically represents a pair of real numbers x and y as (x, y) . In the complex plane we represent a complex number as

$$z = x + iy \quad (6.1)$$

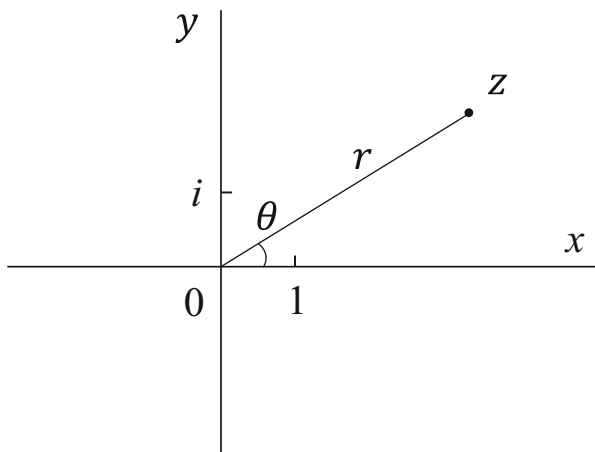
by designating x as an abscissa on the real axis and y as an ordinate on the imaginary axis. The two axes are orthogonal to each other.

An absolute value (or modulus) of z is defined as a real non-negative number

$$|z| = \sqrt{x^2 + y^2}. \quad (6.31)$$

Analogously to a real two-dimensional polar coordinate, we can introduce in the complex plane a non-negative radial coordinate r and an angular coordinate θ

Fig. 6.8 Complex plane where z is expressed in a polar form using a non-negative radial coordinate r and an angular coordinate θ



(Fig. 6.8). In the complex analysis the angular coordinate is called an argument and denoted by $\arg z$ so as to be

$$\arg z = \theta + 2\pi n; n = 0, \pm 1, \pm 2, \dots$$

In Fig. 6.8, x and y are given by

$$x = r \cos \theta \text{ and } y = r \sin \theta.$$

Thus, we have a polar form of z expressed as

$$z = r(\cos \theta + i \sin \theta). \quad (6.32)$$

Using the well-known Euler's formula (or Euler's identity)

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad (6.33)$$

we get

$$z = r e^{i\theta}. \quad (6.34)$$

From (6.32) and (6.34), we have

$$z^n = r^n e^{in\theta} = r^n (\cos \theta + i \sin \theta)^n = r^n (\cos n\theta + i \sin n\theta), \quad (6.35)$$

where the last equality comes from replacing θ with $n\theta$ in (6.33). Comparing both sides with the last equality of (6.35), we have

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta. \quad (6.36)$$

Equation (6.36) is called the de Moivre's theorem. This relation holds with n being integers including zero along with rational numbers including negative numbers. Euler's formula (6.33) immediately leads to the following important formulae:

$$\cos \theta = \frac{1}{2} (e^{i\theta} + e^{-i\theta}), \quad \sin \theta = \frac{1}{2i} (e^{i\theta} - e^{-i\theta}). \quad (6.37)$$

Notice that although in (6.32) and (6.33) we assumed that θ is a real number, (6.33) and (6.37) hold with any complex number θ . Note also that (6.33) results from (6.37) and the following definition of power series expansion of the exponential function

$$e^z \equiv \sum_{k=0}^{\infty} \frac{z^k}{k!}, \quad (6.38)$$

where z is any complex number [5]. In fact, from (6.37) and (6.38) we have the following familiar expressions of power series expansion of the cosine and sine functions

$$\cos z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k)!}, \quad \sin z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!}.$$

These expressions frequently appear in this chapter.

Next, let us think of a pair of complex numbers z and w with w expressed as $w = u + iv$. Then, we have

$$z - w = (x - u) + i(y - v).$$

Using (6.31), we get

$$|z - w| = \sqrt{(x - u)^2 + (y - v)^2}. \quad (6.39)$$

Equation (6.39) represents the “distance” between z and w . If we define a following function $\rho(z, w)$ such that

$$\rho(z, w) \equiv |z - w|,$$

$\rho(z, w)$ defines a metric (or distance function); see Chap. 13. Thus, the metric gives a distance between any arbitrarily chosen pair of elements $z, w \in \mathbb{C}$ and $(\mathbb{C}, \rho)$ can be dealt with as a metric space. This makes it easy to view the complex plane as a topological space. Discussions about the set theory and topology already developed earlier in this chapter basically hold.

In the theory of analytic functions, a subset A depicted in Fig. 6.7 can be regarded as a part of the complex plane. Since the complex plane $\mathbb{C}$ itself represents a topological space, the subset A may be viewed as a subspace of $\mathbb{C}$. A complex function $f(z)$ of a complex variable z of (6.1) is usually defined in a connected open set called a *region* [5]. The said region is of practical use among various subspaces and can be an entire domain of $\mathbb{C}$ or a subset of $\mathbb{C}$. The connectedness of the region can be considered in a manner similar to that described in Sect. 6.1.2.

6.2 Analytic Functions of a Complex Variable

Since the complex number and complex plane have been well-characterized in the previous section, we deal with the complex functions of a complex variable in the complex plane.

Taking the complex conjugate of (6.1), we have

$$z^* = x - iy. \quad (6.40)$$

Combining (6.1) and (6.40) in a matrix form, we get

$$\begin{pmatrix} z & z^* \end{pmatrix} = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}. \quad (6.41)$$

The matrix $\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}$ is non-singular, and so the inverse matrix exists (see Sect. 11.3) and $\begin{pmatrix} x & y \end{pmatrix}$ can be described as

$$\begin{pmatrix} x & y \end{pmatrix} = \frac{1}{2} \begin{pmatrix} z & z^* \end{pmatrix} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \quad (6.42)$$

or with individual components we have

$$x = \frac{1}{2}(z + z^*) \text{ and } y = -\frac{i}{2}(z - z^*), \quad (6.43)$$

which can be understood as the “vector synthesis” (see Fig. 6.9).

Equations (6.41) and (6.42) imply that two sets of variables $\begin{pmatrix} x & y \end{pmatrix}$ and $\begin{pmatrix} z & z^* \end{pmatrix}$ can be dealt with on an equal basis. According to the custom, however, we prioritize the notation using $\begin{pmatrix} x & y \end{pmatrix}$ over that of $\begin{pmatrix} z & z^* \end{pmatrix}$. Hence, the complex function f is described as

$$f(x, y) = u(x, y) + iv(x, y), \quad (6.44)$$

where both $u(x, y)$ and $v(x, y)$ are the real functions of real variables x and y .

Here, let us pause for a second to consider the differentiation of a function. Suppose that a mathematical function is defined on a real domain (i.e., a real number line). Consider whether that function is differentiable at a certain point x_0 of the real number line. On this occasion, we can approach x_0 only from two directions, i.e., from the side of $x < x_0$ (from the left) or from the side of $x_0 < x$ (from the right); see Fig. 6.10a. Meanwhile, suppose that a mathematical function is defined on a complex domain (i.e., a complex plane). Also consider whether the function is differentiable at a certain point z_0 of the complex plane. In this case, we can approach z_0 from continuously varying directions; see Fig. 6.10b where only four directions are depicted.

In this context let us think of a simple example.

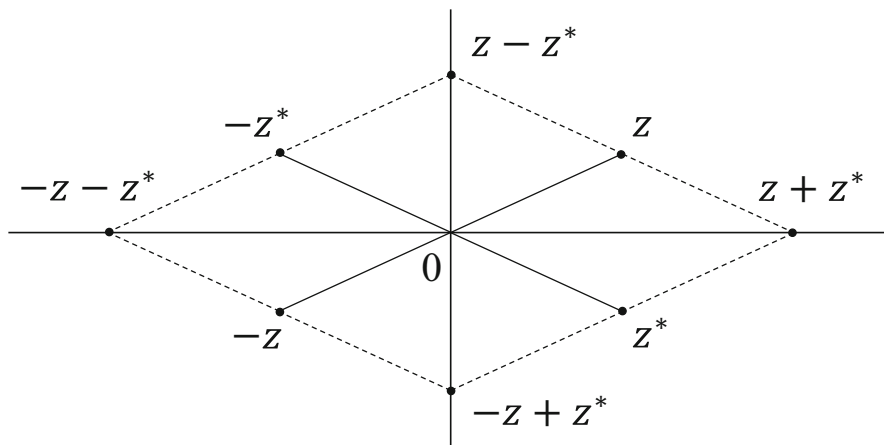


Fig. 6.9 Vector synthesis of z and z^*

Example 6.1 Let $f(x, y)$ be a function described by

$$f(x, y) = 2x + iy = z + x. \quad (6.45)$$

As remarked above, substituting (6.43) for (6.45), we could have

$$\begin{aligned} h(z, z^*) &= 2 \cdot \frac{1}{2}(z + z^*) + i \left[-\frac{i}{2}(z - z^*) \right] = z + z^* + \frac{1}{2}(z - z^*) \\ &= \frac{1}{2}(3z + z^*), \end{aligned}$$

where $f(x, y)$ and $h(z, z^*)$ denote different functional forms. The derivative (6.45) varies depending on a way z_0 is approached. For instance, think of

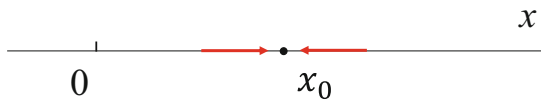
$$\left. \frac{df}{dz} \right|_0 = \lim_{z \rightarrow 0} \frac{f(0 + z) - f(0)}{z} = \lim_{x \rightarrow 0, y \rightarrow 0} \frac{2x + iy}{x + iy} = \lim_{x \rightarrow 0, y \rightarrow 0} \frac{2x^2 + y^2 - ixy}{x^2 + y^2}.$$

Suppose that the differentiation is taken along a straight line in a complex plane represented by $iy = (ik)x$ (k, x, y : real). Then, we have

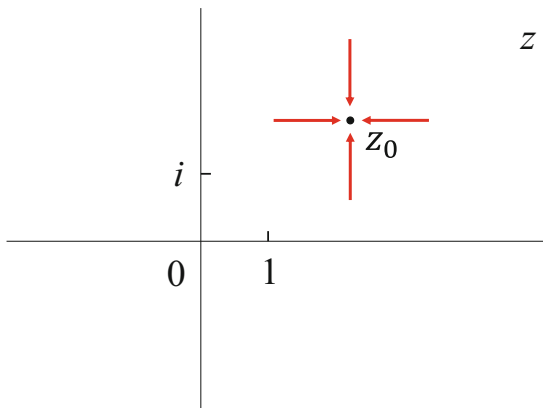
Fig. 6.10 Ways a number (real or complex) is approached. (a) Two ways a number x_0 is approached in a real number line.

(b) Various ways a number z_0 is approached in a complex plane. Here, only four ways are indicated

(a)



(b)



$$\begin{aligned} \frac{df}{dz}\bigg|_0 &= \lim_{x \rightarrow 0, y \rightarrow 0} \frac{2x^2 + k^2x^2 - ikx^2}{x^2 + k^2x^2} = \lim_{x \rightarrow 0, y \rightarrow 0} \frac{2 + k^2 - ik}{1 + k^2} \\ &= \frac{2 + k^2 - ik}{1 + k^2}. \end{aligned} \quad (6.46)$$

However, this means that $\frac{df}{dz}\big|_0$ takes varying values depending upon k . Namely, $\frac{df}{dz}\big|_0$ cannot uniquely be defined but depends on different ways to approach the origin of the complex plane. Thus, we find that the derivative takes different values depending on straight lines along which the differentiation is taken. This means that $f(x, y)$ is not differentiable or analytic at $z = 0$.

Meanwhile, think of $g(z)$ expressed as

$$g(z) = x + iy = z. \quad (6.47)$$

In this case, we get

$$\left. \frac{dg}{dz} \right|_0 = \lim_{z \rightarrow 0} \frac{g(0+z) - g(0)}{z} = \lim_{z \rightarrow 0} \frac{z-0}{z} = 1.$$

As a result, the derivative takes the same value 1, regardless of what straight lines the differentiation is taken along.

Though simple, the above example gives us a heuristic method. As before, let $f(z)$ be a function described as (6.44) such that

$$f(z) = f(x, y) = u(x, y) + iv(x, y), \quad (6.48)$$

where both $u(x, y)$ and $v(x, y)$ possess the first-order partial derivatives with respect to x and y . Then we have a derivative $df(z)/dz$ such that

$$\begin{aligned} \frac{df(z)}{dz} &= \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \\ &= \lim_{\Delta x \rightarrow 0, \Delta y \rightarrow 0} \frac{u(x + \Delta x, y + \Delta y) - u(x, y) + i[v(x + \Delta x, y + \Delta y) - v(x, y)]}{\Delta x + i\Delta y}. \end{aligned} \quad (6.49)$$

We wish to seek the condition that $\frac{df(z)}{dz}$ gives the same result regardless of the order of taking the limit of $\Delta x \rightarrow 0$ and $\Delta y \rightarrow 0$. That is, taking the limit of $\Delta y \rightarrow 0$ first, we have

$$\begin{aligned} \frac{df(z)}{dz} &= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y) + i[v(x + \Delta x, y) - v(x, y)]}{\Delta x} = \frac{\partial u(x, y)}{\partial x} \\ &\quad + i \frac{\partial v(x, y)}{\partial x}. \end{aligned} \quad (6.50)$$

Next, taking the limit of $\Delta x \rightarrow 0$ first, we get

$$\begin{aligned} \frac{df(z)}{dz} &= \lim_{\Delta y \rightarrow 0} \frac{u(x, y + \Delta y) - u(x, y) + i[v(x, y + \Delta y) - v(x, y)]}{i\Delta y} = \\ &\quad -i \frac{\partial u(x, y)}{\partial y} + \frac{\partial v(x, y)}{\partial y}. \end{aligned} \quad (6.51)$$

Consequently, by equating the real and imaginary parts of (6.50) and (6.51) we must have

$$\frac{\partial u(x, y)}{\partial x} = \frac{\partial v(x, y)}{\partial y}, \quad (6.52)$$

$$\frac{\partial v(x, y)}{\partial x} = -\frac{\partial u(x, y)}{\partial y}. \quad (6.53)$$

These relationships (6.52) and (6.53) are called the Cauchy–Riemann conditions. Differentiating (6.52) with respect to x and (6.53) with respect to y and further subtracting one from the other, we get

$$\frac{\partial^2 u(x, y)}{\partial x^2} + \frac{\partial^2 u(x, y)}{\partial y^2} = 0. \quad (6.54)$$

Similarly we have

$$\frac{\partial^2 v(x, y)}{\partial x^2} + \frac{\partial^2 v(x, y)}{\partial y^2} = 0. \quad (6.55)$$

From the above discussion, we draw several important implications.

1. From (6.50), we have

$$\frac{df(z)}{dz} = \frac{\partial f}{\partial x}. \quad (6.56)$$

2. Also from (6.51) we obtain

$$\frac{df(z)}{dz} = -i \frac{\partial f}{\partial y}. \quad (6.57)$$

Meanwhile, we have

$$\frac{\partial}{\partial x} = \frac{\partial z}{\partial x} \frac{\partial}{\partial z} + \frac{\partial z^*}{\partial x} \frac{\partial}{\partial z^*} = \frac{\partial}{\partial z} + \frac{\partial}{\partial z^*}. \quad (6.58)$$

Also, we get

$$\frac{\partial}{\partial y} = \frac{\partial z}{\partial y} \frac{\partial}{\partial z} + \frac{\partial z^*}{\partial y} \frac{\partial}{\partial z^*} = i \left(\frac{\partial}{\partial z} - \frac{\partial}{\partial z^*} \right). \quad (6.59)$$

As in the case of Example 6.1, we rewrite $f(z)$ as

$$f(z) \equiv F(z, z^*), \quad (6.60)$$

where the change in the functional form is due to the variables transformation. Hence, from (6.56) and using (6.58) we have

$$\frac{df(z)}{dz} = \frac{\partial F(z, z^*)}{\partial x} = \left(\frac{\partial}{\partial z} + \frac{\partial}{\partial z^*} \right) F(z, z^*) = \frac{\partial F(z, z^*)}{\partial z} + \frac{\partial F(z, z^*)}{\partial z^*}. \quad (6.61)$$

Similarly, we get

$$\frac{df(z)}{dz} = -i \frac{\partial F(z, z^*)}{\partial y} = \left(\frac{\partial}{\partial z} - \frac{\partial}{\partial z^*} \right) F(z, z^*) = \frac{\partial F(z, z^*)}{\partial z} - \frac{\partial F(z, z^*)}{\partial z^*}. \quad (6.62)$$

Equating (6.61) and (6.62), we obtain

$$\frac{\partial F(z, z^*)}{\partial z^*} = 0. \quad (6.63)$$

This clearly shows that if $F(z, z^*)$ is differentiable with respect to z , $F(z, z^*)$ does not depend on z^* , but only depends upon z . Thus, we may write the differentiable function as

$$F(z, z^*) \equiv f(z).$$

This is an outstanding characteristic of the complex function that is differentiable with respect to the complex variable z .

Taking the contraposition of the above statement, we say that if $F(z, z^*)$ depends on z^* , it is not differentiable with respect to z . Example 6.1 is one of the illustrations.

On the other way around, we consider what can happen if the Cauchy–Riemann conditions are satisfied [5]. Let $f(z)$ be a complex function described by

$$f(z) = u(x, y) + iv(x, y), \quad (6.48)$$

where $u(x, y)$ and $v(x, y)$ satisfy the Cauchy–Riemann conditions and possess continuous first-order partial derivatives with respect to x and y in some region of $\mathbb{C}$. Then, we have

$$u(x + \Delta x, y + \Delta y) - u(x, y) = \frac{\partial u(x, y)}{\partial x} \Delta x + \frac{\partial u(x, y)}{\partial y} \Delta y + \varepsilon_1 \Delta x + \delta_1 \Delta y, \quad (6.64)$$

$$v(x + \Delta x, y + \Delta y) - v(x, y) = \frac{\partial v(x, y)}{\partial x} \Delta x + \frac{\partial v(x, y)}{\partial y} \Delta y + \varepsilon_2 \Delta x + \delta_2 \Delta y, \quad (6.65)$$

where four positive numbers ε_1 , ε_2 , δ_1 , and δ_2 can be made arbitrarily small as Δx and Δy tend to be zero. The relations (6.64) and (6.65) result from the continuity of the first-order partial derivatives of $u(x, y)$ and $v(x, y)$.

Then, we have

$$\begin{aligned} & \frac{f(z + \Delta z) - f(z)}{\Delta z} \\ &= \frac{u(x + \Delta x, y + \Delta y) - u(x, y)}{\Delta z} + \frac{i[v(x + \Delta x, y + \Delta y) - v(x, y)]}{\Delta z} \\ &= \frac{\Delta x}{\Delta z} \left[\frac{\partial u(x, y)}{\partial x} + i \frac{\partial v(x, y)}{\partial x} \right] + \frac{\Delta y}{\Delta z} \left[\frac{\partial u(x, y)}{\partial y} + i \frac{\partial v(x, y)}{\partial y} \right] + \frac{\Delta x}{\Delta z} (\varepsilon_1 + i\varepsilon_2) + \frac{\Delta y}{\Delta z} \\ & \quad \times (\delta_1 + i\delta_2) \\ &= \frac{\Delta x}{\Delta z} \left[\frac{\partial u(x, y)}{\partial x} + i \frac{\partial v(x, y)}{\partial x} \right] + \frac{\Delta y}{\Delta z} \left[-\frac{\partial v(x, y)}{\partial x} + i \frac{\partial u(x, y)}{\partial x} \right] + \frac{\Delta x}{\Delta z} (\varepsilon_1 + i\varepsilon_2) \\ & \quad + \frac{\Delta y}{\Delta z} (\delta_1 + i\delta_2) \\ &= \frac{\Delta x + i\Delta y}{\Delta z} \cdot \frac{\partial u(x, y)}{\partial x} + \frac{(\Delta x + i\Delta y)}{\Delta z} \cdot i \frac{\partial v(x, y)}{\partial x} + \frac{\Delta x}{\Delta z} (\varepsilon_1 + i\varepsilon_2) + \frac{\Delta y}{\Delta z} (\delta_1 + i\delta_2) \\ &= \frac{\partial u(x, y)}{\partial x} + i \frac{\partial v(x, y)}{\partial x} + \frac{\Delta x}{\Delta z} (\varepsilon_1 + i\varepsilon_2) + \frac{\Delta y}{\Delta z} (\delta_1 + i\delta_2), \end{aligned} \quad (6.66)$$

where with the third equality we used the Cauchy–Riemann conditions (6.52) and (6.53). Accordingly, we have

$$\begin{aligned} & \frac{f(z + \Delta z) - f(z)}{\Delta z} - \left[\frac{\partial u(x, y)}{\partial x} + i \frac{\partial v(x, y)}{\partial x} \right] = \frac{\Delta x}{\Delta z} (\varepsilon_1 + i\varepsilon_2) + \frac{\Delta y}{\Delta z} \\ & \quad \times (\delta_1 + i\delta_2). \end{aligned} \quad (6.67)$$

Taking the absolute values of both sides of (6.67), we get

$$\left| \frac{f(z + \Delta z) - f(z)}{\Delta z} - \left[\frac{\partial u(x, y)}{\partial x} + i \frac{\partial v(x, y)}{\partial x} \right] \right| \leq \left| \frac{\Delta x}{\Delta z} (\varepsilon_1 + i\varepsilon_2) \right| + \left| \frac{\Delta y}{\Delta z} (\delta_1 + i\delta_2) \right|, \quad (6.68)$$

where $\left| \frac{\Delta x}{\Delta z} \right| = \frac{|\Delta x|}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} \leq 1$ and $\left| \frac{\Delta y}{\Delta z} \right| = \frac{|\Delta y|}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} \leq 1$. Taking the limit of $\Delta z \rightarrow 0$, by assumption both the terms of RHS of (6.68) approach zero. Thus,

$$\lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \frac{\partial u(x, y)}{\partial x} + i \frac{\partial v(x, y)}{\partial x} = \frac{\partial f}{\partial x}. \quad (6.69)$$

Alternatively, using partial derivatives of $u(x, y)$ and $v(x, y)$ with respect to y we can rewrite (6.69) as

$$\lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = -i \left[\frac{\partial u(x, y)}{\partial y} + i \frac{\partial v(x, y)}{\partial y} \right] = -i \frac{\partial f}{\partial y}. \quad (6.70)$$

Notice that (6.69) and (6.70) are equivalent to the Cauchy–Riemann conditions.

From (6.56) and (6.57), the relations (6.69) and (6.70) imply that for $f(z)$ to be differentiable requires that the first-order partial derivatives of $u(x, y)$ and $v(x, y)$ with respect to x and y should exist. Meanwhile, once $f(z)$ is found to be differentiable (or analytic), its higher order derivatives must be analytic as well (vide infra). This requires, in turn, that the above first-order partial derivatives should be continuous. (Note that the analyticity naturally leads to continuity.) Thus, the following theorem will follow.

Theorem 6.9 Let $f(z)$ be a complex function of complex variables z such that $f(z) = u(x, y) + iv(x, y)$, where x and y are the real variables. Then, a necessary and sufficient condition for $f(z)$ to be differentiable is that the first-order partial derivatives of $u(x, y)$ and $v(x, y)$ with respect to x and y exist and are continuous and that $u(x, y)$ and $v(x, y)$ satisfy the Cauchy–Riemann conditions.

Now, we formally give definitions of the differentiability and analyticity of a complex function of complex variables.

Definition 6.9 Let z_0 be a given point of the complex plane $\mathbb{C}$. Let $f(z)$ be defined in a neighborhood containing z_0 . If $f(z)$ is single-valued and differentiable at all points of this neighborhood containing z_0 , $f(z)$ is said to be analytic at z_0 . A point at which $f(z)$ is analytic is called a regular point of $f(z)$. A point at which $f(z)$ is not analytic is called a singular point of $f(z)$.

We emphasize that the above definition of analyticity at some point z_0 requires the single-valuedness and differentiability at all the points of a neighborhood containing z_0 . This can be understood by Fig. 6.10b and Example 6.1. The analyticity at a point is deeply connected to how and in which direction we take the limitation process.

Thus, we need detailed information about the neighborhood of the point in question to determine whether the function is analytic at that point.

The next definition is associated with a global characteristic of the analytic function.

Definition 6.10 Let $\mathcal{R}$ be a region contained in the complex plane $\mathbb{C}$. Let $f(z)$ be a complex function defined in $\mathcal{R}$. If $f(z)$ is analytic at all points of $\mathcal{R}$, $f(z)$ is said to be analytic in $\mathcal{R}$. In this case $\mathcal{R}$ is called a domain of analyticity. If the domain of analyticity is an entire complex plane $\mathbb{C}$, the function is called an entire function.

To understand various characteristics of the analytic functions, it is indispensable to introduce *Cauchy's integral formula*. In the next section we deal with the integration of complex functions.

6.3 Integration of Analytic Functions: Cauchy's Integral Formula

We can define the complex integration as a natural extension of Riemann integral with regard to a real variable. Suppose that there is a curve C in the complex plane. Both ends of the curve are fixed and located at z_a and z_b . Suppose also that individual points of the curve C are described using a parameter t such that

$$z = z(t) = x(t) + iy(t) \quad (t_0 \equiv t_a \leq t \leq t_n \equiv t_b), \quad (6.71)$$

where $z_0 \equiv z_a = z(t_a)$ and $z_n \equiv z_b = z(t_b)$ together with $z_i = z(t_i)$ ($0 \leq i \leq n$). For this parametrization, we assume that C is subdivided into n pieces designated by $z_0, z_1, \dots, z_n$.

Let $f(z)$ be a complex function defined in a region containing C . In this situation, let us assume the following summation S_n :

$$S_n = \sum_{i=1}^n f(\zeta_i)(z_i - z_{i-1}), \quad (6.72)$$

where ζ_i lies between z_{i-1} and z_i ($1 \leq i \leq n$). Taking the limit $n \rightarrow \infty$ and concomitantly $|z_i - z_{i-1}| \rightarrow 0$, we have

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\zeta_i)(z_i - z_{i-1}). \quad (6.73)$$

If the constant limit exists independent of choice of z_i ($1 \leq i \leq n-1$) and ζ_i ($1 \leq i \leq n$), this limit is called a contour integral of $f(z)$ along the curve C . We write it as

$$I = \lim_{n \rightarrow \infty} S_n \equiv \int_C f(z) dz = \int_{z_a}^{z_b} f(z) dz. \quad (6.74)$$

Using (6.48) and $z_i = x_i + iy_i$, we rewrite (6.72) as

$$\begin{aligned} S_n = & \sum_{i=1}^n [u(\zeta_i)(x_i - x_{i-1}) - v(\zeta_i)(y_i - y_{i-1})] \\ & + \sum_{i=1}^n i[v(\zeta_i)(x_i - x_{i-1}) + u(\zeta_i)(y_i - y_{i-1})]. \end{aligned}$$

Taking the limit of $n \rightarrow \infty$ as well as $|x_i - x_{i-1}| \rightarrow 0$ and $|y_i - y_{i-1}| \rightarrow 0$ ($0 \leq i \leq n$), we have

$$I = \int_C [u(x, y) dx - v(x, y) dy] + i \int_C [v(x, y) dx + u(x, y) dy]. \quad (6.75)$$

Further rewriting (6.75), we get

$$\begin{aligned} I &= \int_C \left[u(x, y) \frac{dx}{dt} dt - v(x, y) \frac{dy}{dt} dt \right] + i \int_C \left[v(x, y) \frac{dx}{dt} dt + u(x, y) \frac{dy}{dt} dt \right] \\ &= \int_C [u(x, y) + iv(x, y)] \frac{dx}{dt} dt + i \int_C [u(x, y) + iv(x, y)] \frac{dy}{dt} dt \\ &= \int_C [u(x, y) + iv(x, y)] \left(\frac{dx}{dt} + i \frac{dy}{dt} \right) dt = \int_C [u(x, y) + iv(x, y)] \frac{dz}{dt} dt, \\ &= \int_C f(z) \frac{dz}{dt} dt = \int_{t_a}^{t_b} f(z) \frac{dz}{dt} dt = \int_{z_a}^{z_b} f(z) dz. \end{aligned} \quad (6.76)$$

Notice that the contour integral I of (6.74), (6.75), or (6.76) depends in general on the paths connecting the points z_a and z_b . If so, (6.76) is merely of secondary importance. However, if $f(z)$ can be described by the derivative of another function, the situation becomes totally different. Suppose that we have

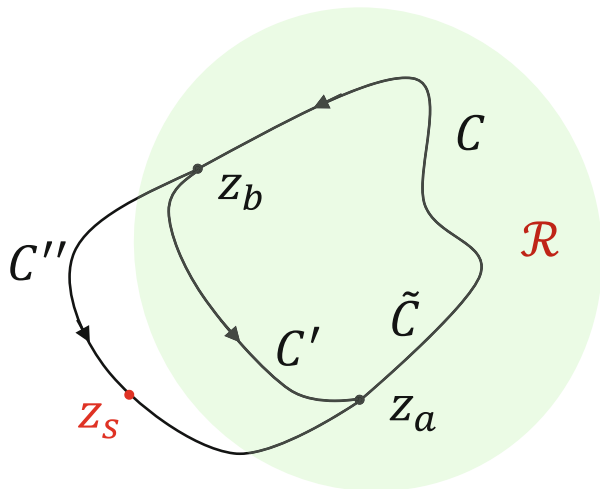
$$f(z) = \frac{d\tilde{F}(z)}{dz}. \quad (6.77)$$

If $\tilde{F}(z)$ is analytic, $f(z)$ is analytic as well. It is because a derivative of an analytic function is again analytic (vide infra). Then, we have

$$f(z) \frac{dz}{dt} = \frac{d\tilde{F}(z)}{dz} \frac{dz}{dt} = \frac{d\tilde{F}[z(t)]}{dt}. \quad (6.78)$$

Inserting (6.78) into (6.76), we get

Fig. 6.11 Region $\mathcal{R}$ and several contours for integration. Regarding the symbols and notations, see text



$$\int_C f(z) \frac{dz}{dt} dt = \int_{z_a}^{z_b} f(z) dz = \int_{t_a}^{t_b} \frac{d\tilde{F}[z(t)]}{dt} dt = \tilde{F}(z_b) - \tilde{F}(z_a). \quad (6.79)$$

In this case $\tilde{F}(z)$ is called a primitive function of $f(z)$. It is obvious from (6.79) that

$$\int_{z_a}^{z_b} f(z) dz = - \int_{z_b}^{z_a} f(z) dz. \quad (6.80)$$

This implies that the contour integral I does not depend on the path C connecting the points z_a and z_b . We must be careful, however, about a situation where there would be a singular point z_s of $f(z)$ in a region $\mathcal{R}$. Then, $f(z)$ is not defined at z_s . If one chooses C' for a contour of the return path in such a way that z_s is on C' (see Fig. 6.11), RHS of (6.80) cannot exist, nor can (6.80) hold. To avoid that situation, we have to “expel” such singularity from the region $\mathcal{R}$ so that $\mathcal{R}$ can wholly be contained in the domain of analyticity of $f(z)$ and that $f(z)$ can be analytic in the entire region $\mathcal{R}$.

Moreover, we must choose $\mathcal{R}$ so that the whole region inside any closed paths within $\mathcal{R}$ can contain only the points that belong to the domain of analyticity of $f(z)$. That is, the region $\mathcal{R}$ must be simply connected in terms of the analyticity of $f(z)$ (see Sect. 6.1). For a region $\mathcal{R}$ to be simply connected ensures that (6.80) holds with any pair of points z_a and z_b in $\mathcal{R}$, so far as the integration path with respect to (6.80) is contained in $\mathcal{R}$.

From (6.80), we further get

$$\int_{z_a}^{z_b} f(z)dz + \int_{z_b}^{z_a} f(z)dz = 0. \quad (6.81)$$

Since the integral does not depend on the paths, we can take a path from z_b to z_a along a curve C' (see Fig. 6.11). Thus, we have

$$\int_C f(z)dz + \int_{C'} f(z)dz = \int_{C+C'} f(z)dz = 0, \quad (6.82)$$

where $C + C'$ constitutes a closed curve $\tilde{C}$ as shown. In this way, in accordance with (6.81) we get

$$\int_{\tilde{C}} f(z)dz = 0, \quad (6.83)$$

where $\tilde{C}$ is followed counter-clockwise according to the custom.

In (6.83) any closed path $\tilde{C}$ can be chosen for the contour within the simply connected region $\mathcal{R}$. Hence, we reach the following important theorem.

Theorem 6.10: Cauchy's Integral Theorem [5] Let $\mathcal{R}$ be a simply connected region and let C be an arbitrary closed curve within $\mathcal{R}$. Let $f(z)$ be analytic on C and within the whole region enclosed by C . Then, we have

$$\oint_C f(z)dz = 0. \quad (6.84)$$

There is a variation in description of Cauchy's integral theorem. For instance, let $f(z)$ be analytic in $\mathcal{R}$ except for a singular point at z_s ; see Fig. 6.12a. Then, the domain of analyticity of $f(z)$ is not identical to $\mathcal{R}$, but is defined as $\mathcal{R} - \{z_s\}$. That is, the domain of analyticity of $f(z)$ is no longer simply connected and, hence, (6.84) is not generally

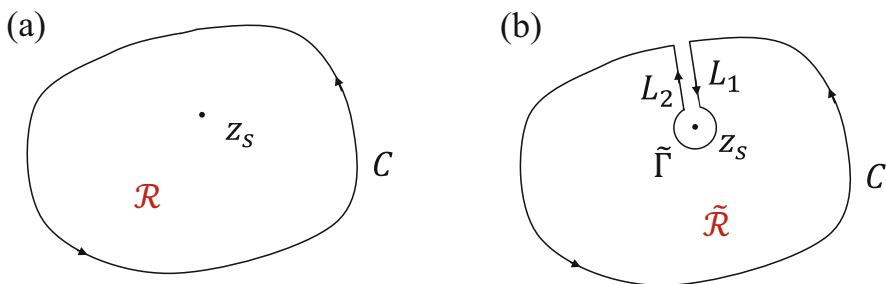


Fig. 6.12 Region $\mathcal{R}$ and contour C for integration. (a) A singular point at z_s is contained within $\mathcal{R}$. (b) The singular point at z_s is not contained within $\tilde{\mathcal{R}}$ so that $f(z)$ can be analytic in a simply connected region $\tilde{\mathcal{R}}$. Regarding the symbols and notations, see text

true. In such a case, we may deform the integration path of $f(z)$ so that its domain of analyticity can be simply connected and that (6.84) can hold. For example, we take $\tilde{\mathcal{R}}$ so that it can be surrounded by curves C and $\tilde{\Gamma}$ as well as lines L_1 and L_2 (Fig. 6.12b). As a result, z_s has been expelled from $\tilde{\mathcal{R}}$ and, at the same time, $\tilde{\mathcal{R}}$ becomes simply connected. Then, as a tangible form of (6.84) we get

$$\oint_{C+\tilde{\Gamma}+L_1+L_2} f(z)dz = 0, \quad (6.85)$$

where $\tilde{\Gamma}$ denotes the clockwise integration (see Fig. 6.12b) and the lines L_1 and L_2 are rendered as closely as possible. In (6.85) the integrations with L_1 and L_2 cancel, because they are taken in the reverse direction. Thus, we have

$$\oint_{C+\tilde{\Gamma}} f(z)dz = 0 \text{ or } \oint_C f(z)dz = \oint_{\Gamma} f(z)dz, \quad (6.86)$$

where Γ stands for the counter-clockwise integration. Equation (6.86) is valid as well, when there are more singular points. In that case, instead of (6.86) we have

$$\oint_C f(z)dz = \sum_i \oint_{\Gamma_i} f(z)dz, \quad (6.87)$$

where Γ_i is taken so that it can encircle individual singular points. Reflecting the nature of singularity, we get different results with (6.87). We will come back to this point later.

On the basis of Cauchy's integral theorem, we show an integral representation of an analytic function that is well-known as Cauchy's integral formula.

Theorem 6.11: Cauchy's Integral Formula Let $f(z)$ be analytic in a simply connected region $\mathcal{R}$. Let C be an arbitrary closed curve within $\mathcal{R}$ that encircles z . Then, we have

$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{\zeta - z} d\zeta, \quad (6.88)$$

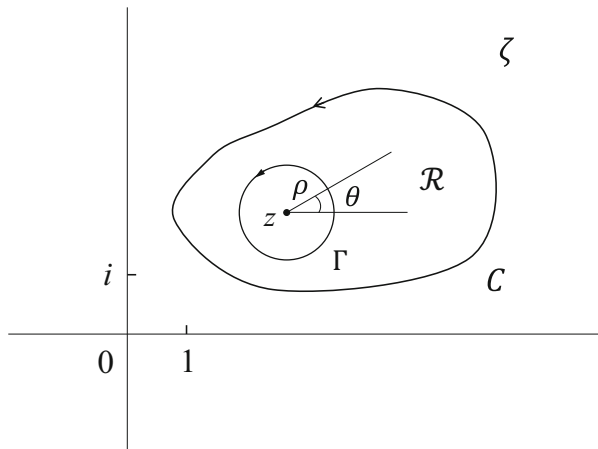
where the contour integration along C is taken in the counter-clockwise direction.

Proof Let Γ be a circle of a radius ρ in the complex plane so that z can be a center of the circle (see Fig. 6.13). Then, $f(\zeta)/(\zeta - z)$ has a singularity at $\zeta = z$. Therefore, we must evaluate (6.88) using (6.86). From (6.86), we have

$$\oint_C \frac{f(\zeta)}{\zeta - z} d\zeta = \oint_{\Gamma} \frac{f(\zeta)}{\zeta - z} d\zeta = f(z) \oint_{\Gamma} \frac{d\zeta}{\zeta - z} + \oint_{\Gamma} \frac{f(\zeta) - f(z)}{\zeta - z} d\zeta. \quad (6.89)$$

An arbitrary point ζ on the circle Γ is described as

Fig. 6.13 Simply connected region $\mathcal{R}$ encircled by a closed contour C in a complex plane ζ . A circle Γ of radius ρ centered at z is contained within $\mathcal{R}$. The contour integration is taken in the counter-clockwise direction along C or Γ



$$\zeta = z + \rho e^{i\theta}. \quad (6.90)$$

Then, taking infinitesimal quantities of (6.90), we have

$$d\zeta = \rho e^{i\theta} i d\theta = (\zeta - z) i d\theta. \quad (6.91)$$

Inserting (6.91) into (6.89), we get

$$\oint_{\Gamma} \frac{d\zeta}{\zeta - z} = \int_0^{2\pi} i d\theta = 2\pi i. \quad (6.92)$$

Rewriting (6.89), we obtain

$$\oint_{\Gamma} \frac{f(\zeta)}{\zeta - z} d\zeta = 2\pi i f(z) + \oint_{\Gamma} \frac{f(\zeta) - f(z)}{\zeta - z} d\zeta. \quad (6.93)$$

Since $f(z)$ is analytic, $f(z)$ is uniformly continuous [6]. Therefore, if we make ρ small enough with an arbitrary positive number ε , we have $|f(\zeta) - f(z)| < \varepsilon$, as $|\zeta - z| = \rho$. Considering this situation, we have

$$\begin{aligned} \left| \frac{1}{2\pi i} \oint_{\Gamma} \frac{f(\zeta)}{\zeta - z} d\zeta - f(z) \right| &= \left| \frac{1}{2\pi i} \oint_{\Gamma} \frac{f(\zeta)}{\zeta - z} d\zeta - f(z) \right| = \left| \frac{1}{2\pi i} \oint_{\Gamma} \frac{f(\zeta) - f(z)}{\zeta - z} d\zeta \right| \\ &\leq \frac{1}{2\pi} \oint_{\Gamma} \left| \frac{f(\zeta) - f(z)}{\zeta - z} \right| \cdot |d\zeta| < \frac{\varepsilon}{2\pi} \oint_{\Gamma} \frac{|d\zeta|}{|\zeta - z|} = \frac{\varepsilon}{2\pi} \int_0^{2\pi} d\theta = \varepsilon. \end{aligned} \quad (6.94)$$

Therefore, we get (6.88) in the limit of $\rho \rightarrow 0$. This completes the proof.

In the above proof, the domain of analyticity $\mathfrak{R}$ of $\frac{f(\zeta)}{\zeta - z}$ (with respect to ζ) is given by $\mathfrak{R} = \mathcal{R} \cap [\mathbb{C} - \{z\}] = \mathcal{R} - \{z\}$. Since both $\mathcal{R}$ and $\mathbb{C} - \{z\}$ are open sets (i.e., $\mathbb{C} - \{z\} = \{z\}^c$ is a complementary set of a closed set $\{z\}$ and, hence, an open set), $\mathfrak{R}$ is an open set as well; see Sect. 6.1. Notice that $\mathfrak{R}$ is not simply connected. For this reason, we had to evaluate (6.88) using (6.89).

Theorem 6.10 (Cauchy's integral theorem) and Theorem 6.11 (Cauchy's integral formula) play a crucial role in the theory of analytic functions.

To derive (6.94), we can equally use the Darboux's inequality [5]. This is intuitively obvious and frequently used in the theory of analytic functions.

Theorem 6.12: Darboux's Inequality [5] Let $f(z)$ be a function for which $|f(z)|$ is bounded on C . Here C is a piecewise continuous path in the complex plane. Then, with the following integral I described by

$$I = \int_C f(z) dz,$$

we have

$$\left| \int_C f(z) dz \right| \leq \max |f| \cdot L, \quad (6.95)$$

where L represents the arc length of the curve C for the contour integration.

Proof As discussed earlier in this section, the integral is the limit of $n \rightarrow \infty$ of the sum described by

$$S_n = \sum_{i=1}^n f(\zeta_i)(z_i - z_{i-1}) \quad (6.72)$$

with

$$I = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\zeta_i)(z_i - z_{i-1}). \quad (6.73)$$

Denoting the maximum modulus of $f(z)$ on C by $\max |f|$, we have

$$|S_n| \leq \sum_{i=1}^n |f(\zeta_i)| \cdot |z_i - z_{i-1}| \leq \max |f| \sum_{i=1}^n |z_i - z_{i-1}|. \quad (6.96)$$

The sum $\sum_{i=1}^n |z_i - z_{i-1}|$ on RHS of inequality (6.96) is the length of a polygon inscribed in the curve C . It is shorter than the arc length L of the curve C . Hence, for all n we have

$$|S_n| \leq \max |f| \cdot L.$$

$$\text{As } n \rightarrow \infty, |S_n| = |I| = \left| \int_C f(z) dz \right| \leq \max |f| \cdot L.$$

This completes the proof.

Applying the Darboux's inequality to (6.94), we get

$$\left| \oint_{\Gamma} \frac{f(\zeta) - f(z)}{\zeta - z} d\zeta \right| = \left| \oint_{\Gamma} \frac{f(\zeta) - f(z)}{\rho e^{i\theta}} d\zeta \right| \leq \max \left| \frac{f(\zeta) - f(z)}{\rho e^{i\theta}} \right| \cdot 2\pi\rho < 2\pi\varepsilon.$$

That is,

$$\left| \frac{1}{2\pi i} \oint_{\Gamma} \frac{f(\zeta) - f(z)}{\zeta - z} d\zeta \right| < \varepsilon. \quad (6.97)$$

So far, we dealt with the differentiation and integration as different mathematical manipulations. However, Cauchy's integral formula (or Cauchy's integral expression) enables us to relate and unify these two manipulations. We have the following proposition for this.

Proposition 6.1 [5] Let C be a piecewise continuous curve of finite length. (The curve C may or may not be closed.) Let $f(z)$ be a continuous function. The contour integration of $f(\zeta)/(\zeta - z)$ gives $\tilde{f}(z)$ such that

$$\tilde{f}(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta. \quad (6.98)$$

Then, $\tilde{f}(z)$ is analytic at any point z that does not lie on C .

Proof We consider the following expression:

$$\Delta \equiv \left| \frac{\tilde{f}(z + \Delta z) - \tilde{f}(z)}{\Delta z} - \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^2} d\zeta \right|, \quad (6.99)$$

where Δ is a real non-negative number. Describing (6.99) by use of (6.98) for $\tilde{f}(z + \Delta z)$ and $\tilde{f}(z)$, we get

$$\Delta = \frac{|\Delta z|}{2\pi} \left| \int_C \frac{f(\zeta)}{(\zeta - z - \Delta z)(\zeta - z)^2} d\zeta \right|. \quad (6.100)$$

To obtain (6.100), in combination with (6.98) we have calculated (6.99) as follows:

$$\begin{aligned}\Delta &= \left| \frac{1}{2\pi i \Delta z} \int_C \frac{f(\zeta)(\zeta - z)^2 - f(\zeta)(\zeta - z - \Delta z)(\zeta - z) - f(\zeta)\Delta z(\zeta - z - \Delta z)}{(\zeta - z - \Delta z)(\zeta - z)^2} d\zeta \right| \\ &= \left| \frac{1}{2\pi i \Delta z} \int_C \frac{f(\zeta)(\Delta z)^2}{(\zeta - z - \Delta z)(\zeta - z)^2} d\zeta \right|.\end{aligned}$$

Since z is not on C , the integrand of (6.100) is bounded. Then, as $\Delta z \rightarrow 0$, $|\Delta z| \rightarrow 0$ and $\Delta \rightarrow 0$ accordingly. From (6.99), this ensures the differentiability of $\tilde{f}(z)$. Then, by definition of the differentiation, $\frac{d\tilde{f}(z)}{dz}$ is given by

$$\frac{d\tilde{f}(z)}{dz} = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^2} d\zeta. \quad (6.101)$$

As $f(\zeta)$ is continuous, $\tilde{f}(z)$ is single-valued. Thus, $\tilde{f}(z)$ is found to be analytic at any point z that does not lie on C . This completes the proof.

If in (6.101) we take C as a closed contour that encircles z , from (6.98) we have

$$\tilde{f}(z) = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{\zeta - z} d\zeta. \quad (6.102)$$

Moreover, if $f(z)$ is analytic in a simply connected region that contains C , from Theorem 6.11 we must have

$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{\zeta - z} d\zeta. \quad (6.103)$$

Comparing (6.102) with (6.103) and taking account of the single-valuedness of $f(z)$, we must have

$$f(z) \equiv \tilde{f}(z). \quad (6.104)$$

Then, from (6.101) we get

$$\frac{df(z)}{dz} = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{(\zeta - z)^2} d\zeta. \quad (6.105)$$

An analogous result holds with the n -th derivative of $f(z)$ such that [5]

$$\frac{d^n f(z)}{dz^n} = \frac{n!}{2\pi i} \oint_C \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta \quad (n : \text{zero or positive integers}). \quad (6.106)$$

Equation (6.106) implies that an analytic function is infinitely differentiable and that the derivatives of all order of an analytic function are again analytic. These prominent properties arise partly from the aforementioned stringent requirement on the differentiability of a function of a complex variable.

So far, we have not assumed the continuity of $df(z)/dz$. However, once we have established (6.106), it assures the presence of, e.g., $d^2 f(z)/dz^2$ and, hence, the continuity of $df(z)/dz$. It is true of $d^n f(z)/dz^n$ with any n (zero or positive integers).

A following theorem is important and intriguing with the analytic functions.

Theorem 6.13: Cauchy–Liouville Theorem A bounded entire function must be a constant.

Proof Using (6.106), we consider the first derivative of an entire function $f(z)$ described as

$$\frac{df}{dz} = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{(\zeta - z)^2} d\zeta.$$

Since $f(z)$ is an entire function, we can arbitrarily choose a large enough circle of radius R centered at z for a closed contour C . On the circle, we have

$$\zeta = z + Re^{i\theta},$$

where θ is a real number changing from 0 to 2π . Then, the above equation can be rewritten as

$$\frac{df}{dz} = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(\zeta)}{(Re^{i\theta})^2} iRe^{i\theta} d\theta = \frac{1}{2\pi R} \int_0^{2\pi} \frac{f(\zeta)}{e^{i\theta}} d\theta. \quad (6.107)$$

Taking an absolute value of both sides, we have

$$\left| \frac{df}{dz} \right| \leq \frac{1}{2\pi R} \int_0^{2\pi} |f(\zeta)| d\theta \leq \frac{M}{2\pi R} \int_0^{2\pi} d\theta = \frac{M}{R}, \quad (6.108)$$

where M is the maximum of $|f(\zeta)|$. As $R \rightarrow \infty$, $|df/dz|$ tends to be zero. This implies that df/dz tends to be zero as well and, hence, that $f(z)$ is constant. This completes the proof.

At the first glance, Cauchy–Liouville theorem looks astonishing in terms of theory of real analysis. It is because we are too familiar with, e.g., $-1 \leq \sin x \leq 1$ for any real number x . Note that $\sin x$ is bounded in a real domain. In fact, in Sect. 8.6 we will show that from (8.98) and (8.99), when $a \rightarrow \pm \infty$ (a : real, $a \neq 0$),

$\sin\left(\frac{\pi}{2} + ia\right)$ takes real values with $\sin\left(\frac{\pi}{2} + ia\right) \rightarrow \infty$. This simple example clearly shows that $\sin\phi$ is an unbounded entire function in a complex domain. As a very familiar example of a bounded entire functions, we show

$$f(z) = \cos^2 z + \sin^2 z \equiv 1, \quad (6.109)$$

which is defined in an entire complex plane.

6.4 Taylor's Series and Laurent's Series

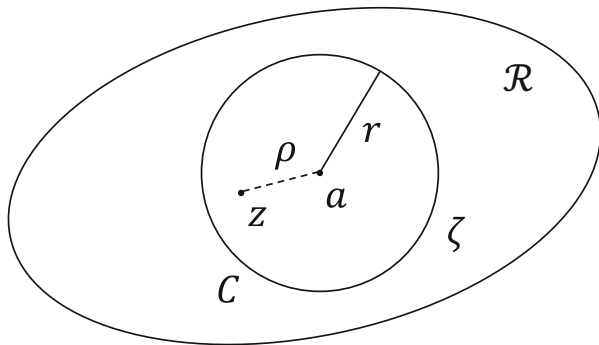
Using Cauchy's integral formula, we study Taylor's series and Laurent's series in relation to the power series expansion of analytic function. Taylor's series and Laurent's series are fundamental tools to study various properties of complex functions in the theory of analytic functions. First, let us examine Taylor's series of an analytic function.

Theorem 6.14 [6] Any analytic function $f(z)$ can be expressed by uniformly convergent power series at an arbitrary regular point of the function in the domain of analyticity $\mathcal{R}$.

Proof Let us assume that we have a circle C of radius r centered at a within $\mathcal{R}$. Choose z inside C with $|z - a| = \rho$ ($\rho < r$) and consider the contour integration of $f(z)$ along C (Fig. 6.14). Then, we have

$$\begin{aligned} \frac{1}{\zeta - z} &= \frac{1}{\zeta - a - (z - a)} = \frac{1}{\zeta - a} \cdot \frac{1}{1 - \frac{z - a}{\zeta - a}} = \frac{1}{\zeta - a} + \frac{z - a}{(\zeta - a)^2} \\ &+ \frac{(z - a)^2}{(\zeta - a)^3} + \cdots, \end{aligned} \quad (6.110)$$

Fig. 6.14 Diagram to explain Taylor's expansion that is assumed to be performed around a . A circle C of radius r centered at a is inside $\mathcal{R}$. Regarding other symbols and notations, see text



where ζ is an arbitrary point on the circle C . To derive (6.110), we used the following formula:

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n,$$

where $x = \frac{z-a}{\zeta-a}$. Since $\left| \frac{z-a}{\zeta-a} \right| = \frac{\rho}{r} < 1$, the geometric series of (6.110) converges. Suppose that $f(\zeta)$ is analytic on C . Then, we have a finite positive number M on C such that

$$|f(\zeta)| < M. \quad (6.111)$$

Using (6.110), we have

$$\frac{f(\zeta)}{\zeta-z} = \frac{f(\zeta)}{\zeta-a} + \frac{(z-a)f(\zeta)}{(\zeta-a)^2} + \frac{(z-a)^2 f(\zeta)}{(\zeta-a)^3} + \dots \quad (6.112)$$

Hence, combining (6.111) and (6.112) we get

$$\begin{aligned} \left| \sum_{\nu=n}^{\infty} \frac{(z-a)^{\nu} f(\zeta)}{(\zeta-a)^{\nu+1}} \right| &\leq \frac{M}{r} \sum_{\nu=n}^{\infty} \left(\frac{\rho}{r} \right)^{\nu} = \frac{M}{r} \left[\frac{1}{1-\frac{\rho}{r}} - \frac{1-\left(\frac{\rho}{r}\right)^n}{1-\frac{\rho}{r}} \right] \\ &= \frac{M}{r-\rho} \left(\frac{\rho}{r} \right)^n. \end{aligned} \quad (6.113)$$

RHS of (6.113) tends to be zero when $n \rightarrow \infty$. Therefore, (6.112) is uniformly convergent with respect to ζ on C . In the above calculations, when we express the infinite series of RHS in (6.112) as Σ and the partial sum of its first n terms as Σ_n , LHS of (6.113) can be expressed as $\Sigma - \Sigma_n (\equiv P_n)$. Since $|P_n| \rightarrow 0$ with $n \rightarrow \infty$, this certainly shows that the series Σ is uniformly convergent [6].

Consequently, we can perform termwise integration [6] of (6.112) on C and subsequently divide the result by $2\pi i$ to get

$$\frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{\zeta-z} d\zeta = \sum_{n=0}^{\infty} \frac{(z-a)^n}{2\pi i} \oint_C \frac{f(\zeta)}{(\zeta-a)^{n+1}} d\zeta. \quad (6.114)$$

From Theorem 6.11, LHS of (6.114) equals $f(z)$. Putting

$$A_n \equiv \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{(\zeta-a)^{n+1}} d\zeta, \quad (6.115)$$

we get

$$f(z) = \sum_{n=0}^{\infty} A_n (z-a)^n. \quad (6.116)$$

This completes the proof.

In the above proof, from (6.106) we have

$$\frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{(\zeta-z)^{n+1}} d\zeta = \frac{1}{n!} \frac{d^n f(z)}{dz^n}. \quad (6.117)$$

Hence, combining (6.117) with (6.115) we get

$$A_n = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{(\zeta-a)^{n+1}} d\zeta = \frac{1}{n!} \left. \frac{d^n f(z)}{dz^n} \right|_{z=a} = \frac{1}{n!} \frac{d^n f(a)}{dz^n}. \quad (6.118)$$

Thus, (6.116) can be rewritten as

$$f(z) = \sum_{n=0}^{\infty} \frac{1}{n!} \frac{d^n f(a)}{dz^n} (z-a)^n.$$

This is the same form as that obtained with respect to a real variable.

Equation (6.116) is a uniformly convergent power series called a Taylor's series or Taylor's expansion of $f(z)$ with respect to $z = a$ with A_n being its coefficient. In Theorem 6.14 we have assumed that a union of a circle C and its inside is simply connected. This topological environment is virtually the same as that of Fig. 6.13. In Theorem 6.11 (Cauchy's integral formula) we have shown the integral representation of an analytic function. On the basis of Cauchy's integral formula, Theorem 6.14 demonstrates that any analytic function is given a tangible functional form.

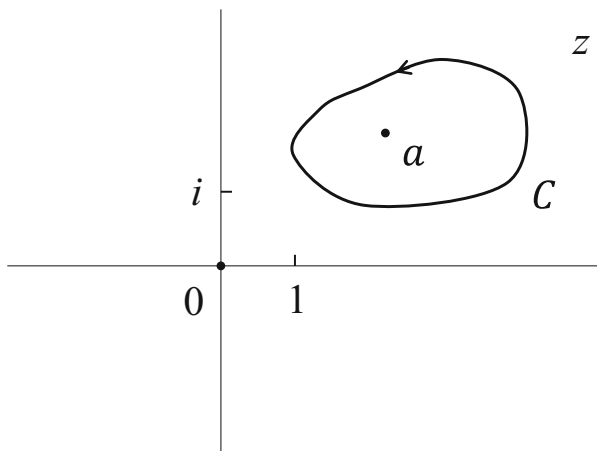
Example 6.2 Let us think of a function $f(z) = 1/z$. This function is analytic except for $z = 0$. Therefore, it can be expanded in the Taylor's series in the region $\mathbb{C} - \{0\}$. We consider the expansion around $z = a$ ($a \neq 0$). Using (6.115), we get the expansion coefficients that are expressed as

$$A_n = \frac{1}{2\pi i} \oint_C \frac{1}{z(z-a)^{n+1}} dz,$$

where C is a closed curve that does not contain $z = 0$ on C or in its inside. Therefore, $1/z$ is analytic in the simply connected region encircled with C so that the point $z = 0$ may not be contained inside C (Fig. 6.15).

Then, from (6.106) we get

Fig. 6.15 Diagram used for calculating the Taylor's series of $f(z) = 1/z$ around $z = a$ ($a \neq 0$). A contour C encircles the point a



$$\frac{1}{2\pi i} \oint_C \frac{1}{z(z-a)^{n+1}} dz = \frac{1}{n!} \left. \frac{d^n(1/z)}{dz^n} \right|_{z=a} = \frac{1}{n!} (-1)^n n! \frac{1}{a^{n+1}} = \frac{(-1)^n}{a^{n+1}}.$$

Hence, from (6.116) we have

$$f(z) = \frac{1}{z} = \sum_{n=0}^{\infty} \frac{(-1)^n}{a^{n+1}} (z-a)^n = \frac{1}{a} \sum_{n=0}^{\infty} \left(1 - \frac{z}{a}\right)^n. \quad (6.119)$$

The series uniformly converges within a convergence circle called a “ball” [7] (vide infra) whose convergence radius r is estimated to be

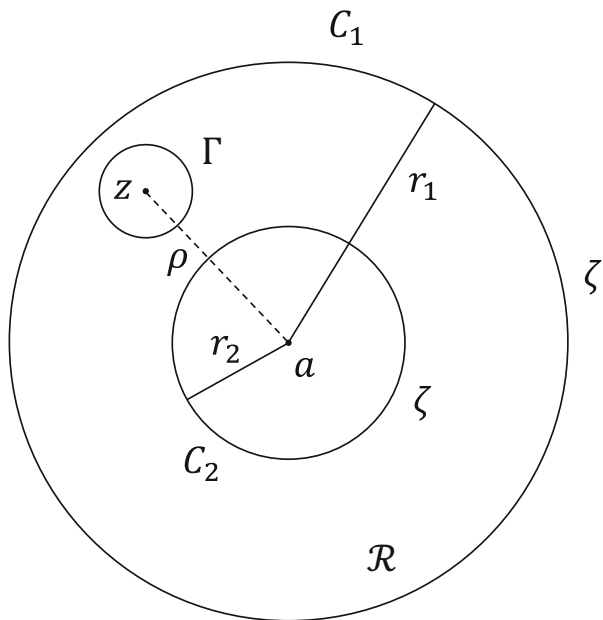
$$\frac{1}{r} = \lim_{n \rightarrow \infty} \sup \sqrt[n]{\left| \frac{(-1)^n}{a^{n+1}} \right|} = \frac{1}{|a|} \sqrt[n]{\frac{1}{|a|}} = \frac{1}{|a|}. \quad (6.120)$$

That is, $r = |a|$. In (6.120) “lim sup” stands for the superior limit and is sometimes said to be “limes superior” [6]. The estimation is due to Cauchy–Hadamard theorem. Readers interested in the theorem and related concepts are referred to appropriate literature [6].

The concept of the convergence circle and radius is very important for defining the analyticity of a function. Formally, the convergence radius is defined as a radius of a convergence circle in a metric space, typically $\mathbb{R}^2$ and $\mathbb{C}$. In $\mathbb{C}$ a region $\mathcal{R}$ within the convergence circle of convergence radius r (i.e., a real positive number) centered at z_0 is denoted by

$$\mathcal{R} = \{z; \exists z_0 \in \mathbb{C}, \rho(z, z_0) < r\}.$$

Fig. 6.16 Diagram to explain Laurent's expansion that is performed around a . The circle Γ containing the point z lies on the annular region bounded by the circles C_1 and C_2



The region $\mathcal{R}$ is an open set by definition (see Sect. 6.1) and the Taylor's series defined in $\mathcal{R}$ is convergent. On the other hand, the Taylor's series may or may not be convergent at z that is on the convergence circle. In Example 6.2, $f(z) = 1/z$ is not defined at $z = 0$, even though $z (=0)$ is on the convergent circle. At other points on the convergent circle, however, the Taylor's series is convergent.

Note that a region $\mathcal{B}^n$ in $\mathbb{R}^n$ similarly defined as

$$\mathcal{B}^n = \{x; x, \exists x_0 \in \mathbb{R}^n, \rho(x, x_0) < r\}$$

is sometimes called a ball or open ball. This concept can readily be extended to any metric space.

Next, we examine Laurent's series of an analytic function.

Theorem 6.15 [6] Let $f(z)$ be analytic in a region $\mathcal{R}$ except for a point a . Then, $f(z)$ can be described by the uniformly convergent power series within a region $\mathcal{R} - \{a\}$.

Proof Let us assume that we have a circle C_1 of radius r_1 centered at a and another circle C_2 of radius r_2 centered at a both within $\mathcal{R}$. Moreover, let another circle Γ be centered at $z (\neq a)$ within $\mathcal{R}$ so that z can be outside of C_2 ; see Fig. 6.16.

In this situation we consider the contour integration along C_1 , C_2 , and Γ . Using (6.87), we have

$$\frac{1}{2\pi i} \oint_{\Gamma} \frac{f(\zeta)}{\zeta - z} d\zeta = \frac{1}{2\pi i} \oint_{C_1} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \oint_{C_2} \frac{f(\zeta)}{\zeta - z} d\zeta. \quad (6.121)$$

From Theorem 6.11, we have

$$\text{LHS of (6.121)} = f(z).$$

Notice that the region containing Γ and its inside form a simply connected region in terms of the analyticity of $f(z)$. With the first term of RHS of (6.121), from Theorem 6.14 we have

$$\frac{1}{2\pi i} \oint_{C_1} \frac{f(\zeta)}{\zeta - z} d\zeta = \sum_{n=0}^{\infty} A_n (z - a)^n \quad (6.122)$$

with

$$A_n = \frac{1}{2\pi i} \oint_{C_1} \frac{f(\zeta)}{(\zeta - a)^{n+1}} d\zeta \quad (n = 0, 1, 2, \dots). \quad (6.123)$$

In fact, if ζ lies on the circle C_1 , the Taylor's series is uniformly convergent as in the case of the proof of Theorem 6.14.

With the second term of RHS of (6.121), however, the situation is different in such a way that z lies outside the circle C_2 . In this case, from Fig. 6.16 we have

$$\left| \frac{\zeta - a}{z - a} \right| = \frac{r_2}{\rho} < 1. \quad (6.124)$$

Then, a geometric series described by

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - a - (z - a)} = \frac{-1}{z - a} \left[1 + \frac{\zeta - a}{z - a} + \frac{(\zeta - a)^2}{(z - a)^2} + \dots \right] \quad (6.125)$$

is uniformly convergent with respect to ζ on C_2 . Accordingly, again we can perform termwise integration [6] of (6.125) on C_2 . Hence, we get

$$\begin{aligned} - \oint_{C_2} \frac{f(\zeta)}{\zeta - z} d\zeta &= \frac{1}{z - a} \oint_{C_2} f(\zeta) d\zeta + \frac{1}{(z - a)^2} \oint_{C_2} f(\zeta)(\zeta - a) d\zeta \\ &\quad + \frac{1}{(z - a)^3} \oint_{C_2} f(\zeta)(\zeta - a)^2 d\zeta + \dots \end{aligned} \quad (6.126)$$

That is, we have

$$-\frac{1}{2\pi i} \oint_{C_2} \frac{f(\zeta)}{\zeta - z} d\zeta = \sum_{n=1}^{\infty} A_{-n} (z - a)^{-n}, \quad (6.127)$$

where

$$A_{-n} = \frac{1}{2\pi i} \oint_{C_2} f(\zeta) (\zeta - a)^{n-1} d\zeta \quad (n = 1, 2, 3, \dots). \quad (6.128)$$

Consequently, from (6.121), with z lying on the annular region bounded by C_1 and C_2 , we get

$$f(z) = \sum_{n=-\infty}^{\infty} A_n (z - a)^n. \quad (6.129)$$

In (6.129), the coefficients A_n are given by (6.123) or (6.128) according to the plus (including zero) or minus sign of n . These complete the proof.

Equation (6.129) is a uniformly convergent power series called a Laurent's series or Laurent's expansion of $f(z)$ with respect to $z = a$ with A_{-n} being its coefficient. Note that the above annular region bounded by C_1 and C_2 is not simply connected, but multiply connected.

We add that if the integration path $\tilde{C}$ is taken in the annular region formed by C_1 and C_2 so that it can be sandwiched in between them, the values of integral expressed as (6.123) and (6.128) remain unchanged in virtue of (6.87). That is, instead of (6.123) and (6.128) we may have a following unified formula

$$A_n = \frac{1}{2\pi i} \oint_{\tilde{C}} \frac{f(\zeta)}{(\zeta - a)^{n+1}} d\zeta \quad (n = 0, \pm 1, \pm 2, \dots) \quad (6.130)$$

that represents the coefficients A_n of the power series

$$f(z) = \sum_{n=-\infty}^{\infty} A_n (z - a)^n. \quad (6.129)$$

6.5 Zeros and Singular Points

We described the fundamental theorems of Taylor's expansion and Laurent's expansion. Next, we explain important concepts of zeros and singular points.

Definition 6.11 Let $f(z)$ be analytic in a region $\mathcal{R}$. If $f(z)$ vanishes at a point $z = a$, the point is called a zero of $f(z)$. When we have

$$f(a) = \left. \frac{df(z)}{dz} \right|_{z=a} = \left. \frac{d^2f(z)}{dz^2} \right|_{z=a} = \cdots = \left. \frac{d^{n-1}f(z)}{dz^{n-1}} \right|_{z=a} = 0, \quad (6.131)$$

but

$$\left. \frac{d^n f(z)}{dz^n} \right|_{z=a} \neq 0, \quad (6.132)$$

the function is said to have a zero of order n at $z = a$.

If (6.131) and (6.132) hold, the first n coefficients in the Taylor's series of the analytic function $f(z)$ at $z = a$ vanish. Hence, we have

$$\begin{aligned} f(z) &= A_n(z-a)^n + A_{n+1}(z-a)^{n+1} + \cdots \\ &= (z-a)^n \sum_{k=0}^{\infty} A_{n+k}(z-a)^k = (z-a)^n h(z), \end{aligned}$$

where we define $h(z)$ as

$$h(z) \equiv \sum_{k=0}^{\infty} A_{n+k}(z-a)^k. \quad (6.133)$$

Then, $h(z)$ is analytic and non-vanishing at $z = a$. From the analyticity $h(z)$ must be continuous at $z = a$ and differ from zero in some finite neighborhood of $z = a$. Consequently, it is also the case with $f(z)$. Therefore, if the set of zeros had an accumulation point at $z = a$, any neighborhood of $z = a$ would contain another zero, in contradiction to the above assumption. To avoid this contradiction, the analytic function $f(z) \equiv 0$ throughout the region $\mathcal{R}$. Taking the contraposition of the above statement, if an analytic function is not identically zero [i.e., $f(z) \not\equiv 0$], the zeros of that function are isolated; see the relevant discussion of Sect. 6.1.2.

Meanwhile, the Laurent's series (6.129) can be rewritten as

$$f(z) = \sum_{k=0}^{\infty} A_k(z-a)^k + \sum_{k=1}^{\infty} A_{-k} \frac{1}{(z-a)^k} \quad (6.134)$$

so that the constitution of the Laurent's series can explicitly be recognized. The second term is called a singular part (or principal part) of $f(z)$ at $z = a$. The singularity is classified as follows:

(1) Removable singularity:

If (6.134) lacks the singular part (i.e., $A_{-k} = 0$ for $k = 1, 2, \cdots$), the singular point is said to be a removable singularity. In this case, we have

$$f(z) = \sum_{k=0}^{\infty} A_k (z-a)^k. \quad (6.135)$$

If in (6.135) we define suitably as

$$f(a) \equiv A_0,$$

$f(z)$ can be regarded as analytic. Examples include the following case where a sin function is expanded as the Taylor's series:

$$\begin{aligned} \sin z &= z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots + \frac{(-1)^{n-1} z^{2n-1}}{(2n-1)!} + \cdots \\ &= \sum_{n=1}^{\infty} \frac{(-1)^{n-1} z^{2n-1}}{(2n-1)!}. \end{aligned} \quad (6.136)$$

Hence, if we define

$$f(z) \equiv \frac{\sin z}{z} \quad \text{and} \quad f(0) \equiv \lim_{z \rightarrow 0} \frac{\sin z}{z} = 1, \quad (6.137)$$

$z = 0$ is a removable singularity of $f(z)$.

(2) Poles:

Suppose that in (6.134) we have a certain positive integer n ($n \geq 1$) such that

$$A_{-n} \neq 0 \quad \text{but} \quad A_{-(n+1)} = A_{-(n+2)} = \cdots = 0. \quad (6.138)$$

Then, the function $f(z)$ is said to have a pole of order n at $z = a$. If, in particular, $n = 1$ in (6.138), i.e.,

$$A_{-1} \neq 0 \quad \text{but} \quad A_{-2} = A_{-3} = \cdots = 0, \quad (6.139)$$

the function $f(z)$ is said to have a *simple pole*. This special case is important in the calculation of various integrals (vide infra).

In the general cases for (6.138), (6.134) can be rewritten as

$$f(z) = \frac{g(z)}{(z-a)^n}, \quad (6.140)$$

where $g(z)$ defined as

$$g(z) \equiv \sum_{k=0}^{\infty} A_{k-n} (z-a)^k \quad (6.141)$$

is analytic and non-vanishing at $z = a$, i.e., $A_{-n} \neq 0$. Explicitly writing (6.140), we have

$$f(z) = \sum_{k=0}^{\infty} A_k (z-a)^k + \sum_{k=1}^n A_{-k} \frac{1}{(z-a)^k}.$$

(3) Essential singularity:

If the singular part of (6.134) comprises an infinite series, the point $z = a$ is called an essential singularity. In that case, $f(z)$ performs a complicated behavior near or at $z = a$. Interested readers are referred to suitable literature [5].

A function $f(z)$ that is analytic in a region within $\mathbb{C}$ except for a set of points where the function has poles is called a *meromorphic* function in the said region. The above definition of meromorphic functions is true of the above Cases (1) and (2), but it is not true of Case (3). Henceforth, we will be dealing with the meromorphic functions.

In the above discussion of Cases (1) and (2), we often deal with a function $f(z)$ that has a single isolated pole at $z = a$. This implies that $f(z)$ is analytic within a certain neighborhood N_a of $z = a$ but is not analytic at $z = a$. More specifically, $f(z)$ is analytic in a region $N_a - \{a\}$. Even though we consider more than one isolated pole, the situation is essentially the same. Suppose that there is another isolated pole at $z = b$. In that case, again take a certain neighborhood N_b of $z = b$ ($\neq a$) and $f(z)$ is analytic in a region $N_b - \{b\}$.

Readers may well wonder why we have to discuss this trifling issue. Nevertheless, think of the situation where a set of poles has an accumulation point. Any neighborhood of the accumulation point contains another pole where the function is not analytic. This is in contradiction to that $f(z)$ is analytic in a region, e.g., $N_a - \{a\}$. Thus, if such a function were present, it would be intractable to deal with.

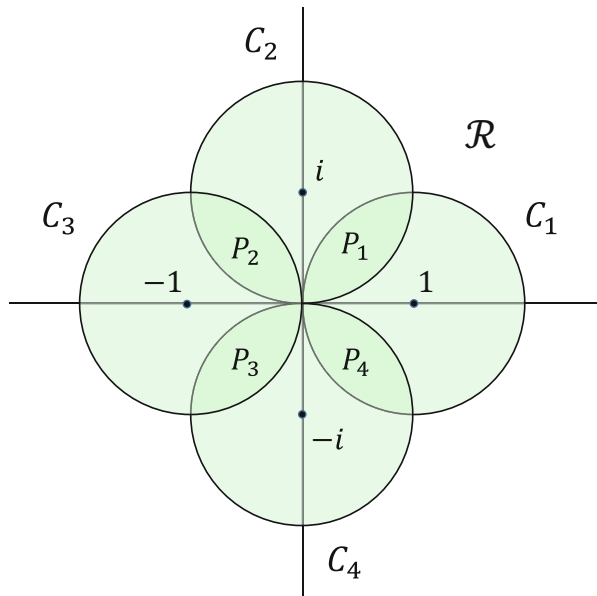
6.6 Analytic Continuation

When we discussed the Cauchy's integral formula (Theorem 6.11), we have known that if a function is analytic in a certain region of $\mathbb{C}$ and on a curve C that encircles the region, the values of the function within the region are determined once the values of the function on C are given. We have the following theorem for this.

Theorem 6.16 Let $f_1(z)$ and $f_2(z)$ be two functions that are analytic within a region $\mathcal{R}$. Suppose that the two functions coincide in a set chosen from among (i) a neighborhood of a point $z \in \mathcal{R}$, or (ii) a segment of a curve lying in $\mathcal{R}$, or (iii) a set of points containing an accumulation point belonging to $\mathcal{R}$. Then, the two functions $f_1(z)$ and $f_2(z)$ coincide throughout $\mathcal{R}$.

Proof Suppose that $f_1(z)$ and $f_2(z)$ coincide on the above set chosen from the three. Then, since $f_1(z) - f_2(z) = 0$ on that set, the set comprises the zeros of $f_1(z) - f_2(z)$. This implies that the zeros are not isolated in $\mathcal{R}$. Thus, as evidenced from the discussion of Sect. 6.5, we conclude that $f_1(z) - f_2(z) \equiv 0$ throughout $\mathcal{R}$. That is, $f_1(z) \equiv f_2(z)$. This completes the proof.

Fig. 6.17 Four convergence circles C_1 , C_2 , C_3 , and C_4 for $1/z$. These convergence circles are centered at 1 , i , -1 , or $-i$



The theorem can be understood as follows. Two different analytic functions cannot coincide in the set chosen from the above three (or more succinctly, a set that contains an accumulation point). In other words, the behavior of an analytic function in $\mathcal{R}$ is uniquely determined by the limited information of the subset belonging to $\mathcal{R}$.

The above statement is reminiscent of the pre-established harmony and, hence, occasionally talked about mysteriously. Instead, we wish to give a simple example.

Example 6.3 In Example 6.2 we described a tangible illustration of the Taylor's expansion of a function $1/z$ as follows:

$$\frac{1}{z} = \sum_{n=0}^{\infty} \frac{(-1)^n}{a^{n+1}} (z-a)^n = \frac{1}{a} \sum_{n=0}^{\infty} \left(1 - \frac{z}{a}\right)^n. \quad (6.119)$$

If we choose 1 , i , -1 , or $-i$ for a , we can draw four convergence circles as depicted in Fig. 6.17. Let each function described by (6.119) be $f_1(z)$, $f_i(z)$, $f_{-1}(z)$, or $f_{-i}(z)$. Let $\mathcal{R}$ be a region that consists of an outer periphery and its inside of the four circles C_1 , C_2 , C_3 , and C_4 (i.e., convergence circles of radius 1). The said region $\mathcal{R}$ is colored pale green in Fig. 6.17. There are four petals as shown.

We can view Fig. 6.17 as follows. For instance, $f_1(z)$ and $f_i(z)$ are defined in C_1 and its inside and C_2 and its inside, respectively, excluding $z = 0$. We have $f_1(z) = f_i(z)$ within the petal P_1 (overlapped part as shown). Consequently, from the statement of Theorem 6.16, $f_1(z) \equiv f_i(z)$ throughout the region encircled by C_1 and C_2 (excluding $z = 0$). Repeating this procedure, we get

$$f_1(z) \equiv f_i(z) \equiv f_{-1}(z) \equiv f_{-i}(z).$$

Thus, we obtain the same functional entity throughout $\mathcal{R}$ except for the origin ($z = 0$). Further continuing similar procedures, we finally reach the entire complex plane except for the origin, i.e., $\mathbb{C} - \{0\}$ regarding the domain of analyticity of $1/z$.

We further compare the above results with those for analytic function defined in the real domain. The Taylor's expansion of $f(x) = 1/x$ (x : real with $x \neq 0$) around a ($a \neq 0$) reads as

$$\begin{aligned} f(x) = \frac{1}{x} &= f(a) + f'(a)(x-a) + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{a^{n+1}}(x-a)^n. \end{aligned} \quad (6.142)$$

This is exactly the same form as that of (6.119) aside from notation of the argument.

The aforementioned procedure that determines the behavior of an analytic function outside the region where that function has been originally defined is called analytic continuation or analytic prolongation. Comparing (6.119) and (6.142), we can see that (6.119) is a consequence of the analytic continuation of $1/x$ from the real domain to the complex domain.

6.7 Calculus of Residues

The theorem of residues and its application to calculus of various integrals are one of central themes of the theory of analytic functions.

The Cauchy's integral theorem (Theorem 6.10) tells us that if $f(z)$ is analytic in a simply connected region $\mathcal{R}$, we have

$$\oint_C f(z) dz = 0, \quad (6.84)$$

where C is a closed curve within $\mathcal{R}$. As we have already seen, this is not necessarily the case if $f(z)$ has singular points in $\mathcal{R}$. Meanwhile, if we look at (6.130), we are aware of a simple but important fact. That is, replacing n with 1, we have

$$A_{-1} = \frac{1}{2\pi i} \oint_C f(\zeta) d\zeta \quad \text{or} \quad \oint_C f(\zeta) d\zeta = 2\pi i A_{-1}. \quad (6.143)$$

This implies that if we could obtain the Laurent's series of $f(z)$ with respect to a singularity located at $z = a$ and encircled by C , we can immediately estimate $\oint_C f(\zeta) d\zeta$ of (6.143) from the coefficient of the term $(z - a)^{-1}$, i.e., A_{-1} . This fact further

implies that even though $f(z)$ has singularities, $\oint_C f(\zeta) d\zeta$ may happen to vanish. Thus, whether (6.84) vanishes depends on the nature of $f(z)$ and its singularities. To examine it, we define a residue of $f(z)$ at $z = a$ (that is encircled by C) as

$$\text{Res } f(a) \equiv \frac{1}{2\pi i} \oint_C f(z) dz \quad \text{or} \quad \oint_C f(z) dz = 2\pi i \text{Res } f(a), \quad (6.144)$$

where $\text{Res } f(a)$ denotes the residue of $f(z)$ at $z = a$. From (6.143) and (6.144), we have

$$A_{-1} = \text{Res } f(a). \quad (6.145)$$

In a formal sense, the point a does not have to be a singular point. If it is a regular point of $f(a)$, we have $\text{Res } f(a) = 0$ trivially. If there is more than one singularity at $z = a_j$, we have

$$\oint_C f(z) dz = 2\pi i \sum_j \text{Res } f(a_j). \quad (6.146)$$

Notice that we assume the isolated singularities with (6.144) and (6.146). These equations are associated with Cases (1) and (2) dealt with in Sect. 6.5. Within this framework, we wish to evaluate the residue of $f(z)$ at a pole of order n located at $z = a$. In Sect. 6.5, we had $f(z)$ described by

$$f(z) = \frac{g(z)}{(z-a)^n}, \quad (6.140)$$

where $g(z)$ is analytic and non-vanishing at $z = a$. Inserting (6.140) into (6.144), we have

$$\begin{aligned} \text{Res } f(a) &= \frac{1}{2\pi i} \oint_C \frac{g(\zeta)}{(\zeta-a)^n} d\zeta = \frac{1}{(n-1)!} \left. \frac{d^{n-1} g(z)}{dz^{n-1}} \right|_{z=a} \\ &= \frac{1}{(n-1)!} \left. \frac{d^{n-1} [f(z)(z-a)^n]}{dz^{n-1}} \right|_{z=a}, \end{aligned} \quad (6.147)$$

where with the second equality we used (6.106).

In the special but important case where $f(z)$ has a simple pole at $z = a$, from (6.147) we get

$$\text{Res } f(a) = f(z)(z-a)|_{z=a}. \quad (6.148)$$

Setting $n = 1$ in (6.147) in combination with (6.140), we obtain

$$\operatorname{Res} f(a) = \frac{1}{2\pi i} \oint_C f(z) dz = \frac{1}{2\pi i} \oint_C \frac{g(z)}{z-a} dz = f(z)(z-a)|_{z=a} = g(z)|_{z=a}.$$

This is nothing but Cauchy's integral formula (6.88) regarding $g(z)$.

Summarizing the above discussion, we can calculate the residue of $f(z)$ at a pole of order n located at $z = a$ using one of the following alternatives, i.e., (i) using (6.147) and (ii) picking up the coefficient A_{-1} in the Laurent's series described by (6.129).

In Sect. 6.3 we mentioned that the closed contour integral (6.86) or (6.87) depends on the nature of singularity. Here we have reached a simple criterion of evaluating that integral. In fact, suppose that we have a Laurent's expansion such that

$$f(z) = \sum_{n=-\infty}^{\infty} A_n (z-a)^n. \quad (6.129)$$

Then, let us calculate the contour integral of $f(z)$ on a closed circle Γ of radius ρ centered at $z = a$. We have

$$\begin{aligned} I &\equiv \oint_{\Gamma} f(\zeta) d\zeta = \sum_{n=-\infty}^{\infty} A_n \oint_{\Gamma} (\zeta-a)^n d\zeta \\ &= \sum_{n=-\infty}^{\infty} iA_n \int_0^{2\pi} \rho^{n+1} e^{i(n+1)\theta} d\theta. \end{aligned} \quad (6.149)$$

The above integral vanishes except for $n = -1$. With $n = -1$ we get

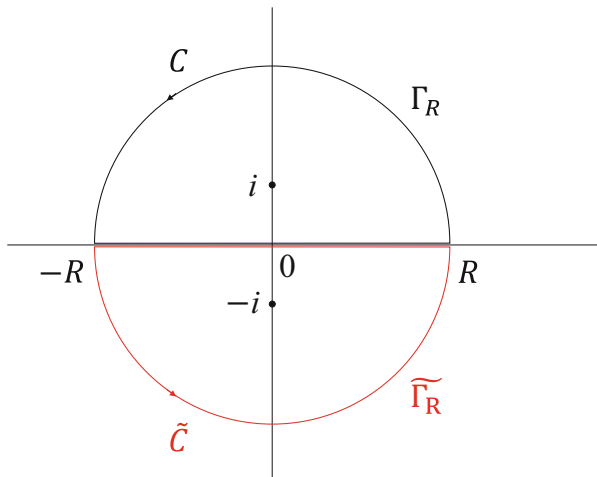
$$I = 2\pi i A_{-1}.$$

Thus, we recover (6.145) in combination of (6.144). To get a Laurent's series of $f(z)$, however, is not necessarily easy. In that case, we estimate a residue by (6.147). Tangible examples can be seen in the next section.

6.8 Examples of Real Definite Integrals

The calculus of residues is of great practical importance. For instance, it is directly associated with the calculations of definite integrals. In particular, even if one finds the real integration to be hard to perform in order to solve a related problem, it is often easy to solve it using complex integration, especially the calculus of residues. Here we study several examples.

Fig. 6.18 Contour for the integration of $1/(1+z^2)$ that appears in Example 6.4. One may equally choose the upper semicircle (denoted by Γ_R) and lower semicircle (denoted by $\tilde{\Gamma}_R$) for contour integration



Example 6.4 Let us consider the following real definite integral:

$$I = \int_{-\infty}^{\infty} \frac{dx}{1+x^2}. \quad (6.150)$$

To this end, we convert the real variable x to the complex variable z and evaluate a contour integral I_C (Fig. 6.18) described by

$$I_C = \int_{-R}^R \frac{dz}{1+z^2} + \int_{\Gamma_R} \frac{dz}{1+z^2}, \quad (6.151)$$

where R is a real positive number enough large ($R \gg 1$); Γ_R denotes an upper semicircle; I_C stands for the contour integral along the closed curve C that comprises the interval $[-R, R]$ and the upper semicircle Γ_R . Simple poles of order 1 are located at $z = \pm i$ as shown. There is only one simple pole at $z = i$ within the upper semicircle of Fig. 6.18. From (6.146), therefore, we have

$$I_C = \oint_C f(z) dz = 2\pi i \operatorname{Res} f(i), \quad (6.152)$$

where $f(z) \equiv \frac{1}{1+z^2}$. Using (6.148), the residue of $f(z)$ at $z = i$ can readily be estimated to be

$$\operatorname{Res} f(i) = f(z)(z-i)|_{z=i} = \frac{1}{2i}. \quad (6.153)$$

To estimate the integral (6.150), we must calculate the second term of (6.151). To this end, we change the variable z such that

$$z = Re^{i\theta}, \quad (6.154)$$

where θ is an argument. Then, using the Darboux's inequality (6.95), we get

$$\begin{aligned} \int_{\Gamma_R} \frac{dz}{1+z^2} &= \left| \int_0^\pi \frac{iRe^{i\theta} d\theta}{R^2 e^{2i\theta} + 1} \right| \leq \int_0^\pi \left| \frac{iRe^{i\theta}}{R^2 e^{2i\theta} + 1} \right| d\theta \\ &= \int_0^\pi \frac{R}{\sqrt{(R^2 e^{2i\theta} + 1)(R^2 e^{-2i\theta} + 1)}} d\theta \\ &= R \int_0^\pi \frac{1}{\sqrt{R^4 + 2R^2 \cos 2\theta + 1}} d\theta \leq R \frac{\pi}{R^2 \left(1 - \frac{1}{R^2}\right)} = \frac{\pi}{R \left(1 - \frac{1}{R^2}\right)}. \end{aligned}$$

Taking $R \rightarrow \infty$, we have

$$\int_{\Gamma_R} \frac{dz}{1+z^2} \rightarrow 0. \quad (6.155)$$

Consequently, if $R \rightarrow \infty$, we have

$$\begin{aligned} \lim_{R \rightarrow \infty} I_C &= \int_{-\infty}^\infty \frac{dx}{1+x^2} + \lim_{R \rightarrow \infty} \int_{\Gamma_R} \frac{dz}{1+z^2} = \int_{-\infty}^\infty \frac{dx}{1+x^2} = I \\ &= 2\pi i \operatorname{Res} f(i) = \pi, \end{aligned} \quad (6.156)$$

where with the first equality we placed the variable back to x ; with the last equality we used (6.152) and (6.153). Equation (6.156) gives an answer to (6.150). Notice in general that if a meromorphic function is given as a quotient of two polynomials, an integral of a type of (6.155) tends to be zero with $R \rightarrow \infty$ in the case where the degree of the denominator of that function is at least two units higher than the degree of the numerator [8].

We may equally choose another contour $\tilde{C}$ (the closed curve comprising the interval $[-R, R]$ and the lower semicircle $\tilde{\Gamma}_R$) for the integration (Fig. 6.18). In that case, we have

$$\begin{aligned} \lim_{R \rightarrow \infty} I_{\tilde{C}} &= \int_{-\infty}^\infty \frac{dx}{1+x^2} + \lim_{R \rightarrow \infty} \int_{\tilde{\Gamma}_R} \frac{dz}{1+z^2} = \int_{-\infty}^\infty \frac{dx}{1+x^2} \\ &= 2\pi i \operatorname{Res} f(-i) = -\pi. \end{aligned} \quad (6.157)$$

Note that in the above equation the contour integral is taken in the counter-clockwise direction and, hence, that the real definite integral has been taken from ∞ to $-\infty$. Notice also that

$$\operatorname{Res} f(-i) = f(z)(z+i)|_{z=-i} = -\frac{1}{2i},$$

because $f(z)$ has a simple pole at $z = -i$ in the lower semicircle (Fig. 6.18) for which the residue has been taken. From (6.157), we get the same result as before such that

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi.$$

Alternatively, we can use the method of partial fraction decomposition. In that case, as the integrand we have

$$\frac{1}{1+z^2} = \frac{1}{2i} \left(\frac{1}{z-i} - \frac{1}{z+i} \right). \quad (6.158)$$

The residue of $\frac{1}{1+z^2}$ at $z = i$ is $\frac{1}{2i}$ [i.e., the coefficient of $\frac{1}{z-i}$ in (6.158)], giving the same result as (6.153).

Example 6.5 Evaluate the following real definite integral:

$$I = \int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^3}. \quad (6.159)$$

As in the case of Example 6.4, we evaluate a contour integration described by

$$I_C = \oint_C \frac{1}{(1+z^2)^3} dz = \int_{-R}^R \frac{dz}{(1+z^2)^3} + \int_{\Gamma_R} \frac{dz}{(1+z^2)^3}, \quad (6.160)$$

where C stands for the closed curve comprising the interval $[-R, R]$ and the upper semicircle Γ_R (see Fig. 6.18).

The function $f(z) \equiv \frac{1}{(1+z^2)^3}$ has two isolated poles at $z = \pm i$. In the present case, however, the poles are of order of 3. Hence, we use (6.147) to estimate the residue. The inside of the upper semicircle contains the pole of order 3 at $z = i$. Therefore, we have

$$\operatorname{Res} f(i) = \frac{1}{2\pi i} \oint_C f(z) dz = \frac{1}{2\pi i} \oint_C \frac{g(z)}{(z-i)^3} dz = \frac{1}{2!} \left. \frac{d^2 g(z)}{dz^2} \right|_{z=i}, \quad (6.161)$$

where

$$g(z) \equiv \frac{1}{(z+i)^3} \quad \text{and} \quad f(z) = \frac{g(z)}{(z-i)^3}.$$

Hence, we get

$$\begin{aligned} \frac{1}{2!} \frac{d^2 g(z)}{dz^2} \Big|_{z=i} &= \frac{1}{2} (-3) \cdot (-4) \cdot (z+i)^{-5} \Big|_{z=i} = \frac{1}{2} (-3) \cdot (-4) \cdot (2i)^{-5} \\ &= \frac{3}{16i} = \text{Res } f(i). \end{aligned} \tag{6.162}$$

For a reason similar to that of Example 6.4, the second term of (6.160) vanishes when $R \rightarrow \infty$. Then, we obtain

$$\begin{aligned} \lim_{R \rightarrow \infty} I_C &= \int_{-\infty}^{\infty} \frac{dz}{(1+z^2)^3} + \lim_{R \rightarrow \infty} \int_{\Gamma_R} \frac{dz}{(1+z^2)^3} = \lim_{R \rightarrow \infty} \oint_C \frac{g(z)}{(z-i)^3} dz = 2\pi i \text{Res } f(i) \\ &= 2\pi i \times \frac{3}{16i} = \frac{3\pi}{8}. \end{aligned}$$

That is,

$$\int_{-\infty}^{\infty} \frac{dz}{(1+z^2)^3} = \int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^3} = I = \frac{3\pi}{8}.$$

If we are able to perform the Laurent's expansion of $f(z)$, we can immediately evaluate the residue using (6.145). To this end, let us utilize the binomial expansion formula (or generalized binomial theorem). We have

$$f(z) = \frac{1}{(1+z^2)^3} = \frac{1}{(z+i)^3(z-i)^3}.$$

Changing the variable $z-i = \zeta$, we have

$$f(z) = \tilde{f}(\zeta) = \frac{1}{(\zeta+2i)^3 \zeta^3} = \frac{1}{\zeta^3} \cdot \frac{1}{(2i)^3} \cdot \left(1 + \frac{\zeta}{2i}\right)^{-3}$$

$$\begin{aligned}
&= \frac{1}{\zeta^3} \cdot \frac{1}{(2i)^3} \cdot \left[1 + (-3) \frac{\zeta}{2i} + \frac{(-3)(-4)}{2!} \left(\frac{\zeta}{2i} \right)^2 + \frac{(-3)(-4)(-5)}{3!} \left(\frac{\zeta}{2i} \right)^3 + \cdots \right] \\
&= -\frac{1}{8i} \left[\frac{1}{\zeta^3} - \frac{3}{2i\zeta^2} - \frac{3}{2\zeta} + \frac{5}{4i} + \cdots \right], \tag{6.163}
\end{aligned}$$

where $\cdots$ of the above equation denotes a power series of ζ . Accompanied by the variable transformation z to ζ , the functional form f has been changed to $\tilde{f}$. Getting back the original functional form, we have

$$f(z) = -\frac{1}{8i} \left[\frac{1}{(z-i)^3} - \frac{3}{2i(z-i)^2} - \frac{3}{2(z-i)} + \frac{5}{4i} + \cdots \right]. \tag{6.164}$$

Equation (6.164) is a Laurent's expansion of $f(z)$. From (6.164) we see that the coefficient of $\frac{1}{z-i}$ of $f(z)$ is $\frac{3}{16i}$, in agreement with (6.145) and (6.162). To obtain the answer of (6.159), once again we may equally choose the contour $\tilde{C}$ (Fig. 6.18) and get the same result as in the case of Example 6.4. In that case, $f(z)$ can be expanded into a Laurent's series around $z = -i$. Readers are encouraged to check it.

We make a few remarks about the (generalized) binomial expansion formula. We described it in (3.181) of Sect 3.6.1 such that

$$(1+x)^{-\lambda} = \sum_{m=0}^{\infty} \binom{-\lambda}{m} x^m, \tag{3.181}$$

where λ is an arbitrary real number. In (3.181), we assumed x is a real number with $|x| < 1$. However, (6.163) suggests that (3.181) holds with x that can be a complex number with $|x| < 1$. Moreover, λ in (3.181) is allowed to be any complex number. Then, on those conditions $(1+x)^{-\lambda}$ is analytic because $1+x \neq 0$, and so $(1+x)^{-\lambda}$ must have a convergent Taylor's series. By the same token, the factor $\left(1 + \frac{\zeta}{2i}\right)^{-3}$ of (6.163) yields a convergent Taylor's series around $\zeta = 0$. In virtue of the factor $\frac{1}{\zeta^3}$, (6.163) gives a convergent Laurent's series around $\zeta = 0$.

Returning to the present issue, we rewrite (3.181) as a Taylor's series such that

$$(1+z)^{-\lambda} = \sum_{m=0}^{\infty} \binom{-\lambda}{m} z^m, \tag{6.165}$$

where λ is any complex number with z also being a complex number of $|z| < 1$. We may view (6.165) as a consequence of the analytic continuation and (6.163) has been dealt with as such indeed.

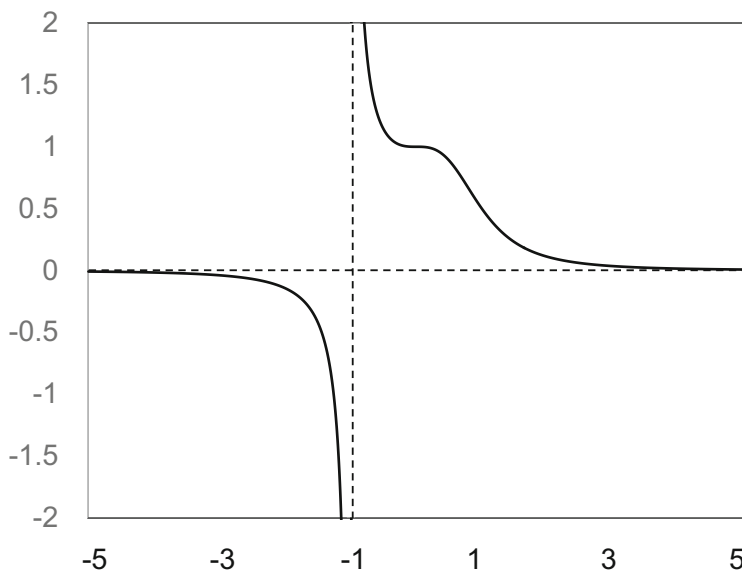


Fig. 6.19 Graphical form of $f(x) \equiv \frac{1}{1+x^3}$. The modulus of $f(x)$ tends to infinity at $x = -1$. Inflection points are present at $x = 0$ and $\sqrt[3]{1/2}$ (≈ 0.7937)

Example 6.6 Evaluate the following real definite integral:

$$I = \int_{-\infty}^{\infty} \frac{dx}{1+x^3}. \quad (6.166)$$

We put

$$f(x) \equiv \frac{1}{1+x^3}. \quad (6.167)$$

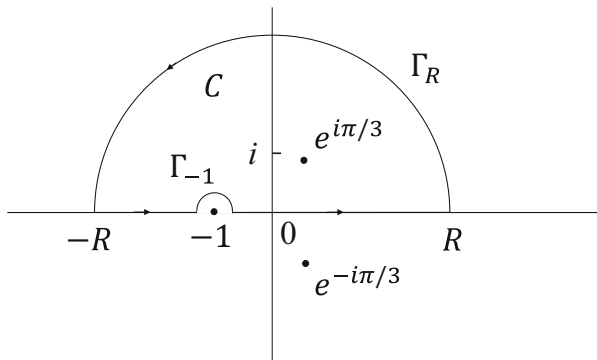
Figure 6.19 depicts a graphical form of $f(x)$. The modulus of $f(x)$ tends to be infinity at $x = -1$. Inflection points are present at $x = 0$ and $\sqrt[3]{1/2}$ (≈ 0.7937).

Now, in the former two examples, the isolated singular points (i.e., poles) existed only in the inside of a closed contour. In the present case, however, a pole is located on the real axis. Bearing in mind this situation, we wish to estimate the integral (6.166).

Rewriting (6.166), we have

$$I = \int_{-\infty}^{\infty} \frac{dx}{(1+x)(x^2-x+1)} \text{ or } I = \int_{-\infty}^{\infty} \frac{dz}{(1+z)(z^2-z+1)}.$$

Fig. 6.20 Contour for the integration of $\frac{1}{1+z^3}$ that appears in Example 6.6



The polynomial of the denominator, i.e., $(1+z)(z^2 - z + 1)$ has a real root at $z = -1$ and complex roots at $z = e^{\pm i\pi/3}$ $\left(= \frac{1 \pm \sqrt{3}i}{2}\right)$. Therefore,

$$f(z) \equiv \frac{1}{(1+z)(z^2 - z + 1)}$$

has three simple poles at $z = -1$ along with $z = e^{\pm i\pi/3}$. This time, we have a contour C depicted in Fig. 6.20, where the contour integration I_C is performed in such a way that

$$\begin{aligned} I_C &= \int_{-R}^{-1-r} \frac{dz}{(1+z)(z^2 - z + 1)} + \int_{\Gamma_{-1}} \frac{dz}{(1+z)(z^2 - z + 1)} \\ &\quad + \int_{-1+r}^R \frac{dz}{(1+z)(z^2 - z + 1)} \\ &\quad + \int_{\Gamma_R} \frac{dz}{(1+z)(z^2 - z + 1)}, \end{aligned} \quad (6.168)$$

where Γ_{-1} stands for a small semicircle of radius r around $z = -1$ as shown. Of the complex roots $z = e^{\pm i\pi/3}$, only $z = e^{i\pi/3}$ is responsible for the contour integration (see Fig. 6.20).

Here we define a principal value P of the integral such that

$$P \int_{-\infty}^{\infty} \frac{dx}{(1+x)(x^2-x+1)} \equiv \lim_{r \rightarrow 0} \left[\int_{-\infty}^{-1-r} \frac{dx}{(1+x)(x^2-x+1)} \right. \\ \left. + \int_{-1+r}^{\infty} \frac{dx}{(1+x)(x^2-x+1)} \right], \quad (6.169)$$

where r is an arbitrary real positive number. As shown in this example, the principal value is a convenient device to traverse the poles on the contour and gives a correct answer in the contour integral. At the same time, getting R to infinity, we obtain

$$\lim_{R \rightarrow \infty} I_C = P \int_{-\infty}^{\infty} \frac{dx}{(1+x)(x^2-x+1)} + \int_{\Gamma_{-1}} \frac{dz}{(1+z)(z^2-z+1)} \\ + \int_{\Gamma_{\infty}} \frac{dz}{(1+z)(z^2-z+1)}. \quad (6.170)$$

Meanwhile, the contour integral I_C is given by (6.146) such that

$$\lim_{R \rightarrow \infty} I_C = \oint_C f(z) dz = 2\pi i \sum_j \text{Res } f(a_j), \quad (6.146)$$

where in the present case $f(z) \equiv \frac{1}{(1+z)(z^2-z+1)}$ and we have only one simple pole at $a_j = e^{i\pi/3}$ within the contour C .

We are going to get the answer by combining (6.170) and (6.146) such that

$$I \equiv P \int_{-\infty}^{\infty} \frac{dx}{(1+x)(x^2-x+1)} \\ = 2\pi i \text{Res } f\left(e^{i\pi/3}\right) - \int_{\Gamma_{-1}} \frac{dz}{(1+z)(z^2-z+1)} \\ - \int_{\Gamma_{\infty}} \frac{dz}{(1+z)(z^2-z+1)}. \quad (6.171)$$

The third term of RHS of (6.171) tends to be zero as in the case of Examples 6.4 and 6.5. Therefore, if we can estimate the second term as well as the residues properly, the integral (6.166) can be adequately determined. Note that in this case the integral is meant by the principal value. To estimate the integral of the second term, we make use of the fact that defining $g(z)$ as

$$g(z) \equiv \frac{1}{z^2 - z + 1}, \quad (6.172)$$

$g(z)$ is analytic at and around $z = -1$. Hence, $g(z)$ can be expanded in the Taylor's series around $z = -1$ such that

$$g(z) = A_0 + \sum_{n=1}^{\infty} A_n (z + 1)^n, \quad (6.173)$$

where $A_0 \neq 0$. In fact,

$$A_0 = g(-1) = 1/3. \quad (6.174)$$

Then, we have

$$\begin{aligned} \int_{\Gamma_{-1}} \frac{dz}{(1+z)(z^2-z+1)} &= \int_{\Gamma_{-1}} \frac{g(z)dz}{1+z} = A_0 \int_{\Gamma_{-1}} \frac{dz}{1+z} \\ &+ \int_{\Gamma_{-1}} \sum_{n=1}^{\infty} A_n (z+1)^{n-1} dz. \end{aligned} \quad (6.175)$$

Changing the variable z to the polar form in RHS such that

$$z = -1 + re^{i\theta}, \quad (6.176)$$

with the first term of RHS we have

$$A_0 \int_{\Gamma_{-1}} \frac{ire^{i\theta} d\theta}{re^{i\theta}} = A_0 \int_{\Gamma_{-1}} id\theta = A_0 \int_{\pi}^0 id\theta = -A_0 \int_0^{\pi} id\theta = -A_0 i\pi. \quad (6.177)$$

Note that in (6.177) the contour integration along Γ_{-1} was performed in the decreasing direction of the argument θ from π to 0. Thus, (6.175) is rewritten as

$$\int_{\Gamma_{-1}} \frac{dz}{(1+z)(z^2-z+1)} = -A_0 i\pi + A_n i \sum_{n=1}^{\infty} r^n \int_{\pi}^0 e^{in\theta} d\theta, \quad (6.178)$$

where with the last equality we exchanged the order of the summation and integration. The second term of RHS of (6.178) vanishes when $r \rightarrow 0$. Thus, (6.171) is further rewritten as

$$I = \lim_{r \rightarrow 0} I = 2\pi i \operatorname{Res} f(e^{i\pi/3}) + A_0 i\pi. \quad (6.179)$$

We have only one simple pole for $f(z)$ at $z = e^{i\pi/3}$ within the semicircle contour C . Then, using (6.148), we have

$$\begin{aligned}\operatorname{Res} f\left(e^{i\pi/3}\right) &= f(z)\left(z - e^{i\pi/3}\right)\Big|_{z=e^{i\pi/3}} \\ &= \frac{1}{(1 + e^{i\pi/3})(e^{i\pi/3} - e^{-i\pi/3})} = -\frac{1}{6}\left(1 + \sqrt{3}i\right).\end{aligned}\quad (6.180)$$

The above calculus is straightforward, but (6.37) can be conveniently used. Thus, finally we get

$$I = -\frac{\pi i}{3}\left(1 + \sqrt{3}i\right) + \frac{\pi i}{3} = \frac{\sqrt{3}\pi}{3} = \frac{\pi}{\sqrt{3}}. \quad (6.181)$$

That is,

$$I \approx 1.8138. \quad (6.182)$$

To avoid any ambiguity of the principal value, we properly define it as [8].

$$P \int_{-\infty}^{\infty} f(x) dx \equiv \lim_{R \rightarrow \infty} \int_{-R}^R f(x) dx.$$

The integral can also be calculated using a lower semicircle including the simple pole located at $z = e^{-\frac{i\pi}{3}} \left(= \frac{1 - \sqrt{3}i}{2}\right)$. That gives the same answer as (6.181). The calculations are left for readers as an exercise.

Example 6.7 [9, 10] It will be interesting to compare the result of Example 6.6 with that of the following real definite integral:

$$I = \int_0^{\infty} \frac{dx}{1 + x^3}. \quad (6.183)$$

To evaluate (6.183), it is convenient to use a sector of circle for contour (see Fig. 6.21a). In this case, the contour integral I_C estimated along a sector is described by

$$I_C = \int_0^R \frac{dz}{1 + z^3} + \int_{\Gamma_R} \frac{dz}{1 + z^3} + \int_L \frac{dz}{1 + z^3}. \quad (6.184)$$

In the third integral of (6.184), we take $\arg z = \theta$ (constant) so that with $z = re^{i\theta}$ we may have $dz = dre^{i\theta}$. Then, that integral is given by

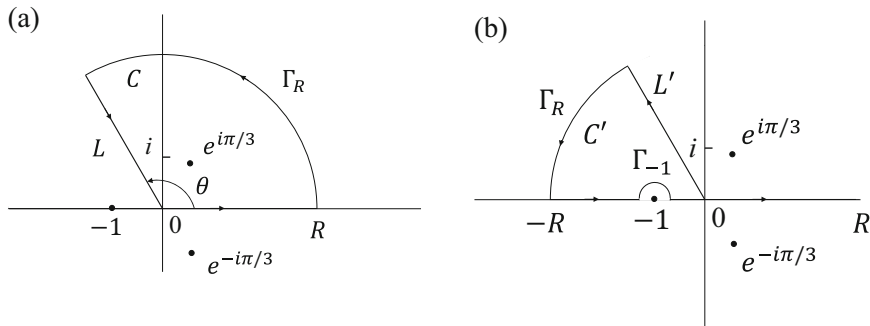


Fig. 6.21 Sector of circle for contour integration of $\frac{1}{1+z^3}$ that appears in Example 6.7. (a) The integration range is $[0, \infty)$. (b) The integration range is $(-\infty, 0]$

$$\int_L \frac{dz}{1+z^3} = \int_L \frac{dr}{1+r^3 e^{3i\theta}}.$$

Setting $3i\theta = 2i\pi$, namely $\theta = 2\pi/3$, we get

$$\int_L \frac{dz}{1+r^3 e^{3i\theta}} = e^{2\pi i/3} \int_R^0 \frac{dr}{1+r^3} = -e^{2\pi i/3} \int_0^R \frac{dr}{1+r^3}.$$

Thus, taking $R \rightarrow \infty$ in (6.184) we have

$$\lim_{R \rightarrow \infty} I_C = \int_0^\infty \frac{dx}{1+x^3} + \int_{\Gamma_\infty} \frac{dz}{1+z^3} - e^{2\pi i/3} \int_0^\infty \frac{dr}{1+r^3}. \quad (6.185)$$

The second term of (6.185) vanishes as before. Changing the variable $r \rightarrow x$, we obtain

$$[\text{RHS of (6.185)}] = \left(1 - e^{2\pi i/3}\right) \int_0^\infty \frac{dx}{1+x^3} = \left(1 - e^{2\pi i/3}\right) I. \quad (6.186)$$

Meanwhile, considering that $f(z)$ has a simple pole at $z = e^{i\pi/3}$ within the contour C , i.e., inside the sector (see Fig. 6.21a), we have

$$[\text{LHS of (6.185)}] = 2\pi i \text{Res} f\left(e^{i\pi/3}\right), \quad (6.187)$$

where $f(x)$ was given in (6.167) of Example 6.6. Then, equating (6.186) with (6.187) we have

$$\begin{aligned}
 I &= 2\pi i \operatorname{Res} f\left(e^{i\pi/3}\right) / \left(1 - e^{2\pi i/3}\right) \\
 &= 2\pi i / \left(1 + e^{i\pi/3}\right) \left(e^{i\pi/3} - e^{-i\pi/3}\right) \left(1 - e^{2\pi i/3}\right).
 \end{aligned}$$

To calculate $\operatorname{Res} f(e^{i\pi/3})$, once again we used (6.148) at $z = e^{i\pi/3}$; see (6.180). Noting that $(1 + e^{i\pi/3})(1 - e^{2\pi i/3}) = 3$ and $e^{i\pi/3} - e^{-i\pi/3} = 2i \sin(\pi/3) = \sqrt{3}i$, we get

$$I = 2\pi i / (3\sqrt{3}i) = \frac{2\pi}{3\sqrt{3}}. \quad (6.188)$$

That is, $I \approx 1.2092$. This number is two-thirds of (6.182).

We further extend the above calculation method to evaluation of real definite integral. For instance, we have a following integral described by

$$\tilde{I} = P \int_{-\infty}^0 \frac{dx}{1+x^3}. \quad (6.189)$$

To evaluate this integral, we define the contour integral $I_{C'}$ as shown in Fig. 6.21b, which is expressed as

$$I_{C'} = P \int_{-R}^0 \frac{dz}{1+z^3} + \int_{\Gamma_{-1}} \frac{dz}{(1+z)(z^2-z+1)} + \int_{L'} \frac{dz}{1+z^3} + \int_{\Gamma_R} \frac{dz}{1+z^3}. \quad (6.190)$$

Notice that combining $I_{C'}$ of (6.190) with I_C of (6.184) makes the contour integration of (6.168). Proceeding as before and noting that there is no singularity inside the contour C' , we have

$$\lim_{R \rightarrow \infty} I_{C'} = 0 = P \int_{-\infty}^0 \frac{dz}{1+z^3} - \frac{\pi i}{3} + e^{2\pi i/3} \int_0^{\infty} \frac{dr}{1+r^3} = P \int_{-\infty}^0 \frac{dz}{1+z^3} - \frac{\pi}{3\sqrt{3}},$$

where we used (6.188) and (6.178) in combination with (6.174). Thus, we get

$$\tilde{I} = \int_{-\infty}^0 \frac{dx}{1+x^3} = \frac{\pi}{3\sqrt{3}}. \quad (6.191)$$

As a matter of course, the result of (6.191) can immediately be obtained by subtracting (6.188) from (6.181).

We frequently encounter real definite integrals including trigonometric functions. In such cases, to make a valid estimation of the integrals it is desired to use complex exponential functions. Jordan's lemma is a powerful tool for this. The proof is given as the following literature [5].

Lemma 6.6: Jordan's Lemma [5] Let Γ_R be a semicircle of radius R centered at the origin in the upper half of the complex plane. Let $f(z)$ be an analytic function that tends uniformly to zero as $|z| \rightarrow \infty$ when $\arg z$ lies in the interval $0 \leq \arg z \leq \pi$. Then, with a non-negative real number α , we have

$$\lim_{R \rightarrow \infty} \int_{\Gamma_R} e^{i\alpha\zeta} f(\zeta) d\zeta = 0.$$

Proof Using polar coordinates, ζ is expressed as

$$\zeta = Re^{i\theta} = R(\cos \theta + i \sin \theta).$$

Then, the integral is denoted by

$$\begin{aligned} I_R &\equiv \int_{\Gamma_R} e^{i\alpha\zeta} f(\zeta) d\zeta = iR \int_0^\pi f(Re^{i\theta}) e^{i\alpha R(\cos \theta + i \sin \theta)} e^{i\theta} d\theta \\ &= iR \int_0^\pi f(Re^{i\theta}) e^{i\alpha R \cos \theta - \alpha R \sin \theta + i\theta} d\theta. \end{aligned}$$

By assumption $f(\zeta)$ tends uniformly to zero as $|\zeta| \rightarrow \infty$, and so we have

$$|f(Re^{i\theta})| < \varepsilon(R),$$

where $\varepsilon(R)$ is a certain positive number that depends only on R and tends to be zero as $|\zeta| \rightarrow \infty$. Therefore, we have

$$|I_R| < \varepsilon(R) R \int_0^\pi e^{-\alpha R \sin \theta} d\theta = 2 \varepsilon(R) R \int_0^{\pi/2} e^{-\alpha R \sin \theta} d\theta,$$

where the last equality results from the symmetry of $\sin \theta$ about $\pi/2$. Meanwhile, we have

$$\sin \theta \geq 2\theta/\pi \text{ when } 0 \leq \theta \leq \pi/2.$$

Hence, we have

$$|I_R| < 2 \varepsilon(R) R \int_0^{\pi/2} e^{-2\alpha R \theta/\pi} d\theta = \frac{\pi \varepsilon(R)}{\alpha} (1 - e^{-\alpha R}).$$

Therefore, we get

$$\lim_{R \rightarrow \infty} I_R = 0.$$

These complete the proof.

Alternatively, for $f(z)$ that is analytic and tends uniformly to zero as $|z| \rightarrow \infty$ when $\arg z$ lies in the interval $\pi \leq \arg z \leq 2\pi$, similarly we have

$$\lim_{R \rightarrow \infty} \int_{\Gamma_R} e^{-i\alpha\zeta} f(\zeta) d\zeta = 0,$$

where α is a certain non-negative real number. In this case, we assume that a semicircle Γ_R of radius R centered at the origin lies in the lower half of the complex plane.

Using the Jordan's lemma, we have several examples.

Example 6.8 Evaluate the following real definite integral:

$$I = \int_{-\infty}^{\infty} \frac{\sin x}{x} dx. \quad (6.192)$$

Rewriting (6.192) as

$$I = \int_{-\infty}^{\infty} \frac{\sin z}{z} dz = \frac{1}{2i} \int_{-\infty}^{\infty} \frac{e^{iz} - e^{-iz}}{z} dz, \quad (6.193)$$

we consider the following contour integral:

$$I_C = P \int_{-R}^R \frac{e^{iz}}{z} dz + \int_{\Gamma_0} \frac{e^{iz}}{z} dz + \int_{\Gamma_R} \frac{e^{iz}}{z} dz, \quad (6.194)$$

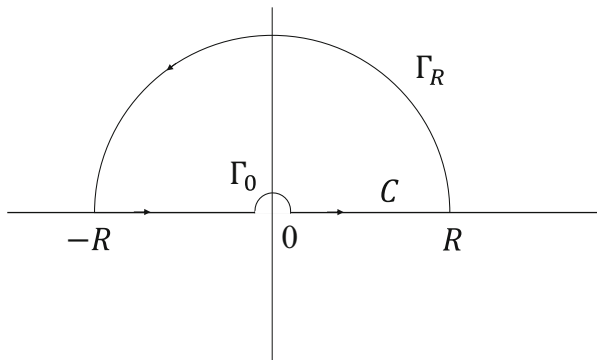
where Γ_0 represents an infinitesimally small semicircle around the origin and Γ_R shows the outer semicircle of radius R centered at the origin (see Fig. 6.22). The contour C consists of the real axis, Γ_0 , and Γ_R as shown.

The third term of (6.194) vanishes when $R \rightarrow \infty$ due to the Jordan's lemma. With the second term, e^{iz} is analytic in the whole complex plane (i.e., entire function) and, hence, can be expanded in the Taylor's series around any complex number. Expanding it around the origin we have

$$e^{iz} = \sum_{n=0}^{\infty} \frac{(iz)^n}{n!} = 1 + \sum_{n=1}^{\infty} \frac{(iz)^n}{n!}. \quad (6.195)$$

As already mentioned in (6.173) through (6.178), among terms of RHS in (6.195) only the first term, i.e., 1, contributes to the integral of the second term of RHS in (6.194). Thus, as in (6.178) we get

Fig. 6.22 Contour for the integration of e^{iz}/z that appears in Example 6.8. The same contour will appear in Example 6.9 as well



$$\int_{\Gamma_0} \frac{e^{iz}}{z} dz = -1 \times i\pi = -i\pi. \quad (6.196)$$

Notice that in (6.196) the argument is decreasing from $\pi \rightarrow 0$ during the contour integration. Within the contour C , there is no singular point, and so $I_C = 0$ due to Theorem 6.10 (Cauchy's integral theorem). Thus, from (6.194) we obtain

$$P \int_{-\infty}^{\infty} \frac{e^{iz}}{z} dz = - \int_{\Gamma_0} \frac{e^{iz}}{z} dz = i\pi. \quad (6.197)$$

Exchanging the variable $z \rightarrow -z$ in (6.197), we have

$$P \left[- \int_{\infty}^{-\infty} \frac{e^{-iz}}{-z} dz \right] = P \left[\int_{-\infty}^{\infty} \frac{e^{-iz}}{-z} dz \right] = i\pi. \quad (6.198)$$

Summing both sides of (6.197) and (6.198), we get

$$P \int_{-\infty}^{\infty} \frac{e^{iz}}{z} dz + P \int_{-\infty}^{\infty} \frac{e^{-iz}}{-z} dz = 2i\pi. \quad (6.199)$$

Rewriting (6.199), we have

$$P \int_{-\infty}^{\infty} \frac{(e^{iz} - e^{-iz})}{z} dz = 2iP \int_{-\infty}^{\infty} \frac{\sin z}{z} dz = 2i\pi. \quad (6.200)$$

That is, the answer is

$$P \int_{-\infty}^{\infty} \frac{\sin z}{z} dz = P \int_{-\infty}^{\infty} \frac{\sin x}{x} dx = I = \pi. \quad (6.201)$$

Notice that as mentioned in (6.137) $z = 0$ is a removable singularity of $(\sin z)/z$, and so the symbol P is superfluous in (6.201).

Example 6.9 Evaluate the following real definite integral:

$$I = \int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx. \quad (6.202)$$

From the trigonometric theorem, we have

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x). \quad (6.203)$$

Then, the integrand of (6.202) with the variable changed is rewritten using (6.37) as

$$\frac{\sin^2 z}{z^2} = \frac{1}{4} \left[\frac{1 - e^{2iz}}{z^2} + \frac{1 - e^{-2iz}}{z^2} \right]. \quad (6.204)$$

Expanding e^{2iz} as before, we have

$$e^{2iz} = \sum_{n=0}^{\infty} \frac{(2iz)^n}{n!} = 1 + 2iz + \sum_{n=2}^{\infty} \frac{(2iz)^n}{n!}. \quad (6.205)$$

Then, we get

$$1 - e^{2iz} = -2iz - \sum_{n=2}^{\infty} \frac{(2iz)^n}{n!},$$

$$\frac{1 - e^{2iz}}{z^2} = -\frac{2i}{z} - \sum_{n=2}^{\infty} \frac{(2i)^n z^{n-2}}{n!}. \quad (6.206)$$

As in Example 6.8, we wish to use (6.206) to estimate the following contour integral I_C :

$$\lim_{R \rightarrow \infty} I_C = P \int_{-\infty}^{\infty} \frac{1 - e^{2iz}}{z^2} dz + \int_{\Gamma_0} \frac{1 - e^{2iz}}{z^2} dz + \int_{\Gamma_{\infty}} \frac{1 - e^{2iz}}{z^2} dz, \quad (6.207)$$

where the contour for the integration is the same as that of Fig. 6.22.

With the second integral of (6.207), as before, only the first term $-2i/z$ of (6.206) contributes to the integral. The result is given as

$$\int_{\Gamma_0} \frac{1 - e^{2iz}}{z^2} dz = -2i \int_{\Gamma_0} \frac{1}{z} dz = -2i \int_{\pi}^0 i d\theta = -2\pi. \quad (6.208)$$

With the third integral of (6.207), we have

$$\int_{\Gamma_{\infty}} \frac{1 - e^{2iz}}{z^2} dz = \int_{\Gamma_{\infty}} \frac{1}{z^2} dz - \int_{\Gamma_{\infty}} \frac{e^{2iz}}{z^2} dz. \quad (6.209)$$

The first term of RHS of (6.209) vanishes similarly to the integral of (6.155) that appeared in Example 6.4. The second term of RHS vanishes as well because of the Jordan's lemma. Within the contour C of (6.207), again there is no singular point, and so $\lim_{R \rightarrow \infty} I_C = 0$. Hence, from (6.207) we have

$$P \int_{-\infty}^{\infty} \frac{1 - e^{2iz}}{z^2} dz = - \int_{\Gamma_0} \frac{1 - e^{2iz}}{z^2} dz = 2\pi. \quad (6.210)$$

Exchanging the variable $z \rightarrow -z$ in (6.210) as before, we have

$$P \left[- \int_{\infty}^{-\infty} \frac{1 - e^{-2iz}}{(-z)^2} dz \right] = P \int_{-\infty}^{\infty} \frac{1 - e^{-2iz}}{z^2} dz = 2\pi. \quad (6.211)$$

Summing both sides of (6.210) and (6.211) in combination with (6.204), we get an answer described by

$$\int_{-\infty}^{\infty} \frac{\sin^2 z}{z^2} dz = \pi. \quad (6.212)$$

Again, the symbol P is superfluous in (6.210) and (6.211).

Example 6.10 Evaluate the following real definite integral:

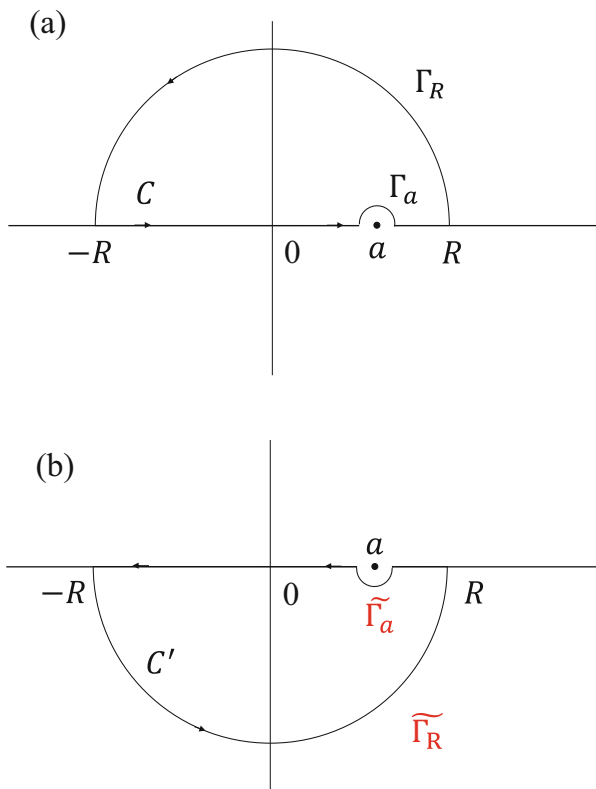
$$I = \int_{-\infty}^{\infty} \frac{\cos x}{(x-a)(x-b)} dx \quad (a \neq b). \quad (6.213)$$

We rewrite the denominator of (6.213) using the method of partial fraction decomposition such that

$$\frac{1}{(x-a)(x-b)} = \frac{1}{a-b} \left(\frac{1}{x-a} - \frac{1}{x-b} \right). \quad (6.214)$$

Then, (6.213) is described by

Fig. 6.23 Contour for the integration of $\frac{\cos z}{(z-a)(z-b)}$ that appears in Example 6.10.
 (a) Along the upper contour C .
 (b) Along the lower contour C'



$$\begin{aligned}
 I &= \frac{1}{a-b} \int_{-\infty}^{\infty} \left(\frac{\cos z}{z-a} - \frac{\cos z}{z-b} \right) dz \\
 &= \frac{1}{2(a-b)} \int_{-\infty}^{\infty} \left(\frac{e^{iz} + e^{-iz}}{z-a} - \frac{e^{iz} + e^{-iz}}{z-b} \right) dz. \quad (6.215)
 \end{aligned}$$

We wish to consider the integration by dividing the integrand and calculate each term individually. First, we estimate

$$\int_{-\infty}^{\infty} \frac{e^{iz}}{z-a} dz. \quad (6.216)$$

To apply the Jordan's lemma to the integration of (6.216), we use the upper contour C that consists of the real axis, Γ_a , and Γ_R ; see Fig. 6.23a. As usual, we consider a following integral:

$$\lim_{R \rightarrow \infty} I_C = P \int_{-\infty}^{\infty} \frac{e^{iz}}{z-a} dz + \int_{\Gamma_a} \frac{e^{iz}}{z-a} dz + \int_{\Gamma_{\infty}} \frac{e^{iz}}{z-a} dz, \quad (6.217)$$

where Γ_a represents an infinitesimally small semicircle around $z = a$. Notice that (6.217) is obtained when we are considering $R \rightarrow \infty$ in Fig. 6.23a. The third term of (6.217) vanishes due to the Jordan's lemma. With the second term of (6.217), we change variable $z - a \rightarrow z$ and rewrite it as

$$\int_{\Gamma_a} \frac{e^{iz}}{z-a} dz = \int_{\Gamma_0} \frac{e^{i(z+a)}}{z} dz = e^{ia} \int_{\Gamma_0} \frac{e^{iz}}{z} dz = -e^{ia} i\pi, \quad (6.218)$$

where with the last equality we used (6.196). There is no singular point within the contour C , and so $\lim_{R \rightarrow \infty} I_C = 0$. Then, we have

$$P \int_{-\infty}^{\infty} \frac{e^{iz}}{z-a} dz = - \int_{\Gamma_a} \frac{e^{iz}}{z-a} dz = e^{ia} i\pi. \quad (6.219)$$

Next, we consider a following integral that appears in (6.215):

$$\int_{-\infty}^{\infty} \frac{e^{-iz}}{z-a} dz. \quad (6.220)$$

To apply the Jordan's lemma to the integration of (6.220), we use the lower contour C' that consists of the real axis, $\tilde{\Gamma}_a$, and $\tilde{\Gamma}_R$ (Figure 6.23b). This time, the integral to be evaluated is given by

$$\lim_{R \rightarrow \infty} I_{C'} = -P \int_{-\infty}^{\infty} \frac{e^{-iz}}{z-a} dz + \int_{\tilde{\Gamma}_a} \frac{e^{-iz}}{z-a} dz + \int_{\tilde{\Gamma}_{\infty}} \frac{e^{-iz}}{z-a} dz \quad (6.221)$$

so that we can trace the contour C' counter-clockwise. Notice that the minus sign is present in front of the principal value. The third term of (6.221) vanishes due to the Jordan's lemma. Evaluating the second term of (6.221) similarly just above, we get

$$\int_{\tilde{\Gamma}_a} \frac{e^{-iz}}{z-a} dz = \int_{\tilde{\Gamma}_0} \frac{e^{-i(z+a)}}{z} dz = e^{-ia} \int_{\tilde{\Gamma}_0} \frac{e^{-iz}}{z} dz = -e^{-ia} i\pi. \quad (6.222)$$

Hence, we obtain

$$P \int_{-\infty}^{\infty} \frac{e^{-iz}}{z-a} dz = \int_{\tilde{\Gamma}_a} \frac{e^{-iz}}{z-a} dz = -e^{-ia} i\pi. \quad (6.223)$$

Notice that in (6.222) the argument is again decreasing from $0 \rightarrow -\pi$ during the contour integration. Summing (6.219) and (6.223), we get

$$\begin{aligned}
P \int_{-\infty}^{\infty} \frac{e^{iz}}{z-a} dz + P \int_{-\infty}^{\infty} \frac{e^{-iz}}{z-a} dz &= P \int_{-\infty}^{\infty} \frac{e^{iz} + e^{-iz}}{z-a} dz \\
&= e^{ia} i\pi - e^{-ia} i\pi = i\pi(e^{ia} - e^{-ia}) = i\pi \cdot 2i \sin a = -2\pi \sin a.
\end{aligned} \tag{6.224}$$

In a similar manner, we also get

$$P \int_{-\infty}^{\infty} \frac{e^{iz} + e^{-iz}}{z-b} dz = -2\pi \sin b.$$

Thus, the final answer is

$$\begin{aligned}
I &= \frac{1}{2(a-b)} \int_{-\infty}^{\infty} \left(\frac{e^{iz} + e^{-iz}}{z-a} - \frac{e^{iz} + e^{-iz}}{z-b} \right) dz = \frac{2\pi(\sin b - \sin a)}{2(a-b)} \\
&= \frac{\pi(\sin b - \sin a)}{a-b}.
\end{aligned}$$

Example 6.11 Evaluate the following real definite integral:

$$I = \int_{-\infty}^{\infty} \frac{\cos x}{1+x^2} dx. \tag{6.225}$$

The calculation is left for readers. Hints: (i) Use $\cos x = (e^{ix} + e^{-ix})/2$ and Jordan's lemma. (ii) Use the contour as shown in Fig. 6.18. Besides the examples listed above, a variety of integral calculations can be seen in literature [9–12].

6.9 Multivalued Functions and Riemann Surfaces

The last topics of the theory of analytic functions are related to multivalued functions. Riemann surfaces play a central role in developing the theory of the multivalued functions.

6.9.1 Brief Outline

We mention a brief outline for notions of multivalued functions and Riemann surfaces. These notions are indispensable for investigating properties of irrational functions and logarithmic functions and performing related calculations, especially integrations with respect to those functions.

Definition 6.12 Let n be a positive integer and let z be an arbitrary complex number. Suppose z can be written as

$$z = w^n. \quad (6.226)$$

Then, with respect to w it is denoted by

$$w = z^{1/n} \quad (6.227)$$

and w is said to be a n -th root of z .

We have expressly shown this simple thing. It is because if we think of real z , $z^{1/n}$ is uniquely determined only when n is odd regardless of whether z is positive or negative. In case we think of even n , (i) if z is positive, we have two n -th root of $\pm z^{1/n}$. (ii) If z is negative, however, we have no n -th root. Meanwhile, if we think of complex z , we have a totally different situation. That is, regardless of whether n is odd or even, we always have n different $z^{1/n}$ for any given complex number z .

As a tangible example, let us think of n -th roots of ± 1 for $n = 2$ and 3 . These are depicted graphically in Fig. 6.24. We can readily understand the statement of the above paragraph. In Fig. 6.24a, if $z = -1$ for $n = 2$, we have either $w = (-1)^{1/2} = (e^{i\pi})^{1/2} = i$ or $w = (-1)^{1/2} = (e^{-i\pi})^{1/2} = -i$ with double root. In Fig. 6.24b, if $z = -1$ for $n = 3$, we have $w = (-1)^{1/3} = -1$; $w = (-1)^{1/3} = (e^{i\pi})^{1/3} = e^{i\pi/3}$; $w = (-1)^{1/3} = (e^{-i\pi})^{1/3} = e^{-i\pi/3}$ with triple root. Moreover, we have $1^{1/2} = 1; -1$. Also, we have $1^{1/3} = 1; e^{2i\pi/3}; e^{-2i\pi/3}$.

Let us extend the above preliminary discussion to the analytic functions. Suppose we are given a following polar form of z such that

$$z = r(\cos \theta + i \sin \theta) \quad (6.32)$$

and

$$w = \rho(\cos \varphi + i \sin \varphi).$$

Then, we have

$$z = w^n = \rho^n(\cos \varphi + i \sin \varphi)^n = \rho^n(\cos n\varphi + i \sin n\varphi), \quad (6.228)$$

where with the last equality we used de Moivre's theorem (6.36). Comparing the real and imaginary parts of (6.32) and (6.228), we have

$$r = \rho^n \text{ or } \rho = r^{1/n} \quad (6.229)$$

and

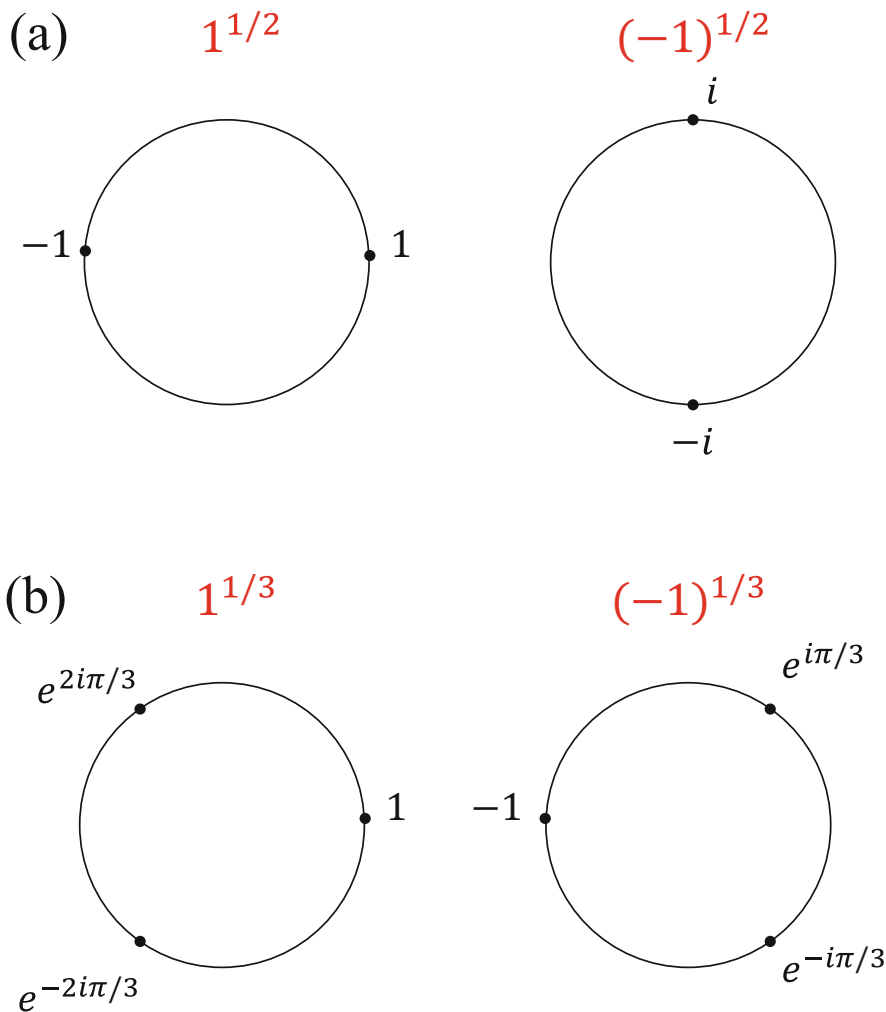


Fig. 6.24 Multiple roots in the complex plane. (a) Square roots of 1 and -1 . (b) Cubic roots of 1 and -1

$$\theta = n\varphi + 2k\pi \quad (k = 0, \pm 1, \pm 2, \dots). \quad (6.230)$$

In (6.229), we define both r and ρ as being positive so that positive ρ can be uniquely determined for any positive integer n (i.e., either even or odd). Recall once again that if n is even, we have two n -th roots for ρ (i.e., both positive and negative) with given positive r . Of these two roots, we discard the negative ρ . Rewriting (6.230), we have

$$\varphi = \frac{1}{n}(\theta - 2k\pi) \quad (k = 0, \pm 1, \pm 2, \dots).$$

Further rewriting (6.227), we get

$$w_k(\theta) = z^{\frac{1}{n}} = r^{\frac{1}{n}} \left[\cos \frac{1}{n}(\theta - 2k\pi) + i \sin \frac{1}{n}(\theta - 2k\pi) \right] \\ (k = 0, 1, 2, \dots, n-1), \quad (6.231)$$

where $w_k(\theta)$ is said to be a branch. Note that we have

$$w_k(\theta - 2n\pi) = w_k(\theta). \quad (6.232)$$

That is, $w_k(\theta)$ is a periodic function of the period $2n\pi$. The number of the total branches is n ; the index k of $w_k(\theta)$ denotes these different branches. In particular, when $k = 0$, we have

$$w_0(\theta) = z^{\frac{1}{n}} = r^{\frac{1}{n}} \left(\cos \frac{1}{n}\theta + i \sin \frac{1}{n}\theta \right). \quad (6.233)$$

Comparing (6.231) and (6.233), we obtain

$$w_k(\theta) = w_0(\theta - 2k\pi). \quad (6.234)$$

From (6.234), we find that $w_k(\theta)$ is obtained by shifting (or rotating) $w_0(\theta)$ by $2k\pi$ toward the positive direction of θ . The branch $w_0(\theta)$ is called a principal branch of the n -th root of z . The value of $w_0(\theta)$ is called the principal value of that branch. The implication of the presence of the branches is as follows: Suppose that the branches besides $w_0(\theta)$ would be absent. Then, from (6.233) we have

$$w_0(2\pi) = z^{\frac{1}{n}} = r^{\frac{1}{n}} \left(\cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n} \right) \neq w_0(0) = r^{\frac{1}{n}}.$$

This causes inconvenience, because $\theta = 0$ and $\theta = 2\pi$ are assumed to be identical in the complex plane. Hence, the single-valuedness would be broken with $w_0(\theta)$. However, in virtue of the presence of $w_1(\theta)$, we have

$$w_1(2\pi) = w_0(2\pi - 2\pi) = w_0(0).$$

This means that the value of $w_0(0)$ is recovered and, hence, the single-valuedness remains intact. After another cycle around the origin, similarly we have

$$w_2(4\pi) = w_0(4\pi - 4\pi) = w_0(0).$$

Thus, in succession we get

$$w_k(2k\pi) = w_0(2k\pi - 2k\pi) = w_0(0).$$

For $k = n$, from (6.231) we have

$$\begin{aligned} w_n(\theta) &= z^{\frac{1}{n}} = r^{\frac{1}{n}} \left[\cos \frac{1}{n}(\theta - 2n\pi) + i \sin \frac{1}{n}(\theta - 2n\pi) \right] \\ &= r^{1/n} \left[\cos \left(\frac{1}{n}\theta - 2\pi \right) + i \sin \left(\frac{1}{n}\theta - 2\pi \right) \right] \\ &= r^{\frac{1}{n}} \left(\cos \frac{1}{n}\theta + i \sin \frac{1}{n}\theta \right) = w_0(\theta). \end{aligned}$$

Thus, we have no more new function. At the same time, the single-valuedness of $w_k(\theta)$ ($k = 0, 1, 2, \dots, n-1$) remains intact during these processes. To summarize the above discussion, if we have n planes (or sheets) as the complex planes and allocate them to $w_0(\theta)$, $w_1(\theta)$, $\dots$, $w_{n-1}(\theta)$ individually, we have a single-valued function $w(\theta)$ as a whole throughout these planes. In a word, the following relation represents the situation:

$$w(\theta) = \begin{cases} w_0(\theta) & \text{for Plane 0,} \\ w_1(\theta) & \text{for Plane 1,} \\ \dots & \dots \\ w_{n-1}(\theta) & \text{for Plane } n-1. \end{cases}$$

This is the essence of the Riemann surface. The superposition of these n planes is called a Riemann surface and each plane is said to be a Riemann sheet [of the function $w(\theta)$]. Each single-valued function $w_0(\theta)$, $w_1(\theta)$, $\dots$, $w_{n-1}(\theta)$ defined on each Riemann sheet is called a branch of $w(\theta)$. In the above discussion, the origin is called a branch point of $w(z)$.

For simplicity, let us think of the function $w(z)$ described by

$$w(z) \equiv z^{1/2} = \sqrt{z}.$$

In this case, for $z = re^{i\theta}$ ($0 \leq \theta \leq 2\pi$) we have two different values w_0 and w_1 given by

$$w_0(\theta) = r^{1/2} e^{i\theta/2},$$

$$w_1(\theta) = r^{1/2} e^{i(\theta-2\pi)/2} = w_0(\theta - 2\pi) = -w_0(\theta). \quad (6.235)$$

Then, we have

$$w(\theta) = \begin{cases} w_0(\theta) & \text{for Plane 0,} \\ w_1(\theta) & \text{for Plane 1.} \end{cases} \quad (6.236)$$

In this case, $w(z)$ is called a “two-valued” function of z and the functions w_0 and w_1 are said to be branches of $w(z)$. Suppose that z makes a counter-clockwise circuit around the origin, starting from, e.g., a real positive number $z = z_0$ to come full circle to the original point $z = z_0$. Then, the argument of z has been increased by 2π . In this situation the arguments of the individual branches w_0 and w_1 are increased by π . Accordingly, w_0 is switched to w_1 and w_1 is switched to w_0 . This situation can be understood more clearly from (6.232) and (6.234). That is, putting $n = 2$ in (6.232), we have

$$w_k(\theta - 4\pi) = w_k(\theta) \quad (k = 0, 1). \quad (6.237)$$

Meanwhile, putting $k = 1$ in (6.234) we get

$$w_1(\theta) = w_0(\theta - 2\pi). \quad (6.238)$$

Replacing θ with $\theta + 2\pi$ in (6.238), we have $w_1(\theta + 2\pi) = w_0(\theta)$. Also replacing θ with $\theta + 4\pi$ in (6.237), we have $w_1(\theta + 4\pi) = w_1(\theta) = w_0(\theta - 2\pi) = w_0(\theta + 2\pi)$, where with the last equality we replaced θ with $\theta + 2\pi$ in (6.237). Rewriting the above, we get

$$w_1(\theta + 2\pi) = w_0(\theta) \text{ and } w_0(\theta + 2\pi) = w_1(\theta).$$

That is, adding 2π to θ , w_0 and w_1 are switched to each other.

Let us further think of the two-valued function of $w(z) = z^{1/2}$. Strictly speaking, an analytic function cannot be a two-valued function. This is because if so, the continuity and differentiability will be lost from the function and, hence, the analyticity would be broken. Then, we must make a suitable device to avoid it. Such a device is called a Riemann surface.

Let us make a kit of the Riemann surface following Fig. 6.25. (i) Take a sheet of paper so that it can represent a complex plane and cut it with scissors along the real axis that starts from the origin so that the cut (or slit) can be made toward the real positive direction. This cut is called a *branch cut* with the origin being a *branch point*. Let us call this sheet Plane I. (ii) Take another sheet of paper and call it Plane II. Also make a cut in Plane II in exactly the same way as that for Plane I; see

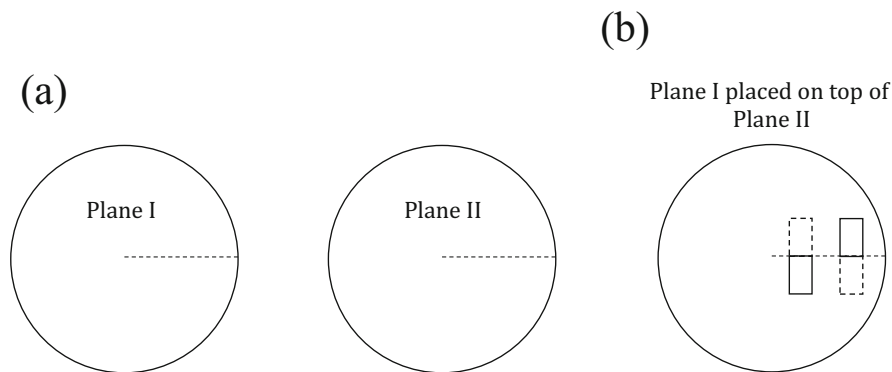


Fig. 6.25 Simple kit that helps visualize the Riemann surface. To make it, follow next procedures: (a) Take two sheets of paper (Planes I and II) and make slits (indicated by dashed lines) as shown. Next, put Plane I on top of Plane II so that the two slits (i.e., branch cuts) can fit in line. (b) Tape together the downside of the cut of Plane I and the foreside of the cut of Plane II. Then, also tape together the downside of the cut of Plane II and the foreside of the cut of Plane I

Fig. 6.25a for the processes (i) and (ii). (iii) Next, put Plane I on top of Plane II so that the two branch cuts can fit in line. (iv) Tape together the downside of the cut of Plane I and the foreside of the cut of Plane II (Fig. 6.25b). (v) Then, also tape together the downside of the cut of Plane II and the foreside of the cut of Plane I (see Fig. 6.25b once again).

Thus, what we see is that starting from, e.g., a real positive number $z = z_0$ of Plane I and coming full circle to the original point $z = z_0$, then we cross the cut to enter Plane II located underneath Plane I. After another cycle within Plane II, we come back to Plane I again by crossing the cut. After all, we come back to the original Plane I after two cycles on the combined planes. This combined plane is called the Riemann surface. In this way, Planes I and II correspond to different branches w_0 and w_1 , respectively, so that each branch can be single-valued. In other words, $w(z) = z^{1/2}$ is a single-valued function that is defined on the whole Riemann surface.

There are a variety of Riemann surfaces according to the nature of the complex functions. Another example of the multivalued function is expressed as

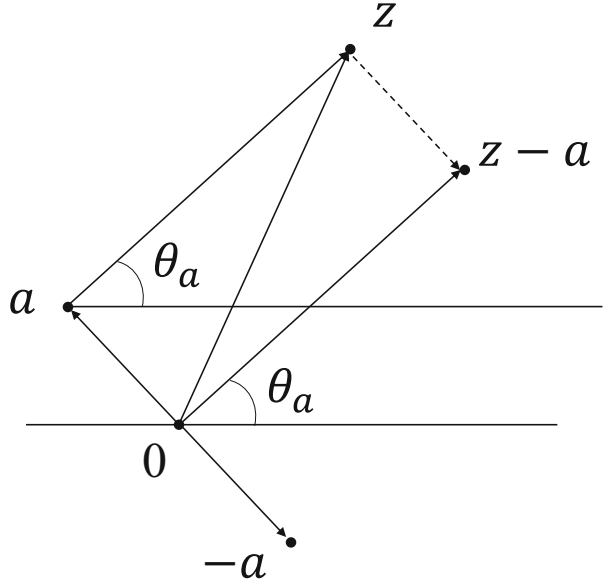
$$w(z) = \sqrt{(z-a)(z-b)},$$

where two branch points are located at $z = a$ and $z = b$. As before, we assume that

$$z - a = r_a e^{i\theta_a} \text{ and } z - b = r_b e^{i\theta_b}.$$

Then, we have

Fig. 6.26 Graphically decided argument θ_a , which is an angle between a line connecting z and a and another line drawn in parallel with the real axis



$$w(\theta_a, \theta_b) = z^{1/2} = \sqrt{r_a r_b} e^{i(\theta_a + \theta_b)/2}. \quad (6.239)$$

In the above, e.g., the argument θ_a can be decided graphically as shown in Fig. 6.26, where θ_a is an angle between a line connecting z and a and another line drawn in parallel with the real axis. We can choose $w(\theta_a, \theta_b)$ of (6.239) for the principal branch and define this as

$$w_0(\theta_a, \theta_b) \equiv \sqrt{r_a r_b} e^{i(\theta_a + \theta_b)/2}. \quad (6.240)$$

Also, we define $w_1(\theta_a, \theta_b)$ as

$$w_1(\theta_a, \theta_b) \equiv \sqrt{r_a r_b} e^{i(\theta_a + \theta_b - 2\pi)/2}. \quad (6.241)$$

Then, as in (6.235) we have

$$w_1(\theta_a, \theta_b) = w_0(\theta_a - 2\pi, \theta_b) = w_0(\theta_a, \theta_b - 2\pi) = -w_0(\theta_a, \theta_b). \quad (6.242)$$

From (6.242), we find that adding 2π to θ_a or θ_b , w_0 and w_1 are switched to each other as before. We also find that after the variable z has come full circle around one of a and b , $w_0(\theta_a, \theta_b)$ changes sign as in the case of (6.235).

We also have

$$\begin{aligned}w_0(\theta_a - 2\pi, \theta_b - 2\pi) &= w_0(\theta_a, \theta_b), \\w_1(\theta_a - 2\pi, \theta_b - 2\pi) &= w_1(\theta_a, \theta_b).\end{aligned}\tag{6.243}$$

From (6.243), however, after the variable z has come full circle around *both* a and b , both $w_0(\theta_a, \theta_b)$ and $w_1(\theta_a, \theta_b)$ keep the original value. This is also the case where the variable z comes full circle without encircling a or b . These behaviors imply that (i) if a contour encircles one of $z = a$ and $z = b$, $w_0(\theta_a, \theta_b)$ and $w_1(\theta_a, \theta_b)$ change sign and switch to each other. Thus, $w_0(\theta_a, \theta_b)$ and $w_1(\theta_a, \theta_b)$ form branches. On the other hand, (ii) if a contour encircles both $z = a$ and $z = b$, $w_0(\theta_a, \theta_b)$ and $w_1(\theta_a, \theta_b)$ remain intact. (iii) If a contour encircles neither $z = a$ nor $z = b$, $w_0(\theta_a, \theta_b)$ and $w_1(\theta_a, \theta_b)$ remain intact as well.

These three cases (i), (ii), and (iii) are depicted in Fig. 6.27a, b, and c, respectively. In Fig. 6.27c after z comes full circle, $\arg z$ relative to $z = a$ returns to the original value keeping $\theta_1 \leq \arg z \leq \theta_2$. This is similarly the case with $\arg z$ relative to $z = b$. Correspondingly, the branch cut(s) are depicted with doubled broken line(s), e.g., as shown in Fig. 6.28. Note that in the cases (ii) and (iii) the branch cut can be chosen so that the circle (or contour) may not cross the branch cut. Although a bit complicated, the Riemann surface can be shaped accordingly. Other choices of the branch cuts are shown in Sect. 6.9.2 (vide infra).

When we consider logarithm functions, we have to deal with rather complicated situation. For polar coordinate representation, we have $z = re^{i\theta}$. That is,

$$\ln z = \ln r + i \arg z = \ln|z| + i \arg z, \tag{6.244}$$

where

$$\arg z = \theta + 2\pi n \quad (n = 0, \pm 1, \pm 2, \dots). \tag{6.245}$$

If $n = 0$ is chosen, $\ln z$ is said to be the principal value. With the logarithm functions, the Riemann surface comprises infinite planes each of which corresponds to the individual branch whose argument is given by (6.245). The point $z = 0$ is called a logarithmic branch point. Meanwhile, the branch point for, e.g., $w(z) = \sqrt{z}$ is said to be an algebraic branch point.

6.9.2 Examples of Multivalued Functions

When we deal with a function having a branch point, we must remind that unless the contour crosses the branch cut (either algebraic or logarithmic), a function we are

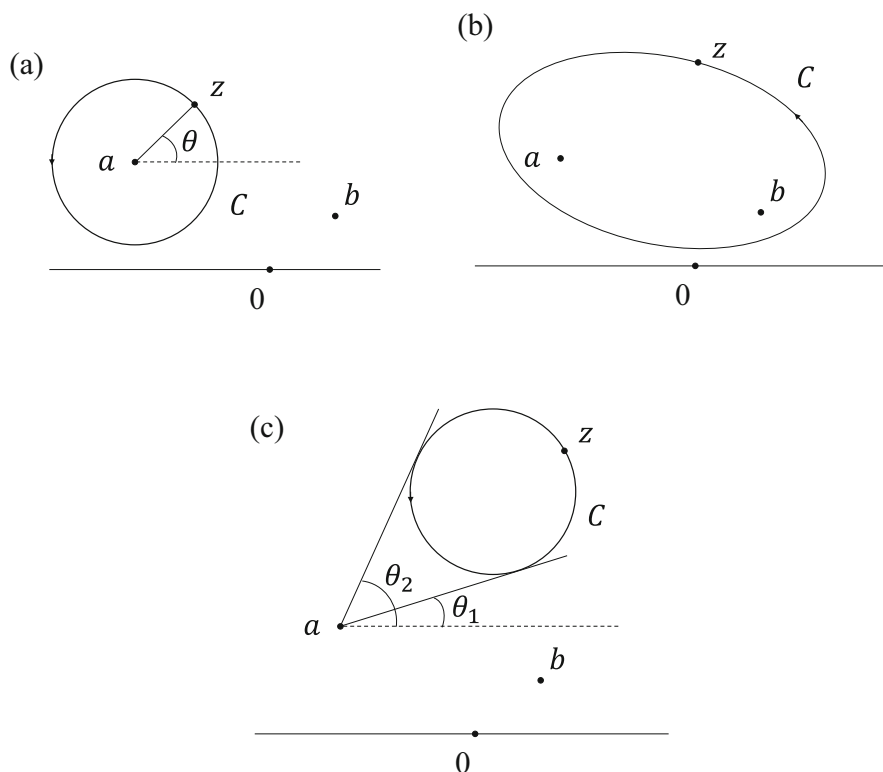


Fig. 6.27 Geometrical relationship between contour C and two branch points (located at $z = a$ and $z = b$) for $\sqrt{(z-a)(z-b)}$. (a) The contour C encircles only $z = a$. (b) C encircles both $z = a$ and $z = b$. (c) C encircles neither $z = a$ nor $z = b$

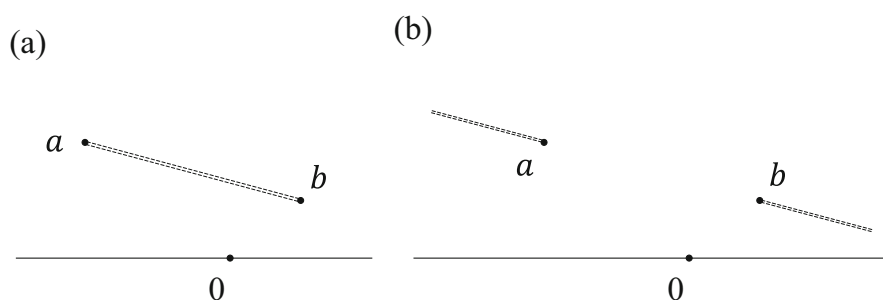
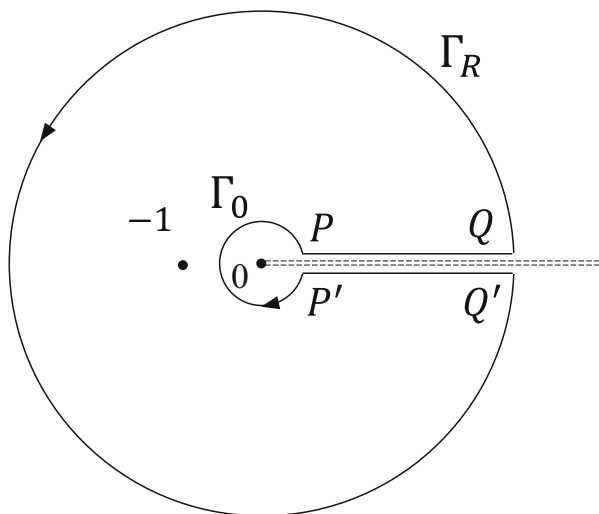


Fig. 6.28 Branch cut(s) for $\sqrt{(z-a)(z-b)}$. (a) A branch cut is shown by a line connecting $z = a$ and $z = b$. (b) Branch cuts are shown by a line connecting $z = a$ and $z = \infty$ and another line connecting $z = b$ and $z = \infty$. In both (a) and (b) the branch cut(s) are depicted with doubled broken line(s)

Fig. 6.29 Branch cut (shown with a doubled broken line) and contour for the integration of $\frac{z^{a-1}}{1+z}$



thinking of is held single-valued and analytic (if that function is originally analytic) and can be differentiated or integrated in a normal way. We give some examples for this.

Example 6.12 [6] Evaluate the following real definite integral:

$$I = \int_0^\infty \frac{x^{a-1}}{1+x} dx \quad (0 < a < 1). \quad (6.246)$$

We rewrite (6.246) as

$$I = \int_0^\infty \frac{z^{a-1}}{1+z} dz. \quad (6.247)$$

We define the integrand of (6.247) as

$$f(z) \equiv \frac{z^{a-1}}{1+z},$$

where z^{a-1} can be further rewritten as

$$z^{a-1} = e^{(a-1)\ln z}.$$

The function $f(z)$ has a branch point at $z = 0$ and a simple pole at $z = -1$. Bearing in mind this situation, we consider a contour for integration (see Fig. 6.29). In Fig. 6.29 we depict the branch cut with a doubled broken line. Lines PQ and $Q'P'$ are contour lines located over and under the branch cut, respectively. We assume that

the lines PQ and $Q'P'$ are illimitably close to the real axis. Thus, starting from the point P the contour integration I_C is described by

$$I_C = \int_{PQ} \frac{z^{a-1}}{1+z} dz + \int_{\Gamma_R} \frac{z^{a-1}}{1+z} dz + \int_{Q'P'} \frac{z^{a-1}}{1+z} dz + \int_{\Gamma_0} \frac{z^{a-1}}{1+z} dz, \quad (6.248)$$

where Γ_R and Γ_0 denote the outer large circle and inner small circle of their radius R ($\gg 1$) and r ($\ll 1$), respectively. Note that with the contour integration Γ_R is traced counter-clockwise but Γ_0 is traced clockwise (see Fig. 6.29). Since the simple pole is present at $z = -1$, the related residue $\text{Res } f(-1)$ is

$$\text{Res } f(-1) = z^{a-1} \Big|_{z=-1} = (-1)^{a-1} = (e^{i\pi})^{a-1} = -e^{ai\pi}.$$

Then, we have

$$I_C = 2\pi i \text{Res } f(-1) = -2\pi i e^{ai\pi}. \quad (6.249)$$

When $R \rightarrow \infty$ and $r \rightarrow 0$, we have

$$\left| \int_{\Gamma_R} \frac{z^{a-1}}{1+z} dz \right| < \frac{R^{a-1}}{R-1} \cdot 2\pi R \rightarrow 0 \text{ and } \left| \int_{\Gamma_0} \frac{z^{a-1}}{1+z} dz \right| < \frac{r^{a-1}}{1-r} \cdot 2\pi r \rightarrow 0. \quad (6.250)$$

Thus, the second and fourth terms of (6.248) vanish.

Meanwhile, we take the principal value of $\ln z$ of (6.244). Since the lines PQ and $Q'P'$ are very close to the real axis, using (6.245) with $n = 0$ we can put $\theta = 0$ on PQ and $\theta = 2\pi$ on $Q'P'$. Then, we have

$$\int_{PQ} \frac{z^{a-1}}{1+z} dz = \int_{PQ} \frac{e^{(a-1)\ln z}}{1+z} dz = \int_{PQ} \frac{e^{(a-1)\ln|z|}}{1+z} dz. \quad (6.251)$$

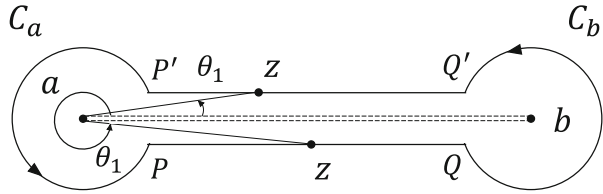
Moreover, we have

$$\begin{aligned} \int_{Q'P'} \frac{z^{a-1}}{1+z} dz &= \int_{Q'P'} \frac{e^{(a-1)\ln z}}{1+z} dz = \int_{Q'P'} \frac{e^{(a-1)(\ln|z|+2\pi i)}}{1+z} dz \\ &= e^{(a-1)2\pi i} \int_{Q'P'} \frac{e^{(a-1)\ln|z|}}{1+z} dz. \end{aligned} \quad (6.252)$$

Notice that the argument underneath the branch cut (i.e., the line $Q'P'$) is increased by 2π relative to that over the branch cut.

Considering (6.249)–(6.252) and taking the limit of $R \rightarrow \infty$ and $r \rightarrow 0$, (6.248) is rewritten by

Fig. 6.30 Branch cut (shown with a doubled broken line) and contour for the integration of $\frac{1}{\sqrt{(z-a)(z-b)}}$ ($a < b$). Only the argument θ_1 is depicted. With θ_2 see text



$$\begin{aligned}
 -2\pi i e^{a\pi i} &= \int_0^\infty \frac{e^{(a-1)\ln|z|}}{1+z} dz + e^{(a-1)2\pi i} \int_\infty^0 \frac{e^{(a-1)\ln|z|}}{1+z} dz \\
 &= [1 - e^{(a-1)2\pi i}] \int_0^\infty \frac{e^{(a-1)\ln|z|}}{1+z} dz = [1 - e^{(a-1)2\pi i}] \int_0^\infty \frac{z^{a-1}}{1+z} dz. \quad (6.253)
 \end{aligned}$$

In (6.253) we have assumed z to be real and positive (see Fig. 6.29), and so we have $e^{(a-1)\ln|z|} = e^{(a-1)\ln z} = z^{a-1}$. Therefore, from (6.253) we get

$$\int_0^\infty \frac{z^{a-1}}{1+z} dz = -\frac{2\pi i e^{a\pi i}}{1 - e^{(a-1)2\pi i}} = -\frac{2\pi i e^{a\pi i}}{1 - e^{2\pi a i}}. \quad (6.254)$$

Using (6.37) for (6.254) and performing simple trigonometric calculations (after multiplying both the numerator and denominator by $1 - e^{-2\pi a i}$), we finally get

$$\int_0^\infty \frac{z^{a-1}}{1+z} dz = \int_0^\infty \frac{x^{a-1}}{1+x} dx = I = \frac{\pi}{\sin a\pi}.$$

Example 6.13 [13] Estimate the following real definite integral:

$$I = \int_a^b \frac{1}{\sqrt{(x-a)(b-x)}} dx \quad (a, b : \text{real with } a < b). \quad (6.255)$$

To this end, we wish to evaluate the following integral described by

$$I_C = \int_C \frac{1}{\sqrt{(z-a)(z-b)}} dz, \quad (6.256)$$

where we define the integrand of (6.256) as

$$f(z) \equiv \frac{1}{\sqrt{(z-a)(z-b)}}.$$

The total contour C comprises C_a , PQ , C_b , and $Q'P'$ as depicted in Fig. 6.30. The function $f(z)$ has two branch points at $z = a$ and $z = b$; otherwise $f(z)$ is analytic. We

draw a branch cut with a doubled broken line as shown. Since the contour C does not cross the branch cut but encircle it, we can evaluate the integral in a normal manner. Starting from the point P' , (6.256) can be expressed by

$$I_C = \int_{C_a} \frac{1}{\sqrt{(z-a)(z-b)}} dz + \int_{PQ} \frac{1}{\sqrt{(z-a)(z-b)}} dz + \int_{C_b} \frac{1}{\sqrt{(z-a)(z-b)}} dz \\ + \int_{Q'P'} \frac{1}{\sqrt{(z-a)(z-b)}} dz. \quad (6.257)$$

We assume that the lines PQ and $Q'P'$ are illimitably close to the real axis. This time, C_a and C_b are both traced counter-clockwise. Putting

$$z - a = r_1 e^{i\theta_1} \quad \text{and} \quad z - b = r_2 e^{i\theta_2},$$

we have

$$f(z) = \frac{1}{\sqrt{r_1 r_2}} e^{-i(\theta_1 + \theta_2)/2}. \quad (6.258)$$

Let C_a and C_b be a small circle of radius ε centered at $z = a$ and $z = b$, respectively. When we are evaluating the integral on C_b , we can put

$$z - b = \varepsilon e^{i\theta_2} \quad (-\pi \leq \theta_2 \leq \pi); \text{ i.e., } r_2 = \varepsilon.$$

We also have $r_1 \approx b - a$; in Fig. 6.30 we assume $b > a$. Hence, we have

$$\left| \int_{C_b} \frac{1}{\sqrt{(z-a)(z-b)}} dz \right| = \left| \int_{-\pi}^{\pi} \frac{i\varepsilon}{\sqrt{r_1}\varepsilon} e^{-i(\theta_1 + \theta_2)/2} d\theta_2 \right| \leq \int_{-\pi}^{\pi} \frac{\sqrt{\varepsilon}}{\sqrt{r_1}} d\theta_2.$$

Therefore, we get

$$\lim_{\varepsilon \rightarrow 0} \left| \int_{C_b} \frac{1}{\sqrt{(z-a)(z-b)}} dz \right| = 0 \quad \text{or} \quad \lim_{\varepsilon \rightarrow 0} \int_{C_b} \frac{1}{\sqrt{(z-a)(z-b)}} dz = 0. \quad (6.259)$$

In a similar manner, we have

$$\lim_{\varepsilon \rightarrow 0} \int_{C_a} \frac{1}{\sqrt{(z-a)(z-b)}} dz = 0.$$

Meanwhile, on the line $Q'P'$ we have

$$z - a = r_1 e^{i\theta_1} \text{ and } z - b = r_2 e^{i\theta_2}. \quad (6.260)$$

In (6.260) we have $r_1 = x - a$, $\theta_1 = 0$ and $r_2 = b - x$, $\theta_2 = \pi$; see Fig. 6.30. Also, we have $dz = dx$. Hence, we have

$$\begin{aligned} \int_{Q'P'} \frac{1}{\sqrt{(z-a)(z-b)}} dz &= \int_b^a \frac{e^{-i\pi/2}}{\sqrt{(x-a)(b-x)}} dx = - \int_b^a \frac{i}{\sqrt{(x-a)(b-x)}} dx \\ &= i \int_a^b \frac{1}{\sqrt{(x-a)(b-x)}} dx. \end{aligned} \quad (6.261)$$

On the line PQ , in turn, we have $r_1 = x - a$ and $\theta_1 = 2\pi$. This is because when going from P' to P , the argument is increased by 2π ; see Fig. 6.30 once again. Also, we have $r_2 = b - x$, $\theta_2 = \pi$, and $dz = dx$. Notice that the argument θ_2 remains unchanged. Then, we get

$$\begin{aligned} \int_{PQ} \frac{1}{\sqrt{(z-a)(z-b)}} dz &= \int_a^b \frac{e^{-3i\pi/2}}{\sqrt{(x-a)(b-x)}} dx \\ &= i \int_a^b \frac{1}{\sqrt{(x-a)(b-x)}} dx. \end{aligned} \quad (6.262)$$

Inserting these results into (6.257), we obtain

$$I_C = \oint_C \frac{1}{\sqrt{(z-a)(z-b)}} dz = 2i \int_a^b \frac{1}{\sqrt{(x-a)(b-x)}} dx. \quad (6.263)$$

The calculation is yet to be finished, however, because (6.263) is not a real definite integral. Then, let us try another contour integration. We choose a contour C of a radius R and centered at the origin. We assume that R is large enough this time. Meanwhile, putting

$$z = \frac{1}{w}, \quad (6.264)$$

we have

$$\sqrt{(z-a)(z-b)} = \frac{1}{w} \sqrt{(1-aw)(1-bw)} \quad \text{with } dz = -\frac{dw}{w^2}. \quad (6.265)$$

Thus, we have

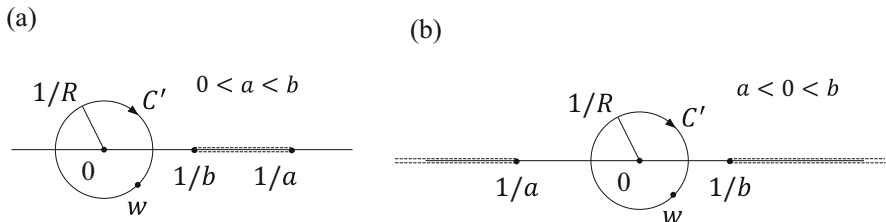


Fig. 6.31 Branch cuts (shown with doubled broken lines) and contour in the w -plane for the integration of $\frac{1}{w\sqrt{(1-aw)(1-bw)}}$. (a) Case of $0 < a < b$. (b) Case of $a < 0 < b$. Notice that in both cases the contour C' is traced clockwise (see text)

$$I_C = \oint_C \frac{1}{\sqrt{(z-a)(z-b)}} dz = - \oint_{C'} \frac{1}{w\sqrt{(1-aw)(1-bw)}} dw, \quad (6.266)$$

where the closed contour C' of diameter $1/R$ comes full circle *clockwise* around the origin of the w -plane (see Fig. 6.31). This is because as in the z -plane the contour C is traced viewing the point $z = \infty$ on the right, the contour C' is traced viewing the corresponding point $w = 0$ on the right. The above clockwise cycle results from such a situation. The integrand of (6.266) has a simple pole at $w = 0$.

Regarding other singularities of the integrand, there are two branch points at $w = 1/a$ and $w = 1/b$. Then, we appropriately choose $R \gg 1$ (or $1/R \ll 1$) so that C' does not cut across the branch cut. To meet this condition, we have two choices. (i) The branch cut connects $w = 1/a$ and $w = 1/b$. (ii) Another choice of the branch cut is that it connects $w = 1/a$ and $w = -\infty$ along with $w = 1/b$ and $w = \infty$ (see Fig. 6.31). If we have $0 < a < b$, we can choose a branch cut as that depicted in Fig. 6.31a corresponding to Fig. 6.28a. In case $a < 0 < b$, the branch cut can be that shown in Fig. 6.31b corresponding to Fig. 6.28b. (When $a < b < 0$, the branch cut geometry may be that similar to Fig. 6.31a.)

Consequently, in both the above cases of (i) and (ii) there is only a simple pole at $w = 0$ without any other singularities inside the contour C' . Therefore, according to (6.148) we have a contour integration described by

$$\begin{aligned} \oint_{C'} \frac{1}{w\sqrt{(1-aw)(1-bw)}} dw &= -2\pi i \operatorname{Res} \frac{1}{w\sqrt{(1-aw)(1-bw)}} \Big|_{w=0} \\ &= -2\pi i \cdot \frac{1}{\sqrt{(1-aw)(1-bw)}} \Big|_{w=0} = -2\pi i, \end{aligned} \quad (6.267)$$

where the above minus sign is due to the clockwise rotation of the contour integration. Hence, from (6.266) we get

$$I_C = \oint_C \frac{1}{\sqrt{(z-a)(z-b)}} dz = 2\pi i. \quad (6.268)$$

Equating (6.263) and (6.268), we finally obtain the answer described by

$$\int_a^b \frac{1}{\sqrt{(x-a)(b-x)}} dx = \pi.$$

Example 6.14 [14] In Chap. 3 we mentioned the analyticity of the function

$$(1 - 2tx + t^2)^{-\lambda}. \quad (6.269)$$

In Chap. 3, we assumed that x is a real number belonging to an interval $[-1, 1]$. Under this condition, we define $f(t)$ in a complex domain as

$$f(t) \equiv 1 - 2tx + t^2. \quad (6.270)$$

We rewrite (6.270) as

$$f(t) = \left[t - \left(x + i\sqrt{1-x^2} \right) \right] \left[t - \left(x - i\sqrt{1-x^2} \right) \right]. \quad (6.271)$$

With $f(t) = 0$ we have two roots at $t_{\pm} = x \pm i\sqrt{1-x^2}$, and so $|t_{\pm}| = 1$ (see Sect. 3.6.1).

When $\lambda = 1/2$, we are dealing with essentially the same problem as that of Example 6.13. For instance, when $x = 0$, the branch points are located at $t = \pm i$. When $x = 1/\sqrt{2}$, the branch points are positioned at $t = (1 \pm i)/\sqrt{2}$. These values of t form the branch points. In Fig. 6.32 we depict these branch points and corresponding branch cuts. In this situation, the function

$$h(t) \equiv (1 - 2tx + t^2)^{-1/2} \quad (6.272)$$

is analytic within a circle $|t| < 1$ of the complex plane t [14] (Fig. 6.32). Therefore, we can freely expand $h(t)$ into a Taylor's series at any point within that circle. If we expand $h(t)$ around $t = 0$, we expect to have a following Taylor's series:

$$h(t) = \sum_{n=0}^{\infty} c_n(x) t^n. \quad (6.273)$$

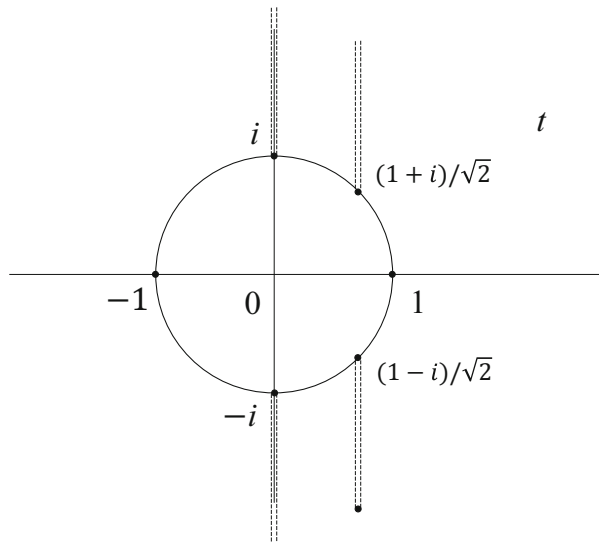
We can readily decide $c_n(x)$ using (6.115) such that

$$c_n(x) = \frac{1}{2\pi i} \oint_C \frac{h(t)}{t^{n+1}} dt, \quad (6.274)$$

where the contour C can be chosen so that C can encircle $t = 0$ in a region $|t| < 1$.

Let us make a substitution such that [14]

Fig. 6.32 Branch points and corresponding branch cuts for $(1 - 2tx + t^2)^{-1/2}$. For instance, when $x = 0$, the branch points are located at $t = \pm i$. When $x = 1/\sqrt{2}$, the branch points are positioned at $t = (1 \pm i)/\sqrt{2}$



$$1 - ut = (1 - 2tx + t^2)^{\frac{1}{2}}. \quad (6.275)$$

Then we have

$$t = \frac{2(u-x)}{u^2-1} \text{ and } dt = 2(1-ut)du/(u^2-1) \text{ with } u \neq \pm 1. \quad (6.276)$$

Using these relations, we rewrite (6.274) as

$$c_n(x) = \frac{1}{2\pi i} \oint_{C'} \frac{(u^2-1)^n}{2^n(u-x)^{n+1}} du = \frac{1}{2\pi i} \oint_{C'} \frac{(-1)^n(1-u^2)^n}{2^n(u-x)^{n+1}} du, \quad (6.277)$$

where the contour C' encircles $u = x$. Here using (6.118), we readily get

$$c_n(x) = \frac{1}{2\pi i} \oint_{C'} \frac{(-1)^n(1-u^2)^n}{2^n(u-x)^{n+1}} du = \frac{(-1)^n}{2^n n!} \left. \frac{d^n(1-u^2)^n}{du^n} \right|_{u=x} \equiv P_n(x). \quad (6.278)$$

The functions $P_n(x)$ are Legendre polynomials that have already appeared in (3.145) and (3.207) of Sects. 3.5 and 3.6. In this manner, the above argument directly confirms that the Legendre polynomials of (3.194) derived from the generating function (3.180) or defined as Rodrigues formula of (3.145) are identical in the complex domain. Originally, (3.145) and (3.207) were defined in the real domain. The identical representations in both the domains real and complex are the consequence of the analytic continuation.

Substituting $x = \pm 1$ for (6.277), we get

$$\begin{aligned}
 c_n(1) &= \frac{1}{2\pi i} \oint_{C'} \frac{(-1)^n (1-u^2)^n}{2^n (u-1)^{n+1}} du = \frac{1}{2^n} \cdot \frac{1}{2\pi i} \oint_{C'} \frac{(u+1)^n}{u-1} du = \frac{1}{2^n} \cdot (u+1)^n \Big|_{u=1} \\
 &= 1
 \end{aligned} \tag{6.279}$$

and

$$\begin{aligned}
 c_n(-1) &= \frac{1}{2\pi i} \oint_{C'} \frac{(-1)^n (1-u^2)^n}{2^n (u+1)^{n+1}} du = \frac{1}{2^n} \cdot \frac{1}{2\pi i} \oint_{C'} \frac{(u-1)^n}{u+1} du = \frac{1}{2^n} \cdot (u-1)^n \Big|_{u=-1} \\
 &= (-1)^n.
 \end{aligned} \tag{6.280}$$

Notice that in (6.279) and (6.280), the integrand has a simple pole at $u = 1$ and $u = -1$, respectively. Hence, we used (6.144) and (6.148) with the contour integration. The contour C' is taken so that it can encircle $u = 1$ in the former case and $u = -1$ in the latter. Hence, we have important relations.

$$P_n(1) = 1 \quad \text{and} \quad P_n(-1) = (-1)^n.$$

Summarizing this chapter, we have explored the theory and applications of the analytic functions on the complex domain. We have studied how the theory of analytic functions produces fruitful results.

References

1. Stoll RR (1979) Set theory and logic. Dover, New York
2. Willard S (2004) General topology. Dover, New York
3. McCarty G (1988) Topology. Dover, New York
4. Mendelson B (1990) Introduction to topology, 3rd edn. Dover, New York
5. Dennery P, Krzywicki A (1996) Mathematics for physicists. Dover, New York
6. Takagi T (2010) Introduction to analysis, Standard edn. Iwanami, Tokyo. (in Japanese)
7. Rudin W (1987) Real and complex analysis, 3rd edn. McGraw-Hill, New York
8. Byron FW Jr, Fuller RW (1992) Mathematics of classical and quantum physics. Dover, New York
9. Arfken GB, Weber HJ, Harris FE (2013) Mathematical methods for physicists, 7th edn. Academic Press, Waltham
10. Hassani S (2006) Mathematical physics. Springer, New York
11. Riley KF, Hobson MP, Bence SJ (2006) Mathematical methods for physics and engineering, 3rd edn. Cambridge University Press, Cambridge
12. Boas ML (2006) Mathematical methods in the physical sciences, 3rd edn. Wiley, New York
13. Omote M (1988) Complex functions. Iwanami, Tokyo. (in Japanese)
14. Lebedev NN (1972) Special functions and their applications. Dover, New York

Part II

Electromagnetism

Electromagnetism is one of the pillars of modern physics, even though it belongs to classical physics along with Newtonian mechanics. Maxwell's equations describe and interpret almost all electromagnetic phenomena and are basis equations of classical physics together with Newton equation. Although after the discovery of relativistic theory Newton equation needed to be modified, Maxwell's equations did not virtually have to be changed. In this part we treat the characteristics of electromagnetic waves that are directly derived from Maxwell's equations.

Electromagnetism becomes connected to quantum mechanics, especially when we deal with emission and absorption of light. In fact, the experiments performed in connection with blackbody radiation led to the discovery of light quanta and establishment of quantum mechanics. These accounts are not only of particular interest from a historical point of view but of great importance in understanding modern physics. To understand the propagation of electromagnetic waves in a dielectric medium is important from a basic aspect of electromagnetism. Moreover, it is deeply connected to optical applications including optical devices such as waveguides and lasers. In light of recent progress of the laser devices, we provide a tangible example of organic lasers as newly occurring laser devices.

The motion of particles as well as spatial and temporal change in, e.g., electromagnetic fields are very often described in terms of differential equations. We describe introductory methods of Green's functions in order to solve those differential equations.

Chapter 7

Maxwell's Equations



Maxwell's equations consist of four first-order partial differential equations. First we deal with basic properties of Maxwell's equations. Next we show how equations of electromagnetic wave motion are derived from Maxwell's equations along vector analysis. It is important to realize that the generation of the electromagnetic wave is a direct consequence of the interplay between the electric field and magnetic field that both change with time. We deal with behaviors of electromagnetic waves in dielectric media where no true charge exists. At a first glance, this restriction seems to narrow a range of application of principles of electromagnetism. In practice, however, such a situation is universalistic; topics cover a wide range of electromagnetic phenomena, e.g., light propagation in dielectrics including water, glass, polymers, etc. Polarized properties characterize the electromagnetic waves. These include linear, circular, and elliptic polarizations. The characteristics are important both from a fundamental aspect and from the point of view of optical applications.

7.1 Maxwell's Equations and Their Characteristics

In this chapter, we first represent Maxwell's equations as vector forms. The equations are represented as a differential form that is consistent with a viewpoint based on "action through medium." The equation of wave motion (or wave equation) is naturally derived from these equations.

Maxwell's equations of electromagnetism are expressed as follows:

$$\operatorname{div} \mathbf{D} = \rho, \quad (7.1)$$

$$\operatorname{div} \mathbf{B} = 0, \quad (7.2)$$

$$\text{rot } \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = \mathbf{0}, \quad (7.3)$$

$$\text{rot } \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t} = \mathbf{i}. \quad (7.4)$$

In (7.3), RHS denotes a zero vector. Let us take some time to get acquainted with physical quantities with their dimension as well as basic ideas and concepts along with definitions of electromagnetism.

The quantity $\mathbf{D}$ is called electric flux density [$\frac{\text{A}\cdot\text{s}}{\text{m}^2} = \frac{\text{C}}{\text{m}^2}$] (or electric displacement); ρ is electric charge density [$\frac{\text{C}}{\text{m}^3}$]. We describe vector quantities $\mathbf{V}$ as in (3.4):

$$\mathbf{V} = (e_1 \ e_2 \ e_3) \begin{pmatrix} V_x \\ V_y \\ V_z \end{pmatrix}. \quad (7.5)$$

The notation div denotes a differential operator such that

$$\text{div } \mathbf{V} \equiv \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}. \quad (7.6)$$

Thus, the div operator converts a vector to a scalar. The quantities $\mathbf{D}$ and the electric field $\mathbf{E}$ [V/m] are associated with the following expression:

$$\mathbf{D} = \varepsilon \mathbf{E}, \quad (7.7)$$

where ε [$\frac{\text{C}^2}{\text{Nm}^2}$] is called a dielectric constant (or permittivity) of the dielectric medium. The dimension can be understood from the following Coulomb's law that describes a force exerted between two charges:

$$F = \frac{1}{4\pi\varepsilon} \frac{QQ'}{r^2}, \quad (7.8)$$

where F is the force; Q and Q' are electric charges of the two charges; r is a distance between the two charges. Equation (7.1) represents Gauss' law of electrostatics.

The electric charge of 1 Coulomb (1 [C]) is defined as follows: Suppose that two point charges having the same electric charge are placed in vacuum 1 m apart. In that situation, if a force F between the two charges is

$$F = \tilde{c}^2 / 10^7 \text{ [N]},$$

where $\tilde{c}$ is a light velocity, then we define the electric charge which each point charge possesses as 1 [C]. Here, note that $\tilde{c}$ is a dimensionless number related to the light velocity in vacuum that is measured in a unit of [m/s]. Notice also that the light velocity c in vacuum is *defined* as

$$c = 299\,792\,458 \text{ [m/s]} \text{ (exact number).}$$

Thus, we have

$$\tilde{c} = 299\,792\,458 \text{ (dimensionless number).}$$

Vacuum is a kind of dielectric media. From (7.8), its dielectric constant ϵ_0 is defined by

$$\epsilon_0 = \frac{10^7}{4\pi\tilde{c}^2} \left[\frac{\text{C}^2}{\text{Nm}^2} \right] \approx 8.854 \times 10^{-12} \left[\frac{\text{C}^2}{\text{Nm}^2} \right]. \quad (7.9)$$

Meanwhile, 1 s (second) has been defined from a certain spectral line emitted from ^{133}Cs . Thus, 1 m (meter) is *defined* by

(distance along which light is propagated in vacuum during 1 s)/299 792 458.

The quantity $\mathbf{B}$ is called magnetic flux density [$\frac{\text{Vs}}{\text{m}^2} = \frac{\text{Wb}}{\text{m}^2}$]. Equation (7.2) represents Gauss's law of magnetostatics. In contrast to (7.1), RHS of (7.2) is zero. This corresponds to the fact that although a true charge exists, a true “magnetic charge” does not exist. (More precisely, such charge has not been detected so far.) The quantity $\mathbf{B}$ is connected with magnetic field $\mathbf{H}$ by the following relation:

$$\mathbf{B} = \mu\mathbf{H}, \quad (7.10)$$

where $\mu \left[\frac{\text{N}}{\text{A}^2} \right]$ is said to be permeability (or magnetic permeability). Thus, $\mathbf{H}$ has a dimension $\left[\frac{\text{A}}{\text{m}} \right]$. Permeability of vacuum μ_0 is defined by

$$\mu_0 = \frac{4\pi}{10^7} \left[\frac{\text{N}}{\text{A}^2} \right]. \quad (7.11)$$

Also we have

$$\mu_0\epsilon_0 = 1/c^2. \quad (7.12)$$

We will encounter this relation later again. Since the quantities c and ϵ_0 have a defined magnitude, so does μ_0 from (7.12).

We often make an issue of a relative magnitude of dielectric constant and magnetic permeability of a dielectric medium. That is, relative permittivity ϵ_r and relative permeability μ_r are defined as follows:

$$\epsilon_r \equiv \epsilon/\epsilon_0 \text{ and } \mu_r \equiv \mu/\mu_0. \quad (7.13)$$

Note that both ϵ_r and μ_r are dimensionless quantities. Those magnitudes are equal to 1 (in the case of vacuum) or larger than 1 (with any other dielectric media).

Equations (7.3) and (7.4) deal with the change in electric and magnetic fields with time. Of these, (7.3) represents Faraday's law of electromagnetic induction due to Michael Faraday (1831). He found that when a permanent magnet was thrust into or out of a closed circuit, the transient current flowed. Moreover, that experiment implied that even without the closed circuit, an electric field was generated around the space that changed the position relative to the permanent magnet. Equation (7.3) is easier to understand if it is rewritten as follows:

$$\text{rot } \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}. \quad (7.14)$$

That is, the electric field $\mathbf{E}$ is generated in such a way that the induced electric field (or induced current) tends to lessen the change in magnetic flux (or magnetic field) produced by the permanent magnet (Lenz's law). The minus sign in RHS indicates that effect.

The rot operator appearing in (7.3) and (7.4) is defined by

$$\begin{aligned} \text{rot } \mathbf{V} = \nabla \times \mathbf{V} &= \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix} \\ &= \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right) \mathbf{e}_1 + \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} \right) \mathbf{e}_2 + \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \right) \mathbf{e}_3. \end{aligned} \quad (7.15)$$

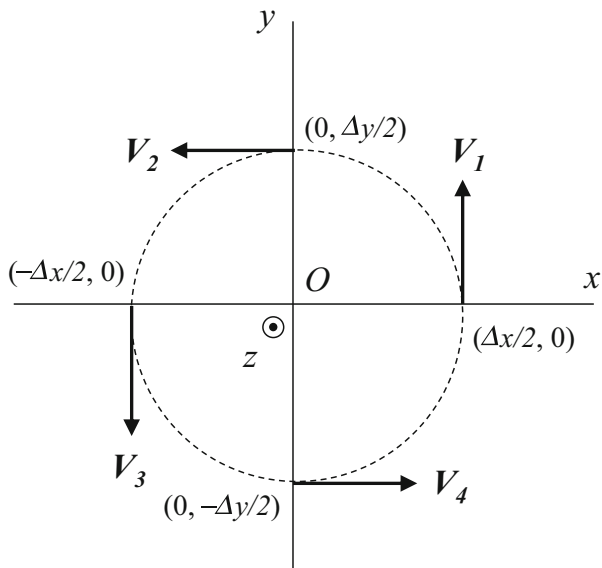
The operator ∇ has already appeared in (3.9). This operator transforms a vector to a vector. Let us think of the meaning of the rot operator. Suppose that there is a vector field that varies with time and spatial positions. Suppose also at some instant the spatial distribution of the field varies as shown in Fig. 7.1, where a spiral vector field $\mathbf{V}$ is present. For a z -component of $\text{rot } \mathbf{V}$ around the origin, we have

$$(\text{rot } \mathbf{V})_z = \frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} = \lim_{\Delta x \rightarrow 0, \Delta y \rightarrow 0} \left[\frac{(V_1)_y - (V_3)_y}{\Delta x} - \frac{(V_2)_x - (V_4)_x}{\Delta y} \right].$$

In the case of Fig. 7.1, $(V_1)_y - (V_3)_y > 0$ and $(V_2)_x - (V_4)_x < 0$ and, hence, we find that $\text{rot } \mathbf{V}$ has a positive z -component. If $V_z = 0$ and $\frac{\partial V_y}{\partial z} = \frac{\partial V_x}{\partial z} = 0$, we find from (7.15) that $\text{rot } \mathbf{V}$ possesses only the z -component. The equation $\frac{\partial V_y}{\partial z} = \frac{\partial V_x}{\partial z} = 0$ implies that the vector field $\mathbf{V}$ is uniform in the direction of the z -axis. Thus, under the above conditions the spiral vector field $\mathbf{V}$ is accompanied by the $\text{rot } \mathbf{V}$ vector field that is directed toward the upper side of the plane of paper (i.e., the positive direction of the z -axis).

Equation (7.4) can be rewritten as

Fig. 7.1 Schematic representation of a spiral vector field $\mathbf{V}$ that yields $\text{rot } \mathbf{V}$



$$\text{rot } \mathbf{H} = \mathbf{i} + \frac{\partial \mathbf{D}}{\partial t}. \quad (7.16)$$

Notice that $\frac{\partial \mathbf{D}}{\partial t}$ has the same dimension as $\left[\frac{\text{A}}{\text{m}^2}\right]$ and is called displacement current. Without this term, we have

$$\text{rot } \mathbf{H} = \mathbf{i}. \quad (7.17)$$

This relation is well known as Ampère's law or Ampère's circuital law (André-Marie Ampère: 1827), which determines a magnetic field yielded by a stationary current. Again with the aid of Fig. 7.1, (7.17) implies that the current given by $\mathbf{i}$ produces spiral magnetic field.

Now, let us think of a change in amount of charges with time in a part of three-dimensional closed space V surrounded by a closed surface S . It is given by

$$\frac{d}{dt} \int_V \rho dV = \int_V \frac{\partial \rho}{\partial t} dV = - \int_S \mathbf{i} \cdot \mathbf{n} dS = - \int_V \text{div } \mathbf{i} dV, \quad (7.18)$$

where $\mathbf{n}$ is an outward-directed normal unit vector; with the last equality we used Gauss's theorem. The Gauss's theorem is described by

$$\int_V \text{div } \mathbf{i} dV = \int_S \mathbf{i} \cdot \mathbf{n} dS.$$

Fig. 7.2 Intuitive diagram that explains the Gauss's theorem

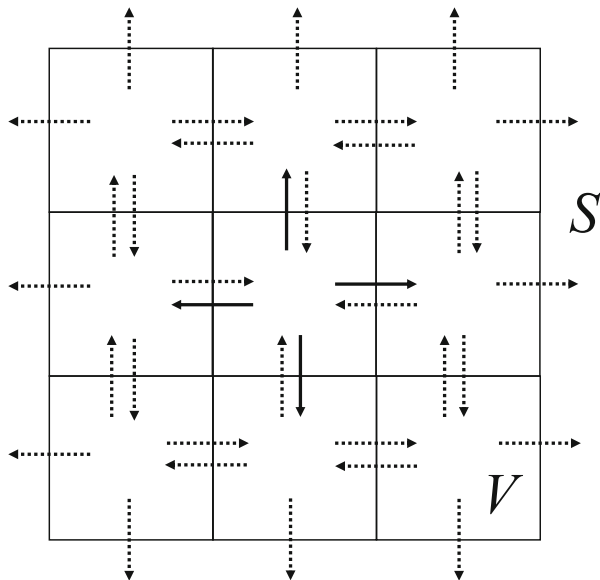


Figure 7.2 shows an intuitive diagram that explains the Gauss's theorem. The diagram shows a cross-section of the closed space V surrounded by a surface S . In this case, imagine a cube or a hexahedron as V . The periphery is the cross-section of the closed surface accordingly. Arrows in the diagram schematically represent $\text{div } \mathbf{i}$ on individual fragments; only those of the center infinitesimal fragment are shown with solid lines. The arrows of adjacent fragments cancel out each other and only the components on the periphery are non-vanishing. Thus, the volume integration of $\text{div } \mathbf{i}$ is converted to the surface integration of $\mathbf{i}$. Readers are referred to appropriate literature with the vector integration [1].

Consequently, from (7.18) we have

$$\int_V \left(\frac{\partial \rho}{\partial t} + \text{div } \mathbf{i} \right) dV = 0. \quad (7.19)$$

Since V is arbitrarily chosen, we get

$$\frac{\partial \rho}{\partial t} + \text{div } \mathbf{i} = 0. \quad (7.20)$$

The relation (7.20) is called a current continuity equation. This relation represents the law of conservation of charge.

Meanwhile, taking div of both sides of (7.17), we have

$$\operatorname{div} \operatorname{rot} \mathbf{H} = \operatorname{div} \mathbf{i}. \quad (7.21)$$

The LHS of (7.21) reads

$$\begin{aligned} \operatorname{div} \operatorname{rot} \mathbf{H} &= \frac{\partial}{\partial x} \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \\ &= \left(\frac{\partial^2}{\partial x \partial y} - \frac{\partial^2}{\partial y \partial x} \right) H_z + \left(\frac{\partial^2}{\partial y \partial z} - \frac{\partial^2}{\partial z \partial y} \right) H_x + \left(\frac{\partial^2}{\partial z \partial x} - \frac{\partial^2}{\partial x \partial z} \right) H_y \\ &= 0. \end{aligned} \quad (7.22)$$

With the last equality of (7.22), we used the fact that if, e.g., $\frac{\partial^2 H_z}{\partial x \partial y}$ and $\frac{\partial^2 H_z}{\partial y \partial x}$ are continuous and differentiable in a certain domain (x, y) , $\frac{\partial^2 H_z}{\partial x \partial y} = \frac{\partial^2 H_z}{\partial y \partial x}$. That is, we assume “ordinary” functions for H_z , H_x , and H_y . Thus from (7.21), we have

$$\operatorname{div} \mathbf{i} = 0. \quad (7.23)$$

From (7.20), we also have

$$\frac{\partial \rho(\mathbf{x}, t)}{\partial t} = 0, \quad (7.24)$$

where we explicitly show that ρ depends upon both $\mathbf{x}$ and t . Note that $\mathbf{x}$ is a position vector described as (3.5). Therefore, (7.24) shows that $\rho(\mathbf{x}, t)$ is temporally constant at a position $\mathbf{x}$, consistent with the stationary current.

Nevertheless, we encounter a problem when $\rho(\mathbf{x}, t)$ is temporally varying. In other words, (7.17) goes against the charge conservation law, when $\rho(\mathbf{x}, t)$ is temporally varying. It was James Clerk Maxwell (1861–1862) that solved the problem by introducing a concept of the displacement current. In fact, taking div of both sides of (7.16), we have

$$\operatorname{div} \operatorname{rot} \mathbf{H} = \operatorname{div} \mathbf{i} + \frac{\partial \operatorname{div} \mathbf{D}}{\partial t} = \operatorname{div} \mathbf{i} + \frac{\partial \rho}{\partial t} = 0, \quad (7.25)$$

where with the first equality we exchanged the order of differentiations with respect to t and $\mathbf{x}$; with the second equality we used (7.1). The last equality of (7.25) results from (7.20). In other words, in virtue of the term of $\frac{\partial \mathbf{D}}{\partial t}$, (7.4) is consistent with the charge conservation law. Thus, the set of Maxwell's Eqs. (7.1) to (7.4) supply us with well-established base in natural science up until the present.

Although the set of these equations describe spatial and temporal changes in electric and magnetic fields in vacuum and matter including metal, in Part II we confine ourselves to the changes in the electric and magnetic fields in a uniform dielectric medium.

7.2 Equation of Wave Motion

If we further confine ourselves to the case where neither electric charge nor electric current is present in a uniform dielectric medium, we can readily obtain equations of wave motion regarding the electric and magnetic fields. That is,

$$\operatorname{div} \mathbf{D} = 0, \quad (7.26)$$

$$\operatorname{div} \mathbf{B} = 0, \quad (7.27)$$

$$\operatorname{rot} \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = \mathbf{0}, \quad (7.28)$$

$$\operatorname{rot} \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t} = \mathbf{0}. \quad (7.29)$$

The relations (7.27) and (7.28) are identical to (7.2) and (7.3), respectively.

Let us start with a formula of vector analysis. First, we introduce a grad operator. We have

$$\operatorname{grad} f = \nabla f = \frac{\partial f}{\partial x} \mathbf{e}_1 + \frac{\partial f}{\partial y} \mathbf{e}_2 + \frac{\partial f}{\partial z} \mathbf{e}_3.$$

That is, the grad operator transforms a scalar to a vector. We have a following formula:

$$\operatorname{rot} \operatorname{rot} \mathbf{V} = \operatorname{grad} \operatorname{div} \mathbf{V} - \nabla^2 \mathbf{V}. \quad (7.30)$$

The operator ∇^2 has already appeared in (1.24). To show (7.30), we compare a x -component of both sides of (7.30). That is,

$$[\text{rot rot } \mathbf{V}]_x = \frac{\partial}{\partial y} \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \right) - \frac{\partial}{\partial z} \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} \right), \quad (7.31)$$

$$\begin{aligned} [\text{grad div } \mathbf{V} - \nabla^2 \mathbf{V}]_x &= \frac{\partial}{\partial x} \left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \right) - \frac{\partial^2 V_x}{\partial x^2} - \frac{\partial^2 V_x}{\partial y^2} - \frac{\partial^2 V_x}{\partial z^2} \\ &= \frac{\partial}{\partial x} \left(\frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \right) - \frac{\partial^2 V_x}{\partial y^2} - \frac{\partial^2 V_x}{\partial z^2}. \end{aligned} \quad (7.32)$$

Again assuming that $\frac{\partial^2 V_y}{\partial y \partial x} = \frac{\partial^2 V_y}{\partial x \partial y}$ and $\frac{\partial^2 V_z}{\partial z \partial x} = \frac{\partial^2 V_z}{\partial x \partial z}$, we have

$$[\text{rot rot } \mathbf{V}]_x = [\text{grad div } \mathbf{V} - \nabla^2 \mathbf{V}]_x. \quad (7.33)$$

Regarding y - and z -components, we have similar relations as well. Thus (7.30) holds.

Taking rot of both sides of (7.28), we have

$$\begin{aligned} \text{rot rot } \mathbf{E} + \text{rot } \frac{\partial \mathbf{B}}{\partial t} &= \text{grad div } \mathbf{E} - \nabla^2 \mathbf{E} + \frac{\partial \text{rot } \mathbf{B}}{\partial t} = -\nabla^2 \mathbf{E} + \mu \varepsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} \\ &= 0, \end{aligned} \quad (7.34)$$

where with the first equality we used (7.30) and for the second equality we used (7.7), (7.10), and (7.29). Thus we have

$$\nabla^2 \mathbf{E} = \mu \varepsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}. \quad (7.35)$$

Similarly, from (7.29) we get

$$\nabla^2 \mathbf{H} = \mu \varepsilon \frac{\partial^2 \mathbf{H}}{\partial t^2}. \quad (7.36)$$

Equations (7.35) and (7.36) are called the equations of wave motions for the electric and magnetic fields.

To consider implications of these equations, let us think of for simplicity a following equation in a one-dimensional space.

$$\frac{\partial^2 y(x, t)}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y(x, t)}{\partial t^2}, \quad (7.37)$$

where y is an arbitrary scalar function that depends on x and t ; v is a constant. Let $f(x, t)$ and $g(x, t)$ be arbitrarily chosen functions. Then, $f(x - vt)$ and $g(x + vt)$ are two solutions of (7.37). In fact, putting $X = x - vt$, we have

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial X} \frac{\partial X}{\partial x} = \frac{\partial f}{\partial X}, \quad \frac{\partial^2 f}{\partial x^2} = \left[\frac{\partial}{\partial X} \left(\frac{\partial f}{\partial X} \right) \right] \frac{\partial X}{\partial x} = \frac{\partial^2 f}{\partial X^2}, \quad (7.38)$$

$$\frac{\partial f}{\partial t} = \frac{\partial f}{\partial X} \frac{\partial X}{\partial t} = (-v) \frac{\partial f}{\partial X}, \quad \frac{\partial^2 f}{\partial t^2} = (-v) \left[\frac{\partial}{\partial X} \left(\frac{\partial f}{\partial X} \right) \right] \frac{\partial X}{\partial t} = (-v)^2 \frac{\partial^2 f}{\partial X^2}. \quad (7.39)$$

Deleting $\partial^2 f / \partial X^2$ from the second equation of (7.38) and the second equation of (7.39), we recover

$$\frac{\partial^2 f}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2}. \quad (7.40)$$

Similarly, we get

$$\frac{\partial^2 g}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 g}{\partial t^2}. \quad (7.41)$$

Therefore, as a general solution $y(x, t)$ we can take a superposition of $f(x, t)$ and $g(x, t)$. That is,

$$y(x, t) = f(x - vt) + g(x + vt). \quad (7.42)$$

The implication of (7.42) is as follows: (i) The function $f(x - vt)$ can be obtained by parallel translation of $f(x)$ by vt in a positive direction of x -axis. In other words, $f(x - vt)$ is obtained by translating $f(x)$ by v in a unit of time in a positive direction of x -axis, or the function represented by $f(x)$ is translated at a rate of v with its form unchanged in time. (ii) The function $g(x + vt)$, however, is translated at a rate of $-v$ with its form unchanged in time as well. (iii) Thus, $y(x, t)$ of (7.42) represents two “waves,” i.e., a forward wave and a backward wave. Propagation velocity of the two waves is $|v|$ accordingly. Usually we choose a positive number for v and v is called a phase velocity.

Comparing (7.35) and (7.36) with (7.41), we have

$$\mu\epsilon = 1/v^2. \quad (7.43)$$

In particular, in a vacuum we recover (7.12).

Notice that f and g can take *any* functional form and, hence, they are not necessarily a periodic wave. Yet, what we are mostly concerned with is a periodic wave such as sinusoidal waves. Thus, we arrive at a following functional form:

$$f(x - vt) = Ae^{i(x - vt)}, \quad (7.44)$$

where A is said to be an amplitude of the wave. The constant A usually takes a positive number, but it may take a complex number including a negative number. An exponent of (7.44) contains a number having a dimension [m]. To make it a dimensionless number, we multiply $(x - vt)$ by a wavenumber k that has been introduced in (1.2) and (1.3).

That is, we have

$$\tilde{f}[k(x - vt)] = Ae^{ik(x - vt)} = Ae^{i(kx - kv t)} = Ae^{i(kx - \omega t)}, \quad (7.45)$$

where $\tilde{f}$ shows the change in the functional from according to the variable transformation. In (7.45), we have

$$kv = k\lambda\nu = \left(\frac{2\pi}{\lambda}\right)\lambda\nu = 2\pi\nu = \omega, \quad (7.46)$$

where ν and ω are said to be frequency and angular frequency, respectively. For a three-dimensional wave f of a scalar function, we have a following form:

$$f = Ae^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} = A[\cos(\mathbf{k} \cdot \mathbf{x} - \omega t) + i \sin(\mathbf{k} \cdot \mathbf{x} - \omega t)], \quad (7.47)$$

where $\mathbf{k}$ is said to be a wavenumber vector. In (7.47),

$$\mathbf{k}^2 = k^2 = k_x^2 + k_y^2 + k_z^2. \quad (7.48)$$

Equation (7.47) is virtually identical to (1.25). When we deal with a problem of classical electromagnetism, we usually take a real part of the results after relevant calculations.

Suppose that (7.47) is a solution of a following wave equation:

$$\nabla^2 f = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2}. \quad (7.49)$$

Substituting LHS of (7.47) for f of (7.49), we have

$$A(-k^2)e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)} = A\frac{1}{v^2}(-\omega^2)e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}. \quad (7.50)$$

Comparing both sides of (7.50), we get

$$k^2v^2 = \omega^2 \quad \text{or} \quad kv = \omega. \quad (7.51)$$

Thus, we recover (7.46).

Here we introduce a unit vector $\mathbf{n}$ as in (1.3) whose direction parallels that of propagation of wave such that

$$\mathbf{k} = k\mathbf{n} = \frac{2\pi}{\lambda}\mathbf{n}, \mathbf{n} = (e_1 \ e_2 \ e_3) \begin{pmatrix} n_x \\ n_y \\ n_z \end{pmatrix}, \quad (7.52)$$

where n_x , n_y , and n_z define direction cosines. Then (7.47) can be rewritten as

$$f = Ae^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}, \quad (7.53)$$

where an exponent is called a phase. Suppose that the phase is fixed at zero. That is,

$$\mathbf{k}\cdot\mathbf{x} - \omega t = 0. \quad (7.54)$$

Since $\omega = kv$, we have

$$\mathbf{n} \cdot \mathbf{x} - vt = 0. \quad (7.55)$$

Equation (7.55) defines a plane in a three-dimensional space and is called a Hesse's normal form. Figure 7.3 schematically represents a plane wave of the field f . The field has the same phase on a plane P . A solid arrow $\mathbf{x}$ represents an arbitrary position vector on the plane and $\mathbf{n}$ is a unit vector perpendicular to the plane P (i.e., parallel to a normal of the plane P). The quantity vt defines a length of a perpendicular that connects the origin O and plane P (i.e., the length of the perpendicular from the origin and a foot of the perpendicular) at a given time t .

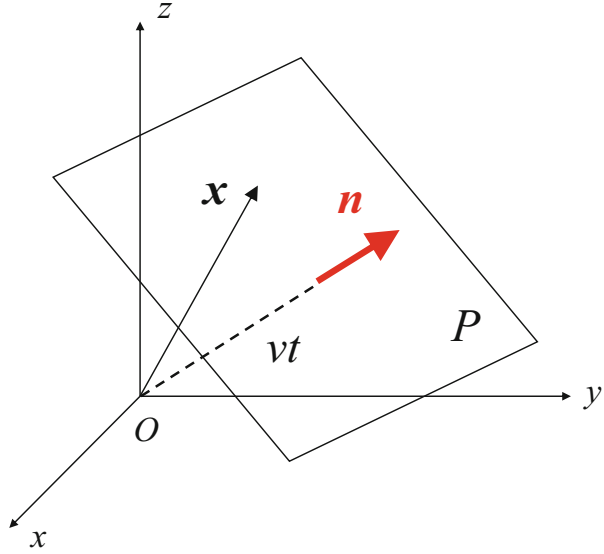
In other words, (7.54) determines a plane in such a way that the wave f has the same phase (zero) at a given time t at position vectors $\mathbf{x}$ on the plane determined by (7.54) or (7.55). That plane is moving in the direction of $\mathbf{n}$ at a phase velocity v . From this situation, a wave f described by (7.53) is called a plane wave.

A refractive index n of a dielectric media is an important index that characterizes its dielectric properties. It is defined as

$$n \equiv c/v = \sqrt{\mu\epsilon/\mu_0\epsilon_0} = \sqrt{\mu_r\epsilon_r}. \quad (7.56)$$

In a non-magnetic substance such as glass and polymer materials, we can assume that $\mu_r \approx 1$. Thus, we get an approximate expression as follows:

Fig. 7.3 Schematic representation of a plane wave. The field has the same phase on a plane P . A solid arrow $\mathbf{x}$ represents an arbitrary position vector on the plane and $\mathbf{n}$ is a unit vector perpendicular to the plane P (i.e., parallel to a normal of the plane P). The quantity v is a phase velocity of the plane wave



$$n \approx \sqrt{\epsilon_r}. \quad (7.57)$$

7.3 Polarized Characteristics of Electromagnetic Waves

As in (7.47), we assume a similar form for a solution of (7.35) such that

$$\mathbf{E} = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} = \mathbf{E}_0 e^{i(\mathbf{k} \mathbf{n} \cdot \mathbf{x} - \omega t)}, \quad (7.58)$$

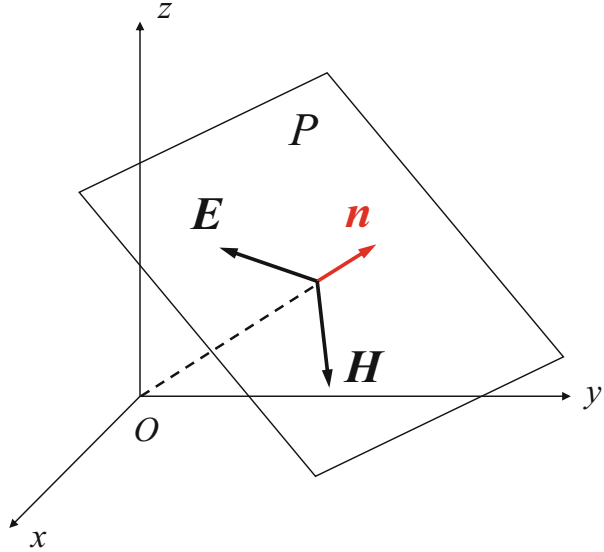
where with the second equality we used (7.52). Similarly, we have

$$\mathbf{H} = \mathbf{H}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} = \mathbf{H}_0 e^{i(\mathbf{k} \mathbf{n} \cdot \mathbf{x} - \omega t)}. \quad (7.59)$$

In (7.58) and (7.59), $\mathbf{E}_0$ and $\mathbf{H}_0$ are constant vectors and may take a complex magnitude. Substituting (7.58) for (7.28) and using (7.10) as well as (7.43) and (7.46), we have

$$i \mathbf{k} \mathbf{n} \times \mathbf{E}_0 e^{i(\mathbf{k} \mathbf{n} \cdot \mathbf{x} - \omega t)} = -(-i\omega) \mu \mathbf{H}_0 e^{i(\mathbf{k} \mathbf{n} \cdot \mathbf{x} - \omega t)} = i v k \mu \mathbf{H}_0 e^{i(\mathbf{k} \mathbf{n} \cdot \mathbf{x} - \omega t)}$$

Fig. 7.4 Mutual geometry of $\mathbf{E}$ and $\mathbf{H}$ for an electromagnetic plane wave in P . $\mathbf{E}$ and $\mathbf{H}$ have the same phase on P at an arbitrary given time. The unit vector $\mathbf{n}$ is perpendicular to P



$$= ik\sqrt{\mu/\epsilon}\mathbf{H}_0 e^{i(\mathbf{k}\cdot\mathbf{x} - \omega t)}.$$

Comparing coefficients of the exponential functions of the first and last sides, we get

$$\mathbf{H}_0 = \mathbf{n} \times \mathbf{E}_0 / \left(\sqrt{\mu/\epsilon} \right). \quad (7.60)$$

Similarly, substituting (7.59) for (7.29) and using (7.7), we get

$$\mathbf{E}_0 = \left(\sqrt{\mu/\epsilon} \right) \mathbf{H}_0 \times \mathbf{n}. \quad (7.61)$$

From (7.26) and (7.27), we have

$$\mathbf{n} \cdot \mathbf{E}_0 = \mathbf{n} \cdot \mathbf{H}_0 = 0. \quad (7.62)$$

This indicates that $\mathbf{E}$ and $\mathbf{H}$ are both perpendicular to $\mathbf{n}$, i.e., the propagation direction of the electromagnetic wave. Thus, the electromagnetic wave is characterized by a transverse wave. The fields $\mathbf{E}$ and $\mathbf{H}$ have the same phase on P at an arbitrary given time. Taking account of (7.60)–(7.62), $\mathbf{E}$, $\mathbf{H}$, and $\mathbf{n}$ are mutually perpendicular to one another. We depict a geometry of $\mathbf{E}$, $\mathbf{H}$, and $\mathbf{n}$ for the electromagnetic plane wave in Fig. 7.4, where a plane P is perpendicular to $\mathbf{n}$.

We find that (7.60) is not independent of (7.61). In fact, taking an outer product from the right with respect to both sides of (7.60), we have

$$\mathbf{H}_0 \times \mathbf{n} = \mathbf{n} \times \mathbf{E}_0 \times \mathbf{n} / \left(\sqrt{\mu/\epsilon} \right) = \mathbf{E}_0 / \left(\sqrt{\mu/\epsilon} \right),$$

where we used (7.61) with the second equality. Thus, we get

$$\mathbf{n} \times \mathbf{E}_0 \times \mathbf{n} = \mathbf{E}_0. \quad (7.63)$$

Meanwhile, vector analysis tells us that [1]

$$\mathbf{C} \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\mathbf{B} \cdot \mathbf{C}) - \mathbf{B}(\mathbf{C} \cdot \mathbf{A}).$$

In the above, putting $\mathbf{B} = \mathbf{C} = \mathbf{n}$, we have

$$\mathbf{n} \times (\mathbf{A} \times \mathbf{n}) = \mathbf{A}(\mathbf{n} \cdot \mathbf{n}) - \mathbf{n}(\mathbf{n} \cdot \mathbf{A}).$$

That is, we have

$$\mathbf{A} = \mathbf{n}(\mathbf{n} \cdot \mathbf{A}) + \mathbf{n} \times (\mathbf{A} \times \mathbf{n}).$$

This relation means that $\mathbf{A}$ can be decomposed into a component parallel to $\mathbf{n}$ and that perpendicular to $\mathbf{n}$. Equation (7.63) shows that $\mathbf{E}_0$ has no component parallel to $\mathbf{n}$. This is another confirmation that $\mathbf{E}$ is perpendicular to $\mathbf{n}$.

In (7.60) and (7.61), $\sqrt{\mu/\epsilon}$ has a dimension $[\Omega]$. Make sure that this can be confirmed by (7.9) and (7.11). Hence, $\sqrt{\mu/\epsilon}$ is said to be characteristic impedance [2]. We denote it by

$$Z \equiv \sqrt{\mu/\epsilon}. \quad (7.64)$$

Thus, we have

$$\mathbf{H}_0 = (\mathbf{n} \times \mathbf{E}_0) / Z.$$

In vacuum we have

$$Z_0 = \sqrt{\mu_0/\epsilon_0} \approx 376.7 \text{ } [\Omega].$$

For the electromagnetic wave to be the transverse wave means that neither $\mathbf{E}$ nor $\mathbf{H}$ has component along $\mathbf{n}$. Choosing the positive direction of the z -axis for $\mathbf{n}$ and ignoring components related to partial differentiation with respect to x and y (i.e., the component related to $\partial/\partial x$ and $\partial/\partial y$), we rewrite (7.28) and (7.29) for individual Cartesian coordinates as

$$\begin{aligned}
-\frac{\partial E_y}{\partial z} + \frac{\partial B_x}{\partial t} &= 0 \quad \text{or} \quad -\frac{\partial D_y}{\partial z} + \mu\epsilon \frac{\partial H_x}{\partial t} = 0, \\
\frac{\partial E_x}{\partial z} + \frac{\partial B_y}{\partial t} &= 0 \quad \text{or} \quad \frac{\partial D_x}{\partial z} + \mu\epsilon \frac{\partial H_y}{\partial t} = 0, \\
-\frac{\partial H_y}{\partial z} - \frac{\partial D_x}{\partial t} &= 0 \quad \text{or} \quad -\frac{\partial B_y}{\partial z} - \mu\epsilon \frac{\partial H_x}{\partial t} = 0, \\
\frac{\partial H_x}{\partial z} - \frac{\partial D_y}{\partial t} &= 0 \quad \text{or} \quad \frac{\partial B_x}{\partial z} - \mu\epsilon \frac{\partial H_y}{\partial t} = 0.
\end{aligned} \tag{7.65}$$

We differentiate the first equation of (7.65) with respect to z to get

$$-\frac{\partial^2 E_y}{\partial z^2} + \frac{\partial^2 B_x}{\partial z \partial t} = 0.$$

Also differentiating the fourth equation of (7.65) with respect to t and multiplying both sides by $-\mu$, we have

$$-\mu \frac{\partial^2 H_x}{\partial t \partial z} + \mu \frac{\partial^2 D_y}{\partial t^2} = 0.$$

Summing both sides of the above equations and arranging terms, we get

$$\frac{\partial^2 E_y}{\partial z^2} = \mu\epsilon \frac{\partial^2 E_y}{\partial t^2}.$$

In a similar manner, we have

$$\frac{\partial^2 E_x}{\partial z^2} = \mu\epsilon \frac{\partial^2 E_x}{\partial t^2}.$$

Similarly, for the magnetic field, we also get

$$\frac{\partial^2 H_x}{\partial z^2} = \mu\epsilon \frac{\partial^2 H_x}{\partial t^2} \quad \text{and} \quad \frac{\partial^2 H_y}{\partial z^2} = \mu\epsilon \frac{\partial^2 H_y}{\partial t^2}.$$

From the above relations, we have two plane electromagnetic waves polarized either the x -axis or y -axis.

What is implied in the above description is that as solutions of (7.35) and (7.36) we have a plane wave characterized by a specific direction defined by $\mathbf{E}_0$ and $\mathbf{H}_0$. This implies that if we observe the electromagnetic wave at a fixed point, both $\mathbf{E}$ and $\mathbf{H}$ oscillate along the mutually perpendicular directions $\mathbf{E}_0$ and $\mathbf{H}_0$. Hence, we say that the “electric wave” is polarized in the direction $\mathbf{E}_0$ and that the “magnetic wave” is polarized in the direction $\mathbf{H}_0$.

To characterize the polarization of the electromagnetic wave, we introduce following unit polarization vectors $\mathbf{e}_e$ and $\mathbf{e}_m$ (with indices e and m related to the electric and magnetic field, respectively) [3]:

$$\mathbf{e}_e = \mathbf{E}_0/E_0 \quad \text{and} \quad \mathbf{e}_m = \mathbf{H}_0/H_0; H_0 = E_0/Z, \quad (7.66)$$

where E_0 and H_0 are said to be amplitude and may be again complex. We have

$$\mathbf{e}_e \times \mathbf{e}_m = \mathbf{n}. \quad (7.67)$$

We call $\mathbf{e}_e$ and $\mathbf{e}_m$ a unit polarization vector of the electric field and magnetic field, respectively. As noted above, $\mathbf{e}_e$, $\mathbf{e}_m$, and $\mathbf{n}$ constitute a right-handed system in this order and are mutually perpendicular to one another.

The phase of $\mathbf{E}$ in the plane wave (7.58) and that of $\mathbf{H}$ in (7.59) are individually the same on all the points of P . From the wave equations of (7.35) and (7.36), however, it is unclear whether $\mathbf{E}$ and $\mathbf{H}$ have the same phase. Suppose that $\mathbf{E}$ and $\mathbf{H}$ would have a different phase such that

$$\mathbf{E} = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$$

and

$$\mathbf{H} = \mathbf{H}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t + \delta)} = \mathbf{H}_0 e^{i\delta} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} = \tilde{\mathbf{H}}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)},$$

where $\tilde{\mathbf{H}}_0 (= \mathbf{H}_0 e^{i\delta})$ is complex and a phase factor $e^{i\delta}$ in the exponent is unknown.

This factor, however, can be set at zero. To show this, let us make qualitative discussion using Fig. 7.5. Figure 7.5 shows the electric field of the plane wave at some instant as a function of phase Φ . Suppose that Φ is taken in the direction of $\mathbf{n}$ in Fig. 7.4. Then, from (7.53) we have

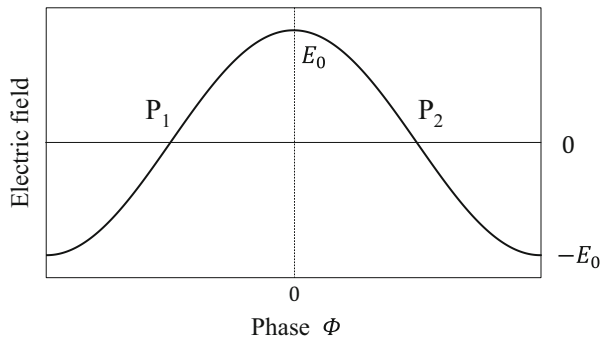
$$\Phi = \mathbf{k} \cdot \mathbf{x} - \omega t = \mathbf{k} \cdot \rho \mathbf{n} - \omega t = k\rho - \omega t,$$

where ρ is distance from the origin. Also we have

$$\mathbf{E} = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} = E_0 \mathbf{e}_e e^{i(k\rho - \omega t)} \quad (E_0 > 0).$$

Suppose furthermore that the phase is measured at $t = 0$ and that the electric field is measured along the $\mathbf{e}_e$ direction. Then, we have

Fig. 7.5 Electric field of a plane wave at some instant as a function of phase. The sinusoidal curve representing the electric field is shifted with time from left to right with its form unchanged



$$\Phi = k\rho \quad \text{and} \quad E = |\mathbf{E}| = E_0 e^{ik\rho}. \quad (7.68)$$

In Fig. 7.5, a real part of E is shown with $E = E_0$ at $\rho = 0$. The sinusoidal curve representing the electric field in Fig. 7.5 is shifted with time from left to right with its form unchanged.

In this situation, the electric field is to be strengthened with a negative value at a phase P_1 with time according to (7.68), whereas at another phase P_2 the electric field is to be strengthened with a positive value with time. As a result, in a region tucked between P_1 and P_2 a spiral magnetic field is generated toward the $\mathbf{e}_m$ direction, i.e., upper side of a plane of paper. The magnitude of the magnetic field is expected to be maximized at a center point of P_1P_2 (i.e., the origin of the coordinate system) where the electric field is maximized as well. Thus, we conclude that $\mathbf{E}$ and $\mathbf{H}$ have the same phase. It is important to realize that the generation of the electromagnetic wave is a direct consequence of the interplay between the electric field and magnetic field that both change with time. It is truly based upon the nature of the electric and magnetic fields that are clearly represented in Maxwell's equations.

Next, we consider the situation where *two* electromagnetic waves are propagated in the same direction but with a different phase. Notice again that we are considering the electromagnetic wave that is propagated in a uniform and infinite dielectric media without BCs.

7.4 Superposition of Two Electromagnetic Waves

Let $\mathbf{E}_1$ and $\mathbf{E}_2$ be two electric waves described such that

$$\mathbf{E}_1 = E_1 \mathbf{e}_1 e^{i(kz - \omega t)} \quad \text{and} \quad \mathbf{E}_2 = E_2 \mathbf{e}_2 e^{i(kz - \omega t + \delta)},$$

where $E_1 (>0)$ and $E_2 (>0)$ are amplitudes and $\mathbf{e}_1$ and $\mathbf{e}_2$ represent unit polarization vectors in the direction of positive x -axis and y -axis; we assume that two waves are

being propagated in the direction of the positive z -axis; δ is a phase difference. The total electric field $\mathbf{E}$ is described as the superposition of $\mathbf{E}_1$ and $\mathbf{E}_2$ such that

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 = E_1 \mathbf{e}_1 e^{i(kz - \omega t)} + E_2 \mathbf{e}_2 e^{i(kz - \omega t + \delta)}. \quad (7.69)$$

Note that we usually discuss the polarization characteristics of electromagnetic wave only by considering electric waves. We emphasize that an electric wave and concomitant magnetic wave share the same phase in a uniform and infinite dielectric media. A reason why the electric wave represents an electromagnetic wave is partly because optical application is mostly made in a non-magnetic substance such as glass, water, plastics, and most of semiconductors.

Let us view temporal change of $\mathbf{E}$ at a fixed point $\mathbf{x} = \mathbf{0}$; $x = y = z = 0$. Then, taking a real part of (7.69), x - and y -components of $\mathbf{E}$; i.e., E_x and E_y are expressed as

$$E_x = E_1 \cos(-\omega t) \text{ and } E_y = E_2 \cos(-\omega t + \delta). \quad (7.70)$$

First, let us briefly think of the case where $\delta = 0$. Eliminating t , we have

$$E_y = \frac{E_2}{E_1} E_x. \quad (7.71)$$

This is an equation of a straight line. The resulting electric field $\mathbf{E}$ is called a linearly polarized light accordingly. That is, when we are observing the electric field of the relevant light at the origin, the field is oscillating along the straight line described by (7.71) with the origin centrally located of the oscillating field. If $\delta = \pi$, we have

$$E_y = -\frac{E_2}{E_1} E_x.$$

This gives a straight line as well. Therefore, if we wish to seek the relationship between E_x and E_y , it suffices to examine it as a function of δ in a region of $-\frac{\pi}{2} \leq \delta \leq \frac{\pi}{2}$.

(i) Case I: $E_1 \neq E_2$.

Let us consider the case where $\delta \neq 0$ in (7.70). Rewriting the second equation of (7.70) and inserting the first equation into it so that we can eliminate t , we have

$$E_y = E_2 (\cos \omega t \cos \delta + \sin \omega t \sin \delta) = E_2 \left(\cos \delta \frac{E_x}{E_1} \pm \sin \delta \sqrt{1 - \frac{E_x^2}{E_1^2}} \right).$$

Rearranging terms of the above equation, we have

$$\frac{E_y}{E_2} - (\cos \delta) \frac{E_x}{E_1} = \pm (\sin \delta) \sqrt{1 - \frac{E_x^2}{E_1^2}}. \quad (7.72)$$

Squaring both sides of (7.72) and arranging the equation, we get

$$\frac{E_x^2}{E_1^2 \sin^2 \delta} - \frac{2(\cos \delta) E_x E_y}{E_1 E_2 \sin^2 \delta} + \frac{E_y^2}{E_2^2 \sin^2 \delta} = 1. \quad (7.73)$$

Using a matrix form, we have

$$\begin{pmatrix} E_x & E_y \end{pmatrix} \begin{pmatrix} \frac{1}{E_1^2 \sin^2 \delta} & -\frac{\cos \delta}{E_1 E_2 \sin^2 \delta} \\ -\frac{\cos \delta}{E_1 E_2 \sin^2 \delta} & \frac{1}{E_2^2 \sin^2 \delta} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix} = 1. \quad (7.74)$$

Note that the above matrix is real symmetric. In that case, to examine properties of the matrix we calculate its determinant along with principal minors. The principal minor means a minor with respect to a diagonal element. In this case, two principal minors are $\frac{1}{E_1^2 \sin^2 \delta}$ and $\frac{1}{E_2^2 \sin^2 \delta}$. Also we have

$$\begin{vmatrix} \frac{1}{E_1^2 \sin^2 \delta} & -\frac{\cos \delta}{E_1 E_2 \sin^2 \delta} \\ -\frac{\cos \delta}{E_1 E_2 \sin^2 \delta} & \frac{1}{E_2^2 \sin^2 \delta} \end{vmatrix} = \frac{1}{E_1^2 E_2^2 \sin^2 \delta}. \quad (7.75)$$

Evidently, two principal minors as well as a determinant are all positive ($\delta \neq 0$). In this case, the (2, 2) matrix of (7.74) is said to be *positive definite*. The related discussion will be given in Part III. The positive definiteness means that in a quadratic form described by (7.74), LHS takes a positive value for *any* real number E_x and E_y except a unique case where $E_x = E_y = 0$, which renders LHS zero. The positive definiteness of a matrix ensures the existence of positive eigenvalues with the said matrix.

Let us consider a real symmetric (2, 2) matrix that has positive principal minors and a positive determinant in a general case. Let such a matrix M be

$$M = \begin{pmatrix} a & c \\ c & b \end{pmatrix},$$

where $a, b > 0$ and $\det M > 0$; i.e., $ab - c^2 > 0$. Let a corresponding quadratic form be Q . Then, we have

$$Q = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} a & c \\ c & b \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = ax^2 + 2cyx + by^2 = a \left[\left(x + \frac{cy}{a} \right)^2 - \frac{c^2 y^2}{a^2} + \frac{aby^2}{a^2} \right]$$

$$= a \left[\left(x + \frac{cy}{a} \right)^2 + \frac{y^2}{a^2} (ab - c^2) \right].$$

Thus, $Q \geq 0$ for any real numbers x and y . We seek a condition under which $Q = 0$. We readily find that with M that has the above properties, only $x = y = 0$ makes $Q = 0$. Thus, M is positive definite. We will deal with this issue from a more general standpoint in Part III.

In general, it is pretty complicated to seek eigenvalues and corresponding eigenvectors in the above case. Yet, we can extract important information from (7.74). The eigenvalues λ of the matrix that appeared in (7.74) are estimated as follows:

$$\lambda = \frac{E_1^2 + E_2^2 \pm \sqrt{(E_1^2 + E_2^2)^2 - 4E_1^2 E_2^2 \sin^2 \delta}}{2E_1^2 E_2^2 \sin^2 \delta}. \quad (7.76)$$

Notice that λ in (7.76) represents two different positive eigenvalues. It is because an inside of the square root is rewritten by

$$(E_1^2 - E_2^2)^2 + 4E_1^2 E_2^2 \cos^2 \delta > 0 \quad (\delta \neq \pm \pi/2).$$

Also we have

$$E_1^2 + E_2^2 > \sqrt{(E_1^2 + E_2^2)^2 - 4E_1^2 E_2^2 \sin^2 \delta}.$$

These clearly show that the quadratic form of (7.74) gives an ellipse (i.e., elliptically polarized light). Because of the presence of the second term of LHS of (7.73), both the major and minor axes of the ellipse are tilted and diverted from the x - and y -axes.

Let us inspect the ellipse described by (7.74). Inserting $E_x = E_1$ obtained at $t = 0$ in (7.70) into (7.73) and solving a quadratic equation with respect to E_y , we get E_y as a double root such that

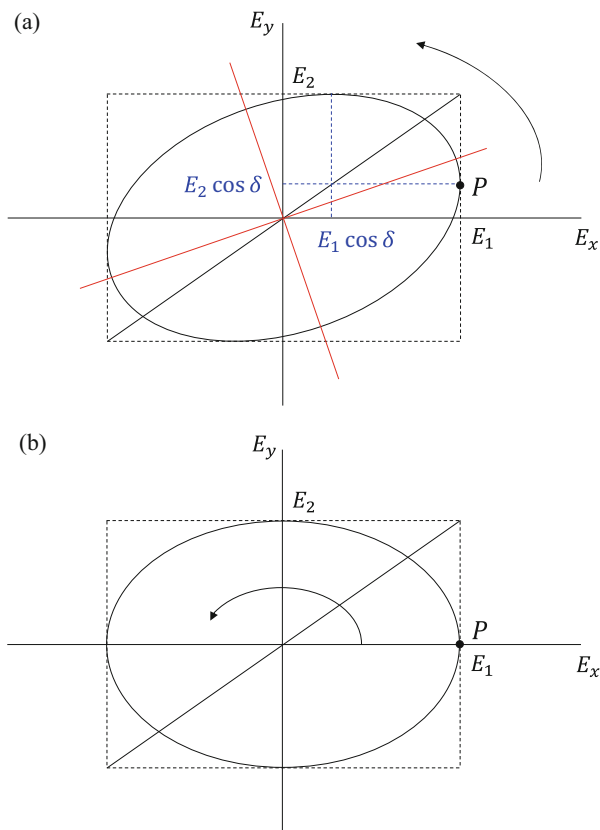
$$E_y = E_2 \cos \delta.$$

Similarly putting $E_y = E_2$ in (7.73), we have

$$E_x = E_1 \cos \delta.$$

These results show that an ellipse described by (7.73) or (7.74) is internally tangent to a rectangle as depicted in Fig. 7.6a. Equation (7.69) shows that the electromagnetic wave is propagated toward the positive direction of the z -axis. Therefore, in Fig. 7.6a we are peeking into the oncoming wave from the bottom of a plane of paper at a certain position of $z = \text{constant}$. We set the constant = 0. Then,

Fig. 7.6 Trace of an electric field of an elliptically polarized light. (a) The trace is internally tangent to a rectangle of $2E_1 \times 2E_2$. In the case of $\delta > 0$, starting from P at $t = 0$, the coordinate point representing the electric field traces the ellipse counter-clockwise with time. (b) The trace of an elliptically polarized light for $\delta = \pi/2$



we find that at $t = 0$ the electric field is represented by the point P ($E_x = E_1$, $E_y = E_2 \cos \delta$); see Fig. 7.6a. From (7.70), if $\delta > 0$, P traces the ellipse counter-clockwise. It reaches a maximum point of $E_y = E_2$ at $t = \delta/2\omega$. Since the trace of electric field forms an ellipse as in Fig. 7.6, the associated light is said to be an elliptically polarized light. If $\delta < 0$ in (7.70), however, P traces the ellipse clockwise.

In a special case of $\delta = \pi/2$, the second term of (7.73) vanishes and we have a simple form described as

$$\frac{E_x^2}{E_1^2} + \frac{E_y^2}{E_2^2} = 1. \quad (7.77)$$

Thus, the principal axes of the ellipse coincide with the x - and y -axes. On the basis of (7.70), we see from Fig. 7.6b that starting from P at $t = 0$, again the coordinate point representing the electric field traces the ellipse counter-clockwise with time; see the curved arrow of Fig. 7.6b. If $\delta < 0$, the coordinate point traces the ellipse clockwise with time.

(ii) Case II: $E_1 = E_2$.

Now, let us consider a simple but important case. When $E_1 = E_2$, (7.73) is simplified to be

$$E_x^2 - 2 \cos \delta E_x E_y + E_y^2 = E_1^2 \sin^2 \delta. \quad (7.78)$$

Using a matrix form, we have

$$\begin{pmatrix} E_x & E_y \end{pmatrix} \begin{pmatrix} 1 & -\cos \delta \\ -\cos \delta & 1 \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix} = E_1^2 \sin^2 \delta. \quad (7.79)$$

We obtain eigenvalues λ of the matrix of (7.79) such that

$$\lambda = 1 \pm |\cos \delta|. \quad (7.80)$$

Setting $-\frac{\pi}{2} \leq \delta \leq \frac{\pi}{2}$, we have

$$\lambda = 1 \pm \cos \delta. \quad (7.81)$$

The corresponding normalized eigenvectors v_1 and v_2 (as a column vector) are

$$v_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} \quad \text{and} \quad v_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}. \quad (7.82)$$

Thus, we have a diagonalizing unitary matrix P such that

$$P = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}. \quad (7.83)$$

Defining the above matrix appearing in (7.79) as A such that

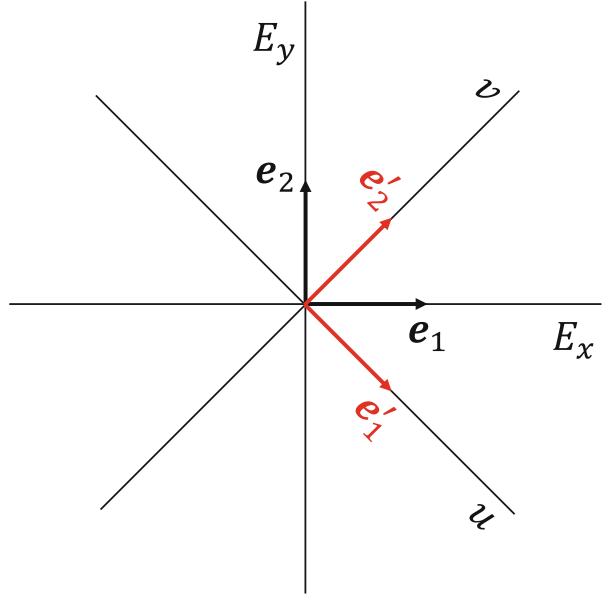
$$A = \begin{pmatrix} 1 & -\cos \delta \\ -\cos \delta & 1 \end{pmatrix}, \quad (7.84)$$

we obtain

$$P^{-1}AP = \begin{pmatrix} 1+\cos \delta & 0 \\ 0 & 1-\cos \delta \end{pmatrix}. \quad (7.85)$$

Notice that eigenvalues $(1 + \cos \delta)$ and $(1 - \cos \delta)$ are both positive as expected. Rewriting (7.79), we have

Fig. 7.7 Relationship between the basis vectors $(\mathbf{e}_1 \mathbf{e}_2)$ and $(\mathbf{e}'_1 \mathbf{e}'_2)$ in the case of $E_1 = E_2$; see text



$$\begin{aligned}
 & \begin{pmatrix} E_x & E_y \end{pmatrix} P P^{-1} \begin{pmatrix} 1 & -\cos \delta \\ -\cos \delta & 1 \end{pmatrix} P P^{-1} \begin{pmatrix} E_x \\ E_y \end{pmatrix} \\
 &= \begin{pmatrix} E_x & E_y \end{pmatrix} P \begin{pmatrix} 1+\cos \delta & 0 \\ 0 & 1-\cos \delta \end{pmatrix} P^{-1} \begin{pmatrix} E_x \\ E_y \end{pmatrix} = E_1^2 \sin^2 \delta. \quad (7.86)
 \end{aligned}$$

Here, let us define new coordinates such that

$$\begin{pmatrix} u \\ v \end{pmatrix} \equiv P^{-1} \begin{pmatrix} E_x \\ E_y \end{pmatrix}. \quad (7.87)$$

This coordinate transformation corresponds to the transformation of basis vectors $(\mathbf{e}_1 \mathbf{e}_2)$ such that

$$(\mathbf{e}_1 \mathbf{e}_2) \begin{pmatrix} E_x \\ E_y \end{pmatrix} = (\mathbf{e}_1 \mathbf{e}_2) P P^{-1} \begin{pmatrix} E_x \\ E_y \end{pmatrix} = (\mathbf{e}'_1 \mathbf{e}'_2) \begin{pmatrix} u \\ v \end{pmatrix}, \quad (7.88)$$

where new basis vectors $(\mathbf{e}'_1 \mathbf{e}'_2)$ are given by

$$(\mathbf{e}'_1 \mathbf{e}'_2) = (\mathbf{e}_1 \mathbf{e}_2) P = \left(\frac{1}{\sqrt{2}} \mathbf{e}_1 - \frac{1}{\sqrt{2}} \mathbf{e}_2 \quad \frac{1}{\sqrt{2}} \mathbf{e}_1 + \frac{1}{\sqrt{2}} \mathbf{e}_2 \right). \quad (7.89)$$

The coordinate system is depicted in Fig. 7.7 along with the basis vectors. The relevant discussion will again appear in Part. III.

Substituting (7.87) for (7.86) and rearranging terms, we get

$$\frac{u^2}{E_1^2(1 - \cos \delta)} + \frac{v^2}{E_1^2(1 + \cos \delta)} = 1. \quad (7.90)$$

Equation (7.90) indicates that a major axis and minor axis are $E_1\sqrt{1 + \cos \delta}$ and $E_1\sqrt{1 - \cos \delta}$, respectively. When $\delta = \pm \pi/2$, (7.90) becomes

$$\frac{u^2}{E_1^2} + \frac{v^2}{E_1^2} = 1. \quad (7.91)$$

This represents a circle. For this reason, the wave described by (7.91) is called a circularly polarized light. In (7.90) where $\delta \neq \pm \pi/2$, the wave is said to be an elliptically polarized light. Thus, we have linearly, elliptically, and circularly polarized lights depending on a magnitude of δ .

Let us closely examine characteristics of the elliptically and circularly polarized lights in the case of $E_1 = E_2$. When $t = 0$, from (7.70) we have

$$E_x = E_1 \quad \text{and} \quad E_y = E_1 \cos \delta. \quad (7.92)$$

This coordinate point corresponds to A_1 whose E_x coordinate is E_1 (see Fig. 7.8a). In the case of $\Delta t = \delta/2\omega$, $E_x = E_y = E_1 \cos(\pm\delta/2)$. This point corresponds to A_2 in Fig. 7.8a. We have

$$\sqrt{E_x^2 + E_y^2} = \sqrt{2}E_1 \cos(\delta/2) = E_1\sqrt{1 + \cos \delta}. \quad (7.93)$$

This is equal to the major axis as anticipated. With $t = \Delta t$,

$$E_x = E_1 \cos(-\omega\Delta t) \quad \text{and} \quad E_y = E_1 \cos(-\omega\Delta t + \delta). \quad (7.94)$$

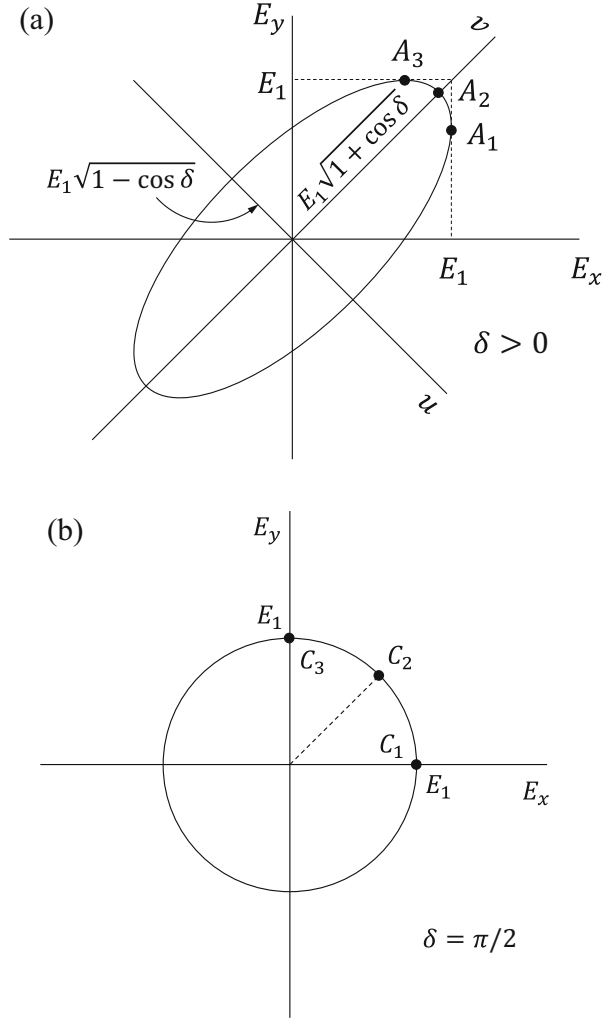
Notice that E_y takes a maximum E_1 when $\Delta t = \delta/\omega$. Consequently, if δ takes a positive value, E_y takes a maximum E_1 for a positive Δt , as is similarly the case with Fig. 7.6a. At that time, $E_x = E_1 \cos(-\delta) < E_1$. This point corresponds to A_3 in Fig. 7.8a. As a result, the electric field traces the ellipse counter-clockwise with time, as in the case of Fig. 7.6. If δ takes a negative value, however, the field traces the ellipse clockwise.

If $\delta = \pm \pi/2$, in (7.94) we have

$$E_x = E_1 \cos(-\omega t) \quad \text{and} \quad E_y = E_1 \cos\left(-\omega t \pm \frac{\pi}{2}\right). \quad (7.95)$$

We examine the case of $\delta = \pi/2$ first. In this case, when $t = 0$, $E_x = E_1$ and $E_y = 0$ (Point C_1 in Fig. 7.8b). If $\omega t = \pi/4$, $E_x = E_y = 1/\sqrt{2}$ (Point C_2 in Fig. 7.8b). In turn, if $\omega t = \pi/2$, $E_x = 0$ and $E_y = E_1$ (Point C_3). Again the electric field traces the circle

Fig. 7.8 Polarized feature of light in the case of $E_1 = E_2$. (a) If $\delta > 0$, the electric field traces an ellipse from A_1 via A_2 to A_3 (see text). (b) If $\delta = \pi/2$, the electric field traces a circle from C_1 via C_2 to C_3 (left-circularly polarized light)



counter-clockwise. In this situation, we see the light from above the z -axis. In other words, we are viewing the light against the direction of its propagation. The wave is said to be *left-circularly polarized* and have *positive* helicity. In contrast, when $\delta = -\pi/2$, starting from Point C_1 the electric field traces the circle clockwise. That light is said to be *right-circularly polarized* and have *negative* helicity.

With the left-circularly polarized light, (7.69) can be rewritten as

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 = E_1(\mathbf{e}_1 + i\mathbf{e}_2)e^{i(kz - \omega t)}. \quad (7.96)$$

Therefore, a complex vector $(\mathbf{e}_1 + i\mathbf{e}_2)$ characterizes the left-circular polarization. On the other hand, $(\mathbf{e}_1 - i\mathbf{e}_2)$ characterizes the right-circular polarization. To

normalize them, it is convenient to use the following vectors as in the case of Sect. 4.3 [3].

$$\mathbf{e}_+ \equiv \frac{1}{\sqrt{2}}(\mathbf{e}_1 + i\mathbf{e}_2) \quad \text{and} \quad \mathbf{e}_- \equiv \frac{1}{\sqrt{2}}(\mathbf{e}_1 - i\mathbf{e}_2). \quad (4.45)$$

In the case of $\delta = 0$, we have a linearly polarized light. For this, the points A_1 , A_2 , and A_3 coalesce to be a point on a straight line of $E_y = E_x$.

References

1. Arfken GB, Weber HJ, Harris FE (2013) Mathematical methods for physicists, 7th edn. Academic Press, Waltham
2. Pain HJ (2005) The physics of vibrations and waves, 6th edn. Wiley, Chichester
3. Jackson JD (1999) Classical electrodynamics, 3rd edn. Wiley, New York

Chapter 8

Reflection and Transmission of Electromagnetic Waves in Dielectric Media



In Chap. 7, we considered the propagation of electromagnetic waves in an infinite uniform dielectric medium. In this chapter, we think of a situation where two (or more) dielectrics are in contact with each other at a plane interface. When two dielectric media adjoin each other with an interface, propagating electromagnetic waves are partly reflected by the interface and partly transmitted beyond the interface. We deal with these phenomena in terms of characteristic impedance of the dielectric media. In the case of an oblique incidence of a wave, we categorize it into a transverse electric (TE) wave and transverse magnetic (TM) wave. If a thin plate of a dielectric is sandwiched by a couple of metal sheets, the electromagnetic wave is confined within the dielectric. In this case, the propagating mode of the wave differs from that of a wave propagating in a free space (i.e., a space filled by a three-dimensionally infinite dielectric medium). If a thin plate of a dielectric having a large refractive index is sandwiched by a couple of dielectrics with a smaller refractive index, the electromagnetic wave is also confined within the dielectric with a larger index. In this case, we have to take account of the total reflection that causes a phase change upon reflection. We deal with such specific modes of the electromagnetic wave propagation. These phenomena are treated both from a basic aspect and from a point of view of device application. The relevant devices are called waveguides in optics.

8.1 Electromagnetic Fields at an Interface

We start with examining a condition of an electromagnetic field at the plane interface. Suppose that two semi-infinite dielectric media D_1 and D_2 are in contact with each other at a plane interface. Let us take a small rectangle S that strides the interface (see Fig. 8.1). Taking a surface integral of both sides of (7.28) over the strip, we have

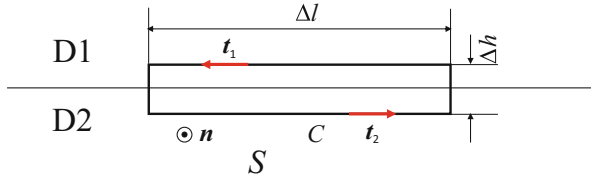
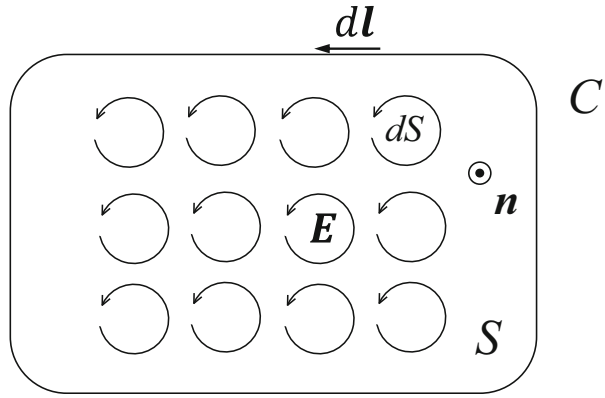


Fig. 8.1 A small rectangle S that strides an interface formed by two semi-infinite dielectric media of D1 and D2. Let a curve C be a closed loop surrounding the rectangle S . A unit vector $\mathbf{n}$ is directed to a normal of S . Unit vectors $\mathbf{t}_1$ and $\mathbf{t}_2$ are directed to a tangential line of the interface plane

Fig. 8.2 Diagram that intuitively explains Stokes' theorem. In the diagram a surface S is encircled by a closed curve C . An infinitesimal portion of C is denoted by $d\mathbf{l}$. The surface S is pertinent to the surface integration. Spiral vector field $\mathbf{E}$ is present on and near S



$$\int_S \text{rot } \mathbf{E} \cdot \mathbf{n} dS + \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot \mathbf{n} dS = 0, \quad (8.1)$$

where $\mathbf{n}$ is a unit vector directed to a normal of S as shown. Applying Stokes' theorem to the first term of (8.1), we get

$$\oint_C \mathbf{E} \cdot d\mathbf{l} + \frac{\partial \mathbf{B}}{\partial t} \cdot \mathbf{n} \Delta l \Delta h = 0. \quad (8.2)$$

With the line integral of the first term, C is a closed loop surrounding the rectangle S and $d\mathbf{l} = \mathbf{t} dl$, where $\mathbf{t}$ is a unit vector directed toward the tangential direction of C (see $\mathbf{t}_1$ and $\mathbf{t}_2$ in Fig. 8.1). The line integration is performed such that C is followed counterclockwise in the direction of $\mathbf{t}$.

Figure 8.2 gives an intuitive diagram that explains Stokes' theorem. The diagram shows an overview of a surface S encircled by a closed curve C . Suppose that we have a spiral vector field $\mathbf{E}$ represented by arrowed circles as shown. In that case, $\text{rot } \mathbf{E}$ is directed toward the upper side of the plane of paper in the individual fragments. A summation of $\text{rot } \mathbf{E} \cdot \mathbf{n} dS$ forms a surface integral covering S . Meanwhile, the arrows of adjacent fragments cancel out each other and only the components on the periphery (i.e., the curve C) are non-vanishing (see Fig. 8.2).

Thus, the surface integral of $\text{rot } \mathbf{E}$ is equivalent to the line integral of $\mathbf{E}$. Accordingly, we get Stokes' theorem described by [1]

$$\int_S \text{rot } \mathbf{E} \cdot \mathbf{n} dS = \oint_C \mathbf{E} \cdot d\mathbf{l}. \quad (8.3)$$

Returning back to Fig. 8.1 and taking $\Delta h \rightarrow 0$, we have $\frac{\partial \mathbf{B}}{\partial t} \cdot \mathbf{n} \Delta l \Delta h \rightarrow 0$. Then, the second term of (8.2) vanishes and we get

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0.$$

This implies that

$$\Delta l (\mathbf{E}_1 \cdot \mathbf{t}_1 + \mathbf{E}_2 \cdot \mathbf{t}_2) = 0,$$

where $\mathbf{E}_1$ and $\mathbf{E}_2$ represent the electric field in the dielectrics D1 and D2 close to the interface, respectively. Considering $\mathbf{t}_2 = -\mathbf{t}_1$ and putting $\mathbf{t}_1 = \mathbf{t}$, we get

$$(\mathbf{E}_1 - \mathbf{E}_2) \cdot \mathbf{t} = 0, \quad (8.4)$$

where $\mathbf{t}$ represents a unit vector in the direction of a tangential line of the interface plane. Equation (8.4) means that the tangential components of the electric field are continuous on both sides of the interface. We obtain a similar result with the magnetic field. This can be shown by taking a surface integral of both sides of (7.29) as well. As a result, we get

$$(\mathbf{H}_1 - \mathbf{H}_2) \cdot \mathbf{t} = 0, \quad (8.5)$$

where $\mathbf{H}_1$ and $\mathbf{H}_2$ represent the magnetic field in D1 and D2 close to the interface, respectively. Hence, from (8.5) the tangential components of the magnetic field are continuous on both sides of the interface as well.

8.2 Basic Concepts Underlying Phenomena

When an electromagnetic wave is incident upon an interface of dielectrics, its reflection and transmission (refraction) take place at the interface. We address a question of how the nature of the dielectrics studied in the previous section is associated with the optical phenomena. When we deal with the problem, we assume non-absorbing media. Notice that the complex wavenumber vector is responsible for an absorbing medium along with a complex index of refraction. Nonetheless, our

approach is useful to discuss related problems in the absorbing media. Characteristic impedance plays a key role in the reflection and transmission of light.

We represent a field (either electric or magnetic) of the incident, reflected, and transmitted (or refracted) waves by $\mathbf{F}_i$, $\mathbf{F}_r$, and $\mathbf{F}_t$, respectively. We call a dielectric of the incidence side (and, hence, reflection side) D1 and another dielectric of the transmission side D2. The fields are described by

$$\mathbf{F}_i = F_i \boldsymbol{\epsilon}_i e^{i(\mathbf{k}_i \cdot \mathbf{x} - \omega t)}, \quad (8.6)$$

$$\mathbf{F}_r = F_r \boldsymbol{\epsilon}_r e^{i(\mathbf{k}_r \cdot \mathbf{x} - \omega t)}, \quad (8.7)$$

$$\mathbf{F}_t = F_t \boldsymbol{\epsilon}_t e^{i(\mathbf{k}_t \cdot \mathbf{x} - \omega t)}, \quad (8.8)$$

where F_i , F_r , and F_t denote an amplitude of the field; $\boldsymbol{\epsilon}_i$, $\boldsymbol{\epsilon}_r$, and $\boldsymbol{\epsilon}_t$ represent a unit vector of polarization direction, i.e., the direction along which the field oscillates; and $\mathbf{k}_i$, $\mathbf{k}_r$, and $\mathbf{k}_t$ are wavenumber vectors such that $\mathbf{k}_i \perp \boldsymbol{\epsilon}_i$, $\mathbf{k}_r \perp \boldsymbol{\epsilon}_r$, and $\mathbf{k}_t \perp \boldsymbol{\epsilon}_t$. These wavenumber vectors represent the propagation directions of individual waves. In (8.6)–(8.8), indices of i , r , and t stand for incidence, reflection, and transmission, respectively.

Let $\mathbf{x}_s$ be an arbitrary position vector at the interface between the dielectrics. Also, let $\mathbf{t}$ be a unit vector paralleling the interface. Thus, tangential components of the field are described as

$$F_{i_t} = F_i (\mathbf{t} \cdot \boldsymbol{\epsilon}_i) e^{i(\mathbf{k}_i \cdot \mathbf{x}_s - \omega t)}, \quad (8.9)$$

$$F_{r_t} = F_r (\mathbf{t} \cdot \boldsymbol{\epsilon}_r) e^{i(\mathbf{k}_r \cdot \mathbf{x}_s - \omega t)}, \quad (8.10)$$

$$F_{t_t} = F_t (\mathbf{t} \cdot \boldsymbol{\epsilon}_t) e^{i(\mathbf{k}_t \cdot \mathbf{x}_s - \omega t)}. \quad (8.11)$$

Note that F_{i_t} and F_{r_t} represent the field in D1 just close to the interface and that F_{t_t} denotes the field in D2 just close to the interface. Thus, in light of (8.4) and (8.5), we have

$$F_{i_t} + F_{r_t} = F_{t_t}. \quad (8.12)$$

Notice that (8.12) holds with any position $\mathbf{x}_s$ and any time t .

Let us think of elementary calculation of exponential functions or exponential polynomials and the relationship between individual coefficients and exponents. With respect to two functions e^{ikx} and $e^{ik'x}$, we have two alternatives according to a value Wronskian takes. Here, Wronskian W is expressed as

$$W = \begin{vmatrix} e^{ikx} & e^{ik'x} \\ (e^{ikx})' & (e^{ik'x})' \end{vmatrix} = -i(k - k')e^{i(k+k')x}. \quad (8.13)$$

(i) $W \neq 0$ if and only if $k \neq k'$. In this case, e^{ikx} and $e^{ik'x}$ are said to be linearly independent. That is, on condition of $k \neq k'$, for any x we have

$$ae^{ikx} + be^{ik'x} = 0 \iff a = b = 0. \quad (8.14)$$

(ii) $W = 0$ if $k = k'$. In that case, we have

$$ae^{ikx} + be^{ik'x} = (a + b)e^{ikx} = 0 \iff a + b = 0.$$

Notice that e^{ikx} never vanishes with any x . To conclude, if we think of an equation of an exponential polynomial

$$ae^{ikx} + be^{ik'x} = 0,$$

we have two alternatives regarding the coefficients. One is a trivial case of $a = b = 0$ and the other is $a + b = 0$.

Next, with respect to e^{ik_1x} , and e^{ik_2x} , and e^{ik_3x} , similarly we have

$$\begin{aligned} W &= \begin{vmatrix} e^{ik_1x} & e^{ik_2x} & e^{ik_3x} \\ (e^{ik_1x})' & (e^{ik_2x})' & (e^{ik_3x})' \\ (e^{ik_1x})'' & (e^{ik_2x})'' & (e^{ik_3x})'' \end{vmatrix} \\ &= -i(k_1 - k_2)(k_2 - k_3)(k_3 - k_1)e^{i(k_1+k_2+k_3)x}, \end{aligned} \quad (8.15)$$

where $W \neq 0$ if and only if $k_1 \neq k_2$, $k_2 \neq k_3$, and $k_3 \neq k_1$. That is, on this condition for any x we have

$$ae^{ik_1x} + be^{ik_2x} + ce^{ik_3x} = 0 \iff a = b = c = 0. \quad (8.16)$$

If the three exponential functions are linearly dependent, at least two of k_1 , k_2 , and k_3 are equal to each other and vice versa. On this condition, again consider a following equation of an exponential polynomial:

$$ae^{ik_1x} + be^{ik_2x} + ce^{ik_3x} = 0. \quad (8.17)$$

Without loss of generality, we assume that $k_1 = k_2$. Then, we have

$$ae^{ik_1x} + be^{ik_2x} + ce^{ik_3x} = (a + b)e^{ik_1x} + ce^{ik_3x} = 0.$$

If $k_1 \neq k_3$, we must have

$$a + b = 0 \quad \text{and} \quad c = 0. \quad (8.18)$$

If, on the other hand, $k_1 = k_3$, i.e., $k_1 = k_2 = k_3$, we have

$$ae^{ik_1x} + be^{ik_2x} + ce^{ik_3x} = (a + b + c)e^{ik_1x} = 0.$$

That is, we have

$$a + b + c = 0. \quad (8.19)$$

Consequently, we must have $k_1 = k_2 = k_3$ so that we can get three nonzero coefficients a , b , and c .

Returning to (8.12), its full description is

$$F_i(\mathbf{t} \cdot \mathbf{e}_i)e^{i(\mathbf{k}_i \cdot \mathbf{x}_s - \omega t)} + F_r(\mathbf{t} \cdot \mathbf{e}_r)e^{i(\mathbf{k}_r \cdot \mathbf{x}_s - \omega t)} - F_t(\mathbf{t} \cdot \mathbf{e}_t)e^{i(\mathbf{k}_t \cdot \mathbf{x}_s - \omega t)} = 0. \quad (8.20)$$

Again, (8.20) must hold with any position $\mathbf{x}_s$ and any time t . Meanwhile, for (8.20) to have a physical meaning, we should have

$$F_i(\mathbf{t} \cdot \mathbf{e}_i) \neq 0, F_r(\mathbf{t} \cdot \mathbf{e}_r) \neq 0, \text{ and } F_t(\mathbf{t} \cdot \mathbf{e}_t) \neq 0. \quad (8.21)$$

On the basis of the above consideration, we must have following two relations:

$$\mathbf{k}_i \cdot \mathbf{x}_s - \omega t = \mathbf{k}_r \cdot \mathbf{x}_s - \omega t = \mathbf{k}_t \cdot \mathbf{x}_s - \omega t$$

or

$$\mathbf{k}_i \cdot \mathbf{x}_s = \mathbf{k}_r \cdot \mathbf{x}_s = \mathbf{k}_t \cdot \mathbf{x}_s, \quad (8.22)$$

and

$$F_i(\mathbf{t} \cdot \mathbf{e}_i) + F_r(\mathbf{t} \cdot \mathbf{e}_r) - F_t(\mathbf{t} \cdot \mathbf{e}_t) = 0$$

or

$$F_i(\mathbf{t} \cdot \mathbf{e}_i) + F_r(\mathbf{t} \cdot \mathbf{e}_r) = F_t(\mathbf{t} \cdot \mathbf{e}_t). \quad (8.23)$$

In this way, we are able to obtain a relation among amplitudes of the fields of incidence, reflection, and transmission. Notice that we get both the relations between exponents and coefficients at once.

First, let us consider (8.22). Suppose that the incident light ($\mathbf{k}_i$) is propagated in a dielectric medium D1 in parallel to the zx -plane and that the interface is the xy -plane (see Fig. 8.3). Also suppose that at the interface the light is reflected partly back to D1 and transmitted (or refracted) partly into another dielectric medium D2. In

Fig. 8.3 Geometry of the incident, reflected, and transmitted lights. We assume that the light is incident from a dielectric medium D1 toward another medium D2. The wavenumber vectors $\mathbf{k}_i$, $\mathbf{k}_r$, and $\mathbf{k}_t$ represent the incident, reflected, and transmitted (or refracted) lights with an angle θ , θ' , and ϕ , respectively. Note here that we did not assume the equality of θ and θ' (see text)

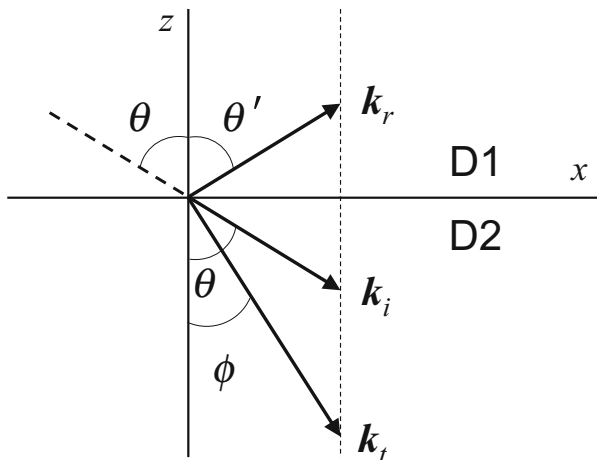


Fig. 8.3, $\mathbf{k}_i$, $\mathbf{k}_r$, and $\mathbf{k}_t$ represent the incident, reflected, and transmitted lights that make an angle θ , θ' , and ϕ with the z -axis, respectively. Then we have

$$\mathbf{k}_i = (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} k_i \sin \theta \\ 0 \\ -k_i \cos \theta \end{pmatrix}, \quad (8.24)$$

$$\mathbf{x}_s = (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}, \quad (8.25)$$

where θ is said to be an incidence angle. A plane formed by $\mathbf{k}_i$ and a normal to the interface is called a plane of incidence (or incidence plane). In Fig. 8.3, the zx -plane forms the incidence plane. From (8.24) and (8.25), we have

$$\mathbf{k}_i \cdot \mathbf{x}_s = k_i x \sin \theta, \quad (8.26)$$

$$\mathbf{k}_r \cdot \mathbf{x}_s = k_{r_x} x + k_{r_y} y, \quad (8.27)$$

$$\mathbf{k}_t \cdot \mathbf{x}_s = k_{t_x} x + k_{t_y} y, \quad (8.28)$$

where $k_i = |\mathbf{k}_i|$; k_{r_x} and k_{r_y} are x and y components of $\mathbf{k}_r$; similarly k_{t_x} and k_{t_y} are x and y components of $\mathbf{k}_t$.

Since (8.22) holds with any x and y , we have

$$k_i \sin \theta = k_{r_x} = k_{t_x}, \quad (8.29)$$

$$k_{r_y} = k_{t_y} = 0. \quad (8.30)$$

From (8.30) neither $\mathbf{k}_r$ nor $\mathbf{k}_t$ has a y component. This means that $\mathbf{k}_i$, $\mathbf{k}_r$, and $\mathbf{k}_t$ are coplanar. That is, the incident, reflected, and transmitted waves are all parallel to the zx -plane. Notice that at the beginning we did not assume the coplanarity of those waves. We did not assume the equality of θ and θ' either (*vide infra*). From (8.29) and Fig. 8.3, however, we have

$$k_i \sin \theta = k_r \sin \theta' = k_t \sin \phi, \quad (8.31)$$

where $k_r = |\mathbf{k}_r|$ and $k_t = |\mathbf{k}_t|$; θ' and ϕ are said to be a reflection angle and a refraction angle, respectively. Thus, the end points of $\mathbf{k}_i$, $\mathbf{k}_r$, and $\mathbf{k}_t$ are connected on a straight line that parallels the z -axis. Figure 8.3 clearly shows it.

Now, we suppose that a wavelength of the electromagnetic wave in D1 is λ_1 and that in D2 is λ_2 . Since the incident light and reflected light are propagated in D1, we have

$$k_i = k_r = 2\pi/\lambda_1. \quad (8.32)$$

From (8.31) and (8.32), we get

$$\sin \theta = \sin \theta'. \quad (8.33)$$

Therefore, we have either $\theta = \theta'$ or $\theta' = \pi - \theta$. Since $0 < \theta, \theta' < \pi/2$, we have

$$\theta = \theta'. \quad (8.34)$$

Then, returning back to (8.31), we have

$$k_i \sin \theta = k_r \sin \theta = k_t \sin \phi. \quad (8.35)$$

This implies that the components tangential to the interface of $\mathbf{k}_i$, $\mathbf{k}_r$, and $\mathbf{k}_t$ are the same.

Meanwhile, we have

$$k_t = 2\pi/\lambda_2. \quad (8.36)$$

Also we have

$$c = \lambda_0 \nu, v_1 = \lambda_1 \nu, v_2 = \lambda_2 \nu, \quad (8.37)$$

where v_1 and v_2 are phase velocities of light in D1 and D2, respectively. Since ν is common to D1 and D2, we have

$$c/\lambda_0 = v_1/\lambda_1 = v_2/\lambda_2$$

or

$$c/v_1 = \lambda_0/\lambda_1 = n_1, c/v_2 = \lambda_0/\lambda_2 = n_2, \quad (8.38)$$

where λ_0 is a wavelength in vacuum and n_1 and n_2 are refractive indices of D1 and D2, respectively. Combining (8.35) with (8.32), (8.36), and (8.38), we have several relations such that

$$\frac{\sin \theta}{\sin \phi} = \frac{k_t}{k_i} = \frac{\lambda_1}{\lambda_2} = \frac{n_2}{n_1} (\equiv n), \quad (8.39)$$

where n is said to be a relative refractive index of D2 relative to D1. The relation (8.39) is called Snell's law. Notice that (8.39) reflects the kinematic aspect of light and that this characteristic comes from the exponents of (8.20).

8.3 Transverse Electric (TE) Waves and Transverse Magnetic (TM) Waves

On the basis of the above argument, we are now in the position to determine the relations among amplitudes of the electromagnetic fields of waves of incidence, reflection, and transmission. Notice that since we are dealing with non-absorbing media, the relevant amplitudes are real (i.e., positive or negative). In other words, when the phase is retained upon reflection, we have a positive amplitude due to $e^{i0} = 1$. When the phase is reversed upon reflection, on the other hand, we will be treating a negative amplitude due to $e^{i\pi} = -1$. Nevertheless, when we consider the total reflection, we deal with a complex amplitude (*vide infra*).

We start with the discussion of the vertical incidence of an electromagnetic wave before the general oblique incidence. In Fig. 8.4a, we depict electric fields $\mathbf{E}$ and magnetic fields $\mathbf{H}$ obtained at a certain moment near the interface. We index, e.g., $\mathbf{E}_i$, for the incident field. There we define unit polarization vectors of the electric field $\mathbf{e}_i$, $\mathbf{e}_r$, and $\mathbf{e}_t$ as identical to be $\mathbf{e}_1$ (a unit vector in the direction of the x -axis). In (8.6), we also define F_i (both electric and magnetic fields) as positive.

We have two cases about a geometry of the fields (see Fig. 8.4). The first case is that all $\mathbf{E}_i$, $\mathbf{E}_r$, and $\mathbf{E}_t$ are directed in the same direction (i.e., the positive direction of the x -axis); see Fig. 8.4a. Another case is that although $\mathbf{E}_i$ and $\mathbf{E}_t$ are directed in the

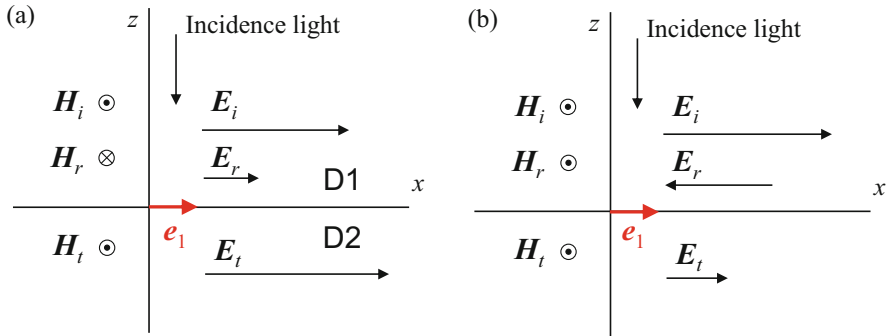


Fig. 8.4 Geometry of the electromagnetic fields near the interface between dielectric media D1 and D2 in the case of vertical incidence. (a) All E_i , E_r , and E_t are directed in the same direction e_1 (i.e., a unit vector in the positive direction of the x -axis). (b) Although E_i and E_t are directed in the same direction, E_r is reversed. In this case, we define E_r as negative

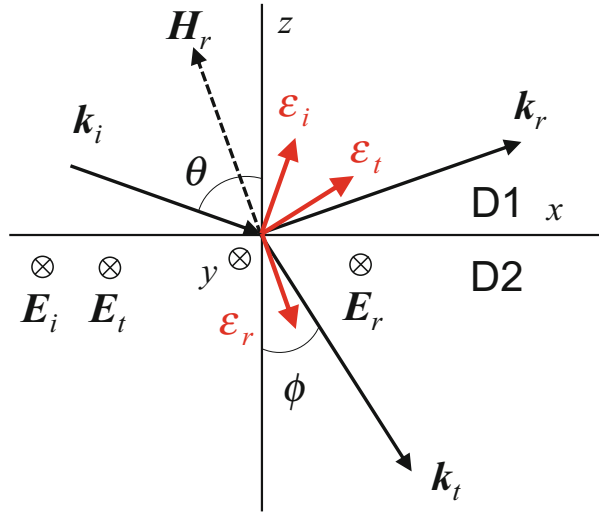
same direction, E_r is reversed (Fig. 8.4b). In this case, we define E_r as negative. Notice that E_i and E_t are always directed in the same direction and that E_r is directed either in the same direction or in the opposite direction according to the nature of the dielectrics. The situation will be discussed soon.

Meanwhile, unit polarization vectors of the magnetic fields are determined by (7.67) for the incident, reflected, and transmitted waves. In Fig. 8.4, the magnetic fields are polarized along the y -axis (i.e., perpendicular to the plane of paper). The magnetic fields H_i and H_t are always directed to the same direction as in the case of the electric fields. On the other hand, if the phase of E_r is conserved, the direction of H_r is reversed and vice versa. This converse relationship with respect to the electric and magnetic fields results solely from the requirement that E , H , and the propagation unit vector n of light must constitute a right-handed system in this order. Notice that n is reversed upon reflection.

Next, let us consider an oblique incidence. With the oblique incidence, electromagnetic waves are classified into two special categories, i.e., transverse electric (TE) waves (or modes) or transverse magnetic (TM) waves (or modes). The TE wave is characterized by the electric field that is perpendicular to the incidence plane, whereas the TM wave is characterized by the magnetic field that is perpendicular to the incidence plane. Here the incidence plane is a plane that is formed by the propagation direction of the incident light and the normal to the interface of the two dielectrics. Since E , H , and n form a right-handed system, in the TE wave, H lies on the incidence plane. For the same reason, in the TM wave, E lies on the incidence plane.

In a general case where a field is polarized in an arbitrary direction, that field can be formed by superimposing two fields corresponding to the TE and TM waves. In other words, if we take an arbitrary field E , it can be decomposed into a component having a unit polarization vector directed perpendicular to the incidence plane and another component having the polarization vector that lies on the incidence plane. These two components are orthogonal to each other.

Fig. 8.5 Geometry of the electromagnetic fields near the interface between dielectric media D1 and D2 in the case of oblique incidence of a TE wave. The electric field $\mathbf{E}$ is polarized along the y -axis (i.e., perpendicular to the plane of paper) with $\mathbf{H}$ polarized in the xz -plane. Unit polarization vectors $\mathbf{e}_i$, $\mathbf{e}_r$, and $\mathbf{e}_t$ are given for $\mathbf{H}$ with red arrows. To avoid complication, neither $\mathbf{H}_i$ nor $\mathbf{H}_t$ is shown



Example 8.1 TE Wave In Fig. 8.5 we depict the geometry of oblique incidence of a TE wave. The xy -plane defines the interface of the two dielectrics and $\mathbf{t}$ of (8.9) lies on that plane. The xz -plane defines the incidence plane. In this case, $\mathbf{E}$ is polarized along the y -axis with $\mathbf{H}$ polarized in the xz -plane. That is, regarding $\mathbf{E}$, we choose polarization direction $\mathbf{e}_i$, $\mathbf{e}_r$, and $\mathbf{e}_t$ of the electric field as $\mathbf{e}_2$ (a unit vector toward the positive direction of the y -axis that is perpendicular to the plane of paper). In Fig. 8.5, the polarization direction of the electric field is denoted by a symbol $\otimes$. Therefore, we have

$$\mathbf{e}_2 \cdot \mathbf{e}_i = \mathbf{e}_2 \cdot \mathbf{e}_r = \mathbf{e}_2 \cdot \mathbf{e}_t = 1. \quad (8.40)$$

For $\mathbf{H}$ we define the direction of unit polarization vectors $\mathbf{e}_i$, $\mathbf{e}_r$, and $\mathbf{e}_t$ so that their direction cosine relative to the x -axis can be positive; see Fig. 8.5. Choosing $\mathbf{e}_1$ (a unit vector in the direction of the x -axis) for $\mathbf{t}$ of (8.9) with regard to $\mathbf{H}$, we have

$$\mathbf{e}_1 \cdot \mathbf{e}_i = \cos \theta, \mathbf{e}_1 \cdot \mathbf{e}_r = \cos \theta, \text{ and } \mathbf{e}_1 \cdot \mathbf{e}_t = \cos \phi. \quad (8.41)$$

Note that we have used the same symbols $\mathbf{e}_i$, $\mathbf{e}_r$, and $\mathbf{e}_t$ to represent the unit polarization vectors of both the electric field $\mathbf{E}$ and magnetic field $\mathbf{H}$. In Fig. 8.5, the unit polarization vectors of $\mathbf{H}$ are depicted with red arrows.

According as $\mathbf{H}_r$ is directed to the same direction as $\mathbf{e}_r$ or the opposite direction to $\mathbf{e}_r$, the amplitude is defined as positive or negative, as in the case of the vertical incidence. In Fig. 8.5, we depict the case where the amplitude H_r is negative. That is, the phase of the magnetic field is reversed upon reflection and, hence, $\mathbf{H}_r$ is in an opposite direction to $\mathbf{e}_r$ in Fig. 8.5. Note that $\mathbf{H}_i$ and $\mathbf{H}_t$ are in the same direction as $\mathbf{e}_i$ and $\mathbf{e}_t$, respectively. To avoid complication, neither $\mathbf{H}_i$ nor $\mathbf{H}_t$ is shown in Fig. 8.5.

Taking account of the above note of caution, we apply (8.23) to both $\mathbf{E}$ and $\mathbf{H}$ to get the following relations:

$$E_i + E_r = E_t, \quad (8.42)$$

$$H_i \cos \theta + H_r \cos \theta = H_t \cos \phi. \quad (8.43)$$

To derive the above equations, we choose $\mathbf{t} = \mathbf{e}_2$ with $\mathbf{E}$ and $\mathbf{t} = \mathbf{e}_1$ with $\mathbf{H}$ for (8.23). Because of the abovementioned converse relationship with $\mathbf{E}$ and $\mathbf{H}$, we have

$$E_r H_r < 0. \quad (8.44)$$

Suppose that we carry out an experiment to determine six amplitudes in (8.42) and (8.43). Out of those quantities, we can freely choose and fix E_i . Then, we have five unknown amplitudes, i.e., E_r , E_t , H_i , H_r , and H_t . Thus, we need three more relations to determine them. Here information about the characteristic impedance Z is useful. It was defined as (7.64). From (8.6) to (8.8) as well as (8.44), we get

$$Z_1 = \sqrt{\mu_1/\varepsilon_1} = E_i/H_i = -E_r/H_r, \quad (8.45)$$

$$Z_2 = \sqrt{\mu_2/\varepsilon_2} = E_t/H_t, \quad (8.46)$$

where ε_1 and μ_1 are permittivity and permeability of D1, respectively, and ε_2 and μ_2 are permittivity and permeability of D2, respectively. As an example, we have

$$\begin{aligned} \mathbf{H}_i &= \mathbf{n} \times \mathbf{E}_i / Z_1 = \mathbf{n} \times \mathbf{E}_i \boldsymbol{\varepsilon}_{i,e} e^{i(\mathbf{k}_i \cdot \mathbf{x} - \omega t)} / Z_1 = \mathbf{E}_i \boldsymbol{\varepsilon}_{i,m} e^{i(\mathbf{k}_i \cdot \mathbf{x} - \omega t)} / Z_1 \\ &= H_i \boldsymbol{\varepsilon}_{i,m} e^{i(\mathbf{k}_i \cdot \mathbf{x} - \omega t)}, \end{aligned} \quad (8.47)$$

where we distinguish polarization vectors of electric field ($\boldsymbol{\varepsilon}_{i,e}$) and magnetic field ($\boldsymbol{\varepsilon}_{i,m}$). With the notation and geometry of the fields $\mathbf{E}_i$ and $\mathbf{H}_i$, see Fig. 7.4. Comparing coefficients of the last relation of (8.47), we get

$$E_i / Z_1 = H_i. \quad (8.48)$$

On the basis of (8.42)–(8.46), we are able to decide E_r , E_t , H_i , H_r , and H_t .

What we wish to determine, however, is a ratio among those quantities. To this end, dividing (8.42) and (8.43) by E_i (>0), we define following quantities:

$$R_E^\perp \equiv E_r/E_i \quad \text{and} \quad T_E^\perp \equiv E_t/E_i, \quad (8.49)$$

where $R_E^\perp$ and $T_E^\perp$ are said to be a reflection coefficient and transmission coefficient with the electric field, respectively; the symbol $\perp$ means a quantity of the TE wave (i.e., electric field oscillating vertically with respect to the incidence plane). Thus rewriting (8.42) and (8.43) and using $R_E^\perp$ and $T_E^\perp$, we have

$$\left. \begin{aligned} R_E^\perp - T_E^\perp &= -1, \\ R_E^\perp \frac{\cos \theta}{Z_1} + T_E^\perp \frac{\cos \phi}{Z_2} &= \frac{\cos \theta}{Z_1}. \end{aligned} \right\} \quad (8.50)$$

Using Cramer's rule of matrix algebra, we have a solution such that

$$R_E^\perp = \frac{\begin{vmatrix} -1 & -1 \\ \frac{\cos \theta}{Z_1} & \frac{\cos \phi}{Z_2} \end{vmatrix}}{\begin{vmatrix} 1 & -1 \\ \frac{\cos \theta}{Z_1} & \frac{\cos \phi}{Z_2} \end{vmatrix}} = \frac{Z_2 \cos \theta - Z_1 \cos \phi}{Z_2 \cos \theta + Z_1 \cos \phi}, \quad (8.51)$$

$$T_E^\perp = \frac{\begin{vmatrix} 1 & -1 \\ \frac{\cos \theta}{Z_1} & \frac{\cos \theta}{Z_1} \end{vmatrix}}{\begin{vmatrix} 1 & -1 \\ \frac{\cos \theta}{Z_1} & \frac{\cos \phi}{Z_2} \end{vmatrix}} = \frac{2Z_2 \cos \theta}{Z_2 \cos \theta + Z_1 \cos \phi}. \quad (8.52)$$

Similarly, defining

$$R_H^\perp \equiv H_r/H_i \quad \text{and} \quad T_H^\perp \equiv H_t/H_i, \quad (8.53)$$

where $R_H^\perp$ and $T_H^\perp$ are said to be a reflection coefficient and transmission coefficient with the magnetic field, respectively, we get

$$R_H^\perp = \frac{Z_1 \cos \phi - Z_2 \cos \theta}{Z_2 \cos \theta + Z_1 \cos \phi}, \quad (8.54)$$

$$T_H^\perp = \frac{2Z_1 \cos \theta}{Z_2 \cos \theta + Z_1 \cos \phi}. \quad (8.55)$$

In this case, rewrite (8.42) as a relation among H_i , H_r , and H_t using (8.45) and (8.46). Derivation of (8.54) and (8.55) is left for readers. Notice also that

Table 8.1 Reflection and transmission coefficients of TE and TM waves

$\perp$ Incidence (TE)	$\parallel$ Incidence (TM)
$R_E^\perp = \frac{Z_2 \cos \theta - Z_1 \cos \phi}{Z_2 \cos \theta + Z_1 \cos \phi}$	$R_E^\parallel = \frac{Z_2 \cos \phi - Z_1 \cos \theta}{Z_1 \cos \theta + Z_2 \cos \phi}$
$R_H^\perp = \frac{Z_1 \cos \phi - Z_2 \cos \theta}{Z_2 \cos \theta + Z_1 \cos \phi} = -R_E^\perp$	$R_H^\parallel = \frac{Z_1 \cos \theta - Z_2 \cos \phi}{Z_1 \cos \theta + Z_2 \cos \phi} = -R_E^\parallel$
$T_E^\perp = \frac{2Z_2 \cos \theta}{Z_2 \cos \theta + Z_1 \cos \phi}$	$T_E^\parallel = \frac{2Z_2 \cos \theta}{Z_1 \cos \theta + Z_2 \cos \phi}$
$T_H^\perp = \frac{2Z_1 \cos \theta}{Z_2 \cos \theta + Z_1 \cos \phi}$	$T_H^\parallel = \frac{2Z_1 \cos \theta}{Z_1 \cos \theta + Z_2 \cos \phi}$
$-R_E^\perp R_H^\perp + T_E^\perp T_H^\perp \frac{\cos \phi}{\cos \theta} = 1 \quad (R^\perp + T^\perp = 1)$	$-R_E^\parallel R_H^\parallel + T_E^\parallel T_H^\parallel \frac{\cos \phi}{\cos \theta} = 1 \quad (R^\parallel + T^\parallel = 1)$

$$R_H^\perp = -R_E^\perp. \quad (8.56)$$

This relation can easily be derived by (8.45).

Example 8.2 TM Wave In a manner similar to that described above, we obtain information about the TM wave. Switching a role of $\mathbf{E}$ and $\mathbf{H}$, we assume that $\mathbf{H}$ is polarized along the y-axis with $\mathbf{E}$ polarized in the zx -plane. Following the aforementioned procedures, we have

$$E_i \cos \theta + E_r \cos \theta = E_t \cos \phi, \quad (8.57)$$

$$H_i + H_r = H_t. \quad (8.58)$$

From (8.57) and (8.58), similarly we get

$$R_E^\parallel = \frac{Z_2 \cos \phi - Z_1 \cos \theta}{Z_1 \cos \theta + Z_2 \cos \phi}, \quad (8.59)$$

$$T_E^\parallel = \frac{2Z_2 \cos \theta}{Z_1 \cos \theta + Z_2 \cos \phi}. \quad (8.60)$$

Also, we get

$$R_H^\parallel = \frac{Z_1 \cos \theta - Z_2 \cos \phi}{Z_1 \cos \theta + Z_2 \cos \phi} = -R_E^\parallel, \quad (8.61)$$

$$T_H^\parallel = \frac{2Z_1 \cos \theta}{Z_1 \cos \theta + Z_2 \cos \phi}. \quad (8.62)$$

The symbol $\parallel$ of (8.59)–(8.62) means a quantity of the TM wave.

In Table 8.1, we list the important coefficients in relation to the reflection and transmission of electromagnetic waves along with their relationship.

In Examples 8.1 and 8.2, we have examined how the reflection and transmission coefficients vary as a function of characteristic impedance as well as incidence and refraction angles. Meanwhile, in a nonmagnetic substance, a refractive index n can be approximated as

$$n \approx \sqrt{\epsilon_r}, \quad (7.57)$$

assuming that $\mu_r \approx 1$. In this case, we have

$$Z = \sqrt{\mu/\epsilon} = \sqrt{\mu_r \mu_0 / \epsilon_r \epsilon_0} \approx Z_0 / \sqrt{\epsilon_r} \approx Z_0 / n.$$

Using this relation, we can readily rewrite the reflection and transmission coefficients as a function of refractive indices of the dielectrics. The derivation is left for the readers.

8.4 Energy Transport by Electromagnetic Waves

Returning to (7.58) and (7.59), let us consider energy transport in a dielectric medium by electromagnetic waves. Let us describe their electric ($\mathbf{E}$) and magnetic ($\mathbf{H}$) fields of the electromagnetic waves in a uniform and infinite dielectric medium such that

$$\mathbf{E} = E \boldsymbol{\epsilon}_e e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, \quad (8.63)$$

$$\mathbf{H} = H \boldsymbol{\epsilon}_m e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, \quad (8.64)$$

where $\boldsymbol{\epsilon}_e$ and $\boldsymbol{\epsilon}_m$ are unit polarization vector; we assume that both E and H are positive. Notice again that $\boldsymbol{\epsilon}_e$, $\boldsymbol{\epsilon}_m$, and $\mathbf{n}$ constitute a right-handed system in this order.

The energy transport is characterized by a Poynting vector $\mathbf{S}$ that is described by

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}. \quad (8.65)$$

Since $\mathbf{E}$ and $\mathbf{H}$ have a dimension $[\frac{\text{V}}{\text{m}}]$ and $[\frac{\text{A}}{\text{m}}]$, respectively, $\mathbf{S}$ has a dimension $[\frac{\text{W}}{\text{m}^2}]$. Hence, $\mathbf{S}$ represents an energy flow per unit time and per unit area with respect to the propagation direction. For simplicity, let us assume that the electromagnetic wave is propagating toward the z -direction. Then we have

$$\mathbf{E} = E \boldsymbol{\epsilon}_e e^{i(kz - \omega t)}, \quad (8.66)$$

$$\mathbf{H} = H\mathbf{e}_m e^{i(kz - \omega t)}. \quad (8.67)$$

To seek a time-averaged energy flow toward the z -direction, it suffices to multiply real parts of (8.66) and (8.67) and integrate it during a period T at a point of $z = 0$. Thus, a time-averaged Poynting vector $\bar{\mathbf{S}}$ is given by

$$\bar{\mathbf{S}} = \mathbf{e}_3 \frac{EH}{T} \int_0^T \cos^2 \omega t dt, \quad (8.68)$$

where $T = 1/\nu = 2\pi/\omega$. Using a trigonometric formula

$$\cos^2 \omega t = \frac{1}{2}(1 + \cos 2\omega t), \quad (8.69)$$

the integration can easily be performed. Thus, we get

$$\bar{\mathbf{S}} = \frac{1}{2} EH \mathbf{e}_3. \quad (8.70)$$

Equivalently, we have

$$\bar{\mathbf{S}} = \frac{1}{2} \mathbf{E} \times \mathbf{H}^*. \quad (8.71)$$

Meanwhile, an energy density W is given by

$$W = \frac{1}{2} (\mathbf{E} \cdot \mathbf{D} + \mathbf{H} \cdot \mathbf{B}), \quad (8.72)$$

where the first and second terms are pertinent to the electric and magnetic fields, respectively. Note in (8.72) that the dimension of $\mathbf{E} \cdot \mathbf{D}$ is $[\frac{\text{V}}{\text{m}} \cdot \frac{\text{C}}{\text{m}^2}] = [\frac{\text{J}}{\text{m}^3}]$ and that the dimension of $\mathbf{H} \cdot \mathbf{B}$ is $[\frac{\text{A}}{\text{m}} \cdot \frac{\text{V}\cdot\text{s}}{\text{m}^2}] = [\frac{\text{W}\cdot\text{s}}{\text{m}^3}] = [\frac{\text{J}}{\text{m}^3}]$. Using (7.7) and (7.10), we have

$$W = \frac{1}{2} (\epsilon E^2 + \mu H^2). \quad (8.73)$$

As in the above case, estimating a time-averaged energy density $\bar{W}$, we get

$$\bar{W} = \frac{1}{2} \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) = \frac{1}{4} \epsilon E^2 + \frac{1}{4} \mu H^2. \quad (8.74)$$

We also get this relation by integrating (8.73) over a wavelength λ at a time of $t = 0$. Using (7.60) and (7.61), we have

$$\epsilon E^2 = \mu H^2. \quad (8.75)$$

This implies that the energy density resulting from the electric field and that due to the magnetic field have the same value. Thus, rewriting (8.74) we have

$$\overline{W} = \frac{1}{2} \epsilon E^2 = \frac{1}{2} \mu H^2. \quad (8.76)$$

Moreover, using (7.43), we have for an impedance

$$Z = E/H = \sqrt{\mu/\epsilon} = \mu v \text{ or } E = \mu v H. \quad (8.77)$$

Using this relation along with (8.75), we get

$$\overline{\mathbf{S}} = \frac{1}{2} v \epsilon E^2 \mathbf{e}_3 = \frac{1}{2} v \mu H^2 \mathbf{e}_3. \quad (8.78)$$

Thus, we have various relations among amplitudes of electromagnetic waves and related physical quantities together with constant of dielectrics.

Returning to Examples 8.1 and 8.2, let us further investigate the reflection and transmission properties of the electromagnetic waves. From (8.51) to (8.55) as well as (8.59) to (8.62), we get in both the cases of TE and TM waves

$$-R_E^\perp R_H^\perp + T_E^\perp T_H^\perp \frac{\cos \phi}{\cos \theta} = 1, \quad (8.79)$$

$$-R_E^\parallel R_H^\parallel + T_E^\parallel T_H^\parallel \frac{\cos \phi}{\cos \theta} = 1. \quad (8.80)$$

In both the TE and TM cases, we define *reflectance* R and *transmittance* T such that

$$R \equiv -R_E R_H = R_E^2 = 2|\overline{\mathbf{S}}_r|/2|\overline{\mathbf{S}}_i| = |\overline{\mathbf{S}}_r|/|\overline{\mathbf{S}}_i|, \quad (8.81)$$

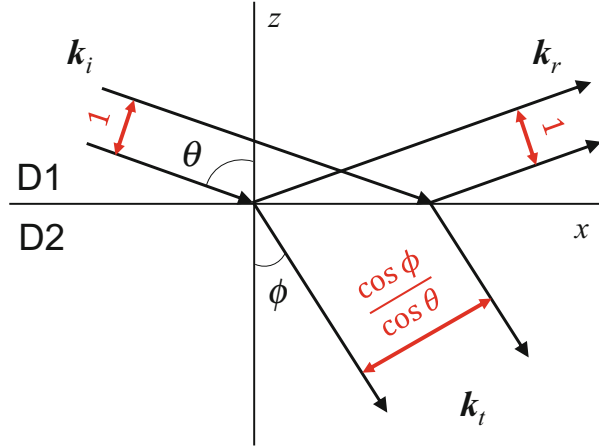
where $\overline{\mathbf{S}}_r$ and $\overline{\mathbf{S}}_i$ are time-averaged Poynting vectors of the reflected wave and incident waves, respectively.

Also, we have

$$T \equiv T_E T_H \frac{\cos \phi}{\cos \theta} = \frac{2|\overline{\mathbf{S}}_t|}{2|\overline{\mathbf{S}}_i|} \frac{\cos \phi}{\cos \theta} = \frac{|\overline{\mathbf{S}}_t|}{|\overline{\mathbf{S}}_i|} \frac{\cos \phi}{\cos \theta}, \quad (8.82)$$

where $\overline{\mathbf{S}}_t$ is a time-averaged Poynting vector of the transmitted wave. Thus, we have

Fig. 8.6 Luminous flux near the interface



$$R + T = 1. \quad (8.83)$$

The relation (8.83) represents the energy conservation. The factor $\frac{\cos \phi}{\cos \theta}$ can be understood by Fig. 8.6 that depicts a luminous flux near the interface. Suppose that we have an incident wave with an irradiance $I \left[\frac{\text{W}}{\text{m}^2} \right]$ whose incidence plane is the zx -plane. Notice that I has the same dimension as a Poynting vector.

Here let us think of the luminous flux that is getting through a unit area (i.e., a unit length square) perpendicular to the propagation direction of the light. Then, this flux illuminates an area on the interface of a unit length (in the y -direction) multiplied by a length of $\frac{\cos \phi}{\cos \theta}$ (in the x -direction). That is, the luminous flux has been widened (or squeezed) by $\frac{\cos \phi}{\cos \theta}$ times after getting through the interface. The irradiance has been weakened (or strengthened) accordingly (see Fig. 8.6). Thus, to take a balance of income and outgo with respect to the luminous flux before and after getting through the interface, the transmission irradiance must be multiplied by a factor $\frac{\cos \phi}{\cos \theta}$.

8.5 Brewster Angles and Critical Angles

In this section and subsequent sections, we deal with nonmagnetic substance as dielectrics; namely, we assume $\mu_r \approx 1$. In that case, as mentioned in Sect. 8.3, we rewrite, e.g., (8.51) and (8.59) as

$$R_E^\perp = \frac{\cos \theta - n \cos \phi}{\cos \theta + n \cos \phi}, \quad (8.84)$$

$$R_E^{\parallel} = \frac{\cos \phi - n \cos \theta}{\cos \phi + n \cos \theta}, \quad (8.85)$$

where $n (=n_2/n_1)$ is a relative refractive index of D2 relative to D1. Let us think of a condition on which $R_E^{\perp} = 0$ or $R_E^{\parallel} = 0$.

First, we consider (8.84). We have

$$\begin{aligned} [\text{numerator of (8.84)}] &= \cos \theta - n \cos \phi = \cos \theta - \frac{\sin \theta}{\sin \phi} \cos \phi \\ &= \frac{\sin \phi \cos \theta - \sin \theta \cos \phi}{\sin \phi} = \frac{\sin(\phi - \theta)}{\sin \phi}, \end{aligned} \quad (8.86)$$

where with the second equality we used Snell's law; the last equality is due to trigonometric formula. Since we assume $0 < \theta < \pi/2$ and $0 < \phi < \pi/2$, we have $-\pi/2 < \phi - \theta < \pi/2$. Therefore, if and only if $\phi - \theta = 0$, $\sin(\phi - \theta) = 0$. Namely, only when $\phi = \theta$, $R_E^{\perp}$ could vanish. For different dielectrics having different refractive indices, only if $\phi = \theta = 0$ (i.e., a vertical incidence), we have $\phi = \theta$. But, in that case we obtain

$$\lim_{\phi \rightarrow 0, \theta \rightarrow 0} \frac{\sin(\phi - \theta)}{\sin \phi} = \frac{0}{0}.$$

This is a limit of indeterminate form. From (8.84), however, we have

$$R_E^{\perp} = \frac{1 - n}{1 + n}, \quad (8.87)$$

for $\phi = \theta = 0$. This implies that $R_E^{\perp}$ does not vanish at $\phi = \theta = 0$. Thus, $R_E^{\perp}$ never vanishes for any θ or ϕ . Note that for this condition, naturally we have

$$R_E^{\parallel} = \frac{1 - n}{1 + n}.$$

This is because with $\phi = \theta = 0$ we have no physical difference between TE and TM waves.

In turn, let us examine (8.85) similarly with the case of TM wave. We have

$$[\text{numerator of (8.85)}] = \cos \phi - n \cos \theta = \cos \phi - \frac{\sin \theta}{\sin \phi} \cos \theta$$

$$= \frac{\sin \phi \cos \phi - \sin \theta \cos \theta}{\sin \phi} = \frac{\sin(\phi - \theta) \cos(\phi + \theta)}{\sin \phi}. \quad (8.88)$$

With the last equality of (8.88), we used a trigonometric formula. From (8.86) we know that $\frac{\sin(\phi - \theta)}{\sin \phi}$ does not vanish. Therefore, for $R_E^\parallel$ to vanish, we need $\cos(\phi + \theta) = 0$. Since $0 < \phi + \theta < \pi$, $\cos(\phi + \theta) = 0$ if and only if

$$\phi + \theta = \pi/2. \quad (8.89)$$

In other words, for particular angles $\theta = \theta_B$ and $\phi = \phi_B$ that satisfy

$$\phi_B + \theta_B = \pi/2, \quad (8.90)$$

we have $R_E^\parallel = 0$, i.e., we do not observe a reflected wave. The particular angle θ_B is said to be the Brewster angle. For θ_B we have

$$\sin \phi_B = \sin\left(\frac{\pi}{2} - \theta_B\right) = \cos \theta_B,$$

$$n = \sin \theta_B / \sin \phi_B = \sin \theta_B / \cos \theta_B = \tan \theta_B \quad \text{or} \quad \theta_B = \tan^{-1} n,$$

$$\phi_B = \tan^{-1} n^{-1}. \quad (8.91)$$

Suppose that we have a parallel plate consisting a dielectric D2 of a refractive index n_2 sandwiched with another dielectric D1 of a refractive index n_1 (Fig. 8.7). Let θ_B be the Brewster angle when the TM wave is incident from D1 to D2. In the above discussion, we defined a relative refractive index n of D2 relative D1 as $n = n_2/n_1$; recall (8.39). The other way around, suppose that the TM wave is incident from D2 to D1. Then, the relative refractive index of D1 relative to D2 is $n_1/n_2 = n^{-1}$. In this situation, another Brewster angle (from D2 to D1) defined as $\tilde{\theta}_B$ is given by

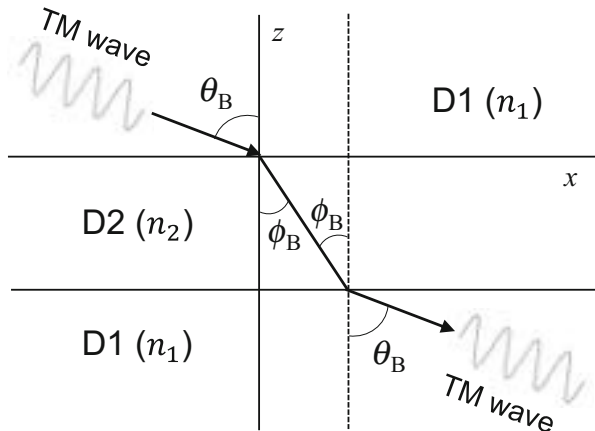
$$\tilde{\theta}_B = \tan^{-1} n^{-1}. \quad (8.92)$$

This number is, however, identical to ϕ_B in (8.91). Thus, we have

$$\tilde{\theta}_B = \phi_B. \quad (8.93)$$

Thus, regarding the TM wave that is propagating in D2 after getting through the interface and is to get back to D1, $\tilde{\theta}_B = \phi_B$ is again the Brewster angle. In this way, the said TM wave is propagating from D1 to D2 and then getting back from D2 to D1 without being reflected by the two interfaces. This conspicuous feature is often utilized for an optical device.

Fig. 8.7 Diagram that explains the Brewster angle. Suppose that a parallel plate consisting of a dielectric D2 of a refractive index n_2 is sandwiched with another dielectric D1 of a refractive index n_1 . The incidence angle θ_B represents the Brewster angle observed when the TM wave is incident from D1 to D2. ϕ_B is another Brewster angle that is observed when the TM wave is getting back from D2 to D1



If an electromagnetic wave is incident from a dielectric of a higher refractive index to that of a lower index, the total reflection takes place. This is equally the case with both TE and TM waves. For the total reflection to take place, θ should be larger than a critical angle θ_c that is defined by

$$\theta_c = \sin^{-1} n. \quad (8.94)$$

This is because at θ_c from the Snell's law we have

$$\frac{\sin \theta_c}{\sin \frac{\pi}{2}} = \sin \theta_c = \frac{n_2}{n_1} (\equiv n). \quad (8.95)$$

From (8.95), we have

$$\tan \theta_c = n / \sqrt{1 - n^2} > n = \tan \theta_B.$$

In the case of the TM wave, therefore, we find that

$$\theta_c > \theta_B. \quad (8.96)$$

The critical angle is always larger than the Brewster angle with TM waves.

8.6 Total Reflection

In Sect. 8.2 we saw that Snell's law results from the kinematical requirement. For this reason, we may consider it as a universal relation that can be extended to complex refraction angles. In fact, for Snell's law to hold with $\theta > \theta_c$, we must have

$$\sin \phi > 1. \quad (8.97)$$

This needs us to extend ϕ to a complex domain. Putting

$$\phi = \frac{\pi}{2} + ia \quad (a : \text{real}, a \neq 0), \quad (8.98)$$

we have

$$\sin \phi \equiv \frac{1}{2i}(e^{i\phi} - e^{-i\phi}) = \frac{1}{2}(e^{-a} + e^a) > 1, \quad (8.99)$$

$$\cos \phi \equiv \frac{1}{2}(e^{i\phi} + e^{-i\phi}) = \frac{i}{2}(e^{-a} - e^a). \quad (8.100)$$

Thus, $\cos \phi$ is pure imaginary.

Now, let us consider a transmitted wave whose electric field is described as

$$\mathbf{E}_t = E \boldsymbol{\epsilon}_t e^{i(\mathbf{k}_t \cdot \mathbf{x} - \omega t)}, \quad (8.101)$$

where $\boldsymbol{\epsilon}_t$ is the unit polarization vector and $\mathbf{k}_t$ is a wavenumber vector of the transmission wave. Suppose that the incidence plane is the zx -plane. Then, we have

$$\mathbf{k}_t \cdot \mathbf{x} = k_{t_x}x + k_{t_z}z = xk_t \sin \phi + zk_t \cos \phi, \quad (8.102)$$

where k_{t_x} and k_{t_z} are x and z components of $\mathbf{k}_t$, respectively; $k_t = |\mathbf{k}_t|$. Putting

$$\cos \phi = ib \quad (b : \text{real}, b \neq 0), \quad (8.103)$$

we have

$$\mathbf{E}_t = E \boldsymbol{\epsilon}_t e^{i(xk_t \sin \phi + ibzk_t - \omega t)} = E \boldsymbol{\epsilon}_t e^{i(xk_t \sin \phi - \omega t)} e^{-bzk_t}. \quad (8.104)$$

With the total reflection, we must have

$$z \rightarrow \infty \Rightarrow e^{-bzk_t} \rightarrow 0. \quad (8.105)$$

To meet this requirement, we have

$$b > 0. \quad (8.106)$$

Meanwhile, we have

$$\cos^2 \phi = 1 - \sin^2 \phi = 1 - \frac{\sin^2 \theta}{n^2} = \frac{n^2 - \sin^2 \theta}{n^2}, \quad (8.107)$$

where notice that $n < 1$, because we are dealing with the incidence of light from a medium with a higher refractive index to a low index medium. When we consider the total reflection, the numerator of (8.107) is negative, and so we have two choices such that

$$\cos \phi = \pm i \frac{\sqrt{\sin^2 \theta - n^2}}{n}. \quad (8.108)$$

From (8.103) and (8.106), we get

$$\cos \phi = i \frac{\sqrt{\sin^2 \theta - n^2}}{n}. \quad (8.109)$$

Hence, inserting (8.109) into (8.84), we have for the TE wave

$$R_E^\perp = \frac{\cos \theta - i \sqrt{\sin^2 \theta - n^2}}{\cos \theta + i \sqrt{\sin^2 \theta - n^2}}. \quad (8.110)$$

Then, we have

$$\begin{aligned} R^\perp &= -R_E^\perp (R_H^\perp)^* = R_E^\perp (R_E^\perp)^* = \frac{\cos \theta - i \sqrt{\sin^2 \theta - n^2}}{\cos \theta + i \sqrt{\sin^2 \theta - n^2}} \\ &\quad \cdot \frac{\cos \theta + i \sqrt{\sin^2 \theta - n^2}}{\cos \theta - i \sqrt{\sin^2 \theta - n^2}} = 1. \end{aligned} \quad (8.111)$$

As for the TM wave, substituting (8.109) for (8.85), we have

$$R_E^\parallel = \frac{-n^2 \cos \theta + i \sqrt{\sin^2 \theta - n^2}}{n^2 \cos \theta + i \sqrt{\sin^2 \theta - n^2}}. \quad (8.112)$$

In this case, we also get

$$R^\parallel = \frac{-n^2 \cos \theta + i \sqrt{\sin^2 \theta - n^2}}{n^2 \cos \theta + i \sqrt{\sin^2 \theta - n^2}} \cdot \frac{-n^2 \cos \theta - i \sqrt{\sin^2 \theta - n^2}}{n^2 \cos \theta - i \sqrt{\sin^2 \theta - n^2}} = 1. \quad (8.113)$$

The relations (8.111) and (8.113) ensure that the energy flow gets back to a higher refractive index medium.

Thus, the total reflection is characterized by the complex reflection coefficient expressed as (8.110) and (8.112) as well as a reflectance of 1. From (8.110) and

(8.112), we can estimate a change in a phase of the electromagnetic wave that takes place by virtue of the total reflection. For this purpose, we put

$$R_E^\perp \equiv e^{i\alpha} \quad \text{and} \quad R_E^\parallel \equiv e^{i\beta}. \quad (8.114)$$

Rewriting (8.110), we have

$$R_E^\perp = \frac{\cos^2 \theta - (\sin^2 \theta - n^2) - 2i \cos \theta \sqrt{\sin^2 \theta - n^2}}{1 - n^2}. \quad (8.115)$$

At a critical angle θ_c , from (8.95) we have

$$\sin \theta_c = n. \quad (8.116)$$

Therefore, we have

$$1 - n^2 = \cos^2 \theta_c. \quad (8.117)$$

Then, as expected, we get

$$R_E^\perp|_{\theta=\theta_c} = 1. \quad (8.118)$$

Note, however, that at $\theta = \pi/2$ (i.e., grazing incidence), we have

$$R_E^\perp|_{\theta=\pi/2} = -1. \quad (8.119)$$

From (8.115), an argument α in a complex plane is given by

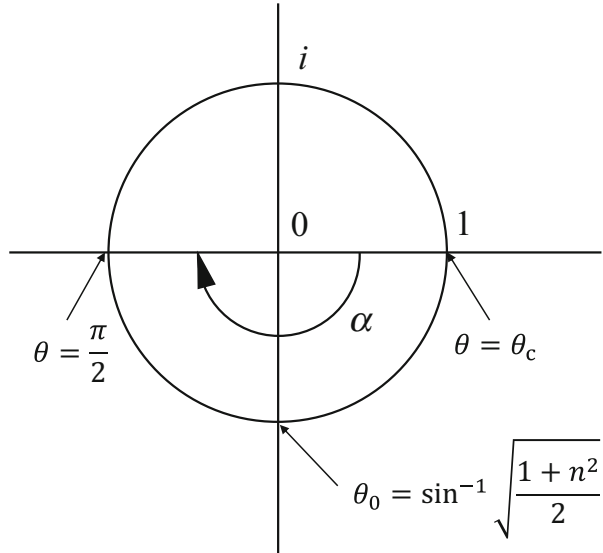
$$\tan \alpha = - \frac{2 \cos \theta \sqrt{\sin^2 \theta - n^2}}{\cos^2 \theta - (\sin^2 \theta - n^2)}. \quad (8.120)$$

The argument α defines a phase shift upon the total reflection. Considering (8.115) and (8.118), we have

$$\alpha|_{\theta=\theta_c} = 0.$$

Since $1 - n^2 > 0$ (i. e., $n < 1$) and in the total reflection region $\sin^2 \theta - n^2 > 0$, the imaginary part of $R_E^\perp$ is negative for any θ (i.e., 0 to $\pi/2$). On the other hand, the real part of $R_E^\perp$ varies from 1 to -1 , as is evidenced from (8.118) and (8.119). At θ_0 that satisfies a following condition:

Fig. 8.8 Phase shift α defined in a complex plane for the total reflection of TE wave. The number n denotes a relative refractive index of D2 relative to D1. At a critical angle θ_c , $\alpha = 0$



$$\sin \theta_0 = \sqrt{\frac{1+n^2}{2}}, \quad (8.121)$$

the real part is zero. Thus, the phase shift α varies from 0 to $-\pi$ as indicated in Fig. 8.8. Comparing (8.121) with (8.116) and taking into account $n < 1$, we have

$$\theta_c < \theta_0 < \pi/2.$$

Similarly, we estimate the phase change for a TM wave. Rewriting (8.112), we have

$$\begin{aligned} R_E^{\parallel} &= \frac{-n^4 \cos^2 \theta + \sin^2 \theta - n^2 + 2in^2 \cos \theta \sqrt{\sin^2 \theta - n^2}}{n^4 \cos^2 \theta + \sin^2 \theta - n^2} \\ &= \frac{-n^4 \cos^2 \theta + \sin^2 \theta - n^2 + 2in^2 \cos \theta \sqrt{\sin^2 \theta - n^2}}{(1-n^2)(\sin^2 \theta - n^2 \cos^2 \theta)}. \end{aligned} \quad (8.122)$$

Then, we have

$$R_E^{\parallel} \Big|_{\theta=\theta_c} = -1. \quad (8.123)$$

Also at $\theta = \pi/2$ (i.e., grazing incidence), we have

$$R_E^{\parallel} \Big|_{\theta=\pi/2} = 1. \quad (8.124)$$

From (8.122), an argument β is given by

$$\tan \beta = \frac{2n^2 \cos \theta \sqrt{\sin^2 \theta - n^2}}{-n^4 \cos^2 \theta + \sin^2 \theta - n^2}. \quad (8.125)$$

Considering (8.122) and (8.123), we have

$$\beta|_{\theta=\theta_c} = \pi. \quad (8.126)$$

In the total reflection region, we have

$$\sin^2 \theta - n^2 \cos^2 \theta > n^2 - n^2 \cos^2 \theta = n^2 (1 - \cos^2 \theta) > 0. \quad (8.127)$$

Therefore, the denominator of (8.122) is positive and, hence, the imaginary part of $R_E^{\parallel}$ is positive as well for any θ (i.e., 0 to $\pi/2$). From (8.123) and (8.124), on the other hand, the real part of $R_E^{\parallel}$ in (8.122) varies from -1 to 1 . At $\tilde{\theta}_0$ that satisfies a following condition:

$$\cos \tilde{\theta}_0 = \sqrt{\frac{1-n^2}{1+n^4}}, \quad (8.128)$$

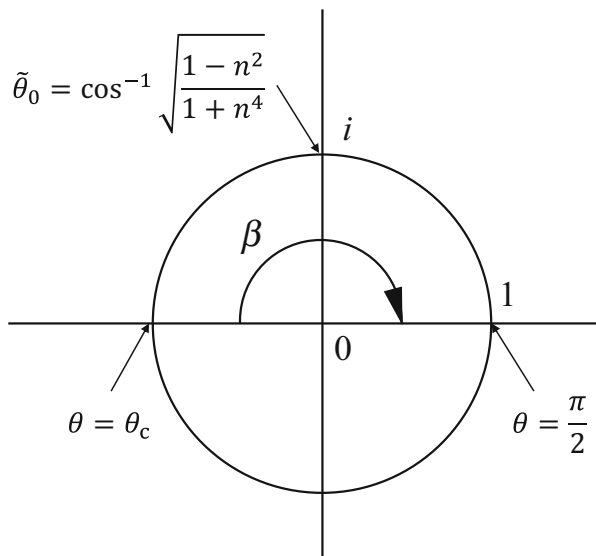
the real part of $R_E^{\parallel}$ is zero. Once again, we have

$$\theta_c < \tilde{\theta}_0 < \pi/2. \quad (8.129)$$

Thus, the phase β varies from π to 0 as depicted in Fig. 8.9.

In this section, we mentioned somewhat peculiar features of complex trigonometric functions such as $\sin \phi > 1$ in light of real functions. As a matter of course, the complex angle ϕ should be determined experimentally from (8.109). In this context readers are referred to Chap. 6 that dealt with the theory of analytic functions [2].

Fig. 8.9 Phase shift β for the total reflection of TM wave. At a critical angle θ_c , $\beta = \pi$



8.7 Waveguide Applications

There are many optical devices based upon light propagation. Among them, waveguide devices utilize the total reflection. We explain their operation principle.

Suppose that we have a thin plate (usually said to be a slab) comprising a dielectric medium that infinitely spreads two-dimensionally and that the plate is sandwiched with another dielectric (or maybe air or vacuum) or metal. In this situation electromagnetic waves are confined within the slab. Moreover, only under a restricted condition those waves are allowed to propagate in parallel to the slab plane. Such electromagnetic waves are usually called propagation modes or simply “modes.” An optical device thus designed is called a *waveguide*. These modes are characterized by repeated total reflection during the propagation. Another mode is an evanescent mode. Because of the total reflection, the energy transport is not allowed to take place vertically to the interface of two dielectrics. For this reason, the evanescent mode is thought to be propagated in parallel to the interface very close to it.

8.7.1 TE and TM Waves in a Waveguide

In a waveguide configuration, propagating waves are classified into TE and TM modes. Quality of materials that constitute a waveguide largely governs the propagation modes within the waveguide.

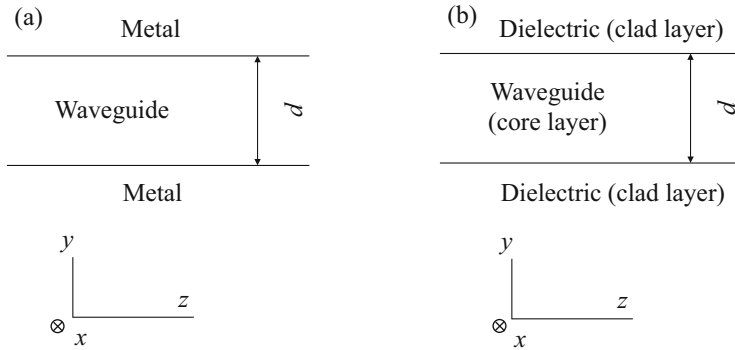


Fig. 8.10 Cross-section of a slab waveguide comprising a dielectric medium. (a) A waveguide is sandwiched between a couple of metal layers. (b) A waveguide is sandwiched between a couple of layers consisting of another dielectric called clad layer. The sandwiched layer is called core layer

Figure 8.10 depicts a cross-section of a slab waveguide. We assume that the electromagnetic wave is propagated toward the positive direction of the z -axis and that a waveguide infinitely spreads toward the z - and x -axes. Suppose that the said waveguide is spatially confined toward the y -axis. Let the thickness of the waveguide be d . From a point of view of material that shapes a waveguide, waveguides are classified into two types. (i) Electromagnetic waves are completely confined within the waveguide layer. This case typically happens when a dielectric forming the waveguide is sandwiched between a couple of metal layers (Fig. 8.10a). This is because the electromagnetic wave is not allowed to exist or propagate inside the metal.

(ii) Electromagnetic waves are not completely confined within the waveguide. This case happens when the dielectric of the waveguide is sandwiched by a couple of other dielectrics. We distinguish this case as the *total internal reflection* from the above case (i). We will further describe it in Sect. 8.7.2. For the total internal reflection to take place, the refractive index of the waveguide must be higher than those of other dielectrics (Fig. 8.10b). The dielectric of the waveguide is called core layer and the other dielectric is called clad layer. In this case, electromagnetic waves are allowed to propagate inside of the clad layer, even though the region is confined very close to the interface between the clad and core layers. Such electromagnetic waves are said to be an evanescent wave. According to these two cases (i) and (ii), we have different conditions under which the allowed modes can exist.

Now, let us return to Maxwell's equations. We have introduced the equations of wave motion (7.35) and (7.36) from Maxwell's equations (7.28) and (7.29) along with (7.7) and (7.10). One of their simplest solutions is a plane wave described by (7.53). The plane wave is characterized by that the wave has the same phase on an infinitely spreading plane perpendicular to the propagation direction (characterized by a wavenumber vector $\mathbf{k}$). In a waveguide, however, the electromagnetic field is confined with respect to the direction parallel to the normal to the slab plane (i.e., the direction of the y -axis in Fig. 8.10). Consequently, the electromagnetic field can no

longer have the same phase in that direction. Yet, as solutions of equations of wave motion, we can have a solution that has the same phase with the direction of the x -axis. Bearing in mind such a situation, let us think of Maxwell's equations in relation to the equations of wave motion.

Ignoring components related to partial differentiation with respect to x (i.e., the component related to $\partial/\partial x$) and rewriting (7.28) and (7.29) for individual Cartesian coordinates, we have [3]

$$\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} + \frac{\partial B_x}{\partial t} = 0, \quad (8.130)$$

$$\frac{\partial E_x}{\partial z} + \frac{\partial B_y}{\partial t} = 0, \quad (8.131)$$

$$- \frac{\partial E_x}{\partial y} + \frac{\partial B_z}{\partial t} = 0, \quad (8.132)$$

$$\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} - \frac{\partial D_x}{\partial t} = 0, \quad (8.133)$$

$$\frac{\partial H_x}{\partial z} - \frac{\partial D_y}{\partial t} = 0, \quad (8.134)$$

$$- \frac{\partial H_x}{\partial y} - \frac{\partial D_z}{\partial t} = 0. \quad (8.135)$$

Of the above equations, we collect those pertinent to E_x and differentiate (8.131), (8.132), and (8.133) with respect to z , y , and t , respectively, to get

$$\frac{\partial^2 E_x}{\partial z^2} + \frac{\partial^2 B_y}{\partial z \partial t} = 0, \quad (8.136)$$

$$\frac{\partial^2 E_x}{\partial y^2} - \frac{\partial^2 B_z}{\partial y \partial t} = 0, \quad (8.137)$$

$$\frac{\partial^2 H_z}{\partial t \partial y} - \frac{\partial^2 H_y}{\partial t \partial z} - \frac{\partial^2 D_x}{\partial t^2} = 0. \quad (8.138)$$

Multiplying (8.138) by μ and further adding (8.136) and (8.137) to it and using (7.7), we get

$$\frac{\partial^2 E_x}{\partial y^2} + \frac{\partial^2 E_x}{\partial z^2} = \mu \varepsilon \frac{\partial^2 E_x}{\partial t^2}. \quad (8.139)$$

This is a two-dimensional equation of wave motion. In a similar manner, from (8.130), (8.134), and (8.135), we have for the magnetic field

$$\frac{\partial^2 H_x}{\partial y^2} + \frac{\partial^2 H_x}{\partial z^2} = \mu \varepsilon \frac{\partial^2 H_x}{\partial t^2}. \quad (8.140)$$

Equations (8.139) and (8.140) are two-dimensional wave equations with respect to the y - and z -coordinates. With the direction of the x -axis, a propagating wave has the same phase. Suppose that we have plane wave solutions for them as in the case of (7.58) and (7.59). Then, we have

$$\begin{aligned} \mathbf{E} &= \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} = \mathbf{E}_0 e^{i(k_n x - \omega t)}, \\ \mathbf{H} &= \mathbf{H}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} = \mathbf{H}_0 e^{i(k_n x - \omega t)}. \end{aligned} \quad (8.141)$$

Note that a plane wave expressed by (7.58) and (7.59) is propagated uniformly in a dielectric medium. In a waveguide, on the other hand, the electromagnetic waves undergo repeated (total) reflections from the two boundaries positioned either side of the waveguide while being propagated.

In a three-dimensional version, the wavenumber vector has three components k_x , k_y , and k_z as expressed in (7.48). In (8.141), in turn, $\mathbf{k}$ has y - and z -components such that

$$\mathbf{k}^2 = k^2 = k_y^2 + k_z^2. \quad (8.142)$$

Equations (8.139) and (8.140) can be rewritten as

$$\frac{\partial^2 E_x}{\partial (\pm y)^2} + \frac{\partial^2 E_x}{\partial (\pm z)^2} = \mu \varepsilon \frac{\partial^2 E_x}{\partial (\pm t)^2}, \quad \frac{\partial^2 H_x}{\partial (\pm y)^2} + \frac{\partial^2 H_x}{\partial (\pm z)^2} = \mu \varepsilon \frac{\partial^2 H_x}{\partial (\pm t)^2}.$$

Accordingly, we have four wavenumber vector components

Fig. 8.11 Four possible propagation directions $\mathbf{k}$ of an electromagnetic wave in a slab waveguide

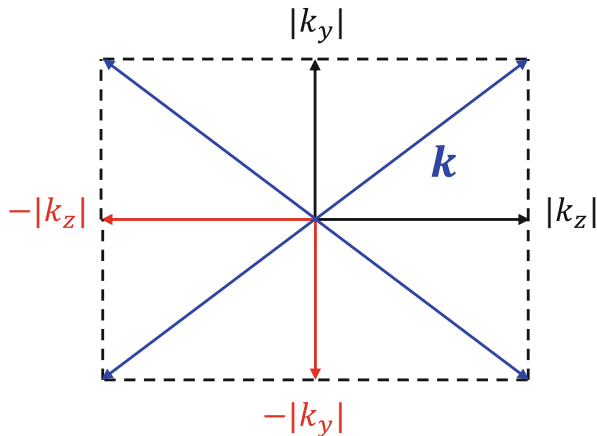
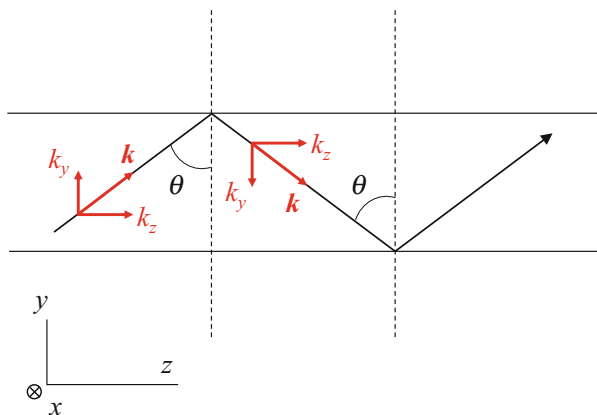


Fig. 8.12 Geometries and propagation of the electromagnetic waves in a slab waveguide



$$k_y = \pm |k_y| \quad \text{and} \quad k_z = \pm |k_z|.$$

Figure 8.11 indicates this situation where an electromagnetic wave can be propagated within a slab waveguide in either one direction out of four choices of $\mathbf{k}$. In this section, we assume that the electromagnetic wave is propagated toward the positive direction of the z -axis, and so we define k_z as positive. On the other hand, k_y can be either positive or negative. Thus, we get

$$k_z = k \sin \theta \quad \text{and} \quad k_y = \pm k \cos \theta. \quad (8.143)$$

Figure 8.12 shows the geometries of the electromagnetic waves within the slab waveguide. The slab plane is parallel to the zx -plane. Let the positions of the two interfaces of the slab waveguide be

$$y=0 \quad \text{and} \quad y=d. \quad (8.144)$$

That is, we assume that the thickness of the waveguide is d .

Since (8.139) describes a wave equation for only one component E_x , (8.139) is suited for representing a TE wave. In (8.140), in turn, a wave equation is given only for H_x , and hence, it is suited for representing a TM wave. With the TE wave, the electric field oscillates parallel to the slab plane and vertical to the propagation direction. With the TM wave, in turn, the magnetic field oscillates parallel to the slab plane and vertical to the propagation direction. In a general case, electromagnetic waves in a slab waveguide are formed by superposition of TE and TM waves. Notice that Fig. 8.12 is applicable to both TE and TM waves.

Let us further proceed with the waveguide analysis. The electric field $\mathbf{E}$ within the waveguide is described by superposition of the incident and reflected waves. Using the first equation of (8.141) and (8.143), we have

$$\mathbf{E}(z, y) = \epsilon_e E_+ e^{i(kz \sin \theta + ky \cos \theta - \omega t)} + \epsilon_e' E_- e^{i(kz \sin \theta - ky \cos \theta - \omega t)}, \quad (8.145)$$

where E_+ (E_-) and ϵ_e (ϵ_e') represent an amplitude and unit polarization vector of the incident (reflected) waves, respectively. The vector ϵ_e (or ϵ_e') is defined in (7.67).

Equation (8.145) is common to both the cases of TE and TM waves. From now, we consider the TE mode case. Suppose that the slab waveguide is sandwiched with a couple of metal sheet of high conductance. Since the electric field must be absent inside the metal, the electric field at the interface must be zero owing to the continuity condition of a tangential component of the electric field. Thus, we require the following condition should be met with (8.145):

$$\begin{aligned} \mathbf{t} \cdot \mathbf{E}(z, 0) = 0 &= \mathbf{t} \cdot \epsilon_e E_+ e^{i(kz \sin \theta - \omega t)} + \mathbf{t} \cdot \epsilon_e' E_- e^{i(kz \sin \theta - \omega t)} \\ &= (\mathbf{t} \cdot \epsilon_e E_+ + \mathbf{t} \cdot \epsilon_e' E_-) e^{i(kz \sin \theta - \omega t)}. \end{aligned} \quad (8.146)$$

Therefore, since $e^{i(kz \sin \theta - \omega t)}$ never vanishes, we have

$$\mathbf{t} \cdot \epsilon_e E_+ + \mathbf{t} \cdot \epsilon_e' E_- = 0, \quad (8.147)$$

where $\mathbf{t}$ is a tangential unit vector at the interface.

Since $\mathbf{E}$ is polarized along the x -axis, setting $\epsilon_e = \epsilon_e' = \mathbf{e}_1$ and taking $\mathbf{t}$ as $\mathbf{e}_1$, we get

$$E_+ + E_- = 0.$$

This means that the reflection coefficient of the electric field is -1 . Denoting $E_+ = -E_- \equiv E_0$ (>0), we have

$$\begin{aligned}
\mathbf{E} &= \mathbf{e}_1 E_0 \left[e^{i(kz \sin \theta + ky \cos \theta - \omega t)} - e^{i(kz \sin \theta - ky \cos \theta - \omega t)} \right] \\
&= \mathbf{e}_1 E_0 \left[(e^{iky \cos \theta} - e^{-iky \cos \theta}) e^{i(kz \sin \theta - \omega t)} \right] \\
&= \mathbf{e}_1 2i E_0 \sin(ky \cos \theta) e^{i(kz \sin \theta - \omega t)}. \tag{8.148}
\end{aligned}$$

Requiring the electric field to vanish at another interface of $y = d$, we have

$$\mathbf{E}(z, d) = \mathbf{0} = \mathbf{e}_1 2i E_0 \sin(kd \cos \theta) e^{i(kz \sin \theta - \omega t)}.$$

Note that in terms of the boundary conditions, we are thinking of Dirichlet conditions (see Sects. 1.3 and 10.3). In this case, we have nodes for the electric field at the interface between metal and a dielectric. For this condition to be satisfied, we must have

$$kd \cos \theta = l\pi \quad (l = 1, 2, \dots). \tag{8.149}$$

From (8.149), we have a following condition for l :

$$l \leq kd/\pi. \tag{8.150}$$

Meanwhile, we have

$$k = nk_0, \tag{8.151}$$

where n is a refractive index of a dielectric that shapes the slab waveguide and the quantity k_0 is a wavenumber of the electromagnetic wave in vacuum. The index n is given by

$$n = c/v, \tag{8.152}$$

where c and v are light velocity in vacuum and the dielectric media, respectively. Here v is meant as a velocity in an infinitely spreading dielectric. Thus, θ is allowed to take several (or more) numbers depending upon k , d , and l .

Since in the z -direction no specific boundary conditions are imposed, we have propagating modes in that direction characterized by a propagation constant (*vide infra*). Looking at (8.148), we notice that $k \sin \theta$ plays a role of a wavenumber in a free space. For this reason, a quantity β defined as

$$\beta = k \sin \theta = nk_0 \sin \theta \quad (\beta > 0) \quad (8.153)$$

is said to be a propagation constant. In (8.153), k_0 is a wavenumber in vacuum. From (8.149) and (8.153), we get

$$\beta = \left(k^2 - \frac{l^2 \pi^2}{d^2} \right)^{1/2}. \quad (8.154)$$

Thus, the allowed TE waves indexed by l are called TE *modes* and represented as TE_l . The phase velocity v_p is given by

$$v_p = \omega / \beta. \quad (8.155)$$

Meanwhile, the group velocity v_g is given by

$$v_g = \frac{d\omega}{d\beta} = \left(\frac{d\beta}{d\omega} \right)^{-1}. \quad (8.156)$$

Using (8.154) and noting that $k^2 = \omega^2/v^2$, we get

$$v_g = v^2 \beta / \omega. \quad (8.157)$$

Thus, we have

$$v_p v_g = v^2. \quad (8.158)$$

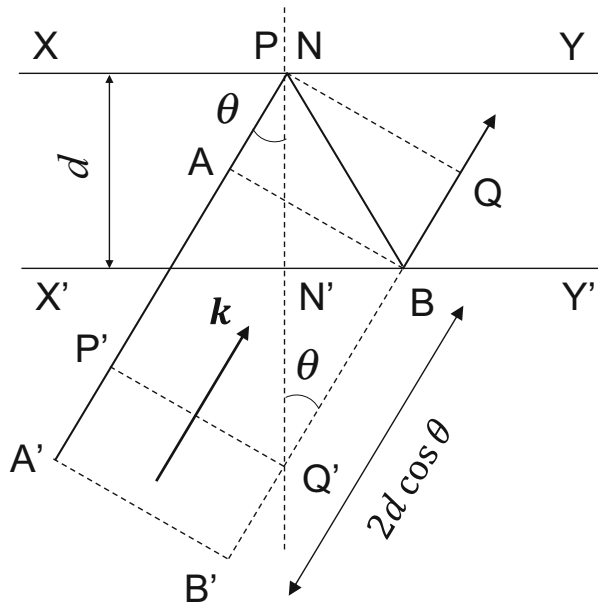
Note that in (1.22) of Sect 1.1, we saw a relationship similar to (8.158).

The characteristics of TM waves can be analyzed in a similar manner by examining the magnetic field H_x . In that case, the reflection coefficient of the magnetic field is +1 and we have antinodes for the magnetic field at the interface. Concomitantly, we adopt Neumann conditions as the boundary conditions (see Sects. 1.3 and 8.3). Regardless of the difference in the boundary conditions, however, discussion including (8.149)–(8.158) applies to the analysis of TM waves. Once H_x is determined, E_y and E_z can be determined as well from (8.134) and (8.135).

8.7.2 Total Internal Reflection and Evanescent Waves

If a slab waveguide shaped by a dielectric is sandwiched by a couple of another dielectric (Fig. 8.10b), the situation differs from a metal waveguide (Fig. 8.10a) we encountered in Sect. 8.7.1. Suppose in Fig. 8.10b that the former dielectric D1 of a refractive index n_1 is sandwiched by the latter dielectric D2 of a refractive index n_2 .

Fig. 8.13 Cross-section of the waveguide where the light is propagated in the direction of $\mathbf{k}$. A dielectric fills a semi-infinite space situated below NXY. We suppose another virtual plane N'X'Y' that is parallel with NXY. We need the plane N'X'Y' to estimate an optical path difference (or phase difference)



Suppose that an electromagnetic wave is being propagated from D1 toward D2. Then, we must have

$$n_1 > n_2 \quad (8.159)$$

so that the total internal reflection can take place at the interface of D1 and D2. In this case, the dielectrics D1 and D2 act as a core layer and a clad layer, respectively.

The biggest difference between the present waveguide and the previous one is that unlike the previous case, the total internal reflection occurs in the present case. Concomitantly, an evanescent wave is present in the clad layer very close to the interface.

First, let us estimate the conditions that are satisfied so that an electromagnetic wave can be propagated within a waveguide. Figure 8.13 depicts a cross-section of the waveguide where the light is propagated in the direction of $\mathbf{k}$. In Fig. 8.13, suppose that we have a normal $\mathbf{N}$ to the plane of paper at P . Then, $\mathbf{N}$ and a straight line XY shape a plane NXY . Also suppose that a dielectric fills a semi-infinite space situated below NXY . Further suppose that there is another virtual plane $N'X'Y'$ that is parallel with NXY as shown. Here $\mathbf{N}'$ is parallel to $\mathbf{N}$. The separation of the two parallel planes is d . We need the virtual plane $N'X'Y'$ just to estimate an optical path difference (or phase difference, more specifically) between two waves, i.e., a propagating wave and a reflected wave.

Let $\mathbf{n}$ be a unit vector in the direction of $\mathbf{k}$, i.e.,

$$\mathbf{n} = \mathbf{k} / |\mathbf{k}| = \mathbf{k} / k. \quad (8.160)$$

Then the electromagnetic wave is described as

$$\mathbf{E} = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} = \mathbf{E}_0 e^{i(k\mathbf{n} \cdot \mathbf{x} - \omega t)}. \quad (8.161)$$

Suppose that we take a coordinate system such that

$$\mathbf{x} = r\mathbf{n} + s\mathbf{u} + t\mathbf{v}, \quad (8.162)$$

where $\mathbf{u}$ and $\mathbf{v}$ represent unit vectors in the direction perpendicular to $\mathbf{n}$. Then (8.161) can be expressed by

$$\mathbf{E} = \mathbf{E}_0 e^{i(kr - \omega t)}. \quad (8.163)$$

Suppose that the wave is propagated starting from a point A to P and reflected at P. Then, the wave is further propagated to B and reflected again to reach Q. The wave front is originally at AB and finally at PQ. Thus, the Z-shaped optical path length APBQ is equal to a separation between A'B' and PQ. Notice that the separation between A'B' and P'Q' is taken so that it is equal to that between AB and PQ. The geometry of Fig. 8.13 implies that two waves starting from AB and A'B' at once reach PQ again at once.

We find the separation between AB and A'B' is

$$2d \cos \theta.$$

Let us tentatively call these waves Wave-AB and Wave-A'B' and describe their electric fields as $\mathbf{E}_{AB}$ and $\mathbf{E}_{A'B'}$, respectively. Then, we denote

$$\mathbf{E}_{AB} = \mathbf{E}_0 e^{i(kr - \omega t)}, \quad (8.164)$$

$$\mathbf{E}_{A'B'} = \mathbf{E}_0 e^{i[k(r+2d \cos \theta) - \omega t]}, \quad (8.165)$$

where k is a wavenumber in the dielectric. Note that since $\mathbf{E}_{A'B'}$ gets behind $\mathbf{E}_{AB}$, a plus sign appears in the first term of the exponent. Therefore, the phase difference between the two waves is

$$2kd \cos \theta. \quad (8.166)$$

Now, let us come back to the actual geometry of the waveguide. That is, the core layer of thickness d is sandwiched by a couple of clad layers (Fig. 8.10b). In this situation, the wave $\mathbf{E}_{AB}$ experiences the total internal reflection two times, which we ignored in the above discussion of the metal waveguide. Since the total internal

reflection causes a complex phase shift, we have to take account of this effect. The phase shift was defined as α of (8.114) for a TE mode and β for a TM mode. Notice that in Fig. 8.13 the electric field oscillates perpendicularly to the plane of paper with the TE mode, whereas it oscillates in parallel with the plane of paper with the TM mode. For both cases the electric field oscillates perpendicularly to $\mathbf{n}$. Consequently, the phase shift due to these reflections has to be added to (8.166). Thus, for the phase commensuration to be obtained, the following condition must be satisfied:

$$2kd \cos \theta + 2\delta = 2l\pi \quad (l = 0, 1, 2, \dots), \quad (8.167)$$

where δ is either δ_{TE} or δ_{TM} defined below according to the case of the TE wave and TM wave, respectively. For a practical purpose, (8.167) is dealt with by a numerical calculation, e.g., to design an optical waveguide.

Unlike (8.149), what is the most important with (8.167) is that the condition $l = 0$ is permitted because of $\delta < 0$ (see just below).

For convenience and according to the custom, we adopt a phase shift notation other than that defined in (8.114). With the TE mode, the phase is retained upon reflection at the critical angle, and so we identify α with an additional component δ_{TE} . In the TM case, on the other hand, the phase is reversed upon reflection at the critical angle (i.e., a π shift occurs). Since this π shift has been incorporated into β , it suffices to consider only an additional component δ_{TM} . That is, we have

$$\delta_{\text{TE}} \equiv \alpha \quad \text{and} \quad \delta_{\text{TM}} \equiv \beta - \pi. \quad (8.168)$$

We rewrite (8.110) as

$$R_E^\perp = e^{i\alpha} = e^{i\delta_{\text{TE}}} \equiv \frac{ae^{-i\sigma}}{ae^{i\sigma}} = e^{-2i\sigma}$$

and

$$\delta_{\text{TE}} = -2\sigma \quad (\sigma > 0), \quad (8.169)$$

where we have

$$ae^{i\sigma} = \cos \theta + i\sqrt{\sin^2 \theta - n^2}. \quad (8.170)$$

Therefore,

$$\tan \sigma = -\tan \frac{\delta_{\text{TE}}}{2} = \frac{\sqrt{\sin^2 \theta - n^2}}{\cos \theta}. \quad (8.171)$$

Meanwhile, rewriting (8.112) we have

$$R_E^{\parallel} = e^{i\beta} = e^{i(\delta_{\text{TM}} + \pi)} \equiv -\frac{be^{-i\tau}}{be^{i\tau}} = -e^{-2i\tau} = e^{i(-2\tau + \pi)}, \quad (8.172)$$

where

$$be^{i\tau} = n^2 \cos \theta + i\sqrt{\sin^2 \theta - n^2}. \quad (8.173)$$

Note that the minus sign with the second last equality in (8.172) is due to the phase reversal upon reflection. From (8.172), we may put

$$\beta = -2\tau + \pi.$$

Comparing this with the second equation of (8.168), we get

$$\delta_{\text{TM}} = -2\tau \quad (\tau > 0). \quad (8.174)$$

Consequently, we get

$$\tan \tau = -\tan \frac{\delta_{\text{TM}}}{2} = \frac{\sqrt{\sin^2 \theta - n^2}}{n^2 \cos \theta}. \quad (8.175)$$

Finally, the additional phase change δ_{TE} and δ_{TM} upon the total reflection is given by [4].

$$\delta_{\text{TE}} = -2 \tan^{-1} \frac{\sqrt{\sin^2 \theta - n^2}}{\cos \theta} \quad \text{and} \quad \delta_{\text{TM}} = -2 \tan^{-1} \frac{\sqrt{\sin^2 \theta - n^2}}{n^2 \cos \theta}. \quad (8.176)$$

We emphasize that in (8.176) both δ_{TE} and δ_{TM} are negative quantities. This phase shift has to be included in (8.167) as a negative quantity δ .

At a first glance, (8.176) seems to differ largely from (8.120) and (8.125). Nevertheless, noting that a trigonometric formula

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

and remembering that δ_{TM} in (8.168) includes π arising from the phase reversal, we find that both the relations are virtually identical.

Evanescent waves are drawing a large attention in the field of basic physics and applied device physics. If the total internal reflection is absent, ϕ is real. But, under the total internal reflection, ϕ is pure imaginary. The electric field of the evanescent wave is described as

$$\begin{aligned}
 \mathbf{E}_t &= E \mathbf{e}_t e^{i(k_t z \sin \phi + k_t y \cos \phi - \omega t)} = E \mathbf{e}_t e^{i(k_t z \sin \phi + k_t y b - \omega t)} \\
 &= E \mathbf{e}_t e^{i(k_t z \sin \phi - \omega t)} e^{-k_t y b}.
 \end{aligned} \tag{8.177}$$

In (8.177), a unit polarization vector $\mathbf{e}_t$ is either perpendicular to the plane of paper of Fig. 8.13 (the TE case) or in parallel to it (the TM case). Notice that the coordinate system is different from that of (8.104). The quantity $k_t \sin \phi$ is the propagation constant. Let $v_p^{(s)}$ and $v_p^{(e)}$ be a phase velocity of the electromagnetic wave in the slab waveguide (i.e., core layer) and evanescent wave in the clad layer, respectively. Then, in virtue of Snell's law, we have

$$v_1 < v_p^{(s)} = \frac{v_1}{\sin \theta} = \frac{\omega}{k \sin \theta} = \frac{\omega}{k_t \sin \phi} = v_p^{(e)} = \frac{v_2}{\sin \phi} < v_2, \tag{8.178}$$

where v_1 and v_2 are light velocity in a free space filled by the dielectrics D1 and D2, respectively. For this, we used a relation described as

$$\omega = v_1 k = v_2 k_t. \tag{8.179}$$

We also used Snell's law with the third equality of (8.178). Notice that $\sin \phi > 1$ in the evanescent region and that $k_t \sin \phi$ is a propagation constant in the clad layer. Also note that $v_p^{(s)}$ is equal to $v_p^{(e)}$ and that these phase velocities are in between the two velocities of the free space. Thus, the evanescent waves must be present, accompanying propagating waves that undergo the total internal reflections in a slab waveguide.

As remarked in (8.105), the electric field of evanescent waves decays exponentially with increasing z . This implies that the evanescent waves exist only in the clad layer very close to an interface of core and clad layers.

8.8 Stationary Waves

So far, we have been dealing with propagating waves either in a free space or in a waveguide. If the dielectric shaping the waveguide is confined in another direction, the propagating waves show specific properties. Examples include optical fibers.

In this section we consider a situation where the electromagnetic wave is propagating in a dielectric medium and reflected by a "wall" formed by metal or another dielectric. In such a situation, the original wave (i.e., a forward wave) causes interference with the backward wave, and a stationary wave is formed as a consequence of the interference.

To approach this issue, we deal with superposition of two waves that have different phases and different amplitudes. To generalize the problem, let ψ_1 and ψ_2 be two cosine functions described as

$$\psi_1 = a_1 \cos b_1 \text{ and } \psi_2 = a_2 \cos b_2. \quad (8.180)$$

Their addition is expressed as

$$\psi = \psi_1 + \psi_2 = a_1 \cos b_1 + a_2 \cos b_2. \quad (8.181)$$

Here we wish to unify (8.181) as a single cosine (or sine) function. To this end, we modify a description of ψ_2 such that

$$\begin{aligned} \psi_2 &= a_2 \cos[b_1 + (b_2 - b_1)] \\ &= a_2 [\cos b_1 \cos(b_2 - b_1) - \sin b_1 \sin(b_2 - b_1)] \\ &= a_2 [\cos b_1 \cos(b_1 - b_2) + \sin b_1 \sin(b_1 - b_2)]. \end{aligned} \quad (8.182)$$

Then, the addition is described by

$$\begin{aligned} \psi &= \psi_1 + \psi_2 \\ &= [a_1 + a_2 \cos(b_1 - b_2)] \cos b_1 + a_2 \sin(b_1 - b_2) \sin b_1. \end{aligned} \quad (8.183)$$

Putting R such that

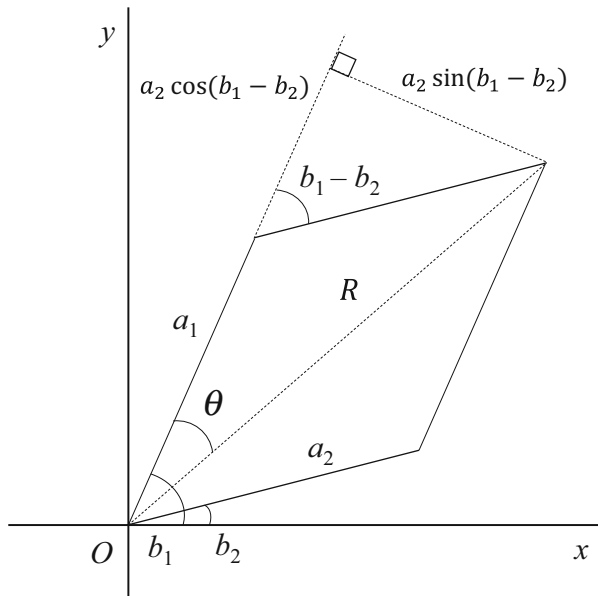
$$\begin{aligned} R &= \sqrt{[a_1 + a_2 \cos(b_1 - b_2)]^2 + a_2^2 \sin^2(b_1 - b_2)} \\ &= \sqrt{a_1^2 + a_2^2 + 2a_1 a_2 \cos(b_1 - b_2)}, \end{aligned} \quad (8.184)$$

we get

$$\psi = R \cos(b_1 - \theta), \quad (8.185)$$

where θ is expressed by

Fig. 8.14 Geometric diagram in relation to the superposition of two waves having different amplitudes (a_1 and a_2) and different phases (b_1 and b_2)



$$\tan \theta = \frac{a_2 \sin(b_1 - b_2)}{a_1 + a_2 \cos(b_1 - b_2)}. \quad (8.186)$$

Figure 8.14 represents a geometric diagram in relation to the superposition of two waves having different amplitudes (a_1 and a_2) and different phases (b_1 and b_2) [5].

To apply (8.185) to the superposition of two electromagnetic waves that are propagating forward and backward to collide head-on with each other, we change the variables such that

$$b_1 = kx - \omega t \quad \text{and} \quad b_2 = -kx - \omega t, \quad (8.187)$$

where the former equation represents a forward wave, whereas the latter a backward wave. Then we have

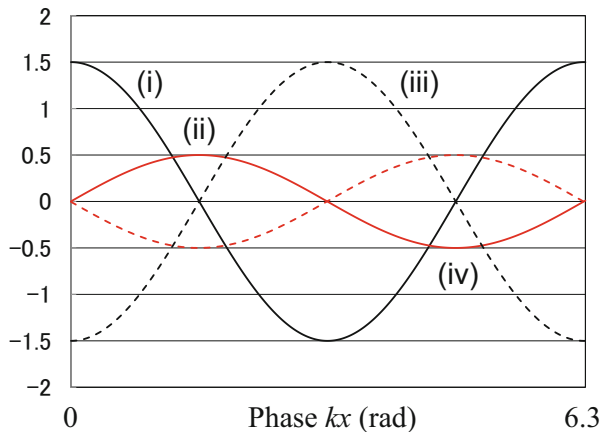
$$b_1 - b_2 = 2kx. \quad (8.188)$$

Equation (8.185) is rewritten as

$$\psi(x, t) = R \cos(kx - \omega t - \theta), \quad (8.189)$$

with

Fig. 8.15 Superposition of two sinusoidal waves. In (8.189) and (8.190), we put $a_1 = 1$, $a_2 = 0.5$, with (i) $t = 0$; (ii) $t = T/4$; (iii) $t = T/2$; (iv) $t = 3T/4$. $\psi(x, t)$ is plotted as a function of phase kx



$$R = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos 2kx} \quad (8.190)$$

and

$$\tan \theta = \frac{a_2 \sin 2kx}{a_1 + a_2 \cos 2kx}. \quad (8.191)$$

Equation (8.189) looks simple, but since both R and θ vary as a function of x , the situation is somewhat complicated unlike a simple sinusoidal wave. Nonetheless, when x takes a special value, (8.189) is expressed by a simple function form. For example, at $t = 0$,

$$\psi(x, 0) = (a_1 + a_2) \cos kx.$$

This corresponds to (i) of Fig. 8.15. If $t = T/2$ (where T is a period, i.e., $T = 2\pi/\omega$), we have

$$\psi\left(x, \frac{T}{2}\right) = -(a_1 + a_2) \cos kx.$$

This corresponds to (iii) of Fig. 8.15. But, the waves described by (ii) or (iv) do not have a simple function form.

We characterize Fig. 8.15 below. If we have

$$2kx = n\pi \quad \text{or} \quad x = n\lambda/4 \quad (8.192)$$

with λ being a wavelength, then $\theta = 0$ or π , and so θ can be eliminated. This situation occurs with every quarter period of a wavelength. Let us put $t = 0$ and examine how the superposed wave looks like. For instance, putting $x = 0$, $x = \lambda/4$, and $x = \lambda/2$, we have

$$\psi(0,0) = |a_1 + a_2|, \psi(\lambda/4,0) = \psi(3\lambda/4,0) = 0, \psi(\lambda/2,0) = -|a_1 + a_2|, \quad (8.193)$$

respectively. Notice that in Fig. 8.15 we took $a_1, a_2 > 0$. At another instant $t = T/4$, we have similarly

$$\begin{aligned} \psi(0, T/4) &= 0, \psi(\lambda/4, T/4) = |a_1 - a_2|, \psi(\lambda/2, T/4) = 0, \\ \psi(3\lambda/4, T/4) &= -|a_1 - a_2|. \end{aligned} \quad (8.194)$$

Thus, the waves that vary with time are characterized by two dram-shaped envelopes that have extremals $|a_1 + a_2|$ and $|a_1 - a_2|$ or those $-|a_1 + a_2|$ and $-|a_1 - a_2|$. An important implication of Fig. 8.15 is that no node is present in the superposed wave. In other words, there is no instant t_0 when $\psi(x, t_0) = 0$ for any x . From the aspect of energy transport of electromagnetic waves, if $|a_1| > |a_2|$ (where a_1 and a_2 represent an amplitude of the forward and backward waves, respectively), the net energy flow takes place in the travelling direction of the forward wave. If, on the other hand, $|a_1| < |a_2|$, the net energy flow takes place in the travelling direction of the backward wave. In this respect, think of Poynting vectors.

In case $|a_1| = |a_2|$, the situation is particularly simple. No net energy flow takes place in this case. Correspondingly, we observe nodes. Such waves are called stationary waves. Let us consider this simple situation for an electromagnetic wave that is incident perpendicularly to the interface between two dielectrics (one of them may be a metal).

Returning back to (7.58) and (7.66), we describe two electromagnetic waves that are propagating in the positive and negative direction of the z -axis such that

$$\mathbf{E}_1 = E_1 \mathbf{e}_e e^{i(kz - \omega t)} \quad \text{and} \quad \mathbf{E}_2 = E_2 \mathbf{e}_e e^{i(-kz - \omega t)}, \quad (8.195)$$

where $\mathbf{e}_e$ is a unit polarization vector arbitrarily fixed so that it can be parallel to the interface, i.e., wall (i.e., perpendicular to the z -axis). The situation is depicted in Fig. 8.16. Notice that in Fig. 8.16 $\mathbf{E}_1$ represents the forward wave (i.e., incident wave) and $\mathbf{E}_2$ the backward wave (i.e., wave reflected at the interface). Thus, a superposed wave is described as

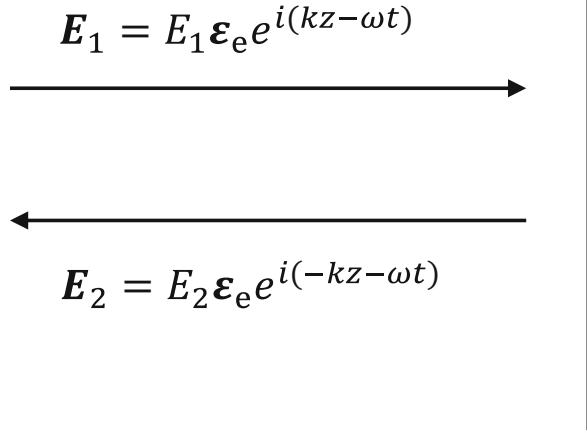
$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2. \quad (8.196)$$

Taking account of the reflection of an electromagnetic wave perpendicularly incident on a wall, let us consider following two cases:

(i) Syn-phase:

The phase of the electric field is retained upon reflection. We assume that $E_1 = E_2 (>0)$. Then, we have

Fig. 8.16 Superposition of electric fields of forward (or incident) wave E_1 and backward (or reflected) wave E_2



$$\begin{aligned}
 E &= E_1 \epsilon_e \left[e^{i(kz - \omega t)} + e^{i(-kz - \omega t)} \right] = E_1 \epsilon_e e^{-i\omega t} (e^{ikz} + e^{-ikz}) \\
 &= 2E_1 \epsilon_e e^{-i\omega t} \cos kz.
 \end{aligned} \tag{8.197}$$

In (8.197), we put $z = 0$ at the interface for convenience. Taking a real part of (8.197), we have

$$E = 2E_1 \epsilon_e \cos \omega t \cos kz. \tag{8.198}$$

Note that in (8.198) variables z and t have been separated. This implies that we have a stationary wave. For this case to be realized, the characteristic impedance of the dielectric of the incident wave side should be smaller enough than that of the other side; see (8.51) and (8.59). In other words, the dielectric constant of the incident side should be large enough. We have nodes at positions that satisfy

$$-kz = \frac{\pi}{2} + m\pi \quad (m=0, 1, 2, \dots) \quad \text{or} \quad -z = \frac{1}{4}\lambda + \frac{m}{2}\lambda. \tag{8.199}$$

Note that we are thinking of the stationary wave in the region of $z < 0$. Equation (8.199) indicates that nodes are formed at a quarter wavelength from the interface and every half wavelength from it. The node means the position where no electric field is present.

Meanwhile, antinodes are observed at positions

$$-kz = m\pi \quad (m = 0, 1, 2, \dots) \quad \text{or} \quad -z = +\frac{m}{2}\lambda.$$

Thus, the nodes and antinodes alternate with every quarter wavelength.

(ii) Antiphase:

The phase of the electric field is reversed. We assume that $E_1 = -E_2 (>0)$. Then, we have

$$\begin{aligned} \mathbf{E} &= E_1 \mathbf{e}_e \left[e^{i(kz - \omega t)} - e^{i(-kz - \omega t)} \right] = E_1 \mathbf{e}_e e^{-i\omega t} (e^{ikz} - e^{-ikz}) \\ &= 2iE_1 \mathbf{e}_e e^{-i\omega t} \sin kz. \end{aligned} \quad (8.200)$$

Taking a real part of (8.197), we have

$$\mathbf{E} = 2E_1 \mathbf{e}_e \sin \omega t \sin kz. \quad (8.201)$$

In (8.201) variables z and t have been separated as well. For this case to be realized, the characteristic impedance of the dielectric of the incident wave side should be larger enough than that of the other side. In other words, the dielectric constant of the incident side should be small enough. Practically, this situation can easily be attained choosing a metal of high reflectance for the wall material. We have nodes at positions that satisfy

$$-kz = m\pi \quad (m = 0, 1, 2, \dots) \quad \text{or} \quad -z = \frac{m}{2}\lambda. \quad (8.202)$$

The nodes are formed at the interface and every half wavelength from it. As in the case of the syn-phase, the antinodes take place with the positions shifted by a quarter wavelength relative to the nodes.

If there is another interface at say $-z = L (>0)$, the wave goes back and forth many times. If an absolute value of the reflection coefficient at the interface is high enough (i.e., close to the unity), attenuation of the wave is ignorable. For both the syn-phase and antiphase cases, we must have

$$kL = m\pi \quad (m = 1, 2, \dots) \quad \text{or} \quad L = \frac{m}{2}\lambda \quad (8.203)$$

so that stationary waves can stand stable. The stationary waves indexed with a positive integer m in (8.203) are said to be longitudinal modes. Note that the index l that features the transverse modes in the waveguide can be zero or a positive integer. Unlike the previous case of (8.167) in Sect. 8.7, the index m of (8.203) is not allowed to be zero.

If L in (8.203) is large enough compared to λ , m is a pretty large number as well, and what is more, many different numbers of m are allowed between given

interfaces. In other words, stationary waves of many different wavelengths are allowed to be present at once between the given interfaces. In such a case, the stationary waves are referred to as a longitudinal multimode.

For a practical purpose, an optical device having interfaces that make the stationary waves stabilize is said to be a *resonator*. Various geometries and constitutions of the resonator are proposed in combination with various dielectrics including semiconductors. Related discussion can be seen in Chap. 9.

References

1. Arfken GB, Weber HJ, Harris FE (2013) Mathematical methods for physicists, 7th edn. Academic Press, Waltham
2. Rudin W (1987) Real and complex analysis, 3rd edn. McGraw-Hill, New York
3. Smith FG, King TA, Wilkins D (2007) Optics and photonics, 2nd edn. Wiley, Chichester
4. Born M, Wolf E (2005) Principles of optics, 7th edn. Cambridge University Press, Cambridge
5. Pain HJ (2005) The physics of vibrations and waves, 6th edn. Wiley, Chichester

Chapter 9

Light Quanta: Radiation and Absorption



So far we discussed propagation of light and its reflection and transmission (or refraction) at an interface of dielectric media. We described characteristics of light from the point of view of an electromagnetic wave. In this chapter, we describe properties of light in relation to quantum mechanics. To this end, we start with Planck's law of radiation that successfully reproduced experimental results related to a blackbody radiation. Before this law had been established, Rayleigh–Jeans law failed to explain the experimental results at a high frequency region of radiation (the ultraviolet catastrophe). The Planck's law of radiation led to the discovery of light quanta. Einstein interpreted Planck's law of radiation on the basis of a model of two-level atoms. This model includes the so-called Einstein A and B coefficients that are important in optics applications, especially lasers. We derive these coefficients from a classical point of view based on a dipole oscillation. We also consider a close relationship between electromagnetic waves confined in a cavity and a motion of a harmonic oscillator.

9.1 Blackbody Radiation

Historically, the relevant theory was first propounded by Max Planck and then Albert Einstein as briefly discussed in Chap. 1. The theory was developed on the basis of the experiments called cavity radiation or blackbody radiation. Here, however, we wish to derive Planck's law of radiation on the assumption of the existence of quantum harmonic oscillators.

As discussed in Chap. 2, the ground state of a quantum harmonic oscillator has an energy $\hbar\omega/2$. Therefore, we measure energies of the oscillator in reference to that state. Let N_0 be the number of oscillators (i.e., light quanta) present in the ground state. Then, according to Boltzmann distribution law the number of oscillators of the first excited state N_1 is

$$N_1 = N_0 e^{-\hbar\omega/k_B T}, \quad (9.1)$$

where k_B is Boltzmann constant and T is absolute temperature. Let N_j be the number of oscillators of the j -th excited state. Then we have

$$N_j = N_0 e^{-j\hbar\omega/k_B T}. \quad (9.2)$$

Let N be the total number of oscillators in the system. Then, we get

$$\begin{aligned} N &= N_0 + N_0 e^{-\hbar\omega/k_B T} + \cdots + N_0 e^{-j\hbar\omega/k_B T} + \cdots \\ &= N_0 \sum_{j=0}^{\infty} e^{-j\hbar\omega/k_B T}. \end{aligned} \quad (9.3)$$

Let E be a total energy of the oscillator system in reference to the ground state. That is, we put a ground state energy at zero. Then we have

$$\begin{aligned} E &= 0 \cdot N_0 + N_0 \hbar\omega e^{-\hbar\omega/k_B T} + \cdots + N_0 j \hbar\omega e^{-j\hbar\omega/k_B T} + \cdots \\ &= N_0 \sum_{j=0}^{\infty} j \hbar\omega e^{-j\hbar\omega/k_B T}. \end{aligned} \quad (9.4)$$

Therefore, an average energy of oscillators $\bar{E}$ is given by

$$\bar{E} = \frac{E}{N} = \hbar\omega \frac{\sum_{j=0}^{\infty} j e^{-j\hbar\omega/k_B T}}{\sum_{j=0}^{\infty} e^{-j\hbar\omega/k_B T}}. \quad (9.5)$$

Putting $x \equiv e^{-\hbar\omega/k_B T}$ [1], we have

$$\bar{E} = \hbar\omega \frac{\sum_{j=0}^{\infty} j x^j}{\sum_{j=0}^{\infty} x^j}. \quad (9.6)$$

Since $x < 1$, we have

$$\begin{aligned} \sum_{j=0}^{\infty} j x^j &= \sum_{j=0}^{\infty} j x^{j-1} \cdot x = \left[\frac{d}{dx} \left(\sum_{j=0}^{\infty} x^j \right) \right] x = \left[\frac{d}{dx} \left(\frac{1}{1-x} \right) \right] x \\ &= \frac{x}{(1-x)^2}, \end{aligned} \quad (9.7)$$

$$\sum_{j=0}^{\infty} x^j = \frac{1}{1-x}. \quad (9.8)$$

Then, we get

$$\bar{E} = \frac{\hbar\omega x}{1-x} = \frac{\hbar\omega e^{-\hbar\omega/k_B T}}{1-e^{-\hbar\omega/k_B T}} = \frac{\hbar\omega}{e^{\hbar\omega/k_B T} - 1}. \quad (9.9)$$

The function

$$\frac{1}{e^{\hbar\omega/k_B T} - 1} \quad (9.10)$$

is a form of Bose–Einstein distribution functions; more specifically it is called the Bose–Einstein distribution function for photons today. If $(\hbar\omega/k_B T) \ll 1$, we have

$$e^{\hbar\omega/k_B T} \approx 1 + (\hbar\omega/k_B T).$$

Therefore, we get

$$\bar{E} \approx k_B T. \quad (9.11)$$

Thus, the relation (9.9) asymptotically agrees with a classical theory. In other words, according to the classical theory related to law of equipartition of energy, energy of $k_B T/2$ is distributed to each of two degrees of freedom of motion, i.e., a kinetic energy and a potential energy of a harmonic oscillator.

9.2 Planck's Law of Radiation and Mode Density of Electromagnetic Waves

Researcher at the time tried to seek the relationship between the energy density inside the cavity and (angular) frequency of radiation. To reach the relationship, let us introduce a concept of mode density of electromagnetic waves related to the blackbody radiation. We define the mode density $D(\omega)$ as the number of *modes* of electromagnetic waves per unit volume per unit angular frequency. We refer to the electromagnetic waves having allowed specific angular frequencies and polarization as modes. These modes must be described as *linearly independent* functions.

Determination of the mode density is related to boundary conditions (BCs) imposed on a physical system. We already dealt with this problem in Chaps. 2, 3, and 8. These BCs often appear when we find solutions of a differential equation. Let us consider a following wave equation:

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}. \quad (9.12)$$

According to the method of separation of variables, we put

$$\psi(x, t) = X(x)T(t). \quad (9.13)$$

Substituting (9.13) for (9.12) and dividing both sides by $X(x)T(t)$, we have

$$\frac{1}{X} \frac{d^2 X}{dx^2} = \frac{1}{v^2} \frac{1}{T} \frac{d^2 T}{dt^2} = -k^2, \quad (9.14)$$

where k is an undetermined (possibly complex) constant. For the x component, we get

$$\frac{d^2 X}{dx^2} + k^2 X = 0. \quad (9.15)$$

Remember that k is supposed to be a complex number for the moment (see [Example 1.1](#)). Modifying [Example 1.1](#) a little bit such that (9.15) is posed in a domain $[0, L]$ and imposing the Dirichlet conditions such that

$$X(0) = X(L) = 0, \quad (9.16)$$

we find a solution of

$$X(x) = a \sin kx, \quad (9.17)$$

where a is a constant. The constant k can be determined to satisfy the BCs; i.e.,

$$kL = m\pi \quad \text{or} \quad k = m\pi/L \quad (m = 1, 2, \dots). \quad (9.18)$$

Thus, we get real numbers for k . Then, we have a solution

$$T(t) = b \sin kvt = b \sin \omega t. \quad (9.19)$$

The overall solution is then

$$\psi(x, t) = c \sin kx \sin \omega t. \quad (9.20)$$

This solution has already appeared in Chap. 8 as a stationary solution. The readers are encouraged to derive these results.

In a three-dimensional case, we have a wave equation

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}. \quad (9.21)$$

In this case, we also assume

$$\psi(x, t) = X(x)Y(y)Z(z)T(t). \quad (9.22)$$

Similarly we get

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = \frac{1}{v^2} \frac{1}{T} \frac{d^2 T}{dt^2} = -k^2. \quad (9.23)$$

Putting

$$\frac{1}{X} \frac{d^2 X}{dx^2} = -k_x^2, \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = -k_y^2, \quad \frac{1}{Z} \frac{d^2 Z}{dz^2} = -k_z^2, \quad (9.24)$$

we have

$$k_x^2 + k_y^2 + k_z^2 = k^2. \quad (9.25)$$

Then, we get a stationary wave solution as in the one-dimensional case such that

$$\psi(x, t) = c \sin k_x x \sin k_y y \sin k_z z \sin \omega t. \quad (9.26)$$

The BCs to be satisfied with $X(x)$, $Y(y)$, and $Z(z)$ are

$$k_x L = m_x \pi, \quad k_y L = m_y \pi, \quad k_z L = m_z \pi \quad (m_x, m_y, m_z = 1, 2, \dots). \quad (9.27)$$

Returning to the main issue, let us deal with the mode density. Think of a cube of each side of L that is placed as shown in Fig. 9.1. Calculating k , we have

$$k = \sqrt{k_x^2 + k_y^2 + k_z^2} = \frac{\pi}{L} \sqrt{m_x^2 + m_y^2 + m_z^2} = \frac{\omega}{c}, \quad (9.28)$$

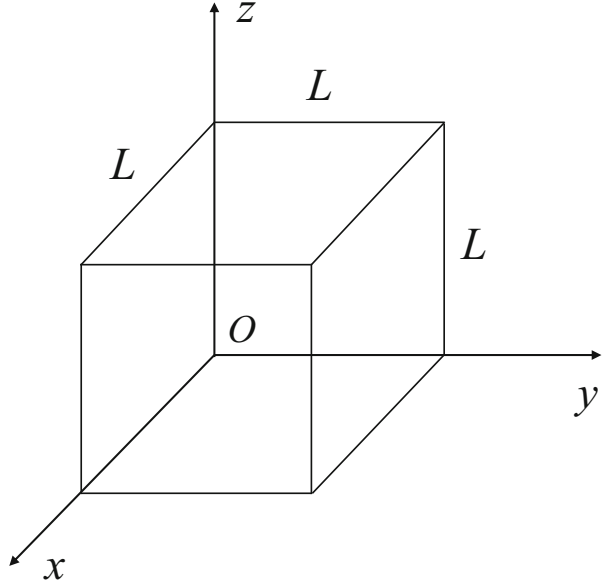
where we assumed that the inside of a cavity is vacuum and, hence, the propagation velocity of light is c . Rewriting (9.28), we have

$$\sqrt{m_x^2 + m_y^2 + m_z^2} = \frac{L\omega}{\pi c}. \quad (9.29)$$

The number m_x , m_y , and m_z represents allowable modes in the cavity; the set of (m_x, m_y, m_z) specifies individual modes. Note that m_x , m_y , and m_z are all positive integers. If, for instance, $-m_x$ were allowed, this would produce $\sin(-k_x x) = -\sin k_x x$; but this function is linearly dependent on $\sin k_x x$. Then, a mode indexed by $-m_x$ should not be regarded as an independent mode. Given a ω , a set (m_x, m_y, m_z) that satisfies (9.29) corresponds to each mode.

Therefore, the number of modes that satisfies a following expression

Fig. 9.1 Cube of each side of L . We use this simple model to estimate mode density



$$\sqrt{m_x^2 + m_y^2 + m_z^2} \leq \frac{L\omega}{\pi c} \quad (9.30)$$

represents those corresponding to angular frequencies equal to or less than given ω . Each mode has one-to-one correspondence with the lattice indexed by (m_x, m_y, m_z) . Accordingly, if $m_x, m_y, m_z \gg 1$, the number of allowed modes approximately equals one-eighth of a volume of a sphere having a radius of $\frac{L\omega}{\pi c}$. Let N_L be the number of modes whose angular frequencies equal to or less than ω . Recalling that there are two independent modes having the same index (m_x, m_y, m_z) but mutually orthogonal polarities, we have

$$N_L = \frac{4\pi}{3} \left(\frac{L\omega}{\pi c} \right)^3 \cdot \frac{1}{8} \cdot 2 = \frac{L^3 \omega^3}{3\pi^2 c^3}. \quad (9.31)$$

Consequently, the mode density $D(\omega)$ is expressed as

$$D(\omega)d\omega = \frac{1}{L^3} \frac{dN_L}{d\omega} d\omega = \frac{\omega^2}{\pi^2 c^3} d\omega, \quad (9.32)$$

where $D(\omega)d\omega$ represents the number of modes per unit volume whose angular frequencies range ω and $\omega + d\omega$.

Now, we introduce a function $\rho(\omega)$ as an energy density per unit angular frequency. Then, combining $D(\omega)$ with (9.9), we get

$$\rho(\omega) = D(\omega) \frac{\hbar\omega}{e^{\hbar\omega/k_B T} - 1} = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{1}{e^{\hbar\omega/k_B T} - 1}. \quad (9.33)$$

The relation (9.33) is called Planck's law of radiation. Notice that $\rho(\omega)$ has a dimension $[\text{J m}^{-3} \text{ s}]$.

To solve (9.15) under the Dirichlet conditions (9.16) is pertinent to analyzing an electric field within a cavity surrounded by a metal husk, because the electric field must be absent at an interface between the cavity and metal. The problem, however, can equivalently be solved using the magnetic field. This is because at the interface the reflection coefficient of electric field and magnetic field has a reversed sign (see Chap. 8). Thus, given an equation for the magnetic field, we may use the Neumann condition. This condition requires differential coefficients to vanish at the boundary (i.e., the interface between the cavity and metal). In a similar manner to the above, we get

$$\psi(x, t) = c \cos kx \cos \omega t. \quad (9.34)$$

By imposing BCs, again we have (9.18) that leads to the same result as the above.

We may also impose the periodic BCs. This type of equation has already been treated in Chap. 3. In that case we have a solution of

$$e^{ikx} \quad \text{and} \quad e^{-ikx}.$$

The BCs demand that $e^0 = 1 = e^{ikL}$. That is,

$$kL = 2\pi m \quad (m = 0, \pm 1, \pm 2, \dots). \quad (9.35)$$

Notice that e^{ikx} and e^{-ikx} are linearly independent and, hence, minus sign for m is permitted. Correspondingly, we have

$$N_L = \frac{4\pi}{3} \left(\frac{L\omega}{2\pi c} \right)^3 \cdot 2 = \frac{L^3 \omega^3}{3\pi^2 c^3}. \quad (9.36)$$

In other words, here we have to consider a whole volume of a sphere of a *half* radius of the previous case. Thus, we reach the same conclusion as before.

If the average energy of an oscillator were described by (9.11), we would obtain a following description of $\rho(\omega)$ such that

$$\rho(\omega) = D(\omega) k_B T = \frac{\omega^2}{\pi^2 c^3} k_B T. \quad (9.37)$$

This relation is well-known as Rayleigh–Jeans law, but (9.37) disagreed with experimental results in that according to Rayleigh–Jeans law, $\rho(\omega)$ diverges toward infinity as ω goes to infinity. The discrepancy between the theory and experimental

results was referred to as “ultraviolet catastrophe.” Planck’s law of radiation described by (9.33), however, reproduces the experimental results well.

9.3 Two-Level Atoms

Although Planck established Planck’s law of radiation, researchers at that time hesitated in professing the existence of light quanta. It was Einstein that derived Planck’s law by assuming two-level atoms in which light quanta play a role.

His assumption comprises the following three postulates: (1) The physical system to be addressed comprises the so-called hypothetical two-level atoms that have only two energy levels. If two-level atoms absorb a light quantum, a ground-state electron is excited up to a higher level (i.e., the stimulated absorption). (2) The higher-level electron may spontaneously lose its energy and return back to the ground state (the spontaneous emission). (3) The higher-level electron may also lose its energy and return back to the ground state. Unlike (2), however, the excited electron has to be stimulated by being irradiated by light quanta having an energy corresponding to the energy difference between the ground state and excited state (the stimulated emission). Figure 9.2 schematically depicts the optical processes of the Einstein model.

Under those postulates, Einstein dealt with the problem probabilistically. Suppose that the ground state and excited state have energies E_1 and E_2 . Einstein assumed that light quanta having an energy equaling $E_2 - E_1$ take part in all the above three transition processes. He also propounded the idea that the light quanta have an energy that is proportional to its (angular) frequency. That is, he thought that the following relation should hold:

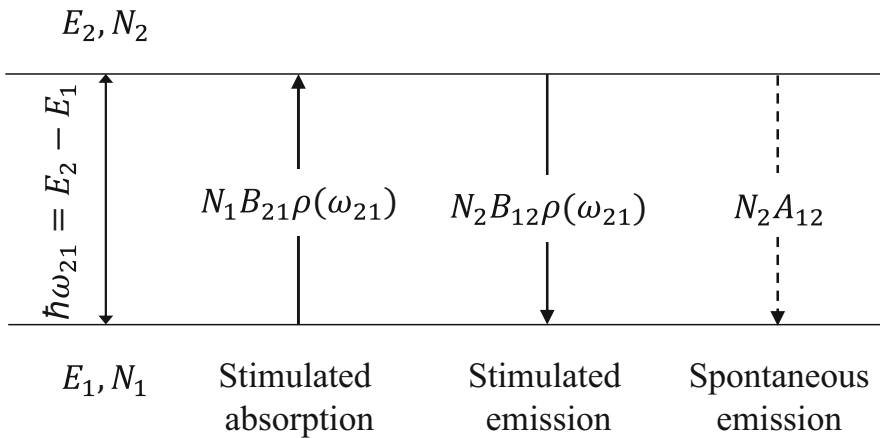


Fig. 9.2 Optical processes of Einstein two-level atom model

$$\hbar\omega_{21} = E_2 - E_1, \quad (9.38)$$

where ω_{21} is an angular frequency of light that takes part in the optical transitions. For the time being, let us follow Einstein's postulates.

1. Stimulated absorption: This process is simply said to be an “absorption.” Let W_a [s^{-1}] be the transition probability that the electron absorbs a light quantum to be excited to the excited state. W_a is described as

$$W_a = N_1 B_{21} \rho(\omega_{21}), \quad (9.39)$$

where N_1 is the number of atoms occupying the ground state; B_{21} is a proportional constant; $\rho(\omega_{21})$ is due to (9.33). Note that in (9.39) we used ω_{21} instead of ω in (9.33). The coefficient B_{21} is called Einstein B coefficient; more specifically one of Einstein B coefficients. Namely, B_{21} is pertinent to the transition from the ground state to excited state.

2. Emission processes: The processes include both the spontaneous and stimulated emissions. Let W_e [s^{-1}] be the transition probability that the electron emits a light quantum and returns back to the ground state. W_e is described as

$$W_e = N_2 B_{12} \rho(\omega_{21}) + N_2 A_{12}, \quad (9.40)$$

where N_2 is the number of atoms occupying the excited state; B_{12} and A_{12} are proportional constants. The coefficient A_{12} is called Einstein A coefficient relevant to the spontaneous emission. The coefficient B_{12} is associated with the stimulated emission and also called Einstein B coefficient together with B_{21} . Here, B_{12} is pertinent to the transition from the excited state to ground state.

Now, we have

$$B_{12} = B_{21}. \quad (9.41)$$

The reasoning for this is as follows: The coefficients B_{12} and B_{21} are proportional to the matrix elements pertinent to the optical transition. Let T be an operator associated with the transition. Then, a matrix element is described using an inner product notation of Chap. 1 by

$$B_{21} = \langle \psi_2 | T | \psi_1 \rangle, \quad (9.42)$$

where ψ_1 and ψ_2 are initial and final states of the system in relation to the optical transition. As a good approximation, we use $e\mathbf{r}$ for T (dipole approximation), where e is an elementary charge and $\mathbf{r}$ is a position operator (see Chap. 1). If (9.42) represents the absorption process (i.e., the transition from the ground state to excited state), the corresponding emission process should be described as a reversed process by

$$B_{12} = \langle \psi_1 | T | \psi_2 \rangle. \quad (9.43)$$

Notice that in (9.43) ψ_2 and ψ_1 are initial and final states.

Taking complex conjugate of (9.42), we have

$$B_{21}^* = \langle \psi_1 | T^\dagger | \psi_2 \rangle, \quad (9.44)$$

where $T^\dagger$ is an operator adjoint to T (see Chap. 1). With an Hermitian operator H , from Sect. 1.4 we have

$$H^\dagger = H. \quad (1.119)$$

Since T is also Hermitian, we have

$$T^\dagger = T. \quad (9.45)$$

Thus, we get

$$B_{21}^* = B_{12}. \quad (9.46)$$

But, as in the cases of Sects. 4.2 and 4.3, ψ_1 and ψ_2 can be represented as real functions. Then, we have

$$B_{21}^* = B_{21} = B_{12}.$$

That is, we assume that the matrix B is real symmetric. In the case of two-level atoms, as a matrix form we get

$$B = \begin{pmatrix} 0 & B_{12} \\ B_{12} & 0 \end{pmatrix}. \quad (9.47)$$

Compare (9.47) with (4.28).

Now, in the thermal equilibrium, we have

$$W_e = W_a. \quad (9.48)$$

That is,

$$N_2 B_{21} \rho(\omega_{21}) + N_2 A_{12} = N_1 B_{21} \rho(\omega_{21}), \quad (9.49)$$

where we used (9.41) for LHS. Assuming Boltzmann distribution law, we get

$$\frac{N_2}{N_1} = \exp[-(E_2 - E_1)/k_B T]. \quad (9.50)$$

Here if moreover we assume (9.38), we get

$$\frac{N_2}{N_1} = \exp(-\hbar\omega_{21}/k_B T). \quad (9.51)$$

Combing (9.49) and (9.51), we have

$$\exp(-\hbar\omega_{21}/k_B T) = \frac{B_{21}\rho(\omega_{21})}{B_{21}\rho(\omega_{21}) + A_{12}}. \quad (9.52)$$

Solving (9.52) with respect to $\rho(\omega_{12})$, we finally get

$$\rho(\omega_{21}) = \frac{A_{12}}{B_{21}} \cdot \frac{\exp(-\hbar\omega_{21}/k_B T)}{1 - \exp(-\hbar\omega_{21}/k_B T)} = \frac{A_{12}}{B_{21}} \cdot \frac{1}{\exp(\hbar\omega_{21}/k_B T) - 1}. \quad (9.53)$$

Assuming that

$$\frac{A_{12}}{B_{21}} = \frac{\hbar\omega_{21}^3}{\pi^2 c^3}, \quad (9.54)$$

we have

$$\rho(\omega_{21}) = \frac{\hbar\omega_{21}^3}{\pi^2 c^3} \cdot \frac{1}{\exp(\hbar\omega_{21}/k_B T) - 1}. \quad (9.55)$$

This is none other than Planck's law of radiation.

9.4 Dipole Radiation

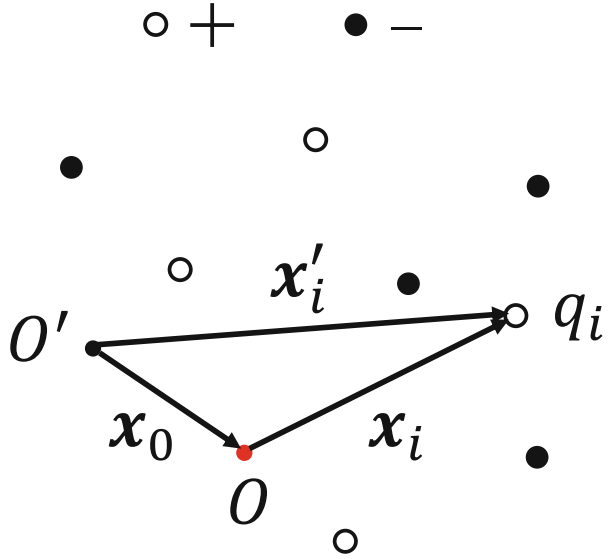
In (9.54) we only know the ratio of A_{12} to B_{21} . To have a good knowledge of these Einstein coefficients, we briefly examine a mechanism of the dipole radiation. The electromagnetic radiation results from an accelerated motion of a dipole.

A dipole moment $\mathbf{p}(t)$ is defined as a function of time t by

$$\mathbf{p}(t) = \int \mathbf{x}' \rho(\mathbf{x}', t) d\mathbf{x}', \quad (9.56)$$

where $\mathbf{x}'$ is a position vector in a Cartesian coordinate; an integral is taken over a whole three-dimensional space; ρ is a charge density appearing in (7.1). If we

Fig. 9.3 Dipole moment viewed from the frame O or O'



consider a system comprising point charges, integration can immediately be carried out to yield

$$\mathbf{p}(t) = \sum_i q_i \mathbf{x}_i, \quad (9.57)$$

where q_i is a charge of each point charge i and $\mathbf{x}_i$ is a position vector of the point charge i . From (9.56) and (9.57), we find that $\mathbf{p}(t)$ depends on how we set up the coordinate system. However, if a total charge of the system is zero, $\mathbf{p}(t)$ does not depend on the coordinate system. Let $\mathbf{p}(t)$ and $\mathbf{p}'(t)$ be a dipole moment viewed from the frame O and O' , respectively (see Fig. 9.3). Then we have

$$\begin{aligned} \mathbf{p}'(t) &= \sum_i q_i \mathbf{x}'_i = \sum_i q_i (\mathbf{x}_0 + \mathbf{x}_i) = \left(\sum_i q_i \right) \mathbf{x}_0 + \sum_i q_i \mathbf{x}_i = \sum_i q_i \mathbf{x}_i \\ &= \mathbf{p}(t). \end{aligned} \quad (9.58)$$

Notice that with the third equality the first term vanishes because the total charge is zero. The system comprising two point charges that have an opposite charge ($\pm q$) is particularly simple but very important. In that case we have

$$\mathbf{p}(t) = q\mathbf{x}_1 + (-q)\mathbf{x}_2 = q(\mathbf{x}_1 - \mathbf{x}_2) = q\tilde{\mathbf{x}}. \quad (9.59)$$

Here we assume that $q > 0$ according to the custom and, hence, $\tilde{\mathbf{x}}$ is a vector directing from the minus point charge to the plus charge.

Figure 9.4 displays geometry of an oscillating dipole and electromagnetic radiation from it. Figure 9.4a depicts the dipole. It is placed at the origin of the coordinate

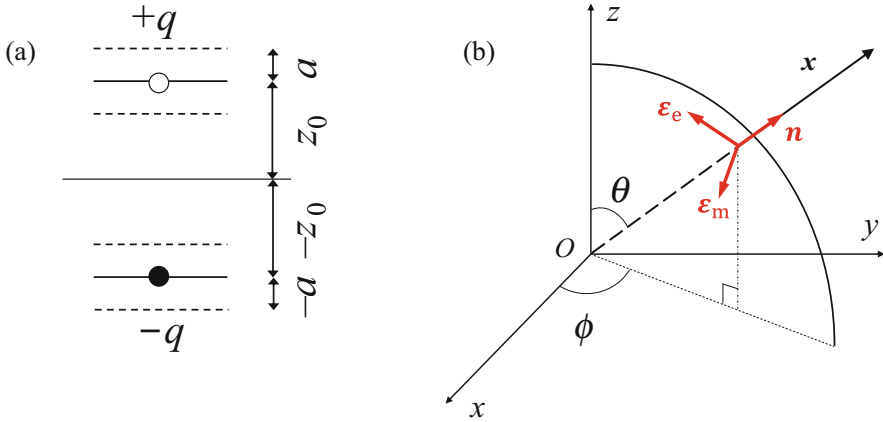


Fig. 9.4 Electromagnetic radiation from an accelerated motion of a dipole. (a) A dipole placed at the origin of the coordinate system is executing harmonic oscillation along the z -direction around an equilibrium position. (b) Electromagnetic radiation from a dipole in a wave zone. $\mathbf{e}_e$ and $\mathbf{e}_m$ are unit polarization vectors of the electric field and magnetic field, respectively. $\mathbf{e}_e$, $\mathbf{e}_m$, and $\mathbf{n}$ form a right-handed system

system and assumed to be of an atomic or molecular scale in extension; we regard a center of the dipole as the origin. Figure 9.4b represents a large-scale geometry of the dipole and surrounding space of it. For the electromagnetic radiation, an accelerated motion of the dipole is of primary importance. The electromagnetic fields produced by $\mathbf{p}(t)$ vary as the inverse of r , where r is a macroscopic distance between the dipole and observation point. Namely, r is much larger compared to the dipole size.

There are other electromagnetic fields that result from the dipole moment. The fields result from $\mathbf{p}(t)$ and $\dot{\mathbf{p}}(t)$. Strictly speaking, we have to include those quantities that are responsible for the electromagnetic fields associated with the dipole radiation. Nevertheless, the fields produced by $\mathbf{p}(t)$ and $\dot{\mathbf{p}}(t)$ vary as a function of the inverse cube and inverse square of r , respectively. Therefore, the surface integral of the square of the fields associated with $\mathbf{p}(t)$ and $\dot{\mathbf{p}}(t)$ asymptotically reaches zero with enough large r with respect to a sphere enveloping the dipole. Regarding $\ddot{\mathbf{p}}(t)$, however, the surface integral of the square of the fields remains finite even with enough large r . For this reason, we refer to the spatial region where $\ddot{\mathbf{p}}(t)$ does not vanish as a *wave zone*.

Suppose that a dipole placed at the origin of the coordinate system is executing harmonic oscillation along the z -direction around an equilibrium position (see Fig. 9.4). Motion of two charges having plus and minus signs is described by

$$\mathbf{z}_+ = z_0 \mathbf{e}_3 + a e^{i\omega t} \mathbf{e}_3 \quad (z_0, a > 0), \quad (9.60)$$

$$\mathbf{z}_- = -z_0 \mathbf{e}_3 - a e^{i\omega t} \mathbf{e}_3, \quad (9.61)$$

where $\mathbf{z}_+$ and $\mathbf{z}_-$ are position vectors of a plus charge and minus charge, respectively; $\mathbf{z}_0$ and $-\mathbf{z}_0$ are equilibrium positions of each charge; a is an amplitude of the harmonic oscillation; ω is an angular frequency of the oscillation. Then, accelerations of the charges are given by

$$\mathbf{a}_+ \equiv \ddot{\mathbf{z}}_+ = -a\omega^2 e^{i\omega t} \mathbf{e}_3, \quad (9.62)$$

$$\mathbf{a}_- \equiv \ddot{\mathbf{z}}_- = a\omega^2 e^{i\omega t} \mathbf{e}_3. \quad (9.63)$$

Meanwhile, we have

$$\mathbf{p}(t) = q\mathbf{z}_+ + (-q)\mathbf{z}_- = q(\mathbf{z}_+ - \mathbf{z}_-) \quad (q > 0). \quad (9.64)$$

Therefore,

$$\ddot{\mathbf{p}}(t) = q(\ddot{\mathbf{z}}_+ - \ddot{\mathbf{z}}_-) = -2qa\omega^2 e^{i\omega t} \mathbf{e}_3. \quad (9.65)$$

The quantity $\ddot{\mathbf{p}}(t)$, i.e., the second derivative of $\mathbf{p}(t)$ with respect to time produces the electric field described by [2]

$$\mathbf{E} = -\ddot{\mathbf{p}}/4\pi\epsilon_0 c^2 r = -qa\omega^2 e^{i\omega t} \mathbf{e}_3 / 2\pi\epsilon_0 c^2 r, \quad (9.66)$$

where r is a distance between the dipole and observation point. In (9.66) we ignored a term proportional to inverse square and cube of r for the aforementioned reason.

As described in (9.66), the strength of the radiation electric field in the wave zone measured at a point away from the oscillating dipole is proportional to a component of the vector of the acceleration motion of the dipole [i.e., $\ddot{\mathbf{p}}(t)$]. The radiation electric field lies in the direction perpendicular to a line connecting the observation point and the point of the dipole (Fig. 9.4). Let $\mathbf{e}_e$ be a unit polarization vector of the electric field in that direction and let $\mathbf{E}^\perp$ be the radiation electric field. Then, we have

$$\mathbf{E}^\perp = -qa\omega^2 e^{i\omega t} (\mathbf{e}_3 \cdot \mathbf{e}_e) \mathbf{e}_e / 2\pi\epsilon_0 c^2 r = -qa\omega^2 e^{i\omega t} \mathbf{e}_e \sin \theta / 2\pi\epsilon_0 c^2 r. \quad (9.67)$$

As shown in Sect. 7.3, $(\mathbf{e}_e \cdot \mathbf{e}_3) \mathbf{e}_e$ in (9.67) “extracts” from $\mathbf{e}_3$ a vector component parallel to $\mathbf{e}_e$. Such an operation is said to be a projection of a vector. The related discussion can be seen in Part III.

It takes a time of r/c for the emitted light from the charge to arrive at the observation point. Consequently, the acceleration of the charge has to be measured at the time when the radiation leaves the charge. Let t be the instant when the electric field is measured at the measuring point. Then, it follows that the radiation leaves the charge at a time of $t - r/c$. Thus, the electric field relevant to the radiation that can be observed far enough away from the oscillating charge is described as [2]

$$\mathbf{E}^\perp(\mathbf{x}, t) = -\frac{qa\omega^2 e^{i\omega(t-\frac{r}{c})} \sin \theta}{2\pi\epsilon_0 c^2 r} \boldsymbol{\epsilon}_e. \quad (9.68)$$

The radiation electric field must necessarily be accompanied by a magnetic field. Writing the radiation magnetic field as $\mathbf{H}^\perp(\mathbf{x}, t)$, we have [3]

$$\begin{aligned} \mathbf{H}^\perp(\mathbf{x}, t) &= -\frac{1}{c\mu_0} \cdot \frac{qa\omega^2 e^{i\omega(t-\frac{r}{c})} \sin \theta}{2\pi\epsilon_0 c^2 r} \mathbf{n} \times \boldsymbol{\epsilon}_e = -\frac{qa\omega^2 e^{i\omega(t-\frac{r}{c})} \sin \theta}{2\pi cr} \mathbf{n} \times \boldsymbol{\epsilon}_e \\ &= -\frac{qa\omega^2 e^{i\omega(t-\frac{r}{c})} \sin \theta}{2\pi cr} \boldsymbol{\epsilon}_m, \end{aligned} \quad (9.69)$$

where $\mathbf{n}$ represents a unit vector in the direction parallel to a line connecting the observation point and the dipole. The $\boldsymbol{\epsilon}_m$ is a unit polarization vector of the magnetic field as defined by (7.67). From the above, we see that the radiation electromagnetic waves in the wave zone are transverse waves that show the properties the same as those of electromagnetic waves in a free space.

Now, let us evaluate a time-averaged energy flux from an oscillating dipole. Using (8.71), we have

$$\begin{aligned} \bar{S}(\theta) &= \frac{1}{2} \mathbf{E} \times \mathbf{H}^* = \frac{1}{2} \cdot \frac{qa\omega^2 e^{i\omega(t-\frac{r}{c})} \sin \theta}{2\pi\epsilon_0 c^2 r} \cdot \frac{qa\omega^2 e^{-i\omega(t-\frac{r}{c})} \sin \theta}{2\pi cr} \mathbf{n} \\ &= \frac{\omega^4 \sin^2 \theta}{8\pi^2 \epsilon_0 c^3 r^2} (qa)^2 \mathbf{n}. \end{aligned} \quad (9.70)$$

If we are thinking of an atom or a molecule in which the dipole consists of an electron and a positive charge that compensates it, q is replaced with $-e$ ($e < 0$). Then (9.70) reads as

$$\bar{S}(\theta) = \frac{\omega^4 \sin^2 \theta}{8\pi^2 \epsilon_0 c^3 r^2} (ea)^2 \mathbf{n}. \quad (9.71)$$

Let us relate the above argument to Einstein A and B coefficients. Since we are dealing with an isolated dipole, we might well suppose that the radiation comes from the spontaneous emission. Let P be a total power of emission from the oscillating dipole that gets through a sphere of radius r . Then we have

$$\begin{aligned}
 P &= \int \bar{\mathbf{S}}(\theta) \cdot \mathbf{n} dS = \int_0^{2\pi} d\phi \int_0^\pi \bar{\mathbf{S}}(\theta) r^2 \sin \theta d\theta \\
 &= \frac{\omega^4}{8\pi^2 \epsilon_0 c^3} (ea)^2 \int_0^{2\pi} d\phi \int_0^\pi \sin^3 \theta d\theta.
 \end{aligned} \tag{9.72}$$

Changing $\cos \theta$ to t , the integral $I \equiv \int_0^\pi \sin^3 \theta d\theta$ can be converted into

$$I = \int_{-1}^1 (1 - t^2) dt = 4/3. \tag{9.73}$$

Thus, we have

$$P = \frac{\omega^4}{3\pi\epsilon_0 c^3} (ea)^2. \tag{9.74}$$

A probability of the spontaneous emission is given by $N_2 A_{12}$. Since we are dealing with a single dipole, N_2 can be put 1. Accordingly, an expected power of emission is $A_{12} \hbar \omega_{21}$. Replacing ω in (9.74) with ω_{21} in (9.55) and equating $A_{12} \hbar \omega_{21}$ to P , we get

$$A_{12} = \frac{\omega_{21}^3}{3\pi\epsilon_0 c^3 \hbar} (ea)^2. \tag{9.75}$$

From (9.54), we also get

$$B_{12} = \frac{\pi}{3\epsilon_0 \hbar^2} (ea)^2. \tag{9.76}$$

In order to relate these results to those of quantum mechanics, we may replace a^2 in the above expressions with a square of an absolute value of the matrix elements of the position operator $\mathbf{r}$. That is, representing $|1\rangle$ and $|2\rangle$ as the quantum states of the ground and excited states of a two-level atom, we define $\langle 1|\mathbf{r}|2\rangle$ as the matrix element of the position operator. Relating $|\langle 1|\mathbf{r}|2\rangle|^2$ to a^2 , we get

$$A_{12} = \frac{\omega_{21}^3 e^2}{3\pi\epsilon_0 c^3 \hbar} \left| \langle 1|\mathbf{r}|2\rangle \right|^2 \quad \text{and} \quad B_{12} = \frac{\pi e^2}{3\epsilon_0 \hbar^2} \left| \langle 1|\mathbf{r}|2\rangle \right|^2. \tag{9.77}$$

From (9.77), we have

$$B_{12} = \frac{\pi e^2}{3\epsilon_0 \hbar^2} \langle 1|\mathbf{r}|2\rangle \langle 1|\mathbf{r}|2\rangle^* = \frac{\pi e^2}{3\epsilon_0 \hbar^2} \langle 1|\mathbf{r}|2\rangle \langle 2|\mathbf{r}^\dagger|1\rangle = \frac{\pi e^2}{3\epsilon_0 \hbar^2} \langle 1|\mathbf{r}|2\rangle \times \langle 2|\mathbf{r}|1\rangle, \quad (9.78)$$

where with the second equality we used (1.116); the last equality comes from that $\mathbf{r}$ is Hermitian. Meanwhile, we have

$$B_{21} = \frac{\pi e^2}{3\epsilon_0 \hbar^2} |\langle 2|\mathbf{r}|1\rangle|^2 = \frac{\pi e^2}{3\epsilon_0 \hbar^2} \langle 2|\mathbf{r}|1\rangle \langle 1|\mathbf{r}^\dagger|2\rangle = \frac{\pi e^2}{3\epsilon_0 \hbar^2} \langle 2|\mathbf{r}|1\rangle \langle 1|\mathbf{r}|2\rangle. \quad (9.79)$$

Hence, we recover (9.41).

9.5 Lasers

9.5.1 Brief Outlook

A concept of the two-level atoms proposed by Einstein is universal and independent of materials and can be utilized for some particular purposes. Actually, in later years many researchers tried to validate that concept and verified its validity. After basic researches of 1950s and 1960s, fundamentals were put into practical use as various optical devices. Typical examples are masers and lasers, abbreviations for “micro-wave amplification by stimulated emission of radiation” and “light amplification by stimulated emission of radiation.” Of these, lasers are common and particularly important nowadays. On the basis of universality of the concept, a lot of materials including semiconductors and dyes are used in gaseous, liquid, and solid states.

Let us consider a rectangular parallelepiped of a laser medium with a length L and cross-section area S (Fig. 9.5). Suppose there are N two-level atoms in the rectangular parallelepiped such that

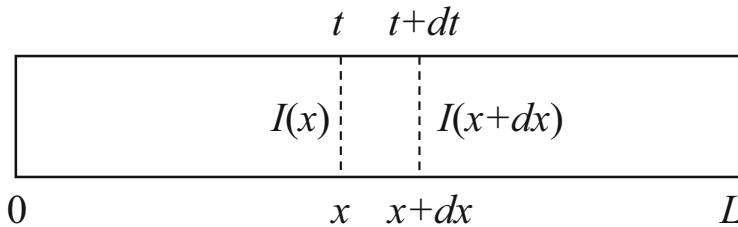


Fig. 9.5 Rectangular parallelepiped of a laser medium with a length L and a cross-section area S (not shown). $I(x)$ denotes irradiance at a point of x from the origin

$$N = N_1 + N_2, \quad (9.80)$$

where N_1 and N_2 represent the number of atoms occupying the ground and excited states, respectively. Suppose that a light is propagated from the left of the rectangular parallelepiped and entering it. Then, we expect that three processes occur simultaneously. One is a stimulated absorption and others are stimulated emission and spontaneous emission. After these process, an increment dE in photon energy of the total system (i.e., the rectangular parallelepiped) during dt is described by

$$dE = \{N_2 [B_{21}\rho(\omega_{21}) + A_{12}] - N_1 B_{21}\rho(\omega_{21})\} \hbar\omega_{21}dt. \quad (9.81)$$

In light of (9.39) and (9.40), a dimensionless quantity $dE/\hbar\omega_{21}$ represents a number of effective events of photons emission that have occurred during dt . Since in lasers the stimulated emission is dominant, we shall forget about the spontaneous emission and rewrite (9.81) as

$$\begin{aligned} dE &= \{N_2 B_{21}\rho(\omega_{21}) - N_1 B_{21}\rho(\omega_{21})\} \hbar\omega_{21}dt \\ &= B_{21}\rho(\omega_{21})(N_2 - N_1) \hbar\omega_{21}dt. \end{aligned} \quad (9.82)$$

Under a thermal equilibrium, we have $N_2 < N_1$ on the basis of Boltzmann distribution law, and so $dE < 0$. In this occasion, therefore, the photon energy decreases. For the light amplification to take place, therefore, we must have a following condition:

$$N_2 > N_1. \quad (9.83)$$

This energy distribution is called inverted distribution or population inversion. Thus, the laser oscillation is a typical non-equilibrium phenomenon. To produce the population inversion, we need an external exciting source using an electrical or optical device.

The essence of lasers rests upon the fact that stimulated emission produces a photon that possesses a wavenumber vector ($\mathbf{k}$) and a polarization ($\boldsymbol{\epsilon}$) both the same as those of an original photon. For this reason, the laser light is monochromatic and highly directional. To understand the fundamental mechanism underlying the related phenomena, interested readers are encouraged to seek appropriate literature of quantum theory of light for further reading [4].

To make a discussion simple and straightforward, let us assume that the light is incident parallel to the long axis of the rectangular parallelepiped. Then, the stimulated emission produces light to be propagated in the same direction. As a result, an irradiance I measured in that direction is described as

$$I = \frac{E}{SL} \cdot c'. \quad (9.84)$$

Note that the light velocity in the laser medium c' is given by

$$c' = c/n, \quad (9.85)$$

where n is a refractive index of the laser medium. Taking an infinitesimal of both sides of (9.84), we have

$$\begin{aligned} dI &= dE \frac{c'}{SL} = B_{21} \rho(\omega_{21}) (N_2 - N_1) \hbar \omega_{21} \frac{c'}{SL} dt \\ &= B_{21} \rho(\omega_{21}) \tilde{N} \hbar \omega_{21} c' dt, \end{aligned} \quad (9.86)$$

where $\tilde{N} = (N_2 - N_1)/SL$ denotes a “net” density of atoms that occupy the excited state.

The energy density $\rho(\omega_{21})$ can be written as

$$\rho(\omega_{21}) = I(\omega_{21}) g(\omega_{21}) / c', \quad (9.87)$$

where $I(\omega_{12})$ [$\text{J s}^{-1} \text{m}^{-2}$] represents an intensity of radiation; $g(\omega_{21})$ is a gain function [s]. The gain function is a measure that shows how favorably (or unfavorably) the transition takes place at the said angular frequency ω_{12} . This is normalized in the emission range such that

$$\int_0^\infty g(\omega) d\omega = 1.$$

The quantity $I(\omega_{21})$ is an energy flux that gets through per unit area per unit time. This flux corresponds to an energy contained in a long and thin rectangular parallelepiped of a length c' and a unit cross-section area. To obtain $\rho(\omega_{21})$, $I(\omega_{12})$ should be divided by c' in (9.87) accordingly. Using (9.87) and replacing $c' dt$ with a distance dx and $I(\omega_{12})$ with $I(x)$ as a function of x , we rewrite (9.86) as

$$dI(x) = \frac{B_{21} g(\omega_{21}) \tilde{N} \hbar \omega_{21}}{c'} I(x) dx. \quad (9.88)$$

Dividing (9.88) by $I(x)$ and integrating both sides, we have

$$\int_{I_0}^I \frac{dI(x)}{I(x)} = \int_{I_0}^I d \ln I(x) = \frac{B_{21} g(\omega_{21}) \tilde{N} \hbar \omega_{21}}{c'} \int_0^x dx, \quad (9.89)$$

where I_0 is an irradiance of light at an instant when the light is entering the laser medium from the left. Thus, we get

$$I(x) = I_0 \exp \left[\frac{B_{21} g(\omega_{21}) \tilde{N} \hbar \omega_{21}}{c'} x \right]. \quad (9.90)$$

Equation (9.90) shows that an irradiance of the laser light is augmented exponentially along the path of the laser light. In (9.90), denoting an exponent as G

$$G \equiv \frac{B_{21}g(\omega_{21})\tilde{N}\hbar\omega_{21}}{c'}, \quad (9.91)$$

we get

$$I(x) = I_0 \exp Gx.$$

The constant G is called a gain constant. This is an index that indicates the laser performance. Large numbers B_{21} , $g(\omega_{21})$, and $\tilde{N}$ yield a high performance of the laser.

In Sects. 8.8 and 9.2, we sought conditions for electromagnetic waves to cause constructive interference. In a one-dimensional dielectric medium, the condition is described as

$$kL = m\pi \quad \text{or} \quad m\lambda = 2L \quad (m = 1, 2, \dots), \quad (9.92)$$

where k and λ denote a wavenumber and wavelength in the dielectric medium, respectively. Indexing k and λ with m that represents a mode, we have

$$k_m L = m\pi \quad \text{or} \quad m\lambda_m = 2L \quad (m = 1, 2, \dots). \quad (9.93)$$

This condition can be expressed by different manners such that

$$\omega_m = 2\pi\nu_m = 2\pi c' / \lambda_m = 2\pi c / n\lambda_m = m\pi c / nL. \quad (9.94)$$

It is often the case that if the laser is a long and thin rod, rectangular parallelepiped, etc., we see that sharply resolved and regularly spaced spectral lines are observed in emission spectrum. These lines are said to be a longitudinal multimode. The separation between two neighboring emission lines is referred to as the *free spectral range* [2]. If adjacent emission lines are clearly resolved so that the free spectral range can easily be recognized, we can derive useful information from the laser oscillation spectra (vide infra).

Rewriting (9.94), as, e.g.,

$$\omega_m n = \frac{\pi c}{L} m, \quad (9.95)$$

and taking differential (or variation) of both sides, we get

$$n\delta\omega_m + \omega_m\delta n = n\delta\omega_m + \omega_m \frac{\delta n}{\delta\omega_m} \delta\omega_m = \left(n + \omega_m \frac{\delta n}{\delta\omega_m} \right) \delta\omega_m = \frac{\pi c}{L} \delta m. \quad (9.96)$$

Therefore, we get

$$\delta\omega_m = \frac{\pi c}{L} \left(n + \omega_m \frac{\delta n}{\delta\omega_m} \right)^{-1} \delta m. \quad (9.97)$$

Equation (9.97) premises the wavelength dispersion of a refractive index of a laser medium. Here, the wavelength dispersion means that the refractive index varies as a function of wavelengths of light in a matter. The laser materials often have a considerably large dispersion and relevant information is indispensable.

From (9.97), we find that

$$n_g \equiv n + \omega_m \frac{\delta n}{\delta\omega_m} \quad (9.98)$$

plays a role of refractive index when the laser material has a wavelength dispersion. The quantity n_g is said to be a *group refractive index* (or group index). Thus, (9.97) is rewritten as

$$\delta\omega = \frac{\pi c}{Ln_g} \delta m, \quad (9.99)$$

where we omitted the index m of ω_m . When we need to distinguish the refractive index n clearly from the group refractive index, we refer to n as a *phase* refractive index. Rewriting (9.98) as a relation of continuous quantities and using differentiation instead of variation, we have [2]

$$n_g = n + \omega \frac{dn}{d\omega} \quad \text{or} \quad n_g = n - \lambda \frac{dn}{d\lambda}. \quad (9.100)$$

To derive the second equation of (9.100), we used the following relations: Namely, taking a variation of $\lambda\omega = 2\pi c$, we have

$$\omega d\lambda + \lambda d\omega = 0 \quad \text{or} \quad \frac{\omega}{d\omega} = -\frac{\lambda}{d\lambda}.$$

Several formulae or relation equations were proposed to describe the wavelength dispersion. One of the famous and useful formulas among them is Sellmeier's dispersion formula [5]. As an example, the Sellmeier's dispersion formula can be described as

$$n = \sqrt{A + \frac{B}{1 - \left(\frac{c}{\lambda}\right)^2}}, \quad (9.101)$$

where A , B , and C are appropriate constants with A and B being dimensionless and C having a dimension [m]. In an actual case, it would be difficult to determine n analytically. However, if we are able to obtain well-resolved spectra, δm can be put as 1 and $\delta\omega_m$ can be determined from the free spectral range. Expressing it as $\delta\omega_{\text{FSR}}$ from (9.99) we have

$$\delta\omega_{\text{FSR}} = \frac{\pi C}{Ln_g} \quad \text{or} \quad n_g = \frac{\pi C}{L(\delta\omega_{\text{FSR}})}. \quad (9.102)$$

Thus, one can determine n_g as a function of wavelengths.

9.5.2 Organic Lasers

By virtue of prominent features of lasers and related phenomena, researchers and engineers have been developing and proposing up to now various device structures and their operation principles in the research field of device physics. Of these, we have organic lasers as newly occurring laser devices. Organic crystals possess the well-defined structure–property relationship and some of them exhibit peculiar light-emitting features. Therefore, those crystals are suited for studying their lasing property. In this section we study the light-emitting properties (especially the lasing property) of the organic crystals in relation to the wavelength dispersion of refractive index of materials. From the point of view of device physics, another key issue lies in designing an efficient diffraction grating or resonator (see Sect. 8.8). We address these fundamental issues from as broad a viewpoint as possible so that we may acquire systematic knowledge about optical device applications of materials of different compositions either inorganic or organic.

In the following tangible examples, we further investigate specific aspects of the light-emitting properties of the organic crystals to pursue fundamental properties of the organic light-emitting materials. The emission properties are examined in a device configuration of (1) thin rectangular parallelepiped or (2) slab waveguide equipped with a diffraction grating. In the former case, we investigate the longitudinal multimode of emissions that arise from an optical resonator shaped by a pair of parallel crystal faces (see Example 9.1). In the latter case, we study the transverse mode of the emissions in addition to the longitudinal mode (Examples 9.2 and 9.3).

Example 9.1 [6] We wish to show and interpret optical emission data of the organic crystals within a framework of the interference theory of electromagnetic waves mentioned in the previous chapter. As an example, Fig. 9.6 [6] displays a broadband emission spectrum obtained under a weak optical excitation regime (using a mercury lamp) with a crystal consisting of an organic semiconductor AC'7. The optical

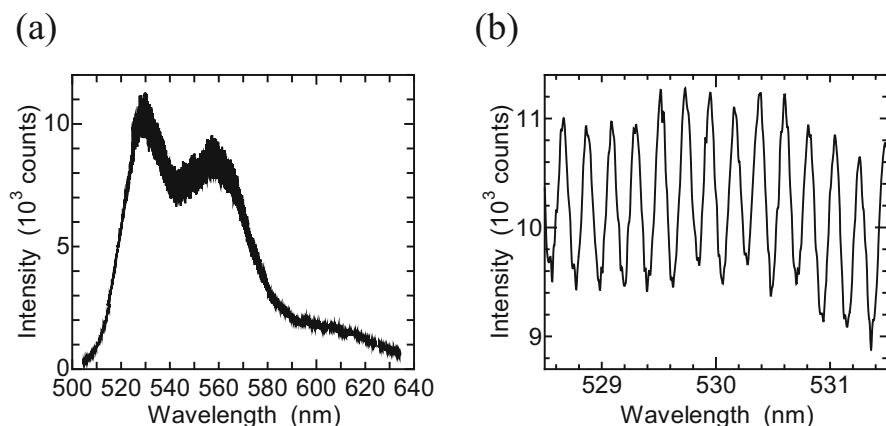
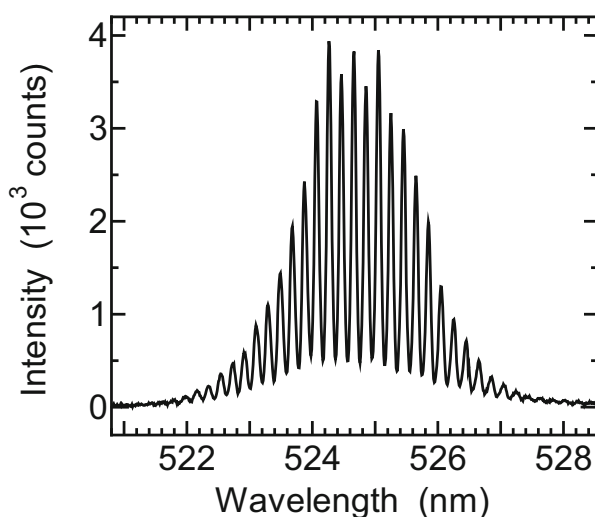


Fig. 9.6 Broadband emission spectra of an organic semiconductor crystal AC'7. (a) Full spectrum. (b) Enlarged profile of the spectrum around 530 nm. (Reproduced from Yamao T, Okuda Y, Makino Y, Hotta S (2011) Dispersion of the refractive indices of thiophene/phenylene co-oligomer single crystals. *J Appl Phys* 110(5): 053113/7 pages [6], with the permission of AIP Publishing. <https://doi.org/10.1063/1.3634117>)

Fig. 9.7 Laser oscillation spectrum of an organic semiconductor crystal AC'7. (Reproduced from Yamao T, Okuda Y, Makino Y, Hotta S (2011) Dispersion of the refractive indices of thiophene/phenylene co-oligomer single crystals. *J Appl Phys* 110(5): 053113/7 pages [6], with the permission of AIP Publishing. <https://doi.org/10.1063/1.3634117>)



measurements were carried out using an AC'7 crystal that had a pair of parallel crystal faces which worked as an optical resonator called Fabry–Pérot resonator. It caused the interference modulation of emissions and, as a result, produced a periodic modulation that was superposed onto the broadband emission. It is clearly seen in Fig. 9.6b.

As another example, Fig. 9.7 [6] displays a laser oscillation spectrum obtained under a strong excitation regime (using a laser beam) with the same AC'7 crystal as the above. The structural formula of AC'7 is shown in Fig. 9.8 together with other

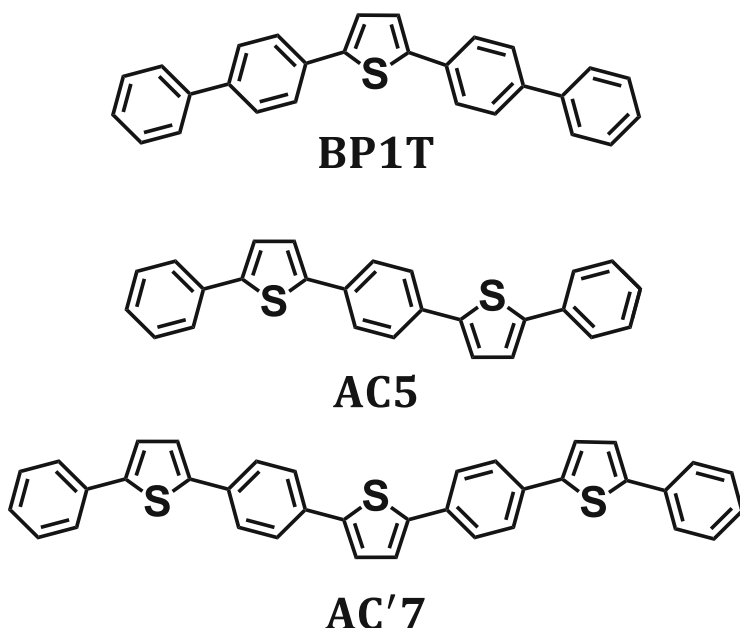


Fig. 9.8 Structural formulae of several organic semiconductors BP1T, AC5, and AC'7

related organic semiconductors. The spectrum again exhibits the periodic modulation accompanied by the longitudinal multimode. Sharply resolved emission lines are clearly noted with a regular spacing in the spectrum. If we compare the regular spacing with that of Fig. 9.6b, we immediately recognize that the regular spacing is virtually the same with the two spectra in spite of a considerably large difference between their overall spectral profile. In terms of the interference, both the broadband spectra and laser emission lines gave the same free spectral range [6].

Thus, both the broadband emission spectra and laser oscillation spectra are equally useful to study the wavelength dispersion of refractive index of the crystal. Choosing a suitable empirical formula of the wavelength dispersion, we can compare it with experimentally obtained results. For example, if we choose, e.g., (9.101) for the empirical formula, what we do next is to determine the constants A , B , and C of (9.101) by comparing the computation results with actual emission data.

If well-resolved interference modulation spectra can be obtained, this enables one to immediately get a precise value of n_g from (9.102). In turn, it leads to information about the wavelength dispersion of n through (9.100) and (9.101). Bearing in mind this situation, Yamao et al. got a following expression [6]:

Table 9.1 Optimized constants of A , B , and C for Sellmeier’s dispersion formula (9.101) with several organic semiconductor crystals^a

Material	A	B	C (nm)
BP1T	5.7	1.04	397
AC5	3.9	1.44	402
AC’7	6.0	1.06	452

^a Reproduced from Yamao T, Okuda Y, Makino Y, Hotta S (2011) Dispersion of the refractive indices of thiophene/phenylene co-oligomer single crystals. J Appl Phys 110(5): 053113/7 pages [6], with the permission of AIP Publishing. <https://doi.org/10.1063/1.3634117>

$$n_g = \frac{A \left[1 - \left(\frac{\epsilon}{\lambda} \right)^2 \right]^2 + B}{\sqrt{\left[1 - \left(\frac{\epsilon}{\lambda} \right)^2 \right]^3} \sqrt{A \left[1 - \left(\frac{\epsilon}{\lambda} \right)^2 \right] + B}}. \tag{9.103}$$

Notice that (9.103) is obtained by inserting (9.101) into (9.100) and expressing n_g as a function of λ . Determining optimum constants A , B , and C , a set of these constants yield a reliable dispersion formula in (9.101). Numerical calculations can be utilized effectively. The procedures are as follows: (1) Tentatively choosing probable numbers for A , B , and C in (9.103), n_g can be expressed as a function of λ . (2) The resulting fitting curve is then compared with n_g data experimentally decided from (9.102). After this procedure, one can choose another set of A , and B , and C and again compare the fitting curve with the experimental data. (3) This procedure can be repeated many times through iterative numerical computations of (9.103) using different sets of A , B , and C .

Thus, we should be able to adjust and determine better and better combination of A , B , and C so that the refined function (9.103) can reproduce the experimental results as precise as one pleases. At the same time, we can determine the most reliable combination of A , B , and C . Optimized constants A , B , and C of (9.101) are listed in Table 9.1 [6]. Using these numbers A , B , and C , the dispersion of the phase refractive indices n can be calculated using (9.101) as a function of wavelengths λ . Figure 9.9 [6] shows several examples of the wavelength dispersion of n_g and n for the organic semiconductor crystals. Once again, it is intriguing to see that n_g associated with the laser oscillation line sits *on* the dispersion curve determined from the broadband emission lines (Fig. 9.9a).

The formulae (9.101) and (9.103) along with associated procedures to determine the constants A , B , and C are expected to apply widely to various laser and light-emitting materials consisting of semiconducting inorganic and organic materials.

Example 9.2 [7] If we wish to construct a laser device, it will be highly desired to equip the laser material with a suitable diffraction grating or resonator [8, 9]. Suppose a situation where we choose a thin crystal for an optically active material equipped with the diffraction grating. In that case, we are to deal with a device consisting of a crystal slab waveguide. We have already encountered such a device configuration in Sect. 8.7 where the transverse modes must be considered on the basis of (8.167). We

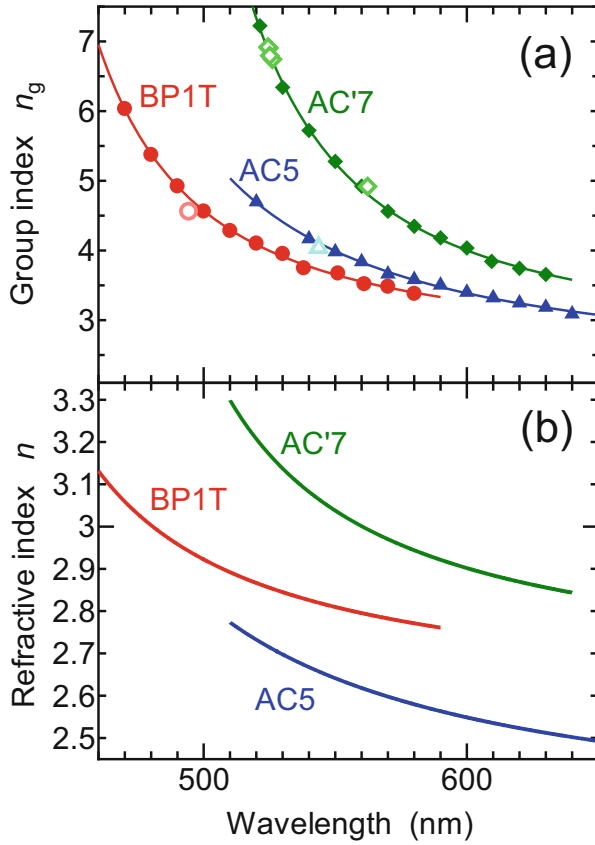


Fig. 9.9 Examples of the wavelength dispersion of refractive indices for several organic semiconductor crystals. **(a)** Dispersion curves of the group indices (n_g). The optimized solid curves are plotted using (9.103). The data plotted by filled circles (BP1T), filled triangles (AC5), and filled diamonds (AC'7) were obtained by the broadband interference modulation of emissions. An open circle, an open triangle, and open diamonds show the group indices estimated from the mode intervals of the multimode laser oscillations. **(b)** Dispersion curves of the phase refractive indices (n). These data are plotted using the Sellmeier's dispersion formula (9.101) with the optimized parameters A , B , and C (listed in Table 9.1). (Reproduced from Yamao T, Okuda Y, Makino Y, Hotta S (2011) Dispersion of the refractive indices of thiophene/phenylene co-oligomer single crystals. *J Appl Phys* 110(5): 053113/7 pages [6], with the permission of AIP Publishing. <https://doi.org/10.1063/1.3634117>)

should also take account of the longitudinal modes that arise from the presence of the diffraction grating. As a result, allowed modes in the optical device are indexed by, e.g., TM_{ml} , TE_{ml} , where m and l denote the longitudinal modes and transverse modes, respectively (see Sects. 8.7 and 8.8).

Besides the information about the dispersion of phase refractive index, we need numerical data of the propagation constant and its dispersion. The propagation constant β has appeared in Sect. 8.7.1 and is defined as

$$\beta = k \sin \theta = nk_0 \sin \theta, \quad (8.153)$$

where θ is identical to an angle denoted by θ that was defined in Fig. 8.12. The propagation constant β (>0) characterizes the light propagation within a slab waveguide where the transverse modes of light are responsible.

As the phase refractive index n has a dispersion, the propagation constant β has a dispersion as well. In optics, $n \sin \theta$ is often referred to as an *effective index*. We denote it by

$$n_{\text{eff}} \equiv n \sin \theta \quad \text{or} \quad \beta = k_0 n_{\text{eff}}. \quad (9.104)$$

As $0 \leq \theta \leq \pi/2$, we have $n \sin \theta \leq n$. In general, it is going to be difficult to obtain analytical description or solution for the dispersion of both the phase refractive index and effective index. As in Example 9.1, we usually obtain the relevant data by the numerical calculations.

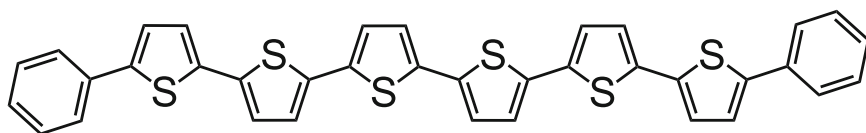
Under these circumstances, to establish a design principle of high-performance organic light-emitting devices Yamao et al. [7] have developed a light-emitting device where an organic crystal of P6T (see structural formula of Fig. 9.10a) was placed onto a diffraction grating (Fig. 9.10b [7]). The compound P6T is known as one of a family of thiophene/phenylene co-oligomers (TPCOs) together with BP1T, AC5, AC'7, etc. that appear in Fig. 9.8 [10, 11].

The diffraction grating was engraved using a focused ion beam (FIB) apparatus on an aluminum-doped zinc oxide (AZO) substrate. The resulting substrate was then laminated with the thin crystal of P6T (Fig. 9.10c). Using those devices, Yamao et al. excited the P6T crystal with ultraviolet light of a mercury lamp and collected the emissions from the crystal and analyzed those emission spectra (vide infra). Detailed analysis of those spectra has revealed that there is a close relationship between the peak location of emission lines and emission direction. Thus, to analyze the angle-dependent emission spectra turns out to be a powerful tool to accurately determine the dispersion of propagation constant of the laser material along with light-emitting materials more generally.

In a light-emitting device, we must consider a problem of out-coupling of light. Seeking efficient out-coupling of light is equivalent to solving equations of electromagnetic wave motion inside and outside the slab crystal under appropriate boundary conditions. The boundary conditions are given such that tangential components of electromagnetic fields are continuous across the interface formed by the slab crystal and air or a device substrate. Moreover, if a diffraction grating is present in an optical device (including the laser), the diffraction grating influences emission characteristics of the device. In other words, regarding the out-coupling of light we must impose the following condition on the device such that [7]

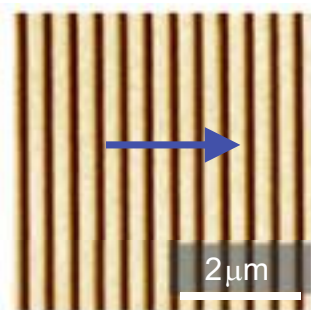
$$\beta - mK = (k_0 \cdot \tilde{e})\tilde{e}, \quad (9.105)$$

(a)



P6T

(b)



(c)

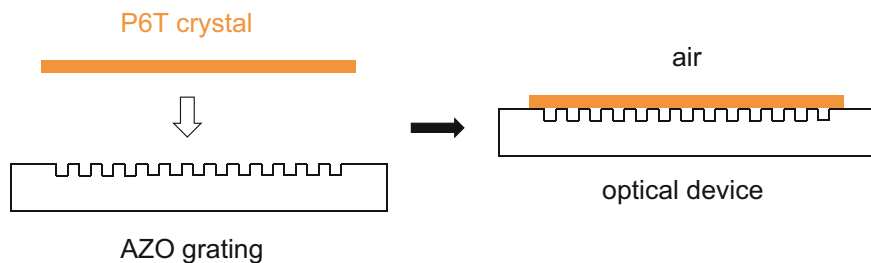
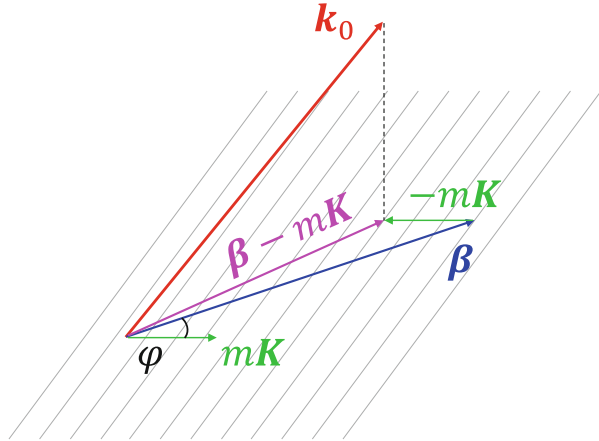


Fig. 9.10 Construction of an organic light-emitting device. (a) Structural formula of P6T. (b) Diffraction grating of an organic device. The purple arrow indicates the direction of grating wavevector K . (Reproduced from Yamao T, Higashihara S, Yamashita S, Sano H, Inada Y, Yamashita K, Ura S, Hotta S (2018) Design principle of high-performance organic single-crystal light-emitting devices. *J Appl Phys* 123(23): 235501/13 pages [7], with the permission of AIP Publishing. <https://doi.org/10.1063/1.5030486>). (c) Cross-section of the device consisting of P6T crystal/AZO substrate

Fig. 9.11 Geometrical relationship among the light propagation in crystal (indicated with β), the propagation outside the crystal (k_0), and the grating wavevector (K). This geometry represents the phase matching between the emission inside the device (organic slab crystal) and out-coupled emission (i.e., the emission outside the device); see text



where β is the propagation vector with $|\beta| \equiv \beta$ that appeared in (8.153). The quantity β is called a propagation constant. In (9.105), furthermore, m is the diffraction order with a natural number (i.e., a positive integer); k_0 denotes the wavenumber vector of an emission in vacuum with

$$|k_0| \equiv k_0 = 2\pi/\lambda_p,$$

where λ_p is the emission peak wavelength in vacuum. Note that k_0 is almost identical to the wavenumber of the emission in air (i.e., the emission to be detected). The unit vector $\tilde{e}$ that is oriented in the direction parallel to $\beta - mK$ within the crystal plane is defined as

$$\tilde{e} = \frac{\beta - mK}{|\beta - mK|}. \quad (9.106)$$

In the above equation, K is the grating wavevector defined as

$$|K| \equiv K = 2\pi/\Lambda,$$

where Λ is the grating period and the direction of K is perpendicular to the grating grooves; that direction is indicated in Fig. 9.10b with a purple arrow. Equation (9.105) can be visualized in Fig. 9.11 as geometrical relationship among the light propagation in crystal (indicated with β), the propagation outside the crystal (k_0), and the grating wavevector (K).

If we are dealing with the emission parallel to the substrate plane, i.e., grazing emission, (9.105) can be reduced to

$$\beta - mK = k_0. \quad (9.107)$$

Equations (9.105) and (9.107) represent the relationship between β and k_0 , i.e., the phase matching conditions between the emission inside the device (organic slab crystal) and the out-coupled emission, namely the emission outside the device (in air).

Thus, the boundary conditions can be restated as (1) the tangential component continuity of electromagnetic fields at both the planes of interface (see Sect. 8.1) and (2) the phase matching of the electromagnetic fields both inside and outside the laser medium. We examine these two factors below.

1. Tangential component continuity of electromagnetic fields (waveguide condition):

Let us suppose that we are dealing with a dielectric medium with anisotropic dielectric constant, but isotropic magnetic permeability. In this situation, Maxwell's equations must be formulated so that we can deal with the electromagnetic properties in an anisotropic medium. Such substances are widely available and organic crystals are counted as typical examples. Among those crystals, P6T crystallizes in the monoclinic system as in many other cases of organic crystals [10, 11]. In this case, the permittivity tensor (or electric permittivity tensor) ϵ is written as [7]

$$\epsilon = \begin{pmatrix} \epsilon_{aa} & 0 & \epsilon_{ac^*} \\ 0 & \epsilon_{bb} & 0 \\ \epsilon_{c^*a} & 0 & \epsilon_{c^*c^*} \end{pmatrix}. \quad (9.108)$$

Notice that in (9.108) the permittivity tensor is described in the orthogonal coordinate of abc^* -system, where a and b coincide with those of the crystallographic a - and b -axes of P6T with the c^* -axis being perpendicular to the ab -plane. Note that in many of organic crystals the c -axis is not perpendicular to the ab -plane. Meanwhile, we assume that the magnetic permeability μ of P6T is isotropic and identical to that of vacuum. That is, in a tensor form we have

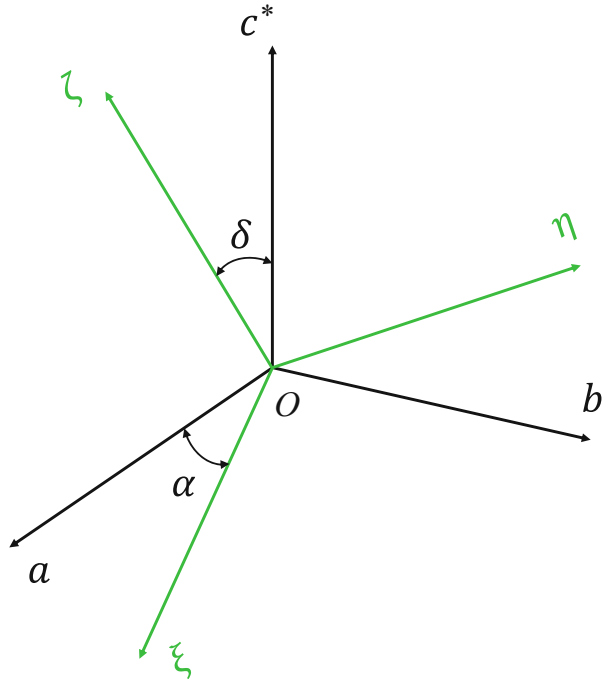
$$\mu = \begin{pmatrix} \mu_0 & 0 & 0 \\ 0 & \mu_0 & 0 \\ 0 & 0 & \mu_0 \end{pmatrix}, \quad (9.109)$$

where μ_0 is the magnetic permeability of vacuum.

Then, the Maxwell's equations in a matrix form read as

$$\text{rot } \mathbf{E} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t}, \quad (9.110)$$

Fig. 9.12 Laboratory coordinate system (the $\xi\eta\zeta$ -system) for the device experiments. Regarding the symbols and notations, see text



$$\text{rot } \mathbf{H} = \varepsilon_0 \begin{pmatrix} \varepsilon_{aa} & 0 & \varepsilon_{ac^*} \\ 0 & \varepsilon_{bb} & 0 \\ \varepsilon_{c^*a} & 0 & \varepsilon_{c^*c^*} \end{pmatrix} \frac{\partial \mathbf{E}}{\partial t}. \quad (9.111)$$

In (9.111) the equation is described in the abc^* -system. But, it will be desired to describe the Maxwell's equations in the laboratory coordinate system (see Fig. 9.12) so that we can readily visualize the light propagation in an anisotropic crystal such as P6T. Figure 9.12 depicts the $\xi\eta\zeta$ -system, where the ξ -axis is in the direction of the light propagation within the crystal, namely the ξ -axis parallels β in (9.107). We assume that the $\xi\eta\zeta$ -system forms an orthogonal coordinate.

Here we define the dielectric constant ellipsoid $\tilde{\mathbf{e}}$ described by

$$(\xi \ \eta \ \zeta) \tilde{\mathbf{e}} \begin{pmatrix} \xi \\ \eta \\ \zeta \end{pmatrix} = 1, \quad (9.112)$$

which represents an ellipsoid in the $\xi\eta\zeta$ -system. If one cuts the ellipsoid by a plane that includes the origin of the coordinate system and is perpendicular to the ξ -axis, its cross-section is an ellipse. In this situation, one can choose the η - and ζ -axes for the principal axis of the ellipse. This implies that when we put $\xi = 0$, we must have

$$(0 \ \eta \ \zeta) \tilde{\mathbf{e}} \begin{pmatrix} 0 \\ \eta \\ \zeta \end{pmatrix} = \varepsilon_{\eta\eta} \eta^2 + \varepsilon_{\zeta\zeta} \zeta^2 = 1. \quad (9.113)$$

In other words, we must have $\tilde{\mathbf{e}}$ in the form of

$$\tilde{\mathbf{e}} = \begin{pmatrix} \varepsilon_{\xi\xi} & \varepsilon_{\xi\eta} & \varepsilon_{\xi\zeta} \\ \varepsilon_{\eta\xi} & \varepsilon_{\eta\eta} & 0 \\ \varepsilon_{\zeta\xi} & 0 & \varepsilon_{\zeta\zeta} \end{pmatrix},$$

where $\tilde{\mathbf{e}}$ is expressed in the $\xi\eta\zeta$ -system. In terms of the matrix algebra, the principal submatrix with respect to the η - and ζ -components should be diagonalized (see Sect. 11.3). On this condition, the electric flux density of the electromagnetic wave is polarized in the η - and ζ -direction. A half of the reciprocal square root of the principal axes, i.e., $1/\sqrt{\varepsilon_{\eta\eta}}$ or $1/\sqrt{\varepsilon_{\zeta\zeta}}$ represents the anisotropic refractive indices.

Suppose that the $\xi\eta\zeta$ -system is reached by two successive rotations starting from the abc^* -system in such a way that we first perform the rotation by α around the c^* -axis that is followed by another rotation by δ around the ξ -axis (Fig. 9.12). Then we have [7]

$$\begin{pmatrix} \varepsilon_{\xi\xi} & \varepsilon_{\xi\eta} & \varepsilon_{\xi\zeta} \\ \varepsilon_{\eta\xi} & \varepsilon_{\eta\eta} & 0 \\ \varepsilon_{\zeta\xi} & 0 & \varepsilon_{\zeta\zeta} \end{pmatrix} = R_{\xi}^{-1}(\delta) R_{c^*}^{-1}(\alpha) \begin{pmatrix} \varepsilon_{aa} & 0 & \varepsilon_{ac^*} \\ 0 & \varepsilon_{bb} & 0 \\ \varepsilon_{c^*a} & 0 & \varepsilon_{c^*c^*} \end{pmatrix} R_{c^*}(\alpha) R_{\xi}(\delta), \quad (9.114)$$

where $R_{c^*}(\alpha)$ and $R_{\xi}(\delta)$ stand for the first and second rotation matrices of the above, respectively. In (9.114) we have

$$R_{c^*}(\alpha) = \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad R_{\xi}(\delta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \delta & -\sin \delta \\ 0 & \sin \delta & \cos \delta \end{pmatrix}. \quad (9.115)$$

With the matrix presentations and related coordinate transformations for (9.114) and (9.115), see Sects. 11.4, 17.1, and 17.4.2. Note that the rotation angle δ cannot be decided independent of α . In the literature [7], furthermore, the quantity $\varepsilon_{\xi\eta} = \varepsilon_{\eta\xi}$ was ignored as small compared to other components. As a result, (9.111) is converted into the following equation described by

$$\text{rot } \mathbf{H} = \varepsilon_0 \begin{pmatrix} \varepsilon_{\xi\xi} & 0 & \varepsilon_{\xi\zeta} \\ 0 & \varepsilon_{\eta\eta} & 0 \\ \varepsilon_{\zeta\xi} & 0 & \varepsilon_{\zeta\zeta} \end{pmatrix} \frac{\partial \mathbf{E}}{\partial t}. \quad (9.116)$$

Notice here that the electromagnetic quantities $\mathbf{H}$ and $\mathbf{E}$ are measured in the $\xi\eta\zeta$ -coordinate system.

Meanwhile, the electromagnetic waves that are propagating within the slab crystal are described as

$$\Phi_\nu \exp[i(\beta\xi - \omega t)], \quad (9.117)$$

where Φ_ν stands for either an electric field or a magnetic field with ν chosen from $\nu = 1, 2, 3$ representing each component of the $\xi\eta\zeta$ -system. Inserting (9.117) into (9.116) and separating the resulting equation into ξ , η , and ζ components, we obtain six equations with respect to six components of $\mathbf{H}$ and $\mathbf{E}$. Of these, we are particularly interested in E_ξ and E_ζ as well as H_η , because we assume that the electromagnetic wave is propagated as a TM mode [7].

Using the set of the above six equations, with H_η in the P6T crystal we get the following second-order linear differential equation (SOLDE):

$$\frac{d^2 H_\eta}{d\zeta^2} + 2in_{\text{eff}}k_0 \frac{\varepsilon_{\xi\xi}}{\varepsilon_{\zeta\zeta}} \frac{dH_\eta}{d\zeta} + \left[\left(\varepsilon_{\xi\xi} - \frac{\varepsilon_{\xi\zeta}^2}{\varepsilon_{\zeta\zeta}} \right) - \frac{\varepsilon_{\xi\xi}}{\varepsilon_{\zeta\zeta}} n_{\text{eff}}^2 \right] k_0^2 H_\eta = 0, \quad (9.118)$$

where $n_{\text{eff}} = n \sin \theta$ in (8.153) is the effective index. Assuming as a solution of (9.118)

$$H_\eta = H_{\text{cryst}} e^{-i\kappa\zeta} \cos(k\zeta + \phi_s) \quad (\kappa \neq 0, k \neq 0, H_{\text{cryst}}, \phi_s : \text{constant}) \quad (9.119)$$

and inserting (9.119) into (9.118), we get a following type of equation described by

$$Ze^{-i\kappa\zeta} \cos(k\zeta + \phi_s) + \Omega e^{-i\kappa\zeta} \sin(k\zeta + \phi_s) = 0. \quad (9.120)$$

In (9.119) H_{cryst} represents the magnetic field within the P6T crystal. The constants Z and Ω in (9.120) can be described using κ and k together with the constant coefficients of $dH_\eta/d\zeta$ and H_η of SOLDE (9.118). Meanwhile, we have

$$\left| \begin{array}{cc} e^{-i\kappa\zeta} \cos(k\zeta + \phi_s) & e^{-i\kappa\zeta} \sin(k\zeta + \phi_s) \\ [e^{-i\kappa\zeta} \cos(k\zeta + \phi_s)]' & [e^{-i\kappa\zeta} \sin(k\zeta + \phi_s)]' \end{array} \right| = ke^{-2i\kappa} \neq 0,$$

where the differentiation is pertinent to ζ . This shows that $e^{-i\kappa\zeta} \cos(k\zeta + \phi_s)$ and $e^{-i\kappa\zeta} \sin(k\zeta + \phi_s)$ are linearly independent. This in turn implies that

$$Z \equiv \Omega \equiv 0 \quad (9.121)$$

in (9.120). Thus, from (9.121) we can determine κ and k such that

$$\kappa = \beta \frac{\varepsilon_{\xi\zeta}}{\varepsilon_{\zeta\zeta}} = n_{\text{eff}} k_0 \frac{\varepsilon_{\xi\zeta}}{\varepsilon_{\zeta\zeta}}, \quad k^2 = \frac{1}{\varepsilon_{\zeta\zeta}^2} (\varepsilon_{\xi\xi} \varepsilon_{\zeta\zeta} - \varepsilon_{\xi\zeta}^2) (\varepsilon_{\zeta\zeta} - n_{\text{eff}}^2) k_0^2. \quad (9.122)$$

Further using the aforementioned six equations to eliminate E_ζ , we obtain

$$E_{\xi} = i \frac{\varepsilon_{\xi\xi}^{\prime\prime}}{\omega \varepsilon_0 (\varepsilon_{\xi\xi}^{\prime} \varepsilon_{\xi\xi}^{\prime\prime} - \varepsilon_{\xi\xi}^{\prime 2})} k H_{\text{cryst}} e^{-ik_{\xi}^{\prime} \zeta} \sin(k_{\xi}^{\prime} \zeta + \phi_s). \quad (9.123)$$

Equations (9.119) and (9.123) define the electromagnetic fields as a function of ζ within the P6T slab crystal.

Meanwhile, the fields in the AZO substrate and air can readily be determined. Assuming that both the substances are isotropic, we have $\varepsilon_{\xi\xi}^{\prime} = 0$ and $\varepsilon_{\xi\xi}^{\prime} = \varepsilon_{\xi\xi}^{\prime} \approx \tilde{n}^2$ and, hence, we get

$$\frac{d^2 H_{\eta}}{d\zeta^2} + (\tilde{n}^2 - n_{\text{eff}}^2) k_0^2 H_{\eta} = 0, \quad (9.124)$$

where $\tilde{n}$ is the refractive index of either the substrate or air. Since

$$1 < \tilde{n} \leq n_{\text{eff}} \leq n, \quad (9.125)$$

where n is the phase refractive index of the P6T crystal, we have $\tilde{n}^2 - n_{\text{eff}}^2 < 0$.

Defining a quantity

$$\gamma \equiv \sqrt{n_{\text{eff}}^2 - \tilde{n}^2} k_0 \quad (9.126)$$

for the substrate (i.e., AZO) and air, from (9.124) we express the electromagnetic fields as

$$H_{\eta} = \tilde{H} e^{\gamma \zeta}, \quad (9.127)$$

where $\tilde{H}$ denotes the magnetic field relevant to the substrate or air. Considering that $\zeta < 0$ for the substrate and $\zeta > d/\cos \delta$ in air (see Fig. 9.13), with the magnetic field we have

$$H_{\eta} = \tilde{H} e^{\gamma \zeta} \quad \text{and} \quad H_{\eta} = \tilde{H} e^{-\gamma(\zeta - \frac{d}{\cos \delta})} \quad (9.128)$$

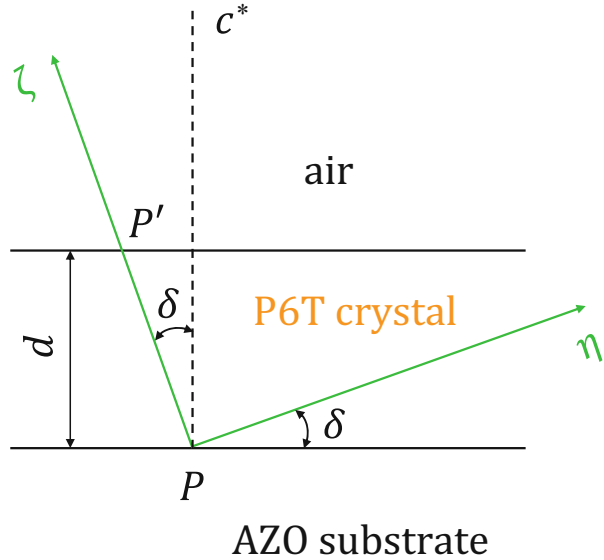
for the substrate and air, respectively. Notice that $H_{\eta} \rightarrow 0$, when $\zeta \rightarrow -\infty$ with the substrate and $\zeta \rightarrow \infty$ for air. Correspondingly, with the electric field we get

$$E_{\xi} = -i \frac{\gamma}{\omega \varepsilon_0 \tilde{n}^2} \tilde{H} e^{\gamma \zeta} \quad \text{and} \quad E_{\xi} = i \frac{\gamma}{\omega \varepsilon_0 \tilde{n}^2} \tilde{H} e^{-\gamma(\zeta - \frac{d}{\cos \delta})} \quad (9.129)$$

for the substrate and air, respectively.

The device geometry is characterized by that the P6T slab crystal is sandwiched by air and the device substrate so as to form the three-layered (air/crystal/substrate) structure (Fig. 9.10c). Therefore, we impose boundary

Fig. 9.13 Geometry of the cross-section of air/P6T crystal/AZO substrate. The angle δ is identical with that of Fig. 9.12. The points P and P' are located at the crystal/substrate interface and air/crystal interface, respectively. The distance between P and P' is $d/\cos \delta$



conditions on H_η in (9.119) and (9.128) along with E_ξ in (9.123) and (9.129) in such a way that their tangential components are continuous across the interfaces between the crystal and the substrate and between the crystal and air.

Figure 9.13 represents the geometry of the cross-section of air/P6T crystal/AZO substrate. From (9.119) and (9.128), (a) the tangential continuity condition for the magnetic field at the crystal/substrate interface is described as

$$H_{\text{cryst}} e^{-ik \cdot 0} \cos(k \cdot 0 + \phi_s) (\cos \delta) = \tilde{H} e^{\gamma \cdot 0} (\cos \delta), \quad (9.130)$$

where the factor of $\cos \delta$ comes from the requirement of the tangential continuity condition. (b) The tangential continuity condition for the electric field at the same interface reads as

$$i \frac{\epsilon_{\zeta\zeta}}{\omega \epsilon_0 (\epsilon_{\xi\xi} \epsilon_{\zeta\zeta} - \epsilon_{\xi\zeta}^2)} k H_{\text{cryst}} e^{-ik \cdot 0} \sin(k \cdot 0 + \phi_s) = -i \frac{\gamma}{\omega \epsilon_0 \tilde{n}^2} \tilde{H} e^{\gamma \cdot 0}. \quad (9.131)$$

Thus, dividing both sides of (9.131) by both sides of (9.130), we get

$$\frac{\epsilon_{\zeta\zeta}}{\epsilon_{\xi\xi} \epsilon_{\zeta\zeta} - \epsilon_{\xi\zeta}^2} k \tan \phi_s = -\frac{\gamma}{\tilde{n}^2} \quad \text{or} \quad \frac{\epsilon_{\zeta\zeta}}{\epsilon_{\xi\xi} \epsilon_{\zeta\zeta} - \epsilon_{\xi\zeta}^2} k \tan(-\phi_s) = \frac{\gamma}{\tilde{n}^2}, \quad (9.132)$$

where γ and $\tilde{n}$ are *substrate related* quantities; see (9.125) and (9.126). Another boundary condition at the air/crystal interface can be obtained in a similar manner. Notice that in that case $\zeta = d/\cos \delta$; see Fig. 9.13. As a result, we have

$$\frac{\varepsilon_{\zeta\zeta}}{\varepsilon_{\xi\xi}\varepsilon_{\zeta\zeta} - \varepsilon_{\xi\zeta}^2} k \tan\left(\frac{kd}{\cos \delta} + \phi_s\right) = \frac{\gamma}{\tilde{n}^2}, \quad (9.133)$$

where γ and $\tilde{n}$ are *air related* quantities. The quantity ϕ_s can be eliminated considering arc tangent of (9.132) and (9.133).

In combination with (9.122), we finally get the following eigenvalue equation for TM modes:

$$\begin{aligned} & k_0 d \sqrt{\frac{1}{\varepsilon_{\zeta\zeta}^2} (\varepsilon_{\xi\xi}\varepsilon_{\zeta\zeta} - \varepsilon_{\xi\zeta}^2) (\varepsilon_{\zeta\zeta} - n_{\text{eff}}^2)} / (\cos \delta) \\ &= l\pi + \tan^{-1} \left[\frac{1}{n_{\text{air}}^2} \sqrt{(\varepsilon_{\xi\xi}\varepsilon_{\zeta\zeta} - \varepsilon_{\xi\zeta}^2) \left(\frac{n_{\text{eff}}^2 - n_{\text{air}}^2}{\varepsilon_{\zeta\zeta} - n_{\text{eff}}^2} \right)} \right] \\ &+ \tan^{-1} \left[\frac{1}{n_{\text{sub}}^2} \sqrt{(\varepsilon_{\xi\xi}\varepsilon_{\zeta\zeta} - \varepsilon_{\xi\zeta}^2) \left(\frac{n_{\text{eff}}^2 - n_{\text{sub}}^2}{\varepsilon_{\zeta\zeta} - n_{\text{eff}}^2} \right)} \right], \end{aligned} \quad (9.134)$$

where $n_{\text{air}} (=1)$ is the refractive index of air, n_{sub} is that of AZO substrate equipped with a diffraction grating, d is the crystal thickness, and l is the order of the transverse mode. The refractive index n_{sub} is estimated as the average of the refractive indices of air and AZO weighted by volume fraction they occupy in the diffraction grating.

Regarding the index l , the condition $l \geq 0$ (i.e., zero or a positive integer) must be satisfied. Note that $l = 0$ is allowed because the total internal reflection (Sects. 8.5 and 8.6) is responsible in the crystal waveguide. Thus, n_{eff} can be determined as a function of k_0 with d and l being parameters. To solve (9.134), iterative numerical computation is needed. The detailed calculation procedures for this can be seen in the literature [7] and supplementary material therein.

Note that (9.134) includes the well-known eigenvalue equation for a waveguide that comprises an isotropic core dielectric medium sandwiched by a couple of clad layers having the same refractive index (symmetric waveguide) [12]. In that case, in fact, the second and third terms in RHS of (9.134) are the same and their sum is identical with $-\delta_{\text{TM}}$ of (8.176) in Sect. 8.7.2. To confirm it, in (9.134) use $\varepsilon_{\xi\xi} = \varepsilon_{\zeta\zeta} \approx n^2$ [see (7.57)] and $\varepsilon_{\xi\zeta} = 0$ together with the relative refractive index given by (8.39).

2. Phase matching of the electromagnetic fields (transverse mode requisition):

Once we have obtained the eigenvalue equation described by (9.134), we will be able to solve the problem from the crystal thickness d . Even though the order of the transverse mode l is yet unknown, l is a discrete value (i.e., zero or a

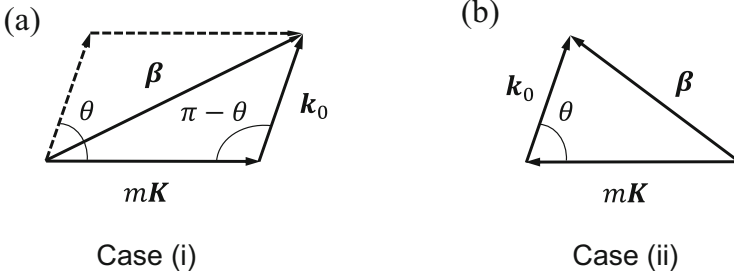


Fig. 9.14 Two cases of the light propagation in the device geometry of slab waveguide. (a) The projection of the vectors $\boldsymbol{\beta}$ and $\mathbf{k}_0$ onto $\mathbf{K}$ is parallel to each other. (b) The projection $\boldsymbol{\beta}$ and $\mathbf{k}_0$ onto $\mathbf{K}$ is antiparallel. For sake of simplicity, the projection of $\boldsymbol{\beta}$ onto $\mathbf{K}$ is parallel to $\mathbf{K}$ for both the cases. The number m is assumed to be a positive integer

positive integer) so that the computation of (9.134) can be simplified. Namely, inserting probable values l into (9.134) we can greatly save time and effort to get a solution.

Meanwhile, the condition of the phase matching is equivalent to determining the index m related to the longitudinal mode. To determine m , we return back to (9.107). Figure 9.14 features following two cases with regard to the light propagation represented by (9.107) in the device geometry of slab waveguide. Case (1): The projection of the vectors $\boldsymbol{\beta}$ and $\mathbf{k}_0$ onto $\mathbf{K}$ is parallel to each other (see Fig. 9.14a). Case (2): The projection $\boldsymbol{\beta}$ and $\mathbf{k}_0$ onto $\mathbf{K}$ are antiparallel (Fig. 9.14b). The angle θ of Fig. 9.14 is defined as that made by $\mathbf{k}_0$ and $\mathbf{K}$. Suppose geometric transformations that convert the vectors $\boldsymbol{\beta}$, $\mathbf{k}_0$, and $\mathbf{K}$ to those having the opposite direction. Then, the mutual geometric arrangement of these vectors remains unchanged together with the form of (9.107) before and after the transformations. Concomitantly, the angle θ is switched to $\theta - \pi$. As a result, if we set θ as $0 \leq \theta \leq \pi/2$, we have $-\pi \leq \theta - \pi \leq -\pi/2$.

Meanwhile, let us think of the reflection with respect to the plane that contains the vector $\mathbf{K}$ and is perpendicular to the plane of paper. In that case, we are uncertain as to whether $\boldsymbol{\beta}$ undergoes the same reflection according to the said reflection operation. It is because the dielectric constant ellipsoid does not necessarily possess the mirror symmetry with respect to that plane in an anisotropic medium. We need the information about the emission data taken at the angle $-\theta$ accordingly. Similarly, setting $-\theta$ as $0 \leq -\theta \leq \pi/2$ (or $-\pi/2 \leq \theta \leq 0$), we have $-3\pi/2 \leq \theta - \pi \leq -\pi$. Combining the above results, the ranges of θ and $\theta - \pi$ cover the emission angles $[-3\pi/2, \pi/2]$. It is equivalent to the whole range $[0, 2\pi]$. Thus, the angle-dependent measurements covering $-\pi/2 \leq \theta \leq \pi/2$ provide us with sufficient data to be analyzed.

Now, care should be taken to determine $\boldsymbol{\beta}$. This is because from an experimental perspective $\mathbf{k}_0$ can be decided accurately but arbitrariness is involved with $\boldsymbol{\beta}$. Defining $\beta \equiv |\boldsymbol{\beta}|$, $k_0 \equiv |\mathbf{k}_0|$, and $K \equiv |\mathbf{K}|$, we examine the relationship among

magnitudes of β , k_0 , and K according to the above two cases. Using an elementary trigonometric formula, we describe the results as follows:

$$\beta^2 = k_0^2 + m^2 K^2 - 2mKk_0 \cos(\pi - \theta) = k_0^2 + m^2 K^2 + 2mKk_0 \cos \theta,$$

or

$$\beta^2 = k_0^2 + m^2 K^2 - 2mKk_0 \cos \theta, \quad (9.135)$$

where we assume that m is a positive integer. In (9.135) the former and latter equations correspond to Case (1) and Case (2), respectively (see Fig. 9.14). Replacing β with $\beta = k_0 n_{\text{eff}}$ and taking the variation of the resulting equation with regard to k_0 and θ , we get

$$\frac{\delta k_0}{\delta \theta} = \frac{\mp mKk_0 \sin \theta}{k_0(n_{\text{eff}}^2 - 1) \mp mK \cos \theta}, \quad (9.136)$$

where the signs $-$ and $+$ are associated with Case (1) and Case (2), respectively. In (9.136) we did not take the variation of n_{eff} , because it was assumed to be weakly dependent on θ compared to k_0 . Note that since k_0 was not allowed to take continuous numbers, we used the sign of variation δ instead of the sign of differentiation d or ∂ . The denominator of (9.136) is positive (readers, check it), and so $\delta k_0 / \delta \theta$ is negative or positive with $0 \leq \theta \leq \pi/2$ for Case (1) and Case (2), respectively. The quantity $\delta k_0 / \delta \theta$ is positive or negative with $-\pi/2 \leq \theta \leq 0$ for Case (1) and Case (2), respectively. For both Case (1) and Case (2), $\delta k_0 / \delta \theta = 0$ at $\theta = 0$, namely, k_0 (or the emission wavelength) takes an extremum at $\theta = 0$. In terms of the emission wavelength, Case (1) and Case (2) are related to the redshift and blueshift in the spectra, respectively. These shifts can be detected and inspected closely by the angle-dependent emission spectra (vide infra).

Meanwhile, taking inner products (see Chap. 13) of both sides of (9.107), we have

$$\langle \beta - mK | \beta - mK \rangle = \langle k_0 | k_0 \rangle.$$

That is, we get

$$\beta^2 + m^2 K^2 - 2mK\beta \cos \varphi = k_0^2, \quad (9.137)$$

where φ is an angle between mK and β (see Fig. 9.11). To precisely decide φ is accompanied by experimental ambiguity, however. To avoid the ambiguity, therefore, it is preferred to deal with a case where all the vectors β , K , and k_0 are (anti)parallel to $\tilde{e}$ in (9.105). From a practical point of view, moreover, it is the most desired to make a device so that it can strongly emit light in the direction parallel to the grating wavevector.

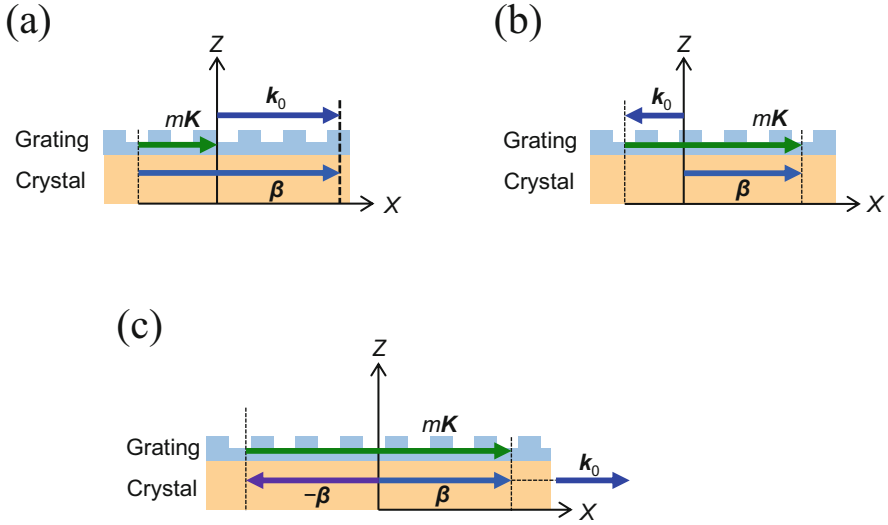


Fig. 9.15 Schematic diagram for the relationship among k_0 , β , and K . This geometry is obtained by substituting $\varphi = 0$ for (9.137). (a) Redshifted case. (b) Blueshifted case. (c) Bragg's condition case. All these diagrams are depicted as a cross section of the devices cut along the K direction. (Reproduced from Inada Y, Kawata Y, Kawai T, Hotta S, Yamao T (2020) Laser oscillation at an intended wavelength from a distributed feedback structure on anisotropic organic crystals. *J Appl Phys* 127(5): 053102/6 pages [13], with the permission of AIP Publishing. <https://doi.org/10.1063/1.5129425>)

In that case, we have $\varphi = 0$ and (9.137) is reduced to

$$(\beta - mK)^2 = k_0^2 \quad \text{or} \quad \beta - mK = \pm k_0.$$

Then, this gives us two choices such that

$$\begin{aligned} (1) \quad & \beta = k_0 + mK = n_{\text{eff}} k_0, \quad k_0 = \frac{2\pi}{\lambda_p} = \frac{mK}{n_{\text{eff}} - 1}, \\ (2) \quad & -\beta = k_0 - mK = -n_{\text{eff}} k_0, \quad k_0 = \frac{2\pi}{\lambda_p} = \frac{mK}{n_{\text{eff}} + 1}. \end{aligned} \quad (9.138)$$

In (9.138), all the physical quantities are assumed to be positive. According to the above two cases, we have the following corresponding geometry such that

1. $\beta, K \parallel k_0$ (β and K are parallel to k_0),
2. $\beta, K \parallel_a k_0$ (β and K are antiparallel to k_0).

The above geometry is depicted in Fig. 9.15 [13]. Figure 9.15 includes a case of the Bragg's condition as well (Fig. 9.15c). Notice that Fig. 9.15a and Fig. 9.15b are special cases of the configurations depicted in Fig. 9.14a and Fig. 9.14b, respectively.

Further rewriting (9.138) to combine the two equations, we get

$$\lambda_p = \frac{2\pi(n_{\text{eff}} \mp 1)}{mK} = \frac{(n_{\text{eff}} \mp 1)\Lambda}{m}, \quad (9.139)$$

where the sign $\mp$ corresponds to the aforementioned Case (1) and Case (2), respectively; we used the relation of $K = 2\pi/\Lambda$. Equation (9.139) is useful for us to guess the magnitude of m . It is because since m is changed stepwise (e.g., $m = 1, 2, 3, \text{etc.}$), we have only to examine a limited number of m so that n_{eff} may satisfy (9.125). When the laser oscillation is responsible, (9.138) and (9.139) should be replaced with the following relation that represents the Bragg's condition [14]:

$$2\beta = mK \quad (9.140)$$

or

$$\lambda_p = \frac{2n_{\text{eff}}\Lambda}{m}. \quad (9.141)$$

Once we find out a suited m , (9.107) enables us to seek the relationship between λ_p and n_{eff} experimentally.

As described above, we have explained two major factors of the out-coupling of emission. In the above discussion, we assumed that the TM modes dominate. Combining (9.107) and (9.134), we successfully assign individual emissions to the specific TM_{ml} modes, in which the indices m and l are longitudinal and transverse modes, respectively. Note that $m \geq 1$ and $l \geq 0$. Whereas the index m is associated with the presence of diffraction grating, l is pertinent to the waveguide condition.

In general, it is quite difficult to precisely predict the phase refractive index and effective index of anisotropic dielectric materials such as organic crystals. This situation leads to the difficulty in developing optical devices based on the organic crystals. In that respect, (9.134) is highly useful in designing high-performance optical devices. At the same time, the angle-dependent emission spectroscopy offers various information necessary to construct the device.

As a typical example, Fig. 9.16a [7] shows the angle-dependent emission spectra of a P6T crystal. For this, the grazing emission was intended. There are two series of progressions in which the emission peak locations are either redshifted or blueshifted with increasing angles $|\theta|$; the angle θ is defined as that made between two vectors $\mathbf{K}$ and $\mathbf{k}_0$; see Figs. 9.14 and 9.16b as well as (9.135). Also, these progressions exhibit the extrema at $\theta = 0$ as expected. Thus, Fig. 9.16 obviously demonstrates that the redshifted and blueshifted progressions result from the device geometry represented by Fig. 9.14a [corresponding to Case (1)] and Fig. 9.14b [Case (2)], respectively.

The first and second equations of (9.138) are associated with the emission spectra recorded at $\theta = 0$ in a series of the redshifted and blueshifted progressions, respectively. That is, the index n_{eff} was determined from (9.138) by replacing λ_p with the peak wavelength observed at $\theta = 0$ of Fig. 9.16a and replacing m with the

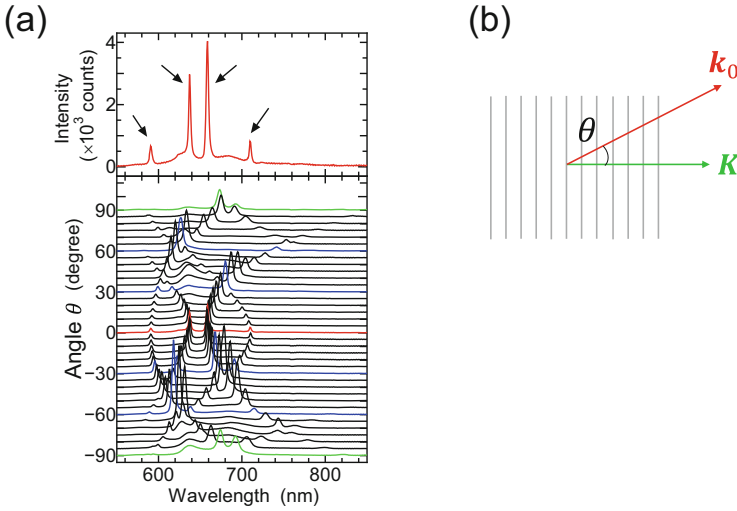


Fig. 9.16 Angle-dependent emission spectra of a P6T crystal. (a) Emission spectra. (Reproduced from Yamao T, Higashihara S, Yamashita S, Sano H, Inada Y, Yamashita K, Ura S, Hotta S (2018) Design principle of high-performance organic single-crystal light-emitting devices. *J Appl Phys* 123(23): 235501/13 pages [7], with the permission of AIP Publishing. <https://doi.org/10.1063/1.5030486>). (b) Geometrical relationship between k_0 and K . The grazing emissions (k_0) were collected at various angles θ made with the grating wavevector direction (K)

most suitable integer m (vide supra). In such a manner, individual experimental emission data were assigned to a specific TM_{ml} mode. Namely, the TM_{ml} mode assignment has been completed by combining (k_0, l) that was determined from (9.134) and (k_0, m) decided by (9.138). Notice that the same number m is held unchanged within a same series of redshifted or blueshifted progression (i.e., $-90^\circ \leq \theta \leq 90^\circ$ in Fig. 9.16a). This allows us to experimentally determine the dispersion of effective indices within the same progression (either redshifted or blueshifted) using (9.107).

The emission data experimentally obtained and assigned above are displayed in Fig. 9.17 [7]. The observed emission peaks are indicated with open circles/triangles or closed circles/triangles with blueshifted or redshifted peaks, respectively. In Fig. 9.17, the TM_{20} (blueshifted) and TM_{10} (redshifted) modes are seamlessly jointed so as to be an upper group of emission peaks. A lower group of emission peaks are assigned to TM_{21} (blueshifted) or TM_{11} (redshifted) modes. For those emission data, once again, the relevant modes are seamlessly jointed. Notice that these modes may have multiple values of n_{eff} according to different $\epsilon_{\xi\xi}$, $\epsilon_{\zeta\zeta}$, and $\epsilon_{\xi\zeta}$ that vary with the different propagation directions of light (i.e., the ξ -axis) within the slab crystal waveguide plane.

The color solid curves represent the results obtained by purely numerical computations based on (9.134). Nonetheless, the dielectric constant ellipsoid elements $\epsilon_{\xi\xi}$, $\epsilon_{\zeta\zeta}$, and $\epsilon_{\xi\zeta}$ reflect the direction of the wavenumber vector k_0 through (9.107)

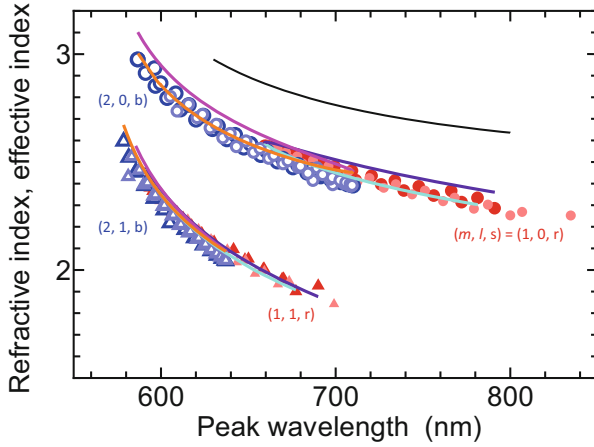


Fig. 9.17 Wavelength dispersion of the effective indices n_{eff} for the P6T crystal waveguide. The open and closed symbols show the data pertinent to the blueshifted and redshifted peaks, respectively. These data were calculated from either (9.105) or (9.107). The colored solid curves represent the dispersions of the effective refractive indices computed from (9.134). The numbers and characters (m, l, s) denote the diffraction order (m), order of transverse mode (l), and shift direction (s) that distinguishes between blueshift (b) and redshift (r). A black solid curve drawn at the upper right indicates the wavelength dispersion of the phase refractive indices (n) of the P6T crystal related to the one principal axis [15]. (Reproduced from Yamao T, Higashihara S, Yamashita S, Sano H, Inada Y, Yamashita K, Ura S, Hotta S (2018) Design principle of high-performance organic single-crystal light-emitting devices. *J Appl Phys* 123(23): 235501/13 pages [7], with the permission of AIP Publishing. <https://doi.org/10.1063/1.5030486>)

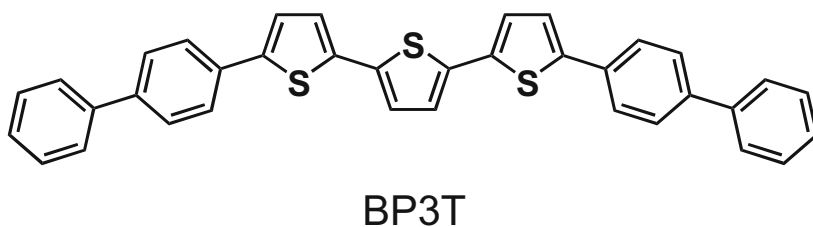
and, hence, (9.134) is associated with the experimental data in that sense. As clearly recognized from Fig. 9.17, agreement between the experiments and numerical computations is satisfactory. These curves are compared with a black solid curve drawn at the upper right that indicates the wavelength dispersion of the phase refractive indices (n) of the P6T crystal related to the one principal axis [15].

In a practical sense, it will be difficult to make a device that causes the laser oscillation at an intended wavelength using an anisotropic crystal as in the case of present example. It is because mainly of difficulty in predicting n_{eff} of the anisotropic crystal. From (9.134), however, we can readily determine the wavelength dispersion of n_{eff} of the crystal. From the Bragg's condition (9.141), in turn, Λ can precisely be designed for a given index m and suitable λ_p (at which n_{eff} can be computed with a crystal of its thickness d). Naturally, it is sensible to set λ_p at a wavelength where high optical gain (or emission gain) is anticipated [16].

Example 9.3 [13] Thus, everything has been arranged for one to construct a high-performance optical device, especially a laser. Using a single crystal of BP3T (see Fig. 9.18a for the structural formula), Inada and co-workers have developed a device that causes the laser oscillation at an intended wavelength [13].

To this end, the BP3T crystal was placed on a silicon oxide substrate and subsequently spin-coated with a photoresist film. The photoresist film was then

(a)



(b)

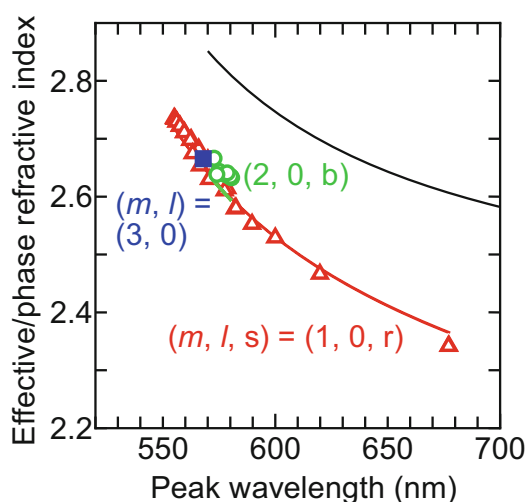


Fig. 9.18 Wavelength dispersions of the effective/phase refractive indices for the BP3T crystal waveguide. (a) Structural formula of BP3T. (b) Dispersion data of the device D1 (see Table 9.2). Colored solid curves (either green or red) are obtained from (9.134). The data of green circles and red triangles are obtained through (9.107) using the emission directions and the emission line locations. The laser oscillation line is indicated with a blue square as a TM_{30} mode occurring at 568.1 nm. The black solid curve drawn at the upper right indicates the dispersion of the phase refractive index of the BP3T crystal pertinent to the one principal axis [15]. (Reproduced from Inada Y, Kawata Y, Kawai T, Hotta S, Yamao T (2020) Laser oscillation at an intended wavelength from a distributed feedback structure on anisotropic organic crystals. J Appl Phys 127(5): 053102/6 pages [13], with the permission of AIP Publishing. <https://doi.org/10.1063/1.5129425>)

engraved with a one-dimensional diffraction grating so that the a -axis of the BP3T crystal was parallel to the grating wavevector. With this device geometry, the angle-dependent spectroscopy was carried out in a weak excitation regime to determine the wavelength dependence of the effective index. At the same time, strong excitation measurement was performed to analyze the laser oscillation data.

Table 9.2 Laser characteristics of BP3T crystal devices^a

Device	Crystal thickness d (nm)	Grating period Λ (nm)	Actual lasing wavelength λ_a (nm)	Intended emission wavelength λ_i (nm)	Effective index n_{eff} at λ_a	Validity V^b
D1	428	319.8	568.1	569.0	2.65 ^c 2.66 ^d	0.994
D2	295	349.2	569.6	569.0	2.48 ^c 2.45 ^d	0.988

^a Reproduced from Inada Y, Kawata Y, Kawai T, Hotta S, Yamao T (2020) Laser oscillation at an intended wavelength from a distributed feedback structure on anisotropic organic crystals. *J Appl Phys* 127(5): 053102/6 pages [13], with the permission of AIP Publishing. <https://doi.org/10.1063/1.5129425>

^b See (9.142) in the text

^c Estimated from $n_{\text{eff}}(\text{WG})$ in (9.134)

^d Estimated from $n_{\text{eff}}(\text{PM})$ in (9.141)

The BP3T crystal exhibited a high emission gain around 570 nm [17]. Accordingly, setting the laser oscillation wavelength at 569.0 nm, Inada and co-workers made the BP3T crystal devices with different crystal thicknesses. As an example, using a crystal of 428 nm in thickness (device D1; Table 9.2), on the basis of (9.134) they estimated the effective index n_{eff} at 569.0 nm in the direction of the a -axis of the BP3T crystal. Meanwhile, using (9.141) and assuming m to be 3 in (9.141), the grating period was decided with the actual period of the resulting device being 319.8 nm (see Table 9.2).

The results of the emission measurements for the device D1 are displayed in Fig. 9.18b. The figure shows the dispersion of the effective indices n_{eff} of the BP3T crystal. Once again, the TM_{20} (blueshifted) and TM_{10} (redshifted) modes are seamlessly jointed. As in the case of Fig. 9.9a, the laser oscillation line shows up as a TM_{30} mode among other modes. That is, the effective index n_{eff} of the laser oscillation line sits on the dispersion curve consisting of other emission lines arising from the weak excitation.

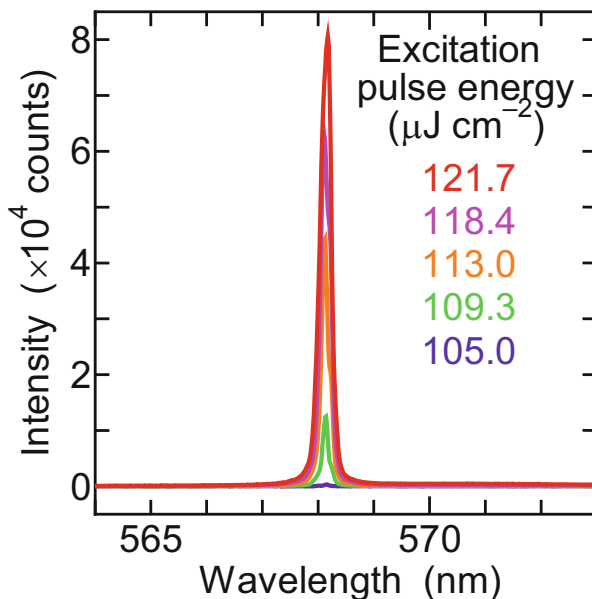
Figure 9.19 shows the distinct single-mode laser oscillation line occurring at 568.1 nm (Table 9.2). This clearly indicates that the laser oscillation has been achieved at a location very close to the designed wavelength (at 569.0 nm). Compare the spectrum of Fig. 9.19 with that of Fig. 9.7. Manifestation of the single-mode line is evident and should be contrasted with the longitudinal multimode laser oscillation of Fig. 9.7.

To quantify the validity of (9.134), the following quantity V is useful [13]:

$$V \equiv 1 - \left| \frac{n_{\text{eff}}(\text{WG}) - n_{\text{eff}}(\text{PM})}{n_{\text{eff}}(\text{WG})} \right|, \quad (9.142)$$

where V stands for validity of (9.134); $n_{\text{eff}}(\text{WG})$ was estimated from the waveguide condition of (9.134) in combination with the emission data obtained with the device; $n_{\text{eff}}(\text{PM})$ was obtained from the phase matching condition described by (9.107),

Fig. 9.19 Single-mode laser oscillation spectrum of an organic semiconductor crystal BP3T (the device D1). A sharply resolved laser oscillation line is noted at 568.1 nm. (Reproduced from Inada Y, Kawata Y, Kawai T, Hotta S, Yamao T (2020) Laser oscillation at an intended wavelength from a distributed feedback structure on anisotropic organic crystals. *J Appl Phys* 127(5): 053102/6 pages [13], with the permission of AIP Publishing. <https://doi.org/10.1063/1.5129425>)



(9.139), or (9.141). We have $0 \leq V \leq 1$ and a larger number of V (close to 1) means that the validity of (9.134) is high (see Table 9.2). Note that the validity estimation by means of (9.142) can equally be applied to both the laser oscillation and non-laser emission lines.

In a word, as long as the thickness of a crystal becomes available along with permittivity tensor of the crystal, one can readily construct a single-mode laser that oscillates at an intended wavelength.

As discussed above in detail, Examples 9.1, 9.2, and 9.3 enable us to design a high-performance laser device using an organic crystal that is characterized by a pretty complicated crystal structure associated with anisotropic refractive indices. From a practical point of view, a laser device is often combined with a diffraction grating. Since organic semiconductors are less robust as compared to inorganic semiconductors, special care should be taken not to damage an organic semiconductor crystal when equipping it with the diffraction grating via rigorous semiconductor processing. In that respect, Inada and co-workers further developed an effective method [18]. The method is based upon the focused ion beam lithography in combination with subsequent plasma etching. They skillfully minimized a damage from which an organic crystal might well suffer.

The above-mentioned design principle enables one to construct effective laser devices that consist of light-emitting materials either organic or inorganic more widely. At the same time, these examples are expected to provide an effective methodology in interdisciplinary fields encompassing solid-state physics and solid-state chemistry as well as device physics.

9.6 Mechanical System

As outlined above, two-level atoms have distinct characteristics in connection with lasers. Electromagnetic waves similarly confined within a one-dimensional cavity exhibit related properties and above all have many features in common with a harmonic oscillator [19].

We have already described several features and properties of the harmonic oscillator (Chap. 2). Meanwhile, we have briefly discussed the formation of electromagnetic stationary waves (Chap. 8). There are several resemblance and correspondence between the harmonic oscillator and electromagnetic stationary waves when we view them as mechanical systems. The point is that in a harmonic oscillator the position and momentum are in-quadrature relationship; i.e., their phase difference is $\pi/2$. For the electromagnetic stationary waves, the electric field and magnetic field are in-quadrature relationship as well.

In Chap. 8, we examine the conditions under which stationary waves are formed. In a dielectric medium both sides of which are equipped with metal layers (or mirrors), the electric field is described as

$$\mathbf{E} = 2E_1 \mathbf{e}_e \sin \omega t \sin kz. \quad (8.201)$$

In (8.201), we assumed that forward and backward electromagnetic waves are propagating in the direction of the z -axis. Here, we assume that the interfaces (or walls) are positioned at $z = 0$ and $z = L$. Within a domain $[0, L]$, the two waves form a stationary wave. Since this expression assumed two waves, the electromagnetic energy was doubled. To normalize the energy so that a single wave is contained, the amplitude E_1 of (8.201) should be divided by $\sqrt{2}$. Therefore, we think of a following description for $\mathbf{E}$:

$$\mathbf{E} = \sqrt{2}E \mathbf{e}_1 \sin \omega t \sin kz \quad \text{or} \quad E_x = \sqrt{2}E \sin \omega t \sin kz,$$

where we designated the polarization vector as a direction of the x -axis. At the same time, we omitted the index from the amplitude. Thus, from the second equation of (7.65) we have

$$\frac{\partial H_y}{\partial t} = -\frac{1}{\mu} \frac{\partial E_x}{\partial z} = -\frac{\sqrt{2}Ek}{\mu} \sin \omega t \cos kz. \quad (9.143)$$

Note that this equation appeared as (8.131) as well. Integrating both sides of (9.143), we get

$$H_y = \frac{\sqrt{2}Ek}{\mu\omega} \cos \omega t \cos kz.$$

Using a relation $\omega = vk$, we have

$$H_y = \frac{\sqrt{2}E}{\mu v} \cos \omega t \cos kz + C,$$

where v is a light velocity in the dielectric medium and C is an integration constant. Removing C and putting

$$H \equiv \frac{E}{\mu v},$$

we have

$$H_y = \sqrt{2}H \cos \omega t \cos kz.$$

Using a vector expression, we have

$$\mathbf{H} = \mathbf{e}_2 \sqrt{2}H \cos \omega t \cos kz.$$

Thus, $\mathbf{E}$ ($\parallel \mathbf{e}_1$), $\mathbf{H}$ ($\parallel \mathbf{e}_2$), and $\mathbf{n}$ ($\parallel \mathbf{e}_3$) form the right-handed system in this order. As noted in Sect. 8.8, at the interface (or wall) the electric field and magnetic field form nodes and antinodes, respectively. Namely, the two fields are in-quadrature.

Let us calculate electromagnetic energy of the dielectric medium within a cavity. In the present case the cavity is meant as the dielectric sandwiched by a couple of metal layer. We have

$$W = W_e + W_m,$$

where W is the total electromagnetic energy; W_e and W_m are electric and magnetic energies, respectively. Let the length of the cavity be L . Then the energy per unit cross-section area is described as

$$W_e = \frac{\epsilon}{2} \int_0^L \mathbf{E}^2 dz \quad \text{and} \quad W_m = \frac{\mu}{2} \int_0^L \mathbf{H}^2 dz.$$

Performing integration, we get

$$\begin{aligned} W_e &= \frac{\epsilon}{2} L E^2 \sin^2 \omega t, \\ W_m &= \frac{\mu}{2} L H^2 \cos^2 \omega t = \frac{\mu}{2} L \left(\frac{E}{\mu v} \right)^2 \cos^2 \omega t = \frac{\epsilon}{2} L E^2 \cos^2 \omega t, \end{aligned} \tag{9.144}$$

where we used $1/v^2 = \epsilon\mu$ with the last equality. Thus, we have

$$W = \frac{\epsilon}{2}LE^2 = \frac{\mu}{2}LH^2. \quad (9.145)$$

Representing an energy per unit volume as $\tilde{W}_e$, $\tilde{W}_m$, and $\tilde{W}$, we have

$$\tilde{W}_e = \frac{\epsilon}{2}E^2 \sin^2 \omega t, \quad \tilde{W}_m = \frac{\epsilon}{2}E^2 \cos^2 \omega t, \quad \tilde{W} = \frac{\epsilon}{2}E^2 = \frac{\mu}{2}H^2. \quad (9.146)$$

In Chap. 2, we treated motion of a harmonic oscillator. There, we had

$$x(t) = \frac{v_0}{\omega} \sin \omega t = x_0 \sin \omega t. \quad (2.7)$$

Here, we have defined an amplitude of the harmonic oscillation as x_0 (>0) given by

$$x_0 \equiv \frac{v_0}{\omega}.$$

Then, momentum is described as

$$p(t) = m\dot{x}(t) = m\omega x_0 \cos \omega t.$$

Defining

$$p_0 \equiv m\omega x_0,$$

we have

$$p(t) = p_0 \cos \omega t.$$

Let a kinetic energy and potential energy of the oscillator be K and V , respectively. Then, we have

$$\begin{aligned} K &= \frac{1}{2m}p(t)^2 = \frac{1}{2m}p_0^2 \cos^2 \omega t, \\ V &= \frac{1}{2}m\omega^2 x(t)^2 = \frac{1}{2}m\omega^2 x_0^2 \sin^2 \omega t, \\ W &= K + V = \frac{1}{2m}p_0^2 = \frac{1}{2}mv_0^2 = \frac{1}{2}m\omega^2 x_0^2. \end{aligned} \quad (9.147)$$

Comparing (9.146) and (9.147), we recognize the following relationship in energy between the electromagnetic fields and harmonic oscillator motion [19]:

$$\sqrt{m\omega}x_0 \leftrightarrow \sqrt{\epsilon}E \quad \text{and} \quad p_0/\sqrt{m} \leftrightarrow \sqrt{\mu}H. \quad (9.148)$$

Thus, there is an elegant contradistinction between the dynamics of electromagnetic fields in cavity and motion of a harmonic oscillator. In fact, quantum electrodynamics (QED) is based upon the treatment of a quantum harmonic oscillator introduced in Chap. 2. Various aspects of QED will be studied in Part V along with the Dirac equation and its handling.

References

1. Moore WJ (1955) Physical chemistry, 3rd edn. Prentice-Hall, Englewood Cliffs
2. Smith FG, King TA, Wilkins D (2007) Optics and photonics, 2nd edn. Wiley, Chichester
3. Sunakawa S (1965) Theoretical electromagnetism. Kinokuniya, Tokyo (in Japanese)
4. Loudon R (2000) The quantum theory of light, 3rd edn. Oxford University Press, Oxford
5. Born M, Wolf E (2005) Principles of optics, 7th edn. Cambridge University Press, Cambridge
6. Yamao T, Okuda Y, Makino Y, Hotta S (2011) Dispersion of the refractive indices of thiophene/phenylene co-oligomer single crystals. *J Appl Phys* 110(5):053113
7. Yamao T, Higashihara S, Yamashita S, Sano H, Inada Y, Yamashita K, Ura S, Hotta S (2018) Design principle of high-performance organic single-crystal light-emitting devices. *J Appl Phys* 123(23):235501
8. Siegman AE (1986) Lasers. University Science Books, Sausalito
9. Palmer C (2005) Diffraction grating handbook, 6th edn. Newport Corporation, New York
10. Hotta S, Yamao T (2011) The thiophene/phenylene co-oligomers: exotic molecular semiconductors integrating high-performance electronic and optical functionalities. *J Mater Chem* 21(5):1295–1304
11. Yamao T, Nishimoto Y, Terasaki K, Akagami H, Katagiri T, Hotta S, Goto M, Azumi R, Inoue M, Ichikawa M, Taniguchi Y (2010) Single crystal growth and charge transport properties of an alternating co-oligomer composed of thiophene and phenylene rings. *Jpn J Appl Phys* 49(4S):04DK20
12. Yariv A, Yeh P (2003) Optical waves in crystals: propagation and control of laser radiation. Wiley, New York
13. Inada Y, Kawata Y, Kawai T, Hotta S, Yamao T (2020) Laser oscillation at an intended wavelength from a distributed feedback structure on anisotropic organic crystals. *J Appl Phys* 127(5):053102
14. Pain HJ (2005) The physics of vibrations and waves, 6th edn. Wiley, Chichester
15. Sakurai Y, Hayashi W, Yamao T, Hotta S (2014) Phase refractive index dispersions of organic oligomer crystals with different molecular alignments. *Jpn J Appl Phys* 53(2S):02BB01
16. Yamao T, Yamamoto K, Hotta S (2009) A high optical-gain organic crystal comprising thiophene/phenylene co-oligomer nanomolecules. *J Nanosci Nanotechnol* 9(4):2582–2585
17. Bisri SZ, Takenobu T, Yomogida Y, Shimotani H, Yamao T, Hotta S, Iwasa Y (2009) High mobility and luminescent efficiency in organic single-crystal light-emitting transistors. *Adv Funct Mater* 19:1728–1735
18. Inada Y, Yamashita S, Murakami S, Takahashi K, Yamao T, Hotta S (2021) Direct fabrication of a diffraction grating onto organic oligomer crystals by focused ion beam lithography followed by plasma etching. *Jpn J Appl Phys* 60(12):120901
19. Fox M (2006) Quantum optics. Oxford University Press, Oxford

Chapter 10

Introductory Green's Functions



In this chapter, we deal with various properties and characteristics of differential equations, especially first-order linear differential equations (FOLDEs) and second-order linear differential equations (SOLDEs). These differential equations are characterized by differential operators and boundary conditions (BCs). Of these, differential operators appearing in SOLDEs are particularly important. Under appropriate conditions, the said operators can be converted to Hermitian operators. The SOLDEs associated with classical orthogonal polynomials play a central role in many fields of mathematical physics including quantum mechanics and electromagnetism. We study the general principle of SOLDEs in relation to several specific SOLDEs we have studied in Part I and examine general features of an eigenvalue problem and an initial-value problem (IVP). In this context, Green's functions provide a powerful tool for solving SOLDEs. For a practical purpose, we deal with actual construction of Green's functions. In Sect. 8.8, we dealt with steady-state characteristics of electromagnetic waves in dielectrics in terms of propagation, reflection, and transmission. When we consider transient characteristics of electromagnetic and optical phenomena, we often need to deal with SOLDEs having constant coefficients. This is well known in connection with a motion of a damped harmonic oscillator. In the latter part of this chapter, we treat the initial-value problem of a SOLDE of this type.

10.1 Second-Order Linear Differential Equations (SOLDEs)

A general form of n -th order linear differential equations has the following form:

Table 10.1 Characteristics of SOLDEs

	Type I	Type II	Type III	Type IV
Equation	Homogeneous	Homogeneous	Inhomogeneous	Inhomogeneous
Boundary conditions	Homogeneous	Inhomogeneous	Homogeneous	Inhomogeneous

$$a_n(x) \frac{d^n u}{dx^n} + a_{n-1}(x) \frac{d^{n-1} u}{dx^{n-1}} + \cdots + a_1(x) \frac{du}{dx} + a_0(x)u = d(x). \quad (10.1)$$

If $d(x) = 0$, the differential equation is said to be homogeneous; otherwise, it is called inhomogeneous. Equation (10.1) is a linear function of u and its derivatives. Likewise, we have a SOLDE such that

$$a(x) \frac{d^2 u}{dx^2} + b(x) \frac{du}{dx} + c(x)u = d(x). \quad (10.2)$$

In (10.2) we assume that the variable x is real. The equation can be solved under appropriate boundary conditions (BCs). A general form of BCs is described as

$$B_1(u) = \alpha_1 u(a) + \beta_1 \left. \frac{du}{dx} \right|_{x=a} + \gamma_1 u(b) + \delta_1 \left. \frac{du}{dx} \right|_{x=b} = \sigma_1, \quad (10.3)$$

$$B_2(u) = \alpha_2 u(a) + \beta_2 \left. \frac{du}{dx} \right|_{x=a} + \gamma_2 u(b) + \delta_2 \left. \frac{du}{dx} \right|_{x=b} = \sigma_2, \quad (10.4)$$

where $\alpha_1, \beta_1, \gamma_1, \delta_1, \sigma_1$, etc. are real constants; $u(x)$ is defined in an interval $[a, b]$, where a and b can be infinity (i.e., $\pm\infty$). The LHS of $B_1(u)$ and $B_2(u)$ is referred to as boundary functionals [1, 2]. If $\sigma_1 = \sigma_2 = 0$, the BCs are called homogeneous; otherwise, the BCs are said to be inhomogeneous. In combination with the inhomogeneous equation expressed as (10.2), Table 10.1 summarizes characteristics of SOLDEs. We have four types of SOLDEs according to homogeneity and inhomogeneity of equations and BCs.

Even though SOLDEs are mathematically tractable, yet it is not easy necessarily to solve them depending upon the nature of $a(x)$, $b(x)$, and $c(x)$ of (10.2). Nonetheless, if those functions are constant coefficients, it can readily be solved. We will deal with SOLDEs of that type in great deal later. Suppose that we find two linearly independent solutions $u_1(x)$ and $u_2(x)$ of a following homogeneous equation:

$$a(x) \frac{d^2 u}{dx^2} + b(x) \frac{du}{dx} + c(x)u = 0. \quad (10.5)$$

Then, any solution $u(x)$ of (10.5) can be expressed as their linear combination such that

$$u(x) = c_1 u_1(x) + c_2 u_2(x), \quad (10.6)$$

where c_1 and c_2 are some arbitrary constants.

In general, suppose that there are arbitrarily chosen n functions; i.e., $f_1(x), f_2(x), \dots, f_n(x)$. Suppose a following equation with those functions:

$$a_1 f_1(x) + a_2 f_2(x) + \dots + a_n f_n(x) = 0, \quad (10.7)$$

where $a_1, a_2, \dots$, and a_n are constants. If $a_1 = a_2 = \dots = a_n = 0$, (10.7) always holds. In this case, (10.7) is said to be a trivial linear relation. If $f_1(x), f_2(x), \dots$, and $f_n(x)$ satisfy a non-trivial linear relation, $f_1(x), f_2(x), \dots$, and $f_n(x)$ are said to be linearly dependent. That is, the non-trivial expression means that in (10.7) at least one of $a_1, a_2, \dots$, and a_n is non-zero. Suppose that $a_n \neq 0$. Then, from (10.7), $f_n(x)$ is expressed as

$$f_n(x) = -\frac{a_1}{a_n} f_1(x) - \frac{a_2}{a_n} f_2(x) - \dots - \frac{a_{n-1}}{a_n} f_{n-1}(x). \quad (10.8)$$

If $f_1(x), f_2(x), \dots$, and $f_n(x)$ are not linearly dependent, they are called linearly independent. In other words, the statement that $f_1(x), f_2(x), \dots$, and $f_n(x)$ are linearly independent is equivalent to that (10.7) holds if and only if $a_1 = a_2 = \dots = a_n = 0$. We will have relevant discussion in Part III.

Now suppose that with the above two linearly independent functions $u_1(x)$ and $u_2(x)$, we have

$$a_1 u_1(x) + a_2 u_2(x) = 0. \quad (10.9)$$

Differentiating (10.9), we have

$$a_1 \frac{du_1(x)}{dx} + a_2 \frac{du_2(x)}{dx} = 0. \quad (10.10)$$

Expressing (10.9) and (10.10) in a matrix form, we get

$$\begin{pmatrix} u_1(x) & u_2(x) \\ \frac{du_1(x)}{dx} & \frac{du_2(x)}{dx} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0. \quad (10.11)$$

Thus, that $u_1(x)$ and $u_2(x)$ are linearly independent is equivalent to that the following expression holds:

$$\begin{vmatrix} u_1(x) & u_2(x) \\ \frac{du_1(x)}{dx} & \frac{du_2(x)}{dx} \end{vmatrix} \equiv W(u_1, u_2) \neq 0, \quad (10.12)$$

where $W(u_1, u_2)$ is called Wronskian of $u_1(x)$ and $u_2(x)$. In fact, if $W(u_1, u_2) = 0$, then we have

$$u_1 \frac{du_2}{dx} - u_2 \frac{du_1}{dx} = 0. \quad (10.13)$$

This implies that there is a functional relationship between u_1 and u_2 . In fact, if we can express as $u_2 = u_2(u_1(x))$, then $\frac{du_2}{dx} = \frac{du_2}{du_1} \frac{du_1}{dx}$. That is,

$$u_1 \frac{du_2}{dx} = u_1 \frac{du_2}{du_1} \frac{du_1}{dx} = u_2 \frac{du_1}{dx} \quad \text{or} \quad u_1 \frac{du_2}{du_1} = u_2 \quad \text{or} \quad \frac{du_2}{u_2} = \frac{du_1}{u_1}, \quad (10.14)$$

where the second equality of the first equation comes from (10.13). The third equation can easily be integrated to yield

$$\ln \frac{u_2}{u_1} = c \quad \text{or} \quad \frac{u_2}{u_1} = e^c \quad \text{or} \quad u_2 = e^c u_1. \quad (10.15)$$

Equation (10.15) shows that $u_1(x)$ and $u_2(x)$ are linearly dependent. It is easy to show if $u_1(x)$ and $u_2(x)$ are linearly dependent, $W(u_1, u_2) = 0$. Thus, we have a following statement:

Two functions are linearly dependent. $\Leftrightarrow W(u_1, u_2) = 0$.

Then, as the contraposition of this statement we have

Two functions are linearly independent. $\Leftrightarrow W(u_1, u_2) \neq 0$.

Conversely, suppose that we have another solution $u_3(x)$ for (10.5) besides $u_1(x)$ and $u_2(x)$. Then, we have

$$\begin{aligned} a(x) \frac{d^2 u_1}{dx^2} + b(x) \frac{du_1}{dx} + c(x) u_1 &= 0, \\ a(x) \frac{d^2 u_2}{dx^2} + b(x) \frac{du_2}{dx} + c(x) u_2 &= 0, \\ a(x) \frac{d^2 u_3}{dx^2} + b(x) \frac{du_3}{dx} + c(x) u_3 &= 0. \end{aligned} \quad (10.16)$$

Again rewriting (10.16) in a matrix form, we have

$$\begin{pmatrix} \frac{d^2 u_1}{dx^2} & \frac{du_1}{dx} & u_1 \\ \frac{d^2 u_2}{dx^2} & \frac{du_2}{dx} & u_2 \\ \frac{d^2 u_3}{dx^2} & \frac{du_3}{dx} & u_3 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0. \quad (10.17)$$

A necessary and sufficient condition to obtain a non-trivial solution (i.e., a solution besides $a = b = c = 0$) is that [1]

$$\begin{vmatrix} \frac{d^2 u_1}{dx^2} & \frac{du_1}{dx} & u_1 \\ \frac{d^2 u_2}{dx^2} & \frac{du_2}{dx} & u_2 \\ \frac{d^2 u_3}{dx^2} & \frac{du_3}{dx} & u_3 \end{vmatrix} = 0. \quad (10.18)$$

Note here that

$$\begin{vmatrix} \frac{d^2 u_1}{dx^2} & \frac{du_1}{dx} & u_1 \\ \frac{d^2 u_2}{dx^2} & \frac{du_2}{dx} & u_2 \\ \frac{d^2 u_3}{dx^2} & \frac{du_3}{dx} & u_3 \end{vmatrix} = - \begin{vmatrix} u_1 & u_2 & u_3 \\ \frac{du_1}{dx} & \frac{du_2}{dx} & \frac{du_3}{dx} \\ \frac{d^2 u_1}{dx^2} & \frac{d^2 u_2}{dx^2} & \frac{d^2 u_3}{dx^2} \end{vmatrix} \equiv -W(u_1, u_2, u_3), \quad (10.19)$$

where $W(u_1, u_2, u_3)$ is Wronskian of $u_1(x)$, $u_2(x)$, and $u_3(x)$. In the above relation, we used the fact that a determinant of a matrix is identical to that of its transposed matrix and that a determinant of a matrix changes the sign after permutation of row vectors. To be short, a necessary and sufficient condition to get a non-trivial solution is that $W(u_1, u_2, u_3)$ vanishes.

This implies that $u_1(x)$, $u_2(x)$, and $u_3(x)$ are linearly dependent. However, we have assumed that $u_1(x)$ and $u_2(x)$ are linearly independent, and so (10.18) and (10.19) mean that $u_3(x)$ must be described as a linear combination of $u_1(x)$ and $u_2(x)$. That is, we have no third linearly independent solution. Consequently, the general solution of (10.5) must be given by (10.6). In this sense, $u_1(x)$ and $u_2(x)$ are said to be a fundamental set of solutions of (10.5).

Next, let us consider the inhomogeneous equation of (10.2). Suppose that $u_p(x)$ is a particular solution of (10.2). Let us think of a following function $v(x)$ such that:

$$u(x) = v(x) + u_p(x). \quad (10.20)$$

Substituting (10.20) for (10.2), we have

$$a(x) \frac{d^2 v}{dx^2} + b(x) \frac{dv}{dx} + c(x)v + a(x) \frac{d^2 u_p}{dx^2} + b(x) \frac{du_p}{dx} + c(x)u_p = d(x). \quad (10.21)$$

Therefore, we have

$$a(x) \frac{d^2 v}{dx^2} + b(x) \frac{dv}{dx} + c(x)v = 0.$$

But $v(x)$ can be described by a linear combination of $u_1(x)$ and $u_2(x)$ as in the case of (10.6). Hence, the general solution of (10.2) should be expressed as

$$u(x) = c_1 u_1(x) + c_2 u_2(x) + u_p(x), \quad (10.22)$$

where c_1 and c_2 are arbitrary (complex) constants.

10.2 First-Order Linear Differential Equations (FOLDEs)

In a discussion that follows, first-order linear differential equations (FOLDEs) supply us with useful information. A general form of FOLDEs is expressed as

$$a(x) \frac{du}{dx} + b(x)u = d(x) \quad [a(x) \neq 0]. \quad (10.23)$$

An associated boundary condition is given by a boundary functional $B(u)$ such that

$$B(u) = \alpha u(a) + \beta u(b) = \sigma, \quad (10.24)$$

where α , β , and σ are real constants; $u(x)$ is defined in an interval $[a, b]$. If in (10.23) $d(x) \equiv 0$, (10.23) can readily be integrated to yield a solution. Let us multiply both sides of (10.23) by $w(x)$. Then we have

$$w(x)a(x) \frac{du}{dx} + w(x)b(x)u = w(x)d(x). \quad (10.25)$$

Here, we define $p(x)$ as follows:

$$p(x) \equiv w(x)a(x), \quad (10.26)$$

where $w(x)$ is called a *weight function*. As mentioned in Sect. 2.4, the weight function is a real and non-negative function within the domain considered. Here we suppose that

$$\frac{dp(x)}{dx} = w(x)b(x). \quad (10.27)$$

Then, (10.25) can be rewritten as

$$\frac{d}{dx} [p(x)u] = w(x)d(x). \quad (10.28)$$

Thus, we can immediately integrate (10.28) to obtain a solution

$$u = \frac{1}{p(x)} \left[\int^x w(x')d(x')dx' + C \right], \quad (10.29)$$

where C is an arbitrary integration constant.

To seek $w(x)$, from (10.26) and (10.27) we have

$$p' = (wa)' = wb = wa \left(\frac{b}{a} \right). \quad (10.30)$$

This can easily be integrated for wa to be expressed as

$$wa = C' \exp \left(\int \frac{b}{a} dx \right) \quad \text{or} \quad w = \frac{C'}{a} \exp \left(\int \frac{b}{a} dx \right), \quad (10.31)$$

where C' is an arbitrary integration constant. The quantity $\frac{C'}{a}$ must be non-negative so that w can be non-negative.

Example 10.1 Let us think of a following FOLDE within an interval $[a, b]$; i.e., $a \leq x \leq b$.

$$\frac{du}{dx} + xu = x. \quad (10.32)$$

A boundary condition is set such that

$$u(a) = \sigma. \quad (10.33)$$

Notice that (10.33) is obtained by setting $\alpha = 1$ and $\beta = 0$ in (10.24). Following the above argument, we obtain a solution described as

$$u(x) = \frac{1}{p(x)} \left[\int_a^x w(x') d(x') dx' + u(a)p(a) \right]. \quad (10.34)$$

Also, we have

$$p(x) = w(x) = \exp \left(\int_a^x x' dx' \right) = \exp \left[\frac{1}{2} (x^2 - a^2) \right]. \quad (10.35)$$

The integration of RHS of (10.34) can be performed as follows:

$$\begin{aligned} \int_a^x w(x') d(x') dx' &= \int_a^x x' \exp \left[\frac{1}{2} (x'^2 - a^2) \right] dx' \\ &= \exp \left(-\frac{a^2}{2} \right) \int_{a^2/2}^{x^2/2} \exp t dt \\ &= \exp \left(-\frac{a^2}{2} \right) \left[\exp \left(\frac{x^2}{2} \right) - \exp \left(\frac{a^2}{2} \right) \right] \\ &= \exp \left(-\frac{a^2}{2} \right) \exp \left(\frac{x^2}{2} \right) - 1 = p(x) - 1, \end{aligned} \quad (10.36)$$

where with the second equality we used an integration by substitution of $\frac{1}{2}x'^2 \rightarrow t$. Considering (10.33) and putting $p(a) = 1$ in (10.34), (10.34) is rewritten as

$$\begin{aligned} u(x) &= \frac{1}{p(x)} [p(x) - 1 + \sigma] = 1 + \frac{\sigma - 1}{p(x)} \\ &= 1 + (\sigma - 1) \exp \left[\frac{1}{2} (a^2 - x^2) \right], \end{aligned} \quad (10.37)$$

where with the last equality we used (10.35).

Notice that when we put $\sigma = 1$ in (10.37), we get $u(x) \equiv 1$. In fact, $u(x) \equiv 1$ is certainly a solution of (10.32) and satisfies the BC of (10.33). Uniqueness of a solution is thus guaranteed.

In the next two examples, we make general discussions.

Example 10.2 Let us consider a following differential operator:

$$L_x = \frac{d}{dx}. \quad (10.38)$$

We think of a following identity using an integration by parts:

$$\int_a^b dx \varphi^* \left(\frac{d}{dx} \psi \right) + \int_a^b dx \left(\frac{d}{dx} \varphi^* \right) \psi = [\varphi^* \psi]_a^b. \quad (10.39)$$

Rewriting this, we get

$$\int_a^b dx \varphi^* \left(\frac{d}{dx} \psi \right) - \int_a^b dx \left[-\frac{d}{dx} \varphi \right]^* \psi = [\varphi^* \psi]_a^b. \quad (10.40)$$

Looking at (10.40), we notice that LHS comprises a difference between two integrals, while RHS referred to as a boundary term (or surface term) does not contain an integral.

Recalling the expression (1.128) and defining

$$\tilde{L}_x \equiv -\frac{d}{dx}$$

in (10.40), we have

$$\langle \varphi | L_x \psi \rangle - \langle \tilde{L}_x \varphi | \psi \rangle = [\varphi^* \psi]_a^b. \quad (10.41)$$

Here, RHS of (10.41) needs to vanish so that we can have

$$\langle \varphi | L_x \psi \rangle = \langle \tilde{L}_x \varphi | \psi \rangle. \quad (10.42)$$

Meanwhile, adopting the expression (1.112) with respect to an adjoint operator, we have a following expression such that

$$\langle \varphi | L_x \psi \rangle = \langle L_x^\dagger \varphi | \psi \rangle. \quad (10.43)$$

Comparing (10.42) and (10.43) and considering that φ and ψ are arbitrary functions, we have $\tilde{L}_x = L_x^\dagger$. Thus, as an operator adjoint to L_x we get

$$L_x^\dagger = \tilde{L}_x = -\frac{d}{dx} = -L_x.$$

Notice that only if the surface term vanishes, the adjoint operator $L_x^\dagger$ can appropriately be defined. We will encounter a similar expression again in Part III.

We add that if with an operator A we have a relation described by

$$A^\dagger = -A, \quad (10.44)$$

the operator A is said to be anti-Hermitian. We have already encountered such an operator in Sects. 1.5 and 3.7.

Let us then examine on what condition the surface term vanishes. The RHS of (10.40) and (10.41) is given by

$$\varphi^*(b)\psi(b) - \varphi^*(a)\psi(a).$$

For this term to vanish, we should have

$$\varphi^*(b)\psi(b) = \varphi^*(a)\psi(a) \quad \text{or} \quad \frac{\varphi^*(b)}{\varphi^*(a)} = \frac{\psi(a)}{\psi(b)}.$$

If $\psi(b) = 2\psi(a)$, then we should have $\varphi(b) = \varphi(a)/2$ for the surface term to vanish. Recalling (10.24), the above conditions are expressed as

$$B(\psi) = 2\psi(a) - \psi(b) = 0, \quad (10.45)$$

$$B'(\varphi) = \frac{1}{2}\varphi(a) - \varphi(b) = 0. \quad (10.46)$$

The boundary functional $B'(\varphi)$ is said to be adjoint to $B(\psi)$. The two boundary functionals are admittedly different. If, however, we set $\psi(b) = \psi(a)$, then we should have $\varphi(b) = \varphi(a)$ for the surface term to vanish. That is

$$B(\psi) = \psi(a) - \psi(b) = 0, \quad (10.47)$$

$$B'(\varphi) = \varphi(a) - \varphi(b) = 0. \quad (10.48)$$

Thus, the two functionals are identical and ψ and φ satisfy homogeneous BCs with respect to these functionals.

As discussed above, a FOLDE is characterized by its differential operator as well as a BC (or boundary functional). This is similarly the case with SOLDEs as well.

Example 10.3 Next, let us consider a following differential operator:

$$L_x = \frac{1}{i} \frac{d}{dx}. \quad (10.49)$$

As in the case of Example 10.2, we have

$$\int_a^b dx \varphi^* \left(\frac{1}{i} \frac{d}{dx} \psi \right) - \int_a^b dx \left[\frac{1}{i} \frac{d}{dx} \varphi \right]^* \psi = \frac{1}{i} [\varphi^* \psi]_a^b. \quad (10.50)$$

Also rewriting (10.50) using an inner product notation, we get

$$\langle \varphi | L_x \psi \rangle - \langle L_x \varphi | \psi \rangle = \frac{1}{i} [\varphi^* \psi]_a^b. \quad (10.51)$$

Apart from the factor $\frac{1}{i}$, RHS of (10.51) is again given by

$$\varphi^*(b)\psi(b) - \varphi^*(a)\psi(a).$$

Repeating a discussion similar to Example 10.2, when the surface term vanishes, we get

$$\langle \varphi | L_x \psi \rangle = \langle L_x \varphi | \psi \rangle. \quad (10.52)$$

Comparing (10.43) and (10.52), we have

$$\langle L_x \varphi | \psi \rangle = \langle L_x^\dagger \varphi | \psi \rangle. \quad (10.53)$$

Again considering that φ and ψ are arbitrary functions, we get

$$L_x^\dagger = L_x. \quad (10.54)$$

As in (10.54), if the differential operator is identical to its adjoint operator, such an operator is called *self-adjoint*. On the basis of (1.119), L_x would apparently be Hermitian. However, we have to be careful to assure that L_x is Hermitian. For a differential operator to be Hermitian, (1) the said operator must be self-adjoint. (2) The two boundary functionals adjoint to each other must be identical. In other words, ψ and φ must satisfy the same homogeneous BCs with respect to these functionals. In this example, we must have the same boundary functionals as those described by (10.47) and (10.48). If and only if the conditions (1) and (2) are satisfied, the operator is said to be Hermitian. It seems somewhat a formal expression. Nonetheless, satisfaction of these conditions is also the case with second-order differential operators so that these operators can be Hermitian. In fact, SOLDEs we studied in Part I are essentially dealt with within the framework of the aforementioned formalism.

10.3 Second-Order Differential Operators

The second-order differential operators are the most common operators and frequently treated in mathematical physics. The general differential operators are described as

$$L_x = a(x) \frac{d^2}{dx^2} + b(x) \frac{d}{dx} + c(x), \quad (10.55)$$

where $a(x)$, $b(x)$, and $c(x)$ can in general be complex functions of a real variable x . Let us think of the following identities [1]:

$$\begin{aligned} v^* a \frac{d^2 u}{dx^2} - u \frac{d^2 (av^*)}{dx^2} &= \frac{d}{dx} \left[av^* \frac{du}{dx} - u \frac{d(av^*)}{dx} \right], \\ v^* b \frac{du}{dx} + u \frac{d(bv^*)}{dx} &= \frac{d}{dx} [buv^*], \\ v^* cu - ucv^* &= 0. \end{aligned} \quad (10.56)$$

Summing both sides of (10.56), we have an identity

$$\begin{aligned} v^* \left\{ a \frac{d^2 u}{dx^2} + b \frac{du}{dx} + cu \right\} - u \left\{ \frac{d^2 (av^*)}{dx^2} - \frac{d(bv^*)}{dx} + cv^* \right\} \\ = \frac{d}{dx} \left[av^* \frac{du}{dx} - u \frac{d(av^*)}{dx} \right] + \frac{d}{dx} [buv^*]. \end{aligned} \quad (10.57)$$

Hence, following the expressions of Sect. 10.2, we define $L_x^\dagger$ such that

$$[L_x^\dagger v]^* \equiv \frac{d^2 (av^*)}{dx^2} - \frac{d(bv^*)}{dx} + cv^*. \quad (10.58)$$

Taking a complex conjugate of both sides, we get

$$L_x^\dagger v = \frac{d^2 (a^* v)}{dx^2} - \frac{d(b^* v)}{dx} + c^* v. \quad (10.59)$$

Considering the differential of a product function, we have as $L_x^\dagger$

$$L_x^\dagger = a^* \frac{d^2}{dx^2} + \left(2 \frac{da^*}{dx} - b^* \right) \frac{d}{dx} + \frac{d^2 a^*}{dx^2} - \frac{db^*}{dx} + c^*. \quad (10.60)$$

Replacing (10.57) with (10.55) and (10.59), we have

$$v^* (L_x u) - [L_x^\dagger v]^* u = \frac{d}{dx} \left[av^* \frac{du}{dx} - u \frac{d(av^*)}{dx} + buv^* \right]. \quad (10.61)$$

Assuming that the relevant SOLDE is defined in $[r, s]$ and integrating (10.61) within that interval, we get

$$\int_r^s dx \left\{ v^* (L_x u) - [L_x^\dagger v]^* u \right\} = \left[av^* \frac{du}{dx} - u \frac{d(av^*)}{dx} + buv^* \right]_r^s. \quad (10.62)$$

Using the definition of an inner product described in (1.128) and rewriting (10.62), we have

$$\langle v | L_x u \rangle - \langle L_x^\dagger v | u \rangle = \left[av^* \frac{du}{dx} - u \frac{d(av^*)}{dx} + buv^* \right]_r^s.$$

Here if RHS of the above (i.e., the surface term of the above expression) vanishes, we get

$$\langle v | L_x u \rangle = \langle L_x^\dagger v | u \rangle.$$

We find that this notation is consistent with (1.112).

Bearing in mind this situation, let us seek a condition under which the differential operator L_x is Hermitian. Suppose here that $a(x)$, $b(x)$, and $c(x)$ are all real and that

$$\frac{da(x)}{dx} = b(x). \quad (10.63)$$

Then, instead of (10.60), we have

$$L_x^\dagger = a(x) \frac{d^2}{dx^2} + b(x) \frac{d}{dx} + c(x) = L_x.$$

Thus, we are successful in constituting a self-adjoint operator L_x . In that case, (10.62) can be rewritten as

$$\int_r^s dx \left\{ v^* (L_x u) - [L_x v]^* u \right\} = \left[a \left(v^* \frac{du}{dx} - u \frac{dv^*}{dx} \right) \right]_r^s. \quad (10.64)$$

Notice that $b(x)$ is eliminated from (10.64). If RHS of (10.64) vanishes, we get

$$\langle v | L_x u \rangle = \langle L_x v | u \rangle.$$

This notation is consistent with (1.119) and the Hermiticity of L_x becomes well-defined.

If we do not have the condition of $\frac{da(x)}{dx} = b(x)$, how can we deal with the problem? The answer is that following the procedures in Sect. 10.2, we can convert L_x to a self-adjoint operator by multiplying L_x by a weight function $w(x)$ introduced in (10.26), (10.27), and (10.31). Replacing $a(x)$, $b(x)$, and $c(x)$ with $w(x)a(x)$, $w(x)b(x)$, and $w(x)c(x)$, respectively, in the identity (10.57), we rewrite (10.57) as

$$\begin{aligned}
& v^* \left\{ aw \frac{d^2 u}{dx^2} + bw \frac{du}{dx} + cwu \right\} - u \left\{ \frac{d^2(awv^*)}{dx^2} - \frac{d(bwv^*)}{dx} + cwv^* \right\} \\
&= \frac{d}{dx} \left[awv^* \frac{du}{dx} - u \frac{d(awv^*)}{dx} \right] + \frac{d}{dx} [bwuv^*].
\end{aligned} \tag{10.65}$$

Let us calculate $\{\cdots\}$ of the second term for LHS of (10.65). Using the conditions of (10.26) and (10.27), we have

$$\begin{aligned}
& \frac{d^2(awv^*)}{dx^2} - \frac{d(bwv^*)}{dx} + cwv^* \\
&= \left[(aw)' v^* + awv^{*'} \right]' - \left[(bw)' v^* + bwv^{*'} \right] + cwv^* \\
&= \left[bwv^{*'} + awv^{*''} \right]' - \left[(bw)' v^* + bwv^{*'} \right] + cwv^* \\
&= (bw)' v^* + bwv^{*''} + (aw)' v^{*'} + awv^{*''} - (bw)' v^* - bwv^{*'} + cwv^* \\
&= (aw)' v^{*'} + awv^{*''} + cwv^* = bwv^{*'} + awv^{*''} + cwv^* \\
&= w \left(av^{*''} + bv^{*'} + cv^* \right) = w \left(a^* v^{*''} + b^* v^{*'} + c^* v^* \right) \\
&= w \left(av'' + bv' + cv \right)^*.
\end{aligned} \tag{10.66}$$

The second last equality of (10.66) is based on the assumption that $a(x)$, $b(x)$, and $c(x)$ are real functions. Meanwhile, for RHS of (10.65) we have

$$\begin{aligned}
& \frac{d}{dx} \left[awv^* \frac{du}{dx} - u \frac{d(awv^*)}{dx} \right] + \frac{d}{dx} [bwuv^*] \\
&= \frac{d}{dx} \left[awv^* \frac{du}{dx} - u(aw)' v^* - uaw \frac{dv^*}{dx} \right] + \frac{d}{dx} [bwuv^*] \\
&= \frac{d}{dx} \left[awv^* \frac{du}{dx} - ubwv^{*'} - uaw \frac{dv^*}{dx} \right] + \frac{d}{dx} [bwuv^*] \\
&= \frac{d}{dx} \left[aw \left(v^* \frac{du}{dx} - u \frac{dv^*}{dx} \right) \right] = \frac{d}{dx} \left[p \left(v^* \frac{du}{dx} - u \frac{dv^*}{dx} \right) \right].
\end{aligned} \tag{10.67}$$

With the last equality of (10.67), we used (10.26). Using (10.66) and (10.67), we rewrite (10.65) once again as

$$\begin{aligned}
& v^* w \left[a \frac{d^2 u}{dx^2} + b \frac{du}{dx} + cu \right] - uw \left[a \frac{d^2 v}{dx^2} + b \frac{dv}{dx} + cv \right]^* \\
&= \frac{d}{dx} \left[p \left(v^* \frac{du}{dx} - u \frac{dv^*}{dx} \right) \right].
\end{aligned} \tag{10.68}$$

Then, integrating (10.68) from r to s , we finally get

$$\int_r^s dx w(x) \{v^*(L_x u) - [L_x v]^* u\} = \left[p \left(v^* \frac{du}{dx} - u \frac{dv^*}{dx} \right) \right]_r^s. \quad (10.69)$$

The relations (10.69) along with (10.62) are called the generalized Green's identity.

We emphasize that as far as the coefficients $a(x)$, $b(x)$, and $c(x)$ in (10.55) are real functions, the associated differential operator L_x can be converted to a self-adjoint form following the procedures of (10.66) and (10.67).

In the above, LHS of the original homogeneous differential equation (10.5) is rewritten as

$$\begin{aligned} & a(x)w(x) \frac{d^2 u}{dx^2} + b(x)w(x) \frac{du}{dx} + c(x)w(x)u \\ &= \frac{d}{dx} \left[p(x) \frac{du}{dx} \right] + cw(x)u. \end{aligned}$$

Rewriting this, we have

$$L_x u = \frac{1}{w(x)} \frac{d}{dx} \left[p(x) \frac{du}{dx} \right] + cu \quad [w(x) > 0], \quad (10.70)$$

where we have

$$p(x) = a(x)w(x) \quad \text{and} \quad \frac{dp(x)}{dx} = b(x)w(x). \quad (10.71)$$

The latter equation of (10.71) corresponds to (10.63) if we assume $w(x) \equiv 1$. When the differential operator L_x is defined as (10.70), L_x is said to be self-adjoint with respect to a weight function of $w(x)$.

Now we examine boundary functionals. The homogeneous adjoint boundary functionals are described as follows:

$$B_1^\dagger(u) = \alpha_1 v^*(r) + \beta_1 \frac{dv^*}{dx} \Big|_{x=r} + \gamma_1 v^*(s) + \delta_1 \frac{dv^*}{dx} \Big|_{x=s} = 0, \quad (10.72)$$

$$B_2^\dagger(u) = \alpha_2 v^*(r) + \beta_2 \frac{dv^*}{dx} \Big|_{x=r} + \gamma_2 v^*(s) + \delta_2 \frac{dv^*}{dx} \Big|_{x=s} = 0. \quad (10.73)$$

In (10.3) putting $\alpha_1 = 1$ and $\beta_1 = \gamma_1 = \delta_1 = 0$, we have

$$B_1(u) = u(r) = \sigma_1. \quad (10.74)$$

Also putting $\gamma_2 = 1$ and $\alpha_2 = \beta_2 = \delta_2 = 0$, we have

$$B_2(u) = u(s) = \sigma_2. \quad (10.75)$$

Further putting

$$\sigma_1 = \sigma_2 = 0, \quad (10.76)$$

we also get homogeneous BCs of

$$B_1(u) = B_2(u) = 0; \quad \text{i.e.,} \quad u(r) = u(s) = 0. \quad (10.77)$$

For RHS of (10.69) to vanish, it suffices to define $B_1^\dagger(u)$ and $B_2^\dagger(u)$ such that

$$B_1^\dagger(u) = v^*(r) \quad \text{and} \quad B_2^\dagger(u) = v^*(s). \quad (10.78)$$

Then, homogeneous adjoint BCs read as

$$v^*(r) = v^*(s) = 0 \quad \text{i.e.,} \quad v(r) = v(s) = 0. \quad (10.79)$$

In this manner, we can readily construct the homogeneous adjoint BCs the same as those of (10.77) so that L_x can be Hermitian.

We list several prescriptions of typical BCs below.

- (1) $u(r) = u(s) = 0$ (Dirichlet conditions),
 - (2) $\frac{du}{dx}\Big|_{x=r} = \frac{du}{dx}\Big|_{x=s} = 0$ (Neumann conditions),
 - (3) $u(r) = u(s)$ and $\frac{du}{dx}\Big|_{x=r} = \frac{du}{dx}\Big|_{x=s}$ (periodic conditions).
- (10.80)

Yet, care should be taken when handling RHS of (10.69), i.e., the surface terms. It is because conditions (1)–(3) are not necessary but sufficient conditions for the surface terms to vanish. Such conditions are not limited to them. Meanwhile, we often have to deal with the non-vanishing surface terms. In that case, we have to start with (10.62) instead of (10.69).

In Sect. 10.2, we mentioned the definition of Hermiticity of the differential operator in such a way that the said operator is self-adjoint and that homogeneous BCs and homogeneous adjoint BCs are the same. In light of the above argument, however, we may relax the conditions for a differential operator to be Hermitian. This is particularly the case when $p(x) = a(x)w(x)$ in (10.69) vanishes at both the endpoints. We will encounter such a situation in Sect. 10.7.

10.4 Green's Functions

Having aforementioned discussions, let us proceed with studies of Green's functions for SOLDEs. Though minimum, we have to mention a bit of formalism.

Given L_x defined by (10.55), let us assume

$$L_x u(x) = d(x) \quad (10.81)$$

under homogeneous BCs with an inhomogeneous term $d(x)$ being an arbitrary function. We also assume that (10.81) is well-defined in a domain $[r, s]$. The numbers r and s can be infinity. Suppose simultaneously that we have

$$L_x^\dagger v(x) = h(x) \quad (10.82)$$

under homogeneous adjoint BCs [1, 2] with an inhomogeneous term $h(x)$ being an arbitrary function as well.

Let us describe the above relations as

$$L|u\rangle = |d\rangle \quad \text{and} \quad L^\dagger|v\rangle = |h\rangle. \quad (10.83)$$

Suppose that there is an inverse operator $L^{-1} \equiv G$ such that

$$GL = LG = E, \quad (10.84)$$

where E is an identity operator. Operating G on (10.83), we have

$$GL|u\rangle = E|u\rangle = |u\rangle = G|d\rangle. \quad (10.85)$$

This implies that (10.81) has been solved and the solution is given by $G|d\rangle$. Since an inverse operation to differentiation is integration, G is expected to be an integral operator.

We have

$$\langle x|LG|y\rangle = L_x \langle x|G|y\rangle = L_x G(x, y). \quad (10.86)$$

Meanwhile, using (10.84) we get

$$\langle x|LG|y\rangle = \langle x|E|y\rangle = \langle x|y\rangle. \quad (10.87)$$

Using a weight function $w(x)$, we generalize an inner product of (1.128) such that

$$\langle g|f\rangle \equiv \int_r^s w(x)g(x)^*f(x)dx. \quad (10.88)$$

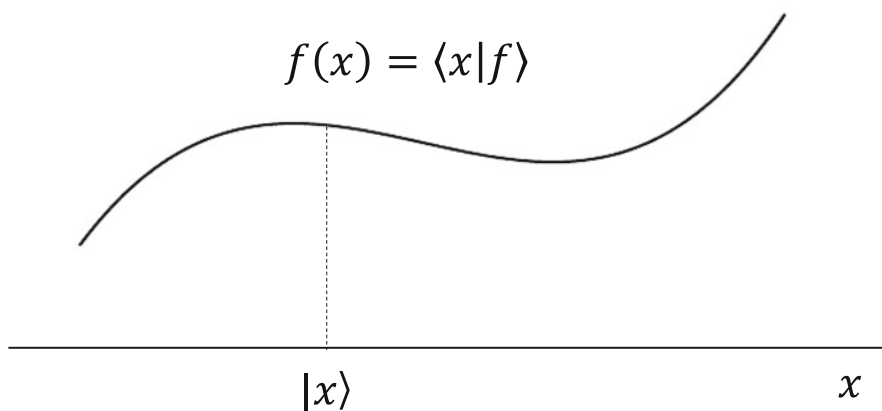


Fig. 10.1 Function $|f\rangle$ and its coordinate representation $f(x)$

As we expand an arbitrary vector using basis vectors, we “expand” an arbitrary function $|f\rangle$ using basis vectors $|x\rangle$. Here, we are treating real numbers as if they formed continuous innumerable basis vectors on a real number line (see Fig. 10.1). Thus, we could expand $|f\rangle$ in terms of $|x\rangle$ such that

$$|f\rangle = \int_r^s dx w(x) f(x) |x\rangle. \quad (10.89)$$

In (10.89) we considered $f(x)$ as if it were an expansion coefficient. The following notation would be reasonable accordingly:

$$f(x) \equiv \langle x|f\rangle. \quad (10.90)$$

In (10.90), $f(x)$ can be viewed as coordinate representation of $|f\rangle$. Thus, from (10.89) we get

$$\langle x'|f\rangle = f(x') = \int_r^s dx w(x) f(x) \langle x'|x\rangle. \quad (10.91)$$

Alternatively, we have

$$f(x') = \int_r^s dx f(x) \delta(x - x'). \quad (10.92)$$

This comes from a property of the δ function [1] described as

$$\int_r^s dx f(x) \delta(x) = f(0). \quad (10.93)$$

Comparing (10.91) and (10.92), we have

$$w(x)\langle x'|x\rangle = \delta(x-x') \quad \text{or} \quad \langle x'|x\rangle = \frac{\delta(x-x')}{w(x)} = \frac{\delta(x'-x)}{w(x)}. \quad (10.94)$$

Thus comparing (10.86) and (10.87) and using (10.94), we get

$$L_x G(x, y) = \frac{\delta(x-y)}{w(x)}. \quad (10.95)$$

In a similar manner, we also have

$$L_x^\dagger g(x, y) = \frac{\delta(x-y)}{w(x)}. \quad (10.96)$$

To arrive at (10.96), we start the discussion assuming an operator $(L_x^\dagger)^{-1}$ such that $(L_x^\dagger)^{-1} \equiv g$ with $gL^\dagger = L^\dagger g = E$.

The function $G(x, y)$ is called a Green's function and $g(x, y)$ is said to be an adjoint Green's function. Handling of Green's functions and adjoint Green's functions is based upon (10.95) and (10.96), respectively. As (10.81) is defined in a domain $r \leq x \leq s$, (10.95) and (10.96) are defined in a domain $r \leq x \leq s$ and $r \leq y \leq s$. Notice that except for the point $x = y$ we have

$$L_x G(x, y) = 0 \quad \text{and} \quad L_x^\dagger g(x, y) = 0. \quad (10.97)$$

That is, $G(x, y)$ and $g(x, y)$ satisfy the homogeneous equation with respect to the variable x . Accordingly, we require $G(x, y)$ and $g(x, y)$ to satisfy the same homogeneous BCs with respect to the variable x as those imposed upon $u(x)$ and $v(x)$ of (10.81) and (10.82), respectively [1].

The relation (10.88) can be obtained as follows: Operating $\langle g|$ on (10.89), we have

$$\langle g|f\rangle = \int_r^s dx w(x) f(x) \langle g|x\rangle = \int_r^s w(x) g(x)^* f(x) dx, \quad (10.98)$$

where for the last equality we used

$$\langle g|x\rangle = \langle x|g\rangle^* = g(x)^*. \quad (10.99)$$

For this, see (1.113) where A is replaced with an identity operator E with regard to a complex conjugate of an inner product of two vectors. Also see (13.2) of Sect. 13.1.

If in (10.69) the surface term (i.e., RHS) vanishes under appropriate conditions e.g., (10.80), we have

$$\int_r^s dx w(x) \left\{ v^* (L_x u) - [L_x^\dagger v]^* u \right\} = 0, \quad (10.100)$$

which is called Green's identity. Since (10.100) is derived from identities (10.56), (10.100) is an identity as well (as a terminology of Green's identity shows). Therefore, (10.100) must hold with any functions u and v so far as they satisfy homogeneous BCs. Thus, replacing v in (10.100) with $g(x, y)$ and using (10.96) together with (10.81), we have

$$\begin{aligned} & \int_r^s dx w(x) \left\{ g^*(x, y) [L_x u(x)] - [L_x^\dagger g(x, y)]^* u(x) \right\} \\ &= \int_r^s dx w(x) \left\{ g^*(x, y) d(x) - \left[\frac{\delta(x-y)}{w(x)} \right]^* u(x) \right\} \\ &= \int_r^s dx w(x) g^*(x, y) d(x) - u(y) = 0, \end{aligned} \quad (10.101)$$

where with the second last equality we used a property of the δ functions. Also notice that $\frac{\delta(x-y)}{w(x)}$ is a real function. Rewriting (10.101), we get

$$u(y) = \int_r^s dx w(x) g^*(x, y) d(x). \quad (10.102)$$

Similarly, replacing u in (10.100) with $G(x, y)$ and using (10.95) together with (10.82), we have

$$v(y) = \int_r^s dx w(x) G^*(x, y) h(x). \quad (10.103)$$

Replacing u and v in (10.100) with $G(x, q)$ and $g(x, t)$, respectively, we have

$$\int_r^s dx w(x) \left\{ g^*(x, t) [L_x G(x, q)] - [L_x^\dagger g(x, t)]^* G(x, q) \right\} = 0. \quad (10.104)$$

Notice that we have chosen q and t for the second argument y in (10.95) and (10.96), respectively. Inserting (10.95) and (10.96) into the above equation after changing arguments, we have

$$\int_r^s dx w(x) \left\{ g^*(x, t) \frac{\delta(x-q)}{w(x)} - \left[\frac{\delta(x-t)}{w(x)} \right]^* G(x, q) \right\} = 0. \quad (10.105)$$

Thus, we get

$$g^*(q, t) = G(t, q) \quad \text{or} \quad g(q, t) = G^*(t, q). \quad (10.106)$$

This implies that $G^*(t, q)$ must satisfy the adjoint BCs with respect to the second argument q . Inserting (10.106) into (10.102), we get

$$u(y) = \int_r^s dx w(x) G(y, x) d(x). \quad (10.107)$$

Or exchanging the arguments x and y , we have

$$u(x) = \int_r^s dy w(y) G(x, y) d(y). \quad (10.108)$$

Similarly, substituting (10.106) for (10.103) we get

$$v(y) = \int_r^s dx w(x) g(y, x) h(x). \quad (10.109)$$

Or, we have

$$v(x) = \int_r^s dy w(y) g(x, y) h(y). \quad (10.110)$$

Equations (10.107)–(10.110) clearly show that homogeneous equations [given by putting $d(x) = h(x) = 0$] have a trivial solution $u(x) \equiv 0$ and $v(x) \equiv 0$ under homogeneous BCs. Note that it is always the case when we are able to construct a Green's function. This in turn implies that we can construct a Green's function if the differential operator is accompanied by *initial conditions*. Conversely, if the homogeneous equation has a non-trivial solution under homogeneous BCs, (10.107)–(10.110) will not work.

If the differential operator L in (10.81) is Hermitian, according to the associated remarks of Sect. 10.2 we must have $L_x = L_x^\dagger$ and $u(x)$ and $v(x)$ of (10.81) and (10.82) must satisfy the same homogeneous BCs. Consequently, in the case of an Hermitian operator we should have

$$G(x, y) = g(x, y). \quad (10.111)$$

From (10.106) and (10.111), if the operator is Hermitian we get

$$G(x, y) = G^*(y, x). \quad (10.112)$$

In Sect. 10.3 we assume that the coefficients $a(x)$, and $b(x)$, and $c(x)$ are real to assure that L_x is Hermitian [1]. On this condition $G(x, y)$ is real as well (vide infra). Then we have

$$G(x, y) = G(y, x). \quad (10.113)$$

That is, $G(x, y)$ is real symmetric with respect to the arguments x and y .

To be able to apply Green's functions to practical use, we will have to estimate a behavior of the Green's function near $x = y$. This is because in light of (10.95) and (10.96) there is a "jump" at $x = y$.

When we deal with a case where a self-adjoint operator is relevant, using a function $p(x)$ of (10.69) we have

$$\frac{a(x)}{p(x)} \frac{\partial}{\partial x} \left(p \frac{\partial G}{\partial x} \right) + c(x)G = \frac{\delta(x-y)}{w(x)}. \quad (10.114)$$

Multiplying both sides by $\frac{p(x)}{a(x)}$, we have

$$\frac{\partial}{\partial x} \left(p \frac{\partial G}{\partial x} \right) = \frac{p(x)}{a(x)} \frac{\delta(x-y)}{w(x)} - \frac{p(x)c(x)}{a(x)} G(x, y). \quad (10.115)$$

Using a property of the δ function expressed by

$$f(x) \delta(x) = f(0) \delta(x) \quad \text{or} \quad f(x) \delta(x-y) = f(y) \delta(x-y), \quad (10.116)$$

we have

$$\frac{\partial}{\partial x} \left(p \frac{\partial G}{\partial x} \right) = \frac{p(y)}{a(y)} \frac{\delta(x-y)}{w(y)} - \frac{p(x)c(x)}{a(x)} G(x, y). \quad (10.117)$$

Integrating (10.117) with respect to x , we get

$$p \frac{\partial G(x, y)}{\partial x} = \frac{p(y)}{a(y)w(y)} \theta(x-y) - \int_r^x dt \frac{p(t)c(t)}{a(t)} G(t, y) + C, \quad (10.118)$$

where C is a constant. The function $\theta(x-y)$ is defined by

$$\theta(x) = \begin{cases} 1 & (x > 0), \\ 0 & (x < 0). \end{cases} \quad (10.119)$$

Note that we have

$$\frac{d\theta(x)}{dx} = \delta(x). \quad (10.120)$$

The function $\theta(x)$ is called a step function or Heaviside step function. In RHS of (10.118) the first term has a discontinuity at $x = y$ because of $\theta(x-y)$, whereas the second term is continuous with respect to y . Thus, we have

$$\begin{aligned}
& \lim_{\varepsilon \rightarrow +0} \left[p(y + \varepsilon) \frac{\partial G(x, y)}{\partial x} \Big|_{x=y+\varepsilon} - p(y - \varepsilon) \frac{\partial G(x, y)}{\partial x} \Big|_{x=y-\varepsilon} \right] \\
&= \lim_{\varepsilon \rightarrow +0} \frac{p(y)}{a(y)w(y)} [\theta(+\varepsilon) - \theta(-\varepsilon)] = \frac{p(y)}{a(y)w(y)}.
\end{aligned} \tag{10.121}$$

Since $p(y)$ is continuous with respect to the argument y , this factor drops off and we get

$$\lim_{\varepsilon \rightarrow +0} \left[\frac{\partial G(x, y)}{\partial x} \Big|_{x=y+\varepsilon} - \frac{\partial G(x, y)}{\partial x} \Big|_{x=y-\varepsilon} \right] = \frac{1}{a(y)w(y)}. \tag{10.122}$$

Thus, $\frac{\partial G(x, y)}{\partial x}$ is accompanied by a discontinuity at $x = y$ by a magnitude of $\frac{1}{a(y)w(y)}$. Since RHS of (10.122) is continuous with respect to the argument y , integrating (10.122) again with respect to x , we find that $G(x, y)$ is continuous at $x = y$. These properties of $G(x, y)$ are useful to calculate Green's functions in practical use. We will encounter several examples in next sections.

Suppose that there are two Green's functions that satisfy the same homogeneous BCs. Let $G(x, y)$ and $\tilde{G}(x, y)$ be such functions. Then, we must have

$$L_x G(x, y) = \frac{\delta(x - y)}{w(x)} \quad \text{and} \quad L_x \tilde{G}(x, y) = \frac{\delta(x - y)}{w(x)}. \tag{10.123}$$

Subtracting both sides of (10.123), we have

$$L_x [\tilde{G}(x, y) - G(x, y)] = 0. \tag{10.124}$$

In virtue of the linearity of BCs, $G(x, y) - \tilde{G}(x, y)$ must satisfy the same homogeneous BCs as well. But (10.124) is a homogeneous equation, and so we must have a trivial solution from the aforementioned constructability of the Green's function such that

$$G(x, y) - \tilde{G}(x, y) \equiv 0 \quad \text{or} \quad G(x, y) \equiv \tilde{G}(x, y). \tag{10.125}$$

This obviously indicates that a Green's function should be unique.

We have assumed in Sect. 10.3 that the coefficients $a(x)$, and $b(x)$, and $c(x)$ are real. Therefore, taking complex conjugate of (10.95) we have

$$L_x G(x, y)^* = \frac{\delta(x - y)}{w(x)}. \tag{10.126}$$

Notice here that both $\delta(x - y)$ and $w(x)$ are real functions. Subtracting (10.95) from (10.126), we have

$$L_x[G(x, y)^* - G(x, y)] = 0.$$

Again, from the uniqueness of the Green's function, we get $G(x, y)^* = G(x, y)$; i.e., $G(x, y)$ is real accordingly. This is independent of specific structures of L_x . In other words, so far as we are dealing with real coefficients $a(x)$, $b(x)$, and $c(x)$, $G(x, y)$ is real whether or not L_x is self-adjoint.

10.5 Construction of Green's Functions

So far we dealt with homogeneous boundary conditions (BCs) with respect to a differential equation

$$a(x) \frac{d^2 u}{dx^2} + b(x) \frac{du}{dx} + c(x)u = d(x), \quad (10.2)$$

where coefficients $a(x)$, $b(x)$, and $c(x)$ are real. In this case, if $d(x) \equiv 0$ in (10.108), namely, the SOLDE is homogeneous equation, we have a solution $u(x) \equiv 0$ under homogeneous boundary conditions on the basis of (10.108). If, however, we have inhomogeneous boundary conditions (BCs), additional terms appear on RHS of (10.108) in both the cases of homogeneous and inhomogeneous equations. In this section, we examine how we can deal with this problem.

Following the remarks made in Sect. 10.3, we start with (10.62) or (10.69). If we deal with a self-adjoint or Hermitian operator, we can apply (10.69) to the problem. In a more general case where the operator is not self-adjoint, (10.62) is useful. In this respect, in Sect. 10.6 we have a good opportunity for this.

In Sect. 10.3, we mentioned that we may relax the definition of Hermiticity of the differential operator in the case where the surface term vanishes. Meanwhile, we should bear in mind that the Green's functions and adjoint Green's functions are constructed using homogeneous BCs regardless of whether we are concerned with a homogeneous equation or inhomogeneous equation. Thus, even if the surface terms do not vanish, we may regard the differential operator as Hermitian. This is because we deal with essentially the same Green's function to solve a problem with both the cases of homogeneous equation and inhomogeneous equations (vide infra). Notice also that whether or not RHS vanishes, we are to use the same Green's function [1]. In this sense, we do not have to be too strict with the definition of Hermiticity.

Now, suppose that for a differential (self-adjoint) operator L_x we are given

$$\int_r^s dx w(x) \{v^* (L_x u) - [L_x v]^* u\} = \left[p(x) \left(v^* \frac{du}{dx} - u \frac{dv^*}{dx} \right) \right]_r^s, \quad (10.69)$$

where $w(x) > 0$ in a domain $[r, s]$ and $p(x)$ is a real function. Note that since (10.69) is an identity, we may choose $G(x, y)$ for v with an appropriate choice of $w(x)$. Then from (10.95) and (10.112), we have

$$\begin{aligned} & \int_r^s dx w(x) \left\{ G(y, x) d(x) - \left[\frac{\delta(x-y)}{w(x)} \right]^* u(x) \right\} \\ &= \int_r^s dx w(x) \left\{ G(y, x) d(x) - \frac{\delta(x-y)}{w(x)} u(x) \right\} \\ &= \left[p(x) \left\{ G(y, x) \frac{du(x)}{dx} - u(x) \frac{\partial G(y, x)}{\partial x} \right\} \right]_{x=r}^s. \end{aligned} \quad (10.127)$$

Note in (10.127) we used the fact that both $\delta(x-y)$ and $w(x)$ are real. Using a property of the δ function, we get

$$\begin{aligned} u(y) &= \int_r^s dx w(x) G(y, x) d(x) \\ &\quad - \left[p(x) \left\{ G(y, x) \frac{du(x)}{dx} - u(x) \frac{\partial G(y, x)}{\partial x} \right\} \right]_{x=r}^s. \end{aligned} \quad (10.128)$$

When the differential operator L_x can be made Hermitian under an appropriate condition of (10.71), as a real Green's function we have

$$G(x, y) = G(y, x). \quad (10.113)$$

The function $G(x, y)$ satisfies homogeneous BCs. Hence, if we assume, e.g., the Dirichlet BCs [see (10.80)], we have

$$G(r, y) = G(s, y) = 0. \quad (10.129)$$

Using the symmetric property of $G(x, y)$ with respect to arguments x and y , from (10.129) we get

$$G(y, r) = G(y, s) = 0. \quad (10.130)$$

Thus, the first term of the surface terms of (10.128) is eliminated to yield

$$u(y) = \int_r^s dx w(x) G(y, x) d(x) + \left[p(x) u(x) \frac{\partial G(y, x)}{\partial x} \right]_{x=r}^s.$$

Exchanging the arguments x and y , we get

$$u(x) = \int_r^s dy w(y) G(x, y) d(y) + \left[p(y) u(y) \frac{\partial G(x, y)}{\partial y} \right]_{y=r}^s. \quad (10.131)$$

Then, (1) substituting surface terms of $u(s)$ and $u(r)$ that are associated with the inhomogeneous BCs described as

$$B_1(u) = \sigma_1 \quad \text{and} \quad B_2(u) = \sigma_2 \quad (10.132)$$

and (2) calculating $\frac{\partial G(x, y)}{\partial y} \Big|_{y=r}$ and $\frac{\partial G(x, y)}{\partial y} \Big|_{y=s}$, we will be able to obtain a unique solution. Notice that even though we have formally the same differential operators, we get different Green's functions depending upon different BCs. We see tangible examples later.

On the basis of the general discussion of Sect. 10.4 and this section, we are in the position to construct the Green's functions. Except for the points of $x = y$, the Green's function $G(x, y)$ must satisfy the following differential equation:

$$L_x G(x, y) = 0, \quad (10.133)$$

where L_x is given by

$$L_x = a(x) \frac{d^2}{dx^2} + b(x) \frac{d}{dx} + c(x). \quad (10.135)$$

The differential equation $L_x u = d(x)$ is defined within an interval $[r, s]$, where r may be $-\infty$ and s may be $+\infty$.

From now on, we regard $a(x)$, $b(x)$, and $c(x)$ as real functions. From (10.133), we expect the Green's function to be described as a linear combination of a fundamental set of solutions $u_1(x)$ and $u_2(x)$. Here the fundamental set of solutions are given by two linearly independent solutions of a homogeneous equation $L_x u = 0$. Then we should be able to express $G(x, y)$ as a combination of $F_1(x, y)$ and $F_2(x, y)$ that are described as

$$\begin{aligned} F_1(x, y) &= c_1 u_1(x) + c_2 u_2(x) \quad \text{for } r \leq x < y, \\ F_2(x, y) &= d_1 u_1(x) + d_2 u_2(x) \quad \text{for } y < x \leq s, \end{aligned} \quad (10.134)$$

where c_1 , c_2 , d_1 , and d_2 are arbitrary (complex) constants to be determined later. These constants are given as a function of y . The combination has to be made such that

$$G(x, y) = \begin{cases} F_1(x, y) & \text{for } r \leq x < y, \\ F_2(x, y) & \text{for } y < x \leq s. \end{cases} \quad (10.135)$$

Thus using $\theta(x)$ function defined as (10.119), we describe $G(x, y)$ as

$$G(x, y) = F_1(x, y)\theta(y - x) + F_2(x, y)\theta(x - y). \quad (10.136)$$

Notice that $F_1(x, y)$ and $F_2(x, y)$ are “ordinary” functions and that $G(x, y)$ is not, because $G(x, y)$ contains the $\theta(x)$ function.

If we have

$$F_2(x, y) = F_1(y, x), \quad (10.137)$$

$$G(x, y) = F_1(x, y)\theta(y - x) + F_1(y, x)\theta(x - y). \quad (10.138)$$

Hence, we get

$$G(x, y) = G(y, x). \quad (10.139)$$

From (10.113), L_x is Hermitian. Suppose that $F_1(x, y) = (x - r)(y - s)$ and $F_2(x, y) = (x - s)(y - r)$. Then, (10.137) is satisfied and, hence, if we can construct the Green's function from $F_1(x, y)$ and $F_2(x, y)$, L_x should be Hermitian. However, if we had, e.g., $F_1(x, y) = x - r$ and $F_2(x, y) = y - s$, $G(x, y) \neq G(y, x)$, and so L_x would not be Hermitian.

The Green's functions must satisfy the homogeneous BCs. That is,

$$B_1(G) = B_2(G) = 0. \quad (10.140)$$

Also, we require continuity condition of $G(x, y)$ at $x = y$ and discontinuity condition of $\partial G(x, y)/\partial x$ at $x = y$ described by (10.122). Thus, we have four conditions including BCs and continuity and discontinuity conditions to be satisfied by $G(x, y)$. Thus, we can determine four constants c_1 , c_2 , d_1 , and d_2 by the four conditions.

Now, let us inspect further details about the Green's functions by an example.

Example 10.4 Let us consider a following differential equation

$$\frac{d^2 u}{dx^2} + u = 1. \quad (10.141)$$

We assume that a domain of the argument x is $[0, L]$. We set boundary conditions such that

$$u(0) = \sigma_1 \quad \text{and} \quad u(L) = \sigma_2. \quad (10.142)$$

Thus, if at least one of σ_1 and σ_2 is not zero, we are dealing with an inhomogeneous differential equation under inhomogeneous BCs.

Next, let us seek conditions that the Green's function satisfies. We also seek a fundamental set of solutions of a homogeneous equation described by

$$\frac{d^2 u}{dx^2} + u = 0. \quad (10.143)$$

This is obtained by putting $a = c = 1$ and $b = 0$ in a general form of (10.5) with a weight function being unity. The differential equation (10.143) is therefore self-adjoint according to the argument of Sect. 10.3. A fundamental set of solutions are given by

$$e^{ix} \quad \text{and} \quad e^{-ix}.$$

Then, we have

$$\begin{aligned} F_1(x, y) &= c_1 e^{ix} + c_2 e^{-ix} \quad \text{for } 0 \leq x < y, \\ F_2(x, y) &= d_1 e^{ix} + d_2 e^{-ix} \quad \text{for } y < x \leq L. \end{aligned} \quad (10.144)$$

The functions $F_1(x, y)$ and $F_2(x, y)$ must satisfy the following BCs such that

$$F_1(0, y) = c_1 + c_2 = 0 \quad \text{and} \quad F_2(L, y) = d_1 e^{iL} + d_2 e^{-iL} = 0. \quad (10.145)$$

Thus, we have

$$F_1(x, y) = c_1 (e^{ix} - e^{-ix}), \quad F_2(x, y) = d_1 (e^{ix} - e^{2iL} e^{-ix}). \quad (10.146)$$

Therefore, at $x = y$ we have

$$c_1 (e^{iy} - e^{-iy}) = d_1 (e^{iy} - e^{2iL} e^{-iy}). \quad (10.147)$$

Discontinuity condition of (10.122) is equivalent to

$$\left. \frac{\partial F_2(x, y)}{\partial x} \right|_{x=y} - \left. \frac{\partial F_1(x, y)}{\partial x} \right|_{x=y} = 1. \quad (10.148)$$

This is because both $F_1(x, y)$ and $F_2(x, y)$ are ordinary functions and supposed to be differentiable at any x . The relation (10.148) then reads as

$$id_1 (e^{iy} + e^{2iL} e^{-iy}) - ic_1 (e^{iy} + e^{-iy}) = 1. \quad (10.149)$$

From (10.147) and (10.149), using Cramer's rule we have

$$c_1 = \frac{\begin{vmatrix} 0 & -e^{iy} + e^{2iL}e^{-iy} \\ i & -e^{iy} - e^{2iL}e^{-iy} \end{vmatrix}}{\begin{vmatrix} e^{iy} - e^{-iy} & -e^{iy} + e^{2iL}e^{-iy} \\ e^{iy} + e^{-iy} & -e^{iy} - e^{2iL}e^{-iy} \end{vmatrix}} = \frac{i(e^{iy} - e^{2iL}e^{-iy})}{2(1 - e^{2iL})}, \quad (10.150)$$

$$d_1 = \frac{\begin{vmatrix} e^{iy} - e^{-iy} & 0 \\ e^{iy} + e^{-iy} & i \end{vmatrix}}{\begin{vmatrix} e^{iy} - e^{-iy} & -e^{iy} + e^{2iL}e^{-iy} \\ e^{iy} + e^{-iy} & -e^{iy} - e^{2iL}e^{-iy} \end{vmatrix}} = \frac{i(e^{iy} - e^{-iy})}{2(1 - e^{2iL})}. \quad (10.151)$$

Substituting these parameters for (10.146), we get

$$F_1(x, y) = \frac{\sin x(e^{2iL}e^{-iy} - e^{iy})}{1 - e^{2iL}}, \quad F_2(x, y) = \frac{\sin y(e^{2iL}e^{-ix} - e^{ix})}{1 - e^{2iL}}. \quad (10.152)$$

Making a denominator real, we have

$$\begin{aligned} F_1(x, y) &= \frac{\sin x[\cos(y - 2L) - \cos y]}{2 \sin^2 L}, \\ F_2(x, y) &= \frac{\sin y[\cos(x - 2L) - \cos x]}{2 \sin^2 L}. \end{aligned} \quad (10.153)$$

Using the $\theta(x)$ function, we get

$$\begin{aligned} G(x, y) &= \frac{\sin x[\cos(y - 2L) - \cos y]}{2 \sin^2 L} \theta(y - x) \\ &+ \frac{\sin y[\cos(x - 2L) - \cos x]}{2 \sin^2 L} \theta(x - y). \end{aligned} \quad (10.154)$$

Thus, $G(x, y) = G(y, x)$ as expected. Notice, however, that if $L = n\pi$ ($n = 1, 2, \dots$) the Green's function cannot be defined as an ordinary function even if $x \neq y$. We return to this point later.

The solution for (10.141) under the homogeneous BCs is then described as

$$\begin{aligned} u(x) &= \int_0^L dy G(x, y) \\ &= \frac{\cos(x - 2L) - \cos x}{2 \sin^2 L} \int_0^x \sin y dy + \frac{\sin x}{2 \sin^2 L} \int_x^L [\cos(y - 2L) - \cos y] dy. \end{aligned} \quad (10.155)$$

This can readily be integrated to yields solution for the inhomogeneous equation such that

$$\begin{aligned} u(x) &= \frac{\cos(x-2L) - \cos x - 2 \sin L \sin x - \cos 2L + 1}{2 \sin^2 L} \\ &= \frac{\cos(x-2L) - \cos x - 2 \sin L \sin x + 2 \sin^2 L}{2 \sin^2 L}. \end{aligned} \quad (10.156)$$

Next, let us consider the surface term. This is given by the second term of (10.131). We get

$$\left. \frac{\partial F_1(x, y)}{\partial x} \right|_{y=L} = \frac{\sin x}{\sin L}, \quad \left. \frac{\partial F_2(x, y)}{\partial x} \right|_{y=0} = \frac{\cos(x-2L) - \cos x}{2 \sin^2 L}. \quad (10.157)$$

Therefore, with the inhomogeneous BCs we have the following solution for the inhomogeneous equation:

$$\begin{aligned} u(x) &= \frac{\cos(x-2L) - \cos x - 2 \sin L \sin x + 2 \sin^2 L}{2 \sin^2 L} \\ &+ \frac{2\sigma_2 \sin L \sin x + \sigma_1 [\cos x - \cos(x-2L)]}{2 \sin^2 L}, \end{aligned} \quad (10.158)$$

where the second term is the surface term. If $\sigma_1 = \sigma_2 = 1$, we have

$$u(x) \equiv 1.$$

Looking at (10.141), we find that $u(x) \equiv 1$ is certainly a solution for (10.141) with inhomogeneous BCs of $\sigma_1 = \sigma_2 = 1$. The uniqueness of the solution then ensures that $u(x) \equiv 1$ is a sole solution under the said BCs.

From (10.154), we find that $G(x, y)$ has a singularity at $L = n\pi$ (n : integers). This is associated with the fact that a homogenous equation (10.143) has a non-trivial solution, e.g., $u(x) = \sin x$ under homogeneous BCs $u(0) = u(L) = 0$. The present situation is essentially the same as that of Example 1.1 of Sect. 1.3. In other words, when $\lambda = 1$ in (1.61), the form of a differential equation is identical to (10.143) with virtually the same Dirichlet conditions. The point is that (10.143) can be viewed as a homogeneous equation and, at the same time, as an eigenvalue equation. In such a case, a Green's function approach will fail.

10.6 Initial-Value Problems (IVPs)

10.6.1 General Remarks

The IVPs frequently appear in mathematical physics. The relevant conditions are dealt with as BCs in the theory of differential equations. With boundary functionals $B_1(u)$ and $B_2(u)$ of (10.3) and (10.4), setting $\alpha_1 = \beta_2 = 1$ and other coefficients as zero, we get

$$B_1(u) = u(p) = \sigma_1 \quad \text{and} \quad B_2(u) = \left. \frac{du}{dx} \right|_{x=p} = \sigma_2. \quad (10.159)$$

In the above, note that we choose $[r, s]$ for a domain of x . The points r and s can be infinity as before. Any point p within the domain $[r, s]$ may be designated as a special point on which the BCs (10.159) are imposed. The initial conditions are particularly prominent among BCs. This is because the conditions are set at one point of the argument. This special condition is usually called initial conditions. In this section, we investigate fundamental characteristics of IVPs.

Suppose that we have

$$u(p) = \left. \frac{du}{dx} \right|_{x=p} = 0, \quad (10.160)$$

with homogeneous BCs. Given a differential operator L_x defined as (10.55), i.e.,

$$L_x = a(x) \frac{d^2}{dx^2} + b(x) \frac{d}{dx} + c(x), \quad (10.55)$$

let a fundamental set of solutions be $u_1(x)$ and $u_2(x)$ for

$$L_x u(x) = 0. \quad (10.161)$$

A general solution $u(x)$ for (10.161) is given by a linear combination of $u_1(x)$ and $u_2(x)$ such that

$$u(x) = c_1 u_1(x) + c_2 u_2(x), \quad (10.162)$$

where c_1 and c_2 are arbitrary (complex) constants. Suppose that we have homogeneous BCs expressed by (10.160). Then, we have

$$\begin{aligned} u(p) &= c_1 u_1(p) + c_2 u_2(p) = 0, \\ u'(p) &= c_1 u'_1(p) + c_2 u'_2(p) = 0. \end{aligned}$$

Rewriting it in a matrix form, we have

$$\begin{pmatrix} u_1(p) & u_2(p) \\ u'_1(p) & u'_2(p) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = 0.$$

Since the matrix represents Wronskian of a fundamental set of solutions $u_1(x)$ and $u_2(x)$, its determinant never vanishes at any point p . That is, we have

$$\begin{vmatrix} u_1(p) & u_2(p) \\ u'_1(p) & u'_2(p) \end{vmatrix} \neq 0. \quad (10.163)$$

Then, we necessarily have $c_1 = c_2 = 0$. From (10.162), we have a trivial solution

$$u(x) \equiv 0$$

under the initial conditions as homogeneous BCs. Thus, as already discussed a Green's function can always be constructed for IVPs.

To seek the Green's functions for IVPs, we return back to the generalized Green's identity described as

$$\int_r^s dx \left\{ v^* (L_x u) - [L_x^\dagger v]^* u \right\} = \left[av^* \frac{du}{dx} - u \frac{d(av^*)}{dx} + buv^* \right]_r^s. \quad (10.62)$$

For the surface term (RHS) to vanish, for homogeneous BCs we have, e.g.,

$$u(s) = \frac{du}{dx} \Big|_{x=s} = 0 \quad \text{and} \quad v(r) = \frac{dv}{dx} \Big|_{x=r} = 0,$$

for the two sets of BCs adjoint to each other. Obviously, these are not identical simply because the former is determined at s and the latter is determined at a different point r . For this reason, the operator L_x is not Hermitian, even though it is formally self-adjoint. In such a case, we would rather use L_x directly than construct a self-adjoint operator because we cannot make the operator Hermitian either way.

Hence, unlike the precedent sections we do not need a weight function $w(x)$. Or, we may regard $w(x) \equiv 1$. Then, we reconsider the conditions which the Green's functions should satisfy. On the basis of the general consideration of Sect. 10.4, especially (10.86), (10.87), and (10.94), we have [2]

$$L_x G(x, y) = \langle x | y \rangle = \delta(x - y). \quad (10.164)$$

Therefore, we have

$$\frac{\partial^2 G(x, y)}{\partial x^2} + \frac{b(x)}{a(x)} \frac{\partial G(x, y)}{\partial x} + \frac{c(x)}{a(x)} G(x, y) = \frac{\delta(x - y)}{a(x)}.$$

Integrating or integrating by parts the above equation, we get

$$\begin{aligned} & \frac{\partial G(x, y)}{\partial x} + \left[\frac{b(x)}{a(x)} G(x, y) \right]_{x_0}^x - \int_{x_0}^x \left[\frac{b(\xi)}{a(\xi)} \right]' G(\xi, y) d\xi + \int_{x_0}^x \frac{c(\xi)}{a(\xi)} G(\xi, y) d\xi \\ &= \frac{\theta(x-y)}{a(y)}. \end{aligned}$$

Noting that the functions other than $\frac{\partial G(x, y)}{\partial x}$ and $\frac{\theta(x-y)}{a(y)}$ are continuous, as before we have

$$\lim_{\varepsilon \rightarrow +0} \left[\frac{\partial G(x, y)}{\partial x} \Big|_{x=y+\varepsilon} - \frac{\partial G(x, y)}{\partial x} \Big|_{x=y-\varepsilon} \right] = \frac{1}{a(y)}. \quad (10.165)$$

10.6.2 Green's Functions for IVPs

From a practical point of view, we may set $r = 0$ in (10.62). Then, we can choose a domain for $[0, s]$ (for $s > 0$) or $[s, 0]$ (for $s < 0$) with (10.2). For simplicity, we use x instead of s . We consider two cases of $x > 0$ and $x < 0$.

1. Case I ($x > 0$): Let $u_1(x)$ and $u_2(x)$ be a fundamental set of solutions. We define $F_1(x, y)$ and $F_2(x, y)$ as before such that

$$\begin{aligned} F_1(x, y) &= c_1 u_1(x) + c_2 u_2(x) \quad \text{for } 0 \leq x < y, \\ F_2(x, y) &= d_1 u_1(x) + d_2 u_2(x) \quad \text{for } 0 < y < x. \end{aligned} \quad (10.166)$$

As before, we set

$$G(x, y) = \begin{cases} F_1(x, y) & \text{for } 0 \leq x < y, \\ F_2(x, y) & \text{for } 0 < y < x. \end{cases}$$

Homogeneous BCs are defined as

$$u(0) = 0 \quad \text{and} \quad u'(0) = 0.$$

Correspondingly, we have

$$F_1(0, y) = 0 \quad \text{and} \quad F_1'(0, y) = 0.$$

This is translated into

$$c_1 u_1(0) + c_2 u_2(0) = 0 \quad \text{and} \quad c_1 u_1'(0) + c_2 u_2'(0) = 0.$$

In a matrix form, we get

$$\begin{pmatrix} u_1(0) & u_2(0) \\ u_1'(0) & u_2'(0) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = 0.$$

As mentioned above, since $u_1(x)$ and $u_2(x)$ are a fundamental set of solutions, we have $c_1 = c_2 = 0$. Hence, we get

$$F_1(x, y) = 0.$$

From the continuity and discontinuity conditions (10.165) imposed upon the Green's functions, we have

$$d_1 u_1(y) + d_2 u_2(y) = 0 \quad \text{and} \quad d_1 u_1'(y) + d_2 u_2'(y) = 1/a(y). \quad (10.167)$$

As before, we get

$$d_1 = -\frac{u_2(y)}{a(y)W(u_1(y), u_2(y))} \quad \text{and} \quad d_2 = \frac{u_1(y)}{a(y)W(u_1(y), u_2(y))},$$

where $W(u_1(y), u_2(y))$ is Wronskian of $u_1(y)$ and $u_2(y)$. Thus, we get

$$F_2(x, y) = \frac{u_2(x)u_1(y) - u_1(x)u_2(y)}{a(y)W(u_1(y), u_2(y))}. \quad (10.168)$$

2. Case II ($x < 0$): Next, we think of the case as below:

$$\begin{aligned} F_1(x, y) &= c_1 u_1(x) + c_2 u_2(x) \quad \text{for } y < x \leq 0, \\ F_2(x, y) &= d_1 u_1(x) + d_2 u_2(x) \quad \text{for } x < y < 0. \end{aligned} \quad (10.169)$$

Similarly proceeding as the above, we have $c_1 = c_2 = 0$. Also, we get

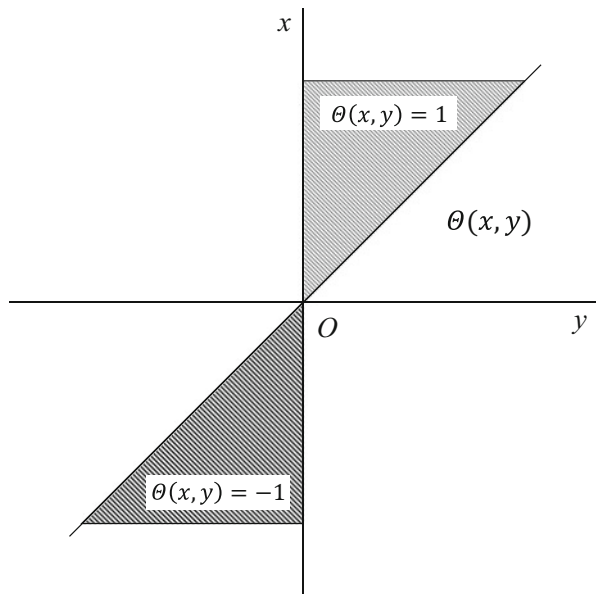
$$F_2(x, y) = \frac{u_1(x)u_2(y) - u_2(x)u_1(y)}{a(y)W(u_1(y), u_2(y))}. \quad (10.170)$$

Here notice that the sign is reversed in (10.170) relative to (10.168). This is because on the discontinuity condition, instead of (10.167) we have to have

$$d_1 u_1'(y) + d_2 u_2'(y) = -1/a(y).$$

This results from the fact that magnitude relationship between the arguments x and y has been reversed in (10.169) relative to (10.166).

Fig. 10.2 Graph of a function $\theta(x, y)$. $\theta(x, y) = 1$ or -1 in hatched areas, otherwise $\theta(x, y) = 0$



Summarizing the above argument, (10.168) is obtained in the domain $0 \leq y < x$; (10.170) is obtained in the domain $x < y < 0$. Noting this characteristic, we define a function such that

$$\theta(x, y) \equiv \theta(x - y)\theta(y) - \theta(y - x)\theta(-y). \quad (10.171)$$

Notice that

$$\theta(x, y) = -\theta(-x, -y).$$

That is, $\theta(x, y)$ is antisymmetric with respect to the origin.

Figure 10.2 shows a feature of $\theta(x, y)$. If the “initial point” is taken at $x = a$, we can use $\theta(x - a, y - a)$ instead; see Fig. 10.3. The function is described as

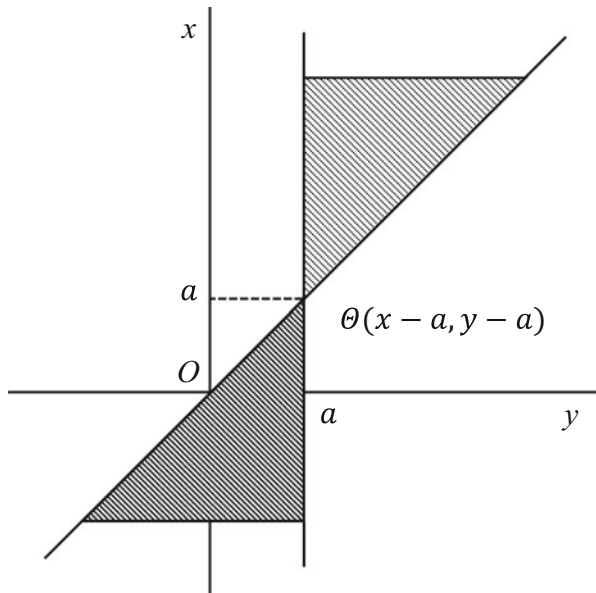
$$\theta(x - a, y - a) = \theta(x - y)\theta(y - a) - \theta(y - x)\theta(a - y).$$

Note that $\theta(x - a, y - a)$ can be obtained by shifting $\theta(x, y)$ toward the positive direction of the x - and y -axes by a (a can be either positive or negative; in Fig. 10.3 we assume $a > 0$). Using the $\theta(x, y)$ function, the Green’s function is described as

$$G(x, y) = \frac{u_2(x)u_1(y) - u_1(x)u_2(y)}{a(y)W(u_1(y), u_2(y))} \theta(x, y). \quad (10.172)$$

Defining a function $\mathcal{F}$ such that

Fig. 10.3 Graph of a function $\Theta(x - a, y - a)$. We assume $a > 0$



$$\mathcal{F}(x, y) \equiv \frac{u_2(x)u_1(y) - u_1(x)u_2(y)}{a(y)W(u_1(y), u_2(y))}, \quad (10.173)$$

we have

$$G(x, y) = \mathcal{F}(x, y)\Theta(x, y). \quad (10.174)$$

Notice that

$$\Theta(x, y) \neq \Theta(y, x) \quad \text{and} \quad G(x, y) \neq G(y, x). \quad (10.175)$$

It is therefore obvious that the differential operator is not Hermitian.

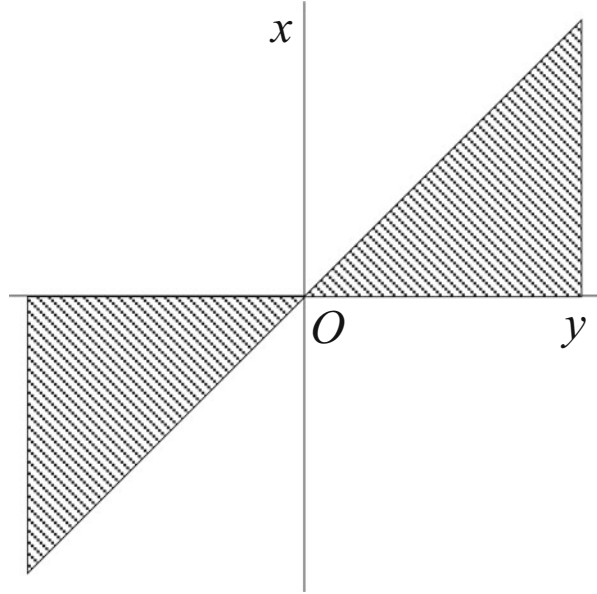
10.6.3 Estimation of Surface Terms

To include the surface term of the inhomogeneous case, we use (10.62).

$$\int_r^s dx \left\{ v^*(L_x u) - [L_x^\dagger v]^* u \right\} = \left[av^* \frac{du}{dx} - u \frac{d(av^*)}{dx} + buv^* \right]_r^s. \quad (10.62)$$

As before, we set $r = 0$ in (10.62). Also, we classify (10.62) into two cases according as $s > 0$ or $s < 0$.

Fig. 10.4 Domain of $G(y, x)$. Areas for which $G(y, x)$ does not vanish are hatched



1. Case I ($x > 0, y > 0$): (10.62) reads as

$$\int_0^\infty dx \left\{ v^*(L_x u) - [L_x^\dagger v]^* u \right\} = \left[av^* \frac{du}{dx} - u \frac{d(av^*)}{dx} + buv^* \right]_0^\infty. \quad (10.176)$$

This time, inserting

$$g(x, y) = G^*(y, x) = G(y, x) \quad (10.177)$$

into v of (10.176) and arranging terms, we have

$$\begin{aligned} u(y) = & \int_0^\infty dx G(y, x) d(x) \\ & - \left[a(x) G(y, x) \frac{du(x)}{dx} - u(x) \frac{da(x)}{dx} G(y, x) - u(x) a(x) \frac{\partial G(y, x)}{\partial x} + b(x) u(x) G(y, x) \right]_{x=0}^\infty. \end{aligned} \quad (10.178)$$

Note that in the above we used $L_x^\dagger g(x, y) = \delta(x - y)$. In Fig. 10.4, we depict a domain of $G(y, x)$ in which $G(y, x)$ does not vanish. Notice that we get the domain by folding back that of $\Theta(x, y)$ (see Fig. 10.2) relative to a straight line $y = x$. Thus, we find that $G(y, x)$ vanishes at $x > y$. So does $\frac{\partial G(y, x)}{\partial x}$; see Fig. 10.4. Namely, the second term of RHS of (10.178) vanishes at $x = \infty$. In other words, $g(x, y)$ and $G(y, x)$ must satisfy the adjoint BCs; i.e.,

$$g(\infty, y) = G(y, \infty) = 0. \quad (10.179)$$

At the same time, the upper limit of integration range of (10.178) can be set at y . Noting the above, we have

$$[\text{the first term of (10.178)}] = \int_0^y dx G(y, x) d(x). \quad (10.180)$$

Also with the second term of (10.178), we get

$$\begin{aligned} & [\text{the second term of (10.178)}] = \\ & + \left[a(x)G(y, x) \frac{du(x)}{dx} - u(x) \frac{da(x)}{dx} G(y, x) - u(x)a(x) \frac{\partial G(y, x)}{\partial x} + b(x)u(x)G(y, x) \right]_{x=0} \\ & = a(0)G(y, 0) \frac{du(0)}{dx} - u(0) \frac{da(0)}{dx} G(y, 0) - u(0)a(0) \frac{\partial G(y, x)}{\partial x} \Big|_{x=0} + b(0)u(0)G(y, 0). \end{aligned} \quad (10.181)$$

If we substitute inhomogeneous BCs

$$u(0) = \sigma_1 \quad \text{and} \quad \frac{du}{dx} \Big|_{x=0} = \sigma_2, \quad (10.182)$$

for (10.181) along with other appropriate values, we should be able to get a unique solution as

$$\begin{aligned} u(y) = & \int_0^y dx G(y, x) d(x) + \left[\sigma_2 a(0) - \sigma_1 \frac{da(0)}{dx} + \sigma_1 b(0) \right] G(y, 0) \\ & - \sigma_1 a(0) \frac{\partial G(y, x)}{\partial x} \Big|_{x=0}. \end{aligned} \quad (10.183)$$

Exchanging arguments x and y , we get

$$\begin{aligned} u(x) = & \int_0^x dy G(x, y) d(y) + \left[\sigma_2 a(0) - \sigma_1 \frac{da(0)}{dy} + \sigma_1 b(0) \right] G(x, 0) \\ & - \sigma_1 a(0) \frac{\partial G(x, y)}{\partial y} \Big|_{y=0}. \end{aligned} \quad (10.184)$$

Here, we consider that $\Theta(x, y) = 1$ in this region and use (10.174). Meanwhile, from (10.174), we have

$$\frac{\partial G(x, y)}{\partial y} = \frac{\partial \mathcal{F}(x, y)}{\partial y} \Theta(x, y) + \frac{\partial \Theta(x, y)}{\partial y} \mathcal{F}(x, y). \quad (10.185)$$

In the second term,

$$\begin{aligned} \frac{\partial \Theta(x, y)}{\partial y} &= \frac{\partial \theta(x-y)}{\partial y} \theta(y) + \theta(x-y) \frac{\partial \theta(y)}{\partial y} - \frac{\partial \theta(y-x)}{\partial y} \theta(-y) - \theta(y-x) \frac{\partial \theta(-y)}{\partial y} \\ &= -\delta(x-y)\theta(y) + \theta(x-y)\delta(y) - \delta(y-x)\theta(-y) + \theta(y-x)\delta(-y) \\ &= -\delta(x-y)[\theta(y) + \theta(-y)] + [\theta(x-y) + \theta(y-x)]\delta(y) \\ &= -\delta(x-y)[\theta(y) + \theta(-y)] + [\theta(x) + \theta(-x)]\delta(y) = -\delta(x-y) + \delta(y), \end{aligned} \quad (10.186)$$

where we used

$$\theta(x) + \theta(-x) \equiv 1 \quad (10.187)$$

as well as

$$f(y)\delta(y) = f(0)\delta(y) \quad (10.188)$$

and

$$\delta(-y) = \delta(y). \quad (10.189)$$

However, the function $-\delta(x-y) + \delta(y)$ is of secondary importance. It is because in (10.184) we may choose $[\varepsilon, x-\varepsilon]$ ($\varepsilon > 0$) for the domain y and put $\varepsilon \rightarrow +0$ after the integration and other calculations related to the surface terms. Therefore, $-\delta(x-y) + \delta(y)$ in (10.186) virtually vanishes.

Thus, we can express (10.185) as

$$\frac{\partial G(x, y)}{\partial y} = \frac{\partial \mathcal{F}(x, y)}{\partial y} \Theta(x, y) = \frac{\partial \mathcal{F}(x, y)}{\partial y}.$$

Then, finally we reach

$$\begin{aligned} u(x) &= \int_0^x dy \mathcal{F}(x, y) d(y) + \left[\sigma_2 a(0) - \sigma_1 \frac{da(0)}{dy} + \sigma_1 b(0) \right] \mathcal{F}(x, 0) \\ &\quad - \sigma_1 a(0) \frac{\partial \mathcal{F}(x, y)}{\partial y} \Big|_{y=0}. \end{aligned} \quad (10.190)$$

2. Case II ($x < 0, y < 0$): Similarly as the above, (10.62) reads as

$$u(y) = \int_{-\infty}^0 dx G(y, x) d(x) - \left[a(x) G(y, x) \frac{du(x)}{dx} - u(x) \frac{da(x)}{dx} G(y, x) - u(x) a(x) \frac{\partial G(y, x)}{\partial x} + b(x) u(x) G(y, x) \right]_{x=-\infty}^0.$$

Similarly as mentioned above, the lower limit of integration range is y . Considering both $G(y, x)$ and $\frac{\partial G(y, x)}{\partial x}$ vanish at $x < y$ (see Fig. 10.4), we have

$$\begin{aligned} u(y) &= \int_y^0 dx G(y, x) d(x) \\ &- \left[a(x) G(y, x) \frac{du(x)}{dx} - u(x) \frac{da(x)}{dx} G(y, x) - u(x) a(x) \frac{\partial G(y, x)}{\partial x} + b(x) u(x) G(y, x) \right]_{x=0} \\ &= - \int_0^y dx G(y, x) d(x) - \left[\sigma_2 a(0) - \sigma_1 \frac{da(0)}{dx} + \sigma_1 b(0) \right] G(y, 0) \\ &+ \sigma_1 a(0) \frac{\partial G(y, x)}{\partial x} \Big|_{x=0}. \end{aligned} \tag{10.191}$$

Comparing (10.191) with (10.183), we recognize that the sign of RHS of (10.191) has been reversed relative to RHS of (10.183). This is also the case after exchanging arguments x and y . Note, however, $\Theta(x, y) = -1$ in the present case. As a result, two minus signs cancel and (10.191) takes exactly the same expression as (10.183). Proceeding with calculations similarly, for both Cases I and II we arrive at a unified solution represented by (10.190) throughout a domain $(-\infty, +\infty)$.

10.6.4 Examples

To deepen understanding of Green's functions, we deal with tangible examples of the IVP below.

Example 10.5 Let us consider a following inhomogeneous differential equation:

$$\frac{d^2 u}{dx^2} + u = 1. \tag{10.192}$$

Note that (10.192) is formally the same differential equation of (10.141). We may encounter (10.192) when we are observing a motion of a charged harmonic oscillator

that is placed under a static electric field. We assume that a domain of the argument x is a whole range of real numbers. We set boundary conditions such that

$$u(0) = \sigma_1 \quad \text{and} \quad u'(0) = \sigma_2. \quad (10.193)$$

As in the case of Example 10.4, a fundamental set of solutions are given by

$$e^{ix} \quad \text{and} \quad e^{-ix}. \quad (10.194)$$

Therefore, following (10.173), we get

$$\mathcal{F}(x, y) = \sin(x - y). \quad (10.195)$$

Also following (10.190) we have

$$\begin{aligned} u(x) &= \int_0^x dy \sin(x - y) + \sigma_2 \sin x - \sigma_1 [-\cos(x - y)] \Big|_{y=0} \\ &= 1 - \cos x + \sigma_2 \sin x + \sigma_1 \cos x. \end{aligned} \quad (10.196)$$

In particular, if we choose $\sigma_1 = 1$ and $\sigma_2 = 0$, we have

$$u(x) \equiv 1. \quad (10.197)$$

This also ensures that this is a unique solution under the inhomogeneous BCs described as $\sigma_1 = 1$ and $\sigma_2 = 0$.

Example 10.6 Damped Oscillator If a harmonic oscillator undergoes friction, the oscillator exerts damped oscillation. Such an oscillator is said to be a damped oscillator. The damped oscillator is often dealt with when we think of bound electrons in a dielectric medium that undergo an effect of a dynamic external field varying with time. This is the case when the electron is placed in an alternating electric field or an electromagnetic wave.

An equation of motion of the damped oscillator is described as

$$m \frac{d^2 u}{dt^2} + r \frac{du}{dt} + ku = d(t), \quad (10.198)$$

where m is a mass of an electron; t denotes time; r is a damping constant; k is a spring constant of the damped oscillator; $d(t)$ represents an external force field (generally time-dependent). To seek a fundamental set of solutions of a homogeneous equation described as

$$m \frac{d^2 u}{dt^2} + r \frac{du}{dt} + ku = 0,$$

putting

$$u(t) = e^{i\rho t} \quad (10.199)$$

and inserting it to (10.198), we have

$$\left(-\rho^2 + \frac{r}{m} i\rho + \frac{k}{m} \right) e^{i\rho t} = 0.$$

Since $e^{i\rho t}$ does not vanish, we have

$$-\rho^2 + \frac{r}{m} i\rho + \frac{k}{m} = 0. \quad (10.200)$$

We call this equation a characteristic quadratic equation. We have three cases for the solution of a quadratic equation of (10.200). Solving (10.200) with respect to ρ , we get

$$\rho = \frac{ir}{2m} \mp \sqrt{-\frac{r^2}{4m^2} + \frac{k}{m}}. \quad (10.201)$$

Regarding (10.201), we have following three cases for the roots ρ : (1) ρ has two different pure imaginary roots; i.e., $-\frac{r^2}{4m^2} + \frac{k}{m} < 0$ (an over damping). (2) ρ has double root $\frac{ir}{2m}$; $-\frac{r^2}{4m^2} + \frac{k}{m} = 0$ (a critical damping). (3) ρ has two complex roots; $-\frac{r^2}{4m^2} + \frac{k}{m} > 0$ (a weak damping). Of these, Case (3) is characterized by an oscillating solution and has many applications in mathematical physics. For the Cases (1) and (2), however, we do not have an oscillating solution.

Case (1):

The characteristic roots are given by

$$\rho = \frac{ir}{2m} \mp i \sqrt{\frac{r^2}{4m^2} - \frac{k}{m}}.$$

Therefore, we have a fundamental set of solutions described by

$$u(t) = \exp\left(-\frac{rt}{2m}\right) \exp\left(\mp \sqrt{\frac{r^2}{4m^2} - \frac{k}{m}} t\right).$$

Then, a general solution is given by

$$u(t) = \exp\left(-\frac{rt}{2m}\right) \left[a \exp\left(\sqrt{\frac{r^2}{4m^2} - \frac{k}{m}} t\right) + b \exp\left(-\sqrt{\frac{r^2}{4m^2} - \frac{k}{m}} t\right) \right].$$

Case (2):

The characteristic roots are given by

$$\rho = \frac{ir}{2m}.$$

Therefore, one of the solutions is

$$u_1(t) = \exp\left(-\frac{rt}{2m}\right).$$

Another solution $u_2(t)$ is given by

$$u_2(t) = c \frac{\partial u_1(t)}{\partial (i\rho)} = c' t \exp\left(-\frac{rt}{2m}\right),$$

where c and c' are appropriate constants. Thus, general solution is given by

$$u(t) = a \exp\left(-\frac{rt}{2m}\right) + bt \exp\left(-\frac{rt}{2m}\right).$$

The most important and interesting feature emerges as a “damped oscillator” in the next Case (3) in many fields of natural science. We are particularly interested in this case.

Case (3):

Suppose that the damping is relatively weak such that the characteristic equation has two complex roots. Let us examine further details of this case following the prescriptions of IVPs. We divide (10.198) by m for the sake of easy handling of the differential equation such that

$$\frac{d^2u}{dt^2} + \frac{r}{m} \frac{du}{dt} + \frac{k}{m} u = \frac{1}{m} d(t).$$

Putting

$$\omega \equiv \sqrt{-\frac{r^2}{4m^2} + \frac{k}{m}}, \quad (10.202)$$

we get a fundamental set of solutions described as

$$u(t) = \exp\left(-\frac{rt}{2m}\right) \exp(\mp i\omega t). \quad (10.203)$$

Given BCs, following (10.172) we get as a Green's function

$$\begin{aligned} G(t, \tau) &= \frac{u_2(t)u_1(\tau) - u_1(t)u_2(\tau)}{W(u_1(\tau), u_2(\tau))} \Theta(t, \tau) \\ &= \frac{1}{\omega} e^{-\frac{r}{2m}(t-\tau)} \sin \omega(t-\tau) \Theta(t, \tau), \end{aligned} \quad (10.204)$$

where $u_1(t) = \exp(-\frac{rt}{2m}) \exp(i\omega t)$ and $u_2(t) = \exp(-\frac{rt}{2m}) \exp(-i\omega t)$.

We examine whether $G(t, \tau)$ is eligible for the Green's function as follows:

$$\begin{aligned} \frac{dG}{dt} &= \frac{1}{\omega} \left[\left(-\frac{r}{2m} \right) e^{-\frac{r}{2m}(t-\tau)} \sin \omega(t-\tau) + \omega e^{-\frac{r}{2m}(t-\tau)} \cos \omega(t-\tau) \right] \Theta(t, \tau) \\ &\quad + \frac{1}{\omega} e^{-\frac{r}{2m}(t-\tau)} \sin \omega(t-\tau) [\delta(t-\tau)\theta(\tau) + \delta(\tau-t)\theta(-\tau)]. \end{aligned} \quad (10.205)$$

The second term of (10.205) vanishes because $\sin \omega(t-\tau)\delta(t-\tau) = 0 \cdot \delta(t-\tau) = 0$. Thus,

$$\begin{aligned} \frac{d^2G}{dt^2} &= \frac{1}{\omega} \left[\left(-\frac{r}{2m} \right)^2 e^{-\frac{r}{2m}(t-\tau)} \sin \omega(t-\tau) - \frac{r}{m} \omega e^{-\frac{r}{2m}(t-\tau)} \cos \omega(t-\tau) \right. \\ &\quad \left. - \omega^2 e^{-\frac{r}{2m}(t-\tau)} \sin \omega(t-\tau) \right] \Theta(t, \tau) + \frac{1}{\omega} \left[\left(-\frac{r}{2m} \right) e^{-\frac{r}{2m}(t-\tau)} \sin \omega(t-\tau) \right. \\ &\quad \left. + \omega e^{-\frac{r}{2m}(t-\tau)} \cos \omega(t-\tau) \right] \times [\delta(t-\tau)\theta(\tau) + \delta(\tau-t)\theta(-\tau)]. \end{aligned} \quad (10.206)$$

In the last term using the property of the δ function and θ function, we get $\delta(t-\tau)$. That is, the following part in (10.206) is calculated such that

$$\begin{aligned} &e^{-\frac{r}{2m}(t-\tau)} \cos \omega(t-\tau) [\delta(t-\tau)\theta(\tau) + \delta(\tau-t)\theta(-\tau)] \\ &= e^{-\frac{r}{2m}0} \cos(\omega \cdot 0) \{ \delta(t-\tau)[\theta(\tau) + \theta(-\tau)] \} \\ &= \delta(t-\tau) [\theta(\tau) + \theta(-\tau)] = \delta(t-\tau). \end{aligned}$$

Thus, rearranging (10.206), we get

$$\begin{aligned}
\frac{d^2 G}{dt^2} &= \delta(t - \tau) + \frac{1}{\omega} \left\{ - \left[\left(-\frac{r}{2m} \right)^2 + \omega^2 \right] e^{-\frac{r}{2m}(t-\tau)} \sin \omega(t - \tau) \right. \\
&\quad \left. - \frac{r}{m} \left[\left(-\frac{r}{2m} \right) e^{-\frac{r}{2m}(t-\tau)} \sin \omega(t - \tau) + \omega e^{-\frac{r}{2m}(t-\tau)} \cos \omega(t - \tau) \right] \right\} \Theta(t, \tau) \\
&= \delta(t - \tau) - \frac{k}{m} G - \frac{r}{m} \frac{dG}{dt},
\end{aligned} \tag{10.207}$$

where we used (10.202) and (10.204) for the last equality. Rearranging (10.207) once again, we have

$$\frac{d^2 G}{dt^2} + \frac{r}{m} \frac{dG}{dt} + \frac{k}{m} G = \delta(t - \tau). \tag{10.208}$$

Defining the following operator

$$L_t \equiv \frac{d^2}{dt^2} + \frac{r}{m} \frac{d}{dt} + \frac{k}{m}, \tag{10.209}$$

we get

$$L_t G = \delta(t - \tau). \tag{10.210}$$

Note that this expression is consistent with (10.164). Thus, we find that (10.210) satisfies the condition (10.123) of the Green's function, where the weight function is identified with unity.

Now, suppose that a sinusoidally changing external field $e^{i\Omega t}$ influences the motion of the damped oscillator. Here we assume that an amplitude of the external field is unity. Then, we have

$$\frac{d^2 u}{dt^2} + \frac{r}{m} \frac{du}{dt} + \frac{k}{m} u = \frac{1}{m} e^{i\Omega t}. \tag{10.211}$$

Thus, as a solution of the homogeneous boundary conditions [i.e., $u(0) = u(\dot{0}) = 0$] we get

$$u(t) = \frac{1}{m\omega} \int_0^t e^{-\frac{r}{2m}(t-\tau)} e^{i\Omega t} \sin \omega(t - \tau) d\tau, \tag{10.212}$$

where t is an arbitrary positive or negative time. Equation (10.212) shows that with the real part we have an external field $\cos \Omega t$ and that with the imaginary part we have an external field $\sin \Omega t$. To calculate (10.212) we use

$$\sin \omega(t - \tau) = \frac{1}{2i} \left[e^{i\omega(t - \tau)} - e^{-i\omega(t - \tau)} \right]. \quad (10.213)$$

Then, the equation can readily be solved by integration of exponential functions, even though we have to do somewhat lengthy (but straightforward) calculations.

Thus for the real part (i.e., the external field is $\cos \Omega t$), we get a solution

$$\begin{aligned} Cu(t) = & \frac{1}{m} \left(\frac{r}{m} \right) \Omega \sin \Omega t + \frac{1}{m} \left(\frac{r}{2m} \right)^2 \cos \Omega t - \frac{1}{m} (\Omega^2 - \omega^2) \cos \Omega t \\ & + \frac{1}{m} e^{-\frac{r}{2m}t} \left[(\Omega^2 - \omega^2) \cos \omega t - \frac{r}{2m} \frac{1}{\omega} (\Omega^2 + \omega^2) \sin \omega t \right. \\ & \left. - \left(\frac{r}{2m} \right)^2 \cos \omega t - \left(\frac{r}{2m} \right)^3 \frac{1}{\omega} \sin \omega t \right], \end{aligned} \quad (10.214)$$

where C is a constant [i.e., a constant denominator of $u(t)$] expressed as

$$C = (\Omega^2 - \omega^2)^2 + \frac{r^2}{2m^2} (\Omega^2 + \omega^2) + \left(\frac{r}{2m} \right)^4. \quad (10.215)$$

For the imaginary part (i.e., the external field is $\sin \Omega t$) we get

$$\begin{aligned} Cu(t) = & -\frac{1}{m} (\Omega^2 - \omega^2) \sin \Omega t - \frac{1}{m} \left(\frac{r}{m} \right) \Omega \cos \Omega t + \frac{1}{m} \left(\frac{r}{2m} \right)^2 \sin \Omega t \\ & + \frac{1}{m} e^{-\frac{r}{2m}t} \left[\frac{\Omega}{\omega} (\Omega^2 - \omega^2) \sin \omega t + \frac{r}{m} \Omega \cos \omega t + \left(\frac{r}{2m} \right)^2 \frac{\Omega}{\omega} \sin \omega t \right]. \end{aligned} \quad (10.216)$$

In Fig. 10.5 we show an example that depicts the positions of a damped oscillator as a function of time. In Fig. 10.5a, an amplitude of an envelope gradually diminishes with time. An enlarged diagram near the origin (Fig. 10.5b) clearly reflects the initial conditions $u(0) = \dot{u}(0) = 0$. In Fig. 10.5, we put $m = 1$ [kg], $\Omega = 1$ [1/s], $\omega = 0.94$ [1/s], and $r = 0.006$ [kg/s]. In the above calculations, if r/m is small enough (i.e., damping is small enough), the third order and fourth order of r/m may be ignored and the approximation is precise enough.

In the case of inhomogeneous BCs, given $\sigma_1 = u(0)$ and $\sigma_2 = \dot{u}(0)$ we can decide additional surface term $S(t)$ such that

$$S(t) = \left(\sigma_2 + \sigma_1 \frac{r}{2m} \right) \frac{1}{\omega} e^{-\frac{r}{2m}t} \sin \omega t + \sigma_1 e^{-\frac{r}{2m}t} \cos \omega t. \quad (10.217)$$

The term $S(t)$ corresponds to the second and third terms of (10.190). Thus, from (10.212) and (10.217),

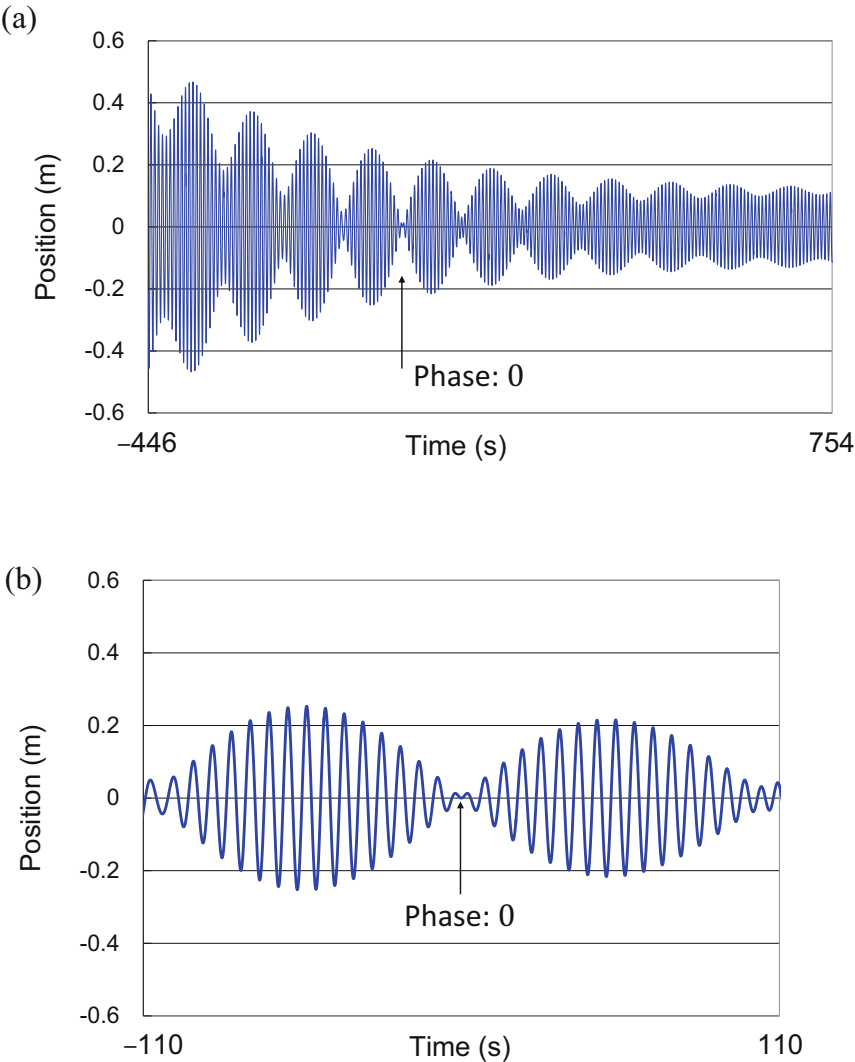


Fig. 10.5 Example of a damped oscillation as a function of t . The data are taken from (10.214). (a) Overall profile. (b) Profile enlarged near the origin

$$u(t) + S(t)$$

gives a unique solution for the SOLDE with inhomogeneous BCs. Notice that $S(t)$ does not depend on the external field.

In the above calculations, use

$$\mathcal{F}(t, \tau) = \frac{1}{\omega} e^{-\frac{r}{2m}(t-\tau)} \sin \omega(t-\tau)$$

together with $a(t) \equiv 1$, $b(t) \equiv r/m$, $d(t) \equiv e^{i\Omega t}/m$ with respect to (10.190). The conformation of (10.217) is left for readers.

10.7 Eigenvalue Problems

We often encounter eigenvalue problems in mathematical physics. Of these, those related to Hermitian differential operators have particularly interesting and important features. The eigenvalue problems we have considered in Part I are typical illustrations. Here we investigate general properties of the eigenvalue problems.

Returning to the case of homogeneous BCs, we consider a following homogeneous SOLDE:

$$a(x) \frac{d^2 u}{dx^2} + b(x) \frac{du}{dx} + c(x)u = 0. \quad (10.5)$$

Defining a following differential operator L_x such that

$$L_x = a(x) \frac{d^2}{dx^2} + b(x) \frac{d}{dx} + c(x), \quad (10.55)$$

we have a homogeneous equation

$$L_x u(x) = 0. \quad (10.218)$$

Putting a constant $-\lambda$ instead of $c(x)$, we have

$$a(x) \frac{d^2 u}{dx^2} + b(x) \frac{du}{dx} - \lambda u = 0. \quad (10.219)$$

If we define a differential operator L_x such that

$$L_x = a(x) \frac{d^2}{dx^2} + b(x) \frac{d}{dx} - \lambda, \quad (10.220)$$

we have a homogeneous equation

$$L_x u = 0 \quad (10.221)$$

to express (10.219). Instead, if we define a differential operator L_x such that

$$L_x = a(x) \frac{d^2}{dx^2} + b(x) \frac{d}{dx},$$

we have the same homogeneous equation

$$L_x u = \lambda u \quad (10.222)$$

to express (10.219).

Equations (10.221) and (10.222) are essentially the same except that the expression is different. The expression using (10.222) is familiar to us as an eigenvalue equation. The difference between (10.5) and (10.219) is that whereas $c(x)$ in (10.5) is a given *fixed* function, λ in (10.219) is constant, but may be *varied* according to the solution of $u(x)$. One of the most essential properties of the eigenvalue problem that is posed in the form of (10.222) is that its solution is not uniquely determined as already studied in various cases of Part I.

Remember that the methods based upon the Green's function are valid for a problem to which a homogeneous differential equation has a trivial solution (i.e., identically zero) under homogeneous BCs. In contrast to this situation, even though the eigenvalue problem is basically posed as a homogeneous equation under homogeneous BCs, non-trivial solutions are expected to be obtained. In this respect, we have seen that in Part I we rejected a trivial solution (i.e., identically zero) because of no physical meaning.

As exemplified in Part I, the eigenvalue problems that appear in mathematical physics are closely connected to the Hermiticity of (differential) operators. This is because in many cases an eigenvalue is required to be real. We have already examined how we can convert a differential operator to the self-adjoint form. That is, if we define $p(x)$ as in (10.26), we have the self-adjoint operator as described in (10.70). As a symbolic description, we have

$$w(x)L_x u = \frac{d}{dx} \left[p(x) \frac{du}{dx} \right] + c(x)w(x)u. \quad (10.223)$$

In the same way, multiplying both sides of (10.222) by $w(x)$, we get

$$w(x)L_x u = \lambda w(x)u. \quad (10.224)$$

For instance, Hermite differential equation that has already appeared as (2.118) in Sects. 2.3 is described as

$$\frac{d^2 u}{dx^2} - 2x \frac{du}{dx} + 2nu = 0. \quad (10.225)$$

If we express (10.225) as in (10.224), multiplying e^{-x^2} on both sides of (10.225) we have

$$\frac{d}{dx} \left(e^{-x^2} \frac{du}{dx} \right) + 2ne^{-x^2} u = 0. \quad (10.226)$$

Notice that the differential operator has been converted to a self-adjoint form according to (10.31) that defines a real and positive weight function e^{-x^2} in the present case. The domain of the Hermite differential equation is $(-\infty, +\infty)$ at the endpoints (i.e., $\pm\infty$) of which the surface term of RHS of (10.69) approaches zero sufficiently rapidly in virtue of e^{-x^2} .

In Sects. 3.4, and 3.5, in turn, we dealt with the (associated) Legendre differential equation (3.127) for which the relevant differential operator is self-adjoint. The surface term corresponding to (10.69) vanishes. It is because $(1 - \xi^2)$ vanishes at the endpoints $\xi = \cos \theta = \pm 1$ (i.e., $\theta = 0$ or π) from (3.107). Thus, the Hermiticity is automatically ensured for the (associated) Legendre differential equation as well as Hermite differential equation. In those cases, even though the differential equations do not satisfy any particular BCs, the Hermiticity is yet ensured.

In the theory of differential equations, the aforementioned properties of the Hermitian operators have been fully investigated as the so-called Sturm–Liouville system (or problem) in the form of a homogeneous differential equation. The related differential equations are connected to classical orthogonal polynomials having personal names such as Hermite, Laguerre, Jacobi, Gegenbauer, Legendre, Tchebichef, etc. These equations frequently appear in quantum mechanics and electromagnetism as typical examples of Sturm–Liouville system. They can be converted to differential equations by multiplying an original form by a weight function. The resulting equations can be expressed as

$$\frac{d}{dx} \left[a(x)w(x) \frac{dY_n(x)}{dx} \right] + \lambda_n w(x) Y_n(x) = 0, \quad (10.227)$$

where $Y_n(x)$ is a collective representation of classical orthogonal polynomials. Equation (10.226) is an example. Conventionally, a following form is adopted instead of (10.227):

$$\frac{1}{w(x)} \frac{d}{dx} \left[a(x)w(x) \frac{dY_n(x)}{dx} \right] + \lambda_n Y_n(x) = 0, \quad (10.228)$$

where we put $(aw)' = bw$. That is, the differential equation is originally described as

$$a(x) \frac{d^2 Y_n(x)}{dx^2} + b(x) \frac{dY_n(x)}{dx} + \lambda_n Y_n(x) = 0. \quad (10.229)$$

In the case of Hermite polynomials, for instance, $a(x) = 1$ and $w(x) = e^{-x^2}$. Since we have $(aw)' = -2xe^{-x^2} = bw$, we can put $b = -2x$. Examples including this case are tabulated in Table 10.2. The eigenvalues λ_n are associated with real numbers

Table 10.2 Classical polynomials and their related SOLDEs

Name of the polynomial	SOLDE form	Weight function: $w(x)$	Domain
Hermite: $H_n(x)$	$\frac{d^2}{dx^2} H_n(x) - 2x \frac{d}{dx} H_n(x) + 2n H_n(x) = 0$	e^{-x^2}	$(-\infty, +\infty)$
Laguerre: $L_n^\nu(x)$	$x \frac{d^2}{dx^2} L_n^\nu(x) + (\nu + 1 - x) \frac{d}{dx} L_n^\nu(x) + n L_n^\nu(x) = 0$	$x^\nu e^{-x} \ (\nu > -1)$	$[0, +\infty)$
Gegenbauer: $C_n^\lambda(x)$	$(1 - x^2) \frac{d^2}{dx^2} C_n^\lambda(x) - (2\lambda + 1)x \frac{d}{dx} C_n^\lambda(x) + n(n + 2\lambda) C_n^\lambda(x) = 0$	$(1 - x^2)^{\lambda - \frac{1}{2}}$ $(\lambda > -\frac{1}{2})$	$[-1, +1]$
Legendre: $P_n(x)$	$(1 - x^2) \frac{d^2}{dx^2} P_n(x) - 2x \frac{d}{dx} P_n(x) + n(n + 1) P_n(x) = 0$	1	$[-1, +1]$

that characterize the individual physical systems. The related fields have wide applications in many branches of natural science.

After having converted the operator to the self-adjoint form; i.e., $L_x = L_x^\dagger$, instead of (10.100) we have

$$\int_r^s dx w(x) \{v^*(L_x u) - [L_x v]^* u\} = 0. \quad (10.230)$$

Rewriting it, we get

$$\int_r^s dx v^* [w(x) L_x u] = \int_r^s dx [w(x) L_x v]^* u. \quad (10.231)$$

If we use an inner product notation described by (10.88), we get

$$\langle v | L_x u \rangle = \langle L_x v | u \rangle. \quad (10.232)$$

Here let us think of two eigenfunctions ψ_i and ψ_j that belong to an eigenvalue λ_i and λ_j , respectively. That is,

$$w(x) L_x \psi_i = \lambda_i w(x) \psi_i \quad \text{and} \quad w(x) L_x \psi_j = \lambda_j w(x) \psi_j. \quad (10.233)$$

Inserting ψ_i and ψ_j into u and v , respectively, in (10.232), we have

$$\begin{aligned} \langle \psi_j | L_x \psi_i \rangle &= \langle \psi_j | \lambda_i \psi_i \rangle = \lambda_i \langle \psi_j | \psi_i \rangle = \lambda_j^* \langle \psi_j | \psi_i \rangle = \langle \lambda_j \psi_j | \psi_i \rangle \\ &= \langle L_x \psi_j | \psi_i \rangle. \end{aligned} \quad (10.234)$$

With the second and third equalities, we have used a rule of the inner product (see Parts I and III). Therefore, we get

$$(\lambda_i - \lambda_j^*) \langle \psi_j | \psi_i \rangle = 0. \quad (10.235)$$

Putting $i = j$ in (10.235), we get

$$(\lambda_i - \lambda_i^*) \langle \psi_i | \psi_i \rangle = 0. \quad (10.236)$$

An inner product $\langle \psi_i | \psi_i \rangle$ vanishes if and only if $|\psi_i\rangle \equiv 0$; see inner product calculation rules of Sect. 13.1. However, $|\psi_i\rangle \equiv 0$ is not acceptable as a physical state. Therefore, we must have $\langle \psi_i | \psi_i \rangle \neq 0$. Thus, we get

$$\lambda_i - \lambda_i^* = 0 \quad \text{or} \quad \lambda_i = \lambda_i^*. \quad (10.237)$$

The relation (10.237) obviously indicates that λ_i is real; i.e., we find that eigenvalues of an Hermitian operator are real. If $\lambda_i \neq \lambda_j = \lambda_j^*$, from (10.235) we get

$$\langle \psi_j | \psi_i \rangle = 0. \quad (10.238)$$

That is, $|\psi_j\rangle$ and $|\psi_i\rangle$ are orthogonal to each other.

We often encounter related orthogonality relationship between vectors and functions. We saw several cases in Part I and will see other cases in Part III.

References

1. Dennery P, Krzywicki A (1996) Mathematics for physicists. Dover, New York
2. Stakgold I (1998) Green's functions and boundary value problems, 2nd edn. Wiley, New York

Part III

Linear Vector Spaces

In this part, we treat vectors and their transformations in linear vector spaces so that we can address various aspects of mathematical physics systematically but intuitively. We outline the general principles of linear vector spaces mostly from an algebraic point of view. Starting with abstract definition and description of vectors, we deal with their transformation in a vector space using a matrix. An inner product is a central concept in the theory of a linear vector space so that two vectors can be associated with each other to yield a scalar. Unlike many books of linear algebra and linear vector spaces, however, we describe canonical forms of matrices before considering the inner product. This is because we can treat the topics in light of the abstract theory of matrices and vector space without a concept of the inner product. Of the canonical forms of matrices, Jordan canonical form is of paramount importance. We study how it is constructed providing a tangible example.

In relation to the inner product space, normal operators such as Hermitian operators and unitary operators frequently appear in quantum mechanics and electromagnetism. From a general aspect, we revisit the theory of Hermitian operators that often appeared in both Parts I and II.

The last part deals with exponential functions of matrices. The relevant topics can be counted as one of the important branches of applied mathematics. Also, the exponential functions of matrices play an essential role in the theory of continuous groups that will be dealt with in Parts IV and V.

Chapter 11

Vectors and Their Transformation



In this chapter, we deal with the theory of finite-dimensional linear vector spaces. Such vector spaces are spanned by a finite number of linearly independent vectors, namely, basis vectors. In conjunction with developing an abstract concept and theory, we mention a notion of mapping among mathematical elements. A linear transformation of a vector is a special kind of mapping. In particular, we focus on endomorphism within a n -dimensional vector space V^n . Here, the endomorphism is defined as a linear transformation: $V^n \rightarrow V^n$. The endomorphism is represented by a (n, n) square matrix. This is most often the case with physical and chemical applications, when we deal with matrix algebra. In this book we focus on this type of transformation.

A non-singular matrix plays an important role in the endomorphism. In this connection, we consider its inverse matrix and determinant. All these fundamental concepts supply us with a sufficient basis for better understanding of the theory of the linear vector spaces. Through these processes, we should be able to get acquainted with connection between algebraic and analytical approaches and gain a broad perspective on various aspects of mathematical physics and related fields.

11.1 Vectors

From both fundamental and practical points of view, it is desirable to define linear vector spaces in an abstract way. Suppose that V is a set of elements denoted by $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, etc. called vectors. The set V is a linear vector space (or simply a vector space), if a sum $\mathbf{a} + \mathbf{b} \in V$ is defined for any pair of vectors $\mathbf{a}$ and $\mathbf{b}$ and the elements of V satisfy the following mathematical relations:

$$(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c}), \quad (11.1)$$

$$\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}, \quad (11.2)$$

$$\mathbf{a} + \mathbf{0} = \mathbf{a}, \quad (11.3)$$

$$\mathbf{a} + (-\mathbf{a}) = \mathbf{0}. \quad (11.4)$$

For the above $\mathbf{0}$ is called the zero vector. Furthermore, for $\mathbf{a} \in V$, $c\mathbf{a} \in V$ is defined (c is a complex number called a scalar) and we assume the following relations among vectors ($\mathbf{a}$ and $\mathbf{b}$) and scalars (c and d):

$$(cd)\mathbf{a} = c(da), \quad (11.5)$$

$$1\mathbf{a} = \mathbf{a}, \quad (11.6)$$

$$c(\mathbf{a} + \mathbf{b}) = c\mathbf{a} + c\mathbf{b}, \quad (11.7)$$

$$(c + d)\mathbf{a} = c\mathbf{a} + d\mathbf{a}. \quad (11.8)$$

On the basis of the above relations, we can construct the following expression called a linear combination:

$$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \cdots + c_n\mathbf{a}_n.$$

If this linear combination is equated to zero, we obtain

$$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \cdots + c_n\mathbf{a}_n = \mathbf{0}. \quad (11.9)$$

If (11.9) holds only in the case where every $c_i = 0$ ($1 \leq i \leq n$), the vectors $\mathbf{a}_1, \mathbf{a}_2, \cdots, \mathbf{a}_n$ are said to be linearly independent. In this case the relation represented by (11.9) is said to be trivial. If the relation is non-trivial (i.e., $\exists c_i \neq 0$), those vectors are said to be linearly dependent.

If in the vector space V the maximum number of linearly independent vectors is n , V is said to be an n -dimensional vector space and sometimes denoted by V^n . In this case any vector $\mathbf{x}$ of V^n is expressed uniquely as a linear combination of linearly independent vectors such that

$$\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n. \quad (11.10)$$

Suppose $\mathbf{x}$ is denoted by

$$\mathbf{x} = x'_1 \mathbf{a}_1 + x'_2 \mathbf{a}_2 + \cdots + x'_n \mathbf{a}_n. \quad (11.11)$$

Subtracting both sides of (11.11) from (11.10), we obtain

$$\mathbf{0} = (x_1 - x'_1) \mathbf{a}_1 + (x_2 - x'_2) \mathbf{a}_2 + \cdots + (x_n - x'_n) \mathbf{a}_n. \quad (11.12)$$

Linear independence of the vectors $\mathbf{a}_1, \mathbf{a}_2, \cdots, \mathbf{a}_n$ implies $x_n - x'_n = 0$; i.e., $x_n = x'_n$ ($1 \leq i \leq n$). These n linearly independent vectors are referred to as basis vectors. A vector space that has a finite number of basis vectors is called finite-dimensional; otherwise, it is infinite-dimensional.

Alternatively, we express (11.10) as

$$\mathbf{x} = (\mathbf{a}_1 \cdots \mathbf{a}_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}. \quad (11.13)$$

A set of coordinates $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ is called a column vector (or a numerical vector) that indicates an “address” of the vector $\mathbf{x}$ with respect to the basis vectors $\mathbf{a}_1, \mathbf{a}_2, \cdots, \mathbf{a}_n$. These vectors are represented as a row vector in (11.13). Any vector in V^n can be expressed as a linear combination of the basis vectors and, hence, we say that V^n is spanned by $\mathbf{a}_1, \mathbf{a}_2, \cdots, \mathbf{a}_n$. This is represented as

$$V^n = \text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \cdots, \mathbf{a}_n\}. \quad (11.14)$$

Let us think of a subset W of V^n (i.e., $W \subset V^n$). If the following relations hold for W , W is said to be a (linear) subspace of V^n .

$$\begin{aligned} \mathbf{a}, \mathbf{b} \in W &\Rightarrow \mathbf{a} + \mathbf{b} \in W, \\ \mathbf{a} \in W &\Rightarrow c\mathbf{a} \in W. \end{aligned} \quad (11.15)$$

These two relations ensure that the relations of (11.1)–(11.8) hold for W as well. The dimension of W is equal to or smaller than n . For instance, $W = \text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \cdots, \mathbf{a}_r\}$ ($r \leq n$) is a subspace of V^n . If $r = n$, $W = V^n$. Suppose that there are two subspaces $W_1 = \text{Span}\{\mathbf{a}_1\}$ and $W_2 = \text{Span}\{\mathbf{a}_2\}$. Note that in this case $W_1 \cup W_2$ is not a subspace, because $W_1 \cup W_2$ does not contain $\mathbf{a}_1 + \mathbf{a}_2$. However, a set U defined by

$$U = \{\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2; \forall \mathbf{x}_1 \in W_1, \forall \mathbf{x}_2 \in W_2\} \quad (11.16)$$

is a subspace of V^n . We denote this subspace by $W_1 + W_2$.

To show this is in fact a subspace, suppose that $\mathbf{x}, \mathbf{y} \in W_1 + W_2$. Then, we may express $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2$ and $\mathbf{y} = \mathbf{y}_1 + \mathbf{y}_2$, where $\mathbf{x}_1, \mathbf{y}_1 \in W_1$; $\mathbf{x}_2, \mathbf{y}_2 \in W_2$. We have $\mathbf{x} + \mathbf{y} = (\mathbf{x}_1 + \mathbf{y}_1) + (\mathbf{x}_2 + \mathbf{y}_2)$, where $\mathbf{x}_1 + \mathbf{y}_1 \in W_1$ and $\mathbf{x}_2 + \mathbf{y}_2 \in W_2$ because both W_1 and W_2 are subspaces. Therefore, $\mathbf{x} + \mathbf{y} \in W_1 + W_2$. Meanwhile, with any scalar

$c, c\mathbf{x} = c\mathbf{x}_1 + c\mathbf{x}_2 \in W_1 + W_2$. By definition (11.15), $W_1 + W_2$ is a subspace accordingly. Suppose here $\mathbf{x}_1 \in W_1$. Then, $\mathbf{x}_1 = \mathbf{x}_1 + \mathbf{0} \in W_1 + W_2$. Then, $W_1 \subset W_1 + W_2$. Similarly, we have $W_2 \subset W_1 + W_2$. Thus, $W_1 + W_2$ contains both W_1 and W_2 . Conversely, let W be an arbitrary subspace that contains both W_1 and W_2 . Then, we have $\forall \mathbf{x}_1 \in W_1 \subset W$ and $\forall \mathbf{x}_2 \in W_2 \subset W$ and, hence, we have $\mathbf{x}_1 + \mathbf{x}_2 \in W$ by definition (11.15). But, from (11.16) $W_1 + W_2 = \{\mathbf{x}_1 + \mathbf{x}_2; \forall \mathbf{x}_1 \in W_1, \forall \mathbf{x}_2 \in W_2\}$. Hence, $W_1 + W_2 \subset W$. Consequently, any subspace necessarily contains $W_1 + W_2$. This implies that $W_1 + W_2$ is the *smallest* subspace that contains both W_1 and W_2 .

Example 11.1 Consider a three-dimensional Cartesian space $\mathbb{R}^3$ (Fig. 11.1). We regard the xy -plane and yz -plane as a subspace W_1 and W_2 , respectively, and $\mathbb{R}^3 = W_1 + W_2$. In Fig. 11.1a, a vector $\overrightarrow{OB}$ (in $\mathbb{R}^3$) is expressed as $\overrightarrow{OA} + \overrightarrow{AB}$ (i.e., a sum of a vector in W_1 and that in W_2). Alternatively, the same vector $\overrightarrow{OB}$ can be expressed as $\overrightarrow{OA'} + \overrightarrow{A'B}$. Conversely, we can designate a subspace in a different way; i.e., in Fig. 11.1b the z -axis is chosen for a subspace W_3 instead of W_2 . We have $\mathbb{R}^3 = W_1 + W_3$ as well. In this case, however, $\overrightarrow{OB}$ is uniquely expressed as $\overrightarrow{OB} = \overrightarrow{OP} + \overrightarrow{PB}$. Notice that in Fig. 11.1a $W_1 \cap W_2 = \text{Span}\{\mathbf{e}_2\}$, where $\mathbf{e}_2$ is a unit vector in the positive direction of the y -axis. In Fig. 11.1b, however, we have $W_1 \cap W_3 = \{\mathbf{0}\}$.

We can generalize this example to the following theorem.

Theorem 11.1 Let W_1 and W_2 be subspaces of V and $V = W_1 + W_2$. Then a vector $\mathbf{x}$ in V is uniquely expressed as

$$\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2; \mathbf{x}_1 \in W_1, \mathbf{x}_2 \in W_2,$$

if and only if $W_1 \cap W_2 = \{\mathbf{0}\}$.

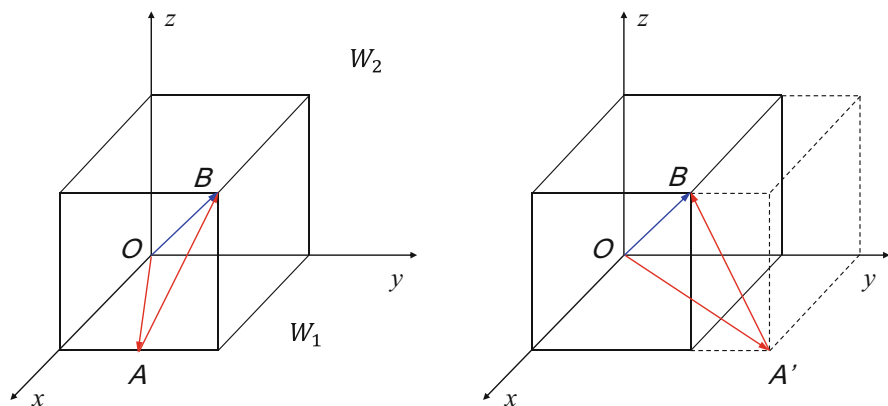
Proof Suppose $W_1 \cap W_2 = \{\mathbf{0}\}$ and $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2 = \mathbf{x}'_1 + \mathbf{x}'_2, \mathbf{x}_1, \mathbf{x}'_1 \in W_1, \mathbf{x}_2, \mathbf{x}'_2 \in W_2$. Then $\mathbf{x}_1 - \mathbf{x}'_1 = \mathbf{x}'_2 - \mathbf{x}_2$. LHS belongs to W_1 and RHS belongs to W_2 . Both sides belong to $W_1 \cap W_2$ accordingly. Hence, from the supposition both the sides should be equal to a zero vector. Therefore, $\mathbf{x}_1 = \mathbf{x}'_1, \mathbf{x}_2 = \mathbf{x}'_2$. This implies that $\mathbf{x}$ is expressed uniquely as $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2$. Conversely, suppose the vector representation ($\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2$) is unique and $\mathbf{x} \in W_1 \cap W_2$. Then, $\mathbf{x} = \mathbf{x} + \mathbf{0} = \mathbf{0} + \mathbf{x}; \mathbf{x}, \mathbf{0} \in W_1$ and $\mathbf{x}, \mathbf{0} \in W_2$. Uniqueness of the representation implies that $\mathbf{x} = \mathbf{0}$. Consequently, $W_1 \cap W_2 = \{\mathbf{0}\}$ follows. This completes the proof.

In case $W_1 \cap W_2 = \{\mathbf{0}\}$, $V = W_1 + W_2$ is said to be a direct sum of W_1 and W_2 or we say that V is decomposed into a direct sum of W_1 and W_2 . We symbolically denote this by

$$V = W_1 \oplus W_2. \quad (11.17)$$

In this case, the following equality holds:

(a)



(b)

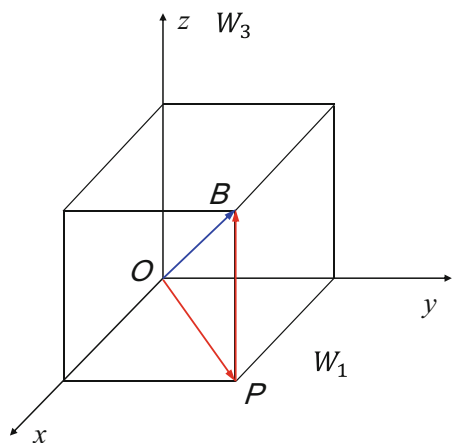


Fig. 11.1 Decomposition of a vector in a three-dimensional Cartesian space $\mathbb{R}^3$ into two subspaces. (a) $\mathbb{R}^3 = W_1 + W_2$; $W_1 \cap W_2 = \text{Span}\{\mathbf{e}_2\}$, where $\mathbf{e}_2$ is a unit vector in the positive direction of the y -axis. (b) $\mathbb{R}^3 = W_1 + W_3$; $W_1 \cap W_3 = \{\mathbf{0}\}$

$$\dim V = \dim W_1 + \dim W_2, \quad (11.18)$$

where “dim” stands for dimension of the vector space considered. To prove (11.18), we suppose that V is a n -dimensional vector space and that W_1 and W_2 are spanned by r_1 and r_2 linearly independent vectors, respectively, such that

$$W_1 = \text{Span}\{e_1^{(1)}, e_2^{(1)}, \dots, e_{r_1}^{(1)}\} \quad \text{and} \quad W_2 = \text{Span}\{e_1^{(2)}, e_2^{(2)}, \dots, e_{r_2}^{(2)}\}. \quad (11.19)$$

This is equivalent to that dimension of W_1 and W_2 is r_1 and r_2 , respectively. If $V = W_1 + W_2$ (here we do not assume that the summation is a direct sum), we have

$$V = \text{Span}\{e_1^{(1)}, e_2^{(1)}, \dots, e_{r_1}^{(1)}, e_1^{(2)}, e_2^{(2)}, \dots, e_{r_2}^{(2)}\}. \quad (11.20)$$

Then, we have $n \leq r_1 + r_2$. This is almost trivial. Suppose $r_1 + r_2 < n$. Then, these $(r_1 + r_2)$ vectors cannot span V , but we need additional vectors for all vectors including the additional vectors to span V . Thus, we should have $n \leq r_1 + r_2$ accordingly. That is,

$$\dim V \leq \dim W_1 + \dim W_2. \quad (11.21)$$

Now, let us assume that $V = W_1 \oplus W_2$. Then, $e_i^{(2)}$ ($1 \leq i \leq r_2$) must be linearly independent of $e_1^{(1)}, e_2^{(1)}, \dots, e_{r_1}^{(1)}$. If not, $e_i^{(2)}$ could be described as a linear combination of $e_1^{(1)}, e_2^{(1)}, \dots, e_{r_1}^{(1)}$. But this would imply that $e_i^{(2)} \in W_1$, i.e., $e_i^{(2)} \in W_1 \cap W_2$, in contradiction to that we have $V = W_1 \oplus W_2$. It is because $W_1 \cap W_2 = \{\mathbf{0}\}$ by assumption. Likewise, $e_j^{(1)}$ ($1 \leq j \leq r_1$) is linearly independent of $e_1^{(2)}, e_2^{(2)}, \dots, e_{r_2}^{(2)}$. Hence, $e_1^{(1)}, e_2^{(1)}, \dots, e_{r_1}^{(1)}, e_1^{(2)}, e_2^{(2)}, \dots, e_{r_2}^{(2)}$ must be linearly independent. Thus, $n \geq r_1 + r_2$. This is because in the vector space V we may well have additional vector(s) that are independent of the above $(r_1 + r_2)$ vectors. Meanwhile, $n \leq r_1 + r_2$ from the above. Consequently, we must have $n = r_1 + r_2$. Thus, we have proven that

$$V = W_1 \oplus W_2 \Rightarrow \dim V = \dim W_1 + \dim W_2.$$

Conversely, suppose $n = r_1 + r_2$. Then, any vector x in V is expressed uniquely as

$$x = (a_1 e_1^{(1)} + \dots + a_{r_1} e_{r_1}^{(1)}) + (b_1 e_1^{(2)} + \dots + b_{r_2} e_{r_2}^{(2)}). \quad (11.22)$$

The vector described by the first term is contained in W_1 and that described by the second term in W_2 . Both the terms are again expressed uniquely. Therefore, we get $V = W_1 \oplus W_2$. This is a proof of

$$\dim V = \dim W_1 + \dim W_2 \Rightarrow V = W_1 \oplus W_2.$$

The above statements are summarized as the following theorem:

Theorem 11.2 Let V be a vector space and let W_1 and W_2 be subspaces of V . Also suppose $V = W_1 + W_2$. Then, the following relation holds:

$$\dim V \leq \dim W_1 + \dim W_2. \quad (11.21)$$

Furthermore, we have

$$\dim V = \dim W_1 + \dim W_2, \quad (11.23)$$

if and only if $V = W_1 \oplus W_2$.

Theorem 11.2 is readily extended to the case where there are three or more subspaces. That is, having $W_1, W_2, \dots, W_m$ so that $V = W_1 + W_2 + \dots + W_m$, we obtain the following relation:

$$\dim V \leq \dim W_1 + \dim W_2 + \dots + \dim W_m. \quad (11.24)$$

The equality of (11.24) holds if and only if $V = W_1 \oplus W_2 \oplus \dots \oplus W_m$.

In light of Theorem 11.2, Example 11.1 says that $3 = \dim \mathbb{R}^3 < 2 + 2 = \dim W_1 + \dim W_2$. But $\dim \mathbb{R}^3 = 2 + 1 = \dim W_1 + \dim W_3$. Therefore, we have $\mathbb{R}^3 = W_1 \oplus W_3$.

11.2 Linear Transformations of Vectors

In the previous section we introduced vectors and their calculation rules in a linear vector space. It is natural and convenient to relate a vector to another vector, as a function f relates a number (either real or complex) to another number such that $y = f(x)$, where x and y are certain two numbers. A linear transformation from the vector space V to another vector space W is a mapping $A : V \rightarrow W$ such that

$$A(ca + db) = cA(a) + dA(b). \quad (11.25)$$

We will briefly discuss the concepts of mapping at the end of this section.

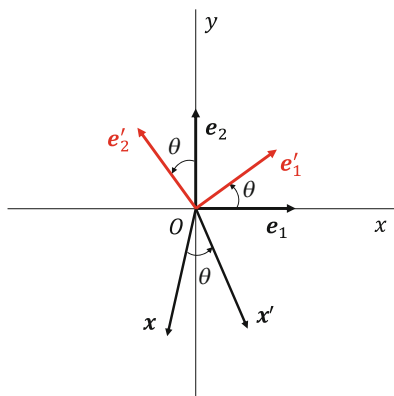
It is convenient to define addition of the linear transformations. It is defined as

$$(A + B)a = Aa + Ba, \quad (11.26)$$

where a is any vector in V .

Since (11.25) is a broad but abstract definition, we begin with a well-known simple example of rotation of a vector within a xy -plane (Fig. 11.2). We denote an arbitrary position vector $\mathbf{x}$ in the xy -plane by

Fig. 11.2 Rotation of a vector $\mathbf{x}$ within a xy -plane



$$\begin{aligned}\mathbf{x} &= x\mathbf{e}_1 + y\mathbf{e}_2 \\ &= (\mathbf{e}_1 \ \mathbf{e}_2) \begin{pmatrix} x \\ y \end{pmatrix},\end{aligned}\tag{11.27}$$

where $\mathbf{e}_1$ and $\mathbf{e}_2$ are unit basis vectors in the xy -plane and x and y are coordinates of the vector $\mathbf{x}$ in reference to $\mathbf{e}_1$ and $\mathbf{e}_2$. The expression (11.27) is consistent with (11.13). The rotation represented in Fig. 11.2 is an example of a linear transformation. We call this rotation R . According to the definition

$$R(x\mathbf{e}_1 + y\mathbf{e}_2) = R(\mathbf{x}) = xR(\mathbf{e}_1) + yR(\mathbf{e}_2).\tag{11.28}$$

Putting

$$R(x\mathbf{e}_1 + y\mathbf{e}_2) = \mathbf{x}', \quad R(\mathbf{e}_1) = \mathbf{e}'_1 \text{ and } R(\mathbf{e}_2) = \mathbf{e}'_2,$$

we have

$$\begin{aligned}\mathbf{x}' &= x\mathbf{e}'_1 + y\mathbf{e}'_2 \\ &= (\mathbf{e}'_1 \ \mathbf{e}'_2) \begin{pmatrix} x \\ y \end{pmatrix}.\end{aligned}\tag{11.29}$$

From Fig. 11.2 we readily obtain

$$\begin{aligned}\mathbf{e}'_1 &= \mathbf{e}_1 \cos \theta + \mathbf{e}_2 \sin \theta, \\ \mathbf{e}'_2 &= -\mathbf{e}_1 \sin \theta + \mathbf{e}_2 \cos \theta.\end{aligned}\tag{11.30}$$

Using a matrix representation,

$$(\mathbf{e}'_1 \ \mathbf{e}'_2) = (\mathbf{e}_1 \ \mathbf{e}_2) \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \quad (11.31)$$

Substituting (11.30) into (11.29), we obtain

$$\mathbf{x}' = (x \cos \theta - y \sin \theta) \mathbf{e}_1 + (x \sin \theta + y \cos \theta) \mathbf{e}_2. \quad (11.32)$$

Meanwhile, $\mathbf{x}'$ can be expressed relative to the original basis vectors $\mathbf{e}_1$ and $\mathbf{e}_2$.

$$\mathbf{x}' = x' \mathbf{e}_1 + y' \mathbf{e}_2. \quad (11.33)$$

Comparing (11.32) and (11.33), uniqueness of the representation ensures that

$$\begin{aligned} x' &= x \cos \theta - y \sin \theta, \\ y' &= x \sin \theta + y \cos \theta. \end{aligned} \quad (11.34)$$

Using a matrix representation once again,

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}. \quad (11.35)$$

Further combining (11.29) and (11.31), we get

$$\mathbf{x}' = R(\mathbf{x}) = (\mathbf{e}_1 \ \mathbf{e}_2) \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}. \quad (11.36)$$

The above example demonstrates that the linear transformation R has a $(2, 2)$ matrix representation shown in (11.36). Moreover, this example obviously shows that if a vector is expressed as a linear combination of the basis vectors, the “coordinates” (represented by a column vector) can be transformed as well by the same matrix.

Regarding an abstract n -dimensional linear vector space V^n , the linear vector transformation A is given by

$$A(\mathbf{x}) = (\mathbf{e}_1 \ \cdots \ \mathbf{e}_n) \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad (11.37)$$

where $\mathbf{e}_1, \mathbf{e}_2, \dots$, and $\mathbf{e}_n$ are basis vectors and $x_1, x_2, \dots$, and x_n are the corresponding coordinates of a vector $\mathbf{x} = \sum_{i=1}^n x_i \mathbf{e}_i$. We assume that the transformation is a mapping $A: V^n \rightarrow V^n$ (i.e., endomorphism). In this case the transformation is represented by a (n, n) matrix. Note that the matrix operates on the basis vectors from the right and that it operates on the coordinates (i.e., a column vector) from the left. In (11.37) we often omit a parenthesis to simply write $A\mathbf{x}$.

Here we mention matrix notation for later convenience. We often identify a linear vector transformation with its representation matrix and denote both transformation and matrix by A . On this occasion, we write

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}, \quad A = (A)_{ij} = (a_{ij}), \text{ etc.}, \quad (11.38)$$

where with the second expression $(A)_{ij}$ and (a_{ij}) represent the matrix A itself; for we frequently use indexed matrix notations such as A^{-1} , $A^\dagger$, $\tilde{A}$, etc. The notation (11.38) can conveniently be used in such cases. Note moreover that a_{ij} represents the matrix A as well.

Equation (11.37) has duality such that the matrix A operates either on the basis vectors or coordinates. This can explicitly be written as

$$\begin{aligned} A(\mathbf{x}) &= \begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_n \end{bmatrix} \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \\ &= (\mathbf{e}_1 \cdots \mathbf{e}_n) \left[\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right]. \end{aligned} \quad (11.39)$$

That is, we assume the associative law with the above expression. Making summation representation, we have

$$\begin{aligned} A(\mathbf{x}) &= \left(\sum_{k=1}^n \mathbf{e}_k a_{k1} \cdots \sum_{k=1}^n \mathbf{e}_k a_{kn} \right) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (\mathbf{e}_1 \cdots \mathbf{e}_n) \begin{pmatrix} \sum_{l=1}^n a_{1l} x_l \\ \vdots \\ \sum_{l=1}^n a_{nl} x_l \end{pmatrix} \\ &= \sum_{k=1}^n \sum_{l=1}^n \mathbf{e}_k a_{kl} x_l \\ &= \sum_{l=1}^n \left(\sum_{k=1}^n \mathbf{e}_k a_{kl} \right) x_l = \sum_{k=1}^n \left(\sum_{l=1}^n a_{kl} x_l \right) \mathbf{e}_k. \end{aligned}$$

That is, the above equation can be viewed in either of two ways, i.e., coordinate transformation with fix vectors or vector transformation with fixed coordinates.

Also, (11.37) can formally be written as

$$A(\mathbf{x}) = (\mathbf{e}_1 \cdots \mathbf{e}_n) A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (\mathbf{e}_1 A \cdots \mathbf{e}_n A) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad (11.40)$$

where we assumed that the distributive law holds with operation of A on $(\mathbf{e}_1 \cdots \mathbf{e}_n)$. Meanwhile, if in (11.37) we put $x_i = 1$, $x_j = 0$ ($j \neq i$), we get

$$A(\mathbf{e}_i) = \sum_{k=1}^n \mathbf{e}_k a_{ki}.$$

Therefore, (11.39) can be rewritten as

$$A(\mathbf{x}) = (A(\mathbf{e}_1) \cdots A(\mathbf{e}_n)) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}. \quad (11.41)$$

Since x_i ($1 \leq i \leq n$) can arbitrarily be chosen, comparing (11.40) and (11.41) we have

$$\mathbf{e}_i A = A(\mathbf{e}_i) \quad (1 \leq i \leq n). \quad (11.42)$$

The matrix representation is unique in reference to the same basis vectors. Suppose that there is another matrix representation of the transformation A such that

$$A(\mathbf{x}) = (\mathbf{e}_1 \cdots \mathbf{e}_n) \begin{pmatrix} a'_{11} & \cdots & a'_{1n} \\ \vdots & \ddots & \vdots \\ a'_{n1} & \cdots & a'_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}. \quad (11.43)$$

Subtracting (11.43) from (11.37), we obtain

$$\begin{aligned} & (\mathbf{e}_1 \cdots \mathbf{e}_n) \left[\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} - \begin{pmatrix} a'_{11} & \cdots & a'_{1n} \\ \vdots & \ddots & \vdots \\ a'_{n1} & \cdots & a'_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right] \\ &= (\mathbf{e}_1 \cdots \mathbf{e}_n) \begin{pmatrix} \sum_{k=1}^n (a_{1k} - a'_{1k})x_k \\ \vdots \\ \sum_{k=1}^n (a_{nk} - a'_{nk})x_k \end{pmatrix} = \mathbf{0}. \end{aligned}$$

On the basis of the linear dependence of the basis vectors, $\sum_{k=1}^n (a_{ik} - a'_{ik})x_k = 0$ ($1 \leq i \leq n$). This relationship holds for any arbitrarily and independently chosen complex numbers x_i ($1 \leq i \leq n$). Therefore, we must have $a_{ik} = a'_{ik}$ ($1 \leq i, k \leq n$), meaning that the matrix representation of A is unique with regard to fixed basis vectors.

Nonetheless, if a set of vectors $\mathbf{e}_1, \mathbf{e}_2, \cdots$, and $\mathbf{e}_n$ does not constitute basis vectors (i.e., those vectors are linearly dependent), the aforementioned uniqueness of the matrix representation loses its meaning. For instance, in V^2 take vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ such that $\mathbf{e}_1 = \mathbf{e}_2$ (i.e., the two vectors are linearly dependent) and let the

transformation matrix be $B = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$. This means that $\mathbf{e}_1$ should be $\mathbf{e}_1$ after the transformation. At the same time, the vector $\mathbf{e}_2 (= \mathbf{e}_1)$ should be converted to $2\mathbf{e}_2 (= 2\mathbf{e}_1)$. It is impossible except for the case of $\mathbf{e}_1 = \mathbf{e}_2 = \mathbf{0}$. The above matrix B in its own right is an object of matrix algebra, of course.

Putting $\mathbf{a} = \mathbf{b} = \mathbf{0}$ and $c = d = 1$ in the definition of (11.25) for the linear transformation, we obtain

$$A(\mathbf{0}) = A(\mathbf{0}) + A(\mathbf{0}).$$

Combining this relation with (11.4) gives

$$A(\mathbf{0}) = \mathbf{0}. \quad (11.44)$$

Then do we have a vector $\mathbf{u} \neq \mathbf{0}$ for which $A(\mathbf{u}) = \mathbf{0}$? An answer is yes. This is because if a $(2, 2)$ matrix of $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ is chosen for R , we get a linear transformation such that

$$(\mathbf{e}'_1 \ \mathbf{e}'_2) = (\mathbf{e}_1 \ \mathbf{e}_2) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = (\mathbf{e}_1 \ \mathbf{0}).$$

That is, we have $R(\mathbf{e}_2) = \mathbf{0}$.

In general, vectors $\mathbf{x} (\in V)$ satisfying $A(\mathbf{x}) = \mathbf{0}$ form a subspace in a vector space V . This is because $A(\mathbf{x}) = \mathbf{0}$ and $A(\mathbf{y}) = \mathbf{0} \Rightarrow A(\mathbf{x} + \mathbf{y}) = A(\mathbf{x}) + A(\mathbf{y}) = \mathbf{0}$, $A(c\mathbf{x}) = cA(\mathbf{x}) = \mathbf{0}$. We call this subspace of a maximum dimension a *null-space* and represent it as $\text{Ker } A$, where Ker stands for “kernel.” In other words, $\text{Ker } A = A^{-1}(\mathbf{0})$. Note that this symbolic notation does not ensure the existence of the inverse transformation A^{-1} (vide infra) but represents a set comprising elements $\mathbf{x}$ that satisfy $A(\mathbf{x}) = \mathbf{0}$. In the above example $\text{Ker } R = \text{Span}\{\mathbf{e}_2\}$.

Let us introduce frequently used terminology and notation. First, suppose that with a vector space V^n an endomorphism A of V^n exists. That is, we have

$$A(V^n) \equiv \{A(\mathbf{x}); \forall \mathbf{x} \in V^n, A(\mathbf{x}) \in V^n\}.$$

Then, $A(V^n)$ forms a subspace of V^n . In fact, for any $\mathbf{x}, \mathbf{y} \in V^n$, we have $A(\mathbf{x}), A(\mathbf{y}) \in A(V^n) \Rightarrow A(\mathbf{x}) + A(\mathbf{y}) = A(\mathbf{x} + \mathbf{y}) \in A(V^n)$ and $cA(\mathbf{x}) = A(c\mathbf{x}) \in A(V^n)$. Since A is an endomorphism, obviously $A(V^n) \subset V^n$. Hence, $A(V^n)$ is a subspace of V^n . The subspace $A(V^n)$ is said to be an image of transformation A and sometimes denoted by $\text{Im } A$.

The number of $\dim A(V^n)$ is said to be a rank of the linear transformation A . That is, we write

$$\dim A(V^n) = \text{rank } A.$$

Also the number of $\dim \text{Ker } A$ is said to be a *nullity* of the linear transformation A . That is, we have

$$\dim \text{Ker } A = \text{nullity } A.$$

With respect to the two subspaces $A(V^n)$ and $\text{Ker } A$, we show an important theorem (the so-called dimension theorem) below.

Theorem 11.3 Dimension Theorem Let V^n be a linear vector space of dimension n . Also let A be an endomorphism: $V^n \rightarrow V^n$. Then we have

$$\dim V^n = \dim A(V^n) + \dim \text{Ker } A. \quad (11.45)$$

Equation (11.45) can be written succinctly as

$$\dim V^n = \text{rank } A + \text{nullity } A.$$

Proof Let $e_1, e_2, \dots$, and e_n be basis vectors of V^n . First, assume

$$A(e_1) = A(e_2) = \dots = A(e_n) = \mathbf{0}.$$

This implies that $\text{nullity } A = n$. Then, $A\left(\sum_{i=1}^n x_i e_i\right) = \sum_{i=1}^n x_i A(e_i) = \mathbf{0}$. Since x_i is arbitrarily chosen, the expression means that $A(\mathbf{x}) = \mathbf{0}$ for $\forall \mathbf{x} \in V^n$. This implies $A = 0$. That is, $\text{rank } A = 0$. Thus, (11.45) certainly holds.

To proceed with proof of the theorem, we think of a linear combination $\sum_{i=1}^n c_i e_i$. Next, assume that $\text{Ker } A = \text{Span}\{e_1, e_2, \dots, e_\nu\}$ ($\nu < n$); $\dim \text{Ker } A = \nu$. After A is operated on the above linear combination, we are left with $\sum_{i=\nu+1}^n c_i A(e_i)$. We put

$$\sum_{i=1}^n c_i A(e_i) = \sum_{i=\nu+1}^n c_i A(e_i) = \mathbf{0}. \quad (11.46)$$

Suppose that the $(n - \nu)$ vectors $A(e_i)$ ($\nu + 1 \leq i \leq n$) are linearly dependent. Then without loss of generality we can assume $c_{\nu+1} \neq 0$. Dividing (11.46) by $c_{\nu+1}$ we obtain

$$\begin{aligned}
A(\mathbf{e}_{\nu+1}) + \frac{c_{\nu+2}}{c_{\nu+1}}A(\mathbf{e}_{\nu+2}) + \cdots + \frac{c_n}{c_{\nu+1}}A(\mathbf{e}_n) &= \mathbf{0}, \\
A(\mathbf{e}_{\nu+1}) + A\left(\frac{c_{\nu+2}}{c_{\nu+1}}\mathbf{e}_{\nu+2}\right) + \cdots + A\left(\frac{c_n}{c_{\nu+1}}\mathbf{e}_n\right) &= \mathbf{0}, \\
A\left(\mathbf{e}_{\nu+1} + \frac{c_{\nu+2}}{c_{\nu+1}}\mathbf{e}_{\nu+2} + \cdots + \frac{c_n}{c_{\nu+1}}\mathbf{e}_n\right) &= \mathbf{0}.
\end{aligned}$$

Meanwhile, the $(\nu + 1)$ vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_\nu$, and $\mathbf{e}_{\nu+1} + \frac{c_{\nu+2}}{c_{\nu+1}}\mathbf{e}_{\nu+2} + \cdots + \frac{c_n}{c_{\nu+1}}\mathbf{e}_n$ are linearly independent, because $\mathbf{e}_1, \mathbf{e}_2, \dots$, and $\mathbf{e}_n$ are basis vectors of V^n . This would imply that the dimension of $\text{Ker } A$ is $\nu + 1$, but this is in contradiction to $\text{Ker } A = \text{Span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_\nu\}$. Thus, the $(n - \nu)$ vectors $A(\mathbf{e}_i)$ ($\nu + 1 \leq i \leq n$) should be linearly independent.

Let $V^{n-\nu}$ be described as

$$V^{n-\nu} = \text{Span}\{A(\mathbf{e}_{\nu+1}), A(\mathbf{e}_{\nu+2}), \dots, A(\mathbf{e}_n)\}. \quad (11.47)$$

Then, $V^{n-\nu}$ is a subspace of V^n , and so $\dim A(V^n) \geq n - \nu = \dim V^{n-\nu}$. Meanwhile, from (11.46) we have

$$A\left(\sum_{i=\nu+1}^n c_i \mathbf{e}_i\right) = \mathbf{0}.$$

From the above discussion, however, this relation holds if and only if $c_{\nu+1} = \cdots = c_n = 0$. This implies that $\text{Ker } A \cap V^{n-\nu} = \{\mathbf{0}\}$. Then, we have

$$V^{n-\nu} + \text{Ker } A = V^{n-\nu} \oplus \text{Ker } A.$$

Furthermore, from Theorem 11.2 we have

$$\dim[V^{n-\nu} \oplus \text{Ker } A] = \dim V^{n-\nu} + \dim \text{Ker } A = (n - \nu) + \nu = n = \dim V^n.$$

Thus, we must have $\dim A(V^n) = n - \nu = \dim V^{n-\nu}$. Since $V^{n-\nu}$ is a subspace of V^n and $V^{n-\nu} \subset A(V^n)$ from (11.47), $V^{n-\nu} = A(V^n)$.

To conclude, we get

$$\begin{aligned}
\dim A(V^n) + \dim \text{Ker } A &= \dim V^n, \\
V^n &= A(V^n) \oplus \text{Ker } A.
\end{aligned} \quad (11.48)$$

This completes the proof.

Comparing Theorem 11.3 with Theorem 11.2, we find that Theorem 11.3 is a special case of Theorem 11.2. Equations (11.45) and (11.48) play an important role

in the theory of linear vector space. We add that Theorem 11.3 holds with a linear transformation that connects two vector spaces with different dimensions [1–3].

As an exercise, we have a following example:

Example 11.2 Let $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4$ be basis vectors of V^4 . Let A be an endomorphism of V^4 and described by

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$

We have

$$(\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4) \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix} = (\mathbf{e}_1 + \mathbf{e}_3 \quad \mathbf{e}_2 + \mathbf{e}_4 \quad \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3 + \mathbf{e}_4 \quad 0).$$

That is,

$$\begin{aligned} A(\mathbf{e}_1) &= \mathbf{e}_1 + \mathbf{e}_3, & A(\mathbf{e}_2) &= \mathbf{e}_2 + \mathbf{e}_4, & A(\mathbf{e}_3) &= \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3 + \mathbf{e}_4, \\ A(\mathbf{e}_4) &= \mathbf{0}. \end{aligned}$$

We have

$$A(-\mathbf{e}_1 - \mathbf{e}_2 + \mathbf{e}_3) = -A(\mathbf{e}_1) - A(\mathbf{e}_2) + A(\mathbf{e}_3) = \mathbf{0}.$$

Then, we find

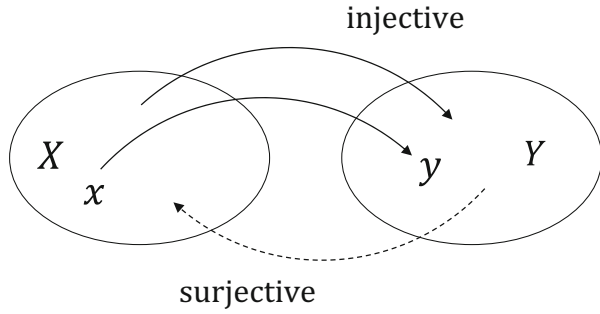
$$A(V^4) = \text{Span}\{\mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_4\}, \quad \text{Ker } A = \text{Span}\{-\mathbf{e}_1 - \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_4\}. \quad (11.49)$$

For any $\mathbf{x} \in V^n$, using scalar c_i ($1 \leq i \leq 4$), we have

$$\begin{aligned} \mathbf{x} &= c_1\mathbf{e}_1 + c_2\mathbf{e}_2 + c_3\mathbf{e}_3 + c_4\mathbf{e}_4 \\ &= \frac{1}{2}(c_1 + c_3)(\mathbf{e}_1 + \mathbf{e}_3) + \frac{1}{2}(2c_2 + c_3 - c_1)(\mathbf{e}_2 + \mathbf{e}_4) \\ &\quad + \frac{1}{2}(c_3 - c_1)(-\mathbf{e}_1 - \mathbf{e}_2 + \mathbf{e}_3) + \frac{1}{2}(c_1 - 2c_2 - c_3 + 2c_4)\mathbf{e}_4. \end{aligned} \quad (11.50)$$

Thus, $\mathbf{x}$ has been uniquely represented as (11.50) with respect to basis vectors $(\mathbf{e}_1 + \mathbf{e}_3)$, $(\mathbf{e}_2 + \mathbf{e}_4)$, $(-\mathbf{e}_1 - \mathbf{e}_2 + \mathbf{e}_3)$, and $\mathbf{e}_4$. The linear independence of these vectors can easily be checked by equating (11.50) with zero. We also confirm that

Fig. 11.3 Concept of mapping from a set X to another set Y



bijjective: injective + surjective

$$V^4 = A(V^4) \oplus \text{Ker } A.$$

In the present case, (11.45) is read as

$$\dim V^4 = \dim A(V^4) + \dim \text{Ker } A = 2 + 2 = 4.$$

Linear transformation is a kind of mapping. Figure 11.3 depicts the concept of mapping. Suppose two sets of X and Y . The mapping f is a correspondence between an element $x (\in X)$ and $y (\in Y)$. The set $f(X) (\subset Y)$ is said to be a range of f . (1) The mapping f is *injective*: If $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$. (2) The mapping is *surjective*: $f(X) = Y$. For $\forall y \in Y$ corresponding element(s) $x \in X$ exist(s). (3) The mapping is *bijjective*: If the mapping f is both injective and surjective, it is said to be bijjective (or reversible mapping or invertible mapping). A mapping that is not invertible is said to be a non-invertible mapping.

If the mapping f is bijjective, a unique element $\exists x \in X$ exists for $\forall y \in Y$ such that $f(x) = y$. In terms of solving an equation, we say that with any given y we can find a unique solution x to the equation $f(x) = y$. In this case x is said to be an inverse element to y and this is denoted by $x = f^{-1}(y)$. The mapping f^{-1} is called an inverse mapping. If the linear transformation is relevant, the mapping is said to be an inverse transformation.

Here we focus on a case where both X and Y form a vector space and the mapping is an endomorphism. Regarding the linear transformation $A: V^n \rightarrow V^n$ (i.e., an endomorphism of V^n), we have a following important theorem:

Theorem 11.4 Let $A: V^n \rightarrow V^n$ be an endomorphism of V^n . A necessary and sufficient condition for the existence of an inverse transformation to A (i.e., A^{-1}) is $A^{-1}(\mathbf{0}) = \{\mathbf{0}\}$.

Proof Suppose $A^{-1}(\mathbf{0}) = \{\mathbf{0}\}$. Then $A(\mathbf{x}_1) = A(\mathbf{x}_2) \Leftrightarrow A(\mathbf{x}_1 - \mathbf{x}_2) = \mathbf{0} \Leftrightarrow \mathbf{x}_1 - \mathbf{x}_2 = \mathbf{0}$; i.e., $\mathbf{x}_1 = \mathbf{x}_2$. This implies that the transformation A is injective. Other way round, suppose that A is injective. If $A^{-1}(\mathbf{0}) \neq \{\mathbf{0}\}$, there should be $\mathbf{b} (\neq \mathbf{0})$ with which $A(\mathbf{b}) = \mathbf{0}$. This is, however, in contradiction to that A is injective. Then we must have $A^{-1}(\mathbf{0}) = \{\mathbf{0}\}$. Thus, $A^{-1}(\mathbf{0}) = \{\mathbf{0}\} \Leftrightarrow A$ is injective.

Meanwhile, $A^{-1}(\mathbf{0}) = \{\mathbf{0}\} \Leftrightarrow \dim A^{-1}(\mathbf{0}) = 0 \Leftrightarrow \dim A(V^n) = n$ (due to Theorem 11.3); i.e., $A(V^n) = V^n$. Then $A^{-1}(\mathbf{0}) = \{\mathbf{0}\} \Leftrightarrow A$ is surjective. Combining this with the above-mentioned statement, we have $A^{-1}(\mathbf{0}) = \{\mathbf{0}\} \Leftrightarrow A$ is bijective. This statement is equivalent to that an inverse transformation exists. This completes the proof.

In the proof of Theorem 11.4, to show that $A^{-1}(\mathbf{0}) = \{\mathbf{0}\} \Leftrightarrow A$ is surjective, we have used the dimension theorem (Theorem 11.3), for which the relevant vector space is finite (i.e., n -dimensional). In other words, that A is surjective is equivalent to that A is injective with a finite-dimensional vector space and vice versa. To conclude, so far as we are thinking of the endomorphism of a finite-dimensional vector space, if we can show it is either injective or surjective, the other necessarily follows and, hence, the mapping is bijective.

11.3 Inverse Matrices and Determinants

The existence of the inverse transformation plays a particularly important role in the theory of linear vector spaces. The inverse transformation is a linear transformation. Let $\mathbf{x}_1 = A^{-1}(\mathbf{y}_1)$, $\mathbf{x}_2 = A^{-1}(\mathbf{y}_2)$. Also, we have $A(c_1\mathbf{x}_1 + c_2\mathbf{x}_2) = c_1A(\mathbf{x}_1) + c_2A(\mathbf{x}_2) = c_1\mathbf{y}_1 + c_2\mathbf{y}_2$. Thus, $c_1\mathbf{x}_1 + c_2\mathbf{x}_2 = A^{-1}(c_1\mathbf{y}_1 + c_2\mathbf{y}_2) = c_1A^{-1}(\mathbf{y}_1) + c_2A^{-1}(\mathbf{y}_2)$, showing that A^{-1} is a linear transformation. As already mentioned, a matrix that represents a linear transformation A is uniquely determined with respect to fixed basis vectors. This should be the case with A^{-1} accordingly. We have an important theorem for this.

Theorem 11.5 [4] The necessary and sufficient condition for the matrix A^{-1} that represents the inverse transformation to A to exist is that $\det A \neq 0$ (“det” means a determinant). Here the matrix A represents the linear transformation A . The matrix A^{-1} is uniquely determined and given by

$$(A^{-1})_{ij} = (-1)^{i+j}(M)_{ji}/(\det A), \quad (11.51)$$

where $(M)_{ij}$ is the *minor* with respect to the element A_{ij} .

Proof First, we suppose that the matrix A^{-1} exists so that it satisfies the following relation:

$$\sum_{k=1}^n (A^{-1})_{ik} (A)_{kj} = \delta_{ij} \quad (1 \leq i, j \leq n). \quad (11.52)$$

On this condition, suppose that $\det A = 0$. From the properties of determinants, this implies that one of the columns of A (let it be the m -th column) can be expressed as a linear combination of the other columns of A such that

$$A_{km} = \sum_{j \neq m} A_{kj} c_j. \quad (11.53)$$

Putting $i = m$ in (11.52), multiplying by c_j , and summing over $j \neq m$, we get

$$\sum_{k=1}^n (A^{-1})_{mk} \sum_{j \neq m} A_{kj} c_j = \sum_{j \neq m} \delta_{mj} c_j = 0. \quad (11.54)$$

From (11.52) and (11.53), however, we obtain

$$\sum_{k=1}^n (A^{-1})_{mk} \sum_{j \neq m} A_{kj} c_j = \left\{ (A^{-1})_{ik} (A)_{km} \right\}_{i=m} = 1. \quad (11.55)$$

There is the inconsistency between (11.54) and (11.55). The inconsistency resulted from the supposition that the matrix A^{-1} exists. Therefore, we conclude that if $\det A = 0$, A^{-1} does not exist. Taking contraposition to the above statement, we say that if A^{-1} exists, $\det A \neq 0$. Suppose next that $\det A \neq 0$. In this case, on the basis of the well-established result, a unique A^{-1} exists and it is given by (11.51). This completes the proof.

Summarizing the characteristics of the endomorphism within a finite-dimensional vector space, we have

$$\text{injective} \Leftrightarrow \text{surjective} \Leftrightarrow \text{bijective} \Leftrightarrow \det A \neq 0.$$

Let a matrix A be

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}. \quad (11.56)$$

The determinant of a matrix A is denoted by $\det A$ or by

$$\begin{vmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{vmatrix}.$$

Here, the determinant is defined as

$$\det A \equiv \sum_{\sigma} \begin{pmatrix} 1 & 2 & \cdots & n \\ i_1 & i_2 & \cdots & i_n \end{pmatrix} \varepsilon(\sigma) a_{1i_1} a_{2i_2} \cdots a_{ni_n}, \quad (11.57)$$

where σ means permutation among $1, 2, \dots, n$ and $\varepsilon(\sigma)$ denotes a sign of $+$ (in the case of even permutations) or $-$ (for odd permutations).

We deal with triangle matrices for future discussion. It is denoted by

$$T = \begin{pmatrix} a_{11} & & & * \\ & \ddots & & \\ & & \ddots & \\ 0 & & & a_{nn} \end{pmatrix}, \quad (11.58)$$

where an asterisk (*) means that upper right off-diagonal elements can take any complex numbers (including zero). A large zero shows that all the lower left off-diagonal elements are zero. Its determinant is given by

$$\det T = a_{11} a_{22} \cdots a_{nn}. \quad (11.59)$$

In fact, focusing on a_{ni_n} we notice that only if $i_n = n$, a_{ni_n} does not vanish. Then we get

$$\det A \equiv \sum_{\sigma} \begin{pmatrix} 1 & 2 & \cdots & n-1 \\ i_1 & i_2 & \cdots & i_{n-1} \end{pmatrix} \varepsilon(\sigma) a_{1i_1} a_{2i_2} \cdots a_{n-1, i_{n-1}} a_{nn}.$$

Repeating this process, we finally obtain (11.59).

The endomorphic linear transformation can be described succinctly as

$$A(\mathbf{x}) = \mathbf{y}, \quad (11.60)$$

where we have vectors such that $\mathbf{x} = \sum_{i=1}^n x_i \mathbf{e}_i$ and $\mathbf{y} = \sum_{i=1}^n y_i \mathbf{e}_i$. In reference to the same set of basis vectors ($\mathbf{e}_1 \cdots \mathbf{e}_n$) and using a matrix representation, we have

$$A(\mathbf{x}) = (\mathbf{e}_1 \cdots \mathbf{e}_n) \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (\mathbf{e}_1 \cdots \mathbf{e}_n) \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}. \quad (11.61)$$

From the unique representation of a vector in reference to the basis vectors, (11.61) is simply expressed as

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}. \quad (11.62)$$

With a shorthand notation, we have

$$y_i = \sum_{k=1}^n a_{ik}x_k \quad (1 \leq i \leq n). \quad (11.63)$$

From the above discussion, for the linear transformation A to be bijective, $\det A \neq 0$. In terms of the system of linear equations, we say that for (11.62) to have a unique solution $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ for a given $\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$, we must have $\det A \neq 0$. Conversely, $\det A = 0$ is equivalent to that (11.62) has indefinite solutions or has no solution. As far as the matrix algebra is concerned, (11.61) is symbolically described by omitting a parenthesis as

$$A\mathbf{x} = \mathbf{y}. \quad (11.64)$$

However, when the vector transformation is explicitly taken into account, the full representation of (11.61) should be borne in mind.

The relations (11.60) and (11.64) can be considered as a set of simultaneous equations. A necessary and sufficient condition for (11.64) to have a unique solution $\mathbf{x}$ for a given $\mathbf{y}$ is $\det A \neq 0$. In that case, the solution $\mathbf{x}$ of (11.64) can be symbolically expressed as

$$\mathbf{x} = A^{-1}\mathbf{y}, \quad (11.65)$$

where A^{-1} represents an inverse matrix of A .

Here, we have a good place to supplement somewhat confusing but important concepts of matrix algebra in relation to the inverse matrix. Let A be a (n, n) matrix expressed as

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}. \quad (11.56)$$

We often need a matrix “derived” from A of (11.56) so that we may obtain (m, m) matrix with $m < n$. Let us call such a matrix derived from A a submatrix $\hat{A}$. Among various types of submatrices $\hat{A}$, we think of a special type of matrix $\hat{A}(i, j)$ that is defined as a submatrix obtained by striking out the i -th row and j -th column. Then, $\hat{A}(i, j)$ is a $(n - 1, n - 1)$ submatrix.

We call $\det \hat{A}(i, j)$ a minor with respect to a_{ij} . Moreover, we define a *cofactor* Δ_{ij} as

$$\Delta_{ij} \equiv (-1)^{i+j} \det \hat{A}(i, j). \quad (11.66)$$

In (11.66), Δ_{ij} is said to be a cofactor of a_{ij} ; see Theorem 11.5. A transposed matrix of a matrix consisting of Δ_{ij} is sometimes referred to as an *adjugate* matrix of A . In relation to Theorem 11.5, if A is non-singular, the adjugate matrix of A is given by $A^{-1}(\det A)$. In that case, the adjugate matrix of A is non-singular as well.

In a particular case, suppose that we strike out the i -th row and i -th column. Then, $\hat{A}(i, i)$ is called a *principal submatrix* and $\det \hat{A}(i, i)$ is said to be a principal minor with respect to a_{ii} , i.e., a diagonal element of A . In this case, we have $\Delta_{ii} \equiv \det \hat{A}(i, i)$. If we strike out several pairs of i -th rows and i -th columns, we can make (m, m) matrices ($m < n$), which are also called principal submatrices (of m -th order). Determinants of the resulting matrices are said to be principal minors of m -th order. Then, a_{ii} ($1 \leq i \leq n$) itself is a principal minor as well.

Related discussion and examples have already appeared in Sect. 7.4. Other examples will appear in Sects. 12.2 and 14.5.

Example 11.3 Think of three-dimensional rotation by θ in $\mathbb{R}^3$ around the z -axis. The relevant transformation matrix is

$$R = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

As $\det R = 1 \neq 0$, the transformation is bijective. This means that for $\forall \mathbf{y} \in \mathbb{R}^3$ there is always a corresponding $\mathbf{x} \in \mathbb{R}^3$. This $\mathbf{x}$ can be found by solving $R\mathbf{x} = \mathbf{y}$; i.e., $\mathbf{x} = R^{-1}\mathbf{y}$. Putting $\mathbf{x} = x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3$ and $\mathbf{y} = x'\mathbf{e}_1 + y'\mathbf{e}_2 + z'\mathbf{e}_3$, a matrix representation is given by

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = R^{-1} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}.$$

Thus, $\mathbf{x}$ can be obtained by rotating $\mathbf{y}$ by $-\theta$.

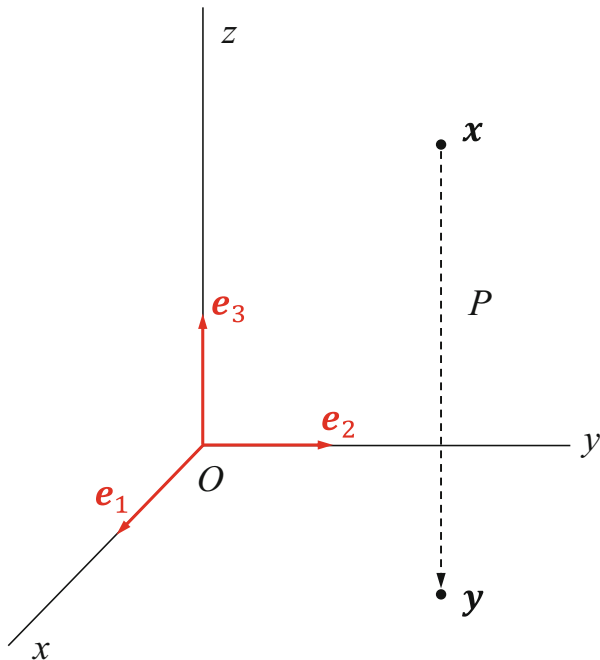
Example 11.4 Think of a following matrix that represents a linear transformation P :

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

This matrix transforms a vector $\mathbf{x} = x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3$ into $\mathbf{y} = x\mathbf{e}_1 + y\mathbf{e}_2$ as follows:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}. \quad (11.67)$$

Fig. 11.4 Example of an endomorphism $P: \mathbb{R}^3 \rightarrow \mathbb{R}^3$



Here, we are thinking of an endomorphism $P: \mathbb{R}^3 \rightarrow \mathbb{R}^3$. Geometrically, it can be viewed as in Fig. 11.4. Let us think of (11.67) from a point of view of solving a system of linear equations and newly consider the next equation. In other words, we are thinking of finding x , y , and z with given a , b , and c in the following equation:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}. \quad (11.68)$$

If $c = 0$, we find a solution of $x = a$, $y = b$, but z can be any (complex) number; we have thus indefinite solutions. If $c \neq 0$, however, we have no solution. The former situation reflects the fact that the transformation represented by P is not injective. Meanwhile, the latter reflects the fact that the transformation is not surjective. Remember that as $\det P = 0$, the transformation is not injective or surjective.

11.4 Basis Vectors and Their Transformations

In the previous sections we show that a vector is uniquely represented as a column vector in reference to a set of the fixed basis vectors. The representation, however, will be changed under a different set of basis vectors.

First let us think of a linear transformation of a set of basis vectors $\mathbf{e}_1, \mathbf{e}_2, \dots$, and $\mathbf{e}_n$. The transformation matrix A representing a linear transformation A is defined as follows:

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}.$$

Notice here that we often denote both a linear transformation and its corresponding matrix by the same character. After the transformation, suppose that the resulting vectors are given by $\mathbf{e}'_1, \mathbf{e}'_2$, and $\mathbf{e}'_n$. This is explicitly described as

$$(\mathbf{e}_1 \cdots \mathbf{e}_n) \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} = (\mathbf{e}'_1 \cdots \mathbf{e}'_n). \quad (11.69)$$

With a shorthand notation, we have

$$\mathbf{e}'_i = \sum_{k=1}^n a_{ki} \mathbf{e}_k \quad (1 \leq i \leq n). \quad (11.70)$$

Care should be taken not to confuse (11.70) with (11.63). Here, a set of vectors $\mathbf{e}'_1, \mathbf{e}'_2$, and $\mathbf{e}'_n$ may or may not be linearly independent. Let us operate both sides of (11.70) from the left on $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ and equate both the sides to zero. That is,

$$(\mathbf{e}_1 \cdots \mathbf{e}_n) \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (\mathbf{e}'_1 \cdots \mathbf{e}'_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \mathbf{0}.$$

Since $\mathbf{e}_1, \mathbf{e}_2, \dots$, and $\mathbf{e}_n$ are the basis vectors, we get

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \mathbf{0}. \quad (11.71)$$

Meanwhile, we must have $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \mathbf{0}$ so that $\mathbf{e}'_1, \mathbf{e}'_2$, and $\mathbf{e}'_n$ can be linearly independent (i.e., so as to be a set of basis vectors). But this means that (11.71) has such a unique (and trivial) solution and, hence, $\det A \neq 0$. If conversely $\det A \neq 0$, (11.71) has a unique trivial solution and $\mathbf{e}'_1, \mathbf{e}'_2$, and $\mathbf{e}'_n$ are linearly independent. Thus, a necessary and sufficient condition for $\mathbf{e}'_1, \mathbf{e}'_2$, and $\mathbf{e}'_n$ to be a set of basis vectors is $\det A \neq 0$. If $\det A = 0$, (11.71) has indefinite solutions (including a trivial solution) and $\mathbf{e}'_1, \mathbf{e}'_2$, and $\mathbf{e}'_n$ are linearly dependent and vice versa. In case $\det A \neq 0$, an inverse matrix A^{-1} exists, and so we have

$$(\mathbf{e}_1 \cdots \mathbf{e}_n) = (\mathbf{e}'_1 \cdots \mathbf{e}'_n) A^{-1}. \quad (11.72)$$

In the previous steps we see how the linear transformation (and the corresponding matrix representation) converts a set of basis vectors $(\mathbf{e}_1 \cdots \mathbf{e}_n)$ to another set of basis vectors $(\mathbf{e}'_1 \cdots \mathbf{e}'_n)$. Is this possible then to find a suitable transformation between two sets of arbitrarily chosen basis vectors? The answer is yes. This is because any vector can be expressed uniquely by a linear combination of any set of basis vectors. A whole array of such linear combinations uniquely define a transformation matrix between the two sets of basis vectors as expressed in (11.69) and (11.72). The matrix has non-zero determinant and has an inverse matrix. A concept of the transformation between basis vectors is important and very often used in various fields of natural science.

Example 11.5 We revisit Example 11.2. The relation (11.49) tells us that the basis vectors of $A(V^4)$ and those of $\text{Ker } A$ span V^4 in total. Therefore, in light of the above argument, there should be a linear transformation R between the two sets of vectors, i.e., $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4$ and $\mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_4, -\mathbf{e}_1 - \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_4$. Moreover, the matrix R associated with the linear transformation must be non-singular (i.e., $\det R \neq 0$). In fact, we find that R is expressed as

$$R = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}.$$

This is because we have a following relation between the two sets of basis vectors:

$$(\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4) \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} = (\mathbf{e}_1 + \mathbf{e}_3 \quad \mathbf{e}_2 + \mathbf{e}_4 \quad -\mathbf{e}_1 - \mathbf{e}_2 + \mathbf{e}_3 \quad \mathbf{e}_4).$$

We have $\det R = 2 \neq 0$ as expected.

Next, let us consider successive linear transformations of vectors. Again we assume that the transformations are endomorphism: $V^n \rightarrow V^n$. We have to take into account transformations of basis vectors along with the targeted vectors. First we choose a transformation by a non-singular matrix (having a non-zero determinant) for the subsequent transformation to have a unique matrix representation (vide supra). The vector transformation by P is expressed as

$$P(\mathbf{x}) = (\mathbf{e}_1 \cdots \mathbf{e}_n) \begin{pmatrix} p_{11} & \cdots & p_{1n} \\ \vdots & \ddots & \vdots \\ p_{n1} & \cdots & p_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad (11.73)$$

where the non-singular matrix P represents the transformation P . Notice here that the transformation and its matrix are represented by the same P . As mentioned in Sect. 11.2, the matrix P can be operated either from the right on the basis vectors or from the left on the column vector. We explicitly write

$$\begin{aligned} P(\mathbf{x}) &= \left[(\mathbf{e}_1 \cdots \mathbf{e}_n) \begin{pmatrix} p_{11} & \cdots & p_{1n} \\ \vdots & \ddots & \vdots \\ p_{n1} & \cdots & p_{nn} \end{pmatrix} \right] \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \\ &= (\mathbf{e}'_1 \cdots \mathbf{e}'_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \end{aligned} \quad (11.74)$$

where $(\mathbf{e}'_1 \cdots \mathbf{e}'_n) = (\mathbf{e}_1 \cdots \mathbf{e}_n)P$ [here P is the non-singular matrix defined in (11.73)]. Alternatively, we have

$$\begin{aligned} P(\mathbf{x}) &= (\mathbf{e}_1 \cdots \mathbf{e}_n) \left[\begin{pmatrix} p_{11} & \cdots & p_{1n} \\ \vdots & \ddots & \vdots \\ p_{n1} & \cdots & p_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right] \\ &= (\mathbf{e}_1 \cdots \mathbf{e}_n) \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \end{pmatrix}, \end{aligned} \quad (11.75)$$

where

$$\begin{pmatrix} x'_1 \\ \vdots \\ x'_n \end{pmatrix} = \begin{pmatrix} p_{11} & \cdots & p_{1n} \\ \vdots & \ddots & \vdots \\ p_{n1} & \cdots & p_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}. \quad (11.76)$$

Equation (11.75) gives a column vector representation regarding the vector that has been obtained by the transformation P and is viewed in reference to the basis vectors $(\mathbf{e}_1 \cdots \mathbf{e}_n)$. Combining (11.74) and (11.75), we get

$$P(\mathbf{x}) = (\mathbf{e}'_1 \cdots \mathbf{e}'_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (\mathbf{e}_1 \cdots \mathbf{e}_n) \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \end{pmatrix}. \quad (11.77)$$

We further make another linear transformation $A : V^n \rightarrow V^n$. In this case a corresponding matrix A may be non-singular (i.e., $\det A \neq 0$) or singular ($\det A = 0$). We have to distinguish the matrix representations of the two cases. It is because the matrix representations are uniquely defined in reference to an individual set of basis vectors; see (11.37) and (11.43). Let us denote the matrices A_O and A' with respect to

the basis vectors $(\mathbf{e}_1 \cdots \mathbf{e}_n)$ and $(\mathbf{e}'_1 \cdots \mathbf{e}'_n)$, respectively. Then, $A[P(\mathbf{x})]$ can be described in two different ways as follows:

$$A[P(\mathbf{x})] = (\mathbf{e}'_1 \cdots \mathbf{e}'_n) A' \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (\mathbf{e}_1 \cdots \mathbf{e}_n) A_O \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \end{pmatrix}. \quad (11.78)$$

This can be rewritten in reference to a linearly independent set of vectors $\mathbf{e}_1, \cdots, \mathbf{e}_n$ as

$$A[P(\mathbf{x})] = [(\mathbf{e}_1 \cdots \mathbf{e}_n) P A'] \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (\mathbf{e}_1 \cdots \mathbf{e}_n) \left[A_O P \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right]. \quad (11.79)$$

As (11.79) is described for a vector $\mathbf{x} = \sum_{i=1}^n x_i \mathbf{e}_i$ arbitrarily chosen in V^n , we get

$$P A' = A_O P. \quad (11.80)$$

Since P is non-singular, we finally obtain

$$A' = P^{-1} A_O P. \quad (11.81)$$

We can see (11.79) from a point of view of successive linear transformations. When the subsequent operation is viewed in reference to the basis vectors $\mathbf{e}'_1, \cdots, \mathbf{e}'_n$ newly reached by the precedent transformation, we make it a rule to write the relevant subsequent operator A' *from the right*. In the case where the subsequent operation is viewed in reference to the original basis vectors, however, we write the subsequent operator A_O *from the left*. Further discussion and examples can be seen in Part IV.

We see (11.81) in a different manner. Suppose we have

$$A(\mathbf{x}) = (\mathbf{e}_1 \cdots \mathbf{e}_n) A_O \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}. \quad (11.82)$$

Note that since the transformation A has been performed in reference to the basis vectors $(\mathbf{e}_1 \cdots \mathbf{e}_n)$, A_O should be used for the matrix representation. This is rewritten as

$$A(\mathbf{x}) = (\mathbf{e}_1 \cdots \mathbf{e}_n) P P^{-1} A_O P P^{-1} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}. \quad (11.83)$$

Meanwhile, any vector $\mathbf{x}$ in V^n can be written as

$$\begin{aligned}
\mathbf{x} &= (\mathbf{e}_1 \cdots \mathbf{e}_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \\
&= (\mathbf{e}_1 \cdots \mathbf{e}_n) P P^{-1} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (\mathbf{e}'_1 \cdots \mathbf{e}'_n) P^{-1} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (\mathbf{e}'_1 \cdots \mathbf{e}'_n) \begin{pmatrix} \tilde{x}_1 \\ \vdots \\ \tilde{x}_n \end{pmatrix}.
\end{aligned} \tag{11.84}$$

In (11.84) we put

$$(\mathbf{e}_1 \cdots \mathbf{e}_n) P = (\mathbf{e}'_1 \cdots \mathbf{e}'_n), \quad P^{-1} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \tilde{x}_1 \\ \vdots \\ \tilde{x}_n \end{pmatrix}. \tag{11.85}$$

Equation (11.84) gives a column vector representation regarding the *same* vector $\mathbf{x}$ viewed in reference to the basis set of vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ or $\mathbf{e}'_1, \dots, \mathbf{e}'_n$. Equation (11.85) should not be confused with (11.76). That is, (11.76) relates the two coordinates $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ and $\begin{pmatrix} x'_1 \\ \vdots \\ x'_n \end{pmatrix}$ of *different vectors* before and after the transformation viewed in reference to the *same set of basis vectors*. The relation (11.85), however, relates two coordinates $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ and $\begin{pmatrix} \tilde{x}_1 \\ \vdots \\ \tilde{x}_n \end{pmatrix}$ of *the same vector* viewed in reference to *different set of basis vectors*. Thus, from (11.83) we have

$$A(\mathbf{x}) = (\mathbf{e}'_1 \cdots \mathbf{e}'_n) P^{-1} A_0 P \begin{pmatrix} \tilde{x}_1 \\ \vdots \\ \tilde{x}_n \end{pmatrix}. \tag{11.86}$$

Meanwhile, viewing the transformation A in reference to the basis vectors $(\mathbf{e}'_1 \cdots \mathbf{e}'_n)$, we have

$$A(\mathbf{x}) = (\mathbf{e}'_1 \cdots \mathbf{e}'_n) A' \begin{pmatrix} \tilde{x}_1 \\ \vdots \\ \tilde{x}_n \end{pmatrix}. \tag{11.87}$$

Equating (11.86) and (11.87),

$$A' = P^{-1} A_0 P. \tag{11.88}$$

Thus, (11.81) is recovered.

The relations expressed by (11.81) and (11.88) are called a similarity transformation on A . The matrices A_0 and A' are said to be similar to each other. As

mentioned earlier, if A_0 (and hence A') is non-singular, the linear transformation A produces a set of basis vectors other than $e'_1, \dots, e'_n$, say $e''_1, \dots, e''_n$. We write this symbolically as

$$(e_1 \cdots e_n)PA = (e'_1 \cdots e'_n)A = (e''_1 \cdots e''_n). \quad (11.89)$$

Therefore, such A defines successive transformations of the basis vectors in conjunction with P defined in (11.73). The successive transformations and resulting basis vectors supply us with important applications in the field of group theory. Topics will be dealt with in Parts IV and V.

References

1. Satake I-O (1975) Linear algebra (Pure and applied mathematics). Marcel Dekker, New York
2. Satake I (1974) Linear algebra. Shokabo, Tokyo. (in Japanese)
3. Hassani S (2006) Mathematical physics. Springer, New York
4. Dennery P, Krzywicki A (1996) Mathematics for physicists. Dover, New York

Chapter 12

Canonical Forms of Matrices



In Sect. 11.4 we saw that the transformation matrices are altered depending on the basis vectors we choose. Then a following question arises. Can we convert a (transformation) matrix to as simple a form as possible by similarity transformation(s)? In Sect. 11.4 we have also shown that if we have two sets of basis vectors in a linear vector space V^n we can always find a non-singular transformation matrix between the two. In conjunction with the transformation of the basis vectors, the matrix undergoes similarity transformation. It is our task in this chapter to find a simple form or a specific form (i.e., canonical form) of a matrix as a result of the similarity transformation. For this purpose, we should first find eigenvalue(s) and corresponding eigenvector(s) of the matrix. Depending upon the nature of matrices, we get various canonical forms of matrices such as a triangle matrix and a diagonal matrix. Regarding any form of matrices, we can treat these matrices under a unified form called the Jordan canonical form.

12.1 Eigenvalues and Eigenvectors

An eigenvalue problem is one of the important subjects of the theory of linear vector spaces. Let A be a linear transformation on V^n . The resulting matrix gets to several different kinds of canonical forms of matrices. A typical example is a diagonal matrix. To reach a satisfactory answer, we start with a so-called eigenvalue problem.

Suppose that after the transformation of $\mathbf{x}$ we have

$$A(\mathbf{x}) = \alpha\mathbf{x}, \quad (12.1)$$

where α is a certain (complex) number. Then we say that α is an eigenvalue and that $\mathbf{x}$ is an eigenvector that corresponds to the eigenvalue α . Using a notation of (11.37) of Sect. 11.2, we have

$$A(\mathbf{x}) = (\mathbf{e}_1 \cdots \mathbf{e}_n) \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \alpha (\mathbf{e}_1 \cdots \mathbf{e}_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

From linear dependence of $\mathbf{e}_1, \mathbf{e}_2, \dots$, and $\mathbf{e}_n$, we simply write

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \alpha \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

If we identify $\mathbf{x}$ with $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ at fixed basis vectors $(\mathbf{e}_1 \cdots \mathbf{e}_n)$, we may naturally rewrite (12.1) as

$$A\mathbf{x} = \alpha\mathbf{x}. \quad (12.2)$$

If $\mathbf{x}_1$ and $\mathbf{x}_2$ belong to the eigenvalue α , so do $\mathbf{x}_1 + \mathbf{x}_2$ and $c\mathbf{x}_1$ (c is an appropriate complex number). Therefore, all the eigenvectors belonging to the eigenvalue α along with $\mathbf{0}$ (a zero vector) form a subspace of A (within V^n) corresponding to the eigenvalue α .

Strictly speaking, we should use terminologies such as a “proper” (or ordinary) eigenvalue, eigenvector, eigenspace, etc. to distinguish them from a “generalized” eigenvalue, eigenvector, eigenspace, etc. We will return to this point later. Further rewriting (12.2) we have

$$(A - \alpha E)\mathbf{x} = \mathbf{0}, \quad (12.3)$$

where E is a (n, n) unit matrix. Equations (12.2) and (12.3) are said to be an eigenvalue equation (or eigenequation). In (12.2) or (12.3) $\mathbf{x} = \mathbf{0}$ always holds (as a trivial solution). Consequently, for $\mathbf{x} \neq \mathbf{0}$ to be a solution we must have

$$|A - \alpha E| = 0. \quad (12.4)$$

In (12.4), $|A - \alpha E|$ stands for $\det(A - \alpha E)$.

Now let us define the following polynomial:

$$f_A(x) = |xE - A| = \begin{vmatrix} x - a_{11} & \cdots & -a_{1n} \\ \vdots & \ddots & \vdots \\ -a_{n1} & \cdots & x - a_{nn} \end{vmatrix}. \quad (12.5)$$

A necessary and sufficient condition for α to be an eigenvalue is that α is a root of $f_A(x) = 0$. The function $f_A(x)$ is said to be a characteristic polynomial and we call $f_A(x) = 0$ a characteristic equation. This is an n -th order polynomial. Putting

$$f_A(x) = x^n + a_1 x^{n-1} + \cdots + a_n, \quad (12.6)$$

we have

$$a_1 = -(a_{11} + a_{22} + \cdots + a_{nn}) \equiv -\text{Tr}A, \quad (12.7)$$

where Tr stands for “trace” that is a summation of diagonal elements. Moreover,

$$a_n = (-1)^n |A|. \quad (12.8)$$

The characteristic equation $f_A(x) = 0$ has n roots including possible multiple roots. Let those roots be $\alpha_1, \cdots, \alpha_n$ (some of them may be identical). Then we have

$$f_A(x) = \prod_{i=1}^n (x - \alpha_i). \quad (12.9)$$

Furthermore, according to relations between roots and coefficients we get

$$\alpha_1 + \cdots + \alpha_n = -a_1 = \text{Tr}A, \quad (12.10)$$

$$\alpha_1 \cdots \alpha_n = (-1)^n a_n = |A|. \quad (12.11)$$

The characteristic equation $f_A(x)$ is invariant under a similarity transformation. In fact,

$$\begin{aligned} f_{P^{-1}AP}(x) &= |xE - P^{-1}AP| = |P^{-1}(xE - A)P| \\ &= |P|^{-1} |xE - A| |P| = |xE - A| = f_A(x). \end{aligned} \quad (12.12)$$

This leads to invariance of the trace under a similarity transformation. That is,

$$\text{Tr}(P^{-1}AP) = \text{Tr}A. \quad (12.13)$$

Let us think of a following triangle matrix:

$$T = \begin{pmatrix} a_{11} & & & * \\ & \ddots & & \\ & & \ddots & \\ 0 & & & a_{nn} \end{pmatrix}. \quad (11.58)$$

The matrix of this type is thought to be one of canonical forms of matrices. Its characteristic equation $f_T(x)$ is

$$f_T(x) = |xE - T| = \begin{vmatrix} x - a_{11} & * & * \\ 0 & \ddots & * \\ 0 & 0 & x - a_{nn} \end{vmatrix}. \quad (12.14)$$

Therefore, we get

$$f_T(x) = \prod_{i=1}^n (x - a_{ii}), \quad (12.15)$$

where we used (11.59). From (11.58) and (12.15), we find that eigenvalues of a triangle matrix are given by its diagonal elements accordingly. Our immediate task will be to examine whether and how a given matrix is converted to a triangle matrix through a similarity transformation. A following theorem is important.

Theorem 12.1 Every (n, n) square matrix can be converted to a triangle matrix by similarity transformation [1].

Proof A triangle matrix is either an “upper” triangle matrix [to which all the lower left off-diagonal elements are zero; see (11.58)] or a “lower” triangle matrix (to which all the upper right off-diagonal elements are zero). In the present case we show the proof for the upper triangle matrix. Regarding the lower triangle matrix the theorem is proven in a similar manner.

The proof is due to mathematical induction. First we show that the theorem is true of a $(2, 2)$ matrix. Suppose that one of the eigenvalues of A_2 is α_1 and that its corresponding eigenvector is $\mathbf{x}_1$. Then we have

$$A_2 \mathbf{x}_1 = \alpha_1 \mathbf{x}_1, \quad (12.16)$$

where we assume that $\mathbf{x}_1$ represents a column vector. Let a non-singular matrix be

$$P_1 = (\mathbf{x}_1 \quad \mathbf{p}_1), \quad (12.17)$$

where $\mathbf{p}_1$ is another column vector chosen in such a way that $\mathbf{x}_1$ and $\mathbf{p}_1$ are linearly independent so that P_1 can be a non-singular matrix. Then, we have

$$P_1^{-1} A_2 P_1 = (P_1^{-1} A_2 \mathbf{x}_1 \quad P_1^{-1} A_2 \mathbf{p}_1). \quad (12.18)$$

Meanwhile,

$$\mathbf{x}_1 = (\mathbf{x}_1 \quad \mathbf{p}_1) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = P_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \quad (12.19)$$

Hence, we have

$$P_1^{-1}A_2\mathbf{x}_1 = \alpha_1 P_1^{-1}\mathbf{x}_1 = \alpha_1 P_1^{-1}P_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \alpha_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ 0 \end{pmatrix}. \quad (12.20)$$

Thus (12.18) can be rewritten as

$$P_1^{-1}A_2P_1 = \begin{pmatrix} \alpha_1 & * \\ 0 & * \end{pmatrix}. \quad (12.21)$$

This shows that Theorem 12.1 is true of a (2, 2) matrix A_2 .

Now let us show that Theorem 12.1 holds in a general case, i.e., for a (n, n) square matrix A_n . Let α_n be one of the eigenvalues of A_n . On the basis of the argument of the (2, 2) matrix case, suppose that after a suitable similarity transformation by a non-singular matrix $\tilde{P}$ we have

$$\tilde{A}_n = (\tilde{P})^{-1}A_n\tilde{P}. \quad (12.22)$$

Then, we can describe $\tilde{A}_n$ as

$$\tilde{A}_n = \begin{pmatrix} \alpha_n & x_1 & \cdots & x_{n-1} \\ 0 & \hline \vdots & A_{n-1} \\ 0 & \hline \end{pmatrix}. \quad (12.23)$$

In (12.23), α_n is one of the eigenvalues of A_n . To show that (12.23) is valid, we have a similar argument as in the case of a (2, 2) matrix. That is, we set $\tilde{P}$ such that

$$\tilde{P} = (\mathbf{a}_n \quad \mathbf{p}_1 \quad \mathbf{p}_2 \quad \cdots \quad \mathbf{p}_{n-1}),$$

where $\mathbf{a}_n$ is an eigenvector corresponding to α_n and $\tilde{P}$ is a non-singular matrix formed by n linearly independent column vectors $\mathbf{a}_n, \mathbf{p}_1, \mathbf{p}_2, \cdots$ and $\mathbf{p}_{n-1}$. Then we have

$$\begin{aligned} \tilde{A}_n &= (\tilde{P})^{-1}A_n\tilde{P} \\ &= \left((\tilde{P})^{-1}A_n\mathbf{a}_n \quad (\tilde{P})^{-1}A_n\mathbf{p}_1 \quad (\tilde{P})^{-1}A_n\mathbf{p}_2 \quad \cdots \quad (\tilde{P})^{-1}A_n\mathbf{p}_{n-1} \right). \end{aligned} \quad (12.24)$$

The vector $\mathbf{a}_n$ can be expressed as

$$\mathbf{a}_n = (\mathbf{a}_n \quad \mathbf{p}_1 \quad \mathbf{p}_2 \quad \cdots \quad \mathbf{p}_{n-1}) \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \tilde{P} \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Therefore, we have

$$(\tilde{P})^{-1} A_n \mathbf{a}_n = (\tilde{P})^{-1} \alpha_n \mathbf{a}_n = (\tilde{P})^{-1} \alpha_n \tilde{P} \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \alpha_n \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} \alpha_n \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Thus, from (12.24) we see that one can express a matrix form of $\tilde{A}_n$ as (12.23).

By a hypothesis of the mathematical induction, we assume that there exists a $(n - 1, n - 1)$ non-singular square matrix P_{n-1} and an upper triangle matrix Δ_{n-1} such that

$$P_{n-1}^{-1} A_{n-1} P_{n-1} = \Delta_{n-1}. \quad (12.25)$$

Let us define a following matrix:

$$P_n = \begin{pmatrix} d & 0 & \cdots & 0 \\ 0 & \hline \vdots & P_{n-1} \\ 0 & \hline \end{pmatrix},$$

where $d \neq 0$. The P_n is (n, n) non-singular square matrix; remember that $\det P_n = d(\det P_{n-1}) \neq 0$. Operating P_n on $\tilde{A}_n$ from the right, we have

$$\begin{aligned}
& \begin{pmatrix} \alpha_n & x_1 & \cdots & x_{n-1} \\ 0 & \hline \vdots & A_{n-1} \\ 0 & \end{pmatrix} \begin{pmatrix} d & 0 & \cdots & 0 \\ 0 & \hline \vdots & P_{n-1} \\ 0 & \end{pmatrix} \\
&= \begin{pmatrix} \alpha_n d & \mathbf{x}_{n-1}^T P_{n-1} \\ 0 & \hline \vdots & A_{n-1} P_{n-1} \\ 0 & \end{pmatrix}, \tag{12.26}
\end{aligned}$$

where $\mathbf{x}_{n-1}^T$ is a transpose of a column vector $\begin{pmatrix} x_1 \\ \vdots \\ x_{n-1} \end{pmatrix}$. Therefore, $\mathbf{x}_{n-1}^T P_{n-1}$ is a $(1, n-1)$ matrix (i.e., a row vector). Meanwhile, we have

$$\begin{aligned}
& \begin{pmatrix} d & 0 & 0 & 0 \\ 0 & \hline \vdots & P_{n-1} \\ 0 & \end{pmatrix} \begin{pmatrix} \alpha_n & \mathbf{x}_{n-1}^T P_{n-1}/d \\ 0 & \hline \vdots & \Delta_{n-1} \\ 0 & \end{pmatrix} \\
&= \begin{pmatrix} \alpha_n d & \mathbf{x}_{n-1}^T P_{n-1} \\ 0 & \hline \vdots & P_{n-1} \Delta_{n-1} \\ 0 & \end{pmatrix}. \tag{12.27}
\end{aligned}$$

From the assumption of (12.25), we have

$$\text{LHS of (12.26)} = \text{LHS of (12.27)}. \tag{12.28}$$

That is,

$$\tilde{A}_n P_n = P_n \Delta_n. \tag{12.29}$$

In (12.29) we define Δ_n such that

$$\Delta_n = \begin{pmatrix} \alpha_n & \mathbf{x}_{n-1}^T P_{n-1} / d \\ 0 & \Delta_{n-1} \\ \vdots & \\ 0 & \end{pmatrix}, \quad (12.30)$$

which has appeared in LHS of (12.27). As Δ_{n-1} is a triangle matrix from the assumption, Δ_n is a triangle matrix as well. Combining (12.22) and (12.29), we finally get

$$\Delta_n = (\tilde{P}P_n)^{-1} A_n \tilde{P}P_n. \quad (12.31)$$

Notice that $\tilde{P}P_n$ is a non-singular matrix, and so $\tilde{P}P_n P_n^{-1} \tilde{P}^{-1} = E$. Hence, $P_n^{-1} \tilde{P}^{-1} = (\tilde{P}P_n)^{-1}$. The equation obviously shows that A_n has been converted to a triangle matrix Δ_n . This completes the proof.

Equation (12.31) implies that eigenvectors are disposed on diagonal positions of a triangle matrix. Triangle matrices can further undergo a similarity transformation.

Example 12.1 Let us think of a following triangle matrix A :

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}. \quad (12.32)$$

Eigenvalues of A are 2 and 1. Remember that diagonal elements of a triangle matrix give eigenvalues. According to (12.20), a vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ can be chosen for an eigenvector (as a column vector representation) corresponding to the eigenvalue 2. Another eigenvector can be chosen to be $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$. This is because for an eigenvalue 1, we obtain

$$\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \mathbf{x} = \mathbf{0}$$

as an eigenvalue equation $(A - E)\mathbf{x} = \mathbf{0}$. Therefore, with an eigenvector $\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$ corresponding to the eigenvalue 1, we get $c_1 + c_2 = 0$. Therefore, we have $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ as a simple form of the eigenvector. Hence, putting $P = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, similarity transformation is carried out such that

$$P^{-1}AP = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}. \quad (12.33)$$

This is a simple example of matrix diagonalization.

Regarding the eigenvalue/eigenvector problems, we have another important theorem.

Theorem 12.2 Eigenvectors corresponding to different eigenvalues of A are linearly independent.

Proof We prove the theorem by mathematical induction. Let α_1 and α_2 be two different eigenvalues of a matrix A and let $\mathbf{a}_1$ and $\mathbf{a}_2$ be eigenvectors corresponding to α_1 and α_2 , respectively.

Let us think of a following equation:

$$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 = \mathbf{0}. \quad (12.34)$$

Suppose that $\mathbf{a}_1$ and $\mathbf{a}_2$ are linearly dependent. Then, without loss of generality we can put $c_1 \neq 0$. Accordingly, we get

$$\mathbf{a}_1 = -\frac{c_2}{c_1}\mathbf{a}_2. \quad (12.35)$$

Operating A from the left of (12.35) we have

$$\alpha_1\mathbf{a}_1 = -\frac{c_2}{c_1}\alpha_2\mathbf{a}_2 = \alpha_2\mathbf{a}_1. \quad (12.36)$$

With the second equality we have used (12.35). From (12.36) we have

$$(\alpha_1 - \alpha_2)\mathbf{a}_1 = \mathbf{0}. \quad (12.37)$$

As $\alpha_1 \neq \alpha_2$, $\alpha_1 - \alpha_2 \neq 0$. This implies $\mathbf{a}_1 = \mathbf{0}$, in contradiction to that $\mathbf{a}_1$ is a eigenvector. Thus, $\mathbf{a}_1$ and $\mathbf{a}_2$ must be linearly independent.

Next we assume that Theorem 12.2 is true of the case where we have $(n-1)$ eigenvalues $\alpha_1, \alpha_2, \dots$, and α_{n-1} that are different from one another and corresponding eigenvectors $\mathbf{a}_1, \mathbf{a}_2, \dots$, and $\mathbf{a}_{n-1}$ are linearly independent. Let us think of a following equation:

$$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \dots + c_{n-1}\mathbf{a}_{n-1} + c_n\mathbf{a}_n = \mathbf{0}, \quad (12.38)$$

where $\mathbf{a}_n$ is an eigenvector corresponding to an eigenvalue α_n . Suppose here that $\mathbf{a}_1, \mathbf{a}_2, \dots$, and $\mathbf{a}_n$ are linearly dependent. If $c_n = 0$, we have

$$c_1 \mathbf{a}_1 + c_2 \mathbf{a}_2 + \cdots + c_{n-1} \mathbf{a}_{n-1} = \mathbf{0}. \quad (12.39)$$

But, from the linear independence of $\mathbf{a}_1, \mathbf{a}_2, \cdots$, and $\mathbf{a}_{n-1}$ we have $c_1 = c_2 = \cdots = c_{n-1} = 0$. Thus, it follows that all the eigenvectors $\mathbf{a}_1, \mathbf{a}_2, \cdots$, and $\mathbf{a}_n$ are linearly independent. However, this is in contradiction to the assumption. We should therefore have $c_n \neq 0$. Accordingly we get

$$\mathbf{a}_n = - \left(\frac{c_1}{c_n} \mathbf{a}_1 + \frac{c_2}{c_n} \mathbf{a}_2 + \cdots + \frac{c_{n-1}}{c_n} \mathbf{a}_{n-1} \right). \quad (12.40)$$

Operating A from the left of (12.38) again, we have

$$\alpha_n \mathbf{a}_n = - \left(\frac{c_1}{c_n} \alpha_1 \mathbf{a}_1 + \frac{c_2}{c_n} \alpha_2 \mathbf{a}_2 + \cdots + \frac{c_{n-1}}{c_n} \alpha_{n-1} \mathbf{a}_{n-1} \right). \quad (12.41)$$

Here we think of two cases of (1) $\alpha_n \neq 0$ and (2) $\alpha_n = 0$.

1. Case I: $\alpha_n \neq 0$.

Multiplying both sides of (12.40) by α_n we have

$$\alpha_n \mathbf{a}_n = - \left(\frac{c_1}{c_n} \alpha_n \mathbf{a}_1 + \frac{c_2}{c_n} \alpha_n \mathbf{a}_2 + \cdots + \frac{c_{n-1}}{c_n} \alpha_n \mathbf{a}_{n-1} \right). \quad (12.42)$$

Subtracting (12.42) from (12.41) we get

$$\mathbf{0} = \frac{c_1}{c_n} (\alpha_n - \alpha_1) \mathbf{a}_1 + \frac{c_2}{c_n} (\alpha_n - \alpha_2) \mathbf{a}_2 + \cdots + \frac{c_{n-1}}{c_n} (\alpha_n - \alpha_{n-1}) \mathbf{a}_{n-1}. \quad (12.43)$$

Since we assume that eigenvalue is different from one another, $\alpha_n \neq \alpha_1, \alpha_n \neq \alpha_2, \cdots, \alpha_n \neq \alpha_{n-1}$. This implies that $c_1 = c_2 = \cdots = c_{n-1} = 0$. From (12.40) we have $\mathbf{a}_n = \mathbf{0}$. This is, however, in contradiction to that $\mathbf{a}_n$ is an eigenvector. This means that our original supposition that $\mathbf{a}_1, \mathbf{a}_2, \cdots$, and $\mathbf{a}_n$ are linearly dependent was wrong. Thus, the eigenvectors $\mathbf{a}_1, \mathbf{a}_2, \cdots$, and $\mathbf{a}_n$ should be linearly independent.

2. Case II: $\alpha_n = 0$.

Suppose again that $\mathbf{a}_1, \mathbf{a}_2, \cdots$, and $\mathbf{a}_n$ are linearly dependent. Since as before $c_n \neq 0$, we get (12.40) and (12.41) once again. Putting $\alpha_n = 0$ in (12.41) we have

$$\mathbf{0} = - \left(\frac{c_1}{c_n} \alpha_1 \mathbf{a}_1 + \frac{c_2}{c_n} \alpha_2 \mathbf{a}_2 + \cdots + \frac{c_{n-1}}{c_n} \alpha_{n-1} \mathbf{a}_{n-1} \right). \quad (12.44)$$

Since eigenvalues are different, we should have $\alpha_1 \neq 0, \alpha_2 \neq 0, \cdots$, and $\alpha_{n-1} \neq 0$. Then, considering that $\mathbf{a}_1, \mathbf{a}_2, \cdots$, and $\mathbf{a}_{n-1}$ are linearly independent, for (12.44) to hold we must have $c_1 = c_2 = \cdots = c_{n-1} = 0$. But, from (12.40) we have $\mathbf{a}_n = \mathbf{0}$, again in contradiction to that $\mathbf{a}_n$ is an eigenvector. Thus, the

eigenvectors $\mathbf{a}_1, \mathbf{a}_2, \dots$, and $\mathbf{a}_n$ should be linearly independent. These complete the proof.

12.2 Eigenspaces and Invariant Subspaces

Equations (12.21), (12.30), and (12.31) show that if we adopt an eigenvector as one of the basis vectors, the matrix representation of the linear transformation A in reference to such basis vectors is obtained so that the leftmost column is zero except for the left top corner on which an eigenvalue is positioned. (Note that if the said eigenvector is zero, the leftmost column is a zero vector.) Meanwhile, neither Theorem 12.1 nor Theorem 12.2 tells about multiplicity of eigenvalues. If the eigenvalues have multiplicity, we have to think about different aspects. This is a major issue of this section.

Let us start with a discussion of invariant subspaces. Let A be a (n, n) square matrix. If a subspace W in V^n is characterized by

$$\mathbf{x} \in W \Rightarrow A\mathbf{x} \in W,$$

W is said to be invariant with respect to A (or simply A -invariant) or an invariant subspace in $V^n = \text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$. Suppose that $\mathbf{x}$ is an eigenvector of A and that its corresponding eigenvalue is α . Then, $\text{Span}\{\mathbf{x}\}$ is an invariant subspace of V^n . It is because $A(c\mathbf{x}) = cA\mathbf{x} = c\alpha\mathbf{x} = \alpha(c\mathbf{x})$ and $c\mathbf{x}$ is again an eigenvector belonging to α . Suppose that $\dim W = m$ ($m \leq n$) and that $W = \text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m\}$. If W is A -invariant, A causes a linear transformation within W . In that case, expressing A in reference to $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m, \mathbf{a}_{m+1}, \dots, \mathbf{a}_n$, we have

$$\begin{aligned} & (\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_m \quad \mathbf{a}_{m+1} \quad \dots \quad \mathbf{a}_n)A \\ &= (\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_m \quad \mathbf{a}_{m+1} \quad \dots \quad \mathbf{a}_n) \begin{pmatrix} A_m & * \\ 0 & * \end{pmatrix}, \end{aligned} \quad (12.45)$$

where A_m is a (m, m) square matrix and “zero” denotes a $(n - m, m)$ zero matrix. Notice that the transformation A makes the remaining $(n - m)$ basis vectors $\mathbf{a}_{m+1}, \mathbf{a}_{m+2}, \dots$, and $\mathbf{a}_n$ in V^n be converted to a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \dots$, and $\mathbf{a}_n$. The triangle matrix Δ_n given in (12.30) and (12.31) is an example to which A_m is a $(1, 1)$ matrix (i.e., simply a complex number).

Let us examine properties of the A -invariant subspace still further. Let $\mathbf{a}$ be any vector in V^n and think of following $(n + 1)$ vectors [2]:

$$\mathbf{a}, A\mathbf{a}, A^2\mathbf{a}, \dots, A^n\mathbf{a}.$$

These vectors are linearly dependent since there are at most n linearly independent vectors in V^n . These vectors span a subspace in V^n . Let us call this subspace M ; i.e.,

$$M \equiv \text{Span}\{\mathbf{a}, A\mathbf{a}, A^2\mathbf{a}, \dots, A^n\mathbf{a}\}.$$

Consider the following equation:

$$c_0\mathbf{a} + c_1A\mathbf{a} + c_2A^2\mathbf{a} + \dots + c_nA^n\mathbf{a} = \mathbf{0}. \quad (12.46)$$

Not all c_i ($0 \leq i \leq n$) are zero, because the vectors are linearly dependent. Suppose that $c_n \neq 0$. Then, from (12.46) we have

$$A^n\mathbf{a} = -\frac{1}{c_n}(c_0\mathbf{a} + c_1A\mathbf{a} + c_2A^2\mathbf{a} + \dots + c_{n-1}A^{n-1}\mathbf{a}).$$

Operating A on the above equation from the left, we have

$$A^{n+1}\mathbf{a} = -\frac{1}{c_n}(c_0A\mathbf{a} + c_1A^2\mathbf{a} + c_2A^3\mathbf{a} + \dots + c_{n-1}A^n\mathbf{a}).$$

Thus, $A^{n+1}\mathbf{a}$ is contained in M . That is, we have $A^{n+1}\mathbf{a} \in \text{Span}\{\mathbf{a}, A\mathbf{a}, A^2\mathbf{a}, \dots, A^n\mathbf{a}\}$. Next, suppose that $c_n = 0$. Then, at least one of c_i ($0 \leq i \leq n-1$) is not zero. Suppose that $c_{n-1} \neq 0$. From (12.46), we have

$$A^{n-1}\mathbf{a} = -\frac{1}{c_{n-1}}(c_0\mathbf{a} + c_1A\mathbf{a} + c_2A^2\mathbf{a} + \dots + c_{n-2}A^{n-2}\mathbf{a}).$$

Operating A^2 on the above equation from the left, we have

$$A^{n+1}\mathbf{a} = -\frac{1}{c_{n-1}}(c_0A^2\mathbf{a} + c_1A^3\mathbf{a} + c_2A^4\mathbf{a} + \dots + c_{n-2}A^n\mathbf{a}).$$

Again, $A^{n+1}\mathbf{a}$ is contained in M . Repeating similar procedures, we reach

$$c_0\mathbf{a} + c_1A\mathbf{a} = \mathbf{0}.$$

If $c_1 = 0$, then we must have $c_0 \neq 0$. If so, $\mathbf{a} = \mathbf{0}$. This is impossible, however, because we should have $\mathbf{a} \neq \mathbf{0}$. Then, we have $c_1 \neq 0$ and, hence,

$$A\mathbf{a} = -\frac{c_0}{c_1}\mathbf{a}.$$

Operating once again A^n on the above equation from the left, we have

$$A^{n+1}\mathbf{a} = -\frac{c_0}{c_1}A^n\mathbf{a}.$$

Thus, once again $A^{n+1}\mathbf{a}$ is contained in M .

In the above discussion, we get $AM \subset M$. Further operating A , $A^2M \subset AM \subset M$, $A^3M \subset A^2M \subset AM \subset M$, $\dots$. Then we have

$$\mathbf{a}, A\mathbf{a}, A^2\mathbf{a}, \dots, A^n\mathbf{a}, A^{n+1}\mathbf{a}, \dots \in \text{Span}\{\mathbf{a}, A\mathbf{a}, A^2\mathbf{a}, \dots, A^n\mathbf{a}\}.$$

That is, M is an A -invariant subspace. We also have

$$m \equiv \dim M \leq \dim V^n = n.$$

There are m basis vectors in M , and so representing A in a matrix form in reference to the n basis vectors of V^n including these m vectors, we have

$$A = \begin{pmatrix} A_m & * \\ 0 & * \end{pmatrix}. \quad (12.47)$$

Note again that in (12.45) V^n is spanned by the m basis vectors in M together with other $(n - m)$ linearly independent vectors.

We happen to encounter a situation where two subspaces W_1 and W_2 are at once A -invariant. Here we can take basis vectors $\mathbf{a}_1, \mathbf{a}_2, \dots$, and $\mathbf{a}_m$ for W_1 and $\mathbf{a}_{m+1}, \mathbf{a}_{m+2}, \dots$, and $\mathbf{a}_n$ for W_2 . In reference to such $\mathbf{a}_1, \mathbf{a}_2, \dots$, and $\mathbf{a}_n$ as basis vector of V^n , we have

$$A = \begin{pmatrix} A_m & 0 \\ 0 & A_{n-m} \end{pmatrix}, \quad (12.48)$$

where A_{n-m} is a $(n - m, n - m)$ square matrix and “zero” denotes either a $(n - m, m)$ or $(m, n - m)$ zero matrix. Alternatively, we denote

$$A = A_m \oplus A_{n-m}.$$

In this situation, the matrix A is said to be reducible.

As stated above, A causes a linear transformation within both W_1 and W_2 . In other words, A_m and A_{n-m} cause a linear transformation within W_1 and W_2 in reference to $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m$ and $\mathbf{a}_{m+1}, \mathbf{a}_{m+2}, \dots, \mathbf{a}_n$, respectively. In this case V^n can be represented as a direct sum of W_1 and W_2 such that

$$\begin{aligned} V^n &= W_1 \oplus W_2 \\ &= \text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m\} \oplus \text{Span}\{\mathbf{a}_{m+1}, \mathbf{a}_{m+2}, \dots, \mathbf{a}_n\}. \end{aligned} \quad (12.49)$$

This is because $\text{Span}\{a_1, a_2, \dots, a_m\} \cap \text{Span}\{a_{m+1}, a_{m+2}, \dots, a_n\} = \{0\}$. In fact, if the two subspaces possess x ($\neq 0$) in common, $a_1, a_2, \dots, a_n$ become linearly dependent, in contradiction.

The vector space V^n may well further be decomposed into subspaces with a lower dimension. For further discussion we need a following theorem and a concept of a generalized eigenvector and generalized eigenspace.

Theorem 12.3 Hamilton–Cayley Theorem [3, 4] Let $f_A(x)$ be the characteristic polynomial pertinent to a linear transformation $A: V^n \rightarrow V^n$. Then $f_A(A)(x) = 0$ for $\forall x \in V^n$. That is, $\text{Ker } f_A(A) = V^n$.

Proof To prove the theorem we use the following well-known property of determinants. Namely, let A be a (n, n) square matrix expressed as

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}.$$

Let $\tilde{A}$ be the cofactor matrix of A , namely

$$\tilde{A} = \begin{pmatrix} \Delta_{11} & \cdots & \Delta_{1n} \\ \vdots & \ddots & \vdots \\ \Delta_{n1} & \cdots & \Delta_{nn} \end{pmatrix},$$

where Δ_{ij} is a cofactor of a_{ij} . Then

$$\tilde{A}^T A = A \tilde{A}^T = |A| E, \quad (12.50)$$

where $\tilde{A}^T$ is a transposed matrix of $\tilde{A}$. We now apply (12.50) to the characteristic polynomial to get a following equation expressed as

$$(\widetilde{x\tilde{E} - A})^T (xE - A) = (xE - A)(\widetilde{x\tilde{E} - A})^T = |xE - A|E = f_A(A)E, \quad (12.51)$$

where $\widetilde{x\tilde{E} - A}$ is the cofactor matrix of $x\tilde{E} - A$. Let the cofactor of the (i, j) -element of $(xE - A)$ be Δ_{ij} . Note in this case that Δ_{ij} is an at most $(n - 1)$ -th order polynomial of x . Let us put accordingly

$$\Delta_{ij} = b_{ij,0}x^{n-1} + b_{ij,1}x^{n-2} + \cdots + b_{ij,n-1}. \quad (12.52)$$

Also, we put $B_k = (b_{ij,k})$. Then we have

$$\begin{aligned}
x\widetilde{E} - A &= \begin{pmatrix} \Delta_{11} & \cdots & \Delta_{1n} \\ \vdots & \ddots & \vdots \\ \Delta_{n1} & \cdots & \Delta_{nn} \end{pmatrix} \\
&= \begin{pmatrix} b_{11,0}x^{n-1} + \cdots + b_{11,n-1} & \cdots & b_{1n,0}x^{n-1} + \cdots + b_{1n,n-1} \\ \vdots & \ddots & \vdots \\ b_{n1,0}x^{n-1} + \cdots + b_{n1,n-1} & \cdots & b_{nn,0}x^{n-1} + \cdots + b_{nn,n-1} \end{pmatrix} \\
&= x^{n-1} \begin{pmatrix} b_{11,0} & \cdots & b_{1n,0} \\ \vdots & \ddots & \vdots \\ b_{n1,0} & \cdots & b_{nn,0} \end{pmatrix} + \cdots + \begin{pmatrix} b_{11,0} & \cdots & b_{1n,n-1} \\ \vdots & \ddots & \vdots \\ b_{n1,0} & \cdots & b_{nn,n-1} \end{pmatrix} \\
&= x^{n-1}B_0 + x^{n-2}B_1 + \cdots + B_{n-1}.
\end{aligned} \tag{12.53}$$

Thus, we get

$$|xE - A|E = (xE - A)(x^{n-1}B_0^T + x^{n-2}B_1^T + \cdots + B_{n-1}^T). \tag{12.54}$$

Replacing x with A and considering that B_l^T ($l=0, \cdots, n-1$) is commutative with A , we have

$$f_A(A) = (A - A)(A^{n-1}B_0^T + A^{n-2}B_1^T + \cdots + B_{n-1}^T) = 0. \tag{12.55}$$

This means that $f_A(A)(\mathbf{x}) = 0$ for $\forall \mathbf{x} \in V^n$. These complete the proof.

In relation to Hamilton–Cayley theorem, we consider an important concept of a minimal polynomial.

Definition 12.1 Let $f(x)$ be a polynomial of x with scalar coefficients such that $f(A) = 0$, where A is a (n, n) matrix. Among $f(x)$, a lowest-order polynomial with the highest-order coefficient of one is said to be a minimal polynomial. We denote it by $\varphi_A(x)$; i.e., $\varphi_A(A) = 0$.

We have an important property for this. Namely, a minimal polynomial $\varphi_A(x)$ is a divisor of $f(A)$. In fact, suppose that we have

$$f(x) = g(x)\varphi_A(x) + h(x).$$

Inserting A into x , we have

$$f(A) = g(A)\varphi_A(A) + h(A) = h(A) = 0.$$

From the above equation, $h(x)$ should be a polynomial whose order is lower than that of $\varphi_A(A)$. But the presence of such $h(x)$ is in contradiction to the definition of the minimal polynomial. This implies that $h(x) \equiv 0$. Thus, we get

$$f(x) = g(x)\varphi_A(x).$$

That is, $\varphi_A(x)$ must be a divisor of $f(A)$.

Suppose that $\varphi'_A(x)$ is another minimal polynomial. If the order of $\varphi'_A(x)$ is lower than that of $\varphi_A(x)$, we can choose $\varphi'_A(x)$ for a minimal polynomial from the beginning. Thus we assume that $\varphi_A(x)$ and $\varphi'_A(x)$ are of the same order. We have

$$f(x) = g(x)\varphi_A(x) = g'(x)\varphi'_A(x).$$

Then, we have

$$\varphi_A(x)/\varphi'_A(x) = g'(x)/g(x) = c, \quad (12.56)$$

where c is a constant, because $\varphi_A(x)$ and $\varphi'_A(x)$ are of the same order. But c should be one, since the highest-order coefficient of the minimal polynomial is one. Thus, $\varphi_A(x)$ is uniquely defined.

12.3 Generalized Eigenvectors and Nilpotent Matrices

Equation (12.4) ensures that an eigenvalue is accompanied by an eigenvector. Therefore, if a matrix $A : V^n \rightarrow V^n$ has different n eigenvalues without multiple roots, the vector space V^n is decomposed to a direct sum of one-dimensional subspaces spanned by those individual linearly independent eigenvectors (see discussion of Sect 12.1). Thus, we have

$$\begin{aligned} V^n &= W_1 \oplus W_2 \oplus \cdots \oplus W_n \\ &= \text{Span}\{\mathbf{a}_1\} \oplus \text{Span}\{\mathbf{a}_2\} \oplus \cdots \oplus \text{Span}\{\mathbf{a}_n\}, \end{aligned} \quad (12.57)$$

where $\mathbf{a}_i$ ($1 \leq i \leq n$) are eigenvectors corresponding to different n eigenvalues. The situation, however, is not always simple. Even though a matrix has eigenvalues of multiple roots, we have yet a simple case as shown in a next example.

Example 12.2 Let us think of a following matrix $A : V^3 \rightarrow V^3$ described by

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}. \quad (12.58)$$

The matrix has a triple root 2. As can easily be seen below, individual eigenvectors $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ form basis vectors of each invariant subspace, indicating that V^3 can be decomposed to a direct sum of the three invariant subspaces as in (12.57) such that

$$(\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3) \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = (2\mathbf{a}_1 \quad 2\mathbf{a}_2 \quad 2\mathbf{a}_3).$$

Let us think of another simple example.

Example 12.3 Let us think of a linear transformation $A : V^2 \rightarrow V^2$ expressed as

$$A(\mathbf{x}) = (\mathbf{a}_1 \quad \mathbf{a}_2) \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad (12.59)$$

where $\mathbf{a}_1$ and $\mathbf{a}_2$ are basis vectors and for $\forall \mathbf{x} \in V^2, \mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2$. This example has a double root 3. We have

$$(\mathbf{a}_1 \quad \mathbf{a}_2) \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix} = (3\mathbf{a}_1 \quad \mathbf{a}_1 + 3\mathbf{a}_2). \quad (12.60)$$

Thus, $\text{Span}\{\mathbf{a}_1\}$ is A -invariant, but this is not the case with $\text{Span}\{\mathbf{a}_2\}$. This implies that $\mathbf{a}_1$ is an eigenvector corresponding to an eigenvalue 3 but $\mathbf{a}_2$ is not. Detailed discussion about matrices of this kind follows below.

Nilpotent matrices play an important role in matrix algebra. These matrices are defined as follows.

Definition 12.2 Let N be a linear transformation in a vector space V^n . Suppose that we have

$$N^k = 0 \quad \text{and} \quad N^{k-1} \neq 0, \quad (12.61)$$

where N is a (n, n) square matrix and $k (\leq 2)$ is a certain natural number. Then, N is said to be a nilpotent matrix of an order k or a k -th order nilpotent matrix. If (12.61) holds with $k = 1$, N is a zero matrix.

Nilpotent matrices have the following properties:

1. Eigenvalues of a nilpotent matrix are zero. Let N be a k -th order nilpotent matrix. Suppose that

$$N\mathbf{x} = \alpha\mathbf{x}, \quad (12.62)$$

where α is an eigenvalue and $\mathbf{x}$ ($\neq \mathbf{0}$) is its corresponding eigenvector. Operating N ($k - 1$) more times from the left of both sides of (12.62), we have

$$N^k\mathbf{x} = \alpha^k\mathbf{x}. \quad (12.63)$$

Meanwhile, $N^k = 0$ by definition, and so $\alpha^k = 0$, namely $\alpha = 0$.

Conversely, suppose that eigenvalues of a (n, n) matrix N are zero. From Theorem 12.1, via a suitable similarity transformation with P we have

$$P^{-1}NP = \tilde{N},$$

where $\tilde{N}$ is a triangle matrix. Then, using (12.12) and (12.15) we have

$$f_{P^{-1}NP}(x) = f_{\tilde{N}}(x) = f_N(x) = x^n.$$

From Theorem 12.3, we have

$$f_N(N) = N^n = 0.$$

Namely, N is a nilpotent matrix. In a trivial case, we have $N = 0$ (zero matrix).

By Definition 12.2, we have $N^k = 0$ with a k -th nilpotent (n, n) matrix N . Then, its minimal polynomial is $\varphi_N(x) = x^k$ ($k \leq n$).

2. A nilpotent matrix N is not diagonalizable (except for a zero matrix). Suppose that N is diagonalizable. Then N can be diagonalized by a non-singular matrix P such that

$$P^{-1}NP = 0.$$

The above equation holds, because N only takes eigenvalues of zero. Operating P from the left of the above equation and P^{-1} from the right, we have

$$N = 0.$$

This means that N would be a zero matrix, in contradiction. Thus, a nilpotent matrix N is not diagonalizable.

Example 12.4 Let N be a matrix of a following form:

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then, we can easily check that $N^2 = 0$. Therefore, N is a nilpotent matrix of a second order. Note that N is an upper triangle matrix, and so eigenvalues are given by diagonal elements. In the present case the eigenvalue is zero (as a double root), consistent with the aforementioned property.

With a nilpotent matrix of an order k , we have at least one vector $\mathbf{x}$ such that $N^{k-1}\mathbf{x} \neq 0$. Here we add that a zero transformation A (or matrix) can be defined as

$$A = 0 \Leftrightarrow A\mathbf{x} = 0 \quad \text{for } \forall \mathbf{x} \in V^n. \quad (12.64)$$

Taking contraposition of this, we have

$$A \neq 0 \Leftrightarrow A\mathbf{x} \neq 0 \quad \text{for } \exists \mathbf{x} \in V^n.$$

In relation to a nilpotent matrix, we have a following important theorem.

Theorem 12.4 If N is a k -th order nilpotent matrix, then for $\exists \mathbf{x} \in V$ we have following linearly independent k vectors:

$$\mathbf{x}, N\mathbf{x}, N^2\mathbf{x}, \dots, N^{k-1}\mathbf{x}.$$

Proof Let us think of a following equation:

$$\sum_{i=0}^{k-1} c_i N^i \mathbf{x} = 0. \quad (12.65)$$

Multiplying (12.65) by N^{k-1} from the left and using $N^k = 0$, we get

$$c_0 N^{k-1} \mathbf{x} = \mathbf{0}. \quad (12.66)$$

This implies that $c_0 = 0$. Thus, we are left with

$$\sum_{i=1}^{k-1} c_i N^i \mathbf{x} = 0. \quad (12.67)$$

Next, multiplying (12.67) by N^{k-2} from the left, we get similarly

$$c_1 N^{k-1} \mathbf{x} = \mathbf{0}, \quad (12.68)$$

implying that $c_1 = 0$. Continuing this procedure, we finally get $c_i = 0$ ($0 \leq i \leq k-1$). This completes the proof.

This immediately tells us that for a k -th order nilpotent matrix $N : V^n \rightarrow V^n$, we should have $k \leq n$. This is because the number of linearly independent vectors is at most n .

In Example 12.4, N causes a transformation of basis vectors in V^2 such that

$$(\mathbf{e}_1 \quad \mathbf{e}_2)N = (\mathbf{e}_1 \quad \mathbf{e}_2) \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = (0 \quad \mathbf{e}_1).$$

That is, $N\mathbf{e}_2 = \mathbf{e}_1$. Then, linearly independent two vectors $\mathbf{e}_2$ and $N\mathbf{e}_2$ correspond to the case of Theorem 12.4.

So far we have shown simple cases where matrices can be diagonalized via similarity transformation. This is equivalent to that the relevant vector space is decomposed to a direct sum of (invariant) subspaces. Nonetheless, if a characteristic polynomial of the matrix has multiple root(s), it remains uncertain whether the vector space is decomposed to such a direct sum. To answer this question, we need a following lemma.

Lemma 12.1 Let $f_1(x), f_2(x), \dots$, and $f_s(x)$ be polynomials without a common factor. Then we have s polynomials $M_1(x), M_2(x), \dots, M_s(x)$ that satisfy the following relation:

$$M_1(x)f_1(x) + M_2(x)f_2(x) + \dots + M_s(x)f_s(x) = 1. \quad (12.69)$$

Proof Let $M_i(x)$ ($1 \leq i \leq s$) be arbitrarily chosen polynomials and deal with a set of $g(x)$ that can be expressed as

$$g(x) = M_1(x)f_1(x) + M_2(x)f_2(x) + \dots + M_s(x)f_s(x). \quad (12.70)$$

Let a whole set of $g(x)$ be $\mathfrak{H}$. Then $\mathfrak{H}$ has the following two properties:

1. $g_1(x), g_2(x) \in \mathfrak{H} \Rightarrow g_1(x) + g_2(x) \in \mathfrak{H}$,
2. $g(x) \in \mathfrak{H}, M(x)$: an arbitrarily chosen polynomial $\Rightarrow M(x)g(x) \in \mathfrak{H}$.

Now let the lowest-order polynomial out of the set $\mathfrak{H}$ be $g_0(x)$. Then $\forall g(x) (\in \mathfrak{H})$ are a multiple of $g_0(x)$. Suppose that dividing $g(x)$ by $g_0(x)$, we have

$$g(x) = M(x)g_0(x) + h(x), \quad (12.71)$$

where $h(x)$ is a certain polynomial. Since $g(x), g_0(x) \in \mathfrak{H}$, we have $h(x) \in \mathfrak{H}$ from the above properties (1) and (2). If $h(x) \neq 0$, the order of $h(x)$ is lower than that of $g_0(x)$ from (12.71). This is, however, in contradiction to the definition of $g_0(x)$. Therefore, $h(x) = 0$. Thus,

$$g(x) = M(x)g_0(x). \quad (12.72)$$

This implies that $\mathfrak{H}$ is identical to a whole set of polynomials comprising multiples of $g_0(x)$. In particular, polynomials $f_i(x)$ ($1 \leq i \leq s$) $\in \mathfrak{H}$. To show this, put $M_i(x) = 1$ with other $M_j(x) = 0$ ($j \neq i$). Hence, the polynomial $g_0(x)$ should be a common factor of $f_i(x)$. Meanwhile, $g_0(x) \in \mathfrak{H}$, and so by virtue of (12.70) we have

$$g_0(x) = M_1(x)f_1(x) + M_2(x)f_2(x) + \cdots + M_s(x)f_s(x). \quad (12.73)$$

On assumption the greatest common factor of $f_1(x), f_2(x), \cdots$, and $f_s(x)$ is 1. This implies that $g_0(x) = 1$. Thus, we finally get (12.69) and complete the proof.

12.4 Idempotent Matrices and Generalized Eigenspaces

In Sect. 12.1 we have shown that eigenvectors corresponding to different eigenvalues of A are linearly independent. Also if those eigenvalues do not possess multiple roots, the vector space comprises a direct sum of the subspaces of corresponding eigenvectors. However, how do we have to treat a situation where eigenvalues possess multiple roots? Even in this case there is at least one eigenvector that corresponds to the eigenvalue. To adequately address the question, we need a concept of generalized eigenvectors and generalized eigenspaces.

For this purpose we extend and generalize the concept of eigenvectors. For a certain natural number l , if a vector $\mathbf{x}$ ($\in V^n$) satisfies a following relation, $\mathbf{x}$ is said to be a generalized eigenvector of rank l that corresponds to an eigenvalue α .

$$(A - \alpha E)^l \mathbf{x} = \mathbf{0}, \quad (A - \alpha E)^{l-1} \mathbf{x} \neq \mathbf{0}.$$

Thus (12.3) implies that an eigenvector of (12.3) may be said to be a generalized eigenvector of rank 1. When we need to distinguish it from generalized eigenvectors of rank l ($l \geq 2$), we call it a “proper” eigenvector. Thus far we have only been concerned with the proper eigenvectors. We have a following theorem related to the generalized eigenvectors.

In this section, idempotent operators play a role. The definition is simple.

Definition 12.3 An operator A is said to be idempotent if $A^2 = A$.

From this simple definition, we can draw several important pieces of information. Let A be an idempotent operator that operates on V^n . Let $\mathbf{x}$ be an arbitrary vector in V^n . Then, $A^2\mathbf{x} = A(A\mathbf{x}) = A\mathbf{x}$. That is,

$$A(A\mathbf{x} - \mathbf{x}) = \mathbf{0}.$$

Then, we have

$$A\mathbf{x} = \mathbf{x} \quad \text{or} \quad A\mathbf{x} = \mathbf{0}. \quad (12.74)$$

From [Theorem 11.3](#) (dimension theorem), for a general case of linear operators we had

$$V^n = A(V^n) \oplus \text{Ker } A,$$

where

$$A(V^n) = \{A\mathbf{x}; \mathbf{x} \in V^n\}, \quad \text{Ker } A = \{\mathbf{x}; \mathbf{x} \in V^n, A\mathbf{x} = \mathbf{0}\}.$$

Consequently, if A is an idempotent operator, we must have

$$A(V^n) = \{\mathbf{x}; \mathbf{x} \in V^n, A\mathbf{x} = \mathbf{x}\}, \quad \text{Ker } A = \{\mathbf{x}; \mathbf{x} \in V^n, A\mathbf{x} = \mathbf{0}\}. \quad (12.75)$$

Thus, we find that an idempotent operator A decomposes V^n into a direct sum of $A(V^n)$ to which $A\mathbf{x} = \mathbf{x}$ and $\text{Ker } A$ to which $A\mathbf{x} = \mathbf{0}$. Conversely, if there exists an operator A that satisfies (12.75), such A must be an idempotent operator.

Meanwhile, we have

$$(E - A)^2 = E - 2A + A^2 = E - 2A + A = E - A.$$

Hence, $E - A$ is an idempotent matrix as well. Moreover, we have

$$A(E - A) = (E - A)A = 0.$$

Putting $E - A \equiv B$ and following a procedure similar to the above

$$B\mathbf{x} - \mathbf{x} = (E - A)\mathbf{x} - \mathbf{x} = \mathbf{x} - A\mathbf{x} - \mathbf{x} = -A\mathbf{x}.$$

Therefore, $B(V^n) = \{\mathbf{x}; \mathbf{x} \in V^n, B\mathbf{x} = \mathbf{x}\}$ is identical to $\text{Ker } A$. Writing

$$W_A = \{\mathbf{x}; \mathbf{x} \in V^n, A\mathbf{x} = \mathbf{x}\}, \quad W_{\bar{A}} = \{\mathbf{x}; \mathbf{x} \in V^n, A\mathbf{x} = \mathbf{0}\}, \quad (12.76)$$

we get

$$W_A = AV^n, \quad W_{\bar{A}} = (E - A)V^n = BV^n = V^n - AV^n = \text{Ker } A.$$

That is,

$$V^n = W_A + W_{\bar{A}}.$$

Suppose that $\exists \mathbf{u} \in W_A \cap W_{\bar{A}}$. Then, from (12.76) $A\mathbf{u} = \mathbf{u}$ and $A\mathbf{u} = \mathbf{0}$, namely $\mathbf{u} = \mathbf{0}$, and so V is a direct sum of W_A and $W_{\bar{A}}$. That is,

$$V^n = W_A \bigoplus W_{\bar{A}}.$$

Notice that if we consider an identity operator E as an idempotent operator, we are thinking of a trivial case. That is, $V^n = V^n \bigoplus \{\mathbf{0}\}$.

The result can immediately be extended to the case where more idempotent operators take part in the vector space. Let us define operators such that

$$A_1 + A_2 + \cdots + A_s = E,$$

where E is a (n, n) identity matrix. Also

$$A_i A_j = A_i \delta_{ij}.$$

Moreover, we define W_i such that

$$W_i = A_i V^n = \{\mathbf{x}; \mathbf{x} \in V^n, A_i \mathbf{x} = \mathbf{x}\}. \quad (12.77)$$

Then

$$V^n = W_1 \bigoplus W_2 \bigoplus \cdots \bigoplus W_s. \quad (12.78)$$

In fact, suppose that $\exists \mathbf{x}(\in V^n) \in W_i, W_j (i \neq j)$. Then $A_i \mathbf{x} = \mathbf{x} = A_j \mathbf{x}$. Operating A_j ($j \neq i$) from the left, $A_j A_i \mathbf{x} = A_j \mathbf{x} = A_j A_j \mathbf{x}$. That is, $\mathbf{0} = \mathbf{x} = A_j \mathbf{x}$, implying that $W_i \cap W_j = \{\mathbf{0}\}$. Meanwhile,

$$\begin{aligned} V^n &= (A_1 + A_2 + \cdots + A_s) V^n = A_1 V^n + A_2 V^n + \cdots + A_s V^n \\ &= W_1 + W_2 + \cdots + W_s. \end{aligned} \quad (12.79)$$

As $W_i \cap W_j = \{\mathbf{0}\} (i \neq j)$, (12.79) is a direct sum. Thus, (12.78) will follow.

Example 12.5 Think of the following transformation A :

$$(\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4)A = (\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = (\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{0} \quad \mathbf{0}).$$

Put $\mathbf{x} = \sum_{i=1}^4 x_i \mathbf{e}_i$. Then, we have

$$A(\mathbf{x}) = \sum_{i=1}^4 x_i A(\mathbf{e}_i) = \sum_{i=1}^2 x_i \mathbf{e}_i = W_A,$$

$$(E - A)(\mathbf{x}) = \mathbf{x} - A(\mathbf{x}) = \sum_{i=3}^4 x_i \mathbf{e}_i = W_{\bar{A}}.$$

In the above,

$$(\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4)(E - A) = (\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4) \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = (0 \quad 0 \quad \mathbf{e}_3 \quad \mathbf{e}_4).$$

Thus, we have

$$W_A = \text{Span}\{\mathbf{e}_1, \mathbf{e}_2\}, \quad W_{\bar{A}} = \text{Span}\{\mathbf{e}_3, \mathbf{e}_4\}, \quad V^4 = W_A \oplus W_{\bar{A}}.$$

The properties of idempotent matrices can easily be checked. It is left for readers.

Using the aforementioned idempotent operators, let us introduce the following theorem.

Theorem 12.5 [3, 4] Let A be a linear transformation $V^n \rightarrow V^n$. Suppose that a vector $\mathbf{x}$ ($\in V^n$) satisfies the following relation:

$$(A - \alpha E)^l \mathbf{x} = \mathbf{0}, \quad (12.80)$$

where l is an enough large natural number. Then a set comprising $\mathbf{x}$ forms an A -invariant subspace that corresponds to an eigenvalue α . Let $\alpha_1, \alpha_2, \dots, \alpha_s$ be eigenvalues of A different from one another.

Then V^n is decomposed to a direct sum of the A -invariant subspaces that correspond individually to $\alpha_1, \alpha_2, \dots, \alpha_s$. This is succinctly expressed as follows:

$$V^n = \tilde{W}_{\alpha_1} \oplus \tilde{W}_{\alpha_2} \oplus \dots \oplus \tilde{W}_{\alpha_s}. \quad (12.81)$$

Here $\tilde{W}_{\alpha_i}$ ($1 \leq i \leq s$) is given by

$$\tilde{W}_{\alpha_i} = \left\{ \mathbf{x}; \mathbf{x} \in V^n, (A - \alpha_i E)^{l_i} \mathbf{x} = \mathbf{0} \right\}, \quad (12.82)$$

where l_i is an enough large natural number. If multiplicity of α_i is n_i , $\dim \tilde{W}_{\alpha_i} = n_i$.

Proof Let us define the aforementioned A -invariant subspaces as $\tilde{W}_{\alpha_k}$ ($1 \leq k \leq s$). Let $f_A(x)$ be a characteristic polynomial of A . Factorizing $f_A(x)$ into a product of powers of first-degree polynomials, we have

$$f_A(x) = \prod_{i=1}^s (x - \alpha_i)^{n_i}, \quad (12.83)$$

where n_i is a multiplicity of α_i . Let us put

$$f_i(x) = f_A(x) / (x - \alpha_i)^{n_i} = \prod_{j \neq i}^s (x - \alpha_j)^{n_j}. \quad (12.84)$$

Then $f_1(x)$, $f_2(x)$, $\dots$, and $f_s(x)$ do not have a common factor. Consequently, Lemma 12.1 tells us that there are polynomials $M_1(x)$, $M_2(x)$, $\dots$, and $M_s(x)$ such that

$$M_1(x)f_1(x) + \dots + M_s(x)f_s(x) = 1. \quad (12.85)$$

Replacing x with a matrix A , we get

$$M_1(A)f_1(A) + \dots + M_s(A)f_s(A) = E. \quad (12.86)$$

Or defining $M_i(A)f_i(A) \equiv A_i$, we have

$$A_1 + A_2 + \dots + A_s = E, \quad (12.87)$$

where E is a (n, n) identity matrix. Moreover we have

$$A_i A_j = A_i \delta_{ij}. \quad (12.88)$$

In fact, if $i \neq j$,

$$\begin{aligned} A_i A_j &= M_i(A)f_i(A)M_j(A)f_j(A) = M_i(A)M_j(A)f_i(A)f_j(A) \\ &= M_i(A)M_j(A) \prod_{k \neq i}^s (A - \alpha_k)^{n_k} \prod_{l \neq j}^s (A - \alpha_l)^{n_l} \\ &= M_i(A)M_j(A)f_A(A) \prod_{k \neq i,j}^s (A - \alpha_k)^{n_k} = 0. \end{aligned} \quad (12.89)$$

The second equality results from the fact that $M_j(A)$ and $f_i(A)$ are commutable since both are polynomials of A . The last equality follows from Hamilton–Cayley theorem. On the basis of (12.87) and (12.89),

$$A_i = A_i E = A_i (A_1 + A_2 + \dots + A_s) = A_i^2. \quad (12.90)$$

Thus, we find that A_i is an idempotent matrix.

Next, let us show that using A_i determined above, $A_i V^n$ is identical to $\tilde{W}_{\alpha_i}$ ($1 \leq i \leq s$). To this end, we define W_i such that

$$W_i = A_i V^n = \{\mathbf{x}; \mathbf{x} \in V^n, A_i \mathbf{x} = \mathbf{x}\}. \quad (12.91)$$

We have

$$(x - \alpha_i)^{n_i} f_i(x) = f_A(x). \quad (12.92)$$

Therefore, from Hamilton–Cayley theorem we have

$$(A - \alpha_i E)^{n_i} f_i(A) = 0. \quad (12.93)$$

Operating $M_i(A)$ from the left and from the fact that $M_i(A)$ commutes with $(A - \alpha_i E)^{n_i}$, we get

$$(A - \alpha_i E)^{n_i} A_i = 0, \quad (12.94)$$

where we used $M_i(A)f_i(A) = A_i$. Operating both sides of this equation on V^n , furthermore, we have

$$(A - \alpha_i E)^{n_i} A_i V^n = 0.$$

This means that

$$A_i V^n \subset \tilde{W}_{\alpha_i} \quad (1 \leq i \leq s). \quad (12.95)$$

Conversely, suppose that $\mathbf{x} \in \tilde{W}_{\alpha_i}$. Then $(A - \alpha_i E)^l \mathbf{x} = \mathbf{0}$ holds for a certain natural number l . If $M_i(x)f_i(x)$ were divided out by $x - \alpha_i$, LHS of (12.85) would be divided out by $x - \alpha_i$ as well, leading to the contradiction. Thus, it follows that $(x - \alpha_i)^l$ and $M_i(x)f_i(x)$ do not have a common factor. Consequently, Lemma 12.1 ensures that we have polynomials $M(x)$ and $N(x)$ such that

$$M(x)(x - \alpha_i)^l + N(x)M_i(x)f_i(x) = 1$$

and, hence,

$$M(A)(A - \alpha_i E)^l + N(A)M_i(A)f_i(A) = E. \quad (12.96)$$

Operating both sides of (12.96) on $\mathbf{x}$, we get

$$M(A)(A - \alpha_i E)^l \mathbf{x} + N(A)M_i(A)f_i(A)\mathbf{x} = N(A)A_i \mathbf{x} = \mathbf{x}. \quad (12.97)$$

Notice that the first term of (12.97) vanishes from (12.82).

As A_i is a polynomial of A , it commutes with $N(A)$. Hence, we have

$$\mathbf{x} = A_i[N(A)\mathbf{x}] \in A_i V^n. \quad (12.98)$$

Thus, we get

$$\tilde{W}_{\alpha_i} \subset A_i V^n \quad (1 \leq i \leq s). \quad (12.99)$$

From (12.95) and (12.99), we conclude that

$$\tilde{W}_{\alpha_i} = A_i V^n \quad (1 \leq i \leq s). \quad (12.100)$$

In other words, W_i defined as (12.91) is identical to $\tilde{W}_{\alpha_i}$ that is defined as (12.82). Thus, we have

$$V^n = W_1 \oplus W_2 \oplus \cdots \oplus W_s$$

or

$$V^n = \tilde{W}_{\alpha_1} \oplus \tilde{W}_{\alpha_2} \oplus \cdots \oplus \tilde{W}_{\alpha_s}. \quad (12.101)$$

This completes the former half of the proof. With the latter half, the proof is as follows.

Suppose that $\dim \tilde{W}_{\alpha_i} = n'_i$. In parallel to the decomposition of V^n to the direct sum of (12.81), A can be reduced as

$$A \sim \begin{pmatrix} A^{(1)} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & A^{(s)} \end{pmatrix}, \quad (12.102)$$

where $A^{(i)}$ ($1 \leq i \leq s$) is a (n'_i, n'_i) matrix and a symbol $\sim$ indicates that A has been transformed by suitable similarity transformation. The matrix $A^{(i)}$ represents a linear transformation that A causes to $\tilde{W}_{\alpha_i}$. We denote a n'_i order identity matrix by $E_{n'_i}$. Equation (12.82) implies that the matrix represented by

$$N_i = A^{(i)} - \alpha_i E_{n'_i} \quad (12.103)$$

is a nilpotent matrix. The order of a nilpotent matrix is at most n'_i (vide supra) and, hence, l_i can be n'_i . With N_i we have

$$\begin{aligned} f_{N_i}(x) &= |xE_{n'_i} - N_i| = |xE_{n'_i} - [A^{(i)} - \alpha_i E_{n'_i}]| = |xE_{n'_i} - A^{(i)} + \alpha_i E_{n'_i}| \\ &= |(x + \alpha_i)E_{n'_i} - A^{(i)}| = x^{n'_i}. \end{aligned} \quad (12.104)$$

The last equality is because eigenvalues of a nilpotent matrix are all zero. Meanwhile,

$$f_{A^{(i)}}(x) = |xE_{n'_i} - A^{(i)}| = f_{N_i}(x - \alpha_i) = (x - \alpha_i)^{n'_i}. \quad (12.105)$$

Equation (12.105) implies that

$$f_A(x) = \prod_{i=1}^s f_{A^{(i)}}(x) = \prod_{i=1}^s (x - \alpha_i)^{n'_i} = \prod_{i=1}^s (x - \alpha_i)^{n_i}. \quad (12.106)$$

The last equality comes from (12.83). Thus, $n'_i = n_i$. At the same time, we may equate l_i in (12.82) to n_i . These procedures complete the proof.

Theorem 12.1 shows that any square matrix can be converted to a (upper) triangle matrix by a similarity transformation. Theorem 12.5 demonstrates that the matrix can further be segmented according to individual eigenvalues. Considering Theorem 12.1 again, $A^{(i)}$ ($1 \leq i \leq s$) can be described as an upper triangle matrix by

$$A^{(i)} \sim \begin{pmatrix} \alpha_i & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \alpha_i \end{pmatrix}. \quad (12.107)$$

Therefore, denoting $N^{(i)}$ such that

$$N^{(i)} = A^{(i)} - \alpha_i E_{n_i} = \begin{pmatrix} 0 & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix}, \quad (12.108)$$

we find that $N^{(i)}$ is nilpotent. This is because all the eigenvalues of $N^{(i)}$ are zero. From (12.108), we have

$$[N^{(i)}]^{\mu_i} = 0 \quad (\mu_i \leq n_i).$$

12.5 Decomposition of Matrix

To investigate canonical forms of matrices, it would be convenient if a matrix can be decomposed into appropriate forms. To this end, the following definition is important.

Definition 12.4 A matrix similar to a diagonal matrix is said to be semi-simple.

In the above definition, if a matrix is related to another matrix by similarity transformation, those matrices are said to be similar to each other. When we have two matrices A and A' , we express it by $A \sim A'$ as stated above. This relation satisfies the equivalence law. That is,

$$(1) A \sim A, (2) A \sim A' \Rightarrow A' \sim A, (3) A \sim A', A' \sim A'' \Rightarrow A \sim A''.$$

Readers, check this. We have a following important theorem with the matrix decomposition.

Theorem 12.6 [3, 4] Any (n, n) square matrix A is expressed uniquely as

$$A = S + N, \quad (12.109)$$

where S is semi-simple and N is nilpotent; S and N are commutable, i.e., $SN = NS$. Furthermore, S and N are polynomials of A with scalar coefficients.

Proof Using (12.86) and (12.87), we write

$$S = \alpha_1 A_1 + \cdots + \alpha_s A_s = \sum_{i=1}^s \alpha_i M_i(A) f_i(A). \quad (12.110)$$

Then, (12.110) is a polynomial of A . From Theorem 12.1 and Theorem 12.5, $A^{(i)}$ ($1 \leq i \leq s$) in (12.102) is characterized by that $A^{(i)}$ is a triangle matrix whose eigenvalues α_i ($1 \leq i \leq s$) are positioned on diagonal positions and that the order of $A^{(i)}$ is identical to the multiplicity of α_i . Since A_i ($1 \leq i \leq s$) is an idempotent matrix, it should be diagonalized through similarity transformation (see Sect. 12.7). In fact, corresponding to (12.102), S is transformed via similarity transformation into

$$S \sim \begin{pmatrix} \alpha_1 E_{n_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \alpha_s E_{n_s} \end{pmatrix}, \quad (12.111)$$

where E_{n_i} ($1 \leq i \leq s$) is an identity matrix of an order n_i that is identical to the multiplicity of α_i . This expression is equivalent to, e.g.,

$$A_1 \sim \begin{pmatrix} E_{n_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix}$$

in (12.110). Thus, S is obviously semi-simple (i.e., diagonalizable). Putting $N = A - S$, N is described after the above transformation as

$$N \sim \begin{pmatrix} N^{(1)} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & N^{(s)} \end{pmatrix}, \quad N^{(i)} = A^{(i)} - \alpha_i E_{n_i}. \quad (12.112)$$

Since each $N^{(i)}$ is nilpotent as stated in Sect. 12.4, N is nilpotent as well. Also (12.112) is a polynomial of A as in the case of S . Therefore, S and N are commutable.

To prove the uniqueness of the decomposition, we show the following:

1. Let S and S' be commutable semi-simple matrices. Then, those matrices are simultaneously diagonalized. That is, with a certain non-singular matrix P , $P^{-1}SP$ and $P^{-1}S'P$ are diagonalized at once. Hence, $S \pm S'$ is semi-simple as well.

2. Let N and N' be commutable nilpotent matrices. Then, $N \pm N'$ is nilpotent as well.
3. A matrix both semi-simple and nilpotent is zero matrix.
1. Let different eigenvalues of S be $\alpha_1, \dots, \alpha_s$. Then, since S is semi-simple, a vector space V^n is decomposed into a direct sum of eigenspaces W_{α_i} ($1 \leq i \leq s$). That is, we have

$$V^n = W_{\alpha_1} \oplus \dots \oplus W_{\alpha_s}.$$

Since S and S' are commutable, with $\exists \mathbf{x} \in W_{\alpha_i}$ we have $SS'\mathbf{x} = S'S\mathbf{x} = S'(\alpha_i\mathbf{x}) = \alpha_i S'\mathbf{x}$. Hence, we have $S'\mathbf{x} \in W_{\alpha_i}$. Namely, W_{α_i} is S' -invariant. Therefore, if we adopt the basis vectors $\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ with respect to the direct sum decomposition, we get

$$S \sim \begin{pmatrix} \alpha_1 E_{n_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \alpha_s E_{n_s} \end{pmatrix}, \quad S' \sim \begin{pmatrix} S'_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & S'_s \end{pmatrix}.$$

Since S' is semi-simple, S'_i ($1 \leq i \leq s$) must be semi-simple as well. Here, let $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ be original basis vectors before the basis vector transformation and let P be a representation matrix of the said transformation. Then, we have

$$(\mathbf{e}_1 \quad \cdots \quad \mathbf{e}_n)P = (\mathbf{a}_1 \quad \cdots \quad \mathbf{a}_n).$$

Thus, we get

$$P^{-1}SP = \begin{pmatrix} \alpha_1 E_{n_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \alpha_s E_{n_s} \end{pmatrix}, \quad P^{-1}S'P = \begin{pmatrix} S'_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & S'_s \end{pmatrix}.$$

This means that both $P^{-1}SP$ and $P^{-1}S'P$ are diagonal. That is, $P^{-1}SP \pm P^{-1}S'P = P^{-1}(S \pm S')P$ is diagonal, indicating that $S \pm S'$ is semi-simple as well.

2. Suppose that $N^\nu = 0$ and $N'^{\nu'} = 0$. From the assumption, N and N' are commutable. Consequently, using binomial theorem we have

$$\begin{aligned} (N \pm N')^m &= N^m \pm mN^{m-1}N' + \cdots + (\pm 1)^i \frac{m!}{i!(m-i)!} N^{m-i}N'^i + \cdots \\ &\quad + (\pm 1)^m N'^m. \end{aligned} \quad (12.113)$$

Putting $m = \nu + \nu' - 1$, if $i \geq \nu'$, $N'^i = 0$ from the supposition. If $i < \nu'$, we have $m - i > m - \nu' = \nu + \nu' - 1 - \nu' = \nu - 1$; i.e., $m - i \geq \nu$. Therefore, $N^{m-i} = 0$. Consequently, we have $N^{m-i}N'^i = 0$ with any i in (12.113). Thus, we get $(N \pm N')^m = 0$, indicating that $N \pm N'$ is nilpotent.

3. Let S be a semi-simple and nilpotent matrix. We describe S as

$$S \sim \begin{pmatrix} \alpha_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \alpha_n \end{pmatrix}, \quad (12.114)$$

where some of α_i ($1 \leq i \leq n$) may be identical. Since S is nilpotent, all α_i ($1 \leq i \leq n$) is zero. We have then $S \sim 0$; i.e., $S = 0$ accordingly.

Now, suppose that a matrix A is decomposed differently from (12.109). That is, we have

$$A = S + N = S' + N' \quad \text{or} \quad S - S' = N' - N. \quad (12.115)$$

From the assumption, S' and N' are commutable. Moreover, since S, S', N , and N' are described by a polynomial of A , they are commutable with one another. Hence, from (1) and (2) along with the second equation of (12.115), $S - S'$ and $N' - N$ are both semi-simple and nilpotent at once. Consequently, from (3) $S - S' = N' - N = 0$. Thus, we finally get $S = S'$ and $N = N'$. That is, the decomposition is unique.

These complete the proof.

Theorem 12.6 implies that the matrix decomposition of (12.109) is unique. This matrix decomposition is said to be Jordan decomposition. On the basis of Theorem 12.6, we investigate Jordan canonical forms of matrices in the next section.

12.6 Jordan Canonical Form

Once the vector space V^n has been decomposed to a direct sum of generalized eigenspaces with the matrix reduced in parallel, we are able to deal with individual eigenspaces $\tilde{W}_{\alpha_i}$ ($1 \leq i \leq s$) and the corresponding $A^{(i)}$ ($1 \leq i \leq s$) given in (12.102) separately.

12.6.1 Canonical Form of Nilpotent Matrix

To avoid complication of notation, we think of a following example where we assume a (n, n) matrix that operates on V^n : Let the nilpotent matrix be N : Suppose that $N^{\nu-1} \neq 0$ and $N^\nu = 0$ ($1 \leq \nu \leq n$). To avoid complex notation, we focus on one of the eigenspaces $\tilde{W}_{\alpha_i}$ ($1 \leq i \leq s$) and regard it as n -dimensional vector space V^n .

If $\nu = 1$, the nilpotent matrix N is a zero matrix. Notice that since a characteristic polynomial is described by $f_N(x) = x^n$, $f_N(N) = N^n = 0$ from Hamilton–Cayley theorem. Let $W^{(i)}$ be given such that

$$W^{(i)} = \{\mathbf{x}; \mathbf{x} \in V^n, N^i \mathbf{x} = \mathbf{0}, N^{i-1} \mathbf{x} \neq \mathbf{0} \ (i \geq 1)\}; \quad W^{(0)} \equiv \{\mathbf{0}\}.$$

Then we have

$$V^n = W^{(\nu)} \supset W^{(\nu-1)} \supset \dots \supset W^{(1)} \supset W^{(0)} \equiv \{\mathbf{0}\}. \quad (12.116)$$

Note that when $\nu = 1$, we have trivially $V^n = W^{(\nu)} \supset W^{(0)} \equiv \{\mathbf{0}\}$. Let us put $\dim W^{(i)} = m_i$, $m_i - m_{i-1} = r_i$ ($1 \leq i \leq \nu$), $m_0 \equiv 0$. Then we can add r_ν linearly independent vectors $\mathbf{a}_1, \mathbf{a}_2, \dots$, and $\mathbf{a}_{r_\nu}$ to the basis vectors of $W^{(\nu-1)}$ so that those r_ν vectors can be basis vectors of $W^{(\nu)}$. Unless $N = 0$ (i.e., zero matrix), we must have at least one such vector; from the supposition with $\exists \mathbf{x} \neq \mathbf{0}$ we have $N^{\nu-1} \mathbf{x} \neq \mathbf{0}$, and so $\mathbf{x} \notin W^{(\nu-1)}$. At least one such vector $\mathbf{x}$ is present and it is eligible for a basis vector of $W^{(\nu)}$. Hence, $r_\nu \geq 1$ and we have

$$W^{(\nu)} = \text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{r_\nu}\} \bigoplus W^{(\nu-1)}. \quad (12.117)$$

Note that (12.117) is expressed as a direct sum. Meanwhile, $N\mathbf{a}_1, N\mathbf{a}_2, \dots, N\mathbf{a}_{r_\nu} \in W^{(\nu-1)}$. In fact, suppose that $\mathbf{x} \in W^{(\nu)}$, i.e., $N^\nu \mathbf{x} = N^{\nu-1}(N\mathbf{x}) = \mathbf{0}$. That is, $N\mathbf{x} \in W^{(\nu-1)}$.

According to a similar reasoning made above, we have

$$\text{Span}\{N\mathbf{a}_1, N\mathbf{a}_2, \dots, N\mathbf{a}_{r_\nu}\} \cap W^{(\nu-2)} = \{\mathbf{0}\}.$$

Moreover, these r_ν vectors $N\mathbf{a}_1, N\mathbf{a}_2, \dots$, and $N\mathbf{a}_{r_\nu}$ are linearly independent. Suppose that

$$c_1 N\mathbf{a}_1 + c_2 N\mathbf{a}_2 + \dots + c_{r_\nu} N\mathbf{a}_{r_\nu} = \mathbf{0}. \quad (12.118)$$

Operating $N^{\nu-2}$ from the left, we have

$$N^{\nu-1}(c_1 \mathbf{a}_1 + c_2 \mathbf{a}_2 + \dots + c_{r_\nu} \mathbf{a}_{r_\nu}) = \mathbf{0}.$$

This would imply that $c_1 \mathbf{a}_1 + c_2 \mathbf{a}_2 + \dots + c_{r_\nu} \mathbf{a}_{r_\nu} \in W^{(\nu-1)}$. On the basis of the above argument, however, we must have $c_1 = c_2 = \dots = c_{r_\nu} = 0$. In other words, if $\exists c_i$ ($1 \leq i \leq r_\nu$) were non-zero, we would have $\mathbf{a}_i \in W^{(\nu-1)}$, in contradiction. From (12.118) this means linear independence of $N\mathbf{a}_1, N\mathbf{a}_2, \dots$, and $N\mathbf{a}_{r_\nu}$.

As $W^{(\nu-1)} \supset W^{(\nu-2)}$, we may well have additional linearly independent vectors within the basis vectors of $W^{(\nu-1)}$. Let those vectors be $\mathbf{a}_{r_\nu+1}, \dots, \mathbf{a}_{r_{\nu-1}}$. Here we assume that the number of such vectors is $r_{\nu-1} - r_\nu$. We have $r_{\nu-1} - r_\nu \geq 0$ accordingly. In this way we can construct basis vectors of $W^{(\nu-1)}$ by including $\mathbf{a}_{r_\nu+1}, \dots$, and $\mathbf{a}_{r_{\nu-1}}$ along with $N\mathbf{a}_1, N\mathbf{a}_2, \dots$, and $N\mathbf{a}_{r_\nu}$. As a result, we get

$$W^{(\nu-1)} = \text{Span}\{Na_1, \dots, Na_{r_\nu}, a_{r_\nu+1}, \dots, a_{r_{\nu-1}}\} \bigoplus W^{(\nu-2)}.$$

We can repeat these processes to construct $W^{(\nu-2)}$ such that

$$W^{(\nu-2)} = \text{Span}\{N^2a_1, \dots, N^2a_{r_\nu}, Na_{r_\nu+1}, \dots, Na_{r_{\nu-1}}, a_{r_{\nu-1}+1}, \dots, a_{r_{\nu-2}}\} \bigoplus W^{(\nu-3)}.$$

For $W^{(i)}$, furthermore, we have

$$\begin{aligned} W^{(i)} = & \text{Span}\{N^{\nu-i}a_1, \dots, N^{\nu-i}a_{r_\nu}, N^{\nu-i-1}a_{r_\nu+1}, \dots, N^{\nu-i-1}a_{r_{\nu-1}}, \dots, a_{r_{i+1}+1}, \dots, a_{r_i}\} \\ & \bigoplus W^{(i-1)}. \end{aligned} \quad (12.119)$$

Further repeating the procedures, we exhaust all the n basis vectors of $W^{(\nu)} = V^n$. These vectors are given as follows:

$$N^k a_{r_{i+1}+1}, \dots, N^k a_{r_i} \quad (1 \leq i \leq \nu; 0 \leq k \leq i-1).$$

At the same time we have

$$0 \equiv r_{\nu+1} < 1 \leq r_\nu \leq r_{\nu-1} \leq \dots \leq r_1. \quad (12.120)$$

Table 12.1 [3, 4] shows the resulting structure of these basis vectors pertinent to Jordan blocks. In Table 12.1, if laterally counting basis vectors, from the top we have $r_\nu, r_{\nu-1}, \dots, r_1$ vectors. Their sum is n . This is the same number as that vertically counted. The dimension n of the vector space V^n is thus given by

$$n = \sum_{i=1}^{\nu} r_i = \sum_{i=1}^{\nu} i(r_i - r_{i+1}). \quad (12.121)$$

Let us examine the structure of Table 12.1 more closely. More specifically, let us inspect the i -layered structures of $(r_i - r_{i+1})$ vectors. Picking up a vector from among $a_{r_{i+1}+1}, \dots, a_{r_i}$, we call it a_ρ . Then, we get following set of vectors $a_\rho, Na_\rho, N^2a_\rho, \dots, N^{i-1}a_\rho$ in the i -layered structure. These i vectors are displayed “vertically” in Table 12.1. These vectors are linearly independent (see Theorem 12.4) and form an i -dimensional N -invariant subspace; i.e., $\text{Span}\{N^{i-1}a_\rho, N^{i-2}a_\rho, \dots, Na_\rho, a_\rho\}$, where $r_{i+1} + 1 \leq \rho \leq r_i$. Matrix representation of the linear transformation N with respect to the set of these i vectors is [3, 4]

Table 12.1 Jordan blocks and structure of basis vectors for a nilpotent matrix^a

	$r_\nu \downarrow$								
$r_\nu \leftarrow$	$a_1, \dots, a_{r_\nu}$		$r_\nu - 1 - r_\nu \downarrow$						
$r_{\nu-1} \leftarrow$	$Na_1, \dots, Na_{r_\nu}$		$a_{r_\nu+1}, \dots, a_{r_\nu-1}$						
$\dots$	$N^2 a_1, \dots, N^2 a_{r_\nu}$		$Na_{r_\nu+1}, \dots, Na_{r_\nu-1}$						
$\dots$	$\dots$		$\dots$	$\dots$	$r_i - r_{i+1} \downarrow$				
$r_i \leftarrow$	$N^{\nu-i} a_1, \dots, N^{\nu-i} a_{r_\nu}$		$N^{\nu-i-1} a_{r_\nu+1}, \dots, N^{\nu-i-1} a_{r_\nu-1}$	$\dots$	$a_{r_{i+1}+1}, \dots, a_{r_i}$				
$r_{i-1} \leftarrow$	$N^{\nu-i-1} a_1, \dots, N^{\nu-i-1} a_{r_\nu}$		$N^{\nu-i-2} a_{r_\nu+1}, \dots, N^{\nu-i-2} a_{r_\nu-1}$	$\dots$	$Na_{r_{i+1}+1}, \dots, Na_{r_{i-1}}$				
$\dots$	$\dots$		$\dots$	$\dots$	$\dots$	$r_2 - r_3 \downarrow$			
$r_2 \leftarrow$	$N^{\nu-2} a_1, \dots, N^{\nu-2} a_{r_\nu}$		$N^{\nu-3} a_{r_\nu+1}, \dots, N^{\nu-3} a_{r_\nu-1}$	$\dots$	$N^{i-2} a_{r_{i+1}+1}, \dots, N^{i-2} a_{r_i}$	$\dots$	$a_{r_3+1}, \dots, a_{r_2}$	$r_1 - r_2 \downarrow$	
$r_1 \leftarrow$	$N^{\nu-1} a_1, \dots, N^{\nu-1} a_{r_\nu}$		$N^{\nu-2} a_{r_\nu+1}, \dots, N^{\nu-2} a_{r_\nu-1}$	$\dots$	$N^{i-1} a_{r_{i+1}+1}, \dots, N^{i-1} a_{r_i}$	$\dots$	$Na_{r_3+1}, \dots, Na_{r_2}$	$a_{r_2+1}, \dots, a_{r_1}$	
$n = \sum_{k=1}^{\nu} r_k$	νr_ν		$(\nu - 1)(r_\nu - 1 - r_\nu)$	$\dots$	$i(r_i - r_{i+1})$	$\dots$	$2(r_2 - r_3)$	$r_1 - r_2$	

^a Adapted from Satake I (1974) Linear algebra (Mathematics Library 1: in Japanese) [4], with the permission of Shokabo Co., Ltd., Tokyo

$$\begin{aligned}
 & (N^{i-1}\mathbf{a}_\rho \ N^{i-2}\mathbf{a}_\rho \cdots N\mathbf{a}_\rho \ \mathbf{a}_\rho)N \\
 &= (N^{i-1}\mathbf{a}_\rho \ N^{i-2}\mathbf{a}_\rho \cdots N\mathbf{a}_\rho \ \mathbf{a}_\rho) \begin{pmatrix} 0 & 1 & & & \\ & 0 & 1 & & \\ & & 0 & \ddots & \\ \vdots & & & \ddots & \ddots \\ & & & & 0 & 1 \\ & & \cdots & & & 0 & 1 \\ & & & & & & 0 \end{pmatrix}. \quad (12.122)
 \end{aligned}$$

These (i, i) matrices of (12.122) are called i -th order Jordan blocks. Equation (12.122) clearly shows that $N^{i-1}\mathbf{a}_\rho$ is a proper eigenvector and that we have $i-1$ generalized eigenvectors $N^{i-2}\mathbf{a}_\rho, \dots, N\mathbf{a}_\rho, \mathbf{a}_\rho$.

Notice that the number of those Jordan blocks is $(r_i - r_{i+1})$. Let us expressly define this number as [3, 4]

$$J_i = r_i - r_{i+1}, \quad (12.123)$$

where J_i is the number of the i -th order Jordan blocks. The total number of Jordan blocks within a whole vector space $V^n = W^{(\nu)}$ is

$$\sum_{i=1}^{\nu} J_i = \sum_{i=1}^{\nu} (r_i - r_{i+1}) = r_1. \quad (12.124)$$

Recalling the dimension theorem mentioned in (11.45), we have

$$\dim V^n = \dim \text{Ker } N^i + \dim N^i(V^n) = \dim \text{Ker } N^i + \text{rank } N^i. \quad (12.125)$$

Meanwhile, since $W^{(i)} = \text{Ker } N^i$, $\dim W^{(i)} = m_i = \dim \text{Ker } N^i$. From (12.116), $m_0 \equiv 0$. Then (11.45) is now read as

$$\dim V^n = m_i + \text{rank } N^i. \quad (12.126)$$

That is

$$n = m_i + \text{rank } N^i \quad (12.127)$$

or

$$m_i = n - \text{rank } N^i. \quad (12.128)$$

Meanwhile, from Table 12.1 [3, 4] we have

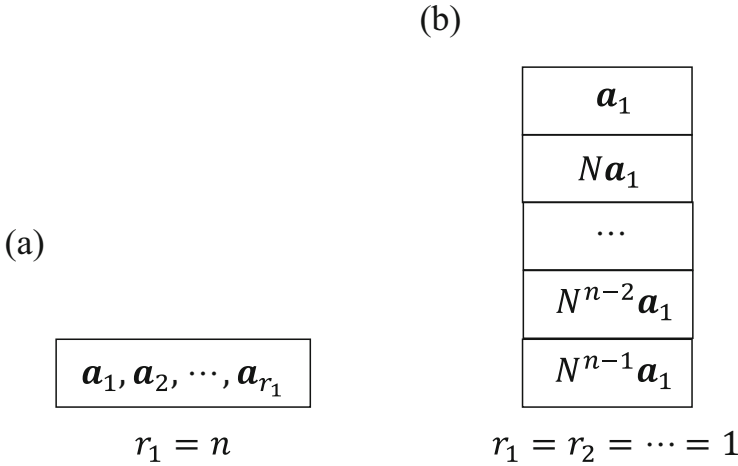


Fig. 12.1 Examples of the structure of Jordan blocks. (a) $r_1 = n$. (b) $r_1 = r_2 = \dots = r_n = 1$

$$\dim W^{(i)} = m_i = \sum_{k=1}^i r_k, \quad \dim W^{(i-1)} = m_{i-1} = \sum_{k=1}^{i-1} r_k. \quad (12.129)$$

Hence, we have

$$r_i = m_i - m_{i-1}. \quad (12.130)$$

Then we get [3, 4]

$$\begin{aligned} J_i &= r_i - r_{i+1} = (m_i - m_{i-1}) - (m_{i+1} - m_i) = 2m_i - m_{i-1} - m_{i+1} \\ &= 2(n - \text{rank } N^i) - (n - \text{rank } N^{i-1}) - (n - \text{rank } N^{i+1}) \\ &= \text{rank } N^{i-1} + \text{rank } N^{i+1} - 2\text{rank } N^i. \end{aligned} \quad (12.131)$$

The number J_i is therefore defined uniquely by N . The total number of Jordan blocks r_1 is also computed using (12.128) and (12.130) as

$$r_1 = m_1 - m_0 = m_1 = n - \text{rank } N = \dim \text{Ker } N, \quad (12.132)$$

where the last equality arises from the dimension theorem expressed as (11.45).

In Table 12.1 [3, 4], moreover, we have two extreme cases. That is, if $\nu = 1$ in (12.116), i.e., $N = 0$, from (12.132) we have $r_1 = n$ and $r_2 = \dots = r_n = 0$; see Fig. 12.1a. Also we confirm $n = \sum_{i=1}^{\nu} r_i$ in (12.121). In this case, all the eigenvectors are proper eigenvectors with multiplicity of n and we have n first-order Jordan blocks. The other is the case of $\nu = n$. In that case, we have

$$r_1 = r_2 = \cdots = r_n = 1. \quad (12.133)$$

In the latter case, we also have $n = \sum_{i=1}^{\nu} r_i$ in (12.121). We have only one proper eigenvector and $(n - 1)$ generalized eigenvectors; see Fig. 12.1b. From (12.132), we have this special case where we have only one n -th order Jordan block, when $\text{rank} N = n - 1$.

12.6.2 Jordan Blocks

Let us think of (12.81) on the basis of (12.102). Picking up $A^{(i)}$ from (12.102) and considering (12.108), we put

$$N_i = A^{(i)} - \alpha_i E_{n_i} \quad (1 \leq i \leq s), \quad (12.134)$$

where E_{n_i} denotes (n_i, n_i) identity matrix. We express a nilpotent matrix as N_i as before. In (12.134) the number n_i corresponds to n in V^n of Sect. 12.6.1. As N_i is a (n_i, n_i) matrix, we have

$$N_i^{n_i} = 0.$$

Here we are speaking of ν -th order nilpotent matrices N_i such that $N_i^{\nu-1} \neq 0$ and $N_i^{\nu} = 0$ ($1 \leq \nu \leq n_i$). We can deal with N_i in a manner fully consistent with the theory we developed in Sect. 12.6.1. Each $A^{(i)}$ comprises one or more Jordan blocks $A^{(\kappa)}$ that is expressed as

$$A^{(\kappa)} = N_{\kappa_i} + \alpha_i E_{\kappa_i} \quad (1 \leq \kappa_i \leq n_i), \quad (12.135)$$

where $A^{(\kappa)}$ denotes the κ -th Jordan block in $A^{(i)}$. In $A^{(\kappa)}$, N_{κ_i} and E_{κ_i} are nilpotent (κ_i, κ_i) -matrix and (κ_i, κ_i) identity matrix, respectively. In N_{κ_i} , κ_i zeros are displayed on the principal diagonal and entries of 1 are positioned on the matrix element next above the principal diagonal. All other entries are zero; see, e.g., a matrix of (12.122). As in the Sect. 12.6.1, the number κ_i is called a dimension of the Jordan block. Thus, $A^{(i)}$ of (12.102) can further be reduced to segmented matrices $A^{(\kappa)}$. Our next task is to find out how many Jordan blocks are contained in individual $A^{(i)}$ and what is the dimension of those Jordan blocks.

Corresponding to (12.122), the matrix representation of the linear transformation by $A^{(\kappa)}$ with respect to the set of κ_i vectors is

$$\begin{aligned}
& \left(\left[A^{(\kappa)} - \alpha_i E_{\kappa_i} \right]^{\kappa_i - 1} \mathbf{a}_\sigma \left[A^{(\kappa)} - \alpha_i E_{\kappa_i} \right]^{\kappa_i - 2} \mathbf{a}_\sigma \cdots \mathbf{a}_\sigma \right) \left[A^{(\kappa)} - \alpha_i E_{\kappa_i} \right] \\
&= \left(\left[A^{(\kappa)} - \alpha_i E_{\kappa_i} \right]^{\kappa_i - 1} \mathbf{a}_\sigma \left[A^{(\kappa)} - \alpha_i E_{\kappa_i} \right]^{\kappa_i - 2} \mathbf{a}_\sigma \cdots \mathbf{a}_\sigma \right) \\
&\quad \times \begin{pmatrix} 0 & 1 & & & & \\ & 0 & 1 & & & \\ & & 0 & \ddots & & \\ & \vdots & & \ddots & \ddots & \vdots \\ & & & 0 & 1 & \\ & & \cdots & & 0 & 1 \\ & & & & & 0 \end{pmatrix}. \tag{12.136}
\end{aligned}$$

A vector $\mathbf{a}_\sigma$ stands for a vector associated with the κ -th Jordan block of $A^{(i)}$. From (12.136) we obtain

$$\left[A^{(\kappa)} - \alpha_i E_{\kappa_i} \right] \left\{ \left[A^{(\kappa)} - \alpha_i E_{\kappa_i} \right]^{\kappa_i - 1} \mathbf{a}_\sigma \right\} = 0. \tag{12.137}$$

Namely,

$$A^{(\kappa)} \left\{ \left[A^{(\kappa)} - \alpha_i E_{\kappa_i} \right]^{\kappa_i - 1} \mathbf{a}_\sigma \right\} = \alpha_i \left\{ \left[A^{(\kappa)} - \alpha_i E_{\kappa_i} \right]^{\kappa_i - 1} \mathbf{a}_\sigma \right\}. \tag{12.138}$$

This shows that $\left[A^{(\kappa)} - \alpha_i E_{\kappa_i} \right]^{\kappa_i - 1} \mathbf{a}_\sigma$ is a proper eigenvector of $A^{(\kappa)}$ that corresponds to an eigenvalue α_i . Conversely, $\mathbf{a}_\sigma$ is a generalized eigenvector of rank κ_i . There are another $(\kappa_i - 2)$ generalized eigenvectors of $\left[A^{(\kappa)} - \alpha_i E_{\kappa_i} \right]^\mu \mathbf{a}_\sigma$ ($1 \leq \mu \leq \kappa_i - 2$). In total, there are κ_i eigenvectors [a proper eigenvector and $(\kappa_i - 1)$ generalized eigenvectors]. Also we see that the sole proper eigenvector can be found for each Jordan block.

In reference to these κ_i eigenvectors as the basis vectors, the (κ_i, κ_i) -matrix $A^{(\kappa)}$ (i.e., a Jordan block) is expressed as

$$A^{(\kappa)} = \begin{pmatrix} \alpha_i & 1 & & & & \\ & \alpha_i & 1 & & & \\ & & \alpha_i & \ddots & & \\ & \vdots & & \ddots & \ddots & \vdots \\ & & & & \alpha_i & 1 \\ & & \cdots & & & \alpha_i & 1 \\ & & & & & & \alpha_i \end{pmatrix}. \quad (12.139)$$

A (n_i, n_i) -matrix $A^{(i)}$ of (12.102) pertinent to an eigenvalue α_i contains a direct sum of Jordan blocks whose dimension ranges from 1 to n_i . The largest possible number of Jordan blocks of dimension d (that satisfies $\lceil \frac{n_i}{2} + 1 \rceil \leq d \leq n_i$, where $\lceil \mu \rceil$ denotes a largest integer that does not exceed μ) is at most one.

An example depicted below is a matrix $A^{(p)}$ ($1 \leq p \leq s$) that explicitly includes two one-dimensional Jordan blocks, a (3, 3) three-dimensional Jordan block, and a (5, 5) five-dimensional Jordan block:

$$A^{(p)} \sim \begin{pmatrix} \alpha_p & 0 & & & & & & & & \\ & \alpha_p & 0 & & & & & & & \\ & & \alpha_p & 1 & & \cdots & & & & \\ & & & \alpha_p & 1 & & & & & \\ & \vdots & & \alpha_p & 0 & & & \vdots & & \\ & & & & \alpha_p & 1 & & & & \\ & & & & & \alpha_p & 1 & & & \\ & & & & \cdots & & \alpha_p & 1 & & \\ & & & & & & & \alpha_p & 1 & \\ & & & & & & & & \alpha_p & 1 \\ & & & & & & & & & \alpha_p \end{pmatrix}.$$

where $A^{(p)}$ is a (10, 10) upper triangle matrix in which an eigenvalue α_p is displayed on the principal diagonal with entries 0 or 1 on the matrix element next above the principal diagonal with all other entries being zero.

Theorem 12.1 shows that every (n, n) square matrix can be converted to a triangle matrix by suitable similarity transformation. Diagonal elements give eigenvalues. Furthermore, Theorem 12.5 ensures that A can be reduced to generalized eigenspaces $\tilde{W}_{\alpha_i}$ ($1 \leq i \leq s$) according to individual eigenvalues. Suppose, for example, that after a suitable similarity transformation a full matrix A is represented as

$$P = \begin{pmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{pmatrix}$$

is a non-singular matrix. The matrix P is to be operated on $A^{(i)}$ so that we can separately perform the similarity transformation with respect to a (4, 4) nilpotent matrix $A^{(i)} - \alpha_i E_4$ following the procedures mentioned Sect. 12.6.1.

Thus, only an α_i -associated segment can be treated with other segments left unchanged. In a similar fashion, we can consecutively deal with matrix segments related to other eigenvalues. In a practical case, however, it is more convenient to seek different eigenvalues and corresponding (generalized) eigenvectors at once and convert the matrix to Jordan canonical form. To make a guess about the structure of a matrix, however, the following argument will be useful. Let us think of an example after that.

Using (12.140), we have

[illegible]

Here we are treating the (n, n) matrix A on V^n . Note that a matrix $A - \alpha_i E$ is not nilpotent as a whole. Suppose that the multiplicity of α_i is n_i ; in (12.140) $n_i = 4$. Since eigenvalues $\alpha_1, \alpha_2, \dots$, and α_s take different values from one another, $\alpha_1 - \alpha_i, \alpha_2 - \alpha_i, \dots$, and $\alpha_s - \alpha_i \neq 0$. In a triangle matrix diagonal elements give eigenvalues and, hence, $\alpha_1 - \alpha_i, \alpha_2 - \alpha_i, \dots$, and $\alpha_s - \alpha_i$ are non-zero eigenvalues of $A - \alpha_i E$. We rewrite (12.140) as

$$A - \alpha_i E \sim \begin{pmatrix} M^{(1)} & & & \\ & M^{(2)} & & \\ & & \ddots & \\ & & & M^{(i)} \\ & & & & \ddots \\ & & & & & M^{(s)} \end{pmatrix}, \quad (12.142)$$

where $M^{(1)} = (\alpha_1 - \alpha_i)$, $M^{(2)} = \begin{pmatrix} \alpha_2 - \alpha_i & * & * \\ & \alpha_2 - \alpha_i & * \\ & & \alpha_2 - \alpha_i \end{pmatrix}, \dots, M^{(i)} =$

$$\begin{pmatrix} 0 & * & * & * \\ & 0 & * & * \\ & & 0 & * \\ & & & 0 \end{pmatrix}, \dots, M^{(s)} = \begin{pmatrix} \alpha_s - \alpha_i & * \\ & \alpha_s - \alpha_i \end{pmatrix}.$$

Thus, $M^{(p)}$ ($p \neq i$) is a non-singular matrix and $M^{(i)}$ is a nilpotent matrix. Note that if we can find μ_i such that $[M^{(i)}]^{\mu_i-1} \neq 0$ and $[M^{(i)}]^{\mu_i} = 0$, with a minimal polynomial $\varphi_{M^{(i)}}(x)$ for $M^{(i)}$, we have $\varphi_{M^{(i)}}(x) = x^{\mu_i}$. Consequently, we get

$$(A - \alpha_i E)^{\mu_i} \sim \begin{pmatrix} [M^{(1)}]^{\mu_i} & & & \\ & [M^{(2)}]^{\mu_i} & & \\ & & \ddots & \\ & & & 0 \\ & & & & \ddots \\ & & & & & [M^{(s)}]^{\mu_i} \end{pmatrix}. \quad (12.143)$$

In (12.143) diagonal elements of non-singular triangle matrices $[M^{(p)}]^{\mu_i}$ ($p \neq i$) are $(\alpha_p - \alpha_i)^{\mu_i}$ ($\neq 0$). Thus, we have a “perforated” matrix $(A - \alpha_i E)^{\mu_i}$, where $[M^{(i)}]^{\mu_i} = 0$ in (12.143). Putting

$$\Phi_A(x) \equiv \prod_{i=1}^s (x - \alpha_i)^{\mu_i},$$

we get

$$\Phi_A(A) \equiv \prod_{i=1}^s (A - \alpha_i E)^{\mu_i} = 0.$$

A polynomial $\Phi_A(x)$ gives a minimal polynomial for $f_A(x)$.

From the above argument, we can choose μ_i for l_i in (12.82). Meanwhile, $M^{(i)}$ in (12.142) is identical to N_i in (12.134). Rewriting (12.134), we get

$$A^{(i)} = M^{(i)} + \alpha_i E_{n_i}. \quad (12.144)$$

Let us think of matrices $[A^{(i)} - \alpha_i E_{n_i}]^k$ and $(A - \alpha_i E)^k$ ($k \leq \mu_i$). From (11.45), we find

$$\dim V^n = n = \dim \text{Ker } (A - \alpha_i E)^k + \text{rank } (A - \alpha_i E)^k \quad (12.145)$$

and

$$\dim \tilde{W}_{\alpha_i} = n_i = \dim \text{Ker } [A^{(i)} - \alpha_i E_{n_i}]^k + \text{rank } [A^{(i)} - \alpha_i E_{n_i}]^k, \quad (12.146)$$

where $\text{rank } (A - \alpha_i E)^k = \dim (A - \alpha_i E)^k(V^n)$ and $\text{rank } [A^{(i)} - \alpha_i E_{n_i}]^k = \dim [A^{(i)} - \alpha_i E_{n_i}]^k(\tilde{W}_{\alpha_i})$.

Noting that

$$\dim \text{Ker } (A - \alpha_i E)^k = \dim \text{Ker } [A^{(i)} - \alpha_i E_{n_i}]^k, \quad (12.147)$$

we get

$$n - \text{rank } (A - \alpha_i E)^k = n_i - \text{rank } [A^{(i)} - \alpha_i E_{n_i}]^k. \quad (12.148)$$

This notable property comes from the non-singularity of $[M^{(p)}]^k$ ($p \neq i$, k : a positive integer); i.e., all the eigenvalues of $[M^{(p)}]^k$ are non-zero. In particular, as $\text{rank } [A^{(i)} - \alpha_i E_{n_i}]^{\mu_i} = 0$, from (12.148) we have

$$\text{rank } (A - \alpha_i E)^l = n - n_i \quad (l \geq \mu_i). \quad (12.149)$$

Meanwhile, putting $k = 1$ in (12.148) and using (12.132) we get

$$\begin{aligned} \dim \text{Ker } [A^{(i)} - \alpha_i E_{n_i}] &= n_i - \text{rank } [A^{(i)} - \alpha_i E_{n_i}] = n - \text{rank } (A - \alpha_i E) \\ &= \dim \text{Ker } (A - \alpha_i E) = r_1^{(i)}. \end{aligned} \quad (12.150)$$

The value $r_1^{(i)}$ gives the number of Jordan blocks with an eigenvalue α_i .

Moreover, we must consider a following situation. We know how the matrix $A^{(i)}$ in (12.134) is reduced to Jordan blocks of lower dimension. To get detailed information about it, however, we have to get the information about (generalized) eigenvalues corresponding to eigenvalues other than α_i . In this context, Eq. (12.148) is useful. Equation (12.131) tells how the number of Jordan blocks in a nilpotent matrix is determined. If we can get this knowledge before we have found out all the (generalized) eigenvectors, it will be easier to address the problem. Let us rewrite (12.131) as

$$\begin{aligned} J_q^{(i)} &= r_q - r_{q+1} \\ &= \text{rank} \left[A^{(i)} - \alpha_i E_{n_i} \right]^{q-1} + \text{rank} \left[A^{(i)} - \alpha_i E_{n_i} \right]^{q+1} - 2 \text{rank} \left[A^{(i)} - \alpha_i E_{n_i} \right]^q, \end{aligned} \quad (12.151)$$

where we define $J_q^{(i)}$ as the number of the q -th order Jordan blocks within $A^{(i)}$. Note that these blocks are expressed as (q, q) matrices. Meanwhile, using (12.148) $J_q^{(i)}$ is expressed as

$$J_q^{(i)} = \text{rank} (A - \alpha_i E)^{q-1} + \text{rank} (A - \alpha_i E)^{q+1} - 2 \text{rank} (A - \alpha_i E)^q. \quad (12.152)$$

This relation can be obtained by replacing k in (12.148) with $q-1$, $q+1$, and q , respectively and deleting n and n_i from these three relations. This enables us to gain access to a whole structure of the linear transformation represented by the (n, n) matrix A without reducing it to subspaces.

To enrich our understanding of Jordan canonical forms, a following tangible example will be beneficial.

12.6.3 Example of Jordan Canonical Form

Let us think of a following matrix A :

$$A = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ -3 & -1 & -2 & 4 \end{pmatrix}. \quad (12.153)$$

The characteristic equation $f_A(x)$ is given by

$$\begin{aligned}
 f_A(x) &= \begin{vmatrix} x-1 & 1 & 0 & 0 \\ -1 & x-3 & 0 & 0 \\ 0 & 0 & x-2 & 0 \\ 3 & 1 & 2 & x-4 \end{vmatrix} \\
 &= (x-4)(x-2)^3.
 \end{aligned} \tag{12.154}$$

Equating (12.154) to zero, we get an eigenvalue 4 as a simple root and an eigenvalue 2 as a triple root. The vector space V^4 is then decomposed to two invariant subspaces. The first is a one-dimensional kernel (or null-space) of the transformation $(A - 4E)$ and the other is a three-dimensional kernel of the transformation $(A - 2E)^3$. We have to seek eigenvectors that span these invariant subspaces with individual eigenvalues of x .

1. Case I ($x = 4$):

An eigenvector belonging to the first invariant subspace must satisfy a proper eigenvalue equation since the eigenvalue 4 is simple root. This equation is expressed as

$$(A - 4E)\mathbf{x} = \mathbf{0}.$$

This reads in a matrix form as

$$\begin{pmatrix} -3 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ -3 & -1 & -2 & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \mathbf{0}. \tag{12.155}$$

This is equivalent to a following set of four equations:

$$\begin{aligned}
 -3c_1 - c_2 &= 0, \\
 c_1 - c_2 &= 0, \\
 -2c_3 &= 0, \\
 -3c_1 - c_2 - 2c_3 &= 0.
 \end{aligned}$$

These are equivalent to that $c_3 = 0$ and $c_1 = c_2 = -3c_1$. Therefore, $c_1 = c_2 = c_3 = 0$ with an arbitrarily chosen number of c_4 , which is chosen as 1 as usual. Hence, designating the proper eigenvector as $\mathbf{e}_1^{(4)}$, its column vector representation is

$$\mathbf{e}_1^{(4)} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

A (4, 4) matrix in (12.155) representing $(A - 4E)$ has a rank 3. The number of Jordan blocks for an eigenvalue 4 is given by (12.150) as

$$r_1^{(4)} = 4 - \text{rank}(A - 4E) = 1. \quad (12.156)$$

In this case, the Jordan block is naturally one-dimensional. In fact, using (12.152) we have

$$\begin{aligned} J_1^{(4)} &= \text{rank}(A - 4E)^0 + \text{rank}(A - 4E)^2 - 2\text{rank}(A - 4E) \\ &= 4 + 3 - 2 \times 3 = 1. \end{aligned} \quad (12.157)$$

In (12.157), $J_1^{(4)}$ gives the number of the first-order Jordan blocks for an eigenvalue 4. We used

$$(A - 4E)^2 = \begin{pmatrix} 8 & 4 & 0 & 0 \\ -4 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 8 & 4 & 4 & 0 \end{pmatrix} \quad (12.158)$$

and confirmed that $\text{rank}(A - 4E)^2 = 3$.

2. Case II ($x = 2$):

The eigenvalue 2 has a triple root. Therefore, we must examine how the invariant subspaces can further be decomposed to subspaces of lower dimension. To this end we first start with a secular equation expressed as

$$(A - 2E)\mathbf{x} = \mathbf{0}. \quad (12.159)$$

The matrix representation is

$$\begin{pmatrix} -1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -3 & -1 & -2 & 2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \mathbf{0}.$$

This is equivalent to a following set of two equations:

$$\begin{aligned} c_1 + c_2 &= 0, \\ -3c_1 - c_2 - 2c_3 + 2c_4 &= 0. \end{aligned}$$

From the above, we can put $c_1 = c_2 = 0$ and $c_3 = c_4 (=1)$. The equations allow the existence of another proper eigenvector. For this we have $c_1 = -c_2 = 1$,

$c_3 = 0$, and $c_4 = 1$. Thus, for the two proper eigenvectors corresponding to an eigenvalue 2, we get

$$\mathbf{e}_1^{(2)} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{e}_2^{(2)} = \begin{pmatrix} 1 \\ -1 \\ 0 \\ 1 \end{pmatrix}.$$

A dimension of the invariant subspace corresponding to an eigenvalue 2 is three (due to the triple root) and, hence, there should be one generalized eigenvector. To determine it, we examine a following matrix equation:

$$(A - 2E)^2 \mathbf{x} = \mathbf{0}. \quad (12.160)$$

The matrix representation is

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -4 & 0 & -4 & 4 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \mathbf{0}.$$

That is,

$$-c_1 - c_3 + c_4 = 0. \quad (12.161)$$

Furthermore, we have

$$(A - 2E)^3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -8 & 0 & -8 & 8 \end{pmatrix}. \quad (12.162)$$

Moreover, $\text{rank}(A - 2E)^l (=1)$ remains unchanged for $l \geq 2$ as expected from (12.149).

It will be convenient to examine a structure of the invariant subspace. For this purpose, we seek the number of Jordan blocks $r_1^{(2)}$ and their order. Using (12.150), we have

$$r_1^{(2)} = 4 - \text{rank}(A - 2E) = 4 - 2 = 2. \quad (12.163)$$

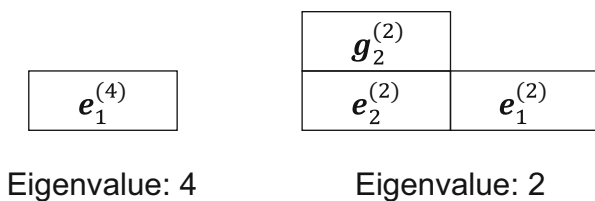
The number of first-order Jordan blocks is

$$\begin{aligned} J_1^{(2)} &= \text{rank}(A - 2E)^0 + \text{rank}(A - 2E)^2 - 2\text{rank}(A - 2E) \\ &= 4 + 1 - 2 \times 2 = 1. \end{aligned} \quad (12.164)$$

In turn, the number of second-order Jordan blocks is

Fig. 12.2 Structure of Jordan blocks of a matrix shown in (12.170)

Jordan blocks



$$\begin{aligned}
 J_2^{(2)} &= \text{rank}(A - 2E) + \text{rank}(A - 2E)^3 - 2\text{rank}(A - 2E)^2 \\
 &= 2 + 1 - 2 \times 1 = 1.
 \end{aligned} \tag{12.165}$$

In the above, $J_1^{(2)}$ and $J_2^{(2)}$ are obtained from (12.152). Thus, Fig. 12.2 gives a constitution of Jordan blocks for eigenvalues 4 and 2. The overall number of Jordan blocks is three; the number of the first-order and second-order Jordan blocks is two and one, respectively.

The proper eigenvector $\mathbf{e}_1^{(2)}$ is related to $J_1^{(2)}$ of (12.164). A set of the proper eigenvector $\mathbf{e}_2^{(2)}$ and the corresponding generalized eigenvector $\mathbf{g}_2^{(2)}$ is pertinent to $J_2^{(2)}$ of (12.165). We must have the generalized eigenvector $\mathbf{g}_2^{(2)}$ in such a way that

$$(A - 2E)^2 \mathbf{g}_2^{(2)} = (A - 2E) \left[(A - 2E) \mathbf{g}_2^{(2)} \right] = (A - 2E) \mathbf{e}_2^{(2)} = \mathbf{0}. \tag{12.166}$$

From (12.161), we can put $c_1 = c_3 = c_4 = 0$ and $c_2 = -1$. Thus, the matrix representation of the generalized eigenvector $\mathbf{g}_2^{(2)}$ is

$$\mathbf{g}_2^{(2)} = \begin{pmatrix} 0 \\ -1 \\ 0 \\ 0 \end{pmatrix}. \tag{12.167}$$

We stress here that $\mathbf{e}_1^{(2)}$ is not eligible for a proper pair with $\mathbf{g}_2^{(2)}$ in J_2 . It is because from (12.166) we have

$$(A - 2E) \mathbf{g}_2^{(2)} = \mathbf{e}_2^{(2)}, \quad (A - 2E) \mathbf{g}_2^{(2)} \neq \mathbf{e}_1^{(2)}. \tag{12.168}$$

Thus, we have determined a set of (generalized) eigenvectors. The matrix representation R for the basis vectors transformation is given by

$$R = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{pmatrix} \sim (\mathbf{e}_1^{(4)} \mathbf{e}_1^{(2)} \mathbf{e}_2^{(2)} \mathbf{g}_2^{(2)}), \quad (12.169)$$

where the symbol $\sim$ denotes the column vector representation; $\mathbf{e}_1^{(4)}$, $\mathbf{e}_1^{(2)}$, and $\mathbf{e}_2^{(2)}$ represent proper eigenvectors and $\mathbf{g}_2^{(2)}$ is a generalized eigenvector. Performing similarity transformation using this R , we get a following Jordan canonical form:

$$\begin{aligned} R^{-1}AR &= \begin{pmatrix} -1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ -1 & -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ -3 & -1 & -2 & 4 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix}. \end{aligned} \quad (12.170)$$

The structure of Jordan blocks is shown in Fig. 12.2. Notice here that the trace of A remains unchanged before and after similarity transformation.

Next, we consider column vector representations. According to (11.37), let us view the matrix A as a linear transformation over V^4 . Then A is given by

$$\begin{aligned} A(\mathbf{x}) &= (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3 \ \mathbf{e}_4) A \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \\ &= (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3 \ \mathbf{e}_4) \begin{pmatrix} 1 & -1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ -3 & -1 & -2 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}, \end{aligned} \quad (12.171)$$

where $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$, and $\mathbf{e}_4$ are basis vectors and x_1 , x_2 , x_3 , and x_4 are corresponding coordinates of a vector $\mathbf{x} = \sum_{i=1}^n x_i \mathbf{e}_i$ ($\in V^4$). We rewrite (12.171) as

$$\begin{aligned}
 A(\mathbf{x}) &= (\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4) R R^{-1} A R R^{-1} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \\
 &= \begin{pmatrix} \mathbf{e}_1^{(4)} & \mathbf{e}_1^{(2)} & \mathbf{e}_2^{(2)} & \mathbf{g}_2^{(2)} \end{pmatrix} \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \\ x'_4 \end{pmatrix}, \tag{12.172}
 \end{aligned}$$

where we have

$$\begin{aligned}
 (\mathbf{e}_1^{(4)} \mathbf{e}_1^{(2)} \mathbf{e}_2^{(2)} \mathbf{g}_2^{(2)}) &= (\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4) R, \\
 \begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \\ x'_4 \end{pmatrix} &= R^{-1} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}. \tag{12.173}
 \end{aligned}$$

After (11.84), we have

$$\begin{aligned}
 \mathbf{x} &= (\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = (\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4) R R^{-1} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \\
 &= \begin{pmatrix} \mathbf{e}_1^{(4)} & \mathbf{e}_1^{(2)} & \mathbf{e}_2^{(2)} & \mathbf{g}_2^{(2)} \end{pmatrix} \begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \\ x'_4 \end{pmatrix}. \tag{12.174}
 \end{aligned}$$

As for V^n in general, let us put $R = (p)_{ij}$ and $R^{-1} = (q)_{ij}$. Also represent j -th (generalized) eigenvectors by a column vector and denote them by $p^{(j)}$. There we display individual (generalized) eigenvectors in the order of $(\mathbf{e}^{(1)} \mathbf{e}^{(2)} \dots \mathbf{e}^{(j)} \dots \mathbf{e}^{(n)})$, where $\mathbf{e}^{(j)}$ ($1 \leq j \leq n$) denotes either a proper eigenvector or a generalized eigenvector according to (12.174). Each $p^{(j)}$ is represented in reference to original basis vectors $(\mathbf{e}_1 \dots \mathbf{e}_n)$. Then we have

$$R^{-1}p^{(j)} = \sum_{k=1}^n q_{ik} p_k^{(j)} = \delta_i^{(j)}, \quad (12.175)$$

where $\delta_i^{(j)}$ denotes a column vector to which only the j -th row is 1, otherwise 0. Thus, a column vector $\delta_i^{(j)}$ is an “address” of $e^{(j)}$ in reference to $(e^{(1)} e^{(2)} \dots e^{(j)} \dots e^{(n)})$ taken as basis vectors.

In our present case, in fact, we have

$$\begin{aligned} R^{-1}p^{(1)} &= \begin{pmatrix} -1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ -1 & -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \\ R^{-1}p^{(2)} &= \begin{pmatrix} -1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ -1 & -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \\ R^{-1}p^{(3)} &= \begin{pmatrix} -1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ -1 & -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \\ R^{-1}p^{(4)} &= \begin{pmatrix} -1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ -1 & -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \end{aligned} \quad (12.176)$$

In (12.176) $p^{(1)}$ is a column vector representation of $e_1^{(4)}$; $p^{(2)}$ is a column vector representation of $e_1^{(2)}$, and so on.

A minimal polynomial $\Phi_A(x)$ is expressed as

$$\Phi_A(x) = (x-4)(x-2)^2.$$

Readers can easily make sure of it.

We remark that a non-singular matrix R pertinent to the similarity transformation is not uniquely determined, but we have arbitrariness. In (12.169), for example, if we adopt $-\mathbf{g}_2^{(2)}$ instead of $\mathbf{g}_2^{(2)}$, we should adopt $-\mathbf{e}_2^{(2)}$ instead of $\mathbf{e}_2^{(2)}$ accordingly. Thus, instead of R in (12.169) we may choose

$$R' = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 \end{pmatrix}. \quad (12.177)$$

In this case, we also get the same Jordan canonical form as before. That is,

$$R'^{-1}AR' = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

Suppose that we choose R'' such that

$$\begin{aligned} (e_1 \ e_2 \ e_3 \ e_4)R'' &= (e_1 \ e_2 \ e_3 \ e_4) \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix} \\ &= (e_1^{(2)} \ e_2^{(2)} \ g_2^{(2)} \ e_1^{(4)}). \end{aligned}$$

In this case, we have

$$R''^{-1}AR'' = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix}.$$

Note that we get a different disposition of the matrix elements from that of (12.172).

Next, we decompose A into a semi-simple matrix and a nilpotent matrix. In (12.172), we had

$$R^{-1}AR = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

Defining

$$S = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix} \quad \text{and} \quad N = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

we have

$$R^{-1}AR = S + N \quad \text{i.e.,} \quad A = R(S + N)R^{-1} = RSR^{-1} + RNR^{-1}.$$

Performing the above matrix calculations and putting $\tilde{S} = RSR^{-1}$ and $\tilde{N} = RNR^{-1}$, we get

$$A = \tilde{S} + \tilde{N}$$

with

$$\tilde{S} = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ -2 & 0 & -2 & 4 \end{pmatrix} \quad \text{and} \quad \tilde{N} = \begin{pmatrix} -1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & -1 & 0 & 0 \end{pmatrix}.$$

That is, we have

$$\begin{aligned} A &= \begin{pmatrix} 1 & -1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ -3 & -1 & -2 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ -2 & 0 & -2 & 4 \end{pmatrix} \\ &\quad + \begin{pmatrix} -1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & -1 & 0 & 0 \end{pmatrix}. \end{aligned} \tag{12.178}$$

Even though matrix forms S and N differ depending on the choice of different matrix forms of similarity transformation R (namely, R' , R'' , etc. represented above), the decomposition (12.178) is unique. That is, $\tilde{S}$ and $\tilde{N}$ are uniquely determined, once a matrix A is given. The matrices $\tilde{S}$ and $\tilde{N}$ are commutable. The confirmation is left for readers as an exercise.

We present another simple example. Let A be a matrix described as

$$A = \begin{pmatrix} 0 & 4 \\ -1 & 4 \end{pmatrix}.$$

Eigenvalues of A are 2 as a double root. According to routine, we have an eigenvector $\mathbf{e}_1^{(2)}$ as a column vector, e.g.,

$$\mathbf{e}_1^{(2)} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Another eigenvector is a generalized eigenvector $\mathbf{g}_1^{(2)}$ of rank 2. This can be decided such that

$$(A - 2E)\mathbf{g}_1^{(2)} = \begin{pmatrix} -2 & 4 \\ -1 & 2 \end{pmatrix} \mathbf{g}_1^{(2)} = \mathbf{e}_1^{(2)}.$$

As an option, we get

$$\mathbf{g}_1^{(2)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Thus, we can choose R for a diagonalizing matrix together with an inverse matrix R^{-1} such that

$$R = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \text{and} \quad R^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Therefore, with a Jordan canonical form we have

$$R^{-1}AR = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}. \quad (12.179)$$

As before, putting

$$S = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \quad \text{and} \quad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

we have

$$R^{-1}AR = S + N, \quad \text{i.e.,} \quad A = R(S + N)R^{-1} = RSR^{-1} + RNR^{-1}.$$

Putting $\tilde{S} = RSR^{-1}$ and $\tilde{N} = RNR^{-1}$, we get

$$A = \tilde{S} + \tilde{N}$$

with

$$\tilde{S} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \quad \text{and} \quad \tilde{N} = \begin{pmatrix} -2 & 4 \\ -1 & 2 \end{pmatrix}. \quad (12.180)$$

We may also choose R' for a diagonalizing matrix together with an inverse matrix R'^{-1} instead of R and R^{-1} , respectively, such that we have, e.g.,

$$R' = \begin{pmatrix} 2 & -3 \\ 1 & -1 \end{pmatrix} \quad \text{and} \quad R'^{-1} = \begin{pmatrix} -1 & 3 \\ -1 & 2 \end{pmatrix}.$$

Using these matrices, we get exactly the same Jordan canonical form and the matrix decomposition as (12.179) and (12.180). Thus, again we find that the matrix decomposition is unique.

Another simple example is a lower triangle matrix

$$A = \begin{pmatrix} 2 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Following now familiar procedures, as a diagonalizing matrix we have, e.g.,

$$R = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad R^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Then, we get

$$R^{-1}AR = S = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Therefore, the “decomposition” is

$$A = RSR^{-1} = \begin{pmatrix} 2 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

where the first term is a semi-simple matrix and the second is a nilpotent matrix (i.e., zero matrix). Thus, the decomposition is once again unique.

12.7 Diagonalizable Matrices

Among canonical forms of matrices, the simplest form is a diagonalizable matrix. Here we define a diagonalizable matrix as a matrix that can be converted to that whose off-diagonal elements are zero. In Sect. 12.5 we have investigated different properties of the matrices. In this section we examine basic properties of diagonalizable matrices.

In Sect 12.6.1 we have shown that $\text{Span}\{N^{i-1}\mathbf{a}_\rho, N^{i-2}\mathbf{a}_\rho, \dots, N\mathbf{a}_\rho, \mathbf{a}_\rho\}$ forms a N -invariant subspace of a dimension i , where $\mathbf{a}_\rho$ satisfies the relations $N^i\mathbf{a}_\rho = \mathbf{0}$ and $N^{i-1}\mathbf{a}_\rho \neq \mathbf{0}$ ($r_{i+1} + 1 \leq j \leq r_i$) as in (12.122). Of these vectors, only $N^{i-1}\mathbf{a}_\rho$ is a sole

proper eigenvector that is accompanied by $(i - 1)$ generalized eigenvectors. Note that only the proper eigenvector can construct one-dimensional N -invariant subspace by itself. This is because regarding other generalized eigenvectors $\mathbf{g}$ (here $\mathbf{g}$ stands for all of generalized eigenvectors), $N\mathbf{g} (\neq 0)$ and $\mathbf{g}$ are linearly independent. Note that with a proper eigenvector $\mathbf{e}$, we have $N\mathbf{e} = \mathbf{0}$. A corresponding Jordan block is represented by a matrix as given in (12.122) in reference to the basis vectors comprising these i eigenvectors.

Therefore, if a (n, n) matrix A has only proper eigenvectors, all Jordan blocks are one-dimensional. This means that A is diagonalizable. That A has only proper eigenvectors is equivalent to that those eigenvectors form a one-dimensional subspace and that V^n is a direct sum of the subspaces spanned by individual proper eigenvectors. In other words, if V^n is a direct sum of subspaces (i.e., eigenspaces) spanned by individual proper eigenvectors of A , A is diagonalizable.

Next, suppose that A is diagonalizable. Then, after an appropriate similarity transformation with a non-singular matrix P , A has a following form:

$$P^{-1}AP = \begin{pmatrix} \alpha_1 & & & & \\ & \ddots & & & \\ & & \alpha_1 & & \\ & & & \alpha_2 & \\ & & & & \ddots \\ & & & & & \alpha_2 \\ & & & & & & \ddots \\ & & & & & & & \alpha_s \\ & & & & & & & & \ddots \\ & & & & & & & & & \alpha_s \end{pmatrix}. \quad (12.181)$$

In this case, let us examine what form a minimal polynomial $\varphi_A(x)$ for A takes. A characteristic polynomial $f_A(x)$ for A is invariant through similarity transformation, so is $\varphi_A(x)$. That is,

$$\varphi_{P^{-1}AP}(x) = \varphi_A(x). \quad (12.182)$$

From (12.181), we find that $A - \alpha_i E$ ($1 \leq i \leq s$) has a “perforated” form such as (12.143) with the diagonalized form unchanged. Then we have

$$(A - \alpha_1 E)(A - \alpha_2 E) \cdots (A - \alpha_s E) = 0. \quad (12.183)$$

This is because a product of matrices having only diagonal elements is merely a product of individual diagonal elements. Meanwhile, in virtue of Hamilton–Cayley theorem,

$$f_A(A) = \prod_{i=1}^s (A - \alpha_i E)^{n_i} = 0.$$

Rewriting this expression, we have

$$(A - \alpha_1 E)^{n_1} (A - \alpha_2 E)^{n_2} \cdots (A - \alpha_s E)^{n_s} = 0. \quad (12.184)$$

In light of (12.183), this implies that a minima polynomial $\varphi_A(x)$ is expressed as

$$\varphi_A(x) = (x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_s). \quad (12.185)$$

Surely $\varphi_A(x)$ of (12.185) has a lowest-order polynomial among those satisfying $f(A) = 0$ and is a divisor of $f_A(x)$. Also $\varphi_A(x)$ has a highest-order coefficient of 1. Thus, $\varphi_A(x)$ should be a minimal polynomial of A and we conclude that $\varphi_A(x)$ does not have a multiple root.

Then let us think how V^n is characterized in case $\varphi_A(x)$ does not have a multiple root. This is equivalent to that $\varphi_A(x)$ is described by (12.185). To see this, suppose that we have two matrices A and B and let $BV^n = W$. We wish to use the following relation [3, 4]:

$$\begin{aligned} \text{rank}(AB) &= \dim ABV^n = \dim AW = \dim W - \dim[A^{-1}(0) \cap W] \\ &\geq \dim W - \dim[A^{-1}(0)] = \dim BV^n - (n - \dim AV^n) \\ &= \text{rank } A + \text{rank } B - n. \end{aligned} \quad (12.186)$$

In (12.186), the third equality comes from the fact that the domain of A is restricted to W . Concomitantly, $A^{-1}(\mathbf{0})$ is restricted to $A^{-1}(\mathbf{0}) \cap W$ as well; notice that $A^{-1}(\mathbf{0}) \cap W$ is a subspace. Considering these situations, we use a relation corresponding to that of (11.45). The fourth equality is due to the dimension theorem of (11.45). Applying (12.186) to (12.183) successively, we have

$$\begin{aligned} 0 &= \text{rank} [(A - \alpha_1 E)(A - \alpha_2 E) \cdots (A - \alpha_s E)] \\ &\geq \text{rank} (A - \alpha_1 E) + \text{rank} [(A - \alpha_2 E) \cdots (A - \alpha_s E)] - n \\ &\geq \text{rank} (A - \alpha_1 E) + \text{rank} (A - \alpha_2 E) + \text{rank} [(A - \alpha_3 E) \cdots (A - \alpha_s E)] - 2n \\ &\geq \cdots \\ &\geq \text{rank} (A - \alpha_1 E) + \cdots + \text{rank} (A - \alpha_s E) - (s - 1)n \\ &= \sum_{i=1}^s \text{rank} [(A - \alpha_i E) - n] + n. \end{aligned}$$

Finally we get

$$\sum_{i=1}^s \text{rank} [n - (A - \alpha_i E)] \geq n. \quad (12.187)$$

As $\text{rank} [n - (A - \alpha_i E)] = \dim W_{\alpha_i}$, we have

$$\sum_{i=1}^s \dim W_{\alpha_i} \geq n. \quad (12.188)$$

Meanwhile, we have

$$\begin{aligned}
 V^n &\supset W_{\alpha_1} \oplus W_{\alpha_2} \oplus \cdots \oplus W_{\alpha_s}, \\
 n &\geq \dim \left(W_{\alpha_1} \oplus W_{\alpha_2} \oplus \cdots \oplus W_{\alpha_s} \right) = \sum_{i=1}^s \dim W_{\alpha_i}.
 \end{aligned}
 \tag{12.189}$$

The equality results from the property of a direct sum. From (12.188) and (12.189),

$$\sum_{i=1}^s \dim W_{\alpha_i} = n. \tag{12.190}$$

Hence,

$$V^n = W_{\alpha_1} \oplus W_{\alpha_2} \oplus \cdots \oplus W_{\alpha_s}. \tag{12.191}$$

Thus, we have proven that if the minimal polynomial does not have a multiple root, V^n is decomposed into direct sum of eigenspaces as in (12.191).

If in turn V^n is decomposed into direct sum of eigenspaces as in (12.191), A can be diagonalized by a similarity transformation. The proof is as follows: Suppose that (12.191) holds. Then, we can take only eigenvectors for the basis vectors of V^n . Suppose that $\dim W_{\alpha_i} = n_i$. Then, we can take vectors $\mathbf{a}_k \left[\left(\sum_{j=1}^{i-1} n_j \right) + 1 \leq k \leq \sum_{j=1}^i n_j \right]$ so that $\mathbf{a}_k$ can be the basis vectors of W_{α_i} . In reference to this basis set, we describe a vector $\mathbf{x} \in V^n$ such that

$$\mathbf{x} = (\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

Operating A on $\mathbf{x}$, we get

$$\begin{aligned}
 A(\mathbf{x}) &= (\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n) A \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = (\mathbf{a}_1 A \ \mathbf{a}_2 A \ \cdots \ \mathbf{a}_n A) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \\
 &= (\alpha_1 \mathbf{a}_1 \ \alpha_2 \mathbf{a}_2 \ \cdots \ \alpha_n \mathbf{a}_n) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \\
 &= (\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n) \begin{pmatrix} \alpha_1 & & & \\ & \alpha_2 & & \\ & & \ddots & \\ & & & \alpha_n \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix},
 \end{aligned}
 \tag{12.192}$$

where with the second equality we used the notation (11.40); with the third equality some of α_i ($1 \leq i \leq n$) may be identical; $\mathbf{a}_i$ is an eigenvector that corresponds to an eigenvalue α_i . Suppose that $\mathbf{a}_1, \mathbf{a}_2, \dots$, and $\mathbf{a}_n$ are obtained by transforming an “original” basis set $\mathbf{e}_1, \mathbf{e}_2, \dots$, and $\mathbf{e}_n$ by R . Then, we have

$$A(\mathbf{x}) = (\mathbf{e}_1 \quad \mathbf{e}_2 \quad \cdots \quad \mathbf{e}_n) R \begin{pmatrix} \alpha_1 & & & \\ & \alpha_2 & & \\ & & \ddots & \\ & & & \alpha_n \end{pmatrix} R^{-1} \begin{pmatrix} x_1^{(0)} \\ x_2^{(0)} \\ \vdots \\ x_n^{(0)} \end{pmatrix}.$$

We denote the transformation A with respect to a basis set $\mathbf{e}_1, \mathbf{e}_2, \dots$, and $\mathbf{e}_n$ by A_0 ; see (11.82) with the notation. Then, we have

$$A(\mathbf{x}) = (\mathbf{e}_1 \quad \mathbf{e}_2 \quad \cdots \quad \mathbf{e}_n) A_0 \begin{pmatrix} x_1^{(0)} \\ x_2^{(0)} \\ \vdots \\ x_n^{(0)} \end{pmatrix}.$$

Therefore, we get

$$R^{-1} A_0 R = \begin{pmatrix} \alpha_1 & & & \\ & \alpha_2 & & \\ & & \ddots & \\ & & & \alpha_n \end{pmatrix}. \quad (12.193)$$

Thus, A is similar to a diagonal matrix as represented in (12.192) and (12.193).

It is obvious to show a minimal polynomial of a diagonalizable matrix has no multiple roots. The proof is left for readers.

Summarizing the above arguments, we have a following theorem:

Theorem 12.7 [3, 4] The following three statements related to A are equivalent:

1. The matrix A is similar to a diagonal matrix (i.e., semi-simple).
2. The minimal polynomial $\varphi_A(x)$ does not have a multiple root.
3. The vector space V^n is decomposed into a direct sum of eigenspaces.

In Example 12.1 we showed the diagonalization of a matrix. There A has two different eigenvalues. Since with a (n, n) matrix having n different eigenvalues its characteristic polynomial does not have a multiple root, the minimal polynomial necessarily has no multiple roots. The above theorem therefore ensures that a matrix having no multiple roots must be diagonalizable.

Another consequence of this theorem is that an idempotent matrix is diagonalizable. The matrix is characterized by $A^2 = A$. Then $A(A - E) = 0$. Taking its determinant, $(\det A)[\det(A - E)] = 0$. Therefore, we have either $\det A = 0$ or $\det(A - E) = 0$. Hence, eigenvalues of A are zero or 1. Think of $f(x) = x(x - 1)$. As $f(A) = 0$, $f(x)$ should be a minimal polynomial. It has no multiple roots, and so the matrix is diagonalizable.

Example 12.6 Let us revisit Example 12.1, where we dealt with

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}. \quad (12.32)$$

From (12.33), $f_A(x) = (x - 2)(x - 1)$. Note that $f_A(x) = f_{P^{-1}AP}(x)$. Let us treat a problem according to Theorem 12.5. Also we use the notation of (12.85). Given $f_1(x) = x - 1$ and $f_2(x) = x - 2$, let us decide $M_1(x)$ and $M_2(x)$ such that these can satisfy

$$M_1(x)f_1(x) + M_2(x)f_2(x) = 1. \quad (12.194)$$

We find $M_1(x) = 1$ and $M_2(x) = -1$. Thus, using the notation of Theorem 12.5, Sect. 12.4 we have

$$\begin{aligned} A_1 &= M_1(A)f_1(A) = A - E = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \\ A_2 &= M_2(A)f_2(A) = -A + 2E = \begin{pmatrix} 0 & -1 \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

We also have

$$A_1 + A_2 = E, \quad A_i A_j = A_j A_i = A_i \delta_{ij}. \quad (12.195)$$

Thus, we find that A_1 and A_2 are idempotent matrices. As both A_1 and A_2 are expressed by a polynomial A , they are commutative with A .

We find that A is represented by

$$A = \alpha_1 A_1 + \alpha_2 A_2, \quad (12.196)$$

where α_1 and α_2 denote eigenvalues 2 and 1, respectively. Thus choosing proper eigenvectors for basis vectors, we have decomposed a vector space V^n into a direct sum of invariant subspaces comprising the proper eigenvectors. Concomitantly, A is represented as in (12.196). The relevant decomposition is always possible for a diagonalizable matrix.

Thus, idempotent matrices play an important role in the linear vector spaces.

Example 12.7 Let us think of a following matrix.

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}. \quad (12.197)$$

This is a triangle matrix, and so diagonal elements give eigenvalues. We have an eigenvalue 1 of double root and that 0 of simple root. The matrix can be diagonalized using P such that

$$\begin{aligned} \tilde{A} = P^{-1}AP &= \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \end{aligned} \quad (12.198)$$

As can be checked easily, $\tilde{A}^2 = \tilde{A}$. We also have

$$E - \tilde{A} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (12.199)$$

where $(E - \tilde{A})^2 = E - \tilde{A}$ holds as well. Moreover, $\tilde{A}(E - \tilde{A}) = (E - \tilde{A})\tilde{A} = 0$. Thus $\tilde{A}$ and $E - \tilde{A}$ behave like A_1 and A_2 of (12.195).

Next, suppose that $\mathbf{x} \in V^n$ is expressed as a linear combination of basis vectors $\mathbf{a}_1$, $\mathbf{a}_2$, $\dots$, and $\mathbf{a}_n$. Then

$$\mathbf{x} = c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \dots + c_{n-1}\mathbf{a}_{n-1} + c_n\mathbf{a}_n. \quad (12.200)$$

Here let us define a following linear transformation $P^{(k)}$ such that $P^{(k)}$ “extracts” the k -th component of $\mathbf{x}$. That is,

$$P^{(k)}(\mathbf{x}) = P^{(k)}\left(\sum_{j=1}^n c_j\mathbf{a}_j\right) = \sum_{j=1}^n \sum_{i=1}^n p_{ij}^{[k]} c_j\mathbf{a}_i = c_k\mathbf{a}_k, \quad (12.201)$$

where $p_{ij}^{[k]}$ is the matrix representation of $P^{(k)}$. In fact, suppose that there is another arbitrarily chosen vector $\mathbf{x}$ such that

$$\mathbf{y} = d_1\mathbf{a}_1 + d_2\mathbf{a}_2 + \dots + d_{n-1}\mathbf{a}_{n-1} + d_n\mathbf{a}_n. \quad (12.202)$$

Then we have

$$P^{(k)}(ax + by) = (ac_k + bd_k)a_k = ac_k a_k + bd_k a_k = aP^{(k)}(x) + bP^{(k)}(y). \quad (12.203)$$

Thus $P^{(k)}$ is a linear transformation. In (12.201), for the third equality to hold, we should have

$$\left(p_{ij}^{[k]}\right) = \delta_i^{(k)} \delta_{(k)}^j, \quad (12.204)$$

where $\delta_i^{(j)}$ has been defined in (12.175). Meanwhile, $\delta_{(k)}^j$ denotes a row vector to which only the k -th column is 1, otherwise 0. Note that $\delta_i^{(k)}$ represents a $(n, 1)$ matrix and that $\delta_{(k)}^j$ denotes a $(1, n)$ matrix. Therefore, $\delta_i^{(k)} \delta_{(k)}^j$ represents a (n, n) matrix whose (k, k) element is 1, otherwise 0. Thus, $P^{(k)}(x)$ is denoted by

$$P^{(k)}(x) = (a_1 \cdots a_n) \begin{pmatrix} 0 & & & & \\ & \ddots & & & \\ & & 0 & & \\ & & & 1 & \\ & & & & 0 \\ & & & & & \ddots \\ & & & & & & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = x_k a_k, \quad (12.205)$$

where only the (k, k) element is 1, otherwise 0. Then $P^{(k)}[P^{(k)}(x)] = P^{(k)}(x)$. That is

$$\left[P^{(k)}\right]^2 = P^{(k)}. \quad (12.206)$$

Also $P^{(k)}[P^{(l)}(x)] = 0$ if $k \neq l$. Meanwhile, we have $P^{(1)}(x) + \cdots + P^{(n)}(x) = x$. Hence, $P^{(1)}(x) + \cdots + P^{(n)}(x) = [P^{(1)} + \cdots + P^{(n)}](x) = x$. Since this relation holds with any $x \in V^n$, we get

$$P^{(1)} + \cdots + P^{(n)} = E. \quad (12.207)$$

As shown above, an idempotent matrix such as $P^{(k)}$ always exists. In particular, if the basis vectors comprise only proper eigenvectors, the decomposition as expressed in (12.196) is possible. In that case, it is described as

$$A = \alpha_1 A_1 + \cdots + \alpha_n A_n, \quad (12.208)$$

where $\alpha_1, \cdots$, and α_n are eigenvalues (some of which may be identical) and $A_1, \cdots$, and A_n are idempotent matrices such as those represented by (12.205).

In fact, the above situation is equivalent to that A is semi-simple. This implies that A can be decomposed into a following form described by

$$R^{-1}AR = \alpha_1 P^{(1)} + \cdots + \alpha_n P^{(n)}, \quad (12.209)$$

where R is a non-singular operator. Rewriting (12.209), we have

$$A = \alpha_1 R P^{(1)} R^{-1} + \cdots + \alpha_n R P^{(n)} R^{-1}. \quad (12.210)$$

Putting $\tilde{A}_k \equiv R P^{(k)} R^{-1}$ ($1 \leq k \leq n$), we obtain

$$A = \alpha_1 \tilde{A}_1 + \cdots + \alpha_n \tilde{A}_n. \quad (12.211)$$

We can readily show that

$$\tilde{A}_k \tilde{A}_l = \delta_{kl}. \quad (12.212)$$

That is, $\tilde{A}_k$ ($1 \leq k \leq n$) is an idempotent matrix.

Yet, we have to be careful to construct idempotent matrices according to a formalism described in Theorem 12.5. It is because we often encounter a situation where different matrices give an identical characteristic polynomial. We briefly mention this in the next example.

Example 12.8 Let us think about following two matrices:

$$A = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}. \quad (12.213)$$

Then, following Theorem 12.5, Sect. 12.4, we have

$$f_A(x) = f_B(x) = (x-3)(x-2)^2, \quad (12.214)$$

with eigenvalues $\alpha_1 = 3$ and $\alpha_2 = 2$. Also we have $f_1(x) = (x-2)^2$ and $f_2(x) = x-3$. Following the procedures of (12.85) and (12.86), we obtain

$$M_1(x) = x-2 \quad \text{and} \quad M_2(x) = -x^2 + 3x - 3. \quad (12.215)$$

Therefore, we have

$$M_1(x)f_1(x) = (x-2)^3, \quad M_2(x)f_2(x) = (x-3)(-x^2 + 3x - 3). \quad (12.216)$$

Hence, we get

$$\begin{aligned} A_1 &\equiv M_1(A)f_1(A) = (A-2E)^3, \\ A_2 &\equiv M_2(A)f_2(A) = (A-3E)(-A^2 + 3A - 3E). \end{aligned} \quad (12.217)$$

Similarly, we get B_1 and B_2 by replacing A with B in (12.217). Thus, we have

$$A_1 = B_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A_2 = B_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (12.218)$$

Notice that we get the same idempotent matrix of (12.218), even though the matrix forms of A and B differ. Also we have

$$A_1 + A_2 = B_1 + B_2 = E. \quad (12.219)$$

Then, we have

$$A = (A_1 + A_2)A = A_1A + A_2A, \quad B = (B_1 + B_2)B = B_1B + B_2B. \quad (12.220)$$

Nonetheless, although $A = 3A_1 + 2A_2$ holds, $B \neq 3B_1 + 2B_2$. That is, the decomposition of the form of (12.208) is not true of B . The decomposition of this kind is possible only with diagonalizable matrices.

In summary, a (n, n) matrix with s ($1 \leq s \leq n$) different eigenvalues has at least s proper eigenvectors. (Note that a diagonalizable matrix has n proper eigenvectors.) In the case of $s < n$, the matrix has multiple root(s) and may have generalized eigenvectors. If the matrix has a generalized eigenvector of rank ν , the matrix is accompanied by $(\nu - 1)$ generalized eigenvectors along with a sole proper eigenvector. Those vectors form an invariant subspace along with the proper eigenvector (s). In total, such n (generalized) eigenvectors span a whole vector space V^n .

With the eigenvalue equation $A(\mathbf{x}) = \alpha\mathbf{x}$, we have an indefinite but non-trivial solution $\mathbf{x} \neq \mathbf{0}$ for only restricted numbers α (i.e., eigenvalues) in a complex plane. However, we have a unique but trivial solution $\mathbf{x} = \mathbf{0}$ for complex numbers α other than eigenvalues. This is characteristic of the eigenvalue problem.

References

1. Mirsky L (1990) An Introduction to linear algebra. Dover, New York
2. Hassani S (2006) Mathematical physics. Springer, New York
3. Satake I-O (1975) Linear algebra (Pure and applied mathematics). Marcel Dekker, New York
4. Satake I (1974) Linear algebra. Shokabo, Tokyo. (in Japanese)

Chapter 13

Inner Product Space



Thus far we have treated the theory of linear vector spaces. The vector spaces, however, were somewhat “structureless,” and so it will be desirable to introduce a concept of metric or measure into the linear vector spaces. We call a linear vector space where the inner product is defined an inner product space. In virtue of a concept of the inner product, the linear vector space is given a variety of structures. For instance, introduction of the inner product to the linear vector space immediately leads to the definition of adjoint operators and Gram matrices.

Above all, the concept of inner product can readily be extended to a functional space and facilitate understanding of, e.g., orthogonalization of functions, as was exemplified in Parts I and II. Moreover, definition of the inner product allows us to relate matrix operators and differential operators. In particular, it is a key issue to understand logical structure of quantum mechanics. This can easily be understood from the fact that Paul Dirac, who was known as one of the prominent founders of quantum mechanics, invented bra and ket vectors to represent an inner product.

13.1 Inner Product and Metric

Inner product relates two vectors to a complex number. To do this, we introduce the notation $|a\rangle$ and $\langle b|$ to represent the vectors. This notation is due to Dirac and widely used in physics and mathematical physics. Usually $|a\rangle$ and $\langle b|$ are called a “ket” vector and a “bra” vector, respectively, again due to Dirac. Alternatively, we may call $\langle a|$ an adjoint vector of $|a\rangle$. Or we denote $\langle a| \equiv |a\rangle^\dagger$. The symbol “ $\dagger$ ” (dagger) means that for a matrix its transposed matrix should be taken with complex conjugate matrix elements. That is, $(a_{ij})^\dagger = (a_{ji}^*)$. If we represent a full matrix, we have

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}, A^\dagger = \begin{pmatrix} a_{11}^* & \cdots & a_{n1}^* \\ \vdots & \ddots & \vdots \\ a_{1n}^* & \cdots & a_{nn}^* \end{pmatrix}. \quad (13.1)$$

We call $A^\dagger$ an adjoint matrix or adjoint to A ; see (1.106). The operation of transposition and complex conjugate is commutable. A further remark will be made below after showing the definition of the inner product. The symbols $|a\rangle$ and $\langle b|$ represent vectors and, hence, we do not need to use bold characters to show that those are vectors.

The definition of the inner product is as follows:

$$\langle b|a\rangle = \langle a|b\rangle^*, \quad (13.2)$$

$$\langle a | (\beta|b\rangle + \gamma|c\rangle) = \beta\langle a|b\rangle + \gamma\langle a|c\rangle, \quad (13.3)$$

$$\langle a|a\rangle \geq 0. \quad (13.4)$$

In (13.4), equality holds if and only if $|a\rangle = 0$. Note here that two vectors are said to be orthogonal to each other if their inner product vanishes, i.e., $\langle b|a\rangle = \langle a|b\rangle = 0$. In particular, if a vector $|a\rangle \in V^n$ is orthogonal to all the vectors in V^n , i.e., $\langle x|a\rangle = 0$ for $\forall |x\rangle \in V^n$, then $|a\rangle = 0$. This is because if we choose $|a\rangle$ for $|x\rangle$, we have $\langle a|a\rangle = 0$. This means that $|a\rangle = 0$. We call a linear vector space to which the inner product is defined an inner product space.

We can create another structure to a vector space. An example is a metric (or distance function). Suppose that there is an arbitrary set Q . If a real non-negative number $\rho(a, b)$ is defined as follows with any arbitrary elements $a, b, c \in Q$, the set Q is called a metric space [1]:

$$\rho(a, b) = \rho(b, a), \quad (13.5)$$

$$\rho(a, b) \geq 0 \text{ for } \forall a, b; \rho(a, b) = 0 \text{ if and only if } a = b, \quad (13.6)$$

$$\rho(a, b) + \rho(b, c) \geq \rho(a, c). \quad (13.7)$$

In our study a vector space is chosen for the set Q . Here let us define a norm for each vector a . The norm is defined as

$$\|a\| = \sqrt{\langle a|a\rangle}. \quad (13.8)$$

If we define $\rho(a, b) \equiv \|a - b\|$, $\|a - b\|$ satisfies the definitions of metric. Equations (13.5) and (13.6) are obvious. For (13.7) let us consider a vector $|c\rangle$ as $|c\rangle = |a\rangle - x\langle b|a\rangle|b\rangle$ with real x . Since $\langle c|c\rangle \geq 0$, we have

$$x^2 \langle a|b \rangle \langle b|a \rangle \langle b|b \rangle - 2x \langle a|b \rangle \langle b|a \rangle + \langle a|a \rangle \geq 0$$

or

$$x^2 |\langle a|b \rangle|^2 \langle b|b \rangle - 2x |\langle a|b \rangle|^2 + \langle a|a \rangle \geq 0. \quad (13.9)$$

The inequality (13.9) related to the quadratic equation in x with real coefficients requires the inequality such that

$$\langle a|a \rangle \langle b|b \rangle \geq \langle a|b \rangle \langle b|a \rangle = |\langle a|b \rangle|^2. \quad (13.10)$$

That is,

$$\sqrt{\langle a|a \rangle} \cdot \sqrt{\langle b|b \rangle} \geq |\langle a|b \rangle|. \quad (13.11)$$

Namely,

$$||a|| \cdot ||b|| \geq |\langle a|b \rangle| \geq \Re \langle a|b \rangle. \quad (13.12)$$

The relations (13.11) and (13.12) are known as Cauchy–Schwarz inequality. Meanwhile, we have

$$||a + b||^2 = \langle a + b|a + b \rangle = ||a||^2 + ||b||^2 + 2\Re \langle a|b \rangle, \quad (13.13)$$

$$(||a|| + ||b||)^2 = ||a||^2 + ||b||^2 + 2||a|| \cdot ||b||. \quad (13.14)$$

Comparing (13.13) and (13.14) and using (13.12), we have

$$||a|| + ||b|| \geq ||a + b||. \quad (13.15)$$

The inequality (13.15) is known as the triangle inequality. In (13.15) replacing $a \rightarrow a - b$ and $b \rightarrow b - c$, we get

$$||a - b|| + ||b - c|| \geq ||a - c||. \quad (13.16)$$

Thus, (13.16) is equivalent to (13.7). At the same time, the norm defined in (13.8) may be regarded as a “length” of a vector a .

As $\beta|b\rangle + \gamma|c\rangle$ represents a vector, we use a shorthand notation for it as

$$|\beta b + \gamma c\rangle \equiv \beta|b\rangle + \gamma|c\rangle. \quad (13.17)$$

According to the definition (13.3)

$$\langle a | \beta b + \gamma c \rangle = \beta \langle a | b \rangle + \gamma \langle a | c \rangle. \quad (13.18)$$

Also from (13.2),

$$\langle \beta b + \gamma c | a \rangle = \langle a | \beta b + \gamma c \rangle^* = [\beta \langle a | b \rangle + \gamma \langle a | c \rangle]^* = \beta^* \langle b | a \rangle + \gamma^* \langle c | a \rangle. \quad (13.19)$$

Comparing the leftmost side and rightmost side of (13.19) and considering $|a\rangle$ can arbitrarily be chosen, we get

$$\langle \beta b + \gamma c | = \beta^* \langle b | + \gamma^* \langle c |. \quad (13.20)$$

Therefore, when we take out a scalar from a bra vector, it should be converted to a complex conjugate. When the scalar is taken out from a ket vector, however, it is unaffected by definition (13.3). To show this, we have $|\alpha a\rangle = \alpha |a\rangle$. Taking its adjoint, $\langle \alpha a | = \alpha^* \langle a |$.

We can view (13.17) as a linear transformation in a vector space. In other words, if we regard $|\cdot\rangle$ as a linear transformation of a vector $a \in V^n$ to $|a\rangle \in \tilde{V}^n$, $|\cdot\rangle$ is that of V^n to $\tilde{V}^n$. Conversely, $\langle \cdot |$ could not be regarded as a linear transformation of $a \in V^n$ to $\langle a | \in \tilde{V}^{n'}$. Sometimes the said transformation is referred to as “antilinear” or “sesquilinear.” From the point of view of formalism, the inner product can be viewed as an operation: $\tilde{V}^{n'} \times \tilde{V}^n \rightarrow \mathbb{C}$. The vector space $\tilde{V}^{n'}$ is said to be a dual vector space (or dual space) of $\tilde{V}^n$. The notion of the dual space possesses important position in the theory of vector space. We will come back to the definition and properties of the dual space in Part V (Chap. 24).

Let us consider $|x\rangle = |y\rangle$ in an inner product space $\tilde{V}^n$. Then $|x\rangle - |y\rangle = 0$. That is, $|x - y\rangle = 0$. Therefore, we have $x = y$, or $|0\rangle = 0$. This means that the linear transformation $|\cdot\rangle$ converts $\mathbf{0} \in V^n$ to $|0\rangle \in \tilde{V}^n$. This is a characteristic of a linear transformation represented in (11.44). Similarly we have $\langle 0 | = 0$. However, we do not have to get into further details about the notation in this book. Also if we have to specify a vector space, we simply do so by designating it as V^n .

13.2 Gram Matrices

Once we have defined an inner product between any pair of vectors $|a\rangle$ and $|b\rangle$ of a vector space V^n , we can define and calculate various quantities related to inner products. As an example, let $|a_1\rangle, \dots, |a_n\rangle$ and $|b_1\rangle, \dots, |b_n\rangle$ be two sets of vectors in V^n . The vectors $|a_1\rangle, \dots, |a_n\rangle$ may or may not be linearly independent. This is true of $|b_1\rangle, \dots, |b_n\rangle$. Let us think of a following matrix M defined as below:

$$M = \begin{pmatrix} \langle a_1 | \\ \vdots \\ \langle a_n | \end{pmatrix} (|b_1\rangle \cdots |b_n\rangle) = \begin{pmatrix} \langle a_1 | b_1 \rangle & \cdots & \langle a_1 | b_n \rangle \\ \vdots & \ddots & \vdots \\ \langle a_n | b_1 \rangle & \cdots & \langle a_n | b_n \rangle \end{pmatrix}. \quad (13.21)$$

We assume the following cases.

1. Suppose that in (13.21) $|b_1\rangle, \dots, |b_n\rangle$ are linearly dependent. Then, without loss of generality we can put

$$|b_1\rangle = c_2|b_2\rangle + c_3|b_3\rangle + \cdots + c_n|b_n\rangle. \quad (13.22)$$

Then we have

$$M = \begin{pmatrix} c_2\langle a_1 | b_2 \rangle + c_3\langle a_1 | b_3 \rangle + \cdots + c_n\langle a_1 | b_n \rangle & \langle a_1 | b_2 \rangle & \cdots & \langle a_1 | b_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ c_2\langle a_n | b_2 \rangle + c_3\langle a_n | b_3 \rangle + \cdots + c_n\langle a_n | b_n \rangle & \langle a_n | b_2 \rangle & \cdots & \langle a_n | b_n \rangle \end{pmatrix}. \quad (13.23)$$

Multiplying the second column, $\dots$, and the n -th column by $(-c_2), \dots$, and $(-c_n)$, respectively, and adding them to the first column, we get

$$\tilde{M} = \begin{pmatrix} 0 & \langle a_1 | b_2 \rangle & \cdots & \langle a_1 | b_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \langle a_n | b_2 \rangle & \cdots & \langle a_n | b_n \rangle \end{pmatrix}. \quad (13.24)$$

Note that in (13.24), $\tilde{M}$ was obtained after giving the above operations to M . We have $\det \tilde{M} = 0$. Since the above operations keep the determinant of a matrix unchanged, again we get $\det M = 0$.

2. Suppose in turn that in (13.21) $|a_1\rangle, \dots, |a_n\rangle$ are linearly dependent. In that case, again, without loss of generality we can put

$$|a_1\rangle = d_2|a_2\rangle + d_3|a_3\rangle + \cdots + d_n|a_n\rangle. \quad (13.25)$$

Focusing attention on individual rows and taking a similar procedure described above, we have

$$\tilde{M} = \begin{pmatrix} 0 & \cdots & 0 \\ \langle a_2 | b_1 \rangle & \cdots & \langle a_2 | b_n \rangle \\ \vdots & \ddots & \vdots \\ \langle a_n | b_1 \rangle & \cdots & \langle a_n | b_n \rangle \end{pmatrix}. \quad (13.26)$$

Again, $\det M = 0$.

Next, let us examine the case where $\det M = 0$. In that case, n column vectors of M in (13.21) are linearly dependent. Without loss of generality, we suppose that the first column is expressed as a linear combination of the other $(n - 1)$ columns such that

$$\begin{pmatrix} \langle a_1 | b_1 \rangle \\ \vdots \\ \langle a_n | b_1 \rangle \end{pmatrix} = c_2 \begin{pmatrix} \langle a_1 | b_2 \rangle \\ \vdots \\ \langle a_n | b_2 \rangle \end{pmatrix} + \cdots + c_n \begin{pmatrix} \langle a_1 | b_n \rangle \\ \vdots \\ \langle a_n | b_n \rangle \end{pmatrix}. \quad (13.27)$$

Rewriting this, we have

$$\begin{pmatrix} \langle a_1 | b_1 - c_2 b_2 - \cdots - c_n b_n \rangle \\ \vdots \\ \langle a_n | b_1 - c_2 b_2 - \cdots - c_n b_n \rangle \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}. \quad (13.28)$$

Multiplying the first row, $\cdots$, and the n -th row of (13.28) by an appropriate complex number p_1^* , $\cdots$, and p_n^* , respectively, we have

$$\begin{aligned} \langle p_1 a_1 | b_1 - c_2 b_2 - \cdots - c_n b_n \rangle &= 0, \\ \cdots, \\ \langle p_n a_n | b_1 - c_2 b_2 - \cdots - c_n b_n \rangle &= 0. \end{aligned}$$

Adding all the above, we get

$$\langle p_1 a_1 + \cdots + p_n a_n | b_1 - c_2 b_2 - \cdots - c_n b_n \rangle = 0.$$

Now, suppose that $|a_1\rangle, \cdots, |a_n\rangle$ are the basis vectors. Then, $|p_1 a_1 + \cdots + p_n a_n\rangle$ represents any vectors in a vector space. This implies that $|b_1 - c_2 b_2 - \cdots - c_n b_n\rangle = 0$; for this see remarks after (13.4). That is,

$$|b_1\rangle = c_2 |b_2\rangle + \cdots + c_n |b_n\rangle. \quad (13.29)$$

Thus, $|b_1\rangle, \cdots, |b_n\rangle$ are linearly dependent.

Meanwhile, $\det M = 0$ implies that n row vectors of M in (13.21) are linearly dependent. In that case, performing similar calculations to the above, we can readily show that if $|b_1\rangle, \cdots, |b_n\rangle$ are the basis vectors, $|a_1\rangle, \cdots, |a_n\rangle$ are linearly dependent.

We summarize the above discussion by a following statement: suppose that we have two sets of vectors $|a_1\rangle, \cdots, |a_n\rangle$ and $|b_1\rangle, \cdots, |b_n\rangle$.

$$\text{At least a set of vectors are linearly dependent.} \Leftrightarrow \det M = 0. \quad (13.30)$$

$$\text{Both the sets of vectors are linearly independent.} \Leftrightarrow \det M \neq 0. \quad (13.31)$$

The latter statement is obtained by considering contraposition of the former statement.

We restate the above in a following theorem:

Theorem 13.1 Let $|a_1\rangle, \cdots, |a_n\rangle$ and $|b_1\rangle, \cdots, |b_n\rangle$ be two sets of vectors defined in a vector space V^n . A necessary and sufficient condition for both these sets of vectors to be linearly independent is that with a matrix M defined as

$$M = \begin{pmatrix} \langle a_1 | \\ \vdots \\ \langle a_n | \end{pmatrix} (|b_1\rangle \cdots |b_n\rangle) = \begin{pmatrix} \langle a_1 | b_1 \rangle & \cdots & \langle a_1 | b_n \rangle \\ \vdots & \ddots & \vdots \\ \langle a_n | b_1 \rangle & \cdots & \langle a_n | b_n \rangle \end{pmatrix},$$

we must have $\det M \neq 0$. ■

Next we consider a norm of a vector expressed in reference to a set of basis vectors $|e_1\rangle, \dots, |e_n\rangle$ of V^n . Let us express a vector $|x\rangle$ in an inner product space as follows as in the case of (11.10) and (11.13):

$$\begin{aligned} |x\rangle &= x_1 |e_1\rangle + x_2 |e_2\rangle + \cdots + x_n |e_n\rangle \\ &= |x_1 e_1 + x_2 e_2 + \cdots + x_n e_n\rangle \\ &= (|e_1\rangle \cdots |e_n\rangle) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}. \end{aligned} \quad (13.32)$$

A bra vector $\langle x|$ is then denoted by

$$\langle x| = (x_1^* \cdots x_n^*) \begin{pmatrix} \langle e_1| \\ \vdots \\ \langle e_n| \end{pmatrix}. \quad (13.33)$$

Thus, we have an inner product described as

$$\begin{aligned} \langle x|x\rangle &= (x_1^* \cdots x_n^*) \begin{pmatrix} \langle e_1| \\ \vdots \\ \langle e_n| \end{pmatrix} (|e_1\rangle \cdots |e_n\rangle) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \\ &= (x_1^* \cdots x_n^*) \begin{pmatrix} \langle e_1|e_1\rangle & \cdots & \langle e_1|e_n\rangle \\ \vdots & \ddots & \vdots \\ \langle e_n|e_1\rangle & \cdots & \langle e_n|e_n\rangle \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}. \end{aligned} \quad (13.34)$$

Here the matrix G expressed as follows is called a Gram matrix [2–4]:

$$G = \begin{pmatrix} \langle e_1|e_1\rangle & \cdots & \langle e_1|e_n\rangle \\ \vdots & \ddots & \vdots \\ \langle e_n|e_1\rangle & \cdots & \langle e_n|e_n\rangle \end{pmatrix}. \quad (13.35)$$

As $\langle e_j|e_i\rangle = \langle e_i|e_j\rangle^*$, we have $G = G^\dagger$. From Theorem 13.1, $\det G \neq 0$. With a shorthand notation, we write $(G)_{ij} = (\langle e_i|e_j\rangle)$. As already mentioned in Sect. 1.4, if for a matrix H we have a relation described by

$$H = H^\dagger, \quad (1.119)$$

it is said to be an Hermitian matrix or a self-adjoint matrix. We often say that such a matrix is Hermitian.

Since the Gram matrices frequently appear in matrix algebra and play a role, their properties are worth examining. Since G is an Hermitian matrix, it can be diagonalized through similarity transformation using a unitary matrix. We will give its proof later (see Sect. 14.3).

Let us deal with (13.34) further. We have

$$\langle x|x \rangle = (x_1^* \cdots x_n^*) U U^\dagger \begin{pmatrix} \langle e_1|e_1 \rangle & \cdots & \langle e_1|e_n \rangle \\ \vdots & \ddots & \vdots \\ \langle e_n|e_1 \rangle & \cdots & \langle e_n|e_n \rangle \end{pmatrix} U U^\dagger \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad (13.36)$$

where U is defined as $U U^\dagger = U^\dagger U = E$ (or $U^{-1} = U^\dagger$). Such a matrix U is called a unitary matrix. We represent a matrix form of U as

$$U = \begin{pmatrix} u_{11} & \cdots & u_{1n} \\ \vdots & \ddots & \vdots \\ u_{n1} & \cdots & u_{nn} \end{pmatrix}, \quad U^\dagger = \begin{pmatrix} u_{11}^* & \cdots & u_{n1}^* \\ \vdots & \ddots & \vdots \\ u_{1n}^* & \cdots & u_{nn}^* \end{pmatrix}. \quad (13.37)$$

Here, putting

$$U^\dagger \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \xi_1 \\ \vdots \\ \xi_n \end{pmatrix}$$

or equivalently

$$\sum_{k=1}^n x_k u_{ki}^* \equiv \xi_i \quad (13.38)$$

and taking its adjoint such that

$$(x_1^* \cdots x_n^*) U = (\xi_1^* \cdots \xi_n^*)$$

or equivalently

$$\sum_{k=1}^n x_k^* u_{ki} = \xi_i^*,$$

we have

$$\langle x|x \rangle = (\xi_1^* \cdots \xi_n^*) U^\dagger \begin{pmatrix} \langle e_1|e_1 \rangle & \cdots & \langle e_1|e_n \rangle \\ \vdots & \ddots & \vdots \\ \langle e_n|e_1 \rangle & \cdots & \langle e_n|e_n \rangle \end{pmatrix} U \begin{pmatrix} \xi_1 \\ \vdots \\ \xi_n \end{pmatrix}. \quad (13.39)$$

We assume that the Gram matrix is diagonalized by a similarity transformation by U . After being diagonalized, similarly to (12.192) and (12.193) the Gram matrix has a following form G' :

$$U^{-1}GU = U^\dagger GU = G' = \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix}. \quad (13.40)$$

Thus, we get

$$\langle x|x \rangle = (\xi_1^* \cdots \xi_n^*) \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix} \begin{pmatrix} \xi_1 \\ \vdots \\ \xi_n \end{pmatrix} = \lambda_1 |\xi_1|^2 + \cdots + \lambda_n |\xi_n|^2. \quad (13.41)$$

From the relation (13.4), $\langle x|x \rangle \geq 0$. This implies that in (13.41) we have

$$\lambda_i \geq 0 \quad (1 \leq i \leq n). \quad (13.42)$$

To show this, suppose that for $\exists \lambda_i, \lambda_i < 0$. Suppose also that for $\exists \xi_i, \xi_i \neq 0$. Then, with $\begin{pmatrix} 0 \\ \vdots \\ \xi_i \\ \vdots \\ 0 \end{pmatrix}$ we have $\langle x|x \rangle = \lambda_i |\xi_i|^2 < 0$, in contradiction.

Since we have $\det G \neq 0$, taking a determinant of (13.40) we have

$$\det G' = \det (U^\dagger GU) = \det (U^\dagger UG) = \det E \det G = \det G = \prod_{i=1}^n \lambda_i \neq 0. \quad (13.43)$$

Combining (13.42) and (13.43), all the eigenvalues λ_i are positive; i.e.,

$$\lambda_i > 0 \quad (1 \leq i \leq n). \quad (13.44)$$

The norm $\langle x|x \rangle = 0$, if and only if $\xi_1 = \xi_2 = \cdots = \xi_n = 0$ which corresponds to $x_1 = x_2 = \cdots = x_n = 0$ from (13.38).

For further study, we generalize the aforementioned feature a little further. That is, if $|e_1\rangle, \cdots$, and $|e_n\rangle$ are linearly dependent, from Theorem 13.1 we get $\det G = 0$. This implies that there is at least one eigenvalue $\lambda_i = 0$ ($1 \leq i \leq n$) and that for a vector $\begin{pmatrix} 0 \\ \vdots \\ \xi_i \\ \vdots \\ 0 \end{pmatrix}$ with $\exists \xi_i \neq 0$ we have $\langle x|x \rangle = 0$. (Suppose that in V^2 we have linearly dependent vectors $|e_1\rangle$ and $|e_2\rangle = | - e_1\rangle$; see Example 13.2 below.)

Let H be an Hermitian matrix [i.e., $(H)_{ij} = (H^*)_{ji}$]. If we have ϕ as a function of complex variables $x_1, \cdots, x_n$ such that

$$\phi(x_1, \cdots, x_n) = \sum_{i,j=1}^n x_i^* (H)_{ij} x_j, \quad (13.45)$$

where H_{ij} is a matrix element of an Hermitian matrix; $\phi(x_1, \cdots, x_n)$ is said to be an Hermitian quadratic form. Suppose that $\phi(x_1, \cdots, x_n)$ satisfies $\phi(x_1, \cdots, x_n) = 0$ if and only if $x_1 = x_2 = \cdots = x_n = 0$, and otherwise $\phi(x_1, \cdots, x_n) > 0$ with any other sets of

x_i ($1 \leq i \leq n$). Then, the said Hermitian quadratic form is called positive definite and we write

$$H > 0. \quad (13.46)$$

If $\phi(x_1, \dots, x_n) \geq 0$ for any x_n ($1 \leq i \leq n$) and $\phi(x_1, \dots, x_n) = 0$ for at least a set of $(x_1, \dots, x_n)$ to which $\exists x_i \neq 0$, $\phi(x_1, \dots, x_n)$ is said to be positive semi-definite or non-negative. In that case, we write

$$H \geq 0. \quad (13.47)$$

From the above argument, a Gram matrix comprising linearly independent vectors is positive definite, whereas that comprising linearly dependent vectors is non-negative. On the basis of the above argument including (13.36)–(13.44), we have

$$\begin{aligned} H > 0 &\Leftrightarrow \lambda_i > 0 \quad (1 \leq i \leq n), \det H > 0; \\ H \geq 0 &\Leftrightarrow \lambda_i \geq 0 \quad \text{with } \exists \lambda_i = 0 \quad (1 \leq i \leq n), \det H = 0. \end{aligned} \quad (13.48)$$

Notice here that eigenvalues λ_i remain unchanged after (unitary) similarity transformation. Namely, the eigenvalues are inherent to H .

We have already encountered several examples of positive definite and non-negative operators. A typical example of the former case is Hamiltonian of quantum-mechanical harmonic oscillator (see Chap. 2). In this case, energy eigenvalues are all positive (i.e., positive definite). Orbital angular momenta L^2 of hydrogen-like atoms, however, are non-negative operators and, hence, an eigenvalue of zero is permitted.

Alternatively, the Gram matrix is defined as $B^\dagger B$, where B is any (n, n) matrix. If we take an orthonormal basis $|\eta_1\rangle, |\eta_2\rangle, \dots, |\eta_n\rangle, |e_i\rangle$ can be expressed as

$$\begin{aligned} |e_i\rangle &= \sum_{j=1}^n b_{ji} |\eta_j\rangle, \\ \langle e_k | e_i \rangle &= \sum_{j=1}^n \sum_{l=1}^n b_{lk}^* b_{ji} \langle \eta_l | \eta_j \rangle = \sum_{j=1}^n \sum_{l=1}^n b_{lk}^* b_{ji} \delta_{lj} = \sum_{j=1}^n b_{jk}^* b_{ji} \\ &= \sum_{j=1}^n (B^\dagger)_{kj} (B)_{ji} = (B^\dagger B)_{ki}. \end{aligned} \quad (13.49)$$

For the second equality of (13.49), we used the orthonormal condition $\langle \eta_l | \eta_j \rangle = \delta_{lj}$. Thus, the Gram matrix G defined in (13.35) can be regarded as identical to $B^\dagger B$.

In Sect. 11.4 we have dealt with a linear transformation of a set of basis vectors $e_1, e_2, \dots$, and e_n by a matrix A defined in (11.69) and examined whether the transformed vectors e'_1, e'_2 , and e'_n are linearly independent. As a result, a necessary and sufficient condition for e'_1, e'_2 , and e'_n to be linearly independent (i.e., to be a set of basis vectors) is $\det A \neq 0$. Thus, we notice that B plays a same role as A of (11.69)

and, hence, $\det B \neq 0$ if and only if the set of vectors $|e_1\rangle, |e_2\rangle, \dots |e_n\rangle$ defined in (13.32) are linearly independent. By the same token as the above, we conclude that eigenvalues of $B^\dagger B$ are all positive, only if $\det B \neq 0$ (i.e., B is non-singular).

Alternatively, if B is singular, $\det B^\dagger B = \det B^\dagger \det B = 0$. In that case at least one of the eigenvalues of $B^\dagger B$ must be zero. The Gram matrices appearing in (13.35) are frequently dealt with in the field of mathematical physics in conjunction with quadratic forms. Further topics can be seen in the next chapter.

Example 13.1 Let us take two vectors $|e_1\rangle$ and $|e_2\rangle$ that are expressed as

$$|e_1\rangle = |e_1\rangle + |e_2\rangle, \quad |e_2\rangle = |e_1\rangle + i|e_2\rangle. \quad (13.50)$$

Here we have $\langle e_i | e_j \rangle = \delta_{ij}$ ($1 \leq i, j \leq 2$). Then we have a Gram matrix expressed as

$$G = \begin{pmatrix} \langle e_1 | e_1 \rangle & \langle e_1 | e_2 \rangle \\ \langle e_2 | e_1 \rangle & \langle e_2 | e_2 \rangle \end{pmatrix} = \begin{pmatrix} 2 & 1+i \\ 1-i & 2 \end{pmatrix}. \quad (13.51)$$

Principal minors of G are $|2| = 2 > 0$ and $\begin{vmatrix} 2 & 1+i \\ 1-i & 2 \end{vmatrix} = 4 - (1+i)(1-i) = 2 > 0$. Therefore, according to Theorem 14.12, $G > 0$ (vide infra).

Let us diagonalize the matrix G . To this end, we find roots of the characteristic equation. That is,

$$\det |G - \lambda E| = \begin{vmatrix} 2-\lambda & 1+i \\ 1-i & 2-\lambda \end{vmatrix} = 0, \quad \lambda^2 - 4\lambda + 2 = 0. \quad (13.52)$$

We have $\lambda = 2 \pm \sqrt{2}$. Then as a diagonalizing unitary matrix U we get

$$U = \begin{pmatrix} \frac{1+i}{2} & \frac{1+i}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}. \quad (13.53)$$

Thus, we get

$$\begin{aligned} U^\dagger G U &= \begin{pmatrix} \frac{1-i}{2} & \frac{\sqrt{2}}{2} \\ \frac{1-i}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} 2 & 1+i \\ 1-i & 2 \end{pmatrix} \begin{pmatrix} \frac{1+i}{2} & \frac{1+i}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} \\ &= \begin{pmatrix} 2+\sqrt{2} & 0 \\ 0 & 2-\sqrt{2} \end{pmatrix}. \end{aligned} \quad (13.54)$$

The eigenvalues $2 + \sqrt{2}$ and $2 - \sqrt{2}$ are real positive as expected. That is, the Gram matrix is positive definite. $\square$

Example 13.2 Let $|e_1\rangle$ and $|e_2\rangle (= -|e_1\rangle)$ be two vectors. Then, we have a Gram matrix expressed as

$$G = \begin{pmatrix} \langle e_1|e_1\rangle & \langle e_1|e_2\rangle \\ \langle e_2|e_1\rangle & \langle e_2|e_2\rangle \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \quad (13.55)$$

Similarly in the case of Example 13.1, we have as an eigenvalue equation

$$\det |G - \lambda E| = \begin{vmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{vmatrix} = 0, \quad \lambda^2 - 2\lambda = 0. \quad (13.56)$$

We have $\lambda = 2$ or 0 . As a diagonalizing unitary matrix U we get

$$U = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}. \quad (13.57)$$

Thus, we get

$$U^\dagger G U = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}. \quad (13.58)$$

As expected, we have a diagonal matrix, one of the eigenvalues for which is zero. That is, the Gram matrix is non-negative.

In the present case, let us think of a following Hermitian quadratic form:

$$\begin{aligned} \phi(x_1, x_2) &= \sum_{i,j=1}^2 x_i^* (G)_{ij} x_j = (x_1^* \ x_2^*) U U^\dagger G U U^\dagger \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ &= (x_1^* \ x_2^*) U \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} U^\dagger \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (\xi_1^* \ \xi_2^*) \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = 2|\xi_1|^2. \end{aligned}$$

Then, if we take $\begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ \xi_2 \end{pmatrix}$ with $\xi_2 \neq 0$ as a column vector, we get $\phi(x_1, x_2) = 0$. With this type of column vector, we have

$$U^\dagger \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ \xi_2 \end{pmatrix}, \text{ i.e., } \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = U \begin{pmatrix} 0 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0 \\ \xi_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \xi_2 \\ \xi_2 \end{pmatrix},$$

where ξ_2 is an arbitrary complex number. Thus, even for $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ we may have $\phi(x_1, x_2) = 0$. To be more specific, if we take, e.g., $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ or $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, we get

$$\phi(1, 1) = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0,$$

$$\phi(1, 2) = \begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = 1 > 0.$$

□

13.3 Adjoint Operators

A linear transformation A is similarly defined as before and A transforms $|x\rangle$ of (13.32) such that

$$A(|x\rangle) = (|e_1\rangle \cdots |e_n\rangle) \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad (13.59)$$

where (a_{ij}) is a matrix representation of A . Defining $A(|x\rangle) = |A(x)\rangle$ and $\langle(x)A^\dagger| = (\langle x|A)^\dagger = [A(|x\rangle)]^\dagger$ to be consistent with (1.117), we have

$$\langle(x)A^\dagger| = (x_1^* \cdots x_n^*) \begin{pmatrix} a_{11}^* & \cdots & a_{n1}^* \\ \vdots & \ddots & \vdots \\ a_{1n}^* & \cdots & a_{nn}^* \end{pmatrix} \begin{pmatrix} \langle e_1| \\ \vdots \\ \langle e_n| \end{pmatrix}. \quad (13.60)$$

Therefore, putting $|y\rangle = \sum_{i=1}^n y_i |e_i\rangle$, we have

$$\begin{aligned} \langle(x)A^\dagger|y\rangle &= (x_1^* \cdots x_n^*) \begin{pmatrix} a_{11}^* & \cdots & a_{n1}^* \\ \vdots & \ddots & \vdots \\ a_{1n}^* & \cdots & a_{nn}^* \end{pmatrix} \begin{pmatrix} \langle e_1| \\ \vdots \\ \langle e_n| \end{pmatrix} (|e_1\rangle \cdots |e_n\rangle) \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \\ &= (x_1^* \cdots x_n^*) A^\dagger G \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}. \end{aligned} \quad (13.61)$$

Meanwhile, we get

$$\begin{aligned} \langle y|A(x)\rangle &= (y_1^* \cdots y_n^*) \begin{pmatrix} \langle e_1| \\ \vdots \\ \langle e_n| \end{pmatrix} (|e_1\rangle \cdots |e_n\rangle) \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \\ &= (y_1^* \cdots y_n^*) GA \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}. \end{aligned} \quad (13.62)$$

Hence, we have

$$\langle y|A(x)\rangle^* = (y_1 \cdots y_n)G^*A^*\begin{pmatrix} x_1^* \\ \vdots \\ x_n^* \end{pmatrix} = (y_1 \cdots y_n)G^T(A^\dagger)^T\begin{pmatrix} x_1^* \\ \vdots \\ x_n^* \end{pmatrix}. \quad (13.63)$$

With the second equality, we used $G^* = (G^\dagger)^T = G^T$ (note that G is Hermitian) and $A^* = (A^\dagger)^T$. A complex conjugate matrix A^* is defined as

$$A^* \equiv \begin{pmatrix} a_{11}^* & \cdots & a_{1n}^* \\ \vdots & \ddots & \vdots \\ a_{n1}^* & \cdots & a_{nn}^* \end{pmatrix}.$$

Comparing (13.61) and (13.63), we find that one is transposed matrix of the other. Also note that $(AB)^T = B^T A^T$, $(ABC)^T = C^T B^T A^T$, etc. Since an inner product can be viewed as a (1, 1) matrix, two mutually transposed (1, 1) matrices are identical. Hence, we get

$$\langle (x)A^\dagger|y\rangle = \langle y|A(x)\rangle^* = \langle A(x)|y\rangle, \quad (13.64)$$

where the second equality is due to (13.2).

The other way around, we may use (13.64) for the definition of an adjoint operator of a linear transformation A . In fact, on the basis of (13.64) we have

$$\begin{aligned} \sum_{i,j,k} x_i^* (A^\dagger)_{ik} (G)_{kj} y_j &= \sum_{i,j,k} y_j (G^*)_{jk} (A^*)_{ki} x_i^* \\ &= \sum_{i,j,k} x_i^* y_j \left[(A^\dagger)_{ik} (G)_{kj} - (G^*)_{jk} (A^*)_{ki} \right] \\ &= \sum_{i,j,k} x_i^* y_j \left[(A^\dagger)_{ik} - (A^*)_{ki} \right] (G)_{kj} = 0. \end{aligned} \quad (13.65)$$

With the third equality of (13.65), we used $(G^*)_{jk} = (G)_{kj}$, i.e., $G^\dagger = G$ (Hermitian matrix). Thanks to the freedom in choice of basis vectors as well as x_i and y_i , we must have

$$(A^\dagger)_{ik} = (A^*)_{ki}. \quad (13.66)$$

Adopting the matrix representation of (13.59) for A , we get [1]

$$(A^\dagger)_{ik} = (a_{ki}^*). \quad (13.67)$$

Thus, we confirm that the adjoint operator $A^\dagger$ is represented by a complex conjugate transposed matrix of A , in accordance with (13.1).

Taking complex conjugate of (13.64), we have

$$\langle (x)A^\dagger|y \rangle^* = \langle y|(A^\dagger)^\dagger(x) \rangle = \langle y|A(x) \rangle.$$

Comparing both sides of the second equality, we get

$$(A^\dagger)^\dagger = A. \quad (13.68)$$

In Chap. 11, we have seen how a matrix representation of a linear transformation A is changed from A_0 to A' by the basis vectors transformation. We have

$$A' = P^{-1}A_0P. \quad (11.88)$$

In a similar manner, taking the adjoint of (11.88) we get

$$(A')^\dagger = P^\dagger A^\dagger (P^{-1})^\dagger = P^\dagger A^\dagger (P^\dagger)^{-1}. \quad (13.69)$$

In (13.69), we denote the adjoint operator before the basis vectors transformation simply by $A^\dagger$ to avoid complicated notation. We also have

$$(A^\dagger)' = (A')^\dagger. \quad (13.70)$$

Meanwhile, suppose that $(A^\dagger)' = Q^{-1}A^\dagger Q$. Then, from (13.69) and (13.70) we have

$$P^\dagger = Q^{-1}.$$

Next let us perform a calculation as below:

$$\begin{aligned} \langle (au + \beta v)A^\dagger|y \rangle &= \langle y|A(au + \beta v) \rangle^* = \alpha^* \langle y|A(u) \rangle^* + \beta^* \langle y|A(v) \rangle^* \\ &= \alpha^* \langle (u)A^\dagger|y \rangle + \beta^* \langle (v)A^\dagger|y \rangle = \langle \alpha(u)A^\dagger + \beta(v)A^\dagger|y \rangle. \end{aligned}$$

As y is an element arbitrarily chosen from a relevant vector space, we have

$$\langle (au + \beta v)A^\dagger | = \langle \alpha(u)A^\dagger + \beta(v)A^\dagger |, \quad (13.71)$$

or

$$(au + \beta v)A^\dagger = \alpha(u)A^\dagger + \beta(v)A^\dagger. \quad (13.72)$$

Equation (13.71) states the equality of two vectors in an inner product space on both sides, whereas (13.72) states that in a vector space where the inner product is

not defined. In either case, both (13.71) and (13.72) show that $A^\dagger$ is indeed a linear transformation. In fact, the matrix representation of (13.66) and (13.67) is independent of the concept of the inner product.

Suppose that there are two (or more) adjoint operators B and C that correspond to A . Then, from (13.64) we have

$$\langle (x)B|y \rangle = \langle (x)C|y \rangle = \langle y|A(x) \rangle^*. \quad (13.73)$$

Also we have

$$\langle (x)B - (x)C|y \rangle = \langle (x)(B - C)|y \rangle = 0. \quad (13.74)$$

As x and y are arbitrarily chosen elements, we get $B = C$, indicating the uniqueness of the adjoint operator.

It is of importance to examine how the norm of a vector is changed by the linear transformation. To this end, let us perform a calculation as below:

$$\begin{aligned} & \langle (x)A^\dagger | A(x) \rangle \\ &= (x_1^* \cdots x_n^*) \begin{pmatrix} a_{11}^* & \cdots & a_{n1}^* \\ \vdots & \ddots & \vdots \\ a_{1n}^* & \cdots & a_{nn}^* \end{pmatrix} \begin{pmatrix} \langle e_1 | \\ \vdots \\ \langle e_n | \end{pmatrix} (|e_1\rangle \cdots |e_n\rangle) \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \\ &= (x_1^* \cdots x_n^*) A^\dagger G A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}. \end{aligned} \quad (13.75)$$

Equation (13.75) gives the norm of vector after its transformation.

We may have a case where the norm is conserved before and after the transformation. Actually, comparing (13.34) and (13.75), we notice that if $A^\dagger G A = G$, $\langle x|x \rangle = \langle (x)A^\dagger | A(x) \rangle$. Let us have a following example for this.

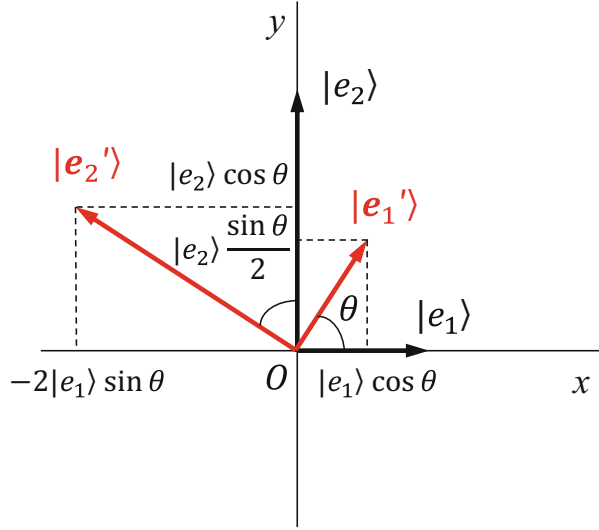
Example 13.3 Let us take two mutually orthogonal vectors $|e_1\rangle$ and $|e_2\rangle$ with $\|e_2\| = 2\|e_1\|$ as basis vectors in the xy -plane (Fig. 13.1). We assume that $|e_1\rangle$ is a unit vector. Hence, a set of vectors $|e_1\rangle$ and $|e_2\rangle$ does not constitute the orthonormal basis. Let $|x\rangle$ be an arbitrary position vector expressed as

$$|x\rangle = (|e_1\rangle \ |e_2\rangle) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}. \quad (13.76)$$

Then we have

$$\langle x|x \rangle = (x_1 \ x_2) \begin{pmatrix} \langle e_1 | \\ \langle e_2 | \end{pmatrix} (|e_1\rangle \ |e_2\rangle) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Fig. 13.1 Basis vectors $|e_1\rangle$ and $|e_2\rangle$ in the xy -plane and their linear transformation by R



$$\begin{aligned}
 &= (x_1 \ x_2) \begin{pmatrix} \langle e_1 | e_1 \rangle & \langle e_1 | e_2 \rangle \\ \langle e_2 | e_1 \rangle & \langle e_2 | e_2 \rangle \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (x_1 \ x_2) \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\
 &= x_1^2 + 4x_2^2.
 \end{aligned} \tag{13.77}$$

Next let us think of a following linear transformation R whose matrix representation is given by

$$R = \begin{pmatrix} \cos \theta & -2 \sin \theta \\ (\sin \theta)/2 & \cos \theta \end{pmatrix}. \tag{13.78}$$

The transformation matrix R is geometrically represented in Fig. 13.1. Following (11.36) we have

$$R(|x\rangle) = (|e_1\rangle \ |e_2\rangle) \begin{pmatrix} \cos \theta & -2 \sin \theta \\ (\sin \theta)/2 & \cos \theta \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}. \tag{13.79}$$

As a result of the transformation R , the basis vectors $|e_1\rangle$ and $|e_2\rangle$ are transformed into $|e'_1\rangle$ and $|e'_2\rangle$, respectively, as in Fig. 13.1 such that

$$(|e'_1\rangle \ |e'_2\rangle) = (|e_1\rangle \ |e_2\rangle) \begin{pmatrix} \cos \theta & -2 \sin \theta \\ (\sin \theta)/2 & \cos \theta \end{pmatrix}.$$

Taking an inner product of (13.79), we have

$$\begin{aligned}
& \langle (x)R^\dagger | R(x) \rangle \\
&= (x_1 \ x_2) \begin{pmatrix} \cos \theta & (\sin \theta)/2 \\ -2 \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} \cos \theta & -2 \sin \theta \\ (\sin \theta)/2 & \cos \theta \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\
&= (x_1 \ x_2) \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1^2 + 4x_2^2. \tag{13.80}
\end{aligned}$$

Putting

$$G = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix},$$

we have

$$R^\dagger G R = G. \tag{13.81}$$

Comparing (13.77) and (13.80), we notice that a norm of $|x\rangle$ remains unchanged after the transformation R . This means that R is virtually a unitary transformation. A somewhat unfamiliar matrix form of R resulted from the choice of basis vectors other than an orthonormal basis. $\square$

13.4 Orthonormal Basis

Now we introduce an orthonormal basis, the simplest and most important basis set in an inner product space. If we choose the orthonormal basis so that $\langle e_i | e_j \rangle = \delta_{ij}$, a Gram matrix $G = E$. Thus, $R^\dagger G R = G$ reads as $R^\dagger R = E$. In that case a linear transformation is represented by a unitary matrix and it conserves a norm of a vector and an inner product with two arbitrary vectors.

So far we assumed that an adjoint operator $A^\dagger$ operates only on a row vector from the right, as is evident from (13.61). At the same time, A operates only on the column vector from the left as in (13.62). To render the notation of (13.61) and (13.62) consistent with the associative law, we have to examine the commutability of $A^\dagger$ with G . In this context, choice of the orthonormal basis enables us to get through such a troublesome situation and largely eases matrix calculation. Thus,

$$\begin{aligned}
& \langle (x)A^\dagger | A(x) \rangle \\
&= (x_1^* \cdots x_n^*) \begin{pmatrix} a_{11}^* & \cdots & a_{n1}^* \\ \vdots & \ddots & \vdots \\ a_{1n}^* & \cdots & a_{nn}^* \end{pmatrix} \begin{pmatrix} \langle e_1 | \\ \vdots \\ \langle e_n | \end{pmatrix} (|e_1\rangle \cdots |e_n\rangle) \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}
\end{aligned}$$

$$= (x_1^* \cdots x_n^*) A^\dagger E A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (x_1^* \cdots x_n^*) A^\dagger A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}. \quad (13.82)$$

At the same time, we adopt a simple notation as below instead of (13.75)

$$\langle x A^\dagger | A x \rangle = (x_1^* \cdots x_n^*) A^\dagger A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}. \quad (13.83)$$

This notation has become now consistent with the associative law. Note that $A^\dagger$ and A operate on either a column or row vector. We can also do without a symbol “|” in (13.83) and express it as

$$\langle x A^\dagger A x \rangle = \langle x A^\dagger | A x \rangle. \quad (13.84)$$

Thus, we can freely operate $A^\dagger$ and A from both the left and right. By the same token, we rewrite (13.62) as

$$\langle y | A x \rangle = \langle y A | x \rangle = (y_1^* \cdots y_n^*) \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (y_1^* \cdots y_n^*) A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}. \quad (13.85)$$

Here, notice that a vector $|x\rangle$ is represented by a column vector with respect to the orthonormal basis. Using (13.64), we have

$$\begin{aligned} \langle y | A x \rangle^* &= \langle x A^\dagger | y \rangle = (x_1^* \cdots x_n^*) A^\dagger \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \\ &= (x_1^* \cdots x_n^*) \begin{pmatrix} a_{11}^* & \cdots & a_{n1}^* \\ \vdots & \ddots & \vdots \\ a_{1n}^* & \cdots & a_{nn}^* \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}. \end{aligned} \quad (13.86)$$

If in (13.83) we put $|y\rangle = |A x\rangle$, $\langle x A^\dagger | A x \rangle = \langle y | y \rangle \geq 0$. Thus, we define a norm of $|A x\rangle$ as

$$\|A x\| = \sqrt{\langle x A^\dagger | A x \rangle}. \quad (13.87)$$

Now we are in a position to construct an orthonormal basis in V^n using n linearly independent vectors $|i\rangle$ ($1 \leq i \leq n$). The following theorem is well-known as the Gram–Schmidt orthonormalization.

Theorem 13.2 Gram-Schmidt Orthonormalization Theorem [5] Suppose that there are a set of linearly independent vectors $|i\rangle$ ($1 \leq i \leq n$) in V^n . Then one can construct an orthonormal basis $|e_i\rangle$ ($1 \leq i \leq n$) so that $\langle e_i | e_j \rangle = \delta_{ij}$ ($1 \leq j \leq n$) and each vector $|e_i\rangle$ can be a linear combination of the vectors $|i\rangle$.

Proof First let us take $|1\rangle$. This can be normalized such that

$$|e_1\rangle = \frac{|1\rangle}{\sqrt{\langle 1|1\rangle}}, \quad \langle e_1|e_1\rangle = 1. \quad (13.88)$$

Next let us take $|2\rangle$ and then make a following vector:

$$|e_2\rangle = \frac{1}{L_2}[2\rangle - \langle e_1|2\rangle|e_1\rangle], \quad (13.89)$$

where L_2 is a normalization constant such that $\langle e_2|e_2\rangle = 1$. Note that $|e_2\rangle$ cannot be a zero vector. This is because if $|e_2\rangle$ were a zero vector, $|2\rangle$ and $|e_1\rangle$ (or $|1\rangle$) would be linearly dependent, in contradiction to the assumption. We have $\langle e_1|e_2\rangle = 0$. Thus, $|e_1\rangle$ and $|e_2\rangle$ are orthonormal.

After this, the proof is based upon mathematical induction. Suppose that the theorem is true of $(n - 1)$ vectors. That is, let $|e_i\rangle$ ($1 \leq i \leq n - 1$) so that $\langle e_i|e_j\rangle = \delta_{ij}$ ($1 \leq j \leq n - 1$) and each vector $|e_i\rangle$ can be a linear combination of the vectors $|i\rangle$. Meanwhile, let us define

$$|\tilde{n}\rangle \equiv |n\rangle - \sum_{j=1}^{n-1} \langle e_j|n\rangle |e_j\rangle. \quad (13.90)$$

Again, the vector $|\tilde{n}\rangle$ cannot be a zero vector as asserted above. We have

$$\langle e_k|\tilde{n}\rangle = \langle e_k|n\rangle - \sum_{j=1}^{n-1} \langle e_j|n\rangle \langle e_k|e_j\rangle = \langle e_k|n\rangle - \sum_{j=1}^{n-1} \langle e_j|n\rangle \delta_{kj} = 0, \quad (13.91)$$

where $1 \leq k \leq n - 1$. The second equality comes from the assumption of the induction. The vector $|\tilde{n}\rangle$ can always be normalized such that

$$|e_n\rangle = \frac{|\tilde{n}\rangle}{\sqrt{\langle \tilde{n}|\tilde{n}\rangle}}, \quad \langle e_n|e_n\rangle = 1. \quad (13.92)$$

Thus, the theorem is proven. ■

We note that in (13.92) a phase factor $e^{i\theta}$ (θ : an arbitrarily chosen real number) can be added such that

$$|e_n\rangle = \frac{e^{i\theta}|\tilde{n}\rangle}{\sqrt{\langle \tilde{n}|\tilde{n}\rangle}}. \quad (13.93)$$

To prove Theorem 13.2, we have used the following simple but important theorem.

Theorem 13.3 Let us have any n vectors $|i\rangle \neq 0$ ($1 \leq i \leq n$) in V^n and let these vectors $|i\rangle$ be orthogonal to one another. Then the vectors $|i\rangle$ are linearly independent.

Proof Let us think of the following equation:

$$c_1|1\rangle + c_2|2\rangle + \cdots + c_n|n\rangle = 0. \quad (13.94)$$

Multiplying (13.94) by $\langle i|$ from the left and considering the orthogonality among the vectors, we have

$$c_i \langle i|i\rangle = 0. \quad (13.95)$$

Since $\langle i|i\rangle \neq 0$, $c_i = 0$. The above is true of any c_i and $|i\rangle$. Then $c_1 = c_2 = \cdots = c_n = 0$. Thus, (13.94) implies that $|1\rangle, |2\rangle, \cdots, |n\rangle$ are linearly independent. ■

It is worth reviewing Theorem 13.2 from the point of view of the similarity transformation. In Sect. 13.2, we have seen that a Gram matrix comprising linearly independent vectors is a positive definite Hermitian matrix. In that case, we had

$$U^{-1}GU = U^\dagger GU = G' = \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix}, \quad (13.40)$$

where $\lambda_i > 0$ ($1 \leq i \leq n$). Now, let us define a following matrix given by

$$D = \begin{pmatrix} 1/\sqrt{\lambda_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1/\sqrt{\lambda_n} \end{pmatrix}. \quad (13.96)$$

Then, we have

$$D^T U^\dagger G U D = \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{pmatrix}, \quad (13.97)$$

where RHS of (13.97) is a (n, n) identity matrix. Thus, we find that (13.97) is equivalent to what Theorem 13.2 implies.

If the vector space we are thinking of is a *real* vector space, the corresponding Gram matrix is real as well and U of (13.97) is an orthogonal matrix. Then, instead of (13.97) we have

$$D^T U^T G U D = (UD)^T G U D = \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{pmatrix}. \quad (13.98)$$

Defining $UD \equiv P$, we have

$$P^T G P = E_n, \quad (13.99)$$

where E_n stands for the (n, n) identity matrix and P is a real non-singular matrix. The transformation of G represented by (13.99) is said to be an equivalence transformation (see Sect. 14.5). Equation (13.99) gives a special case where Theorem 14.11 holds; also see Sect. 14.5.

The real vector space characterized by a Gram matrix E_n is called Euclidean space. Its related topics are discussed in Parts IV and V. In Part V, in particular, we will investigate several properties of vector spaces other than the (real) inner product space in relation to the quantum theory of fields.

References

1. Hassani S (2006) Mathematical physics. Springer, New York
2. Satake I-O (1975) Linear algebra (pure and applied mathematics). Marcel Dekker, New York
3. Satake I (1974) Linear algebra. Shokabo, Tokyo
4. Riley KF, Hobson MP, Bence SJ (2006) Mathematical methods for physics and engineering, 3rd edn. Cambridge University Press, Cambridge
5. Dennery P, Krzywicki A (1996) Mathematics for physicists. Dover, New York

Chapter 14

Hermitian Operators and Unitary Operators



Hermitian operators and unitary operators are quite often encountered in mathematical physics and, in particular, quantum physics. In this chapter we investigate their basic properties. Both Hermitian operators and unitary operators fall under the category of normal operators. The normal matrices are characterized by an important fact that those matrices can be diagonalized by a unitary matrix. Moreover, Hermitian matrices always possess real eigenvalues. This fact largely eases mathematical treatment of quantum mechanics. In relation to these topics, in this chapter we investigate projection operators systematically. We find their important application to physicochemical problems in Part IV. We further investigate Hermitian quadratic forms and real symmetric quadratic forms as an important branch of matrix algebra. In connection with this topic, positive definiteness and non-negative property of a matrix are an important concept. This characteristic is readily applicable to theory of differential operators, thus rendering this chapter closely related to basic concepts of quantum physics.

14.1 Projection Operators

In Chap. 12 we considered the decomposition of a vector space to direct sum of invariant subspaces. We also mentioned properties of idempotent operators. Moreover, we have shown how an orthonormal basis can be constructed from a set of linearly independent vectors. In this section an orthonormal basis set is implied as basis vectors in an inner product space V^n .

Let us start with a concept of an orthogonal complement. Let W be a subspace in V^n . Let us think of a set of vectors $|x\rangle$ such that

$$\left\{ |x\rangle; \langle x|y\rangle = 0 \text{ for } \forall |y\rangle \in W \right\}. \quad (14.1)$$

We name this set $W^\perp$ and call it an orthogonal complement of W . The set $W^\perp$ forms a subspace of V^n . In fact, if $|a\rangle, |b\rangle \in W^\perp$, $\langle a|y\rangle = 0$, $\langle b|y\rangle = 0$. Since $(\langle a| + \langle b|) | y \rangle = \langle a|y\rangle + \langle b|y\rangle = 0$. Therefore, $|a\rangle + |b\rangle \in W^\perp$ and $\langle \alpha a|y\rangle = \alpha^* \langle a|y\rangle = 0$. Hence, $|\alpha a\rangle = \alpha |a\rangle \in W^\perp$. Then, $W^\perp$ is a subspace of V^n .

Theorem 14.1 [1, 2] Let W be a subspace and $W^\perp$ be its orthogonal complement in V^n . Then,

$$V^n = W \oplus W^\perp. \quad (14.2)$$

Proof Suppose that an orthonormal basis comprising $|e_1\rangle, |e_2\rangle, \dots, |e_n\rangle$ spans V^n ;

$$V^n = \text{Span}\{|e_1\rangle, |e_2\rangle, \dots, |e_n\rangle\}.$$

Of the orthonormal basis, let $|e_1\rangle, |e_2\rangle, \dots, |e_r\rangle$ ($r < n$) span W . Let an arbitrarily chosen vector from V^n be $|x\rangle$. Then we have

$$|x\rangle = x_1 |e_1\rangle + x_2 |e_2\rangle + \dots + x_n |e_n\rangle = \sum_{i=1}^n x_i |e_i\rangle. \quad (14.3)$$

Multiplying $\langle e_j|$ on (14.3) from the left, we have

$$\langle e_j|x\rangle = \sum_{i=1}^n x_i \langle e_j|e_i\rangle = \sum_{i=1}^n x_i \delta_{ji} = x_j. \quad (14.4)$$

That is,

$$|x\rangle = \sum_{i=1}^n \langle e_i|x\rangle |e_i\rangle. \quad (14.5)$$

Meanwhile, put

$$|x'\rangle = \sum_{i=1}^r \langle e_i|x\rangle |e_i\rangle. \quad (14.6)$$

Then we have $|x'\rangle \in W$. Also putting $|x''\rangle = |x\rangle - |x'\rangle$ and multiplying $\langle e_i|$ ($1 \leq i \leq r$) on it from the left, we get

$$\langle e_i|x''\rangle = \langle e_i|x\rangle - \langle e_i|x'\rangle = \langle e_i|x\rangle - \langle e_i|x\rangle = 0. \quad (14.7)$$

Taking account of $W = \text{Span}\{|e_1\rangle, |e_2\rangle, \dots, |e_r\rangle\}$, from the definition of the orthogonal complement we get $|x''\rangle \in W^\perp$. That is, for $\forall |x\rangle \in V^n$

$$|x\rangle = |x'\rangle + |x''\rangle. \quad (14.8)$$

This means that $V^n = W + W^\perp$. Meanwhile, we have $W \cap W^\perp = \{0\}$. In fact, suppose that $|x\rangle \in W \cap W^\perp$. Then $\langle x|x\rangle = 0$ because of the orthogonality. However, this implies that $|x\rangle = 0$. Consequently, from Theorem 11.1 we have $V^n = W \oplus W^\perp$. This completes the proof.

The consequence of Theorem 14.1 is that the dimension of $W^\perp$ is $(n - r)$; see Theorem 11.2. In other words, we have

$$\dim V^n = n = \dim W + \dim W^\perp.$$

Moreover, the contents of Theorem 14.1 can readily be generalized to more subspaces such that

$$V^n = W_1 \oplus W_2 \oplus \cdots \oplus W_r, \quad (14.9)$$

where $W_1, W_2, \dots$, and W_r ($r \leq n$) are mutually orthogonal complements. In this case $\forall |x\rangle \in V^n$ can be expressed uniquely as the sum of individual vectors $|w_1\rangle, |w_2\rangle, \dots$, and $|w_r\rangle$ of each subspace; i.e.,

$$|x\rangle = |w_1\rangle + |w_2\rangle + \cdots + |w_r\rangle = |w_1 + w_2 + \cdots + w_r\rangle. \quad (14.10)$$

Let us define the following operators similarly to the case of (12.201):

$$P_i(|x\rangle) = |w_i\rangle \quad (1 \leq i \leq r). \quad (14.11)$$

Thus, the operator P_i extracts a vector $|w_i\rangle$ in a subspace W_i . Then we have

$$(P_1 + P_2 + \cdots + P_r)(|x\rangle) = P_1|x\rangle + P_2|x\rangle + \cdots + P_r|x\rangle = |w_1\rangle + |w_2\rangle + \cdots + |w_r\rangle = |x\rangle. \quad (14.12)$$

Since $|x\rangle$ is an arbitrarily chosen vector, we get

$$P_1 + P_2 + \cdots + P_r = E. \quad (14.13)$$

Moreover,

$$P_i[P_i(|x\rangle)] = P_i(|w_i\rangle) = |w_i\rangle \quad (1 \leq i \leq r). \quad (14.14)$$

Therefore, from (14.11) and (14.14) we have

$$P_i[P_i(|x\rangle)] = P_i(|x\rangle). \quad (14.15)$$

The vector $|x\rangle$ is arbitrarily chosen, and so we get

$$P_i^2 = P_i. \quad (14.16)$$

Choose another arbitrary vector $|y\rangle \in V^n$ such that

$$|y\rangle = |u_1\rangle + |u_2\rangle + \cdots + |u_r\rangle = |u_1 + u_2 + \cdots + u_r\rangle. \quad (14.17)$$

Then, we have

$$\begin{aligned} \langle x|P_i y\rangle &= \langle w_1 + w_2 + \cdots + w_r|P_i|u_1 + u_2 + \cdots + u_r\rangle \\ &= \langle w_1 + w_2 + \cdots + w_r|u_i\rangle = \langle w_i|u_i\rangle. \end{aligned} \quad (14.18)$$

With the last equality, we used the mutual orthogonality of the subspaces. Meanwhile, we have

$$\begin{aligned} \langle y|P_i x\rangle^* &= \langle u_1 + u_2 + \cdots + u_r|P_i|w_1 + w_2 + \cdots + w_r\rangle^* \\ &= \langle u_1 + u_2 + \cdots + u_r|w_i\rangle^* = \langle u_i|w_i\rangle^* = \langle w_i|u_i\rangle. \end{aligned} \quad (14.19)$$

Comparing (14.18) and (14.19), we get

$$\langle x|P_i y\rangle = \langle y|P_i x\rangle^* = \langle x|P_i^\dagger y\rangle, \quad (14.20)$$

where we used (13.64) with the second equality. Since $|x\rangle$ and $|y\rangle$ are arbitrarily chosen, we get

$$P_i^\dagger = P_i. \quad (14.21)$$

Equation (14.21) shows that P_i is Hermitian.

The above discussion parallels that made in Sect. 12.4 with an idempotent operator. We have a following definition about a projection operator.

Definition 14.1 An operator P is said to be a projection operator if $P^2 = P$ and $P^\dagger = P$. That is, an idempotent and Hermitian operator is a projection operator.

As described above, a projection operator is characterized by (14.16) and (14.21). An idempotent operator does not premise the presence of an inner product space, but we only need a direct sum of subspaces. In contrast, if we deal with the projection operator, we are thinking of orthogonal complements as subspaces and their direct sum. The projection operator can adequately be defined in an inner product vector space having an orthonormal basis.

From (14.13) we have

$$\begin{aligned}
(P_1 + P_2 + \cdots + P_r)(P_1 + P_2 + \cdots + P_r) &= \sum_{i=1}^r P_i^2 + \sum_{i \neq j} P_i P_j \\
&= \sum_{i=1}^r P_i + \sum_{i \neq j} P_i P_j = E + \sum_{i \neq j} P_i P_j = E.
\end{aligned} \tag{14.22}$$

In (14.22) we used (14.13) and (14.16). Therefore, we get

$$\sum_{i \neq j} P_i P_j = 0.$$

In particular, we have

$$P_i P_j = 0, \quad P_j P_i = 0 \quad (i \neq j). \tag{14.23}$$

In fact, we have

$$P_i [P_j(|x\rangle)] = P_i(|w_j\rangle) = 0 \quad (i \neq j, \quad 1 \leq i, j \leq n). \tag{14.24}$$

The second equality comes from $W_i \cap W_j = \{0\}$. Notice that in (14.24), indices i and j are interchangeable. Again, $|x\rangle$ is arbitrarily chosen, and so (14.23) holds. Combining (14.16) and (14.23), we write

$$P_i P_j = \delta_{ij}. \tag{14.25}$$

In virtue of the relation (14.23), $P_i + P_j$ ($i \neq j$) is a projection operator as well [3]. In fact, we have

$$(P_i + P_j)^2 = P_i^2 + P_i P_j + P_j P_i + P_j^2 = P_i^2 + P_j^2 = P_i + P_j,$$

where the second equality comes from (14.23). Also we have

$$(P_i + P_j)^\dagger = P_i^\dagger + P_j^\dagger = P_i + P_j.$$

The following notation is often used:

$$P_i = \frac{|w_i\rangle\langle w_i|}{\|w_i\| \cdot \|w_i\|}. \tag{14.26}$$

Then we have

$$P_i |x\rangle = \frac{|w_i\rangle\langle w_i|}{\|w_i\| \cdot \|w_i\|} (|w_1\rangle + |w_2\rangle + \cdots + |w_r\rangle)$$

$$\begin{aligned}
&= [|w_i\rangle(\langle w_i|w_1\rangle + \langle w_i|w_2\rangle + \cdots + \langle w_i|w_r\rangle)]/\langle w_i|w_i\rangle \\
&= [|w_i\rangle\langle w_i|w_i\rangle]/\langle w_i|w_i\rangle = |w_i\rangle.
\end{aligned}$$

Furthermore, we have

$$P_i^2 |x\rangle = P_i |w_i\rangle = |w_i\rangle = P_i |x\rangle. \quad (14.27)$$

Equation (14.16) is recovered accordingly. Meanwhile,

$$(|w_i\rangle\langle w_i|)^\dagger = (\langle w_i|)^\dagger(|w_i\rangle)^\dagger = |w_i\rangle\langle w_i|. \quad (14.28)$$

Hence, we recover

$$P_i^\dagger = P_i. \quad (14.21)$$

In (14.28) we used $(AB)^\dagger = B^\dagger A^\dagger$. In fact, we have

$$\langle x|B^\dagger A^\dagger|y\rangle = \langle xB^\dagger|A^\dagger y\rangle = \langle yA|Bx\rangle^* = \langle y|AB|x\rangle^* = \langle x|(AB)^\dagger|y\rangle. \quad (14.29)$$

With the second equality of (14.29), we used (13.86) where A is replaced with B and $|y\rangle$ is replaced with $A^\dagger|y\rangle$. Since (14.29) holds for arbitrarily chosen vectors $|x\rangle$ and $|y\rangle$, comparing the first and last sides of (14.29) we have

$$(AB)^\dagger = B^\dagger A^\dagger. \quad (14.30)$$

We can express (14.29) alternatively as follows:

$$\langle xB^\dagger|A^\dagger y\rangle = \langle y(A^\dagger)^\dagger|(B^\dagger)^\dagger x\rangle^* = \langle yA|Bx\rangle^* = \langle yABx\rangle^*, \quad (14.31)$$

where with the second equality we used (13.68). Also recall the remarks after (13.83) with the expressions of (14.29) and (14.31). Other notations can be adopted.

We can view a projection operator under a more strict condition. Related operators can be defined as well. As in (14.26), let us define an operator such that

$$\tilde{P}_k = |e_k\rangle\langle e_k|, \quad (14.32)$$

where $|e_k\rangle$ ($k = 1, \dots, n$) are the orthonormal basis set of V^n . Operating $\tilde{P}_k$ on $|x\rangle = x_1|e_1\rangle + x_2|e_2\rangle + \cdots + x_n|e_n\rangle$ from the left, we get

$$\begin{aligned}\tilde{P}_k |x\rangle &= |e_k\rangle \langle e_k | \left(\sum_{j=1}^n x_j |e_j\rangle \right) = |e_k\rangle \left(\sum_{j=1}^n x_j \langle e_k | e_j \rangle \right) = \\ &= |e_k\rangle \left(\sum_{j=1}^n x_j \delta_{kj} \right) = x_k |e_k\rangle.\end{aligned}$$

Thus, we find that $\tilde{P}_k$ plays the same role as $P^{(k)}$ defined in (12.201). Represented by a matrix, $\tilde{P}_k$ has the same structure as that denoted in (12.205). Evidently,

$$(\tilde{P}_k)^2 = \tilde{P}_k, \tilde{P}_k^\dagger = \tilde{P}_k, \tilde{P}_1 + \tilde{P}_2 + \cdots + \tilde{P}_n = E. \quad (14.33)$$

Now let us modify $P^{(k)}$ in (12.201). There $P^{(k)} = \delta_i^{(k)} \delta_{(k)}^j$, where only the (k, k) element is 1, otherwise 0, in the (n, n) matrix. We define a matrix $P_{(m)}^{(k)} = \delta_i^{(k)} \delta_{(m)}^j$. A full matrix representation for it is

$$P_{(m)}^{(k)} = \begin{pmatrix} 0 & & & & \\ & \ddots & & & \\ & & 0 & & 1 \\ & & & 0 & \\ & & & & 0 \\ & & & & & \ddots \\ & & & & & & 0 \end{pmatrix}, \quad (14.34)$$

where only the (k, m) element is 1, otherwise 0.

In an example of (14.34), $P_{(m)}^{(k)}$ is an upper triangle matrix ($k < m$). Therefore, its eigenvalues are all zero, and so $P_{(m)}^{(k)}$ is a nilpotent matrix. If $k > m$, the matrix is a lower triangle matrix and nilpotent as well. Such a matrix is not Hermitian (nor a projection operator), as can be immediately seen from the matrix form of (14.34). Because of the properties of nilpotent matrices mentioned in Sect. 12.3, $P_{(m)}^{(k)}$ ($k \neq m$) is not diagonalizable either.

Various relations can be extracted. As an example, we have

$$P_{(m)}^{(k)} P_{(n)}^{(l)} = \sum_q \delta_i^{(k)} \delta_{(m)}^q \delta_q^{(l)} \delta_{(n)}^j = \delta_i^{(k)} \delta_{ml} \delta_{(n)}^j = \delta_{ml} P_{(n)}^{(k)}. \quad (14.35)$$

Note that $P_{(k)}^{(k)} \equiv P^{(k)}$ defined in (12.201). From (14.35), moreover, we have

$$\begin{aligned}P_{(m)}^{(k)} P_{(n)}^{(m)} &= P_{(n)}^{(k)}, P_{(m)}^{(n)} P_{(n)}^{(m)} = P_{(n)}^{(n)}, P_{(n)}^{(m)} P_{(m)}^{(n)} = P_{(m)}^{(m)}, \\ P_{(m)}^{(k)} P_{(m)}^{(k)} &= \delta_{mk} P_{(m)}^{(k)}, P_{(m)}^{(m)} P_{(m)}^{(m)} = [P_{(m)}^{(m)}]^2 = P_{(m)}^{(m)}, \text{etc.}\end{aligned}$$

These relations remain unchanged after the unitary similarity transformation by U . For instance, taking the first equation of the above, we have

$$U^\dagger P_{(m)}^{(k)} P_{(n)}^{(m)} U = \left[U^\dagger P_{(m)}^{(k)} U \right] \left[U^\dagger P_{(n)}^{(m)} U \right] = U^\dagger P_{(n)}^{(k)} U.$$

Among these operators, only $P_{(k)}^{(k)}$ is eligible for a projection operator. We will encounter further examples in Parts IV and V.

Replacing A in (13.62) with $P_{(k)}^{(k)}$, we obtain

$$\langle y | P_{(k)}^{(k)}(x) \rangle = (y_1^* \cdots y_n^*) G P_{(k)}^{(k)} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}. \quad (14.36)$$

Within a framework of an orthonormal basis where $G = E$, the representation is largely simplified to be

$$\langle y | P_{(k)}^{(k)}(x) \rangle = (y_1^* \cdots y_n^*) P_{(k)}^{(k)} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = y_k^* x_k. \quad (14.37)$$

14.2 Normal Operators

There are a large group of operators called normal operators that play an important role in mathematical physics, especially quantum physics. A normal operator is defined as an operator on an inner product space that commutes with its adjoint operator. That is, let A be a normal operator. Then, we have

$$AA^\dagger = A^\dagger A. \quad (14.38)$$

The normal operators include an Hermitian operator H defined as $H^\dagger = H$ as well as a unitary operator U defined as $UU^\dagger = U^\dagger U = E$.

In this condition let us estimate the norm of $|A^\dagger x\rangle$ together with $|Ax\rangle$ defined by (13.87). If A is a normal operator,

$$\|A^\dagger x\| = \sqrt{\langle x(A^\dagger)^\dagger |A^\dagger x\rangle} = \sqrt{\langle xAA^\dagger x\rangle} = \sqrt{\langle xA^\dagger Ax\rangle} = \|Ax\|. \quad (14.39)$$

The other way around suppose that $\|Ax\| = \|A^\dagger x\|$. Then, since $\|Ax\|^2 = \|A^\dagger x\|^2$, $\langle xA^\dagger Ax\rangle = \langle xAA^\dagger x\rangle$. That is, $\langle x | A^\dagger A - AA^\dagger | x \rangle = 0$ for an arbitrarily chosen vector $|x\rangle$. To assert $A^\dagger A - AA^\dagger = 0$, i.e., $A^\dagger A = AA^\dagger$ on the assumption that $\langle x | A^\dagger A - AA^\dagger | x \rangle = 0$, we need the following theorems:

Theorem 14.2 [4] A linear transformation A on an inner product space is the zero transformation if and only if $\langle y | Ax \rangle = 0$ for any vectors $|x\rangle$ and $|y\rangle$.

Proof If $A = 0$, then $\langle y|Ax\rangle = \langle y|0\rangle = 0$. This is because in (13.3) putting $\beta = 1 = -\gamma$ and $|b\rangle = |c\rangle$, we get $\langle a|0\rangle = 0$. Conversely, suppose that $\langle y|Ax\rangle = 0$ for any vectors $|x\rangle$ and $|y\rangle$. Then, putting $|y\rangle = |Ax\rangle$, $\langle xA^\dagger|Ax\rangle = 0$ and $\langle y|y\rangle = 0$. This implies that $|y\rangle = |Ax\rangle = 0$. For $|Ax\rangle = 0$ to hold for any $|x\rangle$ we must have $A = 0$. Note here that if A is a singular matrix, for some vectors $|x\rangle$, $|Ax\rangle = 0$. However, even though A is singular, for $|Ax\rangle = 0$ to hold for any $|x\rangle$, $A = 0$. This completes the proof.

We have another important theorem under a further restricted condition.

Theorem 14.3 [4] A linear transformation A on an inner product space is the zero transformation if and only if $\langle x|Ax\rangle = 0$ for any vectors $|x\rangle$.

Proof As in the case of Theorem 14.2, a necessary condition is trivial. To prove a sufficient condition, let us consider the following:

$$\langle x + y|A(x + y)\rangle = \langle x|Ax\rangle + \langle y|Ay\rangle + \langle x|Ay\rangle + \langle y|Ax\rangle,$$

$$\langle x|Ay\rangle + \langle y|Ax\rangle = \langle x + y|A(x + y)\rangle - \langle x|Ax\rangle - \langle y|Ay\rangle. \quad (14.40)$$

From the assumption that $\langle x|Ax\rangle = 0$ with any vectors $|x\rangle$, we have

$$\langle x|Ay\rangle + \langle y|Ax\rangle = 0. \quad (14.41)$$

Meanwhile, replacing $|y\rangle$ by $|iy\rangle$ in (14.41), we get

$$\langle x|Aiy\rangle + \langle iy|Ax\rangle = i[\langle x|Ay\rangle - \langle y|Ax\rangle] = 0. \quad (14.42)$$

That is,

$$\langle x|Ay\rangle - \langle y|Ax\rangle = 0. \quad (14.43)$$

Summing both sides of (14.41) and (14.43), we get

$$\langle x|Ay\rangle = 0. \quad (14.44)$$

Theorem 14.2 means that $A = 0$, indicating that the sufficient condition holds. This completes the proof.

Thus returning to the beginning, i.e., remarks made after (14.39), we establish the following theorem:

Theorem 14.4 A necessary and sufficient condition for a linear transformation A on an inner product space to be a normal operator is that a following relation

$$\|A^\dagger x\| = \|Ax\| \quad (14.45)$$

holds.

14.3 Unitary Diagonalization of Matrices

A normal operator has a distinct property. The normal operator can be diagonalized by a similarity transformation by a unitary matrix. The transformation is said to be a unitary similarity transformation. Let us prove the following theorem.

Theorem 14.5 [5] A necessary and sufficient condition for a matrix A to be diagonalized by unitary similarity transformation is that the matrix A is a normal matrix.

Proof To prove the necessary condition, suppose that A can be diagonalized by a unitary matrix U . That is,

$$U^\dagger A U = D, \text{ i.e., } A = U D U^\dagger \text{ and } A^\dagger = U D^\dagger U^\dagger, \quad (14.46)$$

where D is a diagonal matrix. Then

$$\begin{aligned} A A^\dagger &= (U D U^\dagger) (U D^\dagger U^\dagger) = U D D^\dagger U^\dagger = U D^\dagger D U^\dagger = (U D^\dagger U^\dagger) \\ &\times (U D U^\dagger) = A^\dagger A. \end{aligned} \quad (14.47)$$

For the third equality, we used $D D^\dagger = D^\dagger D$ (i.e., D and $D^\dagger$ are commutable). This shows that A is a normal matrix.

To prove the sufficient condition, let us show that a normal matrix can be diagonalized by unitary similarity transformation. The proof is due to mathematical induction, as is the case with [Theorem 12.1](#).

First we show that Theorem is true of a $(2, 2)$ matrix. Suppose that one of the eigenvalues of A_2 is α_1 and that its corresponding eigenvector is $|x_1\rangle$. Following procedures of the proof for [Theorem 12.1](#) and remembering the Gram–Schmidt orthonormalization theorem, we can construct a unitary matrix U_1 such that

$$U_1 = (|x_1\rangle \ |p_1\rangle), \quad (14.48)$$

where $|x_1\rangle$ represents a column vector and $|p_1\rangle$ is another arbitrarily determined column vector. Then we can convert A_2 to a triangle matrix such that

$$\tilde{A}_2 \equiv U_1^\dagger A_2 U_1 = \begin{pmatrix} \alpha_1 & x \\ 0 & y \end{pmatrix}. \quad (14.49)$$

Then we have

$$\begin{aligned} \tilde{A}_2 [\tilde{A}_2]^\dagger &= (U_1^\dagger A_2 U_1) (U_1^\dagger A_2 U_1)^\dagger = (U_1^\dagger A_2 U_1) (U_1^\dagger A_2^\dagger U_1) \\ &= U_1^\dagger A_2 A_2^\dagger U_1 = U_1^\dagger A_2^\dagger A_2 U_1 = (U_1^\dagger A_2^\dagger U_1) (U_1^\dagger A_2 U_1) \\ &= [\tilde{A}_2]^\dagger \tilde{A}_2. \end{aligned} \quad (14.50)$$

With the fourth equality, we used the supposition that A_2 is a normal matrix. Equation (14.50) means that $\tilde{A}_2$ defined in (14.49) is a normal operator. Via simple matrix calculations, we have

$$\widetilde{A_2} [\widetilde{A_2}]^\dagger = \begin{pmatrix} |\alpha_1|^2 + |x|^2 & xy^* \\ x^*y & |y|^2 \end{pmatrix}, \quad [\widetilde{A_2}]^\dagger \widetilde{A_2} = \begin{pmatrix} |\alpha_1|^2 & \alpha_1^* x \\ \alpha_1 x^* & |x|^2 + |y|^2 \end{pmatrix}. \quad (14.51)$$

For (14.50) to hold, we must have $x = 0$ in (14.51). Accordingly, we get

$$\widetilde{A_2} = \begin{pmatrix} \alpha_1 & 0 \\ 0 & y \end{pmatrix}. \quad (14.52)$$

This implies that a normal matrix A_2 has been diagonalized by the unitary similarity transformation.

Now let us examine a general case where we consider a (n, n) square normal matrix A_n . Let α_n be one of the eigenvalues of A_n . On the basis of the argument of the $(2, 2)$ matrix case, after a suitable similarity transformation by a unitary matrix $\tilde{U}$ we first have

$$\tilde{A}_2 = (\tilde{U})^\dagger A_n \tilde{U}, \quad (14.53)$$

where we can put

$$\tilde{A}_n = \begin{pmatrix} \alpha_n & \mathbf{x}^T \\ 0 & B \end{pmatrix}, \quad (14.54)$$

where $\mathbf{x}$ is a column vector of order $(n - 1)$, 0 is a zero column vector of order $(n - 1)$, and B is a $(n - 1, n - 1)$ matrix. Then we have

$$[\widetilde{A}_n]^\dagger = \begin{pmatrix} \alpha_n^* & 0 \\ \mathbf{x}^* & B^\dagger \end{pmatrix}, \quad (14.55)$$

where $\mathbf{x}^*$ is a complex column vector. Performing matrix calculations, we have

$$\begin{aligned} \widetilde{A}_n [\widetilde{A}_n]^\dagger &= \begin{pmatrix} |\alpha_n|^2 + \mathbf{x}^T \mathbf{x}^* & \mathbf{x}^T B^\dagger \\ B \mathbf{x}^* & B B^\dagger \end{pmatrix}, \\ [\widetilde{A}_n]^\dagger [\widetilde{A}_n] &= \begin{pmatrix} |\alpha_n|^2 & \alpha_n^* \mathbf{x}^T \\ \alpha_n \mathbf{x}^* & \mathbf{x}^* \mathbf{x}^T + B^\dagger B \end{pmatrix}. \end{aligned} \quad (14.56)$$

For $\widetilde{A}_n [\widetilde{A}_n]^\dagger = [\widetilde{A}_n]^\dagger [\widetilde{A}_n]$ to hold with (14.56), we must have $\mathbf{x} = \mathbf{0}$. Thus we get

$$\widetilde{A}_n = \begin{pmatrix} \alpha_n & 0 \\ 0 & B \end{pmatrix}. \quad (14.57)$$

Since $\widetilde{A}_n$ is a normal matrix, so is B . According to mathematical induction, let us assume that the theorem holds with a $(n-1, n-1)$ matrix, i.e., B . Then, also from the assumption there exists a unitary matrix C and a diagonal matrix D , both of order $(n-1)$, such that $BC = CD$. Hence,

$$\begin{pmatrix} \alpha_n & 0 \\ 0 & B \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & C \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & C \end{pmatrix} \begin{pmatrix} \alpha_n & 0 \\ 0 & D \end{pmatrix}. \quad (14.58)$$

Here putting

$$\widetilde{C}_n = \begin{pmatrix} 1 & 0 \\ 0 & C \end{pmatrix} \text{ and } \widetilde{D}_n = \begin{pmatrix} \alpha_n & 0 \\ 0 & D \end{pmatrix}, \quad (14.59)$$

we get

$$\widetilde{A}_n \widetilde{C}_n = \widetilde{C}_n \widetilde{D}_n. \quad (14.60)$$

As $\widetilde{C}_n$ is a (n, n) unitary matrix, $[\widetilde{C}_n]^\dagger \widetilde{C}_n = \widetilde{C}_n [\widetilde{C}_n]^\dagger = E$. Hence, $[\widetilde{C}_n]^\dagger \widetilde{A}_n \widetilde{C}_n = \widetilde{D}_n$. Thus, from (14.53) finally we get

$$[\widetilde{C}_n]^\dagger (\widetilde{U})^\dagger A_n \widetilde{U} \widetilde{C}_n = \widetilde{D}_n. \quad (14.61)$$

Putting $\widetilde{U} \widetilde{C}_n = V$, V being another unitary operator,

$$V^\dagger A_n V = \widetilde{D}_n. \quad (14.62)$$

Obviously, from (14.59) $\widetilde{D}_n$ is a diagonal matrix.

These complete the proof.

A direct consequence of Theorem 14.5 is that with any normal matrix we can find a set of orthonormal eigenvectors corresponding to individual eigenvalues whether or not those are degenerate.

In Sect. 12.2 we dealt with a decomposition of a linear vector space and relevant reduction of an operator when discussing canonical forms of matrices. In this context Theorem 14.5 gives a simple and clear criterion for this. Equation (14.57) implies that a $(n - 1, n - 1)$ submatrix B can further be reduced to matrices having lower dimensions. Considering that a diagonal matrix is a special case of triangle matrices, a normal matrix that has been diagonalized by the unitary similarity transformation gives eigenvalues by its diagonal elements.

From a point of view of the aforementioned aspect, let us consider the characteristics of normal matrices, starting with the discussion about the invariant subspaces. We have a following important theorem.

Theorem 14.6 Let A be a normal matrix and let one of its eigenvalues be α . Let W_α be an eigenspace corresponding to α . Then, W_α is both A -invariant and $A^\dagger$ -invariant. Also $W_\alpha^\perp$ is both A -invariant and $A^\dagger$ -invariant.

Proof Theorem 14.5 ensures that a normal matrix is diagonalized by unitary similarity transformation. Therefore, we deal with only “proper” eigenvalues and eigenvectors here. First we show if a subspace W is A -invariant, then its orthogonal complement $W^\perp$ is $A^\dagger$ -invariant. In fact, suppose that $|x\rangle \in W$ and $|x'\rangle \in W^\perp$. Then, from (13.64) and (13.86) we have

$$\langle x'|Ax\rangle = 0 = \langle xA^\dagger|x'\rangle^* = \langle x|A^\dagger x'\rangle^*. \quad (14.63)$$

The first equality comes from the fact that $|x\rangle \in W \Rightarrow A|x\rangle (=|Ax\rangle) \in W$ as W is A -invariant. From the last equality of (14.63), we have

$$A^\dagger|x'\rangle (=|A^\dagger x'\rangle) \in W^\perp. \quad (14.64)$$

That is, $W^\perp$ is $A^\dagger$ -invariant.

Next suppose that $|x\rangle \in W_\alpha$. Then we have

$$AA^\dagger|x\rangle = A^\dagger A|x\rangle = A^\dagger(\alpha|x\rangle) = \alpha A^\dagger|x\rangle. \quad (14.65)$$

Therefore, $A^\dagger|x\rangle \in W_\alpha$. This means that W_α is $A^\dagger$ -invariant. From the above remark, $W_\alpha^\perp$ is $(A^\dagger)^\dagger$ -invariant, and so A -invariant accordingly. This completes the proof.

From Theorem 14.5, we know that the resulting diagonal matrix $\tilde{D}_n$ in (14.62) has a form with n eigenvalues (α_n) some of which may be multiple roots arranged in diagonal elements. After diagonalizing the matrix, those eigenvalues can be sorted out according to different eigenvalues $\alpha_1, \alpha_2, \dots$, and α_s . This can also be done by unitary similarity transformation. The relevant unitary matrix U is represented as

$$U = \begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & & 0 & \cdots & & 1 \\ & & & & 1 & & \\ & & \vdots & & & \ddots & \vdots \\ & & & & & & 1 & 0 \\ & 1 & & \cdots & & & & & 1 & \ddots \\ & & & & & & & & & & 1 \end{pmatrix}, \quad (14.66)$$

where except (i, j) and (j, i) elements equal to 1, all the off-diagonal elements are zero. If operated from the left, U exchanges the i -th and j -th rows of the matrix. If operated from the right, U exchanges the i -th and j -th columns of the matrix. Note that U is at once unitary and Hermitian with eigenvalue 1 or -1 . Note that $U^2 = E$. This is because exchanging two columns (or two rows) two times produces identity transformation.

Thus performing such unitary similarity transformations appropriate times, we get

$$\widetilde{D_n} \sim \begin{pmatrix} \alpha_1 & & & & & & \\ & \ddots & & & & & \\ & & \alpha_1 & & & & \\ & & & \alpha_2 & & & \\ & & & & \ddots & & \\ & & & & & \alpha_2 & \\ & & & & & & \ddots & \\ & & & & & & & \alpha_s & \\ & & & & & & & & \ddots & \\ & & & & & & & & & \alpha_s \end{pmatrix}. \quad (14.67)$$

The matrix is identical to that represented in (12.181).

In parallel, V^n is decomposed to mutually orthogonal subspaces associated with different eigenvalues $\alpha_1, \alpha_2, \dots$, and α_s such that

$$V^n = W_{\alpha_1} \oplus W_{\alpha_2} \oplus \cdots \oplus W_{\alpha_s}. \quad (14.68)$$

This expression is formally identical to that represented in (12.191). Note, however, that in (12.181) orthogonal subspaces are not implied. At the same time, A_n is reduced to

$$A_n \sim \begin{pmatrix} A^{(1)} & & \cdots & \\ \vdots & A^{(2)} & & \\ & & \ddots & \\ & & & A^{(s)} \end{pmatrix}, \quad (14.69)$$

according to the different eigenvalues.

A normal operator has other distinct properties. Following theorems are good examples.

Theorem 14.7 Let A be a normal operator on V^n . Then $|x\rangle$ is an eigenvector of A with an eigenvalue α , if and only if $|x\rangle$ is an eigenvector of $A^\dagger$ with an eigenvalue α^* .

Proof We apply (14.45) for the proof. Both $(A - \alpha E)^\dagger = A^\dagger - \alpha^* E$ and $(A - \alpha E)$ are normal, since A is normal. Consequently, we have $\|(A - \alpha E)x\| = 0$ if and only if $\|(A^\dagger - \alpha^* E)x\| = 0$. Since only the zero vector has a zero norm, we get

$$(A - \alpha E)|x\rangle = 0 \text{ if and only if } (A^\dagger - \alpha^* E)|x\rangle = 0.$$

This completes the proof.

Theorem 14.8 Let A be a normal operator on V^n . Then, eigenvectors corresponding to different eigenvalues are mutually orthogonal.

Proof Let A be a normal operator on V^n . Let $|u\rangle$ be an eigenvector corresponding to an eigenvalue α and $|v\rangle$ be an eigenvector corresponding to an eigenvalue β with $\alpha \neq \beta$. Then we have

$$\alpha \langle v|u\rangle = \langle v|Au\rangle = \langle v|Au\rangle = \langle u|A^\dagger v\rangle^* = \langle u|\beta^* v\rangle^* = \langle \beta^* v|u\rangle = \beta \langle v|u\rangle, \quad (14.70)$$

where with the fourth equality we used Theorem 14.7. Then we get

$$(\alpha - \beta) \langle v|u\rangle = 0.$$

Since $\alpha - \beta \neq 0$, $\langle v|u\rangle = 0$. Namely, the eigenvectors $|u\rangle$ and $|v\rangle$ are mutually orthogonal. This completes the proof.

In (12.208) we mentioned the decomposition of diagonalizable matrices. As for the normal matrices, we have a related matrix decomposition. Let A be a normal operator. Then, according to Theorem 14.5, A can be diagonalized and expressed as (14.67). This is equivalently expressed as a following succinct relation. That is, if we choose U for a diagonalizing unitary matrix, we have

$$U^\dagger A U = \alpha_1 P_1 + \alpha_2 P_2 + \cdots + \alpha_s P_s, \quad (14.71)$$

where $\alpha_1, \alpha_2, \dots$, and α_s are different eigenvalues (possibly degenerate) of A ; P_l ($1 \leq l \leq s$) is described such that, e.g.,

$$P_1 = \begin{pmatrix} E_{n_1} & \cdots & \\ & 0_{n_2} & \vdots \\ & & \ddots & \\ & & \cdots & 0_{n_s} \end{pmatrix}, \quad (14.72)$$

where E_{n_1} stands for a (n_1, n_1) identity matrix with n_1 corresponding to multiplicity of α_1 . A matrix represented by 0_{n_2} is a (n_2, n_2) zero matrix, and so forth. This expression is in accordance with (14.69). From a matrix form (14.72), obviously P_l ($1 \leq l \leq s$) is a projection operator. Thus, operating U and $U^\dagger$ on both sides (14.71) from the left and right of (14.71), respectively, we obtain

$$A = \alpha_1 U P_1 U^\dagger + \alpha_2 U P_2 U^\dagger + \cdots + \alpha_s U P_s U^\dagger. \quad (14.73)$$

Defining $\tilde{P}_l \equiv U P_l U^\dagger$ ($1 \leq l \leq s$), we have

$$A = \alpha_1 \tilde{P}_1 + \alpha_2 \tilde{P}_2 + \cdots + \alpha_s \tilde{P}_s. \quad (14.74)$$

In (14.74), we can easily check that $\tilde{P}_l$ is a projection operator with $\alpha_1, \alpha_2, \dots$, and α_s being different eigenvalues of A . If one of the eigenvalues is zero, in (14.74) we may disregard the term related to the zero eigenvalue. For example, suppose that $\alpha_1 = 0$ in (14.74), we get

$$A = \alpha_2 \tilde{P}_2 + \cdots + \alpha_s \tilde{P}_s.$$

In case, moreover, α_l ($1 \leq l \leq s$) is degenerate, we express $\tilde{P}_l$ as $\tilde{P}_l^\mu$ ($1 \leq \mu \leq m_l$), where m_l is multiplicity of α_l . In that case, we may write

$$\tilde{P}_l = \tilde{P}_l^1 \oplus \cdots \oplus \tilde{P}_l^{m_l}. \quad (14.75)$$

Also, for $k \neq l$ we have

$$\tilde{P}_k \tilde{P}_l = U P_k U^\dagger U P_l U^\dagger = U P_k E P_l U^\dagger = U P_k P_l U^\dagger = 0 \quad (1 \leq k, l \leq s).$$

The last equality comes from (14.23). Similarly, we have

$$\tilde{P}_l \tilde{P}_k = 0.$$

Thus, we have

$$\tilde{P}_k \tilde{P}_l = \delta_{kl}.$$

If the operator is decomposed as in the case of (14.75), we can express

$$\tilde{P}_l^\mu \tilde{P}_l^\nu = \delta_{\mu\nu} \quad (1 \leq \mu, \nu \leq m_l).$$

Conversely, if an operator A is expressed by (14.74), that operator is normal operator. In fact, we have

$$\begin{aligned}
A^\dagger A &= \left(\sum_i \alpha_i \tilde{P}_i \right)^\dagger \left(\sum_j \alpha_j \tilde{P}_j \right) = \sum_{i,j} \alpha_i^* \alpha_j \tilde{P}_i \tilde{P}_j = \sum_{i,j} \alpha_i^* \alpha_j \delta_{ij} \tilde{P}_i = \sum_i |\alpha_i|^2 \tilde{P}_i, \\
AA^\dagger &= \left(\sum_j \alpha_j \tilde{P}_j \right) \left(\sum_i \alpha_i \tilde{P}_i \right)^\dagger = \sum_{i,j} \alpha_i^* \alpha_j \tilde{P}_j \tilde{P}_i = \sum_{i,j} \alpha_i^* \alpha_j \delta_{ji} \tilde{P}_j \\
&= \sum_i |\alpha_i|^2 \tilde{P}_i.
\end{aligned} \tag{14.76}$$

Hence, $A^\dagger A = AA^\dagger$. If projection operators are further decomposed as in the case of (14.75), we have a related expression to (14.76). Thus, a necessary and sufficient condition for an operator to be a normal operator is that the said operator is expressed as (14.74). The relation (14.74) is well-known as a spectral decomposition theorem. Thus, the spectral decomposition theorem is equivalent to Theorem 14.5.

In Sect. 12.7, we have already encountered the matrix decomposition (12.211) similar to that of (14.74). In this context, (12.211) may well be referred to as the spectral decomposition in a broader sense. Notice, however, that in (12.211) A_k ($1 \leq k \leq n$) is not necessarily Hermitian. In Part V, we will encounter interesting examples of matrix decomposition such as the spectral decomposition.

The relations (12.208) and (14.74) are virtually the same, aside from the fact that whereas (14.74) premises an inner product space, (12.208) does not premise it. Correspondingly, whereas the related operators are called projection operators with the case of (14.74), those operators are said to be idempotent operators for (12.208).

Example 14.1 Let us think of a Gram matrix of Example 13.1, as shown below

$$G = \begin{pmatrix} 2 & 1+i \\ 1-i & 2 \end{pmatrix}. \tag{13.51}$$

After a unitary similarity transformation, we got

$$U^\dagger G U = \begin{pmatrix} 2+\sqrt{2} & 0 \\ 0 & 2-\sqrt{2} \end{pmatrix}. \tag{13.54}$$

Putting $\tilde{G} = U^\dagger G U$ and rewriting (13.54), we have

$$\tilde{G} = \begin{pmatrix} 2+\sqrt{2} & 0 \\ 0 & 2-\sqrt{2} \end{pmatrix} = (2+\sqrt{2}) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + (2-\sqrt{2}) \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

By a back calculation of $U \tilde{G} U^\dagger = G$, we get

$$\begin{aligned}
G = & (2 + \sqrt{2}) \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{2}(1+i)}{4} \\ \frac{\sqrt{2}(1-i)}{4} & \frac{1}{2} \end{pmatrix} \\
& + (2 - \sqrt{2}) \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{2}(1+i)}{4} \\ -\frac{\sqrt{2}(1-i)}{4} & \frac{1}{2} \end{pmatrix}. \quad (14.77)
\end{aligned}$$

Putting eigenvalues $\alpha_1 = 2 + \sqrt{2}$ and $\alpha_2 = 2 - \sqrt{2}$ along with

$$A_1 = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{2}(1+i)}{4} \\ \frac{\sqrt{2}(1-i)}{4} & \frac{1}{2} \end{pmatrix}, A_2 = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{2}(1+i)}{4} \\ -\frac{\sqrt{2}(1-i)}{4} & \frac{1}{2} \end{pmatrix}, \quad (14.78)$$

we get

$$G = \alpha_1 A_1 + \alpha_2 A_2. \quad (14.79)$$

In the above, A_1 and A_2 are projection operators. In fact, as anticipated we have

$$A_1^2 = A_1, \quad A_2^2 = A_2, \quad A_1 A_2 = A_2 A_1 = 0, \quad A_1 + A_2 = E. \quad (14.80)$$

Moreover, (14.78) obviously shows that both A_1 and A_2 are Hermitian. Thus, (14.77) and (14.79) are an example of spectral decomposition. The decomposition is unique. $\square$

Example 14.1 can be dealt with in parallel to Example 12.5. In Example 12.5, however, an inner product space is not implied, and so we used an idempotent matrix instead of a projection operator. Note that as can be seen in Example 12.5 that idempotent matrix was not Hermitian.

14.4 Hermitian Matrices and Unitary Matrices

Of normal matrices, Hermitian matrices and unitary matrices play a crucial role both in fundamental and applied science. Let us think of several topics and examples.

In quantum physics, one frequently treats expectation value of an operator. In general, such an operator is Hermitian, more strictly an observable. Moreover, a vector on an inner product space is interpreted as a state on a Hilbert space. Suppose that there is a linear operator (or observable that represents a physical quantity) O that has discrete (or countable) eigenvalues $\alpha_1, \alpha_2, \dots$. The number of the eigenvalues may be a finite number or an infinite number, but here we assume the

finite number; i.e., let us suppose that we have eigenvalues $\alpha_1, \alpha_2, \dots$, and α_s , in consistent with our previous discussion.

In quantum physics, we have a well-known Born probability rule. The rule says the following: Suppose that we carry out a physical measurement on A with respect to a physical state $|u\rangle$. Here we assume that $|u\rangle$ has been normalized. Then, the probability $\wp_l$ that A takes α_l ($1 \leq l \leq s$) is given by

$$\wp_l = \|\tilde{P}_l u\|^2, \quad (14.81)$$

where $\tilde{P}_l$ is a projection operator that projects $|u\rangle$ to an eigenspace W_{α_l} spanned by $|\alpha_l, k\rangle$. Here k ($1 \leq k \leq n_l$) reflects the multiplicity n_l of an eigenvalue α_l . Hence, we express the n_l -dimensional eigenspace W_{α_l} as

$$W_{\alpha_l} = \text{Span}\{|\alpha_l, 1\rangle, |\alpha_l, 2\rangle, \dots, |\alpha_l, n_l\rangle\}. \quad (14.82)$$

Now, we define an expectation value $\langle A \rangle$ of A such that

$$\langle A \rangle \equiv \sum_{l=1}^s \alpha_l \wp_l. \quad (14.83)$$

From (14.81) we have

$$\wp_l = \|\tilde{P}_l u\|^2 = \langle u \tilde{P}_l^\dagger \tilde{P}_l u \rangle = \langle u \tilde{P}_l | \tilde{P}_l u \rangle = \langle u \tilde{P}_l \tilde{P}_l u \rangle = \langle u \tilde{P}_l u \rangle. \quad (14.84)$$

For the third equality we used the fact that $\tilde{P}_l$ is Hermitian; for the last equality we used $\tilde{P}_l^2 = \tilde{P}_l$. Summing (14.84) with the index l , we have

$$\sum_l \wp_l = \sum_l \langle u \tilde{P}_l u \rangle = \left\langle u \left(\sum_l \tilde{P}_l \right) u \right\rangle = \langle u E u \rangle = \langle u | u \rangle = 1,$$

where with the third equality we used (14.33).

Meanwhile, from the spectral decomposition theorem, we have

$$A = \alpha_1 \tilde{P}_1 + \alpha_2 \tilde{P}_2 + \dots + \alpha_s \tilde{P}_s. \quad (14.74)$$

Operating $\langle u|$ and $|u\rangle$ on both sides of (14.74), we get

$$\begin{aligned} \langle u | A | u \rangle &= \alpha_1 \langle u | \tilde{P}_1 | u \rangle + \alpha_2 \langle u | \tilde{P}_2 | u \rangle + \dots + \alpha_s \langle u | \tilde{P}_s | u \rangle \\ &= \alpha_1 \wp_1 + \alpha_2 \wp_2 + \dots + \alpha_s \wp_s. \end{aligned} \quad (14.85)$$

Equating (14.83) and (14.85), we have

$$\langle A \rangle = \langle u | A | u \rangle. \quad (14.86)$$

In quantum physics, a real number is required for an expectation value of an observable (i.e., a physical quantity). To warrant this, we have following theorems.

Theorem 14.9 A linear transformation A on an inner product space is Hermitian if and only if $\langle u|A|u \rangle$ is real for all $|u\rangle$ of that inner product space.

Proof If $A = A^\dagger$, then $\langle u|A|u \rangle^* = \langle u|A^\dagger|u \rangle = \langle u|A|u \rangle$. Therefore, $\langle u|A|u \rangle$ is real. Conversely, if $\langle u|A|u \rangle$ is real for all $|u\rangle$, we have

$$\langle u|A|u \rangle = \langle u|A|u \rangle^* = \langle u|A^\dagger|u \rangle.$$

Hence,

$$\langle u|A - A^\dagger|u \rangle = 0. \quad (14.87)$$

From Theorem 14.3, we get $A - A^\dagger = 0$, i.e., $A = A^\dagger$. This completes the proof.

Theorem 14.10 [4] The eigenvalues of an Hermitian operator A are real.

Proof Let α be an eigenvalue of A and let $|u\rangle$ be a corresponding eigenvector. Then, $A|u\rangle = \alpha|u\rangle$. Operating $\langle u|$ from the left, we have $\langle u|A|u\rangle = \alpha\langle u|u\rangle = \alpha\|u\|^2$. Thus,

$$\alpha = \langle u|A|u\rangle / \|u\|^2. \quad (14.88)$$

Since A is Hermitian, $\langle u|A|u\rangle$ is real. Then, the eigenvalue α is real as well. This completes the proof.

Unitary matrices have a following conspicuous features: (1) An inner product is held invariant under unitary transformation: Suppose that $|x'\rangle = U|x\rangle$ and $|y'\rangle = U|y\rangle$, where U is a unitary operator. Then $\langle y'|x'\rangle = \langle y|U^\dagger U|x\rangle = \langle y|x\rangle$. A norm of any vector is held invariant under unitary transformation as well. This is easily checked by replacing $|y\rangle$ with $|x\rangle$ in the above. (2) Let U be a unitary matrix and suppose that λ is an eigenvalue with $|\lambda\rangle$ being its corresponding eigenvector of that matrix. Then we have

$$|\lambda\rangle = U^\dagger U|\lambda\rangle = U^\dagger(\lambda|\lambda\rangle) = \lambda U^\dagger|\lambda\rangle = \lambda\lambda^*|\lambda\rangle, \quad (14.89)$$

where with the last equality we used Theorem 14.7. Thus

$$(1 - \lambda\lambda^*)|\lambda\rangle = 0. \quad (14.90)$$

As $|\lambda\rangle \neq 0$ is assumed, $1 - \lambda\lambda^* = 0$. That is

$$\lambda\lambda^* = |\lambda|^2 = 1. \quad (14.91)$$

Thus, eigenvalues of a unitary matrix have unit absolute value. Let those eigenvalues be $\lambda_k = e^{i\theta_k}$ ($k = 1, 2, \dots; \theta_k : \text{real}$). Then, from (12.11) we have

$$\det U = \prod_k \lambda_k = e^{i(\theta_1 + \theta_2 + \dots)} \text{ and } |\det U| = \prod_k |\lambda_k| = \prod_k |e^{i\theta_k}| = 1.$$

That is, any unitary matrix has a determinant of unit absolute value.

Example 14.2 Let us think of a following unitary matrix R :

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \quad (14.92)$$

A characteristic equation is

$$\begin{vmatrix} \cos \theta - \lambda & -\sin \theta \\ \sin \theta & \cos \theta - \lambda \end{vmatrix} = \lambda^2 - 2\lambda \cos \theta + 1. \quad (14.93)$$

Solving (14.93), we have

$$\lambda = \cos \theta \pm \sqrt{\cos^2 \theta - 1} = \cos \theta \pm i |\sin \theta|.$$

1. $\theta = 0$: This is a trivial case. The matrix R has automatically been diagonalized to be an identity matrix. Eigenvalues are 1 (double root).
2. $\theta = \pi$: This is a trivial case. Eigenvalues are -1 (double root).
3. $\theta \neq 0, \pi$: Let us think of $0 < \theta < \pi$. Then $\lambda = \cos \theta \pm i \sin \theta$. As a diagonalizing unitary matrix U , we get

$$U = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{pmatrix}, U^\dagger = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \end{pmatrix}.$$

Therefore, we have

$$\begin{aligned} U^\dagger R U &= \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{pmatrix} \\ &= \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}. \end{aligned} \quad (14.94)$$

A trace of the resulting matrix is $2 \cos \theta$. In the case of $\pi < \theta < 2\pi$, we get a diagonal matrix similarly. The conformation is left for readers. $\square$

14.5 Hermitian Quadratic Forms

The Hermitian quadratic forms appeared in, e.g., (13.34) and (13.83) in relation to Gram matrices in Sect. 13.2. The Hermitian quadratic forms have wide applications in the field of mathematical physics and materials science.

Let H be an Hermitian operator and $|x\rangle$ on an inner product space V^n . We define the Hermitian quadratic form in an arbitrary orthonormal basis as follows:

$$\langle x|H|x\rangle \equiv (x_1^* \cdots x_n^*) H \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \sum_{i,j} x_i^* (H)_{ij} x_j,$$

where $|x\rangle$ is represented as a column vector, as already mentioned in Sect. 13.4. Let us start with unitary diagonalization of (13.40), where a Gram matrix is a kind of Hermitian matrix. Following similar procedures, as in the case of (13.36) we obtain a diagonal matrix and an inner product such that

$$\langle x|H|x\rangle = (\xi_1^* \cdots \xi_n^*) \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix} \begin{pmatrix} \xi_1 \\ \vdots \\ \xi_n \end{pmatrix} = \lambda_1 |\xi_1|^2 + \cdots + \lambda_n |\xi_n|^2. \quad (14.95)$$

Notice, however, that the Gram matrix comprising basis vectors (that are linearly independent) is positive definite. Remember that if a Gram matrix is constructed by $A^\dagger A$ (Hermitian as well) according to whether A is non-singular or singular, $A^\dagger A$ is either positive definite or non-negative. The Hermitian matrix we are dealing with here, in general, does not necessarily possess the positive definiteness or non-negative feature. Yet, remember that $\langle x|H|x\rangle$ and eigenvalues $\lambda_1, \dots, \lambda_n$ are real.

Positive definiteness of matrices is an important concept in relation to the Hermitian (and real) quadratic forms (see Sect. 13.2). In particular, in the case where all the matrix elements are real and $|x\rangle$ is defined in a real domain, we are dealing with the quadratic form with respect to a real symmetric matrix. In the case of the real quadratic forms, we sometimes adopt the following notation [1, 2]:

$$A[\mathbf{x}] \equiv \mathbf{x}^T A \mathbf{x} = \sum_{i,j=1}^n a_{ij} x_i x_j; \quad \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

where $A = (a_{ij})$ is a real symmetric matrix and x_i ($1 \leq i \leq n$) are real numbers.

Here, we wish to show that the positive definiteness is invariant under a transformation P^TAP , where P is non-singular. In other words, if $A > 0$, we must have

$$A' = P^TAP > 0. \quad (14.96)$$

In (14.96) the transformation $A \rightarrow A'$ is said to be an equivalence transformation (or equivalent transformation) of A by P . Also, if A and A' satisfy (14.96), A and A' are said to be equivalent and we write [1, 2]

$$A' \simeq A.$$

Note that the equivalence transformation satisfies the equivalence law; readers, check it. To prove the above proposition stated by (14.96), we take in advance the notion of the polar decomposition of a matrix (see Sect. 24.9.1). The point is that any real non-singular matrix P is uniquely decomposed into a product such that $P = OS$, where O is an orthogonal matrix and S is a positive definite real symmetric matrix (see Corollary 24.3).

Then, we have

$$A' = (OS)^T A (OS) = S(O^T A O) S = S(O^{-1} A O) S.$$

Defining $\tilde{A} \equiv O^{-1} A O$, we have $\tilde{A} > 0$. It is because the similarity transformation of A by O holds its eigenvalues unchanged. Next, suppose that S is diagonalized through an orthogonal matrix $\tilde{O}$. It is always possible on the basis of Theorem 14.5. Then, we have

$$\tilde{O}^T A' \tilde{O} = \tilde{O}^T S (\tilde{O} \tilde{O}^T) \tilde{A} (\tilde{O} \tilde{O}^T) S \tilde{O} = (\tilde{O}^T S \tilde{O}) \tilde{O}^T \tilde{A} \tilde{O} (\tilde{O}^T S \tilde{O}) = \hat{D} \hat{A} \hat{D}, \quad (14.97)$$

where D is a diagonal matrix given by

$$D = \tilde{O}^T S \tilde{O} = \begin{pmatrix} d_1 & \cdots & \\ \vdots & \ddots & \vdots \\ & \cdots & d_n \end{pmatrix}$$

with $d_i > 0$ ($i = 1, \dots, n$). Also, we have $\hat{A} = \tilde{O}^T \tilde{A} \tilde{O}$. Since $\tilde{A} > 0$, we have $\hat{A} > 0$.

Meanwhile, let us think of a following inner product $\langle x | \hat{D} \hat{A} \hat{D} | x \rangle$, where x is an arbitrary real vector in V^n . We have

$$\begin{aligned}
\langle x | \widehat{D} \widehat{A} \widehat{D} | x \rangle &= (x_1 \cdots x_n) \widehat{D} \widehat{A} \widehat{D} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = [(x_1 \cdots x_n) \widehat{D}] \widehat{A} \left[\widehat{D} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right] \\
&= (x_1 d_1 \cdots x_n d_n) \widehat{A} \begin{pmatrix} x_1 d_1 \\ \vdots \\ x_n d_n \end{pmatrix}.
\end{aligned} \tag{14.98}$$

Suppose here that $(x_1 \cdots x_n) \neq 0$. That is, at least one of x_i ($i = 1, \cdots, n$) is a non-zero (real) number. Then, at least one of $x_i d_i$ ($i = 1, \cdots, n$) is a non-zero (real) number as well. Since $\widehat{A} > 0$, the rightmost side of (14.98) is positive. This implies that $\langle x | \widehat{D} \widehat{A} \widehat{D} | x \rangle$ is positive with arbitrarily chosen $(x_1 \cdots x_n) \neq 0$. Namely, $\widehat{D} \widehat{A} \widehat{D} > 0$. Thus, from (14.97) and (13.48) we have $A' > 0$. It is because a similarity transformation of A' by $\widehat{O}$ keeps its eigenvalues unchanged.

Thus, we have established the next important theorem.

Theorem 14.11 [1, 2] Let A be a real symmetric matrix with $A > 0$. Then, we have $P^T A P > 0$, where P is a real non-singular matrix.

Since A is a real symmetric matrix, we must have a suitable orthogonal matrix $\widehat{O}$ that diagonalizes A such that

$$\widehat{O}^T A \widehat{O} = \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix},$$

where λ_i ($1 \leq i \leq n$) are the eigenvalues of A .

Since $A > 0$ from the assumption, we should have $\forall \lambda_i > 0$ ($1 \leq i \leq n$) from (13.48). Then, we have

$$\det(\widehat{O}^T A \widehat{O}) = (\det \widehat{O}^T) (\det A) (\det \widehat{O}) = (\pm 1) \det A (\pm 1) = \det A = \prod_{i=1}^n \lambda_i.$$

Notice that we distinguish the equivalence transformation from the similarity transformation $P^{-1} A P$. Nonetheless, if we choose an orthogonal matrix O for P , the two transformations are the same because $O^T = O^{-1}$.

We often encounter real quadratic forms in the field of electromagnetism. Typical example is a trace of electromagnetic fields observed with an elliptically or circularly polarized light (see Sect. 7.3). A permittivity tensor of an anisotropic media such as crystals (either inorganic or organic) is another example. A tangible example for this appeared in Sect. 9.5.2 with an anisotropic organic crystal.

Regarding the real quadratic forms, we have a following important theorem (Theorem 14.12). To prove Theorem 14.12, we need Theorem 14.11.

Theorem 14.12 [1, 2] Let A be a n -dimensional real symmetric matrix $A = (a_{ij})$. Let $A^{(k)}$ be k -dimensional principal submatrices described by

$$A^{(k)} = \begin{pmatrix} a_{i_1 i_1} & a_{i_1 i_2} & \cdots & a_{i_1 i_k} \\ a_{i_2 i_1} & a_{i_2 i_2} & \cdots & a_{i_2 i_k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_k i_1} & a_{i_k i_2} & \cdots & a_{i_k i_k} \end{pmatrix} \quad (1 \leq i_1 < i_2 < \cdots < i_k \leq n),$$

where the principal submatrices mean a matrix made by striking out rows and columns on diagonal elements. Then, we have

$$A > 0 \Leftrightarrow \det A^{(k)} > 0 \quad (1 \leq k \leq n). \quad (14.99)$$

Proof First, suppose that $A > 0$. Then, in a quadratic form $A[\mathbf{x}]$ equating $(n - k)$ variables $x_l = 0$ ($l \neq i_1, i_2, \dots, i_k$), we obtain a quadratic form of

$$\sum_{\mu, \nu=1}^k a_{i_\mu i_\nu} x_{i_\mu} x_{i_\nu}.$$

Since $A > 0$, this (partial) quadratic form is positive definite as well; i.e., we have

$$A^{(k)} > 0.$$

Therefore, we get

$$\det A^{(k)} > 0.$$

This is due to (13.48). Notice that $\det A^{(k)}$ is said to be a principal minor. Thus, we have proven $\Rightarrow$ of (14.99).

To prove $\Leftarrow$, in turn, we use mathematical induction. If $n = 1$, we have a trivial case; i.e., A is merely a real positive number. Suppose that $\Leftarrow$ is true of $n - 1$. Then, we have $A^{(n-1)} > 0$ by supposition. Thus, it follows that it will suffice to show $A > 0$ on condition that $A^{(n-1)} > 0$ and $\det A > 0$ in addition. Let A be a n -dimensional real symmetric non-singular matrix such that

$$A = \begin{pmatrix} A^{(n-1)} & \mathbf{a} \\ \mathbf{a}^T & a_n \end{pmatrix},$$

where $A^{(n-1)}$ is a symmetric matrix and non-singular as well. We define P such that

$$P = \begin{pmatrix} E & A^{(n-1)^{-1}} \mathbf{a} \\ \mathbf{0} & 1 \end{pmatrix},$$

where E is a $(n - 1, n - 1)$ unit matrix. Notice that $\det P = \det E \cdot 1 = 1 \neq 0$, indicating that P is non-singular. We have

$$P^T = \begin{pmatrix} E & \mathbf{0} \\ \mathbf{a}^T A^{(n-1)-1} & 1 \end{pmatrix}.$$

For this expression, consider a non-singular matrix S . Then, we have $SS^{-1} = E$. Taking its transposition, we have $(S^{-1})^T S^T = E$. Therefore, if S is a symmetric matrix $(S^{-1})^T S = E$; i.e., $(S^{-1})^T = S^{-1}$. Hence, an inverse matrix of a symmetric matrix is symmetric as well. Then, for a symmetric matrix $A^{(n-1)}$ we have

$$\left(\mathbf{a}^T A^{(n-1)-1}\right)^T = \left(A^{(n-1)-1}\right)^T (\mathbf{a}^T)^T = A^{(n-1)-1} \mathbf{a}.$$

Therefore, A can be expressed as

$$\begin{aligned} A &= P^T \begin{pmatrix} A^{(n-1)} & \mathbf{0} \\ \mathbf{0} & a_n - A^{(n-1)-1}[\mathbf{a}] \end{pmatrix} P. \\ &= \begin{pmatrix} E & \mathbf{0} \\ \mathbf{a}^T A^{(n-1)-1} & 1 \end{pmatrix} \begin{pmatrix} A^{(n-1)} & \mathbf{0} \\ \mathbf{0} & a_n - A^{(n-1)-1}[\mathbf{a}] \end{pmatrix} \begin{pmatrix} E & A^{(n-1)-1} \mathbf{a} \\ \mathbf{0} & 1 \end{pmatrix}. \end{aligned} \quad (14.100)$$

Now, taking a determinant of (14.100), we have

$$\begin{aligned} \det A &= \det P^T \left[\det A^{(n-1)} \right] \left\{ a_n - A^{(n-1)-1}[\mathbf{a}] \right\} \det P \\ &= \left[\det A^{(n-1)} \right] \left\{ a_n - A^{(n-1)-1}[\mathbf{a}] \right\}. \end{aligned}$$

By supposition, we have $\det A^{(n-1)} > 0$ and $\det A > 0$. Hence, we have

$$a_n - A^{(n-1)-1}[\mathbf{a}] > 0.$$

Putting $\tilde{a}_n \equiv a_n - A^{(n-1)-1}[\mathbf{a}]$ and $\mathbf{x} \equiv \begin{pmatrix} \mathbf{x}^{(n-1)} \\ x_n \end{pmatrix}$, we get

$$\begin{pmatrix} A^{(n-1)} & \mathbf{0} \\ \mathbf{0} & \tilde{a}_n \end{pmatrix} [\mathbf{x}] = A^{(n-1)} [\mathbf{x}^{(n-1)}] + \tilde{a}_n x_n^2.$$

Since $A^{(n-1)} > 0$ and $\tilde{a}_n > 0$, we have

$$\tilde{A} \equiv \begin{pmatrix} A^{(n-1)} & \mathbf{0} \\ \mathbf{0} & \tilde{a}_n \end{pmatrix} > 0.$$

Meanwhile, A is expressed as

$$A = P^T \tilde{A} P \text{ or } A \simeq \tilde{A}.$$

From (14.96), we have $A > 0$. These complete the proof.

We also have a related theorem (the following Theorem 14.13) for an Hermitian quadratic form.

Theorem 14.13 Let $A = (a_{ij})$ be a n -dimensional Hermitian matrix. Let $\tilde{A}^{(k)}$ be k -dimensional principal submatrices. Then, we have

$$A > 0 \Leftrightarrow \det \tilde{A}^{(k)} > 0 \quad (1 \leq k \leq n),$$

where $\tilde{A}^{(k)}$ is described as

$$\tilde{A}^{(k)} = \begin{pmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & \ddots & \vdots \\ a_{k1} & \cdots & a_{kk} \end{pmatrix}.$$

The proof is left for readers.

Example 14.3 Let us consider a following Hermitian (real symmetric) matrix and corresponding Hermitian quadratic form.

$$H = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}, \quad \langle x | H | x \rangle = (x_1 \ x_2) \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}. \quad (14.101)$$

Principal minors of H are 5 (>0) and 1 (>0) and $\det H = 5 - 4 = 1$ (>0). Therefore, from Theorem 14.12 we have $H > 0$. A characteristic equation gives the following eigenvalues; i.e.,

$$\lambda_1 = 3 + 2\sqrt{2}, \quad \lambda_2 = 3 - 2\sqrt{2}.$$

Both the eigenvalues are positive as anticipated. As a diagonalizing matrix R , we get

$$R = \begin{pmatrix} 1+\sqrt{2} & 1-\sqrt{2} \\ 1 & 1 \end{pmatrix}.$$

To obtain a unitary matrix, we have to seek norms of column vectors. Corresponding to λ_1 and λ_2 , we estimate their norms to be $\sqrt{4 + 2\sqrt{2}}$ and $\sqrt{4 - 2\sqrt{2}}$, respectively. Using them, as a unitary matrix U we get

$$U = \begin{pmatrix} \frac{1}{\sqrt{4-2\sqrt{2}}} & -\frac{1}{\sqrt{4+2\sqrt{2}}} \\ \frac{1}{\sqrt{4+2\sqrt{2}}} & \frac{1}{\sqrt{4-2\sqrt{2}}} \end{pmatrix}. \quad (14.102)$$

Thus, performing the matrix diagonalization, we obtain a diagonal matrix D such that

$$D = U^\dagger H U = \begin{pmatrix} 3+2\sqrt{2} & 0 \\ 0 & 3-2\sqrt{2} \end{pmatrix}. \quad (14.103)$$

Let us view the above unitary diagonalization in terms of coordinate transformation. Using the above matrix U and changing (14.101) as in (13.36),

$$\begin{aligned} \langle x|H|x\rangle &= (x_1 \ x_2) U U^\dagger \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix} U U^\dagger \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ &= (x_1 \ x_2) U \begin{pmatrix} 3+2\sqrt{2} & 0 \\ 0 & 3-2\sqrt{2} \end{pmatrix} U^\dagger \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}. \end{aligned} \quad (14.104)$$

Making an argument analogous to that of Sect. 13.2 and using similar notation, we have

$$\langle \tilde{x}|H'|\tilde{x}\rangle = (\tilde{x}_1 \ \tilde{x}_2) H' \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix} = (x_1 \ x_2) U U^\dagger H U U^\dagger \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \langle x|H|x\rangle. \quad (14.105)$$

That is, we have

$$\begin{pmatrix} \widetilde{x_1} \\ \widetilde{x_2} \end{pmatrix} = U^\dagger \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}. \quad (14.106)$$

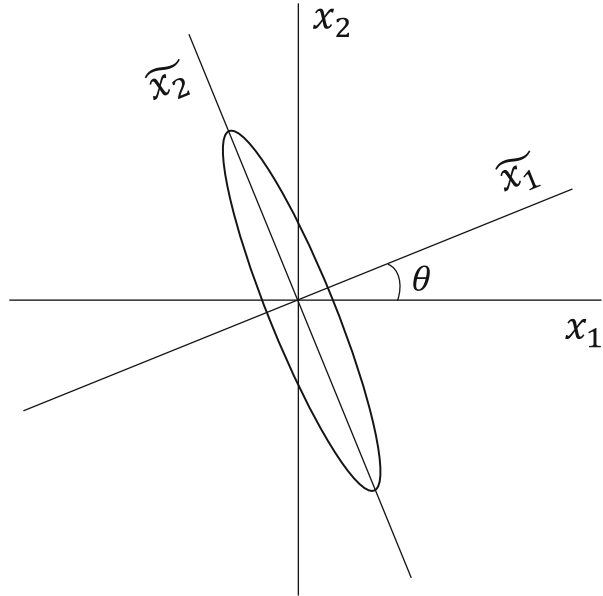
Or, taking an adjoint of (14.106), we have equivalently $(\tilde{x}_1 \ \tilde{x}_2) = (x_1 \ x_2) U$. Notice here that $\begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix}$ and $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ are real and that U is real (i.e., orthogonal matrix). Likewise,

$$H' = U^\dagger H U. \quad (14.107)$$

Thus, it follows that $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ and $\begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix}$ are *different* column vector representations of the *identical* vector that is viewed in reference to two different sets of orthonormal bases (i.e., different coordinate systems).

From (14.102), as an approximation we have

Fig. 14.1 Ellipse obtained through diagonalization of an Hermitian quadratic form. The angle θ is about 22.5°



$$U \cong \begin{pmatrix} 0.92 & -0.38 \\ 0.38 & 0.92 \end{pmatrix} \cong \begin{pmatrix} \cos 22.5^\circ & -\sin 22.5^\circ \\ \sin 22.5^\circ & \cos 22.5^\circ \end{pmatrix}. \quad (14.108)$$

Equating $\langle x|H|x\rangle$ to a constant, we get a hypersurface in a plane. Choosing 1 for a constant, we get an equation of hypersurface (i.e., an ellipse) as a function of $\tilde{x}_1$ and $\tilde{x}_2$ such that

$$\frac{\tilde{x}_1^2}{\left(\sqrt{\frac{1}{3+2\sqrt{2}}}\right)^2} + \frac{\tilde{x}_2^2}{\left(\sqrt{\frac{1}{3-2\sqrt{2}}}\right)^2} = 1. \quad (14.109)$$

Figure 14.1 depicts the ellipse. $\square$

14.6 Simultaneous Eigenstates and Diagonalization

In quantum physics, a concept of simultaneous eigenstate is important and has been briefly mentioned in Sect. 3.3. To rephrase this concept, suppose that there are two operators B and C and ask whether the two operators (or more) possess a common set of eigenvectors. The question is boiled down to whether the two operators commute. To address this question, the following theorem is important:

Theorem 14.14 [4] Two Hermitian matrices B and C commute if and only if there exists a complete orthonormal set of common eigenvectors.

Proof Suppose that there exists a complete orthonormal set of common eigenvectors $\{|x_i; k\rangle \ (1 \leq k \leq m_i)\}$ that span the linear vector space, where i and m_i are a positive integer and that $|x_i\rangle$ corresponds to an eigenvalue b_i of B and c_i of C . Note that if m_i is 1, we say that the spectrum is non-degenerate and that if m_i is equal to two or more, the spectrum is said to be degenerate. Then we have

$$B(|x_i\rangle) = b_i |x_i\rangle, \quad C(|x_i\rangle) = c_i |x_i\rangle. \quad (14.110)$$

Therefore, $BC(|x_i\rangle) = B(c_i |x_i\rangle) = c_i B(|x_i\rangle) = c_i b_i |x_i\rangle$. Similarly, $CB(|x_i\rangle) = c_i b_i |x_i\rangle$. Consequently, $(BC - CB)(|x_i\rangle) = 0$ for any $|x_i\rangle$. As all the set of $|x_i\rangle$ span the vector space, $BC - CB = 0$, namely $BC = CB$.

In turn, assume that $BC = CB$ and that $B(|x_i\rangle) = b_i |x_i\rangle$, where $|x_i\rangle$ are orthonormal. Then, we have

$$CB(|x_i\rangle) = b_i C(|x_i\rangle) \Rightarrow B[C(|x_i\rangle)] = b_i C(|x_i\rangle). \quad (14.111)$$

This implies that $C(|x_i\rangle)$ is an eigenvector of B corresponding to the eigenvalue b_i . We have two cases.

1. The spectrum is non-degenerate: The spectrum is said to be non-degenerate if only one eigenvector belongs to an eigenvalue. In other words, multiplicity of b_i is one. Then, $C(|x_i\rangle)$ must be equal to some constant times $|x_i\rangle$; i.e., $C(|x_i\rangle) = c_i |x_i\rangle$. That is, $|x_i\rangle$ is an eigenvector of C corresponding to an eigenvalue c_i . That is, $|x_i\rangle$ is a common eigenvector to B and C . This completes the proof for the non-degenerate case.
2. The spectrum is degenerate: The spectrum is said to be degenerate if two or more eigenvectors belong to an eigenvalue. Multiplicity of b_i is two or more; here suppose that the multiplicity is m ($m \geq 2$). Let $|x_i^{(1)}\rangle$, $|x_i^{(2)}\rangle$, $\dots$, and $|x_i^{(m)}\rangle$ be linearly independent vectors and belong to the eigenvalue b_i of B . Then, from the assumption we have m eigenvectors

$$C(|x_i^{(1)}\rangle), \quad C(|x_i^{(2)}\rangle), \dots, \quad C(|x_i^{(m)}\rangle)$$

that belong to an eigenvalue b_i of B . This means that individual $C(|x_i^{(\mu)}\rangle)$ ($1 \leq \mu \leq m$) are described by linear combination of $|x_i^{(1)}\rangle$, $|x_i^{(2)}\rangle$, $\dots$, and $|x_i^{(m)}\rangle$. What we want to prove is to show that suitable linear combination of these m vectors constitutes an eigenvector corresponding to an eigenvalue c_μ of C . Here, to avoid complexity we denote the multiplicity by m instead of the above-mentioned m_i .

The vectors $C(|x_i^{(\mu)}\rangle)$ ($1 \leq \mu \leq m$) can be described as

$$C(|x_i^{(1)}\rangle) = \sum_{j=1}^m \alpha_{j1} |x_i^{(j)}\rangle, \quad \dots, \quad C(|x_i^{(m)}\rangle) = \sum_{j=1}^m \alpha_{jm} |x_i^{(j)}\rangle. \quad (14.112)$$

Using full matrix representation, we have

$$\begin{aligned} C\left(\sum_{k=1}^m \gamma_k |x_i^{(k)}\rangle\right) &= \left(|x_i^{(1)}\rangle \cdots |x_i^{(m)}\rangle\right) C \begin{pmatrix} \gamma_1 \\ \vdots \\ \gamma_m \end{pmatrix} \\ &= \left(|x_i^{(1)}\rangle \cdots |x_i^{(m)}\rangle\right) \begin{pmatrix} \alpha_{11} & \cdots & \alpha_{1m} \\ \vdots & \ddots & \vdots \\ \alpha_{m1} & \cdots & \alpha_{mm} \end{pmatrix} \begin{pmatrix} \gamma_1 \\ \vdots \\ \gamma_m \end{pmatrix}. \end{aligned} \quad (14.113)$$

In (14.113), we adopt the notation of (11.37). Since (α_{ij}) is a matrix representation of an Hermitian operator C , (α_{ij}) is Hermitian as well. More specifically, if we take an inner product of a vector expressed in (14.112) with $|x_i^{(l)}\rangle$, then we have

$$\begin{aligned} \langle x_i^{(l)} | C x_i^{(k)} \rangle &= \langle x_i^{(l)} | \sum_{j=1}^m \alpha_{jk} x_i^{(j)} \rangle = \sum_{j=1}^m \alpha_{jk} \langle x_i^{(l)} | x_i^{(j)} \rangle = \sum_{j=1}^m \alpha_{jk} \delta_{lj} \\ &= \alpha_{lk} \quad (1 \leq k, l \leq m), \end{aligned} \quad (14.114)$$

where the third equality comes from the orthonormality of the basis vectors. Meanwhile,

$$\langle x_i^{(l)} | C x_i^{(k)} \rangle = \langle x_i^{(k)} | C^\dagger x_i^{(l)} \rangle^* = \langle x_i^{(k)} | C x_i^{(l)} \rangle^* = \alpha_{kl}^*, \quad (14.115)$$

where the second equality comes from the Hermiticity of C . From (14.114) and (14.115), we get

$$\alpha_{lk} = \alpha_{kl}^* \quad (1 \leq k, l \leq m). \quad (14.116)$$

This indicates that (α_{ij}) is in fact Hermitian.

We are seeking the condition under which linear combinations of the eigenvectors $|x_i^{(k)}\rangle (1 \leq k \leq m)$ for B are simultaneously eigenvectors of C . If the linear combination $\sum_{k=1}^m \gamma_k |x_i^{(k)}\rangle$ is to be an eigenvector of C , we must have

$$C\left(\sum_{k=1}^m \gamma_k |x_i^{(k)}\rangle\right) = c \left(\sum_{j=1}^m \gamma_j |x_i^{(j)}\rangle\right). \quad (14.117)$$

Considering (14.113), we have

$$\left(|x_i^{(1)}\rangle \cdots |x_i^{(m)}\rangle\right) \begin{pmatrix} \alpha_{11} & \cdots & \alpha_{1m} \\ \vdots & \ddots & \vdots \\ \alpha_{m1} & \cdots & \alpha_{mm} \end{pmatrix} \begin{pmatrix} \gamma_1 \\ \vdots \\ \gamma_m \end{pmatrix} = \left(|x_i^{(1)}\rangle \cdots |x_i^{(m)}\rangle\right) c \begin{pmatrix} \gamma_1 \\ \vdots \\ \gamma_m \end{pmatrix}. \quad (14.118)$$

The vectors $|x_i^{(j)}\rangle (1 \leq j \leq m)$ span an invariant subspace (i.e., an eigenspace corresponding to an eigenvalue of b_i). Let us call this subspace W^m . Consequently, in (14.118) we can equate the scalar coefficients of individual $|x_i^{(j)}\rangle (1 \leq j \leq m)$. Then, we get

$$\begin{pmatrix} \alpha_{11} & \cdots & \alpha_{1m} \\ \vdots & \ddots & \vdots \\ \alpha_{m1} & \cdots & \alpha_{mm} \end{pmatrix} \begin{pmatrix} \gamma_1 \\ \vdots \\ \gamma_m \end{pmatrix} = c \begin{pmatrix} \gamma_1 \\ \vdots \\ \gamma_m \end{pmatrix}. \quad (14.119)$$

This is nothing other than an eigenvalue equation. Since (α_{ij}) is an Hermitian matrix, there should be m eigenvalues c_μ some of which may be identical (the degenerate case). Moreover, we can always decide m orthonormal column vectors by solving (14.119). We denote them by $\gamma^{(\mu)}$ ($1 \leq \mu \leq m$) that belong to c_μ . Rewriting (14.119), we get

$$\begin{pmatrix} \alpha_{11} & \cdots & \alpha_{1m} \\ \vdots & \ddots & \vdots \\ \alpha_{m1} & \cdots & \alpha_{mm} \end{pmatrix} \begin{pmatrix} \gamma_1^{(\mu)} \\ \vdots \\ \gamma_m^{(\mu)} \end{pmatrix} = c_\mu \begin{pmatrix} \gamma_1^{(\mu)} \\ \vdots \\ \gamma_m^{(\mu)} \end{pmatrix}. \quad (14.120)$$

Equation (14.120) implies that we can construct a (m, m) unitary matrix from the m orthonormal column vectors $\gamma^{(\mu)}$. Using the said unitary matrix, we will be able to diagonalize (α_{ij}) according to Theorem 14.5.

Having determined m eigenvectors $\gamma^{(\mu)}$ ($1 \leq \mu \leq m$), we can construct a set of eigenvectors such that

$$|y_i^{(\mu)}\rangle = \sum_{k=1}^m \gamma_k^{(\mu)} |x_i^{(k)}\rangle. \quad (14.121)$$

Finally let us confirm that $|y_i^{(\mu)}\rangle (1 \leq \mu \leq m)$ in fact constitute an orthonormal basis. To show this, we have

$$\begin{aligned} \langle y_i^{(\nu)} | y_i^{(\mu)} \rangle &= \left\langle \sum_{k=1}^m \gamma_k^{(\nu)} x_i^{(k)} \middle| \sum_{l=1}^m \gamma_l^{(\mu)} x_i^{(l)} \right\rangle \\ &= \sum_{k=1}^m \sum_{l=1}^m [\gamma_k^{(\nu)}]^* \gamma_l^{(\mu)} \langle x_i^{(k)} | x_i^{(l)} \rangle = \sum_{k=1}^m \sum_{l=1}^m [\gamma_k^{(\nu)}]^* \gamma_l^{(\mu)} \delta_{kl} \\ &= \sum_{k=1}^m [\gamma_k^{(\nu)}]^* \gamma_k^{(\mu)} = \delta_{\nu\mu}. \end{aligned} \quad (14.122)$$

The last equality comes from the fact that a matrix comprising m orthonormal column vectors $\gamma^{(\mu)}$ ($1 \leq \mu \leq m$) forms a unitary matrix. Thus, $|y_i^{(\mu)}\rangle (1 \leq \mu \leq m)$ certainly constitute an orthonormal basis.

The above completes the proof.

Theorem 14.14 can be restated as follows: Two Hermitian matrices B and C can be simultaneously diagonalized by a unitary similarity transformation. As mentioned above, we can construct a unitary matrix U such that

$$U = \begin{pmatrix} \gamma_1^{(1)} & \cdots & \gamma_1^{(m)} \\ \vdots & \ddots & \vdots \\ \gamma_m^{(1)} & \cdots & \gamma_m^{(m)} \end{pmatrix}.$$

Then, using U , matrices B and C are diagonalized such that

$$U^\dagger B U = \begin{pmatrix} b_1 & & \\ & b_i & \\ & & \ddots \\ & & & b_i \end{pmatrix}, \quad U^\dagger C U = \begin{pmatrix} c_1 & & \\ & c_2 & \\ & & \ddots \\ & & & c_m \end{pmatrix}. \quad (14.123)$$

Note that both B and C are represented in an invariant subspace W^m .

As we have already seen in Part I that dealt with an eigenvalue problem of a hydrogen-like atom, squared angular momentum ($\mathbf{L}^2$) and z-component of angular momentum (L_z) possess a mutual set of eigenvectors and, hence, their eigenvalues are determined at once. Related matrix representations to (14.123) were given in (3.159). Conversely, this was not the case with a set of operators L_x , and L_y , and L_z ; see (3.30). Yet, we pointed out the exceptional case where these three operators along with $\mathbf{L}^2$ take an eigenvalue zero in common which an eigenstate $Y_0^0(\theta, \phi) \equiv \sqrt{1/4\pi}$ corresponds to. Nonetheless, no *complete* orthonormal set of common eigenvectors exists with the set of operators L_x , and L_y , and L_z . This fact is equivalent to that these three operators are non-commutative among them. In contrast, $\mathbf{L}^2$ and L_z share a complete orthonormal set of common eigenvectors and, hence, are commutable.

Notice that $C(|x_i^{(1)}\rangle)$, $C(|x_i^{(2)}\rangle)$, $\cdots$, and $C(|x_i^{(m)}\rangle)$ are not necessarily linearly independent (see Sect. 11.4). Suppose that among m eigenvalues c_μ ($1 \leq \mu \leq m$), some $c_\mu = 0$. Then, $\det C = 0$ according to (13.48) and (14.123). This means that C is singular. In that case, $C(|x_i^{(1)}\rangle)$, $C(|x_i^{(2)}\rangle)$, $\cdots$, and $C(|x_i^{(m)}\rangle)$ are linearly dependent. In Sect. 3.3, in fact, we had $|L_z Y_0^0(\theta, \phi)\rangle = |\mathbf{L}^2 Y_0^0(\theta, \phi)\rangle = 0$. But this special situation does not affect the proof of Theorem 14.14.

We know that any matrix A can be decomposed such that

$$A = \frac{1}{2}(A + A^\dagger) + i\left[\frac{1}{2i}(A - A^\dagger)\right], \quad (14.124)$$

where we put $B \equiv \frac{1}{2}(A + A^\dagger)$ and $C \equiv \frac{1}{2i}(A - A^\dagger)$; both B and C are Hermitian. That is, any matrix A can be decomposed to two Hermitian matrices in such a way that

$$A = B + iC. \quad (14.125)$$

Note here that B and C commute if and only if A and $A^\dagger$ commute, that is A is a normal matrix. In fact, from (14.125) we get

$$AA^\dagger - A^\dagger A = 2i(CB - BC). \quad (14.126)$$

From (14.126), if and only if B and C commute, $AA^\dagger - A^\dagger A = 0$, i.e., $AA^\dagger = A^\dagger A$. This indicates that A is a normal matrix.

Taking account of the above context along with Theorem 14.14, we see that we can construct the diagonalizing matrix of A from the complete orthonormal set of eigenvectors common to B and C . This immediately implies that the diagonalizing matrix is unitary. In other words, with respect to a normal operator A on V^n a diagonalizing unitary matrix U must exist. On top of it, such matrix U diagonalizes the Hermitian matrices B and C at once. Then, we have

$$\begin{aligned} U^\dagger A U &= U^\dagger B U + i U^\dagger C U \\ &= \begin{pmatrix} \lambda_1 & \cdots & \\ \vdots & \ddots & \\ & \cdots & \lambda_n \end{pmatrix} + i \begin{pmatrix} \mu_1 & \cdots & \\ \vdots & \ddots & \\ & \cdots & \mu_n \end{pmatrix} = \begin{pmatrix} \lambda_1 + i\mu_1 & \cdots & \\ \vdots & \ddots & \vdots \\ & \cdots & \lambda_n + i\mu_n \end{pmatrix}, \end{aligned} \quad (14.127)$$

where real eigenvalues $\lambda_1, \dots, \lambda_n$ of B as well as those $\mu_1, \dots, \mu_n$ of C are possibly degenerate. The complex eigenvalues $\lambda_1 + i\mu_1, \dots, \lambda_n + i\mu_n$ are possibly degenerate accordingly. Then, (14.127) can be rewritten as

$$U^\dagger A U = \begin{pmatrix} \omega_1 & & & & \\ & \ddots & & & \\ & & \omega_1 & & \\ & & & \omega_2 & \\ & & & & \ddots \\ & & & \omega_2 & \ddots \\ & & & & \ddots & \omega_s \\ & & & & & \ddots & \omega_s \\ & & & & & & \ddots & \omega_s \end{pmatrix}, \quad (14.128)$$

where ω_l ($1 \leq l \leq s \leq n$) denotes different complex eigenvalues possibly with several of them degenerate. Equation (14.128) is equivalent to (14.71). Considering (14.74) and (14.128) represents the spectral decomposition.

Following the argument made in Sect. 14.3, especially including (14.71)–(14.74), we acknowledge that the above discussion is equivalent to the implication of Theorem 14.5. Moreover, (14.12) implies that even with eigenvectors belonging to

a degenerate eigenvalue, we can choose mutually orthogonal (unit) vectors. Thus, we recognize that Theorems 14.5, 14.8, and 14.14 are mutually interrelated. These theorems and related concepts can be interpreted in terms of the projection operators and spectral decomposition. We will survey the related aspects in a broader context from a quantum-mechanical point of view later (see Chap. 24).

In Sects. 14.1 and 14.3, we mentioned the spectral decomposition. There, we showed a case where projection operators commute with one another; see (14.23). Thus, in light of Theorem 14.14, those projection operators can be diagonalized at once to be expressed as, e.g., (14.72). This is a conspicuous feature of the projection operators.

References

1. Satake I-O (1975) Linear algebra (pure and applied mathematics). Marcel Dekker, New York
2. Satake I (1974) Linear algebra. Shokabo, Tokyo
3. Hassani S (2006) Mathematical physics. Springer, New York
4. Byron FW, Fuller RW (1992) Mathematics of classical and quantum physics. Dover, New York
5. Mirsky L (1990) An introduction to linear algebra. Dover, New York

Chapter 15

Exponential Functions of Matrices



In Chap. 12, we dealt with a function of matrices. In this chapter we study several important definitions and characteristics of functions of matrices. If elements of matrices consist of analytic functions of a real variable, such matrices are of particular importance. For instance, differentiation can naturally be defined with the functions of matrices. Of these, exponential functions of matrices are widely used in various fields of mathematical physics. These functions frequently appear in a system of differential equations. In Chap. 10, we showed that SOLDEs with suitable BCs can be solved using Green's functions. In the present chapter, in parallel, we show a solving method based on resolvent matrices. The exponential functions of matrices have broad applications in group theory that we will study in Part IV. In preparation for it, we study how the collection of matrices forms a linear vector space. In accordance with Chap. 13, we introduce basic notions of inner product and norm to the matrices.

15.1 Functions of Matrices

We consider a matrix in which individual components are differentiable with respect to a real variable such that

$$A(t) = (a_{ij}(t)). \quad (15.1)$$

We define the differentiation as

$$A'(t) = \frac{dA(t)}{dt} \equiv (a'_{ij}(t)). \quad (15.2)$$

We have

$$[A(t) + B(t)]' = A'(t) + B'(t), \quad (15.3)$$

$$[A(t)B(t)]' = A'(t)B(t) + A(t)B'(t). \quad (15.4)$$

To get (15.4), putting $A(t)B(t) \equiv C(t)$, we have

$$c_{ij}(t) = \sum_k a_{ik}(t)b_{kj}(t), \quad (15.5)$$

where $C(t) = (c_{ij}(t))$, $B(t) = (b_{ij}(t))$. Differentiating (15.5), we get

$$c'_{ij}(t) = \sum_k a'_{ik}(t)b_{kj}(t) + \sum_k a_{ik}(t)b'_{kj}(t), \quad (15.6)$$

namely, we have $C'(t) = A'(t)B(t) + A(t)B'(t)$, i.e., (15.4) holds.

Analytic functions are usually expanded as an infinite power series. As an analogue, a function of a matrix is expected to be expanded as such. Among various functions of matrices, exponential functions are particularly important and have a wide field of applications. As in the case of a real or complex number x , the exponential function of matrix A can be defined as

$$\exp A \equiv E + A + \frac{1}{2!}A^2 + \cdots + \frac{1}{\nu!}A^\nu + \cdots, \quad (15.7)$$

where A is a (n, n) square matrix and E is a (n, n) identity matrix. Note that E is used instead of a number 1 that appears in the power series expansion of $\exp x$ described as

$$\exp x \equiv 1 + x + \frac{1}{2!}x^2 + \cdots + \frac{1}{\nu!}x^\nu + \cdots.$$

Here we define the convergence of a series of matrix as in the following definition:

Definition 15.1 Let $A_0, A_1, \cdots, A_\nu, \cdots$ [$A_\nu \equiv (a_{ij, \nu})$ ($1 \leq i, j \leq n$)] be a sequence of matrices. Let $a_{ij, 0}, a_{ij, 1}, \cdots, a_{ij, \nu}, \cdots$ be the corresponding sequence of each matrix component. Let us define a series of matrices as $\tilde{A} \equiv (\tilde{a}_{ij}) = A_0 + A_1 + \cdots + A_\nu + \cdots$ and a series of individual matrix components as $\tilde{a}_{ij} \equiv a_{ij, 0} + a_{ij, 1} + \cdots + a_{ij, \nu} + \cdots$. Then, if each $\tilde{a}_{ij}$ converges, we say that $\tilde{A}$ converges.

Regarding the matrix notation in Definition 15.1, readers are referred to (11.38). Now, let us show that the power series of (15.7) converges [1, 2].

Let A be a (n, n) square matrix. Putting $A = (a_{ij})$ and $\max_{i,j} |a_{ij}| = M$ and defining $A^\nu \equiv (a_{ij}^{(\nu)})$, where $a_{ij}^{(\nu)}$ is defined as a (i, j) component of A^ν , we get

$$|a_{ij}^{(\nu)}| \leq n^\nu M^\nu \quad (1 \leq i, j \leq n). \quad (15.8)$$

Note that $A^0 = E$. Hence, $a_{ij}^{(0)} = \delta_{ij}$. Then, if $\nu = 0$, (15.8) routinely holds. When $\nu = 1$, (15.8) obviously holds as well from the definition of M and assuming $n \geq 2$; we are considering (n, n) matrices with $n \geq 2$. We wish to show that (15.8) holds with any $\nu (\geq 2)$ using mathematical induction. Suppose that (15.8) holds with $\nu = \nu - 1$. That is,

$$|a_{ij}^{(\nu-1)}| \leq n^{\nu-1} M^{\nu-1} \quad (1 \leq i, j \leq n).$$

Then, we have

$$(A^\nu)_{ij} \equiv a_{ij}^{(\nu)} = (A^{\nu-1}A)_{ij} = \sum_{k=1}^n (A^{\nu-1})_{ik} (A)_{kj} = \sum_{k=1}^n a_{ik}^{(\nu-1)} a_{kj}. \quad (15.9)$$

Thus, we get

$$\begin{aligned} |a_{ij}^{(\nu)}| &= \left| \sum_{k=1}^n a_{ik}^{(\nu-1)} a_{kj} \right| \leq \sum_{k=1}^n |a_{ik}^{(\nu-1)}| \cdot |a_{kj}| \leq n \cdot n^{\nu-1} M^{\nu-1} \cdot M \\ &= n^\nu M^\nu. \end{aligned} \quad (15.10)$$

Consequently (15.8) holds with $\nu = \nu$.

Meanwhile, we have a next relation such that

$$e^{nM} = \sum_{\nu=0}^{\infty} \frac{1}{\nu!} n^\nu M^\nu. \quad (15.11)$$

The series (15.11) certainly converges; note that nM is not a matrix but a number. From (15.10) we obtain

$$\sum_{\nu=0}^{\infty} \frac{1}{\nu!} |a_{ij}^{(\nu)}| \leq \sum_{\nu=0}^{\infty} \frac{1}{\nu!} n^\nu M^\nu = e^{nM}.$$

This implies that $\sum_{\nu=0}^{\infty} \frac{1}{\nu!} a_{ij}^{(\nu)}$ converges (i.e., absolute convergence [3]). Thus, from Definition 15.1, (15.7) converges.

To further examine the exponential functions of matrices, in the following proposition, we show that a collection of matrices forms a linear vector space.

Proposition 15.1 Let $\mathcal{M}$ be a collection of all (n, n) square matrices. Then, $\mathcal{M}$ forms a linear vector space.

Proof Let $A = (a_{ij}) \in \mathcal{M}$ and $B = (b_{ij}) \in \mathcal{M}$ be (n, n) square matrices. Then, $\alpha A = (\alpha a_{ij})$ and $A + B = (a_{ij} + b_{ij})$ are again (n, n) square matrices, and so we have $\alpha A \in \mathcal{M}$ and $A + B \in \mathcal{M}$. Thus, $\mathcal{M}$ forms a linear vector space.

In Proposition 15.1, $\mathcal{M}$ forms a n^2 -th order vector space V^{n^2} . To define an inner product in this space, we introduce $|P_{(l)}^{(k)}\rangle$ ($k, l = 1, 2, \dots, n$) as a basis set of n^2 vectors. Now, let $A = (a_{ij})$ be a (n, n) square matrices. To explicitly show that A is a vector in the inner product space, we express it as

$$|A\rangle = \sum_{k,l=1}^n a_{kl} |P_{(l)}^{(k)}\rangle.$$

Then, an adjoint matrix of $|A\rangle$ is written as

$$\langle A | \equiv |A\rangle^\dagger = \sum_{k,l=1}^n a_{kl}^* \langle P_{(l)}^{(k)} |, \quad (15.12)$$

where $\langle P_{(l)}^{(k)} |$ is an adjoint matrix of $|P_{(l)}^{(k)}\rangle$. With the notations in (15.12), see Sects. 1.4 and 13.3; for $P_{(l)}^{(k)}$ see Sect. 14.1. Meanwhile, let $B = (b_{ij})$ be another (n, n) square matrix denoted by

$$|B\rangle = \sum_{s,t=1}^n b_{st} |P_{(t)}^{(s)}\rangle.$$

Here, let us define an orthonormal basis set $|P_{(l)}^{(k)}\rangle$ and an inner product between vectors described in a form of matrices such that

$$\langle P_{(l)}^{(k)} | P_{(t)}^{(s)} \rangle \equiv \delta_{ks} \delta_{lt}.$$

Then, from (15.12) the inner product between A and B is given by

$$\begin{aligned} \langle A | B \rangle &\equiv \sum_{k,l,s,t=1}^n a_{kl}^* b_{st} \langle P_{(l)}^{(k)} | P_{(t)}^{(s)} \rangle = \sum_{k,l,s,t=1}^n a_{kl}^* b_{st} \delta_{ks} \delta_{lt} \\ &= \sum_{k,l=1}^n a_{kl}^* b_{kl}. \end{aligned} \quad (15.13)$$

The relation (15.13) leads to a norm already introduced in (13.8). The norm of a matrix is defined as

$$\|A\| \equiv \sqrt{\langle A|A \rangle} = \sqrt{\sum_{i,j=1}^n |a_{ij}|^2}. \quad (15.14)$$

We have the following theorem regarding two matrices.

Theorem 15.1 Let $A = (a_{ij})$ and $B = (b_{ij})$ be two square matrices. Then, we have a following inequality:

$$\|AB\| \leq \|A\| \cdot \|B\|. \quad (15.15)$$

Proof Let $C = AB$. Then, from Cauchy–Schwarz inequality (Sect. 13.1), we get

$$\begin{aligned} \|C\|^2 &= \sum_{i,j} |c_{ij}|^2 = \sum_{i,j} \left| \sum_k a_{ik} b_{kj} \right|^2 \leq \sum_{i,j} \sum_k |a_{ik}|^2 \sum_l |b_{lj}|^2 \\ &= \left(\sum_{i,k} |a_{ik}|^2 \right) \left(\sum_{j,l} |b_{lj}|^2 \right) = \|A\|^2 \cdot \|B\|^2. \end{aligned} \quad (15.16)$$

This leads to (15.15)

15.2 Exponential Functions of Matrices and Their Manipulations

Using Theorem 15.1, we explore important characteristics of exponential functions of matrices. For this we have a next theorem.

Theorem 15.2 [4] Let A and B be mutually commutative matrices, i.e., $AB = BA$. Then, we have

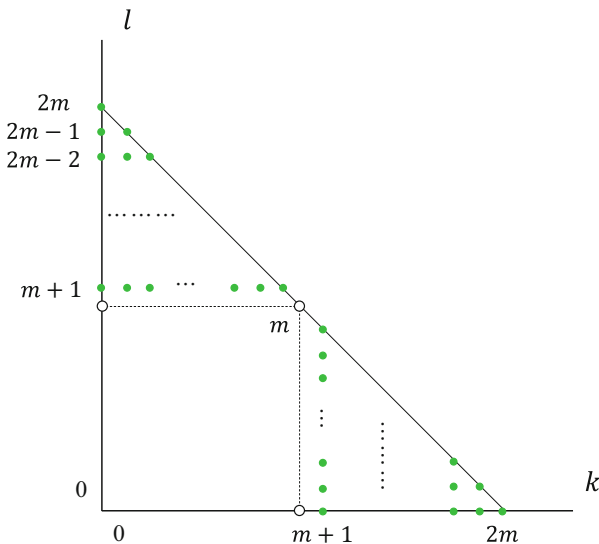
$$\exp(A + B) = \exp A \exp B = \exp B \exp A. \quad (15.17)$$

Proof Since A and B are mutually commutative, $(A + B)^n$ can be expanded through the binomial theorem such that

$$\frac{1}{n!} (A + B)^n = \sum_{k=0}^n \frac{A^k}{k!} \frac{B^{n-k}}{(n-k)!}. \quad (15.18)$$

Meanwhile, with an arbitrary integer m , we have

Fig. 15.1 Number of possible combinations (k, l) . Such combinations are indicated with green points. The number of the green points of upper left triangle is $m(m+1)/2$. That for lower right triangle is $m(m+1)/2$ as well. The total number of the green points is $m(m+1)$ accordingly. Adapted from Yamanouchi T, Sugiura M (1960) Introduction to continuous groups (New Mathematics Series 18; in Japanese) [4], with the permission of Baifukan Co., Ltd., Tokyo



$$\sum_{n=0}^{2m} \frac{1}{n!} (A+B)^n = \left(\sum_{n=0}^m \frac{A^n}{n!} \right) \left(\sum_{n=0}^m \frac{B^n}{n!} \right) + R_m, \quad (15.19)$$

where R_m has a form of $\left(\sum_{(k,l)} A^k B^l \right) / k!l!$, where the summation is taken over all combinations of (k, l) that satisfy $\max(k, l) \geq m+1$ and $k+l \leq 2m$. The number of all the combinations (k, l) is $m(m+1)$ accordingly; see Fig. 15.1 [4]. We remark that if in (15.19) we put $m=0$, (15.19) trivially holds with $R_m=0$ (i.e., zero matrix). The matrix R_m , however, changes with increasing m , and so we must evaluate it at $m \rightarrow \infty$. Putting $\max(\|A\|, \|B\|, 1) = C$ and using Theorem 15.1, we get

$$\|R_m\| \leq \sum_{(k,l)} \frac{\|A\|^k \|B\|^l}{k!l!}. \quad (15.20)$$

In (15.20) $\min(k, l) = 0$, and so $k!l! \geq (m+1)!$. Since $\|A\|^k \|B\|^l \leq C^{k+l} \leq C^{2m}$, we have

$$\sum_{(k,l)} \frac{\|A\|^k \|B\|^l}{k!l!} \leq \frac{m(m+1)C^{2m}}{(m+1)!} = \frac{C^{2m}}{(m-1)!}, \quad (15.21)$$

where $m(m+1)$ in the numerator comes from the number of combinations (k, l) as pointed out above. From Stirling's interpolation formula [5], we have

$$\Gamma(z+1) \approx \sqrt{2\pi} e^{-z} z^{z+\frac{1}{2}}. \quad (15.22)$$

Replacing z with an integer $m-1$, we get

$$\Gamma(m) = (m-1)! \approx \sqrt{2\pi} e^{-(m-1)} (m-1)^{m-\frac{1}{2}}. \quad (15.23)$$

Then, we have

$$\begin{aligned} [\text{RHS of (15.21)}] &= \frac{C^{2m}}{(m-1)!} \approx \frac{e^{m-1}}{\sqrt{2\pi}} \frac{C^{2m}}{(m-1)^{m-\frac{1}{2}}} \\ &= \frac{C}{\sqrt{2\pi}e} \left(\frac{eC^2}{m-1} \right)^{m-\frac{1}{2}} \end{aligned} \quad (15.24)$$

and

$$\lim_{m \rightarrow \infty} \frac{C}{\sqrt{2\pi}e} \left(\frac{eC^2}{m-1} \right)^{m-\frac{1}{2}} = 0. \quad (15.25)$$

Thus, from (15.20) and (15.21), we get

$$\lim_{m \rightarrow \infty} \|R_m\| = 0. \quad (15.26)$$

Remember that $\|R_m\| = 0$ if and only if $R_m = 0$ (zero matrix); see (13.4) and (13.8) in Sect. 13.1. Finally, taking limit ($m \rightarrow \infty$) of both sides of (15.19), we obtain

$$\lim_{m \rightarrow \infty} \sum_{n=0}^{2m} \frac{1}{n!} (A+B)^n = \lim_{m \rightarrow \infty} \left[\left(\sum_{n=0}^m \frac{A^n}{n!} \right) \left(\sum_{n=0}^m \frac{B^n}{n!} \right) + R_m \right].$$

Considering (15.26), we get

$$\lim_{m \rightarrow \infty} \sum_{n=0}^{2m} \frac{1}{n!} (A+B)^n = \lim_{m \rightarrow \infty} \left(\sum_{n=0}^m \frac{A^n}{n!} \right) \left(\sum_{n=0}^m \frac{B^n}{n!} \right). \quad (15.27)$$

This implies that (15.17) holds. It is obvious that if A and B commute, then $\exp A$ and $\exp B$ commute. That is, $\exp A \exp B = \exp B \exp A$. These complete the proof.

From Theorem 15.2, we immediately get the following property.

$$(1) (\exp A)^{-1} = \exp(-A). \quad (15.28)$$

To show this, it suffices to find that A and $-A$ commute. That is, $A(-A) = -A^2 = (-A)A$. Replacing B with $-A$ in (15.17), we have

$$\exp(A - A) = \exp 0 = E = \exp A \exp(-A) = \exp(-A) \exp A. \quad (15.29)$$

This leads to (15.28). Notice here that $\exp 0 = E$ from (15.7). We further have important properties of exponential functions of matrices.

$$(2) P^{-1}(\exp A)P = \exp(P^{-1}AP). \quad (15.30)$$

$$(3) \exp(A^T) = (\exp A)^T. \quad (15.31)$$

$$(4) \exp(A^\dagger) = (\exp A)^\dagger. \quad (15.32)$$

To show (2) we have

$$\begin{aligned} (P^{-1}AP)^n &= (P^{-1}AP)(P^{-1}AP)\cdots(P^{-1}AP) \\ &= P^{-1}A(P P^{-1})A(P P^{-1})\cdots(P P^{-1})AP \\ &= P^{-1}AEAE\cdots EAP = P^{-1}A^n P. \end{aligned} \quad (15.33)$$

Applying (15.33)–(15.7), we have

$$\begin{aligned} \exp(P^{-1}AP) &= E + P^{-1}AP + (P^{-1}AP)^2\left(\frac{1}{2!}\right) + \cdots + (P^{-1}AP)^n\left(\frac{1}{n!}\right) + \cdots \\ &= P^{-1}EP + P^{-1}AP + P^{-1}\frac{A^2}{2!}P + \cdots + P^{-1}\frac{A^n}{n!}P + \cdots \\ &= P^{-1}\left(E + A + \frac{A^2}{2!} + \cdots + \frac{A^n}{n!} + \cdots\right)P = P^{-1}(\exp A)P. \end{aligned} \quad (15.34)$$

With (3) and (4), use the following equations:

$$(A^T)^n = (A^n)^T \text{ and } (A^\dagger)^n = (A^n)^\dagger. \quad (15.35)$$

The confirmation of (15.31) and (15.32) is left for readers. For future purpose, we describe other important theorem and properties of the exponential functions of matrices.

Theorem 15.3 Let $a_1, a_2, \dots, a_n$ be eigenvalues of a (n, n) square matrix A . Then, $\exp a_1, \exp a_2, \dots, \exp a_n$ are eigenvalues of $\exp A$.

Proof From Theorem 12.1 we know that every (n, n) square matrix can be converted to a triangle matrix by similarity transformation. Also, we know that eigenvalues of a triangle matrix are given by its diagonal elements (Sect. 12.1). Let P be a non-singular matrix used for such a similarity transformation. Then, we have

$$P^{-1}AP = \begin{pmatrix} a_1 & & * \\ & \ddots & \\ 0 & & a_n \end{pmatrix}. \quad (15.36)$$

Meanwhile, let T be a triangle matrix described by

$$T = \begin{pmatrix} t_{11} & & * \\ & \ddots & \\ 0 & & t_{nn} \end{pmatrix}. \quad (15.37)$$

Then, from a property of the triangle matrix, we have

$$T^m = \begin{pmatrix} t_{11}^m & & * \\ & \ddots & \\ 0 & & t_{nn}^m \end{pmatrix}.$$

Applying this property to (15.36), we get

$$(P^{-1}AP)^m = \begin{pmatrix} a_1^m & & * \\ & \ddots & \\ 0 & & a_n^m \end{pmatrix} = P^{-1}A^mP, \quad (15.38)$$

where the last equality is due to (15.33). Summing (15.38) over m following (15.34), we obtain another triangle matrix expressed as

$$P^{-1}(\exp A)P = (\exp a_1 \ * \ \cdots \ 0 \ \exp a_n). \quad (15.39)$$

Once again considering that eigenvalues of a triangle matrix are given by its diagonal elements and that the eigenvalues are invariant under a similarity transformation, (15.39) implies that $\exp a_1, \exp a_2, \dots, \exp a_n$ are eigenvalues of $\exp A$. These complete the proof.

A question of whether the eigenvalues are a proper eigenvalue or a generalized eigenvalue is irrelevant to Theorem 15.3. In other words, the eigenvalue may be a proper one or a generalized one (Sect. 12.6). That depends solely upon the nature of A in question.

Theorem 15.3 immediately leads to a next important relation. Taking a determinant of (15.39), we get

$$\begin{aligned} \det[P^{-1}(\exp A)P] &= \det P^{-1} \det(\exp A) \det P = \det(\exp A) \\ &= (\exp a_1)(\exp a_2) \cdots (\exp a_n) = \exp(a_1 + a_2 + \cdots + a_n) \\ &= \exp(\text{Tr} A). \end{aligned} \quad (15.40)$$

To derive (15.40), refer to (11.59) and (12.10) as well. Since (15.40) represents an important property of the exponential functions of matrices, we restate it separately as a following formula [4]:

$$(5) \det(\exp A) = \exp(\text{Tr}A). \quad (15.41)$$

We have examined general aspects until now. Nonetheless, we have not yet studied the relationship between the exponential functions of matrices and specific types of matrices so far. Regarding the applications of exponential functions of matrices to mathematical physics, especially with the transformations of functions and vectors, we frequently encounter orthogonal matrices and unitary matrices. For this purpose, we wish to examine the following properties:

- (6) Let A be a real skew-symmetric matrix. Then, $\exp A$ is a real orthogonal matrix.
 (7) Let A be an anti-Hermitian matrix. Then, $\exp A$ is a unitary matrix.

To show (6), we note that a skew-symmetric matrix is described as

$$A^T = -A \text{ or } A^T + A = 0. \quad (15.42)$$

Then, we have $A^T A = -A^2 = AA^T$ and, hence, A and A^T commute. Therefore, from Theorem 15.2, we have

$$\exp(A + A^T) = \exp 0 = E = \exp A \exp(A^T) = \exp A (\exp A)^T, \quad (15.43)$$

where the last equality comes from Property (3) of (15.31). Equation (15.43) implies that $\exp A$ is a real orthogonal matrix.

Since an anti-Hermitian matrix A is expressed as

$$A^\dagger = -A \text{ or } A^\dagger + A = 0, \quad (15.44)$$

we have $A^\dagger A = -A^2 = AA^\dagger$, and so A and $A^\dagger$ commute. Using (15.32), we have

$$\exp(A + A^\dagger) = \exp 0 = E = \exp A \exp(A^\dagger) = \exp A (\exp A)^\dagger, \quad (15.45)$$

where the last equality comes from Property (4) of (15.32). This implies that $\exp A$ is a unitary matrix.

Regarding Properties (6) and (7), if A is replaced with tA (t : real), the relations (15.42)–(15.45) are held unchanged. Then, Properties (6) and (7) are rewritten as

(6') Let A be a real skew-symmetric matrix. Then, $\exp tA$ (t : real) is a real orthogonal matrix.

(7') Let A be an anti-Hermitian matrix. Then, $\exp tA$ (t : real) is a unitary matrix.

Now, let a function $F(t)$ be expressed as $F(t) = \exp tx$ with real numbers t and x . Then, we have a well-known formula

$$\frac{d}{dt}F(t) = xF(t). \quad (15.46)$$

Next, we extend (15.46) to the exponential functions of matrices. Let us define the differentiation of an exponential function of a matrix with respect to a real parameter t . Then, in concert with (15.46), we have a following important theorem.

Theorem 15.4 [1, 2] Let $F(t) \equiv \exp tA$ with t being a real number and A being a matrix that does not depend on t . Then, we have

$$\frac{d}{dt}F(t) = AF(t). \quad (15.47)$$

In (15.47) we assume that individual matrix components of $F(t)$ are differentiable with respect to t and that the differentiation of a matrix is defined as in (15.2).

Proof We have

$$\begin{aligned} F(t + \Delta t) &= \exp(t + \Delta t)A = \exp(tA + \Delta tA) = (\exp tA)(\exp \Delta tA) \\ &= F(t)F(\Delta t) = (\exp \Delta tA)(\exp tA) = F(\Delta t)F(t), \end{aligned} \quad (15.48)$$

where we considered that tA and ΔtA are commutative and used Theorem 15.2. Therefore, we get

$$\begin{aligned} \frac{1}{\Delta t}[F(t + \Delta t) - F(t)] &= \frac{1}{\Delta t}[F(\Delta t) - E]F(t) = \frac{1}{\Delta t} \left[\sum_{\nu=0}^{\infty} \frac{1}{\nu!} (\Delta tA)^{\nu} - E \right] F(t) \\ &= \left[\sum_{\nu=1}^{\infty} \frac{1}{\nu!} (\Delta t)^{\nu-1} A^{\nu} \right] F(t). \end{aligned} \quad (15.49)$$

Taking the limit of both sides of (15.49), we get

$$\lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} [F(t + \Delta t) - F(t)] \equiv \frac{d}{dt}F(t) = AF(t).$$

Notice that in (15.49) only the first term (i.e., $\nu = 1$) of RHS is non-vanishing with respect to $\Delta t \rightarrow 0$. Then, (15.47) will follow. This completes the proof.

On the basis of the power series expansion of the exponential function of a matrix defined in (15.7), A and $F(t)$ are commutative. Therefore, instead of (15.47) we may write

$$\frac{d}{dt}F(t) = F(t)A. \quad (15.50)$$

Using Theorem 15.4, we show the following important properties. These are converse propositions to Properties (6) and (7).

- (8) Let $\exp tA$ be a real orthogonal matrix with any real number t . Then, A is a real skew-symmetric matrix.
- (9) Let $\exp tA$ be a unitary matrix with any real number t . Then, A is an anti-Hermitian matrix.

To show (8), from the assumption with any real number t , we have

$$\exp tA(\exp tA)^T = \exp tA \exp(tA^T) = E. \quad (15.51)$$

Differentiating (15.51) with respect to t by use of (15.4), we have

$$(A \exp tA) \exp(tA^T) + (\exp tA) A^T \exp(tA^T) = 0. \quad (15.52)$$

Since (15.52) must hold with any real number, (15.52) must hold with $t \rightarrow 0$ as well. On this condition, from (15.52) we get $A + A^T = 0$. That is, A is a real skew-symmetric matrix.

In the case of (9), similarly we have

$$\exp tA(\exp tA)^\dagger = \exp tA \exp(tA^\dagger) = E. \quad (15.53)$$

Differentiating (15.53) with respect to t and taking $t \rightarrow 0$, we get $A + A^\dagger = 0$. That is, A is an anti-Hermitian matrix.

Properties (6)–(9) including Properties (6)' and (7)' are frequently used later in Chap. 20 in relation to Lie groups and Lie algebras.

15.3 System of Differential Equations

In this section we describe important applications of the exponential functions of matrices to systems of differential equations.

15.3.1 Introduction

In Chap. 10 we have dealt with SOLDEs using Green's functions. In the present section, we examine the properties of systems of differential equations. There is a close relationship between the two topics. First we briefly outline it.

We describe an example of the system of differential equations. First, let us think of the following general equation:

$$\dot{x}(t) = a(t)x(t) + b(t)y(t), \quad (15.54)$$

$$y(\dot{t}) = c(t)x(t) + d(t)y(t), \quad (15.55)$$

where $x(t)$ and $y(t)$ vary as a function of t ; $a(t)$, $b(t)$, $c(t)$ and $d(t)$ are coefficients. Differentiating (15.54) with respect to t , we get

$$\begin{aligned} x(\ddot{t}) &= a(\dot{t})x(t) + a(t)x(\dot{t}) + b(\dot{t})y(t) + b(t)y(\dot{t}) \\ &= a(\dot{t})x(t) + a(t)x(\dot{t}) + b(\dot{t})y(t) + b(t)[c(t)x(t) + d(t)y(t)] \\ &= [a(\dot{t}) + b(t)c(t)]x(t) + a(t)x(\dot{t}) + [b(\dot{t}) + b(t)d(t)]y(t), \end{aligned} \quad (15.56)$$

where with the second equality we used (15.55). To delete $y(t)$ from (15.56), we multiply $b(t)$ on both sides of (15.56) and multiply $b(\dot{t}) + b(t)d(t)$ on both sides of (15.54) and further subtract the latter equation from the former. As a result, we obtain

$$b\ddot{x} - [\dot{b} + (a + d)b]\dot{x} + [a(\dot{b} + bd) - b(\dot{a} + bc)]x = 0. \quad (15.57)$$

Upon the above derivation, we assume $b \neq 0$. Solving (15.57) and substituting the resulting solution for (15.54), we can get a solution for $y(t)$. If, on the other hand, $b = 0$, we have $x(\dot{t}) = a(t)x(t)$. This is a simple FOLDE and can readily be integrated to yield

$$x = C \exp\left(\int^t a(t')dt'\right), \quad (15.58)$$

where C is an integration constant.

Meanwhile, differentiating (15.55) with respect to t , we have

$$\begin{aligned} \ddot{y} &= \dot{c}x + c\dot{x} + \dot{d}y + d\dot{y} = \dot{c}x + cax + \dot{d}y + d\dot{y} \\ &= (\dot{c} + ca) C \exp\left(\int^t a(t')dt'\right) + \dot{d}y + d\dot{y}, \end{aligned}$$

where we used (15.58) as well as $\dot{x} = ax$. Thus, we are going to solve a following inhomogeneous SOLDE:

$$\ddot{y} - d\dot{y} - \dot{d}y = (\dot{c} + ca)C \exp\left(\int^t a(t')dt'\right). \quad (15.59)$$

In Sect. 10.2 we have shown how we are able to solve an inhomogeneous FOLDE using a weight function. For a later purpose, let us consider a next FOLDE assuming $a(x) \equiv 1$ in (10.23):

$$\dot{x}(t) + p(t)x = q(t). \quad (15.60)$$

As another method to solve (15.60), we examine the method of variation of constants. The homogeneous equation corresponding to (15.60) is

$$\dot{x}(t) + p(t)x = 0. \quad (15.61)$$

It can be solved as just before such that

$$x(t) = C \exp\left(-\int^t p(t')dt'\right) \equiv u(t). \quad (15.62)$$

Now we assume that a solution of (15.60) can be sought by putting

$$x(t) = k(t)u(t), \quad (15.63)$$

where we assume that the functional form $u(t)$ remains unchanged in the inhomogeneous equation (15.60) and that instead the “constant” $k(t)$ may change as a function of t . Inserting (15.63) into (15.60), we have

$$\begin{aligned} \dot{x} + px &= \dot{k}u + k\dot{u} + pku = \dot{k}u + k(\dot{u} + pu) = \dot{k}u + k(-pu + pu) \\ &= \dot{k}u = q, \end{aligned} \quad (15.64)$$

where with the third equality we used (15.61) and (15.62) to get $\dot{u} = -pu$. In this way, (15.64) can easily be integrated to give

$$k(t) = \int^t \frac{q(t')}{u(t')} dt' + C,$$

where C is an integration constant. Thus, as a solution of (15.60), we have

$$x(t) = k(t)u(t) = u(t) \int^t \frac{q(t')}{u(t')} dt' + Cu(t). \quad (15.65)$$

Comparing (15.65) with (10.29), we notice that these two expressions are related. In other words, the method of variation of constants in (15.65) is essentially the same as that using the weight function in (10.29).

Another important point to bear in mind is that in Chap. 10 we have solved SOLDEs of constant coefficients using the method of Green’s functions. In that case, we have separately estimated a homogeneous term and inhomogeneous term (i.e., surface term). In the present section, we are going to seek a method corresponding to that using Green’s functions. In the above discussion, we see that a system of differential equations with two unknowns can be translated into a SOLDE. Then,

we expect such a system of differential equations of constant coefficients to be solved somehow or other. We will hereinafter describe it in detail giving examples.

15.3.2 *System of Differential Equations in a Matrix Form: Resolvent Matrix*

The equations of (15.54) and (15.55) are unified in a homogeneous single matrix equation such that

$$\begin{pmatrix} \dot{x}(t) & \dot{y}(t) \end{pmatrix} = (x(t) \ y(t)) \begin{pmatrix} a(t) & c(t) \\ b(t) & d(t) \end{pmatrix}. \quad (15.66)$$

In (15.66), $x(t)$ and $y(t)$ and their derivatives represent the functions (of t), and so we describe them as row matrices. With this expression of equation, see Sect. 11.2 and (11.37) therein. Another expression is a transposition of (15.66) described by

$$\begin{pmatrix} \dot{x}(t) \\ \dot{y}(t) \end{pmatrix} = \begin{pmatrix} a(t) & b(t) \\ c(t) & d(t) \end{pmatrix} \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}. \quad (15.67)$$

Notice that the coefficient matrix has been transposed in (15.67) accordingly. We are most interested in an equation of constant coefficient, and hence, we rewrite (15.66) once again explicitly as

$$\begin{pmatrix} \dot{x}(t) & \dot{y}(t) \end{pmatrix} = (x(t) \ y(t)) \begin{pmatrix} a & c \\ b & d \end{pmatrix} \quad (15.68)$$

or

$$\frac{d}{dt}(x(t) \ y(t)) = (x(t) \ y(t)) \begin{pmatrix} a & c \\ b & d \end{pmatrix}. \quad (15.69)$$

Putting $F(t) \equiv (x(t) \ y(t))$, we have

$$\frac{dF(t)}{dt} = F(t) \begin{pmatrix} a & c \\ b & d \end{pmatrix}. \quad (15.70)$$

This is exactly the same as (15.50) if we put

$$A = \begin{pmatrix} a & c \\ b & d \end{pmatrix}. \quad (15.71)$$

Then, we get

$$F(t) = \exp tA. \quad (15.72)$$

As already shown in (15.7) of Sect. 15.1, an exponential function of a matrix, i.e., $\exp tA$, can be defined as

$$\exp tA \equiv E + tA + \frac{1}{2!}(tA)^2 + \cdots + \frac{1}{\nu!}(tA)^\nu + \cdots. \quad (15.73)$$

From Property (5) of (15.41), we have $\det(\exp A) \neq 0$. This is because no matter what number (either real or complex) $\text{Tr}A$ may take, $\exp t(\text{Tr}A)$ never vanishes. That is, $F(t) = \exp A$ is non-singular, and so an inverse matrix of $\exp A$ must exist. This is the case with $\exp tA$ as well; see equation below:

$$(5)' \det(\exp tA) = \exp \text{Tr}(tA) = \exp t(\text{Tr}A). \quad (15.74)$$

Note that if A (or tA) is a (n, n) matrix, $F(t)$ of (15.72) is a (n, n) matrix as well. Since in that general case $F(t)$ is non-singular, it consists of n linearly independent column vectors, i.e., $(n, 1)$ matrices; see Chap. 11. Then, $F(t)$ is symbolically described as

$$F(t) = (\mathbf{x}_1(t) \ \mathbf{x}_2(t) \ \cdots \ \mathbf{x}_n(t)),$$

where $\mathbf{x}_i(t)$ ($1 \leq i \leq n$) represents a i -th column vector. Note that since $F(0) = E$, $\mathbf{x}_i(0)$ has its i -th row component of 1, otherwise 0. Since $F(t)$ is non-singular, n column vectors $\mathbf{x}_i(t)$ ($1 \leq i \leq n$) are linearly independent. In the case of, e.g., (15.66), A is a $(2, 2)$ matrix, which implies that $x(t)$ and $y(t)$ represent two linearly independent column vectors, i.e., $(2, 1)$ matrices. Furthermore, if we look at the constitution of (15.70), $x(t)$ and $y(t)$ represent two linearly independent solutions of (15.70). We will come back to this point later.

On the basis of the method of variation of constants, we deal with inhomogeneous equations and describe below the procedures for solving the system of equations with two unknowns. But, they can be readily conformed to the general case where the equations with n unknowns are dealt with.

Now, we rewrite (15.68) symbolically such that

$$\dot{\mathbf{x}}(t) = \mathbf{x}(t)A, \quad (15.75)$$

where $\mathbf{x}(t) \equiv (x(t) \ y(t))$, $\dot{\mathbf{x}}(t) \equiv (\dot{x}(t) \ \dot{y}(t))$, and $A \equiv \begin{pmatrix} a & c \\ b & d \end{pmatrix}$. Then, the inhomogeneous equation can be described as

$$\dot{\mathbf{x}}(t) = \mathbf{x}(t)A + \mathbf{b}(t), \quad (15.76)$$

where we define the inhomogeneous term as $\mathbf{b} \equiv (p \ q)$. This is equivalent to

$$\begin{pmatrix} \dot{x}(t) & \dot{y}(t) \end{pmatrix} = (x(t) \ y(t)) \begin{pmatrix} a & c \\ b & d \end{pmatrix} + (p \ q). \quad (15.77)$$

Using the same solution $F(t)$ that appears in the homogeneous equation, we assume that the solution of the inhomogeneous equation is described by

$$\mathbf{x}(t) = \mathbf{k}(t)F(t), \quad (15.78)$$

where $\mathbf{k}(t)$ is a variable “constant.” Replacing $\mathbf{x}(t)$ in (15.76) with $\mathbf{x}(t) = \mathbf{k}(t)F(t)$ of (15.78), we obtain

$$\dot{\mathbf{x}} = \dot{\mathbf{k}}F + \mathbf{k}\dot{F} = \dot{\mathbf{k}}F + \mathbf{k}FA = \mathbf{k}FA + \mathbf{b}, \quad (15.79)$$

where with the second equality we used $\dot{F} = FA$ in (15.50). Then, we have

$$\dot{\mathbf{k}}F = \mathbf{b}. \quad (15.80)$$

This can be integrated so that we may have

$$\mathbf{k} = \int^t ds \left[\mathbf{b}(s)F(s)^{-1} \right]. \quad (15.81)$$

As previously noted, $F(s)^{-1}$ exists because $F(s)$ is non-singular. Thus, as a solution we get

$$\mathbf{x}(t) = \mathbf{k}(t)F(t) = \int_{t_0}^t ds \left[\mathbf{b}(s)F(s)^{-1}F(t) \right]. \quad (15.82)$$

Here we define

$$\mathfrak{R}(s, t) \equiv F(s)^{-1}F(t). \quad (15.83)$$

The matrix $\mathfrak{R}(s, t)$ is said to be a *resolvent matrix* [6]. This matrix plays an essential role in solving the system of differential equations both homogeneous and inhomogeneous with either homogeneous or inhomogeneous boundary conditions (BCs). Rewriting (15.82), we get

$$\mathbf{x}(t) = \mathbf{k}(t)F(t) = \int_{t_0}^t ds [\mathbf{b}(s)\mathfrak{R}(s, t)]. \quad (15.84)$$

We summarize principal characteristics of the resolvent matrix.

$$(1) \mathfrak{R}(s, s) = F(s)^{-1}F(s) = E, \quad (15.85)$$

where E is a (2, 2) unit matrix, i.e., $E = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ for an equation having two unknowns.

$$(2) \mathfrak{R}(s, t)^{-1} = [F(s)^{-1}F(t)]^{-1} = F(t)^{-1}F(s) = \mathfrak{R}(t, s). \quad (15.86)$$

$$(3) \mathfrak{R}(s, t)\mathfrak{R}(t, u) = F(s)^{-1}F(t)F(t)^{-1}F(u) = F(s)^{-1}F(u) = \mathfrak{R}(s, u). \quad (15.87)$$

It is easy to include a term in solution related to the inhomogeneous BCs. As already seen in (10.33) of Sect. 10.2, a boundary condition for the FOLDEs is set such that, e.g.,

$$u(a) = \sigma, \quad (15.88)$$

where $u(t)$ is a solution of a FOLDE. In that case, the BC can be translated into the “initial” condition. In the present case, we describe the BCs as

$$\mathbf{x}(t_0) \equiv \mathbf{x}_0 \equiv (\sigma \ \tau), \quad (15.89)$$

where $\sigma \equiv x(t_0)$ and $\tau \equiv y(t_0)$. Then, the said term associated with the inhomogeneous BCs is expected to be written as

$$\mathbf{x}(t_0)\mathfrak{R}(t_0, t).$$

In fact, since $\mathbf{x}(t_0)\mathfrak{R}(t_0, t_0) = \mathbf{x}(t_0)E = \mathbf{x}(t_0)$, this satisfies the BCs. Then, the full expression of the solution of the inhomogeneous equation that takes account of the BCs is assumed to be expressed as

$$\mathbf{x}(t) = \mathbf{x}(t_0)\mathfrak{R}(t_0, t) + \int_{t_0}^t ds [\mathbf{b}(s)\mathfrak{R}(s, t)]. \quad (15.90)$$

Let us confirm that (15.90) certainly gives a proper solution of (15.76). First we give a formula on the differentiation of

$$Q(t) = \frac{d}{dt} \int_{t_0}^t P(t, s) ds, \quad (15.91)$$

where $P(t, s)$ stands for a $(1, 2)$ matrix whose general form is described by $(q(t, s), r(t, s))$, in which $q(t, s)$ and $r(t, s)$ are functions of t and s . We rewrite (15.91) as

$$\begin{aligned} Q(t) &= \lim_{\Delta \rightarrow 0} \frac{1}{\Delta} \left[\int_{t_0}^{t+\Delta} P(t+\Delta, s) ds - \int_{t_0}^t P(t, s) ds \right] \\ &= \lim_{\Delta \rightarrow 0} \frac{1}{\Delta} \left[\int_{t_0}^{t+\Delta} P(t+\Delta, s) ds - \int_{t_0}^t P(t+\Delta, s) ds + \int_{t_0}^t P(t+\Delta, s) ds \right. \\ &\quad \left. - \int_{t_0}^t P(t, s) ds \right] \\ &= \lim_{\Delta \rightarrow 0} \frac{1}{\Delta} \left[\int_t^{t+\Delta} P(t+\Delta, s) ds + \int_{t_0}^t P(t+\Delta, s) ds - \int_{t_0}^t P(t, s) ds \right] \\ &\approx \lim_{\Delta \rightarrow 0} \frac{1}{\Delta} P(t, t) \Delta + \lim_{\Delta \rightarrow 0} \frac{1}{\Delta} \left[\int_{t_0}^t P(t+\Delta, s) ds - \int_{t_0}^t P(t, s) ds \right] \\ &= P(t, t) + \int_{t_0}^t \frac{\partial P(t, s)}{\partial t} ds. \end{aligned} \quad (15.92)$$

Replacing $P(t, s)$ of (15.92) with $\mathbf{b}(s)\mathfrak{R}(s, t)$, we have

$$\begin{aligned} \frac{d}{dt} \int_{t_0}^t \mathbf{b}(s)\mathfrak{R}(s, t) ds &= \mathbf{b}(t)\mathfrak{R}(t, t) + \int_{t_0}^t \mathbf{b}(s) \frac{\partial \mathfrak{R}(s, t)}{\partial t} ds \\ &= \mathbf{b}(t) + \int_{t_0}^t \mathbf{b}(s) \frac{\partial \mathfrak{R}(s, t)}{\partial t} ds, \end{aligned} \quad (15.93)$$

where with the last equality we used (15.85). Considering (15.83) and (15.93) and differentiating (15.90) with respect to t , we get

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{x}(t_0)F(t_0)^{-1}\dot{F}(t) + \int_{t_0}^t ds \left[\mathbf{b}(s)F(s)^{-1}\dot{F}(t) \right] + \mathbf{b}(t) \\ &= \mathbf{x}(t_0)F(t_0)^{-1}F(t)A + \int_{t_0}^t ds \left[\mathbf{b}(s)F(s)^{-1}F(t)A \right] + \mathbf{b}(t) \\ &= \mathbf{x}(t_0)\mathfrak{R}(t_0, t)A + \int_{t_0}^t ds [\mathbf{b}(s)\mathfrak{R}(s, t)A] + \mathbf{b}(t) \end{aligned}$$

$$\begin{aligned}
&= \left\{ \mathbf{x}(t_0)\mathfrak{R}(t_0, t) + \int_{t_0}^t ds [\mathbf{b}(s)\mathfrak{R}(s, t)] \right\} A + \mathbf{b}(t) \\
&= \mathbf{x}(t)A + \mathbf{b}(t),
\end{aligned} \tag{15.94}$$

where with the second equality we used (15.70) and with the last equality we used (15.90). Thus, we have certainly recovered the original differential equation of (15.76). Consequently, (15.90) is the proper solution for the given inhomogeneous equation with inhomogeneous BCs.

The above discussion and formulation equally apply to the general case of the differential equation with n unknowns, even though the calculation procedures become increasingly complicated with the increasing number of unknowns.

15.3.3 Several Examples

To deepen our understanding of the essence and characteristics of the system of differential equations, we deal with several examples regarding the equations of two unknowns with constant coefficients. The constant matrices A of specific types (e.g., anti-Hermitian matrices and skew-symmetric matrices) have a wide field of applications in mathematical physics, especially in Lie groups and Lie algebras (see Chap. 20). In the subsequent examples, however, we deal with various types of matrices.

Example 15.1 Solve the following equation

$$\begin{aligned}
x(\dot{t}) &= x + 2y + 1, \\
y(\dot{t}) &= 2x + y + 2,
\end{aligned} \tag{15.95}$$

under the BCs $x(0) = c$ and $y(0) = d$.

In a matrix form, (15.95) can be written as

$$\begin{pmatrix} x(\dot{t}) & y(\dot{t}) \end{pmatrix} = (x \ y) \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} + (1 \ 2). \tag{15.96}$$

The homogeneous equation corresponding to (15.96) is expressed as

$$\begin{pmatrix} x(\dot{t}) & y(\dot{t}) \end{pmatrix} = (x \ y) \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}. \tag{15.97}$$

As discussed in Sect. 15.3.2, this can readily be solved to produce a solution $F(t)$ such that

$$F(t) = \exp tA, \quad (15.98)$$

where $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$. Since A is a real symmetric (i.e., Hermitian) matrix, it should be diagonalized by the unitary similarity transformation (see Sect. 14.3). For the purpose of getting an exponential function of a matrix, it is easier to estimate $\exp tD$ (where D is a diagonal matrix) than to directly calculate $\exp tA$. That is, we rewrite (15.97) as

$$\begin{pmatrix} \dot{x}(t) & \dot{y}(t) \end{pmatrix} = (x \ y) U U^\dagger \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} U U^\dagger, \quad (15.99)$$

where U is a unitary matrix, i.e., $U^\dagger = U^{-1}$.

Following routine calculation procedures (as in, e.g., Example 14.3), we readily get a following diagonal matrix and the corresponding unitary matrix for the diagonalization. That is, we obtain

$$U = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

and

$$D = U^{-1} \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} U = U^{-1} A U = \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}. \quad (15.100)$$

Or we have

$$A = U D U^{-1}. \quad (15.101)$$

Hence, we get

$$F(t) = \exp tA = \exp(tUDU^{-1}) = U(\exp tD)U^{-1}, \quad (15.102)$$

where with the last equality we used (15.30). From Theorem 15.3, we have

$$\exp tD = \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{3t} \end{pmatrix}. \quad (15.103)$$

In turn, we get

$$\begin{aligned}
 F(t) &= \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{3t} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \\
 &= \frac{1}{2} \begin{pmatrix} e^{-t} + e^{3t} & -e^{-t} + e^{3t} \\ -e^{-t} + e^{3t} & e^{-t} + e^{3t} \end{pmatrix}. \tag{15.104}
 \end{aligned}$$

With an inverse matrix, we have

$$F(t)^{-1} = \frac{1}{2} \begin{pmatrix} e^{-3t} + e^t & e^{-3t} - e^{3t} \\ e^{-3t} - e^{3t} & e^{-3t} + e^t \end{pmatrix}. \tag{15.105}$$

Therefore, as a resolvent matrix, we get

$$\begin{aligned}
 \mathfrak{R}(s, t) &= F(s)^{-1} F(t) \\
 &= \frac{1}{2} \begin{pmatrix} e^{-(t-s)} + e^{3(t-s)} & -e^{-(t-s)} + e^{3(t-s)} \\ -e^{-(t-s)} + e^{3(t-s)} & e^{-(t-s)} + e^{3(t-s)} \end{pmatrix}. \tag{15.106}
 \end{aligned}$$

Thus, as a solution of (15.95), we obtain

$$\mathbf{x}(t) = \mathbf{x}(0)\mathfrak{R}(0, t) + \int_0^t ds [\mathbf{b}(s)\mathfrak{R}(s, t)], \tag{15.107}$$

where $\mathbf{x}(0) = (x(0) \ y(0)) = (c \ d)$ represents the BCs and $\mathbf{b}(s) = (1 \ 2)$ comes from (15.96). This can easily be calculated so that we can have

$$\begin{aligned}
 \mathbf{x}(0)\mathfrak{R}(0, t) &= \left(\frac{1}{2} [(c-d)e^{-t} + (c+d)e^{3t}] \quad \frac{1}{2} [(d-c)e^{-t} + (c+d)e^{3t}] \right) \tag{15.108}
 \end{aligned}$$

and

$$\int_0^t ds [\mathbf{b}(s)\mathfrak{R}(s, t)] = \left(\frac{1}{2} (e^{-t} + e^{3t}) - 1 \quad -\frac{1}{2} (e^{-t} - e^{3t}) \right). \tag{15.109}$$

Note that the first component of (15.108) and (15.109) represents the x component of the solution for (15.95) and that the second component represents the y component. From (15.108), we have

$$\mathbf{x}(0)\mathfrak{R}(0, 0) = \mathbf{x}(0)E = \mathbf{x}(0) = (c \ d). \tag{15.110}$$

Thus, we find that the BCs are certainly satisfied. Putting $t = 0$ in (15.109), we have

$$\int_0^0 ds [\mathbf{b}(s) \mathfrak{R}(s, 0)] = 0, \quad (15.111)$$

confirming that the second term of (15.107) vanishes at $t = 0$.

The summation of (15.108) and (15.109) gives an overall solution of (15.107). Separating individual components of the solution, we describe them as

$$\begin{aligned} x(t) &= \frac{1}{2} [(c - d + 1)e^{-t} + (c + d + 1)e^{3t} - 2], \\ y(t) &= \frac{1}{2} [(d - c - 1)e^{-t} + (c + d + 1)e^{3t}]. \end{aligned} \quad (15.112)$$

Moreover, differentiating (15.112) with respect to t , we recover the original form of (15.95). The confirmation is left for readers.

Remarks on Example 15.1: We become aware of several points in Example 15.1.

1. The resolvent matrix given by (15.106) can be obtained by replacing t of (15.104) with $t - s$. This is one of important characteristics of the resolvent matrix that is derived from an exponential function of a constant matrix.

To see it, if A is a constant matrix, tA and $-sA$ commute. By the definition (15.7) of the exponential function, we have

$$\exp tA = E + tA + \frac{1}{2!} t^2 A^2 + \cdots + \frac{1}{\nu!} t^\nu A^\nu + \cdots, \quad (15.113)$$

$$\exp(-sA) = E + (-sA) + \frac{1}{2!} (-s)^2 A^2 + \cdots + \frac{1}{\nu!} (-s)^\nu A^\nu + \cdots. \quad (15.114)$$

Both (15.113) and (15.114) are polynomials of a constant matrix A and, hence, $\exp tA$ and $\exp(-sA)$ commute as well. Then, from Theorem 15.2 and Property (1) of (15.28), we get

$$\begin{aligned} \exp(tA - sA) &= (\exp tA)[\exp(-sA)] = [\exp(-sA)](\exp tA) \\ &= \exp(-sA + tA) = \exp[(t - s)A]. \end{aligned} \quad (15.115)$$

That is, using (15.28), (15.72), and (15.83), we get

$$\begin{aligned} \mathfrak{R}(s, t) &= F(s)^{-1} F(t) = \exp(sA)^{-1} \exp(tA) = \exp(-sA) \exp(tA) \\ &= \exp[(t - s)A] = F(t - s). \end{aligned} \quad (15.116)$$

This implies that once we get $\exp tA$, we can safely obtain the resolvent matrix by automatically replacing t of (15.72) with $t - s$. Also, $\exp tA$ and $\exp(-sA)$ are commutative. In the present case, therefore, by exchanging the order of products of $F(s)^{-1}$ and $F(t)$ in (15.106), we have the same result as (15.106) such that

$$\begin{aligned}\Re(s, t) &= F(t)F(s)^{-1} \\ &= \frac{1}{2} \begin{pmatrix} e^{-(t-s)} + e^{3(t-s)} & -e^{-(t-s)} + e^{3(t-s)} \\ -e^{-(t-s)} + e^{3(t-s)} & e^{-(t-s)} + e^{3(t-s)} \end{pmatrix}. \end{aligned} \quad (15.117)$$

2. The resolvent matrix of the present example is real symmetric. This is because if A is real symmetric, i.e., $A^T = A$, we have

$$(A^n)^T = (A^T)^n = A^n, \quad (15.118)$$

that is, A^n is real symmetric as well. From (15.7), $\exp A$ is real symmetric accordingly. In a similar manner, if A is Hermitian, $\exp A$ is also Hermitian.

3. Let us think of a case where A is not a constant matrix but varies as a function of t . In that case, we have

$$A(t) = \begin{pmatrix} a(t) & b(t) \\ c(t) & d(t) \end{pmatrix}. \quad (15.119)$$

Also, we have

$$A(s) = \begin{pmatrix} a(s) & b(s) \\ c(s) & d(s) \end{pmatrix}. \quad (15.120)$$

We can easily check that in a general case

$$A(t)A(s) \neq A(s)A(t), \quad (15.121)$$

namely, $A(t)$ and $A(s)$ are not generally commutative. The matrix $tA(t)$ is not commutative with $-sA(s)$, either. In turn, (15.115) does not hold either. Thus, we need an elaborated treatment for this.

Example 15.2 Find the resolvent matrix of the following homogeneous equation:

$$\begin{aligned}\dot{x}(t) &= 0, \\ \dot{y}(t) &= x + y. \end{aligned} \quad (15.122)$$

In a matrix form, (15.122) can be written as

$$\begin{pmatrix} \dot{x}(t) & \dot{y}(t) \end{pmatrix} = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}. \quad (15.123)$$

The matrix $A = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$ is characterized by an idempotent matrix (see Sect. 12.4). As in the case of Example 15.1, we get

$$\begin{pmatrix} x(t) & y(t) \end{pmatrix} = (x \ y) P P^{-1} \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} P P^{-1}, \quad (15.124)$$

where we choose $P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and, hence, $P^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$. Notice that in this example since A was not a symmetric matrix, we did not use a unitary matrix with the similarity transformation. As a diagonal matrix D , we have

$$D = P^{-1} \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} P = P^{-1} A P = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \quad (15.125)$$

Or we have

$$A = P D P^{-1}. \quad (15.126)$$

Hence, we get

$$F(t) = \exp tA = \exp(t P D P^{-1}) = P(\exp tD)P^{-1}, \quad (15.127)$$

where we used (15.30) again. From Theorem 15.3, we have

$$\exp tD = \begin{pmatrix} 1 & 0 \\ 0 & e^t \end{pmatrix}. \quad (15.128)$$

In turn, we get

$$F(t) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & e^t \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 + e^t \\ 0 & e^t \end{pmatrix}. \quad (15.129)$$

With an inverse matrix, we have

$$F(t)^{-1} = \begin{pmatrix} 1 & e^{-t} - 1 \\ 0 & e^{-t} \end{pmatrix}. \quad (15.130)$$

Therefore, as a resolvent matrix, we get

$$\mathfrak{R}(s, t) = F(s)^{-1} F(t) = \begin{pmatrix} 1 & -1 + e^{t-s} \\ 0 & e^{t-s} \end{pmatrix}. \quad (15.131)$$

Using the resolvent matrix, we can include the inhomogeneous term along with BCs (or initial conditions). Once again, by exchanging the order of products of $F(s)^{-1}$

and $F(t)$ in (15.131), we have the same resolvent matrix. This can readily be checked. Moreover, we get $\mathfrak{R}(s, t) = F(t - s)$.

Example 15.3 Solve the following equation

$$\begin{aligned} \dot{x}(t) &= x + c, \\ \dot{y}(t) &= x + y + d, \end{aligned} \quad (15.132)$$

under the BCs $x(0) = \sigma$ and $y(0) = \tau$. In a matrix form, (15.132) can be written as

$$\begin{pmatrix} \dot{x}(t) & \dot{y}(t) \end{pmatrix} = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} c & d \end{pmatrix}. \quad (15.133)$$

The matrix of a type of $M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has been fully investigated in Sect. 12.6 and characterized as a matrix that cannot be diagonalized. In such a case, we directly calculate $\exp tM$. Here, remember that product of triangle matrices of the same kind (i.e., an upper triangle matrix or a lower triangle matrix; see Sect. 12.1) is a triangle matrix of the same kind.

We have

$$M^2 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, M^3 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}, \dots$$

Repeating this, we get

$$M^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}. \quad (15.134)$$

Thus, we have

$$\begin{aligned} \exp tM &= \mathfrak{R}(0, t) = E + tM + \frac{1}{2!}t^2M^2 + \dots + \frac{1}{\nu!}t^\nu M^\nu + \dots \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + t \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} + \frac{1}{2!}t^2 \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} + \dots + \frac{1}{\nu!}t^\nu \begin{pmatrix} 1 & \nu \\ 0 & 1 \end{pmatrix} + \dots \\ &= \begin{pmatrix} e^t & t \left[1 + t + \frac{1}{2!}t^2 + \dots + \frac{1}{(\nu-1)!}t^{\nu-1} + \dots \right] \\ 0 & e^t \end{pmatrix} = \begin{pmatrix} e^t & te^t \\ 0 & e^t \end{pmatrix}. \end{aligned} \quad (15.135)$$

With the resolvent matrix, we get

$$\mathfrak{R}(s, t) = \exp(-sM)\exp tM = \begin{pmatrix} e^{t-s} & (t-s)e^{t-s} \\ 0 & e^{t-s} \end{pmatrix}. \quad (15.136)$$

Using (15.107), we get

$$\mathbf{x}(t) = (\sigma \ \tau)\mathfrak{R}(0, t) + \int_0^t ds[(c \ d)\mathfrak{R}(s, t)]. \quad (15.137)$$

Thus, we obtain the solution described by

$$\begin{aligned} x(t) &= (c + \sigma)e^t - c, \\ y(t) &= (c + \sigma)te^t + (d - c + \tau)e^t + c - d. \end{aligned} \quad (15.138)$$

From (15.138), we find that the original inhomogeneous equation (15.132) and BCs of $x(0) = \sigma$ and $y(0) = \tau$ have been recovered.

Example 15.4 Find the resolvent matrix of the following homogeneous equation:

$$\begin{aligned} \dot{x}(t) &= 0, \\ \dot{y}(t) &= x. \end{aligned} \quad (15.139)$$

In a matrix form, (15.139) can be written as

$$\begin{pmatrix} \dot{x}(t) & \dot{y}(t) \end{pmatrix} = (x \ y) \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \quad (15.140)$$

The matrix $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is characterized by a nilpotent matrix (Sect. 12.3). Since N^2 and matrices of higher order of N vanish, we have

$$\exp tN = E + tN = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}. \quad (15.141)$$

We have a corresponding resolvent matrix such that

$$\mathfrak{R}(s, t) = \begin{pmatrix} 1 & t-s \\ 0 & 1 \end{pmatrix}. \quad (15.142)$$

Using the resolvent matrix, we can include the inhomogeneous term along with BCs.

Example 15.5 As a trivial case, let us consider a following homogeneous equation:

$$\begin{pmatrix} x(\dot{t}) & y(\dot{t}) \end{pmatrix} = (x \ y) \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}, \quad (15.143)$$

where one of a and b can be zero or both of them can be zero. Even though (15.143) merely apposes two FOLDEs, we can deal with such cases in an automatic manner. From Theorem 15.3, as eigenvalues of $\exp A$ where $A = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$, we have e^a and e^b . That is, we have

$$F(t) = \exp tA = \begin{pmatrix} e^{at} & 0 \\ 0 & e^{bt} \end{pmatrix}. \quad (15.144)$$

In particular, if $a = b$ (including $a = b = 0$), we merely have the same FOLDE twice changing the arguments x with y .

Readers may well wonder why we have to argue such a trivial case expressly. That is because the theory we developed in this chapter holds widely and we are able to make a clear forecast about a constitution of the solution for differential equations of a first order. For instance, (15.144) immediately tells us that a fundamental solution for

$$x(\dot{t}) = ax(t) \quad (15.145)$$

is e^{at} . We only have to perform routine calculations with a counterpart of $x(t)$ [or $y(t)$] of $(x(t) \ y(t))$ using (15.107) to solve an inhomogeneous differential equation under BCs.

Returning back to Example 15.1, we had

$$\begin{pmatrix} x(\dot{t}) & y(\dot{t}) \end{pmatrix} = (x \ y) \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} + (1 \ 2). \quad (15.96)$$

This equation can be converted to

$$\begin{pmatrix} x(\dot{t}) & y(\dot{t}) \end{pmatrix} U = (x \ y) U U^{-1} \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} U + (1 \ 2) U, \quad (15.146)$$

where $U = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$. Then, $(x \ y) U = (x \ y) \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$. Defining $(X \ Y)$ as

$$(X \ Y) \equiv (x \ y) U = (x \ y) \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}, \quad (15.147)$$

we have

$$\begin{pmatrix} \dot{X}(t) & \dot{Y}(t) \end{pmatrix} = \begin{pmatrix} X & Y \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix} + (1 \ 2)U. \quad (15.148)$$

Then, according to (15.90), we get a solution

$$\mathbf{X}(t) = \mathbf{X}(t_0)\tilde{\mathfrak{R}}(t_0, t) + \int_{t_0}^t ds [\tilde{\mathbf{b}}(s)\tilde{\mathfrak{R}}(s, t)], \quad (15.149)$$

where $\mathbf{X}(t_0) = (X(t_0) \ Y(t_0))$ and $\tilde{\mathbf{b}}(s) = \mathbf{b}(s)U$. The resolvent matrix $\tilde{\mathfrak{R}}(s, t)$ is given by

$$\tilde{\mathfrak{R}}(s, t) = \begin{pmatrix} e^{-(t-s)} & 0 \\ 0 & e^{3(t-s)} \end{pmatrix}. \quad (15.150)$$

To convert $(X \ Y)$ to $(x \ y)$, operating U^{-1} on both sides of (15.149) from the right, we obtain

$$\mathbf{X}(t)U^{-1} = \mathbf{X}(t_0)U^{-1}U\tilde{\mathfrak{R}}(t_0, t)U^{-1} + \int_{t_0}^t ds [\tilde{\mathbf{b}}(s)U^{-1}U\tilde{\mathfrak{R}}(s, t)]U^{-1}, \quad (15.151)$$

where with the first and second terms we insert $U^{-1}U = E$. Then, we get

$$\mathbf{x}(t) = \mathbf{x}(t_0)[U\tilde{\mathfrak{R}}(t_0, t)U^{-1}] + \int_{t_0}^t ds \mathbf{b}(s)[U\tilde{\mathfrak{R}}(s, t)U^{-1}]. \quad (15.152)$$

We calculate $U\tilde{\mathfrak{R}}(s, t)U^{-1}$ to have

$$\begin{aligned} U\tilde{\mathfrak{R}}(s, t)U^{-1} &= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} e^{-(t-s)} & 0 \\ 0 & e^{3(t-s)} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} e^{-(t-s)} + e^{3(t-s)} & -e^{-(t-s)} + e^{3(t-s)} \\ -e^{-(t-s)} + e^{3(t-s)} & e^{-(t-s)} + e^{3(t-s)} \end{pmatrix} = \mathfrak{R}(s, t). \end{aligned} \quad (15.153)$$

Thus, we recover (15.90), i.e., we have

$$\mathbf{x}(t) = \mathbf{x}(t_0)\mathfrak{R}(t_0, t) + \int_{t_0}^t ds [\mathbf{b}(s)\mathfrak{R}(s, t)]. \quad (15.90)$$

Equation (15.153) shows that $\mathfrak{R}(s, t)$ and $\tilde{\mathfrak{R}}(s, t)$ are connected to each other through the unitary similarity transformation such that

$$\tilde{\mathfrak{R}}(s, t) = U^{-1} \mathfrak{R}(s, t) U = U^\dagger \mathfrak{R}(s, t) U. \quad (15.154)$$

Example 15.6 Let us consider the following system of differential equations:

$$\begin{pmatrix} \dot{x}(t) & \dot{y}(t) \end{pmatrix} = (x \ y) \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix} + (a \ 0). \quad (15.155)$$

The matrix $\begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix}$ is characterized by an anti-Hermitian operator. This type of operator frequently appears in Lie groups and Lie algebras that we will deal with in Chap. 20. Here we define an operator D as

$$D = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \text{ or } \omega D = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix}. \quad (15.156)$$

The calculation procedures will be seen in Chap. 20, and so we only show the result here. That is, we have

$$\exp t\omega D = \begin{pmatrix} \cos \omega t & -\sin \omega t \\ \sin \omega t & \cos \omega t \end{pmatrix}. \quad (15.157)$$

We seek the resolvent matrix $\mathfrak{R}(s, t)$ such that

$$\mathfrak{R}(s, t) = \begin{pmatrix} \cos \omega(t-s) & -\sin \omega(t-s) \\ \sin \omega(t-s) & \cos \omega(t-s) \end{pmatrix}. \quad (15.158)$$

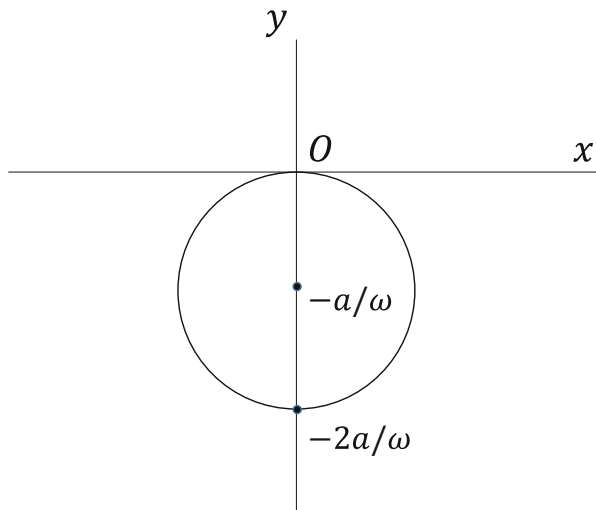
The implication of (15.158) is simple; synthesis of a rotation of ωt and the inverse rotation $-\omega s$ is described by $\omega(t-s)$. Physically, (15.155) represents the motion of a point mass under an external field.

We routinely obtained the above solution using (15.90). If, for example, we solve (15.155) under the BCs for which the point mass is placed at rest at the origin of the xy -plane at $t = 0$, we get a solution described by

$$\begin{aligned} x &= \frac{a}{\omega} \sin \omega t, \\ y &= \frac{a}{\omega} (\cos \omega t - 1). \end{aligned} \quad (15.159)$$

Deleting t from (15.159), we get

Fig. 15.2 Circular motion of a point mass



$$x^2 + \left(y + \frac{a}{\omega}\right)^2 = \left(\frac{a}{\omega}\right)^2. \quad (15.160)$$

In other words, the point mass performs a circular motion (see Fig. 15.2).

In Example 15.1 we mentioned that if the matrix A that defines the system of differential equations varies as a function of t , then the solution method based upon $\exp tA$ does not work. Yet, we have an important case where A is not a constant matrix. Let us consider a next illustration.

15.4 Motion of a Charged Particle in Polarized Electromagnetic Wave

In Sect. 4.5 we dealt with an electron motion under a circularly polarized light. Here we consider the motion of a charged particle in a linearly polarized electromagnetic wave.

Suppose that a charged particle is placed in vacuum. Suppose also that an electromagnetic wave (light) linearly polarized along the x -direction is propagated in the positive direction of the z -axis. Unlike the case of Sect. 4.5, we have to take account of influence from both of the electric field and magnetic field components of Lorentz force.

From (7.58) we have

$$\mathbf{E} = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} = E_0 \mathbf{e}_1 e^{i(kz - \omega t)}, \quad (15.161)$$

where $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ are the unit vector of the x -, y -, and z -direction. (The latter two unit vectors appear just below.) If the extent of motion of the charged particle is narrow enough around the origin compared to a wavelength of the electromagnetic field, we can ignore kz in the exponent (see Sect. 4.5). Then, we have

$$\mathbf{E} \approx E_0 \mathbf{e}_1 e^{-i\omega t}. \quad (15.162)$$

Taking a real part of (15.162), we get

$$\mathbf{E} = E_0 \mathbf{e}_1 \cos \omega t. \quad (15.163)$$

Meanwhile, from (7.59) we have

$$\mathbf{H} = \mathbf{H}_0 e^{i(kz - \omega t)} \approx \mathbf{H}_0 \cos \omega t. \quad (15.164)$$

From (7.60) furthermore, we get

$$\mathbf{H}_0 = \mathbf{e}_3 \times \frac{\mathbf{E}_0}{\sqrt{\frac{\mu_0}{\epsilon_0}}} = \mathbf{e}_3 \times \frac{E_0 \mathbf{e}_1}{\mu_0 c} = \frac{E_0 \mathbf{e}_2}{\mu_0 c}. \quad (15.165)$$

Thus, as the magnetic flux density, we have

$$\mathbf{B} = \mu_0 \mathbf{H} \approx \mu_0 \mathbf{H}_0 \cos \omega t = \frac{E_0 \mathbf{e}_2 \cos \omega t}{c}. \quad (15.166)$$

The Lorentz force $\mathbf{F}$ exerted on the charged particle is then described by

$$\mathbf{F} = e\mathbf{E} + e \dot{\mathbf{x}} \times \mathbf{B}, \quad (15.167)$$

where $\mathbf{x} (=x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3)$ is a position vector of the charged particle having a charge e . In (15.167) the first term represents the electric Lorentz force and the second term shows the magnetic Lorentz force; see Sect. 4.5. Replacing $\mathbf{E}$ and $\mathbf{B}$ in (15.167) with those in (15.163) and (15.166), we get

$$\begin{aligned} \mathbf{F} &= \left(1 - \frac{1}{c} \dot{z}\right) e E_0 \mathbf{e}_1 \cos \omega t + \frac{e E_0}{c} \dot{x} \mathbf{e}_3 \cos \omega t \\ &= m \ddot{\mathbf{x}} = m \ddot{x} \mathbf{e}_1 + m \ddot{y} \mathbf{e}_2 + m \ddot{z} \mathbf{e}_3, \end{aligned} \quad (15.168)$$

where m is a mass of the charged particle. Comparing each vector components, we have

$$\begin{aligned}
m\ddot{x} &= \left(1 - \frac{1}{c}\dot{z}\right)eE_0 \cos \omega t, \\
m\ddot{y} &= 0, \\
m\ddot{z} &= \frac{eE_0}{c}\dot{x} \cos \omega t.
\end{aligned} \tag{15.169}$$

Note that the Lorentz force is not exerted in the direction of the y -axis. Putting $a \equiv eE_0/m$ and $b \equiv eE_0/mc$, we get

$$\begin{aligned}
\ddot{x} &= a \cos \omega t - b\dot{z} \cos \omega t, \\
\ddot{z} &= b\dot{x} \cos \omega t.
\end{aligned} \tag{15.170}$$

Further putting $\dot{x} = \xi$ and $\dot{z} = \zeta$, we obtain

$$\begin{aligned}
\dot{\xi} &= -b\zeta \cos \omega t + a \cos \omega t, \\
\dot{\zeta} &= b\xi \cos \omega t.
\end{aligned} \tag{15.171}$$

Writing (15.171) in a matrix form, we obtain a following system of inhomogeneous differential equations:

$$\begin{pmatrix} \dot{\xi}(t) & \dot{\zeta}(t) \end{pmatrix} = \begin{pmatrix} \xi & \zeta \end{pmatrix} \begin{pmatrix} 0 & b \cos \omega t \\ -b \cos \omega t & 0 \end{pmatrix} + (a \cos \omega t \ 0), \tag{15.172}$$

where the inhomogeneous term is given by $(a \cos \omega t \ 0)$. First, we think of a homogeneous equation described by

$$\begin{pmatrix} \dot{\xi}(t) & \dot{\zeta}(t) \end{pmatrix} = \begin{pmatrix} \xi & \zeta \end{pmatrix} \begin{pmatrix} 0 & b \cos \omega t \\ -b \cos \omega t & 0 \end{pmatrix}. \tag{15.173}$$

Meanwhile, we consider the following equation:

$$\begin{aligned}
\frac{d}{dt} \begin{pmatrix} \cos f(t) & \sin f(t) \\ -\sin f(t) & \cos f(t) \end{pmatrix} &= \begin{pmatrix} -\dot{f}(t) \sin f(t) & \dot{f}(t) \cos f(t) \\ -\dot{f}(t) \cos f(t) & -\dot{f}(t) \sin f(t) \end{pmatrix} \\
&= \begin{pmatrix} \cos f(t) & \sin f(t) \\ -\sin f(t) & \cos f(t) \end{pmatrix} \begin{pmatrix} 0 & \dot{f}(t) \\ -\dot{f}(t) & 0 \end{pmatrix}.
\end{aligned} \tag{15.174}$$

Closely inspecting (15.173) and (15.174), we find that if we can decide $f(t)$ so that $f(t)$ may satisfy

$$f'(t) = b \cos \omega t, \quad (15.175)$$

we might well be able to solve (15.172). In that case, moreover, we expect that two linearly independent column vectors

$$\begin{pmatrix} \cos f(t) \\ -\sin f(t) \end{pmatrix} \text{ and } \begin{pmatrix} \sin f(t) \\ \cos f(t) \end{pmatrix} \quad (15.176)$$

give a fundamental set of solutions of (15.172). A function $f(t)$ that satisfies (15.175) can be given by

$$f(t) = \left(\frac{b}{\omega}\right) \sin \omega t = \tilde{b} \sin \omega t, \quad (15.177)$$

where $\tilde{b} \equiv b/\omega$. Notice that at $t = 0$ two column vectors of (15.176) give

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

if we choose $f(t) = \tilde{b} \sin \omega t$ for $f(t)$ in (15.176).

Thus, defining $F(t)$ as

$$F(t) \equiv \begin{pmatrix} \cos(\tilde{b} \sin \omega t) & \sin(\tilde{b} \sin \omega t) \\ -\sin(\tilde{b} \sin \omega t) & \cos(\tilde{b} \sin \omega t) \end{pmatrix} \quad (15.178)$$

and taking account of (15.174), we recast the homogeneous equation (15.173) as

$$\frac{dF(t)}{dt} = F(t)A, \quad (15.179)$$

$$A \equiv \begin{pmatrix} 0 & b \cos \omega t \\ -b \cos \omega t & 0 \end{pmatrix}.$$

Equation (15.179) is formally the same as (15.47), even though $F(t)$ is not given in the form of $\exp tA$. This enables us to apply the general scheme mentioned in Sect. 15.3.2 to the present case. That is, we are going to address the given problem using the method of variation of constants such that

$$\mathbf{x}(t) = \mathbf{k}(t)F(t), \quad (15.78)$$

where we define $\mathbf{x}(t) \equiv (\xi(t) \ \zeta(t))$ and $\mathbf{k}(t) \equiv (u(t) \ v(t))$ as a variable constant.

Hence, using (15.80)–(15.90), we should be able to find the solution of (15.172) described by

$$\mathbf{x}(t) = \mathbf{x}(t_0)\mathfrak{R}(t_0, t) + \int_{t_0}^t ds [\mathbf{b}(s)\mathfrak{R}(s, t)], \quad (15.90)$$

where $\mathbf{b}(s) \equiv (a \cos \omega t \ 0)$, i.e., the inhomogeneous term in (15.172). The resolvent matrix $\mathfrak{R}(s, t)$ is expressed as

$$\mathfrak{R}(s, t) = F(s)^{-1} F(t) = \begin{pmatrix} \cos[f(t) - f(s)] & \sin[f(t) - f(s)] \\ -\sin[f(t) - f(s)] & \cos[f(t) - f(s)] \end{pmatrix}, \quad (15.180)$$

where $f(t)$ is given by (15.177). With $F(s)^{-1}$, we simply take a transpose matrix of (15.178), because it is an orthogonal matrix. Thus, we obtain

$$F(s)^{-1} = \begin{pmatrix} \cos(\tilde{b} \sin \omega s) & -\sin(\tilde{b} \sin \omega s) \\ \sin(\tilde{b} \sin \omega s) & \cos(\tilde{b} \sin \omega s) \end{pmatrix}.$$

The resolvent matrix (15.180) possesses the properties described as (15.85), (15.86), and (15.87). These can readily be checked. Also, we become aware that $F(t)$ of (15.178) represents a rotation matrix in $\mathbb{R}^2$; see Sect. 11.2. Therefore, $F(t)$ and $F(s)^{-1}$ are commutative. Notice, however, that we do not have a resolvent matrix as a simple form that appears in (15.116). In fact, from (15.178) we find $\mathfrak{R}(s, t) \neq F(t - s)$. It is because the matrix A given as (15.179) is not a constant but depends on t .

Rewriting (15.90) as

$$\mathbf{x}(t) = \mathbf{x}(0)\mathfrak{R}(0, t) + \int_0^t ds [(a \cos \omega t \ 0)\mathfrak{R}(s, t)] \quad (15.181)$$

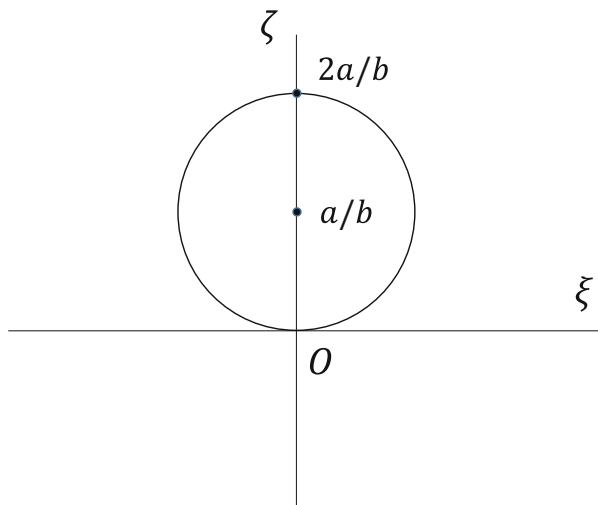
and setting BCs as $\mathbf{x}(0) = (\sigma \ \tau)$, with the individual components, we finally obtain

$$\begin{aligned} \xi(t) &= \sigma \cos(\tilde{b} \sin \omega t) - \tau \sin(\tilde{b} \sin \omega t) + \frac{a}{b} \sin(\tilde{b} \sin \omega t), \\ \zeta(t) &= \sigma \sin(\tilde{b} \sin \omega t) + \tau \cos(\tilde{b} \sin \omega t) - \frac{a}{b} \cos(\tilde{b} \sin \omega t) + \frac{a}{b}. \end{aligned} \quad (15.182)$$

Note that differentiating both sides of (15.182) with respect to t , we recover original equation of (15.172) to be solved. Further setting $\xi(0) = \sigma = \zeta(0) = \tau = 0$ as BCs, we have

$$\begin{aligned} \xi(t) &= \frac{a}{b} \sin(\tilde{b} \sin \omega t), \\ \zeta(t) &= \frac{a}{b} [1 - \cos(\tilde{b} \sin \omega t)]. \end{aligned} \quad (15.183)$$

Fig. 15.3 Velocities ξ and ζ of a charged particle in the x - and z -directions, respectively. The charged particle exerts a periodic motion under the influence of a linearly polarized electromagnetic wave



Notice that the above BCs correspond to that the charged particle is initially placed (at the origin) at rest (i.e., zero velocity). Deleting the argument t , we get

$$\xi^2 + \left(\zeta - \frac{a}{b}\right)^2 = \left(\frac{a}{b}\right)^2. \quad (15.184)$$

Figure 15.3 depicts the particle velocities ξ and ζ in the x - and z -directions, respectively. It is interesting that although the velocity in the x -direction switches from negative to positive, that in the z -direction is kept nonnegative. This implies that the particle is oscillating along the x -direction but continually drifting toward the z -direction while oscillating. In fact, if we integrate $\zeta(t)$, we have a continually drifting term $\frac{a}{b}t$.

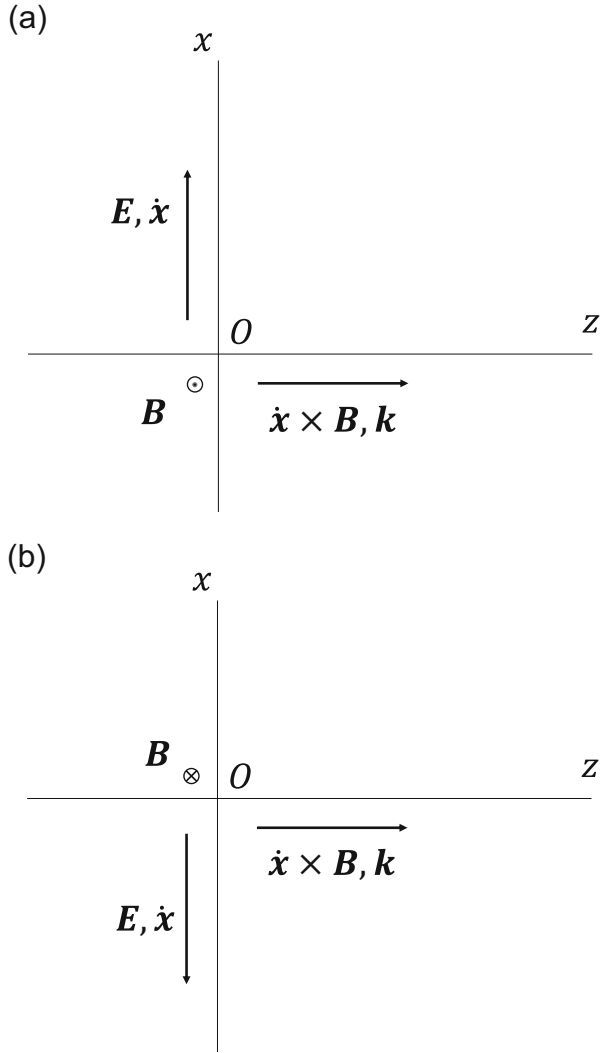
Figure 15.4 shows the Lorentz force $\mathbf{F}$ as a function of time in the case where the charge of the particle is positive. Again, the electric Lorentz force (represented by $\mathbf{E}$) switches from positive to negative in the x -direction, but the magnetic Lorentz force (represented by $\mathbf{\dot{x}} \times \mathbf{B}$) holds nonnegative in the z -direction all the time.

To precisely analyze the positional change of the particle, we need to integrate (15.182) once again. This requires the detailed numerical calculation. For this purpose the following relation is useful [7]:

$$\begin{aligned} \cos(x \sin t) &= \sum_{m=-\infty}^{\infty} J_m(x) \cos mt, \\ \sin(x \sin t) &= \sum_{m=-\infty}^{\infty} J_m(x) \sin mt, \end{aligned} \quad (15.185)$$

where the functions $J_m(x)$ are called the Bessel functions that satisfy the following equation:

Fig. 15.4 Lorentz force as a function of time. In (a) and (b), the phase of electromagnetic fields E and B is reversed. As a result, the electric Lorentz force (represented by E) switches from positive to negative in the x -direction, but the magnetic Lorentz force (represented by $\dot{x} \times B$) holds nonnegative all the time in the z -direction



$$\frac{d^2 J_n(x)}{dx^2} + \frac{1}{x} \frac{dJ_n(x)}{dx} + \left(1 - \frac{n^2}{x^2}\right) J_n(x) = 0. \quad (15.186)$$

Although we do not examine the properties of the Bessel functions in this book, the related topics are dealt with in detail in the literature [5, 7–9]. The Bessel functions are widely investigated as one of special functions in many branches of mathematical physics.

To conclude this section, we have described an example to which the matrix A of (15.179) that characterizes the system of differential equations is not a constant matrix but depends on the parameter t . Unlike the case of the constant matrix A , it

may well be difficult to find a fundamental set of solutions. Once we can find the fundamental set of solutions, however, it is always possible to construct the resolvent matrix and solve the problem using it. In this respect, we have already encountered a similar situation in Chap. 10 where Green's functions were constructed by a fundamental set of solutions.

In this section, a fundamental set of solutions could be found and, as a consequence, we could construct the resolvent matrix. It is, however, not always the case that we can recast the system of differential equation (15.66) as the form of (15.179). Nonetheless, whenever we are successful in finding the fundamental set of solutions, we can construct a resolvent matrix and, hence, solve the problems. This is a generic and powerful tool.

References

1. Satake I-O (1975) Linear algebra (Pure and applied mathematics). Marcel Dekker, New York
2. Satake I (1974) Linear algebra. Shokabo, Tokyo
3. Rudin W (1987) Real and complex analysis, 3rd edn. McGraw-Hill, New York
4. Yamanouchi T, Sugiura M (1960) Introduction to continuous groups. Baifukan, Tokyo
5. Dennery P, Krzywicki A (1996) Mathematics for physicists. Dover, New York
6. Inami T (1998) Ordinary differential equations. Iwanami, Tokyo
7. Riley KF, Hobson MP, Bence SJ (2006) Mathematical methods for physics and engineering, 3rd edn. Cambridge University Press, Cambridge
8. Arfken GB, Weber HJ, Harris FE (2013) Mathematical methods for physicists, 7th edn. Academic Press, Waltham
9. Hassani S (2006) Mathematical physics. Springer, New York

Part IV

Group Theory and Its Chemical Applications

The universe comprises space and matter. These two mutually stipulate their modality of existence. We often comprehend related various aspects as a manifestation of symmetry. In this part we deal with the symmetry from a point of view of group theory. In this part of the book, we outline and emphasize chemical applications of the methods of mathematical physics. This part supplies us with introductory description of group theory. The group theory forms an important field of both pure and applied mathematics. Starting with the definition of groups, we cover a variety of topics related to group theory. Of these, symmetry groups are familiar to chemists, because they deal with a variety of matter and molecules that are characterized by different types of symmetry. The symmetry group is a kind of finite group and called a point group as well. Meanwhile, we have various infinite groups that include rotation group as a typical example. We also mention an introductory theory of the rotation group of $SO(3)$ that deals with an important topic of, e.g., Euler angles. We also treat successive coordinate transformations.

Next, we describe representation theory of groups. Schur's lemmas and related grand orthogonality theorem underlie the representation theory of groups. In parallel, characters and irreducible representations are important concepts that support the representation theory. We present various representations, e.g., regular representation, direct-product representation, symmetric and antisymmetric representations. These have wide applications in the field of quantum mechanics and quantum chemistry, and so forth.

On the basis of the above topics, we highlight quantum chemical applications of group theory in relation to a method of molecular orbitals. As tangible examples, we adopt aromatic molecules and methane.

The last part deals with the theory of continuous groups. The relevant theory has wide applications in many fields of both pure and applied physics and chemistry. We highlight the topics of $SU(2)$ and $SO(3)$ that very often appear there. Tangible examples help understand the essence.

Chapter 16

Introductory Group Theory



A group comprises mathematical elements that satisfy four simple definitions. A bunch of groups exists under these simple definitions. This makes the group theory a discriminating field of mathematics. To get familiar with various concepts of groups, we first show several tangible examples. Group elements can be numbers (both real and complex) and matrices. More abstract mathematical elements can be included as well. Examples include transformation, operation, etc. as already studied in previous parts. Once those mathematical elements form a group, they share several common notions such as classes, subgroups, and direct-product groups. In this context, readers are encouraged to conceive different kinds of groups close to their heart. Mapping is an important concept as in the case of vector spaces. In particular, isomorphism and homomorphism frequently appear in the group theory. These concepts are closely related to the representation theory that is an important pillar of the group theory.

16.1 Definition of Groups

In contrast to a broad range of applications, the definition of the group is simple. Let $\mathcal{G}$ be a set of elements g_ν , where ν is an index either countable (e.g., integers) or uncountable (e.g., real numbers) and the number of elements may be finite or infinite. We denote this by $\mathcal{G} = \{g_\nu\}$. If a group is a finite group, we express it as

$$\mathcal{G} = \{g_1, g_2, \dots, g_n\}, \quad (16.1)$$

where n is said to be an order of the group.

Definition of the group comprises the following four axioms with respect to a well-defined “multiplication” rule between any pair of elements. The multiplication

is denoted by a symbol “ $\diamond$ ” below. Note that the symbol $\diamond$ implies an ordinary multiplication, an ordinary addition, etc.

(A1) If a and b are any elements of the set $\mathcal{G}$, then so is $a \diamond b$. (We sometimes say that the set is “closed” regarding the multiplication.)

(A2) Multiplication is associative, i.e., $a \diamond (b \diamond c) = (a \diamond b) \diamond c$.

(A3) The set $\mathcal{G}$ contains an element e called the identity element such that we have $a \diamond e = e \diamond a = a$ with any element a of $\mathcal{G}$.

(A4) For any a of $\mathcal{G}$, we have an element b such that $a \diamond b = b \diamond a = e$. The element b is said to be the inverse element of a . We denote $b \equiv a^{-1}$.

In the above definitions, we assume that the commutative law does not necessarily hold, that is, $a \diamond b \neq b \diamond a$. In that case the group $\mathcal{G}$ is said to be a non-commutative group. However, we have a case where the commutative law holds, i.e., $a \diamond b = b \diamond a$. If so, the group $\mathcal{G}$ is called a commutative group or an Abelian group.

Let us think of some examples of groups. Henceforth, we follow the convention and write ab to express $a \diamond b$.

Example 16.1 We present several examples of groups below. Examples (1)–(4) are simple, but Example (5) is general.

1. $\mathcal{G} = \{1, -1\}$. The group $\mathcal{G}$ makes a group with respect to the multiplication. This is an example of a finite group.
2. $\mathcal{G} = \{\cdots, -3, -2, -1, 0, 1, 2, 3, \cdots\}$. The group $\mathcal{G}$ makes a group with respect to the addition. This is an infinite group. For instance, take $a(>0)$ and make $a + 1$ and make $(a + 1) + 1$, $[(a + 1) + 1] + 1$, $\cdots$ again and again. Thus, addition is not closed and, hence, we must have an infinite group.

3. Let us start with a matrix $a = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Then, the inverse $a^{-1} = a^3 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

We have $a^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. Its inverse is a^2 itself. These four elements make a group. That is, $\mathcal{G} = \{e, a, a^2, a^3\}$. This is an example of cyclic groups.

4. $\mathcal{G} = \{1\}$. It is a most trivial case, but sometimes the trivial case is very important as well. We will come back later to this point.
5. Let us think of a more general case. In Chap. 11, we discussed endomorphism on a vector space and showed the necessary and sufficient condition for the existence of an inverse transformation. In this relation, consider a set that comprises matrices such that

$$GL(n, \mathbb{C}) \equiv \{A = (a_{ij}); i, j = 1, 2, \cdots, n, a_{ij} \in \mathbb{C}, \det A \neq 0\}.$$

This may be either a finite group or an infinite group. The former can be a symmetry group and the latter can be a rotation group. This group is characterized by a set of invertible and endomorphic linear transformations over a vector space V^n and called a linear transformation group or a general linear group and denoted by $GL(n, \mathbb{C})$, $GL(V^n)$, $GL(V)$, etc. The relevant transformations are bijective. We can

Table 16.1 Multiplication table of $\mathcal{G} = \{e, a, a^2, a^3\}$

g	e	a	$b \equiv a^2$	$c \equiv a^3$
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

readily make sure that axioms (A1) to (A4) are satisfied with $GL(n, \mathbb{C})$. Here, the vector space can be $\mathbb{C}^n$ or a function space.

The structure of a finite group is tabulated in a *multiplication table*. This is made up such that group elements are arranged in a first row and a first column and that an intersection of an element g_i in the row and g_j in the column is designated as a product $g_i \diamond g_j$. Choosing the above (3) for an example, we make its multiplication table (see Table 16.1). There we define $a^2 = b$ and $a^3 = c$.

Having a look at Table 16.1, we notice that in the individual rows and columns each group element appears once and only once. This is well known as a rearrangement theorem.

Theorem 16.1 Rearrangement theorem [1] In each row or each column in the group multiplication table, individual group elements appear once and only once. From this, each row and each column list merely rearranged group elements.

Proof Let a set $\mathcal{G} = \{g_1 \equiv e, g_2, \dots, g_n\}$ be a group. Arbitrarily choosing any element h from $\mathcal{G}$ and multiplying individual elements by h , we obtain a set $\mathfrak{H} = \{hg_1, hg_2, \dots, hg_n\}$. Then, all the group elements of $\mathcal{G}$ appear in $\mathfrak{H}$ once and only once. Choosing any group element g_i , let us multiply g_i by h^{-1} to get $h^{-1}g_i$. Since $h^{-1}g_i$ must be a certain element g_k of $\mathcal{G}$, we put $h^{-1}g_i = g_k$. Multiplying both sides by h , we have $g_i = hg_k$. Therefore, we are able to find this very element hg_k in $\mathfrak{H}$, i.e., g_i in $\mathfrak{H}$. This implies that the element g_i necessarily appears in $\mathfrak{H}$. Suppose in turn that g_i appears more than once. Then, we must have $g_i = hg_k = hg_l$ ($k \neq l$). Multiplying the relation by h^{-1} , we would get $h^{-1}g_i = g_k = g_l$, in contradiction to the supposition. This means that g_i appears in $\mathfrak{H}$ once and only once. This confirms that the theorem is true of each row of the group multiplication table.

A similar argument applies with a set $\mathfrak{H}' = \{g_1h, g_2h, \dots, g_nh\}$. This confirms in turn that the theorem is true of each column of the group multiplication table. These complete the proof.

16.2 Subgroups

As we think of subspaces in a linear vector space, we have subgroups in a group. The definition of the subgroup is that a subset $\mathcal{H}$ of a group makes a group with respect to the multiplication $\diamond$ that is defined for the group $\mathcal{G}$. The identity element makes a

group by itself. Both $\{e\}$ and $\mathcal{G}$ are subgroups as well. We often call subgroups other than $\{e\}$ and $\mathcal{G}$ “proper” subgroups.

A necessary and sufficient condition for the subset $\mathcal{H}$ to be a subgroup is the following:

$$\begin{aligned} (1) \quad & h_i, h_j \in \mathcal{H} \Rightarrow h_i \circ h_j \in \mathcal{H}, \\ (2) \quad & h \in \mathcal{H} \Rightarrow h^{-1} \in \mathcal{H}. \end{aligned}$$

If $\mathcal{H}$ is a subgroup of $\mathcal{G}$, it is obvious that the relations (1) and (2) hold. Conversely, if (1) and (2) hold, $\mathcal{H}$ is a subgroup. In fact, (1) ensures the aforementioned relation (A1). Since $\mathcal{H}$ is a subset of $\mathcal{G}$, this guarantees the associative law (A2). The relation (2) ensures (A4). Finally, in virtue of (1) and (2), $h \circ h^{-1} = e$ is contained in $\mathcal{H}$; this implies that (A3) is satisfied. Thus, $\mathcal{H}$ is a subgroup, because $\mathcal{H}$ satisfies the axioms (A1) to (A4). Of the above examples, (3) has a subgroup $\mathcal{H} = \{e, a^2\}$.

It is important to decompose a set into subsets that do not mutually contain an element (except for a special element) among them. We saw this in Part III when we decomposed a linear vector space into subspaces. In that case the said special element was a zero vector. Here let us consider a related question in its similar aspects.

Let $\mathcal{H} = \{h_1 \equiv e, h_2, \dots, h_s\}$ be a subgroup of $\mathcal{G}$. Also let us consider $a\mathcal{H}$ where $\exists a \in \mathcal{G}$ and $a \notin \mathcal{H}$. Suppose that $a\mathcal{H}$ is a subset of $\mathcal{G}$ such that $a\mathcal{H} = \{ah_1, ah_2, \dots, ah_s\}$. Then, we have another subset $\mathcal{H} + a\mathcal{H}$. If $\mathcal{H}$ contains s elements, so does $a\mathcal{H}$. In fact, if it were not the case, namely, if $ah_i = ah_j$, multiplying both sides by a^{-1} , we would have $h_i = h_j$, in contradiction. Next, let us take b such that $b \notin \mathcal{H}$ and $b \notin a\mathcal{H}$ and make up $b\mathcal{H}$ and $\mathcal{H} + a\mathcal{H} + b\mathcal{H}$ successively. Our question is whether these procedures decompose $\mathcal{G}$ into subsets mutually exclusive and collectively exhaustive.

Suppose that we can succeed in such a decomposition and get

$$\mathcal{G} = g_1\mathcal{H} + g_2\mathcal{H} + \dots + g_k\mathcal{H}, \quad (16.2)$$

where $g_1, g_2, \dots, g_k$ are mutually different elements with g_1 being the identity e . In that case (16.2) is said to be the left coset decomposition of $\mathcal{G}$ by $\mathcal{H}$. Similarly, right coset decomposition can be done to give

$$\mathcal{G} = \mathcal{H}g_1 + \mathcal{H}g_2 + \dots + \mathcal{H}g_k. \quad (16.3)$$

In general, however,

$$g_k\mathcal{H} \neq \mathcal{H}g_k \text{ or } g_k\mathcal{H}g_k^{-1} \neq \mathcal{H}. \quad (16.4)$$

Taking the case of left coset as an example, let us examine whether different cosets mutually contain a common element. Suppose that $g_i\mathcal{H}$ and $g_j\mathcal{H}$ mutually

contain a common element. Then, that element would be expressed as $g_i h_p = g_j h_q$ ($1 \leq i, j \leq n; 1 \leq p, q \leq s$). Thus, we have $g_i h_p h_q^{-1} = g_j$. Since $\mathcal{H}$ is a subgroup of G , $h_p h_q^{-1} \in \mathcal{H}$. This implies that $g_j \in g_i \mathcal{H}$. It is in contradiction to the definition of left coset. Thus, we conclude that different cosets do not mutually contain a common element.

Suppose that the order of G and $\mathcal{H}$ is n and s , respectively. Different k cosets comprise s elements individually and different cosets do not mutually possess a common element and, hence, we must have

$$n = sk, \quad (16.5)$$

where k is called an index of $\mathcal{H}$. We will have many examples afterward.

16.3 Classes

Another method to decompose a group into subsets is called conjugacy classes. A conjugate element is defined as follows: Let a be an element arbitrarily chosen from a group. Then an element gag^{-1} is called a conjugate element or conjugate to a . If c is conjugate to b and b is conjugate to a , then c is conjugate to a . It is because

$$c = g'bg'^{-1}, b = gag^{-1} \Rightarrow c = g'bg'^{-1} = g'gag^{-1}g'^{-1} = g'ga(g'g)^{-1}. \quad (16.6)$$

In the above a set containing a and all the elements conjugate to a is said to be a (conjugate) class of a . Denoting this set by $\mathbb{C}_a$, we have

$$\mathbb{C}_a = \{a, g_2ag_2^{-1}, g_3ag_3^{-1}, \dots, g_nag_n^{-1}\}. \quad (16.7)$$

In $\mathbb{C}_a$ the same element may appear repeatedly. It is obvious that in every group the identity element e forms a class by itself. That is,

$$\mathbb{C}_e = \{e\}. \quad (16.8)$$

As in the case of the decomposition of a group into (left or right) cosets, we can decompose a group to classes. If group elements are not exhausted by a set comprising $\mathbb{C}_e$ or $\mathbb{C}_a$, let us take b such that $b \neq e$ and $b \notin \mathbb{C}_a$ and make $\mathbb{C}_b$ similarly to (16.7). Repeating this procedure, we should be able to decompose a group into classes. In fact, if group elements have not yet exhausted after these procedures, take remaining element z and make a class. If the remaining element is only z in this moment, z can make a class by itself (as in the case of e). Notice that for an Abelian group every element makes a class by itself. Thus, with a finite group, we have a decomposition such that

$$g = \mathbb{C}_e + \mathbb{C}_a + \mathbb{C}_b + \cdots + \mathbb{C}_z. \quad (16.9)$$

To show that (16.9) is really a decomposition, suppose that, for instance, a set $\mathbb{C}_a \cap \mathbb{C}_b$ is not an empty set and that $x \in \mathbb{C}_a \cap \mathbb{C}_b$. Then we must have α and β that satisfy a following relation: $x = \alpha\alpha\alpha^{-1} = \beta\beta\beta^{-1}$, i.e., $b = \beta^{-1}\alpha\alpha\alpha^{-1}\beta = \beta^{-1}\alpha\alpha(\beta^{-1}\alpha)^{-1}$. This implies that b has already been included in $\mathbb{C}_a$, in contradiction to the supposition. Thus, (16.9) is in fact a decomposition of $\mathcal{G}$ into a finite number of classes.

In the above we thought of a class conjugate to a single element. This notion can be extended to a class conjugate to a subgroup. Let $\mathcal{H}$ be a subgroup of $\mathcal{G}$. Let g be an element of $\mathcal{G}$. Let us now consider a set $\mathcal{H}' = g\mathcal{H}g^{-1}$. The set $\mathcal{H}'$ is a subgroup of $\mathcal{G}$ and is called a conjugate subgroup.

In fact, let h_i and h_j be any two elements of $\mathcal{H}$, that is, let $gh_i g^{-1}$ and $gh_j g^{-1}$ be any two elements of $\mathcal{H}'$. Then, we have

$$(gh_i g^{-1})(gh_j g^{-1}) = gh_i h_j g^{-1} = gh_k g^{-1}, \quad (16.10)$$

where $h_k = h_i h_j \in \mathcal{H}$. Hence, $gh_k g^{-1} \in \mathcal{H}'$. Meanwhile, $(gh_i g^{-1})^{-1} = gh_i^{-1} g^{-1} \in \mathcal{H}'$. Thus, conditions (1) and (2) of Sect. 16.2 are satisfied with $\mathcal{H}'$. Therefore, $\mathcal{H}'$ is a subgroup of $\mathcal{G}$. The subgroup $\mathcal{H}'$ has the same order as $\mathcal{H}$. This is because with any two different elements h_i and h_j $gh_i g^{-1} \neq gh_j g^{-1}$.

If for $\forall g \in \mathcal{G}$ and a subgroup $\mathcal{H}$ we have a following equality

$$g^{-1}\mathcal{H}g = \mathcal{H}, \quad (16.11)$$

such a subgroup $\mathcal{H}$ is said to be an invariant subgroup. If (16.11) holds, $\mathcal{H}$ should be a sum of classes (reader, please show this). A set comprising only the identity, i.e., $\{e\}$, forms a class. Therefore, if $\mathcal{H}$ is a proper subgroup, $\mathcal{H}$ must contain two or more classes. The relation (16.11) can be rewritten as

$$g\mathcal{H} = \mathcal{H}g. \quad (16.12)$$

This implies that the left coset is identical to the right coset. Thus, as far as we are dealing with a coset pertinent to an invariant subgroup, we do not have to distinguish left and right cosets.

Now let us anew consider the (left) coset decomposition of $\mathcal{G}$ by an invariant subgroup $\mathcal{H}$

$$\mathcal{G} = g_1\mathcal{H} + g_2\mathcal{H} + \cdots + g_k\mathcal{H}, \quad (16.13)$$

where we have $\mathcal{H} = \{h_1 \equiv e, h_2, \cdots, h_s\}$. Then multiplication of two elements that belong to the cosets $g_i\mathcal{H}$ and $g_j\mathcal{H}$ is expressed as

$$\begin{aligned}
 (g_i h_l)(g_j h_m) &= (g_i g_j g_j^{-1} h_l)(g_j h_m) = g_i g_j (g_j^{-1} h_l g_j) h_m = g_i g_j h_p h_m \\
 &= g_i g_j h_q,
 \end{aligned} \tag{16.14}$$

where the third equality comes from (16.11). That is, we should have $\exists h_p$ such that $g_j^{-1} h_l g_j = h_p$ and $h_p h_m = h_q$. In (16.14) $h_\alpha \in \mathcal{H}$ (α stands for l, m, p, q , etc. with $1 \leq \alpha \leq s$). Note that $g_i g_j h_q \in g_i g_j \mathcal{H}$. Accordingly, a product of elements belonging to $g_i \mathcal{H}$ and $g_j \mathcal{H}$ belongs to $g_i g_j \mathcal{H}$. We rewrite (16.14) as a relation between the sets

$$(g_i \mathcal{H})(g_j \mathcal{H}) = g_i g_j \mathcal{H}. \tag{16.15}$$

Viewing LHS of (16.15) as a product of two cosets, we find that the said product is a coset as well. This implies that a collection of the cosets forms a group. Such a group that possesses cosets as elements is said to be a factor group or quotient group. In this context, the multiplication is a product of cosets. We denote the factor group by

$$\mathcal{G}/\mathcal{H}.$$

An identity element of this factor group is $\mathcal{H}$. This is because in (16.15) putting $g_i = e$, we get $\mathcal{H}(g_j \mathcal{H}) = g_j \mathcal{H}$. Alternatively, putting $g_j = e$, we have $(g_i \mathcal{H})\mathcal{H} = g_i \mathcal{H}$. In (16.15) moreover putting $g_j = g_i^{-1}$, we get

$$(g_i \mathcal{H})(g_i^{-1} \mathcal{H}) = g_i g_i^{-1} \mathcal{H} = \mathcal{H}. \tag{16.16}$$

Hence, $(g_i \mathcal{H})^{-1} = g_i^{-1} \mathcal{H}$. That is, the inverse element of $g_i \mathcal{H}$ is $g_i^{-1} \mathcal{H}$.

16.4 Isomorphism and Homomorphism

As in the case of the linear vector space, we consider the mapping between group elements. Of these, the notion of isomorphism and homomorphism is important.

Definition 16.1 Let $\mathcal{G} = \{x, y, \dots\}$ and $\mathcal{G}' = \{x', y', \dots\}$ be groups and let a mapping $\mathcal{G} \rightarrow \mathcal{G}'$ exist. Suppose that there is a one-to-one correspondence (i.e., injective mapping)

$$x \leftrightarrow x', \quad y \leftrightarrow y', \quad \dots$$

between the elements such that $xy = z$ implies that $x'y' = z'$ and vice versa. Meanwhile, any element in $\mathcal{G}'$ must be the image of some element of $\mathcal{G}$. That is, the mapping is surjective as well and, hence, the mapping is bijective. Then, the two

groups $\mathcal{g}$ and $\mathcal{g}'$ are said to be *isomorphic*. The relevant mapping is called an *isomorphism*. We symbolically denote this relation by

$$\mathcal{g} \cong \mathcal{g}'.$$

Note that the aforementioned groups can be either a finite group or an infinite group. We did not designate identity elements. Suppose that x is the identity e . Then, from the relations $xy = z$ and $x'y' = z'$, we have

$$ey = z = y, x'y' = z' = y'. \quad (16.17)$$

Then, we get

$$x' = e', \text{ i.e., } e \leftrightarrow e'. \quad (16.18)$$

Also let us put $y = x^{-1}$. Then,

$$xx^{-1} = z = e, x'(x^{-1})' = e', x'y' = z' = e'. \quad (16.19)$$

Comparing the second and third equations of (16.19), we get

$$y' = x'^{-1} = (x^{-1})'. \quad (16.20)$$

The bijective character mentioned above can somewhat be loosened in such a way that the one-to-one correspondence is replaced with n -to-one correspondence. We have a following definition.

Definition 16.2 Let $\mathcal{g} = \{x, y, \dots\}$ and $\mathcal{g}' = \{x', y', \dots\}$ be groups and let a mapping $\mathcal{g} \rightarrow \mathcal{g}'$ exist. Also let a mapping $\rho: \mathcal{g} \rightarrow \mathcal{g}'$ exist such that with any arbitrarily chosen two elements, the following relation holds:

$$\rho(x)\rho(y) = \rho(xy). \quad (16.21)$$

Then, the two groups $\mathcal{g}$ and $\mathcal{g}'$ are said to be *homomorphic*. The relevant mapping is called *homomorphism*. We symbolically denote this relation by

$$\mathcal{g} \sim \mathcal{g}'.$$

In this case, we have

$$\rho(e)\rho(e) = \rho(ee) = \rho(e), \text{ i.e., } \rho(e) = e',$$

where e' is an identity element of $\mathcal{g}'$. Also, we have

$$\rho(x)\rho(x^{-1}) = \rho(xx^{-1}) = \rho(e) = e'.$$

Therefore,

$$[\rho(x)]^{-1} = \rho(x^{-1}).$$

The two groups can be either a finite group or an infinite group. Note that in the above the mapping is not injective. The mapping may or may not be surjective. Regarding the identity and inverse elements, we have the same relations as (16.18) and (16.20). From Definitions 16.1 and 16.2, we say that the bijective homomorphism is the isomorphism.

Let us introduce an important notion of a kernel of a mapping. In this regard, we have a following definition:

Definition 16.3 Let $\mathcal{g} = \{e, x, y, \dots\}$ and $\mathcal{g}' = \{e', x', y', \dots\}$ be groups and let e and e' be the identity elements. Suppose that there exists a homomorphic mapping $\rho: \mathcal{g} \rightarrow \mathcal{g}'$. Also let $\mathcal{F}$ be a subset of $\mathcal{g}$ such that

$$\rho(\mathcal{F}) = e'. \quad (16.22)$$

Then, $\mathcal{F}$ is said to be a kernel of ρ .

Regarding the kernel, we have following important theorems.

Theorem 16.2 Let $\mathcal{g} = \{e, x, y, \dots\}$ and $\mathcal{g}' = \{e', x', y', \dots\}$ be groups, where e and e' are identity elements. A necessary and sufficient condition for a surjective and homomorphic mapping $\rho: \mathcal{g} \rightarrow \mathcal{g}'$ to be isomorphic is that a kernel $\mathcal{F} = \{e\}$.

Proof We assume that $\mathcal{F} = \{e\}$. Suppose that $\rho(x) = \rho(y)$. Then, we have

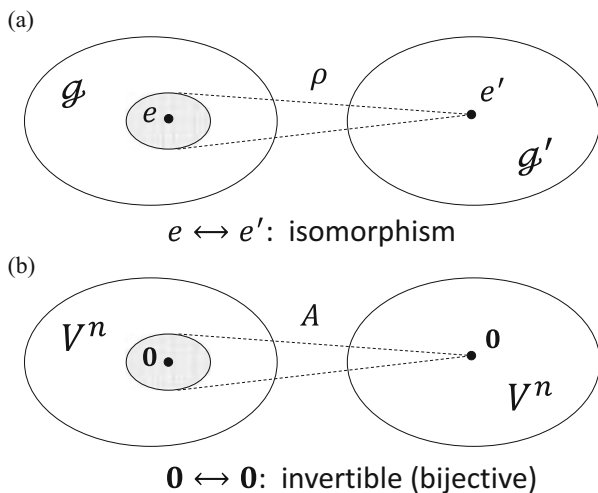
$$\rho(x)[\rho(y)]^{-1} = \rho(x)\rho(y^{-1}) = \rho(xy^{-1}) = e'. \quad (16.23)$$

The first and second equalities result from the homomorphism of ρ . Since $\mathcal{F} = \{e\}$, $xy^{-1} = e$, i.e., $x = y$. Therefore, ρ is injective (i.e., one-to-one correspondence). As ρ is surjective from the assumption, ρ is bijective. The mapping ρ is isomorphic accordingly.

Conversely, suppose that ρ is isomorphic. Also suppose for $\exists x \in \mathcal{g}$ $\rho(x) = e'$. From (16.18), $\rho(e) = e'$. We have $\rho(x) = \rho(e) = e' \Rightarrow x = e$ due to the isomorphism of ρ (i.e., one-to-one correspondence). This implies $\mathcal{F} = \{e\}$. This completes the proof.

We become aware of close relationship between Theorem 16.2 and linear transformation versus kernel already mentioned in Sect. 11.2 of Part III. Figure 16.1 shows this relationship. Figure 16.1a represents homomorphic mapping $\rho: \mathcal{g} \rightarrow \mathcal{g}'$ between two groups, whereas Fig. 16.1b shows linear transformation (endomorphism) $A: V^n \rightarrow V^n$ in a vector space V^n .

Fig. 16.1 Mapping in a group and vector space. (a) Homomorphic mapping $\rho: \mathcal{G} \rightarrow \mathcal{G}'$ between two groups. (b) Linear transformation (endomorphism) $A: V^n \rightarrow V^n$ in a vector space V^n



Theorem 16.3 Suppose that there exists a homomorphic mapping $\rho: \mathcal{G} \rightarrow \mathcal{G}'$, where $\mathcal{G}$ and $\mathcal{G}'$ are groups. Then, a kernel $\mathcal{F}$ of ρ is an invariant subgroup of $\mathcal{G}$.

Proof Let k_i and k_j be any two arbitrarily chosen elements of $\mathcal{F}$. Then,

$$\rho(k_i) = \rho(k_j) = e', \quad (16.24)$$

where e' is the identity element of $\mathcal{G}'$. From (16.21) we have

$$\rho(k_i k_j) = \rho(k_i) \rho(k_j) = e' e' = e'. \quad (16.25)$$

Therefore, $k_i k_j \in \mathcal{F}$. Meanwhile, from (16.20) we have

$$\rho(k_i^{-1}) = [\rho(k_i)]^{-1} = e'^{-1} = e'. \quad (16.26)$$

Then, $k_i^{-1} \in \mathcal{F}$. Thus, $\mathcal{F}$ is a subgroup of $\mathcal{G}$.

Next, for $\forall g \in \mathcal{G}$ we have

$$\rho(g k_i g^{-1}) = \rho(g) \rho(k_i) \rho(g^{-1}) = \rho(g) e' \rho(g^{-1}) = e'. \quad (16.27)$$

Accordingly we have $g k_i g^{-1} \in \mathcal{F}$. Thus, $g \mathcal{F} g^{-1} \subset \mathcal{F}$. Since g is chosen arbitrarily, replacing it with g^{-1} , we have $g^{-1} \mathcal{F} g \subset \mathcal{F}$. Multiplying g and g^{-1} on both sides from the left and right, respectively, we get $\mathcal{F} \subset g \mathcal{F} g^{-1}$. Consequently, we get

$$g \mathcal{F} g^{-1} = \mathcal{F}. \quad (16.28)$$

This implies that $\mathcal{F}$ of ρ is an invariant subgroup of $\mathcal{G}$.

Theorem 16.4 Homomorphism Theorem Let $\mathcal{G} = \{x, y, \dots\}$ and $\mathcal{G}' = \{x', y', \dots\}$ be groups and let a homomorphic (and surjective) mapping $\rho: \mathcal{G} \rightarrow \mathcal{G}'$ exist. Also let $\mathcal{F}$ be a kernel of $\mathcal{G}$. Let us define a surjective mapping $\tilde{\rho}: \mathcal{G}/\mathcal{F} \rightarrow \mathcal{G}'$ such that

$$\tilde{\rho}(g_i\mathcal{F}) = \rho(g_i). \quad (16.29)$$

Then, $\tilde{\rho}$ is an isomorphic mapping.

Proof From (16.15) and (16.21), it is obvious that $\tilde{\rho}$ is homomorphic. The confirmation is left for readers. Let $g_i\mathcal{F}$ and $g_j\mathcal{F}$ be two different cosets. Suppose here that $\rho(g_i) = \rho(g_j)$. Then we have

$$\rho(g_i^{-1}g_j) = \rho(g_i^{-1})\rho(g_j) = [\rho(g_i)]^{-1}\rho(g_j) = [\rho(g_i)]^{-1}\rho(g_i) = e'. \quad (16.30)$$

This implies that $g_i^{-1}g_j \in \mathcal{F}$. That is, we would have $g_j \in g_i\mathcal{F}$. This is in contradiction to the definition of a coset. Thus, we should have $\rho(g_i) \neq \rho(g_j)$. In other words, the different cosets $g_i\mathcal{F}$ and $g_j\mathcal{F}$ have been mapped into different elements $\rho(g_i)$ and $\rho(g_j)$ in $\mathcal{G}'$. That is, $\tilde{\rho}$ is isomorphic; i.e., $\mathcal{G}/\mathcal{F} \cong \mathcal{G}'$.

16.5 Direct-Product Groups

So far we have investigated basic properties of groups. In Sect. 16.4 we examined factor groups. The homomorphism theorem shows that the factor group is characterized by division. In the case of a finite group, an order of the group is reduced. In this section, we study the opposite character, i.e., properties of direct product of groups, or direct-product groups.

Let $\mathcal{H} = \{h_1 \equiv e, h_2, \dots, h_m\}$ and $\mathcal{H}' = \{h'_1 \equiv e, h'_2, \dots, h'_n\}$ be groups of the order of m and n , respectively. Suppose that (1) $\forall h_i (1 \leq i \leq m)$ and $\forall h'_j (1 \leq j \leq n)$ commute, i.e., $h_i h'_j = h'_j h_i$, and that (2) $\mathcal{H} \cap \mathcal{H}' = \{e\}$. Under these conditions let us construct a set $\mathcal{G}$ such that

$$\mathcal{G} = \{h_1 h'_1 \equiv e, h_i h'_j (1 \leq i \leq m, 1 \leq j \leq n)\}. \quad (16.31)$$

In other words, $\mathcal{G}$ is a set comprising mn elements $h_i h'_j$. A product of elements is defined as

$$(h_i h'_j)(h_k h'_l) = h_i h_k h'_j h'_l = h_p h'_q, \quad (16.32)$$

where $h_p = h_i h_k$ and $h'_q = h'_j h'_i$. The identity element is $ee = e$; $eh_i h_k = h_i h_k e$. The inverse element is $(h_i h'_j)^{-1} = h'_j{}^{-1} h_i{}^{-1} = h_i{}^{-1} h'_j{}^{-1}$. Associative law is obvious from $h_i h'_j = h'_j h_i$. Thus, $\mathcal{G}$ forms a group. This is said to be a direct product of groups, or a direct-product group. The groups $\mathcal{H}$ and $\mathcal{H}'$ are called direct factors of $\mathcal{G}$. In this case, we succinctly represent

$$\mathcal{G} = \mathcal{H} \otimes \mathcal{H}'.$$

In the above, the condition (2) is equivalent to that $\forall g \in \mathcal{G}$ is uniquely represented as

$$g = hh'; h \in \mathcal{H}, h' \in \mathcal{H}'. \quad (16.33)$$

In fact, suppose that $\mathcal{H} \cap \mathcal{H}' = \{e\}$ and that g can be represented in two ways such that

$$g = h_1 h'_1 = h_2 h'_2; h_1, h_2 \in \mathcal{H}, h'_1, h'_2 \in \mathcal{H}'. \quad (16.34)$$

Then, we have

$$h_2^{-1} h_1 = h'_2 h'_1{}^{-1}; h_2^{-1} h_1 \in \mathcal{H}, h'_2 h'_1{}^{-1} \in \mathcal{H}'. \quad (16.35)$$

From the supposition, we get

$$h_2^{-1} h_1 = h'_2 h'_1{}^{-1} = e. \quad (16.36)$$

That is, $h_2 = h_1$ and $h'_2 = h'_1$. This means that the representation is unique.

Conversely, suppose that the representation is unique and that $x \in \mathcal{H} \cap \mathcal{H}'$. Then we must have

$$x = xe = ex. \quad (16.37)$$

Thanks to uniqueness of the representation, $x = e$. This implies $\mathcal{H} \cap \mathcal{H}' = \{e\}$.

Now suppose $h \in \mathcal{H}$. Then for $\forall g \in \mathcal{G}$ putting $g = h_\nu h'_\mu$, we have

$$ghg^{-1} = h_\nu h'_\mu h h'_\mu{}^{-1} h_\nu{}^{-1} = h_\nu h h'_\mu h'_\mu{}^{-1} h_\nu{}^{-1} = h_\nu h h_\nu{}^{-1} \in \mathcal{H}. \quad (16.38)$$

Then we have $g\mathcal{H}g^{-1} \subset \mathcal{H}$. Similarly to the proof of Theorem 16.3, we get

$$g\mathcal{H}g^{-1} = \mathcal{H}. \quad (16.39)$$

This shows that $\mathcal{H}$ is an invariant subgroup of $\mathcal{G}$. Similarly, $\mathcal{H}'$ is an invariant subgroup as well.

Regarding the unique representation of the group element of a direct-product group, we become again aware of the close relationship between the direct product and direct sum that was mentioned earlier in Part III.

Reference

1. Cotton FA (1990) Chemical applications of group theory, 3rd edn. Wiley, New York

Chapter 17

Symmetry Groups



We have many opportunities to observe symmetry both macroscopic and microscopic in natural world. First, we need to formulate the symmetry appropriately. For this purpose, we must regard various symmetry operations as mathematical elements and classify these operations under several categories. In Part III we examined various properties of vectors and their transformations. We also showed that the vector transformation can be viewed as the coordinate transformation. On these topics, we focused upon abstract concepts in various ways. On another front, however, we have not paid attention to specific geometric objects, especially molecules. In this chapter, we study the symmetry of these concrete objects. For this, it will be indispensable to correctly understand a variety of symmetry operations. At the same time, we deal with the vector and coordinate transformations as group elements. Among such transformations, rotations occupy a central place in the group theory and related field of mathematics. Regarding the three-dimensional Euclidean space, $SO(3)$ is particularly important. This is characterized by an infinite group in contrast to various symmetry groups (or point groups) we investigate in the former parts of this chapter.

17.1 A Variety of Symmetry Operations

To understand various aspects of symmetry operations, it is convenient and essential to consider a general point that is fixed in a three-dimensional Euclidean space and to examine how this point is transformed in the space. In parallel to the description in Part III, we express the coordinate of the general point P as

$$P = \begin{pmatrix} x \\ y \\ z \end{pmatrix}. \quad (17.1)$$

Note that P may be on or within or outside a geometric object or molecule that we are dealing with. The relevant position vector $\mathbf{x}$ for P is expressed as

$$\begin{aligned} \mathbf{x} &= x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3 \\ &= (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \end{aligned} \quad (17.2)$$

where $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ denote an orthonormal basis vectors pointing to positive directions of x -, y -, and z -axes, respectively. Similarly we denote a linear transformation A by

$$A(\mathbf{x}) = (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}. \quad (17.3)$$

Among various linear transformations that are represented by matrices, orthogonal transformations are the simplest and most widely used. We use orthogonal matrices to represent the orthogonal transformations accordingly.

Let us think of a movement or translation of a geometric object and an operation that causes such a movement. First suppose that the geometric object is fixed on a coordinate system. Then the object is moved (or translate) to another place. If before and after such a movement (or translation) one could not tell whether the object has been moved, we say that the object possesses a “symmetry” in a certain sense. Thus, we have to specify this symmetry. In that context, group theory deals with the symmetry and defines it clearly.

To tell whether the object has been moved (to another place), we usually distinguish it by change in (1) positional relationship and (2) attribute or property. To make the situation simple, let us consider a following example:

Example 17.1 We have two round disks, i.e., Disk A and Disk B. Suppose that Disk A is a solid white disk, whereas Disk B is partly painted black (see Fig. 17.1). In Fig. 17.1 we are thinking of a rotation of an object (e.g., round disk) around an axis standing on its center (C) and stretching perpendicularly to the object plane.

If an arbitrarily chosen positional vector fixed on the object before the rotation is moved to another position that was not originally occupied by the object, then we recognize that the object has certainly been moved. For instance, imagine that a round disk having a through-hole located aside from center is rotating. What about the case where that position was originally occupied by the object, then? We have two possibilities. The first alternative is that we cannot recognize that the object has been moved. The second one is that we can yet recognize that the object has been

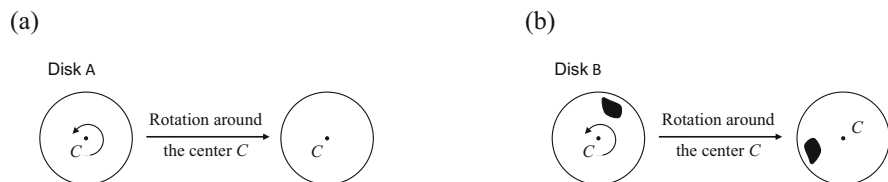


Fig. 17.1 Rotation of an object. (a) Case where we cannot recognize that the object has been moved. (b) Case where we can recognize that the object has been moved because of its attribute (i.e., because the round disk is partly painted black)

moved. According to Fig. 17.1a, b, we have the former case and the latter case, respectively. In the latter case, we have recognized the movement of the object by its attribute, i.e., by that the object is partly painted black. $\square$

In the above example, we do not have to be rigorous. We have a clear intuitive criterion for a judgment of whether a geometric object has been moved. From now on, we assume that the geometric character of an object is pertinent to both its positional relationship and attribute. We define the equivalent (or indistinguishable) disposition of an object and the operation that yields such an equivalent disposition as follows:

Definition 17.1

1. Symmetry operation: A geometric operation that produces an equivalent (or indistinguishable) disposition of an object.
2. Equivalent (or indistinguishable) disposition: Suppose that regarding a geometric operation of an object, we cannot recognize that the object has been moved before and after that geometric operation. In that case, the original disposition of the object and the resulting disposition reached after the geometric operation are referred to as an equivalent disposition. The relevant geometric operation is the symmetry operation. ∇

Here we should clearly distinguish *translation* (i.e., parallel displacement) from the abovementioned symmetry operations. This is because for a geometric object to possess the translation symmetry the object must be infinite in extent, typically an infinite crystal lattice. The relevant discipline is widely studied as space group and has a broad class of applications in physics and chemistry. However, we will not deal with the space group or associated topics, but focus our attention upon symmetry groups in this book.

In the above example, the rotation is a symmetry operation with Fig. 17.1a, but the said geometric operation is not a symmetric operation with Fig. 17.1b.

Let us further inspect properties of the symmetry operation. Let us consider a set $\mathfrak{H}$ consisting of symmetry operations. Let a and b be any two symmetric operations of $\mathfrak{H}$. Then, (1) $a \circ b$ is a symmetric operation as well. (2) Multiplication of successive symmetric operations a, b, c is associative, i.e., $a \circ (b \circ c) = (a \circ b) \circ c$. (3) The set $\mathfrak{H}$ contains an element of e called the identity element such that we have $a \circ e = e \circ a = a$ with any element a of $\mathfrak{H}$. Operating “nothing” should be e . If the

rotation is relevant, 2π rotation is thought to be e . These are intuitively acceptable. (4) For any a of $\mathcal{G}$, we have an element b such that $a \circ b = b \circ a = e$. The element b is said to be the inverse element of a . We denote it by $b \equiv a^{-1}$. The inverse element corresponds to an operation that brings the disposition of a geometric object back to the original disposition. Thus, $\mathfrak{H}$ forms a group. We call $\mathfrak{H}$ satisfying the above criteria a *symmetry group*. A symmetry group is called a point group as well. This is because a point group comprises symmetry operations of geometric objects as group elements, and those objects have at least one fixed point after the relevant symmetry operation. The name of a point group comes from this fact.

As mentioned above, the symmetric operation is best characterized by a (3, 3) orthogonal matrix. In Example 17.1, e.g., the π rotation is represented by an orthogonal matrix A such that

$$A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (17.4)$$

This operation represents a π rotation around the z -axis. Let us think of another symmetric operation described by

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (17.5)$$

This produces a mirror symmetry with respect to the xy -plane. Then, we have

$$C \equiv AB = BA = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (17.6)$$

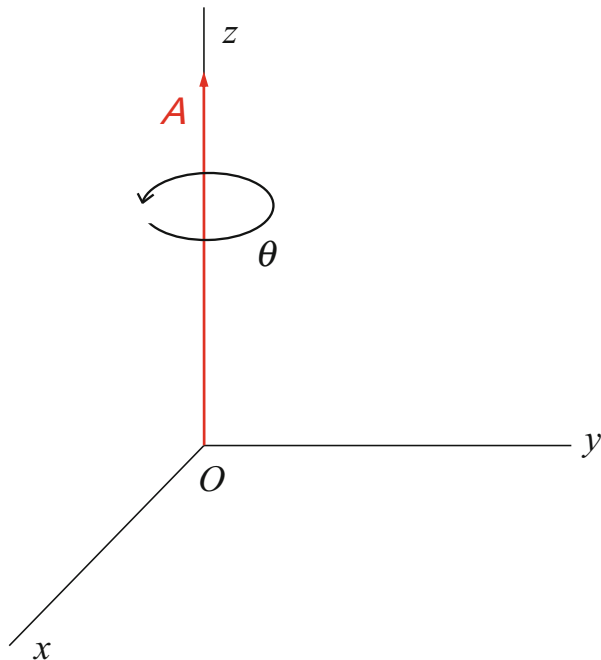
The operation C shows an inversion about the origin. Thus, A , B , and C along with an identity element E form a group. Here E is expressed as

$$E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (17.7)$$

The above group is represented by four three-dimensional diagonal matrices whose elements are 1 or -1 . Therefore, it is evident that an inverse element of A , B , and C is A , B , and C itself, respectively. The said group is commutative (Abelian) and said to be a *four-group* (or Klein four-group, more specifically) [1].

Meanwhile, we have a number of non-commutative groups. From a point of view of a matrix structure, non-commutativity comes from off-diagonal elements of the matrix. A typical example is a rotation matrix of a rotation angles different from zero

Fig. 17.2 Rotation by θ
around the z -axis



or $n\pi$ (n : integer). For later use, let us have a matrix form that expresses a θ rotation around the z -axis. Figure 17.2 depicts a graphical illustration for this.

A matrix R has a following form:

$$R = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (17.8)$$

Note that R has been reduced. This implies that the three-dimensional Euclidean space is decomposed into a two-dimensional subspace (xy -plane) and a one-dimensional subspace (z -axis). The xy -coordinates are not mixed with the z -component after the rotation R . Note, however, that if the rotation axis is oblique against the xy -plane, this is not the case. We will come back to this point later.

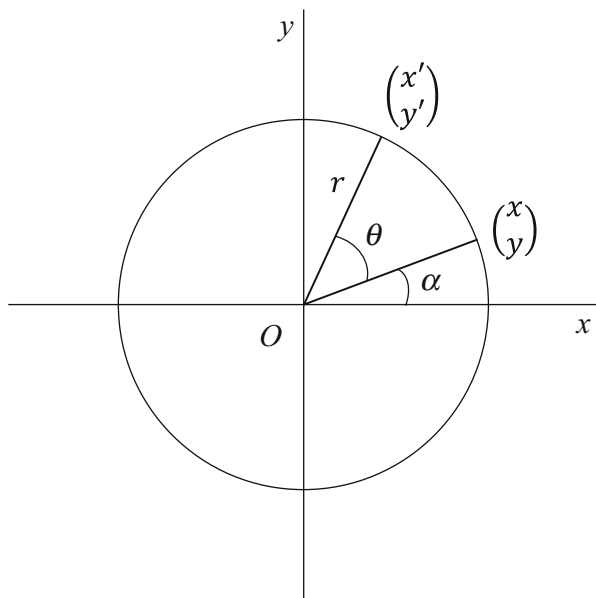
Taking only the xy -coordinates in Fig. 17.3, we make a calculation. Using an addition theorem of trigonometric functions, we get

$$x' = r \cos(\theta + \alpha) = r(\cos \alpha \cos \theta - \sin \alpha \sin \theta) = x \cos \theta - y \sin \theta, \quad (17.9)$$

$$y' = r \sin(\theta + \alpha) = r(\sin \alpha \cos \theta + \cos \alpha \sin \theta) = y \cos \theta + x \sin \theta, \quad (17.10)$$

where we used $x = r \cos \alpha$ and $y = r \sin \alpha$. Combining (17.9) and (17.10), as a matrix form, we get

Fig. 17.3 Transformation of the xy -coordinates by a θ rotation



$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}. \quad (17.11)$$

The matrix of RHS in (17.11) is the same as (11.31) and represents a transformation matrix of a rotation angle θ within the xy -plane. Whereas in Chap. 11 we considered this from the point of view of the transformation of basis vectors, here we deal with the transformations of coordinates in the fixed coordinate system. If $\theta \neq \pi$, off-diagonal elements do not vanish. Including the z -component we get (17.8).

Let us summarize symmetry operations and their (3, 3) matrix representations. The coordinates before and after a symmetry operation are expressed as

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ and } \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}, \text{ respectively.}$$

1. Identity transformation:

To leave a geometric object or a coordinate system unchanged (or unmoved). By convention, we denote it by a capital letter E . It is represented by a (3, 3) identity matrix.

2. Rotation symmetry around a rotation axis:

Here a “proper” rotation is intended. We denote a rotation by a rotation axis and its magnitude (i.e., rotation angle). Thus, we have

$$R_{z\theta} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, R_{y\phi} = \begin{pmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{pmatrix},$$

$$R_{x\varphi} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi & -\sin \varphi \\ 0 & \sin \varphi & \cos \varphi \end{pmatrix}. \quad (17.12)$$

With $R_{y\phi}$ we first consider a following coordinate transformation:

$$\begin{pmatrix} z' \\ x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} z \\ x \\ y \end{pmatrix}. \quad (17.13)$$

This can easily be visualized in Fig. 17.4; consider that cyclic permutation of $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ produces $\begin{pmatrix} z \\ x \\ y \end{pmatrix}$. Shuffling the order of coordinates, we get

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}. \quad (17.14)$$

By convention of the symmetry groups, the following notation C_n is used to denote a rotation. A subscript n of C_n represents the order of the rotation axis. The order means the largest number of n so that the rotation through $2\pi/n$ gives an equivalent configuration. Successive rotations of m times ($m < n$) are denoted by

$$C_n^m.$$

If $m = n$, the successive rotations produce an equivalent configuration the same as the beginning, i.e., $C_n^n = E$. The rotation angles θ , φ , etc. used above are restricted to $2\pi m/n$ accordingly.

3. Mirror symmetry with respect to a plane of mirror symmetry:

We denote a mirror symmetry by a mirror symmetry plane (e.g., xy -plane, yz -plane, etc.). We have

$$M_{xy} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, M_{yz} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, M_{zx} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (17.15)$$

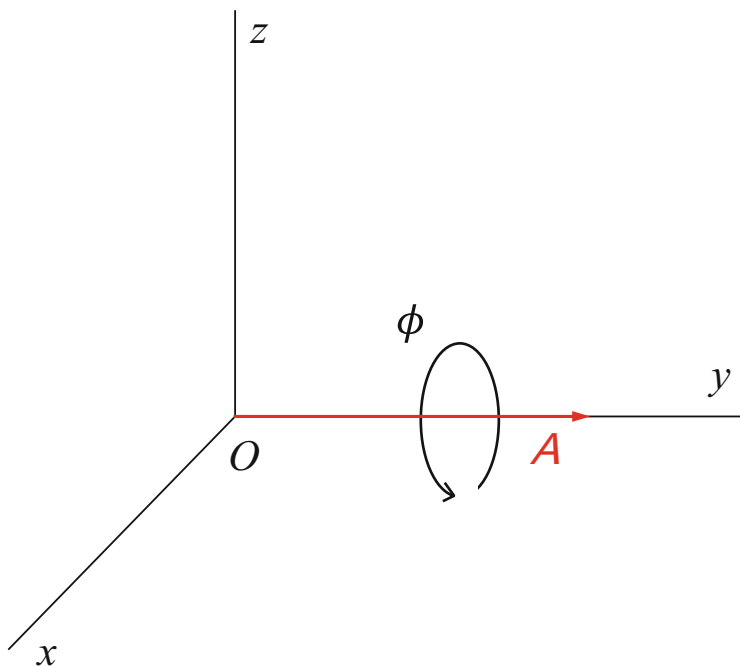


Fig. 17.4 Rotation by ϕ around the y -axis

The mirror symmetry is usually denoted by σ_v , σ_h , and σ_d , whose subscripts stand for “vertical,” “horizontal,” and “dihedral,” respectively. Among these symmetry operations, σ_v and σ_d include a rotation axis in the symmetry plane, while σ_h is perpendicular to the rotation axis if such an axis exists. Notice that a group belonging to C_s symmetry possesses only E and σ_h . Although σ_h can exist by itself as a mirror symmetry, neither σ_v nor σ_d can exist as a mirror symmetry by itself. We will come back to this point later.

4. Inversion symmetry with respect to a center of inversion:

We specify an inversion center if necessary, e.g., an origin of a coordinate system O . The operation of inversion is denoted by

$$I_O = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (17.16)$$

Note that as obviously from the matrix form, I_O is commutable with any other symmetry operations. Note also that I_O can be expressed as successive symmetry operations or product of symmetry operations. For instance, we have

$$I_O = R_{z\pi} M_{xy} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (17.17)$$

Note that $R_{z\pi}$ and M_{xy} are commutable, i.e., $R_{z\pi} M_{xy} = M_{xy} R_{z\pi}$.

5. Improper rotation:

This is a combination of a proper rotation and a reflection by a mirror symmetry plane. That is, rotation around an axis is performed first and then reflection is carried out by a mirror plane that is perpendicular to the rotation axis. For instance, an improper rotation is expressed as

$$\begin{aligned} M_{xy} R_{z\theta} &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \end{aligned} \quad (17.18)$$

As mentioned just above, the inversion symmetry I can be viewed as an improper rotation. Note that in this case the reflection and rotation operations are commutable. However, we will follow a conventional custom that considers the inversion symmetry as an independent symmetry operation. Readers may well wonder why we need to consider the improper rotation. The answer is simple; it solely rests upon the axiom (A1) of the group theory. A group must be closed with respect to the multiplication.

The improper rotations are usually denoted by S_n . A subscript n again stands for an order of rotation.

17.2 Successive Symmetry Operations

Let us now consider successive reflections in different planes and successive rotations about different axes [2]. Figure 17.5a displays two reflections with respect to the planes σ and $\tilde{\sigma}$ both perpendicular to the xy -plane. The said planes make a dihedral angle θ with their intersection line identical to the z -axis. Also, the plane σ is identical with the zx -plane. Suppose that an arrow lies on the xy -plane perpendicularly to the zx -plane. As in (17.15), an operation σ is represented as

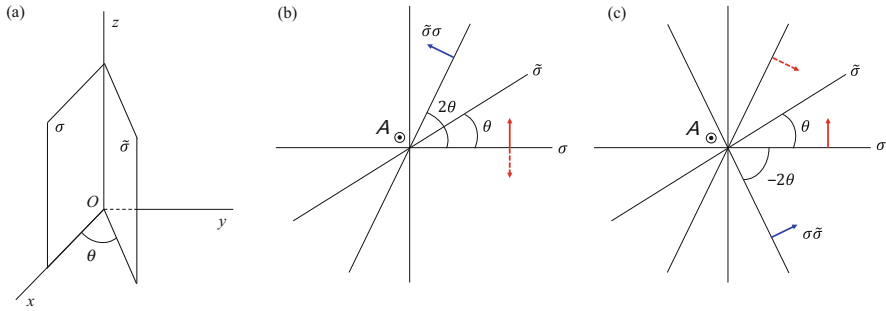


Fig. 17.5 Successive two reflections about two planes σ and $\tilde{\sigma}$ that make an angle θ . (a) Reflections σ and $\tilde{\sigma}$ with respect to two planes. (b) Successive operations of σ and $\tilde{\sigma}$ in this order. The combined operation is denoted by $\tilde{\sigma}\sigma$. The operations result in a 2θ rotation around the z -axis. (c) Successive operations of $\tilde{\sigma}$ and σ in this order. The combined operation is denoted by $\sigma\tilde{\sigma}$. The operations result in a -2θ rotation around the z -axis

$$\sigma = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (17.19)$$

To determine a matrix representation of $\tilde{\sigma}$, we calculate a matrix again as in the above case. As a result we have

$$\begin{aligned} \tilde{\sigma} &= \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \cos 2\theta & \sin 2\theta & 0 \\ \sin 2\theta & -\cos 2\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned} \quad (17.20)$$

Notice that this matrix representation is referred to the original xyz -coordinate system; see discussion of Sect. 11.4. Hence, we describe the successive transformations σ followed by $\tilde{\sigma}$ as

$$\tilde{\sigma}\sigma = \begin{pmatrix} \cos 2\theta & -\sin 2\theta & 0 \\ \sin 2\theta & \cos 2\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (17.21)$$

The expression (17.21) means that the multiplication should be done first by σ and then by $\tilde{\sigma}$; see Fig. 17.5b. Note that σ and $\tilde{\sigma}$ are conjugate to each other (Sect. 16.3). In this case, the combined operations produce a 2θ rotation around the z -axis.

If, on the other hand, the multiplication is made first by $\tilde{\sigma}$ and then by σ , we have a -2θ rotation around the z -axis; see Fig. 17.5c. As a matrix representation, we have

$$\sigma\tilde{\sigma} = \begin{pmatrix} \cos 2\theta & \sin 2\theta & 0 \\ -\sin 2\theta & \cos 2\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (17.22)$$

Thus, successive operations of reflection by the planes that make a dihedral angle θ yield a rotation $\pm 2\theta$ around the z -axis (i.e., the intersection line of σ and $\tilde{\sigma}$). The operation $\sigma\tilde{\sigma}$ is an inverse to $\tilde{\sigma}\sigma$. That is,

$$(\sigma\tilde{\sigma})(\tilde{\sigma}\sigma) = E. \quad (17.23)$$

We have $\det\sigma\tilde{\sigma} = \det\tilde{\sigma}\sigma = 1$.

Meanwhile, putting

$$R_{2\theta} = \begin{pmatrix} \cos 2\theta & -\sin 2\theta & 0 \\ \sin 2\theta & \cos 2\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (17.24)$$

we have

$$\tilde{\sigma}\sigma = R_{2\theta} \text{ or } \tilde{\sigma} = R_{2\theta}\sigma. \quad (17.25)$$

This implies the following: Suppose a plane σ and a straight line on it. Also suppose that one first makes a reflection about σ and then makes a 2θ rotation around the said straight line. Then, the resulting transformation is equivalent to a reflection about $\tilde{\sigma}$ that makes an angle θ with σ . At the same time, the said straight line is an intersection line of σ and $\tilde{\sigma}$. Note that a dihedral angle between the two planes σ and $\tilde{\sigma}$ is half an angle of the rotation. Thus, any two of the symmetry operations related to (17.25) are mutually dependent; any two of them produce the third symmetry operation.

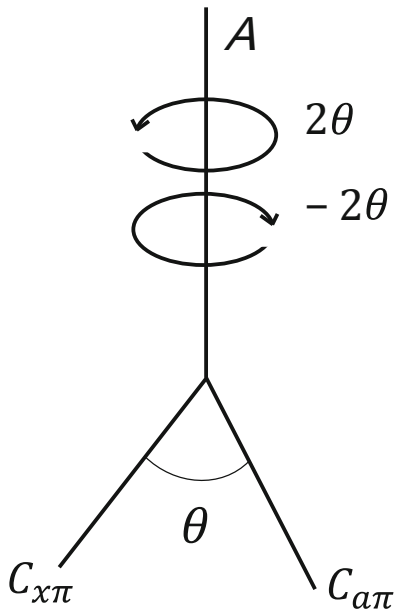
In the above illustration, we did not take account of the presence of a symmetry axis. If the aforementioned axis is a symmetry axis C_n , we must have

$$2\theta = 2\pi/n \text{ or } n = \pi/\theta. \quad (17.26)$$

From a symmetry requirement, there should be n planes of mirror symmetry in combination with the C_n axis. Moreover, an intersection line of these n mirror symmetry planes should coincide with that C_n axis. This can be seen as various C_{nv} groups.

Next we consider another successive symmetry operations. Suppose that there are two C_2 axes ($C_{x\pi}$ and $C_{a\pi}$ as shown) that intersect at an angle θ (Fig. 17.6). There, $C_{x\pi}$ is identical to the x -axis and the other ($C_{a\pi}$) lies on the xy -plane making an angle θ with $C_{x\pi}$. Following procedures similar to the above, we have matrix representations of the successive C_2 operations in reference to the xyz -system such that

Fig. 17.6 Successive two π rotations around two C_2 axes of $C_{x\pi}$ and $C_{a\pi}$ that intersect at an angle θ . Of these, $C_{x\pi}$ is identical to the x -axis (not shown)



$$\begin{aligned}
 C_{x\pi} &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \\
 C_{a\pi} &= \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} \cos 2\theta & \sin 2\theta & 0 \\ \sin 2\theta & -\cos 2\theta & 0 \\ 0 & 0 & -1 \end{pmatrix}, \tag{17.27}
 \end{aligned}$$

where $C_{a\pi}$ can be calculated similarly to (17.20). Again we get

$$\begin{aligned}
 C_{a\pi} C_{x\pi} &= \begin{pmatrix} \cos 2\theta & -\sin 2\theta & 0 \\ \sin 2\theta & \cos 2\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\
 C_{x\pi} C_{a\pi} &= \begin{pmatrix} \cos 2\theta & \sin 2\theta & 0 \\ -\sin 2\theta & \cos 2\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{17.28}
 \end{aligned}$$

Notice that $(C_{x\pi} C_{a\pi})(C_{a\pi} C_{x\pi}) = E$. Once again putting

$$R_{2\theta} = \begin{pmatrix} \cos 2\theta & -\sin 2\theta & 0 \\ \sin 2\theta & \cos 2\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (17.24)$$

we have [1, 2]

$$C_{a\pi}C_{x\pi} = R_{2\theta} \quad \text{or} \quad C_{a\pi} = R_{2\theta}C_{x\pi}. \quad (17.29)$$

Note that in the above two illustrations for the successive symmetry operations, both the relevant operators have been represented in reference to the *original* xyz -system. For this reason, the latter operation was done from the *left* in (17.28).

From a symmetry requirement, once again the aforementioned C_2 axes must be present in combination with the C_n axis. Moreover, those C_2 axes should be perpendicular to the C_n axis. This can be seen in various D_n groups.

Another illustration of successive symmetry operations is an improper rotation. If the rotation angle is π , this causes an inversion symmetry. In this illustration, reflection, rotation, and inversion symmetries coexist. A C_{2h} symmetry is a typical example.

Equations (17.21), (17.22), and (17.24) demonstrate the same relation. Namely, two successive mirror symmetry operations about a couple of planes and two successive π -rotations about a couple of C_2 axes cause the same effect with regard to the geometric transformation. In this relation, we emphasize that two successive reflection operations make a determinant of relevant matrices 1. These aspects cause an interesting effect and we will briefly discuss it in relation to O and T_d groups.

If furthermore the abovementioned mirror symmetry planes and C_2 axes coexist, the symmetry planes coincide with or bisect the C_2 axes and vice versa. If these were not the case, another mirror plane or C_2 axis would be generated from the symmetry requirement and the newly generated plane or axis would be coupled with the original plane or axis. From the above argument, these processes again produce another C_n axis. That must be prohibited.

Next, suppose that a C_n axis intersects *obliquely* with a plane of mirror symmetry. A rotation of $2\pi/n$ around such an axis produces another mirror symmetry plane. This newly generated plane intersects with the original mirror plane and produces a different C_n axis according to the above discussion. Thus, in this situation a mirror symmetry plane cannot coexist with a sole rotation axis. In a geometric object with higher symmetry such as O_h , however, several mirror symmetry planes can coexist with several rotation axes in such a way that the axes intersect with the mirror planes *obliquely*. In case the C_n axis intersects *perpendicularly* to a mirror symmetry plane, that plane can coexist with a sole rotation axis (see Fig. 17.7). This is actually the case with a geometric object having a C_{2h} symmetry. The mirror symmetry plane is denoted by σ_h .

Now, let us examine simple examples of molecules and geometric figures as well as associated symmetry operations.

Fig. 17.7 Mirror symmetry plane σ_h and a sole rotation axis perpendicular to it

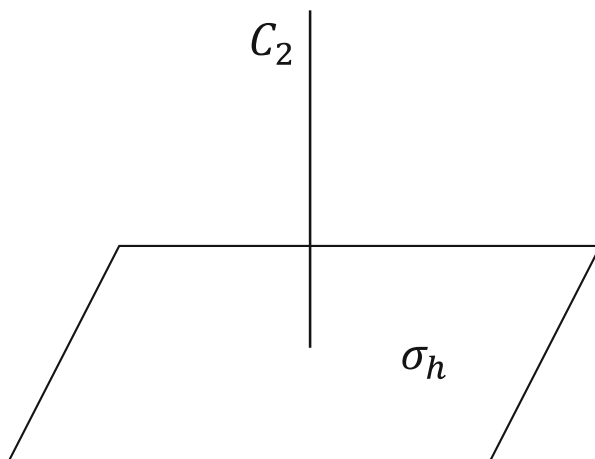
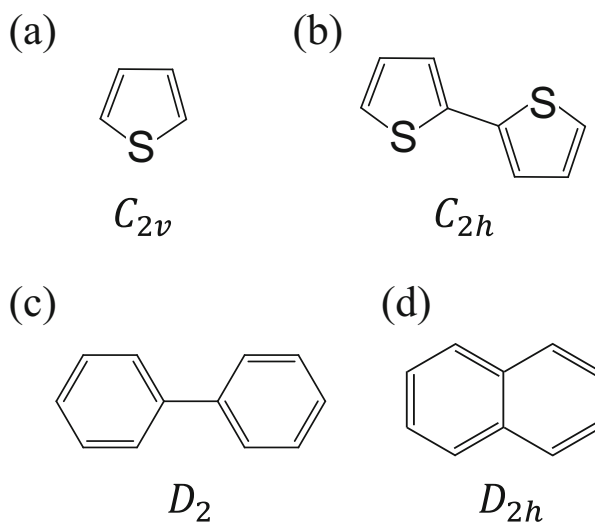


Fig. 17.8 Chemical structural formulae and point groups of (a) thiophene, (b) bithiophene, (c) biphenyl, and (d) naphthalene



Example 17.2 Figure 17.8 shows chemical structural formulae of thiophene, bithiophene, biphenyl, and naphthalene. These molecules belong to C_{2v} , C_{2h} , D_2 , and D_{2h} , respectively. Note that these symbols are normally used to show specific point groups. Notice also that in biphenyl two benzene rings are twisted relative to the molecular axis. As an example, a multiplication table is shown in Table 17.1 for a C_{2v} group. Table 17.1 clearly demonstrates that the group constitution of C_{2v} differs from that of the group appearing in Example 16.1 (3), even though the order is four for both the case. Similar tables are given with C_{2h} and D_2 . This is left for readers as an exercise. We will find that the multiplication tables of these groups have the same structure and that C_{2v} , C_{2h} , and D_2 are all isomorphic to one another as a four-group. Table 17.2 gives matrix representation of symmetry operations for C_{2v} . The

Table 17.1 Multiplication table of C_{2v}

C_{2v}	E	$C_2(z)$	$\sigma_v(zx)$	$\sigma'_v(yz)$
E	E	C_2	σ_v	σ'_v
$C_2(z)$	C_2	E	σ'_v	σ_v
$\sigma_v(zx)$	σ_v	σ'_v	E	C_2
$\sigma'_v(yz)$	σ'_v	σ_v	C_2	E

Table 17.2 Matrix representation of symmetry operations for C_{2v}

	E	C_2 (around z -axis)	$\sigma_v(zx)$	$\sigma'_v(yz)$
Matrix	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Table 17.3 Multiplication table of D_{2h}

D_{2h}	E	$C_2(z)$	$\sigma_v(zx)$	$\sigma'_v(yz)$	i	$\sigma''_v(xy)$	$C'_2(y)$	$C''_2(x)$
E	E	C_2	σ_v	σ'_v	i	σ''_v	C'_2	C''_2
$C_2(z)$	C_2	E	σ'_v	σ_v	σ''_v	i	C'_2	C'_2
$\sigma_v(zx)$	σ_v	σ'_v	E	C_2	C'_2	C''_2	i	σ''_v
$\sigma'_v(yz)$	σ'_v	σ_v	C_2	E	C'_2	C'_2	σ''_v	i
i	i	σ''_v	C'_2	C''_2	E	C_2	σ_v	σ'_v
$\sigma''_v(xy)$	σ''_v	i	C''_2	C'_2	C_2	E	σ'_v	σ_v
$C'_2(y)$	C'_2	C''_2	i	σ''_v	σ_v	σ'_v	E	C_2
$C''_2(x)$	C''_2	C'_2	σ''_v	i	σ'_v	σ_v	C_2	E

representation is defined as the transformation by the symmetry operations of a set of basis vectors $(x \ y \ z)$ in $\mathbb{R}^3$.

Meanwhile, Table 17.3 gives the multiplication table of D_{2h} . We recognize that the multiplication table of C_{2v} appears on upper left and lower right blocks. If we suitably rearrange the order of group elements, we can make another multiplication table so that, e.g., C_{2h} may appear on upper left and lower right blocks. As in the case of Table 17.2, Table 17.4 summarizes the matrix representation of symmetry operations for D_{2h} . There are eight group elements, i.e., an identity, an inversion, three mutually perpendicular C_2 axes, and three mutually perpendicular planes of mirror symmetry (σ). Here, we consider a possibility of constructing subgroups of D_{2h} . The order of the subgroups must be a divisor of eight, and so let us list subgroups whose order is four and examine how many subgroups exist. We have ${}_8C_4 = 70$ combinations, but those allowed should be restricted from the requirement of forming a group. This is because all the groups must contain identity element, and so the number allowed is equal to or no greater than ${}_7C_3 = 35$.

(1) In light of the aforementioned discussion, two C_2 axes mutually intersecting at $\pi/2$ yield another C_2 axis around the normal to a plane defined by the intersecting axes. Thus, three C_2 axes have been chosen and a D_2 symmetry results. In this case, we have only one choice. (2) In the case of C_{2v} , two planes mutually intersecting at

Table 17.4 Matrix representation of symmetry operations for D_{2h}

D_{2h}	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$
Matrix	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Table 17.5 Choice of symmetry operations for construction of subgroups of D_{2h}

Subgroup	E	$C_2(z)$	σ	i	Choice
D_2	1	3	0	0	1
C_{2v}	1	1	2	0	3
C_{2h}	1	1	1	1	3

$\pi/2$ yield a C_2 axis around their line of intersection. There are three possibilities of choosing two axes out of three (i.e., ${}_3C_2 = 3$). (3) If we choose the inversion (i) along with, e.g., one of the three C_2 axes, a σ necessarily results. This is also the case when we first combine a σ with i to obtain a C_2 axis. We have three possibilities (i.e., ${}_3C_1 = 3$) as well. Thus, we have only seven choices to construct subgroups of D_{2h} having an order of four. This is summarized in Table 17.5.

An inverse of any element is that element itself. Therefore, if with any above subgroup one chooses any element out of the remaining four elements and combines it with the identity, one can construct a subgroup C_s , C_2 , or C_i of an order of 2. Since all those subgroups of an order of 4 and 2 are commutative with D_{2h} , these subgroups are invariant subgroups. Thus in terms of a direct-product group, D_{2h} can be expressed as various direct factors. Conversely, we can construct factor groups from coset decomposition. For instance, we have

$$D_{2h}/C_{2v} \cong C_s, D_{2h}/C_{2v} \cong C_2, D_{2h}/C_{2v} \cong C_i. \quad (17.30)$$

In turn, we express direct-product groups as, e.g.,

$$D_{2h} = C_{2v} \otimes C_s, D_{2h} = C_{2v} \otimes C_2, D_{2h} = C_{2v} \otimes C_i. \quad (17.31)$$

□

Example 17.3 Figure 17.9 shows an equilateral triangle placed on the xy -plane of a three-dimensional Cartesian coordinate. An orthonormal basis e_1 and e_2 are designated as shown. As for a chemical species of molecules, we have, e.g., boron trifluoride (BF_3). In the molecule, a boron atom is positioned at a molecular center with three fluorine atoms located at vertices of the equilateral triangle. The boron atom and fluorine atoms form a planar molecule (Fig. 17.10).

A symmetry group belonging to D_{3h} comprises 12 symmetry operations such that

$$D_{3h} = \{E, C_3, C'_3, C_2, C'_2, C''_2, \sigma_h, S_3, S'_3, \sigma_v, \sigma'_v, \sigma''_v\}, \quad (17.32)$$

where symmetry operations of the same species but distinct operation are denoted by a “prime” or “double prime.” When we represent these operations by a matrix, it is straightforward in most cases. For instance, a matrix for σ_v is given by M_{yz} of (17.15). However, we should make some matrix calculations about C'_2 , C''_2 , σ'_v , and σ''_v .

To determine a matrix representation of, e.g., σ'_v in reference to the xyz -coordinate system with orthonormal basis vectors e_1 and e_2 , we consider the $x'y'z'$ -coordinate

Fig. 17.9 Equilateral triangle placed on the xy -plane. Several symmetry operations of D_{3h} are shown. We consider the successive operations (i), (ii), and (iii) to represent σ'_v in reference to the xyz -coordinate system (see text)

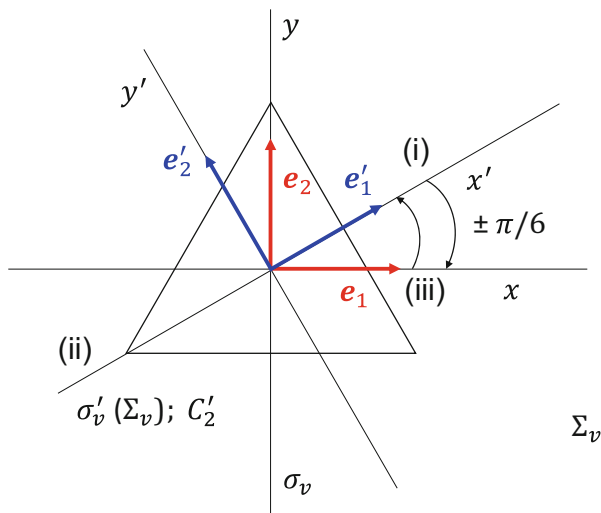
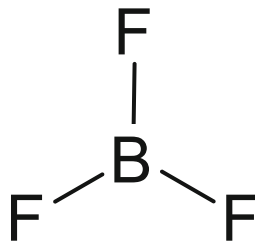


Fig. 17.10 Boron trifluoride (BF_3) belonging to a D_{3h} point group



system with orthonormal basis vectors e'_1 and e'_2 (see Fig. 17.9). A transformation matrix between the two sets of basis vectors is represented by

$$(e'_1 \ e'_2 \ e'_3) = (e_1 \ e_2 \ e_3) R_{\frac{\pi}{6}} = (e_1 \ e_2 \ e_3) \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (17.33)$$

This representation corresponds to (11.69). Let Σ_v be a reflection with respect to the $z'x'$ -plane. This is *the same* operation as σ'_v . However, a matrix representation is *different*. This is because Σ_v is represented in reference to the $x'y'z'$ -system, while σ'_v is in reference to xyz -system. The matrix representation of Σ_v is simple and expressed as

$$\Sigma_v = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (17.34)$$

Referring to (11.80), we have

$$R_{z_6} \Sigma_v = \sigma'_v R_{z_6} \quad \text{or} \quad \sigma'_v = R_{z_6} \Sigma_v [R_{z_6}]^{-1}. \quad (17.35)$$

Thus, we see that in the first equation of (17.35) the order of multiplications is reversed according as the latter operation is expressed in the xyz -system or in the $x'y'z'$ -system. As a full matrix representation, we get

$$\begin{aligned} \sigma'_v &= \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned} \quad (17.36)$$

Notice that this matrix representation is referred to the original xyz -coordinate system as before. Graphically, (17.35) corresponds to the multiplication of the symmetry operations done in the order of (i) $-\pi/6$ rotation, (ii) reflection (denoted by Σ_v), and (iii) $\pi/6$ rotation (Fig. 17.9). The associated matrices are multiplied from the left.

Similarly, with C'_2 we have

$$C'_2 = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (17.37)$$

The matrix form of (17.37) can also be decided in a manner similar to the above according to three successive operations shown in Fig. 17.9. In terms of classes, σ_v, σ'_v , and σ''_v form a conjugacy class and C_2, C'_2 , and C''_2 form another conjugacy class. With regard to the reflection and rotation, we have $\det \sigma'_v = -1$ and $\det C_2 = 1$, respectively. $\square$

17.3 O and T_d Groups

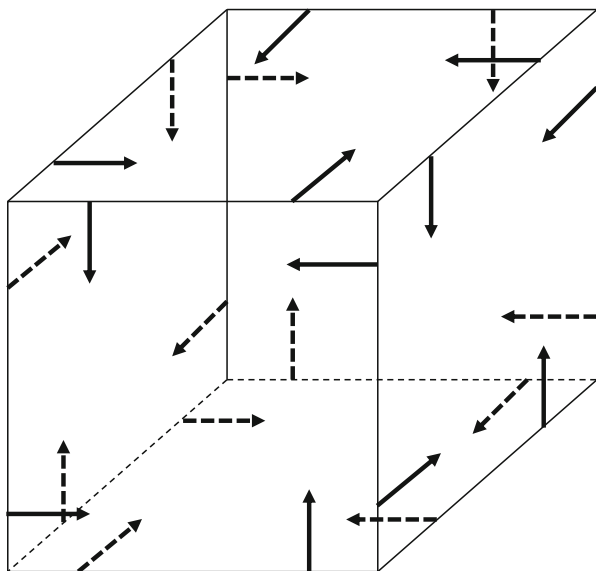
According as a geometric object (or a molecule) has a higher symmetry, we have to deal with many symmetry operations and relationship between them. As an example, we consider O and T_d groups. Both groups have 24 symmetry operations and are isomorphic.

Let us think of the group O first. We start with considering rotations of $\pi/2$ around x -, y -, and z -axis. The matrices representing these rotations are obtained from (17.12) to give

$$R_{x\frac{\pi}{2}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, R_{y\frac{\pi}{2}} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}, R_{z\frac{\pi}{2}} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (17.38)$$

We continue multiplications of these matrices so that the matrices can make up a complete set (i.e., a closure). Counting over those matrices, we have 24 of them and they form a group termed O . The group O is a pure rotation group. Here, the pure rotation group is defined as a group whose group elements only comprise proper rotations (with their determinant of 1). An example is shown in Fig. 17.11, where individual vertices of the cube have three arrows for the cube not to possess mirror symmetries or inversion symmetry. The group O has five conjugacy classes. Figure 17.12 summarizes them. Geometric characteristics of individual classes are sketched as well. These classes are categorized by a *trace* of the matrix. This is

Fig. 17.11 Cube whose individual vertices have three arrows for the cube not to possess mirror symmetries. This object belongs to a point group O called pure rotation group



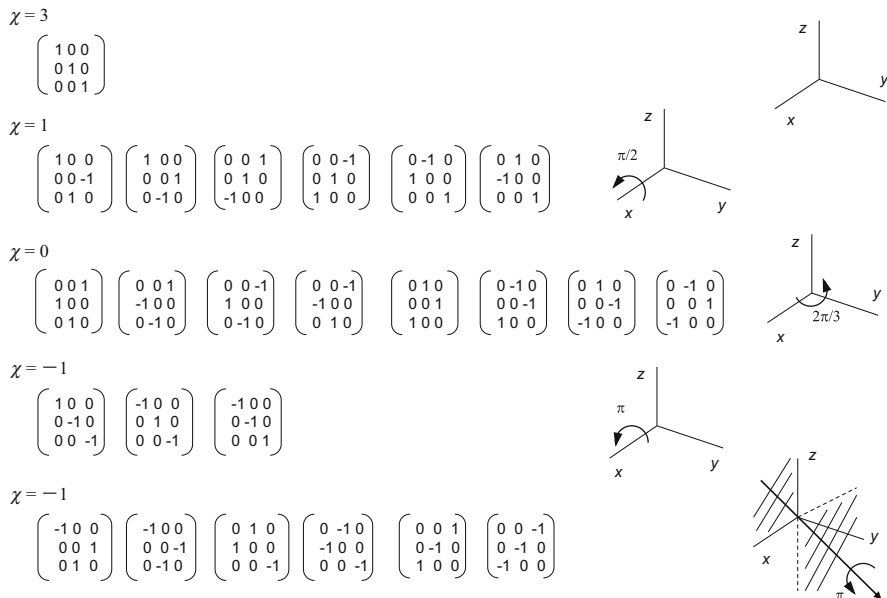


Fig. 17.12 Point group O and its five conjugacy classes. Geometric characteristics of individual classes are briefly sketched

because the trace is kept unchanged by a similarity transformation. (Remember that elements of the same conjugacy class are connected with a similarity transformation.) Having a look at the sketches, we notice that each operation switches the basis vectors $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$, i.e., x -, y -, and z -axes. Therefore, the presence of diagonal elements (either 1 or -1) implies that the matrix takes the basis vector(s) as an eigenvector with respect to the rotation. Corresponding eigenvalue(s) are either 1 or -1 accordingly. This is expected from the fact that the matrix is an orthogonal matrix (i.e., unitary). The trace, namely, a summation of diagonal elements, is closely related to the geometric feature of the operation.

The operations of a π rotation around the x -, y -, and z -axis and those of a π rotation around an axis bisecting any two of three axes have a trace -1 . The former operations take all the basis vectors as an eigenvectors; that is, all the diagonal elements are non-vanishing. With the latter operations, however, only one diagonal element is -1 . This feature comes from that the bisected axes are switched by the rotation, whereas the remaining axis is reversed by the rotation.

Another characteristic is the generation of eight rotation axes that trisect the x -, y -, and z -axes, more specifically a solid angle $\pi/2$ formed by the x -, y -, and z -axes. Since the rotation switches all the x -, y -, and z -axes, the trace is zero. At the same time, we find that this operation belongs to C_3 . This operation is generated by successive two $\pi/2$ rotations around two mutually orthogonal axes. To inspect this situation more closely, we consider a conjugacy class of $\pi/2$ rotation that belongs to the C_4 symmetry and includes six elements, i.e., $R_{x\frac{\pi}{2}}$, $R_{y\frac{\pi}{2}}$, $R_{z\frac{\pi}{2}}$, $R_{x\frac{3\pi}{2}}$, $R_{y\frac{3\pi}{2}}$, and $R_{z\frac{3\pi}{2}}$. With

these notations, e.g., $R_{x\frac{\pi}{2}}$ stands for a $\pi/2$ counterclockwise rotation around the x -axis; $R_{\bar{x}\frac{\pi}{2}}$ denotes a $\pi/2$ counterclockwise rotation around the $-x$ -axis. Consequently, $R_{\bar{x}\frac{\pi}{2}}$ implies a $\pi/2$ *clockwise* rotation around the x -axis and, hence, an inverse element of $R_{x\frac{\pi}{2}}$. Namely, we have

$$R_{\bar{x}\frac{\pi}{2}} = \left(R_{x\frac{\pi}{2}}\right)^{-1}. \quad (17.39)$$

Now let us consider the successive two rotations. This is denoted by the multiplication of matrices that represent the related rotations. For instance, the multiplication of, e.g., $R_{x\frac{\pi}{2}}$ and $R'_{y\frac{\pi}{2}}$ produces the following:

$$\begin{aligned} R_{xyz\frac{2\pi}{3}} &= R_{x\frac{\pi}{2}} R'_{y\frac{\pi}{2}} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \end{aligned} \quad (17.40)$$

In (17.40), we define $R_{xyz\frac{2\pi}{3}}$ as a $2\pi/3$ counterclockwise rotation around an axis that trisects the x -, y -, and z -axes. The prime “'” of $R'_{y\frac{\pi}{2}}$ means that the operation is carried out in reference to the *new* coordinate system reached by the previous operation $R_{x\frac{\pi}{2}}$. For this reason, $R'_{y\frac{\pi}{2}}$ is operated (i.e., multiplied) from the *right* in (17.40). Compare this with the remark made just after (17.29). Changing the order of $R_{x\frac{\pi}{2}}$ and $R'_{y\frac{\pi}{2}}$, we have

$$\begin{aligned} R_{xyz\frac{2\pi}{3}} &= R_{y\frac{\pi}{2}} R'_{x\frac{\pi}{2}} \\ &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{pmatrix}, \end{aligned} \quad (17.41)$$

where $R_{xyz\frac{2\pi}{3}}$ is a $2\pi/3$ counterclockwise rotation around an axis that trisects the x -, y -, and $-z$ -axes. Notice that we used $R'_{x\frac{\pi}{2}}$ this time, because it was performed after $R_{y\frac{\pi}{2}}$. Thus, we notice that there are eight related operations that trisect eight octants of the coordinate system. These operations are further categorized into four sets in which the two elements are an inverse element of each other. For instance, we have

$$R_{\bar{x}\bar{y}\bar{z}\frac{2\pi}{3}} = \left(R_{xyz\frac{2\pi}{3}}\right)^{-1}. \quad (17.42)$$

Notice that a $2\pi/3$ counterclockwise rotation around an axis that trisects the $-x$ -, $-y$ -, and $-z$ -axes is equivalent to a $2\pi/3$ clockwise rotation around an axis that

trisects the x -, y -, and z -axes. Also, we have $R_{\bar{xy}z\frac{2\pi}{3}} = \left(R_{xyz\frac{2\pi}{3}}\right)^{-1}$, etc. Moreover, we have “cyclic” relations such as

$$R_{x\frac{\pi}{2}}R'_{y\frac{\pi}{2}} = R_{y\frac{\pi}{2}}R'_{z\frac{\pi}{2}} = R_{z\frac{\pi}{2}}R'_{x\frac{\pi}{2}} = R_{xyz\frac{2\pi}{3}}. \quad (17.43)$$

Returning back to Sect. 11.4, we had

$$A[P(\mathbf{x})] = [(\mathbf{e}_1 \cdots \mathbf{e}_n)PA'] \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (\mathbf{e}_1 \cdots \mathbf{e}_n) \left[A_O P \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right]. \quad (11.79)$$

Implication of (11.79) is that LHS is related to the transformation of basis vectors while retaining coordinates $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ and that transformation matrices should be operated on the basis vectors from the right. Meanwhile, RHS describes the transformation of coordinates while retaining basis vectors. In that case, transformation matrices should be operated on the coordinates from the left. Thus, the order of operator multiplication is reversed. Following (11.80), we describe

$$R_{x\frac{\pi}{2}}R'_{y\frac{\pi}{2}} = R_O R_{x\frac{\pi}{2}}, \text{ i.e., } R_O = R_{x\frac{\pi}{2}}R'_{y\frac{\pi}{2}} \left(R_{x\frac{\pi}{2}}\right)^{-1}, \quad (17.44)$$

where R_O is viewed in reference to the original (or fixed) coordinate system and conjugate to $R'_{y\frac{\pi}{2}}$. Thus, we have

$$\begin{aligned} R_O &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned} \quad (17.45)$$

Note that (17.45) is identical to a matrix representation of a $\pi/2$ rotation around the z -axis. This is evident from the fact that the y -axis is converted to the original z -axis by $R_{x\frac{\pi}{2}}$; readers, imagine it.

We have two conjugacy classes of π rotation (the C_2 symmetry). One of them includes six elements, i.e., $R_{xy\pi}$, $R_{yz\pi}$, $R_{zx\pi}$, $R_{\bar{xy}\pi}$, $R_{\bar{yz}\pi}$, and $R_{\bar{zx}\pi}$. For these notations a subscript, e.g., xy , stands for an axis that bisects the angle formed by x - and y -axes. A subscript $\bar{xy}$ denotes an axis bisecting the angle formed by $-x$ - and y -axes. Another class includes three elements, i.e., $R_{x\pi}$, $R_{y\pi}$, and $R_{z\pi}$.

As for $R_{x\pi}$, $R_{y\pi}$, and $R_{z\pi}$, a combination of these operations should yield a C_2 rotation axis as discussed in Sect. 17.2. Of these three rotation axes, in fact, any two

produce a C_2 rotation around the remaining axis, as is the case with naphthalene belonging to the D_{2h} symmetry (see Sect. 17.2).

Regarding the class comprising six π rotation elements, a combination of, e.g., $R_{xy\pi}$ and $R_{\bar{xy}\pi}$ crossing each other at a right angle causes a related effect. For the other combinations, the two C_2 axes intersect each other at $\pi/3$; see Fig. 17.6 and put $\theta = \pi/3$ there. In this respect, elementary analytic geometry teaches the positional relationship among planes and straight lines. The argument is as follows: A plane determined by three points $\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$, $\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$, and $\begin{pmatrix} x_3 \\ y_3 \\ z_3 \end{pmatrix}$ that do not sit on a line is expressed by a following equation:

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0. \quad (17.46)$$

Substituting $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, and $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ for $\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$, $\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$, and $\begin{pmatrix} x_3 \\ y_3 \\ z_3 \end{pmatrix}$ in (17.46), respectively, we have

$$x - y + z = 0. \quad (17.47)$$

Taking account of direction cosines and using Hesse's normal form, we get

$$\frac{1}{\sqrt{3}}(x - y + z) = 0, \quad (17.48)$$

where the normal to the plane expressed in (17.48) has direction cosines of $\frac{1}{\sqrt{3}}$, $-\frac{1}{\sqrt{3}}$, and $\frac{1}{\sqrt{3}}$ in relation to the x -, y -, and z -axes, respectively.

Therefore, the normal is given by a straight line connecting the origin and $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$. In other words, a line connecting the origin and a corner of a cube is the normal to the plane described by (17.48). That plane is formed by two intersecting lines, i.e., rotation axes of C_2 and C'_2 (see Fig. 17.13 that depicts a cube of each side of 2). These axes make an angle $\pi/3$; this can easily be checked by taking an inner product between $\begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$. These column vectors are two direction cosines of C_2 and C'_2 . On the basis of the discussion of Sect. 17.2, we must have a rotation axis of C_3 . That is, this axis trisects a solid angle $\pi/2$ shaped by three intersecting sides.

Fig. 17.13 Rotation axes of C_2 and C'_2 (marked red) along with another rotation axis C_3 (marked blue) in a point group O

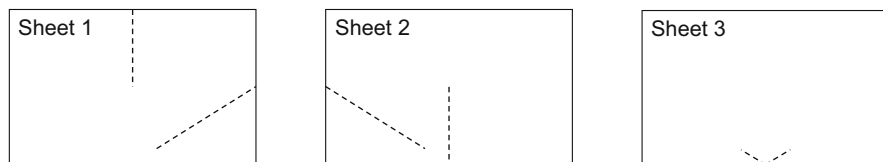
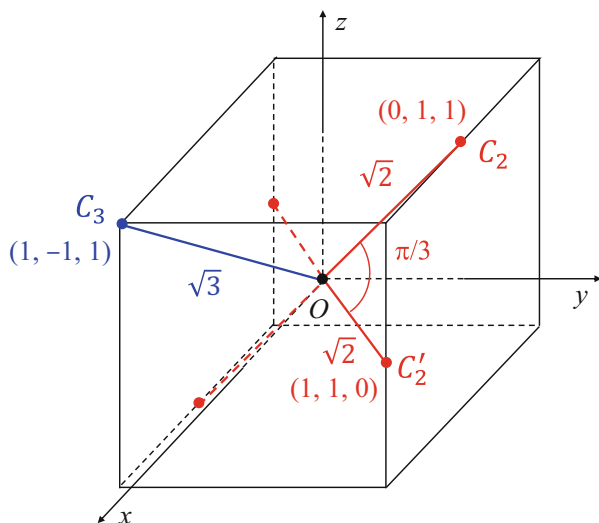


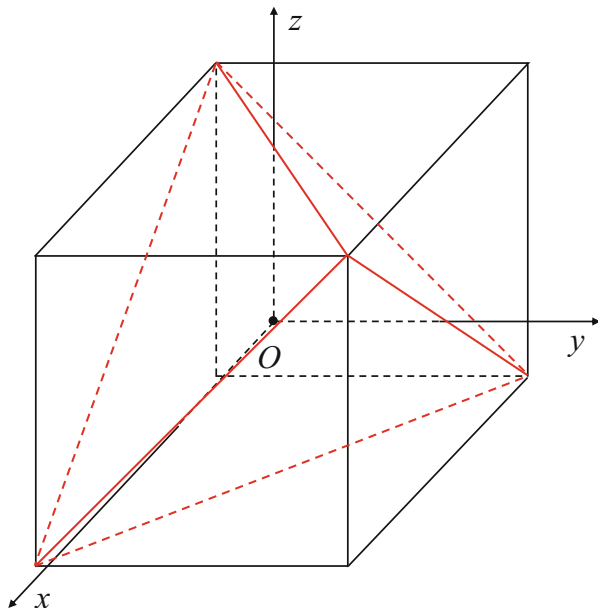
Fig. 17.14 Simple kit that helps visualize the positional relationship among planes and straight lines in three-dimensional space. To make it, follow next procedures: (1) Take three thick sheets of paper and make slits (dashed lines) as shown. (2) Insert Sheet 2 into Sheet 1 so that the two sheets can make a right angle. (3) Insert Sheet 3 into combined Sheets 1 and 2

It is sometimes hard to visualize or envisage the positional relationship among planes and straight lines in three-dimensional space. It will therefore be useful to make a simple kit to help visualize it. Figure 17.14 gives an illustration.

Another typical example having 24 group elements is T_d . A molecule of methane belongs to this symmetry. Table 17.6 collects the relevant symmetry operations and their (3, 3) matrix representations. As in the case of Fig. 17.12, the matrices show how a set of vectors ($x\ y\ z$) are transformed according to the symmetry operations. Comparing it with Fig. 17.12, we immediately recognize that the close relationship between T_d and O exists and that these point groups share notable characteristics.

(1) Both T_d and O consist of five conjugacy classes, each of which contains the same number of symmetry species. (2) Both T_d and O contain a pure rotation group T as a subgroup. The subgroup T consists of 12 group elements E , $8C_3$, and $3C_2$. Other remaining 12 group elements of T_d are symmetry species related to reflection: S_4 and σ_d . The elements $6S_4$ and $6\sigma_d$ correspond to $6C_4$ and $6C_2$ of O , respectively. That is, successive operations of S_4 cause similar effects to those of C_4 of O . Meanwhile, successive operations of σ_d are related to those of $6C_2$ of O .

Fig. 17.15 Regular tetrahedron inscribed in a cube. As an example of symmetry operations, we can choose three pairs of planes of σ_d (six planes in total) given by equations of $x = \pm y$ and $y = \pm z$ and $z = \pm x$



Let us imagine in Fig. 17.15 that a regular tetrahedron is inscribed in a cube. As an example of symmetry operations, suppose that three pairs of planes of σ_d (six planes in total) are given by equations of $x = \pm y$ and $y = \pm z$ and $z = \pm x$. Their Hesse's normal forms are represented as

$$\frac{1}{\sqrt{2}}(x \pm y) = 0, \quad (17.49)$$

$$\frac{1}{\sqrt{2}}(y \pm z) = 0, \quad (17.50)$$

$$\frac{1}{\sqrt{2}}(z \pm x) = 0. \quad (17.51)$$

Then, a dihedral angle α of the two planes is given by

$$\cos \alpha = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} \quad (17.52)$$

or

$$\cos \alpha = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = 0. \quad (17.53)$$

That is, $\alpha = \pi/3$ or $\alpha = \pi/2$. On the basis of the discussion of Sect. 17.2, the intersection of the two planes must be a rotation axis of C_3 or C_2 . Once again, in the case of C_3 , the intersection is a straight line connecting the origin and a vertex of the cube. This can readily be verified as follows: For instance, two planes given by $x = y$ and $y = z$ make an angle $\pi/3$ and produce an intersection line $x = y = z$. This line, in turn, connects the origin and a vertex of the cube.

If we choose, e.g., two planes $x = \pm y$ from the above, these planes make a right angle and their intersection must be a C_2 axis. The three C_2 axes coincide with the x -, y -, and z -axes. In this light, σ_d functions similarly to $6C_2$ of O in that their combinations produce $8C_3$ or $3C_2$. Thus, constitution and operation of T_d and O are related.

Let us more closely inspect the structure and constitution of O and T_d . First we construct mapping ρ between group elements of O and T_d such that

$$\begin{aligned}\rho : g \in O &\leftrightarrow g' \in T_d, \text{ if } g \in T, \\ \rho : g \in O &\leftrightarrow -(g')^{-1} \in T_d, \text{ if } g \notin T.\end{aligned}$$

In the above relation, the minus sign indicates that with an inverse representation matrix R must be replaced with $-R$. Then, $\rho(g) = g'$ is an isomorphic mapping. In fact, comparing Fig. 17.12 and Table 17.6, ρ gives identical matrix representation for O and T_d . For example, taking the first matrix of S_4 of T_d , we have

$$-\begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}^{-1} = -\begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The resulting matrix is identical to the first matrix of C_4 of O . Thus, we find T_d and O are isomorphic to each other.

Both O and T_d consist of 24 group elements and isomorphic to a *symmetric group* S_4 ; do not confuse it with the same symbol S_4 as a group element of T_d . The subgroup T consists of three conjugacy classes E , $8C_3$, and $3C_2$. Since T is constructed only by entire classes, it is an invariant subgroup; in this respect see the discussion of Sect. 16.3. The groups O , T_d , and T along with T_h and O_h form cubic groups [2]. In Table 17.7, we list these cubic groups together with their name and order. Of these, O is a pure rotation subgroup of O_h and T is a pure rotation subgroup of T_d and T_h .

Table 17.7 Several cubic groups and their characteristics

Notation	Group name	Order	Remark
T	Tetrahedral rotation	12	Subgroup of T_h , T_d
T_h , T_d	Tetrahedral	24	
O	Octahedral rotation	24	Subgroup of O_h
O_h	Octahedral	48	

Symmetry groups are related to permutations of n elements (or objects or numbers). The permutation has already appeared in (11.57) when we defined a determinant of a matrix. That was defined as

$$\sigma = \begin{pmatrix} 1 & 2 & \cdots & n \\ i_1 & i_2 & \cdots & i_n \end{pmatrix},$$

where σ means the permutation of the numbers $1, 2, \dots, n$. Therefore, the above symmetry group has $n!$ group elements (i.e., different ways of rearrangements). Although we do not dwell on symmetric groups much, we describe a following important theorem related to finite groups without proof. Interested readers are referred to literature [1].

Theorem 17.1 Cayley's Theorem [1] Every finite group $\mathcal{G}$ of order n is isomorphic to a subgroup (containing a whole group) of the symmetric group S_n . ■

17.4 Special Orthogonal Group $SO(3)$

In Parts III and IV thus far, we have dealt with a broad class of linear transformations. Related groups are finite groups. Here we will describe characteristics of special orthogonal group $SO(3)$, a kind of infinite groups. The $SO(3)$ represents rotations in three-dimensional Euclidean space $\mathbb{R}^3$. Rotations are made around an axis (a line through the origin) with the origin fixed. The rotation is defined by an azimuth of direction and a magnitude (angle) of rotation. The azimuth of direction is defined by two parameters and the magnitude is defined by one parameter, a rotation angle. Hence, the rotation is defined by three independent parameters. Since those parameters are continuously variable, $SO(3)$ is one of the continuous groups.

Two rotations result in another rotation with the origin again fixed. A reverse rotation is unambiguously defined. An identity transformation is naturally defined. An associative law holds as well. Thus, the relevant rotations form a group, i.e., $SO(3)$. The rotation is represented by a real $(3, 3)$ matrix whose determinant is 1. A matrix representation is uniquely determined once an orthonormal basis is set in $\mathbb{R}^3$. Any rotation is represented by a rotation matrix accordingly. The rotation matrix R is defined by

$$R^T R = R R^T = E, \quad (17.54)$$

where R is a real matrix with $\det R = 1$. Notice that we exclude the case where $\det R = -1$. Matrices that satisfy (17.54) with $\det R = \pm 1$ are referred to as orthogonal matrices that cause orthogonal transformation. Correspondingly, orthogonal groups (represented by orthogonal matrices) contain rotation groups as a special case. In other words, the orthogonal groups contain the rotation groups as a

subgroup. An orthogonal group in $\mathbb{R}^3$ denoted $O(3)$ contains $SO(3)$ as a subgroup. By the same token, orthogonal matrices contain rotation matrices as a special case.

In Sect. 17.2 we treated reflection and improper rotation with a determinant of their matrices being -1 . In this section these transformations are excluded and only rotations are dealt with. We focus on geometric characteristics of the rotation groups. Readers are referred to more detailed representation theory of $SO(3)$ in appropriate literature [1].

17.4.1 Rotation Axis and Rotation Matrix

In this section we represent a vector such as $|x\rangle$. We start with showing that any rotation has a unique presence of a rotation axis. The rotation axis is defined by the following: Suppose that there is a rigid body with some point within the body fixed. Here the said rigid body can be that with infinite extent. Then the rigid body exerts rotation. The rotation axis is a line on which every point is unmoved during the rotation. As a matter of course, identity matrix E has three linearly independent rotation axes. (Practically, this represents *no* rotation.)

Theorem 17.2 Any rotation matrix R is accompanied by at least one rotation axis. Unless the rotation matrix is identity, the rotation matrix should be accompanied by one and only one rotation axis.

Proof As R is an orthogonal matrix of a determinant 1, so are R^T and R^{-1} . Then we have

$$(R - E)^T = R^T - E = R^{-1} - E. \quad (17.55)$$

Hence, we get

$$\begin{aligned} \det(R - E) &= \det(R^T - E) = \det(R^{-1} - E) = \det[R^{-1}(E - R)] \\ &= \det(R^{-1})\det(E - R) = \det(E - R) = -\det(R - E). \end{aligned} \quad (17.56)$$

Note here that with any $(3, 3)$ matrix A of $\mathbb{R}^3$,

$$\det A = -(-1)^9 \det A = -\det(-A).$$

This equality holds with $\mathbb{R}^n$ (n : odd), but in $\mathbb{R}^n$ (n : even), we have $\det A = \det(-A)$. Then, (17.56) results in a trivial equation $0 = 0$ accordingly. Therefore, the discussion made below only applies to $\mathbb{R}^n$ (n : odd). Thus, from (17.56) we have

$$\det(R - E) = 0. \quad (17.57)$$

This implies that for $\exists |x_0\rangle \neq 0$

$$(R - E) |x_0\rangle = 0. \quad (17.58)$$

Therefore, we get

$$R(a|x_0\rangle) = a |x_0\rangle, \quad (17.59)$$

where a is an arbitrarily chosen real number. In this case an eigenvalue of R is 1, which an eigenvector $a |x_0\rangle$ corresponds to. Thus, as a rotation axis, we have a straight line expressed as

$$l = \text{Span}\{a|x_0\rangle; a \in \mathbb{R}\}. \quad (17.60)$$

This proves the presence of a rotation axis.

Next suppose that there are two (or more) rotation axes. The presence of two rotation axes naturally implies that there are two linearly independent vectors (i.e., two straight lines that mutually intersect at the fixed point). Suppose that such vectors are $|u\rangle$ and $|v\rangle$. Then we have

$$(R - E) |u\rangle = 0, \quad (17.61)$$

$$(R - E) |v\rangle = 0. \quad (17.62)$$

Let us consider a vector $\forall |y\rangle$ that is chosen from $\text{Span}\{|u\rangle, |v\rangle\}$, to which we assume that

$$\text{Span}\{|u\rangle, |v\rangle\} \equiv \text{Span}\{a|u\rangle, b|v\rangle; a, b \in \mathbb{R}\}. \quad (17.63)$$

That is, $\text{Span}\{|u\rangle, |v\rangle\}$ represents a plane P formed by two mutually intersecting straight lines. Then we have

$$y = s|u\rangle + t|v\rangle, \quad (17.64)$$

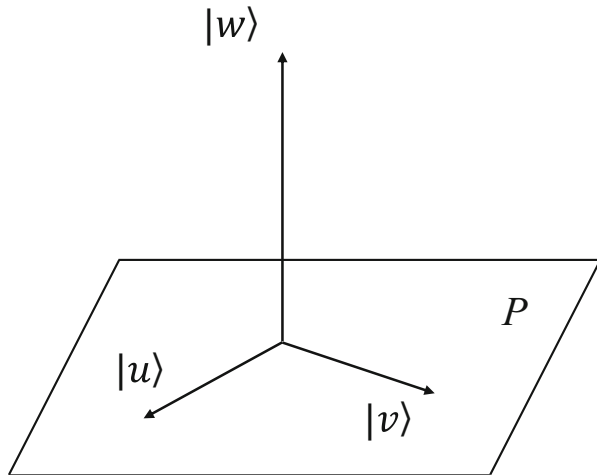
where s and t are some real numbers. Operating $R - E$ on (17.64), we have

$$(R - E) |y\rangle = (R - E)(s|u\rangle + t|v\rangle) = s(R - E) |u\rangle + t(R - E) |v\rangle = 0. \quad (17.65)$$

This indicates that any vectors in P can be an eigenvector of R , implying that an infinite number of rotation axes exist.

Now, take another vector $|w\rangle$ that is perpendicular to the plane P (see Fig. 17.16). Let us consider an inner product $\langle y | R w \rangle$. Since

Fig. 17.16 Plane P formed by two mutually intersecting straight lines represented by $|u\rangle$ and $|v\rangle$. Another vector $|w\rangle$ is perpendicular to the plane P



$$R|u\rangle = |u\rangle, \langle u|R^\dagger = \langle u|R^T = \langle u|. \quad (17.66)$$

Similarly, we have

$$\langle v|R^T = \langle v|. \quad (17.67)$$

Therefore, using the relation (17.64), we get

$$\langle y|R^T = \langle y|. \quad (17.68)$$

Here we are dealing with real numbers and, hence, we have

$$R^\dagger = R^T. \quad (17.69)$$

Now we have

$$\langle y|Rw\rangle = \langle yR^T|Rw\rangle = \langle y|R^TRw\rangle = \langle y|Ew\rangle = \langle y|w\rangle = 0. \quad (17.70)$$

In (17.70) the second equality comes from the associative law; the third is due to (17.54). The last equality comes from that $|w\rangle$ is perpendicular to P .

From (17.70) we have

$$\langle y|(R - E)w\rangle = 0. \quad (17.71)$$

However, we should be careful not to conclude immediately from (17.71) that $(R - E)|w\rangle = 0$; i.e., $R|w\rangle = |w\rangle$. This is because in (17.70) $|y\rangle$ does not represent all vectors in $\mathbb{R}^3$, but merely represents all the vectors in $\text{Span}\{|u\rangle, |v\rangle\}$. Nonetheless, both $|w\rangle$ and $|Rw\rangle$ are perpendicular to P , and so

$$|Rw\rangle = a|w\rangle, \quad (17.72)$$

where a is an arbitrarily chosen real number. From (17.72), we have

$$\langle wR^\dagger | Rw \rangle = \langle wR^T | Rw \rangle = \langle w | w \rangle = |a|^2 \langle w | w \rangle, \text{ i.e., } a = \pm 1,$$

where the first equality comes from that R is an orthogonal matrix. Since $\det R = 1$, $a = 1$. Thus, from (17.72) this time around, we have $|Rw\rangle = |w\rangle$; that is,

$$(R - E)|w\rangle = 0. \quad (17.73)$$

Equations (17.65) and (17.73) imply that for any vector $|x\rangle$ arbitrarily chosen from $\mathbb{R}^3$, we have

$$(R - E)|x\rangle = 0. \quad (17.74)$$

Consequently, we get

$$R - E = 0 \text{ or } R = E. \quad (17.75)$$

The above procedures represented by (17.61)–(17.75) indicate that the presence of two rotation axes necessarily requires a transformation matrix to be identity. This implies that all the vectors in $\mathbb{R}^3$ are an eigenvector of a rotation matrix.

Taking contraposition of the above, unless the rotation matrix is identity, the relevant rotation cannot have two rotation axes. Meanwhile, the proof of the former half ensures the presence at least one rotation axis. Hence, any rotation is characterized by a unique rotation axis except for the identity transformation. This completes the proof. ■

An immediate consequence of Theorem 17.2 is that the rotation matrix should have an eigenvalue 1 which an eigenvector representing the rotation axis corresponds to. This statement includes a trivial fact that all the eigenvalues of the identity matrix are 1. In Sect. 14.4, we calculated eigenvalues of a two-dimensional rotation matrix. The eigenvalues were $e^{i\theta}$ or $e^{-i\theta}$, where θ is a rotation angle.

Let us consider rotation matrices that we dealt with in Sect. 17.1. The matrix representing the rotation around the z -axis by a rotation angle θ is expressed by

$$R = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (17.8)$$

In Sect. 14.4 we treated diagonalization of a rotation matrix. As R is reduced, the diagonalization can be performed in a manner essentially the same as (14.92). That is, as a diagonalizing unitary matrix, we have

$$U = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{i}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}, U^\dagger = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (17.76)$$

As a result of the unitary similarity transformation, we get

$$\begin{aligned} U^\dagger R U &= \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{i}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} e^{i\theta} & 0 & 0 \\ 0 & e^{-i\theta} & 0 \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned} \quad (17.77)$$

Thus, eigenvalues are 1, $e^{i\theta}$, and $e^{-i\theta}$. The eigenvalue 1 results from the existence of the unique rotation axis. When $\theta = 0$, (17.77) gives an identity matrix with all the eigenvalues 1 as expected. When $\theta = \pi$, eigenvalues are -1 , -1 , and 1. The eigenvalue 1 is again associated with the unique rotation axis. The (unitary) similarity transformation keeps a trace unchanged, that is, the trace χ is

$$\chi = 1 + 2 \cos \theta. \quad (17.78)$$

As R is a normal operator, spectral decomposition can be done as in the case of Example 14.1. Here we only show the result below:

$$R = e^{i\theta} \begin{pmatrix} \frac{1}{2} & \frac{i}{2} & 0 \\ -\frac{i}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix} + e^{-i\theta} \begin{pmatrix} \frac{1}{2} & -\frac{i}{2} & 0 \\ \frac{i}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Three matrices of the above equation are projection operators.

17.4.2 Euler Angles and Related Topics

Euler angles are well known and have been used in various fields of science. We wish to connect the above discussion with Euler angles.

In Part III we dealt with successive linear transformations. This can be extended to the case of three or more successive transformations. Suppose that we have three successive transformation R_1 , R_2 , and R_3 and that the coordinate system (a three-dimensional orthonormal basis) is transformed from $O \rightarrow I \rightarrow II \rightarrow III$ accordingly. The symbol “O” stands for the original coordinate system and I, II, and III represent successively transformed systems.

With the discussion that follows, let us denote the transformation by R'_2 , R'_3 , R''_3 , etc. in reference to the coordinate system I. For example, R'_3 means that the third transformation is viewed from the system I. The transformation R''_3 indicates the third transformation is viewed from the system II. That is, the number of primes “'” denotes the number of the coordinate system to distinguish the systems I and II. Let R_2 (without prime) stand for the second transformation viewed from the system O.

Meanwhile, we have

$$R_1 R'_2 = R_2 R_1. \quad (17.79)$$

This notation is in parallel to (11.80). Similarly, we have

$$R'_2 R''_3 = R'_3 R'_2 \quad \text{and} \quad R_1 R'_3 = R_3 R_1. \quad (17.80)$$

Therefore, we get [3]

$$R_1 R'_2 R''_3 = R_1 R'_3 R'_2 = R_3 R_1 R'_2 = R_3 R_2 R_1. \quad (17.81)$$

Also combining (17.79) and (17.80), we have

$$R''_3 = (R_2 R_1)^{-1} R_3 (R_2 R_1). \quad (17.82)$$

Let us call R'_2 , R''_3 , etc. a transformation on a “moving” coordinate system (i.e., the systems I, II, III, $\dots$). On the other hand, we call R_1 , R_2 , etc. a transformation on a “fixed” system (i.e., original coordinate system O). Thus (17.81) shows that the multiplication order is reversed with respect to the moving system and fixed system [3].

For a practical purpose, it would be enough to consider three successive transformations. Let us think of, however, a general case where n successive transformations are involved (n denotes a positive integer). For the purpose of succinct notation, let us define the linear transformations and relevant coordinate systems as those in Fig. 17.17. The diagram shows the transformation of the basis vectors. We define a following orthogonal transformation $R_j^{(i)}$:

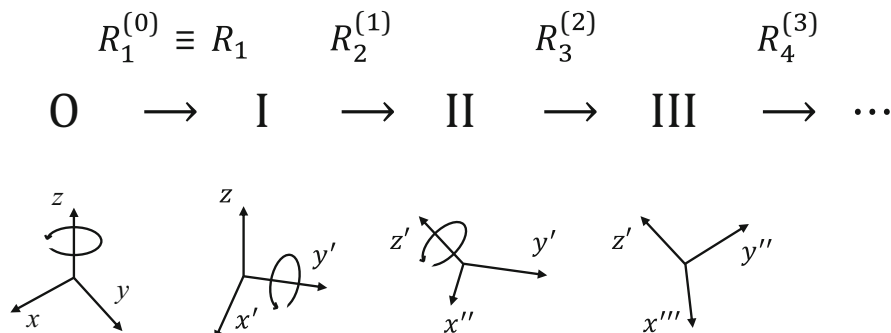


Fig. 17.17 Successive orthogonal transformations and relevant coordinate systems

$$R_j^{(i)} \quad (0 \leq i < j \leq n), \text{ and } R_i \equiv R_i^{(0)}, \quad (17.83)$$

where $R_j^{(i)}$ is defined as a transformation R_j described in reference to the coordinate system i ; $R_i^{(0)}$ means that R_i is referred to the original coordinate system (i.e., the fixed coordinate). Then we have

$$R_{i-1}^{(i-2)} R_k^{(i-1)} = R_k^{(i-2)} R_{i-1}^{(i-2)} \quad (k > i-1). \quad (17.84)$$

Particularly, when $i = 3$

$$R_2^{(1)} R_k^{(2)} = R_k^{(1)} R_2^{(1)} \quad (k > 2). \quad (17.85)$$

For $i = 2$ we have

$$R_1 R_k^{(1)} = R_k R_1 \quad (k > 1). \quad (17.86)$$

We define n time successive transformations on a moving coordinate system as $\tilde{R}_n$ such that

$$\tilde{R}_n \equiv R_1 R_2^{(1)} R_3^{(2)} \cdots R_{n-2}^{(n-3)} R_{n-1}^{(n-2)} R_n^{(n-1)}. \quad (17.87)$$

Applying (17.84) on $R_{n-1}^{(n-2)} R_n^{(n-1)}$ and rewriting (17.87), we have

$$\tilde{R}_n = R_1 R_2^{(1)} R_3^{(2)} \cdots R_{n-2}^{(n-3)} R_n^{(n-2)} R_{n-1}^{(n-2)}. \quad (17.88)$$

Applying (17.84) again on $R_{n-2}^{(n-3)} R_n^{(n-2)}$, we get

$$\tilde{R}_n = R_1 R_2^{(1)} R_3^{(2)} \cdots R_n^{(n-3)} R_{n-2}^{(n-3)} R_{n-1}^{(n-2)}. \quad (17.89)$$

Proceeding similarly, we have

$$\tilde{R}_n = R_1 R_n^{(1)} R_2^{(1)} R_3^{(2)} \cdots R_{n-2}^{(n-3)} R_{n-1}^{(n-2)} = R_n R_1 R_2^{(1)} R_3^{(2)} \cdots R_{n-2}^{(n-3)} R_{n-1}^{(n-2)}, \quad (17.90)$$

where with the last equality we used (17.86). In this case, we have

$$R_1 R_n^{(1)} = R_n R_1. \quad (17.91)$$

To reach RHS of (17.90), we applied (17.84) $(n - 1)$ times in total. Then we repeat the above procedures with respect to $R_{n-1}^{(n-2)}$ another $(n - 2)$ times to get

$$\tilde{R}_n = R_n R_{n-1} R_1 R_2^{(1)} R_3^{(2)} \cdots R_{n-2}^{(n-3)}. \quad (17.92)$$

Further proceeding similarly, we finally get

$$\tilde{R}_n = R_n R_{n-1} R_{n-2} \cdots R_3 R_2 R_1. \quad (17.93)$$

In total, we have applied the permutation of (17.84) $n(n - 1)/2$ times. When n is 2, $n(n - 1)/2 = 1$. This is the case with (17.79). If n is 3, $n(n - 1)/2 = 3$. This is the case with (17.81). Thus, (17.93) once again confirms that the multiplication order is reversed with respect to the moving system and fixed system.

Meanwhile, we define $\tilde{P}$ as

$$\tilde{P} \equiv R_1 R_2^{(1)} R_3^{(2)} \cdots R_{n-2}^{(n-3)} R_{n-1}^{(n-2)}. \quad (17.94)$$

Alternately, we describe

$$\tilde{P} = R_{n-1} R_{n-2} \cdots R_3 R_2 R_1. \quad (17.95)$$

Then, from (17.87) and (17.90), we get

$$\tilde{R}_n = \tilde{P} R_n^{(n-1)} = R_n \tilde{P}. \quad (17.96)$$

Equivalently, we have

$$R_n = \tilde{P} R_n^{(n-1)} \tilde{P}^{-1} \quad (17.97)$$

or

$$R_n^{(n-1)} = \tilde{P}^{-1} R_n \tilde{P}. \quad (17.98)$$

Moreover, we have

$$\begin{aligned} \tilde{P}^T &= \left[R_1 R_2^{(1)} R_3^{(2)} \cdots R_{n-2}^{(n-3)} R_{n-1}^{(n-2)} \right]^T \\ &= \left[R_{n-1}^{(n-2)} \right]^T \left[R_{n-2}^{(n-3)} \right]^T \cdots \left[R_2^{(1)} \right]^T \left[R_1 \right]^T \\ &= \left[R_{n-1}^{(n-2)} \right]^{-1} \left[R_{n-2}^{(n-3)} \right]^{-1} \cdots \left[R_2^{(1)} \right]^{-1} \left[R_1^{-1} \right] \\ &= \left[R_1 R_2^{(1)} R_3^{(2)} \cdots R_{n-2}^{(n-3)} R_{n-1}^{(n-2)} \right]^{-1} = \tilde{P}^{-1}. \end{aligned} \quad (17.99)$$

The third equality of (17.99) comes from the fact that matrices $R_{n-1}^{(n-2)}$, $R_{n-2}^{(n-3)}$, $R_2^{(1)}$, and R_1 are orthogonal matrices. Then, we get

$$\tilde{P}^T \tilde{P} = \tilde{P} \tilde{P}^T = E. \quad (17.100)$$

Thus, $\tilde{P}$ is an orthogonal matrix.

From a point of view of practical application, (17.97) and (17.98) are very useful. This is because R_n and $R_n^{(n-1)}$ are conjugate to each other. Consequently, R_n has the same eigenvalues and trace as $R_n^{(n-1)}$. In light of (11.81), we see that (17.97) and (17.98) relate R_n (i.e., viewed in reference to the original coordinate system) to $R_n^{(n-1)}$ [i.e., the same transformation viewed in reference to the coordinate system reached after the $(n-1)$ transformations]. Since the transformation $R_n^{(n-1)}$ is usually described in a simple form, matrix calculations to compute R_n can readily be done. Now let us consider an example.

Example 17.4 Successive rotations A typical illustration of three successive transformations in moving coordinate systems is well known in association with Euler angles. This contains the following three steps:

1. Rotation by α around the z -axis in the original coordinate system (O)
2. Rotation by β around the y' -axis in the transferred coordinate system (I)
3. Rotation by γ around the z'' -axis (the same as z' -axis) in the transferred coordinate system (II)

The above three steps are represented by matrices of $R_{z\alpha}$, $R'_{y'\beta}$, and $R''_{z''\gamma}$ in (17.12). That is, as a total transformation $\tilde{R}_3$, we have

$$\begin{aligned}
\widetilde{R}_3 &= \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{pmatrix} \begin{pmatrix} \cos \gamma & -\sin \gamma & 0 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} \cos \alpha \cos \beta \cos \gamma - \sin \alpha \sin \gamma & -\cos \alpha \cos \beta \sin \gamma - \sin \alpha \cos \gamma & \cos \alpha \sin \beta \\ \sin \alpha \cos \beta \cos \gamma + \cos \alpha \sin \gamma & -\sin \alpha \cos \beta \sin \gamma + \cos \alpha \cos \gamma & \sin \alpha \sin \beta \\ -\sin \beta \cos \gamma & \sin \beta \sin \gamma & \cos \beta \end{pmatrix}.
\end{aligned} \tag{17.101}$$

This matrix corresponds to (17.87), where $n = 3$. The angles α , β , and γ in (17.101) are called Euler angles and their domains are usually taken as $0 \leq \alpha \leq 2\pi$, $0 \leq \beta \leq \pi$, $0 \leq \gamma \leq 2\pi$. The matrix (17.101) is widely used in quantum mechanics and related fields of natural science. The matrix notation, however, differs from literature to literature, and so care should be taken [3–5]. Using the notation of (17.87), we have

$$R_1 = \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{17.102}$$

$$R_2^{(1)} = \begin{pmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{pmatrix}, \tag{17.103}$$

$$R_3^{(2)} = \begin{pmatrix} \cos \gamma & -\sin \gamma & 0 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{17.104}$$

From (17.94) we also get

$$\tilde{P} \equiv R_1 R_2^{(1)} = \begin{pmatrix} \cos \alpha \cos \beta & -\sin \alpha & \cos \alpha \sin \beta \\ \sin \alpha \cos \beta & \cos \alpha & \sin \alpha \sin \beta \\ -\sin \beta & 0 & \cos \beta \end{pmatrix}. \tag{17.105}$$

Corresponding to (17.97), we have

$$R_3 = \tilde{P} R_3^{(2)} \tilde{P}^{-1} = R_1 R_2^{(1)} R_3^{(2)} \left[R_2^{(1)} \right]^{-1} R_1^{-1}. \tag{17.106}$$

Now matrix calculations are readily performed such that

$$\begin{aligned}
 R_3 &= \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{pmatrix} \begin{pmatrix} \cos \gamma & -\sin \gamma & 0 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 &\quad \times \begin{pmatrix} \cos \beta & 0 & -\sin \beta \\ 0 & 1 & 0 \\ \sin \beta & 0 & \cos \beta \end{pmatrix} \begin{pmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} (\cos^2 \alpha \cos^2 \beta + \sin^2 \alpha) \cos \gamma & \cos \alpha \sin \alpha \sin^2 \beta (1 - \cos \gamma) & \cos \alpha \cos \beta \sin \beta (1 - \cos \gamma) \\ + \cos^2 \alpha \sin^2 \beta & -\cos \beta \sin \gamma & + \sin \alpha \sin \beta \sin \gamma \\ \cos \alpha \sin \alpha \sin^2 \beta (1 - \cos \gamma) & (\sin^2 \alpha \cos^2 \beta + \cos^2 \alpha) \cos \gamma & \sin \alpha \cos \beta \sin \beta (1 - \cos \gamma) \\ + \cos \beta \sin \gamma & + \sin^2 \alpha \sin^2 \beta & -\cos \alpha \sin \beta \sin \gamma \\ \cos \beta \sin \beta \cos \alpha (1 - \cos \gamma) & \sin \alpha \cos \beta \sin \beta (1 - \cos \gamma) & \sin^2 \beta \cos \gamma \\ - \sin \alpha \sin \beta \sin \gamma & + \cos \alpha \sin \beta \sin \gamma & + \cos^2 \beta \end{pmatrix}. \quad (17.107)
 \end{aligned}$$

Notice that in (17.107) we have a trace χ described as

$$\chi = 1 + 2 \cos \gamma. \quad (17.108)$$

The trace is the same as that of $R_3^{(2)}$, as expected from (17.106) and (17.97).

Equation (17.107) apparently seems complicated but has simple and well-defined meaning. The rotation represented by $R_3^{(2)}$ is characterized by a rotation by γ around the z'' -axis. Figure 17.18 represents the orientation of the z'' -axis viewed in reference to the original xyz -system. That is, the z'' -axis (identical to the rotation axis A) is designated by an azimuthal angle α and a zenithal angle β as shown. The operation R_3 is represented by a rotation by γ around the axis A in the xyz -system. The angles α , β , and γ coincide with the Euler angles designated with the same independent parameters α , β , and γ .

From (17.77) and (17.107), a diagonalizing matrix for R_3 is $\tilde{P}U$. That is,

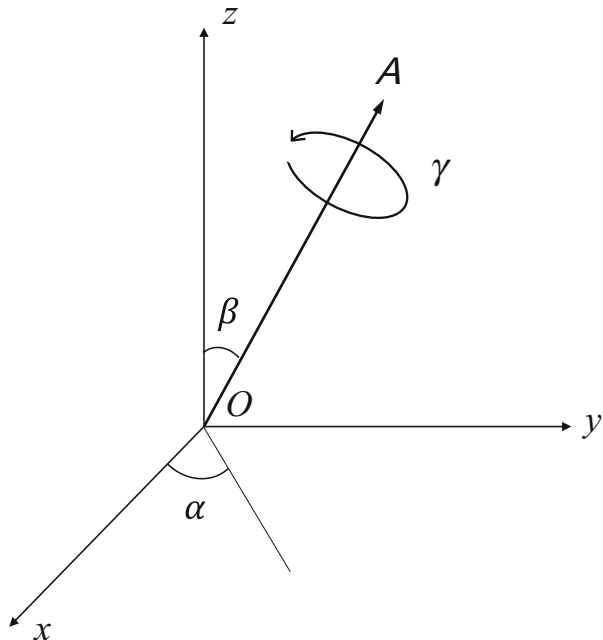
$$U^\dagger \tilde{P}^{-1} R_3 \tilde{P} U = (\tilde{P} U)^\dagger R_3 \tilde{P} U = U^\dagger R_3^{(2)} U = \begin{pmatrix} e^{i\gamma} & 0 & 0 \\ 0 & e^{-i\gamma} & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (17.109)$$

Note that as $\tilde{P}$ is a real matrix, we have

$$\tilde{P}^\dagger = \tilde{P}^T = \tilde{P}^{-1} \quad (17.110)$$

and

Fig. 17.18 Rotation γ around the rotation axis A . The orientation of A is defined by angles α and β as shown



$$\begin{aligned}
 \tilde{P}U &= \begin{pmatrix} \cos \alpha \cos \beta & -\sin \alpha & \cos \alpha \sin \beta \\ \sin \alpha \cos \beta & \cos \alpha & \sin \alpha \sin \beta \\ -\sin \beta & 0 & \cos \beta \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{i}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} \frac{1}{\sqrt{2}} \cos \alpha \cos \beta + \frac{i}{\sqrt{2}} \sin \alpha & \frac{1}{\sqrt{2}} \cos \alpha \cos \beta - \frac{i}{\sqrt{2}} \sin \alpha & \cos \alpha \sin \beta \\ \frac{1}{\sqrt{2}} \sin \alpha \cos \beta - \frac{i}{\sqrt{2}} \cos \alpha & \frac{1}{\sqrt{2}} \sin \alpha \cos \beta + \frac{i}{\sqrt{2}} \cos \alpha & \sin \alpha \sin \beta \\ -\frac{1}{\sqrt{2}} \sin \beta & -\frac{1}{\sqrt{2}} \sin \beta & \cos \beta \end{pmatrix}.
 \end{aligned}
 \tag{17.111}$$

A vector representing the rotation axis corresponds to an eigenvalue 1. The direction cosines of x -, y -, and z -component for the rotation axis A are $\cos \alpha \sin \beta$, $\sin \alpha \sin \beta$, and $\cos \beta$ (see Fig. 3.1), respectively, when viewed in reference to the original xyz -coordinate system. This can directly be shown as follows: The characteristic equation of R_3 is expressed as

$$|R_3 - \lambda E| = 0. \quad (17.112)$$

Using (17.107) we have

$$|R_3 - \lambda E| = \begin{vmatrix} (\cos^2 \alpha \cos^2 \beta + \sin^2 \alpha) \cos \gamma & \cos \alpha \sin \alpha \sin^2 \beta (1 - \cos \gamma) & \cos \alpha \cos \beta \sin \beta (1 - \cos \gamma) \\ + \cos^2 \alpha \sin^2 \beta - \lambda & - \cos \beta \sin \gamma & + \sin \alpha \sin \beta \sin \gamma \\ \cos \alpha \sin \alpha \sin^2 \beta (1 - \cos \gamma) & (\sin^2 \alpha \cos^2 \beta + \cos^2 \alpha) \cos \gamma & \sin \alpha \cos \beta \sin \beta (1 - \cos \gamma) \\ + \cos \beta \sin \gamma & + \sin^2 \alpha \sin^2 \beta - \lambda & - \cos \alpha \sin \beta \sin \gamma \\ \cos \alpha \cos \beta \sin \beta (1 - \cos \gamma) & \sin \alpha \cos \beta \sin \beta (1 - \cos \gamma) & \sin^2 \beta \cos \gamma \\ - \sin \alpha \sin \beta \sin \gamma & + \cos \alpha \sin \beta \sin \gamma & + \cos^2 \beta - \lambda \end{vmatrix}. \quad (17.113)$$

When $\lambda = 1$, we must have the direction cosines of the rotation axis as an eigenvector. That is, we get

$$\begin{pmatrix} (\cos^2 \beta \cos^2 \alpha + \sin^2 \alpha) \cos \gamma & \sin^2 \beta \cos \alpha \sin \alpha (1 - \cos \gamma) & \cos \beta \sin \beta \cos \alpha (1 - \cos \gamma) \\ + \sin^2 \beta \cos^2 \alpha - 1 & - \cos \beta \sin \gamma & + \sin \beta \sin \alpha \sin \gamma \\ \sin^2 \beta \cos \alpha \sin \alpha (1 - \cos \gamma) & (\cos^2 \beta \sin^2 \alpha + \cos^2 \alpha) \cos \gamma & \sin \alpha \cos \beta \sin \beta (1 - \cos \gamma) \\ + \cos \beta \sin \gamma & + \sin^2 \alpha \sin^2 \beta - 1 & - \cos \alpha \sin \beta \sin \gamma \\ \cos \alpha \cos \beta \sin \beta (1 - \cos \gamma) & \sin \alpha \cos \beta \sin \beta (1 - \cos \gamma) & \sin^2 \beta \cos \gamma \\ - \sin \alpha \sin \beta \sin \gamma & + \cos \alpha \sin \beta \sin \gamma & + \cos^2 \beta - 1 \end{pmatrix} \times \begin{pmatrix} \cos \alpha \sin \beta \\ \sin \alpha \sin \beta \\ \cos \beta \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \quad (17.114)$$

The above matrix calculations certainly verify that (17.114) holds. The confirmation is left as an exercise for readers.

As an application of (17.107) to an illustrative example, let us consider a $2\pi/3$ rotation around an axis trisecting a solid angle $\pi/2$ formed by the x -, y -, and z -axes (see Fig. 17.19 and Sect. 17.3). Then we have

$$\cos \alpha = \sin \alpha = 1/\sqrt{2}, \quad \cos \beta = 1/\sqrt{3}, \quad \sin \beta = 2/\sqrt{3},$$

$$\cos \gamma = -1/2, \quad \sin \gamma = \sqrt{3}/2. \quad (17.115)$$

Substituting (17.115) for (17.107), we get

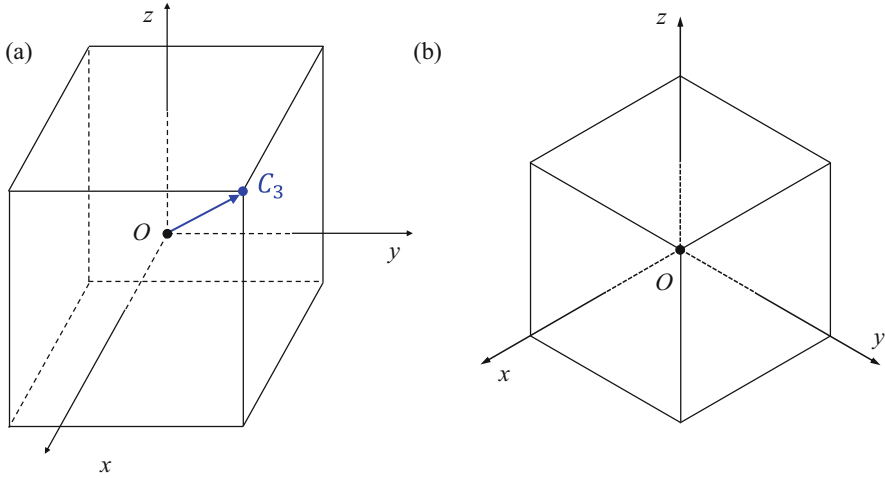


Fig. 17.19 Rotation axis of C_3 that permutes basis vectors. **(a)** The C_3 axis is a straight line that connects the origin and each vertex of a cube. **(b)** Cube viewed along the C_3 axis that connects the origin and vertex

$$R_3 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \quad (17.116)$$

If we write a linear transformation R_3 following (11.37), we get

$$R_3(|x\rangle) = (|e_1\rangle |e_2\rangle |e_3\rangle) \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad (17.117)$$

where $|e_1\rangle$, $|e_2\rangle$, and $|e_3\rangle$ represent unit basis vectors in the direction of x -, y -, and z -axes, respectively. We have $|x\rangle = x_1 |e_1\rangle + x_2 |e_2\rangle + x_3 |e_3\rangle$. Thus, we get

$$R_3(|x\rangle) = (|e_2\rangle |e_3\rangle |e_1\rangle) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}. \quad (17.118)$$

This implies a cyclic permutation of the basis vectors. This is well characterized by Fig. 17.19. Alternatively, (17.117) can be expressed in terms of the column vectors (i.e., coordinate) transformation as

$$R_3(|x\rangle) = (|e_1\rangle |e_2\rangle |e_3\rangle) \begin{pmatrix} x_3 \\ x_1 \\ x_2 \end{pmatrix}. \quad (17.119)$$

Care should be taken on which linear transformation is intended out of the basis vector transformation or coordinate transformation.

As mentioned above, we have compared geometric features on the moving coordinate system and fixed coordinate system. The features apparently seem to differ at a glance, but are essentially the same. $\square$

References

1. Hamermesh M (1989) Group theory and its application to physical problems. Dover, New York
2. Cotton FA (1990) Chemical applications of group theory, 3rd edn. Wiley, New York
3. Edmonds AR (1957) Angular momentum in quantum mechanics. Princeton University Press, Princeton
4. Chen JQ, Ping J, Wang F (2002) Group representation theory for physicists, 2nd edn. World Scientific, Singapore
5. Arfken GB, Weber HJ, Harris FE (2013) Mathematical methods for physicists, 7th edn. Academic Press, Waltham

Chapter 18

Representation Theory of Groups



Representation theory is an important pillar of the group theory. As we shall see soon, a word “representation” and its definition sound a bit daunting. To be short, however, we need “numbers” or “matrices” to do a mathematical calculation. Therefore, we may think of the representation as merely numbers and matrices. Individual representations have their dimension. If that dimension is one, we treat a representation as a number (real or complex). If the dimension is two or more, we are going to deal with matrices; in the case of the n -dimension, it is a (n, n) square matrix. In this chapter we focus on the representation theory of finite groups. In this case, we have an important theorem stating that a representation of any finite group can be converted to a unitary representation by a similarity transformation. That is, group elements of a finite group are represented by a unitary matrix. According to the dimension of the representation, we have the same number of basis vectors. Bearing these things firmly in mind, we can pretty easily understand this important notion of representation.

18.1 Definition of Representation

In Sect. 16.4 we dealt with various aspects of the mapping between group elements. Of these, we studied fundamental properties of isomorphism and homomorphism. In this section we introduce the notion of representation of groups and study it. If we deal with a finite group consisting of n elements, we describe it as $\mathcal{G} = \{g_1 \equiv e, g_2, \dots, g_n\}$ as in the case of Sect. 16.1.

Definition 18.1 Let $\mathcal{G} = \{g_\nu\}$ be a group comprising elements g_ν . Suppose that a (d, d) matrix $D(g_\nu)$ is given for each group element g_ν . Suppose also that in correspondence with

$$g_\mu g_\nu = g_\rho, \quad (18.1)$$

$$D(g_\mu)D(g_\nu) = D(g_\rho) \quad (18.2)$$

holds. Then a set $\mathcal{L}$ consisting of $D(g_\nu)$, that is,

$$\mathcal{L} = \{D(g_\nu)\}$$

is said to be a representation.

Although the definition seems somewhat daunting, the representation is, as already seen, merely a homomorphism.

We call individual matrices $D(g_\nu)$ a representation matrix. A dimension d of the matrix is said to be a dimension of representation as well. In correspondence with $g_\nu e = g_\nu$, we have

$$D(g_\mu)D(e) = D(g_\mu). \quad (18.3)$$

Therefore, we get

$$D(e) = E, \quad (18.4)$$

where E is an identity matrix of dimension d . Also in correspondence with $g_\nu g_\nu^{-1} = g_\nu^{-1} g_\nu = E$, we have

$$D(g_\nu)D(g_\nu^{-1}) = D(g_\nu^{-1})D(g_\nu) = D(e) = E. \quad (18.5)$$

That is,

$$D(g_\nu^{-1}) = [D(g_\nu)]^{-1}. \quad (18.6)$$

Namely, an inverse matrix corresponds to an inverse element. If the representation has one-to-one correspondence, the representation is said to be faithful. In the case of n -to-one correspondence, the representation is homomorphic. In particular, representing all group elements by 1 [as $(1, 1)$ matrix] is called an “identity representation.”

Illustrative examples of the representation have already appeared in Sects. 17.2 and 17.3. In that case the representation was faithful. For instance, in Tables 17.2, 17.4, and 17.6 as well as Fig. 17.12, the number of representation matrices is the same as that of group elements. In Sects. 17.2 and 17.3, in most cases we have used real orthogonal matrices. Such matrices are included among unitary matrices. Representation using unitary matrices is said to be a “unitary representation.” In fact, a representation of a finite group can be converted to a unitary representation.

Theorem 18.1 [1, 2] A representation of any finite group can be converted to a unitary representation by a similarity transformation.

Proof Let $\mathcal{g} = \{g_1, g_2, \dots, g_n\}$ be a finite group of order n . Let $D(g_\nu)$ be a representation matrix of g_ν of dimension d . Here we suppose that $D(g_\nu)$ is not unitary, but non-singular. Using $D(g_\nu)$, we construct the following matrix H such that

$$H = \sum_{i=1}^n D(g_i)^\dagger D(g_i). \quad (18.7)$$

Then, for an arbitrarily chosen group element g_j , we have

$$\begin{aligned} D(g_j)^\dagger H D(g_j) &= \sum_{i=1}^n D(g_j)^\dagger D(g_i)^\dagger D(g_i) D(g_j) \\ &= \sum_{i=1}^n [D(g_i) D(g_j)]^\dagger D(g_i) D(g_j) \\ &= \sum_{i=1}^n [D(g_i g_j)]^\dagger D(g_i g_j) = \sum_{k=1}^n D(g_k)^\dagger D(g_k) \\ &= H, \end{aligned} \quad (18.8)$$

where the third equality comes from homomorphism of the representation and the second last equality is due to rearrangement theorem (see Sect. 16.1).

Note that each matrix $D(g_i)^\dagger D(g_i)$ is an Hermitian Gram matrix constructed by a non-singular matrix and that H is a summation of such Gram matrices. Consequently, on the basis of the argument of Sect. 13.2, H is positive definite and all the eigenvalues of H are positive. Then, using an appropriate unitary matrix U , we can get a diagonal matrix Λ such that

$$U^\dagger H U = \Lambda. \quad (18.9)$$

Here, the diagonal matrix Λ is given by

$$\Lambda = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_d \end{pmatrix}, \quad (18.10)$$

with $\lambda_i > 0$ ($1 \leq i \leq d$). We define $\Lambda^{1/2}$ such that

$$\Lambda^{1/2} = \begin{pmatrix} \sqrt{\lambda_1} & & & \\ & \sqrt{\lambda_2} & & \\ & & \ddots & \\ & & & \sqrt{\lambda_d} \end{pmatrix}, \quad (18.11)$$

$$\left(\Lambda^{1/2}\right)^{-1} = \begin{pmatrix} \sqrt{\lambda_1^{-1}} & & & \\ & \sqrt{\lambda_2^{-1}} & & \\ & & \ddots & \\ & & & \sqrt{\lambda_d^{-1}} \end{pmatrix}. \quad (18.12)$$

Notice that both $\Lambda^{1/2}$ and $(\Lambda^{1/2})^{-1}$ are non-singular. Furthermore, we define a matrix V such that

$$V = U \left(\Lambda^{1/2}\right)^{-1}. \quad (18.13)$$

Then, multiplying both sides of (18.8) by V^{-1} from the left and by V from the right and inserting $VV^{-1} = E$ in between, we get

$$V^{-1}D(g_j)^\dagger V V^{-1}H V V^{-1}D(g_j)V = V^{-1}HV. \quad (18.14)$$

Meanwhile, we have

$$V^{-1}HV = \Lambda^{1/2}U^\dagger H U \left(\Lambda^{1/2}\right)^{-1} = \Lambda^{1/2}\Lambda \left(\Lambda^{1/2}\right)^{-1} = \Lambda. \quad (18.15)$$

With the second equality of (18.15), we used (18.9). Inserting (18.15) into (18.14), we get

$$V^{-1}D(g_j)^\dagger V \Lambda V^{-1}D(g_j)V = \Lambda. \quad (18.16)$$

Multiplying both sides of (18.16) by Λ^{-1} from the left, we get

$$\Lambda^{-1}V^{-1}D(g_j)^\dagger (V\Lambda) \cdot V^{-1}D(g_j)V = E. \quad (18.17)$$

Using (18.13), we have

$$\Lambda^{-1}V^{-1} = (V\Lambda)^{-1} = \left[U\Lambda^{\frac{1}{2}}\right]^{-1} = \left(\Lambda^{\frac{1}{2}}\right)^{-1}U^{-1} = \left(\Lambda^{\frac{1}{2}}\right)^{-1}U^\dagger.$$

Meanwhile, taking adjoint of (18.13) and noting (18.12), we get

$$V^\dagger = \left[\left(\Lambda^{1/2}\right)^{-1}\right]^\dagger U^\dagger = \left(\Lambda^{1/2}\right)^{-1}U^\dagger.$$

Using (18.13) once again, we have

$$V\Lambda = U\Lambda^{1/2}.$$

Also using (18.13) we have

$$(V^{-1})^\dagger = \left\{ \left[U \left(\Lambda^{\frac{1}{2}} \right)^{-1} \right]^{-1} \right\}^\dagger = \left[\Lambda^{\frac{1}{2}} U^\dagger \right]^\dagger = [U^\dagger]^\dagger \left(\Lambda^{\frac{1}{2}} \right)^\dagger = U \Lambda^{\frac{1}{2}},$$

where we used unitarity of U and Hermiticity of $\Lambda^{1/2}$ indicated in (18.11). Using the above relations and rewriting (18.17), finally we get

$$V^\dagger D(g_j)^\dagger (V^{-1})^\dagger \cdot V^{-1} D(g_j) V = E. \quad (18.18)$$

Defining $\tilde{D}(g_j)$ as follows

$$\tilde{D}(g_j) \equiv V^{-1} D(g_j) V, \quad (18.19)$$

and taking adjoint of both sides of (18.19), we get

$$V^\dagger D(g_j)^\dagger (V^{-1})^\dagger = \tilde{D}(g_j)^\dagger. \quad (18.20)$$

Then, from (18.18) we have

$$\tilde{D}(g_j)^\dagger \tilde{D}(g_j) = E. \quad (18.21)$$

Equation (18.21) implies that a representation of any finite group can be converted to a unitary representation by a similarity transformation of (18.19). This completes the proof.

18.2 Basis Functions of Representation

In Part III we considered a linear transformation of a vector. In that case we have defined vectors as abstract elements with which operation laws of (11.1)–(11.8) hold. We assumed that the operation is addition. In this part, so far we have dealt with vectors mostly in $\mathbb{R}^3$. Therefore, vectors naturally possess geometric features. In this section, we extend a notion of vectors so that they can be treated under a wider scope. More specifically, we include mathematical functions treated in analysis as vectors.

To this end, let us think of a basis of a representation. According to the dimension d of the representation, we have d basis vectors. Here we adopt d linearly independent basis vectors for the representation.

Let $\psi_1, \psi_2, \dots$, and ψ_d be linearly independent vectors in a vector space V^d . Let $\mathcal{G} = \{g_1, g_2, \dots, g_n\}$ be a finite group of order n . Here we assume that g_i ($1 \leq i \leq n$) is a linear transformation (or operator) such as a symmetry operation dealt with in Chap. 17. Suppose that the following relation holds with $g_i \in \mathcal{G}$ ($1 \leq i \leq n$):

$$g_i(\psi_\nu) = \sum_{\mu=1}^d \psi_\mu D_{\mu\nu}(g_i) \quad (1 \leq \nu \leq d). \quad (18.22)$$

Here we followed the notation of (11.37) that represented a linear transformation of a vector. Rewriting (18.22) more explicitly, we have

$$g_i(\psi_\nu) = (\psi_1 \ \psi_2 \ \dots \ \psi_d) \begin{pmatrix} D_{11}(g_i) & \dots & D_{1d}(g_i) \\ \vdots & \ddots & \vdots \\ D_{d1}(g_i) & \dots & D_{dd}(g_i) \end{pmatrix}. \quad (18.23)$$

Comparing (18.23) with (11.37), we notice that $\psi_1, \psi_2, \dots$, and ψ_d act as vectors. The corresponding coordinates of the vectors (or a column vector), i.e., $x_1, x_2, \dots$, and x_d , have been omitted.

Now let us make sure that a set $\mathcal{L}$ consisting of $\{D(g_1), D(g_2), \dots, D(g_n)\}$ forms a representation. Operating g_j on (18.22), we have

$$\begin{aligned} g_j[g_i(\psi_\nu)] &= [g_j g_i](\psi_\nu) = g_j \sum_{\mu=1}^d \psi_\mu D_{\mu\nu}(g_i) = \sum_{\mu=1}^d g_j[\psi_\mu D_{\mu\nu}(g_i)] \\ &= \sum_{\mu=1}^d (g_j \psi_\mu) D_{\mu\nu}(g_i) = \sum_{\mu=1}^d \sum_{\lambda=1}^d \psi_\lambda D_{\lambda\mu}(g_j) D_{\mu\nu}(g_i) \\ &= \sum_{\lambda=1}^d \psi_\lambda [D(g_j) D(g_i)]_{\lambda\nu}. \end{aligned} \quad (18.24)$$

Putting $g_j g_i = g_k$ according to multiplication of group elements, we get

$$g_k(\psi_\nu) = \sum_{\lambda=1}^d \psi_\lambda [D(g_j) D(g_i)]_{\lambda\nu}. \quad (18.25)$$

Meanwhile, replacing g_i in (18.22) with g_k , we have

$$g_k(\psi_\nu) = \sum_{\mu=1}^d \psi_\mu D_{\mu\nu}(g_k). \quad (18.26)$$

Comparing (18.25) and (18.26) and considering the uniqueness of vector representation based on linear independence of the basis vectors (see Sect. 11.1), we get

$$[D(g_j)D(g_i)]_{\lambda\nu} = D_{\lambda\nu}(g_k) \equiv [D(g_k)]_{\lambda\nu}, \quad (18.27)$$

where the last identity follows the notation (11.38) of Sect. 11.2. Hence, we have

$$D(g_j)D(g_i) = D(g_k). \quad (18.28)$$

Thus, the set $\mathcal{L}$ consisting of $\{D(g_1), D(g_2), \dots, D(g_n)\}$ is certainly a representation of the group $\mathcal{G} = \{g_1, g_2, \dots, g_n\}$. In such a case, the set $\mathcal{B}$ consisting of linearly independent d functions (i.e., vectors)

$$\mathcal{B} = \{\psi_1, \psi_2, \dots, \psi_d\} \quad (18.29)$$

is said to be a basis functions of the representation D . The number d equals the dimension of representation. Correspondingly, the representation matrix is a (d, d) square matrix. As remarked above, the correspondence between the elements of $\mathcal{L}$ and $\mathcal{G}$ is not necessarily one-to-one (isomorphic), but may be n -to-one (homomorphic).

Definition 18.2 Let D and D' be two representations of $\mathcal{G} = \{g_1, g_2, \dots, g_n\}$. Suppose that these representations are related to each other by similarity transformation such that

$$D'(g_i) = T^{-1}D(g_i)T \quad (1 \leq i \leq n), \quad (18.30)$$

where T is a non-singular matrix. Then, D and D' are said to be equivalent representations, or simply equivalent. If the representations are not equivalent, they are called inequivalent.

Suppose that $\mathcal{B} = \{\psi_1, \psi_2, \dots, \psi_d\}$ is a basis of a representation D of $\mathcal{G} = \{g_1, g_2, \dots, g_n\}$. Then we have (18.22). Let T be a non-singular matrix. Using T , we want to transform a basis of the representation D from $\mathcal{B}$ to a new set $\mathcal{B}' = \{\psi'_1, \psi'_2, \dots, \psi'_d\}$. Individual elements of the new basis are expressed as

$$\psi'_\nu = \sum_{\lambda=1}^d \psi_\lambda T_{\lambda\nu}. \quad (18.31)$$

Since T is non-singular, this ensures that $\psi'_1, \psi'_2, \dots$, and ψ'_d are linearly independent and, hence, that $\mathcal{B}'$ forms another basis set of D (see discussion of Sect. 11.4). Thus, we can describe ψ_μ ($1 \leq \mu \leq n$) in terms of ψ'_ν . That is,

$$\psi_\mu = \sum_{\lambda=1}^d \psi'_\lambda (T^{-1})_{\lambda\mu}. \quad (18.32)$$

Operating g_i on both sides of (18.31), we have

$$\begin{aligned}
g_i(\psi'_\nu) &= \sum_{\lambda=1}^d g_i(\psi_\lambda) T_{\lambda\nu} = \sum_{\lambda=1}^d \sum_{\mu=1}^d \psi_\mu D_{\mu\lambda}(g_i) T_{\lambda\nu} \\
&= \sum_{\lambda=1}^d \sum_{\mu=1}^d \sum_{\kappa=1}^d \psi'_\kappa (T^{-1})_{\kappa\mu} D_{\mu\lambda}(g_i) T_{\lambda\nu} = \sum_{\kappa=1}^d \psi'_\kappa [T^{-1} D(g_i) T]_{\kappa\nu}. \quad (18.33)
\end{aligned}$$

Let D' be a representation of $\mathcal{g}$ in reference to $\mathcal{B}' = \{\psi'_1, \psi'_2, \dots, \psi'_d\}$. Then we have

$$g_i(\psi'_\nu) = \sum_{\kappa=1}^d \psi'_\kappa D'_{\kappa\nu}.$$

Hence, (18.30) follows in virtue of the linear independence of $\psi'_1, \psi'_2, \dots$, and ψ'_d . Thus, we see that the transformation of basis vectors via (18.31) causes similarity transformation between representation matrices. This is in parallel with (11.81) and (11.88).

Below we show several important notions as a definition. We assume that a group is a finite group.

Definition 18.3 Let D and $\tilde{D}$ be two representations of $\mathcal{g} = \{g_1, g_2, \dots, g_n\}$. Let $D(g_i)$ and $\tilde{D}(g_i)$ be two representation matrices of g_i ($1 \leq i \leq n$). If we construct $\Delta(g_i)$ such that

$$\Delta(g_i) = \begin{pmatrix} D(g_i) & \\ & \tilde{D}(g_i) \end{pmatrix}, \quad (18.34)$$

then $\Delta(g_i)$ is a representation as well. The representation $\Delta(g_i)$ is said to be a direct sum of $D(g_i)$ and $\tilde{D}(g_i)$. We denote it by

$$\Delta(g_i) = D(g_i) \oplus \tilde{D}(g_i). \quad (18.35)$$

A dimension $\Delta(g_i)$ is a sum of that of $D(g_i)$ and that of $\tilde{D}(g_i)$.

Definition 18.4 Let D be a representation of $\mathcal{g}$ and let $D(g_i)$ be a representation matrix of a group element g_i . If we can convert $D(g_i)$ to a block matrix such as (18.34) by its similarity transformation, D is said to be a reducible representation, or completely reducible. If the representation is not reducible, it is irreducible.

A reducible representation can be decomposed (or reduced) to a direct sum of plural irreducible representations. Such an operation is called reduction.

Definition 18.5 Let $\mathcal{g} = \{g_1, g_2, \dots, g_n\}$ be a group and let V be a vector space. Then, we denote a representation D of $\mathcal{g}$ operating on V by

$$D : \mathcal{g} \rightarrow GL(V).$$

We call V a representation space (or carrier space) $\mathcal{L}_S$ of D [3, 4].

From Definition 18.5, the dimension of representation is identical with the dimension of V . Suppose that there is a subspace W in V . If W is $D(g_i)$ -invariant (see Sect. 12.2) with all $\forall g_i \in \mathcal{g}$, W is said to be an invariant subspace of V . Here, we say that W is $D(g_i)$ -invariant if we have $|x\rangle \in W \Rightarrow D(g_i)|x\rangle \in W$; see Sect. 12.2. Notice that $|x\rangle$ may well represent a function ψ_ν of (18.22). In this context, we have a following important theorem.

Theorem 18.2 [3] Let $D: \mathcal{g} \rightarrow GL(V)$ be a unitary representation over V . Let W be a $D(g_i)$ -invariant subspace of V , where g_i ($1 \leq i \leq n$) $\in \mathcal{g} = \{g_1, g_2, \dots, g_n\}$. Then, $W^\perp$ is a $D(g_i)$ -invariant subspace of V as well.

Proof Suppose that $|a\rangle \in W^\perp$ and let $|b\rangle \in W$. Then, from (13.86) we have

$$\langle b|D(g_i)|a\rangle = \langle a|D(g_i)^\dagger|b\rangle^* = \langle a|[D(g_i)]^{-1}|b\rangle^* = \langle a|D(g_i^{-1})|b\rangle^*, \quad (18.36)$$

where with the second equality we used the fact that $D(g_i)$ is a unitary matrix; also, see (18.6). But, since W is $D(g_i)$ -invariant from the supposition, $|D(g_i^{-1})|b\rangle \in W$. Here notice that g_i^{-1} must be identical to $\exists g_k$ ($1 \leq k \leq n$) $\in \mathcal{g}$. Therefore, we should have

$$\langle a|D(g_i^{-1})|b\rangle = 0.$$

Hence, from (18.36) we get

$$0 = \langle a|D(g_i^{-1})|b\rangle^* = \langle b|D(g_i)|a\rangle.$$

This implies that $|D(g_i)|a\rangle \in W^\perp$. In turn, this means that $W^\perp$ is a $D(g_i)$ -invariant subspace of V . This completes the proof.

From Theorem 14.1, we have $V = W \oplus W^\perp$. Correspondingly, as in the case of (12.102), $D(g_i)$ is reduced as follows:

$$D(g_i) = \begin{pmatrix} D^{(W)}(g_i) & \\ & D^{(W^\perp)}(g_i) \end{pmatrix},$$

where $D^{(W)}(g_i)$ and $D^{(W^\perp)}(g_i)$ are representation matrices associated with subspaces W and $W^\perp$, respectively. This is the converse of the additive representations that appeared in Definition 18.3. From (11.17) and (11.18), we have

$$\dim V = \dim W + \dim W^\perp. \quad (18.37)$$

If V is a d -dimensional vector space, V is spanned by d linearly independent basis vectors (or functions) ψ_μ ($1 \leq \mu \leq d$). Suppose that the dimension W and $W^\perp$ is $d^{(W)}$ and $d^{(W^\perp)}$, respectively. Then, W and $W^\perp$ are spanned by $d^{(W)}$ linearly independent vectors and $d^{(W^\perp)}$ linearly independent vectors, respectively. The subspaces W and $W^\perp$ may well further be decomposed into orthogonal complements [i.e., $D(g_i)$ -invariant subspaces of V]. In this situation, in general, we write

$$D(g_i) = D^{(1)}(g_i) \bigoplus D^{(2)}(g_i) \bigoplus \cdots \bigoplus D^{(\omega)}(g_i). \quad (18.38)$$

Thus, unitary representation is completely reducible (Definition 18.4), if it is reducible at any rate. This is one of the conspicuous features of the unitary representation. This can readily be understood during the course of the proof of Theorem 18.2. We will develop further detailed discussion in Sect. 18.4.

If the aforementioned decomposition cannot be done, D is irreducible. Therefore, it will be of great importance to examine a dimension of irreducible representations contained in (18.38). This is closely related to the choice of basis vectors and properties of the invariant subspaces. Notice that the same irreducible representation may repeatedly occur in (18.38).

Before advancing to the next section, however, let us think of an example to get used to abstract concepts. This is at the same time for the purpose of taking the contents of Chap. 19 in advance.

Example 18.1 Figure 18.1 shows structural formulae including resonance structures for allyl radical. Thanks to the resonance structures, the allyl radical belongs to C_{2v} ; see the multiplication table in Table 17.1. The molecule lies on the yz -plane and a line connecting C_1 and a bonded H is the C_2 axis (see Fig. 18.1). The C_2 axis is identical to the z -axis. The molecule has mirror symmetries with respect to the yz - and zx -planes. We denote π -orbitals of C_1 , C_2 , and C_3 by ϕ_1 , ϕ_2 , and ϕ_3 , respectively. We suppose that these orbitals extend toward the x -direction with a positive sign in the upper side and a negative sign in the lower side relative to the plane of paper. Notice that we follow custom of the group theory notation with the coordinate setting in Fig. 18.1.

We consider an inner product space V^3 spanned by ϕ_1 , ϕ_2 , and ϕ_3 . To explicitly show that these are vectors of the inner product space, we express them as $|\phi_1\rangle$, $|\phi_2\rangle$, and $|\phi_3\rangle$ in this example. Then, according to (11.19) of Sect. 11.1, we write

$$V^3 = \text{Span}\{|\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle\}.$$

The vector space V^3 is a representation space pertinent to a representation D of the present example. Also, in parallel to (13.32) of Sect. 13.2, we express an arbitrary vector $|\psi\rangle \in V^3$ as

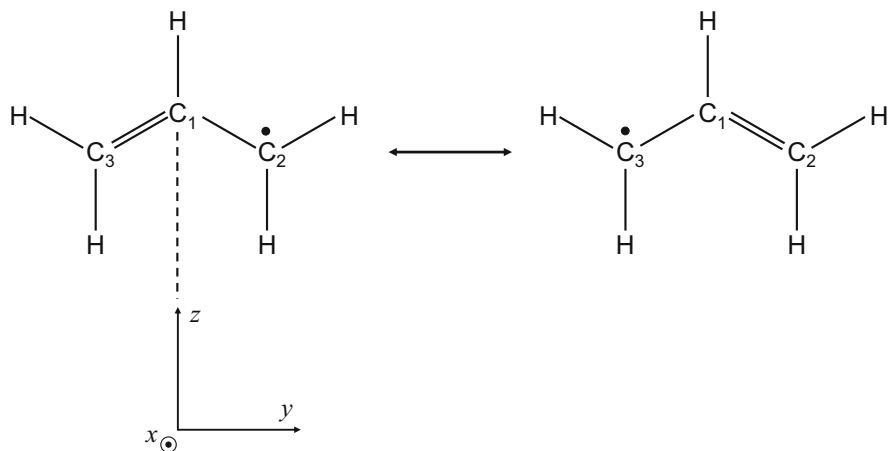


Fig. 18.1 Allyl radical and its resonance structure. The molecule is placed on the yz -plane. The z -axis is identical with a straight line connecting C_1 and H (i.e., C_2 axis)

$$\begin{aligned}
 |\psi\rangle &= c_1|\phi_1\rangle + c_2|\phi_2\rangle + c_3|\phi_3\rangle = |c_1\phi_1 + c_2\phi_2 + c_3\phi_3\rangle \\
 &= (|\phi_1\rangle \quad |\phi_2\rangle \quad |\phi_3\rangle) \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}.
 \end{aligned}$$

Let us now operate a group element of C_{2v} . For example, choosing $C_2(z)$, we have

$$C_2(z)(|\psi\rangle) = (|\phi_1\rangle \quad |\phi_2\rangle \quad |\phi_3\rangle) \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}. \quad (18.39)$$

Thus, $C_2(z)$ is represented by a (3, 3) matrix. Other group elements are represented similarly. These results are collected in Table 18.1, where each matrix is given with respect to $|\phi_1\rangle$, $|\phi_2\rangle$, and $|\phi_3\rangle$ as basis vectors. Notice that the matrix representations differ from those of Table 17.2, where we chose $|x\rangle$, $|y\rangle$, and $|z\rangle$ for the basis vectors. From Table 18.1, we immediately see that the representation matrices are reduced to an upper (1, 1) diagonal matrix (i.e., just a number) and a lower (2, 2) diagonal matrix.

In Table 18.1, we find that all the matrices are Hermitian (as well as unitary). Since C_{2v} is an Abelian group (i.e., commutative), in light of Theorem 14.14, we should be able to diagonalize these matrices by a single unitary similarity transformation all at once. In fact, E and $\sigma'_v(yz)$ are invariant with respect to unitary similarity transformation, and so we only have to diagonalize $C_2(z)$ and $\sigma_v(zx)$ at once. As a characteristic equation of the above (3, 3) matrix of (18.39), we have

Table 18.1 Matrix representation of symmetry operations given with respect to $|\phi_1\rangle$, $|\phi_2\rangle$, and $|\phi_3\rangle$

E	$C_2(z)$	$\sigma_v(zx)$	$\sigma'_v(yz)$
$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$$\begin{vmatrix} -\lambda - 1 & 0 & 0 \\ 0 & -\lambda & -1 \\ 0 & -1 & -\lambda \end{vmatrix} = 0.$$

Solving the above equation, we get $\lambda = 1$ or $\lambda = -1$ (as a double root). Also as a diagonalizing unitary matrix U , we get

$$U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} = U^\dagger. \quad (18.40)$$

Thus, (18.39) can be rewritten as

$$\begin{aligned} C_2(z)(|\psi\rangle) &= (|\phi_1\rangle \quad |\phi_2\rangle \quad |\phi_3\rangle) U U^\dagger \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix} U U^\dagger \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} \\ &= \left(|\phi_1\rangle \quad \frac{1}{\sqrt{2}} |\phi_2 + \phi_3\rangle \quad \frac{1}{\sqrt{2}} |\phi_2 - \phi_3\rangle \right) \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} (c_2 + c_3) \\ \frac{1}{\sqrt{2}} (c_2 - c_3) \end{pmatrix}. \end{aligned}$$

The diagonalization of the representation matrix $\sigma_v(zx)$ using the same U as the above is left for the readers as an exercise. Table 18.2 shows the results of the diagonalized representations with regard to the vectors $|\phi_1\rangle$, $\frac{1}{\sqrt{2}} |\phi_2 + \phi_3\rangle$, and $\frac{1}{\sqrt{2}} |\phi_2 - \phi_3\rangle$. Notice that traces of the matrices remain unchanged in Tables 18.1 and 18.2, i.e., before and after the unitary similarity transformation. From Table 18.2, we find that the vectors are eigenfunctions of the individual symmetry operations. Each diagonal element is a corresponding eigenvalue of those operations. Of the three vectors, $|\phi_1\rangle$ and $\frac{1}{\sqrt{2}} |\phi_2 + \phi_3\rangle$ have the same eigenvalues with respect to individual symmetry operations. With symmetry operations C_2 and $\sigma_v(zx)$, another vector $\frac{1}{\sqrt{2}} |\phi_2 - \phi_3\rangle$ has an eigenvalue of a sign opposite to that of the former two vectors. Thus, we find that we have arrived at “symmetry-adapted” vectors by taking linear combination of original vectors.

Table 18.2 Matrix representation of symmetry operations given with respect to $|\phi_1\rangle$, $\frac{1}{\sqrt{2}}(|\phi_2 + \phi_3\rangle)$, and $\frac{1}{\sqrt{2}}(|\phi_2 - \phi_3\rangle)$

E	$C_2(z)$	$\sigma_v(zx)$	$\sigma'_v(yz)$
$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

Returning back to Table 17.2, we see that the representation matrices have already been diagonalized. In terms of the representation space, we constructed the representation matrices with respect to x , y , and z as symmetry-adapted basis vectors. In the next chapter, we make the most of such vectors constructed by the symmetry-adapted linear combination.

Meanwhile, allocating appropriate numbers to coefficients c_1 , c_2 , and c_3 , we represent any vector in V^3 spanned by $|\phi_1\rangle$, $|\phi_2\rangle$, and $|\phi_3\rangle$. In the next chapter, we deal with molecular orbital (MO) calculations. Including the present case of the allyl radical, we solve the energy eigenvalue problem by appropriately determining those coefficients (i.e., eigenvectors) and corresponding energy eigenvalues in a representation space. A dimension of the representation space depends on the number of molecular orbitals. The representation space is decomposed into several or more (but finite) orthogonal complements according to (18.38). We will come back to the present example in Sect. 19.4.4 and further investigate the problems there.

As can be seen in the above example, the representation space accommodates various types of vectors, e.g., mathematical functions in the present case. If the representation space is decomposed into invariant subspaces, we can choose appropriate basis vectors for each subspace, the number of which is equal to the dimensionality of each irreducible representation [4]. In this context, the situation is related to that of Part III where we have studied how a linear vector space is decomposed into direct sum of invariant subspaces that are spanned by associated basis vectors. In particular, it will be of great importance to construct mutually orthogonal symmetry-adapted vectors through linear combination of original basis vectors in each subspace associated with the irreducible representation. We further study these important subjects in the following several sections.

18.3 Schur's Lemmas and Grand Orthogonality Theorem (GOT)

Pivotal notions of the representation theory of finite groups rest upon Schur's lemmas (first lemma and second lemma).

Schur's First Lemma [1, 2]

Let D and $\tilde{D}$ be two irreducible representations of $\mathcal{g}$. Let dimensions of representations of D and $\tilde{D}$ be m and n , respectively. Suppose that with $\forall g \in \mathcal{g}$ the following relation holds:

$$D(g)M = M\tilde{D}(g), \quad (18.41)$$

where M is a (m, n) matrix. Then we must have

Case (1): $M = 0$ or

Case (2): M is a square matrix (i.e., $m = n$) with $\det M \neq 0$.

In Case (1), the representations of D and $\tilde{D}$ are inequivalent. In Case (2), on the other hand, D and $\tilde{D}$ are equivalent.

Proof

(a) First, suppose that $m > n$. Let $\mathcal{B} = \{\psi_1, \psi_2, \dots, \psi_m\}$ be a basis set of the representation space related to D . Then we have

$$g(\psi_\nu) = \sum_{\mu=1}^m \psi_\mu D_{\mu\nu}(g) \quad (1 \leq \nu \leq m).$$

Next we form a linear combination of $\psi_1, \psi_2, \dots$, and ψ_m such that

$$\phi_\nu = \sum_{\mu=1}^m \psi_\mu M_{\mu\nu} \quad (1 \leq \nu \leq n). \quad (18.42)$$

Operating g on both sides of (18.42), we have

$$\begin{aligned} g(\phi_\nu) &= \sum_{\mu=1}^m g(\psi_\mu) M_{\mu\nu} = \sum_{\mu=1}^m \left[\sum_{\lambda=1}^m \psi_\lambda D_{\lambda\mu}(g) \right] M_{\mu\nu} \\ &= \sum_{\lambda=1}^m \psi_\lambda \left[\sum_{\mu=1}^m D_{\lambda\mu}(g) M_{\mu\nu} \right] = \sum_{\lambda=1}^m \psi_\lambda \left[\sum_{\mu=1}^n M_{\lambda\mu} \tilde{D}_{\mu\nu}(g) \right] \\ &= \sum_{\mu=1}^n \left[\sum_{\lambda=1}^m \psi_\lambda M_{\lambda\mu} \right] \tilde{D}_{\mu\nu}(g) = \sum_{\mu=1}^n \phi_\mu \tilde{D}_{\mu\nu}(g). \end{aligned} \quad (18.43)$$

With the fourth equality of (18.43), we used (18.41). Therefore, $\tilde{\mathcal{B}} = \{\phi_1, \phi_2, \dots, \phi_n\}$ constitutes a basis set of the representation space for $\tilde{D}$.

If $m > n$, we would have been able to construct a basis of the representation D using n ($< m$) functions of ϕ_ν ($1 \leq \nu \leq n$) that are a linear combination of ψ_μ ($1 \leq \mu \leq m$). In that case, we would have obtained a representation of a smaller dimension n for $\tilde{D}$ by taking a linear combination of $\psi_1, \psi_2, \dots$, and ψ_m . As $n < m$ by supposition, this implies that as in (18.34), D would be decomposed into representations whose dimension is smaller than m . It is in contradiction to the supposition that D is irreducible. To avoid this contradiction, we must have $M = 0$. This makes (18.41) trivially hold.

(b) Next we consider a case of $m < n$. Taking a complex conjugate of (18.41) and exchanging both sides, we get

$$\tilde{D}(g)^\dagger M^\dagger = M^\dagger D(g)^\dagger. \quad (18.44)$$

Then, from (18.21) and (18.28), we have

$$D(g)^\dagger D(g) = E \text{ and } D(g)D(g^{-1}) = D(e). \quad (18.45)$$

Therefore, we get

$$D(g^{-1}) = D(g)^{-1} = D(g)^\dagger.$$

Similarly, we have

$$\tilde{D}(g^{-1}) = \tilde{D}(g)^{-1} = \tilde{D}(g)^\dagger.$$

Thus, (18.44) is rewritten as

$$\tilde{D}(g^{-1})M^\dagger = M^\dagger D(g^{-1}). \quad (18.46)$$

The number of rows of $M^\dagger$ is larger than that of columns. This time, dimension n of $\tilde{D}(g^{-1})$ is larger than m of $D(g^{-1})$. A group element g designates an arbitrary element of $\mathcal{g}$, so does g^{-1} . Thus, (18.46) can be treated in parallel with (18.41) where $m > n$ with a (m, n) matrix M . Figure 18.2 graphically shows the magnitude relationship in dimensions of representation between representation matrices and M . Thus, in parallel with the above argument of (a), we exclude the case where $m < n$ as well.

(c) We consider the third case of $m = n$. Similarly as before we make a linear combination of vectors contained in the basis set $\mathcal{B} = \{\psi_1, \psi_2, \dots, \psi_m\}$ such that

$$\phi_\nu = \sum_{\mu=1}^m \psi_\mu M_{\mu\nu} \quad (1 \leq \nu \leq m), \quad (18.47)$$

where M is a (m, m) square matrix. If $\det M = 0$, then $\phi_1, \phi_2, \dots$, and ϕ_m are linearly dependent (see Sect. 11.4). With the number of linearly independent vectors p , we have $p < m$ accordingly. As in (18.43) again, this implies that we would have obtained a representation of a smaller dimension p for $\tilde{D}$, in contradiction to the supposition that $\tilde{D}$ is irreducible. To avoid this contradiction, we must have $\det M \neq 0$. These complete the proof.

Schur's Second Lemma [1, 2]

Let D be a representation of $\mathcal{g}$. Suppose that with $\forall g \in \mathcal{g}$ we have

(a): $m > n$

$$\boxed{\begin{matrix} D(g) \\ (m, m) \end{matrix}} \times \boxed{\begin{matrix} M \\ (m, n) \end{matrix}} = \boxed{\begin{matrix} M \\ (m, n) \end{matrix}} \times \boxed{\begin{matrix} \tilde{D}(g) \\ (n, n) \end{matrix}}$$

(b): $m < n$

$$\boxed{\begin{matrix} \tilde{D}(g^{-1}) \\ (n, n) \end{matrix}} \times \boxed{\begin{matrix} M^\dagger \\ (n, m) \end{matrix}} = \boxed{\begin{matrix} M^\dagger \\ (n, m) \end{matrix}} \times \boxed{\begin{matrix} D(g^{-1}) \\ (m, m) \end{matrix}}$$

Fig. 18.2 Magnitude relationship between dimensions of representation matrices $[D(g)$ and $\tilde{D}(g)]$ and M . (a) $m > n$. (b) $m < n$. The diagram is based on (18.41) and (18.46) of the text

$$D(g)M = MD(g). \quad (18.48)$$

Then, if D is irreducible, $M = cE$ with c being an arbitrary complex number, where E is an identity matrix.

Proof Let c be an arbitrarily chosen complex number. From (18.48) we have

$$D(g)(M - cE) = (M - cE)D(g). \quad (18.49)$$

If D is irreducible, Schur's first lemma implies that we must have either (1) $M - cE = 0$ or (2) $\det(M - cE) \neq 0$. A matrix M has at least one proper eigenvalue λ (Sect. 12.1), and so choosing λ for c in (18.49), we have $\det(M - \lambda E) = 0$. Consequently, only former case is allowed. That is, we have

$$M = cE. \quad (18.50)$$

This completes the proof.

Schur's lemmas lead to important orthogonality theorem that plays a fundamental role in many scientific fields. The orthogonality theorem includes that of matrices and their traces (or characters).

Theorem 18.3 Grand Orthogonality Theorem (GOT) [5] Let $D^{(1)}, D^{(2)}, \dots$ be all inequivalent irreducible representations of a group $\mathcal{G} = \{g_1, g_2, \dots, g_n\}$ of order

n . Let $D^{(\alpha)}$ and $D^{(\beta)}$ be two irreducible representations chosen from among $D^{(1)}, D^{(2)}, \dots$. Then, regarding their matrix representations, we have the following relationship:

$$\sum_g D_{ij}^{(\alpha)}(g)^* D_{kl}^{(\beta)}(g) = \frac{n}{d_\alpha} \delta_{\alpha\beta} \delta_{ik} \delta_{jl}, \quad (18.51)$$

where $\sum_g$ means that the summation should be taken over all n group elements and d_α denotes a dimension of the representation $D^{(\alpha)}$. The symbol $\delta_{\alpha\beta}$ means that $\delta_{\alpha\beta} = 1$ when $D^{(\alpha)}$ and $D^{(\beta)}$ are equivalent and that $\delta_{\alpha\beta} = 0$ when $D^{(\alpha)}$ and $D^{(\beta)}$ are inequivalent.

Proof First we prove the case where $D^{(\alpha)} = D^{(\beta)}$. For the sake of simple expression, we omit a superscript and denote $D^{(\alpha)}$ simply by D . Let us construct a matrix A such that

$$A = \sum_g D(g) X D(g^{-1}), \quad (18.52)$$

where X is an arbitrary matrix. Hence,

$$\begin{aligned} D(g')A &= \sum_g D(g')D(g)XD(g^{-1}) = \sum_g D(g')D(g)XD(g^{-1})D(g'^{-1})D(g') \\ &= \sum_g D(g')D(g)XD(g^{-1}g'^{-1})D(g') \\ &= \sum_g D(g'g)XD[(g'g)^{-1}]D(g'). \end{aligned} \quad (18.53)$$

Thanks to the rearrangement theorem, for fixed g' the element $g'g$ runs through all the group elements as g does so. Therefore, we have

$$\sum_g D(g'g)XD[(g'g)^{-1}] = \sum_g D(g)XD(g^{-1}) = A. \quad (18.54)$$

Thus,

$$D(g)A = AD(g). \quad (18.55)$$

According to Schur's second lemma, we have

$$A = \lambda E. \quad (18.56)$$

The value of a constant λ depends upon the choice of X . Let X be $\delta_i^{(l)} \delta_{(m)}^j$ where all the matrix elements are zero except for the (l, m) -component that takes 1 (Sects. 12.5 and 12.6). Thus from (18.53) we have

$$\sum_{g,p,q} D_{ip}(g) \delta_p^{(l)} \delta_{(m)}^q D_{qj}(g^{-1}) = \sum_g D_{il}(g) D_{mj}(g^{-1}) = \lambda_{lm} \delta_{ij}, \quad (18.57)$$

where λ_{lm} is a constant to be determined. Using the fact that the representation is unitary, from (18.57) we have

$$\sum_g D_{il}(g) D_{jm}(g)^* = \lambda_{lm} \delta_{ij}. \quad (18.58)$$

Next, we wish to determine coefficients λ_{lm} . To this end, setting $i = j$ and summing over i in (18.57), we get for LHS

$$\begin{aligned} \sum_g \sum_i D_{il}(g) D_{mi}(g^{-1}) &= \sum_g [D(g^{-1}) D(g)]_{ml} = \sum_g [D(g^{-1}g)]_{ml} \\ &= \sum_g [D(e)]_{ml} = \sum_g \delta_{ml} = n \delta_{ml}, \end{aligned} \quad (18.59)$$

where n is equal to the order of group. As for RHS, we have

$$\sum_i \lambda_{lm} \delta_{ii} = \lambda_{lm} d, \quad (18.60)$$

where d is equal to a dimension of D . From (18.59) and (18.60), we get

$$\lambda_{lm} d = n \delta_{lm} \quad \text{or} \quad \lambda_{lm} = \frac{n}{d} \delta_{lm}. \quad (18.61)$$

Therefore, from (18.58)

$$\sum_g D_{il}(g) D_{jm}(g)^* = \frac{n}{d} \delta_{lm} \delta_{ij}. \quad (18.62)$$

Specifying a species of the irreducible representation, we get

$$\sum_g D_{il}^{(\alpha)}(g) D_{jm}^{(\alpha)}(g)^* = \frac{n}{d_\alpha} \delta_{lm} \delta_{ij}, \quad (18.63)$$

where d_α is a dimension of $D^{(\alpha)}$.

Next, we examine the relationship between two inequivalent irreducible representations. Let $D^{(\alpha)}$ and $D^{(\beta)}$ be such representations with dimensions d_α and d_β , respectively. Let us construct a matrix B such that

$$B = \sum_g D^{(\alpha)}(g) X D^{(\beta)}(g^{-1}), \quad (18.64)$$

where X is again an arbitrary matrix. Hence,

$$\begin{aligned}
D^{(\alpha)}(g')B &= \sum_g D^{(\alpha)}(g')D^{(\alpha)}(g)XD^{(\beta)}(g^{-1}) \\
&= \sum_g D^{(\alpha)}(g')D^{(\alpha)}(g)XD^{(\beta)}(g^{-1})D^{(\beta)}(g'^{-1})D^{(\beta)}(g') \\
&= \sum_g D^{(\alpha)}(g'g)XD^{(\beta)}\left[(g'g)^{-1}\right]D^{(\beta)}(g') = BD^{(\beta)}(g').
\end{aligned} \tag{18.65}$$

According to Schur's first lemma, we have

$$B = 0. \tag{18.66}$$

Putting $X = \delta_i^{(l)} \delta_{(m)}^j$ as before and rewriting (18.64), we get

$$\sum_g D_{il}^{(\alpha)}(g)D_{jm}^{(\beta)}(g)^* = 0. \tag{18.67}$$

Combining (18.63) and (18.67), we get (18.51). These procedures complete the proof.

18.4 Characters

Representation matrices of a group are square matrices. In Part III we examined properties of a trace, i.e., a sum of diagonal elements of a square matrix. In group theory the trace is called a character.

Definition 18.6 Let D be a (matrix) representation of a group $\mathcal{G} = \{g_1, g_2, \dots, g_n\}$. The sum of diagonal elements $\chi(g)$ is defined as follows:

$$\chi(g) \equiv \text{Tr}D(g) = \sum_{i=1}^d D_{ii}(g), \tag{18.68}$$

where g stands for group elements $g_1, g_2, \dots, g_n$, Tr stands for “trace,” and d is a dimension of the representation D . Let $\mathbb{C}$ be a set defined as

$$\mathbb{C} = \{\chi(g_1), \chi(g_2), \dots, \chi(g_n)\}. \tag{18.69}$$

Then, the set $\mathbb{C}$ is called a character of D . A character of an irreducible representation is said to be an irreducible character.

Let us describe several properties of the character or trace.

1. A character of the identity element $\chi(e)$ is equal to a dimension d of a representation. This is because the identity element is given by a unit matrix.
2. Let P and Q be two square matrices. Then, we have

$$\text{Tr}(PQ) = \text{Tr}(QP). \quad (18.70)$$

This is because

$$\sum_i (PQ)_{ii} = \sum_i \sum_j P_{ij} Q_{ji} = \sum_j \sum_i Q_{ji} P_{ij} = \sum_j (QP)_{jj}. \quad (18.71)$$

Putting $Q = SP^{-1}$ in (18.70), we get

$$\text{Tr}(PSP^{-1}) = \text{Tr}(SP^{-1}P) = \text{Tr}(S). \quad (18.72)$$

Therefore, we have the following property:

3. Characters of group elements that are conjugate to each other are equal. If g_i and g_j are conjugate, these elements are connected by means of a suitable element g such that

$$gg_i g^{-1} = g_j. \quad (18.73)$$

Accordingly, a representation matrix is expressed as

$$D(g)D(g_i)D(g^{-1}) = D(g)D(g_i)[D(g)]^{-1} = D(g_j). \quad (18.74)$$

Taking a trace of both sides of (18.74), from (18.72) we have

$$\chi(g_i) = \chi(g_j). \quad (18.75)$$

4. Any two equivalent representations have the same trace. This immediately follows from (18.30).

There are several orthogonality theorem about a trace. Among them, a following theorem is well known.

Theorem 18.4 A trace of irreducible representations satisfies the following orthogonality relation:

$$\sum_g \chi^{(\alpha)}(g)^* \chi^{(\beta)}(g) = n \delta_{\alpha\beta}, \quad (18.76)$$

where $\chi^{(\alpha)}$ and $\chi^{(\beta)}$ are traces of irreducible representations $D^{(\alpha)}$ and $D^{(\beta)}$, respectively.

Proof In (18.51) putting $i = j$ and $k = l$ in both sides and summing over all i and k , we have

$$\begin{aligned}
\sum_g \sum_{i,k} D_{ii}^{(\alpha)}(g)^* D_{kk}^{(\beta)}(g) &= \sum_{i,k} \frac{n}{d_\alpha} \delta_{\alpha\beta} \delta_{ik} \delta_{ik} = \sum_{i,k} \frac{n}{d_\alpha} \delta_{\alpha\beta} \delta_{ik} = \frac{n}{d_\alpha} \delta_{\alpha\beta} d_\alpha \\
&= n \delta_{\alpha\beta}.
\end{aligned} \tag{18.77}$$

From (18.68) and (18.77), we get (18.76). This completes the proof.

Since a character is identical with group elements belonging to the same conjugacy class K_l , we may write it as $\chi(K_l)$ and rewrite a summation of (18.76) as a summation of the conjugacy classes. Thus, we have

$$\sum_{l=1}^{n_c} \chi^{(\alpha)}(K_l)^* \chi^{(\beta)}(K_l) k_l = n \delta_{\alpha\beta}, \tag{18.78}$$

where n_c denotes the number of conjugacy classes in a group and k_l indicates the number of group elements contained in a class K_l .

We have seen a case where a representation matrix can be reduced to two (or more) block matrices as in (18.34). Also as already seen in Part III, the block matrix decomposition takes place with normal matrices (including unitary matrices). The character is often used to examine a constitution of a reducible representation or reducible matrix.

Alternatively, if a unitary matrix (a normal matrix, more widely) is decomposed into block matrices, we say that the unitary matrix comprises a direct sum of those block matrices. In physics and chemistry dealing with atoms, molecules, crystals, etc., we very often encounter such a situation. Extending (18.35), the relation can generally be summarized as

$$D(g_i) = D^{(1)}(g_i) \oplus D^{(2)}(g_i) \oplus \cdots \oplus D^{(\omega)}(g_i), \tag{18.79}$$

where $D(g_i)$ is a reducible representation for a group element g_i and $D^{(1)}(g_i)$, $D^{(2)}(g_i)$, $\cdots$, and $D^{(\omega)}(g_i)$ are irreducible representations in a group. The notation $D^{(\omega)}(g_i)$ means that $D^{(\omega)}(g_i)$ may be equivalent (or identical) to $D^{(1)}(g_i)$, $D^{(2)}(g_i)$, etc. or may be inequivalent to them. More specifically, the same irreducible representations may well appear several times.

To make the above situation clear, we usually use a following equation instead:

$$D(g_i) = \sum_{\alpha} q_{\alpha} D^{(\alpha)}(g_i), \tag{18.80}$$

where q_{α} is zero or a positive integer and $D^{(\alpha)}$ is different types of irreducible representations. If the same $D^{(\alpha)}$ repeatedly appears in the direct sum, then q_{α} specifies how many times $D^{(\alpha)}$ appears in the direct sum. Unless $D^{(\alpha)}$ appears, q_{α} is zero.

Bearing the above in mind, we take a trace of (18.80). Then we have

$$\chi(g) = \sum_{\alpha} q_{\alpha} \chi^{(\alpha)}(g), \quad (18.81)$$

where $\chi(g)$ is a character of the reducible representation for a group element g , where we omitted a subscript i indicating the i -th group element g_i ($1 \leq i \leq n$). To find q_{α} , let us multiply both sides of (18.81) by $\chi^{(\alpha)}(g)^*$ and take summation over group elements. That is,

$$\sum_g \chi^{(\alpha)}(g)^* \chi(g) = \sum_{\beta} q_{\beta} \sum_g \chi^{(\alpha)}(g)^* \chi^{(\beta)}(g) = \sum_{\beta} q_{\beta} n \delta_{\alpha\beta} = q_{\alpha} n, \quad (18.82)$$

where we used (18.76) with the second equality. Thus, we get

$$q_{\alpha} = \frac{1}{n} \sum_g \chi^{(\alpha)}(g)^* \chi(g). \quad (18.83)$$

The integer q_{α} explicitly gives the number of appearance of $D^{(\alpha)}(g)$ that appears in a reducible representation $D(g)$. The expression pertinent to the classes is

$$q_{\alpha} = \frac{1}{n} \sum_j \chi^{(\alpha)}(K_j)^* \chi(K_j) k_j, \quad (18.84)$$

where K_j and k_j denote the j -th class of the group and the number of elements belonging to K_j , respectively.

18.5 Regular Representation and Group Algebra

Now, the readers may wonder how many different irreducible representations exist for a group. To answer this question, let us introduce a special representation of the *regular representation*.

Definition 18.7 Let $\mathcal{G} = \{g_1, g_2, \dots, g_n\}$ be a group. Let us define a (n, n) square matrix $D^{(R)}(g_{\nu})$ for an arbitrary group element g_{ν} ($1 \leq \nu \leq n$) such that

$$\left[D^{(R)}(g_{\nu}) \right]_{ij} = \delta(g_i^{-1} g_{\nu} g_j) \quad (1 \leq i, j \leq n), \quad (18.85)$$

where

$$\delta(g_{\nu}) = \begin{cases} 1 & \text{for } g_{\nu} = e \text{ (i.e., identity),} \\ 0 & \text{for } g_{\nu} \neq e. \end{cases} \quad (18.86)$$

Let us consider a set

$$\mathcal{R} = \left\{ D^{(R)}(g_1), D^{(R)}(g_2), \dots, D^{(R)}(g_n) \right\}. \quad (18.87)$$

Then, the set $\mathcal{R}$ is said to be a regular representation of the group $\mathcal{G}$.

In fact, $\mathcal{R}$ is a representation. This is confirmed as follows: In (18.85), if $g_i^{-1}g_\nu g_j = e$, $\delta(g_i^{-1}g_\nu g_j) = 1$. This occurs, when $g_\nu g_j = g_i$ (A). Meanwhile, let us consider a situation where $g_j^{-1}g_\mu g_k = e$. This occurs when $g_\mu g_k = g_j$ (B). Replacing g_j in (A) with that in (B), we have

$$g_\nu g_\mu g_k = g_i. \quad (18.88)$$

That is,

$$g_i^{-1}g_\nu g_\mu g_k = e. \quad (18.89)$$

If we choose g_i and g_ν , g_j is uniquely decided from (A). If g_μ is separately chosen, then g_k is uniquely decided from (B) as well, because g_j has already been uniquely decided. Thus, performing the following matrix calculations, we get

$$\begin{aligned} \sum_j \left[D^{(R)}(g_\nu) \right]_{ij} \left[D^{(R)}(g_\mu) \right]_{jk} &= \sum_j \delta(g_i^{-1}g_\nu g_j) \delta(g_j^{-1}g_\mu g_k) = \delta(g_i^{-1}g_\nu g_\mu g_k) \\ &= \left[D^{(R)}(g_\nu g_\mu) \right]_{ik}. \end{aligned} \quad (18.90)$$

Rewriting (18.90) in a matrix product form, we get

$$D^{(R)}(g_\nu) D^{(R)}(g_\mu) = D^{(R)}(g_\nu g_\mu). \quad (18.91)$$

Thus, $D^{(R)}$ is certainly a representation.

To further confirm this, let us think of an example.

Example 18.2 Let us consider a thiophene molecule that we have already examined in Sect. 17.2. In a multiplication table, we arrange E , C_2 , σ_v , $\sigma_{v'}$ in a first column and their inverse element E , C_2 , σ_v , $\sigma_{v'}$ in a first row. In this case, the inverse element is the same as original element itself. Paying attention, e.g., to C_2 , we allocate the number 1 on the place where C_2 appears and the number 0 otherwise. Then, that matrix is a regular representation of C_2 ; see Table 18.3. Thus as $D^{(R)}(C_2)$, we get

Table 18.3 How to make a regular representation of C_{2v}

C_{2v}	E^{-1}	$C_2(z)^{-1}$	$\sigma_v(zx)^{-1}$	$\sigma'_v(yz)^{-1}$
E	E	C_2	σ_v	σ'_v
$C_2(z)$	C_2	E	σ'_v	σ_v
$\sigma_v(zx)$	σ_v	σ'_v	E	C_2
$\sigma'_v(yz)$	σ'_v	σ_v	C_2	E

$$D^{(R)}(C_2) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (18.92)$$

As evidenced in (18.92), the rearrangement theorem ensures that the number 1 appears once and only once in each column and each row in such a way that individual column and row vectors become linearly independent. Thus, at the same time, we confirm that the matrix is unitary.

Another characteristic of the regular representation is that the identity is represented by an identity matrix. In this example, we have

$$D^{(R)}(E) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (18.93)$$

For the other symmetry operations, we have

$$D^{(R)}(\sigma_v) = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, D^{(R)}(\sigma_{v'}) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Let $\chi^{(R)}(g_\nu)$ be a character of the regular representation. Then, according to the definition of (18.85),

$$\chi^{(R)}(g_\nu) = \sum_{i=1}^n \delta(g_i^{-1} g_\nu g_i) = \begin{cases} n & \text{for } g_\nu = e, \\ 0 & \text{for } g_\nu \neq e. \end{cases} \quad (18.94)$$

As can be seen from (18.92), the regular representation is reducible because the matrix is decomposed into block matrices. Therefore, the representation can be reduced to a direct sum of irreducible representations such that

$$D^{(R)} = \sum_{\alpha} q_{\alpha} D^{(\alpha)}, \quad (18.95)$$

Table 18.4 Character table of C_{2v}

C_{2v}	E	$C_2(z)$	$\sigma_v(zx)$	$\sigma'_v(yz)$	
A_1	1	1	1	1	$z; x^2, y^2, z^2$
A_2	1	1	-1	-1	xy
B_1	1	-1	1	-1	$x; zx$
B_2	1	-1	-1	1	$y; yz$

where q_α is a positive integer or zero and $D^{(\alpha)}$ is an irreducible representation. Then, from (18.81), we have

$$\chi^{(R)}(g_\nu) = \sum_\alpha q_\alpha \chi^{(\alpha)}(g_\nu), \quad (18.96)$$

where $\chi^{(\alpha)}$ is a trace of the irreducible representation $D^{(\alpha)}$. Using (18.83) and (18.94),

$$\begin{aligned} q_\alpha &= \frac{1}{n} \sum_g \chi^{(\alpha)}(g)^* \chi^{(R)}(g) = \frac{1}{n} \chi^{(\alpha)}(e)^* \chi^{(R)}(e) = \frac{1}{n} \chi^{(\alpha)}(e)^* n = \chi^{(\alpha)}(e)^* \\ &= d_\alpha. \end{aligned} \quad (18.97)$$

Note that a dimension d_α of the representation $D^{(\alpha)}$ is equal to a trace of its identity matrix. Also notice from (18.95) and (18.97) that $D^{(R)}$ contains every irreducible representation $D^{(\alpha)}$ d_α times. To show it more clearly, Table 18.4 gives a character table of C_{2v} . The regular representation matrices are given in Example 18.2. For this, we have

$$D^{(R)}(C_{2v}) = A_1 + A_2 + B_1 + B_2. \quad (18.98)$$

This relation obviously indicates that all the irreducible representations of C_{2v} are contained one time (that is equal to the dimension of representation of C_{2v}).

To examine general properties of the irreducible representations, we return to (18.94) and (18.96). Replacing q_α with d_α there, we get

$$\sum_\alpha d_\alpha \chi^{(\alpha)}(g_\nu) = \begin{cases} n & \text{for } g_\nu = e, \\ 0 & \text{for } g_\nu \neq e. \end{cases} \quad (18.99)$$

In particular, when $g_\nu = e$, again we have $\chi^{(\alpha)}(e) = d_\alpha$ (d_α is a real number!). That is,

$$\sum_\alpha d_\alpha^2 = n. \quad (18.100)$$

This is very important relation in the representation theory of a finite group in that (18.100) sets an upper limit to the number of irreducible representations and their dimensions. That number cannot exceed the order of a group.

In (18.76) and (18.78), we have shown the orthogonality relationship between traces. We have another important orthogonality relationship between them. To prove this theorem, we need a notion of group algebra [5]. The argument is as follows:

(a) Let us think of a set comprising group elements expressed as

$$\aleph = \sum_g a_g g, \quad (18.101)$$

where g is a group element of a group $\mathcal{G} = \{g_1, g_2, \dots, g_n\}$, a_g is an arbitrarily chosen complex number, and Σ_g means that summation should be taken over group elements. Let $\aleph'$ be another set similarly defined as (18.101). That is,

$$\aleph' = \sum_{g'} a'_{g'} g'. \quad (18.102)$$

Then we can define a following sum:

$$\aleph + \aleph' = \sum_g a_g g + \sum_{g'} a'_{g'} g' = \sum_g (a_g g + a'_{g'} g) = \sum_g (a_g + a'_{g'}) g. \quad (18.103)$$

Also, we get

$$\begin{aligned} \aleph \cdot \aleph' &= \left(\sum_g a_g g \right) \left(\sum_{g'} a'_{g'} g' \right) = \sum_{g'} \left(\sum_g a_g g g' \right) a'_{g'} \\ &= \sum_{g'^{-1}} \left(\sum_g a_g g g'^{-1} \right) a'_{g'^{-1}} = \sum_{g'^{-1}} \left(\sum_{gg'} a_{gg'} g g' g'^{-1} \right) a'_{g'^{-1}} \\ &= \sum_{g'^{-1}} \left(\sum_{gg'} a_{gg'} g \right) a'_{g'^{-1}} = \sum_{gg'} \left(\sum_{g'^{-1}} a_{gg'} a'_{g'^{-1}} \right) g \\ &= \sum_g \left(\sum_{g'} a_{gg'} a'_{g'^{-1}} \right) g, \end{aligned} \quad (18.104)$$

where we used the rearrangement theorem and suitable exchange of group elements. Thus, we see that the above-defined set $\aleph$ is closed under summations (i.e., linear combinations) and multiplications. A set closed under summations and multiplications is said to be an algebra. If the set forms a group, the said set is called a *group algebra*.

So far we have treated calculations as a multiplication between two elements g and g' , i.e., $g \circ g'$ (see Sect. 16.1). Now we start regarding the calculations as summation as well. In that case, group elements act as basis vectors in a vector space. Bearing this in mind, let us further define specific group algebra.

(b) Let K_i be the i -th conjugacy class of a group $\mathcal{G} = \{g_1, g_2, \dots, g_n\}$. Also let K_i be such that

$$K_i = \{A_1^{(i)}, A_2^{(i)}, \dots, A_{k_i}^{(i)}\}, \quad (18.105)$$

where k_i is the number of elements belonging to K_i . Now think of a set $gK_i g^{-1}$ ($\forall g \in \mathcal{G}$). Then, due to the definition of a class, we have

$$gK_i g^{-1} \subset K_i. \quad (18.106)$$

Multiplying g^{-1} from the left and g from the right of both sides, we get $K_i \subset g^{-1}K_i g$. Since g is arbitrarily chosen, replacing g with g^{-1} , we have

$$K_i \subset gK_i g^{-1}. \quad (18.107)$$

Therefore, we get

$$gK_i g^{-1} = K_i. \quad (18.108)$$

Meanwhile, for $A_\alpha^{(i)}, A_\beta^{(i)} \in K_i; A_\alpha^{(i)} \neq A_\beta^{(i)}$ ($1 \leq \alpha, \beta \leq k_i$), we have

$$gA_\alpha^{(i)} g^{-1} \neq gA_\beta^{(i)} g^{-1} \quad (\forall g \in \mathcal{G}). \quad (18.109)$$

This is because if the equality holds with (18.109), we have $A_\alpha^{(i)} = A_\beta^{(i)}$, in contradiction.

(c) Let K be a set collecting several classes and described as

$$K = \sum_i a_i K_i, \quad (18.110)$$

where a_i is a positive integer or zero. Thanks to (18.105) and (18.110), we have

$$gKg^{-1} = K. \quad (18.111)$$

Conversely, if a group algebra K satisfies (18.111), K can be expressed as a sum of classes such as (18.110). Here suppose that K is not expressed by (18.110), but described by

$$K = \sum_i a_i K_i + Q, \quad (18.112)$$

where Q is an “incomplete” set that does not form a class. Then,

$$\begin{aligned} gKg^{-1} &= g \left(\sum_i a_i K_i + Q \right) g^{-1} = g \left(\sum_i a_i K_i \right) g^{-1} + gQg^{-1} = \sum_i a_i K_i \\ &\quad + gQg^{-1} \\ &= K = \sum_i a_i K_i + Q, \end{aligned} \quad (18.113)$$

where the equality before the last comes from (18.111). Thus, from (18.113) we get

$$gQg^{-1} = Q \quad (\forall g \in \mathcal{G}). \quad (18.114)$$

By definition of the classes, this implies that Q must form a “complete” class, in contradiction to the supposition. Thus, (18.110) holds.

In Sect. 16.3, we described several characteristics of an invariant subgroup. As readily seen from (16.11) and (18.111), any invariant subgroup consists of two or more entire classes. Conversely, if a group comprises entire classes, it must be an invariant subgroup.

(d) Let us think of a product of classes. Let K_i and K_j be sets described as (18.105).

We define a product $K_i K_j$ as a set containing products $A_\alpha^{(i)} A_\beta^{(j)}$ ($1 \leq \alpha \leq k_i, 1 \leq \beta \leq k_j$). That is,

$$K_i K_j = \sum_{l=1}^{k_i} \sum_{m=1}^{k_j} A_l^{(i)} A_m^{(j)}. \quad (18.115)$$

Multiplying (18.115) by g ($\forall g \in \mathcal{G}$) from the left and by g^{-1} from the right, we get

$$gK_i K_j g^{-1} = (gK_i g)(g^{-1}K_j g^{-1}) = K_i K_j. \quad (18.116)$$

From the above discussion, we get

$$K_i K_j = \sum_l c_{ijl} K_l, \quad (18.117)$$

where c_{ijl} is a positive integer or zero. In fact, when we take $gK_iK_jg^{-1}$, we merely permute the terms in (18.117).

- (e) If two group elements of the group $\mathcal{G} = \{g_1, g_2, \dots, g_n\}$ are conjugate to each other, their inverse elements are conjugate to each other as well. In fact, suppose that for $g_\mu, g_\nu \in K_i$, we have

$$g_\mu = g g_\nu g^{-1}. \quad (18.118)$$

Then, taking the inverse of (18.118), we get

$$g_\mu^{-1} = g g_\nu^{-1} g^{-1}. \quad (18.119)$$

Thus, given a class K_i , there exists another class $K_{i'}$ that consists of the inverses of the elements of K_i . If $g_\mu \neq g_\nu$, $g_\mu^{-1} \neq g_\nu^{-1}$. Therefore, K_i and $K_{i'}$ are of the same order. Suppose that the number of elements contained in K_i is k_i and that in $K_{i'}$ is $k_{i'}$. Then, we have

$$k_i = k_{i'}. \quad (18.120)$$

If $K_j \neq K_{i'}$ (i.e., K_j is not identical with $K_{i'}$ as a set), $K_i K_j$ does not contain e . In fact, suppose that for $g_\rho \in K_i$ there were a group element $b \in K_j$ such that $bg_\rho = e$. Then, we would have

$$b = g_\rho^{-1} \text{ and } b \in K_{i'}. \quad (18.121)$$

This would mean that $b \in K_j \cap K_{i'}$, implying that $K_j = K_{i'}$ by definition of class. It is in contradiction to $K_j \neq K_{i'}$. Consequently, $K_i K_j$ does not contain e . Taking the product of the classes K_i and $K_{i'}$, we obtain the identity e precisely k_i times. Rewriting (18.117), we have

$$K_i K_j = c_{ij1} K_1 + \sum_{l \neq 1} c_{ijl} K_l, \quad (18.122)$$

where $K_1 = \{e\}$. As mentioned below, we are most interested in the first term in (18.122).

In (18.122), if $K_j = K_{i'}$, we have

$$c_{ij1} = k_i. \quad (18.123)$$

On the other hand, if $K_j \neq K_{i'}$, $c_{ij1} = 0$. Summarizing the above arguments, we get

$$c_{ij1} = \begin{cases} k_i & \text{for } K_j = K_{i'}, \\ 0 & \text{for } K_j \neq K_{i'}. \end{cases} \quad (18.124)$$

Or, symbolically we write

$$c_{ij1} = k_i \delta_{ji'}. \quad (18.125)$$

18.6 Classes and Irreducible Representations

After the aforementioned considerations, we have the following theorem:

Theorem 18.5 Let $\mathcal{G} = \{g_1, g_2, \dots, g_n\}$ be a group. A trace of irreducible representations satisfies the following orthogonality relation:

$$\sum_{\alpha=1}^{n_r} \chi^{(\alpha)}(K_i) \chi^{(\alpha)}(K_j)^* = \frac{n}{k_i} \delta_{ij}, \quad (18.126)$$

where summation α is taken over all inequivalent n_r irreducible representations, K_i and K_j indicate conjugacy classes, and k_i denotes the number of elements contained in the i -th class K_i .

Proof Rewriting (18.108), we have

$$gK_i = K_i g \quad (\forall g \in \mathcal{G}). \quad (18.127)$$

Since a homomorphic correspondence holds between a group element and its representation matrix, a similar correspondence holds as well with (18.127). Let $\widehat{K}_i$ be a sum of k_i matrices of the α -th irreducible representation $D^{(\alpha)}$ and let $\widehat{K}_i$ be expressed as

$$\widehat{K}_i = \sum_{g \in K_i} D^{(\alpha)}. \quad (18.128)$$

Note that in (18.128) $D^{(\alpha)}$ functions as a linear transformation with respect to a group algebra. From (18.127), we have

$$D^{(\alpha)} \widehat{K}_i = \widehat{K}_i D^{(\alpha)} \quad (\forall g \in \mathcal{G}). \quad (18.129)$$

Since $D^{(\alpha)}$ is an irreducible representation, $\widehat{K}_i$ must be expressed on the basis of Schur's second lemma as

$$\widehat{K}_i = \lambda E. \quad (18.130)$$

To determine λ , we take a trace of both sides of (18.130). Then, from (18.128) and (18.130), we get

$$k_i \chi^{(\alpha)}(K_i) = \lambda d_\alpha. \quad (18.131)$$

Thus, we get

$$\widehat{K}_i = \frac{k_i}{d_\alpha} \chi^{(\alpha)}(K_i) E. \quad (18.132)$$

Next, corresponding to (18.122), we have

$$\widehat{K}_i \widehat{K}_j = \sum_l c_{ijl} \widehat{K}_l. \quad (18.133)$$

Replacing $\widehat{K}_l$ in (18.133) with that of (18.132), we get

$$k_i k_j \chi^{(\alpha)}(K_i) \chi^{(\alpha)}(K_j) = d_\alpha \sum_l c_{ijl} k_l \chi^{(\alpha)}(K_l). \quad (18.134)$$

Returning to (18.99) and rewriting it, we have

$$\sum_\alpha d_\alpha \chi^{(\alpha)}(K_i) = n \delta_{i1}, \quad (18.135)$$

where again we have $K_1 = \{e\}$. With respect to (18.134), we sum over all the irreducible representations α . Then we have

$$\begin{aligned} \sum_\alpha k_i k_j \chi^{(\alpha)}(K_i) \chi^{(\alpha)}(K_j) &= \sum_l \left[c_{ijl} k_l \sum_\alpha d_\alpha \chi^{(\alpha)}(K_l) \right] = \sum_l c_{ijl} k_l n \delta_{l1} \\ &= c_{ij1} n. \end{aligned} \quad (18.136)$$

In (18.136) we remark that $k_1 = 1$, meaning that the number of group elements that $K_1 = \{e\}$ contains is 1. Rewriting (18.136), we get

$$k_i k_j \sum_\alpha \chi^{(\alpha)}(K_i) \chi^{(\alpha)}(K_j) = c_{ij1} n = k_i n \delta_{ji'}, \quad (18.137)$$

where we used (18.125) with the last equality. Moreover, using

$$\chi^{(\alpha)}(K_{i'}) = \chi^{(\alpha)}(K_i)^*, \quad (18.138)$$

we get

$$k_i k_j \sum_{\alpha} \chi^{(\alpha)}(K_i) \chi^{(\alpha)}(K_j)^* = c_{ij} n = k_i n \delta_{ji}. \quad (18.139)$$

Rewriting (18.139), we finally get

$$\sum_{\alpha=1}^{n_r} \chi^{(\alpha)}(K_i) \chi^{(\alpha)}(K_j)^* = \frac{n}{k_i} \delta_{ij}. \quad (18.126)$$

These complete the proof.

Equations (18.78) and (18.126) are well known as orthogonality relations. So far, we have no idea about the mutual relationship between the two numbers n_c and n_r in magnitude. In (18.78), let us consider a following set S

$$S = \left\{ \sqrt{k_1} \chi^{(\alpha)}(K_1), \sqrt{k_2} \chi^{(\alpha)}(K_2), \dots, \sqrt{k_{n_c}} \chi^{(\alpha)}(K_{n_c}) \right\}. \quad (18.140)$$

Viewing individual components in S as coordinates of a n_c -dimensional vector, (18.78) can be considered as an inner product as expressed using (complex) coordinates of the two vectors. At the same time, (18.78) represents an orthogonal relationship between the vectors. Since we can obtain at most n_c mutually orthogonal (i.e., linearly independent) vectors in a n_c -dimensional space, for the number (n_r) of such vectors, we have

$$n_r \leq n_c. \quad (18.141)$$

Here n_r is equal to the number of different α , i.e., the number of irreducible representations.

Meanwhile, in (18.126) we consider a following set S'

$$S' = \left\{ \chi^{(1)}(K_i), \chi^{(2)}(K_i), \dots, \chi^{(n_r)}(K_i) \right\}. \quad (18.142)$$

Similarly, individual components in S' can be considered as coordinates of a n_r -dimensional vector. Again, (18.126) implies the orthogonality relation among vectors. Therefore, as for the number (n_c) of mutually orthogonal vectors, we have

$$n_c \leq n_r. \quad (18.143)$$

Thus, from (18.141) and (18.143), we finally reach a simple but very important conclusion about the relationship between n_c and n_r such that

$$n_r = n_c. \quad (18.144)$$

That is, the number (n_r) of inequivalent irreducible representations is equal to that (n_c) of conjugacy classes of the group. An immediate and important consequence of (18.144) together with (18.100) is that the representation of an Abelian group is one-dimensional. This is because in the Abelian group individual group elements constitute a conjugacy class. We have $n_c = n$ accordingly.

18.7 Projection Operators: Revisited

In Sect. 18.2 we have described how basis vectors (or functions) are transformed by a symmetry operation. In that case the symmetry operation is performed by a group element that belongs to a transformation group. More specifically, given a group $\mathcal{G} = \{g_1, g_2, \dots, g_n\}$ and a set of basis vectors $\psi_1, \psi_2, \dots$, and ψ_d , the basis vectors are transformed by $g_i \in \mathcal{G}$ ($1 \leq i \leq n$) such that

$$g_i(\psi_\nu) = \sum_{\mu=1}^d \psi_\mu D_{\mu\nu}(g_i). \quad (18.22)$$

We may well ask then how we can construct such basis vectors. In this section we address this question. A central concept about this is a projection operator. We have already studied the definition and basic properties of the projection operators. In this section, we deal with them bearing in mind that we apply the group theory to molecular science, especially to quantum chemical calculations.

In Sect. 18.6, we examined the permissible number of irreducible representations. We have reached a conclusion that the number (n_r) of inequivalent irreducible representations is equal to that (n_c) of conjugacy classes of the group. According to this conclusion, we modify the above Eq. (18.22) such that

$$g\left(\psi_i^{(\alpha)}\right) = \sum_{j=1}^{d_\alpha} \psi_j^{(\alpha)} D_{ji}^{(\alpha)}(g), \quad (18.145)$$

where α and d_α denote the α -th irreducible representation and its dimension, respectively; the subscript i is omitted from g_i for simplicity. This naturally leads to the next question of how the basis vectors $\psi_j^{(\alpha)}$ are related to $\psi_j^{(\beta)}$ that are basis vectors as well, but belong to a different irreducible representation β .

Suppose that we have an arbitrarily chosen function f . Then, f is assumed to contain various components of different irreducible representations. Thus, let us assume that f is decomposed into the component such that

$$f = \sum_{\alpha} \sum_m c_m^{(\alpha)} \psi_m^{(\alpha)}, \quad (18.146)$$

where $c_m^{(\alpha)}$ is a coefficient of the expansion and $\psi_m^{(\alpha)}$ is the m -th component of α -th irreducible representation.

Now, let us define the following operator:

$$P_{l(m)}^{(\alpha)} \equiv \frac{d_{\alpha}}{n} \sum_g D_{lm}^{(\alpha)}(g)^* g, \quad (18.147)$$

where Σ_g means that the summation should be taken over all n group elements g and d_{α} denotes a dimension of the representation $D^{(\alpha)}$. Operating $P_{l(m)}^{(\alpha)}$ on f , we have

$$\begin{aligned} P_{l(m)}^{(\alpha)} f &= \frac{d_{\alpha}}{n} \sum_g D_{lm}^{(\alpha)}(g)^* g f = \frac{d_{\alpha}}{n} \sum_g D_{lm}^{(\alpha)}(g)^* g \sum_{\nu} \sum_k c_k^{(\nu)} \psi_k^{(\nu)} \\ &= \frac{d_{\alpha}}{n} \sum_g D_{lm}^{(\alpha)}(g)^* \sum_{\nu} \sum_k c_k^{(\nu)} g \psi_k^{(\nu)} \\ &= \frac{d_{\alpha}}{n} \sum_g D_{lm}^{(\alpha)}(g)^* \sum_{\nu} \sum_k c_k^{(\nu)} \sum_{j=1}^{d_{\nu}} \psi_j^{(\nu)} D_{jk}^{(\nu)}(g) \\ &= \frac{d_{\alpha}}{n} \sum_{\nu} \sum_k c_k^{(\nu)} \sum_{j=1}^{d_{\nu}} \psi_j^{(\nu)} \left[\sum_g D_{lm}^{(\alpha)}(g)^* D_{jk}^{(\nu)}(g) \right] \\ &= \frac{d_{\alpha}}{n} \sum_{\nu} \sum_k c_k^{(\nu)} \sum_{j=1}^{d_{\nu}} \psi_j^{(\nu)} \frac{n}{d_{\alpha}} \delta_{\alpha\nu} \delta_{lj} \delta_{mk} = c_m^{(\alpha)} \psi_l^{(\alpha)}, \end{aligned} \quad (18.148)$$

where we used (18.51) for the equality before the last.

Thus, an implication of the operator $P_{l(m)}^{(\alpha)}$ is that if f contains the component $\psi_m^{(\alpha)}$, i.e., $c_m^{(\alpha)} \neq 0$, $P_{l(m)}^{(\alpha)}$ plays a role in *extracting* the component $c_m^{(\alpha)} \psi_m^{(\alpha)}$ from f and then *converting* it to $c_m^{(\alpha)} \psi_l^{(\alpha)}$. If the $\psi_m^{(\alpha)}$ component is not contained in f , that means from (18.148) $P_{l(m)}^{(\alpha)} f = 0$. In that case we choose a more suitable function for f . In this context, we will investigate an example of a quantum chemical calculation later.

In the above case, let us call $P_{l(m)}^{(\alpha)}$ a projection operator *sensu lato* by relaxing Definition 14.1 of the projection operator. In Sect. 14.1, we dealt with several aspects of the projection operators. There we have mentioned that a projection operator should be idempotent and Hermitian in a rigorous sense (Definition 14.1). To address another important aspect of the projection operator, let us prove an important relation in a following theorem.

Theorem 18.6 [1, 2] Let $P_{l(m)}^{(\alpha)}$ and $P_{s(t)}^{(\beta)}$ be projection operators defined in (18.147). Then, the following equation holds:

$$P_{l(m)}^{(\alpha)} P_{s(t)}^{(\beta)} = \delta_{\alpha\beta} \delta_{ms} P_{l(t)}^{(\alpha)}. \quad (18.149)$$

Proof We have

$$\begin{aligned} P_{l(m)}^{(\alpha)} P_{s(t)}^{(\beta)} &= \frac{d_\alpha}{n} \sum_g D_{lm}^{(\alpha)}(g)^* g \frac{d_\beta}{n} \sum_{g'} D_{st}^{(\beta)}(g')^* g' \\ &= \frac{d_\alpha}{n} \sum_g D_{lm}^{(\alpha)}(g)^* g \frac{d_\beta}{n} \sum_{g'} D_{st}^{(\beta)}(g^{-1}g')^* g^{-1}g' \\ &= \frac{d_\alpha}{n} \frac{d_\beta}{n} \sum_g \sum_{g'} D_{lm}^{(\alpha)}(g)^* D_{st}^{(\beta)}(g^{-1}g')^* g g^{-1}g' \\ &= \frac{d_\alpha}{n} \frac{d_\beta}{n} \sum_g \sum_{g'} D_{lm}^{(\alpha)}(g)^* \left\{ \left[D^{(\beta)}(g) \right]^\dagger D^{(\beta)}(g') \right\}_{st}^* e g' \\ &= \frac{d_\alpha}{n} \frac{d_\beta}{n} \sum_g \sum_{g'} D_{lm}^{(\alpha)}(g)^* \left\{ \sum_k D^{(\beta)}(g)_{ks}^* D^{(\beta)}(g')_{kt} \right\}^* g' \\ &= \frac{d_\alpha}{n} \frac{d_\beta}{n} \sum_k \sum_{g'} \left[\sum_g D_{lm}^{(\alpha)}(g)^* D^{(\beta)}(g)_{ks} \right] D^{(\beta)}(g')_{kt}^* g' \\ &= \frac{d_\alpha}{n} \frac{d_\beta}{n} \sum_k \sum_{g'} \frac{n}{d_\alpha} \delta_{\alpha\beta} \delta_{lk} \delta_{ms} D^{(\beta)}(g')_{kt}^* g' = \frac{d_\beta}{n} \sum_{g'} \delta_{\alpha\beta} \delta_{ms} D^{(\beta)}(g')_{lt}^* g' \\ &= \delta_{\alpha\beta} \delta_{ms} \frac{d_\beta}{n} \sum_{g'} D^{(\beta)}(g')_{lt}^* g' = \delta_{\alpha\beta} \delta_{ms} P_{l(t)}^{(\alpha)}. \end{aligned} \quad (18.150)$$

This completes the proof.

In the above proof, we used the rearrangement theorem and grand orthogonality theorem (GOT) as well as the homomorphism and unitarity of the representation matrices.

Comparing (18.146) and (18.148), we notice that the term $c_m^{(\alpha)} \psi_m^{(\alpha)}$ is not extracted entirely, but $c_m^{(\alpha)} \psi_l^{(\alpha)}$ is given instead. This is due to the linearity of $P_{l(m)}^{(\alpha)}$. Nonetheless, this is somewhat inconvenient for a practical purpose. To overcome this inconvenience, we modify (18.149). In (18.149) putting $s = m$, we have

$$P_{l(m)}^{(\alpha)} P_{m(t)}^{(\beta)} = \delta_{\alpha\beta} P_{l(t)}^{(\alpha)}. \quad (18.151)$$

We further modify the relation. Putting $\beta = \alpha$, we have

$$P_{l(m)}^{(\alpha)} P_{m(t)}^{(\alpha)} = P_{l(t)}^{(\alpha)}. \quad (18.152)$$

Putting $t = l$ furthermore,

$$P_{l(m)}^{(\alpha)} P_{m(l)}^{(\alpha)} = P_{l(l)}^{(\alpha)}. \quad (18.153)$$

In particular, putting $m = l$ moreover, we get

$$P_{l(l)}^{(\alpha)} P_{l(l)}^{(\alpha)} = \left[P_{l(l)}^{(\alpha)} \right]^2 = P_{l(l)}^{(\alpha)}. \quad (18.154)$$

In fact, putting $m = l$ in (18.148), we get

$$P_{l(l)}^{(\alpha)} f = c_l^{(\alpha)} \psi_l^{(\alpha)}. \quad (18.155)$$

This means that the term $c_l^{(\alpha)} \psi_l^{(\alpha)}$ has been extracted entirely.

Moreover, in (18.149) putting $\beta = \alpha$, $s = l$, and $t = m$, we have

$$P_{l(m)}^{(\alpha)} P_{l(m)}^{(\alpha)} = \left[P_{l(m)}^{(\alpha)} \right]^2 = \delta_{ml} P_{l(m)}^{(\alpha)}.$$

Therefore, for $P_{l(m)}^{(\alpha)}$ to be an idempotent operator, we must have $m = l$. That is, of various operators $P_{l(m)}^{(\alpha)}$, only $P_{l(l)}^{(\alpha)}$ is eligible for an idempotent operator.

Meanwhile, fully describing $P_{l(l)}^{(\alpha)}$ we have

$$P_{l(l)}^{(\alpha)} = \frac{d_\alpha}{n} \sum_g D_{ll}^{(\alpha)}(g)^* g. \quad (18.156)$$

Taking complex conjugate transposition (i.e., adjoint) of (18.156), we have

$$\begin{aligned} \left[P_{l(l)}^{(\alpha)} \right]^\dagger &= \frac{d_\alpha}{n} \sum_g D_{ll}^{(\alpha)}(g) g^\dagger = \frac{d_\alpha}{n} \sum_{g^{-1}} D_{ll}^{(\alpha)}(g^{-1}) (g^{-1})^\dagger \\ &= \frac{d_\alpha}{n} \sum_g D_{ll}^{(\alpha)}(g)^* g = P_{l(l)}^{(\alpha)}, \end{aligned} \quad (18.157)$$

where we used unitarity of g (with the third equality) and equivalence of summation over g and g^{-1} . Notice that the notation of $g^\dagger$ is less common. This should be interpreted as meaning that $g^\dagger$ operates on a vector constituting a representation space. Thus, notation $g^\dagger$ implies that $g^\dagger$ is equivalent to its unitary representation matrix $D^{(\alpha)}(g)$. Note also that $D_{ll}^{(\alpha)}$ is not a matrix but a (complex) *number*. Namely, in (18.156) $D_{ll}^{(\alpha)}(g)^*$ is a coefficient of an operator g .

Equations (18.154) and (18.157) establish that $P_{l(l)}^{(\alpha)}$ is a projection operator in a rigorous sense; namely, it is an idempotent and Hermitian operator (see Definition 14.1). Let us call $P_{l(l)}^{(\alpha)}$ a projection operator *sensu stricto* accordingly. Also, we notice that $c_m^{(\alpha)}\psi_m^{(\alpha)}$ is entirely extracted from an arbitrary function f including a coefficient. This situation resembles that of (12.205).

Regarding $P_{l(m)}^{(\alpha)}$ ($l \neq m$), on the other hand, we have

$$\begin{aligned} \left[P_{l(m)}^{(\alpha)} \right]^\dagger &= \frac{d_\alpha}{n} \sum_g D_{lm}^{(\alpha)}(g) g^\dagger = \frac{d_\alpha}{n} \sum_{g^{-1}} D_{lm}^{(\alpha)}(g^{-1}) (g^{-1})^\dagger \\ &= \frac{d_\alpha}{n} \sum_g D_{ml}^{(\alpha)}(g)^* g = P_{m(l)}^{(\alpha)}. \end{aligned} \quad (18.158)$$

Hence, $P_{l(m)}^{(\alpha)}$ is not Hermitian. We have many other related operators and equations. For instance,

$$\left[P_{l(m)}^{(\alpha)} P_{m(l)}^{(\alpha)} \right]^\dagger = \left[P_{m(l)}^{(\alpha)} \right]^\dagger \left[P_{l(m)}^{(\alpha)} \right]^\dagger = P_{l(m)}^{(\alpha)} P_{m(l)}^{(\alpha)}. \quad (18.159)$$

Therefore, $P_{l(m)}^{(\alpha)} P_{m(l)}^{(\alpha)}$ is Hermitian, recovering the relation (18.153).

In (18.149) putting $m = l$ and $t = s$, we get

$$P_{l(l)}^{(\alpha)} P_{s(s)}^{(\beta)} = \delta_{\alpha\beta} \delta_{ls} P_{l(s)}^{(\alpha)}. \quad (18.160)$$

As in the case of (18.146), we assume that h is described as

$$h = \sum_\alpha \sum_m d_m^{(\alpha)} \phi_m^{(\alpha)}, \quad (18.161)$$

where $\phi_m^{(\alpha)}$ is transformed in the same manner as that for $\psi_m^{(\alpha)}$ in (18.146), but linearly independent of $\psi_m^{(\alpha)}$. Namely, we have

$$g(\phi_i^{(\alpha)}) = \sum_{j=1}^{d_\alpha} \phi_j^{(\alpha)} D_{ji}^{(\alpha)}(g). \quad (18.162)$$

Tangible examples can be seen in Chap. 19.

Operating $P_{l(l)}^{(\alpha)}$ on both sides of (18.155), we have

$$\left[P_{l(l)}^{(\alpha)} \right]^2 f = P_{l(l)}^{(\alpha)} \left[c_l^{(\alpha)} \psi_l^{(\alpha)} \right] = P_{l(l)}^{(\alpha)} f = c_l^{(\alpha)} \psi_l^{(\alpha)}, \quad (18.163)$$

where with the second equality we used (18.154). That is, we have

$$P_{l(l)}^{(\alpha)} \left[c_l^{(\alpha)} \psi_l^{(\alpha)} \right] = c_l^{(\alpha)} \psi_l^{(\alpha)}. \quad (18.164)$$

This equation means that $c_l^{(\alpha)} \psi_l^{(\alpha)}$ is an eigenfunction corresponding to an eigenvalue 1 of $P_{l(l)}^{(\alpha)}$. In other words, once $c_l^{(\alpha)} \psi_l^{(\alpha)}$ is extracted from f , it belongs to the “position” l of an irreducible representation α . Furthermore, for some constants c and d as well as the functions f and h that appeared in (18.146) and (18.161), we consider a following equation:

$$\begin{aligned} P_{l(l)}^{(\alpha)} \left[c c_l^{(\alpha)} \psi_l^{(\alpha)} + d d_l^{(\alpha)} \phi_l^{(\alpha)} \right] &= c P_{l(l)}^{(\alpha)} c_l^{(\alpha)} \psi_l^{(\alpha)} + d P_{l(l)}^{(\alpha)} d_l^{(\alpha)} \phi_l^{(\alpha)} \\ &= c c_l^{(\alpha)} \psi_l^{(\alpha)} + d d_l^{(\alpha)} \phi_l^{(\alpha)}, \end{aligned} \quad (18.165)$$

where with the last equality we used (18.164).

This means that an arbitrary linear combination of $c_l^{(\alpha)} \psi_l^{(\alpha)}$ and $d_l^{(\alpha)} \phi_l^{(\alpha)}$ again belongs to the position l of an irreducible representation α . If $\psi_l^{(\alpha)}$ and $\phi_l^{(\alpha)}$ are linearly independent, we can construct two orthonormal basis vectors following Theorem 13.2 (Gram–Schmidt orthonormalization theorem). If there are other linearly independent vectors $p_l^{(\alpha)} \xi_l^{(\alpha)}, q_l^{(\alpha)} \varphi_l^{(\alpha)}, \dots$, where $\xi_l^{(\alpha)}, \varphi_l^{(\alpha)}$, etc. belong to the position l of an irreducible representation α , then $c c_l^{(\alpha)} \psi_l^{(\alpha)} + d d_l^{(\alpha)} \phi_l^{(\alpha)} + p p_l^{(\alpha)} \xi_l^{(\alpha)} + q q_l^{(\alpha)} \varphi_l^{(\alpha)} + \dots$ again belongs to the position l of an irreducible representation α . Thus, we can construct orthonormal basis vectors of the representation space according to Theorem 13.2.

Regarding arbitrary functions f and h as vectors and using (18.160), we make an inner product of (18.160) such that

$$\begin{aligned} \left\langle h | P_{l(l)}^{(\alpha)} P_{s(s)}^{(\beta)} | f \right\rangle &= \left\langle h | P_{l(l)}^{(\alpha)} P_{l(l)}^{(\alpha)} P_{s(s)}^{(\beta)} | f \right\rangle = \left\langle h | P_{l(l)}^{(\alpha)} \delta_{\alpha\beta} \delta_{ls} P_{l(s)}^{(\alpha)} | f \right\rangle \\ &= \delta_{\alpha\beta} \delta_{ls} \left\langle h P_{l(l)}^{(\alpha)} | P_{l(s)}^{(\alpha)} | f \right\rangle, \end{aligned} \quad (18.166)$$

where with the first equality we used (18.154). Meanwhile, from (18.148) we have

$$P_{l(s)}^{(\alpha)} | f \rangle = c_s^{(\alpha)} \psi_l^{(\alpha)}. \quad (18.167)$$

Also using (18.155), we get

$$P_{l(l)}^{(\alpha)}|h\rangle = d_l^{(\alpha)}|\phi_l^{(\alpha)}\rangle. \quad (18.168)$$

Taking adjoint of (18.168), we have

$$\langle h | [P_{l(l)}^{(\alpha)}]^\dagger = \langle h | P_{l(l)}^{(\alpha)} = [d_l^{(\alpha)}]^* \langle \phi_l^{(\alpha)} |, \quad (18.169)$$

where we used (18.157). The relation (18.169) is due to the notation of Sect. 13.3. Substituting (18.167) and (18.168) for (18.166),

$$\langle h | P_{l(l)}^{(\alpha)} P_{s(s)}^{(\beta)} | f \rangle = \delta_{\alpha\beta} \delta_{ls} [d_l^{(\alpha)}]^* c_s^{(\alpha)} \langle \phi_l^{(\alpha)} | \psi_s^{(\alpha)} \rangle.$$

Meanwhile, (18.166) can also be described as

$$\langle h | P_{l(l)}^{(\alpha)} P_{s(s)}^{(\beta)} | f \rangle = \langle d_l^{(\alpha)} \phi_l^{(\alpha)} | c_s^{(\beta)} \psi_s^{(\beta)} \rangle = [d_l^{(\alpha)}]^* c_s^{(\beta)} \langle \phi_l^{(\alpha)} | \psi_s^{(\beta)} \rangle. \quad (18.170)$$

For (18.166) and (18.170) to be identical, we must have

$$\delta_{\alpha\beta} \delta_{ls} [d_l^{(\alpha)}]^* c_s^{(\alpha)} \langle \phi_l^{(\alpha)} | \psi_s^{(\alpha)} \rangle = [d_l^{(\alpha)}]^* c_s^{(\beta)} \langle \phi_l^{(\alpha)} | \psi_s^{(\beta)} \rangle.$$

Deleting coefficients, we get

$$\langle \phi_l^{(\alpha)} | \psi_s^{(\beta)} \rangle = \delta_{\alpha\beta} \delta_{ls} \langle \phi_l^{(\alpha)} | \psi_s^{(\alpha)} \rangle. \quad (18.171)$$

The relation (18.171) is frequently used to estimate whether definite integrals vanish. Functional forms depend upon actual problems we encounter in various situations. We will deal with this problem in Chap. 19 in relation to, e.g., discussion on optical transition and evaluation of overlap integrals.

To evaluate (18.170), if $\alpha \neq \beta$ or $l \neq s$, we get simply

$$\langle h | P_{l(l)}^{(\alpha)} P_{s(s)}^{(\beta)} | f \rangle = 0 \quad (\alpha \neq \beta \text{ or } l \neq s). \quad (18.172)$$

That is, under a given condition $\alpha \neq \beta$ or $l \neq s$, $P_{s(s)}^{(\beta)} | f \rangle$ and $P_{l(l)}^{(\alpha)} | h \rangle$ are orthogonal (see Theorem 13.3 of Sect. 13.4). The relation clearly indicates that functions belonging to different irreducible representations ($\alpha \neq \beta$) are mutually orthogonal. Even though the functions belong to the same irreducible representation, the functions are orthogonal if they are allocated to the different “place” as a basis vector designated by l, s , etc. Here the place means the index j of $\psi_j^{(\alpha)}$ in (18.146) or $\phi_j^{(\alpha)}$ in (18.161) that designates the “order” of $\psi_j^{(\alpha)}$ within $\psi_1^{(\alpha)}, \psi_2^{(\alpha)}, \dots, \psi_{d_\alpha}^{(\alpha)}$ or $\phi_j^{(\alpha)}$

within $\phi_1^{(\alpha)}, \phi_2^{(\alpha)}, \dots, \phi_{d_\alpha}^{(\alpha)}$. This takes place if the representation is multidimensional (i.e., dimensionality: d_α).

In (18.160) putting $\alpha = \beta$, we get

$$P_{l(l)}^{(\alpha)} P_{s(s)}^{(\alpha)} = \delta_{ls} P_{l(s)}^{(\alpha)}.$$

In the above relation, furthermore, unless $l = s$ we have

$$P_{l(l)}^{(\alpha)} P_{s(s)}^{(\alpha)} = 0.$$

Therefore, on the basis of the discussion of Sect. 14.1, $P_{l(l)}^{(\alpha)} + P_{s(s)}^{(\alpha)}$ is a projection operator as well in the case of $l \neq s$. Notice, however, that if $l = s$, $P_{l(l)}^{(\alpha)} + P_{s(s)}^{(\alpha)}$ is *not* a projection operator. Readers can readily show it.

Moreover, let us define $P^{(\alpha)}$ as below:

$$P^{(\alpha)} \equiv \sum_{l=1}^{d_\alpha} P_{l(l)}^{(\alpha)}. \quad (18.173)$$

Similarly, $P^{(\alpha)}$ is again a projection operator as well.

From (18.156), $P^{(\alpha)}$ in (18.173) can be rewritten as

$$P^{(\alpha)} = \frac{d_\alpha}{n} \sum_g \sum_{l=1}^{d_\alpha} D_{ll}^{(\alpha)}(g)^* g = \frac{d_\alpha}{n} \sum_g \left[\chi^{(\alpha)}(g) \right]^* g. \quad (18.174)$$

Returning to (18.155) and taking summation over l there, we have

$$\sum_{l=1}^{d_\alpha} P_{l(l)}^{(\alpha)} f = \sum_{l=1}^{d_\alpha} c_l^{(\alpha)} \psi_l^{(\alpha)}.$$

Using (18.173), we get

$$P^{(\alpha)} f = \sum_{l=1}^{d_\alpha} c_l^{(\alpha)} \psi_l^{(\alpha)}. \quad (18.175)$$

Thus, an operator $P^{(\alpha)}$ has a clear meaning. That is, $P^{(\alpha)}$ plays a role in extracting all the vectors (or functions) that belong to the irreducible representation α including their coefficients. Defining those functions as

$$\psi^{(\alpha)} \equiv \sum_{l=1}^{d_\alpha} c_l^{(\alpha)} \psi_l^{(\alpha)}, \quad (18.176)$$

we succinctly rewrite (18.175) as

$$P^{(\alpha)}f = \psi^{(\alpha)}. \quad (18.177)$$

In turn, let us calculate $P^{(\beta)}P^{(\alpha)}$. This can be done with (18.174) as follows:

$$P^{(\beta)}P^{(\alpha)} = \frac{d_\beta}{n} \sum_g \left[\chi^{(\beta)}(g) \right]^* g \frac{d_\alpha}{n} \sum_{g'} \left[\chi^{(\alpha)}(g') \right]^* g'. \quad (18.178)$$

To carry out the calculation, (1) first we replace g' with $g^{-1}g'$ and rewrite the summation over g' as that over $g^{-1}g'$. (2) Using the homomorphism and unitarity of the representation, we rewrite $[\chi^{(\alpha)}(g')]^*$ as

$$\left[\chi^{(\alpha)}(g') \right]^* = \sum_{ij} \left[D^{(\alpha)}(g) \right]_{ji} \left[D^{(\alpha)}(g') \right]_{ji}^*. \quad (18.179)$$

Rewriting (18.178) we have

$$\begin{aligned} P^{(\beta)}P^{(\alpha)} &= \frac{d_\beta}{n} \frac{d_\alpha}{n} \sum_g \sum_{k,i,j} \left[D^{(\beta)}(g) \right]_{kk}^* \left[D^{(\alpha)}(g) \right]_{ji} \sum_{g'} \left[D^{(\alpha)}(g') \right]_{ji}^* g' \\ &= \frac{d_\beta}{n} \frac{d_\alpha}{n} \sum_{k,i,j} \frac{n}{d_\beta} \delta_{\alpha\beta} \delta_{kj} \delta_{ki} \sum_{g'} \left[D^{(\alpha)}(g') \right]_{ji}^* g' = \frac{d_\alpha}{n} \sum_k \sum_{g'} \left[D^{(\alpha)}(g') \right]_{kk}^* g' \\ &= \delta_{\alpha\beta} \frac{d_\alpha}{n} \sum_{g'} \left[\chi^{(\alpha)}(g') \right]^* g' = \delta_{\alpha\beta} P^{(\alpha)}. \end{aligned} \quad (18.180)$$

These relationships are anticipated from the fact that both $P^{(\alpha)}$ and $P^{(\beta)}$ are projection operators.

As in the case of (18.160) and (18.180) is useful to evaluate an inner product of functions. Again taking an inner product of (18.180) with arbitrary functions f and g , we have

$$\left\langle g | P^{(\beta)} P^{(\alpha)} | f \right\rangle = \left\langle g | \delta_{\alpha\beta} P^{(\alpha)} | f \right\rangle = \delta_{\alpha\beta} \left\langle g | P^{(\alpha)} | f \right\rangle. \quad (18.181)$$

If we define P such that

$$P \equiv \sum_{\alpha=1}^{n_r} P^{(\alpha)}, \quad (18.182)$$

then P is once again a projection operator (see Sect. 14.1). Now taking summation in (18.177) over all the irreducible representations α , we have

$$\sum_{\alpha=1}^{n_r} P^{(\alpha)} f = \sum_{\alpha=1}^{n_r} \psi^{(\alpha)} = f. \quad (18.183)$$

The function f has been arbitrarily taken and, hence, we get

$$P = E. \quad (18.184)$$

The relation described in (18.182)–(18.184) is called completeness relation (see Sect. 5.1).

As mentioned in Example 18.1, the concept of the representation space is not only important but also very useful for addressing various problems of physics and chemistry. For instance, regarding molecular orbital methods to be dealt with in Chap. 19, we consider a representation space whose dimension is equal to the number of electrons of a molecule about which we wish to know, e.g., energy eigenvalues of those electrons. In that case, the dimension of representation space is equal to the number of molecular orbitals. According to a symmetry species of the molecule, the representation matrix is decomposed into a direct sum of invariant eigenspaces relevant to individual irreducible representations. If the basis vectors belong to different irreducible representation, such vectors are orthogonal to each other in virtue of (18.171). Even though those vectors belong to the same irreducible representation, they are orthogonal if they are allocated to a different place. However, it is often the case that the vectors belong to the same place of the same irreducible representation. Then, it is always possible according to Theorem 13.2 to make them mutually orthogonal by taking their linear combination. In such a way, we can construct an orthonormal basis set throughout the representation space. In fact, the method is a powerful tool for solving an energy eigenvalue problem and for determining associated eigenfunctions (or molecular orbitals).

18.8 Direct-Product Representation

In Sect. 16.5 we have studied basic properties of direct-product groups. Correspondingly, we examine in this section properties of direct-product representation. This notion is very useful to investigate optical transitions in molecular systems and selection rules relevant to those transitions.

Let $D^{(\alpha)}$ and $D^{(\beta)}$ be two different irreducible representations whose dimensions are d_α and d_β , respectively. Then, operating a group element g on the basis functions, we have

$$g(\psi_i) = \sum_{k=1}^{d_\alpha} \psi_k D_{ki}^{(\alpha)}(g) \quad (1 \leq i \leq d_\alpha), \quad (18.185)$$

$$g(\phi_j) = \sum_{l=1}^{d_\beta} \phi_l D_{lj}^{(\beta)}(g) \quad (1 \leq j \leq d_\beta), \quad (18.186)$$

where ψ_k ($1 \leq k \leq d_\alpha$) and ϕ_l ($1 \leq l \leq d_\beta$) are basis functions of $D^{(\alpha)}$ and $D^{(\beta)}$, respectively. We can construct $d_\alpha d_\beta$ new basis vectors using $\psi_k \phi_l$. These functions are transformed according to g such that

$$\begin{aligned} g(\psi_i \phi_j) &= g(\psi_i) g(\phi_j) = \left[\sum_k \psi_k D_{ki}^{(\alpha)}(g) \right] \left[\sum_l \phi_l D_{lj}^{(\beta)}(g) \right] \\ &= \sum_k \sum_l \psi_k \phi_l D_{ki}^{(\alpha)}(g) D_{lj}^{(\beta)}(g). \end{aligned} \quad (18.187)$$

Here let us define the following matrix $[D^{(\alpha \times \beta)}(g)]_{kl, ij}$ such that

$$[D^{(\alpha \times \beta)}(g)]_{kl, ij} \equiv D_{ki}^{(\alpha)}(g) D_{lj}^{(\beta)}(g). \quad (18.188)$$

Then we have

$$g(\psi_i \phi_j) = \sum_k \sum_l \psi_k \phi_l [D^{(\alpha \times \beta)}(g)]_{kl, ij}. \quad (18.189)$$

The notation using double scripts is somewhat complicated. We notice, however, that in (18.188) the order of subscript of $kilj$ is converted to kl, ij , i.e., the subscripts i and l have been interchanged.

We write $D^{(\alpha)}$ and $D^{(\beta)}$ in explicit forms as follows:

$$D^{(\alpha)}(g) = \begin{pmatrix} d_{1,1}^{(\alpha)} & \cdots & d_{1,d_\alpha}^{(\alpha)} \\ \vdots & \ddots & \vdots \\ d_{d_\alpha,1}^{(\alpha)} & \cdots & d_{d_\alpha,d_\alpha}^{(\alpha)} \end{pmatrix}, D^{(\beta)}(g) = \begin{pmatrix} d_{1,1}^{(\beta)} & \cdots & d_{1,d_\beta}^{(\beta)} \\ \vdots & \ddots & \vdots \\ d_{d_\beta,1}^{(\beta)} & \cdots & d_{d_\beta,d_\beta}^{(\beta)} \end{pmatrix}. \quad (18.190)$$

Thus,

$$D^{(\alpha \times \beta)}(g) = D^{(\alpha)}(g) \otimes D^{(\beta)}(g) = \begin{pmatrix} d_{1,1}^{(\alpha)} D^{(\beta)}(g) & \cdots & d_{1,d_\alpha}^{(\alpha)} D^{(\beta)}(g) \\ \vdots & \ddots & \vdots \\ d_{d_\alpha,1}^{(\alpha)} D^{(\beta)}(g) & \cdots & d_{d_\alpha,d_\alpha}^{(\alpha)} D^{(\beta)}(g) \end{pmatrix}. \quad (18.191)$$

To get familiar with the double-scripted notation, we describe a case of (2, 2) matrices. Denoting

$$D^{(\alpha)}(g) = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \text{ and } D^{(\beta)}(g) = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}, \quad (18.192)$$

we get

$$\begin{aligned} D^{(\alpha \times \beta)}(g) &= D^{(\alpha)}(g) \otimes D^{(\beta)}(g) \\ &= \begin{pmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{pmatrix}. \end{aligned} \quad (18.193)$$

Corresponding to (18.22) and (18.189) describes the transformation of $\psi_i \phi_j$ regarding the double script. At the same time, a set $\{\psi_i \phi_j; (1 \leq i \leq d_\alpha, 1 \leq j \leq d_\beta)\}$ is a basis of $D^{(\alpha \times \beta)}$. In fact,

$$\begin{aligned} \left[D^{(\alpha \times \beta)}(gg') \right]_{kl,ij} &= D_{ki}^{(\alpha)}(gg') D_{lj}^{(\beta)}(gg') \\ &= \left[\sum_\mu D_{k\mu}^{(\alpha)}(g) D_{\mu i}^{(\alpha)}(g') \right] \left[\sum_\nu D_{l\nu}^{(\beta)}(g) D_{\nu j}^{(\beta)}(g') \right] \\ &= \sum_\mu \sum_\nu D_{k\mu}^{(\alpha)}(g) D_{l\nu}^{(\beta)}(g) D_{\mu i}^{(\alpha)}(g') D_{\nu j}^{(\beta)}(g') \\ &= \sum_\mu \sum_\nu \left[D^{(\alpha \times \beta)}(g) \right]_{kl,\mu\nu} \left[D^{(\alpha \times \beta)}(g') \right]_{\mu\nu,ij}. \end{aligned} \quad (18.194)$$

Equation (18.194) shows that the rule of matrix calculation with respect to the double subscript is satisfied. Consequently, we get

$$D^{(\alpha \times \beta)}(gg') = D^{(\alpha \times \beta)}(g) D^{(\alpha \times \beta)}(g'). \quad (18.195)$$

The relation (18.195) indicates that $D^{(\alpha \times \beta)}$ is certainly a representation of a group. This representation is said to be a direct-product representation.

A character of the direct-product representation is given by putting $k = i$ and $l = j$ in (18.188). That is,

$$\left[D^{(\alpha \times \beta)}(g) \right]_{ij,ij} = D_{ii}^{(\alpha)}(g) D_{jj}^{(\beta)}(g). \quad (18.196)$$

Denoting

$$\chi^{(\alpha \times \beta)}(g) \equiv \sum_{ij} \left[D^{(\alpha \times \beta)}(g) \right]_{ij,ij}, \quad (18.197)$$

we have

$$\chi^{(\alpha \times \beta)}(g) = \chi^{(\alpha)}(g) \chi^{(\beta)}(g). \quad (18.198)$$

Even though $D^{(\alpha)}$ and $D^{(\beta)}$ are both irreducible, $D^{(\alpha \times \beta)}$ is not necessarily irreducible. Suppose that

$$D^{(\alpha)}(g) \otimes D^{(\beta)}(g) = \sum_{\gamma} q_{\gamma} D^{(\gamma)}(g), \quad (18.199)$$

where q_{γ} is given by (18.83). Then, we have

$$q_{\gamma} = \frac{1}{n} \sum_g \chi^{(\gamma)}(g)^* \chi^{(\alpha \times \beta)}(g) = \frac{1}{n} \sum_g \chi^{(\gamma)}(g)^* \chi^{(\alpha)}(g) \chi^{(\beta)}(g), \quad (18.200)$$

where n is an order of the group. This relation is often used to perform quantum mechanical or chemical calculations, especially to evaluate optical transitions of matter including atoms and molecular systems. This is also useful to examine whether a definite integral of a product of functions (or an inner product of vectors) vanishes.

In Sect. 16.5, we investigated definition and properties of direct-product groups. Similarly to the case of the direct-product representation, we consider a representation of the direct-product groups. Let us consider two groups $\mathcal{G}$ and $\mathcal{H}$ and assume that a direct-product group $\mathcal{G} \otimes \mathcal{H}$ is defined (Sect. 16.5). Also let $D^{(\alpha)}$ and $D^{(\beta)}$ be d_{α} - and d_{β} -dimensional representations of groups $\mathcal{G}$ and $\mathcal{H}$, respectively. Furthermore, let us define a matrix $D^{(\alpha \times \beta)}(ab)$ as in (18.188) such that

$$\left[D^{(\alpha \times \beta)}(ab) \right]_{kl,ij} \equiv D_{ki}^{(\alpha)}(a) D_{lj}^{(\beta)}(b), \quad (18.201)$$

where a and b are arbitrarily chosen from $\mathcal{G}$ and $\mathcal{H}$, respectively, and $ab \in \mathcal{G} \otimes \mathcal{H}$. Then a set comprising $D^{(\alpha \times \beta)}(ab)$ forms a representation of $\mathcal{G} \otimes \mathcal{H}$. (Readers, please verify it.) A dimension of $D^{(\alpha \times \beta)}$ is $d_{\alpha} d_{\beta}$. The character is given in (18.69), and so in the present case, by putting $i = k$ and $j = l$ in (18.201), we get

$$\begin{aligned} \chi^{(\alpha \times \beta)}(ab) &= \sum_k \sum_l \left[D^{(\alpha \times \beta)}(ab) \right]_{kl,kl} = \sum_k \sum_l D_{kk}^{(\alpha)}(a) D_{ll}^{(\beta)}(b) \\ &= \chi^{(\alpha)}(a) \chi^{(\beta)}(b). \end{aligned} \quad (18.202)$$

Equation (18.202) resembles (18.198). Hence, we should be careful not to confuse them. In (18.198), we were thinking of a direct-product representation within a *sole* group $\mathcal{G}$. In (18.202), however, we are considering a representation

of the direct-product group comprising two *different* groups. In fact, even though a character is computed with respect to a sole group element g in (18.198), in (18.202) we evaluate a character regarding two group elements a and b chosen from different groups $\mathcal{G}$ and $\mathcal{H}$, respectively.

18.9 Symmetric Representation and Antisymmetric Representation

As mentioned in Sect. 18.2, we have viewed a group element g as a linear transformation over a vector space V . There we dealt with widely chosen functions as vectors. In this chapter we introduce other useful ideas so that we can apply them to molecular science and atomic physics.

In the previous section, we defined direct-product representation. In $D^{(\alpha \times \beta)}(g) = D^{(\alpha)}(g) \otimes D^{(\beta)}(g)$, we can freely consider a case where $D^{(\alpha)}(g) = D^{(\beta)}(g)$. Then we have

$$g(\psi_i \phi_j) = g(\psi_i)g(\phi_j) = \sum_k \sum_l \psi_k \phi_l D_{ki}^{(\alpha)}(g) D_{lj}^{(\alpha)}(g). \quad (18.203)$$

Regarding a product function $\psi_j \phi_i$, we can get a similar equation such that

$$g(\psi_j \phi_i) = g(\psi_j)g(\phi_i) = \sum_k \sum_l \psi_k \phi_l D_{kj}^{(\alpha)}(g) D_{li}^{(\alpha)}(g). \quad (18.204)$$

On the basis of the linearity of the relations (18.203) and (18.204), let us construct a linear combination of the product functions. That is,

$$\begin{aligned} g(\psi_i \phi_j \pm \psi_j \phi_i) &= g(\psi_i \phi_j) \pm g(\psi_j \phi_i) \\ &= \sum_k \sum_l \psi_k \phi_l D_{ki}^{(\alpha)}(g) D_{lj}^{(\alpha)}(g) \pm \sum_k \sum_l \psi_k \phi_l D_{kj}^{(\alpha)}(g) D_{li}^{(\alpha)}(g) \\ &= \sum_k \sum_l (\psi_k \phi_l \pm \psi_l \phi_k) D_{ki}^{(\alpha)}(g) D_{lj}^{(\alpha)}(g). \end{aligned} \quad (18.205)$$

Here, defining $\Psi_{ij}^{\pm}$ as

$$\Psi_{ij}^{\pm} \equiv \psi_i \phi_j \pm \psi_j \phi_i, \quad (18.206)$$

we rewrite (18.205) as

$$g\Psi_{ij}^{\pm} = \sum_k \sum_l \Psi_{kl}^{\pm} \left\{ \frac{1}{2} \left[D_{ki}^{(\alpha)}(g) D_{lj}^{(\alpha)}(g) \pm D_{li}^{(\alpha)}(g) D_{kj}^{(\alpha)}(g) \right] \right\}. \quad (18.207)$$

Notice that Ψ_{ij}^+ and Ψ_{ij}^- in (18.206) are symmetric and antisymmetric with respect to the interchange of subscript i and j , respectively. That is,

$$\Psi_{ij}^\pm = \pm \Psi_{ji}^\pm. \quad (18.208)$$

We may naturally ask how we can constitute representation (matrices) with (18.207). To answer this question, we have to carry out calculations by replacing g with gg' in (18.207) and following the procedures of Sect. 18.2. Thus, we have

$$\begin{aligned} gg' \Psi_{ij}^\pm &= \sum_k \sum_l \Psi_{kl}^\pm \left\{ \frac{1}{2} \left[D_{ki}^{(\alpha)}(gg') D_{lj}^{(\alpha)}(gg') \pm D_{li}^{(\alpha)}(gg') D_{kj}^{(\alpha)}(gg') \right] \right\} \\ &= \sum_k \sum_l \Psi_{kl}^\pm \left\{ \frac{1}{2} \left[\sum_\mu D_{k\mu}^{(\alpha)}(g) D_{\mu i}^{(\alpha)}(g') \sum_\nu D_{l\nu}^{(\alpha)}(g) D_{\nu j}^{(\alpha)}(g') \right. \right. \\ &\quad \left. \left. \pm \sum_\mu D_{l\mu}^{(\alpha)}(g) D_{\mu i}^{(\alpha)}(g') \sum_\nu D_{k\nu}^{(\alpha)}(g) D_{\nu j}^{(\alpha)}(g') \right] \right\} \\ &= \sum_k \sum_l \Psi_{kl}^\pm \left\{ \frac{1}{2} \sum_\mu \sum_\nu \left[D_{k\mu}^{(\alpha)}(g) D_{l\nu}^{(\alpha)}(g) \pm D_{l\mu}^{(\alpha)}(g) D_{k\nu}^{(\alpha)}(g) \right] D_{\mu i}^{(\alpha)}(g') D_{\nu j}^{(\alpha)}(g') \right\} \\ &= \sum_k \sum_l \Psi_{kl}^\pm \left[\frac{1}{2} \sum_\mu \sum_\nu \left\{ \left[D^{(\alpha \times \alpha)}(g) \right]_{kl, \mu\nu} \pm \left[D^{(\alpha \times \alpha)}(g) \right]_{lk, \mu\nu} \right\} D_{\mu i}^{(\alpha)}(g') D_{\nu j}^{(\alpha)}(g') \right], \end{aligned} \quad (18.209)$$

where the last equality follows from the definition of a direct-product representation (18.188). We notice that the terms of $[D^{(\alpha \times \alpha)}(gg')]_{kl, \mu\nu} \pm [D^{(\alpha \times \alpha)}(gg')]_{lk, \mu\nu}$ in (18.209) are symmetric and antisymmetric with respect to the interchange of subscripts k and l together with subscripts μ and ν , respectively. Comparing both sides of (18.209), we see that this must be the case with i and j as well. Then, the last factor of (18.209) should be rewritten as:

$$D_{\mu i}^{(\alpha)}(g') D_{\nu j}^{(\alpha)}(g') = \frac{1}{2} \left[D_{\mu i}^{(\alpha)}(g') D_{\nu j}^{(\alpha)}(g') \pm D_{\nu i}^{(\alpha)}(g') D_{\mu j}^{(\alpha)}(g') \right].$$

Now, we define the following notations $D^{[\alpha \times \alpha]}(g)$ and $D^{\{\alpha \times \alpha\}}(g)$ represented by

$$\begin{aligned} \left\{ D^{[\alpha \times \alpha]}(g) \right\}_{kl, \mu\nu} &\equiv \frac{1}{2} \left\{ \left[D^{(\alpha \times \alpha)}(g) \right]_{kl, \mu\nu} + \left[D^{(\alpha \times \alpha)}(g) \right]_{lk, \mu\nu} \right\} \\ &= \frac{1}{2} \left[D_{k\mu}^{(\alpha)}(g) D_{l\nu}^{(\alpha)}(g) + D_{l\mu}^{(\alpha)}(g) D_{k\nu}^{(\alpha)}(g) \right] \end{aligned} \quad (18.210)$$

and

$$\begin{aligned}
\{D^{\{\alpha \times \alpha\}}(g)\}_{kl,\mu\nu} &\equiv \frac{1}{2} \left\{ \left[D^{(\alpha \times \alpha)}(g) \right]_{kl,\mu\nu} - \left[D^{(\alpha \times \alpha)}(g) \right]_{lk,\mu\nu} \right\} \\
&= \frac{1}{2} \left[D_{k\mu}^{(\alpha)}(g) D_{l\nu}^{(\alpha)}(g) - D_{l\mu}^{(\alpha)}(g) D_{k\nu}^{(\alpha)}(g) \right]. \quad (18.211)
\end{aligned}$$

Meanwhile, using $D_{\mu i}^{(\alpha)}(g') D_{\nu j}^{(\alpha)}(g') = [D^{(\alpha \times \alpha)}(g')]_{\mu\nu,ij}$ and considering the exchange of summation with respect to the subscripts μ and ν , we can also define $D^{[\alpha \times \alpha]}(g')$ and $D^{\{\alpha \times \alpha\}}(g)$ according to the symmetric and antisymmetric cases, respectively. Thus for the symmetric case, we have

$$\begin{aligned}
gg' \Psi_{ij}^+ &= \sum_k \sum_l \sum_\mu \sum_\nu \Psi_{kl}^+ \left\{ D^{[\alpha \times \alpha]}(g) \right\}_{kl,\mu\nu} \left\{ D^{[\alpha \times \alpha]}(g') \right\}_{\mu\nu,ij} \\
&= \sum_k \sum_l \Psi_{kl}^+ \left\{ D^{[\alpha \times \alpha]}(g) D^{[\alpha \times \alpha]}(g') \right\}_{kl,ij}. \quad (18.212)
\end{aligned}$$

Using (18.210) and (18.207) can be rewritten as

$$g \Psi_{ij}^+ = \sum_k \sum_l \Psi_{kl}^+ \left\{ D^{[\alpha \times \alpha]}(g) \right\}_{kl,ij}. \quad (18.213)$$

Then we have

$$gg' \Psi_{ij}^+ = \sum_k \sum_l \Psi_{kl}^+ \left\{ D^{[\alpha \times \alpha]}(gg') \right\}_{kl,ij}. \quad (18.214)$$

Comparing (18.212) and (18.214), we finally get

$$D^{[\alpha \times \alpha]}(gg') = D^{[\alpha \times \alpha]}(g) D^{[\alpha \times \alpha]}(g'). \quad (18.215)$$

Similarly, for the antisymmetric case, we have

$$g \Psi_{ij}^- = \sum_k \sum_l \Psi_{kl}^- \left\{ D^{\{\alpha \times \alpha\}}(g) \right\}_{kl,ij}, \quad (18.216)$$

$$gg' \Psi_{ij}^- = \sum_k \sum_l \Psi_{kl}^- \left\{ D^{\{\alpha \times \alpha\}}(g) D^{\{\alpha \times \alpha\}}(g') \right\}_{kl,ij}. \quad (18.217)$$

Hence, we obtain

$$D^{\{\alpha \times \alpha\}}(gg') = D^{\{\alpha \times \alpha\}}(g) D^{\{\alpha \times \alpha\}}(g'). \quad (18.218)$$

Thus, both $D^{[\alpha \times \alpha]}(g)$ and $D^{\{\alpha \times \alpha\}}(g)$ produce well-defined representations.

Letting dimension of the representation α be d_α , we have $d_\alpha(d_\alpha + 1)/2$ functions belonging to the symmetric representation and $d_\alpha(d_\alpha - 1)/2$ functions belonging to

the antisymmetric representation. With the two-dimensional representation, for instance, functions belonging to the symmetric representation are

$$\psi_1\phi_1, \psi_1\phi_2 + \psi_2\phi_1, \text{ and } \psi_2\phi_2. \quad (18.219)$$

A function belonging to the antisymmetric representation is

$$\psi_1\phi_2 - \psi_2\phi_1. \quad (18.220)$$

Note that these vectors have not yet been normalized.

From (18.210) and (18.211), we can readily get useful expressions with characters of symmetric and antisymmetric representations. In (18.210) putting $\mu = k$ and $\nu = l$ and summing over k and l ,

$$\begin{aligned} \chi^{[\alpha \times \alpha]}(g) &= \sum_k \sum_l \left\{ D^{[\alpha \times \alpha]}(g) \right\}_{kl,kl} \left\{ D^{[\alpha \times \alpha]}(g) \right\}_{kl,kl} \\ &= \frac{1}{2} \sum_k \sum_l \left[D_{kk}^{(\alpha)}(g) D_{ll}^{(\alpha)}(g) + D_{lk}^{(\alpha)}(g) D_{kl}^{(\alpha)}(g) \right] \\ &= \frac{1}{2} \left\{ \left[\chi^{(\alpha)}(g) \right]^2 + \left[\chi^{(\alpha)}(g^2) \right] \right\}. \end{aligned} \quad (18.221)$$

Similarly we have

$$\chi^{\{\alpha \times \alpha\}}(g) = \frac{1}{2} \left\{ \left[\chi^{(\alpha)}(g) \right]^2 - \left[\chi^{(\alpha)}(g^2) \right] \right\}. \quad (18.222)$$

References

1. Inui T, Tanabe Y, Onodera Y (1990) Group theory and its applications in physics. Springer, Berlin
2. Inui T, Tanabe Y, Onodera Y (1980) Group theory and its applications in physics. Shokabo, Tokyo
3. Hassani S (2006) Mathematical physics. Springer, New York
4. Chen JQ, Ping J, Wang F (2002) Group representation theory for physicists, 2nd edn. World Scientific, Singapore
5. Hamermesh M (1989) Group theory and its application to physical problems. Dover, New York

Chapter 19

Applications of Group Theory to Physical Chemistry



On the basis of studies of group theory, now in this chapter we apply the knowledge to the molecular orbital (MO) calculations (or quantum chemical calculations). As tangible examples, we adopt aromatic hydrocarbons (ethylene, cyclopropenyl radical, benzene, and ally radical) and methane. The approach is based upon a method of linear combination of atomic orbitals (LCAO). To seek an appropriate LCAO MO, we make the most of a method based on a symmetry-adapted linear combination (SALC). To use projection operators is a powerful tool for this purpose. For the sake of correct understanding, it is desired to consider transformation of functions. To this end, we first show several examples.

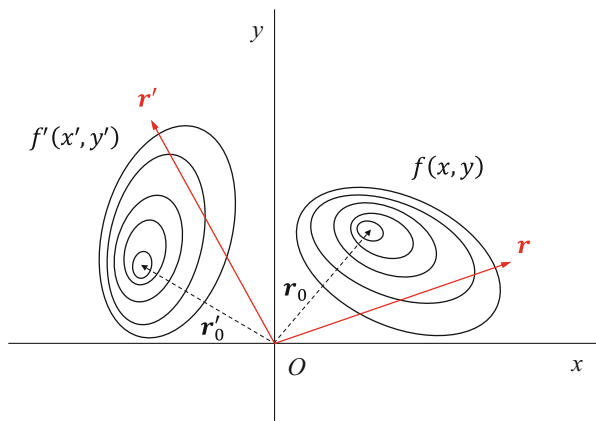
In the process of carrying out MO calculations, we encounter a secular equation as an eigenvalue equation. Using a SALC eases the calculations of the secular equation. Molecular science relies largely on spectroscopic measurements and researchers need to assign individual spectral lines to a specific transition between the relevant molecular states. Representation theory works well in this situation. Thus, the group theory finds a perfect fit with its applications in the molecular science.

19.1 Transformation of Functions

Before showing individual examples, let us consider the transformation of functions (or vectors) by the symmetry operation. Here we consider scalar functions.

For example, let us suppose an arbitrary scalar function $f(x, y)$ on a Cartesian xy -coordinate. Figure 19.1 shows a contour map of $f(x, y) = \text{constant}$. Then suppose that the map is rotated around the origin. More specifically, the position vector $\mathbf{r}_0$ fixed on a “summit” [i.e., a point that gives a maximal value of $f(x, y)$] undergoes a symmetry operation, namely, rotation around the origin. As a result, $\mathbf{r}_0$ is transformed to $\mathbf{r}'_0$. Here, we assume that a “mountain” represented by $f(x, y)$ is a rigid body before and

Fig. 19.1 Contour map of $f(x, y)$ and $f'(x', y')$. The function $f'(x', y')$ is obtained by rotating a map of $f(x, y)$ around the z -axis



after the transformation. Concomitantly, a general point $\mathbf{r}$ (see Sect. 17.1) is transformed to $\mathbf{r}'$ in exactly the same way as $\mathbf{r}_0$.

A new function f' gives a new contour map after the rotation. Let us describe f' as

$$f' \equiv O_R f. \quad (19.1)$$

The RHS of (19.1) means that f' is produced as a result of operating the rotation R on f . We describe the position vector after being transformed as $\mathbf{r}'_0$. Following the notation of Sect. 11.1, $\mathbf{r}'_0$ and $\mathbf{r}'$ are expressed as

$$\mathbf{r}'_0 = R(\mathbf{r}_0) \quad \text{and} \quad \mathbf{r}' = R(\mathbf{r}). \quad (19.2)$$

The matrix representation for R is given by e.g., (11.35). In (19.2) we have

$$\mathbf{r}_0 = (\mathbf{e}_1 \ \mathbf{e}_2) \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} \quad \text{and} \quad \mathbf{r}'_0 = (\mathbf{e}_1 \ \mathbf{e}_2) \begin{pmatrix} x'_0 \\ y'_0 \end{pmatrix}, \quad (19.3)$$

where $\mathbf{e}_1$ and $\mathbf{e}_2$ are orthonormal basis vectors in the xy -plane. Also we have

$$\mathbf{r} = (\mathbf{e}_1 \ \mathbf{e}_2) \begin{pmatrix} x \\ y \end{pmatrix} \quad \text{and} \quad \mathbf{r}' = (\mathbf{e}_1 \ \mathbf{e}_2) \begin{pmatrix} x' \\ y' \end{pmatrix}. \quad (19.4)$$

Meanwhile, the following equation must hold:

$$f'(x', y') = f(x, y). \quad (19.5)$$

Or, using (19.1) we have

$$O_R f(x', y') = f(x, y). \quad (19.6)$$

The above argument can be extended to a three-dimensional (or higher dimensional) space. In that case, similarly we have

$$O_R f(x', y', z') = f(x, y, z), O_R f(\mathbf{r}') = f(\mathbf{r}), \text{ or } f'(\mathbf{r}') = f(\mathbf{r}), \quad (19.7)$$

where

$$\mathbf{r} = (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \text{and} \quad \mathbf{r}' = (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}. \quad (19.8)$$

The last relation of (19.7) comes from (19.1). Using (19.2) and (19.3), we rewrite (19.7) as

$$O_R f[R(\mathbf{r})] = f(\mathbf{r}). \quad (19.9)$$

Replacing $\mathbf{r}$ with $R^{-1}(\mathbf{r})$, we get

$$O_R f\{R[R^{-1}(\mathbf{r})]\} = f[R^{-1}(\mathbf{r})]. \quad (19.10)$$

That is,

$$O_R f(\mathbf{r}) = f[R^{-1}(\mathbf{r})]. \quad (19.11)$$

More succinctly, (19.11) may be rewritten as

$$Rf(\mathbf{r}) = f[R^{-1}(\mathbf{r})]. \quad (19.12)$$

Comparing (19.11) with (19.12) and considering (19.1), we have

$$f' \equiv O_R f \equiv Rf. \quad (19.13)$$

To gain a good understanding of the function transformation, let us think of some examples.

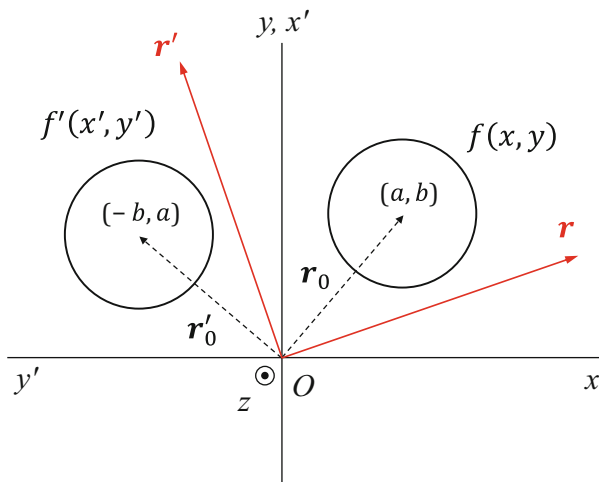
Example 19.1

Let $f(x, y)$ be a function described by

$$f(x, y) = (x - a)^2 + (y - b)^2; a, b > 0. \quad (19.14)$$

A contour is shown in Fig. 19.2. We consider a $\pi/2$ rotation around the z -axis. Then, $f(x, y)$ is transformed into $Rf(x, y) = f'(x, y)$ such that

Fig. 19.2 Contour map of $f(x, y) = (x - a)^2 + (y - b)^2$ and $f'(x', y') = (x' + b)^2 + (y' - a)^2$ before and after a $\pi/2$ rotation around the z -axis



$$Rf(x, y) = f'(x, y) = (x + b)^2 + (y - a)^2. \quad (19.15)$$

We also have

$$\mathbf{r}_0 = (\mathbf{e}_1 \ \mathbf{e}_2) \begin{pmatrix} a \\ b \end{pmatrix},$$

$$\mathbf{r}'_0 = (\mathbf{e}_1 \ \mathbf{e}_2) \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = (\mathbf{e}_1 \ \mathbf{e}_2) \begin{pmatrix} -b \\ a \end{pmatrix},$$

where we define R as

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Similarly,

$$\mathbf{r} = (\mathbf{e}_1 \ \mathbf{e}_2) \begin{pmatrix} x \\ y \end{pmatrix} \quad \text{and} \quad \mathbf{r}' = (\mathbf{e}_1 \ \mathbf{e}_2) \begin{pmatrix} -y \\ x \end{pmatrix}.$$

From (19.15), we have

$$f'(x', y') = (x' + b)^2 + (y' - a)^2 = (-y + b)^2 + (x - a)^2 = f(x, y). \quad (19.16)$$

This ensures that (19.5) holds. The implication of (19.16) combined with (19.5) is that a view of $f'(x', y')$ from $(-b, a)$ is the same as that of $f(x, y)$ from (a, b) . Imagine that if we are standing at $(-b, a)$ of $f'(x', y')$, we are in the bottom of the “valley” of $f'(x', y')$. Exactly in the same manner, if we are standing at (a, b) of $f(x, y)$, we are in

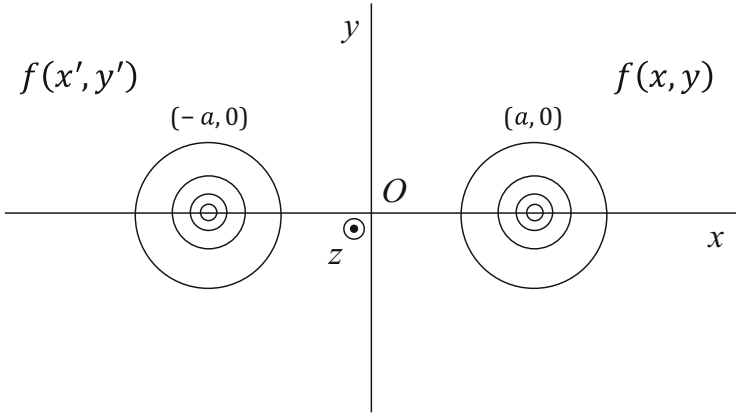


Fig. 19.3 Contour map of $f(\mathbf{r}) = e^{-2[(x-a)^2+y^2+z^2]} + e^{-2[(x+a)^2+y^2+z^2]}$ ($a > 0$). The function form remains unchanged by a reflection with respect to the yz -plane

the bottom of the valley of $f(x, y)$ as well. Notice that for both $f(x, y)$ and $f'(x', y')$, (a, b) and $(-b, a)$ are the lowest point, respectively.

Meanwhile, we have

$$R^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, R^{-1}(\mathbf{r}) = (\mathbf{e}_1 \ \mathbf{e}_2) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = (\mathbf{e}_1 \ \mathbf{e}_2) \begin{pmatrix} y \\ -x \end{pmatrix}.$$

Then we have

$$f[R^{-1}(\mathbf{r})] = [(y) - a]^2 + [(-x) - b]^2 = (x + b)^2 + (y - a)^2 = Rf(\mathbf{r}). \quad (19.17)$$

Thus, (19.12) certainly holds.

Example 19.2

Let $f(\mathbf{r})$ and $g(\mathbf{r})$ be functions described by

$$f(\mathbf{r}) = e^{-2[(x-a)^2+y^2+z^2]} + e^{-2[(x+a)^2+y^2+z^2]} \quad (a > 0), \quad (19.18)$$

$$g(\mathbf{r}) = e^{-2[(x-a)^2+y^2+z^2]} - e^{-2[(x+a)^2+y^2+z^2]} \quad (a > 0). \quad (19.19)$$

Figure 19.3 shows an outline of the contour of $f(\mathbf{r})$. We consider a following symmetry operation in a three-dimensional coordinate system:

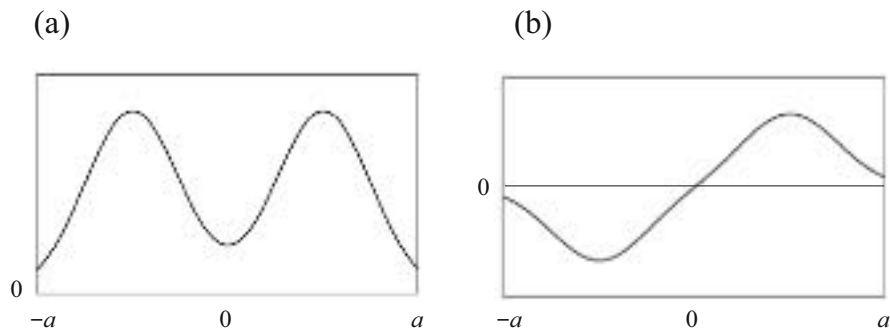


Fig. 19.4 Plots of $f(\mathbf{r}) = e^{-2[(x-a)^2+y^2+z^2]} + e^{-2[(x+a)^2+y^2+z^2]}$ and $g(\mathbf{r}) = e^{-2[(x-a)^2+y^2+z^2]} - e^{-2[(x+a)^2+y^2+z^2]}$ ($a > 0$) as a function of x . The plots have been taken along the x -axis. (a) $f(\mathbf{r})$. (b) $g(\mathbf{r})$

$$R = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad R^{-1} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (19.20)$$

This represents a reflection with respect to the yz -plane. Then we have

$$f[R^{-1}(\mathbf{r})] = Rf(\mathbf{r}) = f(\mathbf{r}), \quad (19.21)$$

$$g[R^{-1}(\mathbf{r})] = Rg(\mathbf{r}) = -g(\mathbf{r}). \quad (19.22)$$

Plotting $f(\mathbf{r})$ and $g(\mathbf{r})$ as a function of x on the x -axis, we depict results in Fig. 19.4.

Looking at (19.21) and (19.22), we find that $f(\mathbf{r})$ and $g(\mathbf{r})$ are solutions of an eigenvalue equation for an operator R . Corresponding eigenvalues are 1 and -1 for $f(\mathbf{r})$ and $g(\mathbf{r})$, respectively. In particular, $f(\mathbf{r})$ holds the functional form after the transformation R . In this case, $f(\mathbf{r})$ is said to be invariant with the transformation R . Moreover, $f(\mathbf{r})$ is invariant with the following eight transformations:

$$R = \begin{pmatrix} \pm 1 & 0 & 0 \\ 0 & \pm 1 & 0 \\ 0 & 0 & \pm 1 \end{pmatrix}. \quad (19.23)$$

These transformations form a group that is isomorphic to D_{2h} . Therefore, $f(\mathbf{r})$ is eligible for a basis function of the totally symmetric representation A_g of D_{2h} . Notice that $f(\mathbf{r})$ is invariant as well with a rotation of an arbitrary angle around the x -axis. On the other hand, $g(\mathbf{r})$ belongs to B_{3u} .

19.2 Method of Molecular Orbitals (MOs)

Bearing in mind these arguments, we examine several examples of quantum chemical calculations. Our approach is based upon the molecular orbital theory. The theory assumes the existence of molecular orbitals (MOs) in a molecule, as the notion of atomic orbitals has been well-established in atomic physics. Furthermore, we assume that the molecular orbitals comprise a linear combination of atomic orbitals (LCAO).

This notion is equivalent to that individual electrons in a molecule are independently moving in a potential field produced by nuclei and other electrons. In other words, we assume that each electron is moving along an MO that is extended over the whole molecule. Electronic state in the molecule is formed by different MOs of various energies that are occupied by electrons. As in the case of an atom, an MO $\psi_i(\mathbf{r})$ occupied by an electron is described as

$$H\psi_i(\mathbf{r}) \equiv \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}) \right] \psi_i(\mathbf{r}) = E_i \psi_i(\mathbf{r}), \quad (19.24)$$

where H is Hamiltonian of a molecule; m is a mass of an electron (note that we do not use a reduced mass μ here); $\mathbf{r}$ is a position vector of the electron; ∇^2 is the Laplacian (Laplace operator); $V(\mathbf{r})$ is a potential of the molecule at $\mathbf{r}$; E_i is an energy of the electron occupying ψ_i and said to be a molecular orbital energy. We assume that $V(\mathbf{r})$ possesses a symmetry the same as that of the molecule.

Let $\mathcal{G}$ be a symmetry group and let a group element arbitrarily chosen from among $\mathcal{G}$ be R . Suppose that an arbitrary position vector $\mathbf{r}$ is moved to another position $\mathbf{r}'$. This transformation is expressed as (19.2). Since $V(\mathbf{r})$ has the same symmetry as the molecule, an electron “feels” the same potential field at $\mathbf{r}'$ as that at $\mathbf{r}$. That is,

$$V(\mathbf{r}) = V'(\mathbf{r}') = V(\mathbf{r}'). \quad (19.25)$$

Or we have

$$V(\mathbf{r})\psi(\mathbf{r}) = V(\mathbf{r}')\psi'(\mathbf{r}'), \quad (19.26)$$

where ψ is an arbitrary function. Defining

$$V(\mathbf{r})\psi(\mathbf{r}) \equiv [V\psi](\mathbf{r}) = V\psi(\mathbf{r}), \quad (19.27)$$

and recalling (19.1) and (19.7), we get

$$[RV]\psi(\mathbf{r}') = R[V\psi(\mathbf{r}')] = V'\psi'(\mathbf{r}') = V'(\mathbf{r}')\psi'(\mathbf{r}') = V(\mathbf{r}')\psi'(\mathbf{r}')$$

$$= V(\mathbf{r}')R\psi(\mathbf{r}') = VR\psi(\mathbf{r}') = [VR]\psi(\mathbf{r}'). \quad (19.28)$$

Comparing the first and last sides and remembering that ψ is an arbitrary function, we get

$$RV = VR. \quad (19.29)$$

Next, let us examine the symmetry of the Laplacian ∇^2 . The Laplacian is defined in Sect. 1.2 as

$$\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}, \quad (1.24)$$

where x , y , and z denote the Cartesian coordinates. Let S be an orthogonal matrix that transforms the xyz -coordinate system to $x'y'z'$ -coordinate system. We suppose that S is expressed as

$$S = \begin{pmatrix} s_{11} & s_{12} & s_{13} \\ s_{21} & s_{22} & s_{23} \\ s_{31} & s_{32} & s_{33} \end{pmatrix} \text{ and } \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} s_{11} & s_{12} & s_{13} \\ s_{21} & s_{22} & s_{23} \\ s_{31} & s_{32} & s_{33} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}. \quad (19.30)$$

Since S is an orthonormal matrix, we have

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} s_{11} & s_{21} & s_{31} \\ s_{12} & s_{22} & s_{32} \\ s_{13} & s_{23} & s_{33} \end{pmatrix} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}. \quad (19.31)$$

The equation is due to

$$SS^T = S^T S = E, \quad (19.32)$$

where E is a unit matrix. Then we have

$$\frac{\partial}{\partial x'} = \frac{\partial x}{\partial x'} \frac{\partial}{\partial x} + \frac{\partial y}{\partial x'} \frac{\partial}{\partial y} + \frac{\partial z}{\partial x'} \frac{\partial}{\partial z} = s_{11} \frac{\partial}{\partial x} + s_{12} \frac{\partial}{\partial y} + s_{13} \frac{\partial}{\partial z}. \quad (19.33)$$

Partially differentiating (19.33), we have

$$\frac{\partial}{\partial x'^2} = s_{11} \frac{\partial}{\partial x'} \frac{\partial}{\partial x} + s_{12} \frac{\partial}{\partial x'} \frac{\partial}{\partial y} + s_{13} \frac{\partial}{\partial x'} \frac{\partial}{\partial z}$$

$$\begin{aligned}
&= s_{11} \left(s_{11} \frac{\partial}{\partial x} + s_{12} \frac{\partial}{\partial y} + s_{13} \frac{\partial}{\partial z} \right) \frac{\partial}{\partial x} + s_{12} \left(s_{11} \frac{\partial}{\partial x} + s_{12} \frac{\partial}{\partial y} + s_{13} \frac{\partial}{\partial z} \right) \frac{\partial}{\partial y} \\
&\quad + s_{13} \left(s_{11} \frac{\partial}{\partial x} + s_{12} \frac{\partial}{\partial y} + s_{13} \frac{\partial}{\partial z} \right) \frac{\partial}{\partial z} \\
&= s_{11}^2 \frac{\partial^2}{\partial x^2} + s_{11}s_{12} \frac{\partial^2}{\partial y \partial x} + s_{11}s_{13} \frac{\partial^2}{\partial z \partial x} \\
&\quad + s_{12}s_{11} \frac{\partial^2}{\partial x \partial y} + s_{12}^2 \frac{\partial^2}{\partial y^2} + s_{12}s_{13} \frac{\partial^2}{\partial z \partial y} \\
&\quad + s_{13}s_{11} \frac{\partial^2}{\partial x \partial z} + s_{13}s_{12} \frac{\partial^2}{\partial y \partial z} + s_{13}^2 \frac{\partial^2}{\partial z^2}. \tag{19.34}
\end{aligned}$$

Calculating terms of $\frac{\partial}{\partial y^2}$ and $\frac{\partial}{\partial z^2}$, we have similar results. Then, collecting all those 27 terms, we get, e.g.,

$$(s_{11}^2 + s_{21}^2 + s_{31}^2) \frac{\partial^2}{\partial x^2} = \frac{\partial^2}{\partial x^2}, \tag{19.35}$$

where we used an orthogonal relationship of S . Cross terms of $\frac{\partial^2}{\partial x \partial y}$, $\frac{\partial^2}{\partial y \partial x}$, etc. all vanish. Consequently, we get

$$\frac{\partial}{\partial x'^2} + \frac{\partial}{\partial y'^2} + \frac{\partial}{\partial z'^2} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}. \tag{19.36}$$

Defining ∇'^2 as

$$\nabla'^2 \equiv \frac{\partial}{\partial x'^2} + \frac{\partial}{\partial y'^2} + \frac{\partial}{\partial z'^2}, \tag{19.37}$$

we have

$$\nabla'^2 = \nabla^2. \tag{19.38}$$

Notice that (19.38) holds with not only the symmetry operation of the molecule, but also any rotation operation in $\mathbb{R}^3$ (see Sect. 17.4).

As in the case of (19.28), we have

$$\begin{aligned} [R\nabla^2]\psi(\mathbf{r}') &= R[\nabla^2\psi(\mathbf{r}')] = \nabla'^2\psi'(\mathbf{r}') = \nabla^2\psi'(\mathbf{r}') \\ &= \nabla^2 R\psi(\mathbf{r}') = [\nabla^2 R]\psi(\mathbf{r}'). \end{aligned} \quad (19.39)$$

Consequently, we get

$$R\nabla^2 = \nabla^2 R. \quad (19.40)$$

Adding both sides of (19.29) and (19.40), we get

$$R(\nabla^2 + V) = (\nabla^2 + V)R. \quad (19.41)$$

From (19.24), we have

$$RH = HR. \quad (19.42)$$

Thus, we confirm that the Hamiltonian H commutes with any symmetry operation R . In other words, H is invariant with the coordinate transformation relevant to the symmetry operation.

Now we consider matrix representation of (19.42). Also let D be an irreducible representation of the symmetry group $\mathcal{G}$. Then, we have

$$D(g)H = HD(g) \quad (g \in \mathcal{G}).$$

Let $\{\psi_1, \psi_2, \dots, \psi_d\}$ be a set of basis vectors that span a representation space $\mathcal{L}_S$ associated with D . Then, on the basis of Schur's Second Lemma of Sect. 18.3, (19.42) immediately leads to an important conclusion that if H is represented by a matrix, we must have

$$H = \lambda E, \quad (19.43)$$

where λ is an arbitrary complex constant. That is, H is represented such that

$$H = \begin{pmatrix} \lambda & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda \end{pmatrix}, \quad (19.44)$$

where the above matrix is (d, d) square diagonal matrix. This implies that H is reduced to

$$H = \lambda D^{(0)} \bigoplus \cdots \bigoplus \lambda D^{(0)},$$

where $D^{(0)}$ denotes the totally symmetric representation; notice that it is given by 1 for any symmetry operation. Thus, the commutability of an operator with all symmetry operation is equivalent to that the operator belongs to the totally symmetric representation. Since H is Hermitian, λ should be real.

Operating both sides of (19.42) on $\{\psi_1, \psi_2, \cdots, \psi_d\}$ from the right, we have

$$\begin{aligned} (\psi_1 \ \psi_2 \ \cdots \ \psi_d)RH &= (\psi_1 \ \psi_2 \ \cdots \ \psi_d)HR \\ &= (\psi_1 \ \psi_2 \ \cdots \ \psi_d) \begin{pmatrix} \lambda & & & \\ & \lambda & & \\ & & \ddots & \\ & & & \lambda \end{pmatrix} R = (\lambda\psi_1 \ \lambda\psi_2 \ \cdots \ \lambda\psi_d)R \\ &= \lambda(\psi_1 \ \psi_2 \ \cdots \ \psi_d)R. \end{aligned} \quad (19.45)$$

In particular, putting $R = E$, we get

$$(\psi_1 \ \psi_2 \ \cdots \ \psi_d)H = \lambda(\psi_1 \ \psi_2 \ \cdots \ \psi_d).$$

Simplifying this equation, we have

$$\psi_i H = \lambda \psi_i \quad (1 \leq i \leq d). \quad (19.46)$$

Thus, λ is found to be an energy eigenvalue of H and ψ_i ($1 \leq i \leq d$) are eigenfunctions belonging to the eigenvalue λ .

Meanwhile, using

$$R(\psi_i) = \sum_{k=1}^d \psi_k D_{ki}(R),$$

we rewrite (19.45) as

$$\begin{aligned} (\psi_1 \ \psi_2 \ \cdots \ \psi_d)RH &= (\psi_1 R \ \psi_2 R \ \cdots \ \psi_d R)H \\ &= \left(\sum_{k=1}^d \psi_k D_{k1}(R) \ \sum_{k=1}^d \psi_k D_{k2}(R) \ \cdots \ \sum_{k=1}^d \psi_k D_{kd}(R) \right) H \\ &= \lambda \left(\sum_{k=1}^d \psi_k D_{k1}(R) \ \sum_{k=1}^d \psi_k D_{k2}(R) \ \cdots \ \sum_{k=1}^d \psi_k D_{kd}(R) \right), \end{aligned}$$

where with the second equality we used the relation (11.42); i.e.,

$$\psi_i R = R(\psi_i).$$

Thus, we get

$$\sum_{k=1}^d \psi_k D_{ki}(R) H = \lambda \sum_{k=1}^d \psi_k D_{ki}(R) \quad (1 \leq i \leq d). \quad (19.47)$$

The relations (19.45)–(19.47) imply that $\psi_1, \psi_2, \dots$, and ψ_d as well as their linear combinations using a representation matrix $D(R)$ are eigenfunctions that belong to the same eigenvalue λ . That is, $\psi_1, \psi_2, \dots$, and ψ_d are said to be degenerate with a multiplication d .

After the remarks of [Theorem 14.5](#), we can construct an orthonormal basis set $\{\tilde{\psi}_1, \tilde{\psi}_2, \dots, \tilde{\psi}_d\}$ as eigenfunctions. These vectors can be constructed via linear combinations of $\psi_1, \psi_2, \dots$, and ψ_d . After transforming the vectors $\tilde{\psi}_i$ ($1 \leq i \leq d$) by R , we get

$$\begin{aligned} \left\langle \sum_{k=1}^d \tilde{\psi}_k D_{ki}(R) \middle| \sum_{l=1}^d \tilde{\psi}_l D_{lj}(R) \right\rangle &= \sum_{k=1}^d \sum_{l=1}^d D_{ki}(R)^* D_{lj}(R) \langle \tilde{\psi}_k | \tilde{\psi}_l \rangle \\ &= \sum_{k=1}^d D_{ki}(R)^* D_{kj}(R) = \sum_{k=1}^d [D^\dagger(R)]_{ik} [D(R)]_{kj} = [D^\dagger(R) D(R)]_{ij} = \delta_{ij}. \end{aligned}$$

With the last equality, we used unitarity of $D(R)$. Thus, the orthonormal basis set is retained after unitary transformation of $\{\tilde{\psi}_1, \tilde{\psi}_2, \dots, \tilde{\psi}_d\}$.

In the above discussion, we have assumed that $\psi_1, \psi_2, \dots$, and ψ_d belong to a certain irreducible representation $D^{(\nu)}$. Since $\tilde{\psi}_1, \tilde{\psi}_2, \dots$, and $\tilde{\psi}_d$ consist of their linear combinations, $\tilde{\psi}_1, \tilde{\psi}_2, \dots$, and $\tilde{\psi}_d$ belong to $D^{(\nu)}$ as well. Also, we have assumed that $\tilde{\psi}_1, \tilde{\psi}_2, \dots$, and $\tilde{\psi}_d$ form an orthonormal basis and, hence, according to [Theorem 11.3](#) these vectors constitute basis vectors that belong to $D^{(\nu)}$. In particular, functions derived using projection operators share these characteristics. In fact, the above principles underlie molecular orbital calculations dealt with in Sects. 19.4 and 19.5. We will go into more details in subsequent sections.

Bearing the aforementioned argument in mind, we make the most of the relation expressed by (18.171) to evaluate inner products of functions (or vectors). In this connection, we often need to calculate matrix elements of an operator. One of the most typical examples is matrix elements of Hamiltonian. In the field of molecular science we have to estimate an overlap integral, Coulomb integral, resonance integral, etc. Other examples include transition matrix elements pertinent to, e.g., electric dipole transition. To this end, we deal with direct-product representation (see Sects. 18.7 and 18.8). Let $O^{(\gamma)}$ be an Hermitian operator belonging to the γ -th irreducible representation. Let us think of a following inner product:

$$\langle \phi_l^{(\beta)} | O^{(\gamma)} \psi_s^{(\alpha)} \rangle, \quad (19.48)$$

where $\psi_s^{(\alpha)}$ and $\phi_l^{(\beta)}$ are the s -th component of α -th irreducible representation and the l -th component of β -th irreducible representation, respectively; see (18.146) and (18.161).

(i) Let us think of first the case where $O^{(\gamma)}$ is Hamiltonian H . Suppose that $\psi_s^{(\alpha)}$ belongs to an eigenvalue λ of H . Then, we have

$$H | \psi_l^{(\alpha)} \rangle = \lambda | \psi_l^{(\alpha)} \rangle.$$

In that case, with (19.48) we get

$$\langle \phi_l^{(\beta)} | H \psi_s^{(\alpha)} \rangle = \lambda \langle \phi_l^{(\beta)} | \psi_s^{(\alpha)} \rangle. \quad (19.49)$$

Having the benefit of (18.171), (19.49) can be rewritten as

$$\langle \phi_l^{(\beta)} | H \psi_s^{(\alpha)} \rangle = \lambda \delta_{\alpha\beta} \delta_{ls} \langle \phi_l^{(\alpha)} | \psi_l^{(\alpha)} \rangle. \quad (19.50)$$

At the first glance this equation seems trivial, but the equation gives us a powerful tool to save us a lot of troublesome calculations. In fact, if we encounter a series of inner product calculations (i.e., definite integrals), we ignore many of them. In (19.50), the matrix elements of Hamiltonian do not vanish only if $\alpha = \beta$ and $l = s$. That is, we only have to estimate $\langle \phi_l^{(\alpha)} | \psi_l^{(\alpha)} \rangle$. Otherwise the inner products vanish.

The functional forms of $\phi_l^{(\alpha)}$ and $\psi_l^{(\alpha)}$ are determined depending upon individual practical problems. We will show tangible examples later (Sects. 19.4 and 19.5).

(ii) Next, let us consider matrix elements of the optical transition. In this case, we are thinking of transition probability between quantum states. Assuming the dipole approximation, we use $\mathbf{e}_e \cdot \mathbf{P}^{(\gamma)}$ for $O^{(\gamma)}$ in $\langle \phi_l^{(\beta)} | O^{(\gamma)} \psi_s^{(\alpha)} \rangle$ of (19.48). The quantity $\mathbf{P}^{(\gamma)}$ represents an electric dipole moment associated with the position vector. Normally, we can readily find it in a character table, in which we examine which irreducible representation γ the position vector components x , y , and z indicated in a rightmost column correspond to. Table 18.4 is an example. If we take a unit polarization vector $\mathbf{e}_e$ in parallel to the position vector component that the character table designates, $\mathbf{e}_e \cdot \mathbf{P}^{(\gamma)}$ is non-vanishing. Suppose that $\psi_s^{(\alpha)}$ and $\phi_l^{(\beta)}$ are an initial state and final state, respectively. At the first glance, we would wish to use (18.200) and count how many times a representation β occurs for a direct-product representation $D^{(\gamma \times \alpha)}$. It is often the case, however, where β does not belong to an irreducible representation.

Even in such a case, if either α or β belongs to a totally symmetric representation, the handling will be easier. This situation corresponds to that an initial electronic configuration or a final electronic configuration forms a closed shell an electronic

state of which is totally symmetric [1]. The former occurs when we consider the optical absorption that takes place, e.g., in a molecule of a ground electronic state. The latter corresponds to an optical emission that ends up with a ground state. Let us consider the former case. Since $O_j^{(\gamma)}$ is Hermitian, we rewrite (19.48) as

$$\langle \phi_l^{(\beta)} | O_j^{(\gamma)} \psi_s^{(\alpha)} \rangle = \langle O_j^{(\gamma)} \phi_l^{(\beta)} | \psi_s^{(\alpha)} \rangle = \langle \psi_s^{(\alpha)} | O_j^{(\gamma)} \phi_l^{(\beta)} \rangle^*, \quad (19.51)$$

where we assume that $\psi_s^{(\alpha)}$ is a ground state having a closed shell electronic configuration. Therefore, $\psi_s^{(\alpha)}$ belongs to a totally symmetric irreducible representation. For (19.48) not to vanish, therefore, we may alternatively state that it is necessary for

$$D^{(\gamma \times \beta)} = D^{(\gamma)} \otimes D^{(\beta)}$$

to contain $D^{(\alpha)}$ belonging to a totally symmetric representation. Note that in group theory we usually write $D^{(\gamma)} \times D^{(\beta)}$ instead of $D^{(\gamma)} \otimes D^{(\beta)}$; see (18.191).

If $\phi_l^{(\beta)}$ belongs to a reducible representation, from (18.80) we have

$$D^{(\beta)} = \sum_{\omega} q_{\omega} D^{(\omega)},$$

where $D^{(\omega)}$ belongs to an irreducible representation ω . Then, we get

$$D^{(\gamma)} \times D^{(\beta)} = \sum_{\omega} q_{\omega} D^{(\gamma)} \times D^{(\omega)}.$$

Thus, applying (18.199) and (18.200), we can obtain a direct sum of irreducible representations. After that, we examine whether an irreducible representation α is contained in $D^{(\gamma)} \times D^{(\beta)}$. Since the totally symmetric representation is one-dimensional, $s = 1$ in (19.51).

19.3 Calculation Procedures of Molecular Orbitals (MOs)

We describe a brief outline of the MO method based on LCAO (LCAOMO). Suppose that a molecule consists of n atomic orbitals and that each MO ψ_i ($1 \leq i \leq n$) comprises a linear combination of those n atomic orbitals ϕ_k ($1 \leq k \leq n$). That is, we have

$$\psi_i = \sum_{k=1}^n c_{ki} \phi_k \quad (1 \leq i \leq n), \quad (19.52)$$

where c_{ki} are complex coefficients. We assume that ϕ_k are normalized. That is,

$$\langle \phi_k | \phi_k \rangle \equiv \int \phi_k^* \phi_k d\tau = 1, \quad (19.53)$$

where $d\tau$ implies that an integration should be taken over $\mathbb{R}^3$. The notation $\langle g|f \rangle$ means an inner product defined in Sect. 13.1. The inner product is usually defined by a definite integral of g^*f whose integration range covers a part or all of $\mathbb{R}^3$ depending on a constitution of a physical system.

First we try to solve Schrödinger equation given as an eigenvalue equation. The said equation is described as

$$H\psi = \lambda\psi \quad \text{or} \quad (H - \lambda\psi) = 0. \quad (19.54)$$

Replacing ψ with (19.52), we have

$$\sum_{k=1}^n c_k (H - \lambda) \phi_k = 0 \quad (1 \leq k \leq n), \quad (19.55)$$

where the subscript i in (19.52) has been omitted for simplicity. Multiplying ϕ_j^* from the left and integrating over whole $\mathbb{R}^3$, we have

$$\sum_{k=1}^n c_k \int \phi_j^* (H - \lambda) \phi_k d\tau = 0. \quad (19.56)$$

Rewriting (19.56), we get

$$\sum_{k=1}^n c_k \int (\phi_j^* H \phi_k - \lambda \phi_j^* \phi_k) d\tau = 0. \quad (19.57)$$

Here let us define following quantities:

$$H_{jk} = \int \phi_j^* H \phi_k d\tau \quad \text{and} \quad S_{jk} = \int \phi_j^* \phi_k d\tau, \quad (19.58)$$

where $S_{ii} = 1$ ($1 \leq i \leq n$) due to a normalized function of ϕ_i . Then we get

$$\sum_{k=1}^n c_k (H_{jk} - \lambda S_{jk}) = 0. \quad (19.59)$$

Rewriting (19.59) in a matrix form, we get

$$\begin{pmatrix} H_{11} - \lambda & \cdots & H_{1n} - \lambda S_{1n} \\ \vdots & \ddots & \vdots \\ H_{n1} - \lambda S_{n1} & \cdots & H_{nn} - \lambda \end{pmatrix} \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = 0. \quad (19.60)$$

Suppose that by solving (19.59) or (19.60), we get λ_i ($1 \leq i \leq n$), some of which may be identical (i.e., the degenerate case), and obtain n different column

eigenvectors corresponding to n eigenvalues λ_i . In light of (12.4) of Sect. 12.1, the following condition must be satisfied for this to get eigenvectors for which not all c_k is zero:

$$\det(H_{jk} - \lambda S_{jk}) = 0$$

or

$$\begin{vmatrix} H_{11} - \lambda & \cdots & H_{1n} - \lambda S_{1n} \\ \vdots & \ddots & \vdots \\ H_{n1} - \lambda S_{n1} & \cdots & H_{nn} - \lambda \end{vmatrix} = 0. \quad (19.61)$$

Equation (19.61) is called a secular equation. This equation is pertinent to a determinant of an order n , and so we are expected to get n roots for this, some of which are identical (i.e., a degenerate case).

In the above discussion, it is useful to introduce the following notation:

$$\langle \phi_j | H \phi_k \rangle \equiv H_{jk} = \int \phi_j^* H \phi_k d\tau \text{ and } \langle \phi_j | \phi_k \rangle \equiv S_{jk} = \int \phi_j^* \phi_k d\tau. \quad (19.62)$$

The above notation has already been introduced in Sect. 1.4. Equation (19.62) certainly satisfies the definition of the inner product described in (13.2)–(13.4); readers, please check it. On the basis of (13.64), we have

$$\langle y | Hx \rangle^* = \langle x H^\dagger | y \rangle. \quad (19.63)$$

Since H is Hermitian, $H^\dagger = H$. That is,

$$\langle y | Hx \rangle^* = \langle x H | y \rangle. \quad (19.64)$$

To solve (19.61) with an enough large number n is usually formidable. However, if we can find appropriate conditions, (19.61) can pretty easily be solved. An essential point rests upon how we deal with off-diagonal elements H_{jk} and S_{jk} . If we are able to appropriately choose basis vectors so that we can get

$$H_{jk} = 0 \quad \text{and} \quad S_{jk} = 0 \quad (j \neq k), \quad (19.65)$$

the secular equation is reduced to a simple form

$$\begin{vmatrix} \tilde{H}_{11} - \lambda \tilde{S}_{11} & & \\ & \tilde{H}_{22} - \lambda \tilde{S}_{22} & \\ & & \ddots \\ & & & \tilde{H}_{nn} - \lambda \tilde{S}_{nn} \end{vmatrix} = 0, \quad (19.66)$$

where all the off-diagonal elements are zero and an eigenvalue λ is given by

$$\lambda_i = \tilde{H}_{ii} / \tilde{S}_{ii}. \quad (19.67)$$

This means that the eigenvalue equation has automatically been solved.

The best way to achieve this is to choose basis functions (vectors) such that the functions conform to the symmetry which the molecule belongs to. That is, we “shuffle” the atomic orbitals as follows:

$$\xi_i = \sum_{k=1}^n d_{ki} \phi_k \quad (1 \leq i \leq n), \quad (19.68)$$

where ξ_i are new functions chosen instead of ψ_i of (19.52). That is, we construct a linear combination of the atomic orbitals that belongs to individual irreducible representation. The said linear combination is called symmetry-adapted linear combination (SALC). We remark that all atomic orbitals are not necessarily included in the SALC. Then we have

$$\tilde{H}_{jk} = \int \xi_j^* H \xi_k d\tau \quad \text{and} \quad \tilde{S}_{jk} = \int \xi_j^* \xi_k d\tau. \quad (19.69)$$

Thus, instead of (19.61), we have a new secular equation of

$$\det(\tilde{H}_{jk} - \lambda \tilde{S}_{jk}) = 0. \quad (19.70)$$

Since the Hamiltonian H is totally symmetric, in terms of the direct-product representation $H\xi_k$ in (19.69) belongs to an irreducible representation which ξ_k belongs to. This can intuitively be understood. However, to assert this, use (18.76) and (18.200) along with the fact that characters of the totally symmetric representation are 1.

In light of (18.171) and (18.181), if ξ_k and ξ_j belong to different irreducible representations, $\tilde{H}_{jk}$ and $\tilde{S}_{jk}$ both vanish at once. From (18.172), at the same time, ξ_k and ξ_j are orthogonal to each other. The most ideal situation is to get (19.66) with all the off-diagonal elements vanishing. Regarding the diagonal elements of (19.70), we always have the direct product of the same representation and, hence, the integrals (19.69) do not vanish. Thus, we get a powerful guideline for the evaluation of (19.70) in an as simplest as possible form.

If the SALC orbitals ξ_j and ξ_k belong to the same irreducible representation, the relevant matrix elements are generally non-vanishing. Even in that case, however, according to Theorem 13.2 we can construct a set of orthonormal vectors by taking appropriate linear combination of ξ_j and ξ_k . The resulting vectors naturally belong to the same irreducible representation. This can be done by solving the secular equation as described below. On top of it, Hermiticity of the Hamiltonian ensures that those vectors can be rendered orthogonal (see Theorem 14.5). If n electrons are present in a molecule, we are to deal with n SALC orbitals which we view as vectors accordingly. In terms of representation theory, these vectors span a representation space where the vectors undergo symmetry operations. Thus, we can construct

orthonormal basis vectors belonging to various irreducible representations throughout the representation space.

To address our problems, we take following procedures: (i) First we have to determine a symmetry species (i.e., point group) of a molecule. (ii) Next, we pick up atomic orbitals contained in a molecule and examine how those orbitals are transformed by symmetry operations. (iii) We examine how those atomic orbitals are transformed according to symmetry operations. Since the symmetry operations are represented by a (n, n) unitary matrix, we can readily decide a trace of the matrix. Generally, that matrix representation is reducible, and so we reduce the representation according to the procedures of (18.81)–(18.83). Thus, we are able to determine how many irreducible representations are contained in the original reducible representation. (iv) After having determined the irreducible representations, we construct SALCs and constitute a secular equation using them. (v) Solving the secular equation, we determine molecular orbital energies and decide functional forms of the corresponding MOs. (vi) We examine physicochemical properties such as optical transition within a molecule.

In the procedure (iii) of the above, if the same irreducible representations appear more than once, we have to solve a secular equation of an order of two or more. Even in that case, we can render a set of resulting eigenvectors orthogonal to one another during the process of solving a problem.

To construct the above-mentioned SALC, we make the most of projection operators that are defined in Sect. 14.1. In (18.147) putting $m = l$, we have

$$P_{l(l)}^{(\alpha)} = \frac{d_\alpha}{n} \sum_g D_{ll}^{(\alpha)}(g)^* g. \quad (19.71)$$

Or we can choose a projection operator $P^{(\alpha)}$ described as

$$P^{(\alpha)} = \frac{d_\alpha}{n} \sum_g \sum_{l=1}^{d_\alpha} D_{ll}^{(\alpha)}(g)^* g = \frac{d_\alpha}{n} \sum_g [\chi^{(\alpha)}(g)]^* g. \quad (18.174)$$

In the one-dimensional representation, $P_{l(l)}^{(\alpha)}$ and $P^{(\alpha)}$ are identical. As expressed in (18.155) and (18.175), these projection operators act on an arbitrary function and extract specific component(s) pertinent to a specific irreducible representation of the point group which the molecule belongs to.

At first glance the definition of $P_{l(l)}^{(\alpha)}$ and $P^{(\alpha)}$ looks daunting, but the use of character tables relieves a calculation task. In particular, all the irreducible representations are one-dimensional (i.e., just a number!) with Abelian groups as mentioned in Sect. 18.6. For this, notice that individual group elements form a class by themselves. Therefore, utilization of character tables becomes easier for Abelian groups. Even though we encounter a case where a dimension of representation is more than one (i.e., the case of non-commutative groups), $D_{ll}^{(\alpha)}$ can be determined without much difficulty.

19.4 MO Calculations Based on π -Electron Approximation

On the basis of the general argument developed in Sects. 19.2 and 19.3, we preform molecular orbital calculations of individual molecules. First, we apply group theory to the molecular orbital calculations about aromatic hydrocarbons such as ethylene, cyclopropenyl radical and cation as well as benzene and allyl radical. In these cases, in addition to adoption of the molecular orbital theory, we adopt the so-called π -electron approximation. With the first three molecules, we will not have to “solve” a secular equation. However, for allyl radical we deal with two SALC orbitals belonging to the same irreducible representations and, hence, the final molecular orbitals must be obtained by solving the secular equation.

19.4.1 Ethylene

We start with one of the simplest examples, ethylene. Ethylene is a planar molecule and belongs to D_{2h} symmetry (see Sect. 17.2). In the molecule two π -electrons extend vertically to the molecular plane toward upper and lower directions (Fig. 19.5). The molecular plane forms a node to atomic $2p_z$ orbitals; that is those atomic orbitals change a sign relative to the molecular plane (i.e., the xy -plane).

In Fig. 19.5, two p_z atomic orbitals of carbon are denoted by ϕ_1 and ϕ_2 . We should be able to construct basis vectors using ϕ_1 and ϕ_2 . Corresponding to the two atomic orbitals, we are dealing with a two-dimensional vector space. Let us consider how ϕ_1 and ϕ_2 are transformed by a symmetry operation. First we examine an operation $C_2(z)$. This operation exchanges ϕ_1 and ϕ_2 . That is, we have

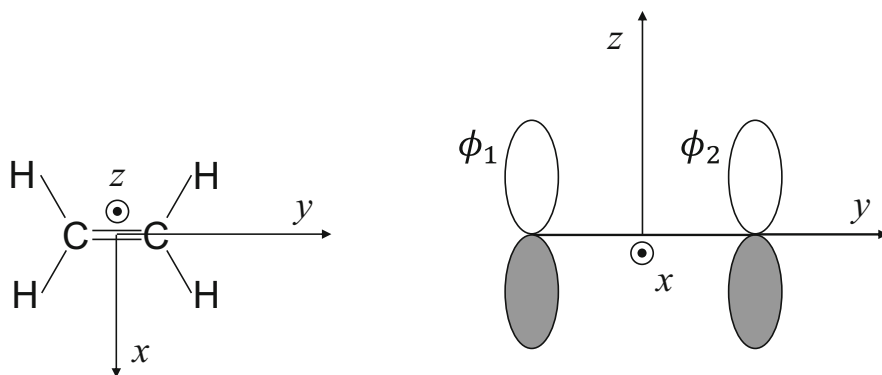


Fig. 19.5 Ethylene molecule placed on the xyz -coordinate system. Two p_z atomic orbitals of carbon are denoted by ϕ_1 and ϕ_2 . The atomic orbitals change a sign relative to the molecular plane (i.e., the xy -plane)

$$C_2(z)(\phi_1) = \phi_2 \quad (19.72)$$

and

$$C_2(z)(\phi_2) = \phi_1. \quad (19.73)$$

Equations (19.72) and (19.73) can be combined into a following equation:

$$(\phi_1 \ \phi_2)C_2(z) = (\phi_2 \ \phi_1). \quad (19.74)$$

Thus, using a matrix representation, we have

$$C_2(z) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (19.75)$$

In Sect. 17.2, we had

$$R_{z\theta} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (19.76)$$

or

$$C_2(z) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (19.77)$$

Whereas (19.76) and (19.77) show the transformation of a position vector in $\mathbb{R}^3$, (19.75) represents the transformation of functions in a two-dimensional vector space. A vector space composed of functions may be finite-dimensional or infinite-dimensional. We have already encountered the latter case in Part I where we dealt with the quantum mechanics of a harmonic oscillator. Such a function space is often referred to as a Hilbert space. An essence of (19.75) is characterized by that a trace (or character) of the matrix is zero.

Let us consider another transformation $C_2(y)$. In this case the situation is different from the above case in that ϕ_1 is converted to $-\phi_1$ by $C_2(y)$ and that ϕ_2 is converted to $-\phi_2$. Notice again that the molecular plane forms a node to atomic p -orbitals. Thus, $C_2(y)$ is represented by

$$C_2(y) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (19.78)$$

The trace of the matrix $C_2(y)$ is -2 . In this way, choosing ϕ_1 and ϕ_2 for the basis functions for the representation of D_{2h} we can determine the characters for individual symmetry transformations of atomic $2p_z$ orbitals in ethylene belonging to D_{2h} . We collect the results in Table 19.1.

Table 19.1 Characters for individual symmetry transformations of atomic $2p_z$ orbitals in ethylene

D_{2h}	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	$\sigma(xy)$	$\sigma(zx)$	$\sigma(yz)$
Γ	2	0	-2	0	0	-2	0	2

Table 19.2 Character table of D_{2h}

D_{2h}	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	$\sigma(xy)$	$\sigma(zx)$	$\sigma(yz)$	
A_g	1	1	1	1	1	1	1	1	x^2, y^2, z^2
B_{1g}	1	1	-1	-1	1	1	-1	-1	xy
B_{2g}	1	-1	1	-1	1	-1	1	-1	zx
B_{3g}	1	-1	-1	1	1	-1	-1	1	yz
A_u	1	1	1	1	-1	-1	-1	-1	
B_{1u}	1	1	-1	-1	-1	-1	1	1	z
B_{2u}	1	-1	1	-1	-1	1	-1	1	y
B_{3u}	1	-1	-1	1	-1	1	1	-1	x

Next, we examine what kind of irreducible representations is contained in our present representation of ethylene. To do this, we need a character table of D_{2h} (see Table 19.2). If a specific kind of irreducible representation is contained, then we want to examine how many times that specific representation takes place. Equation (18.83) is very useful for this purpose. In the present case, $n = 8$ in (18.83). Also taking into account (18.79) and (18.80), we get

$$\Gamma = B_{1u} \oplus B_{3g}, \quad (19.79)$$

where Γ is a reducible representation for a set consisting of two p_z atomic orbitals of ethylene. Equation (19.79) clearly shows that Γ is a direct sum of two irreducible representations B_{1u} and B_{3g} that belong to D_{2h} . In instead of (19.79), we usually simply express the direct product in quantum chemical notation as

$$\Gamma = B_{1u} + B_{3g}. \quad (19.80)$$

As a next step, we are going to find an appropriate basis function belonging to two irreducible representations of (19.80). For this purpose, we use projection operators expressed as

$$P^{(\alpha)} = \frac{d_\alpha}{n} \sum_g [\chi^{(\alpha)}(g)]^* g. \quad (18.174)$$

Taking ϕ_1 , for instance, we apply (18.174) to ϕ_1 . That is,

$$P^{(B_{1u})} \phi_1 = \frac{1}{8} \sum_g [\chi^{(B_{1u})}(g)]^* g \phi_1$$

Table 19.3 Character table of C_2

C_2	E	C_2	
A	1	1	z, x^2, y^2, z^2, xy
B	1	-1	x, y, yz, zx

$$\begin{aligned}
&= \frac{1}{8} [1 \cdot \phi_1 + 1 \cdot \phi_2 + (-1)(-\phi_1) + (-1)(-\phi_2) + (-1)(-\phi_2) \\
&\quad + (-1)(-\phi_1) + 1 \cdot \phi_2 + 1 \cdot \phi_1] = \frac{1}{2} (\phi_1 + \phi_2). \quad (19.81)
\end{aligned}$$

Also with the B_{3g} , we apply (18.174) to ϕ_1 and get

$$\begin{aligned}
P^{(B_{3g})} \phi_1 &= \frac{1}{8} \sum_g [\chi^{(B_{3g})}(g)]^* g \phi_1 \\
&= \frac{1}{8} [1 \cdot \phi_1 + (-1)\phi_2 + (-1)(-\phi_1) + 1(-\phi_2) + 1(-\phi_2) + (-1)(-\phi_1) \\
&\quad + (-1)\phi_2 + 1 \cdot \phi_1] = \frac{1}{2} (\phi_1 - \phi_2). \quad (19.82)
\end{aligned}$$

Thus, after going through routine but sure procedures, we have reached appropriate basis functions that belong to each irreducible representation.

As mentioned earlier (see Table 17.5), D_{2h} can be expressed as a direct-product group and C_2 group is contained in D_{2h} as a subgroup. We can use it for the present analysis. Suppose now that we have a group $\mathcal{G}$ and that $\mathcal{H}$ is a subgroup of $\mathcal{G}$. Let g be an arbitrary element of $\mathcal{G}$ and let $D(g)$ be a representation of g . Meanwhile, let h be an arbitrary element of $\mathcal{H}$. Then with $\forall h \in \mathcal{H}$, a collection of $D(h)$ is a representation of $\mathcal{H}$. We write this relation as

$$D \downarrow \mathcal{H}. \quad (19.83)$$

This representation is called a subduced representation of D to $\mathcal{H}$. Table 19.3 shows a character table of irreducible representations of C_2 . In the present case, we are thinking of $C_2(z)$ as C_2 ; see Table 17.5 and Fig. 19.5. Then we have

$$B_{1u} \downarrow C_2 = A \quad \text{and} \quad B_{3g} \downarrow C_2 = B. \quad (19.84)$$

The expression of (19.84) is called a compatibility relation. Note that in (19.84) C_2 is not a symmetry operation, but means a subgroup of D_{2h} . Thus, (19.81) is reduced to

$$P^{(A)} \phi_1 = \frac{1}{2} \sum_g [\chi^{(A)}(g)]^* g \phi_1 = \frac{1}{2} [1 \cdot \phi_1 + 1 \cdot \phi_2] = \frac{1}{2} (\phi_1 + \phi_2). \quad (19.85)$$

Also, we have

$$P^{(B)}\phi_1 = \frac{1}{2} \sum_g [\chi^{(B)}(g)]^* g\phi_1 = \frac{1}{2} [1 \cdot \phi_1 + (-1)\phi_2] = \frac{1}{2}(\phi_1 - \phi_2). \quad (19.86)$$

The relations of (19.85) and (19.86) are essentially the same as (19.81) and (19.82), respectively.

We can easily construct a character table (see Table 19.3). There should be two irreducible representations. Regarding the totally symmetric representation, we allocate 1 to each symmetry operation. For another representation we allocate 1 to an identity element E and -1 to an element C_2 so that the row and column of the character table are orthogonal to each other.

Readers might well ask why we bother to make circuitous approaches to reaching predictable results such as (19.81) and (19.82) or (19.85) and (19.86). This question seems natural when we are dealing with a case where the number of basis vectors (i.e., a dimension of the vector space) is small, typically 2 as in the present case. With increasing dimension of the vector space, however, to seek and determine appropriate SALCs become increasingly complicated and difficult. Under such circumstances, a projection operator is an indispensable tool to address the problems.

We have a two-dimensional secular equation to be solved such that

$$\begin{vmatrix} \tilde{H}_{11} - \lambda \tilde{S}_{11} & \tilde{H}_{12} - \lambda \tilde{S}_{12} \\ \tilde{H}_{21} - \lambda \tilde{S}_{21} & \tilde{H}_{22} - \lambda \tilde{S}_{22} \end{vmatrix} = 0. \quad (19.87)$$

In the above argument, $\frac{1}{2}(\phi_1 + \phi_2)$ belongs to B_{1u} and $\frac{1}{2}(\phi_1 - \phi_2)$ belongs to B_{3g} . Since they belong to different irreducible representations, we have

$$\tilde{H}_{12} = \tilde{S}_{12} = 0. \quad (19.88)$$

Thus, the secular equation (19.87) is reduced to

$$\begin{vmatrix} \tilde{H}_{11} - \lambda \tilde{S}_{11} & 0 \\ 0 & \tilde{H}_{22} - \lambda \tilde{S}_{22} \end{vmatrix} = 0. \quad (19.89)$$

As expected, (19.89) has automatically been solved to give a solution

$$\lambda_1 = \tilde{H}_{11}/\tilde{S}_{11} \quad \text{and} \quad \lambda_2 = \tilde{H}_{22}/\tilde{S}_{22}. \quad (19.90)$$

The next step is to determine the energy eigenvalue of the molecule. Note here that a role of SALCs is to determine a suitable irreducible representation that corresponds to a “direction” of a vector. As the coefficient keeps the direction of a vector unaltered, it would be of secondary importance. The final form of normalized MOs can be decided last. That procedure includes the normalization of a vector. Thus, we tentatively choose following functions for SALCs, i.e.,

$$\xi_1 = \phi_1 + \phi_2 \quad \text{and} \quad \xi_2 = \phi_1 - \phi_2.$$

Then, we have

$$\begin{aligned} \tilde{H}_{11} &= \int \xi_1^* H \xi_1 d\tau = \int (\phi_1 + \phi_2)^* H (\phi_1 + \phi_2) d\tau \\ &= \int \phi_1^* H \phi_1 d\tau + \int \phi_1^* H \phi_2 d\tau + \int \phi_2^* H \phi_1 d\tau + \int \phi_2^* H \phi_2 d\tau \\ &= H_{11} + H_{12} + H_{21} + H_{22} = H_{11} + 2H_{12} + H_{22}. \end{aligned} \quad (19.91)$$

Similarly, we have

$$\tilde{H}_{22} = \int \xi_2^* H \xi_2 d\tau = H_{11} - 2H_{12} + H_{22}. \quad (19.92)$$

The last equality comes from the fact that we have chosen real functions for ϕ_1 and ϕ_2 as studied in Part I. Moreover, we have

$$\begin{aligned} H_{11} &\equiv \int \phi_1^* H \phi_1 d\tau = \int \phi_1 H \phi_1 d\tau = \int \phi_2 H \phi_2 d\tau = H_{22}, \\ H_{12} &\equiv \int \phi_1^* H \phi_2 d\tau = \int \phi_1 H \phi_2 d\tau = \int \phi_2^* H \phi_1 d\tau = \int \phi_2 H \phi_1 d\tau \\ &= H_{21}. \end{aligned} \quad (19.93)$$

The first equation comes from the fact that both H_{11} and H_{22} are calculated using the same $2p_z$ atomic orbital of carbon. The second equation results from the fact that H is Hermitian. Notice that both ϕ_1 and ϕ_2 are real functions. Following the convention, we denote

$$\alpha \equiv H_{11} = H_{22} \quad \text{and} \quad \beta \equiv H_{12}, \quad (19.94)$$

where α is called Coulomb integral and β is said to be resonance integral. Then, we have

$$\tilde{H}_{11} = 2(\alpha + \beta). \quad (19.95)$$

In a similar manner, we get

$$\tilde{H}_{22} = 2(\alpha - \beta). \quad (19.96)$$

Meanwhile, we have

$$\begin{aligned} \tilde{S}_{11} &= \langle \xi_1 | \xi_1 \rangle = \int (\phi_1 + \phi_2)^* (\phi_1 + \phi_2) d\tau = \int (\phi_1 + \phi_2)^2 d\tau \\ &= \int \phi_1^2 d\tau + \int \phi_2^2 d\tau + 2 \int \phi_1 \phi_2 d\tau = 2 + 2 \int \phi_1 \phi_2 d\tau, \end{aligned} \quad (19.97)$$

where we used the fact that ϕ_1 and ϕ_2 have been normalized. Also following the convention, we denote

$$S \equiv \int \phi_1 \phi_2 d\tau = S_{12}, \quad (19.98)$$

where S is called overlap integral. Thus, we have

$$\tilde{S}_{11} = 2(1 + S). \quad (19.99)$$

Similarly, we get

$$\tilde{S}_{22} = 2(1 - S). \quad (19.100)$$

Substituting (19.95) and (19.96) along with (19.99) and (19.100) for (19.90), we get as the energy eigenvalue

$$\lambda_1 = \frac{\alpha + \beta}{1 + S} \text{ and } \lambda_2 = \frac{\alpha - \beta}{1 - S}. \quad (19.101)$$

From (19.97), we get

$$\|\xi_1\| = \sqrt{\langle \xi_1 | \xi_1 \rangle} = \sqrt{2(1 + S)}. \quad (19.102)$$

Thus, for one of MOs corresponding to an energy eigenvalue λ_1 , we get

$$\Psi_1 = \frac{|\xi_1\rangle}{\|\xi_1\|} = \frac{\phi_1 + \phi_2}{\sqrt{2(1 + S)}}. \quad (19.103)$$

Also, we have

$$\|\xi_2\| = \sqrt{\langle \xi_2 | \xi_2 \rangle} = \sqrt{2(1-S)}. \quad (19.104)$$

For another MO corresponding to an energy eigenvalue λ_2 , we get

$$\psi_2 = \frac{|\xi_2\rangle}{\|\xi_2\|} = \frac{\phi_1 - \phi_2}{\sqrt{2(1-S)}}. \quad (19.105)$$

Note that both normalized MOs and energy eigenvalues depend upon whether we ignore an overlap integral, as being the case with the simplest Hückel approximation. Nonetheless, MO functional forms (in this case either $\phi_1 + \phi_2$ or $\phi_1 - \phi_2$) remain the same regardless of the approximation levels about the overlap integral. Regarding quantitative evaluation of α , β , and S , we will briefly mention it later.

Once we have decided symmetry of MO (or an irreducible representation which the orbital belongs to) and its energy eigenvalue, we will be in a position to examine various physicochemical properties of the molecule. One of them is an optical transition within a molecule, particularly electric dipole transition. In most cases, the most important transition is that occurring among the highest-occupied molecular orbital (HOMO) and lowest-unoccupied molecular orbital (LUMO). In the case of ethylene, those levels are depicted in Fig. 19.6. In a ground state (i.e., the most stable state), two electrons are positioned in a B_{1u} state (HOMO). An excited state is assigned to B_{3g} (LUMO). The ground state that consists only of fully occupied MOs belongs to a totally symmetric representation. In the case of ethylene, the ground state belongs to A_g accordingly.

If a photon is absorbed by a molecule, that molecule is excited by an energy of the photon. In ethylene, this process takes place by exciting an electron from B_{1u} to B_{3g} . The resulting electronic state ends up as an electron remaining in B_{1u} and another excited to B_{3g} (a final state). Thus, the representation of the final state electronic configuration (denoted by Γ_f) is described as

$$\Gamma_f = B_{1u} \times B_{3g}. \quad (19.106)$$

That is, the final excited state is expressed as a direct product of the states associated with the optical transition. To determine the symmetry of the final state, we use (18.198) and (18.200). If Γ in (19.106) is reducible, using (18.200) we can determine the number of times that individual representations take place. Calculating $\chi^{(B_{1u})}(g)\chi^{(B_{3g})}(g)$ for each group element, we can readily get the result. Using a character table, we have

$$\Gamma_f = B_{2u}. \quad (19.107)$$

Thus, we find that the transition is $A_g \rightarrow B_{2u}$, where A_g is called an initial state electronic configuration and B_{2u} is called a final state electronic configuration. This transition is characterized by an electric dipole transition moment operator $\mathbf{P}$. Here

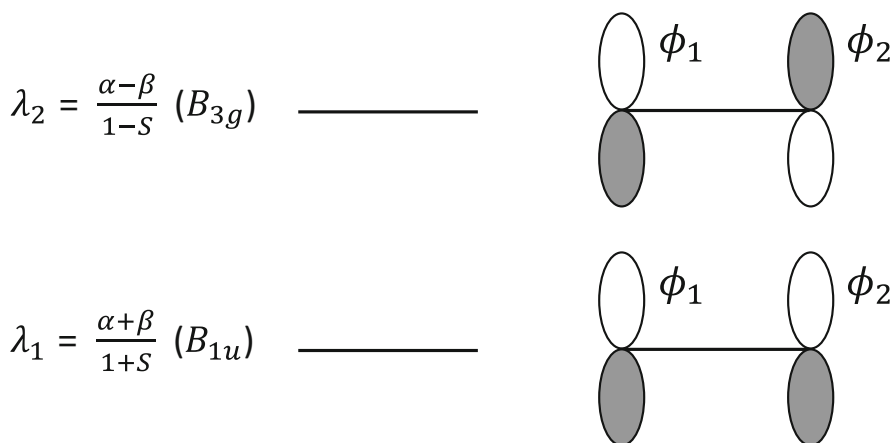


Fig. 19.6 HOMO and LUMO energy levels and their assignments of ethylene

we have an important physical quantity of transition matrix element. This quantity T_{fi} is approximated by

$$T_{fi} \equiv \langle \Theta_f | \epsilon_e \cdot \mathbf{P} | \Theta_i \rangle, \quad (19.108)$$

where ϵ_e is a unit polarization vector of the electric field; Θ_i and Θ_f are the initial state and final state electronic configuration, respectively. The description (19.108) is in parallel with (4.5). Notice that in (4.5) we dealt with a single electron system such as a particle confined in a square-well potential, a sole one-dimensional harmonic oscillator, and an electron in a hydrogen atom.

In the present case, however, we are dealing with a two-electron system, ethylene. Consequently, we cannot describe Θ_f by a simple wave function, but must use a more elaborated function. Nonetheless, when we discuss the optical transition of a molecule, it is often the case that when we study optical absorption or optical emission we first wish to know whether such a phenomenon truly takes place. In such a case, qualitative prediction for this is of great importance. This can be done by judging whether the integral (19.108) vanishes. If the integral does not vanish, the relevant transition is said to be *allowed*. If, however, the integral vanishes, the transition is called *forbidden*. In this context, a systematic approach based on group theory is a powerful tool for this.

Let us consider optical absorption of ethylene. With the transition matrix element for this, we have

$$T_{fi} = \langle \Theta_f^{(B_{2u})} | \epsilon_e \cdot \mathbf{P} | \Theta_i^{(A_g)} \rangle. \quad (19.109)$$

Again, note that a closed-shell electronic configuration belong to a totally symmetric representation [1]. Suppose that the position vector $\mathbf{x}$ belongs to an irreducible representation η . A necessary condition for (19.109) not to vanish is that $D^{(\eta)} \times D^{(A_g)}$ contains the irreducible representation $D^{(B_{2u})}$. In the present case, all the representations are one-dimensional, and so we can use $\chi^{(\omega)}$ instead of $D^{(\omega)}$, where ω shows an arbitrary irreducible representation of D_{2h} . This procedure is straightforward as seen in Sect. 19.2.

However, if the character is real (it is the case with many symmetry groups and with the point group D_{2h} as well), the situation will be easier. Suppose in general that we are examining whether a following matrix element vanishes:

$$M_{fi} = \langle \Phi_f^{(\beta)} | O^{(\gamma)} | \Phi_i^{(\alpha)} \rangle, \quad (19.110)$$

where α , β , and γ stand for irreducible representations and O is an appropriate operator. In this case, (18.200) can be rewritten as

$$\begin{aligned} q_\alpha &= \frac{1}{n} \sum_g \chi^{(\alpha)}(g)^* \chi^{(\gamma \times \beta)}(g) = \frac{1}{n} \sum_g \chi^{(\alpha)}(g) \chi^{(\gamma \times \beta)}(g) \\ &= \frac{1}{n} \sum_g \chi^{(\alpha)}(g) \chi^{(\gamma)}(g) \chi^{(\beta)}(g) = \frac{1}{n} \sum_g \chi^{(\gamma)}(g) \chi^{(\alpha)}(g) \chi^{(\beta)}(g) \\ &= \frac{1}{n} \sum_g \chi^{(\gamma)}(g) \chi^{(\alpha \times \beta)}(g) = \frac{1}{n} \sum_g \chi^{(\gamma)}(g)^* \chi^{(\alpha \times \beta)}(g) = q_\gamma. \end{aligned} \quad (19.111)$$

Consequently, the number of times that $D^{(\gamma)}$ appears in $D^{(\alpha \times \beta)}$ is identical to the number of times that $D^{(\alpha)}$ appears in $D^{(\gamma \times \beta)}$. Thus, it suffices to examine whether $D^{(\gamma \times \beta)}$ contains $D^{(\alpha)}$. In other words, we only have to examine whether $q_\alpha \neq 0$ in (19.111).

Thus, applying (19.111)–(19.109), we examine whether $D^{(B_{2u} \times A_g)}$ contains the irreducible representation $D^{(\eta)}$ that is related to $\mathbf{x}$. We easily get

$$B_{2u} = B_{2u} \times A_g. \quad (19.112)$$

Therefore, if $\epsilon_e \cdot \mathbf{P}$ (or $\mathbf{x}$) belongs to B_{2u} , the transition is allowed. Consulting the character table, we find that y belongs to B_{2u} . In this case, in fact (19.111) reads as

$$q_{B_{2u}} = \frac{1}{8} [1 \times 1 + (-1) \times (-1) + 1 \times 1 + (-1) \times (-1)]$$

$$+(-1) \times (-1) + 1 \times 1 + (-1) \times (-1) + 1 \times 1] = 1.$$

Equivalently, we simply write

$$B_{2u} \times B_{2u} = A_g.$$

This means that if a light polarized along the y -axis is incident, i.e., ϵ_e is parallel to the y -axis, the transition is allowed. In that situation, ethylene is said to be polarized along the y -axis or polarized in the direction of the y -axis. As a molecular axis is parallel to the y -axis, ethylene is polarized in the direction of the molecular axis. This is often the case with aromatic molecules having a well-defined molecular long axis such as ethylene.

We would examine whether ethylene is polarized along, e.g., the x -axis. From a character table of D_{2h} (see Table 19.2), x belongs to B_{3u} . In that case, using (19.111) we have

$$q_{B_{3u}} = \frac{1}{8} [1 \times 1 + (-1) \times (-1) + 1 \times (-1) + (-1) \times 1 \\ + (-1) \times (-1) + 1 \times 1 + (-1) \times 1 + 1 \times (-1)] = 0.$$

This implies that B_{3u} is not contained in $B_{1u} \times B_{3g} (=B_{2u})$.

The above results on the optical transitions are quite obvious. Once we get used to using a character table, quick estimation will be done.

19.4.2 Cyclopropenyl Radical [1]

Let us think of another example, cyclopropenyl radical that has three resonant structures (Fig. 19.7). It is a planar molecule and three carbon atoms form an equilateral triangle. Hence, the molecule belongs to D_{3h} symmetry (see Sect. 17.2). In the molecule three π -electrons extend vertically to the molecular plane toward upper and lower directions. Suppose that cyclopropenyl radical is placed on the xy -plane. Three p_z atomic orbitals of carbons located at vertices of an equilateral triangle are denoted by ϕ_1 , ϕ_2 and ϕ_3 in Fig. 19.8. The orbitals are numbered *clockwise* so that the calculations can be consistent with the conventional notation of a character table (vide infra). We assume that these π -orbitals take positive and negative signs on the upper and lower sides of the plane of paper, respectively, with a nodal plane lying on the xy -plane. The situation is similar to that of ethylene and the problem can be treated in parallel to the case of ethylene.

As in the case of ethylene, we can choose ϕ_1 , ϕ_2 and ϕ_3 as real functions. We construct basis vectors using these vectors. What we want to do to address the problem is as follows: (i) We examine how ϕ_1 , ϕ_2 and ϕ_3 are transformed by the

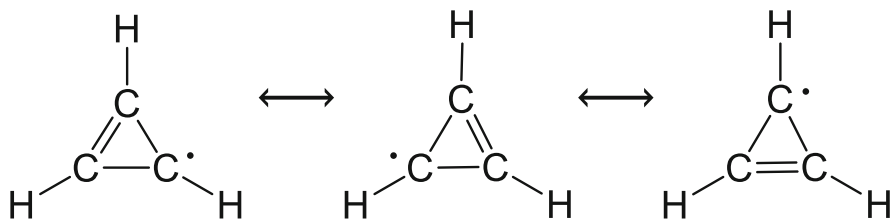
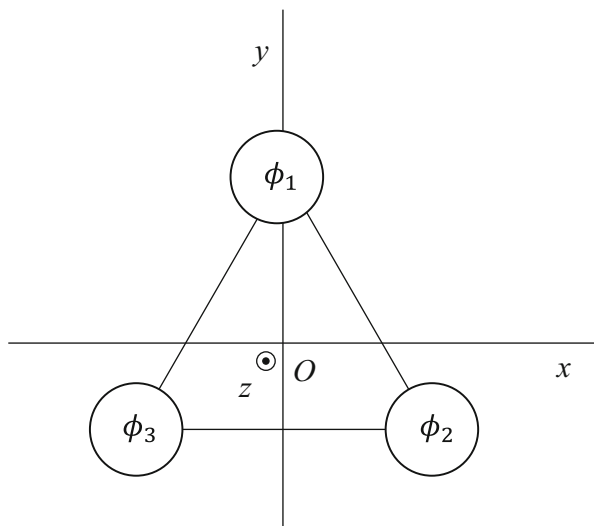


Fig. 19.7 Three resonant structures of cyclopropenyl radical

Fig. 19.8 Three p_z atomic orbitals of carbons for cyclopropenyl radical that is placed on the xy -plane. The carbon atoms are located at vertices of an equilateral triangle. The atomic orbitals are denoted by ϕ_1 , ϕ_2 , and ϕ_3



symmetry operations of D_{3h} . According to the analysis, we can determine what irreducible representations SALCs should be assigned to. (ii) On the basis of knowledge obtained in (i), we construct proper MOs.

In Table 19.4 we list a character table of D_{3h} along with symmetry species. First we examine traces (characters) of representation matrices. Similarly in the case of ethylene, a subgroup C_3 of D_{3h} plays an essential role (vide infra). This subgroup contains three group elements such that

$$C_3 = \{E, C_3, C_3^2\}.$$

In the above, we use the same notation for the group name and group element, and so we should be careful not to confuse them.

They are transformed as follows:

$$C_3(z)(\phi_1) = \phi_3, C_3(z)(\phi_2) = \phi_1, \text{ and } C_3(z)(\phi_3) = \phi_2; \quad (19.113)$$

Table 19.4 Character table of D_{3h}

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$	
A'_1	1	1	1	1	1	1	$x^2 + y^2, z^2$
A'_2	1	1	-1	1	1	-1	
E'	2	-1	0	2	-1	0	$(x, y); (x^2 - y^2, xy)$
A''_1	1	1	1	-1	-1	-1	
A''_2	1	1	-1	-1	-1	1	z
E''	2	-1	0	-2	1	0	(yz, zx)

$$C_3^2(z)(\phi_1) = \phi_2, C_3^2(z)(\phi_2) = \phi_3, \text{ and } C_3^2(z)(\phi_3) = \phi_1. \quad (19.114)$$

Equation (19.113) can be combined into a following form:

$$(\phi_1 \ \phi_2 \ \phi_3)C_3(z) = (\phi_3 \ \phi_1 \ \phi_2). \quad (19.115)$$

Using a matrix representation, we have

$$C_3(z) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}. \quad (19.116)$$

In turn, (19.114) is expressed as

$$C_3^2(z) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \quad (19.117)$$

Both traces of (19.116) and (19.117) are zero.

Similarly let us check the representation matrices of other symmetry species. Of these, e.g., for C_2 related to the y -axis (see Fig. 19.8) we have

$$(\phi_1 \ \phi_2 \ \phi_3)C_2 = (-\phi_1 - \phi_3 - \phi_2). \quad (19.118)$$

Therefore,

$$C_2 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}. \quad (19.119)$$

We have a trace -1 accordingly. Regarding σ_h , we have

$$(\phi_1 \ \phi_2 \ \phi_3)\sigma_h = (-\phi_1 - \phi_2 - \phi_3) \text{ and } \sigma_h = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (19.120)$$

Table 19.5 Characters for individual symmetry transformations of $2p_z$ orbitals in cyclopropenyl radical

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ	3	0	-1	-3	0	1

Table 19.6 Character table of C_3

C_3	E	C_3	C_3^2	$\varepsilon = \exp(i2\pi/3)$
A	1	1	1	$z; x^2 + y^2, z^2$
E	$\begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$	$\begin{Bmatrix} \varepsilon \\ \varepsilon^* \end{Bmatrix}$	$\begin{Bmatrix} \varepsilon^* \\ \varepsilon \end{Bmatrix}$	$(x, y); (x^2 - y^2, xy), (yz, zx)$

In this way we can determine the trace for individual symmetry transformations of basis functions ϕ_1 , ϕ_2 , and ϕ_3 . We collect the results of characters of a reducible representation Γ in Table 19.5. It can be reduced to a summation of irreducible representations according to the procedures given in (18.81)–(18.83) and using a character table of D_{3h} (Table 19.4). As a result, we get

$$\Gamma = A_2'' + E'' \quad (19.121)$$

As in the case of ethylene, we make the best use of the information of a subgroup C_3 of D_{3h} . Let us consider a subduced representation of D of D_{3h} to C_3 . For this, in Table 19.6 we show a character table of irreducible representations of C_3 . We can readily construct this character table. There should be three irreducible representations. Regarding the totally symmetric representation, we allocate 1 to each symmetry operation. Hence, for other representations we allocate 1 to an identity element E and two other triple roots of 1, i.e., ε and ε^* [where $\varepsilon = \exp(i2\pi/3)$] to an element C_3 and C_3^2 as shown so that the row and column vectors of the character table are orthogonal to each other.

Returning to the construction of the subduced representation, we have

$$A_2'' \downarrow C_3 = A \quad \text{and} \quad E'' \downarrow C_3 = 2E \quad (19.122)$$

Then, corresponding to (18.147) we have

$$\begin{aligned} P^{(A)}\phi_1 &= \frac{1}{3} \sum_g [\chi^{(A)}(g)]^* g\phi_1 = \frac{1}{3} [1 \cdot \phi_1 + 1 \cdot \phi_2 + 1 \cdot \phi_3] \\ &= \frac{1}{3} (\phi_1 + \phi_2 + \phi_3). \end{aligned} \quad (19.123)$$

Also, we have

$$\begin{aligned}
 P^{[E^{(1)}]} \phi_1 &= \frac{1}{3} \sum_g \left[\chi^{[E^{(1)}]}(g) \right]^* g \phi_1 = \frac{1}{3} [1 \cdot \phi_1 + \varepsilon^* \phi_3 + (\varepsilon^*)^* \phi_2] \\
 &= \frac{1}{3} (\phi_1 + \varepsilon \phi_2 + \varepsilon^* \phi_3). \tag{19.124}
 \end{aligned}$$

Also, we get

$$\begin{aligned}
 P^{[E^{(2)}]} \phi_1 &= \frac{1}{3} \sum_g \left[\chi^{[E^{(2)}]}(g) \right]^* g \phi_1 = \frac{1}{3} [1 \cdot \phi_1 + (\varepsilon^*)^* \phi_3 + \varepsilon^* \phi_2] \\
 &= \frac{1}{3} (\phi_1 + \varepsilon^* \phi_2 + \varepsilon \phi_3). \tag{19.125}
 \end{aligned}$$

Here is the best place to mention an eigenvalue of a symmetry operator. Let us designate SALCs as follows:

$$\xi_1 = \phi_1 + \phi_2 + \phi_3, \xi_2 = \phi_1 + \varepsilon \phi_2 + \varepsilon^* \phi_3, \text{ and } \xi_3 = \phi_1 + \varepsilon^* \phi_2 + \varepsilon \phi_3. \tag{19.126}$$

Let us choose C_3 for a symmetry operator. Then we have

$$\begin{aligned}
 C_3(\xi_1) &= C_3(\phi_1 + \phi_2 + \phi_3) = C_3\phi_1 + C_3\phi_2 + C_3\phi_3 = \phi_3 + \phi_1 \\
 &\quad + \phi_2 = \xi_1, \tag{19.127}
 \end{aligned}$$

where for the second equality we used the fact that C_3 is a linear operator. That is, regarding a SALC ξ_1 , an eigenvalue of C_3 is 1.

Similarly, we have

$$\begin{aligned}
 C_3(\xi_2) &= C_3(\phi_1 + \varepsilon \phi_2 + \varepsilon^* \phi_3) = C_3\phi_1 + \varepsilon C_3\phi_2 + \varepsilon^* C_3\phi_3 \\
 &= \phi_3 + \varepsilon \phi_1 + \varepsilon^* \phi_2 = \varepsilon(\phi_1 + \varepsilon \phi_2 + \varepsilon^* \phi_3) = \varepsilon \xi_2. \tag{19.128}
 \end{aligned}$$

Furthermore, we get

$$C_3(\xi_3) = \varepsilon^* \xi_3. \tag{19.129}$$

Thus, we find that regarding SALCs ξ_2 and ξ_3 , eigenvalues of C_3 are ε and ε^* , respectively. These pieces of information imply that if we appropriately choose proper functions for basis vectors, a character of a symmetry operation for a one-dimensional representation is identical to an eigenvalue of the said symmetry operation (see Table 19.6).

Regarding the last parts of the calculations, we follow the procedures described in the case of ethylene. Using the above functions ξ_1 , ξ_2 , and ξ_3 , we construct the secular equation such that

$$\begin{vmatrix} \tilde{H}_{11} - \lambda \tilde{S}_{11} & & \\ & \tilde{H}_{22} - \lambda \tilde{S}_{22} & \\ & & \tilde{H}_{33} - \lambda \tilde{S}_{33} \end{vmatrix} = 0. \quad (19.130)$$

Since we have obtained three SALCs that are assigned to individual irreducible representations A , $E^{(1)}$, and $E^{(2)}$, these SALCs span the representation space V^3 . This makes off-diagonal elements of the secular equation vanish and it is simplified as in (19.130). Here, we have

$$\begin{aligned} \tilde{H}_{11} &= \int \xi_1^* H \xi_1 d\tau = \int (\phi_1 + \phi_2 + \phi_3)^* H (\phi_1 + \phi_2 + \phi_3) d\tau \\ &= 3(\alpha + 2\beta), \end{aligned} \quad (19.131)$$

where we used the same α and β as defined in (19.94). Strictly speaking, α and β appearing in (19.131) should be slightly different from those of (19.94), because a Hamiltonian is different. This approximation, however, would be enough for the present studies. In a similar manner, we get.

$$\tilde{H}_{22} = \int \xi_2^* H \xi_2 d\tau = 3(\alpha - \beta) \text{ and } \tilde{H}_{33} = \int \xi_3^* H \xi_3 d\tau = 3(\alpha - \beta). \quad (19.132)$$

Meanwhile, we have

$$\begin{aligned} \tilde{S}_{11} &= \langle \xi_1 | \xi_1 \rangle = \int (\phi_1 + \phi_2 + \phi_3)^* (\phi_1 + \phi_2 + \phi_3) d\tau = \int (\phi_1 + \phi_2 + \phi_3)^2 d\tau \\ &= 3(1 + 2S). \end{aligned} \quad (19.133)$$

Similarly, we get

$$\tilde{S}_{22} = \tilde{S}_{33} = 3(1 - S). \quad (19.134)$$

Readers are urged to verify (19.133) and (19.134).

Substituting (19.131) through (19.134) for (19.130), we get as the energy eigenvalue

$$\lambda_1 = \frac{\alpha + 2\beta}{1 + 2S}, \lambda_2 = \frac{\alpha - \beta}{1 - S}, \text{ and } \lambda_3 = \frac{\alpha - \beta}{1 - S}. \quad (19.135)$$

Notice that two MOs belonging to E'' have the same energy. These MOs are said to be energetically degenerate. This situation is characteristic of a two-dimensional representation. Actually, even though the group C_3 has only one-dimensional representations (because it is an Abelian group), the two complex conjugate representations labelled E behave as if they were a two-dimensional representation [2]. We will again encounter the same situation in a next example, benzene.

From (19.126), we get

$$\|\xi_1\| = \sqrt{\xi_1|\xi_1} = \sqrt{3(1 + 2S)}. \quad (19.136)$$

Thus, as one of MOs corresponding to an energy eigenvalue λ_1 ; i.e., Ψ_1 , we get

$$\Psi_1 = \frac{|\xi_1\rangle}{\|\xi_1\|} = \frac{\phi_1 + \phi_2 + \phi_3}{\sqrt{3(1 + 2S)}}. \quad (19.137)$$

Also, we have

$$\|\xi_2\| = \sqrt{\langle\xi_2|\xi_2\rangle} = \sqrt{3(1 - S)} \text{ and } \|\xi_3\| = \sqrt{\langle\xi_3|\xi_3\rangle} = \sqrt{3(1 - S)}. \quad (19.138)$$

Thus, for another MO corresponding to an energy eigenvalue λ_2 , we get

$$\Psi_2 = \frac{|\xi_2\rangle}{\|\xi_2\|} = \frac{\phi_1 + \varepsilon\phi_2 + \varepsilon^*\phi_3}{\sqrt{3(1 - S)}}. \quad (19.139)$$

Also, with a MO corresponding to λ_3 ($=\lambda_2$), we have

$$\Psi_3 = \frac{|\xi_3\rangle}{\|\xi_3\|} = \frac{\phi_1 + \varepsilon^*\phi_2 + \varepsilon\phi_3}{\sqrt{3(1 - S)}}. \quad (19.140)$$

Equations (19.139) and (19.140) include complex numbers, and so it is inconvenient to computer analysis. In that case, we can convert it to real numbers. In Part III, we examined properties of unitary transformations. Since the unitary transformation keeps a norm of a vector unchanged, this is suited to our present purpose. This can be done using a following unitary matrix U :

$$U = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{pmatrix}. \quad (19.141)$$

Then we have

$$(\Psi_2 \ \Psi_3)U = \left(\frac{1}{\sqrt{2}}(\Psi_2 + \Psi_3) \ \frac{i}{\sqrt{2}}(-\Psi_2 + \Psi_3) \right). \quad (19.142)$$

Thus, defining $\tilde{\Psi}_2$ and $\tilde{\Psi}_3$ as

$$\tilde{\Psi}_2 = \frac{1}{\sqrt{2}}(\Psi_2 + \Psi_3) \quad \text{and} \quad \tilde{\Psi}_3 = \frac{i}{\sqrt{2}}(-\Psi_2 + \Psi_3), \quad (19.143)$$

we get

$$\begin{aligned} \tilde{\Psi}_2 &= \frac{1}{\sqrt{6(1-S)}} [2\phi_1 + (\varepsilon + \varepsilon^*)\phi_2 + (\varepsilon^* + \varepsilon)\phi_3] \\ &= \frac{1}{\sqrt{6(1-S)}} (2\phi_1 - \phi_2 - \phi_3). \end{aligned} \quad (19.144)$$

Also, we have

$$\begin{aligned} \tilde{\Psi}_3 &= \frac{i}{\sqrt{6(1-S)}} [(\varepsilon^* - \varepsilon)\phi_2 + (\varepsilon - \varepsilon^*)\phi_3] = \frac{i}{\sqrt{6(1-S)}} [-i\sqrt{3}\phi_2 + i\sqrt{3}\phi_3] \\ &= \frac{1}{\sqrt{2(1-S)}} (\phi_2 - \phi_3). \end{aligned} \quad (19.145)$$

Thus, we have successfully converted complex functions to real functions. In the above unitary transformation, notice that a norm of the vectors remains unchanged before and after the unitary transformation.

As cyclopropenyl radical has three π electrons, two occupy the lowest energy level of A'' . Another electron occupies a level E'' . Since this level possesses an energy higher than α , the electron occupying this level is anticipated to be unstable. Under such a circumstance, a molecule tends to lose the said electron so as to be a cation. Following the argument given in the previous case of ethylene, it is easy to make sure that the allowed transition of the cyclopropenyl radical takes place when

the light is polarized parallel to the molecular plane (i.e., the xy -plane in Fig. 19.8). The proof is left for readers as an exercise. This polarizing feature is typical of planar molecules with high molecular symmetry.

19.4.3 Benzene

Benzene has structural formula which is shown in Fig. 19.9. It is a planar molecule and six carbon atoms form a regular hexagon. Hence, the molecule belongs to D_{6h} symmetry. In the molecule six π -electrons extend vertically to the molecular plane toward upper and lower directions as in the case of ethylene and cyclopropenyl radical. This is a standard illustration of quantum chemistry and dealt with in many textbooks. As in the case of ethylene and cyclopropenyl radical, the problem can be treated similarly.

As before, six equivalent p_z atomic orbitals of carbon are denoted by ϕ_1 to ϕ_6 in Fig. 19.10. These vectors or their linear combinations span a six-dimensional representation space. We construct basis vectors using these vectors. Following the previous procedures, we construct proper SALC orbitals along with MOs. Similarly as before, a subgroup C_6 of D_{6h} plays an essential role. This subgroup contains six group elements such that

$$C_6 = \{E, C_6, C_3, C_2, C_3^2, C_6^5\}. \quad (19.146)$$

Taking $C_6(z)$ as an example, we have

$$(\phi_1 \ \phi_2 \ \phi_3 \ \phi_4 \ \phi_5 \ \phi_6)C_6(z) = (\phi_6 \ \phi_1 \ \phi_2 \ \phi_3 \ \phi_4 \ \phi_5). \quad (19.147)$$

Using a matrix representation, we have

Fig. 19.9 Structural formula of benzene. It belongs to D_{6h} symmetry

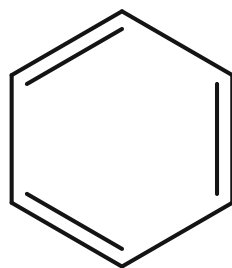


Fig. 19.10 Six equivalent p_z atomic orbitals of carbon of benzene

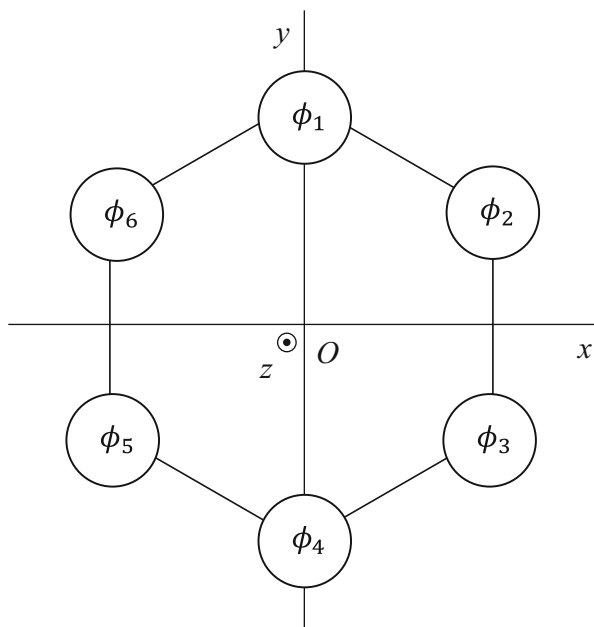


Table 19.7 Characters for individual symmetry transformations of $2p_z$ orbitals in benzene

D_{6h}	E	$2C_6$	$2C_3$	C_2	$3C'_2$	$3C''_2$	i	$2S_3$	$2S_6$	σ_h	$3\sigma_d$	$3\sigma_v$
Γ	6	0	0	0	-2	0	0	0	0	-6	0	2

$$C_6(z) = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (19.148)$$

Once again, we can determine the trace for individual symmetry transformations belonging to D_{6h} . We collect the results in Table 19.7. The representation is reducible and this is reduced as follows using a character table of D_{6h} (Table 19.8). As a result, we get

$$\Gamma = A_{2u} + B_{2g} + E_{1g} + E_{2u}. \quad (19.149)$$

As a subduced representation of D of D_{6h} to C_6 , we have

$$A_{2u} \downarrow C_6 = A, B_{2g} \downarrow C_6 = B, E_{1g} \downarrow C_6 = 2E_1, E_{2u} \downarrow C_6 = 2E_2. \quad (19.150)$$

Here, we used Table 19.9 that shows a character table of irreducible representations of C_6 . Following the previous procedures, as SALCs we have

Table 19.8 Character table of D_{6h}

D_{6h}	E	$2C_6$	$2C_3$	C_2	$3C'_2$	$3C''_2$	i	$2S_3$	$2S_6$	σ_h	$3\sigma_d$	$3\sigma_v$	
A_{1g}	1	1	1	1	1	1	1	1	1	1	1	1	$x^2 + y^2, z^2$
A_{2g}	1	1	1	1	-1	-1	1	1	1	1	-1	-1	
B_{1g}	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1	
B_{2g}	1	-1	1	-1	-1	1	1	-1	1	-1	-1	1	
E_{1g}	2	1	-1	-2	0	0	2	1	-1	-2	0	0	(yz, zx)
E_{2g}	2	-1	-1	2	0	0	2	-1	-1	2	0	0	$(x^2 - y^2, xy)$
A_{1u}	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1	
A_{2u}	1	1	1	1	-1	-1	-1	-1	-1	-1	1	1	z
B_{1u}	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1	
B_{2u}	1	-1	1	-1	-1	1	-1	1	-1	1	1	-1	
E_{1u}	2	1	-1	-2	0	0	-2	-1	1	2	0	0	(x, y)
E_{2u}	2	-1	-1	2	0	0	-2	1	1	-2	0	0	

Table 19.9 Character table of C_6

C_6	E	C_6	C_3	C_2	C_3^2	C_6^5	$\varepsilon = \exp(i\pi/3)$
A	1	1	1	1	1	1	$z; x^2 + y^2, z^2$
B	1	-1	1	-1	1	-1	
E_1	$\begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$	$\begin{Bmatrix} \varepsilon \\ \varepsilon^* \end{Bmatrix}$	$\begin{Bmatrix} -\varepsilon^* \\ -\varepsilon \end{Bmatrix}$	$\begin{Bmatrix} -1 \\ -1 \end{Bmatrix}$	$\begin{Bmatrix} -\varepsilon \\ -\varepsilon^* \end{Bmatrix}$	$\begin{Bmatrix} \varepsilon^* \\ \varepsilon \end{Bmatrix}$	$(x, y); (yz, zx)$
E_2	$\begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$	$\begin{Bmatrix} -\varepsilon^* \\ -\varepsilon \end{Bmatrix}$	$\begin{Bmatrix} -\varepsilon \\ -\varepsilon^* \end{Bmatrix}$	$\begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$	$\begin{Bmatrix} -\varepsilon^* \\ -\varepsilon \end{Bmatrix}$	$\begin{Bmatrix} -\varepsilon \\ -\varepsilon^* \end{Bmatrix}$	$(x^2 - y^2, xy)$

$$6P^{(A)}\phi_1 \equiv \xi_1 = \phi_1 + \phi_2 + \phi_3 + \phi_4 + \phi_5 + \phi_6,$$

$$6P^{(B)}\phi_1 \equiv \xi_2 = \phi_1 - \phi_2 + \phi_3 - \phi_4 + \phi_5 - \phi_6,$$

$$P^{[E_1^{(1)}]}\phi_1 \equiv \xi_3 = \phi_1 + \varepsilon\phi_2 - \varepsilon^*\phi_3 - \phi_4 - \varepsilon\phi_5 + \varepsilon^*\phi_6,$$

$$P^{[E_1^{(2)}]}\phi_1 \equiv \xi_4 = \phi_1 + \varepsilon^*\phi_2 - \varepsilon\phi_3 - \phi_4 - \varepsilon^*\phi_5 + \varepsilon\phi_6,$$

$$P^{[E_2^{(1)}]}\phi_1 \equiv \xi_5 = \phi_1 - \varepsilon^*\phi_2 - \varepsilon\phi_3 + \phi_4 - \varepsilon^*\phi_5 - \varepsilon\phi_6,$$

$$P^{[E_2^{(2)}]}\phi_1 \equiv \xi_6 = \phi_1 - \varepsilon\phi_2 - \varepsilon^*\phi_3 + \phi_4 - \varepsilon\phi_5 - \varepsilon^*\phi_6, \quad (19.151)$$

where $\varepsilon = \exp(i\pi/3)$.

Correspondingly, we have a diagonal secular equation of a sixth-order such that

$$\begin{vmatrix} \tilde{H}_{11} - \lambda \tilde{S}_{11} & & & & & \\ & \tilde{H}_{22} - \lambda \tilde{S}_{22} & & & & \\ & & \tilde{H}_{33} - \lambda \tilde{S}_{33} & & & \\ & & & \tilde{H}_{44} - \lambda \tilde{S}_{44} & & \\ & & & & \tilde{H}_{55} - \lambda \tilde{S}_{55} & \\ & & & & & \tilde{H}_{66} - \lambda \tilde{S}_{66} \end{vmatrix} = 0. \quad (19.152)$$

Here, we have, for example,

$$\tilde{H}_{11} = \int \xi_1^* H \xi_1 d\tau = \langle \xi_1 | H \xi_1 \rangle = 6(\alpha + 2\beta + 2\beta' + \beta''). \quad (19.153)$$

In (19.153), we used the same α and β defined in (19.94) as in the case of cyclopropenyl radical. That is, α is a Coulomb integral and β is a resonance integral between two adjacent $2p_z$ orbital of carbon. Meanwhile, β' is a resonance integral between orbitals of “meta” positions such as ϕ_1 and ϕ_3 . A quantity β'' is a resonance integral between orbitals of “para” positions such as ϕ_1 and ϕ_4 . It is unfamiliar to include such kind resonance integrals of β' and β'' at a simple π -electron approximation level. To ignore such resonance integrals is because of a practical purpose to simplify the calculations. However, we have no reason to exclude them. Or rather, the use of appropriate SALCs makes it feasible to include β' and β'' .

In a similar manner, we get

$$\begin{aligned} \tilde{H}_{22} &= \int \xi_2^* H \xi_2 d\tau = \langle \xi_2 | H \xi_2 \rangle = 6(\alpha - 2\beta + 2\beta' - \beta''), \\ \tilde{H}_{33} &= \tilde{H}_{44} = 6(\alpha + \beta - \beta' - \beta''), \\ \tilde{H}_{55} &= \tilde{H}_{66} = 6(\alpha - \beta - \beta' + \beta''). \end{aligned} \quad (19.154)$$

Meanwhile, we have

$$\tilde{S}_{11} = \langle \xi_1 | \xi_1 \rangle = 6(1 + 2S + 2S' + S''),$$

$$\tilde{S}_{22} = \langle \xi_2 | \xi_2 \rangle = 6(1 - 2S + 2S' - S''),$$

$$\tilde{S}_{33} = \tilde{S}_{44} = 6(1 + S - S' - S''),$$

$$\tilde{S}_{55} = \tilde{S}_{66} = 6(1 - S - S' + S''), \quad (19.155)$$

where S , S' , and S'' are overlap integrals between the ortho, meta, and para positions, respectively. Substituting (19.153) through (19.155) for (19.152), the energy eigenvalues are readily obtained as

$$\lambda_1 = \frac{\alpha + 2\beta + 2\beta' + \beta''}{1 + 2S + 2S' + S''}, \lambda_2 = \frac{\alpha - 2\beta + 2\beta' - \beta''}{1 - 2S + 2S' - S''}, \lambda_3 = \lambda_4 = \frac{\alpha + \beta - \beta' - \beta''}{1 + S - S' - S''},$$

$$\lambda_5 = \lambda_6 = \frac{\alpha - \beta - \beta' + \beta''}{1 - S - S' + S''}. \quad (19.156)$$

Notice that two MOs ξ_3 and ξ_4 as well as ξ_5 and ξ_6 are degenerate. As can be seen, λ_3 and λ_4 are doubly degenerate. So are λ_5 and λ_6 .

From (19.151), we get, for instance,

$$||\xi_1|| = \sqrt{\langle \xi_1 | \xi_1 \rangle} = \sqrt{6(1 + 2S + 2S' + S'')}. \quad (19.157)$$

Thus, for one of the normalized MOs corresponding to an energy eigenvalue λ_1 , we get

$$\psi_1 = \frac{|\xi_1\rangle}{||\xi_1||} = \frac{\phi_1 + \phi_2 + \phi_3 + \phi_4 + \phi_5 + \phi_6}{\sqrt{6(1 + 2S + 2S' + S'')}}. \quad (19.158)$$

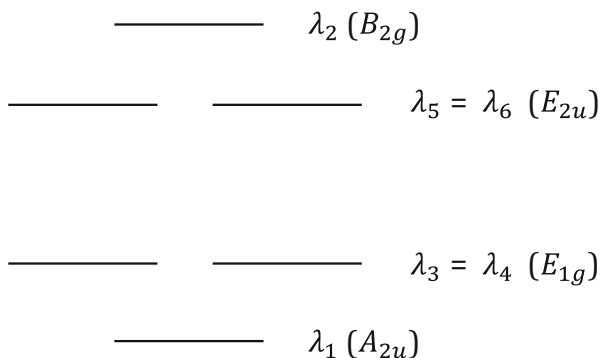
Following the previous examples, we have other normalized MOs. That is, we have

$$\psi_2 = \frac{|\xi_2\rangle}{||\xi_2||} = \frac{\phi_1 - \phi_2 + \phi_3 - \phi_4 + \phi_5 - \phi_6}{\sqrt{6(1 - 2S + 2S' - S'')}},$$

$$\psi_3 = \frac{|\xi_3\rangle}{||\xi_3||} = \frac{\phi_1 + \varepsilon\phi_2 - \varepsilon^*\phi_3 - \phi_4 - \varepsilon\phi_5 + \varepsilon^*\phi_6}{\sqrt{6(1 + S - S' - S'')}},$$

$$\psi_4 = \frac{|\xi_4\rangle}{||\xi_4||} = \frac{\phi_1 + \varepsilon^*\phi_2 - \varepsilon\phi_3 - \phi_4 - \varepsilon^*\phi_5 + \varepsilon\phi_6}{\sqrt{6(1 + S - S' - S'')}}.$$

Fig. 19.11 Energy diagram and MO assignments of benzene. Energy eigenvalues λ_1 to λ_6 are given in (19.156)



$$\Psi_5 = \frac{|\xi_5\rangle}{\|\xi_5\|} = \frac{\phi_1 - \varepsilon^* \phi_2 - \varepsilon \phi_3 + \phi_4 - \varepsilon^* \phi_5 - \varepsilon \phi_6}{\sqrt{6(1 - S - S' + S'')}} ,$$

$$\Psi_6 = \frac{|\xi_6\rangle}{\|\xi_6\|} = \frac{\phi_1 - \varepsilon \phi_2 - \varepsilon^* \phi_3 + \phi_4 - \varepsilon \phi_5 - \varepsilon^* \phi_6}{\sqrt{6(1 - S - S' + S'')}} . \quad (19.159)$$

The eigenfunction Ψ_i ($1 \leq i \leq 6$) corresponds to the eigenvalue λ_i . As in the case of cyclopropenyl radical, Ψ_3 and Ψ_4 can be transformed to $\tilde{\Psi}_3$ and $\tilde{\Psi}_4$, respectively, through a unitary matrix of (19.141). Thus, we get

$$\tilde{\Psi}_3 = \frac{2\phi_1 + \phi_2 - \phi_3 - 2\phi_4 - \phi_5 + \phi_6}{\sqrt{12(1 + S - S' - S'')}} ,$$

$$\tilde{\Psi}_4 = \frac{\phi_2 + \phi_3 - \phi_5 - \phi_6}{2\sqrt{1 + S - S' - S''}} . \quad (19.160)$$

Similarly, transforming Ψ_5 and Ψ_6 to $\tilde{\Psi}_5$ and $\tilde{\Psi}_6$, respectively, we have

$$\tilde{\Psi}_5 = \frac{2\phi_1 - \phi_2 - \phi_3 + 2\phi_4 - \phi_5 - \phi_6}{\sqrt{12(1 - S - S' + S'')}} ,$$

$$\tilde{\Psi}_6 = \frac{\phi_2 - \phi_3 + \phi_5 - \phi_6}{2\sqrt{1 - S - S' + S''}} . \quad (19.161)$$

Figure 19.11 shows an energy diagram and MO assignments of benzene.

A major optical transition takes place among HOMO (E_{1g}) and LUMO (E_{2u}) levels. In the case of optical absorption, an initial electronic configuration is assigned to the totally symmetric representation A_{1g} and the symmetry of the final electronic configuration is described as

$$\Gamma = E_{1g} \times E_{2u}.$$

Therefore, a transition matrix element is expressed as

$$\left\langle \Phi^{(E_{1g} \times E_{2u})} | \epsilon_e \cdot \mathbf{P} | \Phi^{(A_{1g})} \right\rangle, \quad (19.162)$$

where $\Phi^{(A_{1g})}$ stands for the totally symmetric ground state electronic configuration; $\Phi^{(E_{1g} \times E_{2u})}$ denotes an excited state electronic configuration represented by a direct-product representation. This representation is reducible and expressed as a direct sum of irreducible representations such that

$$\Gamma = B_{1u} + B_{2u} + E_{1u}. \quad (19.163)$$

Notice that unlike ethylene a direct-product representation associated with the final state is reducible.

To examine whether (19.162) is non-vanishing, as in the case of (19.109) we estimate whether a direct-product representation $A_{1g} \times E_{1g} \times E_{2u} = E_{1g} \times E_{2u} = B_{1u} + B_{2u} + E_{1u}$ contains an irreducible representation which $\epsilon_e \cdot \mathbf{P}$ belongs to. Consulting a character table of D_{6h} , we find that x and y belong to an irreducible representation E_{1u} and that z belongs to A_{2u} . Since the direct sum Γ in (19.163) contains E_{1u} , benzene is expected to be polarized along both x - and y -axes (see Fig. 19.10). Since (19.163) does not contain A_{2u} , the transition along the z -axis is forbidden. Accordingly, the transition takes place when the light is polarized parallel to the molecular plane (i.e., the xy -plane). This is a common feature among planar aromatic molecules including benzene and cyclopropenyl radical. On the other hand, A_{2u} is not contained in Γ , and so we do not expect the optical transition to occur in the direction of the z -axis.

19.4.4 Allyl Radical [1]

We revisit the allyl radical and perform its MO calculations. As already noted, Tables 18.1 and 18.2 of Example 18.1 collected representation matrices of individual symmetry operations in reference to the basis vectors comprising three atomic orbitals of allyl radical. As usual, we examine traces (or characters) of those matrices. Table 19.10 collects them. The representation is readily reduced according to the character table of C_{2v} (see Table 18.4) so that we have

Table 19.10 Characters for individual symmetry transformations of $2p_z$ orbitals in allyl radical

C_{2v}	E	$C_2(z)$	$\sigma_v(zx)$	$\sigma'_v(yz)$
Γ	3	-1	1	-3

$$\Gamma = A_2 + 2B_1. \quad (19.164)$$

We have two SALC orbitals that belong to the same irreducible representation of B_1 . As noted in Sect. 19.3, the orbitals obtained by a linear combination of these two SALCs belong to B_1 as well. Such a linear combination is given by a unitary transformation. In the present case, it is convenient to transform the basis vectors two times. The first transformation is carried out to get SALCs and the second one will be done in the process of solving a secular equation using the SALCs. Schematically showing the procedures, we have

$$\begin{Bmatrix} \phi_1 \\ \phi_2 \\ \phi_3 \end{Bmatrix} \rightarrow \begin{Bmatrix} \Psi_1 \\ \Psi_2 \\ \Psi_3 \end{Bmatrix} \rightarrow \begin{Bmatrix} \Phi_1 \\ \Phi_2 \\ \Phi_3 \end{Bmatrix},$$

where ϕ_1, ϕ_2, ϕ_3 show the original atomic orbitals; Ψ_1, Ψ_2, Ψ_3 the SALCs; Φ_1, Φ_2, Φ_3 the final Mos. Thus, the three sets of vectors are connected through unitary transformations.

Starting with ϕ_1 and following the previous cases, we have, e.g.,

$$P^{(B_1)}\phi_1 = \frac{1}{4} \sum_g [\chi^{(B_1)}(g)]^* g\phi_1 = \phi_1.$$

Also starting with ϕ_2 , we have

$$P^{(B_1)}\phi_2 = \frac{1}{4} \sum_g [\chi^{(B_1)}(g)]^* g\phi_2 = \frac{1}{2}(\phi_2 + \phi_3).$$

Meanwhile, we get

$$P^{(A_2)}\phi_1 = 0,$$

$$P^{(A_2)}\phi_2 = \frac{1}{4} \sum_g [\chi^{(A_2)}(g)]^* g\phi_2 = \frac{1}{2}(\phi_2 - \phi_3).$$

Thus, we recovered the results of Example 18.1. Notice that ϕ_1 does not participate in A_2 , but take part in B_1 by itself.

Normalized SALCs are given as follows:

$$\Psi_1 = \phi_1, \Psi_2 = (\phi_2 + \phi_3)/\sqrt{2(1+S')}, \Psi_3 = (\phi_2 - \phi_3)/\sqrt{2(1-S')},$$

where we define $S' \equiv \int \phi_2 \phi_3 d\tau$. If Ψ_1 , Ψ_2 , and Ψ_3 belonged to different irreducible representations (as in the cases of previous three examples of ethylene, cyclopropenyl radical, and benzene), the secular equation would be fully reduced to a form of (19.66). In the present case, however, Ψ_1 and Ψ_2 belong to the same irreducible representation B_1 , making the situation a bit complicated. Nonetheless, we can use (19.61) and the secular equation is “partially” reduced.

Defining

$$H_{jk} = \int \Psi_j^* H \Psi_k d\tau \quad \text{and} \quad S_{jk} = \int \Psi_j^* \Psi_k d\tau,$$

we have a following secular equation the same form as (19.61):

$$\det(H_{jk} - \lambda S_{jk}) = 0.$$

More specifically, we have

$$\begin{vmatrix} \alpha - \lambda & \sqrt{\frac{2}{1+S'}}(\beta - S\lambda) & 0 \\ \sqrt{\frac{2}{1+S'}}(\beta - S\lambda) & \frac{\alpha + \beta'}{1+S'} - \lambda & 0 \\ 0 & 0 & \frac{\alpha - \beta'}{1-S'} - \lambda \end{vmatrix} = 0,$$

where α , β , and S are similarly defined as (19.94) and (19.98). The quantities β' is defined as

$$\beta' \equiv \int \phi_2^* H \phi_3 d\tau.$$

Thus, the secular equation is separated into the following two:

$$\begin{vmatrix} \alpha - \lambda & \sqrt{\frac{2}{1+S'}}(\beta - S\lambda) \\ \sqrt{\frac{2}{1+S'}}(\beta - S\lambda) & \frac{\alpha + \beta'}{1+S'} - \lambda \end{vmatrix} = 0 \quad \text{and} \quad \frac{\alpha - \beta'}{1-S'} - \lambda = 0. \quad (19.165)$$

The second equation immediately gives

$$\lambda = \frac{\alpha - \beta'}{1 - S'}.$$

The first equation of (19.165) is somewhat complicated, and so we adopt the next approximation. That is,

$$S' = \beta' = 0. \quad (19.166)$$

This approximation is justified, because two carbon atoms C_2 and C_3 are pretty remote, and so the interaction between them is likely to be weak enough. Thus, we rewrite the first equation of (19.165) as

$$\begin{vmatrix} \alpha - \lambda & \sqrt{2}(\beta - S\lambda) \\ \sqrt{2}(\beta - S\lambda) & \alpha - \lambda \end{vmatrix} = 0. \quad (19.167)$$

Moreover, we assume that since S is a small quantity compared to 1, a square term of $S^2 \ll 1$. Hence, we ignore S^2 .

Using the approximation of (19.166) and assuming $S^2 \approx 0$, from (19.167) we have a following quadratic equation:

$$\lambda^2 - 2(\alpha - 2\beta S)\lambda + \alpha^2 - 2\beta^2 = 0.$$

Solving this equation, we have

$$\lambda = \alpha - 2\beta S \pm \sqrt{2}\beta \sqrt{1 - \frac{2\alpha S}{\beta}} \approx \alpha - 2\beta S \pm \sqrt{2}\beta \left(1 - \frac{\alpha S}{\beta}\right),$$

where the last approximation is based on

$$\sqrt{1 - x} \approx 1 - \frac{1}{2}x$$

with a small quantity x . Thus, we get

$$\lambda_L \approx (\alpha + \sqrt{2}\beta)(1 - \sqrt{2}S) \text{ and } \lambda_H \approx (\alpha - \sqrt{2}\beta)(1 + \sqrt{2}S), \quad (19.168)$$

where $\lambda_L < \lambda_H$. To determine the corresponding eigenfunctions Φ (i.e., MOs), we use a linear combination of two SALCs Ψ_1 and Ψ_2 . That is, putting

$$\Phi = c_1\Psi_1 + c_2\Psi_2 \quad (19.169)$$

and from (19.167), we obtain

$$(\alpha - \lambda)c_1 + \sqrt{2}(\beta - S\lambda)c_2 = 0.$$

Thus, with $\lambda = \lambda_L$ we get $c_1 = c_2$ and for $\lambda = \lambda_H$ we get $c_1 = -c_2$. Consequently, as a normalized eigenfunction Φ_L corresponding to λ_L we get

$$\Phi_L = \frac{1}{\sqrt{2}}(\Psi_1 + \Psi_2) = \frac{1}{2}(1 + \sqrt{2}S)^{-1/2}(\sqrt{2}\phi_1 + \phi_2 + \phi_3). \quad (19.170)$$

Likewise, as a normalized eigenfunction Φ_H corresponding to λ_H we get

$$\Phi_H = \frac{1}{\sqrt{2}}(-\Psi_1 + \Psi_2) = \frac{1}{2}(1 - \sqrt{2}S)^{-1/2}(-\sqrt{2}\phi_1 + \phi_2 + \phi_3). \quad (19.171)$$

As another eigenvalue λ_0 and corresponding eigenfunction Φ_0 , we have

$$\lambda_0 \approx \alpha \quad \text{and} \quad \Phi_0 = \frac{1}{\sqrt{2}}(\phi_2 - \phi_3), \quad (19.172)$$

where we have $\lambda_L < \lambda_0 < \lambda_H$. The eigenfunction Φ_0 does not participate in chemical bonding and, hence, is said to be a non-bonding orbital.

It is worth noting that eigenfunctions Φ_L , Φ_0 , and Φ_H have the same function forms as those obtained by the simple Hückel theory that ignores the overlap integrals S . It is because the interaction between ϕ_2 and ϕ_3 that construct SALCs of Ψ_2 and Ψ_3 is weak.

Notice that within the framework of the simple Hückel theory, two sets of basis vectors $|\phi_1\rangle$, $\frac{1}{\sqrt{2}}|\phi_2 + \phi_3\rangle$, $\frac{1}{\sqrt{2}}|\phi_2 - \phi_3\rangle$ and $|\Phi_L\rangle$, $|\Phi_H\rangle$, $|\Phi_0\rangle$ are connected by a following unitary matrix V :

$$\begin{aligned} (|\Phi_L\rangle \quad |\Phi_H\rangle \quad |\Phi_0\rangle) &= \left(|\phi_1\rangle \quad \frac{1}{\sqrt{2}}|\phi_2 + \phi_3\rangle \quad \frac{1}{\sqrt{2}}|\phi_2 - \phi_3\rangle \right) V \\ &= \left(|\phi_1\rangle \quad \frac{1}{\sqrt{2}}|\phi_2 + \phi_3\rangle \quad \frac{1}{\sqrt{2}}|\phi_2 - \phi_3\rangle \right) \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned} \quad (19.173)$$

Although both SALCs Ψ_1 and Ψ_2 belong to the same irreducible representation B_1 , they are not orthogonal. As can be seen from (19.170) and (19.171), however, we find that Φ_L and Φ_H that are sought by solving the secular equation (19.167) have been mutually orthogonal. Starting from $|\phi_1\rangle$, $|\phi_2\rangle$, and $|\phi_3\rangle$ of Example 18.1, we reached $|\Phi_L\rangle$, $|\Phi_H\rangle$, and $|\Phi_0\rangle$ via two-step unitary transformations (18.40) and (19.173). The combined unitary transformations $W = UV$ are unitary again. That is, we have

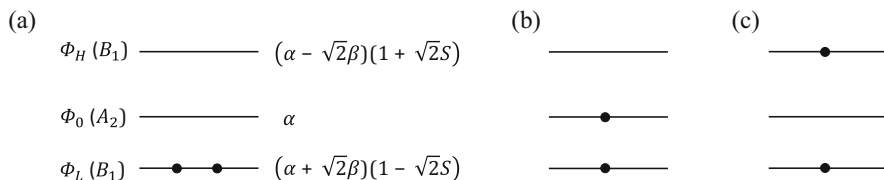
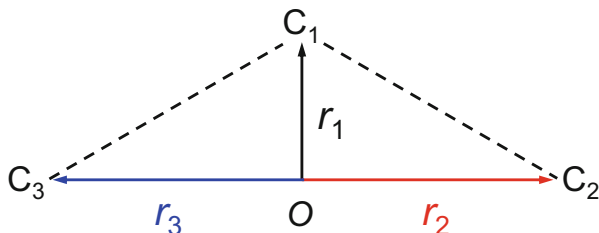


Fig. 19.12 Electronic configurations and symmetry species of individual eigenstates along with their corresponding energy eigenvalues for the allyl cation. (a) Ground state. (b) First excited state. (c) Second excited state

Fig. 19.13 Geometry and position vectors of carbon atoms of the allyl cation. The origin is located at the center of a line segment connecting C_2 and C_3 ($r_2 + r_3 = 0$)



$$\begin{aligned}
 (|\phi_1\rangle \quad |\phi_2\rangle \quad |\phi_3\rangle)UV &= (|\phi_1\rangle \quad |\phi_2\rangle \quad |\phi_3\rangle) \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}} \end{pmatrix} \\
 &= (|\Phi_L\rangle \quad |\Phi_H\rangle \quad |\Phi_0\rangle). \quad (19.174)
 \end{aligned}$$

The optical transition of allyl radical represents general features of the optical transitions of molecules. To make a story simple, let us consider a case of allyl cation. Figure 19.12 shows the electronic configurations together with symmetry species of individual eigenstates and their corresponding energy eigenvalues for the allyl cation. In Fig. 19.13, we redraw its geometry where the origin is located at the center of a line segment connecting C_2 and C_3 ($r_2 + r_3 = 0$). Major optical transitions (or optical absorption) are $\Phi_L \rightarrow \Phi_0$ and $\Phi_L \rightarrow \Phi_H$.

(i) $\Phi_L \rightarrow \Phi_0$: In this case, following (19.109) the transition matrix element T_{fi} is described by

$$T_{fi} = \langle \Theta_f^{(B_1 \times A_2)} | \epsilon_e \cdot \mathbf{P} | \Theta_i^{(A_1)} \rangle. \quad (19.175)$$

In the above equation, we designate the irreducible representation of eigenstates according to (19.164). Therefore, the symmetry of the final electronic configuration is described as

$$\Gamma = B_1 \times A_2 = B_2.$$

Since a direct product of the initial electronic configuration $[\Theta_i^{(A_1)}]$ and the final configuration $\Theta_f^{(B_1 \times A_2)}$ is $B_1 \times A_2 \times A_1 = B_2$. Then, if $\epsilon_e \cdot \mathbf{P}$ belongs to the same irreducible representation B_2 , the associated optical transition should be allowed. Consulting a character table of C_{2v} , we find that y belongs to B_2 . Thus, the allowed optical transition is polarized along the y -direction (see Fig. 18.1).

(ii) $\Phi_L \rightarrow \Phi_H$: In parallel with the above case, T_{fi} is described by

$$T_{fi} = \langle \Theta_f^{(B_1 \times B_1)} | \epsilon_e \cdot \mathbf{P} | \Theta_i^{(A_1)} \rangle. \quad (19.176)$$

Thus, the transition is characterized by

$$A_1 \rightarrow A_1,$$

where the former A_1 indicates the electronic ground state and the latter A_1 indicates the excited state given by $B_1 \times B_1 = A_1$. The direct product of them is simply described as $B_1 \times B_1 \times A_1 = A_1$. Consulting a character table again, we find that z belongs to A_1 . This implies that the allowed transition is polarized along the z -direction (see Fig. 18.1).

Next, we investigate the above optical transition in a semi-quantitative manner. In principle, the transition matrix element T_{fi} should be estimated from (19.175) and (19.176) that use electronic configurations of two-electron system. Nonetheless, a formulation of (4.7) of Sect. 4.1 based upon one-electron states well serves our present purpose. For $\Phi_L \rightarrow \Phi_0$ transition, we have

$$\begin{aligned} T_{fi}(\Phi_L \rightarrow \Phi_0) &= \int \Phi_0^* (\epsilon_e \cdot \mathbf{P}) \Phi_L d\tau = e\epsilon_e \cdot \int \Phi_0^* \mathbf{r} \Phi_L d\tau \\ &= e\epsilon_e \cdot \int \frac{1}{2} \left(\sqrt{2}\phi_1 + \phi_2 + \phi_3 \right) \mathbf{r} \frac{1}{\sqrt{2}} (\phi_2 - \phi_3) d\tau \\ &= \frac{e\epsilon_e}{2\sqrt{2}} \cdot \int \left(\sqrt{2}\phi_1 \mathbf{r} \phi_2 - \sqrt{2}\phi_1 \mathbf{r} \phi_3 + \phi_2 \mathbf{r} \phi_2 + \phi_3 \mathbf{r} \phi_2 - \phi_3 \mathbf{r} \phi_3 \right) d\tau \end{aligned}$$

$$\approx \frac{e\epsilon_e}{2\sqrt{2}} \cdot \int (\phi_2 \mathbf{r} \phi_2 - \phi_3 \mathbf{r} \phi_3) d\tau$$

$$\approx \frac{e\epsilon_e}{2\sqrt{2}} \cdot \left[\mathbf{r}_2 \int |\phi_2|^2 d\tau - \mathbf{r}_3 \int |\phi_3|^2 d\tau \right] = \frac{e\epsilon_e}{2\sqrt{2}} \cdot (\mathbf{r}_2 - \mathbf{r}_3) = \frac{e\epsilon_e}{\sqrt{2}} \cdot \mathbf{r}_2,$$

where with the first near equality we ignored integrals $\int \phi_i \mathbf{r} \phi_j d\tau$ ($i \neq j$); with the last near equality $\mathbf{r} \approx \mathbf{r}_2$ or $\mathbf{r}_3$. For these approximations, we assumed that an electron density is very high near C_2 or C_3 with ignorable density at a place remote from them. Choosing ϵ_e for the direction of $\mathbf{r}_2$, we get

$$T_{fi}(\Phi_L \rightarrow \Phi_0) \approx \frac{e}{\sqrt{2}} |\mathbf{r}_2|.$$

With $\Phi_L \rightarrow \Phi_H$ transition, similarly we have

$$T_{fi}(\Phi_L \rightarrow \Phi_H) = \int \Phi_H^* \epsilon_e \cdot \mathbf{P} \Phi_L d\tau \approx \frac{e\epsilon_e}{4} \cdot (\mathbf{r}_3 - 2\mathbf{r}_1 + \mathbf{r}_2) = -\frac{e\epsilon_e}{2} \cdot \mathbf{r}_1.$$

Choosing ϵ_e for the direction of $\mathbf{r}_1$, we get

$$T_{fi}(\Phi_L \rightarrow \Phi_H) \approx \frac{e}{2} |\mathbf{r}_1|.$$

Transition probability is proportional to a square of an absolute value of T_{fi} . Using $|\mathbf{r}_2| \approx \sqrt{3} |\mathbf{r}_1|$, we have

$$|T_{fi}(\Phi_L \rightarrow \Phi_0)|^2 \approx \frac{e^2}{2} |\mathbf{r}_2|^2 = \frac{3e^2}{2} |\mathbf{r}_1|^2,$$

$$|T_{fi}(\Phi_L \rightarrow \Phi_H)|^2 \approx \frac{e^2}{4} |\mathbf{r}_1|^2.$$

Thus, we obtain

$$|T_{fi}(\Phi_L \rightarrow \Phi_0)|^2 \approx 6 |T_{fi}(\Phi_L \rightarrow \Phi_H)|^2. \quad (19.177)$$

Thus, the transition probability of $\Phi_L \rightarrow \Phi_0$ is about six times that for $\Phi_L \rightarrow \Phi_H$. Note that in the above simple estimation we ignore an overlap integral S .

From the above discussion, we conclude that (i) the $\Phi_L \rightarrow \Phi_0$ transition is polarized along the $\mathbf{r}_2$ direction (i.e., the molecular long axis) and that the $\Phi_L \rightarrow \Phi_H$ transition is polarized along the $\mathbf{r}_1$ direction (i.e., the molecular short axis). (ii) Transition probability of $\Phi_L \rightarrow \Phi_0$ is about six times that of $\Phi_L \rightarrow \Phi_H$. Note

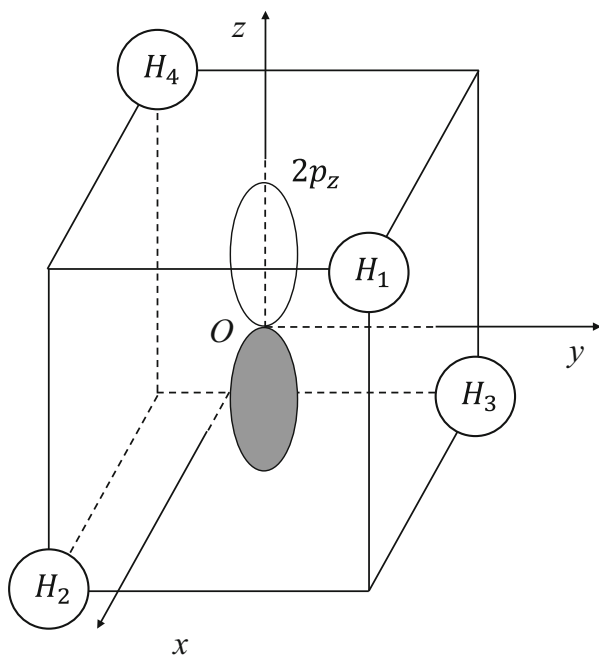
that the polarized characteristics are consistent with those obtained from the discussion based on the group theory. The conclusion reached by the semi-quantitative estimation of a simple molecule of allyl cation well typifies the general optical features of more complicated molecules having a well-defined molecular long axis such as polyenes.

19.5 MO Calculations of Methane

So far, we investigated MO calculations of aromatic molecules based upon π -electron approximation. These are a homogeneous system that has the same quality of electrons. Here we deal with methane that includes a carbon and surrounding four hydrogens. These hydrogen atoms form a regular tetrahedron with the carbon atom positioned at a center of the tetrahedron. It is therefore considered as a heterogeneous system. The calculation principle, however, is consistent, namely, we make the most of projection operators and construct appropriate SALCs of methane.

We deal with four $1s$ electrons of hydrogen along with two $2s$ electrons and two $2p$ electrons of carbon. Regarding basis functions of carbon, however, we consider $2s$ atomic orbital and three $2p$ orbitals (i.e., $2p_x$, $2p_y$, $2p_z$ orbitals). That is, we deal with eight atomic orbitals all together. These are depicted in Fig. 19.14. The dimension of the vector space (i.e., representation space) is eight accordingly.

Fig. 19.14 Four $1s$ atomic orbitals of hydrogen and a $2p_z$ orbital of carbon. The former orbitals are represented by H_1 to H_4 . $2p_x$ and $2p_y$ orbitals of carbon are omitted for simplicity



As before, we wish to determine irreducible representations which individual MOs belong to. As already mentioned in Sect. 17.3, there are 24 symmetry operations in a point group T_d which methane belongs to (see Table 17.6). According to the symmetry operations, we decide transformation matrices related to each operation. For example, C_3^{xyz} transforms basis functions as follows:

$$\begin{aligned} & (H_1 \ H_2 \ H_3 \ H_4 \ C2s \ C2p_x \ C2p_y \ C2p_z) C_3^{xyz} \\ &= (H_1 \ H_3 \ H_4 \ H_2 \ C2s \ C2p_y \ C2p_z \ C2p_x), \end{aligned}$$

where by the above notations we denoted atomic species and molecular orbitals. Hence, as a matrix representation we have

$$C_3^{xyz} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \quad (19.178)$$

where C_3^{xyz} is the same operation as $R_{xyz\frac{2\pi}{3}}$ appeared in Sect. 17.3. Therefore,

$$\chi(C_3^{xyz}) = 2.$$

As another example σ_d^{yz} , we have

$$\begin{aligned} & (H_1 \ H_2 \ H_3 \ H_4 \ C2s \ C2p_x \ C2p_y \ C2p_z) \sigma_d^{yz} \\ &= (H_1 \ H_2 \ H_4 \ H_3 \ C2s \ C2p_x \ C2p_z \ C2p_y), \end{aligned}$$

where σ_d^{yz} represents a mirror symmetry with respect to the plane that includes the x -axis and bisects the angle formed by the y - and z -axes. Thus, we have

$$\sigma_d^{yz} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}. \quad (19.179)$$

Then, we have

$$\chi\left(S_4^{\pi/2}\right)=0.$$

As for the identity matrix E , we have

$$\chi(E)=8.$$

Thus, Table 19.11 collects characters of individual symmetry transformations with respect to hydrogen $1s$ and carbon $2s$ and $2p$ orbitals in methane. From the above examples, we notice that all the symmetry operators R reduce an eight-dimensional representation space V^8 to subspaces of $\text{Span}\{H_1, H_2, H_3, H_4\}$ and $\text{Span}\{C2s, C2p_x, C2p_y, C2p_z\}$. In terms of a notation of Sect. 12.2, we have

$$V^8 = \text{Span}\{H_1, H_2, H_3, H_4\} \oplus \text{Span}\{C2s, C2p_x, C2p_y, C2p_z\}. \quad (19.182)$$

In other words, V^8 is decomposed into the above two R -invariant subspaces (see Part III); one is the hydrogen-related subspace and the other is the carbon-related subspace.

Correspondingly, a representation D comprising the above representation matrices should be reduced to a direct sum of irreducible representations $D^{(\alpha)}$. That is, we should have

$$D = \sum_{\alpha} q_{\alpha} D^{(\alpha)},$$

where q_{α} is a positive integer or zero. Here we are thinking of decomposition of $(8, 8)$ matrices such as (19.178) into submatrices. We estimate q_{α} using (18.83). That is,

$$q_{\alpha} = \frac{1}{n} \sum_g \chi^{(\alpha)}(g)^* \chi(g). \quad (18.83)$$

With an irreducible representation A_1 of T_d , for instance, we have

$$q_{A_1} = \frac{1}{24} \sum_g \chi^{(A_1)}(g)^* \chi(g) = \frac{1}{24} (1 \cdot 8 + 1 \cdot 8 \cdot 2 + 1 \cdot 6 \cdot 4) = 2.$$

As for A_2 , we have

$$q_{A_2} = \frac{1}{24} [1 \cdot 8 + 1 \cdot 8 \cdot 2 + (-1) \cdot 6 \cdot 4] = 0.$$

Regarding T_2 , we get

$$q_{T_2} = \frac{1}{24} (3 \cdot 8 + 1 \cdot 6 \cdot 4) = 2.$$

For other irreducible representations of T_d , we get $q_{\alpha} = 0$. Consequently, we have

$$D = 2D^{(A_1)} + 2D^{(T_2)}. \quad (19.183)$$

Evidently, both for the hydrogen-related representation $D^{(H)}$ and carbon-related representation $D^{(C)}$, we have individually

$$D^{(H)} = D^{(A_1)} + D^{(T_2)} \text{ and } D^{(C)} = D^{(A_1)} + D^{(T_2)}, \quad (19.184)$$

where $D = D^{(H)} + D^{(C)}$.

In fact, taking (19.180), for example, in a subspace $\text{Span}\{H_1, H_2, H_3, H_4\}$, the operator C_2^z is expressed as

$$C_2^z = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Following the routine procedures based on characteristic polynomials, we get eigenvalues $+1$ (as a double root) and -1 (a double root as well). Unitary similarity transformation using a unitary matrix P expressed as

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

yields a diagonal matrix described by

$$P^{-1}C_2^zP = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Note that this diagonal matrix is identical to submatrix of (19.180) for $\text{Span}\{C2s, C2p_x, C2p_y, C2p_z\}$. Similarly, $S_4^{\pi/2}$ of (19.181) gives eigenvalues ± 1 and $\pm i$ for both the subspaces. Notice that since $S_4^{\pi/2}$ is unitary, its eigenvalues take a complex number with an absolute value of 1. Since these symmetry operation matrices are unitary, these matrices must be diagonalized according to Theorem 14.5. Using unitary matrices whose column vectors are chosen from eigenvectors of the matrices, diagonal elements are identical to eigenvalues including their multiplicity. Writing representation matrices of symmetry operations for the hydrogen-associated subspace and carbon-associated subspace as H and C , we find that H and C have the

same eigenvalues in common. Notice here that different types of transformation matrices (e.g., C_2^z , $S_4^{z^2}$, etc.) give a different set of eigenvalues. Via unitary similarity transformation using unitary matrices P and Q , we get

$$P^{-1}HP = Q^{-1}CQ \quad \text{or} \quad (PQ^{-1})^{-1}H(PQ^{-1}) = C.$$

Namely, H and C are similar, i.e., the representation is equivalent. Thus, recalling Schur's First Lemma, (19.184) results.

Our next task is to construct SALCs, i.e., proper basis vectors using projection operators. From the above, we anticipate that the MOs comprise a linear combination of the hydrogen-associated SALC and carbon-associated SALC that belongs to the same irreducible representation (i.e., A_1 or T_2). For this purpose, we first find proper SALCs using projection operators described as

$$P_{l(l)}^{(\alpha)} = \frac{d_\alpha}{n} \sum_g D_{ll}^{(\alpha)}(g)^* g. \quad (18.156)$$

We apply this operator to $\text{Span}\{H_1, H_2, H_3, H_4\}$. Taking, e.g., H_1 and operating both sides of (18.156) on H_1 , as a basis vector corresponding to a one-dimensional representation of A_1 we have

$$\begin{aligned} P_{1(1)}^{(A_1)} H_1 &= \frac{d_{A_1}}{n} \sum_g D_{11}^{(A_1)}(g)^* g H_1 = \frac{1}{24} [(1 \cdot H_1) + (1 \cdot H_1 + 1 \cdot H_1 + 1 \cdot H_3 \\ &+ 1 \cdot H_4 + 1 \cdot H_3 + 1 \cdot H_2 + 1 \cdot H_4 + 1 \cdot H_2) + (1 \cdot H_4 + 1 \cdot H_3 + 1 \cdot H_2) \\ &+ (1 \cdot H_3 + 1 \cdot H_2 + 1 \cdot H_2 + 1 \cdot H_4 + 1 \cdot H_4 + 1 \cdot H_3) \\ &+ (1 \cdot H_1 + 1 \cdot H_4 + 1 \cdot H_1 + 1 \cdot H_2 + 1 \cdot H_1 + 1 \cdot H_3)]. \\ &= \frac{1}{24} (6 \cdot H_1 + 6 \cdot H_2 + 6 \cdot H_3 + 6 \cdot H_4) \\ &= \frac{1}{4} (H_1 + H_2 + H_3 + H_4). \end{aligned} \quad (19.185)$$

The case of C_{2s} is simple, because all the symmetry operations convert C_{2s} to itself. That is, we have

$$P_{1(1)}^{(A_1)} C_{2s} = \frac{1}{24} (24 \cdot C_{2s}) = C_{2s}. \quad (19.186)$$

Regarding C_{2p} , taking C_{2p_x} , for instance, we have

Table 19.12 Character table of T_d

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$	
A_1	1	1	1	1	1	$x^2 + y^2 + z^2$
A_2	1	1	1	-1	-1	
E	2	-1	2	0	0	$(2z^2 - x^2 - y^2, x^2 - y^2)$
T_1	3	0	-1	1	-1	
T_2	3	0	-1	-1	1	$(x, y, z); (xy, yz, zx)$

$$\begin{aligned}
P_{1(1)}^{(A_1)} C2p_x &= \frac{1}{24} \{ (1 \cdot C2p_x) + [1 \cdot C2p_y + 1 \cdot C2p_z + 1 \cdot (-C2p_y) \\
&+ 1 \cdot (-C2p_z) + 1 \cdot (-C2p_y) + 1 \cdot C2p_z + 1 \cdot C2p_y + 1 \cdot (-C2p_z)] \\
&+ [(1 \cdot C2p_x + 1 \cdot (-C2p_x) + 1 \cdot (-C2p_x)) + [1 \cdot (-C2p_x) \\
&+ 1 \cdot (-C2p_x) + 1 \cdot (-C2p_z) + 1 \cdot C2p_z + 1 \cdot C2p_y + 1 \cdot (-C2p_y)] \\
&+ [1 \cdot C2p_y + 1 \cdot C2p_x + 1 \cdot C2p_z + 1 \cdot (-C2p_y) + 1 \cdot C2p_x + 1 \cdot (-C2p_z)] \} \\
&= 0.
\end{aligned}$$

The calculation is somewhat tedious, but it is natural that since $C2s$ is spherically symmetric, it belongs to the totally symmetric representation. Conversely, it is natural to think that $C2p_x$ is totally unlikely to contain a totally symmetric representation. This is also the case with $C2p_y$ and $C2p_z$.

Table 19.12 shows the character table of T_d in which the three-dimensional irreducible representation T_2 is spanned by basis vectors $(x \ y \ z)$. Since in Table 17.6 each $(3, 3)$ matrix is given in reference to the vectors $(x \ y \ z)$, it can directly be utilized to represent T_2 . More specifically, we can directly choose the diagonal elements $(1, 1)$, $(2, 2)$, and $(3, 3)$ of the individual $(3, 3)$ matrices for $D_{11}^{(T_2)}$, $D_{22}^{(T_2)}$, and $D_{33}^{(T_2)}$ elements of the projection operators, respectively. Thus, we can construct SALCs using projection operators explained in Sect. 18.7. For example, using H_1 we obtain SALCs that belong to T_2 such that

$$P_{1(1)}^{(T_2)} H_1 = \frac{3}{24} \sum_g D_{11}^{(T_2)}(g)^* g H_1 = \frac{3}{24} \{ (1 \cdot H_1) \}$$

$$\begin{aligned}
& +[(-1) \cdot H_4 + (-1) \cdot H_3 + 1 \cdot H_2] \\
& +[0 \cdot (-H_3) + 0 \cdot (-H_2) + 0 \cdot (-H_2) + 0 \cdot (-H_4) + (-1) \cdot H_4 + (-1) \cdot H_3] \\
& +\{0 \cdot H_1 + 0 \cdot H_4 + 1 \cdot H_1 + 1 \cdot H_2 + 0 \cdot H_1 + 0 \cdot H_3\} \\
& = \frac{3}{24}[2 \cdot H_1 + 2 \cdot H_2 + (-2) \cdot H_3 + (-2) \cdot H_4] \\
& = \frac{1}{4}(H_1 + H_2 - H_3 - H_4). \tag{19.187}
\end{aligned}$$

Similarly, we have

$$P_{2(2)}^{(T_2)} H_1 = \frac{1}{4}(H_1 - H_2 + H_3 - H_4), \tag{19.188}$$

$$P_{3(3)}^{(T_2)} H_1 = \frac{1}{4}(H_1 - H_2 - H_3 + H_4). \tag{19.189}$$

Now we can easily guess that $P_{1(1)}^{(T_2)} C2p_x$ solely contains $C2p_x$. Likewise, $P_{2(2)}^{(T_2)} C2p_y$ and $P_{3(3)}^{(T_2)} C2p_z$ contain only $C2p_y$ and $C2p_z$, respectively. In fact, we obtain what we anticipate. That is,

$$P_{1(1)}^{(T_2)} C2p_x = C2p_x, P_{2(2)}^{(T_2)} C2p_y = C2p_y, P_{3(3)}^{(T_2)} C2p_z = C2p_z. \tag{19.190}$$

This gives a good example to illustrate the general concept of projection operators and related calculations of inner products discussed in Sect. 18.7. For example, we have a non-vanishing inner product of $\langle P_{1(1)}^{(T_2)} H_1 | P_{1(1)}^{(T_2)} C2p_x \rangle$ and an inner product of, e.g., $\langle P_{2(2)}^{(T_2)} H_1 | P_{1(1)}^{(T_2)} C2p_x \rangle$ is zero. This significantly reduces efforts to solve a secular equation (vide infra). Notice that functions $P_{1(1)}^{(T_2)} H_1$, $P_{1(1)}^{(T_2)} C2p_x$, etc. are linearly independent of one another. Recall also that $P_{1(1)}^{(T_2)} H_1$ and $P_{1(1)}^{(T_2)} C2p_x$ correspond to $\phi_l^{(\alpha)}$ and $\psi_l^{(\alpha)}$ in (18.171), respectively. That is, the former functions are linearly independent, while they belong to the same place “1” of the same three-dimensional irreducible representation T_2 .

Equations (19.187)–(19.190) seem intuitively obvious. This is because if we draw a molecular geometry of methane (see Fig. 19.14), we can immediately recognize the relationship between the directionality in $\mathbb{R}^3$ and “directionality” of

SALCs represented by sums of hydrogen atomic orbitals (or $C2p_x$, $C2p_y$, and $C2p_z$ of carbon).

As stated above, we have successfully obtained SALCs relevant to methane. Therefore, our next task is to solve an eigenvalue problem and construct appropriate MOs. To do this, let us first normalize the SALCs. We assume that carbon-based SALCs have already been normalized as well-studied atomic orbitals. We suppose that all the functions are real. For the hydrogen-based SALCs

$$\begin{aligned}\langle H_1 + H_2 + H_3 + H_4 | H_1 + H_2 + H_3 + H_4 \rangle &= 4\langle H_1 | H_1 \rangle + 12\langle H_1 | H_2 \rangle \\ &= 4 + 12S^{HH} = 4(1 + 3S^{HH}),\end{aligned}\quad (19.191)$$

where the second equality comes from the fact that $|H_1\rangle$, i.e., a 1 s atomic orbital is normalized. We define $\langle H_1 | H_2 \rangle = S^{HH}$, i.e., an overlap integral between two adjacent hydrogen atoms. Note also that $\langle H_i | H_j \rangle$ ($1 \leq i, j \leq 4; i \neq j$) is the same because of the symmetry requirement.

Thus, as a normalized SALC we have

$$H^{(A_1)} \equiv \frac{|H_1 + H_2 + H_3 + H_4\rangle}{2\sqrt{1 + 3S^{HH}}}. \quad (19.192)$$

Defining a denominator as $c \left(= 2\sqrt{1 + 3S^{HH}}\right)$, we have

$$H^{(A_1)} = |H_1 + H_2 + H_3 + H_4\rangle / c. \quad (19.193)$$

Also, we have

$$\begin{aligned}\langle H_1 + H_2 - H_3 - H_4 | H_1 + H_2 - H_3 - H_4 \rangle &= 4\langle H_1 | H_1 \rangle - 4\langle H_1 | H_2 \rangle \\ &= 4 - 4S^{HH} = 4(1 - S^{HH}).\end{aligned}\quad (19.194)$$

Thus, we have

$$H_1^{(T_2)} \equiv \frac{|H_1 + H_2 - H_3 - H_4\rangle}{2\sqrt{1 - S^{HH}}}. \quad (19.195)$$

Also defining a denominator as $d \left(= 2\sqrt{1 - S^{HH}}\right)$, we have

$$H_1^{(T_2)} \equiv |H_1 + H_2 - H_3 - H_4\rangle/d.$$

Similarly, we define other hydrogen-based SALCs as

$$H_2^{(T_2)} \equiv |H_1 - H_2 + H_3 - H_4\rangle/d,$$

$$H_3^{(T_2)} \equiv |H_1 - H_2 - H_3 + H_4\rangle/d.$$

The next step is to construct MOs using the above SALCs. To this end, we make a linear combination using SALCs belonging to the same irreducible representations. In the case of A_1 , we choose $H^{(A_1)}$ and $C2s$. Naturally, we anticipate two linear combinations of

$$a_1 H^{(A_1)} + b_1 C2s, \quad (19.196)$$

where a_1 and b_1 are arbitrary constants. On the basis of the discussions of projection operators in Sect. 18.7, both the above two linear combinations belong to A_1 as well. Similarly, according to the projection operators $P_{1(1)}^{(T_2)}$, $P_{2(2)}^{(T_2)}$, and $P_{3(3)}^{(T_2)}$, we make three sets of linear combinations

$$q_1 P_{1(1)}^{(T_2)} H_1 + r_1 C2p_x; q_2 P_{2(2)}^{(T_2)} H_1 + r_2 C2p_y; q_3 P_{3(3)}^{(T_2)} H_1 + r_3 C2p_z, \quad (19.197)$$

where q_1 , r_1 , etc. are arbitrary constants. These three sets of linear combinations belong to individual “addresses” 1, 2, and 3 of T_2 . What we have to do is to determine coefficients of the above MOs and to normalize them by solving the secular equations. With two different energy eigenvalues, we get two orthogonal (i.e., linearly independent) MOs for the individual four sets of linear combinations of (19.196) and (19.197). Thus, total eight linear combinations constitute MOs of methane.

In light of (19.183), the secular equation can be reduced as follows according to the representations A_1 and T_2 . There we have changed the order of entries in the equation so that we can deal with the equation easily. Then we have

$$\begin{vmatrix} H_{11} - \lambda & H_{12} - \lambda S_{12} & & & & & \\ H_{21} - \lambda S_{21} & H_{22} - \lambda & & & & & \\ & & G_{11} - \lambda & G_{12} - \lambda T_{12} & & & \\ & & G_{21} - \lambda T_{21} & G_{22} - \lambda & & & \\ & & & & F_{11} - \lambda & F_{12} - \lambda V_{12} & \\ & & & & F_{21} - \lambda V_{21} & F_{22} - \lambda & \\ & & & & & & K_{11} - \lambda & K_{12} - \lambda W_{12} \\ & & & & & & K_{21} - \lambda W_{21} & K_{22} - \lambda \end{vmatrix} = 0, \quad (19.198)$$

where off-diagonal elements are all zero except for those explicitly described. This is because of the symmetry requirement (see Sect. 19.2). Thus, in (19.198) the secular equation is decomposed into four (2, 2) blocks. The first block is pertinent to A_1 of a hydrogen-based component and carbon-based component from the left, respectively. Lower three blocks are pertinent to T_2 of hydrogen-based and carbon-based components from the left, respectively, in order of $P_{1(1)}^{(T_2)}$, $P_{2(2)}^{(T_2)}$, and $P_{3(3)}^{(T_2)}$ SALCs from the top. The notations follow those of (19.59).

We compute these equations. The calculations are equivalent to solving the following four two-dimensional secular equations:

$$\begin{vmatrix} H_{11} - \lambda & H_{12} - \lambda S_{12} \\ H_{21} - \lambda S_{21} & H_{22} - \lambda \end{vmatrix} = 0,$$

$$\begin{vmatrix} G_{11} - \lambda & G_{12} - \lambda T_{12} \\ G_{21} - \lambda T_{21} & G_{22} - \lambda \end{vmatrix} = 0,$$

$$\begin{vmatrix} F_{11} - \lambda & F_{12} - \lambda V_{12} \\ F_{21} - \lambda V_{21} & F_{22} - \lambda \end{vmatrix} = 0,$$

$$\begin{vmatrix} K_{11} - \lambda & K_{12} - \lambda W_{12} \\ K_{21} - \lambda W_{21} & K_{22} - \lambda \end{vmatrix} = 0. \quad (19.199)$$

Notice that these four secular equations are the same as a (2, 2) determinant of (19.59) in the form of a secular equation. Note, at the same time, that while (19.59) did not assume SALCs, (19.199) takes account of SALCs. That is, (19.199) expresses a secular equation with respect to two SALCs that belong to the same irreducible representation.

The first equation of (19.199) reads as

$$(1 - S_{12}^2)\lambda^2 - (H_{11} + H_{22} - 2H_{12}S_{12})\lambda + H_{11}H_{22} - H_{12}^2 = 0. \quad (19.200)$$

In (19.200) we define quantities as follows:

$$\begin{aligned} S_{12} &\equiv \int H^{(A_1)} C2s d\tau \\ &\equiv \langle H^{(A_1)} | C2s \rangle = \frac{\langle H_1 + H_2 + H_3 + H_4 | C2s \rangle}{2\sqrt{1 + 3S^{HH}}} = \frac{2\langle H_1 | C2s \rangle}{\sqrt{1 + 3S^{HH}}} \end{aligned}$$

$$= \frac{2S_{A_1}^{CH}}{\sqrt{1 + 3S^{HH}}}. \quad (19.201)$$

In (19.201), an overlap integral between hydrogen atomic orbitals and $C2s$ is identical from the symmetry requirement and it is defined as

$$S_{A_1}^{CH} \equiv \langle H_1 | C2s \rangle. \quad (19.202)$$

Also in (19.200), other quantities are defined as follows:

$$H_{11} \equiv \langle H^{(A_1)} | \mathfrak{H} H^{(A_1)} \rangle = \frac{\alpha^H + 3\beta^{HH}}{1 + 3S^{HH}}, \quad (19.203)$$

$$H_{22} \equiv \langle C2s | \mathfrak{H} C2s \rangle, \quad (19.204)$$

$$H_{12} \equiv \langle H^{(A_1)} | \mathfrak{H} C2s \rangle = \frac{\langle H_1 + H_2 + H_3 + H_4 | \mathfrak{H} C2s \rangle}{2\sqrt{1 + 3S^{HH}}} = \frac{2\langle H_1 | \mathfrak{H} C2s \rangle}{\sqrt{1 + 3S^{HH}}} = \frac{2\beta_{A_1}^{CH}}{\sqrt{1 + 3S^{HH}}}, \quad (19.205)$$

where $\mathfrak{H}$ is a Hamiltonian of a methane molecule. In (19.203) and (19.205), moreover, we define the quantities as

$$\alpha^H \equiv \langle H_1 | \mathfrak{H} H_1 \rangle, \beta^{HH} \equiv \langle H_1 | \mathfrak{H} H_2 \rangle, \beta_{A_1}^{CH} \equiv \langle H_1 | \mathfrak{H} C2s \rangle. \quad (19.206)$$

The quantity of H_{11} is a “Coulomb” integral of the hydrogen-based SALC that involves four hydrogen atoms. Solving the first equation of (19.199), we get

$$\lambda = \frac{H_{11} + H_{22} - 2H_{12}S_{12} \pm \sqrt{(H_{11} - H_{22})^2 + 4[H_{12}^2 + H_{11}H_{22}S_{12}^2 - H_{12}S_{12}(H_{11} + H_{22})]}}{2(1 - S_{12}^2)}. \quad (19.207)$$

Similarly, we obtain related solutions for the latter three eigenvalue equations of (19.199). With the second equation of (19.199), for instance, we have

$$(1 - T_{12}^2)\lambda^2 - (G_{11} + G_{22} - 2G_{12}T_{12})\lambda + G_{11}G_{22} - G_{12}^2 = 0. \quad (19.208)$$

Solving this, we get

$$\lambda = \frac{G_{11} + G_{22} - 2G_{12}T_{12} \pm \sqrt{(G_{11} - G_{22})^2 + 4[G_{12}^2 + G_{11}G_{22}T_{12}^2 - G_{12}T_{12}(G_{11} + G_{22})]}}{2(1 - T_{12}^2)}. \quad (19.209)$$

In (19.208) and (19.209) we define these quantities as follows:

$$\begin{aligned} T_{12} &\equiv \int H_1^{(T_2)} C2p_x d\tau \equiv \langle H_1^{(T_2)} | C2p_x \rangle = \frac{\langle H_1 + H_2 - H_3 - H_4 | C2p_x \rangle}{2\sqrt{1 - S^{HH}}} \\ &= \frac{2\langle H_1 | C2p_x \rangle}{\sqrt{1 - S^{HH}}} = \frac{2S_{T_2}^{CH}}{\sqrt{1 - S^{HH}}}, \end{aligned} \quad (19.210)$$

$$G_{11} \equiv \langle H_1^{(T_2)} | \mathfrak{H} H_1^{(T_2)} \rangle = \frac{\alpha^H - \beta^{HH}}{1 - S^{HH}}, \quad (19.211)$$

$$G_{22} \equiv \langle C2p_x | \mathfrak{H} C2p_x \rangle, \quad (19.212)$$

$$\begin{aligned} G_{12} &\equiv \langle H_1^{(T_2)} | \mathfrak{H} C2p_x \rangle = \frac{\langle H_1 + H_2 - H_3 - H_4 | \mathfrak{H} C2p_x \rangle}{2\sqrt{1 - S^{HH}}} \\ &= \frac{2\langle H_1 | \mathfrak{H} C2p_x \rangle}{\sqrt{1 - S^{HH}}} = \frac{2\beta_{T_2}^{CH}}{\sqrt{1 - S^{HH}}}. \end{aligned} \quad (19.213)$$

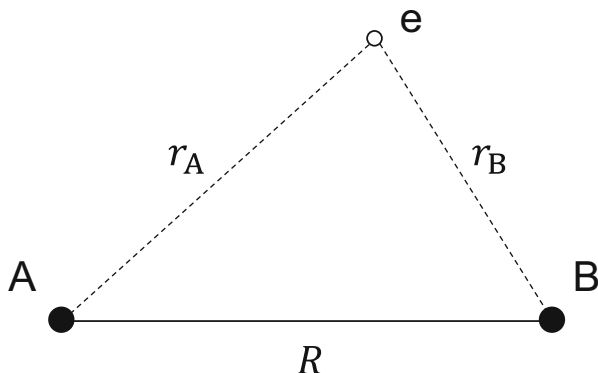
In the above equations, we have further defined integrals such that

$$S_{T_2}^{CH} \equiv \langle H_1 | C2p_x \rangle, \beta_{T_2}^{CH} \equiv \langle H_1 | \mathfrak{H} C2p_x \rangle. \quad (19.214)$$

In (19.210), an overlap integral T_{12} between four hydrogen atomic orbitals and $C2p_x$ is identical again from the symmetry requirement. That is, the integrals of (19.213) are *additive* regarding a product of plus components of a hydrogen atomic orbital of (19.187) and $C2p_x$ as well as another product of minus components of a hydrogen atomic orbital of (19.187) and $C2p_x$; see Fig. 19.14. Notice that $C2p_x$ has a node on yz -plane.

The third and fourth equations of (19.199) give exactly the same eigenvalues as that given in (19.209). This is obvious from the fact that all the latter three equations of (19.199) are associated with a irreducible representation T_2 . The corresponding three MOs are triply degenerate.

Fig. 19.15 Configuration of electron (e) and hydrogen nuclei (A and B). r_A and r_B denote a separation between the electron and A and that between the electron and B, respectively. R denotes a separation between A and B



In (19.207) and (19.209), the plus sign gives a higher orbital energy and the minus sign gives a lower energy. Equations (19.207) and (19.209), however, look somewhat complicated. To simplify the situation, (i) in, e.g., (19.207) let us consider a case where $|H_{11}| \gg |H_{22}|$ or $|H_{11}| \ll |H_{22}|$. In that case, $(H_{11} - H_{22})^2$ dominates inside a square root, and so ignoring $-2H_{12}S_{12}$ and $-S_{12}^2$ we have $\lambda \approx H_{11}$ or $\lambda \approx H_{22}$. Inserting these values into (19.199), we have either $\Psi \approx H^{(A_1)}$ or $\Psi \approx C2s$, where Ψ is a resulting MO. This implies that no interaction would arise between $H^{(A_1)}$ and $C2s$. (ii) If, however, $H_{11} = H_{22}$, we would get

$$\lambda = \frac{H_{11} - H_{12}S_{12} \pm |H_{12} - H_{11}S_{12}|}{1 - S_{12}^2}. \quad (19.215)$$

As $H_{12} - H_{11}S_{12}$ is positive or negative, we have a following alternative:

Case I: $H_{12} - H_{11}S_{12} > 0$. We have $\lambda_H = \frac{H_{11}+H_{12}}{1+S_{12}}$ and $\lambda_L = \frac{H_{11}-H_{12}}{1-S_{12}}$, where $\lambda_H > \lambda_L$.

Case II: $H_{12} - H_{11}S_{12} < 0$. We have $\lambda_H = \frac{H_{11}-H_{12}}{1-S_{12}}$ and $\lambda_L = \frac{H_{11}+H_{12}}{1+S_{12}}$, where $\lambda_H > \lambda_L$.

In the above cases, we would anticipate maximum orbital mixing between $H^{(A_1)}$ and $C2s$. (iii) If H_{11} and H_{22} are moderately different in between the above (i) and (ii), the orbital mixing is likely to be moderate. This is an actual situation. With (19.209) we have eigenvalues expressed similarly to those of (19.207). Therefore, classifications related to the above Cases I and II hold with $G_{12} - G_{11}T_{12}$, $F_{12} - F_{11}V_{12}$, and $K_{12} - K_{11}W_{12}$ as well.

In spite of simplicity of (19.215), the quantities H_{11} , H_{12} , and S_{12} are hard to calculate. In general cases including the present example (i.e., methane), the difficulty in calculating these quantities results essentially from the fact that we are dealing with a many-particle interactions that include electron repulsion. Nonetheless, for the simplest case of a hydrogen molecular ion H_2^+ the estimation is feasible [3]. Let us estimate $H_{12} - H_{11}S_{12}$ quantitatively according to Atkins and Friedman [3].

The Hamiltonian of H_2^+ is described as

$$H = -\frac{\hbar^2}{2m} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r_A} - \frac{e^2}{4\pi\epsilon_0 r_B} + \frac{e^2}{4\pi\epsilon_0 R}, \quad (19.216)$$

where m is an electron rest mass and other symbols are defined in Fig. 19.15; the last term represents the repulsive interaction between the two hydrogen nuclei. To estimate the quantities in (19.215), it is convenient to use dimensionless ellipsoidal coordinates $\begin{pmatrix} \mu \\ \nu \\ \phi \end{pmatrix}$ such that [3].

$$\mu = \frac{r_A + r_B}{R} \quad \text{and} \quad \nu = \frac{r_A - r_B}{R}. \quad (19.217)$$

The quantity ϕ is an azimuthal angle around the molecular axis (i.e., the straight line connecting the two nuclei). Then, we have

$$\begin{aligned} H_{11} &= \langle A | H | A \rangle = \left\langle A \left| -\frac{\hbar^2}{2m} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r_A} \right| A \right\rangle + \left\langle A \left| -\frac{e^2}{4\pi\epsilon_0 r_B} \right| A \right\rangle \\ &\quad + \left\langle A \left| \frac{e^2}{4\pi\epsilon_0 R} \right| A \right\rangle \\ &= E_{1s} - \frac{e^2}{4\pi\epsilon_0} \left\langle A \left| \frac{1}{r_B} \right| A \right\rangle + \frac{e^2}{4\pi\epsilon_0 R}, \end{aligned} \quad (19.218)$$

where E_{1s} is the same as that given in (3.258) with $Z = n = 1$ and μ replaced with m (i.e., $E_{1s} = -\frac{\hbar^2}{2ma^2}$). Using a coordinate representation of (3.301), we have

$$|A\rangle = \sqrt{\frac{1}{\pi}} a^{-3/2} e^{-r_A/a}. \quad (19.219)$$

Moreover considering (19.217), we have

$$\left\langle A \left| \frac{1}{r_B} \right| A \right\rangle = \frac{1}{\pi a^3} \int d\tau e^{-2r_A/a} \frac{1}{r_B}.$$

Converting Cartesian coordinates to ellipsoidal coordinates [3] such that

$$\int d\tau = \left(\frac{R}{2}\right)^3 \int_0^{2\pi} d\phi \int_1^\infty d\mu \int_{-1}^1 d\nu (\mu^2 - \nu^2),$$

we have

$$\begin{aligned}
 \left\langle A \left| \frac{1}{r_B} \right| A \right\rangle &= \frac{R^3}{8\pi a^3} \cdot 2\pi \int_1^\infty d\mu \int_{-1}^1 d\nu (\mu^2 - \nu^2) \frac{e^{-(\mu+\nu)R/a}}{\frac{R}{2}(\mu - \nu)} \\
 &= \frac{R^2}{2a^3} \int_1^\infty d\mu \int_{-1}^1 d\nu (\mu + \nu) e^{-(\mu+\nu)R/a}. \quad (19.220)
 \end{aligned}$$

Putting $I = \int_1^\infty d\mu \int_{-1}^1 d\nu (\mu + \nu) e^{-(\mu+\nu)R/a}$, we obtain

$$I = \int_1^\infty \mu e^{-\mu R/a} d\mu \int_{-1}^1 e^{-\nu R/a} d\nu + \int_1^\infty e^{-\mu R/a} d\mu \int_{-1}^1 \nu e^{-\nu R/a} d\nu. \quad (19.221)$$

The above definite integrals can readily be calculated using the methods described in Sect. 3.7.2; see, e.g., (3.262) and (3.263). For instance, we have

$$\int_{-1}^1 e^{-c\nu} d\nu = \left[\frac{e^{-c\nu}}{-c} \right]_{-1}^1 = \frac{1}{c} (e^c - e^{-c}). \quad (19.222)$$

Differentiating (19.222) with respect to the parameter c , we get

$$\int_{-1}^1 \nu e^{-c\nu} d\nu = \frac{1}{c^2} [(1-c)e^c - (1+c)e^{-c}]. \quad (19.223)$$

In the present case, c is to be replaced with R/a . Other calculations of the definite integrals are left for readers. Thus, we obtain

$$I = \frac{a^3}{R^3} \left[2 - 2 \left(1 + \frac{R}{a} \right) e^{-2R/a} \right].$$

In turn, we have

$$\left\langle A \left| \frac{1}{r_B} \right| A \right\rangle = \frac{R^2}{2a^3} \cdot I = \frac{1}{R} \left[1 - \left(1 + \frac{R}{a} \right) e^{-2R/a} \right].$$

Introducing a symbol j_0 according to Atkins and Friedman given by [3].

$$j_0 \equiv \frac{e^2}{4\pi\epsilon_0},$$

we finally obtain

$$H_{11} = E_{1s} - \frac{j_0}{R} \left[1 - \left(1 + \frac{R}{a} \right) e^{-\frac{2R}{a}} \right] + \frac{j_0}{R}. \quad (19.224)$$

The quantity S_{12} appearing in (19.199) can be obtained as follows:

$$S_{12} = \langle B|A \rangle = \frac{R^3}{8\pi a^3} \int_0^{2\pi} d\phi \int_1^\infty d\mu \int_{-1}^1 d\nu (\mu^2 - \nu^2) e^{-\mu R/a}, \quad (19.225)$$

where we used $|B\rangle = \sqrt{\frac{1}{\pi}} a^{-3/2} e^{-r_B/a}$ and (19.217). Noting that

$$\int_1^\infty d\mu \int_{-1}^1 d\nu (\mu^2 - \nu^2) e^{-\mu R/a} = \int_1^\infty d\mu \left(2\mu^2 - \frac{2}{3} \right) e^{-\mu R/a}$$

and following procedures similar to those described above, we get

$$S_{12} = \left[1 + \frac{R}{a} + \frac{1}{3} \left(\frac{R}{a} \right)^2 \right] e^{-R/a}. \quad (19.226)$$

In turn, for H_{12} in (19.199) we have

$$\begin{aligned} H_{12} &= \langle A|H|B \rangle = \left\langle A \left| -\frac{\hbar^2}{2m} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r_B} \right| B \right\rangle + \left\langle A \left| -\frac{e^2}{4\pi\epsilon_0 r_A} \right| B \right\rangle \\ &\quad + \left\langle A \left| \frac{e^2}{4\pi\epsilon_0 R} \right| B \right\rangle \\ &= E_{1s} \langle A|B \rangle - \frac{e^2}{4\pi\epsilon_0} \left\langle A \left| \frac{1}{r_A} \right| B \right\rangle + \frac{e^2}{4\pi\epsilon_0 R} \langle A|B \rangle. \end{aligned} \quad (19.227)$$

Note that in (19.227) $|B\rangle$ is an eigenfunction belonging to $E_{1s} = -\frac{\hbar^2}{2ma^2}$ that is an eigenvalue of an operator $-\frac{\hbar^2}{2m} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r_B}$. From (3.258), we estimate E_{1s} to be 13.61 eV. Using (19.217) and converting Cartesian coordinates to ellipsoidal coordinates once again, we get

$$\left\langle A \left| \frac{1}{r_A} \right| B \right\rangle = \frac{1}{a} \left(1 + \frac{R}{a} \right) e^{-R/a}.$$

Thus, we have

$$H_{12} = \left(E_{1s} + \frac{j_0}{R} \right) S_{12} - \frac{j_0}{a} \left(1 + \frac{R}{a} \right) e^{-\frac{R}{a}}. \quad (19.228)$$

We are now in the position to evaluate $H_{12} - H_{11}S_{12}$ in (19.215). According to Atkins and Friedman [3], we define the following notations:

$$j' \equiv j_0 \left\langle A \left| \frac{1}{r_B} \right| A \right\rangle \quad \text{and} \quad k' \equiv j_0 \left\langle A \left| \frac{1}{r_A} \right| B \right\rangle.$$

Then, we have

$$H_{12} - H_{11}S_{12} = -k' + j'S_{12}. \quad (19.229)$$

The calculation of this quantity is straightforward. The result is

$$\begin{aligned} & H_{12} - H_{11}S_{12} \\ &= j_0 \left(\frac{1}{R} - \frac{2R}{3a^2} \right) e^{-R/a} - \frac{j_0}{R} \left(1 + \frac{R}{a} \right) \left[1 + \frac{R}{a} + \frac{1}{3} \left(\frac{R}{a} \right)^2 \right] e^{-3R/a} \\ &= \frac{j_0}{a} \left(\frac{a}{R} - \frac{2R}{3a} \right) e^{-R/a} - \frac{j_0}{a} \left(1 + \frac{a}{R} \right) \left[1 + \frac{R}{a} + \frac{1}{3} \left(\frac{R}{a} \right)^2 \right] e^{-3R/a}. \end{aligned} \quad (19.230)$$

In (19.230) we notice that whereas the second term is always negative, the first term may be negative or positive depending upon R . We could not tell a priori whether $H_{12} - H_{11}S_{12}$ is negative accordingly. If we had $R \ll a$, (19.230) would become positive.

Let us then make a quantitative estimation. The Bohr radius a is about 52.9 pm (using an electron rest mass). As an experimental result, R is approximately 106 pm [3]. Hence, for a H_2^+ ion we have

$$R/a \approx 2.0.$$

Using this number, we get

$$H_{12} - H_{11}S_{12} \approx -0.13j_0/a < 0.$$

We estimate $H_{12} - H_{11}S_{12}$ to be ~ -3.5 eV. Therefore, from (19.215) we get

$$\lambda_L = \frac{H_{11} + H_{12}}{1 + S_{12}} \quad \text{and} \quad \lambda_H = \frac{H_{11} - H_{12}}{1 - S_{12}}, \quad (19.231)$$

where λ_L and λ_H indicate the lower and higher energy eigenvalues, respectively. Namely, Case II in the above is more likely. Correspondingly, for MOs we have

$$\Psi_L = \frac{|A\rangle + |B\rangle}{\sqrt{2(1 + S_{12})}} \quad \text{and} \quad \Psi_H = \frac{|A\rangle - |B\rangle}{\sqrt{2(1 - S_{12})}}, \quad (19.232)$$

where Ψ_L and Ψ_H belong to λ_L and λ_H , respectively. The results are virtually the same as those given in (19.101)–(19.105) of Sect. 19.1. In (19.101) and Fig. 19.6, however, we merely assumed that $\lambda_1 = \frac{\alpha+\beta}{1+S}$ is lower than $\lambda_2 = \frac{\alpha-\beta}{1-S}$. Here, we have confirmed that this is truly the case. A chemical bond is formed in such a way that an electron is distributed as much as possible along the molecular axis (see Fig. 19.4 for a schematic) and that in this configuration a minimized orbital energy is achieved.

In our present case, H_{11} in (19.203) and H_{22} in (19.204) should differ. This is also the case with G_{11} in (19.211) and G_{22} in (19.212). Here we return back to (19.199). Suppose that we get a MO by solving, e.g., the first secular equation of (19.199) such that

$$\Psi = a_1 H^{(A_1)} + b_1 C2s.$$

From the secular equation, we get

$$a_1 = -\frac{H_{12} - \lambda S_{12}}{H_{11} - \lambda} b_1, \quad (19.233)$$

where S_{12} and H_{12} were defined as in (19.201) and (19.205), respectively. A normalized MO $\tilde{\Psi}$ is described by

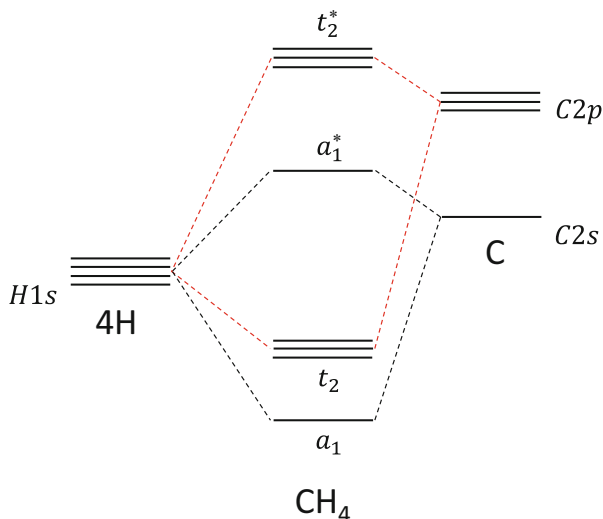
$$\tilde{\Psi} = \frac{a_1 H^{(A_1)} + b_1 C2s}{\sqrt{a_1^2 + b_1^2 + 2a_1 b_1 S_{12}}}. \quad (19.234)$$

Thus, according to two different energy eigenvalues λ we will get two linearly independent MOs. Other three secular equations are dealt with similarly.

From the energetical consideration of H_2^+ we infer that $a_1 b_1 > 0$ with a bonding MO and $a_1 b_1 < 0$ with an anti-bonding MO. Meanwhile, since both $H^{(A_1)}$ and $C2s$ belong to the irreducible representation A_1 , so does $\tilde{\Psi}$ on the basis of the discussion on the projection operators of Sect. 18.7. Thus, we should be able to construct proper MOs that belong to A_1 . Similarly, we get proper MOs belonging to T_2 by solving other three secular equations of (19.199). In this case, three bonding MOs are triply degenerate, so are three anti-bonding MOs. All these six MOs belong to the irreducible representation T_2 . Thus, we can get a complete set of MOs for methane. These eight MOs span the representation space V^8 .

Fig. 19.16 Probable energy diagram and MO symmetry species of methane.

(Adapted from http://www.science.oregonstate.edu/~gablek/CH334/Chapter1/methane_MOs.htm with kind permission of Professor Kevin P. Gable)



To precisely determine the energy levels, we need to take more elaborate approaches to approximate and calculate various parameters that appear in (19.199). At the same time, we need to perform detailed experiments including spectroscopic measurements and interpret those results carefully [4]. Taking account of these situations, Fig. 19.16 [5] displays as an example of MO calculations that give a probable energy diagram and MO symmetry species of methane. The diagram comprises a ground state bonding a_1 state and its corresponding anti-bonding state a_1^* along with triply degenerate bonding t_2 states, and their corresponding anti-bonding state t_2^* .

We emphasize that the said elaborate approaches ensue truly from the “paper-and-pencil” methods based upon group theory. The group theory thus supplies us with a powerful tool and clear guideline for addressing various quantum chemical problems, a few of which we are introducing as an example in this book.

Finally, let us examine the optical transition of methane. In this case, we have to consider electronic configurations of the initial and final states. If we are dealing with optical absorption, the initial state is the ground state A_1 that is described by the totally symmetric representation. The final state, however, will be an excited state, which is described by a direct-product representation related to the two states that are associated with the optical transition. The matrix element is expressed as

$$\left\langle \Theta^{(\alpha \times \beta)} | \mathbf{e}_e \cdot \mathbf{P} | \Theta^{(A_1)} \right\rangle, \quad (19.235)$$

where $\Theta^{(A_1)}$ stands for an electronic configuration of the totally symmetric ground state; $\Theta^{(\alpha \times \beta)}$ denotes an electronic configuration of an excited state represented by a direct-product representation pertinent to irreducible representations α and β ; $\mathbf{P}$ is an electric dipole operator and $\mathbf{e}_e$ is a unit polarization vector of the electric field. From a

character table for T_d , we find that $\epsilon_e \cdot \mathbf{P}$ belongs to the irreducible representation T_2 (see Table 19.12).

The ground state electronic configuration is A_1 (totally symmetric). It is denoted by

$$a_1^2 t_2^2 t_2'^2 t_2''^2,$$

where three T_2 states are distinguished by a prime and double prime. For possible configuration of excited states, we have

$$\begin{aligned} a_1^2 t_2 t_2'^2 t_2''^2 t_2^* (A_1 \rightarrow T_2 \times T_2), a_1^2 t_2 t_2'^2 t_2''^2 a_1^* (A_1 \rightarrow T_2 \times A_1 = T_2), \\ a_1 t_2^2 t_2'^2 t_2''^2 t_2^* (A_1 \rightarrow A_1 \times T_2 = T_2), a_1 t_2^2 t_2'^2 t_2''^2 a_1^* (A_1 \rightarrow A_1 \times A_1 = A_1). \end{aligned} \quad (19.236)$$

In (19.236), we have optical excitations of $t_2 \rightarrow t_2^*$, $t_2 \rightarrow a_1^*$, $a_1 \rightarrow t_2^*$, and $a_1 \rightarrow a_1^*$, respectively. Consequently, the excited states are denoted by $T_2 \times T_2$, T_2 , T_2 , and A_1 , respectively. Since $\epsilon_e \cdot \mathbf{P}$ is Hermitian as seen in (19.51) of Sect. 19.2, we have

$$\left\langle \Theta^{(\alpha \times \beta)} | \epsilon_e \cdot \mathbf{P} | \Theta^{(A_1)} \right\rangle = \left\langle \Theta^{(A_1)} | \epsilon_e \cdot \mathbf{P} | \Theta^{(\alpha \times \beta)} \right\rangle^*. \quad (19.237)$$

Therefore, according to the general theory of Sect. 19.2, we need to examine whether $T_2 \times D^{(\alpha)} \times D^{(\beta)}$ contains A_1 to judge whether the optical transition is allowed. As mentioned above, $D^{(\alpha)} \times D^{(\beta)}$ is chosen from among $T_2 \times T_2$, T_2 , T_2 , and A_1 . The results are given as follows:

$$T_2 \times T_2 \times T_2 = 3T_1 + 4T_2 + 2E + A_2 + A_1, \quad (19.238)$$

$$T_2 \times T_2 = T_1 + T_2 + E + A_1, \quad (19.239)$$

$$T_2 \times A_1 = T_2. \quad (19.240)$$

Admittedly, (19.238) and (19.239) contain A_1 , and so the transition is allowed. As for (19.240), however, the transition is forbidden because it does not contain A_1 . In light of the character table of T_d (Table 19.12), we find that the allowed transitions (i.e., $t_2 \rightarrow t_2^*$, $t_2 \rightarrow a_1^*$, and $a_1 \rightarrow t_2^*$) equally take place in the direction polarized along the x -, y -, z -axes. This is often the case with molecules having higher symmetries such as methane. The transition $a_1 \rightarrow a_1^*$, however, is forbidden.

In the above argument including the energetic consideration, we could not tell magnitude relationship of MO energies or photon energies associated with the

optical transition. Once again, this requires more accurate calculations and experiments. Yet, the discussion we have developed gives us strong guiding principles in the investigation of molecular science.

References

1. Cotton FA (1990) Chemical applications of group theory, 3rd edn. Wiley, New York
2. Hamermesh M (1989) Group theory and its application to physical problems. Dover, New York
3. Atkins P, Friedman R (2005) Molecular quantum mechanics, 4th edn. Oxford University Press, Oxford
4. Anslyn EV, Dougherty DA (2006) Modern physical organic chemistry. University Science Books, Sausalito
5. Gable KP (2014) Molecular orbitals: methane. http://www.science.oregonstate.edu/~gablek/CH334/Chapter1/methane_MOs.htm

Chapter 20

Theory of Continuous Groups



In Chap. 16, we classified the groups into finite groups and infinite groups. We focused mostly on finite groups and their representations in the precedent chapters. Of the infinite groups, continuous groups and their properties have been widely investigated. In this chapter we think of the continuous groups as a collective term that include Lie groups and topological groups. The continuous groups are also viewed as the transformation group. Aside from the strict definition, we will see the continuous group as a natural extension of the rotation group or $SO(3)$ that we studied briefly in Chap. 17. Here we reconstruct $SO(3)$ on the basis of the notion of infinitesimal rotation. We make the most of the exponential functions of matrices the theory of which we have explored in Chap. 15. In this context, we study the basic properties and representations of the special unitary group $SU(2)$ that has close relevance to $SO(3)$. Thus, we focus our attention on the representation theory of $SU(2)$ and $SO(3)$. The results are associated with the (generalized) angular momenta which we dealt with in Chap. 3. In particular, we show that the spherical surface harmonics constitute the basis functions of the representation space of $SO(3)$.

Finally, we study various important properties of $SU(2)$ and $SO(3)$ within a framework of Lie groups and Lie algebras. The last sections comprise the description of abstract ideas, but those are useful for us to comprehend the constitution of the group theory from a higher perspective.

20.1 Introduction: Operators of Rotation and Infinitesimal Rotation

To characterize the continuous transformation appropriately, we have to describe an infinitesimal transformation of rotation. This is because a rotation with a finite angle is attained by an infinite number of infinitesimal rotations.

First let us consider how a function Ψ is transformed by the rotation (see Sect. 19.1). Here we express Ψ by

$$\Psi = (\psi_1 \ \psi_2 \ \cdots \ \psi_d) \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_d \end{pmatrix}, \quad (20.1)$$

where $\{\psi_1, \dots, \psi_d\}$ span the representation space relevant to the rotation; d is a dimension of the representation space; $c_1, c_2, \dots, c_d$ are coefficients (or coordinates). Henceforth, we make it a rule to denote a vector in a representation space as in (20.1). This is a natural extension of (11.13). We assume ψ_ν ($\nu = 1, 2, \dots, d$) as a vector component and ψ_ν is normally expressed as a function of a position vector $\mathbf{x}$ in a real three-dimensional space (see Chap. 18). Considering for simplicity a one-dimensional representation space (i.e., $d = 1$) and omitting the index ν , as a transformation of ψ we have

$$R_\theta \psi(\mathbf{x}) = \psi[R_\theta^{-1}(\mathbf{x})], \quad (20.2)$$

where the index θ indicates the rotation angle in a real space whose dimension is two or three. Note that the dimensions of the real space and representation space may or may not be the same. Equation (20.2) is essentially the same as (19.11).

As a three-dimensional matrix representation of rotation, we have, e.g.,

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (20.3)$$

The matrix R_θ represents a rotation around the z -axis in $\mathbb{R}^3$; see Fig. 11.2 and (11.31). Borrowing the notation of (17.3) and (17.12), we have

$$R_\theta(\mathbf{x}) = (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}. \quad (20.4)$$

Therefore, as an inverse operator we get

$$R_\theta^{-1}(\mathbf{x}) = (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}. \quad (20.5)$$

Let us think of an infinitesimal rotation. Taking a first-order quantity of an infinitesimal angle θ , (20.5) can be approximated as

$$\begin{aligned}
 R_{\theta}^{-1}(\mathbf{x}) &\approx (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} 1 & \theta & 0 \\ -\theta & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = (\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3) \begin{pmatrix} x + \theta y \\ y - \theta x \\ z \end{pmatrix} \\
 &= (x + \theta y)\mathbf{e}_1 + (y - \theta x)\mathbf{e}_2 + z\mathbf{e}_3.
 \end{aligned} \tag{20.6}$$

Putting

$$\psi(\mathbf{x}) = \psi(x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3) \equiv \psi(x, y, z),$$

we have

$$R_{\theta}\psi(\mathbf{x}) = \psi[R_{\theta}^{-1}(\mathbf{x})] \tag{20.2}$$

or

$$\begin{aligned}
 R_{\theta}\psi(x, y, z) &\approx \psi(x + \theta y, y - \theta x, z) \\
 &= \psi(x, y, z) + \frac{\partial \psi}{\partial x} \theta y + \frac{\partial \psi}{\partial y} (-\theta x) \\
 &= \psi(x, y, z) - i\theta(-i) \left[x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right] \psi(x, y, z).
 \end{aligned}$$

Now, using a dimensionless operator $\mathcal{M}_z (=L_z/\hbar)$ such that

$$\mathcal{M}_z \equiv -i \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right), \tag{20.7}$$

we get

$$R_{\theta}\psi(x, y, z) \approx \psi(x, y, z) - i\theta \mathcal{M}_z \psi(x, y, z) = (1 - i\theta \mathcal{M}_z) \psi(x, y, z). \tag{20.8}$$

Note that the operator $\mathcal{M}_z$ has already appeared in Sect. 3.5 [also see (3.19), Sect. 3.2], but in this section we denote it by $\mathcal{M}_z$ to distinguish it from the previous notation M_z . This is because $\mathcal{M}_z$ and M_z are connected through a unitary similarity transformation (vide infra).

From (20.8), we express an infinitesimal rotation θ around the z -axis as

$$R_{\theta} = 1 - i\theta \mathcal{M}_z. \tag{20.9}$$

Next, let us think of a rotation of a finite angle θ . We have

$$R_\theta = (R_{\theta/n})^n, \quad (20.10)$$

where RHS of (20.10) implies that the rotation R_θ of a finite angle θ is attained by n successive infinitesimal rotations of $R_{\theta/n}$ with a large enough n . Taking a limit $n \rightarrow \infty$ [1, 2], we get

$$R_\theta = \lim_{n \rightarrow \infty} (R_{\theta/n})^n = \lim_{n \rightarrow \infty} \left(1 - i \frac{\theta}{n} \mathcal{M}_z\right)^n = \exp(-i\theta \mathcal{M}_z). \quad (20.11)$$

Recall the following definition of an exponential function other than (15.7) [3]:

$$e^x \equiv \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n. \quad (20.12)$$

Note that as in the case of (15.7), (20.12) holds when x represents a matrix. Thus, if a dimension of the representation space d is two or more, (20.9) should be read as

$$R_\theta = E - i\theta \mathcal{M}_z, \quad (20.13)$$

where E is a (d, d) identity matrix; $\mathcal{M}_z$ is represented by a (d, d) square matrix accordingly. As already shown in Chap. 15, an exponential function of a matrix is well-defined. Since $\mathcal{M}_z$ is Hermitian, $-i\mathcal{M}_z$ is anti-Hermitian (see Chaps. 2 and 3). Thus, using (20.11) R_θ can be expressed as

$$R_\theta = \exp(\theta A_z), \quad (20.14)$$

with

$$A_z \equiv -i\mathcal{M}_z. \quad (20.15)$$

Operators of this type play an essential role in continuous groups. This is because in (20.14) the operator A_z represents a Lie algebra, which in turn generates a Lie group. The Lie group is categorized as one of the continuous groups along with a topological group. In this chapter we further explore various characteristics of $SO(3)$ and $SU(2)$, both of which are Lie groups and have been fully investigated from various aspects. The Lie algebras frequently appear in the theory of continuous groups in combination with the Lie groups. Brief outline of the theory of Lie groups and Lie algebras will be given at the end of this chapter.

Let us further consider implications of (20.9) and (20.13). From those equations we have

$$\mathcal{M}_z = (1 - R_\theta)/i\theta \quad (20.16)$$

or

$$\mathcal{M}_z = (E - R_\theta)/i\theta. \quad (20.17)$$

As a very trivial case, we might well be able to think of a one-dimensional real space. Let the space extend along the z -axis and a unit vector of that space be $\mathbf{e}_3$. Then, with any rotation R_θ around the z -axis makes $\mathbf{e}_3$ unchanged (or invariant). That is, we have

$$R_\theta(\mathbf{e}_3) = \mathbf{e}_3.$$

With the three-dimensional case, we have

$$E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad R_\theta = \begin{pmatrix} 1 & -\theta & 0 \\ \theta & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (20.18)$$

Therefore, from (20.17) we get

$$\mathcal{M}_z = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (20.19)$$

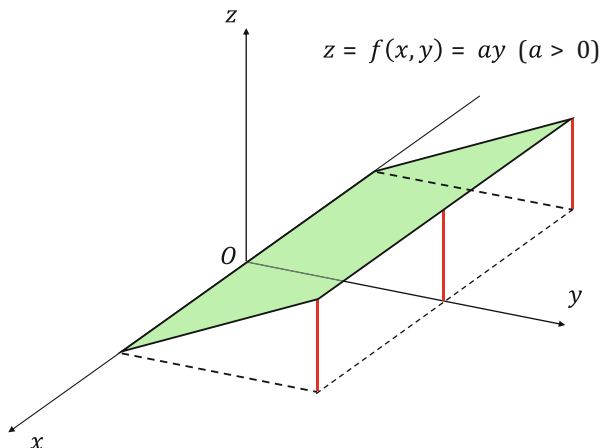
Next, viewing x , y , and z as vector components (i.e., functions), we think of the following equation:

$$\mathcal{M}_z(X) = (x \ y \ z) \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}. \quad (20.20)$$

The notation of (20.20) denotes the vector transformation in the representation space, in accordance with (11.37). In (20.20) we consider $(x \ y \ z)$ as basis vectors as in the case of $(\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3)$ of (17.2) and (17.3). In other words, we are thinking of a vector transformation in a three-dimensional representation space spanned by a set of basis vectors $(x \ y \ z)$. In this case, $\mathcal{M}_z$ represented by (20.19) operates on a vector (i.e., a function) X expressed by

$$X = (x \ y \ z) \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}, \quad (20.21)$$

Fig. 20.1 Function $f(x, y) = ay$ ($a > 0$). The function ay is drawn green. Only a part of $f(x, y) > 0$ is shown



where c_1 , c_2 , and c_3 are coefficients (or coordinates). Again, this notation is a natural extension of (11.13). We emphasize once more that $(x \ y \ z)$ does not represent coordinates but functions. As an example, we depict a function $f(x, y) = ay$ ($a > 0$) in Fig. 20.1. The function ay (drawn green in Fig. 20.1) behaves as a vector with respect to a linear transformation like a basis vector e_2 of the Cartesian coordinate. The “directionality” as a vector can easily be visualized with the function ay .

Since $\mathcal{M}_z$ of (20.19) is Hermitian, it must be diagonalized by unitary similarity transformation (Sect. 14.3). We find a diagonalizing unitary matrix P with respect to $\mathcal{M}_z$ such that

$$P = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{i}{\sqrt{2}} & -\frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (20.22)$$

Using (20.22), we rewrite (20.20) as

$$\begin{aligned}
\mathcal{M}_z(X) &= (x \ y \ z) P \cdot P^{-1} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} P \cdot P^{-1} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} \\
&= (x \ y \ z) \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{i}{\sqrt{2}} & -\frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&\times \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} -\frac{1}{\sqrt{2}}x - \frac{i}{\sqrt{2}}y & \frac{1}{\sqrt{2}}x - \frac{i}{\sqrt{2}}y & z \end{pmatrix} \\
&\times \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -\frac{1}{\sqrt{2}}c_1 + \frac{i}{\sqrt{2}}c_2 \\ \frac{1}{\sqrt{2}}c_1 + \frac{i}{\sqrt{2}}c_2 \\ c_3 \end{pmatrix}. \tag{20.23}
\end{aligned}$$

Here we define $\widetilde{M}_z$ such that

$$\widetilde{M}_z \equiv P^{-1} \mathcal{M}_z P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{20.24}$$

Equation (20.24) was obtained by putting $l = 1$ and shuffling the rows and columns of (3.158) in Chap. 3. That is, choosing a (real) unitary matrix R_3 given by

$$R_3 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \tag{17.116}$$

we have

$$R_3^{-1} \widetilde{M}_z R_3 = R_3^{-1} P^{-1} \mathcal{M}_z P R_3 = (P R_3)^{-1} \mathcal{M}_z (P R_3) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = M_z.$$

Hence, M_z that appeared as a diagonal matrix in (3.158) has been obtained by a unitary similarity transformation of $\mathcal{M}_z$ using a unitary operator PR_3 .

Comparing (20.20) and (20.23), we obtain useful information. If we choose $(x\ y\ z)$ for basis vectors, we could not have a representation that diagonalizes $\mathcal{M}_z$. If, however, we choose $\left(-\frac{1}{\sqrt{2}}x - \frac{i}{\sqrt{2}}y \quad \frac{1}{\sqrt{2}}x - \frac{i}{\sqrt{2}}y \quad z \right)$ for the basis vectors, we get a representation that diagonalizes $\mathcal{M}_z$. These basis vectors are sometimes referred to as a spherical basis. If we carefully look at the latter basis vectors (i.e., spherical basis), we notice that these vectors have the same transformation properties as $(Y_1^1\ Y_1^{-1}\ Y_1^0)$. In fact, using (3.216) and converting the spherical coordinates to Cartesian coordinates in (3.216), we get [4]

$$\begin{pmatrix} Y_1^1 & Y_1^{-1} & Y_1^0 \end{pmatrix} = \sqrt{\frac{3}{4\pi}} \frac{1}{r} \begin{pmatrix} -\frac{1}{\sqrt{2}}(x+iy) & \frac{1}{\sqrt{2}}(x-iy) & z \end{pmatrix}, \quad (20.25)$$

where r is a radial coordinate (i.e., a positive number) that appears in the polar coordinate (r, θ) . Note that the factor r contained in $(Y_1^1\ Y_1^{-1}\ Y_1^0)$ is invariant with respect to the rotation.

We can generalize the results and conform them to the discussion of related characteristics in representation spaces of higher dimensions. The spherical surface harmonics possess an important position for this (vide infra).

From Sect. 3.5, (3.150) we have

$$M_z Y_0^0 = 0,$$

where $Y_0^0(\theta, \phi) = \sqrt{1/4\pi}$. The above relation obviously shows that $Y_0^0(\theta, \phi)$ belongs to an eigenvalue of zero of M_z . This is consistent with the earlier comments made on the one-dimensional real space.

20.2 Rotation Groups: $SU(2)$ and $SO(3)$

Bearing in mind the aforementioned background knowledge, we further develop the theory of the rotation groups $SU(2)$ and $SO(3)$ and explore in detail various properties of them. First, we deal with the topics through conventional approaches. Later we will describe them systematically.

Following the argument developed above, we wish to relate an anti-Hermitian operator to a rotation operator of a finite angle θ represented by (17.12). We express Hermitian operators related to the x - and y -axes as

$$\mathcal{M}_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \quad \text{and} \quad \mathcal{M}_y = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}.$$

Then, anti-Hermitian operators corresponding to (20.15) are described by

$$\begin{aligned} A_z &= \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \\ A_y &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \end{aligned} \tag{20.26}$$

where $A_x \equiv -i\mathcal{M}_x$ and $A_y \equiv -i\mathcal{M}_y$. These are characterized by real skew-symmetric matrices.

Using (15.7), we calculate $\exp(\theta A_z)$ such that

$$\begin{aligned} \exp(\theta A_z) &= E + \theta A_z + \frac{1}{2!}(\theta A_z)^2 + \frac{1}{3!}(\theta A_z)^3 + \frac{1}{4!}(\theta A_z)^4 + \frac{1}{5!}(\theta A_z)^5 \\ &+ \cdots. \end{aligned} \tag{20.27}$$

We note that

$$\begin{aligned} A_z^2 &= \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = A_z^6, \quad A_z^4 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = A_z^8 \text{ etc.}; \\ A_z^3 &= \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = A_z^7, \quad A_z^5 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = A_z^9. \end{aligned}$$

Thus, we get

$$\begin{aligned}
& \exp(\theta A_z) \\
&= E + \frac{1}{2!}(\theta A_z)^2 + \frac{1}{4!}(\theta A_z)^4 + \cdots + \theta A_z + \frac{1}{3!}(\theta A_z)^3 + \frac{1}{5!}(\theta A_z)^5 + \cdots \\
&= \begin{pmatrix} 1 - \frac{1}{2!}\theta^2 + \frac{1}{4!}\theta^4 - \cdots & 0 & 0 \\ 0 & 1 - \frac{1}{2!}\theta^2 + \frac{1}{4!}\theta^4 - \cdots & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
&+ \begin{pmatrix} 0 & -\theta + \frac{1}{3!}\theta^3 - \frac{1}{5!}\theta^5 + \cdots & 0 \\ \theta - \frac{1}{3!}\theta^3 + \frac{1}{5!}\theta^5 + \cdots & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (20.28) \\
&= \begin{pmatrix} \cos \theta & 0 & 0 \\ 0 & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & -\sin \theta & 0 \\ \sin \theta & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.
\end{aligned}$$

Hence, we recover the matrix form of (17.12). Similarly, we get

$$\exp(\varphi A_x) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi & -\sin \varphi \\ 0 & \sin \varphi & \cos \varphi \end{pmatrix}, \quad (20.29)$$

$$\exp(\phi A_y) = \begin{pmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{pmatrix}. \quad (20.30)$$

Of the above operators, using (20.28) and (20.30) we recover (17.101) related to the Euler angles such that

$$\widetilde{R}_3 = \exp(\alpha A_z) \exp(\beta A_y) \exp(\gamma A_z). \quad (20.31)$$

Using exponential functions of matrices, (20.31) gives a transformation matrix containing the Euler angles and supplies us with a direct method to be able to make a matrix representation of $SO(3)$.

Although (20.31) is useful for the matrix representation in the real space, its extension to abstract representation spaces would somewhat be limited. In this respect, the method developed in $SU(2)$ can widely be used to produce representation matrices for a representation space of an arbitrary dimension. Let us start with constructing those representation matrices using $(2, 2)$ complex matrices.

20.2.1 Construction of $SU(2)$ Matrices

In Sect. 16.1 we mentioned a general linear group $GL(n, \mathbb{C})$. There are many subgroups of $GL(n, \mathbb{C})$. Among them, $SU(n)$, in particular $SU(2)$, plays a central role in theoretical physics and chemistry. In this section we examine how to construct $SU(2)$ matrices.

Definition 20.1 A group consisting of (n, n) unitary matrices with a determinant 1 is called a special unitary group of degree n and denoted by $SU(n)$.

In Chap. 14 we mentioned that the “absolute value” of determinant of a unitary matrix is 1. In Definition 20.1, however, there is an additional constraint in that we are dealing with unitary matrices whose determinant is 1.

The $SU(2)$ matrices U have the following general form:

$$U = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad (20.32)$$

where a, b, c , and d are complex numbers. According to Definition 20.1 we try to seek conditions for these numbers. The unitarity condition $UU^\dagger = U^\dagger U = E$ reads as

$$\begin{aligned} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix} &= \begin{pmatrix} |a|^2 + |b|^2 & ac^* + bd^* \\ a^*c + b^*d & |c|^2 + |d|^2 \end{pmatrix} \\ &= \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} |a|^2 + |c|^2 & a^*b + c^*d \\ ab^* + cd^* & |b|^2 + |d|^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned} \quad (20.33)$$

Then we have

$$\begin{aligned} \text{(i)} \quad & |a|^2 + |b|^2 = 1, \quad \text{(ii)} \quad |c|^2 + |d|^2 = 1, \quad \text{(iii)} \quad |a|^2 + |c|^2 = 1, \\ \text{(iv)} \quad & |b|^2 + |d|^2 = 1, \quad \text{(v)} \quad a^*c + b^*d = 0, \quad \text{(vi)} \quad a^*b + c^*d = 0, \\ & \text{(vii)} \quad ad - bc = 1. \end{aligned}$$

Of the above conditions, (vii) comes from $\det U = 1$. Condition (iv) can be eliminated by conditions (i)–(iii). From (v) and (vii), using Cramer’s rule we have

$$c = \frac{\begin{vmatrix} 0 & b^* \\ 1 & a \end{vmatrix}}{\begin{vmatrix} a^* & b^* \\ -b & a \end{vmatrix}} = \frac{-b^*}{|a|^2 + |b|^2} = -b^*, \quad d = \frac{\begin{vmatrix} a^* & 0 \\ -b & 1 \end{vmatrix}}{\begin{vmatrix} a^* & b^* \\ -b & a \end{vmatrix}} = \frac{a^*}{|a|^2 + |b|^2} = a^*.$$
(20.34)

From (20.34) we see that (i)–(iv) are equivalent and that (v) and (vi) are redundant. Thus, as a general form of U we get

$$U = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix} \text{ with } |a|^2 + |b|^2 = 1. \quad (20.35)$$

As a result, we have four freedoms with a and b with a restricting condition of $|a|^2 + |b|^2 = 1$. That is, we finally get three freedoms with choice of parameters. Putting

$$a = p + iq \quad \text{and} \quad b = r + is \quad (p, q, r, s : \text{real}),$$

we have

$$p^2 + q^2 + r^2 + s^2 = 1. \quad (20.36)$$

Therefore, the restricted condition of (20.35) is equivalent to that p, q, r , and s are regarded as coordinates that are positioned on a unit sphere of $\mathbb{R}^4$.

From (20.35), we have

$$U^{-1} = U^\dagger = \begin{pmatrix} a^* & -b \\ b^* & a \end{pmatrix}.$$

Also putting

$$U_1 = \begin{pmatrix} a_1 & b_1 \\ -b_1^* & a_1^* \end{pmatrix} \quad \text{and} \quad U_2 = \begin{pmatrix} a_2 & b_2 \\ -b_2^* & a_2^* \end{pmatrix}, \quad (20.37)$$

we get

$$U_1 U_2 = \begin{pmatrix} a_1 a_2 - b_1 b_2^* & a_1 b_2 + b_1 a_2^* \\ -a_2 b_1^* - a_1^* b_2^* & -b_1^* b_2 + a_1^* a_2^* \end{pmatrix}. \quad (20.38)$$

Thus, both U^{-1} and $U_1 U_2$ satisfy criteria (20.35) and, hence, the unitary matrices having a matrix form of (20.35) constitute a group; i.e., $SU(2)$.

Next we wish to connect the $SU(2)$ matrices to the matrices that represent the angular momentum. In Sect. 3.4 we developed the theory of generalized angular momentum. In the case of $j = 1/2$ we can obtain accurate information about the spin states. Using (3.101) as a starting point and defining the state belonging to the spin angular momentum $\mu = -1/2$ and $\mu = 1/2$ as $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, respectively, we obtain

$$J^{(+)} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad J^{(-)} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \quad (20.39)$$

Note that $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are in accordance with (2.64) as a column vector representation. Using (3.72), we have

$$J_x = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_y = \frac{1}{2} \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \quad J_z = \frac{1}{2} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (20.40)$$

In (20.40) J_z was given so that we may have $J_z \begin{pmatrix} 1 \\ 0 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $J_z \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Deleting the coefficient $\frac{1}{2}$ from (20.40), we get Pauli spin matrices such that

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (20.41)$$

Multiplying $-i$ on (20.40), we have anti-Hermitian operators described by

$$\zeta_x = \frac{1}{2} \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}, \quad \zeta_y = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \zeta_z = \frac{1}{2} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}. \quad (20.42)$$

Notice that these are traceless anti-Hermitian matrices.

Now let us seek rotation matrices in a manner related to that deriving (20.31) but at a more general level. From Property (7) of Chap. 15 (see Sect. 15.2), we know that

$$\exp(tA)$$

with an anti-Hermitian operator A combined with a real number t produces a unitary matrix. Let us apply this to the anti-Hermitian matrices described by (20.42).

Using ζ_z of (20.42), we calculate $\exp(\alpha\zeta_z)$ as in (20.27). We show only the results described by

$$\begin{aligned}\exp(\alpha\zeta_z) &= E \cdot \cos \frac{\alpha}{2} - i\sigma_z \sin \frac{\alpha}{2} = \begin{pmatrix} \cos \frac{\alpha}{2} + i \sin \frac{\alpha}{2} & 0 \\ 0 & \cos \frac{\alpha}{2} - i \sin \frac{\alpha}{2} \end{pmatrix} \\ &= \begin{pmatrix} e^{i\alpha/2} & 0 \\ 0 & e^{-i\alpha/2} \end{pmatrix}.\end{aligned}\quad (20.43)$$

Readers are recommended to derive this. Using ζ_y and applying similar calculation procedures to the above, we also get

$$\exp(\beta\zeta_y) = \begin{pmatrix} \cos \frac{\beta}{2} & \sin \frac{\beta}{2} \\ -\sin \frac{\beta}{2} & \cos \frac{\beta}{2} \end{pmatrix}.\quad (20.44)$$

The representation matrices describe “complex rotations” and (20.43) and (20.44) correspond to (20.28) and (20.30), respectively. We define $D_{\alpha,\beta,\gamma}^{(1/2)}$ as the representation matrix that describes the successive rotations α , β , and γ and corresponds to $\tilde{R}_3$ in (17.101) and (20.31) including Euler angles. As a result, we have

$$\begin{aligned}D_{\alpha,\beta,\gamma}^{(1/2)} &= \exp(\alpha\zeta_z) \exp(\beta\zeta_y) \exp(\gamma\zeta_z) \\ &= \begin{pmatrix} e^{\frac{i}{2}(\alpha+\gamma)} \cos \frac{\beta}{2} & e^{\frac{i}{2}(\alpha-\gamma)} \sin \frac{\beta}{2} \\ -e^{\frac{i}{2}(\gamma-\alpha)} \sin \frac{\beta}{2} & e^{-\frac{i}{2}(\alpha+\gamma)} \cos \frac{\beta}{2} \end{pmatrix},\end{aligned}\quad (20.45)$$

where the superscript $1/2$ of $D_{\alpha,\beta,\gamma}^{(1/2)}$ corresponds to the non-negative generalized angular momentum j ($=1/2$). Equation (20.45) can be obtained by putting $a = e^{\frac{i}{2}(\alpha+\gamma)} \cos \frac{\beta}{2}$ and $b = e^{\frac{i}{2}(\alpha-\gamma)} \sin \frac{\beta}{2}$ in the matrix U of (20.35). Using this general $SU(2)$ representation matrix, we construct representation matrices of higher orders.

20.2.2 $SU(2)$ Representation Matrices: Wigner Formula

Suppose that we have two functions (or vectors) v and u . We regard v and u as linearly independent vectors that undergo a unitary transformation by a unitary matrix U of (20.35). That is, we have

$$(v' \ u') = (v \ u)U = (v \ u) \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix},\quad (20.46)$$

where $(v \ u)$ are transformed into $(v' \ u')$. Namely, we have

$$v' = av - b^*u, u' = bv + a^*u.$$

Let us further define according to Hamermesh [5] an orthonormal set of $2j + 1$ functions such that

$$f_m = \frac{u^{j+m}v^{j-m}}{\sqrt{(j+m)!(j-m)!}} \quad (m = -j, -j+1, \dots, j-1, j), \quad (20.47)$$

where j is an integer or a half-odd-integer. Equation (20.47) implies that $2j + 1$ monomials f_m of $2j$ -degree with respect to v and u constitute the orthonormal basis. Related discussion can be seen in Sect. 20.3.2. The description that follows includes contents related to generalized angular momenta (see Sect. 3.4).

We examine how these functions f_m (or vectors) are transformed by (20.45). We denote this transformation by $\mathfrak{R}(a, b)$. That is, we have

$$\mathfrak{R}(a, b)(f_m) \equiv \sum_{m'} f_{m'} D_{m'm}^{(j)}(a, b), \quad (20.48)$$

where $D_{m'm}^{(j)}(a, b)$ denotes matrix elements with respect to the transformation. Replacing u and v with u' and v' , respectively, and using the binomial theorem we get

$$\begin{aligned} \mathfrak{R}(a, b)f_m &= \frac{1}{\sqrt{(j+m)!(j-m)!}} (bv + a^*u)^{j+m} (av - b^*u)^{j-m} \\ &= \sum_{\mu, \nu} \frac{1}{\sqrt{(j+m)!(j-m)!}} (bv + a^*u)^{j+m} (av - b^*u)^{j-m} \\ &= \sum_{\mu, \nu} \frac{1}{\sqrt{(j+m)!(j-m)!}} \frac{(j+m)!(j-m)!}{(j+m-\mu)!\mu!(j-m-\nu)!\nu!} (a^*u)^{j+m-\mu} (bv)^\mu (-b^*u)^{j-m-\nu} (av)^\nu \\ &= \sum_{\mu, \nu} \frac{\sqrt{(j+m)!(j-m)!}}{(j+m-\mu)!\mu!(j-m-\nu)!\nu!} (a^*)^{j+m-\mu} b^\mu (-b^*)^{j-m-\nu} a^\nu u^{2j-\mu-\nu} v^{\mu+\nu}. \end{aligned} \quad (20.49)$$

Here, to eliminate ν , putting

$$2j - \mu - \nu = j + m', \quad \mu + \nu = j - m',$$

we get

$$\begin{aligned} \mathfrak{R}(a, b)f_m &= \sum_{\mu, m'} \frac{\sqrt{(j+m)!(j-m)!}}{(j+m-\mu)!\mu!(m'-m+\mu)!(j-m'-\mu)!} \\ &\quad (a^*)^{j+m-\mu} b^\mu (-b^*)^{m'-m+\mu} a^{j-m'-\mu} u^{j+m'} v^{j-m'}. \end{aligned} \quad (20.50)$$

Expressing $f_{m'}$ as

$$f_{m'} = \frac{u^{j+m'} v^{j-m'}}{\sqrt{(j+m')!(j-m')!}} (m' = -j, -j+1, \dots, j-1, j), \quad (20.51)$$

we obtain

$$\begin{aligned} \Re(a, b)(f_m) \\ = \sum_{\mu, m'} f_{m'} \frac{\sqrt{(j+m)!(j-m)!(j+m')!(j-m')!}}{(j+m-\mu)! \mu! (m'-m+\mu)! (j-m'-\mu)!} (a^*)^{j+m-\mu} b^\mu (-b^*)^{m'-m+\mu} a^{j-m'-\mu}. \end{aligned} \quad (20.52)$$

Equating (20.52) with (20.48), as the (m', m) component of $D^{(j)}(a, b)$ we get

$$\begin{aligned} D_{m'm}^{(j)}(a, b) \\ = \sum_{\mu} \frac{\sqrt{(j+m)!(j-m)!(j+m')!(j-m')!}}{(j+m-\mu)! \mu! (m'-m+\mu)! (j-m'-\mu)!} (a^*)^{j+m-\mu} b^\mu (-b^*)^{m'-m+\mu} a^{j-m'-\mu}. \end{aligned} \quad (20.53)$$

In (20.53), factorials of negative integers must be avoided; see (3.216). Finally, replacing a (or a^*) and b (or b^*) with $e^{\frac{i}{2}(\alpha+\gamma)} \cos \frac{\beta}{2}$ [or $e^{-\frac{i}{2}(\alpha+\gamma)} \cos \frac{\beta}{2}$] and $e^{\frac{i}{2}(\alpha-\gamma)} \sin \frac{\beta}{2}$ [or $e^{\frac{i}{2}(\gamma-\alpha)} \sin \frac{\beta}{2}$] by use of (20.45), respectively, we get

$$\begin{aligned} D_{m'm}^{(j)}(\alpha, \beta, \gamma) = \sum_{\mu} \frac{(-1)^{m'-m+\mu} \sqrt{(j+m)!(j-m)!(j+m')!(j-m')!}}{(j+m-\mu)! \mu! (m'-m+\mu)! (j-m'-\mu)!} \\ \times e^{-i(am'+\gamma m)} \left(\cos \frac{\beta}{2}\right)^{2j+m-m'-2\mu} \left(\sin \frac{\beta}{2}\right)^{m'-m+2\mu}. \end{aligned} \quad (20.54)$$

In (20.54), the notation $D^{(j)}(a, b)$ has been replaced with $D^{(j)}(\alpha, \beta, \gamma)$, where α, β , and γ represent Euler angles that appeared in (17.101). Equation (20.54) is called Wigner formula [5–7].

To confirm that $D_{m'm}^{(j)}$ is indeed unitary, we carry out the following calculations. From (20.47) we have

$$\sum_{m=-j}^j |f_m|^2 = \sum_{m=-j}^j \frac{|u^{j+m} v^{j-m}|^2}{(j+m)!(j-m)!} = \sum_{m=-j}^j \frac{|u|^{2(j+m)} |v|^{2(j-m)}}{(j+m)!(j-m)!}.$$

Meanwhile, we have

$$\left[|u|^2 + |v|^2\right]^{2j} = \sum_{k=0}^{2j} \frac{(2j)! |u|^{2(2j-k)} |v|^{2k}}{(2j-k)! k!} = \sum_{m=-j}^j \frac{(2j)! |u|^{2(j+m)} |v|^{2(j-m)}}{(j+m)! (j-m)!},$$

where with the last equality k was replaced with $j - m$. Comparing the above two equations, we get

$$\sum_{m=-j}^j |f_m|^2 = \frac{1}{(2j)!} \left[|u|^2 + |v|^2\right]^{2j}. \quad (20.55)$$

Viewing (20.46) as variables transformation and taking adjoint of (20.46), we have

$$\begin{pmatrix} v'^* \\ u'^* \end{pmatrix} = U^\dagger \begin{pmatrix} v^* \\ u^* \end{pmatrix}, \quad (20.56)$$

where we defined $U = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix}$ (i.e., a unitary matrix). Operating (20.56) on both sides of (20.46) from the right, we get

$$(v' \ u') \begin{pmatrix} v'^* \\ u'^* \end{pmatrix} = (v \ u) U U^\dagger \begin{pmatrix} v^* \\ u^* \end{pmatrix} = (v \ u) \begin{pmatrix} v^* \\ u^* \end{pmatrix}.$$

That is, we have a quadratic form described by

$$|u'|^2 + |v'|^2 = |u|^2 + |v|^2. \quad (20.57)$$

From (20.55) and (20.57), we conclude that $\sum_{m=-j}^j |f_m|^2$ (i.e., the length of vector) is invariant under the transformation $D_{m'm}^{(j)}$. This implies that the transformation is unitary and, hence, that $D_{m'm}^{(j)}$ is a unitary matrix [5].

20.2.3 $SO(3)$ Representation Matrices and Spherical Surface Harmonics

Once representation matrices $D^{(j)}(\alpha, \beta, \gamma)$ have been obtained, we are able to get further useful information. We immediately notice that (20.54) is related to (3.216) that is explicitly described by trigonometric functions. Putting $m = 0$ in (20.54) we have

$$\begin{aligned}
D_{m'0}^{(j)}(\alpha, \beta, \gamma) &= D_{m'0}^{(j)}(\alpha, \beta) \\
&= \sum_{\mu} \frac{(-1)^{m'+\mu} j! \sqrt{(j+m')!(j-m')!}}{(j-\mu)! \mu! (m'+\mu)! (j-m'-\mu)!} e^{-iam'} \left(\cos \frac{\beta}{2} \right)^{2j-m'-2\mu} \left(\sin \frac{\beta}{2} \right)^{m'+2\mu}.
\end{aligned} \tag{20.58}$$

Note that (20.58) does not depend on the variable γ , because $m = 0$ leads to $e^{im\gamma} = e^{i \cdot 0 \cdot \gamma} = e^0 = 1$ in (20.54). Notice also that the event of $m = 0$ in (20.54) never happens with the case where j is a half-odd-integer but happens with the case where j is a half-odd-integer but happens with the case where j is an integer l . In fact, $D^{(l)}(\alpha, \beta, \gamma)$ is a $(2l+1)$ -degree representation of $SO(3)$. Later in this section, we will give a full matrix form in the case of $l = 1$. Replacing m' with m in (20.58), we get

$$\begin{aligned}
D_{m0}^{(j)}(\alpha, \beta, \gamma) \\
&= \sum_{\mu} \frac{(-1)^{m+\mu} j! \sqrt{(j+m)! (j-m)!}}{(j-\mu)! \mu! (m+\mu)! (j-m-\mu)!} e^{-iam} \left(\cos \frac{\beta}{2} \right)^{2j-m-2\mu} \left(\sin \frac{\beta}{2} \right)^{m+2\mu}.
\end{aligned} \tag{20.59}$$

For (20.59) to be consistent with (3.216), by changing notations of variables and replacing j with an integer l we further rewrite (20.59) to get

$$\begin{aligned}
D_{m0}^{(l)}(\phi, \theta) \\
&= \sum_r \frac{(-1)^{m+r} l! \sqrt{(l+m)! (l-m)!}}{r! (l-m-r)! (l-r)! (m+r)!} e^{-im\phi} \left(\cos \frac{\theta}{2} \right)^{2l-2r-m} \left(\sin \frac{\theta}{2} \right)^{2r+m}.
\end{aligned} \tag{20.60}$$

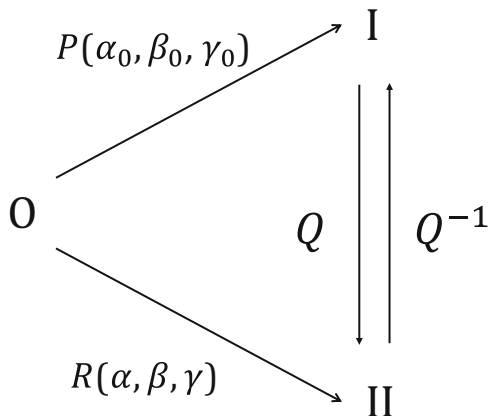
Notice that in (20.60) we deleted the variable γ as it was redundant. Comparing (20.60) with (3.216), we get [5]

$$\left[D_{m0}^{(l)}(\phi, \theta) \right]^* = \sqrt{\frac{4\pi}{2l+1}} Y_l^m(\theta, \phi). \tag{20.61}$$

From a general point of view, let us return to a discussion of $SU(2)$ and introduce a following important theorem in relation to the basis functions of $SU(2)$.

Theorem 20.1 [1] Let $\{\psi_{-j}, \psi_{-j+1}, \dots, \psi_j\}$ be a set of $(2j+1)$ functions. Let $D^{(j)}$ be a representation of the special unitary group $SU(2)$. Suppose that we have a following expression for $\psi_{m,k}$ ($m = -j, \dots, j$) such that

Fig. 20.2 Coordinate systems O, I, and II and their transformation. Regarding the symbols and notations, see text



$$\psi_{m,k}(\alpha, \beta, \gamma) = \left[D_{mk}^{(j)}(\alpha, \beta, \gamma) \right]^*, \quad (20.62)$$

where α , β , and γ are Euler angles that appeared in (17.101). Then, the above functions are basis functions of the representation of $D^{(j)}$.

Proof In Fig. 20.2 we show coordinate systems O, I, and II. Let P , Q , R be operators of transformation (i.e., rotation) between the coordinate system as shown. The transformation Q is defined as the coordinate transformation that transforms I to II. Note that their inverse transformations, e.g., Q^{-1} exist. In Fig. 20.2, P and R are specified as $P(\alpha_0, \beta_0, \gamma_0)$ and $R(\alpha, \beta, \gamma)$, respectively. Then, we have [1, 2]

$$Q^{-1}R = P. \quad (20.63)$$

According to the expression (19.12), we have

$$Q\psi_{m,k}(\alpha, \beta, \gamma) = \psi_{m,k}[Q^{-1}(\alpha, \beta, \gamma)] = \psi_{m,k}(\alpha_0, \beta_0, \gamma_0) = \left[D_{mk}^{(j)}(Q^{-1}R) \right]^*,$$

where the last equality comes from (20.62) and (20.63); (α, β, γ) and $(\alpha_0, \beta_0, \gamma_0)$ stand for the transformations $R(\alpha, \beta, \gamma)$ and $P(\alpha_0, \beta_0, \gamma_0)$, respectively. The matrix element $D_{mk}^{(j)}(Q^{-1}R)$ can be written as

$$D_{mk}^{(j)}(Q^{-1}R) = \sum_n D_{mn}^{(j)}(Q^{-1})D_{nk}^{(j)}(R) = \sum_n \left[D_{nm}^{(j)}(Q) \right]^* D_{nk}^{(j)}(R),$$

where with the last equality we used the unitarity of the representation matrix $D^{(j)}$. Taking complex conjugate of the above expression, we get

$$\begin{aligned} Q\psi_{m,k}(\alpha, \beta, \gamma) &= \psi_{m,k}(\alpha_0, \beta_0, \gamma_0) = \sum_n D_{nm}^{(j)}(Q) \left[D_{nk}^{(j)}(R) \right]^* \\ &= \sum_n D_{nm}^{(j)}(Q) \left[D_{nk}^{(j)}(\alpha, \beta, \gamma) \right]^* = \sum_n \psi_{n,k}(\alpha, \beta, \gamma) D_{nm}^{(j)}(Q), \end{aligned} \quad (20.64)$$

where with the last equality we used the supposition (20.62). Equation (20.64) implies that $\psi_{m,k}$ ($m = -j, \dots, 0, \dots, j$) are basis functions of the representation of $D^{(j)}$. This completes the proof.

If we restrict Theorem 20.1 to $SO(3)$, we get an important result. Replacing j with an integer l and putting $k = 0$ in (20.62) and (20.64), we get

$$\begin{aligned} Q\psi_{m,0}(\alpha, \beta, \gamma) &= \psi_{m,0}(\alpha_0, \beta_0, \gamma_0) = \sum_n \psi_{n,0}(\alpha, \beta, \gamma) D_{nm}^{(l)}(Q), \\ \psi_{m,0}(\alpha, \beta, \gamma) &= \left[D_{m0}^{(l)}(\alpha, \beta, \gamma) \right]^*. \end{aligned} \quad (20.65)$$

This expression shows that the complex conjugates of individual components for the “center” column vector of the representation matrix give the basis functions of the representation of $D^{(l)}$. Removing γ as it is redundant and by the aid of (20.61), we get

$$\psi_{m,0}(\alpha, \beta) \equiv \sqrt{\frac{4\pi}{2l+1}} Y_l^m(\beta, \alpha).$$

Thus, in combination with (20.61), Theorem 20.1 immediately indicates that the spherical surface harmonics $Y_l^m(\theta, \phi)$ ($m = -l, \dots, 0, \dots, l$) span the representation space with respect to $D^{(l)}$. That is, we have

$$QY_l^m(\beta, \alpha) = \sum_n Y_l^n(\beta, \alpha) D_{nm}^{(l)}(Q). \quad (20.66)$$

Now, let us get back to (20.63). From (20.63), we have

$$Q = RP^{-1}. \quad (20.67)$$

Since P represents an arbitrary transformation, we may choose any (orthogonal) coordinate system I in $\mathbb{R}^3$. Meanwhile, a transformation which transforms I into II is necessarily present and given by Q that is expressed as (20.67); see (11.72) and the relevant argument of Sect. 11.4. Consequently, Q represents an arbitrary transformation in $\mathbb{R}^3$ in turn.

Now, in (20.67) putting $P = E$ we get $Q \equiv R(\alpha, \beta, \gamma)$. Then, replacing Q with R in (20.66) as a special case we have

$$R(\alpha, \beta, \gamma)Y_l^m(\beta, \alpha) = \sum_n Y_l^n(\beta, \alpha)D_{nm}^{(l)}(\alpha, \beta, \gamma). \quad (20.68)$$

The LHS of (20.68) is associated with three successive transformations. To describe it appropriately, let us recall (19.12) again. We have

$$Rf(\mathbf{r}) = f[R^{-1}(\mathbf{r})]. \quad (19.12)$$

Now, we are thinking of the case where R in (19.12) is described by $R(\alpha, \beta, \gamma)$ or $\widetilde{R}_3 (= R_{z\alpha}R'_{y'\beta}R''_{z''\gamma})$ in (17.101). That is, we have

$$R(\alpha, \beta, \gamma)f(\mathbf{r}) = f[R(\alpha, \beta, \gamma)^{-1}(\mathbf{r})].$$

Considering the constitution of $\widetilde{R}_3$ of (17.101), we readily get [5]

$$R(\alpha, \beta, \gamma)^{-1} = R(-\gamma, -\beta, -\alpha).$$

Thus, we have

$$R(\alpha, \beta, \gamma)f(\mathbf{r}) = f[R(-\gamma, -\beta, -\alpha)(\mathbf{r})]. \quad (20.69)$$

Applying (20.69) to (20.68), we get the successive (inverse) transformations of the functions such that

$$Y_l^m(\beta, \alpha) \rightarrow Y_l^m(\beta, \alpha - \gamma) \rightarrow Y_l^m(\beta - \beta, \alpha - \gamma) \rightarrow Y_l^m(\beta - \beta, \alpha - \gamma - \alpha).$$

The last entry in the above is [5]

$$Y_l^m(\beta - \beta, \alpha - \gamma - \alpha) = Y_l^m(0, -\gamma).$$

Then, from (20.68) we get

$$Y_l^m(0, -\gamma) = \sum_n Y_l^n(\beta, \alpha)D_{nm}^{(l)}(\alpha, \beta, \gamma). \quad (20.70)$$

Meanwhile, from (20.61) we have

$$Y_l^m(0, -\gamma) = \sqrt{\frac{2l+1}{4\pi}} \left[D_{m0}^{(l)}(-\gamma, 0) \right]^*.$$

Returning to (20.60), if we would have $\sin \frac{\theta}{2}$ factor in (20.60) or in (3.216), the relevant terms vanish with $D_{m0}^{(l)}(-\gamma, 0)$ except for special cases where $2r + m = 0$ in (3.216) or (20.60). But, to avoid having a negative integer inside the factorial in the denominator of (20.60), must have $r + m \geq 0$. Since $r \geq 0$, we must have $2r + m = (r + m) + r \geq 0$ as well. In other words, only if $r = m = 0$ in (20.60), we have $2r + m = 0$, for which $D_{m0}^{(l)}(-\gamma, 0)$ does not vanish. Using these conditions in (20.60), (20.60) is greatly simplified to be

$$D_{m0}^{(l)}(-\gamma, 0) = \delta_{m0}.$$

From (20.61) we get [5]

$$Y_l^m(0, -\gamma) = \sqrt{\frac{2l+1}{4\pi}} \delta_{0m}. \quad (20.71)$$

Notice that this expression is consistent with (3.147) and (3.148). Meanwhile, replacing ϕ and θ in (20.61) with α and β , respectively, multiplying the resulting equation by $D_{mk}^{(l)}(\alpha, \beta, \gamma)$, and further summing over m , we get

$$\begin{aligned} \sum_m \sqrt{\frac{4\pi}{2l+1}} Y_l^m(\beta, \alpha) D_{mk}^{(l)}(\alpha, \beta, \gamma) &= \sum_m \left[D_{m0}^{(l)}(\alpha, \beta, \gamma) \right]^* D_{mk}^{(l)}(\alpha, \beta, \gamma) = \delta_{0k} \\ &= \sqrt{\frac{4\pi}{2l+1}} Y_l^k(0, -\gamma), \end{aligned} \quad (20.72)$$

where with the second equality we used the unitarity of the matrices $D_{nm}^{(l)}(\alpha, \beta, \gamma)$ (see Sect. 20.2.2) and with the last equality we used (20.70) multiplied by the constant $\sqrt{\frac{4\pi}{2l+1}}$. At the same time, we recovered (20.71) comparing the last two sides of (20.72).

Rewriting (20.48) in a $(2j+1, 2j+1)$ matrix form, we get

$$\begin{aligned} &(f_{-j} f_{-j+1} \cdots f_{j-1} f_j) \Re(a, b) \\ &= (f_{-j} f_{-j+1} \cdots f_{j-1} f_j) \begin{pmatrix} D_{-j, -j} & \cdots & D_{-j, j} \\ \vdots & \ddots & \vdots \\ D_{j, -j} & \cdots & D_{j, j} \end{pmatrix}, \end{aligned} \quad (20.73)$$

where f_m ($m = -j, -j+1, \dots, j-1, j$) is described as in (20.47). Though abstract, the matrix on RHS of (20.73) gives $(2j+1)$ -dimensional representation $D^{(j)}$ of

(20.54). If j is an integer l , we can choose $Y_l^m(\beta, \alpha)$ for individual f_m (i.e., basis functions).

To get used to the abstract representation theory in continuous groups, we wish to think of a following trivial case: Putting $j = 0$ in (20.54), (20.54) becomes trivial, but still valid. In that case, we have $m' = m = 0$ so that the inside of the factorial can be non-negative. In turn, we must have $\mu = 0$ as well. Thus, we have

$$D_{00}^{(0)}(\alpha, \beta, \gamma) = 1.$$

Or, inserting $l = m = 0$ into (20.61), we have

$$\left[D_{00}^{(0)}(\phi, \theta) \right]^* = 1 = \sqrt{4\pi} Y_0^0(\theta, \phi), \quad \text{i.e.,} \quad Y_0^0(\theta, \phi) = \sqrt{1/4\pi}.$$

Thus, we recover (3.150).

Next, let us reproduce three-dimensional representation $D^{(1)}$ using (20.54). A full matrix form is expressed as

$$\begin{aligned} D^{(1)} &= \begin{pmatrix} D_{-1,-1} & D_{-1,0} & D_{-1,1} \\ D_{0,-1} & D_{0,0} & D_{0,1} \\ D_{1,-1} & D_{1,0} & D_{1,1} \end{pmatrix} \\ &= \begin{pmatrix} e^{i(\alpha+\gamma)} \cos^2 \frac{\beta}{2} & \frac{1}{\sqrt{2}} e^{i\alpha} \sin \beta & e^{i(\alpha-\gamma)} \sin^2 \frac{\beta}{2} \\ -\frac{1}{\sqrt{2}} e^{i\gamma} \sin \beta & \cos \beta & \frac{1}{\sqrt{2}} e^{-i\gamma} \sin \beta \\ e^{i(\gamma-\alpha)} \sin^2 \frac{\beta}{2} & -\frac{1}{\sqrt{2}} e^{-i\alpha} \sin \beta & e^{-i(\alpha+\gamma)} \cos^2 \frac{\beta}{2} \end{pmatrix}. \end{aligned} \quad (20.74)$$

Equation (20.74) is a special case of (20.73), in which $j = 1$. Note that (20.74) is a unitary matrix. The confirmation is left for readers. The complex conjugate of the column vector $m = 0$ in (20.61) with $l = 1$ is described by

$$\left[D_{m0}^{(1)}(\phi, \theta) \right]^* = \begin{pmatrix} \frac{1}{\sqrt{2}} e^{-i\alpha} \sin \beta \\ \cos \beta \\ -\frac{1}{\sqrt{2}} e^{i\alpha} \sin \beta \end{pmatrix} \quad (20.75)$$

that is related to the spherical surface harmonics (see Sect. 3.6); also see p. 760, Table 15.4 of Ref. [4].

Now, (20.25), (20.61), (20.65), and (20.73) provide a clue to relating $D^{(1)}$ to the rotation matrix of $\mathbb{R}^3$, i.e., R_3 of (17.101) that includes Euler angles. The argument is as follows: As already shown in (20.61) and (20.65), we have related the basis

vectors $(f_{-1} f_0 f_1)$ of $D^{(1)}$ to the spherical harmonics $(Y_1^{-1} Y_1^0 Y_1^1)$. Denoting the spherical basis by

$$\tilde{f}_{-1} \equiv \frac{1}{\sqrt{2}}(x - iy), \quad \tilde{f}_0 \equiv z, \quad \tilde{f}_1 \equiv \frac{1}{\sqrt{2}}(-x - iy), \quad (20.76)$$

we have

$$(\tilde{f}_{-1} \tilde{f}_0 \tilde{f}_1) = (x z y)U = (x z y) \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{i}{\sqrt{2}} & 0 & -\frac{i}{\sqrt{2}} \end{pmatrix}, \quad (20.77)$$

where we define a (unitary) matrix U as

$$U \equiv \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{i}{\sqrt{2}} & 0 & -\frac{i}{\sqrt{2}} \end{pmatrix}.$$

Equation (20.77) describes the relationship among two set of functions $(\tilde{f}_{-1} \tilde{f}_0 \tilde{f}_1)$ and $(x z y)$. We should not regard $(x z y)$ as coordinate in $\mathbb{R}^3$ as mentioned in Sect. 20.1. The $(3, 3)$ matrix of (20.77) is essentially the same as P of (20.22). Following (20.73) and taking account of (20.25), we express $(\tilde{f}_{-1}' \tilde{f}_0' \tilde{f}_1')$ as

$$(\tilde{f}_{-1}' \tilde{f}_0' \tilde{f}_1') \equiv (\tilde{f}_{-1} \tilde{f}_0 \tilde{f}_1) \Re(a, b) = (\tilde{f}_{-1} \tilde{f}_0 \tilde{f}_1) D^{(1)}. \quad (20.78)$$

This means that a set of functions $(\tilde{f}_{-1} \tilde{f}_0 \tilde{f}_1)$ forms the basis vectors of $D^{(1)}$ as well. Meanwhile, the relation (20.77) holds after the transformation, because (20.77) is independent of the choice of specific coordinate systems. That is, we have

$$(\tilde{f}_{-1}' \tilde{f}_0' \tilde{f}_1') = (x' z' y') U.$$

Then, from (20.78) we get

$$\begin{aligned}
 (x' z' y') &= \begin{pmatrix} \tilde{f}_{-1}' & \tilde{f}_0' & \tilde{f}_1' \end{pmatrix} U^{-1} = (\tilde{f}_{-1} \ \tilde{f}_0 \ \tilde{f}_1) D^{(1)} U^{-1} \\
 &= (x \ z \ y) U D^{(1)} U^{-1}.
 \end{aligned} \tag{20.79}$$

Defining a unitary matrix V as

$$V = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

and operating V on both sides of (20.79) from the right, we get

$$(x' z' y') V = (x \ z \ y) V \cdot V^{-1} U D^{(1)} U^{-1} V.$$

That is, we have

$$(x' y' z') = (x \ y \ z) (U^{-1} V)^{-1} D^{(1)} U^{-1} V. \tag{20.80}$$

Note that $(x' z' y') V = (x' y' z')$ and $(x \ z \ y) V = (x \ y \ z)$; that is, V exchanges the order of z and y in $(x \ z \ y)$.

Meanwhile, regarding $(x \ y \ z)$ as the basis vectors in $\mathbb{R}^3$ we have

$$(x' y' z') = (x \ y \ z) \mathcal{R}, \tag{20.81}$$

where $\mathcal{R}$ represents a $(3, 3)$ orthogonal matrix. Equation (20.81) represents a rotation in $SO(3)$. Since, x , y , and z are linearly independent, comparing (20.80) and (20.81) we get

$$\mathcal{R} \equiv (U^{-1} V)^{-1} D^{(1)} U^{-1} V = (U^{-1} V)^\dagger D^{(1)} U^{-1} V. \tag{20.82}$$

More explicitly, using $a = e^{\frac{i}{2}(\alpha+\gamma)} \cos \frac{\beta}{2}$ and $b = e^{\frac{i}{2}(\alpha-\gamma)} \sin \frac{\beta}{2}$ as in the case of (20.45), we describe $\mathcal{R}$ as

$$\mathcal{R} = \begin{pmatrix} \Re(a^2 - b^2) & -\Im(a^2 + b^2) & 2\Re(ab) \\ \Im(a^2 - b^2) & \Re(a^2 + b^2) & 2\Im(ab) \\ -2\Re(ab^*) & 2\Im(ab^*) & |a|^2 - |b|^2 \end{pmatrix}. \tag{20.83}$$

In (20.83), $\Re$ and $\Im$ denote a real part and an imaginary part of a complex number, respectively.

Comparing (20.83) with (17.101), we find out that $\mathcal{R}$ is identical to $\tilde{R}_3$ of (17.101). Equations (20.82) and (20.83) clearly show that $D^{(1)}$ and $\mathcal{R}$ ($\equiv \tilde{R}_3$) are connected to each other through the unitary similarity transformation, even though these matrices differ in that the former matrix is unitary and the latter is a real orthogonal matrix. That is, $D^{(1)}$ and $\tilde{R}_3$ are equivalent in terms of representation; see

Schur's First Lemma of Sect. 18.3. Thus, $D^{(1)}$ is certainly a representation matrix of $SO(3)$. The trace of $D^{(1)}$ and $\tilde{R}_3$ is given by

$$\text{Tr} D^{(1)} = \text{Tr} \tilde{R}_3 = \cos(\alpha + \gamma)(1 + \cos \beta) + \cos \beta.$$

Let us continue the discussion still further. We think of the quantity $(x/r \ y/r \ z/r)$, where r is radial coordinate of (20.25). Operating (20.82) on $(x/r \ y/r \ z/r)$ from the right, we have

$$\begin{aligned} (x/r \ y/r \ z/r) \mathcal{R} &= (x/r \ y/r \ z/r) V^{-1} U D^{(1)} U^{-1} V \\ &= (\widetilde{f_{-1}/r} \ \widetilde{f_0/r} \ \widetilde{f_1/r}) D^{(1)} U^{-1} V. \end{aligned}$$

From (20.25), (20.61), (20.75), and (20.76), we get

$$(\widetilde{f_{-1}/r} \ \widetilde{f_0/r} \ \widetilde{f_1/r}) = \left(\frac{1}{\sqrt{2}} e^{-i\alpha} \sin \beta \quad \cos \beta \quad -\frac{1}{\sqrt{2}} e^{i\alpha} \sin \beta \right). \quad (20.84)$$

Using (20.74) and (20.84), we obtain

$$(\widetilde{f_{-1}/r} \ \widetilde{f_0/r} \ \widetilde{f_1/r}) D^{(1)} = (0 \ 1 \ 0).$$

This is a tangible example of (20.72). Then, we have

$$(\widetilde{f_{-1}/r} \ \widetilde{f_0/r} \ \widetilde{f_1/r}) D^{(1)} U^{-1} V = (x/r \ y/r \ z/r) \mathcal{R} = (0 \ 0 \ 1). \quad (20.85)$$

For the confirmation of (20.85), use the spherical coordinate representation (3.17) to denote x , y , and z in the above equation. This seems to be merely a confirmation of the unitarity of representation matrices; see (20.72). Nonetheless, we emphasize that the simple relation (20.85) holds only with a special case where the parameters α and β in $D^{(1)}$ (or $\mathcal{R}$) are identical with the angular components of spherical coordinates associated with the Cartesian coordinates x , y , and z . Equation (20.85), however, does not hold in a general case where the parameters α and β are different from those angular components. In other words, (20.68) that describes the special case where $Q \equiv R(\alpha, \beta, \gamma)$ leads to (20.85); see Fig. 20.2. Equation (20.66), however, is used for the general case where Q represents an arbitrary transformation in $\mathbb{R}^3$.

Regarding further detailed discussion on the topics, readers are referred to the literature [1, 2, 5, 6].

20.2.4 Irreducible Representations of $SU(2)$ and $SO(3)$

In this section we further explore characteristics of the representation of $SU(2)$ and $SO(3)$ and show the irreducibility of the representation matrices of $SU(2)$ and $SO(3)$.

To prove the irreducibility of the $SU(2)$ representation matrices $D^{(j)}(\alpha, \beta, \gamma)$, we need to use Schur's lemmas already explained in Sect. 18.3. At present, our task is to find a suitable matrix A that commutes with $D^{(j)}(\alpha, \beta, \gamma)$. Since a general form of $D^{(j)}(\alpha, \beta, \gamma)$ has a complicated structure as shown in (20.54), it would be a formidable task to directly deal with $D^{(j)}(\alpha, \beta, \gamma)$ to verify its irreducibility. To gain clear understanding, we impose a restriction on $D^{(j)}(\alpha, \beta, \gamma)$ so that we can readily perform related calculations [5]. To this end, in (20.54) we consider a special case where $\beta = 0$; i.e., $D^{(j)}(\alpha, 0, \gamma)$. The point is that if we can find a matrix A which commutes with $D^{(j)}(\alpha, 0, \gamma)$, the matrix that commutes with $D^{(j)}(\alpha, \beta, \gamma)$ of a general form must be of the type A as well (if such matrix does exist).

We wish to seek conditions under which $D^{(j)}(\alpha, 0, \gamma)$ does not vanish. In (20.54) the exponential term never vanishes. When $\beta = 0$, $\cos \frac{\beta}{2} (= 1)$ does not vanish either. For the factor $(\sin \frac{\beta}{2})^{m' - m + 2\mu}$ not to vanish, we must have

$$m' - m + 2\mu = 0. \quad (20.86)$$

In the denominator of (20.54), to avoid a negative number in factorials we must have $m' - m + \mu \geq 0$ and $\mu \geq 0$. If $\mu > 0$, $m' - m + 2\mu = (m' - m + \mu) + \mu > 0$ as well. Therefore, for (20.86) to hold, we must have $\mu = 0$. From (20.86), in turn, this means $m' - m = 0$, i.e., $m' = m$. Then, we have a greatly simplified expression with the matrix elements $D_{m'm}^{(j)}(\alpha, 0, \gamma)$ such that

$$D_{m'm}^{(j)}(\alpha, 0, \gamma) = \delta_{m'm} e^{-i(\alpha m' + \gamma m)}. \quad (20.87)$$

This implies that $D^{(j)}(\alpha, 0, \gamma)$ is diagonal. Suppose that we have a matrix $A = (A_{m'm})$. If A commutes with $D^{(j)}(\alpha, 0, \gamma)$, it must be diagonal from (20.87). That is, A should have a form $A_{m'm} = a_m \delta_{m'm}$. This can readily be confirmed using simple matrix algebra. The confirmation is left for readers. We seek further conditions for A to commute with $D^{(j)}$ that has a general form of $D^{(j)}(\alpha, \beta, \gamma)$ ($\beta \neq 0$). For this purpose, we consider the (j, m) component of $D^{(j)}(\alpha, \beta, \gamma)$. Putting $m' = j$ in (20.54), we obtain $(j - m' - \mu)! = (-\mu)!$ in the denominator. Then, for (20.54) to be meaningful, we must have $\mu = 0$. Hence, we get

$$D_{jm}^{(j)}(\alpha, \beta, \gamma) = (-1)^{j-m} \sqrt{\frac{(2j)!}{(j+m)!(j-m)!}} e^{-i(\alpha j + \gamma m)} \left(\cos \frac{\beta}{2}\right)^{j+m} \left(\sin \frac{\beta}{2}\right)^{j-m}. \quad (20.88)$$

With respect to (20.88), we think of the following three cases: (i) $m \neq j$, $m \neq -j$; (ii) $m = j$; (iii) $m = -j$. For the case (i), $D_{jm}^{(j)}(\alpha, \beta, \gamma)$ vanishes only when $\beta = 0$ or π . For the case (ii), it vanishes only when $\beta = \pi$; in the case (iii), $D_{jm}^{(j)}(\alpha, \beta, \gamma)$ vanishes only when $\beta = 0$. The matrix element $D_{jm}^{(j)}(\alpha, \beta, \gamma)$ does not vanish other than the above three cases.

Taking account of the above consideration, we continue our discussion. If A commutes with $D^{(j)}(\alpha, \beta, \gamma)$, then with respect to the (j, m) component of $AD^{(j)}(\alpha, \beta, \gamma)$ we have

$$(AD^{(j)})_{jm} = \sum_k A_{jk} D_{km}^{(j)} = \sum_k a_k \delta_{jk} D_{km}^{(j)} = a_j D_{jm}^{(j)}. \quad (20.89)$$

Meanwhile, we obtain

$$(D^{(j)}A)_{jm} = \sum_k D_{jk}^{(j)} A_{km} = \sum_k D_{jk}^{(j)} a_m \delta_{km} = a_m D_{jm}^{(j)}. \quad (20.90)$$

From (20.89) and (20.90), for A to commute with $D^{(j)}$ we must have

$$D_{jm}^{(j)}(\alpha, \beta, \gamma)(a_j - a_m) = 0. \quad (20.91)$$

Considering that the matrix elements $D_{jm}^{(j)}(\alpha, \beta, \gamma)$ in (20.88) do not vanish in a general case for α, β, γ , from (20.91) we must have $a_j - a_m = 0$ for m that satisfies $1 \leq m \leq j - 1$. When $m = j$, we have a trivial equation $0 = 0$. This implies that a_j cannot be determined. Thus, putting $a_m = a_j \equiv a \neq 0$ ($1 \leq m \leq j$), we have

$$A_{m'm} = a \delta_{m'm},$$

where a is an arbitrary complex constant. Then, from Schur's Second Lemma we conclude that $D^{(j)}(\alpha, \beta, \gamma)$ must be irreducible.

We should be careful, however, about the application of Schur's Second Lemma. This is because Schur's Second Lemma described in Sect. 18.3 says the following:

A representation D is irreducible $\Rightarrow$ The matrix M that is commutative with all $D(g)$ is limited to a form of $M = cE$.

Nonetheless, we are uncertain of truth or falsehood of the converse proposition. Fortunately, however, the converse proposition is true if the representation is unitary. We prove the converse proposition by its contraposition such that

A representation D is not irreducible (i.e., reducible) $\Rightarrow$ Of the matrices that are commutative with all $D(g)$, we can find at least one matrix M of the type other than $M = cE$.

In this context, Theorem 18.2 (Sect. 18.2) as well as (18.38) teaches us that the unitary representation $D(g)$ is completely reducible so that we can describe it as a direct sum such that

$$D(g) = D^{(1)}(g) \oplus D^{(2)}(g) \oplus \cdots \oplus D^{(\omega)}(g), \quad (20.92)$$

where g is any group element. Then, we can choose a following matrix A for a linear transformation that is commutative with $D(g)$:

$$A = \alpha^{(1)} E_1 \oplus \alpha^{(2)} E_2 \oplus \cdots \oplus \alpha^{(\omega)} E_\omega, \quad (20.93)$$

where $\alpha^{(1)}, \dots, \alpha^{(\omega)}$ are complex constants that may be different from one another; $E_1, \dots, E_\omega$ are unit matrices having the same dimension as $D^{(1)}(g), \dots, D^{(\omega)}(g)$, respectively. Then, even though A is not a type of $A = \alpha \delta_{m'm}$, A commutes with $D(g)$ for any g .

Here we rephrase Schur's Second Lemma as the following theorem.

Theorem 20.2 Let D be a unitary representation of a group G . Suppose that with $\forall g \in G$ we have

$$D(g)M = MD(g). \quad (18.48)$$

A necessary and sufficient condition for the said unitary representation to be irreducible is that linear transformations commutative with $D(g)$ ($\forall g \in G$) are limited to those of a form described by $A = \alpha \delta_{m'm}$, where α is a (complex) constant.

Next, we examine the irreducibility of the $SO(3)$ representation matrices. We develop the discussion in conjunction with explanation about the important properties of the representation matrices of $SU(2)$ as well as $SO(3)$. We rewrite (20.54) when $j = l$ (l : zero or positive integers) to relate it to the rotation in $\mathbb{R}^3$. Namely, we have

$$\begin{aligned} \left[D^{(l)}(\alpha, \beta, \gamma) \right]_{m'm} &= \sum_{\mu} \frac{(-1)^{m'-m+\mu} \sqrt{(l+m)!(l-m)!(l+m')!(l-m')!}}{(l+m-\mu)! \mu! (m'-m+\mu)! (l-m'-\mu)!} \\ &\quad \times e^{-i(\alpha m' + \gamma m)} \left(\cos \frac{\beta}{2} \right)^{2l+m-m'-2\mu} \left(\sin \frac{\beta}{2} \right)^{m'-m+2\mu}. \end{aligned} \quad (20.94)$$

Meanwhile, as mentioned in Sect. 20.2.3 the transformation $\tilde{R}_3$ (appearing in Sect. 17.4.2) described by

$$\tilde{R}_3 = R_{z\alpha} R'_{y'\beta} R''_{z''\gamma}$$

may be regarded as $R(\alpha, \beta, \gamma)$ of (20.63). Consequently, the representation matrix $D^{(l)}(\alpha, \beta, \gamma)$ is described by

$$D^{(l)}(\alpha, \beta, \gamma) \equiv D^{(l)}(\tilde{R}_3) = D_{\alpha}^{(l)}(R_{z\alpha}) D_{\beta}^{(l)}(R'_{y'\beta}) D_{\gamma}^{(l)}(R''_{z''\gamma}). \quad (20.95)$$

Equation (20.95) is based upon the definition of representation (18.2). This notation is in parallel with (20.45) in which $D_{\alpha, \beta, \gamma}^{(1/2)}$ is described by the product of three exponential functions of matrices each of which was associated with the rotation characterized by Euler angles of (17.101). Denoting $D_{\alpha}^{(l)}(R_{z\alpha}) = D^{(l)}(\alpha, 0, 0)$, $D_{\beta}^{(l)}(R'_{y'\beta}) = D^{(l)}(0, \beta, 0)$, and $D_{\gamma}^{(l)}(R''_{z''\gamma}) = D^{(l)}(0, 0, \gamma)$, we have

$$D^{(l)}(\alpha, \beta, \gamma) = D^{(l)}(\alpha, 0, 0) D^{(l)}(0, \beta, 0) D^{(l)}(0, 0, \gamma), \quad (20.96)$$

where with each representation two out of three Euler angles are zero.

In light of (20.94), we estimate each factor of (20.96). By the same token as before, putting $\beta = \gamma = 0$ in (20.94) let us examine on what conditions $(\sin \frac{\theta}{2})^{m' - m + 2\mu}$ survives. For this factor to survive, we must have $m' - m + 2\mu = 0$ as in (20.87). Then, following the argument as before, we should have $\mu = 0$ and $m' = m$. Thus, $D^{(l)}(\alpha, 0, 0)$ must be a diagonal matrix. This is the case with $D^{(l)}(0, 0, \gamma)$ as well.

Equation (3.216) implies that the functions $Y_l^m(\beta, \alpha)$ ($m = -l, \dots, 0, \dots, l$) are eigenfunctions with regard to the rotation about the z -axis. In other words, we have

$$R_{z\alpha} Y_l^m(\theta, \phi) = Y_l^m(\theta, \phi - \alpha) = e^{-i\alpha} Y_l^m(\theta, \phi).$$

This is because from (3.216) $Y_l^m(\theta, \phi)$ can be described as

$$Y_l^m(\theta, \phi) = F(\theta) e^{im\phi},$$

where $F(\theta)$ is a function of θ . Hence, we have

$$Y_l^m(\theta, \phi - \alpha) = F(\theta) e^{im(\phi - \alpha)} = e^{-i\alpha} F(\theta) e^{im\phi} = e^{-i\alpha} Y_l^m(\theta, \phi).$$

Therefore, with a full matrix representation we get

$$\begin{aligned}
& (Y_l^{-l} Y_l^{-l+1} \dots Y_l^0 \dots Y_l^l) R_{z\alpha} \\
&= (Y_l^{-l} Y_l^{-l+1} \dots Y_l^0 \dots Y_l^l) \begin{pmatrix} e^{-i(-l)\alpha} & & & \\ & e^{-i(-l+1)\alpha} & & \\ & & \ddots & \\ & & & e^{-i\alpha} \end{pmatrix} \quad (20.97) \\
&= (Y_l^{-l} Y_l^{-l+1} \dots Y_l^0 \dots Y_l^l) \begin{pmatrix} e^{i\alpha} & & & \\ & e^{i(l-1)\alpha} & & \\ & & \ddots & \\ & & & e^{-i\alpha} \end{pmatrix}.
\end{aligned}$$

With a shorthand notation, we have

$$D^{(l)}(\alpha, 0, 0) = e^{-im\alpha} \delta_{m'm} \quad (m = -l, \dots, l). \quad (20.98)$$

From (20.96), with the (m', m) matrix elements of $D^{(l)}(\alpha, \beta, \gamma)$ we get

$$\begin{aligned}
& [D^{(l)}(\alpha, \beta, \gamma)]_{m'm} \\
&= \sum_{s,t} [D^{(l)}(\alpha, 0, 0)]_{m's} [D^{(l)}(0, \beta, 0)]_{st} [D^{(l)}(0, 0, \gamma)]_{tm} \\
&= \sum_{s,t} e^{-is\alpha} \delta_{m's} [D^{(l)}(0, \beta, 0)]_{st} e^{-im\gamma} \delta_{tm} \\
&= e^{-im'\alpha} [D^{(l)}(0, \beta, 0)]_{m'm} e^{-im\gamma}. \quad (20.99)
\end{aligned}$$

Meanwhile, from (20.94) we have

$$\begin{aligned}
& [D^{(l)}(0, \beta, 0)]_{m'm} \\
&= \sum_{\mu} \frac{(-1)^{m'-m+\mu} \sqrt{(l+m)! (l-m)! (l+m')! (l-m')!}}{(l+m-\mu)! \mu! (m'-m+\mu)! (l-m'-\mu)!} \left(\cos \frac{\beta}{2} \right)^{2l+m-m'-2\mu} \left(\sin \frac{\beta}{2} \right)^{m'-m+2\mu}. \quad (20.100)
\end{aligned}$$

Thus, we find that $[D^{(l)}(\alpha, \beta, \gamma)]_{m'm}$ of (20.99) has been factorized into three factors. Equation (20.94) has such an implication. The same characteristic is shared with the $SU(2)$ representation matrices (20.54) more generally. This can readily be understood from the aforementioned discussion.

Taking $D^{(1)}$ of (20.74) as an example, we have

$$\begin{aligned}
D^{(1)}(\alpha, \beta, \gamma) &= D^{(1)}(\alpha, 0, 0)D^{(1)}(0, \beta, 0)D^{(1)}(0, 0, \gamma) \\
&= \begin{pmatrix} e^{i\alpha} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{-i\alpha} \end{pmatrix} \begin{pmatrix} \cos^2 \frac{\beta}{2} & \frac{1}{\sqrt{2}} \sin \beta & \sin^2 \frac{\beta}{2} \\ -\frac{1}{\sqrt{2}} \sin \beta & \cos \beta & \frac{1}{\sqrt{2}} \sin \beta \\ \sin^2 \frac{\beta}{2} & -\frac{1}{\sqrt{2}} \sin \beta & \cos^2 \frac{\beta}{2} \end{pmatrix} \begin{pmatrix} e^{i\gamma} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{-i\gamma} \end{pmatrix} \\
&= \begin{pmatrix} e^{i(\alpha+\gamma)} \cos^2 \frac{\beta}{2} & \frac{1}{\sqrt{2}} e^{i\alpha} \sin \beta & e^{i(\alpha-\gamma)} \sin^2 \frac{\beta}{2} \\ -\frac{1}{\sqrt{2}} e^{i\gamma} \sin \beta & \cos \beta & \frac{1}{\sqrt{2}} e^{-i\gamma} \sin \beta \\ e^{i(\gamma-\alpha)} \sin^2 \frac{\beta}{2} & -\frac{1}{\sqrt{2}} e^{-i\alpha} \sin \beta & e^{-i(\alpha+\gamma)} \cos^2 \frac{\beta}{2} \end{pmatrix}.
\end{aligned} \tag{20.74}$$

In this way, we have reproduced the result of (20.74).

Once $D^{(l)}(\alpha, \beta, \gamma)$ has been factorized as in (20.94), we can readily examine whether the representation is irreducible. To know what kinds of matrices commute with $D^{(l)}(\alpha, \beta, \gamma)$, it suffices to examine whether the matrices commute with individual factors. The argument is due to Hamermesh [5].

(i) Let A be a matrix having a dimension the same as that of $D^{(l)}(\alpha, \beta, \gamma)$, namely A is a $(2l+1, 2l+1)$ square matrix with a form of $A = (a_{ij})$. First, we examine the commutativity of $D^{(l)}(\alpha, 0, 0)$ and A . With the matrix elements we have

$$\begin{aligned}
[AD^{(l)}(\alpha, 0, 0)]_{m'm} &= \sum_k a_{m'k} [D^{(l)}(\alpha, 0, 0)]_{km} = \sum_k a_{m'k} e^{-im\alpha} \delta_{km} \\
&= a_{m'm} e^{-im\alpha},
\end{aligned} \tag{20.101}$$

where the second equality comes from (20.98). Also, we have

$$[D^{(l)}(\alpha, 0, 0)A]_{m'm} = \sum_k [D^{(l)}(\alpha, 0, 0)]_{m'k} a_{km} = a_{m'm} e^{-im'\alpha}. \tag{20.102}$$

From (20.101) and (20.102), for a condition of commutativity we have

$$a_{m'm} (e^{-im\alpha} - e^{-im'\alpha}) = 0.$$

If $m' \neq m$, we should have $a_{m'm} = 0$. If $m' = m$, $a_{m'm}$ does not have to vanish but may take an arbitrary complex number. Hence, A is a diagonal matrix, namely

$$A = a_m \delta_{m'm}, \tag{20.103}$$

where a_m is an arbitrary complex number.

(ii) Next we examine the commutativity of $D^{(l)}(0, \beta, 0)$ and A of (20.103). Using (20.66), we have

$$QY_l^m(\beta, \alpha) = \sum_n Y_l^n(\beta, \alpha) D_{nm}^{(l)}(Q). \quad (20.66)$$

Choosing $R(0, \theta, 0)$ for Q and putting $\alpha = 0$ in (20.66), we have

$$R(0, \theta, 0)Y_l^m(\beta, 0) = \sum_n Y_l^n(\beta, 0) D_{nm}^{(l)}(0, \theta, 0). \quad (20.104)$$

Meanwhile, from (20.69) we get

$$\begin{aligned} R(0, \theta, 0)Y_l^m(\beta, 0) &= Y_l^m \left[R(0, \theta, 0)^{-1}(\beta, 0) \right] = Y_l^m [R(0, -\theta, 0)(\beta, 0)] \\ &= Y_l^m(\beta - \theta, 0). \end{aligned} \quad (20.105)$$

Combining (20.104) and (20.105) we get

$$Y_l^m(\beta - \theta, 0) = \sum_n Y_l^n(\beta, 0) D_{nm}^{(l)}(0, \theta, 0). \quad (20.106)$$

Further putting $\beta = 0$ in (20.106), we have

$$Y_l^m(-\theta, 0) = \sum_n Y_l^n(0, 0) D_{nm}^{(l)}(0, \theta, 0). \quad (20.107)$$

Taking account of (20.71), RHS does not vanish in (20.107) only when $n = 0$. On this condition, we obtain

$$Y_l^m(-\theta, 0) = Y_l^0(0, 0) D_{0m}^{(l)}(0, \theta, 0). \quad (20.108)$$

Meanwhile, from (20.71) with $m = 0$, we have

$$Y_l^0(0, 0) = \sqrt{\frac{2l+1}{4\pi}}. \quad (20.109)$$

Inserting this into (20.108), we have

$$Y_l^m(-\theta, 0) = \sqrt{\frac{2l+1}{4\pi}} D_{0m}^{(l)}(0, \theta, 0). \quad (20.110)$$

Replacing θ with $-\theta$ in (20.110), we get

$$D_{0m}^{(l)}(0, -\theta, 0) = \sqrt{\frac{4\pi}{2l+1}} Y_l^m(\theta, 0). \quad (20.111)$$

Compare (20.111) with (20.61) and confirm (20.111) using (20.74) with $l = 1$. Viewing (20.111) as a “row vector,” (20.111) with $l = 1$ can be expressed as

$$\begin{aligned} & \left(D_{0,-1}^{(1)}(0, -\theta, 0) \quad D_{0,0}^{(1)}(0, -\theta, 0) \quad D_{0,1}^{(1)}(0, -\theta, 0) \right) \\ &= \sqrt{\frac{4\pi}{3}} \left(Y_1^{-1}(\theta, 0) \quad Y_1^0(\theta, 0) \quad Y_1^1(\theta, 0) \right) = \left(\frac{1}{\sqrt{2}} \sin \theta \quad \cos \theta \quad -\frac{1}{\sqrt{2}} \sin \theta \right). \end{aligned}$$

The above relation should be compared with (20.74).

Getting back to our topic, with the matrix elements we have

$$\left[AD^{(l)}(0, \theta, 0) \right]_{0m} = \sum_k a_k \delta_{0k} \left[D^{(l)}(0, \theta, 0) \right]_{km} = a_0 \left[D^{(l)}(0, \theta, 0) \right]_{0m}$$

and

$$\left[D^{(l)}(0, \theta, 0) A \right]_{0m} = \sum_k \left[D^{(l)}(0, \theta, 0) \right]_{0k} a_m \delta_{km} = \left[D^{(l)}(0, \theta, 0) \right]_{0m} a_m,$$

where we assume $A = a_m \delta_{m'm}$ in (20.103). Since from (20.108) $[D^{(l)}(0, \theta, 0)]_{0m}$ does not vanish other than specific values of θ , for A and $D^{(l)}(0, \theta, 0)$ to commute we must have

$$a_m = a_0$$

for all $m = -l, -l+1, \dots, 0, \dots, l-1, l$. Then, from (20.103) we get

$$A = a_0 \delta_{m'm}, \quad (20.112)$$

where a_0 is an arbitrary complex number. Thus, a matrix that commutes with both $D^{(l)}(\alpha, 0, 0)$ and $D^{(l)}(0, \beta, 0)$ must be of a form of (20.112). The same discussion holds with $D^{(l)}(\alpha, \beta, \gamma)$ of (20.94) as a whole with regard to the commutativity. On the basis of Schur's Second Lemma (Sect. 18.3), we conclude that the representation $D^{(l)}(\alpha, \beta, \gamma)$ is again irreducible. This is one of the prominent features of $SO(3)$ as well as $SU(2)$.

As can be seen clearly in (20.97), the dimension of representation matrix $D^{(l)}(\alpha, \beta, \gamma)$ is $2l + 1$. Suppose that there is another representation matrix $D^{(l')}(\alpha, \beta, \gamma)$ ($l' \neq l$). Then, $D^{(l)}(\alpha, \beta, \gamma)$ and $D^{(l')}(\alpha, \beta, \gamma)$ are inequivalent (see Sect. 18.3). Meanwhile, the unitary representation of $SO(3)$ is completely reducible and, hence, any reducible unitary representation D can be described by

$$D = \sum_l a_l D^{(l)}(\alpha, \beta, \gamma), \quad (20.113)$$

where a_l is zero or a positive integer. If $a_l \geq 2$, this means that the same representations $D^{(l)}(\alpha, \beta, \gamma)$ repeatedly appear. Equation (20.113) provides a tangible example of (20.92) expressed as

$$D(g) = D^{(1)}(g) \oplus D^{(2)}(g) \oplus \cdots \oplus D^{(\omega)}(g) \quad (20.92)$$

and applies to the representation matrices of $SU(2)$ more generally. We will encounter related discussion in Sect. 20.3 in connection with the direct-product representation.

20.2.5 Parameter Space of $SO(3)$

As already discussed in Sect. 17.4.2, we need three angles α , β , and γ to specify the rotation in $\mathbb{R}^3$. Their domains are usually taken as follows:

$$0 \leq \alpha \leq 2\pi, 0 \leq \beta \leq \pi, 0 \leq \gamma \leq 2\pi.$$

The domains defined above are referred to as a parameter space.

Yet, there are different implications depending upon choosing different coordinate systems. The first choice is a moving coordinate system where α , β , and γ are Euler angles (see Sect. 17.4.2). The second choice is a fixed coordinate system where α is an azimuthal angle, β is a zenithal angle, and γ is a rotation angle. Notice that in the former case α , β , and γ represent equivalent rotation angles. In the latter case, however, although α and β define the orientation of a rotation axis, γ is designated as a rotation angle. In the present section we further discuss characteristics of the parameter space.

First, we study several characteristics of $(3, 3)$ real orthogonal matrices. (i) The $(3, 3)$ real orthogonal matrices have eigenvalues 1, $e^{i\gamma}$, and $e^{-i\gamma}$ ($0 \leq \gamma \leq \pi$). This is a direct consequence of (17.77) and invariance of the characteristic equation of the matrix. The rotation axis is associated with an eigenvector that belongs to the eigenvalue 1.

Let $A = (a_{ij})$ ($1 \leq i, j \leq 3$) be a $(3, 3)$ real orthogonal matrix. Let $\mathbf{u}$ be an eigenvector belonging to the eigenvalue 1. We suppose that when we are thinking of the rotation on the fixed coordinate system, $\mathbf{u}$ is given by a column vector such that

$$\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}.$$

Then, we have

$$A \mathbf{u} = 1 \mathbf{u} = \mathbf{u}. \quad (20.114)$$

Operating A^T on both sides of (20.114), we get

$$A^T A \mathbf{u} = E \mathbf{u} = \mathbf{u} = A^T \mathbf{u}, \quad (20.115)$$

where we used the property of an orthogonal matrix; i.e., $A^T A = E$. From (20.114) and (20.115), we have

$$(A - A^T) \mathbf{u} = 0. \quad (20.116)$$

Writing (20.116) in matrix components, we have [5]

$$\left[\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} - \begin{pmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{pmatrix} \right] \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = 0. \quad (20.117)$$

That is,

$$\begin{aligned} (a_{12} - a_{21})u_2 + (a_{13} - a_{31})u_3 &= 0, \\ (a_{21} - a_{12})u_1 + (a_{23} - a_{32})u_3 &= 0, \\ (a_{31} - a_{13})u_1 + (a_{32} - a_{23})u_2 &= 0. \end{aligned} \quad (20.118)$$

Solving (20.118), we get

$$u_1 : u_2 : u_3 = (a_{32} - a_{23}) : (a_{13} - a_{31}) : (a_{21} - a_{12}). \quad (20.119)$$

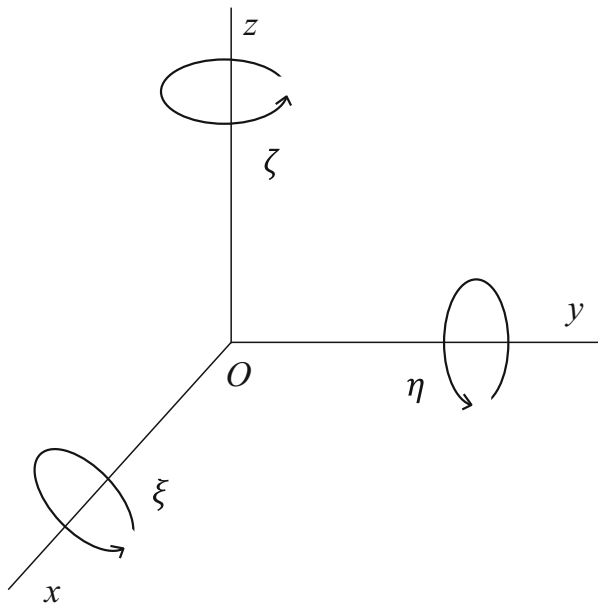
If we normalize $\mathbf{u}$, i.e., $u_1^2 + u_2^2 + u_3^2 = 1$, $\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$ gives direction cosines of the eigenvector, namely the rotation axis. Equation (20.119) applies to any orthogonal matrix. A simple example is (20.3); a more complicated example can be seen in (17.107). Confirmation is left for readers. We use this property soon below.

Let us consider infinitesimal transformations of rotation. As already implied in (20.6), an infinitesimal rotation θ around the z -axis is described by

$$R_{z\theta} = \begin{pmatrix} 1 - \theta & 0 & 0 \\ \theta & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (20.120)$$

Now, we newly introduce infinitesimal rotation operators as below:

Fig. 20.3 Infinitesimal rotations ξ , η , and ζ around the x -, y -, and z -axes, respectively



$$R_{x\xi} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -\xi \\ 0 & \xi & 1 \end{pmatrix}, R_{y\eta} = \begin{pmatrix} 1 & 0 & \eta \\ 0 & 1 & 0 \\ -\eta & 0 & 1 \end{pmatrix},$$

$$R_{z\zeta} = \begin{pmatrix} 1 & -\zeta & 0 \\ \zeta & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (20.121)$$

These operators represent infinitesimal rotations ξ , η , and ζ around the x -, y -, and z -axes, respectively (see Fig. 20.3). Note that these operators commute one another to the first order of infinitesimal quantities ξ , η , and ζ . That is, we have

$$R_{x\xi}R_{y\eta} = R_{y\eta}R_{x\xi}, R_{y\eta}R_{z\zeta} = R_{z\zeta}R_{y\eta}, R_{x\xi}R_{y\eta}R_{z\zeta} = R_{z\zeta}R_{y\eta}R_{x\xi}, \text{ etc.}$$

with, e.g.,

$$R_{x\xi}R_{y\eta}R_{z\zeta} = \begin{pmatrix} 1 & -\zeta & \eta \\ \zeta & 1 & -\xi \\ -\eta & \xi & 1 \end{pmatrix} \quad (20.122)$$

to the first order. Notice that the order of operations $R_{x\xi}$, $R_{y\eta}$, and $R_{z\zeta}$ is disregarded.

Next, let us consider a successive transformation with a finite rotation angle ω that follows the infinitesimal rotations. Readers may well ask why and for what purpose we need to make such elaborate calculations. It is partly because when we were dealing with finite groups in the previous chapters, group elements were finite and

relevant calculations were straightforward accordingly. But now we are thinking of continuous groups that possess an infinite number of group elements and, hence, we have to consider a “density” of group elements. In this respect, we are now thinking of how the density of group elements in the parameter space is changed according to the rotation of a finite angle ω . The results are used for various group calculations including orthogonalization of characters.

Getting back to our subject, we further operate a finite rotation of an angle ω around the z -axis subsequent to the aforementioned infinitesimal rotations. The discussion developed below is due to Hamermesh [5]. We denote this rotation by R_ω . Since we are dealing with a spherically symmetric system, any rotation axis can equivalently be chosen without loss of generality. Notice also that the rotations of the same rotation angle belong to the same conjugacy class [see (17.106) and Fig. 17.18]. Defining $R_{x\xi}R_{y\eta}R_{z\zeta} \equiv R_\Delta$ of (20.122), the rotation R that combines R_Δ and subsequently occurring R_ω is described by

$$\begin{aligned} R = R_\Delta R_\omega &= \begin{pmatrix} 1 & -\zeta & \eta \\ \zeta & 1 & -\xi \\ -\eta & \xi & 1 \end{pmatrix} \begin{pmatrix} \cos \omega & -\sin \omega & 0 \\ \sin \omega & \cos \omega & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \cos \omega - \zeta \sin \omega & -\sin \omega - \zeta \cos \omega & \eta \\ \sin \omega + \zeta \cos \omega & \cos \omega - \xi \sin \omega & -\xi \\ \xi \sin \omega - \eta \cos \omega & \xi \cos \omega + \eta \sin \omega & 1 \end{pmatrix}. \end{aligned} \quad (20.123)$$

Hence, we have

$$\begin{aligned} R_{32} - R_{23} &= \xi \cos \omega + \eta \sin \omega - (-\xi) = \xi(1 + \cos \omega) + \eta \sin \omega, \\ R_{13} - R_{31} &= \eta - (\xi \sin \omega - \eta \cos \omega) = \eta(1 + \cos \omega) - \xi \sin \omega, \\ R_{21} - R_{12} &= \sin \omega + \zeta \cos \omega - (-\sin \omega - \zeta \cos \omega) = 2(\sin \omega + \zeta \cos \omega). \end{aligned} \quad (20.124)$$

Since (20.124) gives a relative directional ratio of the rotation axis, we should normalize a vector whose components are given by (20.124) to get direction cosines of the rotation axis. Note that the direction of the rotation axis related to R of (20.123) should be close to R_ω and that only $R_{12} - R_{21}$ in (20.124) has a term (i.e., $2 \sin \omega$) lacking infinitesimal quantities ξ , η , ζ . Therefore, it suffices to normalize $R_{12} - R_{21}$ to seek the direction cosines to the first order. Hence, dividing (20.124) by $2 \sin \omega$, as components of the direction cosines we get

$$\frac{\xi(1 + \cos \omega)}{2 \sin \omega} + \frac{\eta}{2}, \frac{\eta(1 + \cos \omega)}{2 \sin \omega} - \frac{\xi}{2}, 1. \quad (20.125)$$

Meanwhile, combining (17.78) and (20.123), we can find a rotation angle ω' for R in (20.123). The trace χ of (20.123) is written as

$$\chi = 1 + 2(\cos \omega - \zeta \sin \omega). \quad (20.126)$$

From (17.78) we have

$$\chi = 1 + 2 \cos \omega'. \quad (20.127)$$

Equating (20.126) and (20.127), we obtain

$$\cos \omega' = \cos \omega - \zeta \sin \omega. \quad (20.128)$$

Approximating (20.128), we get

$$1 - \frac{1}{2} \omega'^2 \approx 1 - \frac{1}{2} \omega^2 - \zeta \omega. \quad (20.129)$$

From (20.129), we have a following approximate expression:

$$(\omega' + \omega)(\omega' - \omega) \approx 2\omega(\omega' - \omega) \approx 2\zeta\omega.$$

Hence, we obtain

$$\omega' \approx \omega + \zeta. \quad (20.130)$$

Combining (20.125) and (20.130), we get their product to the first order such that

$$\omega \left[\frac{\xi(1 + \cos \omega)}{2 \sin \omega} + \frac{\eta}{2} \right], \omega \left[\frac{\eta(1 + \cos \omega)}{2 \sin \omega} - \frac{\xi}{2} \right], \omega + \zeta. \quad (20.131)$$

Since these quantities are products of individual direction cosines and the rotation angle $\omega + \zeta$, we introduce variables $\tilde{x}$, $\tilde{y}$, and $\tilde{z}$ as the x -, y -, and z -related quantities, respectively. Namely, we have

$$\tilde{x} = \omega \left[\frac{\xi(1 + \cos \omega)}{2 \sin \omega} + \frac{\eta}{2} \right], \tilde{y} = \omega \left[\frac{\eta(1 + \cos \omega)}{2 \sin \omega} - \frac{\xi}{2} \right], \tilde{z} = \omega + \zeta. \quad (20.132)$$

To evaluate how the density of group elements is changed as a function of the rotation angle ω , we are interested in variations of $\tilde{x}$, $\tilde{y}$, and $\tilde{z}$ that depend on ξ , η , and ζ . To this end, we calculate the Jacobian J as

$$\begin{aligned}
 J = \frac{\partial(\tilde{x}, \tilde{y}, \tilde{z})}{\partial(\xi, \eta, \zeta)} &= \begin{vmatrix} \frac{\partial \tilde{x}}{\partial \xi} & \frac{\partial \tilde{x}}{\partial \eta} & \frac{\partial \tilde{x}}{\partial \zeta} \\ \frac{\partial \tilde{y}}{\partial \xi} & \frac{\partial \tilde{y}}{\partial \eta} & \frac{\partial \tilde{y}}{\partial \zeta} \\ \frac{\partial \tilde{z}}{\partial \xi} & \frac{\partial \tilde{z}}{\partial \eta} & \frac{\partial \tilde{z}}{\partial \zeta} \end{vmatrix} = \begin{vmatrix} \frac{\omega(1 + \cos \omega)}{2 \sin \omega} & \frac{\omega}{2} & 0 \\ -\frac{\omega}{2} & \frac{\omega(1 + \cos \omega)}{2 \sin \omega} & 0 \\ 0 & 0 & 1 \end{vmatrix} \\
 &= \frac{\omega^2}{4} \left[\frac{(1 + \cos \omega)^2}{\sin^2 \omega} + 1 \right] = \frac{\omega^2}{4 \sin^2 \frac{\omega}{2}},
 \end{aligned} \tag{20.133}$$

where we used formulae of trigonometric functions with the last equality. Note that (20.133) does not depend on the $\pm$ signs of ω . This is because J represents the relative volume density ratio between two parameter spaces of the $\tilde{x}\tilde{y}\tilde{z}$ -system and $\xi\eta\zeta$ -system and this ratio is solely determined by the modulus of rotation angle. Taking $\omega \rightarrow 0$, we have

$$\begin{aligned}
 \lim_{\omega \rightarrow 0} J &= \lim_{\omega \rightarrow 0} \frac{\omega^2}{4 \sin^2 \frac{\omega}{2}} = \lim_{\omega \rightarrow 0} \frac{(\omega^2)'}{(\sin^2 \frac{\omega}{2})'} = \frac{1}{4} \lim_{\omega \rightarrow 0} \frac{2\omega}{2 \sin \frac{\omega}{2} \cdot \frac{1}{2} \cos \frac{\omega}{2}} \\
 &= \lim_{\omega \rightarrow 0} \frac{\omega}{\sin \omega} = \lim_{\omega \rightarrow 0} \frac{(\omega)'}{(\sin \omega)'} = \lim_{\omega \rightarrow 0} \frac{1}{\cos \omega} = 1.
 \end{aligned}$$

Thus, the relative volume density ratio in the limit of $\omega \rightarrow 0$ is 1, as expected. Let $dV = d\tilde{x}d\tilde{y}d\tilde{z}$ and $d\Pi = d\xi d\eta d\zeta$ be volume elements of each coordinate system. Then we have

$$dV = J d\Pi. \tag{20.134}$$

We assume that

$$J = \frac{dV}{d\Pi} = \frac{\rho_{\xi\eta\zeta}}{\rho_{\tilde{x}\tilde{y}\tilde{z}}},$$

where $\rho_{\tilde{x}\tilde{y}\tilde{z}}$ and $\rho_{\xi\eta\zeta}$ are a “number density” of group elements in each coordinate system. We suppose that the infinitesimal rotations in the $\xi\eta\zeta$ -coordinate are converted by a finite rotation R_ω of (20.123) into the $\tilde{x}\tilde{y}\tilde{z}$ -coordinate. In this way, J can be viewed as an “expansion” coefficient as a function of rotation angle $|\omega|$. Hence, we have

$$dV = \frac{\rho_{\xi\eta\zeta}}{\rho_{\tilde{x}\tilde{y}\tilde{z}}} d\Pi \quad \text{or} \quad \rho_{\tilde{x}\tilde{y}\tilde{z}} dV = \rho_{\xi\eta\zeta} d\Pi. \quad (20.135)$$

Equation (20.135) implies that total (infinite) number of group elements that are contained in the group $SO(3)$ is invariant with respect to the transformation.

Let us calculate the total volume of $SO(3)$. This can be measured such that

$$\int d\Pi = \int \frac{1}{J} dV. \quad (20.136)$$

Converting dV to the polar coordinate representation, we get

$$\Pi[SO(3)] = \int d\Pi = \int_0^\pi d\tilde{\omega} \int_0^\pi d\theta \int_0^{2\pi} d\phi \frac{4 \sin^2 \frac{\tilde{\omega}}{2}}{\tilde{\omega}^2} \tilde{\omega}^2 \sin \theta = 8\pi^2, \quad (20.137)$$

where we denote the total volume of $SO(3)$ by $\Pi[SO(3)]$. Note that ω in (20.133) is replaced with $\tilde{\omega} \equiv |\omega|$ so that the radial coordinate can be positive. As already noted, (20.133) does not depend on the $\pm$ signs of ω . In other words, among the angles α , β , and ω (usually γ is used instead of ω ; see Sect. 17.4.2), ω is taken as $-\pi \leq \omega < \pi$ (instead of $0 \leq \omega < 2\pi$) so that it can be conformed to the radial coordinate.

We utilize (20.137) for calculation of various functions. Let $f(\phi, \theta, \omega)$ be an arbitrary function on a parameter space. We can readily estimate a “mean value” of $f(\phi, \theta, \omega)$ on that space. That is, we have

$$\overline{f(\phi, \theta, \omega)} \equiv \frac{\int f(\phi, \theta, \omega) d\Pi}{\int d\Pi}, \quad (20.138)$$

where $\overline{f(\phi, \theta, \omega)}$ represents a mean value of the function and it is given by

$$\overline{f(\phi, \theta, \omega)} = \left(4 \int_0^\pi d\tilde{\omega} \int_0^\pi d\theta \int_0^{2\pi} d\phi f(\phi, \theta, \omega) \sin^2 \frac{\tilde{\omega}}{2} \sin \theta \right) / \int d\Pi. \quad (20.139)$$

There would be some inconvenience to use (20.139) because of the mixture of variables ω and $\tilde{\omega}$. Yet, that causes no problem if $f(\phi, \theta, \omega)$ is an even function with respect to ω (see Sect. 20.2.6).

20.2.6 Irreducible Characters of $SO(3)$ and Their Orthogonality

To evaluate (20.139), let us get back to the irreducible representations $D^{(l)}(\alpha, \beta, \gamma)$ of Sect. 20.2.3. Also, we recall how successive coordinate transformations produce changes in the orthogonal matrices of transformation (Sect. 17.4.2). Since the parameters α , β , and γ uniquely specify individual rotation, different sets of α , β , and γ (in this section we use ω instead of γ) cause different irreducible representations. A corresponding rotation matrix is given by (17.101). In fact, the said matrix is a representation of the rotation. Keeping these points in mind, we discuss irreducible representations and their characters particularly in relation to the orthogonality of the irreducible characters.

Figure 20.4 depicts a geometrical arrangement of two rotation axes accompanied by the same rotation angle ω . Let R_ω and R'_ω be such two rotations around the rotation axis A and A' , respectively. Let Q be another rotation that transforms the rotation axis A to A' . Then we have [1, 2]

$$R'_\omega = QR_\omega Q^{-1}. \quad (20.140)$$

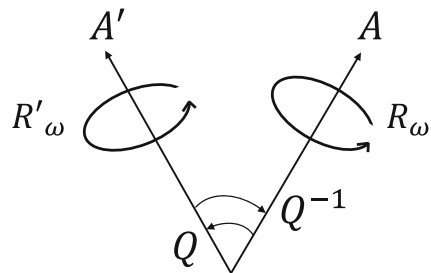
This implies that R_ω and R'_ω belong to the same conjugacy class. To consider the situation more clearly, let us view the two rotations R_ω and R'_ω from two different coordinate systems, say some xyz -coordinate system and another $x'y'z'$ -coordinate system. Suppose also that A coincides with the z -axis and that A' coincides with the z' -axis. Meanwhile, if we describe R_ω in reference to the xyz -system and R'_ω in reference to the $x'y'z'$ -system, the representation of these rotations must be identical in reference to the individual coordinate systems. Let this representation matrix be $\mathfrak{R}_\omega$.

- (i) If we describe R_ω and R'_ω with respect to the xyz -system, we have

$$R_\omega = \mathfrak{R}_\omega, R'_\omega = (Q^{-1})^{-1} \mathfrak{R}_\omega Q^{-1} = Q \mathfrak{R}_\omega Q^{-1}. \quad (20.141)$$

Namely, we reproduce

Fig. 20.4 Geometrical arrangement of two rotation axes A and A' accompanied by the same rotation angle ω



$$R'_\omega = QR_\omega Q^{-1}. \quad (20.140)$$

(ii) If we describe R_ω and R'_ω with respect to the $x'y'z'$ -system, we have

$$R'_\omega = \mathfrak{R}_\omega, R_\omega = Q^{-1}\mathfrak{R}_\omega Q. \quad (20.142)$$

Hence, again we reproduce (20.140). That is, the relation (20.140) is independent of specific choices of the coordinate system.

Taking the representation indexed by l of Sect. 20.2.4 with respect to (20.140), we have

$$\begin{aligned} D^{(l)}(R'_\omega) &= D^{(l)}(QR_\omega Q^{-1}) = D^{(l)}(Q)D^{(l)}(R_\omega)D^{(l)}(Q^{-1}) \\ &= D^{(l)}(Q)D^{(l)}(R_\omega) \left[D^{(l)}(Q) \right]^{-1}, \end{aligned} \quad (20.143)$$

where with the last equality we used (18.6). Operating $D^{(l)}(Q)$ on (20.143) from the right, we get

$$D^{(l)}(R'_\omega)D^{(l)}(Q) = D^{(l)}(Q)D^{(l)}(R_\omega). \quad (20.144)$$

Since both $D^{(l)}(R'_\omega)$ and $D^{(l)}(R_\omega)$ are irreducible, from (18.41) of Schur's First Lemma $D^{(l)}(R'_\omega)$ and $D^{(l)}(R_\omega)$ are equivalent. An infinite number of rotations determined by the orientation of the rotation axis A that is accompanied by an azimuthal angle α ($0 \leq \alpha < 2\pi$) and a zenithal angle β ($0 \leq \beta < \pi$) (see Fig. 17.18) form a conjugacy class with each specific ω shared. Thus, we have classified various representations $D^{(l)}(R_\omega)$ according to ω and l . Meanwhile, we may identify $D^{(l)}(R_\omega)$ with $D^{(l)}(\alpha, \beta, \gamma)$ of Sect. 20.2.4.

In Sect. 20.2.2 we know that the spherical surface harmonics $Y_l^m(\theta, \phi)$ ($m = -l, \dots, 0, \dots, l$) constitute basis functions of $D^{(l)}$ and span the representation space. A dimension of the matrix (or the representation space) is $2l + 1$. Therefore, $D^{(l)}$ and $D^{(l')}$ for different l and l' have a different dimension. From (18.41) of Schur's First Lemma, such $D^{(l)}$ and $D^{(l')}$ are inequivalent.

Returning to (20.139), let us evaluate a mean value of $\overline{f(\phi, \theta, \omega)}$. Let the trace of $D^{(l)}(R'_\omega)$ and $D^{(l)}(R_\omega)$ be $\chi^{(l)}(R'_\omega)$ and $\chi^{(l)}(R_\omega)$, respectively. Remembering (12.13), the trace is invariant under a similarity transformation, and so $\chi^{(l)}(R'_\omega) = \chi^{(l)}(R_\omega)$. Then, we put

$$\chi^{(l)}(\omega) \equiv \chi^{(l)}(R_\omega) = \chi^{(l)}(R'_\omega). \quad (20.145)$$

Consequently, it suffices to evaluate the trace $\chi^{(l)}(\omega)$ using $D^{(l)}(\alpha, 0, 0)$ of (20.96) whose representation matrix is given by (20.97). We have

$$\begin{aligned}
 \chi^{(l)}(\omega) &= \sum_{m=-l}^l e^{im\omega} = \frac{e^{-il\omega} [1 - e^{i\omega(2l+1)}]}{1 - e^{i\omega}} \\
 &= \frac{e^{-il\omega} (1 - e^{-i\omega}) [1 - e^{i\omega(2l+1)}]}{(1 - e^{i\omega})(1 - e^{-i\omega})}, \tag{20.146}
 \end{aligned}$$

where

$$\begin{aligned}
 &[\text{numerator of (20.146)}] \\
 &= e^{-il\omega} + e^{il\omega} - e^{i\omega(l+1)} - e^{-i\omega(l+1)} = 2[\cos l\omega - \cos(l+1)\omega] \\
 &= 4 \sin\left(l + \frac{1}{2}\right) \omega \sin \frac{\omega}{2}
 \end{aligned}$$

and

$$[\text{denominator of (20.146)}] = 2(1 - \cos \omega) = 4 \sin^2 \frac{\omega}{2}.$$

Thus, we get

$$\chi^{(l)}(\omega) = \frac{\sin(l + \frac{1}{2})\omega}{\sin \frac{\omega}{2}}. \tag{20.147}$$

To adapt (18.76) to the present formulation, we rewrite it as

$$\frac{1}{n} \sum_g \chi^{(\alpha)}(g)^* \chi^{(\beta)}(g) = \delta_{\alpha\beta}. \tag{20.148}$$

A summation over a finite number of group element n in (20.148) should be read as integration in the case of the continuous groups. Instead of a finite number of irreducible representations in a finite group, we are thinking of an infinite number of irreducible representations with continuous groups. In (20.139) the denominator $\int d\Pi (= 8\pi^2)$ corresponds to n of (20.148). If $f(\phi, \theta, \omega)$ or $f(\alpha, \beta, \omega)$ is an even function with respect to ω ($-\pi \leq \omega < \pi$) as in the case of (20.147), the numerator of (20.139) can be expressed in a form of

$$\begin{aligned}
 \int f(\alpha, \beta, \omega) d\Pi &= \int_0^\pi d\omega \int_0^\pi d\beta \int_0^{2\pi} d\alpha f(\alpha, \beta, \omega) \frac{4 \sin^2 \frac{\omega}{2}}{\omega^2} \omega^2 \sin \beta \\
 &= 4 \int_0^\pi d\omega \int_0^\pi d\beta \int_0^{2\pi} d\alpha f(\alpha, \beta, \omega) \sin^2 \frac{\omega}{2} \sin \beta. \tag{20.149}
 \end{aligned}$$

If, moreover, $f(\alpha, \beta, \omega)$ depends only on ω , again as in the case of the character described by (20.147), the calculation is further simplified such that

$$\int f(\omega) d\Pi = 16\pi \int_0^\pi d\omega f(\omega) \sin^2 \frac{\omega}{2}. \quad (20.150)$$

Replacing $f(\omega)$ with $[\chi^{(l')}(\omega)]^* \chi^{(l)}(\omega)$, we have

$$\begin{aligned} \int [\chi^{(l')}(\omega)]^* \chi^{(l)}(\omega) d\Pi &= 16\pi \int_0^\pi d\omega [\chi^{(l')}(\omega)]^* \chi^{(l)}(\omega) \sin^2 \frac{\omega}{2} \\ &= 16\pi \int_0^\pi d\omega \frac{\sin(l' + \frac{1}{2})\omega}{\sin \frac{\omega}{2}} \frac{\sin(l + \frac{1}{2})\omega}{\sin \frac{\omega}{2}} \sin^2 \frac{\omega}{2} \\ &= 16\pi \int_0^\pi d\omega \sin\left(l' + \frac{1}{2}\right)\omega \sin\left(l + \frac{1}{2}\right)\omega \\ &= 8\pi \int_0^\pi d\omega [\cos(l' - l)\omega - \cos(l' + l + 1)\omega]. \end{aligned} \quad (20.151)$$

Since $l' + l + 1 > 0$, the integral of $\cos(l' + l + 1)\omega$ term vanishes. Only if $l' - l = 0$, the integral of $\cos(l' - l)\omega$ term does not vanish but takes a value of π . Therefore, we have

$$\int [\chi^{(l')}(\omega)]^* \chi^{(l)}(\omega) d\Pi = 8\pi^2 \delta_{l'l}. \quad (20.152)$$

Finally, with $\overline{[\chi^{(l')}(\omega)]^* \chi^{(l)}(\omega)}$ defined as in (20.139), we get

$$\overline{[\chi^{(l')}(\omega)]^* \chi^{(l)}(\omega)} = \delta_{l'l}. \quad (20.153)$$

This relation gives the orthogonalization condition in concert with (20.148) obtained in the case of a finite group.

In Sect. 18.6 we have shown that the number of inequivalent irreducible representations of a finite group is well-defined and given by the number of conjugacy classes of that group. This is clearly demonstrated in (18.144). In the case of the continuous groups, however, the situation is somewhat complicated and we are uncertain of the number of the inequivalent irreducible representations. We address this problem as follows: Suppose that $\Omega(\alpha, \beta, \omega)$ would be an irreducible

representation that is inequivalent to individual $D^{(l)}(\alpha, \beta, \omega)$ ($l = 0, 1, 2, \dots$). Let $K(\omega)$ be a character of $\Omega(\alpha, \beta, \omega)$. Defining $f(\omega) \equiv [\chi^{(l)}(\omega)]^* K(\omega)$, (20.150) reads as

$$\int [\chi^{(l)}(\omega)]^* K(\omega) d\Pi = 16\pi \int_0^\pi d\omega [\chi^{(l)}(\omega)]^* K(\omega) \sin^2 \frac{\omega}{2}. \quad (20.154)$$

The orthogonalization condition (20.153) demands that the integral of (20.154) vanishes. Similarly, we would have

$$\int [\chi^{(l+1)}(\omega)]^* K(\omega) d\Pi = 16\pi \int_0^\pi d\omega [\chi^{(l+1)}(\omega)]^* K(\omega) \sin^2 \frac{\omega}{2}, \quad (20.155)$$

which must vanish as well. Subtracting RHS of (20.154) from RHS of (20.155) and taking account of the fact that $\chi^{(l)}(\omega)$ is real [see (20.147)], we have

$$\int_0^\pi d\omega [\chi^{(l+1)}(\omega) - \chi^{(l)}(\omega)] K(\omega) \sin^2 \frac{\omega}{2} = 0. \quad (20.156)$$

Invoking a trigonometric formula of

$$\sin a - \sin b = 2 \cos \frac{a+b}{2} \sin \frac{a-b}{2}$$

and applying it to (20.147), (20.156) can be rewritten as

$$2 \int_0^\pi d\omega [\cos(l+1)\omega] K(\omega) \sin^2 \frac{\omega}{2} = 0 \quad (l = 0, 1, 2, \dots). \quad (20.157)$$

Putting $l = 0$ in LHS of (20.154) and from the assumption that the integral of (20.154) vanishes, we also have

$$\int_0^\pi d\omega K(\omega) \sin^2 \frac{\omega}{2} = 0, \quad (20.158)$$

where we used $\chi^{(0)}(\omega) = 1$. Then, (20.157) and (20.158) are combined to give

$$\int_0^\pi d\omega (\cos l\omega) K(\omega) \sin^2 \frac{\omega}{2} = 0 \quad (l = 0, 1, 2, \dots). \quad (20.159)$$

This implies that all the Fourier cosine coefficients of $K(\omega) \sin^2 \frac{\omega}{2}$ vanish. Considering that $\cos l\omega$ ($l = 0, 1, 2, \dots$) forms a complete orthonormal system in the region $[0, \pi]$, we must have

$$K(\omega) \sin^2 \frac{\omega}{2} \equiv 0. \quad (20.160)$$

Requiring $K(\omega)$ to be continuous, (20.160) is equivalent to $K(\omega) \equiv 0$. This implies that there is no other inequivalent irreducible representation than $D^{(l)}(\alpha, \beta, \omega)$ ($l = 0, 1, 2, \dots$). In other words, the representations $D^{(l)}(\alpha, \beta, \omega)$ ($l = 0, 1, 2, \dots$) constitute a complete set of irreducible representations. The spherical surface harmonics $Y_l^m(\theta, \phi)$ ($m = -l, \dots, 0, \dots, l$) constitute basis functions of $D^{(l)}$ and span the representation space of $D^{(l)}(\alpha, \beta, \omega)$ accordingly.

In the above discussion, we assumed that $K(\omega)$ is an even function with respect to ω as in the case of $\chi^{(l)}(\omega)$ in (20.147). Hence, $K(\omega) \sin^2 \frac{\omega}{2}$ is an even function as well. Therefore, we assumed that $K(\omega) \sin^2 \frac{\omega}{2}$ can be expanded to the Fourier cosine series.

20.3 Clebsch–Gordan Coefficients of Rotation Groups

In Sect. 18.8, we studied direct-product representations. We know that even though $D^{(\alpha)}$ and $D^{(\beta)}$ are both irreducible, $D^{(\alpha \times \beta)}$ is not necessarily irreducible. This is a common feature for both finite and infinite groups. Moreover, the reducible unitary representation can be expressed as a direct sum of irreducible representations such that

$$D^{(\alpha)}(g) \otimes D^{(\beta)}(g) = \sum_{\gamma} q_{\gamma} D^{(\gamma)}(g). \quad (18.199)$$

In the infinite groups, typically the continuous groups, this situation often appears when we deal with the addition of two angular momenta. Examples include the addition of two (or more) orbital angular momenta and that of the orbital angular momentum and spin angular momentum [6, 8].

Meanwhile, there is a set of basis vectors that spans the representation space with respect to individual irreducible representations. There, we have a following question: What are the basis vectors that span the total reduced representation space described by (18.199)? An answer for this question is that these vectors must be constructed by the basis vectors relevant to the individual irreducible representations. The Clebsch–Gordan coefficients appear as coefficients with respect to a

linear combination of the basis vectors associated with those irreducible representations.

In a word, to find the Clebsch–Gordan coefficients is equivalent to the calculation of proper coefficients of the basis functions that span the representation space of the direct-product groups. The relevant continuous groups which we are interested in are $SU(2)$ and $SO(3)$.

20.3.1 Direct-Product of $SU(2)$ and Clebsch–Gordan Coefficients

In Sect. 20.2.6 we have explained the orthogonality of irreducible characters of $SO(3)$ by deriving (20.153). It takes advanced theory to show the orthogonality of irreducible characters of $SU(2)$. Fortunately, however, we have an expression similar to (20.153) with $SU(2)$ as well. Readers are encouraged to look up appropriate literature for this [9]. On the basis of this important expression, we further develop the representation theory of the continuous groups. Within this framework, we calculate the Clebsch–Gordan coefficients.

As in the case of the previous section, the character of the irreducible representation $D^{(j)}(\alpha, \beta, \gamma)$ of $SU(2)$ is described only as a function of a rotation angle. Similarly to (20.147), in $SU(2)$ we get

$$\chi^{(j)}(\omega) = \sum_{m=-j}^j e^{im\omega} = \frac{\sin(j + \frac{1}{2})\omega}{\sin \frac{\omega}{2}}, \quad (20.161)$$

where $\chi^{(j)}(\omega)$ is an irreducible character of $D^{(j)}(\alpha, \beta, \gamma)$. It is because we can align the rotation axis in the direction of, e.g., the z -axis and the representation matrix of a rotation angle ω is typified by $D^{(j)}(\omega, 0, 0)$ or $D^{(j)}(0, 0, \omega)$. Or, we can evenly choose any direction of the rotation axis at will.

We are interested in calculating angular momenta of a coupled system, i.e. addition of angular momenta of that system [5, 6]. The word of “coupled system” needs some explanation. This is because on the one hand we may deal with, e.g., a sum of an orbital angular momentum and a spin angular momentum of a “single” electron, but on the other hand we might calculate, e.g., a sum of two angular momenta of “two” electrons. In either case, we will deal with the total angular momenta of the coupled system.

Now, let us consider a direct-product group for this kind of problem. Suppose that one system is characterized by a quantum number j_1 and the other by j_2 . Here these numbers are supposed to be those of (3.89), namely the highest positive number of the generalized angular momentum in the z -direction. That is, the representation matrices of rotation for systems 1 and 2 are assumed to be $D^{(j_1)}(\omega, 0, 0)$ and $D^{(j_2)}(\omega, 0, 0)$, respectively. Then, we are to deal with the direct-product representation $D^{(j_1)} \otimes D^{(j_2)}$ (see Sect. 18.8). As in (18.193), we describe

$$D^{(j_1 \times j_2)}(\omega) = D^{(j_1)}(\omega) \otimes D^{(j_2)}(\omega). \quad (20.162)$$

In (20.162), the group is represented by a rotation angle ω , more strictly the rotation operation of an angle ω . The quantum numbers j_1 and j_2 mean the irreducible representations of the rotations for systems 1 and 2, respectively. According to (20.161), we have

$$\chi^{(j_1 \times j_2)}(\omega) = \chi^{(j_1)}(\omega) \chi^{(j_2)}(\omega),$$

where $\chi^{(j_1 \times j_2)}(\omega)$ gives a character of the direct-product representation; see (18.198). Furthermore, we have [5]

$$\begin{aligned} \chi^{(j_1 \times j_2)}(\omega) &= \left(\sum_{m_1=-j_1}^{j_1} e^{im_1\omega} \right) \left(\sum_{m_2=-j_2}^{j_2} e^{im_2\omega} \right) \\ &= \sum_{m_1=-j_1}^{j_1} \sum_{m_2=-j_2}^{j_2} e^{i(m_1+m_2)\omega} \\ &= \sum_{J=|j_1-j_2|}^{j_1+j_2} \sum_{M=-J}^J e^{iM\omega} = \sum_{J=|j_1-j_2|}^{j_1+j_2} \chi^{(J)}(\omega), \end{aligned} \quad (20.163)$$

where a positive number J can be chosen from among

$$J = |j_1 - j_2|, |j_1 - j_2| + 1, \dots, j_1 + j_2. \quad (20.164)$$

Rewriting (20.163) more explicitly, we have

$$\chi^{(j_1 \times j_2)}(\omega) = \chi^{(|j_1 - j_2|)}(\omega) + \chi^{(|j_1 - j_2|+1)}(\omega) + \dots + \chi^{(j_1+j_2)}(\omega). \quad (20.165)$$

If both j_1 and j_2 are integers, from (20.165) we can immediately derive an important result. In light of (18.199) and (18.200), we multiply $\chi^{(k)}(\omega)^*$ on both sides of (20.165) and then integrate it to get

$$\begin{aligned} &\int \chi^{(k)}(\omega)^* \chi^{(j_1 \times j_2)}(\omega) d\Pi \\ &= \int \chi^{(k)}(\omega)^* \chi^{(|j_1 - j_2|)}(\omega) d\Pi + \int \chi^{(k)}(\omega)^* \chi^{(|j_1 - j_2|+1)}(\omega) d\Pi + \dots \\ &+ \int \chi^{(k)}(\omega)^* \chi^{(j_1+j_2)}(\omega) d\Pi \\ &= 8\pi^2 (\delta_{k,|j_1-j_2|} + \delta_{k,|j_1-j_2|+1} + \dots + \delta_{k,j_1+j_2}), \end{aligned} \quad (20.166)$$

where k is zero or an integer and with the last equality we used (20.152). Dividing both sides by $\int d\Pi$, we get

$$\overline{\chi^{(k)}(\omega)^* \chi^{(j_1 \times j_2)}(\omega)} = \delta_{k, |j_1 - j_2|} + \delta_{k, |j_1 - j_2| + 1} + \cdots + \delta_{k, j_1 + j_2}, \quad (20.167)$$

where we used (20.153). Equation (20.166) implies that only if k is identical to one out of $|j_1 - j_2|$, $|j_1 - j_2| + 1$, $\cdots$, $j_1 + j_2$, $\chi^{(k)}(\omega)^* \chi^{(j_1 \times j_2)}(\omega)$ does not vanish. If, for example, we choose $|j_1 - j_2|$ for k in (20.167), we have

$$\overline{\chi^{(|j_1 - j_2|)}(\omega)^* \chi^{(j_1 \times j_2)}(\omega)} = \delta_{|j_1 - j_2|, |j_1 - j_2|} = 1. \quad (20.168)$$

This relation corresponds to (18.200) where a finite group is relevant. Note that the volume of $SO(3)$, i.e., $\int d\Pi = 8\pi^2$ corresponds to the order n of a finite group in (20.148). Considering (18.199) and (18.200) once again, it follows that in the above case $D^{(|j_1 - j_2|)}$ takes place once and only once in the direct-product representation $D^{(j_1 \times j_2)}$. Thus, we obtain the following important relation:

$$D^{(j_1 \times j_2)} = D^{(j_1)}(\omega) \otimes D^{(j_2)}(\omega) = \sum_{J=|j_1 - j_2|}^{j_1 + j_2} D^{(J)}. \quad (20.169)$$

Any group (finite or infinite) is said to be simply reducible, if each irreducible representation takes place at most once when the direct-product representation of that group in question is reduced. Equation (20.169) clearly shows that $SO(3)$ is simply reducible. With respect to $SU(2)$ we also have the same expression of (20.169) [9].

In Sect. 18.8 we examined the direct-product representation of two (irreducible) representations. Let $D^{(\mu)}(g)$ and $D^{(\nu)}(g)$ be two different irreducible representations of the group G . Here G may be either a finite or infinite group. Note that in Sect. 18.8 we have focused on the finite groups and that now we are thinking of an infinite group, typically $SU(2)$ and $SO(3)$. Let n_μ and n_ν be a dimension of the representation space of the irreducible representations μ and ν , respectively. We assume that each representation space is spanned by following basis functions:

$$\psi_j^{(\mu)} \quad (\mu = 1, \cdots, n_\mu) \text{ and } \phi_l^{(\nu)} \quad (\nu = 1, \cdots, n_\nu).$$

We know from Sect. 18.8 that $n_\mu n_\nu$ new basis functions can be constructed using the functions that are described by $\psi_j^{(\mu)} \phi_l^{(\nu)}$. Our next task will be to classify these $n_\mu n_\nu$ functions into those belonging to different irreducible representations. For this, we need relation of (18.199). According to the custom, we rewrite it as follows:

$$D^{(\mu)}(\omega) \otimes D^{(\nu)}(\omega) = \sum_{\gamma} (\mu\nu\gamma) D^{(\gamma)}(\omega), \quad (20.170)$$

where ω denotes a group element (i.e., rotation angle); $(\mu\nu\gamma)$ is identical with q_γ of RHS of (18.199) and shows how many times the same representation γ takes place in the direct-product representations. Naturally, we have

$$(\mu\nu\gamma) = (\nu\mu\gamma).$$

Then, we get

$$\sum_{\gamma} (\mu\nu\gamma) n_{\gamma} = n_{\mu} n_{\nu}.$$

Meanwhile, we should be able to constitute the functions that have transformation properties the same as those of the irreducible representation γ . In other words, these functions should make the basis functions belonging to γ . Let the function be $\Psi_s^{(\gamma_{\tau_{\gamma}})}$. Then, we have

$$\Psi_s^{(\gamma_{\tau_{\gamma}})} = \sum_{j,l} (\mu j, \nu l | \gamma_{\tau_{\gamma}} s) \psi_j^{(\mu)} \phi_l^{(\nu)}, \quad (20.171)$$

where $\tau_{\gamma} = 1, \dots, (\mu\nu\gamma)$ and s stands for the number that designates the basis functions of the irreducible representation γ ; i.e., $s = 1, \dots, n_{\gamma}$. (We often designate the basis functions with a negative integer. In that case, we choose that negative integer for s ; see Examples 20.1 and 20.2 shown later.) If γ takes place more than once, we label γ as $\gamma_{\tau_{\gamma}}$. The coefficients with respect to the above linear combination

$$(\mu j, \nu l | \gamma_{\tau_{\gamma}} s)$$

are called Clebsch–Gordan coefficients. Readers might well be bewildered by the notations of abstract algebra, but Examples of Sect. 20.3.3 should greatly relieve them of anxiety about it.

As we shall see later in this chapter, the functions $\Psi_s^{(\gamma_{\tau_{\gamma}})}$ are going to be normalized along with the product functions $\psi_j^{(\mu)} \phi_l^{(\nu)}$. The Clebsch–Gordan coefficients must form a unitary matrix accordingly. Notice that a transformation between two orthonormal basis sets is unitary (see Chap. 14). According as there are $n_{\mu} n_{\nu}$ product functions as the basis vectors, the Clebsch–Gordan coefficients are associated with $(n_{\mu} n_{\nu}, n_{\mu} n_{\nu})$ square matrices. The said coefficients can be defined with any group either finite or infinite. Or rather, difficulty in determining the Clebsch–Gordan coefficients arises when τ_{γ} is more than one. This is because we should take account of linear combinations described by

$$\sum_{\tau_{\gamma}} c_{\gamma_{\tau_{\gamma}}} \Psi_s^{(\gamma_{\tau_{\gamma}})}$$

that belong to the irreducible representation γ . This would cause arbitrariness and ambiguity [5]. Nevertheless, we do not need to get into further discussion about this

problem. From now on, we focus on the Clebsch–Gordan coefficients of $SU(2)$ and $SO(3)$, a typical simply reducible group, namely τ_γ is at most one.

For $SU(2)$ and $SO(3)$ to be simply reducible saves us a lot of labor. The argument for this is as follows: Equation (20.169) shows how the dimension of $(2j_1 + 1)(2j_2 + 1)$ for the representation space with the direct product of two representations is redistributed to individual irreducible representations that take place only once. The arithmetic based on this fact is as follows:

$$\begin{aligned}
 & (2j_1 + 1)(2j_2 + 1) \\
 &= \left\{ 2(j_1 - j_2)(2j_2 + 1) + 2 \left[\frac{1}{2} \cdot 2j_2(2j_2 + 1) \right] \right\} + (2j_2 + 1) \\
 &= 2(j_1 - j_2)(2j_2 + 1) + 2(0 + 1 + 2 + \cdots + 2j_2) + (2j_2 + 1) \quad (20.172) \\
 &= [2(j_1 - j_2) + 0] + 2[(j_1 - j_2) + 1] + 2[(j_1 - j_2) + 2] + \cdots \\
 &+ 2[(j_1 - j_2) + 2j_2] + (2j_2 + 1) \\
 &= [2(j_1 - j_2) + 1] + \{2[(j_1 - j_2) + 1] + 1\} + \cdots + [2(j_1 + j_2) + 1],
 \end{aligned}$$

where with the second equality, we used the formula $1 + 2 + \cdots + n = \frac{1}{2}n(n + 1)$; resulting numbers of 0, 1, 2, $\cdots$, and $2j_2$ are distributed to $2j_2 + 1$ terms of the subsequent RHS of (20.172) each; the third term of $2j_2 + 1$ is distributed to $2j_2 + 1$ terms each as well on the rightmost hand side. Equation (20.172) is symmetric with j_1 and j_2 , but if we assume that $j_1 \geq j_2$, each term of the rightmost hand side of (20.172) is positive. Therefore, for convenience we assume $j_1 \geq j_2$ without loss of generality.

To clarify the situation, we list several tables (Tables 20.1, 20.2, and 20.3). Tables 20.1 and 20.2 show specific cases, but Table 20.3 represents a general case. In these tables the topmost layer has a diagram indicated by . This diagram corresponds to the case of $J = |j_1 - j_2|$ and contains different $2|j_1 - j_2| + 1$ quantum states. The lower layers include diagrams indicated by . These diagrams correspond to the case of $J = |j_1 - j_2| + 1, \cdots, j_1 + j_2$. For each J , $2J + 1$ functions (or quantum states) are included and labeled according to different numbers $M \equiv m_1 + m_2 = -J, -J + 1, \cdots, J$. The rightmost column displays $J = |j_1 - j_2|, |j_1 - j_2| + 1, \cdots, j_1 + j_2$ from the topmost row through the bottommost row.

Tables 20.1, 20.2, and 20.3 comprise $2j_2 + 1$ layers, where total $(2j_1 + 1)(2j_2 + 1)$ functions that form basis vectors of $D^{(j_1)}(\omega) \otimes D^{(j_2)}(\omega)$ are redistributed according to the M number. The same numbers that appear from the topmost row through the bottommost row are marked red and connected with dotted aqua lines. These lines form a parallelogram together with the top and bottom horizontal lines as shown. The upper and lower sides of the parallelogram are $2|j_1 - j_2|$ in width. The parallelogram gets flattened with decreasing $|j_1 - j_2|$. If $j_1 = j_2$, the parallelogram coalesces into a line (Table 20.1). Regarding the notations M and J , see Sects. 20.3.2 and 20.3.3 (vide infra).

Bearing in mind the above remarks, we focus on the coupling of two angular momenta as a typical example. Within this framework we deal with the Clebsch–Gordan coefficients with regard to $SU(2)$. We develop the discussion according to

Table 20.1 Summation of angular momenta. $J = 0$, 1, or 2; $M \equiv -J, -J + 1, \dots, J$

$m_2 \backslash m_1$	-1	0	1 ($= j_1$)
-1	-2	-1	0
0	-1	0	1
1 ($= j_2$)	0	1	2

Regarding a dotted aqua line, see text

Table 20.2 Summation of angular momenta. $J = 1, 2$, or 3; $M \equiv -J, -J + 1, \dots, J$

$m_2 \backslash m_1$	-2	-1	0	1	2 ($= j_1$)
-1	-3	-2	-1	0	1
0	-2	-1	0	1	2
1 ($= j_2$)	-1	0	1	2	3

Table 20.3 Summation of angular momenta in a general case

$m_2 \backslash m_1$	$-j_1$	$-j_1 + 1$	$\dots$	$2j_2 - j_1$	$\dots$	$j_1 - 2j_2$	$\dots$	$j_1 - 1$	j_1
$-j_2$	$-j_1 - j_2$	$-j_1 - j_2 + 1$	$\dots$	$j_2 - j_1$	$\dots$	$j_1 - 3j_2$	$\dots$	$j_1 - j_2 - 1$	$j_1 - j_2$
$-j_2 + 1$	$-j_1 - j_2 + 1$	$-j_1 - j_2 + 2$	$\dots$	$j_2 - j_1 + 1$	$\dots$	$j_1 - 3j_2 + 1$	$\dots$	$j_1 - j_2$	$j_1 - j_2 + 1$
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$
$j_2 - 1$	$-j_1 + j_2 - 1$	$-j_1 + j_2$	$\dots$	$3j_2 - j_1 - 1$	$\dots$	$j_1 - j_2 - 1$	$\dots$	$j_1 + j_2 - 2$	$j_1 + j_2 - 1$
j_2	$-j_1 + j_2$	$-j_1 + j_2 + 1$	$\dots$	$3j_2 - j_1$	$\dots$	$j_1 - j_2$	$\dots$	$j_1 + j_2 - 1$	$j_1 + j_2$

We suppose $j_1 > 2j_2$. $J = |j_1 - j_2|, \dots, j_1 + j_2$; $M \equiv -J, -J + 1, \dots, J$

literature [5]. The relevant approach helps address a more complicated situation where the coupling of three or more angular momenta including $j - j$ coupling and $L - S$ coupling is responsible [6].

From (20.47) we know that f_m forms a basis set that spans a $(2j + 1)$ -dimensional representation space associated with $D^{(j)}$ such that

$$f_m = \frac{u^{j+m} v^{j-m}}{\sqrt{(j+m)!(j-m)!}} \quad (m = -j, -j+1, \dots, j-1, j). \quad (20.47)$$

The functions f_m are transformed according to $\mathfrak{R}(a, b)$ so that we have

$$\mathfrak{R}(a, b)(f_m) \equiv \sum_{m'} f_{m'} D_{m'm}^{(j)}(a, b), \quad (20.48)$$

where j can be either an integer or a half-odd-integer. We henceforth work on f_m of (20.47).

20.3.2 Calculation Procedures of Clebsch—Gordan Coefficients

In the previous section we have described how systematically the quantum states are made up of two angular momenta. We express those states in terms of a linear combination of the basis functions related to the direct-product representations of (20.169). To determine those coefficients, we follow calculation procedures due to Hamermesh [5]. The calculation procedures are rather lengthy, and so we summarize each item separately.

1. Invariant quantities A_J and B_J :

We start with seeking invariant quantities under the unitary transformation U of (20.35) expressed as

$$U = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix} \text{ with } |a|^2 + |b|^2 = 1. \quad (20.35)$$

As in (20.46) let us consider a combination of variables $u \equiv (u_2 \ u_1)$ and $v \equiv (v_2 \ v_1)$ that are transformed such that

$$(u'_2 \ u'_1) = (u_2 \ u_1) \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix}, \quad (v'_2 \ v'_1) = (v_2 \ v_1) \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix}. \quad (20.173)$$

For a shorthand notation we have

$$u' = um, \ v' = vm, \quad (20.174)$$

where we define u', v', m as

$$u' \equiv (u'_2 \ u'_1), v' \equiv (v'_2 \ v'_1), m \equiv \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix}. \quad (20.175)$$

As in (20.47) we also define the following functions as

$$\begin{aligned} \psi_{m_1}^{j_1} &= \frac{u_1^{j_1+m_1} u_2^{j_1-m_1}}{\sqrt{(j_1+m_1)!(j_1-m_1)!}} \quad (m_1 = -j_1, -j_1+1, \dots, j_1-1, j_1), \\ \phi_{m_2}^{j_2} &= \frac{v_1^{j_2+m_2} v_2^{j_2-m_2}}{\sqrt{(j_2+m_2)!(j_2-m_2)!}} \quad (m_2 = -j_2, -j_2+1, \dots, j_2-1, j_2). \end{aligned} \quad (20.176)$$

Meanwhile, we also consider a combination of variables $x \equiv (x_2 \ x_1)$ that are transformed as

$$(x'_2 \ x'_1) = (x_2 \ x_1) \begin{pmatrix} a^* & b^* \\ -b & a \end{pmatrix}. \quad (20.177)$$

For a shorthand notation we have

$$x' = xm^*, \quad (20.178)$$

where we have

$$x' \equiv (x'_2 \ x'_1), x \equiv (x_2 \ x_1), m^* = \begin{pmatrix} a^* & b^* \\ -b & a \end{pmatrix}. \quad (20.179)$$

The unitary matrix m^* is said to be a complex conjugate matrix of m . We say that in (20.174) u and v are transformed covariantly and that in (20.178) x is transformed contravariantly. Defining a unitary operator g as

$$g \equiv \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad (20.180)$$

and operating g on both sides of (20.174), we have

$$u'g = u(gg^\dagger)mg = ug(g^\dagger mg) = ugm^*. \quad (20.181)$$

Rewriting (20.181) as

$$u'g = (ug)m^*, \quad (20.182)$$

we find that ug is transformed contravariantly. Taking transposition of (20.178), we have

$$x'^T = (m^*)^T x^T = m^\dagger x^T.$$

Then, we get

$$u'x'^T = (um)(m^\dagger x^T) = u(mm^\dagger)x^T = ux^T = u_2x_2 + u_1x_1 = ux^T, \quad (20.183)$$

where with the third equality we used the unitarity of m . This implies that ux^T is an invariant quantity under the unitary transformation by m . We have

$$v'x'^T = vx^T, \quad (20.184)$$

meaning that vx^T is an invariant as well.

In a similar manner, we have

$$\begin{aligned} u'_1v'_2 - u'_2v'_1 &= v'(u'g)^T = v'g^T u'^T = vmg^T m^T u^T = vg^T u^T = v(ug)^T \\ &= (v_2 \ v_1) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} u_2 \\ u_1 \end{pmatrix} = u_1v_2 - u_2v_1. \end{aligned} \quad (20.185)$$

Thus, $v(ug)^T$ (or $u_1v_2 - u_2v_1$) is another invariant under the unitary transformation by m . In the above discussion we can view (20.183)–(20.185) as a quadratic form of two variables (see Sect. 14.5).

Using these invariants, we define other important invariants A_J and B_J as follows [5]:

$$A_J \equiv (u_1v_2 - u_2v_1)^{j_1+j_2-J} (u_2x_2 + u_1x_1)^{j_1-j_2+J} (v_2x_2 + v_1x_1)^{j_2-j_1+J}, \quad (20.186)$$

$$B_J \equiv (u_2x_2 + u_1x_1)^{2J}. \quad (20.187)$$

The polynomial A_J is of degree $2j_1$ with respect to the covariant variables u_1 and u_2 and is of degree $2j_2$ with v_1 and v_2 . The degree of the contravariant variables x_1 and x_2 is $2J$. Expanding A_J in powers of x_1 and x_2 , we get

$$A_J = \sum_{M=-J}^J W_M^J X_M^J, \quad (20.188)$$

where X_M^J is defined as

$$X_M^J \equiv \frac{x_1^{J+M} x_2^{J-M}}{\sqrt{(J+M)!(J-M)!}} \quad (M = -J, -J+1, \dots, J-1, J) \quad (20.189)$$

and coefficients W_M^J are polynomials of u_1, u_2, v_1 , and v_2 . The coefficients W_M^J will be determined soon in combination with X_M^J .

Meanwhile, we have

$$\begin{aligned}
B_J &= \sum_{k=0}^{2J} \frac{(2J)!}{k!(2J-k)!} (u_1 x_1)^k (u_2 x_2)^{2J-k} \\
&= (2J)! \sum_{M=-J}^J \frac{1}{(J+M)!(J-M)!} (u_1 x_1)^{J+M} (u_2 x_2)^{J-M} \\
&= (2J)! \sum_{M=-J}^J \frac{1}{(J+M)!(J-M)!} u_1^{J+M} u_2^{J-M} x_1^{J+M} x_2^{J-M} \\
&= (2J)! \sum_{M=-J}^J \Phi_M^J X_M^J,
\end{aligned} \tag{20.190}$$

where Φ_M^J is defined as

$$\Phi_M^J \equiv \frac{u_1^{J+M} u_2^{J-M}}{\sqrt{(J+M)!(J-M)!}} \quad (M = -J, -J+1, \dots, J-1, J). \tag{20.191}$$

Implications of the above calculations of abstract algebra are as follows: In Sect. 20.2.2 we determined the representation matrices $D_{m'm}^{(j)}$ of $SU(2)$ using (20.47) and (20.48). We see that the functions f_m ($m = -j, -j+1, \dots, j-1, j$) of (20.47) are transformed as basis vectors by the unitary transformation $D^{(j)}$. Meanwhile, the functions Φ_M^J ($M = -J, -J+1, \dots, J-1, J$) of (20.191) are transformed as the basis vectors by the unitary transformation $D^{(J)}$. Here let us compare (20.188) and (20.190). Then, we see that whereas Φ_M^J associates X_M^J with the invariant B_J , W_M^J associates X_M^J with the invariant A_J . Thus, W_M^J plays the same role as Φ_M^J and, hence, we expect that W_M^J is eligible for the basis vectors for $D^{(J)}$ as well.

2. Binomial expansion of A_J :

To calculate W_M^J , using the binomial theorem we expand A_J such that [5]

$$\begin{aligned}
(u_1 v_2 - u_2 v_1)^{j_1+j_2-J} &= \sum_{\lambda=0}^{j_1+j_2-J} (-1)^\lambda \binom{j_1+j_2-J}{\lambda} (u_1 v_2)^{j_1+j_2-J-\lambda} (u_2 v_1)^\lambda, \\
(u_2 x_2 + u_1 x_1)^{j_1-j_2+J} &= \sum_{\mu=0}^{j_1-j_2+J} \binom{j_1-j_2+J}{\mu} (u_1 x_1)^{j_1-j_2+J-\mu} (u_2 x_2)^\mu, \\
(v_2 x_2 + v_1 x_1)^{j_2-j_1+J} &= \sum_{\nu=0}^{j_2-j_1+J} \binom{j_2-j_1+J}{\nu} (v_1 x_1)^{j_2-j_1+J-\nu} (v_2 x_2)^\nu.
\end{aligned} \tag{20.192}$$

Therefore, we have

$$A_J = \sum_{\lambda, \mu, \nu} (-1)^\lambda \binom{j_1 + j_2 - J}{\lambda} \binom{j_1 - j_2 + J}{\mu} \binom{j_2 - j_1 + J}{\nu} \quad (20.193)$$

$$\times u_1^{2j_1 - \lambda - \mu} u_2^{\lambda + \mu} v_1^{j_2 - j_1 + J - \nu + \lambda} v_2^{j_1 + j_2 - J - \lambda + \nu} x_1^{2J - \mu - \nu} x_2^{\mu + \nu}.$$

Introducing new summation variables $m_1 = j_1 - \lambda - \mu$ and $m_2 = J - j_1 + \lambda - \nu$, we obtain

$$A_J = \sum_{\lambda, \mu, \nu} (-1)^\lambda \binom{j_1 + j_2 - J}{\lambda} \binom{j_1 - j_2 + J}{j_1 - \lambda - m_1} \binom{j_2 - j_1 + J}{j_2 - \lambda + m_2} \quad (20.194)$$

$$\times u_1^{j_1 + m_1} u_2^{j_1 - m_1} v_1^{j_2 + m_2} v_2^{j_2 - m_2} x_1^{J + m_1 + m_2} x_2^{J - m_1 - m_2},$$

where to derive $\binom{j_2 - j_1 + J}{j_2 - \lambda + m_2}$ in RHS we used

$$\binom{a}{b} = \binom{a}{a-b}. \quad (20.195)$$

Further using (20.176) and (20.189), we modify A_J such that

$$A_J = \sum_{m_1, m_2, \lambda} (-1)^\lambda \left[\frac{(j_1 + j_2 - J)!(j_1 - j_2 + J)!(j_2 - j_1 + J)!}{\lambda!(j_1 + j_2 - J - \lambda)!(j_1 - \lambda - m_1)!} \right]$$

$$\times \frac{[(j_1 + m_1)!(j_1 - m_1)!(j_2 + m_2)!(j_2 - m_2)!(J + m_1 + m_2)!(J - m_1 - m_2)!]^{1/2}}{(J - j_2 + \lambda + m_1)!(j_2 - \lambda + m_2)!(J - j_1 + \lambda - m_2)!} \psi_{m_1}^{j_1} \phi_{m_2}^{j_2} X_{m_1 + m_2}^J.$$

Setting $m_1 + m_2 = M$, we have

$$A_J = \sum_{m_1, m_2, \lambda} (-1)^\lambda \left[\frac{(j_1 + j_2 - J)!(j_1 - j_2 + J)!(j_2 - j_1 + J)!}{\lambda!(j_1 + j_2 - J - \lambda)!(j_1 - \lambda - m_1)!} \right]$$

$$\times \frac{[(j_1 + m_1)!(j_1 - m_1)!(j_2 + m_2)!(j_2 - m_2)!(J + M)!(J - M)!]^{1/2}}{(J - j_2 + \lambda + m_1)!(j_2 - \lambda + m_2)!(J - j_1 + \lambda - m_2)!} \psi_{m_1}^{j_1} \phi_{m_2}^{j_2} X_M^J. \quad (20.196)$$

In this way, we get W_M^J for (20.188) expressed as

$$W_M^J = (j_1 + j_2 - J)!(j_1 - j_2 + J)!(j_2 - j_1 + J)! \quad (20.197)$$

$$\times \sum_{m_1, m_2, m_1 + m_2 = M} C_{m_1, m_2}^J \psi_{m_1}^{j_1} \phi_{m_2}^{j_2},$$

where we define C_{m_1, m_2}^J as

$$C_{m_1, m_2}^J \equiv \sum_{\lambda} \frac{(-1)^{\lambda} [(j_1 + m_1)! (j_1 - m_1)! (j_2 + m_2)! (j_2 - m_2)! (J + M)! (J - M)!]^{1/2}}{\lambda! (j_1 + j_2 - J - \lambda)! (j_1 - \lambda - m_1)! (J - j_2 + \lambda + m_1)! (j_2 - \lambda + m_2)! (J - j_1 + \lambda - m_2)!}. \quad (20.198)$$

From (20.197), we assume that the functions W_M^J ($M = -J, -J + 1, \dots, J - 1, J$) are suited for forming basis vectors of $D^{(J)}$. Then, any basis set Λ_M^J should be connected to W_M^J via unitary transformation such that

$$\Lambda_M^J = U W_M^J,$$

where U is a unitary matrix. Then, we get $W_M^J = U^\dagger \Lambda_M^J$. What we have to do is only to normalize W_M^J . If we carefully look at the functional form of (20.197), we become aware that we only have to adjust the first factor of (20.197) that is a function of only J .

Hence, as the proper normalized functions we expect to have

$$\Psi_M^J = C(J) W_M^J = C(J) U^\dagger \Lambda_M^J, \quad (20.199)$$

where $C(J)$ is a constant that depends only on J with given numbers j_1 and j_2 . An implication of (20.199) is that we can get suitably normalized functions Ψ_M^J using arbitrary basis vectors Λ_M^J .

Putting

$$\rho_J \equiv C(J) (j_1 + j_2 - J)! (j_1 - j_2 + J)! (j_2 - j_1 + J)!, \quad (20.200)$$

we get

$$\Psi_M^J = \rho_J \sum_{m_1, m_2, m_1 + m_2 = M} C_{m_1, m_2}^J \psi_{m_1}^{j_1} \phi_{m_2}^{j_2}. \quad (20.201)$$

The combined states are completely decided by J and M . If we fix J at a certain number, (20.201) is expressed as a linear combination of different functions $\psi_{m_1}^{j_1} \phi_{m_2}^{j_2}$ with varying m_1 and m_2 but fixed $m_1 + m_2 = M$. Returning back to Tables 20.1, 20.2, and 20.3, the same M can be found in at most $2j_2 + 1$ places. This implies that Ψ_M^J of (20.201) has at most $2j_2 + 1$ terms on condition that $j_1 \geq j_2$. In (20.201) the Clebsch–Gordan coefficients are given by

$$\rho_J C_{m_1, m_2}^J.$$

The descriptions of the Clebsch–Gordan coefficients are different from literature to literature [5, 6]. We adopt the description due to Hamermesh [5] such that

$$(j_1 m_1 j_2 m_2 | JM) \equiv \rho_J C_{m_1, m_2}^J, \\ \Psi_M^J = \sum_{m_1, m_2, m_1+m_2=M} (j_1 m_1 j_2 m_2 | JM) \psi_{m_1}^{j_1} \phi_{m_2}^{j_2}. \quad (20.202)$$

3. Normalization of Ψ_M^J :

Normalization condition for Ψ_M^J of (20.201) or (20.202) is given by

$$|\rho_J|^2 \sum_{m_1, m_2, m_1+m_2=M} |C_{m_1, m_2}^J|^2 = 1. \quad (20.203)$$

To obtain (20.203), we assume the following normalization condition for the basis vectors $\psi_{m_1}^{j_1} \phi_{m_2}^{j_2}$ that span the representation space. That is, for an appropriate pair of complex variables z_1 and z_2 , we define their inner product as follows [9]:

$$\langle z_1^l z_2^{m-l} | z_1^k z_2^{m-k} \rangle = \sqrt{l!(m-l)!} \sqrt{k!(m-k)!} \delta_{lk}$$

i.e.,

$$\left\langle \frac{z_1^l z_2^{m-l}}{\sqrt{l!(m-l)!}} \middle| \frac{z_1^k z_2^{m-k}}{\sqrt{k!(m-k)!}} \right\rangle = \delta_{lk}. \quad (20.204)$$

Equation (20.204) implies that $m+1$ monomials of m -degree with respect to z_1 and z_2 constitute the orthonormal basis.

Since ρ_J defined in (20.200) is independent of M , we can evaluate it by choosing M conveniently in (20.203). Setting $M (=m_1 + m_2) = J$ and substituting it in $J - j_1 + \lambda - m_2$ of (20.198), we obtain $(m_1 + m_2) - j_1 + \lambda - m_2 = \lambda + m_1 - j_1$. So far as we are dealing with integers, an inside of the factorial must be non-negative, and so we have

$$\lambda + m_1 - j_1 \geq 0 \quad \text{or} \quad \lambda \geq j_1 - m_1. \quad (20.205)$$

Meanwhile, looking at another factorial $(j_1 - \lambda - m_1)!$, we get

$$\lambda \leq j_1 - m_1. \quad (20.206)$$

From the above equations, we obtain only one choice for λ such that [5]

$$\lambda = j_1 - m_1. \quad (20.207)$$

Notice that $j_1 - m_1$ is an integer, whichever j_1 takes out of an integer or a half-odd-integer. Then, we get

$$\begin{aligned}
C_{m_1, m_2}^J &= (-1)^{j_1 - m_1} \frac{\sqrt{(2J)!}}{(J - j_2 + j_1)!(J - j_1 + j_2)!} \cdot \sqrt{\frac{(j_1 + m_1)!(j_2 + m_2)!}{(j_1 - m_1)!(j_2 - m_2)!}} \\
&= (-1)^{j_1 - m_1} \sqrt{\frac{(2J)!}{(J - j_2 + j_1)!(J - j_1 + j_2)!}} \cdot \sqrt{\binom{j_1 + m_1}{j_2 - m_2} \binom{j_2 + m_2}{j_1 - m_1}}.
\end{aligned} \tag{20.208}$$

To confirm (20.208), calculate $\binom{j_1 + m_1}{j_2 - m_2} \binom{j_2 + m_2}{j_1 - m_1}$ and use, e.g.,

$$j_1 - j_2 + m_1 + m_2 = j_1 - j_2 + M = j_1 - j_2 + J. \tag{20.209}$$

Thus, we get

$$\begin{aligned}
&\sum_{m_1, m_2, m_1 + m_2 = M} |C_{m_1, m_2}^J|^2 \\
&= \frac{(2J)!}{(J - j_2 + j_1)!(J - j_1 + j_2)!} \sum_{m_1, m_2, m_1 + m_2 = M} \binom{j_1 + m_1}{j_2 - m_2} \binom{j_2 + m_2}{j_1 - m_1}.
\end{aligned} \tag{20.210}$$

To evaluate the sum, we use the following relations [4, 5, 10]:

$$\Gamma(z)\Gamma(1 - z) = \pi / (\sin \pi z). \tag{20.211}$$

Replacing z with $x + 1$ in (20.211), we have

$$\Gamma(x + 1)\Gamma(-x) = \pi / [\sin \pi(x + 1)]. \tag{20.212}$$

Meanwhile, using gamma functions, we have

$$\begin{aligned}
\binom{x}{y} &= \frac{x!}{y!(x - y)!} = \frac{\Gamma(x + 1)}{y!\Gamma(x - y + 1)} = \frac{\Gamma(x + 1)\Gamma(-x)}{\Gamma(x - y + 1)\Gamma(y - x)} \cdot \frac{\Gamma(y - x)}{y!\Gamma(-x)} \\
&= \frac{\pi}{\sin \pi(x + 1)} \cdot \frac{\sin \pi(x - y + 1)}{\pi} \cdot \frac{\Gamma(y - x)}{y!\Gamma(-x)} = \frac{\sin \pi(x - y + 1)}{\sin \pi(x + 1)} \cdot \frac{\Gamma(y - x)}{y!\Gamma(-x)} \\
&= (-1)^y \cdot \frac{\Gamma(y - x)}{y!\Gamma(-x)}.
\end{aligned} \tag{20.213}$$

Replacing x in (20.213) with $y - x - 1$, we have

$$\binom{y-x-1}{y} = (-1)^y \cdot \frac{\Gamma(x+1)}{y!\Gamma(x-y+1)} = (-1)^y \cdot \binom{x}{y},$$

where with the last equality we used (20.213). That is, we get

$$\binom{x}{y} = (-1)^y \binom{y-x-1}{y}. \quad (20.214)$$

Applying (20.214) to (20.210), we get

$$\begin{aligned} & \sum_{m_1, m_2, m_1+m_2=M} |C_{m_1, m_2}^J|^2 \\ &= \frac{(2J)!}{(J-j_2+j_1)!(J-j_1+j_2)!} \\ & \times \sum_{m_1, m_2, m_1+m_2=M} (-1)^{j_2-m_2} (-1)^{j_1-m_1} \binom{j_2-m_2-j_1-m_1-1}{j_2-m_2} \\ & \times \binom{j_1-m_1-j_2-m_2-1}{j_1-m_1} \\ &= \frac{(-1)^{j_1+j_2-J} (2J)!}{(J-j_2+j_1)!(J-j_1+j_2)!} \sum_{m_1, m_2, m_1+m_2=M} \binom{j_2-j_1-J-1}{j_2-m_2} \binom{j_1-j_2-J-1}{j_1-m_1}. \end{aligned} \quad (20.215)$$

Meanwhile, from the binomial theorem, we have

$$\begin{aligned} (1+x)^r (1+x)^s &= \sum_{\alpha} \binom{r}{\alpha} x^{\alpha} \sum_{\beta} \binom{s}{\beta} x^{\beta} = \sum_{\alpha, \beta} \binom{r}{\alpha} \binom{s}{\beta} x^{\alpha+\beta} \\ &= (1+x)^{r+s} = \sum_{\gamma} \binom{r+s}{\gamma} x^{\gamma}. \end{aligned} \quad (20.216)$$

Comparing the coefficients of the γ -th power monomials of the last three sides, we get

$$\binom{r+s}{\gamma} = \sum_{\alpha, \beta, \alpha+\beta=\gamma} \binom{r}{\alpha} \binom{s}{\beta}. \quad (20.217)$$

Applying (20.217) to (20.215) and employing (20.214) once again, we obtain

$$\begin{aligned}
& \sum_{m_1, m_2, m_1+m_2=M} |C_{m_1, m_2}^J|^2 \\
&= \frac{(-1)^{j_1+j_2-J} (2J)!}{(J-j_2+j_1)!(J-j_1+j_2)!} \binom{-2J-2}{j_1+j_2-J} \\
&= \frac{(-1)^{j_1+j_2-J} (-1)^{j_1+j_2-J} (2J)!}{(J-j_2+j_1)!(J-j_1+j_2)!} \binom{j_1+j_2+J+1}{j_1+j_2-J} \\
&= \frac{(2J)!}{(J-j_2+j_1)!(J-j_1+j_2)!} \binom{j_1+j_2+J+1}{j_1+j_2-J} \\
&= \frac{(2J)!}{(J-j_2+j_1)!(J-j_1+j_2)!} \cdot \frac{(j_1+j_2+J+1)!}{(j_1+j_2-J)!(2J+1)!} \\
&= \frac{(j_1+j_2+J+1)!}{(2J+1)(J-j_2+j_1)!(J-j_1+j_2)!(j_1+j_2-J)!}.
\end{aligned} \tag{20.218}$$

Inserting (20.218) into (20.203), as a positive number ρ_J we have

$$\rho_J = \sqrt{\frac{(2J+1)(J-j_2+j_1)!(J-j_1+j_2)!(j_1+j_2-J)!}{(j_1+j_2+J+1)!}}. \tag{20.219}$$

From (20.200), we find

$$C(J) = \sqrt{\frac{2J+1}{(J-j_2+j_1)!(J-j_1+j_2)!(j_1+j_2-J)!(j_1+j_2+J+1)!}}. \tag{20.220}$$

Thus, as the Clebsch–Gordan coefficients $(j_1 m_1 j_2 m_2 | JM)$, at last we get

$$\begin{aligned}
(j_1 m_1 j_2 m_2 | JM) &\equiv \rho_J C_{m_1, m_2}^J \\
&= \sqrt{\frac{(2J+1)(J-j_2+j_1)!(J-j_1+j_2)!(j_1+j_2-J)!}{(j_1+j_2+J+1)!}} \\
&\times \sum_{\lambda} \frac{(-1)^{\lambda} [(j_1+m_1)!(j_1-m_1)!(j_2+m_2)!(j_2-m_2)!(J+M)!(J-M)!]^{1/2}}{\lambda!(j_1+j_2-J-\lambda)!(j_1-\lambda-m_1)!(J-j_2+\lambda+m_1)!(j_2-\lambda+m_2)!(J-j_1+\lambda-m_2)!}.
\end{aligned} \tag{20.221}$$

During the course of the above calculations, we encountered, e.g., $\Gamma(-x)$ and the factorial involving a negative integer. Under ordinary circumstances, however, such things must be avoided. Nonetheless, it would be convenient for practical use, if we properly recognize that we first choose a number, e.g., $-x$ close to (but not identical with) a negative integer and finally take the limit of $-x$ at a negative integer after the related calculations have been finished.

Choosing, e.g., $M (=m_1 + m_2) = -J$ in (20.203) instead of setting $M = J$ in the above, we will see that only the factor of $\lambda = j_2 + m_2$ in (20.198) survives. Yet we get the same result as the above. The confirmation is left for readers.

20.3.3 Examples of Calculation of Clebsch–Gordan Coefficients

Simple examples mentioned below help us understand the calculation procedures of the Clebsch–Gordan coefficients and their implications.

Example 20.1 As a simplest example, we examine a case of $D^{(1/2)} \otimes D^{(1/2)}$. At the same time, it is an example of the symmetric and antisymmetric representations discussed in Sect. 18.9. Taking two sets of complex valuables $u \equiv (u_2 \ u_1)$ and $v \equiv (v_2 \ v_1)$ and rewriting them according to (20.176), we have

$$\psi_{-1/2}^{1/2} = u_2, \psi_{1/2}^{1/2} = u_1, \phi_{-1/2}^{1/2} = v_2, \phi_{1/2}^{1/2} = v_1,$$

so that we can get the basis functions of the direct-product representation described by

$$(u_2 v_2 \ u_1 v_2 \ u_2 v_1 \ u_1 v_1).$$

Notice that we have four functions above; i.e., $(2j_1 + 1)(2j_2 + 1) = 4, j_1 = j_2 = 1/2$. Using (20.221) we can determine the proper basis functions with respect to Ψ_M^J of (20.202). For example, for Ψ_{-1}^1 we have only one choice to take $m_1 = m_2 = -\frac{1}{2}$ to get $m_1 + m_2 = M = -1$. In (20.221) we have no other choice but to take $\lambda = 0$. In this way, we have

$$\Psi_{-1}^1 = u_2 v_2 = \psi_{-1/2}^{1/2} \phi_{-1/2}^{1/2}.$$

Similarly, we get

$$\Psi_1^1 = u_1 v_1 = \psi_{1/2}^{1/2} \phi_{1/2}^{1/2}.$$

With two other functions, using (20.221) we obtain

$$\Psi_0^1 = \frac{1}{\sqrt{2}}(u_2 v_1 + u_1 v_2) = \frac{1}{\sqrt{2}}(\psi_{-1/2}^{1/2} \phi_{1/2}^{1/2} + \psi_{1/2}^{1/2} \phi_{-1/2}^{1/2}),$$

$$\Psi_0^0 = \frac{1}{\sqrt{2}}(u_1 v_2 - u_2 v_1) = \frac{1}{\sqrt{2}}(\psi_{1/2}^{1/2} \phi_{-1/2}^{1/2} - \psi_{-1/2}^{1/2} \phi_{1/2}^{1/2}).$$

We put the above results in a simple matrix form representing the basis vector transformation such that

$$(u_2 v_2 \ u_1 v_2 \ u_1 v_1 \ u_2 v_1) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 0 & 1 & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{pmatrix} = (\Psi_{-1}^1 \ \Psi_0^1 \ \Psi_1^1 \ \Psi_0^0). \quad (20.222)$$

In this way, (20.222) shows how the basis functions that possess a proper irreducible representation and span the direct-product representation space are constructed using the original product functions. The Clebsch–Gordan coefficients play a role as a linker (i.e., a unitary operator) of those two sets of functions. In this respect these coefficients resemble those appearing in the symmetry-adapted linear combinations (SALCs) mentioned in Sects. 18.2 and 19.3.

Expressing the unitary transformation according to (20.173) and choosing $(\Psi_{-1}^1 \ \Psi_0^1 \ \Psi_1^1 \ \Psi_0^0)$ as the basis vectors, we describe the transformation in a similar manner to (20.48) such that

$$\mathfrak{R}(a, b)(\Psi_{-1}^1 \ \Psi_0^1 \ \Psi_1^1 \ \Psi_0^0) = (\Psi_{-1}^1 \ \Psi_0^1 \ \Psi_1^1 \ \Psi_0^0) D^{(1/2)} \otimes D^{(1/2)}. \quad (20.223)$$

Then, as the matrix representation of $D^{(1/2)} \otimes D^{(1/2)}$ we get

$$D^{(1/2)} \otimes D^{(1/2)} = \begin{pmatrix} a^2 & \sqrt{2}ab & b^2 & 0 \\ -\sqrt{2}ab^* & aa^* - bb^* & \sqrt{2}a^*b & 0 \\ (b^*)^2 & -\sqrt{2}a^*b^* & (a^*)^2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (20.224)$$

Notice that (20.224) is a block matrix, namely (20.224) has been reduced according to the symmetric and antisymmetric representations. Symbolically writing (20.223) and (20.224), we have

$$D^{(1/2)} \otimes D^{(1/2)} = D^{(1)} \oplus D^{(0)}.$$

The block matrix of (20.224) is unitary, and so the (3, 3) submatrix and (1, 1) submatrix (i.e., the number of 1) are unitary accordingly. To show it, use the conditions (20.35) and (20.173). The confirmation is left for readers. The (3, 3) submatrix is a symmetric representation, whose representation space $(\Psi_{-1}^1 \ \Psi_0^1 \ \Psi_1^1)$

span. Note that these functions are symmetric with the exchange $m_1 \leftrightarrow m_2$. Or, we may think that these functions are symmetric with the exchange $\psi \leftrightarrow \phi$. Though trivial, the basis function Ψ_0^0 spans the antisymmetric representation space. The corresponding submatrix is merely the number 1. This is directly related to the fact that $v(ug)^T$ (or $u_1v_2 - u_2v_1$) of (20.185) is an invariant. The function Ψ_0^0 changes the sign (i.e., antisymmetric) with the exchange $m_1 \leftrightarrow m_2$ or $\psi \leftrightarrow \phi$.

Once again, readers might well ask why we need to work out such an elaborate means to pick up a small pebble. With increasing dimension of the vector space, however, to seek an appropriate set of proper basis functions becomes increasingly as difficult as lifting a huge rock. Though simple, a next example gives us a feel for it. Under such situations, a projection operator is an indispensable tool to address the problems.

Example 20.2 We examine a direct product of $D^{(1)} \otimes D^{(1)}$, where $j_1 = j_2 = 1$. This is another example of the symmetric and antisymmetric representations. Taking two sets of complex valuables $u \equiv (u_2 u_1)$ and $v \equiv (v_2 v_1)$ and rewriting them according to (20.176) again, we have nine, i.e., $(2j_1 + 1)(2j_2 + 1)$ product functions expressed as

$$\psi_{-1}^1 \phi_{-1}^1, \psi_0^1 \phi_{-1}^1, \psi_1^1 \phi_{-1}^1, \psi_{-1}^1 \phi_0^1, \psi_0^1 \phi_0^1, \psi_1^1 \phi_0^1, \psi_{-1}^1 \phi_1^1, \psi_0^1 \phi_1^1, \psi_1^1 \phi_1^1. \quad (20.225)$$

These product functions form the basis functions of the direct-product representation. Consequently, we will be able to construct proper eigenfunctions by means of linear combinations of these functions. This method is again related to that based on SALCs discussed in Chap. 19. In the present case, it is accomplished by finding the Clebsch–Gordan coefficients described by (20.221).

As implied in Table 20.1, we need to determine the individual coefficients of nine functions of (20.225) with respect to the following functions that are constituted by linear combinations of the above nine functions:

$$\Psi_{-2}^2, \Psi_{-1}^2, \Psi_0^2, \Psi_1^2, \Psi_2^2, \Psi_{-1}^1, \Psi_0^1, \Psi_1^1, \Psi_0^0.$$

(i) With the first five functions, we have $J = j_1 + j_2 = 2$. In this case, the determination procedures of the coefficients are pretty much simplified. Substituting $J = j_1 + j_2$ for the second factor of the denominator of (20.221), we get $(-\lambda)!$ As well as $\lambda!$ In the first factor of the denominator. Therefore, for the inside of factorials not to be negative we must have $\lambda = 0$. Consequently, we have [5]

$$\rho_J = \sqrt{\frac{(2j_1)!(2j_2)!}{(2J)!}},$$

$$C_{m_1, m_2}^J = \left[\frac{(J+M)!(J-M)!}{(j_1+m_1)!(j_1-m_1)!(j_2+m_2)!(j_2-m_2)!} \right]^{1/2}.$$

With $\Psi_{\pm 2}^2$ we have only one choice for ρ_J and C_{m_1, m_2}^J . That is, we have $m_1 = \pm j_1$ and $m_2 = \pm j_2$. Then, $\Psi_{\pm 2}^2 = \psi_{\pm 1}^1 \phi_{\pm 1}^1$. In the case of $M = -1$, we have two choices in (20.221); i.e., $m_1 = -1$, $m_2 = 0$ and $m_1 = 0$, $m_2 = -1$. Then, we get

$$\Psi_{-1}^2 = \frac{1}{\sqrt{2}} (\psi_{-1}^1 \phi_0^1 + \psi_0^1 \phi_{-1}^1). \quad (20.226)$$

In the case of $M = 1$, similarly we have two choices in (20.221); i.e., $m_1 = 1$, $m_2 = 0$ and $m_1 = 0$, $m_2 = 1$. We get

$$\Psi_1^2 = \frac{1}{\sqrt{2}} (\psi_1^1 \phi_0^1 + \psi_0^1 \phi_1^1). \quad (20.227)$$

Moreover, in the case of $M = 0$, we have three choices in (20.221); i.e., $m_1 = -1$, $m_2 = 1$ and $m_1 = 0$, $m_2 = 0$ along with $m_1 = 1$, $m_2 = -1$ (see Table 20.1). As a result, we get

$$\Psi_0^2 = \frac{1}{\sqrt{6}} (\psi_{-1}^1 \phi_1^1 + 2\psi_0^1 \phi_0^1 + \psi_1^1 \phi_{-1}^1). \quad (20.228)$$

(ii) With $J = 1$, i.e., Ψ_{-1}^1 , Ψ_0^1 , and Ψ_1^1 , we start with (20.221). Noting the $\lambda!$ and $(1 - \lambda)!$ in the first two factors of denominator, we have $\lambda = 0$ or $\lambda = 1$. In the case of $M = -1$, we have only one choice of $m_1 = 0$, $m_2 = -1$ for $\lambda = 0$. Similarly, $m_1 = -1$, $m_2 = 0$ for $\lambda = 1$. Hence, we get

$$\Psi_{-1}^1 = \frac{1}{\sqrt{2}} (\psi_0^1 \phi_{-1}^1 - \psi_{-1}^1 \phi_0^1). \quad (20.229)$$

In a similar manner, for $M = 0$ and $M = 1$, respectively, we have

$$\Psi_0^1 = \frac{1}{\sqrt{2}} (\psi_1^1 \phi_{-1}^1 - \psi_{-1}^1 \phi_1^1), \quad (20.230)$$

$$\Psi_1^1 = \frac{1}{\sqrt{2}} (\psi_1^1 \phi_0^1 - \psi_0^1 \phi_1^1). \quad (20.231)$$

(iii) With $J = 0$ (i.e., Ψ_0^0), we have $J = j_1 - j_2 (=0)$. In this case, we have $J - j_1 + \lambda - m_2 = -j_2 + \lambda - m_2$ in the denominator of (20.221) [5]. Since this factor is inside the factorial, we have $-j_2 + \lambda - m_2 \geq 0$. Also, we have $j_2 - \lambda + m_2 \geq 0$ in the denominator of (20.221). Hence, we have only one choice of $\lambda = j_2 + m_2$. Therefore, we get

$$\rho_J = \sqrt{\frac{(2J+1)!(2j_2)!}{(2j_1+1)!}},$$

$$C_{m_1, m_2}^J = (-1)^{j_2+m_2} \left[\frac{(j_1+m_1)!(j_1-m_1)!}{(J+M)!(J-M)!(j_2+m_2)!(j_2-m_2)!} \right]^{1/2}.$$

In the above, we have three choices of $m_1 = -1, m_2 = 1; m_1 = m_2 = 0; m_1 = 1, m_2 = -1$. As a result, we get

$$\Psi_0^0 = \frac{1}{\sqrt{3}} (\psi_{-1}^1 \phi_1^1 - \psi_0^1 \phi_0^1 + \psi_1^1 \phi_{-1}^1).$$

Summarizing the above results, we have constructed the proper eigenfunctions with respect to the combined angular momenta using the basis functions of the direct-product representation. An example of the relevant matrix representations is described as follows:

$$\begin{aligned} & (\psi_{-1}^1 \phi_{-1}^1 \psi_0^1 \phi_{-1}^1 \psi_1^1 \phi_{-1}^1 \psi_0^1 \phi_1^1 \psi_1^1 \phi_1^1 \psi_{-1}^1 \phi_0^1 \psi_{-1}^1 \phi_1^1 \psi_1^1 \phi_0^1 \psi_0^1 \phi_0^1) \\ & \times \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{6}} & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{3}} \\ 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & -\frac{1}{\sqrt{2}} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{6}} & 0 & 0 & 0 & -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{3}} \\ 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & \frac{2}{\sqrt{6}} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{\sqrt{3}} \end{pmatrix} \quad (20.232) \\ & = (\Psi_{-2}^2 \Psi_{-1}^2 \Psi_0^2 \Psi_1^2 \Psi_2^2 \Psi_{-1}^1 \Psi_0^1 \Psi_1^1 \Psi_0^0). \end{aligned}$$

In (20.232) the matrix elements represent the Clebsch–Gordan coefficients and their combination in (20.232) forms a unitary matrix. In (20.232) we have arbitrariness with the disposition of the elements of both row and column vectors. In other

words, we have the arbitrariness with the unitary similarity transformation of the matrix, but the unitary matrix that underwent the unitary similarity transformation is again a unitary matrix (see Chap. 14). To show that the determinant of (20.232) is -1 , use the expansion of the determinant and use the fact that when (a multiple of) a certain column (or row) is added to another column (or row), the determinant is unchanged (see Sect. 11.3).

We find that the functions belonging to $J = 2$ and $J = 0$ are the basis functions of the symmetric representations. Meanwhile, those belonging to $J = 1$ are the basis functions of the antisymmetric representations (see Sect. 18.9). Expressing the unitary transformation of this example according to (20.173) and choosing $(\Psi_{-2}^2 \Psi_{-1}^2 \Psi_0^2 \Psi_1^2 \Psi_2^2 \Psi_{-1}^1 \Psi_0^1 \Psi_1^1 \Psi_0^0)$ as the basis vectors, we describe the transformation as

$$\begin{aligned} \mathfrak{R}(a, b) (\Psi_{-2}^2 \Psi_{-1}^2 \Psi_0^2 \Psi_1^2 \Psi_2^2 \Psi_{-1}^1 \Psi_0^1 \Psi_1^1 \Psi_0^0) \\ = (\Psi_{-2}^2 \Psi_{-1}^2 \Psi_0^2 \Psi_1^2 \Psi_2^2 \Psi_{-1}^1 \Psi_0^1 \Psi_1^1 \Psi_0^0) D^{(1)} \otimes D^{(1)}. \end{aligned} \quad (20.233)$$

Notice again that the representation matrix of $D^{(1)} \otimes D^{(1)}$ can be converted (or reduced) to block unitary matrices according to the symmetric and antisymmetric representations such that

$$D^{(1)} \otimes D^{(1)} = D^{(2)} \oplus D^{(1)} \oplus D^{(0)}, \quad (20.234)$$

where $D^{(2)}$ and $D^{(0)}$ are the symmetric representations and $D^{(1)}$ is the antisymmetric representation. Recall that $SO(3)$ is simply reducible. To show (20.234), we need somewhat lengthy calculations, but the computation procedures are straightforward.

A set of basis functions $(\Psi_{-2}^2 \Psi_{-1}^2 \Psi_0^2 \Psi_1^2 \Psi_2^2)$ spans a five-dimensional symmetric representation space. In turn, Ψ_0^0 spans a one-dimensional symmetric representation space. Another set of basis functions $(\Psi_{-1}^1 \Psi_0^1 \Psi_1^1)$ spans a three-dimensional antisymmetric representation space.

We add that to seek some special Clebsch–Gordan coefficients would be easy. For example, we can readily find out them in the case of $J = j_1 + j_2 = M$ or $J = j_1 + j_2 = -M$. The final result of the normalized basis functions can be

$$\Psi_{j_1+j_2}^{j_1+j_2} = \psi_{j_1}^{j_1} \phi_{j_2}^{j_2}, \Psi_{-(j_1+j_2)}^{j_1+j_2} = \psi_{-j_1}^{j_1} \phi_{-j_2}^{j_2}.$$

That is, among the related coefficients, only

$$(j_1, m_1 = j_1, j_2, m_2 = j_2 | J = j_1 + j_2, M = J)$$

and

$$(j_1, m_1 = -j_1, j_2, m_2 = -j_2 | J = j_1 + j_2, M = -J)$$

survive and take a value of 1; see the corresponding parts of the matrix elements in (20.221). This can readily be seen in Tables 20.1, 20.2, and 20.3.

20.4 Lie Groups and Lie Algebras

In Sect. 20.2 we have developed a practical approach to dealing with various aspects and characteristics of $SU(2)$ and $SO(3)$. In this section we introduce an elegant theory of Lie groups and Lie algebras that enable us to systematically study $SU(2)$ and $SO(3)$.

20.4.1 Definition of Lie Groups and Lie Algebras: One-Parameter Groups

We start with the following discussion. Let $A(t)$ be a non-singular matrix whose elements vary as a function of a real number t . Here we think of a matrix as a function of a real number t . If under such conditions $A(t)$ constitutes a group, $A(t)$ is said to be a one-parameter group. In particular, we are interested in a situation where we have

$$A(s+t) = A(s)A(t). \quad (20.235)$$

We further get

$$A(s)A(t) = A(s+t) = A(t+s) = A(t)A(s). \quad (20.236)$$

Therefore, any pair of $A(t)$ and $A(s)$ is commutative. Putting $t = s = 0$ in (20.236), we have

$$A(0) = A(0)A(0). \quad (20.237)$$

Since $A(0)$ is non-singular, $A(0)^{-1}$ must exist. Then, multiplying $A(0)^{-1}$ on both sides of (20.237), we obtain

$$A(0) = E, \quad (20.238)$$

where E is an identity matrix. Also, we have

$$A(t)A(-t) = A(0) = E, \quad (20.239)$$

meaning that

$$A(-t) = A(t)^{-1}.$$

Differentiating both sides of (20.235) with respect to s at $s = 0$, we get

$$A'(t) = A'(0)A(t). \quad (20.240)$$

Defining

$$X \equiv A'(0), \quad (20.241)$$

we obtain

$$A'(t) = XA(t). \quad (20.242)$$

Integrating (20.242), we have

$$A(t) = \exp(tX) \quad (20.243)$$

under an initial condition $A(0) = E$. Thus, any one-parameter group $A(t)$ is described by (20.243) on condition that $A(t)$ is represented by (20.235).

In Sect. 20.1 we mentioned the Lie algebra in relation to (20.14). In this context (20.243) shows how the Lie algebra is related to the one-parameter group. For a group to be a continuous group is thus deeply connected to this one-parameter group. Equation (20.45) of Sect. 20.2 is an example of product of one-parameter groups. Bearing in mind the above situation, we describe definitions of the Lie group and Lie algebra.

Definition 20.2 [9] Let G be a subgroup of $GL(n, \mathbb{C})$. Suppose that A_k ($k = 1, 2, \dots$) $\in G$ and that $\lim_{k \rightarrow \infty} A_k = A$ [$\in GL(n, \mathbb{C})$]. If $A \in G$, then G is said to be a linear Lie group of dimension n .

We usually call a linear Lie group simply a Lie group. Note that $GL(n, \mathbb{C})$ has appeared in Sect. 16.1. The definition again is most likely to bewilder chemists in that they read it as though the definition itself including the word of Lie group was written in the air. Nevertheless, a next example is again expected to relieve the chemists of useless anxiety.

Example 20.3 Let G be a group and let $A_k \in G$ with $\det A_k = 1$ such that

$$A_k = \begin{pmatrix} \cos \theta_k & -\sin \theta_k \\ \sin \theta_k & \cos \theta_k \end{pmatrix} (k : \text{real number}). \quad (20.244)$$

Suppose that $\lim_{k \rightarrow \infty} \theta_k = \theta$ and then we have

$$\lim_{k \rightarrow \infty} A_k = \lim_{k \rightarrow \infty} \begin{pmatrix} \cos \theta_k & -\sin \theta_k \\ \sin \theta_k & \cos \theta_k \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \equiv A. \quad (20.245)$$

Certainly, we have $A \in G$, because $\det A_k = \det A = 1$. Such group is called $SO(2)$.

In a word, a lie group is a group whose component consists of continuous and differentiable (or analytic) functions. In that sense the groups such as $SU(2)$ and $SO(3)$ we have already encountered are Lie groups.

Next, the following is in turn a definition of a Lie algebra.

Definition 20.3 [9] Let G be a linear Lie group. If with $\forall t$ (t : real number) we have

$$\exp(tX) \in G, \quad (20.246)$$

all X are called a Lie algebra of the corresponding Lie group G . We denote the Lie algebra of the Lie group G by $\mathfrak{g}$.

The relation (20.246) includes the case where X is a $(1, 1)$ matrix (i.e., merely a complex number) as the trivial case. Most importantly, (20.246) is directly connected to (20.243) and Definition 20.3 shows the direct relevance between the Lie group and Lie algebra. The two theories of Lie group and Lie algebra underlie continuous groups.

The lie algebra has following properties:

- (i) If $X \in \mathfrak{g}$, $aX \in \mathfrak{g}$ as well, where a is any real number.
- (ii) If $X, Y \in \mathfrak{g}$, $X + Y \in \mathfrak{g}$ as well.
- (iii) If $X, Y \in \mathfrak{g}$, $[X, Y] (\equiv XY - YX) \in \mathfrak{g}$ as well.

In fact, $\exp[t(aX)] = \exp[(ta)X]$ and ta is a real number so that we have (i). Regarding (ii) we should examine individual properties of X . As already shown in Property (7)' of Chap. 15 (Sect. 15.2), e.g., a unitary matrix of $\exp(tX)$ is associated with an anti-Hermitian matrix of X ; we are interested in this particular case. In the case of $SU(2)$ X is a traceless anti-Hermitian matrix and as to $SO(3)$ X is a real skew-symmetric matrix. In the former case, for example, traceless anti-Hermitian matrices X and Y can be expressed as

$$X = \begin{pmatrix} ic & -a + ib \\ a + ib & -ic \end{pmatrix}, Y = \begin{pmatrix} ih & -f + ig \\ f + ig & -ih \end{pmatrix}, \quad (20.247)$$

where a, b, c , etc. are real. Then, we have

$$X + Y = \begin{pmatrix} i(c + h) & -(a + f) + i(b + g) \\ (a + f) + i(b + g) & -i(c + h) \end{pmatrix}. \quad (20.248)$$

Thus, $X + Y$ is a traceless anti-Hermitian matrix as well. With the property (iii), we need some explanation. Suppose $X, Y \in \mathfrak{g}$. Then, with any real numbers s and t , $\exp(sX)$, $\exp(tY)$, $\exp(-sX)$, $\exp(-tY) \in G$ by Definition 20.3. Then, their product $\exp(sX) \exp(tY) \exp(-sX) \exp(-tY) \in G$ as well. Taking an infinitesimal transformation of these four products within the first order of s and t , we have

$$\begin{aligned}
& \exp(sX) \exp(tY) \exp(-sX) \exp(-tY) \\
& \approx (1 + sX)(1 + tY)(1 - sX)(1 - tY) \\
& = (1 + tY + sX + stXY)(1 - tY - sX + stXY) \\
& \approx (1 - tY - sX + stXY) + (tY - stYX) + (sX - stXY) + stXY \\
& = 1 + st(XY - YX).
\end{aligned} \tag{20.249}$$

Notice that we ignored the terms having s^2 , t^2 , st^2 , s^2t , and s^2t^2 as a coefficient. Defining

$$[X, Y] \equiv XY - YX, \tag{20.250}$$

we have

$$\exp(sX) \exp(tY) \exp(-sX) \exp(-tY) \approx 1 + st[X, Y] \approx \exp(st[X, Y]). \tag{20.251}$$

Since LHS represents an element of G and st is a real number, by Definition 20.3 $[X, Y] \in \mathfrak{g}$. The quantity $[X, Y]$ is called a commutator. The commutator and its definition appeared in (1.139).

From properties (i) and (ii), the Lie algebra $\mathfrak{g}$ forms a vector space (see Chap. 11). An element of $\mathfrak{g}$ is usually expressed by a matrix. Zero vector corresponds to zero matrix (see Proposition 15.1). As already seen in Chap. 15, we defined an inner product between any $A, B \in \mathfrak{g}$ such that

$$\langle A | B \rangle \equiv \sum_{i,j} a_{ij}^* b_{ij}. \tag{20.252}$$

Thus, $\mathfrak{g}$ constitutes an inner product (vector) space.

20.4.2 Properties of Lie Algebras

Let us further continue our discussion of Lie algebras by thinking of a following proposition about the inner product.

Proposition 20.1 Let $f(t)$ and $g(t)$ be continuous and differentiable functions with respect to a real number t . Then, we have

$$\frac{d}{dt} \langle f(t) | g(t) \rangle = \left\langle \frac{df(t)}{dt} | g(t) \right\rangle + \left\langle f(t) | \frac{dg(t)}{dt} \right\rangle. \tag{20.253}$$

Proof We calculate a following equation:

$$\begin{aligned}
& \langle f(t + \Delta) | g(t + \Delta) \rangle - \langle f(t) | g(t) \rangle \\
&= \langle f(t + \Delta) | g(t + \Delta) \rangle - \langle f(t) | g(t + \Delta) \rangle + \langle f(t) | g(t + \Delta) \rangle - \langle f(t) | g(t) \rangle \\
&= \langle f(t + \Delta) - f(t) | g(t + \Delta) \rangle + \langle f(t) | g(t + \Delta) - g(t) \rangle.
\end{aligned} \tag{20.254}$$

Dividing both sides of (20.254) by a real number Δ and taking a limit as below, we have

$$\begin{aligned}
& \lim_{\Delta \rightarrow 0} \frac{1}{\Delta} [\langle f(t + \Delta) | g(t + \Delta) \rangle - \langle f(t) | g(t) \rangle] \\
&= \lim_{\Delta \rightarrow 0} \left[\left\langle \frac{f(t + \Delta) - f(t)}{\Delta} | g(t + \Delta) \right\rangle + \left\langle f(t) | \frac{g(t + \Delta) - g(t)}{\Delta} \right\rangle \right],
\end{aligned} \tag{20.255}$$

where we used the calculation rules for inner products of (13.3) and (13.20). Then, we get (20.253). This completes the proof.

Now, let us apply (20.253) to a unitary operator $A(t)$ we dealt with in the previous section. Replacing $f(t)$ and $g(t)$ in (20.253) with $\psi A(t)^\dagger$ and $A(t)\chi$, respectively, we have

$$\frac{d}{dt} \langle \psi A(t)^\dagger | A(t) \chi \rangle = \left\langle \psi \frac{dA(t)^\dagger}{dt} | A(t) \chi \right\rangle + \left\langle \psi A(t)^\dagger | \frac{dA(t)}{dt} \chi \right\rangle, \tag{20.256}$$

where ψ and χ are arbitrarily chosen vectors that do not depend on t . Then, we have

$$\text{LHS of (20.256)} = \frac{d}{dt} \langle \psi | A(t)^\dagger A(t) | \chi \rangle = \frac{d}{dt} \langle \psi | E | \chi \rangle = \frac{d}{dt} \langle \psi | \chi \rangle = 0, \tag{20.257}$$

where with the second equality we used the unitarity of $A(t)$. Taking limit $t \rightarrow 0$ in (20.256), we have

$$\lim_{t \rightarrow 0} [\text{RHS of (20.256)}] = \left\langle \psi \frac{dA(0)^\dagger}{dt} | A(0) \chi \right\rangle + \left\langle \psi A(0)^\dagger | \frac{dA(0)}{dt} \chi \right\rangle.$$

Meanwhile, taking a limit of $t \rightarrow 0$ in the following relation, we have

$$\left. \frac{dA(t)^\dagger}{dt} \right|_{t \rightarrow 0} = \left[\left. \frac{dA(t)}{dt} \right|_{t \rightarrow 0} \right]^\dagger = [A'(0)]^\dagger,$$

where we assumed that operations of the adjoint and $t \rightarrow 0$ are commutable. Then, we get

$$\begin{aligned} \lim_{t \rightarrow 0} [\text{RHS of (20.256)}] &= \langle \psi [A'(0)]^\dagger | A(0) \chi \rangle + \left\langle \psi A(0)^\dagger \left| \frac{dA(0)}{dt} \chi \right. \right\rangle \\ &= \langle \psi X^\dagger | \chi \rangle + \langle \psi | X \chi \rangle = \langle \psi (X^\dagger + X) | \chi \rangle, \end{aligned} \quad (20.258)$$

where we used $X \equiv A'(0)$ of (20.241) and $A(0) = A(0)^\dagger = E$. From (20.257) and (20.258), we have

$$0 = \langle \psi (X^\dagger + X) | \chi \rangle.$$

Since ψ and χ are arbitrary, from Theorem 14.2 we get

$$X^\dagger + X \equiv 0 \quad \text{i.e.,} \quad X^\dagger = -X. \quad (20.259)$$

This indicates that X is an anti-Hermitian operator. Let X be expressed as (x_{ij}) . Then, from (20.259) $x_{ij} = -x_{ji}^*$. As for diagonal elements, $x_{ii} = -x_{ii}^*$; i.e., $x_{ii} + x_{ii}^* = 0$. Hence, x_{ii} is a pure imaginary number or zero. We have the following theorem accordingly.

Theorem 20.3 Let $A(t)$ be a one-parameter unitary group with $A(0) = E$ that is described by $A(t) = \exp(tX)$ and satisfies (20.235). Then, $X \equiv A'(0)$ is an anti-Hermitian operator.

Differentiating both sides of $A(t) = \exp(tX)$ with respect to t and using (15.47) of Theorem 15.4 [11, 12], we have

$$A'(t) = X \exp(tX) = XA(t) = A(t)X. \quad (20.260)$$

Putting $t = 0$ in (20.260), we restore $X = A'(0)$. If we require $A(t)$ to be unitary, from Theorem 20.3 again X should be anti-Hermitian. Thus, once we have a one-parameter group in a form of a unitary operator $A(t) = \exp(tX)$, the exponent can be separated into a product of a parameter t and a t -independent constant anti-Hermitian operator X . In fact, all the one-parameter groups that have appeared in Sect. 20.2 are of a type of $\exp(tX)$.

Conversely, let us consider what type of operator $\exp(tX)$ would be if X is anti-Hermitian. Here we assume that X is a (n, n) matrix. With an arbitrary real number t we have

$$\begin{aligned} [\exp(tX)][\exp(tX)]^\dagger &= [\exp(tX)][\exp(tX^\dagger)] = [\exp(tX)][\exp(-tX)] \\ &= [\exp(tX - tX)] = \exp 0 = E, \end{aligned} \quad (20.261)$$

where we used (15.32) and Theorem 15.2 as well as the assumption that X is anti-Hermitian. Note that $\exp(tX)$ and $\exp(-tX)$ are commutative; see (15.29) of Sect. 15.2. Equation (20.261) implies that $\exp(tX)$ is unitary, i.e., $\exp(tX) \in U(n)$. The

notation $U(n)$ means a unitary group. Hence, $X \in \mathfrak{u}(n)$, where $\mathfrak{u}(n)$ means the Lie algebra corresponding to $U(n)$.

Next, let us seek a Lie algebra that corresponds to a special unitary group $SU(n)$. By Definition 20.3, it is obvious that since $U(n) \supset SU(n)$, $\mathfrak{u}(n) \supset \mathfrak{su}(n)$, where $\mathfrak{su}(n)$ means the Lie algebra corresponding to $SU(n)$. It suffices to find the condition under which $\det[\exp(tX)] = 1$ holds with any real number t and the elements $X \in \mathfrak{u}(n)$. From (15.41) [9], we have

$$\det[\exp(tX)] = \exp \operatorname{Tr}(tX) = \exp t[\operatorname{Tr}(X)] = 1, \quad (20.262)$$

where Tr stands for trace (see Chap. 12). This is equivalent to that

$$t[\operatorname{Tr}(X)] = 2mi\pi \quad (m : \text{zero or integers}) \quad (20.263)$$

holds with any real t . For this, we must have $\operatorname{Tr}(X) = 0$ with $m = 0$. This implies that $\mathfrak{su}(n)$ comprises anti-Hermitian matrices with its trace being zero (i.e., traceless). Consequently, the Lie algebra $\mathfrak{su}(n)$ is certainly a subset (or subspace) of $\mathfrak{u}(n)$ that consists of anti-Hermitian matrices.

In relation to (20.256) let us think of a real orthogonal matrix $A(t)$. If $A(t)$ is real and orthogonal, (20.256) can be rewritten as

$$\frac{d}{dt} \langle \psi A(t)^T | A(t) \chi \rangle = \left\langle \psi \frac{dA(t)^T}{dt} | A(t) \chi \right\rangle + \left\langle \psi A(t)^T | \frac{dA(t)}{dt} \chi \right\rangle. \quad (20.264)$$

Following the procedure similar to the case of (20.259), we get

$$X^T + X \equiv 0 \quad \text{i.e.,} \quad X^T = -X. \quad (20.265)$$

In this case we obtain a skew-symmetric matrix. All its diagonal elements are zero and, hence, the skew-symmetric matrix is traceless. Other typical Lie groups are an orthogonal group $O(n)$ and a special orthogonal group $SO(n)$ and the corresponding Lie algebras are denoted by $\mathfrak{o}(n)$ and $\mathfrak{so}(n)$, respectively. Note that both $\mathfrak{o}(n)$ and $\mathfrak{so}(n)$ consist of skew-symmetric matrices.

Conversely, let us consider what type of operator $\exp(tX)$ would be if X is a skew-symmetric matrix. Here we assume that X is a (n, n) matrix. With an arbitrary real number t we have

$$\begin{aligned} [\exp(tX)][\exp(tX)]^T &= [\exp(tX)][\exp(tX^T)] = [\exp(tX)][\exp(-tX)] \\ &= [\exp(tX - tX)] = \exp 0 = E, \end{aligned} \quad (20.266)$$

where we used (15.31). This implies that $\exp(tX)$ is an orthogonal matrix, i.e., $\exp(tX) \in O(n)$. Hence, $X \in \mathfrak{o}(n)$. Notice that $\exp(tX)$ and $\exp(-tX)$ are commutative.

Summarizing the above, we have the following theorem.

Theorem 20.4 The n -th order Lie algebra $\mathfrak{u}(n)$ corresponding to the Lie group $U(n)$ (i.e., unitary group) consists of all the anti-Hermitian (n, n) matrices. The n -th order Lie algebra $\mathfrak{su}(n)$ corresponding to the Lie group $SU(n)$ (i.e., special unitary group) comprises all the anti-Hermitian (n, n) matrices with the trace zero. The n -th order Lie algebras $\mathfrak{o}(n)$ and $\mathfrak{so}(n)$ corresponding to the Lie groups $O(n)$ and $SO(n)$, respectively, are all the real skew-symmetric (n, n) matrices.

Notice that if A and B are anti-Hermitian matrices, so are $A + B$ and cA (c : real number). This is true of skew-symmetric matrices as well. Hence, $\mathfrak{u}(n)$, $\mathfrak{su}(n)$, $\mathfrak{o}(n)$, and $\mathfrak{so}(n)$ form a linear vector space.

In Sect. 20.3.1 we introduced a commutator. The commutators are ubiquitous in quantum mechanics, especially as commutation relations (see Part I). Major properties are as follows:

$$\begin{aligned} \text{(i)} \quad & [X + Y, Z] = [X, Z] + [Y, Z], \\ \text{(ii)} \quad & [aX, Y] = a[X, Y] \quad (a \in \mathbb{R}), \\ \text{(iii)} \quad & [X, Y] = -[Y, X], \\ \text{(iv)} \quad & [X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0. \end{aligned} \tag{20.267}$$

The last equation of (20.267) is well-known as Jacobi's identity. Readers are encouraged to check it.

Since the Lie algebra forms a linear vector space of a finite dimension, we denote its basis vectors by $X_1, X_2, \dots, X_d$. Then, we have

$$\mathfrak{g}(d) = \text{Span}\{X_1, X_2, \dots, X_d\}, \tag{20.268}$$

where $\mathfrak{g}(d)$ stands for a Lie algebra of dimension d . As their commutators belong to $\mathfrak{g}(d)$ as well, with an arbitrary pair of X_i and X_j ($i, j = 1, 2, \dots, d$) we have

$$[X_i, X_j] = \sum_{k=1}^d f_{ijk} X_k, \tag{20.269}$$

where a set of real coefficients f_{ijk} is said to be structure constants, which define the structure of the Lie algebra.

Example 20.4 In (20.42) we write $\zeta_1 \equiv \zeta_x$, $\zeta_2 \equiv \zeta_y$, and $\zeta_3 \equiv \zeta_z$. Rewriting it explicitly, we have

$$\zeta_1 = \frac{1}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \zeta_2 = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \zeta_3 = \frac{1}{2} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}. \tag{20.270}$$

Then, we get

$$[\zeta_i, \zeta_j] = \sum_{k=1}^3 \epsilon_{ijk} \zeta_k, \quad (20.271)$$

where ϵ_{ijk} is called the Levi-Civita symbol [4] and denoted by

$$\epsilon_{ijk} = \begin{cases} +1 & (i, j, k) = (1, 2, 3), (2, 3, 1), (3, 1, 2); \\ -1 & (i, j, k) = (3, 2, 1), (1, 3, 2), (2, 1, 3); \\ 0 & \text{otherwise.} \end{cases} \quad (20.272)$$

Notice that in (20.272) if (i, j, k) represents an even permutation of $(1, 2, 3)$, $\epsilon_{ijk} = 1$, but if (i, j, k) is an odd permutation of $(1, 2, 3)$, $\epsilon_{ijk} = -1$. Otherwise, for e.g., $i = j$, $j = k$, or $k = i$, etc. $\epsilon_{ijk} = 0$. The relation of (20.271) is essentially the same as (3.30) and (3.69) of Chap. 3. We have

$$\mathfrak{su}(2) = \text{Span}\{\zeta_1, \zeta_2, \zeta_3\}. \quad (20.273)$$

Equation (20.270) shows that the trace of ζ_i ($i = 1, 2, 3$) is zero.

Example 20.5 In (20.26) we write $A_1 \equiv A_x$, $A_2 \equiv A_y$, and $A_3 \equiv A_z$. Rewriting it, we have

$$A_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, A_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, A_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (20.274)$$

Again, we have

$$[A_i, A_j] = \sum_{k=1}^3 \epsilon_{ijk} A_k. \quad (20.275)$$

This is of the same form as that of (20.271). Namely,

$$\mathfrak{so}(3) = \mathfrak{o}(3) = \text{Span}\{A_1, A_2, A_3\}. \quad (20.276)$$

Equation (20.274) clearly shows that diagonal elements of A_i ($i = 1, 2, 3$) are zero.

From Examples 20.4 and 20.5, $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ [or $\mathfrak{o}(3)$] structurally resemble each other. This fact is directly related to similarity between $SU(2)$ and $SO(3)$. Note also that the dimensionality of $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ is the same in terms of linear vector spaces.

In Chap. 3, we deal with the generalized angular momentum using the relation (3.30). Using (20.15), we can readily obtain the relation the same as (20.271) and (20.275). That is, from (3.30) defining the following anti-Hermitian operators as

$$\tilde{J}_x \equiv -\frac{iJ_x}{\hbar}, \tilde{J}_y \equiv -\frac{iJ_y}{\hbar}, \tilde{J}_z \equiv -iJ_z/\hbar,$$

we get

$$[\tilde{J}_l, \tilde{J}_m] = \sum_{k=1}^3 \epsilon_{lmn} \tilde{J}_n,$$

where l, m, n stand for x, y, z . The derivation is left for readers.

20.4.3 Adjoint Representation of Lie Groups

In (20.252) of Sect. 20.3.1 we defined an inner product of $\mathfrak{g}$ as

$$\langle A|B \rangle \equiv \sum_{i,j} a_{ij}^* b_{ij}. \quad (20.252)$$

We can readily check that the definition (20.252) satisfies those of the inner product of (13.2), (13.3), and (13.4). In fact, we have

$$\langle B|A \rangle = \sum_{i,j} b_{ij}^* a_{ij} = \sum_{i,j} (a_{ij}^* b_{ij})^* = \left(\sum_{i,j} a_{ij}^* b_{ij} \right)^* = \langle A|B \rangle^*. \quad (20.277)$$

Equation (13.3) is obvious from the calculation rules of a matrix. With (13.4), we get

$$\langle A|A \rangle = \text{Tr}(A^\dagger A) = \sum_{i,j} |a_{ij}|^2 \geq 0, \quad (20.278)$$

where we have $A = (a_{ij})$ and the equality holds if and only if all $a_{ij} = 0$, i.e., $A = 0$. Thus, $\langle A|A \rangle$ gives a positive definite inner product on $\mathfrak{g}$.

From (20.278), we may equally define an inner product of $\mathfrak{g}$ as

$$\langle A|B \rangle = \text{Tr}(A^\dagger B). \quad (20.279)$$

In fact, we have

$$\text{Tr}(A^\dagger B) = \sum_{i,j} (A^\dagger)_{ij} (B)_{ji} = \sum_{i,j} a_{ji}^* b_{ji} \equiv \langle A|B \rangle.$$

In another way, we can readily compare (20.279) with (20.252) using, e.g., a $(2, 2)$ matrix and find that both equations give the same result. It is left for readers as an exercise.

From Theorem 20.4, $\mathfrak{u}(n)$ corresponding to the unitary group $U(n)$ consists of all the anti-Hermitian (n, n) matrices. With these matrices, we have

$$\langle B|A \rangle = \text{Tr}(B^\dagger A) = \sum_{i,j} b_{ji}^* a_{ji} = \sum_{i,j} (-b_{ij}) (-a_{ij}^*) = \sum_{i,j} a_{ij}^* b_{ij} = \langle A|B \rangle. \quad (20.280)$$

Then, comparing (20.277) and (20.280) we have

$$\langle A|B \rangle^* = \langle A|B \rangle.$$

This means that $\langle A|B \rangle$ is real. For example, using $\mathfrak{u}(2)$, let us evaluate an inner product. We denote arbitrary $X_1, X_2 \in \mathfrak{u}(2)$ by

$$X_1 = \begin{pmatrix} ia & c+id \\ -c+id & ib \end{pmatrix}, X_2 = \begin{pmatrix} ip & r+is \\ -r+is & iq \end{pmatrix},$$

where $a, b, c, d; p, q, r, s$ are real. Then, according to the calculation rule of an inner product of the Lie algebra, we have

$$\langle X_1|X_2 \rangle = ap + 2(cr + ds) + bq,$$

which gives a real inner product. Notice that this is the case with $\mathfrak{su}(2)$ as well.

Next, let us think of the following inner product:

$$\langle gXg^{-1}|gYg^{-1} \rangle = \text{Tr}[(gXg^{-1})^\dagger gYg^{-1}], \quad (20.281)$$

where g is any non-singular matrix.

Bearing in mind that we are particularly interested in the continuous groups $U(n)$ [or its subgroup $SU(n)$] and $O(n)$ [or its subgroup $SO(n)$], we assume that g is an element of those groups and represented by a unitary matrix (including an orthogonal matrix); i.e., $g^{-1} = g^\dagger$ (or g^T). Then, if g is a unitary matrix, we have

$$\begin{aligned} \text{Tr}[(gXg^{-1})^\dagger gYg^{-1}] &= \text{Tr}[(gXg^\dagger)^\dagger gYg^\dagger] \\ &= \text{Tr}[gX^\dagger g^\dagger gYg^\dagger] = \text{Tr}(gX^\dagger Yg^\dagger) = \text{Tr}(X^\dagger Y) = \langle X|Y \rangle, \end{aligned} \quad (20.282)$$

where with the second last equality we used (12.13), namely invariance of the trace under the (unitary) similarity transformation. If g is represented by a real orthogonal matrix, instead of (20.282) we have

$$\begin{aligned} \text{Tr}[(gXg^{-1})^\dagger gYg^{-1}] &= \text{Tr}[(gXg^T)^\dagger gYg^T] \\ &= \text{Tr}[(g^T)^\dagger X^\dagger g^\dagger gYg^T] = \text{Tr}(g^* X^\dagger g^T gYg^\dagger) = \text{Tr}(gX^\dagger Yg^T) = \langle X|Y \rangle, \end{aligned} \quad (20.283)$$

where g^* is a complex conjugate matrix of g . Combining (20.281) and (20.282) or (20.283), we have

$$\langle gXg^{-1} | gYg^{-1} \rangle = \langle X | Y \rangle. \quad (20.284)$$

This relation clearly shows that the real inner product of (20.284) remains unchanged by the operation

$$X \mapsto gXg^{-1},$$

where X is an anti-Hermitian (or skew-symmetric) matrix and g is a unitary (or orthogonal) matrix.

Now, we give the following definition and related theorems in relation to both the Lie groups and Lie algebras.

Definition 20.4 Let G be a Lie group chosen from $U(n)$ [including $SU(n)$] and $O(n)$ [including $SO(n)$]. Let $\mathfrak{g}$ be a Lie algebra corresponding to G . Let $g \in G$ and $X \in \mathfrak{g}$. We define the following transformation on $\mathfrak{g}$ such that

$$\text{Ad}[g](X) \equiv gXg^{-1}. \quad (20.285)$$

Then, $\text{Ad}[g]$ is said to be an adjoint representation of G on $\mathfrak{g}$. We write the relation (20.285) as

$$\text{Ad}[g] : \mathfrak{g} \rightarrow \mathfrak{g}.$$

From Definition 20.4, $\text{Ad}[g](X) \in \mathfrak{g}$. The operator $\text{Ad}[g]$ is a kind of mapping (i.e., endomorphism) discussed in Sect. 11.2. Notice that both g and X are represented by (n, n) matrices. The matrix g is either a unitary matrix or an orthogonal matrix. The matrix X is either an anti-Hermitian matrix or a skew-symmetric matrix.

Theorem 20.5 Let g be an element of a Lie group G and $\mathfrak{g}$ be a Lie algebra of G . Then, $\text{Ad}[g]$ is a linear transformation on $\mathfrak{g}$.

Proof Let $X, Y \in \mathfrak{g}$. Then, we have

$$\begin{aligned} \text{Ad}[g](aX + bY) &\equiv g(aX + bY)g^{-1} = a(gXg^{-1}) + b(gYg^{-1}) \\ &= a\text{Ad}[g](X) + b\text{Ad}[g](Y). \end{aligned} \quad (20.286)$$

Thus, we find that $\text{Ad}[g]$ is a linear transformation. By Definition 20.3, $\exp(tX) \in G$ with an arbitrary real number t . Meanwhile, we have

$$\exp\{t\text{Ad}[g](X)\} = \exp(tgXg^{-1}) = \exp(gtXg^{-1}) = g\exp(tX)g^{-1},$$

where with the last equality we used (15.30).

Then, if $g \in G$, $g \exp(tX)g^{-1} \in G$. Again from Definition 20.3, $\text{Ad}[g](X) \in \mathfrak{g}$. That is, $\text{Ad}[g]$ is a linear transformation on $\mathfrak{g}$. This completes the proof.

Notice that in Theorem 20.5, G may be any linear Lie group chosen from among $GL(n, \mathbb{C})$. Lie algebra corresponding to $GL(n, \mathbb{C})$ is denoted by $\mathfrak{gl}(n, \mathbb{C})$. The following theorem shows that $\text{Ad}[g]$ is a representation.

Theorem 20.6 Let g be an element of a Lie group G . Then, $g \mapsto \text{Ad}[g]$ is a representation of G on $\mathfrak{g}$.

Proof From (20.285), we have

$$\begin{aligned} \text{Ad}[g_1 g_2](X) &\equiv (g_1 g_2)X(g_1 g_2)^{-1} = g_1 g_2 X g_2^{-1} g_1^{-1} = g_1 \text{Ad}[g_2](X) g_1^{-1} \\ &= \text{Ad}[g_1](\text{Ad}[g_2](X)) = (\text{Ad}[g_1] \text{Ad}[g_2])(X). \end{aligned} \quad (20.287)$$

Comparing the first and last sides of (20.287), we get

$$\text{Ad}[g_1 g_2] = \text{Ad}[g_1] \text{Ad}[g_2].$$

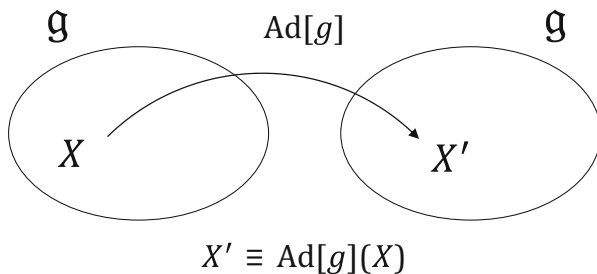
That is, $g \mapsto \text{Ad}[g]$ is a representation of G on $\mathfrak{g}$.

By virtue of Theorem 20.6, we call $g \mapsto \text{Ad}[g]$ an adjoint representation of G on $\mathfrak{g}$. Once again, we remark that

$$X \mapsto \text{Ad}[g](X) \equiv X' \quad \text{or} \quad \text{Ad}[g] : \mathfrak{g} \rightarrow \mathfrak{g} \quad (20.288)$$

is a linear transformation $\mathfrak{g} \rightarrow \mathfrak{g}$, namely an endomorphism that operates on a vector space $\mathfrak{g}$ (see Sect. 11.2). Figure 20.5 schematically depicts it. The notation of the adjoint representation Ad is somewhat confusing. That is, (20.288) shows that $\text{Ad}[g]$ is a linear transformation $\mathfrak{g} \rightarrow \mathfrak{g}$ (i.e., endomorphism). Conversely, Ad is thought to be a mapping $G \rightarrow G'$ where G and G' mean two groups. Namely, we express it symbolically as [9]

Fig. 20.5 Linear transformation $\text{Ad}[g]: \mathfrak{g} \rightarrow \mathfrak{g}$ (i.e., endomorphism). $\text{Ad}[g]$ is an endomorphism that operates on a vector space $\mathfrak{g}$



$$g \mapsto \text{Ad}[g] \quad \text{or} \quad \text{Ad} : G \rightarrow G'. \quad (20.289)$$

The G and G' may or may not be identical. Examples can be seen in, e.g., (20.294) and (20.302) later.

As mentioned above, if $\mathfrak{g}$ is either $\mathfrak{u}(n)$ or $\mathfrak{o}(n)$, $\text{Ad}[g]$ is an orthogonal transformation on $\mathfrak{g}$. The representation of $\text{Ad}[g]$ is real accordingly.

An immediate implication of this is that the transformation of the basis vectors of $\mathfrak{g}$ has a connection with the corresponding Lie groups such as $U(n)$ and $O(n)$. Moreover, it is well known and studied that there is a close relationship between $\text{Ad}[SU(2)]$ and $SO(3)$. First, we wish to seek basis vectors of $\mathfrak{su}(2)$. A general form of $X \in \mathfrak{su}(2)$ is a following traceless anti-Hermitian matrix described by

$$X = \begin{pmatrix} ic & -a + ib \\ a + ib & -ic \end{pmatrix},$$

where a , b , and c are real. We have encountered this type of matrix in (20.42) and (20.270). Using an inner product described in (20.252), we determine an orthonormal basis set of $\mathfrak{su}(2)$ such that

$$e_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}, e_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, e_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad (20.290)$$

where we have $\langle e_i | e_j \rangle = \delta_{ij}$ ($i, j = 1, 2, 3$).

Using the basis set of $\mathfrak{su}(2)$ given by (20.290) for those of the representation space, we wish to seek a tangible representation matrix of $\text{Ad}[g]$ [$g \in SU(2)$]. To construct $\text{Ad}[g]$ [$g \in SU(2)$], we choose $g_\alpha, g_\beta \in SU(2)$ given by

$$g_\alpha \equiv \begin{pmatrix} e^{i\alpha/2} & 0 \\ 0 & e^{-i\alpha/2} \end{pmatrix} \quad \text{and} \quad g_\beta \equiv \begin{pmatrix} \cos \frac{\beta}{2} & \sin \frac{\beta}{2} \\ -\sin \frac{\beta}{2} & \cos \frac{\beta}{2} \end{pmatrix}.$$

The elements g_α and g_β have appeared in (20.43) and (20.44). Then, we have [9]

$$\begin{aligned} \text{Ad}[g_\alpha](e_1) &= g_\alpha e_1 g_\alpha^{-1} = \frac{1}{\sqrt{2}} \begin{pmatrix} e^{i\alpha/2} & 0 \\ 0 & e^{-i\alpha/2} \end{pmatrix} \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} \begin{pmatrix} e^{-i\alpha/2} & 0 \\ 0 & e^{i\alpha/2} \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & \sin \alpha - i \cos \alpha \\ -\sin \alpha - i \cos \alpha & 0 \end{pmatrix} = e_1 \cos \alpha + e_2 \sin \alpha. \end{aligned} \quad (20.291)$$

Similarly, we get

$$\text{Ad}[g_\alpha](e_2) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & \cos\alpha + i\sin\alpha \\ -\cos\alpha + i\sin\alpha & 0 \end{pmatrix} = -e_1 \sin\alpha + e_2 \cos\alpha, \quad (20.292)$$

$$\text{Ad}[g_\alpha](e_3) = \frac{1}{\sqrt{2}} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = e_3. \quad (20.293)$$

Notice that we obtained (20.291)–(20.293) as a product of three diagonal matrices. Thus, we obtain

$$(e_1 \ e_2 \ e_3) \text{Ad}[g_\alpha] = (e_1 \ e_2 \ e_3) \begin{pmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (20.294)$$

From (20.294), as a real representation matrix we get

$$\text{Ad}[g_\alpha] = \begin{pmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (20.295)$$

Calculating $(e_1 \ e_2 \ e_3) \text{Ad}[g_\beta]$ likewise, we get

$$\text{Ad}[g_\beta] = \begin{pmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{pmatrix}. \quad (20.296)$$

Moreover, with

$$g_\gamma = \begin{pmatrix} e^{i\gamma/2} & 0 \\ 0 & e^{-i\gamma/2} \end{pmatrix}, \quad (20.297)$$

we have

$$\text{Ad}[g_\gamma] = \begin{pmatrix} \cos\gamma & -\sin\gamma & 0 \\ \sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (20.298)$$

Finally, let us reproduce (17.101) by calculating

$$\text{Ad}[g_\alpha] \cdot \text{Ad}[g_\beta] \cdot \text{Ad}[g_\gamma] = \text{Ad}[g_\alpha g_\beta g_\gamma], \quad (20.299)$$

where we used the fact that $g_\omega \mapsto \text{Ad}[g_\omega]$ ($\omega = \alpha, \beta, \gamma$) is a representation of $SU(2)$ on $\mathfrak{su}(2)$. To obtain an explicit form of the representation matrix of $\text{Ad}[g]$ [$g \in SU(2)$], we use

$$g_{\alpha\beta\gamma} \equiv g_\alpha g_\beta g_\gamma = \begin{pmatrix} e^{\frac{i}{2}(\alpha+\gamma)} \cos \frac{\beta}{2} & e^{\frac{i}{2}(\alpha-\gamma)} \sin \frac{\beta}{2} \\ -e^{\frac{i}{2}(\gamma-\alpha)} \sin \frac{\beta}{2} & e^{-\frac{i}{2}(\alpha+\gamma)} \cos \frac{\beta}{2} \end{pmatrix}. \quad (20.300)$$

Note that the matrix of (20.300) is identical with (20.45). Using (20.300), we calculate $\text{Ad}[g_{\alpha\beta\gamma}](e_1)$ such that

$$\begin{aligned} \text{Ad}[g_{\alpha\beta\gamma}](e_1) &= (g_{\alpha\beta\gamma})e_1(g_{\alpha\beta\gamma}^{-1}) \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} e^{\frac{i}{2}(\alpha+\gamma)} \cos \frac{\beta}{2} & e^{\frac{i}{2}(\alpha-\gamma)} \sin \frac{\beta}{2} \\ -e^{\frac{i}{2}(\gamma-\alpha)} \sin \frac{\beta}{2} & e^{-\frac{i}{2}(\alpha+\gamma)} \cos \frac{\beta}{2} \end{pmatrix} \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} \begin{pmatrix} e^{-\frac{i}{2}(\alpha+\gamma)} \cos \frac{\beta}{2} & -e^{\frac{i}{2}(\alpha-\gamma)} \sin \frac{\beta}{2} \\ e^{\frac{i}{2}(\gamma-\alpha)} \sin \frac{\beta}{2} & e^{\frac{i}{2}(\alpha+\gamma)} \cos \frac{\beta}{2} \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} -i \sin \beta \cos \gamma & -i(\cos \alpha \cos \beta \cos \gamma - \sin \alpha \sin \gamma) \\ -i(\cos \alpha \cos \beta \cos \gamma - \sin \alpha \sin \gamma) & + \sin \alpha \cos \beta \cos \gamma + \cos \alpha \sin \gamma \\ -(\sin \alpha \cos \beta \cos \gamma + \cos \alpha \sin \gamma) & i \sin \beta \cos \gamma \end{pmatrix} \\ &= (e_1 \ e_2 \ e_3) \begin{pmatrix} \cos \alpha \cos \beta \cos \gamma - \sin \alpha \sin \gamma \\ \sin \alpha \cos \beta \cos \gamma + \cos \alpha \sin \gamma \\ -\sin \beta \cos \gamma \end{pmatrix}. \end{aligned} \quad (20.301)$$

Similarly calculating $\text{Ad}[g_{\alpha\beta\gamma}](e_2)$ and $\text{Ad}[g_{\alpha\beta\gamma}](e_3)$ and combining the results with (20.301), we obtain

$$\begin{aligned} (e_1 \ e_2 \ e_3) \text{Ad}[g_{\alpha\beta\gamma}] &= \\ (e_1 \ e_2 \ e_3) &\begin{pmatrix} \cos \alpha \cos \beta \cos \gamma - \sin \alpha \sin \gamma & -\cos \alpha \cos \beta \sin \gamma - \sin \alpha \cos \gamma & \cos \alpha \sin \beta \\ \sin \alpha \cos \beta \cos \gamma + \cos \alpha \sin \gamma & -\sin \alpha \cos \beta \sin \gamma + \cos \alpha \cos \gamma & \sin \alpha \sin \beta \\ -\sin \beta \cos \gamma & \sin \beta \sin \gamma & \cos \beta \end{pmatrix}. \end{aligned} \quad (20.302)$$

The matrix of RHS of (20.302) is the same as (17.101) that contains Euler angles. The calculations are somewhat lengthy but straightforward. As can be seen, e.g., in (20.294) and (20.302), we may symbolically express the above adjoint representation as [9]

$$\text{Ad} : SU(2) \rightarrow SO(3) \quad (20.303)$$

in parallel to (20.289). Equation (20.303) shows the mapping between different groups $SU(2)$ and $SO(3)$. From (20.302) we may identify $\text{Ad}[g_{\alpha\beta\gamma}]$ with the $SO(3)$ matrix (17.101) and write

$$\text{Ad}[g_{\alpha\beta\gamma}] = \begin{pmatrix} \cos\alpha\cos\beta\cos\gamma - \sin\alpha\sin\gamma & -\cos\alpha\cos\beta\sin\gamma - \sin\alpha\cos\gamma & \cos\alpha\sin\beta \\ \sin\alpha\cos\beta\cos\gamma + \cos\alpha\sin\gamma & -\sin\alpha\cos\beta\sin\gamma + \cos\alpha\cos\gamma & \sin\alpha\sin\beta \\ -\sin\beta\cos\gamma & \sin\beta\sin\gamma & \cos\beta \end{pmatrix}. \quad (20.304)$$

In the above discussion, we make several remarks. (i) $g \in SU(2)$ is described by $(2, 2)$ complex matrices, but $\text{Ad}[SU(2)]$ is expressed by $(3, 3)$ real orthogonal matrices. Thus, the dimension of the matrices is different. (ii) Notice again that $(e_1 \ e_2 \ e_3)$ of (20.302) is an orthonormal basis set of $\mathfrak{su}(2)$. Hence, (20.302) can be viewed as the orthogonal transformation of $(e_1 \ e_2 \ e_3)$. That is, we may regard $\text{Ad}[SU(2)]$ as all the matrix representations of $SO(3)$. From (12.64) and (20.302), we write [9]

$$\text{Ad}[SU(2)] - SO(3) = 0 \quad \text{or} \quad \text{Ad}[SU(2)] = SO(3).$$

Regarding the adjoint representation of G , we have the following important theorem.

Theorem 20.7 The adjoint representation of $\text{Ad}[SU(2)]$ is a surjective homomorphism of $SU(2)$ on $SO(3)$. The kernel $\mathcal{F}$ of the representation is $\{e, -e\} \in SU(2)$. With any rotation $R \in SO(3)$, two elements $\pm g \in SU(2)$ satisfy $\text{Ad}[g] = \text{Ad}[-g] = R$.

Proof We have shown that by (20.302) the adjoint representation of $\text{Ad}[SU(2)]$ is surjective mapping of $SU(2)$ on $SO(3)$. Since $\text{Ad}[g]$ is a representation, namely homomorphism mapping from $SU(2)$ to $SO(3)$, we seek its kernel on the basis of Theorem 16.3 (Sect. 16.4). Let h be an element of kernel of $SU(2)$ such that

$$\text{Ad}[h] = E \in SO(3), \quad (20.305)$$

where E is the identity transformation of $SO(3)$.

Operating (20.305) on $e_3 \in \mathfrak{su}(2)$, from (20.305) we have a following expression such that

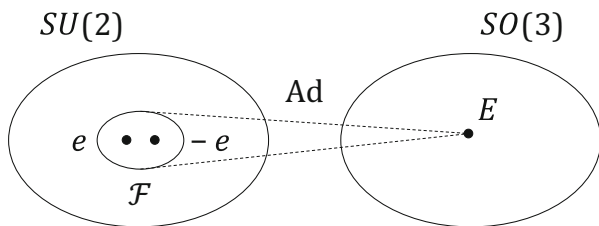
$$\text{Ad}[h](e_3) = he_3h^{-1} = E(e_3) = e_3 \quad \text{or} \quad he_3 = e_3h,$$

where e_3 is denoted by (20.290). Expressing h as $\begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix}$, we get

$$\begin{pmatrix} ih_{11} & -ih_{12} \\ ih_{21} & -ih_{22} \end{pmatrix} = \begin{pmatrix} ih_{11} & ih_{12} \\ -ih_{21} & -ih_{22} \end{pmatrix}.$$

From the above relation, we must have $h_{12} = h_{21} = 0$. As $h \in SU(2)$, $\text{deth} = 1$. Hence, for h we choose $h = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}$, where $a \neq 0$. Moreover, considering $he_2 = e_2h$ we get

Fig. 20.6 Homomorphism mapping $\text{Ad}: SU(2) \rightarrow SO(3)$ with a kernel $\mathcal{F} = \{e, -e\}$. E is the identity transformation of $SO(3)$



$$\begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix} = \begin{pmatrix} 0 & a^{-1} \\ -a & 0 \end{pmatrix}. \quad (20.306)$$

This implies that $a = a^{-1}$, or $a = \pm 1$. That is, $h = \pm e$, where e denotes the identity element of $SU(2)$. (Note that we get no further conditions from $he_1 = e_1h$.) Thus, we obtain

$$\mathcal{F} = \{e, -e\}. \quad (20.307)$$

In fact, with respect to $X \in \mathfrak{su}(2)$ we have

$$\text{Ad}[\pm e](X) = (\pm e)X(\pm e)^{-1} = (\pm e)X(\pm e) = X. \quad (20.308)$$

Therefore, certainly we get

$$\text{Ad}[\pm e] = E. \quad (20.309)$$

Meanwhile, suppose that with $g_1, g_2 \in SU(2)$ we have $\text{Ad}[g_1] = \text{Ad}[g_2] = E \in SO(3)$. Then, we have

$$\begin{aligned} \text{Ad}[g_1 g_2^{-1}] &= \text{Ad}[g_1] \text{Ad}[g_2^{-1}] = \text{Ad}[g_2] \text{Ad}[g_2^{-1}] = \text{Ad}[g_2 g_2^{-1}] \\ &= \text{Ad}[e] = E. \end{aligned} \quad (20.310)$$

Therefore, from (20.307) and (20.309) we get

$$g_1 g_2^{-1} = \pm e \quad \text{or} \quad g_1 = \pm g_2. \quad (20.311)$$

Conversely, we have

$$\text{Ad}[-g] = \text{Ad}[-e] \text{Ad}[g] = \text{Ad}[g], \quad (20.312)$$

where with the first equality we used Theorem 20.6 and with the second equality we used (20.309). The relations (20.311) and (20.312) imply that with any $g \in SU(2)$ there are two elements $\pm g$ that satisfy $\text{Ad}[g] = \text{Ad}[-g]$. These complete the proof.

In the above proof of Theorem 20.7, (20.309) is a simple, but important expression. In view of Theorem 16.4 (Homomorphism theorem), we summarize the above discussion as follows: Let Ad be a homomorphism mapping such that

$$\text{Ad} : SU(2) \rightarrow SO(3) \quad (20.303)$$

with a kernel $\mathcal{F} = \{e, -e\}$. The relation is schematically represented in Fig. 20.6. Notice the resemblance between Figs. 20.6 and 16.1a.

Correspondingly, from Theorem 16.4 of Sect. 16.4 we have an isomorphic mapping $\tilde{\text{Ad}}$ such that

$$\tilde{\text{Ad}} : SU(2)/\mathcal{F} \rightarrow SO(3), \quad (20.313)$$

where $\mathcal{F} = \{e, -e\}$. Symbolically writing (20.313) as in Sect. 16.4, we have $SU(2)/\mathcal{F} \cong SO(3)$. Notice that $\mathcal{F}$ is an invariant subgroup of $SU(2)$. Using a general form of $SU(2)$ of (20.300) and a general form of $SO(3)$ of (20.302), we can readily show $\text{Ad}[g_{\alpha\beta\gamma}] = \text{Ad}[-g_{\alpha\beta\gamma}]$. That is, from (20.300) we have

$$g_{\alpha+2\pi, \beta+2\pi, \gamma+2\pi} = -g_{\alpha\beta\gamma}. \quad (20.314)$$

Both the above two elements $g_{\alpha\beta\gamma}$ and $-g_{\alpha\beta\gamma}$ produce an identical orthogonal matrix given in RHS of (20.302).

To associate the above results of abstract algebra with the tangible matrix form obtained earlier, we rewrite (20.54) by applying a simple trigonometric formula of

$$\sin \beta = 2 \sin \frac{\beta}{2} \cos \frac{\beta}{2}.$$

That is, we get

$$\begin{aligned} D_{m'm}^{(j)}(\alpha, \beta, \gamma) &= \sum_{\mu} \frac{(-1)^{m'-m+\mu} \sqrt{(j+m)!(j-m)!(j+m')!(j-m')!}}{(j+m-\mu)!\mu!(m'-m+\mu)!(j-m'-\mu)!} \\ &\quad \times e^{-i(am'+\gamma m)} \left(\cos \frac{\beta}{2}\right)^{2j} \left[\left(\cos \frac{\beta}{2}\right)^{m-m'-2\mu} \left(\sin \frac{\beta}{2}\right)^{m'-m+2\mu} \right] \\ &= \sum_{\mu} \frac{(-1)^{m'-m+\mu} \sqrt{(j+m)!(j-m)!(j+m')!(j-m')!}}{(j+m-\mu)!\mu!(m'-m+\mu)!(j-m'-\mu)!} \cdot 2^{m-m'-2\mu} \\ &\quad \times e^{-i(am'+\gamma m)} (\sin \beta)^{m'-m+2\mu} \left(\cos \frac{\beta}{2}\right)^{2j+2(m-m'-2\mu)}. \end{aligned} \quad (20.315)$$

From (20.315), we clearly see that (i) if j is zero or a positive integer, so are m and m' . Therefore, (20.315) is a periodic function with 2π with respect to α , β , and γ . But (ii) if j is a half-odd-integer, so are m and m' . Then, (20.315) is a periodic function with 4π . Note that in the case of (ii) $2j = 2n + 1$ ($n = 0.1, 2, \dots$). Therefore, in (20.315) we have

$$\left(\cos \frac{\beta}{2}\right)^{2j+2(m-m'-2\mu)} = \left(\cos \frac{\beta}{2}\right) \cdot \left(\cos \frac{\beta}{2}\right)^{2(n+m-m'-2\mu)}. \quad (20.316)$$

As studied in Sect. 20.2.3, in the former case (i) the spherical surface harmonics span the representation space of $D^{(l)}$ ($l = 0, 1, 2, \dots$). The simplest case for it is $D^{(0)} = 1$; a $(1, 1)$ identity matrix, i.e., merely a number 1. The second simplest case was given by (20.74) that describes $D^{(1)}$. Meanwhile, the simplest case for (ii) is $D^{(1/2)}$ whose matrix representation was given by $D_{\alpha\beta\gamma}^{(1/2)}$ in (20.45) or by $g_{\alpha\beta\gamma}$ in (20.300). With $\pm g_{\alpha\beta\gamma}$, both $\text{Ad}[\pm g_{\alpha\beta\gamma}]$ produce a real orthogonal $(3, 3)$ matrix expressed as (20.302). This representation matrix allowedly belongs to $SO(3)$.

20.5 Connectedness of Lie Groups

As already discussed in Chap. 6, reflecting the topological characteristics the connectedness is an important concept in the Lie groups. If we think of the analytic functions on the complex plane $\mathbb{C}$, the connectedness was easily envisaged geometrically. To deal with the connectedness in general Lie groups, however, we need abstract concepts. In this section we take a down-to-earth approach as much as possible.

20.5.1 Several Definitions and Examples

To discuss this topic, let us give several definitions and examples.

Definition 20.5 Suppose that A is a subset of $GL(n, \mathbb{C})$ and that we have any pair of elements $a, b \in A \subset GL(n, \mathbb{C})$. Meanwhile, let $f(t)$ be a continuous function defined in an interval $0 \leq t \leq 1$ and take values within A ; that is, $f(t) \in A$ with $\forall t$. If $f(0) = a$ and $f(1) = b$, then a and b are said to be connected within A .

Major parts of the discussion that follows are based mostly upon literature [9]. A simple example for the above is given below.

Example 20.6 Let $a = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $b = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ be elements of $SO(2)$. Let $f(x)$ be described by

$$f(t) = \begin{pmatrix} \cos \pi t & -\sin \pi t \\ \sin \pi t & \cos \pi t \end{pmatrix}.$$

Then, $f(t)$ is a continuous matrix function of all real numbers t . We have $f(0) = a$ and $f(1) = b$ and, hence, a and b are connected to each other within $SO(2)$.

For a and b to be connected within A is denoted by $a \sim b$. Then, the symbol $\sim$ satisfies the equivalence relation. That is, we have a following proposition.

Proposition 20.2 We have $a, b, c \in A \subset GL(n, \mathbb{C})$. Then, we have following three relations. (i) $a \sim a$. (ii) If $a \sim b$, $b \sim a$. (iii) If $a \sim b$ and $b \sim c$, then we have $a \sim c$.

Proof (i) Let $f(t) = a$ ($0 \leq t \leq 1$). Then, $f(0) = f(1) = a$ so that we have $a \sim a$. (ii) Let $f(t)$ be a continuous curve that connects a and b such that $f(0) = a$ and $f(1) = b$ ($0 \leq t \leq 1$), and so $a \sim b$. Let $g(t)$ be another continuous curve within A . Then, $g(t) \equiv f(1 - t)$ is also a continuous curve within A . Then, $g(0) = b$ and $g(1) = a$. This implies $b \sim a$. (iii) Let $f(t)$ and $g(t)$ be two continuous curves that connect a and b and b and c , respectively. Meanwhile, we define $h(t)$ as

$$h(t) = \begin{cases} f(2t) & (0 \leq t \leq 1/2) \\ g(2t - 1) & (1/2 \leq t \leq 1) \end{cases},$$

so that $h(t)$ can be a third continuous curve. Then, $h(1/2) = f(1) = g(0)$. From the supposition we have $h(0) = f(0) = a$, $h(1/2) = f(1) = g(0) = b$, and $h(1) = g(1) = c$. This implies if $a \sim b$ and $b \sim c$, then we have $a \sim c$.

Let us give another related definition.

Definition 20.6 Let A be a subset of $GL(n, \mathbb{C})$. Let a point $a \in A$. Then, a collection of points that are connected to a within A is said to be a connected component of a and denoted by $C(a)$.

From Proposition 20.2 (i), $a \in C(a)$. Let $C(a)$ and $C(b)$ be two different connected components. We have two alternatives with $C(a)$ and $C(b)$. (I) $C(a) \cap C(b) = \emptyset$. (II) $C(a) = C(b)$. Suppose $c \in C(a) \cap C(b)$. Then, it follows that $c \sim a$ and $c \sim b$. From Proposition 20.2 (ii) and (iii), we have $a \sim b$. Hence, we get $C(a) = C(b)$. Thus, the subset A is a direct sum of several connected components such that

$$A = C(a) \cup C(b) \cup \cdots \cup C(z) \text{ with any } C(p) \cap C(q) = \emptyset (p \neq q), \quad (20.317)$$

where p, q are taken from among $a, b, \dots, z$. In particular, the connected component containing e (i.e., the identity element) is of major importance.

Definition 20.7 Let A be a subset of $GL(n, \mathbb{C})$. If any pair of elements $a, b (\in A)$ is connected to each other within A , then A is connected or A is called a connected set.

The connected set is nothing but a set that consists of a single connected set. With a connected set, we have the following important theorem.

Theorem 20.8 An image of a connected set A obtained by a continuous mapping f is a connected set as well.

Proof The proof is almost self-evident. Let $f: A \rightarrow B$ be a continuous mapping from a connected set A to B . Suppose $f(A) = B$. Then, we have $\forall a, a' \in A$ and $b, b' \in B$ that

satisfy $f(a) = b$ and $f(a') = b'$. Then, from Definition 20.5 there is a continuous function $g(t)$ ($0 \leq t \leq 1$) that satisfies $g(0) = a$ and $g(1) = a'$.

Now, define a function $h(t) \equiv f[g(t)]$. Then, $h(t)$ is a continuous mapping with $h(0) = f[g(0)] = f(a) = b$ and $h(1) = f[g(1)] = f(a') = b'$. Again from Definition 20.5, $h(t)$ is a continuous curve that connects b and b' within B . Meanwhile, from supposition of $f(A) = B$, with $\forall b, b' \in B$ we must have $\exists a_0, a'_0 \in A$ that satisfy $f(a_0) = b$ and $f(a'_0) = b'$. Consequently, from Definition 20.7, B is a connected set as well. This completes the proof.

Applications of Theorem 20.8 are given below as examples.

Example 20.7 Let $f(x)$ be a real continuous function defined on a subset $A \subset GL(n, \mathbb{R})$, where $x \in A$ with A being a connected set. Suppose that for a pair of $a, b \in A$, $f(a) = p$ and $f(b) = q$ (p, q : real) with $p < q$. Then, $f(x)$ takes all real numbers for its value that exist on the interval $[p, q]$. In other words, we have $f(A) \supset [p, q]$. This is well-known as an intermediate value theorem.

Example 20.8 [9] Suppose that A is a connected subset of $GL(n, \mathbb{R})$. Then, with any pair of elements $a, b \in A \subset GL(n, \mathbb{R})$, we must have a continuous function $g(t)$ ($0 \leq t \leq 1$) such that $g(0) = a$ and $g(1) = b$. Meanwhile, suppose that we can define a real determinant for any $\forall g \in A$ as a real continuous function $\det g(t)$.

Now, suppose that $\det a = p < 0$ and $\det b = q > 0$ with $p < q$. Then, from Theorem 20.8 we would have $\det(A) \supset [p, q]$. Consequently, we must have some $x \in A$ such that $\det(x) = 0$. But this is in contradiction to $A \subset GL(n, \mathbb{R})$, because we must have $\det(A) \neq 0$. Thus, within a connected set the sign of the determinant should be constant, if the determinant is defined as a real continuous function.

In relation to the discussion of the connectedness, we have the following general theorem.

Theorem 20.9 [9] Let G_0 be a connected component within a linear Lie group G that contains the identity element e . Then, G_0 is an invariant subgroup of G . A connected component $C(g)$ containing $g \in G$ is identical with $gG_0 = G_0g$.

Proof Let $a, b \in G_0$. Since $G_0 = C(e)$, from the supposition there are continuous curves $f(t)$ and $g(t)$ within G_0 that connect a with e and b with e , respectively, such that $f(0) = a, f(1) = e$ and $g(0) = b, g(1) = e$. Meanwhile, let $h(t) \equiv f(t)[g(t)]^{-1}$. Then, $h(t)$ is another continuous curve. We have $h(0) = f(0)[g(0)]^{-1} = ab^{-1}$ and $h(1) = f(1)[g(1)]^{-1} = e(e^{-1}) = e \cdot e = e$. That is, ab^{-1} and e are connected, being indicative of $ab^{-1} \in G_0$. This implies that G_0 is a subgroup of G .

Next, with $\forall g \in G$ and $\forall a \in G_0$ we have $f(t)$ in the same sense as the above. Defining $k(t)$ as $k(t) = g f(t) g^{-1}$, $k(t)$ is a continuous curve within G with $k(0) = g a g^{-1}$ and $k(1) = g e g^{-1} = e$. Hence, we have $g a g^{-1} \in G_0$. Since a is an arbitrary point of G_0 , we have $g G_0 g^{-1} \subset G_0$ accordingly. Similarly, g is an arbitrary point of G , and so replacing g with g^{-1} in the above, we get $g^{-1} G_0 g \subset G_0$. Operating g and g^{-1} from the left and right, respectively, we obtain $G_0 \subset g G_0 g^{-1}$. Combining this with the above

relation $gG_0g^{-1} \subset G_0$, we obtain $gG_0g^{-1} = G_0$ or $gG_0 = G_0g$. This means that G_0 is an invariant subgroup of G ; see Sect. 16.3.

Taking $\forall a \in G_0$ and $f(t)$ in the same sense as the above again, $l(t) = g f(t)$ is a continuous curve within G with $l(0) = ga$ and $l(1) = g e = g$. Then, ga and g are connected within G . That is, $ga \in C(g)$. Since a is an arbitrary point of G_0 , $gG_0 \subset C(g)$. Meanwhile, choosing any $\forall p \in C(g)$, we set a continuous curve $m(t)$ ($0 \leq t \leq 1$) that connects p with g within G . This implies that $m(0) = p$ and $m(1) = g$.

Also setting a continuous curve $n(t) = g^{-1}m(t)$, we have $n(0) = g^{-1}m(0) = g^{-1}p$ and $n(1) = g^{-1}m(1) = g^{-1}g = e$. This means that $g^{-1}p \in G_0$. Multiplying g on its both sides, we get $p \in gG_0$. Since p is arbitrarily chosen from $C(g)$, we get $C(g) \subset gG_0$. Combining this with the above relation $gG_0 \subset C(g)$, we obtain $C(g) = gG_0 = G_0g$. These complete the proof.

In the above proof, we add the following statement: In Sect. 16.2 with the necessary and sufficient condition for the subset to be a subgroup, we describe (1) $h_i, h_j \in \mathcal{H} \Rightarrow h_i \circ h_j \in \mathcal{H}$. (2) $h \in \mathcal{H} \Rightarrow h^{-1} \in \mathcal{H}$. In (1) if we replace h_j with h_i^{-1} , $h_i \circ h_j^{-1} = h_i \circ h_i^{-1} = e \in \mathcal{H}$. In turn, if we replace h_i with e , $h_i \circ h_j^{-1} = e \circ h_j^{-1} = h_j^{-1} \in \mathcal{H}$. Finally, if we replace $h_j = h_j^{-1}$, $h_i \circ h_j^{-1} = h_i \circ (h_j^{-1})^{-1} = h_i \circ h_j \in \mathcal{H}$. Thus, $\mathcal{H}$ satisfies the axioms (A1), (A3), and (A4) of Sect. 16.1 and, hence, forms a (sub)group. In other words, the conditions (1) and (2) are combined so as to be

$$h_i \in \mathcal{H}, h_j \in \mathcal{H} \Rightarrow h_i \circ h_j^{-1} \in \mathcal{H}.$$

The above general theorem can immediately be applied to an important class of Lie groups $SO(n)$. We discuss this topic below.

20.5.2 $O(3)$ and $SO(3)$

In Chap. 17 we dealt with finite groups related to $O(3)$ and $SO(3)$, i.e., their subgroups. In this section, we examine several important properties in terms of Lie groups.

The groups $O(3)$ and $SO(3)$ are characterized by their determinant $\det O(3) = \pm 1$ and $\det SO(3) = 1$. Example 20.8 implies that in $O(3)$ there should be two different connected components according to $\det O(3) = \pm 1$. Also Theorem 20.9 tells us that a connected component $C(E)$ is an invariant subgroup G_0 in $O(3)$, where E is the identity element of $O(3)$. Obviously, G_0 is $SO(3)$. In turn, another connected component is $SO(3)^c$, where $SO(3)^c$ denotes a complementary set of $SO(3)$; with the notation see Sect. 6.1. Remembering (20.317), we have

$$O(3) = SO(3) \cup SO(3)^c. \quad (20.318)$$

This can be rewritten as a direct sum such that

$$O(3) = C(E) \cup C(-E) \quad \text{with} \quad C(E) \cap C(-E) = \emptyset, \quad (20.319)$$

where E denotes a $(3, 3)$ unit matrix and $-E = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$. In the above, $SO(3)$ is identical to $C(E)$ with $SO(3)$ itself being a connected set; $SO(3)^c$ is identical with $C(-E)$ that is another connected set.

Another description of $O(3)$ is given by [13]

$$O(3) = SO(3) \otimes \{E, -E\}, \quad (20.320)$$

where the symbol $\otimes$ denotes a direct-product group (Sect. 16.5). An alternative description for this is

$$O(3) = SO(3) \cup \{(-E)SO(3)\}, \quad (20.321)$$

where the direct sum is implied.

In Sect. 20.2.6, we discussed two rotations around different rotation axes with the same rotation angle. These rotations belong to the same conjugacy class; see (20.140). With Q as another rotation that transforms the rotation axis of R_ω to that of R'_ω , the two rotations R_ω and R'_ω having the same rotation angle ω are associated with each other through the following relation:

$$R'_\omega = QR_\omega Q^{-1}. \quad (20.140)$$

Notice that this relation is independent of the choice of specific coordinate system. In Fig. 20.4 let us choose the z -axis for the rotation axis A with respect to the rotation R_ω . Then, the rotation matrix R_ω is expressed in reference to the Cartesian coordinate system as

$$R_\omega = \begin{pmatrix} \cos \omega & -\sin \omega & 0 \\ \sin \omega & \cos \omega & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (20.322)$$

Meanwhile, multiplying $-E$ on both sides of (20.140), we have

$$-R'_\omega = Q(-R_\omega)Q^{-1}. \quad (20.323)$$

Note that $-E$ is commutative with any Q (or Q^{-1}). Also, notice that from (20.323) $-R'_\omega$ and $-R_\omega$ are again connected via a unitary similarity transformation. From (20.322), we have

$$\begin{aligned}
 -R_\omega &= \begin{pmatrix} -\cos \omega & \sin \omega & 0 \\ -\sin \omega & -\cos \omega & 0 \\ 0 & 0 & -1 \end{pmatrix} \\
 &= \begin{pmatrix} \cos(\omega + \pi) & -\sin(\omega + \pi) & 0 \\ \sin(\omega + \pi) & \cos(\omega + \pi) & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad (20.324)
 \end{aligned}$$

where $-R_\omega$ represents an improper rotation of an angle $\omega + \pi$ (see Sect. 17.1).

Referring to Fig. 17.12 and Table 17.6 (Sect. 17.3), as another tangible example we further get

$$\begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (20.325)$$

where the first component of LHS belongs to O (octahedral rotation group) and the RHS belongs to T_d (tetrahedral group) that gives a mirror symmetry. Combining (20.324) and (20.325), we get the following schematic representation:

$$(\text{Rotation}) \times (\text{Inversion}) = (\text{Improper rotation}) \text{ or } (\text{Mirror symmetry}).$$

This is a finite group version of (20.320).

20.5.3 Simply Connected Lie Groups: Local Properties and Global Properties

As a final topic of Lie groups, we outline the characteristic of simply connected groups.

In (20.318), (20.319), and (20.321) we showed three different decompositions of $O(3)$. This can be generalized to $O(n)$. That is, instead of $-E$ we may take F described by [9]

$$F = \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & 1 & \\ & & & & -1 \end{pmatrix}, \quad (20.326)$$

where $\det F = -1$. Then, we have

$$O(n) = SO(n) \cup SO(n)^c = C(E) \cup C(F) = SO(n) + FSO(n), \quad (20.327)$$

where the last side represents the coset decomposition (Sect. 16.2). This is essentially the same as (20.321). In terms of the isomorphism, we have

$$O(n)/SO(n) \cong \{E, F\}, \quad (20.328)$$

where $\{E, F\}$ is a subgroup of $O(n)$.

If in particular n is odd ($n \geq 3$), we can choose $-E$ instead of F , because $\det(-E) = -1$. Then, similarly to (20.320) we have

$$O(n) = SO(n) \otimes \{E, -E\} \quad (n : \text{odd}), \quad (20.329)$$

where E is a (n, n) identity matrix. Note that since $-E$ is commutative with any elements of $O(n)$, the direct-product of (20.329) is allowed.

Of the connected components, that containing the identity element is of particular importance. This is because as already mentioned in Sect. 20.3.1, the “initial” condition of the one-parameter group $A(t)$ is set at $t = 0$ such that

$$A(0) = E. \quad (20.238)$$

Under this condition, it is important to examine how $A(t)$ evolves with t , starting from E at $t = 0$. In this connection we have the following theorems that give good grounds to further discussion of Sect. 20.2. We describe them without proof. Interested readers are encouraged to look up literature [9].

Theorem 20.10 Let G be a linear Lie group. Let G_0 be a connected component of the identity element $e \in G$. Then, G_0 is another linear Lie group and the Lie algebra of G and G_0 is identical.

Theorem 20.10 shows the significance of the connected component of the identity. From this theorem the Lie algebras of, e.g., $\mathfrak{o}(n)$ and $\mathfrak{so}(n)$ are identical, namely both of the Lie algebras are given by real skew-symmetric matrices. This is inherently related to the properties of Lie algebras. The zero matrix (i.e., zero vector) as an element of Lie algebra corresponds to the identity element of the Lie group. This is obvious from the relation

$$e = \exp 0,$$

where e is the identity element of the Lie group and 0 represents the zero matrix. Basically Lie algebras are suited for describing local properties associated with infinitesimal transformations around the identity element e . More specifically, the exponential functions of the real skew-symmetric matrices cannot produce $-E$. Considering that one of the connected components of $O(3)$ is $SO(3)$ that contains the identity element, it is intuitively understandable that $\mathfrak{o}(n)$ and $\mathfrak{so}(n)$ are the same.

Another interesting point is that $SO(3)$ is at once an open set and a closed set (i.e., clopen set; see Sect. 6.1). This is relevant to Theorem 6.3 which says that a necessary and sufficient condition for a subset S of a topological space to be both open and closed at once is $S^b = \emptyset$. Since the determinant of any elements of $O(3)$ is alternatively ± 1 , it is natural that $SO(3)$ has no boundary; if there were boundary, its determinant would be zero, in contradiction to $SO(3) \subset GL(n, \mathbb{R})$; see Example 20.8.

Theorem 20.11 Let G be a connected linear Lie group. (The Lie group G may be G_0 in the sense of Theorem 20.10.) Then, $\forall g \in G$ can be described by

$$g = (\exp t_1 X_1)(\exp t_2 X_2) \cdots (\exp t_d X_d),$$

where X_i ($i = 1, 2, \dots, d$) is a basis set of Lie algebra corresponding to G and t_i ($i = 1, 2, \dots, d$) is an arbitrary real number.

Originally, the notion of Lie algebra has been introduced to deal with infinitesimal transformation near the identity element. Yet, Theorem 20.11 says that any transformation (or element) of a connected linear Lie group can be described by a finite number of elements of the corresponding Lie algebra. That is, the Lie algebra determines the global characteristics of the connected Lie group.

Strengthening the condition of connectedness, we need to deal with simply connected groups. In this context we have the following definitions.

Definition 20.8 Let G be a linear Lie group and let an interval $I = [0, 1]$. Let f be a continuous function such that

$$f : I \rightarrow G \quad \text{or} \quad f(I) \subset G. \quad (20.330)$$

Then, the function f is called a path, in which $f(0)$ and $f(1)$ are an initial point and an end point, respectively. If $f(0) = f(1) = x_0$, f is called a loop (i.e., a closed path) at x_0 [14]. If $f(t) \equiv x_0$ ($0 \leq t \leq 1$), f is said to be a constant loop.

Definition 20.9 Let f and g be two paths. If f and g can continuously be deformed from one to the other, they are said to be homotopic.

To be more specific with Definition 20.9, let us define a function $h(s, t)$ that is continuous with respect to s and t in a region $I \times I = \{(s, t); 0 \leq s \leq 1, 0 \leq t \leq 1\}$. If furthermore $h(0, t) = f(t)$ and $h(1, t) = g(t)$ hold, f and g are homotopic [9].

Definition 20.10 Let G be a connected Lie group. If all the loops at an arbitrary chosen point $\forall x \in G$ are homotopic to a constant loop, G is said to be simply connected.

This would somewhat be an abstract concept. Rephrasing the statement in a word, for G to be simply connected is that loops at any $\forall x \in G$ can continually be contracted to that point x . A next example helps understand the meaning of being simply connected.

Example 20.9 Let S^2 be a spherical surface of $\mathbb{R}^3$ such that

$$S^2 = \{\mathbf{x} \in \mathbb{R}^3; \langle \mathbf{x} | \mathbf{x} \rangle = 1\}. \quad (20.331)$$

Any $\mathbf{x}$ is equivalent in virtue of the spherical symmetry of S^2 . Let us think of any loop at $\mathbf{x}$. This loop can be contracted to that point $\mathbf{x}$. Hence, S^2 is simply connected.

By the same token, a spherical hypersurface (or hypersphere) given by $S^n = \{\mathbf{x} \in \mathbb{R}^{n+1}; \langle \mathbf{x} | \mathbf{x} \rangle = 1\}$ is simply connected. Thus, from (20.36) $SU(2)$ is simply connected. Note that the parameter space of $SU(2)$ is S^3 .

Example 20.10 In Sect. 20.4.2 we showed that $SO(3)$ is a connected set. But it is *not* simply connected. To see this, we return to Sect. 17.4.2 that dealt with the three-dimensional rotation matrices. Rewriting $\tilde{R}_3$ in (17.101) as $\tilde{R}_3(\alpha, \beta, \gamma)$, we have

$$\tilde{R}_3(\alpha, \beta, \gamma) = \begin{pmatrix} \cos \alpha \cos \beta \cos \gamma - \sin \alpha \sin \gamma & -\cos \alpha \cos \beta \sin \gamma - \sin \alpha \cos \gamma & \cos \alpha \sin \beta \\ \sin \alpha \cos \beta \cos \gamma + \cos \alpha \sin \gamma & -\sin \alpha \cos \beta \sin \gamma + \cos \alpha \cos \gamma & \sin \alpha \sin \beta \\ -\sin \beta \cos \gamma & \sin \beta \sin \gamma & \cos \beta \end{pmatrix}. \quad (20.332)$$

Note that in (20.332) we represent the rotation in the moving coordinate system. Putting $\beta = 0$ in (20.332), we get

$$\begin{aligned} & \tilde{R}_3(\alpha, 0, \gamma) \\ &= \begin{pmatrix} \cos \alpha \cos \gamma - \sin \alpha \sin \gamma & -\cos \alpha \sin \gamma - \sin \alpha \cos \gamma & 0 \\ \sin \alpha (\cos 0) \cos \gamma + \cos \alpha \sin \gamma & -\sin \alpha (\cos 0) \sin \gamma + \cos \alpha \cos \gamma & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \cos(\alpha + \gamma) & -\sin(\alpha + \gamma) & 0 \\ \sin(\alpha + \gamma) & \cos(\alpha + \gamma) & 0 \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned} \quad (20.333)$$

If in (20.333) we replace α with $\alpha + \phi_0$ and γ with $\gamma - \phi_0$, we are going to obtain the same result as (20.333). That is, different sets of parameters give the same result (i.e., the same rotation) such that

$$(\alpha, 0, \gamma) \leftrightarrow (\alpha + \phi_0, 0, \gamma - \phi_0). \quad (20.334)$$

Putting $\beta = \pi$ in (20.332), we get

$$\tilde{R}_3(\alpha, \pi, \gamma) = \begin{pmatrix} -\cos(\alpha - \gamma) & -\sin(\alpha - \gamma) & 0 \\ -\sin(\alpha - \gamma) & \cos(\alpha - \gamma) & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (20.335)$$

If, in turn, we replace α with $\alpha + \phi_0$ and γ with $\gamma + \phi_0$ in (20.335), we obtain the same result as (20.335). Again, different sets of parameters give the same result (i.e., the same rotation) such that

$$(\alpha, \pi, \gamma) \leftrightarrow (\alpha + \phi_0, \pi, \gamma + \phi_0). \quad (20.336)$$

Meanwhile, R_3 of (17.107) is expressed as

$$R_3 = \begin{pmatrix} (\cos^2 \alpha \cos^2 \beta + \sin^2 \alpha) \cos \gamma & \cos \alpha \sin \alpha \sin^2 \beta (1 - \cos \gamma) & \cos \alpha \cos \beta \sin \beta (1 - \cos \gamma) \\ + \cos^2 \alpha \sin^2 \beta & - \cos \beta \sin \gamma & + \sin \alpha \sin \beta \sin \gamma \\ \cos \alpha \sin \alpha \sin^2 \beta (1 - \cos \gamma) & (\sin^2 \alpha \cos^2 \beta + \cos^2 \alpha) \cos \gamma & \sin \alpha \cos \beta \sin \beta (1 - \cos \gamma) \\ + \cos \beta \sin \gamma & + \sin^2 \alpha \sin^2 \beta & - \cos \alpha \sin \beta \sin \gamma \\ \cos \beta \sin \beta \cos \alpha (1 - \cos \gamma) & \sin \alpha \cos \beta \sin \beta (1 - \cos \gamma) & \sin^2 \beta \cos \gamma \\ - \sin \alpha \sin \beta \sin \gamma & + \cos \alpha \sin \beta \sin \gamma & + \cos^2 \beta \end{pmatrix}. \quad (20.337)$$

Note in (20.337) we represent the rotation in the fixed coordinate system. In this case, again we have arbitrariness with the choice of parameters. Figure 20.7 shows a geometry of the rotation axis (A) accompanied by a rotation γ ; see Fig. 17.18 once again. In this parameter space, specifying $SO(3)$ the rotation is characterized by azimuthal angle α , zenithal angle β , and magnitude of rotation γ . The domains of variability of the parameters (α, β, γ) are

$$0 \leq \alpha \leq 2\pi, \quad 0 \leq \beta \leq \pi, \quad 0 \leq \gamma \leq 2\pi.$$

Figure 20.7a shows a geometry viewed in parallel to the equator that includes a zenithal angle β as well as a point P (i.e., a point of intersection between the rotation axis and geosphere) and its antipodal point $\tilde{P}$. If β is measured at $\tilde{P}$, β and γ should be replaced with $\pi - \beta$ and $2\pi - \gamma$, respectively. Meanwhile, Fig. 20.7b, c depict an azimuthal angle α viewed from above the north pole (N). If the azimuthal angle α is measured at the antipodal point $\tilde{P}$, α should be replaced with either $\alpha + \pi$ (in the case of $0 \leq \alpha \leq \pi$; see Fig. 20.7b) or $\alpha - \pi$ (in the case of $\pi < \alpha \leq 2\pi$; see Fig. 20.7c). Consequently, we have the following correspondence:

$$(\alpha, \beta, \gamma) \leftrightarrow (\alpha \pm \pi, \pi - \beta, 2\pi - \gamma). \quad (20.338)$$

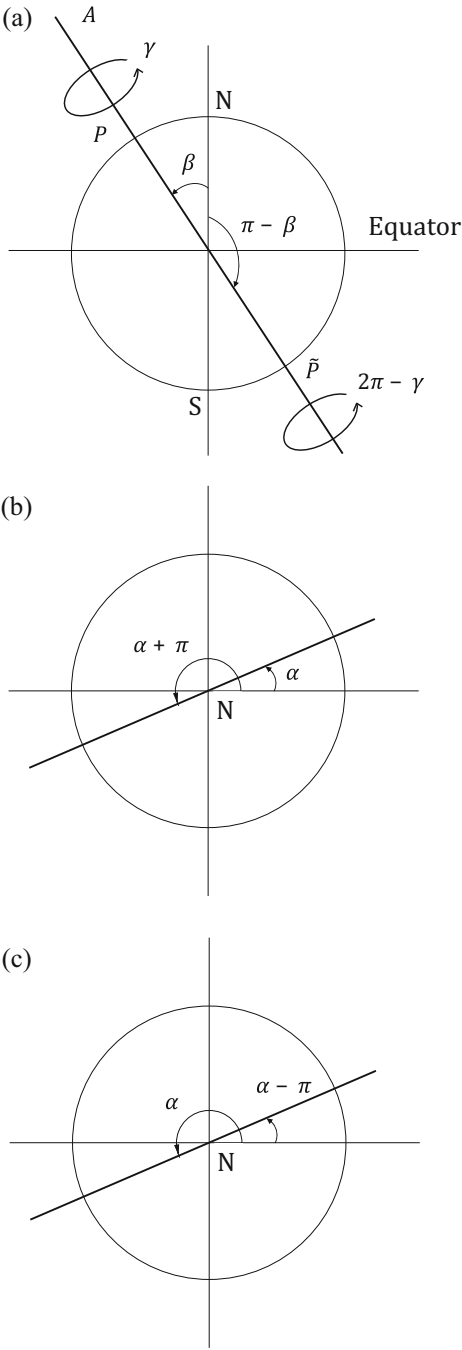
Thus, we confirm once again that the different sets of parameters give the same rotation. In other words, two different points in the parameter space correspond to the same transformation (i.e., rotation). In fact, using these different sets of parameters, the matrix form of (20.337) is held unchanged. The conformation is left for readers.

Because of the aforementioned characteristic of the parameter space, a loop cannot be contracted to a single point in $SO(3)$. For this reason, $SO(3)$ is not simply connected.

In contrast to Example 20.10, the mapping from the parameter space S^3 to $SU(2)$ is bijective in the case of $SU(2)$. That is, any two different points of S^3 give different transformations on $SU(2)$ (i.e., injective). For any element of $SU(2)$ it has a corresponding point in S^3 (i.e., surjective). This can be rephrased such that $SU(2)$ and S^3 are isomorphic.

In this context, let us give important concepts. Let (T, τ) and (S, σ) be two topological spaces. Suppose that $f: (T, \tau) \rightarrow (S, \sigma)$ is a bijective mapping. If in this

Fig. 20.7 Geometry of the rotation axis (A) accompanied by a rotation γ . (a) Geometry viewed in parallel to the equator. (b), (c) Geometry viewed from above the north pole (N). If the azimuthal angle α is measured at the antipodal point $\tilde{P}$, α should be replaced with (b) $\alpha + \pi$ (in the case of $0 \leq \alpha \leq \pi$) or (c) $\alpha - \pi$ (in the case of $\pi < \alpha \leq 2\pi$)



case, furthermore, both f and f^{-1} are continuous mappings, (T, τ) and (S, σ) are said to be *homeomorphic* [14]. In our present case, both $SU(2)$ and S^3 are homeomorphic.

Apart from being simply connected or not, $SU(2)$ and $SO(3)$ resemble in many aspects. The resemblance has already shown up in Chap. 3 when we considered the (orbital) angular momentum and generalized angular momentum. As shown in (3.30) and (3.69), the commutation relation was virtually the same. It is obvious when we compare (20.271) and (20.275). That is, in terms of Lie algebras their structure constants are the same and, eventually, the structure of corresponding Lie groups is related. This became well understood when we considered the adjoint representation of the Lie group G on its Lie algebra $\mathfrak{g}$. Of the groups whose Lie algebras are characterized by the same structure constants, the simply connected group is said to be a universal covering group. For example, among the Lie groups $O(3)$, $SO(3)$, and $SU(2)$, only $SU(2)$ is simply connected and, hence, is a universal covering group.

The understanding of the Lie algebra is indispensable to explicitly describing the elements of the Lie group, especially the connected components of the identity element $e \in G$. Then, we are able to understand the global characteristics of the Lie group. In this chapter, we have focused on $SU(2)$ and $SO(3)$ as a typical example of linear Lie groups.

Finally, though formal, we describe a definition of a topological group.

Definition 20.11 [15] Let G be a set. If G satisfies the following conditions, G is called a topological group.

1. The set G is a group.
2. The set G is a T_1 -space.
3. Group operations (or calculations) of G are continuous. That is, let P and Q be two mappings such that

$$P : G \times G \rightarrow G; P(g, h) = gh \quad (g, h \in G), \quad (20.339)$$

$$Q : G \rightarrow G; Q(g) = g^{-1} \quad (g \in G). \quad (20.340)$$

Then, both the mappings P and Q are continuous.

By Definition 20.11, a topological group combines the structures of group and topological space. Usually, the continuous group is a collective term of Lie groups and topological groups. The continuous groups are very frequently dealt with in various fields of natural science and mathematical physics.

References

1. Inui T, Tanabe Y, Onodera Y (1990) Group theory and its applications in physics. Springer, Berlin

2. Inui T, Tanabe Y, Onodera Y (1980) Group theory and its applications in physics, Expanded edn. Shokabo, Tokyo. (in Japanese)
3. Takagi T (2010) Introduction to analysis, Standard edn. Iwanami, Tokyo. (in Japanese)
4. Arfken GB, Weber HJ, Harris FE (2013) Mathematical methods for physicists, 7th edn. Academic Press, Waltham
5. Hamermesh M (1989) Group theory and its application to physical problems. Dover, New York
6. Rose ME (1995) Elementary theory of angular momentum. Dover, New York
7. Hassani S (2006) Mathematical physics. Springer, New York
8. Edmonds AR (1957) Angular momentum in quantum mechanics. Princeton University Press, Princeton, NJ
9. Yamanouchi T, Sugiura M (1960) Introduction to continuous groups. Baifukan, Tokyo. (in Japanese)
10. Dennery P, Krzywicki A (1996) Mathematics for physicists. Dover, New York
11. Satake I-O (1975) Linear algebra (pure and applied mathematics). Marcel Dekker, New York
12. Satake I (1974) Linear algebra. Shokabo, Tokyo. (in Japanese)
13. Steeb W-H (2007) Continuous symmetries, Lie algebras, differential equations and computer algebra, 2nd edn. World Scientific, Singapore
14. McCarty G (1988) Topology. Dover, New York
15. Murakami S (2004) Foundation of continuous groups, Revised edn. Asakura Publishing, Tokyo. (in Japanese)

Part V

Introduction to the Quantum Theory of Fields

In Part I we described the fundamental feature of quantum mechanics. The theory based upon the Schrödinger equation was inconsistent with the theory of relativity, however. To make the theory consistent with the special theory of relativity, Dirac developed his theory that led to the discovery of the Dirac equation, named after himself. The success by Dirac was followed by the development of the quantum theory of fields.

In this part, we study the basic properties and constitution of the Dirac equation. We examine various properties of the plane wave solutions of the Dirac equation. The quantum theory of fields was first established as quantum electrodynamics (QED), in which the Dirac field and electromagnetic field (or photon field) play the leading role. With the electromagnetic field, we highlight the coexistence of classical theory and quantum theory. Before the quantization of the Dirac field and photon field, however, we start with the canonical quantization of the scalar field (or the Klein–Gordon field) because of simplicity and accessibility as well as the wide applicability to other fields. After studying the quantization of fields, we approach several features of the interaction between the fields, with emphasis upon the practical aspect. The interactions between the fields are presented as nonlinear field equations. Since it is difficult to obtain analytic solutions of the nonlinear equation, suitable approximation must be done. The method for this is well established as the perturbation theory that makes the most of the S -matrix. In this book, we approach the problem by calculating the interaction process in lowest order of the perturbation theory. As a tangible example, we take up Compton scattering and study its characteristics in detail.

In the latter half, we examine basic formalism of the Lorentz group from the point of view of the Lie groups and Lie algebras. In particular, we explore in detail the Lorentz transformation properties of the Dirac equation and its plane wave solutions. We stress the importance and practicability of the matrix representation of the Dirac operators and Dirac spinors with respect to the Dirac equation. We study their transformation properties in terms of the matrix algebra that is connected to the Lorentz group. The Dirac operators associated with the Lorentz transformation are somewhat intractable, however, since the operators are not represented by normal

matrices (e.g., Hermitian matrix or unitary matrix) because of the presence of the Lorentz boost. We examine how these issues are dealt with within the framework of the standard matrix theory. This last part of the book deals with advanced topics of Lie algebra, with its differential representation stressed. Since the theory of vector spaces underlies all parts of the final part, several extended concepts of vector spaces are explained in relation to Part III.

Historically speaking, the Schrödinger equation has obtained citizenship across a wide range of scientific community including physics, chemistry, materials science, and even biology. As for the Dirac equation, on the other hand, it seems to have found the place only in theoretical physics, especially the quantum theory of fields. Nonetheless, in recent years the Dirac equation often shows up as a fundamental equation in, e.g., solid-state physics. Under these circumstances, we wish to investigate the constitution and properties of the Dirac equation as well as the Dirac operators and spinors on a practical level throughout the whole of this part.

Chapter 21

The Dirac Equation



In this chapter we describe the feature of the Dirac equation including its historical background. The Dirac equation is a quantum theory of electron that deals with the electron dynamics within the framework of special theory of relativity. Before the establishment of the theory achieved by Dirac, several attempts to develop the relativistic quantum-mechanical theory of electron were made mainly on the basis of the Klein–Gordon equation. The theory, however, had a serious problem in terms of the probability density. Dirac proposed the Dirac equation, named after himself, to dispel the problem. Most importantly, the Dirac equation opened up a novel route to the quantum theory of fields. The quantum theory of fields has first been established as the quantum electrodynamics (QED) that deals with electron and photons as quantized fields. The theory has later become the foundation of the advanced quantum field theory based on the gauge theory. In this chapter we study and pursue the framework of the Dirac equation with emphasis upon the plane wave solutions to make it a clear starting point for gaining a good understanding of QED. The structure of the Dirac equation can be well-understood using the polar coordinate representation. The results will be used in Chap. 24 as well to further develop the theory. The Dirac equation is originally “relativistic.” To help understand the nature of the Dirac equation, we add several remarks on the special theory of relativity at the beginning of the chapter accordingly.

21.1 Historical Background

The success in establishing the atomic-scale mechanics by the Schrödinger equation prompted many researchers to pursue relativistic formulation of the quantum mechanics. As a first approach for this, the Klein–Gordon equation (occasionally referred to as Klein–Gordon–Fock equation) was proposed [1, 2]. It is expressed as

$$\left[\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 + \left(\frac{mc}{\hbar} \right)^2 \right] \phi(t, \mathbf{x}) = 0, \quad (21.1)$$

where the operator ∇^2 has already appeared in (1.24) and (7.30). More succinctly we have

$$\left[\square + \left(\frac{mc}{\hbar} \right)^2 \right] \phi(t, \mathbf{x}) = 0,$$

where $\square$ is d'Alembertian (or d'Alembert operator) defined by

$$\square \equiv \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2.$$

The Klein–Gordon equation is a relativistic wave equation for a quantum-mechanical free particle. This equation is directly associated with the equivalence theorem of mass and energy. In other words, it is based on the following relationship among energy, momentum, and mass of the particle:

$$E^2 = p^2 c^2 + m^2 c^4, \quad (21.2)$$

where E , p , and m are energy, momentum, and mass of the particle, respectively; see (1.20) and (1.21). Equation (21.1) can be obtained using the correspondence between momentum or energy and differential operators expressed in (1.31) or (1.40).

Nonetheless, the Klein–Gordon equation included a fundamental difficulty in explaining a probability density for the presence of particle. The argument is as follows: The Schrödinger equation is described as

$$i\hbar \frac{\partial \psi}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi. \quad (1.45)$$

Multiplying both sides of (1.45) by ψ^* from the left, we have

$$i\hbar \psi^* \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \psi^* \nabla^2 \psi + V \psi^* \psi. \quad (21.3)$$

Taking complex conjugate of (1.45) and multiplying ψ from the right, we have

$$-i\hbar \frac{\partial \psi^*}{\partial t} \psi = -\frac{\hbar^2}{2m} (\nabla^2 \psi^*) \psi + V \psi^* \psi. \quad (21.4)$$

Subtracting (21.4) from (21.3), we get

$$i\hbar\left(\psi^* \frac{\partial \psi}{\partial t} + \frac{\partial \psi^*}{\partial t} \psi\right) = \frac{\hbar^2}{2m} [(\nabla^2 \psi^*)\psi - \psi^* \nabla^2 \psi]. \quad (21.5)$$

Further rewriting (21.5), we have

$$\frac{\partial}{\partial t}(\psi^* \psi) + \frac{\hbar}{2mi} [\psi^* \nabla^2 \psi - (\nabla^2 \psi^*)\psi] = 0. \quad (21.6)$$

Here, we define ρ and $\mathbf{j}$ as

$$\begin{aligned} \rho(\mathbf{x}, t) &\equiv \psi^*(\mathbf{x}, t)\psi(\mathbf{x}, t), \\ \mathbf{j}(\mathbf{x}, t) &\equiv \frac{\hbar}{2mi} [\psi^* \nabla \psi - (\nabla \psi^*)\psi], \end{aligned} \quad (21.7)$$

where $\mathbf{x}$ is a position vector and ∇ is nabla [defined in (3.10)]. Recasting (21.6) by use of (21.7), we get a continuity equation expressed as

$$\frac{\partial \rho}{\partial t} + \text{div } \mathbf{j} = 0. \quad (21.8)$$

If we deem ρ to be a probability density of the particle and $\mathbf{j}$ to be its flow, (21.8) represents the conservation law of the probability density; see (7.20). Notice that the quantity $\rho(\mathbf{x}, t)$ defined in (21.7) is positive (or zero) anywhere at any time.

Meanwhile, we can similarly define ρ and $\mathbf{j}$ with the Klein–Gordon equation of (21.1) such that

$$\begin{aligned} \rho(\mathbf{x}, t) &\equiv \frac{i\hbar}{2m} \left(\phi^* \frac{\partial \phi}{\partial t} - \frac{\partial \phi^*}{\partial t} \phi \right), \\ \mathbf{j}(\mathbf{x}, t) &\equiv \frac{\hbar}{2mi} [\phi^* \nabla \phi - (\nabla \phi^*)\phi] \end{aligned} \quad (21.9)$$

to derive a continuity equation similar to (21.8). A problem lies, however, in regarding ρ as the probability density. It is because ρ may be either positive or negative. To see this, put, e.g., $\phi = f + ig$ (where f and g are real functions) and calculate the first equation of (21.9). Thus, the conservation of the probability density could not be guaranteed.

Under such circumstances, it was Dirac that settled the problem (1928); see Sect. 1.2. He proposed the Dirac equation and his theory was counted as one of the most successful theories among those describing the equation of motion of a quantum-mechanical particle (i.e., electron) within the framework of the special theory of relativity.

21.2 Several Remarks on the Special Theory of Relativity

21.2.1 Minkowski Space and Lorentz Transformation

The special theory of relativity is based on the principle of constancy of light velocity. We give several brief remarks on the theory together with examples. Regarding the mathematical equations dealt with in the quantum theory of fields, we normally define natural units to represent physical quantities by putting

$$c = 1, \hbar = 1. \quad (21.10)$$

We follow the custom from here on. Using the natural units, as the Klein–Gordon equation we have

$$(\square + m^2)\phi(t, \mathbf{x}) = 0. \quad (21.11)$$

Also, time t and space coordinates $\mathbf{x}$ are denoted altogether by

$$x = x^\mu = \begin{pmatrix} t \\ \mathbf{x} \end{pmatrix} = \begin{pmatrix} x_0^0 \\ x_1^0 \\ x_2^0 \\ x_3^0 \end{pmatrix} \quad (\mu = 0, 1, 2, 3) \quad (21.12)$$

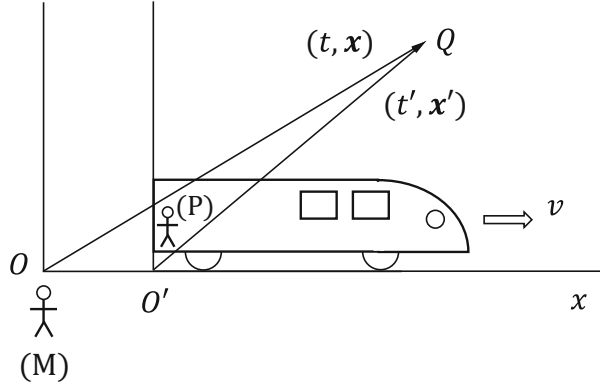
where x^0 represents the time coordinate and x^1, x^2 , and x^3 show the space coordinates. The quantity x is referred to collectively as a space–time coordinate. We follow the custom as well with the superscript notation of physical quantities. The implication will be mentioned later (also, see Sect. 24.1). The quantity x^μ ($\mu = 0, 1, 2, 3$) is called a four-vector; this is often represented by a single letter x . That is, $\phi(x)$ is meant by $\phi(t, \mathbf{x})$.

Example 21.1 Suppose that a train is running on a straight rail track with a constant velocity v toward the positive direction of the x -axis and that a man (M) standing on firm ground is watching the train. Also, suppose that at the moment when the tail end of the train is passing in front of him, he and a passenger (P) on the train adjust their clocks to $t = t' = 0$. Here, t is time for the man on firm ground and t' for the passenger (Fig. 21.1). Suppose moreover that a light is emitted at the tail end of the train at $t = t' = 0$.

Let O be the inertial frame of reference fixed on the man on firm ground and let O' be another inertial frame of reference fixed on the train. Also, let the origin of O be on that man and let that of O' be on the tail end of the train. Let us assume that the emitted light is observed at $t = t$ at a position vector $\mathbf{x} (=xe_1 + ye_2 + ze_3)$ of O and that the same light is observed at $t' = t'$ at a position vector $\mathbf{x}' (=x'e'_1 + y'e'_2 + z'e'_3)$ of O' . In the above, e_1, e_2, e_3 and e'_1, e'_2, e'_3 are individual basis sets for the Cartesian coordinates related to O and O' , respectively.

To make the situation clearer, suppose that a laser beam has been emitted at $t = 0$ and $\mathbf{x} = \mathbf{0}$ in the frame O and that the beam has been emitted at $t' = 0$ and $\mathbf{x}' = \mathbf{0}$ in

Fig. 21.1 Two inertial frames of reference O and O' . In O a man (M) is standing on firm ground and in O' a passenger (P) is standing at the tail end of the train. The train is running on a straight rail track with a constant velocity v in reference to O toward the positive direction of the x -axis. The man and passenger are observing how a light (e.g., a laser beam) is emitted and that light hits a ball after a moment at a certain place Q (see text)



the frame O' . After a moment, the same beam hits a ball at a certain place Q . Let Q be specified as $x = x$ in the frame O and at $x' = x'$ in the frame O' . Let time when the beam hits the ball be $t = t$ in the frame O and $t' = t'$ in the frame O' . Naturally, the man M chooses his standing position for $x = 0$ (i.e., the origin of O) and the passenger P chooses the tail end of the train for $x' = 0$ (i.e., the origin of O'). In this situation, the special theory of relativity tells us that the following relationship must hold:

$$(ct)^2 - x^2 = (ct')^2 - x'^2 = 0.$$

Note that the light velocity c is the same with the two frames of O and O' from the requirement of the special theory of relativity. Using the natural units, the above equation is rewritten as

$$t^2 - x^2 = t'^2 - x'^2 = 0. \quad (21.13)$$

Equation (21.13) represents an invariant (i.e., zero in this case) with respect to the choice of space–time coordinates, or inertial frames of reference.

Before taking a step further to think of the implication of (21.13) and generalize the situation of Example 21.1, let us introduce a mathematical technique. Let $B(y, x)$ be a bilinear form defined as [3, 4]

$$B(y, x) \equiv y^T \eta x = (y^1 \cdots y^n) \eta \begin{pmatrix} x^1 \\ \vdots \\ x^n \end{pmatrix}, \quad (21.14)$$

where x and y represent a set of coordinates of vectors in a n -dimensional vector space. A common basis set is assumed for x and y . The coordinates x and y are defined as a column vector described by

$$x \equiv \begin{pmatrix} x^1 \\ \vdots \\ x^n \end{pmatrix}, y \equiv \begin{pmatrix} y^1 \\ \vdots \\ y^n \end{pmatrix}.$$

Although the coordinates x^i and y^i ($i = 1, \dots, n$) are usually chosen from either real numbers or complex numbers, in this chapter we assume that both x^i and y^i are real. In the Minkowski space we will be dealing with in this chapter, x^i and y^i ($i = 0, 1, 2, 3$) are real numbers. The coordinates x^0 and y^0 denote the time coordinate and x^k and y^k ($k = 1, 2, 3$) represent the space coordinates. We call x^i and y^i ($i = 0, 1, 2, 3$) the space–time coordinates altogether.

In (21.14) η is said to be the *metric tensor* that characterizes the vector space which we are thinking of. In a general case of a n -dimensional vector space, we have

$$\eta = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & & -1 & \\ & & & & \ddots \\ & & & & & -1 \end{pmatrix}, \quad (21.15)$$

where η is a (n, n) matrix having p integers of 1 and q integers of -1 ($p + q = n$) with diagonal elements; all the off-diagonal elements are zero. For this notation, see Sect. 24.1 as well.

In the present specific case, we have

$$\eta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (21.16)$$

The tensor η of (21.16) is called Minkowski metric and the vector space pertinent to the special theory of relativity is known as a four-dimensional Minkowski space accompanied by space–time coordinates represented by (21.12). By virtue of the metric tensor η , we have the following three cases:

$$B(x, x) \begin{cases} < 0 \\ = 0 \\ > 0 \end{cases}. \quad (21.17)$$

We define a new row vector such that

$$(x_0 \ x_1 \ x_2 \ x_3) \equiv (x^0 \ x^1 \ x^2 \ x^3) \eta.$$

Then, the bilinear form $B(x, x)$ given by (21.14) can be rewritten as

$$B(x, x) = (x_0 \ x_1 \ x_2 \ x_3) \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}. \quad (21.18)$$

With respect to individual components x_ν , we have

$$x_\nu = \sum_{\mu=0}^3 x^\mu \eta_{\mu\nu} = \sum_{\mu=0}^3 x^\mu \eta_{\nu\mu}, \quad (21.19)$$

where x^μ is called a contravariant (column) vector and x_ν a covariant (row) vector. These kinds of vectors have already appeared in Sect. 20.3.2 and will be dealt with later in detail in Sect. 24.1 in relation to tensor and tensor space. Then, we have

$$x_0 = x^0, x_1 = -x^1, x_2 = -x^2, x_3 = -x^3. \quad (21.20)$$

Adopting Einstein summation convention, we express (21.18) as

$$B(x, x) = \sum_{\mu=0}^3 x_\mu x^\mu \equiv x_\mu x^\mu, \quad (21.21)$$

where the same subscripts and superscripts are supposed to be summed over 0 to 3. Using this summation convention, (21.13) can be rewritten as

$$x_\mu x^\mu = x'_\mu x'^\mu = 0. \quad (21.22)$$

Notice that η of (21.16) is symmetric matrix. The inverse matrix of η is the same as η . That is, $\eta^{-1} = \eta$ or

$$\eta^{\mu\nu} \eta_{\nu\rho} = \delta^\mu_\rho,$$

where δ^μ_ρ is Kronecker delta. Once again, note that the subscript and superscript ν are summed. The symbols $\eta^{\mu\nu}$ and $\eta_{\nu\rho}$ represent η^{-1} and η , respectively. That is, $(\eta^{\mu\nu}) = (\eta)^{\mu\nu} = (\eta^{-1})^{\mu\nu}$ and $(\eta_{\mu\nu}) = (\eta)_{\mu\nu}$; with these notations see (11.38).

What we immediately see from (21.17) is that unlike the inner product, $B(x, x)$ does not have a positive definite feature. If we put e.g., $x^0 = x_0 = 0$, we have $B(x, x) \leq 0$. For this reason, the Minkowski metric η is said to be an indefinite metric or the Minkowski space is called an indefinite metric space.

Inserting $E = \Lambda^{-1} \Lambda$ in (21.18), we rewrite it as

$$B(x, x) = (x_0 \ x_1 \ x_2 \ x_3) \Lambda^{-1} \Lambda \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}, \quad (21.23)$$

where Λ is a non-singular matrix. Suppose in (21.23) that the contravariant vector is transformed such that

$$\begin{pmatrix} x'^0 \\ x'^1 \\ x'^2 \\ x'^3 \end{pmatrix} = \Lambda \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}. \quad (21.24)$$

Suppose, in turn, that with the transformation of a covariant vector we have

$$(x'_0 \ x'_1 \ x'_2 \ x'_3) = (x_0 \ x_1 \ x_2 \ x_3) \Lambda^{-1}. \quad (21.25)$$

Then, (21.23) can further be rewritten as

$$B(x, x) = (x'_0 \ x'_1 \ x'_2 \ x'_3) \begin{pmatrix} x'^0 \\ x'^1 \\ x'^2 \\ x'^3 \end{pmatrix} = \sum_{\mu=0}^3 x'_\mu x'^\mu = x'_\mu x'^\mu = B(x', x'). \quad (21.26)$$

Equation (21.26) implies that if we demand the contravariant vector x^μ ($\mu = 0, 1, 2, 3$) and the covariant vector x_μ be transformed according to (21.24) and (21.25), respectively, the bilinear form $B(x, x)$ is held invariant before and after the transformation caused by Λ . In fact, using (21.24) and (21.25) we have

$$B(x', x') = x'_\mu x'^\mu = x_\sigma (\Lambda^{-1})^\sigma_\mu \Lambda^\mu_\rho x^\rho = x_\sigma \delta^\sigma_\rho x^\rho = x_\sigma x^\sigma = B(x, x).$$

In this respect, (21.22) is a special case of (21.26). Or rather, the transformation Λ is decided so that $B(x, x)$ can be held invariant. A linear transformation Λ that holds $B(x, x)$ invariant on the Minkowski space is called Lorentz transformation. A tangible form of the matrix Λ will be decided below.

To explicitly show how the covariant vector is transformed as a *column* vector, taking the transposition of (21.25) we get

$$\begin{pmatrix} x'_0 \\ x'_1 \\ x'_2 \\ x'_3 \end{pmatrix} = (\Lambda^{-1})^T \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix}. \quad (21.27)$$

Equation (21.27) indicates that a covariant vector is transformed by a matrix $(\Lambda^{-1})^T$. We have $(\Lambda^{-1})^T = (\Lambda^T)^{-1}$. To confirm this, taking transposition of $\Lambda \Lambda^{-1} = E$, we get $(\Lambda^{-1})^T \Lambda^T = E$. Multiplying both sides of this equation by $(\Lambda^T)^{-1}$ from the right, we obtain $(\Lambda^{-1})^T = (\Lambda^T)^{-1}$.

Regarding the discussion of the bilinear form, we will come back to the issue in Chap. 24.

Remark 21.1 Up to now in this book, we have not thought about the necessity to distinguish the covariant and contravariant vectors except for a few cases (see Sect. 20.3.2). This is because if we think of a vector, e.g., in $\mathbb{R}^n$ (i.e., the n -dimensional Euclidean space), the metric called Euclidean metric is represented by an n -dimensional identity matrix E_n . Instead of dealing with the Lorentz transformation, let us think of a familiar real orthogonal transformation in $\mathbb{R}^3$.

In parallel with (21.14), as a bilinear form we have

$$B(x, x) = (x^1 \ x^2 \ x^3) E_3 \begin{pmatrix} x^1 \\ x^2 \\ x^3 \end{pmatrix} = (x^1 \ x^2 \ x^3) \begin{pmatrix} x^1 \\ x^2 \\ x^3 \end{pmatrix},$$

where E_3 is a $(3, 3)$ identity matrix. In this case the bilinear form $B(x, x)$ is identical to the positive definite inner product $\langle x | x \rangle$. With the transformation of a contravariant vector, we have

$$\begin{pmatrix} x'^1 \\ x'^2 \\ x'^3 \end{pmatrix} = R \begin{pmatrix} x^1 \\ x^2 \\ x^3 \end{pmatrix}. \quad (21.28)$$

With a covariant vector, in turn, similarly to the case of (21.27) we have

$$\begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \end{pmatrix} = (R^{-1})^T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = (R^T)^T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = R \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad (21.29)$$

where with the second equality we used $R^{-1} = R^T$ for an orthogonal matrix. From (21.28) and (21.29), we find that with the orthogonal transformation in $\mathbb{R}^3$ we do not need to distinguish a contravariant vector and a covariant vector. Note, however, that unless the transformation is orthogonal, the contravariant vector and covariant vector should be distinguished. Also, we will come back to this issue in Chap. 24.

21.2.2 Event and World Interval

If we need to specify space–time points where physical phenomena happen, we call such physical phenomena “events.” Suppose, for instance, that we are examining a motion of a baseball game and that we are going to decide when and where the ball is hit with a bat. In that situation the phenomenon of batting the ball is called an event. At the same time, we can specify the space–time coordinates where (or when) the ball is hit in an arbitrarily chosen inertial frame of reference. Bearing in mind such a situation, suppose in Example 21.1 that two events E and F have happened. Also, suppose that in Fig. 21.1 an observer M on the frame O has experienced the events E and F at $x^\mu(E)$ and $x^\mu(F)$, respectively. Meanwhile, another observer P on the

frame O' has experienced the same events E and F at $x'^\mu(E)$ and $x'^\mu(F)$, respectively. In that situation we define a following quantity s .

$$s \equiv \sqrt{[x^0(F) - x^0(E)]^2 - \sum_{k=1}^3 [x^k(F) - x^k(E)]^2}. \quad (21.30)$$

The quantity s is said to be a world interval between the events E and F . The world interval between the two events is classified as follows:

$$s = \begin{cases} \text{timelike } (s : \text{real}) \\ \text{spacelike } (s : \text{imaginary}) \\ \text{lightlike, null } (s : \text{zero}). \end{cases} \quad (21.31)$$

In terms of the special principle of relativity which demands that physical laws be the same in every inertial frame of reference, the world interval s must be an invariant under the Lorentz transformation. The relationship $s = 0$ is associated with the principle of constancy of light velocity. Therefore, we can see (21.31) as a natural extension of the light velocity constancy principle. Historically, this principle has been established on the basis of the precise physical measurements including those performed in the Michelson–Morley experiment [5].

Now we are in the position to determine an explicit form of Λ . To make the argument simple, we revisit Example 21.1.

Example 21.2 We assume that in Fig. 21.1 the frame O' is moving at a velocity of v toward the positive direction of the x^1 -axis of the frame O so that the x^1 - and x'^1 -axes can coincide. In this situation, it suffices to only take account of x^1 as a space coordinate. In other words, we have

$$x'^2 = x^2 \quad \text{and} \quad x'^3 = x^3. \quad (21.32)$$

Then, we assume a following linear transformation [5]:

$$x'^0 = px^0 + qx^1, \quad x'^1 = rx^0 + ux^1. \quad (21.33)$$

Writing (21.33) in a matrix form, we get

$$\begin{pmatrix} x'^0 \\ x'^1 \end{pmatrix} = \begin{pmatrix} p & q \\ r & u \end{pmatrix} \begin{pmatrix} x^0 \\ x^1 \end{pmatrix}. \quad (21.34)$$

Corresponding to (21.30), we have

$$s^2 = [x^0(F) - x^0(E)]^2 - [x^1(F) - x^1(E)]^2 = [x^0(F)]^2 - [x^1(F)]^2,$$

$$s'^2 = [x'^0(F) - x'^0(E)]^2 - [x'^1(F) - x'^1(E)]^2 = [x'^0(F)]^2 - [x'^1(F)]^2. \quad (21.35)$$

In (21.35) we assume $x^0(E) = x^1(E) = 0$ and $x'^0(E) = x'^1(E) = 0$; i.e., as discussed earlier we regard the event E as the fact that the tail end of the train is just passing in front of the man standing on firm ground. Deleting F from (21.35) and on the basis of the Lorentz invariance of s , we must have

$$s'^2 = [x'^0]^2 - [x'^1]^2 = [x^0]^2 - [x^1]^2 = s^2. \quad (21.36)$$

The coefficients p , q , r , and u of (21.33) can be determined from the requirement that (21.36) holds with any x^0 and x^1 . Inserting x'^0 and x'^1 of RHSs of (21.33) into (21.36) and comparing the coefficients of $x^i x^k$ ($i, k = 0, 1; i \leq k$) in (21.36), we get [5]

$$p^2 - r^2 = 1, \quad pq - ru = 0, \quad q^2 - u^2 = -1. \quad (21.37)$$

To decide four parameters, we need another condition different from (21.37). Since we assume that the frame O' is moving at the velocity of v to the x^1 -direction of the frame O , the origin of O' (i.e., $x'^1 = 0$) should correspond to $x^1 = vx^0$ of O . Namely, from (21.33) we have

$$0 = rx^0 + uvx^0 = (r + uv)x^0. \quad (21.38)$$

As (21.38) must hold with any x^0 , we have

$$r + uv = 0. \quad (21.39)$$

Since (21.37) and (21.39) are second-order equations with respect to p , q , r , and u , we do not obtain a unique solution. Solving (21.37) and (21.39), we get $p = \pm 1/\sqrt{1-v^2}$, $u = \pm p$, $q = -vp$, and $r = -vu$.

However, p , q , r , and u must satisfy the following condition such that

$$\lim_{v \rightarrow 0} \begin{pmatrix} p & q \\ r & u \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then, we must have

$$p = u = 1/\sqrt{1-v^2}, \quad r = q = -v/\sqrt{1-v^2}.$$

Inserting these four parameters into (21.33), we rewrite (21.33) as

$$x'^0 = \frac{1}{\sqrt{1-v^2}} (x^0 - vx^1), \quad x'^1 = \frac{1}{\sqrt{1-v^2}} (-vx^0 + x^1).$$

Using a single parameter γ given by

$$\gamma \equiv 1/\sqrt{1-v^2},$$

(21.34) can be rewritten as

$$\begin{pmatrix} x'^0 \\ x'^1 \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma v \\ -\gamma v & \gamma \end{pmatrix} \begin{pmatrix} x^0 \\ x^1 \end{pmatrix}. \quad (21.40)$$

With the reversed form, we obtain

$$\begin{pmatrix} x^0 \\ x^1 \end{pmatrix} = \begin{pmatrix} \gamma & \gamma v \\ \gamma v & \gamma \end{pmatrix} \begin{pmatrix} x'^0 \\ x'^1 \end{pmatrix}. \quad (21.41)$$

Taking account of (21.32), a full (4, 4) matrix form is

$$\begin{pmatrix} x'^0 \\ x'^1 \\ x'^2 \\ x'^3 \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma v & 0 & 0 \\ -\gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}. \quad (21.42)$$

The (4, 4) matrix of (21.42) is a simple example of the Lorentz transformations. These transformations form a group that is called the Lorentz group. If $v = 0$, we have $\gamma = 1$. Then, the Lorentz transformation given by (21.42) becomes the identity matrix (i.e., the identity element of the Lorentz group). The inverse element of the matrix of (21.42) is given by its inverse matrix. The matrix given by (21.42) can conveniently be expressed as follows: Putting

$$\tanh \omega \equiv v \quad (-\infty < \omega < \infty \quad \Leftrightarrow \quad -1 < v < 1) \quad (21.43)$$

and using the formulae of the hyperbolic functions

$$\cosh \omega = \frac{1}{\sqrt{1 - \tanh^2 \omega}}, \quad (21.44)$$

we have

$$\cosh \omega = \frac{1}{\sqrt{1 - v^2}} = \gamma, \quad \sinh \omega = \gamma v = v/\sqrt{1 - v^2}. \quad (21.45)$$

Thus, (21.42) can be rewritten as

$$\begin{pmatrix} \gamma & -\gamma v & 0 & 0 \\ -\gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \cosh \omega & -\sinh \omega & 0 & 0 \\ -\sinh \omega & \cosh \omega & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (21.46)$$

The matrix of (21.46) is Hermitian (or real symmetric) with real positive eigenvalues and the determinant of 1. We will come back to this feature in Chap. 24. If two successive Lorentz transformations are given by $\tanh \omega_1$ and $\tanh \omega_2$, their multiplication is given by

$$\begin{aligned} & \begin{pmatrix} \cosh \omega_2 & -\sinh \omega_2 & 0 & 0 \\ -\sinh \omega_2 & \cosh \omega_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cosh \omega_1 & -\sinh \omega_1 & 0 & 0 \\ -\sinh \omega_1 & \cosh \omega_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \cosh(\omega_2 + \omega_1) & -\sinh(\omega_2 + \omega_1) & 0 & 0 \\ -\sinh(\omega_2 + \omega_1) & \cosh(\omega_2 + \omega_1) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \end{aligned} \quad (21.47)$$

Equation (21.47) obviously shows the closure in terms of the multiplication. The associative law with respect to the multiplication is evident. Thus, the matrices represented by (21.46) satisfy the axioms (A1) to (A4) of Sect. 16.1. Hence, a collection of these matrices forms a group (i.e., the Lorentz group). As clearly seen in (21.46) and (21.47), the constitution of the Lorentz group is well characterized by the variable ω , which is sometimes called “rapidity.” Notice that the matrices given above are not unitary. These matrices are often referred to as the Lorentz boosts.

Example 21.3 Here let us briefly think of the Lorentz contraction. Suppose that a straight stick exists in a certain inertial frame of reference O . The stick may or may not be moving in reference to that frame. Being strictly formal, measuring a length of the stick in that frame comprises (1) to measure and fix a spatial point P (x_P^1, x_P^2, x_P^3) that corresponds to one end of the stick and (2) to measure and fix another spatial point Q (x_Q^1, x_Q^2, x_Q^3) corresponding to the other end. The former and latter operations correspond to the events E and F , respectively. These operations must be done at the *same time* in the frame. Then, the distance between P and Q (P and Q may be either different or identical) defined as d_{PQ} in the frame is given by a following absolute value of the world interval:

$$d_{PQ} \equiv \left| \sqrt{-\sum_{k=1}^3 (x_Q^k - x_P^k)^2} \right|. \quad (21.48)$$

We define the length of the straight stick as d_{PQ} . We have $d_{PQ} \geq 0$; $d_{PQ} = 0$ if and only if P and Q are identical. Notice that the world interval between the events E and F is a pure imaginary from (21.30), because those events happen at the same time.

Bearing in mind the above discussion, suppose that the straight stick is laid along the x^1 -direction with its end in contact with the tail end of the train. Suppose that the length of the stick is L' in the frame O' of Example 21.1. This implies that from (21.40) $x^1 = L'$. (Remember that the man standing on firm ground and the passenger on the train adjusted their watches to $t = t' = 0$, at the moment when the tail end of the train was passing in front of the man.) If we assume that the length of the stick is read off at t (or $x^0 = 0$) in the frame O of Example 21.1, again from (21.40) we have

$$L = x^1 = \sqrt{1 - v^2} x'^1 = L' \sqrt{1 - v^2} = L' / \gamma, \quad (21.49)$$

where L is a length of the stick measured in the frame O . This implies that the man on firm ground judges that the stick has been shortened by $\sqrt{1 - v^2}$ (i.e., the Lorentz contraction).

In an opposite way, suppose that a straight stick of its length of L is laid along the x^1 -direction with its end at the origin of the frame O . Similarly to the above case, $x^1 = L$. If the length of the stick is read off at t' (or $x^0 = 0$) in the frame O' , from (21.41), i.e., the inverse matrix to (21.40), this time we have

$$L' = x'^1 = \sqrt{1 - v^2} x^1 = L \sqrt{1 - v^2} = L / \gamma,$$

where L' is a length of the stick measured in the frame O' . The Lorentz contraction is again observed.

Substituting $x^1 = L$ in (21.40) gives $x'^0 \neq 0$. This observation implies that even though measurements of either end of the stick have been done at the same time in the frame O , the coinstantaneity is lost with the frame O' .

The relateness of the time dilation can be dealt with in a similar manner. The discussion is left for readers.

In the above examples, we have shown the brief outline and constitution of the special theory of relativity. Since the theory is closely related to the Lorentz transformation, or more specifically the Lorentz group, we wish to survey the constitution and properties of the Lorentz group in more detail in Chap. 24.

21.3 Constitution and Solutions of the Dirac Equation

21.3.1 General Form of the Dirac Equation

To address the difficulty in the probabilistic interpretation of (21.9) mentioned in Sect. 21.1, Dirac postulated that the Klein–Gordon Eq. (21.11) could be “factorized” in such a way that

$$(\square + m^2)\phi(x) = (-i\gamma^\mu \partial_\mu - m)(i\gamma^\nu \partial_\nu - m)\phi(x) = 0, \quad (21.50)$$

where γ^μ denotes a certain constant mathematical quantity and ∂_μ is a shorthand notation of $\frac{\partial}{\partial x^\mu}$. The quantity ∂_μ behaves as a covariant vector (see Sect. 21.2). Remember that x represents the four-vector x^μ . Once again note that the same subscripts and superscripts should be summed. To show that ∂_μ is a covariant vector, we have

$$\partial_\nu \equiv \frac{\partial}{\partial x^\nu} = \frac{\partial x'^\mu}{\partial x^\nu} \frac{\partial}{\partial x'^\mu} = \frac{\partial \Lambda^\mu{}_\rho x^\rho}{\partial x^\nu} \frac{\partial}{\partial x'^\mu} = \Lambda^\mu{}_\rho \delta^\rho_\nu \frac{\partial}{\partial x'^\mu} = \Lambda^\mu{}_\nu \frac{\partial}{\partial x'^\mu} = \Lambda^\mu{}_\nu \partial'_\mu. \quad (21.51)$$

Multiplying (21.51) by $(\Lambda^{-1})^\nu{}_\kappa$, we have

$$\partial_\nu (\Lambda^{-1})^\nu{}_\kappa = \frac{\partial}{\partial x^\nu} (\Lambda^{-1})^\nu{}_\kappa = \Lambda^\mu{}_\nu \frac{\partial}{\partial x'^\mu} (\Lambda^{-1})^\nu{}_\kappa = \delta^\mu_\kappa \frac{\partial}{\partial x'^\mu} = \frac{\partial}{\partial x'^\kappa} = \partial'_\kappa. \quad (21.52)$$

Equation (21.52) shows that ∂_μ is transformed as in (21.25). That is, ∂_μ behaves as a covariant vector.

Meanwhile, defining

$$A \equiv i\gamma^\mu \partial_\mu,$$

we rewrite (21.50) as

$$(-A - m)(A - m)\phi(x) = 0. \quad (21.53)$$

This expression leads to a following equation:

$$(A - m)\phi(x) = 0. \quad (21.54)$$

Since $(-A - m)$ and $(A - m)$ are commutative, we have an alternative expression such that

$$(A - m)(-A - m)\phi(x) = 0. \quad (21.55)$$

This naturally produces

$$(-A - m)\phi(x) = 0. \quad (21.56)$$

Equations (21.54) and (21.56) both share a solid footing for detailed analysis of the Dirac equation. According to the custom, however, we solely use (21.54) for our present purposes. As a consequence of the fact that the original Klein–Gordon equation (21.11) is a second-order linear differential equation (SOLDE), we have

another linearly independent solution (see Chap. 10). Letting that solution be $\chi(x)$, we have

$$(A - m)\chi(x) = 0. \quad (21.57)$$

We give a full description of the Dirac equation as

$$(i\gamma^\mu \partial_\mu - m)\phi(x) = 0 \quad \text{and} \quad (i\gamma^\mu \partial_\mu - m)\chi(x) = 0. \quad (21.58)$$

As in the case of Example 1.1 of Sect. 1.3, we choose exponential functions for $\phi(x)$ and $\chi(x)$. More specifically, we assume the solutions described by

$$\phi(x) = u(p)e^{-ipx} \quad \text{and} \quad \chi(x) = v(p)e^{ipx}, \quad (21.59)$$

where $x = \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}$, $p = \begin{pmatrix} p^0 \\ p^1 \\ p^2 \\ p^3 \end{pmatrix}$. Also, remark that

$$px = p^\mu x_\mu = p_\mu x^\mu. \quad (21.60)$$

Notice that in (21.59) neither $u(p)$ nor $v(p)$ contains space–time coordinates. That is, $u(p)$ and $v(p)$ behave as a coefficient of the exponential function.

To determine the form of γ^μ in (21.50), we consider the cross term of $\gamma^\mu \partial_\mu \gamma^\nu \partial_\nu$ ($\mu \neq \nu$). We have

$$\gamma^\mu \partial_\mu \gamma^\nu \partial_\nu = \gamma^\mu \gamma^\nu \partial_\mu \partial_\nu = \gamma^\nu \gamma^\mu \partial_\nu \partial_\mu = \gamma^\nu \gamma^\mu \partial_\mu \partial_\nu \quad (\mu \neq \nu), \quad (21.61)$$

where the last equality was obtained by exchanging ∂_ν for ∂_μ . Therefore, the coefficient of $\partial_\mu \partial_\nu$ ($\mu \neq \nu$) is

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu.$$

Taking account of the fact that the cross terms $\partial_\mu \partial_\nu$ ($\mu \neq \nu$) are not present in the original Klein–Gordon equation (21.11), we must have

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 0 \quad (\mu \neq \nu). \quad (21.62)$$

To satisfy the relation (21.62), γ^μ must have a matrix form. With regard to the terms of ∂_μ^2 , comparing their coefficients with those of (21.11), we get

$$(\gamma^0)^2 = 1, (\gamma^1)^2 = (\gamma^2)^2 = (\gamma^3)^2 = -1. \quad (21.63)$$

Combining (21.62) and (21.63), we have

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2\eta^{\mu\nu}. \quad (21.64)$$

Introducing an anti-commutator

$$\{A, B\} \equiv AB + BA, \quad (21.65)$$

we obtain

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}. \quad (21.66)$$

The quantities γ^μ are represented by (4, 4) matrices and called the gamma matrices. Among several representations of the gamma matrices, the Dirac representation is frequently used. With this representation, we have [1, 2, 6]

$$\gamma^0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \gamma^k = \begin{pmatrix} 0 & \sigma_k \\ -\sigma_k & 0 \end{pmatrix} \quad (k = 1, 2, 3), \quad (21.67)$$

where each component denotes (2, 2) matrix; 1 shows a (2, 2) identity matrix; 0 shows a (2, 2) zero matrix; σ_k ($k = 1, 2, 3$) are the Pauli spin matrices given in (20.41). That is, as a full representation of the gamma matrices we have

$$\begin{aligned} \gamma^0 &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \\ \gamma^1 &= \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \gamma^2 = \begin{pmatrix} 0 & 0 & 0 & i \\ 0 & 0 & -i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}, \\ \gamma^3 &= \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}. \end{aligned} \quad (21.68)$$

Let us determine the solutions of the Dirac equation. Replacing $\phi(x)$ and $\chi(x)$ of (21.58) with those of (21.59), we get

$$(p_\mu \gamma^\mu - m)u(p) = 0 \quad \text{and} \quad (-p_\mu \gamma^\mu - m)v(p) = 0. \quad (21.69)$$

In accordance with (21.68) and (21.69), $u(p)$ and $v(p)$ are (4, 1) matrices, i.e., column vectors or strictly speaking *spinors* to be determined below. If we replaced p of $v(p)$ with $-p$, where

$$p = \begin{pmatrix} E \\ \mathbf{p} \end{pmatrix} = \begin{pmatrix} p_1^0 \\ p_2^0 \\ p_3^0 \end{pmatrix}, \quad -p = \begin{pmatrix} -E \\ -\mathbf{p} \end{pmatrix} = \begin{pmatrix} -p_1^0 \\ -p_2^0 \\ -p_3^0 \end{pmatrix}, \quad (21.70)$$

we would have

$$(p_\mu \gamma^\mu - m)v(-p) = 0. \quad (21.71)$$

Comparing (21.71) with the first equation of (21.69), we have

$$v(-p) = cu(p) \quad (c : \text{constant}). \quad (21.72)$$

So far, the relevant discussion looks parallel with that appeared in Example 1.1 of Sect. 1.3. However, the present case differs from the previous example in the following aspects: (1) Although boundary conditions (BCs) of Dirichlet type were imposed in the previous case, we are dealing with a plane wave solution with an infinite space–time extent. (2) In the previous case (Example 1.1), k could take both positive and negative numbers. In the present case, however, a negative p^0 does not fit naturally into our intuition, even though the Lorentz covariance permits a negative number for p^0 . This is because p^0 represents energy and unlike the momentum $\mathbf{p}$, p^0 does not have “directionality.”

Instead of considering p as the quantities changeable between $-\infty$ and $+\infty$, we single out p^0 and deal with it as a positive parameter ($0 < p^0 < +\infty$) with varying values depending on $|\mathbf{p}|$, while assuming that $\mathbf{p} = \begin{pmatrix} p^1 \\ p^2 \\ p^3 \end{pmatrix}$ are parameters freely changeable between $-\infty$ and $+\infty$. This treatment seems at first glance to violate the Lorentz invariance, but it is not the case. We will discuss the issue in Chap. 22. Thus, we safely assign p^0 (>0) and $-p^0$ (<0) to $u(p)$ and $v(p)$, respectively. Meanwhile, as shown in (1.40) we have a relationship between the time coordinate and energy described by

$$i\hbar \frac{\partial}{\partial t} = i \frac{\partial}{\partial t} = i \frac{\partial}{\partial x^0} \leftrightarrow E \equiv p^0. \quad (1.40)$$

Therefore, once we have defined $p^0 > 0$, $\phi(x) = u(p) e^{-ipx}$ of (21.58) and (21.59) must be assigned to the positive-energy state and $\chi(x) = v(p) e^{ipx}$ of (21.58) and (21.59) to the negative-energy state. To further specify the solutions of (21.69) we restate (21.69) as

$$(p_\mu \gamma^\mu - m)u(\mathbf{p}, h) = 0, \quad (21.73)$$

$$(-p_\mu \gamma^\mu - m)v(\mathbf{p}, h) = 0, \quad (21.74)$$

where (21.73) and (21.74) represent the positive-energy state and negative-energy state, respectively. The index h is called helicity and takes a value either $+1$ or -1 . Notice that we have abandoned $u(p)$ and $v(p)$ in favor of $u(\mathbf{p}, h)$ and $v(\mathbf{p}, h)$. We will give further consideration to the index h in the next section.

21.3.2 Plane Wave Solutions of the Dirac Equation

The Dirac equation is of special importance as a relativistic equation. The Dirac equation has a wide range of applications in physics including atomic physics that deals with the properties of hydrogen-like atoms (see Chap. 3) as one of the major subjects [2]. Among the applications of the Dirac equations, however, its plane wave solutions are most frequently dealt with for a reason mentioned below: In the field of high-energy physics, particles (e.g., electrons) need to be accelerated close to light velocity and in that case relativistic effects become increasingly dominant. What is more, the motion of particles is well approximated by a plane wave besides collisions of the particles that take place during a very short period of time and in a very small extent of space. In this book, we solely deal with the plane wave solutions accordingly.

First, we examine the properties for solutions of $u(\mathbf{p}, h)$ of (21.73). Using the gamma matrices given in (21.68), the matrix equation to be solved is described as

$$\begin{pmatrix} p^0 - m & 0 & p^3 & -p^1 - ip^2 \\ 0 & p^0 - m & -p^1 + ip^2 & -p^3 \\ -p^3 & p^1 + ip^2 & -p^0 - m & 0 \\ p^1 - ip^2 & p^3 & 0 & -p^0 - m \end{pmatrix} \begin{pmatrix} c^0 \\ c^1 \\ c^2 \\ c^3 \end{pmatrix} = 0, \quad (21.75)$$

where $\begin{pmatrix} c^0 \\ c^1 \\ c^2 \\ c^3 \end{pmatrix}$ is a column vector representation of $u(\mathbf{p}, h)$. That is, we have

$$u(\mathbf{p}, h) = \begin{pmatrix} c^0 \\ c^1 \\ c^2 \\ c^3 \end{pmatrix}. \quad (21.76)$$

We must have $u(\mathbf{p}, h) \neq 0$ to obtain a physically meaningful solution with $u(\mathbf{p}, h)$. For this, the determinant of the (4, 4) matrix of (21.75) should be zero. That is,

$$\begin{vmatrix} p^0 - m & 0 & p^3 & -p^1 - ip^2 \\ 0 & p^0 - m & -p^1 + ip^2 & -p^3 \\ -p^3 & p^1 + ip^2 & -p^0 - m & 0 \\ p^1 - ip^2 & p^3 & 0 & -p^0 - m \end{vmatrix} = 0. \quad (21.77)$$

This is a quartic equation with respect to p^μ . The equation leads to a simple form described by

$$\left[(p^0)^2 - \mathbf{p}^2 - m^2 \right]^2 = 0. \quad (21.78)$$

Hence, we have

$$(p^0)^2 - \mathbf{p}^2 - m^2 = 0 \quad \text{or} \quad p^0 = \sqrt{\mathbf{p}^2 + m^2}. \quad (21.79)$$

Remember that p^0 represents the positive energy. Note that (21.79) is consistent with (1.8), (1.9), and (1.20); recall the natural units. The result, however, has already been implied in (21.2). Thus, we have to find another way to solve (21.75). To this end, first let us consider (21.75) in a simpler form. That is, putting $p^1 = p^2 = 0$ in (21.75), (21.75) reads as

$$\begin{pmatrix} p^0 - m & 0 & p^3 & 0 \\ 0 & p^0 - m & 0 & -p^3 \\ -p^3 & 0 & -p^0 - m & 0 \\ 0 & p^3 & 0 & -p^0 - m \end{pmatrix} \begin{pmatrix} c^0 \\ c^1 \\ c^2 \\ c^3 \end{pmatrix} = 0. \quad (21.80)$$

Clearly, the first and second column vectors are linearly independent with each other. Multiplying the first column by $(p^0 + m)$ and the third column by p^3 produces the same column vectors. This implies that the first and third column vectors are linearly dependent with each other. In a similar manner, the second and fourth column vectors are linearly dependent as well. Thus, the rank of the matrix of (21.80) is two. If $p^3 = 0$ in (21.80), i.e., $\mathbf{p} = \mathbf{0}$, from (21.79) we have $p^0 = m (>0)$. Then, (21.80) can be rewritten as

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -2m & 0 \\ 0 & 0 & 0 & -2m \end{pmatrix} \begin{pmatrix} c^0 \\ c^1 \\ c^2 \\ c^3 \end{pmatrix} = 0. \quad (21.81)$$

Equation (21.81) gives the solution of an electron at rest. We will come back to this simple situation in Sect. 24.5. The rank of the matrix of (21.81) is two. In a general case of (21.75) the rank of the matrix is two as well. This means the presence of two linearly independent solutions for (21.75). The index h ($h = \pm 1$) of $u(\mathbf{p}, h)$

represents and distinguishes these two solutions. The situation is the same as the negative-energy solutions of $v(\mathbf{p}, h)$ given in (21.74); see Sect. 21.3.3.

Now, (21.75) can be rewritten in a succinct form as

$$\begin{pmatrix} p^0 - m & -\boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & -p^0 - m \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{pmatrix} = 0. \quad (21.82)$$

Here, we introduce another (4, 4) matrix

$$\mathfrak{H} \equiv \frac{1}{|\mathbf{p}|} \begin{pmatrix} \boldsymbol{\sigma} \cdot \mathbf{p} & 0 \\ 0 & \boldsymbol{\sigma} \cdot \mathbf{p} \end{pmatrix}. \quad (21.83)$$

Since $\boldsymbol{\sigma} \cdot \mathbf{p}$ is Hermitian, $\mathfrak{H}$ is Hermitian as well. Since an Hermitian matrix can be diagonalized by unitary similarity transformation (see Theorem 14.5 of Sect. 14.3), so can both $\boldsymbol{\sigma} \cdot \mathbf{p}$ and $\mathfrak{H}$ be. Let χ be a (2, 1) matrix, i.e., a column vector. As an eigenfunction of (21.83), we assume a following type of (4, 1) matrix:

$$\mathbf{X} \equiv \begin{pmatrix} \chi \\ s\chi \end{pmatrix}, \quad (21.84)$$

where s is a parameter to be determined below. To confirm whether this assumption is valid, let us examine the properties of the following (2, 2) submatrix $\mathfrak{S}$, which is described by

$$\mathfrak{S} \equiv \frac{1}{|\mathbf{p}|} \boldsymbol{\sigma} \cdot \mathbf{p}. \quad (21.85)$$

The factor $\boldsymbol{\sigma} \cdot \mathbf{p}$ is contained in common in the matrix of LHS for (21.82) and $\mathfrak{H}$ of (21.83).

Using the notation of (21.85), we have

$$\mathfrak{H} = \begin{pmatrix} \mathfrak{S} & 0 \\ 0 & \mathfrak{S} \end{pmatrix} = \frac{1}{|\mathbf{p}|} \begin{pmatrix} \boldsymbol{\sigma} \cdot \mathbf{p} & 0 \\ 0 & \boldsymbol{\sigma} \cdot \mathbf{p} \end{pmatrix}. \quad (21.86)$$

We can readily check the unitarity of $\mathfrak{S}$, i.e.,

$$\mathfrak{S}^\dagger \mathfrak{S} = E_2,$$

where E_2 is a (2, 2) identity matrix. Also, we have

$$\det \mathfrak{S} = -1. \quad (21.87)$$

Since the (2, 2) matrix $\mathfrak{S}$ is both Hermitian and unitary, (21.87) immediately implies that the eigenvalues of $\mathfrak{S}$ are ± 1 ; see Chaps. 12 and 14. Representing the

two linearly independent functions (i.e., eigenfunctions belonging to the eigenvalues of $\mathfrak{S}$ of ± 1) as χ_- and χ_+ , we have

$$\mathfrak{S}\chi_- = (-1)\chi_- \text{ and } \mathfrak{S}\chi_+ = (+1)\chi_+ \quad (21.88)$$

where χ_- belongs to the eigenvalue -1 and χ_+ belongs to the eigenvalue $+1$ with respect to the positive-energy solutions.

When we do not specify the eigenvalue, we write

$$\mathfrak{S}\chi = h\chi \quad (h = \pm 1), \quad (21.89)$$

where h is said to be a *helicity*. Thus, we get

$$\mathfrak{H}X = \begin{pmatrix} \mathfrak{S} & 0 \\ 0 & \mathfrak{S} \end{pmatrix} \begin{pmatrix} \chi \\ s\chi \end{pmatrix} = \begin{pmatrix} \mathfrak{S}\chi \\ s\mathfrak{S}\chi \end{pmatrix} = h \begin{pmatrix} \chi \\ s\chi \end{pmatrix} = hX. \quad (21.90)$$

That is, we certainly find that X of (21.84) is an eigenvector of $\mathfrak{H}$ that belongs to an eigenvalue h .

Meanwhile, defining the matrix of (21.82) as

$$\mathfrak{G} \equiv \begin{pmatrix} p^0 - m & -\sigma \mathbf{p} \\ \sigma \mathbf{p} & -p^0 - m \end{pmatrix} = p_\mu \gamma^\mu - m, \quad (21.91)$$

we have

$$\mathfrak{G}u(\mathbf{p}, h) = 0 \quad (21.92)$$

as the equation to be solved. In (21.92) $u(\mathbf{p}, h)$ indicates a $(4, 1)$ column vector of (21.76) or (21.82). We find

$$[\mathfrak{G}, \mathfrak{H}] = 0. \quad (21.93)$$

That is, $\mathfrak{G}$ and $\mathfrak{H}$ commute with each other. Note that (21.93) results naturally from the presence of gamma matrices. We remember that two Hermitian matrices commute if and only if there exists a complete orthonormal set (CONS) of common eigenfunctions (Theorem 14.14). This is *almost* true of $\mathfrak{G}$ and $\mathfrak{H}$. Care should be taken, however, to apply Theorem 14.14 to the present case. This is because $\mathfrak{G}$ is *neither* a normal matrix *nor* an Hermitian matrix except for special cases. Therefore, the matrix $\mathfrak{G}$ is in general impossible to diagonalize by the unitary similarity transformation (Theorem 14.5). From the point of view of the matrix algebra, it is of great interest whether or not $\mathfrak{G}$ can be diagonalized. Moreover, if $\mathfrak{G}$ could be diagonalize, it is important to know what kind of non-singular matrix is used for the similarity transformation (i.e., diagonalization) and what eigenvalues $\mathfrak{G}$ possesses. Yet, (21.92) immediately tells us that $\mathfrak{G}$ possesses an eigenvalue zero. Otherwise, we

are to have $u(\mathbf{p}, h) \equiv 0$. We will come back to this intriguing discussion in Chap. 24 to examine the condition for $\mathfrak{G}$ to be diagonalized.

Here, we examine what kind of eigenfunctions $\mathfrak{G}$ and $\mathfrak{H}$ share as the common eigenvectors. Since $\mathfrak{G}$ and $\mathfrak{H}$ commute, from (21.84) and (21.90) we have

$$\mathfrak{G}\mathfrak{H}X = \mathfrak{H}\mathfrak{G}X = h\mathfrak{G}X. \quad (21.94)$$

This means that $\mathfrak{G}X$ is an eigenvector of $\mathfrak{H}$. Let us tentatively consider that $\mathfrak{G}X$ can be described by

$$\mathfrak{G}X = cX, \quad (21.95)$$

where c is a constant (certainly including zero). This implies that X is an eigenfunction of $\mathfrak{G}$ and that from (21.90) and (21.95) $\mathfrak{G}$ and $\mathfrak{H}$ share a common eigenfunction X .

Using the above results and notations, let us examine the conditions under which $\begin{pmatrix} \chi \\ s\chi \end{pmatrix}$ is a solution of (21.75). Replacing the column vector of (21.75) with $\begin{pmatrix} \chi \\ s\chi \end{pmatrix}$, we have

$$\begin{aligned} \mathfrak{G}X &= \mathfrak{G} \begin{pmatrix} \chi \\ s\chi \end{pmatrix} = \begin{pmatrix} p^0 - m & -\boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & -p^0 - m \end{pmatrix} \begin{pmatrix} \chi \\ s\chi \end{pmatrix} = \begin{pmatrix} p^0 - m & -\mathcal{P}\mathfrak{S} \\ \mathcal{P}\mathfrak{S} & -p^0 - m \end{pmatrix} \begin{pmatrix} \chi \\ s\chi \end{pmatrix} \\ &= \begin{pmatrix} (p^0 - m)\chi - \mathcal{P}s\mathfrak{S}\chi \\ \mathcal{P}\mathfrak{S}\chi - (p^0 + m)s\chi \end{pmatrix} = \begin{pmatrix} (p^0 - m)\chi - \mathcal{P}sh\chi \\ \mathcal{P}h\chi - (p^0 + m)s\chi \end{pmatrix} = 0, \end{aligned} \quad (21.96)$$

where we define a positive number $\mathcal{P}$ such that

$$\mathcal{P} \equiv |\mathbf{p}|. \quad (21.97)$$

For (21.96) to be satisfied, namely, for $\begin{pmatrix} \chi \\ s\chi \end{pmatrix}$ to be an eigenvector of $\mathfrak{G}$ that belongs to the eigenvalue zero, the coefficient of χ of each row for (21.96) must vanish. That is, we must have

$$\begin{aligned} (p^0 - m) - \mathcal{P}sh &= 0, \\ \mathcal{P}h - (p^0 + m)s &= 0. \end{aligned}$$

Then, if we postulate the following relations:

$$s = \frac{\mathcal{P}h}{p^0 + m} = \frac{p^0 - m}{\mathcal{P}h}, \quad (21.98)$$

(21.96) will be satisfied. Equation (21.98) implies that

$$(\mathcal{P}h)^2 = \mathcal{P}^2 h^2 = \mathcal{P}^2 = \mathbf{p}^2 = (p^0 + m)(p^0 - m) = (p^0)^2 - m^2.$$

However, this is equivalent to (21.79). Thus, we have confirmed that $X = \begin{pmatrix} \chi \\ s\chi \end{pmatrix}$ of (21.84) is certainly an eigenfunction of $\mathfrak{G}$, if we choose s as $s = \mathcal{P}h/(p^0 + m)$. At the same time, we find that $X = \begin{pmatrix} \chi \\ s\chi \end{pmatrix}$ is indeed the common eigenfunction shared by $\mathfrak{G}$ and $\mathfrak{H}$ with a definite value s designated by (21.98). This eigenfunction belongs to the eigenvalue h ($h = \pm 1$) of $\mathfrak{H}$ and the eigenvalue 0 of $\mathfrak{G}$. It will be convenient for a purpose of future reference to define a positive number S as

$$S \equiv \frac{\mathcal{P}}{p^0 + m} \quad (21.99)$$

so that we have $s = hS$. The eigenfunction of (21.96) is then expressed as

$$X = \begin{pmatrix} \chi \\ hS\chi \end{pmatrix}. \quad (21.100)$$

Hence, as $u(\mathbf{p}, h)$ given in (21.76) we obtain

$$u(\mathbf{p}, h) = N \begin{pmatrix} \chi \\ hS\chi \end{pmatrix}, \quad (21.101)$$

where N is a normalization constant that will be determined later in Sect. 21.4.

To solve the Dirac equation in an actual form, we wish to choose the polar coordinate representation (see Sect. 3.2). For this, we put

$$\left. \begin{aligned} p^1 &= \mathcal{P} \sin \theta \cos \phi, \\ p^2 &= \mathcal{P} \sin \theta \sin \phi, \\ p^3 &= \mathcal{P} \cos \theta. \end{aligned} \right\}$$

Then, we have

$$\mathfrak{S} = \begin{pmatrix} -\cos \theta & e^{i\phi} \sin \theta \\ e^{-i\phi} \sin \theta & \cos \theta \end{pmatrix}. \quad (21.102)$$

Note that $\det \mathfrak{S} = -1$. Solving the eigenvalue equation (21.89) according to Sects. 12.1 and 14.3, we get normalized eigenfunctions described by

$$\chi_- = \begin{pmatrix} e^{i\phi/2} \cos \frac{\theta}{2} \\ -e^{-i\phi/2} \sin \frac{\theta}{2} \end{pmatrix} \text{ and } \chi_+ = \begin{pmatrix} e^{i\phi/2} \sin \frac{\theta}{2} \\ e^{-i\phi/2} \cos \frac{\theta}{2} \end{pmatrix} \quad (21.103)$$

where χ_- and χ_+ are the eigenfunctions of the operator $\mathfrak{S}$ belonging to the helicity $h = -1$ and $+1$, respectively. Then, as the positive-energy solution of (21.73) we have

$$u(\mathbf{p}, -1) = N \begin{pmatrix} e^{i\phi/2} \cos \frac{\theta}{2} \\ -e^{-i\phi/2} \sin \frac{\theta}{2} \\ -Se^{i\phi/2} \cos \frac{\theta}{2} \\ Se^{-i\phi/2} \sin \frac{\theta}{2} \end{pmatrix}, u(\mathbf{p}, +1) = N \begin{pmatrix} e^{i\phi/2} \sin \frac{\theta}{2} \\ e^{-i\phi/2} \cos \frac{\theta}{2} \\ Se^{i\phi/2} \sin \frac{\theta}{2} \\ Se^{-i\phi/2} \cos \frac{\theta}{2} \end{pmatrix}. \quad (21.104)$$

Since the rank of $\mathfrak{S}$ is two, any positive-energy solution can be expressed as a linear combination of $u(\mathbf{p}, -1)$ and $u(\mathbf{p}, +1)$.

21.3.3 Negative-Energy Solution of the Dirac Equation

Using (21.74), with the negative-energy state $v(\mathbf{p}, h)$ we have the Dirac equation expressed as

$$(-p_\mu \gamma^\mu - m)v(\mathbf{p}, h) = \begin{pmatrix} -p^0 - m & -\boldsymbol{\sigma} \cdot (-\mathbf{p}) \\ \boldsymbol{\sigma} \cdot (-\mathbf{p}) & p^0 - m \end{pmatrix} \begin{pmatrix} \tilde{\chi} \\ \tilde{\chi} \end{pmatrix} = 0. \quad (21.105)$$

In accordance with (21.84) we put

$$\tilde{X} = \begin{pmatrix} \tilde{\chi} \\ \tilde{\chi} \end{pmatrix}. \quad (21.106)$$

Meanwhile, defining $\tilde{\mathfrak{S}}$ as

$$\tilde{\mathfrak{S}} \equiv \frac{1}{|-\mathbf{p}|} \boldsymbol{\sigma} \cdot (-\mathbf{p}) = \frac{1}{p} \boldsymbol{\sigma} \cdot (-\mathbf{p}) = -\mathfrak{S} \quad (21.107)$$

and defining $\tilde{\mathfrak{G}}$ as

$$\tilde{\mathfrak{G}} \equiv -p_\mu \gamma^\mu - m, \quad (21.108)$$

we rewrite (21.105) as

$$\tilde{\mathfrak{S}} \begin{pmatrix} \tilde{\chi} \\ t\tilde{\chi} \end{pmatrix} = \begin{pmatrix} -p^0 - m & -\mathcal{P}\tilde{\mathfrak{S}} \\ \mathcal{P}\tilde{\mathfrak{S}} & p^0 - m \end{pmatrix} \begin{pmatrix} \tilde{\chi} \\ t\tilde{\chi} \end{pmatrix} = 0. \quad (21.109)$$

As in the case of (21.96), we get

$$\tilde{\mathfrak{S}} \begin{pmatrix} \tilde{\chi} \\ t\tilde{\chi} \end{pmatrix} = \begin{pmatrix} -(p^0 + m)\tilde{\chi} - \mathcal{P}t\tilde{\mathfrak{S}}\tilde{\chi} \\ \mathcal{P}\tilde{\mathfrak{S}}\tilde{\chi} + (p^0 - m)t\tilde{\chi} \end{pmatrix} = \begin{pmatrix} -(p^0 + m)\tilde{\chi} - \mathcal{P}th\tilde{\chi} \\ \mathcal{P}h\tilde{\chi} + (p^0 - m)t\tilde{\chi} \end{pmatrix} = 0, \quad (21.110)$$

where we put

$$\tilde{\mathfrak{S}}\tilde{\chi} = h\tilde{\chi}.$$

For (21.110) to be satisfied, we have

$$-(p^0 + m) - \mathcal{P}ht = 0, \quad (21.111)$$

$$h\mathcal{P} + (p^0 - m)t = 0. \quad (21.112)$$

Then, we get

$$t = -\frac{p^0 + m}{\mathcal{P}h} = -\frac{\mathcal{P}h}{p^0 - m}. \quad (21.113)$$

Again, (21.113) is equivalent to (21.79). Hence, (21.106) is represented as

$$\begin{pmatrix} \tilde{\chi} \\ t\tilde{\chi} \end{pmatrix} = \frac{p^0 + m}{\mathcal{P}h} \begin{pmatrix} \mathcal{P}h \\ p^0 + m \end{pmatrix} \tilde{\chi} = \frac{p^0 + m}{\mathcal{P}h} \begin{pmatrix} h\mathcal{S}\tilde{\chi} \\ -\tilde{\chi} \end{pmatrix} = \frac{1}{\mathcal{S}} \begin{pmatrix} \mathcal{S}\tilde{\chi} \\ -h\tilde{\chi} \end{pmatrix}, \quad (21.114)$$

where with the last equality the relations $h^2 = 1$ and (21.99) were used. As a consequence of switching the operator $\mathfrak{S}$ to $\tilde{\mathfrak{S}}$, the eigenvalue equation (21.89) is replaced with

$$\tilde{\mathfrak{S}}\tilde{\chi} = h\tilde{\chi} = -\mathfrak{S}\tilde{\chi}. \quad (21.115)$$

Defining $\tilde{\mathfrak{H}}$ as in the case of (21.86) such that

$$\tilde{\mathfrak{H}} \equiv \begin{pmatrix} \tilde{\mathfrak{S}} & 0 \\ 0 & \tilde{\mathfrak{S}} \end{pmatrix} = -\mathfrak{H} = \frac{1}{|\mathbf{p}|} \begin{pmatrix} -\sigma_{\mathbf{p}} & 0 \\ 0 & -\sigma_{\mathbf{p}} \end{pmatrix}, \quad (21.116)$$

we get

$$\tilde{\mathfrak{H}} \begin{pmatrix} s\tilde{\chi} \\ -h\tilde{\chi} \end{pmatrix} = \begin{pmatrix} \tilde{\mathfrak{S}} & 0 \\ 0 & \tilde{\mathfrak{S}} \end{pmatrix} \begin{pmatrix} s\tilde{\chi} \\ -h\tilde{\chi} \end{pmatrix} = \begin{pmatrix} s\tilde{\mathfrak{S}}\tilde{\chi} \\ -h\tilde{\mathfrak{S}}\tilde{\chi} \end{pmatrix} = \begin{pmatrix} sh\tilde{\chi} \\ -h^2\tilde{\chi} \end{pmatrix} = h \begin{pmatrix} s\tilde{\chi} \\ -h\tilde{\chi} \end{pmatrix}. \quad (21.117)$$

From (21.114) and (21.117), as the negative-energy solution $v(\mathbf{p}, h)$ of (21.105) we get

$$v(\mathbf{p}, h) = \tilde{N} \begin{pmatrix} s\tilde{\chi} \\ -h\tilde{\chi} \end{pmatrix}, \quad (21.118)$$

where $\tilde{N}$ is a normalization constant to be determined later.

From (21.115), the negative-energy state corresponding to the helicity eigenvalue $+h$ of $\tilde{\mathfrak{H}}$ should be translated into the positive-energy state corresponding to the eigenvalue $-h$ of $\mathfrak{H}$. Borrowing the results of (21.103), with the normalized eigenfunctions we have

$$\tilde{\chi}_+ = \begin{pmatrix} -e^{i\phi/2} \cos \frac{\theta}{2} \\ e^{-i\phi/2} \sin \frac{\theta}{2} \end{pmatrix} \quad \text{and} \quad \tilde{\chi}_- = \begin{pmatrix} -e^{i\phi/2} \sin \frac{\theta}{2} \\ -e^{-i\phi/2} \cos \frac{\theta}{2} \end{pmatrix}, \quad (21.119)$$

where $\tilde{\chi}_+$ belongs to the helicity eigenvalue $h = +1$ of $\tilde{\mathfrak{S}}$ and $\tilde{\chi}_-$ belongs to the helicity $h = -1$ of $\tilde{\mathfrak{S}}$ with respect to the negative-energy solutions. Thus, using (21.114) we obtain

$$v(\mathbf{p}, +1) = \tilde{N} \begin{pmatrix} -Se^{i\phi/2} \cos \frac{\theta}{2} \\ Se^{-i\phi/2} \sin \frac{\theta}{2} \\ e^{i\phi/2} \cos \frac{\theta}{2} \\ -e^{-i\phi/2} \sin \frac{\theta}{2} \end{pmatrix}, \quad v(\mathbf{p}, -1) = \tilde{N} \begin{pmatrix} -Se^{i\phi/2} \sin \frac{\theta}{2} \\ -Se^{-i\phi/2} \cos \frac{\theta}{2} \\ -e^{i\phi/2} \sin \frac{\theta}{2} \\ -e^{-i\phi/2} \cos \frac{\theta}{2} \end{pmatrix}. \quad (21.120)$$

The column vectors of (21.104) and (21.120) expressed as $(4, 1)$ matrices are called the Dirac spinors. In both the cases, we retain the freedom to choose a phase factor, because it does not alter the quantum state. Strictly speaking, the factor $e^{\pm i p x}$ of (21.59) is a phase factor, but this factor is invariant under the Lorentz transformation. Ignoring the phase factor, with the second equation of (21.120) we may choose

$$v(\mathbf{p}, -1) = \tilde{N} \begin{pmatrix} Se^{i\phi/2} \sin \frac{\theta}{2} \\ Se^{-i\phi/2} \cos \frac{\theta}{2} \\ e^{i\phi/2} \sin \frac{\theta}{2} \\ e^{-i\phi/2} \cos \frac{\theta}{2} \end{pmatrix}. \quad (21.121)$$

We will revisit this issue in relation to the Dirac adjoint and the charge conjugation (vide infra).

Since the rank of $\tilde{\mathfrak{G}}$ is two, any negative-energy solution can be expressed as a linear combination of $v(\mathbf{p}, +1)$ and $v(\mathbf{p}, -1)$ as in the case of the positive-energy solutions. In the description of the Dirac spinor, a following tangible example helps further realize what is the point.

Example 21.4 In Sect. 21.3.1 we have mentioned that the parameter $\mathbf{p}$ can be freely chosen between $-\infty$ and $+\infty$ in the three-dimensional space. Then, we wish to examine how the Dirac spinor can be described by switching the momentum from $\mathbf{p}$ to $-\mathbf{p}$. Choosing $u(\mathbf{p}, -1)$ for the Dirac spinor, for instance, we had

$$u(\mathbf{p}, -1) = N \begin{pmatrix} e^{i\phi/2} \cos \frac{\theta}{2} \\ -e^{-i\phi/2} \sin \frac{\theta}{2} \\ -Se^{i\phi/2} \cos \frac{\theta}{2} \\ Se^{-i\phi/2} \sin \frac{\theta}{2} \end{pmatrix}. \quad (21.104)$$

Corresponding to $\mathbf{p} \rightarrow -\mathbf{p}$, we have

$$(\phi, \theta) \rightarrow (\phi \pm \pi, \pi - \theta); \quad (20.338)$$

see Sect. 20.5.3. Upon the replacement of (20.338), $u(\mathbf{p}, -1)$ is converted to

$$\begin{aligned}
u'(-\mathbf{p}, +1) &= N \begin{pmatrix} (\pm i)e^{\frac{i\phi}{2}} \sin \frac{\theta}{2} \\ (\pm i)e^{-\frac{i\phi}{2}} \cos \frac{\theta}{2} \\ (\pm i)Se^{\frac{i\phi}{2}} \sin \frac{\theta}{2} \\ (\pm i)Se^{-\frac{i\phi}{2}} \cos \frac{\theta}{2} \end{pmatrix} = (\pm i)N \begin{pmatrix} e^{\frac{i\phi}{2}} \sin \frac{\theta}{2} \\ e^{-\frac{i\phi}{2}} \cos \frac{\theta}{2} \\ Se^{\frac{i\phi}{2}} \sin \frac{\theta}{2} \\ Se^{-\frac{i\phi}{2}} \cos \frac{\theta}{2} \end{pmatrix} \\
&= (\pm i)u(\mathbf{p}, +1),
\end{aligned} \tag{21.122}$$

where u' denotes the change in the functional form; the helicity has been switched from -1 to $+1$ according to (21.89). With the above switching, we have

$$u(\mathbf{p}, -1) \rightarrow (\pm i)u(\mathbf{p}, 1) \tag{21.123}$$

with the energy unaltered. For $u(\mathbf{p}, +1)$, similarly we have

$$u(\mathbf{p}, +1) \rightarrow (\pm i)u(\mathbf{p}, -1). \tag{21.124}$$

With the negative-energy solutions, we obtain a similar result. Then, aside from the phase factor, the momentum switching produces

$$\chi_- \leftrightarrow \chi_+; \quad \tilde{\chi}_+ \leftrightarrow \tilde{\chi}_-; \quad h_- \leftrightarrow h_+,$$

where h_- and h_+ stand for the helicity -1 and $+1$, respectively.

Although in the non-relativistic quantum mechanics (i.e., the Schrödinger equation) the helicity was not an important dynamical variable, it plays an essential role as a conserved quantity in the relativistic Dirac equation. In a quantum-mechanical system, the total angular momentum ($\mathbf{A}$) containing orbital angular momentum ($\mathbf{L}$) and spin angular momentum ($\boldsymbol{\sigma}$) is a conserved quantity. That is, we have

$$\mathbf{A} = \mathbf{L} + \boldsymbol{\sigma}. \tag{21.125}$$

Taking an inner product of (21.125) with $\mathbf{p}$, we obtain

$$\mathbf{A} \cdot \mathbf{p} = \mathbf{L} \cdot \mathbf{p} + \boldsymbol{\sigma} \cdot \mathbf{p}. \tag{21.126}$$

If the angular momentum is measured in the direction of motion (i.e., the direction parallel to $\mathbf{p}$), $\mathbf{L} \cdot \mathbf{p} = 0$. It is because in this situation we can represent

$$\mathbf{L} \cdot \mathbf{p} = (\mathbf{x} \times \mathbf{p}) \cdot \mathbf{p} = \mathbf{0},$$

where $\mathbf{x}$ is a position vector. Thus, we get

$$\mathbf{A} \cdot \mathbf{p} = \boldsymbol{\sigma} \cdot \mathbf{p}. \quad (21.127)$$

Since we are dealing with the free field, we have

$$\mathbf{A} \cdot \mathbf{p} = \boldsymbol{\sigma} \cdot \mathbf{p} = c, \quad (21.128)$$

where c is a constant. That is, the component of the total angular momentum in the direction of motion is conserved and equal to the component of the spin angular momentum in the direction of motion.

Getting back to the definition of $\mathfrak{S}$, we may define it as

$$\mathfrak{S} \equiv \lim_{|\mathbf{p}| \rightarrow 0} \frac{1}{|\mathbf{p}|} \boldsymbol{\sigma} \cdot \mathbf{p}. \quad (21.129)$$

Equation (21.129) implies that $\mathfrak{S}$ can be defined even with an infinitesimally small amount of $|\mathbf{p}|$. We remark that we can observe the motion of the electron from various inertial frames of reference. Depending upon the choice of the inertial frame, we may have varying $\mathbf{p}$ (both of varying directions and magnitudes). In particular, small changes in $\mathbf{p}$ may lead to a positive or a negative with $\boldsymbol{\sigma} \cdot \mathbf{p}$ and the sudden jump in the helicity between ± 1 accordingly. This seems somewhat unnatural. Or rather, it may be natural to assume that an electron inherently carries spin regardless of its movement form (either moving or resting).

Even though in (21.129) $\mathfrak{S}$ seemed to be described as an indeterminate form of limit at $\mathbf{p} \rightarrow \mathbf{0}$, it was defined in (21.102) actually, regardless of the magnitude of momentum $|\mathbf{p}|$. Or rather, it is preferable to assume that the angles ϕ and θ represent the geometric transformation of the basis vectors. We will come back to this point later in Chap. 24.

Readers may well wonder whether we defined an operator $\begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix}$ instead of $\mathfrak{H}$ defined in (21.83). Such an operator, however, would be non-commutative with H_p defined just below (see Sect. 21.3.4). This aspect warrants that the projection of the spin operator $\boldsymbol{\sigma}$ onto the direction of motion should be taken into account. For this reason, from here on, we will not distinguish the spin operator from the helicity operator $\mathfrak{H}$ and $\tilde{\mathfrak{H}}$ when we are thinking of the plane wave solutions of the Dirac equation.

21.3.4 One-Particle Hamiltonian of the Dirac Equation

In this section, we wish to think of how we can specify the quantum state of the plane wave of an electron in the relativistic Dirac field. We know from Chaps. 3 and 14 that it is important to specify a set of mutually commutative physical quantities (i.e., Hermitian operators) to characterize the physical system. Normally, it will suffice to

describe the Hamiltonian of the system and seek the operators that commute with the Hamiltonian.

First, let us think of the one-particle Hamiltonian H_p for the plane wave with the positive energy. It is given by [2]

$$H_p = \gamma^0(\boldsymbol{\gamma} \cdot \mathbf{p} + m). \quad (21.130)$$

The derivation of (21.130) will be performed in Chap. 22 in relation to the quantization of fields. As a matrix representation, we have

$$H_p = \begin{pmatrix} m & \boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & -m \end{pmatrix}. \quad (21.131)$$

Hence, operating H_p on $\begin{pmatrix} \chi \\ hS\chi \end{pmatrix}$, we have

$$H_p \begin{pmatrix} \chi \\ hS\chi \end{pmatrix} = \begin{pmatrix} m & \boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & -m \end{pmatrix} \begin{pmatrix} \chi \\ hS\chi \end{pmatrix} = \begin{pmatrix} m\chi + \boldsymbol{\sigma} \cdot \mathbf{p} hS\chi \\ \boldsymbol{\sigma} \cdot \mathbf{p} \chi - mhS\chi \end{pmatrix} = \begin{pmatrix} p^0 \chi \\ hSp^0 \chi \end{pmatrix} = p^0 \begin{pmatrix} \chi \\ hS\chi \end{pmatrix}. \quad (21.132)$$

For the negative-energy solution, in turn, similarly operating H_{-p} on

$$X' = \begin{pmatrix} S\tilde{\chi} \\ -h\tilde{\chi} \end{pmatrix}, \quad (21.133)$$

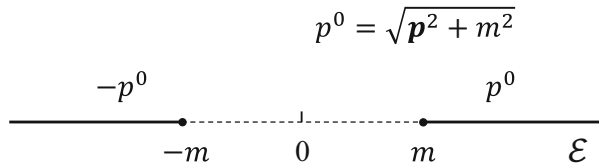
we have

$$H_{-p} \begin{pmatrix} S\tilde{\chi} \\ -h\tilde{\chi} \end{pmatrix} = \begin{pmatrix} m & -\boldsymbol{\sigma} \cdot \mathbf{p} \\ -\boldsymbol{\sigma} \cdot \mathbf{p} & -m \end{pmatrix} \begin{pmatrix} S\tilde{\chi} \\ -h\tilde{\chi} \end{pmatrix} = \begin{pmatrix} mS\tilde{\chi} - \boldsymbol{\sigma} \cdot \mathbf{p} h\tilde{\chi} \\ \boldsymbol{\sigma} \cdot \mathbf{p} S\tilde{\chi} - mh\tilde{\chi} \end{pmatrix} = \begin{pmatrix} -Sp^0 \tilde{\chi} \\ hp^0 \tilde{\chi} \end{pmatrix} = -p^0 \begin{pmatrix} S\tilde{\chi} \\ -h\tilde{\chi} \end{pmatrix}. \quad (21.134)$$

Thus, as expected we have reproduced a positive energy p^0 and negative energy $-p^0$, to which $u(\mathbf{p}, h)$ and $v(\mathbf{p}, h)$ belong to, respectively. The spectra of H_p and H_{-p} are characterized by the continuous spectra as shown in Fig. 21.2. Note that energies $\mathcal{E}$ ranging $-m < \mathcal{E} < m$ are forbidden.

We can readily check that H_p and $\mathfrak{H}$ are commutative with each other. So are H_{-p} and $\mathfrak{H}$. Thus, we find that the Dirac spinors $u(\mathbf{p}, h)$ and $v(\mathbf{p}, h)$ determined in (21.101)

Fig. 21.2 Continuous spectra of H_p and H_{-p} . Energies $\mathcal{E}$ ranging $-m < \mathcal{E} < m$ are forbidden



and (21.118) are the simultaneous eigenstates of the Hamiltonian and Hermitian helicity operator of the Dirac field. Naturally, these Dirac spinors constitute the complete set of the solutions of the Dirac equation. In Table 21.1 we summarize the matrix form of the Hamiltonian and helicity operators relevant to the Dirac field together with their simultaneous eigenstates and their eigenvalues.

It is worth comparing the Dirac equation with the Schrödinger equation from the point of view of an eigenvalue problem. An energy eigenvalue equation based on the Schrödinger equation is usually described as

$$H\phi(\mathbf{x}) = E\phi(\mathbf{x}), \quad (1.55)$$

where H denotes the Hamiltonian. In a hydrogen atom, the total angular momentum operator L^2 is commutative with H ; see (3.15). Since L^2 (or M^2) and L_z (or M_z) have simultaneously been diagonalized, L_z is commutative with H ; see (3.158) and (3.159) as well. Since all H , L^2 , and L_z are Hermitian operators, there must exist a complete orthonormal set of common eigenvectors (Theorem 14.14). Thus, with the hydrogen atom, the solution $\phi(\mathbf{x})$ is represented by (3.300), i.e., $Y_l^m(\theta, \phi)\tilde{R}_l^{(n)}(r)$.

As compared with the Schrödinger equation, the eigenvalue equation of the Dirac equation is given by

$$\mathfrak{G}u(\mathbf{p}, h) = 0 \quad \text{or} \quad \tilde{\mathfrak{G}}v(\mathbf{p}, h) = 0.$$

In other words, we are seeking an eigenfunction $u(\mathbf{p}, h)$ or $v(\mathbf{p}, h)$ that belongs to an eigenvalue of zero.

Since $H_{\mathbf{p}}$ and $\mathfrak{H}$ (or $H_{-\mathbf{p}}$ and $\tilde{\mathfrak{H}}$) are commutative Hermitian operators, according to Theorem 14.14 there must exist a CONS of common eigenvectors as well. Nonetheless, $H_{\pm\mathbf{p}}$ is *not* commutative with $\mathfrak{G}$ or $\tilde{\mathfrak{G}}$ in general. Therefore, the said CONS cannot be a solution of the Dirac equation as an exceptional case, however, $H_{\pm\mathbf{p}}$ is commutative with $\mathfrak{G}$ and $\tilde{\mathfrak{G}}$ if $\mathbf{p} = \mathbf{0}$. (We have already encountered a similar case in Sect. 3.3 where L_x , L_y , and L_z are commutative with one another if the orbital angular momentum $\mathbf{L} = \mathbf{0}$, even though it is not the case with $\mathbf{L} \neq \mathbf{0}$.) In that exceptional case where $\mathbf{p} = \mathbf{0}$, both $\mathfrak{G}$ and $\tilde{\mathfrak{G}}$ are Hermitian (i.e., real diagonalized metrics), and so we have a CONS of common eigenvectors that are solutions of the Dirac equation.

The fact that $H_{\pm\mathbf{p}}$ ($\mathbf{p} \neq \mathbf{0}$) is not commutative with $\mathfrak{G}$ or $\tilde{\mathfrak{G}}$ needs further explanation. Consider an eigenvalue problem $\mathfrak{G}\Omega = \omega\Omega$ (or $\tilde{\mathfrak{G}}\tilde{\Omega} = \tilde{\omega}\tilde{\Omega}$). Then, both the positive-energy solution and negative-energy solution of the Dirac equation can be a solution of the above two types of eigenvalue problems. That is, both Ω and $\tilde{\Omega}$ take both the positive-energy and negative-energy solutions as an eigenfunction. Nonetheless, $H_{\mathbf{p}}$ ($H_{-\mathbf{p}}$) takes only a positive (negative) energy as an eigenvalue. Correspondingly, $H_{\mathbf{p}}$ ($H_{-\mathbf{p}}$) takes only the positive-energy (negative-energy) solution as the eigenfunction. For this reason, $H_{\pm\mathbf{p}}$ ($\mathbf{p} \neq \mathbf{0}$) is not commutative with $\mathfrak{G}$ or $\tilde{\mathfrak{G}}$. For further discussion, see Chap. 24.

Table 21.1 Hamiltonian and helicity operators of the Dirac field

Dirac spinor	Hamiltonian H_p, H_{-p}	Helicity operator $\mathfrak{H}, \tilde{\mathfrak{H}}$	Eigenstate ^a	Eigenvalue	
				Energy	Helicity
$u(p, h)$	$H_p = \begin{pmatrix} m & \boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & -m \end{pmatrix}$	$\mathfrak{H} = \frac{1}{ \mathbf{p} } \begin{pmatrix} \boldsymbol{\sigma} \cdot \mathbf{p} & 0 \\ 0 & \boldsymbol{\sigma} \cdot \mathbf{p} \end{pmatrix}$	$N \begin{pmatrix} \chi \\ h \chi \end{pmatrix}$	p^0	h
$v(p, h)$	$H_{-p} = \begin{pmatrix} m & -\boldsymbol{\sigma} \cdot \mathbf{p} \\ -\boldsymbol{\sigma} \cdot \mathbf{p} & -m \end{pmatrix}$	$\tilde{\mathfrak{H}} = \frac{1}{ \mathbf{p} } \begin{pmatrix} -\boldsymbol{\sigma} \cdot \mathbf{p} & 0 \\ 0 & -\boldsymbol{\sigma} \cdot \mathbf{p} \end{pmatrix}$	$\tilde{N} \begin{pmatrix} \tilde{S} \chi \\ -h \tilde{\chi} \end{pmatrix}$	$-p^0$	h

^aWith the normalization constants N and $\tilde{N}$, we have $N = \tilde{N} = \sqrt{p^0 + m}$; see (21.138) and (21.143)

In a general case where $\mathbf{p} \neq \mathbf{0}$, however, the eigenvectors eligible for the solution of the Dirac equation are not likely to constitute the CONS. As already mentioned in Sect. 21.3.2, the fact that neither $\mathcal{G}$ nor $\tilde{\mathcal{G}}$ is Hermitian in the general case makes the problem more complicated. Hence, we must abandon the familiar orthonormalization method based on the inner product in favor of another way. We develop a different method in the next section accordingly.

21.4 Normalization of the Solutions of the Dirac Equation

First, we introduce the Dirac adjoint $\bar{\psi}$ such that

$$\bar{\psi} \equiv \psi^\dagger \gamma^0. \quad (21.135)$$

Using this notation, we have

$$\begin{aligned} \bar{u}(\mathbf{p}, h) u(\mathbf{p}, h) &= N^* (\chi^\dagger h S \chi^\dagger) \gamma^0 N \begin{pmatrix} \chi \\ h S \chi \end{pmatrix} = |N|^2 (\chi^\dagger - h S \chi^\dagger) \begin{pmatrix} \chi \\ h S \chi \end{pmatrix} \\ &= |N|^2 (\chi^\dagger \cdot \chi - h^2 S^2 \chi^\dagger \cdot \chi), \\ &= |N|^2 (1 - S^2) = |N|^2 \left[1 - \left(\frac{p}{p^0 + m} \right)^2 \right] = |N|^2 \frac{2m}{p^0 + m}, \end{aligned} \quad (21.136)$$

where χ represents either χ_+ or χ_- and $\chi^\dagger \cdot \chi = 1$ in either case; we have $h^2 = 1$; with the last equality we used the relation $(p^0)^2 = p^2 + m^2$. The number N is the normalization constant of (21.101). According to the custom, we wish to get a normalized eigenfunction $u(\mathbf{p}, h)$ such that

$$\bar{u}(\mathbf{p}, h) u(\mathbf{p}, h) = 2m. \quad (21.137)$$

This implies that $|N|^2 = p^0 + m$. Choosing N to be real and positive, we have

$$N = \sqrt{p^0 + m}. \quad (21.138)$$

Then, from (21.101) we get

$$u(\mathbf{p}, h) = \sqrt{p^0 + m} \begin{pmatrix} \chi \\ h S \chi \end{pmatrix}. \quad (21.139)$$

With the eigenfunctions having different helicity h and h' ($h \neq h'$), we have

$$\bar{u}(\mathbf{p}, h) u(\mathbf{p}, h') = 0 \quad (h \neq h'), \quad (21.140)$$

because $\chi_h^\dagger \cdot \chi_{h'} = 0$, namely, χ_h and $\chi_{h'}$ ($h \neq h'$) are orthogonal. Notice that χ_h or $\chi_{h'}$ denotes either χ_- or χ_+ . Combined with the relation $\chi^\dagger \cdot \chi = 1$, we have

$$\chi_h^\dagger \cdot \chi_{h'} = \delta_{hh'}.$$

Further combining (21.137) and (21.140), we get

$$\bar{u}(\mathbf{p}, h) u(\mathbf{p}, h') = 2m\delta_{hh'}. \quad (21.141)$$

With the negative-energy solution, using (21.114) and similarly normalizing the function given in (21.118) we obtain

$$\bar{v}(\mathbf{p}, h) v(\mathbf{p}, h') = -2m\delta_{hh'}, \quad (21.142)$$

where $v(\mathbf{p}, h)$ is expressed as

$$v(\mathbf{p}, h) = \tilde{N} \begin{pmatrix} s\tilde{\chi} \\ -h\tilde{\chi} \end{pmatrix} = \sqrt{p^0 + m} \begin{pmatrix} s\tilde{\chi} \\ -h\tilde{\chi} \end{pmatrix}.$$

That is, the normalization constant given by

$$\tilde{N} = \sqrt{p^0 + m} \quad (21.143)$$

for (21.118) is the same as the constant N of (21.138). Note, however, the minus sign in (21.142). Thus, the normalization of the eigenfunctions differs from that based on the positive definite inner product (see Theorem 13.2: Gram–Schmidt orthonormalization Theorem).

With regard to the normalization of the plane wave solutions of the Dirac equation, we have the following useful relations:

$$\sum_{h=-1, +1} u(\mathbf{p}, h) \bar{u}(\mathbf{p}, h) = p_\mu \gamma^\mu + m, \quad (21.144)$$

$$\sum_{h=+1, -1} v(\mathbf{p}, h) \bar{v}(\mathbf{p}, h) = p_\mu \gamma^\mu - m, \quad (21.145)$$

where the summation with h should be taken as both -1 and $+1$. We will come back to these important expressions for the normalization in Chap. 24 in relation to the projection operators.

We remark the following relationship with respect to the Dirac adjoint. We had

$$\mathfrak{G}u(\mathbf{p}, h) = 0. \quad (21.92)$$

Taking the adjoint of this expression, we have

$$u^\dagger(\mathbf{p}, h)\mathfrak{G}^\dagger = u^\dagger(\mathbf{p}, h)\gamma^0\gamma^0\mathfrak{G}^\dagger = \bar{u}(\mathbf{p}, h)\gamma^0\mathfrak{G}^\dagger = 0. \quad (21.146)$$

In (21.91) we had $\mathfrak{G} = p_\mu\gamma^\mu - m$. Taking the adjoint of both sides and taking account of the fact that p_μ and m are real, we have

$$\mathfrak{G}^\dagger = p_\mu(\gamma^\mu)^\dagger - m = p_\mu\gamma^0\gamma^\mu\gamma^0 - m, \quad (21.147)$$

where notice that [1]

$$(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0. \quad (21.148)$$

Multiplying both sides of (21.147) by γ^0 from both the left and right, we get

$$\gamma^0\mathfrak{G}^\dagger\gamma^0 = p_\mu\gamma^\mu - m = \mathfrak{G}. \quad (21.149)$$

Moreover, multiplying (21.146) by γ^0 from the right, we get

$$\bar{u}(\mathbf{p}, h)\gamma^0\mathfrak{G}^\dagger\gamma^0 = \bar{u}(\mathbf{p}, h)\mathfrak{G} = 0, \quad (21.150)$$

where with the first equality we used (21.149).

Meanwhile, we have

$$(p_\nu\gamma^\nu)\gamma^\mu + \gamma^\mu(p_\nu\gamma^\nu) = p_\nu(\gamma^\nu\gamma^\mu + \gamma^\mu\gamma^\nu) = p_\nu \cdot 2\eta^{\nu\mu} = 2p^\mu.$$

That is,

$$(p_\nu\gamma^\nu)\gamma^\mu + \gamma^\mu(p_\nu\gamma^\nu) = 2p^\mu. \quad (21.151)$$

Multiplying both sides of (21.151) by $\bar{u}(\mathbf{p}, h)$ from the left and $u(\mathbf{p}, h')$ from the right, we obtain

$$\bar{u}(\mathbf{p}, h)[(p_\nu\gamma^\nu)\gamma^\mu + \gamma^\mu(p_\nu\gamma^\nu)]u(\mathbf{p}, h') = 2p^\mu\bar{u}(\mathbf{p}, h)u(\mathbf{p}, h') = 2p^\mu 2m\delta_{hh'}, \quad (21.152)$$

where with the last equality we used (21.141). Using (21.92) and (21.150), we get

$$\text{LHS of (21.152)} = \bar{u}(\mathbf{p}, h)(m\gamma^\mu + \gamma^\mu m)u(\mathbf{p}, h') = 2m\bar{u}(\mathbf{p}, h)\gamma^\mu u(\mathbf{p}, h'). \quad (21.153)$$

Comparing RHS of (21.152) and (21.153), we obtain

$$\bar{u}(\mathbf{p}, h) \gamma^\mu u(\mathbf{p}, h') = 2p^\mu \delta_{hh'}. \quad (21.154)$$

Likewise, we have

$$\bar{v}(\mathbf{p}, h) \gamma^\mu v(\mathbf{p}, h') = 2p^\mu \delta_{hh'}. \quad (21.155)$$

In particular, putting $\mu = 0$ in (21.154) and (21.155) we have

$$\begin{aligned} \bar{u} \gamma^0 u(\mathbf{p}, h') &= [u(\mathbf{p}, h)]^\dagger u(\mathbf{p}, h') = 2p^0 \delta_{hh'}, \\ \bar{v} \gamma^0 v(\mathbf{p}, h') &= [v(\mathbf{p}, h)]^\dagger v(\mathbf{p}, h') = 2p^0 \delta_{hh'}. \end{aligned} \quad (21.156)$$

With respect to the other normalization relations, we have, e.g.,

$$u^\dagger(\mathbf{p}, h) v(-\mathbf{p}, h') = v^\dagger(\mathbf{p}, h) u(-\mathbf{p}, h') = 0. \quad (21.157)$$

Equation (21.157) immediately follows from the fact that $u(\pm\mathbf{p}, h)$ and $v(\pm\mathbf{p}, h)$ belong to the different energy eigenvalues p^0 and $-p^0$, respectively; see the discussion of Sect. 21.3.4.

21.5 Charge Conjugation

As in the case of (21.141) and (21.142), the normalization based upon the Dirac adjoint keeps (21.141) and (21.142) invariant by the multiplication of a phase factor $c = e^{i\theta}$ (θ : real). That is, instead of $u(\mathbf{p}, h)$ and $v(\mathbf{p}, h)$ we may freely use $cu(\mathbf{p}, h)$ and $cv(\mathbf{p}, h)$. Yet, we normally impose a pretty strong constraint on the eigenfunctions $u(\mathbf{p}, h)$ and $v(\mathbf{p}, h)$ (vide infra). Such a constraint is known as the charge conjugation or charge-conjugation transformation, which is a kind of discrete transformation and thought of as the particle–antiparticle transduction. The charge-conjugation transformation of a function ψ is defined by

$$\psi^C \equiv i\gamma^2 \psi^*. \quad (21.158)$$

This transformation is defined as the interchange between a particle and an antiparticle. The notion of antiparticle is characteristic of the quantum theory of fields. If ψ represents the quantum state of the particle, ψ^C corresponds to that of the antiparticle and vice versa. If ψ is associated with the charge Q ($=1$), ψ^C carries the charge of $-Q$ ($=-1$).

The charge conjugation is related to the Dirac adjoint such that

$$\psi^C = C\bar{\psi}^T, \quad (21.159)$$

where C is given by

$$C \equiv i\gamma^2\gamma^0. \quad (21.160)$$

That is, we have

$$\psi^C = C\bar{\psi}^T = i\gamma^2\gamma^0(\gamma^0)^T(\psi^\dagger)^T = i\gamma^2\gamma^0\gamma^0\psi^* = i\gamma^2\psi^* \quad (21.161)$$

to recover the definition of the charge conjugation of (21.158). Writing ψ^* as a column vector representation such that

$$\psi^* = \begin{pmatrix} \psi_1^* \\ \psi_2^* \\ \psi_3^* \\ \psi_4^* \end{pmatrix}, \quad (21.162)$$

we obtain

$$\psi^C = i\gamma^2\psi^* = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \psi_1^* \\ \psi_2^* \\ \psi_3^* \\ \psi_4^* \end{pmatrix} = \begin{pmatrix} -\psi_4^* \\ \psi_3^* \\ \psi_2^* \\ -\psi_1^* \end{pmatrix}, \quad (21.163)$$

where we used the representation of (21.68) with γ^2 . As for the representation of the matrix C , we get

$$C = i\gamma^2\gamma^0 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}. \quad (21.164)$$

The matrix C is anti-Hermitian with eigenvalues of $\pm i$ (each, double root). Also, we have $C = -C^{-1}$. As anticipated, successive charge-conjugation transformation of ψ gives

$$(\psi^C)^C = i\gamma^2 (i\gamma^2 \psi^*)^* = i\gamma^2 (-i) (-\gamma^2) (\psi^*)^* = -(\gamma^2)^2 \psi = -(-E_4)\psi = \psi, \quad (21.165)$$

where with the second to the last equality, we used (21.63).

We apply the notion of charge conjugation to $u(\mathbf{p}, h)$ and $v(\mathbf{p}, h)$ of (21.104) and (21.120) so that we may obtain

$$v(\mathbf{p}, h) = i\gamma^2 u(\mathbf{p}, h)^*. \quad (21.166)$$

Also, we have

$$u(\mathbf{p}, h) = i\gamma^2 v(\mathbf{p}, h)^*. \quad (21.167)$$

Using the column vector representation for (21.166), we get

$$\begin{aligned} i\gamma^2 u(\mathbf{p}, +1)^* &= N \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \left[e^{i\frac{\phi}{2}} \sin \frac{\theta}{2} \right]^* \\ \left[e^{-i\frac{\phi}{2}} \cos \frac{\theta}{2} \right]^* \\ \left[\mathcal{S} e^{i\frac{\phi}{2}} \sin \frac{\theta}{2} \right]^* \\ \left[\mathcal{S} e^{-i\frac{\phi}{2}} \cos \frac{\theta}{2} \right]^* \end{pmatrix} \\ &= \tilde{N} \begin{pmatrix} -\mathcal{S} e^{i\phi/2} \cos \frac{\theta}{2} \\ \mathcal{S} e^{-i\phi/2} \sin \frac{\theta}{2} \\ e^{i\phi/2} \cos \frac{\theta}{2} \\ -e^{-i\phi/2} \sin \frac{\theta}{2} \end{pmatrix} = v(\mathbf{p}, +1), \end{aligned}$$

$$\begin{aligned}
 i\gamma^2 u(\mathbf{p}, -1)^* &= N \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \left[e^{\frac{i\phi}{2}} \cos \frac{\theta}{2} \right]^* \\ \left[-e^{-\frac{i\phi}{2}} \sin \frac{\theta}{2} \right]^* \\ \left[-Se^{\frac{i\phi}{2}} \cos \frac{\theta}{2} \right]^* \\ \left[Se^{-\frac{i\phi}{2}} \sin \frac{\theta}{2} \right]^* \end{pmatrix} \\
 &= \tilde{N} \begin{pmatrix} -Se^{i\phi/2} \sin \frac{\theta}{2} \\ -Se^{-i\phi/2} \cos \frac{\theta}{2} \\ -e^{i\phi/2} \sin \frac{\theta}{2} \\ -e^{-i\phi/2} \cos \frac{\theta}{2} \end{pmatrix} = v(\mathbf{p}, -1).
 \end{aligned} \tag{21.168}$$

In (21.168), we used $N = \tilde{N}$. Notice that (21.168) is exactly the same as the second equation of (21.120) including the phase factor. This feature is essential when we deal with the quantization of the Dirac field (see Chap. 22). Also, note that the charge conjugation holds the helicity invariant. As already mentioned, the charge conjugation transforms a particle carrying the charge Q ($= +1$) into its antiparticle Q ($= -1$) or vice versa. Therefore, the charge conjugation does not cause the change in the movement form of a particle but is responsible for the inner transformation between particle and antiparticle. Then, it is natural that the charge conjugation holds the helicity invariant.

We also derive the charge-conjugation transformation in a general form as follows:

$$(p_\mu \gamma^\mu - m)u(\mathbf{p}, h) = 0,$$

↓ take complex conjugate:

$$\left[p_2(-\gamma^2) + \sum_{\mu \neq 2} (p_\mu \gamma^\mu) - m \right] u(\mathbf{p}, h)^* = 0,$$

↓ insert $i\gamma^2 i\gamma^2 = E_4$ [(4, 4) identity matrix], before $u(\mathbf{p}, h)^*$:

$$\left\{ \left[p_2(-\gamma^2) + \sum_{\mu \neq 2} (p_\mu \gamma^\mu) - m \right] (i\gamma^2) \right\} (i\gamma^2) u(\mathbf{p}, h)^* = 0,$$

↓ use $C\bar{u}(\mathbf{p}, h)^T \equiv i\gamma^2 u(\mathbf{p}, h)^*$:

$$\left[p_2(-\gamma^2)i\gamma^2 + \sum_{\mu \neq 2} (p_\mu \gamma^\mu) i\gamma^2 - m i\gamma^2 \right] C\bar{u}(\mathbf{p}, h)^T = 0,$$

↓ use $\gamma^\mu \gamma^2 = -\gamma^2 \gamma^\mu$ ($\mu \neq 2$) and $(\gamma^2)^2 = -E_4$:

$$\left[i p_2 - i\gamma^2 \sum_{\mu \neq 2} (p_\mu \gamma^\mu) - m i\gamma^2 \right] C\bar{u}(\mathbf{p}, h)^T = 0,$$

↓ multiply $i\gamma^2$ on the above expression from the left:

$$\left[-p_2 \gamma^2 - \sum_{\mu \neq 2} (p_\mu \gamma^\mu) - m \right] C\bar{u}(\mathbf{p}, h)^T = 0.$$

Thus, we obtain

$$(-p_\mu \gamma^\mu - m) C\bar{u}(\mathbf{p}, h)^T = 0. \quad (21.169)$$

Meanwhile, as already seen, we have had

$$(-p_\mu \gamma^\mu - m) v(\mathbf{p}, h) = 0. \quad (21.74)$$

Comparing (21.169) and (21.74), we get

$$v(\mathbf{p}, h) = C\bar{u}(\mathbf{p}, h)^T. \quad (21.170)$$

In an opposite way, we have

$$u(\mathbf{p}, h) = C\bar{v}(\mathbf{p}, h)^T. \quad (21.171)$$

The concepts of the Dirac adjoint and charge conjugation play an essential role in the QED, and so we will come back to this point in Chaps. 23 and 24 to consider these concepts in terms of the matrix algebra.

21.6 Characteristics of the Gamma Matrices

In this section we summarize important expressions and relations of the gamma matrices. The gamma matrices were constructed using the Pauli spin matrices as in (21.67) and their tangible forms were given by (21.68). The gamma matrices are inherently coupled to the Lorentz transformation. The gamma matrices combine the basic importance and practical feature in the quantum theory of fields accordingly. In

terms of the latter aspect, we make the most of various formula of the gamma matrices (see Chaps. 23 and 24). We summarize several of them in the last part of this section.

We started with the discussion of the Dirac equation by introducing the gamma matrices (Sect. 21.3.1). The key equation for this is given by

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}. \quad (21.66)$$

As the frequently used representation of the gamma matrices, we had the Dirac representation described by

$$\gamma^0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \gamma^k = \begin{pmatrix} 0 & \sigma_k \\ -\sigma_k & 0 \end{pmatrix} \quad (k=1, 2, 3). \quad (21.67)$$

In (21.67), γ^0 is an Hermitian operator and γ^k ($k=1, 2, 3$) is anti-Hermitian, while both γ^0 and γ^k are unitary. While sticking to the representation of γ^0 and (anti-) Hermitian property of these operators, we wish to deal with the representation of gamma matrices in a little bit broader context. Suppose that γ^k ($k=1, 2, 3$) is represented by

$$\gamma^k = \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ p & q & r & s \\ t & u & v & w \end{pmatrix}, \quad (21.172)$$

where $a, b, \dots, w$ are arbitrary complex numbers. Taking account of

$$\gamma^0 \gamma^k + \gamma^k \gamma^0 = 0 \quad (k=1, 2, 3) \quad (21.173)$$

and the fact that γ^k is anti-Hermitian, (21.172) is reduced to

$$\gamma^k = -\gamma^{k\dagger} = \begin{pmatrix} 0 & b & c & d \\ -b^* & 0 & g & h \\ -c^* & -g^* & 0 & s \\ -d^* & -h^* & -s^* & 0 \end{pmatrix}. \quad (21.174)$$

From the unitarity of γ^k , we get

$$\begin{aligned} cg^* + dh^* &= 0, & -bg + ds^* &= 0, & bh - cs &= 0, \\ cb^* + hs^* &= 0, & db^* - gs &= 0, & dc^* + hg^* &= 0; \end{aligned}$$

$$\begin{aligned}
 |b|^2 + |c|^2 + |d|^2 &= 1, |b|^2 + |g|^2 + |h|^2 = 1, \\
 |c|^2 + |g|^2 + |s|^2 &= 1, |d|^2 + |h|^2 + |s|^2 = 1.
 \end{aligned}
 \tag{21.175}$$

Putting $b = c = h = s = 0$ in (21.175), we have $|d|^2 = |g|^2 = 1$. Then, using θ_k and ρ_k ($k = 1, 2$) as real parameters, we obtain

$$\gamma^k = \begin{pmatrix} 0 & 0 & 0 & e^{i\theta_k} \\ 0 & 0 & e^{i\rho_k} & 0 \\ 0 & -e^{-i\rho_k} & 0 & 0 \\ -e^{-i\theta_k} & 0 & 0 & 0 \end{pmatrix} \quad (k = 1, 2). \tag{21.176}$$

From (21.66), we have

$$\begin{aligned}
 \{\gamma^1, \gamma^2\} &= \begin{pmatrix} -e^{i(\theta_1 - \theta_2)} & 0 & 0 & 0 \\ 0 & -e^{i(\rho_1 - \rho_2)} & 0 & 0 \\ 0 & 0 & -e^{i(\rho_2 - \rho_1)} & 0 \\ 0 & 0 & 0 & -e^{i(\theta_2 - \theta_1)} \end{pmatrix} \\
 + \begin{pmatrix} -e^{i(\theta_2 - \theta_1)} & 0 & 0 & 0 \\ 0 & -e^{i(\rho_2 - \rho_1)} & 0 & 0 \\ 0 & 0 & -e^{i(\rho_1 - \rho_2)} & 0 \\ 0 & 0 & 0 & -e^{i(\theta_1 - \theta_2)} \end{pmatrix} &= 0.
 \end{aligned}
 \tag{21.177}$$

Hence,

$$2 \cos(\theta_1 - \theta_2) = 2 \cos(\rho_1 - \rho_2) = 0.$$

This is equivalent to $\theta_1 - \theta_2 = \pm \pi/2$ and $\rho_1 - \rho_2 = \pm \pi/2$. Setting $\theta_1 = \rho_1 = 0$ and $\theta_2 = -\rho_2 = \pi/2$ in (21.176), we get γ^1 and γ^2 represented in (21.68). Similarly, putting $b = d = g = s = 0$ in (21.174), we have $|c|^2 = |h|^2 = 1$ from (21.175). Using τ and σ as real parameters, we obtain γ^3 described by

$$\gamma^3 = \begin{pmatrix} 0 & 0 & e^{i\tau} & 0 \\ 0 & 0 & 0 & e^{i\sigma} \\ -e^{-i\tau} & 0 & 0 & 0 \\ 0 & -e^{-i\sigma} & 0 & 0 \end{pmatrix}. \tag{21.178}$$

Setting $\tau = \pi$ and $\sigma = 0$, we get γ^3 represented in (21.68).

There is arbitrariness for choosing the gamma matrices. Multiplying both sides of (21.66) by a unitary operator $U^\dagger$ from the left and U from the right and defining

$$\tilde{\gamma}^\mu \equiv U^\dagger \gamma^\mu U, \quad (21.179)$$

we get

$$\{\tilde{\gamma}^\mu, \tilde{\gamma}^\nu\} = 2\eta^{\mu\nu}. \quad (21.180)$$

That is, newly defined gamma matrices $\tilde{\gamma}^\mu$ satisfy the same condition (21.66) while retaining unitarity and (anti-)Hermiticity.

The characteristic polynomials of γ^k ($k = 1, 2, 3$) given by (21.68) are all

$$(\lambda^2 + 1)^2 = 0. \quad (21.181)$$

That is, $\lambda = \pm i$ (each, doubly degenerate). Their determinants are all 1. With γ^0 , its characteristic polynomial is

$$(\lambda^2 - 1)^2 = 0. \quad (21.182)$$

Hence, $\lambda = \pm 1$ (each, doubly degenerate). Its determinant is 1.

We have another gamma matrix, γ^5 , that is linearly independent of γ^μ ($\mu = 0, 1, 2, 3$). The matrix is defined as

$$\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}. \quad (21.183)$$

The matrix γ^5 is Hermitian and unitary with eigenvalues $\lambda = \pm 1$ (each, doubly degenerate) and γ^5 is anti-commutative with γ^μ ($\mu = 0, 1, 2, 3$). The determinant of γ^5 is 1. In Table 21.2, we summarize major characteristics of the gamma matrices including γ^5 .

Table 21.2 Major characteristics of the gamma matrices

Matrix	Characteristic	Eigenvalue	Determinant
γ^0	Hermitian, unitary	± 1 (each doubly degenerate)	1
γ^i ($i = 1, 2, 3$)	anti-Hermitian, unitary	$\pm i$ (each doubly degenerate)	1
$\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$	Hermitian, unitary	± 1 (each doubly degenerate)	1

So far, we have assumed that the gamma matrices comprise a fourth-order square matrix. But it is not self-evident. Now, we tentatively assume that the gamma matrix is the n -th order square matrix. From (21.62) we have

$$\gamma^\mu \gamma^\nu = -\gamma^\nu \gamma^\mu \quad (\mu \neq \nu).$$

Taking the determinant of the above equation, we have

$$\det \gamma^\mu \det \gamma^\nu = (-1)^n \det \gamma^\nu \det \gamma^\mu. \quad (21.184)$$

Thus, we obtain $(-1)^n = 1$. Namely, n must be an even number. In (21.184) we know that $\det \gamma^\mu \neq 0$ from (21.63). We acknowledge that the linearly independent matrices are E (identity matrix), γ^μ ($\mu = 0, 1, 2, 3, 5$), and $10 \left[= \binom{5}{2} \right]$ products consisting of two different gamma matrices. Therefore, the number of linearly independent matrices is 16. Hence, n of (21.184) is at least 4, because we have $4^2 (=16)$ matrix elements for a $(4, 4)$ matrix. In fact, it suffices to choose a $(4, 4)$ matrix for the gamma matrix. The detailed proof for $n = 4$ is given in literature and website [7, 8].

Other than the characteristics of the gamma matrices summarized in Table 21.2, various properties of the gamma matrices are summarized or tabulated in many textbooks. We list some typical examples of them below. Among them, the contractions and trace calculations are frequently used. Useful applications can be seen in subsequent chapters.

(i) Contractions: [1]

$$\gamma_\lambda \gamma^\lambda = 4E_4, \quad \gamma_\lambda \gamma^\alpha \gamma^\lambda = -2\gamma^\alpha, \quad (21.185)$$

$$\gamma_\lambda \gamma^\alpha \gamma^\beta \gamma^\lambda = 4\eta^{\alpha\beta}, \quad \gamma_\lambda \gamma^\alpha \gamma^\beta \gamma^\rho \gamma^\lambda = -2\gamma^\rho \gamma^\beta \gamma^\alpha, \quad (21.186)$$

$$\gamma_\lambda (A_\sigma \gamma^\sigma) \gamma^\lambda = -2(A_\sigma \gamma^\sigma), \quad \gamma_\lambda (A_\sigma \gamma^\sigma) (B_\sigma \gamma^\sigma) \gamma^\lambda = 4A_\sigma B^\sigma, \quad (21.187)$$

$$\gamma_\lambda (A_\sigma \gamma^\sigma) (B_\sigma \gamma^\sigma) (C_\sigma \gamma^\sigma) \gamma^\lambda = -2(C_\sigma \gamma^\sigma) (B_\sigma \gamma^\sigma) (A_\sigma \gamma^\sigma). \quad (21.188)$$

Note that in the above, E_4 denotes the $(4, 4)$ identity matrix.

(ii) Traces: [1]

$$\text{Tr}[(A_\sigma \gamma^\sigma) (B_\sigma \gamma^\sigma)] = 4A_\sigma B^\sigma, \quad (21.189)$$

$$\begin{aligned} & \text{Tr}[(A_\sigma \gamma^\sigma) (B_\sigma \gamma^\sigma) (C_\sigma \gamma^\sigma) (D_\sigma \gamma^\sigma)] \\ &= 4[(A_\sigma B^\sigma) (C_\sigma D^\sigma) - (A_\sigma C^\sigma) (B_\sigma D^\sigma) + (A_\sigma D^\sigma) (B_\sigma C^\sigma)]. \end{aligned} \quad (21.190)$$

References

1. Mandl F, Shaw G (2010) Quantum field theory, 2nd edn. Wiley, Chichester
2. Itzykson C, Zuber J-B (2005) Quantum field theory. Dover, New York
3. Satake I-O (1975) Linear algebra (pure and applied mathematics). Marcel Dekker, New York
4. Satake I (1974) Linear algebra. Shokabo, Tokyo. (in Japanese)
5. Møller C (1952) The theory of relativity. Oxford University Press, London
6. Kaku M (1993) Quantum field theory. Oxford University Press, New York
7. Sakamoto M (2014) Quantum field theory. Shokabo, Tokyo. (in Japanese)
8. Sakamoto M (2014). https://www.shokabo.co.jp/author/2511/answer/QFT_answer_all.pdf;
http://www.shokabo.co.jp/author/2511/answer/QFT_answer_all.pdf. (in Japanese)

Chapter 22

Quantization of Fields



In Chap. 1 we thought the canonical commutation relation as the key concept of quantum mechanics. The canonical commutation relation is based on the commutation relation between the canonical coordinate and canonical momentum. These quantities are represented as generalized coordinates and generalized momenta so that these fundamental variables can conform to different coordinate systems. In this chapter, we reformulate the canonical commutation relation as the guiding principle of the quantization of fields. This will be done within the framework of the Lagrangian formalism. In this chapter we deal with the scalar field, Dirac field, and photon field (electromagnetic field). The commutation relation is first represented as the equal-time commutation relation, as the Lagrangian formalism singles out the time coordinate, so does the equal-time commutation relation. This looks at first glance inconsistent with the Lorentz invariance (or covariance). However, this superficial inconsistency is worked out by introducing the invariant delta functions. At the same time, by translating the canonical commutation relation in the coordinate space into that in the momentum space, the calculations are eased with respect to the field quantities.

The covariant nature is indispensable to any physical theory that is based on the theory of relativity. In that sense, the invariant delta functions provide us with a good example. The combination of the invariant delta function and time-ordered product yields the Feynman propagator that is fully necessary to evaluate the interaction between the fields. These procedures are directly associated with the construction of the quantum electrodynamics (QED).

22.1 Lagrangian Formalism of the Fields [1]

Analytical mechanics is based upon the introduction of scalar quantities (e.g., Lagrangian and Hamiltonian) that are independent of the choice of individual coordinate system. Of these quantities, Lagrangian and the associated action integral

play a central role in the quantum theory of fields as a powerful tool to develop the theory and to perform calculations. We describe a brief outline of the Lagrangian formalism [1].

For the sake of brevity, suppose that a single particle is moving under the influence of potential U . Then, the kinetic energy T of the particle is described as

$$T = \sum_{k=1}^3 \frac{1}{2} m \dot{x}_k^2 = \sum_{k=1}^3 \frac{1}{2m} p_k^2, \quad (22.1) \leftarrow$$

where m is a mass of the particle; x_k and p_k ($k = 1, 2, 3$) represent the x -, y -, and z -component of the Cartesian coordinate and momentum of the particle in that direction, respectively. In the Cartesian coordinate, we have

$$p_k = m \dot{x}_k \quad (k = 1, 2, 3). \quad (22.2) \leftarrow$$

Here, we may modify (22.2) so that we can use any suited variables to specify the position of the particle instead of the Cartesian coordinate x_i . Examples include the spherical coordinate introduced in Sect. 3.2. Let q_k ($k = 1, 2, 3$) be a set of such variables, which are said to be generalized coordinates. Suppose that x_k is expressed as a function of q_k such that

$$\begin{aligned} x_1 &= x_1(q_1, q_2, q_3), \\ x_2 &= x_2(q_1, q_2, q_3), \\ x_3 &= x_3(q_1, q_2, q_3). \end{aligned}$$

Differentiating x_k with respect to t (i.e., time), we get

$$\dot{x}_k = \sum_{l=1}^3 \frac{\partial x_k}{\partial q_l} \dot{q}_l \quad (k = 1, 2, 3). \quad (22.3) \leftarrow$$

Assuming that $\dot{x}_k$ depends upon both q_1, q_2, q_3 and $\dot{q}_1, \dot{q}_2, \dot{q}_3$, $\dot{x}_k$ can be described by

$$\dot{x}_k = \dot{x}_k(q_1, q_2, q_3, \dot{q}_1, \dot{q}_2, \dot{q}_3). \quad (22.4) \leftarrow$$

Then, from (22.3) we obtain

$$\frac{\partial \dot{x}_k}{\partial \dot{q}_l} = \frac{\partial x_k}{\partial q_l}. \quad (22.5) \leftarrow$$

Comparing (22.1) and (22.2), we may express p_k as

$$p_k = m\dot{x}_k = \frac{\partial T}{\partial \dot{x}_k} \quad (k = 1, 2, 3). \quad (22.6) \leftarrow$$

Since in (22.1) T is described by $\dot{x}_k$ ($k = 1, 2, 3$), T can be described by $q_1, q_2, q_3, \dot{q}_1, \dot{q}_2, \dot{q}_3$ as well through (22.4) such that

$$T = T(\dot{x}_1, \dot{x}_2, \dot{x}_3) = T(q_1, q_2, q_3, \dot{q}_1, \dot{q}_2, \dot{q}_3). \quad (22.7) \leftarrow$$

Thus, in parallel with (22.6) we can define a generalized momentum (or conjugate momentum) $\tilde{p}_k$ such that

$$\tilde{p}_k \equiv \frac{\partial T}{\partial \dot{q}_k} \quad (k = 1, 2, 3). \quad (22.8) \leftarrow$$

Taking account of (22.4) and applying the differentiation of a composite function to (22.7), we have

$$\tilde{p}_k = \sum_{l=1}^3 \frac{\partial T}{\partial \dot{x}_l} \frac{\partial \dot{x}_l}{\partial \dot{q}_k} \quad (k = 1, 2, 3). \quad (22.9) \leftarrow$$

Using (22.5) and (22.6), we obtain

$$\tilde{p}_k = \sum_{l=1}^3 m\dot{x}_l \frac{\partial x_l}{\partial q_k}. \quad (22.10) \leftarrow$$

Further differentiating (22.10) with respect to t , we get

$$\dot{\tilde{p}}_k = \sum_{l=1}^3 m\ddot{x}_l \frac{\partial x_l}{\partial q_k} + \sum_{l=1}^3 m\dot{x}_l \frac{d}{dt} \left(\frac{\partial x_l}{\partial q_k} \right). \quad (22.11) \leftarrow$$

Now, suppose that the single particle we are dealing with undergoes an infinitesimal displacement so that q_1, q_2, q_3 are changed by dq_1, dq_2, dq_3 . This causes $x_k = x_k(q_1, q_2, q_3)$ to be changed by dx_k such that

$$dx_k = \sum_{l=1}^3 \frac{\partial x_k}{\partial q_l} dq_l. \quad (22.12) \leftarrow$$

If we think of the *virtual* displacement δx_k [1], we rewrite (22.12) as

$$\delta x_k = \sum_{l=1}^3 \frac{\partial x_k}{\partial q_l} \delta q_l. \quad (22.13) \leftarrow$$

Suppose that the particle is subject to a force $\mathbf{F}$, which exerts an infinitesimal virtual work δW to the particle described by

$$\delta W = \sum_{k=1}^3 F_k \delta x_k. \quad (22.14) \leftarrow$$

Note that F_k is the x -, y -, and z -component of the original Cartesian coordinate. Namely, $\mathbf{F}$ can be expressed such that

$$\mathbf{F} = \sum_{k=1}^3 F_k \mathbf{e}_k, \quad (22.15) \leftarrow$$

where $\mathbf{e}_k$ ($k = 1, 2, 3$) are orthonormal basis vectors of the Cartesian coordinate that appeared in Sect. 3.2. Also, notice that we have used the word virtual work and symbol δW in correspondence with the virtual displacement δx_k .

Measuring δW in reference to the generalized coordinate, we get

$$\delta W = \sum_{k=1}^3 F_k \delta x_k = \sum_{k=1}^3 F_k \left(\sum_{l=1}^3 \frac{\partial x_k}{\partial q_l} \delta q_l \right) = \sum_{l=1}^3 \left(\sum_{k=1}^3 F_k \frac{\partial x_k}{\partial q_l} \right) \delta q_l, \quad (22.16) \leftarrow$$

where δx_k in (22.14) was replaced with δq_k through (22.13) and the order of summation was exchanged with respect to l and k . In (22.16), we define Q_l as

$$Q_l \equiv \sum_{k=1}^3 F_k \frac{\partial x_k}{\partial q_l}. \quad (22.17) \leftarrow$$

Then, we have

$$\delta W = \sum_{l=1}^3 Q_l \delta q_l. \quad (22.18) \leftarrow$$

Comparing (22.14) and (22.18), we have

$$\sum_{k=1}^3 F_k \delta x_k = \sum_{k=1}^3 Q_k \delta q_k. \quad (22.19)$$

Thus, we notice that Q_k represented in reference to the generalized coordinate system corresponds to F_k represented in reference to the original Cartesian coordinate system. Hence, Q_k is said to be a generalized force.

Describing the Newton's equation of motion in reference to the Cartesian coordinate, we obtain

$$F_l = m\ddot{x}_l. \quad (22.20) \leftarrow$$

Substituting (22.20) for (22.11), we have

$$\dot{p}_k = \sum_{l=1}^3 F_l \frac{\partial x_l}{\partial q_k} + \sum_{l=1}^3 m\dot{x}_l \frac{d}{dt} \left(\frac{\partial x_l}{\partial q_k} \right) = Q_k + \sum_{l=1}^3 m\dot{x}_l \frac{d}{dt} \left(\frac{\partial x_l}{\partial q_k} \right), \quad (22.21) \leftarrow$$

where with the second equality we used (22.17).

Once again, taking account of (22.4) and applying the differentiation of a composite function to (22.7), we have

$$\frac{\partial T}{\partial q_k} = \sum_{l=1}^3 \frac{\partial T}{\partial \dot{x}_l} \frac{\partial \dot{x}_l}{\partial q_k} = \sum_{l=1}^3 m\dot{x}_l \frac{\partial \dot{x}_l}{\partial q_k} \quad (k = 1, 2, 3), \quad (22.22) \leftarrow$$

where with the second equality we used (22.6). Calculating $\frac{\partial \dot{x}_l}{\partial q_k}$ by use of (22.3), we obtain

$$\begin{aligned} \frac{\partial \dot{x}_l}{\partial q_k} &= \sum_{m=1}^3 \frac{\partial x_l^2}{\partial q_k \partial q_m} \dot{q}_m = \sum_{m=1}^3 \frac{\partial x_l^2}{\partial q_m \partial q_k} \dot{q}_m = \sum_{m=1}^3 \frac{\partial}{\partial q_m} \left(\frac{\partial x_l}{\partial q_k} \right) \frac{dq_m}{dt} = \frac{d}{dt} \\ &\times \left(\frac{\partial x_l}{\partial q_k} \right). \end{aligned} \quad (22.23) \leftarrow$$

The last equality of (22.23) is obtained from the differentiation of the composite function $x_l = x_l(q_1, q_2, q_3, \dot{q}_1, \dot{q}_2, \dot{q}_3)$ with respect to t . Substituting (22.23) for RHS of (22.22), we get

$$\frac{\partial T}{\partial q_k} = \sum_{l=1}^3 m\dot{x}_l \frac{d}{dt} \left(\frac{\partial x_l}{\partial q_k} \right) \quad (k = 1, 2, 3). \quad (22.24) \leftarrow$$

Replacing the second term of (22.21) with LHS of (22.24), finally we obtain a succinct expression described by

$$\dot{p}_k = Q_k + \frac{\partial T}{\partial q_k}. \quad (22.25) \leftarrow$$

Using (22.8) that defined the generalized momentum, (22.25) is rewritten as

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_k} \right) = Q_k + \frac{\partial T}{\partial q_k} \quad (k = 1, 2, 3). \quad (22.26) \leftarrow$$

Equation (22.26) can further be transformed to a well-known form based upon the Lagrangian L . Suppose that the force is a conservative force. In that case, Q_k can be described as

$$Q_k = - \frac{\partial U}{\partial q_k}, \quad (22.27) \leftarrow$$

where U is potential. Then, (22.26) is further rewritten as

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_k} \right) - \frac{\partial T}{\partial q_k} + \frac{\partial U}{\partial q_k} = 0 \quad (k = 1, 2, 3). \quad (22.28) \leftarrow$$

We assume that U is independent of $\dot{q}_k$ ($k = 1, 2, 3$). That is, we have

$$\frac{\partial U}{\partial \dot{q}_k} = 0. \quad (22.29) \leftarrow$$

Accordingly, we get

$$\frac{d}{dt} \left[\frac{\partial (T - U)}{\partial \dot{q}_k} \right] - \frac{\partial (T - U)}{\partial q_k} = 0. \quad (22.30) \leftarrow$$

Defining the Lagrangian L as

$$L \equiv T - U, \quad (22.31) \leftarrow$$

we obtain

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) - \frac{\partial L}{\partial q_k} = 0 \quad (k = 1, 2, 3). \quad (22.32) \leftarrow$$

Equation (22.32) is called the Lagrange's equations of motion (or Euler-Lagrange equation) and utilized in various fields of physics and natural science. Although we have assumed a single particle system so far, the number of particles does not need to be limited. If a dynamical system we are thinking of contains n particles, the freedom of that system is $3n$. However, some part of them may be put

under a constraint; for instance, suppose that motion of some particles is restricted to the surface of a sphere, or some particles are subject to free fall under the force of gravity. In such cases the freedom decreases. Hence, the freedom of the dynamical system is normally characterized by a positive integer n . Then, (22.32) is reformulated as follows:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) - \frac{\partial L}{\partial q_k} = 0 \quad (k = 1, 2, \dots, n). \quad (22.33) \leftarrow$$

We remark that on the condition of (22.29), (22.8) can be rewritten as

$$\tilde{p}_k \equiv \frac{\partial L}{\partial \dot{q}_k} = \frac{\partial T}{\partial \dot{q}_k} \quad (k = 1, 2, \dots, n). \quad (22.34) \leftarrow$$

The relations (22.8) and (22.34) make a turning point in our formulation. Through the medium of (22.7) and (22.33), we express the Lagrangian L as

$$L = L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t) \equiv L(q, \dot{q}, t), \quad (22.35) \leftarrow$$

where with the last identity q and $\dot{q}$ represent $q_1, \dots, q_n$ and $\dot{q}_1, \dots, \dot{q}_n$ collectively. Consequently, $\tilde{p}_k$ ($k = 1, 2, \dots, n$) can be described as a function of $q, \dot{q}, t$. In an opposite way, we could describe L as $\tilde{L}(q, \tilde{p}, t)$ or $\tilde{L}(\dot{q}, \tilde{p}, t)$, but this would break the original functional form of L . Instead, we wish to seek another way of describing a physical quantity P as a function of $q, \tilde{p}$, and t , i.e., $P(q, \tilde{p}, t)$. Such satisfactory prescription is called the Legendre transformation [1]. The Legendre transformation is defined by

$$P(q, \tilde{p}, t) = L(q, \dot{q}, t) - \tilde{p}\dot{q}. \quad (22.36) \leftarrow$$

The implication of (22.36) is that the physical quantity P has been described as a function of the independent variables $q, \tilde{p}, t$ instead of $q, \dot{q}, t$. Notice that $P(q, \tilde{p}, t)$ and $L(q, \dot{q}, t)$ are connected to each other through $\tilde{p}$ and $\dot{q}$. The quantities $q, \tilde{p}$ are said to be canonical variables. That is, the Legendre transformation enables us to describe P as a function of the canonical variables. The Hamiltonian $H(q, \tilde{p}, t)$, another important quantity in the analytical mechanics, is given by the Legendre transformation and defined as

$$H(q, \tilde{p}, t) = \sum_{k=1}^n \tilde{p}_k \dot{q}_k - L(q, \dot{q}, t). \quad (22.37) \leftarrow$$

If the kinetic energy T is given by a second-order homogeneous polynomial with respect to $\dot{q}_k$ ($k = 1, 2, \dots, n$) as in (22.1), from (22.34) we have

$$\frac{\partial T}{\partial \dot{q}_k} = \tilde{p}_k = 2T_k / \dot{q}_k. \quad (22.38) \leftarrow$$

Hence, we obtain

$$2T_k = \tilde{p}_k \dot{q}_k.$$

Taking the summation of this expression, we get

$$\sum_{k=1}^n \tilde{p}_k \dot{q}_k = 2 \sum_{k=1}^n T_k = 2T, \quad (22.39) \leftarrow$$

where $\sum_{k=1}^n T_k = T$. Hence, the Hamiltonian $H(q, \tilde{p}, t)$ given in (22.37) is described by

$$H(q, \tilde{p}, t) = 2T - L = 2T - (T - U) = T + U. \quad (22.40) \leftarrow$$

Thus, the Hamiltonian represents the total energy of the dynamical system. The Legendre transformation is often utilized in thermodynamics. Interested readers are referred to literature [1].

Along with the Lagrange's equations of motion, we have another important guiding principle of the Lagrangian formalism. It is referred to as the principle of least action or Hamilton's principle. To formulate the principle properly, we define an action integral S such that [2]

$$S(q_1, \dots, q_n) \equiv \int_{-\infty}^{\infty} dt L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t) \quad (22.41) \leftarrow$$

The principle of least action is simply described as

$$\delta S(q_1, \dots, q_n) = 0. \quad (22.42) \leftarrow$$

The principle of least action says that particles of the dynamical system move in such a way that the action integral takes an extremum. Equations (22.41) and (22.42) lead to the following equation:

$$\delta S(q) = \int_{-\infty}^{\infty} dt [L(q + \delta q, \dot{q} + \delta \dot{q}, t) - L(q, \dot{q}, t)] = 0, \quad (22.43) \leftarrow$$

where q and $\dot{q}$ represent $q_1, \dots, q_n$ and $\dot{q}_1, \dots, \dot{q}_n$ all in one, respectively. We have

$$L(q + \delta q, \dot{q} + \delta \dot{q}, t) = L(q, \dot{q}, t) + \sum_{k=1}^n \left(\delta q_k \frac{\partial L}{\partial q_k} + \delta \dot{q}_k \frac{\partial L}{\partial \dot{q}_k} \right). \quad (22.44) \leftarrow$$

Noting that

$$\delta \dot{q}_k \frac{\partial L}{\partial \dot{q}_k} = \left(\frac{d}{dt} \delta q_k \right) \frac{\partial L}{\partial \dot{q}_k} = -\delta q_k \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) + \frac{d}{dt} \left(\delta q_k \frac{\partial L}{\partial \dot{q}_k} \right), \quad (22.45) \leftarrow$$

we get

$$\begin{aligned} \delta S(q) &= \int_{-\infty}^{\infty} dt \sum_{k=1}^n \left[\delta q_k \frac{\partial L}{\partial q_k} - \delta q_k \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) + \frac{d}{dt} \left(\delta q_k \frac{\partial L}{\partial \dot{q}_k} \right) \right] \\ &= \int_{-\infty}^{\infty} dt \sum_{k=1}^n \left\{ \delta q_k \left[\frac{\partial L}{\partial q_k} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) \right] + \frac{d}{dt} \left(\delta q_k \frac{\partial L}{\partial \dot{q}_k} \right) \right\} \\ &= \int_{-\infty}^{\infty} dt \sum_{k=1}^n \delta q_k \left[\frac{\partial L}{\partial q_k} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) \right] + \int_{-\infty}^{\infty} dt \sum_{k=1}^n \frac{d}{dt} \left(\delta q_k \frac{\partial L}{\partial \dot{q}_k} \right) \\ &= \int_{-\infty}^{\infty} dt \sum_{k=1}^n \delta q_k \left[\frac{\partial L}{\partial q_k} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) \right] + \sum_{k=1}^n \left[\delta q_k \frac{\partial L}{\partial \dot{q}_k} \right]_{-\infty}^{\infty} = 0. \end{aligned}$$

Assuming that infinitesimal variations vanish at $t = \pm \infty$, we obtain

$$\delta S(q) = \int_{-\infty}^{\infty} dt \sum_{k=1}^n \delta q_k \left[\frac{\partial L}{\partial q_k} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) \right] = 0. \quad (22.46) \leftarrow$$

The integrand of (22.46) must vanish so that (22.46) can hold with any arbitrarily chosen variations δq_k . In other words, the Lagrange's equations of motion must be represented by

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) - \frac{\partial L}{\partial q_k} = 0 \quad (k = 1, 2, \dots, n). \quad (22.33) \leftarrow$$

In what follows, we will make the most of the Lagrangian formalism.

22.2 Introductory Fourier Analysis [3]

Fourier analysis is one of the most important mathematical techniques. It contains the Fourier series expansion, Fourier transform, inverse Fourier transform, etc. We dealt with these techniques in this book from place to place (e.g., in Chaps. 5 and 20). Since the Fourier analysis, especially the Fourier transform is indispensable tool in

the quantum theory of fields, we introduce the related techniques rather systematically in this section. This is because the field quantization is associated with the whole space-time. Be that as it may, the major concepts related to the field quantization have already been shown in Parts I and IV. Here, we further extend and improve such concepts.

22.2.1 Fourier Series Expansion

Let us start with the Fourier series expansion. Consider the following SOLDE that appeared in Example 1.1 of Sect. 1.3:

$$\frac{d^2 y(x)}{dx^2} + \lambda y(x) = 0. \quad (1.61)$$

We solved (1.61) under the Dirichlet boundary conditions (see Sect. 10.3) of $y(L) = 0$ and $y(-L) = 0$ ($L > 0$). The normalized eigenfunctions were described by

$$y(x) = \sqrt{\frac{1}{L}} \cos kx \quad \left[kL = \frac{\pi}{2} + m\pi \quad (m = 0, 1, 2, \dots) \right], \quad (1.91)$$

with $\lambda = (2m + 1)^2 \pi^2 / 4L^2$ ($m = 0, 1, 2, \dots$) or

$$y(x) = \sqrt{\frac{1}{L}} \sin kx \quad [kL = n\pi \quad (n = 1, 2, 3, \dots)], \quad (1.92)$$

with $\lambda = (2n)^2 \pi^2 / 4L^2$ ($n = 1, 2, 3, \dots$).

Meanwhile, in Sect. 3.3 we dealt with essentially the same SOLDE

$$-\frac{1}{\Phi(\phi)} \frac{d^2 \Phi(\phi)}{d\phi^2} = \eta. \quad (3.58)$$

Rewriting (3.58), we get

$$\frac{d^2 \Phi(\phi)}{d\phi^2} + \eta \Phi(\phi) = 0 \quad (0 \leq \phi \leq 2\pi). \quad (22.47)$$

Under the periodic boundary conditions (BCs) of $\Phi(0) = \Phi(2\pi)$ and $\Phi'(0) = \Phi'(2\pi)$, we obtained the eigenfunctions described by

$$\tilde{\Phi}(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi} \quad (m = 0, \pm 1, \pm 2, \dots), \quad (3.64)$$

with $\eta = m^2$ ($m = 0, \pm 1, \pm 2, \dots$).

The above two examples show how the different BCs (boundary conditions) produced different eigenfunctions that corresponded to the different eigenvalues, even though we were considering the same differential equation. When we deal with the problems of the quantum field theory, the periodic BCs are more likely to be encountered than the Dirichlet BCs. If we admit that with the quantum-mechanical wave functions ψ , $|\psi|^2$ represents the existence probability (of either particle or field), it is natural to assume that $|\psi(x)|^2 \rightarrow 0$ at $x \rightarrow \pm \infty$, where x denotes the space-time coordinate. In light of the periodic BCs, we may safely say that $\psi(x)$ vanishes “smoothly” at $x \rightarrow \pm \infty$.

Having the aforementioned background knowledge, let us consider the Fourier series expansion of a function $f(x)$ that satisfies the periodic BCs specified by

$$f(-a) = f(a) \quad \text{and} \quad f'(-a) = f'(a) \quad (a > 0). \quad (22.48)$$

With a system of eigenfunctions $\mathcal{F}_n(x)$, we use

$$\mathcal{F}_n(x) = \frac{1}{\sqrt{2a}} e^{in\pi x/a} \quad (n = 0, \pm 1, \pm 2, \dots; a > 0) \quad (22.49)$$

Note that if we put $a = \pi$ in (22.49), (22.49) is of the same form as (3.64). Then, using (22.49) $f(x)$ can be expanded such that

$$f(x) = \frac{1}{2a} \sum_{n=-\infty}^{\infty} c(n) e^{in\pi x/a} \quad (22.50)$$

with

$$c(n) = \int_{-a}^a e^{-in\pi x/a} f(x) dx. \quad (22.51)$$

In the next paragraph, we further explore the characteristics and implications of (22.50) and (22.51).

22.2.2 Fourier Integral Transforms: Fourier Transform and Inverse Fourier Transform

Let us suppose that $a \rightarrow \infty$ in (22.50) and (22.51), keeping the BCs the same as (22.48) to see what happens. Putting

$$\frac{n\pi}{a} = k, \quad (22.52) \leftarrow$$

we think of its difference

$$\frac{\Delta n\pi}{a} = \Delta k. \quad (22.53) \leftarrow$$

Taking $\Delta n = 1$, we have $\frac{1}{2a} = \frac{\Delta k}{2\pi}$. Then, (22.50) is rewritten as

$$f(x) = \frac{\Delta k}{2\pi} \sum_{n=-\infty \leftarrow}^{\infty \leftarrow} c(n) e^{ikx} = \frac{1}{2\pi} \sum_{n=-\infty \leftarrow}^{\infty \leftarrow} c(n) e^{ikx} \Delta k. \quad (22.54) \leftarrow$$

We see from (22.53) that as $a \rightarrow \infty$, $\Delta k \rightarrow 0$. Then, (22.54) can further be converted into

$$f(x) = \frac{1}{2\pi} \int_{-\infty \leftarrow}^{\infty} c(k) e^{ikx} dk, \quad (22.55) \leftarrow$$

where $c(n)$ in (22.50) was changed to $c(k)$ through (22.52). Multiplying both sides of (22.55) by e^{-iqx} and integrating it with respect to x , we have

$$\begin{aligned} \int_{-\infty \leftarrow}^{\infty} e^{-iqx} f(x) dx &= \frac{1}{2\pi} \int_{-\infty \leftarrow}^{\infty} \int_{-\infty \leftarrow}^{\infty} c(k) e^{i(k-q)x} dk dx \\ &= \frac{1}{2\pi} \int_{-\infty \leftarrow}^{\infty} c(k) \left[\int_{-\infty \leftarrow}^{\infty} e^{i(k-q)x} dx \right] dk \\ &= \frac{1}{2\pi} \int_{-\infty \leftarrow}^{\infty} c(k) 2\pi \delta(k-q) dk = c(q). \end{aligned} \quad (22.56) \leftarrow$$

In (22.56) we used a property of the δ function described by

$$\delta(k) = \frac{1}{2\pi} \int_{-\infty \leftarrow}^{\infty} e^{ikx} dx. \quad (22.57) \leftarrow$$

Changing the argument q to k in (22.56), we get

$$c(k) = \int_{-\infty \leftarrow}^{\infty} e^{-ikx} f(x) dx. \quad (22.58) \leftarrow$$

The mathematical manipulation that transforms (22.55) into (22.58) is referred to as the Fourier integral transform. The transformation from (22.58) to (22.55) is said to be the inverse Fourier transform. These transforms are called the Fourier integral transforms as a collective term.

Rewriting (22.55) as

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{c}(k) e^{ikx} dk, \quad (22.59)$$

where $\tilde{c}(k) \equiv \frac{1}{\sqrt{2\pi}} c(k)$, similarly we get

$$\int_{-\infty}^{\infty} e^{-iqx} f(x) dx = \frac{\sqrt{2\pi}}{2\pi} \int_{-\infty}^{\infty} \tilde{c}(k) 2\pi \delta(k-q) dk = \sqrt{2\pi} \tilde{c}(q). \quad (22.60)$$

That is, we have

$$\tilde{c}(q) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iqx} f(x) dx. \quad (22.61)$$

In (22.59) and (22.61), both $f(x)$ and $\tilde{c}(q)$ share $1/\sqrt{2\pi}$ as a coefficient of the integral in common. Writing $\tilde{c}(q)$ as $c(q)$ anew, we obtain

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} c(k) e^{ikx} dk, \quad (22.62)$$

$$c(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} f(x) dx. \quad (22.63)$$

In what follows, we make it a rule to represent the Fourier integral transforms as (22.62) and (22.63), when we deal with the transformation of the canonical coordinates and canonical momenta. Note, however, that when we calculate the invariant delta functions and Feynman propagator, we use the Fourier transform pairs of (22.55) and (22.58). Though non-essential, care should be taken accordingly.

In Sect. 5.1 we mentioned that the eigenvectors $|k\rangle$ ($k = 0, 1, 2, \dots$) (i.e., solutions of an eigenvalue equation) form a complete orthonormal system (CONS) such that

$$P \equiv \sum_k |k\rangle \langle k| = E, \quad (5.15)$$

where E is an identity operator. In (5.15) the number of $|k\rangle$ might be either finite or infinite, but $|k\rangle$ was countable. A dynamical system we will be dealing with in this chapter, however, normally has infinite and uncountable degrees of freedom. To conform (5.15) to our present purpose, (5.15) should be modified as follows:

$$\hat{E} = \int_{-\infty}^{\infty} |k\rangle \langle k| dk, \quad (22.64)$$

where $\hat{E}$ is the identity operator and k is a suitable continuous parameter.

Both (5.15) and (22.64) can be regarded as the completeness of the projection operators (see Sect. 14.1). Multiplying both sides of (22.64) by $\langle x|$ and $|f\rangle$ from the left and from the right, respectively, we get

$$\langle x|f\rangle = \int_{-\infty}^{\infty} \langle x|k\rangle \langle k|f\rangle dk.$$

Using the relation introduced in Sect. 10.4 as

$$f(x) \equiv \langle x|f\rangle \quad (10.90)$$

and further defining

$$c(k) \equiv \langle k|f\rangle, \quad (22.65)$$

we have

$$f(x) = \int_{-\infty}^{\infty} \langle x|k\rangle c(k) dk. \quad (22.66)$$

Comparing (22.62) and (22.66), we obtain the following relationship [4]:

$$\langle x|k\rangle = \frac{1}{\sqrt{2\pi}} e^{ikx}. \quad (22.67)$$

As in the case of (22.64), we have

$$\hat{E} = \int_{-\infty}^{\infty} |x\rangle \langle x| dx. \quad (22.68)$$

Multiplying both sides of (22.68) by $\langle k|$ and $|f\rangle$ from the left and from the right, respectively, we get

$$\langle k|f\rangle = \int_{-\infty}^{\infty} \langle k|x\rangle \langle x|f\rangle dx. \quad (22.69)$$

Thus, we obtain

$$\begin{aligned} c(k) &= \int_{-\infty}^{\infty} \langle k|x\rangle f(x) dx = \int_{-\infty}^{\infty} \langle x|k\rangle^* f(x) dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-ikx} f(x) dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} f(x) dx, \end{aligned} \quad (22.70)$$

where with the second equality we used (13.2). Thus, we reproduced (22.63).

22.3 Quantization of the Scalar Field [5, 6]

22.3.1 Lagrangian Density and Action Integral

Going back to Sects. 1.1 and 21.1, the energy–momentum–mass relationship of a particle is represented in the natural units as

$$E^2 = p^2 + m^2. \quad (22.71) \leftarrow$$

As already seen in Sect. 21.1, the Klein–Gordon equation was based upon this relationship. From (22.71), we have

$$E = \pm \sqrt{p^2 + m^2}, \quad (22.72) \leftarrow$$

to which both positive and negative energies are presumed. The quantization of (neutral) scalar field (or Klein–Gordon field), Dirac field, and electromagnetic field to be dealt with in this chapter is closely associated with this assumption. At the same time, this caused problems at the early stage of the quantum field theory along with the difficulty in the probabilistic interpretation of (21.9). Nevertheless, after Dirac invented the equation named after himself (i.e., the Dirac equation), the Klein–Gordon equation turned out to be valid as well as the Dirac equation as a basis equation describing quantum fields [7].

Historically speaking, the quantum theory of fields was at first established as the theory that dealt with the dynamical system consisting of photons and electrons and referred to as quantum electrodynamics (QED). In this connection, the quantization of scalar field is not necessarily required. In this section, however, we first describe the quantization of the (neutral) scalar fields based on the Klein–Gordon equation. It is because the scalar field quantization is not only the simplest example but also a prototype of the related quantum theories of fields. Hence, we study the quantization of scalar field in detail accordingly.

We start with the formulation of the Lagrangian density of the free Klein–Gordon field. The Lagrangian density $\mathcal{L}$ is given by [5, 8]

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2). \quad (22.73) \leftarrow$$

The action integral $S(\phi)$ follows from (22.73) such that

$$S(\phi) = \int dx \mathcal{L} = \int dx \frac{1}{2} \left\{ \partial_\mu \phi(x) \partial^\mu \phi(x) - m^2 [\phi(x)]^2 \right\}. \quad (22.74) \leftarrow$$

Note that unlike (22.41) where the integration was taken only with the time coordinate, the integration in (22.74) has been taken over the whole space–time. That is

$$\int dx \equiv \int_{-\infty \leftarrow}^{\infty} dx^0 \int_{-\infty \leftarrow}^{\infty} dx^1 \int_{-\infty \leftarrow}^{\infty} dx^2 \int_{-\infty \leftarrow}^{\infty} dx^3. \quad (22.75) \leftarrow$$

Taking variation with respect to ϕ of $S(\phi)$, we have

$$\begin{aligned} \delta S(\phi) &= \delta \int dx \mathcal{L}(\phi) \leftarrow \\ &= \frac{1}{2} \int dx \{ [\partial_\mu \delta \phi(x)] \partial^\mu \phi(x) + \partial_\mu \phi(x) [\partial^\mu \delta \phi(x)] - m^2 \cdot 2\phi(x) \delta \phi(x) \} \leftarrow \\ &= \frac{1}{2} \int dx^{\neq \mu} \{ [\delta \phi(x)] \partial^\mu \phi(x) \}_{x^\mu = -\infty}^{\infty \leftarrow} + \frac{1}{2} \int dx_{\neq \mu} \{ \partial_\mu \phi(x) [\delta \phi(x)] \}_{x_\mu = -\infty \leftarrow}^{\infty \leftarrow} \\ &\quad + \frac{1}{2} \int dx \{ -\partial_\mu \partial^\mu \phi(x) - \partial^\mu \partial_\mu \phi(x) - m^2 \cdot 2\phi(x) \} \delta \phi(x) = 0, \end{aligned} \quad (22.76) \leftarrow$$

where the second to the last equality resulted from the integration by parts; $x^{\neq \mu}$ (or $x_{\neq \mu}$) in the first and second terms means the triple integral with respect to the variables x other than x^μ (or x_μ). We assume that $\delta \phi(x) \rightarrow 0$ at x^μ (x_μ) $\rightarrow \pm \infty$. Hence, with the second to the last equality of (22.76), the first and second integrals vanish. Consequently, for $\delta S(\phi) = 0$ to be satisfied with regard to any arbitrarily taken $\delta \phi(x)$, the integrand of the third integral must vanish. This is equivalent to

$$-\partial_\mu \partial^\mu \phi(x) - \partial^\mu \partial_\mu \phi(x) - m^2 \cdot 2\phi(x) = 0.$$

That is, we obtain

$$[\partial_\mu \partial^\mu \phi(x) + m^2] \phi(x) = 0. \quad (22.77) \leftarrow$$

Note that in the above equation $\partial_\mu \partial^\mu \phi(x) = \partial^\mu \partial_\mu \phi(x)$. Equation (22.77) is identical with (21.11). Thus, we find that the Klein–Gordon equation has followed from the principle of least action. In other words, on the basis of that principle we obtain an equation of motion of the quantum field from the action integral described by

$$S(\phi) = \int dx \mathcal{L}[\phi(x), \partial_\mu \phi(x)] \leftarrow \quad (22.78) \leftarrow$$

Namely, the Euler–Lagrange equation described by [2]

$$\partial_\mu \left[\frac{\partial \mathcal{L} \leftarrow}{\partial (\partial_\mu \phi)} \right] - \frac{\partial \mathcal{L} \leftarrow}{\partial \phi} = 0 \quad (22.79)$$

follows from $\delta S(\phi) = 0$. The Euler–Lagrange equation can be regarded as a relativistic extension of the Lagrange’s equations of motion (22.33). The derivation of (22.79) is left for the readers [2].

The implication of (22.78) and (22.79) is that once $\mathcal{L}[\phi(x), \partial_\mu \phi(x)]$ is properly formulated with a relativistic field, the equation of motion for that field can adequately be derived from the principle of least action. The Lagrangian formalism of the Dirac field and electromagnetic field will be given in Sects. 22.4 and 22.5, respectively.

22.3.2 Equal-Time Commutation Relation and Field Quantization

Once the Lagrangian density $\mathcal{L}$ has been given with the field to be quantized, the quantization of fields is carried out as the equal-time commutation relation between the canonical variables, i.e., generalized coordinate (or canonical coordinate) and generalized momentum (or canonical momentum). That is, if $\mathcal{L}$ is given as $\mathcal{L}[\phi(x), \partial_\mu \phi(x)]$ in (22.78) as a function of generalized coordinates $\phi(x)$ as well as their derivatives $\partial_\mu \phi(x)$, the quantization (or canonical quantization) of the scalar field is postulated as the canonical commutation relations expressed as

$$\begin{aligned} [\phi(t, \mathbf{x}), \pi(t, \mathbf{y})] &= i\delta^3(\mathbf{x} - \mathbf{y}), \\ [\phi(t, \mathbf{x}), \phi(t, \mathbf{y})] &= [\pi(t, \mathbf{x}), \pi(t, \mathbf{y})] = 0. \end{aligned} \quad (22.80)$$

In (22.80), generalized momentum $\pi(x)$ is defined in analogy with (22.34) as

$$\pi \equiv \frac{\partial \mathcal{L}}{\partial \dot{\phi}}. \quad (22.81)$$

Also, we have

$$\delta^3(\mathbf{x} - \mathbf{y}) \equiv \delta(x^1 - y^1)\delta(x^2 - y^2)\delta(x^3 - y^3).$$

If more than one coordinate is responsible for (22.80), that can appropriately be indexed. In that case, the commutation relations are sometimes represented as a matrix form (see Sect. 22.4). The first equation of (22.80) is essentially the same as (1.140) and (1.141), which is rewritten in the natural units (i.e., $\hbar = 1$) as

$$[q, p] = iE, \quad (22.82)$$

where E is the identity operator. Thus, we notice that E of (22.82) is replaced with $\delta^3(\mathbf{x} - \mathbf{y})$ in (22.80). This is because the whole spatial coordinates are responsible for the field quantization.

Equation (22.80) is often said to be the equal-time commutation relation. As is so named, the equal-time commutation relation favors the time coordinate over the other spatial coordinate. For this reason, the Lorentz covariance seems unclear, but the covariance manifests itself later, especially when we deal with, e.g., the invariant delta functions and the Feynman propagator.

With the free Klein–Gordon field, from (22.73) the canonical momentum $\pi(t, \mathbf{x})$ is described as

$$\pi(t, \mathbf{x}) = \frac{\partial \mathcal{L}}{\partial (\partial_0 \phi)} = \partial_0^{\leftarrow} \phi(t, \mathbf{x}) = \dot{\phi}(t, \mathbf{x}). \quad (22.83) \leftarrow$$

Using (22.83), the equal-time commutation relation is written as

$$\begin{aligned} [\phi(t, \mathbf{x}), \pi(t, \mathbf{y})] &= [\phi(t, \mathbf{x}), \dot{\phi}(t, \mathbf{y})] = i\delta^3(\mathbf{x} - \mathbf{y}), \\ [\phi(t, \mathbf{x}), \phi(t, \mathbf{y})] &= [\dot{\phi}(t, \mathbf{x}), \dot{\phi}(t, \mathbf{y})] = 0. \end{aligned} \quad (22.84) \leftarrow$$

Now, we are in the position to do with our task at hand. We wish to seek Fourier integral representation of the free Klein–Gordon field $\phi(x)$. The three-dimensional representation is given by [8]

$$\phi(t, \mathbf{x}) = \int \frac{d^3 \mathbf{k}}{\sqrt{(2\pi)^3}} q_{\mathbf{k}}(t) e^{i\mathbf{k}\mathbf{x}}, \quad (22.85) \leftarrow$$

where $d^3 \mathbf{k} \equiv dk^1 dk^2 dk^3$. Notice that (22.85) is the three-dimensional version of (22.62). The Fourier integral representation of (22.85) differs from that appeared in Sect. 22.2.2 in that the expansion coefficient $q_{\mathbf{k}}(t)$ contains the time coordinate as a parameter. Inserting (22.85) into the Klein–Gordon equation (22.77), we obtain

$$q_{\mathbf{k}}^{\ddot{}}(t) + (\mathbf{k}^2 + m^2) q_{\mathbf{k}}(t) = 0. \quad (22.86) \leftarrow$$

This is the equation of motion of a harmonic oscillator. With a general solution of (22.86), as in the case of (2.4) of Sect. 2.1 we have

$$q_{\mathbf{k}}(t) = q_1(\mathbf{k}) e^{-ik_0 t} + q_2(\mathbf{k}) e^{ik_0 t}, \quad (22.87) \leftarrow$$

with

$$k_0 \equiv \sqrt{\mathbf{k}^2 + m^2} (> 0). \quad (22.88) \leftarrow$$

Notice that in (22.87) each term has been factorized with respect to $\mathbf{k}$ and t so that the exponential factors contain only t as a variable. The quantity k_0 represents the angular frequency of the harmonic oscillator and varies with $|\mathbf{k}|$. Also, note that the

angular frequency and energy have the same dimension in the natural units (vide infra).

Here we assume that $\phi(x)$ represents the neutral scalar field so that $\phi(x)$ is Hermitian. In the present case $\phi(x)$ is represented by a (1, 1) matrix. Whereas it is a real c -number before the field quantization, after the field quantization it should be regarded as a real q -number. Namely, the order of the multiplication must be considered properly between the q -numbers.

Taking adjoint of (22.85), we have

$$\phi^\dagger(x) = \int \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3}} q_{\mathbf{k}}^\dagger(t) e^{-i\mathbf{k}x}. \quad (22.89) \leftarrow$$

Replacing $\mathbf{k}$ with $-\mathbf{k}$ in (22.89), we get

$$\phi^\dagger(x) = \int \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3}} q_{-\mathbf{k}}^\dagger(t) e^{i\mathbf{k}x}. \quad (22.90) \leftarrow$$

For (22.90) we used

$$\begin{aligned} \iint \int_{-\infty}^{+\infty} \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3}} q_{\mathbf{k}}^\dagger(t) e^{-i\mathbf{k}x} &= \iint \int_{+\infty}^{-\infty} \frac{(-1)^3 d^3\mathbf{k}}{\sqrt{(2\pi)^3}} q_{-\mathbf{k}}^\dagger(t) e^{i\mathbf{k}x} \\ &= \iint \int_{-\infty}^{+\infty} \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3}} q_{-\mathbf{k}}^\dagger(t) e^{i\mathbf{k}x}. \end{aligned} \quad (22.91) \leftarrow$$

In (22.91), the integration range covers a domain from $-\infty$ to $+\infty$ with each component of $\mathbf{k}$. Since the domain is symmetric relative to the origin, it suffices for the integral calculation of (22.91) to switch $\mathbf{k}$ to $-\mathbf{k}$ in the integrand.

For $\phi(x)$ to be Hermitian, that is, for $\phi^\dagger(x) = \phi(x)$ to hold, comparing (22.85) and (22.90) we must have

$$q_{-\mathbf{k}}^\dagger(t) = q_{\mathbf{k}}(t). \quad (22.92) \leftarrow$$

Meanwhile, from (22.87) we have

$$q_{-\mathbf{k}}^\dagger(t) = q_1^\dagger(-\mathbf{k}) e^{ik_0 t} + q_2^\dagger(-\mathbf{k}) e^{-ik_0 t}. \quad (22.93) \leftarrow$$

Comparing once again (22.87) and (22.93), we obtain

$$q_1^\dagger(-\mathbf{k}) = q_2(\mathbf{k}) \quad \text{and} \quad q_2^\dagger(-\mathbf{k}) = q_1(\mathbf{k}), \quad (22.94)$$

with the first and second relations of (22.94) being identical. Substituting (22.87) for (22.85), we get

$$\begin{aligned}
 \phi(t, \mathbf{x}) &= \int_{-\infty}^{\infty} \frac{d^3 \mathbf{k}}{\sqrt{(2\pi)^3}} [q_1(\mathbf{k}) e^{-ik_0 t} + q_2(\mathbf{k}) e^{ik_0 t}] e^{i\mathbf{k}\mathbf{x}} \\
 &= \int_{-\infty}^{\infty} \frac{d^3 \mathbf{k}}{\sqrt{(2\pi)^3}} [q_1(\mathbf{k}) e^{-ik_0 t} e^{i\mathbf{k}\mathbf{x}}] + \int_{-\infty}^{\infty} \frac{(-1)^3 d^3 \mathbf{k}}{\sqrt{(2\pi)^3}} [q_2(-\mathbf{k}) e^{ik_0 t} e^{i(-\mathbf{k})\mathbf{x}}] \\
 &= \int_{-\infty}^{\infty} \frac{d^3 \mathbf{k}}{\sqrt{(2\pi)^3}} [q_1(\mathbf{k}) e^{-ik_0 t} e^{i\mathbf{k}\mathbf{x}}] + \int_{-\infty}^{\infty} \frac{(-1)^3 d^3 \mathbf{k}}{\sqrt{(2\pi)^3}} [q_1^\dagger(\mathbf{k}) e^{ik_0 t} e^{i(-\mathbf{k})\mathbf{x}}] \\
 &= \int_{-\infty}^{\infty} \frac{d^3 \mathbf{k}}{\sqrt{(2\pi)^3}} [q_1(\mathbf{k}) e^{-ikx}] + \int_{-\infty}^{\infty} \frac{d^3 \mathbf{k}}{\sqrt{(2\pi)^3}} [q_1^\dagger(\mathbf{k}) e^{ikx}] \\
 &= \int_{-\infty}^{\infty} \frac{d^3 \mathbf{k}}{\sqrt{(2\pi)^3}} [q_1(\mathbf{k}) e^{-ikx} + q_1^\dagger(\mathbf{k}) e^{ikx}],
 \end{aligned} \tag{22.95}$$

where with the second equality $\mathbf{k}$ of the second term has been replaced with $-\mathbf{k}$; with the third equality (22.94) was used in the second term. Note also that in (22.95)

$$kx \equiv k_0 x_0 - \mathbf{k}\mathbf{x} = \sqrt{\mathbf{k}^2 + m^2} t - \mathbf{k}\mathbf{x}. \tag{22.96}$$

Defining $a(\mathbf{k})$ as [8]

$$a(\mathbf{k}) \equiv \sqrt{2k_0} q_1(\mathbf{k}), \tag{22.97}$$

we get

$$\phi(x) = \int \frac{d^3 \mathbf{k}}{\sqrt{(2\pi)^3 (2k_0)}} [a(\mathbf{k}) e^{-ikx} + a^\dagger(\mathbf{k}) e^{ikx}]. \tag{22.98}$$

The origin of the coefficient $\sqrt{2k_0}$ is explained in literature [9] (vide infra). Equation (22.98) clearly shows that $\phi(x)$ is Hermitian, that is $\phi^\dagger(x) = \phi(x)$. We divide (22.98) into the “positive-frequency” part $\phi^+(x)$ and “negative-frequency” part $\phi^-(x)$ such that [5]

$$\begin{aligned}
\phi(x) &= \phi^+(x) + \phi^-(x), \\
\phi^+(x) &\equiv \int \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3(2k_0)}} a(\mathbf{k}) e^{-ikx}, \\
\phi^-(x) &\equiv \int \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3(2k_0)}} a^\dagger(\mathbf{k}) e^{ikx}.
\end{aligned} \tag{22.99}$$

Note that the positive- and negative-frequency correspond to the positive- and negative-energy, respectively. From (22.83) and (22.98) we have

$$\begin{aligned}
\pi(x) = \partial^0 \phi(x) = \dot{\phi}(x) &= \int \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3(2k_0)}} \\
&\times [-ik_0 a(\mathbf{k}) e^{-ikx} + ik_0 a^\dagger(\mathbf{k}) e^{ikx}].
\end{aligned} \tag{22.100}$$

Next, we convert the equal-time commutation relations (22.84) into the commutation relation between $a(\mathbf{k})$ and $a^\dagger(\mathbf{q})$. For this purpose, we solve (22.98) and (22.100) with respect to $a(\mathbf{k})$ and $a^\dagger(\mathbf{k})$. To perform the calculation, we make the most of the Fourier transform developed in Sect. 22.2.2. In particular, the three-dimensional version of (22.57) is useful. The formula is expressed as

$$(2\pi)^3 \delta^3(\mathbf{k}) = \int_{-\infty}^{\infty} e^{ikx} d^3x. \tag{22.101}$$

To apply (22.101) to (22.98) and (22.100), we deform those equations such that

$$\phi(t, \mathbf{x}) = \int \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3(2k_0)}} [a(\mathbf{k}) e^{-ik_0 t} + a^\dagger(-\mathbf{k}) e^{ik_0 t}] e^{i\mathbf{k}\mathbf{x}}, \tag{22.102}$$

$$\dot{\phi}(t, \mathbf{x}) = \int \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3(2k_0)}} (-ik_0) [a(\mathbf{k}) e^{-ik_0 t} - a^\dagger(-\mathbf{k}) e^{ik_0 t}] e^{i\mathbf{k}\mathbf{x}}. \tag{22.103}$$

In (22.102) and (22.103), $\mathbf{k}$ has been replaced with $-\mathbf{k}$ in their second terms of RHS. We want to solve these equations in terms of $a(\mathbf{k}) e^{-ik_0 t}$ and $a^\dagger(-\mathbf{k}) e^{ik_0 t}$. The calculation procedures are as follows: (1) Multiply both sides of (22.102) and (22.103) by $e^{-i\mathbf{q}\mathbf{x}}$ and integrate them with respect to $\mathbf{x}$ by use of (22.101) to get a factor $\delta^3(\mathbf{k} - \mathbf{q})$. (2) Then, we perform the integration with respect to $\mathbf{k}$ to delete $\mathbf{k}$. As a consequence of the integration, $\mathbf{k}$ is replaced with $\mathbf{q}$ and $k_0 = \sqrt{\mathbf{k}^2 + m^2}$ is replaced with $\sqrt{\mathbf{q}^2 + m^2} \equiv q_0$ accordingly. In that way, we convert (22.102) and (22.103) into the equations with respect to $a(\mathbf{q}) e^{-iq_0 t}$ and $a^\dagger(\mathbf{q}) e^{iq_0 t}$. The result is expressed as follows:

$$\begin{aligned}
 a(\mathbf{q})e^{-iq_0t} &= i \int \frac{d^3\mathbf{x}}{\sqrt{(2\pi)^3(2q_0)}} [\dot{\phi}(t, \mathbf{x}) - iq_0\phi(t, \mathbf{x})] e^{-i\mathbf{q}\cdot\mathbf{x}}, \\
 a^\dagger(\mathbf{q})e^{iq_0t} &= -i \int \frac{d^3\mathbf{x}}{\sqrt{(2\pi)^3(2q_0)}} [\dot{\phi}(t, \mathbf{x}) + iq_0\phi(t, \mathbf{x})] e^{i\mathbf{q}\cdot\mathbf{x}}.
 \end{aligned} \tag{22.104} \leftarrow$$

Note that in the second equation of (22.104) $-\mathbf{q}$ was switched back to $\mathbf{q}$ to change $a^\dagger(-\mathbf{q})$ to $a^\dagger(\mathbf{q})$.

(3) Next, we calculate the commutation relation between $a(\mathbf{k})$ and $a^\dagger(\mathbf{q})$. For this, the commutator $[a(\mathbf{k})e^{-ik_0t}, a^\dagger(\mathbf{q})e^{iq_0t}]$ can directly be calculated using the equal-time commutation relations (22.84). The factors e^{-ik_0t} and e^{iq_0t} are gotten rid of in the final stage. The relevant calculations are straightforward and greatly simplified by virtue of the integration of the factor $\delta^3(\mathbf{x} - \mathbf{y})$ that comes from (22.84). The use of (22.101) also eases computations. As a result, we get succinct equation that is described by

$$[a(\mathbf{k})e^{-ik_0t}, a^\dagger(\mathbf{q})e^{iq_0t}] = e^{-i(k_0 - q_0)t} [a(\mathbf{k}), a^\dagger(\mathbf{q})] = \frac{q_0}{\sqrt{k_0q_0}} \delta^3(\mathbf{k} - \mathbf{q}).$$

Since $e^{-i(k_0 - q_0)t}$ never vanishes, we have $[a(\mathbf{k}), a^\dagger(\mathbf{q})] = 0$ if $\mathbf{k} \neq \mathbf{q}$. If $\mathbf{k} = \mathbf{q}$, we obtain $k_0 = \sqrt{\mathbf{k}^2 + \mathbf{m}^2} = \sqrt{\mathbf{q}^2 + \mathbf{m}^2} = q_0$. Consequently, $[a(\mathbf{k}), a^\dagger(\mathbf{q})] = \delta^3(\mathbf{k} - \mathbf{q})$. Meanwhile, we have $[a(\mathbf{k}), a(\mathbf{q})] = [a^\dagger(\mathbf{k}), a^\dagger(\mathbf{q})] = 0$ with any value of $\mathbf{k}$ and $\mathbf{q}$.

Summarizing the above results, with the commutation relations among the annihilation and creation operators we get

$$\begin{aligned}
 [a(\mathbf{k}), a^\dagger(\mathbf{q})] &\stackrel{\leftarrow}{=} \delta^3(\mathbf{k} - \mathbf{q}), \\
 [a(\mathbf{k}), a(\mathbf{q})] &= [a^\dagger(\mathbf{k}), a^\dagger(\mathbf{q})] \stackrel{\leftarrow}{=} 0.
 \end{aligned} \tag{22.105} \leftarrow$$

Note that the factor $\sqrt{2k_0}$ of (22.97) has been introduced so that (22.105) can be expressed in the simplest form possible. Equation (22.105) is free from the inclusion of space-time coordinates and, hence, conveniently used for various calculations, e.g., those of interaction between the quantum fields. Looking at (22.105), we find that we have again reached a result similar to that obtained in Sect. 2.2 expressed as

$$[a, a^\dagger] = 1 \quad \text{and} \quad [a, a^\dagger] = E. \tag{2.24} \leftarrow$$

In fact, characteristics of $a(\mathbf{k})$, $a^\dagger(\mathbf{q})$, and a , $a^\dagger$ are closely related in that $a^\dagger(\mathbf{q})$ and $a^\dagger$ function as the creation operators and that $a(\mathbf{k})$ and a act as the annihilation operators. Thus, the above discussion clearly shows that the quantum-mechanical description of the harmonic oscillators is of fundamental importance in the quantum field theory. The operator $a(\mathbf{k})$ is sometimes given in a simple form such that [8]

$$a(\mathbf{k}) = i \int \frac{d^3\mathbf{x}}{\sqrt{(2\pi)^3(2k_0)}} \left[e^{i\mathbf{k}\cdot\mathbf{x}} \overleftrightarrow{\partial}_0 \phi \right], \quad (22.106)$$

where

$$f \overleftrightarrow{\partial}_0 g \equiv f(\partial_0 g) - (\partial_0 f)g. \quad (22.107)$$

As already shown in Chap. 2, the use of $a(\mathbf{k})$ and $a^\dagger(\mathbf{k})$ enables us to calculate various physical quantities in simple algebraic forms. In the next section, we give the calculation processes for the Hamiltonian of the free Klein–Gordon field. The concept of the Fock space is introduced as well. The results can readily be extended to the calculations regarding, e.g., the Dirac field and photon field.

22.3.3 Hamiltonian and Fock Space

The Hamiltonian is described in a form analogous to that of the classical theory of analytical mechanics. It is given as the spatial integration of the Hamiltonian density $\mathcal{H}$ defined as

$$\mathcal{H}(x) = \pi(x)\dot{\phi}(x) - \mathcal{L}[\phi(x), \partial_\mu\phi(x)] \quad (22.108)$$

In our present case, we have $\pi(x) = \dot{\phi}(t, \mathbf{x})$ and from (22.73) we get

$$\mathcal{H}(x) = \frac{1}{2} \left\{ [\pi(x)]^2 + [\nabla\phi(x)]^2 + [m\phi(x)]^2 \right\}. \quad (22.109)$$

The Hamiltonian H is then given by

$$H = \int d^3\mathbf{x} \mathcal{H} = \int d^3\mathbf{x} \frac{1}{2} \left\{ [\pi(x)]^2 + [\nabla\phi(x)]^2 + [m\phi(x)]^2 \right\}. \quad (22.110)$$

The first term of the above integral is

$$\begin{aligned}
& \frac{1}{2} \int d^3 \mathbf{x} [\pi(t, \mathbf{x})]^2 = \frac{1}{2} \int d^3 \mathbf{x} [\dot{\phi}(t, \mathbf{x})]^2 \\
& = \frac{1}{2} d^3 \mathbf{x} \left\{ \int \frac{d^3 \mathbf{k}}{\sqrt{(2\pi)^3 (2k_0)}} (-ik_0) [a(\mathbf{k}) e^{-ik_0 t} - a^\dagger(-\mathbf{k}) e^{ik_0 t}] e^{i\mathbf{k}\mathbf{x}} \right. \\
& \quad \times \left. \int \frac{d^3 \mathbf{q}}{\sqrt{(2\pi)^3 (2q_0)}} (-iq_0) [a(\mathbf{q}) e^{-iq_0 t} - a^\dagger(-\mathbf{q}) e^{iq_0 t}] e^{i\mathbf{q}\mathbf{x}} \right. \\
& = -\frac{1}{4(2\pi)^3} \int d^3 \mathbf{x} \left\{ \int \frac{d^3 \mathbf{k} d^3 \mathbf{q}}{\sqrt{k_0 q_0}} k_0 q_0 [a(\mathbf{k}) a(\mathbf{q}) e^{-i(k_0+q_0)t} \right. \\
& \quad - a(\mathbf{k}) a^\dagger(-\mathbf{q}) e^{-i(k_0-q_0)t} - a^\dagger(-\mathbf{k}) a(\mathbf{q}) e^{i(k_0-q_0)t} \\
& \quad \left. + a^\dagger(-\mathbf{k}) a^\dagger(-\mathbf{q}) e^{i(k_0+q_0)t}] e^{i(\mathbf{k}+\mathbf{q})\mathbf{x}} \right\} \\
& = -\frac{1}{4(2\pi)^3} \int d^3 \mathbf{x} e^{i(\mathbf{k}+\mathbf{q})\mathbf{x}} \left\{ \int \frac{d^3 \mathbf{k} d^3 \mathbf{q}}{\sqrt{k_0 q_0}} k_0 q_0 [a(\mathbf{k}) a(\mathbf{q}) e^{-i(k_0+q_0)t} \quad [\dagger] \right. \\
& \quad - a(\mathbf{k}) a^\dagger(-\mathbf{q}) e^{-i(k_0-q_0)t} - a^\dagger(-\mathbf{k}) a(\mathbf{q}) e^{i(k_0-q_0)t} \\
& \quad \left. + a^\dagger(-\mathbf{k}) a^\dagger(-\mathbf{q}) e^{i(k_0+q_0)t}] \right\} \\
& = -\frac{1}{4(2\pi)^3} (2\pi)^3 \int \frac{d^3 \mathbf{k} d^3 \mathbf{q}}{\sqrt{k_0 q_0}} \delta(\mathbf{k}+\mathbf{q}) k_0 q_0 [a(\mathbf{k}) a(\mathbf{q}) e^{-i(k_0+q_0)t} \quad [\dagger\dagger] \\
& \quad - a(\mathbf{k}) a^\dagger(-\mathbf{q}) e^{-i(k_0-q_0)t} - a^\dagger(-\mathbf{k}) a(\mathbf{q}) e^{i(k_0-q_0)t} \\
& \quad + a^\dagger(-\mathbf{k}) a^\dagger(-\mathbf{q}) e^{i(k_0+q_0)t}] \\
& = -\frac{1}{4(2\pi)^3} (2\pi)^3 \int \frac{d^3 \mathbf{k}}{\sqrt{k_0 q_0}} k_0 q_0 [a(\mathbf{k}) a(-\mathbf{k}) e^{-i(k_0+q_0)t} \\
& \quad - a(\mathbf{k}) a^\dagger(\mathbf{k}) e^{-i(k_0-q_0)t} - a^\dagger(-\mathbf{k}) a(-\mathbf{k}) e^{i(k_0-q_0)t} \\
& \quad + a^\dagger(-\mathbf{k}) a^\dagger(\mathbf{k}) e^{i(k_0+q_0)t}] \\
& = \frac{1}{4} \int \frac{d^3 \mathbf{k}}{\sqrt{k_0 q_0}} k_0 q_0 [-a(\mathbf{k}) a(-\mathbf{k}) e^{-i(k_0+q_0)t} + a^\dagger(\mathbf{k}) a^\dagger(\mathbf{k}) e^{-i(k_0-q_0)t} \\
& \quad + a^\dagger(-\mathbf{k}) a(-\mathbf{k}) e^{i(k_0-q_0)t} - a^\dagger(-\mathbf{k}) a^\dagger(\mathbf{k}) e^{i(k_0+q_0)t} \\
& = \frac{1}{4} \int d^3 \mathbf{k} k_0 [-a(\mathbf{k}) a(-\mathbf{k}) e^{-2ik_0 t} + a^\dagger(\mathbf{k}) a^\dagger(\mathbf{k}) + a^\dagger(\mathbf{k}) a(\mathbf{k}) \quad [\dagger\dagger\dagger\dagger] \\
& \quad - a^\dagger(-\mathbf{k}) a^\dagger(\mathbf{k}) e^{2ik_0 t}].
\end{aligned}$$

(22.111)

In the above somewhat lengthy calculations, we have several points. Namely, at [†] we exchanged the order of the integrations $d^3\mathbf{x}$ and $d^3\mathbf{k}d^3\mathbf{q}$ with respect to the factor $e^{i(\mathbf{k} + \mathbf{q})\cdot\mathbf{x}}$. This exchange allowed us to use (22.101) at [††]. At [† † †] we switched $-\mathbf{k}$ to $\mathbf{k}$ with regard to the factor $a^\dagger(-\mathbf{k})a(-\mathbf{k})$. Also, we used $k_0 = q_0$. It is because since $k_0 = \sqrt{\mathbf{k}^2 + m^2}$ and $q_0 = \sqrt{\mathbf{q}^2 + m^2}$, we have $k_0 = q_0$ in the case of $\mathbf{k} + \mathbf{q} = 0$, which comes from the factor $\delta(\mathbf{k} + \mathbf{q})$ appearing in [††].

The second and third terms of (22.110) can readily be calculated in a similar manner. We describe the results as follows:

$$\begin{aligned} & \frac{1}{2} \int d^3\mathbf{x} [\nabla\phi(x)]^2 \\ &= \frac{1}{4} \int \frac{d^3\mathbf{k}}{\sqrt{k_0 q_0}} k^2 \left[+\mathfrak{a}(\mathbf{k})a(-\mathbf{k})e^{-2ik_0 t} + \mathfrak{a}(\mathbf{k})a^\dagger(\mathbf{k}) + \mathfrak{a}^\dagger(\mathbf{k})a(\mathbf{k}) \right. \\ & \quad \left. + \mathfrak{a}^\dagger(-\mathbf{k})a^\dagger(\mathbf{k})e^{2ik_0 t} \right], \end{aligned} \quad (22.112) \leftarrow$$

$$\begin{aligned} & \frac{1}{2} \int d^3\mathbf{x} [m\phi(x)]^2 \\ &= \frac{1}{4} \int \frac{d^3\mathbf{k}}{\sqrt{k_0 q_0}} m^2 \left[+\mathfrak{a}(\mathbf{k})a(-\mathbf{k})e^{-2ik_0 t} + \mathfrak{a}(\mathbf{k})a^\dagger(\mathbf{k}) + \mathfrak{a}^\dagger(\mathbf{k})a(\mathbf{k}) \right. \\ & \quad \left. + \mathfrak{a}^\dagger(-\mathbf{k})a^\dagger(\mathbf{k})e^{2ik_0 t} \right]. \end{aligned} \quad (22.113) \leftarrow$$

Combining (22.112) and (22.113), we obtain

$$\begin{aligned} & \frac{1}{2} \left\{ \int d^3\mathbf{x} [\nabla\phi(x)]^2 + [m\phi(x)]^2 \right\} \\ &= \frac{1}{4} \int \frac{d^3\mathbf{k}}{\sqrt{k_0 q_0}} (k^2 + m^2) \\ & \quad \times \left[+a(\mathbf{k})a(-\mathbf{k})e^{-2ik_0 t} + \mathfrak{a}(\mathbf{k})a^\dagger(\mathbf{k}) + \mathfrak{a}^\dagger(\mathbf{k})a(\mathbf{k}) + \mathfrak{a}^\dagger(-\mathbf{k})a^\dagger(\mathbf{k})e^{2ik_0 t} \right] \\ &= \frac{1}{4} \int d^3\mathbf{k} k_0 \\ & \quad \times \left[+a(\mathbf{k})a(-\mathbf{k})e^{-2ik_0 t} + \mathfrak{a}(\mathbf{k})a^\dagger(\mathbf{k}) + \mathfrak{a}^\dagger(\mathbf{k})a(\mathbf{k}) + \mathfrak{a}^\dagger(-\mathbf{k})a^\dagger(\mathbf{k})e^{2ik_0 t} \right], \end{aligned} \quad (22.114) \leftarrow$$

where we used $\mathbf{k}^2 + m^2 = k_0^2 = q_0^2$. The integrals including the terms $\mp a(\mathbf{k})a(-\mathbf{k})e^{-2ik_0 t}$ and $\mp \mathfrak{a}^\dagger(-\mathbf{k})a^\dagger(\mathbf{k})e^{2ik_0 t}$ in (22.111) and (22.114) are canceled out. Taking account of these results, finally we get the Hamiltonian in a simple form described by

$$\begin{aligned}
 H &= \frac{1}{2} \int d^3 \mathbf{k} k_0 [a(\mathbf{k}) a^\dagger(\mathbf{k}) + a^\dagger(\mathbf{k}) a(\mathbf{k})] \leftarrow \\
 &= \frac{1}{2} \int d^3 \mathbf{k} \sqrt{k^2 + m^2} [a(\mathbf{k}) a^\dagger(\mathbf{k}) + a^\dagger(\mathbf{k}) a(\mathbf{k})],
 \end{aligned} \tag{22.115} \leftarrow$$

where we explicitly described the dependence of the integrand on $\mathbf{k}$.

Let us continue the discussion further. Using the commutation relation (22.105), we have

$$\lim_{\boldsymbol{\epsilon} \rightarrow 0} [a(\mathbf{k}), a^\dagger(\mathbf{k} - \boldsymbol{\epsilon})] = \lim_{\boldsymbol{\epsilon} \rightarrow 0} \delta^3(\boldsymbol{\epsilon}),$$

where $\boldsymbol{\epsilon}$ is an infinitesimal three-dimensional real vector. Then, we have

$$H = \frac{1}{2} \int d^3 \mathbf{k} k_0 \left[a^\dagger(\mathbf{k}) a(\mathbf{k}) + a^\dagger(\mathbf{k}) a(\mathbf{k}) + \lim_{\boldsymbol{\epsilon} \rightarrow 0} \delta^3(\boldsymbol{\epsilon}) \right] \leftarrow \tag{22.116} \leftarrow$$

From (22.101), we obtain

$$\lim_{\boldsymbol{\epsilon} \rightarrow 0} \delta^3(\boldsymbol{\epsilon}) = \frac{1}{(2\pi)^3} \lim_{\boldsymbol{\epsilon} \rightarrow 0} \int e^{i\boldsymbol{\epsilon} \cdot \mathbf{x}} d^3 \mathbf{x} \rightarrow \frac{1}{(2\pi)^3} \int d^3 \mathbf{x}.$$

Therefore, we get

$$H = \int d^3 \mathbf{k} k_0 a^\dagger(\mathbf{k}) a(\mathbf{k}) + \frac{1}{2(2\pi)^3} \int k_0 d^3 \mathbf{k} \int d^3 \mathbf{x}. \tag{22.117} \leftarrow$$

Defining the number density operator $n(\mathbf{k})$ as

$$n(\mathbf{k}) \equiv a^\dagger(\mathbf{k}) a(\mathbf{k}), \tag{22.118} \leftarrow$$

we have

$$H = \int d^3 \mathbf{k} k_0 n(\mathbf{k}) + \frac{1}{2(2\pi)^3} \int d^3 \mathbf{k} d^3 \mathbf{x} \left(\frac{1}{2} k_0 \right). \tag{22.119} \leftarrow$$

The second term of (22.119) means a total energy included in the whole phase space. In other words, the quantity $(k_0/2)$ represents the energy density per “volume” $(2\pi)^3$ of phase space. Note here that in the natural units the dimension of time and length is the inverse of mass (mass dimension). The angular frequency and wavenumber as well as energy and momentum have the same dimension of mass accordingly. This is represented straight in (22.88).

Representing (2.21) in the natural units, we have

$$H = \omega a^\dagger a + \frac{1}{2}\omega. \quad (22.120) \leftarrow$$

The analogy between (22.117) and (22.120) is evident. The operator $n(\mathbf{k})$ is closely related to the number operator introduced in Sect. 2.2; see (2.58). The second term of (22.117) corresponds to the zero-point energy and has relevance to the discussion about the harmonic oscillator of Chap. 2. Thus, we appreciate how deeply the quantization of fields is connected to the treatment of the harmonic oscillators.

In accordance with (2.28) of Sect. 2.2, we define the state of vacuum $|0\rangle$ such that

$$a(\mathbf{k})|0\rangle = 0 \quad \text{for } \forall \mathbf{k}. \quad (22.121) \leftarrow$$

Operating (22.119) on $|0\rangle$, we have

$$\begin{aligned} H|0\rangle &\equiv \int d^3\mathbf{k} k_0 n(\mathbf{k})|0\rangle + \frac{1}{(2\pi)^3} \int d^3\mathbf{k} d^3\mathbf{x} \left(\frac{1}{2}k_0\right)|0\rangle \\ &= \frac{1}{(2\pi)^3} \int d^3\mathbf{k} d^3\mathbf{x} \left(\frac{1}{2}k_0\right)|0\rangle. \end{aligned} \quad (22.122) \leftarrow$$

That is, $|0\rangle$ seems to be an “eigenstate” of H that belongs to an “eigenvalue” of $\frac{1}{(2\pi)^3} \int d^3\mathbf{k} d^3\mathbf{x} (\frac{1}{2}k_0)$. Since this value diverges to infinity, it is usually ignored and we plainly state that the energy of $|0\rangle$ (i.e., vacuum) is zero. To discuss the implication of (22.119) and (22.122) is beyond the scope of this book and, hence, we do not get into further details about this issue.

In contrast to the above argument, the following momentum operator P^μ has a well-defined meaning:

$$P^\mu \equiv \int d^3\mathbf{k} k^\mu n(\mathbf{k}) + \frac{1}{(2\pi)^3} \int d^3\mathbf{k} d^3\mathbf{x} \left(\frac{1}{2}k^\mu\right) \quad (\mu = 1, 2, 3). \quad (22.123) \leftarrow$$

The second term vanishes, because k^μ ($\mu = 1, 2, 3$) is an antisymmetric function with respect to k^μ ($\mu = 1, 2, 3$). Then, we get

$$P^\mu = \int d^3\mathbf{k} k^\mu n(\mathbf{k}) \quad (\mu = 1, 2, 3). \quad (22.124) \leftarrow$$

Notice that from (22.119) and (22.124), if we ignore the second term of (22.119) and the index μ is extended to 0, we have $P^0 \equiv H$. Thus, we reformulate P^μ as a four-vector such that

$$P^\mu = \int d^3\mathbf{k} k^\mu n(\mathbf{k}) \quad (\mu = 0, 1, 2, 3). \quad (22.125)$$

Meanwhile, using the creation operators $a^\dagger(\mathbf{k})$, let us consider the following states:

$$|\mathbf{k}_1, k_2, \dots, k_n\rangle \equiv a^\dagger(\mathbf{k}_1)a^\dagger(\mathbf{k}_2)\cdots a^\dagger(\mathbf{k}_n) |0\rangle, \quad (22.126)$$

where the freedom of the dynamical system is specified by an integer i ($i = 1, \dots, n$); k_i denotes the four-momentum of a particle that is present at the i th state.

Operating $a^\dagger(\mathbf{q})a(\mathbf{q})$ on both sides of (22.126) from the left, we have

$$\begin{aligned} a^\dagger(\mathbf{q})a(\mathbf{q}) |\mathbf{k}_1, k_2, \dots, k_n\rangle &= a^\dagger(\mathbf{q})a(\mathbf{q})a^\dagger(\mathbf{k}_1)a^\dagger(\mathbf{k}_2)\cdots a^\dagger(\mathbf{k}_n) |0\rangle \\ &= a^\dagger(\mathbf{q})[a^\dagger(\mathbf{k}_1)a(\mathbf{q}) + \delta^3(\mathbf{q} - \mathbf{k}_1)]a^\dagger(\mathbf{k}_2)\cdots a^\dagger(\mathbf{k}_n) |0\rangle \\ &= a^\dagger(\mathbf{q})a^\dagger(\mathbf{k}_1)a(\mathbf{q})a^\dagger(\mathbf{k}_2)\cdots a^\dagger(\mathbf{k}_n) |0\rangle + \delta^3(\mathbf{q} - \mathbf{k}_1)a^\dagger(\mathbf{q})a^\dagger(\mathbf{k}_2)\cdots a^\dagger(\mathbf{k}_n) |0\rangle \\ &= a^\dagger(\mathbf{q})a^\dagger(\mathbf{k}_1)[a^\dagger(\mathbf{k}_2)a(\mathbf{q}) + \delta^3(\mathbf{q} - \mathbf{k}_2)]\cdots a^\dagger(\mathbf{k}_n) |0\rangle \\ &\quad + \delta^3(\mathbf{q} - \mathbf{k}_1)a^\dagger(\mathbf{q})a^\dagger(\mathbf{k}_2)\cdots a^\dagger(\mathbf{k}_n) |0\rangle \\ &= a^\dagger(\mathbf{q})a^\dagger(\mathbf{k}_1)a^\dagger(\mathbf{k}_2)a(\mathbf{q})\cdots a^\dagger(\mathbf{k}_n) |0\rangle + \delta^3(\mathbf{q} - \mathbf{k}_2)a^\dagger(\mathbf{q})a^\dagger(\mathbf{k}_1)\cdots a^\dagger(\mathbf{k}_n) |0\rangle \\ &\quad + \delta^3(\mathbf{q} - \mathbf{k}_1)a^\dagger(\mathbf{q})a^\dagger(\mathbf{k}_2)\cdots a^\dagger(\mathbf{k}_n) |0\rangle \\ &= \cdots \\ &= a^\dagger(\mathbf{q})a^\dagger(\mathbf{k}_1)a^\dagger(\mathbf{k}_2)\cdots a^\dagger(\mathbf{k}_n)a(\mathbf{q}) |0\rangle + \delta^3(\mathbf{q} - \mathbf{k}_n)a^\dagger(\mathbf{q})a^\dagger(\mathbf{k}_1)\cdots a^\dagger(\mathbf{k}_{n-1}) |0\rangle \\ &\quad + \cdots + \delta^3(\mathbf{q} - \mathbf{k}_1)a^\dagger(\mathbf{q})a^\dagger(\mathbf{k}_2)\cdots a^\dagger(\mathbf{k}_n) |0\rangle \\ &= \delta^3(\mathbf{q} - \mathbf{k}_n)a^\dagger(\mathbf{q})a^\dagger(\mathbf{k}_1)\cdots a^\dagger(\mathbf{k}_{n-1}) |0\rangle \\ &\quad + \cdots + \delta^3(\mathbf{q} - \mathbf{k}_1)a^\dagger(\mathbf{q})a^\dagger(\mathbf{k}_2)\cdots a^\dagger(\mathbf{k}_n) |0\rangle, \end{aligned} \quad (22.127)$$

where with the second to the last equality the first term vanishes because from (22.121) we have $a(\mathbf{q}) |0\rangle = 0$; with the last equality n terms containing the $\delta^3(\mathbf{q} - \mathbf{k}_i)$ ($i = 1, \dots, n$) factor survive.

Thus, from (22.118), (22.125), and (22.127) we get

$$\begin{aligned} P^\mu |\mathbf{k}_1, k_2, \dots, k_n\rangle &= \int d^3\mathbf{q} k^\mu a^\dagger(\mathbf{q})a(\mathbf{q}) |\mathbf{k}_1, k_2, \dots, k_n\rangle \\ &= \int d^3\mathbf{q} k^\mu [\delta^3(\mathbf{q} - \mathbf{k}_n)a^\dagger(\mathbf{q})a^\dagger(\mathbf{k}_1)\cdots a^\dagger(\mathbf{k}_{n-1}) |0\rangle \\ &\quad + \cdots + \delta^3(\mathbf{q} - \mathbf{k}_1)a^\dagger(\mathbf{q})a^\dagger(\mathbf{k}_2)\cdots a^\dagger(\mathbf{k}_n) |0\rangle] \\ &= k_n^\mu a^\dagger(\mathbf{k}_n)a^\dagger(\mathbf{k}_1)\cdots a^\dagger(\mathbf{k}_{n-1}) |0\rangle + \cdots + k_1^\mu a^\dagger(\mathbf{k}_1)a^\dagger(\mathbf{k}_2)\cdots a^\dagger(\mathbf{k}_n) |0\rangle \\ &= (k_1^\mu + \cdots + k_n^\mu) a^\dagger(\mathbf{k}_1)a^\dagger(\mathbf{k}_2)\cdots a^\dagger(\mathbf{k}_n) |0\rangle \\ &= (k_1^\mu + \cdots + k_n^\mu) |\mathbf{k}_1, k_2, \dots, k_n\rangle, \end{aligned} \quad (22.128)$$

where with the second to the last equality we used the second equation of (22.105); with the last equality we used the definition of (22.126). With respect to the energy of j th state, we have

$$k_j^0 \equiv \sqrt{\mathbf{k}_j^2 + m^2} \quad (> 0). \quad (22.129)$$

Therefore, once $\mathbf{k}_j$ ($j = 1, \dots, n$) is determined from (22.128), k_j^0 is also determined from (22.129) and, hence, k_j^μ ($\mu = 0, 1, 2, 3$) is determined accordingly. Thus, we obtain [8]

$$P^\mu |k_1, k_2, \dots, k_n\rangle = (k_1^\mu + \dots + k_n^\mu) |k_1, k_2, \dots, k_n\rangle \quad (\mu = 1, 2, 3). \quad (22.130)$$

From (22.130), we find that $|k_1, k_2, \dots, k_n\rangle$ is the eigenstate of the four-momentum operator P^μ ($\mu = 0, 1, 2, 3$) that belongs to the eigenvalue $(k_1^\mu + \dots + k_n^\mu)$. The state described by $|k_1, k_2, \dots, k_n\rangle$ is called a Fock basis. A Hilbert space spanned by the Fock basis is called a Fock space. As in the case of Chap. 2, the operator $a^\dagger(\mathbf{k})$ is interpreted as an operator that creates a particle having momentum $\mathbf{k}$.

The calculation procedures of (22.127) naturally lead to the normal product (or normal-ordered product), where the annihilation operators $a(\mathbf{k})$ always stand to the right of the creation operators $a^\dagger(\mathbf{k})$. This is because if so, any term that contains $a(\mathbf{k})$ as a factor vanishes in virtue of (22.121). This situation is reminiscent of (2.28), where $|a\psi_0\rangle = 0$ corresponds to $a(\mathbf{k})|0\rangle = 0$.

22.3.4 Invariant Delta Functions of the Scalar Field

We have already mentioned that the Lorentz invariance (or covariance) is not necessarily clear with the equal-time commutation relation. In this section we describe characteristics of the invariant delta functions of the scalar field to explicitly show the Lorentz invariance of these functions. In this section, we deal with the invariant delta functions of the scalar field in detail, since their usage offers a typical example as a prototype of the field quantization and, hence, can be applied to the Dirac field and photon field as well.

The invariant delta function is defined using a commutator as

$$i\Delta(x - y) \equiv [\phi(x), \phi(y)], \quad (22.131)$$

where x and y represent an arbitrary space-time four-coordinate such that $x = \begin{pmatrix} t \\ \mathbf{x} \end{pmatrix}$

and $y = \begin{pmatrix} t' \\ \mathbf{y} \end{pmatrix}$. Several different invariant delta functions will appear soon. Since their definition and notation differ depending on literature, care should be taken. Note that although we have first imposed the condition $t = t'$ on the commutation relation, it is not the case with the present consideration. Since the invariant delta function $\Delta(x - y)$ is a scalar function as we shall see below, we must have a Lorentz

invariant form for it. One of the major purposes of this section is to represent $\Delta(x - y)$ and related functions in the Lorentz invariant form.

Dividing the positive-frequency part and negative-frequency part of $\phi(x)$ and $\phi(y)$ such that [5]

$$\begin{aligned}\phi(x) &= \phi^+(x) + \phi^-(x), \\ \phi(y) &= \phi^+(y) + \phi^-(y),\end{aligned}\tag{22.132} \leftarrow$$

we have

$$\begin{aligned}[\phi(x), \phi(y)] &= [\phi^+(x), \phi^+(y)] + [\phi^+(x), \phi^-(y)] \\ &\quad + [\phi^-(x), \phi^+(y)] + [\phi^-(x), \phi^-(y)].\end{aligned}\tag{22.133} \leftarrow$$

From the expansion formulae of (22.99) as well as the second equation of (22.105), we find that

$$[\phi^+(x), \phi^+(y)] = [\phi^-(x), \phi^-(y)] = 0.\tag{22.134} \leftarrow$$

Then, we have

$$[\phi(x), \phi(y)] = [\phi^+(x), \phi^-(y)] + [\phi^-(x), \phi^+(y)].\tag{22.135} \leftarrow$$

Next, applying the first equation of (22.105) to the commutator consisting of (22.99), we obtain

$$\begin{aligned}[\phi^+(x), \phi^-(y)] &= \int \frac{d^3\mathbf{k} d^3\mathbf{q}}{(2\pi)^3 2\sqrt{k_0 q_0}} [a(\mathbf{k}) e^{-ikx}, a^\dagger(\mathbf{q}) e^{iqy}] \\ &= \int \frac{d^3\mathbf{k} d^3\mathbf{q}}{(2\pi)^3 2\sqrt{k_0 q_0}} e^{-i(kx - qy)} [a(\mathbf{k}), a^\dagger(\mathbf{q})] = \int \frac{d^3\mathbf{k} d^3\mathbf{q}}{(2\pi)^3 2\sqrt{k_0 q_0}} e^{-i(kx - qy)} \delta^3(\mathbf{k} - \mathbf{q}) \\ &= \frac{1}{(2\pi)^3} \int \frac{d^3\mathbf{k}}{2k_0} e^{-ik(x - y)},\end{aligned}\tag{22.136} \leftarrow$$

where with the third equality we used the first equation of (22.105). Also, in (22.136) we used the fact that since $k_0 = \sqrt{\mathbf{k}^2 + m^2}$ and $q_0 = \sqrt{\mathbf{q}^2 + m^2}$ with $\mathbf{k} = \mathbf{q}$, we have $k_0 = q_0$ accordingly (vide supra). We define one of the invariant delta functions, i.e., $\Delta^+(x - y)$ as

$$i\Delta^+(\bar{x}-y) \equiv [\phi^+(\bar{x}), \phi^-(y)] = \frac{1}{(2\pi)^3} \int \frac{d^3\mathbf{k}}{2k_0} e^{-ik(x-y)}. \quad (22.137) \leftarrow$$

Exchanging the coordinate x and y in (22.137) and reversing the order of the operators in the commutator, we get the following expression where another invariant delta function $\Delta^-(x-y)$ is defined as [5]

$$\begin{aligned} i\Delta^-(x-y) &\equiv [\phi^-(x), \phi^+(\bar{y})] = -i\Delta^+(\bar{y}-x) = \\ &= -\frac{1}{(2\pi)^3} \int \frac{d^3\mathbf{k}}{2k_0} e^{-ik(y-x)}. \end{aligned} \quad (22.138) \leftarrow$$

From (22.131), (22.135), (22.137), and (22.138), we have

$$\Delta(x-y) = \Delta^+(\bar{x}-y) + \Delta^-(x-y). \quad (22.139) \leftarrow$$

From (22.137) and (22.138), we can readily get the following expression described by

$$\begin{aligned} (\square_x + \not{m}^2)\Delta^+(\bar{x}-y) &= (\square_x + \not{m}^2)\Delta^-(x-y) = 0, \\ (\square_x + \not{m}^2)\Delta(x-y) &= 0, \end{aligned} \quad (22.140) \leftarrow$$

where $\square_x$ means that the differentiation should be performed with respect to x .

Note that upon defining the invariant delta functions, the notations and equalities vary from literature to literature. In this book, to avoid the complexity of the notation for “contraction,” we have chosen (22.137) and (22.138) for the definitions of the invariant delta functions. The contraction is frequently used for the perturbation calculation, which will appear later in Chap. 23.

The contraction is a convenient notation to describe a time-ordered product (or T-product) that is defined as $\langle 0|T[\phi(x)\phi(y)]|0\rangle$ (see Sect. 22.3.5). The time-ordered product is represented by the following symbol:

$$\langle 0|T[\phi(x)\phi(y)]|0\rangle \equiv \underbrace{\phi(x)\phi(y)}. \quad (22.141) \leftarrow$$

Equation (22.141) is referred to as the Feynman propagator and the discussion about it will be made in the next section.

To show that (22.137) is explicitly Lorentz invariant, we wish to convert the integral with respect to $\mathbf{k}$ into the integral with respect to $k = \begin{pmatrix} k_0 \\ \mathbf{k} \end{pmatrix}$. For this purpose, we use the δ function. We have

$$\begin{aligned}
\delta(k^2 - m^2) &= \delta[k_0^2 - (k^2 + m^2)] \\
&= \frac{1}{2\sqrt{k^2 + m^2}} \left[\delta\left(k_0 - \sqrt{k^2 + m^2}\right) + \delta\left(k_0 + \sqrt{k^2 + m^2}\right) \right].
\end{aligned} \tag{22.142} \leftarrow$$

This relation results from the following property of the δ function [10]. We have

$$\delta[f(x)] = \sum_n \frac{\delta(x - a_n)}{|f'(a_n)|}, \quad \leftarrow \tag{22.143} \leftarrow$$

where $f(a_n) = 0, f'(a_n) \neq 0$. As a special case, we obtain [4]

$$\delta(x^2 - a^2) = \frac{1}{2a} [\delta(x - a) + \delta(x + a)] \quad (a > 0). \tag{22.144} \leftarrow$$

Replacing x with k_0 and a with $\sqrt{k^2 + m^2}$, we get (22.142).

Taking account of $k_0 > 0$ in (22.136), we wish to pick a positive value up in (22.142). Multiplying both sides of (22.142) by $\theta(k_0)$, we get

$$\begin{aligned}
\delta(k^2 - m^2)\theta(k_0) &\leftarrow \\
&= \frac{1}{2\sqrt{k^2 + m^2}} \left[\delta\left(k_0 - \sqrt{k^2 + m^2}\right)\theta(k_0) + \delta\left(k_0 + \sqrt{k^2 + m^2}\right)\theta(k_0) \right]. \leftarrow
\end{aligned} \tag{22.145} \leftarrow$$

Since the argument in the δ function is positive in the second term of (22.145), that term vanishes and we are left with

$$\delta(k^2 - m^2)\theta(k_0) = \frac{1}{2\sqrt{k^2 + m^2}} \delta\left(k_0 - \sqrt{k^2 + m^2}\right). \tag{22.146} \leftarrow$$

Operating an arbitrary function $f(k_0)$ on both sides of (22.146), we obtain

$$\begin{aligned}
\delta(k^2 - m^2)\theta(k_0)f(k_0) &= \frac{1}{2\sqrt{k^2 + m^2}} \delta\left(k_0 - \sqrt{k^2 + m^2}\right)f(k_0) \leftarrow \\
&= \frac{1}{2\sqrt{k^2 + m^2}} \delta\left(k_0 - \sqrt{k^2 + m^2}\right)f\left(\sqrt{k^2 + m^2}\right),
\end{aligned} \tag{22.147} \leftarrow$$

where with the second equality we used the following relation [11]:

$$f(x) \delta(x - a) = f(a) \delta(x - a). \quad (22.148) \leftarrow$$

Integrating (22.147) with respect to k_0 , we obtain

$$\begin{aligned} & \int dk_0 \delta(k^2 - m^2) \theta(k_0) f(k_0) \leftarrow \\ &= \int dk_0 \frac{1}{2\sqrt{k^2 + m^2}} \delta\left(k_0 - \sqrt{k^2 + m^2}\right) f\left(\sqrt{k^2 + m^2}\right) \quad (22.149) \leftarrow \\ &= \frac{1}{2\sqrt{k^2 + m^2}} f\left(\sqrt{k^2 + m^2}\right). \end{aligned}$$

Multiplying both sides of (22.149) by $g(\mathbf{k})$ and further integrating it with respect to $\mathbf{k}$, we get

$$\begin{aligned} & \int d^3k dk_0 \delta(k^2 - m^2) \theta(k_0) f(k_0) g(\mathbf{k}) \leftarrow \\ &= \int d^4k \delta(k^2 - m^2) \theta(k_0) f(k_0) g(\mathbf{k}) = \int d^3k \frac{f\left(\sqrt{k^2 + m^2}\right)}{2\sqrt{k^2 + m^2}} g(\mathbf{k}). \quad (22.150) \leftarrow \end{aligned}$$

We wish to use (22.150) to show that (22.137) is explicitly Lorentz invariant. To this end, we put

$$f(k_0) \equiv e^{-ik_0(x^0 - y^0)} \quad \text{and} \quad g(\mathbf{k}) \equiv e^{i\mathbf{k}(x - y)}. \quad (22.151) \leftarrow$$

Under the condition of $k_0 = \sqrt{k^2 + m^2}$, in RHS of (22.150) we have

$$\frac{f\left(\sqrt{k^2 + m^2}\right)}{2\sqrt{k^2 + m^2}} g(\mathbf{k}) = \frac{1}{2k_0} e^{-ik_0(x^0 - y^0)} e^{i\mathbf{k}(x - y)} = \frac{1}{2k_0} e^{-ik(x - y)}.$$

Applying (22.150) to (22.137), we get

$$\begin{aligned} i\Delta^+(x - y) &= \frac{1}{(2\pi)^3} \int \frac{d^3\mathbf{k}}{2k_0} e^{-ik(x - y)} \leftarrow \\ &= \frac{1}{(2\pi)^3} \int d^4k \delta(k^2 - m^2) \theta(k_0) e^{-ik(x - y)}. \quad (22.152) \leftarrow \end{aligned}$$

Note that with the first equality of (22.152) the factor $e^{-ik(x - y)}$ is given by

$$e^{-ik(x-y)} = e^{-i\sqrt{k^2+m^2}(x^0-y^0)} e^{ik(x-y)} \quad \leftarrow \quad (22.153) \leftarrow$$

with k_0 replaced with $\sqrt{k^2 + m^2}$. With the second equality of (22.152), however, k_0 is changed to an explicit integration variable. Also, notice that in spite of the above change in k_0 , the function $e^{-ik(x-y)}$ is represented in the same form in (22.152) throughout. The important point is that the positive number $\sqrt{k^2 + m^2}$ does not appear in LHS of (22.145) explicitly, but that number is absorbed into the Lorentz invariant form of $\delta(k^2 - m^2)\theta(k_0)$. Thus, (22.137) has successfully been converted into the explicitly Lorentz invariant form.

In a similar way, for $\Delta^-(x-y)$ we obtain another Lorentz invariant form described by

$$i\Delta^-(x-y) = \frac{-1}{(2\pi)^3} \int dk \delta(k^2 - m^2) \theta(k_0) e^{ik(x-y)}. \quad (22.154) \leftarrow$$

From (22.137) to (22.139) as well as (22.152) and (22.154), we have

$$\begin{aligned} i\Delta(x-y) &= i\Delta^+(x-y) + i\Delta^-(x-y) \quad \leftarrow \\ &= [\phi^+(x), \phi^-(y)] + [\phi^-(x), \phi^+(y)] \\ &= \frac{1}{(2\pi)^3} \int d^4k \delta(k^2 - m^2) \theta(k_0) \left[e^{-ik(x-y)} - e^{ik(x-y)} \right] \quad \leftarrow (22.155) \leftarrow \\ &= \frac{-2i}{(2\pi)^3} \int d^4k \delta(k^2 - m^2) \theta(k_0) \sin k(x-y). \end{aligned}$$

Note that as a representation equivalent to (22.155), we have

$$i\Delta(x-y) = \frac{1}{(2\pi)^3} \int d^4k \delta(k^2 - m^2) \varepsilon(k_0) e^{-ik(x-y)}, \quad (22.156) \leftarrow$$

where $\varepsilon(x)$ is defined by

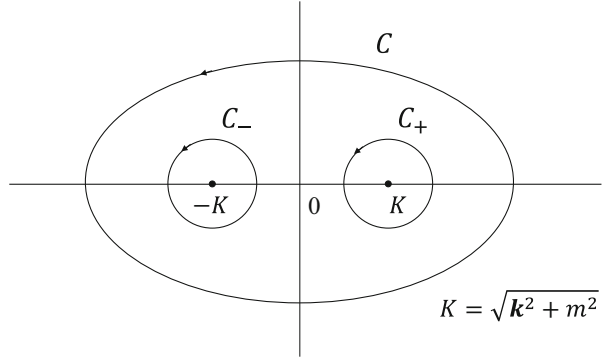
$$\varepsilon(x) \equiv x/|x|, \varepsilon(0) \equiv 0 \quad (x : \text{real}). \quad (22.157) \leftarrow$$

To show that (22.155) and (22.156) are identical, use the fact that $\varepsilon(x)$ is an odd function with respect to x that is expressed as

$$\varepsilon(x) = \theta(x) - \theta(-x). \quad (22.158) \leftarrow$$

Thus, we have shown that all the above functions $\Delta^+(x-y)$, $\Delta^-(x-y)$, and $\Delta(x-y)$ are Lorentz invariant. On account of the Lorentz invariance, these functions are said to be invariant delta functions. Since the invariant delta functions are given

Fig. 22.1 Contours to evaluate the complex integrations with respect to the invariant delta functions $\Delta^\pm(x-y)$ and $\Delta(x-y)$ of the scalar field. The contours C_+ , C_- , and C correspond to $\Delta^+(x-y)$, $\Delta^-(x-y)$, and $\Delta(x-y)$, respectively



as a scalar, they are of special importance in the quantum field theory. This situation is the same with the Dirac field and photon field as well.

In (22.152) and (22.154), we assume that the integration range spans the whole real numbers from $-\infty$ to $+\infty$. As another way to represent the invariant delta functions in the Lorentz invariant form, we may choose the complex variable for k_0 . The contour integration method developed in Chap. 6 is useful for the present purpose. In Fig. 22.1, we draw several contours to evaluate the complex integrations with respect to the invariant delta functions. With these contours, the invariant delta functions are represented as [5]

$$\Delta^\pm(x-y) = \frac{1}{(2\pi)^4} \int_{C_\pm} d^4k \frac{e^{-ik(x-y)}}{m^2 - k^2}. \quad (22.159)$$

To examine whether (22.159) is equivalent to (22.137) or (22.138), we rewrite the denominator of (22.159) as

$$m^2 - k^2 = m^2 + k_0^2 - \mathbf{k}^2. \quad (22.160)$$

Further defining a positive number $K \equiv \sqrt{\mathbf{k}^2 + m^2}$ in accordance with (22.149) and defining the integral $I_{C_\pm}$ as

$$I_{C_\pm} \equiv \int_{C_\pm} d^4k \frac{e^{-ik(x-y)}}{m^2 - k^2}, \quad (22.161)$$

we have

$$\begin{aligned}
 I_{C_{\pm}} &= \int_{C_{\pm}} d^4k \frac{e^{-ik(x-y)}}{m^2 - k^2} = \int_{C_{\pm}} d^4k \frac{1}{2K} e^{-ik(x-y)} \left(\frac{1}{k_0 + K} - \frac{1}{k_0 - K} \right) \\
 &= \int d^3k e^{ik(x-y)} \frac{1}{2K} \int_{C_{\pm}} dk_0 e^{-ik_0(x^0 - y^0)} \left(\frac{1}{k_0 + K} - \frac{1}{k_0 - K} \right). \quad (22.162)
 \end{aligned}$$

The function $\frac{1}{k_0 + K}$ is analytic within the complex domain k_0 encircled by C_+ . With the function $\frac{1}{k_0 - K}$, however, a simple pole is present within the said domain (at $k_0 = K$). Therefore, choosing C_+ for the contour, only the second term contributes to the integral with the integration I_{C_+} . It is given by

$$\begin{aligned}
 I_{C_{++}} &= \int_{C_{++}} d^4k \frac{1}{2K} e^{-ik(x-y)} \left(-\frac{1}{k_0 - K} \right) \\
 &= \int d^3k e^{ik(x-y)} \frac{1}{2K} \int_{C_{++}} dk_0 e^{-ik_0(x^0 - y^0)} \left(-\frac{1}{k_0 - K} \right) \\
 &= \int d^3k e^{ik(x-y)} \frac{1}{2K} \left\{ 2\pi i \text{Res} \left[e^{-ik_0(x^0 - y^0)} \left(-\frac{1}{k_0 - K} \right) \right]_{k_0=K} \right\} \\
 &= 2\pi i \int d^3k e^{ik(x-y)} \frac{1}{2K} \left[-e^{-iK(x^0 - y^0)} \right], \quad (22.163)
 \end{aligned}$$

where with the third equality we used (6.146) and (6.148). Then, from (22.159) Δ^+ is given by [5]

$$\Delta^+(x-y) = \frac{1}{(2\pi)^4} I_{C_{++}} = \frac{-i}{(2\pi)^3} \int d^3k \frac{1}{2K} e^{-ik(x-y)}, \quad (22.164)$$

where we have

$$e^{-ik(x-y)} = e^{-iK(x^0 - y^0)} e^{i\mathbf{k}(x - y)}. \quad (22.165)$$

Thus, $\Delta^+(x-y)$ of (22.159) is obviously identical to (22.137) if C_+ is chosen for the contour. Notice that in (22.164) k_0 is no longer an integration variable, but a real positive number K ($= \sqrt{\mathbf{k}^2 + m^2}$) as in (22.165), independent of k_0 . In turn, the function $\frac{1}{k_0 + K}$ is analytic within the complex domain encircled by C_- . With the function $\frac{1}{k_0 - K}$, a simple pole is present within the said domain. Therefore, choosing C_- for the contour, only the first term of RHS of (22.162) contributes to the integral with the integration I_{C_-} . It is given by

$$\begin{aligned}
I_{C_-} &= \int_{C_-} d^4k \frac{1}{2K} e^{-ik(x-y)} \left(\frac{1}{k_0 + iK} \right) \\
&= \int d^3k e^{ik(x-y)} \frac{1}{2K} \int_{C_-} dk_0 e^{-ik_0(x^0-y^0)} \left(\frac{1}{k_0 + iK} \right) \\
&= \int d^3k e^{ik(x-y)} \frac{1}{2K} \left\{ 2\pi i \text{Res} \left[e^{-ik_0(x^0-y^0)} \left(\frac{1}{k_0 + iK} \right) \right]_{k_0 = -K} \right\} \quad (22.166) \leftarrow \\
&= 2\pi i \int \frac{1}{2K} d^3k e^{-ik(x-y)} e^{iK(x^0-y^0)} \quad \leftarrow \\
&= 2\pi i \int d^3k \frac{1}{2K} e^{ik(x-y)},
\end{aligned}$$

where with the third equality we used (6.148) again. Also, note that with the second to the last equality we used

$$\int d^3k e^{ik(x-y)} = \oint d^3k e^{-ik(x-y)}. \quad (22.167) \leftarrow$$

Then, $\Delta^-(x-y)$ is given by

$$\Delta^-(x-y) = \frac{1}{(2\pi)^4} I_{C_-} = \frac{i}{(2\pi)^3} \int d^3k \frac{1}{2K} e^{ik(x-y)}, \quad (22.168) \leftarrow$$

where $e^{ik(x-y)} = e^{iK(x^0-y^0)} e^{-ik(x-y)}$. This shows that $\Delta^-(x-y)$ of (22.159) is identical with (22.138) if C_- is chosen for the contour.

Then, from (22.139), (22.164), and (22.168), we have [5]

$$\begin{aligned}
\Delta(x-y) &= \frac{-i}{(2\pi)^3} \int d^3k \frac{1}{2K} \left[e^{-ik(x-y)} - e^{ik(x-y)} \right] \quad \leftarrow \\
&= \frac{-1}{(2\pi)^3} \int d^3k \frac{1}{K} \sin k(x-y). \quad (22.169) \leftarrow
\end{aligned}$$

Or from (22.131), we have equivalently

$$[\phi(x), \phi(y)] = \frac{1}{(2\pi)^3} \int d^3k \frac{1}{2K} \left[e^{-ik(x-y)} - e^{ik(x-y)} \right]. \quad \leftarrow \quad (22.170) \leftarrow$$

This relation can readily be derived from (22.137) and (22.138) as well. Returning to the notation (22.159), we get

$$\begin{aligned}
 \Delta(x-y) &= \frac{1}{(2\pi)^4} (I_{C_+} + I_{C_-}) = \frac{1}{(2\pi)^4} \left[\int_{C_+} d^4k \frac{e^{-ik(x-y)}}{m^2 - k^2} + \int_{C_-} d^4k \frac{e^{-ik(x-y)}}{m^2 - k^2} \right] \\
 &= \frac{1}{(2\pi)^4} \int_C d^4k \frac{e^{-ik(x-y)}}{m^2 - k^2},
 \end{aligned} \tag{22.171}$$

where C is a contour that encircles both C_+ and C_- as shown in Fig. 22.1. This is evident from (6.87) and (6.146) that relate the complex integral with residues.

We summarize major properties of the invariant delta functions below. From (22.169), we have

$$\Delta(x) = -i[\phi(x), \phi(0)] = -i \int \frac{d^3\mathbf{k}}{(2\pi)^3 2k_0} (e^{-ikx} - e^{ikx}). \tag{22.172}$$

Therefore, $\Delta(x) = -\Delta(-x)$ (i.e., odd function). Putting $x^0 = 0$ in (22.172), we have

$$\Delta(x)|_{x^0=0} = -i \int \frac{d^3\mathbf{k}}{(2\pi)^3 2k_0} (e^{ik \cdot x} - e^{-ik \cdot x}) = 0, \tag{22.173}$$

where with the second equality we used (22.167). Differentiating (22.172) with respect to x^0 , we get

$$\begin{aligned}
 \dot{\Delta}(x) &\equiv -i \int \frac{d^3\mathbf{k}}{(2\pi)^3 2k_0} [(-ik_0)e^{-ikx} - ik_0 e^{ikx}] = - \int \frac{d^3\mathbf{k}}{(2\pi)^3 2} \\
 &\times (e^{-ikx} + e^{ikx}).
 \end{aligned} \tag{22.174}$$

Putting $x^0 = 0$ in (22.174), we have

$$\dot{\Delta}(x)|_{x^0=0} = - \int \frac{d^3\mathbf{k}}{(2\pi)^3 2} (e^{ik \cdot x} + e^{-ik \cdot x}) = -\delta^3(x),$$

where we used (22.101). As an alternative way, from (22.169) we have

$$\Delta(x) = \frac{-1}{(2\pi)^3} \int d^3\mathbf{k} \frac{1}{K} \sin kx.$$

Using $(\sin kx) = k_0 \cos kx$, we obtain

$$\Delta(x) \equiv \frac{-1}{(2\pi)^3} \int d^3\mathbf{k} \cos kx.$$

Then, we get

$$\Delta(x) \Big|_{x^0=0}^{\leftarrow} = \frac{-1}{(2\pi)^3} \int d^3\mathbf{k} \cos(-\mathbf{k} \cdot \mathbf{x}) = \frac{-1}{(2\pi)^3} \int d^3\mathbf{k} \cos(\mathbf{k} \cdot \mathbf{x}) = -\delta^3(\mathbf{x}).$$

Getting back to the equal-time commutation relation, we had

$$[\phi(t, \mathbf{x}), \phi(t, \mathbf{y})] = 0. \quad (22.84) \leftarrow$$

Using (22.131) and (22.173), however, the above equation can be rewritten as

$$[\phi(t, \mathbf{x}), \phi(t, \mathbf{y})] = i\Delta(0, \mathbf{x} - \mathbf{y}) = 0. \quad (22.175) \leftarrow$$

A world interval between the events that are labeled by $t, \mathbf{x}$ and $t, \mathbf{y}$ is imaginary (i.e., spacelike) as can be seen in (21.31). Since this nature is Lorentz invariant, we conclude that

$$[\phi(x), \phi(y)] = 0 \quad \text{for } (x - y)^2 < 0. \quad (22.176) \leftarrow$$

This clearly states that the fields at any two points x and y with spacelike separation commute [5]. In other words, this statement can be translated into that the physical measurements of the fields at two points with spacelike separation must not interfere with each other. This is known as the microcausality.

22.3.5 Feynman Propagator of the Scalar Field

Once the Lorentz invariant functions have been introduced, this directly leads to an important quantity called the Feynman propagator. It is expressed as a c -number directly connected to the experimental results that are dealt with in the framework of the interaction between fields. The Feynman propagator is defined as a vacuum expectation value of the time-ordered product of $\phi(x)\phi(y)$, i.e.,

$$\langle 0 | T[\phi(x)\phi(y)] | 0 \rangle. \quad (22.177) \leftarrow$$

With the above quantity, the time-ordered product $T[\phi(x)\phi(y)]$ of the neutral scalar field is defined by

$$T[\phi(x)\phi(y)] \equiv \theta(x^0 - y^0)\phi(x)\phi(y) + \theta(y^0 - x^0)\phi(y)\phi(x). \quad (22.178) \leftarrow$$

The time-ordered product (abbreviated as T-product) frequently appears in the quantum field theory together with the normal-ordered product (abbreviated as N-product). Their tangible usage for practical calculations will be seen in the next chapter. Using (22.102), we have

$$\begin{aligned} T[\phi(x)\phi(y)] &= \int \frac{d^3\mathbf{k}d^3\mathbf{q}}{(2\pi)^3 2\sqrt{k_0q_0}} \theta(x^0 - y^0) [a(\mathbf{k})e^{-ikx} + a^\dagger(\mathbf{k})e^{ikx}] [a(\mathbf{q})e^{-iqy} + a^\dagger(\mathbf{q})e^{iqy}] \\ &\quad + \int \frac{d^3\mathbf{k}d^3\mathbf{q}}{(2\pi)^3 2\sqrt{k_0q_0}} \theta(y^0 - x^0) [a(\mathbf{q})e^{-iqy} + a^\dagger(\mathbf{q})e^{iqy}] [a(\mathbf{k})e^{-ikx} + a^\dagger(\mathbf{k})e^{ikx}]. \end{aligned}$$

Each integral contains four terms, only one of which survives after taking the vacuum expectation values $\langle 0 | \cdots | 0 \rangle$; use the equations of $a(\mathbf{k}) | \Phi \rangle \Leftarrow \langle 0 | a^\dagger(\mathbf{k}) = 0$. Then, we have

$$\begin{aligned} \langle 0 | T[\phi(x)\phi(y)] | 0 \rangle &\leftarrow \\ &= \int \frac{d^3\mathbf{k}d^3\mathbf{q}}{(2\pi)^3 2\sqrt{k_0q_0}} \theta(x^0 - y^0) e^{-ikx} e^{iqy} \langle 0 | a(\mathbf{k}) a^\dagger(\mathbf{q}) | 0 \rangle \\ &\quad + \int \frac{d^3\mathbf{k}d^3\mathbf{q}}{(2\pi)^3 2\sqrt{k_0q_0}} \theta(y^0 - x^0) e^{-iqy} e^{ikx} \langle 0 | a(\mathbf{q}) a^\dagger(\mathbf{k}) | 0 \rangle. \end{aligned} \quad (22.179) \leftarrow$$

Using the first equation of (22.105), we have

$$\begin{aligned} a(\mathbf{k}) a^\dagger(\mathbf{q}) &= a^\dagger(\mathbf{q}) a(\mathbf{k}) + \delta^3(\mathbf{k} - \mathbf{q}), \\ a(\mathbf{q}) a^\dagger(\mathbf{k}) &= a^\dagger(\mathbf{k}) a(\mathbf{q}) + \delta^3(\mathbf{q} - \mathbf{k}). \end{aligned}$$

Hence, we obtain

$$\begin{aligned} \langle 0 | a(\mathbf{k}) a^\dagger(\mathbf{q}) | 0 \rangle &= \langle 0 | a^\dagger(\mathbf{q}) a(\mathbf{k}) | 0 \rangle + \langle 0 | \delta^3(\mathbf{k} - \mathbf{q}) | 0 \rangle \\ &= \langle 0 | \delta^3(\mathbf{k} - \mathbf{q}) | 0 \rangle = \delta^3(\mathbf{k} - \mathbf{q}) \langle 0 | 0 \rangle \Leftarrow \delta^3(\mathbf{k} - \mathbf{q}). \end{aligned}$$

Similarly, we have

$$\langle 0 | a(\mathbf{q}) a^\dagger(\mathbf{k}) | 0 \rangle = \delta^3(\mathbf{q} - \mathbf{k}).$$

Substituting the above equations for (22.179) and integrating the resulting equation with respect to $\mathbf{q}$, we get

$$\begin{aligned} \langle 0 | T[\phi(x)\phi(y)] | 0 \rangle &= \int \frac{d^3\mathbf{k}}{(2\pi)^3 2k_0} \\ &\times \left[\theta(x^0 - y^0) e^{-ik(x-y)} + \theta(y^0 - x^0) e^{ik(x-y)} \right], \quad \leftarrow \end{aligned} \quad (22.180) \leftarrow$$

where, e.g., $e^{-ik(x-y)}$ is represented by

$$e^{-ik(x-y)} = e^{-ik_0(x^0 - y^0) + i\mathbf{k}(\mathbf{x} - \mathbf{y})}, \quad k_0 = \sqrt{\mathbf{k}^2 + m^2}. \quad (22.181) \leftarrow$$

We call (22.180) the Feynman propagator and symbolically define it as [8]

$$\Delta_F(x-y) \equiv \langle 0 | T[\phi(x)\phi(y)] | 0 \rangle. \quad (22.182) \leftarrow$$

Notice that the definition and notation of the invariant delta functions and Feynman propagator are different from literature to literature. Caution should be taken accordingly. In this book we *define the Feynman propagator as identical with the vacuum expectation value of the time-ordered product* [8].

Meanwhile, the commutator $[\phi^+(x), \phi^-(y)]$ is a scalar function. Then, we have [5]

$$\begin{aligned} i\Delta^+(\bar{x} - y) &= [\phi^+(\bar{x}), \phi^-(y)] = \langle 0 | [\phi^+(\bar{x}), \phi^-(y)] | 0 \rangle = \langle 0 | \phi^+(\bar{x}) \phi^-(y) | 0 \rangle - \\ &= \langle 0 | \phi(x) \phi(y) | 0 \rangle. \end{aligned} \quad (22.183) \leftarrow$$

This relation is particularly important, because the invariant delta function $\Delta^+(x-y)$ is not only a scalar, but also given by a vacuum expectation value of the product function $\phi(x)\phi(y)$. This aspect will become clearer when we evaluate the Feynman propagator. Switching the arguments x and y and using (22.138), we have

$$i\Delta^+(\bar{y} - x) = -i\Delta^-(x - y) = \langle 0 | \phi(y) \phi(x) | 0 \rangle. \quad (22.184) \leftarrow$$

Using (22.178) and (22.182), we get [5]

$$\Delta_F(x-y) = i[\theta(x^0 - y^0) \Delta^+(\bar{x} - y) - \theta(y^0 - x^0) \Delta^-(x - y)]. \quad \leftarrow (22.185) \leftarrow$$

The Feynman propagator plays an important role in evaluating the interactions between various fields, especially through the perturbation calculations. As in the case of (22.155), we wish somehow to describe (22.180) in the explicitly Lorentz invariant form. To do this, normally starting with (22.180), we arrive at the Lorentz invariant Fourier integral representation. In the present case, however, in an opposite manner we start with the Fourier integral representation of the time-ordered product in the four-momentum space and show how the said representation naturally leads to (22.180).

The Fourier integral representation of the Feynman propagator is described in an invariant form as [8]

$$\Delta_F(x-y) = \frac{-i}{(2\pi)^4} \int d^4k \frac{e^{-ik(x-y)}}{m^2 - k^2 - i\epsilon}, \quad (22.186) \leftarrow$$

where ϵ is an infinitesimal positive constant and we will take the limit $\epsilon \rightarrow 0$ after the whole calculation. When we dealt with the invariant delta functions in the last section, the constant ϵ was not included in the denominator. The presence or absence of ϵ comes from the difference in the integration paths of the complex integral that are chosen for the evaluation of individual invariant functions. We will make a brief discussion on this point at the end of this section.

It is convenient to express the Feynman propagator in the momentum space (momentum space representation). In practice, taking the Fourier transform of (22.186), we have

$$\frac{-i}{m^2 - k^2 - i\epsilon} = \int d^4x \Delta_F(x-y) e^{ik(x-y)}, \quad (22.187) \leftarrow$$

where we applied the relations (22.55) and (22.58) to (22.186). For the derivation, use

$$(2\pi)^4 \delta^4(k) = \int_{-\infty}^{\infty} e^{ikx} d^4x.$$

We define $\widetilde{\Delta}_F(k)$ as

$$\widetilde{\Delta}_F(k) \equiv \frac{-i}{m^2 - k^2 - i\epsilon}. \quad (22.188) \leftarrow$$

The momentum space representation $\widetilde{\Delta}_F(k)$ is frequently used for dealing with problems of the interaction between the fields (see Chap. 23).

We modify (22.186) as follows for the sake of easy handling. Defining a positive constant K as $K \equiv \sqrt{k^2 + m^2}$ as before, we think of an equation

$$\mathcal{A} \equiv \frac{1}{2K} \left[\frac{1}{k_0 - (-K + i\epsilon)} - \frac{1}{k_0 - (K - i\epsilon)} \right]. \quad (22.189) \leftarrow$$

After some calculations, (22.189) is deformed to

$$\mathcal{A} = \frac{1}{2K} \cdot \frac{2K - 2i\epsilon}{K^2 - k_0^2 - 2iK\epsilon + \epsilon^2} = \frac{1 - (i\epsilon/K)}{m^2 - k^2 - 2iK\epsilon + \epsilon^2}. \quad (22.190) \leftarrow$$

In (22.190), $2K\epsilon$ in the denominator is an infinitesimal positive constant as well; we neglect ϵ^2 as usual. The term $-i\epsilon/K$ in the numerator does not affect subsequent calculations and, hence, can be discarded from the beginning. Thus, (22.190) can be regarded as identical to $1/(m^2 - k^2 - i\epsilon)$. Then, instead of calculating (22.186), we are going to evaluate

$$\Delta_F(x-y) \leftarrow \frac{-i}{(2\pi)^4} \int d^4k \frac{1}{2K} e^{-ik(x-y)} \left[\frac{1}{k_0 - (-K + i\epsilon)} - \frac{1}{k_0 - (K - i\epsilon)} \right]. \quad (22.191) \leftarrow$$

Rewriting (22.191), we have

$$\begin{aligned} \Delta_F(x-y) &= \frac{-i}{(2\pi)^4} \int d^3\mathbf{k} e^{ik(x-y)} \left\{ \int_{-\infty \leftarrow}^{+\infty} dk_0 \frac{1}{2K} e^{-ik_0(x^0-y^0)} \right. \leftarrow \\ &\quad \times \left. \left[\frac{1}{k_0 - (-K + i\epsilon)} - \frac{1}{k_0 - (K - i\epsilon)} \right] \right\}. \quad (22.192) \leftarrow \end{aligned}$$

With the above relation, we define I as

$$I \equiv \int_{-\infty \leftarrow}^{+\infty} dk_0 \frac{1}{2K} e^{-ik_0(x^0-y^0)} \left[\frac{\leftarrow 1}{k_0 - (-K + i\epsilon)} - \frac{1}{k_0 - (K - i\epsilon)} \right]. \quad (22.193) \leftarrow$$

Since complex numbers are contained in the denominator of the integrand I of (22.192), we wish to convert I to a complex integration with respect to k_0 (see Chap. 6). From (22.192) we immediately see that the integrand has simple poles at $k_0 = \pm(K - i\epsilon)$. The point is that the presence of the simple poles allows us to use (6.146) described by

$$\oint_C f(z) dz = 2\pi i \sum_j \text{Res} f(a_j). \quad (6.146) \leftarrow$$

Another point is that the functions

$$\frac{1}{k_0 - (-K + i\epsilon)} \quad \text{and} \quad \frac{1}{\leftarrow k_0 - (K - i\epsilon)} \quad \leftarrow$$

tend uniformly to zero with $|k_0| \rightarrow \infty$. This situation allows us to apply Jordan's lemma to the complex integration of (22.192); see Sect 6.8, Lemma 6.6. To perform the complex integration, we have the following two cases.

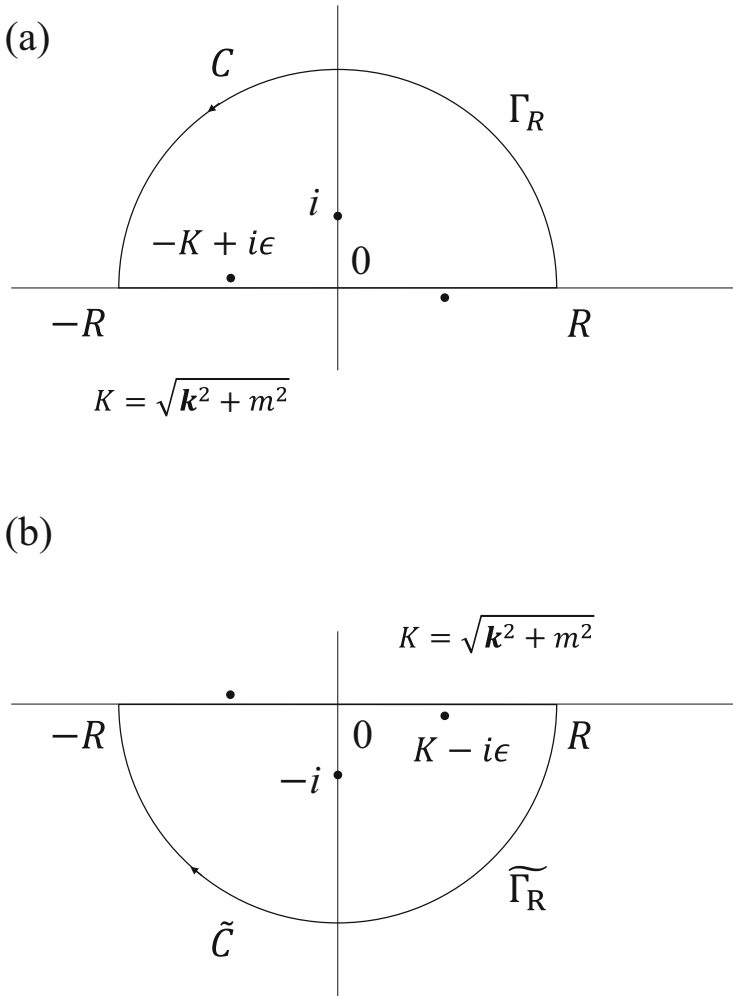


Fig. 22.2 Contours to calculate the complex integrations with respect to the Feynman propagator $\Delta_F(x - y)$ of the scalar field. (a) Contour integration with the upper semicircle C . (b) Contour integration with the lower semicircle $\tilde{C}$

1. Case I:

We choose an upper semicircle C for the contour integration (see Fig. 22.2a). As in the case of Example 6.4, we define I_C as

$$I_C = \oint_C dk_0 \frac{1}{2K} e^{-ik_0(x^0 - y^0)} \left[\frac{1}{k_0 - (-K + i\epsilon)} - \frac{1}{k_0 - (K - i\epsilon)} \right]. \quad \leftarrow (22.194)$$

As the function $\frac{1}{k_0 - (K - i\epsilon)}$ is analytic on and inside the contour C , its associated contour integration vanishes due to the Cauchy's integral theorem (Theorem 6.10). Hence, we are left with

$$I_C = \oint_C dk_0 \frac{1}{2K} e^{-ik_0(x^0 - y^0)} \frac{1}{k_0 - (-K + i\epsilon)}. \quad (22.195) \leftarrow$$

Making the upper semicircle Γ_R large enough, we get

$$\begin{aligned} \lim_{R \rightarrow \infty} I_C &= \int_{-\infty - i\epsilon}^{\infty} dk_0 \frac{1}{2K} e^{-ik_0(x^0 - y^0)} \frac{1}{k_0 - (-K + i\epsilon)} \\ &+ \lim_{R \rightarrow \infty} \int_{\Gamma_R} dk_0 \frac{1}{2K} e^{-ik_0(x^0 - y^0)} \frac{1}{k_0 - (-K + i\epsilon)}. \end{aligned} \quad \leftarrow (22.196) \leftarrow$$

In (22.196), the contour integration is taken counter-clockwise so that $k_0 = -K + i\epsilon$ can be contained inside the closed contour (i.e., the upper semicircle C).

For the contour integral along C to have a physical meaning (i.e., for Jordan's Lemma to be applied to the present case), we must have $x^0 - y^0 < 0$. For this, put $k_0 = Re^{i\theta}$ (R : real positive; $0 < \theta < \pi$) so that we may get

$$e^{-ik_0(x^0 - y^0)} = e^{-iR(x^0 - y^0) \cos \theta} e^{R(x^0 - y^0) \sin \theta}.$$

Taking the absolute value of both sides for the above equation, we obtain

$$\left| e^{-ik_0(x^0 - y^0)} \right| = \left| e^{-iR(x^0 - y^0) \cos \theta} \right| \cdot \left| e^{R(x^0 - y^0) \sin \theta} \right| = \left| e^{R(x^0 - y^0) \sin \theta} \right|.$$

We have $\left| e^{R(x^0 - y^0) \sin \theta} \right| \rightarrow 0$ with $R \rightarrow \infty$. Since the function $\frac{1}{k_0 - (-K + i\epsilon)}$ is analytic on Γ_R and tends uniformly to zero as $R \rightarrow \infty$, the second term of (22.196) vanishes according to Lemma 6.6 (Jordan's Lemma). Thus, we obtain

$$\lim_{R \rightarrow \infty} I_C = \int_{-\infty - i\epsilon}^{\infty} dk_0 \frac{1}{2K} e^{-ik_0(x^0 - y^0)} \frac{1}{k_0 - (-K + i\epsilon)} = I. \quad \leftarrow$$

Applying (6.146) and (6.148) to the present case, we get

$$I = 2\pi i \frac{1}{2K} e^{-ik_0(x^0 - y^0)} \Big|_{k_0 = -K + i\epsilon} \leftarrow = 2\pi i \frac{1}{2K} e^{(iK + \epsilon)(x^0 - y^0)}.$$

Taking $\epsilon \rightarrow 0$, we obtain

$$I = 2\pi i \frac{1}{2K} e^{iK(x^0 - y^0)}. \quad (22.197) \leftarrow$$

2. Case II:

We choose a lower semicircle $\tilde{C}$ for the contour integration (see Fig. 22.2b). As in Case I, we define $I_{\tilde{C}}$ as

$$I_{\tilde{C}} = \oint_{\tilde{C}} dk_0 \frac{1}{2K} e^{-ik_0(x^0 - y^0)} \left[\frac{\leftarrow 1}{k_0 - (-K + i\epsilon)} - \frac{1}{k_0 - (K - i\epsilon)} \right]. \quad \leftarrow$$

This time, as the function $\frac{1}{k_0 - (-K + i\epsilon)}$ is analytic on and inside the contour $\tilde{C}$, its associated contour integration vanishes due to the Cauchy's integral theorem, and so we are left with

$$\begin{aligned} \lim_{R \rightarrow \infty} I_{\tilde{C}} &= \int_{-\infty}^{\infty} dk_0 \frac{1}{2K} e^{-ik_0(x^0 - y^0)} \left[-\frac{1}{k_0 - (K - i\epsilon)} \right] \quad \leftarrow \\ &+ \lim_{R \rightarrow \infty} \int_{\tilde{\Gamma}_R} dk_0 \frac{1}{2K} e^{-ik_0(x^0 - y^0)} \left[-\frac{1}{k_0 - (K - i\epsilon)} \right]. \quad \leftarrow \end{aligned} \quad (22.198) \leftarrow$$

Using Jordan's lemma as in Case I and applying (6.146) and (6.148) to the present case, we get

$$I = -2\pi i \frac{1}{2K} \left[-e^{-ik_0(x^0 - y^0)} \right]_{k_0=K}^{\leftarrow} = 2\pi i \frac{1}{2K} e^{-iK(x^0 - y^0)}. \quad (22.199) \leftarrow$$

In (22.199), the minus sign before $2\pi i$ with the first equality resulted from that the contour integration was taken clockwise so that $k_0 = K - i\epsilon$ was contained inside the closed contour (i.e., the lower semicircle $\tilde{C}$). Note that in this case we must have $x^0 - y^0 > 0$ for Jordan's lemma to be applied.

Combining the results of Case I and Case II and taking account of the fact that we must have $x^0 - y^0 < 0$ ($x^0 - y^0 > 0$) with Case I (Case II), we get

$$I = 2\pi i \frac{1}{2K} \left\{ \theta(-x^0 + y^0) e^{iK(x^0 - y^0)} + \theta(x^0 - y^0) e^{-iK(x^0 - y^0)} \right\}. \quad (22.200) \leftarrow$$

Substituting (22.200) for (22.192), we get

$$\begin{aligned}
\Delta_F(x-y) &= \frac{-i}{(2\pi)^4} \int d^3k I e^{-ik(x-y)} \quad \leftarrow \\
&= \frac{1}{(2\pi)^3} \int \frac{d^3k}{2K} e^{ik(x-y)} \left[\theta(-x^0 + y^0) e^{iK(x^0 - y^0)} + \theta(x^0 - y^0) e^{-iK(x^0 - y^0)} \right] \quad \leftarrow \\
&= \frac{1}{(2\pi)^3} \int \frac{d^3k}{2K} \left[\theta(-x^0 + y^0) e^{ik(x-y)} + \theta(x^0 - y^0) e^{-ik(x-y)} \right] \quad \leftarrow \\
&= \langle 0 | T[\phi(x)\phi(y)] | 0 \rangle,
\end{aligned}
\tag{22.201} \leftarrow$$

where with the second equality we switched the integral variable $\mathbf{k}$ to $-\mathbf{k}$ in the first term; the last equality came from (22.180). Thus, we have shown that (22.186) properly reproduces the definition of the Feynman propagator $\Delta_F(x-y)$ given by (22.182).

To calculate the invariant delta functions, we have encountered several examples of the complex integration. It is worth noting that comparing the contours C of Fig. 22.2a and C_- of Fig. 22.1 their topological features are virtually the same. This is also the case with the topological relationship between the contours $\tilde{C}$ of Fig. 22.2b and C_+ of Fig. 22.1. That is, the former two contours encircle only $k_0 = -K$ (or $k_0 = -K + i\epsilon$), whereas the latter two contours encircle only $k_0 = K$ (or $k_0 = K - i\epsilon$).

Comparing (22.186) with (22.159) and (22.171), we realize that the functional forms of the integrand are virtually the same except for the presence or absence of the term $-i\epsilon$ in the denominator. With the complex integration of k_0 , simple poles $\pm K$ are located *inside* the contour for the integration regarding (22.159) and (22.171). With (22.186), however, if the term $-i\epsilon$ were lacked, the simple poles $\pm K$ would be located *on* the integration path. To avoid this situation, we “raise” or “lower” the pole by adding $+i\epsilon$ ($\epsilon > 0$) or $-i\epsilon$ to its complex coordinate, respectively, so that we can apply Cauchy’s Integral Theorem (see Theorem 6.10) and Cauchy’s Integral Formula (Theorem 6.11) to the contour integration [11]. After completing the relevant calculations, we take $\epsilon \rightarrow 0$ to evaluate the integral properly; see (22.197), for example. Nevertheless, we often do without this mathematical manipulation. An example can be seen in Chap. 23.

22.3.6 General Consideration on the Field Quantization

So far, we have described the quantization procedures of the scalar field. We will subsequently deal with the quantization of the Dirac field and electromagnetic field (or photon field). The quantization processes and resulting formulation have different faces depending on different types of relativistic fields. Nevertheless, the quantization methods can be integrated into a unified approach. This section links the contents of the previous sections to those of the subsequent sections.

In Sect. 22.2.2, we had

$$\hat{E} = \int_{-\infty}^{\infty} |k\rangle \langle k| dk, \quad (22.64)$$

where $\hat{E}$ is the identity operator. We extend a variable k of (22.64) to the momentum four-vector p . That is, we have

$$\hat{E} = \int_{-\infty}^{\infty} |p\rangle \langle p| d^4p, \quad (22.202)$$

where $p = \begin{pmatrix} p_0 \\ \mathbf{p} \end{pmatrix}$, to which p_0 denotes the energy and $\mathbf{p}$ represents the three-dimensional momenta of a relativistic particle. We define the quantity p^2 as

$$p^2 \equiv p_\mu p^\mu.$$

Then, from (21.79) we have

$$p^2 = p_0^2 - \mathbf{p}^2 = m^2 \quad \text{or} \quad p^2 - m^2 = 0.$$

Taking account of this constraint properly, we modify (22.202) in an explicitly invariant form in such a way that [7]

$$\begin{aligned} \hat{E} &= \int_{-\infty}^{\infty} \delta(p^2 - m^2) |p\rangle \langle p| d^4p \\ &= \int_{-\infty}^{\infty} d^3\mathbf{p} \int_{-\infty}^{\infty} dp_0 \delta[p_0^2 - (\mathbf{p}^2 + m^2)] |p\rangle \langle p| \end{aligned} \quad (22.203)$$

To further modify (22.203), using (22.143) where $f(x) = x^2 - a^2$ we obtain

$$\delta(x^2 - a^2) = \frac{\delta(x - a)}{2|x|} + \frac{\delta(x + a)}{2|-a|}.$$

Exchanging x and a and using the fact that the δ function is an even function, we have

$$\delta(x^2 - a^2) = \frac{\delta(x - a)}{2|x|} + \frac{\delta(x + a)}{2|-x|},$$

where x is regarded as the valuable. Replacing x with p_0 and a with $\sqrt{\mathbf{p}^2 + m^2}$, we get

$$\begin{aligned}\delta(p^2 - m^2) &= \delta[p_0^2 - (\mathbf{p}^2 + m^2)] \\ &= \frac{\delta(p_0 - \sqrt{\mathbf{p}^2 + m^2})}{2|p_0|} + \frac{\delta(p_0 + \sqrt{\mathbf{p}^2 + m^2})}{2|-p_0|}.\end{aligned}$$

Inserting this relation into (22.203), we obtain

$$\begin{aligned}\hat{E} &= \int_{-\infty}^{\infty} \delta(p^2 - m^2) |p\rangle \langle p| d^4p \\ &= \int_{-\infty}^{\infty} d^3p \int_{-\infty}^{\infty} dp_0 \left[\frac{\delta(p_0 - \sqrt{\mathbf{p}^2 + m^2})}{2|p_0|} + \frac{\delta(p_0 + \sqrt{\mathbf{p}^2 + m^2})}{2|-p_0|} \right] |p\rangle \langle p|\end{aligned}$$

Defining $|p\rangle \equiv |\mathbf{p}, p_0\rangle$ along with a real positive number $E_p \equiv \sqrt{\mathbf{p}^2 + m^2}$ and using a property of the δ function, we get

$$\hat{E} = \int_{-\infty}^{\infty} d^3p \frac{1}{2E_p} [|p, E_p\rangle \langle p, E_p| + |p, -E_p\rangle \langle p, -E_p|].$$

Operating $\langle x|$ and $|\Phi\rangle$ on both sides of the above equation from the left and right, respectively, we get

$$\begin{aligned}\langle x|\hat{E}|\Phi\rangle &= \langle x|\Phi\rangle \\ &= \int_{-\infty}^{\infty} \frac{d^3p}{2E_p} [\langle x|\mathbf{p}, E_p\rangle \langle \mathbf{p}, E_p|\Phi\rangle + \langle x|\mathbf{p}, -E_p\rangle \langle \mathbf{p}, -E_p|\Phi\rangle],\end{aligned}\tag{22.204}$$

where $|x\rangle$ represents the continuous space-time basis vectors of the Minkowski space (see Fig. 10.1, which shows a one-dimensional case); $|\Phi\rangle$ denotes the quantum state of the relativistic particle (photon, electron, etc.) we are thinking of. According to (10.90), we express $\langle x|\Phi\rangle$ of (22.204) as

$$\Phi(x) \equiv \langle x|\Phi\rangle.$$

Also defining $|x\rangle \equiv |\mathbf{x}, t\rangle$, we rewrite (22.204) as

$$\Phi(x) = \int_{-\infty}^{\infty} \frac{d^3p}{2E_p} [\langle \mathbf{x}, t|\mathbf{p}, E_p\rangle \langle \mathbf{p}, E_p|\Phi\rangle + \langle \mathbf{x}, t|\mathbf{p}, -E_p\rangle \langle \mathbf{p}, -E_p|\Phi\rangle].$$

Using (22.67), we have

$$\langle \mathbf{x}, t | \mathbf{p}, \pm E_p \rangle = \frac{1}{\sqrt{(2\pi)^3}} e^{i\mathbf{p}\mathbf{x}}. \quad (22.205) \leftarrow$$

In the above equation, neither t nor E_p can be regarded as Fourier components because of the presence of $\delta(p^2 - m^2)$ in the integrand of (22.203). In other words, if we choose $\mathbf{p}$ for freely changeable parameter over the range $(-\infty, +\infty)$, we cannot deal with p_0 as an independent parameter on account of the factor $\delta(p^2 - m^2)$ but have to regard it as a dependent parameter $p_0 = \pm \sqrt{\mathbf{p}^2 + m^2}$.

The factor $\langle \mathbf{p}, \pm E_p | \Phi \rangle$ describes the quantum state of the free particle in the positive-energy state and negative-energy state, respectively. The former (latter) state contains $e^{-iE_p t}$ ($e^{iE_p t}$) as a factor accordingly, independent of the particle species. The factor $\langle \mathbf{p}, \pm E_p | \Phi \rangle$ reflects the nature of individual particle species. As already seen in Chap. 21, for example, $\langle \mathbf{p}, \pm E_p | \Phi \rangle$ denotes a spinor with the Dirac field. As for the photon field, $\langle \mathbf{p}, \pm E_p | \Phi \rangle$ represents an electromagnetic four-vector, as we will see later. More importantly, $\langle \mathbf{p}, \pm E_p | \Phi \rangle$ contains a q -number, i.e., operator. Normally, in the quantum field theory, the positive-energy (or negative-energy) state is accompanied by the annihilation (or creation) operator. The factor $\langle \mathbf{p}, \pm E_p | \Phi \rangle$ must properly be “normalized.”

Including the aforementioned characteristics, $\Phi(x)$ is described as

$$\begin{aligned} \Phi(x) &\leftarrow \\ &= \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{2E_p} \frac{1}{\sqrt{(2\pi)^3}} e^{i\mathbf{p}\mathbf{x}} [e^{-iE_p t} \alpha_+(\mathbf{p}) \Phi_+(\mathbf{p}, E_p) + e^{iE_p t} \alpha_-^\dagger(\mathbf{p}) \Phi_-(\mathbf{p}, -E_p)], \end{aligned} \quad (22.206) \leftarrow$$

where $\alpha_+(\mathbf{p})$ and $\alpha_-^\dagger(\mathbf{p})$ are the annihilation and creation operators (i.e., q -numbers), respectively. In (22.206) we have defined

$$\langle \mathbf{p}, E_p | \Phi \rangle \equiv \alpha_+(\mathbf{p}) \Phi_+(\mathbf{p}, E_p) \quad \text{and} \quad \langle \mathbf{p}, -E_p | \Phi \rangle \equiv \alpha_-^\dagger(\mathbf{p}) \Phi_-(\mathbf{p}, -E_p).$$

Even though $\Phi_\pm(\mathbf{p}, \pm E_p)$ contains the representation of spinor, vector, etc., it is a c -number. Further defining the operators such that [8]

$$a_+(\mathbf{p}) \equiv \sqrt{2E_p} \alpha_+(\mathbf{p}) \quad \text{and} \quad a_-^\dagger(\mathbf{p}) \equiv \sqrt{2E_p} \alpha_-^\dagger(\mathbf{p}),$$

we obtain

$$\begin{aligned}
\Phi(x) &\leftarrow \\
&= \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 (2E_p)}} e^{ipx} \left[e^{-iE_p t} a_+(\mathbf{p}) \Phi_+(\mathbf{p}, E_p) + e^{iE_p t} a_-^\dagger(\mathbf{p}) \Phi_-(\mathbf{p}, -E_p) \right].
\end{aligned} \tag{22.207} \leftarrow$$

Then, we get

$$\begin{aligned}
\Phi(x) &= \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 (2E_p)}} \left[e^{ipx} e^{-iE_p t} a_+(\mathbf{p}) \Phi_+(\mathbf{p}, E_p) \right. \\
&\quad \left. + e^{-ipx} e^{iE_p t} a_-^\dagger(-\mathbf{p}) \Phi_-(-\mathbf{p}, -E_p) \right] \\
&= \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 (2E_p)}} \left[e^{-ipx} a_+(\mathbf{p}) \Phi_+(\mathbf{p}, E_p) + e^{ipx} a_-^\dagger(-\mathbf{p}) \Phi_-(-\mathbf{p}, -E_p) \right],
\end{aligned} \tag{22.208} \leftarrow$$

where with the second equality $e^{-ipx} = e^{ipx} e^{-iE_p t}$ and $e^{ipx} = e^{-ipx} e^{iE_p t}$; in the second term of RHS $\mathbf{p}$ was switched to $-\mathbf{p}$. In (22.208), e.g., $b_-^\dagger(\mathbf{p})$ is often used instead of $a_-^\dagger(-\mathbf{p})$ for the sake of distinction and clarity.

The quantization of the relativistic fields is based upon (22.208) in common with regard to the neutral scalar field, Dirac field, and the photon field. The first term of (22.208) is related to the positive-energy state that is characterized by $a_+(\mathbf{p})$ and $\Phi_+(\mathbf{p}, E_p)$. The second term, in turn, is associated with the negative-energy state characterized by $a_-^\dagger(-\mathbf{p})$ and $\Phi_-(-\mathbf{p}, -E_p)$. In Sect. 22.3.2, we established a basic equation of (22.98) as a starting point for the quantization of the scalar field. It was unclear whether (22.98) is Lorentz invariant. Its equivalent expression of (22.208), however, has been derived from the explicitly invariant equation of (22.203). Thus, this enables us to adopt (22.208) as a fundamental equation for the quantization of relativistic fields.

As in (22.208), it is always possible to separate (or decompose) the field operator $\Phi(x)$ into the positive-energy and negative-energy parts. Defining

$$\Phi(x) = \Phi_+(x) + \Phi_-(x),$$

we have

$$\begin{aligned}
\Phi_+(x) &\equiv \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 (2E_p)}} e^{-ipx} a_+(\mathbf{p}) \Phi_+(\mathbf{p}, E_p), \\
\Phi_-(x) &\equiv \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 (2E_p)}} e^{ipx} a_-^\dagger(-\mathbf{p}) \Phi_-(-\mathbf{p}, -E_p).
\end{aligned} \tag{22.209}$$

Using these equations, we obtain various useful representations regarding, e.g., commutation relations, the invariant delta functions, and Feynman propagators (see Sects. 22.3.4 and 22.3.5 as well as the subsequent Sects. 22.4 and 22.5).

In the case of the neutral scalar field (or Klein–Gordon field), we do not have to distinguish $a_\pm(\pm\mathbf{p})$ but we only have to take account of $a(\mathbf{p})$ [and $a^\dagger(\mathbf{p})$]. Also in the neutral scalar case, as a normalization condition to be satisfied in (22.208) we may put

$$\Phi_+(\mathbf{p}, E_p) \equiv \Phi_-(-\mathbf{p}, -E_p) \equiv 1.$$

Thus, we have reproduced (22.98). In the case of the Dirac field, in turn, we have following correspondence:

$$\Phi_+(\mathbf{p}, E_p) \leftrightarrow \psi(\mathbf{p}, h) \quad \text{and} \quad \Phi_-(-\mathbf{p}, -E_p) \leftrightarrow \bar{\psi}(\mathbf{p}, h),$$

where we consider the helicity h as an additional freedom (see Sect. 22.4). Regarding other specific notations, see the subsequent sections.

22.4 Quantization of the Dirac Field [5, 6]

22.4.1 Lagrangian Density and Hamiltonian Density of the Dirac Field

As in the case of the scalar field, we start with the Lagrangian density of the Dirac field. The Lagrangian density $\mathcal{L}(x)$ is given by

$$\mathcal{L}(x) \equiv \bar{\psi}(x) (i\gamma^\mu \partial_\mu - m) \psi(x). \tag{22.210}$$

Then, the action integral $S(\psi)$ follows from (22.210) such that

$$S(\psi) = \int dx \mathcal{L}(x) = \int dx [\bar{\psi}(x) (i\gamma^\mu \partial_\mu - m) \psi(x)] \tag{22.211}$$

Taking variation with respect to ψ of $S(\psi)$ as before, we have

$$\begin{aligned}
\delta S(\psi) &= \int dx \delta \mathcal{L}(x) \leftarrow \\
&= \int dx [\delta \bar{\psi} i \gamma^\mu \partial_\mu \psi + \bar{\psi} i \gamma^\mu (\partial_\mu \delta \psi) - m \delta \bar{\psi} \cdot \psi - m \bar{\psi} \cdot \delta \psi] \\
&= \int dx \delta \bar{\psi} (i \gamma^\mu \partial_\mu \psi - m \psi) + \int dx (\bar{\psi} i \gamma^\mu \partial_\mu - m \bar{\psi}) \delta \psi \\
&= \int dx \delta \bar{\psi} (i \gamma^\mu \partial_\mu \psi - m \psi) + [\bar{\psi} i \gamma^\mu \delta \psi]_{-\infty}^{\infty} + \int dx (-\partial_\mu \bar{\psi} i \gamma^\mu - m \bar{\psi}) \delta \psi \\
&= \int dx \delta \bar{\psi} (i \gamma^\mu \partial_\mu \psi - m \psi) - \int dx (i \partial_\mu \bar{\psi} \gamma^\mu + m \bar{\psi}) \delta \psi,
\end{aligned} \tag{22.212}$$

where with the second to the last equality we used integration by parts and $\delta \psi$ was supposed to vanish at $x = \pm \infty$. Since ψ and $\bar{\psi}$ are regarded as independent variables, for $\delta S(\psi) = 0$ to hold with any arbitrarily chosen $\delta \bar{\psi}$ and $\delta \psi$ we must have

$$(i \gamma^\mu \partial_\mu - m) \psi(x) = 0 \tag{22.213}$$

and

$$i \partial_\mu \bar{\psi}(x) \gamma^\mu + m \bar{\psi}(x) = 0. \tag{22.214}$$

Instead of (22.214), we sometimes write [6]

$$i \left(\overleftarrow{\bar{\psi}} \overleftarrow{\partial}_\mu \right) \gamma^\mu + m \bar{\psi} = 0. \tag{22.215}$$

Equations (22.214) and (22.215) are called the adjoint Dirac equation. Rewriting (22.214), we have

$$i\partial_\mu\psi^\dagger\gamma^0\gamma^\mu + m\psi^\dagger\gamma^0 = 0,$$

↓←take a transposition and rearrange the terms :

$$i(\gamma^\mu)^T(\gamma^0)^T\partial_\mu\psi^* + m\gamma^0\psi^* = 0,$$

↓←use transposition of $\gamma^0\gamma^\mu = 2\eta^{0\mu} - \gamma^\mu\gamma^0$ (21.66) : ←

$$i\left[2\eta^{0\mu} - (\gamma^0)^T(\gamma^\mu)^T\right]\partial_\mu\psi^* + m\gamma^0\psi^* = 0,$$

↓←multiply both sides by $(\gamma^0)^T$ from the left and

use $(\gamma^0)^T(\gamma^0)^T = (\gamma^0)^T\gamma^0 = E$ (identity matrix) : (22.216)←

$$\begin{aligned} & i\left[2\eta^{0\mu}(\gamma^0)^T - (\gamma^\mu)^T\right]\partial_\mu\psi^* + m\psi^* \\ &= i(\gamma^0)^T\partial_0\psi^* - i\sum_{k=1}^3(\gamma^k)^T\partial_k\psi^* + m\psi^* = 0, \end{aligned}$$

↓←take complex conjugate of the above :

$$-i\left(\gamma^0\partial_0\psi + \sum_{k=1}^3\gamma^k\partial_k\psi\right) + m\psi = 0,$$

where we used the fact that γ^0 is an Hermitian operator and that γ^k ($k = 1, 2, 3$) are anti-Hermitian. Rewriting (22.216), finally we recover

$$(i\gamma^\mu\partial_\mu - m)\psi(x) = 0. \quad (22.213)←$$

Thus, (22.213) can naturally be derived from (22.214), which is again equivalent to (21.54).

As in the case of the scalar field, the Hamiltonian density $\mathcal{H}(x)$ of the Dirac field is given by

$$\mathcal{H}(x) = \pi(x)\dot{\psi}(x) - \mathcal{L}[\psi(x), \partial_\mu\psi(x)]. \quad (22.217)←$$

Since $\psi(x)$ represents a spinor field that is described by a $(4, 1)$ matrix (i.e., column vector), $\pi(x)$ is a $(1, 4)$ matrix (row vector) so that $\mathcal{H}$ is a scalar, i.e., a $(1, 1)$ matrix. If we need to distinguish each component, we put an index, e.g., $\psi(x)^a$ and $\pi(x)_b$ ($a, b = 1, 2, 3, 4$). When we need to distinguish the time coordinate from the space coordinates, we describe, e.g., $\pi(x)$ as $\pi(t, \mathbf{x})$. We have

$$\pi(x) = \frac{\partial\mathcal{L}(\mathbf{x})}{\partial[\partial_0\psi(x)]} \stackrel{\leftarrow}{=} \bar{\psi}(x)i\gamma^0 = \psi^\dagger(x)\gamma^0i\gamma^0 = i\psi^\dagger(x). \quad (22.218)←$$

The representation of (22.218) is somewhat symbolic, because $\psi(x)$ involved as the differentiation variable is a matrix. Therefore, we use the following expression to specify individual components of $\psi(x)$ and $\pi(x)$:

$$\begin{aligned}
 \pi_a(x) &= \frac{\partial \mathcal{L}(\psi)}{\partial [\partial_0 \psi^a(x)]} = [\bar{\psi}(x) i \gamma^0]_a = [\psi^\dagger(x) \gamma^0 i \gamma^0]_a = i [\psi^\dagger(x)]_a \\
 &= i \psi^*(x)_a.
 \end{aligned} \tag{22.219}$$

Note that in (22.219) $\psi^\dagger(x)$ is a $(1, 4)$ matrix and that $\psi^*(x)_a$ is its a th component. The derivation of (22.219) is based on a so-called compendium quantization method. Interested readers are referred to literature for further discussion [2, 6, 8]. Using (22.219), the Hamiltonian density $\mathcal{H}$ of (22.217) is expressed as

$$\begin{aligned}
 \mathcal{H}(x) &= i \psi^\dagger(x) \not{\partial} \psi(x) - \bar{\psi}(x) (\not{\partial} - m) \psi(x) \\
 &= i \psi^\dagger(x) \not{\partial} \psi(x) - \psi^\dagger(x) \gamma^0 i \not{\partial} \psi(x) + \bar{\psi}(x) m \psi(x) \\
 &= i \psi^\dagger(x) \not{\partial} \psi(x) - \psi^\dagger(x) \gamma^0 i \not{\partial} \psi(x) + \bar{\psi}(x) m \psi(x) \\
 &= - \sum_{k=1}^3 \psi^\dagger(x) \gamma^0 i \gamma^k \partial_k \psi(x) + \bar{\psi}(x) m \psi(x) \\
 &= \bar{\psi}(x) \left[- \sum_{k=1}^3 i \gamma^k \partial_k + m \right] \psi(x).
 \end{aligned}$$

Defining $\alpha \equiv \gamma^0 \boldsymbol{\gamma}$ [where $\boldsymbol{\gamma} \equiv (\gamma^1, \gamma^2, \gamma^3)$] and $\beta \equiv \gamma^0$ according to the custom, we have

$$\mathcal{H} = \psi^\dagger(x) (-i \boldsymbol{\alpha} \cdot \boldsymbol{\nabla} + m \beta) \psi(x). \tag{22.220}$$

Hence, with the Hamiltonian H we get

$$H = \int d^3 \mathbf{x} \mathcal{H} = \int d^3 \mathbf{x} \psi^\dagger(x) (-i \boldsymbol{\alpha} \cdot \boldsymbol{\nabla} + m \beta) \psi(x). \tag{22.221}$$

According to the classical theory (i.e., the theory before the quantum field theory) based on the one-particle wave function and its probabilistic interpretation, the integrand of (22.221) can be viewed as an expectation value of the one-particle energy if $\psi(x)$ is normalized. Then, the classical Schrödinger equation could be described as

$$\tilde{H} \psi(x) = (-i \boldsymbol{\alpha} \cdot \boldsymbol{\nabla} + m \beta) \psi(x) = i \partial_0 \psi(x), \tag{22.222}$$

where $\tilde{H}$ represents the one-particle Hamiltonian. Then, (22.222) is written as

$$\sum_{k=1}^3 (-i\gamma^0\gamma^k\partial_k + \cancel{m}^0 m - \gamma^0\gamma^0 i\partial_0)\psi(x) = 0. \quad (22.223) \leftarrow$$

Multiplying both sides of (22.223) by γ^0 from the left, we obtain

$$\sum_{k=1}^3 (-i\gamma^k\partial_k + m - i\gamma^0\partial_0)\psi(x) = 0.$$

Rearranging the terms, we get

$$(i\gamma^\mu\partial_\mu - m)\psi(x) = 0. \quad (22.213) \leftarrow$$

Thus, once again we have recovered the Dirac equation. If Schrödinger had invented the gamma matrices, he would have discovered the Dirac equation (22.213).

If we choose $\phi(x)$ of (21.59) for the solution of the Dirac equation, $\tilde{H}$ is identical with H_p of (21.131) and gives a positive one-particle energy p^0 . If, however, $\chi(x)$ of (21.59) is chosen, $\tilde{H}$ is identical with H_{-p} that gives a negative energy $-p^0$, in agreement with the result of Sect. 21.3.3.

Unlike the classical theory, $\psi(x)$ must be dealt with as a quantum-mechanical operator. Instead of the commutation relation given in (22.84) with the scalar field, we use the following equal-time *anti*-commutation relation (see Sect. 21.3.1) described by

$$\begin{aligned} \{\psi(t, \mathbf{x}), \pi(t, \mathbf{y})\} &= \{\psi(t, \mathbf{x}), i\psi^\dagger(t, \mathbf{y})\} \rightleftharpoons i\delta^3(\mathbf{x} - \mathbf{y}), \\ \{\psi(t, \mathbf{x}), \psi^T(t, \mathbf{y})\} &\rightleftharpoons \{\pi^T(t, \mathbf{x}), \pi(t, \mathbf{y})\} = 0. \end{aligned} \quad (22.224) \leftarrow$$

The expression of (22.224) is somewhat symbolic. To deal with the anti-commutators of (22.224) requires caution. It is because although $\psi(t, \mathbf{x})\pi(t, \mathbf{y})$ is a (4, 4) matrix, $\pi(t, \mathbf{y})\psi(t, \mathbf{x})$ is a (1, 1) matrix. To avoid this inconsistency, we define

$$\{\psi(t, \mathbf{x}), \pi(t, \mathbf{y})\} \equiv \psi(t, \mathbf{x})\pi(t, \mathbf{y}) + \{\pi(t, \mathbf{y})^T[\psi(t, \mathbf{x})]^T\}^T. \quad (22.225) \leftarrow$$

In the second term of (22.225), $\pi(t, \mathbf{y})\psi(t, \mathbf{x})$ is replaced with $\pi(t, \mathbf{y})^T[\psi(t, \mathbf{x})]^T$ and the components of $\pi(t, \mathbf{y})$ are placed in front of those of $\psi(t, \mathbf{x})$.

Another word of caution is that both $\psi(t, \mathbf{x})$ and $\pi(t, \mathbf{y})$ are *Grassmann numbers*. Let a and b be Grassmann numbers. Then, we have $ab = -ba$, $a^2 = b^2 = 0$. Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices whose matrix elements are Grassmann numbers. Unlike ordinary c -numbers, we have $a_{ij}b_{kl} \neq b_{kl}a_{ij}$. Hence, the standard matrix multiplication rules must be modified. If numbers we are dealing with are not Grassmann numbers but ordinary c -numbers, we would simply have

$$\{\pi(t, \mathbf{y})^T [\psi(t, \mathbf{x})]^T\}^T = \psi(t, \mathbf{x}) \pi(t, \mathbf{y}).$$

This equation, however, is not true of the present case where both $\psi(t, \mathbf{x})$ and $\pi(t, \mathbf{y})$ are Grassmann numbers. To highlight this feature, let us think of a following simple example.

Example 22.1 Suppose that we have matrices

$$A = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} \quad \text{and} \quad B = (b_1 \ b_2 \ b_3 \ b_4). \quad (22.226) \leftarrow$$

Then, we have

$$AB = \begin{pmatrix} a_1 b_1 & a_1 b_2 & a_1 b_3 & a_1 b_4 \\ a_2 b_1 & a_2 b_2 & a_2 b_3 & a_2 b_4 \\ a_3 b_1 & a_3 b_2 & a_3 b_3 & a_3 b_4 \\ a_4 b_1 & a_4 b_2 & a_4 b_3 & a_4 b_4 \end{pmatrix}, \quad (22.227) \leftarrow$$

$$(B^T A^T)^T = \begin{pmatrix} b_1 a_1 & b_2 a_1 & b_3 a_1 & b_4 a_1 \\ b_1 a_2 & b_2 a_2 & b_3 a_2 & b_4 a_2 \\ b_1 a_3 & b_2 a_3 & b_3 a_3 & b_4 a_3 \\ b_1 a_4 & b_2 a_4 & b_3 a_4 & b_4 a_4 \end{pmatrix}.$$

If all the above numbers are ordinary c -numbers, we have no problem with $AB = (B^T A^T)^T$. But, if we are dealing with the Grassmann numbers, we have $AB \neq (B^T A^T)^T$. If we express the anti-commutator between A and B , symbolically we have

$$\begin{aligned} \{A, B\} &= AB + (B^T A^T)^T \\ &= \begin{pmatrix} a_1 b_1 + \phi_1 a_1 & a_1 b_2 + \phi_2 a_1 & a_1 b_3 + \phi_3 a_1 & a_1 b_4 + \phi_4 a_1 \\ a_2 b_1 + \phi_1 a_2 & a_2 b_2 + \phi_2 a_2 & a_2 b_3 + \phi_3 a_2 & a_2 b_4 + \phi_4 a_2 \\ a_3 b_1 + \phi_1 a_3 & a_3 b_2 + \phi_2 a_3 & a_3 b_3 + \phi_3 a_3 & a_3 b_4 + \phi_4 a_3 \\ a_4 b_1 + \phi_1 a_4 & a_4 b_2 + \phi_2 a_4 & a_4 b_3 + \phi_3 a_4 & a_4 b_4 + \phi_4 a_4 \end{pmatrix}. \end{aligned}$$

What we have done in the above calculation is to compute $a_i b_k + b_k a_i$ ($i, k = 1, 2, 3, 4$) for the individual matrix elements.

Thus, this simple example shows that care should be taken when dealing with the Grassmann numbers, particularly in the case of matrix calculations. $\square$

Getting back to our present issue, we write the first equation of (22.224) in a matrix form to get

$$\begin{aligned} & \begin{pmatrix} \psi^1 \\ \psi^2 \\ \psi^3 \\ \psi^4 \end{pmatrix} (\pi_1 \quad \pi_2 \quad \pi_3 \quad \pi_4) + \left[\begin{pmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \\ \pi_4 \end{pmatrix} (\psi^1 \quad \psi^2 \quad \psi^3 \quad \psi^4) \right]^T \\ &= \begin{pmatrix} i\delta^3(\mathbf{x}-\mathbf{y}) & \leftarrow & & \\ & i\delta^3(\mathbf{x}-\mathbf{y}) & \leftarrow & \\ & & i\delta^3(\mathbf{x}-\mathbf{y}) & \leftarrow \\ & & & i\delta^3(\mathbf{x}-\mathbf{y}) \end{pmatrix}, \end{aligned} \quad (22.228) \leftarrow$$

where with RHS all the off-diagonal elements vanish with the diagonal elements being $i\delta^3(\mathbf{x}-\mathbf{y})$. That is, (22.228) is a diagonal matrix. Also, notice that using (22.218), (22.228) can be rewritten as

$$\begin{aligned} & \begin{pmatrix} \psi^1 \\ \psi^2 \\ \psi^3 \\ \psi^4 \end{pmatrix} (\psi_1^* \quad \psi_2^* \quad \psi_3^* \quad \psi_4^*) + \left[\begin{pmatrix} \psi_1^* \\ \psi_2^* \\ \psi_3^* \\ \psi_4^* \end{pmatrix} (\psi^1 \quad \psi^2 \quad \psi^3 \quad \psi^4) \right]^T \\ &= \begin{pmatrix} \delta^3(\mathbf{x}-\mathbf{y}) & \leftarrow & & \\ & \delta^3(\mathbf{x}-\mathbf{y}) & \leftarrow & \\ & & \delta^3(\mathbf{x}-\mathbf{y}) & \leftarrow \\ & & & \delta^3(\mathbf{x}-\mathbf{y}) \end{pmatrix}. \end{aligned} \quad (22.229) \leftarrow$$

Another way to show (22.228) is to write down the matrix elements in such a way that [8]

$$\{\psi^\alpha(t, \mathbf{x}), \pi_\beta(t, \mathbf{y})\} = i\delta_\beta^\alpha \delta^3(\mathbf{x}-\mathbf{y}) \quad (\alpha, \beta = 1, 2, 3, 4). \quad (22.230) \leftarrow$$

Namely,

$$\psi^\alpha(t, \mathbf{x})\pi_\beta(t, \mathbf{y}) + \pi_\beta(t, \mathbf{y})\psi^\alpha(t, \mathbf{x}) = i\delta_\beta^\alpha \delta^3(\mathbf{x}-\mathbf{y}). \quad (22.231)$$

Note that neither α nor β is an index that indicates the four-vector, but that both α and β are indices of spinors. Then, we specify $\alpha, \beta = 1, 2, 3, 4$ instead of $\alpha, \beta = 0, 1, 2, 3$.

Equations (22.230) and (22.231) imply the presence of a transposed matrix, which has appeared in (22.228) and (22.229). Unlike (22.84) describing the commutation relation of the scalar field, the anti-commutation relation for the Dirac field of (22.224), (22.229), and (22.230) is represented by (4, 4) matrices.

22.4.2 Quantization Procedure of the Dirac Field

The quantization of the Dirac field can be performed basically in parallel with that of the scalar field. That is, (A) the operator $\psi(x)$ needs to be Fourier-expanded in an invariant form. (B) The creation operator and annihilation operator are introduced. (C) The anti-commutation relation between the creation and annihilation operators needs to be established. The calculation processes, however, differ from the scalar field quantization in several points. For instance, (a) $\psi(x)$ does not need to be Hermitian. (b) The calculations are somewhat time-consuming in that we have to do (4, 4) matrices calculations. (But they are straightforward.)

A. We start with the Fourier expansion of $\psi(x)$. In Sect. 22.3.6 we had

$$\begin{aligned} \Phi(x) = & \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 (2E_p)}} \\ & \times [e^{-ipx} a_+(\mathbf{p}) \Phi_+(\mathbf{p}, E_p) + e^{ipx} a_-^\dagger(-\mathbf{p}) \Phi_-(-\mathbf{p}, -E_p)]. \end{aligned} \quad (22.208) \leftarrow$$

In the Dirac field, we must include an additional freedom of the helicity h that takes a number 1 or -1 . Accordingly, Fourier expansion of the identity operator (22.203) is reformulated as

$$\hat{E} = \int_{-\infty}^{\infty} \sum_{h=\pm 1} \delta(p^2 - m^2) |p, h\rangle \langle p, h| dp^4.$$

Hence, (22.208) is rewritten as

$$\begin{aligned} \Phi(x) = & \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 (2E_p)}} \sum_{h=\pm 1} \left[e^{-ipx} a_+(\mathbf{p}, h) \Phi_+(\mathbf{p}, E_p, h) \right. \\ & \left. + e^{ipx} a_-^\dagger(-\mathbf{p}, h) \Phi_-(-\mathbf{p}, -E_p, h) \right]. \end{aligned}$$

- B. To this relation, we wish to introduce the creation operator and annihilation operator. To this end, we rewrite the field quantity $\Phi(x)$ using the following expressions given by [8]

$$a_+(\mathbf{p}, h) \leftrightarrow b(\mathbf{p}, h) \quad \text{and} \quad a_-^\dagger(-\mathbf{p}, h) \leftrightarrow d^\dagger(\mathbf{p}, h) \quad \leftarrow$$

as well as

$$\Phi_+(\mathbf{p}, E_p, h) \leftrightarrow u(\mathbf{p}, h) \quad \text{and} \quad \Phi_-(-\mathbf{p}, -E_p, h) \leftrightarrow v(\mathbf{p}, h).$$

With respect to the Dirac field, using $\psi(x)$ in place of $\Phi(x)$ in the above expression, we get [8]

$$\begin{aligned} \psi(x) &\leftarrow \\ &= \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 2p_0}} \sum_{h=\pm 1} [e^{-ipx} b(\mathbf{p}, h) u(\mathbf{p}, h) + e^{ipx} d^\dagger(\mathbf{p}, h) v(\mathbf{p}, h)], \end{aligned} \quad (22.232) \leftarrow$$

where p_0 was used instead of E_p in (22.208). The operators $b(\mathbf{p}, h)$ and $d^\dagger(\mathbf{p}, h)$ are an annihilation operator of an electron and a creation operator of a positron, respectively.

The introduction of $d^\dagger(\mathbf{p}, h)$ needs some explanation. So far, we permitted the negative-energy state. However, if a particle carrying a negative energy did exist in the real world, we would face an intolerable situation. To get rid of this big hurdle, Dirac proposed the hole theory (1930). Even though several aspects of the hole theory are explained in literature [6, 8], we wish to add a few comments to the theory. Dirac thought that vacuum consisted of electrons that occupied all the negative-energy states (strictly speaking, the energy states with an energy of $-m$ or less where m is an electron mass). If an electron happened to be excited so as to have a positive energy, a created hole within the negative-energy electrons would behave as if it were a particle carrying a positive charge ($|e| = -e$) and a positive energy with a rest mass m , i.e., a positron. The positron predicted by Dirac was indeed soon discovered experimentally by Carl Anderson (1932). Getting back to the present issue, the operator $d^\dagger(\mathbf{p}, h)$ might well be interpreted as an annihilation operator of an electron carrying a negative energy. But once the positron was discovered in the real world, $d^\dagger(\mathbf{p}, h)$ must be interpreted as the creation operator of a positron.

Notice again that according to the custom the annihilation operator $b(\mathbf{p}, h)$ is a “coefficient” of the positive-energy state $u(\mathbf{p}, h)$ and that the creation operator $d^\dagger(\mathbf{p}, h)$ is a coefficient of the negative-energy state $v(\mathbf{p}, h)$. The anti-commutation relations will be introduced because electron is a Fermion particle. At the same time, the anti-commutation relations facilitate various calculations (vide infra). Note that unlike the scalar field $\phi(x)$ described in (22.98), $\psi(x)$ is

asymmetric with respect to the creation and annihilation operators, reflecting the fact that $\psi(x)$ is *not* Hermitian.

- C. Our next task is to translate the canonical anti-commutation relations (22.224) into the anti-commutation relations between the creation and annihilation operators. To this end, we wish first to solve (22.232) with respect to $b(\mathbf{p}, h)$. The calculation procedures for this are as follows:

- (a) Multiply both sides of (22.232) by e^{-iqx} and integrate them with respect to x to apply (22.101). As a result, we obtain

$$\begin{aligned} \int_{-\infty}^{\infty} dx^3 e^{-iqx} \psi(x) &\leftarrow \\ &= \frac{\sqrt{(2\pi)^3}}{\sqrt{2q_0}} \sum_{h'=\pm 1} [e^{-iq_0 t} b(\mathbf{q}, h') u(\mathbf{q}, h') + e^{iq_0 t} d^\dagger(\mathbf{q}, h') v(\mathbf{q}, h')]. \end{aligned} \quad (22.233) \leftarrow$$

- (b) Multiplying both sides of (22.233) by $u^\dagger(\mathbf{q}, h)$ from the left and using (21.156) and (21.157), we delete $d^\dagger(\mathbf{q}, h')$ from (22.233) to get

$$\int_{-\infty}^{\infty} dx^3 e^{-iqx} u^\dagger(\mathbf{q}, h) \psi(x) = \sqrt{(2\pi)^3 2q_0} e^{-iq_0 t} b(\mathbf{q}, h). \quad (22.234) \leftarrow$$

Rewriting (22.234), we have

$$b(\mathbf{q}, h) = \frac{1}{\sqrt{(2\pi)^3 2q_0}} \int_{-\infty}^{\infty} dx^3 e^{iqx} u^\dagger(\mathbf{q}, h) \psi(x). \quad (22.235) \leftarrow$$

- (c) Next, we determine the anti-commutation relations between $b(\mathbf{q}, h)$ and $d(\mathbf{q}, h)$ (and their adjoints). For example,

$$\begin{aligned} \{b(\mathbf{p}, h), b^\dagger(\mathbf{q}, h')\} &\leftarrow \\ &= \frac{1}{(2\pi)^3 2p_0} \left[\int_{-\infty}^{\infty} dx^3 e^{ipx} u^\dagger(\mathbf{p}, h) \psi(x) \int_{-\infty}^{\infty} dy^3 e^{-iqy} \psi^\dagger(y) u(\mathbf{q}, h') \right. \\ &\quad \left. + \int_{-\infty}^{\infty} dy^3 e^{-iqy} \psi^\dagger(y) u(\mathbf{q}, h') \int_{-\infty}^{\infty} dx^3 e^{ipx} u^\dagger(\mathbf{p}, h) \psi(x) \right] \\ &= \frac{1}{(2\pi)^3 2p_0} \times \end{aligned}$$

$$\int_{-\infty \leftarrow}^{\infty} d\mathbf{x}^3 \int_{-\infty \leftarrow}^{\infty} d\mathbf{y}^3 e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{q}\cdot\mathbf{y}} [u^\dagger(\mathbf{p}, h)\psi(x)\psi^\dagger(y)u(\mathbf{q}, h') + \psi^\dagger(y)u(\mathbf{q}, h')u^\dagger(\mathbf{p}, h)\psi(x)]. \quad (22.236) \leftarrow$$

The RHS is a (1, 1) matrix containing Grassmann numbers $(\psi)^\alpha$, $(\psi^\dagger)_\beta$ ($\alpha, \beta = 1, 2, 3, 4$). The quantity $(u)_\alpha$, however, is an ordinary c -number. Note that the matrix multiplication $(1, 4) \times (4, 1) \times (1, 4) \times (4, 1)$ produces a (1, 1) matrix. Matrix calculations are somewhat tedious, but after termwise estimation of (22.236) we are left with a simple form by virtue of the anti-commutation relation (22.224). The calculation processes are as follows: Regarding $u^\dagger(\mathbf{p}, h)\psi(x)\psi^\dagger(y)u(\mathbf{q}, h')$, its full description is given by

$$\begin{aligned} & u^\dagger(\mathbf{p}, h)\psi(x)\psi^\dagger(y)u(\mathbf{q}, h') \quad \leftarrow \\ &= (u_1(\mathbf{p}, h)^* \quad u_2(\mathbf{p}, h)^* \quad u_3(\mathbf{p}, h)^* \quad u_4(\mathbf{p}, h)^*) \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \\ \psi_3(x) \\ \psi_4(x) \end{pmatrix} \quad (22.237) \leftarrow \\ &\quad \times (\psi_1(y)^* \quad \psi_2(y)^* \quad \psi_3(y)^* \quad \psi_4(y)^*) \begin{pmatrix} u_1(\mathbf{q}, h') \\ u_2(\mathbf{q}, h') \\ u_3(\mathbf{q}, h') \\ u_4(\mathbf{q}, h') \end{pmatrix}. \quad \leftarrow \end{aligned}$$

Meanwhile, $\psi^\dagger(y)u(\mathbf{q}, h')u^\dagger(\mathbf{p}, h)\psi(x)$ is similarly expressed as in (22.237). After the calculations, we get

$$\begin{aligned} & u^\dagger(\mathbf{p}, h)\psi(x)\psi^\dagger(y)u(\mathbf{q}, h') + \psi^\dagger(y)u(\mathbf{q}, h')u^\dagger(\mathbf{p}, h)\psi(x) \quad \leftarrow \\ &= \left[\sum_{\alpha=1}^4 u_\alpha u_\alpha^* \right] \delta^3(\mathbf{x} - \mathbf{y}) = 2p_0 \delta_{hh'} \delta^3(\mathbf{x} - \mathbf{y}), \quad (22.238) \leftarrow \end{aligned}$$

where with the last equality we used (21.156) and (22.229). Inserting (22.238) into (22.236), we obtain

$$\begin{aligned} \{b(\mathbf{p}, h), b^\dagger(\mathbf{q}, h')\} &= \frac{2p_0 e^{it(p_0 - q_0)}}{(2\pi)^3 2p_0} \delta_{hh'} \left[\int_{-\infty \leftarrow}^{\infty} d\mathbf{x}^3 \int_{-\infty \leftarrow}^{\infty} d\mathbf{y}^3 e^{i(\mathbf{q}\cdot\mathbf{y} - \mathbf{p}\cdot\mathbf{x})} \delta^3(\mathbf{x} - \mathbf{y}) \right] \quad \leftarrow \\ &= \frac{2p_0 e^{it(p_0 - q_0)}}{(2\pi)^3 2p_0} \delta_{hh'} \left[\int_{-\infty \leftarrow}^{\infty} d\mathbf{x}^3 e^{i\mathbf{x}\cdot(\mathbf{q} - \mathbf{p})} \right] \end{aligned}$$

$$= \frac{2p_0 e^{it(p_0 - q_0)}}{(2\pi)^3 2p_0} \delta_{hh'} (2\pi)^3 \delta^3(\mathbf{q} - \mathbf{p}) = \delta_{hh'} \delta^3(\mathbf{q} - \mathbf{p}), \quad (22.239) \leftarrow$$

where with the second to the last equality we used (22.101); the last equality was due to the inclusion of $\delta^3(\mathbf{q} - \mathbf{p})$, which leads to $q_0 - p_0 = 0$. Thus, the result of (22.239) has largely been simplified. We remark that the above calculation procedures offer an interesting example for a computation of numbers that contain both Grassmann numbers and ordinary c -numbers.

Other anti-commutation relations can be derived similarly. For example, we obtain the following relation as in the case of (22.235):

$$d^\dagger(\mathbf{q}, h) = \frac{1}{\sqrt{(2\pi)^3 2p_0}} \int_{-\infty}^{\infty} d\mathbf{x}^3 e^{-iqx} v^\dagger(\mathbf{q}, h) \psi(x). \quad (22.240) \leftarrow$$

Also, proceeding similarly to (22.239), we have

$$\{d(\mathbf{p}, h), d^\dagger(\mathbf{q}, h')\} = \delta_{hh'} \delta^3(\mathbf{q} - \mathbf{p}). \quad (22.241) \leftarrow$$

With other combinations of the b -related operators and d -related operators, we have the following anti-commutation relations such that

$$\begin{aligned} \{b(\mathbf{p}, h), b(\mathbf{q}, h')\} &= \{b(\mathbf{p}, h), d(\mathbf{q}, h')\} = \{b(\mathbf{p}, h), d^\dagger(\mathbf{q}, h')\} \leftarrow \\ &= \{b^\dagger(\mathbf{p}, h), b^\dagger(\mathbf{q}, h')\} = \{b^\dagger(\mathbf{p}, h), d(\mathbf{q}, h')\} = \{b^\dagger(\mathbf{p}, h), d^\dagger(\mathbf{q}, h')\} \quad \leftarrow (22.242) \leftarrow \\ &= \{d(\mathbf{p}, h), d(\mathbf{q}, h')\} = \{d^\dagger(\mathbf{p}, h), d^\dagger(\mathbf{q}, h')\} = 0. \end{aligned}$$

These results can be viewed in parallel with (22.105) in spite of the difference between the commutation and anti-commutation relations and the difference in presence or absence of the helicity freedom. Since (22.242) does not contain space–time coordinates, it is useful for various calculations in relation to the quantum fields.

22.4.3 Antiparticle: Positron

The Dirac field is characterized by the antiparticles. The antiparticle of electron is the positron. In Sect. 21.4 we have introduced the notion of the charge conjugation. It can be regarded as the particle–antiparticle transformation. Strictly speaking, it is not until we properly describe the field quantization by introducing the creation and annihilation operators that the charge conjugation has its place. Nonetheless, it is of great use in a mathematical approach to introduce the notion of charge conjugation into the spinor field before being quantized. In fact, the formulation of the charge

conjugation developed in Sect. 21.4 can directly be imported along with the creation and annihilation operators into the field quantization of the antiparticle.

In Sect. 21.4 the charge conjugation of the spinor ψ into ψ^C was described as

$$\psi^C = C\bar{\psi}^T = i\gamma^2\psi^*. \quad (21.161) \leftarrow$$

Borrowing this notation and using (22.232), the antiparticle field $\psi_{\text{anti}}(x)$ is expressed as

$$\begin{aligned} \psi_{\text{anti}}(x) &= \psi^C(x) = i\gamma^2\psi^* \\ &= \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 2p_0}} \sum_{h=\pm 1} [e^{ipx} b^\dagger(\mathbf{p}, h) i\gamma^2 u^*(\mathbf{p}, h) + e^{-ipx} d(\mathbf{p}, h) i\gamma^2 v^*(\mathbf{p}, h)]. \end{aligned} \quad (22.243) \leftarrow$$

Further using (21.166) and (21.167), we get

$$\begin{aligned} \psi_{\text{anti}}(x) &\leftarrow \\ &= \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 2p_0}} \sum_{h=\pm 1} [e^{ipx} b^\dagger(\mathbf{p}, h) v(\mathbf{p}, h) + e^{-ipx} d(\mathbf{p}, h) u(\mathbf{p}, h)]. \end{aligned} \quad (22.244) \leftarrow$$

Comparing (22.232) and (22.244), we find that $\psi(x)$ and $\psi_{\text{anti}}(x)$ can be obtained merely by exchanging $b(\mathbf{p}, h)$ and $d(\mathbf{p}, h)$.

The discrimination between the particle and antiparticle is associated with whether the operator $\psi(x)$ is Hermitian or not. Since in the case of the neutral scalar field the operator $\phi(x)$ is Hermitian, we have no discrimination between the particle and antiparticle.

With the Dirac field, however, the operator $\psi(x)$ is not Hermitian, but the particle and antiparticle should be distinguished. We may think that destroying the negative-energy state is equivalent to creating the positive-energy state, or vice versa. Interested readers are referred to appropriate literature with more detailed discussions on the particle and antiparticle as well as their implications [8, 12].

22.4.4 Invariant Delta Functions of the Dirac Field

In this section we introduce the invariant delta functions of the Dirac field. As in the case of the scalar field, these functions are of great importance in studying the interaction of quantum fields. The derivation of these functions is parallel with those of the scalar field.

We have performed the canonical quantization by setting the equal-time anti-commutation relation (22.224). Combining (22.224) with the Dirac field expansion

described by (22.232) and (22.244), we introduced the anti-commutation relations between the creation and annihilation operators expressed as (22.239), (22.241), and (22.242). Further combining the Dirac field expansion with the anti-commutation relations between the creation and annihilation operators, we wish to seek the covariant anti-commutation relations between the Dirac fields. Note that unlike the equal-time anti-commutation relation (22.224) the covariant anti-commutation relations do not contain the (equal) time variable explicitly.

We obtain a following covariant relations at once such that

$$\{\psi(x), \psi(y)\} = \{\bar{\psi}(x), \bar{\psi}(y)\} = 0. \quad (22.245) \leftarrow$$

With this expression see Example 22.1.

To explore other covariant anti-commutation relations of the Dirac field, we decompose the field operators into the positive-frequency and negative-frequency parts as in the case of the scalar field (Sect. 22.3.4). For example, from (22.232) we have

$$\begin{aligned} \psi^+(x) &= \int \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 2p_0}} \sum_{h=\pm 1} e^{-ipx} b(\mathbf{p}, h) u(\mathbf{p}, h), \\ \bar{\psi}^-(x) &= \int \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 2p_0}} \sum_{h=\pm 1} e^{ipx} b^\dagger(\mathbf{p}, h) \bar{u}(\mathbf{p}, h). \end{aligned} \quad (22.246) \leftarrow$$

Similarly, a pair of functions $\psi^-(x)$ and $\bar{\psi}^+(x)$ is obtained such that

$$\begin{aligned} \psi^-(x) &= \int \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 2p_0}} \sum_{h=\pm 1} e^{ipx} d^\dagger(\mathbf{p}, h) v(\mathbf{p}, h), \\ \bar{\psi}^+(x) &= \int \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 2p_0}} \sum_{h=\pm 1} e^{-ipx} d(\mathbf{p}, h) \bar{v}(\mathbf{p}, h). \end{aligned}$$

From (22.246) we get

$$\begin{aligned} &\{\psi^+(x), \bar{\psi}^-(x)\} \quad \leftarrow \\ &= \frac{1}{(2\pi)^3} \int \frac{dq^3 d\mathbf{p}^3}{\sqrt{2p_0 \cdot 2q_0}} \sum_{h, h'=\pm 1} e^{-i(px - qy)} \quad \leftarrow \\ &\quad \times [b(\mathbf{p}, h) b^\dagger(\mathbf{q}, h') u(\mathbf{p}, h) \bar{u}(\mathbf{q}, h') + b^\dagger(\mathbf{q}, h') b(\mathbf{p}, h) \bar{u}(\mathbf{q}, h') u(\mathbf{p}, h)]. \quad \leftarrow \end{aligned} \quad (22.247) \leftarrow$$

As implied in Example 22.1, the factor $\bar{u}(\mathbf{q}, h') u(\mathbf{p}, h)$ in the second term of the integrand of (22.247) must be read as

$$\bar{u}(\mathbf{q}, h') u(\mathbf{p}, h) = \left[\bar{u}(\mathbf{q}, h')^T u(\mathbf{p}, h)^T \right]^T = u(\mathbf{p}, h) \bar{u}(\mathbf{q}, h'). \quad (22.248) \leftarrow$$

Notice that since (22.248) does not contain the Grassmann numbers, unlike (22.227) the second equality of (22.248) holds. Then, (22.247) can be rewritten as

$$\begin{aligned} \{\psi^+(x), \bar{\psi}^-(x)\} &= \frac{1}{(2\pi)^3} \int \frac{d\mathbf{q}^3 d\mathbf{p}^3}{\sqrt{2p_0} \cdot 2q_0} \sum_{h, h' = \pm 1} e^{-i(px - qy)} \quad \leftarrow \\ &\quad \times u(\mathbf{p}, h) \bar{u}(\mathbf{q}, h) [b(\mathbf{p}, h) b^\dagger(\mathbf{q}, h') + b^\dagger(\mathbf{q}, h') b(\mathbf{p}, h)] \quad \leftarrow \\ &= \frac{1}{(2\pi)^3} \int \frac{d\mathbf{q}^3 d\mathbf{p}^3}{\sqrt{2p_0} \cdot 2q_0} \sum_{h, h' = \pm 1} e^{-i(px - qy)} u(\mathbf{p}, h) \bar{u}(\mathbf{q}, h') \{b(\mathbf{p}, h), b^\dagger(\mathbf{q}, h')\} \quad \leftarrow \\ &= \frac{1}{(2\pi)^3} \int \frac{d\mathbf{q}^3 d\mathbf{p}^3}{\sqrt{2p_0} \cdot 2q_0} \sum_{h, h' = \pm 1} e^{-i(px - qy)} u(\mathbf{p}, h) \bar{u}(\mathbf{q}, h') \delta_{hh'} \delta^3(\mathbf{q} - \mathbf{p}) \quad [\dagger] \leftarrow \\ &= \frac{1}{(2\pi)^3} \int \frac{d\mathbf{p}^3}{2p_0} \sum_{h = \pm 1} e^{-ip(x-y)} u(\mathbf{p}, h) \bar{u}(\mathbf{p}, h) \quad \leftarrow \\ &= \frac{1}{(2\pi)^3} \int \frac{d\mathbf{p}^3}{2p_0} e^{-ip(x-y)} (p_\mu \tilde{\gamma}^\mu + m) \quad [\dagger\dagger] \\ &= (i\gamma^\mu \partial_\mu^x + m) \frac{1}{(2\pi)^3} \int \frac{d\mathbf{p}^3}{2p_0} e^{-ip(x-y)} = (i\gamma^\mu \partial_\mu^x + m) i\Delta^+(x-y). \quad [\dagger\dagger\dagger] \end{aligned} \quad (22.249) \leftarrow$$

At $[\dagger]$ in (22.249) we used (22.239); at $[\dagger\dagger]$ we used (21.144); at $[\dagger \dagger \dagger]$ we used (22.164). In (22.249) ∂_μ^x means the partial differentiation with respect to x .

Similarly calculating $\{\psi^-(x), \bar{\psi}^+(y)\}$, we obtain

$$\{\psi^-(x), \bar{\psi}^+(y)\} = (i\gamma^\mu \partial_\mu^x + \not{m}) i\Delta^-(x-y),$$

where $\Delta^-(x-y)$ is identical with that appearing in (22.168). Also, we define

$$\begin{aligned} (i\gamma^\mu \partial_\mu^x + \not{m}) i\Delta^+(x-y) &\equiv iS^+(x-y) = \{\psi^+(x), \bar{\psi}^-(x)\}, \\ (i\gamma^\mu \partial_\mu^x + \not{m}) i\Delta^-(x-y) &\equiv iS^-(x-y) = \{\psi^-(x), \bar{\psi}^+(y)\}. \end{aligned} \quad (22.250) \leftarrow$$

Summing two equations of (22.250), using (22.139), and further defining

$$S(x-y) \equiv S^+(x-y) + S^-(x-y),$$

we obtain [5]

$$\left(i\gamma^\mu \partial_\mu^x + m\right)i\Delta(x-y) \equiv iS(x-y). \quad (22.251) \leftarrow$$

Meanwhile, we have [5]

$$\begin{aligned} \{\psi(x), \bar{\psi}(y)\} &\leftarrow \\ &= \{\psi^+(x), \bar{\psi}^+(x)\} + \{\psi^+(x), \bar{\psi}^-(x)\} + \{\psi^-(x), \bar{\psi}^+(x)\} + \{\psi^-(x), \bar{\psi}^-(x)\} \\ &= \{\psi^+(x), \bar{\psi}^-(x)\} + \{\psi^-(x), \bar{\psi}^+(x)\} = iS^+(x-y) + iS^-(x-y) \leftarrow \\ &= iS(x-y), \end{aligned} \quad (22.252) \leftarrow$$

where with the first equality, the first term and fourth term vanish because of (22.242).

22.4.5 Feynman Propagator of the Dirac Field

Similarly to the case of the scalar field, the Feynman propagator $S_F(x-y)$ is defined as a vacuum expectation value of the T-product such that

$$S_F(x-y) \equiv \langle 0 | T[\psi(x) \bar{\psi}(y)] | 0 \rangle. \quad (22.253) \leftarrow$$

In the case of the Dirac field, the time-ordered product (T-product) $T[\psi(x) \bar{\psi}(y)]$ is defined by

$$T[\psi(x) \bar{\psi}(y)] \equiv \theta(x^0 - y^0) \psi(x) \bar{\psi}(y) - \theta(y^0 - x^0) \bar{\psi}(y) \psi(x). \quad (22.254) \leftarrow$$

Note the minus sign before the second term of (22.254). It comes from the fact that the Fermion operators $\psi(x)$ and $\bar{\psi}(y)$ are anti-commutative. We have to pay attention to the symbolic notation of (22.254) as in the case of the calculation of $\{\psi^+(x), \bar{\psi}^-(x)\}$ and $\{\psi^-(x), \bar{\psi}^+(x)\}$. In fact, $\bar{\psi}(y)\psi(x)$ must be dealt with as a (4, 4) matrix as well. Taking the vacuum expectation value of (22.254), we have

$$\begin{aligned} S_F(x-y) &= \langle 0 | T[\psi(x) \bar{\psi}(y)] | 0 \rangle \leftarrow \\ &= \theta(x^0 - y^0) \langle 0 | \psi(x) \bar{\psi}(y) | 0 \rangle - \theta(y^0 - x^0) \langle 0 | \bar{\psi}(y) \psi(x) | 0 \rangle. \end{aligned} \quad (22.255) \leftarrow$$

Similarly to the scalar field case of (22.183), we have

$$\langle 0 | \psi(x) \bar{\psi}(y) | 0 \rangle = \{\psi^+(x), \bar{\psi}^-(y)\} = iS^+(x-y), \quad (22.256) \leftarrow$$

$$\langle 0 | \bar{\psi}(y) \psi(x) | 0 \rangle = \{\psi^-(x), \bar{\psi}^+(y)\} = iS^-(x-y). \quad (22.257) \leftarrow$$

Substituting (22.256) and (22.257) for (22.255), we get

$$S_F(x-y) = i[\theta(x^0 - y^0)S^+(\bar{x}-y) - \theta(y^0 - x^0)S^-(x-y)]. \quad \leftarrow (22.258) \leftarrow$$

Then, from (22.232), we have

$$\begin{aligned} & T[\psi(x)\bar{\psi}(y)] \\ &= \int_{-\infty \leftarrow}^{\infty} d\mathbf{q}^3 d\mathbf{p}^3 \frac{\theta(x^0 - y^0)}{(2\pi)^3 2p_0} \left\{ \sum_{h=\pm 1}^{\leftarrow} [e^{-ipx} b(\mathbf{p}, h) u(\mathbf{p}, h) + e^{ipx} d^\dagger(\bar{\mathbf{p}}, h) v(\mathbf{p}, h)] \right. \\ & \quad \times \sum_{h'=\pm 1} [e^{ipy} b^\dagger(\mathbf{q}, h') \bar{u}(\mathbf{q}, h') + e^{-iqy} d(\mathbf{q}, h') \bar{v}(\mathbf{q}, h')] \Big\} \\ & \quad - \int_{-\infty \leftarrow}^{\infty} d\mathbf{q}^3 d\mathbf{p}^3 \frac{\theta(y^0 - x^0)}{(2\pi)^3 2p_0} \left\{ \sum_{h'=\pm 1}^{\leftarrow} [e^{ipy} b^\dagger(\mathbf{q}, h') \bar{u}(\mathbf{q}, h') + e^{-iqy} d(\mathbf{q}, h') \bar{v}(\mathbf{q}, h')] \right. \\ & \quad \times \sum_{h=\pm 1} [e^{-ipx} b(\mathbf{p}, h) u(\mathbf{p}, h) + e^{ipx} d^\dagger(\bar{\mathbf{p}}, h) v(\mathbf{p}, h)] \Big\}. \end{aligned} \quad (22.259) \leftarrow$$

Therefore, we obtain

$$\begin{aligned} & \langle 0 | T[\psi(x)\bar{\psi}(y)] | 0 \rangle \quad \leftarrow \\ &= \int_{-\infty \leftarrow}^{\infty} d\mathbf{q}^3 d\mathbf{p}^3 \frac{\theta(x^0 - y^0)}{(2\pi)^3 2p_0} \langle 0 | \left\{ \sum_{h=\pm 1}^{\leftarrow} [e^{-ipx} b(\mathbf{p}, h) u(\mathbf{p}, h) + e^{ipx} d^\dagger(\bar{\mathbf{p}}, h) v(\mathbf{p}, h)] \right. \right. \leftarrow \\ & \quad \times \sum_{h'=\pm 1} [e^{ipy} b^\dagger(\mathbf{q}, h') \bar{u}(\mathbf{q}, h') + e^{-iqy} d(\mathbf{q}, h') \bar{v}(\mathbf{q}, h')] \Big\} | 0 \rangle \\ & \quad - \int_{-\infty \leftarrow}^{\infty} d\mathbf{q}^3 d\mathbf{p}^3 \frac{\theta(y^0 - x^0)}{(2\pi)^3 2p_0} \langle 0 | \left\{ \sum_{h'=\pm 1}^{\leftarrow} [e^{ipy} b^\dagger(\mathbf{q}, h') \bar{u}(\mathbf{q}, h') + e^{-iqy} d(\mathbf{q}, h') \bar{v}(\mathbf{q}, h')] \right. \right. \\ & \quad \times \sum_{h=\pm 1} [e^{-ipx} b(\mathbf{p}, h) u(\mathbf{p}, h) + e^{ipx} d^\dagger(\bar{\mathbf{p}}, h) v(\mathbf{p}, h)] \Big\} | 0 \rangle. \end{aligned} \quad (22.260) \leftarrow$$

Equation (22.260) comprises two integrals and the integrands of each integral comprise $2 \times 2 = 4$ terms. However, we do not need to calculate all of them. This is because in (22.260) we only have to calculate the integrals with respect to the following two integrands that lead to non-vanishing integrals:

$$A \equiv \sum_{h, h'=\pm 1} e^{-i(px - qy)} \langle 0 | b(\bar{\mathbf{p}}, h) u(\mathbf{p}, h) \cdot b^\dagger(\mathbf{q}, h') \bar{u}(\mathbf{q}, h') | 0 \rangle$$

and

$$B \equiv \sum_{h, h' = \pm 1} e^{-i(qy - px)} \langle 0 | d(\bar{\mathbf{q}}, h') \bar{v}(\mathbf{q}, h') \cdot d^\dagger(\mathbf{p}, h) v(\mathbf{p}, h) | 0 \rangle. \quad (22.261) \leftarrow$$

Other terms vanish similarly as in the case of (22.121). It is because we have

$$b(\mathbf{p}, h) | 0 \rangle = d(\mathbf{q}, h') | 0 \rangle = 0$$

and

$$\langle 0 | b^\dagger(\mathbf{q}, h') = \langle 0 | d^\dagger(\mathbf{p}, h) = 0. \quad (22.262) \leftarrow$$

We estimate the above two integrands A and B one by one. First, we rewrite A as

$$\begin{aligned} A &\equiv \sum_{h, h' = \pm 1} e^{-i(px - qy)} \langle 0 | b(\bar{\mathbf{p}}, h) u(\mathbf{p}, h) \cdot b^\dagger(\mathbf{q}, h') \bar{u}(\mathbf{q}, h') | 0 \rangle \\ &= \sum_{h, h' = \pm 1} e^{-i(px - qy)} \langle 0 | [\delta_{hh'}^\dagger \delta^3(\mathbf{q} - \mathbf{p}) - b^\dagger(\mathbf{q}, h') b(\mathbf{p}, h)] u(\mathbf{p}, h) \bar{u}(\mathbf{q}, h') | 0 \rangle \\ &= \sum_{h, h' = \pm 1} e^{-i(px - qy)} \langle 0 | \delta_{hh'}^\dagger \delta^3(\mathbf{q} - \mathbf{p}) u(\mathbf{p}, h) \bar{u}(\mathbf{q}, h') | 0 \rangle \\ &= \sum_{h, h' = \pm 1} e^{-i(px - qy)} \delta_{hh'} \delta^3(\mathbf{q} - \mathbf{p}) \langle 0 | u(\mathbf{p}, h) \bar{u}(\mathbf{q}, h') | 0 \rangle, \end{aligned} \quad (22.263) \leftarrow$$

where the second equality comes from (22.239) and the third equality from (22.262). Paying attention to the fact that $u(\mathbf{p}, h) \bar{u}(\mathbf{q}, h')$ is not a quantum operator but a $(4, 4)$ matrix, we have

$$\langle 0 | u(\mathbf{p}, h) \bar{u}(\mathbf{q}, h') | 0 \rangle = u(\mathbf{p}, h) \bar{u}(\mathbf{q}, h') \langle 0 | 0 \rangle = u(\mathbf{p}, h) \bar{u}(\mathbf{q}, h'). \quad (22.264) \leftarrow$$

Notice that $\langle 0 | 0 \rangle = 1$, i.e., the vacuum state has been normalized. Thus, we obtain

$$A = \sum_{h, h' = \pm 1} e^{-i(px - qy)} \delta_{hh'} \delta^3(\mathbf{q} - \mathbf{p}) u(\mathbf{p}, h) \bar{u}(\mathbf{q}, h'). \quad (22.265) \leftarrow$$

Similarly, we have

$$B = \sum_{h, h' = \pm 1} e^{-i(qy - px)} \delta_{hh'} \delta^3(\mathbf{q} - \mathbf{p}) \bar{v}(\mathbf{q}, h') v(\mathbf{p}, h). \quad (22.266) \leftarrow$$

Neither A nor B is a Grassmann number but ordinary number. Hence, according to the manipulation as noted earlier in (22.248), we obtain

$$\bar{v}(\mathbf{q}, h') v(\mathbf{p}, h) = v(\mathbf{p}, h) \bar{v}(\mathbf{q}, h'). \quad (22.267) \leftarrow$$

Substituting (22.265) and (22.266) for (22.260), by the aid of (22.267) we get

$$\begin{aligned} \langle 0 | T[\psi(x) \bar{\psi}(y)] | 0 \rangle &= \int_{-\infty}^{\infty} d\mathbf{q}^3 d\mathbf{p}^3 \frac{\theta(x^0 - y^0)}{(2\pi)^3 2p_0} A - \int_{-\infty}^{\infty} d\mathbf{q}^3 d\mathbf{p}^3 \frac{\theta(y^0 - x^0)}{(2\pi)^3 2p_0} B \quad \leftarrow \\ &= \int_{-\infty}^{\infty} d\mathbf{q}^3 d\mathbf{p}^3 \frac{\theta(x^0 - y^0)}{(2\pi)^3 2p_0} \sum_{h, h' = \pm 1} e^{-i(px - qy)} \delta_{hh'} \delta^3(\mathbf{q} - \mathbf{p}) u(\mathbf{p}, h) \bar{u}(\mathbf{q}, h') \\ &\quad - \int_{-\infty}^{\infty} d\mathbf{q}^3 d\mathbf{p}^3 \frac{\theta(y^0 - x^0)}{(2\pi)^3 2p_0} \sum_{h, h' = \pm 1} e^{-i(qy - px)} \delta_{hh'} \delta^3(\mathbf{q} - \mathbf{p}) v(\mathbf{p}, h) \bar{v}(\mathbf{q}, h') \leftarrow \\ &\quad (22.268) \leftarrow \end{aligned}$$

Performing the integration with respect to $\mathbf{q}$ and summation with respect to h' , we obtain

$$\begin{aligned} \langle 0 | T[\psi(x) \bar{\psi}(y)] | 0 \rangle &= \int_{-\infty}^{\infty} d\mathbf{p}^3 \frac{\theta(x^0 - y^0)}{(2\pi)^3 2p_0} \sum_{h = \pm 1} e^{-ip(x-y)} u(\mathbf{p}, h) \bar{u}(\mathbf{p}, h) \\ &\quad - \int_{-\infty}^{\infty} d\mathbf{p}^3 \frac{\theta(y^0 - x^0)}{(2\pi)^3 2p_0} \sum_{h = \pm 1} e^{ip(x-y)} v(\mathbf{p}, h) \bar{v}(\mathbf{p}, h). \quad (22.269) \leftarrow \end{aligned}$$

Using (21.144) and (21.145), finally we obtain

$$\begin{aligned} S_F(x-y) &= \langle 0 | T[\psi(x) \bar{\psi}(y)] | 0 \rangle \quad \leftarrow \\ &= \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{(2\pi)^3 2p_0} \left[\theta(x^0 - y^0) e^{-ip(x-y)} (p_\mu \tilde{\gamma}^\mu + m) - \theta(y^0 - x^0) e^{ip(x-y)} (p_\mu \tilde{\gamma}^\mu - m) \right] \\ &= \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{(2\pi)^3 2p_0} \left[\theta(x^0 - y^0) e^{-ip(x-y)} (p_\mu \tilde{\gamma}^\mu + m) + \theta(y^0 - x^0) e^{ip(x-y)} (-p_\mu \tilde{\gamma}^\mu + m) \right]. \quad (22.270) \leftarrow \end{aligned}$$

We get another representation of $S_F(x-y)$ in a way analogous to (22.251). It is described by

$$S_F(x-y) = \left(i\gamma^\mu \partial_\mu^x + \not{m} \right) \Delta_F(x-y), \quad (22.271) \leftarrow$$

where $\Delta_F(x-y)$ was given by (22.201). With the derivation of (22.271), use (22.270) and perform the differentiation with respect to the individual coordinate x^μ . We should be careful in treating the differentiation with respect to x^0 , because it contains the differentiation of $\theta(x^0 - y^0)$.

We show the related calculations as follows:

$$\begin{aligned}
& i\gamma^0 \partial_0^x \Delta_F(x-y) \quad \leftarrow \\
& = \int_{-\infty}^{\infty} \frac{i\gamma^0 d\mathbf{p}^3}{(2\pi)^3 2p_0} \partial_0^x \left[\theta(x^0 - y^0) e^{-ip(x-y)} + \theta(y^0 - x^0) e^{ip(x-y)} \right] \quad \leftarrow \\
& = \int_{-\infty}^{\infty} \frac{i\gamma^0 d\mathbf{p}^3}{(2\pi)^3 2p_0} \left[\delta(x^0 - y^0) e^{-ip(x-y)} + \theta(x^0 - y^0) (-ip_0) e^{-ip(x-y)} \right. \\
& \quad \left. - \delta(y^0 - x^0) e^{ip(x-y)} + \theta(y^0 - x^0) (ip_0) e^{ip(x-y)} \right] \quad \leftarrow \tag{22.272} \\
& = \int_{-\infty}^{\infty} \frac{i\gamma^0 d\mathbf{p}^3}{(2\pi)^3 2p_0} \left[\delta(x^0 - y^0) e^{-ip(x-y)} - \delta(y^0 - x^0) e^{ip(x-y)} \right. \\
& \quad \left. + \theta(x^0 - y^0) (-ip_0) e^{-ip(x-y)} + \theta(y^0 - x^0) (ip_0) e^{ip(x-y)} \right] \quad \leftarrow \\
& = \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3 p_0 \gamma^0}{(2\pi)^3 2p_0} \left[\theta(x^0 - y^0) e^{-ip(x-y)} - \theta(y^0 - x^0) e^{ip(x-y)} \right], \quad \leftarrow
\end{aligned}$$

where with the second to the last equality the first and second integrands canceled out each other. For this, use the properties of δ functions; i.e., $\delta(x^0 - y^0) e^{-ip(x-y)} = \delta(x^0 - y^0) e^{ip(x-y)}$ and $\delta(x^0 - y^0) = \delta(y^0 - x^0)$ along with $\int_{-\infty}^{\infty} e^{ip(x-y)} d^3\mathbf{p} = \int_{-\infty}^{\infty} e^{-ip(x-y)} d^3\mathbf{p}$.

Also, we make use of the following relations:

$$\begin{aligned}
& \sum_{k=1}^3 i\gamma^k \partial_k^x \Delta_F(x-y) \quad \leftarrow \\
& = \sum_{k=1}^3 \int_{-\infty}^{\infty} \frac{i\gamma^k d\mathbf{p}^3}{(2\pi)^3 2p_0} \partial_k^x \left[\theta(x^0 - y^0) e^{-ip(x-y)} + \theta(y^0 - x^0) e^{ip(x-y)} \right] \quad \leftarrow \\
& = \sum_{k=1}^3 \int_{-\infty}^{\infty} \frac{i\gamma^k d\mathbf{p}^3}{(2\pi)^3 2p_0} \left[\theta(x^0 - y^0) (-ip_k) e^{-ip(x-y)} \right. \\
& \quad \left. + \theta(y^0 - x^0) (ip_k) e^{ip(x-y)} \right] \quad \leftarrow \tag{22.273} \\
& = \sum_{k=1}^3 \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3 p_k \gamma^k}{(2\pi)^3 2p_0} \left[\theta(x^0 - y^0) e^{-ip(x-y)} - \theta(y^0 - x^0) e^{ip(x-y)} \right]. \quad \leftarrow
\end{aligned}$$

Adding (22.272) and (22.273) together with $m\Delta_F(x-y)$ and taking account of (22.201), we acknowledge that (22.271) is identical to (22.270).

We wish to relate $S_F(x-y)$ expressed as (22.270) to the momentum representation (or Fourier integral representation) as discussed in Sect. 22.3.5. We are readily aware that the difference between (22.270) and (22.201) is with or without the factor of $(p_\mu \gamma^\mu + m)$ or $(-p_\mu \gamma^\mu + m)$. We derive these factors by modifying the functional form of the integrand of (22.186). As studied in Sect. 22.3.5, (22.201) was based on the residue obtained by the contour integral with respect to k_0 .

In Sect. 6.5, we considered a complex function having a pole of order n at $z = a$ that is represented by $f(z) = g(z)/(z-a)^n$, where $g(z)$ is analytic within a domain we are thinking of. In Sect. 22.3.5, we thought of a type of a function having a simple pole given by $e^{-ikz}/(z-a)$ with $g(z) = e^{-ikz}$. In the present case, we are thinking of a

type of a function described by $e^{-ikz}(cz + b)/(z - a)$, where c and b are certain constants. That is, we have

$$g(z) = e^{-ikz}(cz + \phi).$$

From (6.148), we have

$$\text{Res } f(a) = f(z)(z - a)|_{z=a} = g(a).$$

Therefore, the residue of the present case is given by $g(a) = e^{-ika}(ca + b)$. Taking account of (22.270), we assume the next analytic function described by

$$\begin{aligned} g(p_0, \mathbf{p}) &= e^{ip(x-y)} e^{-ip_0(x^0-y^0)} \left(p_0 \gamma^0 + \sum_{k=1}^3 p_k \gamma^k + m \right) \\ &= e^{-ip(x-y)} (p_\mu \tilde{\gamma}^\mu + \not{m}). \end{aligned} \quad (22.274) \leftarrow$$

Since in (22.274) $e^{ip(x-y)}$ and $\sum_{k=1}^3 p_k \gamma^k$ are constants with respect to p_0 along with m , the $\mathbf{p}$ dependence is denoted by $g(p_0, \mathbf{p})$ in (22.274). The function $g(p_0)$ consists of a product of an exponential function and a linear function of p_0 and is an entire function (Definition 6.10). According to the case where the pole is located at $p_0 = \mp p_0$, we obtain $g(\mp p_0)$, which corresponds to Case I ($-p_0$) and Case II ($+p_0$) of Sect. 22.3.5, respectively.

With Case I, the contour integration produces a residue $g(-p_0, \mathbf{p})$ described by

$$g(-p_0, \mathbf{p}) = e^{ip(x-y)} e^{ip_0(x^0-y^0)} \left(-p_0 \gamma^0 + \sum_{k=1}^3 p_k \gamma^k + \not{m} \right).$$

As in the scalar field case, switching $\mathbf{p}$ to $-\mathbf{p}$ effects no change in an ensuing definite integral. Then, with Case I we obtain a function given by

$$\begin{aligned} g(-p_0, -\mathbf{p}) &= e^{-ip(x-\mathbf{y})} e^{ip_0(x^0-y^0)} \left(-p_0 \gamma^0 - \sum_{k=1}^3 p_k \gamma^k + \not{m} \right) \\ &= e^{ip(x-y)} \left(-p_0 \gamma^0 - \sum_{k=1}^3 p_k \gamma^k + \not{m} \right) \\ &= e^{ip(x-y)} (-p_\mu \gamma^\mu + \not{m}). \end{aligned}$$

Comparing (22.201) and (22.270), we find that instead of $e^{-ik(x-y)}$ in (22.186), $g(p_0, \mathbf{p}) = e^{-ip(x-y)}(p_\mu \tilde{\gamma}^\mu + m)$ produces a proper Feynman propagator of the Dirac field $S_F(x-y)$. It is described by

$$S_F(x-y) = \frac{-i}{(2\pi)^4} \int d^4p \frac{e^{-ip(x-y)} (p_\mu \gamma^\mu + m)}{m^2 - p^2 - i\epsilon}. \quad (22.275) \leftarrow$$

Equation (22.275) is well known as the Lorentz invariant representation of the Feynman propagator for the Dirac field. The momentum space representation $\widetilde{S}_F(p)$ —(or Fourier counterpart of the coordinate representation) of the Feynman propagator is given in accordance with (22.188) as

$$\widetilde{S}_F(p) \equiv \frac{-i(p_\mu \gamma^\mu + m)}{m^2 - p^2 - i\epsilon}. \quad (22.276) \leftarrow$$

We will make the most of the above-mentioned Feynman propagator in the next chapter that deals with the interaction between the quantum fields.

22.5 Quantization of the Electromagnetic Field [5–7]

In Part II we studied the classical electromagnetic theory based on the Maxwell's equations, as in the case of the scalar field and the Dirac field, we deal with the quantization of the electromagnetic field (or photon field). Also, in this case the relativistic description of the electromagnetic field is essential. The quantized theory of both the Dirac field and electromagnetic field directly leads to the construction of the quantum electrodynamics (QED).

22.5.1 Relativistic Formulation of the Electromagnetic Field

To properly define the Lagrangian density and Hamiltonian density of the electromagnetic field, we revisit Part II as a preliminary stage. First, we introduce the Heaviside–Lorentz units (in combination with the natural units) to simplify the description of the Maxwell's equations. It is because the use of the dielectric constant of vacuum (ϵ_0) and permeability of vacuum (μ_0) is often troublesome. To delete these constants, we redefine the electromagnetic quantities as follows:

$$\begin{aligned} E, D &\rightarrow E/\sqrt{\epsilon_0}, \quad \sqrt{\epsilon_0} D; \\ H, B &\rightarrow H/\sqrt{\mu_0}, \quad \sqrt{\mu_0} B; \\ \rho, \mathbf{i} &\rightarrow \sqrt{\epsilon_0} \rho, \quad \sqrt{\epsilon_0} \mathbf{i}. \end{aligned}$$

As a result, Maxwell's equations of electromagnetism (7.1)–(7.4) are recast as follows:

$$\begin{aligned}
\operatorname{div} \mathbf{E} &= \rho, \\
\operatorname{div} \mathbf{B} &= 0, \\
\operatorname{rot} \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= 0, \\
\operatorname{rot} \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} &= \mathbf{i}.
\end{aligned} \tag{22.277} \leftarrow$$

In essence, we have redefined the electromagnetic quantities such that

$$\mathbf{D} = \mathbf{E} \quad \text{and} \quad \mathbf{B} = \mathbf{H}. \tag{22.278} \leftarrow$$

That is, $\mathbf{D} = \varepsilon_0 \mathbf{E}$, e.g., the relationship (7.7) in vacuum is rewritten as

$$\sqrt{\varepsilon_0} \mathbf{D} = \varepsilon_0 (\mathbf{E} / \sqrt{\varepsilon_0}) = \sqrt{\varepsilon_0} \mathbf{E} \quad \text{or} \quad \mathbf{D} = \mathbf{E}.$$

In (22.277) $\mu_0 \varepsilon_0 = 1$ is implied in accordance with the natural units; see (7.12). Next, we introduce vector potential and scalar potential to express the electromagnetic fields. Of the Maxwell's equations, with the magnetic flux density $\mathbf{B}$ we had

$$\operatorname{div} \mathbf{B} = 0. \tag{7.2} \leftarrow$$

We knew a formula of vector analysis described by

$$\operatorname{div} \operatorname{rot} \mathbf{V} = 0, \tag{7.22} \leftarrow$$

for any vector field $\mathbf{V}$. From the above equations, we immediately express $\mathbf{B}$ as

$$\mathbf{B} = \operatorname{rot} \mathbf{A}, \tag{22.279} \leftarrow$$

where $\mathbf{A}$ is a vector field called the vector potential. We know another formula of vector analysis

$$\operatorname{rot} \nabla \Lambda \equiv 0, \tag{22.280} \leftarrow$$

where Λ is any scalar field. The confirmation of (22.280) is left for readers. Therefore, the vector potential $\mathbf{A}$ has arbitrariness of choice so that $\mathbf{B}$ can remain unchanged as follows:

$$\mathbf{A}' \equiv \mathbf{A} - \nabla \Lambda \quad \text{and} \quad \mathbf{B} = \operatorname{rot} \mathbf{A}'. \tag{22.281} \leftarrow$$

Meanwhile, replacing $\mathbf{B}$ in the third equation of (22.277) with (22.279) and exchanging the operating order of the differential operators $\partial/\partial t$ and rot , we obtain

$$\text{rot} \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = \mathbf{0}. \quad (22.282) \leftarrow$$

Using (22.280) we express $\mathbf{E}$ as

$$\mathbf{E} = - \frac{\partial \mathbf{A}}{\partial t} - \nabla \phi, \quad (22.283) \leftarrow$$

where ϕ is another scalar field called the scalar potential. The scalar potential ϕ and vector potential $\mathbf{A}$ are said to be electromagnetic potentials as a collective term. Nonetheless, we have no clear reason to mathematically distinguish ϕ of (22.283) from Λ in (22.280) or (22.281).

Considering that these potentials ϕ and $\mathbf{A}$ have arbitrariness, we wish to set the functional relationship between them by imposing an appropriate constraint. Such constraint on the electromagnetic potentials is called the gauge transformation. The gauge transformation is defined by

$$\begin{aligned} \mathbf{A}'(\mathbf{r}, t) &\equiv \mathbf{A}(\mathbf{r}, t) - \nabla \Lambda(\mathbf{r}, t), \\ \phi'(\mathbf{r}, t) &\equiv \phi(\mathbf{r}, t) + \frac{\partial \Lambda(\mathbf{r}, t)}{\partial t}. \end{aligned} \quad \leftarrow \quad (22.284) \leftarrow$$

Note that $\mathbf{A}'$ and ϕ' denote the change in the functional form caused by the gauge transformation. The gauge transformation holds the electric field $\mathbf{E}$ and magnetic flux density $\mathbf{B}$ unchanged such that

$$\begin{aligned} \mathbf{E}'(\mathbf{r}, t) &\leftarrow \frac{\partial \mathbf{A}'(\mathbf{r}, t)}{\partial t} - \nabla \phi'(\mathbf{r}, t) \quad \leftarrow \\ &= - \frac{\partial \mathbf{A}(\mathbf{r}, t)}{\partial t} + \frac{\partial \nabla \Lambda(\mathbf{r}, t)}{\partial t} - \nabla \phi(\mathbf{r}, t) - \nabla \frac{\partial \Lambda(\mathbf{r}, t)}{\partial t} \quad \leftarrow \\ &= - \frac{\partial \mathbf{A}(\mathbf{r}, t)}{\partial t} - \nabla \phi(\mathbf{r}, t), \end{aligned}$$

where the last equality resulted from the fact that the operating order of $\partial/\partial t$ and grad is exchangeable in the second equality. Also, we have

$$\mathbf{B}'(\mathbf{r}, t) = \text{rot } \mathbf{A}'(\mathbf{r}, t) = \text{rot } \mathbf{A}(\mathbf{r}, t) - \text{rot } [\nabla \Lambda(\mathbf{r}, t)] = \text{rot } \mathbf{A}(\mathbf{r}, t).$$

Having the above-mentioned background knowledge, we recast the Maxwell's equations in the relativistic formulation. For this, we introduce the following anti-symmetric field tensor $F^{\mu\nu}$ (in a contravariant tensor form) expressed as

$$\begin{aligned}
 F^{\mu\nu} &\equiv \begin{pmatrix} F^{00} & F^{01} & F^{02} & F^{03} \\ F^{10} & F^{11} & F^{12} & F^{13} \\ F^{20} & F^{21} & F^{22} & F^{23} \\ F^{30} & F^{31} & F^{32} & F^{33} \end{pmatrix} = \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E^1 & 0 & -B^3 & B^2 \\ E^2 & B^3 & 0 & -B^1 \\ E^3 & -B^2 & B^1 & 0 \end{pmatrix} \\
 &= -F^{\nu\mu},
 \end{aligned} \tag{22.285} \leftarrow$$

where the indices 0, 1, 2, and 3 denote the t -, x -, y -, and z -component, respectively. Note that the field quantities of (22.285) have been taken from (22.277). For later use, we show the covariant form expressed as

$$\begin{aligned}
 F_{\mu\nu} &= \begin{pmatrix} F_{00} & F_{01} & F_{02} & F_{03} \\ F_{10} & F_{11} & F_{12} & F_{13} \\ F_{20} & F_{21} & F_{22} & F_{23} \\ F_{30} & F_{31} & F_{32} & F_{33} \end{pmatrix} \\
 &= \begin{pmatrix} 0 & E^1 & E^2 & E^3 \\ -E^1 & 0 & -B^3 & B^2 \\ -E^2 & B^3 & 0 & -B^1 \\ -E^3 & -B^2 & B^1 & 0 \end{pmatrix} = -F_{\nu\mu}.
 \end{aligned} \tag{22.286} \leftarrow$$

This expression is obtained from

$$F_{\mu\nu} = F^{\rho\sigma} \eta_{\rho\mu} \eta_{\sigma\nu}, \tag{22.287} \leftarrow$$

where $(\eta)_{\rho\mu}$ is the Minkowski metric that has appeared in (21.16). We will discuss the constitution and properties of the tensors in detail in Chap. 24.

Defining the charge–current density four-vector as

$$s^\nu(x) \equiv \begin{pmatrix} \rho(x) \\ \mathbf{i}(x) \end{pmatrix} = \begin{pmatrix} \rho(x) \\ i^1(x) \\ i^2(x) \\ i^3(x) \end{pmatrix}, \tag{22.288} \leftarrow$$

we get the Maxwell's equations that are described in the Lorentz invariant form as

$$\partial_\mu F^{\mu\nu}(x) = s^\nu(x), \tag{22.289} \leftarrow$$

$$\partial^\rho F^{\mu\nu}(x) + \partial^\mu F^{\nu\rho}(x) + \partial^\nu F^{\rho\mu}(x) = 0, \tag{22.290}$$

where x represents the four-vector $\begin{pmatrix} t \\ \mathbf{x} \end{pmatrix}$. Note that we use the notation $\begin{pmatrix} t \\ \mathbf{x} \end{pmatrix}$ to represent the four-vector and that, e.g., the notation $\text{rot } \mathbf{A}(t, \mathbf{x})$ is used to show the dependence of a function on independent variables. In (22.290) the indices ρ, μ , and ν are cyclically permuted. These equations are equivalent to (22.277). For instance, in (22.289) the equation described by

$$\partial_\mu F^{\mu 0}(x) = \partial_0 F^{00}(x) + \partial_1 F^{10}(x) + \partial_2 F^{20}(x) + \partial_3 F^{30}(x) = s^0(x) \leftarrow$$

implies

$$\frac{\partial E^1}{\partial x^1} + \frac{\partial E^2}{\partial x^2} + \frac{\partial E^3}{\partial x^3} = \rho(x).$$

Rewriting the above equation using $(t, \mathbf{x})$, we have

$$\text{div } \mathbf{E} = \rho,$$

i.e., the first equation of (22.277). As an example of (22.290), an equation given by

$$\partial^0 F^{12}(x) + \partial^1 F^{20}(x) + \partial^2 F^{01}(x) = 0$$

means

$$-\frac{\partial B^3}{\partial x^0} - \frac{\partial E^2}{\partial x^1} + \frac{\partial E^1}{\partial x^2} = 0 \quad \text{or} \quad \frac{\partial E^2}{\partial x^1} - \frac{\partial E^1}{\partial x^2} + \frac{\partial B^3}{\partial x^0} = 0.$$

Also, rewriting the above equation using $(t, \mathbf{x})$ we get

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} + \frac{\partial B_z}{\partial t} = 0 \quad \text{or} \quad (\text{rot } \mathbf{E})_z + \left(\frac{\partial \mathbf{B}}{\partial t} \right)_z = 0.$$

That is, the third equation of (22.277) is recovered. The confirmation of other relations is left for readers. In the above derivations of Maxwell's equations, note that

$$\begin{aligned} \partial^0 &= \partial / \partial x_0 = \partial / \partial x^0 = \partial_0, \quad \partial^k = \partial / \partial x_k = -\partial / \partial x^k = \\ &= -\partial_k \quad (k = 1, 2, 3). \end{aligned} \tag{22.291} \leftarrow$$

Since $F^{\mu\nu}$ is the antisymmetric tensor, from (22.289) we have

$$\partial_\mu s^\mu(x) = 0. \tag{22.292}$$

Equation (22.292) is essentially the same as the current continuity equation that has already appeared in (7.20). Using four-vectors, the gauge transformation is succinctly described as

$$A'^{\mu}(x) = A^{\mu}(x) + \partial^{\mu}\Lambda(x), \quad (22.293) \leftarrow$$

where $A^{\mu}(x) = \begin{pmatrix} \phi(x) \\ \mathbf{A}(x) \end{pmatrix}$. Equation (22.293) is equivalent to (22.284). Using $A^{\mu}(x)$, we get

$$F^{\mu\nu}(x) = \partial^{\mu}A^{\nu}(x) - \partial^{\nu}A^{\mu}(x). \quad (22.294) \leftarrow$$

For example, we have

$$F^{01}(x) = \partial^0 A^1(x) - \partial^1 A^0(x),$$

i.e.,

$$-E^1(x) = \frac{\partial A^1(x)}{\partial x^0} - \frac{\partial A^0(x)}{\partial x^1} \quad \text{or} \quad (\mathbf{E})_x = -\left(\frac{\partial \mathbf{A}}{\partial t}\right)_x - (\nabla \phi)_x.$$

Other relations can readily be confirmed. The field strength $F^{\mu\nu}(x)$ is invariant under the gauge transformation (gauge invariance). It can be shown as follows:

$$\begin{aligned} F'^{\mu\nu}(x) &= \partial^{\mu}A'^{\nu}(x) - \partial^{\nu}A'^{\mu}(x) \leftarrow \\ &= \partial^{\mu}[A^{\nu}(x) + \partial^{\nu}\Lambda(x)] - \partial^{\nu}[A^{\mu}(x) + \partial^{\mu}\Lambda(x)] \\ &= \partial^{\mu}A^{\nu}(x) + \partial^{\mu}\partial^{\nu}\Lambda - \partial^{\nu}A^{\mu}(x) - \partial^{\nu}\partial^{\mu}\Lambda = \partial^{\mu}A^{\nu}(x) - \partial^{\nu}A^{\mu}(x) \leftarrow \\ &= F^{\mu\nu}(x), \end{aligned} \quad (22.295) \leftarrow$$

where with the third equality the operating order of ∂^{μ} and ∂^{ν} can be switched. Further substituting (22.294) for (22.289), we obtain

$$\partial_{\mu}[\partial^{\mu}A^{\nu}(x) - \partial^{\nu}A^{\mu}(x)] = \partial_{\mu}\partial^{\mu}A^{\nu}(x) - \partial_{\mu}\partial^{\nu}A^{\mu}(x) = s^{\nu}(x).$$

Exchanging the operating order of ∂_{μ} and ∂^{ν} with the first equality of the above equation, we get

$$\partial_{\mu}\partial^{\mu}A^{\nu}(x) - \partial^{\nu}[\partial_{\mu}A^{\mu}(x)] = s^{\nu}(x).$$

Or using $\partial_{\mu}\partial^{\mu} = \square$ we have

$$\square A^\nu(x) - \partial^\nu [\partial_\mu A^\mu(x)] \equiv s^\nu(x). \quad (22.296) \leftarrow$$

Defining in (22.296)

$$\chi(x) \equiv \partial_\mu A^\mu(x), \quad (22.297) \leftarrow$$

we obtain

$$\square A^\nu(x) - \partial^\nu \chi(x) = s^\nu(x). \quad (22.298) \leftarrow$$

The quantity $A^\mu(x)$ is said to be a four-vector potential and a relativistic field described by such four-vector is called a gauge field. The electromagnetic field is a typical gauge field.

22.5.2 *Lagrangian Density and Hamiltonian Density of the Electromagnetic Field*

Now, we are in the position to describe the Lagrangian density to quantize the free electromagnetic field. Choosing $A^\mu(x)$ for the canonical coordinate, we first define the Lagrangian density $\mathcal{L}(x)$ as

$$\begin{aligned} \mathcal{L}(x) \equiv & -\frac{1}{4} F_{\mu\nu}(x) F^{\mu\nu}(x) = -\frac{1}{4} [\partial_\mu A_\nu(x) - \partial_\nu A_\mu(x)] \\ & \times [\partial^\mu A^\nu(x) - \partial^\nu A^\mu(x)]. \end{aligned} \quad (22.299) \leftarrow$$

Using (22.285) and (22.286) based on the classical electromagnetic fields $\mathbf{E}(t, \mathbf{x})$ and $\mathbf{B}(t, \mathbf{x})$, we obtain

$$\mathcal{L}(x) = \frac{1}{2} \left\{ [\mathbf{E}(t, \mathbf{x})]^2 - [\mathbf{B}(t, \mathbf{x})]^2 \right\}. \quad (22.300) \leftarrow$$

From (22.299), as the corresponding canonical momentum $\pi_\mu(x)$ we have

$$\pi_\mu(x) \equiv \partial \mathcal{L}(x) / \partial \dot{A}^\mu(x) = F_{\mu 0}(x). \quad (22.301) \leftarrow$$

Hamiltonian density $\mathcal{H}(x)$ is then described by

$$\mathcal{H}(x) = \pi_\mu(x) \dot{A}^\mu(x) - \mathcal{L}(x).$$

The first term is given by

$$\pi_\mu(x)\dot{A}^\mu(x) = \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \end{pmatrix} \begin{pmatrix} 0 \\ -E^1 \\ -E^2 \\ -E^3 \end{pmatrix} = \mathbf{E}^2,$$

where we used (22.286) to denote $\pi_\mu(x)$; we also used (22.283), to which $\phi \equiv 0$ is imposed. The condition of $\phi \equiv 0$ is due to the Coulomb gauge that is often used to describe the radiation in vacuum [5]. From (22.300), we get

$$\mathcal{H}(x) = \frac{1}{2} \left\{ [E(t, \mathbf{x})]^2 + [\mathbf{B}(t, \mathbf{x})]^2 \right\}. \quad (22.302) \leftarrow$$

Although (22.300) and (22.302) seem reasonable, (22.301) has an inconvenience. It is because we could not define $\pi_0(x)$, i.e., $\pi_0(x) = F_{00}(x) \equiv 0$. To circumvent this inconvenience, we need to redefine the Lagrangian density. We develop the discussion in accordance with the literature [12]. The point is to use a scalar function $\chi(x)$ of (22.297). That is, we set the Lagrangian density such that [12]

$$\mathcal{L}(\mathfrak{x}) \equiv -\frac{1}{4} F_{\mu\nu}(x) F^{\mu\nu}(x) - \frac{1}{2} \chi^2. \quad (22.303) \leftarrow$$

Rewriting $F_{\mu\nu} F^{\mu\nu}$ as

$$\begin{aligned} -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} &= -\frac{1}{2} (\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A^\mu) \\ &= -\frac{1}{2} \partial_\mu A_\nu \partial^\mu A^\nu + \frac{1}{2} \partial_\mu (A^\nu \partial_\nu A^\mu) - \frac{1}{2} A^\nu \partial_\nu \chi, \end{aligned}$$

we have

$$\mathcal{L}(\mathfrak{x}) \equiv -\frac{1}{2} \partial_\mu A_\nu \partial^\mu A^\nu + \frac{1}{2} \partial_\mu (A^\nu \partial_\nu A^\mu) - \frac{1}{2} A^\nu \partial_\nu \chi - \frac{1}{2} \chi^2. \quad (22.304) \leftarrow$$

We wish to take the variation with respect to A_ν (i.e., δA_ν) termwise in (22.304). Namely, we have

$$\mathcal{L}(\mathfrak{x}) = \mathcal{L}_1(x) + \mathcal{L}_2(x) + \mathcal{L}_3(x) + \mathcal{L}_4(x), \quad (22.305) \leftarrow$$

where $\mathcal{L}_1(x)$, $\cdots$, and $\mathcal{L}_4(x)$ correspond to the first, $\cdots$, and fourth term of (22.304), respectively. We obtain

$$\begin{aligned}
S(A_\nu) &= \int dx \mathcal{L}(\mathfrak{x}) = \left[\int dx \mathcal{L}_1(x) \right] + \cdots + \left[\int dx \mathcal{L}_4(x) \right] \Leftarrow S_1 + \cdots + \mathfrak{S}_4, \\
\delta S(A_\nu) &= \delta \int dx \mathcal{L}(\mathfrak{x}) = \left[\delta \int dx \mathcal{L}_1(x) \right] \Leftarrow \cdots \Leftarrow \left[\delta \int dx \mathcal{L}_4(x) \right] \Leftarrow \\
&= \delta S_1(A_\nu) + \cdots + \mathfrak{S}_4(A_\nu),
\end{aligned} \tag{22.306}$$

where $S_k(A_\nu) = \int dx \mathcal{L}_k(x)$ and $\delta S_k(A_\nu) = \delta \int dx \mathcal{L}_k(x)$ ($k = 1, 2, 3, 4$). We evaluate δS_k one by one. With δS_1 we get

$$\begin{aligned}
\delta S_1(A_\nu) &\Rightarrow \delta \int dx \mathcal{L}_1(x) \Leftarrow \\
&= \int dx \left\{ -\frac{1}{2} [(\partial_\mu \delta A_\nu) \partial^\mu A^\nu + \partial_\mu A_\nu (\partial^\mu \delta A^\nu)] \right\} \Leftarrow \\
&= \int dx \left\{ -\frac{1}{2} [(\partial_\mu \delta A_\nu) \partial^\mu A^\nu + \partial_\mu A^\nu (\partial^\mu \delta A_\nu)] \right\} \Leftarrow \\
&= \frac{1}{2} \int dx \delta A_\nu (\partial_\mu \partial^\mu A^\nu + \partial^\mu \partial_\mu A^\nu) = \int dx \delta A_\nu (\partial_\mu \partial^\mu A^\nu),
\end{aligned} \tag{22.307}$$

where with the third equality the second term was rewritten so that the variation was evaluated with δA_ν . In the derivation of (22.307), we used the integration by parts and assumed that δA_ν vanishes at $x^\mu \rightarrow \pm \infty$ as in the derivation of (22.76). Similarly evaluating $\delta S_2(A_\nu)$, $\delta S_3(A_\nu)$, and $\delta S_4(A_\nu)$, we have

$$\begin{aligned}
\delta S_2(A_\nu) &= \int dx \delta \mathcal{L}_2(x) = -\frac{1}{2} \int dx \delta A_\nu (\partial_\mu \partial^\nu A^\mu), \\
\delta S_3(A_\nu) &= \int dx \delta \mathcal{L}_3(x) = -\frac{1}{2} \int dx \delta A_\nu (\partial^\nu \partial_\mu A^\mu), \\
\delta S_4(A_\nu) &= \int dx \delta \mathcal{L}_4(x) = \int dx \delta A_\nu (\partial^\nu \partial_\mu A^\mu).
\end{aligned} \tag{22.308}$$

Summing $\delta S_1(A_\nu)$, $\delta S_2(A_\nu)$, $\delta S_3(A_\nu)$, and $\delta S_4(A_\nu)$, we obtain a simple result expressed as

$$\delta S(A_\nu) = \delta S_1(A_\nu) = \int dx \delta A_\nu (\partial_\mu \partial^\mu A^\nu). \tag{22.309}$$

Notice that $\delta S_2(A_\nu)$, $\delta S_3(A_\nu)$, and $\delta S_4(A_\nu)$ canceled out one another. Thus, the Euler–Lagrange equation obtained from (22.309) reads as

$$\partial_\mu \partial^\mu A^\nu(x) = 0 \quad \text{or} \quad \square A^\nu(x) = 0. \quad (22.310) \leftarrow$$

For (22.310) to be equivalent to the Maxwell's equations, we need another condition [12]. We seek the condition below.

With a free electromagnetic field without the charge–current density, considering (22.297) and (22.298) on condition that $s^\nu(x) = 0$ we have

$$\partial_\mu \partial^\mu A^\nu(x) - \partial^\nu [\partial_\mu A^\mu(x)] = \partial_\mu \partial^\mu A^\nu(x) - \partial^\nu \chi(x) = 0. \quad (22.311) \leftarrow$$

Then, for (22.310) to be consistent with (22.311) we must have

$$\partial^\nu \chi(x) = 0. \quad (22.312) \leftarrow$$

This leads to

$$\partial_\nu \partial^\nu \chi(x) = 0. \quad (22.313) \leftarrow$$

Putting $\nu = 0$ in (22.312), we have

$$\partial^0 \chi(x) = \partial \chi(x) / \partial t = 0. \quad (22.314) \leftarrow$$

Consequently, if we impose

$$\chi(x) = \partial_\mu A^\mu(x) = 0 \quad \text{at} \quad t = 0, \quad (22.315) \leftarrow$$

from (22.313) we have $\chi(x) \equiv 0$ at all the times [12]. The constraint $\chi(x) = 0$ is called the *Lorentz condition*.

Any gauge in which the Lorentz condition holds is called a *Lorentz gauge* [12]. The significance of the Lorentz condition will be mentioned later in relation to the indefinite metric (Sect. 22.5.5).

Differentiating (22.293) with respect to x^μ , we obtain

$$\partial_\mu A'^\mu(x) = \partial_\mu A^\mu(x) + \partial_\mu \partial^\mu \Lambda(x). \quad (22.316) \leftarrow$$

Using the Lorentz gauge, from (22.315) and (22.316) we have [12]

$$\partial_\mu \partial^\mu \Lambda(x) = 0 \quad \text{or} \quad \square \Lambda(x) = 0. \quad (22.317) \leftarrow$$

When we originally chose $\Lambda(x)$ in (22.280), Λ could be any scalar field. We find, however, that the scalar function $\Lambda(x)$ must satisfy the condition of (22.317) under the Lorentz gauge.

Under the Lorentz gauge, in turn, the last two terms of (22.304) vanish. If we assume that A^ν vanishes at $x = \pm \infty$, the second term of (22.304) vanishes as well. It is because the integration of that term can be regarded the extension of the Gauss's

theorem to the four-dimensional space–time (see Sect. 7.1). Thus, among four terms of (22.304) only the first term survives as the Lagrangian density $\mathcal{L}(x)$ that properly produces the physically meaningful $\pi^0(x)$. That is, we get [12]

$$\mathcal{L}(x) = -\frac{1}{2} \partial_\mu A_\nu \partial^\mu A^\nu. \quad (22.318) \leftarrow$$

The Lagrangian density described by (22.318) is equivalent to that of (22.303) or (22.304) on which the Lorentz condition is imposed and for which the field A^ν is supposed to vanish at $x = \pm \infty$. Using (22.318), as the canonical momentum $\pi^\mu(x)$ we obtain [12]

$$\pi^\mu(x) = \partial \mathcal{L}(x) / \partial \dot{A}_\mu(x) = -\dot{A}^\mu(x). \quad (22.319) \leftarrow$$

Thus, the inconvenience of $\pi^0(x) = 0$ mentioned earlier has been skillfully gotten rid of. Notice that RHS of (22.319) has a minus sign, in contrast to (22.83). What remains to be done is to examine whether or not $\mathcal{L}(x)$ is invariant under the gauge transformation (22.293). We will discuss this issue in Chap. 23.

In the subsequent sections, we perform the canonical quantization of the electromagnetic field using the canonical coordinates and canonical momenta.

22.5.3 Polarization Vectors of the Electromagnetic Field [5]

In Part I and Part II, we discussed the polarization vectors of the electromagnetic fields within the framework of the classical theory. In this section, we revisit the related discussion and rebuild the polarization features of the fields from the relativistic point of view.

Previously, we have mentioned that the electromagnetic waves have two independent polarization states as the transverse modes. Since in this section we are dealing with the electromagnetic fields as the four-vectors, we need to consider two extra polarization freedoms, i.e., the longitudinal mode and scalar mode. Then, we distinguish these modes as the polarization vectors such as

$$\varepsilon^\mu(\mathbf{k}, r) \quad \Leftrightarrow \quad \varepsilon_\mu(\mathbf{k}, r) \quad (\mu, r = 0, 1, 2, 3),$$

where the index μ denotes the component of the four-vector (either contravariant or covariant); $\mathbf{k}$ is a wavenumber vector (see Part II); r represents the four linearly independent polarization states for each $\mathbf{k}$. The modes $\varepsilon^\mu(\mathbf{k}, 1)$ and $\varepsilon^\mu(\mathbf{k}, 2)$ are called transverse polarizations, $\varepsilon^\mu(\mathbf{k}, 3)$ longitudinal polarization, $\varepsilon^\mu(\mathbf{k}, 0)$ scalar polarization. The quantity $\varepsilon_\mu(\mathbf{k}, r)$ is a complex c -number. Also, note that

$$k = \begin{pmatrix} k^0 \\ \mathbf{k} \end{pmatrix} \text{ with } k^0 = k_0 = |\mathbf{k}|. \quad (22.320) \leftarrow$$

Equation (22.320) is a consequence of the fact that photon is massless. The orthonormality and completeness relations are read as [5]

$$\varepsilon_\mu(\mathbf{k}, r)^* \varepsilon^\mu(\mathbf{k}, s) = -\zeta_r \delta_{rs} \quad (r, s = 0, 1, 2, 3), \quad (22.321) \leftarrow$$

$$\sum_{r=0}^3 \zeta_r \varepsilon^\mu(\mathbf{k}, r)^* \varepsilon^\nu(\mathbf{k}, r) = -\eta^{\mu\nu} \quad \text{or} \quad \sum_{r=0}^3 \zeta_r \varepsilon^\mu(\mathbf{k}, r)^* \varepsilon_\nu(\mathbf{k}, r) = -\delta_\nu^\mu \quad (22.322) \leftarrow$$

with

$$\zeta_0 = -1, \quad \zeta_1 = \zeta_2 = \zeta_3 = 1. \quad (22.323) \leftarrow$$

Equation (22.321) means that

$$\varepsilon_\mu(\mathbf{k}, 0)^* \varepsilon^\mu(\mathbf{k}, 0) = 1 > 0, \quad \varepsilon_\mu(\mathbf{k}, r)^* \varepsilon^\mu(\mathbf{k}, r) = -1 < 0 \quad (r = 1, 2, 3). \quad (22.324) \leftarrow$$

If $\varepsilon_\mu(\mathbf{k}, r)$ is real, $\varepsilon^\mu(\mathbf{k}, 0)$ is a timelike vector and that $\varepsilon^\mu(\mathbf{k}, r)$ ($r = 1, 2, 3$) are spacelike vectors in the Minkowski space. We show other important relations in a special frame where $\mathbf{k}$ is directed to the positive direction of the z -axis. We have

$$\varepsilon^\mu(\mathbf{k}, 0) = \begin{pmatrix} 1 \\ \mathbf{0} \end{pmatrix}, \quad \varepsilon^\mu(\mathbf{k}, r) = \begin{pmatrix} 0 \\ \boldsymbol{\varepsilon}(\mathbf{k}, r) \end{pmatrix} \quad (r = 1, 2, 3), \quad (22.325) \leftarrow$$

where $\boldsymbol{\varepsilon}(\mathbf{k}, r)$ denotes a usual three-dimensional vector; to specify it we use a Gothic letter $\boldsymbol{\varepsilon}$. If we wish to represent $\varepsilon^\mu(\mathbf{k}, 0)$ in a general frame, we denote it by a covariant notation [5]

$$\varepsilon^\mu(\mathbf{k}, 0) \equiv n^\mu. \quad (22.326) \leftarrow$$

Also, we have

$$\begin{aligned} \mathbf{k} \cdot \boldsymbol{\varepsilon}(\mathbf{k}, r) &= 0 \quad (r = 1, 2), \quad \boldsymbol{\varepsilon}(\mathbf{k}, 3) = \mathbf{k}/|\mathbf{k}|, \\ \boldsymbol{\varepsilon}(\mathbf{k}, r) \cdot \boldsymbol{\varepsilon}(\mathbf{k}, s) &= \delta_{rs} \quad (r, s = 1, 2, 3). \end{aligned} \quad (22.327) \leftarrow$$

Denoting $\varepsilon^\mu(\mathbf{k}, 3)$ by a covariant form, we have [5]

$$\varepsilon^\mu(\mathbf{k}, 3) = \frac{k^\mu - (kn)n^\mu}{\sqrt{(kn)^2 - k^2}}, \quad (22.328) \leftarrow$$

where the numerator was obtained by subtracting the timelike component from $k = \begin{pmatrix} k^0 \\ \mathbf{k} \end{pmatrix}$; related discussion for the three-dimensional version can be seen in Sect. 7.3.

In (22.321) and (22.322), we used complex numbers for $\varepsilon^\mu(\mathbf{k}, s)$, which is generally used for the circularly (or elliptically) polarized light (Sect. 7.4). With a linearly polarized light, we use a real number for $\varepsilon^\mu(\mathbf{k}, s)$.

All the above discussions relevant to the polarization features of the electromagnetic fields can be dealt with within the framework of the classical theory of electromagnetism.

22.5.4 Canonical Quantization of the Electromagnetic Field

The quantization of the electromagnetic field is inherently represented in a tensor form. This differs from those for the scalar field and the Dirac field. This feature is highly important with both the classical and quantum electromagnetism. As in the case of the scalar field, the canonical quantization is translated into the commutation relations between the creation and annihilation operators. Unlike the scalar field and Dirac field, however, the classical polarization vectors of photon are incorporated into the commutation relations.

The equal-time canonical commutation relation is described as

$$[A^\mu(t, \mathbf{x}), \pi_\nu(t, \mathbf{y})] = i\delta_\nu^\mu \delta^3(\mathbf{x} - \mathbf{y}). \quad (22.329) \leftarrow$$

Equation (22.329) corresponds to (22.84) of the scalar field case and is clearly represented in the tensor form, i.e., a direct product of a contravariant vector $A^\mu(t, \mathbf{x})$ and a covariant vector $\pi_\nu(t, \mathbf{y})$. Multiplying both sides of (22.329) by $\eta^{\nu\rho}$, we obtain

$$[A^\mu(t, \mathbf{x}), \pi^\rho(t, \mathbf{y})] = i\eta^{\mu\rho} \delta^3(\mathbf{x} - \mathbf{y}). \quad (22.330) \leftarrow$$

This is a tensor described as a direct product of two contravariant vectors. Using (22.319), we get

$$[A^\mu(t, \mathbf{x}), \dot{A}_\nu(t, \mathbf{y})] = -i\delta_\nu^\mu \delta^3(\mathbf{x} - \mathbf{y}), \quad (22.331) \leftarrow$$

$$[A^\mu(t, \mathbf{x}), \dot{A}^\rho(t, \mathbf{y})] \leftarrow i\eta^{\mu\rho} \delta^3(\mathbf{x} - \mathbf{y}). \quad (22.332)$$

In (22.331) and (22.332), the minus sign of RHS comes from (22.319). All the other commutators of other combinations vanish. (That is, all the other combinations are commutative.) We have, e.g.,

$$[A^\mu(t, \mathbf{x}), A^\nu(t, \mathbf{y})] = [\dot{A}^\mu(t, \mathbf{x}), \dot{A}^\nu(t, \mathbf{y})] = 0.$$

We perform the quantization of the electromagnetic field in accordance with (22.98) of the scalar field (Sect. 22.3.2). Taking account of the polarization index r ($=0, \dots, 3$), we expand the identity operator $\hat{E}$ such that

$$\hat{E} = \int_{-\infty}^{\infty} \sum_{r=0}^3 \delta(k_0^2 - \mathbf{k}^2) |p, r\rangle \langle p, r| d^4p. \quad (22.333) \leftarrow$$

Notice that in (22.333) photon is massless. As the Fourier expansion of the field operator $A^\mu(x)$ we obtain [2]

$$\begin{aligned} A^\mu(x) &= \int \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3(2k_0)}} \leftarrow \\ &\times \sum_{r=0}^3 [a(\mathbf{k}, r) \varepsilon^\mu(\mathbf{k}, r) e^{-ikx} + a^\dagger(\mathbf{k}, r) \varepsilon^\mu(\mathbf{k}, r)^* e^{ikx}]. \end{aligned} \quad (22.334) \leftarrow$$

In (22.334), the summation with r is pertinent to $a(\mathbf{k}, r)$ [or $a^\dagger(\mathbf{k}, r)$] and $\varepsilon^\mu(\mathbf{k}, r)$ [or $\varepsilon^\mu(\mathbf{k}, r)^*$]. In comparison with (22.208), in (22.334) we use the following correspondence:

$$a_+(\mathbf{p}) \leftrightarrow a(\mathbf{k}, r) \quad \text{and} \quad a^\dagger_-(\mathbf{p}) \leftrightarrow a^\dagger(\mathbf{k}, r) \quad \leftarrow$$

as well as

$$\Phi_+(\mathbf{p}, E_p) \leftrightarrow \varepsilon^\mu(\mathbf{k}, r) \quad \text{and} \quad \Phi_-(-\mathbf{p}, -E_p) \leftrightarrow \varepsilon^\mu(\mathbf{k}, r)^*.$$

Then, as the decomposition of $A^\mu(x)$ into the positive-frequency term $A^{\mu+}(x)$ and negative-frequency term $A^{\mu-}(x)$, we have

$$A^\mu(x) = A^{\mu+}(x) + A^{\mu-}(x) \quad \leftarrow$$

with

$$\begin{aligned}
 A^{\mu+}(\vec{x}) &\equiv \int \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3(2k_0)}} \sum_{r=0-}^3 a(\mathbf{k}, r) \epsilon^\mu(\mathbf{k}, r) e^{-i\mathbf{k}\cdot\vec{x}}, \\
 A^{\mu-}(x) &\equiv \int \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3(2k_0)}} \sum_{r=0-}^3 a^\dagger(\mathbf{k}, r) \epsilon^\mu(\mathbf{k}, r)^* e^{i\mathbf{k}\cdot\vec{x}}.
 \end{aligned}
 \tag{22.335} \leftarrow$$

Note that $A^\mu(x)$ is Hermitian, as is the case with the scalar field $\phi(x)$. With the covariant quantized electromagnetic field $A_\mu(x)$, we have

$$\begin{aligned}
 A_\mu(x) &= \int \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3(2k_0)}} \sum_{r=0-}^3 \\
 &\quad \times [a(\mathbf{k}, r) \epsilon_\mu(\mathbf{k}, r) e^{-i\mathbf{k}\cdot\vec{x}} + a^\dagger(\mathbf{k}, r) \epsilon_\mu(\mathbf{k}, r)^* e^{i\mathbf{k}\cdot\vec{x}}].
 \end{aligned}
 \tag{22.336} \leftarrow$$

The operator $A_\mu(x)$ is also divided into

$$A_\mu(x) = A_\mu^{+}(\vec{x}) + A_\mu^{-}(x),$$

where $A_\mu^{+}(\vec{x})$ and $A_\mu^{-}(x)$ are similarly defined as in the case of (22.335).

Now, we define operators $\mathfrak{B}^\mu(\mathbf{k})$ as

$$\mathfrak{B}^\mu(\mathbf{k}) \equiv \sum_{r=0}^3 a(\mathbf{k}, r) \epsilon^\mu(\mathbf{k}, r). \tag{22.337} \leftarrow$$

Taking the adjoint of (22.337), we get

$$\mathfrak{B}^\mu(\mathbf{k})^\dagger = \sum_{r=0}^3 a^\dagger(\mathbf{k}, r) \epsilon^\mu(\mathbf{k}, r)^*. \tag{22.338} \leftarrow$$

Using the operators $\mathfrak{B}^\mu(\mathbf{k})$ and $\mathfrak{B}^\mu(\mathbf{k})^\dagger$, we may advance discussion of the quantization in parallel with that of the scalar field. Note the following correspondence between the photon field and scalar field:

$$\mathfrak{B}^\mu(\mathbf{k}) \leftrightarrow a(\mathbf{k}), \quad \mathfrak{B}^\mu(\mathbf{k})^\dagger \leftrightarrow a^\dagger(\mathbf{k}).$$

Remember once again that both the photon field operator and real scalar field operator are Hermitian (and bosonic field operators), allowing us to carry out the parallel discussion on the field quantization. As in the case of (22.104), we have

$$\mathfrak{B}^\mu(\mathbf{k})e^{-ik_0t} = i \int \frac{d^3\mathbf{x}}{\sqrt{(2\pi)^3(2k_0)}} \left[\dot{A}^\mu(t, \mathbf{x}) - ik_0 A^\mu(t, \mathbf{x}) \right] e^{-i\mathbf{q}\cdot\mathbf{x}}, \quad (22.339) \leftarrow$$

$$\mathfrak{B}^\mu(\mathbf{k})^\dagger e^{ik_0t} = -i \int \frac{d^3\mathbf{x}}{\sqrt{(2\pi)^3(2k_0)}} \left[\dot{A}^\mu(t, \mathbf{x}) + ik_0 A^\mu(t, \mathbf{x}) \right] e^{i\mathbf{q}\cdot\mathbf{x}}. \quad (22.340) \leftarrow$$

In light of the quantization of the scalar field represented by (22.105), we acknowledge that the following relationship holds:

$$\left[\mathfrak{B}^\mu(\mathbf{k}), \mathfrak{B}_\nu(\mathbf{q})^\dagger \right] = -\delta_\nu^\mu \delta^3(\mathbf{k} - \mathbf{q}), \quad (22.341) \leftarrow$$

where $\mathfrak{B}_\nu(\mathbf{q})^\dagger$ is the covariant form corresponding to $\mathfrak{B}^\nu(\mathbf{q})^\dagger$; the minus sign of the head of RHS in (22.341) comes from (22.319). In fact, we have

$$\begin{aligned} \left[\mathfrak{B}^\mu(\mathbf{k}), \mathfrak{B}_\nu(\mathbf{q})^\dagger \right] &= \left[\sum_{r=0}^3 a(\mathbf{k}, r) \varepsilon^\mu(\mathbf{k}, r), \sum_{s=0}^3 a^\dagger(\mathbf{q}, s) \varepsilon_\nu(\mathbf{q}, s)^* \right] \\ &= \sum_{r=0}^3 \sum_{s=0}^3 \varepsilon^\mu(\mathbf{k}, r) \varepsilon_\nu(\mathbf{q}, s)^* \left[a(\mathbf{k}, r), a^\dagger(\mathbf{q}, s) \right] \leftarrow \\ &= \sum_{r=0}^3 \sum_{s=0}^3 \zeta_r \varepsilon^\mu(\mathbf{k}, r) \varepsilon_\nu(\mathbf{q}, s)^* \zeta_r \left[a(\mathbf{k}, r), a^\dagger(\mathbf{q}, s) \right], \leftarrow \quad (22.342) \leftarrow \end{aligned}$$

where with the last equality we used $\zeta_r^2 = 1$.

For (22.341) and (22.342) to be identical, taking account of (22.322) we must have [5]

$$\zeta_r \left[a(\mathbf{k}, r), a^\dagger(\mathbf{q}, s) \right] = \delta_{rs} \delta^3(\mathbf{k} - \mathbf{q}) \quad \leftarrow \quad (22.343) \leftarrow$$

or

$$\left[a(\mathbf{k}, r), a^\dagger(\mathbf{q}, s) \right] = \zeta_r \delta_{rs} \delta^3(\mathbf{k} - \mathbf{q}). \quad (22.344) \leftarrow$$

Also, as another commutation relation we have

$$\left[a(\mathbf{k}, r), a(\mathbf{q}, s) \right] = \left[a^\dagger(\mathbf{k}, r), a^\dagger(\mathbf{q}, s) \right] = 0.$$

Then, (22.342) becomes

$$\begin{aligned}
\left[\mathfrak{B}^\mu(\mathbf{k}), \mathfrak{B}_\nu(\mathbf{q})^\dagger \right] &= \sum_{r=0}^3 \sum_{s=0}^3 \zeta_r \varepsilon^\mu(\mathbf{k}, r) \varepsilon_\nu(\mathbf{q}, s)^* \delta_{rs} \delta^3(\mathbf{k} - \mathbf{q}) \quad \leftarrow \\
&= \sum_{r=0}^3 \zeta_r \varepsilon^\mu(\mathbf{k}, r) \varepsilon_\nu(\mathbf{q}, r)^* \delta^3(\mathbf{k} - \mathbf{q}) \quad \leftarrow \\
&= \sum_{r=0}^3 \zeta_r \varepsilon^\mu(\mathbf{q}, r) \varepsilon_\nu(\mathbf{q}, r)^* \delta^3(\mathbf{k} - \mathbf{q}) \quad \leftarrow \\
&= -\delta_\nu^\mu \delta^3(\mathbf{k} - \mathbf{q}) \quad \leftarrow
\end{aligned} \tag{22.341}$$

as required. With the second to the last equality of (22.341), we used (10.116), i.e., the property of the δ function; with the last equality we used (22.322). It is worth mentioning that RHS of (22.344) contains ζ_r . This fact implies that the quantization of the radiation field is directly associated with the polarization feature of the classical electromagnetic field.

22.5.5 Hamiltonian and Indefinite Metric

In the last section, we had the commutation relation (22.344) with the creation and annihilation operators as well as the canonical commutation relations (22.331) and (22.332). These equations differ from the corresponding commutation relations (22.84) and (22.105) of the scalar field in that (22.331) has a minus sign and that (22.344) has a factor ζ_r . This feature causes problems with the subsequent discussion.

In relation to this issue, we calculate the Hamiltonian density $\mathcal{H}(x)$ of the electromagnetic field. It is given by

$$\begin{aligned}
\mathcal{H}(x) &\equiv \pi^\mu(x) \dot{A}_\mu(x) - \mathcal{L}(x) \quad \leftarrow \\
&= -\dot{A}^\mu(x) \dot{A}_\mu(x) + \frac{1}{2} \partial_\mu A_\nu \partial^\mu A^\nu, \tag{22.345}
\end{aligned}$$

where with the second equality we used (22.318) and (22.319). As in the case of the scalar field, the Hamiltonian H is given by the triple integral of $\mathcal{H}(x)$ such that

$$H = \int d^3\mathbf{x} \mathcal{H}(x).$$

The calculations are similar to those of the scalar field (Sect. 22.3.3). We estimate the integral termwise. With the first term of (22.345), using (22.334) and (22.336) we have

$$\begin{aligned}
& -\dot{A}^\mu(x)\dot{A}_\mu(x) \leftarrow \\
& = \frac{-1}{2(2\pi)^3} \int \frac{d^3\mathbf{k}}{\sqrt{k_0}} \sum_{r=0}^3 [a(\mathbf{k}, r)\varepsilon^\mu(\mathbf{k}, r)(-ik_0)e^{-ikx} + a^\dagger(\mathbf{k}, r)\varepsilon^\mu(\mathbf{k}, r)^*(ik_0)e^{ikx}] \\
& \quad \times \int \frac{d^3\mathbf{q}}{\sqrt{q_0}} \sum_{s=0}^3 [a(\mathbf{q}, s)\varepsilon_\mu(\mathbf{q}, s)(-iq_0)e^{-iqx} + a^\dagger(\mathbf{q}, s)\varepsilon_\mu(\mathbf{q}, s)^*(iq_0)e^{iqx}] \\
& = \frac{1}{2(2\pi)^3} \int d^3\mathbf{k}\sqrt{k_0} \sum_{r=0}^3 [-a(\mathbf{k}, r)\varepsilon^\mu(\mathbf{k}, r)e^{-ikx} + a^\dagger(\mathbf{k}, r)\varepsilon^\mu(\mathbf{k}, r)^*e^{ikx}] \\
& \quad \times \int d^3\mathbf{q}\sqrt{q_0} \sum_{s=0}^3 [-a(\mathbf{q}, s)\varepsilon_\mu(\mathbf{q}, s)e^{-iqx} + a^\dagger(\mathbf{q}, s)\varepsilon_\mu(\mathbf{q}, s)^*e^{iqx}].
\end{aligned} \tag{22.346} \leftarrow$$

The integrand of (22.346) is given by

$$\begin{aligned}
& [\text{Integrand of (22.346)}] \\
& = \sqrt{k_0q_0} \sum_{r=0}^3 \sum_{s=0}^3 \left[-a(\mathbf{k}, r)\varepsilon^\mu(\mathbf{k}, r)a^\dagger(\mathbf{q}, s)\varepsilon_\mu(\mathbf{q}, s)^*e^{-i(k-q)x} \right. \\
& \quad - a^\dagger(\mathbf{k}, r)\varepsilon^\mu(\mathbf{k}, r)^*a(\mathbf{q}, s)\varepsilon_\mu(\mathbf{q}, s)e^{i(k-q)x} \\
& \quad + a(\mathbf{k}, r)\varepsilon^\mu(\mathbf{k}, r)a(\mathbf{q}, s)\varepsilon_\mu(\mathbf{q}, s)e^{-i(k+q)x} \\
& \quad \left. + a^\dagger(\mathbf{k}, r)\varepsilon^\mu(\mathbf{k}, r)^*a^\dagger(\mathbf{q}, s)\varepsilon_\mu(\mathbf{q}, s)^*e^{i(k+q)x} \right].
\end{aligned}$$

To calculate $\int d^3\mathbf{x} [-\dot{A}^\mu(x)\dot{A}_\mu(x)]$, we exchange the integration order between the variable $\mathbf{x}$ and variable $\mathbf{k}$ (or variable $\mathbf{q}$). That is, performing the integration with respect to $\mathbf{x}$ first, we obtain

$$\begin{aligned}
& \int d^3\mathbf{x} [\text{Integrand of (22.346)}] \\
& = \sqrt{k_0q_0} \sum_{r=0}^3 \sum_{s=0}^3 \left[-a(\mathbf{k}, r)\varepsilon^\mu(\mathbf{k}, r)a^\dagger(\mathbf{q}, s)\varepsilon_\mu(\mathbf{q}, s)^*e^{-i(k_0-q_0)}(2\pi)^3\delta(\mathbf{k}-\mathbf{q}) \right. \\
& \quad - a^\dagger(\mathbf{k}, r)\varepsilon^\mu(\mathbf{k}, r)^*a(\mathbf{q}, s)\varepsilon_\mu(\mathbf{q}, s)e^{i(k_0-q_0)}(2\pi)^3\delta(\mathbf{k}-\mathbf{q}) \leftarrow \\
& \quad + a(\mathbf{k}, r)\varepsilon^\mu(\mathbf{k}, r)a(\mathbf{q}, s)\varepsilon_\mu(\mathbf{q}, s)e^{-i(k_0+q_0)}(2\pi)^3\delta(\mathbf{k}+\mathbf{q}) \leftarrow \\
& \quad \left. + a^\dagger(\mathbf{k}, r)\varepsilon^\mu(\mathbf{k}, r)^*a^\dagger(\mathbf{q}, s)\varepsilon_\mu(\mathbf{q}, s)^*e^{i(k_0+q_0)}(2\pi)^3\delta(\mathbf{k}+\mathbf{q}) \right]. \leftarrow \\
& \tag{22.347} \leftarrow
\end{aligned}$$

Further performing the integration of (22.347) with respect to $\mathbf{q}$, we obtain

$$\begin{aligned}
& \int d^3 \mathbf{q} \quad (22.347) \quad \leftarrow \\
& = (2\pi)^3 k_0 \sum_{r=0}^3 \sum_{s=0}^3 \left[-a(\mathbf{k}, r) \varepsilon^\mu(\mathbf{k}, r) a^\dagger(\mathbf{k}, s) \varepsilon_\mu(\mathbf{k}, s)^* \right. \\
& \quad \left. - a^\dagger(\mathbf{k}, r) \varepsilon^\mu(\mathbf{k}, r)^* a(\mathbf{k}, s) \varepsilon_\mu(\mathbf{k}, s) \right] \quad \leftarrow \\
& \quad + a(\mathbf{k}, r) \varepsilon^\mu(\mathbf{k}, r) a(-\mathbf{k}, s) \varepsilon_\mu(-\mathbf{k}, s) e^{-2ik_0} \\
& \quad + a^\dagger(\mathbf{k}, r) \varepsilon^\mu(\mathbf{k}, r)^* a^\dagger(-\mathbf{k}, s) \varepsilon_\mu(-\mathbf{k}, s)^* e^{2ik_0} \Big]. \quad (22.348) \leftarrow
\end{aligned}$$

Only when $\mathbf{k} = -\mathbf{k}$, the last two terms of (22.348) are physically meaningful. In that case, however, we have $\mathbf{k} = \mathbf{0}$. But this implies no photon state, and so we discard these two terms. Compare the present situation with that of (22.111), i.e., the scalar field case, where $\mathbf{k} = \mathbf{0}$ is physically meaningful. Then, using (22.321) we get

$$\begin{aligned}
& \int d^3 \mathbf{q} \quad (22.347) \quad \leftarrow \\
& = (2\pi)^3 k_0 \sum_{r=0}^3 \sum_{s=0}^3 \left[-a(\mathbf{k}, r) \varepsilon^\mu(\mathbf{k}, s) (-\zeta_r \delta_{rs}) - a^\dagger(\mathbf{k}, r) a(\mathbf{k}, s) (-\zeta_r \delta_{rs}) \right] \\
& = (2\pi)^3 k_0 \sum_{r=0}^3 \zeta_r \left[a(\mathbf{k}, r) a^\dagger(\mathbf{k}, r) + a^\dagger(\mathbf{k}, r) a(\mathbf{k}, r) \right] \quad \leftarrow \\
& = (2\pi)^3 k_0 \sum_{r=0}^3 \zeta_r \left[a^\dagger(\mathbf{k}, r) a(\mathbf{k}, r) + \zeta_r \delta^3(\mathbf{0}) + a^\dagger(\mathbf{k}, r) a(\mathbf{k}, r) \right] \quad \leftarrow \\
& = (2\pi)^3 k_0 \sum_{r=0}^3 \left[\zeta_r 2a^\dagger(\mathbf{k}, r) a(\mathbf{k}, r) + \zeta_r^2 \delta^3(\mathbf{0}) \right] \leftarrow \\
& = (2\pi)^3 k_0 \sum_{r=0}^3 \zeta_r 2a^\dagger(\mathbf{k}, r) a(\mathbf{k}, r) + (2\pi)^3 k_0 4\delta^3(\mathbf{0}),
\end{aligned}$$

where with the third equality we used (22.344); with the last equality we used (22.323). Finally, we get

$$\begin{aligned}
& \int d^3 \mathbf{x} \left[-\dot{A}^\mu(x) \dot{A}_\mu(x) \right] \leftarrow \\
& = \int d^3 \mathbf{k} k_0 \sum_{r=0}^3 \zeta_r a^\dagger(\mathbf{k}, r) a(\mathbf{k}, r) + 2 \int d^3 \mathbf{k} k_0 \delta^3(\mathbf{0}). \quad (22.349) \leftarrow
\end{aligned}$$

In (22.349) the second term leads to the divergence as discussed in Sect. 22.3.3. As before, we discard this term; namely the normal order is implied there.

As for the integration of the second term of (22.345), use the Fourier expansion formulae (22.334) and (22.336) in combination with the δ function formula (22.101). That term will vanish because of the presence of a factor

$$k_\mu k^\mu = k_0 k^0 - |\mathbf{k}|^2 = 0.$$

The confirmation is left for readers. Thus, we obtain the Hamiltonian H described by

$$H = \int d^3\mathbf{k} k_0 \sum_{r=0}^3 \zeta_r a^\dagger(\mathbf{k}, r) a(\mathbf{k}, r). \quad (22.350) \leftarrow$$

In (22.350) the presence of a factor $\zeta_0 = -1$ causes a problem. It is because ζ_0 leads to a negative integrand in (22.350). The argument is as follows: Defining the one-photon state $|1_{\mathbf{q}, s}\rangle$ as [5]

$$|1_{\mathbf{q}, s}\rangle \equiv a^\dagger(\mathbf{q}, s) |0\rangle, \quad (22.351) \leftarrow$$

where $|0\rangle$ denotes the vacuum, its energy is

$$\begin{aligned} H |1_{\mathbf{q}, s}\rangle &= \int d^3\mathbf{k} k_0 \sum_{r=0}^3 \zeta_r a^\dagger(\mathbf{k}, r) a(\mathbf{k}, r) a^\dagger(\mathbf{q}, s) |0\rangle \\ &= \int d^3\mathbf{k} k_0 \sum_{r=0}^3 \zeta_r a^\dagger(\mathbf{k}, r) \left[a^\dagger(\mathbf{q}, s) a(\mathbf{k}, r) + \zeta_r \delta_{rs} \delta^3(\mathbf{k} - \mathbf{q}) \right] |0\rangle \\ &= \int d^3\mathbf{k} k_0 \sum_{r=0}^3 \zeta_r a^\dagger(\mathbf{k}, r) \zeta_r \delta_{rs} \delta^3(\mathbf{k} - \mathbf{q}) |0\rangle \\ &= k_0 (\zeta_s)^2 a^\dagger(\mathbf{q}, s) |0\rangle = k_0 a^\dagger(\mathbf{q}, s) |0\rangle = k_0 |1_{\mathbf{q}, s}\rangle, \end{aligned} \quad (22.352) \leftarrow$$

where with the second equality we used (22.344); with the third equality we used $a(\mathbf{k}, r) |0\rangle = 0$; with the third to the last equality we used (22.323). Thus, (22.352) certainly shows that the energy of the one-photon state is k_0 .

As a square of the norm $\langle 1_{\mathbf{q}, s} | 1_{\mathbf{q}, s} \rangle$ [see (13.8)], we have

$$\begin{aligned} \langle 1_{\mathbf{q}, s} | 1_{\mathbf{q}, s} \rangle &= \langle 0 | a(\mathbf{q}, s) a^\dagger(\mathbf{q}, s) | 0 \rangle = \langle 0 | a^\dagger(\mathbf{q}, s) a(\mathbf{q}, s) + \zeta_s \delta^3(\mathbf{q} - \mathbf{q}) | 0 \rangle \\ &= \zeta_s \delta^3(\mathbf{0}) \langle 0 | 0 \rangle = \zeta_s \delta^3(\mathbf{0}), \end{aligned} \quad (22.353) \leftarrow$$

where with the last equality we assumed that the vacuum state is normalized. With $s = 0$, we have $\zeta_0 = -1$ and, hence,

$$\langle 1_{\mathbf{q},0} | 1_{\mathbf{q},0} \rangle = -\delta^3(\mathbf{0}). \quad (22.354) \leftarrow$$

This implies that the (square of) norm of a scalar photon is negative. The negative norm is said to be an indefinite metric. Admittedly, (22.354) violates the rule of inner product calculations (see Sect. 13.1). In fact, such a metric is intractable, and so it should be appropriately dealt with. So far as the factor $a(\mathbf{k}, 0)a^\dagger(\mathbf{k}, 0)$ is involved, we could not avoid inconvenience. We wish to remove it from quantum-mechanical operators.

At the same time, we have another problem with the Lorentz condition described by

$$\partial_\mu A^\mu(x) = 0. \quad (22.355) \leftarrow$$

The Lorentz condition is incompatible with the equal-time canonical commutation relation. That is, we have [2]

$$\begin{aligned} [A^0(t, \mathbf{x}), \partial_\mu A^\mu(t, \mathbf{x})] &= [A^0(t, \mathbf{x}), \partial_0 A^0(t, \mathbf{x})] + [A^0(t, \mathbf{x}), \partial_k A^k(t, \mathbf{x})] \leftarrow \\ &= [A^0(t, \mathbf{x}), -\pi_0(t, \mathbf{x})] = i\delta^3(\mathbf{x} - \mathbf{y}) \neq 0. \end{aligned} \quad (22.356) \leftarrow$$

This problem was resolved by Gupta and Bleuler by replacing the Lorentz condition (22.355) with a weaker condition. The point is that we divide the Hermitian operator $A^\mu(x)$ into $A^{\mu+}(x)$ and $A^{\mu-}(x)$ such that

$$A^\mu(x) = A^{\mu+}(x) + A^{\mu-}(x) \leftarrow$$

with

$$\begin{aligned} A^{\mu+}(x) &\equiv \int \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3(2k_0)}} \sum_{r=0-}^3 a(\mathbf{k}, r) \varepsilon^\mu(\mathbf{k}, r) e^{-ikx}, \\ A^{\mu-}(x) &\equiv \int \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3(2k_0)}} \sum_{r=0-}^3 a^\dagger(\mathbf{k}, r) \varepsilon^\mu(\mathbf{k}, r)^* e^{ikx}. \end{aligned} \quad (22.335) \leftarrow$$

We also require $A^{\mu+}(x)$ to be

$$\partial_\mu A^{\mu+}(x) | \Psi \rangle = 0, \quad (22.357) \leftarrow$$

where $|\Psi\rangle$ is any physically meaningful state. Taking an adjoint of (22.357), we have

$$\langle \Psi | \partial_\mu A^{\mu-}(x) = 0. \quad (22.358) \leftarrow$$

Accordingly, we obtain

$$\langle \Psi | \partial_\mu A^\mu(x) | \Psi \rangle = \langle \Psi | \partial_\mu A^{\mu-}(x) + \partial_\mu A^{\mu+}(x) | \Psi \rangle = 0. \quad (22.359) \leftarrow$$

Equation (22.359) implies that the Lorentz condition holds with an “expectation value.” Substituting $A^{\mu+}(x)$ of (22.335) for (22.357), we get [2]

$$\sum_{r=0}^3 k_\mu a(\mathbf{k}, r) \varepsilon^\mu(\mathbf{k}, r) | \Psi \rangle = 0. \quad (22.360) \leftarrow$$

With the above equation, notice that the functions e^{-ikx} in (22.335) are linearly independent with respect to individual k . Recalling the special coordinate frame where $\mathbf{k}$ is directed to the positive direction of the z -axis, we have [2]

$$k_\mu = (k_0 \ 0 \ 0 \ -k_0) \ (k_0 > 0), \quad \leftarrow$$

where $k_1 = k_2 = 0$ and $k_3 = -k_0$.

Also, designating $\varepsilon^\mu(\mathbf{k}, r)$ as [2]

$$\begin{aligned} \varepsilon^0(\mathbf{k}, 0) &= \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \equiv \delta_0^{(0)}, \quad \varepsilon^1(\mathbf{k}, 1) = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \equiv \delta_1^{(1)}, \quad \leftarrow \\ \varepsilon^2(\mathbf{k}, 2) &= \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \equiv \delta_2^{(2)}, \quad \varepsilon^3(\mathbf{k}, 3) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \equiv \delta_3^{(3)}, \quad \leftarrow \end{aligned} \quad (22.361) \leftarrow$$

we obtain

$$\begin{aligned} \sum_{r=0}^3 k_\mu a(\mathbf{k}, r) \varepsilon^\mu(\mathbf{k}, r) &= \sum_{r=0}^3 \left[k_0 a(\mathbf{k}, r) \delta_0^{(0)} + k_3 a(\mathbf{k}, r) \delta_3^{(3)} \right] \leftarrow \\ &= k_0 [a(\mathbf{k}, 0) - a(\mathbf{k}, 3)]. \end{aligned}$$

With $\delta_0^{(0)}, \delta_3^{(3)},$ etc. in (22.361), see (12.175). Then, (22.360) is rewritten as

$$k_0 [a(\mathbf{k}, 0) - a(\mathbf{k}, 3)] | \Psi \rangle = 0. \quad \leftarrow \quad (22.362) \leftarrow$$

Since $k_0 \neq 0$, we have

$$[a(\mathbf{k}, 0) - a(\mathbf{k}, 3)] |\Psi\rangle = 0. \quad \leftarrow (22.363) \leftarrow$$

Taking the adjoint of (22.363), we have

$$\langle\Psi| [a^\dagger(\mathbf{k}, 0) - a^\dagger(\mathbf{k}, 3)] = 0. \quad (22.364) \leftarrow$$

Multiplying both sides of (22.363) by $\langle\Psi| a^\dagger(\mathbf{k}, 0)$ or $\langle\Psi| a^\dagger(\mathbf{k}, 3)$ from the left and multiplying both sides of (22.364) by $a(\mathbf{k}, 0)|\Psi\rangle$ or $a(\mathbf{k}, 3)|\Psi\rangle$ from the right, and then summing both sides of all these four equations, we get

$$\langle\Psi| a^\dagger(\mathbf{k}, 0)a(\mathbf{k}, 0) - a^\dagger(\mathbf{k}, 3)a(\mathbf{k}, 3) |\Psi\rangle = 0. \quad (22.365) \leftarrow$$

Further multiplying both sides of (22.365) by k_0 and performing the triple integration with respect to $\mathbf{k}$, we have

$$0 = \int d^3\mathbf{k} k_0 \langle\Psi| a^\dagger(\mathbf{k}, 0)a(\mathbf{k}, 0) - a^\dagger(\mathbf{k}, 3)a(\mathbf{k}, 3) |\Psi\rangle. \quad (22.366) \leftarrow$$

Meanwhile, sandwiching both sides of (22.350) between $\langle\Psi|$ and $|\Psi\rangle$, we obtain

$$\langle\Psi|H|\Psi\rangle = \int d^3\mathbf{k} k_0 \sum_{r=0}^3 \langle\Psi| \zeta_r a^\dagger(\mathbf{k}, r)a(\mathbf{k}, r) |\Psi\rangle \quad (22.367) \leftarrow$$

Finally, summing both sides of (22.366) and (22.367), we get

$$\langle\Psi|H|\Psi\rangle = \int d^3\mathbf{k} k_0 \sum_{r=1}^2 \langle\Psi| a^\dagger(\mathbf{k}, r)a(\mathbf{k}, r) |\Psi\rangle \quad (22.368) \leftarrow$$

Thus, we have successfully gotten rid of the factor $a^\dagger(\mathbf{k}, 0)a(\mathbf{k}, 0)$ together with $a^\dagger(\mathbf{k}, 3)a(\mathbf{k}, 3)$ from the Hamiltonian.

22.5.6 Feynman Propagator of Photon Field

As in the case of both the scalar field and Dirac field, we wish to find the Feynman propagator of the photon field. First, we calculate $[A^\mu(x), A^\nu(y)]$. Using (22.334), we have

$$\begin{aligned}
& [A^\mu(x), A^\nu(y)] \\
&= \int \frac{d^3\mathbf{k} d^3\mathbf{q}}{(2\pi)^3 2\sqrt{k_0 q_0}} \left[\sum_{r=0}^3 \{a(\mathbf{k}, r) \varepsilon^\mu(\mathbf{k}, r) e^{-ikx} \right. \\
&\quad \left. + a^\dagger(\mathbf{k}, r) \varepsilon^\mu(\mathbf{k}, r)^* e^{ikx}\}, \sum_{s=0}^3 \{a(\mathbf{q}, s) \varepsilon^\nu(\mathbf{q}, s) e^{-iqx} + a^\dagger(\mathbf{q}, s) \varepsilon^\nu(\mathbf{q}, s)^* e^{iqx}\} \right] \\
&= \int \frac{d^3\mathbf{k} d^3\mathbf{q}}{(2\pi)^3 2\sqrt{k_0 q_0}} \sum_{r=0}^3 \sum_{s=0}^3 \{ \varepsilon^\mu(\mathbf{k}, r) \varepsilon^\nu(\mathbf{q}, s)^* e^{-ikx} e^{iqx} [a(\mathbf{k}, r), a^\dagger(\mathbf{q}, s)] \leftarrow \\
&\quad + \varepsilon^\mu(\mathbf{k}, r)^* \varepsilon^\nu(\mathbf{q}, s) e^{ikx} e^{-iqx} [a^\dagger(\mathbf{k}, r), a(\mathbf{q}, s)] \} \leftarrow \\
&= \int \frac{d^3\mathbf{k}}{(2\pi)^3 2\sqrt{k_0 q_0}} \sum_{r=0}^3 \left\{ \zeta_r \varepsilon^\mu(\mathbf{k}, r) \varepsilon^\nu(\mathbf{k}, r)^* e^{-i(kx - qx)} - \zeta_r \varepsilon^\mu(\mathbf{k}, r)^* \varepsilon^\nu(\mathbf{k}, r) e^{i(kx - qx)} \right\} \\
&= \int \frac{d^3\mathbf{k} (-\eta^{\mu\nu})}{(2\pi)^3 2k_0} \left[e^{-i\vec{k}(\vec{x} - \vec{y})} - e^{i\vec{k}(\vec{x} - \vec{y})} \right], \leftarrow \tag{22.369}
\end{aligned}$$

where the second to the last equality we used (22.344) and with the last equality we used (22.322). Comparing (22.369) with (22.169), we find that (22.369) differs from (22.169) only by the factor $-\eta^{\mu\nu}$. Also, notice that photon is massless ($m = 0$). Therefore, we have

$$\lim_{m \rightarrow 0} K = \lim_{m \rightarrow 0} \sqrt{\mathbf{k}^2 + m^2} = |\mathbf{k}| = k_0 \quad (k_0 > 0). \tag{22.370} \leftarrow$$

Thus, k_0 corresponds to K of (22.169). In conjunction with (22.131), we define the invariant delta function $D^{\mu\nu}(x - y)$ such that [5, 6]

$$iD^{\mu\nu}(x - y) \equiv [A^\mu(x), A^\nu(y)] = \lim_{m \rightarrow 0} [-i\eta^{\mu\nu} \Delta(x - y)]. \tag{22.371} \leftarrow$$

Once we have gotten the invariant delta function $D^{\mu\nu}(x)$ in the form of (22.369), we can immediately obtain the Feynman propagator with respect to the photon field. Defining the Feynman propagator $D_F^{\mu\nu}(x)$ as

$$D_F^{\mu\nu}(x) \equiv \langle 0 | T[A^\mu(x) A^\nu(y)] | 0 \rangle \tag{22.372} \leftarrow$$

and taking account of (22.369), we get

$$D_F^{\mu\nu}(x) = \lim_{m \rightarrow 0} [-\eta^{\mu\nu} \Delta_F(x - y)].$$

Writing it explicitly, we have

$$\begin{aligned}
 D_F^{\mu\nu}(x-y) &= \frac{-i}{(2\pi)^4} \int d^4k (-\eta^{\mu\nu}) \frac{e^{-ik(x-y)}}{-k^2 - i\epsilon} \\
 &= \frac{-i}{(2\pi)^4} \int d^4k \frac{\eta^{\mu\nu} e^{-ik(x-y)}}{k^2 + i\epsilon}. \quad \leftarrow
 \end{aligned} \tag{22.373}$$

Using the momentum space representation, we have

$$\tilde{D}_F^{\mu\nu}(k) = \frac{-i\eta^{\mu\nu}}{k^2 + i\epsilon} = \frac{i\eta^{\mu\nu}}{-k^2 - i\epsilon} = \lim_{m \rightarrow 0} \left[-\eta^{\mu\nu} \widetilde{\Delta}_F(k) \right] \leftarrow \tag{22.374}$$

Equation (22.374) can further be deformed using (22.322), (22.323), (22.326), and (22.328). That is, (22.322) is rewritten as

$$\begin{aligned}
 \eta^{\mu\nu} &= - \sum_{r=1}^2 \varepsilon^\mu(\mathbf{k}, r)^* \varepsilon^\nu(\mathbf{k}, r) - \varepsilon^\mu(\mathbf{k}, 3)^* \varepsilon^\nu(\mathbf{k}, 3) + \varepsilon^\mu(\mathbf{k}, 0)^* \varepsilon^\nu(\mathbf{k}, 0) \quad \leftarrow \\
 &= - \sum_{r=1}^2 \varepsilon^\mu(\mathbf{k}, r)^* \varepsilon^\nu(\mathbf{k}, r) - \frac{[k^\mu - (kn)n^\mu][k^\nu - (kn)n^\nu]}{(kn)^2 - k^2} + n^\mu n^\nu \quad \leftarrow \\
 &= - \sum_{r=1}^2 \varepsilon^\mu(\mathbf{k}, r)^* \varepsilon^\nu(\mathbf{k}, r) - \frac{[k^\mu k^\nu - (kn)(k^\mu n^\nu + n^\mu k^\nu)] + n^\mu n^\nu k^2}{(kn)^2 - k^2}. \quad \leftarrow
 \end{aligned} \tag{22.375}$$

The quantity $\eta^{\mu\nu}$ in (22.374) can be replaced with that of (22.375) and the resulting $\tilde{D}_F^{\mu\nu}(k)$ can be used for the calculations of interaction among the quantum fields.

Meanwhile, in (22.375) we assume that $k^2 \neq 0$ in general cases. With the real photon we have $k^2 = 0$. Then, we get

$$\eta^{\mu\nu} = - \sum_{r=1}^2 \varepsilon^\mu(\mathbf{k}, r)^* \varepsilon^\nu(\mathbf{k}, r) - \frac{k^\mu k^\nu - (kn)(k^\mu n^\nu + n^\mu k^\nu)}{(kn)^2}. \quad \leftarrow \tag{22.376}$$

Equation (22.376) is rewritten as

$$\sum_{r=1}^2 \varepsilon^\mu(\mathbf{k}, r)^* \varepsilon^\nu(\mathbf{k}, r) = -\eta^{\mu\nu} - \frac{k^\mu k^\nu - (kn)(k^\mu n^\nu + n^\mu k^\nu)}{(kn)^2}. \quad \leftarrow \tag{22.377}$$

Equation (22.377) will be useful in evaluating the matrix elements of the S -matrix including the Feynman amplitude; see the next chapter, especially Sect. 23.8.

References

1. Goldstein H, Poole C, Safko J (2002) Classical mechanics, 3rd edn. Addison Wesley, San Francisco
2. Sakamoto M (2014) Quantum field theory. Shokabo, Tokyo. (in Japanese)
3. Byron FW Jr, Fuller RW (1992) Mathematics of classical and quantum physics. Dover, New York
4. Sunakawa S (1991) Quantum mechanics. Iwanami, Tokyo. (in Japanese)
5. Mandl F, Shaw G (2010) Quantum field theory, 2nd edn. Wiley, Chichester
6. Itzykson C, Zuber J-B (2005) Quantum field theory. Dover, New York
7. Kaku M (1993) Quantum field theory. Oxford University Press, New York
8. Kugo C (1989) Quantum theory of gauge field I. Baifukan, Tokyo. (in Japanese)
9. Greiner W, Reinhardt J (1996) Field quantization. Springer, Berlin
10. Boas ML (2006) Mathematical methods in the physical sciences, 3rd edn. Wiley, New York
11. Dennery P, Krzywicki A (1996) Mathematics for physicists. Dover, New York
12. Schweber SS (1989) An introduction to relativistic quantum field theory. Dover, New York

Chapter 23

Interaction Between Fields



So far, we have focused on the characteristics of the scalar field, Dirac field, and electromagnetic field that behave as free fields without interaction among one another. In this chapter, we deal with the interaction between these fields. The interaction is described as that between the quantized fields, whose important features have been studied in Chap. 22. The interactions between the fields are presented as non-linear field equations. It is difficult to obtain analytic solutions of the non-linear equation, and so suitable approximation must be done. The method for this is well-established as the perturbation theory that makes the most of the S -matrix. Although we have dealt with the perturbation method in Chap. 5, the method we use in the present chapter differs largely from the previous method. Here we focus on the interaction between the Dirac field (i.e., electron) and the electromagnetic field (photon). The physical system that comprises electrons and photons has been fully investigated and established as the quantum electrodynamics (QED). We take this opportunity to get familiar with this major field of quantum mechanics. In the last part of the chapter, we deal with the Compton scattering as a typical example to deepen understanding of the theory [1].

23.1 Lorentz Force and Minimal Coupling

Although in the quantum theory of fields abstract and axiomatic approaches tend to be taken, we wish to adhere to an intuitive approach to the maximum extent possible. To that effect, we get back to the Lorentz force we studied in Chaps. 4 and 15. The Lorentz force and its nature are widely investigated in various areas of natural science. The Lorentz force $\mathbf{F}(t)$ is described by

$$\mathbf{F}(t) = e\mathbf{E}(\mathbf{x}(t)) + e\dot{\mathbf{x}}(t) \times \mathbf{B}(\mathbf{x}(t)), \quad (4.88)$$

where the first term is electric Lorentz force and the second term represents the magnetic Lorentz force.

We wish to relate the Lorentz force to the Hamiltonian of the interacting physical system within the framework of the quantum field theory. To this end, the minimal coupling (or minimal substitution) is an important concept in association with the covariant derivative. The covariant derivative is defined as [1]

$$\begin{aligned} i\frac{\partial}{\partial t} &\rightarrow i\frac{\partial}{\partial t} - q\phi(t, \mathbf{x}), \\ \frac{1}{i}\nabla &\rightarrow \frac{1}{i}\nabla - q\mathbf{A}(t, \mathbf{x}), \end{aligned} \quad (23.1)$$

where q is a charge of the matter field; $\phi(t, \mathbf{x})$ and $\mathbf{A}(t, \mathbf{x})$ are the electromagnetic potentials. Using the four-vector notation, we have

$$\partial_\mu \rightarrow D_\mu \equiv \partial_\mu + iqA_\mu(x), \quad (23.2)$$

where D_μ defines the covariant derivative, which leads to the interaction between the electron and photon. This interaction is referred to as a minimal coupling. To see how the covariant derivative produces the interaction, we replace ∂_μ with D_μ in (22.213) such that

$$(i\gamma^\mu D_\mu - m)\psi(x) = 0. \quad (23.3)$$

Then, we have

$$[i\gamma^\mu (\partial_\mu + ieA_\mu) - m]\psi(x) = 0, \quad (23.4)$$

where e is an elementary charge ($e < 0$), as defined in Chap. 4. Rewriting (23.4), we get

$$i\gamma^0 \partial_0 \psi = \left(-i \sum_{k=1}^3 \gamma^k \partial_k + e\gamma^0 A_0 + e \sum_{k=1}^3 \gamma^k A_k + m \right) \psi. \quad (23.5)$$

Multiplying both sides of (23.5) by γ^0 , we obtain

$$\begin{aligned} i\partial_0 \psi &= \left(-i \sum_{k=1}^3 \gamma^0 \gamma^k \partial_k + e\gamma^0 \gamma^0 A_0 + e \sum_{k=1}^3 \gamma^0 \gamma^k A_k + m\gamma^0 \right) \psi \\ &= \left[\left(-i \sum_{k=1}^3 \gamma^0 \gamma^k \partial_k + m\gamma^0 \right) + \left(eA_0 + e \sum_{k=1}^3 \gamma^0 \gamma^k A_k \right) \right] \psi \end{aligned}$$

$$= \left[\left(\frac{1}{i} \boldsymbol{\alpha} \cdot \boldsymbol{\nabla} + m\beta \right) + (eA_0 - e\boldsymbol{\alpha} \cdot \mathbf{A}) \right] \psi, \quad (23.6)$$

where $\boldsymbol{\alpha} \equiv \gamma^0 \boldsymbol{\gamma}$; $\boldsymbol{\gamma} = \begin{pmatrix} \gamma^1 \\ \gamma^2 \\ \gamma^3 \end{pmatrix}$ and $\beta \equiv \gamma^0$. The operator $(\frac{1}{i} \boldsymbol{\alpha} \cdot \boldsymbol{\nabla} + m\beta)$ is identical with the one-particle Hamiltonian $\tilde{H}$ (of the free Dirac field) defined in (22.222). Therefore, the operator $(eA_0 - e\boldsymbol{\alpha} \cdot \mathbf{A})$ in (23.6) is the interaction Hamiltonian H_{int} that represents the interaction between the Dirac field and electromagnetic field. Symbolically writing (23.6), we have

$$i\partial_0 \psi(x) = (\tilde{H} + H_{\text{int}}) \psi(x), \quad (23.7)$$

where

$$\tilde{H} = \frac{1}{i} \boldsymbol{\alpha} \cdot \boldsymbol{\nabla} + m\beta \quad (23.8)$$

and

$$H_{\text{int}} = eA_0 - e\boldsymbol{\alpha} \cdot \mathbf{A}. \quad (23.9)$$

Defining the one-particle Hamiltonian H as

$$H = \tilde{H} + H_{\text{int}}, \quad (23.10)$$

we get

$$i\partial_0 \psi(x) = H\psi(x). \quad (23.11)$$

Equations (23.7) and (23.11) show the Schrödinger equation of the Dirac field that interacts with the electromagnetic field. The interaction Hamiltonian H_{int} is directly associated with the Lorentz force $\mathbf{F}(t)$ described by (4.88).

Representing $\mathbf{E}$ and $\mathbf{B}$ using the electromagnetic potential as in (22.279) and (22.283), we get

$$\mathbf{F}(t) = -e \left[\boldsymbol{\nabla} \phi(\mathbf{x}(t)) - \frac{\partial \mathbf{A}(\mathbf{x}(t))}{\partial t} \right] + e \dot{\mathbf{x}}(t) \times \text{rot } \mathbf{A}(\mathbf{x}(t)). \quad (23.12)$$

The second term of (23.12) can be rewritten as [2]

$$e \dot{\mathbf{x}}(t) \times \text{rot } \mathbf{A} = e \boldsymbol{\nabla}_A (\mathbf{A} \cdot \dot{\mathbf{x}}(t)) - e (\dot{\mathbf{x}}(t) \cdot \boldsymbol{\nabla}) \mathbf{A}, \quad (23.13)$$

where $\boldsymbol{\nabla}_A$ acts only on $\mathbf{A}$ (i.e., vector potential). Then, (23.12) is further rewritten as

$$\mathbf{F}(t) = -e \left[\nabla \phi(\mathbf{x}(t)) - \frac{\partial \mathbf{A}(\mathbf{x}(t))}{\partial t} \right] + e \nabla_A (\mathbf{A} \cdot \dot{\mathbf{x}}(t)) - e (\dot{\mathbf{x}}(t) \cdot \nabla) \mathbf{A}.$$

Of these terms, the moiety of conservative force $\mathbf{F}_C(t)$ is expressed as

$$\mathbf{F}_C(t) = -e \nabla \phi(\mathbf{x}(t)) + e \nabla_A (\mathbf{A} \cdot \dot{\mathbf{x}}(t)). \quad (23.14)$$

Integrating (23.14) between two spatial points of Q and P , we obtain

$$\begin{aligned} \int_Q^P -\mathbf{F}_C(t) \cdot d\mathbf{l} &= e \int_Q^P \left[\nabla \phi(\mathbf{x}(t)) - \nabla_A (\mathbf{A} \cdot \dot{\mathbf{x}}(t)) \right] \cdot d\mathbf{l} \\ &= e \int_Q^P d\phi(\mathbf{x}(t)) - e \int_Q^P d(\dot{\mathbf{x}}(t) \cdot \mathbf{A}) \\ &= e[\phi(P) - \phi(Q)] - e[\dot{\mathbf{x}}(t) \cdot \mathbf{A}(P) - \dot{\mathbf{x}}(t) \cdot \mathbf{A}(Q)], \end{aligned}$$

where $d\mathbf{l}$ means the line integral; we used $\nabla \phi \cdot d\mathbf{l} = d\phi$ and $\nabla_A (\mathbf{A} \cdot \dot{\mathbf{x}}(t)) \cdot d\mathbf{l} = d(\dot{\mathbf{x}}(t) \cdot \mathbf{A})$. Taking $Q \rightarrow -\infty$ and setting $\phi(-\infty) = \mathbf{A}(-\infty) = 0$, we get

$$\int_Q^P -\mathbf{F}_C(t) \cdot d\mathbf{l} = e\phi(P) - e\dot{\mathbf{x}}(t) \cdot \mathbf{A}(P). \quad (23.15)$$

Comparing (23.15) with (23.6), we find that α corresponds to the classical velocity operator $\dot{\mathbf{x}}(t)$ [3]. Thus, H_{int} of (23.7) corresponds to a work the Lorentz force exerts on an electron. Notice, however, that α of (23.6) does not have a concept of a particle velocity. It is because ∇_A of (23.14) represents a derivative with respect to the vector potential. Hence, the minimal coupling introduced through the covariant derivative defined in (23.2) properly reflects the interaction between the Dirac field and electromagnetic field.

If the interaction is sufficiently weak, namely H_{int} is sufficiently small compared to $\tilde{H}$, we can apply the perturbation method we developed in Chap. 5. It is indeed the case with our problems that follow. Nevertheless, our approaches described in this chapter largely differ from those of Chap. 5. After we present the relevant methods that partly include somewhat abstract notions, we address practical problems such as the Compton scattering which we dealt with in Chap. 1.

23.2 Lagrangian and Field Equation of the Interacting Fields

We continue the aforementioned discussion still further. To represent H_{int} in a scalar form using the four-vector potentials, we rewrite (23.9) as

$$H_{\text{int}} = e\gamma^0\gamma^\mu A_\mu.$$

Assuming that $\psi(x)$ of (23.11) is normalized, we have an expectation value of $\langle H_{\text{int}} \rangle$ such that

$$\begin{aligned}\langle H_{\text{int}}(t) \rangle &= \langle \psi(x) | H_{\text{int}} | \psi(x) \rangle \\ &= \int d^3\mathbf{x} \psi^\dagger(x) e\gamma^0\gamma^\mu A_\mu(x) \psi(x) \\ &= \int d^3\mathbf{x} e\bar{\psi}(x)\gamma^\mu A_\mu(x)\psi(x).\end{aligned}\tag{23.16}$$

Thus, the interaction Hamiltonian density $\mathfrak{H}_{\text{int}}(x)$ is given by the integrand of (23.16) as

$$\mathfrak{H}_{\text{int}}(x) \equiv e\bar{\psi}(x)\gamma^\mu A_\mu(x)\psi(x).\tag{23.17}$$

Then, the interaction Lagrangian density $\mathfrak{L}_{\text{int}}(x)$ is given by

$$\mathfrak{L}_{\text{int}}(x) = -\mathfrak{H}_{\text{int}}(x) = -e\bar{\psi}(x)\gamma^\mu A_\mu(x)\psi(x).\tag{23.18}$$

Note that in (23.18) we have derived the Lagrangian density from the Hamiltonian density in a way opposite to (22.37). This derivation, however, does not cause a problem, because regarding the equation of motion of the interacting system we are considering now, the interaction does not involve derivatives of physical variables (see Sect. 22.1). Or rather, since the kinetic energy is irrelevant to the interaction, from (22.31) and (22.40) we have

$$\mathfrak{L}_{\text{int}}(x) = -\mathfrak{U}_{\text{int}}(x) = -\mathfrak{H}_{\text{int}}(x),$$

where $\mathfrak{U}_{\text{int}}(x)$ denotes the potential energy density.

On the basis of the above discussion and (22.210), (22.318), as well as (23.18), the Lagrangian density $\mathfrak{L}(x)$ of the interacting fields (we assume the interaction between the Dirac field and electromagnetic field) is given by

$$\mathfrak{L}(x) = \mathcal{L}(x) + \mathfrak{L}_{\text{int}}(x)$$

$$= \bar{\psi}(x)(i\gamma^\mu \partial_\mu - m)\psi(x) - \frac{1}{2} \partial_\mu A_\nu(x) \partial^\mu A^\nu(x) - e \bar{\psi}(x) \gamma^\mu A_\mu(x) \psi(x). \quad (23.19)$$

Note that in some textbooks [1] the sign of the third term is reversed. This is due to the difference in the definition of e (i.e., $e < 0$ in this book, whereas $e > 0$ in [1]). Rewriting (23.19) using the covariant derivative, we get

$$\begin{aligned} \mathfrak{L}(x) &= \bar{\psi}(x) \{ i\gamma^\mu [\partial_\mu + ieA_\mu(x)] - m \} \psi(x) - \frac{1}{2} \partial_\mu A_\nu(x) \partial^\mu A^\nu(x) \\ &= \bar{\psi}(x) (i\gamma^\mu D_\mu - m) \psi(x) - \frac{1}{2} \partial_\mu A_\nu(x) \partial^\mu A^\nu(x). \end{aligned} \quad (23.20)$$

The quantum mechanics featured by the Lagrangian density of (23.19) or (23.20) is called quantum electrodynamics (QED). From now on, we focus on QED and describe its characteristics including the fundamental principle and practical problems. For future use, we introduce the charge–current density operator $s^\mu(x)$ as follows [1]:

$$s^\mu(x) \equiv q \bar{\psi}(x) \gamma^\mu \psi(x). \quad (23.21)$$

Notice that in the case of electron, $q = e (< 0)$. Using $s^\mu(x)$, we have

$$\mathfrak{L}_{\text{int}}(x) = -s^\mu(x) A_\mu(x). \quad (23.22)$$

The conserved charge Q is described by

$$\begin{aligned} Q &= \int d^3\mathbf{x} s^0(x) = q \int d^3\mathbf{x} \bar{\psi}(x) \gamma^0 \psi(x) = q \int d^3\mathbf{x} \psi^\dagger(x) \gamma^0 \gamma^0 \psi(x) \\ &= q \int d^3\mathbf{x} \psi^\dagger(x) \psi(x), \end{aligned} \quad (23.23)$$

where with the last equality we used $\gamma^0 \gamma^0 = 1$. Meanwhile, the conserved current $\mathbf{J}$ is represented as

$$\begin{aligned} \mathbf{J} &= \int d^3\mathbf{x} s^k(x) = q \int d^3\mathbf{x} \bar{\psi}(x) \gamma^k \psi(x) = q \int d^3\mathbf{x} \psi^\dagger(x) \gamma^0 \gamma^k \psi(x) \\ &= q \int d^3\mathbf{x} \psi^\dagger(x) \boldsymbol{\alpha} \psi(x), \end{aligned} \quad (23.24)$$

where we used $\boldsymbol{\alpha} \equiv \gamma^0 \boldsymbol{\gamma}$.

To confirm the validity of (23.19), we take the action integral following (22.211) and (22.212). The calculation procedures are almost the same as before. The only difference is about the additional term with respect to $\mathfrak{L}_{\text{int}}(x)$. That is, we have

$$\delta\mathfrak{L}_{\text{int}}(x) = \delta\bar{\psi}(x) [-e\gamma^\mu A_\mu(x)\psi(x)] + [-e\bar{\psi}(x)\gamma^\mu A_\mu(x)]\delta\psi(x).$$

In response to (22.212), we obtain

$$\begin{aligned}\delta S(\phi) &= \int dx \mathfrak{L}(x) = \int dx [\mathcal{L}(x) + \mathfrak{L}_{\text{int}}(x)] \\ &= \int dx \{ \delta\bar{\psi} [i\gamma^\mu \partial_\mu \psi - m\psi - e\gamma^\mu A_\mu(x)\psi(x)] \\ &\quad - \int dx [i\partial_\mu \bar{\psi} \gamma^\mu + m\bar{\psi} + e\bar{\psi}(x)\gamma^\mu A_\mu(x)] \delta\psi \}. \end{aligned} \quad (23.25)$$

On the conditions of $\delta\bar{\psi} = 0$, we get

$$\{i\gamma^\mu [\partial_\mu + ieA_\mu(x)] - m\}\psi(x) = 0 \quad (23.26)$$

or

$$[i\gamma^\mu \partial_\mu - e\gamma^\mu A_\mu(x) - m]\psi(x) = 0. \quad (23.27)$$

Meanwhile, from $\delta\psi = 0$, we obtain the adjoint Dirac equation described by

$$i\partial_\mu \bar{\psi} \gamma^\mu + e\bar{\psi}(x)\gamma^\mu A_\mu(x) + m\bar{\psi} = 0. \quad (23.28)$$

Equation (23.27) is the Dirac equation with the system where electron and photon are interacting with each other. As in the case of Sect. 22.4.1, (23.27) and (23.28) are equivalent. Also, note that (21.150) is the adjoint Dirac equation (without interaction with photon), if we assume that the solution of the Dirac equation is given by $\psi(x) = u(\mathbf{p}, h) e^{-ipx}$.

In conjunction with (23.27), we remark the implication of the charge conjugation. Taking the adjoint of (23.27), we have

$$-i\partial_\mu \psi^\dagger \gamma^{\mu\dagger} - e\psi^\dagger A_\mu^\dagger \gamma^{\mu\dagger} - m\psi^\dagger = 0,$$

↓ multiply γ^0 on the above from the right and insert $\gamma^0 \gamma^0 = 1$:

$$-i\partial_\mu \psi^\dagger \gamma^0 \gamma^{\mu\dagger} \gamma^0 - e\psi^\dagger \gamma^0 A_\mu^\dagger \gamma^{\mu\dagger} \gamma^0 - m\psi^\dagger \gamma^0 = 0,$$

↓ use $\bar{\psi} = \psi^\dagger \gamma^0$, $\gamma^0 \gamma^{\mu\dagger} \gamma^0 = \gamma^\mu$, and $A_\mu^\dagger = A_\mu$ (Hermitian):

$$-i\partial_\mu \bar{\psi} \gamma^\mu - eA_\mu \bar{\psi} \gamma^\mu - m\bar{\psi} = 0,$$

↓ take transposition of the above:

$$-i\partial_\mu\gamma^{\mu T}\bar{\psi}^T - eA_\mu\gamma^{\mu T}\bar{\psi}^T - m\bar{\psi}^T = 0,$$

↓ multiply C on the above from the left and insert $C^{-1}C = 1$:

$$-i\partial_\mu C\gamma^{\mu T}C^{-1}C\bar{\psi}^T - eA_\mu C\gamma^{\mu T}C^{-1}C\bar{\psi}^T - mC\bar{\psi}^T = 0,$$

↓ use $C\gamma^{\mu T}C^{-1} = -\gamma^\mu$ and $C\bar{\psi}^T = \psi^C$:

$$i\partial_\mu\gamma^\mu\psi^C + eA_\mu\gamma^\mu\psi^C - m\psi^C = 0.$$

Hence, we get

$$(i\gamma^\mu\partial_\mu + e\gamma^\mu A_\mu - m)\psi^C = 0. \quad (23.29)$$

Notice that during the above processes that derived (23.29), A_μ commutes with the gamma matrices γ^μ . With the operators ψ^C and C , see (21.161) and (21.164). Comparing (23.27) and (23.29), we find that e is exchanged with $-e$ ($-e > 0$, i.e., positive charge). Thus, (23.29) represents the field equation of the positron.

23.3 Local Phase Transformation and $U(1)$ Gauge Field

In Chap. 22, we mentioned several characteristics of the gauge field and the associated gauge transformation. To give a precise definition of the “gauge field” and “gauge transformation,” however, is beyond the scope of this book. Yet, we can get an intuitive concept of these principles at the core of the covariant derivative. In Chap. 22, we have shown that the field strength $F^{\mu\nu}(x)$ is invariant under the gauge transformation of (22.293).

Meanwhile, we can express an exponential function (or phase factor) as

$$e^{i\Theta(x)}, \quad (23.30)$$

where $\Theta(x)$ is a real function of x . The phase factor (23.30) is different from those that appeared in Part I in that in this chapter the exponent may change depending upon the space-time coordinate x . Let us consider a function transformation described by

$$f'(x) = e^{i\Theta(x)}f(x). \quad (23.31)$$

Also, let us consider a set $\mathfrak{G}(x)$ defined by

$$\mathfrak{G}(x) \equiv \left\{ e^{i\Theta(x)}; \Theta(x) : \text{real function of } x \right\}. \quad (23.32)$$

Then, $\mathfrak{G}(x)$ forms a $U(1)$ group (i.e., continuous transformation group). That is, (A1) closure and (A2) associative law are evident (see Sect. 16.1) by choosing real functions $\Theta_\lambda(x)$ ($\lambda \in \Lambda$), where Λ can be a finite set or infinite set with different cardinalities (see Sect. 6.1). (A3) Identity element corresponds to $\Theta(x) \equiv 0$. (A4) Inverse element $e^{i\Theta(x)}$ is $e^{-i\Theta(x)}$. The functional transformation described by (23.31) and related to $\mathfrak{G}(x)$ is said to be a local phase transformation. The important point is that if we select $\Theta(x)$ appropriately, we can make the covariant derivative of $f(x)$ in (23.31) undergo the same local phase transformation as $f(x)$. In other words, appropriately selecting a functional form as $\Theta(x)$, we may have

$$[D_\mu f(x)]' = e^{i\Theta(x)} [D_\mu f(x)], \quad (23.33)$$

where the prime ($'$) denotes the local phase transformation.

Meanwhile, in Chap. 22 we chose a function $\Lambda(x)$ for the gauge transformation such that

$$A'^\mu(x) = A^\mu(x) + \partial^\mu \Lambda(x). \quad (22.293)$$

We wish to connect $\Theta(x)$ of (23.31) to $\Lambda(x)$. The proper choice is to fix $\Theta(x)$ at

$$\Theta(x) = -q\Lambda(x).$$

We show this as follows:

$$\begin{aligned} [D_\mu f(x)]' &= [\partial_\mu + iqA'^\mu(x)]f'(x) = \partial_\mu f'(x) + iqA'^\mu(x)f'(x) \\ &= \partial_\mu \left[e^{-iq\Lambda(x)} f(x) \right] + iq[A^\mu(x) + \partial^\mu \Lambda(x)] e^{-iq\Lambda(x)} f(x) \\ &= -iqe^{-iq\Lambda(x)} [\partial_\mu \Lambda(x)] f(x) + e^{-iq\Lambda(x)} \partial_\mu f(x) \\ &\quad + iqA^\mu(x) e^{-iq\Lambda(x)} f(x) + iq[\partial_\mu \Lambda(x)] e^{-iq\Lambda(x)} f(x) \\ &= e^{-iq\Lambda(x)} [\partial_\mu + iqA^\mu(x)] f(x) \\ &= e^{-iq\Lambda(x)} D_\mu f(x). \end{aligned} \quad (23.34)$$

In (23.34), the prime ($'$) denotes either the local phase transformation with the Dirac field or the gauge transformation with the electromagnetic field. Applying (23.34) to $\bar{\psi}(x)(i\gamma^\mu D_\mu - m)\psi(x)$, we get

$$\begin{aligned}
[\bar{\psi}(x)(i\gamma^\mu D_\mu - m)\psi(x)]' &= e^{iq\Lambda(x)}\bar{\psi}(x)e^{-iq\Lambda(x)}(i\gamma^\mu D_\mu - m)\psi(x) \\
&= \bar{\psi}(x)(i\gamma^\mu D_\mu - m)\psi(x).
\end{aligned} \tag{23.35}$$

Thus, the first term of $\mathfrak{L}(x)$ of (23.20) is invariant under both the gauge transformation (22.293) and the local phase transformation (23.31). As already seen in (22.318), the second term of $\mathfrak{L}(x)$ of (23.20) represents the Lagrangian density of the free electromagnetic field. From (22.316) and (22.317), $\partial_\mu A^\mu(x)$ is gauge invariant if one uses the Lorentz gauge. It is unclear, however, whether the second term of $\mathfrak{L}(x)$ in (23.20) is gauge invariant. To examine it, from (22.293) we have

$$\begin{aligned}
\partial_\mu A'_\nu(x)\partial^\mu A'^\nu(x) &= \partial_\mu\{[A_\nu + \partial_\nu\Lambda(x)]\}\{\partial^\mu[A^\nu(x) + \partial^\nu\Lambda(x)]\} \\
&= [\partial_\mu A_\nu(x) + \partial_\mu\partial_\nu\Lambda(x)][\partial^\mu A^\nu(x) + \partial^\mu\partial^\nu\Lambda(x)] \\
&= \partial_\mu A_\nu(x)\partial^\mu A^\nu(x) + \partial_\mu A_\nu(x)\partial^\mu\partial^\nu\Lambda(x) \\
&\quad + \partial_\mu\partial_\nu\Lambda(x)\partial^\mu A^\nu(x) + \partial_\mu\partial_\nu\Lambda(x)\partial^\mu\partial^\nu\Lambda(x).
\end{aligned} \tag{23.36}$$

Thus, we see that the term $\partial_\mu A_\nu(x)\partial^\mu A^\nu(x)$ is not gauge invariant. Yet, this does not cause a serious problem. Let us think of the action integral $S(A_\nu)$ of $\partial_\mu A_\nu(x)\partial^\mu A^\nu(x)$. As in (22.306), we have

$$S(A'_\nu) = \int dx \partial_\mu A'_\nu(x)\partial^\mu A'^\nu. \tag{23.37}$$

Taking the variation with respect to A_ν and using integration by parts with the second and third terms of (23.36), we are left with $\partial_\mu\partial^\mu\Lambda(x)$ for them on condition that $A^\nu(x)$ vanishes at $x = \pm\infty$. But, from (22.317), we have

$$\partial_\mu\partial^\mu\Lambda(x) = 0. \tag{23.38}$$

Taking variation with respect to Λ with the last term of (23.36), we also have the same result of (23.38). Thus, only the first term of (23.36) survives on condition that $A^\nu(x)$ vanishes at $x = \pm\infty$ and, hence, we conclude that $\mathfrak{L}(x)$ of (23.20) is virtually unchanged in total by the gauge transformation. That is, $\mathfrak{L}(x)$ is gauge invariant.

In this book, we restrict the gauge field to the electromagnetic vector field $A^\mu(x)$. Correspondingly, we also restrict the gauge transformation to (22.293). By virtue of the local phase transformation (23.31) and the gauge transformation (22.293), we can construct a firm theory of QED. More specifically, we can develop the theory that uniquely determines the interaction between the Dirac field and gauge field (i.e., electromagnetic field).

Before closing this section, it is worth mentioning that the gauge fields must be accompanied by massless particles. Among them, the photon field is the most important both theoretically and practically. Moreover, the photons are only an

entity that has ever been observed experimentally as a massless free particle. Otherwise, gluons are believed to exist, accompanied by the vector bosonic field, but those have never been observed as a massless free particle so far [4, 5]. Also, the massless free particles must move at a light velocity.

We revisit Example 21.1 where the inertial frame of reference O' is moving with a constant velocity v (>0) relative to the frame O . Suppose that the x -axis of O and the x' -axis of O' coincide and that the frame O' is moving in the positive direction of the x' -axis. Also, suppose that a particle is moving along the x -axis (x' -axis, too) at a velocity of u and u' relative to the frame O and O' , respectively. Then, we have [6]

$$u' = \frac{u - v}{1 - uv} \quad \text{and} \quad u = \frac{u' + v}{1 + u'v}. \quad (23.39)$$

If we put $u' = 1$, we have $u = 1$, vice versa. This simple relation clearly demonstrates that if the velocity of a particle is 1 (i.e., light velocity) with respect to an arbitrarily chosen inertial frame of reference, the velocity of that particle must be 1 with respect to any other inertial frames of reference as well. This is the direct confirmation of the principle of constancy of light velocity and an outstanding feature of the gauge field.

23.4 Interaction Picture

In the previous sections, we have seen how the covariant derivative introduces the interaction term into the Dirac equation through minimal coupling. Meanwhile, we also acknowledge that this interaction term results from the variation of the action integral related to the interaction Lagrangian. As a result, we have gotten the proper field equation of (23.27). The magnitude of the interaction relative to the free field, however, is not explicitly represented in (23.27). In this respect, the situation is different from the case of Chap. 5 where the interaction Hamiltonian was quantitatively presented. Thus, we are going to take a different approach in this section. The point is that the Hamiltonian density $\mathcal{H}_{\text{int}}(x)$ is explicitly presented in (23.17). This provides a strong clue to solving the problem. We adopt an interaction picture to address a given issue.

Let us assume the time-dependent total Hamiltonian $H(t)$ as follows:

$$H(t) = H_0(t) + H_{\text{int}}(t), \quad (23.40)$$

where $H_0(t)$ is a free-field Hamiltonian and $H_{\text{int}}(t)$ is an interaction Hamiltonian. All the $H(t)$, $H_0(t)$, and $H_{\text{int}}(t)$ may or may not be time-dependent, but we designate them as, e.g., $H(t)$, where t denotes time. From (23.17), as $H_{\text{int}}(t)$ we have

$$H_{\text{int}}(t) = \int d\mathbf{x}^3 \mathcal{H}_{\text{int}}(x) = \int d\mathbf{x}^3 e\bar{\psi}(x)\gamma^\mu A_\mu(x)\psi(x). \quad (23.41)$$

To deal with the interaction between the quantum fields systematically, we revisit Sect. 1.2. In Chap. 1, we solved the Schrödinger equation by the method of the separation of variables and obtained the solution of (1.48) described by

$$\psi(\mathbf{x}, t) = \exp(-iEt)\phi(\mathbf{x}), \quad (1.60)$$

where we adopted the natural units of $\hbar = 1$. More generally, the solution of (1.47) is given as

$$|\psi(t, \mathbf{x})\rangle = \exp[-iH(t)t]|\psi(0, \mathbf{x})\rangle. \quad (23.42)$$

With the exponential function of the operator $H(t)$, especially that given in a matrix form see Chap. 15. Since $H(t)$ is Hermitian, $-iH(t)t$ is anti-Hermitian. Hence, $\exp[-iH(t)t]$ is unitary; see Sect. 15.2. Thus, (23.42) implies that the state vector $|\psi(t, \mathbf{x})\rangle$ evolves with time through the unitary transformation of $\exp[-iH(t)t]$, starting with $|\psi(0, \mathbf{x})\rangle$.

Now, we wish to generalize the present situation in some degree. We can think of the situation as viewing the same vector in reference to a different set of basis vectors (Sect. 11.4). Suppose that $|\psi(\lambda, \mathbf{x})\rangle$ is an element of a vector space $\mathcal{V}$ (or Hilbert space, more generally) that is spanned by a set of basis vectors $\mathcal{E}_\lambda$ ($\lambda \in \Lambda$), where Λ is an infinite set (see Sect. 6.1). Also, let $|\psi(\lambda, \mathbf{x})\rangle'$ be an element of another vector space $\mathcal{V}'$ spanned by a set of basis vectors $\mathcal{E}'_\lambda$ ($\lambda \in \Lambda$). In general, $\mathcal{V}'$ might be different from the original vector space $\mathcal{V}$, but in this chapter we assume that the mapping caused by an operator is solely an endomorphism (see Sect. 11.2). Suppose that a quantum state ψ is described as

$$\psi = \int d\lambda \mathcal{E}_\lambda |\psi(\lambda, \mathbf{x})\rangle. \quad (23.43)$$

Equation (23.43) is given pursuant to (11.13) so that the vector ψ can be dealt with in a vector space with an infinite dimension. Note, however, if we were *strictly* pursuant to (13.32), we would write instead

$$|\psi\rangle = \int d\lambda |\mathcal{E}_\lambda\rangle \psi(\lambda, \mathbf{x}),$$

where $|\mathcal{E}_\lambda\rangle$ is the basis set of the inner product space with $\psi(\lambda, \mathbf{x})$ being a coordinate of a vector in that space. A reason why we use (23.43) is that we place emphasis on the coordinate representation; the basis set representation is hardly used in the present case.

Operating O on the function ψ , we have

$$O(\psi) = \int d\lambda \mathcal{E}_\lambda O |\psi(\lambda, \mathbf{x})\rangle, \quad (23.44)$$

where O is an arbitrary Hermitian operator. For a future purpose, we assume that the operator O does not depend on λ in a special coordinate system (or frame). Equation (23.44) can be rewritten as

$$O(\psi) = \int d\lambda \mathcal{E}_\lambda V(\lambda) V(\lambda)^\dagger O V(\lambda) V(\lambda)^\dagger |\psi(\lambda, \mathbf{x})\rangle, \quad (23.45)$$

where $V(\lambda)$ is a unitary operator. According to Theorem 14.5, as O is Hermitian, O can be diagonalized via unitary similarity transformation. But in (23.45), we do not necessarily assume the diagonalization. Putting

$$\mathcal{E}'_\lambda \equiv \mathcal{E}_\lambda V(\lambda), O'(\lambda) \equiv V(\lambda)^\dagger O V(\lambda), V(\lambda)^\dagger |\psi(\lambda, \mathbf{x})\rangle \equiv |\psi(\lambda, \mathbf{x})\rangle', \quad (23.46)$$

we rewrite (23.45) such that

$$O(\psi) = \int d\lambda \mathcal{E}'_\lambda O'(\lambda) |\psi(\lambda, \mathbf{x})\rangle'. \quad (23.47)$$

Equation (23.47) shows how the operator $O'(\lambda)$ and the state vector $|\psi(\lambda, \mathbf{x})\rangle'$ change in accordance with the transformation of the basis vectors $\mathcal{E}_\lambda$. We can develop the theory, however, so that $|\psi(\lambda, \mathbf{x})\rangle'$ may not depend upon λ in a specific frame.

To show how it looks in the present case, according to the custom we choose t (time variable) for λ and choose a unitary operator $\exp[-iH(t)t]$ for $V(t)$. Namely,

$$V(t) = \exp[-iH(t)t] \equiv U(t). \quad (23.48)$$

A time-independent state vector $|\psi(0, \mathbf{x})\rangle$ of (23.42) is said to be a state vector of Heisenberg picture (or Heisenberg representation). In other words, we can choose the Heisenberg picture so that the state vector $|\psi(t, \mathbf{x})\rangle'$ may not depend upon t . The other way around, we may choose a special frame where the operator O does not depend on t . That frame is referred to as the Schrödinger frame where a time-dependent state vector $|\psi(t, \mathbf{x})\rangle$ described by (23.42) is called a state vector of Schrödinger picture (or Schrödinger representation). Accordingly, we designate a state vector in the Heisenberg frame as $|\cdot\rangle_H$ and that in the Schrödinger frame as $|\cdot\rangle_S$.

Replacing λ with t , $|\psi(\lambda, \mathbf{x})\rangle$ with $|\psi(t, \mathbf{x})\rangle_S$, and O with O_S in (23.44), we rewrite it as

$$O(\psi) = \int dt \mathcal{E}_t O_S |\psi(t, \mathbf{x})\rangle_S, \quad (23.49)$$

where O_S denotes the operator O in the Schrödinger frame. Equation (23.49) can further be rewritten as

$$\begin{aligned} O(\psi) &= \int dt \mathcal{E}_t U(t) U(t)^\dagger O_S U(t) U(t)^\dagger |\psi(t, \mathbf{x})\rangle_S \\ &= \int dt \mathcal{E}'_t \left[U(t)^\dagger O_S U(t) \right] |\psi(t, \mathbf{x})\rangle'_S \\ &= \int dt \mathcal{E}'_t \left[U(t)^\dagger O_S U(t) \right] |\psi(t, \mathbf{x})\rangle'_S \\ &= \int dt \mathcal{E}'_t O_H(t) |\psi(0, \mathbf{x})\rangle_H, \end{aligned} \quad (23.50)$$

where we define

$$|\psi(t, \mathbf{x})\rangle'_S \equiv U(t)^\dagger |\psi(t, \mathbf{x})\rangle_S, \quad (23.51)$$

$$\mathcal{E}'_t \equiv \mathcal{E}_t U(t),$$

$$|\psi(0, \mathbf{x})\rangle_H \equiv |\psi(t, \mathbf{x})\rangle'_S = U(t)^\dagger |\psi(t, \mathbf{x})\rangle_S, \quad (23.52)$$

$$O_H(t) \equiv U(t)^\dagger O_S U(t). \quad (23.53)$$

Putting $t = 0$ in the above equation, we have

$$|\psi(0, \mathbf{x})\rangle_H = |\psi(0, \mathbf{x})\rangle_S.$$

Equations (23.49) and (23.50) imply the invariance of $O(\psi)$ upon transfer from the Schrödinger frame to the Heisenberg frame. The situation is in parallel to that of Sect. 11.4. We can view (23.51) and (23.52) as the unitary transformation between the two vectors $|\psi(t, \mathbf{x})\rangle_S$ and $|\psi(0, \mathbf{x})\rangle_H$. Correspondingly, (23.53) represents the unitary similarity transformation of the operator O between the two frames.

In (23.53), we assume that $O_H(t)$ is a time-dependent operator, but that O_S is time-independent. In particular, if we choose the Hamiltonian $H(t)$ for O ,

$$H_H(t) = H_H = H_S. \quad (23.54)$$

It is because $H(t)$ and $U(t) = \exp[-iH(t)t]$ are commutative. Thus, the Hamiltonian is a time-independent quantity, regardless of the specific frames chosen.

Differentiating $O_H(t)$ with respect to t in (23.53), we obtain

$$\begin{aligned}
\frac{dO_H(t)}{dt} &= iH(t)e^{iH(t)t}O_S e^{-iH(t)t} + e^{iH(t)t}O_S[-iH(t)]e^{-iH(t)t} \\
&= iH(t)e^{iH(t)t}O_S e^{-iH(t)t} - ie^{iH(t)t}O_S e^{-iH(t)t}H(t) \\
&= i\left[H(t)e^{iH(t)t}O_S e^{-iH(t)t} - e^{iH(t)t}O_S e^{-iH(t)t}H(t)\right] \\
&= i[H(t)O_H(t) - O_H(t)H(t)] = i[H(t), O_H(t)], \tag{23.55}
\end{aligned}$$

where with the second equality, we used the fact that $e^{-iH(t)t}$ and $H(t)$ are commutative. Multiplying both sides of (23.55) by i , we get

$$i\frac{dO_H(t)}{dt} = [O_H(t), H(t)]. \tag{23.56}$$

Equation (23.56) is well-known as the Heisenberg equation of motion. Let us think of a commutator $[O_H(t), P_H(t)]$, where $P_H(t)$ is another physical quantity of the Heisenberg picture. Then, we have

$$[O_H(t), P_H(t)] = U(t)^\dagger [O_S, P_S] U(t), \tag{23.57}$$

where P_S is the physical quantity of the Schrödinger picture that corresponds to $P_H(t)$ of the Heisenberg picture. If, in particular, $[O_S, P_S] = c$ (c is a certain constant), $[O_H(t), P_H(t)] = c$ as well. Thus, we have

$$[q_H(t), p_H(t)] = [q_S, p_S] = i, \tag{23.58}$$

where q and p are the canonical coordinate and canonical momentum, respectively. Thus, the unitary similarity transformation described by (23.53) makes the canonical commutation relation invariant. Such a transformation is called the canonical transformation.

Now, we get back to the major issue related to the interaction picture. A major difficulty in solving the interacting field equation rests partly on the fact that $H_0(t)$ and $H_{\text{int}}(t)$ in general are not commutative. Let us consider another transformation of the state vector and operator different from (23.50). That is, using

$$U_0(t) \equiv \exp[-iH_0(t)t] \tag{23.59}$$

instead of $U(t)$, we have an expression related to (23.51)–(23.53). Namely, we have

$$O(\psi) = \int dt E_t^\dagger O_I(t) |\psi(t, \mathbf{x})\rangle_I, \tag{23.60}$$

where we define

$$|\psi(t, \mathbf{x})\rangle_I \equiv U_0(t)^\dagger |\psi(t, \mathbf{x})\rangle_S = U_0(t)^\dagger U(t) |\psi(0, \mathbf{x})\rangle_H, \quad (23.61)$$

$$\mathcal{E}_t^I \equiv \mathcal{E}_t U_0(t),$$

$$O_I(t) \equiv U_0(t)^\dagger O_S U_0(t). \quad (23.62)$$

In (23.62), $O_I(t)$ is a physical quantity of the interaction picture. This time, $|\psi(t, \mathbf{x})\rangle_I$ is time-dependent.

The previous results (23.45) and (23.46) obtained with the Schrödinger picture and Heisenberg picture basically hold as well between the Schrödinger picture and the interaction picture. Differentiating $|\psi(t, \mathbf{x})\rangle_I$ with respect to t , we get

$$\begin{aligned} \frac{d|\psi(t, \mathbf{x})\rangle_I}{dt} &= iH_0(t)e^{iH_0(t)t}|\psi(t, \mathbf{x})\rangle_S + e^{iH_0(t)t}[-iH(t)]|\psi(t, \mathbf{x})\rangle_S \\ &= iH_0(t)e^{iH_0(t)t}|\psi(t, \mathbf{x})\rangle_S - ie^{iH_0(t)t}[H_0(t) + H_{\text{int}}(t)]|\psi(t, \mathbf{x})\rangle_S \\ &= iH_0(t)e^{iH_0(t)t}|\psi(t, \mathbf{x})\rangle_S - iH_0(t)e^{iH_0(t)t}|\psi(t, \mathbf{x})\rangle_S \\ &\quad - ie^{iH_0(t)t}H_{\text{int}}(t)|\psi(t, \mathbf{x})\rangle_S \\ &= -ie^{iH_0(t)t}H_{\text{int}}(t)e^{-iH_0(t)t}|\psi(t, \mathbf{x})\rangle_I \\ &= -iH_{\text{int}}(t)_I|\psi(t, \mathbf{x})\rangle_I, \end{aligned} \quad (23.63)$$

where with the third equality $e^{iH_0(t)t}$ and $H_0(t)$ are commutative. In (23.63), we defined $H_{\text{int}}(t)_I$ as

$$H_{\text{int}}(t)_I \equiv e^{iH_0(t)t}H_{\text{int}}(t)e^{-iH_0(t)t} = U_0(t)^\dagger H_{\text{int}}(t)U_0(t). \quad (23.64)$$

Also, notice again that in (23.63) $e^{\pm iH_0(t)t}$ and $H_{\text{int}}(t)$ are not commutative in general. Multiplying both sides of (23.63) by i , we obtain

$$i \frac{d|\psi(t, \mathbf{x})\rangle_I}{dt} = H_{\text{int}}(t)_I |\psi(t, \mathbf{x})\rangle_I. \quad (23.65)$$

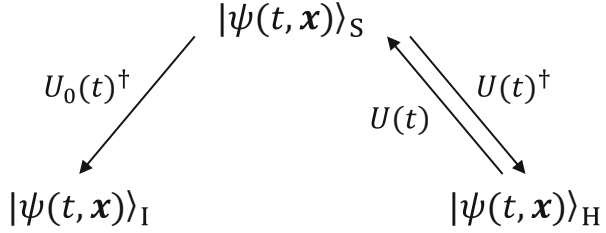
Similarly to (23.56), we have

$$i \frac{dO_I(t)}{dt} = [O_I(t), H_0(t)]. \quad (23.66)$$

Close relationship between (23.66) and (23.56) is evident.

We schematically depict the conversion among the three pictures of Schrödinger, Heisenberg, and interaction in Fig. 23.1. Equation (23.65) formally resembles the time-dependent Schrödinger equation (1.47) and provides a starting point to address a key question of the interacting fields.

Fig. 23.1 State vector $|\psi(t, \mathbf{x})\rangle_I$ of the interaction picture in comparison with those of the Schrödinger picture $|\psi(t, \mathbf{x})\rangle_S$ and the Heisenberg picture $|\psi(0, \mathbf{x})\rangle_H$



23.5 S-Matrix and S-Matrix Expansion

Since (23.65) is a non-linear equation, it is impossible in general to obtain an analytic solution. Instead, we will be approaching the problem by an iteration method. To this end, the S -matrix expansion plays a central role.

From here on, we exclusively adopt the interaction picture that gave (23.65), and so we rewrite (23.65) as follows:

$$i \frac{d|\Psi(t, \mathbf{x})\rangle}{dt} = H_{\text{int}}(t) |\Psi(t, \mathbf{x})\rangle, \quad (23.67)$$

where the subscript I of $|\psi(t, \mathbf{x})\rangle_I$ and $H_{\text{int}}(t)_I$ was omitted. Namely, as we work solely on the interaction picture, we redefine the state vector and Hamiltonian such that

$$|\Psi(t, \mathbf{x})\rangle \equiv |\psi(t, \mathbf{x})\rangle_I \text{ and } H_{\text{int}}(t) \equiv H_{\text{int}}(t)_I. \quad (23.68)$$

If we have no interaction, i.e., $H_{\text{int}}(t) = 0$, we have a state vector $|\Psi(t, \mathbf{x})\rangle$ constant in time. Suppose that the interaction is “switched on” at a certain time t_{on} and that the state vector is in $|\Psi(t_i, \mathbf{x})\rangle$ at an initial time $t = t_i$ well before t_{on} . In that situation, we define the state vector $|\Psi(t_i, \mathbf{x})\rangle$ as

$$|\Psi(t_i, \mathbf{x})\rangle \equiv |i\rangle \approx |\Psi(-\infty, \mathbf{x})\rangle. \quad (23.69)$$

Also, suppose that the interaction is “switched off” at a certain time t_{off} and that the state vector is in $|\Psi(t_f, \mathbf{x})\rangle$ at a final time $t = t_f$ well after t_{off} . Then, the state vector $|\Psi(t_f, \mathbf{x})\rangle$ is approximated by

$$|\Psi(t_f, \mathbf{x})\rangle \approx |\Psi(\infty, \mathbf{x})\rangle. \quad (23.70)$$

Because of the interaction (or collision), the state $|\Psi(\infty, \mathbf{x})\rangle$ may include different final states $|f_\alpha\rangle$ ($\alpha = 1, 2, \dots$), which may involve different species of particles. If we assume that $|f_\alpha\rangle$ form a complete orthonormal system (CONS), we have [1]

$$\sum_{\alpha} |f_{\alpha}\rangle \langle f_{\alpha}| = E, \quad (23.71)$$

where E is an identity operator. Then, we have

$$|\Psi(\infty, \mathbf{x})\rangle = \sum_{\alpha} |f_{\alpha}\rangle \langle f_{\alpha}| \Psi(\infty, \mathbf{x})\rangle. \quad (23.72)$$

- (i) The state $|i\rangle$ is described by the Fock state [see (22.126)] that is characterized by the particle species and definite number of those particles with definite momenta and energies. The particles in $|i\rangle$ are far apart from one another before the collision such that they do not interact with one another. The particles in $|f_{\alpha}\rangle$ are also far apart from one another after the collision.
- (ii) The states $|i\rangle$ and $|f_{\alpha}\rangle$ are described by the plane wave.

Normally, the interaction is assumed to cause in a very narrow space and short time. During the interaction, we cannot get detailed information about $|\Psi(t, \mathbf{x})\rangle$. Before and after the interaction, the species or number of particles may or may not be conserved. The plane wave description is particularly suited for the actual experimental situation where accelerated particle beams collide. The bound states, on the other hand, are neither practical nor appropriate. In what follows we are going to deal with electrons, positrons, and photons within the framework of QED.

The S -matrix S is defined as an operator that relates $|f_{\alpha}\rangle$ to $|i\rangle$ such that

$$\langle f_{\alpha} | \Psi(t_f, \mathbf{x}) \rangle \approx \langle f_{\alpha} | \Psi(\infty, \mathbf{x}) \rangle = \langle f_{\alpha} | S | i \rangle \equiv S_{f_{\alpha}, i} = \langle f_{\alpha} | S | \Psi(-\infty, \mathbf{x}) \rangle, \quad (23.73)$$

where with the first near equality, we used (23.70). Then, (23.72) can be rewritten as

$$|\Psi(\infty, \mathbf{x})\rangle = \sum_{\alpha} S_{f_{\alpha}, i} |f_{\alpha}\rangle. \quad (23.74)$$

Equation (23.74) is a plane wave expansion of $|\Psi(\infty, \mathbf{x})\rangle$. If $|\Psi(t_f, \mathbf{x})\rangle$ and $|i\rangle$ are normalized, we have

$$\langle \Psi(t_f, \mathbf{x}) | \Psi(t_f, \mathbf{x}) \rangle = \langle i | i \rangle = 1 = \langle i | S^{\dagger} S | i \rangle. \quad (23.75)$$

Since the transformation with S does not change the norm, S is unitary.

Integrating (23.67) with respect to t from $-\infty$ to t , we get

$$|\Psi(t, \mathbf{x})\rangle = |\Psi(-\infty, \mathbf{x})\rangle + (-i) \int_{-\infty}^t dt_1 H_{\text{int}}(t_1) |\Psi(t_1, \mathbf{x})\rangle$$

$$= |i\rangle + (-i) \int_{-\infty}^t dt_1 H_{\text{int}}(t_1) |\Psi(t_1, \mathbf{x})\rangle. \quad (23.76)$$

The RSH of (23.76) comprises $|i\rangle$ and an additional second term. Since we can solve (23.67) only iteratively, as a next step replacing $|\Psi(t_1, \mathbf{x})\rangle$ with

$$|i\rangle + (-i) \int_{-\infty}^{t_1} H_{\text{int}}(t_2) |\Psi(t_2, \mathbf{x})\rangle, \quad (23.77)$$

we obtain

$$\begin{aligned} |\Psi(t, \mathbf{x})\rangle &= |i\rangle + (-i) \int_{-\infty}^t dt_1 H_{\text{int}}(t_1) |i\rangle \\ &+ (-i)^2 \int_{-\infty}^t dt_1 \int_{-\infty}^{t_1} dt_2 H_{\text{int}}(t_1) H_{\text{int}}(t_2) |\Psi(t_2, \mathbf{x})\rangle. \end{aligned} \quad (23.78)$$

Repeating the iteration, we get

$$\begin{aligned} &|\Psi(t, \mathbf{x})\rangle \\ &= |i\rangle \\ &+ \left[\sum_{n=1}^r (-i)^n \int_{-\infty}^t dt_1 \cdots \int_{-\infty}^{t_{n-1}} dt_n H_{\text{int}}(t_1) \cdots H_{\text{int}}(t_n) |\Psi(t_n, \mathbf{x})\rangle \right], \end{aligned} \quad (23.79)$$

where $t_0 \equiv t$ when $n = 1$. Although we could further replace $|\Psi(t_r, \mathbf{x})\rangle$ with

$$|i\rangle + (-i) \int_{-\infty}^{t_r} H_{\text{int}}(t_{r+1}) |\Psi(t_{r+1}, \mathbf{x})\rangle,$$

we cut off the iteration process in this stage. Hence, $|\Psi(t, \mathbf{x})\rangle$ of (23.79) is approximated by

$$\begin{aligned} |\Psi(t, \mathbf{x})\rangle &\approx |i\rangle + \left[\sum_{n=1}^r (-i)^n \int_{-\infty}^t dt_1 \cdots \int_{-\infty}^{t_{n-1}} dt_n H_{\text{int}}(t_1) \cdots H_{\text{int}}(t_n) |i\rangle \right] \\ &= |i\rangle + \left[\sum_{n=1}^r (-i)^n \int_{-\infty}^t dt_1 \cdots \int_{-\infty}^{t_{n-1}} dt_n H_{\text{int}}(t_1) \cdots H_{\text{int}}(t_n) \right] |i\rangle \\ &= \left\{ E + \left[\sum_{n=1}^r (-i)^n \int_{-\infty}^t dt_1 \cdots \int_{-\infty}^{t_{n-1}} dt_n H_{\text{int}}(t_1) \cdots H_{\text{int}}(t_n) \right] \right\} |i\rangle. \end{aligned} \quad (23.80)$$

Taking $t, t_1, \dots, t_n \rightarrow \infty$ in (23.80), we have

$$| \Psi(\infty, \mathbf{x}) \rangle \approx \left\{ E + \left[\sum_{n=1}^r (-i)^n \int_{-\infty}^{\infty} dt_1 \cdots \int_{-\infty}^{\infty} dt_n H_{\text{int}}(t_1) \cdots H_{\text{int}}(t_n) \right] \right\} | i \rangle. \quad (23.81)$$

Taking account of (23.73), we have

$$| \Psi(\infty, \mathbf{x}) \rangle = S | i \rangle.$$

Then, from (23.81) we obtain

$$S = E + \left[\sum_{n=1}^r (-i)^n \int_{-\infty}^{\infty} dt_1 \cdots \int_{-\infty}^{\infty} dt_n H_{\text{int}}(t_1) \cdots H_{\text{int}}(t_n) \right].$$

Defining $S^{(0)} \equiv E$ and

$$S^{(r)} \equiv \sum_{n=1}^r (-i)^n \int_{-\infty}^{\infty} dt_1 \cdots \int_{-\infty}^{\infty} dt_n H_{\text{int}}(t_1) \cdots H_{\text{int}}(t_n) \quad (r \geq n \geq 1), \quad (23.82)$$

we get

$$S = \sum_{n=0}^r S^{(n)}. \quad (23.83)$$

The matrix $S^{(n)}$ is said to be the n -th order S -matrix. In (23.83), the number r can be as large as one pleases, even though the related S -matrix calculations become increasingly complicated.

Taking the T-product of (23.82), moreover, we get

$$S^{(r)} = \sum_{n=0}^r \frac{(-i)^n}{n!} \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{\infty} dt_2 \cdots \int_{-\infty}^{\infty} dt_n T[H_{\text{int}}(t_1) \cdots H_{\text{int}}(t_n)]. \quad (23.84)$$

Note that the factorial in the denominator of (23.84) resulted from taking the T-product.

Using the Hamiltonian density $\mathfrak{H}_{\text{int}}(x)$ of (23.17), we rewrite (23.84) as

$$S^{(r)} = \sum_{n=0}^r \frac{(-i)^n}{n!} \int_{-\infty}^{\infty} dx_1 \int_{-\infty}^{\infty} dx_2 \cdots \int_{-\infty}^{\infty} dx_n T[\mathfrak{H}_{\text{int}}(x_1) \cdots \mathfrak{H}_{\text{int}}(x_n)]. \quad (23.85)$$

Combining (23.83) with (23.73) and (23.74), we get

$$| \Psi(\infty, \mathbf{x}) \rangle = \sum_{\alpha} \sum_n \langle f_{\alpha} | S^{(n)} | i \rangle | f_{\alpha} \rangle. \quad (23.86)$$

By virtue of using $\mathfrak{H}_{\text{int}}(x)$, S has been made an explicitly covariant form. Notice that we must add a factor -1 when we reorder Fermion quantities as in the case of the T-product. But, in the above we have not put the factor -1 , because $\mathfrak{H}_{\text{int}}(x)$ contains two Fermion quantities and, hence, exchanging $\mathfrak{H}_{\text{int}}(x_i)$ and $\mathfrak{H}_{\text{int}}(x_j)$ ($x_i \neq x_j$) produces even number permutations of the Fermion quantities. Then, even permutations produce a factor $(-1)^{2n} = 1$ where n is a positive integer.

The S -matrix contains the identity operator E and additional infinite terms and can be expressed as an infinite power series of the fine structure constant α described in the natural units (SI units) by

$$\alpha \equiv e^2/4\pi (e^2/4\pi\epsilon_0\hbar c) \approx 7.29735 \times 10^{-3}. \quad (23.87)$$

23.6 N-Product and T-Product

In Sects. 22.3 and 23.5, we mentioned the normal-ordered product (N-product) and time-ordered product (T-product). The practical calculations based on them frequently appear and occupy an important position in the quantum field theory. We represent them as, for example

$$N(AB) \text{ and } T(ABC\cdots),$$

where the former represents the N-product and the latter denotes the T-product; A , B , C , etc. show the physical quantities (i.e., quantum fields). The N-product and T-product are important not only in the practical calculations but also from the theoretical point of view. In particular, they are useful to clarify the relativistic invariance (or covariance) of the mathematical formulae.

23.6.1 Example of Two Field Operators

We start with the simplest case where two field operators are pertinent. The essence of both the N-product and T-product clearly shows up in this case and the methods developed here provide a good starting point for dealing with more complicated problems. Let $A(x)$ and $B(x)$ be two field operators, where x represents the space-time coordinates. We choose $A(x)$ and $B(x)$ from among the following operators:

$$\phi(x) = \int \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3(2k_0)}} [a(\mathbf{k})e^{-ikx} + a^\dagger(\mathbf{k})e^{ikx}], \quad (22.98)$$

$$\begin{aligned} \psi(x) &= \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 2p_0}} \sum_{h=\pm 1} [b(\mathbf{p}, h)u(\mathbf{p}, h)e^{-ipx} + d^\dagger(\mathbf{p}, h)v(\mathbf{p}, h)e^{ipx}], \\ &\quad (22.232) \end{aligned}$$

$$\begin{aligned} \bar{\psi}(x) &= \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 2p_0}} \sum_{h=\pm 1} [d(\mathbf{p}, h)\bar{v}(\mathbf{p}, h)e^{-ipx} + b^\dagger(\mathbf{p}, h)\bar{u}(\mathbf{p}, h)e^{ipx}], \\ &\quad (23.88) \end{aligned}$$

$$\begin{aligned} \psi_{\text{anti}}(x) &= \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 2p_0}} \sum_{h=\pm 1} [d(\mathbf{p}, h)u(\mathbf{p}, h)e^{-ipx} + b^\dagger(\mathbf{p}, h)v(\mathbf{p}, h)e^{ipx}], \\ &\quad (22.244) \end{aligned}$$

$$\begin{aligned} \bar{\psi}_{\text{anti}}(x) &= \int_{-\infty}^{\infty} \frac{d\mathbf{p}^3}{\sqrt{(2\pi)^3 2p_0}} \sum_{h=\pm 1} [b(\mathbf{p}, h)\bar{v}(\mathbf{p}, h)e^{-ipx} + d^\dagger(\mathbf{p}, h)\bar{u}(\mathbf{p}, h)e^{ipx}], \\ &\quad (23.89) \end{aligned}$$

$$\begin{aligned} A^\mu(x) &= \int \frac{d^3\mathbf{k}}{\sqrt{(2\pi)^3(2k_0)}} \sum_{r=0}^3 [a(\mathbf{k}, r)\varepsilon^\mu(\mathbf{k}, r)e^{-ikx} + a^\dagger(\mathbf{k}, r)\varepsilon^\mu(\mathbf{k}, r)^*e^{ikx}]. \\ &\quad (22.334) \end{aligned}$$

Equations (22.244) and (23.89) represent the operators of a positron. Note that with respect to (22.98) for the real scalar field and (22.334) for the photon field, the field operators $\phi(x)$ and $A^\mu(x)$ are Hermitian. The Dirac field operators, on the other hand, are not Hermitian. For this reason, we need two pairs of operators $b(\mathbf{p}, h)$, $b^\dagger(\mathbf{p}, h)$; $d(\mathbf{p}, h)$, $d^\dagger(\mathbf{p}, h)$ to describe the Dirac field. We see that it is indeed the case when we describe the time-ordered products (vide infra). When QED is relevant, the operators of the Dirac field and photon field are chosen from among $\psi(x)$, $\bar{\psi}(x)$, $\psi_{\text{anti}}(x)$, $\bar{\psi}_{\text{anti}}(x)$, and $A^\mu(x)$.

The above field operators share several features in common. (i) They can be divided into two parts, i.e., the positive-frequency part (containing the factor e^{-ikx})

and the negative-frequency part (containing the factor e^{ikx}) [1]. The positive and negative frequencies correspond to the positive energy and negative energy, respectively. These quantities result from the fact that the relativistic field theory is associated with the Klein–Gordon equation (see Chaps. 21 and 22). (ii) For each operator, the first term, i.e., the positive-frequency part, contains the annihilation operator and the second term (negative-frequency part) contains the creation operator. Accordingly, we express the operators as in, e.g., (22.99) and (22.132) such that [1]

$$\phi(x) = \phi^+(x) + \phi^-(x). \quad (23.90)$$

Care should be taken when dealing with $\bar{\psi}(x)$ [or $\bar{\psi}_{\text{anti}}(x)$]. It is because when taking the Dirac adjoint of $\psi(x)$ [or $\psi_{\text{anti}}(x)$], the positive-frequency part and the negative-frequency part are switched.

Let us get back to the present task. As already mentioned in Sect. 22.3, with respect to the N-product for, e.g., the scalar field, the annihilation operators $a(\mathbf{k})$ always stand to the right of the creation operators $a^\dagger(\mathbf{k})$. Now, we show the computation rule of the N-product as follows:

(i) Boson:

$$\begin{aligned} N[\phi(x)\phi(y)] &= N\{[\phi^+(x) + \phi^-(x)][\phi^+(y) + \phi^-(y)]\} \\ &= N[\phi^+(x)\phi^+(y) + \phi^+(x)\phi^-(y) + \phi^-(x)\phi^+(y) + \phi^-(x)\phi^-(y)] \\ &= \phi^+(x)\phi^+(y) + \phi^-(y)\phi^+(x) + \phi^-(x)\phi^+(y) + \phi^-(x)\phi^-(y). \end{aligned} \quad (23.91)$$

Notice that with the second term in the last equality of (23.91), $\phi^+(x)\phi^-(y)$ has been switched to $\phi^-(y)\phi^+(x)$. Similarly, we obtain

$$\begin{aligned} N[\phi(y)\phi(x)] &= \phi^+(y)\phi^+(x) + \phi^-(x)\phi^+(y) + \phi^-(y)\phi^+(x) + \phi^-(y)\phi^-(x). \end{aligned} \quad (23.92)$$

Applying (22.134) to (23.92), we get

$$\begin{aligned} N[\phi(y)\phi(x)] &= \phi^+(x)\phi^+(y) + \phi^-(x)\phi^+(y) + \phi^-(y)\phi^+(x) + \phi^-(x)\phi^-(y). \end{aligned} \quad (23.93)$$

Comparing (23.91) and (23.93), we have

$$N[\phi(x)\phi(y)] = N[\phi(y)\phi(x)]. \quad (23.94)$$

In this way, in the Boson field case, the N-product is kept unchanged after exchanging the operators involved in the N-product. The expressions related to (23.91) and (23.94) hold with the photon field $A^\mu(x)$ as well.

(ii) Fermion (electron, positron, etc.):

$$\begin{aligned} N[\psi(x)\psi(y)] &= N\{[\psi^+(x) + \psi^-(x)][\psi^+(y) + \psi^-(y)]\} \\ &= N[\psi^+(x)\psi^+(y) + \psi^+(x)\psi^-(y) + \psi^-(x)\psi^+(y) + \psi^-(x)\psi^-(y)] \\ &= \psi^+(x)\psi^+(y) - \psi^-(y)\psi^+(x) + \psi^-(x)\psi^+(y) + \psi^-(x)\psi^-(y). \end{aligned} \quad (23.95)$$

The expression related to (23.92) holds with the Dirac field $\bar{\psi}(x)$, $\psi_{\text{anti}}(x)$, and $\bar{\psi}_{\text{anti}}(x)$. With the above calculations, the positive-frequency part $\psi^+(x)$ always stands to the right of the negative-frequency part $\psi^-(x)$ as in the Boson case. When the multiplication order is switched, however, the sign of the relevant term is reversed (+ to – and – to +) for the Fermion case. The reversal of the sign with respect to $+\psi^+(x)\psi^-(y)$ and $-\psi^-(y)\psi^+(x)$ results from the relations (22.246) and the anti-commutation relations (22.242).

Similarly, we have

$$\begin{aligned} N[\psi(y)\psi(x)] &= \psi^+(y)\psi^+(x) - \psi^-(x)\psi^+(y) + \psi^-(y)\psi^+(x) + \psi^-(y)\psi^-(x). \end{aligned} \quad (23.96)$$

Applying (22.245) to (23.96), we get

$$\begin{aligned} N[\psi(y)\psi(x)] &= -\psi^+(x)\psi^+(y) - \psi^-(x)\psi^+(y) + \psi^-(y)\psi^+(x) - \psi^-(x)\psi^-(y). \end{aligned} \quad (23.97)$$

Comparing (23.95) and (23.97), we have

$$N[\psi(y)\psi(x)] = -N[\psi(x)\psi(y)]. \quad (23.98)$$

Contrast (23.98) with (23.94) with respect to the presence or absence of the minus sign. If we think of $N[\psi(x)\bar{\psi}(y)]$, we can perform the relevant calculations following (23.95)–(23.98) in the same manner.

Thinking of the constitution of the N-product, it is clear that for both the cases of Boson and Fermion, the vacuum expectation value of the N-product is zero. Or rather, the N-product is constructed in such a way that the vacuum expectation value of the product of field operators may vanish.

23.6.2 Calculations Including Both the N-Products and T-Products

We continue dealing with the mathematical expressions of the above example and examine their implications for both the Boson and Fermion cases.

(i) Boson:

Subtracting $N[\phi(x)\phi(y)]$ from $\phi(x)\phi(y)$ in (23.91), we have

$$\phi(x)\phi(y) - N[\phi(x)\phi(y)] = [\phi^+(x), \phi^-(y)] = i\Delta^+(x-y), \quad (23.99)$$

where with the last equality we used (22.137). Exchanging x and y in (23.99), we obtain

$$\phi(y)\phi(x) - N[\phi(y)\phi(x)] = [\phi^+(y), \phi^-(x)] = i\Delta^+(y-x) = -i\Delta^-(x-y), \quad (23.100)$$

where with the last equality we used (22.138). Using (23.94), we get

$$\phi(y)\phi(x) - N[\phi(x)\phi(y)] = -i\Delta^-(x-y). \quad (23.101)$$

Equations (23.99) and (23.101) show that if we subtract the N-product from the product of the field operators, i.e., $\phi(x)\phi(y)$ or $\phi(y)\phi(x)$, we are left with the scalar of $i\Delta^+(x-y)$ or $-i\Delta^-(x-y)$. This implies the relativistic invariance of (23.99) and (23.101). Meanwhile, sandwiching (23.99) between $\langle 0|$ and $|0\rangle$, i.e., taking a vacuum expectation value of (23.99), we obtain

$$\langle 0 | \phi(x)\phi(y) | 0 \rangle - \langle 0 | N[\phi(x)\phi(y)] | 0 \rangle = \langle 0 | [\phi^+(x), \phi^-(y)] | 0 \rangle. \quad (23.102)$$

Taking account of the fact that $\langle 0 | N[\phi(x)\phi(y)] | 0 \rangle$ vanishes, we get

$$\begin{aligned} \langle 0 | \phi(x)\phi(y) | 0 \rangle &= \langle 0 | [\phi^+(x), \phi^-(y)] | 0 \rangle = [\phi^+(x), \phi^-(y)] \langle 0 | 0 \rangle \\ &= [\phi^+(x), \phi^-(y)] = i\Delta^+(x-y), \end{aligned} \quad (23.103)$$

where with the second equality we used the fact that $[\phi^+(x), \phi^-(y)] = i\Delta^+(x-y)$ is a c -number, see (22.137); with the second to the last equality, we used the normalized condition

$$\langle 0 | 0 \rangle = 1. \quad (23.104)$$

Then, using (23.103), we rewrite (23.99) as [1]

$$\phi(x)\phi(y) - N[\phi(x)\phi(y)] = \langle 0 | \phi(x)\phi(y) | 0 \rangle. \quad (23.105)$$

Thus, (23.105) shows the relativistic invariance based on the N-product and the vacuum expectation value of $\phi(x)\phi(y)$.

We wish to derive a more useful expression on the basis of (23.105). To this end, we deal with the mathematical expressions including both the N-products and T-products. Multiplying both sides of (23.99) by $\theta(x^0 - y^0)$, we get

$$\theta(x^0 - y^0)\phi(x)\phi(y) - \theta(x^0 - y^0)N[\phi(x)\phi(y)] = i\theta(x^0 - y^0)\Delta^+(x - y). \quad (23.106)$$

Multiplying both sides of (23.101) by $\theta(y^0 - x^0)$, we obtain

$$\begin{aligned} \theta(y^0 - x^0)\phi(y)\phi(x) - \theta(y^0 - x^0)N[\phi(x)\phi(y)] \\ = -i\theta(y^0 - x^0)\Delta^-(x - y). \end{aligned} \quad (23.107)$$

Adding both sides of (23.106) and (23.107), we get

$$\begin{aligned} \theta(x^0 - y^0)\phi(x)\phi(y) + \theta(y^0 - x^0)\phi(y)\phi(x) \\ - [\theta(x^0 - y^0) + \theta(y^0 - x^0)]N[\phi(x)\phi(y)] \\ = i[\theta(x^0 - y^0)\Delta^+(x - y) - \theta(y^0 - x^0)\Delta^-(x - y)] = \Delta_F(x - y), \end{aligned} \quad (23.108)$$

where the last equality is due to (22.185). Using the definition of the time-ordered product (or T-product) of (22.178) and the relation described by (see Chap. 10)

$$\theta(x^0 - y^0) + \theta(y^0 - x^0) \equiv 1, \quad (23.109)$$

we obtain

$$T[\phi(x)\phi(y)] - N[\phi(x)\phi(y)] = \Delta_F(x - y) = \langle 0 | T[\phi(x)\phi(y)] | 0 \rangle, \quad (23.110)$$

where with the last equality we used (22.182). Comparing (23.110) with (23.105), we find that in (23.110) $T[\phi(x)\phi(y)]$ has replaced $\phi(x)\phi(y)$ of (23.105) except for $\phi(x)\phi(y)$ in $N[\phi(x)\phi(y)]$. Putting it another way, subtracting the N-product from the T-product in (23.110), we are left with the relativistically invariant scalar $\langle 0 | T[\phi(x)\phi(y)] | 0 \rangle$.

(ii) Fermion:

The discussion is in parallel with that of the Boson case mentioned above. From a practical point of view, a product $\psi(x)\bar{\psi}(y)$ is the most important. Subtracting $N[\psi(x)\bar{\psi}(y)]$ from $\psi(x)\bar{\psi}(y)$ in (23.95), we have

$$\psi(x)\bar{\psi}(y) - N[\psi(x)\bar{\psi}(y)] = \{\psi^+(x), \bar{\psi}^-(y)\} = iS^+(x-y) \quad (23.111)$$

↓ multiply both sides by $\theta(x^0 - y^0)$

$$\theta(x^0 - y^0)\psi(x)\bar{\psi}(y) - \theta(x^0 - y^0)N[\psi(x)\bar{\psi}(y)] = i\theta(x^0 - y^0)S^+(x-y). \quad (23.112)$$

Note that with the last equality of (23.111), we used (22.256). In a similar manner, we have

$$\begin{aligned} & \bar{\psi}(y)\psi(x) - N[\bar{\psi}(y)\psi(x)] \\ &= \{\bar{\psi}^+(y), \psi^-(x)\} = \{\psi^-(x), \bar{\psi}^+(y)\} = iS^-(x-y) \end{aligned} \quad (23.113)$$

↓ multiply both sides by $\theta(y^0 - x^0)$

$$\theta(y^0 - x^0)\bar{\psi}(y)\psi(x) - \theta(y^0 - x^0)N[\bar{\psi}(y)\psi(x)] = i\theta(y^0 - x^0)S^-(x-y). \quad (23.114)$$

Subtracting both sides of (23.114) from (23.112) and using the definition of the time-ordered product of (22.254) for the Dirac field, we get

$$\begin{aligned} & T[\psi(x)\bar{\psi}(y)] - [\theta(x^0 - y^0) + \theta(y^0 - x^0)]N[\psi(x)\bar{\psi}(y)] \\ &= T[\psi(x)\bar{\psi}(y)] - N[\psi(x)\bar{\psi}(y)] \\ &= i[\theta(x^0 - y^0)S^+(x-y) - \theta(y^0 - x^0)S^-(x-y)]. \end{aligned} \quad (23.115)$$

where we used the following relationship that holds with the Fermion case:

$$N[\bar{\psi}(y)\psi(x)] = -N[\psi(x)\bar{\psi}(y)]. \quad (23.116)$$

Using (22.258), we rewrite (23.115) as

$$T[\psi(x)\bar{\psi}(y)] - N[\psi(x)\bar{\psi}(y)] = S_F(x-y) = \langle 0|T[\psi(x)\bar{\psi}(y)]|0\rangle. \quad (23.117)$$

Again, we have obtained the expression (23.117) related to (23.110). Collectively writing these equations, we obtain [1]

$$T[A(x)B(y)] - N[A(x)B(y)] = \langle 0|T[A(x)B(y)]|0\rangle. \quad (23.118)$$

In the case where both A and B are Fermion operators, $A(x) = \psi(x)$ [or $\bar{\psi}(x)$] and $B(y) = \bar{\psi}(y)$ [or $\psi(y)$] are intended. In this case, all three terms of (23.118) change the sign upon exchanging $A(x)$ and $B(y)$. Therefore, (23.118) does not change the sign upon the same exchange. Meanwhile, in the case where at least one of A and B is a Boson operator, each term of (23.118) does not change sign upon permutation of $A(x)$ and $B(y)$. Hence, (23.118) holds as well.

In the case of the photon field, the discussion can be developed in parallel to the scalar bosonic field, because the photon is among a Boson family. In both the Boson and Fermion cases, the essence of the perturbation method clearly shows up in the simple expression (23.118). In fact, (23.118) is the simplest case of the Wick's theorem [1, 3]. In the general case, the Wick's theorem is expressed as [1]

$$\begin{aligned} T(ABCD \cdots WXYZ) = & N(ABCD \cdots WXYZ) + N(\underbrace{ABC}_{\text{---}} \cdots YZ) \\ & + N(\underbrace{ABC}_{\text{---}} \cdots \underbrace{YZ}_{\text{---}}) + \cdots + N(\underbrace{ABC}_{\text{---}} \cdots \underbrace{YZ}_{\text{---}}) + N(\underbrace{ABCD}_{\text{---}} \cdots \underbrace{YZ}_{\text{---}}) + \cdots \\ & + N(\underbrace{AB}_{\text{---}} \cdots \underbrace{WXYZ}_{\text{---}}) + \cdots. \end{aligned} \quad (23.119)$$

Equation (23.119) looks complicated, but clearly shows how the Feynman propagators are produced in coexistence of the T-products and N-products. Glancing at (23.110) and (23.117), this feature is once again obvious. In RHS of (23.119), the terms from the second on down are sometimes called the generalized normal products [1]. We have another rule with them. For instance, we have [1]

$$N(\underbrace{ABC}_{\text{---}} \cdots YZ) = (-1)^P \underbrace{AC}_{\text{---}} [N(B \cdots YZ)], \quad (23.120)$$

where P is the number of interchanges of neighboring Fermion operators required to change the order $(ABC \cdots YZ)$ to $(ACB \cdots YZ)$. In this case, the interchange takes place between B and C . If both B and C are Fermion operators, then $P = 1$; otherwise, $P = 0$.

Bearing in mind the future application (e.g., the Compton scattering), let us think of a following example given as

$$N(F_1 B_1 F'_1 F_2 B_2 F'_2), \quad (23.121)$$

where F and F' are Fermion operators; B and B' are Boson operators. Indices 1 and 2 designate the space-time coordinates. Suppose that we have a following contraction within the operators given by

$$\begin{aligned} N(\underbrace{F_1 B_1 F'_1 F_2 B_2 F'_2}_{\text{---}}) &= (-1)^2 \underbrace{F_1 F'_2}_{\text{---}} N(B_1 F'_1 F_2 B_2) \\ &= \underbrace{F_1 F'_2}_{\text{---}} N(B_1 F'_1 F_2 B_2). \end{aligned} \quad (23.122)$$

Meanwhile, suppose that we have

$$\begin{aligned}
N\left(\underbrace{F_2 B_2 F'_2 F_1 B_1 F'_1}\right) &= \underbrace{F'_2 F_1}(-1)^2 N(F_2 B_2 B_1 F'_1) = -\underbrace{F_1 F'_2}(-1)^2 N(F_2 B_2 B_1 F'_1) \\
&= -\underbrace{F_1 F'_2}(-1)^2 (-1) N(B_1 F'_1 F_2 B_2) = \underbrace{F_1 F'_2} N(B_1 F'_1 F_2 B_2). \quad (23.123)
\end{aligned}$$

Comparing (23.122) and (23.123), we get

$$N\left(\underbrace{F_1 B_1 F'_1 F_2 B_2 F'_2}\right) = N\left(\underbrace{F_2 B_2 F'_2 F_1 B_1 F'_1}\right). \quad (23.124)$$

Thus, we have obtained a simple but important relationship with the generalized normal product. We make use of this relationship in the following sections.

23.7 Feynman Rules and Feynman Diagrams in QED

So far, we have developed the general discussion on the interacting fields and how to deal with the interaction using the perturbation methods based upon the S -matrix method. In this section, we develop the methods to apply the general principle to the specific practical problems. For this purpose, we make the most of the Feynman diagrams and associated Feynman rules. In parallel, we represent the mathematical expressions using a symbol or diagram. One of typical examples is a contraction, which we mentioned in the previous section. We summarize frequently appearing contractions below. The contractions represent the Feynman propagators.

$$\underbrace{\phi(x)\phi(y)} = \Delta_F(x-y), \quad (23.125)$$

$$\underbrace{\psi(x)\bar{\psi}(y)} = -\underbrace{\bar{\psi}(y)\psi(x)} = S_F(x-y), \quad (23.126)$$

$$\underbrace{A^\mu(x)A^\nu(y)} = D_F(x-y). \quad (23.127)$$

Among the above relations, (23.126) and (23.127) are pertinent to QED. When we examine the interaction between the fields, it is convenient to describe field operator F such that [1]

$$F = F^+ + F^-, \quad (23.128)$$

where F^+ and F^- represent the positive-frequency part and negative-frequency part, respectively (Sects. 22.3.6 and 23.6.1). Regarding a product of more than one operators F , G , etc., we similarly denote

$$FG\cdots = (F^+ + F^-)(G^+ + G^-)\cdots. \quad (23.129)$$

In our present case of the Hamiltonian density of electromagnetic interaction, we have

$$\mathfrak{H}_{\text{int}}(x) \equiv e[\bar{\psi}^+(x) + \bar{\psi}^-(x)]\left\{\gamma^\mu [A_\mu^+(x) + A_\mu^-(x)]\right\}[\psi^+(x) + \psi^-(x)]. \quad (23.130)$$

The operator $A_\mu(x)$ defined by (22.336) is an Hermitian operator with

$$A_\mu(x) = A_\mu^+(x) + A_\mu^-(x).$$

We have

$$\left[A_\mu^+(x)\right]^\dagger = A_\mu^-(x) \quad \text{and} \quad \left[A_\mu^-(x)\right]^\dagger = A_\mu^+(x). \quad (23.131)$$

Note, however, that neither $\psi(x)$ nor $\bar{\psi}(x)$ is Hermitian. In conjunction to this fact, the first factor of (23.130) contains a creation operator $\bar{\psi}^-$ of electron and an annihilation operator $\bar{\psi}^+$ of positron. The third factor of (23.130), on the other hand, includes a creation operator ψ^- of positron and an annihilation operator ψ^+ of electron. The relevant discussion can be seen in Sect. 23.8.2. According to the prescription of Sect. 23.5, we wish to calculate matrix elements of (23.86) up to the second order of the S -matrix.

For simplicity, we assume that the final state obtained after the interaction between the fields is monochromatic. In other words, the final state of individual particles is assumed to be described by a single plane wave. For instance, if we are observing photons and electrons as the final states after the interaction, those photons and electrons possess definite four-momentum and spin (or polarization). We use f instead of f_α in (23.86) accordingly.

23.7.1 Zeroth- and First-Order S -Matrix Elements

We start with the titled simple cases to show how the S -matrix elements are related to the elementary processes that are associated with the interaction between fields.

(i) Zeroth-order matrix element $\langle f | S^{(0)} | i \rangle$:

We have $S^{(0)} = E$ (identity operator) in this trivial case. We assume

$$|i\rangle = b^\dagger(\mathbf{p}, h)|0\rangle \quad \text{and} \quad |f\rangle = b^\dagger(\mathbf{p}', h')|0\rangle. \quad (23.132)$$

The RHSs of (23.132) imply that an electron with momenta $\mathbf{p}$ (or $\mathbf{p}'$) and helicity h (or h') has been created in vacuum. That is, the elementary process is described by

$$e^- \rightarrow e^-,$$

where e^- denotes an electron. With this process we have

$$\begin{aligned} \langle f | S^{(0)} | i \rangle &= \langle f | E | i \rangle = \langle f | i \rangle = \langle 0 | b(\mathbf{p}', h') b^\dagger(\mathbf{p}, h) | 0 \rangle \\ &= \langle 0 | b(\mathbf{p}', h') b^\dagger(\mathbf{p}, h) | 0 \rangle = \langle 0 | -b^\dagger(\mathbf{p}, h) b(\mathbf{p}', h') + \delta_{h'h} \delta^3(\mathbf{p}' - \mathbf{p}) | 0 \rangle \\ &= \langle 0 | -b^\dagger(\mathbf{p}, h) b(\mathbf{p}', h') | 0 \rangle + \delta_{h'h} \delta^3(\mathbf{p}' - \mathbf{p}) \langle 0 | 0 \rangle = \delta_{h'h} \delta^3(\mathbf{p}' - \mathbf{p}), \end{aligned} \quad (23.133)$$

where with the last equality, we used (22.262). Equation (23.133) implies that $\langle f | S^{(0)} | i \rangle$ vanishes unless we have $h' = h$ or $\mathbf{p}' = \mathbf{p}$. That is, electrons are only moving without changing its helicity or momentum. This means that no interaction occurred.

(ii) First-order matrix element $\langle f | S^{(1)} | i \rangle$ [1]:

We assume that

$$|i\rangle = b^\dagger(\mathbf{p}, h) | 0 \rangle \quad \text{and} \quad |f\rangle = a^\dagger(\mathbf{k}', r') b^\dagger(\mathbf{p}', h') | 0 \rangle. \quad (23.134)$$

The corresponding elementary process would be

$$e^- \rightarrow e^- + \gamma, \quad (23.135)$$

where e^- and γ denote an electron and a photon, respectively. An electron of the initial state $|i\rangle$ in vacuum is converted into an electron and a photon in the final state $|f\rangle$ of (23.135). We pick up the effective normal product from the interaction Hamiltonian (23.17) [1]. Then, we have

$$N[e\bar{\psi}(x)\gamma^\mu A_\mu(x)\psi(x)] = e\bar{\psi}^-(x)\gamma^\mu A_\mu^-(x)\psi^+(x). \quad (23.136)$$

As the matrix element $\langle f | S^{(1)} | i \rangle$, we have

$$\begin{aligned} \langle f | S^{(1)} | i \rangle &= ie \langle 0 | b(\mathbf{p}', h') a(\mathbf{k}', r') \\ &\quad \left[\int d^4x \bar{\psi}^-(x) \gamma^\mu A_\mu^-(x) \psi^+(x) \right] b^\dagger(\mathbf{p}, h) | 0 \rangle. \end{aligned} \quad (23.137)$$

In (23.137), we expand $\bar{\psi}^-(x)$, $A_\mu^-(x)$, and $\psi^+(x)$ using (23.88), (22.334), and (22.232), respectively. That is, we have

$$\bar{\psi}^-(x) = \int_{-\infty}^{\infty} \frac{d^3 \mathbf{w}'}{\sqrt{(2\pi)^3 2w'_0}} \sum_{t'=\pm 1} b^\dagger(\mathbf{w}', t') \bar{u}(\mathbf{w}', t') e^{i\mathbf{w}'x}, \quad (23.138)$$

$$A_\mu^-(x) = \int_{-\infty}^{\infty} \frac{d^3 \mathbf{l}'}{\sqrt{(2\pi)^3 2l'_0}} \sum_{m'=1}^2 a^\dagger(\mathbf{l}', m') \varepsilon_\mu^*(\mathbf{l}', m') e^{i\mathbf{l}'x}, \quad (23.139)$$

$$\psi^+(x) = \int_{-\infty}^{\infty} \frac{d^3 \mathbf{w}}{\sqrt{(2\pi)^3 2w_0}} \sum_{t=\pm 1} b(\mathbf{w}, t) u(\mathbf{w}, t) e^{-i\mathbf{w}x}. \quad (23.140)$$

In (23.139), we took only $m' = 1, 2$ (i.e., transverse mode) with the photon polarization [1]. Using (22.239), we have

$$b(\mathbf{p}', h') b^\dagger(\mathbf{w}', t') = -b^\dagger(\mathbf{w}', t') b(\mathbf{p}', h') + \delta_{h't'} \delta^3(\mathbf{p}' - \mathbf{w}'), \quad (23.141)$$

$$b(\mathbf{w}, t) b^\dagger(\mathbf{p}, h) = -b^\dagger(\mathbf{p}, h) b(\mathbf{w}, t) + \delta_{th} \delta^3(\mathbf{w} - \mathbf{p}). \quad (23.142)$$

Also, using (22.344), we have

$$a(\mathbf{k}', r') a^\dagger(\mathbf{l}', m') = a^\dagger(\mathbf{l}', m') a(\mathbf{k}', r') + \zeta_{r'm'} \delta_{r'm'} \delta^3(\mathbf{k}' - \mathbf{l}'). \quad (23.143)$$

Further using (23.141)–(23.143) and performing the momentum space integration with respect to $d\mathbf{w}'^3$, $d\mathbf{w}^3$, and $d\mathbf{l}'^3$, we get

$$\begin{aligned} & \langle f | S^{(1)} | i \rangle \\ &= ie \frac{1}{\sqrt{(2\pi)^9 \cdot 2p_0 \cdot 2p'_0 \cdot 2k'_0}} \int d^4 x \bar{u}(\mathbf{p}', h') \varepsilon_\mu^*(\mathbf{k}', r') \gamma^\mu \zeta_{r'm'} u(\mathbf{p}, h) e^{i(p'+k'-p)x} \\ &= ie \frac{1}{\sqrt{(2\pi)^9 \cdot 2p_0 \cdot 2p'_0 \cdot 2k'_0}} \bar{u}(\mathbf{p}', h') \varepsilon_\mu^*(\mathbf{k}', r') \gamma^\mu \zeta_{r'm'} u(\mathbf{p}, h) (2\pi)^4 \delta^4(p' + k' - p), \end{aligned} \quad (23.144)$$

where with the last equality, we used the four-dimensional version of (22.101). The four-dimensional δ function means that

$$\delta^4(p' + k' - p) = \delta(p'_0 + k'_0 - p_0) \delta^3(\mathbf{p}' + \mathbf{k}' - \mathbf{p}), \quad (23.145)$$

where $p' = \begin{pmatrix} p'_0 \\ \mathbf{p}' \end{pmatrix}$, $p = \begin{pmatrix} p_0 \\ \mathbf{p} \end{pmatrix}$, etc. From (23.145), $\varepsilon_\mu^*(\mathbf{k}', r')$ in (23.144) can be replaced with $\varepsilon_\mu^*(\mathbf{p} - \mathbf{p}', r')$. Equation (23.145) implies the conservation of four-momentum (or energy-momentum). That is, we have

$$p = p' + k'. \quad (23.146)$$

With the electron, we have

$$p'^2 = p'^0{}^2 - \mathbf{p}'^2 = m^2 \quad \text{and} \quad p^2 = p_0{}^2 - \mathbf{p}^2 = m^2. \quad (23.147)$$

For the photon, we get

$$k'^2 = k_0'^2 - \mathbf{k}'^2 = \omega'^2 - \omega'^2 = 0, \quad (23.148)$$

where ω' is an angular frequency of an emitted photon. But, from (23.146), we have

$$p^2 = p'^2 + 2p'k' + k'^2.$$

This implies $p'k' = 0$. It is, however, incompatible with a real physical process [1]. This can be shown as follows: We have

$$p'k' = p'^0\omega' - \mathbf{p}'\mathbf{k}'.$$

But, we have

$$\mathbf{p}'\mathbf{k}' \leq |\mathbf{p}'| \cdot |\mathbf{k}'| = \sqrt{p'^0{}^2 - m^2} \cdot |\mathbf{k}'| < p'^0\omega'.$$

Hence, $p'k' > p'^0\omega' - p'^0\omega' = 0$. That is, $p'k' > 0$. This is, however, inconsistent with $p'k' = 0$, and so an equality of $p'k' = 0$ is physically unrealistic. The simplest case would be that a single electron is present at rest in the initial state and it emits a photon with the electron recoiling in the final state. Put

$$p = \begin{pmatrix} m \\ \mathbf{0} \end{pmatrix}, p' = \begin{pmatrix} p'^0 \\ \mathbf{p}' \end{pmatrix}, k' = \begin{pmatrix} \omega' \\ \mathbf{k}' \end{pmatrix}.$$

Then, seek a value of $p'k'$ on the basis of the energy–momentum conservation. It is left for readers.

As evidenced from the above illustration, we must have

$$\langle f | S^{(1)} | i \rangle = 0. \quad (23.149)$$

23.7.2 Second-Order S -Matrix Elements and Feynman Diagrams

As already examined above, we had to write down a pretty large number of equations to examine even the simplest cases. It is the Feynman diagram that facilitates understanding the physical meaning of individual processes. A building block comprises three lines jointed at a vertex (see Fig. 23.2); i.e., two electron- or positron-associated lines and a photon-associated line. These lines correspond to the interaction Hamiltonian (23.17) related to the fields. Figure 23.2 is topologically equivalent to the first-order S -matrix $S^{(1)}$ discussed in the last section.

The diagram shows an incoming (or initial) electron (represented by a straight line labelled with p) and an outgoing (final) electron (labelled with p'). A photon line (shown as a wavy line labelled with k or k') intersects with the electron lines at their vertex. The Feynman diagrams comprise such building blocks and the number of the blocks contained in a diagram is determined from the order of the S -matrix. In general, a Feynman diagram consisting of n blocks is related to a n -th order S -matrix $S^{(n)}$. We show several kinds of Feynman diagrams related to $S^{(2)}$ below. Other diagrams representing $S^{(2)}$ processes can readily be constructed using two building blocks, one of which is depicted in Fig. 23.2. Associated Feynman rules will be summarized at the end of this section.

We show examples of the second-order S -matrix $S^{(2)}$. Figure 23.3 summarizes several topologically different diagrams. Individual diagrams correspond to different scattering processes. One of important features that have not appeared in the previous section is that the matrix elements $\langle f | S^{(2)} | i \rangle$ contain the contraction(s). In

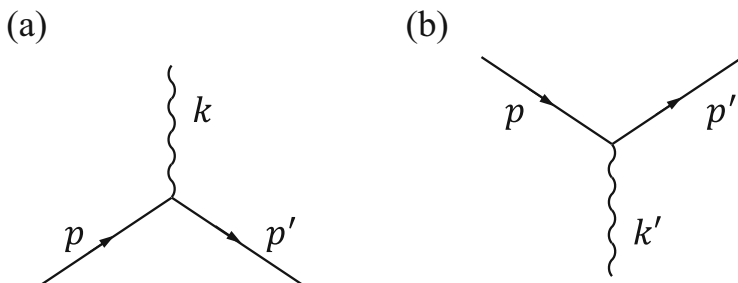


Fig. 23.2 Building blocks of the Feynman diagram. The Feynman diagram comprises two electron- or positron-associated lines (labelled p, p') and a photon-associated line (labelled k, k') that are jointed at a vertex. **(a)** Incoming photon is labelled k . **(b)** Outgoing photon is labelled k'

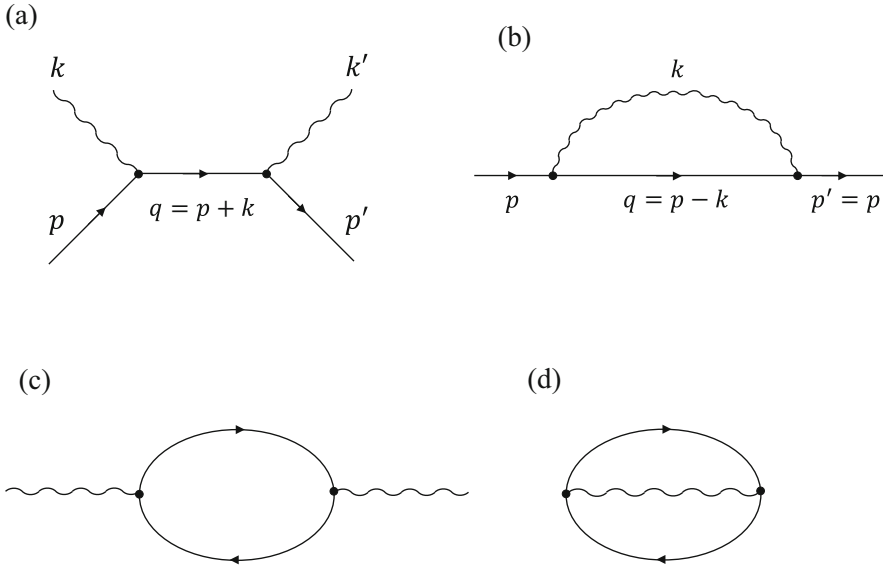


Fig. 23.3 Feynman diagram representing the second-order $S^{(2)}$. The diagram is composed of two building blocks. (a) Feynman diagram containing a single contraction related to electron field. (b) Feynman diagram showing the electron self-energy. (c) Feynman diagram showing the photon self-energy (or vacuum polarization). (d) Feynman diagram showing the vacuum bubble (or vacuum diagram)

QED, each contraction corresponds to the Feynman propagator described by $S_F(x - y)$ or $D_F(x - y)$. On the basis of (23.85), we explicitly show $S^{(2)}$ such that

$$\begin{aligned}
 S^{(2)} &= -\frac{1}{2}e^2 \int_{-\infty}^{\infty} dx_1 \int_{-\infty}^{\infty} dx_2 T[\mathfrak{H}_{\text{int}}(x_1)\mathfrak{H}_{\text{int}}(x_2)] \\
 &= -\frac{1}{2}e^2 \int_{-\infty}^{\infty} dx_1 \int_{-\infty}^{\infty} dx_2 T \\
 &\quad \left\{ [\bar{\psi}(x_1)\gamma^\mu A_\mu(x_1)\psi(x_1)] [\bar{\psi}(x_2)\gamma^\mu A_\mu(x_2)\psi(x_2)] \right\}. \quad (23.150)
 \end{aligned}$$

From (23.150), up to three contractions may be present in $S^{(2)}$. As can be seen in Fig. 23.3, the patterns and number of the contractions determine the topological features of the Feynman diagrams. In other words, the contractions play an essential role in determining how the building blocks are joined in the diagram (see Fig. 23.3).

Before discussing the several features of an example (i.e., Compton scattering) at large in the next section, we wish to briefly mention the characteristics of the Feynman diagrams that have contraction(s). Figure 23.3a shows a diagram having a single contraction. Initial electron and photon are drawn on the left side and final electron and photon are depicted on the right side with the contraction with respect to the electron shown at the center.

Examples that have two contractions are shown in Fig. 23.3b, c, which represent the “electron self-energy” and “photon self-energy,” respectively. The former term causes us to reconsider an electron as a certain entity that is surrounded by “photon cloud.” The photon self-energy, on the other hand, is sometimes referred to as the vacuum polarization. It is because the photon field might modify the distribution of virtual electron–positron pairs in such a way that the photon field polarizes the vacuum like a dielectric [1]. The S -matrix computation of the physical processes represented by Fig. 23.3b, c leads to a divergent integral. We need a notion of *renormalization* to traverse these processes, but we will not go into detail about this issue.

We show another example of the Feynman diagrams in Fig. 23.3d called “vacuum diagram” (or vacuum bubble [7]). It lacks external lines, and so no transitions are thought to occur. Diagrams collected in Fig. 23.3c, d contain two contractions between the Fermions. In those cases, the associated contraction lines go in opposite directions. This implies that both types of Fermions (i.e., electron and positron) take part in the process.

Now, in accordance with the above topological feature of the Feynman diagrams, we list major $S^{(2)}$ processes [1] as follows:

$$S_A^{(2)} = -\frac{e^2}{2} \int dx_1 dx_2 N \left[(\bar{\psi} \gamma^\mu A_\mu \psi)_{x_1} (\bar{\psi} \gamma^\nu A_\nu \psi)_{x_2} \right], \quad (23.151)$$

$$S_B^{(2)} = -\frac{e^2}{2} \int dx_1 dx_2 \left\{ N \left[(\bar{\psi} \gamma^\mu A_\mu \psi)_{x_1} (\bar{\psi} \gamma^\nu A_\nu \psi)_{x_2} \right] + [(\bar{\psi} \gamma^\mu A_\mu \psi)_{x_1} (\bar{\psi} \gamma^\nu A_\nu \psi)_{x_2}] \right\}, \quad (23.152)$$

$$S_C^{(2)} = -\frac{e^2}{2} \int dx_1 dx_2 N \left[(\bar{\psi} \gamma^\mu A_\mu \psi)_{x_1} (\bar{\psi} \gamma^\nu A_\nu \psi)_{x_2} \right], \quad (23.153)$$

$$S_D^{(2)} = -\frac{e^2}{2} \int dx_1 dx_2 \left\{ N \left[(\bar{\psi} \gamma^\mu A_\mu \psi)_{x_1} (\bar{\psi} \gamma^\nu A_\nu \psi)_{x_2} \right] + [(\bar{\psi} \gamma^\mu A_\mu \psi)_{x_1} (\bar{\psi} \gamma^\nu A_\nu \psi)_{x_2}] \right\}, \quad (23.154)$$

$$S_E^{(2)} = -\frac{e^2}{2} \int dx_1 dx_2 N \left[(\bar{\psi} \gamma^\mu A_\mu \psi)_{x_1} (\bar{\psi} \gamma^\nu A_\nu \psi)_{x_2} \right], \quad (23.155)$$

$$S_F^{(2)} = -\frac{e^2}{2} \int dx_1 dx_2 N \left[(\bar{\psi} \gamma^\mu A_\mu \psi)_{x_1} (\bar{\psi} \gamma^\nu A_\nu \psi)_{x_2} \right]. \quad (23.156)$$

Of these equations, $S_B^{(2)}$ contains a single contraction between the Fermions. We describe related important aspects in full detail in the next section with emphasis

upon the Compton scattering. Another process $S_C^{(2)}$ containing a single contraction includes the Møller scattering and Bhabha scattering. Interested readers are referred to the literature [1, 3]. The $S_D^{(2)}$, $S_E^{(2)}$, and $S_F^{(2)}$ processes are associated with the electron self-energy, photon self-energy, and vacuum bubble, respectively. The first two processes contain two contractions, whereas the last process $S_F^{(2)}$ contains three contractions. Thus, we find that the second-order S -matrix elements $S_k^{(2)}$ ($k = A, \dots, F$) reflect major processes of QED.

Among the above equations, (23.151) lacking a contraction is of little importance as in the previous case, because it does not reflect real processes. Equation (23.152) contains two contractions between the Dirac fields ($\bar{\psi}$ and ψ) and related two terms are equivalent. As discussed in Sect. 23.6.2, especially as represented in (23.124), we have a following N-product expressed as

$$N \left[\underbrace{(\bar{\psi} \gamma^\mu A_\mu \psi)_{x_1}}_{\text{contraction}} (\bar{\psi} \gamma^\nu A_\nu \psi)_{x_2} \right] = N \left[(\bar{\psi} \gamma^\mu A_\mu \psi)_{x_2} \underbrace{(\bar{\psi} \gamma^\nu A_\nu \psi)_{x_1}}_{\text{contraction}} \right]. \quad (23.157)$$

Notice that (i) exchanging two factors $(\bar{\psi} \gamma^\mu A_\mu \psi)_{x_1}$ and $(\bar{\psi} \gamma^\mu A_\mu \psi)_{x_2}$ causes even permutations of the Fermion operators ($\bar{\psi}$ and ψ). But, the even permutations do not change the N-products as shown in (23.120). (ii) Since (23.157) needs to be integrated with respect to x_1 and x_2 , these variables can freely be exchanged. Hence, exchanging the integration variables x_1 and x_2 in RHS of (23.157), we obtain

$$N \left[(\bar{\psi} \gamma^\mu A_\mu \psi)_{x_2} \underbrace{(\bar{\psi} \gamma^\nu A_\nu \psi)_{x_1}}_{\text{contraction}} \right] = N \left[\underbrace{(\bar{\psi} \gamma^\mu A_\mu \psi)_{x_1}}_{\text{contraction}} (\bar{\psi} \gamma^\nu A_\nu \psi)_{x_2} \right].$$

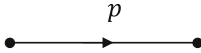
Thus, two terms of (23.152) are identical, and so (23.152) can be equivalently expressed as

$$S_B^{(2)} = -e^2 \int dx_1 dx_2 N \left[\underbrace{(\bar{\psi} \gamma^\mu A_\mu \psi)_{x_1}}_{\text{contraction}} (\bar{\psi} \gamma^\nu A_\nu \psi)_{x_2} \right]. \quad (23.158)$$

As already noted, the contractions represent the Feynman propagators. The contraction in (23.158) corresponds to $S_F(x - y)$ described by (23.126) and contains two uncontracted Fermion operators and two uncontracted photon operators. These uncontracted operators are referred to as external particles (associated with the external lines in the Feynman diagrams) and responsible for absorbing or creating the particles (i.e., electrons or photons). Consequently, (23.158) contributes to many real physical processes (see Sect. 23.8.2).

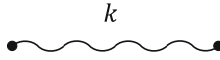
In parallel with the discussion made so far, in Fig. 23.4 we depict several major “components” of the diagram along with their meaning. A set of line drawings accompanied by their physical meaning is usually referred to as Feynman rules. The words “initial” and “final” used in the physical meaning should be taken as formal. In the above discussion, the initial states are associated with the momenta p and k , whereas the final states are pertinent to p' and k' . In this respect, the words initial and

(a)



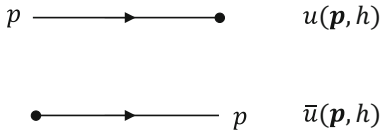
$$\widetilde{S}_F(p) = \frac{i(p_\mu \gamma^\mu + m)}{p^2 - m^2 + i\epsilon}$$

(b)

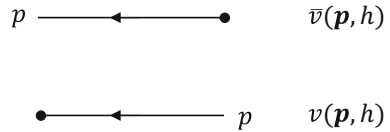


$$\widetilde{D}_F^{\mu\nu}(k) = \frac{-i\eta^{\mu\nu}}{k^2 + i\epsilon}$$

(c)



(d)



(e)

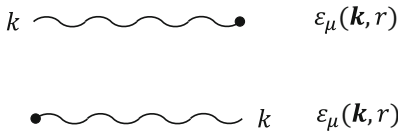


Fig. 23.4 Major components of the Feynman diagrams. (a) Internal (or intermediate) electron line and the corresponding Feynman propagator. (b) Internal (or intermediate) photon line and the corresponding Feynman propagator. (c) External lines of the initial electron [$u(\mathbf{p}, h)$] and final electron [$\bar{u}(\mathbf{p}, h)$] from the top. (d) External lines of the initial positron [$\bar{v}(\mathbf{p}, h)$] and final positron [$v(\mathbf{p}, h)$] from the top. (e) External lines of the initial photon [$\epsilon_\mu(\mathbf{k}, r)$] and final photon [$\epsilon_\mu(\mathbf{k}, r)$] from the top

final are of secondary importance. What is important is a measured (or measurable) quantity of p , k , p' , and k' .

Figure 23.4a, b shows the internal (or intermediate) electron and photon lines labelled by momentum p and k , respectively. Figure 23.4c indicates the external lines of the initial electron [$u(\mathbf{p}, h)$] and final electron [$\bar{u}(\mathbf{p}, h)$] from the top. In Fig. 23.4d, we depict similarly the external lines of initial positron [$\bar{v}(\mathbf{p}, h)$] and final positron [$v(\mathbf{p}, h)$] from the top. Note that the direction of the arrows of Fig. 23.4d is reversed

relative to Fig. 23.4c. Moreover, Fig. 23.4e shows likewise the external lines of photons. They are labelled by the polarization $\varepsilon_\mu(\mathbf{k}, r)$.

23.8 Example: Compton Scattering [1]

23.8.1 Introductory Remarks

To gain better understanding of the interaction between the quantum fields and to get used to dealing with the Feynman diagrams, we take this opportunity to focus upon a familiar example of Compton scattering. In Chap. 1, we dealt with it from a classical point of view (i.e., early-stage quantum theory). Here we revisit this problem to address the major issue from a point of view of quantum field theory. Even though the Compton scattering is only an example among various kinds of interactions of the quantum fields, it represents a general feature of the field interactions and, hence, to explore its characteristics in depth enhances heuristic knowledge.

The elementary process of the Compton scattering is described by

$$\gamma + e^- \rightarrow \gamma + e^-, \quad (23.159)$$

where γ denotes a photon and e^- stands for an electron. Using (23.130), a major part of the interaction Hamiltonian for $S_B^{(2)}$ of (23.158) is described as

$$\begin{aligned} e^2 N \Big\{ [\bar{\psi}^+(x_1) + \bar{\psi}^-(x_1)] \gamma^\mu [A_\mu^+(x_1) + A_\mu^-(x_1)] S_F(x_1 - x_2) \\ \times \gamma^\nu [A_\nu^+(x_2) + A_\nu^-(x_2)] [\psi^+(x_2) + \psi^-(x_2)] \Big\}. \end{aligned} \quad (23.160)$$

Equation (23.160) represents many elementary processes including the Compton scattering (see Sect. 23.8.2). Of the operators in (23.160), $\bar{\psi}^-(x_1)$ and $\psi^+(x_2)$ are responsible for the creation $[b^\dagger(\mathbf{p}, h)]$ and annihilation $[b(\mathbf{p}, h)]$ of electron, respectively. The operators $\psi^-(x_2)$ and $\bar{\psi}^+(x_1)$ are associated with the creation $[d^\dagger(\mathbf{p}, h)]$ and annihilation $[d(\mathbf{p}, h)]$ of positron, respectively. The operators A_μ^- and A_μ^+ are pertinent to the creation and annihilation of photon, respectively. We need to pick appropriate operators up according to individual elementary processes and arrange them in the normal order. In the case of the Compton scattering, for example, the order of operators is either

$$\bar{\psi}^-(x_1) \gamma^\mu A_\mu^-(x_1) \gamma^\nu A_\nu^+(x_2) \psi^+(x_2)$$

or

$$\bar{\psi}^-(x_1)\gamma^\mu A_\nu^-(x_2)\gamma^\nu A_\mu^+(x_1)\psi^+(x_2).$$

Regarding the photon operators, either $A_\mu^+(x_1)$ or $A_\nu^+(x_2)$ is thought to annihilate the “initial” photon. Either $A_\mu^-(x_1)$ or $A_\nu^-(x_2)$ creates the “final” photon accordingly. Note the “initial” and “final” with the double quotation marks written above. Because of the presence of time-ordered products, the words of initial and final are somewhat pointless. Nonetheless, we used them according to the custom. The custom merely arises in the notation $\langle f|S^{(2)}|i\rangle$. That is, in this notation the “initial state” i stands to the right of the “final state” f .

Defining the $S^{(2)}$ relevant to the Compton scattering as $S_{\text{Comp}}^{(2)}$, we have [1]

$$S_{\text{Comp}}^{(2)} = S_a + S_b, \quad (23.161)$$

where S_a and S_b are given by

$$S_a = -e^2 \int d^4x_1 d^4x_2 \bar{\psi}^-(x_1)\gamma^\mu S_F(x_1 - x_2)\gamma^\nu A_\mu^-(x_1)A_\nu^+(x_2)\psi^+(x_2) \quad (23.162)$$

and

$$S_b = -e^2 \int d^4x_1 d^4x_2 \bar{\psi}^-(x_1)\gamma^\mu S_F(x_1 - x_2)\gamma^\nu A_\nu^-(x_2)A_\mu^+(x_1)\psi^+(x_2). \quad (23.163)$$

In (23.162) and (23.163), once again we note that the field operators have been placed in the normal order. In both (23.162) and (23.163), the positive-frequency part ψ^+ and negative-frequency part $\bar{\psi}^-$ of the electron field are assigned to the coordinate x_2 and x_1 , respectively. The normal order of the photon part is given by

$$A_\mu^-(x_1)A_\nu^+(x_2), A_\mu^-(x_1)A_\nu^-(x_2), A_\mu^+(x_1)A_\nu^+(x_2), \text{ and } A_\nu^-(x_2)A_\mu^+(x_1).$$

Of these, the first and last terms are responsible for S_a and S_b , respectively. Other two terms produce no contribution to $S_{\text{Comp}}^{(2)}$.

The term $A_\mu^-(x_1)A_\nu^+(x_2)$ of S_a in (23.162) plays a role in annihilating the initial photon at x_2 and in creating the final photon at x_1 . Meanwhile, the term $A_\nu^-(x_2)A_\mu^+(x_1)$ of S_b in (23.163) plays a role in annihilating the initial photon at x_1 and in creating the final photon at x_2 . Thus, comparing (23.162) and (23.163), we find that the coordinates x_1 and x_2 switch their role in the annihilation and creation of photon. Therefore, it is of secondary importance which “event” has happened prior to another between the event labelled by $A_\nu^+(x_2)$ and that labelled by $A_\mu^+(x_1)$. This is the case with $A_\mu^-(x_1)$ and $A_\nu^-(x_2)$ as well. Such a situation frequently occurs in other kinds of field interactions and their calculations more generally. Normally, we

distinguish the state of the quantum field by momentum and helicity with electron and momentum and polarization with photon.

The related Feynman diagrams and calculations pertinent to the Compton scattering are given and discussed accordingly (vide infra).

23.8.2 *Feynman Amplitude and Feynman Diagrams of Compton Scattering*

The S -matrix $S_B^{(2)}$ of (23.152) represents interaction processes other than the Compton scattering shown in (23.159) as well. These are described as follows:

$$\gamma + e^+ \rightarrow \gamma + e^+, \quad (23.164)$$

$$e^- + e^+ \rightarrow \gamma + \gamma, \quad (23.165)$$

$$\gamma + \gamma \rightarrow e^- + e^+, \quad (23.166)$$

where e^+ stands for a positron. In the $S_B^{(2)}$ types of processes, we are given four particles, i.e.,

$$e^-, e^+, \gamma, \gamma.$$

Using the above four particles, we have four different ways of choosing two of them. That is, we have

$$\gamma, e^-, \gamma, e^+, e^-, e^+, \gamma, \gamma.$$

The above four combinations are the same as the remainder combinations left after the choice. The elementary processes represented by (23.164), (23.165), and (23.166) are called the Compton scattering by positrons, electron–positron pair annihilation, and electron–positron pair creation, respectively. To distinguish those processes unambiguously, the interaction Hamiltonian (23.160) including both the positive frequency-part and negative-frequency part must be written down in full. In the case of the electron–positron pair creation of (23.166), for instance, the second-order S -matrix element is described as [1]

$$S^{(2)} = -e^2 \int d^4x_1 d^4x_2 \bar{\psi}^-(x_1) \gamma^\mu S_F(x_1 - x_2) \gamma^\nu \psi^-(x_2) A_\mu^+(x_1) A_\nu^+(x_2). \quad (23.167)$$

In (23.167), field operators are expressed in the normal order and both the annihilation operators (corresponding to the positive frequency) and creation operators (corresponding to the negative frequency) are easily distinguished.

We have already shown the calculation procedure of $\langle f|S^{(1)}|i\rangle$. Even though it did not describe a real physical process, it helped understand how the calculations are performed on the interacting fields. The calculation processes of $\langle f|S^{(2)}|i\rangle$ are basically the same as those described above. In the present case, instead of (23.134) we assume

$$|i\rangle = a^\dagger(\mathbf{k}, r)b^\dagger(\mathbf{p}, h)|0\rangle \quad \text{and} \quad |f\rangle = a^\dagger(\mathbf{k}', r')b^\dagger(\mathbf{p}', h')|0\rangle.$$

Then, using e.g., (23.161), we have

$$\langle f|S_{\text{Comp}}^{(2)}|i\rangle = \langle f|S_a|i\rangle + \langle f|S_b|i\rangle$$

with

$$\begin{aligned} \langle f|S_a|i\rangle &= \langle 0|b(\mathbf{p}', h')a(\mathbf{k}', r')S_a a^\dagger(\mathbf{k}, r)b^\dagger(\mathbf{p}, h)|0\rangle, \\ \langle f|S_b|i\rangle &= \langle 0|b(\mathbf{p}', h')a(\mathbf{k}', r')S_b a^\dagger(\mathbf{k}, r)b^\dagger(\mathbf{p}, h)|0\rangle. \end{aligned} \quad (23.168)$$

Substituting (23.162) or (23.163) for S_a or S_b in (23.168), respectively, we carry out the calculations as in Sect. 23.7.1. The calculation procedures are summarized as follows: (i) Expand the field operators into the positive- or negative-frequency parts in the three-dimensional momentum space as in Sects. 22.4 and 22.5. (ii) Use the (anti)commutation relation regarding the creation or annihilation operators of photon and electron as in (23.141)–(23.143). (iii) Perform the integration subsequently with respect to the momentum space and further perform the integration with respect to x_1 and x_2 of $S_F(x_1 - x_2)$ that produces δ functions of four-momenta. Then, (iv) further integration of the resulting $S_F(x_1 - x_2)$ in terms of the four-momenta converts it into the Fourier counterpart of $\tilde{S}_F(p)$ denoted by (22.276). Thus, the complicated integration can be reduced to simple algebraic computation.

The above summary would rather hint to us an overwhelming number of computations. As the proverb goes, however, “an attempt is sometimes easier than expected.” We show the major parts of the calculations for $\langle f|S_a|i\rangle$ as below. Similar techniques can be applied to the computation of $\langle f|S_b|i\rangle$.

The S -matrix element is described as

$$\begin{aligned} \langle f|S_a|i\rangle &= (-ie)^2 \frac{\zeta_r \zeta_{r'}}{(2\pi)^6 \sqrt{(2p_0)(2k'_0)(2p_0)(2k_0)}} \\ &\times \left\langle 0 \left| \int d^4x_1 d^4x_2 [\bar{u}(\mathbf{p}', h')\varepsilon_\nu(\mathbf{k}', r')^* \gamma^\nu S_F(x_1 - x_2)\varepsilon_\mu(\mathbf{k}, r)\gamma^\mu u(\mathbf{p}, h)] \right| 0 \right\rangle, \end{aligned} \quad (23.169)$$

where ζ_r and $\zeta_{r'}$ come from (22.344). In (23.169) $S_F(x_1 - x_2)$ is expressed as

$$S_F(x_1 - x_2) = \int \frac{-id^4q}{(2\pi)^4} \frac{e^{-ix_1(q-p'-k')} e^{ix_2(q-p-k)} (q_\mu \gamma^\mu + m)}{m^2 - q^2 - i\epsilon}.$$

Performing the integration of $S_F(x_1 - x_2)$ in (23.169) with respect to x_1 and x_2 , we obtain an expression including δ functions of four-momenta such that

$$\begin{aligned} \langle f | S_a | i \rangle &= (-ie)^2 \frac{\zeta_r \zeta_{r'} \langle 0 | 0 \rangle}{(2\pi)^6 \sqrt{(2p'_0)(2k'_0)(2p_0)(2k_0)}} \times \\ (2\pi)^8 &\left[\bar{u}(\mathbf{p}', h') \varepsilon_\nu(\mathbf{k}', r')^* \gamma^\nu \int \frac{-id^4q}{(2\pi)^4} \frac{\delta^4(q-p'-k') \delta^4(q-p-k) (q_\mu \gamma^\mu + m)}{m^2 - q^2 - i\epsilon} \varepsilon_\mu(\mathbf{k}, r) \gamma^\mu u(\mathbf{p}, h) \right] \\ &= (-ie)^2 (2\pi)^4 \frac{\zeta_r \zeta_{r'} \delta^4(p+k-p'-k')}{(2\pi)^2 \sqrt{(2p'_0)(2k'_0)(2p_0)(2k_0)}} \\ &\times \left[\bar{u}(\mathbf{p}', h') \varepsilon_\nu(\mathbf{k}', r')^* \gamma^\nu \frac{(-i) [(p_\mu + k_\mu) \gamma^\mu + m]}{m^2 - (p+k)^2} \varepsilon_\mu(\mathbf{k}, r) \gamma^\mu u(\mathbf{p}, h) \right] \\ &= \frac{(2\pi)^4 \delta^4(p+k-p'-k')}{(2\pi)^6 \sqrt{(2p'_0)(2k'_0)(2p_0)(2k_0)}} \\ &\times [-e^2 \zeta_r \zeta_{r'} \bar{u}(\mathbf{p}', h') \varepsilon_\nu(\mathbf{k}', r')^* \gamma^\nu \tilde{S}_F(p+k) \varepsilon_\mu(\mathbf{k}, r) \gamma^\mu u(\mathbf{p}, h)]. \end{aligned} \quad (23.170)$$

In (23.170), we used

$$\delta^4(q-p'-k') \delta^4(q-p-k) = \delta^4(p+k-p'-k') \delta^4(q-p-k). \quad (23.171)$$

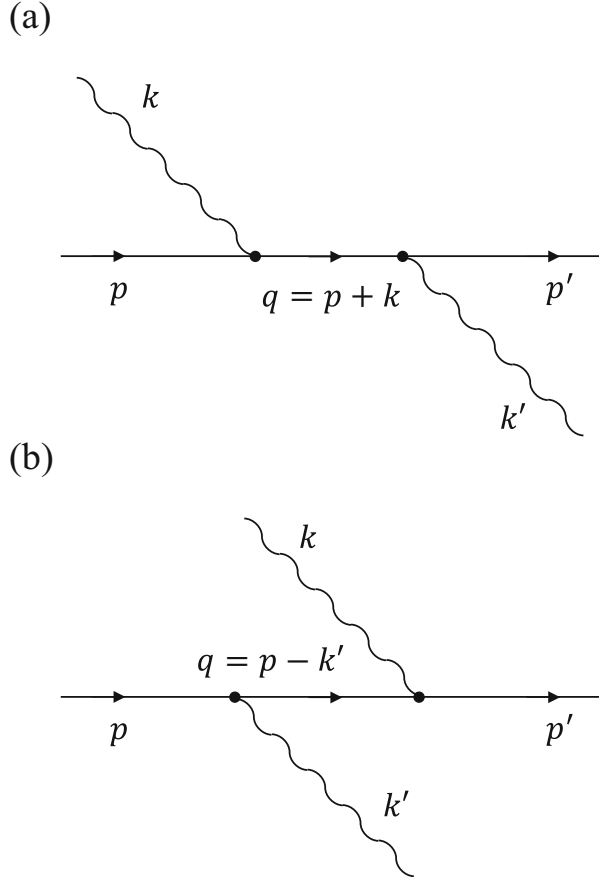
Since the first factor of (23.171) does not contain the variable q , it can be taken out from the integrand. The integration of the integrand containing the second factor produces $\tilde{S}_F(p+k)$ in (23.170). In (23.170), $\tilde{S}_F(p)$ was defined in Sect. 22.4.4 as

$$\tilde{S}_F(p) = \frac{-i(p_\mu \gamma^\mu + m)}{m^2 - p^2 - i\epsilon}. \quad (22.276)$$

Namely, the function $\tilde{S}_F(p)$ is a momentum component (or Fourier component) of $S_F(x)$ defined in (22.275). If in (23.170) we think of only the transverse modes for the photon field, the indices r and r' of ζ_r and $\zeta_{r'}$ run over 1 and 2 [1]. In that case, we have $\zeta_1 = \zeta_2 = 1$, and so we may neglect the factor $\zeta_r \zeta_{r'} (=1)$ accordingly.

On this condition, we define a complex number $\mathcal{M}_a$ as

Fig. 23.5 Feynman diagrams representing the second-order S -matrix of the Compton scattering. (a) Diagram of S_a . (b) Diagram of S_b



$$\mathcal{M}_a \equiv -e^2 \bar{u}(\mathbf{p}', h') \varepsilon_\nu(\mathbf{k}', r')^* \gamma^\nu \tilde{S}_F(p+k) \varepsilon_\mu(\mathbf{k}, r) \gamma^\mu u(\mathbf{p}, h). \quad (23.172)$$

The complex number $\mathcal{M}_a$ is called the Feynman amplitude (or invariant scattering amplitude) associated with the Feynman diagram of S_a (see Fig. 23.5a).

To compute $\tilde{S}_F(p+k)$, we have

$$\begin{aligned} \tilde{S}_F(p+k) &= \frac{(-i)(p_\mu \gamma^\mu + m)}{(p+k)^2 - m^2 + i\epsilon} = \frac{(-i)(p_\mu \gamma^\mu + m)}{p^2 + 2pk + k^2 - m^2 + i\epsilon} \\ &= \frac{(-i)(p_\mu \gamma^\mu + m)}{2pk + i\epsilon} = \frac{(-i)(p_\mu \gamma^\mu + m)}{2pk}, \end{aligned} \quad (23.173)$$

where in the denominator we used $p^2 - m^2 = k^2 = 0$. In (23.173), we dealt with the “square” $(p+k)^2$ as if both p and k were ordinary numbers. Here, it means

$$\begin{aligned}
 (p + k)^2 &\equiv (p_\mu + k_\mu)(p^\mu + k^\mu) = p_\mu p^\mu + p_\mu k^\mu + k_\mu p^\mu + k_\mu k^\mu \\
 &= p_\mu p^\mu + 2p_\mu k^\mu + k_\mu k^\mu = p^2 + 2pk + k^2,
 \end{aligned}$$

where we define $pk \equiv p_\mu k^\mu$. Since it is a convenient notation, it will be used henceforth. Since we had $pk \neq 0$ in the present case, the denominator was not an infinitesimal quantity and, hence, we deleted $i\epsilon$ as redundant. Thus, we get

$$\langle f|S_a|i\rangle = (2\pi)^4 \delta^4(p' + k' - p - k) \frac{1}{(2\pi)^6 \sqrt{(2p'_0)(2k'_0)(2p_0)(2k_0)}} \mathcal{M}_a. \quad (23.174)$$

In a similar manner, with $\langle f|S_b|i\rangle$, we have

$$\begin{aligned}
 \langle f|S_b|i\rangle &= (2\pi)^4 \delta^4(p' + k' - p - k) \frac{1}{(2\pi)^6 \sqrt{(2p'_0)(2k'_0)(2p_0)(2k_0)}} \\
 &\times \left[-e^2 \zeta_r \zeta_{r'} \bar{u}(\mathbf{p}', h') \epsilon_\nu(\mathbf{k}, r) \gamma^\nu \tilde{S}_F(p - k') \epsilon_\mu(\mathbf{k}', r')^* \gamma^\mu u(\mathbf{p}, h) \right]. \quad (23.175)
 \end{aligned}$$

Also, defining a complex number $\mathcal{M}_b$ as

$$\mathcal{M}_b \equiv -e^2 \bar{u}(\mathbf{p}', h') \epsilon_\nu(\mathbf{k}, r) \gamma^\nu \tilde{S}_F(p - k') \epsilon_\mu(\mathbf{k}', r')^* \gamma^\mu u(\mathbf{p}, h), \quad (23.176)$$

we obtain

$$\langle f|S_b|i\rangle = (2\pi)^4 \delta^4(p' + k' - p - k) \frac{1}{(2\pi)^6 \sqrt{(2p'_0)(2k'_0)(2p_0)(2k_0)}} \mathcal{M}_b. \quad (23.177)$$

In (23.174) and (23.177), the presence of $\delta^4(p' + k' - p - k)$ reflects the four-momentum conservation before and after the Compton scattering process. Figure 23.5 represents the Feynman diagrams that correspond to S_a (Fig. 23.5a) and S_b (Fig. 23.5b) related to the Compton scattering. Both the diagrams consist of two building blocks depicted in Fig. 23.2. Even though Fig. 23.5a, b shares a topological feature the same as that of Fig. 23.3a, the connection of the two building blocks has been reversed with S_a and S_b . From the discussions of the present and previous sections, the implications of these Feynman diagrams (Fig. 23.5a, b) are evident.

Equations (23.174) and (23.177) represent a general feature of the field interactions. The factor $(2\pi)^4$ present in both the numerator and denominator could be reduced, but we leave it without change. It is because (23.174) and (23.177) can readily be generalized in the case where initial two particles and N final particles take part in the overall reaction.

In the general two-particle collision, we assume that the reaction is represented by

$$P1 + P2 \rightarrow P'1 + P'2 + \cdots + P'N, \quad (23.178)$$

where $P1$ and $P2$ denote the initial particles and $P'1, P'2, \cdots P'N$ represent the final particles. Then, in accordance with (23.174) and (23.177), the S -matrix element is expressed as

$$\begin{aligned} & \langle f|S|i \rangle \\ &= (2\pi)^4 \delta^4 \left(\sum_{f=1}^N p'_f - \sum_{i=1}^2 p_i \right) \frac{1}{\sqrt{(2\pi)^3 (2E_1) (2\pi)^3 (2E_2)}} \\ & \times \sqrt{\prod_{f=1}^N \frac{1}{(2\pi)^3 (2E'_f)}} \mathcal{M}. \end{aligned} \quad (23.179)$$

In (23.179), $p_i = \begin{pmatrix} E_i \\ \mathbf{p}_i \end{pmatrix}$ ($i = 1, 2$) denotes four-momenta of the initially colliding two particles and $p'_f = \begin{pmatrix} E'_f \\ \mathbf{p}'_f \end{pmatrix}$ ($f = 1, \cdots, N$) represents those of the finally created N particles. Also, in (23.179) we used S and $\mathcal{M}$ instead of S_a , $\mathcal{M}_a$, etc., since a general case is intended. Notice that with the Compton scattering $N = 2$.

23.8.3 Scattering Cross-Section [1]

To connect the general formalism to experimental observations, we wish to relate the S -matrix elements to the transition probability that measures the strength of the interaction between the fields. The relevant discussion is first based on the general expression (23.179), and then, we focus our attention to the Compton scattering.

Let w be the transition probability per unit time and unit volume. This is given by

$$w = |\langle f|S|i \rangle|^2 / TV, \quad (23.180)$$

where T is a time during which the scattering experiment is being done; V is a volume related to the experiment. Usually, TV is taken as the whole space-time, because an elementary particle reaction takes place in an extremely narrow space-time domain.

Conservation of the energy and momentum with respect to (23.179) is read as

$$E_1 + E_2 = \sum_{k=1}^N E'_k, \quad (23.181)$$

$$\mathbf{p}_1 + \mathbf{p}_2 = \sum_{k=1}^N \mathbf{p}'_k. \quad (23.182)$$

From the above equations, we know that all $E'_1, \dots, E'_N$ are not independent valuables. All $\mathbf{p}'_1, \dots, \mathbf{p}'_N$ are not independent valuables, either.

The cross-section σ for the reaction or scattering is described by

$$\sigma = (2\pi)^6 w / v_{\text{rel}}, \quad (23.183)$$

where v_{rel} represents the relative velocity between two colliding particles P1 and P2. In (23.183), v_{rel} is defined by [1]

$$v_{\text{rel}} \equiv \sqrt{(p_1 p_2)^2 - (m_1 m_2)^2} / E_1 E_2, \quad (23.184)$$

where m_1 and m_2 are the rest masses of the colliding particles 1 and 2, respectively. When we have a *colinear* collision of the two particles, (23.184) is reduced to a simple expression. Suppose that at a certain inertial frame of reference Particle 1 and Particle 2 have a velocity $\mathbf{v}_1$ and $\mathbf{v}_2 = a\mathbf{v}_1$ (where a is a real number), respectively.

According to (1.9), we define γ_1 and γ_2 such that

$$\gamma_i \equiv 1 / \sqrt{1 - |\mathbf{v}_i|^2} \quad (i = 1, 2). \quad (23.185)$$

Then, from (1.21), we have

$$E_i = m_i \gamma_i \quad \text{and} \quad \mathbf{p}_i = m_i \gamma_i \mathbf{v}_i \quad (i = 1, 2). \quad (23.186)$$

Also, we obtain

$$\begin{aligned} (p_1 p_2)^2 - (m_1 m_2)^2 &= (m_1 m_2 \gamma_1 \gamma_2 - \mathbf{p}_1 \cdot \mathbf{p}_2)^2 - (m_1 m_2)^2 \\ &= (m_1 m_2 \gamma_1 \gamma_2)^2 \left(1 - 2a|\mathbf{v}_1|^2 + a^2|\mathbf{v}_1|^4 - \frac{1}{\gamma_1^2 \gamma_2^2} \right) \\ &= (m_1 m_2 \gamma_1 \gamma_2)^2 |\mathbf{v}_1|^2 (a - 1)^2. \end{aligned} \quad (23.187)$$

With the above derivation, we used (23.185) and (23.186) along with $\mathbf{v}_2 = a\mathbf{v}_1$ (colinear collision). Then, we get

$$v_{\text{rel}} = |\mathbf{v}_1(a - 1)| = |\mathbf{v}_2 - \mathbf{v}_1|. \quad (23.188)$$

Thus, in the colinear collision case, we have had a relative velocity identical to that in a non-relativistic limit. In a particular instance of $a = 0$ (to which Particle 2 is at rest), $v_{\text{rel}} = |\mathbf{v}_1|$. This is usually the case with a laboratory frame where Particle

1 is caused to collide against Particle 2 at rest. If Particle 1 is photon, in a laboratory frame we have a particularly simple expression of

$$v_{\text{rel}} = 1. \quad (23.189)$$

Substituting (23.180) and (23.184) for (23.183), we obtain

$$\begin{aligned} \sigma = & \frac{(2\pi)^6}{TV} \cdot \frac{E_1 E_2}{\sqrt{(p_1 p_2)^2 - (m_1 m_2)^2}} \\ & \cdot \frac{1}{(2\pi)^6 \cdot 4E_1 E_2} \left| (2\pi)^4 \delta^4 \left(\sum_{f=1}^N p'_f - \sum_{i=1}^2 p_i \right) \right|^2 \\ & \times \prod_f \frac{1}{(2\pi)^3 (2E'_f)} |\mathcal{M}|^2, \end{aligned} \quad (23.190)$$

where $(2\pi)^6$ of the numerator in the first factor is related to the normalization of incident particles 1 and 2 (in the initial state) [8]. In (23.190), we have

$$\left| (2\pi)^4 \delta^4 \left(\sum_{f=1}^N p'_f - \sum_{i=1}^2 p_i \right) \right|^2 = (2\pi)^4 \delta^4(0) \cdot (2\pi)^4 \delta^4 \left(\sum_{f=1}^N p'_f - \sum_{i=1}^2 p_i \right),$$

where we used a property of a δ function expressed as [9]

$$f(x)\delta(x) = f(0)\delta(x). \quad (23.191)$$

But, in analogy with (22.57), we obtain

$$\delta(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i0x} dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} dx \quad \text{or} \quad (2\pi)^4 \delta^4(0) = \int_{-\infty}^{\infty} d^4x \approx TV, \quad (23.192)$$

where $(2\pi)^4 \delta^4(0)$ represents the volume of the whole space-time. Thus, we get

$$\begin{aligned} \sigma = & \frac{(2\pi)^4}{4\sqrt{(p_1 p_2)^2 - (m_1 m_2)^2}} \delta^4 \left(\sum_{f=1}^N p'_f - \sum_{i=1}^2 p_i \right) \\ & \times \left[\prod_{f=1}^N \frac{1}{(2\pi)^3 (2E'_f)} \right] |\mathcal{M}|^2. \end{aligned} \quad (23.193)$$

The cross-section is usually described in terms of differential cross-section $d\sigma$ such that

$$d\sigma = \frac{(2\pi)^4}{4\sqrt{(p_1 p_2)^2 - (m_1 m_2)^2}} \delta^4\left(\sum_{f=1}^N p'_f - \sum_{i=1}^2 p_i\right) \times \left[\prod_{f=1}^N \frac{d\mathbf{p}'_f}{(2\pi)^3 (2E'_f)} \right] |\mathcal{M}|^2, \quad (23.194)$$

where $d\mathbf{p}'_f$ represents an infinitesimal range of momentum from $\mathbf{p}'_f$ to $\mathbf{p}'_f + d\mathbf{p}'_f$ for which the particle P^f ($f = 1, \dots, N$) is found. Then, σ can be expressed in a form of integration of (23.194) as

$$\sigma = \frac{(2\pi)^4}{4\sqrt{(p_1 p_2)^2 - (m_1 m_2)^2}} \delta^4\left(\sum_{f=1}^N p'_f - \sum_{i=1}^2 p_i\right) \times \left[\prod_{f=1}^N \int \frac{d\mathbf{p}'_f}{(2\pi)^3 (2E'_f)} \right] |\mathcal{M}|^2. \quad (23.195)$$

Note that (23.195) is Lorentz invariant. In fact, if in (22.150) we put $f(k_0) \equiv 1$ and $g(\mathbf{k}) \equiv 1$, we have

$$\int d^4 k \delta(k^2 - m^2) \theta(k_0) = \int d^3 \mathbf{k} \frac{1}{2\sqrt{\mathbf{k}^2 + m^2}}.$$

As an equivalent form of (23.195), we obtain

$$\sigma = \frac{(2\pi)^4}{4v_{\text{rel}} E_1 E_2} \delta^4\left(\sum_{f=1}^N p'_f - \sum_{i=1}^2 p_i\right) \left[\prod_{f=1}^N \int \frac{d\mathbf{p}'_f}{(2\pi)^3 (2E'_f)} \right] |\mathcal{M}|^2. \quad (23.196)$$

In particular, if $N = 2$ (to which the Compton scattering is relevant), we obtain

$$d\sigma = \frac{1}{64\pi^2 v_{\text{rel}} E_1 E_2 E'_1 E'_2} \delta^4\left(\sum_{f=1}^2 p'_f - \sum_{i=1}^2 p_i\right) |\mathcal{M}|^2 d^3 \mathbf{p}'_1 d^3 \mathbf{p}'_2. \quad (23.197)$$

Now, we wish to perform practical calculations of the Compton scattering after Mandl and Shaw [1]. Because of the momentum conservation described by

$$\mathbf{p}'_2 = \mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}'_1, \quad (23.198)$$

we regard only $\mathbf{p}'_1$ (of the scattered photon) as an independent variable out of $\mathbf{p}'_1$ and $\mathbf{p}'_2$. The choice is at will, but it is both practical and convenient to choose the momentum of the particle in question for the independent variable. To express the

differential cross-section $d\sigma$ as a function of the scattering angle of the particle, we convert the Cartesian coordinate representation into the polar coordinate representation with respect to the momentum space $\mathbf{p}'_1$ in (23.197). Then, we integrate (23.197) with respect to $\mathbf{p}'_2$ to get

$$d\sigma = \delta^4\left(\sum_{f=1}^2 E'_f - \sum_{i=1}^2 E_i\right) \frac{1}{64\pi^2 E_1 E_2 E'_1 E'_2 v_{\text{rel}}} |\mathcal{M}|^2 |\mathbf{p}'_1|^2 d|\mathbf{p}'_1| d\Omega', \quad (23.199)$$

where $d\Omega'$ denotes a solid angle element, to which the particle $P'1$ is scattered into. Notice that the $\mathbf{p}'_2$ integration converted $\delta^4\left(\sum_{f=1}^2 E'_f - \sum_{i=1}^2 E_i\right)$ into $\delta^4\left(\sum_{f=1}^2 E'_f - \sum_{i=1}^2 E_i\right)$.

Further integration of $d\sigma$ over $|\mathbf{p}'_1|$ produces [1]

$$\int d\sigma d|\mathbf{p}'_1| = d\tilde{\sigma} = \frac{1}{64\pi^2 E_1 E_2 E'_1 E'_2 v_{\text{rel}}} |\mathcal{M}|^2 |\mathbf{p}'_1|^2 \left[\frac{\partial(E'_1 + E'_2)}{\partial|\mathbf{p}'_1|} \right]^{-1} d\Omega', \quad (23.200)$$

where $d\tilde{\sigma}$ means the differential cross-section obtained after the integration of $d\sigma$ over $|\mathbf{p}'_1|$. To derive (23.200), we used the following formula of δ function [1]

$$\begin{aligned} \int f(x, y) \delta[g(x, y)] dx &= \int f(x, y) \delta[g(x, y)] \left(\frac{\partial x}{\partial g} \right)_y dg \\ &= f(x, y) \left[\left(\frac{\partial g}{\partial x} \right)_y \right]^{-1} \bigg|_{g=0}. \end{aligned} \quad (23.201)$$

We make use of (23.200) in Sect. 23.8.6.

23.8.4 Spin and Photon Polarization Sums

We often encounter a situation where a Feynman amplitude $\mathcal{M}$ takes a form [1]

$$\mathcal{M} = c \bar{u}(\mathbf{p}', h') \Gamma u(\mathbf{p}, h), \quad (23.202)$$

where c is a complex number and Γ is usually a $(4, 4)$ matrix (tensor) that contains gamma matrices. The amplitude $\mathcal{M}$ is a complex number as well and we are interested in calculating $|\mathcal{M}|^2$ (a real number).

Hence, we have

$$|\mathcal{M}|^2 = |c|^2 [\bar{u}(\mathbf{p}', h') \Gamma u(\mathbf{p}, h)] [\bar{u}(\mathbf{p}', h') \Gamma u(\mathbf{p}, h)]^*. \quad (23.203)$$

Viewing $\bar{u}(\mathbf{p}', h') \Gamma u(\mathbf{p}, h)$ as a (1, 1) matrix, we have

$$[\bar{u}(\mathbf{p}', h') \Gamma u(\mathbf{p}, h)]^* = [\bar{u}(\mathbf{p}', h') \Gamma u(\mathbf{p}, h)]^\dagger. \quad (23.204)$$

Then, we obtain

$$\begin{aligned} |\mathcal{M}|^2 &= |c|^2 [\bar{u}(\mathbf{p}', h') \Gamma u(\mathbf{p}, h)] [u^\dagger(\mathbf{p}, h) \Gamma^\dagger \bar{u}^\dagger(\mathbf{p}', h')] \\ &= |c|^2 [\bar{u}(\mathbf{p}', h') \Gamma u(\mathbf{p}, h)] [u^\dagger(\mathbf{p}, h) \gamma^0 \gamma^0 \Gamma^\dagger \gamma^0 \gamma^0 \bar{u}^\dagger(\mathbf{p}', h')] \\ &= |c|^2 [\bar{u}(\mathbf{p}', h') \Gamma u(\mathbf{p}, h)] \{ \bar{u}(\mathbf{p}, h) [\gamma^0 \Gamma^\dagger \gamma^0] \gamma^0 [\gamma^0 u(\mathbf{p}', h')] \} \\ &= |c|^2 [\bar{u}(\mathbf{p}', h') \Gamma u(\mathbf{p}, h)] \bar{u}(\mathbf{p}, h) [\gamma^0 \Gamma^\dagger \gamma^0] u(\mathbf{p}', h'), \end{aligned} \quad (23.205)$$

where with the second equality we used $\gamma^0 \gamma^0 = E$; with the third equality, we used $\bar{u}^\dagger = \gamma^0 u$. Defining $\tilde{\Gamma}$ as

$$\tilde{\Gamma} \equiv \gamma^0 \Gamma^\dagger \gamma^0, \quad (23.206)$$

we get

$$|\mathcal{M}|^2 = |c|^2 \bar{u}(\mathbf{p}', h') \Gamma u(\mathbf{p}, h) \bar{u}(\mathbf{p}, h) \tilde{\Gamma} u(\mathbf{p}', h'). \quad (23.207)$$

In (23.207), $u(\mathbf{p}, h)$, $u(\mathbf{p}', h')$, etc. are (4, 1) matrices; Γ and $\tilde{\Gamma}$ as well as $u(\mathbf{p}, h) \bar{u}(\mathbf{p}, h)$ are (4, 4) matrices. Then, $\Gamma u(\mathbf{p}, h) \bar{u}(\mathbf{p}, h) \tilde{\Gamma} u(\mathbf{p}', h')$ is a (4, 1) matrix. Now, we recall (18.70) and (18.71); namely,

$$\text{Tr}(PQ) = \text{Tr}(QP), \quad (18.70)$$

$$\sum_i (PQ)_{ii} = \sum_i \sum_j P_{ij} Q_{ji} = \sum_j \sum_i Q_{ji} P_{ij} = \sum_j (QP)_{jj}. \quad (18.71)$$

Having a look at (18.71), we find (18.70) can be extended to the case where neither P nor Q is a square matrix but PQ is a square matrix. Thus, (23.207) can be rewritten as

$$|\mathcal{M}|^2 = |c|^2 \cdot \text{Tr} [\Gamma u(\mathbf{p}, h) \bar{u}(\mathbf{p}, h) \tilde{\Gamma} u(\mathbf{p}', h') \bar{u}(\mathbf{p}', h')], \quad (23.208)$$

where $[\Gamma u(\mathbf{p}, h) \bar{u}(\mathbf{p}, h) \tilde{\Gamma} u(\mathbf{p}', h') \bar{u}(\mathbf{p}', h')]$ is a (4, 4) matrix. Note, however, that taking traces does not mean sum or average of the spin and photon polarization. It is because in (23.208) the helicity indices h or h' for electron remain without being

summed. The photon polarization indices [hidden in Γ and $\tilde{\Gamma}$ for (23.208)] have not yet been summed either.

The polarization sum and average are completed by taking summation over all the pertinent indices. This will be done in detail in the next section.

23.8.5 Detailed Calculation Procedures of Feynman Amplitude [1]

Now that we have gotten used to the generalities of the scattering theory, we are in the position to carry out the detailed calculations of the Feynman amplitude. In the case of the Compton scattering, the Feynman amplitude $\mathcal{M}_a$ has been given as

$$\mathcal{M}_a \equiv -e^2 \bar{u}(\mathbf{p}', h') \varepsilon_\nu(\mathbf{k}', r')^* \gamma^\nu \tilde{S}_F(p+k) \varepsilon_\mu(\mathbf{k}, r) \gamma^\mu u(\mathbf{p}, h), \quad (23.172)$$

$$\mathcal{M}_b \equiv -e^2 \bar{u}(\mathbf{p}', h') \varepsilon_\nu(\mathbf{k}, r) \gamma^\nu \tilde{S}_F(p-k') \varepsilon_\mu(\mathbf{k}', r')^* \gamma^\mu u(\mathbf{p}, h). \quad (23.176)$$

Comparing (23.172) and (23.176) with (23.202), we find that c in (23.202) is given by

$$c = -e^2 \zeta_r \zeta_{r'}. \quad (23.209)$$

Note that $\zeta_r \zeta_{r'} = 1$ ($r, r' = 1, 2$). The total Feynman amplitude $\mathfrak{M}$ is given by

$$\mathfrak{M} = \mathcal{M}_a + \mathcal{M}_b. \quad (23.210)$$

Then, we have

$$|\mathfrak{M}|^2 = |\mathcal{M}_a|^2 + |\mathcal{M}_b|^2 + \mathcal{M}_a \mathcal{M}_b^* + \mathcal{M}_a^* \mathcal{M}_b. \quad (23.211)$$

From here on, we assume that electrons and/or photons are unpolarized in both their initial and final states. In terms of the experimental techniques, it is more difficult to detect or determine the polarization of helicity of electrons than the polarization of photons. To estimate the polarization sum with the photon field, we need another technique in relation to its gauge transformation (vide infra).

- (i) In the case of the electrons, we first average over initial spins (or helicities) h and sum over final helicities h' . These procedures yield a positive real number $\tilde{X}$ that is described by [1]

$$\tilde{X} = \frac{1}{2} \sum_{h=1}^2 \sum_{h'=1}^2 |\mathfrak{M}|^2. \quad (23.212)$$

(ii) As for the photons, we similarly average over initial polarizations r and sum over final polarizations r' . These procedures yield another positive real number $\mathfrak{X}$ defined as [1]

$$\mathfrak{X} \equiv \frac{1}{2} \sum_{r=1}^2 \sum_{r'=1}^2 \tilde{X} = \frac{1}{4} \sum_{r=1}^2 \sum_{r'=1}^2 \sum_{h=1}^2 \sum_{h'=1}^2 |\mathfrak{M}|^2, \quad (23.213)$$

where $|\mathfrak{M}|^2$ was given in (23.211). We further define $\mathfrak{X}_1$ to $\mathfrak{X}_4$ as

$$\mathfrak{X}_1 \equiv \frac{1}{4} \sum_{r=1}^2 \sum_{r'=1}^2 \sum_{h=1}^2 \sum_{h'=1}^2 |\mathcal{M}_a|^2, \quad (23.214)$$

$$\mathfrak{X}_2 \equiv \frac{1}{4} \sum_{r=1}^2 \sum_{r'=1}^2 \sum_{h=1}^2 \sum_{h'=1}^2 |\mathcal{M}_b|^2, \quad (23.215)$$

$$\mathfrak{X}_3 \equiv \frac{1}{4} \sum_{r=1}^2 \sum_{r'=1}^2 \sum_{h=1}^2 \sum_{h'=1}^2 \mathcal{M}_a \mathcal{M}_b^*, \quad (23.216)$$

$$\mathfrak{X}_4 \equiv \frac{1}{4} \sum_{r=1}^2 \sum_{r'=1}^2 \sum_{h=1}^2 \sum_{h'=1}^2 \mathcal{M}_a^* \mathcal{M}_b. \quad (23.217)$$

As a number $\mathfrak{X}$ we want to compute, we have

$$\mathfrak{X} = \mathfrak{X}_1 + \mathfrak{X}_2 + \mathfrak{X}_3 + \mathfrak{X}_4. \quad (23.218)$$

We calculate four terms of (23.218) one by one. To compute $|\mathcal{M}_a|^2$ and $|\mathcal{M}_b|^2$, we follow (23.214) and (23.215), respectively. We think of $\mathcal{M}_a \mathcal{M}_b^*$ and $\mathcal{M}_a^* \mathcal{M}_b$ afterward. First, with $|\mathcal{M}_a|^2$ we have

$$\Gamma = \frac{\varepsilon_\nu(\mathbf{k}', r')^* \gamma^\nu [(p_\mu + k_\mu) \gamma^\mu + m] \varepsilon_\mu(\mathbf{k}, r) \gamma^\mu}{2pk}. \quad (23.219)$$

Then, from (23.206), we obtain

$$\begin{aligned} \tilde{\Gamma} &= \gamma^0 \frac{(\gamma^0 \gamma^\mu \gamma^0) \varepsilon_\mu(\mathbf{k}, r)^* \gamma^0 [(p_\mu + k_\mu) \gamma^\mu + m] \gamma^0 (\gamma^0 \gamma^\nu \gamma^0) \varepsilon_\nu(\mathbf{k}', r')}{2pk} \gamma^0 \\ &= \frac{\gamma^\mu \varepsilon_\mu(\mathbf{k}, r)^* [(p_\mu + k_\mu) \gamma^\mu + m] \gamma^\nu \varepsilon_\nu(\mathbf{k}', r')}{2pk}. \end{aligned} \quad (23.220)$$

Thereby, we get

$$\begin{aligned}
\mathfrak{X}_1 &= \frac{e^4}{4} \sum_{r=1}^2 \sum_{r'=1}^2 \sum_{h=1}^2 \sum_{h'=1}^2 \text{Tr} [\Gamma u(\mathbf{p}, h) \bar{u}(\mathbf{p}, h) \tilde{\Gamma} u(\mathbf{p}', h') \bar{u}(\mathbf{p}', h')] \\
&= \frac{e^4}{4} \sum_{r, r', h, h'} \text{Tr} \left[\frac{\varepsilon_\nu(\mathbf{k}', r')^* \gamma^\nu [(p_\mu + k_\mu) \gamma^\mu + m] \varepsilon_\lambda(\mathbf{k}, r) \gamma^\lambda}{2pk} u(\mathbf{p}, h) \bar{u}(\mathbf{p}, h) \right. \\
&\quad \times \left. \frac{\gamma^\rho \varepsilon_\rho(\mathbf{k}, r)^* [(p_\mu + k_\mu) \gamma^\mu + m] \varepsilon_\sigma(\mathbf{k}', r') \gamma^\sigma}{2pk} u(\mathbf{p}', h') \bar{u}(\mathbf{p}', h') \right] \quad [\dagger] \\
&= \frac{e^4}{4} \sum_{r, r'} \text{Tr} \left[\frac{\varepsilon_\nu(\mathbf{k}', r')^* \gamma^\nu [(p_\mu + k_\mu) \gamma^\mu + m] \varepsilon_\lambda(\mathbf{k}, r) \gamma^\lambda}{2pk} (p_\alpha \gamma^\alpha + m) \right. \\
&\quad \times \left. \frac{\gamma^\rho \varepsilon_\rho(\mathbf{k}, r)^* [(p_\mu + k_\mu) \gamma^\mu + m] \varepsilon_\sigma(\mathbf{k}', r') \gamma^\sigma}{2pk} (p'_\beta \gamma^\beta + m) \right] \\
&= \frac{e^4}{4} \sum_{r, r'} \text{Tr} \left[\frac{\varepsilon_\nu(\mathbf{k}', r')^* \varepsilon_\sigma(\mathbf{k}', r') \gamma^\nu [(p_\mu + k_\mu) \gamma^\mu + m] \varepsilon_\rho(\mathbf{k}, r)^* \varepsilon_\lambda(\mathbf{k}, r) \gamma^\lambda}{2pk} \right. \\
&\quad \times \left. \frac{(p_\alpha \gamma^\alpha + m) \gamma^\rho [(p_\mu + k_\mu) \gamma^\mu + m] \gamma^\sigma (p'_\beta \gamma^\beta + m)}{2pk} \right] \quad [\dagger\dagger] \\
&= \frac{e^4}{4} \text{Tr} \left[\frac{(-\eta_{\nu\sigma}) \gamma^\nu [(p_\mu + k_\mu) \gamma^\mu + m] (-\eta_{\rho\lambda}) \gamma^\lambda}{2pk} \right. \\
&\quad \times \left. \frac{(p_\alpha \gamma^\alpha + m) \gamma^\rho [(p_\mu + k_\mu) \gamma^\mu + m] \gamma^\sigma (p'_\beta \gamma^\beta + m)}{2pk} \right] \quad [\dagger\dagger\dagger] \\
&= \frac{e^4}{4} \text{Tr} \left[\frac{(-\gamma_\sigma) [(p_\mu + k_\mu) \gamma^\mu + m] (-\gamma_\rho) (p_\alpha \gamma^\alpha + m) \gamma^\rho [(p_\mu + k_\mu) \gamma^\mu + m] \gamma^\sigma (p'_\beta \gamma^\beta + m)}{(2pk)^2} \right] \\
&= \frac{e^4}{4} \text{Tr} \left[\frac{\gamma_\sigma [(p_\mu + k_\mu) \gamma^\mu + m] \gamma_\rho (p_\alpha \gamma^\alpha + m) \gamma^\rho [(p_\mu + k_\mu) \gamma^\mu + m] \gamma^\sigma (p'_\beta \gamma^\beta + m)}{(2pk)^2} \right].
\end{aligned} \tag{23.221}$$

At $[\dagger]$ of (23.221), we used

$$\sum_h u(\mathbf{p}, h) \bar{u}(\mathbf{p}, h) = p_\mu \gamma^\mu + m \text{ and } \sum_{h'} u(\mathbf{p}', h') \bar{u}(\mathbf{p}', h') = p'_\beta \gamma^\beta + m. \quad (21.144)$$

At $[\dagger\dagger]$, we used the fact that $\varepsilon_\sigma(\mathbf{k}', r')$ and $\varepsilon_\rho^*(\mathbf{k}, r)$ can freely be moved in (23.221) because they are not an operator, but merely a complex number. At

$[\dagger \dagger \dagger]$, see discussion made just below based on the fact that the Feynman amplitude $\mathfrak{M}$ must have the gauge invariance.

Suppose that the vector potential $A_\mu(x)$ takes a form [1]

$$A_\mu(x) = \alpha \varepsilon_\mu(\mathbf{k}, r) e^{\pm i k x},$$

where α is a constant.

Also, suppose that in (22.293) we have [1]

$$\Lambda(x) = \alpha \tilde{\Lambda}(k) e^{\pm i k x}.$$

Then, from (22.293), $A^\mu(x)$ should be transformed such that

$$\begin{aligned} A'_\mu(x) &= A_\mu(x) + \partial_\mu \Lambda(x) = A_\mu(x) \pm i k_\mu \tilde{\Lambda}(k) e^{\pm i k x} \\ &= \alpha \varepsilon_\mu(\mathbf{k}, r) e^{\pm i k x} \pm i k_\mu \tilde{\Lambda}(k) e^{\pm i k x} = \alpha [\varepsilon_\mu(\mathbf{k}, r) \pm i k_\mu \tilde{\Lambda}(k) / \alpha] e^{\pm i k x}. \end{aligned} \quad (23.222)$$

Therefore, corresponding to the gauge transformation of $A_\mu(x)$, $\varepsilon_\mu(\mathbf{k}, r)$ is transformed such that [1]

$$\varepsilon'_\mu(\mathbf{k}, r) = \varepsilon_\mu(\mathbf{k}, r) \pm i k_\mu \tilde{\Lambda}(k) / \alpha. \quad (23.223)$$

Taking complex conjugate of (23.223), we have

$$\varepsilon'_\mu(\mathbf{k}, r)^* = \varepsilon_\mu(\mathbf{k}, r)^* \mp i k_\mu \tilde{\Lambda}(k)^* / \alpha^*. \quad (23.224)$$

Using (23.223) and (23.224), we get

$$\begin{aligned} & \sum_{r=1}^2 \varepsilon'_\mu(\mathbf{k}, r)^* \varepsilon'_\nu(\mathbf{k}, r) \\ &= \sum_{r=1}^2 [\varepsilon_\mu(\mathbf{k}, r)^* \mp i k_\mu \tilde{\Lambda}(k)^* / \alpha^*] [\varepsilon_\nu(\mathbf{k}, r) \pm i k_\nu \tilde{\Lambda}(k) / \alpha] \\ &= \sum_{r=1}^2 \varepsilon_\mu(\mathbf{k}, r)^* \varepsilon_\nu(\mathbf{k}, r) \\ &+ \sum_{r=1}^2 \left[\pm i k_\nu \varepsilon_\mu(\mathbf{k}, r)^* \tilde{\Lambda}(k) / \alpha \mp i k_\mu \varepsilon_\nu(\mathbf{k}, r) \tilde{\Lambda}(k)^* / \alpha^* + k_\mu k_\nu |\tilde{\Lambda}(k) / \alpha|^2 \right]. \end{aligned} \quad (23.225)$$

Namely, the gauge transformation of the photon field produces the additional second term in (23.225). Let the (partial) Feynman amplitude $\mathfrak{X}_1$ including the factor $\sum_{r=1}^2 \varepsilon'_\mu(\mathbf{k}, r)^* \varepsilon'_\nu(\mathbf{k}, r)$ be described by

$$\begin{aligned}\mathfrak{X}_1 &= \frac{e^4}{4} \text{Tr} \left\{ \sum_{r'=1}^2 \left[\sum_{r=1}^2 \varepsilon'_\mu(\mathbf{k}, r)^* \varepsilon'_\nu(\mathbf{k}, r) \right] \tilde{\mathcal{M}}^{\cdots\mu\cdots\nu\cdots} \right\} \\ &= \frac{e^4}{4} \text{Tr} \left\{ \sum_{r'=1}^2 \left[\sum_{r=1}^2 \varepsilon_\mu(\mathbf{k}, r)^* \varepsilon_\nu(\mathbf{k}, r) + \Xi(k_\mu, k_\nu) \right] \tilde{\mathcal{M}}^{\cdots\mu\cdots\nu\cdots} \right\},\end{aligned}$$

where $\Xi(k_\mu, k_\nu)$ represents the second term of (23.225). That is, we have

$$\begin{aligned}\Xi(k_\mu, k_\nu) \\ \equiv \sum_{r=1}^2 \left[\pm i k_\nu \varepsilon_\mu(\mathbf{k}, r)^* \tilde{\Lambda}(k) / \alpha \mp i k_\mu \varepsilon_\nu(\mathbf{k}, r) \tilde{\Lambda}(k)^* / \alpha^* + k_\mu k_\nu |\tilde{\Lambda}(k) / \alpha|^2 \right].\end{aligned}\quad (23.226)$$

The factor $\tilde{\mathcal{M}}^{\cdots\mu\cdots\nu\cdots}$ represents a remaining part of $\mathfrak{X}_1$. In the present case, the indices μ and ν of $\tilde{\mathcal{M}}^{\cdots\mu\cdots\nu\cdots}$ indicate those of the gamma matrices γ^μ and γ^ν . Since $\mathfrak{X}_1$ is gauge invariant, $\Xi(k_\mu, k_\nu) \tilde{\mathcal{M}}^{\cdots\mu\cdots\nu\cdots}$ must vanish. As $\Xi(k_\mu, k_\nu)$ is linear with respect to both k_μ and k_ν , it follows that

$$k_\mu \tilde{\mathcal{M}}^{\cdots\mu\cdots\nu\cdots} = k_\nu \tilde{\mathcal{M}}^{\cdots\mu\cdots\nu\cdots} = 0. \quad (23.227)$$

Meanwhile, we had

$$\sum_{r=1}^2 \varepsilon_\mu(\mathbf{k}, r)^* \varepsilon_\nu(\mathbf{k}, r) = -\eta_{\mu\nu} - \frac{k_\mu k_\nu - (kn)(k_\mu n_\nu + n_\mu k_\nu)}{(kn)^2}. \quad (23.228)$$

Note that as pointed out previously, (22.377) holds with the real photon field (i.e., an external photon) as in the present case. Then, we obtain

$$\mathfrak{X}_1 = \frac{e^4}{4} \text{Tr} \left\{ \sum_{r'=1}^2 \left[-\eta_{\mu\nu} - \frac{k_\mu k_\nu - (kn)(k_\mu n_\nu + n_\mu k_\nu)}{(kn)^2} \right] \tilde{\mathcal{M}}^{\cdots\mu\cdots\nu\cdots} \right\}. \quad (23.229)$$

But, from the reason mentioned just above, we have

$$\frac{k_\mu k_\nu - (kn)(k_\mu n_\nu + n_\mu k_\nu)}{(kn)^2} \tilde{\mathcal{M}}^{\cdots\mu\cdots\nu\cdots} = 0, \quad (23.230)$$

with only the first term $-\eta_{\mu\nu} \tilde{\mathcal{M}}^{\cdots\mu\cdots\nu\cdots}$ contributing to $\mathfrak{X}_1$. With regard to another related factor $\sum_{r'=1}^2 \varepsilon_\alpha^*(\mathbf{k}', r') \varepsilon_\beta(\mathbf{k}', r')$ of (23.221), the above argument holds as well.

Thus, the calculation of $\mathfrak{X}_1$ comes down to the trace computation of (23.221). The remaining calculations are straightforward and follow the literature method [1]. We first calculate the numerator of (23.221). To this end, we define a (4, 4) matrix Y included as a numerator of (23.221) as [1]

$$\begin{aligned}
Y &\equiv \gamma_\sigma [(p_\mu + k_\mu)\gamma^\mu + m]\gamma_\rho(p_\alpha\gamma^\alpha + m)\gamma^\rho[(p_\lambda + k_\lambda)\gamma^\lambda + m]\gamma^\sigma \\
&\downarrow \text{ use } \gamma_\rho(p_\alpha\gamma^\alpha)\gamma^\rho = -2(p_\alpha\gamma^\alpha) \text{ and } \gamma_\rho\gamma^\rho = 4: \\
&= \gamma_\sigma [(p_\mu + k_\mu)\gamma^\mu + m](-2p_\alpha\gamma^\alpha + 4m)[(p_\lambda + k_\lambda)\gamma^\lambda + m]\gamma^\sigma \\
&= \gamma_\sigma [-2(p_\mu + k_\mu)\gamma^\mu(p_\alpha\gamma^\alpha)(p_\lambda + k_\lambda)\gamma^\lambda - 2m(p_\mu + k_\mu)\gamma^\mu(p_\alpha\gamma^\alpha) \\
&\quad + 4m(p_\mu + k_\mu)^2 + 4m^2(p_\mu + k_\mu)\gamma^\mu - 2m(p_\alpha\gamma^\alpha)(p_\lambda + k_\lambda)\gamma^\lambda \\
&\quad - 2m^2(p_\alpha\gamma^\alpha) + 4m^2(p_\lambda + k_\lambda)\gamma^\lambda + 4m^3]\gamma^\sigma \\
&= 4(p_\lambda + k_\lambda)\gamma^\lambda(p_\alpha\gamma^\alpha)(p_\mu + k_\mu)\gamma^\mu - 16m(p_\mu + k_\mu)p^\mu \\
&\quad + 16m(p_\mu + k_\mu)^2 - 16m^2(p_\mu + k_\mu)\gamma^\mu + 4m^2(p_\alpha\gamma^\alpha) + 16m^3. \tag{23.231}
\end{aligned}$$

With the last equality of (23.231), we used the following formulae [1]:

$$\gamma_\lambda\gamma^\lambda = 4E_4, \tag{21.185}$$

$$\gamma_\lambda(A_\sigma\gamma^\sigma)\gamma^\lambda = -2(A_\sigma\gamma^\sigma), \gamma_\lambda(A_\sigma\gamma^\sigma)(B_\sigma\gamma^\sigma)\gamma^\lambda = 4A_\sigma B^\sigma, \tag{21.187}$$

$$\gamma_\lambda(A_\sigma\gamma^\sigma)(B_\sigma\gamma^\sigma)(C_\sigma\gamma^\sigma)\gamma^\lambda = -2(C_\sigma\gamma^\sigma)(B_\sigma\gamma^\sigma)(A_\sigma\gamma^\sigma). \tag{21.188}$$

In (21.187) and (21.188), A_α , B_α , and C_α are arbitrary four-vectors.

The spin and polarization average and sum have already been taken in the process of deriving (23.218), an example of which is given as (23.221). Hence, what remains to be done is only the computation of the trace. To this end, we define another real number $\mathfrak{Y}$ as [1]

$$\mathfrak{Y} \equiv \text{Tr} Y(p'_\beta\gamma^\beta + m). \tag{23.232}$$

Then, from (23.221), we have

$$\mathfrak{X}_1 = e^4 \mathfrak{Y} / 16(pk)^2. \tag{23.233}$$

With $\mathfrak{Y}$, we obtain

$$\begin{aligned}
\mathfrak{Y} &= 16 \{ 2[(p_\mu + k_\mu)p^\mu][(p_\rho + k_\rho)p'^\rho] - (p_\mu + k_\mu)(p^\mu + k^\mu)p_\rho p'^\rho \\
&\quad + m^2[-4p_\mu(p^\mu + k^\mu) + 4(p_\mu + k_\mu)(p^\mu + k^\mu)] \\
&\quad + m^2[p_\rho p'^\rho - 4(p_\mu + k_\mu)p'^\mu] + 4m^4 \}. \tag{23.234}
\end{aligned}$$

In deriving (23.234), we used the following formulae [1]:

$$\text{Tr}[(A_\sigma \gamma^\sigma)(B_\sigma \gamma^\sigma)] = 4A_\sigma B^\sigma, \quad (21.189)$$

$$\begin{aligned} & \text{Tr}[(A_\sigma \gamma^\sigma)(B_\sigma \gamma^\sigma)(C_\sigma \gamma^\sigma)(D_\sigma \gamma^\sigma)] \\ &= 4[(A_\sigma B^\sigma)(C_\sigma D^\sigma) - (A_\sigma C^\sigma)(B_\sigma D^\sigma) + (A_\sigma D^\sigma)(B_\sigma C^\sigma)]. \end{aligned} \quad (21.190)$$

Also, we used the fact that the trace of an odd number product of gamma matrices vanishes [1]. Calculations of (23.234) are straightforward using three independent scalars, i.e., m^2 , pk , and pk' ; here notice that pk , pk' , etc. are the abbreviations of

$$pk \equiv p_\mu k^\mu, pk' \equiv p_\mu k'^\mu, \text{etc.} \quad (23.235)$$

We note, for instance,

$$(p + k)p = p^2 + kp = m^2 + kp. \quad (23.236)$$

The conservation of four-momentum is expressed as

$$p + k = p' + k'. \quad (23.237)$$

Taking the square of both sides of (23.237), we have

$$p^2 + 2kp + k^2 = p'^2 + 2k'p' + k'^2. \quad (23.238)$$

Using $p^2 = p'^2 = m^2$ and $k^2 = k'^2 = 0$, we get

$$kp = k'p'. \quad (23.239)$$

From (23.237), we obtain another relation

$$pk' = p'k. \quad (23.240)$$

Rewriting (23.237), in turn, we obtain a relationship

$$p - p' = k' - k. \quad (23.241)$$

Taking the square of both sides of (23.241) again and using $k'k = pk - pk'$, we get

$$pp' = m^2 + pk - pk'. \quad (23.242)$$

Using the above relationship among the scalars, we finally get [1]

$$\mathfrak{V} = 32[m^4 + m^2(pk) + (pk)(pk')]. \quad (23.243)$$

Substituting (23.243) for (23.233), we obtain

$$\mathfrak{X}_1 = \frac{2e^4[m^4 + m^2(pk) + (pk)(pk')]}{(pk)^2}. \quad (23.244)$$

Next, we compute $\mathfrak{X}_2$ of (23.215). From (23.172), with the c -numbers of the photon field, we have

$$|\mathcal{M}_a|^2 \propto \varepsilon_\nu(\mathbf{k}', r')^* \varepsilon_\nu(\mathbf{k}', r') \tilde{S}_F(p+k) \tilde{S}_F^*(p+k) \varepsilon_\mu(\mathbf{k}, r) \varepsilon_\mu(\mathbf{k}, r)^*.$$

Since $\mathfrak{X}_1$ is obtained as a result of averaging the indices r and r' , $\mathfrak{X}_1$ is unaltered by exchanging the indices r and r' . Hence, we obtain

$$\begin{aligned} \mathfrak{X}_1 &\propto \varepsilon_\nu(\mathbf{k}', r)^* \varepsilon_\nu(\mathbf{k}', r) \tilde{S}_F(p+k) \tilde{S}_F^*(p+k) \varepsilon_\mu(\mathbf{k}, r') \varepsilon_\mu(\mathbf{k}, r')^* \\ &= \varepsilon_\nu(-\mathbf{k}', r)^* \varepsilon_\nu(-\mathbf{k}', r) \tilde{S}_F(p+k) \tilde{S}_F^*(p+k) \varepsilon_\mu(-\mathbf{k}, r') \varepsilon_\mu(-\mathbf{k}, r')^*. \end{aligned}$$

Since in the above equation we are thinking of the transverse modes with the polarization, those modes are held unchanged with respect to the exchange of $\mathbf{k} \leftrightarrow -\mathbf{k}$ and $\mathbf{k}' \leftrightarrow -\mathbf{k}'$ as shown in the above equation. Meanwhile, from (23.176), we have

$$\begin{aligned} \mathfrak{X}_2 &\propto |\mathcal{M}_b|^2 \\ &\propto \varepsilon_\nu(\mathbf{k}, r) \varepsilon_\nu(\mathbf{k}, r)^* \tilde{S}_F(p-k') \tilde{S}_F^*(p-k') \varepsilon_\mu(\mathbf{k}', r')^* \varepsilon_\mu(\mathbf{k}', r'). \end{aligned}$$

Comparing the above expressions of $\mathfrak{X}_1$ and $\mathfrak{X}_2$, we notice that $\mathfrak{X}_2$ can be obtained merely by switching the four-momentum $k \leftrightarrow -k'$. Then, changing $k \leftrightarrow -k'$ in (23.244), we immediately get

$$\mathfrak{X}_2 = \frac{2e^4[m^4 - m^2(pk') + (pk)(pk')]}{(pk')^2}. \quad (23.245)$$

To calculate $\mathfrak{X}_3$, we need related but slightly different techniques from that used for the calculation of $\mathfrak{X}_1$. The outline for this is as follows. As similarly to the case of (23.221), we arrive at the following relation described by

$$\begin{aligned} \mathfrak{X}_3 &= \frac{e^4}{4} \text{Tr} \left[\frac{\gamma_\sigma [(p_\rho + k_\rho) \gamma^\rho + m] \gamma_\mu (p_\alpha \gamma^\alpha + m) \gamma^\sigma [(p_\rho - k'_\rho) \gamma^\rho + m] \gamma^\mu (p'_\beta \gamma^\beta + m)}{(2pk)(-2pk')} \right]. \end{aligned} \quad (23.246)$$

Decomposing the numerator of (23.246), we have non-vanishing eight terms that comprise gamma matrices and four-vectors as well as scalars (i.e., an electron mass

m). Of these terms, m is included as either a factor of m^2 or m^4 . Classifying the eight terms according to the power of m , we get

$$m^4 : \text{one term}, m^2 : \text{six terms}, m^0 : \text{one term}. \quad (23.247)$$

We evaluate (23.247) termwise below.

(i) $m^4 : m^4 \gamma_\sigma \gamma^\mu \gamma^\sigma \gamma_\mu$

$$m^4 \gamma_\sigma \gamma^\mu \gamma^\sigma \gamma_\mu = m^4 (\gamma_\sigma \gamma^\mu \gamma^\sigma) \gamma_\mu = m^4 (-2\gamma^\mu) \gamma_\mu = -2m^4 \gamma^\mu \gamma_\mu = -8m^4 E_4, \quad (23.248)$$

where with the second and last equalities, we used (21.185); E_4 denotes the (4, 4) identity matrix. Taking the trace of (23.248), we have

$$\text{Tr}(-8m^4 E_4) = -32m^4. \quad (23.249)$$

Notice that the trace of E_4 is 4.

(ii) m^2 : Six terms, one of which is e.g., $m^2 \gamma_\sigma [(p_\rho + k_\rho) \gamma^\rho] \gamma^\mu [p_\alpha \gamma^\alpha] \gamma^\sigma \gamma_\mu$

$$\begin{aligned} m^2 \gamma_\sigma [(p_\rho + k_\rho) \gamma^\rho] \gamma^\mu [p_\alpha \gamma^\alpha] \gamma^\sigma \gamma_\mu &= m^2 \gamma_\sigma [(p_\rho + k_\rho) \gamma^\rho] \cdot p_\alpha (\gamma^\mu \gamma^\alpha \gamma^\sigma \gamma_\mu) \\ &= m^2 \gamma_\sigma [(p_\rho + k_\rho) \gamma^\rho] \cdot p_\alpha (4\eta^{\alpha\sigma}) \\ &= m^2 \gamma_\sigma [(p_\rho + k_\rho) \gamma^\rho] \cdot (4p^\sigma) = 4m^2 \gamma_\sigma p^\sigma [(p_\rho + k_\rho) \gamma^\rho], \end{aligned} \quad (23.250)$$

where with the second equality, we used the first equation of (21.186). Taking the trace of (23.250), we obtain

$$\text{Tr}\{4m^2 \gamma_\sigma p^\sigma [(p_\rho + k_\rho) \gamma^\rho]\} = 16m^2 p(p+k) = 16m^2 [m^2 + (pk)], \quad (23.251)$$

where with the first equality, we used (21.189). Other five terms of the second power of m are calculated in a similar manner. Summing up the traces of these six matrices, we get

$$96m^4 + 48m^2(pk) - 48m^2(pk'). \quad (23.252)$$

$$\begin{aligned}
\text{(iii)} \quad m^0 : & \gamma_\sigma [(p_\rho + k_\rho) \gamma^\rho] \gamma^\mu [p_\alpha \gamma^\alpha] \gamma^\sigma [(p_\lambda - k'_\lambda) \gamma^\lambda] \gamma_\mu [p'_\beta \gamma^\beta] \\
& \gamma_\sigma [(p_\rho + k_\rho) \gamma^\rho] \gamma^\mu [p_\alpha \gamma^\alpha] \gamma^\sigma [(p_\lambda - k'_\lambda) \gamma^\lambda] \gamma_\mu [p'_\beta \gamma^\beta] \\
& = -2[p_\alpha \gamma^\alpha] \gamma^\mu [(p_\rho + k_\rho) \gamma^\rho] \cdot [(p_\lambda - k'_\lambda) \gamma^\lambda] \gamma_\mu [p'_\beta \gamma^\beta] \\
& = -2[p_\alpha \gamma^\alpha] \cdot \gamma^\mu [(p_\rho + k_\rho) \gamma^\rho] [(p_\lambda - k'_\lambda) \gamma^\lambda] \gamma_\mu \cdot [p'_\beta \gamma^\beta] \\
& = -2[p_\alpha \gamma^\alpha] \cdot 4(p+k)(p-k') \cdot [p'_\beta \gamma^\beta] \\
& = -8(p+k)(p-k')[p_\alpha \gamma^\alpha] [p'_\beta \gamma^\beta], \tag{23.253}
\end{aligned}$$

where with the first equality, we used (21.188) with $B_\sigma \equiv 1$ [or $B \equiv (1 \ 1 \ 1 \ 1)$, where B denotes a covariant four-vector]; with the second to the last equality, we used the second equation of (21.187). Taking the trace of (23.253), we have

$$\begin{aligned}
& \text{Tr} \left\{ -8(p+k)(p-k')[p_\alpha \gamma^\alpha] [p'_\beta \gamma^\beta] \right\} \\
& = -32(p+k)(p-k')pp' \\
& = -32m^2 [m^2 + (pk) - (pk')], \tag{23.254}
\end{aligned}$$

where with the first equality, we used (21.189); the last equality resulted from (23.237), (23.239), and (23.242). Summing (23.249), (23.252), and (23.254), finally we get

$$\begin{aligned}
\mathfrak{X}_3 = & \frac{e^4}{4} \cdot \frac{32m^4 + 16m^2(pk) - 16m^2(pk')}{(2pk)(-2pk')} = \\
& - \frac{e^4 [2m^4 + m^2(pk) - m^2(pk')]}{(pk)(pk')}. \tag{23.255}
\end{aligned}$$

From (23.216) and (23.217), at once we see that $\mathfrak{X}_4$ is obtained as a complex conjugate of $\mathfrak{X}_3$. Also, we realize that (23.255) remains unchanged with respect to the exchange of k with $-k'$. Since $\mathfrak{X}_3$ is real, we obtain

$$\mathfrak{X}_3 = \mathfrak{X}_4 = - \frac{e^4 [2m^4 + m^2(pk) - m^2(pk')]}{(pk)(pk')}. \tag{23.256}$$

Summing up all the above results and substituting them for (23.218), we get $\mathfrak{X}$ of (23.213) as

$$\begin{aligned}
\mathfrak{X} &= \mathfrak{X}_1 + \mathfrak{X}_2 + \mathfrak{X}_3 + \mathfrak{X}_4 = \frac{1}{4} \sum_{r=1}^2 \sum_{r'=1}^2 \sum_{h=1}^2 \sum_{h'=1}^2 |\mathfrak{M}|^2 \\
&= \frac{2e^4 [m^4 + m^2(pk) + (pk)(pk')]}{(pk)^2} + \frac{2e^4 [m^4 - m^2(pk') + (pk)(pk')]}{(pk')^2} \\
&\quad - 2 \times \frac{e^4 [2m^4 + m^2(pk) - m^2(pk')]}{(pk)(pk')} \\
&= 2e^4 \left[m^4 \left(\frac{1}{pk} - \frac{1}{pk'} \right)^2 + 2m^2 \left(\frac{1}{pk} - \frac{1}{pk'} \right) + \left(\frac{pk'}{pk} + \frac{pk}{pk'} \right) \right]. \quad (23.257)
\end{aligned}$$

Once again, notice that (23.257) is unchanged with the exchange between k and $-k'$.

23.8.6 Experimental Observations

Our next task is to connect (23.257) to the experimental results associated with the scattering cross-section that has been expressed as (23.200). In the present situation, we assume a scattering described by (23.178) as

$$P1 + P2 \rightarrow P'1 + P'2. \quad (23.258)$$

It is the case where $N = 2$ in (23.178). In relation to an evaluation of the cross-section of Compton scattering, we frequently encounter an actual case where electrons and/or photons are unpolarized or their polarization is not detected with their initial or final states (i.e., before and after the collision experiments). In this section, we deal with such a case in the general manner possible.

We denote the momentum four-vectors by

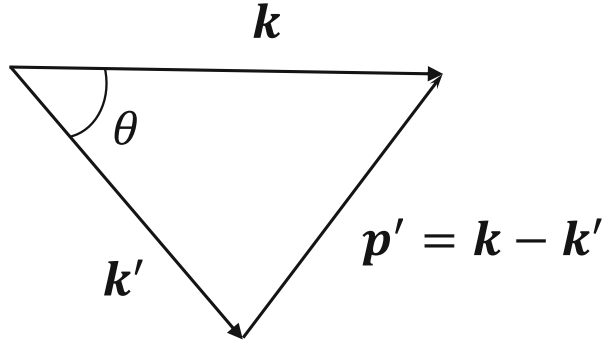
$$p \equiv \begin{pmatrix} E \\ \mathbf{p} \end{pmatrix}, p' \equiv \begin{pmatrix} E' \\ \mathbf{p}' \end{pmatrix}, k \equiv \begin{pmatrix} \omega \\ \mathbf{k} \end{pmatrix}, k' \equiv \begin{pmatrix} \omega' \\ \mathbf{k}' \end{pmatrix}, \quad (23.259)$$

where p and k represent the initial state of the electron and photon field, respectively; p' and k' represent the final state of the electron and photon field, respectively. To use (23.200), we express the quantities of (23.259) in the laboratory frame where a photon beam is incident to the targeted electrons that are nearly stationary. In practice, this is a good approximation. Thereby, p of (23.259) is approximated as

$$p \approx \begin{pmatrix} m \\ \mathbf{0} \end{pmatrix}.$$

Then, the momentum conservation described by $\mathbf{p} + \mathbf{k} = \mathbf{p}' + \mathbf{k}'$ is translated into

Fig. 23.6 Geometry of the photon field. With the meaning of the notation see text



$$\mathbf{p}' = \mathbf{k} - \mathbf{k}'. \quad (23.260)$$

The energy conservation described by $E + \omega = E' + \omega'$ is approximated as

$$m + \omega \approx E' + \omega'. \quad (23.261)$$

Meanwhile, the relation

$$k'k = pk - pk' \quad (23.262)$$

directly derived from the four-momentum conservation is approximated by

$$k'k \approx m\omega - m\omega' = m(\omega - \omega'). \quad (23.263)$$

With the photon field, we have

$$\omega^2 = \mathbf{k}^2, \omega'^2 = \mathbf{k}'^2, \omega = |\mathbf{k}|, \omega' = |\mathbf{k}'|. \quad (23.264)$$

Figure 23.6 shows the geometry of the photon field. We assume that photons are scattered in a radially symmetric direction with respect to the photon momentum $\mathbf{k}$ as a function of the scattering angle θ . Notice that we conform Figs. 23.6 to 1.1. In this geometry, we have

$$k'k = \omega'\omega - \mathbf{k}' \cdot \mathbf{k} = \omega'\omega - \omega'\omega \cos \theta = \omega'\omega(1 - \cos \theta). \quad (23.265)$$

Combining (23.263) and (23.265), we obtain

$$m(\omega - \omega') = \omega\omega'(1 - \cos \theta). \quad (23.266)$$

This is consistent with (1.16), if (1.16) is written in the natural units. In (23.258), we may freely choose P'1 (or P'2) from among electron and photon. For a practical

purpose, however, we choose photon for $P'1$ (i.e., scattered photon) in (23.258). That is, since we are to directly observe the photon scattering in the geometry of the laboratory frame, it will be convenient to measure the energy and momentum of the scattered photon as $P'1$ in the final state. Then, $P'2$ is electron accordingly. To apply (23.200) to the experimental results, we need to express the electron energy E'_2 in terms of the final photon energy ω' . We have

$$E'^2 = \mathbf{p}'^2 + m^2 = (\mathbf{k} - \mathbf{k}')^2 + m^2, \quad (23.267)$$

where with the last equality, we used (23.260). Applying the cosine rule to Fig. 23.6, we have

$$(\mathbf{k} - \mathbf{k}')^2 = |\mathbf{k}|^2 + |\mathbf{k}'|^2 - 2|\mathbf{k}| \cdot |\mathbf{k}'| \cos \theta = \omega^2 + \omega'^2 - 2\omega\omega' \cos \theta. \quad (23.268)$$

Then, we obtain

$$E' = \sqrt{\omega^2 + \omega'^2 - 2\omega\omega' \cos \theta + m^2}. \quad (23.269)$$

Differentiating (23.269) with respect to ω' , we get

$$\frac{\partial E'}{\partial \omega'} = \frac{2\omega' - 2\omega \cos \theta}{2\sqrt{\omega^2 + \omega'^2 - 2\omega\omega' \cos \theta + m^2}}. \quad (23.270)$$

Accordingly, we obtain

$$\begin{aligned} \frac{\partial(E' + \omega')}{\partial \omega'} &= \frac{\omega' - \omega \cos \theta}{\sqrt{\omega^2 + \omega'^2 - 2\omega\omega' \cos \theta + m^2}} + 1 \\ &= \frac{\omega' - \omega \cos \theta + E'}{E'} = \frac{\omega(1 - \cos \theta) + m}{E'}, \end{aligned} \quad (23.271)$$

where with the last equality, we used (23.261). Using (23.266), furthermore, we get

$$\frac{\partial(E' + \omega')}{\partial \omega'} = \frac{m\omega}{E'\omega'}. \quad (23.272)$$

Now, (23.200) is read as

$$d\tilde{\sigma} = \frac{\omega'^2}{64\pi^2 m \omega \omega' E' v_{\text{rel}}} \mathfrak{X} \left[\frac{\partial(\omega' + E')}{\partial \omega'} \right]^{-1} d\Omega'. \quad (23.273)$$

Substituting (23.189), (23.257), and (23.272) for (23.273), we get

$$\begin{aligned} \frac{d\tilde{\sigma}}{d\Omega'} &= \frac{\omega'^2}{64\pi^2 m \omega \omega' E' v_{\text{rel}}} \cdot \frac{E' \omega'}{m \omega} \mathfrak{X} \\ &= \frac{2e^4 \omega'^2}{64\pi^2 m^2 \omega^2} \left[m^4 \left(\frac{1}{pk} - \frac{1}{pk'} \right)^2 + 2m^2 \left(\frac{1}{pk} - \frac{1}{pk'} \right) + \left(\frac{pk'}{pk} + \frac{pk}{pk'} \right) \right]. \end{aligned} \quad (23.274)$$

The energy of the scattered electron was naturally deleted. As mentioned earlier, in (23.274) we assumed $v_{\text{rel}} = 1$ as a good approximation. In the case where the electron is initially at rest, we can use the approximate relations (23.266) as well as $pk = m\omega$ and $pk' = m\omega'$. On these conditions, we obtain a succinct relation described by

$$\frac{d\tilde{\sigma}}{d\Omega'} = \frac{\alpha^2}{2m^2} \left(\frac{\omega'}{\omega} \right)^2 \left(\frac{\omega}{\omega'} + \frac{\omega'}{\omega} - \sin^2 \theta \right), \quad (23.275)$$

where α is the fine structure constant given in (23.87). Moreover, using (23.266) ω' can be eliminated from (23.275) [1].

The polarized measurements of the photon have thoroughly been pursued and the results are well-known as the Klein–Nishina formula. The details can be seen in the literature [1, 3].

23.9 Summary

In this chapter, we have described the features of the interaction among quantum field, especially within the framework of QED. As a typical example, we detailed the calculation procedures of the Compton scattering. It is intriguing to point out that QED adequately explains the various aspects of the Compton scattering, whose experiments precisely opened up a route to QED.

In this chapter, we focused on the S -matrix expansion of the second order. Yet, the essence of QED clearly showed up. Namely, we studied how the second-order S -matrix elements $S_k^{(2)}$ ($k = A, \dots, F$) reflected major processes of QED including the Compton scattering. We have a useful index about whether the integrals relevant to the S -matrix diverge. The index K is given by [1]

$$K = 4 - (3/2)f_e - b_e, \quad (23.276)$$

where f_e and b_e are the number of the external Fermion (i.e., electron) lines and external Boson (photon) lines in the Feynman diagram, respectively. That $K \geq 0$ is a *necessary* condition for the integral to diverge. In the case of the Compton scattering, $f_e = b_e = 2$, then we have $K = -1$. The integrals converge accordingly. We have confirmed that this is really the case. In the cases of the electron self-energy ($K = 1$) and photon self-energy ($K = 2$), the second-order S -matrix integrals diverge [1].

Even though we did not go into detail on those issues, especially the renormalization, we gained a glimpse into the essence of QED that offers prolific aspects on the elementary particle physics.

References

1. Mandl F, Shaw G (2010) Quantum field theory, 2nd edn. Wiley, Chichester
2. Arfken GB, Weber HJ, Harris FE (2013) Mathematical methods for physicists, 7th edn. Academic Press, Waltham
3. Itzykson C, Zuber J-B (2005) Quantum field theory. Dover, New York
4. Kaku M (1993) Quantum field theory. Oxford University Press, New York
5. Peskin ME, Schroeder DV (1995) An introduction to quantum field theory. Westview Press, Boulder
6. Moller C (1952) The theory of relativity. Oxford University Press, London
7. Klauber RD (2013) Student friendly quantum field theory, 2nd edn. Sandtrove Press, Fairfield
8. Sakamoto M (2020) Quantum field theory II. Shokabo, Tokyo. (in Japanese)
9. Dennery P, Krzywicki A (1996) Mathematics for physicists. Dover, New York

Chapter 24

Basic Formalism



We have studied several important aspects of the quantum theory of fields and found that the theory is based in essence upon the theory of relativity. In Chaps. 1 and 21, we briefly surveyed its essence. In this chapter, we explore the basic formalism and fundamental skeleton of the theory more systematically in depth and in breadth. First, we examine the constitution of the Lorentz group from the point of view of Lie algebra. The Lorentz group comprises two components, one of which is virtually the same as the rotation group that we have examined in Part IV. Another component, on the other hand, is called Lorentz boost, which is considerably different from the former one. In particular, whereas the matrices associated with the spatial rotation are unitary, the matrices that represent the Lorentz boost are not unitary. In this chapter, we connect these aspects of the Lorentz group to the matrix representation of the Dirac equation to clarify the constitution of the Dirac equation. Another point of great importance rests on the properties of Dirac operators $\mathcal{G}$ and $\tilde{\mathcal{G}}$ that describe the Dirac equation (Sect. 21.3). We examine their properties from the point of view of matrix algebra. In particular, we study how $\mathcal{G}$ and $\tilde{\mathcal{G}}$ can be diagonalized through the similarity transformation. Remaining parts of this chapter deal with advanced topics of the Lorentz group from the point of view of continuous groups and their Lie algebras.

24.1 Extended Concepts of Vector Spaces

In Part III, we studied the properties and characteristics of the linear vector spaces. We have seen that among the vector spaces, the inner product space and its properties are most widely used and investigated in various areas of natural science, accompanied by the bra-ket notation $\langle ab \rangle$ of inner product due to Dirac (see Chaps. 1 and 13). The inner product space is characterized by the positive definite metric associated with the Gram matrix (Sect. 13.2). In this section, we wish to modify the concept of metric so that it can be applied to wider vector spaces

including the Minkowski space (Chap. 21). In parallel, we formally introduce tensors and related concepts as a typical example of the extended vector spaces.

24.1.1 Bilinear Mapping

First, we introduce a direct product (vector) space formed by the product of two (or more) vector spaces. In Part IV, we introduced a notion of direct product in relation to groups and their representation. In this chapter, we extend this notion to the linear vector spaces. We also study the properties of the bilinear mapping that maps the direct product space to another vector space. First, we have a following definition:

Definition 24.1 Let B be a mapping described by

$$B : W \times V \rightarrow T, \quad (24.1)$$

where V , W , and T are vector spaces over K ; $W \times V$ means the direct product of V and W . Suppose that $B(\cdot, \cdot)$ satisfies the following conditions [1, 2]:

$$\begin{aligned} B(y + y', x) &= B(y, x) + B(y', x), \\ B(y, x + x') &= B(y, x) + B(y, x'), \\ B(ay, x) &= B(y, ax) = aB(y, x) \quad x, x' \in V; y, y' \in W; a \in K, \end{aligned} \quad (24.2)$$

where K is a field usually chosen from a complex number field $\mathbb{C}$ or real number field $\mathbb{R}$. Then, the mapping B is called a bilinear mapping.

The range of B , i.e., $B(\cdot, \cdot)$ is a subspace of T . This is obvious from the definition of (24.2) and (11.15). Since in Definition 24.1 the dimension of V and W can be different, we assume that the dimension of V and W is n and m , respectively. If the direct product $W \times V$ is deemed to be a topological space (Sect. 6.1.2), it is called a direct product space. (However, we do not need to be too strict with the terminology.)

Definition 24.1 does not tell us more detailed structure of $B(\cdot, \cdot)$ with the exception of the fact that $B(\cdot, \cdot)$ forms a subspace of T . To clarify this point, we set further conditions on the bilinear mapping B defined by (24.1) and (24.2). The conditions (B1)–(B3) are given as follows [1, 2]:

(B1) Let V and W be linear vector spaces with $\dim V = n$ and $\dim W = m$. If $x_1, \dots, x_r \in V$ are linearly independent, with $y_1, \dots, y_r \in W$ we have

$$\sum_{i=1}^r B(y_i, x_i) = 0 \quad \Rightarrow \quad y_i = 0 \quad (1 \leq i \leq r), \quad (24.3)$$

where $r \leq n$.

(B2) If $y_1, \dots, y_r \in W$ are linearly independent, for $x_1, \dots, x_r \in V$, we have

$$\sum_{i=1}^r B(y_i, x_i) = 0 \Rightarrow x_i = 0 \quad (1 \leq i \leq r), \quad (24.4)$$

where $r \leq m$.

(B3) The vector space T is spanned by $B(y, x)$ ($x \in V, y \in W$).

Regarding Definition 24.1, we have a following important theorem.

Theorem 24.1 [1, 2] Let B be a bilinear mapping described by $B(\cdot, \cdot) : W \times V \rightarrow T$, where V, W , and T are vector spaces over K . Let $\{e_i; 1 \leq i \leq n\}$ and $\{f_j; 1 \leq j \leq m\}$ be the basis set of V and W , respectively. Then, a necessary and sufficient condition for the statements (B1)–(B3) to hold is that $B(f_j, e_i)$ ($1 \leq i \leq n, 1 \leq j \leq m$) are the basis set of T .

Proof

(i) Sufficient condition: Suppose that $B(f_j, e_i)$ ($1 \leq i \leq n, 1 \leq j \leq m$) are the basis set of T . That is, we have

$$T = \text{Span}\{B(f_j, e_i); 1 \leq i \leq n, 1 \leq j \leq m\}. \quad (24.5)$$

From Definition 24.1 and from the assumption that e_i ($1 \leq i \leq n$) and f_j ($1 \leq j \leq m$) are the basis set of V and W , regarding $\forall x \in V, \forall y \in W, B(y, x)$ is described by a linear combination of $B(f_j, e_i)$. Since that linear combination may take any number (over K) as a coefficient of $B(f_j, e_i)$, the said $B(y, x)$ spans T . It is the assertion of (B3). Suppose that with $x^{(k)} (\in V)$ and $y^{(k)} (\in W)$ we have

$$x^{(k)} = \sum_{i=1}^n \xi_i^{(k)} e_i, y^{(k)} = \sum_{j=1}^m \eta_j^{(k)} f_j \quad (1 \leq k \leq r), \quad (24.6)$$

with $\xi_i^{(k)}, \eta_j^{(k)} \in K$. We assume that in (24.6) $r \leq \max(m, n)$. Then, we have

$$\sum_{k=1}^r B(y^{(k)}, x^{(k)}) = \sum_{j=1}^m \sum_{i=1}^n \left[\sum_{k=1}^r \eta_j^{(k)} \xi_i^{(k)} \right] B(f_j, e_i). \quad (24.7)$$

In (24.7) putting $\sum_{k=1}^r B(y^{(k)}, x^{(k)}) = 0$, we get

$$\sum_{k=1}^r \eta_j^{(k)} \xi_i^{(k)} = 0. \quad (24.8)$$

It is because $B(f_j, e_i)$ ($1 \leq i \leq n, 1 \leq j \leq m$) are the basis set of T and, hence, linearly independent. Equation (24.8) holds with any pair of j and i . Therefore, with a fixed j , from (24.8) we obtain

$$\begin{aligned}
\eta_j^{(1)} \xi_1^{(1)} + \eta_j^{(2)} \xi_1^{(2)} + \cdots + \eta_j^{(r)} \xi_1^{(r)} &= 0, \\
\eta_j^{(1)} \xi_2^{(1)} + \eta_j^{(2)} \xi_2^{(2)} + \cdots + \eta_j^{(r)} \xi_2^{(r)} &= 0, \\
&\vdots \\
\eta_j^{(1)} \xi_n^{(1)} + \eta_j^{(2)} \xi_n^{(2)} + \cdots + \eta_j^{(r)} \xi_n^{(r)} &= 0.
\end{aligned} \tag{24.9}$$

Rewriting (24.9) in a form of column vectors, we get

$$\eta_j^{(1)} \begin{pmatrix} \xi_1^{(1)} \\ \vdots \\ \xi_n^{(1)} \end{pmatrix} + \eta_j^{(2)} \begin{pmatrix} \xi_1^{(2)} \\ \vdots \\ \xi_n^{(2)} \end{pmatrix} + \cdots + \eta_j^{(r)} \begin{pmatrix} \xi_1^{(r)} \\ \vdots \\ \xi_n^{(r)} \end{pmatrix} = 0. \tag{24.10}$$

If $x^{(1)}, \dots, x^{(r)} \in V$ are linearly independent, the corresponding column vectors of (24.10) are linearly independent as well with the fixed basis set of $B(f_j, e_i)$. Consequently, we have

$$\eta_j^{(1)} = \eta_j^{(2)} = \cdots = \eta_j^{(r)} = 0. \tag{24.11}$$

Since (24.11) holds with all j ($1 \leq j \leq m$), from (24.6) we obtain

$$y^{(1)} = y^{(2)} = \cdots = y^{(r)} = 0.$$

The above argument ensures the assertion of (B1). In a similar manner, the statement (B2) is assured. Since $B(f_j, e_i)$ ($1 \leq i \leq n, 1 \leq j \leq m$) are the basis set of T , (B3) is evidently assured.

(ii) Necessary condition: Suppose that the conditions (B1)–(B3) hold. From the argument of the sufficient condition, we have (24.5). Therefore, for $B(f_j, e_i)$ ($1 \leq i \leq n, 1 \leq j \leq m$) to be the basis set of T , we only have to show that $B(f_j, e_i)$ are linearly independent. Suppose that we have

$$\sum_{i,j} \zeta_{ij} B(f_j, e_i) = 0, \zeta_{ij} \in K. \tag{24.12}$$

Putting $y_i = \sum_j \zeta_{ij} f_j$, from (24.12) we obtain

$$\sum_i B(y_i, e_i) = 0. \tag{24.13}$$

Since $\{e_i; 1 \leq i \leq n\}$ are linearly independent, from (B1) $y_i = 0$. Since $\{f_j; 1 \leq j \leq m\}$ are the basis set of W , from $\sum_j \zeta_{ij} f_j = 0$ we must have $\zeta_{ij} = 0$ ($1 \leq i \leq n, 1 \leq j \leq m$). Thus, from (24.12) $B(f_j, e_i)$ are linearly independent. These complete the proof.

Earlier we have mentioned that $B(\cdot, \cdot)$ is a subspace of the vector space T . Practically, however, we may safely identify the range of B with the whole space of T . In other words, we may as well regard the subspace $B(\cdot, \cdot)$ as T itself. Thus, the dimension of T can be regarded as mn .

Abbreviating $B(f_j, e_i)$ as $f_j \cdot e_i$ and omitting superscript k in (24.7), we have

$$B(y, x) = \sum_{j=1}^m \sum_{i=1}^n (f_j \cdot e_i) \eta_j \xi_i. \quad (24.14)$$

Further rewriting (24.14), we obtain

$$B(y, x) = \begin{pmatrix} f_1 \cdot e_1 & \cdots & f_1 \cdot e_n & f_2 \cdot e_1 & \cdots & f_2 \cdot e_n & \cdots & f_m \cdot e_1 & \cdots & f_m \cdot e_n \end{pmatrix} \begin{pmatrix} \eta_1 \xi_1 \\ \vdots \\ \eta_1 \xi_n \\ \eta_2 \xi_1 \\ \vdots \\ \eta_2 \xi_n \\ \vdots \\ \eta_m \xi_1 \\ \vdots \\ \eta_m \xi_n \end{pmatrix}. \quad (24.15)$$

Equations (24.14) and (24.15) clearly show that any elements belonging to T can be expressed as RHS of these equations. Namely, T is spanned by $B(y, x)$ ($x \in V$, $y \in W$). This is what the statement (B3) represents.

Equation (24.15) corresponds to (11.13); in (24.15) mn basis vectors form a row vector and mn “coordinates” form a corresponding column vector.

The next example will help us understand the implication of the statements (B1)–(B3) imposed on the bilinear mapping earlier in this section.

Example 24.1 Let V and W be vector spaces with dimension 3 and 2, respectively. Suppose that $x_i \in V$ ($i = 1, 2, 3$) and $y_j \in W$ ($j = 1, 2, 3$). Let x_i and y_j be given by

$$\begin{aligned} x_1 &= (e_1 \ e_2 \ e_3) \begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix}, x_2 = (e_1 \ e_2 \ e_3) \begin{pmatrix} \xi'_1 \\ \xi'_2 \\ \xi'_3 \end{pmatrix}, x_3 = (e_1 \ e_2 \ e_3) \begin{pmatrix} \xi''_1 \\ \xi''_2 \\ \xi''_3 \end{pmatrix}; \\ y_1 &= (f_1 \ f_2) \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}, y_2 = (f_1 \ f_2) \begin{pmatrix} \eta'_1 \\ \eta'_2 \end{pmatrix}, y_3 = (f_1 \ f_2) \begin{pmatrix} \eta''_1 \\ \eta''_2 \end{pmatrix}, \end{aligned} \quad (24.16)$$

where $(e_1 \ e_2 \ e_3)$ and $(f_1 \ f_2)$ are basis sets of V and W , respectively; ξ_1, η_1 , etc. $\in K$. For simplicity, we write

$$(e_1 \ e_2 \ e_3) \equiv \mathbf{e}, (f_1 \ f_2) \equiv \mathbf{f}. \quad (24.17)$$

We define the bilinear mapping $B(y, x)$ as, e.g.,

$$B(y_1, x_1) = (\mathbf{f} \times \mathbf{e}) \left[\begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} \times \begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix} \right] \equiv (\mathbf{f} \times \mathbf{e}) \begin{pmatrix} \eta_1 \xi_1 \\ \eta_1 \xi_2 \\ \eta_1 \xi_3 \\ \eta_2 \xi_1 \\ \eta_2 \xi_2 \\ \eta_2 \xi_3 \end{pmatrix}, \quad (24.18)$$

where we have

$$(\mathbf{f} \times \mathbf{e}) = (f_1 \cdot e_1 \quad f_1 \cdot e_2 \quad f_1 \cdot e_3 \quad f_2 \cdot e_1 \quad f_2 \cdot e_2 \quad f_2 \cdot e_3). \quad (24.19)$$

The RHS of (24.19) is a row vector comprising the direct products of $f_j e_k$ ($j = 1, 2; k = 1, 2, 3$).

Fixing the basis sets of V and W , we represent the vectors as numerical vectors (or column vectors) [1, 2]. Then, the first equation of the bilinear mapping in (24.3) of (B1) can be read as

$$B(y_1, x_1) + B(y_2, x_2) + B(y_3, x_3) = (\mathbf{f} \times \mathbf{e}) \begin{pmatrix} \eta_1 \xi_1 + \eta'_1 \xi'_1 + \eta''_1 \xi''_1 \\ \eta_1 \xi_2 + \eta'_1 \xi'_2 + \eta''_1 \xi''_2 \\ \eta_1 \xi_3 + \eta'_1 \xi'_3 + \eta''_1 \xi''_3 \\ \eta_2 \xi_1 + \eta'_2 \xi'_1 + \eta''_2 \xi''_1 \\ \eta_2 \xi_2 + \eta'_2 \xi'_2 + \eta''_2 \xi''_2 \\ \eta_2 \xi_3 + \eta'_2 \xi'_3 + \eta''_2 \xi''_3 \end{pmatrix} = 0. \quad (24.20)$$

Omitting the basis set $(\mathbf{f} \times \mathbf{e})$, (24.20) is separated into two components such that

$$\begin{aligned} \eta_1 \begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix} + \eta'_1 \begin{pmatrix} \xi'_1 \\ \xi'_2 \\ \xi'_3 \end{pmatrix} + \eta''_1 \begin{pmatrix} \xi''_1 \\ \xi''_2 \\ \xi''_3 \end{pmatrix} &= 0, \eta_2 \begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix} + \eta'_2 \begin{pmatrix} \xi'_1 \\ \xi'_2 \\ \xi'_3 \end{pmatrix} \\ &+ \eta''_2 \begin{pmatrix} \xi''_1 \\ \xi''_2 \\ \xi''_3 \end{pmatrix} = 0. \end{aligned} \quad (24.21)$$

Assuming that ξ_1, ξ_2 , and ξ_3 ($\in V$) are linearly independent, (24.21) implies that $\eta_1 = \eta'_1 = \eta''_1 = \eta_2 = \eta'_2 = \eta''_2 = 0$. Namely, from (24.16) we have

$$y_1 = y_2 = y_3 = 0. \quad (24.22)$$

This is an assertion of (B1). Meanwhile, think of the following equation:

$$B(y_1, x_1) + B(y_2, x_2) = (f \times e) \begin{pmatrix} \eta_1 \xi_1 + \eta'_1 \xi'_1 \\ \eta_1 \xi_2 + \eta'_1 \xi'_2 \\ \eta_1 \xi_3 + \eta'_1 \xi'_3 \\ \eta_2 \xi_1 + \eta'_2 \xi'_1 \\ \eta_2 \xi_2 + \eta'_2 \xi'_2 \\ \eta_2 \xi_3 + \eta'_2 \xi'_3 \end{pmatrix} = 0. \quad (24.23)$$

This time, (24.23) can be separated into three components such that

$$\begin{aligned} \xi_1 \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} + \xi'_1 \begin{pmatrix} \eta'_1 \\ \eta'_2 \end{pmatrix} &= 0, \xi_2 \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} + \xi'_2 \begin{pmatrix} \eta'_1 \\ \eta'_2 \end{pmatrix} = 0, \xi_3 \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} \\ &+ \xi'_3 \begin{pmatrix} \eta'_1 \\ \eta'_2 \end{pmatrix} = 0. \end{aligned} \quad (24.24)$$

Assuming that y_1 and $y_2 \in W$ are linearly independent, in turn, (24.24) implies that $\xi_1 = \xi'_1 = \xi_2 = \xi'_2 = \xi_3 = \xi'_3 = 0$. Namely, from (24.16), we have

$$x_1 = x_2 = 0. \quad (24.25)$$

This is an assertion of (B2). The statement (B3) is self-evident; see (24.14) and (24.15).

In relation to Theorem 24.1, we give another important theorem.

Theorem 24.2 [1–3] Let V and W be vector spaces over K . Then, a vector space T (over K) and a bilinear mapping $B : W \times V \rightarrow T$ that satisfy the above-mentioned conditions (B1), (B2), and (B3) must exist. Also, let $\mathfrak{T}$ be an arbitrarily chosen vector space over K and let $\mathfrak{B}$ be an arbitrary bilinear mapping expressed as

$$\mathfrak{B} : W \times V \rightarrow \mathfrak{T}. \quad (24.26)$$

Then, a linear mapping $\rho : T \rightarrow \mathfrak{T}$ that satisfies the condition described by

$$\mathfrak{B}(y, x) = \rho[B(y, x)]; x \in V, y \in W \quad (24.27)$$

is uniquely determined.

Proof

- (i) Existence: Suppose that T is a nm -dimensional vector space with the basis set $B(f_j, e_i)$ ($1 \leq i \leq n, 1 \leq j \leq m$), where e_i ($1 \leq i \leq n$) and f_j ($1 \leq j \leq m$) are the basis set of V and W , respectively. We assume that V and W are n - and m -dimensional vector spaces, respectively.

Let $x \in V$ and $y \in W$ be expressed as

$$x = \sum_{i=1}^n \xi_i e_i, y = \sum_{j=1}^m \eta_j f_j. \quad (24.28)$$

We define the bilinear mapping $B(y, x)$ as

$$B(y, x) = \sum_{j=1}^m \sum_{i=1}^n \eta_j \xi_i \epsilon_{ji} \quad (24.29)$$

with $\epsilon_{ji} \equiv B(f_j, e_i)$, so that $\epsilon_{ij} \in T$ ($1 \leq i \leq n, 1 \leq j \leq m$) is the basis set of T . Then, we have $B : W \times V \rightarrow T$. On top of it, a set of (T, B) satisfies the conditions (B1)–(B3) according to Theorem 24.1. Thus, existence of the bilinear mapping of B required by the statement of the theorem must necessarily exist.

(ii) Uniqueness: Suppose that an arbitrary bilinear mapping $\mathfrak{B} : W \times V \rightarrow \mathfrak{T}$ is given. Given such $\mathfrak{B}$ and $\mathfrak{T}$, we define a linear mapping $\rho : T \rightarrow \mathfrak{T}$ that is described by

$$\rho\left(\sum_{i,j} \zeta_{ji} \epsilon_{ji}\right) = \sum_{i,j} \zeta_{ji} \mathfrak{B}(f_j, e_i), \quad (24.30)$$

where ζ_{ji} is an arbitrary number over K . Choosing $\zeta_{ji} = 1$ when $j = k, i = l$, and otherwise $\zeta_{ji} = 0$, we have

$$\rho(\epsilon_{kl}) = \mathfrak{B}(f_k, e_l). \quad (24.31)$$

Meanwhile, we have shown the existence of the bilinear mapping $B(y, x)$ described by (24.29). Operating ρ on both sides of (24.29), we obtain

$$\begin{aligned} \rho[B(y, x)] &= \rho\left(\sum_{j=1}^m \sum_{i=1}^n \eta_j \xi_i \epsilon_{ji}\right) = \sum_{j=1}^m \sum_{i=1}^n \eta_j \xi_i \rho(\epsilon_{ji}) \\ &= \sum_{i,j} \eta_j \xi_i \mathfrak{B}(f_j, e_i) = \sum_{i,j} \mathfrak{B}(\eta_j f_j, \xi_i e_i) = \mathfrak{B}\left(\sum_j \eta_j f_j, \sum_i \xi_i e_i\right) = \mathfrak{B}(y, x), \end{aligned}$$

where with the third equality, we used (24.30) and (24.31); with the fourth and fifth equalities, we used the bilinearity of $\mathfrak{B}$. Thus, the linear mapping ρ defined in (24.30) satisfies (24.27).

The other way around, any linear mapping that satisfies (24.27) can be described as

$$\rho(\epsilon_{ji}) = \rho[B(f_j, e_i)] = \mathfrak{B}(f_j, e_i). \quad (24.32)$$

Multiplying both sides of (24.32) by ζ_{ji} and summing over i and j , we recover (24.30). Thus, the linear mapping ρ is determined uniquely. These complete the proof.

24.1.2 Tensor Product

Despite being apparently craggy and highly abstract, Theorem 24.2 finds extensive applications in the vector space theory. The implication of Theorem 24.2 is that once the set (T, B) is determined for the combination of the vector spaces V and W such that $B : W \times V \rightarrow T$, any other set $(\mathfrak{T}, \mathfrak{B})$ can be derived from (T, B) . In this respect, B defined in Theorem 24.2 is referred to as the *universal* bilinear mapping [3]. Another important point is that the linear mapping $\rho : T \rightarrow \mathfrak{T}$ defined in Theorem 24.2 is bijective (see Fig. 11.3) [1, 2]. That is, the inverse mapping ρ^{-1} exists.

The range of the universal bilinear mapping B is a vector space over K . The vector space T accompanied by the bilinear mapping B , or the set (T, B) is called a tensor product (or Kronecker product) of V and W . Thus, we find that the constitution of tensor product hinges upon Theorem 24.2. The tensor product T is expressed as

$$T \equiv W \otimes V. \quad (24.33)$$

Also, $B(y, x)$ is described as

$$B(y, x) \equiv y \otimes x, \quad (24.34)$$

where $x \in V$, $y \in W$, and $y \otimes x \in T$.

The notion of tensor product produces a fertile plain for the vector space theory. In fact, (24.29) can readily be extended so that various bilinear mappings are included. As a typical example, let us think of linear mappings φ and ψ such that

$$\varphi : V \rightarrow V', \psi : W \rightarrow W', \quad (24.35)$$

where V , V' , W , and W' are vector spaces of the dimension n , n' , m , and m' , respectively. Also, suppose that we have [1, 2]

$$T = W \otimes V \quad \text{and} \quad \mathfrak{T} \equiv W' \otimes V'. \quad (24.36)$$

What we assume in (24.36) is that even though in Theorem 24.2 we have chosen $\mathfrak{T}$ as an arbitrary vector space, in (24.36) we choose $\mathfrak{T}$ as a special vector space (i.e., a tensor product $W' \otimes V'$). From (24.35) and (24.36), using φ and ψ we can construct a bilinear mapping $\mathfrak{B} : W \times V \rightarrow W' \otimes V'$ such that

$$\mathfrak{B}(y, x) = \psi(y) \otimes \varphi(x), \quad (24.37)$$

where $\psi(y) \in W'$, $\varphi(x) \in V'$, and $\psi(y) \otimes \varphi(x) \in \mathfrak{T} = W' \otimes V'$. Then, from Theorem 24.2 and (24.34), a linear mapping $\rho : T \rightarrow \mathfrak{T}$ must uniquely be determined such that

$$\rho(y \otimes x) = \psi(y) \otimes \varphi(x). \quad (24.38)$$

Also, defining this linear mapping ρ as

$$\rho \equiv \psi \otimes \varphi, \quad (24.39)$$

(24.38) can be rewritten as [1, 2]

$$(\psi \otimes \varphi)(y \otimes x) = \psi(y) \otimes \varphi(x). \quad (24.40)$$

Equation (24.40) implies that in response to a tensor product of the vector spaces, we can construct a tensor product of the operators that act on the vector spaces. This is one of the most practical consequences derived from Theorem 24.2.

The constitution of the tensor products is simple and straightforward. Indeed, we have already dealt with the tensor products in connection with the group representation (see Sect. 18.8). Let A and B be matrix representations of φ and ψ , respectively. Following (18.193), the tensor product of $B \otimes A$ is expressed as [4]

$$B \otimes A = \begin{pmatrix} b_{11}A & \cdots & b_{1m'}A \\ \vdots & \ddots & \vdots \\ b_{m'1}A & \cdots & b_{m'm'}A \end{pmatrix}, \quad (24.41)$$

where A and B are given by (a_{ij}) ($1 \leq i \leq n'; 1 \leq j \leq n$) and (b_{ij}) ($1 \leq i \leq m'; 1 \leq j \leq m$), respectively. In RHS of (24.41), A represents a (n', n) full matrix of (a_{ij}) . Thus, the direct product of matrix A and B yields a $(m'n', mn)$ matrix. We write it symbolically as

$$[(m', m) \text{ matrix}] \otimes [(n', n) \text{ matrix}] = [(m'n', mn) \text{ matrix}]. \quad (24.42)$$

The computation rules of (24.41) and (24.42) apply to a column matrix and a row matrix as well. Suppose that we have

$$x = (e_1 \cdots e_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, y = (f_1 \cdots f_m) \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}, \quad (24.43)$$

where $e_1, \dots, e_n$ are the basis vectors of V and $f_1, \dots, f_m$ are the basis vectors of W . Thus, the expression of $y \otimes x$ has already appeared in (24.15) as

$$y \otimes x = (f_1 \cdot e_1 \cdots f_1 \cdot e_n \quad f_2 \cdot e_1 \cdots f_2 \cdot e_n \cdots f_m \cdot e_1 \cdots f_m \cdot e_n) \begin{pmatrix} y_1 x_1 \\ \vdots \\ y_1 x_n \\ y_2 x_1 \\ \vdots \\ y_2 x_n \\ \vdots \\ y_m x_1 \\ \vdots \\ y_m x_n \end{pmatrix}. \quad (24.44)$$

In (24.44), the basis set is $f_p \cdot e_q$ ($1 \leq p \leq m, 1 \leq q \leq n$) and their corresponding coordinates are $y_p x_q$ ($1 \leq p \leq m, 1 \leq q \leq n$). Replacing $\psi \otimes \varphi$ in (24.40) with $B \otimes A$, we obtain

$$(B \otimes A)(y \otimes x) = B(y) \otimes A(x). \quad (24.45)$$

Following the expression of (11.37), we have

$$B(y) = (f'_1 \cdots f'_{m'}) B \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}, \quad A(x) = (e'_1 \cdots e'_{n'}) A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad (24.46)$$

where $e'_1, \dots, e'_{n'}$ are the basis vectors of V' and $f'_1, \dots, f'_{m'}$ are the basis vectors of W' . Thus, (24.45) can be rewritten as

$$\begin{aligned} & \left[(f'_1 \cdots f'_{m'}) B \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} \right] \otimes \left[(e'_1 \cdots e'_{n'}) A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right] \\ &= [(f'_1 \cdots f'_{m'}) \otimes (e'_1 \cdots e'_{n'})] (B \otimes A) \left[\begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} \otimes \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right]. \end{aligned} \quad (24.47)$$

Using (24.41) and (24.44), we get

$$(B \otimes A) \left[\begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} \otimes \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right] = \begin{pmatrix} b_{11}A & \cdots & b_{1m}A \\ \vdots & \ddots & \vdots \\ b_{m'1}A & \cdots & b_{m'm}A \end{pmatrix} \begin{pmatrix} y_1x_1 \\ \vdots \\ y_1x_n \\ y_2x_1 \\ \vdots \\ y_2x_n \\ \vdots \\ y_mx_1 \\ \vdots \\ y_mx_n \end{pmatrix}. \quad (24.48)$$

Notice that since $B \otimes A$ operated on the coordinates from the left with the basis set unaltered, we omitted the basis set $(f'_1 \cdots f'_{m'}) \otimes (e'_1 \cdots e'_n)$ in (24.48). Thus, we only had to perform the numerical calculations based on the standard matrix theory as in RHS of (24.48), where we routinely dealt with a product of a $(m'n', mn)$ matrix and a $(mn, 1)$ matrix, i.e., column vector. The resulting column vector is a $(m'n', 1)$ matrix.

The bilinearity of tensor product of the operators A and B can readily be checked so that we have

$$\begin{aligned} B \otimes (A + A') &= B \otimes A + B \otimes A', \\ (B + B') \otimes A &= B \otimes A + B' \otimes A. \end{aligned} \quad (24.49)$$

The bilinearity of the tensor product of column or row matrices can be checked similarly as well.

Suppose that the dimension of all the vector spaces V , V' , W , and W' is the same n . This simplest case is the most important and frequently appears in many practical applications. We have

$$\begin{aligned} A(x) &= (e_1 \cdots e_n) \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \\ B(y) &= (e_1 \cdots e_n) \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{nn} \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \end{aligned} \quad (24.50)$$

where x and y are given by

$$x = (e_1 \cdots e_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, y = (e_1 \cdots e_n) \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}. \quad (24.51)$$

Now, we are thinking of the endomorphism

$$A : V^n \rightarrow V^n, B : V^n \rightarrow V^n, \quad (24.52)$$

where V^n is a vector space of the dimension n ; A and B are given by

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}, B = \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{nn} \end{pmatrix}. \quad (24.53)$$

Then, replacing φ and ψ of (24.40) with A and B as before, respectively, we obtain

$$(B \otimes A)(y \otimes x) = B(y) \otimes A(x). \quad (24.54)$$

Following (11.37) once again, we rewrite (24.54) such that

$$\begin{aligned} & (B \otimes A)(y \otimes x) \\ &= [(e_1 \cdots e_n) \otimes (e_1 \cdots e_n)][B \otimes A] \left[\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \otimes \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right], \end{aligned} \quad (24.55)$$

where $B \otimes A$ is given by a (n^2, n^2) matrix expressed as

$$B \otimes A = \begin{pmatrix} b_{11}A & \cdots & b_{1n}A \\ \vdots & \ddots & \vdots \\ b_{n1}A & \cdots & b_{nn}A \end{pmatrix}. \quad (24.56)$$

The structure of (24.55) is the same as (24.45) in point of the linear mapping. Also, (24.56) has a same constitution as (24.41), except that the former gives a (n^2, n^2) matrix, whereas the latter is a $(m'n', mn)$ matrix. In other words, instead of (24.48) we have a linear mapping $B \otimes A$ described by

$$B \otimes A : V^{n^2} \rightarrow V^{n^2}. \quad (24.57)$$

In this regard, the tensor product produces an enlarged vector space. As shown above, we use a term of tensor product to represent a direct product of two operators (or matrices) besides a direct product of two vector spaces. In relation to the latter case, we use the same term of tensor product to represent a direct product of row vectors and that of column vectors (vide supra).

A following example helps us understand how to construct a tensor product.

Example 24.2 We have $x = (e_1 \ e_2) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ and $y = (e_1 \ e_2) \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$. Also, let A and B be

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}. \quad (24.58)$$

Then, we get

$$\begin{aligned} (B \otimes A)(y \otimes x) &= [(e_1 \ e_2) \otimes (e_1 \ e_2)][B \otimes A] \left[\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \otimes \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right] \\ &= (e_1 \cdot e_1 \quad e_1 \cdot e_2 \quad e_2 \cdot e_1 \quad e_2 \cdot e_2) \times \\ &\quad \begin{pmatrix} b_{11}a_{11} & b_{11}a_{12} & b_{12}a_{11} & b_{12}a_{12} \\ b_{11}a_{21} & b_{11}a_{22} & b_{12}a_{21} & b_{12}a_{22} \\ b_{21}a_{11} & b_{21}a_{12} & b_{22}a_{11} & b_{22}a_{12} \\ b_{21}a_{21} & b_{21}a_{22} & b_{22}a_{21} & b_{22}a_{22} \end{pmatrix} \begin{pmatrix} y_1x_1 \\ y_1x_2 \\ y_2x_1 \\ y_2x_2 \end{pmatrix}. \end{aligned} \quad (24.59)$$

Omitting the basis set, we have

$$\begin{aligned} \text{RHS of (24.59)} &= \begin{pmatrix} (b_{11}y_1 + b_{12}y_2)(a_{11}x_1 + a_{12}x_2) \\ (b_{21}y_1 + b_{22}y_2)(a_{11}x_1 + a_{12}x_2) \\ (b_{11}y_1 + b_{12}y_2)(a_{21}x_1 + a_{22}x_2) \\ (b_{21}y_1 + b_{22}y_2)(a_{21}x_1 + a_{22}x_2) \end{pmatrix} \\ &= \begin{pmatrix} \sum_{l,j} b_{1l}a_{1j}y_lx_j \\ \sum_{l,j} b_{1l}a_{2j}y_lx_j \\ \sum_{l,j} b_{2l}a_{1j}y_lx_j \\ \sum_{l,j} b_{2l}a_{2j}y_lx_j \end{pmatrix} \equiv \begin{pmatrix} y'_1x'_1 \\ y_1x'_2 \\ y_2x'_1 \\ y_2x'_2 \end{pmatrix}, \end{aligned} \quad (24.60)$$

where the rightmost side denotes the column vector after operating $B \otimes A$ on $y \otimes x$. Thus, we obtain

$$y'_ix'_k = \sum_{j,l} b_{il}a_{kj}y_lx_j. \quad (24.61)$$

Defining $\mathfrak{Y}_{ik} \equiv y_ix_k$, we get

$$\mathfrak{Y}'_{ik} = \sum_{j,l} b_{il}a_{kj}\mathfrak{Y}_{lj}. \quad (24.62)$$

Equation (24.62) has already represented a general feature of the tensor product. We discuss the transformation properties of tensors later.

24.1.3 Bilinear Form

In the previous sections, we have studied several forms of the bilinear mapping. In this section, we introduce a specific form of the bilinear mapping called a bilinear form.

Definition 24.2 Let V and W be n -dimensional and m -dimensional vector spaces over K , respectively. Let B be a mapping described by

$$B : W \times V \rightarrow K, \quad (24.63)$$

where $W \times V$ is a direct product space (see Sect. 24.1.1); K denotes the field (usually chosen from among a complex number field $\mathbb{C}$ or real number field $\mathbb{R}$). Suppose that $B(\cdot, \cdot)$ satisfies the following conditions [1, 2]:

$$\begin{aligned} B(y + y', x) &= B(y, x) + B(y', x), \\ B(y, x + x') &= B(y, x) + B(y, x'), \\ B(ay, x) &= B(y, ax) = aB(y, x), \quad x, x' \in V; y, y' \in W; a \in K. \end{aligned} \quad (24.64)$$

Then, the mapping B is called a bilinear form.

Equation (24.64) is virtually the same as (24.2). Note, however, that although in (24.2) the range of B is a vector space, in (24.64) that of B is K (i.e., simply a number).

Let $x \in V$ and $y \in W$ be expressed as

$$x = \sum_{i=1}^n \xi_i e_i, \quad y = \sum_{j=1}^m \eta_j f_j. \quad (24.28)$$

Then, we have

$$B(y, x) = \sum_{j=1}^m \sum_{i=1}^n \eta_j \xi_i B(f_j, e_i). \quad (24.65)$$

Thus, $B(\cdot, \cdot)$ is uniquely determined by a (m, n) matrix $B(f_j, e_i) (\in K)$.

24.1.4 Dual Vector Space

So far, we have introduced the bilinear mapping and tensor product as well as related notions. The vector spaces, however, were “structureless,” as in the case of the vector spaces where the inner product was not introduced (see Part III). Now, we wish to introduce the concept of the dual (vector) space to equip the vector space with a “metric.”

Definition 24.3 Let V be a vector space over a field K . Then, a collection of arbitrary homogeneous linear functionals f defined on V that map $x \in V$ to K is called a dual vector space (or dual space) of V .

This simple but abstract definition needs some explanation. The dual space of V is written as V^* . With $\forall f \in V^*$ and $\forall g \in V^*$, we define the sum $f + g$ and multiplication αf ($\alpha \in K$) as

$$\begin{aligned}(f + g)(x) &\equiv f(x) + g(x), \\ (\alpha f)(x) &\equiv \alpha \cdot f(x), x \in V.\end{aligned}\tag{24.66}$$

Then, V^* forms a vector space over K with respect to the operation of (24.66). The dimensionality of V^* is the same as that of V [1, 2]. To show this, we make a following argument: Let $\forall x \in V$ be expressed as

$$x = \sum_{i=1}^n \xi^i e_i,$$

where $\xi^i \in K$ and e_i ($i = 1, \dots, n$) is a basis set of a n -dimensional vector space V . Also, for a future purpose, we denote the “coordinate” that corresponds to e_i by a character with a superscript such as ξ^i . The element $x \in V$ is uniquely represented by a set $\{\xi^1, \dots, \xi^n\}$ (see Sect. 11.1). Let $f^i \in V^*$ ($f^i : V \rightarrow K$) be defined as

$$f^i(x) = \xi^i.\tag{24.67}$$

Accordingly, we have

$$x = \sum_{i=1}^n f^i(x) e_i \quad (i = 1, \dots, n).\tag{24.68}$$

Meanwhile, we have

$$f^i(x) = f^i\left(\sum_{k=1}^n \xi^k e_k\right) = \sum_{k=1}^n \xi^k f^i(e_k).\tag{24.69}$$

For (24.67) and (24.69) to be consistent, we get

$$f^i(e_k) = \delta_k^i.\tag{24.70}$$

A set of $\{f^1, \dots, f^n\}$ is a basis set of V^* and called a *dual basis* with respect to $\{e_1, \dots, e_n\}$. As we have already seen in Sect. 11.4, we can choose another basis set of V^* that consists of n linearly independent vectors. Accordingly, the dual basis that satisfies (24.70) is sometimes referred to as a standard basis or canonical basis to distinguish it from other dual bases of V^* . We have

$$V^* = \text{Span} \{f^1, \dots, f^n\}$$

in correspondence with

$$V = \text{Span} \{e_1, \dots, e_n\}.$$

Thus, $\dim V^* = n$. We have $V^* \cong V$ (i.e., isomorphic) accordingly [1, 2]. In this case, we must have a bijective mapping (or isomorphism) between V and V^* , especially between $\{e_1, \dots, e_n\}$ and $\{f^1, \dots, f^n\}$ (Sects. 11.2, 11.4, and 16.4). We define the bijective mapping $\mathfrak{N} : V \rightarrow V^*$ as

$$\mathfrak{N}(e_k) = f^k \quad (k = 1, \dots, n). \quad (24.71)$$

Then, there must be an inverse mapping $\mathfrak{N}^{-1}$ to $\mathfrak{N}$ (see Sect. 11.2). That is, we have

$$e_k = \mathfrak{N}^{-1}(f^k) \quad (k = 1, \dots, n).$$

Now, let $y \in V^*$ be expressed as

$$y = \sum_{i=1}^n \eta_i f^i,$$

where $\eta_i \in K$. As in the case of (24.68), we have

$$y = \sum_{i=1}^n e_i(y) f^i \quad (i = 1, \dots, n), \quad (24.72)$$

with

$$e_i(y) = \eta_i, \quad (24.73)$$

where $e_i \in V$ ($e_i : V^* \rightarrow K$).

As discussed above, we find that the vector spaces V and V^* stand in their own right on an equal footing. Whereas residents in V see V^* as a dream world, those in V^* see V as it is as well. Nonetheless, the two vector spaces are complementary to each other, which can clearly be seen from (24.67), (24.68), (24.72), and (24.73). These situations can be formulated as a recurrence property such that

$$(V^*)^* = V^{**} = V. \quad (24.74)$$

Notice, however, that (24.74) is not necessarily true of an infinite dimensional vector space [1, 2].

A bilinear form is naturally introduced as in (24.64) to the direct product $V^* \times V$. Although the field K can arbitrarily be chosen, from here on we assume $K = \mathbb{R}$ (a real

number field) for a future purpose. Then, as in (24.65), $B(\cdot, \cdot)$ is uniquely determined by a (n, n) matrix $B(\mathfrak{f}^j, e_i) (\in \mathbb{R})$, where $\mathfrak{f}^j (j = 1, \dots, n) (\in V^*)$ are linearly independent in V^* and form a dual basis (vide supra). The other way around, $e_k (k = 1, \dots, n) (\in V)$ are linearly independent in V . To denote these bilinear forms $B(\cdot, \cdot)$, we use a shorthand notation described by

$$[\cdot | \cdot],$$

where an element of V^* and that of V are written in the left-side and right-side “parentheses,” respectively. Also, to clearly specify the nature of a vector species, we use $|\cdot|$ for an element of V and $[\cdot|$ for that of V^* as in the case of a vector of the inner product space (see Chap. 13). Notice, however, that $[\cdot | \cdot]$ is different from the inner product $\langle \cdot | \cdot \rangle$ in the sense that although $\langle \cdot | \cdot \rangle$ is positive definite, it is not necessarily true of $[\cdot | \cdot]$. Using the shorthand notation, e.g., (24.70), can be succinctly expressed as

$$[\mathfrak{f}^j | e_k] = \delta_k^j. \quad (24.75)$$

Because of the isomorphism between V and V^* , $|x|$ is mapped to $[x|$ by the bijective mapping and vice versa. We call $[x|$ a dual vector of $|x|$.

Here we briefly compare the bilinear form with the inner product (Chap. 13). Unlike the inner product space, the basis vectors $[x|$ do not imply the concept of adjoint with respect to the vector spaces we are thinking of now. Although $\langle x | x \rangle$ is positive definite for any $|x| \in V$, the positive definiteness is not necessarily true of $[x | x]$. Think of a following simple example. In case $K = \mathbb{C}$, we have $[x | x] = [ix | -ix] = -[ix | ix]$. Hence, if $[x | x] > 0$, $[ix | ix] < 0$. Also, we may have a case where $[x | x] = 0$ with $|x| \neq 0$ (vide infra). We will come back to this issue later. Hence, it is safe to say that whereas the inner product buys the positive definiteness at the cost of the bilinearity, the bilinear form buys the bilinearity at the cost of the positive definiteness.

The relation represented by (24.70) is somewhat trivial and, hence, it will be convenient to define another bilinear form $B(\cdot, \cdot)$. Meanwhile, in mathematics and physics, we often deal with invariants before and after the vector (or coordinate) transformation (see Sect. 24.1.5). For this purpose, we wish to choose a basis set of V^* more suited than $\mathfrak{f}^k (k = 1, \dots, n)$. Let us define the said basis set of V^* as

$$\{\mathfrak{f}_1, \dots, \mathfrak{f}_n\}.$$

Also, we define a linear mapping $\mathfrak{M}$ such that

$$\mathfrak{M}(e_k) = \mathfrak{f}_k \quad (k = 1, \dots, n), \quad (24.76)$$

where $\mathfrak{M} : V \rightarrow V^*$ is a bijective mapping. Expressing vectors $|x| (\in V)$ and $[x | (\in V^*)$ as

$$|x\rangle = \sum_{i=1}^n \xi^i |e_i\rangle, \quad [x| = \sum_{i=1}^n \xi^i [f_i|, \quad \xi^i \in \mathbb{R}, \quad (24.77)$$

respectively, and using a matrix representation, we express $[x|x]$ as

$$[x|x] = (\xi^1 \cdots \xi^n) \begin{pmatrix} [f_1|e_1] & \cdots & [f_1|e_n] \\ \vdots & \ddots & \vdots \\ [f_n|e_1] & \cdots & [f_n|e_n] \end{pmatrix} \begin{pmatrix} \xi^1 \\ \vdots \\ \xi^n \end{pmatrix}. \quad (24.78)$$

Note that from the definition of $\mathfrak{M}$, we may write

$$\mathfrak{M}|x\rangle = [x|. \quad (24.79)$$

Defining a matrix M as [1, 2]

$$M \equiv \begin{pmatrix} [f_1|e_1] & \cdots & [f_1|e_n] \\ \vdots & \ddots & \vdots \\ [f_n|e_1] & \cdots & [f_n|e_n] \end{pmatrix}, \quad (24.80)$$

we obtain

$$[x|x] = (\xi^1 \cdots \xi^n) M \begin{pmatrix} \xi^1 \\ \vdots \\ \xi^n \end{pmatrix}. \quad (24.81)$$

We rewrite (24.81) as

$$[x|x] = \sum_{i,j} \xi^i M_{ij} \xi^j = \sum_k \xi^k M_{kk} \xi^k + \sum_{l,m; l \neq m} \xi^l M_{lm} \xi^m, \quad (24.82)$$

where we put $M = (M)_{ij}$. The second term of (24.82) can be written as

$$\begin{aligned} \sum_{l,m; l \neq m} \xi^l M_{lm} \xi^m &= \frac{1}{2} \left(\sum_{l,m; l \neq m} \xi^l M_{lm} \xi^m + \sum_{m,l; m \neq l} \xi^m M_{ml} \xi^l \right) \\ &= \frac{1}{2} \sum_{l,m; l \neq m} (M_{lm} + M_{ml}) \xi^l \xi^m. \end{aligned}$$

Therefore, defining $\tilde{M}_{ij} \equiv \frac{1}{2} (M_{ij} + M_{ji})$, we get a symmetric quadratic form described by

$$[x|x] = \sum_k \xi^k \tilde{M}_{kk} \xi^k + \sum_{l,m; l \neq m} \tilde{M}_{lm} \xi^l \xi^m.$$

Then, we assume that M is symmetric from the beginning, i.e., $[f_i|e_j] = [f_j|e_i]$ [1, 2]. Thus, M is real symmetric matrix (note $K = \mathbb{R}$), and so it is to be diagonalized through orthogonal similarity transformation. That is, M can be diagonalized with an orthogonal matrix $P = (P^i_j)$. We have

$$\begin{aligned}
 P^T M P &= \begin{pmatrix} P^1_1 & \cdots & P^n_1 \\ \vdots & \ddots & \vdots \\ P^1_n & \cdots & P^n_n \end{pmatrix} \begin{pmatrix} [f_1|e_1] & \cdots & [f_1|e_n] \\ \vdots & \ddots & \vdots \\ [f_n|e_1] & \cdots & [f_n|e_n] \end{pmatrix} \begin{pmatrix} P^1_1 & \cdots & P^1_n \\ \vdots & \ddots & \vdots \\ P^n_1 & \cdots & P^n_n \end{pmatrix} \\
 &= \begin{pmatrix} \left[\sum_j f_j P^j_1 | \sum_i e_i P^i_1 \right] & \cdots & \left[\sum_j f_j P^j_1 | \sum_i e_i P^i_n \right] \\ \vdots & \ddots & \vdots \\ \left[\sum_j f_j P^j_n | \sum_i e_i P^i_1 \right] & \cdots & \left[\sum_j f_j P^j_n | \sum_i e_i P^i_n \right] \end{pmatrix} \\
 &= \begin{pmatrix} [f'_1|e'_1] & \cdots & [f'_1|e'_n] \\ \vdots & \ddots & \vdots \\ [f'_n|e'_1] & \cdots & [f'_n|e'_n] \end{pmatrix} = \begin{pmatrix} d_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_n \end{pmatrix}, \tag{24.83}
 \end{aligned}$$

where summation with respect to both i and j ranges from 1 to n . In (24.83), we have

$$|e'_k\rangle = P(e_k) = \sum_i P^i_k |e_i\rangle, \quad [f'_k| = P(f_k) = \sum_j P^j_k [f_j|, \quad [f'_j|e'_i] = d_j \delta_{ji}, \tag{24.84}$$

where d_j ($j = 1, \dots, n$) are real eigenvalues of M . Equation (24.84) shows that both the basis sets of $|e_i\rangle$ and $[f_j|$ ($i, j = 1, \dots, n$) of V and V^* , respectively, undergo the orthogonal transformation by P so that it can be converted into new basis sets $|e'_i\rangle$ and $[f'_j|$. The number d_j takes a positive value or a negative value.

Further, interchanging the basis set $|e_i\rangle$ and similarly exchanging $[f_j|$, the resulting diagonalized matrix of (24.83) is converted through orthogonal transformation $\Pi = Q_1 \cdots Q_s$ into

$$\begin{pmatrix} d_{i_1} & & & & & \\ & \ddots & & & & \\ & & d_{i_p} & & & \\ & & & d_{i_{p+1}} & & \\ & & & & \ddots & \\ & & & & & d_{i_{p+q}} \end{pmatrix},$$

where $d_{i_1}, \dots, d_{i_p}$ are positive; $d_{i_{p+1}}, \dots, d_{i_{p+q}}$ are negative; we have $p + q = n$. What we have done in the above process is to rearrange the order of the basis vectors s times so that the positive $d_{i_1}, \dots, d_{i_p}$ can be arranged in the upper diagonal elements and the negative $d_{i_{p+1}}, \dots, d_{i_{p+q}}$ can be arranged in the lower diagonal elements.

Correspondingly, we obtain

$$(|e'_1| \cdots |e'_n|) \Pi = (|e'_{i_1}| \cdots |e'_{i_{p+q}}|), ([f'_1| \cdots |f'_n|) \Pi = ([f'_{i_1}| \cdots |f'_{i_{p+q}}|). \quad (24.85)$$

Assuming a following non-singular matrix D expressed as [1, 2]

$$D = \begin{pmatrix} 1/\sqrt{d_{i_1}} & & & & & \\ & \ddots & & & & \\ & & 1/\sqrt{d_{i_p}} & & & \\ & & & 1/\sqrt{-d_{i_{p+1}}} & & \\ & & & & \ddots & \\ & & & & & 1/\sqrt{-d_{i_{p+q}}} \end{pmatrix}, \quad (24.86)$$

we get

$$\begin{aligned} D^T \Pi^T P^T M P \Pi D &= (P \Pi D)^T M P \Pi D \\ &= \begin{pmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & & -1 & & \\ & & & & \ddots & \\ & & & & & -1 \end{pmatrix}. \end{aligned} \quad (24.87)$$

The numbers p and q are intrinsic to the matrix M . Since M is a real symmetric matrix, M can be diagonalized with the similarity transformation by an orthogonal matrix. But, since the eigenvalues (given by diagonal elements) are unchanged after the similarity transformation, it follows that p (i.e., the number of the positive eigenvalues) and q (the number of the negative eigenvalues) are uniquely determined by M . The integers p and q are called a *signature* [1, 2]. The uniqueness of (p, q) is known as the Sylvester's law of inertia [1, 2].

We add as an aside that if M is a real symmetric singular matrix, M must have an eigenvalue 0. In case the eigenvalue zero is degenerated r -fold, we have $p + q + r = n$. In that case, p , q , and r are also uniquely determined and these integers are called a signature as well. This case is also included as the Sylvester's law of inertia [1, 2].

Note that (24.87) is *not* a similarity transformation, but an equivalence transformation and, hence, that the eigenvalues have been changed [from d_j ($\neq 0$) to ± 1]. Also, notice that unlike the case of the equivalence transformation with an arbitrarily chosen non-singular matrix mentioned in Sect. 14.5, (24.87) represents an equivalence transformation as a special case. That is, the non-singular operator comprises a product of an orthogonal matrix $P \Pi$ and a diagonal matrix D .

The above-mentioned equivalence transformation also causes the transformation of the basis set. Using (24.86), we have

$$\begin{aligned} (|e'_1| \cdots |e'_{i_{p+q}}|) D &= (|e'_{i_1}| / \sqrt{d_{i_1}} \cdots |e'_{i_{p+q}}| / \sqrt{-d_{i_{p+q}}}), \\ ([f'_{i_1}| \cdots [f'_{i_{p+q}}|) D &= ([f'_{i_1}| / \sqrt{d_{i_1}} \cdots [f'_{i_{p+q}}| / \sqrt{-d_{i_{p+q}}}). \end{aligned}$$

Defining

$$|\tilde{e}_{i_k}\rangle \equiv |e'_{i_k}\rangle / \sqrt{\pm d_{i_k}} \text{ and } |\tilde{f}_{i_k}\rangle \equiv |f'_{i_k}\rangle / \sqrt{\pm d_{i_k}} \quad (k=1, \dots, n=p+q),$$

we find that both the basis sets of V and V^* finally undergo the same transformation through the equivalence transformation by $P\Pi D$ such that

$$\begin{aligned} (|\tilde{e}_{i_1}\rangle \cdots |\tilde{e}_{i_n}\rangle) &= (|e_1\rangle \cdots |e_n\rangle) P\Pi D, \\ (|\tilde{f}_{i_1}\rangle \cdots |\tilde{f}_{i_n}\rangle) &= (|f_1\rangle \cdots |f_n\rangle) P\Pi D \end{aligned} \quad (24.88)$$

with

$$|\tilde{f}_j|\tilde{e}_i\rangle = \pm \delta_{ji} \quad (i, j \text{ take a number out of } i_1, \dots, i_p, i_{p+1}, \dots, i_{p+q}). \quad (24.89)$$

Putting $W \equiv P\Pi D$, from (24.81) we have

$$\begin{aligned} [x|x] &= (\xi^1 \cdots \xi^n) (W^T)^{-1} \cdot W^T M W \cdot W^{-1} \begin{pmatrix} \xi^1 \\ \vdots \\ \xi^n \end{pmatrix} \\ &= (\xi^1 \cdots \xi^n) (W^{-1})^T \cdot W^T M W \cdot W^{-1} \begin{pmatrix} \xi^1 \\ \vdots \\ \xi^n \end{pmatrix}. \end{aligned}$$

Just as the basis sets are transformed by W , so the coordinates are transformed according to $W^{-1} = D^{-1}\Pi^{-1}P^{-1} = D^{-1}\Pi^T P^T$. Then, consecutively we have

$$\xi'^k = \sum_i (P^{-1})^k_i \xi^i, \quad \xi'^{i_k} = \sum_{i_m} (\Pi^{-1})^{i_k}_{i_m} \xi'^{i_m},$$

$$\tilde{\xi}^{i_k} = \sum_{i_m} (D^{-1})^{i_k}_{i_m} \xi'^{i_m} = \sum_{i_m} \sqrt{\pm d_{i_k}} \delta^{i_k}_{i_m} \xi'^{i_m} = \sqrt{\pm d_{i_k}} \xi'^{i_k}. \quad (24.90)$$

Collectively writing (24.90), we obtain

$$\begin{pmatrix} \tilde{\xi}^{i_1} \\ \vdots \\ \tilde{\xi}^{i_n} \end{pmatrix} = W^{-1} \begin{pmatrix} \xi^1 \\ \vdots \\ \xi^n \end{pmatrix} = D^{-1}\Pi^{-1}P^{-1} \begin{pmatrix} \xi^1 \\ \vdots \\ \xi^n \end{pmatrix}. \quad (24.91)$$

Thus, (24.88) and (24.91) afford the transformation of the basis vectors and corresponding coordinates. These relations can be compared to (11.85).

Now, if the bilinear form $[y|x]$ ($x \in V, y \in V^*$) described by [1, 2]

$$[y|x] = (\eta^1 \cdots \eta^n) M \begin{pmatrix} \xi^1 \\ \vdots \\ \xi^n \end{pmatrix}$$

satisfies the following conditions, the bilinear form is said to be non-degenerate. That is, (i) if for $\forall x \in V$ we have $[y|x] = 0$, then we must have $y = 0$. Also, (ii) if for $\forall y \in V^*$ we have $[y|x] = 0$, then we must have $x = 0$. The above condition (i) is translated into

$$(\eta^1 \cdots \eta^n)M = 0 \Rightarrow (\eta^1 \cdots \eta^n) = 0.$$

This implies $\det M \neq 0$. Namely, M is non-singular. From the condition (ii) which implies that

$$M \begin{pmatrix} \xi^1 \\ \vdots \\ \xi^n \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} \xi^1 \\ \vdots \\ \xi^n \end{pmatrix} = 0,$$

we get $\det M \neq 0$ as well. The determinant of the matrix for RHS of (24.87) is ± 1 .

Thus, the non-degeneracy of the bilinear form is equivalent to that M has only non-zero eigenvalues. It is worth comparing two vector spaces, the one of which is an inner product space V_{in} and the other of which is a vector space V_{bi} where the bilinear form is defined.

Let $x \in V_{\text{in}}$ and $y \in V_{\text{in}}^*$, where V_{in}^* is a dual space of V_{in} . The inner product can be formulated by a mapping $\mathfrak{I}$ such that

$$\mathfrak{I} : V_{\text{in}}^* \times V_{\text{in}} \rightarrow \mathbb{C}.$$

In that case, we have described a resulting complex number of the above expression as an inner product $\langle y|x \rangle$ and we have defined the computation rule of the inner product as (13.2)–(13.4). Regarding two sets of vectors of V_{in} , we have defined a Gram matrix G as (13.35) in Sect. 13.2. There, we have shown that

$$\text{Both the sets of vectors of } V_{\text{in}} \text{ are linearly independent.} \Leftrightarrow \det G \neq 0. \quad (13.31)$$

As for the vector space V_{bi} , the matrix M defined in (24.80) is a counterpart to the Gram matrix G . With respect to two sets of basis vectors of V_{bi} and its dual space V_{bi}^* , we have an expression related to (13.31). That is, we have [1, 2]

$$\begin{aligned} \text{The bilinear form } B \text{ given by } B : V_{\text{bi}}^* \times V_{\text{bi}} &\rightarrow K \text{ } (K = \mathbb{C} \text{ or } \mathbb{R}) \\ &\text{is non-degenerate.} \Leftrightarrow \det M \neq 0. \end{aligned} \quad (24.92)$$

Let us consider a case where we construct M as in (24.80) from two sets of basis vectors of V_{bi} and V_{bi}^* given by

$$\{|e_1], \cdots, |e_n]; |e_i] \in V_{\text{bi}} \text{ } (i = 1, \cdots, n)\},$$

$$\{[f_1 |, \dots, [f_n | ; [f_i | \in V_{bi}^* (i=1, \dots, n)\}.$$

We assume that these two sets are connected to each other through (24.76). Suppose that we do not know whether these sets are linearly dependent or independent. However, if either of the above two sets of vectors is linearly independent, so is the remainder set. Think of the following relation:

$$\begin{aligned} \xi^1 |e_1] + \dots + \xi^n |e_n] &= 0 \\ \downarrow \mathfrak{M} \uparrow \mathfrak{M}^{-1} \\ \xi^1 [f_1 | + \dots + \xi^n [f_n | &= 0. \end{aligned}$$

In the above relation, $\mathfrak{M}$ has been defined in (24.76). If $|e_1], \dots, |e_n]$ are linearly independent, we obtain $\xi^1 = \dots = \xi^n = 0$. Then, it necessarily follows that $[f_1 |, \dots, [f_n |$ are linearly independent as well, and vice versa. If, the other way around, either of the above two sets of vectors is linearly dependent, so is the remainder set. This can be shown in a manner similar to the above. Nonetheless, even though we construct M using these two basis sets, we do not necessarily have $\det M \neq 0$. To assert $\det M \neq 0$, we need the condition (24.92).

In place of M of (24.80), we use a character $\mathcal{M}$ and define (24.87) anew as

$$\mathcal{M} \equiv \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & & -1 & \\ & & & & \ddots \\ & & & & & -1 \end{pmatrix}. \quad (24.93)$$

Equation (24.93) can be rewritten as

$$\begin{aligned} (\mathcal{M})_{ij} &= -\delta_{ij} \zeta_i = -\delta_{ij} \zeta_j; \zeta_1 =, \dots, = \zeta_p = -\zeta_{p+1} =, \dots, = -\zeta_{p+q} = \\ &-1. \end{aligned} \quad (24.94)$$

Meanwhile, we redefine the notation $\tilde{e}$, $\tilde{f}$, and $\tilde{\xi}$ as e , f , and ξ , respectively, and renumber their subscripts (or superscripts) in (24.88) and (24.90) such that

$$\begin{aligned} \{e_1, \dots, e_n\} &\leftarrow \{\tilde{e}_{i_1}, \dots, \tilde{e}_{i_n}\}, \{f_1, \dots, f_n\} \leftarrow \{\tilde{f}_{i_1}, \dots, \tilde{f}_{i_n}\}, \\ \{\xi^1, \dots, \xi^n\} &\leftarrow \{\tilde{\xi}^{i_1}, \dots, \tilde{\xi}^{i_n}\}. \end{aligned} \quad (24.95)$$

Then, from (24.89) and (24.95), $\mathcal{M}$ can be rewritten as

$$(\mathcal{M})_{ji} = [f_j | e_i] = \pm \delta_{ji} \quad (1 \leq i, j \leq n).$$

The matrix $\mathcal{M}$ is called *metric tensor*. This terminology is, however, somewhat misleading, because $\mathcal{M}$ is merely a matrix consisting of real numbers. Care should be taken accordingly. If all the eigenvalues of $\mathcal{M}$ are positive, the quadratic form $[x|x]$ is positive definite (see Sect. 14.5). In that case, the metric tensor $\mathcal{M}$ can be expressed using an inner product such that

$$(\mathcal{M})_{ji} = [\mathfrak{f}_j | e_i] = \langle \mathfrak{f}_j | e_i \rangle = \delta_{ji} \quad (1 \leq i, j \leq n). \quad (24.96)$$

This is ensured by Theorem 13.2 (Gram–Schmidt orthonormalization Theorem) and features the Euclidean space. That space is characterized by the Euclidean metric given by (24.96). If, on the other hand, $\mathcal{M}$ contains both positive and negative eigenvalues, $[x|x]$ can be positive or negative or zero. In the latter case, $[x|x]$ is said to be an indefinite (Hermitian) quadratic form. As a typical example for this, we have the four-dimensional Minkowski space where the metric tensor $\mathcal{M}$ is represented by

$$\mathcal{M} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (24.97)$$

The matrix $\mathcal{M}$ is identical with η of (21.16) that has already appeared in Chap. 21.

24.1.5 Invariants

The metric tensor $\mathcal{M}$ plays a fundamental role in a n -dimensional real vector space. It is indispensable for studying the structure and properties of the vector space. One of the most important points is that $\mathcal{M}$ has a close relationship to the invariants of the vector field. First, among various operators that operate on a vector, we wish to examine what kinds of operators hold $[x|x]$ invariant after their operation on vectors $|x\rangle$ and $\langle x|$ belonging to V or V^* , respectively.

Let the relevant operator be R . Operating R on

$$|x\rangle = \sum_{i=1}^n \xi^i |e_i\rangle = (|e_1\rangle \cdots |e_n\rangle) \begin{pmatrix} \xi^1 \\ \vdots \\ \xi^n \end{pmatrix} \quad (\in V),$$

we obtain

$$R|x\rangle = |Rx\rangle = \sum_{i,j=1}^n R^i_j \xi^j |e_i\rangle, \quad (24.98)$$

where e_i has been defined in (24.95). Operating $\mathfrak{M}$ of (24.76) on both sides of (24.98), we obtain

$$[Rx | = \sum_{i,j=1}^n R_{ij}^i \xi^j [\mathfrak{f}_i | . \quad (24.99)$$

Then, we get

$$[Rx|Rx] = \sum_{i,j=1}^n R_{ij}^i \xi^j \left(\sum_{k,l=1}^n R_{kl}^k \xi^l [\mathfrak{f}_i | e_k] \right) = \sum_{i,j,k,l=1}^n \xi^j R_{ij}^i \mathcal{M}_{ik} R_{kl}^k \xi^l, \quad (24.100)$$

where with the last equality we used (24.80) and $\mathcal{M}_{ik}$ represents (i, k) -element of $\mathcal{M}$ given in (24.93), i.e., $\mathcal{M}_{ik} = \pm \delta_{ik}$. Meanwhile, we have

$$[x|x] = \sum_{j,l=1}^n \xi^j \mathcal{M}_{jl} \xi^l. \quad (24.101)$$

From (24.100) and (24.101), we have

$$[Rx|Rx] - [x|x] = \sum_{i,j,k,l=1}^n \xi^j (R_{ij}^i \mathcal{M}_{ik} R_{kl}^k - \mathcal{M}_{jl}) \xi^l. \quad (24.102)$$

Since ξ^j and ξ^l can arbitrarily be chosen from $\mathbb{R}$, we must have

$$\sum_{i,k=1}^n R_{ij}^i \mathcal{M}_{ik} R_{kl}^k - \mathcal{M}_{jl} = 0 \quad (24.103)$$

so that $[Rx|Rx] - [x|x] = 0$ (namely, so that $[x|x]$ can be invariant before and after the transformation by R). In a matrix form, (24.103) can be rewritten as

$$R^T \mathcal{M} R = \mathcal{M}. \quad (24.104)$$

Remind that an equation related to (24.104) has already appeared in Sect. 13.3 in relation to the Gram matrix and adjoint operator; see (13.81). If real matrices are chosen for R and G , (13.81) is translated into

$$R^T G R = G.$$

Hence, this expression is virtually the same as (24.104).

Even though (24.103) and (24.104) are equivalent, (24.103) clearly shows a covariant characteristic (vide infra). The transformation R plays a fundamental role in a tensor equation, i.e., an equation which contains tensor(s). In what follows, the transformation of the tensors and tensor equation along with related invariants will form the central theme.

First, we think about the transformation of the basis vectors and its corresponding coordinates. In Sect. 11.4, we described the transformation as

$$\begin{aligned}
 \mathbf{x} &= (\mathbf{e}_1 \cdots \mathbf{e}_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \\
 &= (\mathbf{e}_1 \cdots \mathbf{e}_n) P P^{-1} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (\mathbf{e}'_1 \cdots \mathbf{e}'_n) P^{-1} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (\mathbf{e}'_1 \cdots \mathbf{e}'_n) \begin{pmatrix} \tilde{x}_1 \\ \vdots \\ \tilde{x}_n \end{pmatrix}. \quad (11.84)
 \end{aligned}$$

Here, we rewrite this relationship as

$$\mathbf{x} = (\mathbf{e}_1 \cdots \mathbf{e}_n) \begin{pmatrix} \xi^1 \\ \vdots \\ \xi^n \end{pmatrix} = (\mathbf{e}_1 \cdots \mathbf{e}_n) R^{-1} \cdot R \begin{pmatrix} \xi^1 \\ \vdots \\ \xi^n \end{pmatrix} = (\mathbf{e}'_1 \cdots \mathbf{e}'_n) \begin{pmatrix} \xi'^1 \\ \vdots \\ \xi'^n \end{pmatrix}. \quad (24.105)$$

Note that in (24.105) R was used instead of P^{-1} in (11.84). This is because in this section we place emphasis on the coordinate transformation. Thus, as the transformation of the basis set and its corresponding coordinates of V , we have

$$(\mathbf{e}'_1 \cdots \mathbf{e}'_n) = (\mathbf{e}_1 \cdots \mathbf{e}_n) R^{-1} \quad \text{or} \quad \mathbf{e}'_i = \sum_{j=1}^n \mathbf{e}_j (R^{-1})^j_i. \quad (24.106)$$

Also, we obtain

$$\begin{pmatrix} \xi'^1 \\ \vdots \\ \xi'^n \end{pmatrix} = R \begin{pmatrix} \xi^1 \\ \vdots \\ \xi^n \end{pmatrix} \quad \text{or} \quad \xi'^i = \sum_{j=1}^n R^i_j \xi^j. \quad (24.107)$$

We assume that the transformation of the dual basis $\mathbf{f}^i$ of V^* is described such that

$$\mathbf{f}^i = \sum_{j=1}^n Q^i_j \mathbf{f}^j. \quad (24.108)$$

We wish to decide Q so that it can satisfy (24.70)

$$[\mathbf{f}^i | \mathbf{e}_j] = [\mathbf{f}'^i | \mathbf{e}'_j] = \delta^i_j. \quad (24.109)$$

Inserting $\mathbf{e}'_j$ of (24.106) and $\mathbf{f}'^i$ of (24.108) into (24.109), we get

$$\begin{aligned}
 [\mathbf{f}^i | \mathbf{e}'_j] &= [Q^i_l \mathbf{f}^l | \mathbf{e}_k (R^{-1})^k_j] = Q^i_l (R^{-1})^k_j [\mathbf{f}^l | \mathbf{e}_k] = Q^i_l (R^{-1})^k_j \delta^l_k \\
 &= Q^i_l (R^{-1})^l_j = [\mathbf{f}^i | \mathbf{e}_j] = \delta^i_j. \quad (24.110)
 \end{aligned}$$

Representing (24.110) in a matrix form, we obtain

$$QR^{-1} = E. \quad (24.111)$$

Hence, we get $Q = R$. Thus, we find that f^i undergoes the same transformation as ξ^j . It is said to be a contravariant transformation.

Since both $\mathfrak{N}$ of (24.71) and $\mathfrak{M}$ of (24.76) are bijective mappings and $\{f^1, \dots, f^n\}$ and $\{f_1, \dots, f_n\}$ are two basis sets of V^* , we must have a non-singular matrix that transforms each other (Sect. 11.4). Let T be the said non-singular matrix. Then, we have

$$f_i = \sum_{j=1}^n T_{ij} f^j \quad (i = 1, \dots, n). \quad (24.112)$$

Also, we have

$$\begin{aligned} [x|x] &= \sum_{j,l=1}^n \xi^j [f_j|e_l] \xi^l = \sum_{j,l=1}^n \xi^j \left[\sum_{k=1}^n T_{jk} f^k | e_l \right] \xi^l \\ &= \sum_{j,l,k=1}^n \xi^j T_{jk} [f^k|e_l] \xi^l = \sum_{j,l,k=1}^n \xi^j T_{jk} \delta_l^k \xi^l = \sum_{j,l=1}^n \xi^j T_{jl} \xi^l. \end{aligned} \quad (24.113)$$

Comparing (24.113) with (24.101) and seeing that ξ^j ($j = 1, \dots, n$) can arbitrarily be chosen, we get

$$T_{jl} = \mathcal{M}_{jl} \quad \text{i.e.,} \quad T = \mathcal{M}.$$

Hence, we have

$$f_i = \sum_{j=1}^n \mathcal{M}_{ij} f^j \quad (i = 1, \dots, n). \quad (24.114)$$

Making a bilinear form between e_k and both sides of (24.114), we obtain

$$[f_i|e_k] = \sum_{j=1}^n \mathcal{M}_{ij} [f^j|e_k] = \sum_{j=1}^n \mathcal{M}_{ij} \delta_k^j = \mathcal{M}_{ik}, \quad (24.115)$$

where with the second equality we used (24.109).

Next, we investigate the transformation property of f_i . Let R be a transformation in a vector space V . Operating R^{-1} on both sides of (24.76), we obtain

$$R^{-1} \mathfrak{M}(e_k) = R^{-1} f_k \quad (k = 1, \dots, n). \quad (24.116)$$

Assuming that R^{-1} and $\mathfrak{M}$ are commutative, we have

$$\mathfrak{M}(R^{-1} e_k) = \mathfrak{M}(e'_k) = R^{-1} f_k = f'_k. \quad (24.117)$$

Operating $\mathfrak{M}$ on (24.106), we obtain

$$\mathfrak{M}(e'_i) = \mathfrak{f}'_i = (R^{-1})^j_i \mathfrak{M}(e_j) = \sum_{j=1}^n (R^{-1})^j_i \mathfrak{f}_j, \quad (24.118)$$

where with the first equality we used (24.117); with the last equality we used (24.76). Rewriting (24.118), we get

$$\mathfrak{f}'_i = \sum_{j=1}^n \mathfrak{f}_j (R^{-1})^j_i. \quad (24.119)$$

Namely, $\mathfrak{f}_j$ exhibits the same transformation property as e_j .

We examine the coordinate transformation in a similar manner. To this end, let us define a following quantity ξ_i such that

$$\xi_l \equiv \sum_{j=1}^n \xi^j_l \mathcal{M}_{jl}. \quad (24.120)$$

Using the definition (24.120), (24.101) is rewritten as

$$[x|x] = \sum_{l=1}^n \xi_l \xi^l. \quad (24.121)$$

Further rewriting (24.121), we get

$$[x|x] = \sum_{l=1}^n \xi_l \xi^l = \sum_{k,l=1}^n \xi_k \delta_l^k \xi^l = \sum_{i,k,l=1}^n \xi_k (R^{-1})^k_i R^i_l \xi^l. \quad (24.122)$$

Defining the transformation of ξ_i as

$$\xi'_i \equiv \sum_{k=1}^n \xi_k (R^{-1})^k_i \quad (24.123)$$

and using (24.107), we obtain

$$[x|x] = \sum_{i=1}^n \xi'_i \xi'^i. \quad (24.124)$$

Consequently, $\sum_{l=1}^n \xi_l \xi^l$ is an invariant quantity with regard to the transformation by R . Thus, (24.106), (24.107), (24.108), (24.119), and (24.123) give the transformation properties of the basis vectors and their corresponding coordinates.

Next, we investigate the transformation property of $\mathcal{M}$ of (24.93). Using (24.106) and (24.119), we have

$$[\mathfrak{f}'_i | e'_k] = \sum_{j,l=1}^n [\mathfrak{f}_j | e_l] (R^{-1})^j_i (R^{-1})^l_k. \quad (24.125)$$

Multiplying both sides by $R^i_s R^k_t$ and summing over i and k , we obtain

$$\begin{aligned}
\sum_{i,k=1}^n R^i{}_s R^k{}_t [f'_i|e'_k] &= \sum_{i,k,j,l=1}^n (R^{-1})^j{}_i R^i{}_s (R^{-1})^l{}_k R^k{}_t [f_j|e_l] \\
&= \sum_{j,l=1}^n \delta^j_s \delta^l_t [f_j|e_l] = [f_s|e_t] = \mathcal{M}_{st} = \sum_{i,k=1}^n R^i{}_s \mathcal{M}_{ik} R^k{}_t,
\end{aligned} \tag{24.126}$$

where with the second to the last equality we used (24.115); with the last equality we used (24.103). Putting $[f'_i|e'_k] = \mathcal{M}'_{ik}$ and comparing the leftmost side and rightmost side of (24.126), we have

$$\sum_{i,k=1}^n R^i{}_s \mathcal{M}'_{ik} R^k{}_t = \sum_{i,k=1}^n R^i{}_s \mathcal{M}_{ik} R^k{}_t. \tag{24.127}$$

Rewriting (24.127) in a matrix form, we obtain

$$R^T \mathcal{M}' R = R^T \mathcal{M} R. \tag{24.128}$$

Since R is a non-singular matrix, we get

$$\mathcal{M}' = \mathcal{M}. \tag{24.129}$$

That is, $\mathcal{M}$ is invariant with respect to the basis vector transformation. Hence, $\mathcal{M}$ is called invariant metric tensor. In other words, the transformation R that satisfies (24.128) and (24.129) makes $[x|x]$ invariant and, hence, plays a major role in the vector space characterized by the invariant metric tensor $\mathcal{M}$.

As can be seen immediately from (24.93), $\mathcal{M}$ is represented as a symmetric matrix and an inverse matrix of $\mathcal{M}^{-1}$ is $\mathcal{M}$ itself. We denote $\mathcal{M}^{-1}$ by

$$\mathcal{M}^{-1} \equiv (\mathcal{M}^{-1})^{ij} \equiv (\mathcal{M})^{ij}.$$

Then, we have

$$\sum_{j=1}^n \mathcal{M}^{ij} \mathcal{M}_{jk} = \delta^i_k \quad \text{or} \quad \sum_{k=1}^n \mathcal{M}_{jk} \mathcal{M}^{ki} = \delta^i_j. \tag{24.130}$$

These notations make various expressions simple. For instance, multiplying both sides of (24.120) by $\mathcal{M}^{ik}$ and summing the result over i we have

$$\sum_{i=1}^n \xi_i \mathcal{M}^{ik} \equiv \sum_{i=1}^n \sum_{j=1}^n \xi^j \mathcal{M}_{ji} \mathcal{M}^{ik} = \sum_{j=1}^n \xi^j \delta_j^k = \xi^k.$$

Multiplying both sides of (24.114) by $\mathcal{M}^{il}$ ($= \mathcal{M}^{li}$) and summing over i , we obtain

$$\sum_{i=1}^n \mathfrak{f}_i \mathcal{M}^{il} = \sum_{i=1}^n \sum_{j=1}^n \mathcal{M}^{li} \mathcal{M}_{ij} \mathfrak{f}^j = \sum_{j=1}^n \delta_j^l \mathfrak{f}^j = \mathfrak{f}^l. \quad (24.131)$$

Moreover, we get

$$\begin{aligned} \sum_{i=1}^n \xi^i \mathfrak{f}_i &= \sum_{l=1}^n \sum_{i=1}^n \xi^i \delta_i^l \mathfrak{f}_l = \sum_{l=1}^n \sum_{k=1}^n \sum_{i=1}^n \xi^i \mathcal{M}_{ik} \mathcal{M}^{kl} \mathfrak{f}_l \\ &= \sum_{k=1}^n \xi_k^k \mathfrak{f}^k, \end{aligned} \quad (24.132)$$

where with the second and last equalities we used (24.130).

In this way, we can readily move the indices up and down. With regard to the basis vector transformation of e.g., (24.114), we can carry out the matrix calculation in a similar way. These manipulations are useful in relation to derivation of the invariants.

24.1.6 Tensor Space and Tensors

In Sect. 24.1.1, we showed the definition and a tangible example of the bilinear mapping. Advancing this notion further, we reach the multilinear mapping [1, 2]. Suppose that a mapping $\mathfrak{S}$ from a direct product space $V_s \times \cdots \times V_1$ to a vector space T is described by

$$\mathfrak{S} : V_s \times \cdots \times V_1 \rightarrow T,$$

where $V_1, \cdots, V_s$ are vector spaces and $V_s \times \cdots \times V_1$ is their direct product space. If with $x_{(i)} \in V_i$ ($i = 1, \cdots, s$), $\mathfrak{S}[x_{(s)}, \cdots, x_{(1)}]$ is linear with respect to each $x_{(i)}$ ($i = 1, \cdots, s$), then $\mathfrak{S}$ is called a multilinear mapping. As a natural extension of the tensor product dealt with in Sect. 24.1.2, we can define a tensor product T in relation to the multilinear mapping. It is expressed as

$$T = V_s \otimes \cdots \otimes V_1. \quad (24.133)$$

Then, in accordance with (24.34), we have

$$\mathfrak{S}[x_{(s)}, \cdots, x_{(1)}] = x_{(s)} \otimes \cdots \otimes x_{(1)}.$$

Also, as a natural extension of Theorem 24.2, we have a following theorem. Although the proof based on the mathematical induction can be seen in literature [1, 2], we skip it and show the theorem here without proof.

Theorem 24.3 [1, 2] Let V_i ($i = 1, \cdots, s$) be vector spaces over K . Let $\mathfrak{T}$ be an arbitrary vector space. Let $\tilde{\mathfrak{S}}$ be an arbitrary multilinear mapping expressed as

$$\tilde{\mathfrak{S}} : V_s \times \cdots \times V_1 \rightarrow \mathfrak{T}.$$

Also, a tensor product is defined such that

$$T = V_s \otimes \cdots \otimes V_1.$$

Then, a linear mapping $\rho : V_s \otimes \cdots \otimes V_1 \rightarrow \mathfrak{T}$ that satisfies the condition described by

$$\tilde{\mathfrak{S}}[x_{(s)}, \cdots, x_{(1)}] = \rho[x_{(s)} \otimes \cdots \otimes x_{(1)}]; x_{(i)} \in V_i \ (i = 1, \cdots, s) \quad (24.134)$$

is uniquely determined.

As in the case of the bilinear mapping, we have linear mappings φ_i ($i = 1, \cdots, s$) such that

$$\varphi_i : V_i \rightarrow V'_i \ (i = 1, \cdots, s). \quad (24.135)$$

Also, suppose that we have

$$\mathfrak{T} = V'_s \otimes \cdots \otimes V'_1. \quad (24.136)$$

Notice that (24.136) is an extension of (24.36). Then, from (24.133) to (24.136), we can construct a multilinear mapping $\tilde{\mathfrak{S}} : V_s \times \cdots \times V_1 \rightarrow \mathfrak{T}$ as before such that

$$\tilde{\mathfrak{S}}[x_{(s)}, \cdots, x_{(1)}] = \varphi_s[x_{(s)}] \otimes \cdots \otimes \varphi_1[x_{(1)}]. \quad (24.137)$$

Also defining a linear mapping $\rho : T \rightarrow \mathfrak{T}$ such that

$$\rho \equiv \varphi_s \otimes \cdots \otimes \varphi_1, \quad (24.138)$$

we get

$$(\varphi_s \otimes \cdots \otimes \varphi_1)[x_{(s)} \otimes \cdots \otimes x_{(1)}] = \varphi_s[x_{(s)}] \otimes \cdots \otimes \varphi_1[x_{(1)}]. \quad (24.139)$$

We wish to consider the endomorphism with respect to φ_i ($i = 1, \cdots, s$) within a same vector space V_i as in the case of Sect. 24.1.2. Then, we have

$$\varphi_i : V_i \rightarrow V_i \ (i = 1, \cdots, s). \quad (24.140)$$

In this case, we have

$$\mathfrak{T} = V_s \otimes \cdots \otimes V_1.$$

Accordingly, from (24.133), we get a following endomorphism with the multilinear mapping $\varphi_s \otimes \cdots \otimes \varphi_1$:

$$\varphi_s \otimes \cdots \otimes \varphi_1 : T \longrightarrow T.$$

Replacing φ_i with a (n, n) matrix A_i , (24.55) can readily be generalized such that

$$\begin{aligned} & (A_s \otimes \cdots \otimes A_1) [x_{(s)} \otimes \cdots \otimes x_{(1)}] \\ &= [\mathbf{e}_{(1)} \otimes \cdots \otimes \mathbf{e}_{(s)}] [A_s \otimes \cdots \otimes A_1] [\boldsymbol{\xi}_{(s)} \otimes \cdots \otimes \boldsymbol{\xi}_{(1)}], \end{aligned} \quad (24.141)$$

where $\mathbf{e}_{(i)}$ ($i = 1, \cdots, s$) denotes $(e_{1(i)} \cdots e_{n(i)})$; $\boldsymbol{\xi}_{(i)}$ ($i = 1, \cdots, s$) is an abbreviation for $\begin{pmatrix} \xi_{(i)}^1 \\ \vdots \\ \xi_{(i)}^n \end{pmatrix}$. Also, in (24.141), $A_s \otimes \cdots \otimes A_1$ operates on a vector $\mathbf{e}_{(i)} \cdot \boldsymbol{\xi}_{(i)} \in T$.

In the above, $\mathbf{e}_{(i)} \cdot \boldsymbol{\xi}_{(i)}$ is a symbolic expression meant by

$$\mathbf{e}_{(i)} \cdot \boldsymbol{\xi}_{(i)} = [\mathbf{e}_{(1)} \otimes \cdots \otimes \mathbf{e}_{(s)}] [\boldsymbol{\xi}_{(s)} \otimes \cdots \otimes \boldsymbol{\xi}_{(1)}],$$

where $\mathbf{e}_{(1)} \otimes \cdots \otimes \mathbf{e}_{(s)}$ is represented by a $(1, n^s)$ row matrix; $\boldsymbol{\xi}_{(s)} \otimes \cdots \otimes \boldsymbol{\xi}_{(1)}$ by a $(n^s, 1)$ column matrix. The operator $A_s \otimes \cdots \otimes A_1$ is represented by a (n^s, n^s) matrix. As in the case of (11.39), the operation of $A_s \otimes \cdots \otimes A_1$ is described by

$$\begin{aligned} & (A_s \otimes \cdots \otimes A_1) [x_{(s)} \otimes \cdots \otimes x_{(1)}] \\ &= \{ [\mathbf{e}_{(1)} \otimes \cdots \otimes \mathbf{e}_{(s)}] [A_s \otimes \cdots \otimes A_1] \} [\boldsymbol{\xi}_{(s)} \otimes \cdots \otimes \boldsymbol{\xi}_{(1)}] \\ &= [\mathbf{e}_{(1)} \otimes \cdots \otimes \mathbf{e}_{(s)}] \left\{ [A_s \otimes \cdots \otimes A_1] [\boldsymbol{\xi}_{(s)} \otimes \cdots \otimes \boldsymbol{\xi}_{(1)}] \right\}. \end{aligned}$$

That is, the associative law holds with the matrix multiplication.

We have a more important case where some of vector spaces V_i ($i = 1, \cdots, s$) that are involved in the tensor product are the same space V and the remaining vector spaces are its dual space V^* . The resulting tensor product is described by

$$T = V_s \otimes \cdots \otimes V_1 \quad (V_i = V \text{ or } V^*; i = 1, \cdots, n), \quad (24.142)$$

where T is a tensor product consisting of s vector spaces in total; V^* is a dual space of V . The tensor product described by (24.142) is called a tensor space. If the number of vector spaces V and that of V^* contained in the tensor space are q and r ($q + r = s$), respectively, that tensor space is said to be a (q, r) -type tensor space or a q -th order contravariant— r -th order covariant tensor space.

An element of the tensor space is called a s -th rank tensor. In particular, if all V_i ($i = 1, \dots, s$) are V , the said tensor space is called a s -th rank contravariant tensor space and the elements in that space are said to be a s -th rank contravariant tensor. If all V_i ($i = 1, \dots, s$) are V^* , the tensor space is called a s -th rank covariant tensor space and the elements in that space are said to be a s -th rank covariant tensor accordingly.

As an example, we construct three types of second rank tensors described by

(i)

$$(R \otimes R)[x_{(2)} \otimes x_{(1)}] = [e_{(2)} \otimes e_{(1)}][R \otimes R][\xi_{(2)} \otimes \xi_{(1)}],$$

(ii)

$$\begin{aligned} & [(R^{-1})^T \otimes (R^{-1})^T][y_{(2)} \otimes y_{(1)}] \\ &= [\mathbf{f}_{(2)}^T \otimes \mathbf{f}_{(1)}^T][(R^{-1})^T \otimes (R^{-1})^T][\boldsymbol{\eta}_{(2)}^T \otimes \boldsymbol{\eta}_{(1)}^T], \end{aligned}$$

(iii)

$$\begin{aligned} & [(R^{-1})^T \otimes R][y_{(2)} \otimes x_{(1)}] = [\mathbf{f}_{(2)}^T \otimes e_{(1)}][(R^{-1})^T \otimes R] \\ & \times [\boldsymbol{\eta}_{(2)}^T \otimes \xi_{(1)}]. \end{aligned} \quad (24.143)$$

Equation (24.143) shows (2, 0)-, (0, 2)-, and (1, 1)-type tensors for (i), (ii), and (iii) from the top. In (24.143), with $x_{(i)} \in V$ we denote it by $x_{(i)} = e_{(i)} \cdot \xi_{(i)}$ ($i = 1, 2$); for $y_{(i)} \in V^*$ we express it as $y_{(i)} = \mathbf{f}_{(i)}^T \cdot \boldsymbol{\eta}_{(i)}^T$ ($i = 1, 2$). The vectors $x_{(i)}$ and $y_{(i)}$ are referred to as a contravariant vector and a covariant vector, respectively. Since with a dual vector of V^* , a basis set $\mathbf{f}_{(i)}$ are arranged in a column vector and coordinates are arranged in a row vector, they are transposed so that their arrangement can be consistent with that of a vector of V . Concomitantly, R^{-1} has been transposed as well. Higher rank tensors can be constructed in a similar manner as in (24.143) noted above.

The basis sets $e_{(2)} \otimes e_{(1)}$, $\mathbf{f}_{(2)}^T \otimes \mathbf{f}_{(1)}^T$, and $\mathbf{f}_{(2)}^T \otimes e_{(1)}$ are referred to a standard basis (or canonical basis) of the tensor spaces $V \otimes V$, $V^* \otimes V^*$, and $V^* \otimes V$, respectively. Then, if (24.142) contains the components of V^* , we need to use $\boldsymbol{\eta}_{(i)}^T$ for the coordinate; see (24.132). These basis sets are often disregarded in the computation process of tensors.

Equation (24.143) are rewritten using tensor components (or “coordinates” of a tensor) as

$$(i) \quad \xi'^i{}_{(2)} \xi'^k{}_{(1)} = \sum_{j,l=1}^n R^i{}_l R^k{}_j \xi^l{}_{(2)} \xi^j{}_{(1)},$$

$$(ii) \quad \eta'_{i(2)} \eta'_{k(1)} = \sum_{j,l=1}^n (R^{-1})^l{}_i (R^{-1})^j{}_k \eta_{l(2)} \eta_{j(1)},$$

$$(iii) \quad \eta'_{i(2)} \xi'^k{}_{(1)} = \sum_{j,l=1}^n (R^{-1})^l{}_i R^k{}_j \eta_{l(2)} \xi^j{}_{(1)},$$

where, e.g., $\xi'^i{}_{(2)} \xi'^k{}_{(1)}$ shows the tensor components after being transformed by R . With these relations, see (24.62). Also, notice that the invariant tensor $\left[f_j | e_l \right]$ of (24.125) indicates the transformation property of the tensor components of a second rank tensor of (0, 2)-type. We have another invariant tensor δ^i_j of (1, 1)-type, whose transformation is described by

$$\delta^i_j = (R^{-1})^l{}_j R^i{}_k \delta^k_l = (R^{-1})^l{}_j R^i{}_l = (RR^{-1})^i{}_j = \delta^i_j.$$

Note that in the above relation R can be any non-singular operator. The invariant tensor δ^i_j is exactly the same as $f^i(e_k)$ of (24.70). Although the invariant metric tensor $(\mathcal{M})_{ij}$ of (24.93) is determined by the inherent structure of individual vector spaces that form a tensor product, the invariant tensor δ^i_j is common to any tensor space, regardless of the nature of constituent vector spaces.

To show tensor components, we use a character with super/subscripts such as e.g., Ξ^{ij} ($1 \leq i, j \leq n$), which are components of a second rank tensor of (2, 0)-type. The transformation of Ξ is described by

$$\Xi'^{ik} = \sum_{j,l=1}^n R^i{}_l R^k{}_j \Xi^{lj}, \text{ etc.}$$

We can readily infer the computation rule of the tensor from analogy and generalization of the tensor product calculations shown in Example 24.2.

More generally, let us consider a (q, r) -type tensor space T . We express an element $\mathfrak{T}$ (i.e., a tensor) of T ($\mathfrak{T} \in T$) symbolically as

$$\mathfrak{T} = \Gamma \cdot \Theta \cdot \Psi, \quad (24.144)$$

where Γ denotes a tensor product of the s basis vectors of V or V^* ; Θ denotes a tensor product of the s operators R or $(R^{-1})^T$; Ψ also denotes a tensor product of the s coordinates $\xi^i_{(j)}$ or $\eta_{k(l)}$ where $1 \leq j \leq q$, $1 \leq l \leq r$, $1 \leq i, k \leq n$ (n is a dimension of the vector spaces V and V^*). Individual tensor products are further described by

$$\begin{aligned}
\Gamma &= \mathbf{g}_{(s)} \otimes \mathbf{g}_{(s-1)} \otimes \cdots \otimes \mathbf{g}_{(1)}, \\
\Theta &= Q_{(s)} \otimes Q_{(s-1)} \otimes \cdots \otimes Q_{(1)}, \\
\Psi &= \xi_{(s)} \otimes \xi_{(s-1)} \otimes \cdots \otimes \xi_{(1)}.
\end{aligned} \tag{24.145}$$

In (24.145), Γ represents s basis sets $\mathbf{g}_{(i)}$ ($1 \leq i \leq s$), which denote either $\mathbf{e}_{(i)}$ or $\mathbf{f}_{(i)}^T$ ($1 \leq i \leq s$). The number of $\mathbf{e}_{(i)}$ is q and that of $\mathbf{f}_{(i)}^T$ is r with $q + r = s$. With the tensor product Θ , $Q_{(i)}$ represents R or $(R^{-1})^T$ with the number of R and $(R^{-1})^T$ being q and r , respectively. The tensor product Ψ denotes s coordinates $\xi_{(i)}$ ($1 \leq i \leq s$), which represent either $\xi_{(i)}$ or $\eta_{(i)}^T$. The number of $\xi_{(i)}$ is q and that of $\eta_{(i)}^T$ is r with $q + r = s$.

The symbols $\xi_{(i)}$ and $\eta_{(i)}^T$ represent

$$\xi_{(i)} = \begin{pmatrix} \xi_{(i)}^1 \\ \vdots \\ \xi_{(i)}^n \end{pmatrix} \quad (1 \leq i \leq q), \quad \eta_{(i)}^T = \begin{pmatrix} \eta_{1(i)} \\ \vdots \\ \eta_{n(i)} \end{pmatrix} \quad (1 \leq i \leq r),$$

respectively. Thus, Ψ is described by the tensor products of column vectors. Consequently, Ψ has a dimension of $n^{q+r} = n^s$.

In Sect. 22.5, we have introduced the field tensor $F^{\mu\nu}$, i.e., a (2, 0)-type tensor which is represented by a matrix. To express a physical quantity in a tensor form, Ψ is usually described as

$$\Psi \equiv T_{\eta_{(r)}\eta_{(r-1)}\cdots\eta_{(1)}}^{\xi_{(q)}\xi_{(q-1)}\cdots\xi_{(1)}}. \tag{24.146}$$

Then, Ψ is transformed by R such that

$$\begin{aligned}
\Psi' &= T_{\eta_{(r)}\eta_{(r-1)}\cdots\eta_{(1)}}^{\xi_{(q)}\xi_{(q-1)}\cdots\xi_{(1)}} \\
&= (R^{-1})_{\eta_{(r)}}^{\alpha} (R^{-1})_{\eta_{(r-1)}}^{\beta} \cdots (R^{-1})_{\eta_{(1)}}^{\gamma} R^{\xi_{(q)}}_{\rho} R^{\xi_{(q-1)}}_{\sigma} \cdots R^{\xi_{(1)}}_{\tau} T_{\alpha\beta\cdots\gamma}^{\rho\sigma\cdots\tau}.
\end{aligned} \tag{24.147}$$

As can be seen above, if the tensor products of the basis vectors Γ are fixed (normally at the standard basis) in (24.144), the tensor $\mathfrak{T}$ and their transformation are determined by the part $\Theta \cdot \Psi$ in (24.144). Moreover, since the operator R is related to the invariant metric tensor $\mathcal{M}$ through (24.104), the most important factor of (24.144) is the tensor product Ψ of the coordinates. Hence, it is often the case in practice with physics and natural science that Ψ described by (24.146) is simply referred to as a tensor.

Let us think of the transformation property of a following (1, 1)-type tensor Ξ_i^j :

$$\Xi'^k_i = \sum_{j,l=1}^n (R^{-1})^l_i R^k_j \Xi^j_l. \quad (24.148)$$

Putting $k = i$ in (24.148) and taking the sum over i , we obtain

$$\sum_{i=1}^n \Xi'^i_i = \sum_{i,j,l=1}^n (R^{-1})^l_i R^i_j \Xi^j_l = \sum_{j,l=1}^n \delta^l_j \Xi^j_l = \sum_{l=1}^n \Xi^l_l \equiv \Xi^l_l. \quad (24.149)$$

Thus, (24.149) represents an invariant quantity that does not depend on the coordinate transformation. The above computation rule is known as tensor contraction. In this case, the summation symbol is usually omitted and this rule is referred to as Einstein summation convention (see Chap. 21). We readily infer from (24.149) that a (q, r) -type tensor is converted into a $(q - 1, r - 1)$ -type tensor by the contraction.

24.1.7 Euclidean Space and Minkowski Space

As typical examples of the real vector spaces that are often dealt with in mathematical physics, we have Euclidean space and Minkowski space. Even though the former is familiar enough even at the elementary school level, we wish to revisit and summarize its related topics.

The metric tensor of Euclidean space is given by

$$\mathcal{M} = (\mathcal{M})_{ij} = \delta_{ij}. \quad (24.150)$$

Hence, from (24.104) the transformation R that holds a bilinear form $[x|x]$ invariant satisfies the condition

$$R^T E R = R^T R = E, \quad (24.151)$$

where E is the identity operator of Euclidean space. That is, R must be an orthogonal matrix. From (24.120), we have $\xi_i = \xi^j_j$. Also, from (24.123) we have

$$\begin{aligned} \xi'_i &\equiv \sum_{k=1}^n \xi_k (R^{-1})^k_i = \sum_{k=1}^n \xi_k (R^T)^k_i = \sum_{k=1}^n \xi_k R^i_k \\ &= \sum_{k=1}^n R^i_k \xi_k. \end{aligned} \quad (24.152)$$

From (24.107), ξ_i is the same transformation property as ξ'^i . Therefore, we do not have to distinguish a contravariant vector and covariant vector in Euclidean space. The bilinear form $[x|x]$ is identical with the positive definite inner product $\langle x|x \rangle$ in a real vector space.

In the n -dimensional Minkowski space, the metric tensor $(\mathcal{M})_{ij} = [f_i|e_j]$ is represented by

$$[f_i|e_j] = -\delta_{ij}\zeta_i = -\delta_{ij}\zeta_j = [f_j|e_i]; \zeta_0 = -\zeta_k \quad (k=1, \dots, n-1) = -1. \quad (24.153)$$

In the four-dimensional case, $\mathcal{M}$ is expressed as

$$\mathcal{M} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (24.97)$$

Minkowski space has several properties different from those of Euclidean space because of the difference in the metric tensor. First, we make general discussion concerning the vector spaces, followed by specific topics related to Minkowski space as contrasted with the properties of Euclidean space. The discussion is based on the presence of the bilinear form that has already been mentioned in Sect. 24.1.3. First, we formally give a following definition.

Definition 24.4 [5] Let W be a subspace of a vector space V . Suppose that with $\exists w_0 (\neq 0) \in W$ we have

$$[w_0|w] = 0 \quad \text{for} \quad \forall w \in W. \quad (24.154)$$

Then, the subspace W is said to be *degenerate*. Otherwise, W is called *non-degenerate*. If we have

$$[u|v] = 0 \quad \text{for} \quad \forall u, v \in W,$$

W is called *totally degenerate*.

The statement that the subspaces are degenerate, non-degenerate, or totally degenerate in the above definition is clear from the definition of subspace (Sect. 11.1) and $[\cdot|\cdot]$. The existence of degenerate and totally degenerate subspaces results from the presence of a singular vector, i.e., a vector $u (\neq 0)$ that satisfies $[u|u] = 0$. Note that any subspace of Euclidean space is non-degenerate and that no singular vector is present in Euclidean space except for zero vector. If it was not the case, from (24.154) with $\exists w_0 (\neq 0) \in W$ we would have $[w_0|w_0] = 0$. It is in contradiction to the positive definite inner product of Euclidean space.

In Sect. 14.1, we showed a direct sum decomposition of a vector space into its subspaces. The concept of the direct sum decomposition plays an essential role in this section as well. The definition of an orthogonal complement stated in (14.1), however, needs to be modified in such a way that

$$W^\perp \equiv \left\{ |x]; [x|y] = 0 \text{ for } \forall |y] \in W \right\}. \quad (24.155)$$

In that case, $W^\perp$ is said to be an orthogonal complement of W .

In this context, we have following important proposition and theorems and see how the present situation can be compared with the previous results obtained in the inner product space (Sect. 14.1).

Proposition 24.1 [5] Let V be a vector space where a bilinear form $[\cdot | \cdot]$ is defined. Let W be a subspace of V and let $W^\perp$ be an orthogonal complement of W . A necessary and sufficient condition for $W \cap W^\perp = \{0\}$ to hold is that W is non-degenerate.

Proof

- (i) Sufficient condition: To prove “ W is non-degenerate $\Rightarrow W \cap W^\perp = \{0\}$,” we prove its contraposition. That is, suppose $W \cap W^\perp \neq \{0\}$. Then, we have $0 \neq \exists w_0$ which satisfies $w_0 \in W \cap W^\perp$. From the definition of (24.155) for the orthogonal complement and the assumption of $w_0 \in W^\perp$, we have

$$[w_0 | w] = 0 \text{ for } 0 \neq w_0 \in W, \forall w \in W. \quad (24.156)$$

But, (24.156) implies that W is degenerate.

- (ii) Necessary condition: Let us suppose that when $W \cap W^\perp = \{0\}$, W is degenerate. Then, from the definition of (24.154), (24.156) holds with W . But, from the definition of the orthogonal complement, (24.156) implies $w_0 \in W^\perp$. That is, we have $0 \neq w_0 \in W \cap W^\perp$. This is, however, in contradiction to $W \cap W^\perp = \{0\}$. Then, W must be non-degenerate. These complete the proof.

Theorem 24.4 [5] Let V be a vector space with a dimension n where a bilinear form $[\cdot | \cdot]$ is defined. Let W be a subspace of V with a dimension r . If W is non-degenerate, we have

$$V = W \bigoplus W^\perp. \quad (24.157)$$

Moreover, we have

$$(W^\perp)^\perp = W, \quad (24.158)$$

$$\dim W^\perp = n - r. \quad (24.159)$$

Proof If W is non-degenerate, Theorem 14.1 can be employed. As already seen, we can always construct the orthonormal basis such that [5]

$$[f_i | e_j] = \delta_{ij} \zeta^i = \delta_{ij} \zeta^j \quad (\zeta^j = \pm 1); \quad (24.160)$$

see (24.94) and (24.115). Notice that the terminology of orthonormal basis was borrowed from that of the inner product space and that the existence of the orthonormal basis can be proven in parallel to the proof of Theorem 13.2 (Gram–Schmidt orthonormalization theorem) and Theorem 13.3 [4].

Suppose that with $\forall |x\rangle \in W$, $|x\rangle$ is described as $|x\rangle = \sum_{i=1}^n \xi^i |e_i\rangle$. Then,

$$[f_j|x\rangle = \sum_{i=1}^n \xi^i [f_j|e_i\rangle = \sum_{i=1}^n \xi^i \delta_{ji} \zeta^i = \xi^j \zeta^j. \quad (24.161)$$

Note that the summation with respect to j was not taken. Multiplying both sides of (24.161) by ζ^j , we have

$$\xi^j = \zeta^j [f_j|x\rangle. \quad (24.162)$$

Thus, we obtain

$$|x\rangle = \sum_{j=1}^n \xi^j [f_j|x\rangle |e_j\rangle. \quad (24.163)$$

Since W is a non-degenerate subspace, it is always possible to choose the orthonormal basis for it. Then, we can make $\{e_1, \dots, e_r\}$ the basis set of W . In fact, with $\exists |w_0\rangle \in W$ and $\forall w \in W$, describing

$$|w_0\rangle = \sum_{k=1}^r \eta_0^k |e_k\rangle, \quad |w\rangle = \sum_{j=1}^r \eta^j |e_j\rangle, \quad (24.164)$$

we obtain

$$[w_0|w\rangle = \sum_{k,j=1}^r \eta_0^k \eta^j [f_k|e_j\rangle = \sum_{k,j=1}^r \eta_0^k \eta^j \delta_{kj} \zeta^j = \sum_{j=1}^r \eta_0^j \eta^j \zeta^j. \quad (24.165)$$

For $[w_0|w\rangle = 0$ to hold with $\forall w \in W$, i.e., with $\forall \eta^j \in \mathbb{R}$ ($j = 1, \dots, r$), we must have $\eta_0^j \zeta^j = 0$. As $\zeta^j = \pm 1$ ($\neq 0$), we must have $\eta_0^j = 0$ ($j = 1, \dots, r$). That is, $|w_0\rangle = 0$. This means that W is non-degenerate. Thus, using the orthonormal basis we represent $|x\rangle \in W$ as

$$|x'\rangle = \sum_{j=1}^r \zeta^j [f_j|x\rangle |e_j\rangle, \quad (24.166)$$

where we used (24.162). Putting $|x''\rangle = |x\rangle - |x'\rangle$ and operating $[f_k|$ ($1 \leq k \leq r$) on both sides from the left, we get

$$\begin{aligned} [f_k|x''] &= \sum_{j=1}^n \zeta^j [f_j|x\rangle [f_k|e_j\rangle - \sum_{j=1}^r \zeta^j [f_j|x\rangle [f_k|e_j\rangle \\ &= \sum_{j=1}^r \zeta^j [f_j|x\rangle \delta_{kj} \zeta^j - \sum_{j=1}^r \zeta^j [f_j|x\rangle \delta_{kj} \zeta^j \\ &= (\zeta^k)^2 [f_k|x\rangle - (\zeta^k)^2 [f_k|x\rangle = [f_k|x\rangle - [f_k|x\rangle = 0. \end{aligned} \quad (24.167)$$

Note in (24.167) $[f_k|e_j\rangle = 0$ with $r+1 \leq j \leq n$. Rewriting (24.167), we have

$$[x''|e_k] = 0 \quad (1 \leq k \leq r). \quad (24.168)$$

Since $W = \text{Span} \{e_1, \dots, e_r\}$, (24.168) implies that $[x''|w] = 0$ with $\forall |w\rangle \in W$. Consequently, from the definition of the orthogonal complement, we have $x'' \in W^\perp$. Thus, we obtain

$$|x\rangle = |x'\rangle + |x''\rangle, \quad (24.169)$$

where $|x\rangle \in V$, $|x'\rangle \in W$, and $x'' \in W^\perp$. Then, from Proposition 24.1 we get

$$V = W \oplus W^\perp. \quad (24.157)$$

From Theorem 11.2, we immediately get

$$\dim W^\perp = n - r. \quad (24.159)$$

Also, from the above argument we have $W^\perp = \text{Span} \{e_{r+1}, \dots, e_n\}$. This once again implies that $W^\perp$ is non-degenerate as well. Then, we have

$$V = W^\perp \oplus (W^\perp)^\perp. \quad (24.170)$$

Therefore, we obtain

$$\dim (W^\perp)^\perp = n - (n - r) = r. \quad (24.171)$$

From the definition for the orthogonal complement of (24.155), we have

$$(W^\perp)^\perp = \left\{ |x\rangle; [x|y] = 0 \text{ for } \forall |y\rangle \in W^\perp \right\}. \quad (24.172)$$

Then, if $|x\rangle \in W$, again from the definition such $|x\rangle$ must be contained in $(W^\perp)^\perp$. That is, $(W^\perp)^\perp \supset W$ [1, 2]. But, from the assumption and (24.171), $\dim (W^\perp)^\perp = \dim W$. This implies [1, 2]

$$(W^\perp)^\perp = W. \quad (24.158)$$

These complete the proof.

Getting back to (24.93), we have two non-degenerate subspaces of V where their basis vectors comprise $|e_i\rangle$ ($i = 1, \dots, p$) to which $[f_k|e_i] = \delta_{ki}$ or those comprise $|e_j\rangle$ ($j = p + 1, \dots, p + q = n$) to which $[f_k|e_j] = -\delta_{kj}$. Let us define two such subspaces as

$$W_+ = \text{Span} \{e_1, \dots, e_p\}, W_- = \text{Span} \{e_{p+1}, \dots, e_{p+q}\}. \quad (24.173)$$

Then, from Theorem 24.4 we have

$$V = W_+ \oplus W_-. \quad (24.174)$$

In (24.173) we redefine p and q as

$$p = n_+, q = n_-; n_+ + n_- = n.$$

We have

$$\dim V = \dim W_+ + \dim W_- = n_+ + n_- = n. \quad (24.175)$$

We name W_+ and W_- Euclidean subspace and anti-Euclidean subspace, respectively. With respect to W_+ and W_- , we have a following theorem.

Theorem 24.5 [5] Let V be a vector space with a dimension n . Let W_+ and W_- be Euclidean subspace and anti-Euclidean subspace of V with the dimension n_+ and n_- , respectively ($n_+ + n_- = n$). Then, there must be a totally degenerate subspace W_0 with a dimension n_0 given by

$$n_0 = \min(n_+, n_-). \quad (24.176)$$

Proof From Theorem 24.4, we have $W_+ \cap W_- = \{0\}$. From the definition of the totally degenerate subspace, obviously we have $W_+ \cap W_0 = W_- \cap W_0 = \{0\}$. (Otherwise, W_+ and W_- would be degenerate.) From this, we have

$$n_+ + n_0 \leq n, n_- + n_0 \leq n. \quad (24.177)$$

Then, we obtain

$$n_0 \leq \min(n - n_+, n - n_-) = \min(n_-, n_+) \equiv m. \quad (24.178)$$

Suppose $n_+ = m \leq n_-$. Then, we can choose two sets of orthonormal bases $\{e_{i_1}, \dots, e_{i_m}\}$ for which $[f_{i_k} | e_{i_l}] = \delta_{ikil}$ and $\{e_{j_1}, \dots, e_{j_m}\}$ for which $[f_{j_k} | e_{j_l}] = -\delta_{jkil}$. Let us think of the following set $\mathfrak{W}$ given by

$$\mathfrak{W} = \{e_{i_1} - e_{j_1}, \dots, e_{i_m} - e_{j_m}\}. \quad (24.179)$$

Defining $e_{i_s} - e_{j_s} \equiv \mathfrak{E}_s, f_{i_s} - f_{j_s} \equiv \mathfrak{F}_s$ ($s = 1, \dots, m$), we have

$$[\mathfrak{F}_s | \mathfrak{E}_t] = 0 \quad (s, t = 1, \dots, m). \quad (24.180)$$

Let us consider the following equation:

$$c_1 \mathfrak{E}_1 + \dots + c_m \mathfrak{E}_m = 0. \quad (24.181)$$

Suppose that $\mathfrak{E}_1, \dots, \mathfrak{E}_m$ are linearly dependent. Then, without loss of generality we assume $c_1 \neq 0$. Dividing both sides of (24.181) by c_1 and expressing (24.181) in terms of e_{i_1} , we obtain

$$e_{i_1} = e_{j_1} + \frac{c_2}{c_1} (e_{i_2} - e_{j_2}) + \dots + \frac{c_m}{c_1} (e_{i_m} - e_{j_m}). \quad (24.182)$$

Equation (24.182) shows that e_{i_1} is described by a linear combination of $(2m - 1)$ linearly independent vectors, but it is in contradiction to the fact that $2m$ vectors $e_{i_1}, \dots, e_{i_m}; e_{j_1}, \dots, e_{j_m}$ are linearly independent. Thus, $\mathfrak{E}_1, \dots, \mathfrak{E}_m$ must be linearly independent. Hence, $\text{Span} \{\mathfrak{E}_1, \dots, \mathfrak{E}_m\}$ forms a m -dimensional vector space. Equation (24.180) implies that it is a totally degenerate subspace of V with a dimension of m . This immediately means that

$$W_0 = \text{Span} \{e_{i_1} - e_{j_1}, \dots, e_{i_m} - e_{j_m}\}. \quad (24.183)$$

Then, from the assumption we have $n_0 = m$. Namely, the equality holds with (24.178) so that (24.176) is established.

If, on the other hand, we assume $n_- = m \leq n_+$, proceeding in a similar way as the above discussion we reach the same conclusion, i.e., $n_0 = m$. This completes the proof.

Since we have made general discussion and remarks on the vector spaces where the bilinear form is defined, we wish to show examples of the Minkowski space $\mathbb{M}^4$ and $\mathbb{M}^3$.

Example 24.3 We study several properties of the Minkowski space, taking four-dimensional space $\mathbb{M}^4$ and three-dimensional space $\mathbb{M}^3$ as an example. For both the cases of $\mathbb{M}^4$ and $\mathbb{M}^3$, the dimension of a totally degenerate subspace is 1 (Theorem 24.5). In $\mathbb{M}^4$, n_+ and n_- of (24.176) are 1 and 3, respectively; see (24.97). The space $\mathbb{M}^4$ is denoted by

$$\mathbb{M}^4 = \text{Span} \{e_0, e_1, e_2, e_3\}, \quad (24.184)$$

where e_0, e_1, e_2 , and e_3 are defined according to (24.95) with

$$[f_i|e_j] = -\delta_{ij}\zeta_i = -\delta_{ij}\zeta_j = [f_j|e_i]; \zeta_0 = -\zeta_k \ (k=1, \dots, 3) = -1. \quad (24.185)$$

With the notation $\zeta_k \ (k=0, \dots, 3)$, see (22.323).

According to Theorem 24.4, we have, e.g., a following orthogonal decomposition:

$$\mathbb{M}^4 = \text{Span} \{e_0\} \bigoplus \text{Span} \{e_0\}^\perp,$$

where $\text{Span} \{e_0\}^\perp = \text{Span}\{e_1, e_2, e_3\}$. Since $[f_0|e_0] = 1$, e_0 is a timelike vector. As $[f_1 + f_2 + f_3|e_1 + e_2 + e_3] = -3$, $e_1 + e_2 + e_3$ is a spacelike vector, etc. Meanwhile, according to Theorem 24.5, a dimension n_0 of the totally degenerate subspace W_0 is $n_0 = \min(n_+, n_-) = \min(1, 3) = 1$. Then, for instance, we have

$$W_0 = \text{Span}\{|e_0\rangle + |e_1\rangle\}. \quad (24.186)$$

Proposition 24.1 tells us that it is impossible to get an orthogonal decomposition using a degenerate subspace. In fact, as an orthogonal complement to W_0 we have

$$W_0^\perp = \text{Span} \{e_0 + e_1, e_2, e_3\}. \quad (24.187)$$

Then, $W_0 \cap W_0^\perp = W_0 = |e_0\rangle + |e_1\rangle \neq 0$. Note that $W_0^\perp$ can alternatively be expressed as [5]

$$W_0^\perp = W_0 \bigoplus \text{Span} \{e_2, e_3\}. \quad (24.188)$$

The presence of the totally degenerate subspace W_0 has a special meaning in the theory of special relativity. The collection of all singular vectors forms a *light cone*. Since the dimension of W_0 is one, it is spanned by a unique lightlike vector (i.e., singular vector), e.g., $|e_0\rangle + |e_1\rangle$. It may be chosen from among any singular vectors. Let us denote a singular vector by $u_\lambda \ (\lambda \in \Lambda)$ and the collection of all the singular vectors that form the light cone by L . Then, we have

$$L = \bigcup_{\lambda \in \Lambda} u_\lambda. \quad (24.189)$$

With the above notation, see Sect. 6.1.1. Notice that the set L is *not* a vector space (vide infra). The totally degenerate subspace W_0 can be described by

$$W_0 = \text{Span} \{u_\lambda\},$$

where u_λ is any *single* singular vector.

We wish to consider the coordinate representation of the light cone in the orthogonal coordinate system. In the previous chapters, by the orthogonal coordinate system we meant a two- or three-dimensional Euclidean space, i.e., a real inner

product space accompanied by an orthonormal basis. In this chapter, however, we use the terminologies of orthogonal coordinate system and orthonormal basis, if basis vectors satisfy the relations of (24.153). Let the vector $w \in L$ be expressed as

$$w = \sum_{i=0}^3 \xi^i e_i. \quad (24.190)$$

Then, we obtain

$$\left[\sum_{i=0}^3 \xi^i e_i \middle| \sum_{k=0}^3 \xi^k e_k \right] = (\xi^0)^2 - (\xi^1)^2 - (\xi^2)^2 - (\xi^3)^2 = 0. \quad (24.191)$$

That is,

$$(\xi^1)^2 + (\xi^2)^2 + (\xi^3)^2 = (\xi^0)^2. \quad (24.192)$$

If we regard the time coordinate (represented by ξ^0) as parameter, the light cone can be seen as an expanding (or shrinking) hypersphere in $\mathbb{M}^4$ with elapsing time (both past and future directions).

In Fig. 24.1a, we depict a geometric structure including the light cone of the *three-dimensional* Minkowski space $\mathbb{M}^3$ so that the geometric feature can appeal to eye. There, we have

$$(\xi^1)^2 + (\xi^2)^2 = (\xi^0)^2, \quad (24.193)$$

where ξ^1 and ξ^2 represent the coordinate of the x - and y -axis, respectively; ξ^0 indicates the time coordinate t . Figure 24.1b shows how to make a right circular cone contained in the light cone.

The light cone comprises the whole collection of singular vectors u_λ ($\lambda \in \Lambda$). Geometrically, it consists of a couple of right circular cones that come face-to-face with each other at the vertices, with the axis (i.e., a line that connects the vertex and the center of the bottom circle) shared in common by the two right circular cones in $\mathbb{M}^3$ (see Fig. 24.1a). A lightlike vector (i.e., singular vector) is represented by a straight line obliquely extending from the origin to both the future and past directions in the orthogonal coordinate system. Since individual lightlike vectors form the light cone, it is a kind of ruled surface. A circle expanding (or shrinking) with time ξ^0 according to (24.193) is a conic section of the right circular cone shaping the light cone.

In Fig. 24.1a, the light cone comprises a two-dimensional figure, and so one might well wonder if the light cone is a two-dimensional vector space. Think of, however, e.g., $|e_0\rangle + |e_1\rangle$ and $|e_0\rangle - |e_1\rangle$ within the light cone. If those vectors were basis vectors, $(|e_0\rangle + |e_1\rangle) + (|e_0\rangle - |e_1\rangle) = 2|e_0\rangle$ would be a vector of the light cone as well. That is, however, evidently not the case, because the vector $|e_0\rangle$ is located

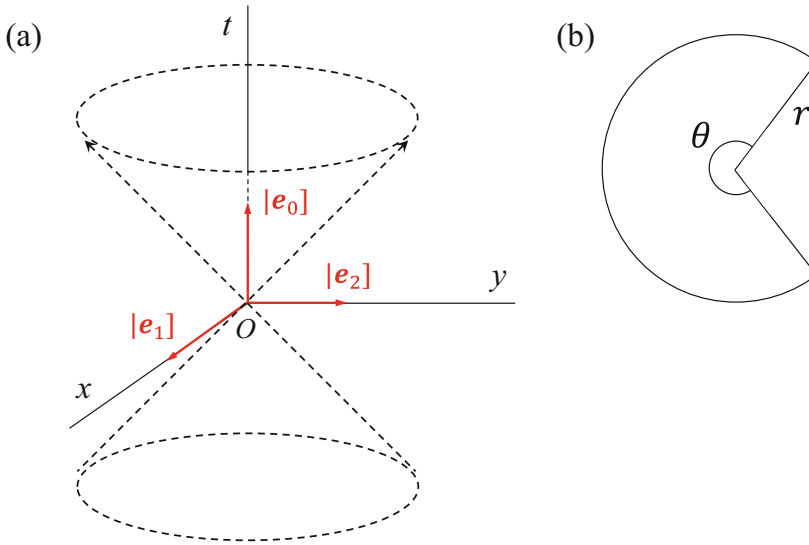


Fig. 24.1 Geometric structure including the light cone of the *three-dimensional* Minkowski space $\mathbb{M}^3$. **(a)** The light cone consists of a couple of right circular cones that come face-to-face with each other at the vertices, with the axis (i.e., a line that connects the vertex and the center of the bottom circle) shared in common. A lightlike vector (i.e., singular vector) is represented by an oblique arrow extending from the origin to both the future and past directions. Individual lightlike vectors form the light cone as a ruled surface. **(b)** Simple kit that helps visualize the light cone. To make it, follow next procedures: (i) Take a thick sheet of paper and cut out two circles to shape them into sectors with a central angle $\theta = \sqrt{2}\pi$ radian $\approx 254.6^\circ$. (ii) Bring together the two sides of the sector so that it can be a right circular cone with its vertical angle of a right angle (i.e., 90°). (iii) Bring together the two right circular cones so that they can be jointed at their vertices with the axis shared in common to shape the light cone depicted in **(a)** of the above

outside the light cone. Whereas a vector “inside” the light cone is a timelike, that “outside” the light cone is spacelike.

We have, e.g.,

$$\mathbb{M}^3 = \text{Span} \{e_0\} \oplus \text{Span} \{e_0\}^\perp, \quad (24.194)$$

where $\text{Span} \{e_0\}^\perp = \text{Span} \{e_1, e_2\}$. Since $[f_0|e_0] = 1$, e_0 is a timelike vector. As $[f_1 + f_2|e_1 + e_2] = -2$, $e_1 + e_2$ is a spacelike vector, etc. Meanwhile, a dimension n_0 of the totally degenerate subspace $\tilde{W}_0$ of $\mathbb{M}^3$ is $n_0 = \min(n_+, n_-) = \min(1, 2) = 1$. Then, for instance, as in (24.186), we have

$$\tilde{W}_0 = \text{Span}\{|e_0\rangle + |e_1\rangle\}. \quad (24.195)$$

Also, as an orthogonal complement to $\tilde{W}_0$, we obtain

$$\tilde{W}_0^\perp = \text{Span} \{e_0 + e_1, e_2\} \quad \text{or} \quad \tilde{W}_0^\perp = \tilde{W}_0 \oplus \text{Span} \{e_2\}. \quad (24.196)$$

The orthogonal complement $\tilde{W}_0^\perp$ forms a tangent space to the light cone L . We have $\tilde{W}_0 \cap \tilde{W}_0^\perp = \tilde{W}_0 = |e_0\rangle + |e_1\rangle \neq 0$.

As described above, we have overviewed the characteristics of the extended concept of vector spaces and studied several properties which the inner product space and Euclidean space lack. Examples included the singular vectors and degenerate subspaces as well as their associated properties. We have already seen that the indefinite metric caused intractability in the field quantization. The presence of the singular vectors is partly responsible for it.

The theory of vector space is a fertile ground for mathematics and physics. Indeed, the theory deals with a variety of vector spaces such as an inner product space (in Chap. 13), Euclidean space, Minkowski space, etc. Besides them a spinor space is also contained, even though we do not go into detail about it. Interested readers are referred to the literature with this topic [5]. Yet, the spinor space naturally makes an appearance in the Dirac equation as a representation space. We will deal with the Dirac equation from a different angle in the following sections.

24.2 Lorentz Group and Lorentz Transformations

In the previous section, we have learned that the transformation R that makes the bilinear form $[\cdot|\cdot]$ invariant must satisfy the condition described by

$$R^T \mathcal{M} R = \mathcal{M}. \quad (24.104)$$

At the same time, (24.104) has given a condition for the metric tensor $\mathcal{M}$ to be invariant with respect to the vector (or coordinate) transformation R in a vector space where $[\cdot|\cdot]$ is defined. In the Minkowski space $\mathbb{M}^4$, R of (24.104) is called the Lorentz transformation and written as Λ according to the custom. Also, $\mathcal{M}$ is denoted by η . Then, we have

$$\eta = \Lambda^T \eta \Lambda. \quad (24.197)$$

Equation (24.197) implies that the matrix η is determined, independent of the choice of the inertial frame of reference. Or rather, the mathematical formulation of the theory of special relativity has been established in such a way that η can be invariant in any inertial frame of reference. This has had its origin in the invariance of the world interval of the events (see Sect. 21.2.2).

As already mentioned in Sect. 21.2, the Lorentz group consists of all the Lorentz transformations Λ . The Lorentz group is usually denoted by $O(3, 1)$. On the basis of (24.197), we define $O(3, 1)$ as a set that satisfies the following relation described by

$$O(3, 1) = \{\Lambda \in GL(4, \mathbb{R}); \Lambda^T \eta \Lambda = \eta\}. \quad (24.198)$$

Suppose that $\Lambda_1, \Lambda_2 \in O(3, 1)$. Then, we have

$$(\Lambda_1 \Lambda_2)^T \eta \Lambda_1 \Lambda_2 = \Lambda_2^T \Lambda_1^T \eta \Lambda_1 \Lambda_2 = \Lambda_2^T (\Lambda_1^T \eta \Lambda_1) \Lambda_2 = \Lambda_2^T \eta \Lambda_2 = \eta. \quad (24.199)$$

That is, $\Lambda_1 \Lambda_2 \in O(3, 1)$. With the identity operator E , we have

$$E^T \eta E = E \eta E = \eta E E = \eta. \quad (24.200)$$

Multiplying both sides of $\Lambda_1^T \eta \Lambda_1 = \eta$ by $(\Lambda_1^T)^{-1}$ from the left and $(\Lambda_1)^{-1}$ from the right, we obtain

$$\eta = (\Lambda_1^T)^{-1} \eta (\Lambda_1)^{-1} = (\Lambda_1^{-1})^T \eta (\Lambda_1)^{-1}, \quad (24.201)$$

where we used $(\Lambda_1^T)^{-1} = (\Lambda_1^{-1})^T$. From (24.199) to (24.201), we find that $O(3, 1)$ certainly satisfies the definitions of group; see Sect. 16.1. Taking determinant of (24.197), we have

$$\det \Lambda^T \det \eta \det \Lambda = \det \Lambda (-1) \det \Lambda = \det \eta = -1, \quad (24.202)$$

where with the first equality we used $\det \Lambda^T = \det \Lambda$ [1, 2]. That is, we obtain

$$(\det \Lambda)^2 = 1 \text{ or } \det \Lambda = \pm 1. \quad (24.203)$$

Lorentz groups are classified according to the sign of determinants and the $(0, 0)$ component (i.e., Λ_0^0) of the Lorentz transformation matrix. Using the notation of tensor elements, (24.197) can be written as

$$\eta_{\rho\sigma} = \eta_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma. \quad (24.204)$$

In (24.204) putting $\rho = \sigma = 0$, we get

$$\eta_{00} (\Lambda_0^0)^2 + \sum_{i=1}^3 \eta_{ii} (\Lambda_i^0)^2 = \eta_{00},$$

namely we have

$$(\Lambda_0^0)^2 = 1 + \sum_{i=1}^3 (\Lambda_i^0)^2. \quad (24.205)$$

Hence,

Table 24.1 Four connected components of the Lorentz group^a

Connected component	Nomenclature	Λ^0_0	$\det \Lambda$
$\mathfrak{L}_0 \equiv SO_0(3, 1)$	Proper orthochronous	$\Lambda^0_0 \geq 1$	+1
$\mathfrak{L}_1$	Improper orthochronous	$\Lambda^0_0 \geq 1$	-1
$\mathfrak{L}_2$	Improper anti-orthochronous	$\Lambda^0_0 \leq -1$	-1
$\mathfrak{L}_3$	Proper anti-orthochronous	$\Lambda^0_0 \leq -1$	+1

^aWith the notations $\mathfrak{L}_0, \mathfrak{L}_1, \mathfrak{L}_2$, and $\mathfrak{L}_3$, see Sect. 24.9.2

$$(\Lambda^0_0)^2 \geq 1. \quad (24.206)$$

Therefore, we obtain either

$$\Lambda^0_0 \geq 1 \quad \text{or} \quad \Lambda^0_0 \leq -1. \quad (24.207)$$

Notice that in (24.207) the case of $\Lambda^0_0 \geq 1$ is not accompanied by the time reversal, but the case of $\Lambda^0_0 \leq -1$ is accompanied by the time reversal. Combining (24.203) and (24.207), we get four combinations for the Lorentz groups and these combinations correspond to individual connected components of the groups (see Chap. 20). Table 24.1 summarizes them [6–8]. The connected component that contains the identity element is an invariant subgroup of $O(3, 1)$; see Theorem 20.9. This component is termed the proper orthochronous Lorentz group and denoted by $SO_0(3, 1)$.

The notation differs from literature to literature, and so care should be taken accordingly. In this book, to denote the proper orthochronous Lorentz group, we use $SO_0(3, 1)$ according to Hirai [6], for which the index 0 means the orthochronous transformation and S stands for the proper transformation (i.e., pure rotation without spatial inversion). In Table 24.1, we distinguish the *improper* Lorentz transformation from the *proper* transformation according as the determinant of the transformation matrix Λ is ± 1 [7]. We will come back to the discussion about the construction of the Lorentz group in Sect. 24.9.2 in terms of the connectedness.

24.2.1 Lie Algebra of the Lorentz Group

The Lie algebra of the Lorentz group is represented by a $(4, 4)$ real matrix. We have an important theorem about the Lie algebra of the Lorentz group.

Theorem 24.6 [8] Let the Lie algebra of the Lorentz group $O(3, 1)$ be denoted as $\mathfrak{o}(3, 1)$. Then, we have

$$\mathfrak{o}(3, 1) = \{X; X^T \eta + \eta X = 0, X \in \mathfrak{gl}(4, \mathbb{R})\}. \quad (24.208)$$

Proof Suppose that $X \in \mathfrak{o}(3, 1)$. Then, from Definition 20.3, $\exp(sX) \in O(3, 1)$ with $\forall s$ (s : real number). Moreover, from (24.198), we have

$$(\exp sX)^T \eta (\exp sX) = \eta. \quad (24.209)$$

Differentiating (24.209) with respect to s , we obtain

$$X^T (\exp sX)^T \eta (\exp sX) + (\exp sX)^T \eta X (\exp sX) = 0. \quad (24.210)$$

Taking the limit $s \rightarrow 0$, we get

$$X^T \eta + \eta X = 0. \quad (24.211)$$

Inversely, suppose that X satisfies (24.211). Then, using (15.31), we obtain

$$(\exp sX)^T \eta = \exp(sX^T) \eta = \sum_{\nu=0}^{\infty} \frac{1}{\nu!} (sX^T)^\nu \eta = \sum_{\nu=0}^{\infty} \frac{s^\nu}{\nu!} (X^T)^\nu \eta. \quad (24.212)$$

In (24.212), we have

$$\begin{aligned} (X^T)^\nu \eta &= (X^T)^{\nu-1} X^T \eta = (X^T)^{\nu-1} (-\eta X) = (X^T)^{\nu-2} X^T (-\eta X) \\ &= (X^T)^{\nu-2} \eta (-1)^2 X^2 = \cdots = \eta (-1)^\nu X^\nu. \end{aligned} \quad (24.213)$$

Substituting (24.213) for (24.212), we get

$$\begin{aligned} (\exp sX)^T \eta &= \sum_{\nu=0}^{\infty} \frac{s^\nu}{\nu!} \eta (-1)^\nu X^\nu = \eta \sum_{\nu=0}^{\infty} \frac{s^\nu}{\nu!} (-X)^\nu = \eta \exp(-sX) \\ &= \eta (\exp sX)^{-1}, \end{aligned} \quad (24.214)$$

where with the last equality we used (15.28). Multiplying both sides of (24.214) by $\exp sX$ from the right, we obtain

$$(\exp sX)^T \eta (\exp sX) = \eta. \quad (24.215)$$

This implies that $\exp sX \in O(3, 1)$. This means $X \in \mathfrak{o}(3, 1)$. These complete the proof.

Let us seek a tangible form of a Lie algebra of the Lorentz group. Suppose that $X \in \mathfrak{o}(3, 1)$ of (24.208). Assuming that

$$X = \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ p & q & r & s \\ t & u & v & w \end{pmatrix} \quad (a \sim w : \text{real}) \quad (24.216)$$

and substituting X of (24.216) for (24.208), we get

$$\begin{pmatrix} 2a & b-e & c-p & d-t \\ b-e & -2f & -(g+q) & -(h+u) \\ c-p & -(g+q) & -2r & -(s+v) \\ d-t & -(h+u) & -(s+v) & -2w \end{pmatrix} = 0. \quad (24.217)$$

Then, we obtain

$$a=f=r=w=0, b=e, c=p, d=t, g=-q, h=-u, s=-v. \quad (24.218)$$

The resulting form of X is

$$X = \begin{pmatrix} 0 & b & c & d \\ b & 0 & g & h \\ c & -g & 0 & s \\ d & -h & -s & 0 \end{pmatrix}. \quad (24.219)$$

Since we have ten conditions for sixteen elements, we must have six independent representations. We show a tangible example of these representations as follows:

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}; \\ \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \quad (24.220)$$

The Lie algebra of the Lorentz group can be expressed as linear combinations of the above matrices (i.e., basis vectors of this Lie algebra). This Lie algebra is a six-dimensional vector space V^6 accordingly. Note that in (24.220) the upper three matrices are anti-Hermitian and that the lower three matrices are Hermitian. We symbolically represent the upper three matrices and the lower three matrices as

$$\mathfrak{A} \equiv (\mathfrak{A}_1, \mathfrak{A}_2, \mathfrak{A}_3) \quad \text{and} \quad \mathfrak{B} \equiv (\mathfrak{B}_1, \mathfrak{B}_2, \mathfrak{B}_3), \quad (24.221)$$

respectively, in the order from the left of (24.220). Regarding the upper three matrices, the (3, 3) principal submatrices excluding the (0, 0) element are identical with A_x , A_y , and A_z in the shown order; see (20.26). In other words, these three matrices represent the spatial rotation in $\mathbb{R}^3$. Moreover, these matrices have been decomposed into a direct sum such that

$$\mathfrak{A}_1 = \{0\} \oplus A_x, \mathfrak{A}_2 = \{0\} \oplus A_y, \mathfrak{A}_3 = \{0\} \oplus A_z, \quad (24.222)$$

where $\{0\}$ denotes the (0, 0) element of $\mathfrak{A}$. The remaining three matrices $\mathfrak{B}$ are called Lorentz boosts.

Expressing a magnitude related to $\mathfrak{A}$ and $\mathfrak{B}$ as $\mathbf{a} \equiv (a_1, a_2, a_3)$ and $\mathbf{b} \equiv (b_1, b_2, b_3)$, respectively, we succinctly describe any Lorentz transformation Λ as

$$\Lambda = \exp(\mathbf{a} \cdot \mathfrak{A}) \exp(\mathbf{b} \cdot \mathfrak{B}). \quad (24.223)$$

The former factor of (24.223) is related to a spatial rotation and represented by a unitary matrix (see Sect. 15.2). The latter factor of (24.223) is pertinent to the Lorentz boost and represented by a Hermitian matrix (Sect. 25.3). Since the Lorentz transformation is represented by a real matrix, a matrix describing the Lorentz boost is symmetric.

For example, with the former factor, we obtain, e.g.,

$$\exp(\theta \mathfrak{A}_3) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (24.224)$$

where we put $a_3 = \theta$. With the Lorentz boost, we have, e.g.,

$$\exp(-\omega \mathfrak{B}_3) = \begin{pmatrix} \cosh \omega & 0 & 0 & -\sinh \omega \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\sinh \omega & 0 & 0 & \cosh \omega \end{pmatrix}, \quad (24.225)$$

where we put $b_3 = -\omega$.

In (24.224), we have $0 \leq \theta \leq 2\pi$, that is, the parameter space θ is bounded. In (24.225), on the other hand, we have $-\infty < \omega < \infty$ where the parameter space ω is unbounded. The former group is called a compact group, whereas the latter is said to be a non-compact group accordingly.

In general, with a compact group, it is possible to construct a unitary representation [8]. For a non-compact group, however, it is not necessarily possible to construct a unitary representation. In fact, although $\exp(\theta \mathfrak{A}_3)$ of (24.224) is unitary [see (7)' of Sect. 15.2], $\exp(-\omega \mathfrak{B}_3)$ in (24.225) is not unitary, but Hermitian. These features of the Lorentz group lead to various intriguing properties (vide infra).

24.2.2 Successive Lorentz Transformations [9]

In the last sections, we have studied how the Lorentz transformation is dealt with using the bilinear form and Lie algebra in the Minkowski space. In the subsequent sections, we examine how the Dirac equation is transformed through the medium of the Lorentz transformation within the framework of standard matrix algebra. To examine the constitution of the Dirac equation appropriately, we need to deal with the successive Lorentz transformations. These transformations can be well understood by the idea of the moving coordinate systems (Sect. 17.4.2).

Let us express a four-vector X (i.e., space-time vector) in the Minkowski space as

$$X = (|e_0\rangle |e_1\rangle |e_2\rangle |e_3\rangle) \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}, \quad (24.226)$$

where $\{|e_0\rangle, |e_1\rangle, |e_2\rangle, |e_3\rangle\}$ are the basis set of Minkowski space defined in Sect. 24.1.7; $\begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}$ are corresponding coordinates. We will use a symbol e_i ($i = 0, 1, 2, 3$) instead of $|e_i\rangle$, in case confusion is unlikely to happen.

Following Sect. 11.4, we specify the transformation in such a way that the said transformation keeps the vector X *unaltered*. In terms of the special theory of relativity, a vector is associated with a certain space-time point where an event has taken place. Consequently, the vector X should be the *same* throughout the inertial frames of reference that are connected to one another via the Lorentz transformations.

In what follows, we express a four-vector X using a shorthand notation as

$$X \equiv \boldsymbol{e}x = (|e_0\rangle |e_1\rangle |e_2\rangle |e_3\rangle) \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix},$$

where $\boldsymbol{e}$ stands for the basis set of $\mathbb{M}^4$ and x denotes the corresponding coordinates (i.e., a column vector). Let a transformation be expressed as P . We rewrite X as

$$X = \boldsymbol{e}x = (\boldsymbol{e}P^{-1})(Px) = \boldsymbol{e}'x', \quad (24.227)$$

where $\boldsymbol{e}$ and $\boldsymbol{e}'$ are the set of basis vectors connected by P with $\boldsymbol{e}' = \boldsymbol{e}P^{-1}$ and $Px = x'$. Once again, we note that in the case where the Lorentz transformation is relevant to the expression, unlike (11.84) of Sect. 11.4 the coordinate representation is placed ahead of the basis vector representation (see Sect. 24.1.5). Next, let us think about the several successive transformations $P_1, \dots, P_n$. We have

$$ex = (eP_1^{-1} \cdots P_n^{-1})(P_n \cdots P_1 x) = e'(P_n \cdots P_1 x) = e'x'. \quad (24.228)$$

From the last equality, we obtain

$$x' = P_n \cdots P_1 x, \quad (24.229)$$

because e' is the basis set of vectors.

Now, we wish to apply the above general point of view to our present case. Let us think of three successive Lorentz transformations. Of these, two transformations are spatial rotations and remaining one is a Lorentz boost. We have four pertinent inertial frames of reference. We denote them by O , O_θ , O_ϕ , and O' . Also, we distinguish the associated basis vectors and corresponding coordinates by

$$e, x(O); \theta, x_\theta(O_\theta); \phi, x_\phi(O_\phi); e', x'(O'). \quad (24.230)$$

We proceed with the calculations from the assumption that an electron is at rest in the frame O , where the Dirac equation can readily be solved. We perform the successive Lorentz transformations that start with O and finally reach O' via O_θ and O_ϕ such that

$$O(e) \rightarrow O_\theta(\theta) \rightarrow O_\phi(\phi) \rightarrow O'(e'). \quad (24.231)$$

Then, the successive transformation of the basis vectors is described by

$$e\Lambda_b\Lambda_\theta^{-1}\Lambda_\phi^{-1} = e', \quad (24.232)$$

where the individual transformations are given in the moving coordinate. Figure 24.2a depicts these transformations similarly as in the case of Fig. 17.17. Equation (24.232) can be rewritten as

$$e = e'\Lambda_\phi\Lambda_\theta\Lambda_b^{-1}. \quad (24.233)$$

In accordance with Fig. 24.2a, Fig. 24.2b shows how the motion of electron looks from the frame O' , where the electron is moving at a velocity $\mathbf{v}$ in the direction defined by a zenithal angle θ and azimuthal angle ϕ . Correspondingly, from (24.229) we have

$$x' = \Lambda_\phi\Lambda_\theta\Lambda_b^{-1}x. \quad (24.234)$$

As the full description of (24.234), we get

$$\begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix} = \Lambda_\phi \Lambda_\theta \Lambda_b^{-1} \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}. \quad (24.235)$$

Thus, we find that three successive Lorentz transformations have produced the total Lorentz transformation Λ described by

$$\Lambda = \Lambda_\phi \Lambda_\theta \Lambda_b^{-1}, \quad (24.236)$$

where Λ_b^{-1} related to the Lorentz boost is the inverse matrix of (24.225). We compute the combined Lorentz transformation Λ such that

$$\begin{aligned} \Lambda &= \Lambda_\phi \Lambda_\theta \Lambda_b^{-1} = \\ &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi & 0 \\ 0 & \sin \phi & \cos \phi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & 0 & \sin \theta \\ 0 & 0 & 1 & 0 \\ 0 & -\sin \theta & 0 & \cos \theta \end{pmatrix} \begin{pmatrix} \cosh \omega & 0 & 0 & \sinh \omega \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh \omega & 0 & 0 & \cosh \omega \end{pmatrix} \\ &= \begin{pmatrix} \cosh \omega & 0 & 0 & \sinh \omega \\ \cos \phi \sin \theta \sinh \omega & \cos \phi \cos \theta & -\sin \phi & \cos \phi \sin \theta \cosh \omega \\ \sin \phi \sin \theta \sinh \omega & \sin \phi \cos \theta & \cos \phi & \sin \phi \sin \theta \cosh \omega \\ \cos \theta \sinh \omega & -\sin \theta & 0 & \cos \theta \cosh \omega \end{pmatrix}. \end{aligned} \quad (24.237)$$

Equation (24.237) describes a general form of the Lorentz transformation with respect to the polar coordinate representation (see Fig. 24.2b). Notice that the transformation Λ expressed as (24.237) satisfies the criterion of (24.197). The confirmation is left for the readers.

The Lorentz boost Λ_b^{-1} has been defined in reference to the frame O . It is worth examining how the said boost is described in reference to the frame O' . For this purpose, we make the most of the general method that was developed in Sect. 17.4.2. It is more convenient to take a reverse procedure to (24.231) so that the general method can directly be applied to the present case. That is, think of the following transformations

$$O' (e') \rightarrow O_\phi (\phi) \rightarrow O_\theta (\theta) \rightarrow O (e) \quad (24.238)$$

in combination with the inverse operators to those shown in Fig. 24.2a. Then, use (17.85) and (17.91) with $k = n = 3$. Deleting $R_3^{(1)}$, we get

$$R_1 R_2^{(1)} R_3^{(2)} \left[R_1 R_2^{(1)} \right]^{-1} = R_3. \quad (24.239)$$

Substituting Λ_ϕ , Λ_θ , and Λ_b^{-1} for R_1 , $R_2^{(1)}$, and $R_3^{(2)}$, respectively, and defining $R_3 \equiv (\tilde{\Lambda}_b)^{-1}$, we obtain

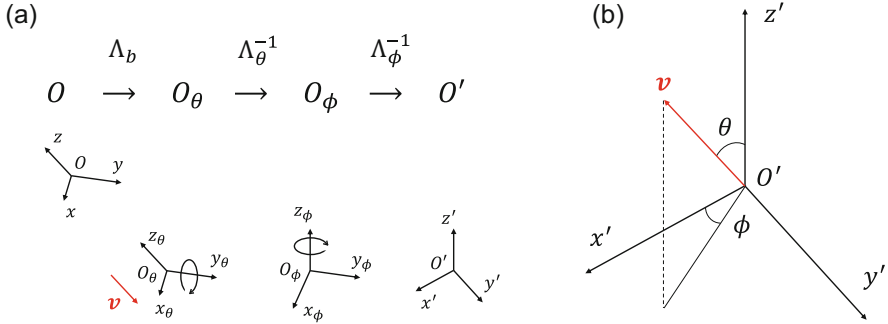


Fig. 24.2 Successive Lorentz transformations and their corresponding coordinate systems. **(a)** Lorentz boost (Λ_b) followed by two spatial rotations (Λ_θ^{-1} and Λ_ϕ^{-1}). The diagram shows the successive three transformations of the basis vectors. **(b)** Geometry of an electron motion. The electron is moving at a velocity $\mathbf{v}$ in the direction defined by a zenithal angle θ and azimuthal angle ϕ in reference to the frame O

$$(\tilde{\Lambda}_b)^{-1} = \Lambda_\phi \Lambda_\theta \Lambda_b^{-1} [\Lambda_\phi \Lambda_\theta]^{-1}. \quad (24.240)$$

Further defining the combined rotation-related unitary transformation Λ_u in the Minkowski space as

$$\Lambda_u \equiv \Lambda_\phi \Lambda_\theta, \quad (24.241)$$

we get

$$\Lambda = \Lambda_u \Lambda_b^{-1}. \quad (24.242)$$

Also, we have

$$(\tilde{\Lambda}_b)^{-1} = \Lambda_u \Lambda_b^{-1} \Lambda_u^\dagger. \quad (24.243)$$

The operator $(\tilde{\Lambda}_b)^{-1}$ represents the Lorentz boost viewed in reference to the frame O' , whereas the operator Λ_b^{-1} is given in reference to O_θ with respect to the same boost. Both the transformations are connected to each other through the unitary similarity transformation and belong to the same conjugacy class, and so they have a same physical meaning. The matrix representation of $(\tilde{\Lambda}_b)^{-1}$ is given by

$$(\tilde{\Lambda}_b)^{-1} = \begin{pmatrix} \cosh\omega & \cos\phi \sin\theta \sinh\omega & \sin\phi \sin\theta \sinh\omega & \cos\theta \sinh\omega \\ \cos\phi \sin\theta \sinh\omega & \cos^2\phi \cos^2\theta + \sin^2\phi + \cos^2\phi \sin^2\theta \cosh\omega & \cos\phi \sin\phi \sin^2\theta \times (\cosh\omega - 1) & \cos\phi \cos\theta \sin\theta \times (\cosh\omega - 1) \\ \sin\phi \sin\theta \sinh\omega & \cos\phi \sin\phi \sin^2\theta \times (\cosh\omega - 1) & \sin^2\phi \cos^2\theta + \cos^2\phi + \sin^2\phi \sin^2\theta \cosh\omega & \sin\phi \cos\theta \sin\theta \times (\cosh\omega - 1) \\ \cos\theta \sinh\omega & \cos\phi \cos\theta \sin\theta \times (\cosh\omega - 1) & \sin\phi \cos\theta \sin\theta \times (\cosh\omega - 1) & \sin^2\theta + \cos^2\theta \cosh\omega \end{pmatrix}. \quad (24.244)$$

The trace χ of (24.244) equals

$$\chi = 2 + \cosh\omega. \quad (24.245)$$

of Λ_b^{-1} , reflecting the fact that the unitary similarity transformation keeps the trace unchanged.

Equation (24.242) gives an interesting example of the decomposition of Lorentz transformation into the unitary part and Hermitian boost part (i.e., polar decomposition). The Lorentz boost is represented by a positive definite Hermitian matrix (or a real symmetric matrix). We will come back to this issue later (Sect. 24.9.1).

24.3 Covariant Properties of the Dirac Equation

24.3.1 General Consideration on the Physical Equation

It is of great importance and interest to examine how the Dirac equation is transformed under the Lorentz transformation. Before addressing this question, we wish to see how physical equations (including the Schrödinger equation, the Klein–Gordon equation, and Dirac equation) undergo the transformation. In general, the physical equations comprise a state function X (scalar, vector, spinor, etc.) and an operator $\mathcal{A}$ that operates on the state function X . A general form of the physical equation is given as

$$\mathcal{A}(X) = 0. \quad (24.246)$$

According to Sect. 11.2, we write X as

$$X = (\xi_1 \cdots \xi_n) \begin{pmatrix} c^1(x) \\ \vdots \\ c^n(x) \end{pmatrix}, \quad (24.247)$$

where $\{\xi_1, \dots, \xi_n\}$ are the basis functions (or basis vectors) of a representation space and $\begin{pmatrix} c^1(x) \\ \vdots \\ c^n(x) \end{pmatrix}$ indicate the corresponding coordinates. In (24.247), we assume that X is an element of the representation space of dimension n that is related to the physical equation. Correspondingly, we write LHS of (24.246) as

$$\mathcal{A}(X) = (\xi_1 \cdots \xi_n) \mathcal{A} \begin{pmatrix} c^1(x) \\ \vdots \\ c^n(x) \end{pmatrix}. \quad (24.248)$$

Also, according to Sect. 11.4, we rewrite (24.248) such that

$$\begin{aligned} \mathcal{A}(X) &= (\xi_1 \cdots \xi_n) R^{-1} R \mathcal{A} R^{-1} R \begin{pmatrix} c^1(x) \\ \vdots \\ c^n(x) \end{pmatrix} \\ &= (\xi'_1 \cdots \xi'_n) R \mathcal{A} R^{-1} \begin{pmatrix} c'^1(x) \\ \vdots \\ c'^n(x) \end{pmatrix} = (\xi'_1 \cdots \xi'_n) \mathcal{A}' \begin{pmatrix} c'^1(x) \\ \vdots \\ c'^n(x) \end{pmatrix}, \end{aligned} \quad (24.249)$$

where R represents a transformation matrix in a general form; it can be e.g., an orthogonal transformation, a Lorentz transformation, etc. Notice again that unlike (11.84) of Sect. 11.4 (see Sect. 24.2.2), the coordinates representation and transformation are placed ahead of those of the basis vectors.

In (24.249), the relation expressed as

$$(\xi'_1 \cdots \xi'_n) = (\xi_1 \cdots \xi_n) R^{-1} \quad (24.250)$$

shows the set of basis functions obtained after the transformation. The relation

$$\begin{pmatrix} c'^1(x) \\ \vdots \\ c'^n(x) \end{pmatrix} = R \begin{pmatrix} c^1(x) \\ \vdots \\ c^n(x) \end{pmatrix} \quad (24.251)$$

represents the corresponding coordinates after being transformed. Also, the operator $\mathcal{A}'$ given by

$$\mathcal{A}' = R \mathcal{A} R^{-1} \quad (24.252)$$

denotes the representation matrix of the operator viewed in reference to $(\xi'_1 \cdots \xi'_n)$.

In general, the collection of the transformations forms a group (i.e., transformation group). In that case, $\{\xi'_1, \dots, \xi'_n\}$ spans the representation space. To stress that

R is the representation matrix of the group, we identify it with $D(R)$. That is, from (18.6) we have

$$D(R) \equiv R \quad \text{and} \quad [D(R)]^{-1} = D(R^{-1}). \quad (24.253)$$

Then, (24.249) is rewritten as

$$\mathcal{A}(X) = (\xi'_1 \cdots \xi'_n) D(R) \mathcal{A} [D(R)]^{-1} \begin{pmatrix} c^1(x) \\ \vdots \\ c^n(x) \end{pmatrix}. \quad (24.254)$$

Instead of the full representation of (24.248) and (24.249), we normally write

$$(\xi_1 \cdots \xi_n) \mathcal{A} = 0 \quad \text{or} \quad (\xi'_1 \cdots \xi'_n) \mathcal{A}' = 0. \quad (24.255)$$

Alternatively, we write

$$\mathcal{A} \begin{pmatrix} c^1(x) \\ \vdots \\ c^n(x) \end{pmatrix} = 0 \quad \text{or} \quad \mathcal{A}' \begin{pmatrix} c'_1(x) \\ \vdots \\ c'_n(x) \end{pmatrix} = 0, \quad (24.256)$$

where

$$\mathcal{A}' = D(R) \mathcal{A} [D(R)]^{-1}, \quad (24.257)$$

$$\begin{pmatrix} c^1(x) \\ \vdots \\ c^n(x) \end{pmatrix} = D(R) \begin{pmatrix} c^1(x) \\ \vdots \\ c^n(x) \end{pmatrix}. \quad (24.258)$$

Equation (24.256) can be viewed as an eigenvalue problem with respect to the operator $\mathcal{A}$ with its corresponding “eigenvector” of $(n, 1)$ matrix. At least one of the eigenvalues is zero (possibly degenerate). The second equation of (24.256) indicates the transformation of the operator $\mathcal{A}$ and its eigenvectors, which are transformed according to (24.257) and (24.258), respectively. Their transformation is accompanied by the transformation of the basis functions (or vectors) $\{\xi_1, \cdots, \xi_n\}$, even though the basis functions (or vectors) are not shown explicitly.

In this section, we solely use (24.256) to conform to the custom. The relevant formalism may well be referred to as the “coordinate representation.” According to the custom, when the formalism is associated with the Lorentz transformation, we express the representation $D(\Lambda)$ as

$$D(\Lambda) \equiv S(\Lambda), \quad (24.259)$$

where Λ indicates a Lorentz transformation; $S(\Lambda)$ denotes a representation of Lorentz group. The notation (Λ) is usually omitted and we simply write S to mean $S(\Lambda)$.

24.3.2 The Klein—Gordon Equation and the Dirac Equation

Now, we wish to see how the tangible physical equations which we are interested in undergo the transformation. We start with the Klein—Gordon equation and then examine the transformation properties of the Dirac equation in detail.

(i) Klein—Gordon equation:

The equation is given by

$$(\partial_\mu \partial^\mu + m^2)\phi(x) = 0. \quad (24.260)$$

Using (24.252), we have

$$\mathcal{A} \equiv \partial_\mu \partial^\mu + m^2, \quad (24.261)$$

$$\mathcal{A}' = S\mathcal{A}S^{-1} = S(\partial_\mu \partial^\mu + m^2)S^{-1}. \quad (24.262)$$

Then, (24.256) can be translated into

$$S(\partial_\mu \partial^\mu + m^2)S^{-1} \cdot S\phi(x) = 0. \quad (24.263)$$

Meanwhile, using (21.25) and (21.51) we obtain

$$\partial_\mu \partial^\mu = \Lambda^\nu{}_\mu \partial'_\nu (\Lambda^{-1})^\mu{}_\rho \partial'^\rho = \delta^\nu_\rho \partial'_\nu \partial'^\rho = \partial'_\rho \partial'^\rho. \quad (24.264)$$

Assuming that S is commutative with both ∂_μ and ∂^μ , (24.263) can be rewritten as

$$(\partial'_\mu \partial'^\mu + m^2)S\phi(x) = 0. \quad (24.265)$$

For (24.265) to be covariant with respect to the Lorentz transformation, we must have

$$(\partial'_\mu \partial'^\mu + m^2)\phi'(x') = 0. \quad (24.266)$$

Since $\phi(x)$ is a scalar function, from the discussion of Sect 19.1 we must have

$$\phi(x) = \phi'(x') = S\phi(x). \quad (24.267)$$

Comparing the first and last sides of (24.267), we get

$$S \equiv 1. \quad (24.268)$$

Thus, the Klein–Gordon equation gives a trivial but important one-dimensional representation with the scalar field.

(ii) Dirac equation:

In the x -system, the Dirac equation has been described as

$$(i\gamma^\mu \partial_\mu - m)\psi(x) = 0. \quad (22.213)$$

Since we assume the four-dimensional representation space with the Dirac spinor $\psi(x)$, we describe $\psi(x)$ as

$$\psi(x) \equiv \begin{pmatrix} c^1(x) \\ c^2(x) \\ c^3(x) \\ c^4(x) \end{pmatrix} \quad (24.269)$$

accordingly. Notice that in (24.269) the superscripts take the integers 1, $\dots$, 4 instead of 0, $\dots$, 3. It is because the representation space for the Dirac equation is not the Minkowski space. Since $\psi(x)$ in (24.269) represents a spinor, the superscripts are not associated with the space-time. Defining

$$\mathcal{A} \equiv i\gamma^\mu \partial_\mu - m, \quad (24.270)$$

(22.213) is read as

$$\mathcal{A}\psi(x) = 0. \quad (24.271)$$

If we were faithful to the notation of (24.248), we would write

$$(\chi_1 \cdots \chi_n) \mathcal{A}\psi(x) = 0.$$

At the moment, however, we only pay attention to the coordinate representation in the representation space. In this case, we may regard the column vector $\psi(x)$, i.e., spinor as the “eigenvector.” The situation is virtually the same as that where we formulated the quantum-mechanical harmonic oscillator in terms of the matrix representation (see Sect. 2.3).

A general form (24.256) is reduced to

$$S(i\gamma^\mu \partial_\mu - m)S^{-1} \cdot S\psi(x) = 0. \quad (24.272)$$

Using (21.51), (24.272) is rewritten as

$$S(i\gamma^\mu \Lambda^\nu{}_\mu \partial'_\nu - m)S^{-1} \cdot S\psi(x) = 0. \quad (24.273)$$

We wish to delete the factor $\Lambda^\nu{}_\mu$ from (24.273) to explicitly describe 22.213 in the Lorentz covariant form. For this, it suffices for the following relation to hold:

$$Si\gamma^\mu \Lambda^\nu{}_\mu \partial'_\nu S^{-1} = i\gamma^\nu \partial'_\nu. \quad (24.274)$$

In (24.274) we have assumed that S is commutative with ∂'_ν . Rewriting (24.274), we get

$$\partial'_\nu (\Lambda^\nu{}_\mu S\gamma^\mu S^{-1} - \gamma^\nu) = 0. \quad (24.275)$$

That is, to maintain the Lorentz covariant form, again, it suffices for the following relation to hold [10, 11]:

$$\Lambda^\nu{}_\mu S\gamma^\mu S^{-1} = \gamma^\nu. \quad (24.276)$$

Notice that in (24.276) neither S nor S^{-1} is in general commutative with γ^μ . Meanwhile, we define the Dirac spinor observed in the x' -system as [10, 11]

$$\psi'(x') \equiv S\psi(x) = S\psi(\Lambda^{-1}x'), \quad (24.277)$$

where $\psi'(x')$ denotes the change in the functional form accompanied by the coordinate transformation. Notice that only in the case where $\psi(x)$ is a scalar function, we have $\psi'(x') = \psi(x)$; see (19.5) and (24.267). Using (24.276), we rewrite (24.273) as

$$(i\gamma^\mu \partial'_\mu - m) \cdot \psi'(x') = 0. \quad (24.278)$$

Thus, on the condition of (24.276), we have gotten the covariant form of the Dirac equation (24.278). Defining

$$\mathcal{A}' \equiv i\gamma^\mu \partial'_\mu - m, \quad (24.279)$$

from (24.273), (24.274), and (24.278), we get

$$\mathcal{A}'\psi'(x') = 0, \quad (24.280)$$

with

$$\mathcal{A}' = S\mathcal{A}S^{-1}. \quad (24.281)$$

Equation (24.281) results from (24.257).

In this way, (24.281) decides the transformation property of the operator $\mathcal{A}$. Needless to say, our next task is to determine a tangible form of S so that (24.276) and (24.281) can hold at once. To do this, (24.276) will provide a good criterion by which to judge whether we have chosen a proper matrix form for S . These will be performed in Sect. 24.4.

A series of equations (24.277)–(24.281) can be viewed as the transformation of the Dirac operators in connection with the Lorentz transformation and is of fundamental importance to solve the Dirac equation. That is, once we successfully find an appropriate form of matrix S in combination with a solution $\psi(x)$ of the Dirac equation in the x -system, from (24.277) we can immediately determine $\psi'(x')$, i.e., a solution of the Dirac equation in the x' -system.

To discuss the transformation of the Dirac equation, in this book we call $\mathfrak{G}$ defined in (21.91) and $\tilde{\mathfrak{G}}$ of (21.108) Dirac operators together with their differential form $\mathcal{A} \equiv i\gamma^\mu \partial_\mu - m$. Note that normally we call the differential operator $(i\gamma^\mu \partial_\mu - m)$ that appeared in Sect. 21.3.1 Dirac operators. When we are dealing with the plane wave solution with the Dirac equation, ∂_μ is to be replaced with $-ip_\mu$ or $+ip_\mu$ (i.e., Fourier components) to give the Dirac operator $\mathfrak{G}$ or $\tilde{\mathfrak{G}}$, respectively.

We also emphasize that (24.277) is contrasted with (24.267), which characterizes the transformation of the scalar function. Comparing the transformation properties of the Klein–Gordon equation and the Dirac equation, we realize that the representation matrix S is closely related to the nature of the representation space. In the representation space, state functions (i.e., scalar, vector, spinor, etc.) and basis functions (e.g., basis vectors in a vector space) as well as the operators undergo the simultaneous transformation by the representation matrix S ; see (24.257) and (24.258). In the next section, we wish to determine a general form of S of the Dirac equation. To this end, we carry the discussion in parallel with that of Sect. 24.2.

Before going into detail, once again we wish to view solving the Dirac equation as an eigenvalue problem. From (21.96), (21.107), and (21.109), we immediately get

$$\mathfrak{G} + \tilde{\mathfrak{G}} = -2mE, \quad (24.282)$$

where E is a $(4, 4)$ identity matrix. Since neither $\mathfrak{G}$ nor $\tilde{\mathfrak{G}}$ is a normal operator (i.e., neither Hermitian nor unitary), it is impossible to diagonalize $\mathfrak{G}$ and $\tilde{\mathfrak{G}}$ through the unitary similarity transformation (see Theorem 14.5). It would therefore be intractable to construct the diagonalizing matrix of the Dirac operators $\mathfrak{G}$ and $\tilde{\mathfrak{G}}$, even though we can find their eigenvalues. Instead, if we are somehow able to find a suitable diagonalizing matrix $S(\Lambda)$, we can at once solve the Dirac equation. Let S be such a matrix that diagonalizes $\mathfrak{G}$. Then, from (24.282) we have

$$S^{-1}\mathfrak{G}S + S^{-1}\tilde{\mathfrak{G}}S = -2mS^{-1}ES = -2mE.$$

Rewriting this equation, we obtain

$$S^{-1}\tilde{\mathfrak{S}}S = -2mE - D,$$

where D is a diagonal matrix given by $D = S^{-1}\mathfrak{S}S$. This implies that $\tilde{\mathfrak{S}}$ can also be diagonalized using the same matrix S .

Bearing in mind this point, in the next section we are going to seek and construct a matrix $S(\Lambda)$ that diagonalizes both $\mathfrak{S}$ and $\tilde{\mathfrak{S}}$.

24.4 Determination of General Form of Matrix $S(\Lambda)$

In the previous section, if the representation matrix $S(\Lambda)$ is properly chosen, $S(i\gamma^\mu\partial_\mu - m)S^{-1}$ and $S\psi(x)$ appropriately represent the Dirac operator and Dirac spinor, respectively. Naturally, then, we are viewing the Dirac operator and spinor in reference to the basis vectors after being transformed. The condition for $S(\Lambda)$ to satisfy is given by (24.276).

The matrices $S(\Lambda)$ of (24.259) and their properties have been fully investigated and available up to now [10, 11]. We will make the most of those results to decide a tangible form of $S(\Lambda)$ for the Dirac equation. Since $S(\Lambda)$ is a representation matrix, from (24.236) we immediately calculate $S(\Lambda)$ such that

$$S(\Lambda) = S(\Lambda_\phi)S(\Lambda_\theta)S(\Lambda_b^{-1}) = S(\Lambda_\phi)S(\Lambda_\theta)[S(\Lambda_b)]^{-1}. \quad (24.283)$$

The matrix $S(\Lambda_\phi)$ produced by the spatial rotation is given as [11]

$$S(\Lambda_\phi) = \exp(\phi\rho_m), \quad (24.284)$$

where ϕ is a rotation angle and ρ_m is given by

$$\rho_m \equiv (-i)J^{kl}; J^{kl} \equiv \frac{i}{4}[\gamma^k, \gamma^l] \quad (k, l, m = 1, 2, 3). \quad (24.285)$$

In (24.285) γ^k is a gamma matrix and k, l, m change cyclic. We have

$$J^{kl} = \frac{1}{2} \begin{pmatrix} \sigma_m & \mathbf{0} \\ \mathbf{0} & \sigma_m \end{pmatrix}, \quad (24.286)$$

where σ_m is a Pauli spin matrix and $\mathbf{0}$ denotes $(2, 2)$ zero matrix. Note that J^{kl} is given by a direct sum of σ_m . That is,

$$J^{kl} = \frac{1}{2} \left[\sigma_m \bigoplus \sigma_m \right]. \quad (24.287)$$

When the rotation by ϕ is made around the z -axis, $S(\Lambda_\phi)$ is given by

$$S(\Lambda_\phi) = \exp(\phi\rho_3) = \begin{pmatrix} e^{i\phi/2} & 0 & 0 & 0 \\ 0 & e^{-i\phi/2} & 0 & 0 \\ 0 & 0 & e^{i\phi/2} & 0 \\ 0 & 0 & 0 & e^{-i\phi/2} \end{pmatrix}. \quad (24.288)$$

With another case where the rotation by θ is made around the y -axis, we have

$$S(\Lambda_\theta) = \exp(\theta\rho_2) = \begin{pmatrix} \cos \frac{\theta}{2} & \sin \frac{\theta}{2} & 0 & 0 \\ -\sin \frac{\theta}{2} & \cos \frac{\theta}{2} & 0 & 0 \\ 0 & 0 & \cos \frac{\theta}{2} & \sin \frac{\theta}{2} \\ 0 & 0 & -\sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}. \quad (24.289)$$

Meanwhile, an example of the transformation matrices $S(\Lambda_b)$ related to the Lorentz boost is expressed as

$$S(\Lambda_b) = \exp(-\omega\beta_k), \quad (24.290)$$

where ω represents the magnitude of boost; β_k is given by

$$\beta_k \equiv (-i) J^{0k}; J^{0k} \equiv \frac{i}{2} \gamma^0 \gamma^k \quad (k = 1, 2, 3). \quad (24.291)$$

If we make the Lorentz boost in the direction of the z -axis, $S(\Lambda_b)$ is expressed as

$$\begin{aligned} S(\Lambda_b) &= \exp(-\omega\beta_3) \\ &= \begin{pmatrix} \cosh \frac{\omega}{2} & 0 & \sinh \frac{\omega}{2} & 0 \\ 0 & \cosh \frac{\omega}{2} & 0 & -\sinh \frac{\omega}{2} \\ \sinh \frac{\omega}{2} & 0 & \cosh \frac{\omega}{2} & 0 \\ 0 & -\sinh \frac{\omega}{2} & 0 & \cosh \frac{\omega}{2} \end{pmatrix}. \end{aligned} \quad (24.292)$$

As in the case of the Lorentz transformation matrices, the generator ρ_m ($m = 1, 2, 3$), which is associated with the spatial rotation, is anti-Hermitian. Hence, $\exp(\phi\rho_3)$ and $\exp(\theta\rho_2)$ are unitary. On the other hand, β_k ($k = 1, 2, 3$), associated with the Lorentz boost, is Hermitian. The matrix $\exp(\omega\beta_3)$ is Hermitian as well; for these see Sects. 15.2 and 25.4. From (24.288), (24.289), and (24.292) we get

$$\begin{aligned}
S(\Lambda) &= S(\Lambda_\phi) S(\Lambda_\theta) [S(\Lambda_b)]^{-1} \\
&= \exp(\phi \rho_3) \exp(\theta \rho_2) \exp(\omega \beta_3) = \\
&\begin{pmatrix} e^{i\phi/2} & 0 & 0 & 0 \\ 0 & e^{-i\phi/2} & 0 & 0 \\ 0 & 0 & e^{i\phi/2} & 0 \\ 0 & 0 & 0 & e^{-i\phi/2} \end{pmatrix} \begin{pmatrix} \cos \frac{\theta}{2} & \sin \frac{\theta}{2} & 0 & 0 \\ -\sin \frac{\theta}{2} & \cos \frac{\theta}{2} & 0 & 0 \\ 0 & 0 & \cos \frac{\theta}{2} & \sin \frac{\theta}{2} \\ 0 & 0 & -\sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} \\
&\quad \times \begin{pmatrix} \cosh \frac{\omega}{2} & 0 & -\sinh \frac{\omega}{2} & 0 \\ 0 & \cosh \frac{\omega}{2} & 0 & \sinh \frac{\omega}{2} \\ -\sinh \frac{\omega}{2} & 0 & \cosh \frac{\omega}{2} & 0 \\ 0 & \sinh \frac{\omega}{2} & 0 & \cosh \frac{\omega}{2} \end{pmatrix} \\
&= \begin{pmatrix} e^{i\phi/2} \cos \frac{\theta}{2} \cosh \frac{\omega}{2} & e^{i\phi/2} \sin \frac{\theta}{2} \cosh \frac{\omega}{2} & -e^{i\phi/2} \cos \frac{\theta}{2} \sinh \frac{\omega}{2} & e^{i\phi/2} \sin \frac{\theta}{2} \sinh \frac{\omega}{2} \\ -e^{-i\phi/2} \sin \frac{\theta}{2} \cosh \frac{\omega}{2} & e^{-i\phi/2} \cos \frac{\theta}{2} \cosh \frac{\omega}{2} & e^{-i\phi/2} \sin \frac{\theta}{2} \sinh \frac{\omega}{2} & e^{-i\phi/2} \cos \frac{\theta}{2} \sinh \frac{\omega}{2} \\ -e^{i\phi/2} \cos \frac{\theta}{2} \sinh \frac{\omega}{2} & e^{i\phi/2} \sin \frac{\theta}{2} \sinh \frac{\omega}{2} & e^{i\phi/2} \cos \frac{\theta}{2} \cosh \frac{\omega}{2} & e^{i\phi/2} \sin \frac{\theta}{2} \cosh \frac{\omega}{2} \\ e^{-i\phi/2} \sin \frac{\theta}{2} \sinh \frac{\omega}{2} & e^{-i\phi/2} \cos \frac{\theta}{2} \sinh \frac{\omega}{2} & -e^{-i\phi/2} \sin \frac{\theta}{2} \cosh \frac{\omega}{2} & e^{-i\phi/2} \cos \frac{\theta}{2} \cosh \frac{\omega}{2} \end{pmatrix}.
\end{aligned} \tag{24.293}$$

Also, we have

$$[S(\Lambda)]^{-1} =$$

$$\begin{pmatrix} e^{-i\phi/2} \cos \frac{\theta}{2} \cosh \frac{\omega}{2} & -e^{i\phi/2} \sin \frac{\theta}{2} \cosh \frac{\omega}{2} & e^{-i\phi/2} \cos \frac{\theta}{2} \sinh \frac{\omega}{2} & -e^{i\phi/2} \sin \frac{\theta}{2} \sinh \frac{\omega}{2} \\ e^{-i\phi/2} \sin \frac{\theta}{2} \cosh \frac{\omega}{2} & e^{i\phi/2} \cos \frac{\theta}{2} \cosh \frac{\omega}{2} & -e^{-i\phi/2} \sin \frac{\theta}{2} \sinh \frac{\omega}{2} & -e^{i\phi/2} \cos \frac{\theta}{2} \sinh \frac{\omega}{2} \\ e^{-i\phi/2} \cos \frac{\theta}{2} \sinh \frac{\omega}{2} & -e^{i\phi/2} \sin \frac{\theta}{2} \sinh \frac{\omega}{2} & e^{-i\phi/2} \cos \frac{\theta}{2} \cosh \frac{\omega}{2} & -e^{i\phi/2} \sin \frac{\theta}{2} \cosh \frac{\omega}{2} \\ -e^{-i\phi/2} \sin \frac{\theta}{2} \sinh \frac{\omega}{2} & -e^{i\phi/2} \cos \frac{\theta}{2} \sinh \frac{\omega}{2} & e^{-i\phi/2} \sin \frac{\theta}{2} \cosh \frac{\omega}{2} & e^{i\phi/2} \cos \frac{\theta}{2} \cosh \frac{\omega}{2} \end{pmatrix}. \quad (24.294)$$

Equation (24.293) gives a general form of $S(\Lambda)$. The construction processes of (24.293) somewhat resemble those already taken to form the unitary transformation matrix of $SU(2)$ in (20.45) or those to construct the transformation matrix of (17.101). Because of the presence of $[S(\Lambda_b)]^{-1}$ in (24.293), however, $S(\Lambda)$ of (24.293) is not unitary. This feature plays an essential role in what follows.

24.5 Transformation Properties of the Dirac Spinors [9]

Now, we are in the position to seek a tangible form of the Dirac spinors. Regarding the plane waves, $\psi(x)$ written in a general form in (24.269) is described by

$$\psi(x) = \begin{pmatrix} c^0(x) \\ c^1(x) \\ c^2(x) \\ c^3(x) \end{pmatrix} = e^{\pm ipx} \begin{pmatrix} c^0 \\ c^1 \\ c^2 \\ c^3 \end{pmatrix} = e^{\pm ipx} w(\mathbf{p}, h), \quad (24.295)$$

where $w(\mathbf{p}, h)$ represents either $u(\mathbf{p}, h)$ or $v(\mathbf{p}, h)$ given in Sect. 21.3. Note that $w(\mathbf{p}, h)$ does not depend on x . Since $px (\equiv p_\mu x^\mu)$ in $e^{\pm ipx}$ is a scalar quantity, it is independent of the choice of an inertial frame of reference, namely, px is invariant with the Lorentz transformation. Therefore, it behaves as a constant in terms of the operation of $S(\Lambda)$.

Let the x -system be set within the inertial frame of reference O where the electron is at rest (see Fig. 24.2). Then, we have

$$\psi(x) = e^{\pm ip_0 x^0} w(\mathbf{0}, h) = e^{\pm imx^0} w(\mathbf{0}, h). \quad (24.296)$$

Meanwhile, (24.277) is rewritten as

$$\psi'(x') = e^{\pm ip'x'} S(\Lambda) w(\mathbf{0}, h) = e^{\pm ip'x'} w(\mathbf{p}', h). \quad (24.297)$$

In (24.297) we assume that x' -system is set within the inertial frame of reference O' where the electron is moving at a velocity of $\mathbf{v}$ (see Fig. 24.2). Note that in (24.297) $p'x' = px = p_0 x^0 = mx^0$ and $p = \begin{pmatrix} p^0 \\ \mathbf{p} \end{pmatrix}; p^0 = p_0, p' = \begin{pmatrix} p'^0 \\ \mathbf{p}' \end{pmatrix}$.

In the remaining parts of this section, we use the variables x and p for the frame O and the variables x' and p' for the frame O' . Note that the frames O and O' are identical to those appearing in Sect. 24.2.2. First, we wish to seek the solution $w(\mathbf{0}, h)$ of the Dirac equation with an electron at rest. In this case, momentum $\mathbf{p}$ of the electron is zero, i.e., $p^1 = p^2 = p^3 = 0$. Then, if we find the solution $w(\mathbf{0}, h)$ in the frame O , we are going to get a solution $w(\mathbf{p}', h)$ in the frame O' such that

$$S(\Lambda)w(\mathbf{0}, h) = w(\mathbf{p}', h). \quad (24.298)$$

Equation (24.298) implies that the helicity is held unchanged by the transformation $S(\Lambda)$. Now, the Dirac operator of the rest frame is given by

$$p_\mu \gamma^\mu - m = p_0 \gamma^0 - m = m \gamma^0 - m = m(\gamma^0 - 1) \equiv A \quad (24.299)$$

or

$$-p_\mu \gamma^\mu - m = -p_0 \gamma^0 - m = -m \gamma^0 - m = m(-\gamma^0 - 1) \equiv B. \quad (24.300)$$

Namely, we have

$$A \equiv \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -2m & 0 \\ 0 & 0 & 0 & -2m \end{pmatrix} \quad (24.301)$$

and

$$B \equiv \begin{pmatrix} -2m & 0 & 0 & 0 \\ 0 & -2m & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (24.302)$$

The operators A and B correspond to the positive-energy solution and the negative-energy solution, respectively. For the former case, we have

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -2m & 0 \\ 0 & 0 & 0 & -2m \end{pmatrix} \begin{pmatrix} c^0 \\ c^1 \\ c^2 \\ c^3 \end{pmatrix} = 0. \quad (24.303)$$

In (24.303) we have $c^2 = c^3 = 0$ and c^1 and c^0 can freely be chosen. Then, with the two independent solutions, we get

$$u_1(\mathbf{0}, -1) = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, u_2(\mathbf{0}, +1) = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}. \quad (24.304)$$

With the plane wave solutions, we have

$$\psi_1(x) = e^{-imx^0} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \psi_2(x) = e^{-imx^0} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad (x^0 \equiv t). \quad (24.305)$$

With the negative-energy solution, on the other hand, we have

$$\begin{pmatrix} -2m & 0 & 0 & 0 \\ 0 & -2m & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} c^0 \\ c^1 \\ c^2 \\ c^3 \end{pmatrix} = 0. \quad (24.306)$$

Therefore, as another two independent solutions we obtain

$$v_1(\mathbf{0}, +1) = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, v_2(\mathbf{0}, -1) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad (24.307)$$

with the plane wave solutions described by

$$\psi_3(x) = e^{imx^0} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \psi_4(x) = e^{imx^0} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \quad (24.308)$$

The above solutions (24.304) can be automatically obtained from the results of Sect. 21.3 (including the normalized constant as well). That is, using (21.104) and putting $\theta = \phi = 0$ as well as $S = 0$ in (21.104), we have the same results as (24.304). Note, however, that in (24.307) we did not take account of the charge conjugation (vide infra). But, using (21.120) and putting $\theta = \phi = 0$ and $S = 0$ once again, we obtain

$$v_1(\mathbf{0}, +1) = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, v_2(\mathbf{0}, -1) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix}. \quad (24.309)$$

Notice that the latter solution of (24.309) has correctly reflected the charge conjugation. Therefore, with the negative-energy plane wave solutions, we must use

$$\psi_3(x) = e^{imx^0} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \psi_4(x) = e^{imx^0} \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix}. \quad (24.310)$$

Even though (24.303) and (24.306) and their corresponding solutions (24.304) and (24.309) look trivial, they provide solid ground for the Dirac equation and its solution of an electron at rest. In our case, these equations are dealt with in the frame O and described by x .

In the above discussion, (24.305) and (24.310) give $\psi(x)$ in (24.296). Thus, substituting (24.304) and (24.309) for $w(\mathbf{0}, h)$ of (24.297) and using $D(\Lambda) = S(\Lambda)$ of (24.293), we get at once the solutions $\psi'(x')$ described by x' with respect to the frame O' . The said solutions are given by

$$\begin{aligned} \psi'_1(x') = e^{-ip'x'} \begin{pmatrix} e^{i\phi/2} \cos \frac{\theta}{2} \cosh \frac{\omega}{2} \\ -e^{-i\phi/2} \sin \frac{\theta}{2} \cosh \frac{\omega}{2} \\ -e^{i\phi/2} \cos \frac{\theta}{2} \sinh \frac{\omega}{2} \\ e^{-i\phi/2} \sin \frac{\theta}{2} \sinh \frac{\omega}{2} \end{pmatrix}, \psi'_2(x') = e^{-ip'x'} \begin{pmatrix} e^{i\phi/2} \sin \frac{\theta}{2} \cosh \frac{\omega}{2} \\ e^{-i\phi/2} \cos \frac{\theta}{2} \cosh \frac{\omega}{2} \\ e^{i\phi/2} \sin \frac{\theta}{2} \sinh \frac{\omega}{2} \\ e^{-i\phi/2} \cos \frac{\theta}{2} \sinh \frac{\omega}{2} \end{pmatrix}, \\ \psi'_3(x') = e^{ip'x'} \begin{pmatrix} -e^{i\phi/2} \cos \frac{\theta}{2} \sinh \frac{\omega}{2} \\ e^{-i\phi/2} \sin \frac{\theta}{2} \sinh \frac{\omega}{2} \\ e^{i\phi/2} \cos \frac{\theta}{2} \cosh \frac{\omega}{2} \\ -e^{-i\phi/2} \sin \frac{\theta}{2} \cosh \frac{\omega}{2} \end{pmatrix}, \psi'_4(x') = e^{ip'x'} \begin{pmatrix} -e^{i\phi/2} \sin \frac{\theta}{2} \sinh \frac{\omega}{2} \\ -e^{-i\phi/2} \cos \frac{\theta}{2} \sinh \frac{\omega}{2} \\ -e^{i\phi/2} \sin \frac{\theta}{2} \cosh \frac{\omega}{2} \\ -e^{-i\phi/2} \cos \frac{\theta}{2} \cosh \frac{\omega}{2} \end{pmatrix}. \end{aligned} \quad (24.311)$$

The individual column vectors of (24.311) correspond to those of (24.293) with the sign for $\psi'_4(x')$ reversed relative to the last column of (24.293). This is because of the charge conjugation. Note that both the positive-energy solution and the negative-energy solution are transformed by the *same* representation matrix S . This is evident from (24.297).

Further rewriting (24.311), we obtain e.g.,

$$\psi'_1(x') = e^{-ip'x'} \cosh \frac{\omega}{2} \begin{pmatrix} e^{i\phi/2} \cos \frac{\theta}{2} \\ -e^{-i\phi/2} \sin \frac{\theta}{2} \\ -e^{i\phi/2} \cos \frac{\theta}{2} \tanh \frac{\omega}{2} \\ e^{-i\phi/2} \sin \frac{\theta}{2} \tanh \frac{\omega}{2} \end{pmatrix}. \quad (24.312)$$

Notice that in (24.312) we normalized $\psi_1(x)$ to unity [instead of $2m$ used in e.g., (21.141)]. Here we wish to associate the hyperbolic functions with the physical quantities of the Dirac fields. From (1.8), (1.9), and (1.21) along with (21.45), we have [12]

$$p'^0 = m\gamma = m \cosh \omega \quad \text{and} \quad \mathbf{p}' = m\gamma \mathbf{v}. \quad (24.313)$$

As important formulae of the hyperbolic functions (associated with the physical quantities), we have

$$\cosh \frac{\omega}{2} = \sqrt{\frac{1 + \cosh \omega}{2}} = \sqrt{\frac{p'^0 + m}{2m}}, \quad \sinh \frac{\omega}{2} = \sqrt{\frac{-1 + \cosh \omega}{2}} = \sqrt{\frac{p'^0 - m}{2m}}, \quad (24.314)$$

$$\tanh \frac{\omega}{2} = \frac{\sinh \frac{\omega}{2}}{\cosh \frac{\omega}{2}} = \sqrt{\frac{p'^0 - m}{p'^0 + m}} = \sqrt{\frac{(p'^0)^2 - m^2}{(p'^0 + m)^2}} = \frac{\mathcal{P}'}{p'^0 + m} = \mathcal{S}', \quad (24.315)$$

where $\mathcal{P}' \equiv |\mathbf{p}'|$ and $\mathcal{S}' \equiv \mathcal{P}' / (p'^0 + m)$ [see (21.97) and (21.99)]. With the second to the last equality of (24.315), we used (21.79) and (21.97); with the last equality we used (21.99). Then, we rewrite (24.312) as

$$\psi'_1(x') = e^{-ip'x'} \sqrt{\frac{p'^0 + m}{2m}} \begin{pmatrix} e^{i\phi/2} \cos \frac{\theta}{2} \\ -e^{-i\phi/2} \sin \frac{\theta}{2} \\ -\mathcal{S}' e^{i\phi/2} \cos \frac{\theta}{2} \\ \mathcal{S}' e^{-i\phi/2} \sin \frac{\theta}{2} \end{pmatrix}. \quad (24.316)$$

The column vector of (24.316) is exactly the same as the first equation of (21.104) including the phase factor (but, aside from the normalization factor $\sqrt{1/2m}$). Other three equations of (24.311) yield the column vectors same as given in (21.104) and (21.120) as well.

Thus, following the procedure of (24.277) we have proven that $S(\Lambda)$ given in (24.293) produces the proper solutions of the Dirac equation. From (24.304) and (24.309), we find that individual column vectors of $S(\Lambda)$ yield four solutions of the Dirac equation, i.e., the eigenvectors of $\mathfrak{G}$ and $\tilde{\mathfrak{G}}$ (aside from the phase factor). This

suggests that $S(\Lambda)$ is a diagonalizing matrix of $\mathfrak{G}$ and $\tilde{\mathfrak{G}}$. For the same reason, the column vectors of $S(\Lambda)$ are the eigenvectors of $\mathfrak{H}$ for (21.86) as well. As already pointed out in Sect. 21.3.2, however, $\mathfrak{G}$ is not Hermitian. Consequently, on the basis of Theorem 14.5, the said eigenvectors shared by $\mathfrak{G}$ and $\mathfrak{H}$ do not constitute the orthonormal system. Hence, we conclude $S(\Lambda)$ is not unitary. This is consistent with the results of Sect. 24.4; see an explicit matrix form of $S(\Lambda)$ indicated by (24.293). That is, the third factor matrix representing the Lorentz boost in LHS of (24.293) is not unitary.

Now, we represent the positive energy solutions and negative-energy solutions as Ψ and $\tilde{\Psi}$ altogether, respectively. Operating both sides of (24.282) on Ψ and $\tilde{\Psi}$ from the left, we get

$$\begin{aligned}\mathfrak{G}\Psi + \tilde{\mathfrak{G}}\Psi &= 0 + \tilde{\mathfrak{G}}\Psi = \tilde{\mathfrak{G}}\Psi = -2m\Psi, \\ \mathfrak{G}\tilde{\Psi} + \tilde{\mathfrak{G}}\tilde{\Psi} &= \mathfrak{G}\tilde{\Psi} + 0 = \mathfrak{G}\tilde{\Psi} = -2m\tilde{\Psi}.\end{aligned}$$

From the above equations, we realize that Ψ is an eigen function that belongs to the eigenvalue 0 of $\mathfrak{G}$ and, at the same time, an eigen function that belongs to the eigenvalue $-2m$ of $\tilde{\mathfrak{G}}$. In turn, $\tilde{\Psi}$ is an eigen function that belongs to the eigenvalue 0 of $\tilde{\mathfrak{G}}$ and, at the same time, an eigen function that belongs to the eigenvalue $-2m$ of $\mathfrak{G}$. Both $\mathfrak{G}$ and $\tilde{\mathfrak{G}}$ possess the eigenvalues 0 and $-2m$ with both doubly degenerate. Thus, the questions raised in Sect. 21.3.2 have been adequately answered.

The point rests upon finding out a proper matrix form of $S(\Lambda)$. We will further pursue the structure of $S(\Lambda)$ in Sect. 24.9 in relation to the polar decomposition of a matrix.

24.6 Transformation Properties of the Dirac Operators [9]

In the previous section, we have derived the functional form of the Dirac spinor from that of the Dirac spinor determined in the rest frame of the electron. In this section we study how the Dirac operator is transformed according to the Lorentz transformation. In the first half of this section, we examine it in reference to the rest frame in accordance with the approach taken in the last section. More generally, however, we need to investigate the transformation property of the Dirac operator in reference to a moving frame of an electron. We deal with this issue in the latter half of the section.

24.6.1 Case I: Single Lorentz Boost

In (24.293) we considered $S(\Lambda) = S(\Lambda_\phi)S(\Lambda_\theta)[S(\Lambda_b)]^{-1}$. To further investigate the operation and properties of $S(\Lambda)$, we consider the spatial rotation factor $S(\Lambda_\phi)S(\Lambda_\theta)$ and the Lorentz boost $[S(\Lambda_b)]^{-1}$ of (24.293) separately.

Defining S_R as

$$S_R \equiv S(\Lambda_u) = S(\Lambda_\phi)S(\Lambda_\theta), \quad (24.317)$$

we have

$$S_R = \begin{pmatrix} e^{i\phi/2} & 0 & 0 & 0 \\ 0 & e^{-i\phi/2} & 0 & 0 \\ 0 & 0 & e^{i\phi/2} & 0 \\ 0 & 0 & 0 & e^{-i\phi/2} \end{pmatrix} \begin{pmatrix} \cos \frac{\theta}{2} & \sin \frac{\theta}{2} & 0 & 0 \\ -\sin \frac{\theta}{2} & \cos \frac{\theta}{2} & 0 & 0 \\ 0 & 0 & \cos \frac{\theta}{2} & \sin \frac{\theta}{2} \\ 0 & 0 & -\sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} \\ = \begin{pmatrix} e^{i\phi/2} \cos \frac{\theta}{2} & e^{i\phi/2} \sin \frac{\theta}{2} & 0 & 0 \\ -e^{-i\phi/2} \sin \frac{\theta}{2} & e^{-i\phi/2} \cos \frac{\theta}{2} & 0 & 0 \\ 0 & 0 & e^{i\phi/2} \cos \frac{\theta}{2} & e^{i\phi/2} \sin \frac{\theta}{2} \\ 0 & 0 & -e^{-i\phi/2} \sin \frac{\theta}{2} & e^{-i\phi/2} \cos \frac{\theta}{2} \end{pmatrix}. \quad (24.318)$$

Also defining a (2, 2) submatrix $\tilde{S}_R$ as

$$\tilde{S}_R \equiv \begin{pmatrix} e^{i\phi/2} \cos \frac{\theta}{2} & e^{i\phi/2} \sin \frac{\theta}{2} \\ -e^{-i\phi/2} \sin \frac{\theta}{2} & e^{-i\phi/2} \cos \frac{\theta}{2} \end{pmatrix}, \quad (24.319)$$

we have

$$S_R = \begin{pmatrix} \tilde{S}_R & \mathbf{0} \\ \mathbf{0} & \tilde{S}_R \end{pmatrix}. \quad (24.320)$$

Then, we obtain

$$S_R A S_R^{-1} = (-2m) \begin{pmatrix} \tilde{S}_R & \mathbf{0} \\ \mathbf{0} & \tilde{S}_R \end{pmatrix} \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & E_2 \end{pmatrix} \begin{pmatrix} \tilde{S}_R^{-1} & \mathbf{0} \\ \mathbf{0} & \tilde{S}_R^{-1} \end{pmatrix} \\ = (-2m) \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \tilde{S}_R E_2 \tilde{S}_R^{-1} \end{pmatrix} = (-2m) \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & E_2 \end{pmatrix} = A, \quad (24.321)$$

where A is that appeared in (24.301). Consequently, the functional form of the Dirac equation for the electron at rest is invariant with respect to the spatial rotation. Corresponding to (24.311), the Dirac spinor is given by e.g.,

$$\psi'_1(x') = S_R \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = e^{-imx^0} \begin{pmatrix} e^{i\phi/2} \cos \frac{\theta}{2} \\ -e^{-i\phi/2} \sin \frac{\theta}{2} \\ 0 \\ 0 \end{pmatrix}, \quad (24.322)$$

$$\psi'_2(x') = S_R \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = e^{-imx^0} \begin{pmatrix} e^{i\phi/2} \sin \frac{\theta}{2} \\ e^{-i\phi/2} \cos \frac{\theta}{2} \\ 0 \\ 0 \end{pmatrix}. \quad (24.323)$$

Similarly, we have

$$\psi'_3(x') = S_R \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = e^{imx^0} \begin{pmatrix} 0 \\ 0 \\ e^{i\phi/2} \cos \frac{\theta}{2} \\ -e^{-i\phi/2} \sin \frac{\theta}{2} \end{pmatrix}, \quad (24.324)$$

$$\psi'_4(x') = S_R \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix} = e^{imx^0} \begin{pmatrix} 0 \\ 0 \\ -e^{i\phi/2} \sin \frac{\theta}{2} \\ -e^{-i\phi/2} \cos \frac{\theta}{2} \end{pmatrix}. \quad (24.325)$$

With the projection operators, we obtain e.g.,

$$|\psi'_1\rangle\langle\psi'_1| = \begin{pmatrix} \cos^2 \frac{\theta}{2} & -e^{i\phi} \cos \frac{\theta}{2} \sin \frac{\theta}{2} & 0 & 0 \\ -e^{-i\phi} \cos \frac{\theta}{2} \sin \frac{\theta}{2} & \sin^2 \frac{\theta}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = (|\psi'_1\rangle\langle\psi'_1|)^2. \quad (24.326)$$

We find that $|\psi'_1\rangle\langle\psi'_1|$ is a Hermitian and idempotent operator (Sects. 12.4 and 14.1); i.e., a projection operator *sensu stricto* (Sect. 18.7). It is the case with other operators $|\psi'_k\rangle\langle\psi'_k|$ ($k = 2, 3, 4$) as well. Similarly calculating other operators, we get

$$\sum_{k=1}^4 |\psi'_k\rangle\langle\psi'_k| = E_4, \quad (24.327)$$

where E_4 is a $(4, 4)$ identity matrix. Other requirements with the projection operators mentioned in Sect. 14.1 are met accordingly (Definition 14.1). Equation (24.327) represents the completeness relation (see Sect. 18.7).

It is worth mentioning that in (24.322)–(24.325) the components of upper two rows and lower two rows are separated with the two pairs of Dirac spinors $\psi_1(x)$, $\psi_2(x)$ and $\psi_3(x)$, $\psi_4(x)$. The spatial rotation does not change the relative movement between the observers standing still on the individual inertial frames of reference. This is the case with the frames O and O_R . Here, we have defined O_R as an inertial frame of reference that is reached by the spatial rotation of O . Consequently, if the observer standing still on O_R is seeing an electron at rest, the other observer on O is seeing that electron at rest. Thus, the physical situation is the same in that sense with those two observers either on O or on O_R .

Meanwhile, defining the Lorentz boost factor $S_\omega \equiv [S(\Lambda_b)]^{-1}$ in (24.293) as

$$S_\omega = \begin{pmatrix} \cosh \frac{\omega}{2} & 0 & -\sinh \frac{\omega}{2} & 0 \\ 0 & \cosh \frac{\omega}{2} & 0 & \sinh \frac{\omega}{2} \\ -\sinh \frac{\omega}{2} & 0 & \cosh \frac{\omega}{2} & 0 \\ 0 & \sinh \frac{\omega}{2} & 0 & \cosh \frac{\omega}{2} \end{pmatrix}, \quad (24.328)$$

we get

$$\begin{aligned} & S_\omega A S_\omega^{-1} \\ &= (-2m) \begin{pmatrix} \cosh \frac{\omega}{2} & 0 & -\sinh \frac{\omega}{2} & 0 \\ 0 & \cosh \frac{\omega}{2} & 0 & \sinh \frac{\omega}{2} \\ -\sinh \frac{\omega}{2} & 0 & \cosh \frac{\omega}{2} & 0 \\ 0 & \sinh \frac{\omega}{2} & 0 & \cosh \frac{\omega}{2} \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & E_2 \end{pmatrix} \\ & \quad \times \begin{pmatrix} \cosh \frac{\omega}{2} & 0 & \sinh \frac{\omega}{2} & 0 \\ 0 & \cosh \frac{\omega}{2} & 0 & -\sinh \frac{\omega}{2} \\ \sinh \frac{\omega}{2} & 0 & \cosh \frac{\omega}{2} & 0 \\ 0 & -\sinh \frac{\omega}{2} & 0 & \cosh \frac{\omega}{2} \end{pmatrix} \end{aligned}$$

$$= (-2m) \begin{pmatrix} -\sinh^2 \frac{\omega}{2} & 0 & -\cosh \frac{\omega}{2} \sinh \frac{\omega}{2} & 0 \\ 0 & -\sinh^2 \frac{\omega}{2} & 0 & \cosh \frac{\omega}{2} \sinh \frac{\omega}{2} \\ \cosh \frac{\omega}{2} \sinh \frac{\omega}{2} & 0 & \cosh^2 \frac{\omega}{2} & 0 \\ 0 & -\cosh \frac{\omega}{2} \sinh \frac{\omega}{2} & 0 & \cosh^2 \frac{\omega}{2} \end{pmatrix}. \quad (24.329)$$

Rewriting (24.329) by use of the formulae of the hyperbolic functions of (24.314), we obtain

$$S_\omega A S_\omega^{-1} = \begin{pmatrix} p'^0 - m & 0 & p'^3 & 0 \\ 0 & p'^0 - m & 0 & -p'^3 \\ -p'^3 & 0 & -p'^0 - m & 0 \\ 0 & p'^3 & 0 & -p'^0 - m \end{pmatrix} (p'^3 > 0). \quad (24.330)$$

In the above computation, we put $\phi = \theta = 0$ in (24.293). From (24.314) we had

$$\cosh \frac{\omega}{2} \sinh \frac{\omega}{2} = \sqrt{\frac{p'^0 + m}{2m}} \sqrt{\frac{p'^0 - m}{2m}} = \frac{|\mathbf{p}'|}{2m} = \frac{p'^3}{2m}.$$

This is identical with RHS of (21.80). The Dirac spinor is given by e.g.,

$$\begin{aligned} \psi'_1(x') &= S_\omega \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = e^{-ip'x'} \begin{pmatrix} \cosh \frac{\omega}{2} \\ 0 \\ -\sinh \frac{\omega}{2} \\ 0 \end{pmatrix} \\ &= e^{-ip'x'} \sqrt{\frac{p'^0 + m}{2m}} \begin{pmatrix} 1 \\ 0 \\ -\frac{|\mathbf{p}'^3|}{p'^0 + m} \\ 0 \end{pmatrix}, \end{aligned} \quad (24.331)$$

$$\begin{aligned}
\psi'_2(x') &= S_\omega \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = e^{-ip'x'} \begin{pmatrix} 0 \\ \cosh \frac{\omega}{2} \\ 0 \\ \sinh \frac{\omega}{2} \end{pmatrix} \\
&= e^{-ip'x'} \sqrt{\frac{p'^0 + m}{2m}} \begin{pmatrix} 0 \\ 1 \\ 0 \\ \frac{|p'^3|}{p'^0 + m} \end{pmatrix}.
\end{aligned} \tag{24.332}$$

Similarly, we have

$$\begin{aligned}
\psi'_3(x') &= S_\omega \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = e^{ip'x'} \begin{pmatrix} -\sinh \frac{\omega}{2} \\ 0 \\ \cosh \frac{\omega}{2} \\ 0 \end{pmatrix} \\
&= e^{ip'x'} \sqrt{\frac{p'^0 + m}{2m}} \begin{pmatrix} -\frac{|p'^3|}{p'^0 + m} \\ 0 \\ 1 \\ 0 \end{pmatrix},
\end{aligned} \tag{24.333}$$

$$\begin{aligned}
\psi'_4(x') &= S_\omega \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix} = e^{ip'x'} \begin{pmatrix} 0 \\ \sinh \frac{\omega}{2} \\ 0 \\ \cosh \frac{\omega}{2} \end{pmatrix} \\
&= e^{ip'x'} \sqrt{\frac{p'^0 + m}{2m}} \begin{pmatrix} 0 \\ -\frac{|p'^3|}{p'^0 + m} \\ 0 \\ -1 \end{pmatrix}.
\end{aligned} \tag{24.334}$$

In the case of the Lorentz boost, we see that from (24.322) to (24.325) the components of upper two rows and lower two rows are intermixed with the two pairs of Dirac spinors $\psi'_1(x'), \psi'_2(x')$ and $\psi'_3(x'), \psi'_4(x')$. This is because whereas one is observing the electron at rest, the other is observing the moving electron.

Using (24.301) and (24.302), we have

$$B = -2m - A. \tag{24.335}$$

Therefore, multiplying both sides of (24.335) by S_ω from the left and S_ω^{-1} from the right and further using (24.330), we get

$$\begin{aligned}
S_\omega B S_\omega^{-1} &= -2m - S_\omega A S_\omega^{-1} \\
&= \begin{pmatrix} -p'^0 - m & 0 & -p'^3 & 0 \\ 0 & -p'^0 - m & 0 & p'^3 \\ p'^3 & 0 & p'^0 - m & 0 \\ 0 & -p'^3 & 0 & p'^0 - m \end{pmatrix}. \quad (24.336)
\end{aligned}$$

Equation (24.336) is the operator that gives the negative-energy solution of the Dirac equation. From (24.293) we see that $S(\Lambda)$ is decomposed into $S(\Lambda) = S_R S_\omega$. Although S_R is unitary, S_ω is not a unitary matrix nor a normal matrix either. Accordingly, S is not a unitary matrix as a whole. Using (24.329) in combination with (24.318), we obtain

$$\begin{aligned}
S A S^{-1} &= S_R (S_\omega A S_\omega^{-1}) S_R^{-1} = S(\Lambda) A [S(\Lambda)]^{-1} = (-2m) \times \\
&\begin{pmatrix} -\sinh^2 \frac{\omega}{2} & 0 & -\cosh \theta \cosh \frac{\omega}{2} \sinh \frac{\omega}{2} & e^{i\phi} \sin \theta \cosh \frac{\omega}{2} \sinh \frac{\omega}{2} \\ 0 & -\sinh^2 \frac{\omega}{2} & e^{-i\phi} \sin \theta \cosh \frac{\omega}{2} \sinh \frac{\omega}{2} & \cosh \theta \cosh \frac{\omega}{2} \sinh \frac{\omega}{2} \\ \cosh \theta \cosh \frac{\omega}{2} \sinh \frac{\omega}{2} & -e^{i\phi} \sin \theta \cosh \frac{\omega}{2} \sinh \frac{\omega}{2} & \cosh^2 \frac{\omega}{2} & 0 \\ -e^{-i\phi} \sin \theta \cosh \frac{\omega}{2} \sinh \frac{\omega}{2} & -\cosh \theta \cosh \frac{\omega}{2} \sinh \frac{\omega}{2} & 0 & \cosh^2 \frac{\omega}{2} \end{pmatrix}. \quad (24.337)
\end{aligned}$$

Using (24.314) and (21.79), (24.337) can be rewritten in the polar coordinate representation as

$$\begin{aligned}
S A S^{-1} &= \begin{pmatrix} p'^0 - m & 0 & |\mathbf{p}'| \cos \theta & -|\mathbf{p}'| e^{i\phi} \sin \theta \\ 0 & p'^0 - m & -|\mathbf{p}'| e^{-i\phi} \sin \theta & -|\mathbf{p}'| \cos \theta \\ -|\mathbf{p}'| \cos \theta & |\mathbf{p}'| e^{i\phi} \sin \theta & -p'^0 - m & 0 \\ |\mathbf{p}'| e^{-i\phi} \sin \theta & |\mathbf{p}'| \cos \theta & 0 & -p'^0 - m \end{pmatrix}. \quad (24.338)
\end{aligned}$$

Converting the polar coordinate into the Cartesian coordinate, we find that (24.338) is identical with the (4, 4) matrix of (21.75). Namely, we have successfully gotten

$$S A S^{-1} = \mathfrak{G}. \quad (24.339)$$

We remark that when deriving (24.338), we may choose $(\tilde{\Lambda}_b)^{-1}$ of (24.240) for the Lorentz boost. Using $S[(\tilde{\Lambda}_b)^{-1}]$ in place of $S(\Lambda) = S(\Lambda_\phi)S(\Lambda_\theta)[S(\Lambda_b)]^{-1}$, instead of $S A S^{-1}$ of (24.338) we get

$$S\left[(\tilde{\Lambda}_b)^{-1}\right] \cdot A \cdot S\left[(\tilde{\Lambda}_b)\right],$$

which yields the same result as SAS^{-1} of (24.338). This is evident in light of (24.321). Corresponding to (24.243), we have

$$\begin{aligned} S\left[(\tilde{\Lambda}_b)^{-1}\right] &= S(\Lambda_u)S(\Lambda_b^{-1})S(\Lambda_u^\dagger) = S(\Lambda_u)S(\Lambda_b^{-1})S(\Lambda_u^{-1}) \\ &= S(\Lambda_u)S(\Lambda_b^{-1})[S(\Lambda_u)]^{-1} = S(\Lambda_u)S(\Lambda_b^{-1})[S(\Lambda_u)]^\dagger \end{aligned}$$

$$= \begin{pmatrix} \cosh \frac{\omega}{2} & 0 & -\cos \theta \sinh \frac{\omega}{2} & e^{i\phi} \sin \theta \sinh \frac{\omega}{2} \\ 0 & \cosh \frac{\omega}{2} & e^{-i\phi} \sin \theta \sinh \frac{\omega}{2} & \cos \theta \sinh \frac{\omega}{2} \\ -\cos \theta \sinh \frac{\omega}{2} & e^{i\phi} \sin \theta \sinh \frac{\omega}{2} & \cosh \frac{\omega}{2} & 0 \\ e^{-i\phi} \sin \theta \sinh \frac{\omega}{2} & \cos \theta \sinh \frac{\omega}{2} & 0 & \cosh \frac{\omega}{2} \end{pmatrix}. \quad (24.340)$$

In (24.340) we obtain the representation $S\left[(\tilde{\Lambda}_b)^{-1}\right]$ as a consequence of viewing the operator $S(\Lambda_b^{-1})$ in reference to the frame O' . The trace of (24.340) is $4 \cosh \frac{\omega}{2}$, which is the same as $[S(\Lambda_b)]^{-1}$, reflecting the fact that the unitary similarity transformation keeps the trace unchanged; see (24.328).

Now, the significance of (24.339) is evident and consistent with (24.281). Tracing the calculation (24.339) reversely, we find that $S^{-1}\mathfrak{G}S = A$. That is, $\mathfrak{G}$ has been diagonalized to yield A through the similarity transformation with S . The operator $\mathfrak{G}$ possesses the eigenvalues of $-2m$ (doubly degenerate) and zero (doubly degenerate). We also find that the rank of $\mathfrak{G}$ is two. From (24.338) and (24.339), we find that the trace of $\mathfrak{G}$ is $-4m$ that is identical with the trace of the matrices A and B of (24.301) and (24.302).

Let us get back to (24.282) once again. Sandwiching both sides of (24.282) between S^{-1} and S from the left and right, respectively, we obtain

$$S^{-1}\mathfrak{G}S + S^{-1}\tilde{\mathfrak{G}}S = -2mS^{-1}ES. \quad (24.341)$$

Using (24.339), $A + S^{-1}\tilde{\mathfrak{G}}S = -2mE$. Further using (24.301) and (24.302), we get

Table 24.2 Eigenvalues and their corresponding Dirac spinors (eigenspinors) of the Dirac operators

Dirac operator ^a		Eigenvalue	Dirac spinor ^b	
Inertial frame			Inertial frame	
O	O'		O	O'
A	$\mathfrak{G}$	0	$\psi_1(x), \psi_2(x)$	$S\psi_1(x), S\psi_2(x)$
		$-2m$	$\psi_3(x), \psi_4(x)$	$S\psi_3(x), S\psi_4(x)$
B	$\tilde{\mathfrak{G}}$	0	$\psi_3(x), \psi_4(x)$	$S\psi_3(x), S\psi_4(x)$
		$-2m$	$\psi_1(x), \psi_2(x)$	$S\psi_1(x), S\psi_2(x)$

^a A and B are taken from (24.301) and (24.302), respectively. $\mathfrak{G}$ and $\tilde{\mathfrak{G}}$ are connected to A and B through similarity transformation by S , respectively; see (24.339) and (24.342)

^b $\psi_1(x)$ and $\psi_2(x)$ are taken from (24.305); $\psi_3(x)$ and $\psi_4(x)$ are taken from (24.310).

$$S^{-1}\tilde{\mathfrak{G}}S = -2mE - A = B \quad \text{or} \quad SBS^{-1} = \tilde{\mathfrak{G}}. \quad (24.342)$$

The operator $\tilde{\mathfrak{G}}$ is found to possess the eigenvalues of $-2m$ (doubly degenerate) and zero (doubly degenerate) as well, in agreement with the argument made at the end of Sect. 24.5. Thus, choosing S of (24.293) for an appropriate diagonalizing matrix, both $\mathfrak{G}$ and $\tilde{\mathfrak{G}}$ have been successfully diagonalized so as to be A and B , respectively.

Table 24.2 summarizes eigenvalues and their corresponding Dirac spinors (eigenspinors) of the Dirac operators. Note that both the eigenvalues 0 and $-2m$ are doubly degenerate. Of the Dirac spinors, those belonging to the eigenvalue 0 are the solutions of the Dirac equation.

Following (24.277), we define

$$S\phi(x) \equiv \phi'(x') \quad \text{and} \quad S\chi(x) \equiv \chi'(x'). \quad (24.343)$$

In (24.343) $\phi(x)$ is chosen from among $\psi_1(x)$ and $\psi_2(x)$ of (24.305) and represents the positive-energy state of an electron at rest; $\chi(x)$ from among $\psi_3(x)$ and $\psi_4(x)$ of (24.310) and for the negative-energy state of an electron at rest. Using (24.301), (24.303) is rewritten as

$$A\phi(x) = 0. \quad (24.344)$$

Then, we have successively

$$AS^{-1} \cdot S\phi(x) = 0, SAS^{-1} \cdot S\phi(x) = 0. \quad (24.345)$$

The second equation of (24.345) can be rewritten as

$$\mathfrak{G}\phi'(x') = 0, \quad (24.346)$$

where $\mathfrak{S} = SAS^{-1}$ and $\phi'(x') = S\phi(x)$. Taking a reverse process of the above three equations, we get

$$\mathfrak{S}\phi'(x') = 0, \mathfrak{S}S \cdot S^{-1}\phi'(x') = 0, S^{-1}\mathfrak{S}S \cdot S^{-1}\phi'(x') = 0. \quad (24.347)$$

The final equation of (24.347) is identical to (24.344), where

$$S^{-1}\mathfrak{S}S = A \quad \text{and} \quad S^{-1}\phi'(x') = \phi(x). \quad (24.348)$$

The implication of (24.348) is that once $\mathfrak{S}$ has been diagonalized with S , $\phi'(x')$ is at once given by operating S on a simple function $\phi(x)$ that represents the state of an electron at rest. This means that the Dirac equation has been solved.

With negative-energy state of an electron, starting with (24.306), we reach a related result described by

$$\tilde{\mathfrak{S}}\chi'(x') = 0 \quad (24.349)$$

with $\tilde{\mathfrak{S}} = SBS^{-1}$ and $\chi'(x') = S\chi(x)$. Evidently, (24.346) and (24.349) correspond to (21.73) and (21.74), respectively. At the same time, we find that both $\mathfrak{S}$ and $\tilde{\mathfrak{S}}$ possess eigenvalues $-2m$ and 0 . It is evident from (24.303) and (24.306).

At the end of this section, we remark that in relation to the charge conjugation mentioned in Sect. 21.5, using the matrix C of (21.164) we obtain following relations:

$$CAC^{-1} = B \quad \text{and} \quad CBC^{-1} = A.$$

24.6.2 Case II: Non-Coaxial Lorentz Boosts

As already seen earlier, the Lorentz transformations include the unitary operators that are related to the spatial rotation and Hermitian operators (or real symmetric operators) called Lorentz boost. In the discussion made thus far, we focused on the transformation that included only a single Lorentz boost. In this section, we deal with the case where the transformation contains complex Lorentz boosts, especially non-coaxial Lorentz boosts. When the non-coaxial Lorentz boosts are involved, we encounter a somewhat peculiar feature.

As described in Sect. 24.2.2, the successive Lorentz transformations consisted of a set of two spatial rotations and a single Lorentz boost. In this section, we further make a step toward constructing the Lorentz transformations. Considering those three successive transformations as a unit, we combine two unit transformations so that the resulting transformation can contain two Lorentz boosts. We have two cases for this: (i) The boosts are coaxial. (ii) The boosts are non-coaxial. Whereas in the former case the boost directions are the same when viewed from a certain inertial frame of reference, in the latter case the boost directions are different.

As can be seen from Fig. 24.2a, if the rotation angle $\theta \neq 0$ in the frame O_θ , the initial Lorentz boost and subsequently occurring boost are non-coaxial. Here, we assume that the second Lorentz boost takes place along the z' -axis of the frame O' . In the sense of the non-coaxial boost, the situation is equivalent to a following case: An observer A1 is watching a moving electron with a constant velocity in the direction of, say the z -axis in a certain inertial frame of reference F1 where the observer A1 stays at rest. Meanwhile, another observer A2 is moving with a constant velocity in the direction of, say along the x -axis relative to the frame F1. On this occasion, we have a question of how the electron is moving in reference to the frame F2 where the observer A2 stays at rest.

In Sect. 24.2.2 we expressed the successive transformations as

$$\Lambda = \Lambda_\phi \Lambda_\theta \Lambda_b^{-1}. \quad (24.236)$$

Let us denote the transformation by (ϕ, θ, b) . Let another transformation be (ϕ', θ', b') that is described by

$$\Lambda' = \Lambda_{\phi'} \Lambda_{\theta'} \Lambda_{b'}^{-1}. \quad (24.350)$$

Suppose that (ϕ', θ', b') takes place subsequently to the initial transformation (ϕ, θ, b) .

The overall transformation Ξ is then described such that

$$\Xi = \Lambda' \Lambda = \Lambda_{\phi'} \Lambda_{\theta'} \Lambda_{b'}^{-1} \Lambda_\phi \Lambda_\theta \Lambda_b^{-1}. \quad (24.351)$$

Note that in (24.351) we are viewing the transformation in the moving coordinate systems. That is, in the second transformation Λ' the boost takes place in the direction of the z' -axis of the frame O' (Fig. 24.2a). As a result, the successive transformations $\Lambda' \Lambda$ cause the non-coaxial Lorentz boosts in the case of $\theta \neq 0$. In a special case where $\theta = 0$ is assumed for the original transformation Λ , however, $\Xi = \Lambda' \Lambda$ produces co-axial boosts.

Let the general form of the Dirac equation be

$$\mathfrak{D}\psi(x) = 0, \quad (24.352)$$

where $\mathfrak{D}$ represents either $\mathfrak{G}$ of (24.339) or $\tilde{\mathfrak{G}}$ of (24.342). Starting with the rest frame of the electron, we have $\mathfrak{D} = A$ or $\mathfrak{D} = B$ and $\psi(x) = e^{\pm imx^0} w(\mathbf{0}, h)$; see (24.296) with the latter equation. Operating $S(\Lambda)$ of (24.293) on (24.352) from the left, we obtain

$$S(\Lambda) \mathfrak{D} [S(\Lambda)]^{-1} S(\Lambda) \psi(x) = 0. \quad (24.353)$$

Defining $\mathfrak{D}' \equiv S(\Lambda) \mathfrak{D} [S(\Lambda)]^{-1}$ and using (24.277), we have

$$\mathfrak{D}'\psi'(x') = 0. \quad (24.354)$$

Further operating $S(\Lambda')$ on (24.354) from the left, we get

$$S(\Lambda')\mathfrak{D}'[S(\Lambda')]^{-1}S(\Lambda')\psi'(x') = 0, \quad (24.355)$$

where Λ' was given in (24.350). Once again, defining $\mathfrak{D}'' \equiv S(\Lambda')\mathfrak{D}'[S(\Lambda')]^{-1}$ and $\psi''(x'') \equiv S(\Lambda')\psi'(x')$, we have

$$\mathfrak{D}''\psi''(x'') = 0. \quad (24.356)$$

We assume that after the successive Lorentz transformations Λ and Λ' , we reach the frame O'' . Also, we have

$$\begin{aligned} \psi''(x'') &= S(\Lambda')\psi'(x') = S(\Lambda')S(\Lambda)\psi(x) = S(\Lambda'\Lambda)\psi(x), \\ \mathfrak{D}'' &= S(\Lambda')\mathfrak{D}'[S(\Lambda')]^{-1} = S(\Lambda')S(\Lambda)\mathfrak{D}[S(\Lambda)]^{-1}[S(\Lambda')]^{-1} \\ &= S(\Lambda')S(\Lambda)\mathfrak{D}[S(\Lambda')S(\Lambda)]^{-1} = S(\Lambda'\Lambda)\mathfrak{D}[S(\Lambda'\Lambda)]^{-1}. \end{aligned} \quad (24.357)$$

In the second equation of (24.357), with the last equality we used the fact that $S(\Lambda)$ is a representation of the Lorentz group. Repeatedly operating $S(\Lambda)$, $S(\Lambda')$, etc., we can obtain a series of successively transformed Dirac spinors and corresponding Dirac operators. Thus, the successive transformations of the Dirac operators can be viewed as a similarity transformation as a whole. In what follows, we focus on a case where two non-coaxial Lorentz boosts take place.

Example 24.4 As a tangible example, we show how the Dirac spinors and Dirac operators are transformed as a result of the successively occurring Lorentz transformations. Table 24.3 shows a (4, 4) matrix of $\Xi = \Lambda'\Lambda$ given in (24.351). Table 24.4 lists another (4, 4) matrix of $S(\Xi) = S(\Lambda'\Lambda) = S(\Lambda')S(\Lambda)$. The matrix $S(\Lambda)$ is given in (24.293). The matrix $S(\Lambda')$ is expressed as

$$\begin{aligned} S(\Lambda') &= S(\Lambda_{\phi'})S(\Lambda_{\theta'})[S(\Lambda_{\theta'})]^{-1} = \\ &= \begin{pmatrix} e^{i\phi'/2} \cos \frac{\theta'}{2} \cosh \frac{\omega'}{2} & e^{i\phi'/2} \sin \frac{\theta'}{2} \cosh \frac{\omega'}{2} & -e^{i\phi'/2} \cos \frac{\theta'}{2} \sinh \frac{\omega'}{2} & e^{i\phi'/2} \sin \frac{\theta'}{2} \sinh \frac{\omega'}{2} \\ -e^{-i\phi'/2} \sin \frac{\theta'}{2} \cosh \frac{\omega'}{2} & e^{-i\phi'/2} \cos \frac{\theta'}{2} \cosh \frac{\omega'}{2} & e^{-i\phi'/2} \sin \frac{\theta'}{2} \sinh \frac{\omega'}{2} & e^{-i\phi'/2} \cos \frac{\theta'}{2} \sinh \frac{\omega'}{2} \\ -e^{i\phi'/2} \cos \frac{\theta'}{2} \sinh \frac{\omega'}{2} & e^{i\phi'/2} \sin \frac{\theta'}{2} \sinh \frac{\omega'}{2} & e^{i\phi'/2} \cos \frac{\theta'}{2} \cosh \frac{\omega'}{2} & e^{i\phi'/2} \sin \frac{\theta'}{2} \cosh \frac{\omega'}{2} \\ e^{-i\phi'/2} \sin \frac{\theta'}{2} \sinh \frac{\omega'}{2} & e^{-i\phi'/2} \cos \frac{\theta'}{2} \sinh \frac{\omega'}{2} & -e^{-i\phi'/2} \sin \frac{\theta'}{2} \cosh \frac{\omega'}{2} & e^{-i\phi'/2} \cos \frac{\theta'}{2} \cosh \frac{\omega'}{2} \end{pmatrix}. \end{aligned} \quad (24.358)$$

Table 24.3 Representation matrix of Lorentz transformation Λ^Λ

$\Xi = \Lambda' \Lambda =$	$-\sin \theta \sinh \omega'$	0	$\cosh \omega' \sinh \omega$ $+\cos \theta \sinh \omega' \cosh \omega$
$\cosh \omega' \cosh \omega$ $+\cos \theta' \sin \theta' \sinh \omega' \cosh \omega$ $+\cos \phi' \cos \theta' \cos \phi \sin \theta \sinh \omega$ $-\sin \phi' \sin \phi \sin \theta \sinh \omega$ $+\cos \phi' \sin \theta' \cos \theta \cosh \omega' \sinh \omega$	$\cos \phi' \cos \theta' \cos \phi \cos \theta$ $-\sin \phi' \sin \phi \cos \theta$ $-\cos \phi' \sin \theta' \sin \theta \cosh \omega'$	$-\cos \phi' \cos \theta' \sin \phi$ $-\sin \phi' \cos \phi$	$\cosh \omega' \sinh \omega$ $+\cos \theta \sinh \omega' \cosh \omega$
$\sin \phi' \sin \theta' \sinh \omega' \cosh \omega$ $+\sin \phi' \cos \theta' \cos \phi \sin \theta \sinh \omega$ $+\cos \phi' \sin \phi \sin \theta \sinh \omega$ $+\sin \phi' \sin \theta' \cos \theta \cosh \omega' \sinh \omega$	$\sin \phi' \cos \theta' \cos \phi \cos \theta$ $+\cos \phi' \sin \phi \cos \theta$ $-\sin \phi' \sin \theta' \sin \theta \cosh \omega'$	$-\sin \phi' \cos \theta' \sin \phi$ $+\cos \phi' \cos \phi$	$\sin \phi' \sin \theta' \sinh \omega' \sinh \omega$ $+\sin \phi' \cos \theta' \cos \phi \sin \theta \cosh \omega$ $+\cos \phi' \sin \phi \sin \theta \cosh \omega$ $+\sin \phi' \sin \theta' \cos \theta \cosh \omega' \cosh \omega$
$\cos \theta' \sinh \omega' \cosh \omega$ $-\sin \theta' \cos \phi \sin \theta \sinh \omega$ $+\cos \theta' \cos \theta \cosh \omega' \sinh \omega$	$-\sin \theta' \cos \phi \cos \theta$ $-\cos \theta' \sin \theta \cosh \omega'$	$\sin \theta' \sin \phi$	$\cos \theta' \sinh \omega' \sinh \omega$ $-\sin \theta' \cos \phi \sin \theta \cosh \omega$ $+\cos \theta' \cos \theta \cosh \omega' \cosh \omega$

Table 24.4 Representation matrix of $S(\Lambda')$

$S(\Lambda' \Lambda) =$			
$-e^{i(\phi'+\phi)/2} \cos \frac{\theta'}{2} \cos \frac{\theta}{2} \cosh \frac{\omega'+\omega}{2}$	$e^{i(\phi'+\phi)/2} \cos \frac{\theta'}{2} \sin \frac{\theta}{2} \cosh \frac{\omega'-\omega}{2}$	$-e^{i(\phi'+\phi)/2} \cos \frac{\theta'}{2} \cos \frac{\theta}{2} \sinh \frac{\omega'+\omega}{2}$	$-e^{i(\phi'+\phi)/2} \cos \frac{\theta'}{2} \sin \frac{\theta}{2} \sinh \frac{\omega'-\omega}{2}$
$-e^{i(\phi'-\phi)/2} \sin \frac{\theta'}{2} \sin \frac{\theta}{2} \cosh \frac{\omega'+\omega}{2}$	$+e^{i(\phi'-\phi)/2} \sin \frac{\theta'}{2} \cos \frac{\theta}{2} \cosh \frac{\omega'-\omega}{2}$	$-e^{i(\phi'-\phi)/2} \sin \frac{\theta'}{2} \sin \frac{\theta}{2} \sinh \frac{\omega'+\omega}{2}$	$+e^{i(\phi'-\phi)/2} \sin \frac{\theta'}{2} \cos \frac{\theta}{2} \sinh \frac{\omega'-\omega}{2}$
$-e^{-i(\phi'-\phi)/2} \sin \frac{\theta'}{2} \cos \frac{\theta}{2} \cosh \frac{\omega'+\omega}{2}$	$-e^{-i(\phi'-\phi)/2} \sin \frac{\theta'}{2} \sin \frac{\theta}{2} \cosh \frac{\omega'-\omega}{2}$	$-e^{-i(\phi'-\phi)/2} \sin \frac{\theta'}{2} \cos \frac{\theta}{2} \sinh \frac{\omega'+\omega}{2}$	$-e^{-i(\phi'-\phi)/2} \sin \frac{\theta'}{2} \sin \frac{\theta}{2} \sinh \frac{\omega'-\omega}{2}$
$-e^{-i(\phi'+\phi)/2} \cos \frac{\theta'}{2} \sin \frac{\theta}{2} \cosh \frac{\omega'+\omega}{2}$	$+e^{-i(\phi'+\phi)/2} \cos \frac{\theta'}{2} \cos \frac{\theta}{2} \cosh \frac{\omega'-\omega}{2}$	$-e^{-i(\phi'+\phi)/2} \cos \frac{\theta'}{2} \sin \frac{\theta}{2} \sinh \frac{\omega'+\omega}{2}$	$+e^{-i(\phi'+\phi)/2} \cos \frac{\theta'}{2} \cos \frac{\theta}{2} \sinh \frac{\omega'-\omega}{2}$
$-e^{i(\phi'+\phi)/2} \cos \frac{\theta'}{2} \cos \frac{\theta}{2} \sinh \frac{\omega'+\omega}{2}$	$-e^{i(\phi'+\phi)/2} \cos \frac{\theta'}{2} \sin \frac{\theta}{2} \sinh \frac{\omega'-\omega}{2}$	$-e^{i(\phi'+\phi)/2} \cos \frac{\theta'}{2} \cosh \frac{\omega'+\omega}{2}$	$-e^{i(\phi'+\phi)/2} \cos \frac{\theta'}{2} \cosh \frac{\omega'-\omega}{2}$
$-e^{i(\phi'-\phi)/2} \sin \frac{\theta'}{2} \sin \frac{\theta}{2} \sinh \frac{\omega'+\omega}{2}$	$+e^{i(\phi'-\phi)/2} \sin \frac{\theta'}{2} \cos \frac{\theta}{2} \sinh \frac{\omega'-\omega}{2}$	$-e^{i(\phi'-\phi)/2} \sin \frac{\theta'}{2} \sinh \frac{\omega'+\omega}{2}$	$+e^{i(\phi'-\phi)/2} \sin \frac{\theta'}{2} \sinh \frac{\omega'-\omega}{2}$
$-e^{-i(\phi'-\phi)/2} \sin \frac{\theta'}{2} \cos \frac{\theta}{2} \sinh \frac{\omega'+\omega}{2}$	$-e^{-i(\phi'-\phi)/2} \sin \frac{\theta'}{2} \sin \frac{\theta}{2} \sinh \frac{\omega'-\omega}{2}$	$-e^{-i(\phi'-\phi)/2} \sin \frac{\theta'}{2} \cosh \frac{\omega'+\omega}{2}$	$-e^{-i(\phi'-\phi)/2} \sin \frac{\theta'}{2} \cosh \frac{\omega'-\omega}{2}$
$-e^{-i(\phi'+\phi)/2} \cos \frac{\theta'}{2} \sin \frac{\theta}{2} \sinh \frac{\omega'+\omega}{2}$	$+e^{-i(\phi'+\phi)/2} \cos \frac{\theta'}{2} \cos \frac{\theta}{2} \sinh \frac{\omega'-\omega}{2}$	$-e^{-i(\phi'+\phi)/2} \cos \frac{\theta'}{2} \sinh \frac{\omega'+\omega}{2}$	$+e^{-i(\phi'+\phi)/2} \cos \frac{\theta'}{2} \sinh \frac{\omega'-\omega}{2}$

Table 24.5 Representation matrix of Dirac operator $\mathfrak{D}''^a$

$\mathfrak{D}'' = S(\Lambda' \Lambda) A [S(\Lambda' \Lambda)]^{-1} = (-2m) \times$			
$\begin{pmatrix} -(\cosh^2 \frac{\omega'}{2} \sinh^2 \frac{\omega}{2} + \sinh^2 \frac{\omega'}{2} \cosh^2 \frac{\omega}{2} + \frac{1}{2} \cos \theta \sinh \omega' \sinh \omega) \\ 0 \\ -\frac{1}{2} \sin \theta' \sinh \theta \cos \theta \sinh \omega \\ + \frac{1}{2} \cos \theta' (\sinh \omega' \cosh \omega + \cos \theta \cosh \omega' \sinh \omega) \\ \frac{1}{2} e^{-i\theta'} \sinh \theta \sinh \omega \left(e^{i\theta} \sin^2 \frac{\theta'}{2} - e^{-i\theta} \cos^2 \frac{\theta'}{2} \right) \\ - \frac{1}{2} e^{-i\theta'} \sinh \theta' (\cos \theta \cosh \omega' \sinh \omega + \sinh \omega' \cosh \omega) \end{pmatrix}$	0 $(1, 1)$ $(4, 1)^*$ $-(3, 1)$	$-(3, 1)$ $-(4, 1)$ $\cosh^2 \frac{\omega'}{2} \cosh^2 \frac{\omega}{2} + \sinh^2 \frac{\omega'}{2} \sinh^2 \frac{\omega}{2} + \frac{1}{2} \cos \theta \sinh \omega' \sinh \omega$ 0	$-(4, 1)^*$ $(3, 1)$ 0 $(3, 3)$

^aIn the above matrix, e.g., (1, 1) means that the (2, 2) matrix element should be replaced with the (1, 1) element

As shown in Sect. 24.5, the Dirac spinors $\psi(x)$ was expressed in reference to the rest frame of the electron as

$$\psi(x) = e^{\pm ip^0 x^0} w(\mathbf{0}, h) = e^{\pm imx^0} w(\mathbf{0}, h). \quad (24.296)$$

Then, we have

$$\psi''(x'') = S(\Xi)\psi(x) = e^{\pm imx^0} S(\Xi)w(\mathbf{0}, h). \quad (24.359)$$

The spinors $w(\mathbf{0}, h)$ were given in (24.305) and (24.310) according to the positive- and negative-energy electron. Thus, individual columns of $S(\Xi)$ in Table 24.4 are directly associated with the four spinors of either the positive- or negative-energy electron, as in the case of (24.293). The orthogonal characteristics hold with the Dirac spinors as in (21.141) and (21.142). Individual matrix elements of $S(\Xi)$ consist of two complex terms. Note, however, in the case of the co-axial boosts (i.e., $\theta = 0$) each matrix element comprises a single term. In both the co-axial and non-coaxial boost cases, individual four columns of $S(\Lambda' \Lambda)$ represent the Dirac spinors with respect to the frame O'' . As mentioned before, the sign for $\psi''_4(x'')$ should be reversed relative to the fourth column of $S(\Lambda' \Lambda)$ because of the charge conjugation.

Meanwhile, Table 24.5 lists individual matrix elements of a (4, 4) matrix of the Dirac operator $\mathfrak{D}''$ with the positive-energy electron. Using $S(\Lambda' \Lambda)$, $\mathfrak{D}''$ viewed in reference to the frame O'' is described by

$$\mathfrak{D}'' = S(\Lambda' \Lambda)A[S(\Lambda' \Lambda)]^{-1} = S(\Xi)A[S(\Xi)]^{-1}, \quad (24.360)$$

where A was given in (24.301). As can be seen in (21.75), the Dirac operator contains all the information of the rest mass of an electron and the energy-momentum of a plane wave electron. This is clearly stated in (24.278). In the x'' -system (i.e., the frame O'') the Dirac equation is described in a similar way such that

$$(i\gamma^\mu \partial''_\mu - m) \cdot \psi''(x'') = 0.$$

The point is that in *any* inertial frame of reference the matrix form of the gamma matrices is held invariant. Or rather, $S(\Xi)$ has been chosen for the gamma matrices to be invariant in such a way that

$$\Lambda^\nu_\mu S\gamma^\mu S^{-1} = \gamma^\nu. \quad (24.276)$$

Readers are encouraged to confirm this relation by calculating a tangible example. Thus, with the plane wave solution (of the positive-energy case) in frame O'' we obtain

$$(p''_{\mu}\gamma^{\mu} - m) \cdot \psi''(x'') = 0,$$

where p''_{μ} is a four-momentum in O'' . Hence, even though the matrix form of Table 24.5 is somewhat complicated, the Dirac equation must uniquely be represented in any inertial frame of reference.

Thus, from Table 24.5 we get the following four-momenta in the frame O'' described by

$$\begin{aligned} p''^0 &= [(1, 1) - (3, 3)]/2, m = [(1, 1) + (3, 3)]/(-2), \\ p''^3 &= (1, 3), p^1 = [-(1, 4) + (4, 1)]/2, p^2 = [(1, 4) + (4, 1)]i/2, \end{aligned} \quad (24.361)$$

where (m, n) to which $1 \leq m, n \leq 4$ indicates the (m, n) -element of the matrix given in Table 24.5. As noted from (21.75), we can determine other matrix elements from the four elements indicated in Table 24.5, in which we certainly confirm that $(1, 1) + (3, 3) = -2m$. This is in agreement with $\mathfrak{G}$ of (21.75) and (21.91). Thus, from (24.361) and Table 24.5 we get

$$\begin{aligned} p''^0 &= m(\cosh \omega' \cosh \omega + \cos \theta \sinh \omega' \sinh \omega), \\ p''^3 &= m \cos \theta' (\sinh \omega' \cosh \omega + \cos \theta \cosh \omega' \sinh \omega) \\ &\quad - m \sin \theta' \sin \theta \cos \phi \sinh \omega, \\ p''^1 &= m[\sin \theta' \cos \phi' (\sinh \omega' \cosh \omega + \cos \theta \cosh \omega' \sinh \omega) \\ &\quad + \cos(\phi' + \phi) \cos^2 \left(\frac{\theta'}{2} \right) \sin \theta \sinh \omega - \cos(\phi' - \phi) \sin^2 \left(\frac{\theta'}{2} \right) \sin \theta \sinh \omega], \\ p''^2 &= m[\sin \theta' \sin \phi' (\sinh \omega' \cosh \omega + \cos \theta \cosh \omega' \sinh \omega) \\ &\quad + \sin(\phi' + \phi) \cos^2 \left(\frac{\theta'}{2} \right) \sin \theta \sinh \omega \\ &\quad - \sin(\phi' - \phi) \sin^2 \left(\frac{\theta'}{2} \right) \sin \theta \sinh \omega]. \end{aligned} \quad (24.362)$$

From the above relations, we recover

$$(p''^0)^2 - \mathbf{p}''^2 - m^2 = 0 \quad \text{or} \quad p''^0 = \sqrt{\mathbf{p}''^2 + m^2}. \quad (21.79)$$

As seen above, the basic construction of the Dirac equation remains unaltered by the operation of successive Lorentz transformations including the non-coaxial boosts.

Example 24.5 We show another simple example of the non-coaxial Lorentz boosts. Suppose that in the moving coordinate systems we have successive non-coaxial boosts Λ described by

$$\begin{aligned}\Lambda &= \begin{pmatrix} \cosh \omega' & -\sinh \omega' & 0 & 0 \\ -\sinh \omega' & \cosh \omega' & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cosh \omega & 0 & 0 & \sinh \omega \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh \omega & 0 & 0 & \cosh \omega \end{pmatrix} \\ &= \begin{pmatrix} \cosh \omega' \cosh \omega & -\sinh \omega' & 0 & \cosh \omega' \sinh \omega \\ -\sinh \omega' \cosh \omega & \cosh \omega' & 0 & -\sinh \omega' \sinh \omega \\ 0 & 0 & 1 & 0 \\ \sinh \omega & 0 & 0 & \cosh \omega \end{pmatrix}. \quad (24.363)\end{aligned}$$

Equation (24.363) corresponds to a special case of (24.351), where $\theta = -\theta' = \pi/2$ and $\phi = \phi' = 0$; see Table 24.3. Readers are encouraged to imagine the geometric feature of the successive non-coaxial boosts in the moving coordinate systems; see Fig. 24.2.

24.7 Projection Operators and Related Operators for the Dirac Equation

In this section, we examine the constitution of the Dirac equation in terms of the projection operator. In the next section, related topics are discussed as well. To this end, we wish to deal with the issue from a somewhat general standpoint. Let G be expressed as

$$G = \begin{pmatrix} pE & \mathbf{H} \\ -\mathbf{H} & qE \end{pmatrix} (p \neq q),$$

where E is a (2, 2) identity matrix; $\mathbf{H}$ is a (2, 2) Hermitian matrix. Then, we have

$$G^\dagger G = \begin{pmatrix} p^2 E + \mathbf{H}^2 & (q-p)\mathbf{H} \\ (q-p)\mathbf{H} & q^2 E + \mathbf{H}^2 \end{pmatrix}, \quad GG^\dagger = \begin{pmatrix} p^2 E + \mathbf{H}^2 & (p-q)\mathbf{H} \\ (p-q)\mathbf{H} & q^2 E + \mathbf{H}^2 \end{pmatrix}. \quad (24.364)$$

For G to be a normal matrix (i.e., $G^\dagger G = GG^\dagger$), we must have $\mathbf{H} \equiv \mathbf{0}$. In our present case, if we choose $\mathfrak{G}$ of (21.91) for G , we have $p = p^0 - m$, $q = -p^0 - m$, and $\mathbf{H} = -\boldsymbol{\sigma} \cdot \mathbf{p}$. Thus, $\mathbf{H} \equiv \mathbf{0}$ is equivalent to $\mathbf{p} \equiv \mathbf{0}$. This means that an electron is at rest. Namely, $\mathfrak{G}$ is normal (and Hermitian) only if the electron is at rest. As long as the electron is moving (in a certain inertial frame of reference), $\mathfrak{G}$ is not normal. In this situation, Theorem 14.5 tells us that $\mathfrak{G}$ could only be diagonalized via similarity

transformation using a non-unitary operator, if $\mathbf{p} \neq \mathbf{0}$. In fact, in (24.339) using a non-unitary operator S , $\mathfrak{G}$ could be diagonalized. This means that although the operator $\mathfrak{G}$ is not normal (nor Hermitian), it is semi-simple (Sect. 12.5). The above discussion is true of the operator $\mathfrak{G}$ of (21.108) as shown in (24.342).

In the discussion of the previous section, we have examined the constitution of the operator S and known that their non-unitarity comes from the Lorentz boost. However, once $\mathfrak{G}$ or $\tilde{\mathfrak{G}}$ is given, the diagonalizing matrix form for S can uniquely be determined (see Sect. 24.9).

Let us start our main theme about the projection operator. From (24.301) and (24.302), we obtain

$$\frac{A}{(-2m)} + \frac{B}{(-2m)} = E, \quad (24.365)$$

where E is the identity operator; i.e., a (4, 4) identity matrix. Operating S from the left and S^{-1} from the right on both sides of (24.365), respectively, we get

$$\frac{SAS^{-1}}{(-2m)} + \frac{SBS^{-1}}{(-2m)} = SES^{-1} = E \quad \text{or} \quad SAS^{-1} + SBS^{-1} = (-2m)E. \quad (24.366)$$

Using (24.339) and (24.342), we restore (24.282). Meanwhile, operating both sides of (24.366) on $\phi'(x')$ of (24.346) from the left and taking account of (24.282), we obtain

$$\tilde{\mathfrak{G}}\phi'(x') = -2m\phi'(x'). \quad (24.367)$$

Operating both sides of (24.366) on $\chi'(x')$ of (24.349) from the left and taking account of (24.282) once again, we have

$$\mathfrak{G}\chi'(x') = -2m\chi'(x'). \quad (24.368)$$

Now, suppose that the general solutions of the plane wave Dirac field $\Psi(x')$ are given by

$$\Psi(x') = \sum_{i=1}^4 \alpha_i \psi'_i(x'), \quad (24.369)$$

where $\psi'_i(x')$ are those of (24.311) and α_i are their suitable coefficients. Then, operating $\tilde{\mathfrak{G}}$ from the left on both sides of (24.369), we get

$$\tilde{\mathfrak{G}}\Psi(x') = \sum_{i=1}^2 (-2m)\alpha_i \psi'_i(x'), \quad (24.370)$$

$$\mathfrak{G}\Psi(x') = \sum_{i=3}^4 (-2m)\alpha_i\psi'_i(x'). \quad (24.371)$$

Rewriting the above equations, we obtain

$$\tilde{\mathfrak{G}}\Psi(x')/(-2m) = \sum_{i=1}^2 \alpha_i\psi'_i(x'), \quad (24.372)$$

$$\mathfrak{G}\Psi(x')/(-2m) = \sum_{i=3}^4 \alpha_i\psi'_i(x'). \quad (24.373)$$

Equations (24.372) and (24.373) mean that $\tilde{\mathfrak{G}}\Psi(x')/(-2m)$ extracts the positive-energy states from $\Psi(x')$ including the coefficients and that $\mathfrak{G}\Psi(x')/(-2m)$ extracts the negative-energy states from $\Psi(x')$ including the coefficients as well. These imply that both the operators $\tilde{\mathfrak{G}}\Psi(x')/(-2m)$ and $\mathfrak{G}\Psi(x')/(-2m)$ act as the projection operators (see Sects. 14.1 and 18.7). Defining $\mathfrak{P}$ and $\mathfrak{Q}$ as

$$\begin{aligned} \mathfrak{P} &\equiv \frac{\tilde{\mathfrak{G}}}{(-2m)} = \frac{-p_\mu\gamma^\mu - m}{(-2m)} = \frac{p_\mu\gamma^\mu + m}{2m}, \\ \mathfrak{Q} &\equiv \frac{\mathfrak{G}}{(-2m)} = \frac{p_\mu\gamma^\mu - m}{(-2m)} = \frac{-p_\mu\gamma^\mu + m}{2m} \end{aligned} \quad (24.374)$$

and referring to (24.282), we obtain

$$\mathfrak{P} + \mathfrak{Q} = E. \quad (24.375)$$

From (24.374), we have

$$\mathfrak{P}^2 = \frac{(p_\mu\gamma^\mu + m)(p_\rho\gamma^\rho + m)}{(2m)(2m)} = \frac{p_\mu\gamma^\mu p_\rho\gamma^\rho + 2mp_\rho\gamma^\rho + m^2}{(2m)(2m)}. \quad (24.376)$$

To calculate (24.376), we take account of the following relation described by [13]

$$\begin{aligned} p_\mu\gamma^\mu p_\rho\gamma^\rho &= p_\mu p_\rho\gamma^\mu\gamma^\rho = \frac{1}{2}p_\mu p_\rho(\gamma^\mu\gamma^\rho + \gamma^\rho\gamma^\mu) = \frac{1}{2}p_\mu p_\rho \cdot 2\eta^{\mu\rho} = p_\mu p^\mu \\ &= p^2 = m^2, \end{aligned} \quad (24.377)$$

where with the third equality we used (21.64). Substituting (24.377) into (24.376), we rewrite it as

$$\mathfrak{P}^2 = \frac{2m(p_\rho\gamma^\rho + m)}{(2m)(2m)} = \frac{p_\mu\gamma^\mu + m}{2m} = \mathfrak{P}, \quad (24.378)$$

where the last equality came from the first equation of (24.374). Or, using the matrix algebra we reach the same result at once. That is, we have

$$\mathfrak{P} = \frac{SBS^{-1}}{(-2m)},$$

$$\mathfrak{P}^2 = \frac{SBS^{-1}}{(-2m)} \cdot \frac{SBS^{-1}}{(-2m)} = \frac{SB^2S^{-1}}{(-2m)^2} = \frac{(-2m)SBS^{-1}}{(-2m)^2} = \frac{SBS^{-1}}{(-2m)} = \mathfrak{P}, \quad (24.379)$$

where with the third equality we used (24.302). Similarly, we obtain

$$\mathfrak{Q}^2 = \mathfrak{Q}. \quad (24.380)$$

Moreover, we have

$$\mathfrak{P}\mathfrak{Q} = \mathfrak{Q}\mathfrak{P} = 0. \quad (24.381)$$

The relations (24.375) through (24.381) imply that $\mathfrak{P}$ and $\mathfrak{Q}$ are projection operators. However, notice that neither $\mathfrak{P}$ nor $\mathfrak{Q}$ is Hermitian. This property originally comes from the involvement of the Lorentz boost term of (24.293). In this regard, $\mathfrak{P}$ and $\mathfrak{Q}$ can be categorized as projection operators *sensu lato* (Sect. 18.7), even though these operators are idempotent as can be seen from (24.378) and (24.380). Equation (24.375) represents the completeness relation.

Let us consider the following example.

Example 24.6 We have an interesting and important example in relation to the projection operators and spectral decomposition dealt with in Sect. 14.3. A projection operator was defined using an adjoint operator (see Sect. 14.1) such that

$$\tilde{P}_k = |w_k\rangle\langle w_k|. \quad (24.382)$$

Notice that (24.382) postulates that $\tilde{P}_k$ is Hermitian. That is,

$$(|w_i\rangle\langle w_i|)^\dagger = (\langle w_i|)^\dagger(|w_i\rangle)^\dagger = |w_i\rangle\langle w_i|. \quad (14.28)$$

To apply this notation to the present case, we define following vectors in relation to (24.304) and (24.309) as:

$$|1\rangle \equiv u_1(\mathbf{0}, -1), |2\rangle \equiv u_2(\mathbf{0}, +1); \quad (24.383)$$

$$|3\rangle \equiv v_1(\mathbf{0}, +1), |4\rangle \equiv v_2(\mathbf{0}, -1). \quad (24.384)$$

Defining

$$\tilde{\mathfrak{P}} \equiv \frac{A}{(-2m)} \quad \text{and} \quad \tilde{\mathfrak{Q}} \equiv \frac{B}{(-2m)}, \quad (24.385)$$

we have

$$\tilde{\mathfrak{P}} + \tilde{\mathfrak{Q}} = E, \tilde{\mathfrak{P}}^2 = \tilde{\mathfrak{P}}, \tilde{\mathfrak{Q}}^2 = \tilde{\mathfrak{Q}}, \tilde{\mathfrak{P}}\tilde{\mathfrak{Q}} = 0. \quad (24.386)$$

Hence, both $\tilde{\mathfrak{P}}$ and $\tilde{\mathfrak{Q}}$ are Hermitian projection operators (i.e., projection operators *sensu stricto*; see Sect. 18.7). This is self-evident, however, notice that neither A nor B in (24.385) involves a Lorentz boost. Then, using (24.383) and (24.384), we obtain

$$|1\rangle\langle 1| + |2\rangle\langle 2| + |3\rangle\langle 3| + |4\rangle\langle 4| = E. \quad (24.387)$$

Rewriting (24.387) in the form of completeness relation, we have

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\ + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = E. \quad (24.388)$$

Though trivial, (24.388) represents the spectral decomposition.

Example 24.7 Another important projection operator can readily be derived from the spin associated operator defined in Chap. 21 as

$$\mathfrak{S} \equiv \frac{1}{|\mathbf{p}|} \boldsymbol{\sigma} \cdot \mathbf{p} \quad (21.85)$$

and

$$\tilde{\mathfrak{S}} \equiv \frac{1}{|-\mathbf{p}|} \boldsymbol{\sigma} \cdot (-\mathbf{p}) = -\mathfrak{S}. \quad (21.107)$$

The above operators were (2, 2) submatrices (Sect. 21.3). The projection operators Σ and $\tilde{\Sigma}$ of (4, 4) full matrix can be defined by use of the direct sum of $\mathfrak{S}$ or $\tilde{\mathfrak{S}}$ such that

$$\Sigma \equiv \frac{1}{2} \left[E + \begin{pmatrix} \mathfrak{S} & 0 \\ 0 & \mathfrak{S} \end{pmatrix} \right], \quad (24.389)$$

$$\tilde{\Sigma} \equiv \frac{1}{2} \left[E + \begin{pmatrix} \tilde{\mathfrak{S}} & 0 \\ 0 & \tilde{\mathfrak{S}} \end{pmatrix} \right], \quad (24.390)$$

where E is the (4, 4) identity matrix. Notice that both Σ and $\tilde{\Sigma}$ are Hermitian. We can readily check that

$$\Sigma + \tilde{\Sigma} = E, \Sigma^2 = \Sigma, \tilde{\Sigma}^2 = \tilde{\Sigma}, \Sigma\tilde{\Sigma} = \tilde{\Sigma}\Sigma = 0. \quad (24.391)$$

More importantly, we have

$$\Sigma^\dagger = \Sigma \text{ and } \tilde{\Sigma}^\dagger = \tilde{\Sigma}. \quad (24.392)$$

In this respect, the operators Σ and $\tilde{\Sigma}$ are the projection operators *sensu stricto*.

In Sect. 21.3.2, as the solution of the Dirac equation (21.96) we have gotten the eigenfunction X described by

$$X = \begin{pmatrix} \chi \\ h s \chi \end{pmatrix}. \quad (21.100)$$

Operating Σ of (24.389) on X of (21.100) from the left, for instance, we obtain

$$\begin{aligned} \Sigma X &= \frac{1}{2} \left[E + \begin{pmatrix} \mathfrak{E} & 0 \\ 0 & \mathfrak{E} \end{pmatrix} \right] \begin{pmatrix} \chi \\ h s \chi \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \chi \\ h s \chi \end{pmatrix} + \frac{1}{2} \begin{pmatrix} \mathfrak{E} \chi \\ h s \mathfrak{E} \chi \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \chi \\ h s \chi \end{pmatrix} + \frac{1}{2} \begin{pmatrix} h \chi \\ h s h \chi \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} \chi \\ h s \chi \end{pmatrix} + \frac{1}{2} h \begin{pmatrix} \chi \\ h s \chi \end{pmatrix}. \end{aligned} \quad (24.393)$$

If in (24.393) $h = 1$, we have $\Sigma X = \begin{pmatrix} \chi \\ h s \chi \end{pmatrix}$. If $h = -1$, on the other hand, $\Sigma X = 0$. Consequently, Σ extracts $\begin{pmatrix} \chi \\ h s \chi \end{pmatrix}$ intactly when $h = 1$, whereas it discards $\begin{pmatrix} \chi \\ h s \chi \end{pmatrix}$ in whole when $h = -1$.

For another example of the projection operator, we introduce

$$\tilde{\Sigma}_h \equiv \frac{1}{2} \left[E + h \begin{pmatrix} \mathfrak{E} & 0 \\ 0 & \mathfrak{E} \end{pmatrix} \right]. \quad (24.394)$$

Operating $\tilde{\Sigma}_h$ of (24.394) on X' of (21.133) from the left, we obtain

$$\begin{aligned} \tilde{\Sigma}_h X' &= \frac{1}{2} \left[E + h \begin{pmatrix} \mathfrak{E} & 0 \\ 0 & \mathfrak{E} \end{pmatrix} \right] \begin{pmatrix} s \tilde{\chi} \\ -h \tilde{\chi} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} s \tilde{\chi} \\ -h \tilde{\chi} \end{pmatrix} + \frac{1}{2} h \begin{pmatrix} s \mathfrak{E} \tilde{\chi} \\ -h \mathfrak{E} \tilde{\chi} \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} s \tilde{\chi} \\ -h \tilde{\chi} \end{pmatrix} + \frac{1}{2} h \begin{pmatrix} s h \tilde{\chi} \\ -h h \tilde{\chi} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} s \tilde{\chi} \\ -h \tilde{\chi} \end{pmatrix} + \frac{1}{2} h^2 \begin{pmatrix} s \tilde{\chi} \\ -h \tilde{\chi} \end{pmatrix} = \begin{pmatrix} s \tilde{\chi} \\ -h \tilde{\chi} \end{pmatrix} = X'. \end{aligned} \quad (24.395)$$

Equation (24.395) looks trivial, but it clearly shows that X' is certainly a proper negative-energy eigenvector that belongs to the helicity h . Similarly, we define

$$\Sigma_h \equiv \frac{1}{2} \left[E + h \begin{pmatrix} \mathfrak{E} & 0 \\ 0 & \mathfrak{E} \end{pmatrix} \right]. \quad (24.396)$$

Also, we have

$$\Sigma_h X = \Sigma_h \begin{pmatrix} \chi \\ hS\chi \end{pmatrix} = X, \quad \Sigma_h X' = \Sigma_h \begin{pmatrix} S\tilde{\chi} \\ -h\tilde{\chi} \end{pmatrix} = 0, \quad \tilde{\Sigma}_h X = 0, \quad (24.397)$$

$$\Sigma_h^2 = \Sigma_h, \quad \tilde{\Sigma}_h^2 = \tilde{\Sigma}_h, \quad \tilde{\Sigma}_h \Sigma_h = \Sigma_h \tilde{\Sigma}_h = 0, \quad \Sigma_h^\dagger = \Sigma_h, \quad \tilde{\Sigma}_h^\dagger = \tilde{\Sigma}_h, \text{ etc.} \quad (24.398)$$

To the above derivation, use (21.89) and (21.115). It is obvious that both Σ_h and $\tilde{\Sigma}_h$ are projection operators *sensu stricto*.

24.8 Spectral Decomposition of the Dirac Operators

In the previous section, we have shown how the Dirac operators $\mathfrak{G}$ and $\tilde{\mathfrak{G}}$ (with the eigenvalues $-2m$ and 0 both doubly degenerate) can be diagonalized. In Sect. 14.3 we discussed the spectral decomposition and its condition. That is, we have shown that a necessary and sufficient condition for an operator to be a normal operator is that the said operator is expressed as the relation (14.74) called the spectral decomposition. This is equivalent to the statement of Theorem 14.5. In this case, relevant projection operators are the projection operators *sensu stricto* (i.e., Hermitian and idempotent).

Meanwhile, in Sects. 12.5 and 12.7 we mentioned that similar decomposition can be done if an operator in question can be diagonalized via similarity transformation (namely, in the case where the said operator is semi-simple). In that case, Definition 14.1 of the projection operator has been relaxed so that the projection operators *sensu lato* can play a role (see Sect. 18.7). The terminology of spectral decomposition should be taken in a broader meaning accordingly. Bearing in mind the above discussion, we examine the conditions for the spectral decomposition of the Dirac operators $\mathfrak{G}$ and $\tilde{\mathfrak{G}}$. Neither $\mathfrak{G}$ nor $\tilde{\mathfrak{G}}$ is Hermitian or normal, but both of them are semi-simple.

Now, operating S and S^{-1} on both sides of (24.387) from the left and right, respectively, we have

$$S(|1\rangle\langle 1| + |2\rangle\langle 2| + |3\rangle\langle 3| + |4\rangle\langle 4|)S^{-1} = SES^{-1} = E. \quad (24.399)$$

If the Lorentz boost is absent, the representation matrix S related to the Lorentz transformation is unitary (i.e., normal). In this case, (24.399) is rewritten as

$$S(|1\rangle\langle 1| + |2\rangle\langle 2| + |3\rangle\langle 3| + |4\rangle\langle 4|)S^\dagger = E. \quad (24.400)$$

Defining

$$S|k\rangle \equiv |k'\rangle \quad (k = 1, 2, 3, 4) \quad (24.401)$$

and taking the adjoint of (24.401), we have

$$\langle k|S^\dagger \equiv \langle k'|. \quad (24.402)$$

Then, from (24.400) we get

$$|1'\rangle\langle 1'| + |2'\rangle\langle 2'| + |3'\rangle\langle 3'| + |4'\rangle\langle 4'| = E. \quad (24.403)$$

Individual projection operators $|k'\rangle\langle k'|$ ($k = 1, 2, 3, 4$) are Hermitian and, hence, these operators are the projection operators *sensu stricto*. We have already seen this case in Sect. 24.7. Notice again that (24.401) represents the transformation of the Dirac spinor caused by the Lorentz transformation.

If the Lorentz boost is present, however, S is not unitary. The spectral decomposition is not straightforward in that case. Instead, we have a following relation with S described by [11]

$$\gamma^0 S^\dagger \gamma^0 = S^{-1}. \quad (24.404)$$

Using (24.404) we rewrite (24.399) as

$$S(|1\rangle\langle 1| + |2\rangle\langle 2| + |3\rangle\langle 3| + |4\rangle\langle 4|)\gamma^0 S^\dagger \gamma^0 = E.$$

Using (24.401), we have

$$(|1'\rangle\langle 1'| + |2'\rangle\langle 2'| + |3'\rangle\langle 3'| + |4'\rangle\langle 4'|)\gamma^0 S^\dagger \gamma^0 = E. \quad (24.405)$$

Note that $|1'\rangle$ and $|2'\rangle$ represent two linearly independent solutions $\phi'(x')$ of (24.346) and that $|3'\rangle$ and $|4'\rangle$ represent two linearly independent solutions $\chi'(x')$ of (24.349).

Using $\langle 1|\gamma^0 = \langle 1|$, $\langle 2|\gamma^0 = \langle 2|$; $\langle 3|\gamma^0 = -\langle 3|$, $\langle 4|\gamma^0 = -\langle 4|$, we get

$$(|1'\rangle\langle 1'| + |2'\rangle\langle 2'| - |3'\rangle\langle 3'| - |4'\rangle\langle 4'|)S^\dagger \gamma^0 = E. \quad (24.406)$$

Using (24.402), we have

$$(|1'\rangle\langle 1'| + |2'\rangle\langle 2'| - |3'\rangle\langle 3'| - |4'\rangle\langle 4'|)\gamma^0 = E. \quad (24.407)$$

Defining

$$\langle k'|\gamma^0 \equiv \overline{\langle k'|}, \quad (24.408)$$

we obtain

$$|1'\rangle\overline{\langle 1'|} + |2'\rangle\overline{\langle 2'|} - |3'\rangle\overline{\langle 3'|} - |4'\rangle\overline{\langle 4'|} = E. \quad (24.409)$$

Note that (24.408) is equivalent to (21.135) that defines the Dirac adjoint. Equation (24.409) represents the completeness relation of the Dirac spinors; be aware of the minus signs of the third and fourth terms. Multiplying $\mathfrak{P}$ of (24.374) on both sides of (24.409) from the left and using (24.349) and (24.367), we have

$$\mathfrak{P}(|1'\rangle\overline{\langle 1'|} + |2'\rangle\overline{\langle 2'|}) = |1'\rangle\overline{\langle 1'|} + |2'\rangle\overline{\langle 2'|} = \mathfrak{P} = \frac{p_\mu\gamma^\mu + m}{2m}. \quad (24.410)$$

In (24.410), we identify $|1'\rangle\overline{\langle 1'|} + |2'\rangle\overline{\langle 2'|}$ with $(p_\mu\gamma^\mu + m)/2m$. This can readily be confirmed using the coordinate representation $\psi'_1(x')$ and $\psi'_2(x')$ of (24.311) for $|1'\rangle$ and $|2'\rangle$, respectively. Readers are encouraged to confirm this.

Multiplying $\mathfrak{Q}$ of (24.374) on both sides of (24.409) from the left and using (24.346) and (24.368), in turn, we obtain

$$\mathfrak{Q}(-|3'\rangle\overline{\langle 3'|} - |4'\rangle\overline{\langle 4'|}) = -|3'\rangle\overline{\langle 3'|} - |4'\rangle\overline{\langle 4'|} = \mathfrak{Q}. \quad (24.411)$$

That is, we get

$$|3'\rangle\overline{\langle 3'|} + |4'\rangle\overline{\langle 4'|} = -\mathfrak{Q} = \frac{p_\mu\gamma^\mu - m}{2m}. \quad (24.412)$$

From (24.410) and (24.412), we recover

$$\mathfrak{P} + \mathfrak{Q} = E. \quad (24.375)$$

Rewriting (24.410), we have

$$-p_\mu\gamma^\mu - m = -2m|1'\rangle\overline{\langle 1'|} + (-2m)|2'\rangle\overline{\langle 2'|}. \quad (24.413)$$

Also, we obtain

$$\begin{aligned}
|1'\rangle\langle 1'| \cdot |1'\rangle\langle 1'| &= |1'\rangle\langle 1'|, |2'\rangle\langle 2'| \cdot |2'\rangle\langle 2'| = |2'\rangle\langle 2'|, \\
|1'\rangle\langle 1'| \cdot |2'\rangle\langle 2'| &= |2'\rangle\langle 2'| \cdot |1'\rangle\langle 1'| = 0.
\end{aligned} \tag{24.414}$$

Similarly, we have

$$\begin{aligned}
p_\mu \gamma^\mu - m &= -2m \left(-|3'\rangle\langle 3'| \right) + (-2m) \left(-|4'\rangle\langle 4'| \right), \\
|3'\rangle\langle 3'| \cdot |3'\rangle\langle 3'| &= -|3'\rangle\langle 3'|, |4'\rangle\langle 4'| \cdot |4'\rangle\langle 4'| = -|4'\rangle\langle 4'|, \\
|3'\rangle\langle 3'| \cdot |4'\rangle\langle 4'| &= |4'\rangle\langle 4'| \cdot |3'\rangle\langle 3'| = 0.
\end{aligned} \tag{24.415}$$

For instance, (24.415) can readily be checked using (24.311) and (24.337). Equation (24.413) can similarly be checked from (24.282).

Looking at (24.413) and (24.415), we find the resemblance between them and (12.196). In fact, both these cases offer examples of the spectral decomposition (see Sects. 12.5 and 12.7). Taking account of the fact that eigenvalues of a matrix remain unchanged after the similarity transformation (Sect. 12.1), we find that the coefficients $-2m$ for RHS of (24.413) and (24.415) are the eigenvalues (doubly degenerate) for the matrix B of (24.302) and A of (24.301).

From (24.414), $|1'\rangle\langle 1'|$ and $|2'\rangle\langle 2'|$ are idempotent operators. Also, from (24.415), $-|3'\rangle\langle 3'|$ and $-|4'\rangle\langle 4'|$ are idempotent as well. The presence of the two idempotent operators in (24.413) and (24.415) reflects that the rank of $(-p_\mu \gamma^\mu - m)$ and $(p_\mu \gamma^\mu - m)$ is 2. Notice that neither the operator $|3'\rangle\langle 3'|$ nor $|4'\rangle\langle 4'|$, on the other hand, is an idempotent operator but, say, “anti-idempotent” operator. That is, for an anti-idempotent operator $\mathfrak{A}$ we have

$$\mathfrak{A}^2 = -\mathfrak{A},$$

where its eigenvalues are 0 or -1 . The operators of both types play an important role in QED. Meanwhile, we have

$$\begin{aligned}
\left(|k'\rangle\langle k'| \right)^\dagger &= (|k'\rangle\langle k'|\gamma^0)^\dagger = (\gamma^0)^\dagger |k'\rangle\langle k'| = \gamma^0 |k'\rangle\langle k'| \\
&\neq |k'\rangle\langle k'|\gamma^0 = |k'\rangle\langle k'|.
\end{aligned}$$

Hence, $|k'\rangle\langle k'|$ is not Hermitian. The operator $-|k'\rangle\langle k'|$ is not Hermitian, either. Thus, the operators $\pm |k'\rangle\langle k'|$ may be regarded as the projection operators *sensu lato* accordingly. The non-Hermiticity of the projection operators results from the fact that S is not unitary; see the discussion made in Sect. 24.7.

As mentioned earlier in this section, however, we have an exceptional case where the Lorentz boost is absent; see Sect. 24.6.1. In this case, we can construct the projection operators *sensu stricto*; see (24.327), (24.387), and (24.403). As can be

seen in these examples, the Dirac operator $\mathfrak{G}$ is Hermitian. Then, as pointed out in Sect. 21.3.4, we must have the complete orthonormal set (CONS) of common eigenfunctions with the operators $\mathfrak{G}$ and $\mathfrak{H}$ according to Theorem 14.14. Then, the CONS constitutes the diagonalizing unitary matrix $S(\Lambda)$. This can easily be checked using (21.86), (21.102), (24.319) and (24.320). With $\mathfrak{H}$, we obtain

$$S_R^{-1} \mathfrak{H} S_R = S_R^\dagger \mathfrak{H} S_R = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

where S_R is given by (24.318); the diagonal elements are eigenvalues -1 and 1 (each doubly degenerate) of $\mathfrak{H}$. Individual columns of (24.318) certainly constitute the CONS. Denoting them from the left as $|j\rangle$ ($j = 1, 2, 3, 4$), we have

$$\langle j|j\rangle = \delta_{ij}, \quad \sum_{j=1}^4 |j\rangle \langle j| = E.$$

The operator $|j\rangle \langle j|$ is a projection operator *sensu stricto*. The above results are essentially the same as those of Example 24.6. With $\mathfrak{G} (=A)$, the result is self-evident.

What about $S^{-1} \mathfrak{H} S$, then? As already seen in Sect. 21.3.2, $\mathfrak{G}$ and $\mathfrak{H}$ commute with each other; see (21.93). Consequently, we expect to have common eigenvectors for $\mathfrak{G}$ and $\mathfrak{H}$. In other words, once $\mathfrak{G}$ has been diagonalized with S as in (24.339), $\mathfrak{H}$ may well be diagonalized with S as well. It is indeed the case. That is, we also have

$$S^{-1} \mathfrak{H} S = S_\omega^{-1} (S_R^{-1} \mathfrak{H} S_R) S_\omega = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (24.416)$$

where S_ω is given by (24.328). Thus, we find that the common eigenvectors for $\mathfrak{G}$ and $\mathfrak{H}$ are designated by the individual column vectors of S given in (24.293). As already shown in Sects. 21.4 and 24.5, these column vectors do not constitute CONS.

Regarding the operator $\tilde{\mathfrak{H}}$ given in (21.116), similarly we get

$$S^{-1} \tilde{\mathfrak{H}} S = -S^{-1} \mathfrak{H} S = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (24.417)$$

We take the normalization as in (21.141) and (21.142) and define the operators $|u_-\rangle, |u_+\rangle$, etc. such that

$$\begin{aligned}
\sqrt{2m} |1'\rangle &\equiv |u_-\rangle, \sqrt{2m} |2'\rangle \equiv |u_+\rangle, \sqrt{2m} |3'\rangle \equiv |v_+\rangle, \sqrt{2m} |4'\rangle \equiv |v_-\rangle; \\
\sqrt{2m} \overline{|1'\rangle} &\equiv \overline{|u_-\rangle}, \sqrt{2m} \overline{|2'\rangle} \equiv \overline{|u_+\rangle}, \sqrt{2m} \overline{|3'\rangle} \equiv \overline{|v_+\rangle}, \sqrt{2m} \overline{|4'\rangle} \equiv \overline{|v_-\rangle},
\end{aligned}
\tag{24.418}$$

where the subscripts $-$ and $+$ denote the helix -1 and $+1$, respectively. Thus, rewriting (24.410) and (24.412) we get the relations described by

$$|u_-\rangle \overline{\langle u_-|} + |u_+\rangle \overline{\langle u_+|} = p_\mu \gamma^\mu + m, \tag{24.419}$$

$$|v_+\rangle \overline{\langle v_+|} + |v_-\rangle \overline{\langle v_-|} = p_\mu \gamma^\mu - m. \tag{24.420}$$

Using (21.141), we have

$$|u_-\rangle \overline{\langle u_-|} \cdot |u_-\rangle \overline{\langle u_-|} = 2m |u_-\rangle \overline{\langle u_-|}, |u_+\rangle \overline{\langle u_+|} \cdot |u_+\rangle \overline{\langle u_+|} = 2m |u_+\rangle \overline{\langle u_+|}.$$

Similarly, using (21.142) we obtain

$$|v_+\rangle \overline{\langle v_+|} \cdot |v_+\rangle \overline{\langle v_+|} = -2m |v_+\rangle \overline{\langle v_+|}, |v_-\rangle \overline{\langle v_-|} \cdot |v_-\rangle \overline{\langle v_-|} = -2m |v_-\rangle \overline{\langle v_-|}.$$

Equivalent expressions of (24.419) and (24.420) are (21.144) and (21.145), respectively. The above relations are frequently used for various calculations including the field interaction (see Chap. 23). Since the operator $p_\mu \gamma^\mu$ in (24.419) or (24.420) is not a normal operator, neither (24.419) nor (24.420) represents a spectral decomposition in a strict sense (see Sect. 14.3). Yet, (24.419) and (24.420) may be referred to as the spectral decomposition in a broader sense (Sect. 12.7).

In the present section and the precedent sections, we have examined the properties of the projection operators and related characteristics of the spectral decomposition in connection with the Dirac equation on the basis of their results, we wish to summarize important aspects below.

- (i) The Dirac operators ($\mathfrak{G}$ and $\tilde{\mathfrak{G}}$) and helicity operators ($\mathfrak{H}$ and $\tilde{\mathfrak{H}}$) are mutually commutative. Both the operators can be diagonalized with the operator $S(\Lambda)$. In other words, individual column vectors of $S(\Lambda)$ are the eigenvectors common to all the operators $\mathfrak{G}$, $\tilde{\mathfrak{G}}$, $\mathfrak{H}$, and $\tilde{\mathfrak{H}}$.
- (ii) Although the Dirac operators are not Hermitian (nor normal) for the case of $\mathbf{p} \neq \mathbf{0}$, the helicity operators are Hermitian. The non-Hermiticity of Dirac operators implies that the diagonalizing matrix $S(\Lambda)$ is not unitary. Consequently, the projection operators formed by the individual column vectors of $S(\Lambda)$ may be termed projection operators *sensu lato*.

Combining the Dirac adjoint with the change in the functional form of the Dirac spinor described by (24.277), we have [10]

$$\begin{aligned}\overline{\psi'(x')} &= \overline{S\psi(x)} = [S\psi(x)]^\dagger \gamma^0 = \psi^\dagger(x) S^\dagger \gamma^0 = \overline{\psi(x)} [\gamma^0]^{-1} S^\dagger \gamma^0 \\ &= \overline{\psi(x)} \gamma^0 S^\dagger \gamma^0 = \overline{\psi(x)} S^{-1},\end{aligned}\tag{24.421}$$

where with the last equality we used (24.404). Multiplying both sides of (24.421) by the Dirac operator $\mathfrak{G}$ from the right, we obtain

$$\overline{\psi'(x')} \mathfrak{G} = \overline{\psi(x)} S^{-1} \mathfrak{G} = \overline{\psi(x)} S^{-1} \cdot SAS^{-1} = \overline{\psi(x)} AS^{-1},$$

where with the second equality we used (24.339). Multiplying both sides of (24.421) by the Dirac operator $\tilde{\mathfrak{G}}$ from the right, we get

$$\overline{\psi'(x')} \tilde{\mathfrak{G}} = \overline{\psi(x)} S^{-1} \tilde{\mathfrak{G}} = \overline{\psi(x)} S^{-1} \cdot SBS^{-1} = \overline{\psi(x)} BS^{-1},$$

where with the second equality we used (24.342). If we choose $\psi_1(x)$ or $\psi_2(x)$ of (24.305) for $\psi(x)$, we have $\overline{\psi(x)}A = 0$. This leads to

$$\overline{\psi'(x')} \mathfrak{G} = 0,\tag{24.422}$$

which reproduces (21.150). Similarly, if we choose $\psi_3(x)$ or $\psi_4(x)$ of (24.308) for $\psi(x)$, in turn, we have $\overline{\psi(x)}B = 0$. Then, we obtain

$$\overline{\psi'(x')} \tilde{\mathfrak{G}} = 0.\tag{24.423}$$

24.9 Connectedness of the Lorentz Group

In this section, we wish to examine the properties of the Lorentz groups in terms of the Lie groups and Lie algebras. In Chap. 20 we studied the structure and properties of $SU(2)$ and $SO(3)$. We mentioned that several different Lie groups share the same Lie algebra. Of those Lie groups, we had a single universal covering group characterized as the simply connected group (Sect. 20.5). As a typical example, only $SU(2)$ is simply connected as the universal covering group amongst the Lie groups such as $O(3)$, $SO(3)$, etc., having the same Lie algebra. In this section, we examine how the Lorentz group is positioned as part of the continuous groups.

24.9.1 Polar Decomposition of a Non-Singular Matrix [14, 15]

We have already studied properties of Lie algebra of the Lorentz group (Sect 24.2). We may ask what kind of properties the Lorentz group has in relation to the above-mentioned characteristics of $SU(2)$ and related continuous groups. In particular, is there a universal covering group for the Lorentz group? To answer this question, we need a fundamentally important theorem about the matrix algebra. The theorem is well-known as the polar decomposition of a matrix.

Theorem 24.7 Polar Decomposition Theorem [15] Any complex non-singular matrix A is uniquely decomposed into a product such that

$$A = UP, \quad (24.424)$$

where U is a unitary matrix and P is a positive definite Hermitian matrix.

Proof Since A is non-singular, $A^\dagger A$ is a positive definite Hermitian matrix (see Sect. 13.2). Hence, $A^\dagger A$ can be diagonalized via unitary similarity transformation. Namely, we have a following equation:

$$W^\dagger A^\dagger A W = \Lambda \equiv \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}, \quad (24.425)$$

where W is a suitable unitary matrix; n is a dimension of the matrix and λ_i ($i = 1, \dots, n$) is a real positive number with all the off-diagonal elements being zero. Defining another positive definite Hermitian matrix $\Lambda^{1/2}$ as

$$\Lambda^{1/2} \equiv \begin{pmatrix} +\sqrt{\lambda_1} & & & \\ & +\sqrt{\lambda_2} & & \\ & & \ddots & \\ & & & +\sqrt{\lambda_n} \end{pmatrix}, \quad (24.426)$$

we rewrite (24.425) such that

$$W^\dagger (A^\dagger A) W = \Lambda^{1/2} \Lambda^{1/2}. \quad (24.427)$$

Further rewriting (24.427), we have

$$A^\dagger A = W \Lambda^{1/2} \Lambda^{1/2} W^\dagger = \left(W \Lambda^{1/2} W^\dagger \right) \left(W \Lambda^{1/2} W^\dagger \right) = \Lambda^{\tilde{1}/2} \cdot \Lambda^{\tilde{1}/2} = \left(\Lambda^{\tilde{1}/2} \right)^2, \quad (24.428)$$

where $\Lambda^{\tilde{1}/2}$ is defined as

$$\Lambda^{\tilde{1}/2} \equiv W \Lambda^{1/2} W^\dagger. \quad (24.429)$$

Since the unitary similarity transformation holds the eigenvalue unchanged, $\Lambda^{\tilde{1}/2}$ is the positive definite Hermitian operator. Meanwhile, we define a following matrix V as

$$V \equiv A \left(\Lambda^{\tilde{1}/2} \right)^{-1}. \quad (24.430)$$

Then, using (24.428) and (24.430) we obtain

$$\begin{aligned} V^\dagger V &= \left[\left(\Lambda^{\tilde{1}/2} \right)^{-1} \right]^\dagger A^\dagger A \left(\Lambda^{\tilde{1}/2} \right)^{-1} = \left[\left(\Lambda^{\tilde{1}/2} \right)^\dagger \right]^{-1} \left(\Lambda^{\tilde{1}/2} \right)^2 \left(\Lambda^{\tilde{1}/2} \right)^{-1} \\ &= \left(\Lambda^{\tilde{1}/2} \right)^{-1} \left(\Lambda^{\tilde{1}/2} \right)^2 \left(\Lambda^{\tilde{1}/2} \right)^{-1} = E, \end{aligned} \quad (24.431)$$

where with the second equality we used $\left[\left(\Lambda^{\tilde{1}/2} \right)^{-1} \right]^\dagger = \left[\left(\Lambda^{\tilde{1}/2} \right)^\dagger \right]^{-1}$ and with the third equality we used the Hermiticity of $\Lambda^{\tilde{1}/2}$. To show the former relation, suppose that we have two non-singular matrices Q and R . We have $(QR)^\dagger = R^\dagger Q^\dagger$. Putting $R = Q^{-1}$, we get $E^\dagger = E = (Q^{-1})^\dagger Q^\dagger$, namely $(Q^\dagger)^{-1} = (Q^{-1})^\dagger$ [14]. Thus, from (24.430) it follows that there exist a unitary matrix V and a positive definite Hermitian matrix $\Lambda^{\tilde{1}/2}$ which satisfy (24.424) such that

$$A = V \cdot \Lambda^{\tilde{1}/2}. \quad (24.432)$$

Next, suppose that there is another combination of unitary matrix U and Hermitian matrix P other than (24.432) and see what happens. From (24.424) we have

$$A^\dagger A = P^\dagger U^\dagger U P = P^2, \quad (24.433)$$

where with the second equality we used $U^\dagger U = E$ (U : unitary) and $P^\dagger = P$ (P : Hermitian). Then, from (24.428) and (24.433), we obtain

$$P^2 = \left(\Lambda^{\tilde{1}/2} \right)^2. \quad (24.434)$$

From (24.429) we get

$$P^2 = \left(\Lambda^{\tilde{1}/2} \right)^2 = W \Lambda W^\dagger. \quad (24.435)$$

Here, assuming $P = W \Omega W^\dagger$, we obtain

$$P^2 = W \Omega^2 W^\dagger = W \Lambda W^\dagger. \quad (24.436)$$

Hence, $\Omega^2 = \Lambda$. Then, we have

$$\Omega = \begin{pmatrix} \pm \sqrt{\lambda_1} & & & \\ & \pm \sqrt{\lambda_2} & & \\ & & \ddots & \\ & & & \pm \sqrt{\lambda_n} \end{pmatrix}. \quad (24.437)$$

From the assumption, however, P is positive definite. Consequently, only the positive sign is allowed in (24.437). Then, we must have $\Omega = \Lambda^{1/2}$, and so from the above assumption of $P = W \Omega W^\dagger$ and (24.429) we get

$$P = W \Lambda^{1/2} W^\dagger = \Lambda^{\tilde{1}/2}. \quad (24.438)$$

Then, from (24.424) we have

$$U = A \left(\Lambda^{\tilde{1}/2} \right)^{-1}. \quad (24.439)$$

From (24.430) we have $U = V$. This implies that the decomposition (24.424) is unique. These complete the proof.

In relation to the proof of Theorem 24.7, we add that including $\Lambda^{1/2}$ we have 2^n matrices $\Lambda_{(\pm)}^{1/2}$ that satisfy

$$\Lambda_{(\pm)}^{1/2} \Lambda_{(\pm)}^{1/2} = \Lambda, \quad (24.440)$$

where $\Lambda_{(\pm)}^{1/2}$ is defined as

$$\Lambda_{(\pm)}^{1/2} = \begin{pmatrix} \pm \sqrt{\lambda_1} & & & \\ & \pm \sqrt{\lambda_2} & & \\ & & \ddots & \\ & & & \pm \sqrt{\lambda_n} \end{pmatrix}. \quad (24.441)$$

Among them, however, only $\Lambda^{1/2}$ is the positive definite operator. The one-dimensional polar representation of a non-zero complex number is the polar coordinate representation expressed as $z = e^{i\theta}r$ [where the unitary part is described first and r (>0) corresponds to a positive definite eigenvalue]. Then, we can safely say that the polar decomposition of a matrix is a multi-dimensional polar representation of the matrix.

Corollary 24.1 [14] Any complex non-singular matrix A is uniquely decomposed into a product such that

$$A = P' U', \quad (24.442)$$

where P' is a positive definite Hermitian matrix and U' is a unitary matrix. (Notice that the product order of P' and U' has been reversed against Theorem 24.7.)

Proof The proof can be carried out following that of Theorem 24.7 except that we perform the calculation using $AA^\dagger$ instead of $A^\dagger A$ used before. That is, we have

$$U'^\dagger (AA^\dagger) U' = \Omega \equiv \begin{pmatrix} \lambda'_1 & & & \\ & \lambda'_2 & & \\ & & \ddots & \\ & & & \lambda'_n \end{pmatrix}, \quad (24.443)$$

where all the off-diagonal elements are zero. Defining $\Omega^{1/2}$ as

$$\Omega^{1/2} \equiv \begin{pmatrix} +\sqrt{\lambda'_1} & & & \\ & +\sqrt{\lambda'_2} & & \\ & & \ddots & \\ & & & +\sqrt{\lambda'_n} \end{pmatrix} \quad (24.444)$$

and proceeding similarly as before, we get

$$AA^\dagger = U' \Omega^{1/2} \Omega^{1/2} U'^\dagger = (U' \Omega^{1/2} U'^\dagger) (U' \Omega^{1/2} U'^\dagger) = \Omega^{\tilde{1}/2} \cdot \Omega^{\tilde{1}/2} = (\Omega^{\tilde{1}/2})^2,$$

where $\Omega^{\tilde{1}/2}$ is defined as $\Omega^{\tilde{1}/2} \equiv U' \Omega^{1/2} U'^\dagger$. Moreover, defining a following unitary matrix W as $W \equiv (\Omega^{\tilde{1}/2})^{-1} A$, we obtain

$$A = \Omega^{\tilde{I}/2} \cdot W.$$

This means the existence of the polar decomposition of (24.442). The uniqueness of the polar decomposition can be shown as before. This completes the proof.

Corollary 24.2 [14] Let A be a non-singular matrix. Then, there exist positive definite Hermitian matrices H_1 and H_2 as well as a unitary matrix U that satisfy

$$A = UH_1 = H_2U. \quad (24.445)$$

If and only if A is a normal matrix, then we have $H_1 = H_2$. (That is, $H_1 = H_2$ and U are commutative.)

Proof From Theorem 24.7, there must exist a positive definite Hermitian matrix H_1 and a unitary matrix U that satisfy the relation $A = UH_1$. Rewriting this, we have

$$A = UH_1U^\dagger U = \tilde{H}_1U, \quad (24.446)$$

where $\tilde{H}_1 \equiv UH_1U^\dagger$. This is a unitary similarity transformation, and so $\tilde{H}_1$ is a positive definite Hermitian matrix. From Corollary 24.1 the decomposition $A = \tilde{H}_1U$ is unique, and so comparing RHSs of (24.445) and (24.446) we can choose H_2 as $H_2 = \tilde{H}_1$. Multiplying this by $U^\dagger$ from the right, we get

$$H_2U = \tilde{H}_1U = UH_1U^\dagger U = UH_1. \quad (24.447)$$

This gives the proof of the former half of Corollary 24.2.

From Corollary 24.1, with respect to (24.447) that U and H_1 commute is equivalent to $H_1 = H_2$. Taking account of this fact, the latter half of the proof is as follows: If U and H_1 commute, we have

$$A^\dagger A = H_1^\dagger U^\dagger UH_1 = H_1^\dagger H_1 = H_1 H_1.$$

Also, we have

$$AA^\dagger = UH_1H_1^\dagger U^\dagger = H_1UU^\dagger H_1^\dagger = H_1H_1^\dagger = H_1H_1, \quad (24.448)$$

where with the second equality we used $UH_1 = H_1U$ and its adjoint $H_1^\dagger U^\dagger = U^\dagger H_1^\dagger$. From the above two equations, A is normal.

Conversely, if A is normal, from (24.447) we have

$$A^\dagger A = H_1^2 = U^\dagger H_2^\dagger H_2 U = U^\dagger H_2^2 U = AA^\dagger = H_2^2.$$

Hence, $H_1^2 = H_2^2$. Since both H_1 and H_2 are positive definite Hermitian, as in the proof of Theorem 24.7 we have $H_1 = H_2$. Namely, U and $H_1 = H_2$ commute. These complete the proof.

Regarding real non-singular matrices, we have the following corollary.

Corollary 24.3 [8] Any real non-singular matrix A is uniquely decomposed into a product such that

$$A = OP, \quad (24.449)$$

where O is an *orthogonal* matrix and P is a positive definite *symmetric* matrix.

Proof The proof can immediately be performed using Theorem 24.7. This is because a real unitary matrix is an orthogonal matrix and a real Hermitian matrix is a symmetric matrix.

Since the Lorentz transformation Λ is represented by a real $(4, 4)$ matrix, from Corollary 24.3 we realize that the matrix Λ is expressed as (24.449); see Sect. 24.2. Tangible examples with respect to the above theorem and corollaries can be seen in the following examples.

Example 24.8 Let us take an example of the polar decomposition for a matrix A expressed as

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}. \quad (24.450)$$

We have

$$A^T A = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}. \quad (24.451)$$

We dealt with the diagonalization of this matrix in Example 14.3. Borrowing U in (14.102), we have

$$D = U^T H U = \begin{pmatrix} 3 + 2\sqrt{2} & 0 \\ 0 & 3 - 2\sqrt{2} \end{pmatrix} = \begin{pmatrix} (\sqrt{2} + 1)^2 & 0 \\ 0 & (\sqrt{2} - 1)^2 \end{pmatrix}. \quad (24.452)$$

Then, we get

$$D^{1/2} = \begin{pmatrix} \sqrt{2} + 1 & 0 \\ 0 & \sqrt{2} - 1 \end{pmatrix} \quad (24.453)$$

as well as

$$\begin{aligned}
D^{\tilde{1}/2} &= UD^{1/2}U^\dagger = UD^{1/2}U^T \\
&= \begin{pmatrix} \frac{1}{\sqrt{4-2\sqrt{2}}} & -\frac{1}{\sqrt{4+2\sqrt{2}}} \\ \frac{1}{\sqrt{4+2\sqrt{2}}} & \frac{1}{\sqrt{4-2\sqrt{2}}} \end{pmatrix} \begin{pmatrix} \sqrt{2}+1 & 0 \\ 0 & \sqrt{2}-1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{4-2\sqrt{2}}} & \frac{1}{\sqrt{4+2\sqrt{2}}} \\ -\frac{1}{\sqrt{4+2\sqrt{2}}} & \frac{1}{\sqrt{4-2\sqrt{2}}} \end{pmatrix} \\
&= \begin{pmatrix} 3/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}. \tag{24.454}
\end{aligned}$$

With the matrix V , we obtain

$$\begin{aligned}
V &= A[D^{\tilde{1}/2}]^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} & 3/\sqrt{2} \end{pmatrix} \\
&= \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}. \tag{24.455}
\end{aligned}$$

Thus, we find that V is certainly an orthogonal matrix. The unique polar decomposition of A is given by

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} = V \cdot D^{\tilde{1}/2} = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 3/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}. \tag{24.456}$$

In (24.456) V and $D^{\tilde{1}/2}$ are commutative, as expected, because A is a real symmetric matrix (i.e., Hermitian and normal). The confirmation that $D^{\tilde{1}/2}$ is positive definite is left for readers.

Example 24.9 Let us take an example of the polar decomposition for a triangle matrix A expressed as

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}. \tag{24.457}$$

Similarly proceeding as the above, we should get a result shown as

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 3/\sqrt{2} \end{pmatrix}. \tag{24.458}$$

In this case, the orthogonal matrix and the positive definite symmetric matrix are not commutative. Finding the other decomposition is left as an exercise for readers.

With respect to the transformation matrix $S(\Lambda)$ of Sect. 24.4, let us think of a following example.

Example 24.10 In the general case of the representation matrix described by (24.293) of the previous section, consider a special case of $\theta = 0$. In that case, from (24.293) we obtain

$$\begin{aligned}
 & \begin{pmatrix} e^{i\phi/2} \cosh \frac{\omega}{2} & 0 & -e^{i\phi/2} \sinh \frac{\omega}{2} & 0 \\ 0 & e^{-i\phi/2} \cosh \frac{\omega}{2} & 0 & e^{-i\phi/2} \sinh \frac{\omega}{2} \\ -e^{i\phi/2} \sinh \frac{\omega}{2} & 0 & e^{i\phi/2} \cosh \frac{\omega}{2} & 0 \\ 0 & e^{-i\phi/2} \sinh \frac{\omega}{2} & 0 & e^{-i\phi/2} \cosh \frac{\omega}{2} \end{pmatrix} \\
 &= \begin{pmatrix} e^{i\phi/2} & 0 & 0 & 0 \\ 0 & e^{-i\phi/2} & 0 & 0 \\ 0 & 0 & e^{i\phi/2} & 0 \\ 0 & 0 & 0 & e^{-i\phi/2} \end{pmatrix} \\
 &\quad \times \begin{pmatrix} \cosh \frac{\omega}{2} & 0 & -\sinh \frac{\omega}{2} & 0 \\ 0 & \cosh \frac{\omega}{2} & 0 & \sinh \frac{\omega}{2} \\ -\sinh \frac{\omega}{2} & 0 & \cosh \frac{\omega}{2} & 0 \\ 0 & \sinh \frac{\omega}{2} & 0 & \cosh \frac{\omega}{2} \end{pmatrix}. \tag{24.459}
 \end{aligned}$$

The LHS of Eq. (24.459) is a normal matrix and the two matrices of RHS are commutative as expected. As in this example, if an axis of the rotation and a direction of the Lorentz boost coincide, such successive operations of the rotation and boost are commutative and, hence, the relevant representation matrix $S(\Lambda)$ is normal according to Corollary 24.2.

Theorem 24.7 helps understand that universal covering group exists in the Lorentz groups. The said group is termed special linear group and abbreviated as $SL(2, \mathbb{C})$. In Sects. 20.2 and 20.5, we have easily shown that $SU(2)$ is isomorphic to S^3 , because we have many constraint conditions imposed upon the parameters of $SU(2)$. Then, we could readily show that since S^3 is simply connected, so is $SU(2)$. With $SL(2, \mathbb{C})$, on the other hand, the number of constraint conditions is limited; we have only two conditions resulting from that the determinant of two-dimensional complex matrices is 1. Therefore, it is not easy to examine whether $SL(2, \mathbb{C})$ is simply connected. In virtue of Theorem 24.7, however, we can readily prove that $SL(2, \mathbb{C})$ is indeed simply connected. It will be one of the main subjects of the next section.

24.9.2 Special Linear Group: $SL(2, \mathbb{C})$

In this section we examine how $SL(2, \mathbb{C})$ is related to the Lorentz group. We start with the most general feature of $SL(2, \mathbb{C})$ on the basis of the discussion of Sect. 20.5. The discussion is based on Definitions 20.8–20.10 (Sect. 20.5) along with the following definition (Definition 24.5).

Definition 24.5 The special linear group $SL(2, \mathbb{C})$ is defined as

$$SL(2, \mathbb{C}) \equiv \{A = (a_{ij}); i, j = 1, 2, a_{ij} \in \mathbb{C}, \det A = 1\}.$$

Theorem 24.8 [8] The group $SL(2, \mathbb{C})$ is simply connected.

Proof Let $g(t, s) \in SL(2, \mathbb{C})$ be an arbitrary continuous matrix function with respect to t in a region $I \times I = \{(t, s); 0 \leq t \leq 1, 0 \leq s \leq 1\}$. Also, let $g(t, 1)$ be an arbitrary loop in $SL(2, \mathbb{C})$. Then, according to Theorem 24.7, $g(t, 1)$ can uniquely be decomposed such that

$$g(t, 1) = U(t)P(t, 1), \quad (24.460)$$

where $U(t)$ is unitary and $P(t, 1)$ is positive definite Hermitian.

Since $U(t) \in SL(2, \mathbb{C})$ is unitary, we must have $U(t) \in SU(2)$. Suppose that the function $P(t, s)$ can be described as

$$P(t, s) = \exp sX(t) \quad (0 \leq s \leq 1), \quad (24.461)$$

where $X(t) \in \mathfrak{sl}(2, \mathbb{C})$; $\mathfrak{sl}(2, \mathbb{C})$ is a Lie algebra of $SL(2, \mathbb{C})$. Since $P(t, 1)$ is Hermitian and s is real, $X(t)$ is Hermitian as well. This can immediately be derived from (15.32). Since we have

$$g(t, 1) = U(t) \exp X(t) \quad \text{and} \quad g(t, 0) = U(t) \quad (24.462)$$

and $\exp sX(t)$ is analytic, $g(t, 1)$ is homotopic to $g(0, t) = U(t)$; see Definition 20.9.

Because $SU(2)$ is simply connected (see Sect. 20.5.3), $U(t)$ is homotopic to any constant loop (Definition 20.10). Consequently, from the above statement $g(t, 1)$ is homotopic to a constant loop. This loop can arbitrarily be chosen and $g(t, 1)$ is an arbitrary loop in $SL(2, \mathbb{C})$. Therefore, $SL(2, \mathbb{C})$ is simply connected.

During the process of the above proof, an important feature of $SL(2, \mathbb{C})$ has already shown up. That is, Hermitian operators are contained in $\mathfrak{sl}(2, \mathbb{C})$. This feature shows a striking contrast with the Lie algebra $\mathfrak{su}(n)$ in that $\mathfrak{su}(n)$ comprises anti-Hermitian operators only.

Theorem 24.9 [8] The Lorentz group $O(3, 1)$ is self-conjugate. That is, if $\Lambda \in O(3, 1)$, then $\Lambda^T \in O(3, 1)$.

Proof From (24.198), for a group element Λ to constitute the Lorentz group $O(3, 1)$ we have

$$\Lambda^T \eta \Lambda = \eta.$$

Multiplying both sides by $\Lambda \eta$ from the left and $\Lambda^{-1} \eta$ from the right, we obtain

$$\Lambda \eta \cdot \Lambda^T \eta \Lambda \cdot \Lambda^{-1} \eta = \Lambda \eta \cdot \eta \cdot \Lambda^{-1} \eta.$$

Using $\eta^2 = 1$, we get

$$\Lambda \eta \Lambda^T = \eta \quad \text{or} \quad (\Lambda^T)^T \eta \Lambda^T = \eta.$$

This implies $\Lambda^T \in O(3, 1)$.

Theorem 24.10 [8] Suppose that with the Lorentz group $O(3, 1)$, individual elements $\Lambda \in O(3, 1)$ are decomposed into

$$\Lambda = UP. \tag{24.463}$$

Then, their factors U and P belong to $O(3, 1)$ as well with P described by

$$P = \exp Y, \tag{24.464}$$

where $Y \in \mathfrak{o}(3, 1)$ is a real symmetric matrix.

Proof From Theorem 24.9, $\Lambda^T \in O(3, 1)$. Then, $\Lambda^T \Lambda \in O(3, 1)$. Since Λ is real, from Corollary 24.3 U is an orthogonal matrix and P is a positive definite symmetric matrix. Meanwhile, we have

$$\Lambda^T \Lambda = P^T U^T U P = P^T P = P^2, \tag{24.465}$$

where with the second equality we used the fact that U is an orthogonal matrix; with the last equality we used the fact that P is symmetric. Then, $P^2 \in O(3, 1)$. Therefore, using $\exists X \in \mathfrak{o}(3, 1)$, P^2 can be expressed as

$$P^2 = \exp X. \tag{24.466}$$

Since P is a positive definite symmetric matrix, so is P^2 . Taking the transposition of (24.466), we have

$$(P^2)^T = P^2 = (\exp X)^T = \exp(X^T), \tag{24.467}$$

where with the last equality we used (15.31). Comparing (24.466) and (24.467), we obtain

$$\exp X = \exp(X^T). \quad (24.468)$$

Since (24.468) equals the positive definite matrix P^2 , a logarithmic function corresponding to (24.468) uniquely exists [8]. Consequently, taking the logarithm of (24.468) we get

$$X^T = X. \quad (24.469)$$

Namely, X is real symmetric. As $X/2 \in \mathfrak{o}(3, 1)$, $\exp(X/2) \in O(3, 1)$. Since P and P^2 are positive definite, $\exp(X/2)$ is uniquely determined such that

$$\exp(X/2) = (P^2)^{1/2} = P. \quad (24.470)$$

Hence, $P \in O(3, 1)$. Putting $X/2 \equiv Y$, we obtain (24.464), to which $Y \in \mathfrak{o}(3, 1)$ is real symmetric. From (24.463) we have $U = \Lambda P^{-1}$, and so $U \in O(3, 1)$. This completes the proof.

Theorem 24.10 shows that $O(3, 1)$ comprises the direct product of a compact subgroup $O(3, 1) \cap O(4)$ and $\exp \tilde{\mathfrak{o}}(3, 1)$, where $\tilde{\mathfrak{o}}(3, 1)$ is a subspace of $\mathfrak{o}(3, 1)$ consisting of real symmetric matrices. We have another two important theorems with the Lorentz group $O(3, 1)$.

Theorem 24.11 [8] The compact subgroup $O(3, 1) \cap O(4)$ is represented as

$$O(3, 1) \cap O(4) = O(3) \oplus O(1). \quad (24.471)$$

That is, any elements Λ of $O(3, 1) \cap O(4)$ can be described by

$$\Lambda = \begin{pmatrix} \pm 1 & 0 & 0 & 0 \\ 0 & \boxed{\Lambda_1} \\ 0 & & & \\ 0 & & & \end{pmatrix} \text{ with } \Lambda_1 \in O(3). \quad (24.472)$$

In an opposite way, any elements that can be expressed as (24.472) belong to $O(3, 1) \cap O(4)$.

Proof We write Λ such that

$$\Lambda = \begin{pmatrix} g_0 & g_1 \\ g_2 & g_3 \end{pmatrix}; g_0 \in \mathbb{R}, g_1 : (1, 3) \text{ matrix}, g_2 : (3, 1) \text{ matrix}, \\ g_3 \in \mathfrak{gl}(3, \mathbb{R}). \quad (24.473)$$

As Λ is an orthogonal matrix, i.e., $\Lambda^T = \Lambda^{-1}$, $\Lambda^T \eta \Lambda = \eta$ is equivalent to

$$\eta\Lambda = \begin{pmatrix} g_0 & g_1 \\ -g_2 & -g_3 \end{pmatrix} = \begin{pmatrix} g_0 & -g_1 \\ g_2 & -g_3 \end{pmatrix} = \Lambda\eta. \quad (24.474)$$

Therefore, we have $g_1 = g_2 = 0$. From $\Lambda^T\eta\Lambda = \eta$ and the above result, we have

$$\begin{aligned} \Lambda^T\eta\Lambda &= \begin{pmatrix} g_0 & g_2^T \\ g_1^T & g_3^T \end{pmatrix} \begin{pmatrix} g_0 & g_1 \\ -g_2 & -g_3 \end{pmatrix} = \begin{pmatrix} g_0^2 & 0 \\ 0 & -g_3^T g_3 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \eta. \end{aligned} \quad (24.475)$$

From the above equation, we have

$$g_0 = \pm 1 \quad \text{and} \quad g_3^T g_3 = 1. \quad (24.476)$$

The second equation of (24.476) implies that g_3 is an orthogonal matrix. In an opposite way, any elements given in a form of (24.472) must belong to $O(3, 1) \cap O(4)$. These complete the proof.

Theorem 24.12 [8] The Lorentz group $O(3, 1)$ consists of four connected components $\mathfrak{L}_0 \equiv SO_0(3, 1)$, $\mathfrak{L}_1$, $\mathfrak{L}_2$, and $\mathfrak{L}_3$ that contain

$$\begin{aligned} E_0 \equiv E &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, E_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \\ E_2 &= \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, E_3 = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \end{aligned} \quad (24.477)$$

respectively.

Proof As in the case of the proof of Theorem 24.8, an arbitrary element Λ of the Lorentz group $O(3, 1)$ can be represented as

$$\Lambda = O(t)P(t, s), \quad (24.478)$$

where $O(t)$ is an orthogonal matrix of $O(4)$; $P(t, s) = \exp sX(t)$ for which $X(t)$ is a real symmetric matrix according to Theorem 24.10. Both $O(t)$ and $P(t, s)$ belong to $O(3, 1)$ (Theorem 24.10). Then, we have $O(t) \in O(3, 1) \cap O(4)$.

Consequently, making the parameter s change continuously from 1 to 0, we find $\forall \Lambda \in O(3, 1)$ is connected to $O(t)$ in $O(3, 1)$. Since $O(t) \in O(3, 1) \cap O(4)$, $O(t)$ must be described in the form of (24.472) according to Theorem 24.11. The elements E_i ($i = 0, 1, 2, 3$) of (24.477) certainly belong to $O(t)$ and, hence, belong to $O(3, 1) \cap O(4)$ as well.

Meanwhile, Example 20.8 tells us that within a connected set the sign of the determinant should be constant, if the determinant is defined as a real continuous function. Also, we have already known that $O(n)$ is decomposed into two connected sets (Sect. 20.5). In the case of $O(3)$, each set is connected to either of the following two matrices according to the determinant ± 1 :

$$\tilde{O} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tilde{O}' = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (24.479)$$

For this, see (20.326) and (20.327). Taking account of $O(1) = \{+1, -1\}$ in (24.471), we have $2 \times 2 = 4$ connected sets in $O(3, 1)$. These four sets originate from the combinations of (24.479) with ± 1 of $O(1)$ and the individual connected sets are represented by E_i ($i = 0, 1, 2, 3$) of (24.477). Suppose that $\mathfrak{L}_0$, $\mathfrak{L}_1$, $\mathfrak{L}_2$, and $\mathfrak{L}_3$ comprise all the elements that are connected to E_0 , E_1 , E_2 , and E_3 , respectively. Then, these $\mathfrak{L}_i$ ($i = 0, 1, 2, 3$) are connected sets and arbitrary elements of $O(3, 1)$ are contained in one of them. Thus, it suffices to show that $\mathfrak{L}_i \cap \mathfrak{L}_j = \emptyset (i \neq j)$.

First, from $\Lambda^T \eta \Lambda = \eta$ we have $(\det \Lambda)^2 = 1$, i.e., $\det \Lambda = \pm 1$. From (24.477) we have $\det E_0 = \det E_3 = 1$ and $\det E_1 = \det E_2 = -1$. Then, all the elements Λ belonging to $\mathfrak{L}_0$ and $\mathfrak{L}_3$ have $\det \Lambda = 1$ and those belonging to $\mathfrak{L}_1$ and $\mathfrak{L}_2$ have $\det \Lambda = -1$. From Example 20.8, the elements of $\mathfrak{L}_0$ cannot be connected to the elements of $\mathfrak{L}_1$ or those of $\mathfrak{L}_2$. Similarly, the elements of $\mathfrak{L}_3$ cannot be connected to the elements of $\mathfrak{L}_1$ or those of $\mathfrak{L}_2$.

Let $\Lambda_0 \in \mathfrak{L}_0$ and let Λ_0 be expressed as $\Lambda_0 = O_0 P_0$ with the polar decomposition. The element Λ_0 is connected to O_0 , and so $O_0 \in \mathfrak{L}_0$ from Theorem 24.10. Since $\mathfrak{L}_0$ is a collection of the elements that are connected to E_0 , so is $O_0 \in [\mathfrak{L}_0 \cap O(4)]$. The element E_0 can be expressed as $E_0 = \tilde{O} \oplus \{+1\}$ so that $\det \tilde{O} = 1$ in (24.479) and $\det \{+1\} = 1$. Meanwhile, let $\Lambda_3 \in \mathfrak{L}_3$ be described by $\Lambda_3 = O_3 P_3$. The element Λ_3 is connected to O_3 , and so $O_3 \in \mathfrak{L}_3$. Similarly to the above case, $O_3 \in [\mathfrak{L}_3 \cap O(4)]$ is connected to E_3 , which is expressed as $E_3 = \tilde{O}' \oplus \{-1\}$ so that $\det \tilde{O}' = -1$ in (24.479) and $\det \{-1\} = -1$. Again, from Example 20.8, within a connected set the sign of the determinant should be constant. Therefore, the elements of $\mathfrak{L}_0$ cannot be connected to the elements of $\mathfrak{L}_3$. The discussion equally applies to the relationship between $\mathfrak{L}_1$ and $\mathfrak{L}_2$.

Thus, after all, we get

$$\begin{aligned}
 O(3, 1) &= C(E_0) \cup C(E_1) \cup C(E_2) \cup C(E_3), \\
 C(E_i) \cap C(E_j) &= \emptyset (i \neq j),
 \end{aligned}
 \tag{24.480}$$

where $C(E_i) \equiv \mathfrak{L}_i$. These complete the proof.

In the above proof, (24.480) is in parallel with (20.318) of $O(3)$, where the number of the connected components is two. The group $\mathfrak{L}_0 \equiv SO_0(3, 1)$ is a connected component that contains the identity element E_0 . Then, according to Theorem 20.9, $SO_0(3, 1)$ is an invariant subgroup of $O(3, 1)$. Thus, following Sect. 16.3 we have a coset decomposition described by

$$O(3, 1) = E_0 SO_0(3, 1) + E_1 SO_0(3, 1) + E_2 SO_0(3, 1) + E_3 SO_0(3, 1). \tag{24.481}$$

Summarizing the above, we classify the connected components such that [8]:

$$\begin{aligned}
 \mathfrak{L}_0 &\equiv SO_0(3, 1) = \{\Lambda; \Lambda \in O(3, 1), \det \Lambda = 1, \Lambda_{00} \geq 1\}, \\
 \mathfrak{L}_1 &= \{\Lambda; \Lambda \in O(3, 1), \det \Lambda = -1, \Lambda_{00} \geq 1\}, \\
 \mathfrak{L}_2 &= \{\Lambda; \Lambda \in O(3, 1), \det \Lambda = -1, \Lambda_{00} \leq -1\}, \\
 \mathfrak{L}_3 &= \{\Lambda; \Lambda \in O(3, 1), \det \Lambda = 1, \Lambda_{00} \leq -1\}.
 \end{aligned}
 \tag{24.482}$$

The set $\{E_0, E_1, E_2, E_3\}$ forms a group, which is isomorphic to the four-group we dealt with in Sects. 17.1 and 17.2. In parallel with (20.328), we have

$$O(3, 1)/SO_0(3, 1) \cong \{E_0, E_1, E_2, E_3\}. \tag{24.483}$$

In theoretical physics, instead of E_1 the following $\tilde{E}_1$ is often used as a representative element:

$$\tilde{E}_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

The element $\tilde{E}_1$ represents the inversion symmetry (i.e., spatial inversion) and is connected to E_1 . This can be shown as in the case of Example 20.6. Suppose a following continuous matrix function:

$$f(t) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \pi t & -\sin \pi t & 0 \\ 0 & \sin \pi t & \cos \pi t & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

Then, $f(t)$ is a continuous matrix function with respect to real t ; $f(t)$ represents product transformations of mirror symmetry with respect to the xy -plane and rotation about the z -axis. We have $f(0) = E_1$ and $f(1) = \tilde{E}_1$. Hence, $f(0)$ and $f(1)$ are connected to each other in $\mathcal{L}_1$.

The element E_1 (or $\tilde{E}_1$) is known as a discrete symmetry along with the element E_2 that represents the time reversal. The matrix C defined in (21.164), on the other hand, is not a Lorentz transformation. This is known from Theorem 24.11. Namely, C is unitary, and so if C belonged to $O(3, 1)$, C would be described in a form of (24.472). But it is not the case. This is associated with the fact that C represents the exchange between a pair of particle and antiparticle. Note that this is not a space-time transformation but an “inner” transformation.

As a result of Theorem 24.12 and on the basis of Theorem 20.10, we reach the following important consequence: The Lie algebras $\mathfrak{o}(3, 1)$ of $O(3, 1)$ and $\mathfrak{so}_0(3, 1)$, of $SO_0(3, 1)$ are identical.

With the modulus of Λ_0^0 , see (24.207). The special theory of relativity is based upon $\mathcal{L}_0 = O_0(3, 1)$ where neither the spatial inversion nor time reversal is assumed. As a tangible example, $\exp(\theta \mathfrak{A}_3)$ of (24.224) and $\exp(-\omega \mathfrak{B}_3)$ of (24.225) satisfy this criterion. Equation (24.223) reflects the polar decomposition. The representation matrix $S(\Lambda)$ of (24.293) with respect to the transformation of the Dirac spinor gives another example of the polar decomposition. In this respect, we show some examples below.

Example 24.11: Lorentz Transformation Matrix In (24.223), $\exp(a \cdot \mathfrak{A})$ and $\exp(b \cdot \mathfrak{B})$ represent an orthogonal matrix and a positive definite symmetric matrix, respectively. From Theorems 24.10 and 24.11, we have

$$SO_0(3, 1) \cap SO(4) = SO(3) \oplus SO(1), \quad (24.484)$$

where $SO(1) \equiv \{1\}$. Any elements $\tilde{\Lambda}$ of $SO_0(3, 1) \cap SO(4)$ can therefore be described by

$$\tilde{\Lambda} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \sqrt{\tilde{\Lambda}_1^2} & 0 & 0 \\ 0 & 0 & \tilde{\Lambda}_1 & 0 \\ 0 & 0 & 0 & \tilde{\Lambda}_1 \end{pmatrix} \quad \text{with } \tilde{\Lambda}_1 \in SO(3). \quad (24.485)$$

In fact, (24.224) is an orthogonal matrix, i.e., a specific type of (24.485), and a unitary matrix component of the polar decomposition (Theorem 24.7 and Corollary 24.3). Equation (24.225) represents an example of a positive definite real symmetric matrix (i.e., Hermitian) component of the polar decomposition. The matrix $\exp(-\omega \mathfrak{B}_3)$ of (24.225) has eigenvalues

$$\cosh \omega + \sinh \omega, \cosh \omega - \sinh \omega, 1, 1. \quad (24.486)$$

Since all these eigenvalues are positive and $\det \exp(-\omega \mathfrak{B}_3) = 1$, $\exp(-\omega \mathfrak{B}_3)$ is positive definite (see Sect. 13.2) and the product of the eigenvalues of (24.486) is 1, as can be easily checked. Consequently, (24.223) is a (unique) polar decomposition.

Example 24.12: Transformation Matrix $S(\Lambda)$ of Dirac Spinor As already shown, (24.293) describes the representation matrix $S(\Lambda)$ with respect to the transformation of the Dirac spinor caused by the Lorentz transformation. The RHS of (24.293) can be decomposed such that

$$S(\Lambda) = \begin{pmatrix} e^{\frac{i\phi}{2}} \cos \frac{\theta}{2} & e^{\frac{i\phi}{2}} \sin \frac{\theta}{2} & 0 & 0 \\ -e^{-\frac{i\phi}{2}} \sin \frac{\theta}{2} & e^{-\frac{i\phi}{2}} \cos \frac{\theta}{2} & 0 & 0 \\ 0 & 0 & e^{\frac{i\phi}{2}} \cos \frac{\theta}{2} & e^{\frac{i\phi}{2}} \sin \frac{\theta}{2} \\ 0 & 0 & -e^{-\frac{i\phi}{2}} \sin \frac{\theta}{2} & e^{-\frac{i\phi}{2}} \cos \frac{\theta}{2} \end{pmatrix} \times \begin{pmatrix} \cosh \frac{\omega}{2} & 0 & -\sinh \frac{\omega}{2} & 0 \\ 0 & \cosh \frac{\omega}{2} & 0 & \sinh \frac{\omega}{2} \\ -\sinh \frac{\omega}{2} & 0 & \cosh \frac{\omega}{2} & 0 \\ 0 & \sinh \frac{\omega}{2} & 0 & \cosh \frac{\omega}{2} \end{pmatrix}. \quad (24.487)$$

The first factor of (24.487) is unitary and the second factor is a real symmetric matrix. The eigenvalues of the second factor are positive. These are given by

$$\begin{aligned} & \cosh \frac{\omega}{2} + \sinh \frac{\omega}{2} \text{ (doubly degenerate), } \cosh \frac{\omega}{2} \\ & - \sinh \frac{\omega}{2} \text{ (doubly degenerate)} \end{aligned} \quad (24.488)$$

with a determinant of 1 (>0). Hence, the second matrix is positive definite. Thus, as in the case of Example 24.10, $S(\Lambda)$ gives a unique polar decomposition.

Given the above tangible examples, we wish to generalize the argument a little bit further. In Theorem 24.7 and Corollary 24.1 of Sect. 24.9.1 we mentioned two ways of the polar decomposition. What about the inverse Lorentz transformation, then? The inverse transformation of (24.424) is described by

$$A^{-1} = P^{-1}U^{-1}. \quad (24.489)$$

Here, suppose that a positive definite Hermitian operator P is represented by a (n, n) square matrix. From Theorem 14.5 using a suitable unitary matrix U , we have a following diagonal matrix D

$$U^\dagger P U = D \equiv \begin{pmatrix} \lambda_1 & \cdots & \\ \vdots & \ddots & \vdots \\ & \cdots & \lambda_n \end{pmatrix}, \quad (24.490)$$

where $\lambda_1, \dots, \lambda_n$ are real positive numbers. Taking the inverse of (24.490), we obtain

$$U^{-1}P^{-1}(U^\dagger)^{-1} = U^\dagger P^{-1}U = \begin{pmatrix} 1/\lambda_1 & \cdots & \\ \vdots & \ddots & \vdots \\ & \cdots & 1/\lambda_n \end{pmatrix}. \quad (24.491)$$

Meanwhile, we have

$$(P^{-1})^\dagger = (P^\dagger)^{-1} = P^{-1}. \quad (24.492)$$

Equations (24.491) and (24.492) imply that P^{-1} is a positive definite Hermitian operator. Thus, the above argument ensures that the inverse transformation A^{-1} of (24.489) can be dealt with on the same footing as the transformation A . In fact, in both Examples 24.11 and 24.12, eigenvalues of the positive definite real symmetric (i.e., Hermitian) matrices are unchanged, or switched to one another (consider e.g., exchange between ω and $-\omega$) after the inversion.

We have known from Theorem 24.8 that an arbitrary continuous function $g(t, s) \in SL(2, \mathbb{C})$ is homotopic to $U(t) \in SU(2)$ and concluded that since $SU(2)$ is simply connected, $SL(2, \mathbb{C})$ is simply connected as well. As for the Lorentz group $O(3, 1)$, on the other hand, $O(3, 1)$ is homotopic to $O(t) \in O(4)$, but since $O(4)$ is not simply connected, we infer that $O(3, 1)$ is not simply connected, either. In the next section, we investigate the representation of $O(3, 1)$, especially that of the proper orthochronous Lorentz group $SO_0(3, 1)$. We further investigate the connectedness of the Lorentz group with this special but important case.

24.10 Representation of the Proper Orthochronous Lorentz Group $SO_0(3, 1)$ [6, 8]

Amongst four connected components of the Lorentz group $O(3, 1)$, the component that is connected to the identity element $E_0 \equiv E$ does not contain the spatial inversion or time reversal. This component itself forms a group termed $SO_0(3, 1)$ and frequently appears in the elementary particle physics.

Let us think of a following set $\mathfrak{V}^4$ represented by

$$\mathfrak{V}^4 = \{H = (h_{ij}); i, j = 1, 2, h_{ij} \in \mathbb{C}, H : \text{Hermitian}\}, \quad (24.493)$$

where (h_{ij}) is a complex $(2, 2)$ Hermitian matrix. That is, $\mathfrak{V}^4$ is a set of all $(2, 2)$ Hermitian matrices. The set $\mathfrak{V}^4$ is a vector space on a real number field $\mathbb{R}$. To show this, we express an arbitrary element H of $\mathfrak{V}^4$ as

$$H = \begin{pmatrix} a & c + id \\ c - id & b \end{pmatrix} \quad (a, b, c, d : \text{real}). \quad (24.494)$$

We put

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_2 = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \sigma_3 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (24.495)$$

Notice that σ_k ($k = 1, 2, 3$) are the Pauli spin matrices that appeared in (20.41). Also, σ_k is expressed as

$$\sigma_k = 2i\zeta_k \quad (k = 1, 2, 3),$$

where ζ_k is given by (20.270). Whereas ζ_k is anti-Hermitian, σ_k is Hermitian. Then, H of (24.494) can be described with Hermitian operators σ_μ ($\mu = 0, 1, 2, 3$) by

$$H = \frac{1}{2}(a + b)\sigma_0 + \frac{1}{2}(-a + b)\sigma_3 + c\sigma_1 + d\sigma_2. \quad (24.496)$$

Equation (24.496) implies that any vector H belonging to $\mathfrak{V}^4$ is expressed as a linear combination of the four basis vectors σ_μ ($\mu = 0, 1, 2, 3$). Since four coefficients of σ_μ ($\mu = 0, 1, 2, 3$) are real, it means that $\mathfrak{V}^4$ is a four-dimensional vector space on a real number field $\mathbb{R}$. If we view $\mathfrak{V}^4$ as an inner product space, the orthonormal set is given by

$$\frac{1}{\sqrt{2}}\sigma_0, \frac{1}{\sqrt{2}}\sigma_1, \frac{1}{\sqrt{2}}\sigma_2, \frac{1}{\sqrt{2}}\sigma_3. \quad (24.497)$$

Notice that among σ_μ ($\mu = 0, 1, 2, 3$), σ_0 is not an element of the Lie algebra $\mathfrak{sl}(2, \mathbb{C})$ since the trace of σ_0 is not zero.

Meanwhile, putting

$$H = \begin{pmatrix} x^0 - x^3 & x^1 + ix^2 \\ x^1 - ix^2 & x^0 + x^3 \end{pmatrix} \quad (x^0, x^1, x^2, x^3 : \text{real}), \quad (24.498)$$

we have

$$H = x^0 \sigma_0 + x^1 \sigma_1 + x^2 \sigma_2 + x^3 \sigma_3 = x^\mu \sigma_\mu. \quad (24.499)$$

That is, (24.498) gives the coordinates x^μ ($\mu = 0, 1, 2, 3$) in reference to the basis vectors σ_μ . Now, we consider the following operation:

$$H \mapsto qHq^\dagger \quad [q \in SL(2, \mathbb{C})], \quad (24.500)$$

with

$$q = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \det q = ad - bc = 1. \quad (24.501)$$

We have $(qHq^\dagger)^\dagger = qH^\dagger q^\dagger = qHq^\dagger$. That is, $qHq^\dagger \in \mathfrak{H}^4$. Defining this mapping as $\varphi[q]$, we write

$$\varphi[q] : \mathfrak{H}^4 \rightarrow \mathfrak{H}^4, \varphi[q](H) \equiv qHq^\dagger. \quad (24.502)$$

The mapping $\varphi[q]$ is a linear transformation within $\mathfrak{H}^4$. It is because we have

$$\varphi[q](H_1 + H_2) = \varphi[q](H_1) + \varphi[q](H_2), \varphi[q](\alpha H) = \alpha \varphi[q](H) \quad (\alpha \in \mathbb{R}). \quad (24.503)$$

Also, we have

$$\begin{aligned} \varphi[qp](H) &= qpH(qp)^\dagger = qpHp^\dagger q^\dagger = q(pHp^\dagger)q^\dagger = \varphi[q][\varphi[p](H)] \\ &= \varphi[q]\varphi[p](H). \end{aligned} \quad (24.504)$$

Thus, we find that $q \mapsto \varphi[q]$ is a real representation of $SL(2, \mathbb{C})$ over $\mathfrak{H}^4$.

Using the notation of Sect. 11.2, we have

$$\begin{aligned} H' &= \varphi[q](H) = (\sigma_0 \ \sigma_1 \ \sigma_2 \ \sigma_3) \varphi[q] \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix} \\ &= (\sigma_0 \varphi[q] \ \sigma_1 \varphi[q] \ \sigma_2 \varphi[q] \ \sigma_3 \varphi[q]) \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}. \end{aligned} \quad (24.505)$$

Then, with e.g., $\sigma_0 \varphi[q]$ we obtain

$$\begin{aligned}
\sigma_0 \varphi[q] &= \varphi[q](\sigma_0) = q \sigma_0 q^\dagger = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix} \\
&= \begin{pmatrix} |a|^2 + |b|^2 & ac^* + bd^* \\ a^*c + b^*d & |c|^2 + |d|^2 \end{pmatrix}.
\end{aligned} \tag{24.506}$$

Using the relation (24.496), after the transformation $\varphi[q](\sigma_0)$ we get

$$\begin{aligned}
H' &= \frac{1}{2} \left(|a|^2 + |b|^2 + |c|^2 + |d|^2 \right) \sigma_0 + \frac{1}{2} \left(|c|^2 + |d|^2 - |a|^2 - |b|^2 \right) \sigma_3 \\
&\quad + \Re(ac^* + bd^*) \sigma_1 + \Im(ac^* + bd^*) \sigma_2.
\end{aligned} \tag{24.507}$$

The coefficients of each σ_i ($i = 0, 1, 2, 3$) act as their coordinates. Meanwhile, from (24.498) we have

$$\det H = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2. \tag{24.508}$$

Equation (24.7) is identical with the bilinear form introduced into the Minkowski space $\mathbb{M}^4$. We know from (24.500) and (24.501) that the transformation $\varphi[q]$ keeps the determinant unchanged. That is, the bilinear form (24.508) is invariant under this transformation. This implies that $\varphi[q]$ is associated with the Lorentz transformation.

To see whether this is really the case, we continue the calculations pertinent to (24.507). In a similar manner, we obtain related results H' after the transformations $\varphi[q](\sigma_1)$, $\varphi[q](\sigma_2)$, and $\varphi[q](\sigma_3)$. The results are shown collectively as [6]

$$\begin{aligned}
&(\sigma_0 \ \sigma_1 \ \sigma_2 \ \sigma_3) \varphi[q] \\
&= (\sigma_0 \ \sigma_1 \ \sigma_2 \ \sigma_3) \times \\
&\begin{pmatrix} \frac{1}{2} \left(|a|^2 + |b|^2 + |c|^2 + |d|^2 \right) & \Re(ab^* + cd^*) & -\Im(ab^* + cd^*) & \frac{1}{2} \left(|b|^2 + |d|^2 - |a|^2 - |c|^2 \right) \\ \Re(ac^* + bd^*) & \Re(ad^* + bc^*) & -\Im(ad^* - bc^*) & \Re(bd^* - ac^*) \\ \Im(ac^* + bd^*) & \Im(ad^* + bc^*) & \Re(ad^* - bc^*) & \Im(bd^* - ac^*) \\ \frac{1}{2} \left(|c|^2 + |d|^2 - |a|^2 - |b|^2 \right) & \Re(cd^* - ab^*) & \Im(ab^* - cd^*) & \frac{1}{2} \left(|a|^2 + |d|^2 - |b|^2 - |c|^2 \right) \end{pmatrix}.
\end{aligned} \tag{24.509}$$

Equation (24.509) clearly gives a matrix representation of $SO_0(3, 1)$. As already noticed, the present method based on (24.500) is in parallel to the derivation of (20.302) containing Euler angles performed by the use of the adjoint representation. In either case, we have represented a vector in the real vector space (either three-dimensional or four-dimensional) and obtained the matrices (20.302) or (24.509) as a set of the expansion coefficients with respect to the basis vectors.

Let us further inspect the properties of (24.509) in the following example.

Example 24.13 We may freely choose the parameters a , b , c , and d under the restriction of (24.501). Let us choose $b = c = 0$ and $a = e^{i\theta/2}$ and $d = e^{-i\theta/2}$ in (24.501). That is,

$$q = \begin{pmatrix} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{pmatrix}. \quad (24.510)$$

Then, we have $ad^* = e^{i\theta}$ to get

$$\varphi[q] = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (24.511)$$

This is just a rotation by θ around the z -axis. Next, we choose $a = d = \cosh(\omega/2)$ and $b = c = -\sinh(\omega/2)$. Namely,

$$q = \begin{pmatrix} \cosh(\omega/2) & -\sinh(\omega/2) \\ -\sinh(\omega/2) & \cosh(\omega/2) \end{pmatrix}. \quad (24.512)$$

Then, we obtain

$$\varphi[q] = \begin{pmatrix} \cosh \omega & -\sinh \omega & 0 & 0 \\ -\sinh \omega & \cosh \omega & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (24.513)$$

Moreover, if we choose $b = c = 0$ and $a = \cosh \frac{\omega}{2} - \sinh \frac{\omega}{2}$ and $d = \cosh \frac{\omega}{2} + \sinh \frac{\omega}{2}$ in (24.501), we have

$$q = \begin{pmatrix} \cosh \frac{\omega}{2} - \sinh \frac{\omega}{2} & 0 \\ 0 & \cosh \frac{\omega}{2} + \sinh \frac{\omega}{2} \end{pmatrix}. \quad (24.514)$$

Then, we get

$$\varphi[q] = \begin{pmatrix} \cosh \omega & 0 & 0 & \sinh \omega \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh \omega & 0 & 0 & \cosh \omega \end{pmatrix}. \quad (24.515)$$

The latter two illustrations represent Lorentz boosts. We find that the $(2, 2)$ matrix of (24.510) belongs to $SU(2)$ and that the $(2, 2)$ matrices of (24.512) and (24.514) are positive definite symmetric matrices. In terms of Theorem 24.7 and Corollary 24.3,

these matrices have *already* been decomposed. If we expressly (or purposely) write in a decomposed form, we may do so with the above examples such that

$$q = \begin{pmatrix} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad (24.516)$$

$$q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \cosh(\omega/2) & -\sinh(\omega/2) \\ -\sinh(\omega/2) & \cosh(\omega/2) \end{pmatrix}. \quad (24.517)$$

The identity matrix of (24.516) behaves as a positive definite Hermitian matrix P in relation to Theorem 24.7. The identity matrix of (24.517) behaves as an orthogonal matrix O in relation to Corollary 24.3.

The parameter θ in (24.510) is bounded. A group characterized by a bounded parameter is called a compact group, to which a unitary representation can be constructed (Theorem 18.1). In contrast, the parameter ω of (24.512) and (24.514) ranges from $-\infty$ to $+\infty$. A group characterized by an unbounded parameter is called a non-compact group and the unitary representation is in general impossible. Equations (24.11) and (24.13) are typical examples.

We investigate the constitution of the Lorentz group a little further. In Sect. 20.4, we have shown that

$$SU(2)/\mathcal{F} \cong SO(3), \mathcal{F} = \{e, -e\}, \quad (24.518)$$

where $\mathcal{F}$ is the kernel and e is the identity element. To show this, it was important to show the existence of the kernel $\mathcal{F}$. In the present case, we wish to show the characteristic of the kernel $\mathcal{N}$ with respect to the homomorphism mapping φ , namely

$$\mathcal{N} = \{q \in SL(2, \mathbb{C}); \varphi[q] = \varphi[\sigma_0]\}. \quad (24.519)$$

First, we show the following corollary.

Corollary 24.4 [8] Let ρ be a continuous representation of a connected linear Lie group G and let e_G be the identity element of G . Let H be another Lie group and let H_0 be a connected component of the identity element e_H of H . We express the image of G by ρ as $\rho(G)$. If $\rho(G)$ is contained in a Lie group H , then $\rho(G)$ is contained in a connected component H_0 .

Proof Since ρ transforms the identity element e_G to e_H , $e_H \in \rho(G)$. The identity element e_H is contained in H . Because $\rho(G)$ is an image of the connected set G , in virtue of Theorem 20.8 we find that $\rho(G)$ is also a connected subset of H . We have $e_H \in \rho(G)$, and so this implies that all the elements of $\rho(G)$ are connected to e_H . This means that $\rho(G) \subset H_0$. This completes the proof.

The representation $\varphi[q]$ keeps the bilinear form given by $\det H$ of (24.508) invariant and $\varphi[q]$, to which $q \in SL(2, \mathbb{C})$, is the representation of $SO_0(3, 1)$. Meanwhile, $SL(2, \mathbb{C})$ is (simply) connected and $SO_0(3, 1)$ is the connected Lie

group that contains the identity element E_0 . Then, according to Corollary 24.4, we have

$$\varphi[SL(2, \mathbb{C})] \subset SO_0(3, 1). \quad (24.520)$$

As already seen in (24.509), $\varphi[q]$ [$q \in SL(2, \mathbb{C})$] has given the matrix representation of the Lorentz group $SO_0(3, 1)$. This seems to imply at first glance that

$$SO_0(3, 1) \subset \varphi[SL(2, \mathbb{C})]. \quad (24.521)$$

Strictly speaking to assert (24.521), however, we need the knowledge of the differential representation and Theorem 25.2 based on it. Equation (24.21) will be proven in Sect. 25.1 accordingly. Taking (24.521) in advance and combining (24.520) with (24.521), we get

$$\varphi[SL(2, \mathbb{C})] = SO_0(3, 1). \quad (24.522)$$

Now, we have a question of what kind of property φ has (see Sect. 16.4). In terms of the mapping, if we have

$$f(x_1) = f(x_2) \quad \Rightarrow \quad x_1 = x_2, \quad (24.523)$$

the mapping f is injective (Sect. 11.2). If (24.523) is not true, f is not injective, and so f is not isomorphic either. Therefore, our question can be translated into whether or not the following proposition is true of φ :

$$\varphi[q] = \varphi[\sigma_0] \quad \Rightarrow \quad q = \sigma_0. \quad (24.524)$$

Meanwhile, $\varphi[\sigma_0]$ gives the identity mapping (Sect. 16.4). In fact, we have

$$\varphi[\sigma_0](H) = \sigma_0 H \sigma_0^\dagger = \sigma_0 H \sigma_0 = H. \quad (24.525)$$

Consequently, our question boils down to the question of whether or not we can find q ($\neq \sigma_0$) for which $\varphi[q]$ gives the identity mapping. If we find such q , $\varphi[q]$ is not isomorphic, but has a non-trivial kernel. Let us get back to our current issue. We have

$$\varphi[-\sigma_0](H) = (-\sigma_0)H(-\sigma_0^\dagger) = \sigma_0 H \sigma_0 = H. \quad (24.526)$$

Then, $\varphi[-\sigma_0]$ gives the identity mapping as well; that is, $\varphi[-\sigma_0] = \varphi[\sigma_0]$. Equation (24.24) does not hold with φ accordingly. Therefore, according to Theorem 16.2 φ is not isomorphic. To fully characterize the representation φ , we must seek element (s) other than $-\sigma_0$ as a candidate for the kernel.

Now, suppose that $q \in SL(2, \mathbb{C})$ is an element of the kernel $\mathcal{N}$, i.e., $q \in \mathcal{N}$. Then, for an arbitrary element $H (\in \mathfrak{V}^4)$ defined in (24.493) we must have

$$\varphi[q](H) = qHq^\dagger = H.$$

In a special case where we choose $\sigma_0 (\in H)$, the following relation must hold:

$$\varphi[q](\sigma_0) = q\sigma_0q^\dagger = qq^\dagger = \sigma_0, \quad (24.527)$$

where the last equality comes from the supposition that $\varphi[q]$ is contained in the kernel $\mathcal{N}$. This means that q is unitary. That is, $q^\dagger = q^{-1}$. Then, for any $H (\in \mathfrak{V}^4)$ we have

$$\varphi[q](H) = qHq^\dagger = qHq^{-1} = H \quad \text{or} \quad qH = Hq. \quad (24.528)$$

Meanwhile, any $(2, 2)$ complex matrices z can be expressed as

$$z = x + iy, x, y \in \mathfrak{V}^4, \quad (24.529)$$

which will be proven in Theorem 24.14 given just below. Then, from both (24.528) and (24.529), q must be commutative with any $(2, 2)$ complex matrices. Let $\mathfrak{C}$ be a set of all the $(2, 2)$ complex matrices described as

$$\mathfrak{C} = \{C = (c_{ij}); i, j = 1, 2, c_{ij} \in \mathbb{C}\}. \quad (24.530)$$

Then, since $\mathfrak{C} \supset SL(2, \mathbb{C})$, it follows that q is commutative with $\forall g \in SL(2, \mathbb{C})$. Consequently, according to Schur's Second Lemma q must be expressed as

$$q = \alpha E [\alpha \in \mathbb{C}, E : (2, 2) \text{ identity matrix}].$$

Since $\det q = 1$, we must have $\alpha^2 = 1$, i.e., $\alpha = \pm 1$. This implies that we have only the possibility that the elements of $\mathcal{N}$ are limited to q that satisfies

$$q = \sigma_0 \quad \text{or} \quad q = -\sigma_0. \quad (24.531)$$

Thus, we obtain

$$\mathcal{N} = \{\sigma_0, -\sigma_0\}. \quad (24.532)$$

Any other elements belonging to $\mathfrak{V}^4$ are excluded from the kernel $\mathcal{N}$. This implies that φ is a two-to-one homomorphic mapping. At the same time, from Theorem 16.4 (Homomorphism Theorem) we obtain

$$SO_0(3, 1) \cong SL(2, \mathbb{C})/\mathcal{N}. \quad (24.533)$$

Equation (24.32) is in parallel to the relationship between $SU(2)$ and $SO(3)$ discussed in Sect. 20.4.3; see (20.314). Thus, we reach a next important theorem.

Theorem 24.13 [8] Suppose that we have a representation $\varphi[q]$ [$q \in SL(2, \mathbb{C})$] on $\mathfrak{V}^4$ described by

$$\varphi[q](H) : \mathfrak{V}^4 \rightarrow \mathfrak{V}^4, \varphi[q](H) = qHq^\dagger. \quad (24.502)$$

Then, φ defined by $\varphi : SL(2, \mathbb{C}) \rightarrow SO_0(3, 1)$ is a two-to-one surjective homomorphic mapping from $SL(2, \mathbb{C})$ to $SO_0(3, 1)$. Namely, we have $SO_0(3, 1) \cong SL(2, \mathbb{C})/\mathcal{N}$, where $\mathcal{N} = \{\sigma_0, -\sigma_0\}$.

Thus, we find that the relationship between $SO_0(3, 1)$ and $SL(2, \mathbb{C})$ can be dealt with in parallel with that between $SO(3)$ and $SU(2)$. As the representation of $SU(2)$ has led to that of $SO(3)$, so the representation of $SL(2, \mathbb{C})$ boils down to that of $SO_0(3, 1)$. We will come back to this issue in Sect. 25.1.

Moreover, we have simple but important theorems on the complex matrices and, more specifically, on $\mathfrak{sl}(2, \mathbb{C})$.

Theorem 24.14 [8] Any $(2, 2)$ complex matrices z can be expressed as

$$z = x + iy \quad (x, y \in \mathfrak{V}^4). \quad (24.534)$$

Proof Let z be described as

$$z = \begin{pmatrix} a + ib & c + id \\ p + iq & r + is \end{pmatrix} \quad (a, b, c, d, p, q, r, s \in \mathbb{R}). \quad (24.535)$$

Then, z is written as

$$\begin{aligned} z = & \frac{1}{2} [(a + r)\sigma_0 + (-a + r)\sigma_3 + (c + p)\sigma_1 + (d - q)\sigma_2] \\ & + \frac{i}{2} [(b + s)\sigma_0 + (-b + s)\sigma_3 + (d + q)\sigma_1 + (-c + p)\sigma_2], \end{aligned} \quad (24.536)$$

where σ_i ($i = 0, 1, 2, 3$) $\in \mathfrak{V}^4$ is given by (24.495). The first and second terms of (24.536) are evidently expressed as x and iy of (24.534), respectively. This completes the proof.

Note in Theorem 24.14 all the $(2, 2)$ complex matrices z form the Lie algebra $\mathfrak{gl}(2, \mathbb{C})$. Therefore, from (24.535) $\mathfrak{gl}(2, \mathbb{C})$ has eight degrees of freedom, namely, $\mathfrak{gl}(2, \mathbb{C})$ is an eight-dimensional vector space.

Theorem 24.15 [8] Let Z be any arbitrary element of $\mathfrak{sl}(2, \mathbb{C})$. Then, z is uniquely described as

$$Z = X + iY \quad [X, Y \in \mathfrak{su}(2)]. \quad (24.537)$$

Proof Let X and Y be described as

$$X = (Z - Z^\dagger)/2, Y = (Z + Z^\dagger)/2i. \quad (24.538)$$

Then, (24.537) certainly holds. Both X and Y are anti-Hermitian. As Z is traceless, so are X and Y . Therefore, $X, Y \in \mathfrak{su}(2)$. Next, suppose that $X' (\neq X)$ and $Y' (\neq Y)$ satisfy (24.538). That is,

$$Z = X' + iY' \quad [X', Y' \in \mathfrak{su}(2)]. \quad (24.539)$$

Subtracting (24.537) from (24.539), we obtain

$$(X' - X) + i(Y' - Y) = 0. \quad (24.540)$$

Taking the adjoint of (24.540), we get

$$-(X' - X) + i(Y' - Y) = 0. \quad (24.541)$$

Adding (24.540) and (24.541), we get

$$2i(Y' - Y) = 0 \quad \text{or} \quad Y' = Y. \quad (24.542)$$

Subtracting (24.541) from (24.540) once again, we have

$$2(X' - X) = 0 \quad \text{or} \quad X' = X. \quad (24.543)$$

Equations (24.40) and (24.41) indicate that the representation (24.537) is unique. These complete the proof.

The fact that an element of $\mathfrak{sl}(2, \mathbb{C})$ is traceless is shown as follows: From (15.41), we have

$$\det(\exp tA) = \exp(\text{Tr}tA); t : \text{real}. \quad (24.544)$$

Suppose $A \in \mathfrak{sl}(2, \mathbb{C})$. Then, $\exp tA \in SL(2, \mathbb{C})$. Therefore, we have

$$\exp(\text{Tr}tA) = \det(\exp tA) = 1, \quad (24.545)$$

where with the first equality we used (15.41) with A replaced with tA ; the last equality came from Definition 24.5. This implies that $\text{Tr}tA = t\text{Tr}A = 0$. That is, A is traceless.

Theorem 24.15 gives an outstanding feature to $\mathfrak{sl}(2, \mathbb{C})$. That is, Theorem 24.15 tells us that the elements belonging to $\mathfrak{sl}(2, \mathbb{C})$ can be decomposed into a matrix belonging to $\mathfrak{su}(2)$ and that belonging to the set $i \cdot \mathfrak{su}(2)$. In turn, the elements of $\mathfrak{sl}(2, \mathbb{C})$ can be synthesized from those of $\mathfrak{su}(2)$. With $\mathfrak{su}(2)$, an arbitrary vector is described as a linear combination on a real field $\mathbb{R}$. In fact, let X be a vector on $\mathfrak{su}(2)$. Then, we have

$$X = \begin{pmatrix} ic & a - ib \\ -a - ib & -ic \end{pmatrix} \quad (a, b, c \in \mathbb{R}). \quad (24.546)$$

Defining the basis vectors of $\mathfrak{su}(2)$ by the use of the matrices defined in (20.270) as

$$\tilde{\zeta}_1 = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}, \tilde{\zeta}_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \tilde{\zeta}_3 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad (24.547)$$

we describe X as

$$X = c\tilde{\zeta}_3 + a\tilde{\zeta}_2 + b\tilde{\zeta}_1 \quad (a, b, c \in \mathbb{R}). \quad (24.548)$$

As for $\mathfrak{sl}(2, \mathbb{C})$, on the other hand, we describe $Z \in \mathfrak{sl}(2, \mathbb{C})$ as

$$Z = \begin{pmatrix} a + ib & c + id \\ p + iq & -a - ib \end{pmatrix} \quad (a, b, c, d, p, q \in \mathbb{R}). \quad (24.549)$$

Using (24.547) for the basis vectors, we write Z as

$$\begin{aligned} Z &= (b - ia)\tilde{\zeta}_3 + \left[\frac{1}{2}(c - p) + \frac{i}{2}(d - q) \right] \tilde{\zeta}_2 \\ &\quad + \left[-\frac{1}{2}(d + q) + \frac{i}{2}(c + p) \right] \tilde{\zeta}_1. \end{aligned} \quad (24.550)$$

Thus, we find that $\mathfrak{sl}(2, \mathbb{C})$ is a complex vector space. If in general a complex Lie algebra $\mathfrak{g}$ has a real Lie subalgebra $\mathfrak{h}$ (i.e., $\mathfrak{h} \subset \mathfrak{g}$) and an arbitrary element $Z \in \mathfrak{g}$ is uniquely expressed as

$$Z = X + iY \quad (X, Y \in \mathfrak{h}), \quad (24.551)$$

then, $\mathfrak{h}$ is said to be a real form of $\mathfrak{g}$. By the same token, $\mathfrak{g}$ is said to be complexification of $\mathfrak{h}$ [8]. In light of (24.551), the relationship between

$\mathfrak{su}(2)$ and $\mathfrak{sl}(2, \mathbb{C})$ is evident. Note that in the case of $\mathfrak{sl}(2, \mathbb{C})$, X is anti-Hermitian and iY is Hermitian. Defining the operators $\tilde{\zeta}_k^*$ such that

$$\tilde{\zeta}_k^* \equiv i\tilde{\zeta}_k \quad (k=1, 2, 3), \quad (24.552)$$

we can construct the useful commutation relations among them.

On the basis of (20.271), we obtain

$$[\tilde{\zeta}_k, \tilde{\zeta}_l] = \sum_{m=1}^3 \epsilon_{klm} \tilde{\zeta}_m, \quad (24.553)$$

$$[\tilde{\zeta}_k^*, \tilde{\zeta}_l^*] = -\sum_{m=1}^3 \epsilon_{klm} \tilde{\zeta}_m^*, \quad (24.554)$$

$$[\tilde{\zeta}_k, \tilde{\zeta}_l^*] = \sum_{m=1}^3 \epsilon_{klm} \tilde{\zeta}_m^*. \quad (24.555)$$

The complexification is an important concept to extend the scope of applications of Lie algebra. Examples of the complexification will be provided in the next chapter.

In the above discussion we mentioned the subalgebra. Its definition is formally given as follows.

Definition 24.6 Let $\mathfrak{g}$ be a Lie algebra. Suppose that $\mathfrak{h}$ is a subspace of $\mathfrak{g}$ and that with $\forall X, Y \in \mathfrak{h}$ we have $[X, Y] \in \mathfrak{h}$. Then, $\mathfrak{h}$ is called a subalgebra of $\mathfrak{g}$.

References

1. Satake I-O (1975) Linear algebra (Pure and applied mathematics). Marcel Dekker, New York
2. Satake I (1974) Linear algebra. Shokabo, Tokyo. (in Japanese)
3. Birkhoff G, Mac Lane S (2010) A survey of modern algebra, 4th edn. CRC Press, Boca Raton
4. Byron FW Jr, Fuller RW (1992) Mathematics of classical and quantum physics. Dover, New York
5. Rosén A (2019) Geometric multivector analysis. Birkhäuser, Cham
6. Hirai T (2001) Linear algebra and representation of groups II. Asakura Publishing, Tokyo. (in Japanese)
7. Kugo C (1989) Quantum theory of gauge field I. Baifukan, Tokyo. (in Japanese)
8. Yamanouchi T, Sugiura M (1960) Introduction to continuous groups. Baifukan, Tokyo. (in Japanese)
9. Hotta S (2023) A method for direct solution of the Dirac equation. Bull Kyoto Inst Technol 15: 27–46
10. Itzykson C, Zuber J-B (2005) Quantum field theory. Dover, New York
11. Sakamoto M (2014) Quantum field theory. Shokabo, Tokyo. (in Japanese)
12. Goldstein H, Poole C, Safko J (2002) Classical mechanics, 3rd edn. Addison Wesley, San Francisco
13. Schweber SS (1989) An introduction to relativistic quantum field theory. Dover, New York
14. Mirsky L (1990) An Introduction to linear algebra. Dover, New York
15. Hassani S (2006) Mathematical physics. Springer, New York

Chapter 25

Advanced Topics of Lie Algebra



Abstract In Chap. 20, we dealt with the Lie groups and Lie algebras within the framework of the continuous groups. With the Lie groups the operation is *multiplicative* (see Part IV), whereas the operation of the Lie algebras is *additive* since the Lie algebra forms a vector space (Part III). It is of importance and interest, however, that the representation is multiplicative in both the Lie groups and Lie algebras. So far, we have examined the properties of the representations of the groups (including the Lie group), but little was mentioned about the Lie algebra. In this chapter, we present advanced topics of the Lie algebra with emphasis upon its representation. Among those topics, the differential representation of Lie algebra is particularly useful. We can form a clear view of the representation theory by contrasting the representation of the Lie algebra with that of the Lie group. Another topic is the complexification of the Lie algebra. We illustrate how to extend the scope of application of Lie algebra through the complexification with actual examples. In the last part, we revisit the topic of the coupling of angular momenta and compare the method with the previous approach based on the theory of continuous groups.

25.1 Differential Representation of Lie Algebra [1]

25.1.1 Overview

As already discussed, the Lie algebra is based on the notion of infinitesimal transformation around the identity element of the Lie groups (Sect. 20.4). The representation of Lie groups and that of Lie algebras should have close relationship accordingly. As the Lie algebra is related to the infinitesimal transformation, so is its representation. We summarize the definition of the representation of Lie algebra. We have a differential representation as another representation specific to Lie algebra. The differential representation is defined in relation to the representation of the Lie group.

Definition 25.1 Let $\sigma(X)$ be a mapping that relates $X \in \mathfrak{g}$ (i.e., X is an element of a Lie algebra $\mathfrak{g}$) to a vector space $\mathfrak{V}$. Suppose that $\sigma(X)$ is a linear transformation and that $\sigma(X)$ satisfies the following relationships (i), (ii), and (iii):

$$(i) \quad \sigma(aX) = a\sigma(X) \quad (a \in \mathbb{R}), \quad (25.1)$$

$$(ii) \quad \sigma(X + Y) = \sigma(X) + \sigma(Y), \quad (25.2)$$

$$(iii) \quad \sigma(XY) = \sigma(X)\sigma(Y). \quad (25.3)$$

Then, the mapping $\sigma(X)$ is called a representation of the Lie algebra $\mathfrak{g}$ on $\mathfrak{V}$.

We schematically write this mapping as

$$\sigma : \mathfrak{g} \rightarrow \mathfrak{V}$$

or

$$\mathfrak{g} \ni X \mapsto \sigma(X) \in \mathfrak{V}. \quad (25.4)$$

The relationship (iii) implies that the $\sigma(X)$ is a homomorphic mapping on $\mathfrak{g}$. In the above definition, if $\mathfrak{V}$ of (25.4) is identical to $\mathfrak{g}$, $\sigma(X)$ is an endomorphism (Chap. 11). Next, we think of the mapping $\sigma(X)$, where X represents a commutator. We have

$$\begin{aligned} \sigma([X, Y]) &= \sigma(XY - YX) = \sigma(XY) - \sigma(YX) = \sigma(X)\sigma(Y) - \sigma(Y)\sigma(X) \\ &= [\sigma(X), \sigma(Y)], \end{aligned}$$

where with the second equality we used (25.2); with the third equality we used (25.3). That is,

$$(iv) \quad \sigma([X, Y]) = [\sigma(X), \sigma(Y)]. \quad (25.5)$$

The reducibility and irreducibility of the representation can similarly be defined as before (see, e.g., Sect. 20.2). The expression (25.5) plays a special role with the representation because it holds the commutation relation unchanged in $\mathfrak{V}$ as well as in $\mathfrak{g}$. More importantly, we have a relationship described by

$$\rho(\exp tX) = \exp[t\sigma(X)] \quad (X \in \mathfrak{g}, t \in \mathbb{R}), \quad (25.6)$$

where ρ is a representation of the Lie group $\mathcal{G}$ corresponding to $\mathfrak{g}$. Differentiating both sides of (25.6) and taking the limit of $t \rightarrow 0$, we get

$$\lim_{t \rightarrow 0} \frac{1}{t} [\rho(\exp tX) - \rho(\exp 0)] = \sigma(X) \lim_{t \rightarrow 0} [\exp t\sigma(X)]. \quad (25.7)$$

From (18.4) we have

$$\rho(\exp 0) = \rho(e) = E, \quad (25.8)$$

where e and E are an identity element of $\mathcal{g}$ and $\rho(\mathcal{g})$, respectively. Then, (25.7) can be rewritten as

$$\sigma(X) = \lim_{t \rightarrow 0} \frac{1}{t} [\rho(\exp tX) - E]. \quad (25.9)$$

Thus, the relationship between the Lie group $\mathcal{g}$ and its representation space $\mathcal{L}_S$ [see (18.36)] can be translated into that between the Lie algebra $\mathfrak{g}$ and the associated vector space $\mathfrak{V}$. In other words, as a whole collection consisting of X forms the Lie algebra $\mathfrak{g}$ (i.e., a vector space $\mathfrak{g}$), so a whole collection consisting of $\sigma(X)$ forms the Lie algebra on the vector space $\mathfrak{V}$. We summarize the above statements as a following definition.

Definition 25.2 Suppose that we have a relationship described by

$$\rho(\exp tX) = \exp[t\sigma(X)] \quad (X \in \mathfrak{g}, t \in \mathbb{R}). \quad (25.6)$$

The representation $\sigma(X)$ of $\mathfrak{g}$ on $\mathfrak{V}$ is called a differential representation for the representation ρ of the Lie group $\mathcal{g}$. In this instance, σ is expressed as

$$\sigma \equiv d\rho. \quad (25.10)$$

To specify the representation space that is associated with the Lie group and Lie algebra, we normally write the combination of the representation σ or ρ and its associated representation space $\mathfrak{V}$ as [1]

$$(\sigma, \mathfrak{V}) \quad \text{or} \quad (\rho, \mathfrak{V}). \quad (25.11)$$

Notice that the same representation space $\mathfrak{V}$ has been used in (25.11) for both the representations σ and ρ relevant to the Lie algebra and Lie group, respectively.

25.1.2 Adjoint Representation of Lie Algebra [1]

One of the most important representations of the Lie algebras is an adjoint representation. In Sect. 20.4.3, we defined the adjoint representation for the Lie group. In the present section, we deal with the adjoint representation for the Lie algebra. It is

defined as the differential representation of the adjoint representation $\text{Ad}[g]$ of the Lie group $\mathcal{G} \ni g$. In this case, the differential representation $\sigma(X)$ is denoted by

$$\sigma(X) \equiv \text{ad}[X], \quad (25.12)$$

where we used an angled bracket with RHS in accordance with Sect. 20.4.3. The operator $\text{ad}[X]$ operates on an element Y of the Lie algebra such that

$$\text{ad}[X](Y). \quad (25.13)$$

With (25.13) we have a simple but important theorem.

Theorem 25.1 [1] With $X, Y \in \mathfrak{g}$ we have

$$\text{ad}[X](Y) = [X, Y]. \quad (25.14)$$

Proof Let $g_X(t)$ be expressed as

$$g_X(t) = \text{Ad}[\exp tX]. \quad (25.15)$$

That is, g_X is an adjoint representation of the Lie group $\mathcal{G}$ on the Lie algebra $\mathfrak{g}$. Notice that we are using (25.15) for $\rho(\exp tX)$ of (25.6). Then, according to (25.6) and (25.9), we have

$$\begin{aligned} \text{ad}[X](Y) &= g'_X(0)Y = \frac{d}{dt} \left[(\exp tX)Y(\exp tX)^{-1} \right]_{t=0} \\ &= \frac{d}{dt} [(\exp tX)Y(\exp -tX)]_{t=0} = XY - YX = [X, Y], \end{aligned} \quad (25.16)$$

where with the third equality we used (15.28) and with the second to the last equality we used (15.4). This completes the proof.

Equation (25.14) can be seen as a definition of the adjoint representation of the Lie algebra. We can directly show that the operator $\text{ad}[X]$ is indeed a representation of $\mathfrak{g}$ as follows: That is, with any arbitrary element $Z \in \mathfrak{g}$ we have

$$\begin{aligned} \text{ad}[X, Y](Z) &= [[X, Y], Z] = -[Z, [X, Y]] = [X, [Y, Z]] - [Y, [X, Z]] \\ &= [X, \text{ad}[Y](Z)] - [Y, \text{ad}[X](Z)] = \text{ad}[X](\text{ad}[Y](Z)) - \text{ad}[Y](\text{ad}[X](Z)) \\ &= \text{ad}[X]\text{ad}[Y](Z) - \text{ad}[Y]\text{ad}[X](Z) = [\text{ad}[X], \text{ad}[Y]](Z), \end{aligned} \quad (25.17)$$

where with the third equality we used Jacobi's identity (20.267). Comparing the first and last sides of (25.17), we get

$$\text{ad}[X, Y](Z) = [\text{ad}[X], \text{ad}[Y]](Z). \quad (25.18)$$

Equation (25.18) holds with any arbitrary element $Z \in \mathfrak{g}$. This implies that

$$\text{ad}[X, Y] = [\text{ad}[X], \text{ad}[Y]]. \quad (25.19)$$

From (25.5), (25.19) satisfies the condition for a representation of the Lie algebra $\mathfrak{g}$.

We know that if $X, Y \in \mathfrak{g}$, $[X, Y] \in \mathfrak{g}$ (see Sect. 20.4). Hence, we have a following relationship described by

$$\mathfrak{g} \ni Y \mapsto \text{ad}[X](Y) = Y' \in \mathfrak{g} \quad (X \in \mathfrak{g}) \quad (25.20)$$

or

$$\text{ad}[X] : \mathfrak{g} \rightarrow \mathfrak{g}. \quad (25.21)$$

That is, $\text{ad}[X]$ is an endomorphism within $\mathfrak{g}$.

Example 25.1 In Example 20.4 we had (20.270) as $\mathfrak{su}(2)$. Among them we use

$\zeta_3 = \frac{1}{2} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \in \mathfrak{su}(2)$ for X in (25.20). We have

$$\text{ad}[X] : \mathfrak{g} \rightarrow \mathfrak{g} \quad [X \in \mathfrak{su}(2), \mathfrak{g} = \mathfrak{su}(2)]. \quad (25.22)$$

Also, we have

$$\text{ad}[\zeta_3](Y) = \zeta_3 Y - Y \zeta_3 \quad [Y \in \mathfrak{su}(2)]. \quad (25.23)$$

Substituting ζ_1 , ζ_2 , and ζ_3 for Y , respectively, we obtain

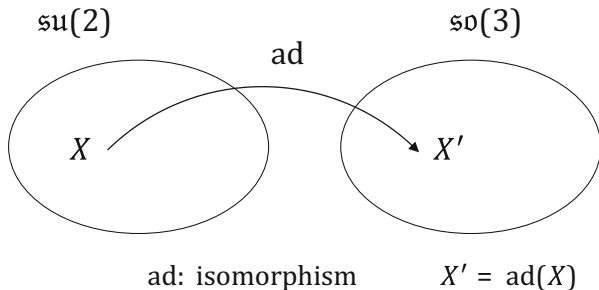
$$\text{ad}[\zeta_3](\zeta_1) = \zeta_2, \quad \text{ad}[\zeta_3](\zeta_2) = -\zeta_1, \quad \text{and} \quad \text{ad}\zeta_3 = 0. \quad (25.24)$$

That is, we get

$$(\zeta_1 \ \zeta_2 \ \zeta_3) \text{ad}[\zeta_3] = (\zeta_1 \ \zeta_2 \ \zeta_3) \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = (\zeta_1 \ \zeta_2 \ \zeta_3) A_3. \quad (25.25)$$

Similarly, we obtain

Fig. 25.1 Isomorphism ad between $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$



$$(\zeta_1 \ \zeta_2 \ \zeta_3) \text{ad}[\zeta_1] = (\zeta_1 \ \zeta_2 \ \zeta_3) \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} = (\zeta_1 \ \zeta_2 \ \zeta_3) A_1, \quad (25.26)$$

$$(\zeta_1 \ \zeta_2 \ \zeta_3) \text{ad}[\zeta_2] = (\zeta_1 \ \zeta_2 \ \zeta_3) \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} = (\zeta_1 \ \zeta_2 \ \zeta_3) A_2. \quad (25.27)$$

The $(3, 3)$ matrices of RHS of (25.25), (25.26), and (25.27) are identical to A_3 , A_1 , and A_2 of (20.274), respectively. Equations (25.25)–(25.27) show that the differential mapping ad transforms the basis vectors ζ_1 , ζ_2 , and ζ_3 of $\mathfrak{su}(2)$ into basis vectors A_1 , A_2 , and A_3 of $\mathfrak{so}(3)$ [= $\mathfrak{o}(3)$], respectively (through the medium of ζ_1 , ζ_2 , and ζ_3). From the discussion of Sect. 11.4, the transformation ad is bijective. Namely, $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ are isomorphic to each other.

In Fig. 25.1 we depict the isomorphism ad between $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ as

$$\text{ad} : \mathfrak{su}(2) \rightarrow \mathfrak{so}(3). \quad (25.28)$$

Notice that the mapping between two Lie algebras $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ represented in (25.28) is in parallel with the mapping between two Lie groups $SU(2)$ and $SO(3)$ that is described by

$$\text{Ad} : SU(2) \rightarrow SO(3). \quad (20.303)$$

Thus, in accordance with (25.11), we have the following correspondence between the differential representation $\text{ad}[X]$ [$X \in \mathfrak{su}(2)$] on a three-dimensional vector space $\mathfrak{V}^3$ (as a representation space) and the representation $\text{Ad}[g]$ [$g \in SU(2)$] on $\mathfrak{V}^3$ such that

$$(\text{ad}, \mathfrak{V}^3) \Leftrightarrow (\text{Ad}, \mathfrak{V}^3). \quad (25.29)$$

In the present example, we have chosen $\mathfrak{su}(2)$ for $\mathfrak{V}^3$; see Sect. 20.4.3.

In (25.6), replacing ρ , t , X , and σ with Ad , α , ζ_3 , and ad , respectively, we get

$$\exp\{\alpha \cdot \text{ad}[\zeta_3]\} = \text{Ad}[\exp(\alpha\zeta_3)] \quad (\zeta_3 \in \mathfrak{g}, \alpha \in \mathbb{R}). \quad (25.30)$$

In fact, comparing (20.28) with (20.295), we find that both sides of the above equation certainly give a same matrix expressed as

$$\begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Notice that [LHS of (25.30)] = (20.28) and [RHS of (25.30)] = (20.295).

As explained above, the present example clearly shows a close relationship between the representations of Lie group and Lie algebra.

We can construct the adjoint representation of the Lie algebras using a representation other than (25.14). In that case it is again helpful to use Jacobi's identity in combination with (20.269) that gives structure constants which define the structure of the Lie algebra (Sect. 20.4.2). We rewrite (20.269) in such a way that the expression given below conforms to the Einstein summation rule

$$[X_i, X_j] = f_{ij}^l X_l. \quad (25.31)$$

Jacobi's identity is described by

$$[X_i, [X_j, X_k]] + [X_j, [X_k, X_i]] + [X_k, [X_i, X_j]] = 0. \quad (25.32)$$

The term $[X_i, [X_j, X_k]]$ can be rewritten by the use of the structure constants such that

$$[X_i, f_{jk}^l X_l] = f_{jk}^l [X_i, X_l] = f_{jk}^l f_{il}^m X_m.$$

Similarly rewriting the second and third terms of (25.32), we obtain

$$(f_{jk}^l f_{il}^m + f_{ki}^l f_{jl}^m + f_{ij}^l f_{kl}^m) X_m = 0.$$

Since X_m ($m = 1, \dots, d$; d is a dimension of the representation) is linearly independent, we must have for each m

$$f_{jk}^l f_{il}^m + f_{ki}^l f_{jl}^m + f_{ij}^l f_{kl}^m = 0. \quad (25.33)$$

Taking account of the fact that f_{ki}^l is antisymmetric with respect to the indices k and i , (25.33) is rewritten as

$$f_{jk}^l f_{il}^m - f_{ik}^l f_{jl}^m = f_{ij}^l f_{lk}^m. \quad (25.34)$$

If in (25.34) we define f_{jk}^l as

$$f_{jk}^l \equiv (\sigma_j)^l_k, \quad (25.35)$$

then the operator $(\sigma_j)^l_k$ is a representation of the Lie algebra. Note that in (25.35) $(\sigma_j)^l_k$ indicates (l, k) element of the matrix σ_j . By the use of (25.35), (25.34) can be rewritten as

$$(\sigma_i)^m_l (\sigma_j)^l_k - (\sigma_j)^m_l (\sigma_i)^l_k = f_{ij}^l (\sigma_l)^m_k. \quad (25.36)$$

That is, we have

$$[\sigma_i, \sigma_j] = f_{ij}^l \sigma_l. \quad (25.37)$$

Equation (25.37) is virtually the same form as (25.31) and (20.269). Hence, we find that (25.35) is the representation of the Lie algebra, whose structure is represented by (25.37) or (20.269). Using the notation (25.14), we get

$$\text{ad}[\sigma_i](\sigma_j) = [\sigma_i, \sigma_j] = f_{ij}^l \sigma_l. \quad (25.38)$$

25.1.3 Differential Representation of Lorentz Algebra [1]

As another important differential representation of the Lie algebra, we wish to introduce the differential representation of the Lorentz algebra. In Sect. 24.10 we examined the properties of the operation described by

$$H \mapsto qHq^\dagger \quad [q \in SL(2, \mathbb{C})], \quad (24.500)$$

with

$$q = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \det q = ad - bc = 1. \quad (24.501)$$

This constituted the representation $(\varphi, \mathfrak{V}^4)$ described by

$$\varphi[q](H) : \mathfrak{V}^4 \rightarrow \mathfrak{V}^4, \quad \varphi[q](H) = qHq^\dagger. \quad (24.502)$$

Now, we wish to construct another differential representation $(d\varphi, \mathfrak{V}^4)$. Following Theorem 25.1 and defining $f(t)$ as

$$f_X(t) \equiv \varphi(\exp tX) \quad [X \in \mathfrak{sl}(2, \mathbb{C})],$$

for $y \in \mathfrak{V}^4 = \{H = (h_{ij}); i, j = 1, 2, h_{ij} \in \mathbb{C}, H : \text{Hermitian}\}$ we have

$$\begin{aligned} d\varphi[X](y) &= f'_X(0)(y) = \frac{d}{dt} [(\exp tX)y(\exp tX)^\dagger]_{t=0} \\ &= \frac{d}{dt} [(\exp tX)y(\exp tX)^\dagger]_{t=0} \\ &= [X(\exp tX)y \exp(tX^\dagger) + (\exp tX)yX^\dagger \exp(tX^\dagger)]_{t=0} \\ &= Xy + yX^\dagger, \end{aligned} \quad (25.39)$$

where with the third equality we used (15.32).

Example 25.2 Following Example 25.1, we compute $d\varphi[X](y)$ with $X \in \mathfrak{sl}(2, \mathbb{C})$ and $y \in \mathfrak{V}^4$. The mapping is expressed as

$$d\varphi[X] : \mathfrak{V}^4 \rightarrow \mathfrak{V}^4; \quad d\varphi[X](y) = Xy + yX^\dagger \quad (25.40)$$

with

$$y \in \mathfrak{V}^4 = \{H = (h_{ij}); i, j = 1, 2, h_{ij} \in \mathbb{C}, H : \text{Hermitian}\},$$

where $\mathfrak{V}^4$ is spanned by $\sigma_0, \sigma_1, \sigma_2$, and σ_3 of (24.497). Note that (25.40) gives an endomorphism on $\mathfrak{V}^4$.

Putting, e.g., $X = \zeta_3 = \frac{1}{2} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$, we have

$$d\varphi[\zeta_3](y) = \zeta_3 y + y\zeta_3^\dagger. \quad (25.41)$$

Substituting $\sigma_0, \sigma_1, \sigma_2$, and σ_3 for y , respectively, we obtain

$$d\varphi[\zeta_3](\sigma_0)=0, \quad d\varphi[\zeta_3](\sigma_1)=\sigma_2, \quad d\varphi[\zeta_3](\sigma_2)=-\sigma_1 \quad \text{and} \quad d\varphi[\zeta_3](\sigma_3)=0.$$

That is, we get

$$(\sigma_0 \ \sigma_1 \ \sigma_2 \ \sigma_3)d\varphi[\zeta_3]=(\sigma_0 \ \sigma_1 \ \sigma_2 \ \sigma_3)\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (25.42)$$

Similarly, we obtain

$$(\sigma_0 \ \sigma_1 \ \sigma_2 \ \sigma_3)d\varphi[\zeta_1]=(\sigma_0 \ \sigma_1 \ \sigma_2 \ \sigma_3)\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad (25.43)$$

$$(\sigma_0 \ \sigma_1 \ \sigma_2 \ \sigma_3)d\varphi[\zeta_2]=(\sigma_0 \ \sigma_1 \ \sigma_2 \ \sigma_3)\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}. \quad (25.44)$$

On the basis of Theorem 24.15, we have known that $\mathfrak{sl}(2, \mathbb{C})$ is spanned by six linearly independent basis vectors. That is, we have

$$\mathfrak{sl}(2, \mathbb{C}) = \text{Span}\{\zeta_1, \zeta_2, \zeta_3, i\zeta_1, i\zeta_2, i\zeta_3\}, \quad (25.45)$$

where $\zeta_1, \zeta_2, \zeta_3$ have been defined in (20.270). Then, corresponding to (25.42), (25.43), and (25.44), we must have related expressions with $i\zeta_1, i\zeta_2$, and $i\zeta_3$. In fact, we obtain

$$(\sigma_0 \ \sigma_1 \ \sigma_2 \ \sigma_3)d\varphi[i\zeta_3]=(\sigma_0 \ \sigma_1 \ \sigma_2 \ \sigma_3)\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad (25.46)$$

$$(\sigma_0 \ \sigma_1 \ \sigma_2 \ \sigma_3)d\varphi[i\zeta_1]=(\sigma_0 \ \sigma_1 \ \sigma_2 \ \sigma_3)\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (25.47)$$

$$(\sigma_0 \ \sigma_1 \ \sigma_2 \ \sigma_3)d\varphi[i\zeta_2] = (\sigma_0 \ \sigma_1 \ \sigma_2 \ \sigma_3) \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (25.48)$$

Equations (25.42), (25.43), and (25.44) are related to the spatial rotation and (25.46), (25.47), and (25.48) to the Lorentz boost. All these six (4, 4) real matrices are identical to those given in (24.220). Equations (25.42)–(25.48) show that the differential mapping $d\varphi$ transforms the basis vectors $\zeta_1, \zeta_2, \zeta_3, i\zeta_1, i\zeta_2, i\zeta_3$ of $\mathfrak{sl}(2, \mathbb{C})$ into those $\mathfrak{A} = (\mathfrak{A}_1, \mathfrak{A}_2, \mathfrak{A}_3), \mathfrak{B} = (\mathfrak{B}_1, \mathfrak{B}_2, \mathfrak{B}_3)$ of $\mathfrak{so}_0(3, 1)$, respectively (through the medium of $\sigma_0, \sigma_1, \sigma_2$, and σ_3). Whereas $\mathfrak{sl}(2, \mathbb{C})$ is a six-dimensional *complex* vector space, $\mathfrak{so}_0(3, 1)$ is a six-dimensional *real* vector space. Once again, from the discussion of Sect. 11.4 the transformation $d\varphi$ is bijective. Namely, $\mathfrak{sl}(2, \mathbb{C})$ and $\mathfrak{so}_0(3, 1)$ [i.e., Lie algebras corresponding to the Lie groups $SL(2, \mathbb{C})$ and $SO_0(3, 1)$] are isomorphic to each other.

Thus, the implication of Examples 25.1 and 25.2 is evident and the differential representation (e.g., ad and $d\varphi$ in the present case) allows us to get a clear view on the representation theory of the Lie algebra. In Sect. 24.1, we had the following relation described by

$$\Lambda = \exp(\mathbf{a} \cdot \mathfrak{A}) \exp(\mathbf{b} \cdot \mathfrak{B}). \quad (24.223)$$

In this section, redefining $\mathbf{a}$ and $\mathbf{b}$ as

$$\mathbf{a} \equiv (\theta_1, \theta_2, \theta_3) \text{ and } \mathbf{b} \equiv (-\omega_1, -\omega_2, -\omega_3). \quad (25.49)$$

We establish the expression the same as (24.223). Notice that the parameters $-\omega_i$ ($i = 1, 2, 3$) have the minus sign. It came from the conventional definition about the direction of the Lorentz boost.

Example 25.3 In Example 25.2, through the complexification we extended the Lie algebra $\mathfrak{su}(2) = \text{Span}\{\zeta_1, \zeta_2, \zeta_3\}$ to $\mathfrak{sl}(2, \mathbb{C})$ such that

$$\mathfrak{sl}(2, \mathbb{C}) = \text{Span}\{\zeta_1, \zeta_2, \zeta_3, i\zeta_1, i\zeta_2, i\zeta_3\}. \quad (25.45)$$

Meanwhile, replacing ρ, t, X , and σ with $\varphi, \omega, i\zeta_3$, and $d\varphi$, respectively, the following equation given by

$$\rho(\exp tX) = \exp[t\sigma(X)] \quad (X \in \mathfrak{g}, t \in \mathbb{R}) \quad (25.6)$$

can be read as

$$\exp\{\omega \cdot d\varphi[i\zeta_3]\} = \varphi[\exp(\omega i\zeta_3)] \quad [i\zeta_3 \in \mathfrak{sl}(2, \mathbb{C}), \quad \omega \in \mathbb{R}]. \quad (25.50)$$

From (20.270) we have

$$i\zeta_3 = \frac{1}{2} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The function $\exp(\omega i\zeta_3)$ can readily be calculated so that we have

$$\exp(\omega i\zeta_3) = \begin{pmatrix} \cosh \frac{\omega}{2} - \sinh \frac{\omega}{2} & 0 \\ 0 & \cosh \frac{\omega}{2} + \sinh \frac{\omega}{2} \end{pmatrix}.$$

Then, from (24.514) and (24.515), we obtain

$$\varphi(\exp \omega i\zeta_3) = \begin{pmatrix} \cosh \omega & 0 & 0 & \sinh \omega \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh \omega & 0 & 0 & \cosh \omega \end{pmatrix}.$$

With respect to RHS of (25.50), in turn, from (25.46) we get

$$\omega d\varphi[i\zeta_3] = \begin{pmatrix} 0 & 0 & 0 & \omega \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \omega & 0 & 0 & 0 \end{pmatrix}.$$

Then, calculating $\exp\{\omega d\varphi[i\zeta_3]\}$ we immediately find that (25.50) holds.

Once again, Examples 25.1 and 25.3 provide tangible illustrations to an abstract relation between the representations of Lie group and Lie algebra that is represented by (25.6). With regard to Examples 25.1 and 25.2, we have a following important theorem in relation to Theorem 24.13.

Theorem 25.2 [1] Let $(\rho, \mathfrak{V})$ be a representation of a linear Lie group G on a vector space $\mathfrak{V}$ and let $(\sigma, \mathfrak{V})$ be a representation of the Lie algebra $\mathfrak{g}$ of G so that σ is a differential representation of ρ ; i.e., $\sigma = d\rho$. Also, let $\mathcal{N}$ be a kernel of ρ . That is, we have

$$\mathcal{N} = \{g \in G; \rho(g) = e\},$$

where e is an identity element. Then, we have

$$(i) \quad G/\mathcal{N} \cong \rho(G).$$

(ii) Let $\mathfrak{n}$ be a Lie algebra of $\mathcal{N}$. Then, $\mathfrak{n}$ is a kernel of the homomorphism mapping $\sigma = d\rho$ of $\mathfrak{g}$. That is,

$$\mathfrak{n} = \{X \in \mathfrak{g}; \sigma(X) = 0\}. \quad (25.51)$$

Proof (i) We have already proven (i) in the proof of Theorem 16.4 (Homomorphism Theorem). With (ii), the proof is as follows: Regarding $\forall X \in \mathfrak{g}$ and any real number t , we have

$$\rho(\exp tX) = \exp[t\sigma(X)] \quad (X \in \mathfrak{g}, t \in \mathbb{R}). \quad (25.6)$$

If $\sigma(X) = 0$, RHS of (25.6) is e for any $t \in \mathbb{R}$. Therefore, according to Definition 16.3 we have $\exp tX \in \mathcal{N}$ with any $t \in \mathbb{R}$. This means that X is an element of the Lie algebra of $\mathcal{N}$. From the definition of the Lie algebra of $\mathcal{N}$, we have $X \in \mathfrak{n}$. In an opposite way, if $X \in \mathfrak{n}$, we have $\exp tX \in \mathcal{N}$ for any $t \in \mathbb{R}$ and, hence, LHS of (25.6) is always e according to Definition 16.3. Consequently, differentiating both sides of (25.6) with respect to t , we have

$$0 = \sigma(X)[\exp t\sigma(X)].$$

Taking the limit $t \rightarrow 0$, we have $0 = \sigma(X)$. That is, $\mathfrak{n}$ comprises the elements X that satisfy (25.51). This completes the proof.

Now, we come back to the proof of Theorem 24.13. We wish to consider $(d\varphi, \mathfrak{V}^4)$ in correspondence with $(\varphi, \mathfrak{V}^4)$. Let us start a discussion about the kernel $\mathcal{N}$ of φ . According to Theorem 24.13, it was given as

$$\mathcal{N} = \{\sigma_0, -\sigma_0\}. \quad (24.532)$$

Since the differential representation $d\varphi$ is related to the neighborhood of the identity element σ_0 , the kernel $\mathfrak{n}$ with respect to $d\varphi$ is only $X = 0$ in light of (25.40).

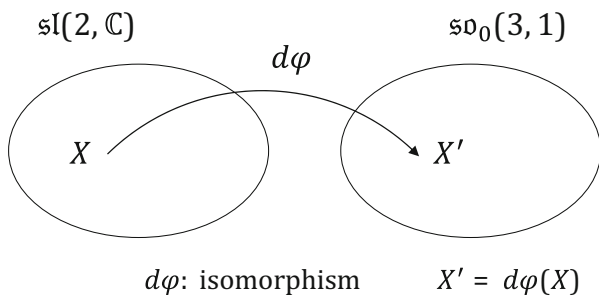
In fact, from Example 25.2, we have

$$d\varphi[a\zeta_1 + b\zeta_3 + c\zeta_3 + di\zeta_1 + ei\zeta_2 + fi\zeta_3] = \begin{pmatrix} 0 & d & e & f \\ d & 0 & -c & b \\ e & c & 0 & -a \\ f & -b & a & 0 \end{pmatrix}.$$

From (25.45), $a\zeta_1 + \cdots + fi\zeta_3$ ($a, \cdots, f$: real) represents all the elements $\mathfrak{sl}(2, \mathbb{C})$. For RHS to be zero element $\mathbf{0}$ of $\mathfrak{so}_0(3, 1)$, we need $a = \cdots = f = 0$. This implies that

$$d\varphi^{-1}(\mathbf{0}') = \{\mathbf{0}\},$$

Fig. 25.2 Isomorphism $d\varphi$ between $\mathfrak{sl}(2, \mathbb{C})$ and $\mathfrak{so}_0(3, 1)$



where $\mathbf{0}$ denotes the zero element of $\mathfrak{sl}(2, \mathbb{C})$; $d\varphi^{-1}$ is the inverse transformation of $d\varphi$. The zero element $\mathbf{0}'$ is represented by a $(4, 4)$ matrix. An explicit form is given by

$$\mathbf{0}' = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The presence of $d\varphi^{-1}$ is equivalent to the statement that

$$d\varphi : \mathfrak{sl}(2, \mathbb{C}) \longrightarrow \mathfrak{so}_0(3, 1) \quad (25.52)$$

is a bijective mapping (see Sect. 11.2). From Theorem 25.2, with respect to the kernel $\mathfrak{n}$ of $d\varphi$ we have

$$\mathfrak{n} = \{X \in \mathfrak{sl}(2, \mathbb{C}); d\varphi(X) = \mathbf{0}'\}.$$

Since $d\varphi$ is bijective, from Theorem 11.4 we get

$$\mathfrak{n} = \{\mathbf{0}\}, \quad \mathbf{0} \equiv \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Consequently, the mapping $d\varphi$ can be characterized as isomorphism expressed as

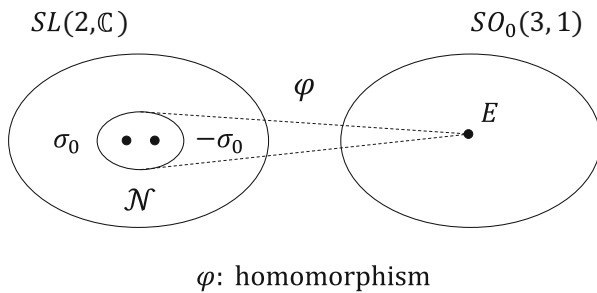
$$\mathfrak{sl}(2, \mathbb{C}) \cong \mathfrak{so}_0(3, 1). \quad (25.53)$$

This relationship is in parallel with that between $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$; see Example 25.1. In Fig. 25.2 we depict the isomorphism $d\varphi$ between $\mathfrak{sl}(2, \mathbb{C})$ and $\mathfrak{so}_0(3, 1)$.

Meanwhile, Theorem 20.11 guarantees that $\forall g \in SO_0(3, 1)$ can be described by

$$g = (\exp t_1 X_1)(\exp t_2 X_2) \cdots (\exp t_6 X_6), \quad (25.54)$$

Fig. 25.3 Homomorphic mapping φ from $SL(2, \mathbb{C})$ to $SO_0(3, 1)$ with the kernel $\mathcal{N} = \{\sigma_0, -\sigma_0\}$



where $X_1, X_2, \dots$, and X_6 can be chosen from among the $(4, 4)$ matrices of, e.g., (25.42), (25.43), (25.44), (25.46), (25.47), and (25.48). With $\forall q \in SL(2, \mathbb{C})$, q can be expressed as

$$q = (\exp t_1 \zeta_1)(\exp t_2 \zeta_2)(\exp t_3 \zeta_3)(\exp t_4 i \zeta_1)(\exp t_5 i \zeta_2)(\exp t_6 i \zeta_3). \quad (25.55)$$

Then, (25.6) is read in the present case as

$$\begin{aligned} \varphi[(\exp t_1 \zeta_1) \cdots (\exp t_6 i \zeta_3)] &= \exp(t_1 d\varphi[\zeta_1]) \cdots \exp(t_6 d\varphi[i \zeta_3]) \\ [\zeta_1, \dots, i \zeta_3] &\in \mathfrak{sl}(2, \mathbb{C}), \quad t_i \ (i = 1, \dots, 6) \in \mathbb{R}. \end{aligned}$$

Since from the above discussion there is the bijective mapping (or isomorphism) between $\mathfrak{sl}(2, \mathbb{C})$ and $\mathfrak{so}_0(3, 1)$, we may identify $X_1, \dots, X_6$ with $d\varphi[\zeta_1], \dots, d\varphi[i \zeta_3]$, respectively; see (25.42)–(25.48). Thus, we get

$$\begin{aligned} \varphi[q] &= \exp(t_1 d\varphi[\zeta_1]) \cdots \exp(t_6 d\varphi[i \zeta_3]) \\ &= (\exp t_1 X_1)(\exp t_2 X_2) \cdots (\exp t_6 X_6) = g. \end{aligned} \quad (25.56)$$

Equation (25.56) implies that $\forall g \in SO_0(3, 1)$ can be described using $\exists q \in SL(2, \mathbb{C})$ given by (25.55). This means that

$$SO_0(3, 1) \subset \varphi[SL(2, \mathbb{C})]. \quad (24.521)$$

Combining (24.520) and (24.521), we get the relation described by

$$\varphi[SL(2, \mathbb{C})] = SO_0(3, 1). \quad (24.522)$$

Figure 25.3 shows the homomorphic mapping φ from $SL(2, \mathbb{C})$ to $SO_0(3, 1)$ with the kernel $\mathcal{N} = \{\sigma_0, -\sigma_0\}$. In correspondence with (25.29) of Example 25.1, we have

Table 25.1 Representation species of Lie algebras

Representation	Representation space	Mapping	Category
Ad	$\mathfrak{su}(2)$ (real)	$SU(2) \rightarrow SO(3)$	Homomorphism
ad	$\mathfrak{su}(2)$ (real)	$\mathfrak{su}(2) \rightarrow \mathfrak{so}(3)$	Isomorphism
φ	$\mathfrak{V}^4$ (real)	$SL(2, \mathbb{C}) \rightarrow SO_0(3, 1)$	Homomorphism
$d\varphi$	$\mathfrak{V}^4$ (real)	$\mathfrak{sl}(2, \mathbb{C}) \rightarrow \mathfrak{so}_0(3, 1)$	Isomorphism

$$(d\varphi, \mathfrak{V}^4) \Leftrightarrow (\varphi, \mathfrak{V}^4). \quad (25.57)$$

Note that in Example 25.1 the representation space $\mathfrak{V}^4$ has been defined as a four-dimensional real vector space given in (24.493).

Comparing Figs. 25.1 and 25.2 with Figs. 25.3 and 20.6, we conclude the following: (i) The differential representations ad and $d\varphi$ exhibit the isomorphic mapping between the Lie algebras, whereas the representations Ad and φ indicate the homomorphic mapping between the Lie groups. (ii) This enables us to deal with the relationship between $SL(2, \mathbb{C})$ and $SO_0(3, 1)$ and that between $SU(2)$ and $SO(3)$ in parallel. Also, we are allowed to deal with the relation between $\mathfrak{sl}(2, \mathbb{C})$ and $\mathfrak{so}_0(3, 1)$ and that between $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ in parallel. Table 25.1 summarizes the properties of several (differential) representations associated with the Lie algebras and corresponding Lie groups.

The Lie algebra is a special kind of vector space. In terms of the vector space, it is worth taking a second look at the basic concept of Chaps. 11 and 20. In Sect. 25.1, we have mentioned the properties of the differential representation of the Lie algebra $\mathfrak{g}$ on a certain vector space $\mathfrak{V}$. That is, we have had

$$\sigma : \mathfrak{g} \rightarrow \mathfrak{V}. \quad (25.4)$$

Equation (25.4) may or may not be an endomorphism.

Meanwhile, the adjoint representation Ad of the Lie group $SU(2)$ was not an endomorphism; see (20.303) denoted by

$$\text{Ad} : SU(2) \rightarrow SO(3). \quad (20.303)$$

These features not only enrich the properties of Lie groups and Lie algebras but also make various representations including differential representation easy to use and broaden the range of applications.

As already seen in Sect. 20.4, $\mathfrak{so}(3)$ and $\mathfrak{su}(2)$ have the same structure constants. Getting back to Theorem 11.3 (Dimension Theorem), we have had

$$\dim V^n = \dim A(V^n) + \dim \text{Ker } A. \quad (11.45)$$

Suppose that there is a mapping $\mathfrak{M}$ from $\mathfrak{su}(2)$ to $\mathfrak{so}(3)$ such that

$$\mathfrak{M} : \mathfrak{su}(2) \rightarrow \mathfrak{so}(3).$$

Equations (20.273) and (20.276) clearly show that both $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ are spanned by three basis vectors, namely the dimension of both $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ is three. Hence, (11.45) should be translated into

$$\dim[\mathfrak{su}(2)] = \dim \mathfrak{M}[\mathfrak{su}(2)] + \dim \text{Ker } \mathfrak{M}. \quad (25.58)$$

Also, we have

$$\dim[\mathfrak{su}(2)] = \dim \mathfrak{M}[\mathfrak{su}(2)] = \dim[\mathfrak{so}(3)] = 3. \quad (25.59)$$

This naturally leads to $\dim \text{Ker } \mathfrak{M} = 0$ in (25.58). From Theorem 11.4, in turn, we conclude that $\mathfrak{M}$ is a bijective mapping, i.e., isomorphism. In this situation, we expect to have the homomorphism between the corresponding Lie groups. This was indeed the case with $SU(2)$ and $SO(3)$ (Theorem 20.7).

By the same token, in this section we have shown that there is one-to-one correspondence between $\mathfrak{sl}(2, \mathbb{C})$ and $\mathfrak{so}_0(3, 1)$. Corresponding to (25.58), we have

$$\dim[\mathfrak{sl}(2, \mathbb{C})] = \dim \tilde{\mathfrak{M}}[\mathfrak{sl}(2, \mathbb{C})] + \dim \text{Ker } \tilde{\mathfrak{M}}, \quad (25.60)$$

where

$$\tilde{\mathfrak{M}} : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \mathfrak{so}_0(3, 1). \quad (25.61)$$

In the above, we have

$$\dim[\mathfrak{sl}(2, \mathbb{C})] = \dim \tilde{\mathfrak{M}}[\mathfrak{sl}(2, \mathbb{C})] = \dim[\mathfrak{so}_0(3, 1)] = 6. \quad (25.62)$$

Again, we have $\dim \text{Ker } \tilde{\mathfrak{M}} = 0$. Notice that $d\varphi$ of (25.52) is a special but important case of $\tilde{\mathfrak{M}}$. Similarly to the case of $\mathfrak{su}(2)$, there has been a homomorphic mapping from $SL(2, \mathbb{C})$ to $SO_0(3, 1)$ (Theorem 24.13).

25.2 Cartan–Weyl Basis of Lie Algebra [1]

In this section, we wish to take the concept of complexification a step further. In this context, we deal with the Cartan–Weyl basis that is frequently used in quantum mechanics and the quantum theory of fields. In our case, the Cartan–Weyl basis has already appeared in Sect. 3.5, even though we did not mention that terminology.

25.2.1 Complexification

We focus on the application of Cartan–Weyl basis to $\mathfrak{so}(3)$ [or $\mathfrak{o}(3)$] to widen the vector space of $\mathfrak{so}(3)$ by their “complexification” (see Sect. 24.10). The Lie algebra $\mathfrak{so}(3)$ is a vector space spanned by three basis vectors whose commutation relation is described by

$$[X_i, X_j] = f_{ij}^k X_k \quad (i, j, k = 1, 2, 3), \quad (25.63)$$

where X_i ($i = 1, 2, 3$) are the basis vectors, which are not commutative with one another. We know that $\mathfrak{so}(3)$ is isomorphic to $\mathfrak{su}(2)$. In the case of $\mathfrak{su}(2)$, the complexification of $\mathfrak{su}(2)$ enables us to deal with many problems within the framework of $\mathfrak{sl}(2, \mathbb{C})$.

Here, we construct the Cartan–Weyl basis from the elements of $\mathfrak{so}(3)$ [or $\mathfrak{o}(3)$] by extending $\mathfrak{so}(3)$ to $\mathfrak{o}(3, \mathbb{C})$ through the complexification. Note that $\mathfrak{o}(3, \mathbb{C})$ comprises complex skew-symmetric matrices. In Sects 20.1 and 20.2 we converted the Hermitian operators to the anti-Hermitian operators belonging to $\mathfrak{so}(3)$ such that, e.g.,

$$A_z \equiv -i\mathcal{M}_z. \quad (20.15)$$

These give a clue to how to address the problem. In an opposite way, we get

$$iA_z = \mathcal{M}_z. \quad (25.64)$$

In light of Theorem 24.15, the Hermitian operator $\mathcal{M}_z$ should now be regarded as due to the complexification of $\mathfrak{so}(3)$ in (25.64). We redefine $A_3 \equiv A_z$ and $\mathcal{M}_3 \equiv \mathcal{M}_z$; see (20.274). In a similar way, we get

$$\mathcal{M}_1 \equiv iA_1 \quad \text{and} \quad \mathcal{M}_2 \equiv iA_2.$$

The operators $\mathcal{M}_1$, $\mathcal{M}_2$, and $\mathcal{M}_3$ construct $\mathfrak{o}(3, \mathbb{C})$ together with A_1 , A_2 , and A_3 . That is, we have

$$\mathfrak{o}(3, \mathbb{C}) = \text{Span} \{A_1, A_2, A_3, \mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3\}. \quad (25.65)$$

A general form of $\mathfrak{o}(3, \mathbb{C})$ is given using real numbers a, b, c, d, f, g as

$$\begin{aligned}
\begin{pmatrix} 0 & a+bi & c+di \\ -(a+bi) & 0 & f+gi \\ -(c+di) & -(f+gi) & 0 \end{pmatrix} &= \begin{pmatrix} 0 & a & c \\ -a & 0 & f \\ -c & -f & 0 \end{pmatrix} + \begin{pmatrix} 0 & bi & di \\ -bi & 0 & gi \\ -di & -gi & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & a & c \\ -a & 0 & f \\ -c & -f & 0 \end{pmatrix} + i \begin{pmatrix} 0 & b & d \\ -b & 0 & g \\ -d & -g & 0 \end{pmatrix}.
\end{aligned} \tag{25.66}$$

Let U be an arbitrary element of $\mathfrak{o}(3, \mathbb{C})$. Then, from (25.66) U is uniquely described as

$$U = V + iW \quad [V, W \in \mathfrak{so}(3)].$$

This implies that $\mathfrak{o}(3, \mathbb{C})$ is a complexification of $\mathfrak{so}(3)$. Or, we may say that $\mathfrak{so}(3)$ is a real form of $\mathfrak{o}(3, \mathbb{C})$.

According to the custom, we define $\mathfrak{F}_+ [\in \mathfrak{o}(3, \mathbb{C})]$ and $\mathfrak{F}_- [\in \mathfrak{o}(3, \mathbb{C})]$ as

$$\mathfrak{F}_+ \equiv A_1 + iA_2 = A_1 + \mathcal{M}_2 \quad \text{and} \quad \mathfrak{F}_- \equiv A_1 - iA_2 = A_1 - \mathcal{M}_2,$$

and choose $\mathfrak{F}_+$ and $\mathfrak{F}_-$ for the basis vectors of $\mathfrak{o}(3, \mathbb{C})$ instead of $\mathcal{M}_1$ and $\mathcal{M}_2$. We further define the operators $J^{(+)}$ and $J^{(-)}$ in accordance with (3.72) such that

$$J^{(+)} \equiv i\mathfrak{F}_+ = iA_1 - A_2 \quad \text{and} \quad J^{(-)} \equiv i\mathfrak{F}_- = iA_1 + A_2.$$

Thus, we have gotten back to the discussion made in Sects. 3.4 and 3.5 to directly apply it to the present situation. Although $J^{(+)}$, $J^{(-)}$, and $\mathcal{M}_3$ are formed by the linear combination of the elements of $\mathfrak{so}(3)$, these operators are not contained in $\mathfrak{so}(3)$ but contained in $\mathfrak{o}(3, \mathbb{C})$. The vectors $J^{(+)}$, $J^{(-)}$, and $\mathcal{M}_3$ form a subspace of $\mathfrak{o}(3, \mathbb{C})$. We term this subspace $\mathfrak{W}$, to which these three vectors constitute a basis set. This basis set is referred to as Cartan–Weyl basis and we have

$$\mathfrak{W} \equiv \text{Span} \left\{ J^{(-)}, \mathcal{M}_3, J^{(+)} \right\}. \tag{25.67}$$

Even though $\mathcal{M}_3$ is Hermitian, $J^{(+)}$ is neither Hermitian nor anti-Hermitian. Nor is $J^{(-)}$, either. Nonetheless, we are well used to dealing with these operators in Sect. 3.5. We have commutation relations described as

$$\begin{aligned}
[\mathcal{M}_3, J^{(+)}] &= J^{(+)}, \\
[\mathcal{M}_3, J^{(-)}] &= -J^{(-)}, \\
[J^{(+)}, J^{(-)}] &= 2\mathcal{M}_3.
\end{aligned} \tag{25.68}$$

Note that the commutation relations of (25.68) have already appeared in (3.74) in the discussion of the generalized angular momentum.

Using $J^{(\pm)} \equiv J^{(\pm)}/\sqrt{2}$, we obtain a “normalized” relations expressed as

$$[\mathcal{M}_3, J^{(\pm)}] = \pm J^{(\pm)} \quad \text{and} \quad [J^{(+)}, J^{(-)}] = \mathcal{M}_3. \tag{25.69}$$

From (25.68) and (25.69), we find that $\mathfrak{W}$ forms a subalgebra of $\mathfrak{o}(3, \mathbb{C})$. Also, at the same time, from (25.14) the differential representation ad with respect to $\mathfrak{W}$ diagonalizes $\text{ad}[\mathcal{M}_3]$. That is, with the three-dimensional representation, using (25.68) or (25.69) we have

$$\begin{pmatrix} J^{(-)} & \mathcal{M}_3 & J^{(+)} \end{pmatrix} \text{ad}[\mathcal{M}_3] = \begin{pmatrix} J^{(-)} & \mathcal{M}_3 & J^{(+)} \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{25.70}$$

Hence, we obtain

$$\text{ad}[\mathcal{M}_3] = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{25.71}$$

In an alternative way, let us consider $\mathfrak{sl}(2, \mathbb{C})$ instead of $\mathfrak{o}(3, \mathbb{C})$. In a similar manner of the above case, we can choose a subspace $\tilde{\mathfrak{W}}$ of $\mathfrak{sl}(2, \mathbb{C})$ defined by

$$\tilde{\mathfrak{W}} \equiv \text{Span}\{i\zeta_1 + \zeta_2, i\zeta_3, i\zeta_1 - \zeta_2\}.$$

At the same time, $\tilde{\mathfrak{W}}$ is a subalgebra of $\mathfrak{sl}(2, \mathbb{C})$. The operators $i\zeta_1 + \zeta_2$, $i\zeta_3$, and $i\zeta_1 - \zeta_2$ correspond to $J^{(-)}$, $\mathcal{M}_3$, and $J^{(+)}$ of the above case, respectively. Then, $\mathfrak{W}$ and $\tilde{\mathfrak{W}}$ are isomorphic and the commutation relations exactly the same as (25.68) hold with the basis vectors $i\zeta_1 + \zeta_2$, $i\zeta_3$, and $i\zeta_1 - \zeta_2$. Corresponding to (25.71), we have

$$\text{ad}[i\zeta_3] = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{25.72}$$

We find that choosing the basis vectors $i\zeta_1 + \zeta_2$, $i\zeta_3$, and $i\zeta_1 - \zeta_2$ contained in $\mathfrak{sl}(2, \mathbb{C})$, $\text{ad}[i\zeta_3]$ can be diagonalized as in (25.71). Thus, in terms of Lie algebra the subalgebra $\mathfrak{W}$ and $\tilde{\mathfrak{W}}$ are equivalent.

From (25.68) and on the basis of the discussions in Sects. 3.4, 3.5, 20.2, and 20.4, we can readily construct an irreducible representation ad of $\mathfrak{M}$ or $\tilde{\mathfrak{M}}$ on the $(2j + 1)$ -dimensional representation space $\mathfrak{V}$, where j is given as positive integers or positive half-odd-integers. The number j is said to be a *highest weight* as defined below.

Definition 25.3 Eigenvalues of the differential representation $\text{ad}[\mathcal{M}_3]$ or $\text{ad}[i\zeta_3]$ are called a weight of the representation ad . Among the weights, the weight that has the largest real part is the highest weight.

In (25.71) and (25.72), for instance, the highest weight of $\mathcal{M}_3$ or $i\zeta_3$ is 1. From the discussion of Sects. 3.4 and 3.5, we immediately obtain the matrix representation of the Cartan–Weyl basis of $J^{(-)}$, $\mathcal{M}_3$, and $J^{(+)}$ or $(i\zeta_1 + \zeta_2, i\zeta_3, \text{ and } i\zeta_1 - \zeta_2)$. Replacing l with j in (3.152), (3.153), and (3.158), we get

$$\mathcal{M}_3 = \begin{pmatrix} -j & & & & & \\ & -j+1 & & & & \\ & & -j+2 & & & \\ & & & \ddots & & \\ & & & & k-j & \\ & & & & & \ddots \\ & & & & & & j \end{pmatrix}, \quad (25.73)$$

where $k = 0, 1, \dots, 2j$ and the highest weight of $\mathcal{M}_3$ is j . Also, we have

$$J^{(-)} = \begin{pmatrix} 0 & \sqrt{2j \cdot 1} & & & & \\ & 0 & \sqrt{(2j-1) \cdot 2} & & & \\ & & 0 & \ddots & & \\ & & & \ddots & 0 & \\ & & & & \sqrt{(2j-k+1) \cdot k} & \\ & & & & 0 & \\ & & & & & \ddots & \ddots \\ & & & & & & 0 & \sqrt{1 \cdot 2j} \\ & & & & & & & 0 \end{pmatrix}, \quad (25.74)$$

$$J^{(+)} = \begin{pmatrix} 0 & & & & & & & \\ \sqrt{2j \cdot 1} & 0 & & & & & & \\ & \sqrt{(2j-1) \cdot 2} & 0 & & & & & \\ & & \ddots & & & & & \\ & & & 0 & & & & \\ & & & \sqrt{(2j-k+1) \cdot k} & 0 & & & \\ & & & & \ddots & \ddots & & \\ & & & & & 0 & & \\ & & & & & \ddots & 0 & \\ & & & & & & \sqrt{1 \cdot 2j} & 0 \end{pmatrix}, \quad (25.75)$$

where $k = 1, 2, \dots, 2j$. In (25.73)–(25.75), the basis vectors $i\zeta_1 + \zeta_2$, $i\zeta_3$, and $i\zeta_1 - \zeta_2$ may equally be used instead of $J^{(-)}$, $\mathcal{M}_3$, and $J^{(+)}$, respectively.

The above matrix representations of the Cartan–Weyl basis of $J^{(-)}$, $\mathcal{M}_3$, and $J^{(+)}$ give important information by directly appealing to the intuition. The matrix $\mathcal{M}_3$ is admittedly a diagonalized matrix. The matrices $J^{(+)}$ and $J^{(-)}$ are lower and upper triangle matrices, respectively. This means that all their eigenvalues are zero (see Sect. 12.1). In turn, this immediately leads to the fact that $J^{(+)}$ and $J^{(-)}$ are $(2j+1)$ th nilpotent $(2j+1, 2j+1)$ square matrices and implies the presence of $(2j+1)$ linearly independent vectors in the representation space $\mathfrak{V}$. Also, we know that neither $J^{(+)}$ nor $J^{(-)}$ can be diagonalized; see Sect. 12.3. Let ad be a representation pertinent to (25.73)–(25.75). Then, the representation $(\text{ad}, \mathfrak{V})$ is irreducible. Summarizing the above results, we have the following theorem.

Theorem 25.3 [1] Let $(\text{ad}, \mathfrak{V})$ be an irreducible representation of $\mathfrak{W}$. Suppose that the highest weight of $(\text{ad}, \mathfrak{V})$ is j (>0). Then, the following relationships holds:

- (i) $\dim \mathfrak{V} = 2j + 1$.
- (ii) The weights are $j, j-1, \dots, -j+1, -j$ with the highest weight j .
- (iii) There exist $(2j+1)$ basis vectors $|\zeta, \mu\rangle$ [$\mu = j, j-1, \dots, -j+1, -j; \zeta = j(j+1)$] that satisfy the following equations:

$$\begin{aligned} \mathcal{M}_3 |\zeta, \mu\rangle &= \mu |\zeta, \mu\rangle \quad (\mu = j, j-1, \dots, -j+1, -j), \\ J^{(+)} |\zeta, \mu\rangle &= \sqrt{(j-\mu)(j+\mu+1)} |\zeta, \mu+1\rangle \quad (\mu = j-1, j-2, \dots, -j), \\ J^{(-)} |\zeta, \mu\rangle &= \sqrt{(j-\mu+1)(j+\mu)} |\zeta, \mu-1\rangle \quad (\mu = j, j-1, \dots, -j+1), \end{aligned} \quad (25.76)$$

where j is chosen from among positive integers or positive half-odd-integers.

Three operators of (25.73)–(25.75) satisfy the commutation relations described by (25.68). Readers are encouraged to confirm it. The dimension of the *representation space* $\mathfrak{V}$ relevant to these irreducible representations is $2j + 1$. Correspondingly, $\mathfrak{V}$ is spanned by the $(2j + 1)$ basis vectors $|\zeta, \mu\rangle$ [$\mu = j, j - 1, \dots, -j + 1, -j; \zeta = j(j + 1)$]. Note, however, that the dimension as the *Lie algebra* of $\mathfrak{W}$ is three; see (25.67). The commutation relations (25.73)–(25.75) equally hold with $i\zeta_1 + \zeta_2$, $i\zeta_3$, and $i\zeta_1 - \zeta_2$, which constitute the basis vectors of $\widetilde{\mathfrak{W}}$.

As explained above, the notion of complexification of the Lie algebra finds wide applications in the field of continuous groups not only from the point of view of abstract algebra, but also with the physical interpretations of, e.g., angular momenta that were dealt with in Chap. 3. In particular, the Cartan–Weyl basis has been constructed through the complexification of $\mathfrak{so}(3)$ or $\mathfrak{su}(2)$. The Lie algebras of $\mathfrak{W}$ and $\widetilde{\mathfrak{W}}$ are subalgebras of $\mathfrak{o}(3, \mathbb{C})$ and $\mathfrak{sl}(2, \mathbb{C})$, respectively.

In relation to the complexification, we add a following condition to the definition of the representation of Lie algebra so that we may extend the concept of the representation to the complex representation. The condition is described by [1]

$$\rho(iX) = i\rho(X), \quad (25.77)$$

where ρ is a representation of the Lie algebra. If ρ satisfies (25.77), ρ is said to be a complex analytic representation of the Lie algebra [1]. Let us formally give a following definition.

Definition 25.4 Let $\mathfrak{g}$ be a complex Lie algebra and let $\mathfrak{V}$ be a vector space. Suppose that there is a complex linear mapping ρ from $\mathfrak{g}$ to $\mathfrak{V}$. If ρ is a representation of $\mathfrak{g}$ on $\mathfrak{V}$ and satisfies the condition (25.77), then such ρ is said to be a complex analytic representation.

By virtue of the condition (25.77), we may extend (i) of Definition 25.1 to

$$(i)' \sigma(aX) = a\sigma(X) \quad (a \in \mathbb{C}). \quad (25.78)$$

Except for (i) of Definition 25.1, the conditions (ii), (iii), and (iv) hold without alterations.

In the context of the complex representation, we have a following general theorem.

Theorem 25.4 [1] Let $\mathfrak{g}$ be a complex Lie algebra and $\mathfrak{h}$ be a real form of $\mathfrak{g}$ (see Sect. 24.10). Let ρ be any representation of $\mathfrak{h}$ on a vector space $\mathfrak{V}$. Moreover, let ρ^c be a representation of $\mathfrak{g}$ on $\mathfrak{V}$. Suppose with ρ^c we have a following relationship:

$$\rho^c(Z) = \rho(X) + i\rho(Y) \quad (Z \in \mathfrak{g}; X, Y \in \mathfrak{h}), \quad (25.79)$$

where $Z = X + iY$. Then, $(\rho^c, \mathfrak{V})$ is a complex analytic representation of $\mathfrak{g}$ on $\mathfrak{V}$.

Proof For the proof, use (25.1), (25.2), and (25.5). The representation ρ^c is a real linear mapping from $\mathfrak{g}$ to $\mathfrak{gl}(\mathfrak{V})$. The representation ρ^c is a complex linear mapping as well. It is because we have

$$\rho^c[i(X + iY)] = \rho^c(iX - Y) = -\rho(Y) + i\rho(X) = i\rho^c(X + iY). \quad (25.80)$$

Meanwhile, we have

$$\begin{aligned} \rho^c([X + iY, U + iW]) &= [\rho(X), \rho(U)] - [\rho(Y), \rho(W)] \\ &\quad + i[\rho(X), \rho(W)] + i[\rho(Y), \rho(U)] \\ &= [\rho^c(X + iY), \rho^c(U + iW)]. \end{aligned}$$

Hence, $(\rho^c, \mathfrak{V})$ is a complex analytic representation of $\mathfrak{g}$. This completes the proof.

In the next section, we make the most of the results obtained in this section.

25.2.2 Coupling of Angular Momenta: Revisited

In Sect. 20.3, we calculated the Clebsch–Gordan coefficients on the basis of the representation theory of the continuous group $SU(2)$. Here, we wish to seek the relevant results using the above-mentioned theory of Lie algebras. We deal with the problems that we considered in Examples 20.1 and 20.2. Since in this section we consider the coupling of the angular momenta, we modify the expression as in Sect. 20.3 such that

$$J = |j_1 - j_2|, |j_1 - j_2| + 1, \dots, j_1 + j_2, \quad (20.164)$$

where J indicates the momenta of the coupled system with j_1 and j_2 being momenta of the individual systems. At the same time, J represents the highest weight. Thus, j used in the previous section should be replaced with J .

We show two examples below and compare the results with those of Sect. 20.3.3. In the examples given below, we adopt the following calculation rule. Let $\mathfrak{D}$ be an operator chosen from among $\mathcal{M}_3$, $J^{(+)}$, and $J^{(-)}$ of (25.76). Suppose that the starting equation we wish to deal with is given by

$$\Xi = \Phi \otimes \Psi,$$

where $\Phi \otimes \Psi$ is a direct product of quantum states Φ and Ψ (or physical states, more generally). Then, we have

$$\mathfrak{D}\Xi = \mathfrak{D}(\Phi \otimes \Psi) = (\mathfrak{D}\Phi) \otimes \Psi + \Phi \otimes (\mathfrak{D}\Psi),$$

where $\mathfrak{D}$ is an operator that is supposed to operate on Ξ .

If we choose $\mathcal{M}_3$ for $\mathfrak{D}$ and both Φ and Ψ are eigenstate of $\mathcal{M}_3$, we have a trivial result with no new result obtained. It can be checked easily. If, however, we choose $J^{(+)}$ or $J^{(-)}$, we can get meaningful results. To show this, we give two examples below.

Example 25.4 We deal with the problem that we considered in Example 20.1. That is, we are examining the case of $D^{(1/2)} \otimes D^{(1/2)}$. According to (20.164), as the highest weight J we have

$$J = 1, 0.$$

(i) $J = 1$

If among the highest weights we choose the largest one, the associated coupled state is always given by the direct product of the individual component state (see Examples 20.1 and 20.2). Thus, we have

$$|1, 3\rangle = |1/2, 2\rangle_A |1/2, 2\rangle_B. \quad (25.81)$$

In (25.81), we adopted the expression $|\mu, \mathbf{2J} + \mathbf{1}\rangle$ with $\mathbf{2J} + \mathbf{1}$ bold faced according to the custom instead of $|\zeta, \mu\rangle$ used in the previous section. The subscripts A and B in RHS of (25.81) designate the individual component systems. The strategy to solve the problem is to operate $J^{(-)}$ of (25.76) on both sides of (25.81) and to diminish μ stepwise with J (or ζ) kept unchanged. That is, we have

$$J^{(-)} |1, 3\rangle = J^{(-)} [|1/2, 2\rangle_A |1/2, 2\rangle_B]. \quad (25.82)$$

Then, we obtain

$$\sqrt{2} |0, 3\rangle = [J^{(-)} |1/2, 2\rangle_A] |1/2, 2\rangle_B + |1/2, 2\rangle_A [J^{(-)} |1/2, 2\rangle_B]. \quad (25.83)$$

Hence, we have

$$|0, 3\rangle = \frac{1}{\sqrt{2}} [|-1/2, 2\rangle_A |1/2, 2\rangle_B + |1/2, 2\rangle_A |-1/2, 2\rangle_B]. \quad (25.84)$$

Further operating $J^{(-)}$ on both sides, we get

$$|-1, 3\rangle = |-1/2, 2\rangle_A |-1/2, 2\rangle_B. \quad (25.85)$$

Thus, we have obtained the symmetric representations (25.81), (25.84), and (25.85). Note that

$$J^{(-)} | -1/2, 2 \rangle_A = J^{(-)} | -1/2, 2 \rangle_B = 0. \quad (25.86)$$

Looking at (25.85), we notice that we can equally start with (25.85) using $J^{(+)}$ instead of starting with (25.81) using $J^{(-)}$.

(ii) $J = 0$

In this case, we determine a suitable solution $|0, 1\rangle$ so that it may become orthogonal to $|0, 3\rangle$ given by (25.84). We can readily find the solution such that

$$|0, 1\rangle = \frac{1}{\sqrt{2}} [| -1/2, 2 \rangle_A | 1/2, 2 \rangle_B - | 1/2, 2 \rangle_A | -1/2, 2 \rangle_B]. \quad (25.87)$$

Compare the above results with those obtained in Example 20.1.

Example 25.5 Next, we consider the case of the direct product $D^{(1)} \otimes D^{(1)}$ in parallel with Example 20.2. In this case the highest weight J is

$$J = 2, 1, 0. \quad (25.88)$$

The calculation processes are a little bit longer than Example 25.4, but the calculation principle is virtually the same.

(i) $J = 2$.

We start with

$$|2, 5\rangle = |1, 3\rangle_A |1, 3\rangle_B. \quad (25.89)$$

Proceeding in a similar manner as before, we get

$$|1, 5\rangle = \frac{1}{\sqrt{2}} (|1, 3\rangle_A |0, 3\rangle_B + |0, 3\rangle_A |1, 3\rangle_B). \quad (25.90)$$

Further operating $J^{(-)}$ on both sides of (25.90), we successively obtain

$$|0, 5\rangle = \frac{1}{\sqrt{6}} (| -1, 3\rangle_A |1, 3\rangle_B + 2|0, 3\rangle_A |0, 3\rangle_B + |1, 3\rangle_A | -1, 3\rangle_B), \quad (25.91)$$

$$| -1, 5\rangle = \frac{1}{\sqrt{2}} (| -1, 3\rangle_A |0, 3\rangle_B + |0, 3\rangle_A | -1, 3\rangle_B), \quad (25.92)$$

$$| -2, 5\rangle = | -1, 3\rangle_A | -1, 3\rangle_B. \quad (25.93)$$

(ii) $J = 1$

In this case, we determine a suitable solution $|1, \mathbf{3}\rangle$ so that it may become orthogonal to $|1, \mathbf{5}\rangle$ given by (25.90). We can readily find the solution described by

$$|1, \mathbf{3}\rangle = \frac{1}{\sqrt{2}}(|1, \mathbf{3}\rangle_A |0, \mathbf{3}\rangle_B - |0, \mathbf{3}\rangle_A |1, \mathbf{3}\rangle_B). \quad (25.94)$$

Then, operating $J^{(-)}$ on both sides of (25.94) successively, we get

$$|0, \mathbf{3}\rangle = \frac{1}{\sqrt{2}}(|1, \mathbf{3}\rangle_A |-1, \mathbf{3}\rangle_B - |-1, \mathbf{3}\rangle_A |0, \mathbf{3}\rangle_B), \quad (25.95)$$

$$|-1, \mathbf{3}\rangle = \frac{1}{\sqrt{2}}(|0, \mathbf{3}\rangle_A |-1, \mathbf{3}\rangle_B - |-1, \mathbf{3}\rangle_A |0, \mathbf{3}\rangle_B). \quad (25.96)$$

(iii) $J = 0$

We determine a suitable solution $|0, \mathbf{1}\rangle$ so that it is orthogonal to $|0, \mathbf{5}\rangle$ given by (25.91) to get the desired solution described by

$$|0, \mathbf{1}\rangle = \frac{1}{\sqrt{3}}(|-1, \mathbf{3}\rangle_A |1, \mathbf{3}\rangle_B - |0, \mathbf{3}\rangle_A |0, \mathbf{3}\rangle_B + |1, \mathbf{3}\rangle_A |-1, \mathbf{3}\rangle_B). \quad (25.97)$$

Notice that if we had chosen for $|0, \mathbf{1}\rangle$ a vector orthogonal to $|0, \mathbf{3}\rangle$ of (25.95), we would not successfully construct a unitary matrix expressed in (20.232). The above is the whole procedure that finds a set of proper solutions. Of these, the solutions obtained with $J = 2$ and 0 possess the symmetric representations, whereas those obtained for $J = 1$ have the antisymmetric representations.

Symbolically writing the above results of the direct product in terms of the direct sum, we have

$$\mathbf{3} \otimes \mathbf{3} = \mathbf{5} \oplus \mathbf{3} \oplus \mathbf{1}. \quad (25.98)$$

If in the above processes we start with $|-2, \mathbf{5}\rangle = |-1, \mathbf{3}\rangle_A |-1, \mathbf{3}\rangle_B$ and operate $J^{(+)}$ on both sides, we will reach the same conclusion as the above. Once again, compare the above results with those obtained in Example 20.2.

As shown in the above examples, we recognize that we have been able to automatically find a whole bunch of solutions compared to the previous calculations carried out in Examples 20.1 and 20.2.

And what is more, as we have seen in this chapter, various problems which the representation theory deals with can be addressed clearly and succinctly by replacing the representation $(\rho, \mathfrak{V})$ of the group G with the representation $(\sigma, \mathfrak{V})$ of its corresponding Lie algebra $\mathfrak{g}$. Thus, the representation theory of the Lie algebras together with the Lie groups serves as a powerful tool to solve various kinds of problems in mathematical physics.

25.3 Decomposition of Lie Algebra

In Sect. 24.10 we have mentioned the decomposition of the complex matrices (Theorem 24.14) and that of $\mathfrak{sl}(2, \mathbb{C})$ (Theorem 24.15) in connection with the discussion that has clarified the relationship between $SL(2, \mathbb{C})$ and $SO_0(3, 1)$. Since the whole collection of the $(2, 2)$ complex matrices forms $\mathfrak{gl}(2, \mathbb{C})$, Theorem 24.14 states the decomposition of $\mathfrak{gl}(2, \mathbb{C})$. Since the Lie algebras form vector spaces, the relevant discussion will come down to the decomposition of the vector spaces.

Let us formally define $\mathfrak{gl}(2, \mathbb{C})$ such that

$$\mathfrak{gl}(2, \mathbb{C}) \equiv \{C = (c_{ij}); i, j = 1, 2, \quad c_{ij} \in \mathbb{C}\}. \quad (25.99)$$

This has already appeared in Sect. 24.10. In fact, $\mathfrak{gl}(2, \mathbb{C})$ is identical to $\mathfrak{C}$ of (24.530).

In Sect. 11.1 we dealt with the decomposition of a vector space into a direct sum of two (or more) subspaces. Meanwhile, in Sect. 14.6 we have shown that any matrix A can be decomposed such that

$$A = B + iC, \quad (14.125)$$

where

$$B \equiv \frac{1}{2}(A + A^\dagger) \text{ and } C \equiv \frac{1}{2i}(A - A^\dagger) \quad (25.100)$$

with both B and C being Hermitian. In other words, any matrix A can be decomposed into the sum of an Hermitian matrix and an anti-Hermitian matrix. Instead of decomposing A as (14.125), we may equally decompose A such that

$$A = B + iC, \quad (14.125)$$

where

$$B \equiv \frac{1}{2}(A - A^\dagger) \text{ and } C \equiv \frac{1}{2i}(A + A^\dagger) \quad (25.101)$$

with both B and C being anti-Hermitian.

If a Lie algebra is formed by a complexification of a subalgebra (see Definition 24.2), such examples are of high practical interest. As a typical example, we had $\mathfrak{sl}(2, \mathbb{C})$ and $\mathfrak{gl}(2, \mathbb{C})$. Both (25.100) and (25.101) are used for the complexification of real Lie algebras. We have already encountered the latter example in Sect. 24.10. There, we have shown that $\mathfrak{sl}(2, \mathbb{C})$ is a complexification of $\mathfrak{su}(2)$, which is a real Lie subalgebra of $\mathfrak{sl}(2, \mathbb{C})$; see Theorem 24.15. In that case, we can choose $\mathfrak{su}(2)$ for a real form $\mathfrak{h}$ of $\mathfrak{sl}(2, \mathbb{C})$. The Lie algebra $\mathfrak{sl}(2, \mathbb{C})$ is decomposed as in (14.125) in

combination with (25.101), where both B and C are anti-Hermitian. Namely, we have

$$Z = B + iC \quad [Z \in \mathfrak{sl}(2, \mathbb{C}); B, C \in \mathfrak{su}(2)].$$

In the case of $\mathfrak{gl}(2, \mathbb{C})$, however, $\mathfrak{gl}(2, \mathbb{C})$ is a complexification of $\mathfrak{V}^4$. The vector space $\mathfrak{V}^4$, which has appeared in (24.493), is evidently a real subalgebra of $\mathfrak{gl}(2, \mathbb{C})$. (The confirmation is left for the readers.) Therefore, we can choose $\mathfrak{V}^4$ for a real form $\mathfrak{h}$ of $\mathfrak{gl}(2, \mathbb{C})$. It is decomposed as in (14.125) in combination with (25.100), where both B and C are Hermitian. In this case, we have

$$Z = B + iC \quad [Z \in \mathfrak{gl}(2, \mathbb{C}); B, C \in \mathfrak{V}^4].$$

Viewing the Lie algebra as a vector space, we have a following theorem with respect to $\mathfrak{V}^4$.

Theorem 25.5 Let the vector space $\mathfrak{V}^4$ be given by

$$\mathfrak{V}^4 = \{H = (h_{ij}); i, j = 1, 2, h_{ij} \in \mathbb{C}, H : \text{Hermitian}\}, \quad (24.493)$$

where (h_{ij}) is a complex $(2, 2)$ Hermitian matrix. Then, the Lie algebra $\mathfrak{gl}(2, \mathbb{C})$ is expressed as a direct sum of $\mathfrak{V}^4$ and $i\mathfrak{V}^4$. That is, we have

$$\mathfrak{gl}(2, \mathbb{C}) = \mathfrak{V}^4 \oplus i\mathfrak{V}^4, \quad (25.102)$$

where $i\mathfrak{V}^4 = \{iH; H \in \mathfrak{V}^4\}$; see (24.493).

Proof It is evident that both $\mathfrak{V}^4$ and $i\mathfrak{V}^4$ form a vector space [i.e., subspace of $\mathfrak{gl}(2, \mathbb{C})$]. Let z be an arbitrarily chosen element of $\mathfrak{gl}(2, \mathbb{C})$. We assume that z can be described as

$$z = \begin{pmatrix} a + ib & c + id \\ p + iq & r + is \end{pmatrix} \quad (a, b, c, d, p, q, r, s \in \mathbb{R}). \quad (24.535)$$

Using σ_i ($i = 0, 1, 2, 3$) $\in \mathfrak{V}^4$ given by (24.495), we have

$$\begin{aligned} z &= \frac{1}{2}[(a+r)\sigma_0 + (-a+r)\sigma_3 + (c+p)\sigma_1 + (d-q)\sigma_2] \\ &\quad + \frac{i}{2}[(b+s)\sigma_0 + (-b+s)\sigma_3 + (d+q)\sigma_1 + (-c+p)\sigma_2]. \end{aligned} \quad (24.536)$$

The element of the first term of (24.536) belongs to $\mathfrak{V}^4$ and that of the second term of (24.536) belongs to $i\mathfrak{V}^4$. Thus, $z \in \mathfrak{V}^4 + i\mathfrak{V}^4$. Since z is arbitrary, this implies that

$$\mathfrak{gl}(2, \mathbb{C}) \subset \mathfrak{V}^4 + i\mathfrak{V}^4. \quad (25.103)$$

Now, suppose that $\exists u \in \mathfrak{gl}(2, \mathbb{C})$ is contained in $\mathfrak{V}^4 \cap i\mathfrak{V}^4$. Then, u can be expressed as

$$\begin{aligned} u &= \alpha\sigma_0 + \beta\sigma_1 + \gamma\sigma_2 + \delta\sigma_3 \quad (\alpha, \beta, \gamma, \delta : \text{arbitrary real numbers}) \\ &= i(\lambda\sigma_0 + \mu\sigma_1 + \nu\sigma_2 + \xi\sigma_3) \quad (\lambda, \mu, \nu, \xi : \text{arbitrary real numbers}). \end{aligned} \quad (25.104)$$

Since $\sigma_0, \sigma_1, \sigma_2$, and σ_3 are basis vectors of $\mathfrak{V}^4$, (25.104) implies that

$$\alpha = i\lambda, \beta = i\mu, \gamma = i\nu, \delta = i\xi. \quad (25.105)$$

Since $\alpha, \beta, \dots, \lambda, \mu$, etc. are real numbers, (25.105) means that

$$\alpha = \beta = \gamma = \delta = \lambda = \mu = \nu = \xi = 0.$$

Namely, $u \equiv \mathbf{0}$. This clearly indicates that

$$\mathfrak{V}^4 \cap i\mathfrak{V}^4 = \{\mathbf{0}\}.$$

Thus, from Theorem 11.1, any vector $w \in \mathfrak{V}^4 + i\mathfrak{V}^4$ is uniquely described by

$$w = w_1 + w_2 \quad (w_1 \in \mathfrak{V}^4, w_2 \in i\mathfrak{V}^4). \quad (25.106)$$

That is, by the definition of the direct sum described by (11.17), $\mathfrak{V}^4 + i\mathfrak{V}^4$ is identical with $\mathfrak{V}^4 \oplus i\mathfrak{V}^4$. Meanwhile, from (25.106) evidently we have

$$\mathfrak{V}^4 \oplus i\mathfrak{V}^4 \subset \mathfrak{gl}(2, \mathbb{C}). \quad (25.107)$$

Using (25.103) and (25.107) along with (25.106), we obtain (25.102). These complete the proof.

Thus, $\mathfrak{gl}(2, \mathbb{C})$ is a complexification of $\mathfrak{V}^4$. In a complementary style, the subalgebra $\mathfrak{V}^4$ is said to be a *real form* of $\mathfrak{gl}(2, \mathbb{C})$. Theorem 25.5 is in parallel with the discussion of Sects. 24.9 and 25.2 on the relationship between $\mathfrak{sl}(2, \mathbb{C})$ and $\mathfrak{su}(2)$. In the latter case, we say that $\mathfrak{sl}(2, \mathbb{C})$ is a complexification of $\mathfrak{su}(2)$ and that $\mathfrak{su}(2)$ is a real form of $\mathfrak{sl}(2, \mathbb{C})$. Caution should be taken in that the real form $\mathfrak{V}^4$ comprises the Hermitian operators, whereas the real form $\mathfrak{su}(2)$ consists of the anti-Hermitian operators. We remark that although $i\mathfrak{V}^4$ is a subspace of $\mathfrak{gl}(2, \mathbb{C})$, it is *not* a subalgebra of $\mathfrak{gl}(2, \mathbb{C})$.

In this book, we chose $\mathfrak{gl}(2, \mathbb{C})$ and $\mathfrak{sl}(2, \mathbb{C})$ as typical examples of the *complex* Lie algebra. A bit confusing, however, these Lie algebras are *real* Lie algebra as well. This is related to the definition of the representation of Lie algebra. In (25.1), we designated a as a *real* number. Suppose that the basis set of the Lie algebra $\mathfrak{g}$ is

$\{\mathbf{e}_1, \dots, \mathbf{e}_d\}$, where d is the dimension of $\mathfrak{g}$ (as a vector space). Then, from (25.1) and (25.2) we obtain

$$\sigma\left(\sum_{k=1}^d a^k \mathbf{e}_k\right) = \sum_{k=1}^d a^k \sigma(\mathbf{e}_k) \quad (a^k : \text{real}). \quad (25.108)$$

Now, we assume that $\sigma : \mathfrak{g} \rightarrow \mathfrak{g}'$. Also, we assume that $\sigma(\mathbf{e}_k)$ ($k = 1, \dots, d$) is the basis set of $\mathfrak{g}'$. Then, $\mathfrak{g}$ and $\mathfrak{g}'$ are isomorphic and σ is a bijective mapping. If σ is endomorphism (i.e., $\mathfrak{g} = \mathfrak{g}'$), σ is merely an identity mapping. Though trivial, (25.108) clearly indicates that σ transforms a real vector $\sum_{k=1}^d a^k \mathbf{e}_k$ into another real vector $\sum_{k=1}^d a^k \sigma(\mathbf{e}_k)$. It follows that both the $\mathfrak{g}$ and $\mathfrak{g}'$ are *real* Lie algebras. Thus, so far as a^k is a real number, we must deal with even a complex Lie algebra as a real Lie algebra. As already discussed, this confusing situation applies to $\mathfrak{gl}(2, \mathbb{C})$ and $\mathfrak{sl}(2, \mathbb{C})$. Note that in contrast, e.g., $\mathfrak{su}(2)$ is *inherently* not a complex algebra, but a real Lie algebra.

Recall (24.550). We have mentioned that $\mathfrak{sl}(2, \mathbb{C})$ is a complex vector space. Yet, if we regard $i\tilde{\zeta}_1$, $i\tilde{\zeta}_2$, and $i\tilde{\zeta}_3$ as the basis vectors of $\mathfrak{sl}(2, \mathbb{C})$, then their coefficients $-a$, $(d - q)/2$, and $(c + p)/2$ of (24.550) are real, and so $\mathfrak{sl}(2, \mathbb{C})$ can be viewed as a real vector space. This situation is the same as that for $\mathfrak{gl}(2, \mathbb{C})$; see (24.536). In this respect, the simplest example is a complex number $z \in \mathbb{C}$. It can be expressed as $z = x + iy$ ($x, y \in \mathbb{R}$). We rewrite it as

$$z = x \cdot 1 + y \cdot i \quad (x, y \in \mathbb{R}). \quad (25.109)$$

We have $z = 0$ if and only if $x = y = 0$. Suppose that we view a whole complex field $\mathbb{C}$ as a two-dimensional vector space. Then, 1 and i constitute the basis vectors of $\mathbb{C}$ with x and y being coefficients. In that sense, $\mathbb{C}$ is a two-dimensional *real* vector space over $\mathbb{R}$. In terms of abstract algebra, $\mathbb{C}$ is referred to as the second-order extension of $\mathbb{R}$.

We will not go into further detail about this issue. Instead, further reading is given at the end of this chapter (see Refs. [2, 3]).

25.4 Further Topics of Lie Algebra

In relation to the advanced topics of the Lie algebra given in this chapter, we wish to mention related topics below. In Sect. 15.2 we have had various properties of the exponential functions of matrices. Here, it is worth adding some more properties of those functions. We have additional important properties below [1].

- (10) Let A be a symmetric matrix. Then, $\exp A$ is a symmetric matrix.
 (11) Let A be an Hermitian matrix. Then, $\exp A$ is an Hermitian matrix.
 (12) Let $\exp A$ be a symmetric matrix. Then, A is a symmetric matrix.
 (13) Let $\exp A$ be an Hermitian matrix. Then, A is an Hermitian matrix.

Properties (10) and (11) are evident from Properties (3) and (4), respectively. To show Property (12), we have

$$(\exp tA)^T = \exp(tA^T) = \exp tA, \quad (25.110)$$

where with the first equality we used Property (3) of (15.31). Differentiating $\exp(tA^T)$ and $\exp tA$ of (25.110) with respect to t and taking a limit of $t \rightarrow 0$ in the derivative, we obtain $A^T = A$. In a similar manner, we can show Property (13).

With $\mathfrak{so}_0(3, 1)$ we have six basis vectors (in a matrix form) as shown in (24.220). Of these, three vectors are real skew-symmetric (or anti-Hermitian) matrices and other three vectors are real symmetric (or Hermitian) matrices. Then, their exponential functions are real orthogonal matrices [from Property (6) of Sect. 15.2] and real symmetric (or Hermitian) matrices [from Properties (12) and (13) as shown above], respectively. Since eigenvalues of the Hermitian matrix are real, the eigenvalues corresponding to its exponential function are positive definite from Theorem 15.3. Thus, naturally we have a good example of the polar decomposition of a matrix with respect to the Lorentz transformation matrix; see Theorem 24.7 and Corollary 24.3. We have a next example for this.

Example 25.6 Returning to the topics dealt with in Sects. 24.1 and 24.2, let us consider the successive Lorentz transformations of the rotation about the z -axis and the subsequent boost in the positive direction of the z -axis. It is a case where the direction of rotation axis and that of boost coincide.

The pertinent elements of the Lie algebra are given by

$$X_r = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, X_b = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad (25.111)$$

where X_r and X_b stand for the rotation and boost, respectively. We express the rotation by θ and the boost by ω as

$$X_r^\theta = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\theta & 0 \\ 0 & \theta & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, X_b^\omega = \begin{pmatrix} 0 & 0 & 0 & -\omega \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -\omega & 0 & 0 & 0 \end{pmatrix}. \quad (25.112)$$

Notice that X_r and X_b are commutative and so are X_r^θ and X_b^ω . Therefore, on the basis of Theorem 15.2 we obtain

$$\exp X_r^\theta \exp X_b^\omega = \exp X_b^\omega \exp X_r^\theta = \exp(X_r^\theta + X_b^\omega). \quad (25.113)$$

Meanwhile, we define X as

$$X \equiv X_r^\theta + X_b^\omega = \begin{pmatrix} 0 & 0 & 0 & -\omega \\ 0 & 0 & -\theta & 0 \\ 0 & \theta & 0 & 0 \\ -\omega & 0 & 0 & 0 \end{pmatrix}. \quad (25.114)$$

Borrowing (24.224) and the related result, as the successive Lorentz transformations Λ we get

$$\begin{aligned} \Lambda &= \exp X_r^\theta \exp X_b^\omega = \exp X_b^\omega \exp X_r^\theta = \exp(X_r^\theta + X_b^\omega) = \exp X \\ &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cosh \omega & 0 & 0 & -\sinh \omega \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\sinh \omega & 0 & 0 & \cosh \omega \end{pmatrix} \\ &= \begin{pmatrix} \cosh \omega & 0 & 0 & -\sinh \omega \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ -\sinh \omega & 0 & 0 & \cosh \omega \end{pmatrix}. \end{aligned} \quad (25.115)$$

It can be readily checked that Λ is normal. Thus, (25.115) provides a good illustration for the polar decomposition. As expected from the discussion of Sect. 24.9.1, two matrices with the second last equality of (25.115) are commutative by virtue of Corollary 24.2. We have

$$\begin{aligned}
\Lambda\Lambda^\dagger &= \Lambda^\dagger\Lambda = (\exp X)(\exp X^\dagger) = (\exp X^\dagger)(\exp X) = \exp(X + X^\dagger) \\
&= \exp(2X_b^\omega) = \begin{pmatrix} \cosh 2\omega & 0 & 0 & -\sinh 2\omega \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\sinh 2\omega & 0 & 0 & \cosh 2\omega \end{pmatrix}. \quad (25.116)
\end{aligned}$$

As shown above, if the successive transformations comprise the rotation around a certain axis (the z -axis in the present case) and the boost in the direction of that same axis, the corresponding representation matrices Λ and $\Lambda^\dagger$ are commutative (i.e., Λ is normal). In this case, the effects of the rotation of Λ and $\Lambda^\dagger$ are cancelled out and we are only left with a doubled effect of the boosts.

25.5 Closing Remarks

In this book we have dealt with classical electromagnetism and quantum mechanics that are bridged by the quantum field theory. Meanwhile, chemistry-oriented topics such as quantum chemistry were studied in a framework of materials science and molecular science including the device physics. The theory of vector spaces has given us a bird's-eye view of all these topics. Theory of analytic functions and group theory supply us with a powerful tool for calculating various physical quantities and invariants that appeared in the mathematical equations. The author is sure that the readers acquire ability to attack various problems in the mathematical physical chemistry.

References

1. Yamanouchi T, Sugiura M (1960) Introduction to continuous groups. Baifukan, Tokyo. (in Japanese)
2. Takeuchi G (1983) Lie algebra and the theory of elementary particles. Shokabo, Tokyo. (in Japanese)
3. Gilmore R (2005) Lie groups, Lie algebras, and some of their applications. Dover, New York

Index

A

- Abelian group, 648, 715, 737, 772, 789
- Abscissa axis, 185, 202
- Absolute convergence, 609
- Absolute temperature, 352
- Absolute value, 122, 129, 141, 144, 202, 211, 222, 349, 366, 591, 804, 809, 837, 943, 1021
- Abstract algebra, 877, 883, 914, 1293, 1301
- Abstract concept, 457, 661, 714, 915, 922
- AC5, 374–377
- AC'7, 376
- Accumulation point, 196–198, 230, 232, 233
- Adherent point, 192, 197
- Adjoint boundary functionals, 411, 415
- Adjoint Dirac equation, 1029, 1081
- Adjoint Green's functions, 419, 424
- Adjoint matrix, 22, 550, 610
- Adjoint operator, 111, 409, 411, 549, 561–566, 1166, 1232
- Algebraic branch point, 264
- Algebraic equation, 15
- Allowed transition, 140, 149, 790, 803, 825
- Allyl radical, 714, 715, 773, 797–805
- Aluminum-doped zinc oxide (AZO), 377, 378, 384–386
- Ampère, A.-M., 281
- Ampère's circuital law, 281
- Amplitude, 287, 293, 294, 308, 310, 313, 315, 316, 321, 336, 344, 345, 347, 364, 396, 398, 445, 446, 1073, 1115–1120, 1124, 1126–1136
- Analytical equation, 3
- Analytical mechanics, 977, 983, 999
- Analytical method, 80, 85, 94
- Analytical solution, 45, 109, 155
- Analytic continuation, 232–234, 241, 273
- Analytic function, 185–274, 330, 607, 608, 898, 1048, 1304
- Analytic prolongation, 234
- Anderson, C., 1036
- Angle-dependent emission spectra, 377, 388, 390, 391
- Angle-dependent spectroscopy, 393
- Angular coordinate, 202, 203
- Angular frequency, 4, 7, 32, 129, 131, 142, 287, 353, 358, 359, 364, 369, 994, 995, 1002, 1107
- Angular momentum
 - generalized, 75–79, 839, 840, 874, 904, 926, 1290
 - operator, 30, 89, 94, 144, 147
 - orbital, 60, 79–110, 124, 144, 874, 926, 959
 - spin, 79, 873, 874, 959
 - total, 139, 959, 960, 962
- Anisotropic
 - crystal, 381, 392
 - dielectric constant, 380
 - medium, 380
- Annihilation operator, 31, 41, 113, 114, 998, 1005, 1026, 1035–1037, 1039–1041, 1061, 1065, 1097, 1104, 1115, 1116
- Anti-commutative, 974, 1043
- Anti-commutator, 947, 1033
- Anti-Hermitian, 29, 111, 119, 409, 830, 901, 968, 972, 974, 1030, 1086, 1191, 1208, 1259, 1267, 1268, 1298, 1299
- Anti-Hermitian operator, 29, 111, 636, 834, 835, 839, 901, 904, 1288, 1300
- Antilinear, 552

- Antiparticle, 967, 970, 1039–1040, 1256
 Anti-phase, 349
 Antipodal point, 924, 925
 Antisymmetric, 435, 750–753, 890–892, 895, 1003, 1053, 1278, 1297
 Approximation method, 155–183
 Arc tangent, 66, 386
 Argument, 30, 57, 68, 156, 192, 203, 234, 238, 245, 251, 255, 261, 263, 264, 267, 268, 270, 273, 313, 328, 330, 365, 407, 416, 420–423, 425, 427, 428, 431, 434, 435, 438, 440, 441, 480, 489, 516, 525, 527, 543, 558, 581, 598, 604, 634, 642, 649, 673, 684, 707, 719, 729, 733, 757, 761, 766, 773, 777, 790, 825, 834, 846, 849, 856, 858, 878, 932, 940, 988, 1003, 1008, 1017, 1068, 1130, 1144, 1156, 1181, 1220, 1257, 1258
 Aromatic hydrocarbons, 755, 773
 Aromatic molecules, 783, 797, 805
 Associated Laguerre polynomials, 59, 110, 118–124
 Associated Legendre differential equation, 85, 94–105, 450
 Associated Legendre functions, 59, 85, 97, 105–109
 Associative law, 24, 466, 566, 567, 650, 658, 689, 692, 943, 1083, 1173
 Asymmetric, 1037
 Atomic orbitals, 761, 768, 771–773, 775, 778, 783, 784, 791, 792, 798, 805, 813
 Attribute, 662, 663
 Azimuthal angle, 30, 72, 700, 819, 861, 869, 924, 925, 1194, 1196
 Azimuthal quantum number, 122, 123, 144, 149
 Azimuth of direction, 689
- B**
 Back calculation, 587
 Backward wave, 286, 343, 347
 Ball, 226, 227, 935, 939
 Basis
 functions, 90, 709–717, 746, 747, 771, 776, 786, 805, 806, 827, 844–846, 849, 869, 873, 874, 876, 877, 880, 890–892, 894, 895, 1198, 1199
 set, 63, 483, 542, 543, 566, 576, 610, 711, 718, 719, 746, 766, 877, 879, 885, 909, 912, 922, 934, 935, 1086, 1143–1148, 1151, 1152, 1154, 1156, 1158, 1160–1162, 1164, 1167, 1174, 1176, 1180, 1193, 1194, 1289, 1300
 vectors, 8, 42, 61, 63, 143, 159, 300, 418, 457, 459, 464, 465, 467, 469–471, 473, 475, 478–484, 495, 497, 501, 503, 514, 516–518, 522, 532–535, 540, 542, 544–546, 554, 558, 562–566, 571, 601, 662, 666, 675, 677, 678, 681, 683, 695, 703, 705, 709, 710, 712, 714, 715, 717, 730, 737, 742, 746, 764, 770, 773, 787, 791, 798, 810, 831, 834, 873, 883, 895, 903, 960, 1025, 1086, 1150, 1160, 1175, 1194, 1198, 1203, 1259, 1261, 1276, 1287–1290, 1302
 BCs, *see* Boundary conditions (BCs)
 Benzene, 674, 755, 773, 789, 791–797, 799
 Bessel functions, 642, 643
 Bijective, 472–474, 476, 648, 655, 924, 926, 1149, 1157, 1158, 1168, 1276, 1281, 1284, 1285, 1287, 1301
 Bijective homomorphism, 655, 656
 Bilinear form, 935, 936, 938, 939, 1155, 1157, 1158, 1162, 1163, 1168, 1177–1179, 1183, 1187, 1193, 1261, 1263
 Bilinearity
 of tensor product, 1152
 Binomial expansion, 240, 241, 883
 Binomial theorem
 generalized, 240
 Biphenyl, 674
 Bithiophene, 674
 Blackbody, 351–353
 Blackbody radiation, 351–353
 Block matrix, 712, 891
 Blueshift, 388
 Blueshifted, 389–392, 394
 Bohr radius
 of hydrogen, 135, 822
 of hydrogen-like atoms, 110
 Boltzmann distribution law, 351, 360
 Born probability rule, 589
 Boron trifluoride, 678
 Bose–Einstein distribution functions, 353
 Boson, 1097–1102, 1139
 Boundary
 functionals, 402, 406, 410, 411, 415, 431
 term, 409
 Boundary conditions (BCs)
 adjoint, 415
 homogeneous, 445
 homogeneous adjoint, 415–417
 inhomogeneous, 424, 623

- Bounded, 25, 29, 77, 81, 212, 219, 221–223, 229, 1192, 1263
 BP1T, 374–377
 BP3T, 392–394
 Bragg's condition, 389, 390, 392
 Branch, 259, 261–264, 643
 Branch cut, 261, 262, 264–269, 271–273
 Branch point, 260, 262, 264–266, 268, 271–273
 Bra vector, 22, 549, 552, 555
 Brewster angle, 322–325
- C**
 Canonical basis, 1156, 1174
 Canonical coordinate, 28, 977, 993, 1055, 1059, 1089
 Canonical forms of matrices, 485–548, 583
 Canonical momentum, 977, 993, 994, 1055, 1059, 1089
 Canonical variables, 983, 993
 Cardinalities, 188, 1083
 Cardinal numbers, 188
 Carrier space, 713
 Cartan–Weyl basis, 1287–1297
 Cartesian coordinate, 8, 47, 111, 175, 291, 361, 677, 762, 819, 821, 834, 852, 919, 978, 980, 1124, 1218
 Cartesian space, 61, 460, 461
 Cauchy–Hadamard theorem, 209–212
 Cauchy–Liouville Theorem, 222
 Cauchy–Riemann conditions, 209–212
 Cauchy–Schwarz inequality, 551, 611
 Cauchy's integral formula, 213–223, 225, 232, 236
 Cauchy's integral theorem, 216, 217, 251, 1021–1023
 Cavity, 33, 351, 353, 355, 357, 396, 397, 399
 Cavity radiation, 351
 Cayley's theorem, 689
 Center of inversion, 668
 Center-of-mass coordinate, 60
 Character, 479, 654, 657, 663, 723, 725, 726, 728, 729, 748–750, 767, 772, 774–777, 782–787, 792, 797, 803, 811, 825, 871, 872, 874, 943, 1156, 1164, 1175
 Characteristic equation, 443, 486, 487, 528, 591, 597, 715, 861
 Characteristic impedance
 of dielectric media, 279, 280, 288, 305–351
 Characteristic polynomial, 486, 508, 540, 543, 547, 974
 Charge conjugation, 967–971, 1039, 1040, 1081, 1209, 1210, 1221, 1227
 Chemical species, 677
 Circular argument, 156
 Circular cone, 1185, 1186
 Circularly polarized light
 left-, 138, 140, 150, 302
 right-, 138, 140
 Clad layer, 332, 339, 340, 343, 386
 Classes, 647, 651–653, 679–681, 683, 685, 688, 725, 726, 731–737, 871
 Classical Hamiltonian, 33
 Classical numbers, 10
 Classical orthogonal polynomials, 49, 97, 401, 450
 Clebsch–Gordan coefficient, 873–895, 1294
 Clopen set, 196, 199, 200, 921
 Closed interval, 201
 Closed-open set, 196
 Closed path, 199, 200, 215, 216, 922
 Closed set, 190, 191, 193–195, 198, 201, 219, 921
 Closed shell, 767, 768, 782
 Closed surface, 281, 282
 Closure, 191–194, 680, 943, 1083
 C-number, 10, 22, 995, 1015, 1026, 1032, 1033, 1038, 1039, 1059, 1099, 1133
 Cofactor, 476, 477, 498
 Cofactor matrix, 498
 Coherent state, 128, 129, 135, 141, 152
 Coinstantaneity, 944
 Column vector, 8, 22, 42, 90, 91, 93, 299, 465, 481, 483, 488, 489, 491, 492, 529, 533–535, 537, 553, 560, 567, 580, 581, 592, 597, 598, 602, 622, 640, 684, 710, 786, 809, 839, 846, 849, 861, 894, 935, 937, 938, 947, 949–953, 957, 968, 969, 1030, 1144–1146, 1152–1154, 1174, 1176, 1193, 1201, 1210–1212, 1239, 1240
 Common factor, 504, 505, 509, 510
 Commutable, 28, 509, 513–515, 537, 550, 580, 603, 668, 669, 900
 Commutation relation
 anti-, 1032, 1035–1039, 1041, 1098, 1116
 canonical, 3, 28–30, 44, 45, 56, 61, 67, 68, 977, 993, 1061, 1065
 equal-time, 977, 993–999, 1005, 1015
 Commutative, 28, 175, 499, 544, 611, 617, 629, 630, 641, 648, 677, 715, 854, 855, 896, 901, 902, 919, 921, 945, 960–962, 1062, 1088–1090, 1168, 1200, 1202, 1240, 1248, 1249, 1265, 1288, 1303, 1304
 Commutative group, 648
 Commutative law, 648

- Commutator, 28–30, 146, 147, 899, 903, 947, 998, 1005–1007, 1017, 1032, 1062, 1089, 1272
- Compact group, 1192, 1263
- Compatibility relation, 776
- Compendium quantization method, 1031
- Complement, 187, 188, 190, 194, 195, 571, 572, 583, 1178, 1179, 1181, 1184, 1186, 1187
- Complementary set, 188, 219, 918
- Completely reducible, 712, 714, 855, 860
- Completeness relation, 746, 1214, 1232, 1233, 1237
- Complete orthonormal system (CONS), 64, 155, 159, 167, 170, 171, 177, 873, 952, 962, 969, 989, 1091, 1239
- Complete system, 159
- Complex amplitude, 313
- Complex analysis, 185, 186, 203
- Complex analytic representation, 1293, 1294
- Complex conjugate, 22, 24, 25, 29, 34, 119, 140, 205, 360, 412, 419, 423, 552, 562, 718, 849, 881, 907, 932, 970, 1030, 1129
- Complex conjugate representation, 846
- Complex conjugate transposed matrix, 22, 550, 740, 789
- Complex domain, 19, 205, 223, 234, 272–274, 326, 1012
- Complex function, 14, 185, 205, 210, 212, 213, 411, 790, 1047
- Complexification, 1268, 1269, 1281, 1288–1294, 1298–1300
- Complex number field, 1142, 1155
- Complex phase shift, 341
- Complex plane, 14, 185, 186, 202–204, 206, 207, 212, 213, 217–219, 223, 249, 250, 258–261, 272, 328, 329, 548, 915
- Complex roots, 243, 442, 443
- Complex variable, 15, 204–213, 222, 237, 557, 886, 1011
- Composite function, 66, 979, 981
- Compton, A., 4, 6, 1075, 1078, 1102, 1109, 1111
- Compton effect, 4
- Compton scattering, 1075, 1078, 1102, 1109, 1111, 1113–1139
- Compton wavelength, 4
- Condon–Shortley phase, 103, 105, 109, 125, 135, 141–143
- Conductance, 336
- Conic section, 1185
- Conjugacy classes, 651, 680, 681, 683, 685, 688, 725, 734, 737
- Conjugate element, 651
- Conjugate momentum, 979
- Conjugate representations, 789
- Connected
 - component, 916–918, 921, 926, 1189, 1255, 1258, 1263
 - set, 916, 917, 919, 923, 1254, 1263
- Connectedness, 199, 200, 204, 915–926, 1241–1258
- CONS, *see* Complete orthonormal system (CONS)
- Conservation law, 283, 933
- Conservation of charge, 282
- Conservation of energy, 4
- Conservative force, 982, 1078
- Conserved quantity, 959
- Constant loop, 922, 1250
- Constant matrix, 629, 630, 643
- Constant term, 173
- Constructability, 423
- Continuity equation
 - current, 282, 1054
- Continuous function, 220, 915, 917, 922, 1258
- Continuous group, 196, 612, 689, 827–926, 1141, 1241, 1242, 1293, 1294
- Continuous spectra, 961
- Contour
 - integral, 213–215, 236–238, 244, 246, 248, 250, 1021, 1047
 - integration, 217–219, 223, 227, 237, 239, 243, 245, 247, 255, 267, 270, 271, 1011, 1020–1023, 1048
 - map, 755, 756, 758, 759
- Contraposition, 210, 230, 404, 474, 503, 554, 693, 854, 1179
- Contravariant
 - tensor, 1051, 1173, 1174
 - vector, 937–939, 1061, 1174, 1177
- Convergence circle, 226, 227, 233
- Convergence radius, 226
- Converse proposition, 617, 854
- Coordinate point, 298, 301
- Coordinate representation, 45–53, 110, 114, 124, 129, 131, 139, 149, 161, 167, 170, 172, 174, 175, 179, 181, 182, 264, 418, 819, 852, 867, 954, 1049, 1086, 1124, 1184, 1193, 1195, 1199, 1201, 1237, 1245
- Coordinate space, 977
- Coordinate transformation, 300, 382, 661, 667, 703, 706, 764, 845, 868, 1167, 1169, 1177, 1187
- Core layer, 300, 382, 661, 667, 703, 764, 845, 868, 1167, 1169, 1177, 1187
- Corpuscular beam, 6, 7
- Correction term, 157–160

- Coset
 - decomposition, 650, 652, 677, 920, 1255
 - left, 650, 651
 - right, 650, 651
- Cosine rule, 1138
- Coulomb gauge, 1056
- Coulomb integral, 766, 778, 794, 816
- Coulomb potential, 59, 69
- Coulomb's law, 278
- Countable, 14, 588, 647, 989
- Covariant
 - derivative, 1076, 1080, 1082, 1083, 1085
 - form, 1052, 1060, 1064, 1095, 1202
 - tensor, 1173, 1174
 - vector, 938, 939, 945, 1061, 1174, 1177
- Covering group
 - universal, 926, 1241, 1242, 1249
- Cramer's rule, 317, 428, 837
- Creation operator, 41, 998, 1004, 1005, 1026, 1035, 1036, 1097, 1104
- Critical angle, 322–325, 328, 329, 331, 341
- Cross-section area, 367, 369, 397
- Crystal
 - anisotropic, 392
 - organic, 372, 377, 380, 389, 390, 393–395, 594
- Cubic groups, 688
- Cyclic group, 648
- Cyclic permutation, 667, 703
- Cyclopropenyl radical, 773, 783–791, 794, 796, 799

- D**
- d'Alembertian, 932
- d'Alembert operator, 932
- Damped oscillator, 441, 443, 445, 446
- Damping
 - constant, 441
 - critical, 442
 - over, 442
 - weak, 442, 443
- Darboux's inequality, 219–220, 238
- de Broglie, L.-V., 6
- de Broglie wave, 6
- Definite integral, 26, 101, 114, 134, 139, 175, 236, 743, 749, 767, 769, 820, 1048
- Degeneracy, 156
- Degenerate
 - doubly, 795, 974, 1212, 1219, 1220, 1235, 1238, 1239, 1257
 - triply, 817, 823, 824
- δ function, 418, 420, 422, 425, 444, 988, 1007, 1008, 1024, 1025, 1047, 1065, 1068, 1106, 1116, 1117, 1122, 1124
- de Moivre's theorem, 203
- de Morgan's law, 189
- Derived set, 196, 197
- Determinant, 43, 61, 296, 405, 432, 457, 473–478, 544, 553, 557, 591, 596, 673, 680, 689, 690, 770, 815, 837, 895, 917, 918, 921, 943, 945, 949, 974, 1163, 1188, 1189, 1249, 1254, 1257
- Device physics, 342, 372, 395, 1304
- Device substrate, 377, 384
- Diagonal elements, 89, 90, 93, 296, 475, 477, 487, 488, 492, 503, 523, 525, 526, 540, 545, 583, 595, 614, 615, 681, 716, 723, 771, 809, 811, 901, 902, 904, 936, 1034, 1160, 1161, 1239
- Diagonalizable, 502, 513, 540, 543, 544
- Diagonalizable matrix, 539–548, 585
- Diagonalization, 493, 543, 592, 598–605, 627, 694, 716, 952, 1087, 1247
- Diagonalized, 44, 502, 504, 513, 540, 542, 545, 556, 571, 580, 581, 583, 585, 591, 593, 603, 605, 627, 632, 716, 717, 809, 832, 951–953, 962, 1087, 1141, 1160, 1161, 1204, 1219–1221, 1229, 1230, 1235, 1239, 1240, 1242, 1290, 1292
- Diagonal matrix, 91, 485, 512, 543, 560, 580, 582, 583, 592, 593, 598, 627, 631, 664, 707, 715, 764, 809, 834, 856, 858, 910, 1034, 1161, 1204, 1258
- Diagonal positions, 492, 513
- Dielectric constant, 278, 279, 348, 349, 380, 1049
- Dielectric constant ellipsoid, 381, 387, 391
- Dielectric media, 279, 280, 288, 294, 305–351
- Differentiable, 67, 205, 207, 210, 212, 222, 283, 428, 607, 617, 898, 899
- Differential cross-section, 1122, 1124
- Differential equations
 - inhomogeneous, 402, 405, 424, 430, 440, 624, 626, 634
 - in a matrix form, 621–626
 - second-order, 941
 - system of, 607, 618–637, 643, 644
 - theory of, 431, 450
- Differential operator, 15, 19, 26, 27, 51, 52, 71, 74, 80, 171, 177, 278, 401, 408, 410, 411, 413, 415, 416, 421, 424–426, 431, 436, 448–450, 549, 571, 932, 1050, 1203

- Diffraction grating, 372, 375–378, 386, 390, 393, 395
 - Diffraction order, 379, 392
 - Dimensionless, 75, 80, 81, 110, 278–280, 287, 368, 372, 819, 829
 - Dimensionless parameter, 51, 156
 - Dimension theorem, 469, 473, 505, 519, 520, 541, 1286
 - Dipole
 - approximation, 129, 144, 359
 - isolated, 365
 - oscillation, 351, 364
 - radiation, 361–367
 - transition, 127–129, 131, 134, 149, 766, 780
 - Dirac, P., 11, 12, 22, 549, 931
 - Dirac adjoint, 958, 964, 965, 967, 971, 1097, 1237, 1240
 - Dirac equation, 12, 399, 931–975, 991, 1032, 1081, 1085, 1141, 1187, 1193, 1194, 1197–1204, 1208, 1210, 1211, 1214, 1218, 1220–1222, 1227–1235, 1240
 - Dirac field, 960, 962, 963, 970, 977, 991, 993, 999, 1005, 1011, 1023, 1026–1049, 1061, 1071, 1075, 1077–1079, 1083, 1084, 1096, 1098, 1101, 1111, 1211
 - Dirac operators, 1141, 1203, 1204, 1208, 1212–1229, 1235–1241
 - Direct factor, 658, 677
 - Direction cosine, 288, 315, 684, 701, 702, 862, 864, 865
 - Direct-product group, 647, 657–659, 677, 746, 749, 750, 874, 919
 - Direct product space, 1142, 1155, 1171
 - Direct sum, 196–198, 460, 462, 497, 500, 501, 504–508, 514–516, 523, 540, 542, 543, 571, 574, 659, 712, 717, 725, 728, 746, 768, 775, 797, 808, 855, 873, 916, 919, 1178, 1192, 1204, 1233, 1298–1300
 - Dirichlet conditions, 15, 19, 26, 73, 354, 357, 416, 430
 - Disconnected, 199, 200
 - Discrete set, 196, 197
 - Discrete topology, 199
 - Disjoint sets, 199, 200
 - Dispersion of refractive index, 373–376
 - Displacement current, 281, 283
 - Distance function, 185, 202, 204, 550
 - Distributive law, 189, 467
 - Divisor, 499, 500, 541, 675
 - Domain
 - of analyticity, 213, 215–217, 219, 223, 234
 - complex, 19, 205, 223, 234, 272–274, 326, 1012
 - of variability, 185
 - Dual basis, 1156, 1158, 1167
 - Dual space, 552, 1156, 1163, 1173
 - Dual vector, 552, 1155–1165, 1174
 - Dynamical system, 982–984, 989, 991, 1004
 - Dynamical variable, 959
- E**
- Effective index, 377, 383, 390–394
 - Eigenenergy, 44, 113, 123, 155–160, 167, 168
 - Eigenequation, 486
 - Eigenfunction, 17–19, 24, 25, 27, 36, 37, 40, 42, 69–71, 74, 89, 112, 114, 139, 156, 177, 452, 716, 742, 746, 765, 766, 796, 800, 801, 821, 856, 892, 894, 951–955, 957, 962, 964, 965, 967, 986, 987, 1234, 1239
 - Eigenspace, 486, 495–500, 505–512, 514, 515, 523, 540, 542, 543, 583, 589, 602, 746
 - Eigenspinor, 1220
 - Eigenstate
 - energy, 167, 168, 180
 - simultaneous, 68, 71, 75, 81, 88, 92, 93, 599–605, 962
 - Eigenvalue
 - equation, 13, 24, 29, 33, 52, 128, 155, 156, 164, 176, 177, 386, 430, 486, 492, 529, 548, 560, 602, 755, 760, 769, 771, 816, 954, 956, 962, 989
 - problem, 14, 16, 19, 21, 24, 122, 448–453, 485, 548, 603, 717, 746, 813, 962, 1199, 1203
 - Eigenvector, 27, 37, 42, 76, 89, 156, 157, 159, 299, 485–495, 498, 500–505, 519–522, 525, 528–534, 537, 538, 540, 542–544, 546, 548, 580, 583, 585, 590, 599–604, 681, 691, 693, 702, 717, 770, 772, 809, 861, 862, 952, 953, 962, 964, 989, 1199, 1201, 1211, 1212, 1234, 1239, 1240
 - Einstein, A., 3, 4, 6, 7, 351, 367
 - Einstein coefficients, 359, 361, 365
 - Einstein summation convention, 937, 1177
 - Electric dipole
 - approximation, 144
 - moment, 129, 169, 170, 767
 - operator, 824
 - transition, 127–129, 131, 134, 149, 766, 780
 - Electric displacement, 278
 - Electric field, 129, 150, 152, 161, 164, 168, 170, 178, 277, 278, 280, 293–295, 297, 298, 301, 302, 307, 313–317, 321, 326, 336, 337, 340–343, 347–349, 357,

- 363–365, 383–385, 396, 397, 441, 637, 781, 824, 1051
 - Electric flux density, 278, 382
 - Electric permittivity tensor, 380
 - Electric wave, 293–295
 - Electromagnetic field, 33, 129, 151, 305–307, 313–315, 332, 363, 377, 380, 384, 386, 398, 594, 643, 977, 991, 993, 1023, 1049–1073, 1075, 1077–1079, 1083, 1084
 - Electromagnetic induction, 280
 - Electromagnetic plane wave, 290
 - Electromagnetic potential, 1051, 1077
 - Electromagnetic wave, 129, 277, 289–303, 305–351, 353–358, 365, 370, 372, 382, 383, 396, 401, 441, 637, 1059
 - Electromagnetism, 170, 277, 278, 287, 450, 594, 1049, 1061, 1304
 - Electronic configuration, 767, 780–782, 797, 802, 803, 824, 825
 - Element, 47, 86, 129, 186, 296, 359, 457, 487, 549, 577, 607, 647, 661, 705, 761, 841, 936, 1032, 1083, 1145, 1271
 - Elementary charge, 60, 129, 151, 359, 1076
 - Elementary particle
 - physics, 1140, 1258
 - reaction, 1120
 - Ellipse, 297, 298, 301, 302, 381, 599
 - Ellipsoidal coordinates, 819
 - Elliptically polarized light, 297, 298, 301, 1061
 - Emission
 - direction, 377, 393
 - gain, 392, 394
 - grazing, 379, 390, 391
 - lines, 370, 374, 375, 377, 393, 394
 - optical, 372, 768
 - spontaneous, 358, 359, 365, 366, 368
 - stimulated, 358, 359, 367, 368
 - Empty set, 188–190, 652
 - Endomorphism, 457, 465, 468, 469, 471–474, 478, 480, 648, 656, 907, 908, 1086, 1152, 1172, 1173, 1272, 1275, 1279, 1286, 1301
 - Energy conservation, 4, 322, 1137
 - Energy eigenvalue, 20, 31, 36, 37, 53, 123, 135, 155, 156, 158, 162, 176, 717, 746, 765, 777, 779, 780, 789, 795, 796, 802, 814, 823, 962, 967
 - Energy level, 41, 157–160, 176, 358, 781, 824
 - Energy-momentum, 991, 1107
 - Energy transport, 319–322, 331, 347
 - Entire function, 213, 222, 223, 250, 1048
 - Equation of motion, 60, 151, 441, 933, 981, 992–994, 1079, 1089
 - Equation of wave motion, 277, 284–289, 334
 - Equilateral triangle, 677, 678, 783, 784
 - Equivalence
 - law, 512, 593
 - relation, 916
 - theorem, 932
 - transformation, 570, 593, 594, 1161, 1162
 - Equivalent transformation, 593
 - Ethylene, 755, 773–784, 786, 788, 790, 791, 797, 799
 - Euclidean space, 1165, 1177–1187
 - Euler angles, 695–704, 836, 840, 842, 845, 849, 856, 861, 911, 1261
 - Euler equation
 - second-order, 173
 - Euler–Lagrange equation, 982, 992, 993, 1057
 - Euler’s formula, 203
 - Euler’s identity, 203
 - Evanescent mode, 331
 - Evanescent wave, 332, 338–343
 - Even function, 19, 51, 134, 161, 867, 873, 1024
 - Even permutations, 475, 904, 1095, 1111
 - Excitation
 - optical, 372, 825
 - strong, 373, 393
 - weak, 393, 394
 - Excited state, 37, 44, 134, 156, 178, 179, 181, 351, 352, 358, 359, 366, 780, 797, 802, 803, 824, 825
 - Existence probability, 21, 32, 987
 - Existence probability density, 142
 - Expansion coefficient, 177, 225, 418, 866, 994, 1261
 - Expectation value, 25, 33–36, 53, 54, 70, 176, 178, 588–590, 1015–1017, 1031, 1043, 1079, 1098–1100
 - Exponential function, 15, 248, 290, 308, 309, 446, 607–644, 827, 830, 836, 856, 921, 946, 1048, 1082, 1086, 1301, 1302
 - Exponential polynomial, 309
 - External force field, 60, 441
 - Extremals, 131, 347
 - Extremum, 180, 388, 984
- F**
- Fabry–Pérot resonator, 373
 - Factor group, 653, 657, 677
 - Faraday, M., 280
 - Faraday’s law of electromagnetic induction, 280
 - Fermion, 1036, 1043, 1095, 1098–1103, 1110, 1111, 1139
 - Feynman
 - amplitude, 1073, 1115–1120, 1124, 1126, 1129

- Feynman (*cont.*)
 diagrams, 1103–1113, 1115–1120, 1139
 propagator, 977, 989, 994, 1007,
 1015–1023, 1028, 1043–1049, 1071,
 1072, 1103, 1109, 1111, 1112
 rules, 1103–1113
 FIB, *see* Focused ion beam (FIB)
 Fine structure constant, 1095, 1139
 First-order approximation, 181, 182
 First-order linear differential equation
 (FOLDE), 45, 46, 401, 406–411, 619,
 620, 624, 634
 First-order partial differential equation, 277
 Fixed coordinate system, 666, 683, 696, 704,
 861, 924
 Flux, 151, 278–280, 322, 365, 369, 382, 638,
 1050, 1051
 Fock
 basis, 1005
 space, 999–1005
 state, 1092
 Focused ion beam (FIB), 377, 395
 FOLDE, *see* First-order linear differential
 equation (FOLDE)
 Forbidden, 127, 781, 797, 825, 961
 Force field
 central, 59, 109
 external, 60, 441
 Forward wave, 286, 343, 345, 347
 Four-group, 664, 674, 1255
 Fourier
 analysis, 985–990
 coefficients, 171
 components, 1026, 1117, 1203
 cosine coefficients, 873
 cosine series, 873
 counterpart, 1049, 1116
 integral transform, 987–990
 series expansion, 985–987
 Fourier transform
 inverse, 985, 987–990
 pairs, 989
 Four-momentum, 1004, 1017, 1104, 1107,
 1119, 1132, 1133, 1137, 1228
 Four-vector, 934, 945, 1003, 1024, 1026, 1035,
 1052–1055, 1059, 1076, 1079, 1131,
 1133, 1135, 1136, 1193
 Fréchet axiom, 200
 Free field, 960, 1075, 1085
 Free particle, 21, 932, 1026, 1085
 Free space, 305, 337, 343, 365
 Functionals, 13, 15, 103, 120, 124, 136, 142,
 165, 171, 173, 206, 210, 225, 234, 241,
 287, 402, 404, 406, 410, 411, 415, 431,
 549, 620, 743, 760, 767, 772, 780, 885,
 959, 983, 1023, 1047, 1051, 1083, 1156,
 1202, 1212, 1214, 1240
 Functional space, 549
 Function space, 25, 649, 774
 Fundamental set of solutions, 15, 405,
 426–428, 431–434, 441–443, 640, 644
 Fundamental solution, 634
- ## G
- Gain function, 369
 Gamma functions, 98–100, 887
 Gamma matrices, 947, 949, 952, 971–975,
 1032, 1082, 1124, 1130, 1132, 1133,
 1204, 1227
 Gauge
 field, 1055, 1082–1085
 invariant, 1084, 1130
 transformation, 1051, 1054, 1082–1084,
 1126, 1129
 Gaussian plane, 185
 Gauss' law, 278
 Gegenbauer polynomials, 97, 102
 Generalized binomial theorem, 98, 240
 Generalized coordinate, 977, 978, 980, 981,
 993
 Generalized eigenspace, 498, 505–512, 515,
 523
 Generalized eigenvalue, 486, 528, 615
 Generalized eigenvector, 498, 500–505, 519,
 521, 522, 525, 528, 531–534, 538, 540,
 548
 Generalized force, 981
 Generalized Green's identity, 432
 Generalized momentum, 979, 982, 993
 General linear group, 648, 837
 General point, 661, 756, 844, 1194
 General solution, 15, 32, 286, 405, 406, 431,
 442, 443, 994, 1230
 Generating function, 97
 Geometric figures, 673
 Geometric object, 661–664, 666, 673, 680
 Geometric series, 224, 228
 Glass, 277, 288, 295
 Global properties, 920–926
 Gram matrix, 555, 556, 558–560, 566, 569,
 570, 587, 592, 707, 1141, 1163, 1166
 Gram–Schmidt orthonormalization, 567, 580,
 742, 965, 1165, 1179
 Grand orthogonality theorem (GOT), 717–723,
 739
 Grassmann numbers, 1032–1034, 1038, 1042,
 1045
 Grating period, 379, 394
 Grating wavevector, 378, 379, 388, 391, 393

Grazing incidence, 328, 330
 Greatest common factor, 505
 Green's functions, 401–453, 607, 618, 620, 621, 644
 Green's identity, 420
 Ground state, 36, 130, 134, 156, 157, 160, 161, 168, 176, 178, 179, 181, 351, 352, 358, 359, 768, 780, 797, 802, 803, 824, 825
 Group
 algebra, 726–734
 element, 647, 649, 651, 659, 675, 680, 685, 688, 689, 705–707, 710, 712, 715, 719, 721, 723–726, 730, 733–735, 737, 738, 746, 750, 761, 772, 780, 784, 791, 855, 863–867, 870, 876, 1250
 finite, 647–649, 651, 654, 655, 657, 689, 705–707, 709, 710, 712, 717, 729, 827, 863, 870, 871, 876, 918, 920
 infinite, 648, 655, 661, 689, 827, 873, 876
 representation, 711, 720, 723, 725–727
 symmetric, 688, 689
 theory, 484, 607, 647–659, 662, 669, 705, 714, 737, 755–826
 Group index, 371, 376
 Group refractive index, 371
 Group velocity, 7, 12, 338

H

Half-integer, 78, 79
 Half-odd-integer, 78, 79, 81, 84, 841, 844, 880, 914, 1291
 Hamilton–Cayley theorem, 498–499, 509, 510, 515, 540
 Hamiltonian, 11, 21, 29, 31, 33, 36, 43, 59–69, 109, 156, 161, 168, 178, 181, 558, 761, 764, 766, 767, 771, 788, 816, 819, 960–964, 977, 983, 984, 999–1005, 1031, 1049, 1065–1071, 1076, 1077, 1079, 1085, 1088, 1091, 1094, 1104, 1105, 1108, 1113, 1115
 Hamiltonian density, 999, 1028–1035, 1049, 1055–1059, 1065, 1079, 1085, 1094, 1104
 Hamilton's principle, 984
 Harmonic oscillator, 30–59, 109–111, 127, 128, 132, 155, 156, 178, 179, 351, 353, 396, 398, 399, 401, 440, 441, 558, 774, 994, 998, 1003, 1201
 Heaviside–Lorentz units, 1049
 Heaviside step function, 422
 Heisenberg, W., 11, 1087–1091
 Heisenberg equation of motion, 1089
 Hermite differential equation, 449, 450

Hermite polynomials, 49, 51, 59, 167, 450
 Hermitian, 7, 34, 70, 132, 156, 360, 401, 555, 571, 616, 707, 765, 830, 943, 995, 1081, 1165, 1278
 Hermitian differential operator, 176, 178, 411, 421, 424, 425, 448
 Hermitian matrix, 24, 133, 555–557, 569, 592, 597, 616, 618, 627, 898, 907, 909, 951, 952, 1192, 1197, 1229, 1242, 1243, 1245–1247, 1259, 1263, 1298, 1299, 1302
 Hermitian operator, 24, 25, 29, 54, 55, 71, 79, 111, 132, 160, 177, 181, 360, 401, 424, 450, 453, 571–605, 636, 741, 766, 834, 835, 839, 901, 904, 960, 962, 972, 1030, 1069, 1087, 1104, 1221, 1243, 1250, 1258, 1259, 1288, 1300
 Hermiticity, 3, 19, 25, 26, 119, 413, 416, 424, 449, 450, 601, 709, 771, 974, 1238, 1240, 1243
 Hesse's normal form, 288, 684, 687
 High-energy physics, 949
 Highest-occupied molecular orbital (HOMO), 780, 781, 797
 Hilbert space, 26, 588, 774, 1005
 Hole theory, 1036
 Homeomorphic, 926
 HOMO, *see* Highest-occupied molecular orbital (HOMO)
 Homogeneous boundary conditions (BCs), 410, 411, 416, 417, 420, 421, 424, 425, 427, 429–433, 445, 448, 449, 623
 Homogeneous equation, 173, 402, 419, 421, 423, 424, 426, 427, 430, 441, 448, 449, 620, 623, 626, 630, 633, 639, 640
 Homomorphic, 654–657, 706, 711, 734, 1265, 1272, 1285–1287
 Homomorphism, 647, 653–657, 705–707, 739, 745, 912–914, 1263, 1283, 1286, 1287
 Homomorphism theorem, 657, 914, 1265
 Homotopic, 922, 1250, 1258
 Hückel approximation, 780
 Hydrogen, 59, 124, 127, 134–144, 167–171, 176, 781, 805, 807–810, 813–819, 962
 Hydrogen-like atoms, 59–125, 127, 134, 143, 144, 155, 157, 558, 603, 949
 Hyperbolic functions, 942, 1211, 1216
 Hypersphere, 923, 1185
 Hypersurface, 599, 923

I

Idempotent matrix, 505–513, 544, 546–548, 588, 631

Identity

element, 648, 649, 651, 653–656, 658, 663, 664, 675, 723, 777, 786, 913, 916–918, 921, 922, 942, 1083, 1189, 1255, 1258, 1263, 1271, 1273, 1282, 1283

mapping, 1264, 1301

matrix, 44, 507, 509, 511, 513, 521, 569, 570, 586, 591, 608, 666, 690, 694, 706, 720, 728, 729, 808, 830, 896, 915, 921, 939, 947, 951, 970, 975, 1030, 1134, 1203, 1214, 1229, 1230, 1233, 1263, 1265

operator, 35, 56, 159, 171, 417, 419, 507, 989, 993, 1024, 1035, 1062, 1092, 1095, 1104, 1177, 1188, 1230

transformation, 666, 689, 693, 912, 913

Image

of transformation, 468

Imaginary axis, 185, 202

Imaginary unit, 185

Improper rotation, 669, 673, 690, 807, 920

Incidence angle, 311, 325

Incidence plane, 311, 314, 315, 317, 322, 326

Incident light, 310, 312, 314

Indeterminate constant, 19

Indeterminate form of limit, 323, 960

Indiscrete topology, 199

Inequivalent, 711, 718, 720–722, 725, 734, 737, 860, 869, 871–873

Inertial frame of reference, 934, 939, 940, 943, 1085, 1121, 1187, 1207, 1222, 1227–1229

Infinite dielectric media, 294, 295, 305, 306

Infinite-dimensional, 459, 1157

Infinite power series, 608, 1095

Infinite series, 163, 224, 232

Inflection point, 242

Inhomogeneous boundary conditions (BCs), 426, 427, 430, 438, 441, 446, 447, 623, 624, 626

Inhomogeneous equation, 402, 405, 424, 430, 623, 624, 633

Initial value problems (IVPs), 401, 431–448

Injective, 472–474, 478, 653, 655, 924, 1264

Inner product

of functions, 745

space, 202, 549–571, 574, 578, 579, 587, 588, 590, 592, 610, 714, 1086, 1141, 1158, 1163, 1179, 1259

of vectors, 419, 601, 749

vector space, 574

In-quadrature, 396, 397

Integral

action, 977, 984, 991–993, 1028, 1080, 1084

line, 306, 307

surface, 305–307, 363

Integrand, 221, 239, 252, 254, 266, 268, 271, 274, 985, 992, 995, 1002, 1019, 1023, 1026, 1031, 1041, 1044, 1045, 1047, 1066, 1068, 1079, 1117

Integration

clockwise, 217

complex, 213, 236, 1011, 1019, 1020, 1023

constant, 151, 152, 397, 407, 619, 620

contour, 217–219, 223, 227, 237, 239, 243, 245, 247, 255, 267, 270, 271, 1011, 1020–1023, 1048

counter-clockwise, 217

definite, 134

by parts, 26, 29, 86, 106, 108, 132, 408, 432, 992, 1029, 1057, 1084

path, 215, 217, 229, 1018, 1023

range, 25, 247, 438, 440, 769, 995, 1011 real, 236

surface, 282, 306

termwise, 224, 228

volume, 282

Interface, 305–308, 310–315, 322, 325, 331, 332, 335–339, 343, 347–351, 357, 377, 380, 385, 386, 396, 397

Interference

constructive, 370

theory, 372

Interference modulation

of emissions, 373, 376

spectra, 374

Interior point, 191

Intermediate state, 160

Intermediate value theorem, 917

Intersection, 188, 195, 649, 669, 671, 677, 688, 924

Invariant, 487, 495–501, 504, 508, 514, 517, 529–531, 539, 540, 544, 548, 571, 583, 590, 593, 602, 603, 615, 652, 656, 659, 677, 688, 713–715, 717, 732, 746, 760, 764, 808, 831, 834, 843, 867, 869, 880, 882, 883, 892, 914, 917, 918, 935, 938, 940, 957, 967, 970, 977, 989, 994, 1005–1015, 1017, 1018, 1023, 1024, 1027, 1028, 1035, 1040–1043, 1049, 1052, 1054, 1059, 1072, 1082, 1084, 1089, 1100, 1118, 1123, 1130, 1158, 1165–1171, 1175–1177, 1187, 1189, 1207, 1214, 1227, 1255, 1261, 1263, 1304

Invariant delta functions, 977, 1005–1015, 1017, 1018, 1023, 1028, 1040–1043, 1072

Invariant scattering amplitude, 1118

Invariant subgroup, 652, 656, 659, 677, 688, 732, 914, 917, 918, 1189, 1255

Invariant subspace, 495–500, 508, 517, 529–531, 539, 540, 544, 548, 571, 583, 602, 603, 713, 714, 717, 808

Inverse element, 472, 648, 653, 655, 658, 664, 682, 706, 727, 733, 942, 1083

Inverse matrix, 43, 205, 457, 476, 479, 480, 538, 596, 622, 628, 631, 706, 937, 942, 944, 1170, 1195

Inverse transformation, 468, 472, 473, 648, 845, 847, 1257, 1258, 1284

Inversion symmetry, 668, 669, 673, 680, 1255

Inverted distribution, 368

Invertible, 472, 648, 656

Irradiance, 322, 367–370

Irrational functions, 256

Irreducible, 712, 714, 717–726, 728, 729, 734–738, 742–746, 749, 764, 766–768, 771–773, 775–777, 780, 782, 784, 786, 788, 792, 797–799, 801, 803, 806, 808, 810–812, 814, 815, 817, 823–825, 853–861, 868–874, 876–878, 891, 1291–1293

Irreducible character, 723, 868–874

Isolated point, 196–199

Isomorphic, 654, 655, 657, 674, 680, 688, 711, 760, 914, 924, 1157, 1249, 1255, 1264, 1276, 1281, 1286, 1288, 1290, 1301

Isomorphism, 647, 653–657, 705, 920, 1157, 1158, 1276, 1284–1287

IVPs, *see* Initial value problems (IVPs)

J

Jacobian, 865

Jacobi's identity, 903, 1274, 1277

j–*j* Coupling, 879

Jordan blocks, 517–528, 530–533, 540

Jordan canonical form, 485, 515–539

Jordan decomposition, 515

Jordan's lemma, 248–256, 1021, 1022

K

Kernel, 468, 529, 655–657, 912–914, 1263–1265, 1283–1285

Ket vector, 22, 549, 552

Klein–Gordon equation, 931–934, 946, 991, 992, 994, 1097, 1197, 1200–1204

Klein–Gordon field, 991, 994, 999, 1028

Kronecker delta, 50, 937

Kronecker product, 1149

L

Laboratory coordinate system, 381

Laboratory frame, 1121, 1136, 1138

Ladder operator, 76, 93, 94, 114

Lagrange's equations of motion, 982, 984, 985, 993

Lagrangian, 977–985, 991–993, 1028–1035, 1049, 1055–1059, 1079–1082, 1084, 1085

Lagrangian density, 991–993, 1028, 1049, 1055–1059, 1079, 1080, 1084

Lagrangian formalism, 977–985, 993

Laguerre polynomials, 59, 110, 118–124

Laplace operator, 8, 761

Laplacian, 8, 761, 762

Laser

- device, 372, 375, 395
- material, 371, 375, 377
- medium, 367–369, 371, 380
- organic, 372–395

Laser oscillation

- multimode, 376, 394
- single-mode, 394, 395

Lasing property, 372

Laurent's expansion, 227, 229, 236, 240, 241

Laurent's series, 223–230, 234, 236, 241

Law of conservation of charge, 282

Law of electromagnetic induction, 280

Law of equipartition of energy, 353

LACO, *see* Linear combination of atomic orbitals (LCAO)

Least action

- principle of, 984, 992, 993

Left-circularly polarized light, 138, 140, 146, 150, 301, 302, 637

Legendre differential equation, 95, 96, 450

Legendre polynomials, 59, 88, 94, 95, 101, 103, 107–109, 273

Legendre transformation, 983, 984

Leibniz rule, 94, 96, 103, 120

Leibniz rule about differentiation, 94, 96

Levi-Civita symbol, 904

Lie algebra, 618, 626, 636, 827, 830, 896–915, 921, 922, 926, 1189–1193, 1241, 1242, 1250, 1256, 1259, 1266, 1269, 1271–1304

Lie group

- linear, 897, 898, 908, 917, 921, 922, 926, 1282

- Light amplification, 367, 368
- Light amplification by stimulated emission of radiation (LASER), 367
- Light-emitting devices, 377, 378, 391, 392
- Light-emitting materials
 - organic, 372
- Light-emitting properties, 372
- Lightlike, 940
- Lightlike vector, 1184–1186
- Light propagation, 151, 277, 331, 377, 379, 381, 387
- Light quantum, 3, 4, 358, 359
- Limes superior, 226
- Limit, 208, 212–214, 218, 219, 226, 233, 267, 323, 438, 440, 613, 617, 729, 830, 866, 889, 900, 960, 1018, 1121, 1190, 1272, 1283, 1302
- Linear algebra, 518
- Linear combination
 - of atomic orbitals (LCAO), 755, 761, 768
 - of basis vectors, 545
 - of functions, 128
 - of a fundamental set of solutions, 426
- Linear combination of atomic orbitals (LCAO), 755, 761, 768
- Linearly dependent, 17, 96, 103, 109, 309, 355, 403–405, 458, 467, 469, 479, 493, 494, 496, 553, 554, 557, 558, 568, 950, 1164, 1183
- Linearly independent, 15, 17, 32, 73, 309, 353, 357, 383, 403–405, 426, 457–459, 461, 462, 470, 479, 482, 489, 493–495, 497, 500, 503–505, 516, 517, 540, 552, 554, 558, 559, 567, 569, 571, 600, 622, 640, 690, 691, 710, 714, 719, 728, 736, 741, 742, 812, 814, 840, 851, 946, 950, 952, 974, 975, 1142–1144, 1146, 1147, 1156, 1158, 1163, 1164, 1183, 1236, 1280
- Linearly polarized light, 144, 146, 295, 303, 1061
- Linear transformation
 - endomorphlic, 475, 648
 - group, 648
 - successive, 480, 482, 695
- Linear vector space
 - finite-dimensional, 457, 473, 474, 1157
 - n -dimensional, 457, 458, 461, 465, 515
- Lithography, 395
- Local properties, 920–926
- Logarithmic branch point, 264
- Logarithmic functions, 256, 1252
- Longitudinal multimode, 350, 370, 374, 394
- Loop, 306, 922, 924, 1250
- Lorentz
 - algebra, 1278–1287
 - boost, 1141, 1192, 1194–1197, 1205, 1206, 1212–1223, 1229, 1230, 1232, 1233, 1235, 1236, 1238, 1249, 1262, 1281
 - condition, 1058, 1059, 1069, 1070
 - contraction, 943, 944
 - force, 150, 637–639, 642, 643, 1075–1079
 - gauge, 1058, 1084
 - group, 942–944, 1141, 1187–1197, 1223, 1241–1258, 1263, 1264
 - invariant, 1006, 1007, 1009–1011, 1015, 1027, 1049, 1052, 1123
 - transformation, 934–940, 942, 943, 957, 971, 1187–1200, 1203, 1205, 1207, 1212, 1221, 1223, 1224, 1228, 1236, 1247, 1256, 1257, 1261, 1302, 1303
- Lowering operator, 93, 110, 114
- Lower left off-diagonal elements, 475
- Lower triangle matrix, 488, 539, 577, 632
- Lowest-unoccupied molecular orbital (LUMO), 780, 781, 797
- L – S Coupling, 879
- Luminous flux, 322
- M**
- Magnetic field, 150, 277, 279–281, 284, 285, 292–294, 307, 313–317, 320, 321, 334, 336, 338, 357, 365, 383–385, 396, 637
- Magnetic flux, 280
- Magnetic flux density, 151, 280
- Magnetic permeability, 279, 380
- Magnetic quantum number, 143, 144
- Magnetic wave, 155, 293, 295
- Magnetostatics, 279
- Magnitude relationship, 434, 720, 825
- Major axis, 301
- Mapping
 - bijjective, 926, 1157, 1158, 1168, 1284, 1285, 1287, 1301
 - bilinear, 1142–1149, 1155, 1172
 - continuous, 916, 917, 926
 - homomorphlic, 655, 656, 1265, 1272, 1285–1287
 - identity, 1264, 1301
 - injective, 653
 - inverse, 472, 1149, 1157
 - invertible, 472
 - isomorphlic, 657, 688
 - multilinear, 1171–1173
 - non-invertible, 472
 - reversible, 472

- surjective, 657, 912
 - universal bilinear, 1149
 - Mathematical induction, 49, 82, 115, 117, 488, 490, 493, 568, 582, 595, 609, 1171
 - Matrix
 - adjugate, 477
 - algebra, 43, 317, 382, 457, 468, 476, 501, 556, 853, 952, 971, 1141, 1193, 1242
 - decomposition, 515, 539, 585, 587, 725
 - equation, 531, 949
 - function, 915, 1250, 1255, 1256
 - mechanics, 11
 - nilpotent, 93, 94, 500–505, 511–521, 526, 536, 539, 577, 633
 - self-adjoint, 555
 - semi-simple, 513, 536, 539
 - singular, 579, 1161
 - skew-symmetric, 616, 618, 626, 902, 903, 907, 921, 1288, 1302
 - symmetric, 133, 592, 594–597, 631, 1160, 1170, 1197, 1247, 1248, 1251–1253, 1262, 1302
 - transposed, 405, 498, 549, 562, 1035
 - Matrix element
 - of electric dipole, 144
 - of Hamiltonian, 766, 767
 - Matter field, 1076
 - Matter wave, 6, 7
 - Maximum number, 458
 - Maxwell, J.C., 332, 333, 380, 381, 1049–1053, 1058
 - Maxwell's equations, 277–303, 332, 333, 380, 381, 1049, 1050, 1052, 1053, 1058
 - Mechanical system, 33, 124, 396–399
 - Meromorphic, 232, 238
 - Meta, 794, 795
 - Methane, 620, 622, 640
 - Method of variation of constants, 685, 755, 805–826
 - Metric
 - Euclidean, 939, 1165
 - indefinite, 476, 937, 1058, 1065–1071, 1187
 - Minkowski, 936, 937, 1052
 - space, 202, 204, 226, 227, 550, 937
 - tensor, 936, 1165, 1170, 1175–1178, 1187
 - Michelson–Morley experiment, 940
 - Microcausality, 1015
 - Minimal coupling, 1075–1078, 1085
 - Minimal substitution, 1076
 - Minkowski space
 - four-dimensional, 936, 1165
 - n -dimensional, 1177
 - Minor, 296, 397, 473, 476, 559, 597
 - Minor axis, 301
 - Mirror symmetry
 - plane of, 667, 671, 673, 675
 - Mode density, 353–358
 - Modulus, 202, 219, 242, 866, 1256
 - Molecular axis, 674, 783, 819, 823
 - Molecular long axis, 783, 804, 805
 - Molecular orbital energy, 761
 - Molecular orbitals (MOs), 717, 746, 761–826
 - Molecular science, 3, 737, 750, 755, 826, 1304
 - Molecular short axis, 804
 - Molecular states, 755
 - Molecular systems, 746, 749
 - Momentum, 3–5, 7, 28–30, 33, 57, 62, 396, 932, 958, 960, 977, 991, 1002, 1005, 1018, 1024, 1073, 1105, 1112, 1116, 1123, 1136
 - Momentum operator, 29–31, 33, 63, 79–94, 147, 1003
 - Momentum space
 - representation, 1017, 1018, 1049, 1073
 - Monoclinic, 380
 - Monomials, 841, 886, 888
 - Moving coordinate system, 695, 696, 698, 861, 923, 1193, 1229
 - Multiple roots, 258, 487, 500, 505, 543, 544, 583
 - Multiplication table, 649, 674, 675, 714, 727
 - Multiplicity, 495, 508, 509, 513, 520, 525, 586, 589, 600, 809
 - Multiply connected, 199, 229
 - Multivalued function, 256–274
- N**
- Nabla, 64, 933
 - Nano capacitor, 161
 - Nano device, 164
 - Naphthalene, 674, 684
 - Natural units, 934, 935, 991, 993, 995, 1002, 1049, 1050, 1086, 1095, 1137
 - Necessary and sufficient condition, 195, 198, 201, 405, 473, 476, 486, 554, 579, 580, 648, 650, 655, 855, 918, 1143, 1179, 1235
 - Negation, 192, 196
 - Negative-energy
 - solution, 951, 955–960, 962, 965, 1209, 1212, 1218
 - state, 948, 949, 1026, 1027, 1036, 1220, 1221, 1231
 - Negative helicity, 302

- Neumann conditions, 27, 338, 416
 - Newtonian equation, 31
 - Newton's equation of motion, 981
 - Nilpotent, 93, 94, 500–505, 511–521, 525, 526, 528, 539, 577, 633, 1292
 - Node, 337, 338, 347–349, 397, 773, 774, 817
 - Non-commutative, 3, 28, 603, 960
 - Non-commutative group, 648, 664, 772
 - Non-compact group, 1192, 1263
 - Non-degenerate, 155–158, 167, 168, 180, 600, 1163, 1178, 1180, 1181
 - Non-degenerate subspace, 1180, 1181
 - Non-equilibrium phenomenon, 368
 - Non-linear equation, 1075, 1091
 - Non-magnetic substance, 288, 295
 - Non-negative, 27, 34, 51, 70, 71, 77, 96, 202, 220, 249, 250, 406, 550, 560, 571, 592, 840, 849, 886
 - Non-negative operator, 76, 558
 - Non-relativistic
 - approximation, 7, 11, 12
 - limit, 1121
 - quantum mechanics, 959
 - Non-singular matrix, 457, 480, 481, 488, 489, 492, 502, 513, 525, 535, 540, 570, 593–596, 614, 707, 711, 896, 906, 938, 952, 1161, 1168, 1170, 1242–1249
 - Non-trivial, 403, 458, 1264
 - Non-trivial solution, 405, 421, 430, 449, 548
 - Non-vanishing, 15, 16, 93, 130, 134, 139, 149, 230, 231, 235, 282, 306, 416, 617, 681, 767, 771, 797, 812, 1044, 1133
 - Non-zero
 - coefficient, 310
 - constant, 27
 - determinant, 480
 - eigenvalue, 525, 527, 1163
 - element, 89, 90
 - vector, 70
 - Norm, 24, 55, 550, 551, 555, 557, 564–567, 578, 585, 590, 607, 610, 789, 790, 1068, 1069, 1092
 - Normalization constant, 37, 46, 53, 114, 568, 954, 957, 965
 - Normalized, 18, 24, 25, 29, 34, 35, 37, 40, 55, 70, 78, 109, 117, 123, 130, 134–136, 158, 369, 568, 753, 769, 779, 813, 885, 1026, 1031, 1045, 1079, 1209, 1211, 1290
 - Normalized eigenfunction, 37, 801
 - Normalized eigenstate, 55
 - Normalized eigenvector, 157
 - Normal operator, 578–581, 585, 604, 694, 1203, 1235
 - Normal-ordered product, 1005, 1016, 1095
 - Normal product, 1005, 1103, 1105
 - N-product, 1016, 1095–1103, 1111
 - n -th Root, 257–259
 - Nullity, 469
 - Null-space, 468, 529
 - Number density operator, 1002
 - Number operator, 41, 1003
 - Numerical vector, 459, 1146
- O**
- $O(3)$, 690, 918, 919, 921
 - $O(3, 1)$, 1187–1190, 1250–1256, 1258
 - $O(n)$, 902, 903, 906, 907, 1254
 - Oblique incidence
 - of wave, 305, 315
 - Observable, 588, 590
 - Octants, 682
 - Odd function, 51, 134, 161, 1014
 - Odd permutation, 475, 904
 - Off-diagonal elements, 475, 488, 539, 584, 664, 666, 770, 771, 788, 815, 936, 1034, 1242, 1245
 - One-dimensional
 - harmonic oscillator, 31, 59, 128, 132
 - infinite potential well, 20
 - position coordinate, 33
 - representation, 772, 787, 828
 - system, 130–134
 - One-parameter group, 896–905, 921
 - One-particle Hamiltonian, 960–964, 1031, 1077
 - One-particle wave function, 1031
 - Open ball, 227
 - Open neighborhood, 201
 - Open set, 190–193, 195, 196, 199, 201, 219
 - Operator, 3, 31, 63, 132, 155, 278, 359, 401, 482, 505, 549, 571, 636, 673, 710, 755, 827, 932, 989, 1077, 1141, 1274
 - Operator method, 33–41, 134
 - Optical device
 - high-performance, 390
 - Optical path, 339, 340
 - Optical process, 127, 358
 - Optical resonator, 372, 373
 - Optical transition, 127–152, 155, 359, 743, 746, 749, 767, 780, 781, 783, 797, 802, 803, 824–826
 - Optical waveguide, 341
 - Optics, 305, 351, 377
 - Orbital energy, 761, 772, 818, 823
 - Order of group, 675, 722
 - Ordinate axis, 131
 - Organic materials, 375
 - Orthochronous Lorentz group

- proper, 1189, 1258
- Orthogonal
 - complement, 571–574, 714, 717, 1178, 1179, 1181, 1184, 1186, 1187
 - coordinate, 202, 381, 1184, 1185
 - decomposition, 1184
 - group, 689–694, 902
 - matrix, 569, 593, 594, 598, 616, 618, 641, 664, 690, 693, 698, 762, 851, 861, 862, 902, 907, 939, 1160, 1161, 1247, 1248, 1251–1253, 1256, 1263
 - similarity transformation, 1160
 - transformation, 662, 689, 695, 696, 912, 939, 1160, 1198
- Orthonormal
 - base, 8, 1182
 - basis set, 63, 571, 576, 610, 746, 766, 877, 972
 - basis vectors, 61, 63, 662, 677, 678, 742, 756, 772, 980
 - eigenfunctions, 40
 - eigenvectors, 157, 583
 - system, 109, 155, 159, 873, 989, 1091, 1212
- Orthonormalization, 567, 580, 742, 964, 965, 1165, 1179
- Oscillating electromagnetic field, 129
- Out-coupling
 - of emission, 379, 380, 390
 - of light, 377
- Over damping, 442
- Overlap integrals, 743, 766, 779, 780, 795, 804, 813, 816, 817
- P**
 - Pair annihilation, 1115
 - Pair creation, 1115
 - Paper-and-pencil, 824
 - Para, 794, 795
 - Parallelepiped, 367, 368, 372
 - Parallelogram, 878
 - Parallel translation, 286
 - Parameter space, 861–867, 923, 924, 1192
 - Parity, 19, 51, 132
 - Partial differential equations, 277
 - Partial differentiation, 9, 10, 291, 333, 1095
 - Partial fraction decomposition
 - method of, 239, 1213
 - Partial sum, 163, 224
 - Particular solution, 173, 405
 - Path, 199, 200, 214–217, 219, 229, 339, 340, 370, 922, 1018, 1023
 - Pauli spin matrices, 839, 947, 971, 1204, 1259
 - Periodic conditions, 27, 416
 - Periodic function, 259, 914
 - Periodic modulation, 373, 374
 - Periodic wave, 287
 - Permeability, 151, 279, 316, 380, 1049
 - Permittivity, 60, 110, 278, 279, 316, 380, 395, 594
 - Permutation, 405, 475, 667, 689, 697, 703, 904, 1095, 1101, 1111
 - Perturbation
 - calculation, 1007, 1017
 - method, 36, 155–176, 183, 1075, 1078, 1102, 1103
 - theory, 176, 1075
 - Perturbed
 - state, 157, 168
 - Phase
 - change, 329, 342
 - difference, 295, 339, 340, 396
 - factor, 18, 40, 76, 79, 86, 122, 135, 143, 293, 568, 957, 959, 967, 970, 1082, 1211
 - matching, 379, 380, 386, 387, 394
 - matching conditions, 380, 394
 - refractive index, 376, 377, 384, 390
 - shift, 328, 329, 331, 341, 342
 - space, 1002
 - velocity, 7, 8, 286, 288, 289, 313, 338, 343
 - Photoabsorption, 134
 - Photoemission, 134
 - Photon
 - absorption, 134, 137, 140
 - emission, 134, 137, 140
 - energy, 368, 825, 1138
 - field, 977, 999, 1005, 1011, 1023, 1026, 1027, 1049, 1063, 1071, 1072, 1084, 1096, 1098, 1102, 1110, 1117, 1129, 1130, 1133, 1136, 1137
 - Physical laws, 940
 - Physicochemical properties, 772, 780
 - Picture
 - Heisenberg, 1087, 1089–1091
 - interaction, 1085–1091
 - Schrödinger, 1087, 1089–1091
 - π -Electron approximation, 773–805
 - Planar molecule, 677, 773, 783, 791
 - Planck, M., 3, 351, 358
 - Planck constant, 3, 4, 129
 - Planck's law of radiation, 351, 353–361
 - Plane of incidence, 311
 - Plane of mirror symmetry, 667, 671, 673
 - Plane wave
 - electromagnetic, 290
 - expansion, 1092
 - solution, 334, 931, 948–955, 960, 965, 1209, 1227
 - Plasma etching, 395

- Plastics, 295
 - Point group, 661, 664, 674, 678, 680, 681, 685, 772, 782, 806
 - Polar coordinate, 30, 47, 65, 149, 175, 202, 249, 1218
 - Polar coordinate representation, 149, 264, 867, 931, 1124, 1195, 1218
 - Polar decomposition
 - of a matrix, 593, 1212, 1242, 1245, 1247, 1302
 - Polar form, 202, 203, 245, 257
 - Polarizability, 168–170, 175
 - Polarization
 - elliptic, 277
 - left-circular, 302
 - longitudinal, 1059
 - photon, 1106, 1124–1126
 - right-circular, 302
 - scalar, 1059
 - transverse, 1059
 - vacuum, 1109, 1110
 - vector, 129, 136, 137, 140, 293, 313–316, 319, 326, 336, 343, 347, 363–365, 396, 767, 781, 824, 1059–1061
 - Pole
 - isolated, 232, 239
 - of order n , 231, 235, 1047
 - simple, 231, 235, 237, 239, 243–245, 247, 266, 267, 271, 274, 1012, 1019, 1023
 - Polyene, 805
 - Polymer, 277, 288
 - Polynomial
 - exponential, 309
 - homogeneous, 983
 - minimal, 499, 500, 502, 526, 527, 535, 540–544
 - Population inversion, 368
 - Portmanteau word, 196
 - Position coordinate, 33
 - Position operator, 29, 30, 57, 359, 366
 - Position vector, 8, 61, 129, 131, 136, 151, 169, 288, 289, 308, 361, 362, 364, 463, 564, 638, 662, 756, 761, 767, 774, 782, 802, 828, 933, 934, 959
 - Positive definite, 17, 27, 34, 36, 296, 297, 558, 559, 569, 571, 592, 593, 595, 707, 905, 937, 939, 965, 1141, 1158, 1165, 1177, 1178, 1197, 1242–1248, 1250–1252, 1256–1258, 1262, 1263, 1302
 - Positive definiteness, 296, 571, 592, 593, 1158
 - Positive-definite operator, 27, 1245
 - Positive-energy
 - solution, 952, 955, 962, 1208, 1210, 1212
 - state, 948, 949, 957, 1026, 1027, 1036, 1040, 1220, 1231
 - Positive helicity, 302
 - Positive semi-definite, 27, 558
 - Positron, 1036, 1039, 1082, 1092, 1096, 1098, 1104, 1108, 1110, 1112, 1113, 1115
 - Power series expansion, 110, 118, 120, 123, 124, 203, 204, 223, 608, 617
 - Poynting vector, 319–322, 347
 - Primitive function, 215
 - Principal axis, 381, 392, 393
 - Principal branch, 259, 263
 - Principal diagonal, 521, 523
 - Principal minor, 296, 477, 559, 595, 597
 - Principal part, 230
 - Principal submatrix, 382, 477
 - Principal value
 - of branch, 259
 - of integral, 243, 244
 - of $\ln z$, 267
 - Principle of constancy of light velocity, 934, 940, 1085
 - Probabilistic interpretation, 944, 991, 1031
 - Probability
 - density, 128, 129, 131, 142, 931–933
 - distribution, 128
 - distribution density, 131
 - Projection operator
 - sensu lato*, 738, 1232, 1235, 1238, 1240
 - sensu stricto*, 741, 1214, 1233–1236, 1238, 1239
 - Proof by contradiction, 195
 - Propagating electromagnetic wave, 305
 - Propagation constant, 337, 338, 343, 376, 377, 379
 - Propagation vector, 379
 - Propagation velocity, 7, 286
 - Proper rotation, 666, 669
 - Proposition
 - converse, 617, 854
 - P6T, 377, 378, 380, 381, 383–385, 390–392
 - Pure imaginary, 29, 326, 342, 442, 901, 944
 - Pure rotation group, 680, 685
- Q**
- QED, *see* Quantum electrodynamics (QED)
 - Q -numbers, 995, 1026
 - Quadratic equation, 180, 297, 442, 551, 800
 - Quadratic form
 - Hermitian, 54, 557, 558, 560, 592–599, 1165
 - real symmetric, 571
 - Quantization
 - canonical, 993, 1040, 1059, 1061–1065
 - field, 986, 991, 993, 995, 1005, 1023–1028, 1035, 1039, 1040, 1063, 1187

- Quantum chemical calculations, 737, 755
 - Quantum chemistry, 791, 1304
 - Quantum electrodynamics (QED), 399, 931, 971, 977, 991, 1049, 1075, 1080, 1084, 1092, 1096, 1103–1113, 1139, 1140, 1238
 - Quantum electromagnetism, 1061
 - Quantum-mechanical, 3, 21–27, 30, 59, 124, 155, 169, 605, 931, 959, 987, 1032, 1069
 - Quantum-mechanical harmonic oscillator, 31–58, 109, 111, 156, 164, 558, 1201
 - Quantum mechanics, 3, 6, 11, 12, 19, 28, 29, 31, 59, 60, 155–183, 351, 366, 401, 450, 549, 571, 699, 774, 903, 959, 977, 1075, 1080, 1304
 - Quantum numbers
 - azimuthal, 122, 149
 - magnetic, 143, 144
 - orbital angular momentum, 110, 124, 144
 - principal, 122–124
 - Quantum operator, 29, 1045
 - Quantum state, 21, 32, 54, 77, 93, 114, 123, 128, 129, 134, 144, 145, 148, 155–164, 167, 168, 171, 176, 366, 767, 878, 880, 957, 960, 967, 1026, 1086, 1294
 - Quantum theory of light, 368
 - Quartic equation, 148, 950
 - Quotient group, 653
- R**
- Radial coordinate, 202, 834, 852, 867
 - Radial wave function, 109–124
 - Radiation field, 129, 139, 140, 1065
 - Raising and lowering operators, 93
 - Raising operator, 76, 93
 - Rank, 468, 469, 505, 522, 527, 530–532, 548, 950, 955, 958, 1174, 1175, 1219, 1238
 - Rapidity, 943
 - Rational number, 203
 - Rayleigh–Jeans law, 351, 357
 - Real analysis, 186, 222
 - Real axis, 18, 37, 185, 202, 242, 250, 254, 255, 263, 267, 269
 - Real domain, 205, 222, 234, 273, 592
 - Real form, 1268, 1289, 1293, 1298–1300
 - Real function, 51, 130, 143, 144, 205, 330, 414, 420, 423, 425, 426, 778, 783, 790, 933, 1082, 1083
 - Real number
 - field, 1142, 1155, 1259
 - line, 207, 418
 - Real space, 828, 834, 837
 - Real symmetric matrix, 592–594, 597, 1160, 1197, 1251
 - Real variable, 14, 25, 205, 212, 213, 225, 237, 411, 607
 - Rearrangement theorem, 649, 707, 721, 728, 730, 739
 - Rectangular parallelepiped, 367, 368, 372
 - Recurrence relation, 86
 - Redshift, 4, 388, 392
 - Redshifted, 389–392, 394
 - Reduced mass, 60, 61, 110, 135, 761
 - Reduced Planck constant, 4, 129
 - Reducible, 497, 712, 714, 725, 726, 728, 768, 772, 775, 780, 786, 792, 797, 854, 855, 860, 873, 876, 878, 895
 - Reduction, 583, 712
 - Reflectance, 321, 328, 349
 - Reflected light, 312
 - Reflected wave, 321, 324, 336, 339, 348
 - Reflection
 - angle, 312
 - coefficient, 317, 328, 336, 338, 349, 357
 - Refraction angle, 312, 319, 326
 - Refractive index
 - of dielectric, 288, 305, 324, 325, 332, 337, 338
 - group, 371
 - of laser medium, 369, 371
 - phase, 376, 377, 384, 390, 393
 - relative, 313, 323, 329
 - Region, 151, 204, 210, 213–218, 221, 225–230, 232–234, 294, 295, 328, 330, 332, 343, 351, 363, 438, 873, 922, 1250
 - Regular hexagon, 791
 - Regular point, 212, 223
 - Regular singular point, 173
 - Regular tetrahedron, 687, 805
 - Relations between roots and coefficients, 487
 - Relative coordinates, 59–61
 - Relative permeability, 279
 - Relative permittivity, 279
 - Relative refractive index, 313, 323, 324, 329
 - Relativistic effect, 949
 - Relativistic field, 993, 1023, 1027, 1055, 1097
 - Relativistic quantum mechanics, 959
 - Relativity
 - special principle of, 940
 - Renormalization, 1110, 1140
 - Representation
 - adjoint, 905–915, 1261, 1273–1278, 1286
 - antisymmetric, 750–753, 890–892, 895, 1297
 - complex conjugate, 789, 846
 - conjugate, 651

Representation (*cont.*)

coordinate, 3, 31, 45–53, 110, 114, 124,
129, 131, 134, 139, 149, 161, 167, 170,
172, 174, 175, 179, 181, 182, 264, 418,
819, 852, 867, 931, 934, 936, 954, 1023,
1049, 1086, 1124, 1184, 1193, 1195,
1198, 1199, 1201, 1237, 1245
differential, 1264, 1271–1287, 1290, 1291
dimension of, 706, 711, 713, 729, 746, 772,
860
Dirac, 947
direct-product, 746–751, 766, 767, 797,
824, 873–876, 880, 890–892, 894
of groups, 705, 749
Heisenberg, 1087
irreducible, 712, 714, 717, 720–722,
724–726, 728, 729, 734, 735, 737, 738,
742–746, 766–768, 771–773, 775–777,
780, 782, 784, 786, 797–799, 801, 803,
806, 808, 810–812, 814, 815, 817,
823–825, 853–861, 868, 870, 871, 873,
876–878, 891, 1291–1293
matrix, 3, 31, 34, 42–45, 89, 92, 93, 464,
465, 467, 475, 477, 480–482, 495, 517,
521, 530, 531, 545, 561–564, 577, 601,
603, 666, 670, 675–679, 685, 689, 717,
721, 756, 764, 772, 785, 828, 837, 856,
891, 894, 912, 961, 1141, 1150, 1196,
1261, 1291, 1292
one-dimensional, 768, 772, 782, 787, 789,
810, 828, 1201, 1245
reducible, 712, 725, 726, 728, 768, 772,
775, 786
regular, 726–734
Schrödinger, 1087
space, 1018, 1049, 1073
subduced, 776, 786, 792
symmetric, 750–753, 760, 765, 767, 768,
771, 777, 780, 786, 797, 811, 824, 891
theory, 690, 705–753, 755, 771, 827, 849,
874, 1281, 1294, 1297
three-dimensional, 831, 849, 1290
totally symmetric, 760, 765, 767, 768, 771,
777, 780, 786, 797, 811, 824
two-dimensional, 753, 789
unitary, 705, 706, 709, 713, 714, 741, 855,
860, 1192, 1263
Representative element, 1255
Residue, 234–237, 239, 240, 244, 267, 1014,
1047, 1048
Resolvent, 607, 621–626, 628–633, 635, 636,
641, 644
Resolvent matrix, 621–626, 628–633, 635, 636,
641, 644
Resonance integral, 778, 794

Resonance structures, 714, 715
Resonant structures, 784
Resonator, 350, 372, 373, 375
Riemann sheet, 260
Riemann surface, 256–274
Right-handed system, 293, 314, 319, 397
Rigid body, 690, 755
Rodrigues formula, 88, 95, 97, 102, 118, 273
Root
complex, 243, 442, 443
double, 257, 297, 442, 501, 503, 537, 545,
591, 716, 809, 968
multiple, 258, 487, 500, 504, 505, 541, 543,
544, 548, 583
simple, 529, 545
square, 258, 297, 382, 818
triple, 257, 501, 529–531, 786
Rotation
angle, 664, 666, 667, 673, 693, 828, 861,
863–866, 868, 874–876, 919, 1204,
1222
axis, 665–668, 673, 674, 683–685, 688,
690–694, 700–703, 861, 862, 864, 868,
869, 874, 919, 924, 925, 1302
group, 680, 685, 689, 690, 827, 834–895,
920, 1141
improper, 669, 673, 690, 807, 920
matrix, 641, 664, 689–694, 849, 868, 919
proper, 666, 669
symmetry, 666
transformation, 661
Row vector, 8, 405, 459, 491, 546, 554, 566,
567, 728, 860, 936, 937, 1030, 1146,
1174
Ruled surface, 1185, 1186

S

SALC, *see* Symmetry-adapted linear
combination (SALC)
Scalar, 278, 284, 458, 459, 471, 499, 513, 552,
602, 977, 1017, 1028, 1030, 1059, 1069,
1079, 1099, 1100, 1102, 1132, 1133,
1197, 1203, 1207
Scalar field, 977, 991–1028, 1030, 1032, 1035,
1036, 1040, 1041, 1043, 1048–1051,
1058, 1061–1065, 1067, 1071, 1075,
1096, 1097, 1201
Scalar function, 286, 287, 755, 1005, 1017,
1056, 1058, 1200, 1202, 1203
Scalar potential, 1050, 1051
Scattering
Bhabha, 1111
cross-section, 1120–1124, 1136
Moller, 1111

- process, 4, 5, 176, 1108, 1111, 1119
- Schrödinger, E., 8
- Schrödinger equation
 - of motion, 60
 - time-evolved, 128
- Schur's First Lemma, 717, 720, 723, 810, 852, 869
- Schur's lemmas, 717–723, 853
- Schur's Second Lemma, 717, 719, 721, 734, 764, 854, 855, 860, 1265
- Second-order differential operator, 411–416
- Second-order extension, 1301
- Second-order linear differential equation (SOLDE), 3, 14, 15, 33, 45, 72, 73, 85, 383, 401–406, 410–412, 417, 424, 447, 448, 451, 607, 618–620, 945, 986
- Secular equation, 530, 755, 770–773, 777, 788, 794, 798, 799, 801, 812, 814, 815, 823
- Selection rule, 127–152, 746
- Self-adjoint, 411, 415, 416, 424, 432, 449, 450, 452, 555
- Self-adjoint operator, 413, 422, 424, 449
- Self-energy
 - electron, 1109, 1139
 - photon, 1109–1111, 1139
- Sellmeier's dispersion formula, 371, 375, 376
- Semicircle, 237–239, 243, 245, 246, 249, 250, 255, 1020–1022
- Semiclassical, 127
- Semiclassical theory, 150
- Semiconductor
 - crystal, 372, 373, 375, 376, 395
 - organic, 372, 373, 375, 376, 395
 - processing, 395
- Semi-infinite dielectric media, 305, 306
- Semi-simple, 512–515, 536, 539, 543, 546, 1230, 1235
- Separation axiom, 200, 210
- Separation condition, 200
- Separation of variables, 12, 69–74, 110, 127, 353, 1086
- Sesquilinear, 552
- Set
 - difference, 188
 - finite, 188, 1083, 1086
 - infinite, 188, 1083, 1086
 - of points, 232
 - theory, 185, 186, 202, 204, 274
- Signature, 1161
- Similarity transformation, 485, 487–489, 492, 502, 504, 511–513, 523, 525, 533, 535, 537, 540, 542, 556, 577, 580, 581, 583, 584, 587, 593, 603, 614, 627, 631, 635, 681, 694, 705, 707, 709, 712, 834, 851, 869, 895, 906, 919, 951, 952, 1087–1089, 1141, 1160, 1161, 1196, 1197, 1203, 1219, 1220, 1223, 1235, 1238, 1242, 1243, 1246
- Simple harmonic motion, 32
- Simply connected, 199, 200, 215–219, 221, 225, 228, 229, 234, 920–926, 1241, 1249, 1250, 1258, 1263
- Simply reducible, 876, 878, 895
- Simultaneous eigenfunction, 89
- Singleton, 201
- Single-valuedness, 212, 221, 259, 260
- Singularity
 - essential, 232
 - removable, 230, 252
- Singular part, 230, 232
- Singular point
 - isolated, 242
 - regular, 173
- Sinusoidal oscillation, 129, 131, 135
- Skew-symmetric, 616, 618, 626, 902, 903, 907, 921, 1288, 1302
- S-matrix
 - element, 1104–1108, 1111, 1115, 1116, 1120
 - expansion, 1091–1095, 1139
- Snell's law, 313, 323, 325, 326, 343
- $SO_0(3, 1)$, 1189, 1255, 1256, 1258, 1261, 1263, 1264, 1266, 1281, 1284–1287, 1298
- Solid angle, 681, 684, 702, 1124
- Solid-state chemistry, 395
- Solid-state physics, 395
- Space coordinates, 127, 934, 936, 1030
- Space group, 663
- Spacelike, 940, 1015, 1186
- Spacelike separation, 1015
- Spacelike vector, 1060, 1184, 1186
- Space-time coordinates, 934–936, 946, 987, 998, 1039, 1082, 1102
- Space-time point, 939, 1193
- Span, 459–462, 468–470, 480, 495–498, 501, 517, 529, 539, 548, 572, 600, 602, 691, 693, 764, 771, 788, 791, 808–810, 823, 828, 846, 869, 873, 874, 886, 891, 915, 1143, 1157, 1181–1184, 1186, 1187, 1288, 1289
- Special functions, 59, 109, 643
- Special linear group, 1249–1258
- Special orthogonal group
 - $SO(3)$, 661, 689–704, 827, 830, 834–874, 876, 878, 895, 896, 898, 904, 909, 911–915, 918, 919, 921, 923, 924, 926, 1241, 1265, 1266, 1276, 1286, 1287
 - $SO(n)$, 902, 903, 906, 907, 918
- Special solution, 8
- Special theory of relativity, 931, 934–944, 1256

- Special unitary group
 $SU(2)$, 827, 830, 834–880, 883, 896, 898,
 904, 909–914, 923, 924, 926, 1207,
 1241, 1242, 1249, 1250, 1258, 1262,
 1265, 1266, 1276, 1286, 1287, 1294
 $SU(n)$, 837, 902, 903, 906, 907
 Spectral decomposition, 587–589, 604, 605,
 694, 1232, 1233, 1235–1241
 Spectral lines, 129, 279, 370, 755
 Spherical basis, 834, 850
 Spherical coordinate, 60, 62, 63, 834, 852, 978
 Spherical surface harmonics, 89, 94–105, 109,
 171, 846, 869, 873, 915
 Spherical symmetry, 59, 60, 922
 Spinor
 Dirac, 957, 958, 961–963, 1201, 1202,
 1207–1212, 1214–1217, 1220, 1223,
 1227, 1236, 1237, 1256, 1257
 space, 1187
 Spin state, 123
 Spring constant, 31
 Square-integrable, 26
 Square matrix, 457, 488, 490, 495, 497, 498,
 501, 512, 513, 523, 608, 609, 614, 711,
 718, 719, 723, 830, 858, 975, 1125,
 1258
 Standard deviation, 54, 57
 Stark effect, 176
 State vector, 129, 133, 135, 157, 1086, 1087,
 1089, 1091
 Stationary current, 283
 Stationary wave, 343–350, 355, 396
 Step function, 422
 Stimulated absorption, 358, 359, 368
 Stirling's interpolation formula, 612
 Stokes' theorem, 306, 307
 Strictly positive, 17
 Structure constants, 903, 926, 1095, 1139,
 1277, 1286
 Structure–property relationship, 372
 Strum–Liouville system, 450
 Subalgebra, 1268, 1269, 1290, 1293,
 1298–1300
 Subgroups, 647, 649–652, 656, 659, 675, 677,
 685, 688–690, 732, 776, 784, 786, 791,
 897, 906, 914, 917, 918, 921, 1252,
 1255
 Submatrix, 382, 476, 477, 583, 809, 891, 892,
 951, 1213
 Subset, 186–198, 200, 201, 204, 233, 459,
 649–651, 655, 902, 915–918, 921, 1263
 Subspace, 190, 199, 200, 204, 459–463,
 468–470, 486, 495–501, 504, 505, 508,
 514, 517, 528–531, 539, 540, 544, 548,
 571–574, 583, 584, 602, 603, 649, 650,
 665, 713, 714, 717, 808, 809, 902, 1142,
 1145, 1178–1184, 1186, 1187, 1252,
 1269, 1289, 1290, 1298–1300
 Superior limit, 226
 Superposed wave, 346, 347
 Superposition
 of waves, 294–303, 336, 344–346, 348
 Surface term, 409–411, 413, 416, 419,
 424–426, 430, 432, 436–440, 446, 450,
 621
 Surjective, 472–474, 478, 653, 655, 657, 912,
 924, 1266
 Sylvester's law of inertia, 1161
 Symmetric, 59, 60, 69, 133, 175, 179, 296, 360,
 386, 422, 425, 571, 592–597, 616, 618,
 626, 627, 630, 631, 663, 664, 688, 689,
 750–753, 760, 765, 767, 768, 771, 777,
 780, 786, 797, 811, 824, 825, 835, 864,
 878, 890–892, 895, 898, 902, 903, 907,
 921, 937, 943, 995, 1051, 1137,
 1159–1161, 1170, 1192, 1197, 1221,
 1247, 1248, 1251–1253, 1256–1258,
 1262, 1288, 1296, 1297, 1302
 Symmetry
 discrete, 1256
 groups, 648, 661–704, 761, 764, 782
 operations, 661, 663–669, 671, 673–680,
 685–687, 710, 716, 717, 728, 737, 755,
 759, 763–765, 771–773, 776, 777, 784,
 787, 797, 806, 809, 810
 requirement, 179, 671, 673, 813, 815–817
 species, 685, 746, 772, 784, 785, 802, 824
 transformation, 774, 775, 786, 792, 798,
 807, 808
 Symmetry-adapted linear combination (SALC),
 755, 771–773, 777–778, 784, 787, 788,
 791, 792, 794, 798, 800, 801, 805, 810,
 811, 813–816, 891, 892
 Syn-phase, 347, 349
 System of differential equations, 607–637, 643,
 644
 System of linear equations, 476, 478
- T**
 Tangent space, 1187
 Taylor's expansion, 223, 225, 229
 Taylor's series, 99, 223–229, 231, 241, 245,
 250, 272
 T_d , 673, 680–689, 806–808, 811, 825, 920
 TE mode, *see* Transverse electric (TE) mode

Tensor

- contraction, 1177
- electric permittivity, 380
- equation, 1166
- invariant, 1175
- permittivity, 380, 395, 594
- product, 1149–1155, 1171–1173, 1175, 1176

Termwise integration, 224, 228

TE wave, *see* Transverse electric (TE) wave

θ function, 444

Thiophene, 674, 727

Thiophene/phenylene co-oligomers (TPCOs), 373, 375–377

Three-dimensional, 8, 59–61, 134–144, 168, 287, 288, 334, 354, 361, 460, 461, 477, 523, 529, 661, 664, 665, 677, 685, 689, 757, 759, 811, 828, 831, 849, 895, 958, 994, 1002, 1060, 1061, 1116, 1183–1186, 1276, 1290

Three-dimensional Cartesian coordinate, 8, 677

Time-averaged Poynting vector, 320–322

Time coordinate, 934–936, 939, 946, 948, 977, 987, 994, 998, 1030, 1039, 1082, 1102, 1185

Time dilation, 944

Timelike, 940, 1061, 1186

Timelike vector, 1060, 1084, 1186

Time-ordered product (T-product), 977, 1007, 1015–1017, 1043, 1094–1103, 1114, 1146

TM mode, *see* Transverse magnetic (TM) mode

TM wave, *see* Transverse magnetic (TM) wave

Topological group, 196, 827, 830, 926

Topological space, 185, 189–191, 194, 196, 199, 200, 204, 921, 926, 1142

Topology, 185–204

Torus, 199

Total internal reflection, 332, 338–343, 386

Totally symmetric, 760, 765, 767, 768, 771, 777, 780, 786, 797, 811, 824

Totally symmetric ground state, 797, 824

Total reflection, 305, 313, 325–331, 334, 342

TPCOs, *see* Thiophene/phenylene co-oligomers (TPCOs)

T-product, *see* Time-ordered product (T-product)

Trace, 255, 298, 301, 302, 487, 533, 592, 594, 680, 681, 694, 698, 700, 716, 720, 723–725, 729, 734, 735, 772, 774, 784–786, 792, 797, 852, 864, 869, 902–904, 906, 975, 1125, 1130–1132, 1134, 1135, 1197, 1219, 1259

Traceless, 839, 898, 902, 909, 1267

Transformation

canonical, 1089

charge-conjugation, 967, 968, 970

contravariant, 1168

coordinate, 300, 382, 661, 667, 703, 704, 764, 845, 868, 1167, 1169, 1177, 1187

discrete, 967

equivalence, 593, 594, 1161, 1162

group, 648, 737, 837, 1083

linear, 457, 463–473, 475–477, 479–482, 484, 485, 495, 497, 498, 501, 508, 511, 517, 521, 528, 533, 545, 546, 552, 558, 561–563, 565, 578, 579, 590, 648, 655, 656, 662, 695, 703, 704, 709, 710, 734, 832, 855, 907, 908, 938, 940, 1260, 1272

local phase, 1082–1085

matrix, 468, 477, 479, 480, 485, 565, 693, 836, 1188, 1198, 1207, 1248, 1256–1258, 1302

particle–antiparticle, 1039

similarity, 485, 487–489, 492, 502, 504, 511–513, 523, 525, 533, 535, 537, 540, 542, 556, 577, 580, 581, 583, 584, 587, 593, 603, 614, 627, 631, 635, 681, 694, 705, 707, 709, 712, 834, 851, 869, 895, 906, 919, 951, 952, 1087–1089, 1141, 1160, 1161, 1196, 1197, 1203, 1219, 1220, 1223, 1235, 1238, 1242, 1243, 1246

successive, 484, 695, 696, 847, 863, 1193, 1194, 1221–1223, 1304

Transition dipole moment, 129, 134

Transition electric dipole moment, 129

Transition matrix element, 133, 766, 781, 797, 802, 803

Transition probability, 129, 135, 141, 176, 359, 767, 804, 1120

Translation symmetry, 663

Transmission

coefficient, 317, 318

of electromagnetic wave, 305–350

irradiance, 322

of light, 308

Transmittance, 321

Transpose

complex conjugate, 22

Transposition, 550, 596, 621, 740, 881, 938, 1030, 1081, 1251

Transverse electric (TE) mode, 336, 338, 341

Transverse electric (TE) wave, 305, 313–319, 327, 329, 331–338, 341

Transverse magnetic (TM) mode, 341, 383, 386, 390
 Transverse magnetic (TM) wave, 305, 313–319, 321, 323–325, 327, 329, 331–338, 341
 Transverse mode, 349, 375, 377, 386, 392, 1059, 1106, 1117, 1133
 Transverse wave, 290, 291, 365
 Trial function, 178, 179
 Triangle inequality, 551
 Triangle matrix
 lower, 488, 539, 577, 632
 upper, 488, 490, 502, 512, 523, 577, 632
 Trigonometric formula, 104, 132, 320, 323, 324, 342, 388, 872, 914
 Trigonometric function, 143, 248, 665, 843, 866
 Triple integral, 992, 1065
 Triply degenerate, 817, 824
 Trivial, 21, 26, 199, 309, 403, 421, 423, 432, 449, 458, 462, 479, 486, 502, 507, 548, 579, 591, 595, 633, 634, 648, 690, 693, 767, 831, 849, 854, 892, 898, 1104, 1158, 1201, 1210, 1233, 1234, 1264, 1301
 Trivial solution, 421, 423, 432, 449, 479, 486, 548
 Trivial topology, 199
 T_1 -space, 199–202, 926
 Two-level atoms, 351, 358–361, 366, 367, 396

U

$U(1)$, 1082–1085
 Ultraviolet catastrophe, 351, 358
 Ultraviolet light, 377
 Unbounded, 25, 29, 223, 1192, 1263
 Uncertainty principle, 29, 53–58
 Uncountable, 647, 989
 Underlying set, 190
 Undetermined constant, 76, 78, 86
 Uniformly convergent, 224, 225, 227–229
 Uniqueness
 of decomposition, 513
 of Green's function, 424
 of representation, 460, 465, 467, 658, 710
 of solution, 408, 430
 Unique solution, 438, 441, 447, 472, 476, 941
 Unitarity, 709, 739, 741, 745, 766, 837, 846, 848, 852, 882, 900, 951, 972, 974, 1230
 Unitary
 diagonalization, 580–588, 592, 598
 group, 827, 837, 844, 901–903, 906

 matrix, 138, 299, 556, 559, 560, 566, 571, 580–583, 585, 590, 591, 597, 602–604, 616, 618, 627, 631, 694, 705, 707, 713, 716, 725, 772, 789, 796, 801, 809, 832, 833, 843, 849–851, 877, 881, 885, 894, 895, 898, 906, 907, 1192, 1218, 1239, 1242, 1243, 1245–1247, 1256, 1258, 1297
 operator, 571–605, 834, 881, 891, 900, 901, 974, 1087, 1221, 1230
 representation, 705–707, 709, 713, 714, 741, 855, 860, 1192, 1263
 transformation, 138, 142, 566, 590, 766, 789, 790, 801, 840, 880, 882, 883, 885, 891, 895, 1086, 1088, 1196, 1207
 Unitary similarity transformation, 577, 580, 583, 584, 587, 603, 627, 635, 694, 834, 851, 895, 906, 919, 951, 1087–1089, 1197, 1203, 1242, 1243, 1246
 Unit polarization vector
 of electric field, 129, 293, 313, 363, 364, 781, 824
 of magnetic field, 314, 365
 Unit set, 201
 Universal covering group, 926, 1241, 1242, 1249
 Universal set, 186–190
 Unknowns, 157, 293, 316, 386, 622, 624, 626
 Unperturbed
 state, 157, 162
 system, 156, 157
 Upper right off-diagonal elements, 475, 488

V

Vacuum expectation value, 1016, 1017, 1043, 1098–1100
 Variable transformation, 48, 80, 142, 241
 Variance, 53–58
 Variance operator, 53
 Variational method, 155, 176–183
 Variational principle, 36, 112
 Vector
 analysis, 284, 291, 1050
 field, 280, 281, 306, 1050, 1084, 1165
 potential, 1050, 1051, 1055, 1077–1079, 1129
 singular, 1178, 1184–1187
 transformation, 465, 466, 476, 480, 514, 661, 704, 831, 1170, 1171
 unit, 4, 131, 132, 281, 288–290, 306–308, 313–315, 336, 339, 340, 365, 460, 461, 564, 605, 638, 831

Vector space

- complex, 138, 1268, 1301
- finite-dimensional, 473
- linear, 457, 465, 469, 471, 473, 485, 544, 549, 583, 600, 609, 610, 649, 650, 653, 717, 903, 1141, 1142
- n -dimensional, 457, 458, 461, 515, 936, 1156
- theory of, 552, 1187, 1304

Venn diagram, 186, 187, 189, 191

Vertex, 688, 703, 1108, 1185, 1186

Vertical incidence, 313, 314, 323

Virtual displacement, 979, 980

Virtual work

- infinitesimal, 980

W

Wave equation, 8, 277, 287, 293, 334, 336, 353, 354, 932

Wave front, 340

Waveguide, 305, 331–343, 349, 372, 375, 377, 380, 386, 390–394

Wavelength, 4, 6, 7, 152, 312, 313, 321, 346, 348–350, 370–376, 388–395, 638

Wavelength dispersion, 371, 372, 374–376, 392, 393

Wavenumber, 4, 7, 287, 337, 338, 340, 370, 379, 1002

Wavenumber vector, 4, 5, 287, 307, 308, 311, 326, 332, 334, 368, 379, 391, 1059

Wavevector, 378, 379, 388, 391, 393

Wave zone, 363–365

Weight function, 51, 406, 413, 415, 417, 428, 432, 445, 450, 620

Wick's theorem, 1102

Wigner formula, 840–843

World interval, 939–944, 1015, 1187

Wronskian, 15, 308, 404, 405, 432, 434

X

X-ray, 4–6

Y

Yukawa potential, 109

Z

Zenithal angle, 72, 700, 861, 869, 924, 1194, 1196

Zero

element, 1283, 1284

matrix, 28, 495, 497, 501, 502, 514–516, 539, 586, 612, 613, 921, 947, 1204

vector, 24, 278, 458, 460, 486, 495, 568, 585, 650, 899, 921, 1178

Zero-point energy, 1003

Zeros, 19, 229–232, 521